Mathematical Statistics Notes

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Chapter 2

Probability

Master guide: which tool does a probability problem need?

- **Combine or translate events** ("and / or / not / at least one")? ⇒ set algebra and De Morgan (§1).
- **Compute one probability from given probabilities**, areas, or counts? ⇒ the axioms, the complement rule, inclusion–exclusion (§2).
- **"Given that…"**, a process in **stages**, or **updating** a belief after evidence? ⇒ conditional probability, probability trees, total probability, Bayes (§3).
- Events that **do not influence each other** / repeated independent trials? ⇒ independence: multiply (§4).
- **"How many ways…"** or equally likely outcomes to **count**? ⇒ counting — permutations vs. combinations (§5) — then form $\mathbb{P} = \dfrac{\text{favourable}}{\text{total}}$ (§6).

2.1 Sample Spaces and the Algebra of Events

Before any probability number can be written down, the situation must be translated into sets. An English sentence such as "at least one defective," "neither A nor B," or "exactly one of the two" becomes a precise subset of the sample space, and every later rule — the axioms, conditioning, independence, counting — operates on these sets, so accuracy here is not optional.

Which rule of this section does the problem need?

Read the English description of the event and match its *logical connective* to a set operation.

- "**and** / both / simultaneously" ⇒ **intersection** $A \cap B$.
- "**or** / at least one of" (inclusive) ⇒ **union** $A \cup B$.
- "**not** / fails / none / does not happen" ⇒ **complement** A^{c}.
- "**cannot both** happen / at most one is in common" ⇒ test **mutually exclusive**: is $A \cap B = \varnothing$?
- "**whenever** A happens, B happens / A forces B" ⇒ **subset** $A \subseteq B$ (implication).
- A **negated compound** — "not (this *or* that)," "not (this *and* that)," "**neither**," "**at least one** fails" — is the cue for **De Morgan**: push the complement inside and *flip* the connective ($\cup \leftrightarrow \cap$).
- An event built as "**this, together with either** that or the other" — one operation distributed over a second — is the cue for a **distributive law**; use it to factor or to split into disjoint pieces.
- Reordering or regrouping a chain of unions/intersections without changing its value is **commutativity / associativity**; these are silent bookkeeping, rarely the crux of a problem.

Why competing rules do not apply: the connective decides everything. "Or" is never an intersection; "and" is never a union. A bare "not A" is a single complement, *not* De Morgan — De Morgan is triggered only when the complement sits *outside a compound*. Disjointness is a yes/no property of a pair, not a way to *build* a new event, so it answers "can these co-occur?" rather than "what is the combined event?"

2.1.1 Vocabulary: experiment, sample space, outcome, event

Definition 2.1.1 (The objects of probability)**.** An **experiment** (or **trial**) is any procedure that can in principle be repeated and whose result is not known in advance. The set of *all* results that the experiment could produce is the **sample space**, written S. A single element $s \in S$ is an **outcome** (or sample point). An **event** is any subset $A \subseteq S$; we say A **occurs** on a given run when the realized outcome s lies in A. The whole space S is the **certain** event (it always occurs) and the empty set $\varnothing$ is the **impossible** event (it never occurs).

The sample space must be chosen so that the outcomes are *mutually exclusive* (one run produces exactly one outcome) and *exhaustive* (every possible result appears). Once S is fixed, an event is named either by listing its outcomes or by stating the property its outcomes share.

Example 2.1.2 (Listing a sample space and naming events: two dice)**.** Roll one red die and one green die, recording the pair (red, green). The experiment has 36 equally detailed outcomes,

$$S = \{(i, j) : i, j \in \{1, 2, 3, 4, 5, 6\}\}, \qquad |S| = 36.$$

Name the event $A =$ "the total of the two faces equals 8" as an explicit subset.

Which rule, and why? The phrase "the total equals 8" describes a *property* the outcome must have, so we are merely *selecting* the points of S that satisfy it — the vocabulary task of turning an event-description into a subset. No union/intersection of two named events is requested, so the algebra laws are not yet in play; we are still building the raw objects they act on.

Solution.

Step 1: scan the grid. The pairs (i, j) with $i + j = 8$ are exactly the diagonal shown shaded below.

r\g	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	[8]
3	4	5	6	7	[8]	9
4	5	6	7	[8]	9	10
5	6	7	[8]	9	10	11
6	7	[8]	9	10	11	12

Boxed cells [8] mark the event $A = \{i + j = 8\}$.

Step 2: list the event. $A = \{(2, 6), (3, 5), (4, 4), (5, 3), (6, 2)\}$, so $|A| = 5$.

Step 3 (count, for later use). With 36 equally likely points, A contains 5 of them; we record $|A| = 5$ out of $|S| = 36$. Final answer: A is the 5-point set listed above.

Example 2.1.3 (Sample space with unequal-size outcomes: a spinner)**.** A spinner is divided into four sectors painted red, blue, green, and yellow, occupying 40%, 30%, 20%, and 10% of the dial. Spin once and record the colour. Give S, the outcome set, and the event "a primary colour" (red, blue, or yellow).

Which rule, and why? We are asked only to *describe* the space and pick out a sub-collection of its outcomes; this is the definitional step. Note the outcomes are *not* equally likely — that is fine, the sample space is a set, not a list of equiprobable points; the unequal sizes belong to the probability function of the next section, not to the algebra here.

Solution.

Step 1: outcomes. The recorded result is a colour, so $S = \{\text{red}, \text{blue}, \text{green}, \text{yellow}\}$, $|S| = 4$.

Step 2: read the event. The primary colours among the four are red, blue, and yellow; green is a secondary colour and is excluded. Selecting the points of S with the stated property gives $A = \{\text{red}, \text{blue}, \text{yellow}\}$.

Step 3: report. The event is the 3-element subset $A = \{\text{red}, \text{blue}, \text{yellow}\} \subseteq S$. (Its probability, $0.40 + 0.30 + 0.10 = 0.80$, is deferred to the probability function; here we only fix the set.)

Example 2.1.4 (A countably infinite sample space)**.** Flip a fair coin repeatedly until the first head appears, then stop. Record the number of flips used. Describe S and the event "the experiment ends after an even number of flips."

Which rule, and why? The trigger is "until the first head": the experiment has no fixed length, so no finite grid can list its outcomes. Recognising that the natural sample space is a *set of counts* (possibly unbounded) is again the vocabulary step — choosing the right S. We are not combining two events, so no algebra law is invoked yet.

Solution.

Step 1: outcomes. The number of flips can be any positive integer, so $S = \{1, 2, 3, \dots\} = \mathbb{N}$, a countably infinite sample space.

Step 2: name the event. "Ends after an even number of flips" selects the even counts: $A = \{2, 4, 6, \dots\}$.

Step 3: report. $A = \{2k : k \in \mathbb{N}\} \subseteq S$. The space is infinite but each outcome is still a single, well-defined sample point.

Example 2.1.5 (Certain and impossible events)**.** A box holds 5 chips numbered 1 through 5; draw one chip. Identify, within this experiment, one certain event and one impossible event, and explain.

Which rule, and why? The words "certain" and "impossible" are direct cues for the two extreme events S and $\varnothing$. This is pure vocabulary: we are matching descriptions to the boundary subsets of S, not operating on events.

Solution.

Step 1: fix S. $S = \{1, 2, 3, 4, 5\}$.

Step 2: a certain event. "The drawn number is at most 5" is satisfied by *every* chip, so it equals S itself — the certain event, which always occurs.

Step 3: an impossible event. "The drawn number is 6" is satisfied by no chip, so it equals $\varnothing$ — the impossible event, which never occurs.

Conclusion. $S = \{1, \dots, 5\}$ is certain; $\{x \in S : x = 6\} = \varnothing$ is impossible.

Example 2.1.6 (Sample space of ordered outcomes: drawing two chips)**.** An urn holds four chips marked a, b, c, d. Draw two chips one after the other *without* replacement, recording the ordered pair (first, second). List S and the event $A =$ "chip a is drawn at some point."

Which rule, and why? We are fixing the raw outcome set and then *selecting* the points with a stated property — the definitional step, not an operation on two named events. The word "ordered" forces us to treat (a, b) and (b, a) as *distinct* sample points, which is a choice about how to build S, made before any algebra is applied.

Solution.

Step 1: build S. Each outcome is an ordered pair of distinct chips, so there are $4 \times 3 = 12$ of them:

$$S = \{ab, ac, ad, ba, bc, bd, ca, cb, cd, da, db, dc\}, \qquad |S| = 12,$$

writing xy for "first x, then y."

Step 2: select the event. The pairs that contain a are $A = \{ab, ac, ad, ba, ca, da\}$, so $|A| = 6$.

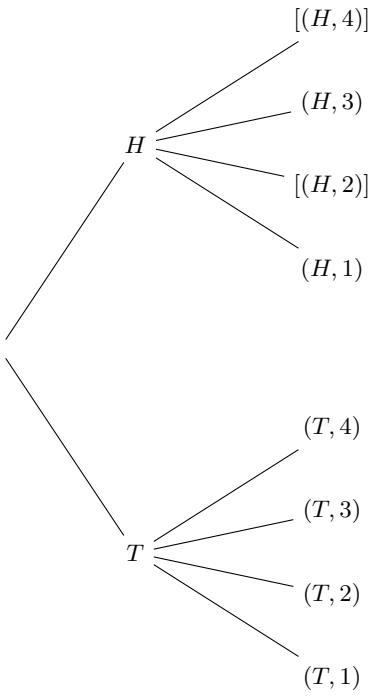
Step 3: report. A is the 6-point subset above; exactly half of the ordered draws involve chip a.

Example 2.1.7 (Product sample space: a two-stage experiment)**.** First toss a coin, then roll a four-sided die showing 1–4. Record the pair (coin, die). Give S and the event $A =$ "heads with an even die."

Which rule, and why? A two-stage experiment has a *product* sample space: one coordinate per stage, with S the set of all combinations. Listing it and then picking the points with a property is the vocabulary step. The tree below organises the enumeration; we are not yet combining two events with a set operation.

Solution.

Step 1: enumerate by a tree.



Boxed leaves mark $A =$ "heads with an even die."

Step 2: write S. $S = \{H, T\} \times \{1, 2, 3, 4\}$, so $|S| = 2 \times 4 = 8$.

Step 3: select the event. Heads with an even die keeps $(H, 2)$ and $(H, 4)$, so $A = \{(H, 2), (H, 4)\}$, the two boxed leaves.

Example 2.1.8 (Choosing the right grain for the sample space)**.** A quality auditor inspects three soldered joints in order and notes each as good (g) or bad (b). One could record the whole pattern, or only the *number* of bad joints. Give both sample spaces, and express "at least two bad joints" in each.

Which rule, and why? The same physical experiment supports several sample spaces; the modelling decision is *how finely* to record an outcome. Both choices are legitimate provided the outcomes stay mutually exclusive and exhaustive. This is the definitional layer — selecting an event as a subset of whichever S we adopt — not an algebraic combination of events.

Solution.

Step 1: the fine space. Recording each joint gives $S_1 = \{g, b\}^3$, the $2^3 = 8$ ordered patterns $ggg, ggb, \ldots, bbb$.

Step 2: the coarse space. Recording only the count of bad joints gives $S_2 = \{0, 1, 2, 3\}$.

Step 3: name the event in each. "At least two bad" is

$$A_1 = \{gbb, bgb, bbg, bbb\} \subseteq S_1 \ (|A_1| = 4), \qquad A_2 = \{2, 3\} \subseteq S_2.$$

Step 4: caution. The four patterns of S_1 are equally likely for a fair process, but the two values of A_2 are *not* — so S_1 is the safer grain when probabilities are wanted. The set A is correct in either space; only the later probability arithmetic cares which grain we picked.

2.1.2 The set operations: union, intersection, complement, disjointness, subset

Definition 2.1.9 (Operations on events)**.** Let $A, B \subseteq S$.

- **Union:** $A \cup B = \{s \in S : s \in A \text{ or } s \in B\}$, read "$A$ or B" (inclusive "or"). It occurs when at least one of A, B occurs.
- **Intersection:** $A \cap B = \{s \in S : s \in A \text{ and } s \in B\}$, read "$A$ and B." It occurs when both occur.
- **Complement:** $A^{\mathsf{c}} = \{s \in S : s \notin A\}$, read "not A." It occurs exactly when A does not.
- **Difference:** $A \setminus B = A \cap B^{\mathsf{c}}$, "$A$ but not B."

Two events are **mutually exclusive** (disjoint) when $A \cap B = \varnothing$: they cannot both occur. We say A **implies** B, written $A \subseteq B$, when every outcome in A also lies in B: whenever A occurs, B must occur too.

Proposition 2.1.10 (Basic identities tying the operations together)**.** *For all events $A, B \subseteq S$:* (i) $A \cup A^{\mathsf{c}} = S$ *and* $A \cap A^{\mathsf{c}} = \varnothing$*;* (ii) $(A^{\mathsf{c}})^{\mathsf{c}} = A$*;* (iii) *if $A \subseteq B$ then $B^{\mathsf{c}} \subseteq A^{\mathsf{c}}$;* (iv) $A \subseteq B$ iff $A \cap B = A$ iff $A \cup B = B$.

Proof. (i) Every $s \in S$ either lies in A or does not, so it lies in $A \cup A^{\mathsf{c}}$; and no s can both lie in and fail to lie in A, so $A \cap A^{\mathsf{c}}$ is empty. (ii) "Not (not A)" holds iff s fails to be outside A, i.e. iff $s \in A$. (iii) If $A \subseteq B$ and $s \in B^{\mathsf{c}}$ then $s \notin B$, hence $s \notin A$ (else $A \subseteq B$ would force $s \in B$), so $s \in A^{\mathsf{c}}$. (iv) If $A \subseteq B$, every point of A is in B, so $A \cap B = A$; conversely if $A \cap B = A$ then each $s \in A$ satisfies $s \in B$. The union form is the same statement with the roles of the smaller and larger set exchanged. □

Examples for intersection ($A \cap B$, the "and" operation).

Example 2.1.11 (Intersection by direct listing)**.** Roll the red/green dice (the 36-point space above). Let $A =$ "red die shows an even number" and $B =$ "the total is at least 10." Find $A \cap B$ and its size.

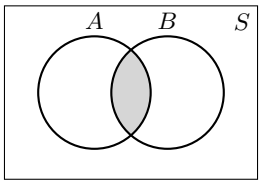
Which rule, and why? The target event is "red even *and* total ≥ 10" — the connective is "and," so we intersect. It is not a union (we need *both* conditions, not at least one) and not a complement (nothing is negated).

Solution.

Step 1: write each event. $A = \{(i, j) : i \in \{2, 4, 6\}\}$ (red even). $B = \{(i, j) : i + j \geq 10\} = \{(4, 6), (5, 5), (6, 4), (5, 6), (6, 5), (6, 6)\}$.

Step 2: keep the points common to both. From B's six points, retain those whose red value i is even: $(4, 6), (6, 4), (6, 5), (6, 6)$. The points $(5, 5)$ and $(5, 6)$ drop out because $i = 5$ is odd.

Step 3: report. $A \cap B = \{(4, 6), (6, 4), (6, 5), (6, 6)\}$, so $|A \cap B| = 4$.



$A \cap B$

Example 2.1.12 (Intersection of attribute events)**.** A factory line produces gadgets; each is classified by **colour** (silver/black) and by **grade** (pass/fail inspection). Let $A =$ "silver" and $B =$ "fail." Of 200 gadgets, 120 are silver, 30 fail, and 18 are silver-and-fail. Express "silver and failing" as a set operation and give its count.

Which rule, and why? "Silver *and* failing" joins two attributes with "and," the intersection cue. We are given $|A \cap B|$ directly, so the only task is to recognise that the requested event *is* $A \cap B$. A union would over-count (it would also include silver passes and black fails), which is not what "and" means.

Solution.

Step 1: translate. "Silver and failing" $= A \cap B$.

Step 2: read off the count. The problem states $|A \cap B| = 18$.

Step 3: report. The event "silver and failing" contains 18 of the 200 gadgets; as a set it is $A \cap B$.

Example 2.1.13 (Intersection of three events)**.** Survey of 50 students: let A, B, C be "owns a bike," "owns a laptop," "has a campus job." Suppose 10 students have all three. Identify the event "bike and laptop and job" and its count, and state in one line why it is an intersection of three sets.

Which rule, and why? Three conditions are chained with "and," so the event is the triple intersection $A \cap B \cap C$. Each additional "and" adds another $\cap$; the associativity law (next subsection) guarantees the grouping $(A \cap B) \cap C$ versus $A \cap (B \cap C)$ does not matter.

Solution.

Step 1: translate. "Bike and laptop and job" $= A \cap B \cap C$.

Step 2: count. Given: exactly 10 students lie in all three sets, so $|A \cap B \cap C| = 10$.

Step 3: report. The event is $A \cap B \cap C$ with 10 members out of 50.

Example 2.1.14 (Intersection that is empty: signalling disjointness)**.** Draw one card from a standard 52-card deck. Let $A =$ "the card is a heart" and $B =$ "the card is a club." Find $A \cap B$.

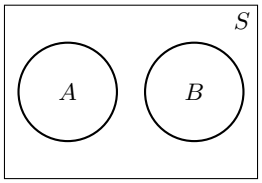
Which rule, and why? Again "heart *and* club" is an "and," so we intersect. The point of the example is that the intersection can come out *empty*: a single card has one suit, so no card is simultaneously a heart and a club. Recognising an empty intersection is exactly how we detect mutual exclusivity.

Solution.

Step 1: intersect. $A \cap B = \{\text{cards that are both a heart and a club}\}$.

Step 2: evaluate. No card carries two suits, so the set has no members: $A \cap B = \varnothing$.

Step 3: interpret. Because $A \cap B = \varnothing$, the events A and B are **mutually exclusive**; they cannot both occur on one draw.



$A \cap B = \varnothing$

Example 2.1.15 (Intersection on the dice grid: "and" as a smaller region)**.** Roll the red/green dice. Let $A =$ "the total is even" and $B =$ "the maximum of the two faces is 4." Find $A \cap B$ and its size.

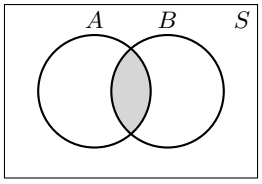
Which rule, and why? "Even total *and* maximum 4" is an "and," so we intersect: keep only the outcomes meeting *both* numerical conditions. A union would wrongly admit even-total cells whose maximum is not 4, so "or" is the wrong connective here.

Solution.

Step 1: list B. "Maximum equals 4" means both faces are ≤ 4 and at least one is 4: $B = \{(1,4),(2,4),(3,4),(4,4),(4,3),(4,2),(4,1)\}$, $|B| = 7$.

Step 2: keep the even totals. Among these, the even totals ($i + j$ even) are $(2,4),(4,4),(4,2)$ — totals $6, 8, 6$. The others have odd totals and drop out.

Step 3: report. $A \cap B = \{(2,4),(4,4),(4,2)\}$, so $|A \cap B| = 3$.



$A \cap B$

Example 2.1.16 (Intersection by complement: "and not" on a survey)**.** Of 80 commuters, 50 ride the bus (A) and 35 ride the subway (B); 20 ride *both*. How many ride the bus *but not* the subway, as a set operation and a count?

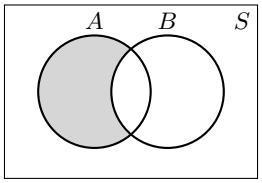
Which rule, and why? "Bus and not subway" is the intersection of A with the complement of B, i.e. $A \cap B^{\mathsf{c}} = A \setminus B$. The "and" fixes it as an intersection; the "not" attaches to the second event only, so we complement B before intersecting — this is not a bare complement of a single event, and not a union.

Solution.

Step 1: translate. "Bus but not subway" $= A \cap B^{\mathsf{c}} = A \setminus B$.

Step 2: split A. The 50 bus riders divide into those who also ride the subway ($|A \cap B| = 20$) and those who do not.

Step 3: subtract. $|A \cap B^{\mathsf{c}}| = |A| - |A \cap B| = 50 - 20 = 30$. Thirty commuters ride the bus but not the subway.



$A \setminus B = A \cap B^{\mathsf{c}}$

Example 2.1.17 (Intersection of overlapping divisibility events)**.** Draw a ticket numbered 1–60. Let $A =$ "divisible by 4" and $B =$ "divisible by 6." Find $A \cap B$, and note the common trap.

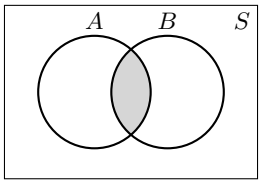
Which rule, and why? "Divisible by 4 *and* by 6" is an "and," hence an intersection. The trap is to multiply $4 \times 6 = 24$: the intersection is "divisible by the *least common multiple* 12," not by 24, because 4 and 6 share the factor 2.

Solution.

Step 1: identify the common condition. A number divisible by both 4 and 6 is divisible by $\operatorname{lcm}(4, 6) = 12$, not by 24.

Step 2: list. Multiples of 12 up to 60: $A \cap B = \{12, 24, 36, 48, 60\}$.

Step 3: count. $|A \cap B| = 5$. (Multiplying $4 \cdot 6$ would have falsely given "multiples of 24," only $\{24, 48\}$, undercounting the event.)



$A \cap B$

Examples for union ($A \cup B$, the "or" operation).

Example 2.1.18 (Union by listing)**.** Roll the two dice. Let $A =$ "doubles" (both faces equal) and $B =$ "the total is 4." Find $A \cup B$ and its size.

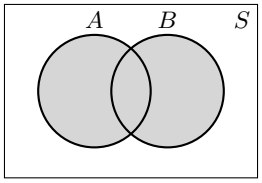
Which rule, and why? "Doubles *or* total 4" uses inclusive "or," so we form the union — every outcome in either event qualifies. We must be careful to list shared outcomes only once; that single-counting is exactly what the set union does.

Solution.

Step 1: list each. $A = \{(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)\}$. $B = \{(1,3), (2,2), (3,1)\}$.

Step 2: merge, dropping repeats. The outcome $(2,2)$ is in both. The union is $A \cup B = \{(1,1),(2,2),(3,3),(4,4),(5,5),(6,6),(1,3),(3,1)\}$.

Step 3: count. $|A| = 6$, $|B| = 3$, shared $= 1$, so $|A \cup B| = 6 + 3 - 1 = 8$, matching the 8 listed points.



$A \cup B$

Example 2.1.19 (Union of mutually exclusive events — counts simply add)**.** A bag holds tickets numbered 1–30. Let $A =$ "a multiple of 10" and $B =$ "an odd number." Find $A \cup B$ and explain why its size is just $|A| + |B|$.

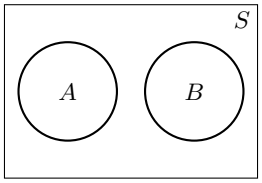
Which rule, and why? "Multiple of 10 *or* odd" is a union. Here the two events are disjoint (every multiple of 10 is even, hence never odd), so the union's size is the plain sum $|A| + |B|$ with no overlap to subtract — the special, easiest case of a union.

Solution.

Step 1: list. $A = \{10, 20, 30\}$ ($|A| = 3$); $B = \{1, 3, 5, \ldots, 29\}$ ($|B| = 15$).

Step 2: check disjointness. A multiple of 10 is even, so $A \cap B = \varnothing$.

Step 3: union. Because there is no overlap, $|A \cup B| = |A| + |B| = 3 + 15 = 18$, and $A \cup B = \{1, 3, 5, \ldots, 29, 10, 20, 30\}$.



$A \cap B = \varnothing$

Example 2.1.20 (Union phrased as "at least one")**.** Two components, a pump and a valve, are tested. Let $A =$ "the pump fails" and $B =$ "the valve fails." Express "at least one component fails" as a set operation.

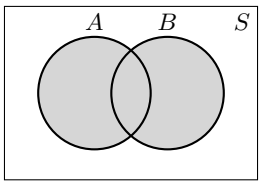
Which rule, and why? "At least one fails" is the canonical English for the inclusive union $A \cup B$: it is satisfied by a pump failure alone, a valve failure alone, or both. It is emphatically *not* the intersection (that would be "both fail"), and it is the complement of "neither fails," a link we exploit with De Morgan shortly.

Solution.

Step 1: translate. "At least one of A, B" $= A \cup B$.

Step 2: name the complement, for contrast. "At least one fails" is the opposite of "both work," i.e. $A^{\mathsf{c}} \cap B^{\mathsf{c}}$; equivalently $A \cup B = (A^{\mathsf{c}} \cap B^{\mathsf{c}})^{\mathsf{c}}$.

Step 3: report. The event is $A \cup B$.



$A \cup B$

Example 2.1.21 (Union of three events, in words)**.** A committee seat goes to whoever is nominated by department X, Y, or Z. Let A, B, C be the nomination events. Write "the candidate is nominated by at least one department" and note why a union of three sets is needed.

Which rule, and why? "By X, Y, *or* Z" and "at least one" both flag a union; three sources means a triple union $A \cup B \cup C$. The grouping is irrelevant by associativity, so we may write it without parentheses.

Solution.

Step 1: translate. Nominated by at least one $= A \cup B \cup C$.

Step 2: meaning. An outcome qualifies if it belongs to any one (or more) of A, B, C.

Step 3: report. The event is $A \cup B \cup C$.

Example 2.1.22 (Union with overlap on the dice grid)**.** Roll the two dice. Let $A =$ "the total is 7" and $B =$ "the red die shows 5." Find $A \cup B$ and its size, watching the overlap.

Which rule, and why? "Total 7 *or* red 5" is an inclusive union. Because a single cell can satisfy *both* (a red 5 that totals 7), the events overlap, so the count is $|A| + |B|$ minus the shared cell — not a plain sum, which would double-count that cell.

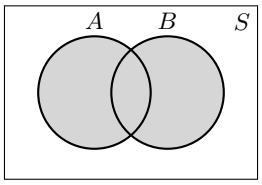
Solution.

Step 1: list each.

$$A = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}, \quad |A| = 6;$$
$$B = \{(5,1),(5,2),(5,3),(5,4),(5,5),(5,6)\}, \quad |B| = 6.$$

Step 2: find the overlap. The only cell in both is $(5,2)$ (red 5, total 7), so $|A \cap B| = 1$.

Step 3: count the union. $|A \cup B| = |A| + |B| - |A \cap B| = 6 + 6 - 1 = 11$. Subtracting the shared $(5,2)$ once is exactly what the union does.



$A \cup B$

Example 2.1.23 (Union as "or" for a single drawn card)**.** Draw one card. Let $A =$ "a spade" and $B =$ "a king." Find $|A \cup B|$, the number of cards that are a spade or a king.

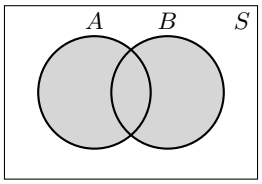
Which rule, and why? "Spade *or* king" is an inclusive union. The king of spades lies in both events, so the events are not disjoint; we must subtract that one shared card so it is counted once, the inclusion–exclusion form of a union.

Solution.

Step 1: counts. $|A| = 13$ spades; $|B| = 4$ kings.

Step 2: overlap. The king of spades is the single common card, $|A \cap B| = 1$.

Step 3: union. $|A \cup B| = 13 + 4 - 1 = 16$ cards are a spade or a king.



$A \cup B$

Example 2.1.24 (Union expressed by its complement: "at least one six")**.** Roll the two dice. Let $A =$ "the red die is 6" and $B =$ "the green die is 6." Express "at least one six" as a union and count it the easy way.

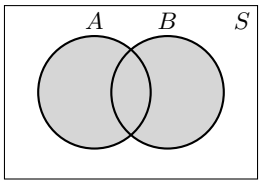
Which rule, and why? "At least one six" is the union $A \cup B$. Rather than add and subtract the overlap, we count its complement "no six on either die," a single clean condition — the "at least one" / "none" shortcut that De Morgan will formalise below. A direct intersection would describe "both sixes," the wrong event.

Solution.

Step 1: translate. "At least one six" $= A \cup B$.

Step 2: count the complement. "No six at all" means each die shows one of $\{1, \ldots, 5\}$: $5 \times 5 = 25$ cells, so $|(A \cup B)^{\mathsf{c}}| = 25$.

Step 3: subtract. $|A \cup B| = 36 - 25 = 11$. (Check by inclusion–exclusion: $|A| + |B| - |A \cap B| = 6 + 6 - 1 = 11$. Agrees.)



$A \cup B$

Examples for complement (A^{c}, the "not" operation).

Example 2.1.25 (Complement by counting)**.** Roll a single fair die, $S = \{1, 2, 3, 4, 5, 6\}$. Let $A =$ "the face is at least 3." Find A^{c} and verify $|A| + |A^{\mathsf{c}}| = |S|$.

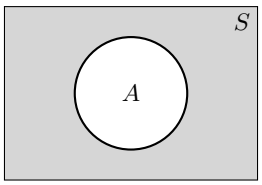
Which rule, and why? "Not (at least 3)" negates a single event, so it is a plain complement A^{c} — a lone "not," not a negated compound, so De Morgan is unnecessary. The complement is taken *relative to* S.

Solution.

Step 1: list A. $A = \{3, 4, 5, 6\}$, $|A| = 4$.

Step 2: complement. $A^{\mathsf{c}} = S \setminus A = \{1, 2\}$, i.e. "the face is at most 2."

Step 3: check. $|A| + |A^{\mathsf{c}}| = 4 + 2 = 6 = |S|$. Consistent.



A^{c}

Example 2.1.26 (Complement to simplify "at least one")**.** An inspector pulls 4 items off a line; let $A =$ "at least one item is defective." Describe A^{c} and argue it is easier to handle.

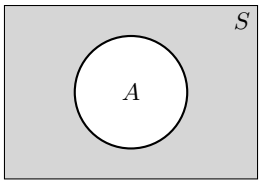
Which rule, and why? The negation cue is structural: the opposite of "at least one defective" is "*zero* defective," a single tidy condition. Whenever "at least one" appears, its complement "none" is usually simpler, so we reach for A^{c} rather than enumerate the many ways to have one, two, three, or four defectives.

Solution.

Step 1: form the complement. A^{c} = "no item is defective" = "all 4 items are good."

Step 2: why simpler. A unfolds into four disjoint cases (exactly $1, 2, 3$, or 4 defective); A^{c} is the single case "all good."

Step 3: relation. $A = (A^{\mathsf{c}})^{\mathsf{c}}$, so once A^{c} is found, A follows by one subtraction (made precise by the complement rule of the next section).



A^{c}

Example 2.1.27 (Complement of a subset event)**.** From the deck, let $A =$ "the card is a face card" (J, Q, K of any suit). Describe A^{c} and count both.

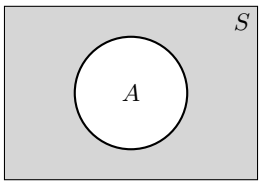
Which rule, and why? "Not a face card" is a single negation, hence one complement. Because face cards form a clean subset of the 52 cards, the complement is just "everything else."

Solution.

Step 1: count A. 3 face ranks $\times$ 4 suits = 12 face cards, $|A| = 12$.

Step 2: complement. A^{c} = "ace through ten" $= 52 - 12 = 40$ cards.

Step 3: check. $12 + 40 = 52 = |S|$. Good.



A^{c}

Example 2.1.28 (Complement on an interval sample space)**.** A pointer lands at a random position on a dial marked 0 to 100, so $S = [0, 100]$. Let $A =$ "the reading is below 25" $= [0, 25)$. Describe A^{c}.

Which rule, and why? "*Not* below 25" negates the one condition defining A, so it is a plain complement A^{c}, taken relative to the whole dial $S = [0, 100]$. There is no compound to break apart, so De Morgan plays no role; we simply take everything in S outside A.

Solution.

Step 1: fix S and A. $S = [0, 100]$ and $A = [0, 25)$ (readings strictly below 25).

Step 2: complement. $A^{\mathsf{c}} = S \setminus A = [25, 100]$, i.e. "the reading is at least 25." The boundary value 25 belongs to A^{c} because A excluded it.

Step 3: sanity check. $A \cup A^{\mathsf{c}} = [0, 25) \cup [25, 100] = [0, 100] = S$ and $A \cap A^{\mathsf{c}} = \varnothing$, exactly as the complement must satisfy.

Example 2.1.29 (Difference as "A but not B")**.** Tickets 1–20. Let $A =$ "even" and $B =$ "a multiple of 4." Find the event "even but not a multiple of 4" and give its size.

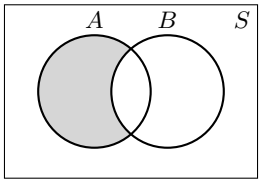
Which rule, and why? "Even *but not* a multiple of 4" combines a positive condition with a negation: "but not" is the set-difference cue, $A \setminus B = A \cap B^{\mathsf{c}}$. The complement is taken *inside* an intersection, which is what makes this a difference rather than a bare complement.

Solution.

Step 1: list. $A = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$, $|A| = 10$. $B = \{4, 8, 12, 16, 20\}$, $|B| = 5$, and $B \subseteq A$.

Step 2: subtract. $A \setminus B = A \cap B^{\mathsf{c}} = \{2, 6, 10, 14, 18\}$.

Step 3: count. $|A \setminus B| = |A| - |A \cap B| = 10 - 5 = 5$.



$A \setminus B = A \cap B^{\mathsf{c}}$

Example 2.1.30 (Complement to count "at least one repeat")**.** A three-digit code is formed by choosing each digit freely from 0–9 (so 1000 codes, including leading zeros). Let $A =$ "some digit is repeated." Use the complement to count A.

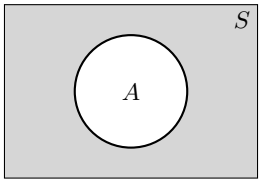
Which rule, and why? "At least one repeat" splits into awkward cases (one pair, or all three equal), so we negate it: $A^{\mathsf{c}} =$ "all three digits distinct" is one clean condition. The complement rule converts the hard count into "total minus all-distinct." This is a single "not" on one event, so no De Morgan is needed.

Solution.

Step 1: total. $|S| = 10 \times 10 \times 10 = 1000$ codes.

Step 2: count the complement. "All distinct" fills the positions in order with no reuse: $10 \times 9 \times 8 = 720$.

Step 3: subtract. $|A| = |S| - |A^{\mathsf{c}}| = 1000 - 720 = 280$ codes have a repeated digit.



A^{c}

Example 2.1.31 (Complement of "at least one woman" on a committee)**.** From 4 men and 3 women, a committee of 3 is chosen at random. Let $A =$ "at least one woman serves." Describe and count A through its complement.

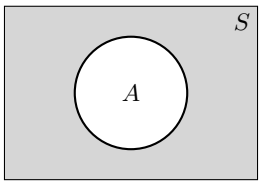
Which rule, and why? "At least one woman" is again the "at least one" cue: its complement "*no* woman, i.e. all men" is the single easy case. We count A^{c} and subtract, rather than enumerate committees with one, two, or three women.

Solution.

Step 1: total committees. $\binom{7}{3} = 35$ equally likely committees.

Step 2: complement. "All men" chooses 3 of the 4 men: $\binom{4}{3} = 4$ committees, so $|A^{\mathsf{c}}| = 4$.

Step 3: subtract. $|A| = 35 - 4 = 31$ committees include at least one woman.



A^{c}

Example 2.1.32 (Complement on a two-dimensional sample space)**.** A dart lands uniformly on the unit square $S = [0, 1] \times [0, 1]$. Let $A =$ "the dart lands in the lower-left triangle $x + y \leq 1$." Describe A^{c} and give its area.

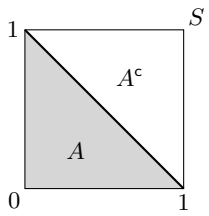
Which rule, and why? "*Not* in the triangle" is a single negation, so A^{c} is a plain complement taken relative to the whole square S. There is no compound to dismantle, so De Morgan is irrelevant; we simply take everything in S outside the triangle.

Solution.

Step 1: fix S and A. S is the unit square (area 1); A is the triangle below the line $x + y = 1$.

Step 2: complement. $A^{\mathsf{c}} = S \setminus A = \{(x, y) \in S : x + y > 1\}$, the upper-right triangle.

Step 3: area check. By symmetry the diagonal splits the square into two equal triangles, so $\text{area}(A) = \text{area}(A^{\mathsf{c}}) = \frac{1}{2}$, and the two add to $1 = \text{area}(S)$.



Shaded $A = \{x + y \leq 1\}$; the unshaded upper-right triangle is A^c.

Examples for mutual exclusivity and subset/implication.

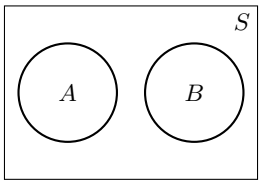
Example 2.1.33 (Detecting mutual exclusivity)**.** Draw one card. Let $A =$ "an ace" and $B =$ "a king." Are A, B mutually exclusive?

Which rule, and why? The question is "can both occur at once?" — a yes/no disjointness test, not a request to build a new event. We answer it by computing $A \cap B$ and checking whether it is empty.

Solution.

Step 1: intersect. A single card has one rank, so it cannot be both an ace and a king: $A \cap B = \varnothing$.

Step 2: conclude. Since the intersection is empty, A and B are mutually exclusive.



$A \cap B = \varnothing$

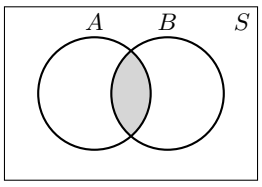
Example 2.1.34 (Not mutually exclusive — a counterexample)**.** Draw one card. Let $A =$ "a heart" and $B =$ "a queen." Decide whether they are mutually exclusive.

Which rule, and why? Same disjointness test, opposite outcome. We look for a single outcome lying in both events; finding even one proves the intersection is nonempty and the events are *not* disjoint.

Solution.

Step 1: hunt for a common point. The queen of hearts is a heart *and* a queen, so it lies in $A \cap B$.

Step 2: conclude. $A \cap B = \{\text{queen of hearts}\} \neq \varnothing$, so A, B are *not* mutually exclusive.



$A \cap B$

Example 2.1.35 (Mutual exclusivity from a partition of outcomes)**.** Draw one ball from an urn holding balls numbered 1–12. Let $A =$ "the number is at most 4," $B =$ "the number is between 5 and 8," and $D =$ "the number is at least 9." Are these three events pairwise mutually exclusive?

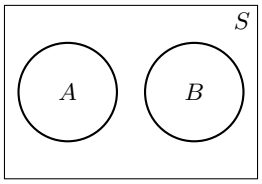
Which rule, and why? The question asks whether any two can occur together, so it is a pairwise disjointness test: form each pairwise intersection and check for emptiness. Because the three ranges of numbers do not overlap, we expect every pairwise intersection to be empty.

Solution.

Step 1: list the events. $A = \{1, 2, 3, 4\}$, $B = \{5, 6, 7, 8\}$, $D = \{9, 10, 11, 12\}$.

Step 2: intersect each pair. $A \cap B = \varnothing$ (no number is both ≤ 4 and in 5–8); $A \cap D = \varnothing$; $B \cap D = \varnothing$.

Step 3: conclude. All three pairwise intersections are empty, so A, B, D are pairwise mutually exclusive. (They also exhaust S, so they form a partition — a fact the later addition rule will exploit.)



$A \cap B = \varnothing$

Example 2.1.36 (Subset / implication)**.** Roll one die. Let $A =$ "the face is 6" and $B =$ "the face is even." Show $A \subseteq B$ and interpret.

Which rule, and why? "Whenever a 6 comes up, the face is even" is an implication: *A forces B*. The cue "whenever A then B" is the subset relation $A \subseteq B$, established by checking that every outcome of A lies in B.

Solution.

Step 1: list. $A = \{6\}$, $B = \{2, 4, 6\}$.

Step 2: check containment. The only element of A, namely 6, is in B; hence $A \subseteq B$.

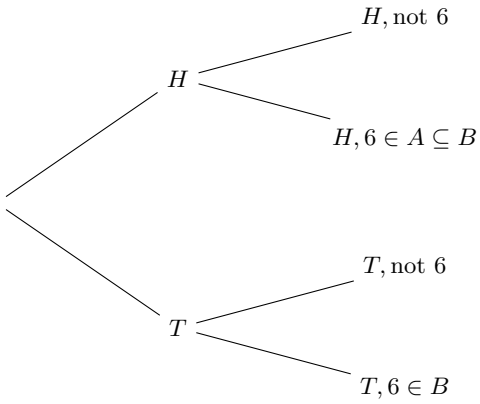
Step 3: interpret. A implies B: rolling a 6 guarantees an even face. By the subset identities above, this also gives $A \cap B = A = \{6\}$ and $A \cup B = B = \{2, 4, 6\}$.

Example 2.1.37 (Chaining implications across two trials)**.** Flip a coin and then roll a die. Let $A =$ "heads then a 6" and $B =$ "the die shows 6." Is $A \subseteq B$? Use a tree to see the structure.

Which rule, and why? A specifies *more* than B (it also pins down the coin), so every outcome in A already satisfies the looser condition B — the "more specific implies more general" pattern, i.e. $A \subseteq B$. It is not equality, since B also contains "tails then 6."

Solution.

Step 1: enumerate via a tree.



Step 2: identify the sets. $A = \{(H, 6)\}$; $B = \{(H, 6), (T, 6)\}$.

Step 3: conclude. Every point of A (just $(H, 6)$) lies in B, so $A \subseteq B$; the containment is strict because $(T, 6) \in B \setminus A$.

Example 2.1.38 (Implication between numerical conditions)**.** Draw a ticket numbered 1–30. Let $A =$ "the number is a multiple of 12" and $B =$ "the number is even." Show that $A \subseteq B$ and use the subset identities to read off $A \cap B$ and $A \cup B$.

Which rule, and why? "Every multiple of 12 is even" is a logical implication $A \Rightarrow B$, i.e. the subset relation $A \subseteq B$. The cue is that one condition (divisible by 12) is *stronger* than the other (even) and forces it. We confirm containment elementwise, not by intersecting to build a new event.

Solution.

Step 1: list. $A = \{12, 24\}$ (multiples of 12 up to 30); $B = \{2, 4, 6, \ldots, 30\}$, the 15 even numbers.

Step 2: check containment. Both 12 and 24 are even, hence in B; so every element of A lies in B and $A \subseteq B$.

Step 3: apply the subset identities. From Proposition (the basic identities), $A \subseteq B$ gives $A \cap B = A = \{12, 24\}$ and $A \cup B = B = \{2, 4, \ldots, 30\}$. The smaller set contributes nothing new to the union and nothing is lost in the intersection.

Example 2.1.39 (Disjointness from a partition of the dice grid by total)**.** Roll the two dice. Let $A =$ "the total is at most 4," $B =$ "the total is 7," and $D =$ "the total is at least 10." Are A, B, D pairwise mutually exclusive?

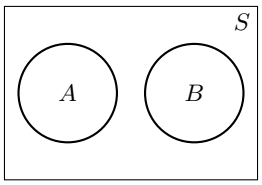
Which rule, and why? The question "can any two occur together?" is a pairwise disjointness test: intersect each pair and check for emptiness. A single roll has *one* total, and the three ranges ≤ 4, $= 7$, ≥ 10 do not overlap, so every pairwise intersection should be empty.

Solution.

Step 1: note the totals are exclusive. A roll yields exactly one total, so it can satisfy at most one of the three disjoint ranges.

Step 2: check the pairs. $A \cap B = \varnothing$ (a total cannot be both ≤ 4 and $= 7$); $A \cap D = \varnothing$; $B \cap D = \varnothing$ (7 is not ≥ 10).

Step 3: conclude. All three pairwise intersections are empty, so A, B, D are pairwise mutually exclusive. (They are *not* exhaustive — totals $5, 6, 8, 9$ lie outside all three — so they do not form a partition of S.)



$A \cap B = \varnothing$

Example 2.1.40 (A subtle non-exclusivity: "not obviously overlapping" events)**.** Draw a ticket numbered 1–50. Let $A =$ "a perfect square" and $B =$ "a multiple of 4." Are A, B mutually exclusive?

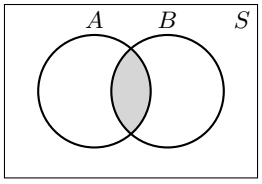
Which rule, and why? It is tempting to call squares and multiples of 4 unrelated, but mutual exclusivity is decided only by $A \cap B$, not by intuition. We hunt for one common point; a single hit, such as 4 or 16, proves the intersection is nonempty and the events are *not* disjoint.

Solution.

Step 1: list. $A = \{1, 4, 9, 16, 25, 36, 49\}$; $B = \{4, 8, 12, \ldots, 48\}$.

Step 2: intersect. The numbers in both are $4, 16, 36$ (squares that are also multiples of 4), so $A \cap B = \{4, 16, 36\} \neq \varnothing$.

Step 3: conclude. Since $A \cap B$ is nonempty, A and B are *not* mutually exclusive — a reminder that overlap must be *checked*, never assumed.



$A \cap B$

Example 2.1.41 (Subset versus equality: a strict implication)**.** Roll the two dice. Let $A =$ "both faces equal 3" and $B =$ "the total is 6." Decide whether $A \subseteq B$, and whether the containment is strict.

Which rule, and why? "Whenever both faces are 3, the total is 6" is an implication $A \Rightarrow B$, hence the subset relation $A \subseteq B$, checked elementwise. To test whether it is *strict* (proper), we look for a point of B outside A; finding one shows $A \neq B$.

Solution.

Step 1: list. $A = \{(3,3)\}$; $B = \{(1,5),(2,4),(3,3),(4,2),(5,1)\}$.

Step 2: containment. The single point $(3,3)$ of A lies in B, so $A \subseteq B$.

Step 3: strictness. B also contains $(1,5)$, which is not in A, so $A \subsetneq B$ is *proper*. The subset identities then give $A \cap B = A = \{(3,3)\}$ and $A \cup B = B$.

2.1.3 De Morgan's laws

Theorem 2.1.42 (De Morgan's laws)**.** *For any events $A, B \subseteq S$,*

$$(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}} \qquad \text{and} \qquad (A \cap B)^{\mathsf{c}} = A^{\mathsf{c}} \cup B^{\mathsf{c}}.$$

More generally, for any (finite or countable) family $\{A_i\}$, $\left(\bigcup_i A_i\right)^{\mathsf{c}} = \bigcap_i A_i^{\mathsf{c}}$ and $\left(\bigcap_i A_i\right)^{\mathsf{c}} = \bigcup_i A_i^{\mathsf{c}}$: complementing a union/intersection flips it to an intersection/union of complements.

Proof. We prove the first law by showing each side contains the other.

($\subseteq$). Take $s \in (A \cup B)^{\mathsf{c}}$. Then $s \notin A \cup B$, so s is in neither A nor B. Hence $s \in A^{\mathsf{c}}$ and $s \in B^{\mathsf{c}}$, i.e. $s \in A^{\mathsf{c}} \cap B^{\mathsf{c}}$.

($\supseteq$). Take $s \in A^{\mathsf{c}} \cap B^{\mathsf{c}}$. Then $s \notin A$ and $s \notin B$, so s belongs to no part of $A \cup B$, giving $s \in (A \cup B)^{\mathsf{c}}$. The two inclusions give equality. The second law follows by applying the first to $A^{\mathsf{c}}, B^{\mathsf{c}}$ in place of A, B and using $(X^{\mathsf{c}})^{\mathsf{c}} = X$: $A^{\mathsf{c}} \cap B^{\mathsf{c}} = (A \cup B)^{\mathsf{c}}$ complemented on both sides yields $(A^{\mathsf{c}} \cap B^{\mathsf{c}})^{\mathsf{c}} = A \cup B$, then rename. The general family version is the same element-chase: "not in any A_i" $\iff$ "in every A_i^{c}." □

Remark. De Morgan is the workhorse for "at least one" / "none" conversions: $(A_1 \cup \cdots \cup A_n)^{\mathsf{c}} = A_1^{\mathsf{c}} \cap \cdots \cap A_n^{\mathsf{c}}$ says "*none* of the events happen" is the same as "*not at least one* happens." This is why computing "no defective" is the standard route to "at least one defective."

Example 2.1.43 ("Neither" is the complement of a union)**.** A traveler's trip succeeds unless a flight is delayed or luggage is lost. Let $A =$ "flight delayed" and $B =$ "luggage lost." Write "neither problem occurs" and simplify.

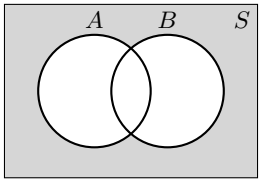
Which rule, and why? "Neither A nor B" is a *negated union*: "not (delayed or lost)." A complement sitting outside a union is the exact trigger for the first De Morgan law, turning it into "not delayed and not lost."

Solution.

Step 1: translate. "Neither problem" $= (A \cup B)^{\mathsf{c}}$.

Step 2: apply De Morgan. $(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}$.

Step 3: read back. The event is "flight on time *and* luggage arrives," $A^{\mathsf{c}} \cap B^{\mathsf{c}}$ — a clean conjunction of two "good" events.



$(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}$

Example 2.1.44 ("Not both" is the complement of an intersection)**.** Two switches must each be closed for a lamp to light. Let $A, B =$ "switch 1 closed," "switch 2 closed." Write "the lamp is off" and simplify.

Which rule, and why? The lamp lights only when *both* switches are closed, $A \cap B$; "off" negates that, "not (both closed)," a *negated intersection*. That structure is the second De Morgan law, which converts it into "at least one switch is open."

Solution.

Step 1: translate. Lamp on $= A \cap B$; lamp off $= (A \cap B)^{\mathsf{c}}$.

Step 2: apply De Morgan. $(A \cap B)^{\mathsf{c}} = A^{\mathsf{c}} \cup B^{\mathsf{c}}$.

Step 3: read back. The lamp is off iff "switch 1 open *or* switch 2 open" — at least one switch open. Note the "and" became "or."

Example 2.1.45 (De Morgan with three events, by counting)**.** Of 60 tourists, 25 visited the museum (A), 30 the gallery (B), 20 the castle (C); the count visiting at least one site is 45. How many visited *none* of the three?

Which rule, and why? "None of the three" is $(A \cup B \cup C)^{\mathsf{c}}$, the complement of a triple union; the extended De Morgan law rewrites it as $A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}$ and, just as importantly, lets us get its count as "total minus at-least-one."

Solution.

Step 1: translate. Visited none $= (A \cup B \cup C)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}$.

Step 2: use the count of the union. Given $|A \cup B \cup C| = 45$ out of 60.

Step 3: subtract. $|(A \cup B \cup C)^{\mathsf{c}}| = 60 - 45 = 15$. Fifteen tourists visited no site.

Example 2.1.46 (Verifying De Morgan numerically on the dice grid)**.** Roll the two dice. Let $A =$ "red is 1" and $B =$ "green is 1." Check directly that $(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}$ by counting outcomes.

Which rule, and why? We are asked to *confirm* a negated-union identity, so De Morgan is the rule under test. The check is a count of the grid: build both sides and see the sizes agree.

Solution.

Step 1: left side. $A \cup B =$ "red is 1 or green is 1" = first row plus first column of the grid. Each has 6 cells, sharing $(1, 1)$, so $|A \cup B| = 6 + 6 - 1 = 11$; thus $|(A \cup B)^{\mathsf{c}}| = 36 - 11 = 25$.

Step 2: right side. $A^{\mathsf{c}} =$ "red $\neq 1$" (30 cells), $B^{\mathsf{c}} =$ "green $\neq 1$" (30 cells); their intersection is "red $\neq 1$ and green $\neq 1$," the 5×5 block, of size 25.

Step 3: compare. Both sides have 25 outcomes (the 5×5 block where neither die is 1), confirming $(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}$.

r\g	1	2	3	4	5	6
1	•	•	•	•	•	•
2	•	○	○	○	○	○
3	•	○	○	○	○	○
4	•	○	○	○	○	○
5	•	○	○	○	○	○
6	•	○	○	○	○	○

Dots • $= A \cup B$ (red or green is 1); circles ○ form the 5×5 block $A^{\mathsf{c}} \cap B^{\mathsf{c}}$.

Example 2.1.47 ("At least one failure" via its complement)**.** A machine runs only if *both* of its two belts hold. Let $A =$ "belt 1 breaks" and $B =$ "belt 2 breaks." Rewrite "the machine keeps running" using De Morgan, and rewrite "at least one belt breaks" as well.

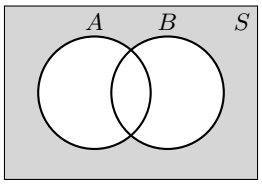
Which rule, and why? "Keeps running" is "*neither* belt breaks" $= (A \cup B)^{\mathsf{c}}$: a complement outside a union, the first De Morgan law, which flips it to $A^{\mathsf{c}} \cap B^{\mathsf{c}}$ ("both hold"). "At least one breaks" is the complement of that, the union $A \cup B$ — the same identity read in reverse.

Solution.

Step 1: translate "running." Running = neither breaks $= (A \cup B)^{\mathsf{c}}$.

Step 2: apply De Morgan. $(A \cup B)^c = A^c \cap B^c$ — "belt 1 holds *and* belt 2 holds," a conjunction of the two "good" events.

Step 3: the opposite event. "At least one breaks" is the complement of "running": $(A^c \cap B^c)^c = A \cup B$. The "and" of holds became the "or" of breaks.



$(A \cup B)^c = A^c \cap B^c$

Example 2.1.48 (De Morgan turning a negated "and" into an "or")**.** A loan is approved only if the applicant passes the credit check (A) *and* the income check (B). Describe "the loan is denied" and simplify with De Morgan.

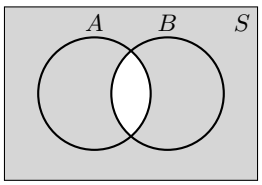
Which rule, and why? Approval is $A \cap B$; "denied" negates it, $(A \cap B)^c$ — a complement outside an intersection. That is the second De Morgan law, converting it to $A^c \cup B^c$: the "and" becomes "or," so denial means failing *at least one* check, not necessarily both.

Solution.

Step 1: translate. Approved $= A \cap B$; denied $= (A \cap B)^c$.

Step 2: apply De Morgan. $(A \cap B)^c = A^c \cup B^c$.

Step 3: read back. Denied iff "credit fails *or* income fails" — a single failed check is enough. The common error of reading denial as "both fail" ($A^c \cap B^c$) describes a strictly smaller event.



$(A \cap B)^c = A^c \cup B^c$

Shaded everything but the lens: denial is failing *at least one* check.

Example 2.1.49 (De Morgan with three events, verified by counting)**.** Roll the two dice. Let $A =$ "red ≤ 2," $B =$ "green ≤ 2." Check by counting that $(A \cup B)^c = A^c \cap B^c$, and read off "both dice are at least 3."

Which rule, and why? We are testing a negated-union identity, so De Morgan is the rule under examination. "Neither A nor B" $= (A \cup B)^c$ should equal "red ≥ 3 and green ≥ 3" $= A^c \cap B^c$; we confirm the two counts match on the grid.

Solution.

Step 1: left side. $A \cup B =$ "red ≤ 2 or green ≤ 2." Here $|A| = 2 \times 6 = 12$, $|B| = 6 \times 2 = 12$, and $|A \cap B| = 2 \times 2 = 4$, so $|A \cup B| = 12 + 12 - 4 = 20$; thus $|(A \cup B)^c| = 36 - 20 = 16$.

Step 2: right side. $A^c =$ "red ≥ 3" and $B^c =$ "green ≥ 3"; their intersection is the 4×4 block of cells with both faces in $\{3, 4, 5, 6\}$, of size 16.

Step 3: compare. Both sides have 16 outcomes — exactly the 4×4 block where each die is at least 3 — confirming $(A \cup B)^c = A^c \cap B^c$.

2.1.4 The distributive laws (with commutative and associative laws)

Theorem 2.1.50 (Distributive laws)**.** *For all events $A, B, C \subseteq S$,*

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C), \qquad A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

Proof. For the first law, chase an arbitrary outcome s:

$$s \in A \cap (B \cup C) \iff s \in A \text{ and } (s \in B \text{ or } s \in C) \iff (s \in A \text{ and } s \in B) \text{ or } (s \in A \text{ and } s \in C),$$

where the middle step is the logical distributivity of "and" over "or." The right side says $s \in (A \cap B) \cup (A \cap C)$, so the two events have identical members. The second law is the dual: replace "and"↔"or" throughout, which is legitimate because the same logical distributivity holds for "or" over "and." □

Remark (Commutative and associative laws). Order and grouping never affect a union or an intersection:

$$A \cup B = B \cup A, \quad A \cap B = B \cap A \quad \text{(commutativity)},$$

$$(A \cup B) \cup C = A \cup (B \cup C), \quad (A \cap B) \cap C = A \cap (B \cap C) \quad \text{(associativity)}.$$

Both follow instantly because "or" and "and" in logic are themselves commutative and associative. These laws justify dropping parentheses in chains such as $A \cup B \cup C$ and reordering events freely; they are routine bookkeeping rather than problem-solving rules.

Example 2.1.51 (Distribute "and" over "or")**.** Draw one card. Let $A =$ "red" (heart or diamond), $B =$ "a face card," $C =$ "an ace." Show "red, and (a face card or an ace)" equals "(red face card) or (red ace)," and count.

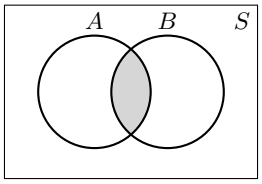
Which rule, and why? The event is one condition ("red") applied to a bracketed "or" ("face or ace"): $A \cap (B \cup C)$. That single-times-bracketed-sum shape is the distributive cue; we expand it to $(A \cap B) \cup (A \cap C)$ to count the two simpler pieces.

Solution.

Step 1: distribute. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Step 2: count each piece. $A \cap B =$ red face cards $= 2$ suits $\times 3 = 6$. $A \cap C =$ red aces $= 2$. They are disjoint (an ace is not a face card).

Step 3: combine. $|(A \cap B) \cup (A \cap C)| = 6 + 2 = 8$ red cards that are a face card or an ace.



$A \cap B$

Example 2.1.52 (Distribute "or" over "and")**.** Tickets 1–24. Let $A =$ "a multiple of 6," $B =$ "even," $C =$ "a multiple of 3." Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ by listing.

Which rule, and why? The shape is "A or (B and C)," a union wrapped around a bracketed "and": $A \cup (B \cap C)$. This is the second distributive law; we expand and confirm both sides list the same tickets.

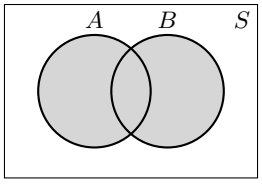
Solution.

Step 1: inner pieces. $A = \{6, 12, 18, 24\}$; $B \cap C =$ "even and multiple of 3" = "multiple of 6" $= \{6, 12, 18, 24\}$.

Step 2: left side. $A \cup (B \cap C) = \{6, 12, 18, 24\}$.

Step 3: right side. $A \cup B =$ all evens $\cup \{6, 12, 18, 24\} =$ all evens $= \{2, 4, \ldots, 24\}$. $A \cup C =$ multiples of $3 = \{3, 6, 9, 12, 15, 18, 21, 24\}$. Their intersection is the numbers that are even *and* a multiple of 3 $= \{6, 12, 18, 24\}$.

Step 4: compare. Both sides equal $\{6, 12, 18, 24\}$. Verified.



$A \cup B$

Example 2.1.53 (Distribution to split into disjoint pieces)**.** Roll the two dice. Let $A =$ "red is even," $B =$ "total is 5," $C =$ "total is 9." Compute $|A \cap (B \cup C)|$ via distribution.

Which rule, and why? We want "red even, and (total 5 or total 9)" $= A \cap (B \cup C)$. Distributing turns one messy intersection into two clean ones, $A \cap B$ and $A \cap C$, which here are disjoint (a total cannot be both 5 and 9), so their counts simply add.

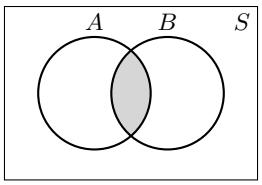
Solution.

Step 1: distribute. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Step 2: list $A \cap B$. Total 5: $(1,4),(2,3),(3,2),(4,1)$; keep red even: $(2,3),(4,1)$, so $|A \cap B| = 2$.

Step 3: list $A \cap C$. Total 9: $(3,6),(4,5),(5,4),(6,3)$; keep red even: $(4,5),(6,3)$, so $|A \cap C| = 2$.

Step 4: add (disjoint). $|A \cap (B \cup C)| = 2 + 2 = 4$.



$A \cap B$

Example 2.1.54 (Distribution behind inclusion of a common factor)**.** Three relays A, B, C feed a circuit. Express "relay A works, and at least one of B, C works," then factor the common A.

Which rule, and why? "A works *and* (at least one of B, C)" is $A \cap (B \cup C)$: a shared condition A multiplying a union. Recognising the common factor A lets us either expand (to count cases) or keep it factored (to read the structure); both are the distributive law.

Solution.

Step 1: translate. Event $= A \cap (B \cup C)$.

Step 2: distribute. $= (A \cap B) \cup (A \cap C)$ — "(A and B) or (A and C)."

Step 3: interpret. Either relays A, B both work, or relays A, C both work (possibly both pairs). The factored form $A \cap (B \cup C)$ and the expanded form are the same event, by the distributive law.

Example 2.1.55 (Commutativity and associativity in a regrouping)**.** Simplify $\big[(A \cup B) \cup C\big] \cap \big[C \cup (B \cup A)\big]$.

Which rule, and why? Nothing is negated and no single factor multiplies a bracketed sum in an essential way; the only moves available are *reordering* ($A \cup B = B \cup A$) and *regrouping* ($(A \cup B) \cup C = A \cup (B \cup C)$). Spotting that both brackets are the same set after reordering is the commutativity/associativity cue.

Solution.

Step 1: flatten each bracket. By associativity, $(A \cup B) \cup C = A \cup B \cup C$; by commutativity, $C \cup (B \cup A) = A \cup B \cup C$ as well.

Step 2: both brackets coincide. The expression is $(A \cup B \cup C) \cap (A \cup B \cup C)$.

Step 3: idempotence. A set intersected with itself is itself, so the value is $A \cup B \cup C$.

Example 2.1.56 (Distribute "and" over "or" on the dice grid)**.** Roll the two dice. Let $A =$ "red is 3," $B =$ "the total is 5," $C =$ "the total is 9." Compute $|A \cap (B \cup C)|$ by distributing.

Which rule, and why? The event "red 3, and (total 5 or total 9)" is $A \cap (B \cup C)$: one condition multiplying a bracketed "or." Distributing gives $(A \cap B) \cup (A \cap C)$, two clean intersections; since a total cannot be both 5 and 9 the pieces are disjoint and their counts add.

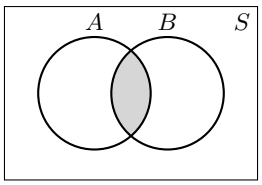
Solution.

Step 1: distribute. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Step 2: count $A \cap B$. With red $= 3$, total 5 forces green $= 2$: just $(3,2)$, so $|A \cap B| = 1$.

Step 3: count $A \cap C$. With red $= 3$, total 9 forces green $= 6$: just $(3,6)$, so $|A \cap C| = 1$.

Step 4: add (disjoint). $|A \cap (B \cup C)| = 1 + 1 = 2$, namely $\{(3,2),(3,6)\}$.



$A \cap B$

Example 2.1.57 (Distribute "or" over "and" and verify by listing)**.** Tickets 1–36. Let $A =$ "a multiple of 9," $B =$ "a multiple of 3," $C =$ "even." Verify $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ by listing both sides.

Which rule, and why? The shape "A or (B and C)" is a union wrapped around a bracketed "and," $A \cup (B \cap C)$ — the second distributive law. We expand and confirm both descriptions produce the same ticket list, which is the law's content.

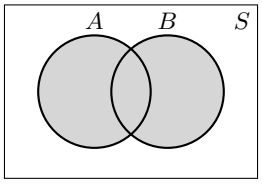
Solution.

Step 1: inner pieces. $A = \{9, 18, 27, 36\}$; $B \cap C =$ "multiple of 3 and even" = "multiple of 6" = $\{6, 12, 18, 24, 30, 36\}$.

Step 2: left side. $A \cup (B \cap C) = \{6, 9, 12, 18, 24, 27, 30, 36\}$.

Step 3: right side. $A \cup B =$ multiples of $3 = \{3, 6, 9, \ldots, 36\}$; $A \cup C =$ "multiple of 9 or even." Their intersection keeps the numbers that are a multiple of 3 *and* (a multiple of 9 or even), which works out to $\{6, 9, 12, 18, 24, 27, 30, 36\}$.

Step 4: compare. Both sides are $\{6, 9, 12, 18, 24, 27, 30, 36\}$. Verified.



$A \cup B$

Example 2.1.58 (Distribution to factor a shared condition)**.** Draw one card. Let $A =$ "a red card," $B =$ "a face card," $C =$ "a ten." Write "red, and (a face card or a ten)," then both distribute it and count.

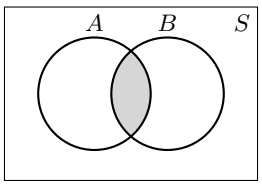
Which rule, and why? "Red *and* (face or ten)" is $A \cap (B \cup C)$: the shared condition red multiplies a union. Distributing to $(A \cap B) \cup (A \cap C)$ turns it into two easy counts; the pieces are disjoint because a ten is not a face card.

Solution.

Step 1: distribute. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Step 2: count each piece. $A \cap B =$ red face cards $= 2$ red suits $\times\, 3 = 6$. $A \cap C =$ red tens $= 2$.

Step 3: add (disjoint). A card is never both a ten and a face card, so $|A \cap (B \cup C)| = 6 + 2 = 8$ red cards that are a face card or a ten.



$A \cap B$

2.1.5 One worked synthesis

Example 2.1.59 (Putting the algebra together)**.** Roll the two dice. Let $A =$ "red is even," $B =$ "green is odd." Use the algebra to describe and count "exactly one of A, B occurs."

Which rule, and why? "Exactly one" is neither a plain union nor intersection; it is "(A and not B) or (not A and B)," $(A \cap B^{\mathsf{c}}) \cup (A^{\mathsf{c}} \cap B)$ — complements inside an "or." The two pieces are disjoint, so once each is counted we simply add.

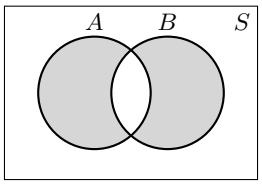
Solution.

Step 1: counts of the base events. A (red even): $3 \times 6 = 18$. B (green odd): $6 \times 3 = 18$. $A \cap B$ (red even and green odd): $3 \times 3 = 9$.

Step 2: the two disjoint pieces. $A \cap B^{\mathsf{c}}$ (red even, green even) $= 3 \times 3 = 9$; $A^{\mathsf{c}} \cap B$ (red odd, green odd) $= 3 \times 3 = 9$.

Step 3: add. "Exactly one" $= (A \cap B^{\mathsf{c}}) \cup (A^{\mathsf{c}} \cap B)$ has $9 + 9 = 18$ outcomes out of 36.

Cross-check. "At least one" $= |A \cup B| = 18 + 18 - 9 = 27$; subtract "both" $|A \cap B| = 9$ to get exactly one $= 27 - 9 = 18$. Agrees.



$(A \setminus B) \cup (B \setminus A)$

2.2 The Probability Function: Axioms and Their Consequences

A **probability function** $\mathbb{P}$ attaches to each event $A \subseteq S$ a real number $\mathbb{P}(A)$ measuring how likely A is to occur. We do not say where these numbers come from — relative frequencies, symmetry, or subjective judgement all qualify — we only demand that they obey three rules, the **Kolmogorov axioms**. Every formula in this section is forced out of those three rules; nothing extra is ever assumed.

Which rule of this section does the problem need?

Read the question and match its *verbal structure* to one of six tools.

- **Kolmogorov axioms (the raw definition).** Use them when you must *verify* that a proposed assignment is a legitimate probability, or when the events named are **mutually exclusive** ("cannot both happen," "disjoint," "no overlap") so their probabilities simply add. Trigger words: *is this a valid probability?*, *disjoint*, *cannot occur together*.
- **Complement rule** $\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A)$. Use it whenever the event is phrased as *at least one*, *not*, *none*, *fails to*, or *at most* — anything where the *opposite* event is easier to count than the event itself.
- **Monotonicity / bounds** $0 \le \mathbb{P}(A) \le 1$ and $A \subseteq B \Rightarrow \mathbb{P}(A) \le \mathbb{P}(B)$. Use it when one event *implies* another ("if A happens then B must," "A is a special case of B"), or to sanity-check that an answer is a legal probability.
- **Inclusion–exclusion, two events** $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. Use it for *or / union* questions when the two events *can overlap*. The tell-tale sign is that you are given $\mathbb{P}(A)$, $\mathbb{P}(B)$, and $\mathbb{P}(A \cap B)$ (the "both") and asked for "A or B," or the reverse.
- **Inclusion–exclusion, three events** — same idea with three pieces: add the singles, subtract the three pairwise overlaps, add back the triple. Triggered by "at least one of three" or any union of three overlapping events.
- **Equally-likely model** $\mathbb{P}(A) = |A|/|S|$. Use it when the outcomes are *symmetric* — *fair* dice/coins, a *well-shuffled* deck, a person/ball drawn *at random*. The whole problem reduces to counting $|A|$ and $|S|$.

Quick discriminators. "At least one" ⇒ complement. "A or B" with a given overlap ⇒ inclusion–exclusion. "Drawn at random / fair" ⇒ equally likely. "Disjoint / mutually exclusive" ⇒ the overlap term is 0, so the union is a plain sum (Axiom 3).

2.2.1 The Kolmogorov axioms

Axiom 1 (Nonnegativity). $\mathbb{P}(A) \geq 0$ for every event $A \subseteq S$.

Axiom 2 (Normalization). $\mathbb{P}(S) = 1$.

Axiom 3 (Countable additivity). If $A_1, A_2, A_3, \ldots$ is a sequence of pairwise mutually exclusive events (so $A_i \cap A_j = \varnothing$ whenever $i \neq j$), then

$$\mathbb{P}\Big(\bigcup_{i=1}^{\infty} A_i\Big) = \sum_{i=1}^{\infty} \mathbb{P}(A_i).$$

Axiom 1 says probabilities are never negative; Axiom 2 fixes the scale by giving the certain event S the value 1; Axiom 3, **countable additivity**, is the workhorse: the probability of a disjoint union is the sum of the parts. Taking all but finitely many A_i equal to $\varnothing$ recovers **finite additivity**: for disjoint $A_1, \ldots, A_n$, $\mathbb{P}(A_1 \cup \cdots \cup A_n) = \mathbb{P}(A_1) + \cdots + \mathbb{P}(A_n)$. We will use the finite form constantly.

Remark. A finite or countable list of nonnegative weights summing to 1 over the outcomes always *defines* a valid probability: set $\mathbb{P}(A) = \sum_{s \in A} \mathbb{P}(\{s\})$. Checking the axioms then reduces to checking that the weights are ≥ 0 and total 1.

Example 2.2.1. A spinner stops in one of four coloured sectors, red, blue, green, yellow, with proposed probabilities 0.30, 0.25, 0.15, 0.30 respectively. Is this a legitimate probability assignment?

Which rule, and why? The question literally asks whether a proposed assignment is a *valid probability*, so we test it directly against the three axioms; no derived formula applies yet. The single colours are mutually exclusive (the spinner shows exactly one), so Axiom 3 reduces to: do the weights add to 1?

Solution.

Check Axiom 1 (nonnegativity). Every listed weight $0.30, 0.25, 0.15, 0.30$ is ≥ 0. Pass.

Check Axioms 2–3 (normalization via additivity). The four colours partition the sample space, so by finite additivity $\mathbb{P}(S)$ must equal their sum:

$$0.30 + 0.25 + 0.15 + 0.30 = 1.00.$$

The total is exactly 1, so $\mathbb{P}(S) = 1$. Pass. All three axioms hold, so the assignment is a *valid* probability function.

Example 2.2.2. A different spinner has sectors with proposed probabilities 0.4, 0.3, 0.2, 0.2. Is this a valid probability?

Which rule, and why? Again a "is this valid?" question ⇒ test the axioms. Nonnegativity is fine, so the only thing that can fail is normalization.

Solution. The weights are all ≥ 0, but

$$0.4 + 0.3 + 0.2 + 0.2 = 1.1 \neq 1.$$

Normalization fails ($\mathbb{P}(S) = 1.1 > 1$, which also violates the bound $\mathbb{P} \leq 1$ proved below), so this is *not* a valid probability function. (Notice it would also be invalid if the sum were 0.9: the axioms demand the total be exactly 1, not merely ≤ 1.)

Example 2.2.3. A loaded six-sided die has $\mathbb{P}(\{k\}) = ck$ for $k = 1, \ldots, 6$, where c is a constant. Find c and then the probability of rolling an even number.

Which rule, and why? We are handed an unnormalized rule for the outcome weights and must make it a genuine probability. Axiom 2 (total = 1) pins down the unknown constant; Axiom 3 then lets us add the weights of the even faces. This is the axioms used as the definition, not a derived rule.

Solution.

Solve for c using Axiom 2. The six faces are disjoint and exhaust S, so their weights must sum to 1:

$$\sum_{k=1}^{6} ck = c(1+2+3+4+5+6) = 21c = 1 \quad \Longrightarrow \quad c = \tfrac{1}{21}.$$

All weights $ck = k/21$ are positive, so Axiom 1 holds.

Add the even faces using Axiom 3. The event $E = \{2, 4, 6\}$ is a disjoint union of single outcomes, so

$$\mathbb{P}(E) = \mathbb{P}(\{2\}) + \mathbb{P}(\{4\}) + \mathbb{P}(\{6\}) = \frac{2}{21} + \frac{4}{21} + \frac{6}{21} = \frac{12}{21} = \frac{4}{7} \approx 0.571.$$

Example 2.2.4. A board game card is drawn from a deck whose cards are coloured. The events "red," "blue," "green" are mutually exclusive and exhaust the deck, with $\mathbb{P}(\text{red}) = 0.5$ and $\mathbb{P}(\text{blue}) = 0.2$. A player asks for $\mathbb{P}(\text{red or green})$.

Which rule, and why? The three colours are stated to be *mutually exclusive* and to exhaust the space, so probabilities add directly (Axiom 3) — there is no overlap term, hence no need for inclusion–exclusion. We first recover the missing $\mathbb{P}(\text{green})$ from normalization.

Solution.

Find $\mathbb{P}(\textit{green})$ via Axiom 2. Since the three colours partition S,

$$\mathbb{P}(\text{red}) + \mathbb{P}(\text{blue}) + \mathbb{P}(\text{green}) = 1 \;\Rightarrow\; 0.5 + 0.2 + \mathbb{P}(\text{green}) = 1 \;\Rightarrow\; \mathbb{P}(\text{green}) = 0.3.$$

Add disjoint events (Axiom 3). "Red" and "green" are disjoint, so

$$\mathbb{P}(\text{red or green}) = \mathbb{P}(\text{red}) + \mathbb{P}(\text{green}) = 0.5 + 0.3 = 0.8.$$

No subtraction is needed precisely because the overlap $\mathbb{P}(\text{red} \cap \text{green}) = 0$.

Example 2.2.5. A digit 0–9 is generated by a process that assigns the proposed weights 0.10, 0.20, 0.30, 0.25, 0.10, 0.05, 0, 0, 0, 0 to the digits $0, 1, 2, \ldots, 9$. Is this a legitimate probability assignment?

Which rule, and why? A direct "is this valid?" question ⇒ test the three axioms, not a derived rule. A weight of exactly 0 is allowed (Axiom 1 demands only ≥ 0); the only thing that can fail here is the total.

Solution.

Check Axiom 1. All ten weights are ≥ 0 (the four trailing zeros are fine: they just say those digits never occur). Pass.

Check Axioms 2–3. The digits are mutually exclusive and exhaust S, so normalization requires their sum to be 1:

$$0.10 + 0.20 + 0.30 + 0.25 + 0.10 + 0.05 + 0 + 0 + 0 + 0 = 1.00.$$

The total is exactly 1, so the assignment is a *valid* probability function.

Example 2.2.6. In a population, a person's blood type is one of O, A, B, AB (mutually exclusive and exhaustive). A geneticist records $\mathbb{P}(\text{O}) = 0.44$, $\mathbb{P}(\text{A}) = 0.42$, $\mathbb{P}(\text{B}) = 0.10$. Use the axioms to recover $\mathbb{P}(\text{AB})$, and then $\mathbb{P}(\text{not O})$.

Which rule, and why? The four types partition S, so Axiom 2 (total = 1) pins down the one missing weight by subtraction — the axioms used as the *definition*. Once the weights are known, "not O" adds the three disjoint remaining weights (Axiom 3); no overlap arises, so inclusion–exclusion is irrelevant.

Solution.

Recover the missing weight via Axiom 2. The four disjoint types sum to 1:

$$0.44 + 0.42 + 0.10 + \mathbb{P}(\text{AB}) = 1 \implies \mathbb{P}(\text{AB}) = 0.04.$$

All four weights are ≥ 0, so Axiom 1 holds.

Add disjoint weights via Axiom 3. "Not O" is the disjoint union of A, B, AB:

$$\mathbb{P}(\text{not O}) = 0.42 + 0.10 + 0.04 = 0.56.$$

(Consistency check against the complement rule below: $1 - 0.44 = 0.56$. Same answer.)

Example 2.2.7. A loaded four-sided die (a tetrahedron, faces $1, 2, 3, 4$) is built so that $\mathbb{P}(\{k\}) = c\,k^2$ for a constant c. Find c and the probability of rolling an odd-numbered face.

Which rule, and why? We are handed an unnormalized weight rule and must turn it into a genuine probability: Axiom 2 fixes the constant, then Axiom 3 adds the relevant faces. This is the axioms-as-definition, not a shortcut formula; nothing here is a union of overlapping events.

Solution.

Solve for c using Axiom 2. The four faces are disjoint and exhaust S:

$$\sum_{k=1}^{4} c\,k^2 = c\,(1 + 4 + 9 + 16) = 30c = 1 \implies c = \tfrac{1}{30}.$$

Every weight $k^2/30$ is positive, so Axiom 1 holds.

Add the odd faces using Axiom 3. The odd faces are $\{1, 3\}$, a disjoint union:

$$\mathbb{P}(\{1, 3\}) = \frac{1^2}{30} + \frac{3^2}{30} = \frac{1 + 9}{30} = \frac{10}{30} = \frac{1}{3} \approx 0.333.$$

Example 2.2.8. A traffic-light camera vendor advertises that, at a given approach, $\mathbb{P}(\text{green}) = 0.55$, $\mathbb{P}(\text{yellow}) = 0.10$, $\mathbb{P}(\text{red}) = 0.40$, treating the three light states as mutually exclusive and exhaustive. Is this a valid probability assignment?

Which rule, and why? Another "is this valid?" check ⇒ go straight to the axioms. Nonnegativity is fine, so only normalization can fail; because the three states partition S, their weights must total exactly 1.

Solution. The weights are all ≥ 0, but

$$0.55 + 0.10 + 0.40 = 1.05 \neq 1.$$

Normalization fails ($\mathbb{P}(S) = 1.05 > 1$), so this is *not* a valid probability function. The overcount of 0.05 cannot be waved away: the axioms require the total to be exactly 1.

2.2.2 The complement rule and the empty event

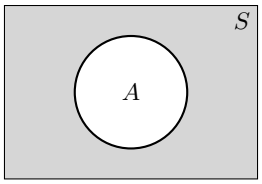
Theorem 2.2.9 (Complement rule)**.** *For every event A, $\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A)$. In particular $\mathbb{P}(\varnothing) = 0$.*

Proof. A and its complement A^{c} are mutually exclusive and their union is all of S. By finite additivity (Axiom 3) and normalization (Axiom 2),

$$\mathbb{P}(A) + \mathbb{P}(A^{\mathsf{c}}) = \mathbb{P}(A \cup A^{\mathsf{c}}) = \mathbb{P}(S) = 1,$$

which rearranges to $\mathbb{P}(A^{\mathsf{c}}) = 1 - \mathbb{P}(A)$. Taking $A = S$ gives $\mathbb{P}(\varnothing) = \mathbb{P}(S^{\mathsf{c}}) = 1 - \mathbb{P}(S) = 1 - 1 = 0$. □

The shaded region below is A^{c}: everything in S outside A. Its area-share is 1 minus the share of A.



A^c

Example 2.2.10. A fair coin is tossed 5 times. Find the probability of *at least one* head.

Which rule, and why? The phrase *at least one* is the classic complement trigger: "at least one head" is the opposite of the single, easy-to-count event "no heads at all." Counting the many ways to get one-or-more heads directly would be far more work.

Solution. Let $A = \{\text{at least one head}\}$. Then $A^c = \{\text{all five tosses are tails}\}$, a single outcome out of $2^5 = 32$ equally likely sequences, so $\mathbb{P}(A^c) = \frac{1}{32}$. Therefore

$$\mathbb{P}(A) = 1 - \mathbb{P}(A^c) = 1 - \frac{1}{32} = \frac{31}{32} \approx 0.969.$$

Example 2.2.11. A batch contains components of which 8% are defective. Three components are chosen (the batch is large, so treat the three picks as independent each with defect probability 0.08). Find the probability that *at least one* of the three is defective.

Which rule, and why? "At least one defective" again screams *complement*: its opposite is "all three are good," a clean product of three numbers. Trying to add the cases "exactly one," "exactly two," "exactly three" defective would be three computations instead of one.

Solution. Each component is good with probability $1 - 0.08 = 0.92$. The opposite event "all three good" has probability

$$\mathbb{P}(\text{all good}) = (0.92)^3 = 0.778688.$$

Hence

$$\mathbb{P}(\text{at least one defective}) = 1 - (0.92)^3 = 1 - 0.778688 = 0.221312 \approx 0.221.$$

Example 2.2.12. A multiple-choice quiz has 10 questions, each with 4 options; a student guesses every answer at random. Find the probability of getting *at least one* question right.

Which rule, and why? "At least one correct" over many independent trials ⇒ complement; the opposite "every single guess is wrong" is one tidy product. The equally-likely model gives the per-question wrong-probability, but the structural rule that cracks the problem is the complement rule.

Solution. Each guess is wrong with probability $\frac{3}{4}$. The questions are independent, so "all ten wrong" has probability $\left(\frac{3}{4}\right)^{10}$. Therefore

$$\mathbb{P}(\text{at least one right}) = 1 - \left(\frac{3}{4}\right)^{10} = 1 - \frac{59{,}049}{1{,}048{,}576} = \frac{989{,}527}{1{,}048{,}576} \approx 0.944.$$

Example 2.2.13. In a survey, the probability that a randomly chosen respondent owns a car is $\mathbb{P}(A) = 0.62$. What is the probability the respondent does *not* own a car?

Which rule, and why? The single word *not* maps the event onto its complement; there is nothing to count, just one subtraction. No union or overlap appears, so inclusion–exclusion is irrelevant.

Solution.

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A) = 1 - 0.62 = 0.38.$$

Example 2.2.14. Two fair dice are rolled. Find the probability that the sum is *at most* 10.

Which rule, and why? "At most 10" covers sums $2, 3, \ldots, 10$ — nine values — whereas its complement "11 or 12" covers only two. The complement rule turns a nine-case count into a two-case count.

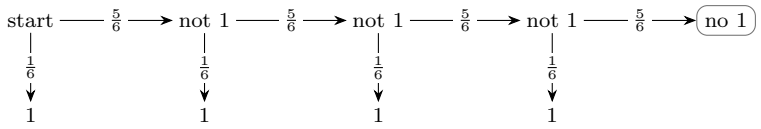
Solution. Of the 36 equally likely outcomes, the sum is 11 in 2 ways ($(5,6),(6,5)$) and 12 in 1 way ($(6,6)$), so

$$\mathbb{P}(\text{sum} > 10) = \frac{2+1}{36} = \frac{3}{36} = \frac{1}{12}, \qquad \mathbb{P}(\text{sum} \le 10) = 1 - \frac{1}{12} = \frac{11}{12} \approx 0.917.$$

Example 2.2.15. Four fair dice are rolled. Find the probability that *at least one* of them shows a 1.

Which rule, and why? "At least one" across several independent rolls is the textbook complement trigger: its opposite, "no die shows a 1," is a single clean product. Listing the ways to get one, two, three, or four 1s would be four separate counts; the complement collapses them into one.

Solution. Each die avoids 1 with probability $\frac{5}{6}$, and the four dice are independent, so "no 1 at all" has probability $\left(\frac{5}{6}\right)^4$. The tree below traces the only branch that the complement keeps — "not 1" four times in a row — against everything else (the event we want).



Following the all-"not 1" branch for all four dice,

$$\mathbb{P}(\text{no 1}) = \left(\frac{5}{6}\right)^4 = \frac{625}{1296}, \qquad \mathbb{P}(\text{at least one 1}) = 1 - \frac{625}{1296} = \frac{671}{1296} \approx 0.518.$$

Example 2.2.16. A spinner is flicked three times, each flick landing on one of the digits 0 through 9 with equal chance, independently. Find the probability that *at least one* flick shows a 7.

Which rule, and why? "At least one 7" over independent trials ⇒ complement: the opposite "no 7 on any flick" is one product. The equally-likely model supplies the per-flick miss probability $\frac{9}{10}$, but it is the complement rule that turns three overlapping "a 7 here" events into a single subtraction.

Solution. Each flick avoids 7 with probability $\frac{9}{10}$; the three flicks are independent, so

$$\mathbb{P}(\text{no 7}) = \left(\frac{9}{10}\right)^3 = \frac{729}{1000}, \qquad \mathbb{P}(\text{at least one 7}) = 1 - \frac{729}{1000} = \frac{271}{1000} = 0.271.$$

Example 2.2.17. A committee of 4 is drawn at random from a club of 6 women and 4 men. Find the probability the committee includes *at least one* man.

Which rule, and why? "At least one man" is awkward head-on (it splits into one, two, three, or four men), so we flip to the complement "all four are women" — a single combination count. The draw is "at random," so the equally-likely model supplies the two counts.

Solution. There are $\binom{10}{4} = 210$ equally likely committees. The complement "no men," i.e. all four chosen from the 6 women, numbers $\binom{6}{4} = 15$. Hence

$$\mathbb{P}(\text{all women}) = \frac{15}{210} = \frac{1}{14}, \qquad \mathbb{P}(\text{at least one man}) = 1 - \frac{1}{14} = \frac{13}{14} \approx 0.929.$$

Example 2.2.18. Two fair dice are rolled. Find the probability that *at least one* die shows an even number.

Which rule, and why? "At least one even" inverts cleanly to "both odd," which is a single product over the two independent dice; the complement rule beats counting the even-bearing outcomes one by one. The equally-likely model gives each die's odd-probability $\frac{3}{6}$.

Solution. Each die is odd with probability $\frac{3}{6} = \frac{1}{2}$, so "both odd" has probability $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$. Therefore

$$\mathbb{P}(\text{at least one even}) = 1 - \frac{1}{4} = \frac{3}{4} = 0.75.$$

Direct check: of 36 outcomes, $3 \times 3 = 9$ have both dice odd, leaving 27 with at least one even, and $27/36 = 3/4$. Agrees.

2.2.3 Monotonicity and the bounds $0 \le \mathbb{P} \le 1$

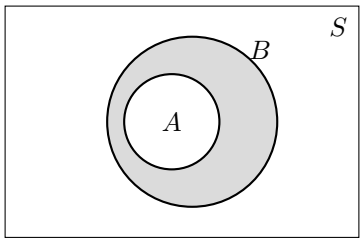
Theorem 2.2.19 (Monotonicity and bounds)**.** *If $A \subseteq B$ then $\mathbb{P}(A) \le \mathbb{P}(B)$, and in fact $\mathbb{P}(B) - \mathbb{P}(A) = \mathbb{P}(B \cap A^{\mathsf{c}})$. Consequently $0 \le \mathbb{P}(A) \le 1$ for every event A.*

Proof. When $A \subseteq B$, write B as the disjoint union of the part inside A and the part of B outside A: $B = A \cup (B \cap A^{\mathsf{c}})$, with $A \cap (B \cap A^{\mathsf{c}}) = \varnothing$. By finite additivity,

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \cap A^{\mathsf{c}}).$$

The second term is ≥ 0 by Axiom 1, so $\mathbb{P}(B) \ge \mathbb{P}(A)$, and rearranging gives the stated difference. For the bounds: $\mathbb{P}(A) \ge 0$ is Axiom 1, and since $A \subseteq S$ monotonicity gives $\mathbb{P}(A) \le \mathbb{P}(S) = 1$. □

In the picture A sits inside B; the extra shaded crescent is $B \cap A^{\mathsf{c}}$, whose probability is exactly the gap $\mathbb{P}(B) - \mathbb{P}(A)$.



Example 2.2.20. A card is drawn from a standard 52-card deck. Let $A = \{\text{ace of spades}\}$ and $B = \{\text{spade}\}$. Without listing every spade, argue $\mathbb{P}(A) \le \mathbb{P}(B)$ and give both values.

Which rule, and why? The event A *implies* B: every ace of spades *is* a spade, i.e. $A \subseteq B$. "A is a special case of B" is the monotonicity trigger; we conclude $\mathbb{P}(A) \le \mathbb{P}(B)$ before computing anything.

Solution. Since the single card "ace of spades" is one of the 13 spades, $A \subseteq B$, so by monotonicity $\mathbb{P}(A) \le \mathbb{P}(B)$. Concretely, with 52 equally likely cards,

$$\mathbb{P}(A) = \frac{1}{52} \approx 0.019, \qquad \mathbb{P}(B) = \frac{13}{52} = \frac{1}{4} = 0.25,$$

and indeed $0.019 \le 0.25$. The gap $\mathbb{P}(B) - \mathbb{P}(A) = \frac{12}{52}$ is the probability of "a spade other than the ace," i.e. $\mathbb{P}(B \cap A^{\mathsf{c}})$.

Example 2.2.21. An analyst claims that for two events, $\mathbb{P}(A) = 0.7$, $\mathbb{P}(B) = 0.4$, and $\mathbb{P}(A \cap B) = 0.55$. Show the claim is impossible.

Which rule, and why? $A \cap B \subseteq B$ always, so monotonicity forces $\mathbb{P}(A \cap B) \le \mathbb{P}(B)$. This is a *consistency check*: whenever an intersection probability is quoted, it can never exceed either single probability.

Solution. Because $A \cap B \subseteq B$, monotonicity requires $\mathbb{P}(A \cap B) \le \mathbb{P}(B)$. But the claim has

$$\mathbb{P}(A \cap B) = 0.55 > 0.40 = \mathbb{P}(B),$$

a contradiction. (Against $\mathbb{P}(A) = 0.7$ the bound is satisfied since $0.55 \le 0.7$, but violating it against $\mathbb{P}(B)$ alone already makes the assignment impossible.) So no probability function can have these values.

Example 2.2.22. Someone proposes that a certain event has probability $\mathbb{P}(A) = 1.3$. Why is this nonsense, and what is the largest a probability can be?

Which rule, and why? Any single number offered as a probability must satisfy the universal bound $0 \le \mathbb{P} \le 1$ from Theorem (monotonicity). This is the bound used purely as a validity test.

Solution. Since $A \subseteq S$, monotonicity gives $\mathbb{P}(A) \le \mathbb{P}(S) = 1$. Thus $\mathbb{P}(A) = 1.3$ violates the upper bound: no event can be "more than certain." Every probability lies in $[0, 1]$, with 0 for the impossible event and 1 for the certain event.

Example 2.2.23. A weather model assigns $\mathbb{P}(\text{rain}) = 0.3$ and, for a particular town, $\mathbb{P}(\text{heavy rain})$ where "heavy rain" implies "rain." Give the tightest bounds the axioms place on $\mathbb{P}(\text{heavy rain})$.

Which rule, and why? "Heavy rain implies rain" is exactly $A \subseteq B$ with $A = \{\text{heavy}\}$, $B = \{\text{rain}\}$. Monotonicity caps $\mathbb{P}(A)$ above by $\mathbb{P}(B)$, and Axiom 1 caps it below by 0. This is a bounding problem, not an exact computation.

Solution. Let $A = \{\text{heavy rain}\} \subseteq B = \{\text{rain}\}$. Then

$$0 \leq \mathbb{P}(A) \leq \mathbb{P}(B) = 0.3.$$

So heavy rain can be anywhere from impossible up to as likely as rain itself, but no more; we cannot pin down a single value without extra information.

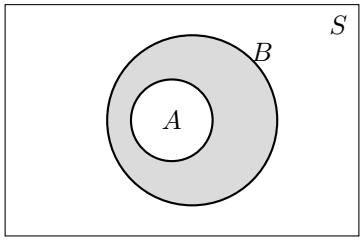
Example 2.2.24. A card is drawn from a standard 52-card deck. Let $A = \{\text{a red king}\}$ and $B = \{\text{a king}\}$. Argue $\mathbb{P}(A) \leq \mathbb{P}(B)$ from the structure, then give both values.

Which rule, and why? Every red king *is* a king, so $A \subseteq B$: "A is a special case of B" is the monotonicity trigger, and we may assert $\mathbb{P}(A) \leq \mathbb{P}(B)$ before any arithmetic. This is not a union question (no "or"), so inclusion–exclusion does not enter.

Solution. The two red kings ($\heartsuit, \diamondsuit$) are among the four kings, so $A \subseteq B$ and monotonicity gives $\mathbb{P}(A) \leq \mathbb{P}(B)$. With 52 equally likely cards,

$$\mathbb{P}(A) = \frac{2}{52} = \frac{1}{26} \approx 0.038, \qquad \mathbb{P}(B) = \frac{4}{52} = \frac{1}{13} \approx 0.077,$$

and indeed $0.038 \leq 0.077$. The gap $\mathbb{P}(B) - \mathbb{P}(A) = \frac{2}{52}$ is $\mathbb{P}(B \cap A^{\mathsf{c}})$, the chance of a *black* king. The nesting is shown below: A sits wholly inside B, and the shaded crescent is exactly that gap.



Example 2.2.25. A quality report claims that for two events $\mathbb{P}(A) = 0.30$, $\mathbb{P}(B) = 0.50$, and $\mathbb{P}(A \cup B) = 0.45$. Show this is impossible.

Which rule, and why? For any events, $B \subseteq A \cup B$, so monotonicity forces $\mathbb{P}(A \cup B) \geq \mathbb{P}(B)$: a union can never be *less* likely than one of its parts. This is a consistency check on a quoted union, not a value to compute.

Solution. Since $B \subseteq A \cup B$, monotonicity requires $\mathbb{P}(A \cup B) \geq \mathbb{P}(B)$. But the claim has

$$\mathbb{P}(A \cup B) = 0.45 < 0.50 = \mathbb{P}(B),$$

a contradiction — the union came out smaller than B alone, which no probability function allows. So these three numbers cannot coexist.

Example 2.2.26. Two events satisfy $\mathbb{P}(A) = 0.85$ and $\mathbb{P}(B) = 0.75$. Without knowing the overlap, find the smallest possible value of $\mathbb{P}(A \cap B)$.

Which rule, and why? We want a guaranteed lower bound on an overlap from the singles alone. Combine monotonicity with the bound $\mathbb{P}(A \cup B) \leq 1$: because the union cannot exceed 1, the intersection cannot be too small. (This "squeeze" is the Bonferroni inequality, a direct child of the axioms, not the equally-likely model.)

Solution. Rearrange two-event inclusion–exclusion and use $\mathbb{P}(A \cup B) \leq 1$:

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) \;\geq\; \mathbb{P}(A) + \mathbb{P}(B) - 1 = 0.85 + 0.75 - 1 = 0.60.$$

So at least 60% of the time *both* occur; the overlap can be as large as 0.75 (if $A \subseteq B$) but never below 0.60. The bound is forced purely by "probabilities cannot exceed 1."

2.2.4 Inclusion–exclusion for two events

Theorem 2.2.27 (Inclusion–exclusion, two events)**.** *For any events A, B,*

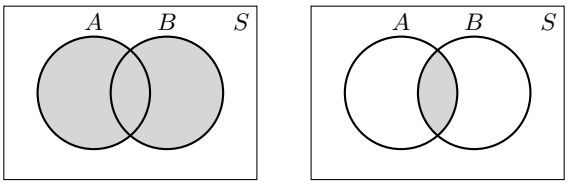
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Proof. Decompose the union into three disjoint pieces: A-only, B-only, and both:

$$A \cup B = (A \cap B^{\mathsf{c}}) \ \cup \ (A^{\mathsf{c}} \cap B) \ \cup \ (A \cap B),$$

all pairwise disjoint. By finite additivity, $\mathbb{P}(A \cup B) = \mathbb{P}(A \cap B^{\mathsf{c}}) + \mathbb{P}(A^{\mathsf{c}} \cap B) + \mathbb{P}(A \cap B)$. Now $\mathbb{P}(A) = \mathbb{P}(A \cap B^{\mathsf{c}}) + \mathbb{P}(A \cap B)$ and $\mathbb{P}(B) = \mathbb{P}(A^{\mathsf{c}} \cap B) + \mathbb{P}(A \cap B)$, so adding $\mathbb{P}(A) + \mathbb{P}(B)$ counts the overlap $\mathbb{P}(A \cap B)$ *twice*. Subtracting it once corrects the double-count, giving the formula. □

The reason a bare sum fails is visible in the Venn picture: $\mathbb{P}(A) + \mathbb{P}(B)$ paints the lens $A \cap B$ twice, so we subtract it once.



$A \cup B$ $\qquad$ $A \cap B$

Example 2.2.28. In a town, a household subscribes to the newspaper with probability 0.55, to a streaming service with probability 0.65, and to *both* with probability 0.40. Find the probability a household subscribes to *at least one* of the two.

Which rule, and why? We want "newspaper *or* streaming" (the union) and the two events clearly *overlap* (some households have both, $0.40 \neq 0$). Both singles and the overlap are given, which is the exact data signature of two-event inclusion–exclusion. The complement rule is not simplest here because the overlap is supplied directly.

Solution. Let $N = \{\text{newspaper}\}$ and $S = \{\text{streaming}\}$. Then

$$\mathbb{P}(N \cup S) = \mathbb{P}(N) + \mathbb{P}(S) - \mathbb{P}(N \cap S) = 0.55 + 0.65 - 0.40 = 0.80.$$

A plain sum $0.55 + 0.65 = 1.20$ would exceed 1; subtracting the double-counted overlap restores a legal 0.80.

Example 2.2.29. Draw one card from a standard deck. Find $\mathbb{P}(\text{king or heart})$.

Which rule, and why? "King *or* heart" is a union, and the two events overlap in the single card *king of hearts*, so a raw sum would double-count it. Two-event inclusion–exclusion with the equally-likely model supplies each piece.

Solution. With $|S| = 52$: there are 4 kings, 13 hearts, and 1 card that is both (the king of hearts). So

$$\mathbb{P}(\text{king} \cup \text{heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = \frac{16}{52} = \frac{4}{13} \approx 0.308.$$

Direct count check: the favourable cards are the 13 hearts plus the 3 non-heart kings, 16 in all — matching.

Example 2.2.30. For two events, $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.3$, and $\mathbb{P}(A \cup B) = 0.6$. Find $\mathbb{P}(A \cap B)$ and then $\mathbb{P}(\text{exactly one of } A, B)$.

Which rule, and why? All four quantities in the two-event formula appear, three known and one unknown, so we solve inclusion–exclusion *backwards* for the overlap. "Exactly one" then comes from removing the double overlap from the union.

Solution.

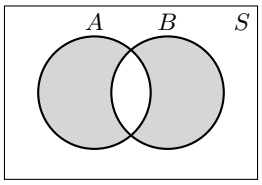
Solve for the overlap. Rearranging the formula,

$$\mathbb{P}(A \cap B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cup B) = 0.5 + 0.3 - 0.6 = 0.2.$$

Exactly one. "Exactly one of A, B" is the union minus the both-part:

$$\mathbb{P}(\text{exactly one}) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.6 - 0.2 = 0.4.$$

Equivalently "exactly one" is the symmetric difference shaded here:



$(A \setminus B) \cup (B \setminus A)$

Example 2.2.31. A fair die is rolled. Let $A = \{\text{even}\} = \{2, 4, 6\}$ and $B = \{\text{at least } 5\} = \{5, 6\}$. Find $\mathbb{P}(A \cup B)$.

Which rule, and why? "Even *or* at least 5" is a union with a genuine overlap: the face 6 lies in both. Two-event inclusion–exclusion under equal likelihood. (Adding $3/6 + 2/6$ blindly would count the 6 twice.)

Solution. With $|S| = 6$: $\mathbb{P}(A) = \frac{3}{6}$, $\mathbb{P}(B) = \frac{2}{6}$, and $A \cap B = \{6\}$ so $\mathbb{P}(A \cap B) = \frac{1}{6}$. Hence

$$\mathbb{P}(A \cup B) = \frac{3}{6} + \frac{2}{6} - \frac{1}{6} = \frac{4}{6} = \frac{2}{3} \approx 0.667.$$

Check: $A \cup B = \{2, 4, 5, 6\}$, four outcomes, $4/6$. Agrees.

Example 2.2.32. 30% of students take a statistics course, 20% take an economics course, and 12% take both. What fraction take *neither*?

Which rule, and why? This is two rules in sequence. "Or" with overlap ⇒ inclusion–exclusion for "at least one course"; then "neither" is the *complement* of "at least one." Recognising the two-step union-then-complement structure is the key.

Solution.

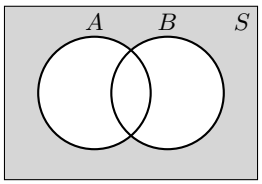
At least one course.

$$\mathbb{P}(\text{stats} \cup \text{econ}) = 0.30 + 0.20 - 0.12 = 0.38.$$

Neither, by complement.

$$\mathbb{P}(\text{neither}) = 1 - \mathbb{P}(\text{stats} \cup \text{econ}) = 1 - 0.38 = 0.62.$$

So 62% take neither course. The region "neither" is the unshaded exterior of both circles:



$(A \cup B)^{\mathsf{c}} = A^{\mathsf{c}} \cap B^{\mathsf{c}}$

Example 2.2.33. In a cafe survey, $\mathbb{P}(\text{a customer orders coffee}) = 0.70$, $\mathbb{P}(\text{orders a pastry}) = 0.45$, and $\mathbb{P}(\text{orders both}) = 0.30$. Find the probability a customer orders coffee *or* a pastry.

Which rule, and why? We want the union "coffee or pastry" and the two events overlap (some order both, $0.30 \neq 0$). All three inputs — two singles and the both — are handed to us, the exact signature of two-event inclusion–exclusion. A bare sum would double-count the both-orderers.

Solution.

$$\mathbb{P}(\text{coffee} \cup \text{pastry}) = 0.70 + 0.45 - 0.30 = 0.85.$$

The naive sum $0.70 + 0.45 = 1.15$ exceeds 1; subtracting the once-double-counted overlap 0.30 restores the legal value 0.85.

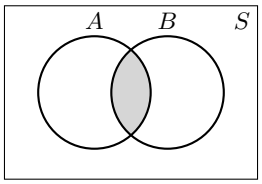
Example 2.2.34. Two fair dice are rolled. Let $A = \{\text{sum is even}\}$ and $B = \{\text{sum} \geq 9\}$. Find $\mathbb{P}(A \cup B)$.

Which rule, and why? "Sum even *or* sum ≥ 9" is a union whose events overlap (e.g. a sum of 10 is both even and ≥ 9), so a raw sum double-counts. Two-event inclusion–exclusion with the equally-likely model over the 36 outcomes supplies each count.

Solution. Of 36 equally likely pairs: a sum is even in $|A| = 18$ ways; a sum is ≥ 9 (i.e. $9, 10, 11, 12$) in $4 + 3 + 2 + 1 = 10$ ways, so $|B| = 10$; and "even *and* ≥ 9" means the sum is 10 or 12, giving $3 + 1 = 4$ ways, so $|A \cap B| = 4$. Hence

$$\mathbb{P}(A \cup B) = \frac{18}{36} + \frac{10}{36} - \frac{4}{36} = \frac{24}{36} = \frac{2}{3} \approx 0.667.$$

The shaded lens below is the double-counted overlap $A \cap B$, removed once:



$A \cap B$

Example 2.2.35. A card is drawn from a standard deck. Find $\mathbb{P}(\text{king or red card})$.

Which rule, and why? "King *or* red" is a union, and the events overlap in the two red kings, so a plain sum over-counts those cards. Two-event inclusion–exclusion with the equally-likely model gives each piece from a direct card count.

Solution. With $|S| = 52$: there are 4 kings, 26 red cards, and 2 cards that are both (kings of hearts and diamonds). So

$$\mathbb{P}(\text{king} \cup \text{red}) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52} = \frac{28}{52} = \frac{7}{13} \approx 0.538.$$

Direct check: the 26 red cards plus the 2 black kings give 28 favourable cards — matching.

Example 2.2.36. For two events, $\mathbb{P}(A) = 0.60$, $\mathbb{P}(A \cup B) = 0.80$, and $\mathbb{P}(A \cap B) = 0.25$. Find $\mathbb{P}(B)$, then $\mathbb{P}(\text{exactly one of } A, B)$.

Which rule, and why? Three of the four quantities in the two-event formula are given, so we solve it *backwards* for the fourth, $\mathbb{P}(B)$. "Exactly one" is then the union with the both-part stripped out. The overlap is supplied, so the complement route offers no shortcut.

Solution.

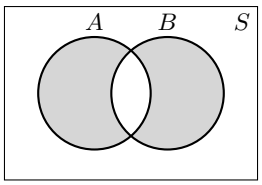
Solve for $\mathbb{P}(B)$. Rearranging $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$,

$$\mathbb{P}(B) = \mathbb{P}(A \cup B) - \mathbb{P}(A) + \mathbb{P}(A \cap B) = 0.80 - 0.60 + 0.25 = 0.45.$$

Exactly one. Remove the overlap from the union:

$$\mathbb{P}(\text{exactly one}) = \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.80 - 0.25 = 0.55.$$

This is the symmetric difference shaded here:



$(A \setminus B) \cup (B \setminus A)$

2.2.5 Inclusion–exclusion for three events

Theorem 2.2.37 (Inclusion–exclusion, three events)**.** *For any events A, B, C,*

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).$$

Proof. Group $A \cup B \cup C = (A \cup B) \cup C$ and apply the two-event rule twice. First,

$$\mathbb{P}\big((A \cup B) \cup C\big) = \mathbb{P}(A \cup B) + \mathbb{P}(C) - \mathbb{P}\big((A \cup B) \cap C\big).$$

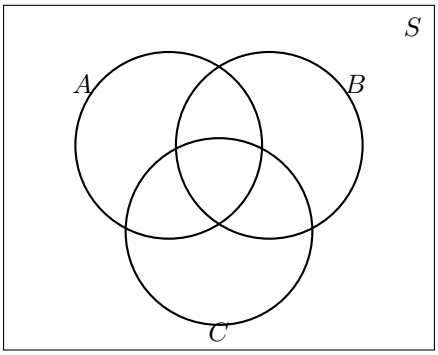
Expand $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$. For the last term, distribute: $(A \cup B) \cap C = (A \cap C) \cup (B \cap C)$, so by the two-event rule again

$$\mathbb{P}\big((A \cup B) \cap C\big) = \mathbb{P}(A \cap C) + \mathbb{P}(B \cap C) - \mathbb{P}(A \cap B \cap C),$$

using $(A \cap C) \cap (B \cap C) = A \cap B \cap C$. Substituting both expansions and collecting terms yields the stated identity. □

Remark (Why the alternating signs). Count how often each region is tallied. A point in exactly one event is added once and never subtracted: weight 1. A point in exactly two events is added twice, subtracted once: weight $2 - 1 = 1$. A point in all three is added three times, subtracted three times, added once: $3 - 3 + 1 = 1$. Every point of the union is counted exactly once — which is what the formula must do.

A three-circle Venn diagram makes the seven regions explicit; the central petal $A \cap B \cap C$ is the one added back at the end.



Example 2.2.38. A streaming subscriber base reports: $\mathbb{P}(A) = 0.50$ watch dramas, $\mathbb{P}(B) = 0.40$ comedies, $\mathbb{P}(C) = 0.30$ documentaries; pairwise, $\mathbb{P}(A \cap B) = 0.20$, $\mathbb{P}(A \cap C) = 0.15$, $\mathbb{P}(B \cap C) = 0.10$; and $\mathbb{P}(A \cap B \cap C) = 0.05$ watch all three genres. What fraction watch *at least one* of the three genres?

Which rule, and why? "At least one of *three* overlapping events" is the three-event union, and the problem hands us all seven inputs (three singles, three pairs, one triple) — the full data signature of three-event inclusion–exclusion. The complement would require "watches none," which is not given directly.

Solution. Substitute into the three-event formula:

$$\mathbb{P}(A \cup B \cup C) = 0.50 + 0.40 + 0.30 - 0.20 - 0.15 - 0.10 + 0.05.$$

Singles sum to 1.20; pairwise sum to 0.45; triple is 0.05. So

$$\mathbb{P}(A \cup B \cup C) = 1.20 - 0.45 + 0.05 = 0.80.$$

Thus 80% watch at least one genre (and, by complement, 20% watch none).

Example 2.2.39. Draw one card from a standard deck. Find $\mathbb{P}$(spade, or face card, or ace). (Face cards are J, Q, K.)

Which rule, and why? Three properties joined by *or*, all able to coincide on a single card (e.g. the king of spades is a spade *and* a face card), so three-event inclusion–exclusion with counts. Note "ace" and "face card" are disjoint, which will zero one pairwise term.

Solution. Let $A =$ spade (13), $B =$ face card (12), $C =$ ace (4); $|S| = 52$.

Pairwise overlaps. Spades that are face cards: J,Q,K of spades $= 3$, so $|A \cap B| = 3$. Spades that are aces: ace of spades $= 1$, so $|A \cap C| = 1$. Face cards that are aces: none (an ace is not a face card), so $|B \cap C| = 0$.

Triple overlap. A spade that is both a face card and an ace: impossible, so $|A \cap B \cap C| = 0$.

Assemble.

$$|A \cup B \cup C| = 13 + 12 + 4 - 3 - 1 - 0 + 0 = 25, \qquad \mathbb{P} = \frac{25}{52} \approx 0.481.$$

Example 2.2.40. Three fair dice are rolled. Find the probability that *at least one* die shows a 6, using inclusion–exclusion (so as to compare with the complement method).

Which rule, and why? "At least one six among three dice" is the union of $A_i = \{\text{die } i \text{ shows } 6\}$ for $i = 1, 2, 3$. With three overlapping events we may use three-event inclusion–exclusion; here all the pieces are easy products. (The complement $(5/6)^3$ is quicker, and we will confirm the two agree.)

Solution. Each $\mathbb{P}(A_i) = \frac{1}{6}$; each pair $\mathbb{P}(A_i \cap A_j) = \frac{1}{36}$ (two specified dice both 6); the triple $\mathbb{P}(A_1 \cap A_2 \cap A_3) = \frac{1}{216}$. Then

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = 3 \cdot \frac{1}{6} - 3 \cdot \frac{1}{36} + \frac{1}{216} = \frac{108}{216} - \frac{18}{216} + \frac{1}{216} = \frac{91}{216} \approx 0.421.$$

Cross-check by complement. $\mathbb{P}(\text{at least one 6}) = 1 - \left(\frac{5}{6}\right)^3 = 1 - \frac{125}{216} = \frac{91}{216}$. The two methods agree.

Example 2.2.41. Among the integers 1 through 100 a number is chosen at random. Find the probability it is divisible by 2, 3, or 5.

Which rule, and why? "Divisible by 2 *or* 3 *or* 5" is a three-event union; the events overlap (e.g. 30 is divisible by all three), so three-event inclusion–exclusion is needed. Counting multiples (and multiples of products) gives every term; the equally-likely model turns counts into probabilities.

Solution. Let A, B, C be "divisible by $2, 3, 5$." Count multiples in $\{1, \ldots, 100\}$ with floor division:

$$|A| = 50, \quad |B| = 33, \quad |C| = 20;$$

$$|A \cap B| = \lfloor 100/6 \rfloor = 16, \quad |A \cap C| = \lfloor 100/10 \rfloor = 10, \quad |B \cap C| = \lfloor 100/15 \rfloor = 6;$$

$$|A \cap B \cap C| = \lfloor 100/30 \rfloor = 3.$$

Therefore

$$|A \cup B \cup C| = 50 + 33 + 20 - 16 - 10 - 6 + 3 = 74, \qquad \mathbb{P} = \frac{74}{100} = 0.74.$$

Example 2.2.42. A language survey of a region reports: $\mathbb{P}(A) = 0.45$ speak Spanish, $\mathbb{P}(B) = 0.30$ French, $\mathbb{P}(C) = 0.20$ German; pairwise $\mathbb{P}(A \cap B) = 0.12$, $\mathbb{P}(A \cap C) = 0.08$, $\mathbb{P}(B \cap C) = 0.06$; and $\mathbb{P}(A \cap B \cap C) = 0.03$ speak all three. What fraction speak *at least one* of the three languages?

Which rule, and why? "At least one of *three* overlapping events" is the three-event union, and all seven inputs (three singles, three pairs, one triple) are supplied — the full data signature of three-event inclusion–exclusion. "Speaks none" is not given, so the complement is not a shortcut here.

Solution. Substitute into the three-event formula:

$$\mathbb{P}(A \cup B \cup C) = 0.45 + 0.30 + 0.20 - 0.12 - 0.08 - 0.06 + 0.03.$$

Singles sum to 0.95; pairwise to 0.26; triple is 0.03. So

$$\mathbb{P}(A \cup B \cup C) = 0.95 - 0.26 + 0.03 = 0.72.$$

Thus 72% speak at least one language (and 28% speak none of the three).

Example 2.2.43. Draw one card from a standard deck. Find $\mathbb{P}$(spade, or heart, or face card). (Face cards are J, Q, K.)

Which rule, and why? Three properties joined by *or*, with some genuine overlaps (e.g. the jack of spades is a spade *and* a face card), so three-event inclusion–exclusion with counts. A useful wrinkle: "spade" and "heart" are *disjoint*, which zeroes one pairwise term and the triple term.

Solution. Let $A =$ spade (13), $B =$ heart (13), $C =$ face card (12); $|S| = 52$.

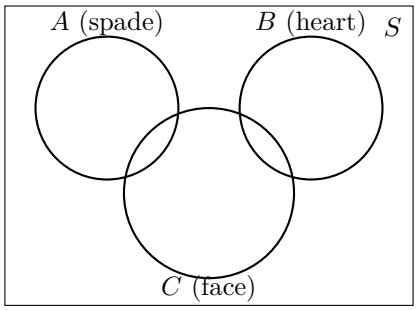
Pairwise overlaps. A card cannot be both a spade and a heart, so $|A \cap B| = 0$. Spade face cards (J,Q,K of spades) number 3, so $|A \cap C| = 3$; heart face cards likewise give $|B \cap C| = 3$.

Triple overlap. A card that is a spade *and* a heart is impossible, so $|A \cap B \cap C| = 0$.

Assemble.

$$|A \cup B \cup C| = 13 + 13 + 12 - 0 - 3 - 3 + 0 = 32, \qquad \mathbb{P} = \frac{32}{52} = \frac{8}{13} \approx 0.615.$$

In the three-circle picture, the A and B circles are drawn apart (disjoint), each meeting only the face-card circle C:



Example 2.2.44. Three fair dice are rolled. Find the probability that *at least one* die shows a 5 or a 6, using inclusion–exclusion (and check against the complement).

Which rule, and why? "At least one die in $\{5, 6\}$" is the union of $A_i = \{\text{die } i \text{ shows 5 or 6}\}$ over $i = 1, 2, 3$. Three overlapping events $\Rightarrow$ three-event inclusion–exclusion; here each $\mathbb{P}(A_i) = \frac{2}{6}$ and the overlaps are clean products. The complement $(4/6)^3$ is faster and will confirm the answer.

Solution. Each $\mathbb{P}(A_i) = \frac{2}{6} = \frac{1}{3}$; each pair $\mathbb{P}(A_i \cap A_j) = \left(\frac{1}{3}\right)^2 = \frac{1}{9}$; the triple $\mathbb{P}(A_1 \cap A_2 \cap A_3) = \left(\frac{1}{3}\right)^3 = \frac{1}{27}$. Then

$$\mathbb{P}(A_1 \cup A_2 \cup A_3) = 3 \cdot \frac{1}{3} - 3 \cdot \frac{1}{9} + \frac{1}{27} = \frac{27}{27} - \frac{9}{27} + \frac{1}{27} = \frac{19}{27} \approx 0.704.$$

Cross-check by complement. $1 - \left(\frac{4}{6}\right)^3 = 1 - \left(\frac{2}{3}\right)^3 = 1 - \frac{8}{27} = \frac{19}{27}$. The two methods agree.

2.2.6 The equally-likely (classical) model

Theorem 2.2.45 (Equally-likely outcomes)**.** *Suppose S is finite with $|S| = N$ outcomes, all equally likely. Then each single outcome has probability $1/N$, and for every event A,*

$$\mathbb{P}(A) = \frac{|A|}{|S|} = \frac{\textit{number of outcomes favourable to } A}{\textit{total number of outcomes}}.$$

Proof. Let the outcomes be $s_1, \ldots, s_N$ with common probability p. They are disjoint and their union is S, so by additivity and normalization $Np = \mathbb{P}(S) = 1$, giving $p = 1/N$. For any event $A = \{s_{i_1}, \ldots, s_{i_k}\}$ with $k = |A|$ outcomes, additivity gives $\mathbb{P}(A) = kp = k/N = |A|/|S|$. □

Remark. This model applies *only* when symmetry justifies equal likelihood — a *fair* coin or die, a *well-shuffled* deck, a draw made *at random*. If outcomes are weighted (a loaded die, biased coin), revert to the axioms and add the actual weights. The whole task becomes counting $|A|$ and $|S|$, which is why combinatorics (covered later) is the natural sequel.

Example 2.2.46. Two fair dice are rolled. Find the probability that the sum equals 8.

Which rule, and why? "Two *fair* dice" guarantees 36 equally likely outcomes ⇒ the classical model $|A|/|S|$. We simply count the outcomes giving a sum of 8. No complement or union structure is present.

Solution. The 36 ordered pairs are equally likely. The pairs summing to 8 are $(2,6)$, $(3,5)$, $(4,4)$, $(5,3)$, $(6,2)$, i.e. five of them. So

$$\mathbb{P}(\text{sum} = 8) = \frac{5}{36} \approx 0.139.$$

The two-dice grid below shows the diagonal of 8's (marked •); count the dots, divide by 36.

+	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	•
3	4	5	6	7	•	9
4	5	6	7	•	9	10
5	6	7	•	9	10	11
6	7	•	9	10	11	12

Example 2.2.47. An urn holds 5 red, 4 blue, and 3 green balls. One ball is drawn at random. Find the probability it is *not* red.

Which rule, and why? "Drawn at random" from a finite urn ⇒ equally likely over the 12 balls. "Not red" is a complement, but the count is so direct we can read $|A|$ off immediately — either route gives the same fraction $|A|/|S|$.

Solution. Total balls $|S| = 5 + 4 + 3 = 12$. Non-red balls number $4 + 3 = 7$. So

$$\mathbb{P}(\text{not red}) = \frac{7}{12} \approx 0.583.$$

(As a complement check: $\mathbb{P}(\text{red}) = \frac{5}{12}$, and $1 - \frac{5}{12} = \frac{7}{12}$.)

Example 2.2.48. A committee of 3 is chosen at random from 7 people, of whom 3 are biologists and 4 are chemists. Find the probability the committee is *all chemists*.

Which rule, and why? "Chosen at random" from a fixed group ⇒ equally likely over all unordered committees, so $\mathbb{P} = |A|/|S|$ with $|S| = \binom{7}{3}$. Order does not matter, which fixes the use of combinations in both counts.

Solution. The number of committees is $|S| = \binom{7}{3} = 35$. An all-chemist committee chooses 3 of the 4 chemists: $|A| = \binom{4}{3} = 4$. Hence

$$\mathbb{P}(\text{all chemists}) = \frac{\binom{4}{3}}{\binom{7}{3}} = \frac{4}{35} \approx 0.114.$$

Example 2.2.49. A box contains 20 light bulbs, 4 of them defective. Two bulbs are drawn at random without replacement. Find the probability that *exactly one* is defective.

Which rule, and why? "At random without replacement" from a finite box ⇒ equally likely over the $\binom{20}{2}$ unordered pairs. "Exactly one defective" is a counting event: choose one defective and one good. Pure classical model.

Solution. Total pairs $|S| = \binom{20}{2} = 190$. "Exactly one defective" picks 1 of the 4 defectives and 1 of the 16 good bulbs:

$$|A| = \binom{4}{1}\binom{16}{1} = 4 \cdot 16 = 64, \qquad \mathbb{P} = \frac{64}{190} = \frac{32}{95} \approx 0.337.$$

Example 2.2.50. A fair coin is tossed 3 times, recording the ordered sequence of heads and tails. Find the probability of getting exactly 2 heads.

Which rule, and why? A *fair* coin tossed a fixed number of times with the ordered sequence recorded gives $2^3 = 8$ equally likely outcomes $\Rightarrow$ classical model. We count sequences with exactly two H's. No overlap or complement structure is needed.

Solution. The 8 equally likely sequences are HHH, HHT, HTH, THH, HTT, THT, TTH, TTT. Exactly two heads occur in HHT, HTH, THH — three sequences. Therefore

$$\mathbb{P}(\text{exactly 2 heads}) = \frac{3}{8} = 0.375.$$

The favourable count $3 = \binom{3}{2}$ is the number of ways to place the two heads among the three tosses.

Example 2.2.51. Two fair dice are rolled. Find the probability that the sum equals 6.

Which rule, and why? "Two *fair* dice" gives 36 equally likely ordered outcomes $\Rightarrow$ the classical model $|A|/|S|$; we just count the pairs that sum to 6. No "or" or "at least" structure appears, so neither inclusion–exclusion nor the complement is called for.

Solution. The pairs summing to 6 are $(1,5), (2,4), (3,3), (4,2), (5,1)$ — five of them. So

$$\mathbb{P}(\text{sum} = 6) = \frac{5}{36} \approx 0.139.$$

The grid marks the anti-diagonal of 6's (•); count the dots and divide by 36:

+	1	2	3	4	5	6
1	2	3	4	5	•	7
2	3	4	5	•	7	8
3	4	5	•	7	8	9
4	5	•	7	8	9	10
5	•	7	8	9	10	11
6	7	8	9	10	11	12

Example 2.2.52. A committee of 4 is chosen at random from 9 people, of whom 5 are seniors and 4 are juniors. Find the probability the committee is *all seniors*.

Which rule, and why? "Chosen at random" from a fixed group $\Rightarrow$ equally likely over all unordered committees, so $\mathbb{P} = |A|/|S|$ with $|S| = \binom{9}{4}$. Order does not matter (a committee is a set), which fixes combinations in both counts.

Solution. The number of committees is $|S| = \binom{9}{4} = 126$. An all-senior committee chooses 4 of the 5 seniors: $|A| = \binom{5}{4} = 5$. Hence

$$\mathbb{P}(\text{all seniors}) = \frac{\binom{5}{4}}{\binom{9}{4}} = \frac{5}{126} \approx 0.040.$$

Example 2.2.53. An urn holds 6 red and 4 white balls. Two are drawn at random without replacement. Find the probability that *both* are red.

Which rule, and why? "At random without replacement" from a finite urn $\Rightarrow$ equally likely over the $\binom{10}{2}$ unordered pairs. "Both red" is a pure counting event (choose 2 of the 6 reds), so the classical model applies with no complement or union needed.

Solution. Total pairs $|S| = \binom{10}{2} = 45$. Both-red pairs number $\binom{6}{2} = 15$. So

$$\mathbb{P}(\text{both red}) = \frac{\binom{6}{2}}{\binom{10}{2}} = \frac{15}{45} = \frac{1}{3} \approx 0.333.$$

The stars-and-balls sketch shows the pool we draw from — 6 shaded (red) and 4 open (white); both picks must come from the shaded group:



Example 2.2.54. Two digits are dialled at random and independently, each uniform on 0–9. Find the probability that the two digits are *equal*.

Which rule, and why? Two independent uniform digits give $10 \times 10 = 100$ equally likely ordered pairs $\Rightarrow$ classical model. "Equal" is a counting event (the diagonal pairs), so we just tally $|A|$ over $|S| = 100$.

Solution. Of the 100 equally likely pairs (d_1, d_2), the equal ones are $(0,0), (1,1), \ldots, (9,9)$ — exactly 10. So

$$\mathbb{P}(\text{equal}) = \frac{10}{100} = \frac{1}{10} = 0.1.$$

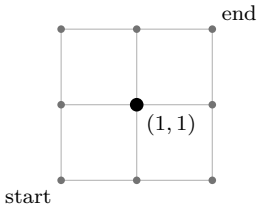
Example 2.2.55. On a 2×2 city grid, a courier walks from the bottom-left corner to the top-right corner using only unit steps right (R) or up (U); every such shortest route is equally likely. Find the probability the route passes through the centre lattice point.

Which rule, and why? "Every shortest route equally likely" $\Rightarrow$ classical model over the set of monotone lattice paths; we count favourable paths over total paths, $|A|/|S|$. Each path is a rearrangement of two R's and two U's, so the counts are binomial coefficients — a clean combinatorial equally-likely problem, not a union or complement.

Solution. A shortest route is any arrangement of 2 R's and 2 U's, so $|S| = \binom{4}{2} = 6$. To pass through the centre point $(1,1)$, the route must reach $(1,1)$ in 2 steps (one R, one U: $\binom{2}{1} = 2$ ways) and then finish to $(2,2)$ in 2 steps (one R, one U: $\binom{2}{1} = 2$ ways), giving $|A| = 2 \times 2 = 4$. Hence

$$\mathbb{P}(\text{through centre}) = \frac{4}{6} = \frac{2}{3} \approx 0.667.$$

The grid shows the lattice; the marked centre point $(1,1)$ is the one every favourable route must visit:



2.3 Conditional Probability, Total Probability, and Bayes

A single idea underlies everything that follows: learning that an event has occurred shrinks the relevant sample space and forces us to rescale. From that one definition flow the multiplication rule (a product of stage-by-stage chances), the law of total probability (a weighted sum over disjoint cases), and Bayes' rule (running the conditioning backwards from evidence to cause). The practical skill is recognising *which* of those four moves a given problem is asking for.

Which rule does this problem need?

Read the problem and ask, in order:

- **Are you told one thing happened and asked the chance of another, in the same trial?** Phrases like "given that," "among those who," "if we know," "restricted to" signal **conditional probability**: $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$. You divide by the probability of the thing you were told.
- **Do you want the chance that several things ALL happen, described as a sequence of stages?** Words like "and then," "followed by," "both," "all three" point to the **multiplication / chain rule**: multiply the chance of the first stage by the conditional chance of the next, and so on. You build $\mathbb{P}(A \cap B)$ from $\mathbb{P}(A)\mathbb{P}(B \mid A)$.

- **Does the experiment split into mutually exclusive cases, and you want the overall chance of some outcome?** A setup such as "one of several machines / urns / routes is chosen, then ..." with the case unknown calls for the **law of total probability**: sum $\mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$ over the cases A_i.
- **Are you given the forward chances $\mathbb{P}(B \mid A_i)$ but asked for a backward one $\mathbb{P}(A_j \mid B)$?** The tell is that the conditioning is reversed from how the data are stated: "a positive test came back — what is the chance of disease?" or "the part was defective — which supplier likely made it?" That is **Bayes' rule**: posterior $\propto$ prior $\times$ likelihood.

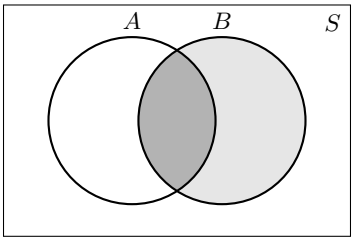
Quick contrast. Total probability and Bayes use the *same* ingredients (a partition plus conditional probabilities); the only difference is the question. If you want $\mathbb{P}(B)$, use total probability. If you already imagine B has happened and want to know which case caused it, use Bayes — and total probability is exactly the denominator you will need.

2.3.1 Definition of conditional probability

Definition 2.3.1. Let B be an event with $\mathbb{P}(B) > 0$. The **conditional probability** of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

The picture: conditioning on B throws away everything outside B and treats B as the new universe of outcomes. The numerator $\mathbb{P}(A \cap B)$ keeps only the part of A that survives inside B; the denominator $\mathbb{P}(B)$ renormalises so that the new universe has total probability 1.



Conditioning on B makes the B-circle the new universe (light); the darker lens $A \cap B$ is the surviving part of A, rescaled by $\mathbb{P}(B)$ so that $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$.

For a fixed conditioning event B, the map $A \mapsto \mathbb{P}(A \mid B)$ satisfies all three axioms (it is nonnegative, gives B probability 1, and is countably additive), so every earlier theorem holds for it. In particular $\mathbb{P}(A^{\mathsf{c}} \mid B) = 1 - \mathbb{P}(A \mid B)$, and inclusion–exclusion works inside the conditional world. When outcomes are equally likely, conditioning is just recounting: $\mathbb{P}(A \mid B) = |A \cap B|/|B|$.

Example 2.3.2 (Reading a die roll)**.** A fair six-sided die is rolled once. Let $A = \{\text{the result is prime}\}$, namely $\{2, 3, 5\}$, and $B = \{\text{the result is at least 4}\}$, namely $\{4, 5, 6\}$. Find $\mathbb{P}(A \mid B)$.

Which rule, and why? The phrase "at least 4" is handed to us as known information — it restricts attention to the three outcomes $\{4, 5, 6\}$ — and then we ask the chance of a prime *within* that restricted world. That "probability of one thing within a known other thing, same single roll" is the signature of conditional probability. There are no stages to multiply (one roll), and no cases to sum over, so the multiplication rule and total probability do not apply.

Solution.

Step 1: identify the events as subsets. $B = \{4, 5, 6\}$ and $A \cap B = \{5\}$ (the only prime that is at least 4).

Step 2: count inside the conditioned space. Outcomes are equally likely, so

$$\mathbb{P}(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{1}{3}.$$

Check via the definition. $\mathbb{P}(A \cap B) = 1/6$ and $\mathbb{P}(B) = 3/6$, so $\mathbb{P}(A \mid B) = (1/6)/(3/6) = 1/3$. The plain (unconditional) chance of a prime was $3/6 = 1/2$; knowing the result is large pulled it down to $1/3$.

Example 2.3.3 (A two-way table of survey responses)**.** A club of 200 members is classified by whether they own a bicycle and whether they commute by car:

	Commutes by car	Does not	Total
Owns a bicycle	30	70	100
Does not own one	80	20	100
Total	110	90	200

A member is chosen at random. Given that the member commutes by car, what is the probability they own a bicycle?

Which rule, and why? "Given that the member commutes by car" fixes a sub-population: we look only at the 110 car commuters, then ask what fraction of *them* own a bicycle. A two-way count table with a "given that" question is a textbook cue for conditional probability; the conditioning event becomes the new denominator.

Solution.

Step 1: the conditioning event. Let $C = \{\text{commutes by car}\}$ and $K = \{\text{owns a bicycle}\}$. The column total gives $\mathbb{P}(C) = 110/200$, and the cell gives $\mathbb{P}(K \cap C) = 30/200$.

Step 2: apply the definition.

$$\mathbb{P}(K \mid C) = \frac{\mathbb{P}(K \cap C)}{\mathbb{P}(C)} = \frac{30/200}{110/200} = \frac{30}{110} = \frac{3}{11} \approx 0.273.$$

Equivalently, read straight off the table: of 110 car commuters, 30 own a bicycle.

Example 2.3.4 (Conditioning changes the answer)**.** Two fair coins are tossed. Let A be "at least one head" and B "the two coins match" (both heads or both tails). Find $\mathbb{P}(B \mid A)$ and compare with $\mathbb{P}(B)$.

Which rule, and why? We are asked for the chance of "matching" *restricted to* the outcomes that already have at least one head. The known event A reshapes the sample space; we want a probability inside it. Single experiment, no sequence of independent stages to multiply, so this is conditioning, not the multiplication rule.

Solution. The equally likely outcomes are $\{HH, HT, TH, TT\}$.

Step 1: list the relevant events. $A = \{HH, HT, TH\}$ so $\mathbb{P}(A) = 3/4$; $B = \{HH, TT\}$; and $A \cap B = \{HH\}$ so $\mathbb{P}(A \cap B) = 1/4$.

Step 2: divide.

$$\mathbb{P}(B \mid A) = \frac{1/4}{3/4} = \frac{1}{3}.$$

Unconditionally $\mathbb{P}(B) = 2/4 = 1/2$. Learning there is at least one head rules out TT, which removes one of the two "matching" outcomes, dropping the chance of a match from $1/2$ to $1/3$.

Example 2.3.5 (Cards drawn without replacement)**.** From a standard 52-card deck, two cards are dealt without replacement. Given that the first card is a heart, what is the probability the second card is also a heart?

Which rule, and why? The first card's suit is stated as known, and we want the second card's chance in that light — a single conditional probability, not a product. (Contrast the next subsection, where we instead want the chance that *both* are hearts; that "both" version is the multiplication rule. Here only the second card's conditional chance is requested.)

Solution.

Step 1: update the deck. Once a heart is removed and known, 51 cards remain, 12 of them hearts.

Step 2: equally likely second card.

$$\mathbb{P}(\text{2nd heart} \mid \text{1st heart}) = \frac{12}{51} = \frac{4}{17} \approx 0.235.$$

The information "first was a heart" both shrinks the deck and depletes the hearts, so the conditional chance $12/51$ sits just below the unconditional $13/52 = 1/4$.

Example 2.3.6 (Two dice, conditioned on a five appearing)**.** Two fair dice are rolled. Given that at least one die shows a five, what is the probability that the sum of the two dice is at least nine?

Which rule, and why? "At least one die shows a five" is handed to us as known information, restricting the 36 equally likely ordered pairs to the eleven that contain a five; we then ask the chance of a large sum *within* that restricted set. Probability of one event inside a known other event, in the same single roll of the pair, is conditional probability. There is no sequence of stages to multiply and no partition to sum over.

Solution.

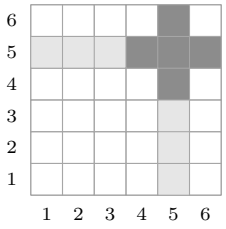
Step 1: list the conditioning event. Let $B = \{\text{a five appears}\}$. The ordered pairs in B are the five rows $(5, j)$, the five columns $(i, 5)$, with $(5, 5)$ counted once, giving $|B| = 11$.

Step 2: count the favourable outcomes inside B. Among those eleven, the ones with sum ≥ 9 are $(5, 4), (4, 5), (5, 5), (5, 6), (6, 5)$, so $|A \cap B| = 5$.

Step 3: divide counts in the conditioned world.

$$\mathbb{P}(A \mid B) = \frac{|A \cap B|}{|B|} = \frac{5}{11} \approx 0.455.$$

light = a five appears; dark = also sum ≥ 9



Unconditionally, $\mathbb{P}(\text{sum} \geq 9) = 10/36 \approx 0.28$; learning that a five is present shifts the odds upward, since a five is a large face that helps build a large sum.

Example 2.3.7 (A committee, conditioned on having a man)**.** A club has 5 women and 3 men. Two members are chosen at random to form a subcommittee. Given that the subcommittee includes at least one man, what is the probability that both members are men?

Which rule, and why? "Includes at least one man" is told to us and shrinks the sample space to those pairs containing a man; we then ask for the chance of an all-male pair *within* that group. That is conditional probability on equally likely two-person subsets. We are not multiplying ordered stages and not partitioning into weighted cases.

Solution.

Step 1: total and the conditioning event. There are $\binom{8}{2} = 28$ equally likely pairs. The all-women pairs number $\binom{5}{2} = 10$, so the event $B = \{\text{at least one man}\}$ has $|B| = 28 - 10 = 18$.

Step 2: favourable pairs inside B. Both men: $\binom{3}{2} = 3$ pairs, and every such pair already lies in B, so $|A \cap B| = 3$.

Step 3: divide.

$$\mathbb{P}(A \mid B) = \frac{3}{18} = \frac{1}{6} \approx 0.167.$$

Counting the complement of B (no men) was the quick route to $|B|$; trying to list "at least one man" directly is the kind of detour the complement trick avoids.

Example 2.3.8 (A drawn ticket, conditioned on parity)**.** A ticket is drawn at random from those numbered 1 through 20. Given that the number drawn is even, what is the probability that it is a multiple of three?

Which rule, and why? "Even" is the given information, narrowing the twenty equally likely tickets to the ten even ones; we then ask the chance of a multiple of three *among the evens*. One draw, one piece of known information, a probability computed inside it: conditional probability by recounting. No stages, no weighted partition.

Solution.

Step 1: the conditioned space. The even tickets are $\{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$, so $|B| = 10$.

Step 2: favourable evens. Multiples of three that are even (i.e. multiples of six) up to 20 are $\{6, 12, 18\}$, so $|A \cap B| = 3$.

Step 3: divide.

$$\mathbb{P}(A \mid B) = \frac{3}{10} = 0.30.$$

The unconditional chance of a multiple of three among 1–20 is $6/20 = 0.30$ as well: here the conditioning happens to leave the proportion unchanged, a reminder that conditioning need not move the answer.

2.3.2 The multiplication rule and the chain rule

Clearing the denominator in the definition gives a formula for the probability that two events both occur.

Theorem 2.3.9 (Multiplication rule)**.** *If $\mathbb{P}(B) > 0$ then $\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$, and symmetrically, if $\mathbb{P}(A) > 0$, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B \mid A)$.*

Proof. Multiply both sides of $\mathbb{P}(A \mid B) = \mathbb{P}(A \cap B)/\mathbb{P}(B)$ by $\mathbb{P}(B)$ to obtain $\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B)$. Swapping the roles of A and B gives the second form, since $A \cap B = B \cap A$. □

Iterating the rule one stage at a time yields the chain rule for any number of events.

Theorem 2.3.10 (Chain rule)**.** *For events $A_1, \ldots, A_n$ with $\mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$,*

$$\mathbb{P}(A_1 \cap A_2 \cap \cdots \cap A_n) = \mathbb{P}(A_1)\,\mathbb{P}(A_2 \mid A_1)\,\mathbb{P}(A_3 \mid A_1 \cap A_2) \cdots \mathbb{P}(A_n \mid A_1 \cap \cdots \cap A_{n-1}).$$

Proof. Induct on n. The case $n = 2$ is the multiplication rule. Assume the formula for $n - 1$ events and set $C = A_1 \cap \cdots \cap A_{n-1}$. By the multiplication rule, $\mathbb{P}(C \cap A_n) = \mathbb{P}(C)\,\mathbb{P}(A_n \mid C)$. Replacing $\mathbb{P}(C)$ by its $(n-1)$-factor expansion (valid because $\mathbb{P}(C) \geq \mathbb{P}(A_1 \cap \cdots \cap A_{n-1}) > 0$ keeps every intermediate conditioning event of positive probability) gives the n-factor formula. □

The chain rule is exactly what a probability **tree** computes: the probability of a leaf is the product of the branch probabilities along the path to it, each branch carrying the conditional probability of that step given everything earlier on the path.

Example 2.3.11 (Both cards hearts)**.** Two cards are dealt without replacement from a 52-card deck. What is the probability that both are hearts?

Which rule, and why? The word "both" asks for $\mathbb{P}(\text{1st heart} \cap \text{2nd heart})$ — an intersection built from two dependent stages. That is the multiplication rule. We are not told either card in advance (so it is not a bare conditional probability), and the two stages are dependent (no replacement), so we must use the *conditional* chance for the second factor, not a plain product of marginals.

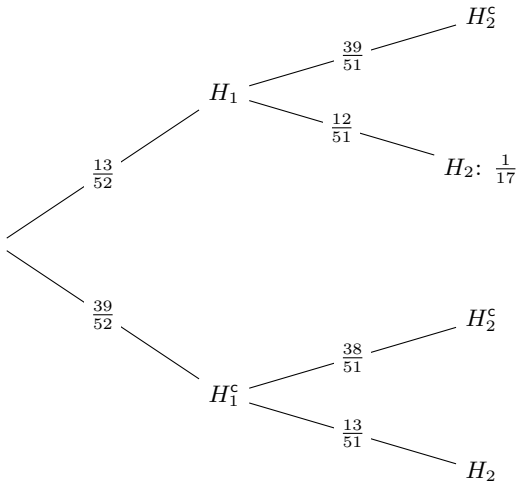
Solution.

Step 1: chance of the first stage. $\mathbb{P}(\text{1st heart}) = 13/52 = 1/4$.

Step 2: conditional chance of the second. Given a heart is gone, $\mathbb{P}(\text{2nd heart} \mid \text{1st heart}) = 12/51$.

Step 3: multiply.

$$\mathbb{P}(\text{both hearts}) = \frac{13}{52} \cdot \frac{12}{51} = \frac{1}{4} \cdot \frac{12}{51} = \frac{12}{204} = \frac{1}{17} \approx 0.0588.$$



(H_i = "card i is a heart." The top path multiplies $\frac{13}{52} \cdot \frac{12}{51}$.)

Example 2.3.12 (Three defectives in a row)**.** A box contains 20 widgets, of which 5 are defective. Three are drawn at random without replacement. What is the probability that all three are defective?

Which rule, and why? "All three" is an intersection across three dependent stages, so it is the chain rule. Each draw changes the contents of the box, so the stages are dependent and each factor must be conditioned on the earlier draws — a plain $(5/20)^3$ would be wrong because it ignores depletion.

Solution. Let $D_i = \{\text{the } i\text{th draw is defective}\}$.

$$\mathbb{P}(D_1 \cap D_2 \cap D_3) = \mathbb{P}(D_1)\,\mathbb{P}(D_2 \mid D_1)\,\mathbb{P}(D_3 \mid D_1 \cap D_2) = \frac{5}{20} \cdot \frac{4}{19} \cdot \frac{3}{18}.$$

Compute.

$$= \frac{5 \cdot 4 \cdot 3}{20 \cdot 19 \cdot 18} = \frac{60}{6840} = \frac{1}{114} \approx 0.00877.$$

Each successful draw both shrinks the box and removes a defective, so every factor falls.

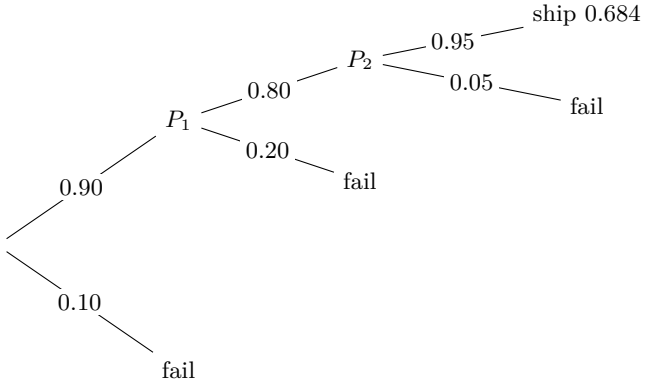
Example 2.3.13 (An assembly line that must clear three checks)**.** A part must pass three independent-quality stages to ship. Stage 1 passes 90% of parts. Of the parts that clear stage 1, stage 2 passes 80%. Of those that clear both, stage 3 passes 95%. What fraction of parts ship?

Which rule, and why? Each percentage is stated *conditionally on surviving the previous stage* ("of the parts that clear stage 1…"), and we want the chance of clearing all three — an intersection of sequential, dependent-in-description stages. That is the chain rule: multiply the stagewise conditional pass rates.

Solution. Let $P_i = \{\text{clears stage } i\}$.

$$\mathbb{P}(P_1 \cap P_2 \cap P_3) = \mathbb{P}(P_1)\,\mathbb{P}(P_2 \mid P_1)\,\mathbb{P}(P_3 \mid P_1 \cap P_2) = 0.90 \times 0.80 \times 0.95.$$

Compute. $0.90 \times 0.80 = 0.72$, and $0.72 \times 0.95 = 0.684$. So about **68.4**% of parts ship.



Example 2.3.14 (Drawing a specific ordered hand)**.** A bag holds 4 red, 3 green, and 2 blue marbles. Three marbles are drawn one at a time without replacement. What is the probability the colours come out red, then green, then red, in that order?

Which rule, and why? A specified *ordered* sequence of dependent draws is precisely a single path through a tree, and the chain rule gives a path's probability as the product of its branch probabilities. The order is fixed, so we do not sum over arrangements; the draws deplete the bag, so each factor is conditioned on the earlier ones.

Solution. Start with 9 marbles (4R, 3G, 2B).

$$\mathbb{P}(R, G, R) = \frac{4}{9} \cdot \frac{3}{8} \cdot \frac{3}{7},$$

where after one red the reds drop to 3 of 8, and after the green the reds are still 3 but only 7 marbles remain.

Compute.

$$= \frac{4 \cdot 3 \cdot 3}{9 \cdot 8 \cdot 7} = \frac{36}{504} = \frac{1}{14} \approx 0.0714.$$

Example 2.3.15 (Both cards aces)**.** Two cards are dealt without replacement from a standard 52-card deck. What is the probability that both are aces?

Which rule, and why? "Both" asks for the intersection $\mathbb{P}(\text{1st ace} \cap \text{2nd ace})$ across two dependent draws, which the multiplication rule builds as a product. Neither card is revealed in advance, so it is not a bare conditional probability; and because the first ace is not replaced, the second factor must be conditioned, not a plain $(4/52)^2$.

Solution.

Step 1: first stage. $\mathbb{P}(\text{1st ace}) = 4/52 = 1/13$.

Step 2: conditional second stage. With one ace gone, $\mathbb{P}(\text{2nd ace} \mid \text{1st ace}) = 3/51$.

Step 3: multiply.

$$\mathbb{P}(\text{both aces}) = \frac{4}{52} \cdot \frac{3}{51} = \frac{12}{2652} = \frac{1}{221} \approx 0.00452.$$

Example 2.3.16 (Red then blue from a bag)**.** A bag holds 5 red and 4 blue marbles. Two are drawn one at a time without replacement. What is the probability the first is red and the second is blue?

Which rule, and why? A specified ordered outcome (red, then blue) over two dependent draws is one path through a tree, and the chain rule makes a path's probability the product of its branch chances. The order is fixed, so we do not sum over arrangements; the draw is without replacement, so the second factor is conditioned on the first.

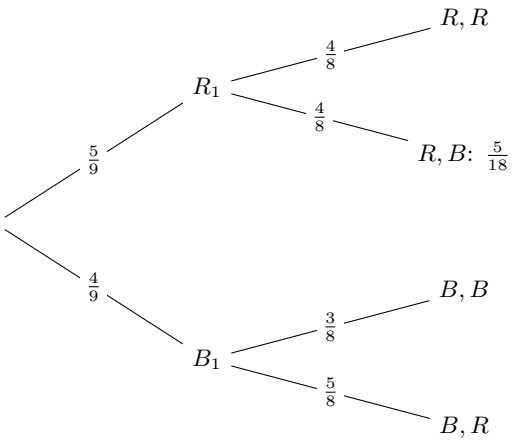
Solution.

Step 1: first stage. $\mathbb{P}(\text{1st red}) = 5/9$.

Step 2: conditional second stage. A red is gone, leaving 8 marbles of which 4 are blue, so $\mathbb{P}(\text{2nd blue} \mid \text{1st red}) = 4/8 = 1/2$.

Step 3: multiply.

$$\mathbb{P}(R \text{ then } B) = \frac{5}{9} \cdot \frac{1}{2} = \frac{5}{18} \approx 0.278.$$



(R_1, B_1 = colour of the first marble. The top path multiplies $\frac{5}{9} \cdot \frac{4}{8} = \frac{5}{18}$.)

Example 2.3.17 (At least one bad fuse: the complement trick)**.** A box of 12 fuses contains 3 defective ones. Four fuses are drawn at random without replacement. What is the probability that at least one of the four is defective?

Which rule, and why? "At least one" over several dependent draws is awkward to attack head-on, but its complement — "none defective" — is a single ordered survival path, exactly what the chain rule multiplies. So we use the multiplication rule on the complement, then subtract from 1. A direct case-by-case sum would be the slow, error-prone route.

Solution.

Step 1: probability that no draw is defective. The 9 good fuses must be drawn each time:

$$\mathbb{P}(\text{none bad}) = \frac{9}{12} \cdot \frac{8}{11} \cdot \frac{7}{10} \cdot \frac{6}{9} = \frac{3024}{11{,}880} = \frac{14}{55}.$$

Step 2: complement.

$$\mathbb{P}(\text{at least one bad}) = 1 - \frac{14}{55} = \frac{41}{55} \approx 0.745.$$

Subtracting one easy product beats summing the "exactly one," "exactly two," and "exactly three" cases separately.

Example 2.3.18 (Four reds in a row)**.** Four cards are dealt without replacement from a 52-card deck. What is the probability that all four are red?

Which rule, and why? "All four" is an intersection across four dependent stages, so it is the chain rule: each card removed changes the deck, and each factor is conditioned on the cards already drawn. The naive $(26/52)^4 = (1/2)^4$ would wrongly ignore that reds are depleted as we go.

Solution. Let $R_i = \{\text{card } i \text{ is red}\}$. There are 26 reds at the start.

$$\mathbb{P}(R_1 \cap R_2 \cap R_3 \cap R_4) = \frac{26}{52} \cdot \frac{25}{51} \cdot \frac{24}{50} \cdot \frac{23}{49}.$$

Compute.

$$= \frac{26 \cdot 25 \cdot 24 \cdot 23}{52 \cdot 51 \cdot 50 \cdot 49} = \frac{358{,}800}{6{,}497{,}400} = \frac{46}{833} \approx 0.0552.$$

The honest answer 0.0552 sits below the careless $(1/2)^4 = 0.0625$, precisely because each red drawn makes the next red slightly less likely.

2.3.3 The law of total probability

When an experiment naturally breaks into mutually exclusive cases, we can find the probability of any event by handling each case separately and weighting by how likely that case is.

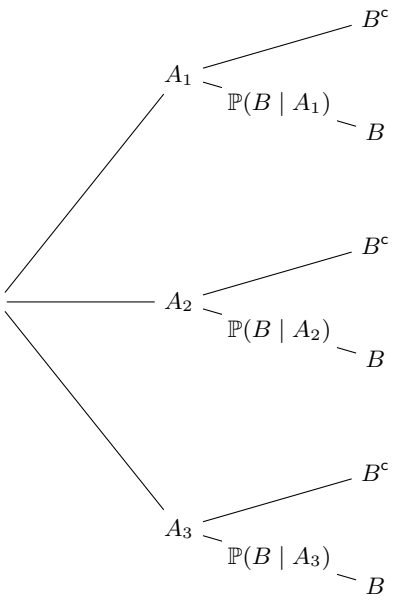
Definition 2.3.19. A **partition** of S is a collection of events $A_1, \ldots, A_n$ that are pairwise disjoint ($A_i \cap A_j = \varnothing$ for $i \neq j$), exhaust the space ($\bigcup_i A_i = S$), and satisfy $\mathbb{P}(A_i) > 0$ for each i.

Theorem 2.3.20 (Law of total probability)**.** *If $A_1, \ldots, A_n$ partition S, then for every event B,*

$$\mathbb{P}(B) = \sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\, \mathbb{P}(A_i).$$

Proof. Because the A_i partition S, the events $B \cap A_i$ are pairwise disjoint and their union is $B \cap S = B$. By countable additivity (Axiom 3), $\mathbb{P}(B) = \sum_i \mathbb{P}(B \cap A_i)$. The multiplication rule rewrites each term as $\mathbb{P}(B \cap A_i) = \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$, giving the stated sum. □

In tree language: each first-stage branch is a case A_i with weight $\mathbb{P}(A_i)$; from it a branch to B carries $\mathbb{P}(B \mid A_i)$; the probability of B is the sum of the path products that end in B.



$\mathbb{P}(B) = \sum_i \mathbb{P}(A_i)\, \mathbb{P}(B \mid A_i)$ sums the B-leaves.

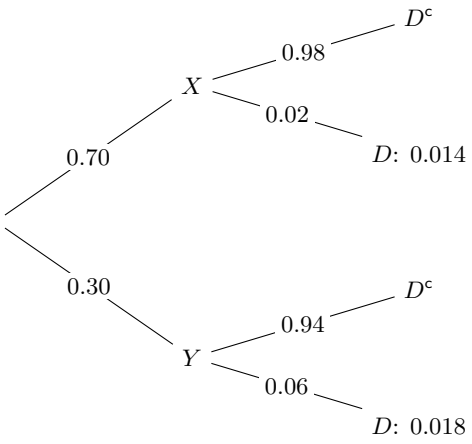
Example 2.3.21 (Two suppliers feeding a line)**.** A factory buys bolts from two suppliers. Supplier X provides 70% of the bolts and 2% of X's bolts are defective; supplier Y provides the other 30% and 6% of Y's bolts are defective. A bolt is picked at random from the combined stock. What is the probability it is defective?

Which rule, and why? The stock splits into two exhaustive, mutually exclusive cases (came from X, came from Y) with known weights, and the defect rate is given *per case*. We want the overall defect chance with the source unknown — a weighted sum across the cases, i.e. the law of total probability. It is not Bayes because the question runs forward (overall chance of defective), not backward (which supplier).

Solution. Let $D = \{\text{defective}\}$, with cases X and Y.

$$\mathbb{P}(D) = \mathbb{P}(D \mid X)\mathbb{P}(X) + \mathbb{P}(D \mid Y)\mathbb{P}(Y) = (0.02)(0.70) + (0.06)(0.30).$$

Compute. $= 0.014 + 0.018 = 0.032$. So $\mathbb{P}(D) = \mathbf{0.032}$ (a 3.2% overall defect rate).



$$\mathbb{P}(D) = 0.014 + 0.018 = 0.032.$$

Example 2.3.22 (Pick an urn, then a chip)**.** Urn I holds 2 red and 3 white chips; urn II holds 4 red and 1 white. A fair coin decides which urn to use, then one chip is drawn. What is the probability the chip is red?

Which rule, and why? The choice of urn partitions the experiment into two equally likely cases, and the red-chip chance is given *conditional on the urn*. With the urn unknown, the overall red chance is the weighted sum — total probability. The two-stage "choose a case, then observe" structure is the classic trigger.

Solution. Let $R = \{\text{red}\}$.

$$\mathbb{P}(R) = \mathbb{P}(R \mid \text{I})\mathbb{P}(\text{I}) + \mathbb{P}(R \mid \text{II})\mathbb{P}(\text{II}) = \frac{2}{5} \cdot \frac{1}{2} + \frac{4}{5} \cdot \frac{1}{2}.$$

Compute. $= \frac{1}{5} + \frac{2}{5} = \frac{3}{5} = 0.6$. So $\mathbb{P}(R) = \mathbf{3/5}$.

Example 2.3.23 (Weather and a late train)**.** On a given morning the weather is rainy with probability 0.3 and clear with probability 0.7. The commuter's train is late 40% of rainy mornings but only 10% of clear mornings. What is the probability the train is late on a random morning?

Which rule, and why? "Rainy" and "clear" form a partition of the morning, and lateness is given per weather case. We want the unconditional chance of lateness, so we average the case-by-case lateness rates by their weather weights: the law of total probability. No reversal is requested, so this is not Bayes (yet).

Solution. Let $L = \{\text{late}\}$, with cases Rain and Clear.

$$\mathbb{P}(L) = \mathbb{P}(L \mid \text{Rain})\mathbb{P}(\text{Rain}) + \mathbb{P}(L \mid \text{Clear})\mathbb{P}(\text{Clear}) = (0.40)(0.30) + (0.10)(0.70).$$

Compute. $= 0.12 + 0.07 = 0.19$. So the train is late with probability $\mathbf{0.19}$.

Example 2.3.24 (Three machines of unequal output)**.** Three machines produce, respectively, 50%, 30%, and 20% of a day's output. Their defect rates are 1%, 3%, and 4%. A finished item is selected at random. What is the probability it is defective?

Which rule, and why? A three-case partition (which machine made it) with per-machine defect rates and an overall question is total probability with three terms. The presence of three exhaustive weighted cases, not two, does not change the rule — just the number of summands.

Solution. Let $D = \{\text{defective}\}$ and M_1, M_2, M_3 the machines.

$$\mathbb{P}(D) = \sum_{i=1}^{3} \mathbb{P}(D \mid M_i)\mathbb{P}(M_i) = (0.01)(0.50) + (0.03)(0.30) + (0.04)(0.20).$$

Compute. $= 0.005 + 0.009 + 0.008 = 0.022$. So $\mathbb{P}(D) = \mathbf{0.022}$.

Machine	Share $\mathbb{P}(M_i)$	Defect rate $\mathbb{P}(D \mid M_i)$	Contribution
M_1	0.50	0.01	0.005
M_2	0.30	0.03	0.009
M_3	0.20	0.04	0.008
		Total $\mathbb{P}(D)$	0.022

Example 2.3.25 (Two routes to work)**.** A commuter takes the highway 60% of mornings and the back road the other 40%. The highway makes her late 15% of the time; the back road makes her late 30% of the time. What is the probability she is late on a random morning?

Which rule, and why? The morning splits into two exhaustive, mutually exclusive cases (highway or back road) with known weights, and lateness is given *per route*. With the route unaveraged, the overall late chance is the weighted sum, the law of total probability. It is not Bayes: the question runs forward to the overall rate, not backward from lateness to a route.

Solution. Let $L = \{\text{late}\}$ with cases H (highway) and R (back road).

$$\mathbb{P}(L) = \mathbb{P}(L \mid H)\mathbb{P}(H) + \mathbb{P}(L \mid R)\mathbb{P}(R) = (0.15)(0.60) + (0.30)(0.40).$$

Compute. $= 0.09 + 0.12 = 0.21$. So $\mathbb{P}(L) = \mathbf{0.21}$.

Example 2.3.26 (Three prize boxes)**.** A game-show contestant picks one of three boxes at random with probabilities 0.30, 0.45, and 0.25. Box 1 contains a prize with chance 0.20, box 2 with chance 0.50, and box 3 with chance 0.90. What is the probability the chosen box holds a prize?

Which rule, and why? Which box is chosen forms a three-case partition with given weights, and the prize chance is stated *conditional on the box*. With the box left general, the overall prize chance is the weighted sum over the three cases, the law of total probability. Three weighted cases instead of two changes only the number of summands, not the rule.

Solution. Let $W = \{\text{prize}\}$ and B_1, B_2, B_3 the boxes.

$$\mathbb{P}(W) = \sum_{i=1}^{3} \mathbb{P}(W \mid B_i)\mathbb{P}(B_i) = (0.20)(0.30) + (0.50)(0.45) + (0.90)(0.25).$$

Compute. $= 0.06 + 0.225 + 0.225 = 0.51$. So $\mathbb{P}(W) = \mathbf{0.51}$.

Box	Weight $\mathbb{P}(B_i)$	Prize rate $\mathbb{P}(W \mid B_i)$	Contribution
B_1	0.30	0.20	0.060
B_2	0.45	0.50	0.225
B_3	0.25	0.90	0.225
		Total $\mathbb{P}(W)$	0.510

Example 2.3.27 (Draw a card, then flip a coin)**.** A card is drawn from a 52-card deck. If it is a face card (jack, queen, or king), a coin that lands heads with probability 0.7 is flipped; otherwise a fair coin is flipped. What is the probability the flip is heads?

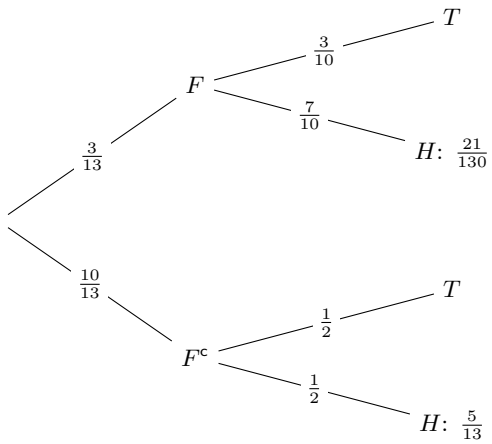
Which rule, and why? The card splits the experiment into two exhaustive cases — face card or not — with known weights 12/52 and 40/52, and the heads chance is given *per case*. The coin's bias depends on the hidden card, so we average over the cases: the law of total probability. The forward "overall chance of heads" question rules out Bayes.

Solution. Let $F = \{\text{face card}\}$, so $\mathbb{P}(F) = 12/52 = 3/13$ and $\mathbb{P}(F^{\mathsf{c}}) = 40/52 = 10/13$.

$$\mathbb{P}(H) = \mathbb{P}(H \mid F)\mathbb{P}(F) + \mathbb{P}(H \mid F^{\mathsf{c}})\mathbb{P}(F^{\mathsf{c}}) = \frac{7}{10} \cdot \frac{3}{13} + \frac{1}{2} \cdot \frac{10}{13}.$$

Compute.

$$= \frac{21}{130} + \frac{5}{13} = \frac{21}{130} + \frac{50}{130} = \frac{71}{130} \approx 0.546.$$



$$\mathbb{P}(H) = \tfrac{21}{130} + \tfrac{50}{130} = \tfrac{71}{130}.$$

Example 2.3.28 (Two kinds of seed)**.** A seed packet is 40% early-variety seeds, which germinate with probability 0.90, and 60% late-variety seeds, which germinate with probability 0.60. A seed is planted at random. What is the probability it germinates?

Which rule, and why? The variety splits the seeds into two exhaustive weighted cases, with germination given *per variety*. With the variety unknown, the overall germination chance is the weighted average across the two cases, the law of total probability. The question asks the overall forward rate, not which variety a sprouted seed was, so it is not Bayes.

Solution. Let $G = \{\text{germinates}\}$ with cases E (early) and L (late).

$$\mathbb{P}(G) = \mathbb{P}(G \mid E)\mathbb{P}(E) + \mathbb{P}(G \mid L)\mathbb{P}(L) = (0.90)(0.40) + (0.60)(0.60).$$

Compute. $= 0.36 + 0.36 = 0.72$. So $\mathbb{P}(G) = \mathbf{0.72}$.

2.3.4 Bayes' rule

Total probability builds $\mathbb{P}(B)$ forward from the cases. Bayes' rule runs the same machine backward: given that B occurred, it tells us how to revise our belief about which case A_j we are in.

Theorem 2.3.29 (Bayes' rule)**.** *If $A_1, \ldots, A_n$ partition S and $\mathbb{P}(B) > 0$, then for each j,*

$$\mathbb{P}(A_j \mid B) = \frac{\mathbb{P}(B \mid A_j)\,\mathbb{P}(A_j)}{\sum_{i=1}^{n} \mathbb{P}(B \mid A_i)\,\mathbb{P}(A_i)}.$$

Proof. By the definition of conditional probability, $\mathbb{P}(A_j \mid B) = \mathbb{P}(A_j \cap B)/\mathbb{P}(B)$. The multiplication rule turns the numerator into $\mathbb{P}(A_j \cap B) = \mathbb{P}(B \mid A_j)\mathbb{P}(A_j)$, and the law of total probability expands the denominator into $\mathbb{P}(B) = \sum_i \mathbb{P}(B \mid A_i)\mathbb{P}(A_i)$. Substituting both gives the formula. □

Read the structure as **posterior** $\propto$ **prior** $\times$ **likelihood**: the *prior* $\mathbb{P}(A_j)$ (belief before evidence) is multiplied by the *likelihood* $\mathbb{P}(B \mid A_j)$ (how well case j explains the evidence), and the denominator merely normalises the revised numbers so they sum to 1 across the cases.

Example 2.3.30 (A diagnostic test and the base rate)**.** A disease is present in 1% of a population. A screening test is 95% sensitive (it returns positive for 95% of truly ill people) and has a 5% false-positive rate (it returns positive for 5% of healthy people). A randomly chosen person tests positive. What is the probability they actually have the disease?

Which rule, and why? The data are forward likelihoods — chance of a positive test *given* disease status — but the question reverses them: chance of disease *given* a positive test. Reversed conditioning with a known prior (the 1% base rate) is the unmistakable signature of Bayes' rule. We will need total probability for $\mathbb{P}(+)$ as the denominator.

Solution. Let $D = \{\text{disease}\}$, $D^{\mathsf{c}} = \{\text{healthy}\}$, $+ = \{\text{positive}\}$. Prior $\mathbb{P}(D) = 0.01$, likelihoods $\mathbb{P}(+ \mid D) = 0.95$ and $\mathbb{P}(+ \mid D^{\mathsf{c}}) = 0.05$.

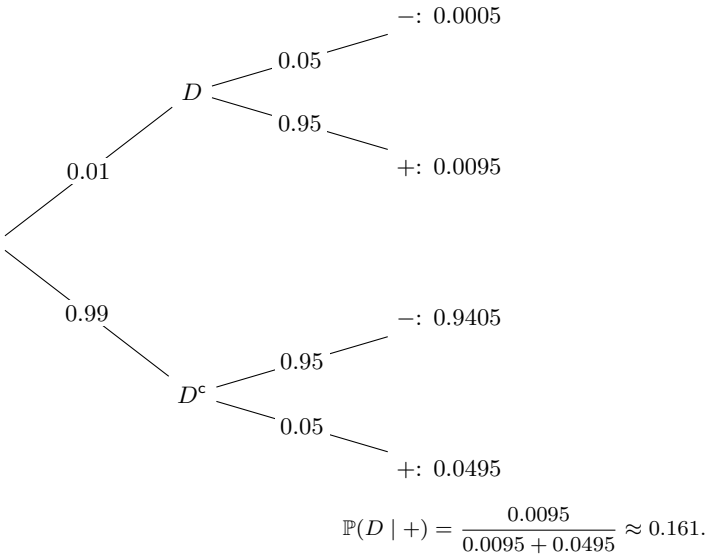
Step 1: denominator by total probability.

$$\mathbb{P}(+) = \mathbb{P}(+ \mid D)\mathbb{P}(D) + \mathbb{P}(+ \mid D^{\mathsf{c}})\mathbb{P}(D^{\mathsf{c}}) = (0.95)(0.01) + (0.05)(0.99) = 0.0095 + 0.0495 = 0.059.$$

Step 2: Bayes.

$$\mathbb{P}(D \mid +) = \frac{\mathbb{P}(+ \mid D)\mathbb{P}(D)}{\mathbb{P}(+)} = \frac{0.0095}{0.059} \approx 0.161.$$

Despite a fairly accurate test, a positive result carries only about a **16**% chance of true disease, because the disease is rare: most positives are false positives drawn from the large healthy majority. This is the base-rate lesson.



Example 2.3.31 (Which supplier made the defective bolt?)**.** Using the earlier factory: supplier X provides 70% of bolts at a 2% defect rate, and supplier Y provides 30% at a 6% defect rate. A randomly chosen bolt turns out to be defective. What is the probability it came from supplier Y?

Which rule, and why? We are told the bolt *is* defective and asked to trace it back to a supplier — conditioning reversed from the forward defect rates. Backward inference toward the hidden cause is Bayes. We already computed the needed denominator $\mathbb{P}(D) = 0.032$ via total probability.

Solution.

Step 1: recall the denominator. $\mathbb{P}(D) = (0.02)(0.70) + (0.06)(0.30) = 0.014 + 0.018 = 0.032$.

Step 2: Bayes for supplier Y.

$$\mathbb{P}(Y \mid D) = \frac{\mathbb{P}(D \mid Y)\mathbb{P}(Y)}{\mathbb{P}(D)} = \frac{(0.06)(0.30)}{0.032} = \frac{0.018}{0.032} = 0.5625.$$

Although Y supplies only 30% of bolts, its higher defect rate makes it the more likely source of a defective one: $\mathbb{P}(Y \mid D) = \mathbf{0.5625}$ versus $\mathbb{P}(X \mid D) = 0.014/0.032 = 0.4375$.

Example 2.3.32 (Updating the urn after seeing red)**.** Urn I holds 2 red and 3 white chips; urn II holds 4 red and 1 white. A fair coin picks the urn, and the drawn chip is red. What is the probability the chip came from urn II?

Which rule, and why? The draw outcome (red) is observed, and we want to revise which urn was used — reverse conditioning from a forward draw probability. That is Bayes. The denominator $\mathbb{P}(R) = 3/5$ is the total-probability result from the matching example above.

Solution.

Step 1: denominator. $\mathbb{P}(R) = \frac{2}{5} \cdot \frac{1}{2} + \frac{4}{5} \cdot \frac{1}{2} = \frac{3}{5}$.

Step 2: Bayes for urn II.

$$\mathbb{P}(\text{II} \mid R) = \frac{\mathbb{P}(R \mid \text{II})\mathbb{P}(\text{II})}{\mathbb{P}(R)} = \frac{\frac{4}{5} \cdot \frac{1}{2}}{\frac{3}{5}} = \frac{2/5}{3/5} = \frac{2}{3}.$$

Seeing red shifts belief toward the red-rich urn II, from a prior 1/2 up to a posterior $\mathbf{2/3}$.

Example 2.3.33 (Three machines, tracing a defect)**.** Recall the three machines making 50%, 30%, 20% of output with defect rates 1%, 3%, 4%. A defective item is found. Which machine most probably produced it, and with what probability?

Which rule, and why? A defective item is in hand and we want the responsible machine — reverse inference over a three-case partition, hence Bayes with three competing numerators sharing the same denominator $\mathbb{P}(D)$. Total probability supplied that denominator (0.022); Bayes distributes it.

Solution.

Step 1: denominator. From before, $\mathbb{P}(D) = 0.005 + 0.009 + 0.008 = 0.022$.

Step 2: posterior for each machine.

$$\mathbb{P}(M_1 \mid D) = \frac{0.005}{0.022} \approx 0.227, \quad \mathbb{P}(M_2 \mid D) = \frac{0.009}{0.022} \approx 0.409, \quad \mathbb{P}(M_3 \mid D) = \frac{0.008}{0.022} \approx 0.364.$$

The posteriors sum to 1. The most probable culprit is $\mathbf{M_2}$ at about 0.409: although M_1 makes the most items, its low defect rate makes it the least likely source of a defect.

Machine	Prior $\mathbb{P}(M_i)$	Likelihood $\mathbb{P}(D \mid M_i)$	Joint	Posterior $\mathbb{P}(M_i \mid D)$
M_1	0.50	0.01	0.005	0.227
M_2	0.30	0.03	0.009	0.409
M_3	0.20	0.04	0.008	0.364
		Total	0.022	1.000

Example 2.3.34 (Two coins, one biased)**.** A drawer has two coins that look identical: a fair coin and a trick coin that lands heads with probability 0.8. One coin is chosen at random and flipped twice, giving two heads. What is the probability the chosen coin was the trick coin?

Which rule, and why? The evidence (two heads) is observed and we want to update which coin was drawn — reversed conditioning, so Bayes. The likelihood of the evidence under each coin must be computed first (two flips, so square the per-flip head chance), then combined with the 1/2 prior. Total probability gives the denominator $\mathbb{P}(HH)$.

Solution. Let $T = \{\text{trick coin}\}$, $F = \{\text{fair coin}\}$, each with prior 1/2.

Step 1: likelihood of two heads under each coin. $\mathbb{P}(HH \mid F) = (1/2)^2 = 0.25$ and $\mathbb{P}(HH \mid T) = (0.8)^2 = 0.64$.

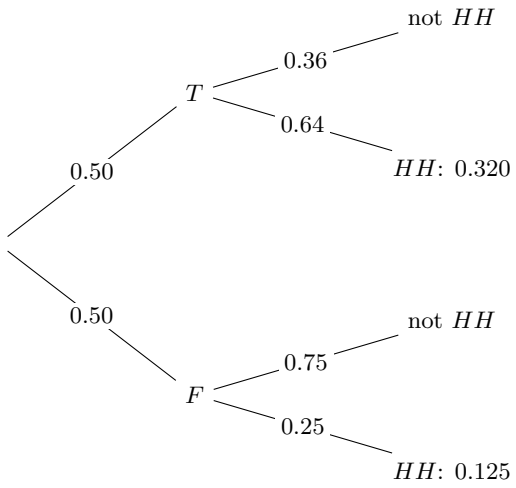
Step 2: denominator by total probability.

$$\mathbb{P}(HH) = (0.64)(0.5) + (0.25)(0.5) = 0.32 + 0.125 = 0.445.$$

Step 3: Bayes.

$$\mathbb{P}(T \mid HH) = \frac{(0.64)(0.5)}{0.445} = \frac{0.32}{0.445} \approx 0.719.$$

Two heads push belief in the trick coin from a prior 0.5 to a posterior of about $\mathbf{0.719}$, since the biased coin explains the heads better than the fair one.



$$\mathbb{P}(T \mid HH) = \frac{0.320}{0.320 + 0.125} \approx 0.719.$$

Example 2.3.35 (A spam filter, base rate matters)**.** Of all incoming email, 30% is spam. A filter flags 95% of spam messages but also flags 2% of legitimate messages. A message has just been flagged. What is the probability it is actually spam?

Which rule, and why? The data are forward likelihoods — chance of being flagged *given* the message's true type — but the question reverses them to the chance of spam *given* a flag. Reversed conditioning with a known prior (the 30% spam rate) is Bayes. Total probability supplies the denominator $\mathbb{P}(\text{flag})$.

Solution. Let $\text{Sp} = \{\text{spam}\}$, $\text{Sp}^{\text{c}} = \{\text{legitimate}\}$, and $\text{Fl} = \{\text{flagged}\}$, with prior $\mathbb{P}(\text{Sp}) = 0.30$.

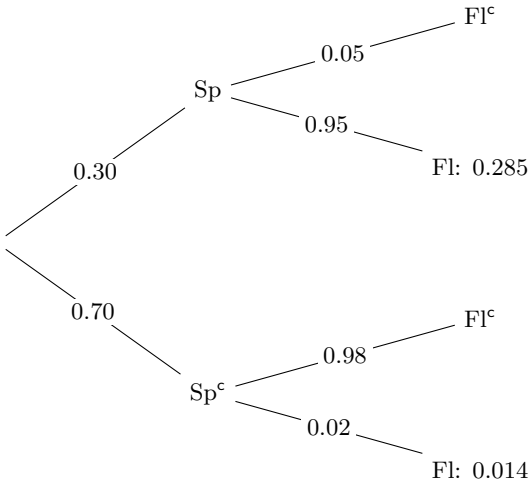
Step 1: denominator by total probability.

$$\mathbb{P}(\text{Fl}) = (0.95)(0.30) + (0.02)(0.70) = 0.285 + 0.014 = 0.299.$$

Step 2: Bayes.

$$\mathbb{P}(\text{Sp} \mid \text{Fl}) = \frac{(0.95)(0.30)}{0.299} = \frac{0.285}{0.299} \approx 0.953.$$

Here the prior is large and the false-flag rate small, so a flag is strong evidence: belief in spam rises from 0.30 to about **0.953** — the mirror image of the rare-disease example, where a small prior kept the posterior low.



$$\mathbb{P}(\text{Sp} \mid \text{Fl}) = \frac{0.285}{0.285 + 0.014} \approx 0.953.$$

Example 2.3.36 (Which factory made the defective bulb?)**.** A retailer stocks light bulbs from two factories. Factory A supplies 80% of the bulbs at a 2% defect rate; factory B supplies 20% at a 10% defect rate. A purchased bulb turns out to be defective. What is the probability it came from factory B?

Which rule, and why? We are told the bulb *is* defective and asked to trace it back to a factory — conditioning reversed from the forward defect rates. Backward inference to the hidden cause is Bayes. We first need the overall defect chance $\mathbb{P}(D)$ from total probability as the denominator.

Solution. Let $D = \{\text{defective}\}$ with cases A and B.

Step 1: denominator by total probability.

$$\mathbb{P}(D) = (0.02)(0.80) + (0.10)(0.20) = 0.016 + 0.020 = 0.036.$$

Step 2: Bayes for factory B.

$$\mathbb{P}(B \mid D) = \frac{(0.10)(0.20)}{0.036} = \frac{0.020}{0.036} = \frac{5}{9} \approx 0.556.$$

Though B supplies only one bulb in five, its high defect rate makes it the more probable source of a bad bulb: $\mathbb{P}(B \mid D) = \mathbf{5/9}$ against $\mathbb{P}(A \mid D) = 4/9$.

Example 2.3.37 (Which prize box, given a prize)**.** Recall the game show: boxes chosen with probabilities 0.30, 0.45, 0.25 contain a prize with chances 0.20, 0.50, 0.90. A contestant opens her box and finds a prize. Which box most probably was it, and with what probability?

Which rule, and why? A prize is in hand and we want the responsible box — reverse inference over a three-case partition, hence Bayes with three competing numerators over the shared denominator $\mathbb{P}(W)$. Total probability already gave that denominator (0.51); Bayes splits it among the boxes.

Solution.

Step 1: recall the denominator. $\mathbb{P}(W) = 0.060 + 0.225 + 0.225 = 0.51$.

Step 2: posterior for each box.

$$\mathbb{P}(B_1 \mid W) = \frac{0.060}{0.51} \approx 0.118, \quad \mathbb{P}(B_2 \mid W) = \frac{0.225}{0.51} \approx 0.441, \quad \mathbb{P}(B_3 \mid W) = \frac{0.225}{0.51} \approx 0.441.$$

The posteriors sum to 1. Boxes 2 and 3 tie at about $\mathbf{0.441}$: box 3 is likelier to hold a prize but is chosen less often, and the two effects exactly cancel, so a found prize points equally at them.

Box	Prior $\mathbb{P}(B_i)$	Likelihood $\mathbb{P}(W \mid B_i)$	Joint	Posterior $\mathbb{P}(B_i \mid W)$
B_1	0.30	0.20	0.060	0.118
B_2	0.45	0.50	0.225	0.441
B_3	0.25	0.90	0.225	0.441
		Total	0.510	1.000

Example 2.3.38 (A cheating flag and the base rate)**.** In an online game, 5% of players cheat. An anti-cheat system flags 90% of cheaters but also flags 4% of honest players. A randomly chosen player has been flagged. What is the probability the player actually cheats?

Which rule, and why? The flag rates are forward likelihoods — chance of a flag *given* whether a player cheats — but the question reverses them to the chance of cheating *given* a flag, with a known 5% prior. That reversal is Bayes; total probability gives the denominator $\mathbb{P}(\text{flag})$. The small base rate is the trap the rule resolves.

Solution. Let $C = \{\text{cheats}\}$ with prior $\mathbb{P}(C) = 0.05$, and $\text{Fl} = \{\text{flagged}\}$.

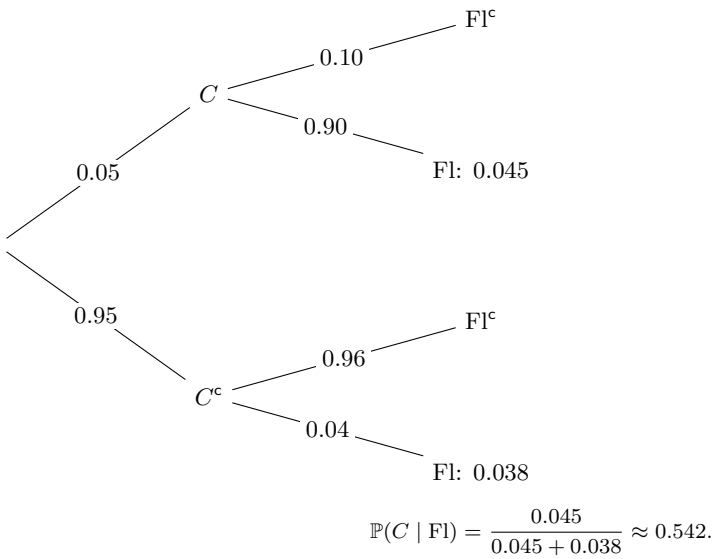
Step 1: denominator by total probability.

$$\mathbb{P}(\text{Fl}) = (0.90)(0.05) + (0.04)(0.95) = 0.045 + 0.038 = 0.083.$$

Step 2: Bayes.

$$\mathbb{P}(C \mid \mathrm{Fl}) = \frac{(0.90)(0.05)}{0.083} = \frac{0.045}{0.083} \approx 0.542.$$

Even a fairly sharp detector yields only about a **54**% chance of true cheating behind a flag: because honest players vastly outnumber cheaters, their 4% false-flag rate produces almost as many flags as the cheaters do. Once more the base rate, not the detector alone, drives the answer.



$$\mathbb{P}(C \mid \mathrm{Fl}) = \frac{0.045}{0.045 + 0.038} \approx 0.542.$$

2.4 Independence

Two events are **independent** when the occurrence of one carries no information about the other: learning that B happened leaves the probability of A exactly where it was. The algebraic test for this is the **product rule**, $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. Its consequences include the independence of complements, the gap between **pairwise** and **mutual** independence for three or more events, and the "at least one" complement formula.

Which independence rule does this problem need?

Read the structure, not just the words. Use this checklist.

- **Product rule for two events** (Definition, the master test). Triggers: separate physical mechanisms (two different coins, draws *with* replacement, components in different machines), or the phrase "independent" handed to you. You either *verify* independence by checking $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, or you *assume* it and multiply to get $\mathbb{P}(A \cap B)$. If draws are *without* replacement, or the same object is reused, suspect dependence and fall back to the multiplication rule $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B \mid A)$ of the previous section instead.
- **Independence of complements** (Proposition). Triggers: the problem gives you $A \perp\!\!\!\perp B$ but asks about "A and not B," "neither," or a non-occurrence. Flip the needed events to complements and keep multiplying — no new computation of the intersection is required.
- **Mutual vs. pairwise independence of n events** (Definition + counterexample). Triggers: three or more events, especially one defined *in terms of* the others ("the two tosses match," "the sum is even," a parity or agreement event). Do not assume that checking pairs is enough; the triple (or higher) product can fail even when every pair passes.
- **"At least one" complement formula** (Corollary of the product rule). Triggers: "at least one," "some," "the system works if any unit works," or any union of independent events where the direct inclusion–exclusion would be messy. Compute $1 - \prod_i \big(1 - \mathbb{P}(A_i)\big)$ instead.

Picture warning. A Venn diagram is the wrong tool here; see the remark below on why independence cannot be drawn as a region.

Remark (Independence is invisible on a Venn diagram). A Venn diagram encodes only *which* outcomes lie in A, in B, in $A \cap B$, and outside both — it shows set membership, nothing about *how much* probability sits in each region. Independence is the numerical relation $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$, a statement about *areas*, not about whether the circles overlap. Two pictures with identical overlap can be one independent and one dependent, depending solely on the probabilities assigned. (Indeed, two events that are independent and both have positive probability *must* overlap, since $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B) > 0$ — so the overlapping Venn picture is mandatory but says nothing.) For that reason this section uses **trees** and **two-way tables**, which display the actual probabilities, and never a Venn diagram.

2.4.1 Rule 1: The product rule (definition of independence)

Definition 2.4.1. Events A and B are **independent**, written $A \perp\!\!\!\perp B$, if

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B).$$

Otherwise they are **dependent**.

When $\mathbb{P}(B) > 0$ this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$: dividing both sides by $\mathbb{P}(B)$ turns the product rule into the statement that conditioning on B does not move the probability of A. The product form is the official definition because it is *symmetric* in A and B and stays meaningful even when a probability equals zero (where the conditional form is undefined). To **verify** independence you compute all three numbers $\mathbb{P}(A \cap B)$, $\mathbb{P}(A)$, $\mathbb{P}(B)$ and check the equation; to **use** an assumed independence you simply multiply.

Remark (Independent $\neq$ mutually exclusive). These opposite ideas are constantly confused. If A, B are mutually exclusive (disjoint) and both have positive probability, then $\mathbb{P}(A \cap B) = 0$ while $\mathbb{P}(A)\mathbb{P}(B) > 0$, so they are *dependent*: knowing A occurred tells you B did *not*. Disjointness is maximal dependence, not independence.

Example 2.4.2 (Two separate dice: verifying independence)**.** Roll a fair red die and a fair white die. Let $A = \{\text{red die shows an even number}\}$ and $B = \{\text{white die shows a 5 or 6}\}$. Are A and B independent?

Which rule, and why? Two *physically separate* dice are rolled, and each event names a feature of a *different* die — the classic signature of the product rule as a thing to *verify*. There is no "without replacement" and no shared object, so the multiplication rule with a conditional is not needed; we just check Definition 2.4.1 directly on the 6×6 grid of equally likely outcomes.

Solution. The sample space is the 36 equally likely pairs (r, w). We read each probability off the grid; shaded cells mark the events.

$r \backslash w$	1	2	3	4	5	6
1					$\bullet$	$\bullet$
2	$\circ$	$\circ$	$\circ$	$\circ$	$\otimes$	$\otimes$
3					$\bullet$	$\bullet$
4	$\circ$	$\circ$	$\circ$	$\circ$	$\otimes$	$\otimes$
5					$\bullet$	$\bullet$
6	$\circ$	$\circ$	$\circ$	$\circ$	$\otimes$	$\otimes$

Here $\circ$ marks A (red even), $\bullet$ marks B (white in $\{5, 6\}$), and $\otimes$ marks $A \cap B$.

Step 1 (marginals). $\mathbb{P}(A) = \dfrac{18}{36} = \dfrac{1}{2}$ (three even red values $\times$ six white values), and $\mathbb{P}(B) = \dfrac{12}{36} = \dfrac{1}{3}$ (six red values $\times$ two white values $5, 6$).

Step 2 (joint). The $\otimes$ cells number $3 \times 2 = 6$, so $\mathbb{P}(A \cap B) = \dfrac{6}{36} = \dfrac{1}{6}$.

Step 3 (test). $\mathbb{P}(A)\mathbb{P}(B) = \dfrac{1}{2} \cdot \dfrac{1}{3} = \dfrac{1}{6} = \mathbb{P}(A \cap B)$. The product rule holds, so $A \perp\!\!\!\perp B$.

Example 2.4.3 (Within one die: independence can still hold)**.** Roll a single fair six-sided die once. Let $A = \{\text{outcome is even}\} = \{2, 4, 6\}$ and $B = \{\text{outcome is } \leq 4\} = \{1, 2, 3, 4\}$. Independent?

Which rule, and why? Only one die is rolled, so there is no second mechanism — yet both events live on the *same* outcome. Independence is *not* automatic and must be tested, not assumed, so we again go straight to Definition 2.4.1. (This warns against the lazy rule "same die ⇒ dependent.")

Solution.

Step 1. $\mathbb{P}(A) = \dfrac{3}{6} = \dfrac{1}{2}$ and $\mathbb{P}(B) = \dfrac{4}{6} = \dfrac{2}{3}$.

Step 2. $A \cap B = \{2, 4\}$, so $\mathbb{P}(A \cap B) = \dfrac{2}{6} = \dfrac{1}{3}$.

Step 3. $\mathbb{P}(A)\mathbb{P}(B) = \dfrac{1}{2} \cdot \dfrac{2}{3} = \dfrac{1}{3} = \mathbb{P}(A \cap B)$. They *are* independent, even though both are properties of the one roll. Independence is an arithmetic coincidence of the probabilities, not a fact about separate experiments.

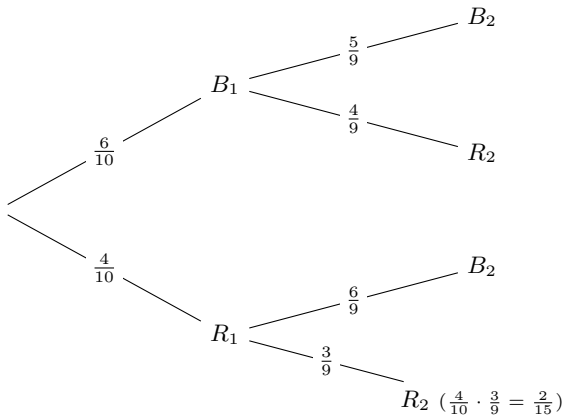
Example 2.4.4 (Drawing without replacement: dependence)**.** A box holds 4 red and 6 blue marbles. Two marbles are drawn one after another *without* replacement. Let $A = \{\text{first is red}\}$ and $B = \{\text{second is red}\}$. Are A and B independent?

Which rule, and why? The trigger is **without replacement**: the first draw changes the contents of the box, so the mechanism for the second draw depends on the first. This is the warning sign that the product rule will *fail* and that the multiplication rule with a conditional governs the joint probability. We confirm the failure numerically.

Solution.

Step 1 (marginals). $\mathbb{P}(A) = \dfrac{4}{10} = \dfrac{2}{5}$. By symmetry the second marble is equally likely to be any of the 10, so $\mathbb{P}(B) = \dfrac{4}{10} = \dfrac{2}{5}$ as well.

Step 2 (joint, via the tree).



The top path gives $\mathbb{P}(A \cap B) = \dfrac{4}{10} \cdot \dfrac{3}{9} = \dfrac{12}{90} = \dfrac{2}{15}$.

Step 3 (test). $\mathbb{P}(A)\mathbb{P}(B) = \dfrac{2}{5} \cdot \dfrac{2}{5} = \dfrac{4}{25} = 0.16$, while $\mathbb{P}(A \cap B) = \dfrac{2}{15} \approx 0.133$. Since $\dfrac{2}{15} \neq \dfrac{4}{25}$, the events are **dependent**. Removing a red marble makes a second red *less* likely, so A lowers the chance of B.

Example 2.4.5 (Spinner and coin: using assumed independence to multiply)**.** A spinner lands on one of three equal sectors colored green, green, yellow (so green has probability $\frac{2}{3}$, yellow $\frac{1}{3}$); separately a fair coin is flipped. Let $G = \{\text{spinner shows green}\}$ and $H = \{\text{coin shows heads}\}$. Find $\mathbb{P}(G \cap H)$ and $\mathbb{P}(\text{green and tails})$.

Which rule, and why? Two unrelated devices — a spinner and a coin — run side by side, so we are *given* independence and may simply multiply marginals. This is the "use" direction of the product rule, the reverse of the verifying examples above.

Solution.

Step 1. $\mathbb{P}(G) = \dfrac{2}{3}$, $\mathbb{P}(H) = \dfrac{1}{2}$.

Step 2. By independence $\mathbb{P}(G \cap H) = \dfrac{2}{3} \cdot \dfrac{1}{2} = \dfrac{1}{3}$.

Step 3. For "green and tails," independence with the complement of H gives $\mathbb{P}(G \cap H^{\mathsf{c}}) = \dfrac{2}{3} \cdot \dfrac{1}{2} = \dfrac{1}{3}$ (justified by Rule 2 below).

Example 2.4.6 (One card, two features: heart vs. face card)**.** Draw one card from a standard 52-card deck. Let $A = \{\text{the card is a heart}\}$ and $B = \{\text{the card is a face card (J, Q, K)}\}$. Are A and B independent?

Which rule, and why? A single draw, with both events read off the *same* card, so there is no second mechanism and independence is not handed to us — it must be *verified* against Definition 2.4.1. There is no "without replacement" (only one card is drawn), so the conditional multiplication rule is not in play; we just count cells in the 4×13 grid of suits $\times$ ranks.

Solution. The 52 cards are equally likely; the grid below has one column per rank and one row per suit. A column for J, Q, K is shaded (B), the heart row is shaded (A), and the three overlap cells carry $\otimes$.

suit\rank	A	2	3	4	5	6	7	8	9	10	J	Q	K
♠											•	•	•
♡	∘	∘	∘	∘	∘	∘	∘	∘	∘	∘	⊗	⊗	⊗
♢											•	•	•
♣											•	•	•

Here $\circ$ marks A (hearts), $\bullet$ marks B (face cards), $\otimes$ marks $A \cap B$.

Step 1 (marginals). $\mathbb{P}(A) = \dfrac{13}{52} = \dfrac{1}{4}$ and $\mathbb{P}(B) = \dfrac{12}{52} = \dfrac{3}{13}$.

Step 2 (joint). The heart face cards are J, Q, K of hearts, three cells, so $\mathbb{P}(A \cap B) = \dfrac{3}{52}$.

Step 3 (test). $\mathbb{P}(A)\mathbb{P}(B) = \dfrac{1}{4} \cdot \dfrac{3}{13} = \dfrac{3}{52} = \mathbb{P}(A \cap B)$. The product rule holds, so $A \perp\!\!\!\perp B$: knowing the card is a heart leaves the chance it is a face card at $\frac{3}{13}$.

Example 2.4.7 (Same die, but *not* independent: even vs. prime)**.** Roll one fair six-sided die. Let $A = \{\text{outcome is even}\} = \{2, 4, 6\}$ and $B = \{\text{outcome is prime}\} = \{2, 3, 5\}$. Independent?

Which rule, and why? As in the "even and ≤ 4" example, one roll forces us to *test* Definition 2.4.1 rather than assume anything. The twist is that here the test *fails*: this is the companion warning to the lazy rule "same die $\Rightarrow$ dependent" — sometimes the same die really does give dependence, and only the arithmetic tells you which.

Solution.

Step 1. $\mathbb{P}(A) = \dfrac{3}{6} = \dfrac{1}{2}$ and $\mathbb{P}(B) = \dfrac{3}{6} = \dfrac{1}{2}$.

Step 2. The only even prime is 2, so $A \cap B = \{2\}$ and $\mathbb{P}(A \cap B) = \dfrac{1}{6}$.

Step 3. $\mathbb{P}(A)\mathbb{P}(B) = \dfrac{1}{2} \cdot \dfrac{1}{2} = \dfrac{1}{4} \neq \dfrac{1}{6} = \mathbb{P}(A \cap B)$. The product rule *fails*, so A and B are **dependent**: an even outcome is less likely than usual to be prime (only 2 qualifies), so the events carry information about each other.

Example 2.4.8 (Three repeated draws with replacement: chaining the product rule)**.** A drawer holds 4 identical-feeling keys, exactly one of which opens a lock. You draw a key, test it, *return* it, and repeat, three times in all. What is the probability that all three draws pick the correct key?

Which rule, and why? Drawing *with replacement* restores the drawer to its original state each time, so the three draws are separate, identical mechanisms and the product rule applies in its "use and multiply" direction across all three. (If the keys were *not* replaced the draws would be dependent and the conditional multiplication rule would govern instead — this is the contrast with the earlier without-replacement marble example.)

Solution.

Step 1. Each draw independently hits the correct key with probability $\mathbb{P}(C) = \dfrac{1}{4}$.

Step 2. By the product rule for independent repetitions, $\mathbb{P}(C_1 \cap C_2 \cap C_3) = \dfrac{1}{4} \cdot \dfrac{1}{4} \cdot \dfrac{1}{4} = \dfrac{1}{64}$.

Step 3 (sanity check). The lone all-correct path is one branch of a tree whose three levels each split $\frac{1}{4}$ versus $\frac{3}{4}$; its probability $\frac{1}{64} \approx 0.016$ is suitably small, as repeatedly guessing right should be.

2.4.2 Rule 2: Independence of complements

If A is independent of B, then the *non*-occurrence of B is equally uninformative about A. The next proposition makes this precise and lets us swap any event for its complement while still multiplying.

Proposition 2.4.9. *If $A \perp\!\!\!\perp B$, then also $A \perp\!\!\!\perp B^{\mathsf{c}}$, $A^{\mathsf{c}} \perp\!\!\!\perp B$, and $A^{\mathsf{c}} \perp\!\!\!\perp B^{\mathsf{c}}$.*

Proof. Split A along B and its complement: $A = (A \cap B) \cup (A \cap B^{\mathsf{c}})$, a disjoint union, so by additivity $\mathbb{P}(A \cap B^{\mathsf{c}}) = \mathbb{P}(A) - \mathbb{P}(A \cap B)$. Using $A \perp\!\!\!\perp B$,

$$\mathbb{P}(A \cap B^{\mathsf{c}}) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\big(1 - \mathbb{P}(B)\big) = \mathbb{P}(A)\,\mathbb{P}(B^{\mathsf{c}}),$$

which is exactly $A \perp\!\!\!\perp B^{\mathsf{c}}$. Applying this same step with the roles of the events exchanged gives $A^{\mathsf{c}} \perp\!\!\!\perp B$, and applying it once more to the pair $A \perp\!\!\!\perp B^{\mathsf{c}}$ gives $A^{\mathsf{c}} \perp\!\!\!\perp B^{\mathsf{c}}$. □

Example 2.4.10 (Hits and misses on two targets)**.** An archer hits target I with probability 0.8 and target II with probability 0.6, the two shots independent. Find the probability of hitting I but missing II.

Which rule, and why? We are told the shots are independent ($A \perp\!\!\!\perp B$) but the question mixes an occurrence ("hit I") with a *non*-occurrence ("miss II"). That mismatch is the cue for Proposition 2.4.9: replace "hit II" by its complement and keep multiplying, rather than recomputing any joint probability from scratch.

Solution. Let $A = \{\text{hit I}\}$, $B = \{\text{hit II}\}$ with $\mathbb{P}(A) = 0.8$, $\mathbb{P}(B) = 0.6$.

Step 1. The event is $A \cap B^{\mathsf{c}}$, with $\mathbb{P}(B^{\mathsf{c}}) = 1 - 0.6 = 0.4$.

Step 2. By Proposition 2.4.9, $A \perp\!\!\!\perp B^{\mathsf{c}}$, so $\mathbb{P}(A \cap B^{\mathsf{c}}) = \mathbb{P}(A)\mathbb{P}(B^{\mathsf{c}}) = 0.8 \times 0.4 = 0.32$.

Example 2.4.11 (Neither machine jams)**.** Two photocopiers run independently; copier X jams on a given day with probability 0.1 and copier Y with probability 0.15. What is the probability that *neither* jams?

Which rule, and why? "Neither" is the conjunction of two non-occurrences, $A^{\mathsf{c}} \cap B^{\mathsf{c}}$. With independence given, Proposition 2.4.9 ($A^{\mathsf{c}} \perp\!\!\!\perp B^{\mathsf{c}}$) lets us multiply the two complement probabilities directly.

Solution.

Step 1. $\mathbb{P}(A^{\mathsf{c}}) = 1 - 0.1 = 0.9$, $\mathbb{P}(B^{\mathsf{c}}) = 1 - 0.15 = 0.85$.

Step 2. $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}}) = \mathbb{P}(A^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}}) = 0.9 \times 0.85 = 0.765$.

Example 2.4.12 (Filling a two-way table from independence)**.** A factory's components pass an electrical test with probability 0.9 and a visual test with probability 0.7, the two tests independent. Build the full 2×2 table of joint probabilities (pass/fail × pass/fail).

Which rule, and why? One independence assumption must populate all four cells, three of which involve at least one "fail." Each fail is a complement, so every cell is a product of a marginal and/or its complement — a direct, repeated application of Proposition 2.4.9. A two-way table is the natural display because it shows the four probabilities explicitly (a Venn diagram could not).

Solution. Let $E = \{\text{pass electrical}\}$, $V = \{\text{pass visual}\}$, $\mathbb{P}(E) = 0.9$, $\mathbb{P}(V) = 0.7$. Each cell is the product of the corresponding row and column marginals because $E \perp\!\!\!\perp V$ and, by Proposition 2.4.9, every complement pairing is independent too.

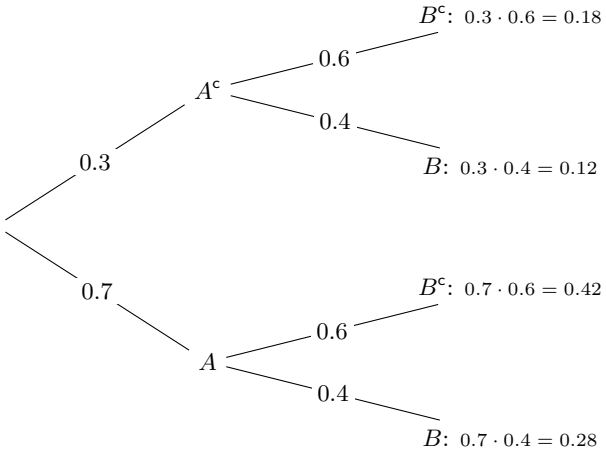
	V (pass)	V^{c} (fail)	row total
E (pass)	$0.9 \cdot 0.7 = 0.63$	$0.9 \cdot 0.3 = 0.27$	0.90
E^{c} (fail)	$0.1 \cdot 0.7 = 0.07$	$0.1 \cdot 0.3 = 0.03$	0.10
column total	0.70	0.30	1.00

Check. The four cells sum to $0.63 + 0.27 + 0.07 + 0.03 = 1.00$, and each margin reproduces the stated single-test probabilities — the signature of independence in table form: every interior cell equals (its row total)$\times$(its column total).

Example 2.4.13 (At least one complement, via a tree)**.** Two independent sensors trip an alarm. Sensor A trips with probability 0.7, sensor B with probability 0.4. Find the probability that exactly one of them trips.

Which rule, and why? "Exactly one" is $(A \cap B^{\mathsf{c}}) \cup (A^{\mathsf{c}} \cap B)$, a disjoint union of two mixed occurrence/non-occurrence events. Proposition 2.4.9 makes each piece a product; we add the two pieces. A tree organizes the four leaves cleanly.

Solution.



Step 1. "Exactly one" collects the two middle leaves: $\mathbb{P}(A \cap B^{\mathsf{c}}) = 0.42$ and $\mathbb{P}(A^{\mathsf{c}} \cap B) = 0.12$.

Step 2. They are disjoint, so add: $0.42 + 0.12 = 0.54$.

Example 2.4.14 (One occurs, the other does not: two independent tests)**.** A part must pass a strength test and, separately, a corrosion test; the two tests are independent. It passes strength with probability 0.75 and corrosion with probability 0.6. Find the probability it passes strength but *fails* corrosion.

Which rule, and why? Independence is given, but the target mixes an occurrence (pass strength) with a *non*-occurrence (fail corrosion) — exactly the mismatch Proposition 2.4.9 resolves. We flip "pass corrosion" to its complement and keep multiplying; no fresh joint probability needs computing, and the "at least one" formula does not apply because this is a single specified combination, not a union.

Solution. Let $S = \{\text{pass strength}\}$, $K = \{\text{pass corrosion}\}$ with $\mathbb{P}(S) = 0.75$, $\mathbb{P}(K) = 0.6$.

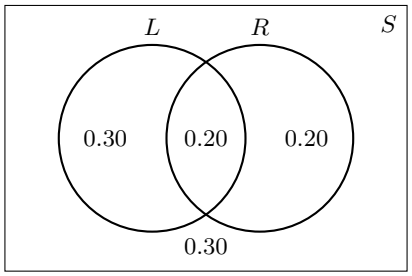
Step 1. The event is $S \cap K^{\mathsf{c}}$, with $\mathbb{P}(K^{\mathsf{c}}) = 1 - 0.6 = 0.4$.

Step 2. By Proposition 2.4.9, $S \perp\!\!\!\perp K^{\mathsf{c}}$, so $\mathbb{P}(S \cap K^{\mathsf{c}}) = \mathbb{P}(S)\mathbb{P}(K^{\mathsf{c}}) = 0.75 \times 0.4 = 0.30$.

Example 2.4.15 (All four regions from one independence assumption)**.** Two independent warning lamps are wired to a panel. The left lamp is lit with probability 0.5 and the right lamp with probability 0.4. Using only complements, find the probability of each of the four lit/unlit patterns, and mark them on the two-circle diagram.

Which rule, and why? One independence fact ($L \perp\!\!\!\perp R$) has to fill all four patterns, three of which contain an "unlit" (a complement). Proposition 2.4.9 makes every complement pairing independent, so each pattern is a single product. We can label the four Venn regions here *because* we already know the four numbers — the diagram displays set membership, while the products supply the probabilities the picture cannot show.

Solution. Let $L = \{\text{left lit}\}$, $R = \{\text{right lit}\}$, $\mathbb{P}(L) = 0.5$, $\mathbb{P}(R) = 0.4$.



Both lit. $\mathbb{P}(L \cap R) = 0.5 \times 0.4 = 0.20$.

Left only. $\mathbb{P}(L \cap R^{\mathsf{c}}) = 0.5 \times 0.6 = 0.30$.

Right only. $\mathbb{P}(L^{\mathsf{c}} \cap R) = 0.5 \times 0.4 = 0.20$.

Neither. $\mathbb{P}(L^{\mathsf{c}} \cap R^{\mathsf{c}}) = 0.5 \times 0.6 = 0.30$.

Check. The four add to $0.20 + 0.30 + 0.20 + 0.30 = 1.00$, and the left circle totals $0.30 + 0.20 = 0.50 = \mathbb{P}(L)$, as it must.

Example 2.4.16 (A complement-trick chain: at most one of three)**.** Three independent smoke detectors sound on detecting smoke with probabilities 0.9, 0.8, and 0.7. During a (false-alarm-free) test, find the probability that *at most one* sounds. Use complements throughout.

Which rule, and why? "At most one" means zero or one detector sounds. Each piece is an intersection mixing "sounds" with "stays silent" (a complement), so Proposition 2.4.9 turns every piece into a product of marginals and complements. This is the harder, multi-term cousin of the "exactly one" tree example; the "at least one" shortcut does not directly fit because we want a *small* count, not a union.

Solution. Let the sound events be A, B, C with $\mathbb{P}(A) = 0.9$, $\mathbb{P}(B) = 0.8$, $\mathbb{P}(C) = 0.7$, so the silent complements have probabilities 0.1, 0.2, 0.3.

Step 1 (none sound). By independence of the complements, $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}) = 0.1 \times 0.2 \times 0.3 = 0.006$.

Step 2 (exactly one sounds). Add the three patterns with one "sound" and two "silent":

$$\underbrace{0.9 \cdot 0.2 \cdot 0.3}_{A \text{ only}=0.054} + \underbrace{0.1 \cdot 0.8 \cdot 0.3}_{B \text{ only}=0.024} + \underbrace{0.1 \cdot 0.2 \cdot 0.7}_{C \text{ only}=0.014} = 0.092.$$

Step 3 (combine). "At most one" is the disjoint union of these, $0.006 + 0.092 = 0.098$.

2.4.3 Rule 3: Mutual vs. pairwise independence of n events

For three or more events the product rule on *pairs* alone is too weak. Genuine independence demands the product rule for *every* sub-collection at once.

Definition 2.4.17. Events $A_1, \ldots, A_n$ are **mutually independent** if for *every* index subset $\{i_1 < \cdots < i_k\} \subseteq \{1, \ldots, n\}$ with $k \geq 2$,

$$\mathbb{P}\big(A_{i_1} \cap \cdots \cap A_{i_k}\big) = \mathbb{P}(A_{i_1}) \cdots \mathbb{P}(A_{i_k}).$$

They are merely **pairwise independent** if this holds for all subsets of size $k = 2$ but possibly not for larger ones.

For n events this is a system of $2^n - n - 1$ equations (one per subset of size ≥ 2): for $n = 3$ that is $\binom{3}{2} + \binom{3}{3} = 3 + 1 = 4$ equations — the three pairwise equations *plus* the one triple equation $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Pairwise independence supplies only the first three; the triple equation is independent of them and can fail on its own.

Theorem 2.4.18 (Pairwise does not imply mutual)**.** *There exist events A, B, C that are pairwise independent but not mutually independent.*

Proof by explicit counterexample. Toss a fair coin twice; the four outcomes HH, HT, TH, TT are equally likely, each of probability $\frac{1}{4}$. Define

$$A = \{\text{first toss is } H\} = \{HH, HT\},$$
$$B = \{\text{second toss is } H\} = \{HH, TH\},$$
$$C = \{\text{the two tosses agree}\} = \{HH, TT\}.$$

Each event contains two of the four outcomes, so $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$. The two-way layout of the four outcomes makes the events visible:

	2nd $= H$ (B)	2nd $= T$
1st $= H$ (A)	HH (A, B, C)	HT (A)
1st $= T$	TH (B)	TT (C)

Pairwise check. Each pairwise intersection is a single outcome of probability $\frac{1}{4}$:

$$\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = \tfrac{1}{4} = \tfrac{1}{2} \cdot \tfrac{1}{2}, \quad \mathbb{P}(A \cap C) = \mathbb{P}(\{HH\}) = \tfrac{1}{4}, \quad \mathbb{P}(B \cap C) = \mathbb{P}(\{HH\}) = \tfrac{1}{4},$$

and every right-hand side is $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$. So all three pairs are independent.

Triple check. But $A \cap B \cap C = \{HH\}$ as well, so $\mathbb{P}(A \cap B \cap C) = \frac{1}{4}$, whereas $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$. Since $\frac{1}{4} \neq \frac{1}{8}$, the triple equation fails and the three events are *not* mutually independent. (Intuitively, any two of A, B, C determine the third: knowing the first toss and whether the tosses agree pins down the second toss.) □

Example 2.4.19 (Verifying full mutual independence of three devices)**.** Three independent indicator lights each turn on with probability $\frac{1}{2}$ (A, B, C for lights $1, 2, 3$). Confirm mutual independence and compute $\mathbb{P}(A \cap B \cap C)$.

Which rule, and why? Three *separate* devices with no event defined in terms of the others is the clean case where mutual independence genuinely holds. Because they are physically independent mechanisms, all four equations of Definition 2.4.17 hold automatically; contrast this with the counterexample, where C was *built from* A, B.

Solution.

Pairs. Each pair multiplies: $\mathbb{P}(A \cap B) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$, and likewise for the other two pairs.

Triple. $\mathbb{P}(A \cap B \cap C) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$. All four equations hold, so the three are mutually independent and $\mathbb{P}(A \cap B \cap C) = \frac{1}{8}$.

Example 2.4.20 (A second pairwise-but-not-mutual counterexample)**.** Roll a fair four-sided die with faces labeled $1, 2, 3, 4$, each outcome probability $\frac{1}{4}$. Let $A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{1, 4\}$. Show these are pairwise but not mutually independent.

Which rule, and why? Three events on one experiment, each of probability $\frac{1}{2}$, with a single shared outcome — exactly the structure that produces pairwise success but triple failure. The cue is that the events overlap only at the lone outcome 1, so every pairwise intersection has probability $\frac{1}{4}$ but so does the triple, which is too large.

Solution.

Marginals. $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{2}{4} = \frac{1}{2}$.

Pairs. Each pair meets only at outcome 1: $\mathbb{P}(A \cap B) = \mathbb{P}(\{1\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, and the same for $A \cap C$ and $B \cap C$. Pairwise independent.

Triple. $A \cap B \cap C = \{1\}$, so $\mathbb{P}(A \cap B \cap C) = \frac{1}{4}$, but $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{8}$. Since $\frac{1}{4} \neq \frac{1}{8}$, not mutually independent.

Example 2.4.21 (Mutual independence simplifies a longer "and")**.** A fair coin is tossed four times independently. Find the probability of the exact sequence heads, heads, tails, heads.

Which rule, and why? Four tosses of one coin are mutually independent (separate flips), so the joint probability of a specified sequence is the product over all four positions — the practical payoff of Definition 2.4.17 for $n = 4$.

Solution.

Step 1. Each toss has probability $\frac{1}{2}$ for its specified face.

Step 2. By mutual independence, $\mathbb{P}(HHTH) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{16}$.

Example 2.4.22 (When you may *not* multiply the triple)**.** In the four-sided-die setup ($A = \{1, 2\}$, $B = \{1, 3\}$, $C = \{1, 4\}$ above), a student wants $\mathbb{P}(A \cap B \cap C)$ and writes $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{8}$. Why is this wrong, and what is the correct value?

Which rule, and why? This is the diagnostic case: the events are only *pairwise* independent, so the multiplication shortcut for three events is *not* licensed. The presence of an event overlap structure (all three share outcome 1) is the red flag that the triple equation of Definition 2.4.17 must be checked, not assumed.

Solution.

Step 1. Multiplying marginals is valid only under *mutual* independence, which fails here (shown above). So $\frac{1}{8}$ is not justified.

Step 2. Compute directly: $A \cap B \cap C = \{1\}$, hence $\mathbb{P}(A \cap B \cap C) = \frac{1}{4}$, not $\frac{1}{8}$.

Example 2.4.23 (Two dice and a parity trap: pairwise but not mutual)**.** Roll a fair red die and a fair white die. Let $A = \{\text{red is even}\}$, $B = \{\text{white is even}\}$, and $C = \{\text{the sum is even}\}$. Show that A, B, C are pairwise independent but not mutually independent.

Which rule, and why? The first two events live on *different* dice, but C is defined *in terms of both* (parity of the sum) — precisely the "one event built from the others" signature that Definition 2.4.17 warns about. So we must check the triple equation separately and not trust the two genuinely-separate dice into thinking everything multiplies.

Solution. All 36 pairs (r, w) are equally likely. The grid marks A (even r, rows $2, 4, 6$), B (even w, columns $2, 4, 6$), and C (sum even, i.e. r, w the same parity); a cell in all three carries $\otimes$.

$r \backslash w$	1	2	3	4	5	6
1	C		C		C	
2		$\otimes$		$\otimes$		$\otimes$
3	C		C		C	
4		$\otimes$		$\otimes$		$\otimes$
5	C		C		C	
6		$\otimes$		$\otimes$		$\otimes$

The $\otimes$ cells are exactly "r even and w even," which here also forces the sum even — so on this grid $A \cap B \subseteq C$, the source of the trouble.

Marginals. $\mathbb{P}(A) = \mathbb{P}(B) = \dfrac{18}{36} = \dfrac{1}{2}$, and the sum is even on the 18 cells where r, w share parity, so $\mathbb{P}(C) = \dfrac{18}{36} = \dfrac{1}{2}$.

Pairs. $\mathbb{P}(A \cap B) = \dfrac{9}{36} = \dfrac{1}{4} = \dfrac{1}{2} \cdot \dfrac{1}{2}$; $\mathbb{P}(A \cap C)$ counts "red even and sum even" = "red even and white even" $= \dfrac{9}{36} = \dfrac{1}{4}$; and likewise $\mathbb{P}(B \cap C) = \dfrac{1}{4}$. Every pair matches the product, so all three are pairwise independent.

Triple. $A \cap B \cap C = A \cap B$ (the $\otimes$ cells), so $\mathbb{P}(A \cap B \cap C) = \dfrac{9}{36} = \dfrac{1}{4}$, whereas $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \dfrac{1}{2} \cdot \dfrac{1}{2} \cdot \dfrac{1}{2} = \dfrac{1}{8}$. Since $\frac{1}{4} \neq \frac{1}{8}$, the three are **not** mutually independent: once you know A and B, event C is forced, so C adds no freedom.

Example 2.4.24 (Mutual independence genuinely holds: three with-replacement draws)**.** Draw a card from a full deck, record whether it is a heart, return it and reshuffle, and repeat for a total of three draws. Let H_1, H_2, H_3 be "draw i is a heart." Are these three mutually independent, and what is $\mathbb{P}(H_1 \cap H_2 \cap H_3)$?

Which rule, and why? Three draws *with replacement* restore the deck each time, so no event is built from the others and every mechanism is identical and separate — the clean case where all four equations of Definition 2.4.17 hold automatically. This contrasts with the parity trap above, where the offending event was a function of the others.

Solution. Each draw is a heart with probability $\mathbb{P}(H_i) = \dfrac{13}{52} = \dfrac{1}{4}$, independent of the rest because the deck is reset.

Pairs. $\mathbb{P}(H_i \cap H_j) = \dfrac{1}{4} \cdot \dfrac{1}{4} = \dfrac{1}{16}$ for each of the three pairs.

Triple. $\mathbb{P}(H_1 \cap H_2 \cap H_3) = \dfrac{1}{4} \cdot \dfrac{1}{4} \cdot \dfrac{1}{4} = \dfrac{1}{64}$. All four equations hold, so the three are mutually independent.

Example 2.4.25 (Diagnosing the structure before multiplying)**.** Toss a fair coin twice. Let $A = \{\text{first is } H\}$, $B = \{\text{second is } H\}$, and $D = \{\text{the two tosses differ}\} = \{HT, TH\}$. A student claims $\mathbb{P}(A \cap B \cap D) = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(D) = \frac{1}{8}$. Check each independence relation and find the true value.

Which rule, and why? Two of the events (A, B) come from separate tosses, but D is an *agreement-type* event built from both — the same red flag as in the original two-coin counterexample. We must test the triple equation of Definition 2.4.17 directly; the pairwise checks alone cannot license the product.

Solution. The four outcomes HH, HT, TH, TT each have probability $\frac{1}{4}$, and $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(D) = \frac{1}{2}$.

Pairs. $\mathbb{P}(A \cap B) = \mathbb{P}(\{HH\}) = \frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$; $\mathbb{P}(A \cap D) = \mathbb{P}(\{HT\}) = \frac{1}{4}$; $\mathbb{P}(B \cap D) = \mathbb{P}(\{TH\}) = \frac{1}{4}$. Pairwise independent.

Triple. But $A \cap B = \{HH\}$ requires the tosses to *agree*, while D requires them to *differ*, so $A \cap B \cap D = \varnothing$ and $\mathbb{P}(A \cap B \cap D) = 0$, not $\frac{1}{8}$. The student's product is unjustified because mutual independence fails; the true value is 0.

2.4.4 Rule 4: The "at least one" complement formula

The hardest-looking union — "at least one of many independent events occurs" — becomes a one-line product once you complement it. "At least one occurs" is the negation of "none occurs," and "none occurs" is a pure intersection of complements, which independence lets us multiply.

Corollary 2.4.26. *If $A_1, \ldots, A_n$ are mutually independent, then*

$$\mathbb{P}(\textit{at least one } A_i \textit{ occurs}) = 1 - \mathbb{P}(\textit{none occur}) = 1 - \prod_{i=1}^{n} \big(1 - \mathbb{P}(A_i)\big).$$

In particular, if all $\mathbb{P}(A_i) = p$, this is $1 - (1-p)^n$.

Proof. "At least one" is the union $\bigcup_i A_i$; its complement is $\bigcap_i A_i^{\mathsf{c}}$ (none occur), so $\mathbb{P}(\bigcup_i A_i) = 1 - \mathbb{P}(\bigcap_i A_i^{\mathsf{c}})$. By Proposition 2.4.9 the complements $A_1^{\mathsf{c}}, \ldots, A_n^{\mathsf{c}}$ are mutually independent, so their intersection probability factors: $\mathbb{P}(\bigcap_i A_i^{\mathsf{c}}) = \prod_i \mathbb{P}(A_i^{\mathsf{c}}) = \prod_i (1 - \mathbb{P}(A_i))$. Substituting gives the formula. □

Example 2.4.27 (At least one six in four rolls)**.** Roll a fair die four times independently. What is the probability of getting at least one six?

Which rule, and why? "At least one" over independent repetitions is the textbook trigger for Corollary 2.4.26. Computing it as a union directly would require inclusion–exclusion over four overlapping events; the complement turns it into a single power.

Solution.

Step 1. Per roll, $\mathbb{P}(\text{six}) = \frac{1}{6}$, so $\mathbb{P}(\text{not six}) = \frac{5}{6}$.

Step 2. "None is a six" has probability $\left(\frac{5}{6}\right)^4 = \dfrac{625}{1296}$.

Step 3. $\mathbb{P}(\text{at least one six}) = 1 - \dfrac{625}{1296} = \dfrac{671}{1296} \approx 0.518$.

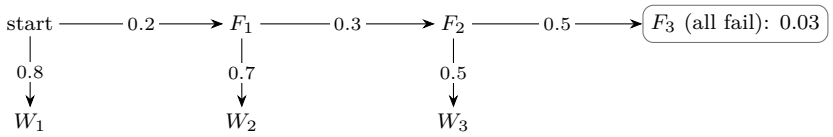
Example 2.4.28 (Redundant components: system works if any works)**.** A backup system has three pumps that fail independently, with failure probabilities 0.2, 0.3, and 0.5. The system functions if at least one pump works. Find the probability the system functions.

Which rule, and why? "Functions if at least one works" is the union of the three "works" events; their failure probabilities are given, so Corollary 2.4.26 applies with *unequal* probabilities (use the product form, not the $1-(1-p)^n$ shortcut).

Solution.

Step 1. "System fails" means all three pumps fail simultaneously. By independence, $\mathbb{P}(\text{all fail}) = 0.2 \times 0.3 \times 0.5 = 0.03$.

Step 2. $\mathbb{P}(\text{functions}) = 1 - 0.03 = 0.97$.



Only the single fully-failed path has probability 0.03; the system works on all other paths, total 0.97.

Example 2.4.29 (How many tries to be 95% sure?)**.** Each independent attempt to open a stuck valve succeeds with probability 0.3. How many attempts guarantee at least a 0.95 chance of opening it at least once?

Which rule, and why? "At least once" across independent equal-probability trials is Corollary 2.4.26 with all p equal, but now the unknown is n: invert the formula $1-(1-p)^n \geq 0.95$.

Solution.

Step 1. We need $1-(0.7)^n \geq 0.95$, i.e. $(0.7)^n \leq 0.05$.

Step 2. Take logarithms: $n \geq \dfrac{\ln 0.05}{\ln 0.7} = \dfrac{-2.996}{-0.357} \approx 8.39$.

Step 3. Since n is an integer, $n = 9$ attempts. (Check: $(0.7)^9 \approx 0.040 \leq 0.05$, while $(0.7)^8 \approx 0.058 > 0.05$.)

Example 2.4.30 (At least one match: a small birthday-style count)**.** Five people each independently pick one of the days Monday through Friday uniformly at random. What is the probability that at least two pick the *same* day? Use the complement.

Which rule, and why? "At least one coincidence" inverts to "all distinct," a pure intersection that factors under independence — the same complement move as Corollary 2.4.26, applied to the event "all five choices are distinct." Direct counting of matches would be far messier.

Solution.

Step 1. "All distinct" assigns the five people distinct days among five, so the favorable choices are the 5! orderings out of 5^5 total equally likely assignments: $\mathbb{P}(\text{all distinct}) = \dfrac{5!}{5^5} = \dfrac{120}{3125} = 0.0384$.

Step 2. $\mathbb{P}(\text{at least one match}) = 1 - 0.0384 = 0.9616$.

Example 2.4.31 (At least one head in five tosses)**.** Toss a fair coin five times independently. Find the probability of getting at least one head.

Which rule, and why? "At least one" over independent equal-probability repetitions is the cleanest trigger for Corollary 2.4.26 with all $p = \frac{1}{2}$. Inclusion–exclusion over five overlapping "head on toss i" events would be a five-term mess; the complement collapses it to one power.

Solution.

Step 1. "No heads" means tails every time, probability $\left(\frac{1}{2}\right)^5 = \dfrac{1}{32}$.

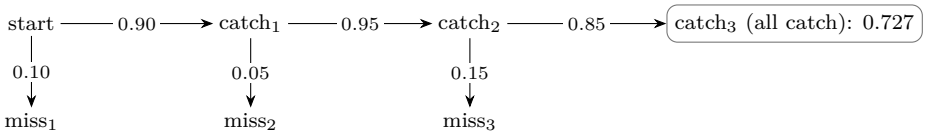
Step 2. $\mathbb{P}(\text{at least one head}) = 1 - \dfrac{1}{32} = \dfrac{31}{32} \approx 0.969$.

Example 2.4.32 (Unequal failure rates: at least one spam slips through)**.** Three independent filters scan an incoming message; they each *miss* a genuine spam with probabilities 0.10, 0.05, and 0.15. The spam reaches the inbox if *at least one* filter misses it. Find that probability.

Which rule, and why? "Reaches the inbox if at least one misses" is the union of three "miss" events with *unequal* probabilities, so Corollary 2.4.26 applies in its general product form $1 - \prod(1 - p_i)$, not the $1 - (1-p)^n$ shortcut (the misses are not equally likely). The tree shows that only the all-caught path avoids the inbox.

Solution.

Step 1 (all three catch it). "Caught" probabilities are 0.90, 0.95, 0.85, so by independence $\mathbb{P}(\text{all catch}) = 0.90 \times 0.95 \times 0.85 = 0.72675$.



Step 2 (complement). $\mathbb{P}(\text{reaches inbox}) = 1 - 0.72675 = 0.27325$. Only the single top path is spam-free; every other path has at least one miss.

Example 2.4.33 (How many rolls for a near-certain six?)**.** A fair die is rolled repeatedly and independently. How many rolls are needed so that the probability of getting at least one six is at least 0.90?

Which rule, and why? "At least once" across independent equal-probability trials is Corollary 2.4.26 with all $p = \frac{1}{6}$, but the unknown is the count n: we invert $1 - \left(\frac{5}{6}\right)^n \geq 0.90$ rather than evaluate it at a fixed n.

Solution.

Step 1. We need $1 - \left(\frac{5}{6}\right)^n \geq 0.90$, i.e. $\left(\frac{5}{6}\right)^n \leq 0.10$.

Step 2. Take logarithms: $n \geq \dfrac{\ln 0.10}{\ln(5/6)} = \dfrac{-2.3026}{-0.1823} \approx 12.6$.

Step 3. Since n must be an integer, $n = 13$ rolls. (Check: $\left(\frac{5}{6}\right)^{13} \approx 0.093 \leq 0.10$, while $\left(\frac{5}{6}\right)^{12} \approx 0.112 > 0.10$.)

Remark (The unifying thread)*.* Rules 2 and 4 are both consequences of Rule 1. Independence of complements (Proposition 2.4.9) is the product rule with one event flipped; the "at least one" formula (Corollary 2.4.26) is the product rule applied to all the complements at once. Rule 3 is the warning that for three or more events you must check *every* subset, not just pairs, before you are allowed to multiply. Throughout, the right pictures are trees and tables — never a Venn diagram, which shows only regions and hides the probabilities that independence is actually about.

2.5 Counting (Combinatorics)

When every outcome of an experiment is equally likely, a probability is just a ratio of two counts: favourable outcomes over total outcomes. The whole art is therefore counting without listing. Every method in this

section is built from a single idea — the multiplication principle — and differs only in how it corrects for two questions: *does order matter?* and *is repetition allowed?*

Which counting tool does the problem need?

Read the problem and answer two questions in order.

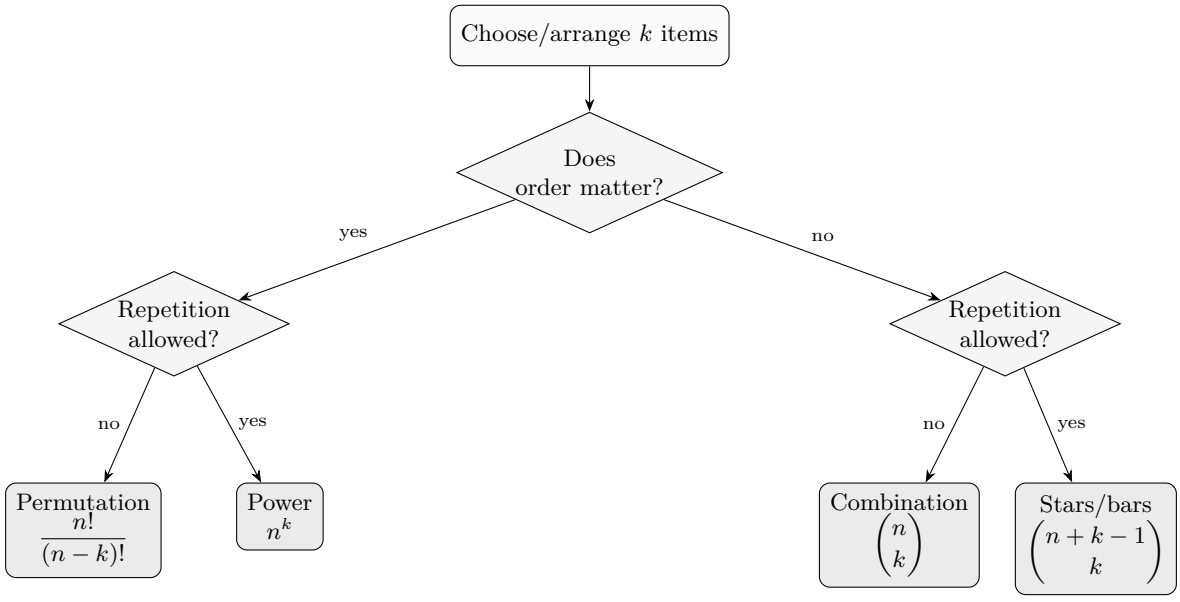
1. **Does order matter?** If rearranging the chosen items gives a genuinely different outcome (rankings, sequences, assigned roles, lining up, passwords), order matters and you are *permuting*. If only the *set* of chosen items matters (a committee, a hand of cards, a subset, a team with no titles), order does not matter and you are *combining*.
2. **Is repetition allowed?** Can the same item be reused (digits in a PIN, flavours that can repeat), or is each item used at most once (people, distinct cards)?

This gives four cases for choosing k from n:

	order matters	**order does not matter**
no repetition	$P(n,k) = \dfrac{n!}{(n-k)!}$	$\dbinom{n}{k} = \dfrac{n!}{k!\,(n-k)!}$
repetition allowed	n^k	$\dbinom{n+k-1}{k}$ (stars and bars)

Two refinements sit on top of these. If you are arranging *all* n objects but some are identical, use the **multinomial coefficient** $n!/(n_1!\cdots n_r!)$. If the arrangement is in a *circle* with no fixed starting seat, divide an ordinary permutation by the number of rotations. When a problem has several stages, count each stage and *multiply* (multiplication principle); when it splits into disjoint cases, count each case and *add*.

The decision is mechanical once you train the two questions. The flowchart below encodes exactly the reasoning above.



Remark. All four boxes assume the items themselves are distinct. If they are not — some are copies of one another — and you must arrange *all* of them, the multinomial coefficient (Theorem 2.5.27) takes over. Identical-items-with-order is the one case the four-box table does not cover directly.

Rule 1: the multiplication principle

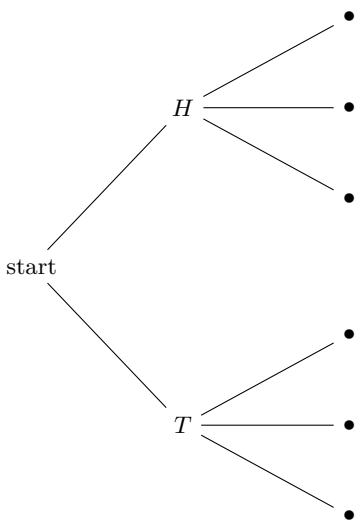
Theorem 2.5.1 (Multiplication principle)**.** *Suppose a procedure is carried out in k successive stages, and that stage i can be performed in exactly n_i ways no matter how the earlier stages turned out. Then the entire procedure can be performed in*

$$n_1\, n_2 \cdots n_k$$

distinct ways.

Proof. Argue by induction on the number of stages k. For $k = 1$ the claim is the definition: there are n_1 ways. Assume the result for $k-1$ stages, so the first $k-1$ stages produce $n_1 \cdots n_{k-1}$ distinct partial outcomes. Each such partial outcome can be completed by stage k in n_k ways, and different partial outcomes (or different stage-k choices) yield different full outcomes. Grouping the full outcomes by their common partial outcome partitions them into $n_1 \cdots n_{k-1}$ blocks of n_k each, for a total of $(n_1 \cdots n_{k-1})\, n_k = n_1 \cdots n_k$. This completes the induction. □

The hypothesis "no matter how the earlier stages turned out" is essential: the *count* n_i at stage i may not depend on earlier choices, even though the specific options may. Picking a shirt then a tie is fine even if which ties match depends on the shirt, provided the *number* of matching ties is the same for every shirt. A tree diagram makes the principle visible: the number of leaves is the product of the branch counts at each level.



coin: 2 ways die-trio: 3 ways $2 \cdot 3 = 6$ leaves

Example 2.5.2 (Building a meal)**.** A diner offers 4 appetisers, 7 main courses, and 3 desserts. How many distinct three-course meals (one of each) can a customer assemble?

Which rule, and why? The meal is assembled in three independent stages and we want the number of *ways to build the whole thing*, not to pick an unordered set. The count of choices at each stage (4, then 7, then 3) does not depend on the earlier picks, which is exactly the setup for the multiplication principle. There is no selecting-without-order here, so combinations are irrelevant.

Solution.

Stage counts. Appetiser: 4. Main: 7. Dessert: 3.

Multiply. $4 \cdot 7 \cdot 3 = 84$. There are $\boxed{84}$ possible meals.

Example 2.5.3 (Licence-plate format with repetition)**.** A licence plate consists of 3 letters (A–Z, 26 options each) followed by 4 digits (0–9). Letters and digits may repeat. How many plates are possible?

Which rule, and why? Each of the 7 positions is a separate stage, and because symbols may repeat the number of options is the same (26 or 10) at every position regardless of earlier choices. This is the multiplication principle in its purest "power" form, n^k, which arises precisely when order matters and repetition is allowed. Permutations would apply only if symbols could not repeat.

Solution.

Letter block. $26 \cdot 26 \cdot 26 = 26^3 = 17{,}576$.

Digit block. $10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10{,}000$.

Multiply the blocks. $26^3 \cdot 10^4 = 17{,}576 \cdot 10{,}000 = 175{,}760{,}000$. There are $\boxed{175{,}760{,}000}$ plates.

Example 2.5.4 (Subsets via a yes/no decision per element)**.** A pizza shop offers 9 distinct toppings. How many different pizzas can be ordered if any subset of toppings (including none and all) is allowed?

Which rule, and why? The trigger is "any subset" of a fixed list. Build the subset by making one independent decision per topping: *in* or *out*. That is 9 stages of 2 options each, so the multiplication principle gives 2^9. This is the slick way to see that an n-element set has 2^n subsets; summing $\binom{9}{0} + \cdots + \binom{9}{9}$ would give the same answer but is far more work.

Solution.

One binary choice per topping. 9 toppings, each either chosen or not: 2 ways each.

Multiply. $2^9 = 512$. There are $\boxed{512}$ possible pizzas (including the plain pizza with no toppings).

Example 2.5.5 (Stage counts that vary but stay constant in number)**.** A traveller plans a 4-leg trip visiting 4 *different* cities chosen from a list of 10, in order. How many ordered itineraries are possible?

Which rule, and why? Here the *set* of available cities shrinks after each leg (no city is revisited), yet the *number* of available cities at each stage is fixed: 10, then 9, then 8, then 7. The multiplication principle still applies because the count per stage is predetermined. Notice this is order-matters-without-repetition, so it coincides with the permutation $P(10, 4)$; we compute it here directly from stages to show the principle underneath.

Solution.

Stage counts. Leg 1: 10 cities. Leg 2: 9. Leg 3: 8. Leg 4: 7.

Multiply. $10 \cdot 9 \cdot 8 \cdot 7 = 5040$. There are $\boxed{5040}$ itineraries.

Example 2.5.6 (An outfit with no repeated category)**.** A wardrobe has 5 shirts, 4 pairs of trousers, 3 pairs of shoes, and 2 jackets. How many complete outfits use exactly one item from each category?

Which rule, and why? The outfit is built in four independent stages, and the number of options at each stage (5, 4, 3, 2) is fixed regardless of earlier picks. That is the multiplication principle. We are not selecting an unordered subset of clothing, so combinations do not apply; and each category contributes exactly one item, so no single power n^k fits either.

Solution.

Stage counts. Shirts: 5; trousers: 4; shoes: 3; jackets: 2.

Multiply. $5 \cdot 4 \cdot 3 \cdot 2 = 120$. There are $\boxed{120}$ outfits.

Example 2.5.7 (A four-digit PIN with repetition allowed)**.** A keypad PIN is 4 digits long, each digit from 0–9, and digits may repeat. How many PINs are possible?

Which rule, and why? Each of the 4 positions is a separate stage with the same 10 options regardless of earlier digits, because repetition is allowed. Order matters (a PIN is a sequence) and repetition is permitted, which is exactly the power form $n^k = 10^4$. A permutation $P(10, 4)$ would be wrong here: it forbids repeats, undercounting PINs like `7007`.

Solution.

Per-position count. Each of the 4 slots: 10 digits.

Multiply. $10 \cdot 10 \cdot 10 \cdot 10 = 10^4 = 10{,}000$. There are $\boxed{10{,}000}$ PINs.

Example 2.5.8 (Counting outcomes on an equally-likely grid)**.** Two fair dice, one white and one black, are rolled. Treating the white roll as the first stage and the black roll as the second, how many ordered outcomes are there in total, and how many give a sum that is a multiple of 3?

Which rule, and why? The roll is two independent stages of 6 options each, so the multiplication principle gives $6 \cdot 6 = 36$ equally-likely ordered pairs — the sample space. To count a sub-event we do *not* need a new rule; we just tally the favourable cells of the 6×6 grid. This is the typical bridge from counting to probability: a count over a count.

Solution.

Total outcomes. $6 \cdot 6 = 36$ ordered pairs.

Tally the grid. Shade every cell whose coordinate sum is $3, 6, 9$, or 12.

black die ↑						
7	8	9	10	11	12	
6	7	8	9	10	11	
5	6	7	8	9	10	
4	5	6	7	8	9	
3	4	5	6	7	8	
2	3	4	5	6	7	

white die →

Count. The shaded cells number 12, so 12 of the 36 outcomes (a probability of $12/36 = \frac{1}{3}$) have a sum divisible by 3. There are $\boxed{12}$ favourable outcomes.

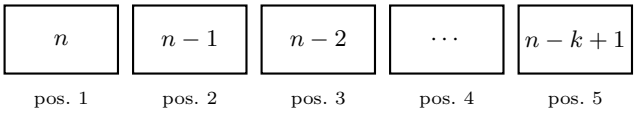
Rule 2: permutations $P(n, k)$

Theorem 2.5.9 (Permutations)**.** *The number of* ***ordered*** *arrangements of k objects selected* without repetition *from a pool of n distinct objects is*

$$P(n, k) = n(n-1)(n-2)\cdots(n-k+1) = \frac{n!}{(n-k)!}.$$

In particular, the number of ways to order all n objects is $P(n, n) = n!$, using the convention $0! = 1$.

Proof. Think of k ordered positions to be filled left to right (slot diagram below). The first position can be filled by any of the n objects. Whatever is placed there is used up, so the second position has $n-1$ choices, the third $n-2$, and in general the ith position has $n-(i-1)$ choices. By the multiplication principle the total is $n(n-1)\cdots(n-k+1)$. Multiplying and dividing by $(n-k)!$ telescopes this product into $n!/(n-k)!$. □



Example 2.5.10 (Ranked finishers)**.** In a race of 8 runners (no ties), how many different outcomes are possible for the gold, silver, and bronze medals?

Which rule, and why? The three medals are *distinct roles*, so the order in which we name the three finishers matters: "Ann gold, Bo silver" differs from "Bo gold, Ann silver." No runner can take two medals, so there is no repetition. Order matters, no repetition ⇒ permutation $P(8, 3)$. A combination $\binom{8}{3}$ would wrongly treat the three medallists as an unordered set.

Solution.

Apply the formula. $P(8, 3) = \dfrac{8!}{(8-3)!} = 8 \cdot 7 \cdot 6.$

Compute. $8 \cdot 7 \cdot 6 = 336$. There are $\boxed{336}$ medal outcomes.

Example 2.5.11 (Seating in a row)**.** Six distinct books are placed left to right on a shelf. In how many orders can all six be arranged?

Which rule, and why? We arrange *all* the objects and every left-to-right order is a different shelf, so order matters and $k = n = 6$. This is the full permutation $P(6, 6) = 6!$. The phrase "in how many orders" is the giveaway that we are not selecting a subset.

Solution.

Full factorial. $P(6, 6) = 6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$.

Compute. $6! = 720$. There are $\boxed{720}$ arrangements.

Example 2.5.12 (Forming a four-letter code)**.** How many four-letter codes can be made from the 26 letters if *no letter repeats*?

Which rule, and why? Position matters in a code (`LMNO` differs from `OMNL`), and the explicit "no letter repeats" forbids reuse. Order matters, no repetition ⇒ permutation $P(26, 4)$. Contrast the earlier plate example, where repetition *was* allowed and the count was 26^4 instead.

Solution.

Slot product. $P(26, 4) = 26 \cdot 25 \cdot 24 \cdot 23$.

Compute. $26 \cdot 25 = 650$; $650 \cdot 24 = 15{,}600$; $15{,}600 \cdot 23 = 358{,}800$. There are $\boxed{358{,}800}$ codes.

Example 2.5.13 (Assigning distinct jobs to a subset of people)**.** A manager has 12 employees and must assign 5 *different* tasks (each task to a different person, each person at most one task). How many assignments are possible?

Which rule, and why? The five tasks are distinguishable, so assigning Task 1 to Dana and Task 2 to Lee differs from the swap; order (which person gets which task) matters. Each person takes at most one task, so no repetition. Order matters, no repetition ⇒ permutation $P(12, 5)$. If the five chosen people were just a pool with no task labels, it would instead be $\binom{12}{5}$.

Solution.

Apply the formula. $P(12, 5) = \dfrac{12!}{7!} = 12 \cdot 11 \cdot 10 \cdot 9 \cdot 8$.

Compute. $12 \cdot 11 = 132$; $132 \cdot 10 = 1320$; $1320 \cdot 9 = 11{,}880$; $11{,}880 \cdot 8 = 95{,}040$. There are $\boxed{95{,}040}$ assignments.

Example 2.5.14 (Permutations with a forced position)**.** Five distinct keynote speakers are scheduled in five time slots, but the headline speaker must occupy the final slot. How many schedules are possible?

Which rule, and why? Order matters (the slots are an ordered sequence) and no speaker repeats, so this is a permutation problem with a *constraint*. Handle the constraint by fixing the forced choice first, then permuting what remains. The constraint does not change the rule; it just reduces the free positions.

Solution.

Place the forced speaker. The headline speaker is fixed in the last slot: 1 way.

Arrange the rest. The remaining 4 speakers fill the first 4 slots in $P(4, 4) = 4! = 24$ ways.

Multiply. $1 \cdot 24 = 24$. There are $\boxed{24}$ schedules.

Example 2.5.15 (Top three from a club)**.** A photography club of 7 members elects a president, a vice-president, and a treasurer (three distinct offices, no person holds two). How many ways can the three offices be filled?

Which rule, and why? The three offices are distinct roles, so naming Ali president and Bo vice-president differs from the swap: order matters. No member holds two offices, so there is no repetition. Order matters, no repetition ⇒ permutation $P(7, 3)$. A combination $\binom{7}{3}$ would lump the three officers into a title-less set and undercount by a factor of 3!.

Solution.

Slot product. $P(7, 3) = 7 \cdot 6 \cdot 5$.

Compute. $7 \cdot 6 = 42$; $42 \cdot 5 = 210$. There are $\boxed{210}$ ways.

Example 2.5.16 (Anagrams of a word with all distinct letters)**.** How many distinguishable strings can be formed by rearranging all the letters of `MATH`?

Which rule, and why? All four letters are distinct and we arrange *all* of them, so every ordering is a new string: $k = n = 4$ and the answer is the full permutation $P(4, 4) = 4!$. There are no repeated letters, so this is a plain factorial rather than the multinomial coefficient used for words like `PEPPER`.

Solution.

Full factorial. $P(4, 4) = 4! = 4 \cdot 3 \cdot 2 \cdot 1$.

Compute. $4! = 24$. There are $\boxed{24}$ anagrams.

Example 2.5.17 (Permutation with a forbidden position — complement trick)**.** Six distinct runners line up at the start; the slowest runner refuses to be placed in lane 1. How many starting line-ups are possible?

Which rule, and why? A line-up is an ordered arrangement of all 6 distinct runners, so the base count is the permutation $P(6, 6) = 6!$. The single restriction (one runner barred from one lane) is cleanest by *complementary counting*: take all arrangements and subtract the forbidden ones, rather than splitting into many allowed cases. Subtraction beats casework whenever the banned set is small and easy to count.

Solution.

All line-ups. $6! = 720$.

Bad line-ups (slow runner in lane 1). Fix that runner in lane 1 and permute the other 5: $5! = 120$.

Subtract. $720 - 120 = 600$. There are $\boxed{600}$ allowed line-ups.

Rule 3: combinations $\binom{n}{k}$

Theorem 2.5.18 (Combinations)**.** *The number of* ***unordered*** *subsets of size k that can be chosen from n distinct objects (without repetition) is*

$$\binom{n}{k} = \frac{P(n, k)}{k!} = \frac{n!}{k!\,(n-k)!}.$$

Proof. Count the ordered selections two ways. On one hand there are $P(n, k)$ of them. On the other hand, every ordered selection is obtained by first choosing the underlying *set* of k objects — say there are C such sets — and then ordering that set, which can be done in $k!$ ways. By the multiplication principle the number of ordered selections is also $C \cdot k!$. Equating, $C \cdot k! = P(n, k)$, so $C = P(n, k)/k! = n!/(k!\,(n-k)!)$. □

Two facts follow at once and save arithmetic. The **symmetry** $\binom{n}{k} = \binom{n}{n-k}$ holds because choosing the k to include is the same act as choosing the $n - k$ to exclude. And the **binomial theorem** $(x + y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$ shows $\binom{n}{k}$ counts the terms with k factors of x. The single most reliable test for "combination versus permutation" is to ask whether *relabelling the chosen items as an ordered list* produces a new, valid outcome; if it does not, you are combining.

Example 2.5.19 (Committee with no titles)**.** From a department of 15 people, how many ways are there to form a committee of 4 if all members are equal (no chair, no titles)?

Which rule, and why? A committee is a *set* of people: listing the same four in a different order does not create a new committee. Order does not matter, no person is chosen twice, so this is a combination $\binom{15}{4}$. The absence of role words (chair, president, ranking) is the signal that order is irrelevant.

Solution.

Apply the formula. $\binom{15}{4} = \dfrac{15 \cdot 14 \cdot 13 \cdot 12}{4!} = \dfrac{15 \cdot 14 \cdot 13 \cdot 12}{24}$.

Compute the numerator. $15 \cdot 14 = 210$; $210 \cdot 13 = 2730$; $2730 \cdot 12 = 32{,}760$.

Divide. $32{,}760/24 = 1365$. There are $\boxed{1365}$ committees.

Example 2.5.20 (A poker-style hand)**.** How many 5-card hands can be dealt from a standard 52-card deck (order of receipt irrelevant)?

Which rule, and why? A hand is the *set* of cards held; the order in which they were dealt does not change the hand. Cards are distinct and none repeats. Order does not matter, no repetition ⇒ combination $\binom{52}{5}$. Using $P(52,5)$ would count each hand 5! times, once per dealing order.

Solution.

Apply the formula. $\displaystyle\binom{52}{5} = \frac{52\cdot 51\cdot 50\cdot 49\cdot 48}{5!}$.

Numerator. $52\cdot 51 = 2652$; $\cdot 50 = 132{,}600$; $\cdot 49 = 6{,}497{,}400$; $\cdot 48 = 311{,}875{,}200$.

Divide by $5! = 120$. $311{,}875{,}200/120 = 2{,}598{,}960$. There are $\boxed{2{,}598{,}960}$ hands.

Example 2.5.21 (Choosing a mixed group with the multiplication principle)**.** A club has 7 seniors and 9 juniors. How many ways are there to pick a delegation of 3 seniors and 2 juniors?

Which rule, and why? Each of the two groups is an unordered selection (a delegation, no titles), so each is a combination. The two selections are made in *stages* — first the seniors, then the juniors — so by the multiplication principle we *multiply* the two combination counts. The clue for multiplying is the word "and" joining two independent sub-choices.

Solution.

Choose seniors. $\displaystyle\binom{7}{3} = \frac{7\cdot 6\cdot 5}{6} = 35.$

Choose juniors. $\displaystyle\binom{9}{2} = \frac{9\cdot 8}{2} = 36.$

Multiply. $35\cdot 36 = 1260$. There are $\boxed{1260}$ delegations.

Example 2.5.22 (Complement and "at least" counting)**.** A box holds 10 good widgets and 4 defective ones. A sample of 4 is drawn at once. In how many ways does the sample contain *at least one* defective widget?

Which rule, and why? The sample is an unordered subset, so combinations apply. "At least one" is the trigger for *complementary counting*: instead of summing the cases (exactly 1, 2, 3, 4 defectives), count the total and subtract the lone unwanted case of *zero* defectives. This is far less work than adding four combination products.

Solution.

Total samples. $\displaystyle\binom{14}{4} = \frac{14\cdot 13\cdot 12\cdot 11}{24} = 1001.$

Samples with no defective (all from the 10 *good).* $\displaystyle\binom{10}{4} = \frac{10\cdot 9\cdot 8\cdot 7}{24} = 210.$

Subtract. $1001 - 210 = 791$. There are $\boxed{791}$ samples with at least one defective.

Example 2.5.23 (Lattice paths as combinations)**.** A courier moves on a grid from the southwest corner to a point 5 blocks east and 3 blocks north, taking only unit steps East (E) or North (N). How many shortest routes are there?

Which rule, and why? Every shortest route uses exactly 5 E-steps and 3 N-steps, 8 steps in all; a route is completely determined by *which* 3 *of the* 8 *step-slots are the* N*-steps.* That is an unordered choice of positions, hence $\binom{8}{3}$. The reframing "choose which positions hold a given symbol" is the standard bridge from arrangements to combinations.

Solution.

Total steps. $5 + 3 = 8$ steps.

Choose the North positions. $\displaystyle\binom{8}{3} = \frac{8\cdot 7\cdot 6}{6} = 56.$

Result. There are $\boxed{56}$ shortest routes. (Equivalently $\binom{8}{5} = 56$ by symmetry, choosing the East positions instead.)

Example 2.5.24 (Handshakes around a table)**.** Eleven delegates meet, and every pair shakes hands exactly once. How many handshakes occur?

Which rule, and why? A handshake is an unordered pair of delegates: Ana–Bo is the same handshake as Bo–Ana, and nobody shakes their own hand (no repetition). Choosing an unordered pair from 11 people is the combination $\binom{11}{2}$. Using $P(11, 2)$ would count each handshake twice, once per order, doubling the answer.

Solution.

Apply the formula. $\displaystyle\binom{11}{2} = \frac{11 \cdot 10}{2} = \frac{110}{2}.$

Compute. $110/2 = 55$. There are $\boxed{55}$ handshakes.

Example 2.5.25 (Selecting from two pools by "and")**.** A trivia squad needs 2 history buffs chosen from 6 and 3 science buffs chosen from 8. How many distinct squads are possible?

Which rule, and why? Each pool contributes an *unordered* selection (a squad has no internal ranking), so each is a combination. The two selections happen in stages joined by "and," so the multiplication principle multiplies the two counts. The word "and" linking two independent sub-choices is the cue to multiply rather than add.

Solution.

Choose history buffs. $\displaystyle\binom{6}{2} = \frac{6 \cdot 5}{2} = 15.$

Choose science buffs. $\displaystyle\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{6} = 56.$

Multiply. $15 \cdot 56 = 840$. There are $\boxed{840}$ squads.

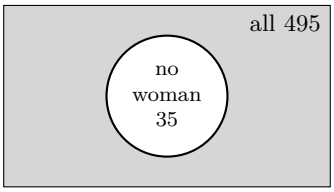
Example 2.5.26 ("At least one" committee — complement plus a set picture)**.** A board has 5 women and 7 men. A committee of 4 is chosen with all members equal. In how many ways does the committee include *at least one* woman?

Which rule, and why? A committee is an unordered subset, so combinations apply. "At least one woman" is the cue for *complementary counting*: rather than summing the cases of exactly 1, 2, 3, or 4 women, count all committees and subtract the single unwanted block of all-men committees. The Venn picture shows why: the wanted region is everything outside the "no women" disk.

Solution.

All committees. $\displaystyle\binom{12}{4} = \frac{12 \cdot 11 \cdot 10 \cdot 9}{24} = 495.$

Committees with no woman (all 4 from the 7 men). $\displaystyle\binom{7}{4} = \binom{7}{3} = \frac{7 \cdot 6 \cdot 5}{6} = 35.$



shaded = at least one woman = $495 - 35 = 460$

Subtract. $495 - 35 = 460$. There are $\boxed{460}$ committees with at least one woman.

Rule 4: arrangements of non-distinct objects (the multinomial coefficient)

Theorem 2.5.27 (Multinomial coefficient)**.** *Suppose n objects are to be arranged in a row, but they come in r types: n_1 identical copies of type 1, n_2 of type 2, …, n_r of type r, where $n_1 + n_2 + \cdots + n_r = n$. The number of* distinguishable *arrangements is*

$$\binom{n}{n_1, n_2, \ldots, n_r} = \frac{n!}{n_1!\, n_2! \cdots n_r!}.$$

Proof. Temporarily make the copies distinct (paint serial numbers on them). Then all n objects are different and there are $n!$ orderings. Now erase the serial numbers. Two of the $n!$ orderings collapse to the same visible arrangement exactly when they differ only by permuting copies *within* a type; there are $n_1!\, n_2! \cdots n_r!$ such internal permutations for each visible arrangement. Hence each distinguishable arrangement was counted $n_1! \cdots n_r!$ times, and dividing gives $n!/(n_1! \cdots n_r!)$. □

When $r = 2$ this reduces to a single binomial: $\dfrac{n!}{k!\,(n-k)!} = \binom{n}{k}$, since choosing which k of the n slots receive the type-1 object *is* a multinomial with two types. The multinomial coefficient is thus the natural generalisation of $\binom{n}{k}$ to more than two categories.

Example 2.5.28 (Rearranging a word with repeats)**.** How many distinguishable letter strings can be formed by rearranging all the letters of the word `PEPPER` (three P's, two E's, one R)?

Which rule, and why? We arrange *all* the letters (so it is a full permutation of 6 symbols), but some symbols are identical, so straight 6! overcounts. The signal is "rearrange the letters of a word with repeated letters." Divide by the factorial of each repeat count: this is the multinomial coefficient.

Solution.

Counts. $n = 6$ with 3 P's, 2 E's, 1 R.

Apply the formula. $\dfrac{6!}{3!\,2!\,1!} = \dfrac{720}{6 \cdot 2 \cdot 1} = \dfrac{720}{12} = 60$. There are $\boxed{60}$ distinguishable strings.

Example 2.5.29 (Splitting people into labelled groups)**.** Twelve interns are to be split into three project teams: 5 on team Alpha, 4 on team Beta, and 3 on team Gamma. How many ways can this be done?

Which rule, and why? We are partitioning all 12 people into groups of *fixed, named* sizes. Each person ends up wearing one of three labels (Alpha/Beta/Gamma), so this is exactly arranging 12 items of which 5, 4, 3 are of each type — a multinomial coefficient. Because the teams are *labelled* (distinct names), we do *not* divide by 3! for reordering the teams.

Solution.

Apply the formula. $\dbinom{12}{5, 4, 3} = \dfrac{12!}{5!\,4!\,3!}$.

Reduce stepwise. $\dfrac{12!}{5!\,4!\,3!} = \dbinom{12}{5}\dbinom{7}{4}\dbinom{3}{3} = 792 \cdot 35 \cdot 1$.

Compute. $792 \cdot 35 = 27{,}720$. There are $\boxed{27{,}720}$ ways.

Example 2.5.30 (Distributing distinct prizes by count)**.** Eight distinct trophies are to be handed out so that 4 go in a display case, 3 go to the archive, and 1 goes to the lobby. How many distributions are possible?

Which rule, and why? Each trophy is assigned one of three destinations, with destination *counts* fixed at $4, 3, 1$. This is the same structure as arranging 8 items into three labelled types, so it is a multinomial coefficient. The destinations are distinct places, so they are labelled and there is no extra division.

Solution.

Apply the formula. $\dbinom{8}{4, 3, 1} = \dfrac{8!}{4!\,3!\,1!} = \dfrac{40{,}320}{24 \cdot 6 \cdot 1} = \dfrac{40{,}320}{144}$.

Compute. $40{,}320/144 = 280$. There are $\boxed{280}$ distributions.

Example 2.5.31 (Arrangement with a non-adjacency constraint)**.** How many distinguishable arrangements of the letters in `LEVEL` (two L's, two E's, one V) have the two L's *not* adjacent?

Which rule, and why? The base count is a multinomial (rearranging a word with repeats). The *not adjacent* condition calls for complementary counting on top of the multinomial: total arrangements minus those with the L's together. Treating "the L's together" as a single glued block reduces that sub-count to another multinomial.

Solution.

Total arrangements. $\dfrac{5!}{2!\,2!\,1!} = \dfrac{120}{4} = 30.$

Glue the two L's into one block `[LL]`. Now arrange 4 items $\{[LL], E, E, V\}$ with two E's identical: $\dfrac{4!}{2!} = \dfrac{24}{2} = 12.$

Subtract. $30 - 12 = 18$. There are $\boxed{18}$ arrangements with the L's apart.

Example 2.5.32 (Rearranging a longer word with several repeats)**.** How many distinguishable strings come from rearranging all the letters of `BANANA` (one B, three A's, two N's)?

Which rule, and why? We arrange *all* 6 letters, but three are identical A's and two are identical N's, so a plain 6! overcounts every string. The signal is "rearrange a word with repeated letters," which is the multinomial coefficient: divide 6! by the factorial of each repeat count. A combination does not fit, because we are ordering all letters, not selecting a subset.

Solution.

Counts. $n = 6$ with 1 B, 3 A's, 2 N's.

Apply the formula. $\dfrac{6!}{1!\,3!\,2!} = \dfrac{720}{1 \cdot 6 \cdot 2} = \dfrac{720}{12} = 60.$ There are $\boxed{60}$ distinguishable strings.

Example 2.5.33 (Equal-sized labelled teams)**.** Nine volunteers are split into three labelled crews — Red, Green, Blue — of 3 each. How many splittings are possible?

Which rule, and why? Every volunteer receives one of three labels, with the label *counts* fixed at $3, 3, 3$. That is arranging 9 items into three labelled types: a multinomial coefficient. Because the crews carry *distinct names*, we do *not* divide by 3! for reordering the crews — that extra division would only apply if the three groups were interchangeable and unlabelled.

Solution.

Apply the formula. $\dbinom{9}{3,3,3} = \dfrac{9!}{3!\,3!\,3!} = \dbinom{9}{3}\dbinom{6}{3}\dbinom{3}{3} = 84 \cdot 20 \cdot 1.$

Compute. $84 \cdot 20 = 1680$. There are $\boxed{1680}$ splittings.

Example 2.5.34 (A long word and a multinomial probability)**.** The letters of `STATISTICS` (three S's, three T's, two I's, one A, one C) are arranged at random in a row, all distinguishable arrangements equally likely. **(a)** How many distinguishable arrangements are there? **(b)** What fraction of them begin with the letter A?

Which rule, and why? Part (a) is a pure multinomial: arrange all 10 letters with repeat counts $3, 3, 2, 1, 1$. Part (b) turns the count into a probability — favourable arrangements over total — and the favourable ones are themselves a smaller multinomial (fix A first, then arrange the remaining 9 letters). Combinations do not apply because we order every letter, not choose a subset.

Solution.

(a) Total arrangements. $\dfrac{10!}{3!\,3!\,2!\,1!\,1!} = \dfrac{3{,}628{,}800}{6 \cdot 6 \cdot 2} = \dfrac{3{,}628{,}800}{72} = 50{,}400.$ There are $\boxed{50{,}400}$ arrangements.

(b) Fix A in position one. The remaining 9 letters (three S's, three T's, two I's, one C) arrange in $\dfrac{9!}{3!\,3!\,2!\,1!} = \dfrac{362{,}880}{6 \cdot 6 \cdot 2} = 5040$ ways.

Form the ratio. $\dfrac{5040}{50{,}400} = \dfrac{1}{10}$. A fraction of $\boxed{1/10}$ of the arrangements begin with A. (This matches intuition: A is one of 10 letter-slots equally likely to lead, and there is just one A.)

Two short add-ons: circular permutations and stars and bars

Circular permutations. When n distinct objects are arranged in a *circle* and only the relative order matters (no fixed seat, so rotations are regarded as the same arrangement), the count is $(n-1)!$ rather than $n!$. The reason: every circular arrangement corresponds to n different linear arrangements (one per choice of which object we read first), so $n!$ overcounts by a factor of n, giving $n!/n = (n-1)!$. If, in addition, reflections (clockwise versus counterclockwise, as for an unmarked necklace) are considered identical, divide again by 2 to get $(n-1)!/2$ for $n \geq 3$.

Example 2.5.35 (Round table)**.** In how many ways can 6 distinct guests be seated around a circular table, counting two seatings as the same when one is a rotation of the other?

Which rule, and why? The phrase "around a circular table" with rotations identified is the exact trigger for circular permutations. We fix one guest as a reference point (killing the rotational symmetry) and arrange the rest linearly, giving $(n-1)!$.

Solution.

Fix one guest, arrange the other 5. $(6-1)! = 5!$.

Compute. $5! = 120$. There are $\boxed{120}$ distinct circular seatings.

Example 2.5.36 (Beads on a bracelet)**.** Seven distinct beads are strung on a bracelet. Two arrangements count as the same if one can be obtained from the other by rotating the bracelet *or* by flipping it over. How many distinct bracelets are there?

Which rule, and why? Beads on a closed loop with no fixed start is the circular-permutation setup, so the rotational count is $(n-1)!$. The extra phrase "flipping it over" identifies clockwise and counterclockwise readings, a reflection symmetry, so we divide once more by 2. This is the $(n-1)!/2$ refinement, used precisely when an arrangement and its mirror image are indistinguishable.

Solution.

Rotational count. $(7-1)! = 6! = 720$.

Identify reflections. Each bracelet equals its flip, so divide by 2: $720/2 = 360$. There are $\boxed{360}$ distinct bracelets.

Example 2.5.37 (Couples alternating around a table)**.** Four married couples (eight people) are seated around a round table so that the genders alternate, with rotations regarded as identical. How many seatings are possible? (Treat one gender group as "set A" and the other as "set B"; alternation means the seats strictly alternate A,B,A,B,... around the circle.)

Which rule, and why? The base structure is circular (rotations identified), but a constraint (alternation) splits the seating into two interleaved stages. We first seat one group to break the rotational symmetry, then drop the other group into the gaps. Fixing one group's circular arrangement uses $(n-1)!$; the second group then fills labelled gaps as an ordinary linear permutation.

Solution.

Seat the A-group circularly. The 4 people of set A around the circle: $(4-1)! = 3! = 6$ ways. This also fixes the rotation.

Seat the B-group in the four gaps. The 4 gaps are now distinct (each sits between two fixed A-people), so the B-group fills them in $4! = 24$ ways.

Multiply. $6 \cdot 24 = 144$. There are $\boxed{144}$ alternating seatings.

Example 2.5.38 (Round table with two people forced together)**.** Five distinct delegates sit around a circular table (rotations identified), but two specific delegates insist on sitting next to each other. How many seatings satisfy this?

Which rule, and why? This is a circular permutation with an adjacency constraint, the rotational analogue of the glued-block trick. Treat the inseparable pair as a single block, count the circular arrangements of the blocks, then multiply by the internal orderings of the block. Gluing reduces the object count from 5 to 4 for the circular step.

Solution.

Glue the pair into one block. Now 4 objects sit in a circle: $(4-1)! = 3! = 6$ ways.

Order inside the block. The two glued delegates can sit in $2! = 2$ internal orders.

Multiply. $6 \cdot 2 = 12$. There are $\boxed{12}$ seatings with the pair together.

Example 2.5.39 (Eight at a round table)**.** Eight distinct delegates sit around a circular table. Two seatings are the same when one is a rotation of the other. How many distinct seatings are there?

Which rule, and why? "Around a circular table" with rotations identified is the plain circular permutation, so the count is $(n-1)!$ rather than $n!$. There is no reflection clause (the table is not flipped over), so we do *not* divide by 2 here, unlike a necklace or bracelet.

Solution.

Fix one delegate, arrange the rest. $(8-1)! = 7!$.

Compute. $7! = 5040$. There are $\boxed{5040}$ distinct seatings.

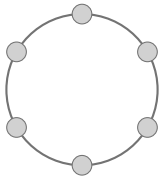
Example 2.5.40 (Charms on a keychain — rotations and flips)**.** Six distinct charms are placed on a circular keychain. Two keychains are identical if one is a rotation *or* a flip of the other. How many distinct keychains are possible?

Which rule, and why? Charms on a closed loop with no fixed start is the circular setup, giving the rotational count $(n-1)!$. "Or a flip" identifies a keychain with its mirror image, a reflection symmetry, so we divide once more by 2. This is the $(n-1)!/2$ refinement, applied exactly when an arrangement and its reflection are indistinguishable. The grey sketch shows the loop has no marked first bead.

Solution.

Rotational count. $(6-1)! = 5! = 120$.

Identify reflections. Each keychain equals its flip, so divide by 2: $120/2 = 60$.



no fixed start; a flip is the same keychain

Result. There are $\boxed{60}$ distinct keychains.

Example 2.5.41 (Round table where two people refuse to be neighbours)**.** Six distinct friends sit around a round table (rotations identified), but two of them, who are quarrelling, refuse to sit next to each other. How many seatings satisfy this?

Which rule, and why? This is a circular permutation with an *anti*-adjacency constraint. The clean route is complementary counting: take all circular seatings, then subtract those in which the quarrelling pair *is* adjacent (found with the glued-block trick). Subtraction beats listing the many allowed cases, just as the complement trick did for linear arrangements.

Solution.

All circular seatings. $(6-1)! = 5! = 120$.

Seatings with the pair together. Glue the pair into one block, leaving 5 objects in a circle: $(5-1)! = 4! = 24$ ways; the block has $2! = 2$ internal orders, giving $24 \cdot 2 = 48$.

Subtract. $120 - 48 = 72$. There are $\boxed{72}$ seatings with the pair apart.

Stars and bars. To count the ways of choosing k items from n types *with* repetition and *without* regard to order — equivalently, the number of nonnegative integer solutions of $x_1 + x_2 + \cdots + x_n = k$ — picture k identical stars in a row and insert $n-1$ bars to split them into n groups (the group sizes are the x_i). A configuration is any arrangement of k stars and $n-1$ bars, and choosing which of the $k+(n-1)$ positions hold the bars gives

$$\binom{n+k-1}{k} = \binom{n+k-1}{n-1}.$$

$$\star\star \mid \mid \star\star\star$$

$x_1 = 2, \quad x_2 = 0, \quad x_3 = 3$ (here $n = 3$ types, $k = 5$ stars, $n - 1 = 2$ bars)

Example 2.5.42 (Buying doughnuts of several kinds)**.** A shop sells 4 kinds of doughnut. A customer buys a box of 9 doughnuts; only how many of each kind matters, repeats are allowed, and a kind may be skipped. How many possible boxes are there?

Which rule, and why? The signals are: repetition allowed (multiple of the same kind), order irrelevant (a box is a multiset, not a sequence), and we are distributing a fixed total 9 across $n = 4$ type-counts. That is precisely nonnegative integer solutions of $x_1 + x_2 + x_3 + x_4 = 9$ — a stars-and-bars problem. Plain combinations $\binom{4}{9}$ are meaningless here (we pick more than the number of types), which itself flags that repetition must be in play.

Solution.

Set up. $n = 4$ types, $k = 9$ doughnuts, so $n - 1 = 3$ bars.

Apply the formula. $\binom{n+k-1}{k} = \binom{4+9-1}{9} = \binom{12}{9} = \binom{12}{3}$.

Compute. $\binom{12}{3} = \dfrac{12 \cdot 11 \cdot 10}{6} = 220$. There are $\boxed{220}$ possible boxes.

Example 2.5.43 (Distributing identical tokens to people)**.** In how many ways can 20 identical reward tokens be handed out to 5 employees, if any employee may receive any number (including zero)?

Which rule, and why? The tokens are *identical* (only the count each person gets matters) and an employee may receive zero or many, so this is the number of nonnegative integer solutions of $x_1 + \cdots + x_5 = 20$. Identical items into distinct recipients with repetition allowed is the stars-and-bars signature, with $k = 20$ stars and $n - 1 = 4$ bars. Because the tokens are indistinguishable, ordinary permutations or n^k do not apply.

Solution.

Set up. $n = 5$ recipients, $k = 20$ tokens, so $n - 1 = 4$ bars.

Apply the formula. $\binom{n+k-1}{k} = \binom{5+20-1}{20} = \binom{24}{20} = \binom{24}{4}$.

Compute. $\binom{24}{4} = \dfrac{24 \cdot 23 \cdot 22 \cdot 21}{24} = 23 \cdot 22 \cdot 21 = 10{,}626$. There are $\boxed{10{,}626}$ distributions.

Example 2.5.44 (A positivity constraint: every recipient gets at least one)**.** A philanthropist splits 12 identical grants among 4 charities so that *each* charity receives *at least one* grant. How many splits are possible?

Which rule, and why? Identical grants into distinct charities is stars and bars, but the "at least one each" condition forbids the zero counts that the basic formula allows. The standard fix is to pre-assign one grant to each charity, then distribute the remainder freely with zeros now permitted. After pre-assigning, count nonnegative solutions of a smaller total.

Solution.

Pre-assign one to each. Give each of the 4 charities one grant: 4 used, $12 - 4 = 8$ remain.

Distribute the rest freely. Nonnegative solutions of $y_1 + y_2 + y_3 + y_4 = 8$ with $n = 4$, $k = 8$: $\binom{4+8-1}{8} = \binom{11}{8} = \binom{11}{3}$.

Compute. $\binom{11}{3} = \dfrac{11 \cdot 10 \cdot 9}{6} = 165$. There are $\boxed{165}$ valid splits.

Example 2.5.45 (Counting nondecreasing sequences)**.** How many nondecreasing length-3 sequences ($a_1 \le a_2 \le a_3$) can be formed using values from $\{1, 2, 3, 4, 5, 6\}$, with repeats allowed?

Which rule, and why? A nondecreasing sequence is determined entirely by *how many times each value is used*, not by any ordering — once the multiset of values is fixed, there is exactly one way to write it in nondecreasing order. So this is choosing a size-3 multiset from 6 types with repetition: stars and bars with $n = 6$ types and $k = 3$ picks. Treating it as a plain permutation would overcount, since rearranging the values would break the order condition.

Solution.

Reframe as a multiset. Choose $k = 3$ values from $n = 6$ types, repetition allowed, order irrelevant.

Apply the formula. $\binom{n+k-1}{k} = \binom{6+3-1}{3} = \binom{8}{3}$.

Compute. $\binom{8}{3} = \dfrac{8 \cdot 7 \cdot 6}{6} = 56$. There are $\boxed{56}$ nondecreasing sequences.

Example 2.5.46 (Scoops from six flavours)**.** An ice-cream shop has 6 flavours. A bowl holds 10 scoops; only the count of each flavour matters, flavours may repeat, and a flavour may be skipped. How many distinct bowls are possible?

Which rule, and why? The signals are: repetition allowed (a flavour may appear several times), order irrelevant (a bowl is a multiset, not a sequence), and a fixed total 10 spread over $n = 6$ flavour-counts. That is nonnegative integer solutions of $x_1 + \cdots + x_6 = 10$, a stars-and-bars problem with $k = 10$ stars and $n - 1 = 5$ bars. A plain combination $\binom{6}{10}$ is meaningless (we take more scoops than there are flavours), which flags that repetition is in play. The sketch shows one bowl: 2 vanilla, 0 mint, … with bars separating the flavour groups.

Solution.

Set up. $n = 6$ flavours, $k = 10$ scoops, so $n - 1 = 5$ bars.

$$\star\star \mid \mid \star\star\star \mid \star \mid \star\star \mid \star$$

10 stars (scoops) and 5 bars split into 6 flavour groups: $2, 0, 3, 1, 2, 1$

Apply the formula. $\binom{n+k-1}{k} = \binom{6+10-1}{10} = \binom{15}{10} = \binom{15}{5}$.

Compute. $\binom{15}{5} = \dfrac{15 \cdot 14 \cdot 13 \cdot 12 \cdot 11}{120} = 3003$. There are $\boxed{3003}$ possible bowls.

Example 2.5.47 (Identical balls into labelled boxes)**.** In how many ways can 15 identical balls be placed into 4 distinct boxes, where a box may hold any number (including none)?

Which rule, and why? The balls are *identical* (only the per-box count matters) and a box may hold zero or many, so this is the number of nonnegative integer solutions of $x_1 + x_2 + x_3 + x_4 = 15$. Identical items into distinct containers with repetition allowed is the stars-and-bars signature, $k = 15$ stars and $n - 1 = 3$ bars. Because the balls are indistinguishable, n^k or permutations would badly overcount.

Solution.

Set up. $n = 4$ boxes, $k = 15$ balls, so $n - 1 = 3$ bars.

Apply the formula. $\binom{n+k-1}{k} = \binom{4+15-1}{15} = \binom{18}{15} = \binom{18}{3}$.

Compute. $\binom{18}{3} = \dfrac{18 \cdot 17 \cdot 16}{6} = 816$. There are $\boxed{816}$ placements.

Example 2.5.48 (Every type used at least once)**.** A gift basket must contain exactly 7 pieces of fruit drawn from 3 types (apples, pears, plums), with *at least one* of each type. How many baskets are possible?

Which rule, and why? Identical fruit of each type into 3 named categories is stars and bars, but the "at least one of each" clause bars the zero counts the basic formula allows. The standard fix is to pre-assign one fruit to each type, then distribute the remainder freely with zeros now permitted. After pre-assigning, count nonnegative solutions of a smaller total — the same positivity manoeuvre used in the grants example.

Solution.

Pre-assign one of each. Three fruits used, $7 - 3 = 4$ remain.

Distribute the rest freely. Nonnegative solutions of $y_1 + y_2 + y_3 = 4$ with $n = 3$, $k = 4$: $\binom{3+4-1}{4} = \binom{6}{4} = \binom{6}{2}$.

Compute. $\binom{6}{2} = \dfrac{6 \cdot 5}{2} = 15$. There are $\boxed{15}$ possible baskets.

2.6 Combinatorial Probability

Counting tools become probability when every outcome of a suitably chosen sample space is equally likely. This section isolates three rules that together solve most finite-sample problems: the classical equally-likely model, the hypergeometric model for sampling without replacement, and its mirror image, the binomial model for sampling with replacement.

How to recognise which rule this section needs

Read the problem and answer three questions in order.

1. **Is every outcome equally likely?** If the wording says "at random," "well shuffled," "fair," or "each is equally likely," you may use the **classical model** $\mathbb{P}(A) = |A|/|S|$. The whole job is then two counts: the size of a sample space of equally likely outcomes, and the size of the favorable event. Every rule below is a packaged instance of this one.
2. **Are you drawing a sample without replacement from a pool of two kinds of object, and asking how many of one kind you get?** Trigger words: "draws n *without replacement*," "selects a committee/sample/hand," a fixed finite population split into "r of type A and w of type B," and a question about the count of type A in the sample. Use the **hypergeometric model**. The draws are *not* independent: each removal changes the composition of what remains.
3. **Is each trial repeated independently with the same success probability?** Trigger words: "with replacement," "each trial," "repeated independently," tossing a coin or rolling a die n times, or a population so large that removing a few items barely changes the proportion. Use the **binomial model**. Here the trials *are* independent and p is constant.

The decisive contrast between (2) and (3) is replacement: *without* replacement and a finite pool means hypergeometric; *with* replacement (or an effectively infinite pool) means binomial. When a sample is small relative to a huge population the two nearly agree, but only the hypergeometric is exact for a finite pool.

2.6.1 Rule 1: the classical equally-likely model

Theorem 2.6.1 (Classical probability)**.** *Let S be a finite sample space whose outcomes are equally likely, so each singleton has probability $1/|S|$. Then for every event $A \subseteq S$,*

$$\mathbb{P}(A) = \frac{|A|}{|S|} = \frac{\textit{number of favorable outcomes}}{\textit{number of possible outcomes}}.$$

Proof. Write $S = \{s_1, \ldots, s_m\}$ with $m = |S|$. Equal likelihood means $\mathbb{P}(\{s_i\}) = c$ for a common constant c. The singletons are disjoint and their union is S, so by the additivity axiom $1 = \mathbb{P}(S) = \sum_{i=1}^{m} \mathbb{P}(\{s_i\}) = mc$, forcing $c = 1/m$. Any event A is the disjoint union of its $|A|$ singletons, hence $\mathbb{P}(A) = \sum_{s \in A} \mathbb{P}(\{s\}) = |A| \cdot \frac{1}{m} = |A|/|S|$. □

The model has exactly one hazard: the count $|S|$ and the count $|A|$ must be taken over the *same* kind of outcome, and those outcomes must really be equally likely. If you count $|S|$ with ordered outcomes you must count $|A|$ with ordered outcomes too; the ordering factors then cancel.

Example 2.6.2 (A fair pair of dice)**.** Two distinguishable fair dice (one red, one blue) are rolled. Find the probability that the two faces sum to 8.

Which rule, and why? The dice are **fair** and the rolls are physical and symmetric, so all ordered pairs of faces are **equally likely**; this is the bare classical model. No sampling-without-replacement is involved (the dice do not deplete one another), so the hypergeometric rule is irrelevant, and we are not counting repeated independent successes of a single labelled event, so the binomial rule is not what the question asks for.

Solution.

Step 1: sample space. An outcome is an ordered pair (red face, blue face). By the multiplication principle $|S| = 6 \times 6 = 36$, and each is equally likely.

Step 2: favorable outcomes. List the pairs summing to 8: $(2,6), (3,5), (4,4), (5,3), (6,2)$ — read off the marked anti-diagonal of the grid below. There are $|A| = 5$.

Step 3: probability. $\mathbb{P}(\text{sum} = 8) = \dfrac{5}{36} \approx 0.139$.

red \ blue	1	2	3	4	5	6
1	2	3	4	5	6	7
2	3	4	5	6	7	**8**
3	4	5	6	7	**8**	9
4	5	6	7	**8**	9	10
5	6	7	**8**	9	10	11
6	7	**8**	9	10	11	12

Example 2.6.3 (A random three-digit code)**.** A combination lock displays a 3-digit code, each digit independently and uniformly chosen from 0 through 9 (repeats allowed). What is the probability the code has all three digits distinct?

Which rule, and why? Each digit is chosen **uniformly**, so all 10^3 ordered codes are equally likely — the classical model on an ordered sample space. We are counting arrangements, not drawing two-type objects from a finite urn, so this is not hypergeometric; and although the digits are independent, the event "all distinct" is a single combinatorial count rather than "k successes in n trials," so the binomial formula is not the tool here.

Solution.

Step 1: total codes. Three positions, 10 choices each: $|S| = 10^3 = 1000$.

Step 2: codes with distinct digits. Fill the positions left to right: 10 ways for the first, 9 for the second (must differ), 8 for the third. By the multiplication principle $|A| = 10 \cdot 9 \cdot 8 = 720$.

Step 3: probability. $\mathbb{P}(\text{all distinct}) = \dfrac{720}{1000} = 0.72$.

Example 2.6.4 (Seating a committee in a row)**.** Four delegates — two from the North party and two from the South party — are seated in a random order along a bench of four seats, all 4! orderings equally likely. Find the probability that the two parties alternate (no two same-party delegates adjacent).

Which rule, and why? "Random order, all orderings equally likely" is the classical model on **permutations**. We count favorable *arrangements*, so order matters and we work on an ordered sample space; there is no draw of objects from an urn (hypergeometric) and no sequence of independent identical trials (binomial).

Solution.

Step 1: sample space. The delegates are distinct people, so $|S| = 4! = 24$ equally likely seatings.

Step 2: favorable arrangements. Alternation forces one of two party patterns, $NSNS$ or $SNSN$. Within a fixed pattern the two North delegates fill their two slots in 2! ways and the two South delegates in 2! ways. Thus $|A| = 2 \cdot 2! \cdot 2! = 2 \cdot 2 \cdot 2 = 8$.

Step 3: probability. $\mathbb{P}(\text{alternate}) = \dfrac{8}{24} = \dfrac{1}{3} \approx 0.333.$

Example 2.6.5 (Matching birthdays in a small group)**.** Five people have birthdays falling independently and uniformly on the 365 days of a non-leap year. Find the probability that at least two share a birthday.

Which rule, and why? "Independently and uniformly" over 365 days makes all 365^5 ordered birthday-lists equally likely — the classical model. The clean route is the complement "all five distinct." We are arranging days into ordered slots (a permutation count), not sampling two-type objects (hypergeometric) and not counting k labelled successes (binomial).

Solution.

Step 1: sample space. An outcome is an ordered list of five birthdays: $|S| = 365^5$.

Step 2: complement (all distinct). Assign distinct days left to right: $365 \cdot 364 \cdot 363 \cdot 362 \cdot 361$ ways. So $\mathbb{P}(\text{all distinct}) = \dfrac{365 \cdot 364 \cdot 363 \cdot 362 \cdot 361}{365^5}$.

Step 3: evaluate. The product of the five ratios is $1 \cdot \frac{364}{365} \cdot \frac{363}{365} \cdot \frac{362}{365} \cdot \frac{361}{365} \approx 0.9729$.

Step 4: take the complement. $\mathbb{P}(\text{at least one match}) = 1 - 0.9729 \approx 0.0271$.

Example 2.6.6 (At least one six on two dice)**.** Two distinguishable fair dice are rolled. Find the probability that at least one die shows a six.

Which rule, and why? The dice are **fair**, so all 36 ordered face-pairs are **equally likely** — the classical model. "At least one" is fastest through the complement "no six on either die," a single block of the grid. Nothing is drawn from a depleting pool (not hypergeometric), and the question is one combinatorial count on a 6×6 grid rather than a tally of k labelled successes (not binomial).

Solution.

Step 1: sample space. Ordered pairs (first die, second die): $|S| = 6 \times 6 = 36$, each equally likely.

Step 2: complement (no six). Each die avoids the six in 5 ways, so the pairs with no six number $5 \times 5 = 25$ — the unshaded 5×5 block at the upper left of the grid.

Step 3: probability. $\mathbb{P}(\text{at least one six}) = 1 - \dfrac{25}{36} = \dfrac{11}{36} \approx 0.306$. The shaded last row and last column below mark the 11 favorable pairs.

(1, 1)	(2, 1)	(3, 1)	(4, 1)	(5, 1)	(6, 1)
(1, 2)	(2, 2)	(3, 2)	(4, 2)	(5, 2)	(6, 2)
(1, 3)	(2, 3)	(3, 3)	(4, 3)	(5, 3)	(6, 3)
(1, 4)	(2, 4)	(3, 4)	(4, 4)	(5, 4)	(6, 4)
(1, 5)	(2, 5)	(3, 5)	(4, 5)	(5, 5)	(6, 5)
(1, 6)	(2, 6)	(3, 6)	(4, 6)	(5, 6)	(6, 6)

Example 2.6.7 (A random lattice path through a checkpoint)**.** A delivery robot walks from corner $(0, 0)$ to corner $(4, 3)$ on a street grid, each step going one block East or one block North, and it chooses one of the shortest routes uniformly at random. Find the probability that its route passes through the intersection $(2, 1)$.

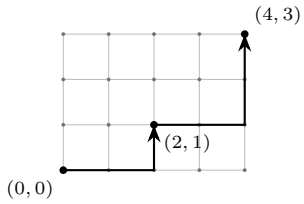
Which rule, and why? "Uniformly at random" over the shortest routes makes every monotone lattice path **equally likely** — the classical model, with $|S|$ and $|A|$ both counted as path counts. Each path is a single arrangement of E/N steps, so we count combinations of step positions, not draws from a two-type urn (hypergeometric) and not k successes in repeated trials (binomial).

Solution.

Step 1: sample space. A route is a sequence of 4 E and 3 N steps; choosing which 3 of the 7 steps are N gives $|S| = \binom{7}{3} = 35$ equally likely routes.

Step 2: favorable routes. A route through $(2, 1)$ splits into a leg $(0, 0) \to (2, 1)$ — 2 E and 1 N, $\binom{3}{1} = 3$ ways — followed by a leg $(2, 1) \to (4, 3)$ — 2 E and 2 N, $\binom{4}{2} = 6$ ways. By the multiplication principle $|A| = 3 \cdot 6 = 18$.

Step 3: probability. $\mathbb{P}(\text{through } (2, 1)) = \dfrac{18}{35} \approx 0.514$.



Example 2.6.8 (A PIN containing a chosen digit)**.** A four-digit PIN has each digit chosen independently and uniformly from 0 through 9 (repeats allowed). Find the probability that the digit 7 appears at least once.

Which rule, and why? Each digit is **uniform**, so all 10^4 ordered PINs are **equally likely** — the classical model. "At least once" is cleanest by the complement "no 7 in any position." We arrange digits into ordered slots (a permutation-style count), not draw two-type objects from a finite pool (hypergeometric); and although "at least one 7" could be expanded as a binomial sum over positions, the complement keeps it a single equally-likely count, which is the intended tool here.

Solution.

Step 1: sample space. Four positions, 10 digits each: $|S| = 10^4 = 10{,}000$.

Step 2: complement (no seven). Each position avoids 7 in 9 ways, so $9^4 = 6561$ PINs contain no 7, giving $\mathbb{P}(\text{no } 7) = \dfrac{6561}{10{,}000} = 0.6561$.

Step 3: take the complement. $\mathbb{P}(\text{at least one } 7) = 1 - 0.6561 = 0.3439$.

2.6.2 Rule 2: the hypergeometric model (sampling without replacement)

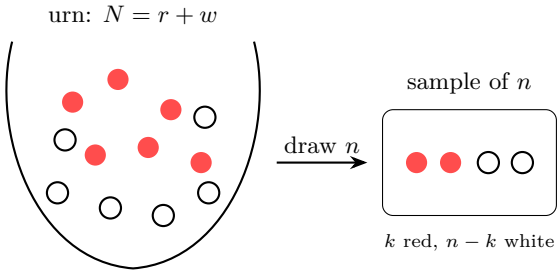
Theorem 2.6.9 (Hypergeometric model)**.** *An urn holds N objects of which r are of type A ("red") and $w = N - r$ are of type B ("white"). Draw n objects at random without replacement. The probability that exactly k of the drawn objects are red is*

$$\mathbb{P}(K = k) = \frac{\binom{r}{k}\binom{w}{n-k}}{\binom{N}{n}}, \qquad \max(0,\, n - w) \le k \le \min(n, r).$$

Proof. Because the draw is at random and order within the sample carries no information, take as sample space the collection of all unordered n-subsets of the N objects; there are $\binom{N}{n}$ of them and, by symmetry,

they are equally likely. To build a favorable subset — one with exactly k red — choose which k of the r red objects are included, in $\binom{r}{k}$ ways, and independently which $n-k$ of the w white objects are included, in $\binom{w}{n-k}$ ways. By the multiplication principle the number of favorable subsets is $\binom{r}{k}\binom{w}{n-k}$. Divide by $\binom{N}{n}$ and apply the classical model (Rule 1). The stated range of k is exactly where both binomial coefficients are nonzero: at least 0 and at most r red can be taken, and the $n-k$ whites require $0 \le n-k \le w$. □

The signature of this model is a single finite pool, split into two kinds, sampled *without* replacement, with the question asking how many of one kind land in the sample.



Example 2.6.10 (A small lottery)**.** A lottery machine holds 40 balls numbered 1 through 40. Six are drawn at random without replacement and constitute the winning set. A player has bought a single ticket marking six numbers. Find the probability that the player matches exactly four of the six drawn numbers.

Which rule, and why? There is one finite pool of 40 balls, split into two types relative to the player's ticket: 6 "player's numbers" and 34 "other numbers." Six balls are drawn **without replacement**, and we want the count of player's numbers in the draw — the exact template of the hypergeometric model. It is not binomial because the draws deplete the machine (no replacement, so trials are dependent and the success probability shifts after each draw).

Solution.

Step 1: identify the parameters. Pool $N = 40$; type-A (player's numbers) $r = 6$; type-B (others) $w = 34$; sample size $n = 6$; target $k = 4$.

Step 2: favorable count. Choose 4 of the player's 6 numbers, $\binom{6}{4}$ ways, and 2 of the other 34, $\binom{34}{2}$ ways: $\binom{6}{4}\binom{34}{2} = 15 \cdot 561 = 8415$.

Step 3: sample-space size. $\binom{40}{6} = 3\,838\,380$.

Step 4: probability. $\mathbb{P}(K=4) = \dfrac{8415}{3\,838\,380} \approx 0.00219$, roughly 1 in 456.

Example 2.6.11 (A bipartisan committee)**.** A department has 7 statisticians and 5 biologists. A committee of 4 is chosen at random (all $\binom{12}{4}$ committees equally likely). Find the probability the committee contains exactly 2 statisticians and 2 biologists.

Which rule, and why? One pool of 12 people, split into two types (statisticians, biologists), selected **without replacement** (a person can sit on the committee only once), with a question about how many of one type are chosen — hypergeometric. "Choose a committee" is the canonical unordered without-replacement draw; binomial would require independent repeated trials with a fixed success probability, which committee selection does not provide.

Solution.

Step 1: parameters. $N = 12$, type-A statisticians $r = 7$, type-B biologists $w = 5$, sample $n = 4$, target $k = 2$.

Step 2: favorable count. $\binom{7}{2}\binom{5}{2} = 21 \cdot 10 = 210$.

Step 3: total committees. $\binom{12}{4} = 495$.

Step 4: probability. $\mathbb{P}(2 \text{ of each}) = \dfrac{210}{495} = \dfrac{14}{33} \approx 0.424$.

Example 2.6.12 (A full house in poker)**.** Five cards are dealt at random from a standard 52-card deck. Find the probability of a **full house**: three cards of one rank and two of another.

Which rule, and why? A deal of 5 cards from 52 is sampling **without replacement** from a finite pool, and the equally-likely sample space is the set of $\binom{52}{5}$ hands. The counting is a layered hypergeometric/multiplication argument on the 13 ranks and 4 suits. It is decidedly not binomial — cards are not replaced and the deck is finite, so successive cards are dependent.

Solution.

Step 1: sample space. $|S| = \binom{52}{5} = 2\,598\,960$ equally likely hands.

Step 2: count full houses. Choose the rank that appears three times: 13 ways. Choose 3 of its 4 suits: $\binom{4}{3} = 4$. Choose a *different* rank for the pair: 12 ways. Choose 2 of its 4 suits: $\binom{4}{2} = 6$. By the multiplication principle $|A| = 13 \cdot 4 \cdot 12 \cdot 6 = 3744$.

Step 3: probability. $\mathbb{P}(\text{full house}) = \dfrac{3744}{2\,598\,960} \approx 0.00144$, about 1 in 694.

Example 2.6.13 (At least one defective, by complement)**.** A carton holds 25 memory chips, of which 4 are defective. A buyer inspects a random sample of 5 without replacement. Find the probability that the sample contains at least one defective chip.

Which rule, and why? A finite pool of 25 chips split into defective/good, sampled **without replacement**, with a question about the number of defectives — hypergeometric. "At least one" is cleanest via the complement "zero defectives," still a hypergeometric term. Replacement is absent, so this is not the binomial model, though the two would nearly coincide if the carton were enormous.

Solution.

Step 1: parameters. $N = 25$, defectives $r = 4$, good $w = 21$, sample $n = 5$.

Step 2: complement (zero defectives, $k = 0$). $\mathbb{P}(K = 0) = \dfrac{\binom{4}{0}\binom{21}{5}}{\binom{25}{5}} = \dfrac{1 \cdot 20{,}349}{53{,}130} \approx 0.383.$

Step 3: take the complement. $\mathbb{P}(K \geq 1) = 1 - 0.383 \approx 0.617$.

Example 2.6.14 (Capture–recapture estimate)**.** To gauge a pond's fish population, a biologist nets 20 fish, tags them, and releases them. Later she nets a fresh sample of 15 fish and finds 3 tagged. Treating the pond as holding $N = 200$ fish, find the probability that exactly 3 of the 15 recaptured fish are tagged.

Which rule, and why? At the time of the second netting the pond is a fixed finite pool of $N = 200$ fish split into two types — 20 tagged and 180 untagged — and the 15 recaptured fish are a sample **without replacement** (no fish is caught twice in one haul). Asking how many of the sample are tagged is exactly hypergeometric. With replacement we would use binomial, but a single net haul removes fish from availability, so the draws are dependent.

Solution.

Step 1: parameters. $N = 200$, tagged $r = 20$, untagged $w = 180$, sample $n = 15$, target $k = 3$.

Step 2: favorable count. $\binom{20}{3}\binom{180}{12}$.

Step 3: assemble the probability.

$$\mathbb{P}(K = 3) = \frac{\binom{20}{3}\binom{180}{12}}{\binom{200}{15}} \approx 0.129.$$

Step 4: interpret. Under $N = 200$ the expected number of tagged fish in the haul is $n\,r/N = 15 \cdot 20/200 = 1.5$, so the observed 3 sits in the upper tail; the most likely single count is 1. The outcome $K = 3$ is therefore above average yet far from rare, carrying a probability near 0.13. (This calculation underlies the maximum-likelihood population estimate: one chooses N to make the observed count most probable, and the small population sizes that would make $K = 3$ *most* likely are exactly those near $N \approx 100$, where the tagged fraction $20/N$ best matches the sample fraction $3/15$.)

Example 2.6.15 (One red pen in a handful)**.** A drawer holds 12 identical-feeling pens: 4 write red and 8 write blue. Three pens are grabbed at random (all $\binom{12}{3}$ handfuls equally likely). Find the probability that exactly one of the three is red.

Which rule, and why? One finite pool of 12 pens, split into two types (red, blue), grabbed **without replacement** — a pen is in the hand or not, never twice — with a question about how many reds appear: the hypergeometric template. It is not binomial, since each pen removed changes the red fraction among those still in the drawer, so the draws are dependent.

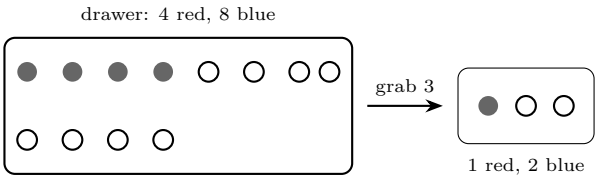
Solution.

Step 1: parameters. $N = 12$, type-A red $r = 4$, type-B blue $w = 8$, sample $n = 3$, target $k = 1$.

Step 2: favorable count. $\binom{4}{1}\binom{8}{2} = 4 \cdot 28 = 112$.

Step 3: sample-space size. $\binom{12}{3} = 220$.

Step 4: probability. $\mathbb{P}(K = 1) = \dfrac{112}{220} = \dfrac{28}{55} \approx 0.509$.



Example 2.6.16 (Exactly two aces in a five-card hand)**.** Five cards are dealt at random from a standard 52-card deck. Find the probability the hand holds exactly two aces.

Which rule, and why? A deal of 5 from 52 is sampling **without replacement** from a finite pool, equally likely over the $\binom{52}{5}$ hands. Split the deck into two types — 4 aces and 48 non-aces — and ask how many aces land in the hand: pure hypergeometric. Binomial would need independent repeated trials with fixed p, but each card dealt depletes the deck and shifts the chance of an ace on the next card.

Solution.

Step 1: parameters. $N = 52$, aces $r = 4$, non-aces $w = 48$, sample $n = 5$, target $k = 2$.

Step 2: favorable count. $\binom{4}{2}\binom{48}{3} = 6 \cdot 17{,}296 = 103\,776$.

Step 3: sample-space size. $\binom{52}{5} = 2\,598\,960$.

Step 4: probability. $\mathbb{P}(K = 2) = \dfrac{103\,776}{2\,598\,960} \approx 0.0399$, about 1 in 25.

Example 2.6.17 (A raffle, by complement)**.** A charity sells 50 raffle tickets and will draw 10 distinct winning tickets at random. You hold 5 of the 50 tickets. Find the probability that at least one of your tickets wins.

Which rule, and why? The 50 tickets are one finite pool split into "yours" (5) and "others" (45); the 10 winners are drawn **without replacement** (a ticket cannot win twice), and we count how many of yours are among them — hypergeometric. "At least one" is cleanest by the complement "none of yours wins," the $k = 0$ term. No replacement occurs, so the binomial model does not apply, though for a far larger ticket pool the two would nearly agree.

Solution.

Step 1: parameters. $N = 50$, yours $r = 5$, others $w = 45$, sample $n = 10$.

Step 2: complement (none of yours, $k = 0$). $\mathbb{P}(K = 0) = \dfrac{\binom{5}{0}\binom{45}{10}}{\binom{50}{10}} = \dfrac{3\,190\,187\,286}{10\,272\,278\,170} \approx 0.311$.

Step 3: take the complement. $\mathbb{P}(K \geq 1) = 1 - 0.311 \approx 0.689$.

2.6.3 Rule 3: the binomial model (sampling with replacement)

The hypergeometric model assumes the pool shrinks with each draw. Replace each object before the next draw — or sample from a pool so vast that removal is negligible — and the draws become independent with a constant success probability. The count of successes then follows the binomial model.

Theorem 2.6.18 (Binomial model)**.** *Perform n independent trials, each a "success" with the same probability p and a "failure" with probability $q = 1 - p$. The probability of exactly k successes is*

$$\mathbb{P}(X = k) = \binom{n}{k} p^k q^{n-k}, \qquad k = 0, 1, \ldots, n.$$

Proof. Fix one particular sequence of outcomes having k successes and $n - k$ failures in specified positions. By independence its probability is the product of the per-trial probabilities, $p^k q^{n-k}$, and this value is the same for *every* sequence with k successes, since it depends only on the counts. The number of such sequences is the number of ways to choose which k of the n positions are the successes, namely $\binom{n}{k}$. These sequences are mutually exclusive outcomes, so by additivity their probabilities sum, giving $\binom{n}{k} p^k q^{n-k}$. □

Remark (Why replacement changes the rule). Sampling *with* replacement from an urn of r red and $w = N - r$ white makes each draw an independent success (= red) with constant $p = r/N$; the red count is binomial(n, p). Sampling *without* replacement makes it hypergeometric. The two share the same mean np, but the binomial allows any k from 0 to n regardless of r, whereas the hypergeometric cannot return more reds than the urn holds. When N is huge relative to n, removing a few objects barely moves p, and the hypergeometric probabilities converge to the binomial ones — which is why "large population" problems are routinely modelled as binomial.

Example 2.6.19 (Repeated draws with replacement)**.** An urn holds 3 red and 7 white balls. A ball is drawn, its color noted, and **returned**; this is repeated 5 times. Find the probability of getting exactly 2 red balls.

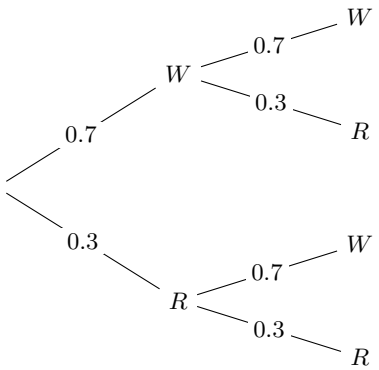
Which rule, and why? The ball is **returned** after each draw, so the urn's composition never changes: each of the 5 draws is an independent trial with the *same* success probability $p = 3/10$. Constant p and independence are the binomial signature. Had the balls *not* been replaced, the pool would shrink and we would use the hypergeometric model instead.

Solution.

Step 1: parameters. Trials $n = 5$, success = red with $p = 3/10 = 0.3$, failure $q = 0.7$, target $k = 2$.

Step 2: apply the formula. $\mathbb{P}(X = 2) = \binom{5}{2}(0.3)^2(0.7)^3 = 10 \cdot 0.09 \cdot 0.343$.

Step 3: evaluate. $10 \cdot 0.09 \cdot 0.343 = 0.3087$. The two-stage structure for one ordering (e.g. red, red, then three whites) is shown below; the binomial coefficient counts the $\binom{5}{2} = 10$ orderings, each contributing the same $p^2 q^3$.



Example 2.6.20 (A multiple-choice quiz)**.** A student guesses on each of 8 multiple-choice questions, each having 4 options with one correct answer, guesses independent. Find the probability of getting exactly 3 correct.

Which rule, and why? Each question is an **independent** trial with the *same* success probability $p = 1/4$, and there is no finite pool being depleted — guessing one question right does not change the odds on the next. Constant p plus independence over a fixed number of trials is the binomial model, not the hypergeometric (nothing is sampled without replacement here).

Solution.

Step 1: parameters. $n = 8$, $p = 1/4 = 0.25$, $q = 0.75$, $k = 3$.

Step 2: apply the formula. $\mathbb{P}(X = 3) = \binom{8}{3}(0.25)^3(0.75)^5 = 56 \cdot 0.015625 \cdot 0.2373$.

Step 3: evaluate. $56 \cdot 0.015625 = 0.875$, and $0.875 \cdot 0.2373 \approx 0.208$.

Example 2.6.21 (Defect rate on a production line)**.** A machine produces items independently, each defective with probability $p = 0.05$. In a run of 10 items, find the probability of at least one defective.

Which rule, and why? Items are produced **independently** with a *constant* defect probability $p = 0.05$ — effectively sampling with replacement from an inexhaustible process, so the binomial model applies. "At least one" is handled by the complement "none." There is no fixed finite pool of items to deplete, so the hypergeometric model does not fit.

Solution.

Step 1: parameters. $n = 10$, $p = 0.05$, $q = 0.95$.

Step 2: complement (zero defectives). $\mathbb{P}(X = 0) = \binom{10}{0}(0.05)^0(0.95)^{10} = (0.95)^{10} \approx 0.5987$.

Step 3: take the complement. $\mathbb{P}(X \geq 1) = 1 - 0.5987 \approx 0.401$.

Example 2.6.22 (Free throws)**.** A player sinks each free throw independently with probability $p = 0.8$. In a sequence of 6 attempts, find the probability of making at least 5.

Which rule, and why? Each attempt is an **independent** trial with the *same* make probability 0.8; nothing is consumed or depleted from a finite pool, so the binomial model governs the count of makes. "At least 5" means $k = 5$ or $k = 6$, summed. The hypergeometric model would require sampling without replacement from a finite two-type pool, which this is not.

Solution.

Step 1: parameters. $n = 6$, $p = 0.8$, $q = 0.2$.

Step 2: the two relevant terms. $\mathbb{P}(X = 5) = \binom{6}{5}(0.8)^5(0.2)^1 = 6 \cdot 0.32768 \cdot 0.2 = 0.393216$; $\mathbb{P}(X = 6) = (0.8)^6 = 0.262144$.

Step 3: add. $\mathbb{P}(X \geq 5) = 0.393216 + 0.262144 \approx 0.655$.

Example 2.6.23 (Hypergeometric vs. binomial: the same urn, two protocols)**.** An urn holds 30 red and 70 white balls ($N = 100$). Three balls are taken. Compute $\mathbb{P}$(exactly 2 red) under (a) sampling *without* replacement and (b) sampling *with* replacement, and compare.

Which rule, and why? This problem deliberately fixes the scenario and varies only the **protocol** to isolate the contrast. Protocol (a) "without replacement" depletes the finite pool, so $\mathbb{P}$ is hypergeometric. Protocol (b) "with replacement" restores the urn each time, making the draws independent with constant $p = 30/100 = 0.3$, so $\mathbb{P}$ is binomial. The decisive cue is the single word replacement.

Solution.

Part (a): hypergeometric. $N = 100$, $r = 30$, $w = 70$, $n = 3$, $k = 2$:

$$\mathbb{P}(K = 2) = \frac{\binom{30}{2}\binom{70}{1}}{\binom{100}{3}} = \frac{435 \cdot 70}{161{,}700} = \frac{30{,}450}{161{,}700} \approx 0.188.$$

Part (b): binomial. $n = 3$, $p = 0.3$, $q = 0.7$, $k = 2$:

$$\mathbb{P}(X = 2) = \binom{3}{2}(0.3)^2(0.7)^1 = 3 \cdot 0.09 \cdot 0.7 = 0.189.$$

Compare. The two answers, 0.188 and 0.189, nearly coincide because the sample ($n = 3$) is tiny relative to the pool ($N = 100$): removing two or three balls barely changes the red fraction, so the without-replacement process behaves almost like the constant-p with-replacement process. Shrink the pool toward the sample size and the gap widens; only the hypergeometric value stays exact for a finite urn.

Example 2.6.24 (Six heads in ten tosses)**.** A fair coin is tossed 10 times, the tosses independent. Find the probability of getting exactly 6 heads.

Which rule, and why? Each toss is an **independent** trial with the *same* success probability $p = 1/2$; the coin is not consumed, so nothing depletes between tosses. Fixed n, constant p, independent trials — the binomial model. There is no finite two-type pool sampled without replacement, so the hypergeometric model is not the tool.

Solution.

Step 1: parameters. $n = 10$, $p = 0.5$, $q = 0.5$, $k = 6$.

Step 2: apply the formula. $\mathbb{P}(X = 6) = \binom{10}{6}(0.5)^6(0.5)^4 = \binom{10}{6}(0.5)^{10} = 210 \cdot \dfrac{1}{1024}$.

Step 3: evaluate. $\mathbb{P}(X = 6) = \dfrac{210}{1024} \approx 0.205$.

Example 2.6.25 (Exactly two sixes in four rolls)**.** A fair die is rolled 4 times independently. Find the probability of rolling a six exactly twice.

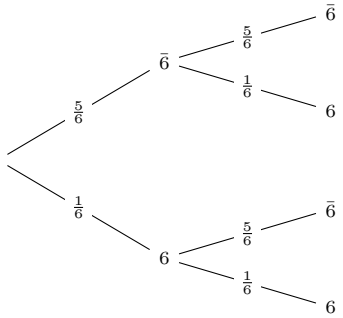
Which rule, and why? Each roll is an **independent** trial with constant success probability $p = 1/6$ for "six"; a die does not wear out, so p never changes between rolls. Constant p plus independence over a fixed number of trials is the binomial signature. Nothing is sampled without replacement from a depleting pool, so the hypergeometric model does not apply.

Solution.

Step 1: parameters. $n = 4$, $p = 1/6$, $q = 5/6$, $k = 2$.

Step 2: apply the formula. $\mathbb{P}(X = 2) = \binom{4}{2}\left(\dfrac{1}{6}\right)^2\left(\dfrac{5}{6}\right)^2 = 6 \cdot \dfrac{1}{36} \cdot \dfrac{25}{36}$.

Step 3: evaluate. $\mathbb{P}(X = 2) = \dfrac{6 \cdot 25}{1296} = \dfrac{150}{1296} \approx 0.116$. The tree below traces one favorable ordering (six, six, then two non-sixes); the coefficient $\binom{4}{2} = 6$ counts the orderings, each contributing the same $(1/6)^2(5/6)^2$.



Example 2.6.26 (A spam filter, by complement)**.** Incoming messages are independently spam with probability $p = 0.1$. Among the next 20 messages, find the probability of at most one spam message.

Which rule, and why? Messages arrive **independently**, each spam with a *constant* $p = 0.1$ — effectively sampling with replacement from an endless stream, so the binomial model governs the spam count. "At most one" means $k = 0$ or $k = 1$, summed. There is no fixed finite pool being depleted, so the hypergeometric model does not fit.

Solution.

Step 1: parameters. $n = 20$, $p = 0.1$, $q = 0.9$.

Step 2: the two relevant terms. $\mathbb{P}(X=0) = (0.9)^{20} \approx 0.1216$; $\mathbb{P}(X=1) = \binom{20}{1}(0.1)(0.9)^{19} = 20 \cdot 0.1 \cdot 0.1351 \approx 0.2702$.

Step 3: add. $\mathbb{P}(X \le 1) = 0.1216 + 0.2702 \approx 0.392$.

2.7 Exercises

All problems below are original. Full solutions begin on page 96.

A. Sample spaces and events

A1. A spinner has four equal sectors labelled $1, 2, 3, 4$. It is spun twice and the two labels are recorded in order. Write the sample space S and give $|S|$. With $A =$ "the two labels are equal" and $B =$ "the labels sum to 5," list A, B, and $A \cap B$, and state whether A and B are mutually exclusive.

A2. For three events A, B, C, write a set expression for each described event: (a) at least one of the three occurs; (b) exactly one of the three occurs; (c) at most one occurs; (d) A occurs but neither B nor C does.

A3. Prove the identity $A \cup (A^{\mathsf{c}} \cap B) = A \cup B$ both by a Venn diagram and by a membership argument.

A4. Use the distributive and De Morgan laws to simplify $\big((A \cap B) \cup (A \cap B^{\mathsf{c}})\big)^{\mathsf{c}}$ to a single complemented event.

A5. A box holds chips numbered 1 through 9. One chip is drawn. Let $E =$ "the number is even," $T =$ "the number exceeds 5," and $D =$ "the number is divisible by 3." List the outcomes in $E \cap T$, in $E \cup D$, and in $T^{\mathsf{c}} \cap D$.

A6. A fair coin is tossed repeatedly until it shows two heads in a row or until five tosses have been made, whichever happens first. List the sample space S.

A7. Show that the events $A \cap B$, $A \cap B^{\mathsf{c}}$, and A^{c} are pairwise disjoint and that their union is the whole sample space S.

A8. Translate into plain words: (a) $A \subseteq B$; (b) $A \cap B = \varnothing$; (c) $A \cup B = S$. Then give a single roll-of-a-die example of events A, B for which all three statements would force $A = B^{\mathsf{c}}$.

A9. A red die and a green die are rolled. Describe the sample space and give $|S|$. With $G =$ "the green die shows an even number" and $H =$ "the two dice show the same number," list $G \cap H$ and decide whether G and H are mutually exclusive.

A10. Two distinct letters are chosen, one after the other, from $\{a, b, c, d\}$ (order matters, no repeats). Write the sample space and give $|S|$. List the event $L =$ "the letter a appears."

A11. For events A, B, C, write a set expression for: (a) at least two of the three occur; (b) none of the three occurs; (c) exactly two occur; (d) C occurs but at most one of A, B does.

A12. A spinner is spun until it first lands on red or until it has been spun three times, whichever comes first. Each spin shows red (R) or blue (B). List the sample space S.

A13. Simplify $(A \cup B) \cap (A \cup B^{\mathsf{c}})$ to a single event using the distributive law, and confirm the result with a Venn diagram.

A14. A point (x, y) is chosen from the integer grid $\{1, 2, 3\} \times \{1, 2, 3\}$. Let $U =$ "$x = y$," $V =$ "$x + y = 4$," and $W =$ "$x < y$." List the outcomes in each event and state which two of U, V, W are mutually exclusive.

A15. Prove the identity $A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$ by a membership argument, and illustrate the left-hand side with a Venn picture.

B. The probability function

B1. Given $\mathbb{P}(A) = 0.55$, $\mathbb{P}(B) = 0.40$, and $\mathbb{P}(A \cap B) = 0.25$, find $\mathbb{P}(A \cup B)$, $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}})$, $\mathbb{P}$(exactly one of A, B), and $\mathbb{P}(A^{\mathsf{c}} \cap B)$.

B2. In a survey of 200 households, 120 subscribe to a streaming service, 90 own a game console, and 55 have both. How many households have at least one, and how many have neither?

B3. A loaded four-sided die assigns probabilities proportional to the face value: face k ($k = 1, 2, 3, 4$) has probability ck. Find c, then compute $\mathbb{P}(\text{face is odd})$ and $\mathbb{P}(\text{face} \geq 3)$.

B4. Prove that for any events A, B, $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$, and state exactly when equality holds.

B5. Events A, B, C satisfy $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 0.4$, all three pairwise intersections equal 0.15, and $\mathbb{P}(A \cap B \cap C) = 0.05$. Find $\mathbb{P}(A \cup B \cup C)$ and $\mathbb{P}(\text{none of the three})$.

B6. Show that if $\mathbb{P}(A) = 0.7$ and $\mathbb{P}(B) = 0.6$ then $\mathbb{P}(A \cap B) \geq 0.3$. What is the largest value $\mathbb{P}(A \cap B)$ could possibly take?

B7. Two fair dice are rolled. Find $\mathbb{P}(\text{sum is 7 or 11})$ and $\mathbb{P}(\text{at least one die shows a 6})$.

B8. A point is chosen uniformly at random in the unit square $[0, 1] \times [0, 1]$. Find the probability that its coordinates (x, y) satisfy $x + y \leq 1$, and the probability that they satisfy $y \leq x^2$.

B9. Given $\mathbb{P}(A) = 0.50$ and $\mathbb{P}(B) = 0.30$ with A, B disjoint, find $\mathbb{P}(A \cup B)$, $\mathbb{P}(A^{\mathsf{c}})$, and $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}})$.

B10. A loaded four-sided die assigns to face k ($k = 1, 2, 3, 4$) a probability proportional to k^2. Find the proportionality constant, then compute $\mathbb{P}(\text{face is odd})$ and $\mathbb{P}(\text{face} = 4)$.

B11. Three fair dice are rolled. Find $\mathbb{P}(\text{the sum equals 10})$ by counting the ordered outcomes.

B12. A point (x, y) is chosen uniformly from the rectangle $[0, 2] \times [0, 3]$. Find $\mathbb{P}(x + y \geq 2)$.

B13. For three events with $\mathbb{P}(A) = 0.5$, $\mathbb{P}(B) = 0.4$, $\mathbb{P}(C) = 0.3$, all pairwise intersections equal to 0.1, and $\mathbb{P}(A \cap B \cap C) = 0.05$, find $\mathbb{P}(A \cup B \cup C)$.

B14. Two numbers X, Y are chosen independently and uniformly from $[0, 1]$. Find $\mathbb{P}\big(\min(X, Y) \leq \frac{1}{3}\big)$.

B15. Prove that for any three events, $\mathbb{P}(A \cup B \cup C) \leq \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C)$ (Boole's inequality for three events).

C. Conditional probability and Bayes

C1. Two fair dice are rolled. Find $\mathbb{P}(\text{sum} = 8 \mid \text{the first die shows an even number})$.

C2. An urn holds 4 green and 6 yellow marbles. Two are drawn without replacement. Find $\mathbb{P}(\text{both green})$ and $\mathbb{P}(\text{second is green} \mid \text{first is green})$.

C3. A factory uses three machines M_1, M_2, M_3 that produce 40%, 35%, 25% of all parts, with defect rates 1%, 2%, 4% respectively. Find the overall defect rate. Given that a randomly chosen part is defective, find the probability it was made by M_3.

C4. A medical screening test for a condition present in 2% of a population has sensitivity 0.95 (it is positive given the condition is present) and specificity 0.90 (it is negative given the condition is absent). A person tests positive. Find the probability the person actually has the condition.

C5. A bag contains two ordinary coins and one trick coin that lands heads on both faces. A coin is drawn at random and tossed twice, landing heads both times. Find the probability the trick coin was drawn.

C6. Show that $\mathbb{P}(A \mid B) + \mathbb{P}(A^{\mathsf{c}} \mid B) = 1$ for any B with $\mathbb{P}(B) > 0$, and give a concrete die example in which $\mathbb{P}(A \mid B) + \mathbb{P}(A \mid B^{\mathsf{c}}) \neq 1$.

C7. Cards are drawn one at a time without replacement from a standard 52-card deck. Find $\mathbb{P}(\text{the third card is the first ace})$.

C8. In a group, 60% of people exercise regularly. Of those who exercise, 20% have high blood pressure; of those who do not, 45% do. A randomly chosen person has high blood pressure. Find the probability that the person exercises regularly.

C9. A standard deck is shuffled and two cards are drawn without replacement. Find $\mathbb{P}(\text{both kings})$ and $\mathbb{P}(\text{second is a king} \mid \text{first is a king})$.

C10. Three urns are chosen with equal probability. Urn 1 holds 2 red and 3 white balls, urn 2 holds 4 red and 2 white, and urn 3 holds 1 red and 5 white. An urn is picked at random and one ball is drawn. Find $\mathbb{P}(\text{red})$, and given the ball is red, find the probability it came from urn 2.

C11. A condition is present in 1% of a population. A test has sensitivity 0.90 and specificity 0.95, and is applied *twice independently* to one person. Both results are positive. Find the probability the person has the condition.

C12. A biased coin lands heads with probability 0.4. It is tossed repeatedly. Find the probability that the first head occurs on exactly the third toss.

C13. A box holds 3 defective and 7 good fuses. Two are drawn without replacement. Given that at least one of the two is defective, find the probability that both are defective.

C14. For events with $\mathbb{P}(B) > 0$, prove the chain rule $\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$, and use it to find $\mathbb{P}(A \cap B)$ when $\mathbb{P}(B) = 0.5$ and $\mathbb{P}(A \mid B) = 0.6$.

C15. A message passes through two relay nodes. Node 1 forwards it correctly with probability 0.9 and node 2, independently, with probability 0.8; the message arrives correctly only if both forward it correctly. Find $\mathbb{P}(\text{arrives correctly})$ and $\mathbb{P}(\text{at least one node forwarded correctly})$.

D. Independence

D1. A and B are independent with $\mathbb{P}(A) = 0.7$ and $\mathbb{P}(B) = 0.4$. Find $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$, and $\mathbb{P}(\text{exactly one of } A, B)$.

D2. Roll one fair die. Let $A = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6\}$. Determine whether A and B are independent. Repeat with $A = \{1, 2, 3, 4\}$ and $C = \{3, 4, 5, 6\}$.

D3. A relay link works if at least one of three independent cables carries the signal. The cables fail independently with probabilities 0.2, 0.1, and 0.05. Find the probability the link works.

D4. Roll two fair dice. Show that "the second die is odd" and "the sum is 7" are independent, but "the second die is odd" and "the sum is 8" are not.

D5. A fair coin is tossed five times. Find the probability of getting exactly three heads, and the probability that the first and last tosses are both heads.

D6. Suppose A and B are independent. Prove that A and B^{c} are also independent.

D7. Three switches operate independently, each closed with probability p. Current flows if switch 1 is closed *and* at least one of switches $2, 3$ is closed. Write the probability current flows as a function of p, and evaluate it at $p = 0.8$.

D8. Give an example of three events A, B, C that are pairwise independent but not mutually independent, using two tosses of a fair coin. Verify both claims.

D9. A and B are independent with $\mathbb{P}(A) = 0.5$ and $\mathbb{P}(B) = 0.2$. Find $\mathbb{P}(\text{exactly one of } A, B)$ and $\mathbb{P}(\text{neither } A \text{ nor } B)$.

D10. Four components operate independently, each working with probability 0.9. The system works only if *all four* work (a series system). Find the probability the system works.

D11. Three components operate independently, each failing with probability 0.3. The system works if *at least one* works (a parallel system). Find the probability the system works.

D12. A fair coin is tossed n times. Find the smallest n for which the probability of at least one head is at least 0.95.

D13. Roll two fair dice. Let $A =$ "the first die is at most 2" and $B =$ "the sum is 7." Determine whether A and B are independent.

D14. Suppose A and B are independent. Prove that A^{c} and B^{c} are also independent.

E. Combinatorics

E1. How many distinguishable arrangements are there of the letters of the word `BANANA`? Of the word `MISSISSIPPI`?

E2. A committee of 4 is to be chosen from 7 women and 5 men. How many committees have exactly 2 women? How many have at least 1 man?

E3. How many four-digit numbers (no leading zero) have all distinct digits? How many of these are even?

E4. In how many ways can 6 distinct books be placed on a shelf if two particular books must stand next to each other?

E5. A standard deck is dealt into a five-card hand. How many hands are flushes (all five of the same suit, including straight flushes)? How many are exactly one pair (two cards of one rank and three cards of three other distinct ranks)?

E6. How many solutions in nonnegative integers does $x_1 + x_2 + x_3 + x_4 + x_5 = 15$ have? How many have every $x_i \geq 1$?

E7. In how many ways can 10 distinct people be split into a group of 4, a group of 3, and a group of 3, where the two groups of 3 are unlabelled (interchangeable)?

E8. Using a combinatorial (counting) argument, prove Pascal's rule $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$ for $1 \leq k \leq n-1$.

E9. How many distinguishable arrangements are there of the letters of the word `BALLOON`? Of the word `STATISTICS`?

E10. A committee of 3 is chosen from 6 men and 4 women. How many committees have at least 2 women?

E11. How many five-card poker hands are full houses (three cards of one rank and two of another)? How many are two pair (two cards of one rank, two of a second rank, and one of a third)?

E12. How many ways can 8 distinct people be seated around a round table (two seatings are the same if one is a rotation of the other)?

E13. In how many ways can a lattice path go from the lower-left to the upper-right corner of a 5×4 grid of city blocks, moving only right or up along the streets?

E14. How many solutions in positive integers does $x_1 + x_2 + x_3 + x_4 = 12$ have? How many solutions in nonnegative integers does $x_1 + x_2 + x_3 = 20$ have subject to each $x_i \leq 10$?

E15. A subcommittee of 5 is chosen from a club of 20 members. How many subcommittees include two particular members (the president and the treasurer)?

F. Combinatorial probability

F1. An urn has 6 red and 4 white balls. Three are drawn without replacement. Find $\mathbb{P}(\text{all three red})$ and $\mathbb{P}(\text{exactly two red})$.

F2. Five cards are dealt from a standard deck. Find $\mathbb{P}$(exactly one ace) and $\mathbb{P}$(at least one ace).

F3. A lottery draws 5 distinct numbers from 1 to 40. A player buys one ticket. Find the probability of matching all 5 numbers, and of matching exactly 3.

F4. Eight people, including a particular pair of friends, are seated at random around a round table. Find the probability the two friends sit next to each other.

F5. A box holds 10 items, of which 3 are defective. A quality inspector draws 4 items at random without replacement. Find $\mathbb{P}$(no defectives) and $\mathbb{P}$(at least one defective).

F6. Twelve balls, numbered 1 through 12, are placed in a row in random order. Find the probability that ball 1 comes somewhere before ball 2 in the row.

F7. A class has 5 first-years, 4 second-years, and 3 third-years. A team of 4 is chosen at random. Find the probability the team contains at least one student from each year.

F8. Find the probability that, among a group of 5 randomly chosen people, at least two were born on the same day of the week (assume the 7 days are equally likely and independent).

F9. Five cards are dealt from a standard deck. Find $\mathbb{P}$(exactly two hearts).

F10. A box holds 5 black and 3 white marbles. Three are drawn without replacement. Find $\mathbb{P}$(at least one white).

F11. Ten people, including Alice and Bob, stand in a random line. Find the probability that Alice and Bob stand next to each other.

F12. Five cards are dealt from a standard deck. Find $\mathbb{P}$(the hand is a flush), that is, all five cards of the same suit (count straight flushes as flushes).

F13. Six fair dice are rolled. Find the probability that all six show different faces.

F14. A bag holds 3 red, 3 blue, and 3 green chips. Three chips are drawn without replacement. Find the probability that all three are of different colours.

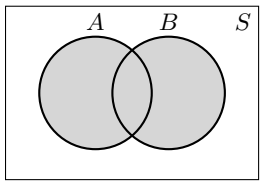
2.8 Solutions

A. Sample spaces and events

A1. $S = \{(i,j) : 1 \le i,j \le 4\}$, so $|S| = 16$. Then $A = \{(1,1),(2,2),(3,3),(4,4)\}$ and $B = \{(1,4),(2,3),(3,2),(4,1)\}$. Since the two coordinates of any point of A are equal, their sum is even, never 5; hence $A \cap B = \varnothing$ and A, B *are* mutually exclusive.

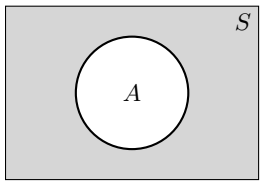
A2. (a) $A \cup B \cup C$; (b) $(A \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}) \cup (A^{\mathsf{c}} \cap B \cap C^{\mathsf{c}}) \cup (A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C)$; (c) $A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}$ together with the three "exactly one" terms of part (b), i.e. the complement of "at least two occur": $(A \cap B) \cup (A \cap C) \cup (B \cap C)^{\mathsf{c}}$; (d) $A \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}$.

A3. Membership: if $s \in A \cup (A^{\mathsf{c}} \cap B)$ then either $s \in A$ or ($s \notin A$ and $s \in B$); in both cases $s \in A \cup B$. Conversely if $s \in A \cup B$ and $s \notin A$ then $s \in B$, so $s \in A^{\mathsf{c}} \cap B$; thus $s \in A \cup (A^{\mathsf{c}} \cap B)$. The two sides are equal. The picture is the union region:



$A \cup B$

A4. $(A \cap B) \cup (A \cap B^{\mathsf{c}}) = A \cap (B \cup B^{\mathsf{c}}) = A \cap S = A$ by distributivity, so the complement is simply A^{c}:



A^{c}

A5. $E = \{2,4,6,8\}$, $T = \{6,7,8,9\}$, $D = \{3,6,9\}$. Then $E \cap T = \{6,8\}$; $E \cup D = \{2,3,4,6,8,9\}$; and $T^{\mathsf{c}} = \{1,2,3,4,5\}$ so $T^{\mathsf{c}} \cap D = \{3\}$.

A6. Each outcome is a string of H's and T's that either ends at the first occurrence of HH (length at most 5), or has length exactly 5 with no HH. The outcomes that stop early at the first HH are

$$HH,\ THH,\ HTHH,\ TTHH,\ HTTHH,\ THTHH,\ TTTHH,$$

and the length-5 outcomes with no two consecutive heads are

$$HTHTH,\ HTHTT,\ HTTHT,\ HTTTH,\ HTTTT,\ THTHT,\ THTTH,$$
$$THTTT,\ TTHTH,\ TTHTT,\ TTTHT,\ TTTTH,\ TTTTT.$$

Their union is S (note the trials need not all have the same length).

A7. $A \cap B$ and $A \cap B^{\mathsf{c}}$ are disjoint because B and B^{c} are; each is contained in A, hence disjoint from A^{c}. So the three are pairwise disjoint. Their union is $\big[(A \cap B) \cup (A \cap B^{\mathsf{c}})\big] \cup A^{\mathsf{c}} = A \cup A^{\mathsf{c}} = S$.

A8. (a) every outcome in A is also in B; (b) A and B cannot both occur; (c) every outcome lies in A or in B (or both). Roll a die with $A = \{1,2,3\}$, $B = \{4,5,6\}$: then $A \cap B = \varnothing$ and $A \cup B = S$, which together say exactly that $B = A^{\mathsf{c}}$ (equivalently $A = B^{\mathsf{c}}$). The condition $A \subseteq B$ is then incompatible unless $A = \varnothing$, so the three statements force the partition $A = B^{\mathsf{c}}$.

A9. $S = \{(r,g) : 1 \le r, g \le 6\}$, so $|S| = 36$. Here $G \cap H$ requires the two dice equal *and* the green one even, i.e. $(2,2), (4,4), (6,6)$, so $G \cap H = \{(2,2),(4,4),(6,6)\} \neq \varnothing$. Hence G and H are *not* mutually exclusive.

A10. $S = \{(x,y) : x, y \in \{a,b,c,d\},\ x \neq y\}$, an ordered pair of distinct letters, so $|S| = 4 \cdot 3 = 12$. The event L ("a appears") is $\{(a,b),(a,c),(a,d),(b,a),(c,a),(d,a)\}$, six outcomes.

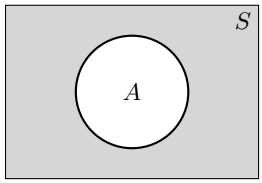
A11. (a) at least two occur: $(A \cap B) \cup (A \cap C) \cup (B \cap C)$; (b) none occur: $A^{\mathsf{c}} \cap B^{\mathsf{c}} \cap C^{\mathsf{c}}$; (c) exactly two occur: $(A \cap B \cap C^{\mathsf{c}}) \cup (A \cap B^{\mathsf{c}} \cap C) \cup (A^{\mathsf{c}} \cap B \cap C)$; (d) C occurs and at most one of A, B: $C \cap (A \cap B)^{\mathsf{c}} = C \cap (A^{\mathsf{c}} \cup B^{\mathsf{c}})$.

A12. The spinner stops at the first red or after three spins. The outcomes are

$$R, \quad BR, \quad BBR, \quad BBB,$$

so $S = \{R,\ BR,\ BBR,\ BBB\}$.

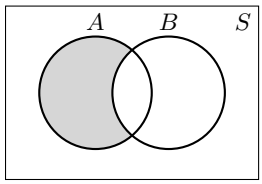
A13. By distributivity, $(A \cup B) \cap (A \cup B^{\mathsf{c}}) = A \cup (B \cap B^{\mathsf{c}}) = A \cup \varnothing = A$. The common region of the two unions is exactly A:



A^{c}

A14. $U = \{(1,1),(2,2),(3,3)\}$; $V = \{(1,3),(2,2),(3,1)\}$; $W = \{(1,2),(1,3),(2,3)\}$. Now $U \cap W = \varnothing$ (one cannot have $x = y$ and $x < y$), so U and W are mutually exclusive. ($U \cap V = \{(2,2)\}$ and $V \cap W = \{(1,3)\}$ are nonempty.)

A15. Membership: $s \in A \setminus (B \cup C)$ iff $s \in A$ and $s \notin B \cup C$, i.e. $s \in A$ and $s \notin B$ and $s \notin C$. That says $s \in A \setminus B$ and $s \in A \setminus C$, i.e. $s \in (A \setminus B) \cap (A \setminus C)$. Each step is reversible, so the sets are equal. The left-hand side is A with its overlap with $B \cup C$ removed:



$A \setminus B = A \cap B^{\mathsf{c}}$

B. The probability function

B1. $\mathbb{P}(A \cup B) = 0.55 + 0.40 - 0.25 = 0.70$; $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}}) = 1 - 0.70 = 0.30$; exactly one $= \mathbb{P}(A \cup B) - \mathbb{P}(A \cap B) = 0.70 - 0.25 = 0.45$; $\mathbb{P}(A^{\mathsf{c}} \cap B) = \mathbb{P}(B) - \mathbb{P}(A \cap B) = 0.40 - 0.25 = 0.15$.

B2. At least one $= 120 + 90 - 55 = 155$; neither $= 200 - 155 = 45$.

B3. $\sum_k ck = c(1+2+3+4) = 10c = 1$, so $c = 0.1$. Odd faces $\{1,3\}$: $0.1(1) + 0.1(3) = 0.4$. Faces ≥ 3, i.e. $\{3,4\}$: $0.1(3) + 0.1(4) = 0.7$.

B4. By inclusion–exclusion $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$; since $\mathbb{P}(A \cap B) \ge 0$, $\mathbb{P}(A \cup B) \le \mathbb{P}(A) + \mathbb{P}(B)$. Equality holds iff $\mathbb{P}(A \cap B) = 0$ (in particular when A, B are disjoint).

B5. Inclusion–exclusion: $3(0.4) - 3(0.15) + 0.05 = 1.2 - 0.45 + 0.05 = 0.80$. None of the three $= 1 - 0.80 = 0.20$.

B6. $\mathbb{P}(A\cap B)=\mathbb{P}(A)+\mathbb{P}(B)-\mathbb{P}(A\cup B)\ge 0.7+0.6-1=0.3$ since $\mathbb{P}(A\cup B)\le 1$. The largest possible value is $\min\{\mathbb{P}(A),\mathbb{P}(B)\}=0.6$, attained when $A\subseteq B$.

B7. Sum 7: 6 outcomes; sum 11: 2 outcomes; disjoint, so $\mathbb{P}=8/36=2/9$. At least one 6: complement "no 6" has $5\times 5=25$ outcomes, so $\mathbb{P}=1-25/36=11/36$.

B8. The region $x+y\le 1$ is the lower triangle of the unit square, area $\frac{1}{2}$, so the probability is $\frac{1}{2}$. For $y\le x^2$, the area under $y=x^2$ over $[0,1]$ is $\int_0^1 x^2\,dx=\frac{1}{3}$, so the probability is $\frac{1}{3}$.

B9. Since A,B are disjoint, $\mathbb{P}(A\cup B)=0.50+0.30=0.80$. Then $\mathbb{P}(A^{\mathsf{c}})=1-0.50=0.50$, and $\mathbb{P}(A^{\mathsf{c}}\cap B^{\mathsf{c}})=\mathbb{P}(A\cup B^{\mathsf{c}})=1-0.80=0.20$.

B10. Set $\mathbb{P}(k)=ck^2$. Then $c(1+4+9+16)=30c=1$, so $c=\frac{1}{30}$. Odd faces $\{1,3\}$: $\frac{1}{30}(1+9)=\frac{10}{30}=\frac{1}{3}$. Face 4: $\frac{16}{30}=\frac{8}{15}$.

B11. There are $6^3=216$ equally likely ordered triples. The number summing to 10 is 27 (for example $(1,3,6),(1,4,5),(2,2,6),(2,4,4),(3,3,4)$ and their orderings), so $\mathbb{P}(\text{sum}=10)=\dfrac{27}{216}=\dfrac{1}{8}$.

B12. The rectangle has area $2\cdot 3=6$. The region $x+y<2$ inside it is the triangle with vertices $(0,0),(2,0),(0,2)$, of area $\frac{1}{2}\cdot 2\cdot 2=2$. Hence $\mathbb{P}(x+y\ge 2)=1-\dfrac{2}{6}=\dfrac{2}{3}$.

B13. Inclusion–exclusion: $\mathbb{P}(A\cup B\cup C)=(0.5+0.4+0.3)-3(0.1)+0.05=1.2-0.3+0.05=0.95$.

B14. Use the complement: $\min(X,Y)>\frac{1}{3}$ iff both $X>\frac{1}{3}$ and $Y>\frac{1}{3}$, which has probability $\left(\frac{2}{3}\right)^2=\frac{4}{9}$. Thus $\mathbb{P}\big(\min(X,Y)\le\frac{1}{3}\big)=1-\frac{4}{9}=\frac{5}{9}$.

B15. Apply the two-event bound twice. $\mathbb{P}(A\cup B\cup C)=\mathbb{P}\big((A\cup B)\cup C\big)\le\mathbb{P}(A\cup B)+\mathbb{P}(C)\le\mathbb{P}(A)+\mathbb{P}(B)+\mathbb{P}(C)$, since $\mathbb{P}(A\cup B)\le\mathbb{P}(A)+\mathbb{P}(B)$.

C. Conditional probability and Bayes

C1. "First die even" has 18 equally likely outcomes. Among them, "sum = 8" needs first die $\in\{2,4,6\}$ with second $\in\{6,4,2\}$: $(2,6),(4,4),(6,2)$, three outcomes. Conditional probability $=3/18=1/6$.

C2. $\mathbb{P}(\text{both green})=\dfrac{4}{10}\cdot\dfrac{3}{9}=\dfrac{12}{90}=\dfrac{2}{15}$. After a green is removed, 3 greens remain among 9, so $\mathbb{P}(\text{second green}\mid\text{first green})=3/9=1/3$.

C3. Overall defect rate $=0.40(0.01)+0.35(0.02)+0.25(0.04)=0.004+0.007+0.010=0.021$. By Bayes, $\mathbb{P}(M_3\mid\text{def})=\dfrac{0.010}{0.021}=\dfrac{10}{21}\approx 0.476$.

C4. Let D be "has the condition." $\mathbb{P}(D)=0.02$, $\mathbb{P}(+\mid D)=0.95$, $\mathbb{P}(+\mid D^{\mathsf{c}})=1-0.90=0.10$. Then $\mathbb{P}(+)=0.02(0.95)+0.98(0.10)=0.019+0.098=0.117$, and $\mathbb{P}(D\mid+)=\dfrac{0.019}{0.117}\approx 0.162$.

C5. Priors $\frac{1}{3}$ (trick), $\frac{2}{3}$ (ordinary). $\mathbb{P}(HH\mid\text{trick})=1$, $\mathbb{P}(HH\mid\text{ordinary})=\frac{1}{4}$. So $\mathbb{P}(HH)=\frac{1}{3}(1)+\frac{2}{3}(\frac{1}{4})=\frac{1}{3}+\frac{1}{6}=\frac{1}{2}$, and $\mathbb{P}(\text{trick}\mid HH)=\dfrac{\frac{1}{3}(1)}{\frac{1}{2}}=\dfrac{2}{3}$.

C6. Conditioning on B yields a legitimate probability function on B, for which A and A^{c} are complementary, so the two conditional probabilities sum to 1. Die example: $A=\{1\}$, $B=\{\text{even}\}=\{2,4,6\}$. Then $\mathbb{P}(A\mid B)=0$ while $\mathbb{P}(A\mid B^{\mathsf{c}})=\mathbb{P}(\{1\}\mid\{1,3,5\})=\frac{1}{3}$, summing to $\frac{1}{3}\neq 1$.

C7. "First two are non-aces and the third is an ace": $\dfrac{48}{52}\cdot\dfrac{47}{51}\cdot\dfrac{4}{50}=\dfrac{48\cdot 47\cdot 4}{52\cdot 51\cdot 50}=\dfrac{9024}{132{,}600}\approx 0.0681$.

C8. Let X be "exercises." $\mathbb{P}(X) = 0.6$, $\mathbb{P}(H \mid X) = 0.20$, $\mathbb{P}(H \mid X^{\mathsf{c}}) = 0.45$. $\mathbb{P}(H) = 0.6(0.20) + 0.4(0.45) = 0.12 + 0.18 = 0.30$, so $\mathbb{P}(X \mid H) = \dfrac{0.12}{0.30} = 0.40$.

C9. $\mathbb{P}(\text{both kings}) = \dfrac{4}{52} \cdot \dfrac{3}{51} = \dfrac{12}{2652} = \dfrac{1}{221}$. After one king is removed, 3 kings remain among 51 cards, so $\mathbb{P}(\text{second king} \mid \text{first king}) = \dfrac{3}{51} = \dfrac{1}{17}$.

C10. Each urn has prior $\frac{1}{3}$. The red-draw probabilities are $\frac{2}{5}, \frac{4}{6}, \frac{1}{6}$, so

$$\mathbb{P}(\text{red}) = \tfrac{1}{3}\big(\tfrac{2}{5} + \tfrac{4}{6} + \tfrac{1}{6}\big) = \tfrac{1}{3}\big(\tfrac{12}{30} + \tfrac{20}{30} + \tfrac{5}{30}\big) = \tfrac{1}{3} \cdot \tfrac{37}{30} = \tfrac{37}{90}.$$

By Bayes, $\mathbb{P}(\text{urn 2} \mid \text{red}) = \dfrac{\frac{1}{3} \cdot \frac{4}{6}}{\frac{37}{90}} = \dfrac{\frac{20}{90}}{\frac{37}{90}} = \dfrac{20}{37} \approx 0.541$.

C11. Let D be "has the condition," $\mathbb{P}(D) = 0.01$. The two tests are conditionally independent, each positive with probability 0.90 if D holds and $1 - 0.95 = 0.05$ if D does not. So $\mathbb{P}(++ \mid D) = 0.90^2 = 0.81$ and $\mathbb{P}(++ \mid D^{\mathsf{c}}) = 0.05^2 = 0.0025$. Then

$$\mathbb{P}(D \mid ++) = \frac{0.01(0.81)}{0.01(0.81) + 0.99(0.0025)} = \frac{0.0081}{0.0081 + 0.002475} = \frac{0.0081}{0.010575} = \frac{36}{47} \approx 0.766.$$

C12. The first head on the third toss means tail, tail, head: $\mathbb{P} = (0.6)(0.6)(0.4) = 0.144 = \dfrac{18}{125}$.

C13. There are $\binom{10}{2} = 45$ equally likely pairs. "At least one defective" is the complement of "both good": $45 - \binom{7}{2} = 45 - 21 = 24$. "Both defective" has $\binom{3}{2} = 3$ pairs. Since both-defective implies at least one defective,

$$\mathbb{P}(\text{both def} \mid \text{at least one def}) = \frac{3/45}{24/45} = \frac{3}{24} = \frac{1}{8}.$$

C14. By definition $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$; multiplying by $\mathbb{P}(B) > 0$ gives $\mathbb{P}(A \cap B) = \mathbb{P}(B)\,\mathbb{P}(A \mid B)$. Here $\mathbb{P}(A \cap B) = 0.5 \cdot 0.6 = 0.30$.

C15. The nodes act independently, so $\mathbb{P}(\text{arrives correctly}) = 0.9 \cdot 0.8 = 0.72$. For "at least one forwarded correctly," use the complement "both failed": $1 - (1 - 0.9)(1 - 0.8) = 1 - (0.1)(0.2) = 1 - 0.02 = 0.98$.

D. Independence

D1. $\mathbb{P}(A \cap B) = 0.7(0.4) = 0.28$; $\mathbb{P}(A \cup B) = 0.7 + 0.4 - 0.28 = 0.82$; exactly one $= 0.82 - 0.28 = 0.54$.

D2. $\mathbb{P}(A) = \frac{4}{6} = \frac{2}{3}$, $\mathbb{P}(B) = \frac{3}{6} = \frac{1}{2}$, $A \cap B = \{2, 4\}$ so $\mathbb{P}(A \cap B) = \frac{2}{6} = \frac{1}{3} = \frac{2}{3} \cdot \frac{1}{2}$: *independent*. For $C = \{3, 4, 5, 6\}$, $\mathbb{P}(C) = \frac{2}{3}$ and $A \cap C = \{3, 4\}$ so $\mathbb{P}(A \cap C) = \frac{1}{3}$, while $\mathbb{P}(A)\mathbb{P}(C) = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9} \neq \frac{1}{3}$: *dependent*.

D3. Link fails only if all three cables fail: $\mathbb{P}(\text{fail}) = 0.2 \cdot 0.1 \cdot 0.05 = 0.001$, so $\mathbb{P}(\text{works}) = 1 - 0.001 = 0.999$.

D4. "Second die odd" has probability $\frac{1}{2}$. Sum 7 has probability $\frac{1}{6}$; their intersection requires second die odd and sum 7: $(6, 1), (4, 3), (2, 5)$, probability $\frac{3}{36} = \frac{1}{12} = \frac{1}{2} \cdot \frac{1}{6}$: independent. Sum 8 has probability $\frac{5}{36}$; its outcomes with the second die odd are $(3, 5)$ and $(5, 3)$ (since $(7, 1)$ is impossible on a six-sided die), giving probability $\frac{2}{36} = \frac{1}{18}$. But $\frac{1}{2} \cdot \frac{5}{36} = \frac{5}{72} \neq \frac{1}{18} = \frac{4}{72}$: *not* independent.

D5. Exactly three heads in five tosses: $\binom{5}{3}/2^5 = 10/32 = 5/16$. First and last both heads: the two fixed tosses each have probability $\frac{1}{2}$ and are independent, so $\frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$.

D6. Since A, B are independent, $\mathbb{P}(A \cap B^{\mathsf{c}}) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\big(1 - \mathbb{P}(B)\big) = \mathbb{P}(A)\mathbb{P}(B^{\mathsf{c}})$. Hence A and B^{c} are independent.

D7. $\mathbb{P}(\text{flows}) = \mathbb{P}(\text{sw 1}) \cdot \mathbb{P}(\text{sw 2 or 3}) = p\big(1-(1-p)^2\big) = p(2p-p^2) = 2p^2-p^3$. At $p=0.8$: $2(0.64)-0.512 = 1.28-0.512 = 0.768$.

D8. Toss a fair coin twice; $S = \{HH, HT, TH, TT\}$, each $\frac{1}{4}$. Let A = "first is H," B = "second is H," C = "the two tosses differ." Then $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = \frac{1}{2}$. Pairwise: $A \cap B = \{HH\}$, $A \cap C = \{HT\}$, $B \cap C = \{TH\}$, each of probability $\frac{1}{4} = \frac{1}{2} \cdot \frac{1}{2}$, so the events are pairwise independent. But $A \cap B \cap C = \varnothing$ has probability 0, while $\mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C) = \frac{1}{8} \neq 0$: not mutually independent.

D9. Exactly one of A, B occurs is $\mathbb{P}(A \cap B^{\mathsf{c}}) + \mathbb{P}(A^{\mathsf{c}} \cap B) = \mathbb{P}(A)\mathbb{P}(B^{\mathsf{c}}) + \mathbb{P}(A^{\mathsf{c}})\mathbb{P}(B) = 0.5(0.8) + 0.5(0.2) = 0.4 + 0.1 = 0.5$. Neither occurs: $\mathbb{P}(A^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}}) = 0.5(0.8) = 0.40$.

D10. A series system works iff all four work, so by independence $\mathbb{P}(\text{works}) = 0.9^4 = 0.6561$.

D11. A parallel system fails iff all three fail, with probability $0.3^3 = 0.027$. Hence $\mathbb{P}(\text{works}) = 1 - 0.027 = 0.973$.

D12. $\mathbb{P}(\text{at least one head}) = 1-(1/2)^n$. We need $1-(1/2)^n \geq 0.95$, i.e. $(1/2)^n \leq 0.05$. Since $(1/2)^4 = 0.0625$ and $(1/2)^5 = 0.03125$, the smallest such n is $\boxed{5}$.

D13. A (first die ≤ 2) has probability $\frac{12}{36} = \frac{1}{3}$; B (sum 7) has probability $\frac{6}{36} = \frac{1}{6}$. Their intersection requires first die $\in \{1,2\}$ and sum 7: $(1,6), (2,5)$, probability $\frac{2}{36} = \frac{1}{18}$. Since $\frac{1}{3} \cdot \frac{1}{6} = \frac{1}{18}$, the events *are* independent.

D14. Independence of A, B gives $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. From D6, A and B^{c} are independent; applying the same fact again to B^{c} and A shows A^{c} and B^{c} are independent. Directly: $\mathbb{P}(A^{\mathsf{c}} \cap B^{\mathsf{c}}) = \mathbb{P}(A \cup B^{\mathsf{c}}) = 1 - \mathbb{P}(A) - \mathbb{P}(B) + \mathbb{P}(A)\mathbb{P}(B) = \big(1-\mathbb{P}(A)\big)\big(1-\mathbb{P}(B)\big) = \mathbb{P}(A^{\mathsf{c}})\mathbb{P}(B^{\mathsf{c}})$.

E. Combinatorics

E1. `BANANA`: 6 letters with A×3, N×2, B×1: $\dfrac{6!}{3!\,2!} = 60$. `MISSISSIPPI`: 11 letters with I×4, S×4, P×2, M×1: $\dfrac{11!}{4!\,4!\,2!} = 34{,}650$.

E2. Exactly 2 women (hence 2 men): $\binom{7}{2}\binom{5}{2} = 21 \cdot 10 = 210$. At least one man: total $\binom{12}{4} = 495$ minus all-women $\binom{7}{4} = 35$, giving $495 - 35 = 460$.

E3. All distinct digits, no leading zero: first digit 9 choices (1–9), then choose the remaining three from the other 9 digits in order: $9 \cdot 9 \cdot 8 \cdot 7 = 4536$. Even ones: split on the last digit. If last digit = 0: first three digits are an ordered selection of 3 from the remaining 9 nonzero-allowed digits: $9 \cdot 8 \cdot 7 = 504$. If last digit $\in \{2,4,6,8\}$ (4 choices): first digit has 8 choices (1–9 minus the chosen last digit), then 8 and 7 for the middle two: $4 \cdot 8 \cdot 8 \cdot 7 = 1792$. Total even $= 504 + 1792 = 2296$.

E4. Glue the two particular books into one block (internally orderable 2 ways); arrange the block with the other 4 books, i.e. 5 objects: $5! \cdot 2 = 240$.

E5. Flushes: choose a suit (4) and any 5 of its 13 cards $\binom{13}{5} = 1287$: $4 \cdot 1287 = 5148$. One pair: choose the pair's rank (13) and its two cards $\binom{4}{2} = 6$; choose 3 other ranks $\binom{12}{3} = 220$ and one card of each suit $4^3 = 64$: $13 \cdot 6 \cdot 220 \cdot 64 = 1{,}098{,}240$.

E6. Nonnegative solutions (stars and bars): $\binom{15+5-1}{5-1} = \binom{19}{4} = 3876$. With every $x_i \geq 1$, substitute $y_i = x_i - 1 \geq 0$ summing to 10: $\binom{10+5-1}{5-1} = \binom{14}{4} = 1001$.

E7. Choose the group of 4: $\binom{10}{4} = 210$. Split the remaining 6 into two unlabelled groups of 3: $\dfrac{1}{2}\binom{6}{3} = \dfrac{20}{2} = 10$. Total $210 \cdot 10 = 2100$.

E8. Both sides count the k-subsets of an n-set $\{1, \ldots, n\}$. Fix element n. A k-subset either contains n — then choose the other $k-1$ elements from the remaining $n-1$, giving $\binom{n-1}{k-1}$ ways — or it does not, giving $\binom{n-1}{k}$ ways. These cases are disjoint and exhaustive, so $\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$.

E9. `BALLOON`: 7 letters with L×2, O×2, the rest single: $\dfrac{7!}{2!\,2!} = \dfrac{5040}{4} = 1260$. `STATISTICS`: 10 letters with S×3, T×3, I×2, A×1, C×1: $\dfrac{10!}{3!\,3!\,2!} = \dfrac{3{,}628{,}800}{72} = 50{,}400$.

E10. "At least 2 women" means 2 or 3 women. With 2 women (and 1 man): $\binom{4}{2}\binom{6}{1} = 6 \cdot 6 = 36$. With 3 women: $\binom{4}{3}\binom{6}{0} = 4$. Total $36 + 4 = 40$.

E11. Full house: choose the triple's rank (13) and its 3 cards $\binom{4}{3} = 4$; then the pair's rank (12 remaining) and its 2 cards $\binom{4}{2} = 6$: $13 \cdot 4 \cdot 12 \cdot 6 = 3744$. Two pair: choose the two ranks $\binom{13}{2} = 78$ and two cards of each $\binom{4}{2}^2 = 36$; then the fifth card from the remaining 11 ranks and any of its 4 suits, 44: $78 \cdot 36 \cdot 44 = 123{,}552$.

E12. Fix one person to break rotational symmetry; the remaining 7 fill the other seats in $7! = 5040$ ways. (Equivalently $8!/8 = 5040$.)

E13. A path needs 5 right-moves and 4 up-moves in some order, so it is determined by choosing which 4 of the 9 moves are "up": $\binom{9}{4} = 126$.

E14. Positive solutions of $x_1 + \cdots + x_4 = 12$: substitute $y_i = x_i - 1 \geq 0$ summing to 8, giving $\binom{8+4-1}{4-1} = \binom{11}{3} = 165$. For $x_1 + x_2 + x_3 = 20$ with each $x_i \leq 10$, start from all nonnegative solutions $\binom{22}{2} = 231$ and subtract those with some $x_i \geq 11$ (set $x_i' = x_i - 11$, sum 9): $3\binom{11}{2} = 3 \cdot 55 = 165$; no term can exceed 10 twice, so the answer is $231 - 165 = 66$.

E15. Both named members are forced in; choose the remaining 3 from the other 18 members: $\binom{18}{3} = 816$.

F. Combinatorial probability

F1. $|S| = \binom{10}{3} = 120$. All red: $\dfrac{\binom{6}{3}}{\binom{10}{3}} = \dfrac{20}{120} = \dfrac{1}{6}$. Exactly two red: $\dfrac{\binom{6}{2}\binom{4}{1}}{\binom{10}{3}} = \dfrac{15 \cdot 4}{120} = \dfrac{60}{120} = \dfrac{1}{2}$.

F2. Exactly one ace: $\dfrac{\binom{4}{1}\binom{48}{4}}{\binom{52}{5}} = \dfrac{4 \cdot 194{,}580}{2{,}598{,}960} \approx 0.299$. At least one ace: $1 - \dfrac{\binom{48}{5}}{\binom{52}{5}} = 1 - \dfrac{1{,}712{,}304}{2{,}598{,}960} \approx 1 - 0.659 = 0.341$.

F3. $|S| = \binom{40}{5} = 658{,}008$. Match all 5: $1/658{,}008 \approx 1.5 \times 10^{-6}$. Match exactly 3: choose 3 of the 5 drawn and 2 of the 35 not drawn: $\dfrac{\binom{5}{3}\binom{35}{2}}{\binom{40}{5}} = \dfrac{10 \cdot 595}{658{,}008} = \dfrac{5950}{658{,}008} \approx 0.00904$.

F4. Seat the first friend anywhere; the second takes one of the 7 remaining seats, of which 2 are adjacent: $\mathbb{P} = \dfrac{2}{7}$.

F5. No defectives: $\dfrac{\binom{7}{4}}{\binom{10}{4}} = \dfrac{35}{210} = \dfrac{1}{6}$. At least one defective: $1 - \dfrac{1}{6} = \dfrac{5}{6}$.

F6. By symmetry, ball 1 is equally likely to fall before or after ball 2 in the ordering (no ties), so the probability is $\dfrac{1}{2}$.

F7. $|S| = \binom{12}{4} = 495$. "At least one from each year" with a team of 4 from groups of sizes $5, 4, 3$ means the year counts are a permutation of $(2, 1, 1)$. The doubled year can be first-years $\binom{5}{2}\binom{4}{1}\binom{3}{1} = 10 \cdot 4 \cdot 3 = 120$; second-years $\binom{5}{1}\binom{4}{2}\binom{3}{1} = 5 \cdot 6 \cdot 3 = 90$; third-years $\binom{5}{1}\binom{4}{1}\binom{3}{2} = 5 \cdot 4 \cdot 3 = 60$. Total favourable $= 120 + 90 + 60 = 270$, so $\mathbb{P} = \dfrac{270}{495} = \dfrac{6}{11} \approx 0.545$.

F8. Complement: all 5 birth-days distinct. $\mathbb{P}(\text{all distinct}) = \dfrac{7\cdot 6\cdot 5\cdot 4\cdot 3}{7^5} = \dfrac{2520}{16{,}807} \approx 0.150$, so $\mathbb{P}(\text{some match}) = 1 - 0.150 = 0.850$.

F9. $|S| = \binom{52}{5} = 2{,}598{,}960$. Exactly two hearts: choose 2 of the 13 hearts and 3 of the 39 non-hearts:

$$\mathbb{P} = \frac{\binom{13}{2}\binom{39}{3}}{\binom{52}{5}} = \frac{78\cdot 9139}{2{,}598{,}960} = \frac{712{,}842}{2{,}598{,}960} = \frac{9139}{33{,}320} \approx 0.274.$$

F10. $|S| = \binom{8}{3} = 56$. "At least one white" is the complement of "all black": $\mathbb{P} = 1 - \dfrac{\binom{5}{3}}{\binom{8}{3}} = 1 - \dfrac{10}{56} = \dfrac{46}{56} = \dfrac{23}{28} \approx 0.821$.

F11. Treat Alice–Bob as a block (orderable 2 ways) among the line of 10. Favourable arrangements: $2\cdot 9!$; total: $10!$. So $\mathbb{P} = \dfrac{2\cdot 9!}{10!} = \dfrac{2}{10} = \dfrac{1}{5}$.

F12. Choose a suit (4 ways) and any 5 of its 13 cards $\binom{13}{5} = 1287$, over all 5-card hands: $\mathbb{P} = \dfrac{4\cdot 1287}{\binom{52}{5}} = \dfrac{5148}{2{,}598{,}960} = \dfrac{33}{16{,}660} \approx 0.00198$.

F13. The 6 rolls give $6^6 = 46{,}656$ equally likely outcomes; "all different" means a permutation of the six faces, $6! = 720$. So $\mathbb{P} = \dfrac{720}{46{,}656} = \dfrac{5}{324} \approx 0.0154$.

F14. $|S| = \binom{9}{3} = 84$. "All three different colours" means one chip of each colour: $3\cdot 3\cdot 3 = 27$ choices. So $\mathbb{P} = \dfrac{27}{84} = \dfrac{9}{28} \approx 0.321$.

Chapter 3

Random Variables

Master guide: which tool does a random-variable problem need?

- **Counting successes** in a fixed number of independent yes/no trials? ⇒ **binomial**; if the draws are *without replacement* from a finite pool, **hypergeometric** (§1).
- A variable that takes **isolated values** with stated probabilities? ⇒ a **discrete** random variable: build its **probability mass function** p_X and **cdf** F_X (§2).
- A variable measured on a **continuum** (time, length, weight)? ⇒ a **continuous** random variable: integrate a **density** f_X; probabilities are areas, and $F_X(x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t$ (§3).
- Need a single **"typical value"**? ⇒ the **expected value** $\mathbb{E}(X)$ (§4). Need to measure **spread** about that value? ⇒ the **variance** $\mathrm{Var}(X)$ and standard deviation (§5).
- **Two variables together** — joint behaviour, marginals, or whether they are **independent**? ⇒ a **joint density** and its marginals (§6).
- A **function of a variable** ($Y = g(X)$) or a **sum/combination** of variables? ⇒ **transformation** methods and the linearity / addition rules for $\mathbb{E}$ and Var (§7, §8).
- The **largest, smallest, or k-th largest** of several observations? ⇒ **order statistics** (§9).
- Updating a distribution after **partial information** about a related variable? ⇒ a **conditional density** (§10).
- Need **moments**, or to identify the distribution of a **sum of independents**? ⇒ the **moment-generating function** (§11).

3.1 Binomial and Hypergeometric Probabilities

The two distributions in this section both count successes in a sequence of yes/no observations, but they differ in one decisive way: whether the trials are independent with a fixed success probability (the **binomial** setting) or are draws from a finite pool that shrinks as you sample (the **hypergeometric** setting). Getting this distinction right is the whole game, because the same word problem ("how many defective parts in a sample of 10?") is binomial when you sample with replacement or from an effectively infinite population, and hypergeometric when you sample without replacement from a small one. We build both pmfs from first principles, state their means and variances, and learn when one safely approximates the other.

Which rule of this section does the problem need?

- **One yes/no trial only**, and you want its mean or variance? ⇒ a **Bernoulli** trial: mean p, variance $p(1-p)$.

- **Fixed number n of independent trials, each success with the same probability p**, counting total successes? $\Rightarrow$ **binomial**, $X \sim \mathrm{Bin}(n, p)$, with $p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}$.
- Sampling **without replacement** from a finite population of N items of which r are "successes," counting successes in a sample of size n? $\Rightarrow$ **hypergeometric**, $X \sim \mathrm{Hyper}(N, r, n)$, with $p_X(k) = \binom{r}{k}\binom{N-r}{n-k}/\binom{N}{n}$.
- The phrase **"at least k"** or **"at most k"**? $\Rightarrow$ a **cumulative** sum of the relevant pmf; use the complement ($\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0)$) whenever it is shorter.
- Population is large and the sample is a *small* fraction of it (rule of thumb $n \leq 0.05\,N$)? $\Rightarrow$ the **binomial** with $p = r/N$ *approximates* the hypergeometric; you may ignore the without-replacement correction.

Why not the alternatives? If p changes from trial to trial (because the pool shrinks), the trials are not independent and the binomial is wrong — use hypergeometric. Conversely, if each trial genuinely resets to the same p (a coin, a fixed defect rate, a random digit), the population is effectively infinite and the hypergeometric correction is unnecessary.

3.1.1 Bernoulli trials

Definition 3.1.1 (Bernoulli trial)**.** A **Bernoulli trial** is a single experiment with exactly two outcomes, labelled *success* (coded 1) and *failure* (coded 0). If X is the indicator of success with $\mathbb{P}(X = 1) = p$ and $\mathbb{P}(X = 0) = 1 - p$, we write $X \sim \mathrm{Bernoulli}(p)$ and

$$p_X(1) = p, \qquad p_X(0) = 1 - p, \qquad 0 \leq p \leq 1.$$

The Bernoulli trial is the atom from which the binomial is built: a binomial count is a sum of n independent Bernoulli trials. Its mean and variance are worth memorising because the binomial's follow immediately.

Proposition 3.1.2 (Mean and variance of a Bernoulli trial)**.** *If $X \sim \mathrm{Bernoulli}(p)$ then $\mathbb{E}(X) = p$ and $\mathrm{Var}(X) = p(1-p)$.*

Proof. Since X takes only the values 0 and 1,

$$\mathbb{E}(X) = 0 \cdot (1-p) + 1 \cdot p = p.$$

Because $X^2 = X$ (as $0^2 = 0$ and $1^2 = 1$), we have $\mathbb{E}(X^2) = \mathbb{E}(X) = p$, so

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = p - p^2 = p(1-p). \qquad \square$$

Example 3.1.3 (A single defective inspection)**.** A machine produces a part that is defective with probability 0.03. Let $X = 1$ if one randomly chosen part is defective and $X = 0$ otherwise. Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

Which rule, and why? There is exactly *one* two-outcome trial, so this is a Bernoulli trial, not a binomial count (there is no n). Apply the Bernoulli mean and variance directly with $p = 0.03$.

Solution.

Step 1: Mean. $\mathbb{E}(X) = p = 0.03$.

Step 2: Variance. $\mathrm{Var}(X) = p(1-p) = 0.03 \cdot 0.97$.

$$\mathbb{E}(X) = 0.03, \qquad \mathrm{Var}(X) = 0.0291.$$

3.1.2 The binomial setting and pmf

Definition 3.1.4 (Binomial setting)**.** The **binomial setting** requires four conditions, sometimes abbreviated BINS:

- **B**inary: each trial is a success/failure Bernoulli trial.
- **I**ndependent: the trials do not influence one another.
- **N**umber: the number of trials n is fixed in advance.

- **S**ame probability: each trial has the same success probability p.

If X counts the successes in such a sequence, then $X \sim \text{Bin}(n, p)$.

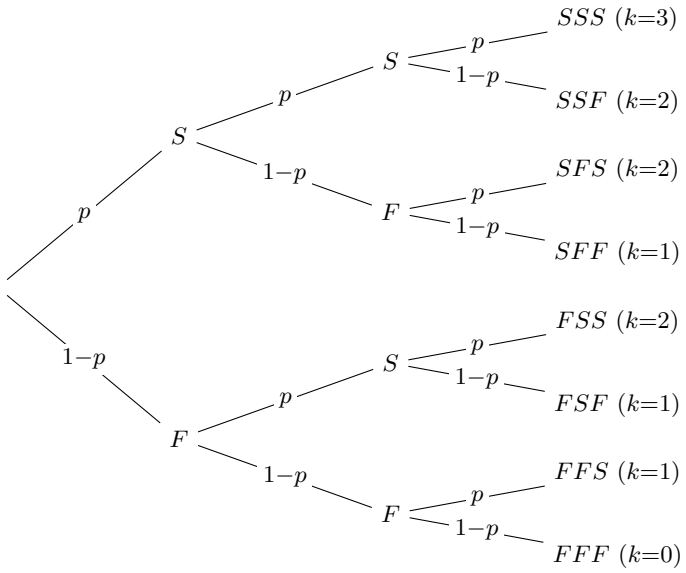
Theorem 3.1.5 (Binomial pmf)**.** *If $X \sim \text{Bin}(n, p)$ then for $k = 0, 1, \ldots, n$,*

$$p_X(k) = \mathbb{P}(X = k) = \binom{n}{k} p^k (1-p)^{n-k},$$

and $p_X(k) = 0$ otherwise.

Proof. Fix a particular sequence of outcomes with exactly k successes and $n - k$ failures. By independence its probability is $p^k(1-p)^{n-k}$, and this value is the same for every such sequence. The number of sequences with k successes among n positions is the number of ways to choose which k positions are successes, namely $\binom{n}{k}$. Summing the equal probabilities over these mutually exclusive sequences gives $\binom{n}{k} p^k (1-p)^{n-k}$. □

The binomial coefficient is exactly a path count. The tree below enumerates the $n = 3$ case: each root-to-leaf path is one outcome sequence, every path with k successes carries the same probability $p^k(1-p)^{3-k}$, and the number of such paths is $\binom{3}{k}$ $(1, 3, 3, 1$ for $k = 0, 1, 2, 3)$.



Three of the eight leaves show $k = 2$ (namely FSS, SFS, SSF), matching $\binom{3}{2} = 3$; each contributes $p^2(1-p)$, so $\mathbb{P}(X = 2) = 3p^2(1-p) = \binom{3}{2} p^2 (1-p)^1$.

Example 3.1.6 (Free-throw streak)**.** A basketball player makes each free throw independently with probability $p = 0.8$. She attempts $n = 6$ free throws. What is the probability she makes exactly 5?

Which rule, and why? Fixed number of trials ($n = 6$), each independent with the same success probability ($p = 0.8$), counting makes — the BINS conditions hold, so $X \sim \text{Bin}(6, 0.8)$ and we want $p_X(5)$.

Solution.

Step 1: Identify parameters. $n = 6$, $p = 0.8$, $k = 5$.

Step 2: Apply the pmf.

$$\mathbb{P}(X = 5) = \binom{6}{5} (0.8)^5 (0.2)^1 = 6 \cdot 0.32768 \cdot 0.2.$$

$$\mathbb{P}(X = 5) = 0.393216 \approx 0.3932.$$

Example 3.1.7 (Defective parts in a production run)**.** Parts come off a line defective with probability $p = 0.05$, independently. In a batch of $n = 12$ parts, find the probability that exactly 2 are defective.

Which rule, and why? The defect rate is a fixed 0.05 for every part and parts are independent, so the count of defectives is binomial, $X \sim \text{Bin}(12, 0.05)$; we want $p_X(2)$. (This is a large "population" with a constant rate, so binomial — not hypergeometric — is correct.)

Solution.

Step 1: Parameters. $n = 12$, $p = 0.05$, $k = 2$.

Step 2: Apply the pmf.

$$\mathbb{P}(X = 2) = \binom{12}{2}(0.05)^2(0.95)^{10} = 66 \cdot 0.0025 \cdot (0.95)^{10}.$$

Since $(0.95)^{10} \approx 0.598737$,

$$\mathbb{P}(X = 2) \approx 66 \cdot 0.0025 \cdot 0.598737 \approx 0.0988.$$

Example 3.1.8 (Random multiple-choice guessing)**.** A quiz has $n = 8$ multiple-choice questions, each with 4 options. A student guesses every answer uniformly at random and independently. Find the probability of getting exactly 3 correct.

Which rule, and why? Each question is an independent trial with the same success probability $p = 1/4$ under pure guessing, and there are a fixed 8 of them; so $X \sim \text{Bin}(8, 1/4)$ and we want $p_X(3)$.

Solution.

Step 1: Parameters. $n = 8$, $p = 1/4$, $1 - p = 3/4$, $k = 3$.

Step 2: Apply the pmf.

$$\mathbb{P}(X = 3) = \binom{8}{3}\left(\tfrac{1}{4}\right)^3\left(\tfrac{3}{4}\right)^5 = 56 \cdot \frac{1}{64} \cdot \frac{243}{1024} = \frac{56 \cdot 243}{65536}.$$

$$\mathbb{P}(X = 3) = \frac{13608}{65536} \approx 0.2076.$$

Example 3.1.9 (Emails flagged as spam)**.** A filter independently flags each incoming email as spam with probability $p = 0.30$. Over the next $n = 5$ emails, find the probability that exactly 1 is flagged.

Which rule, and why? Five independent emails, constant flag probability 0.30, counting flags: a binomial count $X \sim \text{Bin}(5, 0.30)$, target $p_X(1)$.

Solution.

Step 1: Parameters. $n = 5$, $p = 0.30$, $k = 1$.

Step 2: Apply the pmf.

$$\mathbb{P}(X = 1) = \binom{5}{1}(0.30)^1(0.70)^4 = 5 \cdot 0.30 \cdot 0.2401.$$

$$\mathbb{P}(X = 1) = 0.360150 \approx 0.3602.$$

3.1.3 Mean and variance of the binomial

Theorem 3.1.10 (Binomial mean and variance)**.** *If $X \sim \text{Bin}(n, p)$ then*

$$\mathbb{E}(X) = np, \qquad \text{Var}(X) = np(1 - p), \qquad \text{SD}(X) = \sqrt{np(1 - p)}.$$

Proof. Write $X = X_1 + \cdots + X_n$ where the X_i are independent Bernoulli(p) indicators of success on each trial. By linearity, $\mathbb{E}(X) = \sum_{i=1}^{n} \mathbb{E}(X_i) = np$. Because the X_i are independent, variances add: $\text{Var}(X) = \sum_{i=1}^{n} \text{Var}(X_i) = np(1 - p)$, using $\mathbb{E}(X_i) = p$ and $\text{Var}(X_i) = p(1 - p)$ from the Bernoulli proposition. □

Example 3.1.11 (Expected makes and spread)**.** For the free-throw shooter with $p = 0.8$ over $n = 6$ attempts, find the mean, variance, and standard deviation of the number of makes.

Which rule, and why? $X \sim \text{Bin}(6, 0.8)$, so use the binomial mean np and variance $np(1-p)$ directly.

Solution.

Step 1: Mean. $\mathbb{E}(X) = np = 6 \cdot 0.8 = 4.8$.

Step 2: Variance and SD. $\text{Var}(X) = np(1-p) = 6 \cdot 0.8 \cdot 0.2 = 0.96$, so $\text{SD}(X) = \sqrt{0.96} \approx 0.98$.

$$\mathbb{E}(X) = 4.8, \qquad \text{Var}(X) = 0.96, \qquad \text{SD}(X) \approx 0.98.$$

Example 3.1.12 (Defectives, summarised)**.** A shipment of $n = 200$ independent components each fails inspection with probability $p = 0.02$. Find the expected number that fail and the standard deviation.

Which rule, and why? Constant failure rate, independent components, fixed count: $X \sim \text{Bin}(200, 0.02)$. Use $\mathbb{E} = np$ and $\text{SD} = \sqrt{np(1-p)}$.

Solution.

Step 1: Mean. $\mathbb{E}(X) = np = 200 \cdot 0.02 = 4$.

Step 2: SD. $\text{Var}(X) = 200 \cdot 0.02 \cdot 0.98 = 3.92$, so $\text{SD}(X) = \sqrt{3.92} \approx 1.98$.

$$\mathbb{E}(X) = 4, \qquad \text{Var}(X) = 3.92, \qquad \text{SD}(X) \approx 1.98.$$

3.1.4 Cumulative probabilities: "at least" and "at most"

A request for "at most k" successes means the cumulative sum $\mathbb{P}(X \le k) = \sum_{j=0}^{k} p_X(j)$, and "at least k" means $\mathbb{P}(X \ge k) = \sum_{j=k}^{n} p_X(j)$. When "at least one" is wanted, the complement $\mathbb{P}(X \ge 1) = 1 - \mathbb{P}(X = 0)$ is almost always the shortest route.

Example 3.1.13 (At least one defective)**.** In the batch of $n = 12$ parts with defect probability $p = 0.05$, find $\mathbb{P}$(at least one defective).

Which rule, and why? "At least one" with a binomial count $X \sim \text{Bin}(12, 0.05)$: use the complement, since $\mathbb{P}(X \ge 1) = 1 - \mathbb{P}(X = 0)$ needs only one term.

Solution.

Step 1: Complement. $\mathbb{P}(X \ge 1) = 1 - \mathbb{P}(X = 0) = 1 - \binom{12}{0}(0.05)^0(0.95)^{12}$.

Step 2: Evaluate. $(0.95)^{12} \approx 0.540360$, so

$$\mathbb{P}(X \ge 1) \approx 1 - 0.5404 = 0.4596.$$

Example 3.1.14 (At most two flagged emails)**.** With the spam filter ($n = 5$, $p = 0.30$), find the probability that at most 2 of the next 5 emails are flagged.

Which rule, and why? "At most 2" is a cumulative sum $\mathbb{P}(X \le 2) = p_X(0) + p_X(1) + p_X(2)$ for $X \sim \text{Bin}(5, 0.30)$.

Solution.

Step 1: Compute each term.

$$\mathbb{P}(X = 0) = \binom{5}{0}(0.7)^5 = 0.16807,$$
$$\mathbb{P}(X = 1) = \binom{5}{1}(0.3)(0.7)^4 = 0.36015,$$
$$\mathbb{P}(X = 2) = \binom{5}{2}(0.3)^2(0.7)^3 = 10 \cdot 0.09 \cdot 0.343 = 0.30870.$$

Step 2: Add.

$$\mathbb{P}(X \le 2) = 0.16807 + 0.36015 + 0.30870 = 0.83692 \approx 0.8369.$$

Example 3.1.15 (At least three correct guesses)**.** For the guessing student ($n = 8$, $p = 1/4$), find $\mathbb{P}(X \ge 3)$.

Which rule, and why? "At least 3" for $X \sim \mathrm{Bin}(8, 1/4)$. Here the complement $\mathbb{P}(X \geq 3) = 1 - \mathbb{P}(X \leq 2)$ requires only three terms ($k = 0, 1, 2$) instead of six.

Solution.

Step 1: Lower-tail terms. With $p = 1/4$, $q = 3/4$,

$$\mathbb{P}(X = 0) = (3/4)^8 = \tfrac{6561}{65536} \approx 0.100113,$$
$$\mathbb{P}(X = 1) = \binom{8}{1}(1/4)(3/4)^7 = 8 \cdot \tfrac{2187}{65536} \approx 0.266968,$$
$$\mathbb{P}(X = 2) = \binom{8}{2}(1/4)^2(3/4)^6 = 28 \cdot \tfrac{729}{65536} \approx 0.311462.$$

Step 2: Complement.

$$\mathbb{P}(X \geq 3) = 1 - (0.100113 + 0.266968 + 0.311462) = 1 - 0.678543 \approx 0.3215.$$

Example 3.1.16 (Quality gate: pass if at most one failure)**.** A lot of $n = 20$ independent items each fails with probability $p = 0.10$. The lot passes a gate if at most 1 item fails. Find the pass probability.

Which rule, and why? Independent items, constant failure rate, fixed count 20: $X \sim \mathrm{Bin}(20, 0.10)$, and "at most 1" is $\mathbb{P}(X \leq 1) = p_X(0) + p_X(1)$.

Solution.

Step 1: Two terms. With $q = 0.90$,

$$\mathbb{P}(X = 0) = (0.9)^{20} \approx 0.121577, \qquad \mathbb{P}(X = 1) = 20(0.1)(0.9)^{19} \approx 0.270171.$$

Step 2: Add.

$$\mathbb{P}(X \leq 1) \approx 0.121577 + 0.270171 = 0.391748 \approx 0.3917.$$

3.1.5 The hypergeometric setting and pmf

When sampling is done *without replacement* from a finite population, the success probability changes after each draw, so the trials are not independent and the binomial no longer applies. The hypergeometric distribution handles this exactly.

Definition 3.1.17 (Hypergeometric setting)**.** A population has N items, of which r are "successes" and $N - r$ are "failures." A simple random sample of n items is drawn *without replacement*. Let X count the successes in the sample. Then $X \sim \mathrm{Hyper}(N, r, n)$.

Theorem 3.1.18 (Hypergeometric pmf)**.** *If* $X \sim \mathrm{Hyper}(N, r, n)$ *then*

$$p_X(k) = \mathbb{P}(X = k) = \frac{\binom{r}{k}\binom{N-r}{n-k}}{\binom{N}{n}},$$

for integers k *with* $\max(0,\, n - (N - r)) \leq k \leq \min(n, r)$*, and* $p_X(k) = 0$ *otherwise.*

Proof. Every choice of n items from N is equally likely, and there are $\binom{N}{n}$ such samples. A sample with exactly k successes is formed by choosing k of the r successes — in $\binom{r}{k}$ ways — and $n - k$ of the $N - r$ failures — in $\binom{N-r}{n-k}$ ways. Multiplying these independent choices and dividing by the total count gives the displayed ratio. The range restriction simply records that you cannot draw more successes than exist ($k \leq r$) or fewer than the failures permit ($n - k \leq N - r$). □

Example 3.1.19 (Defectives in a small lot)**.** A box holds $N = 15$ memory chips, of which $r = 4$ are defective. You draw $n = 3$ chips at random without replacement. Find the probability that exactly 1 is defective.

Which rule, and why? Drawing *without replacement* from a small finite pool of 15 means p changes after each draw, so this is hypergeometric, $X \sim \text{Hyper}(15, 4, 3)$, not binomial; we want $p_X(1)$.

Solution.

Step 1: Plug into the pmf.

$$\mathbb{P}(X = 1) = \frac{\binom{4}{1}\binom{11}{2}}{\binom{15}{3}} = \frac{4 \cdot 55}{455} = \frac{220}{455}.$$

Step 2: Simplify.

$$\mathbb{P}(X = 1) = \frac{220}{455} = \frac{44}{91} \approx 0.4835.$$

Example 3.1.20 (Committee from a department)**.** A department of $N = 20$ faculty has $r = 8$ statisticians and 12 non-statisticians. A committee of $n = 5$ is chosen at random. Find the probability it contains exactly 3 statisticians.

Which rule, and why? People are selected without replacement (no one serves twice) from a fixed pool, so the count of statisticians is hypergeometric, $X \sim \text{Hyper}(20, 8, 5)$; target $p_X(3)$.

Solution.

Step 1: Apply the pmf.

$$\mathbb{P}(X = 3) = \frac{\binom{8}{3}\binom{12}{2}}{\binom{20}{5}} = \frac{56 \cdot 66}{15504} = \frac{3696}{15504}.$$

Step 2: Simplify.

$$\mathbb{P}(X = 3) = \frac{3696}{15504} = \frac{77}{323} \approx 0.2384.$$

Example 3.1.21 (Capture-recapture flavour: tagged fish)**.** A pond contains $N = 50$ fish, of which $r = 10$ were previously tagged. A researcher nets $n = 8$ fish without replacement. Find the probability that exactly 2 are tagged.

Which rule, and why? Netting without replacement from a finite, fairly small population: hypergeometric, $X \sim \text{Hyper}(50, 10, 8)$. (The sample is 16% of the pond, well above the 5% threshold, so the binomial approximation would be noticeably off — use the exact pmf.)

Solution.

Step 1: Apply the pmf.

$$\mathbb{P}(X = 2) = \frac{\binom{10}{2}\binom{40}{6}}{\binom{50}{8}} = \frac{45 \cdot 3{,}838{,}380}{536{,}878{,}650}.$$

Step 2: Evaluate.

$$\mathbb{P}(X = 2) = \frac{172{,}727{,}100}{536{,}878{,}650} \approx 0.3217.$$

Example 3.1.22 (Drawing aces from a deck)**.** From a standard deck ($N = 52$, $r = 4$ aces), you draw $n = 5$ cards without replacement. Find the probability of getting exactly 2 aces.

Which rule, and why? Cards are dealt without replacement, so successive draws are dependent: hypergeometric, $X \sim \text{Hyper}(52, 4, 5)$, target $p_X(2)$.

Solution.

Step 1: Apply the pmf.

$$\mathbb{P}(X = 2) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}} = \frac{6 \cdot 17{,}296}{2{,}598{,}960} = \frac{103{,}776}{2{,}598{,}960}.$$

Step 2: Simplify.

$$\mathbb{P}(X = 2) = \frac{103{,}776}{2{,}598{,}960} \approx 0.0399.$$

3.1.6 Mean and variance of the hypergeometric

Theorem 3.1.23 (Hypergeometric mean and variance)**.** *If $X \sim \text{Hyper}(N, r, n)$, write the population success fraction $p = r/N$. Then*

$$\mathbb{E}(X) = n\,\frac{r}{N} = np, \qquad \text{Var}(X) = n\,\frac{r}{N}\Big(1 - \frac{r}{N}\Big)\frac{N-n}{N-1} = np(1-p)\,\frac{N-n}{N-1}.$$

The mean is identical to the binomial's, np. The variance is the binomial's $np(1-p)$ multiplied by the **finite-population correction** $\dfrac{N-n}{N-1}$, which is at most 1 (and equals 1 when $n = 1$). Sampling without replacement therefore produces *less* variability than sampling with replacement.

Example 3.1.24 (Mean and variance of tagged fish)**.** For the netting example, $X \sim \text{Hyper}(50, 10, 8)$. Find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Which rule, and why? Hypergeometric mean $n(r/N)$ and variance $np(1-p)\frac{N-n}{N-1}$ with $N = 50, r = 10, n = 8$.

Solution.

Step 1: Mean. $p = r/N = 10/50 = 0.2$, so $\mathbb{E}(X) = np = 8 \cdot 0.2 = 1.6$.

Step 2: Variance with correction.

$$\text{Var}(X) = 8 \cdot 0.2 \cdot 0.8 \cdot \frac{50-8}{50-1} = 1.28 \cdot \frac{42}{49} = 1.28 \cdot \frac{6}{7}.$$

$$\mathbb{E}(X) = 1.6, \qquad \text{Var}(X) = \frac{7.68}{7} \approx 1.097.$$

Example 3.1.25 (Committee composition, summarised)**.** For the committee, $X \sim \text{Hyper}(20, 8, 5)$. Find $\mathbb{E}(X)$ and $\text{Var}(X)$.

Which rule, and why? Apply the hypergeometric mean and variance with $N = 20, r = 8, n = 5, p = 8/20 = 0.4$.

Solution.

Step 1: Mean. $\mathbb{E}(X) = np = 5 \cdot 0.4 = 2$.

Step 2: Variance.

$$\text{Var}(X) = 5 \cdot 0.4 \cdot 0.6 \cdot \frac{20-5}{20-1} = 1.2 \cdot \frac{15}{19} = \frac{18}{19} \approx 0.947.$$

$$\mathbb{E}(X) = 2, \qquad \text{Var}(X) \approx 0.947.$$

3.1.7 Binomial versus hypergeometric

The two distributions answer the same kind of question; the difference is replacement.

	Binomial $\text{Bin}(n, p)$	**Hypergeometric** $\text{Hyper}(N, r, n)$
Sampling	with replacement / infinite pool	without replacement / finite pool
Trials	independent, constant p	dependent, p shifts each draw
$\mathbb{P}(X = k)$	$\binom{n}{k}p^k(1-p)^{n-k}$	$\dfrac{\binom{r}{k}\binom{N-r}{n-k}}{\binom{N}{n}}$
Mean	np	$n\frac{r}{N}$
Variance	$np(1-p)$	$np(1-p)\dfrac{N-n}{N-1}$

Proposition 3.1.26 (Binomial approximates hypergeometric)**.** *Fix n and $p = r/N$. As $N \to \infty$ with $r/N \to p$, the hypergeometric pmf converges to the binomial pmf:*

$$\frac{\binom{r}{k}\binom{N-r}{n-k}}{\binom{N}{n}} \longrightarrow \binom{n}{k} p^k (1-p)^{n-k}.$$

In practice the binomial with $p = r/N$ is an excellent approximation whenever the sample is a small fraction of the population, $n \le 0.05\,N$, because then the finite-population correction $\frac{N-n}{N-1} \approx 1$ and removing one item barely changes p.

Example 3.1.27 (When the approximation is safe, and when it is not)**.** A warehouse holds $N = 10{,}000$ light bulbs of which $r = 400$ are defective ($p = 0.04$). You sample $n = 20$ without replacement and count defectives. (a) Find the exact hypergeometric $\mathbb{P}(X = 0)$. (b) Compare with the binomial approximation. (c) Contrast with a case where the approximation fails.

Which rule, and why? Sampling is without replacement, so the truth is hypergeometric $X \sim \text{Hyper}(10{,}000, 400, 20)$; but $n/N = 20/10{,}000 = 0.002 \le 0.05$, so a binomial $\text{Bin}(20, 0.04)$ should approximate it closely. We check both.

Solution.

Step 1: Exact hypergeometric.

$$\mathbb{P}(X = 0) = \frac{\binom{400}{0}\binom{9600}{20}}{\binom{10000}{20}} = \prod_{j=0}^{19} \frac{9600 - j}{10000 - j} \approx 0.44216.$$

Step 2: Binomial approximation.

$$\mathbb{P}(X = 0) \approx \binom{20}{0}(0.04)^0(0.96)^{20} = (0.96)^{20} \approx 0.44200.$$

The two agree to four decimals — the approximation is excellent because $n \le 0.05N$.

Step 3: A case where it fails. If instead $N = 40$, $r = 16$ ($p = 0.4$), $n = 20$, then $n/N = 0.5 \gg 0.05$. The hypergeometric variance carries the factor $\frac{N-n}{N-1} = \frac{20}{39} \approx 0.513$, so the without-replacement spread is barely half the binomial's; here the binomial would badly overstate the variability and the exact hypergeometric pmf must be used.

(a) 0.4422, (b) 0.4420, (c) approximation invalid when $n \not\le 0.05N$.

Example 3.1.28 (Same numbers, two sampling schemes)**.** An urn has $N = 10$ balls, $r = 4$ red and 6 white. Draw $n = 3$ and let X count reds. Compute $\mathbb{P}(X = 2)$ (a) without replacement and (b) with replacement, and explain the difference.

Which rule, and why? Without replacement is hypergeometric $\text{Hyper}(10, 4, 3)$; with replacement each draw resets to $p = 4/10$, making the draws independent and the count binomial $\text{Bin}(3, 0.4)$. The same scenario splits along the replacement question.

Solution.

Step 1: Without replacement (hypergeometric).

$$\mathbb{P}(X = 2) = \frac{\binom{4}{2}\binom{6}{1}}{\binom{10}{3}} = \frac{6 \cdot 6}{120} = \frac{36}{120} = 0.30.$$

Step 2: With replacement (binomial).

$$\mathbb{P}(X = 2) = \binom{3}{2}(0.4)^2(0.6)^1 = 3 \cdot 0.16 \cdot 0.6 = 0.288.$$

Step 3: Interpret. The two probabilities differ (0.300 vs. 0.288) because removing balls changes the composition of the urn; with $n/N = 0.3$ the sample is a large fraction of the population, so the schemes are genuinely distinct.

$$\mathbb{P}_{\text{w/o}}(X = 2) = 0.300, \qquad \mathbb{P}_{\text{with}}(X = 2) = 0.288.$$

3.2 Discrete Random Variables

A **random variable** attaches a number to each outcome of an experiment, turning messy verbal outcomes ("heads then tails," "the third sample was the first clean one") into ordinary numbers we can add, average, and plot. When those numbers form a finite or listable set — counts, scores, dice totals — the variable is **discrete**, and its entire probabilistic behaviour is captured by two functions: the **probability mass function** p_X, which says how much probability sits on each value, and the **cumulative distribution function** F_X, which accumulates that probability from the left. This section builds both from scratch and shows how to read one off the other.

Which tool does a discrete problem need?

- Asked for the chance of **one exact value**, $\mathbb{P}(X = k)$, or to tabulate every value? $\Rightarrow$ work with the **pmf** $p_X(k) = \mathbb{P}(X = k)$.
- Given a formula for p_X with an **unknown constant** c? $\Rightarrow$ force the masses to be nonnegative and to **sum to** 1, then solve for c.
- Asked for a **cumulative** chance — "at most," "less than," "no more than" — or handed a graph that rises in steps? $\Rightarrow$ use the **cdf** $F_X(x) = \mathbb{P}(X \le x)$.
- Asked for the probability of a **half-open interval** $\mathbb{P}(a < X \le b)$ from a known F_X? $\Rightarrow$ subtract: $F_X(b) - F_X(a)$. Mind whether each endpoint is included.
- Handed a **word problem** (dice totals, matching, defects)? $\Rightarrow$ list the sample space, assign each outcome a number, and collect equal values into p_X.

Why not the continuous tools (§3)? A discrete variable puts positive mass on *isolated* points, so $\mathbb{P}(X = k)$ can be nonzero and F_X jumps; there is no density to integrate. If instead $\mathbb{P}(X = x) = 0$ for every single x and F_X is continuous, the variable is continuous, not discrete.

3.2.1 Random variables and the probability mass function

Definition 3.2.1 (Random variable; discrete)**.** A **random variable** on a sample space S is a function $X\colon S \to \mathbb{R}$ assigning a real number $X(s)$ to each outcome s. It is **discrete** if its set of possible values, the **range** (or support) $\mathcal{R}_X = \{X(s) : s \in S\}$, is finite or countably infinite. For a discrete X the **probability mass function** (pmf) is

$$p_X(k) = \mathbb{P}(X = k) = \mathbb{P}\big(\{s \in S : X(s) = k\}\big), \qquad k \in \mathbb{R}.$$

The pmf records, for each value k, the total probability of all outcomes that X sends to k. Two requirements pin down exactly which functions can be pmfs.

Theorem 3.2.2 (Defining properties of a pmf)**.** *A function p_X is the pmf of some discrete random variable if and only if*

$$\textit{(i)}\quad p_X(k) \ge 0 \ \textit{ for all } k, \qquad \textit{(ii)}\quad \sum_{k \in \mathcal{R}_X} p_X(k) = 1.$$

Proof. Each $p_X(k) = \mathbb{P}(X = k)$ is a probability, hence nonnegative, giving (i). The events $\{X = k\}$, as k ranges over $\mathcal{R}_X$, are disjoint and exhaust S, so by countable additivity $\sum_k \mathbb{P}(X = k) = \mathbb{P}(X \in \mathcal{R}_X) = \mathbb{P}(S) = 1$, giving (ii). Conversely any nonnegative function summing to 1 can be realised by placing mass $p_X(k)$ at each k. □

Example 3.2.3 (Reading a pmf off a table)**.** A quality inspector grades a circuit board $X \in \{0, 1, 2, 3\}$ by its number of faulty solder joints, with

$$p_X(0) = 0.50, \quad p_X(1) = 0.30, \quad p_X(2) = 0.15, \quad p_X(3) = 0.05.$$

Verify this is a valid pmf and find $\mathbb{P}(X \ge 2)$ and $\mathbb{P}(1 \le X < 3)$.

Which rule, and why? The values are isolated counts $0, 1, 2, 3$ with listed probabilities, so X is discrete and we work directly with p_X. "At least" and a bounded range are just sums of the relevant masses — no cdf machinery is needed for so few values.

Solution.

Step 1: Check the two pmf properties. Each entry is ≥ 0, and

$$0.50 + 0.30 + 0.15 + 0.05 = 1.00,$$

so both (i) and (ii) hold; this is a legitimate pmf.

Step 2: Add the masses in each requested set.

$$\mathbb{P}(X \geq 2) = p_X(2) + p_X(3) = 0.15 + 0.05 = 0.20,$$

$$\mathbb{P}(1 \leq X < 3) = p_X(1) + p_X(2) = 0.30 + 0.15 = 0.45.$$

Example 3.2.4 (From outcomes to a pmf: two coin flips)**.** A fair coin is flipped twice; let X be the number of heads. Find p_X.

Which rule, and why? We are handed an experiment, not a table, so we first build the sample space, map each outcome to its number of heads, then collect equal values. This "list, label, collect" routine is the definition of a pmf in action.

Solution.

Step 1: List outcomes and their values. The sample space is $S = \{HH, HT, TH, TT\}$, each outcome having probability $\frac{1}{4}$, with

$$X(HH) = 2, \quad X(HT) = X(TH) = 1, \quad X(TT) = 0.$$

Step 2: Collect outcomes by value.

$$p_X(0) = \tfrac{1}{4}, \qquad p_X(1) = \tfrac{2}{4} = \tfrac{1}{2}, \qquad p_X(2) = \tfrac{1}{4}.$$

These sum to 1, as required.

Example 3.2.5 (An infinite range: first head)**.** A biased coin lands heads with probability $p = \frac{1}{3}$ on each independent toss. Let X be the toss on which the first head appears. Show p_X is a valid pmf.

Which rule, and why? The value X can be any positive integer, so the range is countably infinite; we must verify the masses still sum to 1, which now means summing a geometric series rather than a short list.

Solution.

Step 1: Write the mass of each value. The first head on toss k means $k-1$ tails then a head, so with $q = 1 - p = \frac{2}{3}$,

$$p_X(k) = q^{k-1}p = \left(\tfrac{2}{3}\right)^{k-1}\tfrac{1}{3}, \qquad k = 1, 2, 3, \ldots$$

(This is $X \sim \text{Geom}(p)$.) Each term is positive.

Step 2: Sum the geometric series.

$$\sum_{k=1}^{\infty} q^{k-1}p = p\sum_{j=0}^{\infty} q^j = \frac{p}{1-q} = \frac{p}{p} = 1.$$

Property (ii) holds, so p_X is a valid pmf.

3.2.2 Finding an unknown constant in a pmf

When a pmf is specified only up to a multiplicative constant, the normalisation $\sum_k p_X(k) = 1$ determines that constant uniquely; nonnegativity then fixes its sign.

Example 3.2.6 (Constant times a count)**.** A help desk models the number $X \in \{1, 2, 3, 4\}$ of tickets a new agent closes in an hour by $p_X(k) = c\,k$ for those four values and 0 otherwise. Find c and $\mathbb{P}(X \geq 3)$.

Which rule, and why? A symbolic constant c multiplies the masses, so the cue is "unknown constant in a pmf." Set the total mass to 1 to solve for c; nonnegativity confirms $c > 0$ is the admissible choice.

Solution.

Step 1: Impose $\sum p_X(k) = 1$.

$$\sum_{k=1}^{4} c\,k = c(1 + 2 + 3 + 4) = 10c = 1 \implies c = \tfrac{1}{10}.$$

Since each $p_X(k) = k/10 > 0$, property (i) holds.

Step 2: Use the explicit pmf.

$$\mathbb{P}(X \geq 3) = p_X(3) + p_X(4) = \tfrac{3}{10} + \tfrac{4}{10} = \tfrac{7}{10}.$$

Example 3.2.7 (Constant times a square)**.** Let $p_X(k) = c\,k^2$ for $k \in \{1, 2, 3\}$. Find c and $\mathbb{P}(X \leq 2)$.

Which rule, and why? Same cue — an undetermined c scaling the masses — so normalise. The only change from the previous example is which weights we add.

Solution.

Step 1: Normalise.

$$c(1^2 + 2^2 + 3^2) = c(1 + 4 + 9) = 14c = 1 \implies c = \tfrac{1}{14}.$$

Step 2: Accumulate the requested values.

$$\mathbb{P}(X \leq 2) = p_X(1) + p_X(2) = \tfrac{1}{14} + \tfrac{4}{14} = \tfrac{5}{14}.$$

Example 3.2.8 (Constant in front of an infinite series)**.** A model for the number $X \in \{0, 1, 2, \dots\}$ of typos on a page is $p_X(k) = c\,/\,3^k$. Find c.

Which rule, and why? The constant again sits in front of every mass, but the range is infinite, so normalisation requires summing a geometric series. The cue "find c" still points to $\sum p_X = 1$.

Solution.

Step 1: Sum the geometric series.

$$\sum_{k=0}^{\infty} \frac{c}{3^k} = c \sum_{k=0}^{\infty} \left(\tfrac{1}{3}\right)^k = c \cdot \frac{1}{1 - \frac{1}{3}} = c \cdot \tfrac{3}{2}.$$

Step 2: Set equal to 1 *and solve.*

$$\tfrac{3}{2}\,c = 1 \implies c = \tfrac{2}{3}.$$

Example 3.2.9 (Two-piece pmf with one free value)**.** A loyalty program assigns a customer $X \in \{0, 1, 2, 3\}$ reward stars with $p_X(0) = 0.1$, $p_X(1) = 0.3$, $p_X(3) = 0.2$, and $p_X(2) = a$ left unspecified. Find a and $\mathbb{P}(X \geq 1)$.

Which rule, and why? Most masses are given and one slot is symbolic, so the missing value is forced by the requirement that everything sum to 1 — the same normalisation idea, applied to a single unknown entry.

Solution.

Step 1: Make the total 1.

$$0.1 + 0.3 + a + 0.2 = 1 \implies a = 1 - 0.6 = 0.4.$$

Since $0 \leq a \leq 1$, the pmf is valid.

Step 2: Complement is quickest.

$$\mathbb{P}(X \geq 1) = 1 - p_X(0) = 1 - 0.1 = 0.9.$$

3.2.3 The cumulative distribution function

Definition 3.2.10 (Cumulative distribution function)**.** The **cdf** of a random variable X is

$$F_X(x) = \mathbb{P}(X \le x), \qquad x \in \mathbb{R}.$$

For a discrete X it accumulates mass to the left of x:

$$F_X(x) = \sum_{k \le x} p_X(k).$$

For discrete X the cdf is a **step function**: flat between consecutive values of the range, with an upward jump of height $p_X(k)$ at each value k. The next result lists the properties every cdf shares.

Theorem 3.2.11 (Properties of a cdf)**.** *For any random variable, F_X satisfies*

(a) $0 \le F_X(x) \le 1$; *(b)* F_X *is nondecreasing*; *(c)* $\lim_{x\to-\infty} F_X(x) = 0$, $\lim_{x\to+\infty} F_X(x) = 1$;

(d) F_X *is right-continuous:* $\lim_{t\downarrow x} F_X(t) = F_X(x)$.

For discrete X the jump at a point equals its mass: $p_X(k) = F_X(k) - \lim_{t\uparrow k} F_X(t)$.

Proof. (a) holds since $F_X(x)$ is a probability. If $x \le y$ then $\{X \le x\} \subseteq \{X \le y\}$, so $F_X(x) \le F_X(y)$, giving (b). As $x \to -\infty$ the event $\{X \le x\}$ shrinks to the impossible event and as $x \to +\infty$ it grows to all of S, giving (c). Right-continuity (d) follows because $\{X \le x\} = \bigcap_n \{X \le x + \frac{1}{n}\}$, and continuity of probability from above gives $F_X(x + \frac{1}{n}) \downarrow F_X(x)$. Finally the leftover after removing the strict-inequality part is exactly $\mathbb{P}(X = k)$, the jump height. □

Remark. The jumps include their top endpoint: at a value k, $F_X(k)$ already counts $p_X(k)$ (the solid dot sits at the top of the riser), while the open dot sits at the bottom. This is the graphical face of right-continuity.

Example 3.2.12 (Build and graph a cdf)**.** Returning to the solder-joint count with $p_X(0) = 0.50$, $p_X(1) = 0.30$, $p_X(2) = 0.15$, $p_X(3) = 0.05$, write out F_X and sketch it.

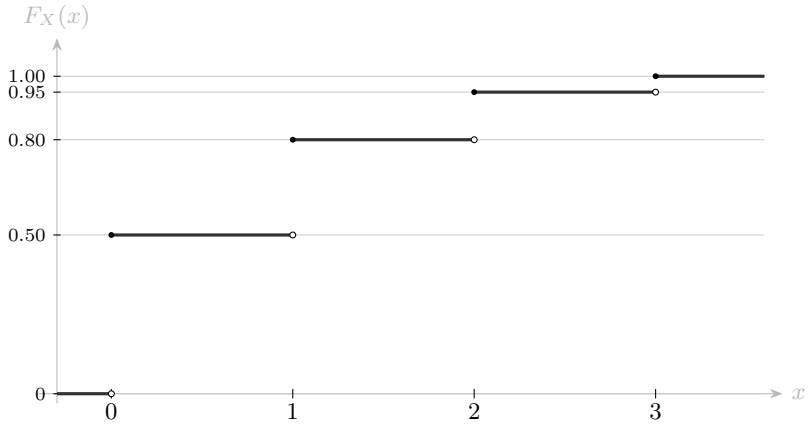
Which rule, and why? We want the accumulated "$\le x$" probabilities and a picture that rises in steps, which is precisely the cdf of a discrete variable: a step function jumping by each mass.

Solution.

Step 1: Accumulate the masses left to right.

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.50, & 0 \le x < 1, \\ 0.80, & 1 \le x < 2, \\ 0.95, & 2 \le x < 3, \\ 1, & x \ge 3. \end{cases}$$

Step 2: Plot the steps. Each riser has height equal to the mass at that value $(0.50, 0.30, 0.15, 0.05)$; a filled dot marks the included value, an open dot the excluded one.



Example 3.2.13 (From cdf back to pmf)**.** A vending machine dispenses X extra free items with cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.4, & 0 \le x < 1, \\ 0.7, & 1 \le x < 2, \\ 1, & x \ge 2. \end{cases}$$

Recover the pmf.

Which rule, and why? We are given F_X and want the masses; since a discrete cdf's jump at each value equals that value's mass, we read off jump heights.

Solution.

Step 1: Locate the jumps. The cdf jumps at $x = 0, 1, 2$.

Step 2: Subtract to get jump heights.

$$p_X(0) = 0.4 - 0 = 0.4, \quad p_X(1) = 0.7 - 0.4 = 0.3, \quad p_X(2) = 1 - 0.7 = 0.3.$$

These sum to 1, confirming the recovery.

3.2.4 Recovering interval probabilities from F_X

Because F_X already stores all cumulative probabilities, any interval probability is a difference of two cdf values. The half-open interval is the clean case.

Theorem 3.2.14 (Interval rule)**.** *For any $a < b$,*

$$\mathbb{P}(a < X \le b) = F_X(b) - F_X(a).$$

Proof. The events $\{X \le a\}$ and $\{a < X \le b\}$ are disjoint with union $\{X \le b\}$, so $F_X(a) + \mathbb{P}(a < X \le b) = F_X(b)$; rearrange. □

To switch which endpoints are included, adjust by the masses at a and b:

$$\mathbb{P}(a \le X \le b) = F_X(b) - F_X(a) + p_X(a), \qquad \mathbb{P}(a < X < b) = F_X(b) - F_X(a) - p_X(b).$$

Example 3.2.15 (Half-open interval)**.** For the solder-joint cdf above ($F_X(0) = 0.50$, $F_X(1) = 0.80$, $F_X(2) = 0.95$, $F_X(3) = 1$), find $\mathbb{P}(0 < X \le 2)$.

Which rule, and why? The question is a half-open interval and F_X is already known, so the interval rule $F_X(b) - F_X(a)$ applies directly with no endpoint correction needed.

Solution.

Step 1: Apply the rule with $a = 0$, $b = 2$.

$$\mathbb{P}(0 < X \leq 2) = F_X(2) - F_X(0) = 0.95 - 0.50 = 0.45.$$

Step 2: Sanity check against masses. $p_X(1) + p_X(2) = 0.30 + 0.15 = 0.45$. Agreed.

Example 3.2.16 (Closed interval needs a correction)**.** Same cdf. Find $\mathbb{P}(1 \leq X \leq 3)$.

Which rule, and why? The lower endpoint is now *included*, so the bare difference would drop $p_X(1)$; the cue "$\leq$ on the left" means add back the mass at the lower endpoint.

Solution.

Step 1: Write the closed interval via the rule.

$$\mathbb{P}(1 \leq X \leq 3) = F_X(3) - F_X(1) + p_X(1).$$

Step 2: Substitute, using $p_X(1) = F_X(1) - F_X(0) = 0.30$.

$$= 1 - 0.80 + 0.30 = 0.50.$$

Check: $p_X(1) + p_X(2) + p_X(3) = 0.30 + 0.15 + 0.05 = 0.50$.

Example 3.2.17 (Open interval drops the top mass)**.** A call-centre logs X, the number of dropped calls in a shift, with

$$F_X(x) = \begin{cases} 0, & x < 0, \\ 0.55, & 0 \leq x < 1, \\ 0.85, & 1 \leq x < 2, \\ 0.97, & 2 \leq x < 3, \\ 1, & x \geq 3. \end{cases}$$

Find $\mathbb{P}(0 < X < 3)$.

Which rule, and why? The upper endpoint is *excluded* (strict $<$), so we must remove $p_X(3)$ from the basic difference; the cue is "$<$ on the right."

Solution.

Step 1: Read the needed cdf values and the top mass.

$$F_X(3) = 1, \quad F_X(0) = 0.55, \quad p_X(3) = F_X(3) - \lim_{t \uparrow 3} F_X(t) = 1 - 0.97 = 0.03.$$

Step 2: Apply the open-interval correction.

$$\mathbb{P}(0 < X < 3) = F_X(3) - F_X(0) - p_X(3) = 1 - 0.55 - 0.03 = 0.42.$$

Check: $p_X(1) + p_X(2) = (0.85 - 0.55) + (0.97 - 0.85) = 0.30 + 0.12 = 0.42$.

Example 3.2.18 (A single point from the cdf)**.** For the call-centre cdf, find $\mathbb{P}(X = 2)$.

Which rule, and why? A single value's probability is the jump in F_X at that point; the cue is "one exact value," read as a one-step difference of the cdf.

Solution.

Step 1: Jump height at $x = 2$.

$$\mathbb{P}(X = 2) = F_X(2) - \lim_{t \uparrow 2} F_X(t) = 0.97 - 0.85 = 0.12.$$

3.2.5 Building a pmf from a word problem

The recurring routine is: enumerate the sample space, assign each outcome its numerical value, then collect outcomes of equal value into the pmf.

Example 3.2.19 (Sum of two fair dice)**.** Two fair six-sided dice are rolled; let T be the sum of the faces. Find p_T and $\mathbb{P}(T \geq 10)$.

Which rule, and why? The total T is a number built from an experiment, so we use "list, label, collect." The 36 ordered pairs are equally likely, which makes each mass a count over 36.

Solution.

Step 1: Count the ordered pairs giving each sum. Of the 36 equally likely ordered pairs (d_1, d_2), the number summing to t is $6 - |t - 7|$ for $t = 2, \ldots, 12$:

t	2	3	4	5	6	7	8	9	10	11	12
$\#\{(d_1, d_2) : d_1 + d_2 = t\}$	1	2	3	4	5	6	5	4	3	2	1

Step 2: Divide each count by 36. Thus $p_T(t) = \dfrac{6 - |t - 7|}{36}$. The eleven masses $1, 2, \ldots, 6, \ldots, 2, 1$ over 36 sum to $36/36 = 1$.

Step 3: Add the tail.

$$\mathbb{P}(T \geq 10) = p_T(10) + p_T(11) + p_T(12) = \frac{3 + 2 + 1}{36} = \frac{6}{36} = \frac{1}{6}.$$

Example 3.2.20 (The matching problem)**.** Three distinct keys are returned at random to their three owners (each of the $3! = 6$ assignments equally likely). Let M be the number of owners who get their own key. Find p_M and $\mathbb{P}(M \geq 1)$.

Which rule, and why? M counts matches in a random assignment, a classic "list, label, collect": enumerate all 6 permutations, count fixed points of each, and collect.

Solution.

Step 1: Enumerate permutations of $(1, 2, 3)$ *and count fixed points.*

permutation	matches M
123	3
132	1
213	1
231	0
312	0
321	1

(231 and 312 are the two derangements, with no match; note $M = 2$ is impossible, since fixing two keys forces the third.)

Step 2: Collect equal values over the 6 *permutations.*

$$p_M(0) = \tfrac{2}{6} = \tfrac{1}{3}, \quad p_M(1) = \tfrac{3}{6} = \tfrac{1}{2}, \quad p_M(2) = 0, \quad p_M(3) = \tfrac{1}{6}.$$

These sum to $\frac{1}{3} + \frac{1}{2} + 0 + \frac{1}{6} = 1$.

Step 3: Use the complement for "at least one."

$$\mathbb{P}(M \geq 1) = 1 - p_M(0) = 1 - \tfrac{1}{3} = \tfrac{2}{3}.$$

Example 3.2.21 (Defectives drawn without replacement)**.** A drawer holds 7 batteries, of which 3 are dead. Two are drawn without replacement; let D be the number of dead ones drawn. Find p_D.

Which rule, and why? D counts "special" items in a sample drawn without replacement from a finite pool, so the masses are hypergeometric counts $\binom{3}{k}\binom{4}{2-k}/\binom{7}{2}$ — still built by collecting outcomes of equal value.

Solution.

Step 1: Total equally likely samples. $\binom{7}{2} = 21$ unordered pairs.

Step 2: Count favourable samples for each k.

$$p_D(0) = \frac{\binom{3}{0}\binom{4}{2}}{21} = \frac{6}{21}, \quad p_D(1) = \frac{\binom{3}{1}\binom{4}{1}}{21} = \frac{12}{21}, \quad p_D(2) = \frac{\binom{3}{2}\binom{4}{0}}{21} = \frac{3}{21}.$$

Step 3: Confirm normalisation.

$$\frac{6+12+3}{21} = \frac{21}{21} = 1.$$

In lowest terms $p_D(0) = \frac{2}{7}$, $p_D(1) = \frac{4}{7}$, $p_D(2) = \frac{1}{7}$.

Example 3.2.22 (Larger of two dice)**.** Two fair dice are rolled; let $L = \max(d_1, d_2)$ be the larger face (ties count that face). Find p_L and F_L.

Which rule, and why? L is a number defined on the 36-outcome experiment, so we "list, label, collect"; a slick shortcut counts via the cdf, since $L \le \ell$ exactly when both dice are $\le \ell$.

Solution.

Step 1: Get F_L from independence. $L \le \ell$ iff both faces are $\le \ell$, and each face is $\le \ell$ with probability $\ell/6$, so for $\ell \in \{1, \dots, 6\}$,

$$F_L(\ell) = \mathbb{P}(L \le \ell) = \left(\frac{\ell}{6}\right)^2 = \frac{\ell^2}{36}.$$

Step 2: Differences give the pmf. For $\ell = 1, \dots, 6$,

$$p_L(\ell) = F_L(\ell) - F_L(\ell - 1) = \frac{\ell^2 - (\ell-1)^2}{36} = \frac{2\ell - 1}{36}.$$

Explicitly the masses are $\frac{1}{36}, \frac{3}{36}, \frac{5}{36}, \frac{7}{36}, \frac{9}{36}, \frac{11}{36}$, which sum to $\frac{1+3+5+7+9+11}{36} = \frac{36}{36} = 1$.

3.3 Continuous Random Variables

When a random variable can land anywhere in an interval — a waiting time, a length, a voltage, a weight — no individual value carries positive probability, and a probability mass function is the wrong tool. Instead the variable is described by a **density** f_X, a nonnegative curve whose total area is 1; probabilities become *areas* under that curve, and the **cumulative distribution function** F_X accumulates area from the left. This section sets up densities and cdfs, computes probabilities and percentiles by integration, and names the three workhorse continuous models: the uniform, exponential, and normal.

Which continuous-density tool does the problem need?

- A formula for f_X with an **unknown constant**, or a piecewise curve you must "make legal"? ⇒ impose the two **density properties**: $f_X \ge 0$ and $\int_{-\infty}^{\infty} f_X = 1$, and solve for the constant (§3.1).
- Asked for $\mathbb{P}(a < X < b)$, $\mathbb{P}(X > b)$, or $\mathbb{P}(X < a)$? ⇒ probability is **area**, $\mathbb{P}(a < X < b) = \int_a^b f_X(x)\,\mathrm{d}x$. Endpoints do not matter because $\mathbb{P}(X = x) = 0$ for a continuous variable (§3.2).
- Given f_X and asked for the **cdf** F_X, or given F_X and asked for the density? ⇒ integrate ($F_X(x) = \int_{-\infty}^{x} f_X$) or differentiate ($f_X = F_X'$); the two are inverse operations (§3.3).
- Asked for a **percentile**, the **median**, or a value c with $\mathbb{P}(X \le c) = p$? ⇒ solve $F_X(c) = p$ for c (§3.4).
- The scenario is "equally likely over an interval", "time until a memoryless event", or "bell-shaped measurement error"? ⇒ reach for the named densities **uniform**, **exponential**, **normal** (§3.5).

Why not a pmf? A pmf assigns positive weight to isolated points and *sums* to 1. A continuous variable spreads its probability over a continuum, so every single point has probability 0 and we *integrate* a density instead of summing a mass function.

3.3.1 Density functions and their two properties

Definition 3.3.1 (Probability density function)**.** A random variable X is **continuous** with **density** (or **pdf**) f_X if f_X is a function on $\mathbb{R}$ such that for every $a \le b$,

$$\mathbb{P}(a \le X \le b) = \int_a^b f_X(x)\,\mathrm{d}x.$$

A function f_X is a legitimate density exactly when it satisfies the two **density properties**:

$$\text{(D1)} \quad f_X(x) \ge 0 \ \text{ for all } x, \qquad\qquad \text{(D2)} \quad \int_{-\infty}^{\infty} f_X(x)\,\mathrm{d}x = 1.$$

A density is *not* a probability: $f_X(x)$ can exceed 1. It is a probability *per unit length*, and only its integral over an interval is a probability. The two properties are the continuous analogue of "$p_X(k) \ge 0$ and $\sum_k p_X(k) = 1$": summation is replaced by integration.

Proposition 3.3.2 (Point probabilities vanish)**.** *If X has a density f_X, then $\mathbb{P}(X = c) = 0$ for every real c. Consequently $\mathbb{P}(a < X < b) = \mathbb{P}(a \le X \le b)$, and including or excluding endpoints never changes a probability.*

Proof. $\mathbb{P}(X = c) = \int_c^c f_X(x)\,\mathrm{d}x = 0$ since the integral over a single point is zero. Splitting $\mathbb{P}(a \le X \le b) = \mathbb{P}(X = a) + \mathbb{P}(a < X < b) + \mathbb{P}(X = b) = \mathbb{P}(a < X < b)$ gives the second claim. □

Example 3.3.3 (Reading a probability off a triangular density)**.** A sensor's calibration error X (in millivolts) has density

$$f_X(x) = \begin{cases} \tfrac{1}{2}x, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Verify f_X is a valid density and find $\mathbb{P}(X > 1.5)$.

Which rule, and why? We are handed a candidate density and asked to confirm legitimacy, then compute a probability. Confirming legitimacy means checking (D1) and (D2); the probability is an *area*, hence an integral of f_X — not a sum, because X ranges over a continuum.

Solution.

Step 1: Check (D1). On $[0, 2]$, $\tfrac{1}{2}x \ge 0$; elsewhere $f_X = 0$. Nonnegative everywhere.

Step 2: Check (D2). Integrate over the support:

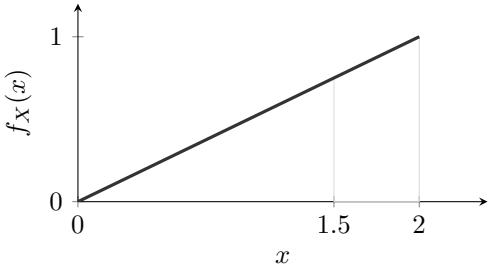
$$\int_0^2 \tfrac{1}{2}x\,\mathrm{d}x = \tfrac{1}{2} \cdot \frac{x^2}{2}\Big|_0^2 = \tfrac{1}{2} \cdot 2 = 1.$$

Both hold, so f_X is a valid density.

Step 3: Compute the area to the right of 1.5*.*

$$\mathbb{P}(X > 1.5) = \int_{1.5}^2 \tfrac{1}{2}x\,\mathrm{d}x = \tfrac{1}{4}x^2\Big|_{1.5}^2 = \tfrac{1}{4}(4 - 2.25) = \tfrac{1.75}{4} = 0.4375.$$

$$\boxed{\mathbb{P}(X > 1.5) = 0.4375}$$



Example 3.3.4 (A density that exceeds 1)**.** X has density $f_X(x) = 3x^2$ for $0 \le x \le 1$ and 0 otherwise. Find $\mathbb{P}(0.4 < X < 0.7)$ and note the largest value of f_X.

Which rule, and why? Again a probability is requested over an interval, so integrate the density. The appearance $f_X(1) = 3 > 1$ is harmless: a density is a rate, not a probability.

Solution.

Step 1: Integrate over the interval.

$$\mathbb{P}(0.4 < X < 0.7) = \int_{0.4}^{0.7} 3x^2 \,\mathrm{d}x = x^3\Big|_{0.4}^{0.7} = 0.343 - 0.064 = 0.279.$$

Step 2: Peak value. On $[0,1]$ the maximum of $3x^2$ is $f_X(1) = 3$, well above 1, yet $\int_0^1 3x^2 \,\mathrm{d}x = 1$ as required.

$$\boxed{\mathbb{P}(0.4 < X < 0.7) = 0.279, \qquad \max f_X = 3}$$

Example 3.3.5 (Endpoints are irrelevant)**.** A bus arrives at a uniformly random time X minutes after the hour, $0 \le X \le 10$, with $f_X(x) = \frac{1}{10}$ on $[0, 10]$. Compare $\mathbb{P}(3 \le X \le 7)$, $\mathbb{P}(3 < X < 7)$, and $\mathbb{P}(X = 5)$.

Which rule, and why? A single point versus an interval: because X is continuous, $\mathbb{P}(X = 5) = 0$ and open/closed endpoints give the same area. This is the point-probability proposition in action.

Solution.

Step 1: The two interval probabilities coincide.

$$\mathbb{P}(3 \le X \le 7) = \mathbb{P}(3 < X < 7) = \int_3^7 \tfrac{1}{10} \,\mathrm{d}x = \tfrac{7-3}{10} = 0.4.$$

Step 2: The point probability. $\mathbb{P}(X = 5) = \int_5^5 \frac{1}{10} \,\mathrm{d}x = 0$.

$$\boxed{\mathbb{P}(3 \le X \le 7) = \mathbb{P}(3 < X < 7) = 0.4, \qquad \mathbb{P}(X = 5) = 0}$$

3.3.2 Finding the normalising constant

When a density is given only up to a multiplicative constant c, property (D2) pins down c: the total area must equal 1, so c is forced.

Example 3.3.6 (Power-law density)**.** The fraction X of a tank's capacity in use at a random instant has $f_X(x) = c(1-x)^2$ for $0 \le x \le 1$ and 0 elsewhere. Find c, then $\mathbb{P}(X < \frac{1}{2})$.

Which rule, and why? An unknown constant in front of a density signals (D2): force the total integral to 1. Only after c is known can we compute the requested area.

Solution.

Step 1: Set the total area to 1*.*

$$\int_0^1 c(1-x)^2 \,\mathrm{d}x = c \cdot \left[-\frac{(1-x)^3}{3}\right]_0^1 = c \cdot \frac{1}{3} = 1 \implies c = 3.$$

Step 2: Use $f_X(x) = 3(1-x)^2$ *to get the probability.*

$$\mathbb{P}\big(X < \tfrac{1}{2}\big) = \int_0^{1/2} 3(1-x)^2 \,\mathrm{d}x = \Big[-(1-x)^3\Big]_0^{1/2} = -\big(\tfrac{1}{2}\big)^3 + 1 = 1 - \tfrac{1}{8} = \tfrac{7}{8}.$$

$$\boxed{c = 3, \qquad \mathbb{P}(X < \tfrac{1}{2}) = \tfrac{7}{8} = 0.875}$$

Example 3.3.7 (Constant on an unbounded support)**.** Battery lifetimes X (in years) follow $f_X(x) = c\,e^{-x/4}$ for $x \geq 0$ and 0 for $x < 0$. Find c.

Which rule, and why? The support runs to ∞, but (D2) is unchanged: integrate over the whole real line, which here means $\int_0^\infty$. A convergent improper integral fixes c.

Solution.

Step 1: Impose (D2) as an improper integral.

$$\int_0^\infty c\,e^{-x/4}\,\mathrm{d}x = c\Big[-4e^{-x/4}\Big]_0^\infty = c\big(0-(-4)\big) = 4c = 1 \implies c = \tfrac{1}{4}.$$

$$\boxed{c = \tfrac{1}{4}}$$

This is an $\mathrm{Exp}(\frac{1}{4})$ density, anticipating §3.5.

Example 3.3.8 (Two-piece (continuous) density)**.** A machine's setup error X on $[0,2]$ has $f_X(x) = cx$ on $[0,1]$ and $f_X(x) = c(2-x)$ on $[1,2]$ (a symmetric "tent"), and 0 elsewhere. Find c.

Which rule, and why? A piecewise density still obeys one global constraint (D2): add the areas of all pieces and set the sum to 1. Solve the single resulting equation for c.

Solution.

Step 1: Integrate each piece.

$$\int_0^1 cx\,\mathrm{d}x = \tfrac{c}{2}, \qquad \int_1^2 c(2-x)\,\mathrm{d}x = c\Big[2x - \tfrac{x^2}{2}\Big]_1^2 = c\big(2-\tfrac{3}{2}\big) = \tfrac{c}{2}.$$

Step 2: Sum and solve.

$$\tfrac{c}{2} + \tfrac{c}{2} = c = 1 \implies c = 1.$$

$$\boxed{c = 1}$$

Example 3.3.9 (Reciprocal-square density)**.** X has $f_X(x) = \dfrac{c}{x^2}$ for $x \geq 1$ and 0 for $x < 1$. Find c and $\mathbb{P}(X > 4)$.

Which rule, and why? Unknown constant on an unbounded support: impose (D2) over $[1,\infty)$. Check the improper integral converges (it does, since $1/x^2$ decays fast enough), then read off the tail probability.

Solution.

Step 1: Normalise.

$$\int_1^\infty \frac{c}{x^2}\,\mathrm{d}x = c\Big[-\frac{1}{x}\Big]_1^\infty = c(0+1) = c = 1 \implies c = 1.$$

Step 2: Tail probability.

$$\mathbb{P}(X > 4) = \int_4^\infty \frac{1}{x^2}\,\mathrm{d}x = \Big[-\frac{1}{x}\Big]_4^\infty = \tfrac{1}{4}.$$

$$\boxed{c = 1, \qquad \mathbb{P}(X > 4) = \tfrac{1}{4}}$$

3.3.3 The cumulative distribution function

Definition 3.3.10 (Cdf and its link to the density)**.** The **cumulative distribution function** of X is

$$F_X(x) = \mathbb{P}(X \le x) = \int_{-\infty}^{x} f_X(t)\,\mathrm{d}t.$$

For a continuous variable F_X is continuous and nondecreasing, with $F_X(-\infty) = 0$ and $F_X(+\infty) = 1$. Wherever f_X is continuous, the Fundamental Theorem of Calculus gives

$$f_X(x) = F_X'(x).$$

Probabilities of intervals come straight from F_X: $\mathbb{P}(a < X \le b) = F_X(b) - F_X(a)$.

So density and cdf are two encodings of the same information: *integrate* the density to get the cdf, *differentiate* the cdf to recover the density.

Example 3.3.11 (Cdf of the triangular density)**.** For $f_X(x) = \frac{1}{2}x$ on $[0, 2]$ (the sensor of §3.1), find F_X on all of $\mathbb{R}$ and use it to recompute $\mathbb{P}(X > 1.5)$.

Which rule, and why? We want the accumulated area as a function of the upper limit, i.e. F_X. Integrate the density piecewise; the requested probability is then a difference of cdf values.

Solution.

Step 1: Integrate on the support. For $0 \le x \le 2$,

$$F_X(x) = \int_0^x \tfrac{1}{2}t\,\mathrm{d}t = \tfrac{1}{4}x^2.$$

Step 2: Extend by the limiting values.

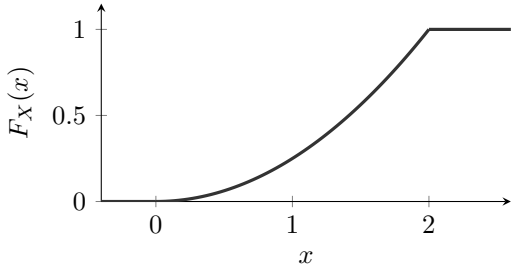
$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{4}x^2, & 0 \le x \le 2, \\ 1, & x > 2. \end{cases}$$

Step 3: Probability via F_X.

$$\mathbb{P}(X > 1.5) = 1 - F_X(1.5) = 1 - \tfrac{1}{4}(1.5)^2 = 1 - 0.5625 = 0.4375,$$

matching the direct integral from §3.1.

$$\boxed{F_X(x) = \tfrac{1}{4}x^2 \text{ on } [0,2], \qquad \mathbb{P}(X > 1.5) = 0.4375}$$



Example 3.3.12 (Recovering a density from a cdf)**.** A delay X (in seconds) has cdf

$$F_X(x) = \begin{cases} 0, & x < 0, \\ x^3, & 0 \le x \le 1, \\ 1, & x > 1. \end{cases}$$

Find the density f_X and $\mathbb{P}(0.2 < X \le 0.5)$.

Which rule, and why? We are given F_X and asked for f_X: differentiate. Interval probabilities then come from differences of F_X (no integral needed).

Solution.

Step 1: Differentiate on each piece. For $0 < x < 1$, $f_X(x) = \frac{d}{dx}x^3 = 3x^2$; elsewhere F_X is flat so $f_X = 0$.

$$f_X(x) = \begin{cases} 3x^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases}$$

Step 2: Interval probability from F_X.

$$\mathbb{P}(0.2 < X \le 0.5) = F_X(0.5) - F_X(0.2) = 0.125 - 0.008 = 0.117.$$

$$\boxed{f_X(x) = 3x^2 \text{ on } [0,1], \qquad \mathbb{P}(0.2 < X \le 0.5) = 0.117}$$

Example 3.3.13 (Cdf on an unbounded support)**.** Battery lifetimes have $f_X(x) = \frac{1}{4}e^{-x/4}$ for $x \ge 0$ (from §3.2). Find F_X and $\mathbb{P}(2 < X < 6)$.

Which rule, and why? Density to cdf on $[0, \infty)$: integrate from 0 to x. The two-sided probability is then $F_X(6) - F_X(2)$.

Solution.

Step 1: Integrate. For $x \ge 0$,

$$F_X(x) = \int_0^x \tfrac{1}{4}e^{-t/4}\,\mathrm{d}t = \Big[-e^{-t/4}\Big]_0^x = 1 - e^{-x/4},$$

and $F_X(x) = 0$ for $x < 0$.

Step 2: The interval.

$$\mathbb{P}(2 < X < 6) = F_X(6) - F_X(2) = \big(1 - e^{-1.5}\big) - \big(1 - e^{-0.5}\big) = e^{-0.5} - e^{-1.5}.$$

Numerically $e^{-0.5} = 0.6065$, $e^{-1.5} = 0.2231$, so the difference is 0.3834.

$$\boxed{F_X(x) = 1 - e^{-x/4} \ (x \ge 0), \qquad \mathbb{P}(2 < X < 6) \approx 0.3834}$$

Example 3.3.14 (Piecewise cdf, both directions)**.** X has cdf $F_X(x) = 0$ for $x < 0$, $F_X(x) = x - \frac{x^2}{4}$ for $0 \le x \le 2$, and $F_X(x) = 1$ for $x > 2$. Find f_X and confirm it integrates to 1.

Which rule, and why? Given F_X, differentiate to get f_X; then sanity-check (D2). Differentiation and integration here are inverse, so the area must return to 1.

Solution.

Step 1: Differentiate. For $0 < x < 2$,

$$f_X(x) = \frac{d}{dx}\Big(x - \tfrac{x^2}{4}\Big) = 1 - \tfrac{x}{2},$$

and 0 outside $[0, 2]$.

Step 2: Check (D2).

$$\int_0^2 \big(1 - \tfrac{x}{2}\big)\,\mathrm{d}x = \Big[x - \tfrac{x^2}{4}\Big]_0^2 = 2 - 1 = 1.$$

$$\boxed{f_X(x) = 1 - \tfrac{x}{2} \text{ on } [0,2]}$$

3.3.4 Quantiles, percentiles, and the median

Definition 3.3.15 (Quantile / percentile)**.** For $0 < p < 1$, a **p-quantile** (or $100p$-th **percentile**) of X is a value x_p with

$$F_X(x_p) = \mathbb{P}(X \le x_p) = p.$$

When F_X is continuous and strictly increasing on the support, x_p is unique and $x_p = F_X^{-1}(p)$. The **median** is the 0.5-quantile.

To find a percentile, set the cdf equal to the target probability and solve for the value: percentiles run the cdf *backwards*.

Example 3.3.16 (Median of the triangular density)**.** For $f_X(x) = \frac{1}{2}x$ on $[0, 2]$, with $F_X(x) = \frac{1}{4}x^2$, find the median.

Which rule, and why? "Median" is the value with half the area to its left, i.e. solve $F_X(m) = 0.5$. This is a percentile problem, so invert the cdf rather than integrate again.

Solution.

Step 1: Set $F_X(m) = \frac{1}{2}$.

$$\tfrac{1}{4}m^2 = \tfrac{1}{2} \implies m^2 = 2 \implies m = \sqrt{2} \approx 1.414.$$

We take the root in $[0, 2]$.

$$\boxed{m = \sqrt{2} \approx 1.414}$$

Example 3.3.17 (90th percentile of a lifetime)**.** Battery lifetimes have $F_X(x) = 1 - e^{-x/4}$ $(x \ge 0)$. Find the 90th percentile $x_{0.90}$.

Which rule, and why? A percentile request: solve $F_X(x) = 0.90$. Because F_X involves $e^{-x/4}$, the inversion uses a logarithm.

Solution.

Step 1: Solve $1 - e^{-x/4} = 0.90$.

$$e^{-x/4} = 0.10 \implies -\tfrac{x}{4} = \ln(0.10) \implies x = -4\ln(0.10) = 4\ln 10.$$

Step 2: Evaluate. $\ln 10 = 2.3026$, so $x_{0.90} = 4(2.3026) = 9.21$ years.

$$\boxed{x_{0.90} = 4\ln 10 \approx 9.21 \text{ years}}$$

Example 3.3.18 (Quartiles of a uniform variable)**.** $X \sim \text{Uniform}(0, 10)$ has $F_X(x) = x/10$ on $[0, 10]$. Find the first quartile Q_1, the median, and the third quartile Q_3.

Which rule, and why? Quartiles are the $0.25, 0.50, 0.75$ quantiles; set F_X equal to each in turn. The uniform cdf is linear, so each inversion is immediate.

Solution.

Step 1: Invert the linear cdf. $F_X(x) = x/10 = p \Rightarrow x = 10p$.

Step 2: Plug in the three probabilities.

$$Q_1 = 10(0.25) = 2.5, \qquad \text{median} = 10(0.5) = 5, \qquad Q_3 = 10(0.75) = 7.5.$$

$$\boxed{Q_1 = 2.5, \quad \text{median} = 5, \quad Q_3 = 7.5}$$

Example 3.3.19 (Percentile of a power density)**.** The tank-usage fraction has $F_X(x) = 1 - (1 - x)^3$ on $[0, 1]$ (from $f_X = 3(1 - x)^2$). What usage level is exceeded only 20% of the time?

Which rule, and why? "Exceeded 20% of the time" means $\mathbb{P}(X > c) = 0.20$, i.e. $F_X(c) = 0.80$: a percentile (the 80th). Solve the cubic cdf equation for c.

Solution.

Step 1: Translate to the cdf. $\mathbb{P}(X > c) = 0.20 \Rightarrow F_X(c) = 0.80$.

Step 2: Solve $1 - (1 - c)^3 = 0.80$.

$$(1 - c)^3 = 0.20 \implies 1 - c = 0.20^{1/3} = 0.5848 \implies c = 0.4152.$$

$$\boxed{c \approx 0.415}$$

3.3.5 The named continuous models

Three densities cover most applications. Each is introduced here by its formula, support, and (stated) mean and variance; later sections derive these moments by integration.

Definition 3.3.20 (Uniform, exponential, normal)**.**

Model	Density $f_X(x)$	$\mathbb{E}(X)$	$\mathrm{Var}(X)$
Uniform Uniform(a, b)	$\dfrac{1}{b-a},\ a \le x \le b$	$\dfrac{a+b}{2}$	$\dfrac{(b-a)^2}{12}$
Exponential Exp(λ)	$\lambda e^{-\lambda x},\ x \ge 0\ (\lambda > 0)$	$\dfrac{1}{\lambda}$	$\dfrac{1}{\lambda^2}$
Normal N(μ, σ^2)	$\dfrac{1}{\sigma\sqrt{2\pi}}\, e^{-(x-\mu)^2/(2\sigma^2)},\ x \in \mathbb{R}$	μ	σ^2

The exponential cdf is $F_X(x) = 1 - e^{-\lambda x}$ $(x \ge 0)$; the uniform cdf is $F_X(x) = \frac{x-a}{b-a}$ on $[a, b]$. The normal cdf has no elementary closed form and is written Φ for the **standard** normal $\mathrm{N}(0, 1)$, with the **standardisation** $Z = \frac{X-\mu}{\sigma} \sim \mathrm{N}(0, 1)$.

Example 3.3.21 (Uniform: equally likely arrival)**.** A file finishes downloading at a time $X \sim \mathrm{Uniform}(0, 20)$ seconds. Find $\mathbb{P}(X > 12)$, the density value, and the median.

Which rule, and why? "Equally likely over an interval" is the uniform signature: constant density $1/(b - a)$. Probabilities are just proportions of the interval length, and the median is its midpoint.

Solution.

Step 1: Density and probability. Here $f_X(x) = \frac{1}{20}$ on $[0, 20]$, so

$$\mathbb{P}(X > 12) = \int_{12}^{20} \tfrac{1}{20}\, \mathrm{d}x = \frac{20 - 12}{20} = 0.40.$$

Step 2: Median. By symmetry the median is the midpoint $\frac{0+20}{2} = 10$ (equivalently $F_X(m) = m/20 = 0.5$).

$$\boxed{f_X = \tfrac{1}{20}, \qquad \mathbb{P}(X > 12) = 0.40, \qquad \text{median} = 10}$$

Example 3.3.22 (Exponential: memoryless waiting)**.** Calls to a help line arrive so that the gap X between consecutive calls is Exp(λ) with mean 5 minutes. Find λ, $\mathbb{P}(X > 8)$, and verify the **memoryless** identity $\mathbb{P}(X > 8 \mid X > 3) = \mathbb{P}(X > 5)$.

Which rule, and why? "Time until the next event" with a stated mean is the exponential. Since $\mathbb{E}(X) = 1/\lambda$, the mean gives λ. Tail probabilities use $\mathbb{P}(X > t) = e^{-\lambda t}$, and memorylessness is a direct conditional-probability check.

Solution.

Step 1: Rate from the mean. $\mathbb{E}(X) = 1/\lambda = 5 \Rightarrow \lambda = \frac{1}{5} = 0.2$.

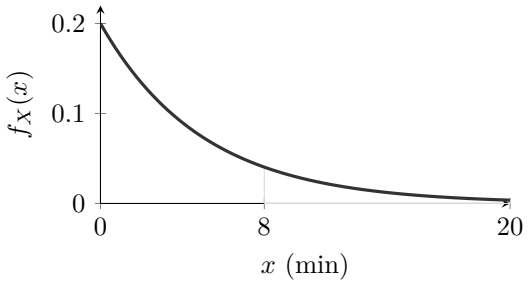
Step 2: Tail probability.

$$\mathbb{P}(X > 8) = e^{-\lambda \cdot 8} = e^{-0.2 \cdot 8} = e^{-1.6} \approx 0.2019.$$

Step 3: Memoryless check. Using $\mathbb{P}(X > t) = e^{-0.2t}$,

$$\mathbb{P}(X > 8 \mid X > 3) = \frac{\mathbb{P}(X > 8)}{\mathbb{P}(X > 3)} = \frac{e^{-1.6}}{e^{-0.6}} = e^{-1.0} = \mathbb{P}(X > 5).$$

$$\boxed{\lambda = 0.2, \quad \mathbb{P}(X > 8) \approx 0.2019, \quad \mathbb{P}(X > 8 \mid X > 3) = \mathbb{P}(X > 5) = e^{-1}}$$



Example 3.3.23 (Normal: standardising a measurement)**.** Bottle fill volumes are $X \sim \mathrm{N}(500, 4)$ (mean 500 mL, variance 4, so $\sigma = 2$ mL). Find $\mathbb{P}(X < 497)$ and $\mathbb{P}(498 < X < 503)$ using $Z = (X - \mu)/\sigma$ and $\Phi(0.5) = 0.6915$, $\Phi(1) = 0.8413$, $\Phi(1.5) = 0.9332$.

Which rule, and why? A bell-shaped measurement with a mean and variance is normal. Normal probabilities have no closed form, so standardise to $Z \sim \mathrm{N}(0, 1)$ and read Φ. Use $\Phi(-z) = 1 - \Phi(z)$ for left tails.

Solution.

Step 1: Standardise the first event. $\sigma = \sqrt{4} = 2$, so for $X < 497$,

$$Z < \frac{497 - 500}{2} = -1.5, \qquad \mathbb{P}(X < 497) = \Phi(-1.5) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668.$$

Step 2: Standardise the interval. For $498 < X < 503$,

$$\frac{498 - 500}{2} = -1, \qquad \frac{503 - 500}{2} = 1.5,$$

$$\mathbb{P}(498 < X < 503) = \Phi(1.5) - \Phi(-1) = 0.9332 - (1 - 0.8413) = 0.9332 - 0.1587 = 0.7745.$$

$$\boxed{\mathbb{P}(X < 497) = 0.0668, \qquad \mathbb{P}(498 < X < 503) = 0.7745}$$

Example 3.3.24 (Normal: an inverse (percentile) problem)**.** Exam scores are $X \sim \mathrm{N}(72, 81)$ ($\sigma = 9$). The top 10% earn distinction. Using $\Phi(1.28) = 0.90$, find the cutoff score.

Which rule, and why? This runs the normal *backwards*: we know the probability (0.90 below the cutoff) and want the value, a percentile. Find the z with $\Phi(z) = 0.90$, then de-standardise via $x = \mu + z\sigma$.

Solution.

Step 1: Find the standard-normal quantile. $\Phi(z) = 0.90 \Rightarrow z = 1.28$.

Step 2: De-standardise.

$$x = \mu + z\sigma = 72 + 1.28 \cdot 9 = 72 + 11.52 = 83.52.$$

$$\boxed{\text{cutoff} \approx 83.5 \text{ points}}$$

Example 3.3.25 (Uniform: a conditional probability)**.** A pointer stops at a uniform angle $X \sim \mathrm{Uniform}(0, 360)$ degrees. Given $X < 180$ (upper half), find $\mathbb{P}(X < 90)$ (first quadrant).

Which rule, and why? Uniform again (equally likely over $[0, 360]$), now with conditioning. For a uniform variable, conditional probability is just the ratio of sub-interval lengths — areas of a flat density.

Solution.

Step 1: Write the conditional probability.

$$\mathbb{P}(X < 90 \mid X < 180) = \frac{\mathbb{P}(X < 90)}{\mathbb{P}(X < 180)}.$$

Step 2: Use proportions of length. $\mathbb{P}(X < 90) = \frac{90}{360} = \frac{1}{4}$ and $\mathbb{P}(X < 180) = \frac{180}{360} = \frac{1}{2}$, so

$$\mathbb{P}(X < 90 \mid X < 180) = \frac{1/4}{1/2} = \tfrac{1}{2}.$$

$$\boxed{\mathbb{P}(X < 90 \mid X < 180) = \tfrac{1}{2}}$$

3.4 Expected Values

The **expected value** $\mathbb{E}(X)$ is the single number that best summarises a random variable's location: it is the long-run average of the values X produces, weighted by how likely each value is. Mechanically it is a center of mass — balance a rod loaded with the probability mass and $\mathbb{E}(X)$ is the fulcrum. This section computes $\mathbb{E}(X)$ for discrete and continuous variables, computes $\mathbb{E}\big(g(X)\big)$ for a function of X without first finding the distribution of $g(X)$, exploits the linearity rule $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$, and records the means of the named families.

Which expectation rule does the problem need?

- Want the average of X itself, and X takes **isolated values**? ⇒ **discrete mean** $\mathbb{E}(X) = \sum_k k\, p_X(k)$.
- Want the average of X itself, and X has a **density**? ⇒ **continuous mean** $\mathbb{E}(X) = \int_{-\infty}^{\infty} x\, f_X(x)\, \mathrm{d}x$.
- Want the average of a **function** $g(X)$ (a payoff, a square, a cost)? ⇒ the **law of the unconscious statistician**: weight $g(k)$ by $p_X(k)$, or integrate $g(x)f_X(x)$. Do *not* first find the pmf/pdf of $g(X)$.
- The variable is a **linear rescaling** $aX + b$, or a **sum** of pieces whose means you know? ⇒ **linearity**: $\mathbb{E}(aX + b) = a\mathbb{E}(X) + b$ and $\mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n)$, with no independence needed.
- X belongs to a **named family**? ⇒ quote the mean (binomial np, Poisson λ, geometric $1/p$, uniform $(a + b)/2$, exponential $1/\lambda$, gamma α/λ) rather than re-summing.

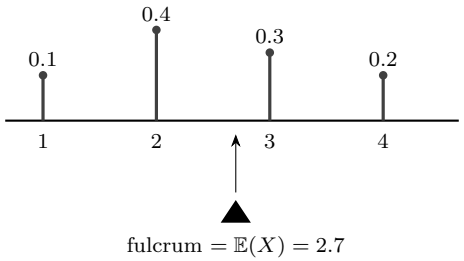
Why not the others. Squaring or any nonlinear g breaks linearity, so $\mathbb{E}\big(g(X)\big) \neq g\big(\mathbb{E}(X)\big)$ in general — use LOTUS, not linearity. And a median or mode answers "typical value" too, but only $\mathbb{E}(X)$ is the probability-weighted average that linearity acts on.

3.4.1 Expected value of a discrete variable

Definition 3.4.1 (Discrete expected value)**.** Let X be discrete with pmf p_X and range $\mathcal{R}$. Its **expected value** (or **mean**) is

$$\mathbb{E}(X) = \sum_{k \in \mathcal{R}} k\, p_X(k),$$

provided $\sum_k |k|\, p_X(k) < \infty$. The sum is each value times its probability; it is the center of mass of the mass diagram and need not itself be an attainable value of X.



Example 3.4.2 (Defective boards on a tray)**.** A tray holds 4 circuit boards. Let X be the number that fail a burn-in test, with pmf

$$p_X(0) = 0.1, \quad p_X(1) = 0.4, \quad p_X(2) = 0.3, \quad p_X(3) = 0.2.$$

Find the expected number of failures.

Which rule, and why? X takes the isolated values $0, 1, 2, 3$ with listed probabilities, so the average is the discrete mean $\sum_k k\, p_X(k)$. There is no density and no function of X to transform, so neither the integral form nor LOTUS is needed.

Solution.

Step 1: multiply each value by its probability.

$$0(0.1) + 1(0.4) + 2(0.3) + 3(0.2) = 0 + 0.4 + 0.6 + 0.6.$$

Step 2: add.

$$\mathbb{E}(X) = 1.6 \text{ failures.}$$

The mean 1.6 is not an attainable count; it is the balance point of the four masses.

Example 3.4.3 (A fair-looking but skewed wager)**.** A spinner pays you a dollar amount X with

$$p_X(0) = \tfrac{1}{2}, \quad p_X(2) = \tfrac{1}{3}, \quad p_X(9) = \tfrac{1}{6}.$$

What is the average payout per spin?

Which rule, and why? Isolated dollar values with given probabilities: discrete mean again. The values are already the quantity of interest (dollars), so no transformation is applied.

Solution.

Step 1: confirm the pmf sums to 1. $\tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{3+2+1}{6} = 1$, so the weights are legitimate.

Step 2: weight each payout.

$$\mathbb{E}(X) = 0 \cdot \tfrac{1}{2} + 2 \cdot \tfrac{1}{3} + 9 \cdot \tfrac{1}{6} = \tfrac{2}{3} + \tfrac{9}{6} = \tfrac{4}{6} + \tfrac{9}{6} = \tfrac{13}{6}.$$

$$\mathbb{E}(X) = \tfrac{13}{6} \approx 2.17 \text{ dollars per spin.}$$

Example 3.4.4 (Sum of two fair dice)**.** Roll two fair dice and let S be their sum, $S \in \{2, \ldots, 12\}$ with the familiar triangular pmf $p_S(k) = \dfrac{6 - |k-7|}{36}$. Find $\mathbb{E}(S)$.

Which rule, and why? S is discrete on $\{2, \ldots, 12\}$; we want its plain average, so it is the discrete mean. Symmetry will shortcut the arithmetic but the rule applied is still $\sum_k k\, p_S(k)$.

Solution.

Step 1: note the symmetry. The pmf is symmetric about $k = 7$: $p_S(7-j) = p_S(7+j)$. A symmetric mass diagram balances at its center.

Step 2: read off (or verify) the mean. By symmetry $\mathbb{E}(S) = 7$. As a check, pair terms equidistant from 7: $(2{+}12)\tfrac{1}{36} + (3{+}11)\tfrac{2}{36} + \cdots + (6{+}8)\tfrac{5}{36} + 7 \cdot \tfrac{6}{36} = 14 \cdot \tfrac{1+2+3+4+5}{36} + \tfrac{42}{36} = \tfrac{210+42}{36}$.

$$\mathbb{E}(S) = \tfrac{252}{36} = 7.$$

Example 3.4.5 (A pmf with a parameter)**.** A discrete variable X on $\{1, 2, 3\}$ has $p_X(k) = c\,k$ for a constant c. Find c, then $\mathbb{E}(X)$.

Which rule, and why? Two tasks: first force the pmf property $\sum_k p_X(k) = 1$ to pin down c (this is a normalising step, not an expectation), then apply the discrete mean.

Solution.

Step 1: normalise. $\sum_{k=1}^{3} c\,k = c(1+2+3) = 6c = 1$, so $c = \frac{1}{6}$ and $p_X(1) = \frac{1}{6}$, $p_X(2) = \frac{2}{6}$, $p_X(3) = \frac{3}{6}$.

Step 2: take the mean.

$$\mathbb{E}(X) = 1 \cdot \tfrac{1}{6} + 2 \cdot \tfrac{2}{6} + 3 \cdot \tfrac{3}{6} = \tfrac{1+4+9}{6} = \tfrac{14}{6} = \tfrac{7}{3}.$$

$$\mathbb{E}(X) = \tfrac{7}{3} \approx 2.33.$$

3.4.2 Expected value of a continuous variable

Definition 3.4.6 (Continuous expected value)**.** Let X have density f_X. Its **expected value** is

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} x\, f_X(x)\, \mathrm{d}x,$$

provided $\int_{-\infty}^{\infty} |x|\, f_X(x)\, \mathrm{d}x < \infty$. The discrete weight $p_X(k)$ is replaced by the infinitesimal mass $f_X(x)\,\mathrm{d}x$, and summation by integration.

Example 3.4.7 (A triangular density)**.** Let X have density $f_X(x) = 2x$ on $0 \le x \le 1$ and 0 elsewhere. Find $\mathbb{E}(X)$.

Which rule, and why? X has a density, so its mean is $\int x f_X(x)\,\mathrm{d}x$, integrated over the support $[0, 1]$ where f_X is nonzero. We average X itself, so no LOTUS transformation enters.

Solution.

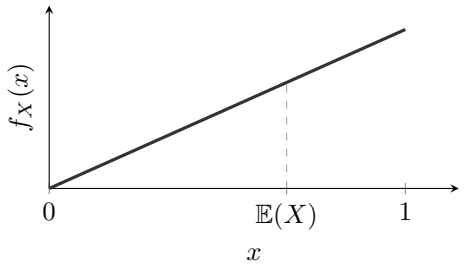
Step 1: set up the integral on the support.

$$\mathbb{E}(X) = \int_0^1 x\,(2x)\,\mathrm{d}x = \int_0^1 2x^2\,\mathrm{d}x.$$

Step 2: integrate.

$$\mathbb{E}(X) = \left[\tfrac{2}{3}x^3\right]_0^1 = \tfrac{2}{3}.$$

The mass leans toward 1, so the balance point $\frac{2}{3}$ sitting right of the midpoint $\frac{1}{2}$ is the expected sign.



Example 3.4.8 (Uniform lifetime)**.** A bus arrives at a uniformly random time X minutes after the hour, $X \sim \mathrm{Uniform}(0, 30)$ with $f_X(x) = \frac{1}{30}$ on $[0, 30]$. Find the expected arrival time.

Which rule, and why? A density on a continuum, so the mean is the integral $\int x f_X(x)\,\mathrm{d}x$. (We will re-derive the uniform mean here; later we simply quote $(a+b)/2$.)

Solution.

Step 1: integrate over the support.

$$\mathbb{E}(X) = \int_0^{30} x \cdot \tfrac{1}{30}\,\mathrm{d}x = \tfrac{1}{30}\Big[\tfrac{x^2}{2}\Big]_0^{30} = \tfrac{1}{30}\cdot\tfrac{900}{2}.$$

Step 2: simplify.

$$\mathbb{E}(X) = \tfrac{450}{30} = 15 \text{ minutes},$$

the midpoint of $[0, 30]$, as symmetry demands.

Example 3.4.9 (Exponential waiting time)**.** The time X (hours) until the next support call is $X \sim \mathrm{Exp}(\lambda)$ with $f_X(x) = \lambda e^{-\lambda x}$ on $x \geq 0$. Show $\mathbb{E}(X) = 1/\lambda$, then evaluate it for $\lambda = 4$ calls per hour.

Which rule, and why? A density on $[0, \infty)$; the mean is $\int_0^\infty x f_X(x)\,\mathrm{d}x$. The integrand needs integration by parts, the standard move for $xe^{-\lambda x}$.

Solution.

Step 1: integrate by parts. With $u = x,\ \mathrm{d}v = \lambda e^{-\lambda x}\,\mathrm{d}x$, so $\mathrm{d}u = \mathrm{d}x$ and $v = -e^{-\lambda x}$,

$$\mathbb{E}(X) = \int_0^\infty x\,\lambda e^{-\lambda x}\,\mathrm{d}x = \Big[-xe^{-\lambda x}\Big]_0^\infty + \int_0^\infty e^{-\lambda x}\,\mathrm{d}x.$$

Step 2: evaluate the pieces. The boundary term is 0 (as $x \to \infty$, $xe^{-\lambda x} \to 0$; at $x = 0$ it is 0), and $\int_0^\infty e^{-\lambda x}\,\mathrm{d}x = \frac{1}{\lambda}$. Hence

$$\mathbb{E}(X) = \tfrac{1}{\lambda}, \qquad \text{so for } \lambda = 4, \quad \mathbb{E}(X) = \tfrac{1}{4} \text{ hour} = 15 \text{ minutes}.$$

Example 3.4.10 (Density supported away from the origin)**.** Let X have density $f_X(x) = \dfrac{3}{x^4}$ for $x \geq 1$ and 0 otherwise (a Pareto-type tail). Find $\mathbb{E}(X)$.

Which rule, and why? A density on $[1, \infty)$; the mean is the improper integral $\int_1^\infty x f_X(x)\,\mathrm{d}x$. We must check it converges before reporting a finite value.

Solution.

Step 1: set up and integrate.

$$\mathbb{E}(X) = \int_1^\infty x \cdot \tfrac{3}{x^4}\,\mathrm{d}x = \int_1^\infty 3x^{-3}\,\mathrm{d}x = \Big[-\tfrac{3}{2}x^{-2}\Big]_1^\infty.$$

Step 2: take the limit. As $x \to \infty$ the term $-\frac{3}{2}x^{-2} \to 0$; at $x = 1$ it is $-\frac{3}{2}$. So

$$\mathbb{E}(X) = 0 - \big(-\tfrac{3}{2}\big) = \tfrac{3}{2}.$$

(The integral converges because the integrand decays like x^{-3}; a heavier tail such as $f_X(x) = 2/x^3$ would make $\int x f_X$ diverge and leave $\mathbb{E}(X)$ undefined.)

3.4.3 The law of the unconscious statistician

To average a function $g(X)$ — a payoff, a cost, a square — you do *not* need the distribution of $Y = g(X)$. Weight g by the distribution of X directly.

Theorem 3.4.11 (Law of the unconscious statistician (LOTUS))**.** *For a function g,*

$$\mathbb{E}\big(g(X)\big) = \sum_k g(k)\,p_X(k) \quad (\textit{discrete}), \qquad \mathbb{E}\big(g(X)\big) = \int_{-\infty}^{\infty} g(x)\,f_X(x)\,\mathrm{d}x \quad (\textit{continuous}),$$

whenever the sum/integral converges absolutely.

Proof. (Discrete case.) Let $Y = g(X)$ with range $\{y_j\}$. Group the k with $g(k) = y_j$:

$$\mathbb{E}(Y) = \sum_j y_j\, p_Y(y_j) = \sum_j y_j \sum_{k:\, g(k)=y_j} p_X(k) = \sum_j \sum_{k:\, g(k)=y_j} g(k)\, p_X(k) = \sum_k g(k)\, p_X(k),$$

since every k lands in exactly one group. The continuous case follows by the same regrouping applied to the density. □

Remark. A consequence worth stating: nonlinearity means $\mathbb{E}\big(g(X)\big) \neq g\big(\mathbb{E}(X)\big)$ in general. For instance $\mathbb{E}(X^2)$ almost never equals $\big(\mathbb{E}(X)\big)^2$; the gap is exactly the variance of the next section.

Example 3.4.12 (Squaring a die)**.** Let X be the result of one fair die, $p_X(k) = \frac{1}{6}$ for $k = 1, \ldots, 6$. Find $\mathbb{E}(X^2)$.

Which rule, and why? We want the average of $g(X) = X^2$, not of X, so apply LOTUS: weight k^2 by $p_X(k)$. Computing $\big(\mathbb{E}X\big)^2$ would be the wrong quantity.

Solution.

Step 1: sum the squared values.

$$\mathbb{E}(X^2) = \sum_{k=1}^{6} k^2 \cdot \tfrac{1}{6} = \tfrac{1}{6}(1 + 4 + 9 + 16 + 25 + 36) = \tfrac{91}{6}.$$

Step 2: report.

$$\mathbb{E}(X^2) = \tfrac{91}{6} \approx 15.17.$$

Note $\big(\mathbb{E}X\big)^2 = (3.5)^2 = 12.25 \neq \frac{91}{6}$, exactly as the remark warns.

Example 3.4.13 (Expected area of a random square)**.** A side length $X \sim \text{Uniform}(0, 2)$ has $f_X(x) = \frac{1}{2}$ on $[0, 2]$. The square has area $A = X^2$. Find $\mathbb{E}(A)$.

Which rule, and why? $A = g(X) = X^2$ is a function of a continuous variable, so LOTUS gives $\mathbb{E}(A) = \int g(x) f_X(x)\, \mathrm{d}x$ directly — no need to derive the density of A.

Solution.

Step 1: integrate x^2 against the density.

$$\mathbb{E}(A) = \int_0^2 x^2 \cdot \tfrac{1}{2}\, \mathrm{d}x = \tfrac{1}{2}\Big[\tfrac{x^3}{3}\Big]_0^2 = \tfrac{1}{2} \cdot \tfrac{8}{3}.$$

Step 2: simplify.

$$\mathbb{E}(A) = \tfrac{4}{3} \approx 1.33.$$

The square of the average side is $(\mathbb{E}X)^2 = 1^2 = 1$, again smaller than $\mathbb{E}(A)$.

Example 3.4.14 (A capped insurance payout)**.** A loss $X \sim \text{Exp}(\lambda)$ with $\lambda = \frac{1}{2}$ (so mean loss 2, in thousands) is paid by a policy that caps the payout at 3: the payout is $g(X) = \min(X, 3)$. Find the expected payout.

Which rule, and why? The payout is a nonlinear function $g(X) = \min(X, 3)$ of a continuous variable. LOTUS handles it as a single integral, splitting where the cap kicks in at $x = 3$.

Solution.

Step 1: split the integral at the cap. With $f_X(x) = \frac{1}{2}e^{-x/2}$,

$$\mathbb{E}\big(g(X)\big) = \int_0^3 x \cdot \tfrac{1}{2}e^{-x/2}\, \mathrm{d}x + \int_3^\infty 3 \cdot \tfrac{1}{2}e^{-x/2}\, \mathrm{d}x.$$

Step 2: evaluate each piece. For the first, integration by parts gives $\int_0^3 x \cdot \frac{1}{2}e^{-x/2}\,\mathrm{d}x = \big[-xe^{-x/2}\big]_0^3 + \int_0^3 e^{-x/2}\,\mathrm{d}x = -3e^{-3/2} + \big[-2e^{-x/2}\big]_0^3 = -3e^{-3/2} - 2e^{-3/2} + 2 = 2 - 5e^{-3/2}$. The second is $3\big[-e^{-x/2}\big]_3^{\infty} = 3e^{-3/2}$.

Step 3: add.

$$\mathbb{E}\big(g(X)\big) = \big(2 - 5e^{-3/2}\big) + 3e^{-3/2} = 2 - 2e^{-3/2} \approx 2 - 2(0.2231) = 1.554.$$

The cap pulls the expected payout below the uncapped mean of 2 (thousand), as it should.

Example 3.4.15 (Expected dollars from a lottery ticket)**.** A scratch ticket reveals a symbol $X \in \{A, B, C\}$ with $\mathbb{P}(A) = 0.7$, $\mathbb{P}(B) = 0.25$, $\mathbb{P}(C) = 0.05$, paying $g(A) = 0$, $g(B) = 10$, $g(C) = 100$ dollars. The ticket costs \$8. Find the expected net gain.

Which rule, and why? The prize is a function g of the symbol drawn, so LOTUS gives $\mathbb{E}\big(g(X)\big)$; the fixed \$8 cost is then handled by linearity ($\mathbb{E}(g(X) - 8) = \mathbb{E}(g(X)) - 8$).

Solution.

Step 1: expected prize via LOTUS.

$$\mathbb{E}\big(g(X)\big) = 0(0.7) + 10(0.25) + 100(0.05) = 0 + 2.5 + 5 = 7.5.$$

Step 2: subtract the cost by linearity.

$$\mathbb{E}(\text{net}) = \mathbb{E}\big(g(X)\big) - 8 = 7.5 - 8 = -0.5 \text{ dollars}.$$

On average each ticket loses 50 cents.

3.4.4 Linearity of expectation

Theorem 3.4.16 (Linearity)**.** *For constants a, b and any random variable X with finite mean,*

$$\mathbb{E}(aX + b) = a\,\mathbb{E}(X) + b.$$

More generally, for any random variables $X_1, \ldots, X_n$ with finite means,

$$\mathbb{E}(X_1 + \cdots + X_n) = \mathbb{E}(X_1) + \cdots + \mathbb{E}(X_n),$$

regardless of whether the X_i are independent.

Proof. Take the continuous case of the first identity; the discrete case is identical with sums. By LOTUS with $g(x) = ax + b$,

$$\mathbb{E}(aX + b) = \int (ax + b) f_X(x)\,\mathrm{d}x = a \int x f_X(x)\,\mathrm{d}x + b \int f_X(x)\,\mathrm{d}x = a\,\mathbb{E}(X) + b,$$

using $\int f_X = 1$. The additivity over several variables follows by writing the expectation as an integral against the joint density and splitting the sum term by term; no factorisation (hence no independence) is used. □

Example 3.4.17 (Fahrenheit from Celsius)**.** A reactor temperature C has $\mathbb{E}(C) = 80$ degrees Celsius. The Fahrenheit reading is $F = \frac{9}{5}C + 32$. Find $\mathbb{E}(F)$.

Which rule, and why? F is an affine function $aC + b$ of C, so linearity applies directly: $\mathbb{E}(aC + b) = a\mathbb{E}(C) + b$. No distribution of C is needed — only its mean.

Solution.

Step 1: apply $\mathbb{E}(aC + b) = a\mathbb{E}(C) + b$.

$$\mathbb{E}(F) = \tfrac{9}{5}\,\mathbb{E}(C) + 32 = \tfrac{9}{5}(80) + 32.$$

Step 2: arithmetic.

$$\mathbb{E}(F) = 144 + 32 = 176 \text{ degrees Fahrenheit}.$$

Example 3.4.18 (Total of three indicators)**.** Three independent components fail during a test with probabilities 0.2, 0.3, 0.5. Let N be the number that fail. Find $\mathbb{E}(N)$ using indicators.

Which rule, and why? Write $N = \mathbb{1}_{F_1} + \mathbb{1}_{F_2} + \mathbb{1}_{F_3}$ as a sum of indicator variables and use additivity of expectation. Linearity needs no independence, which makes the indicator trick clean and general.

Solution.

Step 1: indicators and their means. Let $\mathbb{1}_{F_i}$ be 1 if component i fails. Then $\mathbb{E}(\mathbb{1}_{F_i}) = \mathbb{P}(\text{fail}_i) = p_i$, so $\mathbb{E}(\mathbb{1}_{F_1}) = 0.2$, $\mathbb{E}(\mathbb{1}_{F_2}) = 0.3$, $\mathbb{E}(\mathbb{1}_{F_3}) = 0.5$.

Step 2: add the means.

$$\mathbb{E}(N) = \mathbb{E}(\mathbb{1}_{F_1}) + \mathbb{E}(\mathbb{1}_{F_2}) + \mathbb{E}(\mathbb{1}_{F_3}) = 0.2 + 0.3 + 0.5 = 1.0.$$

Exactly one failure on average.

Example 3.4.19 (Net revenue)**.** Daily demand D at a kiosk has $\mathbb{E}(D) = 140$ units. Each unit sells for \$3 and the kiosk has a fixed daily cost of \$120. Net revenue is $R = 3D - 120$. Find $\mathbb{E}(R)$.

Which rule, and why? $R = aD + b$ with $a = 3$, $b = -120$: an affine transform, so linearity gives $\mathbb{E}(R) = 3\mathbb{E}(D) - 120$ from the mean of D alone.

Solution.

Step 1: apply linearity.

$$\mathbb{E}(R) = 3\,\mathbb{E}(D) - 120 = 3(140) - 120.$$

Step 2: simplify.

$$\mathbb{E}(R) = 420 - 120 = 300 \text{ dollars per day.}$$

Example 3.4.20 (Expected sum of dice, by linearity)**.** Roll two fair dice; let $S = X_1 + X_2$ be their sum. Find $\mathbb{E}(S)$ without the triangular pmf.

Which rule, and why? S is a sum of two variables whose means are easy, so additivity gives $\mathbb{E}(S) = \mathbb{E}(X_1) + \mathbb{E}(X_2)$ instantly — far cheaper than summing the 11-value pmf of S.

Solution.

Step 1: mean of one die. $\mathbb{E}(X_i) = \frac{1}{6}(1 + 2 + 3 + 4 + 5 + 6) = \frac{21}{6} = 3.5$.

Step 2: add.

$$\mathbb{E}(S) = \mathbb{E}(X_1) + \mathbb{E}(X_2) = 3.5 + 3.5 = 7,$$

matching the symmetry computation earlier.

3.4.5 Means of the named distributions

The following means are used constantly; quote them rather than re-summing. Each can be derived by the definition above (binomial via indicators; the others by the matching sum or integral).

Distribution	Parameters	$\mathbb{E}(X)$
$X \sim \text{Bin}(n, p)$	n trials, success prob. p	np
$X \sim \text{Poisson}(\lambda)$	rate $\lambda > 0$	λ
$X \sim \text{Geom}(p)$	success prob. p (trials to 1st success)	$1/p$
$X \sim \text{Uniform}(a, b)$	on $[a, b]$	$(a + b)/2$
$X \sim \text{Exp}(\lambda)$	rate $\lambda > 0$	$1/\lambda$
$X \sim \text{Gamma}(\alpha, \lambda)$	shape α, rate λ	α/λ

Proposition 3.4.21 (Binomial mean by indicators)**.** *If $X \sim \text{Bin}(n, p)$ then $\mathbb{E}(X) = np$.*

Proof. Write $X = \mathbb{1}_1 + \cdots + \mathbb{1}_n$, where $\mathbb{1}_i = 1$ if trial i is a success. Each $\mathbb{E}(\mathbb{1}_i) = p$, so by linearity $\mathbb{E}(X) = \sum_{i=1}^{n} p = np$. (No independence is needed for the mean, only that each trial has success probability p.) □

Proposition 3.4.22 (Poisson mean)**.** *If* $X \sim \text{Poisson}(\lambda)$ *then* $\mathbb{E}(X) = \lambda$.

Proof. With $p_X(k) = e^{-\lambda}\lambda^k/k!$,

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k\, e^{-\lambda}\frac{\lambda^k}{k!} = \sum_{k=1}^{\infty} e^{-\lambda}\frac{\lambda^k}{(k-1)!} = \lambda e^{-\lambda}\sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = \lambda e^{-\lambda}e^{\lambda} = \lambda,$$

where $j = k - 1$ and $\sum_j \lambda^j/j! = e^{\lambda}$. □

Example 3.4.23 (Binomial: defective chips)**.** A machine produces chips independently with defect probability $p = 0.03$. In a box of $n = 500$ chips let X be the number defective. Find $\mathbb{E}(X)$.

Which rule, and why? Fixed n independent yes/no trials with constant p is the binomial setting, so $X \sim \text{Bin}(500, 0.03)$ and $\mathbb{E}(X) = np$ — quote it, do not sum 501 terms.

Solution.

Step 1: identify n and p. $n = 500$, $p = 0.03$.

Step 2: apply $\mathbb{E}(X) = np$.

$$\mathbb{E}(X) = 500(0.03) = 15 \text{ defective chips.}$$

Example 3.4.24 (Poisson: arrivals at a help desk)**.** Tickets arrive at a help desk as $X \sim \text{Poisson}(\lambda)$ with $\lambda = 6$ per hour. Find the expected number in one hour, and in a 20-minute window.

Which rule, and why? Counts of rare independent events at a constant rate are Poisson, so $\mathbb{E}(X) = \lambda$. The shorter window scales the rate, then linearity is not even needed — just the new λ.

Solution.

Step 1: one-hour mean. $\mathbb{E}(X) = \lambda = 6$ tickets.

Step 2: scale the rate to 20 minutes $= \frac{1}{3}$ hour. The window has $\lambda' = 6 \cdot \frac{1}{3} = 2$, so

$$\mathbb{E}(X_{20\text{ min}}) = 2 \text{ tickets.}$$

Example 3.4.25 (Geometric: attempts until success)**.** A salesperson closes a call with probability $p = 0.2$, independently per call. Let X be the number of calls up to and including the first close, $X \sim \text{Geom}(0.2)$. Find $\mathbb{E}(X)$.

Which rule, and why? "Number of independent trials until the first success" is geometric (support $1, 2, 3, \ldots$), so $\mathbb{E}(X) = 1/p$. We use the convention counting the successful trial itself.

Solution.

Step 1: identify p. $p = 0.2$.

Step 2: apply $\mathbb{E}(X) = 1/p$.

$$\mathbb{E}(X) = \tfrac{1}{0.2} = 5 \text{ calls on average.}$$

Example 3.4.26 (Gamma: total service time)**.** The time to serve $\alpha = 3$ customers, each $\text{Exp}(\lambda)$ with $\lambda = 2$ per minute and independent, is $X \sim \text{Gamma}(3, 2)$. Find $\mathbb{E}(X)$.

Which rule, and why? A sum of α independent rate-λ exponentials is $\text{Gamma}(\alpha, \lambda)$, with $\mathbb{E}(X) = \alpha/\lambda$. Equivalently, additivity gives $\mathbb{E}(X) = \alpha \cdot (1/\lambda)$ — the same number, which is a good consistency check.

Solution.

Step 1: apply $\mathbb{E}(X) = \alpha/\lambda$.

$$\mathbb{E}(X) = \tfrac{\alpha}{\lambda} = \tfrac{3}{2} = 1.5 \text{ minutes.}$$

Step 2: cross-check by additivity. Three independent $\text{Exp}(2)$ service times each have mean $\frac{1}{2}$, and $3 \cdot \frac{1}{2} = 1.5$, confirming the gamma mean.

3.5 The Variance

The expected value pins down where a random variable sits on average, but it says nothing about how widely the variable scatters around that center. Two machines can each produce parts of mean length 10 cm and yet one be reliably near 10 while the other swings between 5 and 15. The **variance** measures that scatter: it is the average squared distance from the mean. Its square root, the **standard deviation**, restores the original units and is the everyday measure of spread. This section defines the variance, proves the algebraic shortcut that makes it computable, records how it behaves under shifting and rescaling, tabulates the variances of the named distributions, and closes with Chebyshev's inequality, which converts a variance into a guaranteed bound on probability.

Which variance rule does the problem need?

- Asked for **spread, scatter, variability, or risk** about the mean? ⇒ compute the **variance** $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$. Almost always use the **shortcut formula**: get $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$, then subtract.
- Want the answer in the **original units** (cm, dollars, minutes)? ⇒ report the **standard deviation** $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$, not the variance.
- The variable is **shifted or rescaled**, $Y = aX + b$ (change of units, recentering)? ⇒ use $\mathrm{Var}(aX + b) = a^2 \mathrm{Var}(X)$. The additive constant b *drops out*; only the multiplier a matters, and it enters *squared*.
- The variable is one of the **named distributions** (Bin, Poisson, Geom, Uniform, Exp, N, ...)? ⇒ quote its variance from the table rather than re-integrating.
- Need a **probability bound** from only the mean and variance, with the distribution otherwise unknown? ⇒ **Chebyshev's inequality**.

Why not just use the mean? The mean alone cannot distinguish a steady process from a volatile one; the deviation $X - \mu$ averages to zero by definition, so we square it before averaging. *Why square rather than take absolute value?* Squaring is analytically tractable (it expands, differentiates, and adds nicely under independence), which is exactly why the variance, not the mean absolute deviation, is the standard measure.

3.5.1 Definition and the shortcut formula

Definition 3.5.1 (Variance and standard deviation)**.** Let X be a random variable with finite mean $\mu = \mathbb{E}(X)$. The **variance** of X is the expected squared deviation from the mean,

$$\mathrm{Var}(X) = \mathbb{E}\big[(X - \mu)^2\big],$$

provided this expectation is finite. The **standard deviation** is $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$. Variance is measured in the square of the units of X; the standard deviation is in the same units as X.

Because $(X - \mu)^2 \geq 0$, the variance is always nonnegative, and $\mathrm{Var}(X) = 0$ exactly when X equals the constant μ with probability 1. Computing the variance straight from the definition means evaluating $\mathbb{E}[(X - \mu)^2]$, which for a discrete variable is $\sum_k (k - \mu)^2 \, p_X(k)$ and for a continuous one is $\int (x - \mu)^2 f_X(x) \, \mathrm{d}x$. The next result gives a faster route.

Theorem 3.5.2 (Shortcut formula)**.** *For any random variable X with finite mean $\mu = \mathbb{E}(X)$ and finite second moment $\mathbb{E}(X^2)$,*

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \mathbb{E}(X^2) - \mu^2.$$

Proof. Expand the square inside the expectation and use linearity, remembering that μ is a constant:

$$\mathbb{E}[(X - \mu)^2] = \mathbb{E}[X^2 - 2\mu X + \mu^2] = \mathbb{E}(X^2) - 2\mu\,\mathbb{E}(X) + \mu^2 = \mathbb{E}(X^2) - 2\mu^2 + \mu^2 = \mathbb{E}(X^2) - \mu^2. \qquad \square$$

The shortcut is almost always the practical method: find $\mathbb{E}(X)$, then find $\mathbb{E}(X^2)$ by the law of the unconscious statistician (averaging x^2 against the pmf or density), and subtract the square of the first from the second. The two quantities $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$ are the first two **moments**; the variance is the gap between the second moment and the squared first moment.

Example 3.5.3 (Spread of a three-sided spinner)**.** A board game uses a spinner that lands on 1, 2, or 5 points with probabilities $p_X(1) = 0.5$, $p_X(2) = 0.3$, $p_X(5) = 0.2$. Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

Which rule, and why? The variable takes a few *isolated values* with listed probabilities, and we want its spread; this is a discrete variance. Use the shortcut $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2$ rather than the definition, since both moments are quick sums.

Solution.

Step 1: first moment.

$$\mathbb{E}(X) = 1(0.5) + 2(0.3) + 5(0.2) = 0.5 + 0.6 + 1.0 = 2.1.$$

Step 2: second moment. Average k^2 against the same pmf:

$$\mathbb{E}(X^2) = 1^2(0.5) + 2^2(0.3) + 5^2(0.2) = 0.5 + 1.2 + 5.0 = 6.7.$$

Step 3: subtract.

$$\mathrm{Var}(X) = 6.7 - (2.1)^2 = 6.7 - 4.41 = 2.29, \qquad \mathrm{SD}(X) = \sqrt{2.29} \approx 1.513.$$

Example 3.5.4 (A fair die)**.** Let X be the score of one roll of a fair six-sided die, so $p_X(k) = \frac{1}{6}$ for $k = 1, \ldots, 6$. Compute $\mathrm{Var}(X)$.

Which rule, and why? Equally likely isolated values; a discrete variance via the shortcut. The closed forms $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ and $\sum_{k=1}^{n} k^2 = \frac{n(n+1)(2n+1)}{6}$ make the two moments exact.

Solution.

Step 1: first moment.

$$\mathbb{E}(X) = \frac{1}{6}\sum_{k=1}^{6} k = \frac{1}{6} \cdot \frac{6 \cdot 7}{2} = \frac{21}{6} = \frac{7}{2} = 3.5.$$

Step 2: second moment.

$$\mathbb{E}(X^2) = \frac{1}{6}\sum_{k=1}^{6} k^2 = \frac{1}{6} \cdot \frac{6 \cdot 7 \cdot 13}{6} = \frac{91}{6}.$$

Step 3: subtract.

$$\mathrm{Var}(X) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = \frac{91}{6} - \frac{49}{4} = \frac{182 - 147}{12} = \frac{35}{12} \approx 2.917.$$

Example 3.5.5 (A continuous variance by integration)**.** A sensor's reading X has density $f_X(x) = 2x$ for $0 \le x \le 1$ and 0 otherwise. Find the variance and standard deviation.

Which rule, and why? The variable lives on a continuum with a given density, so moments are integrals $\mathbb{E}(X^k) = \int x^k f_X(x)\,\mathrm{d}x$; then apply the shortcut. The definition-based integral $\int (x-\mu)^2 f$ would also work but is messier.

Solution.

Step 1: first moment.

$$\mathbb{E}(X) = \int_0^1 x \cdot 2x\,\mathrm{d}x = \int_0^1 2x^2\,\mathrm{d}x = \left[\tfrac{2}{3}x^3\right]_0^1 = \frac{2}{3}.$$

Step 2: second moment.

$$\mathbb{E}(X^2) = \int_0^1 x^2 \cdot 2x\,\mathrm{d}x = \int_0^1 2x^3\,\mathrm{d}x = \left[\tfrac{1}{2}x^4\right]_0^1 = \frac{1}{2}.$$

Step 3: subtract.

$$\mathrm{Var}(X) = \frac{1}{2} - \left(\frac{2}{3}\right)^2 = \frac{1}{2} - \frac{4}{9} = \frac{9-8}{18} = \frac{1}{18} \approx 0.0556, \qquad \mathrm{SD}(X) = \frac{1}{\sqrt{18}} \approx 0.236.$$

Example 3.5.6 (Number of defective chips in a small batch)**.** A technician inspects 3 chips. Let X count the defectives. From past data $p_X(0) = 0.70$, $p_X(1) = 0.21$, $p_X(2) = 0.06$, $p_X(3) = 0.03$. Find the standard deviation.

Which rule, and why? A discrete variable with a full pmf; we want spread in the original unit "defectives," so compute the variance by the shortcut and then take the square root for the standard deviation.

Solution.

Step 1: check and take the first moment. The probabilities sum to $0.70 + 0.21 + 0.06 + 0.03 = 1$. Then

$$\mathbb{E}(X) = 0(0.70) + 1(0.21) + 2(0.06) + 3(0.03) = 0 + 0.21 + 0.12 + 0.09 = 0.42.$$

Step 2: second moment.

$$\mathbb{E}(X^2) = 0^2(0.70) + 1^2(0.21) + 2^2(0.06) + 3^2(0.03) = 0 + 0.21 + 0.24 + 0.27 = 0.72.$$

Step 3: subtract and root.

$$\mathrm{Var}(X) = 0.72 - (0.42)^2 = 0.72 - 0.1764 = 0.5436, \qquad \mathrm{SD}(X) = \sqrt{0.5436} \approx 0.737.$$

Example 3.5.7 (Variance of an exponential waiting time)**.** The time X (in minutes) until the next email arrives is $X \sim \mathrm{Exp}(\lambda)$ with $\lambda = \frac{1}{2}$, density $f_X(x) = \frac{1}{2}e^{-x/2}$ for $x \geq 0$. Find $\mathrm{Var}(X)$ directly from the moments.

Which rule, and why? Continuous variable on $[0, \infty)$ with an exponential density; the moments are gamma-type integrals. We confirm the table entry $\mathrm{Var}(X) = 1/\lambda^2$ by direct calculation using $\int_0^\infty x^n e^{-x/2}\,\mathrm{d}x$.

Solution.

Step 1: first moment. With the substitution $u = x/2$, or by the standard result $\int_0^\infty x^n e^{-\lambda x}\,\mathrm{d}x = n!/\lambda^{n+1}$,

$$\mathbb{E}(X) = \int_0^\infty x \cdot \tfrac{1}{2}e^{-x/2}\,\mathrm{d}x = \tfrac{1}{2} \cdot \frac{1!}{(1/2)^2} = \tfrac{1}{2} \cdot 4 = 2 = \frac{1}{\lambda}.$$

Step 2: second moment.

$$\mathbb{E}(X^2) = \int_0^\infty x^2 \cdot \tfrac{1}{2}e^{-x/2}\,\mathrm{d}x = \tfrac{1}{2} \cdot \frac{2!}{(1/2)^3} = \tfrac{1}{2} \cdot 16 = 8 = \frac{2}{\lambda^2}.$$

Step 3: subtract.

$$\mathrm{Var}(X) = 8 - 2^2 = 8 - 4 = 4 = \frac{1}{\lambda^2}.$$

So $\mathrm{SD}(X) = 2$ minutes, equal to the mean — a signature feature of the exponential.

3.5.2 Standard deviation and the units of spread

The variance is in squared units, which is awkward to interpret: a variance of "4 square minutes" has no everyday meaning. The standard deviation $\mathrm{SD}(X) = \sqrt{\mathrm{Var}(X)}$ returns to the variable's own scale and answers "by how much does a typical observation miss the mean?" It is the number to report alongside $\mathbb{E}(X)$ in any summary.

Example 3.5.8 (Comparing the steadiness of two suppliers)**.** Two suppliers deliver bolts whose lengths (mm) have the same mean 20. Supplier A has Var = 0.09; supplier B has Var = 0.64. Which is more consistent, and by how much in real units?

Which rule, and why? Both have equal means, so the mean cannot rank them; consistency is a question of spread. Convert each variance to a standard deviation to compare in millimeters.

Solution.

Step 1: standard deviations.

$$\mathrm{SD}_A = \sqrt{0.09} = 0.30 \text{ mm}, \qquad \mathrm{SD}_B = \sqrt{0.64} = 0.80 \text{ mm}.$$

Step 2: interpret. A typical A-bolt misses 20 mm by about 0.30 mm, while a typical B-bolt misses by about 0.80 mm — more than twice as much. Supplier A is the more consistent. Note how the variance ratio $0.64/0.09 \approx 7.1$ exaggerates the gap; the honest comparison is the standard-deviation ratio $0.80/0.30 \approx 2.7$.

3.5.3 Variance under shifting and scaling

Adding a constant slides the whole distribution without changing its shape, so it cannot change the spread; multiplying by a constant stretches every deviation, and since deviations enter squared, the variance picks up the square of the multiplier.

Theorem 3.5.9 (Affine transformation)**.** *For constants a, b and any random variable X with finite variance,*

$$\mathrm{Var}(aX+b) = a^2\ \mathrm{Var}(X), \qquad \textit{equivalently} \qquad \mathrm{SD}(aX+b) = |a|\ \mathrm{SD}(X).$$

In particular $\mathrm{Var}(X+b) = \mathrm{Var}(X)$ *and* $\mathrm{Var}(aX) = a^2\,\mathrm{Var}(X)$.

Proof. Let $Y = aX + b$. By linearity of expectation $\mathbb{E}(Y) = a\mu + b$, so the deviation is $Y - \mathbb{E}(Y) = aX + b - (a\mu + b) = a(X - \mu)$. Therefore

$$\mathrm{Var}(Y) = \mathbb{E}\big[(Y - \mathbb{E}(Y))^2\big] = \mathbb{E}\big[a^2(X-\mu)^2\big] = a^2\,\mathbb{E}[(X-\mu)^2] = a^2\,\mathrm{Var}(X).$$

Taking square roots gives $\mathrm{SD}(Y) = |a|\,\mathrm{SD}(X)$. □

The additive constant b vanishing is the key practical point: recentering data leaves the spread untouched. The squared multiplier is the second: doubling a variable quadruples its variance.

Example 3.5.10 (Changing temperature units)**.** Daily high temperatures C (in degrees Celsius) at a site have $\mathbb{E}(C) = 18$ and $\mathrm{Var}(C) = 16$. The Fahrenheit reading is $F = \frac{9}{5}C + 32$. Find $\mathrm{Var}(F)$ and $\mathrm{SD}(F)$.

Which rule, and why? F is an affine function $aC + b$ of C with $a = \frac{9}{5}$, $b = 32$. Use $\mathrm{Var}(aX+b) = a^2\,\mathrm{Var}(X)$; the additive 32 drops out and only the multiplier matters.

Solution.

Step 1: apply the rule.

$$\mathrm{Var}(F) = \left(\tfrac{9}{5}\right)^2 \mathrm{Var}(C) = \frac{81}{25}\cdot 16 = \frac{1296}{25} = 51.84.$$

Step 2: standard deviation.

$$\mathrm{SD}(F) = \left|\tfrac{9}{5}\right|\ \mathrm{SD}(C) = \tfrac{9}{5}\cdot 4 = 7.2\ ^\circ\mathrm{F}.$$

The mean shifts to $\frac{9}{5}(18) + 32 = 64.4$, but the $+32$ has no effect on the spread.

Example 3.5.11 (Markup applied to a random cost)**.** A workshop's raw material cost X per order has $\mathbb{E}(X) = \$40$ and $\mathrm{SD}(X) = \$6$. It charges the customer $P = 1.5X + 10$ dollars (a 50% markup plus a fixed \$10 handling fee). Find the standard deviation of the price.

Which rule, and why? Price is affine in cost, $P = aX + b$ with $a = 1.5$, $b = 10$. The handling fee is a constant shift and cannot affect spread; scale the standard deviation by $|a|$.

Solution.

Step 1: variance of the cost. $\mathrm{Var}(X) = \mathrm{SD}(X)^2 = 6^2 = 36$.

Step 2: scale.

$$\mathrm{Var}(P) = 1.5^2 \cdot 36 = 2.25 \cdot 36 = 81, \qquad \mathrm{SD}(P) = \sqrt{81} = \$9.$$

Equivalently $\mathrm{SD}(P) = |1.5| \cdot 6 = \$9$; the \$10 fee is irrelevant to the spread.

Example 3.5.12 (Standardizing a variable)**.** Let X have mean $\mu = 50$ and standard deviation $\sigma = 8$. Define the standardized variable $Z = \dfrac{X-\mu}{\sigma} = \dfrac{X-50}{8}$. Show that $\mathbb{E}(Z) = 0$ and $\mathrm{Var}(Z) = 1$.

Which rule, and why? Z is affine in X, namely $Z = aX + b$ with $a = \frac{1}{8}$ and $b = -\frac{50}{8}$. Use linearity for the mean and the scaling rule for the variance; this is the construction behind every "z-score."

Solution.

Step 1: mean. $\mathbb{E}(Z) = \dfrac{\mathbb{E}(X) - 50}{8} = \dfrac{50 - 50}{8} = 0.$

Step 2: variance. With $a = \frac{1}{8}$,

$$\operatorname{Var}(Z) = \left(\tfrac{1}{8}\right)^2 \operatorname{Var}(X) = \frac{1}{64} \cdot 8^2 = \frac{64}{64} = 1.$$

So standardizing always yields mean 0 and variance 1, regardless of the original scale.

Example 3.5.13 (Negating and recentering)**.** A variable X has $\operatorname{Var}(X) = 5$. Let $Y = 7 - 2X$. Find $\operatorname{Var}(Y)$.

Which rule, and why? $Y = aX + b$ with $a = -2$, $b = 7$. The negative multiplier is squared away and the constant 7 drops out; do not let the minus sign or the shift mislead you.

Solution.

$$\operatorname{Var}(Y) = (-2)^2 \operatorname{Var}(X) = 4 \cdot 5 = 20.$$

The sign of a never affects the variance, since a appears squared.

3.5.4 Variances of the named distributions

For the standard families the variance is known in closed form; quote these rather than re-deriving. The table collects the means alongside for reference.

Distribution	Parameters	Mean $\mathbb{E}(X)$	Variance $\operatorname{Var}(X)$
$\operatorname{Bin}(n, p)$	$n \in \mathbb{N},\ 0 \le p \le 1$	np	$np(1-p)$
$\operatorname{Geom}(p)$	$0 < p \le 1$ (trials to first success)	$1/p$	$(1-p)/p^2$
$\operatorname{Poisson}(\lambda)$	$\lambda > 0$	λ	λ
$\operatorname{Uniform}(a, b)$	$a < b$	$(a+b)/2$	$(b-a)^2/12$
$\operatorname{Exp}(\lambda)$	$\lambda > 0$	$1/\lambda$	$1/\lambda^2$
$\mathrm{N}(\mu, \sigma^2)$	$\sigma > 0$	μ	σ^2

Two of these deserve a short derivation, since they show the shortcut formula at work; the others follow the same pattern.

Proposition 3.5.14 (Variance of the uniform)**.** *If* $X \sim \operatorname{Uniform}(a, b)$ *with density* $f_X(x) = \frac{1}{b-a}$ *on* $[a, b]$, *then* $\operatorname{Var}(X) = \dfrac{(b-a)^2}{12}$.

Proof. The mean is $\mu = (a+b)/2$. For the second moment,

$$\mathbb{E}(X^2) = \int_a^b \frac{x^2}{b-a}\,\mathrm{d}x = \frac{1}{b-a} \cdot \frac{b^3 - a^3}{3} = \frac{b^2 + ab + a^2}{3},$$

using $b^3 - a^3 = (b-a)(b^2 + ab + a^2)$. Then

$$\begin{aligned} \operatorname{Var}(X) &= \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{4(a^2 + ab + b^2) - 3(a^2 + 2ab + b^2)}{12} \\ &= \frac{a^2 - 2ab + b^2}{12} = \frac{(b-a)^2}{12}. \end{aligned}$$

□

Proposition 3.5.15 (Variance of the Poisson)**.** *If* $X \sim \operatorname{Poisson}(\lambda)$ *with* $p_X(k) = e^{-\lambda}\lambda^k/k!$ *for* $k = 0, 1, 2, \ldots$, *then* $\operatorname{Var}(X) = \lambda$.

Proof. Use the **factorial moment** $\mathbb{E}[X(X-1)]$, which is easy for the Poisson:

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\, e^{-\lambda}\frac{\lambda^k}{k!} = \lambda^2 e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda}e^{\lambda} = \lambda^2.$$

Since $\mathbb{E}[X(X-1)] = \mathbb{E}(X^2) - \mathbb{E}(X)$ and $\mathbb{E}(X) = \lambda$, we get $\mathbb{E}(X^2) = \lambda^2 + \lambda$. Therefore

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda. \qquad \square$$

Example 3.5.16 (Variance of a binomial count)**.** A free-throw shooter makes each shot independently with probability $p = 0.8$ and takes $n = 15$ shots. Let X be the number made. Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

Which rule, and why? Fixed number of independent success/failure trials with constant p: this is $\mathrm{Bin}(n, p)$. Quote $\mathrm{Var}(X) = np(1-p)$ from the table rather than summing over all 16 outcomes.

Solution.

Step 1: apply the formula.

$$\mathrm{Var}(X) = np(1-p) = 15(0.8)(0.2) = 2.4.$$

Step 2: standard deviation.

$$\mathrm{SD}(X) = \sqrt{2.4} \approx 1.549 \text{ made shots.}$$

For reference the mean is $np = 12$, so the count typically lands within about 1.5 shots of 12.

Example 3.5.17 (Variance of a Poisson count)**.** Calls to a help desk arrive as $X \sim \mathrm{Poisson}(\lambda)$ with $\lambda = 6$ per hour. Give the variance and standard deviation of the hourly count.

Which rule, and why? Counts of independent events in a fixed interval at a constant rate: a Poisson variable. Its defining feature is mean equal to variance, both λ.

Solution.

$$\mathrm{Var}(X) = \lambda = 6, \qquad \mathrm{SD}(X) = \sqrt{6} \approx 2.449 \text{ calls.}$$

Thus a typical hour sees about 6 calls, give or take roughly 2.4.

Example 3.5.18 (Variance of a uniform service time)**.** A clerk's service time X is uniform on $[2, 8]$ minutes, $X \sim \mathrm{Uniform}(2, 8)$. Find the variance, standard deviation, and verify by the moment integrals.

Which rule, and why? Equally likely values across a continuous interval: a uniform variable. Use $\mathrm{Var} = (b-a)^2/12$, then cross-check with the direct integrals to build confidence.

Solution.

Step 1: table value.

$$\mathrm{Var}(X) = \frac{(8-2)^2}{12} = \frac{36}{12} = 3, \qquad \mathrm{SD}(X) = \sqrt{3} \approx 1.732.$$

Step 2: cross-check. The density is $f_X(x) = \frac{1}{6}$ on $[2, 8]$, so $\mathbb{E}(X) = (2+8)/2 = 5$ and

$$\mathbb{E}(X^2) = \int_2^8 \frac{x^2}{6}\,\mathrm{d}x = \frac{1}{6}\Big[\frac{x^3}{3}\Big]_2^8 = \frac{1}{18}(512 - 8) = \frac{504}{18} = 28.$$

Then $\mathrm{Var}(X) = 28 - 5^2 = 28 - 25 = 3$, matching the table.

Example 3.5.19 (Variance of a geometric count)**.** A quality line tests items until the first defective appears; each item is defective independently with probability $p = 0.1$. Let X be the number of items tested up to and including the first defective, $X \sim \mathrm{Geom}(0.1)$. Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

Which rule, and why? Number of independent trials *up to and including* the first success: the geometric distribution (support $1, 2, 3, \dots$), with variance $(1-p)/p^2$.

Solution.

Step 1: apply the formula.

$$\mathrm{Var}(X) = \frac{1-p}{p^2} = \frac{0.9}{0.01} = 90.$$

Step 2: standard deviation.

$$\mathrm{SD}(X) = \sqrt{90} \approx 9.49 \text{ items.}$$

The mean is $1/p = 10$; the large standard deviation reflects the long right tail of the geometric.

3.5.5 The shortcut formula in practice

When a problem hands you a pmf or density with no named family, the workflow is mechanical: build a small table of k, $p_X(k)$, $k\,p_X(k)$, $k^2 p_X(k)$ (or the two integrals), sum the last two columns to get $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$, and subtract. A diagram of the pmf with the mean marked makes the spread visible.

Example 3.5.20 (A worked pmf table)**.** A small parking lot has X cars waiting to exit, with the pmf below. Compute $\mathbb{E}(X)$, $\mathbb{E}(X^2)$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$.

k	0	1	2	3	4
$p_X(k)$	0.15	0.30	0.30	0.20	0.05

Which rule, and why? A bare pmf with several values and no named family: the table method for the shortcut formula. Accumulate $k\,p_X(k)$ and $k^2 p_X(k)$, then subtract the squared mean.

Solution.

Step 1: tabulate the moment contributions.

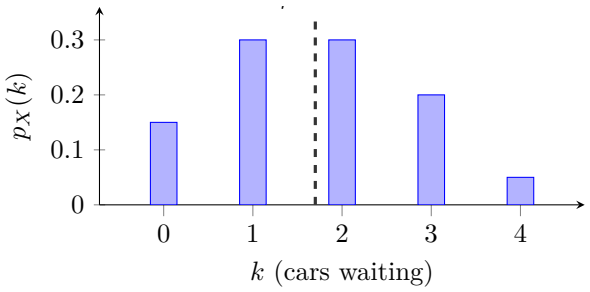
k	0	1	2	3	4	sum
$p_X(k)$	0.15	0.30	0.30	0.20	0.05	1.00
$k\,p_X(k)$	0	0.30	0.60	0.60	0.20	1.70
$k^2 p_X(k)$	0	0.30	1.20	1.80	0.80	4.10

Step 2: read off the moments. $\mathbb{E}(X) = 1.70$ and $\mathbb{E}(X^2) = 4.10$.

Step 3: subtract.

$$\mathrm{Var}(X) = 4.10 - (1.70)^2 = 4.10 - 2.89 = 1.21, \qquad \mathrm{SD}(X) = \sqrt{1.21} = 1.1.$$

A picture of this pmf, with the mean $\mu = 1.70$ marked, shows where the mass sits and how far it typically strays:



Example 3.5.21 (Shortcut on a continuous density)**.** A component's standardized wear X has density $f_X(x) = \frac{3}{2}x^2$ for $-1 \le x \le 1$. Find $\mathrm{Var}(X)$.

Which rule, and why? Continuous variable with a symmetric density on $[-1, 1]$; compute the moments by integration and subtract. Symmetry about 0 will make the mean vanish, but we verify it rather than assume it.

Solution.

Step 1: confirm it is a density and find the mean. It is nonnegative, and $\int_{-1}^{1} \frac{3}{2}x^2 \, \mathrm{d}x = \frac{3}{2} \cdot \frac{2}{3} = 1$. The integrand $x \cdot \frac{3}{2}x^2 = \frac{3}{2}x^3$ is odd, so

$$\mathbb{E}(X) = \int_{-1}^{1} \tfrac{3}{2}x^3 \, \mathrm{d}x = 0.$$

Step 2: second moment. The integrand $x^2 \cdot \frac{3}{2}x^2 = \frac{3}{2}x^4$ is even:

$$\mathbb{E}(X^2) = \int_{-1}^{1} \tfrac{3}{2}x^4 \, \mathrm{d}x = 2 \cdot \tfrac{3}{2} \int_{0}^{1} x^4 \, \mathrm{d}x = 3 \cdot \frac{1}{5} = \frac{3}{5}.$$

Step 3: subtract.

$$\mathrm{Var}(X) = \frac{3}{5} - 0^2 = \frac{3}{5} = 0.6, \qquad \mathrm{SD}(X) = \sqrt{0.6} \approx 0.775.$$

Example 3.5.22 (Spread of a discrete payoff with a rare large value)**.** A scratch ticket pays \$0 with probability 0.90, \$5 with probability 0.09, and \$100 with probability 0.01. Let X be the payoff. Find $\mathbb{E}(X)$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$, and comment on why the spread is large.

Which rule, and why? A discrete payoff with a small but extreme value; use the shortcut. The rare \$100 contributes little to the mean but dominates $\mathbb{E}(X^2)$, which is exactly how a heavy tail inflates variance.

Solution.

Step 1: first moment.

$$\mathbb{E}(X) = 0(0.90) + 5(0.09) + 100(0.01) = 0 + 0.45 + 1.00 = 1.45.$$

Step 2: second moment.

$$\mathbb{E}(X^2) = 0^2(0.90) + 5^2(0.09) + 100^2(0.01) = 0 + 2.25 + 100 = 102.25.$$

Step 3: subtract.

$$\mathrm{Var}(X) = 102.25 - (1.45)^2 = 102.25 - 2.1025 = 100.1475, \qquad \mathrm{SD}(X) \approx \$10.01.$$

The \$100 outcome adds only \$1 to the mean but 100 to $\mathbb{E}(X^2)$, so the standard deviation ($\approx$ \$10) towers over the mean (\$1.45): rare large values are what make spread big.

3.5.6 Chebyshev's inequality

If only the mean and variance are known — not the full distribution — the variance still forces a limit on how often a variable can stray far from its mean. This is Chebyshev's inequality, a universal (if loose) guarantee.

Theorem 3.5.23 (Chebyshev's inequality)**.** *Let X have finite mean μ and finite variance $\sigma^2 = \mathrm{Var}(X)$. For any $k > 0$,*

$$\mathbb{P}\big(|X - \mu| \geq k\sigma\big) \leq \frac{1}{k^2},$$

and equivalently, for any $c > 0$, $\mathbb{P}\big(|X - \mu| \geq c\big) \leq \sigma^2/c^2$.

Proof. Let $c > 0$ and let A be the event $\{|X - \mu| \geq c\}$. On A we have $(X - \mu)^2 \geq c^2$, so $(X - \mu)^2 \geq c^2 \mathbb{1}_A$ pointwise. Taking expectations and using monotonicity,

$$\sigma^2 = \mathbb{E}[(X - \mu)^2] \;\geq\; c^2 \, \mathbb{E}(\mathbb{1}_A) = c^2 \, \mathbb{P}(A).$$

Dividing by c^2 gives $\mathbb{P}(|X - \mu| \geq c) \leq \sigma^2/c^2$. Setting $c = k\sigma$ yields the $1/k^2$ form. □

The bound depends on nothing but the mean and variance, so it holds for every distribution with finite variance; the price of that generality is that it is usually far from tight. It is most useful as a worst-case guarantee.

Example 3.5.24 (At least how often within two standard deviations?)**.** A variable X has mean $\mu = 50$ and standard deviation $\sigma = 4$, distribution otherwise unknown. Give a guaranteed lower bound on $\mathbb{P}(42 < X < 58)$.

Which rule, and why? We know only μ and σ and want a probability guarantee, so Chebyshev applies. The interval $(42, 58)$ is exactly $\mu \pm 2\sigma$, so take $k = 2$.

Solution.

Step 1: identify k. The interval is $50 \pm 8 = 50 \pm 2\sigma$, so $k = 2$.

Step 2: bound the tail, then complement.

$$\mathbb{P}(|X - 50| \geq 8) \leq \frac{1}{2^2} = \frac{1}{4},$$

hence

$$\mathbb{P}(42 < X < 58) = 1 - \mathbb{P}(|X - 50| \geq 8) \geq 1 - \frac{1}{4} = \frac{3}{4}.$$

So at least 75% of the mass lies within two standard deviations of the mean, whatever the distribution.

Example 3.5.25 (Bounding a tail from mean and variance)**.** The daily number of support tickets X has mean $\mu = 120$ and variance $\sigma^2 = 400$. Bound the probability that a day sees 160 or more tickets.

Which rule, and why? Distribution unknown, only mean and variance given, and we want a one-quantity tail bound; Chebyshev with $c = |160 - 120| = 40$. (Chebyshev bounds the two-sided event, so the result also covers the lower tail.)

Solution.

Step 1: distance from the mean. $c = 160 - 120 = 40$, and $\sigma^2 = 400$.

Step 2: apply the inequality.

$$\mathbb{P}(X \geq 160) \leq \mathbb{P}(|X - 120| \geq 40) \leq \frac{\sigma^2}{c^2} = \frac{400}{1600} = \frac{1}{4}.$$

So at most 25% of days reach 160 tickets. The true probability is typically far smaller; Chebyshev only certifies the worst case.

Example 3.5.26 (How wide an interval guarantees 90 percent?)**.** A filling machine dispenses a mean of $\mu = 500$ g with standard deviation $\sigma = 5$ g. Using Chebyshev, find an interval centered at 500 that is guaranteed to contain at least 90% of fills.

Which rule, and why? We want the half-width $k\sigma$ that forces the within-bound probability to be at least 0.90; invert Chebyshev by solving $1/k^2 \leq 0.10$ for k.

Solution.

Step 1: solve for k. We need $1/k^2 \leq 0.10$, i.e. $k^2 \geq 10$, so $k \geq \sqrt{10} \approx 3.162$.

Step 2: build the interval. The half-width is $k\sigma = \sqrt{10} \cdot 5 \approx 15.81$ g, giving

$$500 \pm 15.81 \text{ g}, \quad \text{i.e. about } (484.2,\ 515.8) \text{ g},$$

inside which at least 90% of fills must fall, regardless of the fill distribution.

Example 3.5.27 (When Chebyshev is loose: a known distribution)**.** Let $X \sim \text{Poisson}(25)$, so $\mu = 25$ and $\sigma = 5$. Compare the Chebyshev bound for $\mathbb{P}(|X - 25| \geq 10)$ (that is, $k = 2$) with the rough true value.

Which rule, and why? Here the distribution *is* known, so Chebyshev is not needed for an answer — the point is to see how loose the universal bound is against a concrete distribution.

Solution.

Step 1: Chebyshev bound. With $k = 10/\sigma = 2$,

$$\mathbb{P}(|X - 25| \geq 10) \leq \frac{1}{2^2} = 0.25.$$

Step 2: compare. For Poisson(25) the event $|X - 25| \geq 10$ means $X \leq 15$ or $X \geq 35$; summing the Poisson pmf gives a true probability of roughly 0.05. The guaranteed bound 0.25 is correct but about five times too large — the usual price of a distribution-free guarantee.

3.6 Joint Densities

So far each random variable has lived alone. Often two quantities are produced by the *same* experiment — the height and weight of a patient, the number of defective chips and the number of late shipments in a day — and we want to describe how they move together. The tool is a **joint distribution**: a single function of two arguments that assigns probability to pairs (X, Y). From it we recover each variable's own distribution (its **marginal**), we test whether the two vary independently, and we compute the probability that (X, Y) lands in a region of the plane.

Which joint-density rule does the problem need?

- Asked for the probability of a **specific pair** of discrete values, or to fill a two-way table? $\Rightarrow$ the **joint pmf** $p_{X,Y}(x, y) = \mathbb{P}(X = x, Y = y)$; every entry is nonnegative and the whole table sums to 1.
- Asked for the probability that the pair lands in a **region** (a rectangle, a triangle, "$X + Y \leq 1$")? $\Rightarrow$ a **double sum** of the pmf, or a **double integral** of the joint density $f_{X,Y}$ over that region. Draw the region first.
- Asked for the distribution of **one variable on its own**, ignoring the other? $\Rightarrow$ a **marginal**: sum (or integrate) the other variable out.
- Asked whether the two are **independent**? $\Rightarrow$ check whether the joint **factors** as $f_{X,Y}(x, y) = f_X(x)\, f_Y(y)$ *for all* (x, y), with a support that is a product set. One offending cell or a slanted support kills independence.

Why not a one-variable tool? A single pmf or density cannot answer questions that mention *both* variables at once ("$\mathbb{P}(X > Y)$", "are they independent?"). You need the joint object; the one-variable marginals are a *consequence* of it, never a substitute.

3.6.1 Joint pmf and joint pdf

Definition 3.6.1 (Joint pmf)**.** Let X, Y be discrete random variables. Their **joint probability mass function** is

$$p_{X,Y}(x, y) = \mathbb{P}(X = x,\ Y = y).$$

It satisfies $p_{X,Y}(x, y) \geq 0$ for every pair, and $\sum_x \sum_y p_{X,Y}(x, y) = 1$, the sum running over all values the pair can take.

Definition 3.6.2 (Joint pdf)**.** A function $f_{X,Y} : \mathbb{R}^2 \to \mathbb{R}$ is a **joint density** for the pair (X, Y) if $f_{X,Y}(x, y) \geq 0$ everywhere, $\displaystyle\iint_{\mathbb{R}^2} f_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y = 1$, and for every region $A \subseteq \mathbb{R}^2$,

$$\mathbb{P}\big((X, Y) \in A\big) = \iint_A f_{X,Y}(x, y)\, \mathrm{d}x\, \mathrm{d}y.$$

As in one dimension, $f_{X,Y}$ is not a probability; it is a probability *per unit area*, and individual points carry probability 0.

Example 3.6.3 (A two-way table of counts)**.** A small clinic logs, for each weekday, X = the number of walk-in patients $(0, 1, 2)$ and Y = the number of those who need a follow-up $(0, 1, 2)$, with joint pmf

$p_{X,Y}$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.20	0	0
$x = 1$	0.15	0.20	0
$x = 2$	0.10	0.15	0.20

Verify this is a valid joint pmf, and find $\mathbb{P}(X = Y)$ and $\mathbb{P}(Y < X)$.

Which rule, and why? The data are isolated integer pairs, so this is a **joint pmf**: validity is "all entries ≥ 0 and the table sums to 1". Probabilities of events like $\{X = Y\}$ are *sums of the cells* the event covers — no integral, because the variables are discrete.

Solution.

Step 1: Validity. Every entry is ≥ 0. The cells sum to

$$0.20 + 0.15 + 0.20 + 0.10 + 0.15 + 0.20 = 1.00,$$

(the three zero cells contribute nothing), so $p_{X,Y}$ is a valid joint pmf. Note follow-ups cannot exceed walk-ins, which is why every cell with $y > x$ is 0.

Step 2: $\mathbb{P}(X = Y)$. Add the diagonal cells $(0, 0), (1, 1), (2, 2)$:

$$\mathbb{P}(X = Y) = 0.20 + 0.20 + 0.20 = 0.60.$$

Step 3: $\mathbb{P}(Y < X)$. Add the cells strictly below the diagonal, $(1, 0), (2, 0), (2, 1)$:

$$\mathbb{P}(Y < X) = 0.15 + 0.10 + 0.15 = 0.40.$$

$$\mathbb{P}(X = Y) = 0.60, \qquad \mathbb{P}(Y < X) = 0.40.$$

As a check, $\mathbb{P}(Y > X) = 0$ here, and $0.60 + 0.40 + 0 = 1$.

Example 3.6.4 (Reading a region off a joint pmf)**.** Two fair four-sided dice (faces $1, 2, 3, 4$) are rolled. Let X be the first die and Y the second, so each of the 16 pairs has probability $\frac{1}{16}$. Find $\mathbb{P}(X + Y \leq 4)$ and $\mathbb{P}(\max(X, Y) = 3)$.

Which rule, and why? Equally likely pairs give the uniform joint pmf $p_{X,Y}(x, y) = \frac{1}{16}$ on the 4×4 grid. A probability over a region is just (number of grid points in the region) $\times \frac{1}{16}$ — counting, not integrating.

Solution.

Step 1: Count $X + Y \leq 4$. The pairs are $(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)$ — six of them.

$$\mathbb{P}(X + Y \leq 4) = \frac{6}{16} = \frac{3}{8}.$$

Step 2: Count $\max(X, Y) = 3$. The larger coordinate equals 3 exactly for the five pairs $(3, 1)$, $(3, 2)$, $(3, 3)$, $(1, 3)$, $(2, 3)$.

$$\mathbb{P}(\max(X, Y) = 3) = \frac{5}{16}.$$

Example 3.6.5 (Normalising a joint density on a square)**.** Let (X, Y) have $f_{X,Y}(x, y) = c\,(x + y)$ for $0 \leq x \leq 1$, $0 \leq y \leq 1$, and 0 elsewhere. Find c, then $\mathbb{P}(X + Y \leq 1)$.

Which rule, and why? Continuous pair, so this is a **joint pdf**: the constant is fixed by forcing the double integral over the support to equal 1. The region $\{X + Y \leq 1\}$ is a triangle, so $\mathbb{P}$ is a double integral over that triangle.

Solution.

Step 1: Normalise. Integrate over the unit square:

$$\iint_{[0,1]^2} c(x+y)\,\mathrm{d}x\,\mathrm{d}y = c\int_0^1\!\!\int_0^1 (x+y)\,\mathrm{d}x\,\mathrm{d}y = c\int_0^1 \left(\tfrac{1}{2}+y\right)\mathrm{d}y = c\left(\tfrac{1}{2}+\tfrac{1}{2}\right) = c.$$

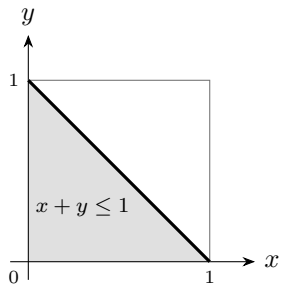
Setting this to 1 gives $c=1$, so $f_{X,Y}(x,y) = x+y$ on the unit square.

Step 2: Integrate over the triangle $x+y\le 1$. For $0\le x\le 1$, y runs from 0 to $1-x$:

$$\begin{aligned}\mathbb{P}(X+Y\le 1) &= \int_0^1\!\!\int_0^{1-x} (x+y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \left[xy+\tfrac{y^2}{2}\right]_0^{1-x}\mathrm{d}x \\ &= \int_0^1 \left(x(1-x)+\tfrac{(1-x)^2}{2}\right)\mathrm{d}x = \int_0^1 \left(\tfrac{1}{2}-\tfrac{x^2}{2}\right)\mathrm{d}x = \tfrac{1}{2}-\tfrac{1}{6} = \tfrac{1}{3}.\end{aligned}$$

$$c=1, \qquad \mathbb{P}(X+Y\le 1) = \frac{1}{3}.$$

The triangle of integration in Step 2:



Example 3.6.6 (A density on a triangular support)**.** Let $f_{X,Y}(x,y) = c\,x$ on the triangle $0<y<x<1$, and 0 elsewhere. Find c and $\mathbb{P}\big(Y<\tfrac{1}{2}\big)$.

Which rule, and why? The support is a slanted triangle, not a rectangle, so the inner limits depend on the outer variable. Fix c by integrating over the triangle; the constant is determined the same way regardless of the support's shape.

Solution.

Step 1: Normalise. On the triangle y runs from 0 to x, then x from 0 to 1:

$$\int_0^1\!\!\int_0^x c\,x\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 c\,x\cdot x\,\mathrm{d}x = c\int_0^1 x^2\,\mathrm{d}x = \frac{c}{3} = 1 \implies c=3.$$

Step 2: $\mathbb{P}(Y<\tfrac{1}{2})$. Split the triangle at $y=\tfrac{1}{2}$. Easier: integrate over $\{0<y<x<1,\ y<\tfrac{1}{2}\}$ with x outer. For $0<x\le\tfrac{1}{2}$, y runs 0 to x; for $\tfrac{1}{2}<x<1$, y runs 0 to $\tfrac{1}{2}$:

$$\begin{aligned}\mathbb{P}\big(Y<\tfrac{1}{2}\big) &= \int_0^{1/2}\!\!\int_0^x 3x\,\mathrm{d}y\,\mathrm{d}x + \int_{1/2}^1\!\!\int_0^{1/2} 3x\,\mathrm{d}y\,\mathrm{d}x \\ &= \int_0^{1/2} 3x^2\,\mathrm{d}x + \int_{1/2}^1 3x\cdot\tfrac{1}{2}\,\mathrm{d}x = \Big[x^3\Big]_0^{1/2} + \tfrac{3}{2}\Big[\tfrac{x^2}{2}\Big]_{1/2}^1 \\ &= \tfrac{1}{8}+\tfrac{3}{4}\Big(1-\tfrac{1}{4}\Big) = \tfrac{1}{8}+\tfrac{3}{4}\cdot\tfrac{3}{4} = \tfrac{1}{8}+\tfrac{9}{16} = \tfrac{11}{16}.\end{aligned}$$

$$c=3, \qquad \mathbb{P}\big(Y<\tfrac{1}{2}\big) = \frac{11}{16}.$$

3.6.2 Marginal distributions

Definition 3.6.7 (Marginals)**.** The **marginal** distribution of one variable is its own distribution, found by removing the other. In the discrete case,

$$p_X(x) = \sum_y p_{X,Y}(x,y), \qquad p_Y(y) = \sum_x p_{X,Y}(x,y);$$

in the continuous case,

$$f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\,\mathrm{d}x.$$

The name comes from writing the row and column totals in the *margins* of a two-way table.

Example 3.6.8 (Margins of the clinic table)**.** For the clinic pmf of Example 6.3, find the marginal pmfs of X and Y and the expected number of follow-ups $\mathbb{E}(Y)$.

Which rule, and why? "Distribution of X alone" is a **marginal**: sum each row to collapse Y. For Y alone, sum each column. Then $\mathbb{E}(Y)$ is the ordinary one-variable mean of the marginal p_Y.

Solution.

Step 1: Row sums give p_X.

$$p_X(0) = 0.20, \quad p_X(1) = 0.15 + 0.20 = 0.35, \quad p_X(2) = 0.10 + 0.15 + 0.20 = 0.45.$$

Step 2: Column sums give p_Y.

$$p_Y(0) = 0.20 + 0.15 + 0.10 = 0.45, \quad p_Y(1) = 0.20 + 0.15 = 0.35, \quad p_Y(2) = 0.20.$$

Each marginal sums to 1, as it must.

Step 3: $\mathbb{E}(Y)$.

$$\mathbb{E}(Y) = 0(0.45) + 1(0.35) + 2(0.20) = 0.35 + 0.40 = 0.75.$$

$$\mathbb{E}(Y) = 0.75 \text{ follow-ups per day.}$$

Example 3.6.9 (Marginals from the triangular density)**.** With $f_{X,Y}(x,y) = 3x$ on $0 < y < x < 1$ (Example 6.6), find both marginal densities.

Which rule, and why? A marginal integrates the *other* variable out — but only over the part of the support that exists for the fixed value. Getting the inner limits from the support $0 < y < x < 1$ is the whole task.

Solution.

Step 1: f_X (integrate out y). For fixed $x \in (0,1)$, y ranges over $(0,x)$:

$$f_X(x) = \int_0^x 3x\,\mathrm{d}y = 3x \cdot x = 3x^2, \qquad 0 < x < 1.$$

Check: $\int_0^1 3x^2\,\mathrm{d}x = 1$. So X has density $3x^2$, concentrated near 1.

Step 2: f_Y (integrate out x). For fixed $y \in (0,1)$, x ranges over $(y,1)$:

$$f_Y(y) = \int_y^1 3x\,\mathrm{d}x = \tfrac{3}{2}\big(1 - y^2\big), \qquad 0 < y < 1.$$

Check: $\int_0^1 \frac{3}{2}(1-y^2)\,\mathrm{d}y = \frac{3}{2}\big(1 - \frac{1}{3}\big) = 1$.

$$f_X(x) = 3x^2 \ (0 < x < 1), \qquad f_Y(y) = \tfrac{3}{2}(1-y^2) \ (0 < y < 1).$$

Example 3.6.10 (Marginals on the unit square)**.** With $f_{X,Y}(x,y) = x + y$ on $[0,1]^2$ (Example 6.5), find f_X and confirm it is a valid density.

Which rule, and why? Rectangular support, so the inner limits are constants 0 to 1 — the marginal is a clean single integral. A marginal must itself integrate to 1, a built-in check.

Solution.

Step 1: Integrate out y from 0 to 1.

$$f_X(x) = \int_0^1 (x+y)\,\mathrm{d}y = x + \tfrac{1}{2}, \qquad 0 \le x \le 1.$$

Step 2: Check.

$$\int_0^1 \left(x + \tfrac{1}{2}\right)\mathrm{d}x = \tfrac{1}{2} + \tfrac{1}{2} = 1. \qquad \blacksquare$$

By symmetry $f_Y(y) = y + \frac{1}{2}$ on $[0,1]$.

$$f_X(x) = x + \tfrac{1}{2}, \qquad f_Y(y) = y + \tfrac{1}{2} \quad (0 \le x, y \le 1).$$

Example 3.6.11 (A marginal that is exponential)**.** A two-stage process has $f_{X,Y}(x,y) = 2e^{-x}e^{-2y}$ for $x > 0,\ y > 0$, and 0 otherwise. Find the marginal of X and identify its distribution.

Which rule, and why? The support is the whole first quadrant (a product set), so integrate y from 0 to ∞ with x held fixed. Recognising the resulting one-variable density as a named distribution is the payoff.

Solution.

Step 1: Integrate out y.

$$f_X(x) = \int_0^\infty 2e^{-x}e^{-2y}\,\mathrm{d}y = e^{-x}\int_0^\infty 2e^{-2y}\,\mathrm{d}y = e^{-x}\cdot 1 = e^{-x}, \qquad x > 0.$$

Step 2: Identify. The density e^{-x} on $(0,\infty)$ is exactly Exp(1), so $X \sim \text{Exp}(1)$. (Likewise $f_Y(y) = 2e^{-2y}$, so $Y \sim \text{Exp}(2)$.)

$$f_X(x) = e^{-x}\ (x > 0), \qquad X \sim \text{Exp}(1).$$

3.6.3 Independence

Definition 3.6.12 (Independent random variables)**.** X and Y are **independent**, written $X \perp\!\!\!\perp Y$, if their joint factors into the product of the marginals for *every* pair of values:

$$p_{X,Y}(x,y) = p_X(x)\,p_Y(y) \ \text{ (discrete)}, \qquad f_{X,Y}(x,y) = f_X(x)\,f_Y(y) \ \text{ (continuous)}.$$

Proposition 3.6.13 (Factorisation test)**.** *$X \perp\!\!\!\perp Y$ if and only if the joint can be written as a product $g(x)\,h(y)$ of a function of x alone and a function of y alone,* and *the support is a rectangle (a product set $\{x \in A\} \times \{y \in B\}$, possibly infinite). If the support is slanted — so that the allowed range of y depends on x — the variables are **not** independent, whatever the formula looks like.*

Proof. If $f_{X,Y} = g(x)h(y)$ on a rectangle, then $f_X(x) = g(x)\int h$ and $f_Y(y) = h(y)\int g$, and since $\left(\int g\right)\left(\int h\right) = \iint f_{X,Y} = 1$ the product $f_X(x)f_Y(y)$ equals $g(x)h(y) = f_{X,Y}(x,y)$. Conversely if the joint factors into the marginals it is visibly of the form $g \cdot h$. A non-rectangular support cannot factor: there are pairs (x,y) with $f_X(x) > 0$ and $f_Y(y) > 0$ yet $f_{X,Y}(x,y) = 0$, so the product rule fails there. □

Example 3.6.14 (Independent by inspection)**.** Check whether X and Y with $f_{X,Y}(x,y) = 2e^{-x}e^{-2y}$ on $x, y > 0$ (Example 6.10) are independent.

Which rule, and why? The support is the full quadrant (a product set) and the formula already splits as $(e^{-x})(2e^{-2y})$. That is exactly the factorisation test, so we can read off independence without further work.

Solution.

Step 1: Compare to the marginals. From Example 6.10, $f_X(x) = e^{-x}$ and $f_Y(y) = 2e^{-2y}$. Their product is

$$f_X(x)f_Y(y) = e^{-x}\cdot 2e^{-2y} = 2e^{-x}e^{-2y} = f_{X,Y}(x,y)$$

for all $x, y > 0$.

Step 2: Conclude. The joint equals the product of the marginals on a rectangular (here, quadrant) support, so $X \perp\!\!\!\perp Y$.

$$X \perp\!\!\!\perp Y, \qquad X \sim \text{Exp}(1),\ Y \sim \text{Exp}(2).$$

Example 3.6.15 (Slanted support breaks independence)**.** Are X and Y with $f_{X,Y}(x, y) = 3x$ on $0 < y < x < 1$ (Example 6.6) independent?

Which rule, and why? The formula $3x$ looks like a function of x alone, which tempts a "yes". But the *support* is the triangle $y < x$, not a rectangle, so the factorisation test forbids independence; we confirm by comparing to the marginals.

Solution.

Step 1: Multiply the marginals. From Example 6.8, $f_X(x) = 3x^2$ and $f_Y(y) = \frac{3}{2}(1 - y^2)$. So

$$f_X(x) f_Y(y) = 3x^2 \cdot \tfrac{3}{2}(1 - y^2) = \tfrac{9}{2}x^2(1 - y^2),$$

which is *not* equal to $3x$.

Step 2: Pinpoint the failure. Take $(x, y) = (\frac{1}{4}, \frac{1}{2})$: here $f_X(\frac{1}{4}) > 0$ and $f_Y(\frac{1}{2}) > 0$, so the product is positive, yet $y > x$ means $f_{X,Y} = 0$. The joint and the product disagree.

$$X \text{ and } Y \text{ are NOT independent.}$$

Example 3.6.16 (Independent discrete pair)**.** Each of two assembly lines, independently, produces a defective unit with probability 0.1. Let $X, Y \in \{0, 1\}$ indicate a defect on line 1, line 2. The joint pmf is

$p_{X,Y}$	$y = 0$	$y = 1$
$x = 0$	0.81	0.09
$x = 1$	0.09	0.01

Verify $X \perp\!\!\!\perp Y$ from the table.

Which rule, and why? Independence of a discrete pair is checked cell by cell against the row/column totals: $X \perp\!\!\!\perp Y$ exactly when every cell equals (its row total)$\times$(its column total).

Solution.

Step 1: Marginals. Row sums: $p_X(0) = 0.90,\ p_X(1) = 0.10$. Column sums: $p_Y(0) = 0.90,\ p_Y(1) = 0.10$.

Step 2: Check every cell.

$$p_X(0)p_Y(0) = 0.81, \quad p_X(0)p_Y(1) = 0.09, \quad p_X(1)p_Y(0) = 0.09, \quad p_X(1)p_Y(1) = 0.01,$$

which match the four table entries exactly. Hence $X \perp\!\!\!\perp Y$.

$$p_{X,Y}(x, y) = p_X(x)\, p_Y(y) \text{ for all } x, y \implies X \perp\!\!\!\perp Y.$$

Example 3.6.17 (Dependent discrete pair)**.** Return to the clinic table of Example 6.3. Are X (walk-ins) and Y (follow-ups) independent?

Which rule, and why? Use the same cell-by-cell test. A single cell that violates (row total)$\times$(column total) is enough to declare dependence — no need to check the rest.

Solution.

Step 1: One revealing cell. From the marginals (Example 6.7), $p_X(0) = 0.20$ and $p_Y(2) = 0.20$, so independence would require

$$p_{X,Y}(0, 2) = p_X(0)\, p_Y(2) = 0.20 \times 0.20 = 0.04.$$

Step 2: Compare. The actual entry is $p_{X,Y}(0, 2) = 0$, since a day with 0 walk-ins cannot have 2 follow-ups. Because $0 \neq 0.04$, the test fails.

$$X \text{ and } Y \text{ are dependent (more walk-ins allow more follow-ups).}$$

3.6.4 Probability over a region

When the variables are continuous, the probability that (X, Y) falls in a region A is the double integral of the joint density over A. The art is choosing the order of integration and the limits so that they trace out A; sketching A first almost always settles them.

Example 3.6.18 (Probability that one exceeds the other)**.** With the independent exponentials $f_{X,Y}(x, y) = 2e^{-x}e^{-2y}$ on $x, y > 0$ (Example 6.10), find $\mathbb{P}(X < Y)$.

Which rule, and why? "$\mathbb{P}(X < Y)$" names a region of the quadrant, so it is a **double integral** over $\{0 < x < y\}$. Integrate the inner variable across the wedge, then the outer one out to ∞.

Solution.

Step 1: Set up over the wedge $x < y$. Hold y fixed and let x run from 0 to y:

$$\mathbb{P}(X < Y) = \int_0^\infty \int_0^y 2e^{-x}e^{-2y}\,\mathrm{d}x\,\mathrm{d}y = \int_0^\infty 2e^{-2y}\big(1 - e^{-y}\big)\,\mathrm{d}y.$$

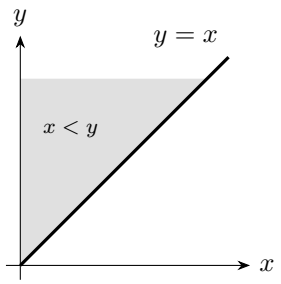
Step 2: Integrate out y.

$$\int_0^\infty \big(2e^{-2y} - 2e^{-3y}\big)\,\mathrm{d}y = 2 \cdot \tfrac{1}{2} - 2 \cdot \tfrac{1}{3} = 1 - \tfrac{2}{3} = \tfrac{1}{3}.$$

$$\mathbb{P}(X < Y) = \frac{1}{3}.$$

(Intuitively Y, with rate 2, dies off faster, so it tends to be the smaller; a value below $\frac{1}{2}$ is right.)

The wedge $0 < x < y$ over which we integrated:



Example 3.6.19 (A rectangle inside the square)**.** With $f_{X,Y}(x, y) = x + y$ on $[0, 1]^2$ (Example 6.5), find $\mathbb{P}\big(X < \frac{1}{2},\ Y > \frac{1}{2}\big)$.

Which rule, and why? The event is a sub-rectangle of the support, so both inner and outer limits are constants. A region that is itself a rectangle is the easiest double integral — no x-dependent limits.

Solution.

Step 1: Set the limits. $x \in (0, \frac{1}{2})$, $y \in (\frac{1}{2}, 1)$:

$$\mathbb{P}\big(X < \tfrac{1}{2},\, Y > \tfrac{1}{2}\big) = \int_0^{1/2} \int_{1/2}^1 (x + y)\,\mathrm{d}y\,\mathrm{d}x.$$

Step 2: Inner integral.

$$\int_{1/2}^1 (x + y)\,\mathrm{d}y = x \cdot \tfrac{1}{2} + \left[\tfrac{y^2}{2}\right]_{1/2}^1 = \tfrac{x}{2} + \left(\tfrac{1}{2} - \tfrac{1}{8}\right) = \tfrac{x}{2} + \tfrac{3}{8}.$$

Step 3: Outer integral.

$$\int_0^{1/2} \left(\tfrac{x}{2} + \tfrac{3}{8}\right)\mathrm{d}x = \tfrac{1}{2} \cdot \tfrac{(1/2)^2}{2} + \tfrac{3}{8} \cdot \tfrac{1}{2} = \tfrac{1}{16} + \tfrac{3}{16} = \tfrac{4}{16} = \tfrac{1}{4}.$$

$$\mathbb{P}\big(X < \tfrac{1}{2},\, Y > \tfrac{1}{2}\big) = \frac{1}{4}.$$

Example 3.6.20 (Region under a line, slanted support)**.** With $f_{X,Y}(x,y) = 3x$ on $0 < y < x < 1$ (Example 6.6), find $\mathbb{P}\big(X > \frac{1}{2}\big)$.

Which rule, and why? The event $X > \frac{1}{2}$ slices the triangular support; the inner limit on y is still 0 to x (the support), while the outer limit on x is now $\frac{1}{2}$ to 1. Drawing the support keeps the limits straight.

Solution.

Step 1: Set up.

$$\mathbb{P}\big(X > \tfrac{1}{2}\big) = \int_{1/2}^{1}\int_{0}^{x} 3x \,\mathrm{d}y\,\mathrm{d}x = \int_{1/2}^{1} 3x^2\,\mathrm{d}x.$$

Step 2: Evaluate.

$$\int_{1/2}^{1} 3x^2\,\mathrm{d}x = \Big[x^3\Big]_{1/2}^{1} = 1 - \tfrac{1}{8} = \tfrac{7}{8}.$$

$$\mathbb{P}\big(X > \tfrac{1}{2}\big) = \frac{7}{8}.$$

This matches $\int_{1/2}^{1} f_X(x)\,\mathrm{d}x$ with $f_X(x) = 3x^2$, a useful cross-check: a region that constrains only X can be done from the marginal of X alone.

Example 3.6.21 (Uniform on a triangle, region below a curve)**.** Let (X,Y) be uniform on the triangle $T = \{0 < x < 1,\ 0 < y < x\}$, so $f_{X,Y}(x,y) = 2$ on T (its area is $\frac{1}{2}$, and 1/area = 2). Find $\mathbb{P}(Y < X^2)$.

Which rule, and why? Uniform on a region means constant density 1/(area), and any probability is (area of the sub-region)/(area of T) — or equivalently a double integral of the constant 2. The event $Y < X^2$ is bounded by a parabola, so we integrate.

Solution.

Step 1: Describe the sub-region. Inside T we need $y < x^2$. For each $x \in (0,1)$, since $x^2 < x$, the constraint $y < x^2$ is tighter than $y < x$, so y runs from 0 to x^2:

$$\mathbb{P}(Y < X^2) = \int_{0}^{1}\int_{0}^{x^2} 2\,\mathrm{d}y\,\mathrm{d}x.$$

Step 2: Evaluate.

$$\int_{0}^{1} 2x^2\,\mathrm{d}x = \tfrac{2}{3}.$$

$$\mathbb{P}(Y < X^2) = \frac{2}{3}.$$

(Sanity: the parabola $y = x^2$ lies below the line $y = x$ throughout T, so most of the triangle satisfies $y < x^2$; a value above $\frac{1}{2}$ is expected.)

3.6.5 Joint expectation and covariance from a table

For any function $g(X,Y)$, the expectation extends the one-variable rule to a double sum or double integral:

$$\mathbb{E}\big[g(X,Y)\big] = \sum_{x}\sum_{y} g(x,y)\,p_{X,Y}(x,y) \quad \text{or} \quad \iint g(x,y)\,f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

Taking $g(x,y) = xy$ and subtracting the product of the means gives the **covariance** $\mathrm{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$, a first measure of how the two move together (developed fully in a later section). Independence forces $\mathrm{Cov}(X,Y) = 0$.

Example 3.6.22 (Covariance from the clinic table)**.** For the clinic pmf (Example 6.3), compute $\mathbb{E}(XY)$ and $\mathrm{Cov}(X,Y)$.

Which rule, and why? $\mathbb{E}(XY)$ is a **joint expectation**: sum xy against each cell. Then $\text{Cov} = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ uses the marginal means. Because the variables are dependent (Example 6.13), a nonzero covariance is plausible.

Solution.

Step 1: $\mathbb{E}(XY)$. Only cells with $x \geq 1$ and $y \geq 1$ contribute:

$$\mathbb{E}(XY) = \underbrace{(1)(1)(0.20)}_{(1,1)} + \underbrace{(2)(1)(0.15)}_{(2,1)} + \underbrace{(2)(2)(0.20)}_{(2,2)} = 0.20 + 0.30 + 0.80 = 1.30.$$

Step 2: Marginal means. From $p_X = (0.20, 0.35, 0.45)$ and $p_Y = (0.45, 0.35, 0.20)$:

$$\mathbb{E}(X) = 0(0.20) + 1(0.35) + 2(0.45) = 1.25, \qquad \mathbb{E}(Y) = 0.75 \text{ (Example 6.7)}.$$

Step 3: Covariance.

$$\text{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 1.30 - (1.25)(0.75) = 1.30 - 0.9375 = 0.3625.$$

$$\mathbb{E}(XY) = 1.30, \qquad \text{Cov}(X, Y) = 0.3625 > 0,$$

positive, as expected: busier days tend to bring more follow-ups.

Example 3.6.23 (Zero covariance under independence)**.** For the independent defect indicators of Example 6.12, confirm $\text{Cov}(X, Y) = 0$ directly.

Which rule, and why? Independence guarantees $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$, hence Cov = 0. Computing $\mathbb{E}(XY)$ from the table and comparing is a concrete check of that theorem.

Solution.

Step 1: $\mathbb{E}(XY)$. The only cell with $xy \neq 0$ is $(1, 1)$:

$$\mathbb{E}(XY) = (1)(1)(0.01) = 0.01.$$

Step 2: Product of means. $\mathbb{E}(X) = 0(0.90) + 1(0.10) = 0.10$, and likewise $\mathbb{E}(Y) = 0.10$, so $\mathbb{E}(X)\mathbb{E}(Y) = 0.01$.

Step 3: Covariance.

$$\text{Cov}(X, Y) = 0.01 - 0.01 = 0.$$

$$\text{Cov}(X, Y) = 0, \quad \text{consistent with } X \perp\!\!\!\perp Y.$$

(The converse fails in general: zero covariance alone does not prove independence, but here we already established independence from the table.)

3.7 Transforming and Combining Random Variables

Given a random variable X whose distribution we already know, we very often care about some derived quantity: the square $Y = X^2$, an exponential reading $Y = e^X$, a calibration $Y = \sqrt{X}$, or the sum $Z = X_1 + X_2$ of two independent measurements. This section answers the single question raised by all of these: *if we know the distribution of the input, what is the distribution of the output?* Three tools do the work — the **cdf method** for any function g, the **Jacobian (change-of-variable) formula** for a smooth *monotone* g, and **convolution** for the sum of two independents.

Which transformation rule does the problem need?

- One variable, $Y = g(X)$, and g is **not one-to-one** on the support of X (e.g. $Y = X^2$ with X ranging over both signs, $Y = |X|$, $Y = \sin X$)? $\Rightarrow$ always-safe **cdf method**: write $F_Y(y) = \mathbb{P}(g(X) \leq y)$, solve the inequality for X, read off probabilities from F_X, then differentiate.

- One variable, $Y = g(X)$, with g **strictly monotone** (increasing or decreasing) and differentiable on the support of X? $\Rightarrow$ the **Jacobian formula** $f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy}g^{-1}(y)\right|$. It is just the cdf method packaged into one line; use it for speed once monotonicity is checked.
- Two **independent** variables and you want the distribution of their **sum** $Z = X + Y$? $\Rightarrow$ **convolution**: $f_Z(z) = \int f_X(x)\, f_Y(z - x)\,\mathrm{d}x$ (or the analogous sum in the discrete case).

Why not the others? The Jacobian formula *requires* a single-valued inverse; applying it to $Y = X^2$ on $(-a, a)$ silently drops the second branch and undercounts the density. The cdf method never fails but costs an extra differentiation. Convolution is for *sums of independents* only — for a nonlinear function of one variable it does not apply.

3.7.1 The cdf method for $Y = g(X)$

The most reliable route to the distribution of $Y = g(X)$ goes through the cdf. Whatever g does, the event $\{Y \le y\}$ is the same as $\{g(X) \le y\}$, an event about X whose probability we can read from F_X.

Definition 3.7.1 (cdf method)**.** Let X have cdf F_X and let $Y = g(X)$. For each real y,

$$F_Y(y) = \mathbb{P}(Y \le y) = \mathbb{P}\big(g(X) \le y\big) = \mathbb{P}\big(X \in g^{-1}((-\infty, y])\big),$$

where $g^{-1}((-\infty, y]) = \{x : g(x) \le y\}$. If Y is continuous, differentiate to get the density: $f_Y(y) = F_Y'(y)$.

The whole skill is turning the inequality $g(X) \le y$ into a statement about X (an interval, or a union of intervals) and then evaluating F_X at the endpoints.

Example 3.7.2 (Square of a uniform reading)**.** A sensor returns a voltage $X \sim \mathrm{Uniform}(0, 2)$, so $f_X(x) = \frac{1}{2}$ for $0 \le x \le 2$. The power dissipated is proportional to $Y = X^2$. Find the density of Y.

Which rule, and why? One variable and a function of it, $Y = X^2$. On the support $(0, 2)$ the map $x \mapsto x^2$ *is* increasing, so the Jacobian formula would also work; but the cdf method is just as quick and makes the support of Y obvious, so we use it.

Solution.

Step 1: cdf. The support of Y is $(0, 4)$. For $0 \le y \le 4$, since $X \ge 0$,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(0 \le X \le \sqrt{y}) = \int_0^{\sqrt{y}} \tfrac{1}{2}\,\mathrm{d}x = \tfrac{\sqrt{y}}{2}.$$

Step 2: differentiate. For $0 < y < 4$,

$$f_Y(y) = \frac{d}{dy}\,\frac{\sqrt{y}}{2} = \frac{1}{4\sqrt{y}}, \qquad \text{and } f_Y(y) = 0 \text{ otherwise.}$$

Step 3: sanity check. $\int_0^4 \frac{1}{4\sqrt{y}}\,\mathrm{d}y = \big[\frac{\sqrt{y}}{2}\big]_0^4 = 1$.

$$f_Y(y) = \frac{1}{4\sqrt{y}} \quad (0 < y < 4).$$

Example 3.7.3 (Square of a standard normal: the chi-square)**.** Let $X \sim \mathrm{N}(0, 1)$ with density $\varphi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$. Find the density of $Y = X^2$.

Which rule, and why? $Y = X^2$ but now X takes *both* signs, so $x \mapsto x^2$ is **not** one-to-one on the support. The Jacobian formula in its plain form would miss the negative branch; the cdf method handles both branches at once.

Solution.

Step 1: cdf, both branches. For $y > 0$, $\{X^2 \le y\} = \{-\sqrt{y} \le X \le \sqrt{y}\}$, so

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}) = 2\Phi(\sqrt{y}) - 1,$$

using the symmetry $\Phi(-a) = 1 - \Phi(a)$, where Φ is the standard-normal cdf.

Step 2: differentiate. With $\Phi' = \varphi$ and the chain rule,

$$f_Y(y) = 2\varphi(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \frac{1}{\sqrt{y}} \cdot \frac{1}{\sqrt{2\pi}} e^{-y/2} = \frac{1}{\sqrt{2\pi\, y}}\, e^{-y/2}, \qquad y > 0.$$

Step 3: name it. This is the Gamma$(\frac{1}{2}, \frac{1}{2})$ density, also called the **chi-square distribution with one degree of freedom**.

$$f_Y(y) = \frac{1}{\sqrt{2\pi\, y}}\, e^{-y/2} \quad (y > 0).$$

Example 3.7.4 (A two-branch fold)**.** Let $X \sim \text{Uniform}(-1, 2)$, so $f_X(x) = \frac{1}{3}$ on $(-1, 2)$. Find the density of $Y = X^2$.

Which rule, and why? $Y = X^2$ with X straddling 0; for small y both $+\sqrt{y}$ and $-\sqrt{y}$ lie in the support, but for larger y only the positive root does. The support of X is asymmetric, so we must split by where $\sqrt{y}$ sits. The cdf method makes this bookkeeping automatic.

Solution.

Step 1: locate the regimes. X^2 ranges over $[0, 4)$. The negative end reaches $x = -1$, i.e. $y = 1$; the positive end reaches $x = 2$, i.e. $y = 4$. For $0 \le y \le 1$ both roots are in support: $\{X^2 \le y\} = \{-\sqrt{y} \le X \le \sqrt{y}\}$. For $1 < y < 4$ only the positive root is in support: $\{X^2 \le y\} = \{-1 \le X \le \sqrt{y}\}$.

Step 2: cdf on each regime.

$$F_Y(y) = \begin{cases} \dfrac{2\sqrt{y}}{3}, & 0 \le y \le 1, \\[2mm] \dfrac{\sqrt{y} + 1}{3}, & 1 < y < 4. \end{cases}$$

Step 3: differentiate.

$$f_Y(y) = \begin{cases} \dfrac{1}{3\sqrt{y}}, & 0 < y < 1, \\[2mm] \dfrac{1}{6\sqrt{y}}, & 1 < y < 4. \end{cases}$$

Step 4: check. $\displaystyle\int_0^1 \frac{\mathrm{d}y}{3\sqrt{y}} + \int_1^4 \frac{\mathrm{d}y}{6\sqrt{y}} = \Big[\tfrac{2\sqrt{y}}{3}\Big]_0^1 + \Big[\tfrac{\sqrt{y}}{3}\Big]_1^4 = \tfrac{2}{3} + \tfrac{2-1}{3} = 1$. The density jumps down at $y = 1$, exactly where the lower branch runs out.

Example 3.7.5 (A linear transformation and a threshold)**.** Daily demand at a kiosk is $X \sim \text{Exp}(\frac{1}{4})$ (mean 4), $f_X(x) = \frac{1}{4}e^{-x/4}$ for $x \ge 0$. Stock is priced so revenue is $Y = 3X + 5$. Find F_Y and $\mathbb{P}(Y > 20)$.

Which rule, and why? $g(x) = 3x + 5$ is strictly increasing and affine, the easiest case of the cdf method; solving $3X + 5 \le y$ for X is immediate.

Solution.

Step 1: invert the inequality. For $y \ge 5$ (the smallest value of Y, at $X = 0$),

$$F_Y(y) = \mathbb{P}(3X + 5 \le y) = \mathbb{P}\Big(X \le \tfrac{y-5}{3}\Big) = 1 - e^{-(y-5)/12},$$

using $F_X(t) = 1 - e^{-t/4}$ and $\frac{1}{4} \cdot \frac{y-5}{3} = \frac{y-5}{12}$.

Step 2: tail probability.

$$\mathbb{P}(Y > 20) = 1 - F_Y(20) = e^{-(20-5)/12} = e^{-15/12} = e^{-1.25} \approx 0.2865.$$

$$F_Y(y) = 1 - e^{-(y-5)/12} \ \ (y \ge 5), \qquad \mathbb{P}(Y > 20) = e^{-5/4} \approx 0.287.$$

A scaled-and-shifted exponential is again exponential in shape, here with mean 12 shifted to start at 5.

3.7.2 The Jacobian (change-of-variable) formula

When g is strictly monotone there is exactly one x with $g(x) = y$, namely $x = g^{-1}(y)$, and the cdf method collapses into a single formula. This is the workhorse for "smooth, one-to-one" transforms.

Theorem 3.7.6 (Change of variable for a monotone g)**.** *Let X be continuous with density f_X, and let g be strictly monotone and differentiable on the support of X, with inverse g^{-1}. Then $Y = g(X)$ has density*

$$f_Y(y) = f_X\big(g^{-1}(y)\big)\left|\frac{d}{dy}\,g^{-1}(y)\right|$$

for y in the range of g, and $f_Y(y) = 0$ elsewhere.

Proof. If g is increasing, $\{g(X) \le y\} = \{X \le g^{-1}(y)\}$, so $F_Y(y) = F_X(g^{-1}(y))$; differentiating with the chain rule gives $f_Y(y) = f_X(g^{-1}(y))\,\frac{d}{dy}g^{-1}(y)$, and $\frac{d}{dy}g^{-1}(y) > 0$. If g is decreasing, $\{g(X) \le y\} = \{X \ge g^{-1}(y)\}$, so $F_Y(y) = 1 - F_X(g^{-1}(y))$ and $f_Y(y) = -f_X(g^{-1}(y))\,\frac{d}{dy}g^{-1}(y)$, with $\frac{d}{dy}g^{-1}(y) < 0$. In both cases the result is $f_X(g^{-1}(y))$ times the absolute value of $\frac{d}{dy}g^{-1}(y)$. □

Example 3.7.7 (Exponential of a uniform)**.** Let $X \sim \text{Uniform}(0,1)$, $f_X(x) = 1$ on $(0,1)$. Find the density of $Y = e^X$.

Which rule, and why? $g(x) = e^x$ is strictly increasing and differentiable, so the Jacobian formula applies directly; the inverse $g^{-1}(y) = \ln y$ is single-valued.

Solution.

Step 1: inverse and its derivative. $x = g^{-1}(y) = \ln y$, so $\frac{d}{dy}\ln y = \frac{1}{y}$. Since $X \in (0,1)$, the range of Y is $(1, e)$.

Step 2: apply the formula. On $1 < y < e$,

$$f_Y(y) = f_X(\ln y)\cdot\left|\tfrac{1}{y}\right| = 1\cdot\tfrac{1}{y} = \frac{1}{y}.$$

Step 3: check. $\int_1^e \frac{1}{y}\,\mathrm{d}y = [\ln y]_1^e = 1$.

$$f_Y(y) = \frac{1}{y} \quad (1 < y < e).$$

Example 3.7.8 (Square root of an exponential)**.** The squared length of a flaw, $X \sim \text{Exp}(1)$, has density $f_X(x) = e^{-x}$ for $x \ge 0$. The actual length is $Y = \sqrt{X}$. Find its density and identify the family.

Which rule, and why? $g(x) = \sqrt{x}$ is strictly increasing on $[0,\infty)$ with single-valued inverse $x = y^2$, so the Jacobian formula applies.

Solution.

Step 1: inverse and derivative. $x = g^{-1}(y) = y^2$, $\frac{d}{dy}y^2 = 2y$, and $y \ge 0$.

Step 2: apply the formula. For $y \ge 0$,

$$f_Y(y) = f_X(y^2)\cdot|2y| = e^{-y^2}\cdot 2y = 2y\,e^{-y^2}.$$

Step 3: identify and check. This is the **Rayleigh** density. Its integral is $\int_0^\infty 2y\,e^{-y^2}\,\mathrm{d}y = \big[-e^{-y^2}\big]_0^\infty = 1$.

$$f_Y(y) = 2y\,e^{-y^2} \quad (y \ge 0).$$

Example 3.7.9 (A decreasing transform)**.** Let $X \sim \text{Uniform}(0,1)$. The waiting odds are recorded as $Y = -\ln X$. Find the density of Y.

Which rule, and why? $g(x) = -\ln x$ is strictly *decreasing* on $(0,1)$. The Jacobian formula covers decreasing g too — the absolute value on the derivative takes care of the sign.

Solution.

Step 1: inverse and derivative. Solving $y = -\ln x$ gives $x = g^{-1}(y) = e^{-y}$, and $\frac{d}{dy}e^{-y} = -e^{-y}$. As x runs over $(0,1)$, y runs over $(0,\infty)$.

Step 2: apply the formula. For $y > 0$,

$$f_Y(y) = f_X(e^{-y}) \cdot \left|-e^{-y}\right| = 1 \cdot e^{-y} = e^{-y}.$$

Step 3: identify. So $Y \sim \text{Exp}(1)$ — this is the standard recipe for generating an exponential from a uniform random number.

$$f_Y(y) = e^{-y} \quad (y > 0), \qquad Y \sim \text{Exp}(1).$$

Example 3.7.10 (Affine scaling of a normal)**.** Let $X \sim \text{N}(0,1)$. Show that $Y = \sigma X + \mu$ (with $\sigma > 0$) is $\text{N}(\mu, \sigma^2)$.

Which rule, and why? $g(x) = \sigma x + \mu$ is strictly increasing and affine; the Jacobian formula turns the standard normal into the general normal in two lines.

Solution.

Step 1: inverse and derivative. $x = g^{-1}(y) = \dfrac{y-\mu}{\sigma}$, so $\frac{d}{dy}g^{-1}(y) = \dfrac{1}{\sigma}$.

Step 2: apply the formula.

$$f_Y(y) = \frac{1}{\sqrt{2\pi}} \exp\Big(-\tfrac{1}{2}\Big(\tfrac{y-\mu}{\sigma}\Big)^2\Big) \cdot \frac{1}{\sigma} = \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{(y-\mu)^2}{2\sigma^2}\Big).$$

This is exactly the $\text{N}(\mu, \sigma^2)$ density, confirming the location-scale structure of the normal family.

3.7.3 Sums of independent variables: convolution

The cdf and Jacobian methods both transform a *single* variable. To combine *two* independent variables additively we slice the event $\{X + Y \le z\}$ over all ways the total z can be split. The result is the convolution integral (or sum).

Theorem 3.7.11 (Convolution formula)**.** *Let X and Y be independent.*

- Discrete: *if X, Y take integer values, then $Z = X + Y$ has pmf* $p_Z(k) = \displaystyle\sum_j p_X(j)\, p_Y(k-j)$.
- Continuous: *if X, Y have densities, then $Z = X + Y$ has density* $f_Z(z) = \displaystyle\int_{-\infty}^{\infty} f_X(x)\, f_Y(z-x)\, \mathrm{d}x$.

Proof sketch (continuous case). By independence the joint density factors as $f_X(x)f_Y(y)$. Then

$$F_Z(z) = \mathbb{P}(X + Y \le z) = \iint_{x+y\le z} f_X(x) f_Y(y)\, \mathrm{d}y\, \mathrm{d}x = \int_{-\infty}^{\infty} f_X(x)\, F_Y(z-x)\, \mathrm{d}x.$$

Differentiating under the integral in z replaces $F_Y(z-x)$ by $f_Y(z-x)$, giving the stated formula. The only subtlety in practice is the *limits*: keep x inside the support of X and $z - x$ inside the support of Y simultaneously. □

Example 3.7.12 (Sum of two dice rolls (discrete convolution))**.** Roll two fair four-sided dice independently, X, Y each uniform on $\{1,2,3,4\}$. Find the pmf of the total $Z = X + Y$.

Which rule, and why? A *sum of two independent* variables, so convolution; here the discrete sum $p_Z(k) = \sum_j p_X(j)p_Y(k-j)$ just counts the ordered pairs $(j, k-j)$ that land on each total.

Solution.

Step 1: count compatible pairs. Each die value has probability $\frac{1}{4}$, so each ordered pair has probability $\frac{1}{16}$. For total k we need $1 \le j \le 4$ and $1 \le k-j \le 4$; the number of such j is $p_Z(k) \cdot 16$:

k	2	3	4	5	6	7	8
#pairs	1	2	3	4	3	2	1

Step 2: divide by 16.

$$p_Z(k) = \frac{\#\text{pairs}}{16}, \qquad \big(p_Z(2), \dots, p_Z(8)\big) = \tfrac{1}{16}(1, 2, 3, 4, 3, 2, 1).$$

Step 3: check. The counts sum to $1+2+3+4+3+2+1 = 16$, so $\sum_k p_Z(k) = 1$. The distribution is the discrete "triangle" peaking at the central total 5.

Example 3.7.13 (Sum of two independent uniforms (the triangular law))**.** Two independent rounding errors $X, Y \sim \text{Uniform}(0, 1)$ accumulate. Find the density of $Z = X + Y$.

Which rule, and why? Sum of two independent continuous variables, so the convolution integral; the careful part is the overlap of the supports, which makes f_Z piecewise.

Solution.

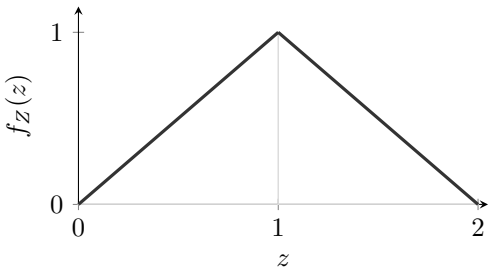
Step 1: set up the integral. $f_X(x) = 1$ on $(0, 1)$ and $f_Y(z-x) = 1$ when $0 < z - x < 1$, i.e. $z - 1 < x < z$. Intersect with $0 < x < 1$:

$$f_Z(z) = \int_{\max(0,\, z-1)}^{\min(1,\, z)} 1 \, \mathrm{d}x = \min(1, z) - \max(0, z-1).$$

Step 2: split into regimes. The support is $0 < z < 2$:

$$f_Z(z) = \begin{cases} z, & 0 < z \le 1, \\ 2 - z, & 1 < z < 2, \\ 0, & \text{otherwise.} \end{cases}$$

Step 3: check. Area $= \int_0^1 z \, \mathrm{d}z + \int_1^2 (2-z) \, \mathrm{d}z = \frac{1}{2} + \frac{1}{2} = 1$. The density is the symmetric "tent" peaking at $z = 1$, shown below.



Example 3.7.14 (Sum of two iid exponentials (a gamma))**.** Two independent service stages each take $\text{Exp}(\lambda)$ time, $f(t) = \lambda e^{-\lambda t}$ for $t \ge 0$. Find the density of the total time $Z = X + Y$.

Which rule, and why? Sum of two independent variables on $[0, \infty)$, so convolution; the support constraint here is simply $0 < x < z$ (both pieces must be nonnegative).

Solution.

Step 1: set up with the support constraint. We need $x > 0$ and $z - x > 0$, i.e. $0 < x < z$. For $z > 0$,

$$f_Z(z) = \int_0^z \lambda e^{-\lambda x}\, \lambda e^{-\lambda(z-x)}\, \mathrm{d}x = \lambda^2 e^{-\lambda z} \int_0^z \mathrm{d}x.$$

Step 2: integrate. The exponentials combine to the constant $e^{-\lambda z}$ (the x cancels), so $\int_0^z \mathrm{d}x = z$ and

$$f_Z(z) = \lambda^2 z\, e^{-\lambda z}, \qquad z > 0.$$

Step 3: identify and check. This is the Gamma$(2, \lambda)$ density. Its integral is $\int_0^\infty \lambda^2 z e^{-\lambda z}\, \mathrm{d}z = \lambda^2 \cdot \frac{1}{\lambda^2} = 1$ (using $\int_0^\infty z e^{-\lambda z}\, \mathrm{d}z = 1/\lambda^2$). A sum of n iid Exp(λ) is Gamma(n, λ).

$$f_Z(z) = \lambda^2 z\, e^{-\lambda z} \quad (z > 0).$$

Example 3.7.15 (Sum of two independent Poissons)**.** Calls to a help desk arrive from two independent regions, $X \sim \text{Poisson}(2)$ and $Y \sim \text{Poisson}(3)$. Find the distribution of the combined count $Z = X + Y$.

Which rule, and why? Sum of two independent *discrete* variables, so the discrete convolution $p_Z(k) = \sum_j p_X(j) p_Y(k-j)$; the binomial theorem collapses the sum.

Solution.

Step 1: write the convolution. With $p_X(j) = e^{-2}2^j/j!$ and $p_Y(k-j) = e^{-3}3^{k-j}/(k-j)!$, for $k \geq 0$,

$$p_Z(k) = \sum_{j=0}^{k} \frac{e^{-2}2^j}{j!} \cdot \frac{e^{-3}3^{k-j}}{(k-j)!} = \frac{e^{-5}}{k!} \sum_{j=0}^{k} \binom{k}{j} 2^j 3^{k-j}.$$

Step 2: apply the binomial theorem. The sum is $(2+3)^k = 5^k$, so

$$p_Z(k) = \frac{e^{-5}5^k}{k!}.$$

Step 3: identify. That is the Poisson(5) pmf: a sum of independent Poissons is Poisson with the rates added, $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$.

$$Z \sim \text{Poisson}(5).$$

Example 3.7.16 (Sum of two independent normals)**.** Two independent measurement errors are $X \sim \text{N}(0, \sigma_1^2)$ and $Y \sim \text{N}(0, \sigma_2^2)$. State the distribution of $Z = X + Y$ and verify the mean and variance.

Which rule, and why? A sum of independents again; convolving two Gaussians is a standard integral, but the cleaner argument is that means and variances of independent sums add (the full convolution is carried out with moment-generating functions in the MGF section).

Solution.

Step 1: mean. $\mathbb{E}(Z) = \mathbb{E}(X) + \mathbb{E}(Y) = 0 + 0 = 0$ by linearity of expectation, which holds with or without independence.

Step 2: variance. Because X and Y are independent, $\text{Cov}(X, Y) = 0$, so

$$\text{Var}(Z) = \text{Var}(X) + \text{Var}(Y) = \sigma_1^2 + \sigma_2^2.$$

Step 3: identify the family. The convolution of two normal densities is again a normal density, so

$$Z \sim \text{N}\big(0,\ \sigma_1^2 + \sigma_2^2\big).$$

More generally $\text{N}(\mu_1, \sigma_1^2) + \text{N}(\mu_2, \sigma_2^2) = \text{N}(\mu_1 + \mu_2,\ \sigma_1^2 + \sigma_2^2)$ for independent summands.

Remark. The pattern across the last four examples is worth memorising as a table of **closure properties**: a sum of independent variables stays in the same family for the Poisson (rates add), the normal (means and variances add), and the gamma with a common rate ($\text{Gamma}(\alpha_1, \lambda) + \text{Gamma}(\alpha_2, \lambda) = \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$, which made the two exponentials a Gamma$(2, \lambda)$). It does *not* stay uniform — two uniforms gave a triangle, not a wider uniform.

3.8 Further Properties of the Mean and Variance

Sections 4 and 5 handled one variable at a time. Here we collect the rules that govern *sums* of random variables. Two facts do almost all the work. Expectation is always linear: $\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y)$ with no assumptions whatsoever. Variance is *not* additive in general — it acquires a cross term $2\operatorname{Cov}(X,Y)$ measuring how the two variables move together — but that cross term vanishes when the variables are independent. From these two facts flow the covariance, the correlation coefficient, and the mean and variance of a sample mean, which underlie nearly all of statistics.

Which rule of this section does the problem need?

- Asked for the **mean of a sum or of a linear combination** $aX+bY+c$? ⇒ **linearity of expectation**: it holds with *no* independence assumption, so use it freely even when X and Y are dependent.
- Asked for $\mathbb{E}(XY)$, or to test whether the product factors? ⇒ $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ **only when** $X \perp\!\!\!\perp Y$. If they are dependent you must compute $\mathbb{E}(XY)$ directly from the joint distribution, then read off the **covariance** $\operatorname{Cov}(X,Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$.
- Asked how strongly two variables **move together**, on a scale free of units? ⇒ the **correlation** $\operatorname{Corr}(X,Y) = \operatorname{Cov}(X,Y)/[\operatorname{SD}(X)\operatorname{SD}(Y)]$, which always lies in $[-1,1]$.
- Asked for the **variance of a sum** $X+Y$ (or of $aX+bY$)? ⇒ $\operatorname{Var}(X+Y) = \operatorname{Var}(X)+\operatorname{Var}(Y)+2\operatorname{Cov}(X,Y)$. If $X \perp\!\!\!\perp Y$ the covariance is 0 and variances simply add; *do not* drop the cross term otherwise.
- Asked for the mean or variance of an **average** $\bar{X} = \frac{1}{n}\sum X_i$ of independent identically distributed observations? ⇒ the **sample-mean** formulas $\mathbb{E}(\bar{X}) = \mu$ and $\operatorname{Var}(\bar{X}) = \sigma^2/n$.

Why competing rules do not apply. Independence is needed for $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ and for variances to add, but it is *never* needed for expectations of *sums*. Conversely, $\operatorname{Cov} = 0$ does not imply independence — it is a weaker condition — so do not conclude $X \perp\!\!\!\perp Y$ merely because their covariance vanishes.

3.8.1 Linearity of expectation

Theorem 3.8.1 (Expectation of a sum)**.** *For any random variables X and Y on the same sample space, regardless of whether they are independent,*

$$\mathbb{E}(X+Y) = \mathbb{E}(X) + \mathbb{E}(Y),$$

and more generally $\mathbb{E}(a_1X_1 + \cdots + a_nX_n + b) = a_1\mathbb{E}(X_1) + \cdots + a_n\mathbb{E}(X_n) + b$ for constants a_i, b, provided each $\mathbb{E}(X_i)$ exists.

Proof. Take the continuous case with joint density $f(x,y)$; the discrete case replaces integrals by sums. By the law of the unconscious statistician applied to $g(x,y) = x+y$,

$$\mathbb{E}(X+Y) = \iint (x+y)\, f(x,y)\,\mathrm{d}x\,\mathrm{d}y = \iint x\, f(x,y)\,\mathrm{d}x\,\mathrm{d}y + \iint y\, f(x,y)\,\mathrm{d}x\,\mathrm{d}y.$$

Integrating out the unused variable turns the first term into $\int x f_X(x)\,\mathrm{d}x = \mathbb{E}(X)$ and the second into $\int y f_Y(y)\,\mathrm{d}y = \mathbb{E}(Y)$. The general linear statement follows by induction and by $\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$ from Section 4. □

Example 3.8.2 (Two loaded dice, summed)**.** A red die and a blue die are weighted so that each shows a 6 with probability 1/4 and each of $1,\ldots,5$ with probability 3/20. The dice are rolled together (they need not be independent — the weighting machine might link them). Let T be the total of the two faces. Find $\mathbb{E}(T)$.

Which rule, and why? We want the mean of a *sum* $T = R + B$. Linearity of expectation applies with no independence assumption, so we never need the joint distribution — only each die's own mean.

Solution.

Step 1: mean of one die. With $\mathbb{P}(k) = 3/20$ for $k = 1, \dots, 5$ and $\mathbb{P}(6) = 1/4 = 5/20$,

$$\mathbb{E}(R) = \frac{3}{20}(1+2+3+4+5) + 6 \cdot \frac{5}{20} = \frac{3}{20} \cdot 15 + \frac{30}{20} = \frac{45+30}{20} = \frac{75}{20} = \frac{15}{4}.$$

Step 2: add the two means. By symmetry $\mathbb{E}(B) = 15/4$ as well, so linearity gives

$$\mathbb{E}(T) = \mathbb{E}(R) + \mathbb{E}(B) = \frac{15}{4} + \frac{15}{4} = \frac{15}{2} = 7.5.$$

Example 3.8.3 (Expected number of matches: indicators)**.** Five distinct letters are placed at random, one per envelope, into five pre-addressed envelopes (a uniformly random permutation). Let M be the number of letters that land in their own correct envelope. Find $\mathbb{E}(M)$.

Which rule, and why? M is a sum of dependent yes/no variables. Computing the pmf of M directly is painful, but linearity lets us add the (easy) means of the indicators even though the indicators are *not* independent.

Solution.

Step 1: write M as a sum of indicators. Let $I_j = \mathbb{1}\{\text{letter } j \text{ is in its own envelope}\}$, so $M = I_1 + \cdots + I_5$.

Step 2: mean of one indicator. Letter j is equally likely to occupy any of the five envelopes, so $\mathbb{P}(I_j = 1) = 1/5$ and $\mathbb{E}(I_j) = 1/5$.

Step 3: sum. The I_j are dependent, but linearity does not care:

$$\mathbb{E}(M) = \sum_{j=1}^{5} \mathbb{E}(I_j) = 5 \cdot \frac{1}{5} = 1.$$

On average exactly one letter lands correctly, for *any* number of envelopes.

Example 3.8.4 (Mean of a binomial by indicators)**.** A factory line produces chips that are defective independently with probability $p = 0.03$. In a batch of $n = 200$ chips, let D be the number of defectives. Use linearity to find $\mathbb{E}(D)$ without summing the binomial pmf.

Which rule, and why? $D \sim \text{Bin}(200, 0.03)$ counts successes, so it is a sum of n Bernoulli indicators. Linearity turns the mean into n copies of one indicator's mean — much faster than $\sum_k k\binom{n}{k}p^k(1-p)^{n-k}$.

Solution.

Step 1: decompose. Write $D = \sum_{i=1}^{200} I_i$ where $I_i = \mathbb{1}\{\text{chip } i \text{ defective}\}$ and $\mathbb{E}(I_i) = p = 0.03$.

Step 2: add.

$$\mathbb{E}(D) = \sum_{i=1}^{200} \mathbb{E}(I_i) = 200(0.03) = 6.$$

This reproduces the stated binomial mean np.

Example 3.8.5 (A linear combination of dependent revenues)**.** A cafe's morning revenue X and afternoon revenue Y (in hundreds of dollars) are correlated through the day's weather, with $\mathbb{E}(X) = 4$ and $\mathbb{E}(Y) = 7$. The owner's daily profit is $P = X + Y - 3$ (a fixed \$300 of costs). Find $\mathbb{E}(P)$.

Which rule, and why? We need the mean of a linear combination. Linearity handles the constant shift and the sum together; the correlation between X and Y is irrelevant to the *mean*.

Solution.

Step 1: apply linearity.

$$\mathbb{E}(P) = \mathbb{E}(X) + \mathbb{E}(Y) - 3 = 4 + 7 - 3 = 8.$$

So expected profit is \$800, no matter how the weather links X and Y.

3.8.2 Expectation of a product; covariance

Theorem 3.8.6 (Product rule under independence)**.** *If $X \perp\!\!\!\perp Y$ and both means exist, then $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$.*

Proof. Independence factors the joint density, $f(x,y) = f_X(x)f_Y(y)$. Hence

$$\mathbb{E}(XY) = \iint xy\, f_X(x)f_Y(y)\,\mathrm{d}x\,\mathrm{d}y = \Big(\int x f_X(x)\,\mathrm{d}x\Big)\Big(\int y f_Y(y)\,\mathrm{d}y\Big) = \mathbb{E}(X)\mathbb{E}(Y),$$

the double integral splitting because each factor depends on only one variable. The discrete case is identical with sums. □

Definition 3.8.7 (Covariance)**.** The **covariance** of X and Y is

$$\mathrm{Cov}(X,Y) = \mathbb{E}\big[(X-\mathbb{E}X)(Y-\mathbb{E}Y)\big] = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y).$$

It is positive when X and Y tend to lie on the same side of their means, negative when on opposite sides. Note $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$.

The two forms are equal by expanding the product and using linearity: $\mathbb{E}[(X-\mu_X)(Y-\mu_Y)] = \mathbb{E}(XY) - \mu_X\mathbb{E}(Y) - \mu_Y\mathbb{E}(X) + \mu_X\mu_Y = \mathbb{E}(XY) - \mu_X\mu_Y$. The right-hand "shortcut" form $\mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$ is almost always the one to compute with.

Proposition 3.8.8 (Properties of covariance)**.** *For random variables and constants a, b, c, d:*

(i) Symmetry: $\mathrm{Cov}(X,Y) = \mathrm{Cov}(Y,X)$.

(ii) Bilinearity: $\mathrm{Cov}(aX+b,\, cY+d) = ac\,\mathrm{Cov}(X,Y)$*; constants shift nothing and scalars pull out.*

(iii) Additivity in a slot: $\mathrm{Cov}(X+Y,\, Z) = \mathrm{Cov}(X,Z) + \mathrm{Cov}(Y,Z)$.

(iv) Independence ⇒ zero: *if $X \perp\!\!\!\perp Y$ then* $\mathrm{Cov}(X,Y) = 0$*. The converse is* false.

Proof. (i) is clear from symmetry of the product. (ii): adding constants does not change $X - \mathbb{E}X$ or $Y - \mathbb{E}Y$, and $(aX - a\mathbb{E}X)(cY - c\mathbb{E}Y) = ac(X-\mathbb{E}X)(Y-\mathbb{E}Y)$; take expectations. (iii): expand $\mathbb{E}[(X+Y)Z] - \mathbb{E}(X+Y)\mathbb{E}(Z)$ by linearity. (iv): independence gives $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$, so the shortcut form is 0. □

Remark. That the converse of (iv) fails is worth a concrete witness. Let X take values $-1, 0, 1$ each with probability $1/3$ and set $Y = X^2$. Then $\mathbb{E}(X) = 0$ and $\mathbb{E}(XY) = \mathbb{E}(X^3) = 0$, so $\mathrm{Cov}(X,Y) = 0$; yet Y is completely determined by X, the opposite of independent. Zero covariance only rules out a *linear* relationship.

Example 3.8.9 (Covariance from a joint table)**.** A vending machine dispenses snacks (X) and drinks (Y) sold in the same hour, with joint pmf $p(x,y) = \mathbb{P}(X=x, Y=y)$ given below.

$x \backslash y$	0	1	2	$p_X(x)$
0	0.10	0.05	0.05	0.20
1	0.10	0.20	0.10	0.40
2	0.05	0.15	0.20	0.40
$p_Y(y)$	0.25	0.40	0.35	1.00

Find $\mathrm{Cov}(X,Y)$.

Which rule, and why? We have a discrete joint distribution and want how X, Y co-move, so we need the covariance. We compute $\mathbb{E}(XY)$ over the table and the two marginal means, then subtract — the shortcut form is far easier than averaging $(X-\mu_X)(Y-\mu_Y)$ cell by cell.

Solution.

Step 1: marginal means. From the margins,

$$\mathbb{E}(X) = 0(0.20) + 1(0.40) + 2(0.40) = 1.20, \qquad \mathbb{E}(Y) = 0(0.25) + 1(0.40) + 2(0.35) = 1.10.$$

Step 2: $\mathbb{E}(XY)$ *over the table.* Only cells with $x \geq 1$ and $y \geq 1$ contribute:

$$\begin{aligned}\mathbb{E}(XY) &= (1)(1)(0.20) + (1)(2)(0.10) + (2)(1)(0.15) + (2)(2)(0.20) \\ &= 0.20 + 0.20 + 0.30 + 0.80 = 1.50.\end{aligned}$$

Step 3: subtract.

$$\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y) = 1.50 - (1.20)(1.10) = 1.50 - 1.32 = 0.18.$$

The positive sign says busy snack hours tend to be busy drink hours.

Example 3.8.10 (A product that does not factor)**.** Draw two cards without replacement from a tiny deck of three cards numbered $1, 2, 3$. Let X be the first number drawn and Y the second. Find $\mathrm{Cov}(X, Y)$ and confirm X and Y are dependent.

Which rule, and why? Sampling *without* replacement makes the draws dependent, so $\mathbb{E}(XY) \neq \mathbb{E}(X)\mathbb{E}(Y)$ in general; we must compute $\mathbb{E}(XY)$ from the joint distribution and then use the shortcut covariance form.

Solution.

Step 1: the joint distribution. The six ordered pairs (x, y) with $x \neq y$ are equally likely, each with probability $1/6$. By symmetry each of X and Y is uniform on $\{1, 2, 3\}$, so

$$\mathbb{E}(X) = \mathbb{E}(Y) = \frac{1 + 2 + 3}{3} = 2.$$

Step 2: $\mathbb{E}(XY)$. Sum xy over the six pairs $(1,2), (2,1), (1,3), (3,1), (2,3), (3,2)$:

$$\mathbb{E}(XY) = \frac{1}{6}\big(2 + 2 + 3 + 3 + 6 + 6\big) = \frac{22}{6} = \frac{11}{3}.$$

Step 3: covariance.

$$\mathrm{Cov}(X, Y) = \frac{11}{3} - 2 \cdot 2 = \frac{11}{3} - \frac{12}{3} = -\frac{1}{3}.$$

The negative value is intuitive: a large first card leaves smaller numbers for the second. Since $\mathrm{Cov} \neq 0$, the variables cannot be independent.

Example 3.8.11 (Independence forces the product to factor)**.** A fair coin is flipped and, independently, a fair four-sided die is rolled. Let $X = 1$ if the coin shows heads and $X = 0$ otherwise, and let Y be the die face $(1, 2, 3, 4)$. Verify $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$ and hence $\mathrm{Cov}(X, Y) = 0$.

Which rule, and why? The two devices are stated independent, so the product rule must hold; this is a check that the machinery is consistent, and a reminder that independence delivers $\mathrm{Cov} = 0$ for free.

Solution.

Step 1: separate means. $\mathbb{E}(X) = \frac{1}{2}$ and $\mathbb{E}(Y) = \frac{1}{4}(1 + 2 + 3 + 4) = \frac{10}{4} = 2.5$.

Step 2: $\mathbb{E}(XY)$ *directly.* Because $X \in \{0, 1\}$, $XY = Y$ exactly when $X = 1$:

$$\mathbb{E}(XY) = \sum_{y=1}^{4} (1)(y)\, \mathbb{P}(X = 1)\mathbb{P}(Y = y) = \mathbb{P}(X = 1)\, \mathbb{E}(Y) = \tfrac{1}{2}(2.5) = 1.25.$$

Step 3: compare. $\mathbb{E}(X)\mathbb{E}(Y) = \frac{1}{2}(2.5) = 1.25 = \mathbb{E}(XY)$, so

$$\mathrm{Cov}(X, Y) = 1.25 - 1.25 = 0,$$

exactly as independence predicts.

Example 3.8.12 (Covariance of a variable with a sum)**.** Let U and V be independent with $\mathrm{Var}(U) = 4$ and $\mathrm{Var}(V) = 9$. Define $X = U + V$ and $Y = U - V$. Find $\mathrm{Cov}(X, Y)$ and decide whether X and Y are correlated.

Which rule, and why? Both X and Y are linear combinations of the *same* underlying variables, so use bilinearity and additivity of covariance rather than any joint table; independence of U, V kills the cross covariances.

Solution.

Step 1: expand by bilinearity.

$$\mathrm{Cov}(X,Y) = \mathrm{Cov}(U+V,\, U-V) = \mathrm{Cov}(U,U) - \mathrm{Cov}(U,V) + \mathrm{Cov}(V,U) - \mathrm{Cov}(V,V).$$

Step 2: simplify. Since $U \perp\!\!\!\perp V$, the mixed terms vanish, and $\mathrm{Cov}(U,U) = \mathrm{Var}(U)$, $\mathrm{Cov}(V,V) = \mathrm{Var}(V)$:

$$\mathrm{Cov}(X,Y) = \mathrm{Var}(U) - \mathrm{Var}(V) = 4 - 9 = -5.$$

The covariance is nonzero, so although X and Y are each built from independent pieces, they are themselves *negatively correlated* through their shared dependence on U and V.

3.8.3 The correlation coefficient

Definition 3.8.13 (Correlation)**.** The **correlation coefficient** of X and Y (when both variances are positive) is the standardised covariance

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\mathrm{SD}(X)\,\mathrm{SD}(Y)} = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}\,.$$

It is dimensionless and unchanged by a change of units.

Proposition 3.8.14 (Range of the correlation)**.** *For any X, Y with positive variances, $-1 \le \mathrm{Corr}(X,Y) \le 1$. Moreover $\mathrm{Corr}(X,Y) = \pm 1$ if and only if $Y = aX + b$ for constants a, b with a having the same sign as the correlation (a perfect straight-line relationship).*

Proof. Let $\rho = \mathrm{Corr}(X,Y)$ and standardise: $X^* = (X - \mu_X)/\sigma_X$, $Y^* = (Y - \mu_Y)/\sigma_Y$, so each has mean 0 and variance 1, and $\mathbb{E}(X^*Y^*) = \rho$. Then

$$0 \le \mathrm{Var}(X^* \pm Y^*) = \mathrm{Var}(X^*) + \mathrm{Var}(Y^*) \pm 2\,\mathrm{Cov}(X^*, Y^*) = 2 \pm 2\rho,$$

giving $-1 \le \rho \le 1$. Equality forces a variance of zero, i.e. $X^* \pm Y^*$ is constant, which rearranges to $Y = aX + b$. □

Example 3.8.15 (Correlation from covariance and variances)**.** For the snack/drink machine of the earlier table we found $\mathrm{Cov}(X,Y) = 0.18$. Suppose also $\mathrm{Var}(X) = 0.56$ and $\mathrm{Var}(Y) = 0.59$. Find $\mathrm{Corr}(X,Y)$.

Which rule, and why? We are asked for the unit-free strength of association, which is exactly the correlation: divide the covariance by the product of the standard deviations.

Solution.

Step 1: standard deviations. $\mathrm{SD}(X) = \sqrt{0.56} \approx 0.7483$ and $\mathrm{SD}(Y) = \sqrt{0.59} \approx 0.7681$.

Step 2: divide.

$$\mathrm{Corr}(X,Y) = \frac{0.18}{0.7483 \times 0.7681} \approx \frac{0.18}{0.5748} \approx 0.313.$$

A modest positive correlation: snack and drink sales rise together, but loosely.

Example 3.8.16 (Correlation is unaffected by units)**.** Daily temperature X (in degrees Celsius) and ice-cream sales Y (in dollars) satisfy $\mathrm{Cov}(X,Y) = 30$, $\mathrm{Var}(X) = 25$, $\mathrm{Var}(Y) = 400$. A colleague re-expresses temperature in Fahrenheit, $W = \frac{9}{5}X + 32$, and sales in cents, $Z = 100Y$. Show $\mathrm{Corr}(W,Z) = \mathrm{Corr}(X,Y)$.

Which rule, and why? Both new variables are affine rescalings of the originals. Covariance scales by the product of the slopes while each standard deviation scales by its own slope, so the ratio — the correlation — is invariant. This is the whole point of standardising.

Solution.

Step 1: original correlation.

$$\operatorname{Corr}(X,Y) = \frac{30}{\sqrt{25}\sqrt{400}} = \frac{30}{5\cdot 20} = \frac{30}{100} = 0.30.$$

Step 2: transformed quantities. With slopes $a = 9/5$ and $c = 100$,

$$\operatorname{Cov}(W,Z) = ac\,\operatorname{Cov}(X,Y) = \tfrac{9}{5}(100)(30), \quad \operatorname{SD}(W) = \tfrac{9}{5}\operatorname{SD}(X), \quad \operatorname{SD}(Z) = 100\,\operatorname{SD}(Y).$$

Step 3: form the ratio.

$$\operatorname{Corr}(W,Z) = \frac{\frac{9}{5}\cdot 100\cdot \operatorname{Cov}(X,Y)}{\frac{9}{5}\operatorname{SD}(X)\cdot 100\,\operatorname{SD}(Y)} = \frac{\operatorname{Cov}(X,Y)}{\operatorname{SD}(X)\operatorname{SD}(Y)} = \operatorname{Corr}(X,Y) = 0.30.$$

The positive slopes leave even the sign untouched.

Example 3.8.17 (A perfect linear relationship)**.** Let X be any random variable with $\operatorname{Var}(X) = 9 > 0$, and set $Y = 5 - 2X$. Find $\operatorname{Corr}(X,Y)$.

Which rule, and why? Y is an exact affine function of X, so we expect Corr $= \pm 1$; the sign should follow the slope -2. We confirm with the bilinearity rules.

Solution.

Step 1: covariance and variance.

$$\operatorname{Cov}(X,Y) = \operatorname{Cov}(X,\,5-2X) = -2\,\operatorname{Cov}(X,X) = -2\operatorname{Var}(X) = -18,$$

$$\operatorname{Var}(Y) = (-2)^2\operatorname{Var}(X) = 4(9) = 36.$$

Step 2: correlation.

$$\operatorname{Corr}(X,Y) = \frac{-18}{\sqrt{9}\sqrt{36}} = \frac{-18}{3\cdot 6} = \frac{-18}{18} = -1.$$

A perfect negative correlation, as the negative slope demands.

3.8.4 Variance of a sum

Theorem 3.8.18 (Variance of a sum)**.** *For any X, Y with finite variances and constants a, b,*

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y),$$

and more generally

$$\operatorname{Var}(aX+bY) = a^2\operatorname{Var}(X) + b^2\operatorname{Var}(Y) + 2ab\,\operatorname{Cov}(X,Y).$$

If $X \perp\!\!\!\perp Y$ (or merely $\operatorname{Cov}(X,Y) = 0$) the cross term drops and $\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y)$. For n pairwise-uncorrelated variables, $\operatorname{Var}\big(\sum_i X_i\big) = \sum_i \operatorname{Var}(X_i)$.

Proof. Write $\mu_X = \mathbb{E}X, \mu_Y = \mathbb{E}Y$. Then $\mathbb{E}(aX+bY) = a\mu_X + b\mu_Y$, so

$$\operatorname{Var}(aX+bY) = \mathbb{E}\big[(a(X-\mu_X) + b(Y-\mu_Y))^2\big].$$

Expand the square and take expectations term by term: $a^2\mathbb{E}[(X-\mu_X)^2] + b^2\mathbb{E}[(Y-\mu_Y)^2] + 2ab\,\mathbb{E}[(X-\mu_X)(Y-\mu_Y)]$, which is exactly $a^2\operatorname{Var}(X) + b^2\operatorname{Var}(Y) + 2ab\operatorname{Cov}(X,Y)$. Setting $a = b = 1$ gives the first identity; Cov $= 0$ removes the last term. □

Remark. The slopes in the cross term are $2ab$, *not* $2|ab|$. Thus for a *difference* $X - Y$ the cross term flips sign: $\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y)$. Subtraction does not reduce variability when the variables are independent — you still *add* the variances.

Example 3.8.19 (Variance of a sum of independents)**.** A morning commute has a bus leg with mean time 20 min and variance 16 (min^2) and a walking leg with mean 8 min and variance 9, the two legs independent. Find the mean and variance of the total trip time $T = B + W$ and its standard deviation.

Which rule, and why? Two independent variables are added, so the means add and — crucially because Cov $= 0$ — the variances add with no cross term.

Solution.

Step 1: mean. $\mathbb{E}(T) = \mathbb{E}(B) + \mathbb{E}(W) = 20 + 8 = 28$ minutes.

Step 2: variance. Independence gives

$$\mathrm{Var}(T) = \mathrm{Var}(B) + \mathrm{Var}(W) = 16 + 9 = 25 \quad (\mathrm{min}^2).$$

Step 3: standard deviation. $\mathrm{SD}(T) = \sqrt{25} = 5$ minutes.

Example 3.8.20 (The cross term cannot be ignored)**.** For the snack/drink machine, $\mathrm{Var}(X) = 0.56$, $\mathrm{Var}(Y) = 0.59$, and $\mathrm{Cov}(X, Y) = 0.18$. Find the variance of the combined item count $X + Y$, and compare with the (wrong) answer that ignores dependence.

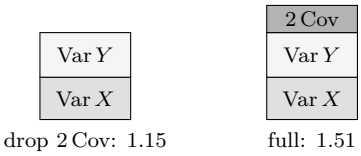
Which rule, and why? The two counts are positively dependent, so the full sum formula is required; dropping $2\,\mathrm{Cov}$ would understate the true variability.

Solution.

Step 1: full formula.

$$\mathrm{Var}(X + Y) = 0.56 + 0.59 + 2(0.18) = 1.15 + 0.36 = 1.51.$$

Step 2: the naive value. Ignoring covariance would give only $0.56 + 0.59 = 1.15$, understating the spread by the 0.36 contributed by positive co-movement. The picture below contrasts the two.



Example 3.8.21 (Variance of a difference)**.** Two independent measurements of the same rod are taken with calibrated gauges. Gauge A reads with variance 0.04 mm^2 and gauge B with variance 0.09 mm^2. The discrepancy $D = A - B$ is used to check calibration. Find $\mathrm{Var}(D)$.

Which rule, and why? This is a *difference* of independents. The cross term is $-2\,\mathrm{Cov}$, but Cov $= 0$ here, so we still add the two variances — subtraction does not cancel variability.

Solution.

Step 1: apply the formula with $a = 1, b = -1$.

$$\mathrm{Var}(D) = \mathrm{Var}(A) + (-1)^2\,\mathrm{Var}(B) - 2(1)(1)\,\mathrm{Cov}(A, B) = 0.04 + 0.09 - 0 = 0.13\ \mathrm{mm}^2.$$

Step 2: standard deviation. $\mathrm{SD}(D) = \sqrt{0.13} \approx 0.36$ mm. Even though we subtracted, the discrepancy is *more* variable than either gauge alone.

Example 3.8.22 (A weighted portfolio)**.** An investor splits money between two assets with returns X and Y, holding fraction 0.6 in the first and 0.4 in the second, so the portfolio return is $R = 0.6X + 0.4Y$. The returns have $\mathrm{Var}(X) = 0.10$, $\mathrm{Var}(Y) = 0.04$, and $\mathrm{Corr}(X, Y) = -0.5$. Find $\mathrm{Var}(R)$.

Which rule, and why? A linear combination $aX+bY$ with $a=0.6, b=0.4$. We need the general two-coefficient variance rule, and because correlation is given (not covariance) we first recover Cov from Corr.

Solution.

Step 1: covariance from correlation.

$$\mathrm{Cov}(X,Y) = \mathrm{Corr}(X,Y)\ \mathrm{SD}(X)\ \mathrm{SD}(Y) = (-0.5)\sqrt{0.10}\sqrt{0.04}.$$

Since $\sqrt{0.10} \approx 0.3162$ and $\sqrt{0.04} = 0.2$, $\mathrm{Cov}(X,Y) = (-0.5)(0.3162)(0.2) \approx -0.03162$.

Step 2: portfolio variance.

$$\begin{aligned}\mathrm{Var}(R) &= 0.6^2\,\mathrm{Var}(X) + 0.4^2\,\mathrm{Var}(Y) + 2(0.6)(0.4)\,\mathrm{Cov}(X,Y)\\ &= 0.36(0.10) + 0.16(0.04) + 0.48(-0.03162)\\ &= 0.036 + 0.0064 - 0.01518 \approx 0.0272.\end{aligned}$$

The negative correlation acts as a diversifier, pulling the portfolio variance well below the $0.036 + 0.0064 = 0.0424$ it would be if the assets were uncorrelated.

3.8.5 Mean and variance of a sample mean

Theorem 3.8.23 (Sample mean)**.** *Let $X_1, \dots, X_n$ be independent and identically distributed with common mean μ and common variance σ^2, and let $\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ be their average. Then*

$$\mathbb{E}(\bar{X}) = \mu, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n}, \qquad \mathrm{SD}(\bar{X}) = \frac{\sigma}{\sqrt{n}}.$$

Proof. By linearity $\mathbb{E}(\bar{X}) = \frac{1}{n}\sum_i \mathbb{E}(X_i) = \frac{1}{n}(n\mu) = \mu$. For the variance, the scaling rule and independence (so the variance of the sum is the sum of the variances) give

$$\mathrm{Var}(\bar{X}) = \frac{1}{n^2}\,\mathrm{Var}\Big(\sum_{i=1}^{n} X_i\Big) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{1}{n^2}(n\sigma^2) = \frac{\sigma^2}{n}.$$

Taking square roots gives $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$. □

The mean of the average matches the population mean, but its spread *shrinks* like $1/\sqrt{n}$: averaging more independent observations gives a more reliable estimate. To *halve* the standard deviation you must *quadruple* the sample size.

Example 3.8.24 (Averaging blood-pressure readings)**.** A patient's true systolic pressure is $\mu = 120$ with single-reading variance $\sigma^2 = 36$ (mmHg^2). A nurse takes $n = 9$ independent readings and averages them. Find $\mathbb{E}(\bar{X})$, $\mathrm{Var}(\bar{X})$, and $\mathrm{SD}(\bar{X})$.

Which rule, and why? An average of n iid measurements is exactly the sample-mean setting; apply $\mathbb{E}(\bar{X}) = \mu$ and $\mathrm{Var}(\bar{X}) = \sigma^2/n$ directly.

Solution.

Step 1: mean. $\mathbb{E}(\bar{X}) = 120$ mmHg.

Step 2: variance and SD.

$$\mathrm{Var}(\bar{X}) = \frac{36}{9} = 4, \qquad \mathrm{SD}(\bar{X}) = \sqrt{4} = 2 \text{ mmHg}.$$

A single reading scatters by 6 mmHg; the nine-reading average by only 2.

Example 3.8.25 (How many readings to reach a target precision?)**.** With the same single-reading variance $\sigma^2 = 36$, how large must n be for the average to have standard deviation at most 1 mmHg?

Which rule, and why? We invert the sample-mean SD formula $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$, solving for n; the $1/\sqrt{n}$ shrinkage tells us each halving of SD costs a quadrupling of n.

Solution.

Step 1: set up the inequality.

$$\frac{\sigma}{\sqrt{n}} \le 1 \iff \sqrt{n} \ge \sigma = 6 \iff n \ge 36.$$

Step 2: conclude. At least $n = 36$ readings are needed. Compared with the 9 that gave SD = 2, quadrupling the sample halved the standard deviation, as the $1/\sqrt{n}$ law predicts.

Example 3.8.26 (Total versus average)**.** A shipment contains $n = 100$ independent castings, each weighing on average $\mu = 2.5$ kg with variance $\sigma^2 = 0.16$ kg^2. Find the mean and variance of (a) the total weight $S = \sum X_i$ and (b) the average weight $\bar{X} = S/100$.

Which rule, and why? The total is a *sum* of independents (variances add), while the average divides that sum by n (variance divided by n^2). Keeping the two straight is the whole exercise — they scale oppositely in n.

Solution.

Step 1: total. By linearity and independence,

$$\mathbb{E}(S) = 100(2.5) = 250 \text{ kg}, \qquad \mathrm{Var}(S) = 100(0.16) = 16 \text{ kg}^2.$$

Step 2: average.

$$\mathbb{E}(\bar{X}) = 2.5 \text{ kg}, \qquad \mathrm{Var}(\bar{X}) = \frac{0.16}{100} = 0.0016 \text{ kg}^2.$$

The total's variance *grows* with n while the average's *shrinks* — the sum is the riskier quantity, the average the more stable one.

Example 3.8.27 (A non-identical, still-independent sum)**.** A relay race has three legs run by independent runners with leg-time variances 4, 1, and 9 (sec^2) and means 50, 48, and 55 seconds. The team time is $T = L_1 + L_2 + L_3$. Find $\mathbb{E}(T)$ and $\mathrm{Var}(T)$, and explain why the σ^2/n formula does *not* apply.

Which rule, and why? The legs are independent but *not* identically distributed, so this is a sum of independents (means add, variances add), not a sample mean. We use the additive rules directly and avoid σ^2/n, which assumes a common variance.

Solution.

Step 1: mean. $\mathbb{E}(T) = 50 + 48 + 55 = 153$ seconds.

Step 2: variance. Independence makes the cross covariances zero, so

$$\mathrm{Var}(T) = \mathrm{Var}(L_1) + \mathrm{Var}(L_2) + \mathrm{Var}(L_3) = 4 + 1 + 9 = 14 \text{ sec}^2.$$

Step 3: why not σ^2/n. That formula requires a *common* variance and an *average*; here the variances differ and we want a total, so only the general additivity of variance for uncorrelated variables applies. $\mathrm{SD}(T) = \sqrt{14} \approx 3.74$ s.

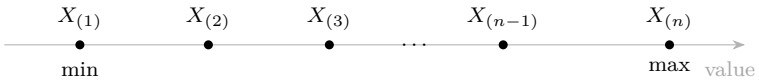
3.9 Order Statistics

Take n independent observations $X_1, \ldots, X_n$ drawn from the same parent distribution and sort them from smallest to largest. The sorted values $X_{(1)} \le X_{(2)} \le \cdots \le X_{(n)}$ are the **order statistics** of the sample: $X_{(1)}$ is the minimum, $X_{(n)}$ is the maximum, and $X_{(k)}$ is the k-th smallest. These quantities answer practical questions — when does the *first* of several components fail, how long until the *last* customer is served, what is the *median* of a batch of measurements — and although the X_i are independent, the sorted values are not, so we compute their distributions directly from the parent cdf F and density f.

Which order-statistic rule does the problem need?

- Asking about the **largest** value, "all of them are below x", or "the last one"? $\Rightarrow$ the **maximum** $X_{(n)}$. Its cdf is $F_{(n)}(x) = [F(x)]^n$, because the max is $\leq x$ exactly when *every* observation is $\leq x$ (§3.9.1).
- Asking about the **smallest** value, "at least one is below x", or "the first failure/arrival"? $\Rightarrow$ the **minimum** $X_{(1)}$, with cdf $F_{(1)}(x) = 1 - [1 - F(x)]^n$, since the min exceeds x exactly when every observation does (§3.9.1).
- Asking about the k**-th smallest** for some interior k (the second largest, the third smallest, etc.)? $\Rightarrow$ the general k**-th order statistic** density (§3.9.2).
- Asking about the **middle** value of the sample? $\Rightarrow$ the **sample median**, a special order statistic (§3.9.3).
- Asking about the **spread** from smallest to largest? $\Rightarrow$ the **sample range** $R = X_{(n)} - X_{(1)}$ (§3.9.4).
- *Why not just use the parent F?* A single X_i has cdf F, but a *sorted* value is a maximum/minimum/k-th smallest of n correlated events; you must combine the n independent observations through the counting argument, not read F off directly.

Throughout, $X_1, \ldots, X_n$ are independent and identically distributed (iid) with common cdf F and (when continuous) density f. We picture the sorted sample as n points dropped on a line and then read left to right:



3.9.1 The maximum and the minimum

Definition 3.9.1 (Sample maximum and minimum)**.** The **sample maximum** is the largest observation, $X_{(n)} = \max\{X_1, \ldots, X_n\}$, and the **sample minimum** is the smallest, $X_{(1)} = \min\{X_1, \ldots, X_n\}$.

Theorem 3.9.2 (Cdf and density of the maximum)**.** *If $X_1, \ldots, X_n$ are iid with cdf F and density f, then*

$$F_{(n)}(x) = [F(x)]^n, \qquad f_{(n)}(x) = n\,[F(x)]^{\,n-1} f(x).$$

Proof. The largest observation is at most x exactly when *all* n observations are at most x. By independence,

$$F_{(n)}(x) = \mathbb{P}(X_{(n)} \leq x) = \mathbb{P}(X_1 \leq x, \ldots, X_n \leq x) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq x) = [F(x)]^n.$$

Differentiating and using the chain rule gives $f_{(n)}(x) = n[F(x)]^{n-1} F'(x) = n[F(x)]^{n-1} f(x)$. □

Theorem 3.9.3 (Cdf and density of the minimum)**.** *Under the same assumptions,*

$$F_{(1)}(x) = 1 - [1 - F(x)]^n, \qquad f_{(1)}(x) = n\,[1 - F(x)]^{\,n-1} f(x).$$

Proof. The smallest observation exceeds x exactly when *every* observation exceeds x, so

$$\mathbb{P}(X_{(1)} > x) = \mathbb{P}(X_1 > x, \ldots, X_n > x) = [1 - F(x)]^n,$$

and $F_{(1)}(x) = 1 - \mathbb{P}(X_{(1)} > x) = 1 - [1 - F(x)]^n$. Differentiating, $f_{(1)}(x) = -n[1 - F(x)]^{n-1}\,(-f(x)) = n[1 - F(x)]^{n-1} f(x)$. □

Example 3.9.4 (Strongest of four uniform links)**.** A chain has four links whose breaking strengths (in suitable units) are iid Uniform$(0, 1)$. Let M be the strongest link's strength. Find $\mathbb{P}(M \leq 0.8)$ and the density of M.

Which rule, and why? The phrase "strongest" is the **maximum** of the four strengths, so use $F_{(n)} = F^n$. The minimum rule would answer "weakest", and the parent F alone ignores that we are taking the largest of four.

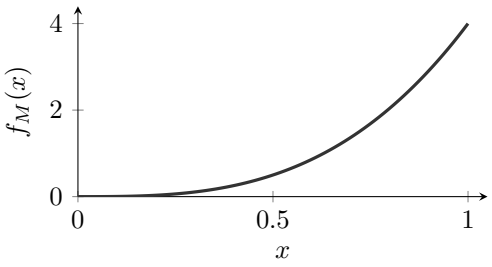
Solution.

Step 1: parent cdf. For Uniform$(0, 1)$, $F(x) = x$ on $[0, 1]$.

Step 2: cdf of the max. With $n = 4$, $F_M(x) = [F(x)]^4 = x^4$ for $0 \le x \le 1$. Hence

$$\mathbb{P}(M \le 0.8) = 0.8^4 = 0.4096.$$

Step 3: density. Differentiate: $f_M(x) = 4x^3$ on $[0, 1]$.



The density piles up near 1: the strongest of four links is usually close to the top of the range.

Example 3.9.5 (First component to fail)**.** Six identical sensors run in parallel; each lifetime (in years) is iid Exp(λ) with $\lambda = 1/5$ (mean 5 years). The array keeps working until the *first* sensor fails. Find the distribution of the time T to the first failure and its mean.

Which rule, and why? "First to fail" is the **minimum** of the six lifetimes, so use $F_{(1)} = 1 - (1 - F)^n$. Recognising the minimum of exponentials is worth memorising: it is again exponential.

Solution.

Step 1: parent. For Exp(λ), $F(x) = 1 - e^{-\lambda x}$, so $1 - F(x) = e^{-\lambda x}$ for $x \ge 0$.

Step 2: cdf of the min. With $n = 6$,

$$F_T(x) = 1 - [1 - F(x)]^6 = 1 - \left(e^{-\lambda x}\right)^6 = 1 - e^{-6\lambda x}, \qquad x \ge 0.$$

Step 3: identify the distribution. This is the cdf of an exponential with rate 6λ, so $T \sim \text{Exp}(6\lambda)$. With $\lambda = 1/5$ the rate is $6/5$ and

$$\mathbb{E}(T) = \frac{1}{6\lambda} = \frac{5}{6} \approx 0.833 \text{ years.}$$

The minimum of n iid Exp(λ) is Exp$(n\lambda)$ — failures arrive n times faster when n components race.

Example 3.9.6 (Longest of three download times)**.** Three files download independently, each taking $X_i \sim \text{Exp}(\lambda)$ minutes with $\lambda = 1$. The page is ready only when the *last* file finishes. Find $\mathbb{E}(L)$ where $L = \max(X_1, X_2, X_3)$.

Which rule, and why? "Last to finish" is the **maximum**, so use $f_{(n)} = nF^{n-1}f$. Unlike the minimum, the maximum of exponentials is *not* exponential, so we integrate.

Solution.

Step 1: density of the max. With $F(x) = 1 - e^{-x}$, $f(x) = e^{-x}$, $n = 3$,

$$f_L(x) = 3\,[1 - e^{-x}]^2 e^{-x}, \qquad x \ge 0.$$

Step 2: expand. Write $[1 - e^{-x}]^2 = 1 - 2e^{-x} + e^{-2x}$, so

$$f_L(x) = 3\big(e^{-x} - 2e^{-2x} + e^{-3x}\big).$$

Step 3: integrate term by term. Using $\int_0^\infty x\,e^{-ax}\,\mathrm{d}x = 1/a^2$,

$$\mathbb{E}(L) = 3\left(\frac{1}{1^2} - \frac{2}{2^2} + \frac{1}{3^2}\right) = 3\big(1 - \tfrac{1}{2} + \tfrac{1}{9}\big) = 3 \cdot \frac{11}{18} = \frac{11}{6} \approx 1.833 \text{ min.}$$

This matches the general fact $\mathbb{E}(X_{(n)}) = \frac{1}{\lambda}\sum_{i=1}^{n} \frac{1}{i}$ for exponentials; here $\sum_{i=1}^{3} 1/i = 1 + \frac{1}{2} + \frac{1}{3} = \frac{11}{6}$.

Example 3.9.7 (At least one defective in a batch)**.** A machine's part diameters are iid with cdf F. A part is **undersized** if its diameter is below the target t, and $F(t) = 0.05$. In a batch of $n = 20$ parts, find the probability that the smallest diameter is below t, i.e. that at least one part is undersized.

Which rule, and why? "At least one below t" is the event $\{X_{(1)} < t\}$ about the **minimum**, since the smallest is below t iff some observation is. Use $F_{(1)} = 1 - (1 - F)^n$. ("All below t" would instead be the maximum event.)

Solution.

Step 1: translate the event. "At least one undersized" $= \{X_{(1)} < t\}$.

Step 2: apply the min cdf.

$$\mathbb{P}(X_{(1)} < t) = F_{(1)}(t) = 1 - [1 - F(t)]^{20} = 1 - (0.95)^{20}.$$

Step 3: evaluate. $(0.95)^{20} \approx 0.3585$, so

$$\mathbb{P}(\text{at least one undersized}) \approx 1 - 0.3585 = 0.6415.$$

Even a 5% per-part defect rate makes a defective batch more likely than not once $n = 20$.

Example 3.9.8 (Maximum of a uniform on a general interval)**.** Let $X_1, \ldots, X_5$ be iid Uniform$(0, 20)$ waiting times. Find the density and mean of the longest, $M = X_{(5)}$.

Which rule, and why? "Longest" is the **maximum**; use $F_{(n)} = F^n$ with the parent cdf of Uniform$(0, 20)$. Take care that F now divides by the interval length 20.

Solution.

Step 1: parent cdf. For Uniform$(0, 20)$, $F(x) = x/20$ on $[0, 20]$, with density $f(x) = 1/20$.

Step 2: density of the max. With $n = 5$,

$$f_M(x) = 5\,[F(x)]^4 f(x) = 5\left(\frac{x}{20}\right)^4 \frac{1}{20} = \frac{5x^4}{20^5}, \qquad 0 \le x \le 20.$$

Step 3: mean. For Uniform$(0, L)$ the max has $\mathbb{E}(X_{(n)}) = \frac{n}{n+1}L$; here

$$\mathbb{E}(M) = \frac{5}{6} \cdot 20 = \frac{100}{6} \approx 16.67 \text{ s.}$$

(Direct check: $\mathbb{E}(M) = \int_0^{20} x\,\frac{5x^4}{20^5}\,\mathrm{d}x = \frac{5}{20^5} \cdot \frac{20^6}{6} = \frac{5 \cdot 20}{6} = \frac{100}{6}$.)

3.9.2 The general k-th order statistic

The max and min are the cases $k = n$ and $k = 1$. For an interior rank we count: the k-th smallest sits at x when one observation lands in $[x, x + \mathrm{d}x]$, exactly $k - 1$ of the others fall below x, and the remaining $n - k$ fall above.

Theorem 3.9.9 (Density of the k-th order statistic)**.** *If $X_1, \ldots, X_n$ are iid with cdf F and density f, then for $1 \le k \le n$*

$$f_{(k)}(x) = \frac{n!}{(k-1)!\,(n-k)!}\,[F(x)]^{\,k-1}\,[1 - F(x)]^{\,n-k}\,f(x).$$

Proof. Partition the n observations into three labelled groups: $k-1$ that fall in $(-\infty, x)$, one that falls in the infinitesimal window $[x, x+\mathrm{d}x]$, and $n-k$ that fall in (x, ∞). The number of ways to assign the n distinct observations to these groups is the multinomial coefficient $\frac{n!}{(k-1)!\,1!\,(n-k)!}$. Each such assignment has probability $[F(x)]^{k-1}\,[f(x)\,\mathrm{d}x]\,[1-F(x)]^{n-k}$ by independence. Summing over the assignments and dividing by $\mathrm{d}x$ gives $f_{(k)}(x)$. □

Remark. The coefficient $\frac{n!}{(k-1)!\,(n-k)!}$ equals $n\binom{n-1}{k-1}$: choose which of the n observations is the one sitting at x (n ways), then which $k-1$ of the remaining $n-1$ fall below it. Setting $k=n$ recovers $nF^{n-1}f$ and $k=1$ recovers $n(1-F)^{n-1}f$.

Example 3.9.10 (Second-largest uniform bid)**.** Five sealed bids are iid Uniform$(0,1)$ (in millions). In a second-price auction the winner pays the *second-highest* bid, $X_{(4)}$. Find its density and expected value.

Which rule, and why? "Second highest of five" is the k-th order statistic with $k=4$, $n=5$ (the 4th smallest is the 2nd largest). Use the general formula, not the pure-max rule, because the rank is interior.

Solution.

Step 1: plug into the formula. With $F(x)=x$, $f(x)=1$, $n=5$, $k=4$,

$$f_{(4)}(x) = \frac{5!}{3!\,1!}\,x^3(1-x)^1 = 20\,x^3(1-x), \qquad 0 \le x \le 1.$$

Step 2: recognise the Beta form. This is the Beta$(4,2)$ density (see the proposition below), whose mean is $\frac{4}{4+2} = \frac{2}{3}$.

Step 3: confirm by integrating.

$$\mathbb{E}(X_{(4)}) = \int_0^1 x \cdot 20x^3(1-x)\,\mathrm{d}x = 20\big(\tfrac{1}{5} - \tfrac{1}{6}\big) = 20 \cdot \tfrac{1}{30} = \frac{2}{3}.$$

The winner pays about \$0.67 million on average.

Proposition 3.9.11 (Uniform order statistics are Beta)**.** *If $U_1, \ldots, U_n$ are iid* Uniform$(0,1)$*, then $U_{(k)} \sim$* Beta$(k,\, n-k+1)$*, with*

$$f_{(k)}(u) = \frac{n!}{(k-1)!\,(n-k)!}\,u^{\,k-1}(1-u)^{\,n-k}, \quad 0 \le u \le 1,$$

$$\mathbb{E}(U_{(k)}) = \frac{k}{n+1}, \qquad \mathrm{Var}(U_{(k)}) = \frac{k\,(n-k+1)}{(n+1)^2(n+2)}.$$

Proof. Substitute $F(u)=u$ and $f(u)=1$ into the k-th order statistic density to get the stated $f_{(k)}$, which is exactly the Beta$(k, n-k+1)$ density. The mean and variance are the standard Beta moments $\frac{\alpha}{\alpha+\beta}$ and $\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ with $\alpha=k$, $\beta=n-k+1$, so $\alpha+\beta=n+1$. □

Example 3.9.12 (Expected positions of all order statistics)**.** Seven independent points are dropped uniformly on $[0,1]$. Where do we expect the 1st, 4th, and 7th smallest to land?

Which rule, and why? This is the uniform case, so the Beta result gives $\mathbb{E}(U_{(k)}) = k/(n+1)$ directly — no integration needed once you recognise the parent is Uniform$(0,1)$.

Solution.

Step 1: apply $\mathbb{E}(U_{(k)}) = k/(n+1)$ with $n=7$.

$$\mathbb{E}(U_{(1)}) = \tfrac{1}{8}, \quad \mathbb{E}(U_{(4)}) = \tfrac{4}{8} = \tfrac{1}{2}, \quad \mathbb{E}(U_{(7)}) = \tfrac{7}{8}.$$

Step 2: interpret. The $n=7$ order statistics chop $[0,1]$ into 8 gaps of equal expected width $1/8$; the expected positions are evenly spaced at $k/(n+1)$. The median ($k=4$) sits at $1/2$ as symmetry demands.

Example 3.9.13 (Third-smallest exponential service time)**.** Four customers have iid service times $X_i \sim \text{Exp}(\lambda)$ with $\lambda = 1$ (minutes). Find $\mathbb{E}\big(X_{(3)}\big)$, the expected time of the third-to-finish.

Which rule, and why? Interior rank ($k = 3$, $n = 4$), exponential parent: use the general density and integrate, or the exponential identity below. The Beta shortcut applies only to uniform parents, so it does not help directly here.

Solution.

Step 1: use the exponential spacing identity. For iid $\text{Exp}(\lambda)$, the order-statistic means telescope to

$$\mathbb{E}\big(X_{(k)}\big) = \frac{1}{\lambda} \sum_{j=n-k+1}^{n} \frac{1}{j}.$$

Step 2: evaluate with $n = 4$, $k = 3$. The sum runs $j = 2, 3, 4$:

$$\mathbb{E}\big(X_{(3)}\big) = \frac{1}{1}\left(\tfrac{1}{2} + \tfrac{1}{3} + \tfrac{1}{4}\right) = \frac{6+4+3}{12} = \frac{13}{12} \approx 1.083 \text{ min}.$$

Step 3: sanity check the ranks. $\mathbb{E}(X_{(1)}) = \frac{1}{4}$, $\mathbb{E}(X_{(2)}) = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$, $\mathbb{E}(X_{(4)}) = \sum_{j=1}^{4} \frac{1}{j} = \frac{25}{12}$: the means increase with rank, as they must.

Example 3.9.14 (A probability for an interior rank)**.** Six measurements are iid Uniform$(0, 1)$. Find the probability that the second-smallest, $X_{(2)}$, is below 0.3.

Which rule, and why? We need a **cdf** of an interior order statistic. Rather than integrate the density, use the counting identity: $X_{(2)} \le x$ iff *at least* 2 of the six observations are $\le x$, which is a binomial tail.

Solution.

Step 1: reduce to a binomial count. Let $Y = \#\{i : X_i \le 0.3\}$. Each observation is ≤ 0.3 with probability $F(0.3) = 0.3$, independently, so $Y \sim \text{Bin}(6, 0.3)$. The event $\{X_{(2)} \le 0.3\}$ is $\{Y \ge 2\}$.

Step 2: complement.

$$\mathbb{P}(X_{(2)} \le 0.3) = \mathbb{P}(Y \ge 2) = 1 - \mathbb{P}(Y = 0) - \mathbb{P}(Y = 1).$$

Step 3: compute. With $p = 0.3$, $q = 0.7$,

$$\mathbb{P}(Y = 0) = 0.7^6 = 0.117649, \qquad \mathbb{P}(Y = 1) = 6(0.3)(0.7)^5 = 0.302526,$$

$$\mathbb{P}(X_{(2)} \le 0.3) = 1 - 0.117649 - 0.302526 \approx 0.5798.$$

In general $F_{(k)}(x) = \mathbb{P}\big(\text{Bin}(n, F(x)) \ge k\big)$ — a clean alternative to integrating the density.

3.9.3 The sample median

Definition 3.9.15 (Sample median)**.** For an **odd** sample size $n = 2m+1$, the **sample median** is the middle order statistic $X_{(m+1)}$. (For even $n = 2m$ it is the average $\frac{1}{2}(X_{(m)} + X_{(m+1)})$; we focus on the odd case, a single order statistic.)

The median is the order statistic with $k = m + 1$, so its density is just the general formula at that rank; it estimates the center of the parent distribution and is far less sensitive to extreme values than the sample mean.

Example 3.9.16 (Median of five uniform readings)**.** Five sensor readings are iid Uniform$(0, 1)$. Find the density, mean, and variance of the sample median $X_{(3)}$.

Which rule, and why? Odd $n = 5$, so the median is the single order statistic $X_{(3)}$ ($k = 3$). Use the general density; since the parent is uniform the Beta moments finish it.

Solution.

Step 1: density. With $F(x) = x$, $f(x) = 1$, $n = 5$, $k = 3$,

$$f_{(3)}(x) = \frac{5!}{2!\,2!}\,x^2(1-x)^2 = 30\,x^2(1-x)^2, \qquad 0 \le x \le 1,$$

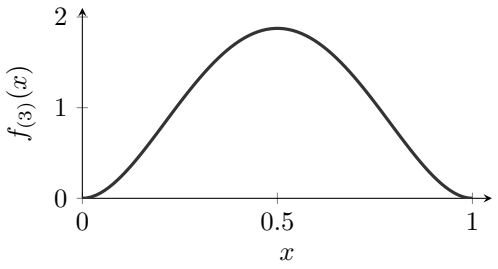
which is the Beta$(3, 3)$ density — symmetric about $1/2$.

Step 2: mean. By symmetry (or $k/(n+1) = 3/6$), $\mathbb{E}(X_{(3)}) = \frac{1}{2}$.

Step 3: variance. Using $\mathrm{Var}(U_{(k)}) = \frac{k(n-k+1)}{(n+1)^2(n+2)}$,

$$\mathrm{Var}(X_{(3)}) = \frac{3 \cdot 3}{6^2 \cdot 7} = \frac{9}{252} = \frac{1}{28} \approx 0.0357.$$

For comparison, a single Uniform$(0, 1)$ reading has variance $1/12 \approx 0.0833$: the median of five is markedly more concentrated about $1/2$.



Example 3.9.17 (Median of three exponential lifetimes)**.** Three light bulbs have iid lifetimes $X_i \sim \mathrm{Exp}(1)$ (in thousands of hours). Find the probability that the sample median exceeds $\ln 2 \approx 0.693$ thousand hours.

Which rule, and why? Odd $n = 3$ makes the median $X_{(2)}$ ($k = 2$). For a **tail probability** use the binomial-count identity $\mathbb{P}(X_{(2)} \le t) = \mathbb{P}(\mathrm{Bin}(3, F(t)) \ge 2)$ rather than integrating.

Solution.

Step 1: parent value at the threshold. $F(t) = 1 - e^{-t}$, and at $t = \ln 2$, $e^{-t} = \frac{1}{2}$, so $F(\ln 2) = \frac{1}{2}$.

Step 2: binomial count. Let $Y \sim \mathrm{Bin}(3, \frac{1}{2})$ count how many lifetimes are $\le \ln 2$. Then

$$\mathbb{P}(X_{(2)} \le \ln 2) = \mathbb{P}(Y \ge 2) = \binom{3}{2}\left(\tfrac{1}{2}\right)^3 + \binom{3}{3}\left(\tfrac{1}{2}\right)^3 = \frac{3+1}{8} = \frac{1}{2}.$$

Step 3: take the complement.

$$\mathbb{P}(X_{(2)} > \ln 2) = 1 - \tfrac{1}{2} = \tfrac{1}{2}.$$

Because $\ln 2$ is the median of Exp(1) ($F = 1/2$ there), the *sample* median of three is equally likely to land on either side of it — a tidy consistency check.

Example 3.9.18 (Median of seven uniform points)**.** Seven independent points are dropped uniformly on $[0, 1]$. Give the density and variance of the sample median.

Which rule, and why? Odd $n = 7$, median is $X_{(4)}$ ($k = 4$). Uniform parent, so the Beta result gives the moments immediately.

Solution.

Step 1: density.

$$f_{(4)}(x) = \frac{7!}{3!\,3!}\,x^3(1-x)^3 = 140\,x^3(1-x)^3, \qquad 0 \le x \le 1 \quad (\mathrm{Beta}(4, 4)).$$

Step 2: mean. $\mathbb{E}(X_{(4)}) = 4/8 = \frac{1}{2}$ by symmetry.

Step 3: variance.

$$\mathrm{Var}(X_{(4)}) = \frac{4 \cdot 4}{8^2 \cdot 9} = \frac{16}{576} = \frac{1}{36} \approx 0.0278.$$

A larger sample sharpens the median: the variance fell from $1/28$ at $n = 5$ to $1/36$ at $n = 7$.

3.9.4 The sample range

Definition 3.9.19 (Sample range)**.** The **sample range** is $R = X_{(n)} - X_{(1)}$, the spread between the largest and smallest observations.

The range is a function of the two extreme order statistics. Because $X_{(1)}$ and $X_{(n)}$ are not independent, we use the joint density of the min and max,

$$f_{(1),(n)}(u, v) = n(n-1)\,[F(v) - F(u)]^{\,n-2} f(u) f(v), \qquad u \le v,$$

which counts: one observation at the low end u, one at the high end v, and the other $n - 2$ strictly between (each contributing factor $F(v) - F(u)$).

Example 3.9.20 (Range of four uniform observations)**.** Four observations are iid Uniform$(0, 1)$. Find the expected sample range $\mathbb{E}(R)$.

Which rule, and why? "Spread from smallest to largest" is the **range** $R = X_{(n)} - X_{(1)}$. The fastest route is linearity of expectation together with the known means of the max and min, avoiding the joint density.

Solution.

Step 1: means of the extremes. For Uniform$(0, 1)$ with $n = 4$, $\mathbb{E}(X_{(4)}) = \frac{4}{5}$ and $\mathbb{E}(X_{(1)}) = \frac{1}{5}$ (Beta means $k/(n+1)$).

Step 2: linearity.

$$\mathbb{E}(R) = \mathbb{E}(X_{(4)}) - \mathbb{E}(X_{(1)}) = \tfrac{4}{5} - \tfrac{1}{5} = \tfrac{3}{5} = 0.6.$$

Step 3: general pattern. For Uniform$(0, 1)$, $\mathbb{E}(R) = \frac{n}{n+1} - \frac{1}{n+1} = \frac{n-1}{n+1}$, which rises toward 1 as n grows: more points stretch the observed spread toward the full interval.

Example 3.9.21 (Range density for two uniforms)**.** Two observations are iid Uniform$(0, 1)$. Find the density of the range R and confirm its mean.

Which rule, and why? Smallest sample with a nontrivial range ($n = 2$); here the joint extreme density is simple, so derive f_R directly and check against $\mathbb{E}(R) = \frac{n-1}{n+1}$.

Solution.

Step 1: joint density of min and max. With $n = 2$, $F(x) = x$, $f(x) = 1$,

$$f_{(1),(2)}(u, v) = 2 \cdot 1 \cdot [v - u]^0 \cdot 1 \cdot 1 = 2, \qquad 0 \le u \le v \le 1.$$

Step 2: change to $R = v - u$. For fixed range $r = v - u \in (0, 1)$, the low end u ranges over $[0, 1 - r]$, an interval of length $1 - r$. Integrating the constant joint density 2 over that segment,

$$f_R(r) = \int_0^{1-r} 2 \,\mathrm{d}u = 2(1 - r), \qquad 0 \le r \le 1.$$

Step 3: mean.

$$\mathbb{E}(R) = \int_0^1 r \cdot 2(1 - r)\,\mathrm{d}r = 2\big(\tfrac{1}{2} - \tfrac{1}{3}\big) = 2 \cdot \tfrac{1}{6} = \tfrac{1}{3},$$

matching $\frac{n-1}{n+1} = \frac{1}{3}$ for $n = 2$.

Example 3.9.22 (Range of exponential failure times)**.** Three components have iid lifetimes $X_i \sim \mathrm{Exp}(\lambda)$. Find $\mathbb{E}(R)$ for the spread between the first and last failures.

Which rule, and why? Again the **range** via linearity: subtract the known mean of the min from the mean of the max. The min of exponentials is exponential, and the max uses the harmonic-sum identity.

Solution.

Step 1: mean of the min. $X_{(1)} \sim \text{Exp}(3\lambda)$, so $\mathbb{E}(X_{(1)}) = \frac{1}{3\lambda}$.

Step 2: mean of the max. $\mathbb{E}(X_{(3)}) = \frac{1}{\lambda}\big(1 + \frac{1}{2} + \frac{1}{3}\big) = \frac{11}{6\lambda}$.

Step 3: subtract.

$$\mathbb{E}(R) = \mathbb{E}(X_{(3)}) - \mathbb{E}(X_{(1)}) = \frac{11}{6\lambda} - \frac{1}{3\lambda} = \frac{11-2}{6\lambda} = \frac{9}{6\lambda} = \frac{3}{2\lambda}.$$

For $\lambda = 1$ this is 1.5 time units between the first and last failure.

Remark (How to choose the rule, in one line). Largest $\Rightarrow F^n$; smallest $\Rightarrow 1-(1-F)^n$; an interior rank $\Rightarrow$ the $\frac{n!}{(k-1)!(n-k)!}F^{k-1}(1-F)^{n-k}f$ density (or the binomial-tail cdf $\mathbb{P}(\text{Bin}(n, F(x)) \geq k)$ for probabilities); the middle of an odd sample $\Rightarrow$ that same density at $k = \frac{n+1}{2}$; the spread $\Rightarrow X_{(n)} - X_{(1)}$ via linearity of the extreme means.

3.10 Conditional Densities

When two random variables travel together, learning the value of one reshapes the distribution of the other. The **conditional density** (or conditional pmf) of Y given $X = x$ is the slice of the joint distribution at that fixed x, renormalised so it is again a legitimate distribution in y. From it we read off **conditional expectations** $\mathbb{E}(Y \mid X = x)$ — the best one-number prediction of Y once x is known — and through the **law of total expectation** we recover the plain mean $\mathbb{E}(Y)$ by averaging those predictions. These tools turn a joint table or a region-shaped joint density into a machine for updating on partial information.

Which rule of this section does the problem need?

- Asked for "the distribution of Y *given* that $X = x$" (a fixed value), or for a probability/pmf/pdf with a condition on a *related* variable? $\Rightarrow$ form the **conditional pmf/pdf** $f_{Y|X}(y \mid x) = f(x,y)/f_X(x)$. Divide the joint by the **marginal of the conditioning variable**, evaluated at the observed x.
- Need to check that your conditional law is valid, or to use it as an ordinary density? $\Rightarrow$ it **sums/integrates to** 1 in y for each fixed x; treat it like any pmf/pdf.
- Want the **expected value of** Y **after observing** $X = x$, or a prediction of Y from X? $\Rightarrow$ the **conditional expectation** $\mathbb{E}(Y \mid X = x) = \sum_y y\, f_{Y|X}(y \mid x)$ or $\int y\, f_{Y|X}\, \mathrm{d}y$.
- Want a plain (unconditional) mean $\mathbb{E}(Y)$ that is awkward directly, but easy once X is fixed — e.g. a count whose rate is itself random, or a two-stage experiment? $\Rightarrow$ the **law of total expectation** $\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big]$.

Why not the competing tools? The plain marginal f_Y (§6) ignores the observed information and answers a different question; independence factorisation $f(x,y) = f_X f_Y$ is only the *special case* where conditioning changes nothing ($f_{Y|X} = f_Y$). Use a conditional density precisely when X and Y are **dependent** and a value of one has been revealed.

3.10.1 The conditional pmf and pdf

Definition 3.10.1 (Conditional pmf and conditional pdf)**.** Let X and Y have joint pmf $p(x,y)$ (discrete) or joint pdf $f(x,y)$ (continuous), with marginal p_X or f_X. For any value x with $p_X(x) > 0$ (resp. $f_X(x) > 0$), the **conditional pmf / conditional pdf** of Y given $X = x$ is

$$p_{Y|X}(y \mid x) = \frac{p(x,y)}{p_X(x)}, \qquad f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)}.$$

In the discrete case this is the elementary conditional probability $\mathbb{P}(Y = y \mid X = x) = \mathbb{P}(X = x, Y = y)/\mathbb{P}(X = x)$. In the continuous case $\mathbb{P}(X = x) = 0$, so the ratio is defined as a limit and interpreted as a density in y.

The recipe is always the same: *joint over marginal of the thing you condition on.* The conditioning value x is held fixed throughout; y is the live variable.

Proposition 3.10.2 (A conditional density is a genuine density)**.** *For each fixed x with $f_X(x) > 0$, the function $y \mapsto f_{Y|X}(y \mid x)$ is nonnegative and integrates to 1:*

$$\int_{-\infty}^{\infty} f_{Y|X}(y \mid x)\,\mathrm{d}y = \int_{-\infty}^{\infty} \frac{f(x,y)}{f_X(x)}\,\mathrm{d}y = \frac{1}{f_X(x)} \int_{-\infty}^{\infty} f(x,y)\,\mathrm{d}y = \frac{f_X(x)}{f_X(x)} = 1.$$

The discrete statement is identical with $\int$ replaced by $\sum_y$, using $\sum_y p(x,y) = p_X(x)$.

Proof. Nonnegativity is immediate since $f \geq 0$ and $f_X(x) > 0$. The inner integral is exactly the definition of the marginal $f_X(x)$ (integrate the joint over y), which cancels the denominator. Hence every conditional slice is a bona fide probability density, and we may use all the machinery of §3–§5 on it. □

Example 3.10.3 (Defective chips: a discrete joint table)**.** A tray holds chips made on two lines. Let $X \in \{0,1\}$ record the line (0 = line A, 1 = line B) and $Y \in \{0,1,2\}$ the number of solder defects on a randomly chosen chip. The joint pmf $p(x,y) = \mathbb{P}(X = x, Y = y)$ is

	$y=0$	$y=1$	$y=2$
$x=0$	0.30	0.15	0.05
$x=1$	0.20	0.20	0.10

Find the conditional pmf of Y given the chip came from line B ($X = 1$), and confirm it is a valid pmf.

Which rule, and why? We are told the value of one variable ($X = 1$) and asked about the distribution of the other (Y). That is a conditional pmf: divide the joint row by the marginal of the conditioning variable. The plain marginal p_Y would ignore the line we observed.

Solution.

Step 1: Marginal of the conditioning variable. Sum the $x = 1$ row:

$$p_X(1) = 0.20 + 0.20 + 0.10 = 0.50.$$

Step 2: Divide the joint entries by that marginal.

$$p_{Y|X}(0 \mid 1) = \frac{0.20}{0.50} = 0.40, \quad p_{Y|X}(1 \mid 1) = \frac{0.20}{0.50} = 0.40, \quad p_{Y|X}(2 \mid 1) = \frac{0.10}{0.50} = 0.20.$$

Step 3: Check. The three values are nonnegative and sum to $0.40 + 0.40 + 0.20 = 1$, so this is a valid pmf.

$$p_{Y|X}(\cdot \mid X = 1) = (0.40,\ 0.40,\ 0.20) \text{ on } \{0,1,2\}.$$

Example 3.10.4 (The same table, conditioning the other way)**.** For the chip table above, find the conditional pmf of the line X given that the chip has exactly one defect ($Y = 1$).

Which rule, and why? Again one variable is observed ($Y = 1$) and we want the other; the roles of "joint" and "marginal" just swap. Now we divide a *column* by the marginal $p_Y(1)$.

Solution.

Step 1: Marginal of Y at $y = 1$. Sum the $y = 1$ column: $p_Y(1) = 0.15 + 0.20 = 0.35$.

Step 2: Divide.

$$p_{X|Y}(0 \mid 1) = \frac{0.15}{0.35} = \frac{3}{7} \approx 0.4286, \qquad p_{X|Y}(1 \mid 1) = \frac{0.20}{0.35} = \frac{4}{7} \approx 0.5714.$$

These sum to 1. Given a single defect, the chip is somewhat more likely to be from line B.

$$p_{X|Y}(\cdot \mid Y = 1) = \left(\tfrac{3}{7},\ \tfrac{4}{7}\right).$$

Example 3.10.5 (A two-die spread, conditioning a count on a total)**.** Roll two fair four-sided dice. Let S be their sum and M the number of dice showing a 4. Find the conditional pmf of M given $S = 5$.

Which rule, and why? We observe the total $S = 5$ and want the distribution of a related count M: a conditional pmf. Restrict to the outcomes consistent with the condition and renormalise.

Solution.

Step 1: Outcomes with $S = 5$. The equally likely ordered pairs summing to 5 are $(1, 4)$, $(4, 1)$, $(2, 3)$, $(3, 2)$ — four of the 16 outcomes, so $\mathbb{P}(S = 5) = 4/16 = 1/4$.

Step 2: Value of M on each. Pairs $(1, 4)$ and $(4, 1)$ each show one 4, so $M = 1$; pairs $(2, 3), (3, 2)$ show no 4, so $M = 0$. No outcome with $S = 5$ shows two 4s.

Step 3: Renormalise over the four outcomes.

$$p_{M|S}(0 \mid 5) = \frac{2}{4} = \tfrac{1}{2}, \qquad p_{M|S}(1 \mid 5) = \frac{2}{4} = \tfrac{1}{2}, \qquad p_{M|S}(2 \mid 5) = 0.$$

The slice sums to 1, as a conditional pmf must.

Example 3.10.6 (Emails and attachments: a built joint table)**.** On a workday a small mailbox receives $X \in \{0, 1, 2\}$ priority emails with pmf $p_X = (0.5, 0.3, 0.2)$. Each priority email independently carries an attachment with probability $\frac{1}{2}$; let Y be the number of attachments. Build the joint pmf and find the conditional pmf of X given $Y = 1$.

Which rule, and why? The number of attachments is observed ($Y = 1$) and we want the hidden count X — a conditional pmf. We first need the joint, obtained from $p(x, y) = p_X(x)\,\mathbb{P}(Y = y \mid X = x)$ with $Y \mid X = x \sim \mathrm{Bin}(x, \frac{1}{2})$.

Solution.

Step 1: Conditional counts. Given $X = x$, $Y \sim \mathrm{Bin}(x, \frac{1}{2})$: $\mathbb{P}(Y = 1 \mid X = 0) = 0$, $\mathbb{P}(Y = 1 \mid X = 1) = \frac{1}{2}$, $\mathbb{P}(Y = 1 \mid X = 2) = 2(\frac{1}{2})(\frac{1}{2}) = \frac{1}{2}$.

Step 2: Joint entries at $y = 1$.

$$p(1, 1) = 0.3 \cdot \tfrac{1}{2} = 0.15, \qquad p(2, 1) = 0.2 \cdot \tfrac{1}{2} = 0.10, \qquad p(0, 1) = 0.$$

Step 3: Marginal and divide. $p_Y(1) = 0 + 0.15 + 0.10 = 0.25$, so

$$p_{X|Y}(1 \mid 1) = \frac{0.15}{0.25} = 0.6, \qquad p_{X|Y}(2 \mid 1) = \frac{0.10}{0.25} = 0.4.$$

Given exactly one attachment, there were two priority emails with probability 0.4.

$$p_{X|Y}(\cdot \mid Y = 1) = (0,\ 0.6,\ 0.4) \text{ on } \{0, 1, 2\}.$$

3.10.2 Conditional densities for continuous variables

For continuous variables the same formula divides the joint pdf by the marginal of the conditioning variable; the only new wrinkle is keeping the **support** straight — fixing x usually shrinks the range of y.

Example 3.10.7 (Uniform on a triangle)**.** Let (X, Y) be uniform on the triangle $\{0 < y < x < 1\}$, so the joint pdf is

$$f(x, y) = \begin{cases} 2, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(The constant is 2 because the triangle has area $\frac{1}{2}$.) Find $f_{Y|X}(y \mid x)$ for a fixed $x \in (0, 1)$, and identify the conditional law of Y.

Which rule, and why? We fix $X = x$ and want the density of Y on that vertical slice: a conditional pdf. Divide the joint by the marginal f_X. The slice is a vertical segment, so its y-support is trimmed to $(0, x)$ — watch the limits.

Solution.

Step 1: Marginal of X. For $0 < x < 1$, integrate y from 0 up to the diagonal $y = x$:

$$f_X(x) = \int_0^x 2 \, \mathrm{d}y = 2x.$$

Step 2: Divide. For $0 < y < x$,

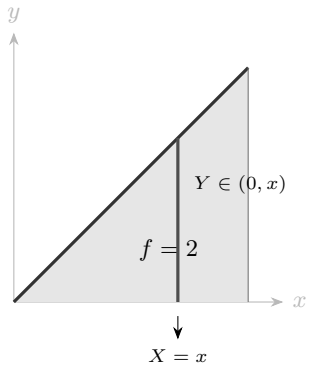
$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)} = \frac{2}{2x} = \frac{1}{x},$$

and 0 otherwise.

Step 3: Identify. A constant $1/x$ on $(0, x)$ is the Uniform$(0, x)$ density. So

$$Y \mid X = x \ \sim \ \text{Uniform}(0, x).$$

The shaded slice below is the region over which we average; fixing x confines Y to $(0, x)$.



Example 3.10.8 (Conditioning the other variable in the triangle)**.** For the same triangle density $f(x, y) = 2$ on $\{0 < y < x < 1\}$, find $f_{X|Y}(x \mid y)$ for a fixed $y \in (0, 1)$.

Which rule, and why? Now we fix $Y = y$ and read off a horizontal slice: divide by the marginal f_Y. On that slice x must satisfy $y < x < 1$, so the x-support is $(y, 1)$.

Solution.

Step 1: Marginal of Y. For $0 < y < 1$, integrate x from the diagonal $x = y$ up to 1:

$$f_Y(y) = \int_y^1 2 \, \mathrm{d}x = 2(1 - y).$$

Step 2: Divide. For $y < x < 1$,

$$f_{X|Y}(x \mid y) = \frac{2}{2(1-y)} = \frac{1}{1-y},$$

i.e. $X \mid Y = y \sim \text{Uniform}(y, 1)$. The conditional support starts at $x = y$ and runs to 1.

$$X \mid Y = y \ \sim \ \text{Uniform}(y, 1).$$

Example 3.10.9 (A nonuniform joint density)**.** Let

$$f(x,y) = \begin{cases} 6xy^2, & 0 < x < 1, \ 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

First verify it is a density, then find $f_{Y|X}(y \mid x)$ and decide whether X and Y are independent.

Which rule, and why? We want the law of Y on the slice $X = x$: a conditional pdf, joint over marginal. The independence question is then settled by comparing $f_{Y|X}$ with the marginal f_Y: equal for all x means independence.

Solution.

Step 1: Check normalisation.

$$\int_0^1 \int_0^1 6xy^2 \,\mathrm{d}x \,\mathrm{d}y = 6\Big(\int_0^1 x \,\mathrm{d}x\Big)\Big(\int_0^1 y^2 \,\mathrm{d}y\Big) = 6 \cdot \tfrac{1}{2} \cdot \tfrac{1}{3} = 1.$$

Step 2: Marginal of X. For $0 < x < 1$,

$$f_X(x) = \int_0^1 6xy^2 \,\mathrm{d}y = 6x \cdot \tfrac{1}{3} = 2x.$$

Step 3: Conditional pdf. For $0 < y < 1$,

$$f_{Y|X}(y \mid x) = \frac{6xy^2}{2x} = 3y^2.$$

Step 4: Independence. The conditional $3y^2$ does not depend on x, and indeed $f_Y(y) = \int_0^1 6xy^2 \,\mathrm{d}x = 3y^2$, so $f_{Y|X} = f_Y$. Hence

$$X \perp\!\!\!\perp Y, \qquad f_{Y|X}(y \mid x) = 3y^2 = f_Y(y).$$

Example 3.10.10 (Lifetime conditioned on an installed level)**.** A sensor's noise floor X is uniform on $(0, 1)$, and given $X = x$ the time Y until the first spurious reading is exponential with rate x, so the joint pdf on $0 < x < 1,\ y > 0$ is

$$f(x, y) = f_X(x)\, f_{Y|X}(y \mid x) = 1 \cdot xe^{-xy} = xe^{-xy}.$$

Find $f_{Y|X}(y \mid x)$ (read it back off) and $\mathbb{P}(Y > 2 \mid X = x)$.

Which rule, and why? We are handed a model already in conditional form ($Y \mid X = x$ exponential), so the conditional density is built in; the tail probability is then an ordinary exponential computation on that slice.

Solution.

Step 1: Conditional pdf. Dividing by $f_X(x) = 1$ leaves

$$f_{Y|X}(y \mid x) = xe^{-xy}, \qquad y > 0,$$

which is the $\mathrm{Exp}(x)$ density — so $Y \mid X = x \sim \mathrm{Exp}(x)$, as posited.

Step 2: Tail. For an exponential with rate x, $\mathbb{P}(Y > t \mid X = x) = e^{-xt}$:

$$\mathbb{P}(Y > 2 \mid X = x) = e^{-2x}.$$

A noisier sensor (larger x) fails sooner, as expected.

3.10.3 Conditional expectation

Definition 3.10.11 (Conditional expectation)**.** The **conditional expectation** of Y given $X = x$ is the mean of the conditional distribution:

$$\mathbb{E}(Y \mid X = x) = \sum_y y\, p_{Y|X}(y \mid x) \ \text{ (discrete)}, \qquad \mathbb{E}(Y \mid X = x) = \int y\, f_{Y|X}(y \mid x) \,\mathrm{d}y \ \text{ (continuous)}.$$

It is a number for each fixed x. Letting x vary, $g(x) = \mathbb{E}(Y \mid X = x)$ is a function of x; the random variable $g(X) = \mathbb{E}(Y \mid X)$ is the best predictor of Y from X in the mean-square sense.

Example 3.10.12 (Predicted defects on each line)**.** Using the chip table, compute $\mathbb{E}(Y \mid X = 1)$ (line B) and $\mathbb{E}(Y \mid X = 0)$ (line A).

Which rule, and why? We want a one-number summary of Y once the line is known: a conditional expectation, i.e. the mean of the conditional pmf we already built.

Solution.

Step 1: Line B. From Example with $p_{Y|X}(\cdot \mid 1) = (0.40, 0.40, 0.20)$,

$$\mathbb{E}(Y \mid X = 1) = 0(0.40) + 1(0.40) + 2(0.20) = 0.80.$$

Step 2: Line A. The $x = 0$ row is $(0.30, 0.15, 0.05)$ with marginal $p_X(0) = 0.50$, so the conditional pmf is $(0.60, 0.30, 0.10)$ and

$$\mathbb{E}(Y \mid X = 0) = 0(0.60) + 1(0.30) + 2(0.10) = 0.50.$$

Line B chips average 0.80 defects versus 0.50 on line A.

Example 3.10.13 (Conditional mean from the triangle)**.** For $f(x, y) = 2$ on $\{0 < y < x < 1\}$, compute $\mathbb{E}(Y \mid X = x)$.

Which rule, and why? We found $Y \mid X = x \sim \text{Uniform}(0, x)$; its mean is the conditional expectation. We could integrate directly, but recognising the uniform shortcuts the work.

Solution.

Step 1: Use the conditional law. Since $Y \mid X = x \sim \text{Uniform}(0, x)$, its mean is the midpoint:

$$\mathbb{E}(Y \mid X = x) = \frac{0 + x}{2} = \frac{x}{2}.$$

Step 2: Confirm by integration.

$$\int_0^x y \cdot \frac{1}{x}\,\mathrm{d}y = \frac{1}{x} \cdot \frac{x^2}{2} = \frac{x}{2}. \blacksquare$$

The prediction $\mathbb{E}(Y \mid X = x) = x/2$ rises with x, which makes sense: a larger x allows a wider range for Y.

Example 3.10.14 (Conditional mean of the exponential-rate model)**.** In the sensor model $Y \mid X = x \sim \text{Exp}(x)$, find $\mathbb{E}(Y \mid X = x)$.

Which rule, and why? The slice law is exponential with rate x, and an exponential's mean is the reciprocal of its rate — a conditional expectation read straight off the named distribution.

Solution.

$$\mathbb{E}(Y \mid X = x) = \frac{1}{x}, \qquad 0 < x < 1.$$

A lower noise floor (small x) means a much longer expected time to the first spurious reading.

Example 3.10.15 (Nonuniform slice, computed by integration)**.** With $f(x, y) = 6xy^2$ on the unit square, we found $f_{Y|X}(y \mid x) = 3y^2$ on $(0, 1)$. Compute $\mathbb{E}(Y \mid X = x)$.

Which rule, and why? A conditional expectation of a continuous slice: integrate y against the conditional density. Because the slice does not depend on x (independence), the answer will be constant.

Solution.

$$\mathbb{E}(Y \mid X = x) = \int_0^1 y \cdot 3y^2\,\mathrm{d}y = 3\int_0^1 y^3\,\mathrm{d}y = 3 \cdot \frac{1}{4} = \frac{3}{4}.$$

Since $X \perp\!\!\!\perp Y$, the conditional mean equals the unconditional mean $\mathbb{E}(Y) = 3/4$ for every x.

3.10.4 The law of total expectation

The conditional means in the last subsection are the raw material for a clean route to the plain mean: average the predictions over the conditioning variable.

Theorem 3.10.16 (Law of total expectation)**.** *If* $\mathbb{E}|Y| < \infty$*, then*

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big] = \begin{cases} \displaystyle\sum_x \mathbb{E}(Y \mid X = x)\, p_X(x), & \text{discrete } X, \\ \displaystyle\int_{-\infty}^{\infty} \mathbb{E}(Y \mid X = x)\, f_X(x)\,\mathrm{d}x, & \text{continuous } X. \end{cases}$$

Proof. Take the continuous case. Write $g(x) = \mathbb{E}(Y \mid X = x) = \int y\, f_{Y|X}(y \mid x)\,\mathrm{d}y$ and substitute $f_{Y|X} = f(x,y)/f_X(x)$:

$$\begin{aligned} \int g(x) f_X(x)\,\mathrm{d}x &= \iint y\, \frac{f(x,y)}{f_X(x)} f_X(x)\,\mathrm{d}y\,\mathrm{d}x = \iint y\, f(x,y)\,\mathrm{d}y\,\mathrm{d}x \\ &= \int y \Big(\int f(x,y)\,\mathrm{d}x \Big)\,\mathrm{d}y = \int y\, f_Y(y)\,\mathrm{d}y = \mathbb{E}(Y). \end{aligned}$$

The $f_X(x)$ cancels, the inner x-integral rebuilds the marginal f_Y, and what remains is the definition of $\mathbb{E}(Y)$. The discrete proof is identical with sums. □

Example 3.10.17 (Mean of Y in the triangle, two ways)**.** For $f(x,y) = 2$ on $\{0 < y < x < 1\}$, find $\mathbb{E}(Y)$ using total expectation, and check directly.

Which rule, and why? The conditional mean $\mathbb{E}(Y \mid X = x) = x/2$ is simple, so averaging it over X via total expectation is easier than integrating y against the marginal from scratch.

Solution.

Step 1: Marginal of X. From Example, $f_X(x) = 2x$ on $(0,1)$, so $\mathbb{E}(X) = \int_0^1 x \cdot 2x\,\mathrm{d}x = 2/3$.

Step 2: Average the conditional mean.

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big] = \mathbb{E}\Big(\frac{X}{2}\Big) = \frac{1}{2}\mathbb{E}(X) = \frac{1}{2} \cdot \frac{2}{3} = \frac{1}{3}.$$

Step 3: Direct check. The marginal of Y is $f_Y(y) = \int_y^1 2\,\mathrm{d}x = 2(1-y)$, so

$$\mathbb{E}(Y) = \int_0^1 y \cdot 2(1-y)\,\mathrm{d}y = \int_0^1 (2y - 2y^2)\,\mathrm{d}y = 1 - \frac{2}{3} = \frac{1}{3}. \blacksquare$$

Example 3.10.18 (Random number of claims with a random rate)**.** An insurer's daily claim count Y given the day's risk level $X = x$ is Poisson with mean x, and the risk level itself is exponential with mean 4: $X \sim \mathrm{Exp}(\frac{1}{4})$. Find $\mathbb{E}(Y)$.

Which rule, and why? The rate of the Poisson is itself random — a textbook mixture. The plain mean $\mathbb{E}(Y)$ is hard to attack directly but trivial once X is fixed, so use total expectation.

Solution.

Step 1: Conditional mean. A Poisson with mean x has $\mathbb{E}(Y \mid X = x) = x$.

Step 2: Average over X. Since $X \sim \mathrm{Exp}(\frac{1}{4})$ has $\mathbb{E}(X) = 1/(\frac{1}{4}) = 4$,

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big] = \mathbb{E}(X) = 4.$$

On an average day the insurer expects 4 claims.

Example 3.10.19 (Two-stage experiment: random sample size)**.** A quality auditor inspects N items in a batch, where $N \sim \mathrm{Poisson}(10)$. Each inspected item is nonconforming independently with probability 0.05; let Y be the number of nonconforming items found. Find $\mathbb{E}(Y)$.

Which rule, and why? Given the sample size $N = n$, the count Y is binomial; but N is random. Condition on N, take the binomial mean, then average — the law of total expectation.

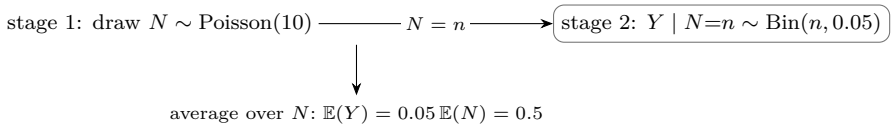
Solution.

Step 1: Conditional mean. Given $N = n$, $Y \sim \mathrm{Bin}(n, 0.05)$, so $\mathbb{E}(Y \mid N = n) = 0.05\,n$.

Step 2: Average over N. With $\mathbb{E}(N) = 10$,

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid N)\big] = \mathbb{E}(0.05\,N) = 0.05 \cdot 10 = 0.5.$$

The auditor expects half a nonconforming item per batch.



Example 3.10.20 (Total expectation from a discrete table)**.** Recover $\mathbb{E}(Y)$ for the chip table by total expectation, and verify against the marginal.

Which rule, and why? We already have both conditional means $\mathbb{E}(Y \mid X = x)$ and the marginal p_X, so total expectation assembles $\mathbb{E}(Y)$ from them; the direct marginal computation is the check.

Solution.

Step 1: Combine conditional means. With $\mathbb{E}(Y \mid X = 0) = 0.50$, $\mathbb{E}(Y \mid X = 1) = 0.80$, and $p_X(0) = p_X(1) = 0.50$,

$$\mathbb{E}(Y) = 0.50(0.50) + 0.80(0.50) = 0.25 + 0.40 = 0.65.$$

Step 2: Check via the marginal of Y. Columns give $p_Y = (0.50, 0.35, 0.15)$, so

$$\mathbb{E}(Y) = 0(0.50) + 1(0.35) + 2(0.15) = 0.35 + 0.30 = 0.65.\blacksquare$$

Remark. The same averaging idea extends to probabilities: $\mathbb{P}(A) = \mathbb{E}\big[\mathbb{P}(A \mid X)\big]$ (the law of total probability), since a probability is the expectation of an indicator. For instance, in the sensor model $\mathbb{P}(Y > 2) = \mathbb{E}\big[e^{-2X}\big] = \int_0^1 e^{-2x}\,\mathrm{d}x = \frac{1}{2}(1 - e^{-2}) \approx 0.432$, obtained by averaging the conditional tail e^{-2x} over $X \sim \mathrm{Uniform}(0, 1)$.

3.11 Moment-Generating Functions

The **moment-generating function** (mgf) of a random variable X is the single function $M_X(t) = \mathbb{E}\big(e^{tX}\big)$ that, when it exists near $t = 0$, encodes the entire distribution. Two jobs make it indispensable. First, its derivatives at $t = 0$ hand back the **moments** $\mathbb{E}(X), \mathbb{E}(X^2), \ldots$ one at a time, so a mean and variance drop out of a single function instead of separate sums or integrals. Second, because the mgf determines the distribution *uniquely*, and because the mgf of a sum of independents is the *product* of the mgfs, it is the cleanest tool for identifying the law of a sum — showing, for instance, that a sum of independent Poissons is again Poisson, and a sum of independent normals is again normal.

Which mgf rule does the problem need?

- Asked to **find an mgf from a distribution**? ⇒ evaluate the **defining expectation** $M_X(t) = \mathbb{E}(e^{tX})$: a sum $\sum_k e^{tk} p_X(k)$ in the discrete case, an integral $\int e^{tx} f_X(x)\,\mathrm{d}x$ in the continuous case. Note the range of t for which it converges.
- Given an mgf and asked for a **mean, variance, or higher moment**? ⇒ the **moment property**: $\mathbb{E}(X^r) = M_X^{(r)}(0)$. In particular $\mathbb{E}(X) = M_X'(0)$ and $\mathrm{Var}(X) = M_X''(0) - \big[M_X'(0)\big]^2$.
- Handed an mgf and asked **which distribution** it is? ⇒ the **uniqueness property**: match it against the catalogue of known mgfs; equal mgfs (on an interval around 0) force equal distributions.
- Want the distribution of a **sum of independent** variables? ⇒ the **product rule** $M_{X_1 + \cdots + X_n}(t) = \prod_i M_{X_i}(t)$; then read off the answer by uniqueness.

- X is an **affine transform** $aX+b$? $\Rightarrow$ the **scaling rule** $M_{aX+b}(t) = e^{bt}M_X(at)$.

Why not the others. The moment property answers "what is $\mathbb{E}(X^k)$," not "which distribution"; for the latter you must invoke uniqueness. The product rule needs *independence*: without it $\mathbb{E}(e^{t(X+Y)}) \neq \mathbb{E}(e^{tX})\mathbb{E}(e^{tY})$ in general. And if the defining expectation diverges for every $t \neq 0$ (heavy-tailed laws), the mgf does not exist and none of these tools apply — a characteristic function would be needed instead.

3.11.1 Definition and existence

Definition 3.11.1 (Moment-generating function)**.** The **moment-generating function** of a random variable X is

$$M_X(t) = \mathbb{E}\big(e^{tX}\big) = \begin{cases} \displaystyle\sum_k e^{tk}\, p_X(k), & X \text{ discrete}, \\ \displaystyle\int_{-\infty}^{\infty} e^{tx}\, f_X(x)\,\mathrm{d}x, & X \text{ continuous}, \end{cases}$$

defined for those real t at which the sum or integral converges. Always $M_X(0) = \mathbb{E}(e^0) = 1$. We say the mgf **exists** if it is finite for all t in some open interval $(-h, h)$ about 0; everything below assumes this.

The exponential weight e^{tx} is what makes the mgf "generate" moments: expanding $e^{tX} = \sum_{r\geq 0} t^r X^r/r!$ and taking expectations term by term gives $M_X(t) = \sum_{r\geq 0} \mathbb{E}(X^r)\, t^r/r!$, so the moments $\mathbb{E}(X^r)$ are exactly the Taylor coefficients of M_X at 0 (scaled by $r!$). The next subsection turns that observation into a differentiation recipe.

Example 3.11.2 (mgf of a single Bernoulli trial)**.** Let X be 1 with probability p and 0 with probability $1-p$ (one success/failure trial). Find $M_X(t)$.

Which rule, and why? We are asked for an mgf from a two-point distribution, so evaluate the defining sum $\sum_k e^{tk} p_X(k)$ directly. No moments or sums of variables are involved yet.

Solution.

Step 1: write the two-term sum. Only $k = 0$ and $k = 1$ carry mass, so

$$M_X(t) = e^{t\cdot 0}(1-p) + e^{t\cdot 1}p = (1-p) + p\,e^t.$$

Step 2: note convergence. The sum is finite for every real t, so the mgf exists on all of $\mathbb{R}$:

$$M_X(t) = 1 - p + p\,e^t, \qquad t \in \mathbb{R}.$$

Example 3.11.3 (mgf of a Poisson count)**.** Let $X \sim \text{Poisson}(\lambda)$ with $p_X(k) = e^{-\lambda}\lambda^k/k!$ for $k = 0, 1, 2, \ldots$. Find $M_X(t)$.

Which rule, and why? A discrete distribution with a known pmf: evaluate $\sum_k e^{tk} p_X(k)$. The factorial in the pmf signals that the exponential series $\sum_k z^k/k! = e^z$ will collapse it.

Solution.

Step 1: insert the pmf and collect the powers.

$$M_X(t) = \sum_{k=0}^{\infty} e^{tk}\, e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{(\lambda e^t)^k}{k!}.$$

Step 2: sum the exponential series. The inner sum is $\exp(\lambda e^t)$, so

$$M_X(t) = e^{-\lambda}\, e^{\lambda e^t} = \exp\big(\lambda(e^t - 1)\big), \qquad t \in \mathbb{R}.$$

Example 3.11.4 (mgf of an exponential lifetime)**.** Let $X \sim \text{Exp}(\lambda)$ with density $f_X(x) = \lambda e^{-\lambda x}$ on $x \geq 0$. Find $M_X(t)$ and state where it converges.

Which rule, and why? A continuous density, so compute $\int_0^\infty e^{tx} f_X(x)\,\mathrm{d}x$. The integrand $e^{(t-\lambda)x}$ only decays if $t < \lambda$, which is what restricts the domain of the mgf.

Solution.

Step 1: combine the exponentials.

$$M_X(t) = \int_0^\infty e^{tx}\,\lambda e^{-\lambda x}\,\mathrm{d}x = \lambda \int_0^\infty e^{-(\lambda - t)x}\,\mathrm{d}x.$$

Step 2: integrate, requiring $\lambda - t > 0$. For $t < \lambda$ the integral is $1/(\lambda - t)$, while for $t \geq \lambda$ it diverges. Hence

$$M_X(t) = \frac{\lambda}{\lambda - t}, \qquad t < \lambda.$$

Example 3.11.5 (mgf of a uniform variable)**.** Let $X \sim \mathrm{Uniform}(a, b)$ with density $f_X(x) = 1/(b-a)$ on $[a, b]$. Find $M_X(t)$.

Which rule, and why? Continuous density on a finite interval, so the defining integral is elementary and converges for every t. The only subtlety is the removable issue at $t = 0$, handled separately.

Solution.

Step 1: integrate e^{tx} over $[a, b]$. For $t \neq 0$,

$$M_X(t) = \frac{1}{b-a}\int_a^b e^{tx}\,\mathrm{d}x = \frac{1}{b-a}\cdot\frac{e^{tb} - e^{ta}}{t}.$$

Step 2: fill in the value at $t = 0$. As $t \to 0$ this ratio tends to 1 (it is $M_X(0)$), so

$$M_X(t) = \frac{e^{tb} - e^{ta}}{t\,(b-a)} \quad (t \neq 0), \qquad M_X(0) = 1, \qquad t \in \mathbb{R}.$$

3.11.2 Generating moments by differentiation

Theorem 3.11.6 (Moment property)**.** *If M_X exists on an interval about 0, then it is infinitely differentiable there, the operations of differentiation and expectation may be exchanged, and for every integer $r \geq 1$*

$$M_X^{(r)}(0) = \mathbb{E}\big(X^r\big).$$

In particular $\mathbb{E}(X) = M_X'(0)$, $\mathbb{E}(X^2) = M_X''(0)$, and

$$\mathrm{Var}(X) = M_X''(0) - \big[M_X'(0)\big]^2.$$

Proof. Differentiating $M_X(t) = \mathbb{E}(e^{tX})$ under the expectation, $M_X^{(r)}(t) = \mathbb{E}\big(X^r e^{tX}\big)$ (legitimate because the mgf exists in a neighbourhood of 0). Setting $t = 0$ gives $M_X^{(r)}(0) = \mathbb{E}(X^r)$. The variance formula is then the shortcut $\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = M_X''(0) - [M_X'(0)]^2$. □

Remark. Because $M_X(t) = \sum_{r \geq 0} \mathbb{E}(X^r)t^r/r!$, you can also just read $\mathbb{E}(X^r)$ off the Taylor coefficient of t^r, multiplied by $r!$. For some mgfs (e.g. the Poisson) the series route is quicker than repeated differentiation; both give the same numbers.

Example 3.11.7 (Mean and variance of a Poisson from its mgf)**.** With $M_X(t) = \exp\big(\lambda(e^t - 1)\big)$ for $X \sim \mathrm{Poisson}(\lambda)$, recover $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

Which rule, and why? We have the mgf and want moments, so differentiate and evaluate at 0: the moment property gives $\mathbb{E}(X) = M_X'(0)$ and $\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2$.

Solution.

Step 1: first derivative. By the chain rule, $M_X'(t) = \exp\big(\lambda(e^t - 1)\big) \cdot \lambda e^t$, so

$$M_X'(0) = e^0 \cdot \lambda e^0 = \lambda = \mathbb{E}(X).$$

Step 2: second derivative. Using the product rule on $M_X'(t) = \lambda e^t\, M_X(t)$,

$$M_X''(t) = \lambda e^t M_X(t) + \lambda e^t\big(\lambda e^t M_X(t)\big) = \lambda e^t M_X(t)\big(1 + \lambda e^t\big).$$

At $t = 0$, $M_X(0) = 1$ gives $M_X''(0) = \lambda(1 + \lambda) = \lambda + \lambda^2 = \mathbb{E}(X^2)$.
Step 3: assemble the variance.

$$\mathrm{Var}(X) = M_X''(0) - [M_X'(0)]^2 = (\lambda + \lambda^2) - \lambda^2 = \lambda.$$

So $\mathbb{E}(X) = \mathrm{Var}(X) = \lambda$, the familiar Poisson identity.

Example 3.11.8 (Mean and variance of an exponential from its mgf)**.** With $M_X(t) = \lambda/(\lambda - t)$ for $t < \lambda$ ($X \sim \mathrm{Exp}(\lambda)$), find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

Which rule, and why? An mgf is given and moments are wanted, so apply the moment property. Writing M_X as $\lambda(\lambda - t)^{-1}$ makes the power-rule derivatives clean.
Solution.
Step 1: differentiate twice. With $M_X(t) = \lambda(\lambda - t)^{-1}$,

$$M_X'(t) = \lambda(\lambda - t)^{-2}, \qquad M_X''(t) = 2\lambda(\lambda - t)^{-3}.$$

Step 2: evaluate at 0.

$$\mathbb{E}(X) = M_X'(0) = \lambda \cdot \lambda^{-2} = \frac{1}{\lambda}, \qquad \mathbb{E}(X^2) = M_X''(0) = 2\lambda \cdot \lambda^{-3} = \frac{2}{\lambda^2}.$$

Step 3: variance.

$$\mathrm{Var}(X) = \frac{2}{\lambda^2} - \Big(\frac{1}{\lambda}\Big)^2 = \frac{1}{\lambda^2}.$$

Example 3.11.9 (Moments of a Bernoulli trial)**.** For X Bernoulli(p) with $M_X(t) = 1 - p + p\,e^t$, find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$ from the mgf.

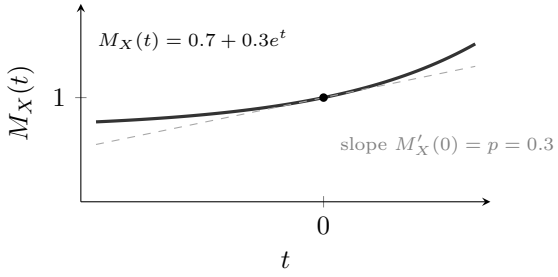
Which rule, and why? Given the mgf, the moment property applies. Here every derivative of $1 - p + pe^t$ is just pe^t, so the arithmetic is immediate — a good sanity check on the method.
Solution.
Step 1: derivatives. $M_X'(t) = p\,e^t$ and $M_X''(t) = p\,e^t$.
Step 2: evaluate at 0.

$$\mathbb{E}(X) = M_X'(0) = p, \qquad \mathbb{E}(X^2) = M_X''(0) = p.$$

Step 3: variance.

$$\mathrm{Var}(X) = p - p^2 = p(1 - p),$$

matching the known Bernoulli mean p and variance $p(1 - p)$.



The dashed line is the tangent at $t = 0$; its height 1 is $M_X(0)$ and its slope is $M_X'(0) = \mathbb{E}(X) = p$.

Example 3.11.10 (Mean of a uniform from its mgf, via the series)**.** For $X \sim \text{Uniform}(0,1)$ the mgf is $M_X(t) = (e^t - 1)/t$. Use the Taylor route to confirm $\mathbb{E}(X) = \frac{1}{2}$ and $\mathbb{E}(X^2) = \frac{1}{3}$.

Which rule, and why? Differentiating $(e^t - 1)/t$ directly is messy near $t = 0$, so use the remark: expand M_X in powers of t and read $\mathbb{E}(X^r)$ off the coefficient of $t^r/r!$.

Solution.

Step 1: expand the numerator. $e^t - 1 = t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots$, so dividing by t,

$$M_X(t) = \frac{e^t - 1}{t} = 1 + \frac{t}{2!} + \frac{t^2}{3!} + \frac{t^3}{4!} + \cdots .$$

Step 2: match coefficients to $\mathbb{E}(X^r)t^r/r!$. The coefficient of t^1 is $\frac{1}{2} = \mathbb{E}(X)/1!$, so $\mathbb{E}(X) = \frac{1}{2}$. The coefficient of t^2 is $\frac{1}{6} = \mathbb{E}(X^2)/2!$, so $\mathbb{E}(X^2) = 2! \cdot \frac{1}{6} = \frac{1}{3}$. Then

$$\text{Var}(X) = \tfrac{1}{3} - \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{3} - \tfrac{1}{4} = \tfrac{1}{12},$$

the standard mean, second moment, and variance of $\text{Uniform}(0,1)$.

3.11.3 The catalogue of mgfs

The mgfs of the named families are worth memorising; each follows from the defining sum/integral exactly as in the worked examples above. The domain column lists the t for which the mgf is finite.

Distribution	$M_X(t)$	valid for
Bernoulli(p)	$1 - p + p\,e^t$	$t \in \mathbb{R}$
$X \sim \text{Bin}(n,p)$	$\big(1 - p + p\,e^t\big)^n$	$t \in \mathbb{R}$
$X \sim \text{Geom}(p)$ (on $1, 2, \dots$)	$\dfrac{p\,e^t}{1 - (1-p)e^t}$	$t < -\ln(1-p)$
$X \sim \text{Poisson}(\lambda)$	$\exp\big(\lambda(e^t - 1)\big)$	$t \in \mathbb{R}$
$X \sim \text{Uniform}(a,b)$	$\dfrac{e^{tb} - e^{ta}}{t\,(b-a)}$	$t \in \mathbb{R}$
$X \sim \text{Exp}(\lambda)$	$\dfrac{\lambda}{\lambda - t}$	$t < \lambda$
$X \sim \text{Gamma}(\alpha, \lambda)$	$\left(\dfrac{\lambda}{\lambda - t}\right)^{\alpha}$	$t < \lambda$
$X \sim \text{N}(\mu, \sigma^2)$	$\exp\big(\mu t + \frac{1}{2}\sigma^2 t^2\big)$	$t \in \mathbb{R}$

Proposition 3.11.11 (mgf of the normal distribution)**.** *If $X \sim \text{N}(\mu, \sigma^2)$ then $M_X(t) = \exp\big(\mu t + \frac{1}{2}\sigma^2 t^2\big)$ for all t.*

Proof. With density $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\big(-(x-\mu)^2/(2\sigma^2)\big)$,

$$M_X(t) = \int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(tx - \frac{(x-\mu)^2}{2\sigma^2}\Big)\,\mathrm{d}x.$$

Complete the square in the exponent: $tx - \frac{(x-\mu)^2}{2\sigma^2} = \mu t + \frac{1}{2}\sigma^2 t^2 - \frac{1}{2\sigma^2}\big(x - (\mu + \sigma^2 t)\big)^2$. The remaining integral is that of a $\text{N}(\mu + \sigma^2 t, \sigma^2)$ density, which equals 1, leaving $M_X(t) = \exp(\mu t + \frac{1}{2}\sigma^2 t^2)$. □

Example 3.11.12 (Standard normal moments from the mgf)**.** For $Z \sim \text{N}(0,1)$, $M_Z(t) = e^{t^2/2}$. Find $\mathbb{E}(Z)$, $\mathbb{E}(Z^2)$, and $\mathbb{E}(Z^3)$.

Which rule, and why? The mgf is given and moments are wanted, so use the moment property; the clean form $e^{t^2/2}$ makes successive derivatives manageable, and its even symmetry foretells odd moments vanish.

Solution.

Step 1: first two derivatives. $M_Z'(t) = t\,e^{t^2/2}$ and $M_Z''(t) = e^{t^2/2} + t^2 e^{t^2/2} = (1+t^2)e^{t^2/2}$. Hence

$$\mathbb{E}(Z) = M_Z'(0) = 0, \qquad \mathbb{E}(Z^2) = M_Z''(0) = 1,$$

so $\mathrm{Var}(Z) = 1 - 0^2 = 1$.

Step 2: third derivative. Differentiating $M_Z''(t) = (1+t^2)e^{t^2/2}$ gives $M_Z'''(t) = 2t\,e^{t^2/2} + (1+t^2)t\,e^{t^2/2} = (3t+t^3)e^{t^2/2}$, so

$$\mathbb{E}(Z^3) = M_Z'''(0) = 0.$$

As anticipated, the symmetric standard normal has all odd moments equal to 0.

Example 3.11.13 (Gamma mean and variance from its mgf)**.** For $X \sim \mathrm{Gamma}(\alpha, \lambda)$, $M_X(t) = \big(\lambda/(\lambda - t)\big)^{\alpha}$, $t < \lambda$. Find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

Which rule, and why? Given the mgf, apply the moment property. Writing $M_X(t) = \lambda^\alpha(\lambda - t)^{-\alpha}$ turns the derivatives into routine power-rule steps.

Solution.

Step 1: differentiate.

$$M_X'(t) = \alpha\lambda^\alpha(\lambda - t)^{-\alpha-1}, \qquad M_X''(t) = \alpha(\alpha+1)\lambda^\alpha(\lambda - t)^{-\alpha-2}.$$

Step 2: evaluate at 0.

$$\mathbb{E}(X) = M_X'(0) = \alpha\lambda^\alpha\lambda^{-\alpha-1} = \frac{\alpha}{\lambda}, \qquad \mathbb{E}(X^2) = M_X''(0) = \alpha(\alpha+1)\lambda^\alpha\lambda^{-\alpha-2} = \frac{\alpha(\alpha+1)}{\lambda^2}.$$

Step 3: variance.

$$\mathrm{Var}(X) = \frac{\alpha(\alpha+1)}{\lambda^2} - \frac{\alpha^2}{\lambda^2} = \frac{\alpha}{\lambda^2}.$$

Setting $\alpha = 1$ recovers the exponential's $1/\lambda$ and $1/\lambda^2$, a useful check.

3.11.4 The scaling rule and uniqueness

Proposition 3.11.14 (Affine scaling)**.** *For constants a, b, $M_{aX+b}(t) = e^{bt}\,M_X(at)$.*

Proof. $M_{aX+b}(t) = \mathbb{E}\big(e^{t(aX+b)}\big) = \mathbb{E}\big(e^{bt}e^{(at)X}\big) = e^{bt}\,\mathbb{E}\big(e^{(at)X}\big) = e^{bt}M_X(at)$, pulling the constant e^{bt} outside the expectation. □

Theorem 3.11.15 (Uniqueness)**.** *If two random variables have mgfs that are finite and equal on some open interval containing* 0*, then they have the same distribution. Equivalently, an mgf (when it exists near* 0*) determines the distribution.*

Uniqueness is what lets us *identify* a distribution from its mgf: compute a candidate's mgf, match it to an entry in the catalogue, and conclude the distributions agree — no inversion integral required.

Example 3.11.16 (Standardising a normal via the scaling rule)**.** Let $X \sim \mathrm{N}(\mu, \sigma^2)$ and $Z = (X - \mu)/\sigma$. Use mgfs to identify the distribution of Z.

Which rule, and why? $Z = aX + b$ with $a = 1/\sigma$, $b = -\mu/\sigma$ is an affine transform, so the scaling rule $M_{aX+b}(t) = e^{bt}M_X(at)$ produces M_Z; uniqueness then names the distribution.

Solution.

Step 1: apply the scaling rule. With $M_X(s) = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$, $a = 1/\sigma$, $b = -\mu/\sigma$,

$$M_Z(t) = e^{bt} M_X(at) = e^{-\mu t/\sigma} \exp\Big(\mu \cdot \tfrac{t}{\sigma} + \tfrac{1}{2}\sigma^2 \cdot \tfrac{t^2}{\sigma^2}\Big).$$

Step 2: simplify the exponent. The $-\mu t/\sigma$ and $+\mu t/\sigma$ cancel, leaving

$$M_Z(t) = \exp\big(\tfrac{1}{2}t^2\big).$$

Step 3: identify by uniqueness. This is the catalogue mgf of $\mathrm{N}(0,1)$, so $Z \sim \mathrm{N}(0,1)$: the familiar standardisation, proved through mgfs.

Example 3.11.17 (Doubling an exponential)**.** Let $X \sim \mathrm{Exp}(3)$ and $Y = 2X$. Identify the distribution of Y from its mgf.

Which rule, and why? $Y = 2X$ is a scaling, so $M_Y(t) = M_X(2t)$ (here $b = 0$). Matching the result to the exponential/gamma row of the catalogue identifies Y by uniqueness.

Solution.

Step 1: scale. With $M_X(s) = 3/(3-s)$ for $s < 3$, set $s = 2t$:

$$M_Y(t) = M_X(2t) = \frac{3}{3-2t} = \frac{3/2}{(3/2)-t}, \qquad 2t < 3.$$

Step 2: identify. This is exactly $\lambda/(\lambda - t)$ with $\lambda = 3/2$, valid for $t < 3/2$. By uniqueness

$$Y \sim \mathrm{Exp}(3/2).$$

The new rate is half the old, consistent with $\mathbb{E}(Y) = 2\mathbb{E}(X) = 2/3 = 1/(3/2)$.

3.11.5 Sums of independent variables

Theorem 3.11.18 (mgf of a sum of independents)**.** *If $X_1, \ldots, X_n$ are independent, then*

$$M_{X_1+\cdots+X_n}(t) = \prod_{i=1}^{n} M_{X_i}(t)$$

on the intersection of their domains. If, in addition, the X_i are identically distributed, $M_{S_n}(t) = \big[M_{X_1}(t)\big]^n$.

Proof. Let $S = \sum_i X_i$. Then $e^{tS} = \prod_i e^{tX_i}$, and independence of the X_i makes the e^{tX_i} independent too, so the expectation of the product is the product of expectations:

$$M_S(t) = \mathbb{E}\Big(\prod_i e^{tX_i}\Big) = \prod_i \mathbb{E}\big(e^{tX_i}\big) = \prod_i M_{X_i}(t). \qquad \square$$

Combined with uniqueness, the product rule is the workhorse for finding the law of a sum: multiply the mgfs, then recognise the product in the catalogue.

Example 3.11.19 (Sum of independent Poissons is Poisson)**.** Let $X \sim \mathrm{Poisson}(\lambda)$ and $Y \sim \mathrm{Poisson}(\mu)$ be independent. Identify the distribution of $S = X + Y$.

Which rule, and why? A sum of two *independent* variables, so multiply their mgfs; the resulting form is matched against the catalogue by uniqueness. Convolving the pmfs would work but is far slower.

Solution.

Step 1: multiply the mgfs.

$$M_S(t) = M_X(t)\, M_Y(t) = \exp\big(\lambda(e^t - 1)\big) \exp\big(\mu(e^t - 1)\big).$$

Step 2: combine exponents.

$$M_S(t) = \exp\big((\lambda+\mu)(e^t-1)\big).$$

Step 3: identify. This is the Poisson mgf with parameter $\lambda+\mu$, so by uniqueness

$$S = X + Y \sim \text{Poisson}(\lambda+\mu).$$

The rates simply add — two independent arrival streams merge into one.

Example 3.11.20 (Sum of independent normals is normal)**.** Let $X \sim \text{N}(\mu_1, \sigma_1^2)$ and $Y \sim \text{N}(\mu_2, \sigma_2^2)$ be independent. Find the distribution of $S = X + Y$.

Which rule, and why? Independent sum again, so the product rule applies; the product of two normal mgfs is itself a normal mgf, and uniqueness reads off the parameters.

Solution.

Step 1: multiply the mgfs.

$$M_S(t) = \exp\big(\mu_1 t + \tfrac{1}{2}\sigma_1^2 t^2\big)\exp\big(\mu_2 t + \tfrac{1}{2}\sigma_2^2 t^2\big).$$

Step 2: add the exponents.

$$M_S(t) = \exp\Big((\mu_1+\mu_2)t + \tfrac{1}{2}(\sigma_1^2+\sigma_2^2)t^2\Big).$$

Step 3: identify. This is the mgf of $\text{N}(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2)$, so

$$S = X + Y \sim \text{N}\big(\mu_1+\mu_2,\ \sigma_1^2+\sigma_2^2\big).$$

Means add and, because of independence, so do variances.

Example 3.11.21 (Sum of independent Bernoullis is binomial)**.** Let $X_1, \ldots, X_n$ be independent Bernoulli(p) trials and $S_n = X_1 + \cdots + X_n$. Identify the distribution of S_n.

Which rule, and why? An iid independent sum, so the mgf is the nth power of a single Bernoulli mgf; the power form matches the binomial row of the catalogue.

Solution.

Step 1: raise the single-trial mgf to the nth power. Each $M_{X_i}(t) = 1 - p + p\,e^t$, so by independence

$$M_{S_n}(t) = \big(1 - p + p\,e^t\big)^n.$$

Step 2: identify. This is exactly the catalogue mgf of $\text{Bin}(n,p)$, so by uniqueness

$$S_n \sim \text{Bin}(n,p),$$

recovering the fact that a binomial count is a sum of n independent Bernoulli trials.

Example 3.11.22 (Sum of independent exponentials is gamma)**.** Let $T_1, \ldots, T_k$ be independent Exp(λ) service times and $W = T_1 + \cdots + T_k$. Identify the distribution of W, then state $\mathbb{E}(W)$ and $\text{Var}(W)$.

Which rule, and why? An iid independent sum, so $M_W = [M_{T_1}]^k$; the power of the exponential mgf is the gamma mgf, and the catalogue moments follow at once.

Solution.

Step 1: power of the exponential mgf. With $M_{T_i}(t) = \lambda/(\lambda - t)$ for $t < \lambda$,

$$M_W(t) = \Big(\frac{\lambda}{\lambda - t}\Big)^k, \qquad t < \lambda.$$

Step 2: identify. This is the Gamma(α, λ) mgf with shape $\alpha = k$, so by uniqueness

$$W \sim \text{Gamma}(k, \lambda).$$

Step 3: moments from the gamma formulas.

$$\mathbb{E}(W) = \frac{k}{\lambda}, \qquad \text{Var}(W) = \frac{k}{\lambda^2},$$

i.e. k times the mean and variance of one exponential, as additivity predicts.

Example 3.11.23 (Total daily demand: a numerical merge)**.** A warehouse receives orders from two independent regions; region A's daily count is $X \sim \text{Poisson}(4)$ and region B's is $Y \sim \text{Poisson}(7)$. Let $S = X + Y$ be the total. Identify the law of S and compute $\mathbb{P}(S = 10)$.

Which rule, and why? Independent Poisson sum, so the product rule plus uniqueness names S; once the parameter is known, the Poisson pmf gives the requested probability. No joint convolution needed.

Solution.

Step 1: merge the rates. By the previous result, $S \sim \text{Poisson}(4 + 7) = \text{Poisson}(11)$.

Step 2: evaluate the pmf at $k = 10$.

$$\mathbb{P}(S = 10) = e^{-11}\frac{11^{10}}{10!}.$$

Step 3: numerical value. Since $11^{10} = 25{,}937{,}424{,}601$, $10! = 3{,}628{,}800$, and $e^{-11} \approx 1.6702 \times 10^{-5}$,

$$\mathbb{P}(S = 10) = e^{-11}\frac{25{,}937{,}424{,}601}{3{,}628{,}800} \approx 0.1194.$$

A direct check: the mean total is 11, and $S = 10$ sits just below it, so a probability near 0.12 is the right order of magnitude.

3.12 Exercises

All problems below are original. Full solutions begin on page 201.

A. Binomial and hypergeometric probabilities

A1. A barista at a coffee kiosk knows that, independently for each customer, the probability that a customer orders an oat-milk drink is 0.3. During one quiet stretch exactly 6 customers arrive. Let X be the number of those 6 who order an oat-milk drink, so $X \sim \text{Bin}(6, 0.3)$.

(a) Compute $\mathbb{P}(X = 2)$.

(b) Compute $\mathbb{E}(X)$ and $\text{Var}(X)$.

A2. A factory test stand checks circuit boards one at a time. Each board passes independently with probability 0.9. A batch of 10 boards is tested; let Y be the number that pass, so $Y \sim \text{Bin}(10, 0.9)$.

(a) Find the probability that all 10 boards pass.

(b) Find the probability that exactly 9 pass.

(c) Find the probability that at least 9 pass.

A3. A drawer holds 12 batteries, of which 4 are dead and 8 are good. You grab 3 batteries at random, all at once (sampling without replacement). Let X be the number of dead batteries among the 3 you grabbed.

(a) Identify the distribution of X, including its parameters.

(b) Compute $\mathbb{P}(X = 1)$.

(c) Compute $\mathbb{P}(X \geq 1)$.

A4. A multiple-choice quiz has 8 questions, each with 5 choices and exactly one correct answer. A student guesses every question independently and uniformly at random. Let W be the number of correct answers.

(a) State the distribution of W.

(b) Find the probability the student gets at most 1 question right.

(c) The quiz is passed only with 6 or more correct. Find the probability of passing by pure guessing.

A5. A wildlife crew tags fish in a small pond. The pond contains 50 fish, of which 15 have been tagged. On a later survey the crew nets 8 distinct fish (without replacement). Let T be the number of tagged fish in the netted sample.

(a) Identify the distribution of T and write its pmf.

(b) Compute the expected number of tagged fish, $\mathbb{E}(T)$.

(c) Compute $\mathbb{P}(T = 3)$, leaving the answer as a ratio of binomial coefficients and as a decimal rounded to four places.

A6. A regional shipment contains 2000 memory modules, of which 40 are defective (2%). A retailer buys 25 modules drawn at random without replacement. Let D be the number of defective modules the retailer receives.

(a) Write the exact distribution of D and the exact pmf for $\mathbb{P}(D = k)$.

(b) Explain why D is well approximated by a binomial distribution, and give the approximating $\text{Bin}(n, p)$ parameters.

(c) Using the binomial approximation, estimate $\mathbb{P}(D = 0)$ and $\mathbb{P}(D \geq 1)$ to four decimal places.

A7. An archer hits the bullseye on each independent shot with probability $p = 0.4$. She will take $n = 12$ shots. Let $H \sim \text{Bin}(12, 0.4)$ be the number of bullseyes.

(a) Show that the most likely value (the mode) of H is 5, by finding all k for which $\mathbb{P}(H = k) \geq \mathbb{P}(H = k - 1)$.

(b) Given that she scores at least one bullseye, find the probability she scores exactly one. (That is, compute $\mathbb{P}(H = 1 \mid H \geq 1)$.)

B. Discrete random variables: pmf and cdf

B1. A bakery grades each morning by the number X of cakes returned as defective, with

$$p_X(0) = 0.45, \quad p_X(1) = 0.30, \quad p_X(2) = 0.15, \quad p_X(3) = 0.07, \quad p_X(4) = 0.03.$$

(a) Verify this is a valid pmf. (b) Find $\mathbb{P}(X \geq 2)$ and $\mathbb{P}(1 \leq X \leq 3)$.

B2. A delivery van makes $X \in \{0, 1, 2, 3\}$ wrong-address stops on a route, modelled by $p_X(k) = c\,(k+1)$ for those four values and 0 otherwise. Find the constant c, then compute $\mathbb{P}(X \leq 1)$ and $\mathbb{P}(X > 2)$.

B3. A weather station reports the number X of days until the next measurable rainfall, $X \in \{1, 2, 3, \dots\}$, modelled by $p_X(k) = c\,(0.6)^k$. (a) Find c. (b) Find $\mathbb{P}(X \leq 3)$ and $\mathbb{P}(X \geq 2)$.

B4. An online tutoring service models the number X of follow-up sessions a new student books, $X \in \{1, 2, 3, \dots\}$, by $p_X(k) = c\,k\,(1/2)^k$. (a) Find the constant c. (b) Compute $\mathbb{P}(X = 1)$, $\mathbb{P}(X = 2)$, and $\mathbb{P}(X = 3)$. *(You may use $\sum_{k \geq 1} k\,r^k = r/(1-r)^2$ for $|r| < 1$.)*

B5. Two fair six-sided dice are rolled; let $W = \min(d_1, d_2)$ be the smaller of the two faces (ties count that face). (a) Find $\mathbb{P}(W \geq w)$ for $w = 1, \dots, 6$ and use it to derive the pmf p_W. (b) Write out the cdf F_W. (c) Find $\mathbb{P}(2 \leq W \leq 4)$.

B6. A four-question true/false quiz is answered by pure guessing, each question correct independently with probability $\frac{1}{2}$. Let X be the number of correct answers, with $p_X(k) = c\binom{4}{k}$ for $k = 0, 1, 2, 3, 4$. (a) Find c and identify the named distribution of X. (b) Write out the cdf F_X. (c) Use F_X to compute $\mathbb{P}(1 < X \leq 3)$.

B7. A technician retests a flaky sensor until it first passes, each independent test passing with probability 0.2. Let X be the number of the test on which it first passes, $X \in \{1, 2, 3, \dots\}$. (a) Give the pmf p_X and show that the cdf is $F_X(n) = 1 - (0.8)^n$ for integers $n \geq 1$. (b) Using F_X, find $\mathbb{P}(3 \leq X \leq 5)$. (c) Find the smallest n for which $\mathbb{P}(X \leq n) \geq 0.95$.

C. Continuous random variables and densities

C1. The depth X (in centimetres) of overnight snowfall at a weather station has density

$$f_X(x) = \begin{cases} \frac{1}{8}x, & 0 \leq x \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

Verify that f_X is a legitimate density, then compute $\mathbb{P}(X > 3)$.

C2. A robot vacuum's battery charge X (as a fraction of full, so $0 \leq x \leq 1$) at a random instant has density $f_X(x) = c\,x^3$ on $[0, 1]$ and 0 elsewhere. Find the constant c, and then find $\mathbb{P}(X < \frac{1}{2})$.

C3. The time X (in hours) until a freshly painted wall is dry has density $f_X(x) = c\,e^{-x/3}$ for $x \geq 0$ and 0 for $x < 0$.

(a) Find c.

(b) Find the cumulative distribution function F_X.

(c) Use F_X to compute $\mathbb{P}(1 < X < 4)$, leaving the answer in terms of e and then as a decimal.

C4. The fraction X of a reservoir that is full one week after the rainy season has cumulative distribution function

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \sqrt{x}, & 0 \leq x \leq 1, \\ 1, & x > 1. \end{cases}$$

Find the density f_X, verify it integrates to 1, and find the median of X.

C5. The lifetime X (in years) of a budget hard drive has density $f_X(x) = \dfrac{4}{(1+x)^5}$ for $x \geq 0$ and 0 otherwise.

(a) Find the cumulative distribution function F_X.

(b) Find the median lifetime.

(c) Find the value $x_{0.90}$ below which 90% of these drives fail (the 90th percentile).

C6. A bus is equally likely to arrive at any moment in the 15-minute window after its scheduled time, so the delay $X \sim \text{Uniform}(0, 15)$ minutes.

(a) Write down the density and the cdf of X.

(b) A rider arrives exactly on schedule and will give up and walk if the bus has not come within 10 minutes. Find the probability the rider waits for the bus (the bus arrives within 10 minutes).

(c) Given that the bus is already more than 6 minutes late, find the probability it arrives within the next 3 minutes (i.e. $\mathbb{P}(X < 9 \mid X > 6)$).

C7. A circuit board has its single defect located at a distance X (in centimetres) from the left edge of a 10-centimetre board. The defect is more likely toward the middle: the density rises linearly from 0 at $x = 0$ to a peak at the centre $x = 5$ and falls linearly back to 0 at $x = 10$ (a symmetric "tent"). Concretely,

$$f_X(x) = \begin{cases} k\,x, & 0 \leq x \leq 5, \\ k(10 - x), & 5 \leq x \leq 10, \\ 0, & \text{otherwise}, \end{cases}$$

for a constant $k > 0$.

(a) Find k.

(b) Find $\mathbb{P}(X \leq 5)$ without computing an integral, and explain the answer in one sentence.

(c) Find $\mathbb{P}(X \leq 3)$.

(d) Given that the defect lies in the left half of the board ($X \leq 5$), find the probability it lies within 3 centimetres of the left edge, i.e. $\mathbb{P}(X \leq 3 \mid X \leq 5)$.

D. Expected values

D1. A small bakery sells a "surprise box" each morning. The number of pastries X in a box follows the pmf

$$p_X(2) = 0.2, \qquad p_X(3) = 0.5, \qquad p_X(4) = 0.2, \qquad p_X(5) = 0.1.$$

Find the expected number of pastries $\mathbb{E}(X)$ in a box.

D2. A street performer earns a tip X (in dollars) from each passerby, where X takes the value 0 with probability 0.6, the value 2 with probability 0.3, and the value 5 with probability 0.1. If the performer's nightly costs are a flat \$4, let $W = X_1 + X_2 + \cdots + X_{50} - 4$ be the net profit after 50 independent passersby (each tip distributed like X). Find $\mathbb{E}(W)$.

D3. A loaded spinner lands on a real-valued angle X measured in radians, where X is uniform on the interval $(\pi,\, 3\pi)$. Using the mean of a uniform random variable, find $\mathbb{E}(X)$, and then use linearity to find $\mathbb{E}\big(\frac{1}{\pi}X - 2\big)$.

D4. The lifetime T (in years) of a budget laptop battery is exponential with rate $\lambda = \frac{1}{3}$ per year, so $f_T(t) = \frac{1}{3}e^{-t/3}$ for $t \geq 0$. A warranty pays the customer a fixed \$90 if the battery is replaced, and replacement happens at time T. The store discounts that payout to present value $g(T) = 90\,e^{-0.1\,T}$. Compute the expected discounted payout $\mathbb{E}\big(g(T)\big) = \mathbb{E}\big(90\,e^{-0.1\,T}\big)$.

D5. A continuous random variable X has density

$$f_X(x) = \begin{cases} c\,x(2-x), & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

First find the constant c, then compute $\mathbb{E}(X)$.

D6. At a help desk, the number of calls N arriving in an hour is Poisson with mean $\lambda = 4$. Each call, independently, is logged as "resolved" with probability 0.75. Let R be the number of resolved calls in the hour. Find $\mathbb{E}(R)$. *(Hint: condition on N; given $N = n$, R is binomial. Use $\mathbb{E}(R \mid N = n) = 0.75\,n$ and then take the expected value over N.)*

D7. A target is a disk of radius 2 (in meters). A dart lands at a point whose distance X from the center has density $f_X(x) = \frac{x}{2}$ for $0 \le x \le 2$ (and 0 otherwise). The score awarded is $g(X) = 10 - X^2$ points. Find the expected score $\mathbb{E}\big(10 - X^2\big)$ by first computing $\mathbb{E}(X^2)$ and then applying linearity.

E. Variance and standard deviation

E1. A vending machine dispenses X cans per customer, with pmf

k	1	2	3	4
$p_X(k)$	0.40	0.35	0.15	0.10

Compute $\mathbb{E}(X)$, $\mathbb{E}(X^2)$, $\mathrm{Var}(X)$, and $\mathrm{SD}(X)$.

E2. A meter reads the voltage error X (in millivolts) of a circuit, with density $f_X(x) = \frac{3}{4}(1 - x^2)$ for $-1 \le x \le 1$ and 0 otherwise. The density is symmetric about 0. Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

E3. Monthly demand at a kiosk has mean $\mathbb{E}(D) = 240$ units and standard deviation $\mathrm{SD}(D) = 30$ units. The kiosk's profit (in dollars) is $P = 8D - 500$, where \$8 is the margin per unit and \$500 is a fixed monthly cost. Find $\mathrm{Var}(P)$ and $\mathrm{SD}(P)$.

E4. Defects on a roll of fabric occur as $X \sim \mathrm{Poisson}(\lambda)$ with $\lambda = 3.5$ per roll. A second inspection on the same roll independently finds $Y \sim \mathrm{Poisson}(2.5)$ further defects. The two counts are independent. Using the variance of a Poisson and the fact that variances of independent variables add, find $\mathrm{SD}(X)$, $\mathrm{SD}(Y)$, and $\mathrm{SD}(X + Y)$.

E5. A daily rainfall amount X (in cm) has density $f_X(x) = \lambda e^{-\lambda x}$ for $x \ge 0$ with $\lambda = \frac{1}{5}$, so $X \sim \mathrm{Exp}(\frac{1}{5})$. A reservoir gains $W = 1.2\,X$ million liters from each day's rain. Find $\mathbb{E}(W)$, $\mathrm{Var}(W)$, and $\mathrm{SD}(W)$, and state how $\mathrm{SD}(W)$ compares to $\mathbb{E}(W)$.

E6. An archer's score on one shot is X rings, with pmf $p_X(k) = c\,k$ for $k = 1, 2, 3, 4, 5$ and 0 otherwise.

(a) Find the constant c.

(b) Compute $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$.

(c) Find $\mathrm{Var}(X)$ and $\mathrm{SD}(X)$.

E7. A particle's position X (in meters) along a track has density $f_X(x) = \frac{3}{8}\,x^2$ for $0 \le x \le 2$ and 0 otherwise.

(a) Verify that f_X integrates to 1, then find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$.

(b) The track is recalibrated to $Y = 3X - 1$ (a stretch plus an offset). Find $\mathrm{Var}(Y)$ and $\mathrm{SD}(Y)$.

(c) The position is otherwise unknown except for $\mathbb{E}(X)$ and $\mathrm{SD}(X)$ from part (a). Use Chebyshev's inequality to give a guaranteed lower bound on $\mathbb{P}\big(|X - \mathbb{E}(X)| < 2\ \mathrm{SD}(X)\big)$.

F. Joint densities, marginals, and independence

F1. A vending machine dispenses a snack (X) and a drink (Y) on each use. Over many uses the joint pmf of (X, Y), where $X \in \{0, 1\}$ counts whether the snack jammed and $Y \in \{0, 1\}$ counts whether the drink jammed, is

$p_{X,Y}(x, y)$	$y = 0$	$y = 1$
$x = 0$	0.72	0.08
$x = 1$	0.12	0.08

(a) Find the marginal pmfs p_X and p_Y. (b) Compute $\mathbb{P}(X + Y \geq 1)$, the probability that at least one of the two jams. (c) Are X and Y independent? Justify with the factorisation test.

F2. Each morning a clinic logs the number of walk-in patients X and the number of those who need a follow-up appointment Y (so $0 \leq Y \leq X$). The joint pmf is

$$p_{X,Y}(x, y) = c\,\frac{1}{x}, \qquad x \in \{1, 2, 3\}, \quad y \in \{1, 2, \ldots, x\}.$$

(a) Find the constant c. (b) Find the marginal pmf p_X and identify it as a named distribution. (c) Find $p_Y(1)$.

F3. Let (X, Y) have joint density

$$f_{X,Y}(x, y) = k\,(x + 2y), \qquad 0 < x < 1, \quad 0 < y < 1,$$

and 0 otherwise. (a) Find the constant k. (b) Find the marginal density $f_X(x)$ for $0 < x < 1$. (c) Compute $\mathbb{P}(Y > X)$.

F4. A factory's two independent sensors report drift amounts X and Y (in mm), where $X \sim \text{Exp}(2)$ and $Y \sim \text{Exp}(3)$ are independent. (a) Write the joint density $f_{X,Y}(x, y)$ on $x > 0,\ y > 0$. (b) Compute $\mathbb{P}(X > 1,\ Y > 1)$. (c) Compute $\mathbb{P}(X < Y)$.

F5. Points (X, Y) are dropped uniformly on the triangle $T = \{(x, y) : 0 < x < y < 2\}$, so $f_{X,Y}(x, y) = c$ on T and 0 otherwise. (a) Find c. (b) Find the marginal density $f_X(x)$ for $0 < x < 2$ and the marginal $f_Y(y)$ for $0 < y < 2$. (c) Are X and Y independent? Explain in one sentence using the support. (d) Compute $\mathbb{P}(Y > 1)$.

F6. A robotic arm's settling time X (seconds) and overshoot Y (mm) have joint density

$$f_{X,Y}(x, y) = 2e^{-x}e^{-2y}, \qquad x > 0,\ y > 0,$$

and 0 otherwise. (a) Show that X and Y are independent and name the marginal distributions. (b) Compute $\mathbb{E}(XY)$. (c) Compute $\mathbb{P}(Y < X)$.

F7. Let (X, Y) have joint density

$$f_{X,Y}(x, y) = 8xy, \qquad 0 < x < y < 1,$$

and 0 otherwise. (a) Verify the density integrates to 1. (b) Find both marginal densities f_X and f_Y. (c) Determine whether X and Y are independent. (d) Compute $\mathbb{E}(X)$ and $\mathbb{E}(Y)$.

G. Transforming and combining random variables

G1. A sensor reports a normalized reading $X \sim \text{Uniform}(0, 1)$. The display value is the linear rescaling $Y = 8X - 3$. Identify the distribution of Y and write its density $f_Y(y)$, stating the range on which it is positive.

G2. The radius (in cm) of a randomly chosen ball bearing is $X \sim \text{Uniform}(0, 2)$. Its cross-sectional "size index" is $Y = X^2$. Use the cdf method to find the density $f_Y(y)$ of Y, and state the values of y for which it is positive.

G3. The lifetime (in years) of a relay is $X \sim \text{Exp}(\lambda)$ with $\lambda > 0$. An engineer records the transformed quantity $Y = \sqrt{X}$. Using the change-of-variable (Jacobian) formula for a monotone transformation, find $f_Y(y)$ and give its support.

G4. A signal's exponent is $X \sim \text{Uniform}(0, \ln 4)$ and the measured intensity is $Y = e^X$. (a) Find the density $f_Y(y)$ and its support. (b) Compute $\mathbb{E}(Y)$ directly from f_Y, and verify it equals $\mathbb{E}(e^X)$ computed from f_X.

G5. The processing delay of a queue is $X \sim \text{Exp}(2)$ (rate 2 per second). A performance score is defined by $Y = e^{-X}$, which lies in $(0, 1)$. Find the density $f_Y(y)$ of the score, and identify the resulting distribution.

G6. Two independent calibration errors $X_1, X_2 \sim \text{Uniform}(0, 1)$ are added to form the total error $S = X_1 + X_2$. Use convolution to derive the density $f_S(s)$ for all s, and state the value of $\mathbb{P}(S \le 1)$.

G7. Two independent stages of a pipeline take times $X \sim \text{Exp}(1)$ and $Y \sim \text{Exp}(2)$ (rates 1 and 2). Let $S = X + Y$ be the total time. Use the convolution formula to find $f_S(s)$ for $s > 0$, verify it integrates to 1, and compute $\mathbb{E}(S)$.

H. Covariance and the mean and variance of sums

H1. A bakery's daily croissant sales X and daily coffee sales Y (in dozens) satisfy $\mathbb{E}(X) = 9$ and $\mathbb{E}(Y) = 14$, and they are linked through the morning rush so they are not independent. The day's flour-and-bean cost is fixed at 5 (hundreds of dollars). Daily profit is $P = 2X + 3Y - 5$ (hundreds of dollars). Find $\mathbb{E}(P)$.

H2. Two independent timing legs of an automated assembly cell have means and variances

$$\mathbb{E}(A) = 12 \text{ s},\ \text{Var}(A) = 5 \text{ s}^2, \qquad \mathbb{E}(B) = 7 \text{ s},\ \text{Var}(B) = 11 \text{ s}^2.$$

Let the cycle time be $C = A + B$ and the lead-lag difference be $D = A - B$. Find $\mathbb{E}(C)$, $\text{Var}(C)$, $\mathbb{E}(D)$, and $\text{Var}(D)$.

H3. A warehouse logs, for a random shift, the number of forklifts in use X and the number of loading docks occupied Y. Their joint pmf $p(x, y) = \mathbb{P}(X = x, Y = y)$ is

$x \backslash y$	1	2	3	$p_X(x)$
1	0.10	0.05	0.05	0.20
2	0.10	0.20	0.10	0.40
3	0.05	0.15	0.20	0.40
$p_Y(y)$	0.25	0.40	0.35	1.00

Find $\text{Cov}(X, Y)$, and state whether X and Y move together or in opposition.

H4. For the warehouse of H3 it can be checked that $\text{Var}(X) = 0.56$ and $\text{Var}(Y) = 0.59$.

(a) Using $\text{Cov}(X, Y)$ from H3, find $\text{Corr}(X, Y)$.

(b) Find $\text{Var}(X + Y)$, and compare it with the value one would get by wrongly ignoring the covariance.

H5. Let U and V be independent with $\text{Var}(U) = 6$ and $\text{Var}(V) = 10$. Define $S = U + 2V$ and $T = 3U - V$. Using the bilinearity and additivity properties of covariance, find $\text{Cov}(S, T)$, and then $\text{Corr}(S, T)$.

H6. A field technician measures soil pH at $n = 16$ independent locations in one plot. Each reading has the same mean $\mu = 6.4$ and the same variance $\sigma^2 = 0.64$. Let $\bar{X} = \frac{1}{16} \sum_{i=1}^{16} X_i$ be the average reading.

(a) Find $\mathbb{E}(\bar{X})$, $\text{Var}(\bar{X})$, and $\text{SD}(\bar{X})$.

(b) How many independent readings would be needed for the average to have standard deviation at most 0.10?

H7. A delivery van makes n independent stops in one neighbourhood. At each stop a parcel is signed for, independently, with probability $p = 0.8$; let N be the number of signed parcels, so $N \sim \text{Bin}(n, p)$. The driver's tip pool is $W = 4N + 2(n - N) = 2N + 2n$ dollars (a signed parcel earns \$4, an unsigned one \$2). Take $n = 25$.

(a) Using indicators, write N as a sum and find $\mathbb{E}(N)$ and $\text{Var}(N)$.

(b) Find $\mathbb{E}(W)$ and $\text{Var}(W)$.

(c) Find $\text{Cov}(N, W)$ and $\text{Corr}(N, W)$, and explain the value of the correlation.

I. Order statistics

I1. Four sensors each report a calibration error that is independent and Uniform$(0, 1)$. Let $X_{(1)} \le X_{(2)} \le X_{(3)} \le X_{(4)}$ be the ordered errors. (a) Find $\mathbb{P}(X_{(4)} \le 0.7)$, the probability that the largest error is at most 0.7. (b) Find $\mathbb{P}(X_{(1)} > 0.2)$, the probability that the smallest error exceeds 0.2. (c) Find $\mathbb{P}(X_{(1)} \le 0.2)$.

I2. Five independent germination times (in days) are modelled as Uniform$(0, 1)$ random variables. Let $M = X_{(5)}$ be the maximum. (a) Write the cdf $F_M(x)$ and density $f_M(x)$ of M. (b) Compute $\mathbb{E}(M)$. (c) Compute $\mathbb{P}(M \le 0.9)$.

I3. Six identical pumps run in parallel; each lifetime (in years) is independent Exp(λ) with mean 4 years, so $\lambda = \frac{1}{4}$. The system fails as soon as the first pump fails, i.e. at the minimum $X_{(1)}$. (a) Identify the distribution of $X_{(1)}$ and give its rate. (b) Find $\mathbb{E}\big(X_{(1)}\big)$. (c) Find $\mathbb{P}\big(X_{(1)} > 1\big)$, the probability the system survives past one year.

I4. Three independent measurements are taken from a Uniform$(0, \theta)$ parent, where $\theta > 0$ is unknown. The sample median is the middle value $X_{(2)}$. (a) Derive the density $f_{X_{(2)}}(x)$ on $(0, \theta)$. (b) Show that $\mathbb{E}\big(X_{(2)}\big) = \theta/2$, so the sample median is an unbiased estimate of the population midpoint. (c) Find $\text{Var}\big(X_{(2)}\big)$.

I5. Four independent reaction times are Uniform$(0, 1)$. Consider the third order statistic $X_{(3)}$ (the second-largest value). (a) Use the general order-statistic density formula to write $f_{X_{(3)}}(x)$ for $0 < x < 1$. (b) Compute $\mathbb{E}\big(X_{(3)}\big)$ and $\text{Var}\big(X_{(3)}\big)$.

I6. Three independent points are dropped uniformly on the interval $[0, 1]$, giving Uniform$(0, 1)$ values. The sample **range** is $R = X_{(3)} - X_{(1)}$, the spread between the largest and smallest points. (a) Show that the density of R is $f_R(r) = 6r(1-r)$ for $0 < r < 1$. (b) Compute $\mathbb{E}(R)$ and verify it equals $\mathbb{E}\big(X_{(3)}\big) - \mathbb{E}\big(X_{(1)}\big)$. (c) Find $\mathbb{P}\big(R > \frac{1}{2}\big)$.

I7. Four light bulbs are switched on at time 0; their lifetimes (in thousands of hours) are independent Exp(λ) with $\lambda = \frac{1}{2}$ (mean 2). Burnt-out bulbs are not replaced. Let $X_{(1)} \le X_{(2)} \le X_{(3)} \le X_{(4)}$ be the failure times. (a) Using the memoryless property, argue that the gap $X_{(1)}$ is Exp(4λ) and the gap $X_{(2)} - X_{(1)}$ is Exp(3λ), and that these gaps are independent. (b) Find $\mathbb{E}\big(X_{(1)}\big)$ and $\mathbb{E}\big(X_{(2)}\big)$, the expected time of the first and second failures.

J. Conditional densities and conditional expectation

J1. A vending machine each morning holds a number X of bottles of a sports drink, where X takes the values $1, 2, 3$ with $\mathbb{P}(X = 1) = \frac{1}{4}$, $\mathbb{P}(X = 2) = \frac{1}{2}$, $\mathbb{P}(X = 3) = \frac{1}{4}$. Given that $X = x$ bottles are stocked, the number Y sold that day is equally likely to be any integer in $\{0, 1, \dots, x\}$. Find the conditional pmf $p_{Y|X}(y \mid 2)$, the conditional expectation $\mathbb{E}(Y \mid X = 2)$, and the unconditional mean $\mathbb{E}(Y)$ using the law of total expectation.

J2. The lifetimes (in years) of a battery and the backup battery in a smoke detector have joint density

$$f(x,y) = e^{-x-y}, \qquad x>0,\ y>0,$$

and are otherwise 0. Show that the conditional density $f_{Y\mid X}(y\mid x)$ does not depend on x, identify the named distribution of $Y \mid X=x$, and state $\mathbb{E}(Y \mid X=x)$.

J3. Let (X,Y) have joint density $f(x,y)=2$ on the triangle $0<x<y<1$ and 0 elsewhere. Find the marginal $f_X(x)$, the conditional density $f_{Y\mid X}(y\mid x)$, identify the conditional distribution of Y given $X=x$, and compute $\mathbb{E}(Y \mid X=x)$ and $\mathrm{Var}(Y \mid X=x)$.

J4. A point is dropped on the unit interval at a location $X \sim \mathrm{Uniform}(0,1)$. Given $X=x$, a second point Y is placed uniformly on $(0,x)$, so $f_{Y\mid X}(y\mid x) = 1/x$ for $0<y<x$. Write down the joint density $f(x,y)$, find the marginal density $f_Y(y)$ of the second point for $0<y<1$, and use it to evaluate $\mathbb{P}(Y<\frac{1}{2})$.

J5. For the same setup as J4 (so $X \sim \mathrm{Uniform}(0,1)$ and $Y \mid X=x \sim \mathrm{Uniform}(0,x)$), find $\mathbb{E}(Y \mid X=x)$, then use the law of total expectation to find $\mathbb{E}(Y)$. Next find the conditional density $f_{X\mid Y}(x\mid y)$ of the first point given the second, and use it to compute $\mathbb{E}(X \mid Y=y)$.

J6. Claims arriving at an insurance desk in one hour follow $N \sim \mathrm{Poisson}(\Lambda)$, but the hourly rate Λ itself varies by hour with density $f_\Lambda(\lambda) = e^{-\lambda}$ for $\lambda>0$ (an Exp(1) rate). Given $\Lambda=\lambda$, the count N has pmf $p_{N\mid\Lambda}(n\mid\lambda) = e^{-\lambda}\lambda^n/n!$. Find the unconditional pmf $p_N(n)$ for $n=0,1,2,\dots$, identify the resulting distribution, and find $\mathbb{E}(N)$ and $\mathrm{Var}(N)$ using the laws of total expectation and total variance.

J7. Let (X,Y) have joint density

$$f(x,y) = \frac{1}{x}\,e^{-y/x}\,e^{-x}, \qquad 0<x<\infty,\ 0<y<\infty,$$

and 0 otherwise. Verify that f integrates to 1, find the conditional density $f_{Y\mid X}(y\mid x)$ and identify the conditional distribution of Y given $X=x$. Then find $\mathbb{E}(Y \mid X=x)$ and use the law of total expectation to compute $\mathbb{E}(Y)$.

K. Moment-generating functions

K1. A charity raffle ticket pays \$0 with probability 0.9 and \$10 with probability 0.1. Let X be the payout of one ticket. Write down the moment-generating function $M_X(t) = \mathbb{E}(e^{tX})$ as a function of t, then differentiate it to recover $\mathbb{E}(X)$ and $\mathbb{E}(X^2)$, and confirm $\mathrm{Var}(X)$ matches a direct computation.

K2. A discrete random variable X has moment-generating function

$$M_X(t) = \frac{1}{6} + \frac{1}{3}\,e^{t} + \frac{1}{2}\,e^{3t}.$$

Read off the pmf of X directly from M_X, and then find $\mathbb{E}(X)$ and $\mathrm{Var}(X)$ both from the pmf and by differentiating M_X at $t=0$.

K3. The waiting time (in minutes) for a help-desk callback is $T \sim \mathrm{Exp}(\frac{1}{5})$, i.e. an exponential with rate $\lambda=\frac{1}{5}$ and mean 5 minutes. Derive its moment-generating function $M_T(t)$ from the definition, state the values of t for which the integral converges, and use M_T to find $\mathbb{E}(T)$ and $\mathbb{E}(T^2)$.

K4. The number of typos X on a randomly chosen manuscript page is $\mathrm{Poisson}(\lambda)$, with moment-generating function $M_X(t) = e^{\lambda(e^t-1)}$. Using M_X alone (no pmf sums), compute $\mathbb{E}(X)$, $\mathbb{E}(X^2)$, and $\mathrm{Var}(X)$, and then find the third moment $\mathbb{E}(X^3)$.

K5. Daily revenue (in thousands of dollars) at a kiosk is modeled as $X \sim \mathrm{N}(\mu,\sigma^2)$ with moment-generating function $M_X(t) = \exp\!\big(\mu t + \frac{1}{2}\sigma^2 t^2\big)$. Suppose a tax-adjusted profit is $Y = 0.8\,X - 2$. Use the rule $M_{aX+b}(t) = e^{bt}M_X(at)$ to find $M_Y(t)$, identify the distribution of Y, and state its mean and variance.

K6. Two independent assembly stations contribute defect counts $X \sim \text{Poisson}(3)$ and $Y \sim \text{Poisson}(5)$ to a daily batch. Using moment-generating functions and the fact that the MGF of a sum of independent variables is the product of their MGFs, find the MGF of the total $S = X + Y$, identify the exact distribution of S, and state $\mathbb{E}(S)$ and $\text{Var}(S)$.

K7. A relay system has three independent backup components whose lifetimes (in years) are T_1, T_2, T_3, each $\text{Exp}(\lambda)$ with rate $\lambda = 2$, so each has MGF $M_{T_i}(t) = \dfrac{\lambda}{\lambda - t}$ for $t < \lambda$. The total time the system survives when the components run one after another is $S = T_1 + T_2 + T_3$. Find $M_S(t)$, identify the named distribution of S (give its parameters), and use M_S to compute $\mathbb{E}(S)$ and $\text{Var}(S)$. (You may use that a $\text{Gamma}(\alpha, \lambda)$ variable has MGF $\big(\lambda/(\lambda - t)\big)^{\alpha}$, mean α/λ, and variance α/λ^2.)

3.13 Solutions

A. Binomial and hypergeometric probabilities

A1. $X \sim \text{Bin}(6, 0.3)$, so $p_X(k) = \binom{6}{k}(0.3)^k(0.7)^{6-k}$.

(a) $\mathbb{P}(X = 2) = \binom{6}{2}(0.3)^2(0.7)^4 = 15\,(0.09)(0.2401) = 0.324135$.

(b) For a binomial, $\mathbb{E}(X) = np = 6(0.3) = 1.8$ and $\text{Var}(X) = np(1-p) = 6(0.3)(0.7) = 1.26$.

A2. $Y \sim \text{Bin}(10, 0.9)$, so $p_Y(k) = \binom{10}{k}(0.9)^k(0.1)^{10-k}$.

(a) $\mathbb{P}(Y = 10) = (0.9)^{10} = 0.348678$.

(b) $\mathbb{P}(Y = 9) = \binom{10}{9}(0.9)^9(0.1) = 10(0.387420\ldots)(0.1) = 0.387420$.

(c) $\mathbb{P}(Y \geq 9) = \mathbb{P}(Y = 9) + \mathbb{P}(Y = 10) = 0.387420 + 0.348678 = 0.736099$.

A3. Sampling $n = 3$ items without replacement from $N = 12$ batteries containing $r = 4$ "successes" (dead) is the hypergeometric setting.

(a) $X \sim \text{Hyper}(N = 12,\, r = 4,\, n = 3)$, with $p_X(k) = \dfrac{\binom{4}{k}\binom{8}{3-k}}{\binom{12}{3}}$ for $k = 0, 1, 2, 3$. Here $\binom{12}{3} = 220$.

(b) $\mathbb{P}(X = 1) = \dfrac{\binom{4}{1}\binom{8}{2}}{\binom{12}{3}} = \dfrac{4 \cdot 28}{220} = \dfrac{112}{220} = \dfrac{28}{55} \approx 0.5091$.

(c) $\mathbb{P}(X = 0) = \dfrac{\binom{4}{0}\binom{8}{3}}{\binom{12}{3}} = \dfrac{1 \cdot 56}{220} = \dfrac{56}{220} = \dfrac{14}{55}$, so $\mathbb{P}(X \geq 1) = 1 - \dfrac{14}{55} = \dfrac{41}{55} \approx 0.7455$.

A4. Each guess is correct independently with probability $p = 1/5 = 0.2$, and there are $n = 8$ questions.

(a) $W \sim \text{Bin}(8, 0.2)$, so $p_W(k) = \binom{8}{k}(0.2)^k(0.8)^{8-k}$.

(b) $\mathbb{P}(W \leq 1) = \mathbb{P}(W = 0) + \mathbb{P}(W = 1) = (0.8)^8 + \binom{8}{1}(0.2)(0.8)^7 = 0.167772 + 0.335544 = 0.503316$.

(c) $\mathbb{P}(W \geq 6) = \mathbb{P}(W = 6) + \mathbb{P}(W = 7) + \mathbb{P}(W = 8)$. Now $\mathbb{P}(W = 6) = \binom{8}{6}(0.2)^6(0.8)^2 = 28(0.000064)(0.64) = 0.00114688$, $\mathbb{P}(W = 7) = \binom{8}{7}(0.2)^7(0.8) = 8(0.0000128)(0.8) = 0.00008192$, and $\mathbb{P}(W = 8) = (0.2)^8 = 0.00000256$. Summing, $\mathbb{P}(W \geq 6) = 0.00123136 \approx 0.0012$. Passing by guessing is very unlikely.

A5. The crew nets $n = 8$ fish without replacement from $N = 50$ fish containing $r = 15$ tagged fish; this is hypergeometric.

(a) $T \sim \text{Hyper}(N = 50,\, r = 15,\, n = 8)$ with $p_T(k) = \dfrac{\binom{15}{k}\binom{35}{8-k}}{\binom{50}{8}}$, for $\max(0,\, 8 - 35) \leq k \leq \min(8, 15)$, i.e. $0 \leq k \leq 8$.

(b) For the hypergeometric, $\mathbb{E}(T) = n \cdot \dfrac{r}{N} = 8 \cdot \dfrac{15}{50} = 8(0.3) = 2.4$.

(c) $\mathbb{P}(T = 3) = \dfrac{\binom{15}{3}\binom{35}{5}}{\binom{50}{8}} = \dfrac{455 \cdot 324{,}632}{536{,}878{,}650} = \dfrac{147{,}707{,}560}{536{,}878{,}650} \approx 0.2751$.

A6. Drawing $n = 25$ modules without replacement from $N = 2000$ with $r = 40$ defective is exactly hypergeometric, but the sample is a tiny fraction of the population, so the binomial is an excellent approximation.

(a) $D \sim \text{Hyper}(N = 2000,\, r = 40,\, n = 25)$ with $p_D(k) = \dfrac{\binom{40}{k}\binom{1960}{25-k}}{\binom{2000}{25}}$, for $0 \leq k \leq 25$.

(b) The sample size is only $n/N = 25/2000 = 1.25\%$ of the population, so removing drawn modules barely changes the defective fraction from draw to draw; the success probability stays near $p = r/N = 40/2000 = 0.02$ throughout. Hence $D \approx \text{Bin}(n, p) = \text{Bin}(25,\, 0.02)$.

(c) Under $\mathrm{Bin}(25, 0.02)$, $\mathbb{P}(D = 0) = (0.98)^{25} = 0.6035$ (to four places, 0.6035), and $\mathbb{P}(D \geq 1) = 1 - (0.98)^{25} = 0.3965$.

A7. $H \sim \mathrm{Bin}(12, 0.4)$, so $p_H(k) = \binom{12}{k}(0.4)^k(0.6)^{12-k}$.

(a) Consider the ratio of consecutive terms,

$$\frac{\mathbb{P}(H = k)}{\mathbb{P}(H = k-1)} = \frac{\binom{12}{k}(0.4)^k(0.6)^{12-k}}{\binom{12}{k-1}(0.4)^{k-1}(0.6)^{13-k}} = \frac{12-k+1}{k} \cdot \frac{0.4}{0.6} = \frac{13-k}{k} \cdot \frac{2}{3}.$$

This ratio is ≥ 1 exactly when $2(13-k) \geq 3k$, i.e. $26 \geq 5k$, i.e. $k \leq 5.2$. So $\mathbb{P}(H = k) \geq \mathbb{P}(H = k-1)$ for $k = 1, 2, 3, 4, 5$ and the terms strictly decrease for $k \geq 6$. The pmf therefore increases up to $k = 5$ and decreases afterward, so the mode is $H = 5$ (with $\mathbb{P}(H = 5) = \binom{12}{5}(0.4)^5(0.6)^7 \approx 0.2270$).

(b) First $\mathbb{P}(H = 0) = (0.6)^{12} = 0.002177$, so $\mathbb{P}(H \geq 1) = 1 - 0.002177 = 0.997823$. Also $\mathbb{P}(H = 1) = \binom{12}{1}(0.4)(0.6)^{11} = 12(0.4)(0.003628) = 0.017414$. Therefore

$$\mathbb{P}(H = 1 \mid H \geq 1) = \frac{\mathbb{P}(H = 1)}{\mathbb{P}(H \geq 1)} = \frac{0.017414}{0.997823} \approx 0.01745.$$

B. Discrete random variables: pmf and cdf

B1. (a) Each listed value is ≥ 0, and

$$0.45 + 0.30 + 0.15 + 0.07 + 0.03 = 1.00,$$

so both pmf properties hold and this is a valid pmf. **(b)** Add the relevant masses:

$$\mathbb{P}(X \geq 2) = p_X(2) + p_X(3) + p_X(4) = 0.15 + 0.07 + 0.03 = 0.25,$$

$$\mathbb{P}(1 \leq X \leq 3) = p_X(1) + p_X(2) + p_X(3) = 0.30 + 0.15 + 0.07 = 0.52.$$

B2. Normalise: $\sum_{k=0}^{3} c(k+1) = c(1+2+3+4) = 10c = 1$, so $c = \frac{1}{10}$. The masses are $p_X(0) = \frac{1}{10}$, $p_X(1) = \frac{2}{10}$, $p_X(2) = \frac{3}{10}$, $p_X(3) = \frac{4}{10}$, all positive. Then

$$\mathbb{P}(X \leq 1) = p_X(0) + p_X(1) = \tfrac{1}{10} + \tfrac{2}{10} = \tfrac{3}{10}, \qquad \mathbb{P}(X > 2) = p_X(3) = \tfrac{4}{10} = \tfrac{2}{5}.$$

B3. (a) The range is infinite, so normalise with a geometric series (starting at $k = 1$):

$$\sum_{k=1}^{\infty} c\,(0.6)^k = c \cdot \frac{0.6}{1 - 0.6} = c \cdot \frac{0.6}{0.4} = \tfrac{3}{2}c = 1 \implies c = \tfrac{2}{3}.$$

So $p_X(k) = \frac{2}{3}(0.6)^k = 0.4\,(0.6)^{k-1}$, i.e. $X \sim \mathrm{Geom}(0.4)$. **(b)** Sum the first three masses:

$$\mathbb{P}(X \leq 3) = \tfrac{2}{3}\big(0.6 + 0.36 + 0.216\big) = \tfrac{2}{3}(1.176) = 0.784.$$

For the tail, $\mathbb{P}(X \geq 2) = 1 - p_X(1) = 1 - \frac{2}{3}(0.6) = 1 - 0.4 = 0.6$.

B4. (a) With $r = \frac{1}{2}$, the given identity gives

$$\sum_{k=1}^{\infty} c\,k\,(1/2)^k = c \cdot \frac{1/2}{(1 - 1/2)^2} = c \cdot \frac{1/2}{1/4} = 2c = 1 \implies c = \tfrac{1}{2}.$$

So $p_X(k) = \frac{1}{2}\,k\,(1/2)^k = k\,(1/2)^{k+1}$, and each mass is positive. **(b)** Evaluate directly:

$$\mathbb{P}(X = 1) = 1 \cdot (1/2)^2 = \tfrac{1}{4}, \quad \mathbb{P}(X = 2) = 2 \cdot (1/2)^3 = \tfrac{1}{4}, \quad \mathbb{P}(X = 3) = 3 \cdot (1/2)^4 = \tfrac{3}{16}.$$

B5. **(a)** The event $\{W \geq w\}$ means both faces are $\geq w$; each face is $\geq w$ with probability $(7-w)/6$, and the faces are independent, so for $w = 1, \ldots, 6$

$$\mathbb{P}(W \geq w) = \Big(\frac{7-w}{6}\Big)^2 = \frac{(7-w)^2}{36}.$$

The pmf is the difference of consecutive tail probabilities: for $w = 1, \ldots, 6$,

$$p_W(w) = \mathbb{P}(W \geq w) - \mathbb{P}(W \geq w+1) = \frac{(7-w)^2 - (6-w)^2}{36} = \frac{13-2w}{36}.$$

Explicitly the masses are

$$p_W(1) = \tfrac{11}{36},\ p_W(2) = \tfrac{9}{36},\ p_W(3) = \tfrac{7}{36},\ p_W(4) = \tfrac{5}{36},\ p_W(5) = \tfrac{3}{36},\ p_W(6) = \tfrac{1}{36},$$

which sum to $\frac{11+9+7+5+3+1}{36} = \frac{36}{36} = 1$. **(b)** Accumulating the masses left to right (or using $F_W(w) = 1 - \mathbb{P}(W \geq w+1) = 1 - \frac{(6-w)^2}{36}$):

$$F_W(x) = \begin{cases} 0, & x < 1, \\ \frac{11}{36}, & 1 \leq x < 2, \\ \frac{20}{36}, & 2 \leq x < 3, \\ \frac{27}{36}, & 3 \leq x < 4, \\ \frac{32}{36}, & 4 \leq x < 5, \\ \frac{35}{36}, & 5 \leq x < 6, \\ 1, & x \geq 6. \end{cases}$$

(c) With the included lower endpoint, use $F_W(4) - F_W(2) + p_W(2)$, or simply add the masses:

$$\mathbb{P}(2 \leq W \leq 4) = p_W(2) + p_W(3) + p_W(4) = \frac{9+7+5}{36} = \frac{21}{36} = \frac{7}{12}.$$

B6. **(a)** Normalise: $\sum_{k=0}^{4} c\binom{4}{k} = c \cdot 2^4 = 16c = 1$, so $c = \frac{1}{16}$. Then $p_X(k) = \binom{4}{k}(\frac{1}{2})^4 = \binom{4}{k}(\frac{1}{2})^k(\frac{1}{2})^{4-k}$, which is exactly the pmf of $X \sim \mathrm{Bin}(4, \frac{1}{2})$. The masses are $\frac{1}{16}, \frac{4}{16}, \frac{6}{16}, \frac{4}{16}, \frac{1}{16}$. **(b)** Accumulate the masses:

$$F_X(x) = \begin{cases} 0, & x < 0, \\ \frac{1}{16}, & 0 \leq x < 1, \\ \frac{5}{16}, & 1 \leq x < 2, \\ \frac{11}{16}, & 2 \leq x < 3, \\ \frac{15}{16}, & 3 \leq x < 4, \\ 1, & x \geq 4. \end{cases}$$

(c) The interval is half-open ($1 < X \leq 3$), so the bare difference applies:

$$\mathbb{P}(1 < X \leq 3) = F_X(3) - F_X(1) = \tfrac{15}{16} - \tfrac{5}{16} = \tfrac{10}{16} = \tfrac{5}{8}.$$

Check against masses: $p_X(2) + p_X(3) = \frac{6}{16} + \frac{4}{16} = \frac{10}{16}$.

B7. **(a)** The first pass on test k means $k-1$ failures (each probability 0.8) then a pass (probability 0.2):

$$p_X(k) = (0.8)^{k-1}(0.2), \qquad k = 1, 2, 3, \ldots,$$

so $X \sim \mathrm{Geom}(0.2)$. The cdf accumulates a finite geometric sum: for an integer $n \geq 1$,

$$F_X(n) = \sum_{k=1}^{n} (0.8)^{k-1}(0.2) = 0.2 \cdot \frac{1-(0.8)^n}{1-0.8} = 0.2 \cdot \frac{1-(0.8)^n}{0.2} = 1 - (0.8)^n.$$

(Equivalently, $F_X(n) = 1 - \mathbb{P}(X > n) = 1 - (0.8)^n$, since $X > n$ means the first n tests all fail.) **(b)** Use the interval rule on the half-open form $\{2 < X \le 5\} = \{3 \le X \le 5\}$:

$$\mathbb{P}(3 \le X \le 5) = F_X(5) - F_X(2) = \big(1 - (0.8)^5\big) - \big(1 - (0.8)^2\big) = (0.8)^2 - (0.8)^5.$$

Numerically $(0.8)^2 = 0.64$ and $(0.8)^5 = 0.32768$, so

$$\mathbb{P}(3 \le X \le 5) = 0.64 - 0.32768 = 0.31232.$$

(c) We need the smallest integer n with $F_X(n) = 1 - (0.8)^n \ge 0.95$, i.e. $(0.8)^n \le 0.05$. Taking logs, $n \ge \dfrac{\ln 0.05}{\ln 0.8} = \dfrac{-2.9957}{-0.2231} \approx 13.43$. Since $(0.8)^{13} \approx 0.0550 > 0.05$ but $(0.8)^{14} \approx 0.0440 \le 0.05$, the smallest such n is $\boxed{n = 14}$.

C. Continuous random variables and densities

C1. (D1) On $[0, 4]$ we have $\frac{1}{8}x \ge 0$, and $f_X = 0$ elsewhere, so $f_X \ge 0$ everywhere. **(D2)**

$$\int_0^4 \tfrac{1}{8}x \, \mathrm{d}x = \tfrac{1}{8} \cdot \frac{x^2}{2}\Big|_0^4 = \tfrac{1}{16} \cdot 16 = 1,$$

so f_X is a legitimate density. The requested probability is the area to the right of 3:

$$\mathbb{P}(X > 3) = \int_3^4 \tfrac{1}{8}x \, \mathrm{d}x = \tfrac{1}{16}x^2\Big|_3^4 = \tfrac{1}{16}(16 - 9) = \tfrac{7}{16} = 0.4375.$$

$$\boxed{\mathbb{P}(X > 3) = \tfrac{7}{16} = 0.4375}$$

C2. The unknown constant is fixed by (D2): force the total area to 1.

$$\int_0^1 c\,x^3 \, \mathrm{d}x = c \cdot \frac{x^4}{4}\Big|_0^1 = \frac{c}{4} = 1 \implies c = 4.$$

With $f_X(x) = 4x^3$,

$$\mathbb{P}\big(X < \tfrac{1}{2}\big) = \int_0^{1/2} 4x^3 \, \mathrm{d}x = x^4\Big|_0^{1/2} = \big(\tfrac{1}{2}\big)^4 = \tfrac{1}{16} = 0.0625.$$

$$\boxed{c = 4, \qquad \mathbb{P}(X < \tfrac{1}{2}) = \tfrac{1}{16} = 0.0625}$$

C3. (a) Impose (D2) over $[0, \infty)$:

$$\int_0^\infty c\,e^{-x/3} \, \mathrm{d}x = c\Big[-3e^{-x/3}\Big]_0^\infty = c(0 + 3) = 3c = 1 \implies c = \tfrac{1}{3}.$$

So $X \sim \mathrm{Exp}(\frac{1}{3})$, with $f_X(x) = \frac{1}{3}e^{-x/3}$ for $x \ge 0$.

(b) For $x \ge 0$,

$$F_X(x) = \int_0^x \tfrac{1}{3}e^{-t/3} \, \mathrm{d}t = \Big[-e^{-t/3}\Big]_0^x = 1 - e^{-x/3},$$

and $F_X(x) = 0$ for $x < 0$.

(c)

$$\mathbb{P}(1 < X < 4) = F_X(4) - F_X(1) = \big(1 - e^{-4/3}\big) - \big(1 - e^{-1/3}\big) = e^{-1/3} - e^{-4/3}.$$

Numerically $e^{-1/3} = 0.7165$ and $e^{-4/3} = 0.2636$, so the difference is 0.4529.

$$\boxed{c = \tfrac{1}{3}, \qquad F_X(x) = 1 - e^{-x/3} \ (x \ge 0), \qquad \mathbb{P}(1 < X < 4) = e^{-1/3} - e^{-4/3} \approx 0.4529}$$

C4. We are given F_X and want f_X: differentiate. For $0 < x < 1$,

$$f_X(x) = \frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}},$$

and $f_X(x) = 0$ outside $(0,1)$ (where F_X is flat). Check (D2):

$$\int_0^1 \frac{1}{2\sqrt{x}}\,\mathrm{d}x = \sqrt{x}\,\Big|_0^1 = 1,$$

as required (the integrand blows up at 0, but the improper integral converges). The median m solves $F_X(m) = \frac{1}{2}$:

$$\sqrt{m} = \tfrac{1}{2} \implies m = \tfrac{1}{4} = 0.25.$$

$$\boxed{f_X(x) = \frac{1}{2\sqrt{x}} \text{ on } (0,1), \qquad \text{median} = \tfrac{1}{4}}$$

C5. (a) Integrate the density from 0 to x (for $x \geq 0$):

$$F_X(x) = \int_0^x \frac{4}{(1+t)^5}\,\mathrm{d}t = 4 \cdot \frac{(1+t)^{-4}}{-4}\Big|_0^x = -\Big[(1+t)^{-4}\Big]_0^x = 1 - \frac{1}{(1+x)^4},$$

and $F_X(x) = 0$ for $x < 0$. (Check: $F_X(\infty) = 1$.)

(b) The median m solves $F_X(m) = \frac{1}{2}$:

$$1 - \frac{1}{(1+m)^4} = \tfrac{1}{2} \implies (1+m)^4 = 2 \implies 1 + m = 2^{1/4} \implies m = 2^{1/4} - 1 \approx 0.1892 \text{ years.}$$

(c) The 90th percentile solves $F_X(x) = 0.90$:

$$\frac{1}{(1+x)^4} = 0.10 \implies (1+x)^4 = 10 \implies 1 + x = 10^{1/4} \implies x_{0.90} = 10^{1/4} - 1 \approx 0.7783 \text{ years.}$$

$$\boxed{F_X(x) = 1 - \frac{1}{(1+x)^4}, \quad \text{median} = 2^{1/4} - 1 \approx 0.189, \quad x_{0.90} = 10^{1/4} - 1 \approx 0.778}$$

C6. (a) A uniform variable on $[0, 15]$ has constant density and a linear cdf:

$$f_X(x) = \tfrac{1}{15} \text{ on } [0,15], \qquad F_X(x) = \frac{x}{15} \text{ on } [0,15]$$

($F_X = 0$ for $x < 0$ and $F_X = 1$ for $x > 15$).

(b) "Arrives within 10 minutes" is $\mathbb{P}(X < 10)$; for a uniform variable this is the proportion of the interval:

$$\mathbb{P}(X < 10) = \frac{10}{15} = \frac{2}{3} \approx 0.6667.$$

(c) Conditional probability is a ratio of sub-interval lengths:

$$\mathbb{P}(X < 9 \mid X > 6) = \frac{\mathbb{P}(6 < X < 9)}{\mathbb{P}(X > 6)} = \frac{(9-6)/15}{(15-6)/15} = \frac{3}{9} = \frac{1}{3}.$$

$$\boxed{\mathbb{P}(X < 10) = \tfrac{2}{3}, \qquad \mathbb{P}(X < 9 \mid X > 6) = \tfrac{1}{3}}$$

C7. (a) Impose (D2). By symmetry the two triangular pieces have equal area, so

$$\int_0^{10} f_X \,\mathrm{d}x = 2\int_0^{5} k\,x\,\mathrm{d}x = 2\cdot k\cdot \frac{x^2}{2}\Big|_0^5 = 2\cdot\frac{25k}{2} = 25k = 1 \implies k = \frac{1}{25} = 0.04.$$

(b) By the symmetry of the tent about $x = 5$, exactly half the area lies on each side:

$$\mathbb{P}(X \le 5) = \tfrac{1}{2}.$$

(The density is a mirror image about the centre, so left and right halves carry equal probability.)

(c) Since $3 \le 5$, only the rising piece $f_X(x) = \frac{1}{25}x$ is involved:

$$\mathbb{P}(X \le 3) = \int_0^3 \tfrac{1}{25}x\,\mathrm{d}x = \tfrac{1}{25}\cdot\frac{x^2}{2}\Big|_0^3 = \frac{9}{50} = 0.18.$$

(d) Because $\{X \le 3\} \subset \{X \le 5\}$,

$$\mathbb{P}(X \le 3 \mid X \le 5) = \frac{\mathbb{P}(X \le 3)}{\mathbb{P}(X \le 5)} = \frac{0.18}{0.5} = 0.36.$$

$$\boxed{k = \tfrac{1}{25},\quad \mathbb{P}(X \le 5) = \tfrac{1}{2},\quad \mathbb{P}(X \le 3) = 0.18,\quad \mathbb{P}(X \le 3 \mid X \le 5) = 0.36}$$

D. Expected values

D1. Apply the discrete definition $\mathbb{E}(X) = \sum_k k\,p_X(k)$:

$$\mathbb{E}(X) = 2(0.2) + 3(0.5) + 4(0.2) + 5(0.1) = 0.4 + 1.5 + 0.8 + 0.5 = 3.2.$$

So a box holds $\mathbb{E}(X) = 3.2$ pastries on average.

D2. First the per-passerby mean:

$$\mathbb{E}(X) = 0(0.6) + 2(0.3) + 5(0.1) = 0 + 0.6 + 0.5 = 1.1.$$

The total tips $S = X_1 + \cdots + X_{50}$ has, by linearity (no independence needed for the mean), $\mathbb{E}(S) = 50\,\mathbb{E}(X) = 50(1.1) = 55$. Then $W = S - 4$, so again by linearity

$$\mathbb{E}(W) = \mathbb{E}(S) - 4 = 55 - 4 = \$\,51.$$

D3. For $X \sim \text{Uniform}(a, b)$ the mean is $(a + b)/2$. Here $a = \pi$, $b = 3\pi$, so

$$\mathbb{E}(X) = \frac{\pi + 3\pi}{2} = \frac{4\pi}{2} = 2\pi.$$

Now apply linearity $\mathbb{E}(aX + b) = a\,\mathbb{E}(X) + b$ with $a = \frac{1}{\pi}$, $b = -2$:

$$\mathbb{E}\big(\tfrac{1}{\pi}X - 2\big) = \tfrac{1}{\pi}\,\mathbb{E}(X) - 2 = \tfrac{1}{\pi}(2\pi) - 2 = 2 - 2 = 0.$$

D4. This is the law of the unconscious statistician: $\mathbb{E}\big(g(T)\big) = \int_0^\infty g(t)\,f_T(t)\,\mathrm{d}t$. With $g(t) = 90\,e^{-0.1\,t}$ and $f_T(t) = \frac{1}{3}e^{-t/3}$,

$$\mathbb{E}\big(90\,e^{-0.1\,T}\big) = \int_0^\infty 90\,e^{-0.1\,t}\cdot\tfrac{1}{3}e^{-t/3}\,\mathrm{d}t = 30\int_0^\infty e^{-(0.1+\frac{1}{3})t}\,\mathrm{d}t.$$

The exponent rate is $0.1 + \frac{1}{3} = \frac{3}{30} + \frac{10}{30} = \frac{13}{30}$, and $\int_0^\infty e^{-rt}\,\mathrm{d}t = \frac{1}{r}$. Hence

$$\mathbb{E}\big(90\,e^{-0.1\,T}\big) = 30\cdot\frac{1}{13/30} = 30\cdot\frac{30}{13} = \frac{900}{13} \approx \$\,69.23.$$

D5. A density must integrate to 1:

$$\int_0^2 c\,x(2-x)\,\mathrm{d}x = c\int_0^2 (2x-x^2)\,\mathrm{d}x = c\Big[x^2 - \tfrac{x^3}{3}\Big]_0^2 = c\Big(4-\tfrac{8}{3}\Big) = c\cdot\tfrac{4}{3} = 1,$$

so $c = \frac{3}{4}$. Then

$$\mathbb{E}(X) = \int_0^2 x\cdot\tfrac{3}{4}x(2-x)\,\mathrm{d}x = \tfrac{3}{4}\int_0^2 (2x^2-x^3)\,\mathrm{d}x = \tfrac{3}{4}\Big[\tfrac{2x^3}{3} - \tfrac{x^4}{4}\Big]_0^2 = \tfrac{3}{4}\Big(\tfrac{16}{3}-4\Big) = \tfrac{3}{4}\cdot\tfrac{4}{3} = 1.$$

(This matches the symmetry of f_X about $x = 1$.) So $\mathbb{E}(X) = 1$.

D6. Condition on the number of calls $N \sim \text{Poisson}(4)$. Given $N = n$, each call is resolved independently with probability 0.75, so $R \mid (N = n) \sim \text{Bin}(n, 0.75)$ and $\mathbb{E}(R \mid N = n) = 0.75\,n$. Taking the expectation over N (law of total expectation),

$$\mathbb{E}(R) = \mathbb{E}\big[\mathbb{E}(R \mid N)\big] = \mathbb{E}(0.75\,N) = 0.75\,\mathbb{E}(N) = 0.75(4) = 3.$$

So $\mathbb{E}(R) = 3$ resolved calls per hour. (Indeed $R \sim \text{Poisson}(3)$ by Poisson thinning, with mean 3.)

D7. Use LOTUS to get $\mathbb{E}(X^2) = \int_0^2 x^2 f_X(x)\,\mathrm{d}x$ with $f_X(x) = \frac{x}{2}$:

$$\mathbb{E}(X^2) = \int_0^2 x^2\cdot\tfrac{x}{2}\,\mathrm{d}x = \tfrac{1}{2}\int_0^2 x^3\,\mathrm{d}x = \tfrac{1}{2}\Big[\tfrac{x^4}{4}\Big]_0^2 = \tfrac{1}{2}\cdot\tfrac{16}{4} = \tfrac{1}{2}\cdot 4 = 2.$$

Then by linearity $\mathbb{E}\big(10 - X^2\big) = 10 - \mathbb{E}(X^2) = 10 - 2 = 8$. The expected score is 8 points.

E. Variance and standard deviation

E1. Tabulate the moment contributions:

k	1	2	3	4	sum
$p_X(k)$	0.40	0.35	0.15	0.10	1.00
$k\,p_X(k)$	0.40	0.70	0.45	0.40	1.95
$k^2 p_X(k)$	0.40	1.40	1.35	1.60	4.75

So $\mathbb{E}(X) = 1.95$ and $\mathbb{E}(X^2) = 4.75$. By the shortcut formula,

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = 4.75 - (1.95)^2 = 4.75 - 3.8025 = 0.9475,$$

and $\mathrm{SD}(X) = \sqrt{0.9475} \approx 0.973$ cans.

E2. The density is symmetric about 0, and the integrand $x\,f_X(x) = \frac{3}{4}(x - x^3)$ is odd, so $\mathbb{E}(X) = 0$ (one checks $\int_{-1}^1 \frac{3}{4}(x-x^3)\,\mathrm{d}x = 0$). For the second moment the integrand $x^2 f_X(x) = \frac{3}{4}(x^2 - x^4)$ is even:

$$\mathbb{E}(X^2) = \int_{-1}^1 \tfrac{3}{4}\big(x^2 - x^4\big)\,\mathrm{d}x = 2\cdot\tfrac{3}{4}\int_0^1 \big(x^2 - x^4\big)\,\mathrm{d}x = \tfrac{3}{2}\Big(\tfrac{1}{3} - \tfrac{1}{5}\Big) = \tfrac{3}{2}\cdot\tfrac{2}{15} = \frac{1}{5}.$$

Therefore

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \frac{1}{5} - 0^2 = \frac{1}{5} = 0.2, \qquad \mathrm{SD}(X) = \sqrt{0.2} \approx 0.447 \text{ mV}.$$

E3. Profit is an affine function $P = aD + b$ with $a = 8$ and $b = -500$. The additive constant does not affect spread, and the multiplier enters squared:

$$\mathrm{Var}(P) = a^2\,\mathrm{Var}(D) = 8^2\cdot\mathrm{SD}(D)^2 = 64\cdot 30^2 = 64\cdot 900 = 57{,}600,$$

$$\mathrm{SD}(P) = |a|\,\mathrm{SD}(D) = 8\cdot 30 = 240 \text{ dollars}.$$

The fixed \$500 cost shifts the mean profit but leaves the standard deviation unchanged.

E4. For a Poisson variable the variance equals the rate. Thus

$$\mathrm{Var}(X) = \lambda_X = 3.5, \qquad \mathrm{SD}(X) = \sqrt{3.5} \approx 1.871,$$

$$\mathrm{Var}(Y) = \lambda_Y = 2.5, \qquad \mathrm{SD}(Y) = \sqrt{2.5} \approx 1.581.$$

Because X and Y are independent, their variances add:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) = 3.5 + 2.5 = 6, \qquad \mathrm{SD}(X+Y) = \sqrt{6} \approx 2.449.$$

(Consistently, the sum of independent Poissons is Poisson$(3.5 + 2.5)$ = Poisson(6), whose variance is indeed 6.)

E5. With $\lambda = \frac{1}{5}$, the exponential has mean $1/\lambda$ and variance $1/\lambda^2$:

$$\mathbb{E}(X) = \frac{1}{\lambda} = 5, \qquad \mathrm{Var}(X) = \frac{1}{\lambda^2} = 25.$$

Since $W = 1.2X$ is a pure scaling (no additive constant),

$$\mathbb{E}(W) = 1.2\,\mathbb{E}(X) = 1.2 \cdot 5 = 6,$$

$$\mathrm{Var}(W) = 1.2^2\ \mathrm{Var}(X) = 1.44 \cdot 25 = 36, \qquad \mathrm{SD}(W) = \sqrt{36} = 6.$$

So $\mathrm{SD}(W) = 6 = \mathbb{E}(W)$: the standard deviation equals the mean, the signature of an exponential (scaling preserves that equality because both the mean and the standard deviation scale by the same factor $|a|$).

E6. *(a) Normalizing constant.* The probabilities must sum to 1:

$$\sum_{k=1}^{5} c\,k = c(1+2+3+4+5) = 15c = 1 \implies c = \frac{1}{15}.$$

(b) Moments. Using $\sum_{k=1}^{5} k^2 = 55$ and $\sum_{k=1}^{5} k^3 = 225$,

$$\mathbb{E}(X) = \sum_{k=1}^{5} k \cdot \frac{k}{15} = \frac{1}{15}\sum_{k=1}^{5} k^2 = \frac{55}{15} = \frac{11}{3} \approx 3.667,$$

$$\mathbb{E}(X^2) = \sum_{k=1}^{5} k^2 \cdot \frac{k}{15} = \frac{1}{15}\sum_{k=1}^{5} k^3 = \frac{225}{15} = 15.$$

(c) Variance and standard deviation.

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = 15 - \left(\frac{11}{3}\right)^2 = 15 - \frac{121}{9} = \frac{135-121}{9} = \frac{14}{9} \approx 1.556,$$

$$\mathrm{SD}(X) = \sqrt{\frac{14}{9}} = \frac{\sqrt{14}}{3} \approx 1.247 \text{ rings}.$$

E7. *(a) Check, mean, variance.* The density integrates to 1:

$$\int_0^2 \frac{3}{8}x^2\,\mathrm{d}x = \frac{3}{8}\Big[\frac{x^3}{3}\Big]_0^2 = \frac{3}{8}\cdot\frac{8}{3} = 1.$$

The moments are

$$\mathbb{E}(X) = \int_0^2 x \cdot \frac{3}{8}x^2\,\mathrm{d}x = \frac{3}{8}\Big[\frac{x^4}{4}\Big]_0^2 = \frac{3}{8}\cdot\frac{16}{4} = \frac{3}{8}\cdot 4 = \frac{3}{2},$$

$$\mathbb{E}(X^2) = \int_0^2 x^2 \cdot \frac{3}{8}x^2 \,\mathrm{d}x = \frac{3}{8}\Big[\frac{x^5}{5}\Big]_0^2 = \frac{3}{8}\cdot\frac{32}{5} = \frac{12}{5} = 2.4,$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = 2.4 - \Big(\frac{3}{2}\Big)^2 = 2.4 - 2.25 = 0.15,$$

with $\mathrm{SD}(X) = \sqrt{0.15} \approx 0.387$ m.

(b) Affine transformation. For $Y = 3X - 1$ the constant -1 drops out and the multiplier 3 is squared:

$$\mathrm{Var}(Y) = 3^2\ \mathrm{Var}(X) = 9 \cdot 0.15 = 1.35, \qquad \mathrm{SD}(Y) = |3|\ \mathrm{SD}(X) = 3\sqrt{0.15} \approx 1.162 \text{ m}.$$

(c) Chebyshev bound. The interval $|X - \mathbb{E}(X)| < 2\ \mathrm{SD}(X)$ is the $k = 2$ case of Chebyshev's inequality. The tail bound gives

$$\mathbb{P}\big(|X - \mathbb{E}(X)| \geq 2\ \mathrm{SD}(X)\big) \leq \frac{1}{2^2} = \frac{1}{4},$$

so by taking the complement,

$$\mathbb{P}\big(|X - \mathbb{E}(X)| < 2\ \mathrm{SD}(X)\big) \geq 1 - \frac{1}{4} = \frac{3}{4}.$$

At least 75% of the mass lies within two standard deviations of the mean.

F. Joint densities, marginals, and independence

F1. **(a)** Sum each row for p_X and each column for p_Y:

$$p_X(0) = 0.72 + 0.08 = 0.80, \quad p_X(1) = 0.12 + 0.08 = 0.20,$$

$$p_Y(0) = 0.72 + 0.12 = 0.84, \quad p_Y(1) = 0.08 + 0.08 = 0.16.$$

(b) The event $\{X + Y \geq 1\}$ is the complement of $\{X = 0, Y = 0\}$, so

$$\mathbb{P}(X + Y \geq 1) = 1 - p_{X,Y}(0,0) = 1 - 0.72 = 0.28.$$

(c) Test the corner $(0,0)$: $p_X(0)\,p_Y(0) = 0.80 \times 0.84 = 0.672 \neq 0.72 = p_{X,Y}(0,0)$. Since factorisation fails at one cell, X and Y are *not* independent.

F2. **(a)** The total mass is

$$\sum_{x=1}^{3}\sum_{y=1}^{x}\frac{c}{x} = \sum_{x=1}^{3}\frac{c}{x}\cdot x = \sum_{x=1}^{3} c = 3c = 1 \quad\Longrightarrow\quad c = \tfrac{1}{3}.$$

(b) Sum out y (there are x equal terms, each c/x):

$$p_X(x) = \sum_{y=1}^{x}\frac{c}{x} = c = \tfrac{1}{3}, \qquad x \in \{1,2,3\},$$

so $X \sim \mathrm{Uniform}\{1,2,3\}$ (discrete uniform on three values). **(c)** Outcome $y = 1$ is possible for every $x \in \{1,2,3\}$:

$$p_Y(1) = \sum_{x=1}^{3}\frac{c}{x} = \tfrac{1}{3}\Big(1 + \tfrac{1}{2} + \tfrac{1}{3}\Big) = \tfrac{1}{3}\cdot\tfrac{11}{6} = \frac{11}{18}.$$

F3. **(a)** Require the double integral over the unit square to equal 1:

$$\int_0^1\int_0^1 k(x+2y)\,\mathrm{d}x\,\mathrm{d}y = k\int_0^1\Big(\tfrac{1}{2} + 2y\Big)\,\mathrm{d}y = k\Big(\tfrac{1}{2} + 1\Big) = \tfrac{3}{2}k = 1,$$

so $k = \frac{2}{3}$. **(b)** Integrate out y over $0 < y < 1$:

$$f_X(x) = \int_0^1 \tfrac{2}{3}(x + 2y)\,\mathrm{d}y = \tfrac{2}{3}\big(x + 1\big), \qquad 0 < x < 1.$$

(c) Integrate over the region $y > x$ inside the square:

$$\mathbb{P}(Y > X) = \int_0^1 \int_x^1 \tfrac{2}{3}(x + 2y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^1 \tfrac{2}{3}\big[xy + y^2\big]_{y=x}^{1}\,\mathrm{d}x.$$

The bracket equals $(x + 1) - (x^2 + x^2) = x + 1 - 2x^2$, so

$$\mathbb{P}(Y > X) = \tfrac{2}{3}\int_0^1 \big(x + 1 - 2x^2\big)\,\mathrm{d}x = \tfrac{2}{3}\Big(\tfrac{1}{2} + 1 - \tfrac{2}{3}\Big) = \tfrac{2}{3} \cdot \tfrac{5}{6} = \frac{5}{9}.$$

F4. **(a)** For independent variables the joint density factors. With $f_X(x) = 2e^{-2x}$ and $f_Y(y) = 3e^{-3y}$,

$$f_{X,Y}(x, y) = 2e^{-2x} \cdot 3e^{-3y} = 6e^{-2x-3y}, \qquad x > 0,\ y > 0.$$

(b) By independence the joint tail factors:

$$\mathbb{P}(X > 1, Y > 1) = \mathbb{P}(X > 1)\,\mathbb{P}(Y > 1) = e^{-2(1)}\,e^{-3(1)} = e^{-5} \approx 0.00674.$$

(c) Integrate over $x < y$:

$$\mathbb{P}(X < Y) = \int_0^\infty \int_0^y 6e^{-2x}e^{-3y}\,\mathrm{d}x\,\mathrm{d}y = \int_0^\infty 3e^{-3y}\big(1 - e^{-2y}\big)\,\mathrm{d}y = 1 - \frac{3}{5} = \frac{2}{5}.$$

(Used $\int_0^\infty 3e^{-3y}\,\mathrm{d}y = 1$ and $\int_0^\infty 3e^{-5y}\,\mathrm{d}y = \frac{3}{5}$.)

F5. **(a)** The triangle $T = \{0 < x < y < 2\}$ has area $\frac{1}{2}(2)(2) = 2$, so a uniform density on it is $c = \frac{1}{2}$. **(b)** For fixed $x \in (0, 2)$ the slice runs $y \in (x, 2)$:

$$f_X(x) = \int_x^2 \tfrac{1}{2}\,\mathrm{d}y = \tfrac{1}{2}(2 - x), \qquad 0 < x < 2.$$

For fixed $y \in (0, 2)$ the slice runs $x \in (0, y)$:

$$f_Y(y) = \int_0^y \tfrac{1}{2}\,\mathrm{d}x = \tfrac{y}{2}, \qquad 0 < y < 2.$$

(c) Not independent: the support is the triangle $x < y$, not a rectangle, so the allowed values of X depend on Y; equivalently $f_X(x)f_Y(y) \neq f_{X,Y}(x, y)$. **(d)** Using the marginal of Y,

$$\mathbb{P}(Y > 1) = \int_1^2 \tfrac{y}{2}\,\mathrm{d}y = \tfrac{1}{4}\big[y^2\big]_1^2 = \tfrac{1}{4}(4 - 1) = \frac{3}{4}.$$

F6. **(a)** The density factors as a function of x times a function of y on a rectangular $(x > 0, y > 0)$ support:

$$f_{X,Y}(x, y) = \big(e^{-x}\big)\big(2e^{-2y}\big).$$

Each factor is a valid density, so $X \sim \mathrm{Exp}(1)$ and $Y \sim \mathrm{Exp}(2)$ are independent. **(b)** By independence $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y)$. With means $1/\lambda$,

$$\mathbb{E}(X) = \frac{1}{1} = 1, \quad \mathbb{E}(Y) = \frac{1}{2}, \quad \Longrightarrow \quad \mathbb{E}(XY) = 1 \cdot \tfrac{1}{2} = \frac{1}{2}.$$

(c) Integrate over $y < x$:

$$\mathbb{P}(Y < X) = \int_0^\infty \int_0^x 2e^{-x}e^{-2y}\,\mathrm{d}y\,\mathrm{d}x = \int_0^\infty e^{-x}\big(1 - e^{-2x}\big)\,\mathrm{d}x = 1 - \frac{1}{3} = \frac{2}{3}.$$

F7. **(a)** Integrate over $0 < x < y < 1$:

$$\int_0^1 \int_0^y 8xy \, \mathrm{d}x \, \mathrm{d}y = \int_0^1 8y \cdot \frac{y^2}{2} \, \mathrm{d}y = \int_0^1 4y^3 \, \mathrm{d}y = \big[y^4\big]_0^1 = 1. \blacksquare$$

(b) For fixed $x \in (0,1)$ the slice is $y \in (x,1)$:

$$f_X(x) = \int_x^1 8xy \, \mathrm{d}y = 8x \cdot \frac{1-x^2}{2} = 4x\big(1-x^2\big), \qquad 0 < x < 1.$$

For fixed $y \in (0,1)$ the slice is $x \in (0,y)$:

$$f_Y(y) = \int_0^y 8xy \, \mathrm{d}x = 8y \cdot \frac{y^2}{2} = 4y^3, \qquad 0 < y < 1.$$

(c) Not independent: the product $f_X(x) f_Y(y) = 4x(1-x^2) \cdot 4y^3 = 16xy^3(1-x^2)$ is not equal to $8xy$ on $0 < x < y < 1$; moreover the support $\{x < y\}$ is non-rectangular, which already rules out independence.
(d) Using the marginals,

$$\mathbb{E}(X) = \int_0^1 x \cdot 4x(1-x^2) \, \mathrm{d}x = \int_0^1 \big(4x^2 - 4x^4\big) \, \mathrm{d}x = \frac{4}{3} - \frac{4}{5} = \frac{8}{15},$$

$$\mathbb{E}(Y) = \int_0^1 y \cdot 4y^3 \, \mathrm{d}y = \int_0^1 4y^4 \, \mathrm{d}y = \frac{4}{5}.$$

G. Transforming and combining random variables

G1. A strictly increasing affine map of a uniform variable is again uniform. As X ranges over $(0,1)$, $Y = 8X - 3$ ranges over $(8 \cdot 0 - 3,\ 8 \cdot 1 - 3) = (-3, 5)$, so $Y \sim \mathrm{Uniform}(-3,5)$. Either by the Jacobian (inverse $x = (y+3)/8$, $|\,\mathrm{d}x/\,\mathrm{d}y| = 1/8$, times $f_X = 1$) or directly from the uniform formula,

$$f_Y(y) = \frac{1}{5 - (-3)} = \frac{1}{8}, \qquad -3 \le y \le 5,$$

and $f_Y(y) = 0$ otherwise.

G2. Here $X \sim \mathrm{Uniform}(0,2)$ has $f_X(x) = \frac{1}{2}$ on $(0,2)$, so $Y = X^2$ takes values in $(0,4)$. For $0 < y < 4$ the cdf method gives, since $X \ge 0$,

$$F_Y(y) = \mathbb{P}(X^2 \le y) = \mathbb{P}(0 \le X \le \sqrt{y}) = \frac{\sqrt{y}}{2}.$$

Differentiating,

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y}\Big(\tfrac{1}{2} y^{1/2}\Big) = \frac{1}{4\sqrt{y}}, \qquad 0 < y < 4,$$

and 0 otherwise. (Check: $\int_0^4 \tfrac{1}{4} y^{-1/2} \, \mathrm{d}y = \tfrac{1}{4} \cdot 2\sqrt{y}\big|_0^4 = \tfrac{1}{4} \cdot 4 = 1$.)

G3. The map $y = \sqrt{x}$ is strictly increasing on $(0,\infty)$ with inverse $x = y^2$, so $\dfrac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y) = 2y$. With $f_X(x) = \lambda e^{-\lambda x}$ for $x > 0$, the change-of-variable formula $f_Y(y) = f_X\big(g^{-1}(y)\big)\,\big|\frac{\mathrm{d}}{\mathrm{d}y} g^{-1}(y)\big|$ gives, for $y > 0$,

$$f_Y(y) = \lambda e^{-\lambda y^2} \cdot 2y = 2\lambda y \, e^{-\lambda y^2}, \qquad y > 0,$$

and 0 otherwise. (This is a Rayleigh density; check $\int_0^\infty 2\lambda y \, e^{-\lambda y^2} \, \mathrm{d}y = \big[-e^{-\lambda y^2}\big]_0^\infty = 1$.)

G4. The map $y = e^x$ is strictly increasing with inverse $x = \ln y$ and $\dfrac{\mathrm{d}}{\mathrm{d}y}\ln y = \dfrac{1}{y}$. As x ranges over $(0, \ln 4)$, y ranges over $(e^0, e^{\ln 4}) = (1, 4)$. With $f_X(x) = \dfrac{1}{\ln 4}$ on $(0, \ln 4)$:

(a) Density.

$$f_Y(y) = f_X(\ln y)\left|\frac{1}{y}\right| = \frac{1}{\ln 4}\cdot\frac{1}{y} = \frac{1}{y\ln 4}, \qquad 1 < y < 4,$$

and 0 otherwise. (Check: $\int_1^4 \frac{1}{y\ln 4}\,\mathrm{d}y = \frac{\ln 4 - \ln 1}{\ln 4} = 1$.)

(b) Mean. From f_Y,

$$\mathbb{E}(Y) = \int_1^4 y\cdot\frac{1}{y\ln 4}\,\mathrm{d}y = \frac{1}{\ln 4}\int_1^4 1\,\mathrm{d}y = \frac{4-1}{\ln 4} = \frac{3}{\ln 4}.$$

Directly from f_X (law of the unconscious statistician),

$$\mathbb{E}(e^X) = \int_0^{\ln 4} e^x\cdot\frac{1}{\ln 4}\,\mathrm{d}x = \frac{1}{\ln 4}\left[e^x\right]_0^{\ln 4} = \frac{4-1}{\ln 4} = \frac{3}{\ln 4},$$

which agrees. Numerically $\ln 4 \approx 1.3863$, so $\mathbb{E}(Y) \approx 2.164$.

G5. The map $y = e^{-x}$ is strictly decreasing on $(0,\infty)$ with inverse $x = -\ln y$ and $\dfrac{\mathrm{d}}{\mathrm{d}y}(-\ln y) = -\dfrac{1}{y}$, so the absolute Jacobian is $1/y$. As x ranges over $(0,\infty)$, y ranges over $(0,1)$. With $f_X(x) = 2e^{-2x}$,

$$f_Y(y) = f_X(-\ln y)\left|\frac{-1}{y}\right| = 2e^{-2(-\ln y)}\cdot\frac{1}{y} = 2e^{2\ln y}\cdot\frac{1}{y} = 2y^2\cdot\frac{1}{y} = 2y, \qquad 0 < y < 1,$$

and 0 otherwise. Thus Y has density $f_Y(y) = 2y$ on $(0,1)$, the Beta$(2,1)$ density. (Check: $\int_0^1 2y\,\mathrm{d}y = 1$.)

G6. Since X_1, X_2 are independent with $f(x) = 1$ on $(0,1)$, the convolution density of $S = X_1 + X_2$ is

$$f_S(s) = \int_{-\infty}^{\infty} f_{X_1}(t)\,f_{X_2}(s-t)\,\mathrm{d}t = \int \mathbb{1}\{0 < t < 1\}\,\mathbb{1}\{0 < s - t < 1\}\,\mathrm{d}t,$$

where the integrand is 1 exactly when $\max(0, s-1) < t < \min(1, s)$. The total error S lies in $(0,2)$.

Case $0 \le s \le 1$. Then t runs over $(0, s)$, of length s, so $f_S(s) = s$.

Case $1 < s \le 2$. Then t runs over $(s-1, 1)$, of length $2 - s$, so $f_S(s) = 2 - s$.

$$f_S(s) = \begin{cases} s, & 0 \le s \le 1, \\ 2 - s, & 1 < s \le 2, \\ 0, & \text{otherwise}, \end{cases}$$

the triangular density on $(0,2)$. By symmetry about $s = 1$,

$$\mathbb{P}(S \le 1) = \int_0^1 s\,\mathrm{d}s = \frac{s^2}{2}\Big|_0^1 = \frac{1}{2}.$$

G7. With $f_X(x) = e^{-x}$ for $x > 0$ and $f_Y(y) = 2e^{-2y}$ for $y > 0$, independence gives the convolution, for $s > 0$ (the integrand is supported on $0 < t < s$):

$$f_S(s) = \int_0^s f_X(t)\,f_Y(s-t)\,\mathrm{d}t = \int_0^s e^{-t}\cdot 2e^{-2(s-t)}\,\mathrm{d}t = 2e^{-2s}\int_0^s e^{t}\,\mathrm{d}t.$$

Since $\int_0^s e^t\,\mathrm{d}t = e^s - 1$,

$$f_S(s) = 2e^{-2s}\big(e^s - 1\big) = 2\big(e^{-s} - e^{-2s}\big), \qquad s > 0,$$

and 0 otherwise.

Integrates to 1.

$$\int_0^\infty 2\big(e^{-s} - e^{-2s}\big)\,\mathrm{d}s = 2\Big(1 - \tfrac{1}{2}\Big) = 2 \cdot \tfrac{1}{2} = 1.\blacksquare$$

Mean. Either integrate, or use $\mathbb{E}(S) = \mathbb{E}(X) + \mathbb{E}(Y) = \dfrac{1}{1} + \dfrac{1}{2}$. Directly,

$$\mathbb{E}(S) = \int_0^\infty s \cdot 2\big(e^{-s} - e^{-2s}\big)\,\mathrm{d}s = 2\Big(\int_0^\infty s\,e^{-s}\,\mathrm{d}s - \int_0^\infty s\,e^{-2s}\,\mathrm{d}s\Big) = 2\Big(1 - \tfrac{1}{4}\Big) = \frac{3}{2},$$

using $\int_0^\infty s\,e^{-as}\,\mathrm{d}s = 1/a^2$. Both methods give $\mathbb{E}(S) = \frac{3}{2}$.

H. Covariance and the mean and variance of sums

H1. Linearity of expectation holds with no independence assumption, so the correlation between X and Y is irrelevant to the mean. Pulling out constants and adding,

$$\mathbb{E}(P) = 2\mathbb{E}(X) + 3\mathbb{E}(Y) - 5 = 2(9) + 3(14) - 5 = 18 + 42 - 5 = 55.$$

Expected daily profit is 55 hundred dollars, i.e. \$5500.

H2. The two legs are independent, so means add for both the sum and the difference, and the variances add in *both* cases (the cross term $\pm 2\,\mathrm{Cov}$ vanishes because $\mathrm{Cov}(A, B) = 0$).

$$\mathbb{E}(C) = \mathbb{E}(A) + \mathbb{E}(B) = 12 + 7 = 19 \text{ s}, \qquad \mathrm{Var}(C) = \mathrm{Var}(A) + \mathrm{Var}(B) = 5 + 11 = 16 \text{ s}^2.$$

$$\mathbb{E}(D) = \mathbb{E}(A) - \mathbb{E}(B) = 12 - 7 = 5 \text{ s}, \qquad \mathrm{Var}(D) = \mathrm{Var}(A) + (-1)^2\,\mathrm{Var}(B) = 5 + 11 = 16 \text{ s}^2.$$

Note the difference D is just as variable as the sum C: subtracting independent quantities does not cancel variability.

H3. Use the shortcut $\mathrm{Cov}(X, Y) = \mathbb{E}(XY) - \mathbb{E}(X)\mathbb{E}(Y)$.

Step 1: marginal means. From the margin columns,

$$\mathbb{E}(X) = 1(0.20) + 2(0.40) + 3(0.40) = 2.20, \qquad \mathbb{E}(Y) = 1(0.25) + 2(0.40) + 3(0.35) = 2.10.$$

Step 2: $\mathbb{E}(XY)$ over all nine cells.

$$\begin{aligned}\mathbb{E}(XY) &= 1(0.10) + 2(0.05) + 3(0.05) \\ &\quad + 2(0.10) + 4(0.20) + 6(0.10) \\ &\quad + 3(0.05) + 6(0.15) + 9(0.20) \\ &= 0.10 + 0.10 + 0.15 + 0.20 + 0.80 + 0.60 + 0.15 + 0.90 + 1.80 = 4.80.\end{aligned}$$

Step 3: subtract.

$$\mathrm{Cov}(X, Y) = 4.80 - (2.20)(2.10) = 4.80 - 4.62 = 0.18.$$

The covariance is positive, so X and Y move together: shifts that put more forklifts in use also tend to occupy more docks.

H4. *Part (a): correlation.* Standardise the covariance by the two standard deviations:

$$\mathrm{Corr}(X,Y) = \frac{\mathrm{Cov}(X,Y)}{\sqrt{\mathrm{Var}(X)\ \mathrm{Var}(Y)}} = \frac{0.18}{\sqrt{0.56}\,\sqrt{0.59}}.$$

Since $\sqrt{0.56} \approx 0.7483$ and $\sqrt{0.59} \approx 0.7681$, their product is ≈ 0.5748, so

$$\mathrm{Corr}(X,Y) \approx \frac{0.18}{0.5748} \approx 0.313.$$

A modest positive correlation.

Part (b): variance of the sum. The full rule keeps the cross term:

$$\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y) = 0.56 + 0.59 + 2(0.18) = 1.15 + 0.36 = 1.51.$$

Dropping the covariance would give only $0.56 + 0.59 = 1.15$, understating the true variance by $2(0.18) = 0.36$ — the positive co-movement of X and Y genuinely inflates the spread of their sum.

H5. Both S and T are linear combinations of the same underlying U, V, so expand by bilinearity and additivity of covariance; independence of U, V kills the mixed terms $\mathrm{Cov}(U,V) = 0$.

Step 1: expand $\mathrm{Cov}(S,T)$.

$$\begin{aligned}\mathrm{Cov}(S,T) &= \mathrm{Cov}(U+2V,\ 3U-V)\\ &= 3\,\mathrm{Cov}(U,U) - \mathrm{Cov}(U,V) + 6\,\mathrm{Cov}(V,U) - 2\,\mathrm{Cov}(V,V)\\ &= 3\,\mathrm{Var}(U) - 0 + 0 - 2\,\mathrm{Var}(V)\\ &= 3(6) - 2(10) = 18 - 20 = -2.\end{aligned}$$

Step 2: variances of S and T. With $\mathrm{Cov}(U,V) = 0$,

$$\begin{aligned}\mathrm{Var}(S) &= \mathrm{Var}(U) + 2^2\,\mathrm{Var}(V) = 6 + 4(10) = 46,\\ \mathrm{Var}(T) &= 3^2\,\mathrm{Var}(U) + \mathrm{Var}(V) = 9(6) + 10 = 64.\end{aligned}$$

Step 3: correlation.

$$\mathrm{Corr}(S,T) = \frac{\mathrm{Cov}(S,T)}{\sqrt{\mathrm{Var}(S)\ \mathrm{Var}(T)}} = \frac{-2}{\sqrt{46\cdot 64}} = \frac{-2}{\sqrt{2944}} \approx \frac{-2}{54.26} \approx -0.0369.$$

Even though S and T are built from independent pieces, they share U and V and so are slightly negatively correlated.

H6. This is the iid sample-mean setting, with $\mu = 6.4$ and $\sigma^2 = 0.64$.

Part (a): mean, variance, SD of $\bar{X}$.

$$\mathbb{E}(\bar{X}) = \mu = 6.4, \qquad \mathrm{Var}(\bar{X}) = \frac{\sigma^2}{n} = \frac{0.64}{16} = 0.04, \qquad \mathrm{SD}(\bar{X}) = \sqrt{0.04} = 0.2.$$

A single reading scatters by $\sigma = \sqrt{0.64} = 0.8$; the 16-reading average by only 0.2.

Part (b): readings for $\mathrm{SD}(\bar{X}) \le 0.10$. Invert $\mathrm{SD}(\bar{X}) = \sigma/\sqrt{n}$:

$$\frac{\sigma}{\sqrt{n}} \le 0.10 \iff \sqrt{n} \ge \frac{0.8}{0.10} = 8 \iff n \ge 64.$$

At least $n = 64$ readings are required. Compared with the 16 that gave SD = 0.2, quadrupling the sample halved the standard deviation, exactly as the $1/\sqrt{n}$ law predicts.

H7. Here $N \sim \text{Bin}(25, 0.8)$ and $W = 2N + 2n = 2N + 50$ (since $n = 25$).

Part (a): mean and variance of N. Write $N = \sum_{i=1}^{25} I_i$, where $I_i = \mathbb{1}\{\text{parcel } i \text{ signed}\}$ has mean 0.8, and the I_i are independent. Then

$$\mathbb{E}(N) = \sum_{i=1}^{25} \mathbb{E}(I_i) = 25(0.8) = 20,$$
$$\text{Var}(N) = \sum_{i=1}^{25} \text{Var}(I_i) = 25\,p(1-p) = 25(0.8)(0.2) = 4.$$

Part (b): mean and variance of W. Since $W = 2N + 50$ is an affine function of N,

$$\mathbb{E}(W) = 2\mathbb{E}(N) + 50 = 2(20) + 50 = 90, \qquad \text{Var}(W) = 2^2\,\text{Var}(N) = 4(4) = 16.$$

(The additive constant 50 shifts the mean but leaves the variance unchanged.)

Part (c): covariance and correlation of N and W. Using bilinearity,

$$\text{Cov}(N, W) = \text{Cov}(N,\, 2N + 50) = 2\,\text{Cov}(N, N) = 2\,\text{Var}(N) = 2(4) = 8,$$

and therefore

$$\text{Corr}(N, W) = \frac{\text{Cov}(N, W)}{\text{SD}(N)\ \text{SD}(W)} = \frac{8}{\sqrt{4}\,\sqrt{16}} = \frac{8}{2 \cdot 4} = 1.$$

The correlation is exactly $+1$ because W is an increasing (positive-slope) affine function of N: the two encode the same information up to rescaling and shifting, a perfect linear relationship.

I. Order statistics

I1. For a Uniform$(0, 1)$ parent, $F(x) = x$ on $(0, 1)$, and with $n = 4$ iid observations the extremes have $F_{X_{(n)}}(x) = F(x)^n$ and $\mathbb{P}\big(X_{(1)} > x\big) = \big(1 - F(x)\big)^n$.

(a) $\mathbb{P}\big(X_{(4)} \le 0.7\big) = F(0.7)^4 = (0.7)^4 = 0.2401$.

(b) All four exceed 0.2 iff each does, so $\mathbb{P}\big(X_{(1)} > 0.2\big) = \big(1 - 0.2\big)^4 = (0.8)^4 = 0.4096$.

(c) $\mathbb{P}\big(X_{(1)} \le 0.2\big) = 1 - (0.8)^4 = 1 - 0.4096 = 0.5904$.

I2. Here $n = 5$ and $F(x) = x$, $f(x) = 1$ on $(0, 1)$.

(a) The maximum is at most x iff every observation is, so

$$F_M(x) = F(x)^5 = x^5, \qquad f_M(x) = F_M'(x) = 5x^4, \qquad 0 < x < 1.$$

(b) $\mathbb{E}(M) = \displaystyle\int_0^1 x \cdot 5x^4\,\mathrm{d}x = 5\int_0^1 x^5\,\mathrm{d}x = \frac{5}{6}$. (In general $\mathbb{E}\big(X_{(n)}\big) = n/(n+1)$ for a Uniform$(0, 1)$ parent.)

(c) $\mathbb{P}(M \le 0.9) = F_M(0.9) = (0.9)^5 = 0.59049$.

I3. Each lifetime has cdf $F(x) = 1 - e^{-\lambda x}$ with $\lambda = \frac{1}{4}$, and $n = 6$.

(a) For the minimum of iid exponentials,

$$\mathbb{P}\big(X_{(1)} > x\big) = \big(1 - F(x)\big)^6 = \big(e^{-\lambda x}\big)^6 = e^{-6\lambda x},$$

which is the survival function of an exponential with rate $6\lambda = 6 \cdot \frac{1}{4} = \frac{3}{2}$. Thus $X_{(1)} \sim \text{Exp}\big(\frac{3}{2}\big)$.

(b) $\mathbb{E}\big(X_{(1)}\big) = \dfrac{1}{6\lambda} = \dfrac{1}{3/2} = \dfrac{2}{3} \approx 0.667$ years.

(c) $\mathbb{P}\big(X_{(1)} > 1\big) = e^{-6\lambda \cdot 1} = e^{-3/2} \approx 0.2231$.

I4. For Uniform$(0,\theta)$, $F(x)=x/\theta$ and $f(x)=1/\theta$ on $(0,\theta)$. With $n=3$ the median is the $k=2$ order statistic.

(a) The general formula gives

$$f_{X_{(2)}}(x)=\frac{n!}{(k-1)!\,(n-k)!}\,F(x)^{k-1}\big(1-F(x)\big)^{n-k}f(x)=\frac{3!}{1!\,1!}\cdot\frac{x}{\theta}\cdot\Big(1-\frac{x}{\theta}\Big)\cdot\frac{1}{\theta}=\frac{6x(\theta-x)}{\theta^3},$$

for $0<x<\theta$.

(b) By symmetry of the density about $\theta/2$, or directly,

$$\mathbb{E}\big(X_{(2)}\big)=\int_0^\theta x\cdot\frac{6x(\theta-x)}{\theta^3}\,\mathrm{d}x=\frac{6}{\theta^3}\int_0^\theta(\theta x^2-x^3)\,\mathrm{d}x=\frac{6}{\theta^3}\Big(\frac{\theta^4}{3}-\frac{\theta^4}{4}\Big)=\frac{6}{\theta^3}\cdot\frac{\theta^4}{12}=\frac{\theta}{2}.$$

(c) Compute the second moment:

$$\mathbb{E}\big(X_{(2)}^2\big)=\frac{6}{\theta^3}\int_0^\theta(\theta x^3-x^4)\,\mathrm{d}x=\frac{6}{\theta^3}\Big(\frac{\theta^5}{4}-\frac{\theta^5}{5}\Big)=\frac{6}{\theta^3}\cdot\frac{\theta^5}{20}=\frac{3\theta^2}{10}.$$

Then

$$\mathrm{Var}\big(X_{(2)}\big)=\frac{3\theta^2}{10}-\Big(\frac{\theta}{2}\Big)^2=\frac{3\theta^2}{10}-\frac{\theta^2}{4}=\frac{6\theta^2-5\theta^2}{20}=\frac{\theta^2}{20}.$$

I5. For Uniform$(0,1)$, $F(x)=x$, $f(x)=1$. With $n=4$ and $k=3$:

(a)

$$f_{X_{(3)}}(x)=\frac{4!}{(3-1)!\,(4-3)!}\,x^2(1-x)^1\cdot1=\frac{24}{2\cdot1}\,x^2(1-x)=12x^2(1-x)=12x^2-12x^3,$$

for $0<x<1$.

(b) Means and the second moment:

$$\mathbb{E}\big(X_{(3)}\big)=\int_0^1 x\big(12x^2-12x^3\big)\,\mathrm{d}x=12\Big(\frac{1}{4}-\frac{1}{5}\Big)=12\cdot\frac{1}{20}=\frac{3}{5},$$

$$\mathbb{E}\big(X_{(3)}^2\big)=\int_0^1 x^2\big(12x^2-12x^3\big)\,\mathrm{d}x=12\Big(\frac{1}{5}-\frac{1}{6}\Big)=12\cdot\frac{1}{30}=\frac{2}{5}.$$

Hence

$$\mathrm{Var}\big(X_{(3)}\big)=\frac{2}{5}-\Big(\frac{3}{5}\Big)^2=\frac{2}{5}-\frac{9}{25}=\frac{10-9}{25}=\frac{1}{25}.$$

(These match the general Uniform$(0,1)$ results $\mathbb{E}\big(X_{(k)}\big)=k/(n+1)$ and $\mathrm{Var}\big(X_{(k)}\big)=k(n-k+1)/\big[(n+1)^2(n+2)\big]$ with $n=4,k=3$.)

I6. Take Uniform$(0,1)$ with $F(x)=x$, $f(x)=1$, and $n=3$.

(a) Fix the smallest value at $X_{(1)}=u$ and the range at $R=r$, so the largest is at $u+r$ and the one remaining point lies between them. The joint density of $\big(X_{(1)},X_{(3)}\big)=(u,\,u+r)$ for a sample of size n is

$$f_{X_{(1)},X_{(3)}}(u,v)=n(n-1)\big(F(v)-F(u)\big)^{n-2}f(u)f(v),\qquad u<v,$$

which for our case is $3\cdot2\,(v-u)^1=6(v-u)$. Substituting $v=u+r$ and integrating out the position $u\in(0,1-r)$,

$$f_R(r)=\int_0^{1-r}6\big((u+r)-u\big)\,\mathrm{d}u=\int_0^{1-r}6r\,\mathrm{d}u=6r(1-r),\qquad 0<r<1.$$

(b)

$$\mathbb{E}(R)=\int_0^1 r\cdot6r(1-r)\,\mathrm{d}r=6\int_0^1(r^2-r^3)\,\mathrm{d}r=6\Big(\frac{1}{3}-\frac{1}{4}\Big)=6\cdot\frac{1}{12}=\frac{1}{2}.$$

For Uniform$(0,1)$ with $n=3$, $\mathbb{E}\big(X_{(3)}\big)=3/4$ and $\mathbb{E}\big(X_{(1)}\big)=1/4$, so $\mathbb{E}\big(X_{(3)}\big)-\mathbb{E}\big(X_{(1)}\big)=3/4-1/4=1/2=\mathbb{E}(R)$, as expected from linearity.

(c)

$$\mathbb{P}\Big(R>\tfrac12\Big)=\int_{1/2}^{1}6r(1-r)\,\mathrm{d}r=6\Big[\frac{r^2}{2}-\frac{r^3}{3}\Big]_{1/2}^{1}=6\Big[\Big(\tfrac12-\tfrac13\Big)-\Big(\tfrac18-\tfrac{1}{24}\Big)\Big]=6\Big[\tfrac16-\tfrac{1}{12}\Big]=6\cdot\frac{1}{12}=\frac12.$$

I7. Lifetimes are iid Exp(λ) with $\lambda=\frac12$ and $n=4$.

(a) The first failure is $X_{(1)}=\min$ of four iid exponentials, whose survival function is $\big(e^{-\lambda x}\big)^4=e^{-4\lambda x}$; hence $X_{(1)}\sim\mathrm{Exp}(4\lambda)$. At the instant $X_{(1)}$ of the first failure, the three surviving bulbs are still "as good as new" by the **memoryless property**, so their remaining lifetimes are again iid Exp(λ) and independent of $X_{(1)}$. The next failure is the minimum of these three exponentials measured from $X_{(1)}$, so the spacing $X_{(2)}-X_{(1)}\sim\mathrm{Exp}(3\lambda)$ and is independent of $X_{(1)}$. (In general the i-th spacing is $\mathrm{Exp}\big((n-i+1)\lambda\big)$.)

(b) Using $\mathbb{E}\big(\mathrm{Exp}(r)\big)=1/r$ and adding independent gaps,

$$\mathbb{E}\big(X_{(1)}\big)=\frac{1}{4\lambda}=\frac{1}{4\cdot\frac12}=\frac12\quad\text{(i.e. 500 hours)},$$

$$\mathbb{E}\big(X_{(2)}\big)=\mathbb{E}\big(X_{(1)}\big)+\mathbb{E}\big(X_{(2)}-X_{(1)}\big)=\frac{1}{4\lambda}+\frac{1}{3\lambda}=\frac12+\frac{1}{3\cdot\frac12}=\frac12+\frac23=\frac76\quad\text{(i.e. }\approx1167\text{ hours)}.$$

J. Conditional densities and conditional expectation

J1. Given $X=2$, Y is uniform on $\{0,1,2\}$, so the conditional pmf is

$$p_{Y\mid X}(y\mid 2)=\tfrac13,\qquad y\in\{0,1,2\}.$$

Hence

$$\mathbb{E}(Y\mid X=2)=\frac{0+1+2}{3}=1.$$

In general, given $X=x$ the variable Y is uniform on $\{0,1,\dots,x\}$ (that is $x+1$ equally likely values), so $\mathbb{E}(Y\mid X=x)=\dfrac{0+1+\cdots+x}{x+1}=\dfrac{x}{2}$. Thus $\mathbb{E}(Y\mid X=1)=\frac12$, $\mathbb{E}(Y\mid X=2)=1$, $\mathbb{E}(Y\mid X=3)=\frac32$. By the law of total expectation,

$$\mathbb{E}(Y)=\sum_x\mathbb{E}(Y\mid X=x)\,\mathbb{P}(X=x)=\tfrac12\cdot\tfrac14+1\cdot\tfrac12+\tfrac32\cdot\tfrac14=\tfrac18+\tfrac12+\tfrac38=1.$$

Equivalently $\mathbb{E}(Y)=\mathbb{E}(X/2)=\frac12\mathbb{E}(X)=\frac12\cdot2=1$, since $\mathbb{E}(X)=\frac14+\frac12\cdot2+\frac14\cdot3=2$.

J2. The marginal density of X is

$$f_X(x)=\int_0^\infty e^{-x-y}\,\mathrm{d}y=e^{-x}\int_0^\infty e^{-y}\,\mathrm{d}y=e^{-x},\qquad x>0,$$

so $X\sim\mathrm{Exp}(1)$. Then for $x>0$,

$$f_{Y\mid X}(y\mid x)=\frac{f(x,y)}{f_X(x)}=\frac{e^{-x-y}}{e^{-x}}=e^{-y},\qquad y>0,$$

which is free of x. Hence $Y\mid X=x\sim\mathrm{Exp}(1)$ for every x, i.e. X and Y are independent, and

$$\mathbb{E}(Y\mid X=x)=\frac11=1.$$

J3. The marginal of X: for $0 < x < 1$ the slice runs over $x < y < 1$, so

$$f_X(x) = \int_x^1 2\,\mathrm{d}y = 2(1-x), \qquad 0 < x < 1.$$

Therefore, for $0 < x < 1$,

$$f_{Y\mid X}(y \mid x) = \frac{2}{2(1-x)} = \frac{1}{1-x}, \qquad x < y < 1,$$

which is the Uniform$(x, 1)$ density. Consequently

$$\mathbb{E}(Y \mid X = x) = \frac{x+1}{2}, \qquad \mathrm{Var}(Y \mid X = x) = \frac{(1-x)^2}{12},$$

using the mean $(a+b)/2$ and variance $(b-a)^2/12$ of Uniform(a, b) with $a = x$, $b = 1$.

J4. The joint density is the product of the marginal of X (which is 1 on $(0, 1)$) and the conditional of Y:

$$f(x, y) = f_X(x)\, f_{Y\mid X}(y \mid x) = 1 \cdot \frac{1}{x} = \frac{1}{x}, \qquad 0 < y < x < 1,$$

and 0 otherwise. The marginal of Y integrates out x over $x > y$ (and $x < 1$):

$$f_Y(y) = \int_y^1 \frac{1}{x}\,\mathrm{d}x = \big[\ln x\big]_y^1 = -\ln y = \ln\frac{1}{y}, \qquad 0 < y < 1.$$

(As a check, $\int_0^1 (-\ln y)\,\mathrm{d}y = \big[y - y\ln y\big]_0^1 = 1$.) Then

$$\mathbb{P}\big(Y < \tfrac{1}{2}\big) = \int_0^{1/2} (-\ln y)\,\mathrm{d}y = \big[y - y\ln y\big]_0^{1/2} = \tfrac{1}{2} - \tfrac{1}{2}\ln\tfrac{1}{2} = \tfrac{1}{2} + \tfrac{1}{2}\ln 2 \approx 0.847.$$

J5. Since $Y \mid X = x \sim$ Uniform$(0, x)$,

$$\mathbb{E}(Y \mid X = x) = \frac{x}{2}.$$

By the law of total expectation, with $X \sim$ Uniform$(0, 1)$,

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big] = \mathbb{E}\left(\frac{X}{2}\right) = \frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4}.$$

For the reverse conditional, use $f(x, y) = 1/x$ on $0 < y < x < 1$ and the marginal $f_Y(y) = -\ln y$:

$$f_{X\mid Y}(x \mid y) = \frac{f(x,y)}{f_Y(y)} = \frac{1/x}{-\ln y} = \frac{1}{-x\ln y}, \qquad y < x < 1.$$

Hence

$$\mathbb{E}(X \mid Y = y) = \int_y^1 x \cdot \frac{1}{-x\ln y}\,\mathrm{d}x = \frac{1}{-\ln y}\int_y^1 \mathrm{d}x = \frac{1-y}{-\ln y} = \frac{1-y}{\ln(1/y)}.$$

J6. For each fixed n,

$$p_N(n) = \int_0^\infty p_{N\mid\Lambda}(n \mid \lambda)\, f_\Lambda(\lambda)\,\mathrm{d}\lambda = \int_0^\infty \frac{e^{-\lambda}\lambda^n}{n!}\, e^{-\lambda}\,\mathrm{d}\lambda = \frac{1}{n!}\int_0^\infty \lambda^n e^{-2\lambda}\,\mathrm{d}\lambda.$$

Using $\int_0^\infty \lambda^n e^{-2\lambda}\,\mathrm{d}\lambda = \dfrac{n!}{2^{\,n+1}}$ (the Gamma integral),

$$p_N(n) = \frac{1}{n!}\cdot\frac{n!}{2^{\,n+1}} = \left(\frac{1}{2}\right)^{n+1} = \frac{1}{2}\left(\frac{1}{2}\right)^n, \qquad n = 0, 1, 2, \ldots$$

This is the geometric distribution on $\{0, 1, 2, \dots\}$ counting failures before the first success with success probability $p = \frac{1}{2}$. For the moments, condition on Λ where $\mathbb{E}(N \mid \Lambda) = \mathrm{Var}(N \mid \Lambda) = \Lambda$ (Poisson) and $\Lambda \sim \mathrm{Exp}(1)$ has $\mathbb{E}(\Lambda) = \mathrm{Var}(\Lambda) = 1$:

$$\mathbb{E}(N) = \mathbb{E}\big[\mathbb{E}(N \mid \Lambda)\big] = \mathbb{E}(\Lambda) = 1,$$

$$\mathrm{Var}(N) = \mathbb{E}\big[\mathrm{Var}(N \mid \Lambda)\big] + \mathrm{Var}\big[\mathbb{E}(N \mid \Lambda)\big] = \mathbb{E}(\Lambda) + \mathrm{Var}(\Lambda) = 1 + 1 = 2.$$

These agree with the geometric formulas $\mathbb{E}(N) = (1-p)/p = 1$ and $\mathrm{Var}(N) = (1-p)/p^2 = 2$.

J7. First the normalisation. For fixed $x > 0$, $\int_0^\infty \frac{1}{x} e^{-y/x}\,\mathrm{d}y = 1$ (an $\mathrm{Exp}(1/x)$ density integrates to 1), so

$$\int_0^\infty \int_0^\infty f(x,y)\,\mathrm{d}y\,\mathrm{d}x = \int_0^\infty e^{-x}\left(\int_0^\infty \frac{1}{x} e^{-y/x}\,\mathrm{d}y\right)\mathrm{d}x = \int_0^\infty e^{-x}\,\mathrm{d}x = 1.$$

The marginal of X is therefore $f_X(x) = e^{-x}$ on $x > 0$, and

$$f_{Y|X}(y \mid x) = \frac{f(x,y)}{f_X(x)} = \frac{\frac{1}{x} e^{-y/x} e^{-x}}{e^{-x}} = \frac{1}{x} e^{-y/x}, \qquad y > 0,$$

so $Y \mid X = x \sim \mathrm{Exp}(1/x)$, the exponential with rate $1/x$ and mean x. Hence

$$\mathbb{E}(Y \mid X = x) = x.$$

By the law of total expectation, with $X \sim \mathrm{Exp}(1)$ so $\mathbb{E}(X) = 1$,

$$\mathbb{E}(Y) = \mathbb{E}\big[\mathbb{E}(Y \mid X)\big] = \mathbb{E}(X) = 1.$$

K. Moment-generating functions

K1. The payout X equals 0 with probability 0.9 and 10 with probability 0.1, so by the definition $M_X(t) = \mathbb{E}(e^{tX}) = \sum_x e^{tx} p_X(x)$,

$$M_X(t) = 0.9\, e^{0 \cdot t} + 0.1\, e^{10t} = 0.9 + 0.1\, e^{10t}.$$

Differentiating,

$$M_X'(t) = 0.1 \cdot 10\, e^{10t} = e^{10t}, \qquad M_X''(t) = 10\, e^{10t}.$$

Evaluating at $t = 0$ gives the moments

$$\mathbb{E}(X) = M_X'(0) = 1, \qquad \mathbb{E}(X^2) = M_X''(0) = 10.$$

Hence

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = 10 - 1 = 9.$$

This matches the direct computation $\mathbb{E}(X) = 0.1 \cdot 10 = 1$, $\mathbb{E}(X^2) = 0.1 \cdot 10^2 = 10$, and $\mathrm{Var}(X) = 0.1 \cdot 0.9 \cdot 10^2 = 9$ (a two-point variance $p(1-p)(b-a)^2$ with $b - a = 10$).

K2. An MGF of a discrete variable is a weighted sum $\sum_x p_X(x) e^{tx}$, so the coefficient of e^{tx} is the probability that $X = x$. Reading off the three terms $\frac{1}{6} e^{0 \cdot t}$, $\frac{1}{3} e^{1 \cdot t}$, $\frac{1}{2} e^{3 \cdot t}$ gives

$$p_X(0) = \tfrac{1}{6}, \qquad p_X(1) = \tfrac{1}{3}, \qquad p_X(3) = \tfrac{1}{2}$$

(these sum to 1, as required). From the pmf,

$$\mathbb{E}(X) = 0 \cdot \tfrac{1}{6} + 1 \cdot \tfrac{1}{3} + 3 \cdot \tfrac{1}{2} = \tfrac{1}{3} + \tfrac{3}{2} = \tfrac{11}{6},$$

$$\mathbb{E}(X^2) = 0 \cdot \tfrac{1}{6} + 1 \cdot \tfrac{1}{3} + 9 \cdot \tfrac{1}{2} = \tfrac{1}{3} + \tfrac{9}{2} = \tfrac{29}{6},$$

$$\mathrm{Var}(X) = \tfrac{29}{6} - \left(\tfrac{11}{6}\right)^2 = \tfrac{174}{36} - \tfrac{121}{36} = \tfrac{53}{36}.$$

By differentiation,

$$M_X'(t) = \tfrac{1}{3} e^t + \tfrac{3}{2} e^{3t}, \qquad M_X''(t) = \tfrac{1}{3} e^t + \tfrac{9}{2} e^{3t},$$

so $M_X'(0) = \frac{1}{3} + \frac{3}{2} = \frac{11}{6} = \mathbb{E}(X)$ and $M_X''(0) = \frac{1}{3} + \frac{9}{2} = \frac{29}{6} = \mathbb{E}(X^2)$, in agreement.

K3. With $\lambda = \frac{1}{5}$ the density is $f_T(x) = \lambda e^{-\lambda x}$ for $x > 0$. By definition,

$$M_T(t) = \int_0^\infty e^{tx}\, \lambda e^{-\lambda x}\, \mathrm{d}x = \lambda \int_0^\infty e^{-(\lambda - t)x}\, \mathrm{d}x.$$

The integral converges precisely when $\lambda - t > 0$, i.e. for $t < \lambda = \frac{1}{5}$, and then

$$M_T(t) = \lambda \cdot \frac{1}{\lambda - t} = \frac{\lambda}{\lambda - t} = \frac{1/5}{1/5 - t}, \qquad t < \tfrac{1}{5}.$$

Differentiating,

$$M_T'(t) = \frac{\lambda}{(\lambda - t)^2}, \qquad M_T''(t) = \frac{2\lambda}{(\lambda - t)^3}.$$

Evaluating at $t = 0$,

$$\mathbb{E}(T) = M_T'(0) = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda} = 5, \qquad \mathbb{E}(T^2) = M_T''(0) = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2} = 2 \cdot 25 = 50.$$

(As a check, $\mathrm{Var}(T) = 50 - 5^2 = 25 = 1/\lambda^2$.)

K4. Write $M_X(t) = e^{\lambda(e^t - 1)}$. Since $\dfrac{d}{dt}\,\lambda(e^t - 1) = \lambda e^t$, the chain rule gives

$$M_X'(t) = \lambda e^t\, M_X(t).$$

Differentiating again with the product rule,

$$M_X''(t) = \lambda e^t M_X(t) + \lambda e^t M_X'(t) = \lambda e^t M_X(t)\big(1 + \lambda e^t\big).$$

At $t = 0$ we have $M_X(0) = 1$, so

$$\mathbb{E}(X) = M_X'(0) = \lambda, \qquad \mathbb{E}(X^2) = M_X''(0) = \lambda(1 + \lambda) = \lambda + \lambda^2,$$

$$\mathrm{Var}(X) = \mathbb{E}(X^2) - [\mathbb{E}(X)]^2 = \lambda + \lambda^2 - \lambda^2 = \lambda.$$

For the third moment, differentiate $M_X'' = \lambda e^t M_X + \lambda^2 e^{2t} M_X$ once more (using $\frac{d}{dt} e^{2t} = 2e^{2t}$):

$$M_X'''(t) = \lambda e^t M_X + \lambda e^t M_X' + 2\lambda^2 e^{2t} M_X + \lambda^2 e^{2t} M_X'.$$

At $t = 0$, with $M_X(0) = 1$ and $M_X'(0) = \lambda$,

$$\mathbb{E}(X^3) = M_X'''(0) = \lambda + \lambda^2 + 2\lambda^2 + \lambda^3 = \lambda^3 + 3\lambda^2 + \lambda.$$

K5. Apply the affine rule $M_{aX+b}(t) = e^{bt} M_X(at)$ with $a = 0.8$ and $b = -2$:

$$M_Y(t) = e^{-2t}\, M_X(0.8t) = e^{-2t} \exp\Big(\mu(0.8t) + \tfrac{1}{2}\sigma^2 (0.8t)^2\Big).$$

Collecting the linear and quadratic terms in t,

$$M_Y(t) = \exp\Big((0.8\mu - 2)\, t + \tfrac{1}{2}\,(0.64\,\sigma^2)\, t^2\Big).$$

This has exactly the form $\exp\big(\mu_Y t + \frac{1}{2}\sigma_Y^2 t^2\big)$ of a normal MGF, so by the uniqueness of MGFs

$$Y \sim \mathrm{N}\big(0.8\mu - 2,\ 0.64\,\sigma^2\big),$$

with mean $\mathbb{E}(Y) = 0.8\mu - 2$ and variance $\mathrm{Var}(Y) = 0.64\,\sigma^2$. (These also follow from $\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$ and $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$ with $a = 0.8$.)

K6. The Poisson MGF is $M_X(t) = e^{3(e^t-1)}$ and $M_Y(t) = e^{5(e^t-1)}$. Because X and Y are independent, the MGF of the sum is the product:

$$M_S(t) = M_X(t)\, M_Y(t) = e^{3(e^t-1)}\, e^{5(e^t-1)} = e^{8(e^t-1)}.$$

This is the MGF of a Poisson with parameter 8, so by uniqueness $S \sim \text{Poisson}(8)$. Therefore

$$\mathbb{E}(S) = 8, \qquad \text{Var}(S) = 8.$$

(Consistently, $\mathbb{E}(S) = \mathbb{E}(X) + \mathbb{E}(Y) = 3 + 5 = 8$ and, by independence, $\text{Var}(S) = \text{Var}(X) + \text{Var}(Y) = 3 + 5 = 8$.)

K7. Each lifetime has $M_{T_i}(t) = \dfrac{\lambda}{\lambda - t} = \dfrac{2}{2 - t}$ for $t < 2$. The three lifetimes are independent, so the MGF of the sum is the product:

$$M_S(t) = \prod_{i=1}^{3} \frac{\lambda}{\lambda - t} = \left(\frac{\lambda}{\lambda - t}\right)^3 = \left(\frac{2}{2 - t}\right)^3, \qquad t < 2.$$

This is the MGF of a Gamma(α, λ) variable with shape $\alpha = 3$ and rate $\lambda = 2$, so by uniqueness

$$S \sim \text{Gamma}(3, 2).$$

Using the stated gamma moments (or, equivalently, differentiating M_S),

$$\mathbb{E}(S) = \frac{\alpha}{\lambda} = \frac{3}{2}, \qquad \text{Var}(S) = \frac{\alpha}{\lambda^2} = \frac{3}{4}.$$

(Consistently, $\mathbb{E}(S) = 3 \cdot \frac{1}{\lambda} = 3 \cdot \frac{1}{2} = \frac{3}{2}$ and, by independence, $\text{Var}(S) = 3 \cdot \frac{1}{\lambda^2} = 3 \cdot \frac{1}{4} = \frac{3}{4}$.)

Chapter 4

Special Distributions

Master guide: which special distribution does the problem call for?

- **Counting rare events** over a fixed window of time, area, or volume, occurring at a constant average rate and independently? ⇒ **Poisson** with mean λ (§1). It is also the large-n, small-p limit of the binomial.
- A continuous measurement that clusters **symmetrically and bell-shaped** about a centre? ⇒ **Normal** $\mathrm{N}(\mu, \sigma^2)$; convert to the standard scale with $Z = \frac{X-\mu}{\sigma}$ and read Φ (§2).
- Approximating a **binomial** count when n is large, or the **average/sum of many** independent pieces? ⇒ the **normal approximation** (continuity correction) and the **Central Limit Theorem** (§3).
- **Waiting for the first success** in repeated independent yes/no trials? ⇒ **geometric** (§4). **Waiting for the r-th success**? ⇒ **negative binomial** (§5).
- A **continuous waiting time** until the r-th event of a Poisson process, or a **sum of independent exponentials**? ⇒ **gamma** (§6); the exponential is the shape-1 case and the chi-square is another special case.

The two most common mix-ups. Poisson counts *how many* events in a fixed window; the gamma/exponential measures *how long* until events occur — same process, but one is a discrete count and the other a continuous time. And "number of trials to a success" is discrete (geometric / negative binomial), whereas "time to an event" is continuous (exponential / gamma).

4.1 The Poisson Distribution

The Poisson distribution models the number of times a rare event occurs in a fixed window of time, length, area, or volume, when the event happens at a constant average rate and separate occurrences do not influence one another. A single number, the mean rate λ, controls the whole distribution: it is at once the expected count and the variance. The same distribution arises as the limit of a binomial $\mathrm{Bin}(n, p)$ when there are very many trials ($n \to \infty$), each with a tiny success probability ($p \to 0$), but a moderate expected number of successes ($np \to \lambda$) — which is exactly why it is the natural law for counting rare events.

Which rule of this section does the problem need?

- The problem asks for the probability of a **count** of events — "how many" arrivals, flaws, calls, or hits — in a **fixed window**, and you are told (or can assume) the events occur at a **constant average rate**, **independently**? Use **Poisson** with $p_X(k) = e^{-\lambda}\lambda^k/k!$, λ the expected count for that window.
- The window changes size (per hour vs. per day, per metre vs. per kilometre)? **Scale the rate**: λ = (rate per unit) × (number of units). The Poisson parameter is always the mean count for the window you are actually asked about.

- You want the chance of **at least one** event? Take the **complement**: $\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-\lambda}$. This avoids summing an infinite tail.
- You have a binomial $\text{Bin}(n, p)$ with **large** n **and small** p (rule of thumb $n \geq 100$, $p \leq 0.05$, np moderate)? Approximate by **Poisson** with $\lambda = np$.
- *Why not the others?* If you are counting *successes in a fixed number n of trials* with a non-small p, that is genuinely **binomial**, not Poisson. If you are asked *how long until* an event (a continuous time), that is **exponential / gamma** (§6), not Poisson. Poisson is for discrete counts in a window.

4.1.1 The Poisson pmf

Definition 4.1.1 (Poisson distribution)**.** A random variable X has the **Poisson distribution** with parameter $\lambda > 0$, written $X \sim \text{Poisson}(\lambda)$, if its support is $\mathcal{R}(X) = \{0, 1, 2, \dots\}$ and its pmf is

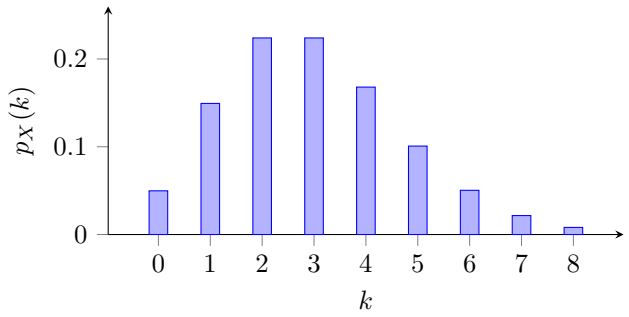
$$p_X(k) = \mathbb{P}(X = k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0, 1, 2, \dots$$

The parameter λ is the mean number of events expected in the window under study.

These are genuine probabilities: each term is positive, and by the Taylor series $e^{\lambda} = \sum_{k=0}^{\infty} \lambda^k/k!$ they sum to one,

$$\sum_{k=0}^{\infty} \frac{e^{-\lambda}\lambda^k}{k!} = e^{-\lambda} \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} = e^{-\lambda} e^{\lambda} = 1.$$

The shape is a right-skewed cluster of bars centred near λ. The bar chart below is the case $\lambda = 3$.



Poisson pmf with $\lambda = 3$. The mass peaks at $k = 2$ and $k = 3$ (both ≈ 0.224) and the right tail decays quickly.

Example 4.1.2 (Typos on a page)**.** A long-form copyeditor finds that a particular author averages $\lambda = 2.4$ typographical errors per page, scattered at random through the text. Let X be the number of typos on a single page. Find $\mathbb{P}(X = 3)$.

Which rule, and why? We are counting how many of a rare event (typos) fall in a fixed window (one page), the events occur at a roughly constant rate and do not cluster, and we are given a mean count. That is the Poisson signature; there is no fixed number of independent "trials" with a fixed p, so this is not binomial.

Solution.

Step 1: Identify the model. $X \sim \text{Poisson}(\lambda)$ with $\lambda = 2.4$ typos per page.

Step 2: Apply the pmf. With $k = 3$,

$$\mathbb{P}(X = 3) = \frac{e^{-2.4}(2.4)^3}{3!} = \frac{e^{-2.4}\,(13.824)}{6} = e^{-2.4}\,(2.304) \approx 0.2090.$$

About a 20.9% chance the page has exactly three typos.

Example 4.1.3 (Phone calls in a window)**.** A small clinic's front desk receives calls at a constant average rate of 3 calls per 10-minute interval. Let X be the number of calls in one such interval. Find $\mathbb{P}(X = 4)$.

Which rule, and why? A count of arrivals in a fixed time window at a constant rate, independent of one another: Poisson with λ equal to the mean count for that window.

Solution.

Step 1: Identify the model. $X \sim \text{Poisson}(\lambda)$ with $\lambda = 3$ calls per 10 minutes.

Step 2: Apply the pmf.

$$\mathbb{P}(X = 4) = \frac{e^{-3}\,3^4}{4!} = \frac{e^{-3}\,(81)}{24} = e^{-3}\,(3.375) \approx 0.1680.$$

Example 4.1.4 (Cumulative probability)**.** For the same author of Example 4.1.2 ($\lambda = 2.4$ typos per page), find the probability that a page has *at most one* typo.

Which rule, and why? Still a Poisson count; "at most one" means $X \le 1$, so add the two smallest-count probabilities $\mathbb{P}(X = 0) + \mathbb{P}(X = 1)$.

Solution.

Step 1: Write the event as a finite sum. $\mathbb{P}(X \le 1) = \mathbb{P}(X = 0) + \mathbb{P}(X = 1)$.

Step 2: Compute each term.

$$\mathbb{P}(X = 0) = e^{-2.4} = 0.0907, \qquad \mathbb{P}(X = 1) = e^{-2.4}(2.4) = 0.2177.$$

Step 3: Add.

$$\mathbb{P}(X \le 1) = 0.0907 + 0.2177 \approx 0.3084.$$

Example 4.1.5 (A heavier tail event)**.** Network packets arrive at a sensor at $\lambda = 3$ per second on average, independently. In one particular second, what is the probability that at least five packets arrive?

Which rule, and why? Count of arrivals in a fixed window at a constant rate; "at least five" is the upper tail $X \ge 5$, best handled as one minus the complementary finite sum $X \le 4$.

Solution.

Step 1: Use the complement. $\mathbb{P}(X \ge 5) = 1 - \mathbb{P}(X \le 4) = 1 - \sum_{k=0}^{4} \dfrac{e^{-3}3^k}{k!}$.

Step 2: Sum the first five terms ($\lambda = 3$).

$$\mathbb{P}(X \le 4) = e^{-3}\Big(1 + 3 + \tfrac{9}{2} + \tfrac{27}{6} + \tfrac{81}{24}\Big) = 0.0498 + 0.1494 + 0.2240 + 0.2240 + 0.1680 = 0.8153.$$

Step 3: Subtract.

$$\mathbb{P}(X \ge 5) = 1 - 0.8153 \approx 0.1847.$$

4.1.2 Poisson as the limit of the binomial

The Poisson law is not an arbitrary formula: it is what the binomial $\text{Bin}(n, p)$ becomes when the number of trials grows without bound while the expected number of successes is held fixed. This is why it governs "many opportunities, each tiny chance" situations.

Theorem 4.1.6 (Binomial-to-Poisson limit)**.** *Fix $\lambda > 0$. For each n let $p_n = \lambda/n$, so that $X_n \sim \text{Bin}(n, p_n)$ has mean $np_n = \lambda$. Then for every fixed $k \in \{0, 1, 2, \dots\}$,*

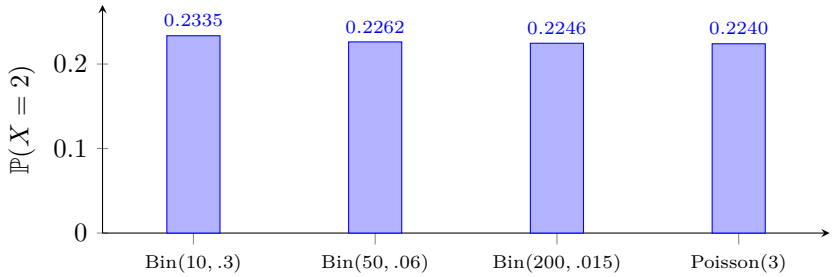
$$\lim_{n\to\infty} \mathbb{P}(X_n = k) = \frac{e^{-\lambda}\lambda^k}{k!}.$$

Proof. Write the binomial pmf with $p_n = \lambda/n$ and separate the factors:

$$\mathbb{P}(X_n = k) = \binom{n}{k}\Big(\frac{\lambda}{n}\Big)^k\Big(1 - \frac{\lambda}{n}\Big)^{n-k} = \frac{\lambda^k}{k!} \cdot \underbrace{\frac{n!}{(n-k)!\,n^k}}_{A_n} \cdot \underbrace{\Big(1 - \frac{\lambda}{n}\Big)^{n}}_{B_n} \cdot \underbrace{\Big(1 - \frac{\lambda}{n}\Big)^{-k}}_{C_n}.$$

As $n \to \infty$ with k fixed: $A_n = \dfrac{n(n-1)\cdots(n-k+1)}{n^k} \to 1$ (a product of k factors each $\to 1$); $B_n = \left(1 - \frac{\lambda}{n}\right)^n \to e^{-\lambda}$ by the standard limit; and $C_n = \left(1 - \frac{\lambda}{n}\right)^{-k} \to 1$. Hence $\mathbb{P}(X_n = k) \to \dfrac{\lambda^k}{k!} e^{-\lambda}$, the Poisson pmf. □

The picture below shows the convergence at $k = 2$: binomials with a fixed mean $np = 3$ but growing n (and shrinking p) creep toward the Poisson value $e^{-3}3^2/2! = 0.2240$.



As n grows with $np = 3$ fixed, $\mathbb{P}(X = 2)$ for the binomial approaches the Poisson value 0.2240.

4.1.3 Model assumptions

Definition 4.1.7 (Poisson model assumptions)**.** A count X over a fixed window is well modelled by Poisson(λ) when the underlying process satisfies, to good approximation:

- **Constant rate.** Events occur at a steady average rate; the expected number in a sub-window is proportional to the size of that sub-window.
- **Independence.** Disjoint sub-windows are independent; one event neither encourages nor blocks another.
- **No simultaneity (rarity).** In a vanishingly small sub-window, the chance of two or more events is negligible compared with the chance of one.

Under these conditions the count in a window of "size" t (time, length, area) is Poisson(νt), where ν is the rate per unit.

Remark. When the rate genuinely varies (rush-hour calls, seasonal demand) or events arrive in bursts (clustered, not independent), the plain Poisson model is suspect. A tell-tale sign in data is that the sample variance is far from the sample mean, since a true Poisson forces them equal (next subsection).

4.1.4 Mean and variance: both equal λ

Theorem 4.1.8 (Mean and variance of a Poisson)**.** *If* $X \sim \text{Poisson}(\lambda)$ *then* $\mathbb{E}(X) = \lambda$ *and* $\text{Var}(X) = \lambda$*; hence* $\text{SD}(X) = \sqrt{\lambda}$*.*

Proof. For the mean, the $k = 0$ term contributes nothing, and reindexing with $j = k - 1$,

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k \frac{e^{-\lambda}\lambda^k}{k!} = \sum_{k=1}^{\infty} \frac{e^{-\lambda}\lambda^k}{(k-1)!} = \lambda \sum_{j=0}^{\infty} \frac{e^{-\lambda}\lambda^j}{j!} = \lambda \cdot 1 = \lambda.$$

For the variance, first compute the factorial moment $\mathbb{E}\big[X(X-1)\big]$ (the $k = 0, 1$ terms vanish, then set $j = k - 2$):

$$\mathbb{E}\big[X(X-1)\big] = \sum_{k=2}^{\infty} k(k-1)\frac{e^{-\lambda}\lambda^k}{k!} = \lambda^2 \sum_{j=0}^{\infty} \frac{e^{-\lambda}\lambda^j}{j!} = \lambda^2.$$

Therefore $\mathbb{E}(X^2) = \mathbb{E}[X(X-1)] + \mathbb{E}(X) = \lambda^2 + \lambda$, and

$$\text{Var}(X) = \mathbb{E}(X^2) - \big(\mathbb{E}X\big)^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda.$$ □

Example 4.1.9 (Reading off mean, spread, and a probability)**.** A warehouse scanner logs misreads at $\lambda = 4$ per shift. Give the expected number of misreads per shift, the standard deviation, and the probability of exactly the expected number.

Which rule, and why? A Poisson count; the parameter is simultaneously the mean and the variance, so the spread comes for free from λ, and the "exactly k" probability is the pmf.

Solution.

Step 1: Mean and standard deviation. $\mathbb{E}(X) = \lambda = 4$ and $\mathrm{SD}(X) = \sqrt{\lambda} = \sqrt{4} = 2$ misreads.

Step 2: Probability of exactly four.

$$\mathbb{P}(X = 4) = \frac{e^{-4}4^4}{4!} = \frac{e^{-4}\,(256)}{24} = e^{-4}\,\tfrac{32}{3} \approx 0.1954.$$

Even at the mean, the single most likely count carries under 20% of the probability — a useful reminder that "expected" is not "almost certain".

Example 4.1.10 (Comparing two processes by spread)**.** Process A logs defects at $\lambda_A = 9$ per batch; process B at $\lambda_B = 1$ per batch. Each is Poisson. Which process has the larger *relative* variability, measured by the coefficient of variation $\mathrm{SD}(X)/\mathbb{E}(X)$?

Which rule, and why? Both counts are Poisson, so $\mathrm{SD} = \sqrt{\lambda}$ and $\mathbb{E} = \lambda$; the coefficient of variation is $\sqrt{\lambda}/\lambda = 1/\sqrt{\lambda}$, which we just evaluate.

Solution.

Step 1: Coefficient of variation for each.

$$\frac{\mathrm{SD}}{\mathbb{E}} = \frac{\sqrt{\lambda}}{\lambda} = \frac{1}{\sqrt{\lambda}}.$$

Step 2: Evaluate. For A, $1/\sqrt{9} = 1/3 \approx 0.333$; for B, $1/\sqrt{1} = 1$.

Process B is relatively more variable (CV = 1 vs. 0.333).

A rarer process (smaller λ) is relatively more erratic batch to batch.

4.1.5 Scaling the rate to the window

Because disjoint sub-windows are independent and the rate is constant, the Poisson parameter is *proportional to the size of the window.* If events occur at rate ν per unit, then over t units the count is Poisson(νt). Always rescale λ to match the window the question asks about.

Example 4.1.11 (Calls over a longer interval)**.** Recall the clinic of Example 4.1.3: calls arrive at rate 3 per 10 minutes. Over a full 30-minute stretch, what is the probability of *two or fewer* calls?

Which rule, and why? Same Poisson process, but a window three times as long, so scale the rate: $\lambda = 3 \times 3 = 9$. Then "two or fewer" is the finite sum $\mathbb{P}(X \le 2)$.

Solution.

Step 1: Scale the rate. Rate = 3 per 10 min $\Rightarrow$ over 30 min, $\lambda = 3 \cdot \frac{30}{10} = 9$, so $X \sim \mathrm{Poisson}(9)$.

Step 2: Sum the three smallest-count terms.

$$\mathbb{P}(X \le 2) = e^{-9}\Big(1 + 9 + \tfrac{81}{2}\Big) = e^{-9}\,(50.5) \approx 0.0062.$$

Over a longer window the mean is large, so very small counts become unlikely.

Example 4.1.12 (Shrinking the window)**.** Radioactive decays are detected at a constant average rate of 12 per hour. What is the probability that a given 5-minute window registers no decays at all?

Which rule, and why? Poisson process; the window is smaller than the stated rate's unit, so scale down: 5 minutes is 1/12 of an hour. "No decays" is the single term $\mathbb{P}(X = 0)$.

Solution.

Step 1: Scale the rate. $\lambda = 12 \cdot \frac{5}{60} = 1$ decay expected in 5 minutes, so $X \sim \text{Poisson}(1)$.

Step 2: Probability of zero.

$$\mathbb{P}(X = 0) = e^{-1} \approx 0.3679.$$

Example 4.1.13 (Defects along a length)**.** A textile loom produces flaws at a rate of 0.05 flaws per metre of cloth. A bolt is 8 metres long. Find the probability that the bolt contains at least two flaws.

Which rule, and why? Counting rare events along a continuous length at a constant rate per metre: Poisson, with the rate scaled to the 8-metre window. "At least two" is most cheaply done by complement.

Solution.

Step 1: Scale the rate. $\lambda = 0.05 \cdot 8 = 0.4$ flaws expected per bolt, so $X \sim \text{Poisson}(0.4)$.

Step 2: Complement of "zero or one".

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) = 1 - e^{-0.4} - e^{-0.4}(0.4) = 1 - e^{-0.4}(1.4).$$

Step 3: Evaluate. $e^{-0.4} = 0.6703$, so

$$\mathbb{P}(X \geq 2) = 1 - 0.6703(1.4) = 1 - 0.9384 \approx 0.0616.$$

Example 4.1.14 (Accidents over two weeks)**.** Minor accidents at a depot occur as a Poisson process at $\lambda = 1.5$ per week. Find the probability of *four or more* accidents over a two-week period.

Which rule, and why? Poisson over a fixed window; scale to two weeks, then "four or more" is the upper tail, best as one minus $\mathbb{P}(X \leq 3)$.

Solution.

Step 1: Scale the rate. Over two weeks $\lambda = 1.5 \cdot 2 = 3$, so $X \sim \text{Poisson}(3)$.

Step 2: Complement of $X \leq 3$ ($\lambda = 3$).

$$\mathbb{P}(X \leq 3) = e^{-3}\left(1 + 3 + \tfrac{9}{2} + \tfrac{27}{6}\right) = 0.0498 + 0.1494 + 0.2240 + 0.2240 = 0.6472.$$

Step 3: Subtract.

$$\mathbb{P}(X \geq 4) = 1 - 0.6472 \approx 0.3528.$$

4.1.6 "At least one" via the complement

The event "at least one" is the single most common Poisson question, and it has a one-line answer. Since $\{X \geq 1\}$ is the complement of $\{X = 0\}$,

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-\lambda}.$$

There is never any need to sum the infinite tail $\mathbb{P}(X = 1) + \mathbb{P}(X = 2) + \cdots$.

Example 4.1.15 (At least one outage)**.** A server cluster suffers unplanned outages as a Poisson process averaging 0.7 outages per day. What is the probability of at least one outage on a given day?

Which rule, and why? Poisson count over a one-day window; "at least one" is the complement of "none".

Solution.

Step 1: Apply the complement formula.

$$\mathbb{P}(X \geq 1) = 1 - e^{-\lambda} = 1 - e^{-0.7}.$$

Step 2: Evaluate. $e^{-0.7} = 0.4966$, so $\mathbb{P}(X \geq 1) = 1 - 0.4966 \approx 0.5034$.

Example 4.1.16 (Designing a window for a target probability)**.** A vineyard's frost sensors trip as a Poisson process at $\lambda = 0.2$ trips per hour. The grower wants the probability of *at least one* trip over a watch of h hours to be at least 0.95. What is the smallest whole number of hours h that achieves this?

Which rule, and why? Poisson over a window of length h; scale the rate to $\lambda_h = 0.2h$ and solve the "at least one" complement inequality for h.

Solution.

Step 1: Set up the inequality. Over h hours, $\lambda_h = 0.2h$ and

$$\mathbb{P}(X \geq 1) = 1 - e^{-0.2h} \geq 0.95 \iff e^{-0.2h} \leq 0.05.$$

Step 2: Solve for h. Take logs: $-0.2h \leq \ln(0.05) = -2.9957$, so

$$h \geq \frac{2.9957}{0.2} = 14.98.$$

Step 3: Round up.

$$h = 15 \text{ hours}.$$

(At $h = 15$, $1 - e^{-3} = 0.9502 \geq 0.95$; at $h = 14$, $1 - e^{-2.8} = 0.9392 < 0.95$.)

Example 4.1.17 (At least one over a scaled window)**.** Email arrives in an inbox at $\lambda = 0.5$ messages per minute. Over a 6-minute coffee break, what is the probability of receiving at least one message? And exactly five?

Which rule, and why? Poisson; scale the rate to the 6-minute window, then use the complement for "at least one" and the pmf for "exactly five".

Solution.

Step 1: Scale the rate. $\lambda = 0.5 \cdot 6 = 3$, so $X \sim \text{Poisson}(3)$.

Step 2: At least one. $\mathbb{P}(X \geq 1) = 1 - e^{-3} = 1 - 0.0498 \approx 0.9502$.

Step 3: Exactly five.

$$\mathbb{P}(X = 5) = \frac{e^{-3}3^5}{5!} = \frac{e^{-3}\,(243)}{120} = e^{-3}\,(2.025) \approx 0.1008.$$

4.1.7 Using Poisson to approximate a binomial

When n is large and p small, computing binomial coefficients like $\binom{n}{k}$ is awkward, but the binomial-to-Poisson limit says we may replace $\text{Bin}(n, p)$ by $\text{Poisson}(\lambda)$ with $\lambda = np$. A practical rule of thumb is $n \geq 100$ and $p \leq 0.05$ (so that np is moderate); the smaller p is, the better the approximation.

Example 4.1.18 (Defective chips on a wafer)**.** A fabrication line produces chips with an independent defect probability $p = 0.004$ each. A wafer holds $n = 500$ chips. Approximate the probability that a wafer has exactly two defective chips.

Which rule, and why? This is exactly binomial ($n = 500$ fixed trials, fixed $p = 0.004$), but n is large and p is tiny with $np = 2$ moderate, so the Poisson approximation with $\lambda = np$ applies and avoids the unwieldy $\binom{500}{2}$.

Solution.

Step 1: Set the Poisson parameter. $\lambda = np = 500 \cdot 0.004 = 2$.

Step 2: Apply the Poisson pmf.

$$\mathbb{P}(X = 2) \approx \frac{e^{-2}2^2}{2!} = \frac{e^{-2}\,(4)}{2} = 2e^{-2} \approx 0.2707.$$

Step 3: Sanity check against the exact binomial. The exact value is $\binom{500}{2}(0.004)^2(0.996)^{498} = 0.2712$, so the approximation is off by only 0.0005.

Example 4.1.19 (At least one event, rare and many)**.** A mass mailing sends $n = 2000$ letters; each independently has probability $p = 0.001$ of triggering a reply. Approximate the probability that at least one reply comes back.

Which rule, and why? Large n, small p, moderate $np = 2$: Poisson-approximate the binomial, then use the "at least one" complement.

Solution.

Step 1: Set the parameter. $\lambda = np = 2000 \cdot 0.001 = 2$.

Step 2: Complement.

$$\mathbb{P}(X \geq 1) \approx 1 - e^{-2} = 1 - 0.1353 \approx 0.8647.$$

(The exact binomial gives $1 - (0.999)^{2000} = 0.8648$.)

Example 4.1.20 (Typo-free run of pages)**.** Returning to the author with 2.4 typos per page, suppose a typo is really one of $n = 1200$ character positions on a page being wrong, each independently with probability $p = 0.002$. Approximate the probability that a page is typo-free, and compare with the direct Poisson model.

Which rule, and why? The per-character error count is binomial with large n and small p; $np = 2.4$ recovers the page rate, so Poisson with $\lambda = 2.4$ approximates it — the same number the direct Poisson model uses.

Solution.

Step 1: Set the parameter. $\lambda = np = 1200 \cdot 0.002 = 2.4$.

Step 2: Probability of zero.

$$\mathbb{P}(X = 0) \approx e^{-2.4} \approx 0.0907.$$

This matches the direct Poisson model of Example 4.1.2 exactly, illustrating that the page-level Poisson rate $\lambda = 2.4$ is the limit of the character-level binomial.

Example 4.1.21 (Rare disease in a cohort)**.** A condition occurs independently in the population with probability $p = 0.0005$. In a cohort of $n = 8000$ people, approximate the probability that *at most one* person is affected.

Which rule, and why? Large n, very small p, moderate $np = 4$: Poisson approximation with $\lambda = np$; "at most one" is the finite sum $\mathbb{P}(X = 0) + \mathbb{P}(X = 1)$.

Solution.

Step 1: Set the parameter. $\lambda = np = 8000 \cdot 0.0005 = 4$.

Step 2: Sum the two terms.

$$\mathbb{P}(X \leq 1) \approx e^{-4} + e^{-4}(4) = e^{-4}(5) \approx 0.0916.$$

With an expected four cases, having zero or one is quite unlikely (about 9%).

4.2 The Normal Distribution

The normal distribution is the bell-shaped curve that describes a continuous measurement clustering symmetrically about a central value, with most of the mass near the centre and thinning tails on either side. Heights, measurement errors, standardised test scores, and — by the Central Limit Theorem of the next section — sums and averages of many small independent contributions are all modelled well by it. Every normal variable is a shifted, rescaled copy of a single **standard normal** $\mathrm{N}(0, 1)$, so all probabilities reduce to reading one function, the standard-normal cdf Φ.

Which rule of this section does the problem need?

- A continuous quantity, stated as "normally distributed" (or bell-shaped, symmetric) with a mean μ and a standard deviation σ? $\Rightarrow$ model it as $X \sim \mathrm{N}(\mu, \sigma^2)$ and **standardise**: $Z = \frac{X-\mu}{\sigma}$.

- Asked for $\mathbb{P}(X \le b)$, $\mathbb{P}(X > a)$, or $\mathbb{P}(a < X < b)$ (a probability, given the cutoffs)? $\Rightarrow$ convert each cutoff to a z-score and read Φ: $\mathbb{P}(a < X < b) = \Phi\big(\frac{b-\mu}{\sigma}\big) - \Phi\big(\frac{a-\mu}{\sigma}\big)$.
- Given a probability and asked for a *cutoff* (a percentile, a threshold, "the top 5% start where?")? $\Rightarrow$ this is the **inverse problem**: find z with $\Phi(z) = q$, then untransform $x = \mu + z\sigma$.
- A quick "roughly what fraction lies within 1/2/3 SDs?" $\Rightarrow$ the 68–95–99.7 **rule**; no table needed.
- A negative z, or a right tail? $\Rightarrow$ use **symmetry** $\Phi(-z) = 1 - \Phi(z)$ and $\mathbb{P}(Z > z) = 1 - \Phi(z)$ rather than hunting for a separate value.

Why competing tools do not apply. If the quantity is a *count* of events (discrete, $0, 1, 2, \dots$) the Poisson or binomial governs it, not the normal — the normal is for a *continuous* measurement. The normal does *re-enter* as an approximation to a binomial count or to a sample mean, but that is the Central Limit Theorem of §3, applied on purpose; a genuinely continuous bell-shaped measurement is handled here directly.

4.2.1 The normal density and its bell shape

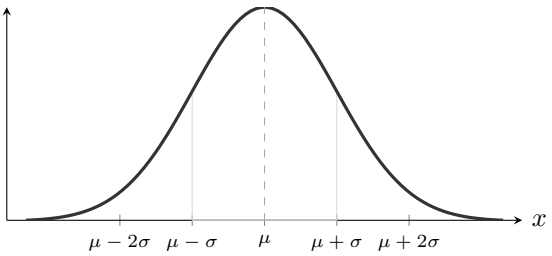
Definition 4.2.1 (Normal distribution)**.** A random variable X is **normal** with mean μ and variance σ^2, written $X \sim \mathrm{N}(\mu, \sigma^2)$ (with $\sigma > 0$), if it has density

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\!\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \qquad -\infty < x < \infty.$$

The graph is the symmetric bell: it peaks at $x = \mu$, is symmetric about $x = \mu$, and has its two points of inflection at $x = \mu \pm \sigma$. One checks $\mathbb{E}(X) = \mu$, $\mathrm{Var}(X) = \sigma^2$, and $\mathrm{SD}(X) = \sigma$. The special case $\mu = 0,\ \sigma = 1$ is the **standard normal** $Z \sim \mathrm{N}(0, 1)$ with density $\varphi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and cdf

$$\Phi(z) = \mathbb{P}(Z \le z) = \int_{-\infty}^{z} \frac{1}{\sqrt{2\pi}}\, e^{-t^2/2}\, \mathrm{d}t.$$

Remark. The density has no elementary antiderivative, so Φ is tabulated (or computed numerically) rather than written in closed form. Two facts make a single table of Φ enough for every normal problem: every normal standardises to Z (next subsection), and Φ is symmetric (below). All numerical answers in this section quote the standard values $\Phi(1) = 0.8413$, $\Phi(1.28) = 0.8997$, $\Phi(1.5) = 0.9332$, $\Phi(1.645) = 0.9500$, $\Phi(1.96) = 0.9750$, $\Phi(2) = 0.9772$, $\Phi(2.33) = 0.9901$, $\Phi(2.576) = 0.9950$, and the symmetry relation $\Phi(-z) = 1 - \Phi(z)$.



The bell curve, peaked and symmetric about μ; the shaded band is the central $\mu \pm \sigma$, which holds about 68% of the area.

Example 4.2.2 (Reading off centre and spread)**.** A lab calibrates a sensor whose reading X (in millivolts) is normal with $\mathbb{E}(X) = 50$ and $\mathrm{Var}(X) = 16$. Where is the curve centred, how wide is one standard deviation, and between which two readings do the inflection points sit?

Which rule, and why? The problem only asks us to interpret the parameters of a stated normal, so we read μ from the mean and σ from the square root of the variance — no probability is requested, so no Φ is needed yet.

Solution.

Step 1: Identify μ and σ. The mean is $\mu = 50$, and $\sigma = \sqrt{\mathrm{Var}(X)} = \sqrt{16} = 4$ mV.

Step 2: Locate the features. The bell peaks and is symmetric about $x = \mu = 50$. The inflection points are at $\mu \pm \sigma$, i.e.

$$\mu - \sigma = 46 \text{ mV} \quad \text{and} \quad \mu + \sigma = 54 \text{ mV}.$$

Example 4.2.3 (Same centre, different spread)**.** Two normal models for fill weight (grams) share the mean $\mu = 200$ but have $\sigma_A = 2$ and $\sigma_B = 5$. Which density is taller at its peak, and why?

Which rule, and why? Both are normal with the same centre; the only difference is σ. The peak height of a normal density is its value at $x = \mu$, namely $f_X(\mu) = \frac{1}{\sigma\sqrt{2\pi}}$, which depends only on σ.

Solution.

Step 1: Peak heights. $f_A(\mu) = \frac{1}{2\sqrt{2\pi}} \approx 0.1995$ and $f_B(\mu) = \frac{1}{5\sqrt{2\pi}} \approx 0.0798$.

Step 2: Interpret. Density A is taller (smaller σ concentrates the same total area 1 into a narrower, taller bell), while B is shorter and more spread out.

$$f_A(\mu) \approx 0.1995 > 0.0798 \approx f_B(\mu).$$

4.2.2 Standardisation and the standard normal

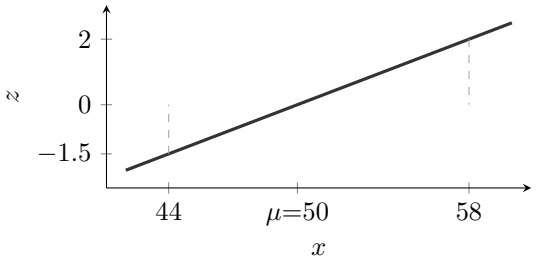
Theorem 4.2.4 (Standardisation)**.** *If $X \sim \mathrm{N}(\mu, \sigma^2)$, then*

$$Z = \frac{X - \mu}{\sigma} \sim \mathrm{N}(0, 1),$$

and consequently for any $a < b$,

$$\mathbb{P}(a < X < b) = \mathbb{P}\left(\frac{a-\mu}{\sigma} < Z < \frac{b-\mu}{\sigma}\right) = \Phi\left(\frac{b-\mu}{\sigma}\right) - \Phi\left(\frac{a-\mu}{\sigma}\right).$$

Proof. A linear transform of a normal is normal, so $Z = (X-\mu)/\sigma$ is normal; it has mean $\mathbb{E}(Z) = \frac{\mathbb{E}(X)-\mu}{\sigma} = 0$ and variance $\mathrm{Var}(Z) = \frac{\mathrm{Var}(X)}{\sigma^2} = 1$, hence $Z \sim \mathrm{N}(0,1)$. For the probability, $a < X < b$ holds exactly when $\frac{a-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}$, and $\mathbb{P}(Z \le z) = \Phi(z)$ by definition; subtracting the two cdf values gives the displayed formula. Because X is continuous, strict and non-strict inequalities give the same probability. □



Standardisation $z = \frac{x-\mu}{\sigma}$ for $\mu = 50,\ \sigma = 4$: a straight-line relabelling of the x-axis into z-scores. Here $x = 44 \mapsto z = -1.5$ and $x = 58 \mapsto z = 2$.

Example 4.2.5 (A single z-score)**.** For the calibrated sensor $X \sim \mathrm{N}(50, 16)$, how many standard deviations above the mean is a reading of 58 mV?

Which rule, and why? "How many standard deviations from the mean" is precisely the z-score, so we apply $z = \frac{x-\mu}{\sigma}$ directly — no area is asked for, only the standardised position.

Solution.

Step 1: Apply the formula. With $\mu = 50,\ \sigma = 4$,

$$z = \frac{58 - 50}{4} = 2.$$

Step 2: Interpret. The reading 58 mV sits exactly 2 standard deviations above the mean.

Example 4.2.6 (Below-the-mean reading)**.** For the same sensor, standardise the reading 44 mV.

Which rule, and why? Same standardisation; the only new feature is that the reading is below the mean, which must produce a *negative* z-score.

Solution.

Step 1: Apply the formula.

$$z = \frac{44 - 50}{4} = \frac{-6}{4} = -1.5.$$

Step 2: Interpret. The reading is 1.5 standard deviations below the mean.

Example 4.2.7 (Comparing two scales)**.** Student Mara scores 82 on an exam with $\mu = 70,\ \sigma = 8$; student Devin scores 55 on a different exam with $\mu = 48,\ \sigma = 5$. Relative to their own exams, who did better?

Which rule, and why? The raw scores are on incomparable scales, so we standardise each to a z-score; the larger z is the better relative performance. Standardisation is exactly the tool that puts both on the common $\mathrm{N}(0, 1)$ scale.

Solution.

Step 1: z-score for Mara. $z_M = \frac{82-70}{8} = \frac{12}{8} = 1.5$.

Step 2: z-score for Devin. $z_D = \frac{55-48}{5} = \frac{7}{5} = 1.4$.

Step 3: Compare. Since $z_M = 1.5 > 1.4 = z_D$, Mara stands slightly higher within her exam.

$$z_M = 1.5 > z_D = 1.4.$$

Example 4.2.8 (Recovering a raw value)**.** On a normal scale with $\mu = 120,\ \sigma = 15$, what raw value x corresponds to $z = -0.8$?

Which rule, and why? We are given the standardised position and asked for the raw value, so we invert the standardisation: $x = \mu + z\sigma$. This is the algebraic step every percentile problem ends with.

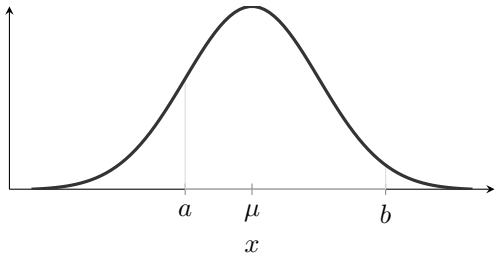
Solution.

Step 1: Untransform. $x = \mu + z\sigma = 120 + (-0.8)(15) = 120 - 12 = 108$.

$$x = 108.$$

4.2.3 Computing $\mathbb{P}(a < X < b)$, tails, and "between"

Once standardised, every probability is a difference of Φ values. The three shapes are: *left tail* $\mathbb{P}(X \le b) = \Phi\big(\frac{b-\mu}{\sigma}\big)$; *right tail* $\mathbb{P}(X > a) = 1 - \Phi\big(\frac{a-\mu}{\sigma}\big)$; and *between* $\mathbb{P}(a < X < b) = \Phi\big(\frac{b-\mu}{\sigma}\big) - \Phi\big(\frac{a-\mu}{\sigma}\big)$.



$\mathbb{P}(a < X < b)$ is the shaded area, computed as $\Phi\big(\frac{b-\mu}{\sigma}\big) - \Phi\big(\frac{a-\mu}{\sigma}\big)$.

Example 4.2.9 (A left tail)**.** Daily water use at a small plant is $X \sim \mathrm{N}(50, 16)$ (thousands of litres). What is the probability a day uses at most 58 thousand litres?

Which rule, and why? A continuous measurement, stated normal, with a probability asked below a single cutoff: this is a left-tail probability $\mathbb{P}(X \le b)$, so we standardise the cutoff and read Φ once.

Solution.

Step 1: Standardise the cutoff. $z = \frac{58-50}{4} = 2$.

Step 2: Read Φ. Using $\Phi(2) = 0.9772$,

$$\mathbb{P}(X \le 58) = \Phi(2) = 0.9772.$$

Example 4.2.10 (A right tail via symmetry)**.** For the same $X \sim \mathrm{N}(50, 16)$, find the probability a day uses more than 56 thousand litres.

Which rule, and why? "More than" a single cutoff is a right tail, $\mathbb{P}(X > a) = 1 - \Phi(z)$; this is where we must subtract from 1 rather than read Φ directly.

Solution.

Step 1: Standardise. $z = \frac{56-50}{4} = 1.5$.

Step 2: Complement. Using $\Phi(1.5) = 0.9332$,

$$\mathbb{P}(X > 56) = 1 - \Phi(1.5) = 1 - 0.9332 = 0.0668.$$

Example 4.2.11 (A "between" probability)**.** For $X \sim \mathrm{N}(50, 16)$, find the probability a day uses between 46 and 58 thousand litres.

Which rule, and why? Two finite cutoffs with the area *between* them: standardise both ends and subtract the two Φ values. The lower z is negative here, so symmetry $\Phi(-z) = 1 - \Phi(z)$ comes in.

Solution.

Step 1: Standardise both ends.

$$z_{\mathrm{lo}} = \frac{46-50}{4} = -1, \qquad z_{\mathrm{hi}} = \frac{58-50}{4} = 2.$$

Step 2: Convert the negative end. By symmetry, $\Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$.

Step 3: Subtract. Using $\Phi(2) = 0.9772$,

$$\mathbb{P}(46 < X < 58) = \Phi(2) - \Phi(-1) = 0.9772 - 0.1587 = 0.8185.$$

Example 4.2.12 (Both cutoffs below the mean)**.** For $X \sim \mathrm{N}(50, 16)$, find $\mathbb{P}(42 < X < 46)$.

Which rule, and why? A "between" probability whose *both* endpoints fall below the mean, so both z-scores are negative; we still subtract two Φ values, converting each with $\Phi(-z) = 1 - \Phi(z)$.

Solution.

Step 1: Standardise.

$$z_{\mathrm{lo}} = \frac{42-50}{4} = -2, \qquad z_{\mathrm{hi}} = \frac{46-50}{4} = -1.$$

Step 2: Convert both via symmetry. $\Phi(-2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$ and $\Phi(-1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$.

Step 3: Subtract.

$$\mathbb{P}(42 < X < 46) = \Phi(-1) - \Phi(-2) = 0.1587 - 0.0228 = 0.1359.$$

Example 4.2.13 (A symmetric central band)**.** For a process $Y \sim \mathrm{N}(0, 1)$ already on the standard scale, find $\mathbb{P}(-1.96 < Y < 1.96)$.

Which rule, and why? Already standard normal, with a symmetric band about 0: the symmetric structure lets us write the answer as $2\Phi(z) - 1$, the workhorse formula behind confidence intervals.

Solution.

Step 1: Use symmetry. For a band $(-z, z)$,

$$\mathbb{P}(-z < Y < z) = \Phi(z) - \Phi(-z) = \Phi(z) - \big(1 - \Phi(z)\big) = 2\Phi(z) - 1.$$

Step 2: Evaluate. Using $\Phi(1.96) = 0.9750$,

$$\mathbb{P}(-1.96 < Y < 1.96) = 2(0.9750) - 1 = 0.9500.$$

4.2.4 Inverse problems: percentiles and thresholds

The previous subsection went from a cutoff to a probability. The **inverse problem** runs the other way: *given* a probability q, find the cutoff x with $\mathbb{P}(X \le x) = q$ (the $100q$-th percentile). The recipe is to find the standard-normal quantile z_q with $\Phi(z_q) = q$, then untransform: $x = \mu + z_q\,\sigma$.

Example 4.2.14 (An upper percentile)**.** Bottle fill weight is $X \sim \mathrm{N}(500, 9)$ grams. Below which weight do 97.5% of bottles fall (the 97.5th percentile)?

Which rule, and why? We are given a probability (0.975) and asked for the *cutoff*, so this is an inverse problem: solve $\Phi(z) = 0.975$ for z, then convert to the x-scale.

Solution.

Step 1: Find the quantile. We need z with $\Phi(z) = 0.975$; the standard value is $z = 1.96$ (since $\Phi(1.96) = 0.9750$).

Step 2: Untransform. With $\mu = 500,\ \sigma = \sqrt{9} = 3$,

$$x = \mu + z\sigma = 500 + 1.96(3) = 500 + 5.88 = 505.88 \text{ g}.$$

Example 4.2.15 (A lower percentile, negative quantile)**.** For the same $X \sim \mathrm{N}(500, 9)$, find the 5th percentile.

Which rule, and why? Given probability 0.05, find the cutoff. Because $0.05 < 0.5$ the cutoff is *below* the mean, so the quantile is negative and we lean on $\Phi(-z) = 1 - \Phi(z)$.

Solution.

Step 1: Find the quantile. We need $\Phi(z) = 0.05$. Since $\Phi(1.645) = 0.9500$, symmetry gives $\Phi(-1.645) = 1 - 0.9500 = 0.0500$, so $z = -1.645$.

Step 2: Untransform.

$$x = 500 + (-1.645)(3) = 500 - 4.935 = 495.065 \text{ g}.$$

Example 4.2.16 (Setting a warranty threshold)**.** The lifetime of a component is $L \sim \mathrm{N}(1{,}000, 2500)$ hours. The manufacturer wants a warranty length w such that only 1% of components fail before w. Find w.

Which rule, and why? We want the cutoff with a tail probability of 0.01 *below* it, i.e. the 1st percentile — an inverse problem with $q = 0.01$. The quantile is negative because $0.01 < 0.5$.

Solution.

Step 1: Find the quantile. We need $\Phi(z) = 0.01$. Since $\Phi(2.33) = 0.9901 \approx 0.99$, symmetry gives $\Phi(-2.33) = 1 - 0.9901 = 0.0099 \approx 0.01$, so $z = -2.33$.

Step 2: Untransform. With $\mu = 1{,}000,\ \sigma = \sqrt{2500} = 50$,

$$w = 1{,}000 + (-2.33)(50) = 1{,}000 - 116.5 = 883.5 \text{ hours}.$$

So a warranty of about 883.5 hours leaves roughly 1% early failures.

Example 4.2.17 (A symmetric tolerance interval)**.** Machined rods have diameter $D \sim \mathrm{N}(12, 0.04)$ mm. Find a symmetric interval $(12 - c,\ 12 + c)$ about the mean that contains 95% of rods.

Which rule, and why? A central probability (0.95) is given and a symmetric half-width c is wanted: use $\mathbb{P}(-z < Z < z) = 2\Phi(z) - 1 = 0.95$ to get the quantile, then scale by σ to convert z into a distance c in millimetres.

Solution.

Step 1: Quantile from the central probability. Solve $2\Phi(z) - 1 = 0.95$, i.e. $\Phi(z) = 0.975$, giving $z = 1.96$ (from $\Phi(1.96) = 0.9750$).

Step 2: Convert to a distance. With $\sigma = \sqrt{0.04} = 0.2$ mm,

$$c = z\,\sigma = 1.96(0.2) = 0.392 \text{ mm},$$

so about 95% of rods have diameter in (11.608, 12.392) mm.

Example 4.2.18 (The top group)**.** Aptitude scores are $S \sim \mathrm{N}(100, 225)$. A program admits the top 10% of scorers. What is the minimum qualifying score?

Which rule, and why? "Top 10%" means $\mathbb{P}(S > x) = 0.10$, equivalently $\mathbb{P}(S \le x) = 0.90$ — the 90th percentile. Given the probability, solve for the cutoff: an inverse problem with $q = 0.90$.

Solution.

Step 1: Find the quantile. We need $\Phi(z) = 0.90$; the standard value is $z = 1.28$ (since $\Phi(1.28) = 0.8997 \approx 0.90$).

Step 2: Untransform. With $\mu = 100,\ \sigma = \sqrt{225} = 15$,

$$x = 100 + 1.28(15) = 100 + 19.2 = 119.2.$$

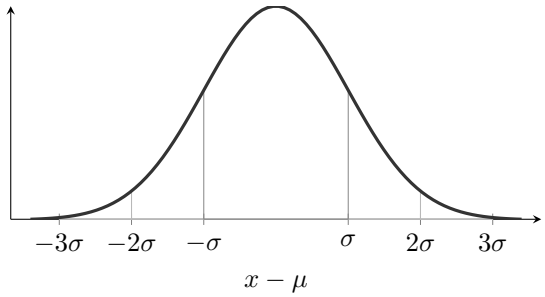
So a score of about 119.2 (round up to 120 in practice) is the cutoff.

4.2.5 The empirical 68–95–99.7 rule

Proposition 4.2.19 (Empirical rule)**.** *For $X \sim \mathrm{N}(\mu, \sigma^2)$,*

$$\mathbb{P}(\mu - \sigma < X < \mu + \sigma) \approx 0.68, \quad \mathbb{P}(\mu - 2\sigma < X < \mu + 2\sigma) \approx 0.95, \quad \mathbb{P}(\mu - 3\sigma < X < \mu + 3\sigma) \approx 0.997.$$

Proof. Standardise: each band becomes $(-k, k)$ for $k = 1, 2, 3$, and $\mathbb{P}(-k < Z < k) = 2\Phi(k) - 1$. Then $2\Phi(1) - 1 = 2(0.8413) - 1 = 0.6826$, $2\Phi(2) - 1 = 2(0.9772) - 1 = 0.9544$, and using $\Phi(3) = 0.9987$, $2\Phi(3) - 1 = 2(0.9987) - 1 = 0.9974$. Rounding gives 0.68, 0.95, 0.997. □



Nested central bands: about 68% within $\pm\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

Example 4.2.20 (A quick band estimate)**.** Adult resting heart rate in a cohort is $X \sim \mathrm{N}(72, 36)$ bpm. Without a table, estimate the fraction of people with rates between 60 and 84 bpm.

Which rule, and why? The bounds are exactly $\mu \pm 2\sigma$, a "1/2/3 SD" band, so the empirical rule answers it instantly — no Φ lookup needed.

Solution.

Step 1: Express the bounds in SDs. With $\mu = 72,\ \sigma = \sqrt{36} = 6$,

$$60 = \mu - 2\sigma, \qquad 84 = \mu + 2\sigma.$$

Step 2: Apply the rule. A $\pm 2\sigma$ band holds about 95%:

$$\mathbb{P}(60 < X < 84) \approx 0.95.$$

Example 4.2.21 (An outer-tail estimate)**.** For the same heart-rate model $X \sim \mathrm{N}(72, 36)$, estimate the fraction of people with rate above 90 bpm.

Which rule, and why? The cutoff $90 = \mu + 3\sigma$ sits at three SDs, so the empirical rule plus symmetry of the two tails gives a fast estimate: 99.7% lies inside $\pm 3\sigma$, leaving 0.3% split evenly between the two outer tails.

Solution.

Step 1: Locate the cutoff. $90 = 72 + 3(6) = \mu + 3\sigma$.

Step 2: Split the leftover. Outside $\pm 3\sigma$ is about $1 - 0.997 = 0.003$, and by symmetry the upper tail is half of that:

$$\mathbb{P}(X > 90) \approx \frac{0.003}{2} = 0.0015.$$

(The table value, using $\Phi(3) = 0.9987$, is $1 - 0.9987 = 0.0013$ — the rule's 0.0015 is the quick estimate.)

Example 4.2.22 (Working backward from a fraction)**.** A teacher reports that "about 68%" of project scores fell between 74 and 86. Assuming a normal model centred in that interval, estimate μ and σ.

Which rule, and why? "About 68% within an interval" is the signature of the $\pm\sigma$ band, so we read the interval as $\mu \pm \sigma$: the midpoint is μ and the half-width is σ.

Solution.

Step 1: Centre. The midpoint is $\mu = \frac{74+86}{2} = 80$.

Step 2: Spread. The half-width is $\sigma = \frac{86-74}{2} = 6$.

$$\mu = 80, \qquad \sigma = 6.$$

4.2.6 Symmetry of Φ

Proposition 4.2.23 (Reflection identity)**.** *For every real z,*

$$\Phi(-z) = 1 - \Phi(z), \qquad \textit{and equivalently} \qquad \mathbb{P}(Z > z) = 1 - \Phi(z).$$

Proof. The standard-normal density $\varphi(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$ is even, $\varphi(-t) = \varphi(t)$. Hence the area to the left of $-z$ equals the area to the right of z: $\Phi(-z) = \mathbb{P}(Z \le -z) = \mathbb{P}(Z \ge z) = 1 - \mathbb{P}(Z < z) = 1 - \Phi(z)$, using continuity for the last step. The second identity is the definition of the right tail. □

Example 4.2.24 (A right tail with a negative cutoff)**.** For $Z \sim \mathrm{N}(0, 1)$, find $\mathbb{P}(Z > -1)$.

Which rule, and why? A right tail past a negative cutoff. Rather than seek a separate value, reflect: $\mathbb{P}(Z > -1) = \Phi(1)$, turning a negative-cutoff question into a single tabulated value.

Solution.

Step 1: Reflect. $\mathbb{P}(Z > -1) = 1 - \Phi(-1) = 1 - \big(1 - \Phi(1)\big) = \Phi(1)$.

Step 2: Evaluate. Using $\Phi(1) = 0.8413$,

$$\mathbb{P}(Z > -1) = 0.8413.$$

Example 4.2.25 (Two-sided tail probability)**.** For $Z \sim \mathrm{N}(0, 1)$, find $\mathbb{P}(|Z| > 1.96)$, the probability of landing in either outer tail beyond 1.96.

Which rule, and why? $|Z| > 1.96$ means $Z < -1.96$ or $Z > 1.96$, two symmetric tails; symmetry lets us compute one tail and double it. This is the complement of the central 95% band.

Solution.

Step 1: One tail. $\mathbb{P}(Z > 1.96) = 1 - \Phi(1.96) = 1 - 0.9750 = 0.0250$.

Step 2: Double by symmetry.

$$\mathbb{P}(|Z| > 1.96) = 2(0.0250) = 0.0500.$$

Example 4.2.26 (Mixing the scale back in)**.** For $X \sim \mathrm{N}(20, 25)$, find $\mathbb{P}(X < 14)$ using symmetry.

Which rule, and why? Standardising gives a negative cutoff, so the left-tail value is not in a one-sided table directly; symmetry $\Phi(-z) = 1 - \Phi(z)$ converts it to a tabulated positive value.

Solution.

Step 1: Standardise. With $\mu = 20$, $\sigma = \sqrt{25} = 5$, $z = \frac{14-20}{5} = -1.2$.

Step 2: Reflect. $\Phi(-1.2) = 1 - \Phi(1.2)$, and the standard value is $\Phi(1.2) = 0.8849$, so

$$\mathbb{P}(X < 14) = \Phi(-1.2) = 1 - 0.8849 = 0.1151.$$

(Value used: $\Phi(1.2) = 0.8849$.)

Example 4.2.27 (A balanced two-cutoff check)**.** For $Z \sim \mathrm{N}(0, 1)$, verify directly that $\mathbb{P}(Z < -1) + \mathbb{P}(Z > 1)$ equals the total mass outside $(-1, 1)$, and give its value.

Which rule, and why? Two symmetric tails again; symmetry says the two pieces are equal, so we compute one and double, and cross-check against $1 - (2\Phi(1) - 1)$.

Solution.

Step 1: Each tail. $\mathbb{P}(Z > 1) = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$, and by symmetry $\mathbb{P}(Z < -1) = \Phi(-1) = 1 - \Phi(1) = 0.1587$.

Step 2: Sum.

$$\mathbb{P}(Z < -1) + \mathbb{P}(Z > 1) = 0.1587 + 0.1587 = 0.3174.$$

Step 3: Cross-check. The central band is $2\Phi(1) - 1 = 0.6826$, and $1 - 0.6826 = 0.3174$, matching.

4.3 The Normal Approximation and the Central Limit Theorem

Two facts make the normal distribution the workhorse of probability. First, a binomial count $\mathrm{Bin}(n, p)$ with many trials looks like a bell curve, so we can read off otherwise unwieldy sums of binomial terms from Φ. Second, and far more general, the **Central Limit Theorem** says that an average (or a sum) of many independent pieces is approximately normal *whatever the shape of the individual pieces*, as long as they have finite mean and variance. This is why measurements built from many small independent contributions — and the averages we form from samples — are so often bell-shaped.

Which rule of this section does the problem need?

- A **count of successes** in n independent yes/no trials, with n large? $\Rightarrow$ **normal approximation to the binomial**. Use $\mu = np$, $\sigma = \sqrt{np(1-p)}$, and a **continuity correction** (± 0.5) because you are replacing a discrete count by a continuous curve. Check $np \geq 10$ and $n(1-p) \geq 10$ first.
- An **average** $\bar{X}_n$ or a **sum** S_n of many independent, identically distributed pieces (any shape)? $\Rightarrow$ **Central Limit Theorem**: $\bar{X}_n \approx \mathrm{N}(\mu, \sigma^2/n)$ and $S_n \approx \mathrm{N}(n\mu, n\sigma^2)$. Standardise and read Φ.
- An **exact** small-n binomial probability, or $np < 10$? $\Rightarrow$ do *not* use the normal approximation; sum the binomial pmf directly (or use the Poisson approximation from §1 when p is tiny). The bell curve is too crude there.
- A probability about **one** observation from a known normal population, not an average? $\Rightarrow$ that is plain standardisation from §2; no CLT and no continuity correction.

The continuity-correction trap. The binomial is discrete and the normal is continuous, so "$X = k$" has positive binomial probability but zero normal probability. Always widen an integer event to the interval that surrounds it: $\mathbb{P}(X = k) \approx \mathbb{P}(k - 0.5 < Y < k + 0.5)$, $\mathbb{P}(X \leq k) \approx \mathbb{P}(Y < k + 0.5)$, $\mathbb{P}(X \geq k) \approx \mathbb{P}(Y > k - 0.5)$. Forgetting the half is the single most common error here. The CLT for a *continuous* average needs no such correction.

4.3.1 The normal approximation to the binomial

Theorem 4.3.1 (DeMoivre–Laplace)**.** *Let $X \sim \mathrm{Bin}(n,p)$ with mean $\mu = np$ and standard deviation $\sigma = \sqrt{np(1-p)}$. For large n the standardised count is approximately standard normal: for every fixed z,*

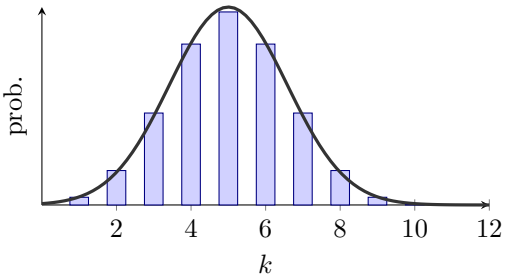
$$\mathbb{P}\left(\frac{X-np}{\sqrt{np(1-p)}} \le z\right) \longrightarrow \Phi(z) \qquad \text{as } n \to \infty.$$

Equivalently X is approximately $\mathrm{N}\big(np,\, np(1-p)\big)$.

Proof. Write $X = \sum_{i=1}^{n} I_i$ where the I_i are independent Bernoulli(p) indicators, each with mean p and variance $p(1-p)$. This is exactly a sum of n iid pieces with finite mean and variance, so the Central Limit Theorem (next subsection) applies and the standardised sum tends to $\mathrm{N}(0,1)$. The DeMoivre–Laplace result is the special case of the CLT for Bernoulli summands. □

Remark (When is it reasonable?). The approximation is good when the bell curve fits comfortably between 0 and n without spilling past either end. A practical rule is $np \ge 10$ *and* $n(1-p) \ge 10$. When p is very small and np stays modest, use the **Poisson** approximation of §1 instead; the normal one is poor in that skewed regime.

The continuity correction matters because we approximate a sum of bars by an area under a smooth curve. Each integer k owns the bar from $k-0.5$ to $k+0.5$; replace the discrete event by that interval of width 1.



The bars are $\mathrm{Bin}(10, \frac{1}{2})$ (so $np = 5$, $np(1-p) = 2.5$); the curve is $\mathrm{N}(5, 2.5)$ laid over them. Each bar's area is well matched by the strip of curve above its half-unit interval.

Example 4.3.2 (Defective chips: a "between" count)**.** A wafer yields $n = 400$ chips, each independently defective with probability $p = 0.05$. Approximate the probability that between 15 and 25 chips inclusive are defective.

Which rule, and why? A count of successes in many independent yes/no trials, so binomial; with $n = 400$ this is too big to sum by hand. We approximate by a normal. (Poisson would also be tenable here since p is small, but $np = 20 \ge 10$, so the normal approximation is fine and is what is asked.)

Solution.

Step 1: Parameters and the adequacy check. $X \sim \mathrm{Bin}(400, 0.05)$, so

$$\mu = np = 400(0.05) = 20, \qquad \sigma = \sqrt{np(1-p)} = \sqrt{400(0.05)(0.95)} = \sqrt{19} = 4.3589.$$

Here $np = 20 \ge 10$ and $n(1-p) = 380 \ge 10$, so the approximation is reasonable.

Step 2: Continuity correction. "$15 \le X \le 25$" is the integer block $15, 16, \ldots, 25$; widen it to the interval $(14.5,\, 25.5)$.

Step 3: Standardise.

$$z_{\mathrm{lo}} = \frac{14.5 - 20}{4.3589} = -1.262, \qquad z_{\mathrm{hi}} = \frac{25.5 - 20}{4.3589} = 1.262.$$

Step 4: Read Φ. Using $\Phi(1.26) = 0.8962$ and the symmetry $\Phi(-z) = 1 - \Phi(z)$,

$$\mathbb{P}(15 \le X \le 25) \approx \Phi(1.26) - \Phi(-1.26) = 2(0.8962) - 1 = 0.7924.$$

$$\boxed{\mathbb{P}(15 \le X \le 25) \approx 0.7924}$$

Example 4.3.3 (Exactly k: a single bar)**.** With the same wafer, $X \sim \text{Bin}(400, 0.05)$, approximate $\mathbb{P}(X = 20)$, the chance of exactly 20 defectives.

Which rule, and why? An "exactly equal" event for a binomial count. Under the normal approximation a single point has zero area, so we MUST use the continuity correction: the bar at 20 is the interval $(19.5, 20.5)$.

Solution.

Step 1: Reuse parameters. $\mu = 20$, $\sigma = \sqrt{19} = 4.3589$ as above.

Step 2: Correct and standardise.

$$z_{\text{lo}} = \frac{19.5 - 20}{4.3589} = -0.1147, \qquad z_{\text{hi}} = \frac{20.5 - 20}{4.3589} = 0.1147.$$

Step 3: Read Φ*.* Using $\Phi(0.11) = 0.5438$,

$$\mathbb{P}(X = 20) \approx \Phi(0.11) - \Phi(-0.11) = 2(0.5438) - 1 = 0.0876.$$

For comparison the exact binomial value is 0.0883; the half-unit strip recovers it to within 0.001.

$$\boxed{\mathbb{P}(X = 20) \approx 0.0876}$$

Example 4.3.4 (A one-sided tail: poll majority)**.** In a referendum 52% of the electorate truly favour a measure. A random sample of $n = 1000$ voters is polled. Approximate the probability that the sample shows a majority *against* — i.e. fewer than 500 of the 1000 say "favour".

Which rule, and why? Each sampled voter independently favours with probability $p = 0.52$; the number who favour is $\text{Bin}(1000, 0.52)$. With $n = 1000$ we approximate normally. "Fewer than 500" is a discrete tail, so correct the boundary.

Solution.

Step 1: Parameters.

$$\mu = np = 1000(0.52) = 520, \qquad \sigma = \sqrt{1000(0.52)(0.48)} = \sqrt{249.6} = 15.799.$$

Both $np = 520$ and $n(1-p) = 480$ exceed 10.

Step 2: Continuity correction. "$X < 500$" means $X \le 499$ (integers), which widens to $Y < 499.5$.

Step 3: Standardise.

$$z = \frac{499.5 - 520}{15.799} = \frac{-20.5}{15.799} = -1.298.$$

Step 4: Read Φ*.* Using $\Phi(1.30) = 0.9032$,

$$\mathbb{P}(X < 500) \approx \Phi(-1.30) = 1 - 0.9032 = 0.0968.$$

So even with a true 52–48 split there is nearly a 10% chance a sample of 1000 shows the wrong majority.

$$\boxed{\mathbb{P}(X < 500) \approx 0.0968}$$

Example 4.3.5 (Upper tail: free-throw streak)**.** A basketball player makes each free throw independently with probability $p = 0.80$. Over a season she attempts $n = 150$ free throws. Approximate the probability she makes *at least* 130 of them.

Which rule, and why? Made shots are a binomial count, $\text{Bin}(150, 0.80)$. Large n, so normal approximation. "At least 130" is $X \ge 130$, an upper tail; the boundary 130 is included, so widen *downward* to 129.5.

Solution.

Step 1: Parameters.

$$\mu = np = 150(0.80) = 120, \qquad \sigma = \sqrt{150(0.80)(0.20)} = \sqrt{24} = 4.8990.$$

$np = 120$ and $n(1-p) = 30$, both ≥ 10.

Step 2: Correct. $X \geq 130 \Rightarrow Y > 129.5$.

Step 3: Standardise.

$$z = \frac{129.5 - 120}{4.8990} = \frac{9.5}{4.8990} = 1.939.$$

Step 4: Read Φ. Using $\Phi(1.94) = 0.9738$,

$$\mathbb{P}(X \geq 130) \approx 1 - \Phi(1.94) = 1 - 0.9738 = 0.0262.$$

$$\boxed{\mathbb{P}(X \geq 130) \approx 0.0262}$$

Example 4.3.6 (When NOT to use it: rare events)**.** A factory ships $n = 500$ units; each is independently defective with probability $p = 0.004$. Estimate $\mathbb{P}(X \leq 1)$, at most one defective. Decide whether the normal approximation is appropriate.

Which rule, and why? Binomial count, but check the adequacy rule first. Here $np = 2 < 10$, so the normal approximation is poor (the distribution is sharply right-skewed and crowds against 0). This is the small-p, modest-np regime where the **Poisson** approximation of §1 is the right tool.

Solution.

Step 1: Recognise the regime. $np = 500(0.004) = 2$. Since $np < 10$, do not use the normal curve; use Poisson(λ) with $\lambda = np = 2$.

Step 2: Poisson tail.

$$\mathbb{P}(X \leq 1) \approx e^{-2}\left(\frac{2^0}{0!} + \frac{2^1}{1!}\right) = e^{-2}(1+2) = 3e^{-2}.$$

Step 3: Evaluate. With $e^{-2} = 0.1353$,

$$\mathbb{P}(X \leq 1) \approx 3(0.1353) = 0.4060.$$

$$\boxed{\mathbb{P}(X \leq 1) \approx 0.4060 \quad \text{(Poisson, not normal)}}$$

4.3.2 The Central Limit Theorem

Theorem 4.3.7 (Central Limit Theorem)**.** *Let $X_1, X_2, \ldots$ be independent and identically distributed with finite mean $\mu = \mathbb{E}(X_i)$ and finite variance $\sigma^2 = \mathrm{Var}(X_i) > 0$. Let $S_n = X_1 + \cdots + X_n$ and $\bar{X}_n = S_n/n$. Then as $n \to \infty$ the standardised sum converges in distribution to standard normal:*

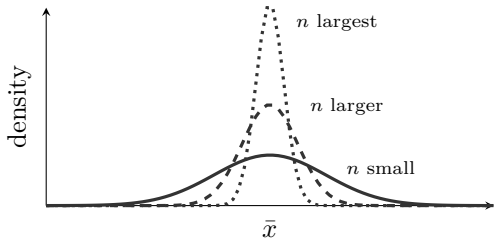
$$\frac{S_n - n\mu}{\sigma\sqrt{n}} \;=\; \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \;\xrightarrow{\;d\;}\; \mathrm{N}(0,1).$$

For large n, therefore,

$$S_n \;\approx\; \mathrm{N}\big(n\mu,\, n\sigma^2\big), \qquad \bar{X}_n \;\approx\; \mathrm{N}\left(\mu,\, \frac{\sigma^2}{n}\right).$$

Why the variances are what they are. Independence and identical distribution give $\mathbb{E}(S_n) = n\mu$ and $\mathrm{Var}(S_n) = n\sigma^2$ (variances of independent pieces add). Dividing by n, $\mathbb{E}(\bar{X}_n) = \mu$ and $\mathrm{Var}(\bar{X}_n) = \mathrm{Var}(S_n)/n^2 = \sigma^2/n$. So the centring and scaling above produce a quantity with mean 0 and variance 1; the theorem asserts its limiting *shape* is the standard normal, regardless of the common distribution of the X_i. A full proof uses characteristic functions and is beyond these notes. □

Remark (Reading the two key facts). The spread of the sample mean shrinks like $\sigma/\sqrt{n}$: quadruple the sample size to halve the standard error. The shape becomes normal even when the individual X_i are skewed or discrete — a larger n is needed when they are strongly skewed, but for many everyday distributions $n \geq 30$ already gives a good bell.



The sampling distribution of $\bar{X}_n$ centres on μ for every n but narrows as n grows, with standard deviation $\sigma/\sqrt{n}$. The three curves have variances σ^2/n in ratio $1 : \frac{1}{4} : \frac{1}{16}$, i.e. n increasing by factors of 4.

Example 4.3.8 (Average call duration)**.** Customer-service calls have mean duration $\mu = 4.0$ minutes and standard deviation $\sigma = 3.0$ minutes, with a strongly right-skewed (non-normal) shape. A shift handles $n = 64$ independent calls. Approximate the probability that the *average* call length exceeds 4.5 minutes.

Which rule, and why? We are asked about an average $\bar{X}$ of many independent pieces, not a single call. The individual distribution is skewed, but the CLT makes $\bar{X}$ approximately normal; we do not need the shape of one call. No continuity correction — the average is continuous.

Solution.

Step 1: Sampling distribution of the mean. By the CLT,

$$\bar{X} \approx \mathrm{N}\left(\mu, \frac{\sigma^2}{n}\right), \qquad \frac{\sigma}{\sqrt{n}} = \frac{3.0}{\sqrt{64}} = \frac{3.0}{8} = 0.375.$$

Step 2: Standardise.

$$z = \frac{4.5 - 4.0}{0.375} = \frac{0.5}{0.375} = 1.333.$$

Step 3: Read Φ. Using $\Phi(1.33) = 0.9082$,

$$\mathbb{P}(\bar{X} > 4.5) \approx 1 - \Phi(1.33) = 1 - 0.9082 = 0.0918.$$

$$\boxed{\mathbb{P}(\bar{X} > 4.5) \approx 0.0918}$$

Example 4.3.9 (A total: elevator load)**.** Adult passengers have weight mean $\mu = 170$ lb and standard deviation $\sigma = 30$ lb. An elevator is rated for 2500 lb. If $n = 14$ adults board, approximate the probability their *total* weight exceeds the rating.

Which rule, and why? The quantity of interest is the sum S_n of 14 independent weights. Use the CLT for a sum: $S_n \approx \mathrm{N}(n\mu, n\sigma^2)$. Weights are continuous, so no correction.

Solution.

Step 1: Sum parameters.

$$\mathbb{E}(S_n) = n\mu = 14(170) = 2380, \qquad \mathrm{SD}(S_n) = \sigma\sqrt{n} = 30\sqrt{14} = 30(3.7417) = 112.25.$$

Step 2: Standardise the threshold.

$$z = \frac{2500 - 2380}{112.25} = \frac{120}{112.25} = 1.069.$$

Step 3: Read Φ. Using $\Phi(1.07) = 0.8577$,

$$\mathbb{P}(S_n > 2500) \approx 1 - \Phi(1.07) = 1 - 0.8577 = 0.1423.$$

$$\boxed{\mathbb{P}(S_n > 2500) \approx 0.1423}$$

Example 4.3.10 (A "between" average: fill weights)**.** A machine fills bags with mean $\mu = 500$ g and standard deviation $\sigma = 8$ g per bag. A quality check averages $n = 25$ bags. Approximate the probability the sample mean lands between 498 and 503 g.

Which rule, and why? An average of 25 independent fills, so CLT: $\bar{X} \approx \mathrm{N}(\mu, \sigma^2/n)$. "Between" means standardise both endpoints and subtract. Continuous, so no half-unit correction.

Solution.

Step 1: Standard error.

$$\frac{\sigma}{\sqrt{n}} = \frac{8}{\sqrt{25}} = \frac{8}{5} = 1.6.$$

Step 2: Standardise both ends.

$$z_{\mathrm{lo}} = \frac{498 - 500}{1.6} = -1.25, \qquad z_{\mathrm{hi}} = \frac{503 - 500}{1.6} = 1.875.$$

Step 3: Read Φ. Using $\Phi(1.25) = 0.8944$ and $\Phi(1.88) = 0.9699$ (rounding 1.875 to 1.88),

$$\begin{aligned} \mathbb{P}(498 < \bar{X} < 503) &\approx \Phi(1.875) - \Phi(-1.25) \\ &= 0.9699 - (1 - 0.8944) \\ &= 0.9699 - 0.1056 = 0.8643. \end{aligned}$$

$$\boxed{\mathbb{P}(498 < \bar{X} < 503) \approx 0.8643}$$

Example 4.3.11 (Sizing the sample: a design problem)**.** Measurements of a part's length have $\sigma = 0.50$ mm. How many independent parts n must be averaged so that the sample mean is within 0.10 mm of the true mean μ with probability 0.95?

Which rule, and why? This inverts the CLT: we know the target tail probability and solve for n. The width 0.10 is a half-interval around μ, so it should equal z^* standard errors with $z^* = 1.96$ (the value with $\Phi(1.96) = 0.9750$, leaving 0.025 in each tail).

Solution.

Step 1: Set up the requirement. We need

$$\mathbb{P}\big(|\bar{X} - \mu| \le 0.10\big) = 0.95 \quad \Longleftrightarrow \quad \frac{0.10}{\sigma/\sqrt{n}} = z^* = 1.96.$$

Here $z^* = 1.96$ because $\Phi(1.96) = 0.9750$ puts 0.025 in each tail.

Step 2: Solve for $\sqrt{n}$.

$$\sqrt{n} = \frac{1.96\,\sigma}{0.10} = \frac{1.96(0.50)}{0.10} = \frac{0.98}{0.10} = 9.8.$$

Step 3: Square and round up.

$$n = (9.8)^2 = 96.04 \implies n = 97.$$

We round *up* so the probability is at least 0.95.

$$\boxed{n = 97}$$

Example 4.3.12 (Dice sum: a discrete summand)**.** Roll a fair six-sided die $n = 70$ times and let S be the total of the faces. A single roll has mean $\mu = 3.5$ and variance $\sigma^2 = \frac{35}{12} = 2.9167$. Approximate $\mathbb{P}(S \ge 260)$.

Which rule, and why? S is a sum of 70 iid rolls; the CLT applies even though one roll is discrete and far from normal. Because the summands are integer-valued and we want a discrete tail on the sum, a continuity correction sharpens the answer: $S \ge 260$ widens to $S > 259.5$.

Solution.

Step 1: Sum parameters.

$$\mathbb{E}(S) = n\mu = 70(3.5) = 245,$$
$$\mathrm{Var}(S) = n\sigma^2 = 70(2.9167) = 204.17, \qquad \mathrm{SD}(S) = \sqrt{204.17} = 14.289.$$

Step 2: Continuity correction and standardise.

$$z = \frac{259.5 - 245}{14.289} = \frac{14.5}{14.289} = 1.015.$$

Step 3: Read Φ. Using $\Phi(1.01) = 0.8438$,

$$\mathbb{P}(S \geq 260) \approx 1 - \Phi(1.01) = 1 - 0.8438 = 0.1562.$$

$$\boxed{\mathbb{P}(S \geq 260) \approx 0.1562}$$

Remark (Putting the two halves together). The normal approximation to the binomial is just the CLT applied to a sum of Bernoulli indicators, so both halves of this section are one idea: *add up many independent pieces and the total is approximately normal.* Use a continuity correction precisely when the thing you are approximating is integer-valued (binomial counts, integer-summand totals like dice); skip it for genuinely continuous averages and sums.

4.4 The Geometric Distribution

Run the same yes/no experiment over and over, independently, with a fixed success probability p on each try, and count how many tries it takes to get the first success. That count is a **geometric** random variable. It is the natural model for "how long do I wait" questions phrased in whole trials: the number of cold calls before the first sale, the number of plates screened before the first positive well, the number of dice rolls before the first six. The single parameter p controls everything — a small p pushes the wait far out into a long, slowly decaying tail.

Which rule of this section does the problem need?

- Counting **trials up to and including the first success** in independent identical yes/no trials? $\Rightarrow$ **geometric** pmf $p_X(k) = (1-p)^{k-1}p$ (§4.4.1).
- Asked for **"at least k trials," "more than k," or "within the first k"**? $\Rightarrow$ use the closed-form **survival/cdf**: $\mathbb{P}(X > k) = (1-p)^k$, $F_X(k) = 1 - (1-p)^k$ (§4.4.2).
- Need the **expected wait or its spread**? $\Rightarrow$ mean $1/p$, variance $(1-p)/p^2$ (§4.4.3).
- Told the success **has not happened yet** and asked about the additional wait? $\Rightarrow$ the **memoryless** property: the future is a fresh geometric, $\mathbb{P}(X > m+n \mid X > m) = \mathbb{P}(X > n)$ (§4.4.4).

Why competing tools do not apply. If the number of trials is *fixed in advance* and you count successes, that is **binomial**, not geometric — geometric fixes the number of *successes* (exactly one) and lets the number of trials vary. Waiting for the *r-th* success instead of the first is **negative binomial** (§5); the geometric is the $r = 1$ case. And if the trials run in *continuous* time rather than in discrete attempts, the analogous wait is **exponential** (§6), the continuous cousin that shares the same memoryless property.

4.4.1 The geometric pmf

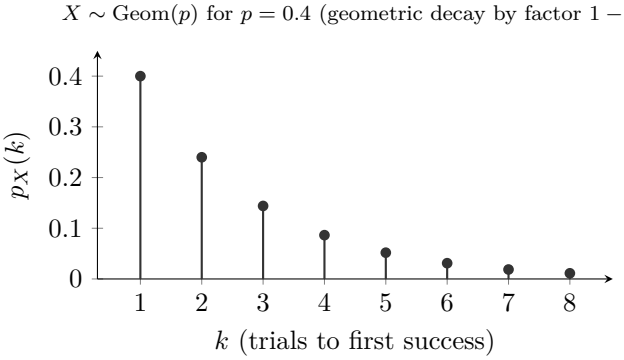
Definition 4.4.1 (Geometric distribution)**.** Let independent trials each succeed with probability p (where $0 < p \leq 1$) and fail with probability $q = 1 - p$. Let X be the number of trials up to and including the first success. Then X is **geometric** with parameter p, written $X \sim \mathrm{Geom}(p)$, with support $\mathcal{R}(X) = \{1, 2, 3, \dots\}$ and probability mass function

$$p_X(k) = \mathbb{P}(X = k) = (1-p)^{k-1}p, \qquad k = 1, 2, 3, \dots$$

Why this is the pmf. For $X = k$ the first $k-1$ trials must all fail and the k-th must succeed. By independence the probability is the product $\underbrace{(1-p)\cdots(1-p)}_{k-1 \text{ failures}} \cdot p = (1-p)^{k-1}p$. This is a valid pmf: the masses are nonnegative and, summing the geometric series,

$$\sum_{k=1}^{\infty}(1-p)^{k-1}p = p\sum_{j=0}^{\infty}(1-p)^{j} = p\cdot\frac{1}{1-(1-p)} = p\cdot\frac{1}{p} = 1.$$

Remark. This is the **trials-count** convention (support starting at 1); a common alternative counts only the *failures* before the first success (support starting at 0), with pmf $(1-p)^k p$ and mean $(1-p)/p$. We use the trials-count form throughout, so the smallest possible value is $X = 1$.



Example 4.4.2 (First six on a die)**.** A fair six-sided die is rolled until a six appears. What is the probability the first six appears on the fourth roll?

Which rule, and why? Repeated independent trials, each "six / not six" with constant probability $p = \frac{1}{6}$, and we want the *trial number* of the *first* success. That is exactly $X \sim \text{Geom}(1/6)$, asking for $\mathbb{P}(X = 4)$. It is not binomial because the number of rolls is not fixed in advance.

Solution.

Step 1: identify the parameter. $p = \frac{1}{6}$, so $q = 1 - p = \frac{5}{6}$.

Step 2: apply the pmf with $k = 4$. Three failures then a success:

$$\mathbb{P}(X = 4) = \left(\tfrac{5}{6}\right)^3\left(\tfrac{1}{6}\right) = \frac{125}{216}\cdot\frac{1}{6} = \frac{125}{1296} \approx 0.0965.$$

Example 4.4.3 (Defective fuses)**.** A machine produces fuses that are defective independently with probability 0.08. An inspector tests fuses one at a time. Find the probability that the first defective fuse is the third one tested.

Which rule, and why? "First defective," independent items, constant defect rate $p = 0.08$: a geometric wait for the first success (here "success" = "defective"). We want $\mathbb{P}(X = 3)$.

Solution.

Step 1: set p. $p = 0.08$, $q = 0.92$.

Step 2: pmf at $k = 3$.

$$\mathbb{P}(X = 3) = (0.92)^2(0.08) = 0.8464 \cdot 0.08 = 0.067712 \approx 0.0677.$$

Example 4.4.4 (Free-throw streak start)**.** A basketball player makes each free throw independently with probability 0.75. Find the probability she needs exactly two attempts to record her first make.

Which rule, and why? Independent shots, constant make probability $p = 0.75$, waiting for the first success: $X \sim \text{Geom}(0.75)$, want $\mathbb{P}(X = 2)$.

Solution.

Step 1. $p = 0.75$, $q = 0.25$.

Step 2: pmf at $k = 2$ — one miss, then a make.

$$\mathbb{P}(X = 2) = (0.25)^1(0.75) = 0.1875.$$

Example 4.4.5 (Most likely wait, and a small comparison)**.** For the die of Example 4.4, which single value of X is most probable, and how much more likely is a first six on roll 1 than on roll 7?

Which rule, and why? Still $X \sim \text{Geom}(1/6)$. The pmf $(1-p)^{k-1}p$ is strictly decreasing in k whenever $0 < p < 1$, because each successive term multiplies by $q = 1 - p < 1$, so the mode of any geometric is always $k = 1$.

Solution.

Step 1: identify the mode. Since masses fall by the constant factor $q = \frac{5}{6}$ at each step, the largest mass is at $k = 1$:

$$\mathbb{P}(X = 1) = p = \tfrac{1}{6} \approx 0.1667.$$

Step 2: ratio of $k = 1$ to $k = 7$. Using $p_X(k)/p_X(1) = q^{\,k-1}$,

$$\frac{\mathbb{P}(X = 1)}{\mathbb{P}(X = 7)} = \frac{1}{q^6} = \left(\tfrac{6}{5}\right)^6 = \frac{46656}{15625} \approx 2.986.$$

A first six on the opening roll is about 3 times as likely as having to wait until the seventh roll.

4.4.2 The cdf and survival function

Theorem 4.4.6 (Geometric tail and cdf)**.** *If $X \sim \text{Geom}(p)$ with $q = 1 - p$, then for every integer $k \geq 0$ the **survival probability** and **cumulative distribution function** are*

$$\mathbb{P}(X > k) = q^k, \qquad F_X(k) = \mathbb{P}(X \leq k) = 1 - q^k.$$

Proof. The event $\{X > k\}$ means the first k trials all fail, which by independence has probability q^k. Hence $F_X(k) = \mathbb{P}(X \leq k) = 1 - \mathbb{P}(X > k) = 1 - q^k$. (Algebraically the same falls out of summing the pmf: $\sum_{j=1}^{k} q^{j-1}p = p \cdot \frac{1-q^k}{1-q} = 1 - q^k$.) □

These closed forms make "at least," "more than," and "within" questions one-liners — no summation needed. Note $\mathbb{P}(X \geq k) = \mathbb{P}(X > k - 1) = q^{k-1}$, the probability of needing *at least* k trials.

Example 4.4.7 (At least so many calls)**.** A salesperson closes each independent call with probability $p = 0.2$. Find the probability that the first sale takes more than 5 calls, and the probability it occurs within the first 4 calls.

Which rule, and why? Independent trials, constant $p = 0.2$, first success: $X \sim \text{Geom}(0.2)$. "More than 5" is a survival probability $\mathbb{P}(X > 5)$; "within the first 4" is the cdf $F_X(4) = \mathbb{P}(X \leq 4)$.

Solution.

Step 1: survival at $k = 5$. With $q = 0.8$,

$$\mathbb{P}(X > 5) = q^5 = 0.8^5 = 0.32768 \approx 0.3277.$$

Step 2: cdf at $k = 4$.

$$F_X(4) = 1 - q^4 = 1 - 0.8^4 = 1 - 0.4096 = 0.5904.$$

Example 4.4.8 (Screening wells)**.** In a drug screen, each well is a "hit" independently with probability $p = 0.03$. What is the probability that the first hit appears within the first 20 wells? At least 50 wells?

Which rule, and why? Independent wells, constant hit rate $p = 0.03$, waiting for the first hit: $X \sim \text{Geom}(0.03)$, $q = 0.97$. "Within 20" is $F_X(20)$; "at least 50" is $\mathbb{P}(X \geq 50) = \mathbb{P}(X > 49) = q^{49}$.

Solution.

Step 1: within the first 20*.*

$$F_X(20) = 1 - 0.97^{20} = 1 - 0.54379 = 0.45621 \approx 0.4562.$$

Step 2: at least 50 *wells.*

$$\mathbb{P}(X \geq 50) = q^{49} = 0.97^{49} = 0.22624 \approx 0.2262.$$

Example 4.4.9 (How many trials to be 90% sure)**.** With per-trial success probability $p = 0.15$, how many trials k guarantee at least a 90% chance of seeing the first success by trial k?

Which rule, and why? $X \sim \text{Geom}(0.15)$, $q = 0.85$. We want the smallest integer k with $F_X(k) = 1 - q^k \geq 0.90$, i.e. an inverse-cdf (percentile) question solved with the closed-form tail.

Solution.

Step 1: set up the inequality.

$$1 - 0.85^k \geq 0.90 \iff 0.85^k \leq 0.10.$$

Step 2: solve for k with logarithms. Since $\ln(0.85) < 0$ the inequality flips on dividing:

$$k \geq \frac{\ln(0.10)}{\ln(0.85)} = \frac{-2.302585}{-0.162519} \approx 14.17.$$

Step 3: round up. Take $k = 15$. Check: $0.85^{15} \approx 0.0874 \leq 0.10$, while $0.85^{14} \approx 0.1028 > 0.10$, so 15 trials is the smallest that works.

$$\boxed{k = 15}$$

Example 4.4.10 (Exactly in a window)**.** For the salesperson of Example 4.7 ($p = 0.2$), find the probability the first sale lands on call 3, 4, or 5.

Which rule, and why? $X \sim \text{Geom}(0.2)$. "On call 3, 4, or 5" is $\mathbb{P}(3 \leq X \leq 5)$, a window probability; differencing the cdf is fastest, since $\mathbb{P}(a \leq X \leq b) = F_X(b) - F_X(a-1)$.

Solution.

Step 1: difference the cdf. With $q = 0.8$,

$$\mathbb{P}(3 \leq X \leq 5) = F_X(5) - F_X(2) = (1 - q^5) - (1 - q^2) = q^2 - q^5.$$

Step 2: evaluate.

$$q^2 - q^5 = 0.64 - 0.32768 = 0.31232 \approx 0.3123.$$

(Check against the pmf: $q^2p + q^3p + q^4p = 0.128 + 0.1024 + 0.08192 = 0.31232$. Agree.)

4.4.3 Mean and variance

Theorem 4.4.11 (Mean and variance of a geometric)**.** *If $X \sim \text{Geom}(p)$ with $q = 1 - p$, then*

$$\mathbb{E}(X) = \frac{1}{p}, \qquad \text{Var}(X) = \frac{1-p}{p^2}, \qquad \text{SD}(X) = \frac{\sqrt{1-p}}{p}.$$

Derivation of the mean. Compute $\mathbb{E}(X) = \sum_{k=1}^{\infty} k\, q^{k-1} p = p \sum_{k=1}^{\infty} k\, q^{k-1}$. Differentiating the geometric series $\sum_{k=0}^{\infty} x^k = \dfrac{1}{1-x}$ term by term for $|x| < 1$ gives

$$\sum_{k=1}^{\infty} k\, x^{k-1} = \frac{d}{dx} \frac{1}{1-x} = \frac{1}{(1-x)^2}.$$

Setting $x = q$ (so $1 - x = p$),

$$\mathbb{E}(X) = p \cdot \frac{1}{(1-q)^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

The variance follows from $\mathbb{E}[X(X-1)] = \sum_k k(k-1)q^{k-1}p = \dfrac{2q}{p^2}$ (a second derivative of the same series), giving $\mathbb{E}(X^2) = \dfrac{2q}{p^2} + \dfrac{1}{p}$ and $\mathrm{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}X)^2 = \dfrac{2q}{p^2} + \dfrac{1}{p} - \dfrac{1}{p^2} = \dfrac{q}{p^2} = \dfrac{1-p}{p^2}$. □

The mean $1/p$ is the intuitive "one success per $1/p$ trials on average"; the spread blows up as $p \to 0$, matching the long tail of rare-success waits.

Example 4.4.12 (Average rolls for a six)**.** On average, how many rolls of a fair die are needed to get the first six, and what is the standard deviation?

Which rule, and why? $X \sim \mathrm{Geom}(1/6)$; "average number of trials" is $\mathbb{E}(X) = 1/p$ and the spread is $\mathrm{SD}(X)$.

Solution.

Step 1: mean. $\mathbb{E}(X) = \dfrac{1}{1/6} = 6$ rolls.

Step 2: standard deviation. With $q = \frac{5}{6}$,

$$\mathrm{Var}(X) = \frac{q}{p^2} = \frac{5/6}{(1/6)^2} = \frac{5/6}{1/36} = 30, \qquad \mathrm{SD}(X) = \sqrt{30} \approx 5.477.$$

Example 4.4.13 (Expected calls to a sale)**.** The salesperson closes each call with probability $p = 0.2$. Find the expected number of calls to the first sale and the variance.

Which rule, and why? $X \sim \mathrm{Geom}(0.2)$; expected wait $= 1/p$, variance $= (1-p)/p^2$.

Solution.

Step 1: mean. $\mathbb{E}(X) = \dfrac{1}{0.2} = 5$ calls.

Step 2: variance.

$$\mathrm{Var}(X) = \frac{1 - 0.2}{0.2^2} = \frac{0.8}{0.04} = 20, \qquad \mathrm{SD}(X) = \sqrt{20} \approx 4.472.$$

Example 4.4.14 (Designing a target rate)**.** A quality engineer wants the expected number of units inspected before the first defect to be 40. If defects occur independently at a constant rate, what defect probability p does that require?

Which rule, and why? "Expected number of trials to the first success = 40" is $\mathbb{E}(X) = 1/p = 40$. Solve for p (an inverse-mean design problem).

Solution.

Step 1: set the mean. $\dfrac{1}{p} = 40.$

Step 2: solve.

$$p = \frac{1}{40} = 0.025.$$

As a sanity check, the standard deviation of the wait is then $\mathrm{SD}(X) = \sqrt{q}/p = \sqrt{0.975}/0.025 \approx 39.5$ units — nearly as large as the mean, the signature of a rare-success geometric.

Example 4.4.15 (Mean and a probability together)**.** A test for a rare allele flags each sampled individual as a carrier with probability $p = 0.04$, independently. Find the expected number sampled until the first carrier, and the probability that the actual wait *exceeds* that expected value.

Which rule, and why? $X \sim \text{Geom}(0.04)$. First $\mathbb{E}(X) = 1/p$; then $\mathbb{P}(X > \mathbb{E}X)$ uses the survival function $\mathbb{P}(X > k) = q^k$ at the (rounded) mean.

Solution.

Step 1: mean. $\mathbb{E}(X) = \dfrac{1}{0.04} = 25$ individuals.

Step 2: probability of exceeding the mean. With $q = 0.96$ and integer threshold $k = 25$,

$$\mathbb{P}(X > 25) = q^{25} = 0.96^{25} = 0.36040 \approx 0.3604.$$

So even though 25 is the average wait, there is about a 36% chance of waiting strictly longer — a reminder that the geometric is right-skewed, with mean pulled above the median.

4.4.4 The memoryless property

Theorem 4.4.16 (Memorylessness)**.** *If $X \sim \text{Geom}(p)$, then for all nonnegative integers m, n,*

$$\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n).$$

That is, given that the first m trials have already failed, the additional *number of trials needed is again* $\text{Geom}(p)$*, exactly as if no trials had occurred. The geometric is the only discrete distribution on $\{1, 2, \dots\}$ with this property.*

Proof. Using $\mathbb{P}(X > k) = q^k$ and the definition of conditional probability, with the event $\{X > m + n\} \subseteq \{X > m\}$,

$$\mathbb{P}(X > m + n \mid X > m) = \frac{\mathbb{P}(X > m + n)}{\mathbb{P}(X > m)} = \frac{q^{m+n}}{q^m} = q^n = \mathbb{P}(X > n). \qquad \square$$

The intuition is plain: the trials are independent, so past failures carry no information about how soon the next success comes. The coin has no memory of its misses.

Example 4.4.17 (The die does not "warm up")**.** You have rolled a fair die 4 times with no six. What is the probability you will need more than 3 additional rolls (i.e. more than 7 total) to finally get a six?

Which rule, and why? $X \sim \text{Geom}(1/6)$ and we are told $X > 4$ (four failures so far). The conditional question $\mathbb{P}(X > 7 \mid X > 4)$ is tailor-made for memorylessness, which collapses it to a fresh $\mathbb{P}(X > 3)$.

Solution.

Step 1: apply memorylessness. With $m = 4$, $n = 3$,

$$\mathbb{P}(X > 7 \mid X > 4) = \mathbb{P}(X > 3).$$

Step 2: evaluate the survival function. With $q = \frac{5}{6}$,

$$\mathbb{P}(X > 3) = q^3 = \left(\tfrac{5}{6}\right)^3 = \frac{125}{216} \approx 0.5787.$$

The four wasted rolls are irrelevant; the remaining wait behaves like a brand-new search.

Example 4.4.18 (Verifying it the long way)**.** For the salesperson with $p = 0.2$, compute $\mathbb{P}(X > 9 \mid X > 6)$ directly from the definition and confirm it equals $\mathbb{P}(X > 3)$.

Which rule, and why? $X \sim \text{Geom}(0.2)$. The point is to *check* memorylessness by brute force: form the conditional probability from $\mathbb{P}(X > k) = q^k$ rather than quoting the theorem.

Solution.

Step 1: direct conditional. With $q = 0.8$,

$$\mathbb{P}(X > 9 \mid X > 6) = \frac{\mathbb{P}(X > 9)}{\mathbb{P}(X > 6)} = \frac{q^9}{q^6} = q^3.$$

Step 2: numerical value and comparison.

$$q^3 = 0.8^3 = 0.512 = \mathbb{P}(X > 3).$$

The cancellation $q^9/q^6 = q^3$ is exactly the memoryless identity at work.

Example 4.4.19 (A gambler's fallacy, defused)**.** Each spin of a roulette-style wheel lands on a chosen number with probability $p = 1/38$, independently. After 20 spins with no hit, a player insists the number is "due." By how much has the expected number of *additional* spins to the first hit changed?

Which rule, and why? $X \sim \text{Geom}(1/38)$, conditioned on $X > 20$. By memorylessness the additional wait $X - 20 \mid X > 20$ is again Geom(1/38), so its expectation is unchanged. We compare the conditional expected extra wait to the original mean.

Solution.

Step 1: original mean. $\mathbb{E}(X) = \dfrac{1}{1/38} = 38$ spins.

Step 2: conditional expected additional wait. Memorylessness gives $X - 20 \mid \{X > 20\} \sim \text{Geom}(1/38)$, so

$$\mathbb{E}(X - 20 \mid X > 20) = \frac{1}{1/38} = 38.$$

Step 3: the change. The expected remaining wait is still 38 spins — a change of 0. The number is not "due"; the wheel has no memory of the 20 misses.

Example 4.4.20 (Conditional pmf is a shifted geometric)**.** A component fails each cycle with probability $p = 0.05$ ("success" = failure). Given that it has survived its first 10 cycles, find the probability it fails on exactly the 13th cycle overall.

Which rule, and why? $X \sim \text{Geom}(0.05)$ conditioned on $X > 10$. We want $\mathbb{P}(X = 13 \mid X > 10)$. By memorylessness the residual wait $X - 10 \mid X > 10$ is geometric, so $\mathbb{P}(X = 13 \mid X > 10) = \mathbb{P}(X = 3)$ for a fresh geometric.

Solution.

Step 1: reduce by memorylessness. The extra wait is $X - 10 \sim \text{Geom}(0.05)$, and $X = 13$ corresponds to an extra wait of 3:

$$\mathbb{P}(X = 13 \mid X > 10) = \mathbb{P}(X = 3) = q^2 p.$$

Step 2: evaluate. With $q = 0.95$,

$$\mathbb{P}(X = 3) = (0.95)^2(0.05) = 0.9025 \cdot 0.05 = 0.045125 \approx 0.0451.$$

Step 3: cross-check directly.

$$\mathbb{P}(X = 13 \mid X > 10) = \frac{\mathbb{P}(X = 13)}{\mathbb{P}(X > 10)} = \frac{q^{12}p}{q^{10}} = q^2 p = 0.045125. \quad \blacksquare$$

Remark. Memorylessness is what makes the geometric the discrete twin of the **exponential** distribution (§6): both forget their past. It is precisely why "waiting" problems with independent identical trials never reward patience — the clock effectively resets after every failure.

4.5 The Negative Binomial Distribution

The geometric distribution counts the trials needed to get *one* success. The negative binomial distribution generalises this: it counts how many independent yes/no trials are needed to collect the r-th success. Whenever a problem asks "how many attempts until I have r of something," with each attempt succeeding independently with the same probability p, the answer is negative binomial. Because reaching the r-th success means stringing together r separate "wait for the next success" episodes, the negative binomial is exactly a sum of r independent geometric random variables — and that single fact hands us its mean and variance for free.

Which rule of this section does the problem need?

- The problem fixes a **target number of successes** r and asks for the **number of trials** (or the number of failures) needed to reach it, with independent trials and a constant success probability p? $\Rightarrow$ **negative binomial** $\text{NegBin}(r, p)$, pmf $p_X(k) = \binom{k-1}{r-1}p^r(1-p)^{k-r}$ for $k = r, r+1, \dots$
- If $r = 1$ (waiting for the *first* success), the negative binomial collapses to the **geometric** $\text{Geom}(p)$ of §4 — use that simpler form.
- The number of *trials* n is **fixed in advance** and you want the number of successes among them? That is the **binomial** $\text{Bin}(n, p)$, not the negative binomial. The tell is which quantity is fixed and which is random: binomial fixes the trials and lets the success count vary; negative binomial fixes the success count and lets the trial count vary.
- Need the mean or variance? Do **not** re-derive from the pmf. Write $X = G_1 + \cdots + G_r$ with the G_i independent $\text{Geom}(p)$ and use $\mathbb{E}(X) = r/p$, $\text{Var}(X) = r(1-p)/p^2$.

Convention used throughout. Here X counts **trials up to and including the r-th success**, so $X \in \{r, r+1, r+2, \dots\}$. Some books instead let the variable count only the *failures* before the r-th success (support $\{0, 1, 2, \dots\}$); that variable is $X - r$. Watch which one a formula assumes — it shifts every answer by r.

4.5.1 The pmf and where it comes from

Definition 4.5.1 (Negative binomial distribution)**.** Run independent trials, each a success with probability p and a failure with probability $1-p$, with $0 < p \le 1$. Let X be the number of trials up to and including the r-th success, where $r \ge 1$ is a fixed integer. Then $X \sim \text{NegBin}(r, p)$ with

$$p_X(k) = \binom{k-1}{r-1}p^r(1-p)^{k-r}, \qquad k = r,\, r+1,\, r+2, \dots$$

Theorem 4.5.2 (Why the pmf has this form)**.** *The probability that the r-th success lands exactly on trial k is $\binom{k-1}{r-1}p^r(1-p)^{k-r}$.*

Proof. For the r-th success to occur on trial k, two things must happen and they are independent. First, trial k itself is a success: probability p. Second, among the earlier $k-1$ trials there must be exactly $r-1$ successes and $(k-1)-(r-1) = k-r$ failures, in some order. The number of orders is $\binom{k-1}{r-1}$, and each such arrangement has probability $p^{r-1}(1-p)^{k-r}$. Multiplying,

$$\mathbb{P}(X = k) = \underbrace{\binom{k-1}{r-1}p^{r-1}(1-p)^{k-r}}_{\text{first } k-1 \text{ trials}} \cdot \underbrace{p}_{\text{trial } k} = \binom{k-1}{r-1}p^r(1-p)^{k-r}. \qquad \square$$

Remark. The constraint is automatic: $\binom{k-1}{r-1} = 0$ for $k < r$, so the pmf correctly vanishes when there are fewer than r trials. Setting $r = 1$ gives $\binom{k-1}{0}p(1-p)^{k-1} = (1-p)^{k-1}p$, the geometric pmf of §4 — the negative binomial really is the geometric with the success target raised from 1 to r.

Example 4.5.3 (Door-to-door survey: third willing household)**.** A surveyor knocks on doors one at a time. Each household independently agrees to take part with probability $p = 0.4$. Let X be the number of doors knocked up to and including the third household that agrees. Find $\mathbb{P}(X = 7)$.

Which rule, and why? We fix the success target ($r = 3$ agreements) and ask for the random number of trials (doors). The trial count varies while the success count is pinned, so this is negative binomial, not binomial. It is not geometric because we want the *third* success, not the first.

Solution.

Step 1: identify the parameters. Each door is an independent trial with $p = 0.4$, $1-p = 0.6$, and we wait for the $r = 3$rd success, so $X \sim \text{NegBin}(3, 0.4)$.

Step 2: apply the pmf at $k = 7$. The 7th door is the third agreement, so the first 6 doors contain exactly 2 agreements and 4 refusals:

$$p_X(7) = \binom{7-1}{3-1}(0.4)^3(0.6)^{7-3} = \binom{6}{2}(0.4)^3(0.6)^4.$$

Step 3: evaluate. Here $\binom{6}{2} = 15$, $(0.4)^3 = 0.064$, and $(0.6)^4 = 0.1296$:

$$p_X(7) = 15 \cdot 0.064 \cdot 0.1296 = 0.124416 \approx 0.1244.$$

Example 4.5.4 (Quality lab: second defective unit)**.** A lab tests units off a line one at a time; each is independently defective with probability $p = 0.25$. Let X be the number of units tested up to and including the second defective. Find $\mathbb{P}(X = 5)$.

Which rule, and why? The success of interest is "defective," the target is $r = 2$ of them, and we want the trial on which the second arrives. Trial count random, success count fixed: $\text{NegBin}(2, 0.25)$.

Solution.

Step 1: parameters. $r = 2$, $p = 0.25$, $1 - p = 0.75$, so $X \sim \text{NegBin}(2, 0.25)$.

Step 2: pmf at $k = 5$. The fifth unit is the second defective, so among the first four units there is exactly one defective:

$$p_X(5) = \binom{4}{1}(0.25)^2(0.75)^3 = 4 \cdot 0.0625 \cdot 0.421875.$$

Step 3: evaluate.

$$p_X(5) = 4 \cdot 0.0625 \cdot 0.421875 = 0.105469 \approx 0.1055.$$

Example 4.5.5 (Basketball: fourth made free throw)**.** A player makes each free throw independently with probability $p = 0.6$. Let X be the number of free throws attempted up to and including the fourth make. Find $\mathbb{P}(X = 6)$.

Which rule, and why? Fixed target of $r = 4$ makes, random number of attempts: negative binomial with $p = 0.6$. Not binomial, because the number of attempts is what we are solving for.

Solution.

Step 1: parameters. $r = 4$, $p = 0.6$, $1 - p = 0.4$, so $X \sim \text{NegBin}(4, 0.6)$.

Step 2: pmf at $k = 6$. The sixth attempt is the fourth make, so the first five attempts hold exactly three makes and two misses:

$$p_X(6) = \binom{5}{3}(0.6)^4(0.4)^2 = 10 \cdot 0.1296 \cdot 0.16.$$

Step 3: evaluate.

$$p_X(6) = 10 \cdot 0.1296 \cdot 0.16 = 0.20736 \approx 0.2074.$$

Example 4.5.6 (Rare-allele screen: third carrier)**.** Volunteers are genotyped one at a time; each independently carries a rare allele with probability $p = 0.2$. Let X be the number genotyped up to and including the third carrier. Find $\mathbb{P}(X = 10)$.

Which rule, and why? Target $r = 3$ carriers, random number screened: $\text{NegBin}(3, 0.2)$. The small p pushes the typical wait out to many trials, which is exactly the regime the negative binomial handles.

Solution.

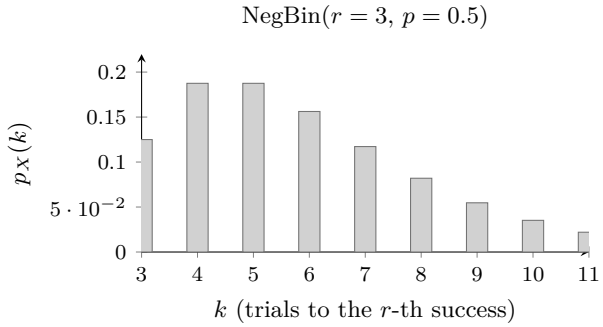
Step 1: parameters. $r = 3$, $p = 0.2$, $1 - p = 0.8$, so $X \sim \text{NegBin}(3, 0.2)$.

Step 2: pmf at $k = 10$. The tenth volunteer is the third carrier, so the first nine contain exactly two carriers and seven non-carriers:

$$p_X(10) = \binom{9}{2}(0.2)^3(0.8)^7 = 36 \cdot 0.008 \cdot 0.2097152.$$

Step 3: evaluate.

$$p_X(10) = 36 \cdot 0.008 \cdot 0.2097152 = 0.060398 \approx 0.0604.$$



The bars start at $k = r = 3$ (you cannot reach 3 successes in fewer than 3 trials), rise to a peak, then taper. Compare with the geometric ($r = 1$), which peaks at its very first bar and only decreases: stacking r geometrics shifts the mass to the right and rounds off the front into a hump.

4.5.2 Cumulative and tail probabilities

There is no simple closed form for the negative binomial cdf, so $\mathbb{P}(X \le m)$ and tail probabilities are computed by summing the pmf over the relevant values of k. Use the complement $\mathbb{P}(X > m) = 1 - \mathbb{P}(X \le m)$ to keep the sums short.

Example 4.5.7 (Survey within budget)**.** With $p = 0.4$ per household as before, the surveyor needs $r = 2$ agreements and has the budget to knock on at most 4 doors. What is the probability the target is met within budget, i.e. $\mathbb{P}(X \le 4)$ for $X \sim \text{NegBin}(2, 0.4)$?

Which rule, and why? "Within budget" is a cumulative event over the negative binomial variable. Sum the pmf from the smallest possible value $k = r = 2$ up to $k = 4$.

Solution.

Step 1: list the terms. The second success can land on trial 2, 3, or 4:

$$p_X(2) = \binom{1}{1}(0.4)^2(0.6)^0 = 0.16, \quad p_X(3) = \binom{2}{1}(0.4)^2(0.6)^1 = 0.192,$$

$$p_X(4) = \binom{3}{1}(0.4)^2(0.6)^2 = 3 \cdot 0.16 \cdot 0.36 = 0.1728.$$

Step 2: add.

$$\mathbb{P}(X \le 4) = 0.16 + 0.192 + 0.1728 = 0.5248 \approx 0.5248.$$

Example 4.5.8 (Overrun risk via the complement)**.** Same surveyor, $\text{NegBin}(2, 0.4)$. What is the probability it takes *more than* 6 doors to collect the two agreements, $\mathbb{P}(X > 6)$?

Which rule, and why? A right-tail event for a negative binomial. The complement $\mathbb{P}(X > 6) = 1 - \mathbb{P}(X \le 6)$ turns an infinite tail into a finite sum of five terms.

Solution.

Step 1: cumulate to 6. Adding $p_X(2)$ through $p_X(6)$ for $\text{NegBin}(2, 0.4)$,

$$0.16 + 0.192 + 0.1728 + 0.13824 + 0.0995328 = 0.76672.$$

(The last two are $p_X(5) = \binom{4}{1}(0.4)^2(0.6)^3 = 0.13824$ and $p_X(6) = \binom{5}{1}(0.4)^2(0.6)^4 = 0.0995328$.)

Step 2: take the complement.

$$\mathbb{P}(X > 6) = 1 - 0.76672 = 0.23328 \approx 0.2333.$$

Example 4.5.9 (Hiring: filling three positions)**.** Candidates are interviewed in sequence; each independently accepts an offer with probability $p = 0.2$. The team needs $r = 3$ acceptances. What is the probability it takes *at least* 15 interviews, $\mathbb{P}(X \geq 15)$ for $X \sim \text{NegBin}(3, 0.2)$?

Which rule, and why? "At least 15" is the right tail $\mathbb{P}(X \geq 15) = 1 - \mathbb{P}(X \leq 14)$. With p small the mean wait is $r/p = 15$, so a wait of 15 or more is roughly a coin flip — a good sanity check.

Solution.

Step 1: set up the complement.

$$\mathbb{P}(X \geq 15) = 1 - \mathbb{P}(X \leq 14) = 1 - \sum_{k=3}^{14} \binom{k-1}{2} (0.2)^3 (0.8)^{k-3}.$$

Step 2: sum the 12 *terms.* Carrying out the sum,

$$\sum_{k=3}^{14} \binom{k-1}{2} (0.2)^3 (0.8)^{k-3} = 0.551949.$$

Step 3: subtract.

$$\mathbb{P}(X \geq 15) = 1 - 0.551949 = 0.448051 \approx 0.4481.$$

This sits just below 0.5, consistent with the mean wait being exactly 15.

4.5.3 Mean and variance: the negative binomial as a sum of geometrics

Theorem 4.5.10 (Decomposition into geometrics)**.** *If $X \sim \text{NegBin}(r, p)$ then $X = G_1 + G_2 + \cdots + G_r$, where the G_i are independent and each $G_i \sim \text{Geom}(p)$ counts the trials from just after the $(i-1)$-th success up to and including the i-th success.*

Proof. Cut the trial sequence at each success. Let G_1 be the number of trials to the first success, G_2 the number of further trials to the second success, and so on, with G_i counting from one past the $(i-1)$-th success through the i-th. Each G_i is the waiting time for the next success in independent trials with the same p, hence $G_i \sim \text{Geom}(p)$; and because the trials are independent, disjoint blocks of trials give independent G_i. The blocks tile the whole sequence up to the r-th success, so $X = G_1 + \cdots + G_r$. □

Corollary 4.5.11 (Mean and variance)**.** *For $X \sim \text{NegBin}(r, p)$,*

$$\mathbb{E}(X) = \frac{r}{p}, \qquad \text{Var}(X) = \frac{r(1-p)}{p^2}, \qquad \text{SD}(X) = \frac{\sqrt{r(1-p)}}{p}.$$

Proof. Each geometric has $\mathbb{E}(G_i) = 1/p$ and $\text{Var}(G_i) = (1-p)/p^2$ (§4). The mean of a sum is the sum of the means, so $\mathbb{E}(X) = r \cdot (1/p) = r/p$. Because the G_i are independent, the variance of the sum is the sum of the variances, so $\text{Var}(X) = r \cdot (1-p)/p^2 = r(1-p)/p^2$. Taking the square root gives the standard deviation. □

Remark. This is why you should never grind out $\mathbb{E}(X) = \sum_k k\, p_X(k)$ by hand. The decomposition replaces a messy series with one line of arithmetic, and it makes the scaling transparent: doubling the target r doubles both the expected number of trials and the variance.

Example 4.5.12 (Expected effort and spread for the survey)**.** With $p = 0.4$ and a target of $r = 3$ agreements ($X \sim \text{NegBin}(3, 0.4)$), find the expected number of doors and the standard deviation.

Which rule, and why? A mean/spread question — use the geometric decomposition, not the pmf.

Solution.

Step 1: mean.

$$\mathbb{E}(X) = \frac{r}{p} = \frac{3}{0.4} = 7.5 \text{ doors}.$$

Step 2: variance and standard deviation.

$$\mathrm{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3 \cdot 0.6}{0.16} = 11.25, \qquad \mathrm{SD}(X) = \sqrt{11.25} \approx 3.354.$$

So the surveyor expects about 7.5 doors, give or take roughly 3.4.

Example 4.5.13 (Rare carriers: how long a screen to expect)**.** For the rare-allele screen with $p = 0.2$ and target $r = 2$ carriers ($X \sim \mathrm{NegBin}(2, 0.2)$), find the mean, variance, and standard deviation of the number screened.

Which rule, and why? Mean/variance for a negative binomial; apply r/p and $r(1-p)/p^2$ directly.

Solution.

Step 1: mean.

$$\mathbb{E}(X) = \frac{2}{0.2} = 10.$$

Step 2: variance and SD.

$$\mathrm{Var}(X) = \frac{2 \cdot 0.8}{0.2^2} = \frac{1.6}{0.04} = 40, \qquad \mathrm{SD}(X) = \sqrt{40} \approx 6.325.$$

The large spread (SD over half the mean) is typical when p is small: rare successes make the wait both long and highly variable.

Example 4.5.14 (Matching mean to a design target)**.** A reliability test stops at the $r = 4$th failure. If each item independently fails with probability $p = 0.25$ ($X \sim \mathrm{NegBin}(4, 0.25)$), how many items do we expect to test, and what is the variance?

Which rule, and why? Fixed target of $r = 4$ failures, random number of items: negative binomial. The question asks for $\mathbb{E}(X)$ and $\mathrm{Var}(X)$, so use the decomposition formulas.

Solution.

Step 1: mean.

$$\mathbb{E}(X) = \frac{4}{0.25} = 16 \text{ items}.$$

Step 2: variance.

$$\mathrm{Var}(X) = \frac{4 \cdot 0.75}{0.25^2} = \frac{3}{0.0625} = 48, \qquad \mathrm{SD}(X) = \sqrt{48} \approx 6.928.$$

Example 4.5.15 (Reading off r and p from the summary numbers)**.** A negative binomial waiting time has mean 20 and variance 180. Recover r and p.

Which rule, and why? We are given $\mathbb{E}(X)$ and $\mathrm{Var}(X)$ and must invert the two formulas $\mathbb{E} = r/p$ and $\mathrm{Var} = r(1-p)/p^2$ for the two unknowns.

Solution.

Step 1: take the ratio to isolate $1 - p$. Dividing,

$$\frac{\mathrm{Var}(X)}{\mathbb{E}(X)} = \frac{r(1-p)/p^2}{r/p} = \frac{1-p}{p} = \frac{180}{20} = 9 \implies 1 - p = 9p \implies p = \tfrac{1}{10} = 0.1.$$

Step 2: solve for r. From $\mathbb{E}(X) = r/p = 20$,

$$r = 20\,p = 20 \cdot 0.1 = 2.$$

$$X \sim \mathrm{NegBin}(r = 2,\ p = 0.1).$$

(Check: $\mathrm{Var} = 2 \cdot 0.9/0.01 = 180$, as given.)

4.5.4 Putting it together

Example 4.5.16 (Full profile of a clinical-recruitment wait)**.** A trial recruits eligible patients one at a time; each independently consents with probability $p = 0.35$, and the trial needs $r = 3$ consents. Let $X \sim \text{NegBin}(3, 0.35)$. Find (a) the expected number approached and the standard deviation, (b) $\mathbb{P}(X = 8)$, and (c) the chance it takes *at least* 12 approaches, $\mathbb{P}(X \geq 12)$.

Which rule, and why? One fixed success target ($r = 3$ consents) with a random number of trials — negative binomial throughout. Use the decomposition for (a), the pmf for (b), and the complement of a cumulative sum for (c).

Solution.

Step 1 (a): mean and spread.

$$\mathbb{E}(X) = \frac{3}{0.35} \approx 8.571, \qquad \text{Var}(X) = \frac{3 \cdot 0.65}{0.35^2} = \frac{1.95}{0.1225} \approx 15.918,$$

$$\text{SD}(X) = \sqrt{15.918} \approx 3.990.$$

Step 2 (b): pmf at $k = 8$. The eighth patient is the third consent, so the first seven hold exactly two consents and five declines:

$$p_X(8) = \binom{7}{2}(0.35)^3(0.65)^5 = 21 \cdot 0.042875 \cdot 0.1160290625 \approx 0.10447,$$

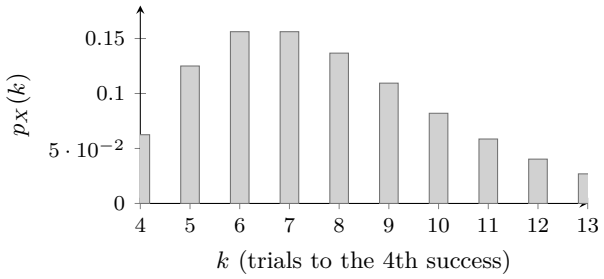
so $\mathbb{P}(X = 8) \approx 0.1045$.

Step 3 (c): right tail by complement.

$$\mathbb{P}(X \geq 12) = 1 - \sum_{k=3}^{11} \binom{k-1}{2}(0.35)^3(0.65)^{k-3} = 1 - 0.799871 = 0.200129,$$

so $\mathbb{P}(X \geq 12) \approx 0.2001$. About one recruitment drive in five runs to 12 or more approaches, even though the average is under 9 — the long right tail of the negative binomial at work.

NegBin($r = 4$, $p = 0.5$): a larger target shifts and spreads the hump



Raising the target to $r = 4$ (same $p = 0.5$) moves the first possible value out to $k = 4$ and pushes the peak to the right, near the mean $r/p = 8$. Each extra required success adds one more geometric to the pile, so the hump drifts right and broadens — the visual signature of $\mathbb{E}(X) = r/p$ and $\text{Var}(X) = r(1-p)/p^2$ both growing with r.

4.6 The Gamma Distribution

The gamma distribution measures a **continuous waiting time**: how long you wait until the α-th event of a process whose events arrive at a constant average rate λ. It is the continuous twin of the negative binomial (which *counts trials* to the r-th success), and it contains the exponential ($\alpha = 1$, the wait to the *first* event)

and the chi-square ($\alpha = n/2,\ \lambda = \frac{1}{2}$) as special cases. Its two parameters are a **shape** $\alpha > 0$, which controls how many events you are waiting for, and a **rate** $\lambda > 0$, the events-per-unit-time of the underlying Poisson process. Before the density itself we need the gamma function, which supplies the normalising constant.

Which rule of this section does the problem need?

- Asked to **evaluate Γ at a number** (or simplify a Γ expression)? $\Rightarrow$ use the **recursion** $\Gamma(\alpha+1) = \alpha\,\Gamma(\alpha)$ together with $\Gamma(n) = (n-1)!$ and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$. No probability is involved yet.
- A **continuous waiting time until the α-th event** of a constant-rate (Poisson) process, or a **sum of α independent exponential** waits with common rate λ? $\Rightarrow$ **gamma**, $X \sim \text{Gamma}(\alpha, \lambda)$, density $f_X(x) = \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$. Mean α/λ, variance α/λ^2.
- The wait to the **very first** event (or any single memoryless lifetime)? $\Rightarrow$ **exponential**, the $\alpha = 1$ case, with the clean closed forms $f(x) = \lambda e^{-\lambda x}$ and $\mathbb{P}(X > t) = e^{-\lambda t}$.
- A **sum of squares of independent standard normals**, or a problem that hands you "χ^2 with n degrees of freedom"? $\Rightarrow$ **chi-square**, which is Gamma$(n/2, \frac{1}{2})$; mean n, variance $2n$.
- **Why competing tools do not apply.** If the quantity is a discrete *count* of events in a fixed window, that is **Poisson**, not gamma. If it is a discrete count of *trials* to the r-th success, that is **negative binomial**. The gamma is for a *continuous time*, taking any positive real value.

One integral does almost everything. Every gamma computation reduces to recognising $\int_0^\infty u^{c-1} e^{-su}\,\mathrm{d}u = \Gamma(c)/s^c$ for $c > 0,\ s > 0$; learn this one fact and the mean, variance, and tail integrals all fall out.

4.6.1 The gamma function

Definition 4.6.1 (Gamma function)**.** For $\alpha > 0$,

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t}\,\mathrm{d}t.$$

The integral converges for every $\alpha > 0$ (near 0 the factor $t^{\alpha-1}$ is integrable, and e^{-t} kills the tail at ∞). It is the continuous interpolation of the factorial.

Theorem 4.6.2 (Key properties of Γ)**.** *For every $\alpha > 0$ and every positive integer n,*

$$\textit{(i)}\ \ \Gamma(\alpha+1) = \alpha\,\Gamma(\alpha), \qquad \textit{(ii)}\ \ \Gamma(n) = (n-1)!, \qquad \textit{(iii)}\ \ \Gamma\left(\tfrac{1}{2}\right) = \sqrt{\pi}.$$

Proof. (i) Integrate by parts in $\Gamma(\alpha+1) = \int_0^\infty t^\alpha e^{-t}\,\mathrm{d}t$ with $u = t^\alpha,\ \mathrm{d}v = e^{-t}\,\mathrm{d}t$, so $\mathrm{d}u = \alpha t^{\alpha-1}\,\mathrm{d}t$ and $v = -e^{-t}$:

$$\Gamma(\alpha+1) = \Big[-t^\alpha e^{-t}\Big]_0^\infty + \alpha \int_0^\infty t^{\alpha-1} e^{-t}\,\mathrm{d}t = 0 + \alpha\,\Gamma(\alpha).$$

The boundary term vanishes at both ends because $t^\alpha e^{-t} \to 0$ as $t \to \infty$ and equals 0 at $t = 0$. (ii) $\Gamma(1) = \int_0^\infty e^{-t}\,\mathrm{d}t = 1 = 0!$, and applying (i) repeatedly, $\Gamma(n) = (n-1)\Gamma(n-1) = \cdots = (n-1)!\,\Gamma(1) = (n-1)!$. (iii) Substitute $t = u^2,\ \mathrm{d}t = 2u\ \mathrm{d}u$: $\Gamma(\frac{1}{2}) = \int_0^\infty t^{-1/2} e^{-t}\,\mathrm{d}t = 2\int_0^\infty e^{-u^2}\,\mathrm{d}u = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$, using the standard Gaussian integral $\int_0^\infty e^{-u^2}\,\mathrm{d}u = \sqrt{\pi}/2$. □

Example 4.6.3 (Plain factorial values)**.** Evaluate $\Gamma(5)$ and $\Gamma(1)$.

Which rule, and why? Both arguments are positive integers, so the factorial identity $\Gamma(n) = (n-1)!$ applies directly; no integration is needed.

Solution.

Step 1: apply $\Gamma(n) = (n-1)!$. $\Gamma(5) = (5-1)! = 4! = 24$ and $\Gamma(1) = (1-1)! = 0! = 1$.

$$\Gamma(5) = 24, \qquad \Gamma(1) = 1.$$

Example 4.6.4 (Half-integer arguments)**.** Evaluate $\Gamma\left(\frac{5}{2}\right)$.

Which rule, and why? The argument is a half-integer, so peel factors off with the recursion $\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$ until you hit the anchor value $\Gamma(\frac{1}{2})=\sqrt{\pi}$.

Solution.

Step 1: strip one factor. $\Gamma\left(\frac{5}{2}\right)=\frac{3}{2}\,\Gamma\left(\frac{3}{2}\right)$.

Step 2: strip another. $\Gamma\left(\frac{3}{2}\right)=\frac{1}{2}\,\Gamma\left(\frac{1}{2}\right)=\frac{1}{2}\sqrt{\pi}$.

Step 3: combine. $\Gamma\left(\frac{5}{2}\right)=\frac{3}{2}\cdot\frac{1}{2}\sqrt{\pi}=\frac{3}{4}\sqrt{\pi}$.

$$\Gamma\left(\tfrac{5}{2}\right)=\tfrac{3}{4}\sqrt{\pi}\approx 1.3293.$$

Example 4.6.5 (Simplifying a ratio)**.** Simplify $\dfrac{\Gamma(8)}{\Gamma(5)}$ without computing each factorial separately.

Which rule, and why? A ratio of gamma values at integers spaced apart: telescope it with the recursion so the common tail cancels, rather than evaluating two large factorials.

Solution.

Step 1: expand the numerator down to $\Gamma(5)$. Applying $\Gamma(\alpha+1)=\alpha\Gamma(\alpha)$ three times, $\Gamma(8)=7\cdot 6\cdot 5\cdot\Gamma(5)$.

Step 2: cancel.

$$\frac{\Gamma(8)}{\Gamma(5)}=7\cdot 6\cdot 5=210.$$

Example 4.6.6 (A disguised gamma integral)**.** Evaluate $I=\displaystyle\int_0^\infty x^4e^{-3x}\,\mathrm{d}x$.

Which rule, and why? This is the master integral $\int_0^\infty u^{c-1}e^{-su}\,\mathrm{d}u=\Gamma(c)/s^c$ in disguise, not a gamma *function* value to look up: here $c-1=4$ and $s=3$.

Solution.

Step 1: match the pattern. With $c-1=4$ we have $c=5$ and $s=3$, so $I=\Gamma(5)/3^5$.

Step 2: plug in $\Gamma(5)=24$.

$$I=\frac{\Gamma(5)}{3^5}=\frac{24}{243}=\frac{8}{81}\approx 0.0988.$$

Example 4.6.7 (Reverse engineering an argument)**.** Find the positive number a for which $\Gamma(a+1)=12\,\Gamma(a-1)$.

Which rule, and why? An equation *in* a gamma argument: express everything in terms of one anchor $\Gamma(a-1)$ using the recursion, then solve the resulting polynomial.

Solution.

Step 1: lower the left side twice. $\Gamma(a+1)=a\,\Gamma(a)=a(a-1)\,\Gamma(a-1)$.

Step 2: equate and cancel $\Gamma(a-1)$. $a(a-1)=12$, i.e. $a^2-a-12=0$, so $(a-4)(a+3)=0$.

Step 3: keep the admissible root. We need $a-1>0$, so $a=4$ (the root $a=-3$ is rejected).

$$a=4.$$

4.6.2 The gamma density, mean, and variance

Definition 4.6.8 (Gamma distribution, rate form)**.** A random variable X has the **gamma distribution** with shape $\alpha>0$ and rate $\lambda>0$, written $X\sim\mathrm{Gamma}(\alpha,\lambda)$, if its density is

$$f_X(x)=\frac{\lambda^\alpha}{\Gamma(\alpha)}\,x^{\alpha-1}e^{-\lambda x},\qquad x>0,$$

and $f_X(x)=0$ for $x\le 0$. (Some books use the **scale** $\theta=1/\lambda$; we use the rate λ throughout, so a larger λ means events arrive faster and waits are shorter.)

Proposition 4.6.9 (It is a valid density)**.** $\displaystyle\int_0^\infty f_X(x)\,\mathrm{d}x = 1.$

Proof. Substitute $u = \lambda x,\ \mathrm{d}x = \mathrm{d}u/\lambda$:

$$\int_0^\infty \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\,\mathrm{d}x = \frac{\lambda^\alpha}{\Gamma(\alpha)} \int_0^\infty \left(\frac{u}{\lambda}\right)^{\alpha-1} e^{-u}\,\frac{\mathrm{d}u}{\lambda} = \frac{1}{\Gamma(\alpha)} \int_0^\infty u^{\alpha-1} e^{-u}\,\mathrm{d}u = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1. \qquad \square$$

Theorem 4.6.10 (Mean and variance)**.** *If* $X \sim \mathrm{Gamma}(\alpha, \lambda)$ *then*

$$\mathbb{E}(X) = \frac{\alpha}{\lambda}, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2}, \qquad \textit{so} \qquad \mathrm{SD}(X) = \frac{\sqrt{\alpha}}{\lambda}.$$

Proof. Both moments come from the master integral $\int_0^\infty u^{c-1} e^{-su}\,\mathrm{d}u = \Gamma(c)/s^c$. For the mean (here $c = \alpha+1$, $s = \lambda$):

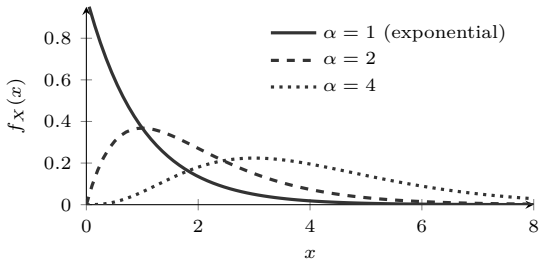
$$\mathbb{E}(X) = \int_0^\infty x\,\frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}\,\mathrm{d}x = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+1)}{\lambda^{\alpha+1}} = \frac{\Gamma(\alpha+1)}{\lambda\,\Gamma(\alpha)} = \frac{\alpha}{\lambda},$$

using $\Gamma(\alpha+1) = \alpha\Gamma(\alpha)$. For the second moment (here $c = \alpha + 2$):

$$\mathbb{E}(X^2) = \frac{\lambda^\alpha}{\Gamma(\alpha)} \cdot \frac{\Gamma(\alpha+2)}{\lambda^{\alpha+2}} = \frac{(\alpha+1)\alpha\,\Gamma(\alpha)}{\lambda^2\,\Gamma(\alpha)} = \frac{\alpha(\alpha+1)}{\lambda^2}.$$

Hence $\mathrm{Var}(X) = \mathbb{E}(X^2) - \mathbb{E}(X)^2 = \dfrac{\alpha(\alpha+1)}{\lambda^2} - \dfrac{\alpha^2}{\lambda^2} = \dfrac{\alpha}{\lambda^2}.$ $\square$

Remark. The shape α sets the silhouette of the curve and the rate λ only rescales the horizontal axis. For $\alpha \le 1$ the density is strictly decreasing with a spike (or vertical asymptote when $\alpha < 1$) at 0; for $\alpha > 1$ it rises from 0, peaks at the mode $x = (\alpha - 1)/\lambda$, then decays. As α grows the curve becomes more symmetric and bell-like — a preview of the central limit theorem, since a $\mathrm{Gamma}(\alpha, \lambda)$ with integer α is a sum of α independent exponentials.



Gamma densities at rate $\lambda = 1$ for shapes $\alpha = 1, 2, 4$. Larger α shifts mass right (mean α/λ) and makes the curve more symmetric.

Example 4.6.11 (Reading off mean, variance, and a probability)**.** A microbrewery's bottling line jams at random; jams occur as a Poisson process at rate $\lambda = 0.5$ per hour. Let X be the time (in hours) until the *third* jam. Find $\mathbb{E}(X)$, $\mathrm{SD}(X)$, and $\mathbb{P}(X \le 4)$.

Which rule, and why? "Time until the third event" of a constant-rate process is gamma with shape equal to the event number, $\alpha = 3$, and rate $\lambda = 0.5$. It is continuous time, not a count, so Poisson does not apply; we wait for the 3rd event, not the 1st, so it is not exponential.

Solution.

Step 1: identify parameters. $X \sim \mathrm{Gamma}(\alpha = 3, \lambda = 0.5)$.

Step 2: mean and SD from the formulas.

$$\mathbb{E}(X) = \frac{\alpha}{\lambda} = \frac{3}{0.5} = 6 \text{ hours}, \qquad \mathrm{SD}(X) = \frac{\sqrt{\alpha}}{\lambda} = \frac{\sqrt{3}}{0.5} = 2\sqrt{3} \approx 3.464 \text{ hours}.$$

Step 3: a probability via the Poisson-link cdf. For integer shape, $\mathbb{P}(X \le t) = 1 - \sum_{k=0}^{\alpha-1} e^{-\lambda t}\dfrac{(\lambda t)^k}{k!}$ (derived below). With $\lambda t = 0.5 \cdot 4 = 2$ and $\alpha = 3$,

$$\mathbb{P}(X \le 4) = 1 - e^{-2}\Big(1 + 2 + \tfrac{2^2}{2}\Big) = 1 - 5e^{-2} = 1 - 5(0.135335) \approx 0.3233.$$

Example 4.6.12 (Solving for the rate from the mean)**.** Calls to a help desk arrive as a Poisson process. An operator must field 5 calls before a break; the average wait until the 5th call is 20 minutes. Treating the wait as gamma, find the arrival rate λ and the variance of the wait.

Which rule, and why? Waiting time to the 5th event ⇒ gamma with $\alpha = 5$; the mean α/λ is given, so invert it for λ.

Solution.

Step 1: use the mean. $\mathbb{E}(X) = \alpha/\lambda = 20$ with $\alpha = 5$ gives $\lambda = 5/20 = 0.25$ calls per minute.

Step 2: variance.

$$\mathrm{Var}(X) = \frac{\alpha}{\lambda^2} = \frac{5}{0.25^2} = \frac{5}{0.0625} = 80 \text{ min}^2, \qquad \mathrm{SD}(X) = \sqrt{80} \approx 8.944 \text{ min}.$$

$$\lambda = 0.25 \text{ per min}, \qquad \mathrm{Var}(X) = 80 \text{ min}^2.$$

Example 4.6.13 (A normalising constant)**.** A continuous lifetime T (years) has density $f_T(t) = c\,t^2 e^{-t/3}$ for $t > 0$. Find c, then $\mathbb{E}(T)$.

Which rule, and why? The shape $t^2 e^{-t/3}$ matches the gamma kernel $x^{\alpha-1}e^{-\lambda x}$ with $\alpha - 1 = 2$ and $\lambda = 1/3$; so c must be the gamma normaliser $\lambda^\alpha/\Gamma(\alpha)$.

Solution.

Step 1: read off the parameters. $\alpha - 1 = 2 \Rightarrow \alpha = 3$, and $\lambda = 1/3$. Thus $T \sim \mathrm{Gamma}(3, \frac{1}{3})$.

Step 2: the constant must be the density's leading factor.

$$c = \frac{\lambda^\alpha}{\Gamma(\alpha)} = \frac{(1/3)^3}{\Gamma(3)} = \frac{1/27}{2!} = \frac{1}{54}.$$

Step 3: mean. $\mathbb{E}(T) = \alpha/\lambda = 3/(1/3) = 9$ years.

$$c = \frac{1}{54}, \qquad \mathbb{E}(T) = 9 \text{ years}.$$

Example 4.6.14 (Half-integer shape)**.** A measurement W has the $\mathrm{Gamma}\big(\frac{5}{2}, 2\big)$ distribution. Find $\mathbb{E}(W)$ and the value of the density at its mode.

Which rule, and why? Direct use of the gamma formulas with a half-integer shape; the only twist is needing $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$ and the mode formula $(\alpha - 1)/\lambda$ for $\alpha > 1$.

Solution.

Step 1: mean. $\mathbb{E}(W) = \alpha/\lambda = (5/2)/2 = 5/4 = 1.25$.

Step 2: locate the mode. $x^\star = \dfrac{\alpha - 1}{\lambda} = \dfrac{3/2}{2} = 0.75$.

Step 3: evaluate the density there. With $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi} \approx 1.329340$ and $\lambda^\alpha = 2^{5/2} = 4\sqrt{2} \approx 5.656854$,

$$f_W(0.75) = \frac{\lambda^\alpha}{\Gamma(\alpha)}(x^\star)^{\alpha-1}e^{-\lambda x^\star} = \frac{5.656854}{1.329340}\,(0.75)^{3/2}e^{-1.5} \approx 4.2554 \cdot 0.649519 \cdot 0.223130 \approx 0.6167.$$

4.6.3 Gamma as a waiting time and as a sum of exponentials

Theorem 4.6.15 (Sum of exponentials)**.** *If $T_1, \dots, T_r$ are independent, each* $\text{Exp}(\lambda)$ *(that is,* $\text{Gamma}(1, \lambda)$*), then*

$$S = T_1 + \cdots + T_r \ \sim \ \text{Gamma}(r, \lambda).$$

Consequently the wait to the r-th event of a rate-λ Poisson process — the sum of r independent exponential interarrival gaps — is $\text{Gamma}(r, \lambda)$*.*

Proof. It suffices to show $T_1 + T_2 \sim \text{Gamma}(2, \lambda)$ when each is $\text{Exp}(\lambda)$; the general case follows by induction (adding one more independent $\text{Exp}(\lambda)$ raises the shape by 1). For $s > 0$, convolve:

$$f_{T_1+T_2}(s) = \int_0^s \lambda e^{-\lambda u}\, \lambda e^{-\lambda(s-u)}\, \mathrm{d}u = \lambda^2 e^{-\lambda s} \int_0^s \mathrm{d}u = \lambda^2 s\, e^{-\lambda s} = \frac{\lambda^2}{\Gamma(2)} s^{2-1} e^{-\lambda s},$$

which is exactly the $\text{Gamma}(2, \lambda)$ density. The inductive step convolves a $\text{Gamma}(k, \lambda)$ with a $\text{Gamma}(1, \lambda)$ and reproduces a $\text{Gamma}(k+1, \lambda)$ by the same calculation. □

Corollary 4.6.16 (Mean and variance, the easy way)**.** *Because $S = \sum_{i=1}^{r} T_i$ with each $T_i \sim \text{Exp}(\lambda)$ having $\mathbb{E}(T_i) = 1/\lambda$ and $\text{Var}(T_i) = 1/\lambda^2$, linearity and independence give $\mathbb{E}(S) = r/\lambda$ and $\text{Var}(S) = r/\lambda^2$, matching the integral computation with $\alpha = r$.*

Theorem 4.6.17 (Gamma–Poisson link (cdf for integer shape))**.** *Let $N(t)$ be the number of events by time t in a Poisson process of rate λ, so $N(t) \sim \text{Poisson}(\lambda t)$, and let X be the time of the r-th event, $X \sim \text{Gamma}(r, \lambda)$ with r a positive integer. The events "$X \le t$" and "$N(t) \ge r$" are the* same *event, hence*

$$F_X(t) = \mathbb{P}(X \le t) = \mathbb{P}\big(N(t) \ge r\big) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t} \frac{(\lambda t)^k}{k!}.$$

Proof. The r-th event has happened by time t if and only if at least r events have occurred by time t: $\{X \le t\} = \{N(t) \ge r\}$. Taking probabilities and using the Poisson pmf with mean λt, $\mathbb{P}(N(t) \ge r) = 1 - \mathbb{P}(N(t) \le r-1) = 1 - \sum_{k=0}^{r-1} e^{-\lambda t}(\lambda t)^k/k!$. □

Example 4.6.18 (A bus and the third passenger)**.** Passengers reach a kiosk as a Poisson process at rate $\lambda = 2$ per minute. Let X be the time until the third passenger arrives. Find $\mathbb{P}(X > 3)$ two ways: by the Poisson link and as a sanity check on the parameters.

Which rule, and why? "Time until the third arrival" is $\text{Gamma}(3, 2)$. Since the shape is an integer we may convert the gamma tail into a Poisson sum about $N(3) \sim \text{Poisson}(\lambda t = 6)$, which avoids integrating the density.

Solution.

Step 1: translate the tail. $\{X > 3\} = \{N(3) \le 2\}$ with $N(3) \sim \text{Poisson}(6)$.

Step 2: sum the Poisson pmf.

$$\mathbb{P}(X > 3) = e^{-6}\Big(1 + 6 + \tfrac{6^2}{2}\Big) = e^{-6}(1 + 6 + 18) = 25\, e^{-6}.$$

Step 3: numeric value. With $e^{-6} = 0.00247875$,

$$\mathbb{P}(X > 3) = 25(0.00247875) \approx 0.0620.$$

Example 4.6.19 (Reliability: spare components)**.** A sensor lasts an $\text{Exp}(\lambda)$ time with mean 1.5 years; when it dies it is instantly replaced by an identical spare, and there are 2 spares (so 3 sensors are used in succession). Let X be the total time the position stays manned. Find the distribution of X, its mean, and $\mathbb{P}(X < 3)$.

Which rule, and why? X is a *sum of* 3 *independent exponential lifetimes* with a common rate, which by the sum-of-exponentials theorem is gamma with shape 3. We do not use the negative binomial: nothing here is counted in discrete trials — the lifetimes are continuous.

Solution.

Step 1: parameters. Each lifetime has mean $1/\lambda = 1.5$, so $\lambda = 2/3$ per year, and $X \sim \text{Gamma}(3, \frac{2}{3})$.

Step 2: mean. $\mathbb{E}(X) = \alpha/\lambda = 3/(2/3) = 4.5$ years.

Step 3: $\mathbb{P}(X < 3)$ *via the Poisson link.* Here $\lambda t = \frac{2}{3} \cdot 3 = 2$, $r = 3$:

$$\mathbb{P}(X < 3) = 1 - e^{-2}\Big(1 + 2 + \tfrac{2^2}{2}\Big) = 1 - 5e^{-2} = 1 - 5(0.135335) \approx 0.3233.$$

Example 4.6.20 (Pooling two gamma waits at the same rate)**.** At a clinic, the wait for the first available nurse is Gamma$(2, 1)$ hours and, independently, the subsequent wait for a doctor is Gamma$(3, 1)$ hours (both rates equal 1). Find the distribution, mean, and variance of the total wait X.

Which rule, and why? Sums of *independent gammas with a common rate* add their shapes: $\text{Gamma}(\alpha_1, \lambda) + \text{Gamma}(\alpha_2, \lambda) = \text{Gamma}(\alpha_1 + \alpha_2, \lambda)$ (each gamma is a bundle of exponentials at the same rate, so the bundles merge). This fails if the rates differ.

Solution.

Step 1: add shapes. $X \sim \text{Gamma}(2 + 3,\, 1) = \text{Gamma}(5, 1)$.

Step 2: moments.

$$\mathbb{E}(X) = \frac{\alpha}{\lambda} = \frac{5}{1} = 5 \text{ hours}, \qquad \text{Var}(X) = \frac{\alpha}{\lambda^2} = \frac{5}{1} = 5 \text{ hours}^2.$$

$$X \sim \text{Gamma}(5, 1), \quad \mathbb{E}(X) = 5, \quad \text{Var}(X) = 5.$$

Example 4.6.21 (At least one more event)**.** Earthquakes above a threshold strike a region as a Poisson process at rate $\lambda = 0.4$ per year. Let X be the time to the 2nd such quake. Find $\mathbb{P}(X \le 5)$.

Which rule, and why? Time to the 2nd event $\Rightarrow$ Gamma$(2, 0.4)$; integer shape, so use the Poisson link with $N(5) \sim \text{Poisson}(\lambda t = 2)$.

Solution.

Step 1: set up. $\{X \le 5\} = \{N(5) \ge 2\}$, $N(5) \sim \text{Poisson}(2)$.

Step 2: complement.

$$\mathbb{P}(X \le 5) = 1 - \mathbb{P}(N(5) \le 1) = 1 - e^{-2}(1 + 2) = 1 - 3e^{-2}.$$

Step 3: numeric. With $e^{-2} = 0.135335$, $\mathbb{P}(X \le 5) = 1 - 3(0.135335) \approx 0.5940$.

$$\mathbb{P}(X \le 5) \approx 0.5940.$$

4.6.4 Special case 1: the exponential ($\alpha = 1$)

Definition 4.6.22 (Exponential distribution)**.** The exponential distribution is the gamma with shape $\alpha = 1$: $X \sim \text{Exp}(\lambda) = \text{Gamma}(1, \lambda)$ has density and tail

$$f_X(x) = \lambda e^{-\lambda x} \ (x > 0), \qquad F_X(x) = 1 - e^{-\lambda x}, \qquad \mathbb{P}(X > t) = e^{-\lambda t} \ (t \ge 0),$$

with $\mathbb{E}(X) = 1/\lambda$, $\text{Var}(X) = 1/\lambda^2$. It models the wait to the *first* event of a rate-λ Poisson process and is **memoryless**: $\mathbb{P}(X > s + t \mid X > s) = \mathbb{P}(X > t)$.

Example 4.6.23 (Tail and memorylessness)**.** The time between machine breakdowns is Exp(λ) with mean 40 hours. (a) Find the chance the machine runs more than 60 hours without a breakdown. (b) Given it has already run 30 hours, find the chance it lasts at least 60 more.

Which rule, and why? A single continuous waiting time with a constant hazard ⇒ exponential, the $\alpha = 1$ gamma; the "given it already survived" twist is exactly what the memoryless property handles.

Solution.

Step 1: rate. $\mathbb{E}(X) = 1/\lambda = 40 \Rightarrow \lambda = 1/40 = 0.025$ per hour.

Step 2: part (a), a plain tail.

$$\mathbb{P}(X > 60) = e^{-\lambda \cdot 60} = e^{-60/40} = e^{-1.5} \approx 0.2231.$$

Step 3: part (b), use memorylessness.

$$\mathbb{P}(X > 30 + 60 \mid X > 30) = \mathbb{P}(X > 60) = e^{-1.5} \approx 0.2231.$$

The extra 30 hours of survival are forgotten; the answer equals part (a).

Example 4.6.24 (Comparing two exponential clocks)**.** Two independent servers have failure times $X \sim \text{Exp}(\lambda_1)$ and $Y \sim \text{Exp}(\lambda_2)$ with $\lambda_1 = 1/5$ and $\lambda_2 = 1/3$ (per day). Find $\mathbb{P}(X < Y)$, the chance server 1 fails first.

Which rule, and why? Each lifetime is a single memoryless wait ⇒ exponential. "Which of two independent exponentials is smaller" has the clean rule $\mathbb{P}(X < Y) = \lambda_1/(\lambda_1 + \lambda_2)$.

Solution.

Step 1: the competing-risks formula. For independent $X \sim \text{Exp}(\lambda_1)$, $Y \sim \text{Exp}(\lambda_2)$, $\mathbb{P}(X < Y) = \dfrac{\lambda_1}{\lambda_1 + \lambda_2}$ (integrate $\int_0^\infty \lambda_1 e^{-\lambda_1 x} e^{-\lambda_2 x}\, \mathrm{d}x$ over the region $Y > x$).

Step 2: plug in.

$$\mathbb{P}(X < Y) = \frac{1/5}{1/5 + 1/3} = \frac{1/5}{8/15} = \frac{3}{8} = 0.375.$$

Example 4.6.25 (Quantile / percentile)**.** Battery life is $\text{Exp}(\lambda)$ with mean 12 hours. The manufacturer wants to advertise a "guaranteed minimum" time t_0 that 90% of batteries exceed. Find t_0.

Which rule, and why? A single exponential lifetime, and we want the time with a specified survival probability — invert the closed-form tail $\mathbb{P}(X > t) = e^{-\lambda t}$.

Solution.

Step 1: rate. $\lambda = 1/12$ per hour.

Step 2: set the survival probability and solve. $\mathbb{P}(X > t_0) = e^{-t_0/12} = 0.90 \Rightarrow -t_0/12 = \ln 0.90 \Rightarrow t_0 = -12 \ln 0.90$.

Step 3: evaluate ($\ln 0.90 = -0.105361$)*.*

$$t_0 = -12(-0.105361) \approx 1.264 \text{ hours.}$$

Example 4.6.26 (From exponential gaps back to a gamma)**.** Customers enter a shop as a Poisson process; the gaps between arrivals are independent $\text{Exp}(\lambda)$ with mean 4 minutes. What is the distribution of the time from opening until the 4th customer, and what is its standard deviation?

Which rule, and why? Independent exponential gaps summed ⇒ gamma. The 4th arrival is the sum of 4 gaps, so shape $\alpha = 4$ at the same rate.

Solution.

Step 1: rate of one gap. Mean gap $1/\lambda = 4 \Rightarrow \lambda = 0.25$ per minute.

Step 2: the sum. Time to the 4th customer $= T_1 + T_2 + T_3 + T_4 \sim \text{Gamma}(4, 0.25)$.

Step 3: standard deviation.

$$\text{SD} = \frac{\sqrt{\alpha}}{\lambda} = \frac{\sqrt{4}}{0.25} = \frac{2}{0.25} = 8 \text{ minutes.}$$

4.6.5 Special case 2: the chi-square (Gamma$(n/2, \frac{1}{2})$)

Definition 4.6.27 (Chi-square distribution)**.** The **chi-square** distribution with n degrees of freedom, χ^2_n, is the gamma with shape $\alpha = n/2$ and rate $\lambda = \frac{1}{2}$:

$$f(x) = \frac{(1/2)^{n/2}}{\Gamma(n/2)}\, x^{n/2-1} e^{-x/2}, \qquad x > 0.$$

Its mean and variance follow from the gamma formulas:

$$\mathbb{E} = \frac{\alpha}{\lambda} = \frac{n/2}{1/2} = n, \qquad \mathrm{Var} = \frac{\alpha}{\lambda^2} = \frac{n/2}{1/4} = 2n.$$

The key fact (justified in a later chapter) is that if $Z_1, \ldots, Z_n$ are independent standard normals, then $Z_1^2 + \cdots + Z_n^2 \sim \chi^2_n$; the sum of squares of normals is a gamma.

Example 4.6.28 (Mean and variance from the link)**.** Confirm directly that $X \sim \chi^2_6$ has $\mathbb{E}(X) = 6$ and $\mathrm{Var}(X) = 12$ using the gamma parameters.

Which rule, and why? "χ^2 with n degrees of freedom" is precisely Gamma$(n/2, \frac{1}{2})$, so feed $\alpha = n/2$ and $\lambda = \frac{1}{2}$ into the gamma mean/variance formulas.

Solution.

Step 1: parameters for $n = 6$. $\alpha = n/2 = 3$, $\lambda = \frac{1}{2}$.

Step 2: moments.

$$\mathbb{E}(X) = \frac{\alpha}{\lambda} = \frac{3}{1/2} = 6, \qquad \mathrm{Var}(X) = \frac{\alpha}{\lambda^2} = \frac{3}{1/4} = 12.$$

These match the general $\mathbb{E} = n$, $\mathrm{Var} = 2n$ with $n = 6$.

Example 4.6.29 (One squared standard normal)**.** Let $Z \sim \mathrm{N}(0, 1)$ and $X = Z^2$. Identify the distribution of X and give its mean and variance.

Which rule, and why? The square of a *single* standard normal is the $n = 1$ case of the sum-of-squares fact, i.e. $\chi^2_1 = \mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$. The half-integer shape $\alpha = \frac{1}{2}$ is why $\Gamma(\frac{1}{2}) = \sqrt{\pi}$ shows up in its density.

Solution.

Step 1: name it. $X = Z^2 \sim \chi^2_1 = \mathrm{Gamma}(\frac{1}{2}, \frac{1}{2})$.

Step 2: moments via $n = 1$.

$$\mathbb{E}(X) = n = 1, \qquad \mathrm{Var}(X) = 2n = 2.$$

(Sanity check: $\mathbb{E}(Z^2) = \mathrm{Var}(Z) = 1$, consistent with $\mathbb{E}(X) = 1$.)

Example 4.6.30 (Adding independent chi-squares)**.** A statistic is $W = X_1 + X_2$, where $X_1 \sim \chi^2_4$ and $X_2 \sim \chi^2_7$ are independent. Find the distribution of W and $\mathbb{P}(W \le 11)$ stated as a gamma-tail probability.

Which rule, and why? Chi-squares are gammas at the common rate $\frac{1}{2}$, so independent ones add their shapes (equivalently, their degrees of freedom add). With integer shape we can write the tail via the Poisson link evaluated at $\lambda t = \frac{1}{2} \cdot t$.

Solution.

Step 1: add degrees of freedom. $\alpha_1 = 2$, $\alpha_2 = 7/2$, sum $= 11/2$, so $W \sim \mathrm{Gamma}(11/2, \frac{1}{2}) = \chi^2_{11}$ ($4 + 7 = 11$ d.f.).

Step 2: mean as a check. $\mathbb{E}(W) = n = 11$, matching $4 + 7$.

Step 3: the requested probability. W has non-integer shape $11/2$, so there is no finite Poisson sum; the probability is the gamma integral

$$\mathbb{P}(W \le 11) = \int_0^{11} \frac{(1/2)^{11/2}}{\Gamma(11/2)}\, w^{9/2} e^{-w/2}\, \mathrm{d}w,$$

which equals $\mathbb{P}(\chi^2_{11} \le 11) \approx 0.5430$ (from a chi-square table; note 11 is the mean, so the value being just above 0.5 is sensible because the chi-square is right-skewed). We report the table value and the digits used: $\mathbb{P}(\chi^2_{11} \le 11) = 0.5430$.

Example 4.6.31 (Scaling a chi-square back to a general gamma)**.** Show that if $X \sim \chi^2_n$ then $Y = X/(2\lambda) \sim \text{Gamma}(n/2, \lambda)$ for any rate $\lambda > 0$, and use this to find $\mathbb{E}(Y)$ when $n = 8$ and $\lambda = 2$.

Which rule, and why? Rescaling a gamma by a positive constant keeps it gamma and only changes the rate: multiplying the variable by c divides the rate by c. This connects the fixed-rate chi-square to an arbitrary-rate gamma.

Solution.

Step 1: rescaling rule. If $X \sim \text{Gamma}(\alpha, \lambda_0)$ and $Y = cX$ with $c > 0$, then $Y \sim \text{Gamma}(\alpha, \lambda_0/c)$ (same shape, rate divided by c). Here $X \sim \chi^2_n = \text{Gamma}(n/2, \frac{1}{2})$ and $c = 1/(2\lambda)$, so the new rate is $\frac{1}{2} \div \frac{1}{2\lambda} = \lambda$. Hence $Y \sim \text{Gamma}(n/2, \lambda)$.

Step 2: the requested mean ($n = 8,\ \lambda = 2$). $Y \sim \text{Gamma}(4, 2)$, so

$$\mathbb{E}(Y) = \frac{\alpha}{\lambda} = \frac{4}{2} = 2.$$

Remark (The whole family at a glance)*.* Every distribution in this section is a gamma. Counting up by shape from the same rate: $\text{Gamma}(1, \lambda)$ is the exponential (one event), $\text{Gamma}(r, \lambda)$ for integer r is the **Erlang** wait to the r-th event, and $\text{Gamma}(n/2, \frac{1}{2})$ is the chi-square. Two slogans keep them straight: *shapes add at a common rate*, and *a discrete count in a fixed window is Poisson while the continuous time until those events is gamma.*

4.7 Exercises

All problems below are original. Full solutions begin on page 270.

A. The Poisson distribution

A1. A help desk receives email tickets at a constant average rate of 2.5 per hour, arriving independently. Let X be the number of tickets in a given one-hour window, so $X \sim \text{Poisson}(2.5)$.

(a) Write the pmf $p_X(k)$ and compute $\mathbb{P}(X = 0)$.

(b) Find $\mathbb{P}(X \geq 1)$, the probability that at least one ticket arrives.

A2. At a coffee bar, customers join the line according to a Poisson process at a rate of 0.6 per minute. Let X be the number of customers who join during a fixed 5-minute stretch.

(a) State the parameter of X and explain how it is obtained from the rate and the window length.

(b) Compute $\mathbb{P}(X = 4)$.

A3. A textbook page contains, on average, 1.8 typographical errors, and errors occur independently across the page. Let $X \sim \text{Poisson}(1.8)$ be the number of errors on one page. Compute $\mathbb{P}(X \leq 2)$.

A4. A factory ships 500 memory chips. Each chip is defective independently with probability 0.004. Let Y be the number of defective chips in the shipment.

(a) Identify the exact distribution of Y and the Poisson distribution that approximates it; state why the approximation is appropriate here.

(b) Use the Poisson approximation to estimate $\mathbb{P}(Y = 3)$.

A5. Two independent kinds of alerts hit a monitoring system: "latency" alerts at a Poisson rate of 4 per hour and "error" alerts at a Poisson rate of 3 per hour. Let T be the total number of alerts (of either kind) in one hour.

(a) Name the distribution of T and its parameter, and justify the merging step.

(b) Compute $\mathbb{P}(T = 5)$.

(c) Given that exactly 5 alerts occurred in the hour, find the probability that exactly 2 of them were latency alerts.

A6. Phone calls arrive at a small clinic's front desk as a Poisson process with mean $\lambda = 1.5$ calls per ten-minute period. In a particular ten-minute period it is known that at least one call arrived. Given this, find the probability that at least two calls arrived. Express the answer as a decimal rounded to four places.

A7. Over a long shift, a busy bakery sells croissants in a way well modeled by a Poisson count with mean $\lambda = 49$ per shift. Because λ is large, use the normal approximation to the Poisson (with a continuity correction) to answer the following. State the standard value of Φ you use in each part.

(a) Estimate $\mathbb{P}(X \leq 60)$, the probability of selling at most 60 croissants.

(b) Find the smallest integer c such that $\mathbb{P}(X \leq c) \geq 0.95$ (the approximate 95th percentile of the shift's sales).

B. The normal distribution

B1. A machine fills bags of trail mix with weight $X \sim \text{N}(250, 16)$ grams (so the mean is 250 g and the variance is 16 g^2). A bag passes inspection if it weighs at most 254 g. What is the probability that a randomly chosen bag passes?

B2. Daily water use at a small greenhouse is $X \sim \mathrm{N}(500, 100^2)$ litres. The supply line is rated to deliver up to 700 litres per day before it strains. Find the probability that a given day's use exceeds 700 litres.

B3. The reaction time (in milliseconds) of a sensor is $X \sim \mathrm{N}(20, 25)$. The manufacturer labels a sensor "in spec" when its reaction time lies strictly between 10 and 30 ms. What fraction of sensors are in spec?

B4. Scores on a standardized placement test are modelled by $X \sim \mathrm{N}(100, 225)$. Find $\mathbb{P}(85 < X < 129.4)$.

B5. Monthly demand for a replacement part is $X \sim \mathrm{N}(260, 400)$ units. The warehouse wants to stock enough so that demand is met in 95% of months. How many units should it stock (the 95th percentile of demand)?

B6. A laboratory thermostat holds temperature at $X \sim \mathrm{N}(\mu, 64)$ degrees, where the mean μ is unknown but the standard deviation is fixed at 8 degrees. An alarm, which trips when the temperature exceeds 40 degrees, is observed to trip on only 0.99% of readings. Assuming the model is correct, determine μ.

B7. The fill volume of a bottling line is $X \sim \mathrm{N}(\mu, \sigma^2)$. Engineers want the central 99% of bottles to fall within the tolerance band [498, 512] mL, with the band centred on the mean.

(a) Find the values of μ and σ that achieve this exactly.

(b) For your fitted distribution, compute $\mathbb{P}(X > 512)$.

(c) Given that a bottle already exceeds 505 mL, find the probability that it also exceeds 512 mL.

C. Normal approximation and the Central Limit Theorem

C1. A fair eight-sided die is rolled once. Let X be the result, so X is uniform on $\{1, 2, \ldots, 8\}$ with $\mathbb{E}(X) = 4.5$ and $\mathrm{Var}(X) = 5.25$. The die is rolled $n = 84$ times independently and S_{84} is the total. Using the Central Limit Theorem (no continuity correction), approximate $\mathbb{P}(S_{84} > 399)$. State which Φ value you use.

C2. At a regional blood bank, each donor independently has type O-negative blood with probability $p = 0.08$. On a day when $n = 275$ donors arrive, let X be the number of O-negative donors. First check that the normal approximation is reasonable ($np \geq 10$ and $n(1-p) \geq 10$), then use the normal approximation *with continuity correction* to estimate $\mathbb{P}(X \leq 17)$. State which Φ value you use.

C3. A multiple-choice quiz has 396 independent true/false questions. A student guesses each answer, so the number correct is $X \sim \mathrm{Bin}(396, 0.5)$. Use the normal approximation with continuity correction to find $\mathbb{P}(179 \leq X \leq 217)$. State which Φ values you use.

C4. The time (in minutes) a customer-service agent spends on one call has mean $\mu = 6$ and standard deviation $\sigma = 4$; successive call times are independent. Over a shift the agent handles $n = 64$ calls. Let $\bar{X}_{64}$ be the average call length. Using the Central Limit Theorem, find the value c such that $\mathbb{P}(\bar{X}_{64} \leq c) = 0.95$. State which z value (with its Φ value) you use.

C5. Defects on a production line occur as a Poisson process at rate 3 per hour. A plant runs an 11-hour operating day, so the number of defects Y in one day is $Y \sim \mathrm{Poisson}(33)$ (mean 33, variance 33). Treating Y as approximately normal with continuity correction, estimate the probability that a day produces 45 or more defects. State which Φ value you use.

C6. A factory ships bolts in boxes. Each bolt weighs, independently, a random amount with mean $\mu = 50$ grams and standard deviation $\sigma = 2$ grams. A box contains $n = 100$ bolts. The shipping label guarantees the total bolt weight is between 4,960 and 5,040 grams. Using the Central Limit Theorem, approximate the probability that a box meets the guarantee. Then find the largest symmetric tolerance $\pm d$ grams about 5,000 for which the total falls in $[5000 - d,\ 5000 + d]$ with probability 0.99. State which Φ and z values you use.

C7. An online retailer finds that each visitor, independently, makes a purchase with probability $p = 0.20$. On a day with n visitors, the retailer wants the number of purchases $X \sim \mathrm{Bin}(n, 0.20)$ to be at least $0.18n$ with probability at least 0.975. Using the normal approximation to the binomial (you may omit

the continuity correction since n is large), find the smallest n that achieves this. State which z value (with its Φ value) you use.

D. The geometric distribution

D1. A vending machine dispenses a defective snack pack with probability 0.05, independently for each pack, and otherwise dispenses a good one. Let X be the number of the pack on which the first defective one appears (counting that pack). Identify the distribution of X and give its pmf, its mean, and its standard deviation.

D2. In a fixed deep-sky survey, each telescope pointing independently catches a transient flare with probability $p = 0.2$. Let X be the number of pointings up to and including the first flare. Compute $\mathbb{P}(X = 4)$ and $\mathbb{P}(X \le 3)$.

D3. A lab assay returns a valid reading on any given run with probability 0.4, independently across runs. Let X be the run on which the first valid reading occurs. Find the smallest integer m such that $\mathbb{P}(X \le m) \ge 0.90$.

D4. A door-to-door fundraiser secures a pledge at each house with probability 0.12, independently. Using the convention X = number of houses visited up to and including the first pledge, find $\mathbb{P}(X > 10)$. Then, given that the first ten houses gave no pledge, find the probability that more than five additional houses must be visited before the first pledge. State which property you used.

D5. For $X \sim \text{Geom}(p)$ with the trials-to-first-success convention, derive the mean $\mathbb{E}(X) = 1/p$ from the pmf. (Use the identity $\sum_{k=1}^{\infty} k\, r^{k-1} = (1-r)^{-2}$ for $|r| < 1$, and prove that identity by differentiating a geometric series.)

D6. A quality-control inspector tests welds one at a time; each weld fails independently with probability p. Let X be the number of welds tested up to and including the first failure. It is observed that a failure is "early" — meaning it occurs on or before the third weld — exactly half the time, i.e. $\mathbb{P}(X \le 3) = 0.5$. Find p (give the exact value and a 4-decimal approximation), and then compute $\mathbb{E}(X)$.

D7. Two independent searchers scan radio channels for a beacon. Searcher A checks one channel per minute and finds the beacon on any given channel with probability 0.10; searcher B checks one channel per minute and finds it with probability 0.15. Let $X_A \sim \text{Geom}(0.10)$ and $X_B \sim \text{Geom}(0.15)$ be their respective numbers of minutes to the first detection, and assume $X_A \perp\!\!\!\perp X_B$. Let $M = \min(X_A, X_B)$ be the first minute at which *either* searcher detects the beacon. Show that M is geometric, identify its parameter, and compute $\mathbb{E}(M)$ and $\mathbb{P}(M > 4)$.

E. The negative binomial distribution

Throughout, use the convention $X \sim \text{NegBin}(r, p)$ with X = the number of independent Bernoulli(p) trials up to and including the r-th success, so

$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, r+2, \ldots,$$

with $\mathbb{E}(X) = r/p$ and $\text{Var}(X) = r(1-p)/p^2$.

E1. A quality inspector tests circuit boards one at a time off a line; each board is defective independently with probability 0.20. Let X be the number of boards she must test to find the 3rd defective board.

(a) Name the distribution of X and give its pmf, stating the range of k.

(b) Compute $\mathbb{P}(X = 5)$.

(c) Compute $\mathbb{E}(X)$ and $\text{Var}(X)$.

E2. A door-to-door volunteer keeps knocking until 2 people agree to sign a petition. Each household independently signs with probability $p = 0.25$, and let X be the number of households visited up to and including the 2nd signature.

(a) Find $\mathbb{P}(X = 4)$.

(b) Find $\mathbb{P}(X \leq 3)$.

E3. A biologist sets identical traps overnight; each trap independently catches the target species with probability 0.15. Traps are checked in a fixed order and reset is not needed. Let X be the number of traps inspected to obtain the 4th catch. Compute $\mathbb{P}(X = 10)$ and the most probable single value $\mathbb{P}(X = 4)$ for comparison, and state which of those two outcomes is more likely.

E4. Let $X \sim \text{NegBin}(r, p)$ under the trials-count convention above. Starting only from the fact that $X = G_1 + G_2 + \cdots + G_r$, where the G_i are independent $\text{Geom}(p)$ waiting times (each with mean $1/p$ and variance $(1-p)/p^2$), derive $\mathbb{E}(X)$ and $\text{Var}(X)$. Then for the specific case $r = 5$, $p = 0.4$, give the numerical mean and standard deviation.

E5. A free-throw shooter makes each attempt independently with probability $p = 0.8$. Let X be the number of attempts up to and including her 6th make.

(a) What is the probability that she needs exactly 8 attempts to reach 6 makes?

(b) Given that she has already taken 7 attempts without yet reaching her 6th make (i.e. at most 5 makes in the first 7 attempts), is the conditional pmf of the *remaining* attempts still negative binomial? Explain briefly using the independence of trials, and identify the distribution of the number of additional attempts needed.

E6. A botanist germinates seeds one batch at a time; each batch independently succeeds with probability $p = 0.30$. She will stop after the r-th successful batch. She wants the *expected* number of batches she must prepare to be at most 20. What is the largest integer r for which this holds, and what is the variance of the number of batches for that r?

E7. A telemarketing system places independent calls, each resulting in a completed sale with probability $p = 0.05$. Let X be the number of calls up to and including the 40th sale. Use a normal approximation, justified by the fact that X is a sum of 40 independent identically distributed geometric waiting times, to estimate $\mathbb{P}(X \leq 923)$. Apply a continuity correction. State which standard value of Φ you use.

F. The gamma distribution

F1. Let $X \sim \text{Gamma}(\alpha, \lambda)$ with the rate-form density $f_X(x) = \dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$ for $x > 0$. A certain electronic relay has lifetime $X \sim \text{Gamma}(3, \frac{1}{2})$ (time in years). (a) Write the density explicitly, simplifying the constant. (b) State $\mathbb{E}(X)$ and $\text{Var}(X)$ using the formulas α/λ and α/λ^2. (c) What is the standard deviation of the lifetime?

F2. Evaluate the following using the recursion $\Gamma(\alpha + 1) = \alpha\,\Gamma(\alpha)$, the fact $\Gamma(n) = (n-1)!$ for positive integers n, and $\Gamma(\frac{1}{2}) = \sqrt{\pi}$: (a) $\Gamma(6)$; (b) $\Gamma(\frac{7}{2})$; (c) the value of the constant $\dfrac{\lambda^\alpha}{\Gamma(\alpha)}$ when $\alpha = \frac{5}{2}$ and $\lambda = 2$.

F3. A density is given by $f(x) = c\,x^4 e^{-3x}$ for $x > 0$ and 0 otherwise. (a) Identify the gamma parameters α and λ by matching the rate form. (b) Find the normalizing constant c. (c) Without integrating again, write down $\mathbb{E}(X)$ and $\text{Var}(X)$.

F4. At a help desk, support tickets arrive as a Poisson process at rate $\lambda = 4$ per hour, independently. Let T be the waiting time (in hours) from opening until the 3rd ticket of the day arrives. (a) Name the distribution of T and give its parameters. (b) Compute $\mathbb{E}(T)$ and $\text{SD}(T)$. (c) Using the relationship between the gamma waiting time and a Poisson count, find the probability that the 3rd ticket has *not* arrived within the first hour, i.e. $\mathbb{P}(T > 1)$.

F5. Let $X_1, X_2, \ldots, X_5$ be independent, each Exp(λ) with $\lambda = \frac{1}{3}$ per minute (so each has mean 3 minutes). Let $S = X_1 + \cdots + X_5$. (a) State the distribution of S and justify it from the gamma's additivity over independent exponentials. (b) Find $\mathbb{E}(S)$ and $\mathrm{Var}(S)$. (c) For a sum of independent gammas with a *common rate*, the shapes add. If additionally $Y \sim \mathrm{Gamma}(2, \frac{1}{3})$ is independent of S, state the distribution of $S + Y$.

F6. A chi-square random variable with n degrees of freedom is Gamma($n/2, \frac{1}{2}$) in the rate form. (a) Write the density of a chi-square variable W with $n = 4$ degrees of freedom and simplify the constant. (b) Recognize this density as that of a familiar one-parameter distribution and name it (with its parameter). (c) Using that recognition, compute $\mathbb{P}(W > 2)$ in closed form, then give a decimal ($e^{-1} \approx 0.3679$).

F7. Calls to a small clinic's after-hours line follow a Poisson process at rate $\lambda = 2$ per hour. Let T be the time (in hours) until the 2nd call arrives, so $T \sim \mathrm{Gamma}(2, 2)$. (a) Derive the cdf $F_T(t) = \mathbb{P}(T \le t)$ for $t > 0$ by using the Poisson-count relationship $\mathbb{P}(T > t) = \mathbb{P}(\text{fewer than 2 events in } [0, t])$. (b) Differentiate your F_T to recover the gamma density and confirm it matches the rate form. (c) Find the probability that the 2nd call arrives between 30 and 60 minutes after the line opens, i.e. $\mathbb{P}(0.5 < T < 1)$; give a decimal using $e^{-1} \approx 0.3679$ and $e^{-2} \approx 0.1353$.

4.8 Solutions

A. The Poisson distribution

A1. For $X \sim \text{Poisson}(2.5)$ the pmf is

$$p_X(k) = \mathbb{P}(X = k) = e^{-2.5}\,\frac{2.5^{\,k}}{k!}, \qquad k = 0, 1, 2, \ldots$$

(a) $\mathbb{P}(X = 0) = e^{-2.5}\,\dfrac{2.5^0}{0!} = e^{-2.5} = 0.0821.$

(b) Using the complement,

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-2.5} = 1 - 0.0821 = 0.9179.$$

A2. **(a)** Poisson rates scale with the length of the window: a rate of 0.6 per minute over a 5-minute window gives mean $\lambda = 0.6 \times 5 = 3$, so $X \sim \text{Poisson}(3)$. (The process is assumed stationary and independent across disjoint sub-intervals, so the counts in five one-minute pieces are independent Poisson(0.6) variables whose sum is Poisson(3).)

(b)

$$\mathbb{P}(X = 4) = e^{-3}\,\frac{3^4}{4!} = e^{-3}\,\frac{81}{24} = e^{-3}\,(3.375) = 0.1680.$$

A3. For $X \sim \text{Poisson}(1.8)$,

$$p_X(k) = e^{-1.8}\,\frac{1.8^{\,k}}{k!}.$$

Summing the first three terms,

$$\begin{aligned}
\mathbb{P}(X = 0) &= e^{-1.8} = 0.1653,\\
\mathbb{P}(X = 1) &= e^{-1.8}(1.8) = 0.2975,\\
\mathbb{P}(X = 2) &= e^{-1.8}\,\tfrac{1.8^2}{2} = 0.2678.
\end{aligned}$$

Hence

$$\mathbb{P}(X \leq 2) = 0.1653 + 0.2975 + 0.2678 = 0.7306.$$

A4. **(a)** Exactly, $Y \sim \text{Bin}(500,\, 0.004)$. The Poisson approximation to the binomial applies when n is large and p is small with np moderate; here $n = 500$ is large, $p = 0.004$ is small, and $\lambda = np = 500 \times 0.004 = 2$, so $Y \approx \text{Poisson}(2)$.

(b)

$$\mathbb{P}(Y = 3) \approx e^{-2}\,\frac{2^3}{3!} = e^{-2}\,\frac{8}{6} = e^{-2}\,(1.3333) = 0.1805.$$

(The exact binomial value is 0.1808, confirming the approximation is excellent.)

A5. **(a)** The superposition (merging) of independent Poisson processes with rates $\lambda_1 = 4$ and $\lambda_2 = 3$ is again a Poisson process whose rate is the sum: the total count in one hour is $T \sim \text{Poisson}(\lambda_1 + \lambda_2) = \text{Poisson}(7)$. (Reason: if $U \sim \text{Poisson}(4)$ and $V \sim \text{Poisson}(3)$ are independent, then $U + V \sim \text{Poisson}(7)$, since the sum of independent Poissons adds the means.)

(b)

$$\mathbb{P}(T = 5) = e^{-7}\,\frac{7^5}{5!} = e^{-7}\,\frac{16807}{120} = e^{-7}\,(140.06) = 0.1277.$$

(c) Conditional on a total count, each event is independently a latency alert with probability $\dfrac{\lambda_1}{\lambda_1 + \lambda_2} = \dfrac{4}{7}$. Thus the number of latency alerts among the 5 events is $\text{Bin}\big(5, \tfrac{4}{7}\big)$, and

$$\mathbb{P}(\text{exactly 2 latency} \mid T = 5) = \binom{5}{2}\left(\frac{4}{7}\right)^2\left(\frac{3}{7}\right)^3 = 10 \cdot \frac{16}{49} \cdot \frac{27}{343} = \frac{4320}{16807} = 0.2571.$$

A6. Let $X \sim \text{Poisson}(1.5)$. We need the conditional probability

$$\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{\mathbb{P}(X \geq 2)}{\mathbb{P}(X \geq 1)},$$

since $\{X \geq 2\} \subset \{X \geq 1\}$. Compute the two tail probabilities via complements:

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X = 0) = 1 - e^{-1.5} = 1 - 0.2231 = 0.7769,$$

$$\mathbb{P}(X \geq 2) = 1 - \mathbb{P}(X = 0) - \mathbb{P}(X = 1) = 1 - e^{-1.5} - e^{-1.5}(1.5) = 1 - 0.2231 - 0.3347 = 0.4422.$$

Therefore

$$\mathbb{P}(X \geq 2 \mid X \geq 1) = \frac{0.4422}{0.7769} = 0.5692.$$

A7. Here $X \sim \text{Poisson}(49)$, so $\mu = \lambda = 49$ and $\sigma = \sqrt{\lambda} = \sqrt{49} = 7$. Because $\lambda = 49$ is large, approximate X by $\text{N}(49, 7^2)$ and apply the continuity correction (a count $\leq k$ becomes the region up to $k + 0.5$).

(a) With the continuity correction, $\mathbb{P}(X \leq 60) \approx \mathbb{P}\big(N \leq 60.5\big)$ where $N \sim \text{N}(49, 49)$. Standardise:

$$z = \frac{60.5 - 49}{7} = \frac{11.5}{7} = 1.643 \approx 1.645.$$

Using the standard value $\Phi(1.645) = 0.9500$,

$$\mathbb{P}(X \leq 60) \approx \Phi(1.645) = 0.9500.$$

(b) We want the smallest integer c with $\mathbb{P}(X \leq c) \geq 0.95$. With the continuity correction this reads $\Phi\left(\dfrac{c + 0.5 - 49}{7}\right) \geq 0.95$. Since $\Phi(1.645) = 0.9500$, set the standardised boundary equal to 1.645:

$$\frac{c + 0.5 - 49}{7} = 1.645 \implies c + 0.5 = 49 + 1.645 \times 7 = 49 + 11.515 = 60.515 \implies c = 60.015.$$

The smallest integer count meeting the requirement is therefore $c = 60$ (consistent with part (a), where $\mathbb{P}(X \leq 60) \approx 0.95$). The standard value used is $\Phi(1.645) = 0.9500$.

B. The normal distribution

B1. Here $\mu = 250$ and $\sigma = \sqrt{16} = 4$. Standardize $Z = (X - \mu)/\sigma$:

$$\mathbb{P}(X \leq 254) = \mathbb{P}\left(Z \leq \frac{254 - 250}{4}\right) = \mathbb{P}(Z \leq 1) = \Phi(1) = 0.8413.$$

Used $\Phi(1) = 0.8413$. So about 84.13% of bags pass.

B2. Here $\mu = 500$, $\sigma = 100$, so $(700 - 500)/100 = 2$:

$$\mathbb{P}(X > 700) = 1 - \mathbb{P}(X \leq 700) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228.$$

Used $\Phi(2) = 0.9772$. The supply is exceeded on about 2.28% of days.

B3. Here $\mu = 20$, $\sigma = \sqrt{25} = 5$. The endpoints standardize to $(10 - 20)/5 = -2$ and $(30 - 20)/5 = 2$, so

$$\mathbb{P}(10 < X < 30) = \Phi(2) - \Phi(-2).$$

By symmetry $\Phi(-2) = 1 - \Phi(2) = 1 - 0.9772 = 0.0228$, hence

$$\mathbb{P}(10 < X < 30) = 0.9772 - 0.0228 = 0.9544.$$

Used $\Phi(2) = 0.9772$ and $\Phi(-z) = 1 - \Phi(z)$. About 95.44% are in spec.

B4. Here $\mu = 100$, $\sigma = \sqrt{225} = 15$. Standardize each endpoint:

$$\frac{85 - 100}{15} = -1, \qquad \frac{129.4 - 100}{15} = \frac{29.4}{15} = 1.96.$$

Therefore

$$\mathbb{P}(85 < X < 129.4) = \Phi(1.96) - \Phi(-1) = \Phi(1.96) - \big(1 - \Phi(1)\big).$$

With $\Phi(1.96) = 0.9750$ and $\Phi(1) = 0.8413$:

$$= 0.9750 - (1 - 0.8413) = 0.9750 - 0.1587 = 0.8163.$$

Used $\Phi(1.96) = 0.9750$, $\Phi(1) = 0.8413$, and $\Phi(-z) = 1 - \Phi(z)$.

B5. We want the 95th percentile $x_{0.95}$, i.e. the value with $\mathbb{P}(X \le x) = 0.95$. The corresponding standard-normal quantile satisfies $\Phi(z) = 0.95$, which gives $z = 1.645$. Then

$$x_{0.95} = \mu + z\sigma = 260 + 1.645\sqrt{400} = 260 + 1.645(20) = 260 + 32.9 = 292.9.$$

Used $\Phi(1.645) = 0.9500$. Stocking 293 units (rounding up) meets demand in at least 95% of months.

B6. Here $\sigma = \sqrt{64} = 8$ and $\mathbb{P}(X > 40) = 0.0099$, so

$$\mathbb{P}(X \le 40) = 1 - 0.0099 = 0.9901.$$

We need the quantile with $\Phi(z) = 0.9901$, namely $z = 2.33$. Hence

$$\frac{40 - \mu}{8} = 2.33 \quad \Longrightarrow \quad \mu = 40 - 2.33(8) = 40 - 18.64 = 21.36.$$

Used $\Phi(2.33) = 0.9901$. The mean temperature is $\mu = 21.36$ degrees.

B7. **(a)** "Central 99% within $[498, 512]$, centred on the mean" forces μ to be the midpoint of the band:

$$\mu = \frac{498 + 512}{2} = 505.$$

The central 99% corresponds to $\mathbb{P}(-z < Z < z) = 0.99$, i.e. each tail has probability 0.005 and $\Phi(z) = 0.995$, giving $z = 2.576$. The half-width of the band is $512 - 505 = 7 = z\,\sigma$, so

$$\sigma = \frac{7}{2.576} = 2.717 \text{ mL (to 3 d.p.)}.$$

Used $\Phi(2.576) = 0.9950$.

(b) With $\mu = 505$ and $\sigma = 7/2.576$, the point 512 standardizes to $(512 - 505)/\sigma = 7/(7/2.576) = 2.576$, so

$$\mathbb{P}(X > 512) = 1 - \Phi(2.576) = 1 - 0.9950 = 0.0050.$$

(This is exactly the upper tail of the symmetric 99% band, as expected.) Used $\Phi(2.576) = 0.9950$.

(c) Since $\mu = 505$, we have $\mathbb{P}(X > 505) = 0.5$. The event $\{X > 512\}$ is contained in $\{X > 505\}$, so by the definition of conditional probability

$$\mathbb{P}(X > 512 \mid X > 505) = \frac{\mathbb{P}(X > 512)}{\mathbb{P}(X > 505)} = \frac{0.0050}{0.5} = 0.0100.$$

Used the result of part (b).

C. Normal approximation and the Central Limit Theorem

C1. By the Central Limit Theorem the sum S_{84} of 84 iid rolls is approximately normal with

$$\mathbb{E}(S_{84}) = 84 \cdot 4.5 = 378, \qquad \mathrm{Var}(S_{84}) = 84 \cdot 5.25 = 441, \qquad \mathrm{SD}(S_{84}) = \sqrt{441} = 21.$$

Standardising,

$$\mathbb{P}(S_{84} > 399) \approx \mathbb{P}\left(Z > \frac{399 - 378}{21}\right) = \mathbb{P}(Z > 1) = 1 - \Phi(1).$$

Using $\Phi(1) = 0.8413$,

$$\mathbb{P}(S_{84} > 399) \approx 1 - 0.8413 = 0.1587.$$

C2. Here $X \sim \mathrm{Bin}(275, 0.08)$. Check the rule of thumb: $np = 275 \cdot 0.08 = 22 \geq 10$ and $n(1-p) = 275 \cdot 0.92 = 253 \geq 10$, so the normal approximation is reasonable. The matching normal has

$$\mu = np = 22, \qquad \sigma = \sqrt{np(1-p)} = \sqrt{275 \cdot 0.08 \cdot 0.92} = \sqrt{20.24} = 4.499.$$

For $\mathbb{P}(X \leq 17)$ the continuity correction replaces 17 by 17.5:

$$\mathbb{P}(X \leq 17) \approx \mathbb{P}\left(Z \leq \frac{17.5 - 22}{4.499}\right) = \mathbb{P}\left(Z \leq \frac{-4.5}{4.499}\right) = \mathbb{P}(Z \leq -1.00) = 1 - \Phi(1).$$

Using $\Phi(1) = 0.8413$ and $\Phi(-z) = 1 - \Phi(z)$,

$$\mathbb{P}(X \leq 17) \approx 1 - 0.8413 = 0.1587.$$

C3. Here $X \sim \mathrm{Bin}(396, 0.5)$, so

$$\mu = np = 396 \cdot 0.5 = 198, \qquad \sigma = \sqrt{np(1-p)} = \sqrt{396 \cdot 0.25} = \sqrt{99} = 9.9499.$$

For $\mathbb{P}(179 \leq X \leq 217)$ the continuity correction widens the interval to [178.5, 217.5]:

$$\mathbb{P}(179 \leq X \leq 217) \approx \mathbb{P}\left(\frac{178.5 - 198}{9.9499} \leq Z \leq \frac{217.5 - 198}{9.9499}\right) = \mathbb{P}(-1.96 \leq Z \leq 1.96),$$

since $19.5/9.9499 = 1.96$. Therefore

$$\mathbb{P}(179 \leq X \leq 217) \approx \Phi(1.96) - \Phi(-1.96) = \Phi(1.96) - \big(1 - \Phi(1.96)\big) = 2\Phi(1.96) - 1.$$

Using $\Phi(1.96) = 0.9750$,

$$\mathbb{P}(179 \leq X \leq 217) \approx 2(0.9750) - 1 = 0.9500.$$

C4. By the Central Limit Theorem the sample mean $\bar{X}_{64}$ is approximately normal with

$$\mathbb{E}(\bar{X}_{64}) = \mu = 6, \qquad \mathrm{SD}(\bar{X}_{64}) = \frac{\sigma}{\sqrt{n}} = \frac{4}{\sqrt{64}} = \frac{4}{8} = 0.5.$$

We need c with $\mathbb{P}(\bar{X}_{64} \leq c) = 0.95$, i.e.

$$\Phi\left(\frac{c - 6}{0.5}\right) = 0.95.$$

The 95th percentile of the standard normal is $z = 1.645$, since $\Phi(1.645) = 0.9500$. Hence

$$\frac{c - 6}{0.5} = 1.645 \quad \Longrightarrow \quad c = 6 + 1.645 \cdot 0.5 = 6 + 0.8225 = 6.8225.$$

So $c \approx 6.82$ minutes.

C5. Here $Y \sim \text{Poisson}(33)$, so $\mu = 33$ and $\sigma = \sqrt{33} = 5.745$. Approximating Y by a normal with these parameters and applying the continuity correction (for $\mathbb{P}(Y \geq 45)$ the boundary 45 becomes 44.5):

$$\mathbb{P}(Y \geq 45) \approx \mathbb{P}\left(Z \geq \frac{44.5 - 33}{5.745}\right) = \mathbb{P}\left(Z \geq \frac{11.5}{5.745}\right) = \mathbb{P}(Z \geq 2.00) = 1 - \Phi(2).$$

Using $\Phi(2) = 0.9772$,

$$\mathbb{P}(Y \geq 45) \approx 1 - 0.9772 = 0.0228.$$

C6. Let T be the total weight of the 100 bolts. By the Central Limit Theorem T is approximately normal with

$$\mathbb{E}(T) = 100 \cdot 50 = 5000, \qquad \text{SD}(T) = \sigma\sqrt{n} = 2\sqrt{100} = 20.$$

Guarantee. For $\mathbb{P}(4960 \leq T \leq 5040)$,

$$\mathbb{P}(4960 \leq T \leq 5040) \approx \mathbb{P}\left(\frac{4960 - 5000}{20} \leq Z \leq \frac{5040 - 5000}{20}\right) = \mathbb{P}(-2 \leq Z \leq 2) = 2\Phi(2) - 1.$$

Using $\Phi(2) = 0.9772$,

$$\mathbb{P}(4960 \leq T \leq 5040) \approx 2(0.9772) - 1 = 0.9544.$$

Tolerance for 0.99. We want d with $\mathbb{P}(5000 - d \leq T \leq 5000 + d) = 0.99$, i.e.

$$\mathbb{P}(-d/20 \leq Z \leq d/20) = 2\Phi(d/20) - 1 = 0.99,$$

so $\Phi(d/20) = 0.995$. The corresponding critical value is $z = 2.576$, since $\Phi(2.576) = 0.9950$. Hence

$$\frac{d}{20} = 2.576 \quad \Longrightarrow \quad d = 2.576 \cdot 20 = 51.52 \text{ grams}.$$

So a tolerance of about ± 51.5 grams captures the total weight with probability 0.99.

C7. Here $X \sim \text{Bin}(n, 0.20)$, with $\mu = 0.2n$ and $\sigma = \sqrt{n(0.2)(0.8)} = \sqrt{0.16\,n} = 0.4\sqrt{n}$. By the normal approximation,

$$\begin{aligned}\mathbb{P}(X \geq 0.18n) &\approx \mathbb{P}\left(Z \geq \frac{0.18n - 0.2n}{0.4\sqrt{n}}\right) = \mathbb{P}\left(Z \geq \frac{-0.02n}{0.4\sqrt{n}}\right) \\ &= \mathbb{P}\big(Z \geq -0.05\sqrt{n}\big) = \Phi\big(0.05\sqrt{n}\big)\,,\end{aligned}$$

using $\mathbb{P}(Z \geq -a) = \Phi(a)$. We require $\Phi(0.05\sqrt{n}) \geq 0.975$. Since $\Phi(1.96) = 0.9750$, this means

$$0.05\sqrt{n} \geq 1.96 \quad \Longrightarrow \quad \sqrt{n} \geq \frac{1.96}{0.05} = 39.2 \quad \Longrightarrow \quad n \geq 39.2^2 = 1536.64.$$

The smallest integer is $n = 1537$. (Check: $0.05\sqrt{1537} = 1.9602 \geq 1.96$, while $0.05\sqrt{1536} = 1.9596 < 1.96$.) So the retailer needs at least $n = 1{,}537$ visitors.

D. The geometric distribution

D1. Each pack is an independent trial with "success" = defective, probability $p = 0.05$, and X counts trials up to and including the first success, so $X \sim \text{Geom}(0.05)$ with

$$p_X(k) = (1-p)^{k-1}p = (0.95)^{k-1}(0.05), \qquad k = 1, 2, \ldots$$

The mean is $\mathbb{E}(X) = 1/p = 1/0.05 = 20$ packs. The variance is $\text{Var}(X) = (1-p)/p^2 = 0.95/0.0025 = 380$, so the standard deviation is $\text{SD}(X) = \sqrt{380} = 19.4936 \approx 19.49$ packs.

D2. Here $X \sim \text{Geom}(0.2)$, so $q = 1 - p = 0.8$. Using the pmf,

$$\mathbb{P}(X = 4) = (0.8)^3(0.2) = (0.512)(0.2) = 0.1024.$$

Using the cdf $F_X(k) = \mathbb{P}(X \leq k) = 1 - (1-p)^k$,

$$\mathbb{P}(X \leq 3) = 1 - (0.8)^3 = 1 - 0.512 = 0.488.$$

D3. With $p = 0.4$ and $q = 0.6$ we need the smallest integer m with

$$\mathbb{P}(X \le m) = 1 - (0.6)^m \ge 0.90 \quad \Longleftrightarrow \quad (0.6)^m \le 0.10.$$

Taking logs (both $\ln 0.6$ and $\ln 0.1$ are negative, so the inequality flips):

$$m \ge \frac{\ln 0.10}{\ln 0.60} = \frac{-2.302585}{-0.510826} = 4.5074.$$

Hence $m = 5$. Check: $(0.6)^5 = 0.07776 \le 0.10$ while $(0.6)^4 = 0.1296 > 0.10$, so $m = \boxed{5}$ is indeed the smallest such integer.

D4. Here $p = 0.12$, $q = 0.88$. The right-tail probability of a geometric variable is

$$\mathbb{P}(X > n) = (1-p)^n = (0.88)^n,$$

since $X > n$ means the first n houses all failed to pledge. Thus

$$\mathbb{P}(X > 10) = (0.88)^{10} = 0.2785.$$

For the conditional part, use the **memoryless property** $\mathbb{P}(X > m + n \mid X > m) = \mathbb{P}(X > n)$. Given no pledge in the first ten houses, the number of *additional* houses needed restarts as a fresh Geom(0.12) count, so

$$\mathbb{P}(X > 15 \mid X > 10) = \mathbb{P}(X > 5) = (0.88)^5 = 0.5277.$$

(The property used is memorylessness of the geometric distribution.)

D5. Let $q = 1 - p$. From the pmf $p_X(k) = q^{k-1}p$,

$$\mathbb{E}(X) = \sum_{k=1}^{\infty} k\, q^{k-1} p = p \sum_{k=1}^{\infty} k\, q^{k-1}.$$

Step 1: prove the summation identity. For $|r| < 1$ the geometric series is $\sum_{k=0}^{\infty} r^k = \dfrac{1}{1-r}$. Differentiating both sides with respect to r (valid term by term inside the radius of convergence) gives

$$\frac{\mathrm{d}}{\mathrm{d}r} \sum_{k=0}^{\infty} r^k = \sum_{k=1}^{\infty} k\, r^{k-1}, \qquad \frac{\mathrm{d}}{\mathrm{d}r} \frac{1}{1-r} = \frac{1}{(1-r)^2},$$

so $\displaystyle\sum_{k=1}^{\infty} k\, r^{k-1} = \frac{1}{(1-r)^2}$.

Step 2: substitute $r = q$. With $r = q$ and $1 - q = p$,

$$\mathbb{E}(X) = p \cdot \frac{1}{(1-q)^2} = p \cdot \frac{1}{p^2} = \frac{1}{p}.$$

$$\boxed{\mathbb{E}(X) = \tfrac{1}{p}.}$$

D6. Using the cdf, the condition $\mathbb{P}(X \le 3) = 1 - (1-p)^3 = 0.5$ gives

$$(1-p)^3 = 0.5 \quad \Longrightarrow \quad 1 - p = 0.5^{1/3} = 2^{-1/3}.$$

Hence the exact value is

$$p = 1 - 2^{-1/3} = 1 - 0.793701 = 0.206299 \approx 0.2063.$$

Then

$$\mathbb{E}(X) = \frac{1}{p} = \frac{1}{0.206299} = 4.8474 \text{ welds} \quad \Big(\text{exactly } \tfrac{1}{1-2^{-1/3}}\Big).$$

D7. Recall for a geometric variable $\mathbb{P}(X > k) = (1-p)^k$. By independence of the two searchers, for every integer $k \geq 0$

$$\mathbb{P}(M > k) = \mathbb{P}(X_A > k,\ X_B > k) = \mathbb{P}(X_A > k)\,\mathbb{P}(X_B > k) = (0.90)^k(0.85)^k = (0.765)^k.$$

A variable whose right tail is $\mathbb{P}(M > k) = (1-p_M)^k$ for all $k \geq 0$ is geometric with success parameter p_M where $1 - p_M = 0.765$, i.e.

$$M \sim \text{Geom}(p_M), \qquad p_M = 1 - 0.765 = 0.235.$$

(Equivalently, $1 - p_M = (1-0.10)(1-0.15)$: a "joint failure" in a minute means both searchers miss.) Therefore

$$\mathbb{E}(M) = \frac{1}{p_M} = \frac{1}{0.235} = 4.2553 \text{ minutes},$$

$$\mathbb{P}(M > 4) = (0.765)^4 = 0.3425.$$

E. The negative binomial distribution

Throughout, $X \sim \text{NegBin}(r, p)$ counts trials up to and including the r-th success:

$$p_X(k) = \binom{k-1}{r-1} p^r (1-p)^{k-r}, \qquad k = r, r+1, \ldots, \qquad \mathbb{E}(X) = \frac{r}{p}, \quad \text{Var}(X) = \frac{r(1-p)}{p^2}.$$

E1. **(a)** Trials are independent with constant success ("defective") probability $p = 0.20$ and we wait for the $r = 3$rd success, so $X \sim \text{NegBin}(3, 0.20)$ with

$$p_X(k) = \binom{k-1}{2} (0.20)^3 (0.80)^{k-3}, \qquad k = 3, 4, 5, \ldots$$

(b) With $k = 5$,

$$\mathbb{P}(X = 5) = \binom{4}{2} (0.20)^3 (0.80)^2 = 6 \cdot (0.008) \cdot (0.64) = 0.03072.$$

(c)

$$\mathbb{E}(X) = \frac{r}{p} = \frac{3}{0.20} = 15, \qquad \text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{3(0.80)}{(0.20)^2} = \frac{2.4}{0.04} = 60.$$

(So $\text{SD}(X) = \sqrt{60} \approx 7.75$ boards.)

E2. Here $r = 2$, $p = 0.25$, so $X \sim \text{NegBin}(2, 0.25)$ with

$$p_X(k) = \binom{k-1}{1} (0.25)^2 (0.75)^{k-2} = (k-1)(0.0625)(0.75)^{k-2}.$$

(a) With $k = 4$,

$$\mathbb{P}(X = 4) = \binom{3}{1} (0.25)^2 (0.75)^2 = 3 \cdot (0.0625) \cdot (0.5625) = 0.10546875 \approx 0.1055.$$

(b) The smallest possible value is $k = 2$, so $\mathbb{P}(X \leq 3) = \mathbb{P}(X = 2) + \mathbb{P}(X = 3)$:

$$\mathbb{P}(X = 2) = \binom{1}{1} (0.25)^2 (0.75)^0 = 0.0625, \qquad \mathbb{P}(X = 3) = \binom{2}{1} (0.25)^2 (0.75)^1 = 0.09375,$$

$$\mathbb{P}(X \leq 3) = 0.0625 + 0.09375 = 0.15625.$$

E3. Each trap is an independent trial with success probability $p = 0.15$, and we wait for the $r = 4$th success, so $X \sim \text{NegBin}(4, 0.15)$ with $p_X(k) = \binom{k-1}{3}(0.15)^4(0.85)^{k-4}$.

For $k = 10$,

$$\mathbb{P}(X = 10) = \binom{9}{3}(0.15)^4(0.85)^6 = 84 \cdot (0.00050625) \cdot (0.85)^6.$$

Now $(0.85)^6 = 0.377150$ (since $0.85^2 = 0.7225$, $0.85^3 = 0.614125$, and $0.614125^2 = 0.377150$), so

$$\mathbb{P}(X = 10) = 84 \cdot 0.00050625 \cdot 0.377150 \approx 0.01604.$$

The smallest value is $k = 4$ (four catches in the first four traps):

$$\mathbb{P}(X = 4) = \binom{3}{3}(0.15)^4(0.85)^0 = (0.15)^4 = 0.00050625.$$

Comparison: $\mathbb{P}(X = 10) \approx 0.01604 > 0.00050625 = \mathbb{P}(X = 4)$, so reaching the 4th catch on exactly the 10th trap is far more likely than the run of four immediate catches. (Getting all four successes with no misses is improbable, which is why the minimum value carries so little probability.)

E4. Mean. Since $X = G_1 + \cdots + G_r$ and expectation is linear (independence is not even needed for the mean),

$$\mathbb{E}(X) = \sum_{i=1}^{r} \mathbb{E}(G_i) = \sum_{i=1}^{r} \frac{1}{p} = \frac{r}{p}.$$

Variance. Because the G_i are independent, variance adds:

$$\text{Var}(X) = \sum_{i=1}^{r} \text{Var}(G_i) = \sum_{i=1}^{r} \frac{1-p}{p^2} = \frac{r(1-p)}{p^2}.$$

These match the stated formulas. For $r = 5$, $p = 0.4$:

$$\mathbb{E}(X) = \frac{5}{0.4} = 12.5, \qquad \text{Var}(X) = \frac{5(0.6)}{(0.4)^2} = \frac{3.0}{0.16} = 18.75, \qquad \text{SD}(X) = \sqrt{18.75} \approx 4.33.$$

E5. Here $r = 6$, $p = 0.8$, $1 - p = 0.2$, so $X \sim \text{NegBin}(6, 0.8)$.

(a) With $k = 8$,

$$\mathbb{P}(X = 8) = \binom{7}{5}(0.8)^6(0.2)^2 = 21 \cdot (0.262144) \cdot (0.04) = 21 \cdot 0.01048576 \approx 0.2202.$$

(b) Yes, the process restarts. The trials are independent and identically distributed, so conditioning on the first 7 attempts (which produced at most 5 makes) tells us nothing about future attempts. If she still needs j more makes to reach her 6th, the number of *additional* attempts is again a sum of j independent geometric waiting times, i.e. $\text{NegBin}(j, 0.8)$. (In particular, if exactly $m \le 5$ makes occurred in the first 7 attempts, the additional attempts needed are $\text{NegBin}(6 - m,\ 0.8)$.) The original $\text{NegBin}(6, 0.8)$ does not describe the remaining attempts because some successes have already been banked; memorylessness of the underlying geometric trials is what makes the tail a fresh negative binomial.

E6. With $p = 0.30$, $X \sim \text{NegBin}(r, 0.30)$ has $\mathbb{E}(X) = r/0.30$. We need

$$\frac{r}{0.30} \le 20 \iff r \le 0.30 \cdot 20 = 6.$$

The largest integer is $r = 6$ (giving mean exactly $6/0.30 = 20$). Its variance is

$$\text{Var}(X) = \frac{r(1-p)}{p^2} = \frac{6(0.70)}{(0.30)^2} = \frac{4.2}{0.09} = 46.\overline{6} \approx 46.67.$$

(For $r = 7$ the mean would be $7/0.30 \approx 23.33 > 20$, so $r = 6$ is the cutoff.)

E7. Here $r = 40$, $p = 0.05$, so $X \sim \text{NegBin}(40, 0.05)$ with

$$\mathbb{E}(X) = \frac{40}{0.05} = 800,$$
$$\text{Var}(X) = \frac{40(0.95)}{(0.05)^2} = \frac{38}{0.0025} = 15{,}200,$$
$$\text{SD}(X) = \sqrt{15{,}200} \approx 123.29.$$

Because X is a sum of 40 independent identically distributed geometric waiting times, the Central Limit Theorem makes X approximately N(800, 15,200). Apply the continuity correction to the integer event $\{X \le 923\}$ by using 923.5:

$$\mathbb{P}(X \le 923) \approx \Phi\left(\frac{923.5 - 800}{123.29}\right) = \Phi\left(\frac{123.5}{123.29}\right) = \Phi(1.0017) \approx \Phi(1.00).$$

Using the standard value $\Phi(1) = 0.8413$,

$$\mathbb{P}(X \le 923) \approx 0.8413.$$

F. The gamma distribution

F1. Here $\alpha = 3$, $\lambda = \frac{1}{2}$. (a) The constant is $\dfrac{\lambda^\alpha}{\Gamma(\alpha)} = \dfrac{(1/2)^3}{\Gamma(3)} = \dfrac{1/8}{2!} = \dfrac{1/8}{2} = \dfrac{1}{16}$, so

$$f_X(x) = \frac{1}{16}\, x^2 e^{-x/2}, \qquad x > 0.$$

(b) $\mathbb{E}(X) = \dfrac{\alpha}{\lambda} = \dfrac{3}{1/2} = 6$ years, and $\text{Var}(X) = \dfrac{\alpha}{\lambda^2} = \dfrac{3}{(1/2)^2} = \dfrac{3}{1/4} = 12$. (c) $\text{SD}(X) = \sqrt{12} = 2\sqrt{3} \approx 3.46$ years.

F2. (a) $\Gamma(6) = (6-1)! = 5! = 120$.

(b) Apply the recursion down to $\Gamma(\frac{1}{2}) = \sqrt{\pi}$:

$$\Gamma(\tfrac{7}{2}) = \tfrac{5}{2}\,\Gamma(\tfrac{5}{2}) = \tfrac{5}{2} \cdot \tfrac{3}{2}\,\Gamma(\tfrac{3}{2}) = \tfrac{5}{2} \cdot \tfrac{3}{2} \cdot \tfrac{1}{2}\,\Gamma(\tfrac{1}{2}) = \frac{15}{8}\sqrt{\pi}.$$

(c) With $\alpha = \frac{5}{2}$, $\lambda = 2$: $\lambda^\alpha = 2^{5/2} = 4\sqrt{2}$, and $\Gamma(\frac{5}{2}) = \frac{3}{2} \cdot \frac{1}{2}\,\Gamma(\frac{1}{2}) = \frac{3}{4}\sqrt{\pi}$. Hence

$$\frac{\lambda^\alpha}{\Gamma(\alpha)} = \frac{4\sqrt{2}}{\frac{3}{4}\sqrt{\pi}} = \frac{16\sqrt{2}}{3\sqrt{\pi}}.$$

F3. (a) Match $f(x) = c\,x^4 e^{-3x}$ to $\dfrac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x}$. The exponent of x gives $\alpha - 1 = 4$, so $\alpha = 5$; the exponential gives $\lambda = 3$.

(b) The constant must equal $\dfrac{\lambda^\alpha}{\Gamma(\alpha)}$, so

$$c = \frac{3^5}{\Gamma(5)} = \frac{243}{4!} = \frac{243}{24} = \frac{81}{8}.$$

(Check: $\int_0^\infty c\,x^4 e^{-3x}\,\mathrm{d}x = c \cdot \dfrac{\Gamma(5)}{3^5} = \dfrac{81}{8} \cdot \dfrac{24}{243} = 1$.)

(c) $\mathbb{E}(X) = \dfrac{\alpha}{\lambda} = \dfrac{5}{3}$ and $\text{Var}(X) = \dfrac{\alpha}{\lambda^2} = \dfrac{5}{9}$.

F4. (a) The waiting time to the r-th event of a Poisson process of rate λ is Gamma(r, λ). Here $r = 3$, $\lambda = 4$, so $T \sim \text{Gamma}(3, 4)$.

(b) $\mathbb{E}(T) = \dfrac{r}{\lambda} = \dfrac{3}{4} = 0.75$ hours; $\text{Var}(T) = \dfrac{r}{\lambda^2} = \dfrac{3}{16}$, so $\text{SD}(T) = \sqrt{3}/4 \approx 0.433$ hours.

(c) Let $N(1)$ be the number of tickets in the first hour, $N(1) \sim \text{Poisson}(\lambda \cdot 1) = \text{Poisson}(4)$. The 3rd ticket has not arrived by time 1 exactly when fewer than 3 have arrived:

$$\mathbb{P}(T > 1) = \mathbb{P}\big(N(1) \le 2\big) = \sum_{k=0}^{2} e^{-4}\frac{4^k}{k!} = e^{-4}\left(1 + 4 + \frac{16}{2}\right) = 13\, e^{-4}.$$

With $e^{-4} \approx 0.0183$, $\mathbb{P}(T > 1) \approx 13(0.0183) = 0.238$.

F5. (a) A Gamma(α, λ) variable is the sum of α independent Exp(λ) variables when α is a positive integer (equivalently, independent gammas with the same rate add their shapes). Each $X_i \sim \text{Exp}(\frac{1}{3}) = \text{Gamma}(1, \frac{1}{3})$, so the sum of the 5 of them is

$$S \sim \text{Gamma}\big(5, \tfrac{1}{3}\big)\,.$$

(b) $\mathbb{E}(S) = \dfrac{5}{1/3} = 15$ minutes (also $5 \times 3 = 15$, as each exponential has mean 3); $\text{Var}(S) = \dfrac{5}{(1/3)^2} = 5 \cdot 9 = 45$.

(c) Same rate $\lambda = \frac{1}{3}$, so shapes add: $S + Y \sim \text{Gamma}\big(5 + 2, \frac{1}{3}\big) = \text{Gamma}\big(7, \frac{1}{3}\big)$.

F6. (a) With $n = 4$: $W \sim \text{Gamma}(2, \frac{1}{2})$, so $\alpha = 2$, $\lambda = \frac{1}{2}$. The constant is $\dfrac{(1/2)^2}{\Gamma(2)} = \dfrac{1/4}{1!} = \dfrac{1}{4}$, giving

$$f_W(x) = \frac{1}{4}\, x\, e^{-x/2}, \qquad x > 0.$$

(b) This is Gamma$(2, \frac{1}{2})$; since the shape is the integer 2, it is the sum of two independent Exp$(\frac{1}{2})$ variables. But part (c) only needs the cdf, and in fact a cleaner recognition is available: a chi-square with 2 degrees of freedom is Gamma$(1, \frac{1}{2}) = \text{Exp}(\frac{1}{2})$. Here $n = 4$, which is Gamma$(2, \frac{1}{2})$ (an Erlang of shape 2), *not* a single exponential. So the familiar one-parameter object it builds from is the Exp$(\frac{1}{2})$ distribution, and W is the sum of two such independent exponentials.

(c) Using the Erlang/Poisson relation with rate $\frac{1}{2}$ and shape 2,

$$\begin{aligned}\mathbb{P}(W > 2) &= \mathbb{P}\big(\text{fewer than 2 events of a Poisson}(\tfrac{1}{2} \cdot 2) \text{ count}\big) \\ &= \sum_{k=0}^{1} e^{-1}\frac{1^k}{k!} = e^{-1}(1 + 1) = 2e^{-1}.\end{aligned}$$

With $e^{-1} \approx 0.3679$, $\mathbb{P}(W > 2) \approx 0.7358$.

F7. (a) Let $N(t) \sim \text{Poisson}(2t)$ count calls in $[0, t]$. The 2nd call has not arrived by time t iff fewer than 2 calls have occurred:

$$\mathbb{P}(T > t) = \mathbb{P}\big(N(t) \le 1\big) = e^{-2t}(1 + 2t)\,,$$

so $F_T(t) = 1 - e^{-2t}(1 + 2t)$ for $t > 0$.

(b) Differentiate:

$$F_T'(t) = -\Big[-2e^{-2t}(1 + 2t) + e^{-2t} \cdot 2\Big] = 2e^{-2t}(1 + 2t) - 2e^{-2t} = 2e^{-2t} \cdot 2t = 4t\, e^{-2t}.$$

The rate form with $\alpha = 2$, $\lambda = 2$ is $\dfrac{2^2}{\Gamma(2)} t^{2-1} e^{-2t} = \dfrac{4}{1} t\, e^{-2t} = 4t\, e^{-2t}$, which matches.

(c) Using $\mathbb{P}(T > t) = e^{-2t}(1 + 2t)$,

$$\mathbb{P}(0.5 < T < 1) = \mathbb{P}(T > 0.5) - \mathbb{P}(T > 1) = e^{-1}(1 + 1) - e^{-2}(1 + 2) = 2e^{-1} - 3e^{-2}.$$

With $e^{-1} \approx 0.3679$ and $e^{-2} \approx 0.1353$: $2(0.3679) - 3(0.1353) = 0.7358 - 0.4059 = 0.3299 \approx 0.330$.

Chapter 5

Estimation

Estimation is the reverse of the previous chapters: there the distribution was fully known and we computed probabilities; here we *observe data* from a distribution with an *unknown parameter* and work backwards to a good guess for that parameter, together with an honest statement of how uncertain the guess is.

Master guide: which estimation tool does the problem call for?

- Asked for a **single best guess** of a parameter from data? ⇒ a **point estimate**: **maximum likelihood** (maximise the likelihood of the data) or the **method of moments** (match sample moments to model moments) (§1).
- Asked for a **range** that plausibly contains the parameter, with a stated confidence? ⇒ an **interval estimate** (confidence interval) for a mean, proportion, or variance (§2).
- Comparing two estimators, or asking whether one is "good"? ⇒ check **unbiasedness** and **efficiency** (smaller variance / smaller mean squared error) (§3); the smallest variance any unbiased estimator can have is the **Cramér–Rao lower bound** (§4).
- Does a statistic capture **all the information** the sample carries about the parameter? ⇒ **sufficiency** and the factorisation criterion (§5).
- Does the estimator **home in on the truth** as the sample grows? ⇒ **consistency** (§6).
- Do you have **prior beliefs** about the parameter to combine with the data? ⇒ **Bayesian estimation**: prior × likelihood → posterior (§7).

Keep straight: a *point* estimate is one number, an *interval* estimate is a range; *unbiased* is about the estimator's average, *consistent* is about its limit, and *efficient* is about its variance — a single estimator can be any combination of these.

5.1 Point Estimation: Maximum Likelihood and the Method of Moments

A **point estimator** is a single number computed from the data $X_1, \ldots, X_n$ that we offer as our best guess of an unknown parameter θ. Two general recipes produce such guesses. **Maximum likelihood** asks: of all values of θ, which one makes the data we actually observed most probable? **The method of moments** asks: which value of θ makes the model's theoretical moments match the moments we see in the sample? Both turn a modelling assumption plus data into a formula $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$; the rest of this section builds those formulas and checks them on the standard families.

Which point-estimation rule does the problem need?

- You are handed a **parametric model** $f(x;\theta)$ and an iid sample, and asked for the "best" single value of θ ⇒ write the **likelihood** $L(\theta) = \prod_i f(x_i;\theta)$, take logs, and maximise. This is the **maximum likelihood estimator (MLE)**.

- The density is **smooth in θ on an open interval** (Bernoulli, Poisson, exponential, normal) $\Rightarrow$ find the MLE by solving $\ell'(\theta) = 0$ and checking $\ell'' < 0$ (a max).
- The parameter sits at a **boundary of the support** – e.g. Uniform$(0, \theta)$, where every $x_i \le \theta$ is required – $\Rightarrow$ the derivative test fails; reason about the constraint directly. Here $\widehat{\theta} = \max_i X_i$.
- You want a quick estimator with **no maximisation**, or the likelihood is awkward $\Rightarrow$ use the **method of moments (MoM)**: set population moments equal to sample moments, $\mathbb{E}(X^k) = \frac{1}{n}\sum X_i^k$, and solve for θ.
- Why not the other tool? MoM ignores how peaked the likelihood is, so it can be inefficient or even fall outside the parameter space; MLE needs a workable likelihood and, at a boundary, cannot be found by differentiating. When the model is a one-parameter exponential family with mean $= \theta$, the two methods *coincide*; otherwise expect them to differ.

Keep straight: the **likelihood** is the same product as the joint density, but read as a function of θ with the data *fixed*; maximising it does not make it a probability distribution over θ (that is the Bayesian posterior of §7).

5.1.1 The likelihood and log-likelihood of a sample

Definition 5.1.1 (Likelihood and log-likelihood)**.** Let $X_1, \dots, X_n$ be iid with density or mass function $f(x; \theta)$ for an unknown parameter θ in a parameter space Θ. Given observed values $x_1, \dots, x_n$, the **likelihood function** is the joint density read as a function of θ,

$$L(\theta) = \prod_{i=1}^{n} f(x_i; \theta),$$

and the **log-likelihood** is $\ell(\theta) = \log L(\theta) = \sum_{i=1}^{n} \log f(x_i; \theta)$. The **maximum likelihood estimator** is the maximiser

$$\widehat{\theta} = \arg\max_{\theta \in \Theta} \; L(\theta) = \arg\max_{\theta \in \Theta} \; \ell(\theta).$$

Remark. We maximise ℓ rather than L because log is strictly increasing (so the maximiser is the same) and turns the product into a sum, which differentiates cleanly. The equation $\ell'(\theta) = 0$ is the **likelihood equation**; a solution is an MLE only after checking it is a maximum (second-derivative or boundary argument), not a minimum or saddle.

Example 5.1.2 (Reading a likelihood off three coin tosses)**.** A bent coin lands heads with unknown probability p. We toss it $n = 3$ times and observe H, T, H, i.e. data $x_1 = 1, x_2 = 0, x_3 = 1$ with $X_i \sim \text{Bin}(1, p)$ (Bernoulli). Write the likelihood and the log-likelihood, and read off where the peak is.

Which rule, and why? We are given a parametric model (Bin$(1, p)$) and asked how plausible each value of p is for the data on hand. That is exactly the likelihood; with only one parameter and a smooth density we will later maximise it, but first we just assemble $L(p)$.

Solution.

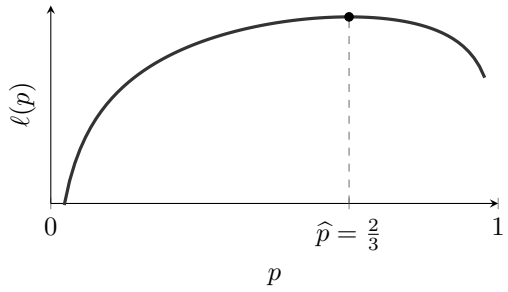
Step 1: write each factor. For a Bernoulli, $f(x; p) = p^x(1-p)^{1-x}$. So

$$L(p) = p^1(1-p)^0 \cdot p^0(1-p)^1 \cdot p^1(1-p)^0 = p^2(1-p), \qquad 0 \le p \le 1.$$

Step 2: take logs. With two heads and one tail, $\ell(p) = 2\log p + \log(1-p)$.

Step 3: locate the peak. Setting $\ell'(p) = \frac{2}{p} - \frac{1}{1-p} = 0$ gives $2(1-p) = p$, so $p = \frac{2}{3}$. Since $\ell''(p) = -2/p^2 - 1/(1-p)^2 < 0$ everywhere, this is the maximum.

$$\widehat{p} = \tfrac{2}{3} = 0.6667 = \frac{\text{number of heads}}{\text{number of tosses}}.$$



The single hump peaking at $p = \frac{2}{3}$ is the picture behind every smooth MLE below: find the top of this curve.

5.1.2 MLEs for the standard families (smooth, interior maximum)

For all of the following, $X_1, \ldots, X_n$ are iid and the maximiser is interior to Θ, so we solve $\ell' = 0$ and confirm $\ell'' < 0$.

Theorem 5.1.3 (MLEs for Bernoulli/binomial, Poisson, exponential, and normal)**.** *Let $\overline{X} = \frac{1}{n}\sum_i X_i$. Then:*

1. $X_i \sim \mathrm{Bin}(1, p)$*:* $\quad \widehat{p} = \overline{X}$ *(the sample proportion of successes).*
2. $X_i \sim \mathrm{Poisson}(\lambda)$*:* $\quad \widehat{\lambda} = \overline{X}$.
3. $X_i \sim \mathrm{Exp}(\lambda)$ *with density* $\lambda e^{-\lambda x}$*:* $\quad \widehat{\lambda} = 1/\overline{X}$.
4. $X_i \sim \mathrm{N}(\mu, \sigma^2)$*:* $\quad \widehat{\mu} = \overline{X}$ *and* $\widehat{\sigma}^2 = \frac{1}{n}\sum_i (X_i - \overline{X})^2$.

Proof. Bernoulli. With $s = \sum x_i$ successes, $L(p) = p^s(1-p)^{n-s}$ and $\ell(p) = s\log p + (n-s)\log(1-p)$. Then $\ell'(p) = \frac{s}{p} - \frac{n-s}{1-p} = 0$ gives $s(1-p) = (n-s)p$, i.e. $s = np$, so $\widehat{p} = s/n = \overline{X}$; $\ell'' < 0$.

Poisson. $f(x;\lambda) = e^{-\lambda}\lambda^x/x!$, so $\ell(\lambda) = -n\lambda + \left(\sum x_i\right)\log\lambda - \sum \log(x_i!)$ and $\ell'(\lambda) = -n + \frac{\sum x_i}{\lambda} = 0$ gives $\widehat{\lambda} = \overline{X}$; $\ell'' = -\sum x_i/\lambda^2 < 0$.

Exponential. $\ell(\lambda) = n\log\lambda - \lambda\sum x_i$, so $\ell'(\lambda) = \frac{n}{\lambda} - \sum x_i = 0$ gives $\widehat{\lambda} = n/\sum x_i = 1/\overline{X}$; $\ell'' = -n/\lambda^2 < 0$.

Normal. $\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\sum(x_i - \mu)^2$. Setting $\partial\ell/\partial\mu = \frac{1}{\sigma^2}\sum(x_i - \mu) = 0$ gives $\widehat{\mu} = \overline{X}$. Then $\partial\ell/\partial\sigma^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum(x_i - \overline{X})^2 = 0$ gives $\widehat{\sigma}^2 = \frac{1}{n}\sum(x_i - \overline{X})^2$. □

Remark. A binomial count $Y \sim \mathrm{Bin}(m, p)$ from a single experiment of m trials gives $\ell(p) = \mathrm{const} + y\log p + (m-y)\log(1-p)$, so $\widehat{p} = y/m$: same formula with the role of "n" played by the number of trials. The Bernoulli sample above is the case $m = 1$ repeated n times, and $\sum x_i$ is again the total number of successes over the total number of trials.

Example 5.1.4 (Poisson: defects per circuit board)**.** An assembly line yields surface defects per board modelled as Poisson(λ). Ten boards give the counts $2, 0, 3, 1, 2, 4, 1, 0, 2, 5$. Estimate λ by maximum likelihood.

Which rule, and why? Counts of independent rare events on each unit are the signature of a Poisson model; the mass function is smooth in $\lambda > 0$, so the MLE comes from $\ell'(\lambda) = 0$, giving $\widehat{\lambda} = \overline{X}$.

Solution.

Step 1: total and mean. $\sum x_i = 2 + 0 + 3 + 1 + 2 + 4 + 1 + 0 + 2 + 5 = 20$, $n = 10$, so $\overline{X} = 20/10 = 2$.

Step 2: apply the MLE. For a Poisson sample $\widehat{\lambda} = \overline{X}$.

$$\widehat{\lambda} = 2.0 \text{ defects per board.}$$

Check. $\ell'(\lambda) = -n + (\sum x_i)/\lambda = -10 + 20/\lambda$, which is 0 at $\lambda = 2$ and changes sign $+ \to -$ there: a genuine maximum.

Example 5.1.5 (Exponential: time between server requests)**.** Idle gaps (seconds) between requests hitting a server are modelled as $\mathrm{Exp}(\lambda)$ with mean $1/\lambda$. Eight gaps are recorded: 0.4, 1.2, 0.9, 2.1, 0.6, 1.5, 0.8, 1.0 (so $\sum x_i = 8.5$). Find the MLE of the rate λ and of the mean gap $1/\lambda$.

Which rule, and why? Waiting/inter-event times with the memoryless flavour point to the exponential model. The log-likelihood $n \log \lambda - \lambda \sum x_i$ is smooth and concave in λ, so set the derivative to zero: $\widehat{\lambda} = 1/\overline{X}$.

Solution.

Step 1: sample mean. $\overline{X} = 8.5/8 = 1.0625$ s.

Step 2: rate MLE. $\widehat{\lambda} = 1/\overline{X} = 1/1.0625 \approx 0.9412$ requests per second.

Step 3: mean by invariance. The MLE of the mean is $1/\widehat{\lambda} = \overline{X} = 1.0625$ s; MLEs carry through one-to-one transformations, so we did not need to re-maximise.

$$\widehat{\lambda} \approx 0.9412 \text{ s}^{-1}, \qquad \widehat{1/\lambda} = 1.0625 \text{ s}.$$

Example 5.1.6 (Normal: assay readings, both parameters unknown)**.** Six replicate concentration readings (mg/L) of a calibrated standard are $5.1, 4.8, 5.3, 4.9, 5.0, 4.9$. Assuming $\mathrm{N}(\mu, \sigma^2)$, find the MLEs of μ and σ^2.

Which rule, and why? Continuous measurements clustered symmetrically about a centre, with both location and spread unknown, call for the normal model. The two likelihood equations decouple: $\widehat{\mu} = \overline{X}$, then $\widehat{\sigma}^2$ is the mean squared deviation about $\overline{X}$ – note the divisor n, not $n-1$.

Solution.

Step 1: mean. $\sum x_i = 5.1 + 4.8 + 5.3 + 4.9 + 5.0 + 4.9 = 30.0$, so $\overline{X} = 30.0/6 = 5.0$ mg/L, hence $\widehat{\mu} = 5.0$.

Step 2: squared deviations. Deviations $x_i - \overline{X}$ are $0.1, -0.2, 0.3, -0.1, 0, -0.1$; their squares are $0.01, 0.04, 0.09, 0.01, 0, 0.01$, summing to 0.16.

Step 3: variance MLE.

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_i (x_i - \overline{X})^2 = \frac{0.16}{6} \approx 0.0267 \text{ (mg/L)}^2, \qquad \widehat{\sigma} \approx 0.163 \text{ mg/L}.$$

For contrast, the unbiased sample variance uses $n-1 = 5$: $S^2 = 0.16/5 = 0.032$. The MLE is the smaller; §3 explains why dividing by n makes $\widehat{\sigma}^2$ slightly biased downward.

Example 5.1.7 (Bernoulli: germination rate)**.** Of $n = 40$ planted seeds, 34 germinate, modelled as iid $\mathrm{Bin}(1, p)$. Find $\widehat{p}$ and report the maximised log-likelihood as a sanity check that the peak is interior.

Which rule, and why? A fixed number of independent yes/no trials with a common success probability is the Bernoulli/binomial setting; the smooth log-likelihood gives $\widehat{p} =$ (successes)/(trials).

Solution.

Step 1: MLE. $\widehat{p} = s/n = 34/40 = 0.85$.

Step 2: confirm interior maximum. $\ell'(p) = \frac{34}{p} - \frac{6}{1-p}$ equals 0 at $p = 0.85$ and is positive just left, negative just right – a maximum strictly inside $(0, 1)$.

$$\widehat{p} = 0.85.$$

Had every seed germinated ($s = 40$), the maximiser would slide to the boundary $p = 1$ and the derivative test would no longer locate it – a foretaste of the uniform case next.

5.1.3 The boundary case: Uniform$(0, \theta)$

Theorem 5.1.8 (MLE for Uniform$(0, \theta)$)**.** *Let $X_1, \ldots, X_n$ be iid* Uniform$(0, \theta)$*, density $f(x; \theta) = \frac{1}{\theta}$ for $0 \le x \le \theta$ and 0 otherwise. Then the MLE is the sample maximum,*

$$\widehat{\theta} = X_{(n)} = \max_i X_i,$$

which is not *obtainable by setting a derivative to zero.*

Proof. The joint density is θ^{-n} when every $x_i \le \theta$ (equivalently $\theta \ge \max_i x_i$) and 0 if any $x_i > \theta$. So

$$L(\theta) = \begin{cases} \theta^{-n}, & \theta \ge \max_i x_i, \\ 0, & \theta < \max_i x_i. \end{cases}$$

On $\theta \ge \max_i x_i$ the function θ^{-n} is strictly *decreasing*, so L is largest at the smallest admissible θ, namely $\theta = \max_i x_i$. Differentiating gives $\ell'(\theta) = -n/\theta \neq 0$ everywhere: there is no stationary point, and the maximum is forced by the constraint $\theta \ge \max x_i$, not by calculus. □

Example 5.1.9 (Uniform: longest call sets the dial)**.** A meter reads call durations (minutes) thought to be Uniform$(0, \theta)$. Five calls last $3.2, 7.8, 5.1, 6.4, 2.9$. Find the MLE of the maximum possible duration θ.

Which rule, and why? The parameter is the *upper edge* of the support, and the data must all lie below it. Any θ below an observed value gives likelihood 0, so the likelihood is maximised at the boundary $\theta = \max x_i$ – a constraint argument, not a derivative.

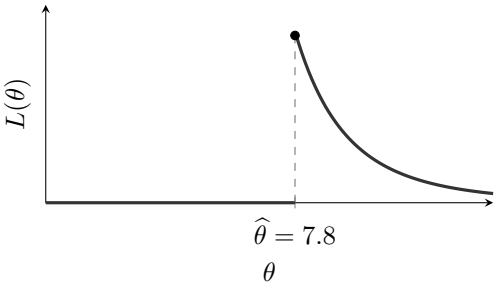
Solution.

Step 1: identify the largest observation. $\max_i x_i = 7.8$.

Step 2: read off the MLE. Any $\theta < 7.8$ is impossible (it would assign probability 0 to the 7.8 call); among admissible $\theta \ge 7.8$, the likelihood θ^{-5} decreases, so

$$\widehat{\theta} = \max_i x_i = 7.8 \text{ minutes}.$$

Note. $\widehat{\theta}$ is biased low – it can never exceed the true θ, so on average it underestimates ($\mathbb{E}\, X_{(n)} = \frac{n}{n+1}\theta$); the bias-corrected version $\frac{n+1}{n} X_{(n)}$ appears in §3.



The likelihood jumps up at $\theta = \max x_i$ and slides down after it: the maximum is at the corner, where no derivative vanishes.

5.1.4 The method of moments

Definition 5.1.10 (Method of moments)**.** Suppose the model has k unknown parameters $\theta_1, \dots, \theta_k$. The **$j$-th population moment** is $\mu_j(\theta) = \mathbb{E}(X^j)$ and the **j-th sample moment** is $m_j = \frac{1}{n}\sum_{i=1}^{n} X_i^j$. The **method-of-moments estimators** solve the k equations

$$\mu_1(\theta) = m_1, \quad \mu_2(\theta) = m_2, \quad \dots, \quad \mu_k(\theta) = m_k$$

for $\theta_1, \dots, \theta_k$. For one parameter this is just $\mathbb{E}(X) = \overline{X}$.

Remark. MoM is mechanical: equate as many low-order theoretical moments to sample moments as there are unknowns and solve. It needs no maximisation and often gives a tidy closed form, but it uses only a few moments and can be inefficient – or land outside Θ – compared with the MLE, which uses the whole likelihood.

Example 5.1.11 (MoM for an exponential rate)**.** Reuse the eight server gaps with $\overline{X} = 1.0625$ s, model Exp(λ). Estimate λ by the method of moments.

Which rule, and why? One unknown parameter and an easy first moment $\mathbb{E}(X) = 1/\lambda$: set it equal to $\overline{X}$ and solve. No maximisation is needed, which is the appeal of MoM.

Solution.

Step 1: match the first moment. For $\text{Exp}(\lambda)$, $\mathbb{E}(X) = 1/\lambda$. Setting $1/\lambda = \overline{X}$ gives $\widehat{\lambda} = 1/\overline{X}$.

Step 2: evaluate. $\widehat{\lambda} = 1/1.0625 \approx 0.9412$ s^{-1}.

$$\widehat{\lambda}_{\text{MoM}} = 1/\overline{X} \approx 0.9412 = \widehat{\lambda}_{\text{MLE}}.$$

Here MoM and MLE *agree exactly* – both reduce to $1/\overline{X}$ – because the exponential is a one-parameter family with $\mathbb{E}(X)$ a one-to-one function of λ.

Example 5.1.12 (MoM for a normal pair)**.** Using the six assay readings ($\overline{X} = 5.0$, $\frac{1}{n}\sum(x_i - \overline{X})^2 = 0.16/6 \approx 0.0267$), estimate μ and σ^2 for $\text{N}(\mu, \sigma^2)$ by the method of moments.

Which rule, and why? Two unknowns, so match two moments: the first moment gives μ, and the centred second moment (the variance) gives σ^2. The algebra collapses to sample mean and sample (uncorrected) variance.

Solution.

Step 1: first moment. $\mathbb{E}(X) = \mu = \overline{X} = 5.0$, so $\widehat{\mu} = 5.0$.

Step 2: second moment. $\mathbb{E}(X^2) = \sigma^2 + \mu^2$, matched to $m_2 = \frac{1}{n}\sum x_i^2$. Subtracting $\widehat{\mu}^2$ leaves $\widehat{\sigma}^2 = m_2 - \overline{X}^2 = \frac{1}{n}\sum(x_i - \overline{X})^2 \approx 0.0267$.

$$\widehat{\mu}_{\text{MoM}} = 5.0, \qquad \widehat{\sigma}^2_{\text{MoM}} \approx 0.0267 = \widehat{\sigma}^2_{\text{MLE}}.$$

Again MoM matches the MLE; for the normal, both give $\overline{X}$ and the divisor-n variance.

Example 5.1.13 (MoM for Uniform$(0, \theta)$: where MoM and MLE disagree)**.** Reuse the five call durations $3.2, 7.8, 5.1, 6.4, 2.9$ (so $\overline{X} = 5.08$), model Uniform$(0, \theta)$. Estimate θ by the method of moments and compare with the MLE $\widehat{\theta} = 7.8$.

Which rule, and why? The MLE here is a boundary statistic ($\max x_i$), but MoM does not care about the support's edge – it matches the mean. For Uniform$(0, \theta)$, $\mathbb{E}(X) = \theta/2$, so set $\theta/2 = \overline{X}$. This will generally differ from $\max x_i$.

Solution.

Step 1: match the mean. $\mathbb{E}(X) = \theta/2$, so $\widehat{\theta}_{\text{MoM}} = 2\overline{X}$.

Step 2: evaluate. $\overline{X} = (3.2 + 7.8 + 5.1 + 6.4 + 2.9)/5 = 25.4/5 = 5.08$, so $\widehat{\theta}_{\text{MoM}} = 2(5.08) = 10.16$.

$$\widehat{\theta}_{\text{MoM}} = 10.16 \quad \text{vs} \quad \widehat{\theta}_{\text{MLE}} = 7.8.$$

Compare. The two disagree, and they can disagree badly: MoM can even produce a value *below* $\max x_i$ (impossible for the true θ) if the largest observation is an outlier, whereas the MLE $\max x_i$ is always admissible. This is the canonical example of MoM and MLE parting ways.

Example 5.1.14 (MoM beyond mean and variance: a shifted exponential)**.** Lifetimes (hours) of a component never fail before a guarantee time τ and thereafter behave exponentially: density $f(x) = \lambda e^{-\lambda(x-\tau)}$ for $x \geq \tau$, with two unknowns $\tau \geq 0$ and $\lambda > 0$. A sample has $\overline{X} = 14.0$ and sample (uncorrected) variance $\frac{1}{n}\sum(x_i - \overline{X})^2 = 9.0$. Estimate τ and λ by the method of moments.

Which rule, and why? The likelihood again has a boundary ($\tau \leq \min x_i$), making the MLE a corner statistic; MoM sidesteps that by matching the first two moments, which for this shifted-exponential family have clean closed forms. Two unknowns, so use $\mathbb{E}(X)$ and $\text{Var}(X)$.

Solution.

Step 1: model moments. If $Y = X - \tau \sim \text{Exp}(\lambda)$ then $\mathbb{E}(X) = \tau + 1/\lambda$ and $\text{Var}(X) = \text{Var}(Y) = 1/\lambda^2$.

Step 2: match variance first. $1/\lambda^2 = 9.0 \Rightarrow \widehat{\lambda} = 1/3 \approx 0.3333$ per hour.

Step 3: match the mean. $\tau + 1/\lambda = 14.0 \Rightarrow \widehat{\tau} = 14.0 - 3.0 = 11.0$ hours.

$$\widehat{\tau}_{\text{MoM}} = 11.0 \text{ h}, \qquad \widehat{\lambda}_{\text{MoM}} = \tfrac{1}{3} \approx 0.3333 \text{ h}^{-1}.$$

With two parameters MoM needs the second moment; matching the variance pinned λ, then the mean pinned the shift τ.

5.1.5 When the two methods agree, and when they part

Proposition 5.1.15 (Agreement on mean-parameter families)**.** *If the model is a one-parameter family in which $\mathbb{E}(X) = g(\theta)$ for a strictly monotone g, and the MLE satisfies the likelihood equation in the form $\mathbb{E}(X) = \overline{X}$ (as for Bernoulli, Poisson, and exponential after reparametrising by the mean), then the MoM and MLE estimators coincide.*

Proof. MoM solves $g(\theta) = \overline{X}$. For these families the likelihood equation $\ell'(\theta) = 0$ rearranges to the same equation $g(\theta) = \overline{X}$ (e.g. Poisson: $\lambda = \overline{X}$; exponential: $1/\lambda = \overline{X}$). With g strictly monotone the solution is unique, so the two estimators are equal. When θ sits at a support boundary (Uniform$(0, \theta)$) the likelihood equation is replaced by a constraint and this argument breaks, which is exactly when the methods differ. □

Remark (Takeaways). • **Same answer:** Bernoulli p, Poisson λ, exponential λ, normal (μ, σ^2) – here MLE and MoM both reduce to sample mean (and divisor-n variance).

- **Different answers:** Uniform$(0, \theta)$ – MLE $= \max X_i$ (admissible, slightly biased low), MoM $= 2\overline{X}$ (can exceed or fall short, occasionally impossible). Prefer the MLE here.
- **Default to the MLE** when the likelihood is workable: it uses all the data and is generally more efficient (§4). Reach for MoM as a quick estimate, a starting value for numerical maximisation, or when the likelihood is intractable.

5.2 Interval Estimation

A point estimate is a single number, so it is almost never exactly right; an **interval estimate** reports a range of plausible values together with a stated **confidence level** $1 - \alpha$. The recipe is always the same: start from an estimator whose sampling distribution we know (a sample mean, a sample proportion, a scaled sample variance), find a **pivot** — a function of the data and the parameter whose distribution is free of the parameter — and invert a probability statement about that pivot into a statement about the parameter. The width of the resulting interval is the **margin of error**, and we can shrink it by collecting more data.

Which interval does the problem need?

- Estimating a **mean** and the population **standard deviation σ is known** (or n is large and you are handed σ)? $\Rightarrow$ the **z-interval** $\overline{X} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$. The pivot $(\overline{X} - \mu)/(\sigma/\sqrt{n})$ is exactly standard normal.
- Estimating a **mean** but σ is **unknown** and must be estimated by S? $\Rightarrow$ the **t-interval** $\overline{X} \pm t_{\alpha/2,\, n-1}\, S/\sqrt{n}$. Replacing σ by S costs you degrees of freedom, so the normal is replaced by t_{n-1} (fatter tails).
- Estimating a **proportion** p from counts of successes? $\Rightarrow$ the **Wald interval** $\widehat{p} \pm z_{\alpha/2}\sqrt{\widehat{p}(1-\widehat{p})/n}$, valid when $n\widehat{p}$ and $n(1-\widehat{p})$ are both moderately large.
- Estimating a **variance** or standard deviation of a normal population? $\Rightarrow$ the **χ^2-interval** built from $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. This distribution is asymmetric, so the interval is *not* symmetric about S^2.
- Asked **how big a sample** guarantees a target margin? $\Rightarrow$ set the margin-of-error formula equal to the target and solve for n (rounding up).

Do not confuse the cases: use z only when σ is genuinely known; otherwise the honest choice is t. A confidence statement is about the *procedure*, not about one computed interval (see below).

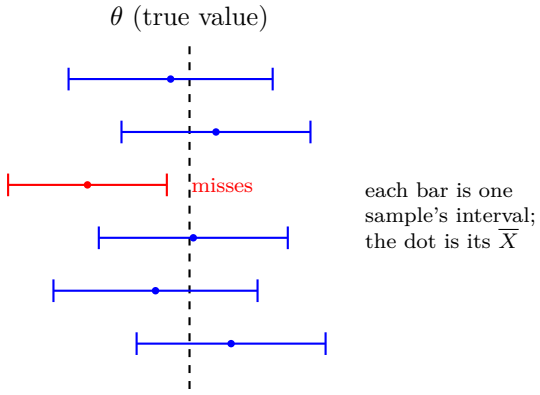
5.2.1 What "confidence" means

Definition 5.2.1 (Confidence interval)**.** Let $X_1, \dots, X_n$ be a sample from a distribution with unknown parameter θ. A $1-\alpha$ **confidence interval** for θ is a pair of statistics $L = L(X_1, \dots, X_n)$ and $U = U(X_1, \dots, X_n)$ such that

$$\mathbb{P}\big(L \le \theta \le U\big) = 1 - \alpha \qquad \text{for every value of } \theta.$$

The number $1-\alpha$ is the **confidence level**; common choices are 0.90, 0.95, 0.99, giving $\alpha = 0.10, 0.05, 0.01$.

Remark (The confidence is a property of the procedure). The endpoints L and U are random, the parameter θ is fixed. The probability $1-\alpha$ refers to the *random interval before the data are seen*: over many independent samples, about $1-\alpha$ of the intervals produced by the procedure will contain θ. Once you plug in numbers and obtain, say, $[3.1,\ 4.7]$, that fixed interval either contains θ or it does not — there is no longer any randomness, so it is wrong to say "θ has a 95% chance of lying in $[3.1, 4.7]$." The correct reading is: "the method that produced $[3.1, 4.7]$ succeeds 95% of the time."



Six intervals from six samples: five cover θ, one (red) misses. At the 95% level roughly one interval in twenty misses in the long run.

5.2.2 Mean of a normal population, σ known

Theorem 5.2.2 (z-interval for a mean)**.** *Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2)$ with σ known. Since $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$, the pivot*

$$Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1)$$

has a distribution free of μ. From $\mathbb{P}(-z_{\alpha/2} \le Z \le z_{\alpha/2}) = 1-\alpha$, where $z_{\alpha/2}$ is the value with $\Phi(z_{\alpha/2}) = 1-\alpha/2$, inverting gives the $1-\alpha$ interval

$$\overline{X} \pm z_{\alpha/2}\,\frac{\sigma}{\sqrt{n}}.$$

Proof. $\mathbb{P}\Big(-z_{\alpha/2} \le \frac{\overline{X}-\mu}{\sigma/\sqrt{n}} \le z_{\alpha/2}\Big) = 1-\alpha$. Multiply through by $\sigma/\sqrt{n}$ and subtract $\overline{X}$, then negate (which flips the inequalities), to get $\mathbb{P}\Big(\overline{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu \le \overline{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\Big) = 1-\alpha$. □

The critical values used below: $z_{0.05} = 1.645$ ($\Phi(1.645) = 0.9500$), $z_{0.025} = 1.96$ ($\Phi(1.96) = 0.9750$), $z_{0.005} = 2.576$ ($\Phi(2.576) = 0.9950$).

Example 5.2.3 (Fill volume of a bottling line)**.** A machine fills bottles with a target volume; long experience fixes the process standard deviation at $\sigma = 2.4$ mL. A sample of $n = 16$ bottles has mean $\overline{X} = 501.0$ mL. Give a 95% confidence interval for the mean fill volume μ.

Which rule, and why? The data are a normal mean and σ is *known* from process history, not estimated from this sample. That is exactly the z-interval; a t-interval would waste the known σ, and proportion/variance intervals answer different questions.

Solution.

Step 1: pick the critical value. For 95%, $\alpha = 0.05$ and $z_{\alpha/2} = z_{0.025} = 1.96$.

Step 2: standard error and margin. The standard error is $\sigma/\sqrt{n} = 2.4/\sqrt{16} = 2.4/4 = 0.60$ mL, so the margin of error is $E = 1.96 \times 0.60 = 1.176$ mL.

Step 3: assemble. 501.0 ± 1.176.

$$\big[\,499.82,\ 502.18\,\big] \text{ mL}.$$

Example 5.2.4 (A wider confidence level)**.** Same line, same sample ($\overline{X} = 501.0$, $\sigma = 2.4$, $n = 16$). Now report a 99% interval and compare its width with the 95% interval.

Which rule, and why? Still a known-σ normal mean, so still the z-interval; only the confidence level (hence $z_{\alpha/2}$) changes.

Solution.

Step 1: critical value. For 99%, $\alpha = 0.01$ and $z_{0.005} = 2.576$.

Step 2: margin. $E = 2.576 \times 0.60 = 1.5456$ mL.

Step 3: assemble and compare. 501.0 ± 1.5456, i.e. [499.45, 502.55]. Its full width $2(1.5456) = 3.091$ mL exceeds the 95% width $2(1.176) = 2.352$ mL: *more confidence costs a wider interval.*

$$\big[\,499.45,\ 502.55\,\big] \text{ mL}.$$

Example 5.2.5 (Large-sample CI by the CLT)**.** A population is *not* normal, but $\sigma = 15$ is known and the sample is large, $n = 100$, with $\overline{X} = 72.4$. Give a 90% interval for μ.

Which rule, and why? The parent is non-normal, but n is large, so the Central Limit Theorem makes $\overline{X}$ approximately normal and the z-pivot approximately standard normal. With σ supplied, the z-interval applies (approximately).

Solution.

Step 1: critical value. For 90%, $\alpha = 0.10$ and $z_{0.05} = 1.645$.

Step 2: margin. $\sigma/\sqrt{n} = 15/\sqrt{100} = 15/10 = 1.5$, so $E = 1.645 \times 1.5 = 2.4675$.

Step 3: assemble. 72.4 ± 2.4675.

$$\big[\,69.93,\ 74.87\,\big].$$

Example 5.2.6 (One-sided lower bound)**.** A coating's mean thickness μ matters only from below (too thin is the failure mode). With $\sigma = 0.8$ microns known, $n = 25$, and $\overline{X} = 12.6$, give a 95% *lower* confidence bound for μ.

Which rule, and why? A one-sided bound puts all of α in one tail, so the critical value is z_α, not $z_{\alpha/2}$. The data are a known-σ normal mean, so the z machinery still applies; we simply keep only the lower endpoint.

Solution.

Step 1: critical value. All $\alpha = 0.05$ in the lower tail uses $z_{0.05} = 1.645$.

Step 2: margin. $\sigma/\sqrt{n} = 0.8/\sqrt{25} = 0.8/5 = 0.16$, so the one-sided margin is $1.645 \times 0.16 = 0.2632$.

Step 3: lower bound. $L = \overline{X} - z_\alpha\,\sigma/\sqrt{n} = 12.6 - 0.2632$.

$$\mu \geq 12.337 \text{ microns} \quad \text{(with 95\% confidence)}.$$

5.2.3 Mean of a normal population, σ unknown

When σ is unknown we replace it by the sample standard deviation $S = \sqrt{\frac{1}{n-1}\sum(X_i - \overline{X})^2}$. The pivot then follows a Student t-distribution rather than a normal.

Theorem 5.2.7 (t-interval for a mean)**.** *Let $X_1, \ldots, X_n \nVdash \mathrm{N}(\mu, \sigma^2)$ with both parameters unknown. Then*

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1},$$

a t-distribution with $n-1$ degrees of freedom, and the $1-\alpha$ interval for μ is

$$\overline{X} \pm t_{\alpha/2,\, n-1}\, \frac{S}{\sqrt{n}},$$

where $t_{\alpha/2,\, n-1}$ is the upper-$\alpha/2$ critical value of t_{n-1}.

Remark (Why t and not z). Estimating σ by the random S injects extra variability, so the pivot has heavier tails than $\mathrm{N}(0,1)$; this forces a larger critical value and a wider interval. As $n \to \infty$, $S \to \sigma$ and $t_{n-1} \to \mathrm{N}(0,1)$, so the two intervals merge.

Example 5.2.8 (Reaction times, small sample)**.** Ten measured reaction times (ms) have $\overline{X} = 248.0$ and $S = 12.0$. Assuming approximate normality, give a 95% interval for the mean reaction time.

Which rule, and why? A normal mean with σ *unknown* (only S is available) and a small n — the defining setting for the t-interval. Using z here would understate the margin.

Solution.

Step 1: degrees of freedom and critical value. $n - 1 = 9$; for 95% we use $t_{0.025,9} = 2.262$.

Step 2: standard error and margin. $S/\sqrt{n} = 12.0/\sqrt{10} = 12.0/3.1623 = 3.7947$, so $E = 2.262 \times 3.7947 = 8.583$.

Step 3: assemble. 248.0 ± 8.583.

$$\big[\,239.42,\ 256.58\,\big] \text{ ms.}$$

Example 5.2.9 (The t versus z gap)**.** Using the data of the previous example, compute the (incorrect) interval you would get by treating S as if it were a known σ and using $z_{0.025} = 1.96$, and compare.

Which rule, and why? This is a deliberate contrast: the correct tool is t because σ is unknown; we compute the z-version only to see how much it under-covers.

Solution.

Step 1: z-margin. $E_z = 1.96 \times 3.7947 = 7.438$, giving [240.56, 255.44].

Step 2: compare. The z-interval is narrower by $2(8.583 - 7.438) = 2.29$ ms total. Because $t_{0.025,9} = 2.262 > 1.96$, the z-interval is too optimistic; its true coverage is below 95%.

$$t\text{-interval } [239.42, 256.58] \text{ wider than } z\text{-interval } [240.56, 255.44].$$

Example 5.2.10 (Moderate sample)**.** A sample of $n = 25$ battery lifetimes (hours) has $\overline{X} = 41.6$ and $S = 5.0$. Give a 90% interval for the mean lifetime.

Which rule, and why? Normal mean, σ estimated by S: t-interval again, now with 24 degrees of freedom and a 90% level.

Solution.

Step 1: critical value. $n - 1 = 24$; for 90%, $t_{0.05,24} = 1.711$.

Step 2: standard error and margin. $S/\sqrt{n} = 5.0/\sqrt{25} = 5.0/5 = 1.0$, so $E = 1.711 \times 1.0 = 1.711$.

Step 3: assemble. 41.6 ± 1.711.

$$\big[\,39.89,\ 43.31\,\big] \text{ hours.}$$

Example 5.2.11 (Computing S first)**.** Five pH readings are 7.1, 7.4, 7.0, 7.3, 7.2. Give a 95% interval for the true mean pH.

Which rule, and why? Raw data, σ unknown, small n: compute $\overline{X}$ and S, then apply the t-interval with 4 degrees of freedom.

Solution.

Step 1: sample mean. The sum is $7.1 + 7.4 + 7.0 + 7.3 + 7.2 = 36.0$, so $\overline{X} = 36.0/5 = 7.20$.

Step 2: sample variance. Deviations from 7.20 are $-0.1, 0.2, -0.2, 0.1, 0.0$; their squares are $0.01, 0.04, 0.04, 0.01, 0.00$, summing to 0.10. Then $S^2 = 0.10/(5-1) = 0.025$ and $S = 0.1581$.

Step 3: critical value, margin, assemble. $n - 1 = 4$, $t_{0.025,4} = 2.776$; $S/\sqrt{n} = 0.1581/\sqrt{5} = 0.07071$, so $E = 2.776 \times 0.07071 = 0.1963$. Then 7.20 ± 0.1963.

$$\big[\,7.004,\ 7.396\,\big].$$

5.2.4 Proportion

Theorem 5.2.12 (Large-sample interval for a proportion)**.** *Let $X \sim \mathrm{Bin}(n, p)$ and $\widehat{p} = X/n$ the sample proportion. By the Central Limit Theorem, for large n,*

$$\frac{\widehat{p} - p}{\sqrt{\widehat{p}(1-\widehat{p})/n}} \approx \mathrm{N}(0, 1),$$

giving the $1 - \alpha$ ***Wald interval***

$$\widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}.$$

The approximation is reliable when $n\widehat{p} \geq 10$ and $n(1-\widehat{p}) \geq 10$.

Example 5.2.13 (Approval poll)**.** In a survey of $n = 400$ voters, 232 approve of a measure. Give a 95% interval for the true approval proportion p.

Which rule, and why? A success/failure count out of n trials estimates a *proportion*; with $n\widehat{p}$ and $n(1-\widehat{p})$ both well above 10, the Wald z-interval applies.

Solution.

Step 1: point estimate and check. $\widehat{p} = 232/400 = 0.58$; $n\widehat{p} = 232 \geq 10$ and $n(1-\widehat{p}) = 168 \geq 10$, so the normal approximation is safe.

Step 2: standard error. $\sqrt{\widehat{p}(1-\widehat{p})/n} = \sqrt{0.58 \cdot 0.42/400} = \sqrt{0.2436/400} = \sqrt{0.000609} = 0.024678$.

Step 3: margin and assemble. $z_{0.025} = 1.96$, so $E = 1.96 \times 0.024678 = 0.04837$; then 0.58 ± 0.04837.

$$\big[\,0.5316,\ 0.6284\,\big].$$

Example 5.2.14 (Defect rate, 99%)**.** A batch of $n = 500$ parts contains 35 defectives. Give a 99% interval for the defect rate p.

Which rule, and why? Counts of defectives out of a total estimate a proportion; $n\widehat{p} = 35$ and $n(1-\widehat{p}) = 465$ both exceed 10, so the Wald interval holds at 99%.

Solution.

Step 1: point estimate. $\widehat{p} = 35/500 = 0.07$.

Step 2: standard error. $\sqrt{0.07 \cdot 0.93/500} = \sqrt{0.0651/500} = \sqrt{0.0001302} = 0.011411$.

Step 3: margin and assemble. $z_{0.005} = 2.576$, so $E = 2.576 \times 0.011411 = 0.029396$; then 0.07 ± 0.029396.

$$\big[\,0.0406,\ 0.0994\,\big].$$

Example 5.2.15 (Germination rate)**.** Of 150 seeds planted, 129 germinate. Give a 90% interval for the germination probability p.

Which rule, and why? A germinate/not count out of n seeds is a proportion; with $n\widehat{p} = 129$ and $n(1-\widehat{p}) = 21$ both above 10, use the Wald interval at 90%.

Solution.

Step 1: point estimate. $\widehat{p} = 129/150 = 0.86$.

Step 2: standard error. $\sqrt{0.86 \cdot 0.14/150} = \sqrt{0.1204/150} = \sqrt{0.00080267} = 0.028331$.

Step 3: margin and assemble. $z_{0.05} = 1.645$, so $E = 1.645 \times 0.028331 = 0.046604$; then 0.86 ± 0.046604.

$$\big[\,0.8134,\ 0.9066\,\big].$$

Example 5.2.16 (Choosing n for a poll)**.** You want a 95% interval for an unknown proportion with margin no larger than 0.03. With no prior guess for p, how many respondents are needed?

Which rule, and why? This is a sample-size problem for a proportion: set the margin $z_{\alpha/2}\sqrt{p(1-p)/n} = E$ and solve for n. With no guess, use the worst case $p = 0.5$, which maximises $p(1-p)$.

Solution.

Step 1: worst-case variance. $p(1-p)$ is largest at $p = 0.5$, giving $0.5 \cdot 0.5 = 0.25$.

Step 2: solve the margin equation. From $z_{\alpha/2}\sqrt{p(1-p)/n} \le E$ we get $n \ge \dfrac{z_{\alpha/2}^2\, p(1-p)}{E^2} = \dfrac{(1.96)^2(0.25)}{(0.03)^2} = \dfrac{3.8416 \cdot 0.25}{0.0009} = \dfrac{0.9604}{0.0009} = 1067.1$.

Step 3: round up. Always round a sample size up.

$$n = 1068.$$

5.2.5 Variance of a normal population

Theorem 5.2.17 (χ^2-interval for a variance)**.** *Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathrm{N}(\mu, \sigma^2)$. Then*

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}.$$

Writing $\chi^2_{\alpha/2,\,n-1}$ and $\chi^2_{1-\alpha/2,\,n-1}$ for the upper- and lower-tail critical values (so a χ^2_{n-1} variable exceeds $\chi^2_{\alpha/2,n-1}$ with probability $\alpha/2$), the $1-\alpha$ interval for σ^2 is

$$\left[\,\frac{(n-1)S^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\,n-1}}\,\right].$$

A confidence interval for σ follows by taking square roots of the endpoints.

Proof. $\mathbb{P}\Big(\chi^2_{1-\alpha/2,n-1} \le \frac{(n-1)S^2}{\sigma^2} \le \chi^2_{\alpha/2,n-1}\Big) = 1-\alpha$. Invert each inequality for σ^2; because we divide by the pivot, the *larger* critical value lands in the *lower* endpoint. □

Remark (The interval is asymmetric). Because χ^2_{n-1} is skewed, the two endpoints are not equidistant from S^2, and the interval is never of the form "estimate ± margin." Build it directly from the two critical values.

Example 5.2.18 (Variability of a filling process)**.** From $n = 10$ bottles, $S^2 = 4.0\ \mathrm{mL}^2$. Give a 95% interval for the process variance σ^2 (assume normality).

Which rule, and why? The target is a *variance* of a normal population, so the pivot is $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$; no mean interval applies.

Solution.

Step 1: critical values. $n - 1 = 9$; for 95%, $\chi^2_{0.975,9} = 2.700$ (lower tail) and $\chi^2_{0.025,9} = 19.023$ (upper tail).

Step 2: numerator. $(n-1)S^2 = 9 \times 4.0 = 36.0$.

Step 3: assemble. Lower endpoint $36.0/19.023 = 1.8925$; upper endpoint $36.0/2.700 = 13.333$.

$$\sigma^2 \in \big[\,1.893,\ 13.333\,\big]\ \mathrm{mL}^2.$$

Example 5.2.19 (From variance to standard deviation)**.** Using the previous data, give a 95% interval for the standard deviation σ.

Which rule, and why? A CI for σ is obtained by taking square roots of the endpoints of the σ^2 interval — the square root is monotone, so coverage is preserved.

Solution.

Step 1: reuse the variance interval. [1.8925, 13.333].

Step 2: square-root the endpoints. $\sqrt{1.8925} = 1.3757$ and $\sqrt{13.333} = 3.6515$.

$$\sigma \in \big[\,1.376,\ 3.652\,\big] \text{ mL}.$$

Example 5.2.20 (Larger sample, 90%)**.** A machinist measures $n = 20$ rod diameters and finds $S^2 = 0.0025$ mm^2. Give a 90% interval for σ^2.

Which rule, and why? Variance of a normal population again; only the degrees of freedom (19) and the level (90%) change.

Solution.

Step 1: critical values. $n - 1 = 19$; for 90%, $\chi^2_{0.95,19} = 10.117$ (lower tail) and $\chi^2_{0.05,19} = 30.144$ (upper tail).

Step 2: numerator. $(n-1)S^2 = 19 \times 0.0025 = 0.0475$.

Step 3: assemble. Lower endpoint $0.0475/30.144 = 0.0015758$; upper endpoint $0.0475/10.117 = 0.0046951$.

$$\sigma^2 \in \big[\,0.0015758,\ 0.0046951\,\big] \text{ mm}^2.$$

Example 5.2.21 (One-sided upper bound on a variance)**.** A precision instrument must keep σ^2 small. From $n = 15$ readings, $S^2 = 0.80$. Give a 95% *upper* confidence bound on σ^2.

Which rule, and why? A one-sided upper bound for a variance puts all of α in the lower tail of the χ^2 pivot, so the bound uses the single critical value $\chi^2_{1-\alpha,\,n-1}$.

Solution.

Step 1: critical value. $n - 1 = 14$; all $\alpha = 0.05$ in the lower tail gives $\chi^2_{0.95,14} = 6.571$.

Step 2: numerator. $(n-1)S^2 = 14 \times 0.80 = 11.2$.

Step 3: upper bound. $\sigma^2 \le (n-1)S^2/\chi^2_{1-\alpha,n-1} = 11.2/6.571$.

$$\sigma^2 \le 1.7044 \quad \text{(with 95\% confidence)}.$$

5.2.6 Margin of error and choosing the sample size

For a known-σ mean, the margin of error at level $1-\alpha$ is $E = z_{\alpha/2}\,\sigma/\sqrt{n}$. Solving for n gives the smallest sample meeting a target margin:

$$n \ge \left(\frac{z_{\alpha/2}\,\sigma}{E}\right)^2,$$

rounded up to the next integer. The margin shrinks like $1/\sqrt{n}$, so *halving* the margin requires *quadrupling* the sample.

Example 5.2.22 (Sample size for a mean)**.** A chemist needs a 95% interval for a mean concentration with margin at most 0.5 units. The process standard deviation is known, $\sigma = 2.0$. How many measurements are needed?

Which rule, and why? A sample-size problem for a known-σ mean: set $z_{\alpha/2}\sigma/\sqrt{n} = E$ and solve for n.

Solution.

Step 1: critical value. 95% gives $z_{0.025} = 1.96$.

Step 2: solve. $n \ge \left(\dfrac{1.96 \times 2.0}{0.5}\right)^2 = \left(\dfrac{3.92}{0.5}\right)^2 = (7.84)^2 = 61.47$.

Step 3: round up.

$$n = 62.$$

Example 5.2.23 (Quadrupling to halve the margin)**.** With $\sigma = 2.0$ and 95% confidence, how many measurements halve the margin of the previous example (target $E = 0.25$)?

Which rule, and why? Same sample-size formula; halving E should quadruple n, a useful sanity check.

Solution.

Step 1: solve. $n \geq \left(\dfrac{1.96 \times 2.0}{0.25}\right)^2 = (15.68)^2 = 245.86.$

Step 2: round up and compare. $n = 246$, almost exactly $4 \times 62 = 248$ (the small gap is rounding); halving the margin roughly quadruples the sample.

$$n = 246.$$

Example 5.2.24 (Higher confidence needs more data)**.** Repeat the first sample-size example ($\sigma = 2.0$, $E = 0.5$) but at 99% confidence.

Which rule, and why? Identical formula; only the critical value grows from $z_{0.025}$ to $z_{0.005}$, which inflates n.

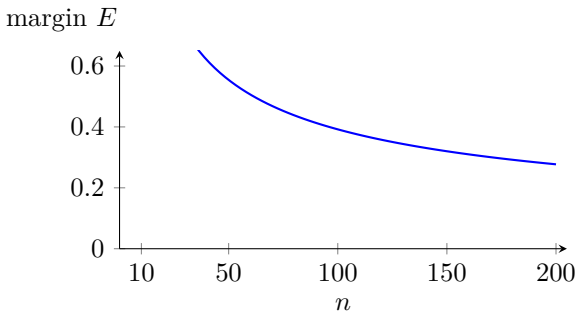
Solution.

Step 1: critical value. 99% gives $z_{0.005} = 2.576$.

Step 2: solve. $n \geq \left(\dfrac{2.576 \times 2.0}{0.5}\right)^2 = \left(\dfrac{5.152}{0.5}\right)^2 = (10.304)^2 = 106.17.$

Step 3: round up.

$$n = 107.$$



The 95% margin $E = 1.96\,(2.0)/\sqrt{n}$ as a function of n: it falls steeply at first, then flattens, so each additional unit of precision costs disproportionately more data — the $1/\sqrt{n}$ law of diminishing returns.

5.3 Properties of Estimators: Unbiasedness and Efficiency

A point estimator $\widehat{\theta} = \widehat{\theta}(X_1, \ldots, X_n)$ is a random variable: feed it a fresh sample and out comes a fresh number. Two questions decide whether it is any good. First, does it hit the target *on average* – is $\mathbb{E}(\widehat{\theta}) = \theta$? That is **unbiasedness**. Second, when it misses, how far does it scatter – how small is $\mathrm{Var}(\widehat{\theta})$? That is **efficiency**. When the two pull against each other we judge by the single combined yardstick $\mathrm{MSE}(\widehat{\theta}) = \mathbb{E}[(\widehat{\theta} - \theta)^2] = \mathrm{Var}(\widehat{\theta}) + [\mathrm{Bias}(\widehat{\theta})]^2$, and discover that the unbiased estimator is not always the winner.

Which property is the problem asking about?

- "Is $\widehat{\theta}$ **unbiased**?" or "what divisor / constant makes it unbiased?" $\Rightarrow$ compute $\mathbb{E}(\widehat{\theta})$ and compare to θ. The **bias** is $\mathrm{Bias}(\widehat{\theta}) = \mathbb{E}(\widehat{\theta}) - \theta$; unbiased means this is 0 for every θ.
- "Which of these *unbiased* estimators is better?" $\Rightarrow$ compare **variances**; the smaller-variance one is more **efficient**. The **relative efficiency** of $\widehat{\theta}_2$ to $\widehat{\theta}_1$ is $\mathrm{Var}(\widehat{\theta}_1)/\mathrm{Var}(\widehat{\theta}_2)$.

- "Compare estimators when one is *biased*" or "is it ever worth accepting bias?" ⇒ you cannot use variance alone; use **mean squared error** $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$. A biased estimator with much smaller variance can win.
- Combining several independent unbiased estimators? ⇒ a **weighted average** is again unbiased; the variance is minimised by weighting *inversely to the variances*.

Do not confuse: unbiased is about the *centre* of the sampling distribution; efficient is about its *spread*; consistency (§6) is about its *limit*. An estimator can be unbiased yet hopelessly noisy, or biased yet tightly concentrated near θ.

5.3.1 Bias and unbiased estimators

Definition 5.3.1 (Bias and unbiasedness)**.** For an estimator $\widehat{\theta}$ of a parameter θ, the **bias** is

$$\mathrm{Bias}(\widehat{\theta}) = \mathbb{E}(\widehat{\theta}) - \theta.$$

We call $\widehat{\theta}$ **unbiased** for θ if $\mathrm{Bias}(\widehat{\theta}) = 0$, i.e. $\mathbb{E}(\widehat{\theta}) = \theta$ for every value of θ. The expectation is taken over the sampling distribution of $\widehat{\theta}$, so unbiasedness is a statement about the *long-run average* of the estimate over many samples, not about any single sample.

Theorem 5.3.2 (The sample mean is unbiased for μ)**.** *Let $X_1, \ldots, X_n$ be iid with mean $\mathbb{E}(X_i) = \mu$. Then $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$ satisfies $\mathbb{E}(\overline{X}) = \mu$, so $\overline{X}$ is unbiased for μ whatever the underlying distribution.*

Proof. By linearity of expectation, $\mathbb{E}(\overline{X}) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{1}{n}\cdot n\mu = \mu$. No assumption beyond a finite common mean is used. □

Example 5.3.3 (A new pipette dispenses, on average, the labelled volume)**.** A micro-pipette is set to 50 μL. Six independent dispenses (in μL) give

$$49.2, \quad 51.1, \quad 50.4, \quad 48.7, \quad 50.9, \quad 49.7.$$

Treat them as iid with unknown mean μ. Estimate μ and confirm the estimator is unbiased.

Which rule, and why? We want a single best guess of the mean dispensed volume; the natural estimator is $\overline{X}$, and the question "is it unbiased?" asks us to check $\mathbb{E}(\overline{X}) = \mu$. No interval or variance comparison is requested, so this is a pure bias check.

Solution.

Step 1: Compute the estimate. The sum is $49.2 + 51.1 + 50.4 + 48.7 + 50.9 + 49.7 = 300.0$, so

$$\widehat{\mu} = \overline{X} = \frac{300.0}{6} = 50.0 \ \mu\mathrm{L}.$$

Step 2: Confirm unbiasedness in general. With $\mathbb{E}(X_i) = \mu$, $\mathbb{E}(\overline{X}) = \frac{1}{6}\sum_{i=1}^{6}\mathbb{E}(X_i) = \frac{1}{6}(6\mu) = \mu$. The estimate 50.0 is one draw from a distribution centred exactly at the true μ.

$$\boxed{\widehat{\mu} = 50.0 \ \mu\mathrm{L}, \qquad \mathrm{Bias}(\overline{X}) = 0}$$

Example 5.3.4 (The uncorrected "variance" is biased)**.** For iid $X_1, \ldots, X_n$ with variance σ^2, define the plug-in spread $V = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$ (the MLE of σ^2 under normality). Show V is biased and find its bias.

Which rule, and why? We are handed a candidate estimator of σ^2 and asked whether it is centred correctly, so we compute $\mathbb{E}(V)$. The key identity is that $\sum(X_i - \overline{X})^2$ loses one degree of freedom because $\overline{X}$ is itself estimated from the data.

Solution.

Step 1: Expand the sum of squared deviations. A standard identity is

$$\sum_{i=1}^{n}(X_i - \overline{X})^2 = \sum_{i=1}^{n}(X_i - \mu)^2 - n(\overline{X} - \mu)^2.$$

Step 2: Take expectations term by term. Since $\mathbb{E}[(X_i - \mu)^2] = \sigma^2$ and $\mathbb{E}[(\overline{X} - \mu)^2] = \mathrm{Var}(\overline{X}) = \sigma^2/n$,

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = n\sigma^2 - n \cdot \frac{\sigma^2}{n} = (n-1)\sigma^2.$$

Step 3: Divide by n.

$$\mathbb{E}(V) = \frac{1}{n}(n-1)\sigma^2 = \frac{n-1}{n}\,\sigma^2, \qquad \mathrm{Bias}(V) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}.$$

The estimator systematically *under*-shoots by σ^2/n; the bias vanishes only as $n \to \infty$.

$$\boxed{\mathbb{E}(V) = \tfrac{n-1}{n}\sigma^2, \qquad \mathrm{Bias}(V) = -\sigma^2/n}$$

Example 5.3.5 (Why S^2 uses the divisor $n-1$)**.** Define the sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$. Show it is unbiased for σ^2, and explain why no smaller divisor will do.

Which rule, and why? The previous example shows the sum of squared deviations has expectation $(n-1)\sigma^2$; to make an *unbiased* estimator we must divide by exactly $n-1$, not n. This is the canonical "which constant makes it unbiased?" calculation.

Solution.

Step 1: Reuse the expectation of the sum. From the previous example, $\mathbb{E}\big[\sum_i (X_i - \overline{X})^2\big] = (n-1)\sigma^2$.

Step 2: Choose the divisor to cancel the $(n-1)$.

$$\mathbb{E}(S^2) = \frac{1}{n-1}\,\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \overline{X})^2\right] = \frac{1}{n-1}\,(n-1)\sigma^2 = \sigma^2.$$

Step 3: Interpret the $n-1$. Because $\overline{X}$ is fitted from the same data, the deviations $X_i - \overline{X}$ are forced to sum to zero – only $n-1$ of them are free. Dividing by $n-1$ (the **degrees of freedom**) corrects exactly for the one constraint. Dividing by n gives the biased V of the previous example.

$$\boxed{\mathbb{E}(S^2) = \sigma^2 \text{ (unbiased)}}$$

Remark. Unbiasedness does not survive nonlinear transformation. Even though S^2 is unbiased for σ^2, the sample standard deviation S is *not* unbiased for σ: by Jensen's inequality $\mathbb{E}(S) = \mathbb{E}(\sqrt{S^2}) \le \sqrt{\mathbb{E}(S^2)} = \sigma$, so S tends to underestimate σ.

Example 5.3.6 (Correcting a biased estimator by rescaling)**.** Let $X_1, \ldots, X_n$ be iid Uniform$(0, \theta)$ and let $M = \max_i X_i$. The MLE of θ is M. Show M is biased and build an unbiased estimator from it.

Which rule, and why? Here the estimator is the sample maximum, whose density is known in closed form, so we can compute $\mathbb{E}(M)$ exactly. Finding it is short of θ tells us to rescale by a constant to remove the bias – the same "which constant?" move as for S^2.

Solution.

Step 1: Distribution of the maximum. For $0 \le m \le \theta$ the cdf of M is $\mathbb{P}(M \le m) = \mathbb{P}(\text{all } X_i \le m) = (m/\theta)^n$, so the density is $f_M(m) = n\,m^{n-1}/\theta^n$.

Step 2: Mean of M.

$$\mathbb{E}(M) = \int_0^\theta m \cdot \frac{n\, m^{n-1}}{\theta^n}\, \mathrm{d}m = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\,\theta.$$

Thus $\mathrm{Bias}(M) = \dfrac{n}{n+1}\theta - \theta = -\dfrac{\theta}{n+1} < 0$: the maximum can never exceed θ, so it always sits a little short.

Step 3: Rescale. Multiplying by $(n+1)/n$ undoes the shrink:

$$\widehat{\theta} = \frac{n+1}{n}\, M \quad \Longrightarrow \quad \mathbb{E}(\widehat{\theta}) = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta.$$

$$\boxed{\mathbb{E}(M) = \tfrac{n}{n+1}\theta \text{ (biased)}, \qquad \widehat{\theta} = \tfrac{n+1}{n} M \text{ (unbiased)}}$$

5.3.2 Comparing estimators by variance; relative efficiency

Among unbiased estimators – all centred on θ – the better one is the one that scatters less. Variance is the right comparison because, for unbiased estimators, MSE = Var.

Definition 5.3.7 (Efficiency and relative efficiency)**.** Given two unbiased estimators $\widehat{\theta}_1, \widehat{\theta}_2$ of the same θ, $\widehat{\theta}_1$ is **more efficient** than $\widehat{\theta}_2$ if $\mathrm{Var}(\widehat{\theta}_1) \le \mathrm{Var}(\widehat{\theta}_2)$ (strictly, for some θ). Their **relative efficiency** (of $\widehat{\theta}_2$ relative to $\widehat{\theta}_1$) is

$$\mathrm{eff}(\widehat{\theta}_2, \widehat{\theta}_1) = \frac{\mathrm{Var}(\widehat{\theta}_1)}{\mathrm{Var}(\widehat{\theta}_2)}.$$

A value below 1 means $\widehat{\theta}_2$ is worse (more variable) than $\widehat{\theta}_1$.

Example 5.3.8 (Mean of all data versus mean of a subsample)**.** A lab can measure a serum concentration with two protocols, both unbiased for the true mean μ with the same per-measurement variance σ^2. Protocol A averages all $n = 12$ replicates; protocol B (cheaper) averages only $m = 4$ of them. Compare their efficiency.

Which rule, and why? Both estimators are sample means, hence unbiased for μ, so the contest is decided by variance alone – this is a relative-efficiency comparison, not an MSE one.

Solution.

Step 1: Variances. For an average of k iid observations, $\mathrm{Var} = \sigma^2/k$. Thus $\mathrm{Var}(\widehat{\mu}_A) = \sigma^2/12$ and $\mathrm{Var}(\widehat{\mu}_B) = \sigma^2/4$.

Step 2: Relative efficiency of B to A.

$$\mathrm{eff}(\widehat{\mu}_B, \widehat{\mu}_A) = \frac{\mathrm{Var}(\widehat{\mu}_A)}{\mathrm{Var}(\widehat{\mu}_B)} = \frac{\sigma^2/12}{\sigma^2/4} = \frac{4}{12} = \frac{1}{3} \approx 0.33.$$

Protocol B carries three times the variance; equivalently, A is three times as efficient. The extra eight measurements are not wasted.

$$\boxed{\mathrm{eff}(\widehat{\mu}_B, \widehat{\mu}_A) = 1/3}$$

Example 5.3.9 (Sample mean versus sample median for normal data)**.** For $X_1, \ldots, X_n$ iid $\mathrm{N}(\mu, \sigma^2)$ both the sample mean $\overline{X}$ and the sample median $\widetilde{X}$ are (asymptotically) unbiased for μ. For large n, $\mathrm{Var}(\overline{X}) = \sigma^2/n$ and $\mathrm{Var}(\widetilde{X}) \approx \dfrac{\pi}{2} \cdot \dfrac{\sigma^2}{n}$. Which is more efficient?

Which rule, and why? Both estimate the centre μ and both are unbiased here, so once again variance decides. We are told the (large-sample) variances, so we just form the ratio.

Solution.

Step 1: Form the relative efficiency.

$$\mathrm{eff}(\widetilde{X}, \overline{X}) = \frac{\mathrm{Var}(\overline{X})}{\mathrm{Var}(\widetilde{X})} = \frac{\sigma^2/n}{(\pi/2)\,\sigma^2/n} = \frac{2}{\pi} \approx 0.637.$$

Step 2: Read it off. The median has about $1/0.637 \approx 1.57$ times the variance of the mean; to match the precision of $\overline{X}$ you would need roughly 57% more data using the median. For *normal* data the mean is the more efficient estimator of centre. (The median wins only for heavier-tailed data, where it resists outliers.)

$$\boxed{\operatorname{eff}(\widetilde{X}, \overline{X}) = 2/\pi \approx 0.637}$$

Theorem 5.3.10 (Best linear combination of independent unbiased estimators)**.** *Let $\widehat{\theta}_1, \widehat{\theta}_2$ be independent and each unbiased for θ, with variances v_1, v_2. Every weighted average $\widehat{\theta}_w = w\,\widehat{\theta}_1 + (1-w)\,\widehat{\theta}_2$ is unbiased, and $\operatorname{Var}(\widehat{\theta}_w)$ is minimised at*

$$w^\star = \frac{v_2}{v_1 + v_2}, \qquad \operatorname{Var}(\widehat{\theta}_{w^\star}) = \frac{v_1 v_2}{v_1 + v_2} = \frac{1}{1/v_1 + 1/v_2}.$$

The optimal weights are inversely proportional to the variances.

Proof. Unbiasedness: $\mathbb{E}(\widehat{\theta}_w) = w\theta + (1-w)\theta = \theta$. By independence $\operatorname{Var}(\widehat{\theta}_w) = w^2 v_1 + (1-w)^2 v_2$. Differentiating in w and setting to zero gives $2wv_1 - 2(1-w)v_2 = 0$, so $w^\star = v_2/(v_1+v_2)$; the second derivative $2(v_1+v_2) > 0$ confirms a minimum. Substituting yields the stated minimum variance. □

Example 5.3.11 (Pooling two independent assays optimally)**.** Two independent labs each report an unbiased estimate of a compound's potency θ. Lab 1's estimator has variance $v_1 = 4$; lab 2's has variance $v_2 = 9$ (same units^2). Find the variance-minimising pooled estimator and its variance.

Which rule, and why? We are combining two independent unbiased estimators, so any weighted average stays unbiased and the only question is the weight that minimises variance – the inverse-variance weighting theorem applies directly.

Solution.

Step 1: Optimal weight on lab 1.

$$w^\star = \frac{v_2}{v_1 + v_2} = \frac{9}{4+9} = \frac{9}{13}.$$

So $\widehat{\theta} = \frac{9}{13}\widehat{\theta}_1 + \frac{4}{13}\widehat{\theta}_2$; the more precise lab 1 gets the larger weight, as it should.

Step 2: Resulting variance.

$$\operatorname{Var}(\widehat{\theta}) = \frac{v_1 v_2}{v_1 + v_2} = \frac{4 \cdot 9}{13} = \frac{36}{13} \approx 2.77.$$

This is smaller than either input variance (4 and 9) – pooling beats trusting either lab alone.

$$\boxed{w^\star = 9/13, \qquad \operatorname{Var}(\widehat{\theta}) = 36/13 \approx 2.77}$$

Example 5.3.12 (A weighted estimator that throws away efficiency)**.** Same two labs, but a manager naively averages the estimates with equal weights, $\widehat{\theta}_{\text{eq}} = \frac{1}{2}\widehat{\theta}_1 + \frac{1}{2}\widehat{\theta}_2$. How much efficiency is lost versus the optimal pooling?

Which rule, and why? Both the equal-weight and optimal-weight estimators are unbiased, so we compare them by variance / relative efficiency. We computed the optimal variance above; here we just evaluate the equal-weight variance and divide.

Solution.

Step 1: Variance of the equal-weight estimator.

$$\operatorname{Var}(\widehat{\theta}_{\text{eq}}) = \tfrac{1}{4}v_1 + \tfrac{1}{4}v_2 = \tfrac{1}{4}(4+9) = \frac{13}{4} = 3.25.$$

Step 2: Relative efficiency versus the optimal estimator.

$$\operatorname{eff}(\widehat{\theta}_{\text{eq}}, \widehat{\theta}) = \frac{\operatorname{Var}(\widehat{\theta})}{\operatorname{Var}(\widehat{\theta}_{\text{eq}})} = \frac{36/13}{13/4} = \frac{144}{169} \approx 0.852.$$

The equal-weight average is about 85% as efficient as inverse-variance pooling – a modest but real loss for ignoring that lab 1 is more precise.

$$\boxed{\mathrm{Var}(\widehat{\theta}_{\mathrm{eq}}) = 3.25, \qquad \mathrm{eff} = 144/169 \approx 0.85}$$

5.3.3 Mean squared error and the bias–variance tradeoff

When estimators differ in *both* centre and spread, variance alone cannot rank them. The right scalar score is the mean squared error, which charges for both kinds of error at once.

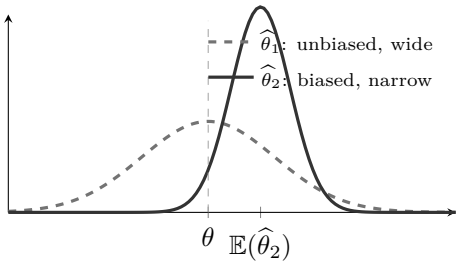
Definition 5.3.13 (Mean squared error)**.** The **mean squared error** of $\widehat{\theta}$ is $\mathrm{MSE}(\widehat{\theta}) = \mathbb{E}\big[(\widehat{\theta} - \theta)^2\big]$.

Theorem 5.3.14 (Bias–variance decomposition)**.** *For any estimator with finite second moment,*

$$\mathrm{MSE}(\widehat{\theta}) = \mathrm{Var}(\widehat{\theta}) + [\mathrm{Bias}(\widehat{\theta})]^2.$$

Proof. Write $\widehat{\theta} - \theta = (\widehat{\theta} - \mathbb{E}\widehat{\theta}) + (\mathbb{E}\widehat{\theta} - \theta)$. The second bracket is the constant $\mathrm{Bias}(\widehat{\theta})$. Squaring and taking expectations, the cross term is $2\,\mathrm{Bias}(\widehat{\theta})\,\mathbb{E}(\widehat{\theta} - \mathbb{E}\widehat{\theta}) = 0$, leaving $\mathbb{E}[(\widehat{\theta} - \mathbb{E}\widehat{\theta})^2] + [\mathrm{Bias}(\widehat{\theta})]^2 = \mathrm{Var}(\widehat{\theta}) + [\mathrm{Bias}(\widehat{\theta})]^2$. □

For an unbiased estimator the bias term vanishes and MSE = Var; that is why variance was the right yardstick in the previous subsection. Once bias is allowed, a little bias can buy a large drop in variance and a smaller MSE overall. The picture below contrasts an unbiased-but-wide estimator with a biased-but-tight one; the tight one spends more of its probability near θ.



The dashed curve is centred exactly on θ but spread out; the solid curve is shifted slightly right (a small bias) yet far more concentrated. If the variance reduction outweighs the squared bias, the solid estimator has the smaller MSE despite missing the centre.

Example 5.3.15 (Shrinking the variance estimator below S^2)**.** For $X_1, \ldots, X_n$ iid $\mathrm{N}(\mu, \sigma^2)$, consider estimators of σ^2 of the form $T_c = \frac{1}{c}\sum_{i=1}^{n}(X_i - \overline{X})^2$. Three choices stand out: $c = n - 1$ (the unbiased S^2), $c = n$ (the MLE), and $c = n + 1$. Compare their MSE. Use the normal-theory facts that $W = \sum_i (X_i - \overline{X})^2$ satisfies $W/\sigma^2 \sim \chi^2_{n-1}$, so $\mathbb{E}(W) = (n-1)\sigma^2$ and $\mathrm{Var}(W) = 2(n-1)\sigma^4$.

Which rule, and why? The three estimators differ in bias (only $c = n - 1$ is unbiased), so variance cannot rank them – we need $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$. The χ^2 facts give $\mathbb{E}(W)$ and $\mathrm{Var}(W)$ in closed form, making every term computable.

Solution.

Step 1: Bias and variance of T_c. Since $T_c = W/c$,

$$\mathbb{E}(T_c) = \frac{(n-1)\sigma^2}{c}, \qquad \mathrm{Bias}(T_c) = \Big(\frac{n-1}{c} - 1\Big)\sigma^2, \qquad \mathrm{Var}(T_c) = \frac{\mathrm{Var}(W)}{c^2} = \frac{2(n-1)\sigma^4}{c^2}.$$

Step 2: Assemble the MSE.

$$\mathrm{MSE}(T_c) = \frac{2(n-1)\sigma^4}{c^2} + \Big(\frac{n-1}{c} - 1\Big)^2\sigma^4 = \sigma^4\left[\frac{2(n-1) + (n-1)^2}{c^2} - \frac{2(n-1)}{c} + 1\right].$$

Step 3: Minimise over c. The bracket has the form $A/c^2 - B/c + 1$ with $A = (n-1)(n+1)$ and $B = 2(n-1)$; its derivative $-2A/c^3 + B/c^2 = 0$ gives

$$c^\star = \frac{2A}{B} = \frac{2(n-1)(n+1)}{2(n-1)} = n+1.$$

So among all divisors the MSE is smallest at $c = n+1$ – *neither* the unbiased $n-1$ nor the MLE's n.

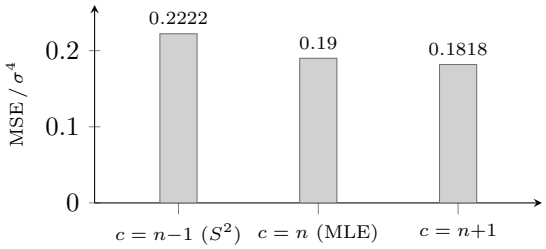
Step 4: Numbers at $n = 10$ (in units of σ^4). Here $n - 1 = 9$, so $A = 9 \cdot 11 = 99$ and $B = 18$:

$$\mathrm{MSE}(S^2) = \frac{99}{81} - \frac{18}{9} + 1 = \frac{2}{9} \approx 0.2222, \quad \mathrm{MSE}(\widehat{\sigma}^2_{\mathrm{MLE}}) = \frac{99}{100} - \frac{18}{10} + 1 = \frac{19}{100} = 0.1900,$$

$$\mathrm{MSE}(T_{n+1}) = \frac{99}{121} - \frac{18}{11} + 1 = \frac{2}{11} \approx 0.1818.$$

The slightly biased T_{n+1} beats the unbiased S^2 in MSE by about 18%, and even edges out the MLE.

$$\boxed{c^\star = n+1; \quad \mathrm{MSE}(T_{n+1}) < \mathrm{MSE}(\widehat{\sigma}^2_{\mathrm{MLE}}) < \mathrm{MSE}(S^2)}$$



At $n = 10$ the MSE drops as the divisor grows from $n-1$ to $n+1$: accepting a small downward bias in exchange for a smaller variance lowers the total error.

Example 5.3.16 (A shrinkage estimator of a mean beats the raw mean in MSE)**.** Suppose $X_1, \ldots, X_n$ are iid $\mathrm{N}(\mu, \sigma^2)$ with σ^2 known, and consider the shrinkage estimator $\widehat{\mu}_\lambda = \lambda \overline{X}$ for a constant $0 < \lambda \leq 1$ (it pulls the estimate toward 0). For which λ is $\mathrm{MSE}(\widehat{\mu}_\lambda) < \mathrm{MSE}(\overline{X})$?

Which rule, and why? $\widehat{\mu}_\lambda$ is biased whenever $\lambda \neq 1$ (its mean is $\lambda\mu$), so the ranking against the unbiased $\overline{X}$ must use MSE, not variance. We write MSE as a function of λ and minimise.

Solution.

Step 1: Bias and variance. $\mathbb{E}(\widehat{\mu}_\lambda) = \lambda\mu$ gives $\mathrm{Bias} = \lambda\mu - \mu = (\lambda - 1)\mu$, and $\mathrm{Var}(\widehat{\mu}_\lambda) = \lambda^2\sigma^2/n$. Hence

$$\mathrm{MSE}(\widehat{\mu}_\lambda) = \frac{\lambda^2\sigma^2}{n} + (\lambda - 1)^2\mu^2.$$

Step 2: Minimise over λ. Differentiate and set to zero:

$$\frac{2\lambda\sigma^2}{n} + 2(\lambda - 1)\mu^2 = 0 \implies \lambda^\star = \frac{\mu^2}{\mu^2 + \sigma^2/n} \in (0, 1).$$

Since $\lambda^\star < 1$, some shrinkage strictly lowers MSE; the optimal λ shrinks more (is smaller) when the noise σ^2/n is large relative to μ^2.

Step 3: Concrete numbers. Take $\mu = 2$, $\sigma^2 = 9$, $n = 9$, so $\sigma^2/n = 1$ and $\mu^2 = 4$. Then $\lambda^\star = 4/(4+1) = 0.8$, and

$$\mathrm{MSE}(\widehat{\mu}_{0.8}) = 0.8^2 \cdot 1 + (0.8 - 1)^2 \cdot 4 = 0.64 + 0.16 = 0.80, \qquad \mathrm{MSE}(\overline{X}) = \frac{\sigma^2}{n} = 1.$$

Shrinking by 20% cuts the MSE from 1 to 0.80. (In practice μ is unknown, so $\lambda^\star$ must be estimated – which is exactly the idea behind empirical-Bayes and James–Stein shrinkage.)

$$\boxed{\lambda^\star = \frac{\mu^2}{\mu^2 + \sigma^2/n}; \quad \mathrm{MSE} = 0.80 < 1 = \mathrm{MSE}(\overline{X}) \text{ at } \mu = 2, \sigma^2 = 9, n = 9}$$

Example 5.3.17 (MSE picks the better divisor for a uniform endpoint)**.** Return to $X_1, \dots, X_n$ iid Uniform$(0, \theta)$ with $M = \max_i X_i$. Compare three estimators of θ by MSE: the MLE M, the unbiased $\frac{n+1}{n}M$, and $\frac{n+2}{n+1}M$. Use $\mathbb{E}(M) = \frac{n}{n+1}\theta$ and $\mathbb{E}(M^2) = \frac{n}{n+2}\theta^2$.

Which rule, and why? The three estimators have different biases (only the middle one is unbiased), so MSE is the only fair comparison. The two moments of M make every $\mathrm{MSE}(cM) = c^2\mathbb{E}(M^2) - 2c\theta\mathbb{E}(M) + \theta^2$ computable.

Solution.

Step 1: MSE of cM. With $\mathbb{E}(M^2) = \frac{n}{n+2}\theta^2$ and $\mathbb{E}(M) = \frac{n}{n+1}\theta$,

$$\mathrm{MSE}(cM) = c^2\,\frac{n}{n+2}\theta^2 - 2c\,\frac{n}{n+1}\theta^2 + \theta^2.$$

Minimising over c gives $c^\star = \dfrac{\mathbb{E}(M)/\theta}{\mathbb{E}(M^2)/\theta^2} = \dfrac{n/(n+1)}{n/(n+2)} = \dfrac{n+2}{n+1}$, the third estimator.

Step 2: Evaluate at $n = 4$ (units of θ^2). Here $\mathbb{E}(M^2) = \frac{4}{6}\theta^2 = \frac{2}{3}\theta^2$ and $\mathbb{E}(M) = \frac{4}{5}\theta$:

$$\mathrm{MSE}(M) = \tfrac{2}{3} - 2\cdot\tfrac{4}{5} + 1 = \tfrac{1}{15} \approx 0.0667, \qquad \mathrm{MSE}\Big(\tfrac{5}{4}M\Big) = \Big(\tfrac{5}{4}\Big)^2\tfrac{2}{3} - 2\cdot\tfrac{5}{4}\cdot\tfrac{4}{5} + 1 = \tfrac{1}{24} \approx 0.0417,$$

$$\mathrm{MSE}\Big(\tfrac{6}{5}M\Big) = \Big(\tfrac{6}{5}\Big)^2\tfrac{2}{3} - 2\cdot\tfrac{6}{5}\cdot\tfrac{4}{5} + 1 = \tfrac{1}{25} = 0.0400.$$

The unbiased $\frac{5}{4}M$ already crushes the MLE's MSE (it removes a large negative bias), and the slightly biased $\frac{6}{5}M$ nudges it down further.

$$\boxed{\mathrm{MSE}\big(\tfrac{6}{5}M\big) = \tfrac{1}{25} < \mathrm{MSE}\big(\tfrac{5}{4}M\big) = \tfrac{1}{24} < \mathrm{MSE}(M) = \tfrac{1}{15}}$$

Example 5.3.18 (When unbiased is the right call after all)**.** Two unbiased estimators of μ, namely $\overline{X}$ ($\mathrm{Var} = \sigma^2/n$) and a single observation X_1 ($\mathrm{Var} = \sigma^2$), are offered alongside the biased constant $\widehat{\mu}_0 \equiv 0$ (zero variance). At $\mu = 5$, $\sigma^2 = 4$, $n = 16$, rank all three by MSE.

Which rule, and why? One candidate is wildly biased (it ignores the data), so variance is meaningless and MSE is mandatory. This guards against the trap of thinking "smallest variance always wins": the zero-variance estimator has enormous squared bias.

Solution.

Step 1: MSE of each. For the two unbiased estimators MSE = Var:

$$\mathrm{MSE}(\overline{X}) = \frac{\sigma^2}{n} = \frac{4}{16} = 0.25, \qquad \mathrm{MSE}(X_1) = \sigma^2 = 4.$$

For $\widehat{\mu}_0 \equiv 0$: Var $= 0$ but Bias $= 0 - \mu = -5$, so

$$\mathrm{MSE}(\widehat{\mu}_0) = 0 + (-5)^2 = 25.$$

Step 2: Rank. $0.25 < 4 < 25$, so $\overline{X}$ wins decisively. The zero-variance estimator is the *worst* despite minimal variance: a confidently wrong guess is punished hardest by MSE. Here accepting bias does *not* pay, because the available bias is huge while the variance saving is total but irrelevant.

$$\boxed{\mathrm{MSE}(\overline{X}) = 0.25 < \mathrm{MSE}(X_1) = 4 < \mathrm{MSE}(\widehat{\mu}_0) = 25}$$

Remark. The pattern across these examples: introduce bias only when it buys a *disproportionate* cut in variance (the variance-estimator divisor $n+1$, modest shrinkage of a mean). When the variance is already small or the bias would be large, the unbiased estimator – or the one with smallest variance among unbiased estimators – is the safe and often optimal choice. The smallest variance achievable by *any* unbiased estimator is quantified by the Cramer–Rao lower bound in the next section.

5.4 Minimum-Variance Estimators and the Cramér–Rao Bound

Among all unbiased estimators of a parameter θ, the best one is the one whose sampling distribution is most tightly concentrated — that is, the one with the smallest variance. A remarkable fact is that there is a hard floor on how small that variance can be: no unbiased estimator built from an iid sample of size n can have variance below $1/[n\mathcal{I}(\theta)]$, where $\mathcal{I}(\theta)$ is the **Fisher information**, a measure of how sharply the data's distribution responds to changes in θ. This section defines $\mathcal{I}(\theta)$, proves the **Cramér–Rao lower bound** (CRLB), computes both for the standard families, and calls an unbiased estimator **efficient** when it attains the floor.

Which rule of this section does the problem need?

- Asked *how much information* one observation carries about θ, or to set up a variance floor? ⇒ compute the **Fisher information** $\mathcal{I}(\theta)$, by either the curvature formula $-\mathbb{E}[\partial_\theta^2 \log f]$ or the score-variance formula $\mathbb{E}[(\partial_\theta \log f)^2]$ — they agree under regularity (§4.1).
- Asked for the *smallest possible variance* of an unbiased estimator, or whether a given variance is even achievable? ⇒ the **Cramér–Rao lower bound** $\mathrm{Var}(\widehat{\theta}) \geq 1/[n\mathcal{I}(\theta)]$ (§4.2).
- Asked whether a specific unbiased estimator is *best possible*? ⇒ check **efficiency**: compare its variance to the CRLB; it is efficient iff it equals the bound, and then it is the **minimum-variance unbiased estimator** (MVUE) (§4.3).
- Estimating a *function* $\tau(\theta)$ rather than θ itself? ⇒ use the generalised bound with the factor $[\tau'(\theta)]^2$ in the numerator (§4.3).

Why competing tools do not apply. Plain unbiasedness (§3) tells you an estimator is centred correctly but says nothing about whether a *lower*-variance unbiased competitor exists; the CRLB answers exactly that. The bound governs *unbiased* estimators — a biased estimator (or one in a non-regular family such as Uniform$(0,\theta)$, whose support depends on θ) can legitimately sit below the floor, so always confirm unbiasedness and regularity before quoting it.

5.4.1 Fisher information

Definition 5.4.1 (Score and Fisher information)**.** Let X have density or mass function $f(x;\theta)$ depending on a scalar parameter θ. The **score** is the θ-derivative of the log-density,

$$S(\theta) = \frac{\partial}{\partial\theta} \log f(X;\theta).$$

The **Fisher information** carried by one observation is the variance of the score, which under mild regularity equals the expected negative curvature of the log-density:

$$\mathcal{I}(\theta) \;=\; \mathbb{E}\left[\left(\tfrac{\partial}{\partial\theta} \log f(X;\theta)\right)^2\right] \;=\; -\,\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(X;\theta)\right].$$

For an iid sample $X_1, \ldots, X_n$ the information adds: the sample carries $n\mathcal{I}(\theta)$.

Theorem 5.4.2 (The two formulas agree; the score has mean zero)**.** *Under the regularity conditions that the support of f does not depend on θ and that we may differentiate under the integral sign, $\mathbb{E}[S(\theta)] = 0$ and the two displayed formulas for $\mathcal{I}(\theta)$ are equal.*

Proof. Since $\int f(x;\theta)\,\mathrm{d}x = 1$ for every θ, differentiating in θ gives $\int \partial_\theta f \,\mathrm{d}x = 0$. Writing $\partial_\theta f = f\,\partial_\theta \log f$ turns this into $\mathbb{E}[\partial_\theta \log f] = \mathbb{E}[S] = 0$. Differentiating $\int \partial_\theta f \,\mathrm{d}x = 0$ once more,

$$0 = \int \frac{\partial^2 f}{\partial\theta^2}\,\mathrm{d}x = \int \Big(\frac{\partial^2 \log f}{\partial\theta^2}\, f + \big(\partial_\theta \log f\big)^2 f\Big)\,\mathrm{d}x = \mathbb{E}\big[\partial_\theta^2 \log f\big] + \mathbb{E}\big[(\partial_\theta \log f)^2\big],$$

using $\partial_\theta^2 f = f\big[\partial_\theta^2 \log f + (\partial_\theta \log f)^2\big]$. Rearranging gives $\mathbb{E}[(\partial_\theta \log f)^2] = -\mathbb{E}[\partial_\theta^2 \log f]$, and because $\mathbb{E}[S] = 0$ the left side is exactly $\mathrm{Var}(S) = \mathcal{I}(\theta)$. □

Remark. The curvature formula is usually the quicker one: differentiate $\log f$ twice, take the negative expectation. A large $\mathcal{I}(\theta)$ means the log-density is sharply peaked in θ, so the data pin the parameter down well — which is precisely why $1/[n\mathcal{I}(\theta)]$ will turn out to be a variance floor.

Example 5.4.3 (Fisher information for Bernoulli p)**.** A single trial $X \sim \text{Bin}(1, p)$ records success ($X = 1$) or failure ($X = 0$). Find $\mathcal{I}(p)$.

Which rule, and why? One observation, a parameter p in the open interval $(0, 1)$ with fixed support $\{0, 1\}$: the family is regular, so either Fisher formula applies. We use the curvature formula.

Solution.

Step 1: Log-mass function. For $x \in \{0, 1\}$, $f(x; p) = p^x(1-p)^{1-x}$, so

$$\log f(x; p) = x \log p + (1 - x) \log(1 - p).$$

Step 2: Differentiate twice in p.

$$\frac{\partial}{\partial p} \log f = \frac{x}{p} - \frac{1-x}{1-p}, \qquad \frac{\partial^2}{\partial p^2} \log f = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}.$$

Step 3: Negative expectation, using $\mathbb{E}[X] = p$.

$$\mathcal{I}(p) = -\,\mathbb{E}\left[-\frac{X}{p^2} - \frac{1-X}{(1-p)^2}\right] = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p}.$$

$$\mathcal{I}(p) = \frac{(1-p)+p}{p(1-p)} = \frac{1}{p(1-p)}.$$

Information is large when p is near 0 or 1 (each trial is very informative) and smallest at $p = \frac{1}{2}$, where $\mathcal{I} = 4$.

Example 5.4.4 (Fisher information for Poisson λ)**.** A count $X \sim \text{Poisson}(\lambda)$ has $f(x; \lambda) = e^{-\lambda}\lambda^x/x!$ for $x = 0, 1, 2, \ldots$. Find $\mathcal{I}(\lambda)$.

Which rule, and why? Discrete count with parameter $\lambda > 0$ and θ-free support $\{0, 1, 2, \ldots\}$; regular family, curvature formula applies.

Solution.

Step 1: Log-mass function. $\log f(x; \lambda) = -\lambda + x \log \lambda - \log(x!)$.

Step 2: Differentiate twice in λ.

$$\frac{\partial}{\partial \lambda} \log f = -1 + \frac{x}{\lambda}, \qquad \frac{\partial^2}{\partial \lambda^2} \log f = -\frac{x}{\lambda^2}.$$

Step 3: Negative expectation, using $\mathbb{E}[X] = \lambda$.

$$\mathcal{I}(\lambda) = -\,\mathbb{E}\left[-\frac{X}{\lambda^2}\right] = \frac{\mathbb{E}[X]}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

Example 5.4.5 (Fisher information for the normal mean)**.** Let $X \sim \text{N}(\mu, \sigma^2)$ with σ^2 known. Find $\mathcal{I}(\mu)$.

Which rule, and why? Continuous family, support all of $\mathbb{R}$ (independent of μ); regular, so use the curvature formula in μ with σ^2 held fixed.

Solution.

Step 1: Log-density (drop μ-free constants).

$$\log f(x; \mu) = -\tfrac{1}{2} \log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

Step 2: Differentiate twice in μ.

$$\frac{\partial}{\partial \mu}\log f = \frac{x-\mu}{\sigma^2}, \qquad \frac{\partial^2}{\partial \mu^2}\log f = -\frac{1}{\sigma^2}.$$

Step 3: Negative expectation. The second derivative is constant, so

$$\mathcal{I}(\mu) = -\,\mathbb{E}\left[-\frac{1}{\sigma^2}\right] = \frac{1}{\sigma^2}.$$

Information rises as the noise σ^2 shrinks — a tighter bell pins μ down more sharply.

Example 5.4.6 (Fisher information for an exponential rate, two ways)**.** Let $X \sim \text{Exp}(\lambda)$ with density $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$. Find $\mathcal{I}(\lambda)$ by both formulas to confirm they agree.

Which rule, and why? Continuous family; the support $(0,\infty)$ does not depend on the *rate* λ, so the family is regular and either formula is valid. (Contrast Uniform$(0,\theta)$, whose support *does* move with θ: there the regularity conditions fail and no Fisher-based bound applies.)

Solution.

Step 1: Log-density. $\log f(x;\lambda) = \log\lambda - \lambda x$.

Step 2: Curvature formula.

$$\frac{\partial}{\partial \lambda}\log f = \frac{1}{\lambda} - x, \qquad \frac{\partial^2}{\partial \lambda^2}\log f = -\frac{1}{\lambda^2} \implies \mathcal{I}(\lambda) = -\,\mathbb{E}\left[-\frac{1}{\lambda^2}\right] = \frac{1}{\lambda^2}.$$

Step 3: Score-variance formula as a check. The score is $S = \frac{1}{\lambda} - X$, and since $\mathbb{E}[X] = 1/\lambda$ we have $\mathbb{E}[S] = 0$ (as the theorem promises) and

$$\mathcal{I}(\lambda) = \text{Var}(S) = \text{Var}\left(\tfrac{1}{\lambda} - X\right) = \text{Var}(X) = \frac{1}{\lambda^2},$$

using $\text{Var}(X) = 1/\lambda^2$ for the exponential. Both routes give $\mathcal{I}(\lambda) = 1/\lambda^2$.

Example 5.4.7 (Reparametrisation: information for the exponential mean)**.** Often the natural target is the *mean* $\theta = 1/\lambda$ rather than the rate. Writing the exponential density in terms of θ, find $\mathcal{I}(\theta)$.

Which rule, and why? The quantity of interest is the mean $\theta = \mathbb{E}[X]$, so re-express the density and differentiate in θ. (Equivalently one could rescale by $(\,\mathrm{d}\lambda/\,\mathrm{d}\theta)^2$, but doing it directly is cleaner and self-checking.)

Solution.

Step 1: Density in the mean parametrisation. With $\lambda = 1/\theta$, $f(x;\theta) = \frac{1}{\theta}e^{-x/\theta}$, so $\log f = -\log\theta - x/\theta$.

Step 2: Differentiate twice in θ.

$$\frac{\partial}{\partial \theta}\log f = -\frac{1}{\theta} + \frac{x}{\theta^2}, \qquad \frac{\partial^2}{\partial \theta^2}\log f = \frac{1}{\theta^2} - \frac{2x}{\theta^3}.$$

Step 3: Negative expectation, using $\mathbb{E}[X] = \theta$.

$$\mathcal{I}(\theta) = -\,\mathbb{E}\left[\frac{1}{\theta^2} - \frac{2X}{\theta^3}\right] = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = -\frac{1}{\theta^2} + \frac{2}{\theta^2} = \frac{1}{\theta^2}.$$

So $\mathcal{I}(\theta) = 1/\theta^2$; this will let $\overline{X}$ attain the bound for the mean (Example 4.12).

5.4.2 The Cramér–Rao lower bound

Theorem 5.4.8 (Cramér–Rao lower bound)**.** *Let $X_1, \dots, X_n$ be iid with density $f(x;\theta)$ satisfying the regularity conditions (support free of θ; differentiation under the integral permitted). If $\widehat{\theta} = \widehat{\theta}(X_1, \dots, X_n)$ is an* unbiased *estimator of θ, then*

$$\operatorname{Var}(\widehat{\theta}) \;\geq\; \frac{1}{n\,\mathcal{I}(\theta)}.$$

More generally, if $\mathbb{E}[\widehat{\theta}] = \tau(\theta)$ for a differentiable τ, then

$$\operatorname{Var}(\widehat{\theta}) \;\geq\; \frac{[\tau'(\theta)]^2}{n\,\mathcal{I}(\theta)}.$$

Proof. Let $S_n = \sum_{i=1}^{n} \partial_\theta \log f(X_i;\theta)$ be the total score. The single-observation scores are iid with mean 0 and variance $\mathcal{I}(\theta)$, so $\mathbb{E}[S_n] = 0$ and $\operatorname{Var}(S_n) = n\mathcal{I}(\theta)$. Differentiating the unbiasedness identity $\mathbb{E}[\widehat{\theta}] = \int \widehat{\theta}(x) \prod_i f(x_i;\theta)\,\mathrm{d}x = \tau(\theta)$ under the integral sign, and using $\partial_\theta \prod_i f = (\prod_i f)\, S_n$, gives

$$\tau'(\theta) = \int \widehat{\theta}(x)\, S_n \prod_i f(x_i;\theta)\,\mathrm{d}x = \mathbb{E}[\widehat{\theta}\, S_n] = \operatorname{Cov}(\widehat{\theta}, S_n),$$

the last step because $\mathbb{E}[S_n] = 0$. By the Cauchy–Schwarz (correlation) inequality $[\operatorname{Cov}(\widehat{\theta}, S_n)]^2 \leq \operatorname{Var}(\widehat{\theta})\operatorname{Var}(S_n)$, hence

$$[\tau'(\theta)]^2 \leq \operatorname{Var}(\widehat{\theta}) \cdot n\mathcal{I}(\theta) \quad\Longrightarrow\quad \operatorname{Var}(\widehat{\theta}) \geq \frac{[\tau'(\theta)]^2}{n\mathcal{I}(\theta)}.$$

For $\tau(\theta) = \theta$ we have $\tau' = 1$, giving the first display. □

Remark. Equality holds exactly when $\widehat{\theta}$ is a linear function of the total score S_n (the Cauchy–Schwarz equality case). That is why a handful of "nice" estimators — the sample mean for p, λ, and μ — hit the floor, while most estimators sit strictly above it.

Example 5.4.9 (CRLB for a Bernoulli proportion)**.** From n independent trials with success probability p we estimate p by the sample proportion $\widehat{p} = \overline{X} = \frac{1}{n}\sum X_i$. State the Cramér–Rao floor and compare.

Which rule, and why? We want the smallest variance any *unbiased* estimator of p can have. $\widehat{p}$ is unbiased ($\mathbb{E}[\widehat{p}] = p$), and the Bernoulli family is regular, so the CRLB with $\tau(\theta) = \theta$ applies; plug in $\mathcal{I}(p) = 1/[p(1-p)]$ from Example 4.2.

Solution.

Step 1: The floor.

$$\frac{1}{n\mathcal{I}(p)} = \frac{1}{n} \cdot p(1-p) = \frac{p(1-p)}{n}.$$

Step 2: Variance of $\widehat{p}$. Since $\operatorname{Var}(X_i) = p(1-p)$,

$$\operatorname{Var}(\widehat{p}) = \frac{1}{n^2}\sum_{i=1}^{n} \operatorname{Var}(X_i) = \frac{p(1-p)}{n}.$$

Step 3: Compare. The variance equals the bound, so $\widehat{p}$ is efficient:

$$\operatorname{Var}(\widehat{p}) = \frac{p(1-p)}{n} = \frac{1}{n\mathcal{I}(p)}.$$

Numerically: with $n = 400$ and true $p = 0.30$, the floor is $0.30 \cdot 0.70/400 = 0.000525$, i.e. a standard error of $\sqrt{0.000525} = 0.0229$ — no unbiased rule can do better.

Example 5.4.10 (CRLB for a Poisson rate)**.** A call centre logs the number of arrivals in each of n one-minute windows, modelled as iid Poisson(λ). The estimator is $\widehat{\lambda} = \overline{X}$. Find the floor and check efficiency.

Which rule, and why? Target λ is the mean of a regular Poisson family; $\overline{X}$ is unbiased for it ($\mathbb{E}[\overline{X}] = \lambda$), so the CRLB applies with $\mathcal{I}(\lambda) = 1/\lambda$ from Example 4.3.

Solution.

Step 1: The floor.

$$\frac{1}{n\mathcal{I}(\lambda)} = \frac{1}{n} \cdot \lambda = \frac{\lambda}{n}.$$

Step 2: Variance of $\overline{X}$. Since $\mathrm{Var}(X_i) = \lambda$,

$$\mathrm{Var}(\overline{X}) = \frac{\mathrm{Var}(X_1)}{n} = \frac{\lambda}{n}.$$

Step 3: Compare. Equal — $\overline{X}$ is efficient for λ:

$$\mathrm{Var}(\overline{X}) = \frac{\lambda}{n} = \frac{1}{n\mathcal{I}(\lambda)}.$$

Numerically: with $n = 50$ windows and $\lambda = 4$ calls/min, the floor is $4/50 = 0.08$, a standard error of 0.283 calls/min.

Example 5.4.11 (CRLB for the normal mean)**.** Measure a fixed voltage n times with iid $\mathrm{N}(\mu, \sigma^2)$ readings, σ^2 known. The estimator is $\widehat{\mu} = \overline{X}$. Find the floor and check efficiency.

Which rule, and why? Estimating the mean μ of a regular normal family with known variance; $\overline{X}$ is unbiased, so the CRLB applies with $\mathcal{I}(\mu) = 1/\sigma^2$ from Example 4.4.

Solution.

Step 1: The floor.

$$\frac{1}{n\mathcal{I}(\mu)} = \frac{1}{n} \cdot \sigma^2 = \frac{\sigma^2}{n}.$$

Step 2: Variance of $\overline{X}$. $\mathrm{Var}(\overline{X}) = \sigma^2/n$ directly.

Step 3: Compare. They coincide, so $\overline{X}$ is the minimum-variance unbiased estimator:

$$\mathrm{Var}(\overline{X}) = \frac{\sigma^2}{n} = \frac{1}{n\mathcal{I}(\mu)}.$$

Numerically: with $\sigma = 0.5$ V and $n = 25$ readings, the floor is $0.25/25 = 0.01$, a standard error of 0.10 V; halving it to 0.05 V would require quadrupling n to 100.

Example 5.4.12 (CRLB for the exponential mean)**.** Inter-failure times of a component are iid $\mathrm{Exp}(\lambda)$; the quantity of interest is the mean lifetime $\theta = 1/\lambda$, estimated by $\widehat{\theta} = \overline{X}$. Find the floor and check efficiency.

Which rule, and why? Here the target is the mean $\theta = 1/\lambda$, so we use the information in the *mean* parametrisation, $\mathcal{I}(\theta) = 1/\theta^2$ from Example 4.6. $\overline{X}$ is unbiased for θ ($\mathbb{E}[\overline{X}] = \theta$), so the plain CRLB applies.

Solution.

Step 1: The floor.

$$\frac{1}{n\mathcal{I}(\theta)} = \frac{1}{n} \cdot \theta^2 = \frac{\theta^2}{n}.$$

Step 2: Variance of $\overline{X}$. For the exponential, $\mathrm{Var}(X_i) = \theta^2$, so

$$\mathrm{Var}(\overline{X}) = \frac{\mathrm{Var}(X_1)}{n} = \frac{\theta^2}{n}.$$

Step 3: Compare. Equal again — $\overline{X}$ is efficient for the exponential mean:

$$\mathrm{Var}(\overline{X}) = \frac{\theta^2}{n} = \frac{1}{n\mathcal{I}(\theta)}.$$

Numerically: if the true mean lifetime is $\theta = 500$ hours and $n = 20$ components are tested, the floor is $500^2/20 = 12{,}500$ hours^2, a standard error of $\sqrt{12{,}500} = 111.8$ hours.

Remark (A caution: the bound is for unbiased, regular families). The MLE of the exponential *rate* is $\widehat{\lambda} = 1/\overline{X}$, which is *biased* ($\mathbb{E}[1/\overline{X}] = \frac{n}{n-1}\lambda \neq \lambda$), so its variance is not constrained by the plain CRLB — the bound governs unbiased estimators only. Likewise the Uniform$(0, \theta)$ family has support $(0, \theta)$ that moves with θ, breaking regularity; there the MVUE $\frac{n+1}{n}\max_i X_i$ has variance $\theta^2/[n(n+2)]$, which falls *below* the naive "θ^2/n" a Fisher calculation would suggest. Always confirm unbiasedness and regularity before quoting the floor.

5.4.3 Efficiency, the MVUE, and estimating a function of θ

Definition 5.4.13 (Efficiency and the MVUE)**.** For an unbiased estimator $\widehat{\theta}$, its **efficiency** is the ratio of the Cramér–Rao floor to its actual variance,

$$e(\widehat{\theta}) = \frac{1/[n\mathcal{I}(\theta)]}{\mathrm{Var}(\widehat{\theta})} \in (0, 1].$$

$\widehat{\theta}$ is **efficient** when $e(\widehat{\theta}) = 1$, i.e. it attains the bound; an efficient estimator is necessarily a **minimum-variance unbiased estimator** (MVUE), since nothing unbiased can beat the floor.

Example 5.4.14 (An unbiased estimator that is *not* efficient)**.** From iid $\mathrm{N}(\mu, \sigma^2)$ data (σ^2 known, $n = 9$), a lazy analyst estimates μ by $\widehat{\mu}_{\mathrm{med}}$ = the sample median instead of $\overline{X}$. For large n the median of normal data has variance approximately $\frac{\pi\sigma^2}{2n}$. Compute its efficiency.

Which rule, and why? Both estimators are (asymptotically) unbiased for μ in a regular family, so we can rank them by the CRLB; the floor is σ^2/n from Example 4.10, and we compare the median's variance to it.

Solution.

Step 1: The floor and the median's variance.

$$\frac{1}{n\mathcal{I}(\mu)} = \frac{\sigma^2}{n}, \qquad \mathrm{Var}(\widehat{\mu}_{\mathrm{med}}) \approx \frac{\pi\sigma^2}{2n}.$$

Step 2: Efficiency ratio.

$$e(\widehat{\mu}_{\mathrm{med}}) = \frac{\sigma^2/n}{\pi\sigma^2/(2n)} = \frac{2}{\pi} \approx 0.637.$$

$$e(\widehat{\mu}_{\mathrm{med}}) \approx 0.64 < 1.$$

The median wastes about 36% of the available information: to match the precision of $\overline{X}$ from $n = 9$ readings it would need roughly $9/0.637 \approx 14$ readings. The sample mean is the efficient choice here.

Example 5.4.15 (Generalised bound: estimating a function of θ)**.** From iid Poisson(λ) counts (n observations) we estimate the *mean rate per hour* $\tau(\lambda) = 60\lambda$ when λ is the rate per minute. The estimator is $\widehat{\tau} = 60\,\overline{X}$. Find the CRLB for τ and check whether $\widehat{\tau}$ attains it.

Which rule, and why? The target is a smooth function $\tau(\lambda) = 60\lambda$ of the parameter, so use the generalised bound with the factor $[\tau'(\lambda)]^2$; the underlying information is $\mathcal{I}(\lambda) = 1/\lambda$ from Example 4.3.

Solution.

Step 1: Derivative of the target. $\tau'(\lambda) = 60$.

Step 2: The floor.

$$\frac{[\tau'(\lambda)]^2}{n\mathcal{I}(\lambda)} = \frac{60^2}{n/\lambda} = \frac{3600\,\lambda}{n}.$$

Step 3: Variance of $\widehat{\tau}$. Since $\mathrm{Var}(\overline{X}) = \lambda/n$,

$$\mathrm{Var}(\widehat{\tau}) = 60^2\ \mathrm{Var}(\overline{X}) = 3600 \cdot \frac{\lambda}{n} = \frac{3600\,\lambda}{n}.$$

Step 4: Compare. The variance equals the generalised floor, so $\widehat{\tau} = 60\overline{X}$ is efficient for 60λ:

$$\operatorname{Var}(\widehat{\tau}) = \frac{3600\,\lambda}{n} = \frac{[\tau'(\lambda)]^2}{n\mathcal{I}(\lambda)}.$$

A linear rescaling of an efficient estimator stays efficient.

Example 5.4.16 (Comparing two unbiased estimators of a normal mean against the floor)**.** From iid $\mathrm{N}(\mu, \sigma^2)$ data with $n = 4$ and known σ^2, consider two unbiased estimators of μ: the ordinary mean $\widehat{\mu}_A = \frac{1}{4}(X_1 + X_2 + X_3 + X_4)$, and a weighted average $\widehat{\mu}_B = 0.4X_1 + 0.2X_2 + 0.2X_3 + 0.2X_4$. Which is more efficient, and is either one efficient?

Which rule, and why? Both are linear, hence unbiased ($\mathbb{E} = \mu$ since the weights sum to 1); in a regular family we rank unbiased estimators by variance and benchmark against the CRLB $\sigma^2/n = \sigma^2/4$.

Solution.

Step 1: Variance of each (weights w_i, $\operatorname{Var} = \sigma^2 \sum w_i^2$*).*

$$\operatorname{Var}(\widehat{\mu}_A) = \sigma^2\Big(4 \cdot \tfrac{1}{16}\Big) = \frac{\sigma^2}{4}, \qquad \operatorname{Var}(\widehat{\mu}_B) = \sigma^2\big(0.4^2 + 3 \cdot 0.2^2\big) = \sigma^2(0.16 + 0.12) = 0.28\,\sigma^2.$$

Step 2: Efficiencies against the floor $\sigma^2/4 = 0.25\sigma^2$.

$$e(\widehat{\mu}_A) = \frac{0.25\sigma^2}{0.25\sigma^2} = 1, \qquad e(\widehat{\mu}_B) = \frac{0.25\sigma^2}{0.28\sigma^2} = \frac{25}{28} \approx 0.893.$$

$$\widehat{\mu}_A \text{ is efficient (the MVUE)}; \qquad e(\widehat{\mu}_B) \approx 0.89 < 1.$$

Equal weights minimise $\sum w_i^2$ subject to $\sum w_i = 1$, so the plain mean attains the floor and the unequal weighting throws away about 11% of the precision.

Example 5.4.17 (A floor for a tail probability)**.** Survival times are iid $\mathrm{Exp}(\lambda)$ (rate λ). We want the one-year survival probability $\tau(\lambda) = \mathbb{P}(X > 1) = e^{-\lambda}$. Using the variance of the score, give the Cramér–Rao floor for any unbiased estimator of $\tau(\lambda)$ based on n observations.

Which rule, and why? The estimand is a nonlinear smooth function $\tau(\lambda) = e^{-\lambda}$, so the generalised CRLB with $[\tau'(\lambda)]^2$ is the right tool; the rate information is $\mathcal{I}(\lambda) = 1/\lambda^2$ from Example 4.5.

Solution.

Step 1: Derivative of the target. $\tau'(\lambda) = -e^{-\lambda}$, so $[\tau'(\lambda)]^2 = e^{-2\lambda}$.

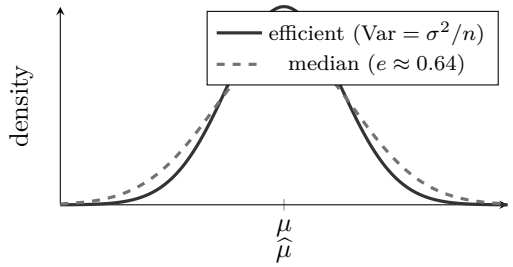
Step 2: Assemble the bound.

$$\operatorname{Var}(\widehat{\tau}) \;\geq\; \frac{[\tau'(\lambda)]^2}{n\mathcal{I}(\lambda)} = \frac{e^{-2\lambda}}{n \cdot (1/\lambda^2)} = \frac{\lambda^2 e^{-2\lambda}}{n}.$$

Step 3: Numerical floor. If $\lambda = 0.5$ per year and $n = 30$, then $\lambda^2 e^{-2\lambda} = 0.25\,e^{-1} = 0.25 \cdot 0.3679 = 0.09197$, so the floor is $0.09197/30 = 0.003066$, i.e. a standard error of at least $\sqrt{0.003066} = 0.0554$ on a probability estimate near $e^{-0.5} = 0.607$.

5.4.4 Picturing the variance floor

The Fisher information sets how sharply the log-likelihood bends at its peak: more curvature means a narrower likelihood and a smaller attainable variance. The picture below contrasts an *efficient* estimator sitting exactly on the floor with an *inefficient* one (the normal median) sitting above it; both are centred at the true μ, but the efficient one is tighter.



Remark (Reading the bound). The CRLB is a statement about the *best case*: it says where the floor is, not that any particular estimator reaches it. When an estimator does reach it — as $\overline{X}$ does for the Bernoulli, Poisson, normal-mean, and exponential-mean problems above — you have proved it is the minimum-variance unbiased estimator and can stop looking for anything better. When it does not (the median, the weighted average), the gap to the floor is exactly the precision you are leaving on the table.

5.5 Sufficient Estimators

Two different people analyse the same sample. The first writes down all n raw observations $x_1, \dots, x_n$; the second throws the raw numbers away and keeps only a single summary $T = T(x_1, \dots, x_n)$, say the total $\sum x_i$. When can the second person estimate the unknown θ *just as well* as the first — losing no information at all? The answer is: exactly when T is a **sufficient statistic** for θ. The defining feature is that once T is known, the conditional distribution of the full data no longer involves θ: the leftover randomness is pure noise that carries no signal about the parameter. The practical test for sufficiency is the **factorisation theorem**, which reads sufficiency straight off the shape of the joint density.

Which rule of this section does the problem need?

- Asked whether a statistic T **captures all the information** the sample carries about θ (so that conditioning on T erases θ)? $\Rightarrow$ the **definition of sufficiency**: show $f(\mathbf{x} \mid T = t; \theta)$ does not depend on θ.
- Asked to **find** a sufficient statistic, or to **verify** a candidate, from a density? $\Rightarrow$ the **Factorisation Theorem**: split the joint density as $g\big(T(\mathbf{x}), \theta\big)\, h(\mathbf{x})$, where h is free of θ and g touches the data only through T.
- The support of X **depends on** θ (e.g. Uniform$(0, \theta)$)? $\Rightarrow$ still the Factorisation Theorem, but the parameter enters through an **indicator** $\mathbb{1}(\max x_i \le \theta)$; the sufficient statistic is an order statistic, not a sum.
- Have an unbiased but **crude** estimator and want to **improve** it without losing unbiasedness? $\Rightarrow$ **Rao–Blackwell**: replace it by its conditional expectation given a sufficient statistic; the variance can only go down.

Why competing tools do not apply. Sufficiency is not about whether an estimator is *unbiased* (§3) or *efficient* (§4) — a sufficient statistic need not even be an estimator of θ. It is about *information content*: a biased, badly scaled function of the data can still be sufficient. Conversely, the MLE is (almost always) a function of a sufficient statistic, which is the deep reason maximum likelihood "uses all the data".

5.5.1 Sufficiency: the definition

Definition 5.5.1 (Sufficient statistic)**.** Let $X_1, \dots, X_n$ be a sample from a distribution with parameter θ, and let $T = T(X_1, \dots, X_n)$ be a statistic. Then T is **sufficient for** θ if the conditional distribution of the sample given $T = t$ does not depend on θ; that is,

$$\mathbb{P}\big(X_1 = x_1, \dots, X_n = x_n \mid T = t\big) \quad \text{(or the conditional density)} \quad \text{is free of } \theta.$$

Equivalently: knowing the value of T makes the rest of the data uninformative about θ.

Remark. A statistic is just any function of the data, so T is itself a random variable; its observed value is a number. "Sufficient" is a property of T relative to a *family* of distributions indexed by θ, not of a single distribution. The whole sample $(X_1, \dots, X_n)$ is trivially sufficient (keep everything); the content of the theory is that a much smaller summary often suffices.

Example 5.5.2 (Two Bernoulli trials: the total is sufficient, the order is not)**.** Let X_1, X_2 be independent $\mathrm{Bin}(1, p)$ (i.e. Bernoulli(p)) trials, and put $T = X_1 + X_2$. Show directly from the definition that T is sufficient for p.

Which rule, and why? We are handed a *candidate* T and asked to confirm it captures all the information about p. The cleanest check at this small size is the definition itself: compute the conditional distribution of (X_1, X_2) given T and watch p cancel. (The Factorisation Theorem below does the same job faster, but seeing p vanish once builds the intuition.)

Solution.

Step 1: the only interesting case is $T = 1$. If $T = 0$ then $(X_1, X_2) = (0, 0)$ with conditional probability 1; if $T = 2$ then $(X_1, X_2) = (1, 1)$ with conditional probability 1. Neither degenerate case involves p. The case $T = 1$ splits between $(1, 0)$ and $(0, 1)$.

Step 2: condition on $T = 1$. Using independence,

$$\mathbb{P}(X_1 = 1, X_2 = 0) = p(1-p), \qquad \mathbb{P}(X_1 = 0, X_2 = 1) = (1-p)p, \qquad \mathbb{P}(T = 1) = 2p(1-p).$$

Therefore

$$\mathbb{P}\big(X_1 = 1, X_2 = 0 \mid T = 1\big) = \frac{p(1-p)}{2p(1-p)} = \tfrac{1}{2},$$

and likewise $\mathbb{P}(X_1 = 0, X_2 = 1 \mid T = 1) = \frac{1}{2}$.

Step 3: read off sufficiency. Every conditional probability equals $\frac{1}{2}$ (or 1), with no p anywhere:

$$\mathbb{P}\big((X_1, X_2) = \cdot \mid T = t\big) \text{ is free of } p \;\Rightarrow\; T = X_1 + X_2 \text{ is sufficient for } p.$$

Once you know the *count* of successes, *which* trial succeeded is a fair coin flip that tells you nothing more about p.

5.5.2 The Factorisation Theorem

Checking the definition is fine for tiny samples but hopeless in general. The next theorem turns sufficiency into pattern-matching on the joint density.

Theorem 5.5.3 (Fisher–Neyman Factorisation Theorem)**.** *Let $X_1, \dots, X_n$ have joint density (or pmf) $f(\mathbf{x}; \theta)$. A statistic $T = T(\mathbf{X})$ is sufficient for θ if and only if there exist nonnegative functions g and h with*

$$f(x_1, \dots, x_n; \theta) \;=\; g\big(T(x_1, \dots, x_n),\, \theta\big)\; h(x_1, \dots, x_n),$$

where h does not depend on θ and g depends on the data only through the value of T.

Why it works (discrete case sketch). If such a factorisation holds, then for any $\mathbf{x}$ with $T(\mathbf{x}) = t$,

$$\mathbb{P}(\mathbf{X} = \mathbf{x} \mid T = t) = \frac{f(\mathbf{x}; \theta)}{\sum_{\mathbf{y}:\, T(\mathbf{y}) = t} f(\mathbf{y}; \theta)} = \frac{g(t, \theta)\, h(\mathbf{x})}{g(t, \theta) \sum_{\mathbf{y}:\, T(\mathbf{y}) = t} h(\mathbf{y})} = \frac{h(\mathbf{x})}{\sum_{\mathbf{y}:\, T(\mathbf{y}) = t} h(\mathbf{y})},$$

where $\mathbf{y}$ ranges over data vectors with $T(\mathbf{y}) = t$. The common factor $g(t, \theta)$ cancels, leaving an expression free of θ: T is sufficient. Conversely, if T is sufficient, write $f(\mathbf{x}; \theta) = \mathbb{P}(T = t; \theta)\, \mathbb{P}(\mathbf{X} = \mathbf{x} \mid T = t)$; the first factor is a function of (t, θ) and the second is free of θ, which is exactly the required factorisation with $g = \mathbb{P}(T = t; \theta)$ and $h = \mathbb{P}(\mathbf{X} = \mathbf{x} \mid T = t)$. □

Remark (How to use it). Write the joint density, then *collect every place θ appears* and check that those pieces touch the data only through a single summary T. Whatever is left — factorials, $\mathbb{1}(x_i \geq 0)$, constants — is the harmless $h(\mathbf{x})$. The summary T that emerges is a sufficient statistic.

Example 5.5.4 (Bernoulli/binomial proportion p)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bin}(1, p)$. Find a sufficient statistic for p.

Which rule, and why? We want to find (not just verify) a sufficient statistic from a pmf. That is the home turf of the Factorisation Theorem: write the joint pmf and isolate p.

Solution.

Step 1: joint pmf. With each $x_i \in \{0, 1\}$,

$$f(\mathbf{x}; p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}\,(1-p)^{\,n-\sum x_i}.$$

Step 2: factorise. Set $T = \sum_{i=1}^{n} x_i$. Then

$$f(\mathbf{x}; p) = \underbrace{p^{T}(1-p)^{\,n-T}}_{g(T,p)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

Step 3: conclude. g touches the data only through T and $h \equiv 1$ is free of p, so

$$T = \sum_{i=1}^{n} X_i \quad \text{is sufficient for } p$$

(equivalently $\widehat{p} = \overline{X} = T/n$, a one-to-one function of T, is sufficient).

Example 5.5.5 (Poisson rate λ)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Find a sufficient statistic for λ.

Which rule, and why? Count data with one parameter and we want its sufficient summary — Factorisation Theorem again. Watch the factorials: they are data-only and belong in h.

Solution.

Step 1: joint pmf. For $x_i \in \{0, 1, 2, \dots\}$,

$$f(\mathbf{x}; \lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = e^{-n\lambda}\,\lambda^{\sum x_i}\,\frac{1}{\prod_{i=1}^{n} x_i!}.$$

Step 2: factorise. With $T = \sum x_i$,

$$f(\mathbf{x}; \lambda) = \underbrace{e^{-n\lambda}\,\lambda^{T}}_{g(T,\lambda)} \cdot \underbrace{\frac{1}{\prod_i x_i!}}_{h(\mathbf{x})}.$$

Step 3: conclude. The factor with λ depends on the data only through T, and h is free of λ, so

$$T = \sum_{i=1}^{n} X_i \quad \text{is sufficient for } \lambda.$$

Example 5.5.6 (Exponential rate λ (waiting times))**.** A sensor logs $n = 5$ inter-arrival times $X_1, \dots, X_5 \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$ with density $f(x; \lambda) = \lambda e^{-\lambda x}$ for $x > 0$. Find a sufficient statistic for λ.

Which rule, and why? Continuous positive data, one rate parameter, support $(0, \infty)$ does *not* depend on λ, so the parameter rides entirely in the exponent — straightforward Factorisation.

Solution.

Step 1: joint density. For all $x_i > 0$,

$$f(\mathbf{x}; \lambda) = \prod_{i=1}^{5} \lambda e^{-\lambda x_i} = \lambda^5 \exp\Big(-\lambda \sum_{i=1}^{5} x_i\Big).$$

Step 2: factorise. With $T = \sum x_i$,

$$f(\mathbf{x};\lambda) = \underbrace{\lambda^5 e^{-\lambda T}}_{g(T,\lambda)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

Step 3: conclude.

$$T = \sum_{i=1}^{5} X_i \ \text{ is sufficient for } \lambda$$

(so is the total time; the individual gaps add nothing once their sum is known). Numerically, if the five gaps are $2.1, 0.4, 3.7, 1.0, 2.8$ seconds then the *observed* value of the sufficient statistic is $T = 10.0$ s, and any estimator of λ should depend on the data only through this 10.0.

Example 5.5.7 (Normal mean μ with known variance)**.** Let $X_1, \dots, X_n \nVdash \mathrm{N}(\mu, \sigma_0^2)$ with σ_0^2 *known.* Find a sufficient statistic for μ.

Which rule, and why? One unknown parameter (μ) in a density whose support is all of $\mathbb{R}$ (independent of μ). Expand the square and track which data-summary carries μ — Factorisation Theorem, being careful to dump the μ-free pieces into h.

Solution.

Step 1: joint density.

$$f(\mathbf{x};\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi}\,\sigma_0} \exp\Big(- \frac{(x_i-\mu)^2}{2\sigma_0^2} \Big) = (2\pi\sigma_0^2)^{-n/2} \exp\Big(- \frac{1}{2\sigma_0^2} \sum_{i=1}^{n} (x_i-\mu)^2 \Big).$$

Step 2: expand the sum of squares.

$$\sum_{i=1}^{n} (x_i-\mu)^2 = \sum_{i=1}^{n} x_i^2 - 2\mu \sum_{i=1}^{n} x_i + n\mu^2.$$

Only the middle term $-2\mu \sum x_i$ (and the constant $n\mu^2$) involves μ.

Step 3: factorise. With $T = \sum x_i$,

$$f(\mathbf{x};\mu) = \underbrace{\exp\Big(\frac{\mu}{\sigma_0^2} T - \frac{n\mu^2}{2\sigma_0^2} \Big)}_{g(T,\mu)} \cdot \underbrace{(2\pi\sigma_0^2)^{-n/2} \exp\Big(- \frac{1}{2\sigma_0^2} \sum_i x_i^2 \Big)}_{h(\mathbf{x})}.$$

The factor h contains $\sum x_i^2$ but *no* μ, so it is legitimate.

Step 4: conclude.

$$T = \sum_{i=1}^{n} X_i \ \text{ (equivalently } \overline{X}\text{) is sufficient for } \mu.$$

Example 5.5.8 (Normal with both parameters unknown: a sufficient *pair*)**.** Let $X_1, \dots, X_n \nVdash \mathrm{N}(\mu, \sigma^2)$ with *both* μ and σ^2 unknown. Find a jointly sufficient statistic for $\theta = (\mu, \sigma^2)$.

Which rule, and why? Two unknown parameters, so we expect a two-dimensional sufficient statistic. The Factorisation Theorem applies verbatim with a vector θ and a vector T; keep every term that contains either μ or σ^2 inside g.

Solution.

Step 1: joint density, sum of squares expanded.

$$f(\mathbf{x};\mu,\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\Big(- \frac{1}{2\sigma^2} \Big[\textstyle\sum_i x_i^2 - 2\mu \sum_i x_i + n\mu^2 \Big] \Big).$$

Step 2: identify the data-summaries carrying the parameters. Both $\sum x_i$ and $\sum x_i^2$ now multiply parameter-dependent factors (σ^2 multiplies $\sum x_i^2$). Set

$$T_1 = \sum_{i=1}^{n} x_i, \qquad T_2 = \sum_{i=1}^{n} x_i^2.$$

Step 3: factorise.

$$f(\mathbf{x};\mu,\sigma^2) = \underbrace{(2\pi\sigma^2)^{-n/2}\exp\Big(-\frac{T_2 - 2\mu T_1 + n\mu^2}{2\sigma^2}\Big)}_{g((T_1,T_2),\,(\mu,\sigma^2))} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

Step 4: conclude.

$$(T_1, T_2) = \Big(\sum_i X_i,\ \sum_i X_i^2\Big) \text{ is jointly sufficient for } (\mu,\sigma^2).$$

Equivalently $(\overline{X}, S^2)$ is jointly sufficient, since $(\overline{X}, S^2)$ is a one-to-one function of (T_1, T_2) via $\overline{X} = T_1/n$ and $S^2 = \big(T_2 - T_1^2/n\big)/(n-1)$.

5.5.3 When the support depends on the parameter: Uniform$(0,\theta)$

If the *range* of X moves with θ, the parameter sneaks in through indicator functions rather than through an exponent. The Factorisation Theorem still applies — you just have to factor the indicators correctly.

Example 5.5.9 (Uniform on $(0,\theta)$: the maximum is sufficient)**.** Let $X_1,\dots,X_n \overset{\text{iid}}{\sim} \text{Uniform}(0,\theta)$, so $f(x;\theta) = \frac{1}{\theta}\mathbb{1}(0 \le x \le \theta)$. Find a sufficient statistic for θ.

Which rule, and why? The support $[0,\theta]$ depends on θ, so the MLE-by-derivative trick fails and so does "put θ in the exponent". Here θ lives in the indicator $\mathbb{1}(x \le \theta)$. The key algebra is $\prod_i \mathbb{1}(x_i \le \theta) = \mathbb{1}(\max_i x_i \le \theta)$: *all* observations lie below θ iff the *largest* one does.

Solution.

Step 1: joint density.

$$f(\mathbf{x};\theta) = \prod_{i=1}^{n} \frac{1}{\theta}\mathbb{1}(0 \le x_i \le \theta) = \theta^{-n}\prod_{i=1}^{n}\mathbb{1}(0 \le x_i)\mathbb{1}(x_i \le \theta).$$

Step 2: collapse the indicators. Split each indicator into a lower part ($x_i \ge 0$, free of θ) and an upper part ($x_i \le \theta$, carrying θ). Then

$$\prod_{i=1}^{n}\mathbb{1}(x_i \le \theta) = \mathbb{1}\big(\max_i x_i \le \theta\big), \qquad \prod_{i=1}^{n}\mathbb{1}(x_i \ge 0) = \mathbb{1}\big(\min_i x_i \ge 0\big).$$

Step 3: factorise. With the maximum $T = \max_i x_i = x_{(n)}$,

$$f(\mathbf{x};\theta) = \underbrace{\theta^{-n}\mathbb{1}\big(T \le \theta\big)}_{g(T,\theta)} \cdot \underbrace{\mathbb{1}\big(\min_i x_i \ge 0\big)}_{h(\mathbf{x})}.$$

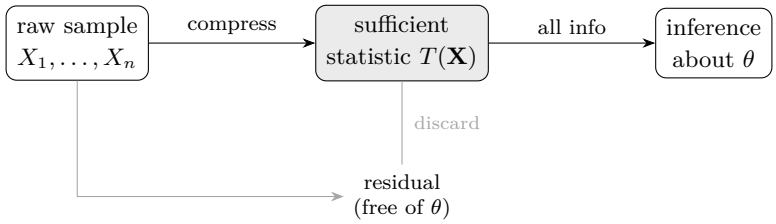
Step 4: conclude. g depends on the data only through $T = \max x_i$, and h is free of θ, so

$$T = \max_{1\le i\le n} X_i = X_{(n)} \text{ is sufficient for } \theta.$$

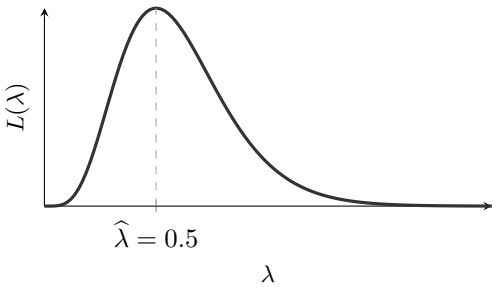
This matches the MLE $\widehat{\theta} = \max X_i$ from §1: maximum likelihood automatically lands on a function of the sufficient statistic.

Reading sufficiency off the picture

The factorisation says the data influence the parameter *only* through the channel T. The schematic below makes the bottleneck visible: the full sample feeds into T, and only T carries forward to any inference about θ; the discarded residual is independent of θ.



A concrete read of the bottleneck: in the exponential example above, the five raw gaps $2.1, 0.4, 3.7, 1.0, 2.8$, and any other five gaps summing to 10.0, lead to the very same likelihood for λ, namely $L(\lambda) = \lambda^5 e^{-10\lambda}$. The plot below shows that single curve, which depends on the data only through $T = 10.0$.



The peak sits at $\widehat{\lambda} = 5/T = 5/10 = 0.5$: maximising $\lambda^5 e^{-10\lambda}$ gives $\ell'(\lambda) = 5/\lambda - 10 = 0 \Rightarrow \lambda = 0.5$, again a function of T alone.

5.5.4 Improving an estimator: the Rao–Blackwell idea

Sufficiency pays off concretely: if you have any unbiased estimator, conditioning it on a sufficient statistic produces a new estimator that is still unbiased but has variance no larger — often strictly smaller.

Theorem 5.5.10 (Rao–Blackwell)**.** *Let $\widehat{\theta}$ be an estimator with $\mathbb{E}(\widehat{\theta}^2) < \infty$, and let T be sufficient for θ. Define the conditioned estimator*

$$\widehat{\theta}^{\,*} = \mathbb{E}\big(\widehat{\theta} \mid T\big).$$

Then $\widehat{\theta}^{\,*}$ is a statistic (it does not depend on θ, because T is sufficient), and

$$\mathbb{E}\big(\widehat{\theta}^{\,*}\big) = \mathbb{E}\big(\widehat{\theta}\big) \qquad \textit{and} \qquad \operatorname{Var}\big(\widehat{\theta}^{\,*}\big) \le \operatorname{Var}\big(\widehat{\theta}\big).$$

Why it works. Because T is sufficient, the conditional law of the data given T is free of θ, so $\widehat{\theta}^{\,*} = \mathbb{E}(\widehat{\theta} \mid T)$ is a genuine statistic. The mean is preserved by the tower property, $\mathbb{E}(\widehat{\theta}^{\,*}) = \mathbb{E}\big[\mathbb{E}(\widehat{\theta} \mid T)\big] = \mathbb{E}(\widehat{\theta})$; in particular bias is unchanged. For the variance, the law of total variance gives

$$\operatorname{Var}(\widehat{\theta}) = \underbrace{\operatorname{Var}\big(\mathbb{E}(\widehat{\theta} \mid T)\big)}_{=\operatorname{Var}(\widehat{\theta}^{\,*})} + \underbrace{\mathbb{E}\big[\operatorname{Var}(\widehat{\theta} \mid T)\big]}_{\ge 0} \;\ge\; \operatorname{Var}(\widehat{\theta}^{\,*}).$$

The discarded term $\mathbb{E}[\operatorname{Var}(\widehat{\theta} \mid T)]$ is exactly the variance you remove by averaging out the θ-free noise. $\square$

Example 5.5.11 (Rao–Blackwellising a crude Poisson estimator)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$ and suppose we want to estimate $\theta = \mathbb{P}(X = 0) = e^{-\lambda}$ (the chance a unit interval is event-free). Start from the crude unbiased estimator

$$\widehat{\theta} = \mathbb{1}(X_1 = 0) = \begin{cases} 1, & X_1 = 0, \\ 0, & X_1 \geq 1, \end{cases}$$

and improve it with Rao–Blackwell.

Which rule, and why? $\widehat{\theta} = \mathbb{1}(X_1 = 0)$ is unbiased ($\mathbb{E}\,\mathbb{1}(X_1 = 0) = \mathbb{P}(X_1 = 0) = e^{-\lambda}$) but absurdly wasteful — it ignores $X_2, \dots, X_n$ entirely. We have a sufficient statistic $T = \sum X_i$ on hand, and we want a better unbiased estimator: that is precisely the Rao–Blackwell setup, $\widehat{\theta}^* = \mathbb{E}(\widehat{\theta} \mid T)$.

Solution.

Step 1: write the improved estimator as a conditional probability.

$$\widehat{\theta}^* = \mathbb{E}\big(\mathbb{1}(X_1 = 0) \mid T\big) = \mathbb{P}\big(X_1 = 0 \mid \textstyle\sum_i X_i = T\big).$$

Step 2: use the conditional law of one Poisson given the sum. For iid Poisson variables, the conditional distribution of X_1 given $\sum_i X_i = t$ is $\text{Bin}\big(t, \frac{1}{n}\big)$. (Each of the t total events is independently assigned to one of the n equally-likely intervals, so the count in interval 1 is binomial with success probability $1/n$.) Hence

$$\mathbb{P}\big(X_1 = 0 \mid \textstyle\sum_i X_i = t\big) = \binom{t}{0}\Big(\frac{1}{n}\Big)^0\Big(1 - \frac{1}{n}\Big)^t = \Big(1 - \frac{1}{n}\Big)^t.$$

Step 3: assemble the estimator.

$$\widehat{\theta}^* = \Big(1 - \frac{1}{n}\Big)^T, \qquad T = \sum_{i=1}^{n} X_i.$$

Step 4: sanity-check unbiasedness. Using the Poisson moment generating function $\mathbb{E}\,s^T = e^{n\lambda(s-1)}$ with $s = 1 - \frac{1}{n}$,

$$\mathbb{E}\big(\widehat{\theta}^*\big) = \mathbb{E}\Big[\Big(1 - \tfrac{1}{n}\Big)^T\Big] = \exp\Big(n\lambda\big[(1 - \tfrac{1}{n}) - 1\big]\Big) = \exp\Big(n\lambda \cdot (-\tfrac{1}{n})\Big) = e^{-\lambda} = \theta.$$

So $\widehat{\theta}^*$ is unbiased, exactly as Rao–Blackwell promises, and (being a function of the whole sample through T) has strictly smaller variance than the one-observation $\mathbb{1}(X_1 = 0)$. For example with $n = 4$ and observed $T = 3$ events, the crude estimator gives either 0 or 1, while the improved estimate is $(1 - \frac{1}{4})^3 = (0.75)^3 \approx 0.4219$ — a far more sensible guess for $e^{-\lambda}$.

Example 5.5.12 (Rao–Blackwell with a Bernoulli mean)**.** Let $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Bin}(1, p)$ and estimate p itself. Start from the throw-away estimator $\widehat{p} = X_1$ (use only the first trial) and Rao–Blackwellise on the sufficient statistic $T = \sum X_i$.

Which rule, and why? $\widehat{p} = X_1$ is unbiased ($\mathbb{E}X_1 = p$) but discards $n - 1$ observations. With the sufficient statistic $T = \sum X_i$ available, Rao–Blackwell should recover the natural pooled estimator.

Solution.

Step 1: conditional expectation.

$$\widehat{p}^* = \mathbb{E}(X_1 \mid T) = \mathbb{P}\big(X_1 = 1 \mid \textstyle\sum_i X_i = T\big).$$

Step 2: by symmetry of exchangeable Bernoulli trials. Given that exactly T of the n trials are successes, each particular trial is equally likely to be one of the successes, so $\mathbb{P}(X_1 = 1 \mid \sum_i X_i = t) = t/n$. Hence

$$\widehat{p}^* = \frac{T}{n} = \overline{X}.$$

Step 3: compare variances. The crude estimator has $\mathrm{Var}(X_1) = p(1-p)$, while the Rao–Blackwellised estimator has $\mathrm{Var}(\overline{X}) = p(1-p)/n$. Conditioning on the sufficient statistic shrinks the variance by the full factor n:

$$\mathrm{Var}(\widehat{p}^{\,*}) = \frac{p(1-p)}{n} \leq p(1-p) = \mathrm{Var}(\widehat{p}).$$

Rao–Blackwell has turned "look at one trial" into "average all n" — the estimator every analyst would have written down anyway, now *derived* as the variance-minimising improvement.

Remark (The big picture). A sufficient statistic is a lossless compression of the data *for the purpose of learning* θ. The Factorisation Theorem finds it by inspection; the MLE is (essentially always) a function of it; and Rao–Blackwell shows that no good estimator ever needs to depend on the data in any other way. This is why, throughout these notes, the estimators that appear — the sample total, the sample mean, the sample maximum — are exactly the sufficient statistics of their families.

5.6 Consistency

The previous sections asked whether an estimator is centred correctly (unbiased) or has small variance (efficient) *at a fixed sample size*. **Consistency** asks a different, large-sample question: as we collect more and more data, does the estimator settle down onto the true value θ? Formally we want the estimator $\widehat{\theta}_n$, built from n observations, to converge in probability to θ — the probability that $\widehat{\theta}_n$ misses θ by more than any fixed tolerance must shrink to 0 as $n \to \infty$. The workhorse for proving this is a single, easily checked sufficient condition: if the mean squared error goes to 0, the estimator is consistent.

Which rule of this section does the problem need?

- Asked to **show an estimator is consistent**? ⇒ the easiest route is almost always the **MSE criterion**: show $\mathrm{Bias}(\widehat{\theta}_n) \to 0$ *and* $\mathrm{Var}(\widehat{\theta}_n) \to 0$, so $\mathrm{MSE} \to 0$, which forces convergence in probability by **Chebyshev**. Spell out both pieces.
- The estimator is a **sample mean** of iid terms with finite mean? ⇒ quote the **Weak Law of Large Numbers** directly: $\overline{X}_n \to \mu$ in probability. (The WLLN is itself the MSE argument applied to $\overline{X}_n$.)
- Asked to find the **limit a (possibly biased) estimator converges to**? ⇒ combine the WLLN with the **continuous mapping** idea: if $\widehat{\theta}_n \to c$ in probability and g is continuous at c, then $g(\widehat{\theta}_n) \to g(c)$.
- Given an **MLE** and asked about its large-sample behaviour? ⇒ under the usual regularity conditions the MLE is consistent; verify by the MSE criterion or by writing it as a continuous function of a sample mean.
- **Do not confuse** consistency with unbiasedness. An estimator can be biased at every finite n yet consistent (bias $\to 0$); it can be unbiased at every n yet *inconsistent* (variance that does not vanish, e.g. using only X_1 to estimate μ). Consistency is a statement about the *limit*, not about any single sample size.

5.6.1 Convergence in probability and the definition of consistency

Definition 5.6.1 (Convergence in probability)**.** A sequence of random variables $Y_1, Y_2, \ldots$ converges in probability to a constant c, written $Y_n \xrightarrow{P} c$, if for every tolerance $\varepsilon > 0$,

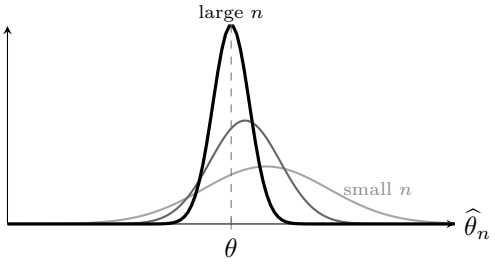
$$\lim_{n\to\infty} \mathbb{P}\big(\,|Y_n - c| \geq \varepsilon\,\big) = 0.$$

Equivalently, $\mathbb{P}(|Y_n - c| < \varepsilon) \to 1$: eventually almost all of the probability mass of Y_n sits within ε of c, no matter how small ε is.

Definition 5.6.2 (Consistent estimator)**.** Let $\widehat{\theta}_n$ be an estimator of θ based on $X_1, \ldots, X_n$. We call $\widehat{\theta}_n$ a **consistent** estimator of θ if

$$\widehat{\theta}_n \xrightarrow{P} \theta \qquad \text{for every value of } \theta \text{ in the parameter space.}$$

The geometric picture is a sequence of sampling distributions that pile up ever more tightly over θ: the spread collapses and the centre slides onto the truth.



5.6.2 The MSE criterion (the main tool)

Checking the definition directly is awkward. The following theorem replaces it by two routine calculations — a bias limit and a variance limit.

Theorem 5.6.3 (Mean-squared-error criterion for consistency)**.** *If*

$$\lim_{n\to\infty} \mathrm{MSE}(\widehat{\theta}_n) = \lim_{n\to\infty} \Big(\mathrm{Var}(\widehat{\theta}_n) + [\mathrm{Bias}(\widehat{\theta}_n)]^2\Big) = 0,$$

that is, if $\mathrm{Bias}(\widehat{\theta}_n) \to 0$ *and* $\mathrm{Var}(\widehat{\theta}_n) \to 0$*, then* $\widehat{\theta}_n \xrightarrow{P} \theta$*, so* $\widehat{\theta}_n$ *is consistent.*

Proof. Recall $\mathrm{MSE}(\widehat{\theta}_n) = \mathbb{E}\big[(\widehat{\theta}_n - \theta)^2\big]$, and the algebraic identity $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2$. Fix $\varepsilon > 0$. Chebyshev's inequality in its mean-square form (Markov's inequality applied to the nonnegative variable $(\widehat{\theta}_n - \theta)^2$) gives

$$\mathbb{P}\big(|\widehat{\theta}_n - \theta| \ge \varepsilon\big) = \mathbb{P}\big((\widehat{\theta}_n - \theta)^2 \ge \varepsilon^2\big) \le \frac{\mathbb{E}[(\widehat{\theta}_n - \theta)^2]}{\varepsilon^2} = \frac{\mathrm{MSE}(\widehat{\theta}_n)}{\varepsilon^2}.$$

By hypothesis the right-hand side $\to 0$ as $n \to \infty$, and since the left-hand side is nonnegative it is squeezed to 0. As $\varepsilon > 0$ was arbitrary, $\widehat{\theta}_n \xrightarrow{P} \theta$. □

Remark. The condition is *sufficient*, not necessary: an estimator can be consistent yet have infinite or undefined variance at every n (so MSE never goes to 0). But for the estimators met in this chapter the MSE criterion settles the question immediately, and it is the first thing to try.

Example 5.6.4 (Sample mean for a Poisson rate)**.** A call centre logs the number of dropped calls per hour. Model the counts as $X_1, \dots, X_n \overset{\text{iid}}{\sim} \mathrm{Poisson}(\lambda)$ and estimate λ by the sample mean $\widehat{\lambda}_n = \overline{X}_n$. Is $\widehat{\lambda}_n$ consistent?

Which rule, and why? We want a limit statement as n grows, so this is a consistency question, not a fixed-n efficiency question. The estimator is a sample mean of iid terms, so either the WLLN or the MSE criterion applies; we use the MSE criterion to display both pieces. A Cramér–Rao bound would answer "how small can the variance be", a different question.

Solution.

Step 1: Bias. For a Poisson, $\mathbb{E}(X_i) = \lambda$, so $\mathbb{E}(\widehat{\lambda}_n) = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}(X_i) = \lambda$. Hence $\mathrm{Bias}(\widehat{\lambda}_n) = \lambda - \lambda = 0$ for every n.

Step 2: Variance. For a Poisson, $\mathrm{Var}(X_i) = \lambda$, and the X_i are independent, so

$$\mathrm{Var}(\widehat{\lambda}_n) = \frac{1}{n^2}\sum_{i=1}^{n} \mathrm{Var}(X_i) = \frac{n\lambda}{n^2} = \frac{\lambda}{n}.$$

Step 3: MSE limit. Therefore $\mathrm{MSE}(\widehat{\lambda}_n) = \dfrac{\lambda}{n} + 0^2 = \dfrac{\lambda}{n} \to 0$ as $n \to \infty$.

$$\widehat{\lambda}_n = \overline{X}_n \xrightarrow{P} \lambda, \qquad \text{so } \overline{X}_n \text{ is consistent for } \lambda.$$

Example 5.6.5 (Sample mean for any finite-variance population: the Weak Law)**.** Daily water usage at a plant has unknown mean μ and a finite variance σ^2 (the distribution is otherwise unspecified). Show that $\overline{X}_n$ is consistent for μ using the WLLN, and identify exactly where finiteness of σ^2 is used.

Which rule, and why? The estimator is the mean of iid terms with finite mean, the exact hypothesis of the **Weak Law of Large Numbers**; that is the cleanest tool. We do not need normality or any specific family — consistency of $\overline{X}_n$ holds for every finite-variance population.

Solution.

Step 1: Quote the WLLN. If $X_1, \ldots, X_n$ are iid with $\mathbb{E}(X_i) = \mu$ finite, then $\overline{X}_n \xrightarrow{P} \mu$.

Step 2: Prove it by the MSE criterion (this is the WLLN's proof). $\mathbb{E}(\overline{X}_n) = \mu$, so the bias is 0, and with finite σ^2,

$$\mathrm{Var}(\overline{X}_n) = \frac{\sigma^2}{n} \xrightarrow[n\to\infty]{} 0.$$

Thus $\mathrm{MSE}(\overline{X}_n) = \sigma^2/n \to 0$, and Chebyshev gives convergence in probability.

Step 3: Where $\sigma^2 < \infty$ enters. Finiteness of σ^2 is exactly what makes $\mathrm{Var}(\overline{X}_n) = \sigma^2/n$ vanish; without it the MSE bound is vacuous. (The WLLN in fact holds under the weaker assumption of a finite mean alone, but the slick MSE proof needs $\sigma^2 < \infty$.)

$$\overline{X}_n \xrightarrow{P} \mu.$$

Example 5.6.6 (Sample proportion)**.** In a quality audit, each inspected unit is defective independently with unknown probability p. Let $\widehat{p}_n = \overline{X}_n = \frac{1}{n}\sum_{i=1}^n X_i$ where $X_i \nvDash \mathrm{Bin}(1, p)$ (Bernoulli). Show $\widehat{p}_n$ is consistent for p, and give the tolerance bound at $n = 400$, $\varepsilon = 0.05$ when $p = 0.1$.

Which rule, and why? A sample proportion is a sample mean of Bernoulli indicators, so consistency follows from the same MSE/WLLN machinery. The extra numeric part is a direct use of the Chebyshev bound from the proof, which gives a concrete, distribution-free guarantee.

Solution.

Step 1: Bias. $\mathbb{E}(X_i) = p$, so $\mathbb{E}(\widehat{p}_n) = p$ and the bias is 0.

Step 2: Variance. $\mathrm{Var}(X_i) = p(1-p)$, so $\mathrm{Var}(\widehat{p}_n) = \dfrac{p(1-p)}{n}$.

Step 3: MSE limit. $\mathrm{MSE}(\widehat{p}_n) = \dfrac{p(1-p)}{n} \to 0$, so $\widehat{p}_n \xrightarrow{P} p$.

Step 4: Numeric Chebyshev bound. With $p = 0.1$, $n = 400$, $\varepsilon = 0.05$,

$$\mathbb{P}\big(|\widehat{p}_n - p| \geq 0.05\big) \leq \frac{\mathrm{Var}(\widehat{p}_n)}{\varepsilon^2} = \frac{0.1 \cdot 0.9/400}{0.05^2} = \frac{0.000225}{0.0025} = 0.09.$$

So at least a 91% chance the estimate is within 0.05 of p — and this bound itself $\to 0$ as n grows, confirming consistency.

$$\widehat{p}_n \xrightarrow{P} p.$$

Example 5.6.7 (An unbiased but INCONSISTENT estimator)**.** A lab will estimate a process mean μ (variance σ^2, iid data) but, to save effort, proposes using only the *first* measurement: $\widehat{\mu}_n = X_1$, ignoring the rest. Is it unbiased? Is it consistent?

Which rule, and why? This separates the two ideas the strategy box warned about. Unbiasedness is a fixed-n average; consistency is a limit. Checking the variance limit is the decisive test.

Solution.

Step 1: Bias. $\mathbb{E}(\widehat{\mu}_n) = \mathbb{E}(X_1) = \mu$, so $\widehat{\mu}_n$ is *unbiased* for every n.

Step 2: Variance. $\mathrm{Var}(\widehat{\mu}_n) = \mathrm{Var}(X_1) = \sigma^2$, which does *not* depend on n and so does not go to 0.

Step 3: Conclusion. $\mathrm{MSE}(\widehat{\mu}_n) = \sigma^2 \not\to 0$. More directly, the sampling distribution of X_1 never tightens, so $\mathbb{P}(|X_1 - \mu| \geq \varepsilon)$ stays bounded away from 0 for small ε. The estimator is unbiased but **not consistent**.

$$\widehat{\mu}_n = X_1 \overset{P}{\not\to} \mu.$$

The moral: collecting more data must actually be used; an estimator that ignores the growing sample cannot be consistent.

5.6.3 Consistent through vanishing bias: biased-but-consistent estimators

Consistency tolerates bias, provided the bias dies out. Here the MSE criterion shows its worth: both the bias and the variance go to 0, so their combination does too.

Example 5.6.8 (MLE of a uniform upper endpoint)**.** Rod lengths from a cutting machine are modelled as $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$ with θ the unknown maximum setting. The MLE is the largest observation, $\widehat{\theta}_n = M_n = \max_i X_i$ (it maximises the likelihood by a boundary argument, not a derivative). Show M_n is consistent for θ, even though it is biased.

Which rule, and why? This is the boundary-case MLE from the point-estimation section. It is biased at every n (it can never exceed θ, so it underestimates), so we cannot lean on unbiasedness; the MSE criterion is exactly built for "biased but vanishing".

Solution.

Step 1: Distribution of the maximum. For $0 \le m \le \theta$, $\mathbb{P}(M_n \le m) = \prod_{i=1}^{n} \mathbb{P}(X_i \le m) = (m/\theta)^n$, so the density is $f_{M_n}(m) = \dfrac{n\, m^{n-1}}{\theta^n}$ on $[0, \theta]$.

Step 2: Mean and bias.

$$\mathbb{E}(M_n) = \int_0^\theta m \cdot \frac{n\, m^{n-1}}{\theta^n}\, \mathrm{d}m = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\, \theta, \qquad \text{Bias}(M_n) = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}.$$

The bias is negative (underestimate) but $\text{Bias}(M_n) = -\theta/(n+1) \to 0$.

Step 3: Variance. First $\mathbb{E}(M_n^2) = \displaystyle\int_0^\theta m^2 \frac{n\, m^{n-1}}{\theta^n}\, \mathrm{d}m = \frac{n}{\theta^n} \cdot \frac{\theta^{n+2}}{n+2} = \frac{n}{n+2}\theta^2$, so

$$\text{Var}(M_n) = \frac{n}{n+2}\theta^2 - \left(\frac{n}{n+1}\right)^2 \theta^2 = \theta^2\, \frac{n(n+1)^2 - n^2(n+2)}{(n+2)(n+1)^2} = \frac{n\,\theta^2}{(n+1)^2(n+2)}.$$

(The numerator simplifies as $n(n+1)^2 - n^2(n+2) = n[(n^2+2n+1) - (n^2+2n)] = n$.) This is of order $\theta^2/n^2 \to 0$.

Step 4: MSE limit.

$$\begin{aligned} \text{MSE}(M_n) &= \frac{n\,\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{\theta^2}{(n+1)^2}\left(\frac{n}{n+2} + 1\right) \\ &= \frac{2(n+1)\,\theta^2}{(n+1)^2(n+2)} = \frac{2\theta^2}{(n+1)(n+2)} \to 0. \end{aligned}$$

$$M_n \xrightarrow{P} \theta, \qquad \text{consistent though biased.}$$

Remark. The bias-corrected version $\widehat{\theta}_n = \frac{n+1}{n} M_n$ is unbiased; since $\frac{n+1}{n} \to 1$ and $M_n \xrightarrow{P} \theta$, it is also consistent. Two different estimators, same limit.

Example 5.6.9 (MLE variance with divisor n)**.** For $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{N}(\mu, \sigma^2)$ the MLE of σ^2 (with μ unknown) is $\widehat{\sigma}_n^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X}_n)^2$, which uses divisor n rather than $n-1$ and is therefore biased. Show it is nonetheless consistent for σ^2.

Which rule, and why? Again a biased estimator, so we cannot use unbiasedness; the MSE criterion handles the vanishing bias. It is cleanest to relate the MLE to the unbiased sample variance S^2, whose moments are standard.

Solution.

Step 1: Relate to S^2. Write $S^2 = \frac{1}{n-1}\sum(X_i - \overline{X}_n)^2$ (unbiased), so $\widehat{\sigma}_n^2 = \frac{n-1}{n}\, S^2$.

Step 2: Bias. Since $\mathbb{E}(S^2) = \sigma^2$,

$$\mathbb{E}(\widehat{\sigma}_n^2) = \frac{n-1}{n}\sigma^2, \qquad \text{Bias}(\widehat{\sigma}_n^2) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n} \to 0.$$

Step 3: Variance. For a normal sample, $\mathrm{Var}(S^2) = \dfrac{2\sigma^4}{n-1}$ (from $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, whose variance is $2(n-1)$). Hence

$$\mathrm{Var}(\widehat{\sigma}_n^2) = \left(\frac{n-1}{n}\right)^2 \mathrm{Var}(S^2) = \frac{(n-1)^2}{n^2} \cdot \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2} \to 0.$$

Step 4: MSE limit. Both pieces vanish, so $\mathrm{MSE}(\widehat{\sigma}_n^2) = \dfrac{2(n-1)\sigma^4}{n^2} + \dfrac{\sigma^4}{n^2} = \dfrac{(2n-1)\sigma^4}{n^2} \to 0.$

$$\widehat{\sigma}_n^2 \xrightarrow{P} \sigma^2.$$

(The same limit holds for S^2 itself, since $\frac{n-1}{n} \to 1$: both estimators of σ^2 are consistent.)

5.6.4 Consistency via continuous functions of a sample mean

Many estimators are not means themselves but smooth functions of a mean. The next principle lets us inherit consistency through that function.

Theorem 5.6.10 (Continuous mapping for consistency)**.** *If $\widehat{\theta}_n \xrightarrow{P} c$ and g is a function continuous at c, then $g(\widehat{\theta}_n) \xrightarrow{P} g(c)$. In particular, if $\widehat{\theta}_n$ is consistent for θ and g is continuous at θ, then $g(\widehat{\theta}_n)$ is consistent for $g(\theta)$.*

Proof sketch. Fix $\varepsilon > 0$. By continuity at c there is $\delta > 0$ with $|y - c| < \delta \Rightarrow |g(y) - g(c)| < \varepsilon$. Hence $\{|g(\widehat{\theta}_n) - g(c)| \geq \varepsilon\} \subseteq \{|\widehat{\theta}_n - c| \geq \delta\}$, so $\mathbb{P}(|g(\widehat{\theta}_n) - g(c)| \geq \varepsilon) \leq \mathbb{P}(|\widehat{\theta}_n - c| \geq \delta) \to 0$. □

Example 5.6.11 (Estimating an exponential rate by the reciprocal mean)**.** Service times are $X_1, \ldots, X_n \nVdash \mathrm{Exp}(\lambda)$ (mean $1/\lambda$). The MLE of the rate is $\widehat{\lambda}_n = 1/\overline{X}_n$. Show it is consistent for λ.

Which rule, and why? Here $\widehat{\lambda}_n$ is *biased* (Jensen: $\mathbb{E}(1/\overline{X}_n) > 1/\mathbb{E}(\overline{X}_n) = \lambda$), so the bias formula is awkward. But $\widehat{\lambda}_n = g(\overline{X}_n)$ with $g(t) = 1/t$ continuous away from 0, and $\overline{X}_n$ is a consistent estimator of the mean — continuous mapping is the efficient route.

Solution.

Step 1: The inner mean is consistent. $\mathbb{E}(X_i) = 1/\lambda$ is finite, so by the WLLN $\overline{X}_n \xrightarrow{P} 1/\lambda$.

Step 2: Apply the continuous function $g(t) = 1/t$. Since $1/\lambda > 0$, g is continuous at $t = 1/\lambda$. By the continuous mapping theorem,

$$\widehat{\lambda}_n = \frac{1}{\overline{X}_n} = g(\overline{X}_n) \xrightarrow{P} g\left(\tfrac{1}{\lambda}\right) = \frac{1}{1/\lambda} = \lambda.$$

$$\widehat{\lambda}_n \xrightarrow{P} \lambda, \qquad \text{consistent (despite finite-}n\text{ bias).}$$

Example 5.6.12 (Method-of-moments estimator of a Bernoulli odds ratio)**.** A clinic estimates the *odds* $\psi = p/(1-p)$ of a positive screening result from $X_1, \ldots, X_n \nVdash \mathrm{Bin}(1, p)$ with $0 < p < 1$. The plug-in estimator is $\widehat{\psi}_n = \dfrac{\widehat{p}_n}{1 - \widehat{p}_n}$, $\widehat{p}_n = \overline{X}_n$. Show it is consistent for ψ.

Which rule, and why? The estimator is a ratio, hence biased and messy to handle by moments. But it is a continuous function $g(t) = t/(1-t)$ of the consistent estimator $\widehat{p}_n$, so continuous mapping gives the limit in one line.

Solution.

Step 1: Inner consistency. From the sample-proportion example, $\widehat{p}_n \xrightarrow{P} p$.

Step 2: Continuity. $g(t) = t/(1-t)$ is continuous at $t = p$ because $p < 1$ (denominator nonzero). Apply the continuous mapping theorem:

$$\widehat{\psi}_n = g(\widehat{p}_n) \xrightarrow{P} g(p) = \frac{p}{1-p} = \psi.$$

$$\widehat{\psi}_n \xrightarrow{P} \psi.$$

5.6.5 Putting it together: a worked consistency check from scratch

Example 5.6.13 (Second-moment estimator of a uniform width)**.** Tolerances on a part are modelled as $X_1, \dots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$. Instead of the maximum, an engineer uses the second sample moment to build $\widehat{\theta}_n = \sqrt{\dfrac{3}{n}\sum_{i=1}^{n} X_i^2}$. Show this is a consistent estimator of θ.

Which rule, and why? This is a method-of-moments-style estimator: it is a continuous function of the sample average of the X_i^2. So the plan is WLLN on $\frac{1}{n}\sum X_i^2$, then continuous mapping by $g(u) = \sqrt{3u}$.

Solution.

Step 1: Population second moment. For $X \sim \text{Uniform}(0, \theta)$, $\mathbb{E}(X^2) = \int_0^\theta x^2 \frac{1}{\theta}\,\mathrm{d}x = \dfrac{\theta^2}{3}$, which is finite.

Step 2: WLLN on the averaged squares. The X_i^2 are iid with finite mean $\theta^2/3$, so

$$\overline{X_n^2} := \frac{1}{n}\sum_{i=1}^{n} X_i^2 \xrightarrow{P} \frac{\theta^2}{3}.$$

Step 3: Continuous mapping. With $g(u) = \sqrt{3u}$, continuous for $u \geq 0$,

$$\widehat{\theta}_n = g\big(\overline{X_n^2}\big) \xrightarrow{P} g\left(\frac{\theta^2}{3}\right) = \sqrt{3 \cdot \frac{\theta^2}{3}} = \theta.$$

$$\widehat{\theta}_n \xrightarrow{P} \theta.$$

Remark (How fast? a Chebyshev tolerance). Consistency only promises the limit; for a concrete sample size use the MSE bound. For the Poisson mean estimator with $\lambda = 2$, to guarantee $\mathbb{P}(|\overline{X}_n - 2| \geq 0.1) \leq 0.05$ it suffices that $\dfrac{\lambda/n}{0.1^2} = \dfrac{2}{0.01\,n} \leq 0.05$, i.e. $n \geq \dfrac{2}{0.0005} = 4000$. Chebyshev is deliberately crude — a normal approximation needs far fewer — but it makes the "shrinking-to-zero" guarantee explicit and rigorous.

5.6.6 Summary

To show $\widehat{\theta}_n$ consistent	**Do this**
$\widehat{\theta}_n$ is a sample mean	Quote the WLLN: $\overline{X}_n \xrightarrow{P} \mu$ (finite mean).
$\widehat{\theta}_n$ may be biased	MSE criterion: show Bias $\to 0$ and Var $\to 0$, then Chebyshev.
$\widehat{\theta}_n = g$(a consistent estimator)	Continuous mapping: g continuous at the limit $\Rightarrow g(\widehat{\theta}_n) \xrightarrow{P} g(\cdot)$.
Want a finite-n guarantee	Chebyshev: $\mathbb{P}(\lvert\widehat{\theta}_n - \theta\rvert \geq \varepsilon) \leq \text{MSE}\,/\varepsilon^2$.

Consistency is the minimal large-sample sanity check on an estimator: with enough data it lands on the truth. It is weaker than unbiasedness in one direction (a consistent estimator may be biased at every n) and unrelated in another (an unbiased estimator can fail to be consistent). The single condition $\text{MSE}(\widehat{\theta}_n) \to 0$ secures it for essentially every estimator in this chapter.

5.7 Bayesian Estimation

Every method so far has treated the unknown parameter θ as a fixed but unknown constant, and let only the data be random. Bayesian estimation makes the opposite modelling choice: it encodes our uncertainty about θ *before* seeing data as a **prior** distribution $\pi(\theta)$, treats the data through the **likelihood** $L(\theta) = f(x \mid \theta)$, and combines them by Bayes' rule into a **posterior** distribution $\pi(\theta \mid x)$ that is our updated belief. The whole method is the single proportionality

$$\pi(\theta \mid x) \;\propto\; \pi(\theta)\, L(\theta), \qquad \text{posterior} \;\propto\; \text{prior} \times \text{likelihood},$$

and a point estimate is then read off the posterior — almost always its mean.

Which rule of this section does the problem need?

- You are given a **prior** on the parameter and asked to update it with data? ⇒ form the **posterior** $\propto$ prior $\times$ likelihood (drop constants not involving θ, then recognise the standard family).
- Counting **successes in** n **trials** (a binomial p) with a **Beta** prior? ⇒ the **Beta–Binomial** conjugate update; posterior is again Beta.
- Counting **events over time/space** (a Poisson λ) with a **Gamma** prior? ⇒ the **Gamma–Poisson** conjugate update; posterior is again Gamma.
- Estimating a **normal mean** μ with known σ^2 and a **normal** prior? ⇒ the **Normal–Normal** update; posterior precision adds, posterior mean is a precision-weighted average.
- Asked for a single number? ⇒ the **Bayes estimator**, the **posterior mean**. Asked for a range? ⇒ a **credible interval**, a central $1-\alpha$ chunk of the posterior.

Why not the frequentist tools? A confidence interval (§2) and the MLE (§1) use the data only; they cannot incorporate the external information a prior carries. Use Bayes when a genuine prior is available (past studies, physical bounds, expert input) and you want a probability statement *about* θ *itself*.

5.7.1 Prior, likelihood, posterior

Definition 5.7.1 (Prior, likelihood, posterior)**.** Let X have density (or mass function) $f(x \mid \theta)$ depending on an unknown parameter θ with parameter space Θ. A **prior** $\pi(\theta)$ is a density on Θ expressing belief about θ before data. After observing $X = x$ the **posterior** is

$$\pi(\theta \mid x) = \frac{f(x \mid \theta)\,\pi(\theta)}{m(x)}, \qquad m(x) = \int_\Theta f(x \mid \theta)\,\pi(\theta)\,\mathrm{d}\theta,$$

where the **marginal** $m(x)$ does not depend on θ. Hence, as a function of θ,

$$\pi(\theta \mid x) \;\propto\; f(x \mid \theta)\,\pi(\theta) \;=\; L(\theta)\,\pi(\theta).$$

The constant $m(x)$ is recovered at the end by requiring $\int \pi(\theta \mid x)\,\mathrm{d}\theta = 1$.

Remark. The proportionality is the workhorse. Carry only the factors that contain θ; collect the rest into the normalising constant. If the surviving θ-shape matches a named family, you can write down the posterior without doing the integral $m(x)$ at all. That is exactly what a **conjugate** prior buys you: a prior family that, paired with a given likelihood, returns a posterior in the *same* family.

Definition 5.7.2 (Conjugate prior)**.** A family $\mathcal{P}$ of priors is **conjugate** to a likelihood $f(x \mid \theta)$ if for every prior in $\mathcal{P}$ the posterior also lies in $\mathcal{P}$. Conjugacy turns Bayesian updating into bookkeeping on the family's parameters.

Example 5.7.3 (A discrete prior on a coin: updating by hand)**.** A bent coin has heads-probability θ, and a friend tells you θ is one of $\{0.3, 0.5, 0.7\}$ with equal prior weight $\frac{1}{3}$ each. You toss it $n = 4$ times and see $x = 3$ heads. Find the posterior on θ and the posterior mean.

Which rule, and why? The parameter takes finitely many values, so the "posterior $\propto$ prior $\times$ likelihood" rule is a discrete weighting — no integral, no conjugacy needed. Binomial likelihood because we count heads in a fixed number of independent tosses.

Solution.

Step 1: likelihood at each candidate. With $L(\theta) \propto \theta^3(1-\theta)^1$ (the $\binom{4}{3}$ is a constant we may drop),

$$L(0.3) = 0.3^3(0.7) = 0.0189, \quad L(0.5) = 0.5^3(0.5) = 0.0625, \quad L(0.7) = 0.7^3(0.3) = 0.1029.$$

Step 2: multiply by the equal prior and normalise. Prior weight $\frac{1}{3}$ cancels in the ratio, so the posterior weights are proportional to the likelihoods. Their sum is $0.0189 + 0.0625 + 0.1029 = 0.1843$, giving

$$\pi(0.3 \mid x) = \tfrac{0.0189}{0.1843} = 0.1025, \quad \pi(0.5 \mid x) = \tfrac{0.0625}{0.1843} = 0.3391, \quad \pi(0.7 \mid x) = \tfrac{0.1029}{0.1843} = 0.5583.$$

Step 3: posterior mean.

$$\mathbb{E}[\theta \mid x] = 0.3(0.1025) + 0.5(0.3391) + 0.7(0.5583) = 0.0308 + 0.1696 + 0.3908 = 0.5911.$$

The data have pulled belief toward the larger values; $\theta = 0.7$ is now the most probable.

5.7.2 Beta prior for a binomial p (Beta–Binomial)

Definition 5.7.4 (Beta distribution)**.** $p \sim \text{Beta}(a, b)$ with $a, b > 0$ has density on $(0, 1)$

$$\pi(p) = \frac{1}{\text{B}(a,b)}\, p^{\,a-1}(1-p)^{\,b-1}, \qquad \text{B}(a,b) = \frac{\Gamma(a)\Gamma(b)}{\Gamma(a+b)},$$

with mean $\mathbb{E}[p] = \dfrac{a}{a+b}$ and variance $\text{Var}(p) = \dfrac{ab}{(a+b)^2(a+b+1)}$. Taking $a = b = 1$ gives the flat Uniform$(0, 1)$ prior.

Theorem 5.7.5 (Beta–Binomial conjugacy)**.** *If $X \mid p \sim \text{Bin}(n, p)$ and the prior is $p \sim \text{Beta}(a, b)$, then after observing $X = x$,*

$$p \mid x \;\sim\; \text{Beta}(a + x,\; b + n - x).$$

The posterior mean is $\dfrac{a+x}{a+b+n}$.

Proof. The likelihood is $L(p) \propto p^x(1-p)^{n-x}$ and the prior is $\pi(p) \propto p^{a-1}(1-p)^{b-1}$. Multiplying,

$$\pi(p \mid x) \propto p^{(a+x)-1}(1-p)^{(b+n-x)-1},$$

which is the Beta$(a + x, b + n - x)$ kernel; the missing constant is forced by normalisation. The mean formula is the Beta mean with parameters $a + x$ and $b + n - x$. □

Remark. The posterior mean is a weighted average of the prior mean and the sample proportion. Writing $\widehat{p}_{\text{MLE}} = x/n$ and prior mean $a/(a + b)$,

$$\frac{a+x}{a+b+n} = \underbrace{\frac{a+b}{a+b+n}}_{w_{\text{prior}}} \cdot \frac{a}{a+b} + \underbrace{\frac{n}{a+b+n}}_{w_{\text{data}}} \cdot \frac{x}{n}.$$

The prior acts like $a + b$ "pseudo-observations" (a prior successes, b prior failures); as $n \to \infty$ the data weight $\to 1$ and the prior is washed out.

Example 5.7.6 (Germination rate with a mildly informative prior)**.** A seed supplier claims a germination probability around $\frac{1}{2}$. You encode mild belief as a Beta$(2, 2)$ prior (mean 0.5, but only $a + b = 4$ pseudo-trials of weight). You plant $n = 20$ seeds and $x = 13$ germinate. Find the posterior, the Bayes estimate, and compare with the MLE.

Which rule, and why? Successes out of a fixed number of independent trials is a binomial count; a Beta prior on the success probability makes this the Beta–Binomial conjugate update.

Solution.

Step 1: update the parameters. $a + x = 2 + 13 = 15$ and $b + n - x = 2 + 7 = 9$, so

$$p \mid x \sim \text{Beta}(15,\, 9).$$

Step 2: Bayes estimate (posterior mean).

$$\mathbb{E}[p \mid x] = \frac{15}{15+9} = \frac{15}{24} = 0.625.$$

Step 3: compare with the MLE and check the weighted-average view. The MLE is $13/20 = 0.65$ and the prior mean is 0.5. With $w_{\text{prior}} = 4/24 = \frac{1}{6}$ and $w_{\text{data}} = 20/24 = \frac{5}{6}$,

$$\tfrac{1}{6}(0.5) + \tfrac{5}{6}(0.65) = 0.0833 + 0.5417 = 0.625. \blacksquare$$

The Bayes estimate sits just below the MLE, shrunk a little toward the prior mean 0.5.

Example 5.7.7 (A flat prior reproduces a familiar number)**.** With no real prior knowledge you use the uniform prior $p \sim \text{Beta}(1,1)$. In a survey $x = 37$ of $n = 50$ respondents approve. Give the posterior mean.

Which rule, and why? Binomial count again; the uniform prior is the special Beta case $a = b = 1$, so the same conjugate update applies.

Solution.

Step 1: update. $p \mid x \sim \text{Beta}(1+37,\ 1+13) = \text{Beta}(38, 14)$.

Step 2: posterior mean.

$$\mathbb{E}[p \mid x] = \frac{38}{52} = 0.7308.$$

This is the "add one success and one failure" (Laplace) estimate $(x+1)/(n+2) = 38/52$; for a flat prior the Bayes estimate is the smoothed proportion, close to the MLE $37/50 = 0.74$.

Example 5.7.8 (A strong prior dominates a small sample)**.** A manufacturing line has historically run at a 2% defect rate, encoded as a strong prior $p \sim \text{Beta}(4, 196)$ (prior mean $4/200 = 0.02$, weight 200 pseudo-units). A new batch of $n = 25$ has $x = 2$ defects (8% raw). Update.

Which rule, and why? Defects in a fixed number of items is binomial; a Beta prior makes it Beta–Binomial. The point of the example is the prior weight, not a new method.

Solution.

Step 1: update. $p \mid x \sim \text{Beta}(4+2,\ 196+23) = \text{Beta}(6, 219)$.

Step 2: posterior mean.

$$\mathbb{E}[p \mid x] = \frac{6}{225} = 0.0267.$$

Step 3: read the shrinkage. The raw rate $2/25 = 0.08$ barely moves the estimate: with $w_{\text{data}} = 25/225 = 0.111$ the posterior mean is $0.889(0.02) + 0.111(0.08) = 0.0178 + 0.0089 = 0.0267$. A heavy prior plus a small sample yields strong shrinkage toward the prior mean.

Example 5.7.9 (Credible interval for the germination rate)**.** For the Beta(15, 9) posterior of the germination example, give a 95% credible interval using a normal approximation, and state how to get the exact interval.

Which rule, and why? A "range with probability 0.95 for p" asks for a credible interval — a chunk of the posterior, not a confidence interval.

Solution.

Step 1: posterior mean and SD. Mean $= 15/24 = 0.625$. With $a = 15, b = 9, a+b = 24$,

$$\text{Var}(p \mid x) = \frac{ab}{(a+b)^2(a+b+1)} = \frac{15 \cdot 9}{24^2 \cdot 25} = \frac{135}{14{,}400} = 0.009375, \quad \text{SD} = 0.0968.$$

Step 2: central 95% via the normal approximation. Using $z_{0.025} = 1.96$,

$$0.625 \pm 1.96(0.0968) = 0.625 \pm 0.190 = (0.435,\ 0.815).$$

Step 3: exact interval. The exact 95% interval uses the 0.025 and 0.975 quantiles of Beta(15, 9), approximately (0.430, 0.804); the normal approximation is close because the posterior is fairly symmetric here. *Interpretation:* given the data and prior, there is a 0.95 posterior probability that p lies in this interval.

5.7.3 Gamma prior for a Poisson λ (Gamma–Poisson)

Definition 5.7.10 (Gamma distribution (shape–rate))**.** $\lambda \sim \text{Gamma}(\alpha, \beta)$ with shape $\alpha > 0$, rate $\beta > 0$ has density on $(0, \infty)$

$$\pi(\lambda) = \frac{\beta^\alpha}{\Gamma(\alpha)} \lambda^{\alpha-1} e^{-\beta\lambda}, \qquad \mathbb{E}[\lambda] = \frac{\alpha}{\beta}, \qquad \text{Var}(\lambda) = \frac{\alpha}{\beta^2}.$$

Theorem 5.7.11 (Gamma–Poisson conjugacy)**.** *If $X_1, \dots, X_n \mid \lambda$ are iid* Poisson(λ) *and the prior is $\lambda \sim \text{Gamma}(\alpha, \beta)$, then with $s = \sum_{i=1}^n x_i$,*

$$\lambda \mid x \;\sim\; \text{Gamma}(\alpha + s,\ \beta + n), \qquad \mathbb{E}[\lambda \mid x] = \frac{\alpha + s}{\beta + n}.$$

Proof. The Poisson likelihood is $L(\lambda) = \prod_i \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} \propto \lambda^s e^{-n\lambda}$. Times the prior $\propto \lambda^{\alpha-1} e^{-\beta\lambda}$,

$$\pi(\lambda \mid x) \propto \lambda^{(\alpha+s)-1}\, e^{-(\beta+n)\lambda},$$

the Gamma$(\alpha + s, \beta + n)$ kernel. The mean is shape over rate. □

Remark. Again the posterior mean is a weighted average. Prior mean α/β, MLE $\widehat{\lambda} = \bar{x} = s/n$, and

$$\frac{\alpha + s}{\beta + n} = \frac{\beta}{\beta + n} \cdot \frac{\alpha}{\beta} + \frac{n}{\beta + n} \cdot \frac{s}{n}.$$

The prior rate β counts as "pseudo-exposure" and α as "pseudo-events."

Example 5.7.12 (Calls per minute at a help desk)**.** Past data suggest about 3 calls per minute, encoded as $\lambda \sim \text{Gamma}(6, 2)$ (prior mean $6/2 = 3$). Over $n = 4$ separate minutes you log $5, 2, 4, 3$ calls. Update and give the Bayes estimate.

Which rule, and why? Counts of events in fixed time windows are Poisson; a Gamma prior on the rate gives the Gamma–Poisson conjugate update.

Solution.

Step 1: summarise the data. $s = 5 + 2 + 4 + 3 = 14$, $n = 4$.

Step 2: update.

$$\lambda \mid x \sim \text{Gamma}(6 + 14,\ 2 + 4) = \text{Gamma}(20,\ 6).$$

Step 3: posterior mean.

$$\mathbb{E}[\lambda \mid x] = \frac{20}{6} = 3.333.$$

The data mean is $14/4 = 3.5$ and the prior mean 3; the posterior 3.333 is between them, leaning toward the data ($w_{\text{data}} = 4/6 = 0.667$).

Example 5.7.13 (Typos per page with a weak prior)**.** You believe roughly 2 typos per page but are unsure, so you use a weak prior Gamma$(1, 0.5)$ (mean $1/0.5 = 2$, low weight $\beta = 0.5$). Across $n = 10$ pages you find $s = 27$ typos. Update.

Which rule, and why? Typo counts per page are Poisson; Gamma prior on the rate ⇒ Gamma–Poisson.

Solution.

Step 1: update.

$$\lambda \mid x \sim \text{Gamma}(1 + 27,\ 0.5 + 10) = \text{Gamma}(28,\ 10.5).$$

Step 2: posterior mean.

$$\mathbb{E}[\lambda \mid x] = \frac{28}{10.5} = 2.667.$$

The MLE is $27/10 = 2.7$; with such a weak prior the Bayes estimate is essentially the MLE, nudged only slightly toward 2.

Example 5.7.14 (Defects per circuit board)**.** Historical data give $\lambda \sim \text{Gamma}(8, 4)$ (prior mean 2 defects/board). You inspect $n = 6$ boards and count $3, 1, 2, 0, 4, 2$ defects. Find the posterior and a Bayes estimate.

Which rule, and why? Defect counts per board are Poisson; a Gamma prior makes it Gamma–Poisson.

Solution.

Step 1: summarise. $s = 3 + 1 + 2 + 0 + 4 + 2 = 12$, $n = 6$.

Step 2: update. $\lambda \mid x \sim \text{Gamma}(8 + 12,\ 4 + 6) = \text{Gamma}(20,\ 10)$.

Step 3: posterior mean.

$$\mathbb{E}[\lambda \mid x] = \frac{20}{10} = 2.0.$$

Here the data mean $12/6 = 2$ equals the prior mean 2, so the posterior mean is also 2; the data sharpen the posterior (variance $20/10^2 = 0.2$, SD $= 0.447$) without shifting it.

Example 5.7.15 (Credible interval for a Poisson rate)**.** For the help-desk posterior Gamma$(20, 6)$, give a 95% credible interval by a normal approximation.

Which rule, and why? A probability range for λ is a credible interval; for a Gamma posterior with moderate shape we may approximate by a normal centred at the posterior mean.

Solution.

Step 1: posterior mean and SD. Mean $= 20/6 = 3.333$; Var $= \alpha/\beta^2 = 20/36 = 0.5556$, SD $= 0.745$.

Step 2: central 95%. With $z_{0.025} = 1.96$,

$$3.333 \pm 1.96(0.745) = 3.333 \pm 1.461 = (1.872,\ 4.794).$$

Step 3: caveat. For small shape the exact Gamma quantiles (here roughly $(2.04, 4.83)$) are right-skewed; the normal approximation widens the left tail too much but is acceptable as a quick interval. There is posterior probability 0.95 that λ lies in this range.

5.7.4 Normal prior for a normal mean (Normal–Normal)

Theorem 5.7.16 (Normal–Normal mean update, σ^2 known)**.** *Let $X_1, \dots, X_n \mid \mu$ be iid $\text{N}(\mu, \sigma^2)$ with σ^2 known, and put a prior $\mu \sim \text{N}(\mu_0, \tau^2)$. Then the posterior is normal,*

$$\mu \mid x \ \sim\ \text{N}\big(\mu_n,\ \tau_n^2\big)\,, \qquad \frac{1}{\tau_n^2} = \frac{1}{\tau^2} + \frac{n}{\sigma^2}, \qquad \mu_n = \tau_n^2\left(\frac{\mu_0}{\tau^2} + \frac{n\,\overline{X}}{\sigma^2}\right).$$

Precisions (reciprocal variances) add; the posterior mean is the precision-weighted average of the prior mean and the sample mean.

Proof. Only $\overline{X}$ matters: $L(\mu) \propto \exp\big(-\frac{n}{2\sigma^2}(\mu - \overline{X})^2\big)$. Times the prior $\propto \exp\big(-\frac{1}{2\tau^2}(\mu - \mu_0)^2\big)$, the exponent is $-\frac{1}{2}\big[(\frac{1}{\tau^2} + \frac{n}{\sigma^2})\mu^2 - 2(\frac{\mu_0}{\tau^2} + \frac{n\overline{X}}{\sigma^2})\mu\big] + \text{const}$. Completing the square in μ gives a normal kernel with precision $\frac{1}{\tau^2} + \frac{n}{\sigma^2}$ and the stated mean. □

Remark. Write the data weight $w = \dfrac{n/\sigma^2}{1/\tau^2 + n/\sigma^2}$; then $\mu_n = (1 - w)\mu_0 + w\,\overline{X}$. A tight prior (small τ^2) or little data (small n) keeps w small and the estimate near the prior; abundant data ($n \to \infty$) sends $w \to 1$ and recovers the sample mean.

Example 5.7.17 (Calibrating a scale against a known standard)**.** A lab scale reads weights with known measurement variance $\sigma^2 = 100$ (grams^2, so $\sigma = 10$). Engineering specs suggest the true mean reading of a test object is about $\mu_0 = 100$ with prior variance $\tau^2 = 25$, i.e. $\mu \sim \text{N}(100, 25)$. You take $n = 16$ readings averaging $\overline{X} = 112$. Update.

Which rule, and why? A normal mean with a *known* variance and a normal prior ⇒ Normal–Normal update. The known σ^2 is what makes the conjugacy clean (no need to model σ^2).

Solution.

Step 1: precisions.

$$\frac{1}{\tau^2} = \frac{1}{25} = 0.04, \qquad \frac{n}{\sigma^2} = \frac{16}{100} = 0.16.$$

Step 2: posterior variance.

$$\frac{1}{\tau_n^2} = 0.04 + 0.16 = 0.20 \;\Rightarrow\; \tau_n^2 = 5, \quad \tau_n = \sqrt{5} = 2.236.$$

Step 3: posterior mean.

$$\mu_n = 5\,(100(0.04) + 112(0.16)) = 5(4 + 17.92) = 5(21.92) = 109.6.$$

Check via weights: $w = 0.16/0.20 = 0.8$, so $\mu_n = 0.2(100) + 0.8(112) = 20 + 89.6 = 109.6$. ■The data pull the estimate 9.6 above the prior mean toward $\overline{X} = 112$.

Example 5.7.18 (Average daily yield with a strong prior)**.** A chemical process has prior mean yield $\mu_0 = 50$ with a fairly confident prior variance $\tau^2 = 4$; the daily yield has known variance $\sigma^2 = 36$. Five days give $\overline{X} = 44$ ($n = 5$). Update.

Which rule, and why? Normal mean, known σ^2, normal prior ⇒ Normal–Normal update.

Solution.

Step 1: precisions. $1/\tau^2 = 1/4 = 0.25$ and $n/\sigma^2 = 5/36 = 0.1389$.

Step 2: posterior variance. $1/\tau_n^2 = 0.25 + 0.1389 = 0.3889 \Rightarrow \tau_n^2 = 2.571$, $\tau_n = 1.604$.

Step 3: posterior mean.

$$\mu_n = 2.571\,(50(0.25) + 44(0.1389)) = 2.571(12.5 + 6.111) = 2.571(18.611) = 47.86.$$

The strong prior ($w = 0.1389/0.3889 = 0.357$) holds the estimate up near 50 despite the low data mean 44: $\mu_n = 0.643(50) + 0.357(44) = 32.14 + 15.71 = 47.86$. ■

Example 5.7.19 (Test scores: a single observation)**.** Prior belief about a student's true ability is $\mu \sim \mathrm{N}(70,\, 9)$ ($\mu_0 = 70, \tau^2 = 9$). One exam (known measurement variance $\sigma^2 = 36$) scores $\overline{X} = 88$, $n = 1$. Update.

Which rule, and why? One normal observation with known variance and a normal prior is still the Normal–Normal update with $n = 1$.

Solution.

Step 1: precisions. $1/\tau^2 = 1/9 = 0.1111$ and $n/\sigma^2 = 1/36 = 0.02778$.

Step 2: posterior variance. $1/\tau_n^2 = 0.1111 + 0.02778 = 0.1389 \Rightarrow \tau_n^2 = 7.2$, $\tau_n = 2.683$.

Step 3: posterior mean.

$$\mu_n = 7.2\,(70(0.1111) + 88(0.02778)) = 7.2(7.778 + 2.444) = 7.2(10.222) = 73.6.$$

With only one noisy score the prior dominates ($w = 0.02778/0.1389 = 0.2$): $\mu_n = 0.8(70) + 0.2(88) = 56 + 17.6 = 73.6$. ■The single high score nudges the estimate up only modestly.

Example 5.7.20 (Credible interval for the scale's mean)**.** For the scale example, posterior $\mu \mid x \sim \mathrm{N}(109.6,\, 5)$. Give a 95% credible interval and contrast it with a frequentist confidence interval.

Which rule, and why? A probability statement about μ given the data is a credible interval; for a normal posterior it is mean $\pm\, z_{\alpha/2}\times$ posterior SD.

Solution.

Step 1: central 95%. Posterior SD $= \sqrt{5} = 2.236$; with $z_{0.025} = 1.96$,

$$109.6 \pm 1.96(2.236) = 109.6 \pm 4.383 = (105.22,\ 113.98).$$

Step 2: contrast with the frequentist CI. The frequentist z-interval ignores the prior and centres on $\overline{X} = 112$ with $\sigma/\sqrt{n} = 10/4 = 2.5$:

$$112 \pm 1.96(2.5) = 112 \pm 4.9 = (107.10,\ 116.90).$$

The credible interval is narrower (the prior adds information, lowering the posterior variance from $2.5^2 = 6.25$ to 5) and shifted toward the prior mean. *Interpretation differs:* the credible interval says "$\mathbb{P}(\mu \in (105.22, 113.98) \mid \text{data}) = 0.95$", a probability about μ; the confidence interval says "95% of intervals built this way cover the fixed μ," a property of the procedure (see §2).

5.7.5 The Bayes estimator and credible intervals

Definition 5.7.21 (Bayes estimator; posterior mean)**.** The **Bayes estimator** under squared-error loss is the posterior mean

$$\widehat{\theta}_{\mathrm{B}} = \mathbb{E}[\theta \mid x] = \int \theta\, \pi(\theta \mid x)\, \mathrm{d}\theta,$$

because the posterior mean is the constant c minimising the posterior expected loss $\mathbb{E}[(\theta - c)^2 \mid x]$. (Under absolute-error loss the minimiser is the posterior median.)

Definition 5.7.22 (Credible interval)**.** A $100(1-\alpha)\%$ **credible interval** is any set C with posterior probability $\mathbb{P}(\theta \in C \mid x) = 1 - \alpha$. The **central** (equal-tailed) version cuts the lower $\alpha/2$ and upper $\alpha/2$ posterior tails:

$$C = \big(q_{\alpha/2},\, q_{1-\alpha/2}\big),$$

the $\alpha/2$ and $1-\alpha/2$ posterior quantiles.

Remark (Credible vs. confidence — one sentence to keep straight)*.* A **credible** interval is a direct probability statement about θ given the observed data and prior; a **confidence** interval makes no probability claim about θ (which is fixed) — its $1-\alpha$ is the long-run coverage rate of the random procedure.

Example 5.7.23 (Bayes estimate as a decision)**.** Returning to the germination posterior Beta$(15, 9)$: report the squared-error Bayes estimate and the absolute-error Bayes estimate.

Which rule, and why? "Single best number" under a stated loss selects a posterior summary: mean for squared-error, median for absolute-error.

Solution.

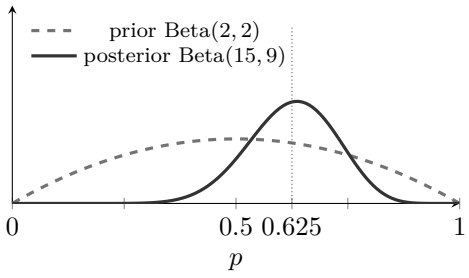
Step 1: squared-error (posterior mean). $\mathbb{E}[p \mid x] = 15/24 = 0.625$.

Step 2: absolute-error (posterior median). The Beta$(15, 9)$ median is approximately 0.628 (for Beta with both parameters > 1 the median is close to the mean; here just above it). For practical purposes both point estimates round to 0.63.

$$\widehat{p}_{\mathrm{B}} = 0.625 \text{ (squared-error)}, \qquad \approx 0.628 \text{ (absolute-error)}.$$

5.7.6 A prior-to-posterior picture

The Beta–Binomial germination example makes the update visible: the Beta$(2, 2)$ prior is broad and centred at 0.5; after 20 trials with 13 successes the Beta$(15, 9)$ posterior is much tighter and has shifted toward the data proportion 0.65.



The dotted line at $p = 0.625$ marks the posterior mean (the Bayes estimate). The posterior is both *shifted* toward the data and *sharper* than the prior — exactly the effect of combining prior belief with 20 informative trials.

Remark (Take-aways for the section). (i) Every Bayesian answer flows from posterior $\propto$ prior $\times$ likelihood. (ii) For the three conjugate pairs you update parameters by inspection: Beta$(a+x, b+n-x)$, Gamma$(\alpha+s, \beta+n)$, and the precision-weighted normal mean. (iii) The Bayes point estimate is the posterior mean, always a weighted average of prior mean and data, with the data weight growing with sample size. (iv) A credible interval is a posterior-probability range about θ — a different object from a confidence interval, even when their endpoints happen to be similar.

5.8 Exercises

All problems below are original. Full solutions begin on page 335.

A. Maximum likelihood and the method of moments

A1. A quality engineer tests 12 independently produced relays and finds that 9 of them pass a stress test; model each test as $\mathrm{Bin}(1, p)$ with unknown pass probability $p \in (0, 1)$. Write the likelihood $L(p)$ and the log-likelihood $\ell(p)$ for this sample, solve $\ell'(p) = 0$, and report the maximum likelihood estimate $\widehat{p}$. Confirm the critical point is a maximum.

A2. Over five consecutive nights a small observatory records the number of meteor streaks brighter than a fixed threshold: 2, 0, 3, 1, 4. Model the nightly counts as iid Poisson(λ). Find the MLE $\widehat{\lambda}$ by solving $\ell'(\lambda) = 0$, then find the method-of-moments estimate by equating the first population moment to the first sample moment. State whether the two estimates agree, and explain in one sentence why.

A3. The times (in hours) until first failure of 6 identical pumps are recorded; their sum is $\sum_{i=1}^{6} x_i = 15$. Model the failure times as iid $\mathrm{Exp}(\lambda)$ with density $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$ (so λ is the rate). Derive the MLE $\widehat{\lambda}$ from the log-likelihood, give its numerical value, and report the estimated mean time to failure $1/\widehat{\lambda}$.

A4. A sample of n iid $\mathrm{N}(\mu, \sigma^2)$ measurements is taken, both parameters unknown. Starting from the joint log-likelihood, derive the MLEs $\widehat{\mu}$ and $\widehat{\sigma}^2$. Then show that $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2$ is biased for σ^2 by computing $\mathbb{E}(\widehat{\sigma}^2)$, and state the simple correction that removes the bias.

A5. Five reaction times (in seconds), assumed iid $\mathrm{Uniform}(0, \theta)$ with density $f(x;\theta) = 1/\theta$ for $0 < x < \theta$, come out as 0.3, 0.7, 0.5, 0.9, 0.6. Find the MLE $\widehat{\theta}$ (note that it is *not* obtained by setting a derivative to zero) and the method-of-moments estimate $\widehat{\theta}_{\mathrm{MoM}}$, and give both numbers. Then compute $\mathbb{E}(\widehat{\theta})$ for the MLE and state the bias-corrected version $\frac{n+1}{n}\widehat{\theta}$.

A6. A reliability lab models the time between successive software faults as iid $\mathrm{Gamma}(\alpha, \beta)$ with density $f(x;\alpha,\beta) = \dfrac{x^{\alpha-1}e^{-x/\beta}}{\beta^{\alpha}\Gamma(\alpha)}$, $x > 0$, so that $\mathbb{E}(X) = \alpha\beta$ and $\mathrm{Var}(X) = \alpha\beta^2$. From $n = 10$ observations the summaries are $\sum_i x_i = 40$ and $\sum_i x_i^2 = 180$. Use the method of moments (matching the first two moments) to estimate α and β. Why is method of moments, rather than maximum likelihood, the natural first choice for this two-parameter shape problem?

A7. Lifetimes follow a **shifted exponential** with density $f(x;\theta) = e^{-(x-\theta)}$ for $x \geq \theta$ (and 0 otherwise), where $\theta \in \mathbb{R}$ is an unknown threshold below which failures cannot occur. From $n = 8$ units the smallest observed lifetime is $\min_i x_i = 2.1$ and the sample mean is $\overline{X} = 3.4$.

(a) Find the MLE $\widehat{\theta}$ by a boundary argument on the likelihood, and the method-of-moments estimate $\widehat{\theta}_{\mathrm{MoM}}$; give both numbers and note that they differ.

(b) Using the invariance of MLEs, estimate the probability $\mathbb{P}(X > 3) = e^{-(3-\theta)}$ that a unit survives past time 3.

B. Interval estimation

B1. A calibration lab weighs the same reference mass $n = 16$ times on a balance whose measurement error is known to be $\mathrm{N}(\mu, \sigma^2)$ with *known* standard deviation $\sigma = 6$ micrograms. The 16 readings have sample mean $\overline{X} = 82.5$ micrograms. Construct a 95% confidence interval for the true mean reading μ, and state in one sentence what "95% confidence" refers to (the procedure, not the single interval).

B2. A nutritionist measures the vitamin C content (mg per serving) of $n = 10$ randomly chosen cereal boxes and obtains sample mean $\overline{X} = 12.4$ and sample standard deviation $S = 2.1$. The contents are modeled as iid $\mathrm{N}(\mu, \sigma^2)$ with *both* parameters unknown. Build a 95% confidence interval for the mean content μ, and explain in one phrase why a t critical value is used here rather than a z value.

B3. In a survey of $n = 400$ independently sampled commuters, 92 report that they would switch to a new express bus line. Treating the responses as iid $\mathrm{Bin}(1, p)$, give the point estimate $\widehat{p}$ and construct the large-sample 95% confidence interval $\widehat{p} \pm z_{\alpha/2}\sqrt{\widehat{p}(1-\widehat{p})/n}$ for the true proportion p of all commuters who would switch.

B4. The transit agency in B3 wants to repeat the survey and estimate p to within a margin of error of 0.03 at the 95% confidence level.

(a) How many commuters must be sampled if no prior information about p is used (the conservative, worst-case choice)?

(b) How many are needed if a pilot study suggests $p \approx 0.20$, so that $\widehat{p}(1-\widehat{p})$ can be planned at $p = 0.20$? Comment in one sentence on why the second number is smaller.

B5. A machinist records the inside diameters (in mm) of $n = 12$ bored cylinders, modeled as iid $\mathrm{N}(\mu, \sigma^2)$. The sample variance is $S^2 = 18.5$. Construct a 90% confidence interval for the population variance σ^2 using the χ^2 pivot $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$, then convert it to a 90% confidence interval for the standard deviation σ.

B6. Customers arrive at a help desk and the times between successive arrivals (in minutes) are modeled as iid $\mathrm{Exp}(\lambda)$ with density $f(x;\lambda) = \lambda e^{-\lambda x}$, $x > 0$ (so λ is the arrival *rate*). From $n = 50$ inter-arrival times the sample mean is $\overline{X} = 4.0$ minutes.

(a) Give the MLE $\widehat{\lambda}$ and, using the Fisher information $\mathcal{I}(\lambda) = 1/\lambda^2$, the large-sample standard error of $\widehat{\lambda}$.

(b) Construct the large-sample (Wald) 95% confidence interval for λ based on the asymptotic normality of the MLE.

B7. The lifetimes (in hours) of $n = 8$ identical LED modules are modeled as iid exponential with *mean* θ, density $f(x;\theta) = \frac{1}{\theta}e^{-x/\theta}$, $x > 0$. The observed sample mean is $\overline{X} = 210$ hours.

(a) Show that $Q = \dfrac{2\sum_{i=1}^{n} X_i}{\theta} = \dfrac{2n\overline{X}}{\theta}$ has a χ^2_{2n} distribution, and use this exact pivot to build a 90% confidence interval for the mean lifetime θ.

(b) For a warranty the lab needs a 95% one-sided *lower* confidence bound for θ (a value L with $\mathbb{P}(\theta \geq L) = 0.95$). Give L.

C. Unbiasedness and efficiency

C1. Two independent gauges measure the same true level μ of a chemical. Gauge A returns T_1 with $\mathbb{E}(T_1) = \mu$ and $\mathrm{Var}(T_1) = 0.5$; gauge B returns T_2 with $\mathbb{E}(T_2) = \mu$ and $\mathrm{Var}(T_2) = 1.2$. A lab combines them as $\widehat{\mu} = a\,T_1 + b\,T_2$. (a) State the condition on a, b that makes $\widehat{\mu}$ unbiased for μ. (b) Among unbiased combinations of this form, is the simple average $a = b = \frac{1}{2}$ unbiased, and what is its variance?

C2. Let $X_1, \ldots, X_n$ be iid Poisson(λ) and let $\overline{X}$ be the sample mean. (a) Show $\overline{X}$ is unbiased for λ and find $\mathrm{Var}(\overline{X})$. (b) A colleague proposes using a single observation X_1 as the estimator. It is also unbiased; compute the efficiency of X_1 relative to $\overline{X}$ for a sample of size $n = 8$ and interpret it.

C3. From an iid sample X_1, X_2, X_3 with mean μ and variance σ^2, compare the two estimators

$$\widehat{\mu}_1 = \overline{X} = \tfrac{1}{3}(X_1 + X_2 + X_3), \qquad \widehat{\mu}_2 = \tfrac{1}{4}X_1 + \tfrac{1}{2}X_2 + \tfrac{1}{4}X_3.$$

Show both are unbiased, find each variance, and compute the efficiency of $\widehat{\mu}_2$ relative to $\widehat{\mu}_1$. Which is the better estimator?

C4. Let $X_1, \dots, X_n$ be iid with mean μ and variance σ^2, and define the "divide-by-n" variance estimator

$$\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

(a) Find $\mathbb{E}(\widehat{\sigma}^2)$ and hence its bias for σ^2. (b) Explain why the sample variance S^2 uses the divisor $n-1$, and evaluate the bias of $\widehat{\sigma}^2$ when $n = 5$ and $\sigma^2 = 20$.

C5. Let $X_1, \dots, X_n$ be iid Uniform$(0, \theta)$. Consider two unbiased estimators of θ:

$$\widehat{\theta}_1 = 2\overline{X}, \qquad \widehat{\theta}_2 = \frac{n+1}{n}\,M, \quad M = \max_i X_i.$$

You may use $\mathbb{E}(M) = \frac{n}{n+1}\theta$ and $\mathrm{Var}(M) = \frac{n}{(n+1)^2(n+2)}\theta^2$. Verify both are unbiased, find $\mathrm{Var}(\widehat{\theta}_1)$ and $\mathrm{Var}(\widehat{\theta}_2)$, and show that the efficiency of $\widehat{\theta}_2$ relative to $\widehat{\theta}_1$ is $(n+2)/3$. Comment on what this says for $n = 10$.

C6. Let $X_1, \dots, X_n$ be iid $\mathrm{N}(\mu, \sigma^2)$ and write $W = \sum_{i=1}^{n}(X_i - \overline{X})^2$, so $W/\sigma^2 \sim \chi^2_{n-1}$. For a constant $c > 0$ consider the family of estimators $\widehat{\sigma}^2_c = c\,W$ of σ^2. (a) Using $\mathbb{E}(W) = (n-1)\sigma^2$ and $\mathrm{Var}(W) = 2(n-1)\sigma^4$, write $\mathrm{MSE}(\widehat{\sigma}^2_c)$ as a function of c. (b) Find the value of c that minimizes the MSE, and show it is neither the unbiased choice $c = \frac{1}{n-1}$ (giving S^2) nor the MLE choice $c = \frac{1}{n}$. (c) For $n = 10$ and $\sigma^2 = 1$, compare the MSEs of the three estimators numerically.

C7. You have three independent, unbiased estimators T_1, T_2, T_3 of a parameter θ, with $\mathrm{Var}(T_1) = 4$, $\mathrm{Var}(T_2) = 9$, $\mathrm{Var}(T_3) = 36$. Consider pooled estimators of the form $\widehat{\theta} = w_1T_1 + w_2T_2 + w_3T_3$ with $w_1 + w_2 + w_3 = 1$. (a) Show that any such $\widehat{\theta}$ is unbiased. (b) Find the weights w_i that minimize $\mathrm{Var}(\widehat{\theta})$ and the resulting minimum variance. (Hint: minimize $\sum w_i^2\sigma_i^2$ subject to $\sum w_i = 1$; the optimum is inverse-variance weighting.) (c) Compute the efficiency of the optimally pooled estimator relative to the simple average $\frac{1}{3}(T_1 + T_2 + T_3)$.

D. Fisher information and the Cramer-Rao bound

D1. A quality monitor records whether each circuit board passes a stress test, modelling the boards as iid $\mathrm{Bin}(1, p)$ (Bernoulli) with success probability p.

(a) Compute the Fisher information $\mathcal{I}(p)$ in a single observation.

(b) State the Cramer-Rao lower bound on the variance of any unbiased estimator of p based on a sample of size n.

(c) The sample proportion $\widehat{p} = \overline{X}$ is unbiased for p. Find its variance and decide whether it attains the bound (i.e. whether it is efficient).

D2. In a one-hour window a help desk logs the number of incoming tickets per minute, modelled as iid Poisson(λ).

(a) Show that the Fisher information in one observation is $\mathcal{I}(\lambda) = 1/\lambda$.

(b) Give the Cramer-Rao bound for an unbiased estimator of λ from n observations, and explain why $\widehat{\lambda} = \overline{X}$ is efficient.

(c) From $n = 40$ minutes the observed mean is $\overline{x} = 2.5$ tickets. Report the estimated variance floor (the bound with λ replaced by $\overline{x}$).

D3. The time (in hours) between machine faults is modelled as iid $\mathrm{Exp}(\lambda)$ with density $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$, where $\lambda > 0$ is the fault rate.

(a) Compute $\mathcal{I}(\lambda)$ for a single observation.

(b) Write down the Cramer-Rao lower bound for an unbiased estimator of λ based on n observations.

(c) The MLE is $\widehat{\lambda} = 1/\overline{X}$. Using $\mathbb{E}(1/\overline{X}) = \frac{n}{n-1}\lambda$, explain why the Cramer-Rao bound does not directly certify $\widehat{\lambda}$ as efficient.

D4. A calibrated scale produces readings modelled as iid $\mathrm{N}(\mu, \sigma^2)$ with the variance σ^2 KNOWN; only the true weight μ is unknown.

(a) Show that $\mathcal{I}(\mu) = 1/\sigma^2$.

(b) State the Cramer-Rao bound for an unbiased estimator of μ from a sample of size n, and verify that $\overline{X}$ attains it.

(c) With $\sigma^2 = 9$ and $n = 25$, give the smallest possible variance of any unbiased estimator of μ.

D5. Background noise voltages are modelled as iid $\mathrm{N}(\mu, \sigma^2)$ where the mean μ is KNOWN (the instrument is zeroed) and the unknown parameter is the variance $\theta = \sigma^2$.

(a) Treating $\theta = \sigma^2$ as the parameter, show that the Fisher information in one observation is $\mathcal{I}(\sigma^2) = \dfrac{1}{2\sigma^4}$.

(b) State the Cramer-Rao bound for an unbiased estimator of σ^2 from n observations.

(c) Show that $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$ is unbiased and has variance $2\sigma^4/n$, hence is efficient. (You may use that $\sum(X_i - \mu)^2/\sigma^2 \sim \chi^2_n$, whose variance is $2n$.)

D6. Daily counts of rare equipment failures at a plant are iid Poisson(λ). The quantity of interest is the probability of a failure-free day, $\tau = g(\lambda) = e^{-\lambda} = \mathbb{P}(X = 0)$.

(a) Using the parameter-transformation form of the bound, $\mathrm{Var}(\widehat{\tau}) \geq \dfrac{[g'(\lambda)]^2}{n\,\mathcal{I}(\lambda)}$ for any unbiased estimator of τ, derive the lower bound in terms of λ and n.

(b) Evaluate the bound for $\lambda = 2$ and $n = 50$. Use $e^{-4} = 0.018316$.

D7. Lifetimes (in thousands of cycles) of a bearing are iid exponential, but parametrised by the *mean* lifetime $\theta = \mathbb{E}(X) > 0$, so the density is $f(x;\theta) = \frac{1}{\theta}e^{-x/\theta}$ for $x > 0$.

(a) Compute $\mathcal{I}(\theta)$ for a single observation and state the Cramer-Rao bound for an unbiased estimator of θ from n observations.

(b) Show that $\overline{X}$ is unbiased for θ and attains the bound.

(c) A colleague proposes the cheap estimator $\widehat{\theta}_1 = X_1$ (use only the first bearing). Show it is unbiased, find its variance, and compute the efficiency of $\widehat{\theta}_1$ relative to $\overline{X}$. With $n = 20$ and observed $\overline{x} = 5$, also report the estimated variance floor $\overline{x}^2/n$.

E. Sufficient statistics

E1. A quality engineer inspects $n = 6$ circuit boards from a production line. Each board is independently defective with unknown probability $p \in (0, 1)$. Let $X_i = 1$ if board i is defective and $X_i = 0$ otherwise, so the X_i are iid $\mathrm{Bin}(1, p)$. The observed data are

$$0,\ 1,\ 0,\ 0,\ 1,\ 0.$$

Write the joint mass function $f(x_1, \ldots, x_6; p)$ as a function of $t = \sum_{i=1}^{6} x_i$, and use the factorisation theorem to identify a one-dimensional sufficient statistic T for p. State its observed value.

E2. Let $X_1, \ldots, X_n$ be iid Poisson(λ) counts of typos per page, $\lambda > 0$ unknown. Using the factorisation theorem, show that $T = \sum_{i=1}^{n} X_i$ is sufficient for λ. Identify the functions $g(T, \lambda)$ and $h(x)$ explicitly, and state whether the sample mean $\overline{X}$ is also sufficient.

E3. Service times (in minutes) at a help desk are modelled as iid Exp(λ) with density $f(x;\lambda) = \lambda e^{-\lambda x}$ for $x > 0$, where $\lambda > 0$ is the unknown rate. From a sample of size n:

(a) Find a one-dimensional sufficient statistic for λ by factorisation.

(b) Show that the MLE $\widehat{\lambda} = n/\sum X_i$ is a function of that sufficient statistic, and explain in one sentence why every MLE here must be.

E4. Let $X_1, \dots, X_n$ be iid Uniform$(0, \theta)$ with $\theta > 0$ unknown. Write the joint density using indicator functions and apply the factorisation theorem to show that $T = \max_i X_i = X_{(n)}$ is sufficient for θ. Explain why the derivative-based approach used for Poisson or exponential families does NOT produce the sufficient statistic here, and why $\sum X_i$ is *not* sufficient for θ.

E5. Let $X_1, \dots, X_n$ be iid N(μ, σ^2) with BOTH μ and σ^2 unknown. Expand the joint density and use the factorisation theorem to show that the two-dimensional statistic

$$T = \left(\sum_{i=1}^{n} X_i,\ \sum_{i=1}^{n} X_i^2 \right)$$

is jointly sufficient for (μ, σ^2). Then show that the pair $(\overline{X},\ S^2)$ carries exactly the same information, i.e. is also sufficient.

E6. A geneticist counts the number of trials up to and including the first success in each of n independent runs; the counts $X_1, \dots, X_n$ are iid Geom(p) with mass function $f(x; p) = (1-p)^{x-1}p$ for $x \in \{1, 2, 3, \dots\}$, where $p \in (0, 1)$ is unknown.

(a) Use the factorisation theorem to find a one-dimensional sufficient statistic T for p.

(b) For $n = 4$ with observed counts $3, 1, 2, 2$, give the observed value of T and the resulting MLE $\widehat{p}$. (You may use that the MLE for the Geom(p) model is $\widehat{p} = n/\sum X_i$.)

E7. Let $X_1, \dots, X_n$ be iid Poisson(λ), and consider estimating λ. The single observation X_1 is an unbiased estimator of λ, but it ignores the rest of the data. Let $T = \sum_{i=1}^{n} X_i$, the sufficient statistic from E2.

(a) Show that X_1 is unbiased for λ and compute its variance.

(b) Apply the Rao–Blackwell construction: compute the Rao–Blackwellised estimator $\widehat{\lambda}^{*} = \mathbb{E}[X_1 \mid T = t]$. You may use that, conditional on $\sum_{i=1}^{n} X_i = t$, the vector $(X_1, \dots, X_n)$ is Multinomial$(t;\ 1/n, \dots, 1/n)$.

(c) Compute $\mathrm{Var}(\widehat{\lambda}^{*})$ and verify that $\widehat{\lambda}^{*}$ has variance no larger than X_1, confirming the Rao–Blackwell improvement.

F. Consistency

F1. A bottling line fills cans whose actual volumes are iid with mean μ and finite variance σ^2. An inspector estimates μ by the sample mean $\overline{X}_n$ of n cans. Using the Weak Law of Large Numbers, state precisely what it means for $\overline{X}_n$ to be a consistent estimator of μ, and explain in one sentence why finiteness of σ^2 is enough (though not necessary) to guarantee it.

F2. In a quality audit, each inspected weld either passes (1) or fails (0) independently with pass-probability p. After inspecting n welds the auditor uses the sample proportion $\widehat{p}_n = \overline{X}_n$. Compute $\mathrm{Bias}(\widehat{p}_n)$ and $\mathrm{Var}(\widehat{p}_n)$, then use the sufficient condition $\mathrm{MSE}(\widehat{p}_n) \to 0$ to prove $\widehat{p}_n$ is consistent for p. State the upper bound on $\mathrm{Var}(\widehat{p}_n)$ that holds for every $p \in (0, 1)$.

F3. Service times at a help desk are iid Exp(λ) (mean $1/\lambda$), and the analyst estimates the mean service time $\tau = 1/\lambda$ by $\overline{X}_n$. Show $\overline{X}_n$ is consistent for τ. Then, treating $\widehat{\lambda}_n = 1/\overline{X}_n$ as the estimator of the rate λ, argue that $\widehat{\lambda}_n$ is consistent for λ, naming the property of limits in probability that you use.

F4. Lifetimes (in thousands of hours) of a sensor are iid Uniform$(0, \theta)$. The MLE of θ from n units is $\widehat{\theta}_n = \max\{X_1, \dots, X_n\} = X_{(n)}$. Find the exact distribution function of $X_{(n)}$, compute $\mathbb{E}(X_{(n)})$ and $\mathrm{Var}(X_{(n)})$, and use $\mathrm{MSE} \to 0$ to prove $\widehat{\theta}_n$ is consistent for θ. Is $\widehat{\theta}_n$ unbiased?

F5. Counts of typos per page in a manuscript are iid Poisson(λ). A reviewer is interested in the probability that a page is typo-free, $\psi = e^{-\lambda}$, and proposes two estimators: (a) $\widehat{\psi}_n^{(1)} = e^{-\overline{X}_n}$, and (b) $\widehat{\psi}_n^{(2)} =$ the fraction of the n sampled pages that have zero typos. Show that both are consistent for ψ, and state the general theorem about continuous transformations that justifies (a).

F6. An estimator sequence $\widehat{\theta}_n$ of $\theta > 0$ has $\mathbb{E}(\widehat{\theta}_n) = \theta + \dfrac{3}{n}$ and $\mathrm{Var}(\widehat{\theta}_n) = \dfrac{5\theta^2}{\sqrt{n}}$. (i) Decide whether $\widehat{\theta}_n$ is consistent, justifying with the MSE criterion. (ii) A competing estimator $\widetilde{\theta}_n$ is unbiased with $\mathrm{Var}(\widetilde{\theta}_n) = \dfrac{c}{\ln n}$ for a constant $c > 0$; is it consistent? (iii) Construct an estimator that is *asymptotically unbiased* (bias $\to 0$) yet *not* consistent, and explain why the two notions differ.

F7. Observations are iid $\mathrm{N}(\mu, \sigma^2)$ with both parameters unknown. Consider the plug-in variance estimator $\widehat{\sigma}_n^2 = \dfrac{1}{n}\sum_{i=1}^n (X_i - \overline{X}_n)^2$ (the MLE) and the unbiased estimator $S_n^2 = \dfrac{1}{n-1}\sum_{i=1}^n (X_i - \overline{X}_n)^2$. Using $\mathbb{E}(\widehat{\sigma}_n^2) = \dfrac{n-1}{n}\sigma^2$ and $\mathrm{Var}(S_n^2) = \dfrac{2\sigma^4}{n-1}$, prove that both $\widehat{\sigma}_n^2$ and S_n^2 are consistent for σ^2. State which of the two is unbiased, and confirm both have MSE $\to 0$.

G. Bayesian estimation

G1. A quality engineer models the probability p that a randomly chosen solar panel passes a stress test. Before seeing data she uses a uniform prior $p \sim \mathrm{Beta}(1, 1)$ on $[0, 1]$. She then tests 12 panels independently and 9 pass. Write the posterior density of p (name the distribution and its parameters), and give the posterior mean $\mathbb{E}(p \mid \text{data})$.

G2. A help-desk receives calls at rate λ per hour. The analyst's prior is $\lambda \sim \mathrm{Gamma}(3, 2)$ with density $\pi(\lambda) \propto \lambda^2 e^{-2\lambda}$ (shape $a = 3$, rate $b = 2$, so prior mean $a/b = 1.5$). Over one 4-hour shift she observes $X_1, \ldots, X_4$ independent Poisson counts that sum to $\sum x_i = 10$. Find the posterior distribution of λ, its parameters, and the posterior mean.

G3. A lab measures the true concentration θ (in mg/L) of a reagent with an instrument of known measurement standard deviation $\sigma = 4$. Prior belief is $\theta \sim \mathrm{N}(50,\, \tau^2)$ with $\tau^2 = 16$. A single new reading gives $x = 58$. The Normal–Normal posterior is again normal. Find the posterior mean and posterior variance, and explain in one line why the posterior mean lies between 50 and 58.

G4. Continue Problem G1 (posterior $\mathrm{Beta}(10, 4)$). Show that the posterior mean is a weighted average of the prior mean $1/2$ and the sample proportion $9/12$, and identify the two weights. Then state which estimate (prior mean, sample proportion, or posterior mean) you would report and why, in one sentence.

G5. In a Gamma–Poisson model the prior on a defect rate λ is $\mathrm{Gamma}(a, b)$ (density $\propto \lambda^{a-1}e^{-b\lambda}$) and we observe iid Poisson counts $X_1, \ldots, X_n$ with $\sum x_i = s$. (a) Derive the posterior of λ from posterior $\propto$ prior $\times$ likelihood. (b) Show the posterior mean can be written as a weighted average $w \cdot \frac{a}{b} + (1 - w) \cdot \bar{x}$ of the prior mean and the sample mean, and give w. (c) Show $w \to 0$ as $n \to \infty$ and interpret.

G6. For the reagent model of Problem G3 (now treat it generally): a single observation x with known sampling variance σ^2 and prior $\theta \sim \mathrm{N}(\mu_0, \tau^2)$. (a) Show the posterior is $\mathrm{N}(\mu_1, \sigma_1^2)$ with $\sigma_1^2 = \left(\frac{1}{\tau^2} + \frac{1}{\sigma^2}\right)^{-1}$ and $\mu_1 = \sigma_1^2\left(\frac{\mu_0}{\tau^2} + \frac{x}{\sigma^2}\right)$ by completing the square in the exponent. (b) Using $\sigma = 4$, $\tau^2 = 16$, $\mu_0 = 50$, $x = 58$, construct a 95% equal-tailed credible interval for θ and state the z value used.

G7. A pharmacologist studies the probability p that a candidate compound is soluble. She uses a $\mathrm{Beta}(2, 2)$ prior (mean $1/2$, but more concentrated than uniform) and screens $n = 20$ compounds, of which 14 are soluble. (a) Give the posterior distribution and posterior mean. (b) The posterior variance of a $\mathrm{Beta}(\alpha, \beta)$ law is $\dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$; compute it here and take its square root to report the posterior standard deviation. (c) Form an *approximate* central 95% credible interval as posterior mean $\pm 1.96\times$(posterior SD), state the z value, and note in one line why this normal approximation is reasonable for these numbers.

5.9 Solutions

A. Maximum likelihood and the method of moments

A1. With $n = 12$ Bernoulli trials and $s = 9$ successes, each observation $x_i \in \{0, 1\}$ has mass $p^{x_i}(1-p)^{1-x_i}$, so

$$L(p) = \prod_{i=1}^{12} p^{x_i}(1-p)^{1-x_i} = p^{\,s}(1-p)^{\,12-s} = p^9(1-p)^3.$$

The log-likelihood and its derivative are

$$\ell(p) = 9\ln p + 3\ln(1-p), \qquad \ell'(p) = \frac{9}{p} - \frac{3}{1-p}.$$

Setting $\ell'(p) = 0$ gives $9(1-p) = 3p$, i.e. $9 = 12p$, so

$$\widehat{p} = \frac{s}{n} = \frac{9}{12} = \frac{3}{4} = 0.75.$$

Since $\ell''(p) = -9/p^2 - 3/(1-p)^2 < 0$ for all $p \in (0,1)$, the log-likelihood is strictly concave and the critical point is the unique maximum.

A2. The counts are $2, 0, 3, 1, 4$ with $n = 5$, $\sum_i x_i = 10$, $\overline{X} = 2$. For Poisson(λ) the mass is $e^{-\lambda}\lambda^{x_i}/x_i!$, so

$$\ell(\lambda) = -n\lambda + \Big(\textstyle\sum_i x_i\Big)\ln\lambda - \sum_i \ln(x_i!), \qquad \ell'(\lambda) = -n + \frac{\sum_i x_i}{\lambda}.$$

Setting $\ell'(\lambda) = 0$ gives $\widehat{\lambda} = \dfrac{\sum_i x_i}{n} = \overline{X} = 2$. (The second derivative $-\sum_i x_i/\lambda^2 < 0$ confirms a maximum.)

Method of moments. The first population moment is $\mathbb{E}(X) = \lambda$; matching it to the first sample moment $\overline{X}$ gives $\widehat{\lambda}_{\text{MoM}} = \overline{X} = 2$.

The two estimates *agree*: for the Poisson family the single parameter λ equals the population mean, so equating $\mathbb{E}(X)$ to $\overline{X}$ produces exactly the same equation as the score equation $\ell'(\lambda) = 0$.

A3. For iid Exp(λ), $f(x;\lambda) = \lambda e^{-\lambda x}$, the log-likelihood is

$$\ell(\lambda) = n\ln\lambda - \lambda\sum_{i=1}^{n} x_i, \qquad \ell'(\lambda) = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i.$$

Setting $\ell'(\lambda) = 0$ gives

$$\widehat{\lambda} = \frac{n}{\sum_i x_i} = \frac{1}{\overline{X}} = \frac{6}{15} = 0.4 \text{ per hour},$$

and $\ell''(\lambda) = -n/\lambda^2 < 0$ confirms a maximum. The estimated mean time to failure is $1/\widehat{\lambda} = \overline{X} = 15/6 = 2.5$ hours.

A4. For iid $\mathrm{N}(\mu, \sigma^2)$,

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Estimate μ. $\dfrac{\partial \ell}{\partial \mu} = \dfrac{1}{\sigma^2}\sum_i (x_i - \mu) = 0 \;\Rightarrow\; \widehat{\mu} = \overline{X}$.

Estimate σ^2. Substituting $\widehat{\mu} = \overline{X}$,

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_i (x_i - \overline{X})^2 = 0 \;\Rightarrow\; \widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2.$$

Bias. Since $\sum_i (X_i - \overline{X})^2/\sigma^2 \sim \chi^2_{n-1}$ has mean $n-1$,

$$\mathbb{E}(\widehat{\sigma}^2) = \frac{1}{n}\,\mathbb{E}\Big[\sum_i (X_i - \overline{X})^2\Big] = \frac{(n-1)\sigma^2}{n} = \sigma^2 - \frac{\sigma^2}{n},$$

so $\mathrm{Bias}(\widehat{\sigma}^2) = -\sigma^2/n \neq 0$: the MLE underestimates σ^2 on average. Dividing by $n-1$ instead of n removes the bias, giving the unbiased $S^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2$ with $\mathbb{E}(S^2) = \sigma^2$.

A5. The data are $0.3, 0.7, 0.5, 0.9, 0.6$ with $n = 5$, $\sum_i x_i = 3.0$, $\overline{X} = 0.6$, and $\max_i x_i = 0.9$. The likelihood is

$$L(\theta) = \prod_{i=1}^{5} \frac{1}{\theta}\,\mathbb{1}\{0 < x_i < \theta\} = \frac{1}{\theta^5}\,\mathbb{1}\{\theta > \max_i x_i\}.$$

MLE. For $\theta \geq \max_i x_i$ the likelihood is θ^{-5}, which *decreases* in θ; it is 0 for $\theta < \max_i x_i$. Hence it is largest at the smallest admissible θ:

$$\widehat{\theta} = \max_i x_i = 0.9.$$

A derivative is useless here because the maximizer sits at the boundary of the parameter region, not at an interior stationary point.

Method of moments. $\mathbb{E}(X) = \theta/2$; matching to $\overline{X}$ gives $\widehat{\theta}_{\mathrm{MoM}} = 2\overline{X} = 2(0.6) = 1.2$.

Bias of the MLE. $M = \max_i X_i$ has cdf $(x/\theta)^n$ and density nx^{n-1}/θ^n on $(0, \theta)$, so

$$\mathbb{E}(\widehat{\theta}) = \int_0^\theta x\,\frac{nx^{n-1}}{\theta^n}\,\mathrm{d}x = \frac{n}{n+1}\,\theta = \frac{5}{6}\theta,$$

which is biased low. The bias-corrected estimator is $\dfrac{n+1}{n}\,\widehat{\theta} = \dfrac{6}{5}(0.9) = 1.08$.

A6. With $n = 10$, $\sum_i x_i = 40$ and $\sum_i x_i^2 = 180$, the sample moments are

$$m_1 = \overline{X} = \frac{40}{10} = 4, \qquad m_2 = \frac{1}{n}\sum_i x_i^2 = \frac{180}{10} = 18,$$

and the (MoM) sample variance is $v = m_2 - m_1^2 = 18 - 16 = 2$.

Match the moments. The population moments are $\mathbb{E}(X) = \alpha\beta$ and $\mathrm{Var}(X) = \alpha\beta^2$. Setting

$$\alpha\beta = m_1 = 4, \qquad \alpha\beta^2 = v = 2.$$

Solve. Dividing the second equation by the first eliminates α:

$$\beta = \frac{\alpha\beta^2}{\alpha\beta} = \frac{v}{m_1} = \frac{2}{4} = \frac{1}{2}, \qquad \alpha = \frac{m_1}{\beta} = \frac{m_1^2}{v} = \frac{16}{2} = 8.$$

So $\widehat{\alpha} = 8$, $\widehat{\beta} = 0.5$. (Check: $\alpha\beta = 8(0.5) = 4$ and $\alpha\beta^2 = 8(0.25) = 2$.) Method of moments is the natural first tool because the MLE for the shape parameter α has no closed form – the score equation involves the digamma function $\psi(\alpha) = \Gamma'(\alpha)/\Gamma(\alpha)$ and must be solved numerically – whereas matching the first two moments gives the simple algebraic solution above.

A7. The shifted-exponential density is $f(x;\theta) = e^{-(x-\theta)}$ for $x \geq \theta$.

(a) MLE. The likelihood is

$$L(\theta) = \prod_{i=1}^{n} e^{-(x_i-\theta)}\,\mathbb{1}\{x_i \geq \theta\} = \exp\Big(n\theta - \sum_i x_i\Big)\,\mathbb{1}\{\theta \leq \min_i x_i\}.$$

On the admissible region $\theta \le \min_i x_i$ this is $e^{n\theta - \sum x_i}$, which *increases* in θ; it is 0 once θ exceeds some x_i. So the likelihood is maximized at the largest admissible value,

$$\widehat{\theta} = \min_i x_i = 2.1.$$

For the method of moments, with $Y = X - \theta \sim \text{Exp}(1)$ we have $\mathbb{E}(X) = \theta + \mathbb{E}(Y) = \theta + 1$; matching to $\overline{X}$ gives

$$\widehat{\theta}_{\text{MoM}} = \overline{X} - 1 = 3.4 - 1 = 2.4.$$

The two estimates *differ* (2.1 vs. 2.4): the MLE keys on the sample minimum (the most informative statistic about a left threshold), while the MoM uses the sample mean.

(b) Invariance. The MLE of a function $g(\theta)$ is $g(\widehat{\theta})$. With $g(\theta) = \mathbb{P}(X > 3) = e^{-(3-\theta)}$ (valid for $\theta < 3$, which holds since $\widehat{\theta} = 2.1$),

$$\widehat{\mathbb{P}(X > 3)} = e^{-(3-\widehat{\theta})} = e^{-(3-2.1)} = e^{-0.9} \approx 0.407.$$

B. Interval estimation

B1. With σ known the pivot is $\dfrac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \text{N}(0,1)$, so the $1-\alpha$ interval is $\overline{X} \pm z_{\alpha/2}\,\sigma/\sqrt{n}$. For 95% use $z_{0.025} = 1.96$. The margin of error is

$$z_{0.025}\,\frac{\sigma}{\sqrt{n}} = 1.96 \cdot \frac{6}{\sqrt{16}} = 1.96 \cdot \frac{6}{4} = 1.96(1.5) = 2.94.$$

The interval is

$$82.5 \pm 2.94 = (79.56,\ 85.44) \text{ micrograms.}$$

Interpretation: the procedure "compute $\overline{X} \pm 2.94$" produces an interval that covers the true μ for 95% of all samples; the confidence is a property of the long-run method, not a probability statement about this one realized interval (which either contains μ or does not).

B2. With σ unknown, replace σ by S; the pivot $\dfrac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$ now follows a t distribution with $n - 1 = 9$ degrees of freedom because the extra randomness in S makes the standardized mean heavier-tailed than $\text{N}(0,1)$. The interval is $\overline{X} \pm t_{\alpha/2,\,n-1}\,S/\sqrt{n}$. For 95% with 9 d.f. use $t_{0.025,9} = 2.262$. The margin of error is

$$t_{0.025,9}\,\frac{S}{\sqrt{n}} = 2.262 \cdot \frac{2.1}{\sqrt{10}} = 2.262 \cdot \frac{2.1}{3.1623} = 2.262(0.6641) = 1.502.$$

The interval is

$$12.4 \pm 1.502 = (10.898,\ 13.902) \text{ mg.}$$

B3. The point estimate is

$$\widehat{p} = \frac{x}{n} = \frac{92}{400} = 0.23.$$

Its estimated standard error is

$$\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = \sqrt{\frac{0.23(0.77)}{400}} = \sqrt{\frac{0.1771}{400}} = \sqrt{0.00044275} = 0.02104.$$

With $z_{0.025} = 1.96$ the margin of error is $1.96(0.02104) = 0.04124$, so the 95% interval is

$$0.23 \pm 0.04124 = (0.1888,\ 0.2712).$$

(The large-sample normal approximation is reasonable here since $n\widehat{p} = 92$ and $n(1-\widehat{p}) = 308$ are both well above 10.)

B4. The margin of error for a proportion is $m = z_{\alpha/2}\sqrt{p(1-p)/n}$; solving for n gives $n = \dfrac{z_{\alpha/2}^2\, p(1-p)}{m^2}$. Use $z_{0.025} = 1.96$ and $m = 0.03$. **(a)** With no prior information, take the worst case $p(1-p) = 0.25$ (maximized at $p = \frac{1}{2}$):

$$n = \frac{(1.96)^2(0.25)}{(0.03)^2} = \frac{3.8416(0.25)}{0.0009} = \frac{0.9604}{0.0009} = 1067.1,$$

so round *up*: $n = 1068$. **(b)** Planning at $p = 0.20$ gives $p(1-p) = 0.16$:

$$n = \frac{(1.96)^2(0.16)}{(0.03)^2} = \frac{3.8416(0.16)}{0.0009} = \frac{0.614656}{0.0009} = 682.95,$$

so $n = 683$. The second number is smaller because $p(1-p)$ is largest at $p = \frac{1}{2}$; moving the planning value away from $\frac{1}{2}$ shrinks the variance the sample must overcome, so fewer observations meet the same margin.

B5. The pivot $\dfrac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$ with $n - 1 = 11$ gives the interval

$$\left(\frac{(n-1)S^2}{\chi^2_{\alpha/2,\,n-1}},\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\,n-1}} \right),$$

where $\chi^2_{p,k}$ denotes the value with area p to its right. For 90% ($\alpha = 0.10$) with 11 d.f. use

$$\chi^2_{0.05,11} = 19.675 \text{ (upper)}, \qquad \chi^2_{0.95,11} = 4.575 \text{ (lower)}.$$

The numerator is $(n-1)S^2 = 11(18.5) = 203.5$. Thus

$$\sigma^2 \in \left(\frac{203.5}{19.675},\ \frac{203.5}{4.575} \right) = (10.343,\ 44.483).$$

Taking square roots gives the interval for σ:

$$\sigma \in \left(\sqrt{10.343},\ \sqrt{44.483} \right) = (3.216,\ 6.670) \text{ mm}.$$

(The interval for σ^2 is asymmetric about $S^2 = 18.5$ because the χ^2 distribution is skewed.)

B6. For $\text{Exp}(\lambda)$, $\log f = \log\lambda - \lambda x$ gives $\ell'(\lambda) = \frac{n}{\lambda} - \sum x_i = 0$, so the MLE is the reciprocal of the sample mean. **(a)**

$$\widehat{\lambda} = \frac{1}{\overline{X}} = \frac{1}{4.0} = 0.25 \text{ per minute}.$$

The asymptotic variance of the MLE is $1/[n\mathcal{I}(\lambda)] = \lambda^2/n$, so the standard error, evaluated at $\widehat{\lambda}$, is

$$\text{SD}(\widehat{\lambda}) \approx \frac{\widehat{\lambda}}{\sqrt{n}} = \frac{0.25}{\sqrt{50}} = \frac{0.25}{7.0711} = 0.035355.$$

(b) By asymptotic normality $\widehat{\lambda} \approx \text{N}\big(\lambda,\, \lambda^2/n\big)$, so the Wald 95% interval is $\widehat{\lambda} \pm z_{0.025}\,\widehat{\lambda}/\sqrt{n}$ with $z_{0.025} = 1.96$:

$$0.25 \pm 1.96(0.035355) = 0.25 \pm 0.069296 = (0.1807,\ 0.3193) \text{ per minute}.$$

B7. **(a)** Each X_i is exponential with mean θ, i.e. $\text{Gamma}(1, \theta)$, so $X_i/\theta \sim \text{Gamma}(1,1) = \text{Exp}(1)$ and $2X_i/\theta \sim \text{Gamma}(1,2) = \chi^2_2$. Sums of independent χ^2 add their degrees of freedom, so

$$Q = \frac{2\sum_{i=1}^{n} X_i}{\theta} = \sum_{i=1}^{n} \frac{2X_i}{\theta} \sim \chi^2_{2n} = \chi^2_{16}.$$

Because Q is a pivot (its distribution does not depend on θ),

$$\mathbb{P}\left(\chi^2_{0.95,16} \le \frac{2n\overline{X}}{\theta} \le \chi^2_{0.05,16}\right) = 0.90,$$

where $\chi^2_{p,k}$ has area p to its right. For 16 d.f. use

$$\chi^2_{0.95,16} = 7.962, \qquad \chi^2_{0.05,16} = 26.296.$$

Inverting the inequality for θ (and noting it reverses, since θ is in the denominator) gives

$$\frac{2n\overline{X}}{\chi^2_{0.05,16}} \le \theta \le \frac{2n\overline{X}}{\chi^2_{0.95,16}}.$$

Here $2n\overline{X} = 2(8)(210) = 3360$, so

$$\theta \in \left(\frac{3360}{26.296},\ \frac{3360}{7.962}\right) = (127.78,\ 422.02) \text{ hours}.$$

(b) Since $\theta = 2n\overline{X}/Q$ is *decreasing* in Q, a lower bound on θ corresponds to an upper bound on Q: the event $\{\theta \ge L\}$ is the event $\{Q \le 2n\overline{X}/L\}$. We therefore put all 5% in the upper tail of Q, using $\chi^2_{0.05,16} = 26.296$ (the 95th percentile, area 0.05 to its right), so that $\mathbb{P}\big(Q \le \chi^2_{0.05,16}\big) = 0.95$. Setting $2n\overline{X}/L = \chi^2_{0.05,16}$,

$$L = \frac{2n\overline{X}}{\chi^2_{0.05,16}} = \frac{3360}{26.296} = 127.78 \text{ hours},$$

i.e. with 95% confidence the mean lifetime is at least 127.8 hours. (This matches the lower endpoint of the two-sided 90% interval, because both place 0.05 in the upper tail of Q.)

C. Unbiasedness and efficiency

C1. **(a)** By linearity, $\mathbb{E}(\widehat{\mu}) = a\,\mathbb{E}(T_1) + b\,\mathbb{E}(T_2) = (a+b)\mu$. This equals μ for every μ iff

$$a + b = 1.$$

(b) The simple average has $a = b = \frac{1}{2}$, so $a + b = 1$ and it is unbiased. Since T_1, T_2 are independent,

$$\mathrm{Var}\left(\tfrac{1}{2}T_1 + \tfrac{1}{2}T_2\right) = \tfrac{1}{4}\,\mathrm{Var}(T_1) + \tfrac{1}{4}\,\mathrm{Var}(T_2) = \tfrac{1}{4}(0.5) + \tfrac{1}{4}(1.2) = \tfrac{1}{4}(1.7) = 0.425.$$

(The simple average is unbiased but not the best linear unbiased combination; inverse-variance weighting, used in C7, does better.)

C2. **(a)** For Poisson(λ), $\mathbb{E}(X_i) = \lambda$ and $\mathrm{Var}(X_i) = \lambda$. Then

$$\mathbb{E}(\overline{X}) = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}(X_i) = \frac{n\lambda}{n} = \lambda, \qquad \mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum_{i=1}^{n}\mathrm{Var}(X_i) = \frac{n\lambda}{n^2} = \frac{\lambda}{n}.$$

So $\overline{X}$ is unbiased with variance λ/n; its bias is 0.

(b) X_1 is unbiased ($\mathbb{E}(X_1) = \lambda$) with $\mathrm{Var}(X_1) = \lambda$. The relative efficiency of X_1 with respect to $\overline{X}$ is the ratio of variances (smaller is better, so we put the better estimator's variance on top):

$$\text{eff}(X_1 \text{ rel. to } \overline{X}) = \frac{\mathrm{Var}(\overline{X})}{\mathrm{Var}(X_1)} = \frac{\lambda/n}{\lambda} = \frac{1}{n}.$$

For $n = 8$ this is $1/8 = 0.125$: a single observation has 8 times the variance of the sample mean, so it carries only 12.5% of the precision per estimate.

C3. Both estimators are weighted sums whose weights sum to 1: $\frac{1}{3}+\frac{1}{3}+\frac{1}{3}=1$ and $\frac{1}{4}+\frac{1}{2}+\frac{1}{4}=1$. Hence $\mathbb{E}(\widehat{\mu}_1)=\mathbb{E}(\widehat{\mu}_2)=\mu$, so both are unbiased.

By independence, $\mathrm{Var}\left(\sum a_i X_i\right)=\sigma^2\sum a_i^2$. Thus

$$\mathrm{Var}(\widehat{\mu}_1)=\sigma^2\Big(\tfrac{1}{9}+\tfrac{1}{9}+\tfrac{1}{9}\Big)=\frac{\sigma^2}{3},\qquad \mathrm{Var}(\widehat{\mu}_2)=\sigma^2\Big(\tfrac{1}{16}+\tfrac{1}{4}+\tfrac{1}{16}\Big)=\sigma^2\cdot\frac{6}{16}=\frac{3\sigma^2}{8}.$$

The efficiency of $\widehat{\mu}_2$ relative to $\widehat{\mu}_1$ is

$$\mathrm{eff}(\widehat{\mu}_2 \text{ rel. to } \widehat{\mu}_1)=\frac{\mathrm{Var}(\widehat{\mu}_1)}{\mathrm{Var}(\widehat{\mu}_2)}=\frac{\sigma^2/3}{3\sigma^2/8}=\frac{8}{9}\approx 0.889.$$

Since this is less than 1, $\widehat{\mu}_2$ is less efficient: $\widehat{\mu}_1=\overline{X}$ is the better estimator. (Indeed equal weights minimize $\sum a_i^2$ subject to $\sum a_i=1$, so $\overline{X}$ is the minimum-variance estimator in this linear unbiased class.)

C4. **(a)** A standard identity gives $\mathbb{E}\big[\sum_{i=1}^n (X_i-\overline{X})^2\big]=(n-1)\sigma^2$. Therefore

$$\mathbb{E}(\widehat{\sigma}^2)=\frac{1}{n}\,\mathbb{E}\Big[\sum_{i=1}^n (X_i-\overline{X})^2\Big]=\frac{n-1}{n}\,\sigma^2,\qquad \mathrm{Bias}(\widehat{\sigma}^2)=\frac{n-1}{n}\sigma^2-\sigma^2=-\frac{\sigma^2}{n}.$$

So $\widehat{\sigma}^2$ underestimates σ^2 on average.

(b) Because $\mathbb{E}\big[\sum (X_i-\overline{X})^2\big]=(n-1)\sigma^2$, dividing the sum of squares by $n-1$ rather than n exactly cancels the factor and yields $\mathbb{E}(S^2)=\sigma^2$; the $n-1$ corrects for the one degree of freedom used in estimating μ by $\overline{X}$. For $n=5$, $\sigma^2=20$,

$$\mathrm{Bias}(\widehat{\sigma}^2)=-\frac{20}{5}=-4.$$

C5. Both estimators have weights/normalizations chosen to make them unbiased.

(a) For Uniform$(0,\theta)$, $\mathbb{E}(X_i)=\theta/2$, so $\mathbb{E}(\widehat{\theta}_1)=2\,\mathbb{E}(\overline{X})=2\cdot\frac{\theta}{2}=\theta$. And $\mathbb{E}(\widehat{\theta}_2)=\frac{n+1}{n}\,\mathbb{E}(M)=\frac{n+1}{n}\cdot\frac{n}{n+1}\theta=\theta$. Both unbiased.

(b) Variances. Since $\mathrm{Var}(X_i)=\theta^2/12$,

$$\mathrm{Var}(\widehat{\theta}_1)=4\ \mathrm{Var}(\overline{X})=4\cdot\frac{\theta^2/12}{n}=\frac{\theta^2}{3n}.$$

For the maximum,

$$\mathrm{Var}(\widehat{\theta}_2)=\Big(\frac{n+1}{n}\Big)^2\mathrm{Var}(M)=\frac{(n+1)^2}{n^2}\cdot\frac{n}{(n+1)^2(n+2)}\theta^2=\frac{\theta^2}{n(n+2)}.$$

(c) Efficiency of $\widehat{\theta}_2$ relative to $\widehat{\theta}_1$:

$$\mathrm{eff}(\widehat{\theta}_2 \text{ rel. to } \widehat{\theta}_1)=\frac{\mathrm{Var}(\widehat{\theta}_1)}{\mathrm{Var}(\widehat{\theta}_2)}=\frac{\theta^2/(3n)}{\theta^2/[n(n+2)]}=\frac{n(n+2)}{3n}=\frac{n+2}{3}.$$

For $n=10$ this is $12/3=4$: the max-based estimator $\widehat{\theta}_2$ has one-quarter the variance of $2\overline{X}$, because $\max_i X_i$ uses the boundary information that the mean throws away. (Both are unbiased, so this is also the ratio of their MSEs.)

C6. Write $\widehat{\sigma}_c^2=cW$ with $\mathbb{E}(W)=(n-1)\sigma^2$ and $\mathrm{Var}(W)=2(n-1)\sigma^4$.

(a) $\mathrm{MSE}(cW)=\mathrm{Var}(cW)+\big[\mathbb{E}(cW)-\sigma^2\big]^2$. Now

$$\mathbb{E}(W^2)=\mathrm{Var}(W)+[\mathbb{E}(W)]^2=2(n-1)\sigma^4+(n-1)^2\sigma^4=(n-1)(n+1)\sigma^4.$$

Hence, expanding $\mathrm{MSE}(cW) = \mathbb{E}\big[(cW-\sigma^2)^2\big] = c^2\mathbb{E}(W^2) - 2c\sigma^2\mathbb{E}(W) + \sigma^4$,

$$\mathrm{MSE}(c) = \big[(n-1)(n+1)c^2 - 2(n-1)c + 1\big]\sigma^4.$$

(b) Differentiate in c and set to zero:

$$\frac{\mathrm{d}}{\mathrm{d}c}\,\mathrm{MSE}(c) = \big[2(n-1)(n+1)c - 2(n-1)\big]\sigma^4 = 0 \quad\Longrightarrow\quad c^\star = \frac{1}{n+1}.$$

The second derivative $2(n-1)(n+1)\sigma^4 > 0$, so this is a minimum. Since $\frac{1}{n+1} < \frac{1}{n} < \frac{1}{n-1}$, the MSE-optimal constant is strictly smaller than both the MLE choice $1/n$ and the unbiased choice $1/(n-1)$: shrinking the sum of squares more than "unbiased" would trades a little bias for a larger reduction in variance.

(c) Substituting each c into $\mathrm{MSE}(c)/\sigma^4 = (n-1)(n+1)c^2 - 2(n-1)c + 1$:

$$\begin{aligned} S^2\ \big(c=\tfrac{1}{n-1}\big): &\quad \mathrm{MSE} = \frac{2}{n-1}\sigma^4, \\ \mathrm{MLE}\ \big(c=\tfrac{1}{n}\big): &\quad \mathrm{MSE} = \frac{2n-1}{n^2}\sigma^4, \\ \mathrm{optimal}\ \big(c=\tfrac{1}{n+1}\big): &\quad \mathrm{MSE} = \frac{2}{n+1}\sigma^4. \end{aligned}$$

For $n = 10$, $\sigma^2 = 1$ (so $\sigma^4 = 1$):

$$\mathrm{MSE}(S^2) = \frac{2}{9} = 0.2222, \qquad \mathrm{MSE}(\mathrm{MLE}) = \frac{19}{100} = 0.1900, \qquad \mathrm{MSE}(\mathrm{opt}) = \frac{2}{11} = 0.1818.$$

The biased $c = 1/(n+1)$ estimator has the smallest MSE, beating both the unbiased S^2 and the MLE – a clean instance of the bias-variance tradeoff.

C7. **(a)** $\mathbb{E}(\widehat{\theta}) = \sum_i w_i\,\mathbb{E}(T_i) = \theta\sum_i w_i = \theta$ whenever $\sum_i w_i = 1$, so every such pooled estimator is unbiased regardless of the particular weights.

(b) By independence $\mathrm{Var}(\widehat{\theta}) = \sum_i w_i^2\sigma_i^2$ with $\sigma_1^2 = 4, \sigma_2^2 = 9, \sigma_3^2 = 36$. Minimize $\sum_i w_i^2\sigma_i^2$ subject to $\sum_i w_i = 1$. With a Lagrange multiplier λ, set $\frac{\partial}{\partial w_i}\big(\sum w_i^2\sigma_i^2 - \lambda(\sum w_i - 1)\big) = 2w_i\sigma_i^2 - \lambda = 0$, giving $w_i \propto 1/\sigma_i^2$. Normalizing,

$$w_i = \frac{1/\sigma_i^2}{\sum_j 1/\sigma_j^2}.$$

Here $\sum_j 1/\sigma_j^2 = \frac{1}{4} + \frac{1}{9} + \frac{1}{36} = \frac{9+4+1}{36} = \frac{14}{36} = \frac{7}{18}$, so

$$w_1 = \frac{1/4}{7/18} = \frac{9}{14}, \qquad w_2 = \frac{1/9}{7/18} = \frac{4}{14} = \frac{2}{7}, \qquad w_3 = \frac{1/36}{7/18} = \frac{1}{14},$$

which sum to 1. The minimum variance is

$$\mathrm{Var}(\widehat{\theta}_{\mathrm{opt}}) = \frac{1}{\sum_j 1/\sigma_j^2} = \frac{18}{7} \approx 2.571.$$

(c) The simple average $\frac{1}{3}(T_1+T_2+T_3)$ has variance

$$\mathrm{Var} = \frac{1}{9}(\sigma_1^2+\sigma_2^2+\sigma_3^2) = \frac{4+9+36}{9} = \frac{49}{9} \approx 5.444.$$

Its efficiency relative to the optimal pooling is

$$\mathrm{eff}(\text{simple avg rel. to opt}) = \frac{\mathrm{Var}(\widehat{\theta}_{\mathrm{opt}})}{\mathrm{Var}(\text{simple avg})} = \frac{18/7}{49/9} = \frac{162}{343} \approx 0.472,$$

equivalently the optimal estimator is about $\frac{49/9}{18/7} = \frac{343}{162} \approx 2.12$ times as efficient. Down-weighting the noisy T_3 (which has $9\times$ the variance of T_1) more than halves the variance of the pooled estimate.

D. Fisher information and the Cramer-Rao bound

D1. For one $\mathrm{Bin}(1,p)$ observation, $f(x;p) = p^x(1-p)^{1-x}$, so $\log f = x\log p + (1-x)\log(1-p)$ and

$$\frac{\partial}{\partial p}\log f = \frac{x}{p} - \frac{1-x}{1-p}, \qquad \frac{\partial^2}{\partial p^2}\log f = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}.$$

(a) Taking $-\mathbb{E}$ with $\mathbb{E}(X) = p$:

$$\mathcal{I}(p) = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

(b) The Cramer-Rao bound for an unbiased estimator of p is

$$\mathrm{Var}(\widehat{p}) \geq \frac{1}{n\mathcal{I}(p)} = \frac{p(1-p)}{n}.$$

(c) $\widehat{p} = \overline{X} = \frac{1}{n}\sum X_i$ with $\mathrm{Var}(X_i) = p(1-p)$ gives $\mathrm{Var}(\widehat{p}) = \frac{p(1-p)}{n}$, which equals the bound exactly. Hence $\overline{X}$ is **efficient**.

D2. For $\mathrm{Poisson}(\lambda)$, $f(x;\lambda) = e^{-\lambda}\lambda^x/x!$, so $\log f = -\lambda + x\log\lambda - \log x!$ and

$$\frac{\partial}{\partial \lambda}\log f = -1 + \frac{x}{\lambda}, \qquad \frac{\partial^2}{\partial \lambda^2}\log f = -\frac{x}{\lambda^2}.$$

(a) With $\mathbb{E}(X) = \lambda$,

$$\mathcal{I}(\lambda) = -\mathbb{E}\left[-\frac{X}{\lambda^2}\right] = \frac{\mathbb{E}(X)}{\lambda^2} = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}.$$

(b) The bound is $\mathrm{Var}(\widehat{\lambda}) \geq \dfrac{1}{n\mathcal{I}(\lambda)} = \dfrac{\lambda}{n}$. Since $\widehat{\lambda} = \overline{X}$ is unbiased with $\mathrm{Var}(\overline{X}) = \mathrm{Var}(X)/n = \lambda/n$, it attains the bound and is efficient. **(c)** Replacing λ by $\overline{x} = 2.5$ at $n = 40$:

$$\frac{\overline{x}}{n} = \frac{2.5}{40} = 0.0625.$$

D3. For $\mathrm{Exp}(\lambda)$, $\log f = \log\lambda - \lambda x$, so

$$\frac{\partial}{\partial \lambda}\log f = \frac{1}{\lambda} - x, \qquad \frac{\partial^2}{\partial \lambda^2}\log f = -\frac{1}{\lambda^2}.$$

(a) This second derivative is constant in x, so

$$\mathcal{I}(\lambda) = -\mathbb{E}\left[-\frac{1}{\lambda^2}\right] = \frac{1}{\lambda^2}.$$

(b) The Cramer-Rao bound for an unbiased estimator of λ is

$$\mathrm{Var}(\widehat{\lambda}) \geq \frac{1}{n\mathcal{I}(\lambda)} = \frac{\lambda^2}{n}.$$

(c) The bound is a statement about *unbiased* estimators. Here $\mathbb{E}(1/\overline{X}) = \frac{n}{n-1}\lambda \neq \lambda$, so $\widehat{\lambda} = 1/\overline{X}$ is biased (it overshoots by the factor $n/(n-1)$). The Cramer-Rao inequality therefore does not apply to it as stated, so it cannot be used to certify $1/\overline{X}$ as efficient. (A bias-corrected version $\frac{n-1}{n}\cdot\frac{1}{\overline{X}}$ is unbiased and is the right object to compare to the bound.)

D4. For $\mathrm{N}(\mu,\sigma^2)$ with σ^2 known, $\log f = -\frac{1}{2}\log(2\pi\sigma^2) - \dfrac{(x-\mu)^2}{2\sigma^2}$, so

$$\frac{\partial}{\partial \mu}\log f = \frac{x-\mu}{\sigma^2}, \qquad \frac{\partial^2}{\partial \mu^2}\log f = -\frac{1}{\sigma^2}.$$

(a) The second derivative is constant, hence

$$\mathcal{I}(\mu) = -\mathbb{E}\left[-\frac{1}{\sigma^2}\right] = \frac{1}{\sigma^2}.$$

(b) The bound is $\mathrm{Var}(\widehat{\mu}) \geq \dfrac{1}{n\mathcal{I}(\mu)} = \dfrac{\sigma^2}{n}$. Since $\overline{X}$ is unbiased with $\mathrm{Var}(\overline{X}) = \sigma^2/n$, it attains the bound and is efficient. **(c)** With $\sigma^2 = 9$, $n = 25$:

$$\frac{\sigma^2}{n} = \frac{9}{25} = 0.36.$$

D5. Write $\theta = \sigma^2$. For $\mathrm{N}(\mu, \theta)$ with μ known, $\log f = -\frac{1}{2}\log(2\pi) - \frac{1}{2}\log\theta - \dfrac{(x-\mu)^2}{2\theta}$, so

$$\frac{\partial}{\partial\theta}\log f = -\frac{1}{2\theta} + \frac{(x-\mu)^2}{2\theta^2}, \qquad \frac{\partial^2}{\partial\theta^2}\log f = \frac{1}{2\theta^2} - \frac{(x-\mu)^2}{\theta^3}.$$

(a) Using $\mathbb{E}[(X-\mu)^2] = \theta$,

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{1}{2\theta^2} - \frac{(X-\mu)^2}{\theta^3}\right] = -\frac{1}{2\theta^2} + \frac{\theta}{\theta^3} = -\frac{1}{2\theta^2} + \frac{1}{\theta^2} = \frac{1}{2\theta^2} = \frac{1}{2\sigma^4}.$$

(b) The Cramer-Rao bound for an unbiased estimator of σ^2 is

$$\mathrm{Var}(\widehat{\sigma}^2) \geq \frac{1}{n\mathcal{I}(\sigma^2)} = \frac{2\sigma^4}{n}.$$

(c) Let $\widehat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \mu)^2$ and set $W = \sum(X_i-\mu)^2/\sigma^2 \sim \chi^2_n$, so $\widehat{\sigma}^2 = \frac{\sigma^2}{n}W$. Then

$$\mathbb{E}(\widehat{\sigma}^2) = \frac{\sigma^2}{n}\mathbb{E}(W) = \frac{\sigma^2}{n}\cdot n = \sigma^2 \quad \text{(unbiased)},$$

$$\mathrm{Var}(\widehat{\sigma}^2) = \frac{\sigma^4}{n^2}\mathrm{Var}(W) = \frac{\sigma^4}{n^2}\cdot 2n = \frac{2\sigma^4}{n},$$

using $\mathbb{E}(\chi^2_n) = n$ and $\mathrm{Var}(\chi^2_n) = 2n$. This equals the bound, so $\widehat{\sigma}^2$ is efficient.

D6. From D2, $\mathcal{I}(\lambda) = 1/\lambda$, and $g(\lambda) = e^{-\lambda}$ has $g'(\lambda) = -e^{-\lambda}$, so $[g'(\lambda)]^2 = e^{-2\lambda}$. **(a)** The transformation form of the bound gives, for any unbiased estimator of τ,

$$\mathrm{Var}(\widehat{\tau}) \geq \frac{[g'(\lambda)]^2}{n\mathcal{I}(\lambda)} = \frac{e^{-2\lambda}}{n\cdot(1/\lambda)} = \frac{\lambda\, e^{-2\lambda}}{n}.$$

(b) At $\lambda = 2$, $n = 50$, with $e^{-4} = 0.018316$:

$$\frac{\lambda e^{-2\lambda}}{n} = \frac{2\cdot e^{-4}}{50} = \frac{2(0.018316)}{50} = \frac{0.036631}{50} = 0.000733.$$

D7. Here $f(x;\theta) = \frac{1}{\theta}e^{-x/\theta}$ (exponential with mean θ), so $\log f = -\log\theta - \dfrac{x}{\theta}$ and

$$\frac{\partial}{\partial\theta}\log f = -\frac{1}{\theta} + \frac{x}{\theta^2}, \qquad \frac{\partial^2}{\partial\theta^2}\log f = \frac{1}{\theta^2} - \frac{2x}{\theta^3}.$$

(a) With $\mathbb{E}(X) = \theta$,

$$\mathcal{I}(\theta) = -\mathbb{E}\left[\frac{1}{\theta^2} - \frac{2X}{\theta^3}\right] = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = -\frac{1}{\theta^2} + \frac{2}{\theta^2} = \frac{1}{\theta^2},$$

so the Cramer-Rao bound is $\mathrm{Var}(\widehat{\theta}) \geq \dfrac{1}{n\mathcal{I}(\theta)} = \dfrac{\theta^2}{n}$. **(b)** For the mean-parametrised exponential, $\mathbb{E}(X) = \theta$ and $\mathrm{Var}(X) = \theta^2$. Thus $\overline{X}$ is unbiased and

$$\mathrm{Var}(\overline{X}) = \frac{\mathrm{Var}(X)}{n} = \frac{\theta^2}{n},$$

which equals the bound; $\overline{X}$ is efficient. **(c)** $\widehat{\theta}_1 = X_1$ has $\mathbb{E}(X_1) = \theta$ (unbiased) and $\mathrm{Var}(X_1) = \theta^2$. The efficiency of $\widehat{\theta}_1$ relative to $\overline{X}$ is

$$\mathrm{eff} = \frac{\mathrm{Var}(\overline{X})}{\mathrm{Var}(\widehat{\theta}_1)} = \frac{\theta^2/n}{\theta^2} = \frac{1}{n},$$

so for $n = 20$ it is $1/20 = 0.05$: using one bearing is only 5% as efficient as using all twenty. The estimated variance floor with $\overline{x} = 5$, $n = 20$ is

$$\frac{\overline{x}^2}{n} = \frac{25}{20} = 1.25.$$

E. Sufficient statistics

E1. Each X_i has mass $p^{x_i}(1-p)^{1-x_i}$ for $x_i \in \{0, 1\}$, so the joint mass of the iid sample is

$$f(x_1, \ldots, x_6; p) = \prod_{i=1}^{6} p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{6-\sum x_i} = p^t(1-p)^{6-t}, \qquad t = \sum_{i=1}^{6} x_i.$$

This already depends on the data only through t. To match the factorisation theorem $f = g(T(x), p)\, h(x)$, take

$$g(t, p) = p^t(1-p)^{6-t}, \qquad h(x_1, \ldots, x_6) = 1.$$

Since g depends on the sample only through t and h does not involve p, the statistic $T = \sum_{i=1}^{6} X_i$ is **sufficient** for p. For the observed data $0, 1, 0, 0, 1, 0$ the observed value is

$$t = 0 + 1 + 0 + 0 + 1 + 0 = 2.$$

E2. For iid Poisson(λ) the joint mass is

$$f(x_1, \ldots, x_n; \lambda) = \prod_{i=1}^{n} \frac{e^{-\lambda}\lambda^{x_i}}{x_i!} = e^{-n\lambda}\, \lambda^{\sum x_i} \frac{1}{\prod_i x_i!}, \qquad x_i \in \{0, 1, 2, \ldots\}.$$

With $t = \sum_{i=1}^{n} x_i$ this factors as $f = g(t, \lambda)\, h(x)$ with

$$g(t, \lambda) = e^{-n\lambda}\, \lambda^t, \qquad h(x_1, \ldots, x_n) = \frac{1}{\prod_{i=1}^{n} x_i!}.$$

Here g depends on the data only through t, and h is free of λ, so by the **factorisation theorem** $T = \sum_{i=1}^{n} X_i$ is sufficient for λ.

The sample mean $\overline{X} = T/n$ is a one-to-one function of T (for fixed n), and any one-to-one function of a sufficient statistic is itself sufficient. Hence $\overline{X}$ is also sufficient for λ.

E3. (a) The joint density of the iid Exp(λ) sample is

$$f(x_1, \ldots, x_n; \lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \lambda^n\, e^{-\lambda \sum_{i=1}^{n} x_i}, \qquad x_i > 0.$$

With $t = \sum_{i=1}^{n} x_i$, this is $g(t, \lambda)\, h(x)$ with

$$g(t, \lambda) = \lambda^n e^{-\lambda t}, \qquad h(x_1, \ldots, x_n) = \mathbf{1}\{x_i > 0 \text{ for all } i\}.$$

The factor g involves the data only through t, so $T = \sum_{i=1}^n X_i$ is sufficient for λ. (Equivalently $\overline{X} = T/n$ is sufficient.)

(b) The MLE is

$$\ell(\lambda) = n \ln \lambda - \lambda \sum_i x_i, \qquad \ell'(\lambda) = \frac{n}{\lambda} - \sum_i x_i = 0 \implies \widehat{\lambda} = \frac{n}{\sum_i X_i} = \frac{n}{T}.$$

Thus $\widehat{\lambda} = n/T$ is a function of the sufficient statistic T. In general the MLE maximises the likelihood, which by factorisation depends on the data only through T (the factor $h(x)$ does not affect the maximiser in λ); hence any MLE is automatically a function of any sufficient statistic.

E4. The joint density of the iid Uniform$(0, \theta)$ sample is

$$f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^n \frac{1}{\theta} \, \mathbf{1}\{0 < x_i < \theta\} = \frac{1}{\theta^n} \, \mathbf{1}\{0 < x_i \text{ for all } i\} \, \mathbf{1}\{\max_i x_i < \theta\}.$$

The key step is that $\prod_i \mathbf{1}\{x_i < \theta\} = \mathbf{1}\{\max_i x_i < \theta\}$: all observations are below θ exactly when the largest one is. Writing $t = \max_i x_i = x_{(n)}$, this factors as $g(t, \theta)\, h(x)$ with

$$g(t, \theta) = \frac{1}{\theta^n} \, \mathbf{1}\{t < \theta\}, \qquad h(x_1, \ldots, x_n) = \mathbf{1}\{x_i > 0 \text{ for all } i\}.$$

So by factorisation $T = X_{(n)} = \max_i X_i$ is sufficient for θ.

Why no derivative. The parameter θ enters only through the *support* (the range of x), not smoothly through the height of a density on a fixed support. The log-likelihood is $-n \ln \theta$ for $\theta > \max_i x_i$ and $-\infty$ otherwise; it is increasing as θ decreases toward $\max_i x_i$ and has no interior stationary point. Setting $\ell'(\theta) = -n/\theta$ to zero gives no solution, so the sufficient statistic and the MLE come from the boundary $\theta \geq \max_i x_i$, not from a derivative.

Why $\sum X_i$ is not sufficient. The likelihood does not depend on the data through $\sum x_i$ at all: it depends only on $\max_i x_i$ (through the indicator $\mathbf{1}\{\max_i x_i < \theta\}$). Two samples with the same sum but different maxima give different likelihood functions of θ, so $\sum X_i$ does not capture all the information about θ and is not sufficient.

E5. For iid N(μ, σ^2) the joint density is

$$f(x; \mu, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi}\,\sigma} \exp\!\left(-\frac{(x_i - \mu)^2}{2\sigma^2}\right) = (2\pi\sigma^2)^{-n/2} \exp\!\left(-\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2\right).$$

Expand the sum in the exponent:

$$\sum_{i=1}^n (x_i - \mu)^2 = \sum_{i=1}^n x_i^2 - 2\mu \sum_{i=1}^n x_i + n\mu^2.$$

Let $T_1 = \sum_i x_i$ and $T_2 = \sum_i x_i^2$. Then

$$f(x; \mu, \sigma^2) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\!\left(-\frac{1}{2\sigma^2}\big(T_2 - 2\mu T_1 + n\mu^2\big)\right)}_{g(T_1, T_2; \mu, \sigma^2)} \cdot \underbrace{1}_{h(x)}.$$

The first factor depends on the data only through (T_1, T_2) and the second is free of the parameters, so by the factorisation theorem

$$T = \Big(\textstyle\sum_i X_i, \ \sum_i X_i^2\Big)$$

is jointly sufficient for (μ, σ^2).

Same information for $(\overline{X}, S^2)$. The map from (T_1, T_2) to $(\overline{X}, S^2)$ is one-to-one (for fixed n):

$$\overline{X} = \frac{T_1}{n}, \qquad S^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2 = \frac{1}{n-1}\Big(T_2 - \tfrac{1}{n}T_1^2\Big),$$

and conversely $T_1 = n\overline{X}$, $T_2 = (n-1)S^2 + n\overline{X}^2$. A one-to-one function of a (jointly) sufficient statistic is again sufficient, so $(\overline{X}, S^2)$ is jointly sufficient for (μ, σ^2) as well.

E6. **(a)** The joint mass of the iid Geom(p) sample is

$$f(x_1, \dots, x_n; p) = \prod_{i=1}^{n} (1-p)^{x_i - 1} p = p^n (1-p)^{\sum_i x_i - n}, \qquad x_i \in \{1, 2, 3, \dots\}.$$

With $t = \sum_{i=1}^n x_i$ this factors as $g(t, p)\, h(x)$ with

$$g(t, p) = p^n (1-p)^{\,t-n}, \qquad h(x_1, \dots, x_n) = \mathbf{1}\{x_i \in \{1, 2, \dots\} \text{ for all } i\}.$$

Hence $T = \sum_{i=1}^n X_i$ is sufficient for p.

(b) For $n = 4$ with counts $3, 1, 2, 2$,

$$t = 3 + 1 + 2 + 2 = 8, \qquad \widehat{p} = \frac{n}{\sum_i X_i} = \frac{4}{8} = \frac{1}{2}.$$

E7. **(a)** For $X_1 \sim \text{Poisson}(\lambda)$ we have $\mathbb{E}(X_1) = \lambda$, so X_1 is unbiased for λ. Its variance is the Poisson variance,

$$\text{Var}(X_1) = \lambda.$$

(b) By symmetry of the iid sample, conditional on $T = \sum_{i=1}^n X_i = t$ the vector $(X_1, \dots, X_n)$ is Multinomial$(t; 1/n, \dots, 1/n)$, so each coordinate is marginally Bin$(t, 1/n)$ and

$$\mathbb{E}[X_1 \mid T = t] = t \cdot \frac{1}{n} = \frac{t}{n}.$$

Therefore the Rao–Blackwellised estimator is

$$\widehat{\lambda}^{\,*} = \mathbb{E}[X_1 \mid T] = \frac{T}{n} = \overline{X}.$$

Conditioning the crude estimator X_1 on the sufficient statistic T returns the sample mean.

(c) Since $T = \sum_i X_i \sim \text{Poisson}(n\lambda)$, we have $\text{Var}(T) = n\lambda$ and

$$\text{Var}(\widehat{\lambda}^{\,*}) = \text{Var}\left(\frac{T}{n}\right) = \frac{1}{n^2}\,\text{Var}(T) = \frac{n\lambda}{n^2} = \frac{\lambda}{n}.$$

Also $\mathbb{E}(\widehat{\lambda}^{\,*}) = \mathbb{E}(T)/n = n\lambda/n = \lambda$, so $\widehat{\lambda}^{\,*}$ is unbiased. Comparing variances,

$$\text{Var}(\widehat{\lambda}^{\,*}) = \frac{\lambda}{n} \;\le\; \lambda = \text{Var}(X_1) \qquad \text{(strictly smaller for } n \ge 2),$$

which confirms the **Rao–Blackwell** improvement: conditioning an unbiased estimator on a sufficient statistic preserves unbiasedness and cannot increase the variance.

F. Consistency

F1. Definition and the WLLN. The estimator $\overline{X}_n$ is **consistent** for μ if it converges in probability to μ: for every $\varepsilon > 0$,

$$\lim_{n\to\infty} \mathbb{P}\big(\big|\overline{X}_n - \mu\big| \geq \varepsilon \big) = 0.$$

The Weak Law of Large Numbers states exactly this for the sample mean of iid variables with a finite mean. Finiteness of σ^2 is sufficient because it lets us bound the tail probability by Chebyshev:

$$\mathbb{P}\big(\big|\overline{X}_n - \mu\big| \geq \varepsilon\big) \leq \frac{\mathrm{Var}(\overline{X}_n)}{\varepsilon^2} = \frac{\sigma^2/n}{\varepsilon^2} = \frac{\sigma^2}{n\varepsilon^2} \xrightarrow[n\to\infty]{} 0.$$

It is not necessary: the WLLN holds whenever the mean is finite (e.g. a distribution with finite mean but infinite variance still has $\overline{X}_n$ consistent), so $\sigma^2 < \infty$ is a convenient sufficient condition, not a requirement.

F2. Sample proportion (Bernoulli). Here $X_i \sim \mathrm{Bin}(1, p)$ iid, so $\mathbb{E}(X_i) = p$ and $\mathrm{Var}(X_i) = p(1-p)$. Since $\widehat{p}_n = \overline{X}_n = \frac{1}{n}\sum X_i$,

$$\mathbb{E}(\widehat{p}_n) = p \implies \mathrm{Bias}(\widehat{p}_n) = 0, \qquad \mathrm{Var}(\widehat{p}_n) = \frac{p(1-p)}{n}.$$

By the MSE criterion,

$$\mathrm{MSE}(\widehat{p}_n) = \mathrm{Var}(\widehat{p}_n) + [\mathrm{Bias}(\widehat{p}_n)]^2 = \frac{p(1-p)}{n} + 0 \xrightarrow[n\to\infty]{} 0,$$

so $\widehat{p}_n \to p$ in probability and $\widehat{p}_n$ is consistent for p. Because the function $p \mapsto p(1-p)$ is maximised at $p = \frac{1}{2}$ with value $\frac{1}{4}$, for every $p \in (0, 1)$

$$\mathrm{Var}(\widehat{p}_n) = \frac{p(1-p)}{n} \leq \frac{1}{4n}.$$

This uniform bound makes the convergence rate independent of the unknown p.

F3. Exponential mean and rate. The service times $X_i \sim \mathrm{Exp}(\lambda)$ have $\mathbb{E}(X_i) = 1/\lambda = \tau$ and $\mathrm{Var}(X_i) = 1/\lambda^2 = \tau^2 < \infty$. Hence

$$\mathbb{E}(\overline{X}_n) = \tau, \qquad \mathrm{Var}(\overline{X}_n) = \frac{\tau^2}{n}, \qquad \mathrm{MSE}(\overline{X}_n) = \frac{\tau^2}{n} \xrightarrow[n\to\infty]{} 0,$$

so $\overline{X}_n$ is consistent for τ (equivalently, by the WLLN).

For the rate, write $\widehat{\lambda}_n = g(\overline{X}_n)$ with $g(t) = 1/t$, which is continuous at the point $t = \tau = 1/\lambda > 0$. The **continuous mapping theorem** (convergence in probability is preserved by a function continuous at the limit) gives

$$\overline{X}_n \xrightarrow{\ \mathbb{P}\ } \tau \implies \widehat{\lambda}_n = \frac{1}{\overline{X}_n} \xrightarrow{\ \mathbb{P}\ } \frac{1}{\tau} = \lambda,$$

so $\widehat{\lambda}_n$ is consistent for λ. (The estimator is biased for finite n by Jensen's inequality, but bias plays no role in this argument.)

F4. Maximum of a uniform sample. For $0 \leq x \leq \theta$, independence gives

$$F_{X_{(n)}}(x) = \mathbb{P}(X_{(n)} \leq x) = \prod_{i=1}^{n} \mathbb{P}(X_i \leq x) = \left(\frac{x}{\theta}\right)^n, \qquad f_{X_{(n)}}(x) = \frac{n\, x^{n-1}}{\theta^n}.$$

The moments are

$$\mathbb{E}(X_{(n)}) = \int_0^\theta x\, \frac{n x^{n-1}}{\theta^n}\, \mathrm{d}x = \frac{n}{n+1}\,\theta, \qquad \mathbb{E}(X_{(n)}^2) = \int_0^\theta x^2\, \frac{n x^{n-1}}{\theta^n}\, \mathrm{d}x = \frac{n}{n+2}\,\theta^2,$$

so

$$\mathrm{Var}(X_{(n)}) = \frac{n}{n+2}\theta^2 - \Big(\frac{n}{n+1}\Big)^2\theta^2 = \frac{n\,\theta^2}{(n+1)^2(n+2)}.$$

The bias is $\mathrm{Bias}(\widehat{\theta}_n) = \dfrac{n}{n+1}\theta - \theta = -\dfrac{\theta}{n+1}$, hence

$$\mathrm{MSE}(\widehat{\theta}_n) = \frac{n\,\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{\theta^2}{(n+1)^2}\cdot\frac{n+(n+2)}{n+2} = \frac{2\,\theta^2}{(n+1)(n+2)} \xrightarrow[n\to\infty]{} 0.$$

Therefore $\widehat{\theta}_n$ is consistent for θ. It is *not* unbiased (bias $= -\theta/(n+1) \neq 0$), though the bias vanishes as $n \to \infty$; the unbiased version is $\frac{n+1}{n}X_{(n)}$.

F5. Two estimators of a Poisson zero-probability. Let $X_i \sim \mathrm{Poisson}(\lambda)$ iid, so $\psi = \mathbb{P}(X_i = 0) = e^{-\lambda}$.

(a) Plug-in estimator. By the WLLN, $\overline{X}_n \xrightarrow{\mathbb{P}} \mathbb{E}(X_i) = \lambda$. The map $t \mapsto e^{-t}$ is continuous everywhere, so by the **continuous mapping theorem**

$$\widehat{\psi}_n^{(1)} = e^{-\overline{X}_n} \xrightarrow{\mathbb{P}} e^{-\lambda} = \psi,$$

giving consistency. (This is the general statement: if $\widehat{\theta}_n \xrightarrow{\mathbb{P}} \theta$ and g is continuous at θ, then $g(\widehat{\theta}_n) \xrightarrow{\mathbb{P}} g(\theta)$.)

(b) Empirical zero-frequency. Define $Y_i = \mathbb{1}\{X_i = 0\}$, which are iid Bernoulli with success probability $\mathbb{P}(X_i = 0) = e^{-\lambda} = \psi$. Then $\widehat{\psi}_n^{(2)} = \frac{1}{n}\sum Y_i$ is a sample proportion, so by F2 it is unbiased with $\mathrm{Var} = \psi(1-\psi)/n \to 0$; hence MSE $\to 0$ and $\widehat{\psi}_n^{(2)} \xrightarrow{\mathbb{P}} \psi$. Both estimators are consistent for ψ.

F6. Diagnosing consistency from bias and variance.

(i) The bias is $\mathrm{Bias}(\widehat{\theta}_n) = \mathbb{E}(\widehat{\theta}_n) - \theta = \dfrac{3}{n} \to 0$ and the variance is $\mathrm{Var}(\widehat{\theta}_n) = \dfrac{5\theta^2}{\sqrt{n}} \to 0$, so

$$\mathrm{MSE}(\widehat{\theta}_n) = \frac{5\theta^2}{\sqrt{n}} + \frac{9}{n^2} \xrightarrow[n\to\infty]{} 0,$$

and $\widehat{\theta}_n$ *is* consistent for θ.

(ii) For $\widetilde{\theta}_n$ the bias is 0 and $\mathrm{Var}(\widetilde{\theta}_n) = c/\ln n \to 0$ as $n \to \infty$ (since $\ln n \to \infty$), so MSE $\to 0$ and $\widetilde{\theta}_n$ *is* consistent. The convergence is extremely slow (only logarithmic), but the MSE criterion only requires the limit to be 0.

(iii) Take iid X_i with mean θ and variance $\sigma^2 > 0$, and let $\overline{\theta}_n = X_1$ for every n (use only the first observation). Then $\mathbb{E}(\overline{\theta}_n) = \theta$ for all n, so the bias is identically 0 and in particular $\to 0$: the estimator is asymptotically (indeed exactly) unbiased. But $\mathrm{Var}(\overline{\theta}_n) = \sigma^2$ does *not* shrink, so $\mathbb{P}(|\overline{\theta}_n - \theta| \geq \varepsilon)$ stays bounded away from 0 and there is no convergence in probability. Thus $\overline{\theta}_n$ is not consistent. The two notions differ because asymptotic unbiasedness only controls the *center* of the sampling distribution, whereas consistency also requires its *spread* (variance) to collapse to 0.

F7. Consistency of the two normal-variance estimators. Write $\widehat{\sigma}_n^2 = \frac{n-1}{n}S_n^2$, where $\mathbb{E}(S_n^2) = \sigma^2$ and $\mathrm{Var}(S_n^2) = \dfrac{2\sigma^4}{n-1}$.

Unbiased estimator S_n^2. Its bias is 0, and

$$\mathrm{MSE}(S_n^2) = \mathrm{Var}(S_n^2) = \frac{2\sigma^4}{n-1} \xrightarrow[n\to\infty]{} 0,$$

so $S_n^2 \xrightarrow{\mathbb{P}} \sigma^2$: consistent. S_n^2 is the unbiased one (the $n-1$ divisor).

MLE $\widehat{\sigma}_n^2$. Its mean is $\mathbb{E}(\widehat{\sigma}_n^2) = \dfrac{n-1}{n}\sigma^2$, so

$$\mathrm{Bias}(\widehat{\sigma}_n^2) = \frac{n-1}{n}\sigma^2 - \sigma^2 = -\frac{\sigma^2}{n} \xrightarrow[n\to\infty]{} 0,$$

and, since $\widehat{\sigma}_n^2 = \frac{n-1}{n} S_n^2$,

$$\operatorname{Var}(\widehat{\sigma}_n^2) = \Big(\frac{n-1}{n}\Big)^2 \operatorname{Var}(S_n^2) = \frac{(n-1)^2}{n^2} \cdot \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2} \xrightarrow[n\to\infty]{} 0.$$

Hence

$$\operatorname{MSE}(\widehat{\sigma}_n^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{\sigma^4}{n^2} = \frac{(2n-1)\sigma^4}{n^2} \xrightarrow[n\to\infty]{} 0,$$

so $\widehat{\sigma}_n^2 \xrightarrow{\mathbb{P}} \sigma^2$ as well. Both estimators are consistent for σ^2; they differ only by the asymptotically negligible factor $\frac{n-1}{n} \to 1$, with S_n^2 being the unbiased member of the pair.

G. Bayesian estimation

G1. The likelihood from 9 passes out of 12 Bernoulli trials is $L(p) \propto p^9(1-p)^3$. The prior is $\text{Beta}(1,1)$, i.e. $\pi(p) \propto p^0(1-p)^0 = 1$. Hence

$$\pi(p \mid \text{data}) \;\propto\; p^0(1-p)^0 \cdot p^9(1-p)^3 \;=\; p^9(1-p)^3,$$

which is the kernel of a $\text{Beta}(\alpha, \beta)$ with $\alpha = 10$, $\beta = 4$. So the posterior is

$$p \mid \text{data} \;\sim\; \text{Beta}(10, 4), \qquad \mathbb{E}(p \mid \text{data}) = \frac{\alpha}{\alpha+\beta} = \frac{10}{14} = \frac{5}{7} \approx 0.7143.$$

(Beta–Binomial conjugacy: $\text{Beta}(\alpha_0, \beta_0)$ prior with k successes in n trials gives $\text{Beta}(\alpha_0 + k,\ \beta_0 + n - k)$.)

G2. The Poisson likelihood for counts summing to $s = \sum x_i = 10$ is $L(\lambda) \propto e^{-n\lambda}\lambda^s = e^{-4\lambda}\lambda^{10}$. The prior is $\text{Gamma}(3, 2)$ with $\pi(\lambda) \propto \lambda^{3-1}e^{-2\lambda} = \lambda^2 e^{-2\lambda}$. Then

$$\pi(\lambda \mid \text{data}) \;\propto\; \lambda^2 e^{-2\lambda} \cdot \lambda^{10} e^{-4\lambda} \;=\; \lambda^{12}\, e^{-6\lambda},$$

the kernel of a Gamma with shape $a' = 13$ and rate $b' = 6$. Thus

$$\lambda \mid \text{data} \;\sim\; \text{Gamma}(13, 6), \qquad \mathbb{E}(\lambda \mid \text{data}) = \frac{a'}{b'} = \frac{13}{6} \approx 2.167.$$

(Gamma–Poisson conjugacy: a $\text{Gamma}(a, b)$ prior with data sum s over n points gives $\text{Gamma}(a + s,\ b + n)$.)

G3. Normal–Normal update with known sampling variance $\sigma^2 = 16$, prior variance $\tau^2 = 16$, prior mean $\mu_0 = 50$, and one reading $x = 58$. The posterior is $\text{N}(\mu_1, \sigma_1^2)$ with

$$\frac{1}{\sigma_1^2} = \frac{1}{\tau^2} + \frac{1}{\sigma^2} = \frac{1}{16} + \frac{1}{16} = \frac{2}{16} = \frac{1}{8} \implies \sigma_1^2 = 8,$$

$$\mu_1 = \sigma_1^2\Big(\frac{\mu_0}{\tau^2} + \frac{x}{\sigma^2}\Big) = 8\Big(\frac{50}{16} + \frac{58}{16}\Big) = 8 \cdot \frac{108}{16} = 8 \cdot 6.75 = 54.$$

So $\theta \mid x \sim \text{N}(54,\ 8)$. The posterior mean is a precision-weighted average of $\mu_0 = 50$ and $x = 58$; since the prior and the datum carry equal precision here ($\tau^2 = \sigma^2$), it sits exactly midway, at 54, hence strictly between 50 and 58.

G4. From G1 the posterior mean is $\dfrac{10}{14}$. Write the Beta–Binomial posterior mean with prior $\text{Beta}(\alpha_0, \beta_0) = \text{Beta}(1, 1)$ and $k = 9$ successes in $n = 12$:

$$\frac{\alpha_0 + k}{\alpha_0 + \beta_0 + n} = \frac{\alpha_0 + \beta_0}{\alpha_0 + \beta_0 + n} \cdot \frac{\alpha_0}{\alpha_0 + \beta_0} + \frac{n}{\alpha_0 + \beta_0 + n} \cdot \frac{k}{n}.$$

Here $\alpha_0 + \beta_0 = 2$ and $n = 12$, so the prior mean is $\frac{\alpha_0}{\alpha_0+\beta_0} = \frac{1}{2}$ and the sample proportion is $\frac{k}{n} = \frac{9}{12} = \frac{3}{4}$. The weights are

$$w_{\text{prior}} = \frac{2}{2+12} = \frac{1}{7}, \qquad w_{\text{data}} = \frac{12}{2+12} = \frac{6}{7}.$$

Check: $\frac{1}{7}\cdot\frac{1}{2}+\frac{6}{7}\cdot\frac{3}{4}=\frac{1}{14}+\frac{18}{28}=\frac{2}{28}+\frac{18}{28}=\frac{20}{28}=\frac{10}{14}$. Report the *posterior mean* $10/14\approx 0.714$: it uses both the prior information and all 12 observations, and is the optimal point estimate under squared-error loss.

G5. *(a)* Likelihood $L(\lambda)\propto e^{-n\lambda}\lambda^{s}$; prior $\pi(\lambda)\propto\lambda^{a-1}e^{-b\lambda}$. Multiplying,

$$\pi(\lambda\mid\text{data})\ \propto\ \lambda^{a-1}e^{-b\lambda}\cdot\lambda^{s}e^{-n\lambda}=\lambda^{(a+s)-1}\,e^{-(b+n)\lambda},$$

the kernel of Gamma$(a+s,\ b+n)$. *(b)* The posterior mean is

$$\frac{a+s}{b+n}=\frac{b}{b+n}\cdot\frac{a}{b}+\frac{n}{b+n}\cdot\frac{s}{n}=w\cdot\frac{a}{b}+(1-w)\,\bar{x},\qquad w=\frac{b}{b+n},$$

since $\bar{x}=s/n$ and $\frac{b}{b+n}\cdot\frac{a}{b}+\frac{n}{b+n}\cdot\frac{s}{n}=\frac{a}{b+n}+\frac{s}{b+n}=\frac{a+s}{b+n}$. *(c)* As $n\to\infty$, $w=\dfrac{b}{b+n}\to 0$, so the posterior mean $\to\bar{x}$: with enough data the likelihood dominates the prior, and the Bayes estimate converges to the sample mean (the MLE).

G6. *(a)* Up to constants in θ, the posterior exponent is

$$-\frac{(x-\theta)^2}{2\sigma^2}-\frac{(\theta-\mu_0)^2}{2\tau^2}=-\frac{1}{2}\Big[\Big(\tfrac{1}{\sigma^2}+\tfrac{1}{\tau^2}\Big)\theta^2-2\Big(\tfrac{x}{\sigma^2}+\tfrac{\mu_0}{\tau^2}\Big)\theta\Big]+\text{const.}$$

Let $A=\frac{1}{\sigma^2}+\frac{1}{\tau^2}$ and $B=\frac{x}{\sigma^2}+\frac{\mu_0}{\tau^2}$. Completing the square, $A\theta^2-2B\theta=A\big(\theta-\frac{B}{A}\big)^2-\frac{B^2}{A}$, so the posterior is normal with

$$\sigma_1^2=\frac{1}{A}=\left(\frac{1}{\tau^2}+\frac{1}{\sigma^2}\right)^{-1},\qquad \mu_1=\frac{B}{A}=\sigma_1^2\left(\frac{\mu_0}{\tau^2}+\frac{x}{\sigma^2}\right),$$

i.e. $\theta\mid x\sim\mathrm{N}(\mu_1,\sigma_1^2)$. *(b)* With $\sigma^2=16$, $\tau^2=16$, $\mu_0=50$, $x=58$: from G3, $\sigma_1^2=8$ and $\mu_1=54$, so $\sigma_1=\sqrt{8}=2.8284$. A 95% equal-tailed credible interval uses $z_{0.025}=1.96$:

$$\mu_1\pm z_{0.025}\,\sigma_1=54\pm 1.96(2.8284)=54\pm 5.544=(48.46,\ 59.54).$$

(Posterior probability 0.95 that θ lies in this interval, given the data and model.)

G7. *(a)* Prior Beta$(2,2)$ with $k=14$ successes in $n=20$ trials gives posterior Beta$(2+14,\ 2+(20-14))=$ Beta$(16,8)$, with mean

$$\frac{\alpha}{\alpha+\beta}=\frac{16}{24}=\frac{2}{3}\approx 0.6667.$$

(b) With $\alpha=16$, $\beta=8$, $\alpha+\beta=24$:

$$\text{Var}=\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}=\frac{16\cdot 8}{24^2\cdot 25}=\frac{128}{14{,}400}=0.008889,\quad \text{SD}=\sqrt{0.008889}=0.09428.$$

(c) Using $z_{0.025}=1.96$, the approximate central 95% credible interval is

$$\frac{2}{3}\pm 1.96(0.09428)=0.6667\pm 0.1848=(0.4819,\ 0.8515).$$

The normal approximation is reasonable because the posterior Beta$(16,8)$ has both shape parameters comfortably above 1 and a mean away from 0 and 1, so the density is unimodal and roughly symmetric.

Chapter 6

Hypothesis Testing

Hypothesis testing turns a question about an unknown parameter into a *decision*. We state two competing claims — a **null hypothesis** H_0 (the default, the "nothing interesting is happening" claim) and an **alternative** H_1 (what we would need evidence to believe) — then compute a **test statistic** from the data and ask whether its value is so extreme, *if* H_0 *were true*, that H_0 becomes untenable. Every test in this chapter is the same machine with different parts swapped in: choose the statistic whose null distribution you know (Z, t, χ^2), mark off the **rejection region** of values too extreme to credit, and reject H_0 exactly when the data land there.

Master guide: which test does the problem call for?

- First, **what parameter is in question, and what is the parent distribution?** That pair fixes the test statistic.
- Testing a **mean** μ with the population σ **known** (or n large and σ supplied)? $\Rightarrow$ the z**-test**: reject when $|(\overline{X} - \mu_0)/(\sigma/\sqrt{n})|$ is too large (§3).
- Testing a **mean** μ with σ **unknown**, estimated by S? $\Rightarrow$ the t**-test** with $n-1$ degrees of freedom (§4).
- Testing a **proportion** p from a count of successes? $\Rightarrow$ the **binomial / large-sample** z**-test** for p (§5).
- Testing a **variance** σ^2 of a normal population? $\Rightarrow$ the χ^2**-test** built from $(n-1)S^2/\sigma_0^2$ (§6).
- **One-sided or two-sided?** If H_1 points in one direction ($\mu > \mu_0$ or $\mu < \mu_0$) put all of α in one tail; if H_1 is "$\neq$", split α between two tails. The wording ("has increased", "differs from") tells you which.
- Asked for the **strength of evidence**, not a fixed accept/reject? $\Rightarrow$ report the p**-value** (§3, §7): reject at level α iff p-value $\leq \alpha$.
- Asked about the **chance of a wrong decision**, or to design the sample size? $\Rightarrow$ **Type I error** (α, rejecting a true H_0), **Type II error** (β, failing to reject a false H_0), and **power** $1-\beta$ (§2).
- Asked for the **best** test, or a test where no off-the-shelf statistic exists? $\Rightarrow$ the **generalized likelihood ratio test** (§7), the general recipe that the named tests are special cases of.

Keep straight: H_0 is what you assume while computing; you never "prove" H_0, you only *fail to reject* it. Significance α is chosen *before* seeing data; the p-value is computed *from* the data. A small p-value means the data are surprising under H_0 — it is not the probability that H_0 is true.

6.1 The Decision Rule: Anatomy of a Hypothesis Test

Before any formula, a hypothesis test is a piece of *logic*: you name a default claim, you decide in advance how surprising the data would have to be to abandon that claim, and then you let the data decide. This section dissects that logic into its parts — the null hypothesis H_0 and its alternative H_1, the **test statistic** that compresses the data into one number, the **rejection region** C of values too extreme to credit, and

the **significance level** α that fixes how large C is. Recognizing a test means, first of all, reading a word problem and writing down the correct H_0 and H_1: deciding which claim is the default, and whether the alternative points one way or both. The single most important habit, and the one this section drills hardest, is the asymmetry of the verdict: we assume H_0 while we compute, and at the end we either *reject* H_0 or *fail to reject* it — we never "accept" or "prove" it. Get the setup right and every named test in later sections is the same machine with a different statistic dropped in.

Which hypothesis goes where and which way H_1 points

- The claim of **no effect / no change / status quo** — the one carrying an equality ($=$, or $\leq$, or $\geq$) — is H_0. It gets the benefit of the doubt.
- The claim you would need **evidence** to believe — the new drug works, the machine has drifted, the rate changed — is H_1. $\Rightarrow$ build C so that landing there is strong evidence for H_1.
- Wording "**has increased**", "exceeds", "is more than" $\Rightarrow$ one-sided, $H_1 : \mu > \mu_0$, all of α in the *right* tail.
- Wording "**has decreased**", "is at most", "is below" $\Rightarrow$ one-sided, $H_1 : \mu < \mu_0$, all of α in the *left* tail.
- Wording "**differs from**", "is not equal to", "has changed" (no direction named) $\Rightarrow$ two-sided, $H_1 : \mu \neq \mu_0$, split α into two tails of $\alpha/2$ each.
- A hypothesis pinning the parameter to a **single value** is *simple*; one allowing a **range** of values is *composite*.

Do not confuse failing to reject with proving H_0: "fail to reject" means the data were not surprising enough, not that H_0 is true. A jury returns "not guilty," never "innocent."

6.1.1 The two hypotheses, and why H_0 carries the equality

Definition 6.1.1 (Null and alternative hypotheses)**.** A **statistical hypothesis** is a claim about the value of an unknown parameter θ of the parent distribution. In a test we name two complementary claims: the **null hypothesis** H_0, the default we provisionally assume, and the **alternative hypothesis** H_1, the claim we will adopt only if the data give strong evidence against H_0. By convention H_0 contains the equality, so that "H_0 is true" fixes (or bounds) the parameter sharply enough to compute the distribution of the test statistic.

Remark. Two conventions earn their keep. First, H_0 is the claim you would feel foolish abandoning on weak evidence — the status quo — so the test is conservative: it protects H_0 unless the data clearly contradict it. Second, H_0 takes the equality because to compute "how surprising is this data *if H_0 holds*" we need a definite sampling distribution, and $\theta = \mu_0$ supplies one. The boundary case $\theta = \mu_0$ is the hardest case for the data to overturn, so controlling the error there controls it across the whole one-sided null.

Definition 6.1.2 (Simple vs. composite)**.** A hypothesis is **simple** if it specifies the distribution completely (e.g. $\mu = 12$ with σ known — one distribution), and **composite** if it leaves a range (e.g. $\mu > 12$, or $\mu \neq 12$ — many distributions).

Example 6.1.3 (Recognizing the default claim)**.** A municipal lab certifies that bottled water from a plant has mean nitrate concentration $\mu = 10$ mg/L. A watchdog group suspects the true mean is *higher*. State H_0 and H_1 and classify each as simple or composite.

Which rule, and why? The plant's certified value is the status quo, so it is H_0 and carries the equality. The watchdog must produce evidence, so its suspicion is H_1. "Higher" names a direction $\Rightarrow$ one-sided to the right.

Solution.

Step 1: identify the default. The certified claim $\mu = 10$ is what holds unless the data overturn it; it becomes the null.

Step 2: write the pair.

$$H_0 : \ \mu = 10 \quad (\text{or } \mu \leq 10),$$
$$H_1 : \ \mu > 10.$$

Step 3: classify. $H_0 : \mu = 10$ pins one value (with σ known) = simple; $H_1 : \mu > 10$ allows a range = composite.

$$\boxed{H_0 : \mu = 10 \text{ (simple)}, \quad H_1 : \mu > 10 \text{ (composite, one-sided right)}}$$

Example 6.1.4 (Direction from the verb, part 1)**.** A textbook publisher's old edition had a mean reading-ease score of 64. After a rewrite they ask whether the score has *changed*. Set up the hypotheses.

Which rule, and why? "Changed" names no direction — worse or better both count — so this is two-sided. The old value is the reference, hence H_0.

Solution.

Step 1: reference value is the null. The established score 64 is μ_0.

Step 2: read the verb. "Has changed" = "$\neq$" $\Rightarrow$ two-sided.

$$\boxed{H_0 : \mu = 64, \qquad H_1 : \mu \neq 64}$$

Example 6.1.5 (Direction from the verb, part 2: "at most")**.** A supplier guarantees that the mean impurity in a chemical batch is **at most** 0.5%. A buyer wants to test the guarantee, ready to reject the shipment only if the impurity is too high. State the hypotheses from the *buyer's* standpoint.

Which rule, and why? The guarantee "at most 0.5" is the claim the buyer will keep trusting unless contradicted, so it is $H_0 : \mu \leq 0.5$. Rejection should happen only for *high* impurity $\Rightarrow$ one-sided to the right.

Solution.

Step 1: the guarantee is the null. "At most 0.5" is $\mu \leq 0.5$; the equality boundary $\mu = 0.5$ is the hardest case to overturn.

Step 2: alternative is the buyer's worry. The buyer rejects only when impurity exceeds the promise.

$$\boxed{H_0 : \mu \leq 0.5, \qquad H_1 : \mu > 0.5}$$

6.1.2 The test statistic and the standard normal yardstick

Definition 6.1.6 (Test statistic)**.** A **test statistic** $T = T(X_1, \ldots, X_n)$ is a function of the sample whose distribution is known *when H_0 is true*. We choose T so that values far from its null-typical range are evidence against H_0. The running statistic of this chapter, for a normal mean with σ known, is the standardized sample mean

$$Z \;=\; \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}, \qquad \text{which is } \mathrm{N}(0,1) \text{ when } \mu = \mu_0.$$

Remark. The point of standardizing is to put every problem on one universal ruler. Whatever the units of $\overline{X}$, the statistic Z measures "how many standard errors is the observed mean from the hypothesized mean," and its null distribution is always $\mathrm{N}(0,1)$. So the same critical values 1.645, 1.96, 2.576 serve every z-problem in the book.

Example 6.1.7 (Computing the observed statistic)**.** A filling line is set to dispense $\mu_0 = 500$ mL with known $\sigma = 4$ mL. A sample of $n = 16$ bottles averages $\overline{X} = 502.1$ mL. Compute the observed standardized statistic z_{obs}.

Which rule, and why? The mean is in question and σ is *known*, so the standard-normal statistic $Z = (\overline{X} - \mu_0)/(\sigma/\sqrt{n})$ is the right yardstick — not a t, which is for unknown σ.

Solution.

Step 1: standard error. $\sigma/\sqrt{n} = 4/\sqrt{16} = 4/4 = 1$ mL.

Step 2: standardize.

$$z_{\text{obs}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{502.1 - 500}{1} = \frac{2.1}{1} = 2.1.$$

$$\boxed{z_{\text{obs}} = 2.1}$$

6.1.3 The rejection region and the significance level

Definition 6.1.8 (Rejection region and significance level)**.** The **rejection region** (or critical region) C is the set of test-statistic values for which we reject H_0. The **significance level** α is the probability, computed *under* H_0, that the statistic falls in C:

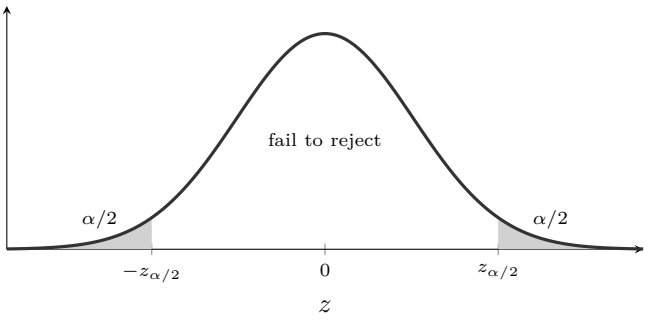
$$\alpha \;=\; \mathbb{P}\big(T \in C \mid H_0 \text{ true}\big).$$

Choosing α (before seeing data) fixes the **critical value(s)** that bound C. The **decision rule** is then simply: *reject* H_0 *if and only if* $T \in C$.

Proposition 6.1.9 (Critical values for a Z-statistic)**.** *For the standard-normal statistic* Z *at level* α*:*

$$\begin{aligned}
H_1 : \mu > \mu_0 \;&\Rightarrow\; C = \{\, Z \geq z_\alpha \,\} && \textit{(right tail, area } \alpha\textit{)},\\
H_1 : \mu < \mu_0 \;&\Rightarrow\; C = \{\, Z \leq -z_\alpha \,\} && \textit{(left tail, area } \alpha\textit{)},\\
H_1 : \mu \neq \mu_0 \;&\Rightarrow\; C = \{\, |Z| \geq z_{\alpha/2} \,\} && \textit{(two tails, area } \alpha/2 \textit{ each)}.
\end{aligned}$$

Proof. The level is the null probability of C. For the right-sided case, $\mathbb{P}(Z \geq c \mid \mu = \mu_0) = \alpha$ defines $c = z_\alpha$ by the definition of the upper α-quantile. For the two-sided case symmetry of $\mathrm{N}(0,1)$ splits the mass equally, so each tail must carry $\alpha/2$, giving the cut-off $z_{\alpha/2}$. The total area in C is α in every case, which is exactly the required level. □



Two-sided rejection region for a Z-statistic: reject H_0 when $|Z| \geq z_{\alpha/2}$ (the shaded tails, total area α); fail to reject in the central region.

Example 6.1.10 (Critical value, one-sided)**.** For a right-sided test of a normal mean at $\alpha = 0.05$, find the critical value and write the rejection region.

Which rule, and why? One-sided right alternative ⇒ put all of $\alpha = 0.05$ in the right tail; the cut-off is the upper 0.05 quantile of $\mathrm{N}(0,1)$.

Solution.

Step 1: locate the quantile. The upper 0.05 point of the standard normal is $z_{0.05} = 1.645$.

Step 2: write C*.* Reject when the statistic is at least that large.

$$\boxed{C = \{\, Z \geq 1.645 \,\}}$$

Example 6.1.11 (Critical values, two-sided)**.** For a two-sided test of a normal mean at $\alpha = 0.01$, find the critical values and the rejection region.

Which rule, and why? Two-sided alternative ⇒ split $\alpha = 0.01$ into two tails of 0.005 each; the cut-offs are the $\pm z_{0.005}$ quantiles.

Solution.

Step 1: per-tail area. $\alpha/2 = 0.01/2 = 0.005$.

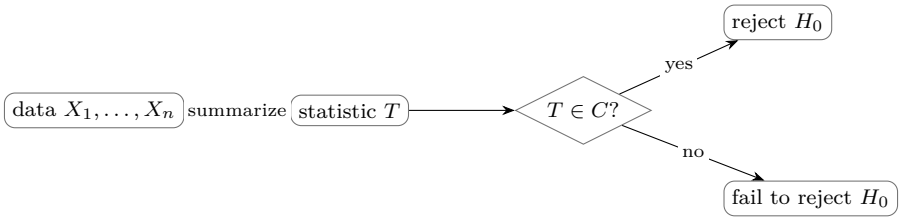
Step 2: quantile. The upper 0.005 point of $\mathrm{N}(0,1)$ is $z_{0.005} = 2.576$.

Step 3: region. Reject when Z is beyond either cut-off.

$$\boxed{C = \{\, |Z| \geq 2.576 \,\} = \{\, Z \leq -2.576 \,\} \cup \{\, Z \geq 2.576 \,\}}$$

6.1.4 The decision rule in one picture

A decision rule is the whole machine assembled: data flow into the statistic, the statistic is checked against C, and out comes a verdict. The schematic below names the pieces in order.



The decision rule: reject H_0 if and only if the test statistic lands in the rejection region C; otherwise fail to reject. The verdict is never "accept."

Definition 6.1.12 (Decision rule)**.** A **decision rule** for a level-α test is the prescription: compute T from the data; *reject H_0 if $T \in C$, otherwise fail to reject H_0*, where C is chosen so that $\mathbb{P}(T \in C \mid H_0) = \alpha$.

Example 6.1.13 (Applying a given rule: reject)**.** A test uses the rule "reject $H_0 : \mu = 80$ in favor of $H_1 : \mu \neq 80$ when $|Z| \geq 1.96$." A study yields $z_{\text{obs}} = -2.4$. State the decision.

Which rule, and why? We are handed the rule and a statistic; recognition here is just checking membership of z_{obs} in $C = \{|Z| \geq 1.96\}$.

Solution.

Step 1: compute $|z_{obs}|$. $|-2.4| = 2.4$.

Step 2: compare to cut-off. $2.4 \geq 1.96$, so $z_{\text{obs}} \in C$.

Step 3: verdict. The statistic is in the rejection region.

$$\boxed{\text{Reject } H_0 \text{ at } \alpha = 0.05.}$$

Example 6.1.14 (Applying a given rule: fail to reject)**.** Same rule (reject when $|Z| \geq 1.96$), but now the data give $z_{\text{obs}} = 1.42$. State the decision and word it carefully.

Which rule, and why? Again pure membership-checking. The subtle part is the *wording* of a non-rejection.

Solution.

Step 1: compare. $|1.42| = 1.42 < 1.96$, so $z_{\text{obs}} \notin C$.

Step 2: verdict — and its phrasing. The data are not extreme enough to overturn H_0. We do *not* conclude $\mu = 80$; we conclude only that there is insufficient evidence against it.

$$\boxed{\text{Fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

Example 6.1.15 (CONTRAST: the asymmetry trap)**.** In the previous example a colleague writes the conclusion as "the test *proves* the mean is 80." Explain why this is wrong, and what the correct claim is.

Which rule, and why? This is the classic mistake the asymmetry of testing is designed to prevent: treating "fail to reject" as "accept" or "prove." Naming the error *is* the recognition skill here.

Solution.

Step 1: what the test controls. The level α bounds only the probability of *wrongly rejecting* a true H_0. It says nothing, by itself, about the chance of missing a false H_0 (that is the Type II error β of §2, often large).

Step 2: why "prove" fails. A small $|Z|$ is consistent with $\mu = 80$ but equally consistent with $\mu = 80.5$, $\mu = 79.3$, and many nearby values — the data cannot single out 80. Absence of strong evidence against H_0 is not evidence for it.

Step 3: the correct statement. "At $\alpha = 0.05$ the data do not provide sufficient evidence to conclude that $\mu \neq 80$."

$$\boxed{\text{Correct: fail to reject } H_0\text{; we never "prove" } H_0.}$$

Example 6.1.16 (Same data, two levels: a reject flips to fail-to-reject)**.** A two-sided test of $H_0 : \mu = 100$ produces $z_{\text{obs}} = 2.20$. Reach the decision at $\alpha = 0.05$ and again at $\alpha = 0.01$, and comment.

Which rule, and why? One observed statistic, two significance levels. Recognizing that the *decision is a function of* α — not a fixed property of the data — is the lesson.

Solution.

Step 1: cut-off at $\alpha = 0.05$. Two-sided $\Rightarrow z_{0.025} = 1.96$. Compare: $2.20 \geq 1.96$, so $z_{\text{obs}} \in C$.

Step 2: decision at 0.05. Reject H_0.

Step 3: cut-off at $\alpha = 0.01$. Two-sided $\Rightarrow z_{0.005} = 2.576$. Compare: $2.20 < 2.576$, so $z_{\text{obs}} \notin C$.

Step 4: decision at 0.01. Fail to reject H_0.

Reject at $\alpha = 0.05$, fail to reject at $\alpha = 0.01$.

The same evidence is "significant" at the looser standard and "not significant" at the stricter one — which is exactly why α must be fixed *before* looking at the data, and why reporting a p-value (the boundary α, here near 0.028) is more informative than a bare reject/fail verdict.

6.1.5 The whole machine once, end to end

Example 6.1.17 (End-to-end, one-sided: a process drift)**.** A bolt-cutting machine is specified to produce bolts of mean length $\mu_0 = 50.0$ mm with known process standard deviation $\sigma = 0.6$ mm. Maintenance fears the machine has *drifted longer*. A sample of $n = 36$ bolts has $\overline{X} = 50.18$ mm. Test at $\alpha = 0.05$.

Which rule, and why? A normal mean with σ *known* $\Rightarrow$ the Z-statistic. "Drifted longer" names a direction $\Rightarrow$ one-sided right; all of α in the right tail.

Solution.

Step 1: hypotheses.

$$H_0 : \mu = 50.0 \ (\mu \leq 50.0), \qquad H_1 : \mu > 50.0.$$

Step 2: statistic. Standard error $\sigma/\sqrt{n} = 0.6/\sqrt{36} = 0.6/6 = 0.1$ mm, so

$$z_{\text{obs}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{50.18 - 50.0}{0.1} = \frac{0.18}{0.1} = 1.8.$$

Step 3: rejection region. One-sided right at $\alpha = 0.05$: $C = \{Z \geq z_{0.05}\} = \{Z \geq 1.645\}$.

Step 4: decide. $z_{\text{obs}} = 1.8 \geq 1.645$, so $z_{\text{obs}} \in C$.

Reject H_0: evidence at $\alpha = 0.05$ that the mean length exceeds 50.0 mm.

Example 6.1.18 (End-to-end, two-sided: a calibration check)**.** A laboratory scale is certified to read a 200.0 g reference mass with no bias; readings have known $\sigma = 0.5$ g. A technician weighs the reference $n = 25$ times and gets $\overline{X} = 199.78$ g. Is the scale *out of calibration* (either direction)? Use $\alpha = 0.05$.

Which rule, and why? Normal mean, σ known $\Rightarrow$ Z-statistic. "Out of calibration, either direction" $\Rightarrow$ two-sided; split α into two tails.

Solution.

Step 1: hypotheses.

$$H_0 : \mu = 200.0, \qquad H_1 : \mu \neq 200.0.$$

Step 2: statistic. $\sigma/\sqrt{n} = 0.5/\sqrt{25} = 0.5/5 = 0.1$ g, so

$$z_{\text{obs}} = \frac{199.78 - 200.0}{0.1} = \frac{-0.22}{0.1} = -2.2.$$

Step 3: rejection region. Two-sided at $\alpha = 0.05$: $C = \{|Z| \geq z_{0.025}\} = \{|Z| \geq 1.96\}$.

Step 4: decide. $|z_{\text{obs}}| = 2.2 \geq 1.96$, so $z_{\text{obs}} \in C$.

Reject H_0: evidence at $\alpha = 0.05$ that the scale is out of calibration.

Example 6.1.19 (End-to-end where the verdict is fail-to-reject)**.** A polling firm's turnout model assumes mean wait-time at a precinct is $\mu_0 = 8.0$ minutes, with known $\sigma = 2.0$ minutes. An observer samples $n = 64$ voters and finds $\overline{X} = 8.35$ minutes. Does the mean wait *differ* from 8.0? Test at $\alpha = 0.01$.

Which rule, and why? Normal mean, σ known $\Rightarrow Z$. "Differ from" $\Rightarrow$ two-sided. A small-looking sample mean shift can still fail to reject once standardized.

Solution.

Step 1: hypotheses.

$$H_0 : \mu = 8.0, \qquad H_1 : \mu \neq 8.0.$$

Step 2: statistic. $\sigma/\sqrt{n} = 2.0/\sqrt{64} = 2.0/8 = 0.25$ min, so

$$z_{\text{obs}} = \frac{8.35 - 8.0}{0.25} = \frac{0.35}{0.25} = 1.4.$$

Step 3: rejection region. Two-sided at $\alpha = 0.01$: $C = \{|Z| \geq z_{0.005}\} = \{|Z| \geq 2.576\}$.

Step 4: decide. $|z_{\text{obs}}| = 1.4 < 2.576$, so $z_{\text{obs}} \notin C$.

Fail to reject H_0: at $\alpha = 0.01$ no evidence the mean wait differs from 8.0.

Example 6.1.20 (Reading direction wrong: a CONTRAST in setup)**.** For the bolt-drift study ("drifted longer," $z_{\text{obs}} = 1.8$, $\alpha = 0.05$), suppose an analyst mistakenly runs a *two-sided* test. Show how the verdict can change, and which setup is correct.

Which rule, and why? This contrasts a correct one-sided setup against a look-alike two-sided one. The recognition point: the alternative's direction is dictated by the question ("longer"), not chosen for convenience, and it changes the cut-off.

Solution.

Step 1: the wrong (two-sided) cut-off. Two-sided at $\alpha = 0.05$ uses $z_{0.025} = 1.96$, so $C = \{|Z| \geq 1.96\}$.

Step 2: wrong verdict. $|z_{\text{obs}}| = 1.8 < 1.96 \Rightarrow$ fail to reject.

Step 3: the correct (one-sided) cut-off. The question asks only about *longer*, so $H_1 : \mu > 50.0$ and $C = \{Z \geq 1.645\}$; then $1.8 \geq 1.645 \Rightarrow$ reject.

Step 4: resolve. The mistaken two-sided test wastes half of α on a tail the question never asked about, raising the cut-off from 1.645 to 1.96 and discarding real evidence.

Correct one-sided test rejects; the spurious two-sided test does not.

This is why the verb in the prompt — "longer," "differs," "at most" — must be read before any arithmetic: it fixes H_1, and H_1 fixes C.

6.2 Type I and Type II Errors, Power, and Sample Size

A test can be wrong in exactly two ways, and the two are not symmetric. We can **reject** a null hypothesis H_0 that is actually true — a false alarm — or we can **fail to reject** an H_0 that is actually false — a miss. The first mistake has probability α, the significance level we fix in advance; the second has a probability β that we do *not* get to fix, because it depends on *how* false H_0 is. The quantity $1 - \beta$, the chance of correctly rejecting a false H_0, is the **power** of the test, and it is the lens through which we judge whether a test is worth running at all. This section is about computing α and β for an explicit decision rule, tracing how power responds when we push the true parameter away from the null, enlarge the sample, or loosen α, and inverting that relationship to *design* a study: find the smallest n that buys a target power. Recognize this material whenever a problem asks not "is the result significant?" but "what is the chance my decision is wrong?" or "how big a sample do I need?"

Which error quantity does the problem want?

- Asked for the chance of a **false alarm** — rejecting a true H_0? $\Rightarrow$ that is **Type I error**, probability α. It is whatever you set the level to; for a given rule, $\alpha = \mathbb{P}(\text{statistic} \in C \mid H_0)$.
- Asked for the chance of a **miss** — failing to reject when a *specific* alternative μ_1 is true? $\Rightarrow$ that is **Type II error**, probability $\beta(\mu_1)$. You must standardize the statistic under μ_1, not under μ_0.
- Asked for the chance of **correctly** catching a false H_0? $\Rightarrow$ **power** $1 - \beta(\mu_1)$. The whole function $\mu \mapsto \text{power}(\mu)$ is the **power function**.
- Asked to **design the sample size** for a target power at a stated alternative? $\Rightarrow$ solve $n \geq \big((z_\alpha + z_\beta)\sigma/(\mu_1 - \mu_0)\big)^2$ and round *up*.
- Comparing a borderline β across rules? $\Rightarrow$ remember the levers: power rises with the **effect size** $|\mu_1 - \mu_0|$, with n, and with α.

Do not confuse α with β, or either with the p-value: α is fixed before data and computed under H_0; β is computed under a *particular* H_1 value and is left free; the p-value is computed from the data after the fact. And a large p-value is *not* evidence that H_0 is true — if power was low, a miss was likely no matter what.

6.2.1 The two errors and their probabilities

Definition 6.2.1 (Type I and Type II error)**.** Fix a test with rejection region C. A **Type I error** occurs when H_0 is true but the data fall in C, so H_0 is rejected. Its probability is the **significance level**

$$\alpha \;=\; \mathbb{P}(\text{reject } H_0 \mid H_0 \text{ true}).$$

A **Type II error** occurs when H_0 is false but the data fall outside C, so H_0 is not rejected. When the truth is a particular alternative value its probability is

$$\beta \;=\; \mathbb{P}(\text{fail to reject } H_0 \mid H_1 \text{ true}).$$

Remark. The asymmetry is deliberate. We pin α down to a small number (commonly 0.05 or 0.01) because a false alarm is the error we have decided to guard against; H_0 enjoys the benefit of the doubt. Then β falls where it may, and one job of the statistician is to make sure it is not embarrassingly large. The four outcomes:

reject H_0, H_0 true:	Type I error (α),
reject H_0, H_0 false:	correct ($1 - \beta$),
fail to reject, H_0 true:	correct ($1 - \alpha$),
fail to reject, H_0 false:	Type II error (β).

Definition 6.2.2 (Power and the power function)**.** The **power** of a test against a specific alternative is $1 - \beta$, the probability of correctly rejecting a false H_0. Letting the true parameter μ range over all its possible values, the **power function** is

$$\text{power}(\mu) \;=\; \mathbb{P}\big(\text{reject } H_0 \mid \text{true mean} = \mu\big).$$

At $\mu = \mu_0$ it equals α (the chance of rejecting when H_0 holds); as μ moves into the alternative it climbs toward 1.

Example 6.2.3 (A coin-acceptance rule: computing α directly)**.** A vending-machine supplier accepts a batch of mechanical coin counters only if a calibration test passes. The rule: out of $n = 20$ independent trial coins, reject the batch (declare it miscalibrated, H_1) if 4 or more are misread; otherwise accept (H_0: the misread rate is the nominal $p = 0.05$). What is the Type I error probability of this rule?

Which rule, and why? We want $\mathbb{P}(\text{reject} \mid H_0 \text{ true})$, and "true" fixes the count of misreads as $\text{Bin}(20, 0.05)$. This is a direct binomial tail, not a z-approximation: n is small and p is far from $1/2$, so we sum exact probabilities.

Solution.

Step 1: name the null distribution. Under H_0, the number of misreads $X \sim \text{Bin}(20, 0.05)$. The rejection region is $C = \{X \geq 4\}$.

Step 2: compute the tail. Use $\alpha = \mathbb{P}(X \geq 4) = 1 - \mathbb{P}(X \leq 3)$ with $q = 0.95$:

$$\begin{aligned}
\mathbb{P}(X = 0) &= 0.95^{20} = 0.35849, \\
\mathbb{P}(X = 1) &= 20(0.05)(0.95^{19}) = 0.37736, \\
\mathbb{P}(X = 2) &= \tbinom{20}{2}(0.05^2)(0.95^{18}) = 0.18868, \\
\mathbb{P}(X = 3) &= \tbinom{20}{3}(0.05^3)(0.95^{17}) = 0.05958.
\end{aligned}$$

Step 3: subtract. $\mathbb{P}(X \leq 3) = 0.35849 + 0.37736 + 0.18868 + 0.05958 = 0.98411$, so $\alpha = 1 - 0.98411 = 0.01589$.

$$\boxed{\alpha \approx 0.0159}$$

Example 6.2.4 (Type II error for the same rule)**.** Suppose the counters are in fact badly miscalibrated, with true misread rate $p = 0.20$. For the "reject if $X \geq 4$ out of 20" rule above, what is the probability the test misses the defect (a Type II error)?

Which rule, and why? Now "true" means the alternative value $p = 0.20$, so we re-standardize: the count is $\text{Bin}(20, 0.20)$, *not* $\text{Bin}(20, 0.05)$. Type II error is the probability the statistic lands *outside* the rejection region under that alternative.

Solution.

Step 1: distribution under H_1. With $p = 0.20$, $X \sim \text{Bin}(20, 0.20)$. A miss means $X \leq 3$.

Step 2: sum the four terms. With $q = 0.80$:

$$\begin{aligned}
\mathbb{P}(X = 0) &= 0.80^{20} = 0.01153, \\
\mathbb{P}(X = 1) &= 20(0.20)(0.80^{19}) = 0.05765, \\
\mathbb{P}(X = 2) &= \tbinom{20}{2}(0.20^2)(0.80^{18}) = 0.13691, \\
\mathbb{P}(X = 3) &= \tbinom{20}{3}(0.20^3)(0.80^{17}) = 0.20536.
\end{aligned}$$

Step 3: add. $\beta = \mathbb{P}(X \leq 3) = 0.01153 + 0.05765 + 0.13691 + 0.20536 = 0.41145$. The power against $p = 0.20$ is $1 - \beta = 0.58855$.

$$\boxed{\beta \approx 0.411, \qquad \text{power} \approx 0.589}$$

6.2.2 Deriving β and power for the one-sided z-test

For the rest of the section we work with the cleanest setting — a normal population with known σ — because there the error probabilities have closed forms and every lever is visible. The setup: observe $X_1, \ldots, X_n$ i.i.d. $\text{N}(\mu, \sigma^2)$ with σ known, and test

$$H_0 : \mu = \mu_0 \quad \text{versus} \quad H_1 : \mu > \mu_0.$$

Reject for large $\overline{X}$. The level-α rule rejects when $z_{\text{obs}} = (\overline{X} - \mu_0)/(\sigma/\sqrt{n}) \geq z_\alpha$, equivalently when $\overline{X}$ exceeds the **critical value**

$$c \;=\; \mu_0 + z_\alpha \, \frac{\sigma}{\sqrt{n}}.$$

Theorem 6.2.5 (Type II error and power of the one-sided z-test)**.** *For the rule "reject when $\overline{X} \geq c$" with $c = \mu_0 + z_\alpha \, \sigma/\sqrt{n}$, the Type II error probability at a true mean $\mu_1 > \mu_0$ is*

$$\beta(\mu_1) \;=\; \Phi\!\left(z_\alpha - \frac{(\mu_1 - \mu_0)\sqrt{n}}{\sigma} \right),$$

and the power is

$$\text{power}(\mu_1) \;=\; 1 - \beta(\mu_1) \;=\; 1 - \Phi\!\left(z_\alpha - \frac{(\mu_1 - \mu_0)\sqrt{n}}{\sigma} \right).$$

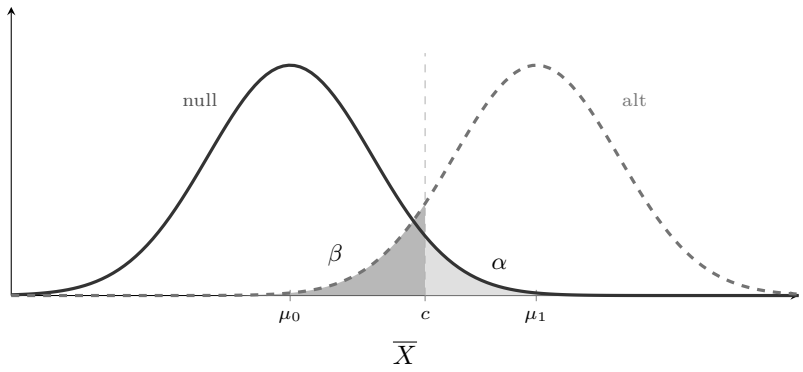
Proof. A miss is the event $\overline{X} < c$. Under H_1 the true mean is μ_1, so $\overline{X} \sim \mathrm{N}(\mu_1, \sigma^2/n)$ and $Z = (\overline{X} - \mu_1)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$. Standardize the boundary using μ_1, not μ_0:

$$\begin{aligned}\beta(\mu_1) &= \mathbb{P}(\overline{X} < c \mid \mu = \mu_1) = \mathbb{P}\left(Z < \frac{c-\mu_1}{\sigma/\sqrt{n}}\right) \\ &= \Phi\left(\frac{\mu_0 + z_\alpha\,\sigma/\sqrt{n} - \mu_1}{\sigma/\sqrt{n}}\right) = \Phi\left(z_\alpha - \frac{(\mu_1-\mu_0)\sqrt{n}}{\sigma}\right).\end{aligned}$$

Power is the complement. □

Remark. The single quantity $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma$ controls everything: it grows with the raw effect $\mu_1 - \mu_0$, with $\sqrt{n}$, and shrinks with σ. A larger d pushes $z_\alpha - d$ further left, so $\Phi(z_\alpha - d) = \beta$ drops and power rises. Raising α lowers z_α, which also lowers β. These three sentences are the whole qualitative story; the examples make them numerical.

The error-probability picture below shows the two densities and the two shaded areas for a one-sided rule.



Solid curve: $\overline{X}$ under H_0 (mean μ_0). Dashed curve: $\overline{X}$ under H_1 (mean μ_1). The light area past c is α; the dark area to the left of c under the alternative is β. Moving μ_1 rightward slides the dashed curve away and shrinks β.

Example 6.2.6 (Critical value, then β at one alternative)**.** A fill line is supposed to put $\mu_0 = 500$ mL in each bottle; overfill wastes product, so quality control tests $H_0 : \mu = 500$ against $H_1 : \mu > 500$ at level $\alpha = 0.05$ using $n = 25$ bottles, with known $\sigma = 8$ mL. Find the critical value c in millilitres, then find β if the line is really running at $\mu_1 = 504$ mL.

Which rule, and why? Known σ, normal data, one-directional alternative ("overfill") ⇒ one-sided upper z-test. The critical value comes from α under H_0; β comes from re-standardizing the same boundary under $\mu_1 = 504$.

Solution.

Step 1: standard error and critical value. $\sigma/\sqrt{n} = 8/\sqrt{25} = 1.6$. With $z_{0.05} = 1.645$,

$$c = \mu_0 + z_{0.05}\,\frac{\sigma}{\sqrt{n}} = 500 + 1.645(1.6) = 500 + 2.632 = 502.632 \text{ mL}.$$

Step 2: standardize the boundary under $\mu_1 = 504$.

$$\frac{c-\mu_1}{\sigma/\sqrt{n}} = \frac{502.632 - 504}{1.6} = \frac{-1.368}{1.6} = -0.855.$$

Step 3: read off β. $\beta = \Phi(-0.855) = 0.1963$. Equivalently, via the formula, $z_\alpha - d = 1.645 - (504-500)(5)/8 = 1.645 - 2.5 = -0.855$, the same number.

$c \approx 502.63$ mL,	$\beta \approx 0.196$,	power ≈ 0.804

Example 6.2.7 (Power climbs as the alternative moves away)**.** For the same fill-line test ($\mu_0 = 500$, $\sigma = 8$, $n = 25$, $\alpha = 0.05$, so $c = 502.632$ and $d = (\mu_1 - 500)(5)/8$), tabulate the power at $\mu_1 = 501$, 502, 503, 504, 506 mL.

Which rule, and why? Same one-sided z-test; we are sweeping the alternative to see the power function rise. Only μ_1 changes, so only $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma$ changes.

Solution.

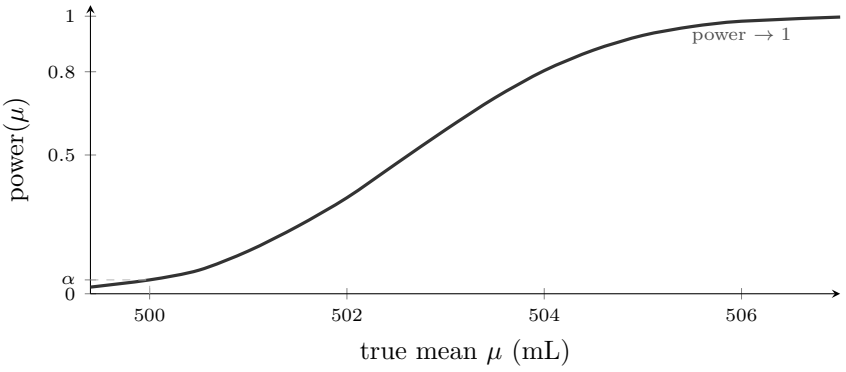
Step 1: compute $z_\alpha - d$ at each μ_1. With $z_\alpha = 1.645$ and $d = (\mu_1 - 500) \cdot 0.625$:

$$\begin{aligned}
\mu_1 = 501 &: d = 0.625, \quad z_\alpha - d = 1.020, \quad \text{power} = 1 - \Phi(1.020) = 0.154, \\
\mu_1 = 502 &: d = 1.250, \quad z_\alpha - d = 0.395, \quad \text{power} = 1 - \Phi(0.395) = 0.346, \\
\mu_1 = 503 &: d = 1.875, \quad z_\alpha - d = -0.230, \quad \text{power} = 1 - \Phi(-0.230) = 0.591, \\
\mu_1 = 504 &: d = 2.500, \quad z_\alpha - d = -0.855, \quad \text{power} = 1 - \Phi(-0.855) = 0.804, \\
\mu_1 = 506 &: d = 3.750, \quad z_\alpha - d = -2.105, \quad \text{power} = 1 - \Phi(-2.105) = 0.982.
\end{aligned}$$

Step 2: read the trend. Power rises monotonically from near α at the null toward 1 as the true mean departs from 500.

$$\boxed{\text{power}(501, \ldots, 506) \approx 0.154,\ 0.346,\ 0.591,\ 0.804,\ 0.982}$$

The shape of that table is the power curve, drawn next.



Power as a function of the true mean for the fill-line test. At $\mu = \mu_0 = 500$ the power equals $\alpha = 0.05$; it rises smoothly toward 1 as the true mean overshoots the target.

Example 6.2.8 (Power grows with the sample size)**.** Keep the fill-line alternative fixed at $\mu_1 = 503$ mL ($\mu_0 = 500$, $\sigma = 8$, $\alpha = 0.05$). Compare the power at $n = 25$, $n = 64$, and $n = 144$.

Which rule, and why? Same one-sided z-test, same effect $\mu_1 - \mu_0 = 3$; we are now turning the n lever. Since $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma$, only $\sqrt{n}$ changes.

Solution.

Step 1: compute d at each n. With $(\mu_1 - \mu_0)/\sigma = 3/8 = 0.375$:

$$\begin{aligned}
n = 25 &: d = 0.375 \cdot 5 = 1.875, \quad z_\alpha - d = -0.230, \quad \text{power} = 0.591, \\
n = 64 &: d = 0.375 \cdot 8 = 3.000, \quad z_\alpha - d = -1.355, \quad \text{power} = 0.912, \\
n = 144 &: d = 0.375 \cdot 12 = 4.500, \quad z_\alpha - d = -2.855, \quad \text{power} = 0.998.
\end{aligned}$$

Step 2: interpret. Quadrupling n from 25 to 144 (so $\sqrt{n}$ from 5 to 12) lifts power from about 0.59 to essentially certain detection of a 3-mL overfill.

$$\boxed{\text{power} = 0.591\ (n{=}25),\ 0.912\ (n{=}64),\ 0.998\ (n{=}144)}$$

Example 6.2.9 (Power grows with α)**.** Once more with the fill-line test ($\mu_0 = 500$, $\sigma = 8$, $n = 25$, alternative $\mu_1 = 503$, so $d = 1.875$). Compare the power at $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.10$.

Which rule, and why? Same one-sided z-test, same n and same effect; we now move the α lever. Raising α lowers the critical z_α, which lowers $z_\alpha - d$ and raises power — at the cost of more false alarms.

Solution.

Step 1: the relevant z_α. $z_{0.01} = 2.326$, $z_{0.05} = 1.645$, $z_{0.10} = 1.282$.

Step 2: $z_\alpha - d$ and power, with $d = 1.875$.

$$\begin{aligned}
&\alpha = 0.01 : z_\alpha - d = 0.451, \ \text{power} = 1 - \Phi(0.451) = 0.326,\\
&\alpha = 0.05 : z_\alpha - d = -0.230, \ \text{power} = 1 - \Phi(-0.230) = 0.591,\\
&\alpha = 0.10 : z_\alpha - d = -0.593, \ \text{power} = 1 - \Phi(-0.593) = 0.723.
\end{aligned}$$

Step 3: note the trade-off. Going from $\alpha = 0.01$ to 0.10 raises power from 0.33 to 0.72, but tenfold more true batches would be wrongly rejected. This is the α-versus-β tension, isolated next.

$$\boxed{\text{power} = 0.326\ (\alpha{=}.01),\ 0.591\ (\alpha{=}.05),\ 0.723\ (\alpha{=}.10)}$$

6.2.3 The α-versus-β trade-off

Proposition 6.2.10 (At fixed n, shrinking α inflates β)**.** *For the one-sided z-test with n and the alternative μ_1 held fixed, $\beta(\mu_1) = \Phi(z_\alpha - d)$ with $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma$ fixed. As α decreases, z_α increases, so $z_\alpha - d$ increases and β increases. The only way to drive both α and β down at once is to increase n (raise d).*

Proof. $z_\alpha = \Phi^{-1}(1-\alpha)$ is strictly decreasing in α, and Φ is strictly increasing, so $\alpha \downarrow \Rightarrow\ z_\alpha \uparrow \Rightarrow\ \Phi(z_\alpha - d) \uparrow$ with d held constant. Increasing n raises d, lowering $\Phi(z_\alpha - d)$ without touching α. □

Example 6.2.11 (The see-saw at fixed n)**.** A lab tests $H_0 : \mu = 100$ versus $H_1 : \mu > 100$ with $n = 16$, $\sigma = 10$, against the fixed alternative $\mu_1 = 105$ (so $d = (105 - 100)\sqrt{16}/10 = 2.000$). Show explicitly that tightening α from 0.10 to 0.01 raises β.

Which rule, and why? One-sided z-test, fixed n and fixed effect; we vary only α and read both α and β off the same boundary. This isolates the trade-off Proposition just proved.

Solution.

Step 1: tabulate $\beta = \Phi(z_\alpha - 2)$.

$$\begin{aligned}
&\alpha = 0.10 : z_\alpha = 1.282,\ z_\alpha - d = -0.718,\ \beta = \Phi(-0.718) = 0.236,\\
&\alpha = 0.05 : z_\alpha = 1.645,\ z_\alpha - d = -0.355,\ \beta = \Phi(-0.355) = 0.361,\\
&\alpha = 0.01 : z_\alpha = 2.326,\ z_\alpha - d = \ \ 0.326,\ \beta = \Phi(0.326) = 0.628.
\end{aligned}$$

Step 2: read the see-saw. Cutting the false-alarm rate from 0.10 to 0.01 nearly triples the miss rate, from 0.24 to 0.63, with n untouched.

$$\boxed{\beta = 0.236\ (\alpha{=}.10)\ \rightarrow\ 0.628\ (\alpha{=}.01) \text{ at fixed } n}$$

6.2.4 Designing the sample size

If we demand both a small α *and* a small β at a stated alternative, neither is negotiable, so the sample size n becomes the unknown to solve for. Set the critical boundary from α and from β simultaneously.

Theorem 6.2.12 (Sample size for a one-sided z-test)**.** *To test $H_0 : \mu = \mu_0$ against $H_1 : \mu > \mu_0$ at level α with power at least $1 - \beta$ at the specific alternative $\mu_1 > \mu_0$ (known σ), the required sample size is the smallest integer n with*

$$n \ \geq\ \left(\frac{(z_\alpha + z_\beta)\,\sigma}{\mu_1 - \mu_0} \right)^2 .$$

Proof. The boundary $c = \mu_0 + z_\alpha\,\sigma/\sqrt{n}$ comes from the level constraint. The power constraint $\beta(\mu_1) \le \beta$ means $\Phi(z_\alpha - d) \le \beta$, i.e. $z_\alpha - d \le -z_\beta$, i.e. $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma \ge z_\alpha + z_\beta$. Solving for n,

$$\sqrt{n} \ge \frac{(z_\alpha + z_\beta)\sigma}{\mu_1 - \mu_0} \quad \Longrightarrow \quad n \ge \left(\frac{(z_\alpha + z_\beta)\sigma}{\mu_1 - \mu_0}\right)^2.$$

Round up to the next integer, since n must be a whole number and the inequality must hold. □

Remark. Two-sided version: replace z_α by $z_{\alpha/2}$ (a small extra term from the far tail is negligible), giving $n \ge \big((z_{\alpha/2} + z_\beta)\sigma/(\mu_1 - \mu_0)\big)^2$. Either way, the cost of detecting a *smaller* effect $\mu_1 - \mu_0$ is quadratic: halving the effect you must detect quadruples the sample.

Example 6.2.13 (Sizing a clinical trial)**.** A new analgesic is meant to lower mean systolic blood pressure. Investigators will test $H_0 : \mu = 140$ versus $H_1 : \mu < 140$ (mmHg) at $\alpha = 0.05$, and they want power 0.90 to detect a genuine drop to $\mu_1 = 135$. Historical $\sigma = 12$ mmHg. How many patients are needed?

Which rule, and why? Known σ, one-sided alternative, a target power ⇒ the sample-size formula. The direction is "<", but $|\mu_1 - \mu_0| = 5$ is what enters; use z_α (one-sided), not $z_{\alpha/2}$.

Solution.

Step 1: gather the z values. $\alpha = 0.05 \Rightarrow z_{0.05} = 1.645$; power $0.90 \Rightarrow \beta = 0.10 \Rightarrow z_{0.10} = 1.282$.

Step 2: plug in. $\sigma = 12$, $|\mu_1 - \mu_0| = |135 - 140| = 5$:

$$n \ge \left(\frac{(1.645 + 1.282)\cdot 12}{5}\right)^2 = \left(\frac{2.927 \cdot 12}{5}\right)^2 = \left(\frac{35.124}{5}\right)^2 = (7.0248)^2 = 49.35.$$

Step 3: round up. Need $n = 50$ patients.

$$\boxed{n = 50}$$

Example 6.2.14 (Two-sided design: calibrating a scale)**.** A precision scale is recalibrated and we want to detect any drift, in either direction, from its target reading $\mu_0 = 200.0$ g. Test $H_0 : \mu = 200.0$ versus $H_1 : \mu \ne 200.0$ at $\alpha = 0.01$, with power 0.95 to detect a drift of 0.5 g, given $\sigma = 1.0$ g. Find n.

Which rule, and why? The alternative is "≠" (drift either way) ⇒ two-sided, so use $z_{\alpha/2}$ in the sample-size formula. The effect to detect is $|\mu_1 - \mu_0| = 0.5$.

Solution.

Step 1: the z values. Two-sided $\alpha = 0.01 \Rightarrow z_{0.005} = 2.576$; power $0.95 \Rightarrow \beta = 0.05 \Rightarrow z_{0.05} = 1.645$.

Step 2: plug in. $\sigma = 1.0$, $|\mu_1 - \mu_0| = 0.5$:

$$n \ge \left(\frac{(2.576 + 1.645)\cdot 1.0}{0.5}\right)^2 = \left(\frac{4.221}{0.5}\right)^2 = (8.442)^2 = 71.27.$$

Step 3: round up. Need $n = 72$ weighings.

$$\boxed{n = 72}$$

Example 6.2.15 (How much does halving the detectable effect cost?)**.** For the same scale ($\mu_0 = 200.0$, $\sigma = 1.0$, two-sided $\alpha = 0.01$, power 0.95), suppose engineers now insist on detecting a finer drift of just 0.25 g. Recompute n and compare with the 0.5-g case.

Which rule, and why? Identical two-sided design; only the effect size $|\mu_1 - \mu_0|$ in the denominator changes, from 0.5 to 0.25. Since n scales as $1/(\text{effect})^2$, halving the effect should roughly quadruple n.

Solution.

Step 1: same numerator, smaller denominator. The numerator $(2.576 + 1.645)(1.0) = 4.221$ is unchanged.

$$n \ge \left(\frac{4.221}{0.25}\right)^2 = (16.884)^2 = 285.07.$$

Step 2: round up and compare. Need $n = 286$, versus 72 before: the ratio $286/72 \approx 3.97$, the expected near-quadrupling.

$$\boxed{n = 286 \text{ (about } 4\times \text{ the 0.5-g requirement)}}$$

Example 6.2.16 (Achieved power after rounding)**.** The fill-line designers ($\mu_0 = 500$, $\sigma = 8$, $\alpha = 0.05$ one-sided) want power 0.90 to detect $\mu_1 = 503$. Find the required n, then report the *actual* power the rounded n delivers.

Which rule, and why? Sample-size formula for the design n, then the power Theorem to audit the integer n we actually use — the rounded-up n over-delivers slightly, so the achieved power exceeds the target.

Solution.

Step 1: design n. $z_{0.05} = 1.645$, $z_{0.10} = 1.282$, $|\mu_1 - \mu_0| = 3$:

$$n \geq \left(\frac{(1.645 + 1.282)\cdot 8}{3}\right)^2 = \left(\frac{2.927 \cdot 8}{3}\right)^2 = \left(\frac{23.416}{3}\right)^2 = (7.805)^2 = 60.92.$$

So take $n = 61$.

Step 2: achieved power at $n = 61$. $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma = 3\sqrt{61}/8 = 3(7.8102)/8 = 2.929$. Then $z_\alpha - d = 1.645 - 2.929 = -1.284$, and power $= 1 - \Phi(-1.284) = 0.900$.

Step 3: confirm. The achieved power is 0.9005, just above the 0.90 target, as rounding up guarantees.

$$\boxed{n = 61, \quad \text{achieved power} \approx 0.900}$$

6.2.5 Reading errors correctly

Remark. Power is what makes a "fail to reject" interpretable. If a test had high power against the effects you cared about and still did not reject, that is genuine evidence the effect is small. But if power was low, failing to reject tells you almost nothing — you would likely have missed a real effect anyway. "Absence of evidence is not evidence of absence" is precisely the statement that a non-rejection under low power is uninformative.

Example 6.2.17 (Two-sided power for an environmental test)**.** A river's dissolved-oxygen level should average $\mu_0 = 8.0$ mg/L; either a drop (pollution) or a spike (sensor fault) matters, so we test $H_0 : \mu = 8.0$ versus $H_1 : \mu \neq 8.0$ at $\alpha = 0.05$ with $n = 16$ samples, $\sigma = 0.6$ mg/L. What is the power if the true mean is $\mu_1 = 8.4$?

Which rule, and why? Two-sided z-test, so the rejection region is *both* tails and we use $z_{\alpha/2} = z_{0.025} = 1.96$. Power against a one-sided shift $\mu_1 > \mu_0$ comes almost entirely from the upper tail; we standardize the upper boundary under μ_1 and may neglect the far lower tail.

Solution.

Step 1: standard error and d. $\sigma/\sqrt{n} = 0.6/4 = 0.15$; $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma = (0.4)(4)/0.6 = 2.667$.

Step 2: upper-tail contribution. Reject above when $\overline{X} > \mu_0 + 1.96(0.15)$, so the upper-tail power is

$$1 - \Phi(z_{\alpha/2} - d) = 1 - \Phi(1.96 - 2.667) = 1 - \Phi(-0.707) = 0.760.$$

Step 3: lower-tail contribution (negligible). The lower critical value sits at $z = -1.96$ under H_0; under μ_1 that boundary standardizes to $-z_{\alpha/2} - d = -1.96 - 2.667 = -4.627$, and $\Phi(-4.627) \approx 0.0000$. Total power ≈ 0.760.

$$\boxed{\text{power}(8.4) \approx 0.760}$$

Example 6.2.18 (Contrast: misreading a large p-value as "H_0 is true")**.** A startup screens a supplement for an effect on reaction time, testing $H_0 : \mu = 250$ ms versus $H_1 : \mu < 250$ at $\alpha = 0.05$ with only $n = 9$ subjects, $\sigma = 30$ ms. They observe $\overline{X} = 243$, get a p-value of 0.24, fail to reject, and announce "the supplement has no effect." Diagnose the error: compute the test's power against the clinically meaningful alternative $\mu_1 = 235$ ms, and explain why the announcement is unwarranted.

Which rule, and why? The *tempting wrong move* is to treat "fail to reject" as "H_0 confirmed." The right diagnostic is power: a one-sided z-test against $\mu_1 = 235$. If power is low, the non-rejection is uninformative and the no-effect claim does not follow.

Solution.

Step 1: the claimed conclusion. They computed $z_{\text{obs}} = (243 - 250)/(30/3) = -7/10 = -0.70$, which is not past $-z_{0.05} = -1.645$, hence no rejection. So far the arithmetic is fine; the *interpretation* is where it goes wrong.

Step 2: power against the alternative they care about. $d = (\mu_0 - \mu_1)\sqrt{n}/\sigma = (250-235)(3)/30 = 45/30 = 1.500$. Power

$$= 1 - \Phi(z_\alpha - d) = 1 - \Phi(1.645 - 1.500) = 1 - \Phi(0.145) = 1 - 0.5576 = 0.4424.$$

Step 3: the verdict. With power only ≈ 0.44, the test would miss a real 15-ms improvement more than half the time. A non-rejection from a study this underpowered is almost uninformative: it is consistent both with no effect *and* with a sizable effect that the small sample simply could not detect. The correct statement is "we did not detect an effect," *not* "there is no effect."

power ≈ 0.44 $\Rightarrow$ fail-to-reject is **not** evidence that H_0 is true

Example 6.2.19 (Fixing the underpowered study)**.** Continuing the supplement example ($\mu_0 = 250$, $\sigma = 30$, one-sided $\alpha = 0.05$, target alternative $\mu_1 = 235$), how many subjects would the startup have needed for power 0.90? Contrast with their $n = 9$.

Which rule, and why? Same one-sided z-test; now invert for n with the sample-size formula. This turns the diagnosis of the previous example into a concrete remedy.

Solution.

Step 1: z values and effect. $z_{0.05} = 1.645$, $z_{0.10} = 1.282$, $|\mu_1 - \mu_0| = 15$, $\sigma = 30$.

Step 2: plug in.

$$n \geq \left(\frac{(1.645 + 1.282) \cdot 30}{15}\right)^2 = \big(2.927 \cdot 2\big)^2 = (5.854)^2 = 34.27.$$

Step 3: round up and compare. Need $n = 35$, nearly four times the 9 they used. The original design never had a real chance of detecting the effect it was meant to find.

$n = 35$ (vs. the 9 actually run)

Example 6.2.20 (One-sided versus two-sided power at the same alternative)**.** For a process with $\mu_0 = 50$, $\sigma = 5$, $n = 25$, $\alpha = 0.05$, and true mean $\mu_1 = 52$, compare the power of the one-sided upper test ($H_1 : \mu > 50$) with that of the two-sided test ($H_1 : \mu \neq 50$).

Which rule, and why? Same data and same alternative; the only change is the alternative's *direction*, which changes the critical z from $z_{0.05} = 1.645$ (one-sided) to $z_{0.025} = 1.96$ (two-sided). The one-sided test, spending all α in the relevant tail, must be the more powerful of the two against an upward shift.

Solution.

Step 1: common quantities. $\sigma/\sqrt{n} = 5/5 = 1$; $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma = (2)(5)/5 = 2.000$.

Step 2: one-sided power. $1 - \Phi(1.645 - 2.000) = 1 - \Phi(-0.355) = 0.639$.

Step 3: two-sided power. Upper tail: $1 - \Phi(1.96 - 2.000) = 1 - \Phi(-0.040) = 0.516$. Lower tail: $\Phi(-1.96 - 2.000) = \Phi(-3.96) \approx 0.00004$, negligible. Total ≈ 0.516.

Step 4: compare. The one-sided test wins, 0.639 versus 0.516 — the price of the two-sided test's protection against shifts in the unanticipated direction.

one-sided 0.639 $>$ two-sided 0.516

6.3 Testing a Mean: the z-Test (σ Known)

This is the first concrete test, and it is the cleanest one in the chapter because the null distribution of its statistic is *exactly* standard normal. Suppose a measurement X is normal with an unknown mean μ but a population standard deviation σ that is *known* — pinned down by a long calibration history, a manufacturer's specification, or a physical constant. We draw $X_1, \ldots, X_n$ and ask whether μ equals some target value μ_0. The sample mean $\overline{X}$ estimates μ; the question is whether $\overline{X}$ sits far enough from μ_0, *measured in standard errors*, to make the target untenable. The statistic that measures that distance is the z-statistic, and under H_0 it follows the standard normal law no matter what n is. The same machine also runs, approximately, when the parent is *not* normal, provided n is large enough for the Central Limit Theorem to flatten the sampling distribution of $\overline{X}$ into a bell — as long as σ is still supplied rather than estimated. Recognizing this last caveat is the whole game: the moment σ is replaced by a sample estimate S, the exact normality breaks and you must move to the t-test of §4.

Is this a z-test for a mean?

- The parameter in question is a **population mean** μ, and the data are measurements (not counts of successes) $\Rightarrow$ a test for a mean.
- The population standard deviation σ is **known** — given to you as a fixed number, not computed from this sample $\Rightarrow$ use the z**-test**: $z_{\text{obs}} = (\overline{X} - \mu_0)/(\sigma/\sqrt{n})$, compared to standard normal.
- The parent is normal? Then $z_{\text{obs}} \sim \text{N}(0,1)$ *exactly* under H_0, for any n. The parent is not normal but n is large and σ is supplied? Then $z_{\text{obs}} \sim \text{N}(0,1)$ *approximately*, by the CLT.
- Direction of H_1: "differs from / changed" $\Rightarrow$ two-sided, split α between both tails; "increased / exceeds" or "decreased / below" $\Rightarrow$ one-sided, all of α in the matching tail.

Do not confuse a known σ with an estimated one: if the standard deviation number was computed *from the same data* as $\overline{X}$ (i.e. it is really S), the statistic is a t, not a z. Using z there understates the spread and inflates your false-alarm rate.

6.3.1 The statistic and its null distribution

Definition 6.3.1 (The one-sample z-statistic)**.** Let $X_1, \ldots, X_n$ be independent with mean μ and known standard deviation σ, and let $\overline{X} = \frac{1}{n}\sum_i X_i$. To test $H_0 : \mu = \mu_0$ the z**-statistic** is

$$z_{\text{obs}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}.$$

The denominator $\sigma/\sqrt{n}$ is the **standard error** of $\overline{X}$: it is how much $\overline{X}$ typically wanders from the true mean. So z_{obs} reports the gap $\overline{X} - \mu_0$ in units of standard error.

Theorem 6.3.2 (Null distribution)**.** *If the parent is* $\text{N}(\mu_0, \sigma^2)$*, then under* H_0

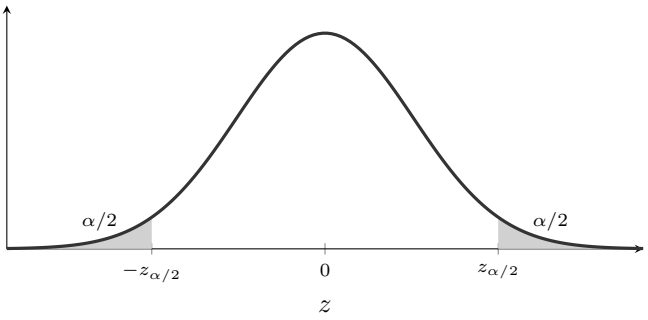
$$z_{obs} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \sim \text{N}(0,1)$$

exactly, for every sample size n*. If the parent is not normal but* σ *is known and* n *is large, the same holds approximately.*

Proof. For a normal parent, $\overline{X}$ is itself exactly normal with $\mathbb{E}[\overline{X}] = \mu_0$ (under H_0) and $\text{Var}(\overline{X}) = \sigma^2/n$. Standardizing a normal variable by subtracting its mean and dividing by its standard deviation $\sigma/\sqrt{n}$ yields a $\text{N}(0,1)$ variable; that standardized variable is exactly z_{obs}. For a non-normal parent, the Central Limit Theorem makes $\overline{X}$ approximately $\text{N}(\mu_0, \sigma^2/n)$ once n is large, and the same standardization gives an approximate $\text{N}(0,1)$. Because σ is a fixed constant, no extra estimation error enters the denominator — which is exactly what fails in the t-case. □

6.3.2 The two-sided test

To test $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$ at level α, reject when $\overline{X}$ is far from μ_0 in *either* direction, i.e. when $|z_{\text{obs}}|$ is large. Split α between the two tails: the critical region is $|z_{\text{obs}}| \geq z_{\alpha/2}$, where $z_{\alpha/2}$ is the value with $\mathbb{P}(Z \geq z_{\alpha/2}) = \alpha/2$. For $\alpha = 0.05$ this is $z_{0.025} = 1.96$; for $\alpha = 0.01$ it is $z_{0.005} = 2.576$.



Two-sided rejection region: reject H_0 when z_{obs} lands in either shaded tail, $|z_{\text{obs}}| \geq z_{\alpha/2}$. Each tail holds area $\alpha/2$, so the total false-alarm probability is α.

Example 6.3.3 (Fill weight of a bottling line)**.** A bottling line is set to dispense $\mu_0 = 500$ mL. From the pump's engineering specification the fill standard deviation is $\sigma = 4$ mL, treated as known. A quality engineer samples $n = 25$ bottles and finds $\overline{X} = 502.1$ mL. At $\alpha = 0.05$, is the line off target in either direction?

Which rule, and why? A mean is in question, σ is given by specification (not estimated from these 25 bottles), and "off target in either direction" is a two-sided alternative. That is the two-sided z-test, not a t-test (we did not compute the spread from the data) and not a one-sided test (no preferred direction).

Solution.

Step 1: Hypotheses. $H_0 : \mu = 500$ versus $H_1 : \mu \neq 500$, level $\alpha = 0.05$.

Step 2: Standard error. $\sigma/\sqrt{n} = 4/\sqrt{25} = 4/5 = 0.8$ mL.

Step 3: Statistic.

$$z_{\text{obs}} = \frac{502.1 - 500}{0.8} = \frac{2.1}{0.8} = 2.625.$$

Step 4: Critical value and decision. The two-sided cutoff is $z_{0.025} = 1.96$. Since $|z_{\text{obs}}| = 2.625 \geq 1.96$, z_{obs} falls in the rejection region.

Reject H_0: the line is off target ($z_{\text{obs}} = 2.63,\ \alpha = 0.05$).

Example 6.3.4 (Battery life claimed unchanged)**.** A battery's mean run time has historically been $\mu_0 = 10.0$ hours with a known process standard deviation $\sigma = 0.6$ hour. After a minor chemistry tweak the manufacturer claims the mean is *unchanged*; a reviewer tests $n = 36$ cells and gets $\overline{X} = 10.15$ hours. Test the no-change claim at $\alpha = 0.05$.

Which rule, and why? Mean, σ known, and "unchanged" (could move up or down) is a two-sided alternative $\Rightarrow$ two-sided z-test.

Solution.

Step 1: Hypotheses. $H_0 : \mu = 10.0$ versus $H_1 : \mu \neq 10.0$, $\alpha = 0.05$.

Step 2: Standard error. $\sigma/\sqrt{n} = 0.6/\sqrt{36} = 0.6/6 = 0.1$ hour.

Step 3: Statistic.

$$z_{\text{obs}} = \frac{10.15 - 10.0}{0.1} = \frac{0.15}{0.1} = 1.50.$$

Step 4: Decision. $|z_{\text{obs}}| = 1.50 < 1.96 = z_{0.025}$, so z_{obs} is not in the rejection region.

Fail to reject H_0: no evidence the mean changed ($z_{\text{obs}} = 1.50$).

Note we do not *accept* that the mean is exactly 10.0; we simply lack evidence against it.

6.3.3 The one-sided tests

When the question points in one direction, all of α goes in the matching tail. For $H_1 : \mu > \mu_0$ ("increased", "exceeds") reject when $z_{\text{obs}} \geq z_\alpha$; for $H_1 : \mu < \mu_0$ ("decreased", "below") reject when $z_{\text{obs}} \leq -z_\alpha$. For $\alpha = 0.05$ the one-sided cutoff is $z_{0.05} = 1.645$, smaller in magnitude than the two-sided 1.96 — a one-sided test spends its whole budget on one side and so detects a true shift in the anticipated direction more readily.

Example 6.3.5 (A fertilizer that has increased yield)**.** Without fertilizer, plots of a wheat variety yield a mean of $\mu_0 = 58.0$ bushels/acre with a known field standard deviation $\sigma = 6.0$. An agronomist applies a new fertilizer to $n = 40$ plots and records $\overline{X} = 60.3$. Does the fertilizer *increase* mean yield at $\alpha = 0.05$?

Which rule, and why? Mean, σ known, and "increase" names a single direction $\Rightarrow$ one-sided (upper-tail) z-test with $H_1 : \mu > 58$. All of α goes in the upper tail.

Solution.

Step 1: Hypotheses. $H_0 : \mu = 58.0$ versus $H_1 : \mu > 58.0$, $\alpha = 0.05$.

Step 2: Standard error. $\sigma/\sqrt{n} = 6.0/\sqrt{40} = 6.0/6.3246 = 0.9487$.

Step 3: Statistic.

$$z_{\text{obs}} = \frac{60.3 - 58.0}{0.9487} = \frac{2.3}{0.9487} = 2.424.$$

Step 4: Decision. Upper-tail cutoff $z_{0.05} = 1.645$. Since $z_{\text{obs}} = 2.424 \geq 1.645$, reject.

Reject H_0: the fertilizer raised mean yield ($z_{\text{obs}} = 2.42$).

Example 6.3.6 (Tensile strength below spec)**.** A cable must meet a mean tensile strength of at least $\mu_0 = 850$ N; the supplier's known process standard deviation is $\sigma = 20$ N. An inspector suspects a bad lot and tests $n = 16$ cables, finding $\overline{X} = 841$ N. Is the lot mean *below* the 850 N spec at $\alpha = 0.01$?

Which rule, and why? Mean, σ known, and "below spec" is a single downward direction $\Rightarrow$ one-sided (lower-tail) z-test with $H_1 : \mu < 850$.

Solution.

Step 1: Hypotheses. $H_0 : \mu = 850$ versus $H_1 : \mu < 850$, $\alpha = 0.01$.

Step 2: Standard error. $\sigma/\sqrt{n} = 20/\sqrt{16} = 20/4 = 5$ N.

Step 3: Statistic.

$$z_{\text{obs}} = \frac{841 - 850}{5} = \frac{-9}{5} = -1.80.$$

Step 4: Decision. Lower-tail cutoff is $-z_{0.01} = -2.326$. Since $z_{\text{obs}} = -1.80 > -2.326$, z_{obs} is *not* past the cutoff.

Fail to reject H_0 at $\alpha = 0.01$ ($z_{\text{obs}} = -1.80$).

The data lean low but not far enough to clear the strict 1% bar; at $\alpha = 0.05$ (cutoff -1.645) the same data *would* reject. The chosen α matters.

6.3.4 The p-value

A fixed-α decision throws away information: $z_{\text{obs}} = 2.0$ and $z_{\text{obs}} = 4.0$ both "reject at 0.05", yet the second is far more surprising. The p-value measures that surprise directly.

Definition 6.3.7 (p-value)**.** The **p-value** is the probability, computed *under* H_0, of getting a test statistic at least as extreme as the one observed, where "extreme" is read in the direction(s) named by H_1. Writing $Z \sim \text{N}(0, 1)$ and Φ for its cdf:

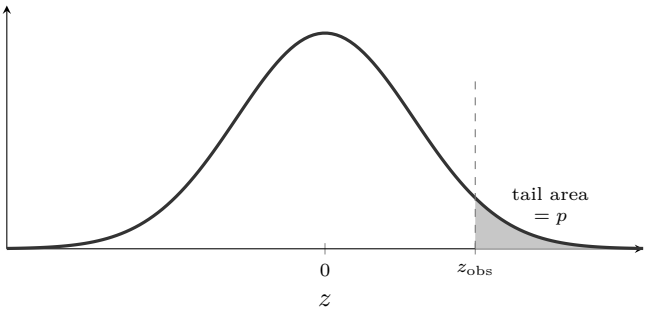
$$\begin{aligned}
H_1 : \mu \neq \mu_0 : &\quad p\text{-value} = \mathbb{P}(|Z| \geq |z_{\text{obs}}|) = 2\big(1 - \Phi(|z_{\text{obs}}|)\big), \\
H_1 : \mu > \mu_0 : &\quad p\text{-value} = \mathbb{P}(Z \geq z_{\text{obs}}) = 1 - \Phi(z_{\text{obs}}), \\
H_1 : \mu < \mu_0 : &\quad p\text{-value} = \mathbb{P}(Z \leq z_{\text{obs}}) = \Phi(z_{\text{obs}}).
\end{aligned}$$

Proposition 6.3.8 (Equivalence of the two decision forms)**.** *At level α, the p-value rule and the critical-value rule give the same decision:*

$$\textit{reject } H_0 \quad \Longleftrightarrow \quad \textit{p-value} \le \alpha.$$

Proof. Take the upper-tail case. The critical-value rule rejects iff $z_{\text{obs}} \ge z_\alpha$. Because $1 - \Phi$ is strictly decreasing, $z_{\text{obs}} \ge z_\alpha$ holds iff $1 - \Phi(z_{\text{obs}}) \le 1 - \Phi(z_\alpha) = \alpha$, and the left side is exactly the p-value. The two- and lower-tail cases follow by the same monotonicity applied to $2(1 - \Phi(|z_{\text{obs}}|))$ and $\Phi(z_{\text{obs}})$ respectively. □

Remark. The p-value is *not* the probability that H_0 is true. It is a statement about the data — "how unusual is a result this extreme *if* H_0 holds" — not about the hypothesis. A p-value of 0.03 means data this extreme occur 3% of the time when H_0 is true; it says nothing directly about $\mathbb{P}(H_0)$.



One-sided p-value picture: with the observed z_{obs} marked, the shaded area to its right is $1 - \Phi(z_{\text{obs}})$. **That shaded tail area IS the p-value** — the chance, under H_0, of a statistic at least as extreme as the one seen. For a two-sided test you shade both matching tails and the area doubles.

Example 6.3.9 (Reporting a two-sided p-value)**.** Reusing the bottling line (Example with $\mu_0 = 500$, $\sigma = 4$, $n = 25$, $\overline{X} = 502.1$, so $z_{\text{obs}} = 2.625$), report the two-sided p-value and interpret it.

Which rule, and why? Two-sided H_1, so "extreme" means far in either tail: p-value $= 2(1 - \Phi(|z_{\text{obs}}|))$.

Solution.

Step 1: Upper-tail area. From a standard normal table, $\Phi(2.625) \approx 0.99567$, so $1 - \Phi(2.625) \approx 0.00433$.

Step 2: Double it.

$$p\text{-value} = 2(0.00433) = 0.00866 \approx 0.0087.$$

Step 3: Interpret. If the line were truly on target, a sample mean this far from 500 (in either direction) would occur under 1% of the time. Since $0.0087 \le 0.05$, reject H_0 — consistent with the earlier critical-value decision.

$$\boxed{p\text{-value} \approx 0.0087 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

Example 6.3.10 (A one-sided p-value that does not reach significance)**.** A call center's known mean handle time is $\mu_0 = 6.0$ minutes with $\sigma = 1.5$ minutes. After a training program, management hopes the mean *dropped*. A sample of $n = 50$ calls gives $\overline{X} = 5.7$ minutes. Report the one-sided p-value and state the conclusion at $\alpha = 0.05$.

Which rule, and why? Mean, σ known, "dropped" is one direction ⇒ lower-tail z-test; the p-value is the lower-tail area $\Phi(z_{\text{obs}})$.

Solution.

Step 1: Statistic. $\sigma/\sqrt{n} = 1.5/\sqrt{50} = 1.5/7.0711 = 0.2121$, so

$$z_{\text{obs}} = \frac{5.7 - 6.0}{0.2121} = \frac{-0.3}{0.2121} = -1.414.$$

Step 2: Lower-tail area. p-value $= \Phi(-1.414) \approx 0.0786$.

Step 3: Decide. Since $0.0786 > 0.05$, do not reject.

$$\boxed{p\text{-value} \approx 0.079 > 0.05 \;\Rightarrow\; \text{fail to reject } H_0.}$$

The improvement is in the hoped-for direction but could plausibly be chance; evidence is suggestive, not significant.

Example 6.3.11 (Two-sided p-value, borderline case)**.** A thermostat is specced at $\mu_0 = 20.0°\text{C}$ with known $\sigma = 0.5°\text{C}$. An auditor takes $n = 64$ readings and finds $\overline{X} = 20.12°\text{C}$. Compute the two-sided p-value.

Which rule, and why? "Specced at" with a deviation either way is two-sided; σ is a known instrument tolerance $\Rightarrow$ two-sided z-test, $p = 2(1 - \Phi(|z_{\text{obs}}|))$.

Solution.

Step 1: Statistic. $\sigma/\sqrt{n} = 0.5/\sqrt{64} = 0.5/8 = 0.0625$, so

$$z_{\text{obs}} = \frac{20.12 - 20.00}{0.0625} = \frac{0.12}{0.0625} = 1.92.$$

Step 2: p-value. $\Phi(1.92) \approx 0.97257$, so $1 - \Phi(1.92) \approx 0.02743$ and

$$p\text{-value} = 2(0.02743) = 0.05486 \approx 0.055.$$

Step 3: Interpret. At $\alpha = 0.05$, $0.055 > 0.05$: fail to reject, but only just. The result sits right on the fence, which is exactly the kind of nuance a bare "reject/accept" would hide.

$$\boxed{p\text{-value} \approx 0.055 \;\Rightarrow\; \text{fail to reject at 0.05 (barely).}}$$

6.3.5 The large-sample CLT version

When the parent is not normal but n is large and σ is supplied, the same z-statistic is approximately standard normal by the Central Limit Theorem. The recognition cue is that you are handed a known σ even though the data are plainly skewed or discrete — counts of events, waiting times, ratings — and n is in the tens or more.

Example 6.3.12 (Mean number of defects per panel (skewed parent))**.** The number of surface defects per manufactured panel is right-skewed (roughly Poisson-like). A process is claimed to average $\mu_0 = 3.0$ defects/panel, and from extensive history the standard deviation is known to be $\sigma = 1.8$. A new shift inspects $n = 100$ panels and gets $\overline{X} = 3.42$. Test, at $\alpha = 0.05$, whether the mean defect count has *risen*.

Which rule, and why? The parent is clearly non-normal (skewed counts), but σ is supplied from history and $n = 100$ is large, so the CLT makes z_{obs} approximately standard normal: a large-sample one-sided z-test. We do *not* need a normal parent here.

Solution.

Step 1: Hypotheses. $H_0 : \mu = 3.0$ versus $H_1 : \mu > 3.0$, $\alpha = 0.05$.

Step 2: Standard error. $\sigma/\sqrt{n} = 1.8/\sqrt{100} = 1.8/10 = 0.18$.

Step 3: Statistic.

$$z_{\text{obs}} = \frac{3.42 - 3.0}{0.18} = \frac{0.42}{0.18} = 2.333.$$

Step 4: p-value and decision. Upper-tail p-value $= 1 - \Phi(2.333) \approx 0.0098$. Since $0.0098 \leq 0.05$ (equivalently $z_{\text{obs}} \geq z_{0.05} = 1.645$), reject.

$$\boxed{\text{Reject } H_0\text{: defect rate rose } (z_{\text{obs}} = 2.33,\; p \approx 0.010).}$$

Example 6.3.13 (Mean session length on an app (large sample, two-sided))**.** App session lengths are heavily right-skewed. The historical mean is $\mu_0 = 14.0$ minutes with a known $\sigma = 9.0$ minutes. After a redesign, an analyst samples $n = 400$ sessions and finds $\overline{X} = 14.95$ minutes and asks whether engagement *changed* (either way).

Which rule, and why? Skewed parent but large $n = 400$ and known $\sigma \Rightarrow$ large-sample z-test by the CLT; "changed either way" makes it two-sided.

Solution.

Step 1: Standard error. $\sigma/\sqrt{n} = 9.0/\sqrt{400} = 9.0/20 = 0.45$.

Step 2: Statistic.

$$z_{\text{obs}} = \frac{14.95 - 14.0}{0.45} = \frac{0.95}{0.45} = 2.111.$$

Step 3: p-value. Two-sided: $\Phi(2.111) \approx 0.98262$, so p-value $= 2(1 - 0.98262) = 2(0.01738) = 0.0348$.

Step 4: Decision. $0.0348 \leq 0.05$, so reject.

$$\boxed{\text{Reject } H_0\text{: session length changed } (z_{\text{obs}} = 2.11,\ p \approx 0.035).}$$

Example 6.3.14 (Environmental: mean lead concentration not above limit)**.** Drinking-water lead readings are skewed. A utility's regulatory limit is $\mu_0 = 15$ μg/L; the assay's known standard deviation is $\sigma = 4.5$. A reviewer wants evidence the true mean is *above* the limit and tests $n = 81$ samples, getting $\overline{X} = 15.6$. Use a one-sided test at $\alpha = 0.05$.

Which rule, and why? Skewed parent, large $n = 81$, known assay $\sigma \Rightarrow$ large-sample one-sided (upper) z-test.

Solution.

Step 1: Standard error. $\sigma/\sqrt{n} = 4.5/\sqrt{81} = 4.5/9 = 0.5$.

Step 2: Statistic.

$$z_{\text{obs}} = \frac{15.6 - 15.0}{0.5} = \frac{0.6}{0.5} = 1.20.$$

Step 3: p-value and decision. Upper-tail p-value $= 1 - \Phi(1.20) \approx 0.1151$. Since $0.1151 > 0.05$, do not reject.

$$\boxed{p\text{-value} \approx 0.115 > 0.05 \ \Rightarrow\ \text{fail to reject } H_0.}$$

There is not enough evidence to conclude the mean exceeds the limit.

Example 6.3.15 (Polling-style mean rating, two-sided p-value)**.** Customers rate a service 1–10; the long-run mean is $\mu_0 = 7.0$ with a known $\sigma = 2.0$ (the rating spread is stable across surveys). A new sample of $n = 64$ ratings averages $\overline{X} = 6.55$. Report the two-sided p-value.

Which rule, and why? Bounded discrete ratings are non-normal, but $n = 64$ is large and σ is known $\Rightarrow$ large-sample two-sided z-test.

Solution.

Step 1: Statistic. $\sigma/\sqrt{n} = 2.0/\sqrt{64} = 2.0/8 = 0.25$, so

$$z_{\text{obs}} = \frac{6.55 - 7.0}{0.25} = \frac{-0.45}{0.25} = -1.80.$$

Step 2: p-value. $\Phi(1.80) \approx 0.96407$, so p-value $= 2(1 - 0.96407) = 2(0.03593) = 0.0719$.

Step 3: Interpret. $0.0719 > 0.05$: fail to reject at 5%. The mean rating dipped, but the dip is within ordinary sampling noise.

$$\boxed{p\text{-value} \approx 0.072 \ \Rightarrow\ \text{fail to reject at } 0.05.}$$

6.3.6 Contrast: using z when σ was really estimated

The single most common misuse of the z-test is plugging a *sample* standard deviation into the σ slot and calling the result a z. When the spread was estimated from the same small sample, the extra uncertainty in that estimate fattens the true sampling distribution into a t with heavier tails; the z procedure then uses cutoffs that are too small and tail areas that are too thin, so it overstates significance and *understates the margin of error.*

Example 6.3.16 (CONTRAST: a z-test where a t-test was required)**.** A lab measures the boiling point of a solvent $n = 9$ times and reports $\overline{X} = 80.6°\text{C}$ with a *sample* standard deviation $s = 1.2°\text{C}$ computed from those same 9 readings. The target is $\mu_0 = 80.0°\text{C}$. A technician runs a two-sided test by treating s as if it were a known σ — a z-test. Show what that mistakenly concludes, and contrast it with the correct t-test of §4.

Which rule, and why? This is the look-alike trap: the standard deviation was *estimated from the $n = 9$ data*, so the correct statistic is a t with $n - 1 = 8$ degrees of freedom, not a z. Using z here is wrong; we run it only to expose the error.

Solution.

Step 1: The (incorrect) z computation. With $s/\sqrt{n} = 1.2/\sqrt{9} = 1.2/3 = 0.4$,

$$z_{\text{obs}} = \frac{80.6 - 80.0}{0.4} = \frac{0.6}{0.4} = 1.50.$$

Two-sided z p-value: $2(1 - \Phi(1.50)) = 2(0.0668) = 0.1336$. Cutoff $z_{0.025} = 1.96$.

Step 2: The correct t computation. The statistic is numerically the same number, $t_{\text{obs}} = 1.50$, but it must be compared to a t with 8 df, whose two-sided cutoff is $t_{0.025,8} = 2.306$ — noticeably larger than 1.96. The t-based two-sided p-value for $t_{\text{obs}} = 1.50$ on 8 df is about 0.172, larger than the z value 0.1336.

Step 3: What the error costs. Both happen to fail to reject here, but the z p-value (0.134) is smaller than the honest t p-value (0.172): the z shortcut makes the evidence look stronger than it is. Equivalently, a 95% interval built with z uses $\pm 1.96(0.4) = \pm 0.78$, whereas the correct t interval uses $\pm 2.306(0.4) = \pm 0.92$ — the z margin is understated by roughly 18%. With a borderline z_{obs} near 1.96, that gap is exactly what flips a correct "fail to reject" into a false "reject".

s estimated from the data $\Rightarrow$ use t_8, not z; z understates the margin.

Remark. The fix is mechanical: ask whether the number in the denominator's standard deviation came from *outside* this sample (a spec, a calibration, a known constant — then z) or from *inside* it (computed from the same X_i — then t). When n is large the two cutoffs nearly coincide ($t_{0.025,\infty} = 1.96 = z_{0.025}$), which is why the large-sample tests above can use z even with an estimated spread; the danger is concentrated in small samples.

6.4 Testing a Mean: the t-Test (σ Unknown)

In practice you almost never know the population standard deviation σ. The z-test of the previous section assumed it was handed to you; here we face the realistic case where σ is unknown and must be estimated from the very same sample we are testing. The natural move is to replace σ by the sample standard deviation S in the standardizing ratio, giving the statistic $t = (\overline{X} - \mu_0)/(S/\sqrt{n})$. The price for estimating σ is that this ratio no longer has a standard normal distribution: it has *more* spread, because S wobbles from sample to sample on top of $\overline{X}$, and that extra variability fattens the tails. The exact reference distribution is the **Student t distribution** with $n - 1$ degrees of freedom. Recognize this test by three cues at once: the parameter in question is a *mean*, the parent population is (at least approximately) *normal*, and the population σ is *not supplied* — only the data, from which you will compute S yourself.

Is this a t-test — and which tail?

- Testing a **mean** μ, parent roughly **normal**, and σ **unknown** (you only have the data)? $\Rightarrow$ the **one-sample t-test**: $t_{\text{obs}} = (\overline{X} - \mu_0)/(S/\sqrt{n})$, compared to t_{n-1}.
- σ is actually **given** (or n is large and a known σ is supplied)? $\Rightarrow$ that is the **z-test** of §3, not this section.
- Two measurements on each subject (before/after, left/right, two methods on one unit)? $\Rightarrow$ form the **differences** d_i and run a one-sample t-test on them against $\mu_0 = 0$.
- **One-sided or two-sided?** "has increased" / "exceeds" $\Rightarrow$ upper tail, reject when $t_{\text{obs}} \geq t_{\alpha,\, n-1}$. "has decreased" / "is below" $\Rightarrow$ lower tail, $t_{\text{obs}} \leq -t_{\alpha,\, n-1}$. "differs" / "is not equal" $\Rightarrow$ both tails, $|t_{\text{obs}}| \geq t_{\alpha/2,\, n-1}$.

Do not confuse the t with the z: for the same α the t critical value is *larger* than the z value (and shrinks toward it as n grows). Using z when σ was estimated understates how extreme your data must be — it is anti-conservative and inflates the true Type I error.

6.4.1 The statistic and its distribution

Definition 6.4.1 (Sample standard deviation)**.** For a sample $X_1, \ldots, X_n$ with mean $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, the **sample variance** is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2 = \frac{1}{n-1}\left(\sum_{i=1}^{n} X_i^2 - n\,\overline{X}^{\,2}\right),$$

and the **sample standard deviation** is $S = \sqrt{S^2}$. The divisor $n-1$ (not n) makes S^2 an unbiased estimator of σ^2; the right-hand "computing form" is the convenient one for hand calculation.

Theorem 6.4.2 (Reference distribution for the t statistic)**.** *Let $X_1, \ldots, X_n$ be independent $\mathrm{N}(\mu, \sigma^2)$. Then*

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{\,n-1},$$

the Student t distribution with $n-1$ degrees of freedom, whatever the value of σ. *Consequently, if $H_0 : \mu = \mu_0$ is true, the test statistic $t_{obs} = (\overline{X} - \mu_0)/(S/\sqrt{n})$ has the t_{n-1} distribution.*

Proof. Write the ratio as

$$\frac{\overline{X} - \mu}{S/\sqrt{n}} = \frac{(\overline{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{(n-1)S^2/\sigma^2/(n-1)}} = \frac{Z}{\sqrt{V/(n-1)}}.$$

The numerator $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal. The quantity $V = (n-1)S^2/\sigma^2$ is χ^2_{n-1}, and for a normal sample $\overline{X}$ and S^2 are independent, so Z and V are independent. A standard normal over the square root of an independent chi-square divided by its degrees of freedom is, by definition, t_{n-1}. □

Remark. Why fatter tails? In the z-statistic only the numerator $\overline{X}$ is random; the denominator $\sigma/\sqrt{n}$ is a fixed constant. In the t-statistic the denominator $S/\sqrt{n}$ is also random and, in small samples, can land well below $\sigma/\sqrt{n}$, occasionally producing a large ratio even when $\overline{X} - \mu_0$ is modest. That extra source of variation pushes probability into the tails. As $n \to \infty$, $S \to \sigma$ in probability, the denominator stops wobbling, and $t_{n-1} \to \mathrm{N}(0,1)$. By $n \approx 30$ the difference is already small; for $n = 10$ it is not (compare $t_{0.025,9} = 2.262$ against $z_{0.025} = 1.96$).

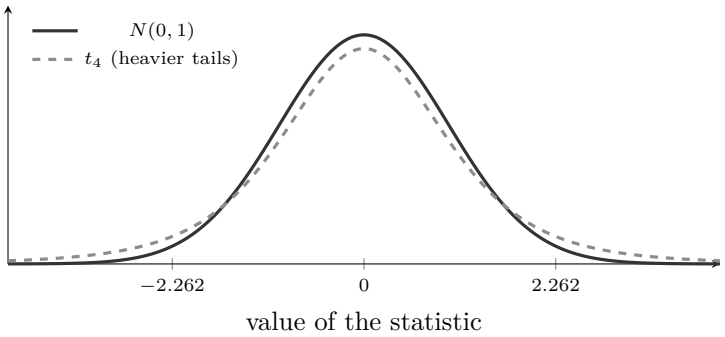


Figure: the t_4 density (dashed) plotted against the standard normal. The t curve is lower in the center and higher in the tails — the cost of estimating σ. The t_4 curve is its exact density, normalized by the constant $\Gamma(\frac{5}{2})/(\sqrt{4\pi}\,\Gamma(2)) = 0.375$; for larger degrees of freedom the two curves become nearly indistinguishable.

Definition 6.4.3 (One-sample t-test)**.** To test $H_0 : \mu = \mu_0$ for a normal population with unknown σ, at level α, compute $t_{\text{obs}} = (\overline{X} - \mu_0)/(S/\sqrt{n})$ and reject H_0 when:

$$\begin{aligned} H_1 : \mu > \mu_0 \quad &\Longrightarrow \quad t_{\text{obs}} \geq t_{\alpha,\, n-1}, \\ H_1 : \mu < \mu_0 \quad &\Longrightarrow \quad t_{\text{obs}} \leq -\,t_{\alpha,\, n-1}, \\ H_1 : \mu \neq \mu_0 \quad &\Longrightarrow \quad |t_{\text{obs}}| \geq t_{\alpha/2,\, n-1}. \end{aligned}$$

Here $t_{\alpha,\, n-1}$ is the upper-α critical value of t_{n-1}.

6.4.2 Two-sided tests

Example 6.4.4 (Fill weight of cereal boxes)**.** A machine is set to fill boxes to a nominal $\mu_0 = 500$ grams. A quality engineer weighs a sample of $n = 10$ boxes and finds $\overline{X} = 496.2$ g with sample standard deviation $S = 5.4$ g. Is there evidence at $\alpha = 0.05$ that the machine is off target (in either direction)? Take $t_{0.025,9} = 2.262$.

Which rule, and why? A mean is in question, weights are approximately normal, and σ is not given — S was computed from the sample. That is the one-sample t-test with $n - 1 = 9$ degrees of freedom. "Off target in either direction" makes it two-sided.

Solution.

Step 1: hypotheses. $H_0 : \mu = 500$ versus $H_1 : \mu \neq 500$, two-sided, $\alpha = 0.05$.

Step 2: standard error. $S/\sqrt{n} = 5.4/\sqrt{10} = 5.4/3.162 = 1.708$ g.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{496.2 - 500}{1.708} = \frac{-3.8}{1.708} = -2.224.$$

Step 4: critical value and decision. The two-sided 5% cutoff is $t_{0.025,9} = 2.262$. Since $|t_{\text{obs}}| = 2.224 < 2.262$, we do not cross the boundary.

> $|t_{\text{obs}}| = 2.224 < 2.262 \;\Rightarrow$ do not reject H_0;
> no significant evidence the machine is off target at $\alpha = 0.05$.

Note how close this is: with the slightly smaller z cutoff 1.96 we *would* have rejected. The next contrast example makes that danger explicit.

Example 6.4.5 (Calibration of a laboratory scale)**.** A reference mass of $\mu_0 = 100.0$ mg is weighed $n = 16$ times on a scale that should be unbiased. The readings give $\overline{X} = 100.37$ mg and $S = 0.62$ mg. Test at $\alpha = 0.05$ whether the scale reads differently from the true mass. Use $t_{0.025,15} = 2.131$.

Which rule, and why? Mean of a normal measurement process, σ unknown (estimated by S), two-sided wording "reads differently" — one-sample two-sided t-test, $df = 15$.

Solution.

Step 1: hypotheses. $H_0 : \mu = 100.0$ versus $H_1 : \mu \neq 100.0$.

Step 2: standard error. $S/\sqrt{n} = 0.62/\sqrt{16} = 0.62/4 = 0.155$ mg.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{100.37 - 100.0}{0.155} = \frac{0.37}{0.155} = 2.387.$$

Step 4: decision. Compare $|t_{\text{obs}}| = 2.387$ with $t_{0.025,15} = 2.131$. Since $2.387 > 2.131$, the statistic falls in the rejection region.

$$\boxed{|t_{\text{obs}}| = 2.387 > 2.131 \;\Rightarrow\; \text{reject } H_0;\ \text{the scale reads significantly high at } \alpha = 0.05.}$$

6.4.3 One-sided tests

Example 6.4.6 (Has a fertilizer raised yield?)**.** Historically a plot variety yields $\mu_0 = 42$ bushels per acre. An agronomist applies a new fertilizer to $n = 12$ plots and records $\overline{X} = 44.8$ bushels with $S = 4.1$. Test at $\alpha = 0.05$ whether the fertilizer *increased* mean yield. Use $t_{0.05,11} = 1.796$.

Which rule, and why? Mean yield, normal, σ unknown (S from the data). The claim "increased" points in one direction, so this is an *upper-tail* one-sample t-test, $df = 11$, with all of α in the right tail.

Solution.

Step 1: hypotheses. $H_0 : \mu = 42$ versus $H_1 : \mu > 42$ (upper tail).

Step 2: standard error. $S/\sqrt{n} = 4.1/\sqrt{12} = 4.1/3.464 = 1.184$.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{44.8 - 42}{1.184} = \frac{2.8}{1.184} = 2.365.$$

Step 4: decision. For an upper-tail test put all 5% in the right tail: $t_{0.05,11} = 1.796$. Since $t_{\text{obs}} = 2.365 > 1.796$, reject.

$$\boxed{t_{\text{obs}} = 2.365 > 1.796 \;\Rightarrow\; \text{reject } H_0;\ \text{evidence the fertilizer increased yield at } \alpha = 0.05.}$$

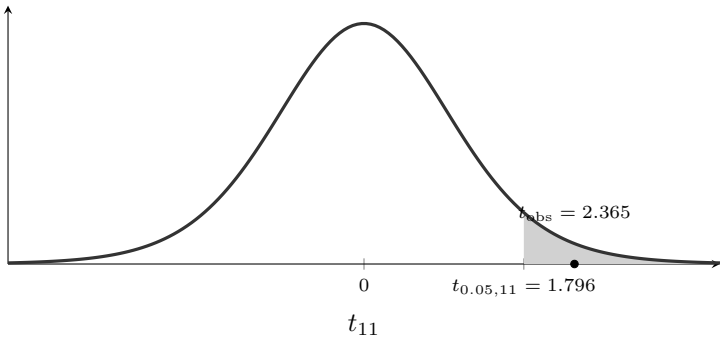


Figure: the upper-tail rejection region of the t_{11} test. All of $\alpha = 0.05$ sits to the right of 1.796 (shaded); the observed $t_{obs} = 2.365$ (dot) lands inside it, so we reject. The curve is the t_{11} density up to a normalizing constant.

Example 6.4.7 (Is the impurity level below the limit?)**.** A regulation caps mean impurity at $\mu_0 = 3.0$ parts per million; a supplier claims its process runs *below* the cap. From $n = 9$ batches: $\overline{X} = 2.74$ ppm, $S = 0.30$ ppm. Test at $\alpha = 0.05$ whether the mean is below 3.0. Use $t_{0.05,8} = 1.860$.

Which rule, and why? Mean, normal, σ unknown. The directional claim "below the cap" is a *lower-tail* one-sample t-test, $df = 8$; reject only for sufficiently negative t_{obs}.

Solution.

Step 1: hypotheses. $H_0 : \mu = 3.0$ versus $H_1 : \mu < 3.0$ (lower tail).

Step 2: standard error. $S/\sqrt{n} = 0.30/\sqrt{9} = 0.30/3 = 0.100$ ppm.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{2.74 - 3.0}{0.100} = \frac{-0.26}{0.100} = -2.60.$$

Step 4: decision. Lower-tail cutoff is $-t_{0.05,8} = -1.860$. Since $t_{\text{obs}} = -2.60 < -1.860$, reject.

$$\boxed{t_{\text{obs}} = -2.60 < -1.860 \;\Rightarrow\; \text{reject } H_0;\ \text{the mean impurity is significantly below 3.0 ppm.}}$$

6.4.4 Starting from raw data: computing $\overline{X}$ and S

Example 6.4.8 (Reaction times from raw measurements)**.** A psychologist measures simple reaction time (in milliseconds) for $n = 8$ participants and suspects the mean differs from the textbook value $\mu_0 = 250$ ms. The raw data are

$$242,\ 268,\ 255,\ 231,\ 260,\ 248,\ 271,\ 253.$$

Compute $\overline{X}$ and S from scratch, then test $H_0 : \mu = 250$ versus $H_1 : \mu \neq 250$ at $\alpha = 0.05$. Use $t_{0.025,7} = 2.365$.

Which rule, and why? Mean of an approximately normal population, σ unknown and *not even S given* — we must build S from the raw numbers, then run a two-sided one-sample t-test, $df = 7$.

Solution.

Step 1: sample mean. Sum the data:

$$\sum x_i = 242 + 268 + 255 + 231 + 260 + 248 + 271 + 253 = 2028, \qquad \overline{X} = \frac{2028}{8} = 253.5.$$

Step 2: deviations and squared deviations. With $\overline{X} = 253.5$:

$$(-11.5,\ 14.5,\ 1.5,\ -22.5,\ 6.5,\ -5.5,\ 17.5,\ -0.5) \quad (\text{the } x_i - \overline{X}),$$
$$132.25 + 210.25 + 2.25 + 506.25 + 42.25 + 30.25 + 306.25 + 0.25 = 1230.0.$$

Step 3: sample variance and SD. Divide the sum of squares by $n - 1 = 7$:

$$S^2 = \frac{1230.0}{7} = 175.71, \qquad S = \sqrt{175.71} = 13.26 \text{ ms}.$$

Step 4: standard error and statistic. $S/\sqrt{n} = 13.26/\sqrt{8} = 13.26/2.828 = 4.688$,

$$t_{\text{obs}} = \frac{253.5 - 250}{4.688} = \frac{3.5}{4.688} = 0.747.$$

Step 5: decision. Compare $|t_{\text{obs}}| = 0.747$ with $t_{0.025,7} = 2.365$. Since $0.747 < 2.365$, do not reject.

$$\boxed{\begin{array}{l} |t_{\text{obs}}| = 0.747 < 2.365 \;\Rightarrow\; \text{do not reject } H_0; \\ \text{no evidence the mean reaction time differs from 250 ms.} \end{array}}$$

Example 6.4.9 (Sum-of-squares shortcut on raw data)**.** Five thickness gauges of a coating (in microns) read 20, 23, 19, 25, 18. The target is $\mu_0 = 20$ microns. Using the computing form $S^2 = \frac{1}{n-1}(\sum x_i^2 - n\overline{X}^2)$, test $H_0 : \mu = 20$ versus $H_1 : \mu \neq 20$ at $\alpha = 0.05$. Use $t_{0.025,4} = 2.776$.

Which rule, and why? Same recognition — mean, normal, σ unknown — but here we practice the computing form of S^2, which avoids listing every deviation. Two-sided t-test, $df = 4$.

Solution.

Step 1: the two sums.

$$\sum x_i = 20 + 23 + 19 + 25 + 18 = 105, \qquad \overline{X} = \frac{105}{5} = 21,$$
$$\sum x_i^2 = 400 + 529 + 361 + 625 + 324 = 2239.$$

Step 2: variance by the computing form.

$$S^2 = \frac{1}{4}\big(2239 - 5 \cdot 21^2\big) = \frac{1}{4}\big(2239 - 2205\big) = \frac{34}{4} = 8.5, \qquad S = \sqrt{8.5} = 2.915.$$

Step 3: statistic. $S/\sqrt{n} = 2.915/\sqrt{5} = 2.915/2.236 = 1.304$,

$$t_{\text{obs}} = \frac{21 - 20}{1.304} = \frac{1}{1.304} = 0.767.$$

Step 4: decision. $|t_{\text{obs}}| = 0.767 < t_{0.025,4} = 2.776$.

$$\boxed{|t_{\text{obs}}| = 0.767 < 2.776 \;\Rightarrow\; \text{do not reject } H_0;\ \text{thickness consistent with the 20-micron target.}}$$

6.4.5 Contrast: t versus z

Example 6.4.10 (The look-alike z-test gives the wrong call)**.** A coffee dispenser should pour $\mu_0 = 240$ mL. A sample of $n = 10$ cups gives $\overline{X} = 243.1$ mL and $S = 4.6$ mL; the true σ is *not* known. A careless analyst standardizes with S but then compares the result to the *normal* cutoff $z_{0.025} = 1.96$ instead of the t cutoff $t_{0.025,9} = 2.262$. Test $H_0 : \mu = 240$ versus $H_1 : \mu \neq 240$ at $\alpha = 0.05$ both ways and show the calls disagree.

Which rule, and why? Because σ is unknown and replaced by S, the correct reference distribution is t_9, *not* $\mathrm{N}(0, 1)$. The z comparison is the classic look-alike error; we compute both to expose it.

Solution.

Step 1: the (common) test statistic. Both analysts compute the same number:

$$\text{stat} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{243.1 - 240}{4.6/\sqrt{10}} = \frac{3.1}{4.6/3.162} = \frac{3.1}{1.455} = 2.131.$$

Step 2: the wrong (anti-conservative) z call. Comparing to $z_{0.025} = 1.96$: $|2.131| > 1.96$, so the z-user *rejects* H_0 and declares the dispenser off target.

Step 3: the correct t call. The proper cutoff is $t_{0.025,9} = 2.262 > 1.96$. Comparing the same 2.131 to 2.262: $|2.131| < 2.262$, so the t-test does *not* reject.

Step 4: why the t cutoff is larger. Estimating σ by S adds variability, fattening the tails, so the threshold for "too extreme" must move outward from 1.96 to 2.262. The z-test ignores this and rejects too readily — its true Type I error exceeds the advertised 5%.

$$\boxed{\begin{array}{l} z \text{ says reject, } t \text{ says do not; the } t\text{-test is correct} \\ \text{because } \sigma \text{ was estimated, and } t_{0.025,9} = 2.262 > 1.96 = z_{0.025}. \end{array}}$$

Remark. The gap between the cutoffs shrinks with sample size: $t_{0.025,9} = 2.262$, $t_{0.025,24} = 2.064$, $t_{0.025,60} = 2.000$, all closing on $z_{0.025} = 1.96$. For large n the two tests essentially agree, which is why the z-test is a defensible large-sample approximation — but in small samples, as above, it can flip the decision.

6.4.6 Paired differences

Example 6.4.11 (Before-and-after training scores)**.** Seven analysts each take a skills test before and after a training course; the paired scores are

Analyst	1	2	3	4	5	6	7
Before	71	65	80	74	69	78	72
After	76	70	79	81	75	80	78

Test at $\alpha = 0.05$ whether the training *raised* mean scores. Use $t_{0.05,6} = 1.943$.

Which rule, and why? Two measurements on each *same* analyst makes the observations paired, not two independent samples. Reduce each pair to its difference $d_i = \text{After} - \text{Before}$, then test whether the mean difference is zero with a one-sample t-test on the d_i, $df = 6$. "Raised" is upper-tail.

Solution.

Step 1: form the differences.

$$d_i: \quad 5,\ 5,\ -1,\ 7,\ 6,\ 2,\ 6.$$

Step 2: mean of differences. $\sum d_i = 5 + 5 - 1 + 7 + 6 + 2 + 6 = 30$, so $\bar{d} = 30/7 = 4.286$.

Step 3: SD of differences. Deviations $d_i - \bar{d}$: $(0.714, 0.714, -5.286, 2.714, 1.714, -2.286, 1.714)$; squared and summed,

$$\sum (d_i - \bar{d})^2 = 0.510 + 0.510 + 27.94 + 7.366 + 2.938 + 5.224 + 2.938 = 47.43,$$

$$S_d^2 = \frac{47.43}{6} = 7.905, \qquad S_d = \sqrt{7.905} = 2.812.$$

Step 4: statistic. Here $\mu_0 = 0$; $S_d/\sqrt{n} = 2.812/\sqrt{7} = 2.812/2.646 = 1.063$,

$$t_{\text{obs}} = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{4.286}{1.063} = 4.032.$$

Step 5: decision. Upper-tail cutoff $t_{0.05,6} = 1.943$. Since $4.032 > 1.943$, reject.

$$\boxed{t_{\text{obs}} = 4.032 > 1.943 \ \Rightarrow\ \text{reject } H_0;\ \text{the training significantly raised mean scores.}}$$

Example 6.4.12 (Two instruments on the same specimens (two-sided paired))**.** Two thermometers, A and B, each measure the same $n = 6$ specimens. The differences $d_i = A - B$ (in °C) come out to 0.3, −0.1, 0.4, 0.2, 0.5, 0.1. Is there evidence at $\alpha = 0.05$ that the instruments *disagree on average* (either direction)? Use $t_{0.025,5} = 2.571$.

Which rule, and why? Same specimens measured twice means paired data. Test the mean difference against 0 with a one-sample t-test on the d_i; "disagree on average" with no direction makes it two-sided, $df = 5$.

Solution.

Step 1: mean difference. $\sum d_i = 0.3 - 0.1 + 0.4 + 0.2 + 0.5 + 0.1 = 1.4$, $\bar{d} = 1.4/6 = 0.2333$.

Step 2: SD of differences (computing form). $\sum d_i^2 = 0.09 + 0.01 + 0.16 + 0.04 + 0.25 + 0.01 = 0.56$,

$$S_d^2 = \frac{1}{5}\big(0.56 - 6 \cdot 0.2333^2\big) = \frac{1}{5}\big(0.56 - 0.3266\big) = \frac{0.2334}{5} = 0.04668, \quad S_d = 0.2161.$$

Step 3: statistic. $S_d/\sqrt{n} = 0.2161/\sqrt{6} = 0.2161/2.449 = 0.08824$,

$$t_{\text{obs}} = \frac{0.2333 - 0}{0.08824} = 2.644.$$

Step 4: decision. $|t_{\text{obs}}| = 2.644 > t_{0.025,5} = 2.571$, so reject.

$$\boxed{|t_{\text{obs}}| = 2.644 > 2.571 \ \Rightarrow\ \text{reject } H_0;\ \text{the two thermometers disagree on average at } \alpha = 0.05.}$$

6.4.7 The p-value with the t distribution

Definition 6.4.13 (p-value of a t-test)**.** The p-value is the probability, computed under H_0, of a test statistic at least as extreme as the observed t_{obs}, in the direction of H_1. With $T \sim t_{n-1}$:

$$\begin{aligned} H_1 : \mu > \mu_0 : &\quad p = \mathbb{P}(T \geq t_{\text{obs}}), \\ H_1 : \mu < \mu_0 : &\quad p = \mathbb{P}(T \leq t_{\text{obs}}), \\ H_1 : \mu \neq \mu_0 : &\quad p = 2\,\mathbb{P}\big(T \geq |t_{\text{obs}}|\big). \end{aligned}$$

Reject H_0 at level α iff $p \leq \alpha$. Because t tables are coarse, by hand one usually *brackets* the p-value between adjacent table columns.

Example 6.4.14 (Bracketing a one-sided p-value)**.** A battery vendor claims mean life exceeds $\mu_0 = 500$ hours. From $n = 16$ batteries, $\overline{X} = 512$ h and $S = 24$ h. Report the p-value for $H_1 : \mu > 500$. Relevant t_{15} upper-tail values: $t_{0.05,15} = 1.753$, $t_{0.025,15} = 2.131$, $t_{0.01,15} = 2.602$.

Which rule, and why? Mean, normal, σ unknown; the question asks for *strength of evidence*, not a fixed accept/reject, so we report a p-value from t_{15}. "Exceeds" is upper-tail.

Solution.

Step 1: test statistic. $S/\sqrt{n} = 24/\sqrt{16} = 24/4 = 6$,

$$t_{\text{obs}} = \frac{512 - 500}{6} = \frac{12}{6} = 2.00.$$

Step 2: locate t_{obs} among the critical values. We have $1.753 < 2.00 < 2.131$, i.e. $t_{0.05,15} < t_{\text{obs}} < t_{0.025,15}$. The upper-tail area to the right of t_{obs} therefore lies between the corresponding tail probabilities.

Step 3: bracket the p-value. Since the area right of 1.753 is 0.05 and the area right of 2.131 is 0.025, and $t_{\text{obs}} = 2.00$ sits between them,

$$0.025 < p < 0.05.$$

Step 4: decision and reading. The whole bracket is below 0.05, so we reject at $\alpha = 0.05$, but the evidence is moderate, not strong: we would *not* reject at $\alpha = 0.025$ or 0.01.

$$\boxed{p \in (0.025,\ 0.05);\ \text{reject at } \alpha = 0.05,\ \text{retain at } \alpha = 0.025.}$$

Example 6.4.15 (Two-sided p-value, doubled tail)**.** A process targets pH $\mu_0 = 7.00$. A sample of $n = 25$ readings gives $\overline{X} = 7.06$ and $S = 0.14$. Report the two-sided p-value for $H_1 : \mu \neq 7.00$. Relevant t_{24} values: $t_{0.05,24} = 1.711$, $t_{0.025,24} = 2.064$, $t_{0.01,24} = 2.492$.

Which rule, and why? Mean, normal, σ unknown, two-sided alternative — a one-sample t-test with $df = 24$, and we want the evidence strength, so we double the one-tail area.

Solution.

Step 1: test statistic. $S/\sqrt{n} = 0.14/\sqrt{25} = 0.14/5 = 0.028$,

$$t_{\text{obs}} = \frac{7.06 - 7.00}{0.028} = \frac{0.06}{0.028} = 2.143.$$

Step 2: bracket the one-tail area. With $|t_{\text{obs}}| = 2.143$ and $2.064 < 2.143 < 2.492$, the one-tail area $\mathbb{P}(T \geq 2.143)$ lies between 0.01 and 0.025.

Step 3: double for the two-sided p-value.

$$p = 2\,\mathbb{P}(T \geq 2.143) \in (2 \cdot 0.01,\ 2 \cdot 0.025) = (0.02,\ 0.05).$$

Step 4: decision. The whole bracket is at or below 0.05, so we reject at $\alpha = 0.05$; but since it straddles values above 0.02, we cannot claim significance at $\alpha = 0.02$.

$$\boxed{p \in (0.02,\ 0.05);\ \text{reject at } \alpha = 0.05.}$$

Example 6.4.16 (When the evidence is weak: a large p-value)**.** A nutrition label claims a snack bar averages $\mu_0 = 200$ calories. A consumer lab measures $n = 12$ bars: $\overline{X} = 203$ calories, $S = 14$. Report the two-sided p-value and decide at $\alpha = 0.05$. Relevant t_{11} values: $t_{0.10,11} = 1.363$, $t_{0.05,11} = 1.796$.

Which rule, and why? Mean, normal, σ unknown, two-sided question about strength of evidence — one-sample t-test, $df = 11$, p-value by doubling the tail.

Solution.

Step 1: test statistic. $S/\sqrt{n} = 14/\sqrt{12} = 14/3.464 = 4.041$,

$$t_{\text{obs}} = \frac{203 - 200}{4.041} = \frac{3}{4.041} = 0.742.$$

Step 2: bracket. Here $|t_{\text{obs}}| = 0.742$ is even smaller than $t_{0.10,11} = 1.363$, so the one-tail area exceeds 0.10; doubling, $p > 0.20$.

Step 3: decision. Far above $\alpha = 0.05$, so do not reject. The 3-calorie difference is well within sampling noise for this sample size.

$$\boxed{p > 0.20 \;\Rightarrow\; \text{do not reject } H_0;\ \text{no evidence the label is wrong.}}$$

Remark. The p-value and the fixed-α rule are two views of the same decision. Rejecting because $|t_{\text{obs}}| \geq t_{\alpha/2,n-1}$ is identical to rejecting because $p \leq \alpha$ — the critical value is just the statistic whose tail area equals α. Reporting the p-value (or a bracket for it) carries strictly more information than the bare accept/reject, since the reader can apply whatever α they prefer.

6.5 Testing a Proportion (Binomial Data)

Often the unknown parameter is not a mean but a **success probability** p: the fraction of voters who approve, the defect rate of a process, the cure rate of a drug, the pass rate of a shipment. We observe a count X of successes in n independent trials, model $X \sim \text{Bin}(n, p)$, and want to test $H_0 : p = p_0$ against a one- or two-sided alternative. There are two honest routes. When n is small we can read tail probabilities *exactly* off the $\text{Bin}(n, p_0)$ probability mass function — this is the **exact binomial test**. When n is large the De Moivre–Laplace central limit theorem makes $\widehat{p} = X/n$ approximately normal, and we use the **large-sample z-test** $z_{\text{obs}} = (\widehat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$. The single most important thing to get right is the standard error: in a *test* you plug in p_0, the null value, because under H_0 that is the true variance of $\widehat{p}$; this is *not* the $\sqrt{\widehat{p}(1-\widehat{p})/n}$ you would use for a confidence interval. The two differ, and the difference is the classic exam trap.

Which proportion test does the problem call for?

- Counting **successes out of n trials** and testing $p = p_0$? $\Rightarrow$ a test for a binomial proportion; pick exact or large-sample by the next two items.
- n **small**, or p_0 near 0 or 1, or you want an exact answer? $\Rightarrow$ the **exact binomial test**: sum $\text{Bin}(n, p_0)$ pmf values in the tail at least as extreme as the observed count.
- n **large** with $np_0 \geq 10$ and $n(1 - p_0) \geq 10$? $\Rightarrow$ the **large-sample z-test** with $z_{\text{obs}} = (\widehat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$.
- **One-sided** ("has risen", "has dropped", "exceeds the target") puts all of α in one tail; **two-sided** ("differs from", "$\neq$") splits it.

Do not confuse the test SE with the interval SE: the test statistic divides by $\sqrt{p_0(1-p_0)/n}$ (the null value p_0), *never* by $\sqrt{\widehat{p}(1-\widehat{p})/n}$. Using $\widehat{p}$ in the denominator is a confidence-interval move, and it gives a different, wrong z.

6.5.1 The large-sample z-test for a proportion

Definition 6.5.1 (Large-sample test for p)**.** Let $X \sim \mathrm{Bin}(n, p)$ and $\widehat{p} = X/n$. To test $H_0 : p = p_0$ use

$$z_{\mathrm{obs}} = \frac{\widehat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}.$$

Under H_0, z_{obs} is approximately $\mathrm{N}(0,1)$. Reject at level α when: $z_{\mathrm{obs}} \geq z_\alpha$ for $H_1 : p > p_0$; $z_{\mathrm{obs}} \leq -z_\alpha$ for $H_1 : p < p_0$; and $|z_{\mathrm{obs}}| \geq z_{\alpha/2}$ for $H_1 : p \neq p_0$.

Why the null value goes in the standard error. If H_0 is true then $X \sim \mathrm{Bin}(n, p_0)$, so $\mathbb{E}[\widehat{p}] = p_0$ and $\mathrm{Var}(\widehat{p}) = p_0(1-p_0)/n$ *exactly*. Standardizing $\widehat{p}$ by its true null mean and standard deviation gives a statistic with mean 0 and variance 1, and the CLT makes it approximately normal. The variance is a known number once p_0 is fixed — there is nothing to estimate — so we must use p_0, not $\widehat{p}$, in the denominator. The interval SE $\sqrt{\widehat{p}(1-\widehat{p})/n}$ is correct in a different problem (estimating an unknown p with no null in hand), but it is the wrong yardstick here. □

Remark. The validity condition $np_0 \geq 10$ and $n(1-p_0) \geq 10$ keeps the binomial close enough to normal that the z approximation is trustworthy. If either fails — small n, or a p_0 pressed against 0 or 1 — abandon the z-test and run the exact binomial test of the next subsection.

Example 6.5.2 (Approval rating, two-sided)**.** A city claims that exactly half its residents support a new transit levy. A poll of $n = 400$ residents finds 232 in favor. At $\alpha = 0.05$, does support differ from 50%?

Which rule, and why? We count successes (232 of 400) and test $p = p_0 = 0.50$. Both $np_0 = 200$ and $n(1-p_0) = 200$ exceed 10, so the large-sample z-test applies; the word "differ" makes it two-sided.

Solution.

Step 1: estimate. $\widehat{p} = 232/400 = 0.58$.

Step 2: test SE with p_0.

$$\mathrm{SE}_0 = \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.5 \cdot 0.5}{400}} = \sqrt{0.000625} = 0.025.$$

Step 3: statistic. $z_{\mathrm{obs}} = (0.58 - 0.50)/0.025 = 0.08/0.025 = 3.20$.

Step 4: compare. Two-sided cutoff $z_{0.025} = 1.96$; since $|3.20| \geq 1.96$ we reject.

$$\boxed{z_{\mathrm{obs}} = 3.20 > 1.96 \;\Rightarrow\; \text{reject } H_0\text{: support differs from 50\%.}}$$

Example 6.5.3 (Defect rate has dropped, one-sided)**.** A line historically runs a 6% defect rate. After a retooling, an audit of $n = 500$ units finds 21 defective. At $\alpha = 0.05$, is there evidence the defect rate has *dropped* below 6%?

Which rule, and why? Counting defectives out of 500 and testing $p = 0.06$; $np_0 = 30$ and $n(1-p_0) = 470$ both exceed 10, so the large-sample z-test applies. "Dropped below" is a left-sided alternative $H_1 : p < 0.06$.

Solution.

Step 1: estimate. $\widehat{p} = 21/500 = 0.042$.

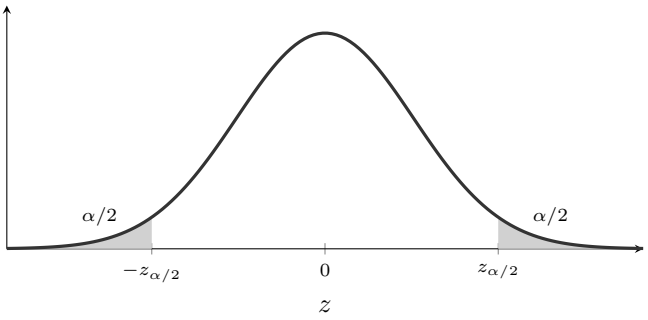
Step 2: test SE.

$$\mathrm{SE}_0 = \sqrt{\frac{0.06 \cdot 0.94}{500}} = \sqrt{\frac{0.0564}{500}} = \sqrt{0.0001128} = 0.010621.$$

Step 3: statistic. $z_{\mathrm{obs}} = (0.042 - 0.060)/0.010621 = -0.018/0.010621 = -1.695$.

Step 4: compare. Left-sided cutoff $-z_{0.05} = -1.645$; since $-1.695 \leq -1.645$ we reject.

$$\boxed{z_{\mathrm{obs}} = -1.695 \leq -1.645 \;\Rightarrow\; \text{reject } H_0\text{: the defect rate has dropped.}}$$



Two-sided rejection region for the proportion z-statistic: reject H_0 when z_{obs} lands in either shaded tail, each carrying area $\alpha/2$.

Example 6.5.4 (Vaccine response rate has risen, one-sided)**.** An old vaccine seroconverts 80% of recipients. A reformulated dose is given to $n = 150$ volunteers and 129 seroconvert. At $\alpha = 0.01$, is the new rate *higher* than 80%?

Which rule, and why? Successes out of trials, testing $p = 0.80$; $np_0 = 120$ and $n(1 - p_0) = 30$ exceed 10, so the z-test applies. "Higher than" is a right-sided alternative $H_1 : p > 0.80$ at $\alpha = 0.01$.

Solution.

Step 1: estimate. $\widehat{p} = 129/150 = 0.86$.

Step 2: test SE.

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{0.80 \cdot 0.20}{150}} = \sqrt{\frac{0.16}{150}} \\ &= \sqrt{0.0010667} = 0.032660. \end{aligned}$$

Step 3: statistic. $z_{\text{obs}} = (0.86 - 0.80)/0.032660 = 0.06/0.032660 = 1.837$.

Step 4: compare. Right-sided cutoff $z_{0.01} = 2.326$; since $1.837 < 2.326$ we do *not* reject.

$$\boxed{z_{\text{obs}} = 1.837 < 2.326 \;\Rightarrow\; \text{fail to reject: no significant rise at 1\%.}}$$

The lift is suggestive but, at the stringent 1% level, not conclusive.

6.5.2 The contrast: null SE versus estimate SE

Example 6.5.5 (The wrong denominator vs. the right one)**.** Take the approval poll again: $X = 232$ of $n = 400$, $H_0 : p = 0.50$ versus $p \neq 0.50$. Compute the test statistic two ways — once with the (wrong) interval SE that uses $\widehat{p}$, once with the (right) test SE that uses p_0 — and compare.

Which rule, and why? This is a deliberate **contrast**. The correct hypothesis-test statistic standardizes by the null standard error $\sqrt{p_0(1 - p_0)/n}$. Substituting the confidence-interval SE $\sqrt{\widehat{p}(1 - \widehat{p})/n}$ is the classic mistake; we run both to see how far apart they land.

Solution.

Step 1: the (wrong) estimate SE.

$$\begin{aligned} \mathrm{SE}_{\widehat{p}} &= \sqrt{\frac{\widehat{p}(1 - \widehat{p})}{n}} = \sqrt{\frac{0.58 \cdot 0.42}{400}} \\ &= \sqrt{0.0006090} = 0.024678, \end{aligned}$$

giving the look-alike statistic $z_{\text{wrong}} = (0.58 - 0.50)/0.024678 = 3.242$.

Step 2: the (right) null SE. From the earlier example $\mathrm{SE}_0 = 0.025$, so $z_{\text{obs}} = 0.08/0.025 = 3.200$.

Step 3: compare. The two differ:

$$z_{\text{wrong}} = 3.242 \quad \text{(interval SE, uses } \widehat{p}\text{)},$$
$$z_{\text{obs}} = 3.200 \quad \text{(test SE, uses } p_0\text{)}.$$

Here both clear 1.96 and the verdict is unchanged, but the numbers are not equal, and when $\widehat{p}$ sits far from p_0 the gap widens enough to flip a borderline decision. Always report z_{obs} with p_0 in the denominator.

$$\boxed{\text{Test SE uses } p_0\text{: } z_{\text{obs}} = 3.200, \text{ not } 3.242.}$$

Remark. Why does $z_{\text{wrong}} > z_{\text{obs}}$ here? Because $\widehat{p} = 0.58$ is farther from $1/2$ than $p_0 = 0.50$, so $\widehat{p}(1-\widehat{p}) = 0.2436 < 0.25 = p_0(1-p_0)$, making the wrong denominator smaller and the wrong statistic larger. The direction of the error depends on which side of $1/2$ the estimate falls; the lesson is simply that the two are not interchangeable.

6.5.3 The exact binomial test (small n)

Definition 6.5.6 (Exact binomial test)**.** To test $H_0\colon p = p_0$ from $X \sim \text{Bin}(n,p)$ without a normal approximation, compute the tail probability under $\text{Bin}(n,p_0)$ of outcomes at least as extreme as the observed x. For $H_1\colon p > p_0$ the p-value is $\mathbb{P}(X \geq x)$; for $H_1\colon p < p_0$ it is $\mathbb{P}(X \leq x)$; each is

$$\mathbb{P}(X \geq x) = \sum_{k=x}^{n} \binom{n}{k} p_0^k (1-p_0)^{n-k},$$

and the analogous lower sum for the left tail. Reject when the p-value $\leq \alpha$.

Example 6.5.7 (Coin fairness, small-n exact one-sided test)**.** A gambler suspects a coin is biased toward heads. In $n = 12$ tosses it shows 10 heads. With $H_0\colon p = 0.5$ against $H_1\colon p > 0.5$, compute the exact p-value and decide at $\alpha = 0.05$.

Which rule, and why? Only $n = 12$ trials, so the normal approximation is shaky ($np_0 = 6$ is on the edge); use the **exact binomial test**. "Biased toward heads" is a right-sided alternative, so the tail is $\mathbb{P}(X \geq 10)$ under $\text{Bin}(12, 0.5)$.

Solution.

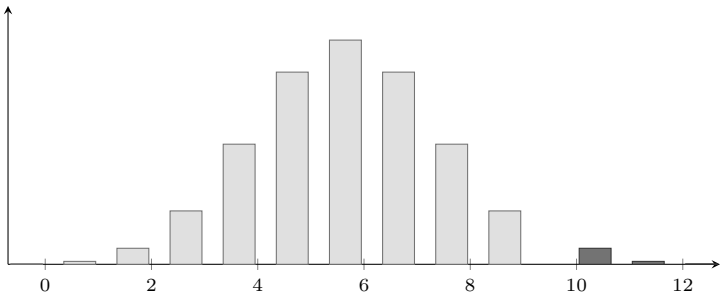
Step 1: identify the tail. Extreme-toward-H_1 means $X \geq 10$.

Step 2: sum the pmf. With $\binom{12}{10} = 66$, $\binom{12}{11} = 12$, $\binom{12}{12} = 1$ and $0.5^{12} = 1/4096$,

$$\begin{aligned}\mathbb{P}(X \geq 10) &= \big(66 + 12 + 1\big) \cdot \tfrac{1}{4096} \\ &= \frac{79}{4096} = 0.01929.\end{aligned}$$

Step 3: decide. Since $0.01929 \leq 0.05$ we reject H_0.

$$\boxed{p\text{-value} = 79/4096 = 0.0193 \leq 0.05 \;\Rightarrow\; \text{reject } H_0.}$$



The $\mathrm{Bin}(12, 0.5)$ *pmf. The dark bars (*$k = 10, 11, 12$*) are the right-tail outcomes at least as extreme as the observed* 10 *heads; their total height* 0.0193 *is the exact p-value.*

Example 6.5.8 (Rare adverse event, exact test near a boundary)**.** A drug's label says serious reactions occur in at most 2% of patients. In a fresh cohort of $n = 80$ patients, 5 have a serious reaction. Test $H_0 : p = 0.02$ against $H_1 : p > 0.02$ exactly, at $\alpha = 0.05$.

Which rule, and why? Here $np_0 = 80 \cdot 0.02 = 1.6 < 10$: the normal approximation is invalid because p_0 is tiny. Use the **exact binomial test**, right tail $\mathbb{P}(X \geq 5)$ under $\mathrm{Bin}(80, 0.02)$.

Solution.

Step 1: complement. $\mathbb{P}(X \geq 5) = 1 - \mathbb{P}(X \leq 4) = 1 - \sum_{k=0}^{4} \binom{80}{k} 0.02^k 0.98^{80-k}$.

Step 2: term by term. With $0.98^{80} = 0.198587$ as the base,

$$\begin{aligned}
\mathbb{P}(X = 0) &= 0.198587, \\
\mathbb{P}(X = 1) &= 80 \cdot \tfrac{0.02}{0.98}\, \mathbb{P}(X = 0) = 0.324224, \\
\mathbb{P}(X = 2) &= \tfrac{79}{2} \cdot \tfrac{0.02}{0.98}\, \mathbb{P}(X = 1) = 0.261364, \\
\mathbb{P}(X = 3) &= \tfrac{78}{3} \cdot \tfrac{0.02}{0.98}\, \mathbb{P}(X = 2) = 0.138723, \\
\mathbb{P}(X = 4) &= \tfrac{77}{4} \cdot \tfrac{0.02}{0.98}\, \mathbb{P}(X = 3) = 0.054520.
\end{aligned}$$

Step 3: sum and complement.

$$\begin{aligned}
\mathbb{P}(X \leq 4) &= 0.198587 + 0.324224 + 0.261364 \\
&\qquad + 0.138723 + 0.054520 = 0.977418, \\
\mathbb{P}(X \geq 5) &= 1 - 0.977418 = 0.022582.
\end{aligned}$$

Step 4: decide. Since $0.0226 \leq 0.05$ we reject.

$$\boxed{p\text{-value} = 0.0226 \leq 0.05 \;\Rightarrow\; \text{reject } H_0\text{: rate exceeds 2\%.}}$$

Example 6.5.9 (Two-sided exact test)**.** A geneticist expects a $1 : 1$ phenotype ratio. Of $n = 14$ offspring, 11 show the dominant trait. Test $H_0 : p = 0.5$ against $H_1 : p \neq 0.5$ exactly at $\alpha = 0.05$.

Which rule, and why? Small $n = 14$ and a two-sided question. For an exact two-sided test of a symmetric $p_0 = 0.5$, double the smaller one-tailed probability: the observed 11 is high, so use $2\,\mathbb{P}(X \geq 11)$ under $\mathrm{Bin}(14, 0.5)$.

Solution.

Step 1: upper tail. With $0.5^{14} = 1/16384$ and $\binom{14}{11} = 364$, $\binom{14}{12} = 91$, $\binom{14}{13} = 14$, $\binom{14}{14} = 1$,

$$\begin{aligned}
\mathbb{P}(X \geq 11) &= \frac{364 + 91 + 14 + 1}{16384} \\
&= \frac{470}{16384} = 0.028687.
\end{aligned}$$

Step 2: double it. By symmetry the two-sided p-value is $2 \cdot 0.028687 = 0.057373$.

Step 3: decide. Since $0.0574 > 0.05$ we do *not* reject.

$$\boxed{p\text{-value} = 0.0574 > 0.05 \;\Rightarrow\; \text{fail to reject the 1:1 ratio.}}$$

Note how the same count that rejects one-sided (the upper tail alone is $0.0287 \leq 0.05$) fails two-sided once we pay for both tails.

6.5.4 p-values and continuity correction

Example 6.5.10 (Large-sample two-sided p-value)**.** A snack maker advertises that 25% of bags contain a bonus token. A consumer group opens $n = 600$ bags and finds 129 tokens. Report the two-sided p-value for $H_0 : p = 0.25$.

Which rule, and why? Large n with $np_0 = 150$ and $n(1 - p_0) = 450$, so the z-test is valid; "does the true rate differ from 25%" is two-sided, and the question asks for strength of evidence, i.e. the p-value rather than a fixed accept/reject.

Solution.

Step 1: estimate and SE. $\widehat{p} = 129/600 = 0.215$ and

$$\mathrm{SE}_0 = \sqrt{\frac{0.25 \cdot 0.75}{600}} = \sqrt{0.0003125} = 0.017678.$$

Step 2: statistic. $z_{\mathrm{obs}} = (0.215 - 0.250)/0.017678 = -0.035/0.017678 = -1.980$.

Step 3: two-sided p-value. Using $\mathbb{P}(Z \le -1.98) = 0.0239$,

$$p\text{-value} = 2 \cdot 0.0239 = 0.0478.$$

Step 4: read it. At $\alpha = 0.05$ this rejects ($0.0478 \le 0.05$); at $\alpha = 0.01$ it does not. The evidence is borderline.

$$\boxed{p\text{-value} = 2\,\mathbb{P}(Z \le -1.98) = 0.0478.}$$

Example 6.5.11 (Same data with continuity correction)**.** Recompute the token problem's one-sided lower p-value with a **continuity correction**, which sharpens the normal approximation to the discrete binomial.

Which rule, and why? Still the large-sample test for $p = 0.25$, but we adjust $\widehat{p}$ by $\frac{1}{2}$ a count toward p_0 before standardizing, because a continuous normal is approximating a discrete count. We take the lower tail since $\widehat{p} < p_0$.

Solution.

Step 1: shift the count. Observed $x = 129$; for the lower tail use $x + \frac{1}{2} = 129.5$, so $\widehat{p}_c = 129.5/600 = 0.215833$.

Step 2: corrected statistic.

$$\begin{aligned} z_c &= \frac{0.215833 - 0.25}{0.017678} \\ &= \frac{-0.034167}{0.017678} = -1.933. \end{aligned}$$

Step 3: p-value. $\mathbb{P}(Z \le -1.933) = 0.0266$, slightly larger (less significant) than the uncorrected 0.0239, as the correction intends.

$$\boxed{z_c = -1.933, \quad p\text{-value (one-sided)} = 0.0266.}$$

Remark. The continuity correction nudges the test toward being more conservative and is most worth doing when n is moderate. As n grows the $\pm\frac{1}{2}$ count becomes a vanishing fraction of n and the correction melts away. For very large n you may safely skip it.

6.5.5 Comparing an observed rate to a contractual target

Example 6.5.12 (Pass rate vs. a contractual target)**.** A supplier's contract guarantees at least a 95% inspection pass rate. The buyer samples $n = 300$ delivered parts and finds 276 pass. At $\alpha = 0.05$, is there evidence the supplier is *below* the 95% target?

Which rule, and why? Counting passes out of 300 against the contractual $p_0 = 0.95$; $np_0 = 285$ and $n(1-p_0) = 15$ both exceed 10, so the large-sample z-test is valid. "Below the target" is a left-sided alternative $H_1 : p < 0.95$.

Solution.

Step 1: estimate. $\widehat{p} = 276/300 = 0.92$.

Step 2: test SE.

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{0.95 \cdot 0.05}{300}} = \sqrt{\frac{0.0475}{300}} \\ &= \sqrt{0.00015833} = 0.012583. \end{aligned}$$

Step 3: statistic. $z_{\text{obs}} = (0.92 - 0.95)/0.012583 = -0.03/0.012583 = -2.384$.

Step 4: decide. Left-sided cutoff $-z_{0.05} = -1.645$; since $-2.384 \leq -1.645$ we reject.

$$\boxed{z_{\text{obs}} = -2.384 \leq -1.645 \;\Rightarrow\; \text{reject } H_0 \text{: below the 95\% target.}}$$

Example 6.5.13 (The same target, but is the SE built right?)**.** For the contract above, suppose a junior analyst standardizes with $\widehat{p} = 0.92$ in the denominator instead of the contractual $p_0 = 0.95$. What statistic do they get, and would the verdict change?

Which rule, and why? Another **contrast**. The contractual target $p_0 = 0.95$ is the null value and belongs in the SE. Swapping in $\widehat{p} = 0.92$ is the interval-SE mistake; we compute it to see the size of the error.

Solution.

Step 1: wrong SE.

$$\begin{aligned} \mathrm{SE}_{\widehat{p}} &= \sqrt{\frac{0.92 \cdot 0.08}{300}} = \sqrt{\frac{0.0736}{300}} \\ &= \sqrt{0.00024533} = 0.015663. \end{aligned}$$

Step 2: wrong statistic. $z_{\text{wrong}} = -0.03/0.015663 = -1.915$.

Step 3: compare verdicts. The right value was $z_{\text{obs}} = -2.384$; the wrong one is -1.915. Both still clear -1.645, so the reject/accept verdict survives here, but the wrong SE *understates* the evidence (a 20% smaller magnitude). On a closer call it could push a true rejection over to "fail to reject".

$$\boxed{z_{\text{wrong}} = -1.915 \neq -2.384 = z_{\text{obs}};\ \text{use } p_0 = 0.95 \text{ in the SE.}}$$

Example 6.5.14 (Quality target met exactly at the margin)**.** A solar-panel maker must keep its field-failure rate at or below 3%. A field study of $n = 1000$ panels records 42 failures. At $\alpha = 0.05$, is the failure rate significantly *above* 3%?

Which rule, and why? Large-sample z-test for $p = 0.03$; $np_0 = 30$ and $n(1-p_0) = 970$ both exceed 10. "Above 3%" is right-sided, $H_1 : p > 0.03$.

Solution.

Step 1: estimate. $\widehat{p} = 42/1000 = 0.042$.

Step 2: test SE.

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{0.03 \cdot 0.97}{1000}} = \sqrt{\frac{0.0291}{1000}} \\ &= \sqrt{0.0000291} = 0.005394. \end{aligned}$$

Step 3: statistic. $z_{\text{obs}} = (0.042 - 0.030)/0.005394 = 0.012/0.005394 = 2.225$.

Step 4: decide. Right-sided cutoff $z_{0.05} = 1.645$; since $2.225 \geq 1.645$ we reject. (For reference, $\mathbb{P}(Z \geq 2.225) = 0.0130$, the one-sided p-value.)

$$\boxed{z_{\text{obs}} = 2.225 \geq 1.645 \;\Rightarrow\; \text{reject } H_0 \text{: failure rate exceeds 3\%.}}$$

Example 6.5.15 (When the count is so small the exact test is mandatory)**.** A boutique winery claims no more than 1% of its bottles are corked. A critic opens $n = 40$ bottles and finds 3 corked. Test $H_0 : p = 0.01$ against $H_1 : p > 0.01$ at $\alpha = 0.05$.

Which rule, and why? Here $np_0 = 40 \cdot 0.01 = 0.4$, nowhere near 10, so the z-test is invalid. Use the **exact binomial test**, right tail $\mathbb{P}(X \geq 3)$ under $\text{Bin}(40, 0.01)$.

Solution.

Step 1: complement. $\mathbb{P}(X \geq 3) = 1 - \mathbb{P}(X \leq 2)$.

Step 2: lower terms. With $0.99^{40} = 0.668971$,

$$\begin{aligned}\mathbb{P}(X = 0) &= 0.668971,\\ \mathbb{P}(X = 1) &= 40 \cdot \tfrac{0.01}{0.99}\,\mathbb{P}(X = 0) = 0.270291,\\ \mathbb{P}(X = 2) &= \tfrac{39}{2} \cdot \tfrac{0.01}{0.99}\,\mathbb{P}(X = 1) = 0.053234.\end{aligned}$$

Step 3: sum and complement.

$$\begin{aligned}\mathbb{P}(X \leq 2) &= 0.668971 + 0.270291 + 0.053234 = 0.992496,\\ \mathbb{P}(X \geq 3) &= 1 - 0.992496 = 0.007504.\end{aligned}$$

Step 4: decide. Since $0.0075 \leq 0.05$ we reject.

$$\boxed{p\text{-value} = 0.0075 \leq 0.05 \;\Rightarrow\; \text{reject } H_0\text{: corked rate exceeds 1\%.}}$$

Example 6.5.16 (Borderline large-sample call read three ways)**.** A polling firm tests whether incumbent support has slipped below 52%. In $n = 900$ likely voters, 459 back the incumbent. Give z_{obs}, the one-sided p-value, and the verdicts at $\alpha = 0.05$ and $\alpha = 0.01$.

Which rule, and why? Large n, $np_0 = 468$ and $n(1 - p_0) = 432$, so the z-test holds; "slipped below" is left-sided $H_1 : p < 0.52$.

Solution.

Step 1: estimate and SE.

$$\widehat{p} = 459/900 = 0.51,$$

$$\text{SE}_0 = \sqrt{\frac{0.52 \cdot 0.48}{900}} = \sqrt{0.00027733} = 0.016653.$$

Step 2: statistic. $z_{\text{obs}} = (0.51 - 0.52)/0.016653 = -0.01/0.016653 = -0.6005$.

Step 3: p-value. $\mathbb{P}(Z \leq -0.60) = 0.2743$.

Step 4: verdicts. Since $0.2743 > 0.05 > 0.01$, we fail to reject at both levels: a 1-point dip in a sample this size is well within noise.

$$\boxed{z_{\text{obs}} = -0.60, \quad p\text{-value} = 0.274 \;\Rightarrow\; \text{fail to reject at 5\% and 1\%.}}$$

6.6 Testing a Variance: the χ^2 Test

A variance or a standard deviation measures *spread*, and spread is often the quantity that actually matters: a fill machine that on average dispenses the right amount but with wildly varying output is still broken, and a gauge that reads correctly on average but scatters its readings is still useless. To test a claim about the spread of a normal population we cannot use Z or t — those test a *mean*. Instead we build the statistic $(n-1)S^2/\sigma_0^2$, whose distribution under $H_0 : \sigma^2 = \sigma_0^2$ is **chi-square** with $n - 1$ degrees of freedom. The single feature you must never forget is that the χ^2 distribution is **asymmetric** (right-skewed): unlike the z- and t-tests, a two-sided variance test is *not* of the tidy "estimate ± margin" form, and its two critical values are two *different* table look-ups, one in each tail. Recognise this test by its target (a variance, a

standard deviation, a precision, a tolerance) and by the normality of the parent population, which here is not a convenience but a genuine requirement.

Which spread test does the problem need?

- Hypothesis names a **variance** σ^2 or **standard deviation** σ of a single **normal** population? $\Rightarrow$ the **χ^2-test** on $(n-1)S^2/\sigma_0^2$ with $n-1$ degrees of freedom.
- Hypothesis is stated about a **standard deviation** ($\sigma = \sigma_0$)? $\Rightarrow$ square both sides first: test $\sigma^2 = \sigma_0^2$. The statistic is the same.
- H_1 says spread **differs** ($\sigma^2 \neq \sigma_0^2$)? $\Rightarrow$ two-sided: reject if the statistic is below the lower critical value $\chi^2_{1-\alpha/2,\,n-1}$ *or* above the upper critical value $\chi^2_{\alpha/2,\,n-1}$ (two *different* numbers).
- H_1 says spread has **increased** ($\sigma^2 > \sigma_0^2$)? $\Rightarrow$ upper-tail one-sided: reject if the statistic exceeds $\chi^2_{\alpha,\,n-1}$.
- H_1 says spread has **decreased** / precision **improved** ($\sigma^2 < \sigma_0^2$)? $\Rightarrow$ lower-tail one-sided: reject if the statistic falls below $\chi^2_{1-\alpha,\,n-1}$.

Do not confuse the tails: because χ^2 is skewed, the lower critical value is *not* the negative of the upper one. You cannot recycle a single positive cutoff with a $\pm$; you must look up each tail separately, and the upper-α point is far from the lower-α point.

6.6.1 The statistic and its null distribution

Theorem 6.6.1 (Distribution of the scaled sample variance)**.** *Let $X_1, \ldots, X_n$ be independent draws from a $\mathrm{N}(\mu, \sigma^2)$ population, and let $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$ be the sample variance. Then*

$$\frac{(n-1)S^2}{\sigma^2} \;\sim\; \chi^2_{n-1},$$

a chi-square distribution with $n-1$ degrees of freedom, and this statistic is independent of $\overline{X}$.

Proof. Write $Z_i = (X_i - \mu)/\sigma$, so $\sum Z_i^2 \sim \chi^2_n$. Algebraically, $\sum(X_i-\mu)^2 = \sum(X_i-\overline{X})^2 + n(\overline{X}-\mu)^2$; dividing by σ^2 gives $\sum Z_i^2 = (n-1)S^2/\sigma^2 + \big((\overline{X}-\mu)/(\sigma/\sqrt{n})\big)^2$. The last term is the square of a standard normal, hence χ^2_1, and it is independent of $(n-1)S^2/\sigma^2$ (mean and variance are independent for a normal sample). A χ^2_n variable split into an independent χ^2_1 piece leaves a χ^2_{n-1} piece, by matching moment generating functions. This needs the normality of the X_i; without it the result fails. □

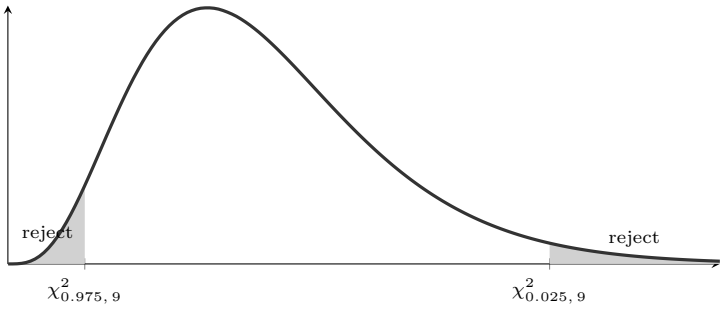
Definition 6.6.2 (The χ^2 test for a variance)**.** To test $H_0\colon \sigma^2 = \sigma_0^2$ for a normal population, compute the **test statistic**

$$\chi^2_{\mathrm{obs}} = \frac{(n-1)S^2}{\sigma_0^2},$$

which has the χ^2_{n-1} distribution when H_0 is true. Let $\chi^2_{q,\,k}$ denote the point with area q to its *right* under the χ^2_k curve, so $\mathbb{P}(\chi^2_k > \chi^2_{q,\,k}) = q$. At significance level α:

- **Two-sided** $H_1\colon \sigma^2 \neq \sigma_0^2$: reject H_0 if $\chi^2_{\mathrm{obs}} < \chi^2_{1-\alpha/2,\,n-1}$ or $\chi^2_{\mathrm{obs}} > \chi^2_{\alpha/2,\,n-1}$.
- **Upper one-sided** $H_1\colon \sigma^2 > \sigma_0^2$: reject H_0 if $\chi^2_{\mathrm{obs}} > \chi^2_{\alpha,\,n-1}$.
- **Lower one-sided** $H_1\colon \sigma^2 < \sigma_0^2$: reject H_0 if $\chi^2_{\mathrm{obs}} < \chi^2_{1-\alpha,\,n-1}$.

Remark. Because the curve is skewed, the equal-tail two-sided test (area $\alpha/2$ in each tail) is the standard convention but is not literally the shortest rejection region; it is the one the tables support, so we use it. The key practical point is that $\chi^2_{1-\alpha/2,\,n-1} \neq -\chi^2_{\alpha/2,\,n-1}$: both cutoffs are positive and they are not mirror images.



Two-sided rejection regions for $n = 10$ ($df = 9$), $\alpha = 0.05$: shade area 0.025 in each tail. The two critical values 2.700 and 19.023 are not symmetric about the peak, because the χ^2 curve is right-skewed.

6.6.2 Two-sided tests

Example 6.6.3 (Fill weights: is the variability off target?)**.** A cereal plant guarantees its filling process has variance $\sigma^2 = 4\,\mathrm{g}^2$ (standard deviation 2 g). A quality engineer pulls $n = 10$ boxes and finds a sample standard deviation $S = 2.9$ g, so $S^2 = 8.41\,\mathrm{g}^2$. The boxes overfilling *or* underfilling erratically are both bad, so test $H_0\colon \sigma^2 = 4$ against $H_1\colon \sigma^2 \neq 4$ at $\alpha = 0.05$.

Which rule, and why? The claim is about a *variance* of a normal population and H_1 is two-sided ("off target" in either direction), so this is the two-sided χ^2 test on $(n-1)S^2/\sigma_0^2$ with $n - 1 = 9$ df — not a t-test, which would test the *mean* weight.

Solution.

Step 1: statistic. With $n = 10$, $S^2 = 8.41$, $\sigma_0^2 = 4$,

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{9 \cdot 8.41}{4} = \frac{75.69}{4} = 18.92.$$

Step 2: critical values. Equal tails of 0.025 on 9 df:

$$\text{lower: } \chi^2_{0.975,\,9} = 2.700,$$
$$\text{upper: } \chi^2_{0.025,\,9} = 19.023.$$

Step 3: decision. Reject if $\chi^2_{\text{obs}} < 2.700$ or > 19.023. Here $18.92 < 19.023$ and $18.92 > 2.700$, so the statistic lands *inside* the acceptance region, barely.

$$\boxed{\chi^2_{\text{obs}} = 18.92 \in (2.700,\ 19.023) \ \Rightarrow\ \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The variability is close to the edge but not formally excessive at this level.

Example 6.6.4 (Contrast: the wrong $\pm$ shortcut)**.** Take the same data ($\chi^2_{\text{obs}} = 18.92$, df $= 9$, $\alpha = 0.05$). A student, used to the z-test, reasons: "the two-sided 5% cutoff is the upper $\chi^2_{0.025,\,9} = 19.023$; the lower cutoff is its mirror image, so reject if the statistic is outside ± 19.023, i.e. if $|\chi^2_{\text{obs}}| > 19.023$." Critique this.

Which rule, and why? This treats χ^2 like a symmetric z or t statistic. The χ^2 statistic is a sum of squares: it is always positive and its curve is skewed, so the lower critical value is a *separate* positive table value, not the negative of the upper one.

Solution.

Step 1: spot the error. A negative cutoff -19.023 is meaningless — the statistic $(n-1)S^2/\sigma_0^2$ can never be negative, so "$|\chi^2_{\text{obs}}| > 19.023$" simply collapses to the upper-tail rule and silently *drops the entire lower tail.*

Step 2: what the lower tail is for. The correct lower cutoff is $\chi^2_{0.975,\,9} = 2.700$, which catches a *suspiciously small* sample variance (a process that is too consistent to be real, or mislabelled units). The mirror-image shortcut would never reject for small spread.

Step 3: numerical consequence here. For these data both methods happen to "not reject," but for a data set with $\chi^2_{\text{obs}} = 1.5$ the correct test rejects (since $1.5 < 2.700$) while the bogus $\pm$ rule does not. The shortcut is wrong in general.

Use two distinct positive cutoffs $\chi^2_{1-\alpha/2,\,n-1}$ and $\chi^2_{\alpha/2,\,n-1}$; never a $\pm$ on one value.

Example 6.6.5 (Lab assay: variance differs from a published value?)**.** A reference method for a blood analyte has a published variance of $\sigma^2 = 0.36$ (units^2). A hospital validates a new analyzer on $n = 20$ control samples and obtains $S^2 = 0.21$. Does the new analyzer's variance differ from the published 0.36 at $\alpha = 0.10$?

Which rule, and why? Variance of a (normal) measurement population, H_1 is "differs" $\Rightarrow$ two-sided χ^2 test, df $= 19$. We are comparing to a fixed published σ_0^2, so it is a one-sample variance test, not an F-test (which would compare *two* sample variances).

Solution.

Step 1: statistic.

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{19 \cdot 0.21}{0.36} = \frac{3.99}{0.36} = 11.08.$$

Step 2: critical values. Equal tails of 0.05 on 19 df:

$$\text{lower: } \chi^2_{0.95,\,19} = 10.117,$$
$$\text{upper: } \chi^2_{0.05,\,19} = 30.144.$$

Step 3: decision. The acceptance region is (10.117, 30.144). Since $11.08 > 10.117$ and $11.08 < 30.144$, the statistic lies inside.

$\chi^2_{\text{obs}} = 11.08 \in (10.117,\, 30.144) \;\Rightarrow\;$ fail to reject at $\alpha = 0.10$.

It is close to the lower edge, hinting the new analyzer may be *more* precise — a one-sided test (next subsection) would address that directly.

6.6.3 One-sided tests: spread too large

Example 6.6.6 (Machining: has the variability increased?)**.** A lathe is specified to hold a shaft diameter with standard deviation $\sigma_0 = 0.05$ mm ($\sigma_0^2 = 0.0025$). After a bearing wears, an operator suspects parts are scattering more. A sample of $n = 10$ shafts gives $S = 0.071$ mm, so $S^2 = 0.005041$. Test $H_0\colon \sigma^2 = 0.0025$ against $H_1\colon \sigma^2 > 0.0025$ at $\alpha = 0.025$.

Which rule, and why? "Suspects parts are scattering *more*" fixes a one-sided upper-tail alternative on the variance, so the entire α goes in the upper tail of χ^2_9. A two-sided test would waste half of α on a decrease nobody is claiming.

Solution.

Step 1: statistic.

$$\chi^2_{\text{obs}} = \frac{9 \cdot 0.005041}{0.0025} = \frac{0.045369}{0.0025} = 18.15.$$

Step 2: critical value. All of $\alpha = 0.025$ in the upper tail: $\chi^2_{0.025,\,9} = 19.023$.

Step 3: decision. Reject only if $\chi^2_{\text{obs}} > 19.023$. Here $18.15 < 19.023$.

$\chi^2_{\text{obs}} = 18.15 < 19.023 \;\Rightarrow\;$ fail to reject H_0; no significant increase at $\alpha = 0.025$.

Note how the *same* cutoff 19.023 that was the *upper* two-sided cutoff in the first example is now the *whole* one-sided cutoff — but because all α sits in one tail the test is more willing to reject for an increase than the two-sided test was.

Example 6.6.7 (Environmental monitoring: pollutant readings too erratic?)**.** A regulator requires a monitoring station's daily ozone readings to have variance no more than $\sigma^2 = 9\,(\text{ppb})^2$. An audit of $n = 16$ days gives $S^2 = 15.5\,(\text{ppb})^2$. At $\alpha = 0.05$, is there evidence the station's variance exceeds the limit?

Which rule, and why? The regulatory concern is excessive spread only, so $H_1 : \sigma^2 > 9$ is upper one-sided on χ^2_{15}. The mean reading is irrelevant here; this is a precision/tolerance question, hence the variance test.

Solution.

Step 1: statistic. With $n = 16$ (df $= 15$),

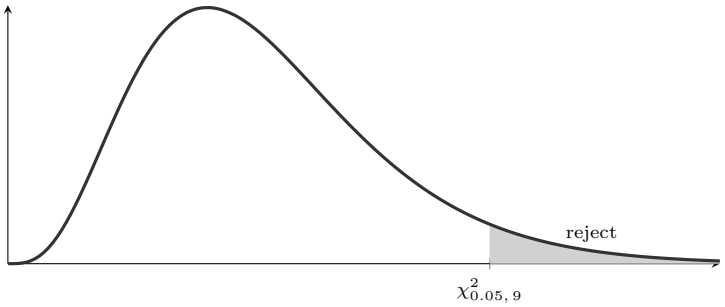
$$\chi^2_{\text{obs}} = \frac{15 \cdot 15.5}{9} = \frac{232.5}{9} = 25.83.$$

Step 2: critical value. $\chi^2_{0.05,\,15} = 24.996$.

Step 3: decision. Reject if $\chi^2_{\text{obs}} > 24.996$. Here $25.83 > 24.996$.

$$\boxed{\chi^2_{\text{obs}} = 25.83 > 24.996 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

The station's variability significantly exceeds the limit.



χ^2_9 statistic

Upper one-sided rejection region for df $= 9$, $\alpha = 0.05$: all of α in the right tail past $\chi^2_{0.05,\,9} = 16.919$. A claim that variance has increased *puts the whole rejection region here.*

6.6.4 One-sided tests: spread too small

Example 6.6.8 (Manufacturing improvement: did precision get better?)**.** A resistor line historically has resistance variance $\sigma^2 = 25\,(\Omega)^2$ ($\sigma_0 = 5\,\Omega$). After installing tighter process controls, an engineer hopes the spread has *shrunk*. A sample of $n = 20$ resistors gives $S^2 = 12.0\,(\Omega)^2$. Test $H_0 : \sigma^2 = 25$ against $H_1 : \sigma^2 < 25$ at $\alpha = 0.05$.

Which rule, and why? The claim is that precision *improved* (variance *decreased*), so this is a lower-tail one-sided χ^2 test on χ^2_{19} — the whole α goes in the *lower* tail, where a too-small statistic signals an unusually small sample variance.

Solution.

Step 1: statistic.

$$\chi^2_{\text{obs}} = \frac{19 \cdot 12.0}{25} = \frac{228.0}{25} = 9.12.$$

Step 2: critical value. For a lower-tail test we need the point with area α to its *left*, i.e. area $1 - \alpha = 0.95$ to its right: $\chi^2_{0.95,\,19} = 10.117$.

Step 3: decision. Reject if $\chi^2_{\text{obs}} < 10.117$. Here $9.12 < 10.117$.

$$\boxed{\chi^2_{\text{obs}} = 9.12 < 10.117 \;\Rightarrow\; \text{reject } H_0;\ \text{the variance has significantly decreased at } \alpha = 0.05.}$$

The process controls produced a statistically significant gain in precision.

Example 6.6.9 (Caution with the lower tail: a near miss)**.** Reuse the resistor setup ($\sigma_0^2 = 25$, $n = 20$, df $= 19$, $\alpha = 0.05$, lower-tail) but suppose instead $S^2 = 14.0$. Does the conclusion change?

Which rule, and why? Same lower-tail χ^2 test; only the data moved. The point is to see how sensitive the decision is to the cutoff and to reinforce that the lower critical value $\chi^2_{0.95,\,19}$ is a fixed table value, not derived from the upper one.

Solution.

Step 1: statistic.

$$\chi^2_{\text{obs}} = \frac{19 \cdot 14.0}{25} = \frac{266.0}{25} = 10.64.$$

Step 2: compare. The cutoff is still $\chi^2_{0.95,\,19} = 10.117$, and now $10.64 > 10.117$.

Step 3: decision.

$$\boxed{\chi^2_{\text{obs}} = 10.64 > 10.117 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

A modest drop in S^2 (from 14 to 12) flipped the decision: the lower-tail test rejects only when the sample variance is convincingly small.

6.6.5 Testing a standard deviation by squaring

Example 6.6.10 (Archery scores: a claim about a standard deviation)**.** A coach claims a shooter's grouping has standard deviation $\sigma = 3.0$ cm. A skeptical analyst believes the true spread is larger and records $n = 12$ shots, finding $S = 4.4$ cm. Test the coach's claim against "the standard deviation is larger" at $\alpha = 0.05$.

Which rule, and why? The hypothesis is phrased on σ, but the χ^2 test is built on the *variance*; square the hypothesized value to get $\sigma_0^2 = 9$ and proceed. "Larger spread" is an upper-tail one-sided alternative on χ^2_{11}.

Solution.

Step 1: convert to a variance hypothesis. $H_0\colon \sigma = 3.0 \Leftrightarrow \sigma^2 = 9$; $H_1\colon \sigma > 3.0 \Leftrightarrow \sigma^2 > 9$. Sample: $S = 4.4 \Rightarrow S^2 = 19.36$.

Step 2: statistic.

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{11 \cdot 19.36}{9} = \frac{212.96}{9} = 23.66.$$

Step 3: critical value and decision. Upper-tail: $\chi^2_{0.05,\,11} = 19.675$. Since $23.66 > 19.675$, reject.

$$\boxed{\chi^2_{\text{obs}} = 23.66 > 19.675 \;\Rightarrow\; \text{reject } H_0;\ \text{the spread is significantly larger than 3.0 cm at } \alpha = 0.05.}$$

Example 6.6.11 (Tolerance audit: a two-sided claim on a standard deviation)**.** A machined pin's diameter is required to have standard deviation exactly $\sigma_0 = 0.020$ mm; both too-loose and suspiciously-tight scatter trigger an investigation. A sample of $n = 10$ pins yields $S = 0.013$ mm. Test $H_0\colon \sigma = 0.020$ against $H_1\colon \sigma \neq 0.020$ at $\alpha = 0.05$.

Which rule, and why? The claim is on a standard deviation, two-sided $\Rightarrow$ square to σ_0^2 and run the two-sided χ^2 test with df $= 9$ (two different cutoffs).

Solution.

Step 1: convert. $\sigma_0^2 = (0.020)^2 = 4.0 \times 10^{-4}$; $S^2 = (0.013)^2 = 1.69 \times 10^{-4}$.

Step 2: statistic.

$$\chi^2_{\text{obs}} = \frac{9 \cdot 1.69 \times 10^{-4}}{4.0 \times 10^{-4}} = \frac{1.521 \times 10^{-3}}{4.0 \times 10^{-4}} = 3.80.$$

Step 3: critical values and decision. Equal tails on 9 df: $\chi^2_{0.975,\,9} = 2.700$, $\chi^2_{0.025,\,9} = 19.023$. Acceptance region (2.700, 19.023); here 3.80 lies inside.

$$\boxed{\chi^2_{\text{obs}} = 3.80 \in (2.700,\ 19.023) \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The pins are tighter than nominal but not significantly so against this two-sided test.

6.6.6 The p-value for a variance test

Proposition 6.6.12 (p-value rules)**.** *With observed statistic $c = \chi^2_{obs}$ on $k = n - 1$ df:*

$$\begin{aligned} H_1 : \sigma^2 > \sigma_0^2 : &\quad p = \mathbb{P}(\chi_k^2 > c), \\ H_1 : \sigma^2 < \sigma_0^2 : &\quad p = \mathbb{P}(\chi_k^2 < c), \\ H_1 : \sigma^2 \neq \sigma_0^2 : &\quad p = 2\min\{\mathbb{P}(\chi_k^2 < c),\, \mathbb{P}(\chi_k^2 > c)\}. \end{aligned}$$

Remark. "As extreme" is *one-tailed* for a one-sided test: only deviations in the direction of H_1 count. The two-sided convention doubles the smaller tail because the skew makes a true mirror-area definition awkward; doubling the nearer tail is the standard, table-friendly choice.

Example 6.6.13 (p-value for an upper-tail variance test)**.** Return to the ozone audit: $n = 16$ (df $= 15$), $\chi^2_{\text{obs}} = 25.83$, upper-tail $H_1 : \sigma^2 > 9$. Report the p-value and decide at $\alpha = 0.05$.

Which rule, and why? One-sided upper-tail test, so the p-value is the single right-tail area $\mathbb{P}(\chi^2_{15} > 25.83)$ — do *not* double it.

Solution.

Step 1: locate the statistic against table points. On 15 df, $\chi^2_{0.05,\,15} = 24.996$ and $\chi^2_{0.025,\,15} = 27.488$. Our value 25.83 sits between them, so the right-tail area is between 0.025 and 0.05.

Step 2: report. Interpolating, $\mathbb{P}(\chi^2_{15} > 25.83) \approx 0.040$.

$$\boxed{p\text{-value} = \mathbb{P}(\chi^2_{15} > 25.83) \approx 0.040 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

This agrees with the earlier fixed-level decision: $25.83 > 24.996$ rejected at $\alpha = 0.05$, and indeed $p \approx 0.040 < 0.05$.

Example 6.6.14 (p-value for a two-sided variance test)**.** Return to the fill-weight data: $n = 10$ (df $= 9$), $\chi^2_{\text{obs}} = 18.92$, two-sided $H_1 : \sigma^2 \neq 4$. Report the p-value.

Which rule, and why? Two-sided alternative on a skewed statistic, so take the nearer tail and double it; the statistic is large, so the right tail is the smaller one.

Solution.

Step 1: which tail is nearer. On 9 df, $\chi^2_{0.025,\,9} = 19.023$ and $\chi^2_{0.05,\,9} = 16.919$. Since 18.92 sits just below 19.023, the right-tail area is just above 0.025, say $\mathbb{P}(\chi^2_9 > 18.92) \approx 0.026$.

Step 2: double the smaller tail.

$$\mathbb{P}(\chi^2_9 > 18.92) \approx 0.026,$$
$$p\text{-value} = 2 \times 0.026 = 0.052.$$

Step 3: decide.

$$\boxed{p\text{-value} \approx 0.052 > 0.05 \;\Rightarrow\; \text{fail to reject at } \alpha = 0.05,}$$

consistent with Step 3 of the first example, where 18.92 fell just inside 19.023.

6.6.7 When the test breaks: sensitivity to non-normality

Remark. The χ^2 variance test is **not robust** to departures from normality. The t-test for a mean leans on the central limit theorem, so for moderate n it tolerates mild non-normality of the parent population. The χ^2 test has no such cushion: the distribution of $(n-1)S^2/\sigma^2$ is exactly χ^2_{n-1} *only* when the data are normal, and it is driven by the parent's *fourth moment* (kurtosis). Heavy tails or skew can make the true sampling distribution of S^2 far from χ^2_{n-1}, so the actual Type I error rate can be much larger or smaller than the nominal α. Always check normality (a plot, a formal test) before trusting a χ^2 variance conclusion.

Example 6.6.15 (Contrast: a misleading "significant" variance result)**.** An analyst tests $H_0\colon \sigma^2 = 1$ against $H_1\colon \sigma^2 > 1$ at $\alpha = 0.05$ with $n = 25$ from a population that is actually heavy-tailed (occasional large outliers), not normal. A single outlier inflates S^2 to 1.9, giving $\chi^2_{\text{obs}} = 24 \cdot 1.9/1 = 45.6$, far past $\chi^2_{0.05,\,24} = 36.415$, so H_0 is "rejected." Is the conclusion trustworthy?

Which rule, and why? Mechanically this is the upper-tail χ^2 test, and the arithmetic is correct — but the test's validity *assumes normality*, and here the parent is heavy-tailed, so the nominal α is not the true error rate.

Solution.

Step 1: the computation, taken at face value.

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{24 \cdot 1.9}{1} = 45.6 > 36.415 = \chi^2_{0.05,\,24}.$$

Formally, reject H_0.

Step 2: why this is unsafe. Under a heavy-tailed parent, S^2 has a much larger variance than the normal theory predicts, so even when $\sigma^2 = 1$ is true the statistic exceeds 36.415 far more than 5% of the time. The single outlier, not a genuine increase in spread, drove the rejection. The true Type I error rate here might be 15% or more, not 5%.

Step 3: the lesson. Contrast this with a t-test for a mean on the same heavy-tailed data: for $n = 25$ the central limit theorem keeps the t-test's error rate near nominal, so the t-test would be far more trustworthy. The variance test has no such protection.

A "significant" χ^2 result on non-normal data may be an artifact; verify normality first.

6.7 The Generalized Likelihood Ratio Test and the Test–Confidence-Interval Duality

Every named test in this chapter — the z-test, the t-test, the proportion test, the variance χ^2-test — is a special case of one general recipe, the **generalized likelihood ratio test** (GLRT). The idea is simple to state. The null hypothesis pins the parameter to a restricted set; the alternative lets it roam over the whole parameter space. We ask: how much better can the data be explained if we drop the null restriction? Concretely we form the ratio of the *best* (maximized) likelihood under H_0 to the *best* likelihood over everything, call it λ. If the null restriction costs us almost nothing (λ near 1) the data are consistent with H_0; if dropping it lets the likelihood shoot up (λ near 0) the data prefer the unrestricted fit and we reject. The same chapter also closes a loop that has run through every section: a two-sided level-α test of $H_0\colon \theta = \theta_0$ and a $1-\alpha$ confidence interval for θ are two readings of one object. The value θ_0 survives the test exactly when it lies inside the interval. That duality is the cleanest way to run a two-sided test once you have a confidence interval in hand, and it explains why "the interval misses θ_0" and "the test rejects θ_0" are always the same event.

Is this a GLRT or a duality problem – and which way do you run it?

- No off-the-shelf statistic, or you are asked to **derive** or **justify** a test from first principles? $\Rightarrow$ build the **GLRT**: write the likelihood, maximize it over the null set ω and over the full space Ω, form $\lambda = \sup_\omega L / \sup_\Omega L$, and reject when $\lambda \le \lambda^*$.
- The null fixes **one** parameter and the model has d free parameters constrained to 0 under H_0, and n is large? $\Rightarrow$ use **Wilks' theorem**: under H_0, $-2\ln\lambda \,\dot\sim\, \chi^2_d$. Read the cutoff and the approximate p-value off the χ^2_d table.
- You already have a $1-\alpha$ **confidence interval** and want a two-sided test of $\theta = \theta_0$ at level α? $\Rightarrow$ use the **duality**: reject iff θ_0 is **outside** the interval. No new computation needed.
- You have a **family of level-α tests** (one for each candidate θ_0) and want an interval? $\Rightarrow$ **invert** them: the interval is the set of θ_0 **not** rejected.

Do not confuse the two-sided pairing with a one-sided one: the duality matches a *two-sided* test with a *two-sided* (finite) interval. A one-sided test of $\theta \le \theta_0$ pairs with a one-sided confidence *bound*, not with a two-sided interval — mixing them throws the error level off by a factor related to α vs. $\alpha/2$.

6.7.1 The likelihood ratio and the rejection rule

Definition 6.7.1 (Generalized likelihood ratio)**.** Let the data $X_1, \dots, X_n$ have joint density (or mass function) $f(\mathbf{x}; \theta)$ with θ ranging over a parameter space Ω, and let the null hypothesis confine θ to a subset $\omega \subset \Omega$. Write the likelihood $L(\theta) = \prod_{i=1}^{n} f(x_i; \theta)$. The **generalized likelihood ratio** is

$$\lambda \;=\; \frac{\sup_{\theta \in \omega} L(\theta)}{\sup_{\theta \in \Omega} L(\theta)}.$$

The **generalized likelihood ratio test** (GLRT) of $H_0 \colon \theta \in \omega$ rejects H_0 iff $\lambda \le \lambda^*$, where $\lambda^* \in (0, 1)$ is chosen so that the test has level α, i.e. $\sup_{\theta \in \omega} \mathbb{P}_\theta(\lambda \le \lambda^*) = \alpha$.

Remark. Because $\omega \subset \Omega$, the numerator maximizes over a smaller set than the denominator, so the numerator can never exceed the denominator and both are positive: $0 < \lambda \le 1$. The ratio is the fraction of the achievable "fit" (peak likelihood) that the null restriction still allows. A λ near 1 says the best null parameter explains the data almost as well as the best parameter of all — no reason to abandon H_0. A small λ says the unrestricted model fits far better, which is evidence against H_0. Hence *small* λ, not large, triggers rejection.

Theorem 6.7.2 (Wilks' theorem, heuristic form)**.** *Suppose H_0 constrains d of the model's free parameters (for instance fixes a single parameter to a value, $d = 1$) and the usual regularity conditions hold. Then under H_0, as $n \to \infty$,*

$$-2 \ln \lambda \;\dot{\sim}\; \chi^2_d,$$

i.e. $-2 \ln \lambda$ is approximately chi-square with d degrees of freedom. Consequently a level-α GLRT rejects H_0 when $-2 \ln \lambda \ge \chi^2_{\alpha, d}$ (the upper-α χ^2_d point), and the approximate p-value is the upper-tail χ^2_d probability beyond the observed $-2 \ln \lambda$.

Remark. The map $\lambda \mapsto -2 \ln \lambda$ is decreasing, so "λ small" is the same event as "$-2 \ln \lambda$ large"; Wilks' theorem simply tells us *how* large is large by giving the null distribution of $-2 \ln \lambda$. The degrees of freedom d is the number of restrictions H_0 imposes — the drop in dimension from Ω to ω. The theorem is what lets us calibrate a GLRT even when the exact distribution of λ is intractable.

The single-peaked picture below is the whole story: the likelihood (drawn as a log-likelihood, which is concave for the standard families) peaks at the unrestricted MLE $\widehat{\theta}$; the null confines us to one point θ_0 off the peak; the ratio compares the two heights.

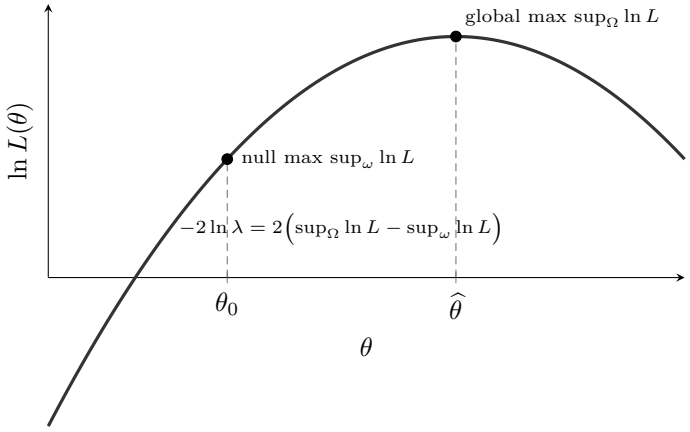


Figure. The log-likelihood peaks at the unrestricted MLE $\widehat{\theta}$; the null pins θ to θ_0. The vertical gap between the two dashed heights is exactly $-\frac{1}{2}$ of $-2 \ln \lambda$. A big gap (data much better explained off the null) means small λ and rejection.

6.7.2 The named tests are GLRTs

Example 6.7.3 (GLRT for a normal mean recovers the two-sided z-test)**.** Let $X_1, \dots, X_n \sim \mathrm{N}(\mu, \sigma^2)$ with σ known. Test $H_0 \colon \mu = \mu_0$ against $H_1 \colon \mu \neq \mu_0$. Derive the GLRT and show its rejection region is $|z_{\mathrm{obs}}| \geq z_{\alpha/2}$.

Which rule, and why? Here $\omega = \{\mu_0\}$ is a single point and $\Omega = \mathbb{R}$, so $d = 1$ parameter is constrained. The GLRT is the principled test; we expect it to reproduce the familiar z-test because the normal likelihood is exactly Gaussian in μ.

Solution.

Step 1: write the likelihood. With σ fixed,

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{\sigma\sqrt{2\pi}} \exp\Big(-\frac{(x_i-\mu)^2}{2\sigma^2}\Big) = (2\pi\sigma^2)^{-n/2} \exp\Big(-\frac{1}{2\sigma^2}\sum_i (x_i-\mu)^2\Big).$$

Step 2: maximize over $\Omega = \mathbb{R}$. The sum $\sum_i (x_i-\mu)^2$ is minimized at $\mu = \overline{X}$, so $\sup_\Omega L = L(\overline{X})$.

Step 3: maximize over $\omega = \{\mu_0\}$. Only one value is allowed, so $\sup_\omega L = L(\mu_0)$.

Step 4: form the ratio. The $(2\pi\sigma^2)^{-n/2}$ prefactors cancel:

$$\lambda = \frac{L(\mu_0)}{L(\overline{X})} = \exp\Big(-\frac{1}{2\sigma^2}\Big[\textstyle\sum_i (x_i-\mu_0)^2 - \sum_i (x_i-\overline{X})^2\Big]\Big).$$

Step 5: simplify the bracket. The identity $\sum_i (x_i-\mu_0)^2 = \sum_i (x_i-\overline{X})^2 + n(\overline{X}-\mu_0)^2$ collapses the difference to $n(\overline{X}-\mu_0)^2$, so

$$\lambda = \exp\Big(-\frac{n(\overline{X}-\mu_0)^2}{2\sigma^2}\Big).$$

Step 6: turn $\lambda \leq \lambda^*$ into a z-statement. Taking logs,

$$\begin{aligned} \lambda \leq \lambda^* &\iff -\frac{n(\overline{X}-\mu_0)^2}{2\sigma^2} \leq \ln \lambda^* \\ &\iff \frac{n(\overline{X}-\mu_0)^2}{\sigma^2} \geq -2\ln\lambda^* \\ &\iff \Big|\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}\Big| \geq \sqrt{-2\ln\lambda^*}. \end{aligned}$$

Step 7: calibrate. Under H_0, $z_{\mathrm{obs}} = (\overline{X}-\mu_0)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$, so choosing the cutoff $\sqrt{-2\ln\lambda^*} = z_{\alpha/2}$ gives level α. The GLRT rejects exactly when

$$\boxed{|z_{\mathrm{obs}}| \geq z_{\alpha/2}}$$

which is the two-sided z-test verbatim.

Remark. With σ **unknown** the same derivation, now maximizing over (μ, σ^2) in the denominator and over $\{\mu_0\} \times (0,\infty)$ in the numerator, produces $\lambda = (1 + t_{\mathrm{obs}}^2/(n-1))^{-n/2}$, where $t_{\mathrm{obs}} = (\overline{X}-\mu_0)/(S/\sqrt{n})$. Since λ is a strictly decreasing function of $|t_{\mathrm{obs}}|$, the rule $\lambda \leq \lambda^*$ is $|t_{\mathrm{obs}}| \geq c$; under H_0, $t_{\mathrm{obs}} \sim t_{n-1}$, so $c = t_{\alpha/2,n-1}$. The GLRT is the two-sided t-test. The moral: the named tests are not separate inventions, they are the likelihood ratio worked out for each model.

Example 6.7.4 (GLRT for an exponential rate, written explicitly)**.** Service-call durations (hours) are modeled as $X_1, \dots, X_n \sim \mathrm{Exp}(\lambda)$ (rate λ, mean $1/\lambda$), density $\lambda e^{-\lambda x}$. A vendor's contract assumes the rate is $\lambda_0 = 2$ per hour. With $n = 20$ calls and total time $\sum_i x_i = 14$ hours, write λ for $H_0 \colon \lambda = 2$ versus $H_1 \colon \lambda \neq 2$ explicitly and evaluate it.

Which rule, and why? One parameter λ, null fixes it to a point ($d = 1$); no canned "exponential rate test" is tabulated, so we build the GLRT from the likelihood directly.

Solution.

Step 1: likelihood and its log. With $T = \sum_i x_i$,

$$L(\lambda) = \lambda^n e^{-\lambda T}, \qquad \ln L(\lambda) = n \ln \lambda - \lambda T.$$

Step 2: unrestricted MLE. Setting $\dfrac{d}{d\lambda} \ln L = \dfrac{n}{\lambda} - T = 0$ gives $\widehat{\lambda} = n/T$. Here $\widehat{\lambda} = 20/14 = 1.4286$.

Step 3: form the ratio. The null allows only λ_0:

$$\lambda = \frac{L(\lambda_0)}{L(\widehat{\lambda})} = \frac{\lambda_0^n e^{-\lambda_0 T}}{\widehat{\lambda}^n e^{-\widehat{\lambda} T}} = \Big(\frac{\lambda_0}{\widehat{\lambda}}\Big)^n e^{-(\lambda_0 - \widehat{\lambda})T}.$$

Step 4: plug in. With $\lambda_0 = 2$, $\widehat{\lambda} = 1.4286$, $n = 20$, $T = 14$:

$$\frac{\lambda_0}{\widehat{\lambda}} = \frac{2}{1.4286} = 1.4, \qquad (\lambda_0 - \widehat{\lambda})T = (2 - 1.4286)(14) = 8.0,$$
$$\lambda = (1.4)^{20} e^{-8.0} = 836.7 \times 3.355 \times 10^{-4} = 0.2807.$$

$$\boxed{\lambda = \big(\lambda_0/\widehat{\lambda}\big)^n e^{-(\lambda_0 - \widehat{\lambda})T} = 0.281}$$

Whether this is small enough to reject is settled by the cutoff in the next example.

Example 6.7.5 (Using $-2 \ln \lambda$ and Wilks for an approximate p-value)**.** Continue the exponential service-call problem: $\lambda = 0.281$ from $n = 20$ observations, one constrained parameter. Get an approximate p-value and decide at $\alpha = 0.05$.

Which rule, and why? Exact distribution of λ here is awkward, but $n = 20$ is reasonably large and $d = 1$ parameter is constrained, so Wilks' theorem applies: $-2 \ln \lambda \dot{\sim} \chi^2_1$.

Solution.

Step 1: transform. $-2 \ln \lambda = -2 \ln(0.281) = -2(-1.2695) = 2.539$.

Step 2: reference distribution. Under H_0 this is approximately χ^2_1. The upper-5% point is $\chi^2_{0.05,1} = 3.841$.

Step 3: compare / p-value. Since $2.539 < 3.841$, we do **not** reject at $\alpha = 0.05$. The approximate p-value is $\mathbb{P}(\chi^2_1 \geq 2.539)$. Because $\chi^2_1 = Z^2$, this equals $2\,\mathbb{P}(Z \geq \sqrt{2.539}) = 2\,\mathbb{P}(Z \geq 1.594) \approx 2(0.0555) = 0.111$.

$$\boxed{-2 \ln \lambda = 2.54, \quad p\text{-value} \approx 0.11, \quad \text{do not reject } H_0.}$$

The data ($\widehat{\lambda} = 1.43$ vs. $\lambda_0 = 2$) lean toward a slower rate but not decisively.

Example 6.7.6 (GLRT for a Poisson mean (count data))**.** A radioactivity counter is supposed to register $\mu_0 = 5$ decays per second. Over $n = 12$ independent one-second windows the counts $X_i \sim \text{Poisson}(\mu)$ sum to $\sum_i x_i = 78$. Build the GLRT statistic $-2 \ln \lambda$ for $H_0 \colon \mu = 5$ versus $H_1 \colon \mu \neq 5$ and test at $\alpha = 0.05$.

Which rule, and why? Single Poisson mean, null fixes it ($d = 1$); the GLRT plus Wilks gives a clean χ^2_1 test where no exact small-sample table is at hand.

Solution.

Step 1: log-likelihood. With $T = \sum_i x_i = 78$ and $n = 12$, $\ln L(\mu) = -n\mu + T \ln \mu - \sum_i \ln(x_i!)$. The last term is free of μ and cancels in the ratio.

Step 2: MLE. $\dfrac{d}{d\mu} \ln L = -n + T/\mu = 0 \Rightarrow \widehat{\mu} = T/n = 78/12 = 6.5$.

Step 3: $-2 \ln \lambda$. The constant terms cancel, leaving

$$-2 \ln \lambda = 2\big[\ln L(\widehat{\mu}) - \ln L(\mu_0)\big] = 2\Big[-n\widehat{\mu} + T \ln \widehat{\mu} - (-n\mu_0 + T \ln \mu_0)\Big].$$

Step 4: substitute. Using $T = n\widehat{\mu}$ this simplifies to $-2\ln\lambda = 2\big[T\ln(\widehat{\mu}/\mu_0) - n(\widehat{\mu}-\mu_0)\big]$:

$$\begin{aligned} T\ln(\widehat{\mu}/\mu_0) &= 78\ln(6.5/5) = 78\ln(1.3) = 78(0.26236) = 20.464,\\ n(\widehat{\mu}-\mu_0) &= 12(6.5-5) = 18.0,\\ -2\ln\lambda &= 2(20.464-18.0) = 2(2.464) = 4.928. \end{aligned}$$

Step 5: decide. Against $\chi^2_{0.05,1} = 3.841$: since $4.928 > 3.841$ we reject. The approximate p-value is $\mathbb{P}(\chi^2_1 \geq 4.928) = 2\,\mathbb{P}(Z \geq 2.220) \approx 2(0.0132) = 0.0264$.

$$\boxed{-2\ln\lambda = 4.93 > 3.841, \quad p\text{-value} \approx 0.026,\ \text{reject } H_0.}$$

Example 6.7.7 (GLRT for a Bernoulli/binomial proportion)**.** A new assembly line claims a defect rate of $p_0 = 0.10$. In $n = 200$ units inspected, $X = 30$ are defective. Construct $-2\ln\lambda$ for $H_0\colon p = 0.10$ versus $H_1\colon p \neq 0.10$ and test at $\alpha = 0.01$.

Which rule, and why? Single Bernoulli parameter, null fixes it ($d = 1$). The GLRT here is the likelihood-ratio cousin of the large-sample proportion z-test of §5; for $n = 200$ Wilks gives χ^2_1.

Solution.

Step 1: log-likelihood. For $X \sim \mathrm{Bin}(n,p)$, dropping the $\binom{n}{X}$ constant, $\ln L(p) = X\ln p + (n-X)\ln(1-p)$.

Step 2: MLE. $\widehat{p} = X/n = 30/200 = 0.15$.

Step 3: $-2\ln\lambda$. With $n - X = 170$,

$$-2\ln\lambda = 2\Big[X\ln\frac{\widehat{p}}{p_0} + (n-X)\ln\frac{1-\widehat{p}}{1-p_0}\Big].$$

Step 4: substitute.

$$\begin{aligned} X\ln(\widehat{p}/p_0) &= 30\ln(0.15/0.10) = 30\ln(1.5) = 30(0.405465) = 12.164,\\ (n-X)\ln\tfrac{1-\widehat{p}}{1-p_0} &= 170\ln(0.85/0.90) = 170\ln(0.94444) = 170(-0.057158) = -9.717,\\ -2\ln\lambda &= 2(12.164-9.717) = 2(2.447) = 4.894. \end{aligned}$$

Step 5: decide. Against $\chi^2_{0.01,1} = 6.635$: since $4.894 < 6.635$, do not reject at $\alpha = 0.01$. The approximate p-value is $\mathbb{P}(\chi^2_1 \geq 4.894) = 2\,\mathbb{P}(Z \geq 2.212) \approx 0.027$.

$$\boxed{-2\ln\lambda = 4.89 < 6.635,\ \text{do not reject at } \alpha = 0.01\ (p \approx 0.027).}$$

At the looser $\alpha = 0.05$ we *would* reject — the conclusion is level-sensitive here.

Example 6.7.8 (CONTRAST: forgetting that small λ, not large, rejects)**.** A student computes $\lambda = 0.92$ in a one-parameter problem and announces "λ is close to 1, the unrestricted fit barely helps, so the data strongly contradict H_0 — reject." Identify the error and give the correct reading.

Which rule, and why? This tests the direction of the rule. The recognition cue: rejection comes from λ *small*, equivalently $-2\ln\lambda$ *large*. The student inverted it.

Solution.

Step 1: restate the rule. Reject iff $\lambda \leq \lambda^*$ with λ^* a small number near 0. A value $\lambda = 0.92$ is near the **no-evidence** end, not the **strong-evidence** end.

Step 2: check via Wilks. $-2\ln\lambda = -2\ln(0.92) = 0.1671$. Against χ^2_1 this is tiny; p-value $= \mathbb{P}(\chi^2_1 \geq 0.167) \approx 0.68$.

Step 3: correct conclusion. The null restriction cost almost nothing, so the data are **entirely consistent** with H_0. The correct action is **fail to reject**.

$$\boxed{\lambda = 0.92 \Rightarrow -2\ln\lambda = 0.17,\ p \approx 0.68:\ \text{fail to reject.}}$$

The phrase "barely helps to drop the null" is precisely the case *for* H_0, not against it.

6.7.3 The test–confidence-interval duality

Theorem 6.7.9 (Duality of two-sided tests and confidence intervals)**.** *Fix $\alpha \in (0,1)$. Suppose that for each candidate value θ_0 we have a level-α test of $H_0\colon \theta = \theta_0$ against $H_1\colon \theta \neq \theta_0$, with acceptance region $A(\theta_0) = \{\mathbf{x} : H_0 \text{ not rejected}\}$. Define the random set*

$$C(\mathbf{X}) = \{\theta_0 : \mathbf{X} \in A(\theta_0)\} = \{\theta_0 : H_0\colon \theta = \theta_0 \text{ is not rejected}\}.$$

Then $C(\mathbf{X})$ is a $1-\alpha$ confidence set for θ. Conversely, if $C(\mathbf{X})$ is a $1-\alpha$ confidence set, then "reject $H_0\colon \theta = \theta_0$ iff $\theta_0 \notin C(\mathbf{X})$" is a level-$\alpha$ test. In short:

$$\theta_0 \in C(\mathbf{X}) \iff H_0\colon \theta = \theta_0 \text{ is not rejected at level } \alpha.$$

Proof. For the first direction, fix the true value θ. By definition of a level-α test, $\mathbb{P}_\theta(\mathbf{X} \notin A(\theta)) \le \alpha$, i.e. the test of the *true* value wrongly rejects with probability at most α. But $\theta \in C(\mathbf{X})$ exactly when $\mathbf{X} \in A(\theta)$, so $\mathbb{P}_\theta(\theta \in C(\mathbf{X})) = \mathbb{P}_\theta(\mathbf{X} \in A(\theta)) \ge 1-\alpha$. That coverage statement, holding for every θ, is precisely what makes $C(\mathbf{X})$ a $1-\alpha$ confidence set. The converse reverses the same equivalence: if $\mathbb{P}_\theta(\theta \in C) \ge 1-\alpha$ for all θ, then declaring rejection when $\theta_0 \notin C$ has size $\mathbb{P}_{\theta_0}(\theta_0 \notin C) \le \alpha$. □

Remark. The everyday case: a z- or t-interval $\overline{X} \pm c \cdot (\mathrm{SE})$ *is* the inversion of the two-sided test, because $|\theta_0 - \overline{X}|/\mathrm{SE} \le c$ (the acceptance event) rearranges to $\theta_0 \in [\overline{X} - c \cdot \mathrm{SE},\ \overline{X} + c \cdot \mathrm{SE}]$. So checking "is θ_0 in the interval?" and running the two-sided test are the same arithmetic written two ways. This is why, once you have reported a confidence interval, a two-sided test needs no further work.

The picture: the confidence interval is a horizontal bar; the null value either sits inside it (consistent — fail to reject) or outside it (rejected).

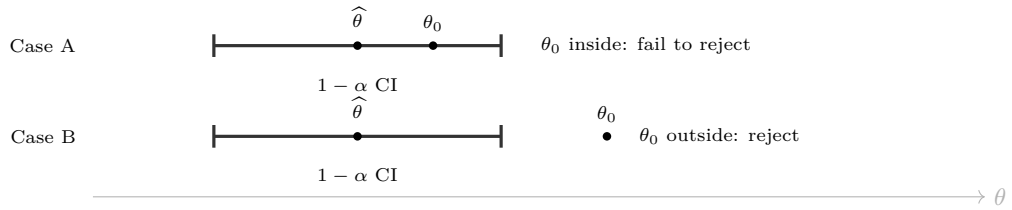


Figure. Same $1-\alpha$ interval in both rows. In Case A the null value θ_0 lies inside, so the matched two-sided level-α test fails to reject; in Case B it lies outside, so the test rejects. The interval *is* the set of non-rejected nulls.

Example 6.7.10 (Using a CI to run a two-sided test — means)**.** A lab reports that a 95% confidence interval for the mean shelf life μ (days) of a reagent is [182.4, 189.6]. A supplier claims the mean is $\mu_0 = 185$ days; a regulator claims it is $\mu_0 = 181$ days. Test each two-sided claim at $\alpha = 0.05$ without any new computation.

Which rule, and why? We already have the $95\% = 1-\alpha$ interval and want two-sided tests at the *matched* $\alpha = 0.05$. By duality, just check membership — no t-statistic needed.

Solution.

Step 1: supplier's 185. Is $185 \in [182.4, 189.6]$? Yes ($182.4 \le 185 \le 189.6$).

Step 2: regulator's 181. Is $181 \in [182.4, 189.6]$? No ($181 < 182.4$).

Step 3: read off decisions. Inside ⇒ fail to reject; outside ⇒ reject.

$\mu_0 = 185$: fail to reject;	$\mu_0 = 181$: reject (at 0.05).

The supplier's value is compatible with the data; the regulator's is not.

Example 6.7.11 (Using a CI to run a two-sided test — a proportion)**.** A poll of $n = 1000$ voters yields a 95% confidence interval [0.467, 0.533] for the proportion p favoring a measure. A pundit asserts the race is tied, $p = 0.50$; a campaign claims a lead, $p = 0.54$. Test each two-sided claim at $\alpha = 0.05$ by duality.

Which rule, and why? The reported 95% interval pairs exactly with the two-sided 0.05 test of any single p_0; membership decides each.

Solution.

Step 1: tied, 0.50. $0.50 \in [0.467, 0.533]$? Yes. Fail to reject "$p = 0.50$".

Step 2: lead, 0.54. $0.54 \in [0.467, 0.533]$? No ($0.54 > 0.533$). Reject "$p = 0.54$".

$$\boxed{p = 0.50 : \text{ fail to reject;} \qquad p = 0.54 : \text{ reject.}}$$

The data cannot rule out a tie but are inconsistent with a 54% share.

Example 6.7.12 (Inverting a family of tests to BUILD the set of non-rejected values)**.** From $n = 16$ normal observations with sample mean $\overline{X} = 50.0$ and known $\sigma = 8.0$, construct the set of all μ_0 that a two-sided z-test at $\alpha = 0.05$ would **not** reject, and identify it.

Which rule, and why? This is the inversion direction: we collect every null value that survives the test. The standard error is known, so the acceptance event is a clean inequality in μ_0.

Solution.

Step 1: acceptance event. The two-sided z-test fails to reject μ_0 iff $|z_{\text{obs}}| \le z_{\alpha/2}$, i.e.

$$\left|\frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}}\right| \le z_{0.025} = 1.96.$$

Step 2: standard error. $\sigma/\sqrt{n} = 8.0/\sqrt{16} = 8.0/4 = 2.0$.

Step 3: solve for μ_0.

$$\begin{aligned} |50.0 - \mu_0| &\le 1.96 \times 2.0 = 3.92 \\ -3.92 \le 50.0 - \mu_0 &\le 3.92 \\ 46.08 \le \mu_0 &\le 53.92. \end{aligned}$$

Step 4: recognize it. The set of non-rejected μ_0 is the interval $[46.08, 53.92]$, which is exactly the 95% confidence interval $\overline{X} \pm 1.96\,(\sigma/\sqrt{n})$.

$$\boxed{\{\mu_0 : \text{not rejected}\} = [46.08,\ 53.92] = 95\%\text{ CI for } \mu.}$$

Building the non-rejection set and inverting to a CI are the same act.

Example 6.7.13 (Two-sided GLRT and its CI, agreeing by duality)**.** With the same data ($n = 16$, $\overline{X} = 50.0$, $\sigma = 8.0$), test $H_0\colon \mu = 54$ against $\mu \neq 54$ at $\alpha = 0.05$ two ways — directly by the z-test, and by duality against the interval from the previous example — and confirm they agree.

Which rule, and why? A single two-sided test plus the matched 95% interval; duality guarantees the same verdict, which is the consistency check we want to see.

Solution.

Step 1: direct z-test. $z_{\text{obs}} = (50.0 - 54)/2.0 = -4.0/2.0 = -2.0$. Since $|-2.0| = 2.0 \ge 1.96$, reject.

Step 2: duality check. Is $54 \in [46.08, 53.92]$? No ($54 > 53.92$). Reject.

Step 3: compare. Both routes reject $\mu = 54$, as the theorem requires. (The margin is thin: 54 is just past the upper endpoint 53.92, matching $z_{\text{obs}} = -2.0$ just past -1.96.)

$$\boxed{\text{Direct: reject } (|z_{\text{obs}}| = 2.0); \quad \text{Duality: } 54 \notin [46.08, 53.92].}$$

Example 6.7.14 (CONTRAST: pairing a one-sided test with a two-sided interval)**.** An analyst wants a one-sided test of $H_0\colon \mu \le 54$ versus $H_1\colon \mu > 54$ at $\alpha = 0.05$, and reasons: "54 is below the lower endpoint 46.08? no — 54 is near the upper endpoint, and it is inside $[46.08, 53.92]$? no, it is just outside, so I reject." Using the 95% *two-sided* interval to run this *one-sided* test, what goes wrong?

Which rule, and why? The duality theorem matches a two-sided test with a two-sided interval at the *same* α. A one-sided test at α pairs with a one-sided *bound* at α, equivalently with the two-sided interval at confidence $1-2\alpha$. Mixing levels is the trap.

Solution.

Step 1: name the mismatch. The two-sided 95% interval uses $z_{0.025} = 1.96$ in each tail. The correct one-sided test of $\mu > \mu_0$ at $\alpha = 0.05$ uses $z_{0.05} = 1.645$ in a single tail.

Step 2: do the one-sided test correctly. Reject $H_0\colon \mu \le 54$ iff $z_{\text{obs}} = (\overline{X} - 54)/2.0 \ge 1.645$. Here $z_{\text{obs}} = (50.0 - 54)/2.0 = -2.0$, which is **far below** 1.645, so we **fail to reject** — the data point the wrong way for "$\mu > 54$" anyway.

Step 3: contrast with the misuse. The analyst's "just outside the two-sided interval" logic would (after fixing the direction) reject, but that event is calibrated to $\alpha/2 = 0.025$ in the relevant tail, not to $\alpha = 0.05$, and it ignores that H_1 points *upward* while the data sit below 54. The matched object for a one-sided 5% test is the one-sided 95% bound $\mu \ge \overline{X} - 1.645\,(\sigma/\sqrt{n}) = 50.0 - 3.29 = 46.71$, and $54 \ge 46.71$, so $H_0\colon \mu \le 54$ is **not** rejected.

Correct one-sided: fail to reject; the two-sided interval is the wrong ruler.

A two-sided interval answers two-sided questions; one-sided questions need a one-sided bound.

Example 6.7.15 (Duality with a t-interval, σ unknown)**.** A 90% confidence t-interval for a process mean is [12.1, 13.7] (from $n = 10$, $\overline{X} = 12.9$, $S/\sqrt{n} = 0.442$, $t_{0.05,9} = 1.833$). Test $H_0\colon \mu = 13.0$ against $\mu \ne 13.0$ at the matched level, and state that level.

Which rule, and why? The interval is a 90% *two-sided* t-interval, so it pairs with a two-sided t-test at $\alpha = 1 - 0.90 = 0.10$. Membership decides it.

Solution.

Step 1: matched level. $1 - \alpha = 0.90 \Rightarrow \alpha = 0.10$. This is a two-sided 0.10 test.

Step 2: membership. Is $13.0 \in [12.1, 13.7]$? Yes ($12.1 \le 13.0 \le 13.7$). Fail to reject.

Step 3: cross-check by the statistic. $t_{\text{obs}} = (12.9 - 13.0)/0.442 = -0.226$, and $|-0.226| = 0.226 < t_{0.05,9} = 1.833$, so fail to reject — the same verdict.

$\alpha = 0.10:\ 13.0 \in [12.1, 13.7] \Rightarrow$ fail to reject ($\lvert t_{\text{obs}}\rvert = 0.23$).

Example 6.7.16 (GLRT for a normal variance recovers the χ^2-test)**.** For $X_1, \ldots, X_n \sim \mathrm{N}(\mu, \sigma^2)$ with μ unknown, sketch why the GLRT of $H_0\colon \sigma^2 = \sigma_0^2$ (vs. $\ne$) is the variance χ^2-test of §6, and note its duality with the χ^2-interval.

Which rule, and why? Here $\omega = \{(\mu, \sigma_0^2) : \mu \in \mathbb{R}\}$ and $\Omega = \{(\mu, \sigma^2)\}$, so $d = 1$ (only σ^2 is constrained). The GLRT machinery should reproduce the statistic $(n-1)S^2/\sigma_0^2$.

Solution.

Step 1: profile out μ. In both numerator and denominator the best μ is $\overline{X}$, leaving a function of σ^2 only. The denominator's MLE is $\widehat{\sigma^2} = \frac{1}{n}\sum_i (x_i - \overline{X})^2$; the numerator fixes $\sigma^2 = \sigma_0^2$.

Step 2: the ratio depends on W. Writing $W = \sum_i (x_i - \overline{X})^2/\sigma_0^2$, one finds $\lambda = (W/n)^{n/2} e^{(n-W)/2}$, a function that decreases as W moves away from its center in either direction. Thus $\lambda \le \lambda^*$ becomes "W too small or too large".

Step 3: calibrate. Under H_0, $W = (n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$, so the two cutoffs are $\chi^2_{1-\alpha/2,n-1}$ and $\chi^2_{\alpha/2,n-1}$ — exactly the two-sided variance χ^2-test.

Step 4: duality. Inverting that two-sided test gives the standard $1-\alpha$ interval for σ^2,

$$\left[(n-1)S^2/\chi^2_{\alpha/2,n-1},\ (n-1)S^2/\chi^2_{1-\alpha/2,n-1}\right],$$

and σ_0^2 is rejected iff it falls outside.

GLRT = two-sided χ^2 variance test; CI = inverted test.

Once again a named test and its interval are one likelihood-ratio construction seen from two sides.

6.8 Exercises

All problems below are original. Full solutions begin on page 409.

A. Setting up tests and the decision rule

A1. A coffee roaster's bagging line is set to a mean fill of 340 g. A quality auditor suspects the line is now *under*-filling bags and wants to flag it only if the mean has dropped. For each setting below, write H_0 and H_1, say whether the test is one- or two-sided, and classify each hypothesis as simple or composite (take σ known throughout).

(a) The auditor's under-filling concern just described ($\mu_0 = 340$ g).

(b) A separate consumer group only cares whether the mean fill "differs from" the advertised 340 g, in either direction.

(c) A supplier guarantees the mean grind size of its beans is "at most" 0.80 mm, and a buyer will reject a lot only if the grind is too coarse.

A2. In each scenario, name the parameter under test, state H_0 and H_1, and decide one- vs. two-sided from the wording. (Do not compute anything.)

(a) Historically 62% of a clinic's patients keep their follow-up appointment. After a reminder-text program, an administrator asks whether the show-up rate "has increased."

(b) A bridge cable is rated for a mean breaking load of 50 tons. An inspector tests whether the true mean breaking load "is below" the rating.

(c) A city claims the mean response time of its ambulances "has not changed" from last year's 9.0 minutes; a reporter wants to check this.

A3. A normal mean is tested with σ known. For each $(H_1,\ \alpha)$ pair, write the rejection region for the standardized statistic $Z = (\overline{X} - \mu_0)/(\sigma/\sqrt{n})$ using the standard critical values.

(a) $H_1 : \mu > \mu_0,\ \ \alpha = 0.05$.

(b) $H_1 : \mu \neq \mu_0,\ \ \alpha = 0.05$.

(c) $H_1 : \mu < \mu_0,\ \ \alpha = 0.01$.

(d) $H_1 : \mu \neq \mu_0,\ \ \alpha = 0.01$.

A4. A laboratory thermostat is specified to hold a mean temperature $\mu_0 = 37.0$ degrees C, with known process standard deviation $\sigma = 0.4$ degrees C. An engineer fears the unit now runs *warm*. A sample of $n = 25$ independent readings averages $\overline{X} = 37.14$ degrees C. Carry out the whole test at $\alpha = 0.05$: state H_0 and H_1, name the test statistic and its null distribution, compute z_{obs}, give the critical value and rejection region, and state the decision in plain words.

A5. A handed-down decision rule reads: "reject $H_0 : \mu = 120$ in favor of $H_1 : \mu \neq 120$ exactly when $|Z| \geq 1.96$." For each separately observed statistic, apply the rule and state the decision and the significance level it operates at.

(a) $z_{\text{obs}} = 2.05$.

(b) $z_{\text{obs}} = -1.70$.

(c) $z_{\text{obs}} = -3.10$.

A6. A two-sided test of $H_0 : \mu = 15.0$ (mg, σ known) on a vitamin tablet yields $z_{\text{obs}} = 2.40$.

(a) Reach the decision at $\alpha = 0.05$.

(b) Reach the decision at $\alpha = 0.01$.

(c) Explain in one or two sentences why the two verdicts differ, and why this shows α must be fixed before the data are seen.

A7. After running the thermostat test of problem A4, a technician summarizes a different study that *failed to reject* $H_0 : \mu = 37.0$ by writing: "the test proves the thermostat holds exactly 37.0 degrees."

(a) Explain why "proves" is the wrong word, in terms of what a failure to reject does and does not establish.

(b) State what quantity α actually controls, and name the error it does *not* control.

(c) Give a correct one-sentence wording of the conclusion.

B. Type I and Type II errors, power, and sample size

B1. A shipping warehouse inspects incoming pallets for damaged units. Out of $n = 15$ sampled cartons a pallet is rejected (H_1: the supplier is below standard) if 3 or more cartons are damaged; otherwise it is accepted (H_0: the damage rate is the nominal $p = 0.05$). Treating the number of damaged cartons as $\mathrm{Bin}(15, p)$:

(a) Find the Type I error probability α of this acceptance rule.

(b) If a bad supplier actually ships at a damage rate $p = 0.30$, find the Type II error probability β and the power of the rule against that alternative.

B2. A bottling plant fills cans to a target of $\mu_0 = 355$ mL. Because overfilling wastes product, quality control tests $H_0 : \mu = 355$ against $H_1 : \mu > 355$ at level $\alpha = 0.05$ using $n = 16$ cans, with known $\sigma = 4$ mL. The rule rejects when $\overline{X} \geq c$.

(a) Find the critical value c in millilitres.

(b) Find the Type II error probability β if the line is really running at $\mu_1 = 357$ mL.

(c) State the power of the test against $\mu_1 = 357$ in plain words.

B3. A materials lab tests whether a new alloy's mean tensile strength has risen above the old grade's $\mu_0 = 400$ MPa. They use a one-sided upper z-test with $\sigma = 20$ MPa, $n = 25$, and $\alpha = 0.05$. Compute the power of the test at each of the alternatives $\mu_1 = 405$, 410, 415, 420 MPa, and describe in one sentence how the power behaves as the true mean moves away from the null.

B4. An agronomist wants to detect whether a fertilizer raises the mean yield above $\mu_0 = 50$ bushels/acre. The effect she cares about is fixed at the alternative $\mu_1 = 53$ bushels/acre, with known $\sigma = 6$ and a one-sided test at $\alpha = 0.05$. Compute the power of the test for sample sizes $n = 16$, $n = 36$, and $n = 64$, and state what this shows about the effect of enlarging the sample.

B5. A polling firm tests $H_0 : \mu = 100$ against $H_1 : \mu > 100$ (a satisfaction index) with $n = 36$, known $\sigma = 15$, against the fixed alternative $\mu_1 = 105$.

(a) Compute the power at the three significance levels $\alpha = 0.01$, $\alpha = 0.05$, and $\alpha = 0.10$.

(b) Explain in one sentence the trade-off the firm faces when it raises α to gain power.

B6. A pharmaceutical team is designing a study to detect whether a drug lowers mean resting heart rate below the placebo value $\mu_0 = 72$ bpm. They will run a one-sided lower z-test at $\alpha = 0.05$ and require power 0.90 to detect a genuine drop to $\mu_1 = 68$ bpm. Historical data give $\sigma = 10$ bpm. Using $n \geq \big((z_\alpha + z_\beta)\sigma/(\mu_1 - \mu_0)\big)^2$, find the smallest sample size n that meets the requirement, then report the actual power that the rounded-up n delivers.

B7. A small biotech screens a compound for an effect on mean enzyme activity, testing $H_0 : \mu = 20$ units against $H_1 : \mu > 20$ at $\alpha = 0.05$ with only $n = 9$ samples and known $\sigma = 9$ units. They observe $\overline{X} = 22.5$, obtain a p-value of about 0.20, fail to reject, and write in their report "the compound has no effect on enzyme activity."

(a) Compute the power of their test against the scientifically meaningful alternative $\mu_1 = 26$ units.

(b) Explain why the large p-value (failing to reject H_0) is *not* proof that H_0 is true, given the power you found.

(c) Find the sample size n that would have given power 0.90 against $\mu_1 = 26$, and contrast it with the 9 they used.

C. The z-test for a mean and p-values

C1. A foundry casts brake rotors to a target mean diameter of $\mu_0 = 320.0$ mm. The casting process has a known standard deviation $\sigma = 1.2$ mm, fixed by the mold tooling. A quality engineer measures $n = 36$ rotors and finds $\overline{X} = 320.5$ mm. At $\alpha = 0.05$, test whether the mean diameter is off target in either direction. State H_0 and H_1, compute z_{obs}, give the critical value, and state the decision in words.

C2. A municipal water authority claims its treatment plant releases effluent with a mean biochemical oxygen demand (BOD) of $\mu_0 = 25.0$ mg/L. The assay's standard deviation is known from calibration to be $\sigma = 3.0$ mg/L. An inspector who suspects the plant is *exceeding* the claimed level measures $n = 25$ grab samples and gets $\overline{X} = 26.4$ mg/L.

(a) State the hypotheses and explain why the test is one-sided (upper tail).

(b) Compute z_{obs} and carry out the test at $\alpha = 0.05$.

(c) Would the decision change at the stricter level $\alpha = 0.01$?

C3. A pharmacy compounds vials to a labeled mean concentration of $\mu_0 = 50.0$ mg/mL, with a known fill standard deviation $\sigma = 0.8$ mg/mL. A regulator worried about *under-filling* draws $n = 16$ vials and measures $\overline{X} = 49.6$ mg/mL. Test, at $\alpha = 0.05$, whether the true mean is *below* the label. State the hypotheses, compute z_{obs}, give the lower-tail critical value, and state the conclusion in plain words.

C4. Reconsider the foundry of Problem C1 ($\mu_0 = 320.0$, $\sigma = 1.2$, $n = 36$, $\overline{X} = 320.5$, so $z_{\text{obs}} = 2.50$).

(a) Report the two-sided p-value, using $\Phi(2.50) \approx 0.99379$.

(b) Interpret the p-value in one sentence about how often such data arise when the process is on target.

(c) Confirm that the p-value decision at $\alpha = 0.05$ agrees with the critical-value decision found in Problem C1.

C5. A streaming service reports that the historical mean number of titles a subscriber adds to their watchlist per month is $\mu_0 = 8.0$; the per-subscriber counts are right-skewed (roughly Poisson-like), but the standard deviation is known from extensive history to be $\sigma = 3.5$. After a recommendation-engine redesign, an analyst samples $n = 49$ subscribers and finds $\overline{X} = 8.9$ titles. The analyst wants evidence that the mean has *increased*.

(a) Explain why a z-test is appropriate even though the parent is not normal.

(b) Compute z_{obs} and report the one-sided p-value, using $\Phi(1.80) \approx 0.96407$.

(c) State the decision at $\alpha = 0.05$.

C6. An online retailer's checkout times are heavily right-skewed. The historical mean is $\mu_0 = 42.0$ seconds with a known $\sigma = 15.0$ seconds. After a one-click redesign, an engineer samples $n = 225$ checkouts and finds $\overline{X} = 39.6$ seconds, and asks whether the mean checkout time *changed* (in either direction). Carry out a two-sided large-sample z-test at $\alpha = 0.05$ and report the two-sided p-value, using $\Phi(2.40) \approx 0.99180$.

C7. A graduate student runs a two-sided z-test of $H_0 : \mu = 100$ against $H_1 : \mu \neq 100$ and obtains a p-value of 0.03. For each statement below, say whether it is *true* or *false* and give a one-sentence justification.

(a) "There is a 3% probability that H_0 is true."

(b) "If H_0 were true, data at least this extreme would occur about 3% of the time."

(c) "The p-value of 0.03 means there is a 97% probability that H_1 is true."

(d) "Because $0.03 \leq 0.05$, we reject H_0 at the 5% level."

(e) "A p-value of 0.03 proves the effect is large and practically important."

D. The t-test for a mean (σ unknown)

D1. A bottling line is supposed to deliver a nominal $\mu_0 = 355$ mL per can. A technician draws $n = 16$ cans and finds $\overline{X} = 352.4$ mL with sample standard deviation $S = 4.0$ mL; the true σ is not known. Test $H_0 : \mu = 355$ against $H_1 : \mu \neq 355$ at $\alpha = 0.05$. Use $t_{0.025,15} = 2.131$. State the standard error, the value of t_{obs}, the critical value, and your decision in words.

D2. A regional hatchery reports that its trout historically reach a mean weight of $\mu_0 = 1.80$ kg at harvest. After switching feed, a biologist weighs $n = 10$ fish and obtains $\overline{X} = 1.92$ kg with $S = 0.21$ kg.

(a) The biologist suspects the new feed *increased* mean weight. State H_0 and H_1, and say why this is an upper-tail t-test and how many degrees of freedom it has.

(b) Carry out the test at $\alpha = 0.05$ using $t_{0.05,9} = 1.833$. Report t_{obs} and the decision.

(c) Would your decision change if the alternative had instead been the two-sided $H_1 : \mu \neq 1.80$ (using $t_{0.025,9} = 2.262$)? Explain.

D3. An environmental rule caps the mean lead concentration in a city's drinking water at $\mu_0 = 15$ parts per billion. A utility samples $n = 12$ taps and measures $\overline{X} = 13.6$ ppb with $S = 2.4$ ppb, and wishes to demonstrate the mean is genuinely *below* the cap. Test $H_0 : \mu = 15$ against $H_1 : \mu < 15$ at $\alpha = 0.05$, using $t_{0.05,11} = 1.796$. Give the test statistic and state the conclusion in plain words.

D4. Eight castings from a new die are measured for hardness (Rockwell C). The raw readings are

$$58,\ 61,\ 55,\ 63,\ 60,\ 57,\ 62,\ 56.$$

The specification calls for a mean hardness of $\mu_0 = 60$.

(a) Compute $\overline{X}$ and the sample standard deviation S from the raw data (show the sum of squared deviations).

(b) Test $H_0 : \mu = 60$ against $H_1 : \mu \neq 60$ at $\alpha = 0.05$, using $t_{0.025,7} = 2.365$. Report t_{obs} and the decision.

D5. A coaching service claims its program raises a student's score on a standardized exam. Six students each take the exam before and after the program; their paired scores are listed below.

Student	1	2	3	4	5	6
Before	540	610	575	590	620	555
After	580	625	600	585	660	590

Treat the data as paired. Test at $\alpha = 0.05$ whether the program *raised* mean scores. Use $t_{0.05,5} = 2.015$. Form the differences $d_i = \text{After} - \text{Before}$, compute $\bar{d}$ and S_d, and give the decision.

D6. Two laboratory pH meters, X and Y, are each used to read the same $n = 7$ buffer solutions. The differences $d_i = X - Y$ come out to

$$0.05,\ -0.02,\ 0.08,\ 0.03,\ 0.06,\ -0.01,\ 0.07.$$

Is there evidence at $\alpha = 0.05$ that the two meters *disagree on average* (in either direction)? Treat the readings as paired, test the mean difference against 0, and use $t_{0.025,6} = 2.447$. Report t_{obs} and your conclusion.

D7. A precision lathe is set to cut shafts to a target diameter of $\mu_0 = 25.00$ mm. A sample of $n = 10$ shafts gives $\overline{X} = 25.18$ mm and $S = 0.27$ mm; the population σ is *not* known. An inexperienced analyst standardizes with S but then compares the result to the normal cutoff $z_{0.025} = 1.96$ rather than the correct t cutoff $t_{0.025,9} = 2.262$.

(a) Compute the common statistic and carry out the (incorrect) z-comparison: what does the analyst conclude about $H_0 : \mu = 25.00$ versus $H_1 : \mu \neq 25.00$ at $\alpha = 0.05$?

(b) Carry out the correct t-comparison. Do the two calls agree?

(c) Explain which decision is right and why the t cutoff exceeds the z cutoff here.

E. Tests for a proportion (binomial data)

E1. A municipal election board claims that exactly 40% of registered voters cast a ballot early. A survey of $n = 500$ registered voters finds 224 who voted early. At $\alpha = 0.05$, test $H_0 : p = 0.40$ against $H_1 : p \neq 0.40$. State the hypotheses, build the standard error from the null value, compute z_{obs}, and decide.

E2. A hospital reports that historically 70% of a certain surgery's patients are discharged within 48 hours. After a new post-op protocol, 322 of $n = 440$ patients are discharged within 48 hours. At $\alpha = 0.05$, is there evidence the discharge rate has *risen* above 70%? Use a one-sided large-sample z-test.

E3. A bottling line is supposed to underfill at most 5% of its bottles. A quality audit of $n = 600$ bottles finds 21 underfilled.

(a) Check the validity conditions $np_0 \geq 10$ and $n(1-p_0) \geq 10$ for the large-sample z-test of $H_0 : p = 0.05$.

(b) Test $H_0 : p = 0.05$ against $H_1 : p < 0.05$ at $\alpha = 0.05$, computing z_{obs} with the null standard error.

(c) State the decision in plain words.

E4. A streaming service believes 30% of new trial users convert to paid plans. In a fresh batch of $n = 750$ trial users, 246 convert.

(a) Compute the test statistic z_{obs} for $H_0 : p = 0.30$ versus $H_1 : p \neq 0.30$ using the correct test standard error SE_0 built from p_0.

(b) A colleague instead divides by the interval standard error $\text{SE}_{\widehat{p}}$ built from $\widehat{p}$, obtaining a different number. Compute that look-alike statistic z_{wrong}.

(c) Explain which one is correct for a hypothesis test and report the two-sided verdict at $\alpha = 0.05$.

E5. A seed company advertises a germination rate of 90%. A gardener plants $n = 15$ seeds and only 10 germinate. Suspecting the true rate is *below* 90%, test $H_0 : p = 0.90$ against $H_1 : p < 0.90$ at $\alpha = 0.05$. Because n is small and p_0 is near 1, use the exact binomial test: compute the lower-tail p-value $\mathbb{P}(X \leq 10)$ under $\text{Bin}(15, 0.90)$ and decide.

E6. An online retailer states that 20% of orders include a gift wrap. A logistics team samples $n = 400$ orders and finds 68 with gift wrap. Report the two-sided p-value for $H_0 : p = 0.20$ against $H_1 : p \neq 0.20$, then state whether you would reject at $\alpha = 0.05$ and at $\alpha = 0.01$. (Use $\mathbb{P}(Z \leq -1.50) = 0.0668$.)

E7. A delivery contract guarantees that at least 98% of parcels arrive undamaged. A client inspects $n = 250$ delivered parcels and finds 238 undamaged.

(a) Confirm the large-sample z-test is valid for $H_0 : p = 0.98$.

(b) Test $H_0 : p = 0.98$ against $H_1 : p < 0.98$ at $\alpha = 0.05$, building the standard error from the contractual target $p_0 = 0.98$.

(c) Is there significant evidence the supplier falls below the 98% contractual target? State the decision in plain words.

F. Tests for a variance (χ^2)

F1. A vaccine manufacturer guarantees that the dose volume delivered by its auto-injector has variance $\sigma^2 = 0.04\,\text{mL}^2$. A regulator pulls $n = 16$ injectors and finds a sample variance $S^2 = 0.075\,\text{mL}^2$. Both excessive and suspiciously small scatter would prompt an investigation. Assume dose volumes are normal. Test $H_0 : \sigma^2 = 0.04$ against $H_1 : \sigma^2 \neq 0.04$ at $\alpha = 0.05$. State the two critical values, the test statistic, and your decision.

F2. A foundry's casting line is rated to hold a part mass with variance $\sigma^2 = 2.5\,\mathrm{g}^2$. After a maintenance shutdown an engineer fears the line is now producing more variable parts. A sample of $n = 21$ castings gives $S^2 = 4.3\,\mathrm{g}^2$. Assuming masses are normal, test $H_0\colon \sigma^2 = 2.5$ against $H_1\colon \sigma^2 > 2.5$ at $\alpha = 0.05$.

F3. A pharmacy compounding robot historically dispenses a liquid with variance $\sigma^2 = 0.09\,\mathrm{mL}^2$. After a firmware update the lead pharmacist hopes the machine is now *more* consistent. A sample of $n = 25$ fills yields $S^2 = 0.045\,\mathrm{mL}^2$. Assuming fills are normal, test $H_0\colon \sigma^2 = 0.09$ against $H_1\colon \sigma^2 < 0.09$ at $\alpha = 0.05$. Does the update significantly improve precision?

F4. A precision-optics supplier claims its lens-curvature gauge reads with standard deviation $\sigma = 0.40$ micrometers. A skeptical buyer suspects the true spread is larger and records $n = 13$ readings, finding $S = 0.62$ micrometers. Assume readings are normal.

(a) Convert the claim about σ into a hypothesis about σ^2.

(b) State H_0 and H_1 and identify the test and its degrees of freedom.

(c) Carry out the test at $\alpha = 0.05$ and state the conclusion in plain words.

F5. A climate station's daily temperature anomalies are required to be recorded with variance no greater than $\sigma^2 = 1.2\,(^\circ\mathrm{C})^2$. An audit of $n = 24$ days gives $S^2 = 2.05\,(^\circ\mathrm{C})^2$. Assume the anomalies are normal and test the upper-tail alternative $H_1\colon \sigma^2 > 1.2$.

(a) Compute the test statistic and its degrees of freedom.

(b) Using the table points $\chi^2_{0.05,\,23} = 35.172$ and $\chi^2_{0.025,\,23} = 38.076$, bracket the p-value.

(c) State the decision at $\alpha = 0.05$ using the p-value.

F6. A 3-D printer is advertised to deposit bead widths with standard deviation $\sigma = 0.030$ mm; the quality team flags any deviation, too coarse or too fine, for recalibration. A sample of $n = 10$ beads gives $S = 0.021$ mm. Assume bead widths are normal.

(a) Convert to a two-sided hypothesis on σ^2 and give the test statistic.

(b) Using $\chi^2_{0.975,\,9} = 2.700$ and $\chi^2_{0.025,\,9} = 19.023$, state the acceptance region and the decision at $\alpha = 0.05$.

F7. An analyst applies the χ^2 variance test to a sample of $n = 30$ insurance-claim amounts, testing $H_0\colon \sigma^2 = 4$ against $H_1\colon \sigma^2 > 4$ at $\alpha = 0.05$. The data are strongly right-skewed with occasional very large claims — a single huge claim pushes S^2 to 7.1, giving a statistic far past the cutoff, so the analyst "rejects" H_0 and reports that claim variability has increased. The arithmetic is correct. Explain why the conclusion may nonetheless be untrustworthy, what assumption is violated, how it affects the true Type I error rate, and how a t-test for a mean on the same data would compare in robustness.

G. Likelihood ratio tests and the test–CI duality

G1. A microbrewery bottles a sparkling water whose fill volume (mL) is modeled as $X_1, \dots, X_n \sim \mathrm{N}(\mu, \sigma^2)$ with $\sigma = 4.0$ *known*. The target is $\mu_0 = 500$. From $n = 25$ bottles the sample mean is $\overline{X} = 502.0$. We test $H_0\colon \mu = 500$ against $H_1\colon \mu \neq 500$.

(a) Starting from the normal likelihood $L(\mu)$, show that the generalized likelihood ratio reduces to $\lambda = \exp\!\big(-n(\overline{X} - \mu_0)^2/(2\sigma^2)\big)$.

(b) Show that the rule $\lambda \le \lambda^*$ is equivalent to $|z_{\mathrm{obs}}| \ge c$ for a constant c, i.e. the GLRT *is* the two-sided z-test. Identify c at $\alpha = 0.05$.

(c) Evaluate λ and z_{obs} for the data and state the decision at $\alpha = 0.05$.

G2. Time-to-failure for a batch of LED drivers (thousands of hours) is modeled as $X_1, \dots, X_n \sim \mathrm{Exp}(\lambda)$ (rate λ, mean $1/\lambda$). The supplier's spec is $\lambda_0 = 0.5$ failures per thousand hours. From $n = 15$ drivers the total running time is $T = \sum_i x_i = 24$ (thousand hours). Test $H_0\colon \lambda = 0.5$ against $H_1\colon \lambda \neq 0.5$.

(a) Write the likelihood $L(\lambda) = \lambda^n e^{-\lambda T}$, find the unrestricted MLE $\widehat{\lambda}$, and evaluate it.

(b) Write $\lambda = (\lambda_0/\widehat{\lambda})^n e^{-(\lambda_0 - \widehat{\lambda})T}$ and evaluate it.

(c) Form $-2\ln\lambda$, refer it to the Wilks χ^2_1 distribution, give an approximate p-value, and decide at $\alpha = 0.05$.

G3. A call center claims its hold-time complaint rate is $p_0 = 0.20$ of all calls. An audit of $n = 150$ randomly sampled calls finds $X = 39$ that generated a complaint. Using the Bernoulli/binomial GLRT, compute

$$-2\ln\lambda = 2\Big[X\ln\tfrac{\widehat{p}}{p_0} + (n-X)\ln\tfrac{1-\widehat{p}}{1-p_0}\Big],$$

refer it to χ^2_1, and test $H_0\colon p = 0.20$ versus $H_1\colon p \neq 0.20$ at $\alpha = 0.05$. Report an approximate p-value.

G4. A wildlife camera grid is designed to register a mean of $\mu_0 = 3$ animal crossings per night. Over $n = 20$ independent nights the counts $X_i \sim \text{Poisson}(\mu)$ total $\sum_i x_i = 84$. For $H_0\colon \mu = 3$ versus $H_1\colon \mu \neq 3$:

(a) Find the MLE $\widehat{\mu}$ and write $-2\ln\lambda = 2\big[T\ln(\widehat{\mu}/\mu_0) - n(\widehat{\mu} - \mu_0)\big]$ with $T = \sum_i x_i$.

(b) Evaluate $-2\ln\lambda$, compare with $\chi^2_{0.05,1} = 3.841$, and state the decision.

(c) Give an approximate p-value.

G5. A reliability engineer reports a 95% confidence interval [118.3, 124.7] for the mean burst pressure μ (psi) of a hose fitting. Without computing any test statistic, decide each two-sided claim at the matched level $\alpha = 0.05$:

(a) A vendor states $\mu_0 = 120$ psi.

(b) A safety standard requires $\mu_0 = 125$ psi.

(c) A skeptic guesses $\mu_0 = 117$ psi.

For each, state whether H_0 is rejected and why.

G6. From $n = 36$ measurements of a coating thickness, the sample mean is $\overline{X} = 80.0$ microns and the population standard deviation is known to be $\sigma = 9.0$ microns. Construct the set of all μ_0 that a two-sided z-test at $\alpha = 0.05$ would *not* reject. Identify this set explicitly and say what familiar object it equals.

G7. Answer each conceptual item briefly, with a one-line justification.

(a) A colleague has a 90% two-sided confidence interval for a process mean and wants to run a two-sided test of $H_0\colon \mu = \mu_0$. At what significance level α does "μ_0 lies outside the interval" correctly perform the test, and why?

(b) Explain in one or two sentences why a one-sided test of $H_0\colon \mu \le \mu_0$ versus $H_1\colon \mu > \mu_0$ at level α should be paired with a one-sided confidence *bound*, not with a two-sided interval. What goes wrong if you read the one-sided test off the two-sided $1-\alpha$ interval?

(c) In a GLRT a student finds $\lambda = 0.88$ and concludes "λ is close to 1, so the evidence against H_0 is strong — reject." State the correct reading of $\lambda = 0.88$ and the correct decision direction.

6.9 Solutions

A. Setting up tests and the decision rule

A1. In every part the certified or advertised value is the status quo, so it carries the equality and is H_0; the claim needing evidence is H_1. With σ known, an equality fixes one distribution (simple); a range allows many (composite).

(a) "Under-filling" names a direction (lower), so the test is one-sided to the left.

$$H_0: \ \mu = 340 \ \ (\text{or } \mu \geq 340) \quad (\text{simple at the boundary}),$$
$$H_1: \ \mu < 340 \quad (\text{composite, one-sided left}).$$

(b) "Differs from" names no direction, so the test is two-sided.

$$\boxed{H_0 : \mu = 340 \text{ (simple)}, \qquad H_1 : \mu \neq 340 \text{ (composite, two-sided)}}$$

(c) The guarantee "at most 0.80" is the trusted default; the buyer rejects only for a *too-coarse* (large) grind, so one-sided right.

$$\boxed{H_0 : \mu \leq 0.80 \text{ (composite, boundary 0.80)}, \qquad H_1 : \mu > 0.80 \text{ (composite, one-sided right)}}$$

A2. The reference/historical value is the null; the verb fixes the direction of H_1.

(a) Parameter: the show-up proportion p. "Has increased" $\Rightarrow$ one-sided right.

$$H_0: \ p = 0.62 \ (\text{or } p \leq 0.62), \qquad H_1: \ p > 0.62.$$

(b) Parameter: the mean breaking load μ. "Is below" $\Rightarrow$ one-sided left.

$$H_0: \ \mu = 50 \ (\text{or } \mu \geq 50), \qquad H_1: \ \mu < 50.$$

(c) Parameter: the mean response time μ. "Has not changed" is the default claim; the reporter wants to detect any change, so the alternative is two-sided.

$$\boxed{H_0: \ \mu = 9.0, \qquad H_1: \ \mu \neq 9.0 \ \ (\text{two-sided})}$$

A3. One-sided alternatives put all of α in the tail H_1 points to; two-sided alternatives split α into two tails of $\alpha/2$. Standard cut-offs: $z_{0.05} = 1.645$, $z_{0.025} = 1.96$, $z_{0.01} = 2.326$, $z_{0.005} = 2.576$.

(a) Right tail, area 0.05: $C = \{\, Z \geq 1.645 \,\}$.

(b) Two tails, 0.025 each: $C = \{\, |Z| \geq 1.96 \,\}$.

(c) Left tail, area 0.01: $C = \{\, Z \leq -2.326 \,\}$.

(d) Two tails, 0.005 each: $C = \{\, |Z| \geq 2.576 \,\}$.

$$\boxed{\text{(a) } Z \geq 1.645; \quad \text{(b) } |Z| \geq 1.96; \quad \text{(c) } Z \leq -2.326; \quad \text{(d) } |Z| \geq 2.576}$$

A4. A normal mean with σ *known* calls for the Z-statistic, which is $\mathrm{N}(0,1)$ when $\mu = \mu_0$. "Runs warm" names a direction (higher), so the test is one-sided right.

Step 1: hypotheses.

$$H_0 : \mu = 37.0 \ (\mu \leq 37.0), \qquad H_1 : \mu > 37.0.$$

Step 2: statistic and null distribution. With σ known,

$$Z = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} \ \sim \ \mathrm{N}(0,1) \ \ \text{under } H_0.$$

Step 3: compute z_{obs}. Standard error $\sigma/\sqrt{n} = 0.4/\sqrt{25} = 0.4/5 = 0.08$, so

$$z_{\text{obs}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{37.14 - 37.0}{0.08} = \frac{0.14}{0.08} = 1.75.$$

Step 4: critical value and rejection region. One-sided right at $\alpha = 0.05$: $z_{0.05} = 1.645$, so $C = \{\, Z \geq 1.645 \,\}$.

Step 5: decide. $z_{\text{obs}} = 1.75 \geq 1.645$, so $z_{\text{obs}} \in C$.

Reject H_0 at $\alpha = 0.05$: significant evidence that the mean temperature exceeds 37.0 degrees C.

A5. The rule is $C = \{\, |Z| \geq 1.96 \,\}$, a two-sided region with each tail carrying 0.025, so it operates at $\alpha = 0.05$. Applying the rule is pure membership-checking.

(a) $|2.05| = 2.05 \geq 1.96 \Rightarrow$ in C: reject H_0 at $\alpha = 0.05$.

(b) $|-1.70| = 1.70 < 1.96 \Rightarrow$ not in C: fail to reject H_0 at $\alpha = 0.05$.

(c) $|-3.10| = 3.10 \geq 1.96 \Rightarrow$ in C: reject H_0 at $\alpha = 0.05$.

(a) reject; (b) fail to reject; (c) reject — all at $\alpha = 0.05$.

A6. One observed statistic $z_{\text{obs}} = 2.40$, two significance levels. Two-sided cut-offs: $z_{0.025} = 1.96$ at $\alpha = 0.05$ and $z_{0.005} = 2.576$ at $\alpha = 0.01$.

(a) At $\alpha = 0.05$: $C = \{|Z| \geq 1.96\}$ and $|2.40| = 2.40 \geq 1.96$, so reject H_0.

(b) At $\alpha = 0.01$: $C = \{|Z| \geq 2.576\}$ and $|2.40| = 2.40 < 2.576$, so fail to reject H_0.

(c) The same evidence is "significant" at the looser standard but "not significant" at the stricter one: the verdict is a function of α, not a fixed property of the data. If α were chosen *after* seeing $z_{\text{obs}} = 2.40$ one could pick whichever level delivers the preferred verdict, so α must be fixed before the data are seen.

Reject at $\alpha = 0.05$; fail to reject at $\alpha = 0.01$.

A7. This is the asymmetry trap: "fail to reject" is not "accept" or "prove."

(a) Failing to reject means only that the data were not surprising enough, under H_0, to overturn it. A small $|z_{\text{obs}}|$ is consistent with $\mu = 37.0$ but equally consistent with $\mu = 37.05$, $\mu = 36.96$, and many nearby values, so the data cannot single out 37.0. Absence of strong evidence against H_0 is not evidence for it.

(b) α controls the probability of a **Type I error** — wrongly rejecting a true H_0. It does *not* control the **Type II error** β — failing to reject a false H_0 — which can be large, so a non-rejection carries no guarantee that H_0 is true.

(c) Correct wording: "At $\alpha = 0.05$ the data do not provide sufficient evidence to conclude that the mean temperature differs from 37.0 degrees C."

Fail to reject H_0; we never "prove" H_0.

B. Type I and Type II errors, power, and sample size

B1. Let X be the number of damaged cartons among the 15 sampled. The rule rejects (H_1) when $X \geq 3$ and accepts ($H_0 : p = 0.05$) when $X \leq 2$.

(a) *Type I error.* "True H_0" fixes $X \sim \text{Bin}(15, 0.05)$. A Type I error is rejecting when H_0 holds, i.e. $X \geq 3$. With $q = 0.95$,

$$\mathbb{P}(X = 0) = 0.95^{15} = 0.46329,$$
$$\mathbb{P}(X = 1) = 15(0.05)(0.95^{14}) = 0.36576,$$
$$\mathbb{P}(X = 2) = \tbinom{15}{2}(0.05^2)(0.95^{13}) = 0.13475.$$

So $\mathbb{P}(X \leq 2) = 0.96380$ and

$$\alpha = \mathbb{P}(X \geq 3) = 1 - 0.96380.$$

$$\boxed{\alpha \approx 0.0362}$$

(b) *Type II error.* Now "true" is the alternative $p = 0.30$, so re-standardize: $X \sim \text{Bin}(15, 0.30)$. A miss means the rule fails to reject, i.e. $X \leq 2$. With $q = 0.70$,

$$\mathbb{P}(X = 0) = 0.70^{15} = 0.00475,$$
$$\mathbb{P}(X = 1) = 15(0.30)(0.70^{14}) = 0.03052,$$
$$\mathbb{P}(X = 2) = \tbinom{15}{2}(0.30^2)(0.70^{13}) = 0.09156.$$

$$\beta = \mathbb{P}(X \leq 2) = 0.00475 + 0.03052 + 0.09156 = 0.12683.$$

$$\boxed{\beta \approx 0.127, \qquad \text{power} = 1 - \beta \approx 0.873}$$

The rule catches a 30%-defect supplier about 87% of the time.

B2. Known σ, normal data, one-directional alternative ("overfill") $\Rightarrow$ one-sided upper z-test. $H_0 : \mu = 355$, $H_1 : \mu > 355$.

(a) *Critical value.* $\sigma/\sqrt{n} = 4/\sqrt{16} = 1$. With $z_{0.05} = 1.645$,

$$c = \mu_0 + z_{0.05}\,\frac{\sigma}{\sqrt{n}} = 355 + 1.645(1) = 356.645 \text{ mL}.$$

$$\boxed{c \approx 356.65 \text{ mL}}$$

(b) *Type II error at $\mu_1 = 357$.* Standardize the boundary under μ_1:

$$\frac{c - \mu_1}{\sigma/\sqrt{n}} = \frac{356.645 - 357}{1} = -0.355,$$

so $\beta = \Phi(-0.355) = 0.3613$. (Equivalently $z_\alpha - d = 1.645 - (357 - 355)(4)/4 = 1.645 - 2 = -0.355$.)

$$\boxed{\beta \approx 0.361}$$

(c) *Power.* $1 - \beta \approx 0.639$: if the line truly overfills to 357 mL, this rule catches it only about 64% of the time.

$$\boxed{\text{power} \approx 0.639}$$

B3. One-sided upper z-test, $H_0 : \mu = 400$ vs $H_1 : \mu > 400$, known $\sigma = 20$, $n = 25$, $z_{0.05} = 1.645$. The effect lever is $d = (\mu_1 - 400)\sqrt{25}/20 = (\mu_1 - 400)/4$, and $\text{power}(\mu_1) = 1 - \Phi(z_\alpha - d)$:

$$\mu_1 = 405 : d = 1.250,\ z_\alpha - d = 0.395,\ \text{power} = 1 - \Phi(0.395) = 0.346,$$
$$\mu_1 = 410 : d = 2.500,\ z_\alpha - d = -0.855,\ \text{power} = 1 - \Phi(-0.855) = 0.804,$$
$$\mu_1 = 415 : d = 3.750,\ z_\alpha - d = -2.105,\ \text{power} = 1 - \Phi(-2.105) = 0.982,$$
$$\mu_1 = 420 : d = 5.000,\ z_\alpha - d = -3.355,\ \text{power} = 1 - \Phi(-3.355) = 1.000.$$

Power rises monotonically toward 1 as the true mean moves farther above the null — the bigger the real improvement, the more reliably the test detects it.

$$\boxed{\text{power}(405, 410, 415, 420) \approx 0.346,\ 0.804,\ 0.982,\ 1.000}$$

B4. One-sided upper z-test, $H_0 : \mu = 50$ vs $H_1 : \mu > 50$, fixed effect $\mu_1 - \mu_0 = 3$, $\sigma = 6$, $z_{0.05} = 1.645$. Only $\sqrt{n}$ changes in $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma = 3\sqrt{n}/6 = \sqrt{n}/2$:

$$\begin{aligned} n = 16 &: d = 2.000,\ z_\alpha - d = -0.355,\ \text{power} = 1 - \Phi(-0.355) = 0.639, \\ n = 36 &: d = 3.000,\ z_\alpha - d = -1.355,\ \text{power} = 1 - \Phi(-1.355) = 0.912, \\ n = 64 &: d = 4.000,\ z_\alpha - d = -2.355,\ \text{power} = 1 - \Phi(-2.355) = 0.991. \end{aligned}$$

Holding the effect and α fixed, enlarging the sample drives the power up: detection of a 3-bushel gain climbs from about 0.64 at $n = 16$ to near certainty at $n = 64$.

$$\boxed{\text{power} = 0.639\ (n{=}16),\ 0.912\ (n{=}36),\ 0.991\ (n{=}64)}$$

B5. One-sided upper z-test, $H_0 : \mu = 100$ vs $H_1 : \mu > 100$, $\sigma = 15$, $n = 36$, alternative $\mu_1 = 105$. The effect lever is fixed: $d = (105 - 100)\sqrt{36}/15 = (5)(6)/15 = 2.000$.

(a) With $z_{0.01} = 2.326$, $z_{0.05} = 1.645$, $z_{0.10} = 1.282$ and power $= 1 - \Phi(z_\alpha - 2)$:

$$\begin{aligned} \alpha = 0.01 &: z_\alpha - d = 0.326,\ \text{power} = 1 - \Phi(0.326) = 0.372, \\ \alpha = 0.05 &: z_\alpha - d = -0.355,\ \text{power} = 1 - \Phi(-0.355) = 0.639, \\ \alpha = 0.10 &: z_\alpha - d = -0.718,\ \text{power} = 1 - \Phi(-0.718) = 0.764. \end{aligned}$$

$$\boxed{\text{power} = 0.372\ (\alpha{=}.01),\ 0.639\ (\alpha{=}.05),\ 0.764\ (\alpha{=}.10)}$$

(b) Raising α lowers the critical z_α, which raises power — but at the price of more false alarms: the firm would wrongly reject more truly-at-100 batches of opinion. Power is bought with Type I risk.

B6. Known σ, one-sided alternative (a drop), target power $\Rightarrow$ the sample-size formula. $H_0 : \mu = 72$, $H_1 : \mu < 72$, $\alpha = 0.05$, power 0.90, so $z_\alpha = z_{0.05} = 1.645$ and $\beta = 0.10 \Rightarrow z_\beta = z_{0.10} = 1.282$. The direction is "<", but $|\mu_1 - \mu_0| = |68 - 72| = 4$ is what enters:

$$n \geq \left(\frac{(z_\alpha + z_\beta)\,\sigma}{\mu_1 - \mu_0} \right)^2 = \left(\frac{(1.645 + 1.282) \cdot 10}{4} \right)^2 = \left(\frac{29.27}{4} \right)^2 = (7.3175)^2 = 53.55.$$

Round up: $n = 54$ patients. Auditing the integer n, with $d = 4\sqrt{54}/10 = 2.939$,

$$\text{power} = 1 - \Phi(z_\alpha - d) = 1 - \Phi(1.645 - 2.939) = 1 - \Phi(-1.294) = 0.902,$$

just above the 0.90 target, as rounding up guarantees.

$$\boxed{n = 54, \quad \text{achieved power} \approx 0.902}$$

B7. One-sided upper z-test, $H_0 : \mu = 20$ vs $H_1 : \mu > 20$, $\sigma = 9$, $n = 9$, $\alpha = 0.05$. Their arithmetic: $z_{\text{obs}} = (22.5 - 20)/(9/\sqrt{9}) = 2.5/3 = 0.833$, which is below $z_{0.05} = 1.645$, giving the p-value $1 - \Phi(0.833) \approx 0.20$ and no rejection. The arithmetic is fine; the *interpretation* is the issue.

(a) *Power against* $\mu_1 = 26$. $d = (\mu_1 - \mu_0)\sqrt{n}/\sigma = (26 - 20)(3)/9 = 2.000$,

$$\text{power} = 1 - \Phi(z_\alpha - d) = 1 - \Phi(1.645 - 2.000) = 1 - \Phi(-0.355) = 0.639.$$

$$\boxed{\text{power against } \mu_1 = 26 \text{ is } \approx 0.639}$$

(b) With power only about 0.64, even a real jump to $\mu_1 = 26$ would be missed roughly a third of the time. A non-rejection from a test this size is therefore consistent *both* with no effect *and* with a genuine effect the small sample failed to catch, so the large p-value does not single out H_0 as true. "Absence of evidence is not evidence of absence": the honest statement is "we did not detect an effect," not "the compound has no effect."

(c) *Sample size for power* 0.90 *at* $\mu_1 = 26$. $z_{0.05} = 1.645$, $z_{0.10} = 1.282$, $|\mu_1 - \mu_0| = 6$, $\sigma = 9$:

$$n \geq \left(\frac{(1.645 + 1.282) \cdot 9}{6}\right)^2 = \left(\frac{26.343}{6}\right)^2 = (4.3905)^2 = 19.28.$$

Round up: $n = 20$, more than double the 9 actually run. The original design never had a real chance of detecting the effect it was meant to find.

$$\boxed{n = 20 \text{ (vs. the 9 used)}}$$

C. The z-test for a mean and p-values

C1. A mean is in question, $\sigma = 1.2$ is given by the tooling (not estimated from these 36 rotors), and "off target in either direction" is a two-sided alternative: a two-sided z-test.

Hypotheses. $H_0 : \mu = 320.0$ versus $H_1 : \mu \neq 320.0$, $\alpha = 0.05$.

Standard error and statistic.

$$\frac{\sigma}{\sqrt{n}} = \frac{1.2}{\sqrt{36}} = \frac{1.2}{6} = 0.20 \text{ mm},$$
$$z_{\text{obs}} = \frac{\overline{X} - \mu_0}{\sigma/\sqrt{n}} = \frac{320.5 - 320.0}{0.20} = \frac{0.5}{0.20} = 2.50.$$

Decision. The two-sided cutoff is $z_{0.025} = 1.96$. Since $|z_{\text{obs}}| = 2.50 \geq 1.96$, z_{obs} lands in the rejection region.

$$\boxed{\text{Reject } H_0\text{: the mean diameter is off target } (z_{\text{obs}} = 2.50,\ \alpha = 0.05).}$$

C2. A mean with $\sigma = 3.0$ known from calibration; "exceeding" names one direction.

(a) $H_0 : \mu = 25.0$ versus $H_1 : \mu > 25.0$, $\alpha = 0.05$. The inspector cares only whether the plant is *above* the claimed BOD, so all of α goes in the upper tail.

(b) Standard error and statistic:

$$\frac{\sigma}{\sqrt{n}} = \frac{3.0}{\sqrt{25}} = \frac{3.0}{5} = 0.60 \text{ mg/L},$$
$$z_{\text{obs}} = \frac{26.4 - 25.0}{0.60} = \frac{1.4}{0.60} = 2.333.$$

Upper-tail cutoff $z_{0.05} = 1.645$. Since $z_{\text{obs}} = 2.333 \geq 1.645$, reject.

$$\boxed{\text{Reject } H_0\text{: evidence the plant exceeds 25 mg/L } (z_{\text{obs}} = 2.33,\ \alpha = 0.05).}$$

(c) At $\alpha = 0.01$ the upper-tail cutoff is $z_{0.01} = 2.326$. Since $z_{\text{obs}} = 2.333 \geq 2.326$ (just barely), we still reject. The decision does not change, though the margin is now razor-thin.

C3. A mean with $\sigma = 0.8$ known from the fill process; "below the label" is a single downward direction, so a one-sided (lower-tail) z-test.

Hypotheses. $H_0 : \mu = 50.0$ versus $H_1 : \mu < 50.0$, $\alpha = 0.05$.

Standard error and statistic.

$$\frac{\sigma}{\sqrt{n}} = \frac{0.8}{\sqrt{16}} = \frac{0.8}{4} = 0.20 \text{ mg/mL},$$
$$z_{\text{obs}} = \frac{49.6 - 50.0}{0.20} = \frac{-0.4}{0.20} = -2.00.$$

Decision. The lower-tail cutoff is $-z_{0.05} = -1.645$. Since $z_{\text{obs}} = -2.00 \le -1.645$, z_{obs} is past the cutoff.

Reject H_0: the mean is below the 50 mg/mL label ($z_{\text{obs}} = -2.00,\ \alpha = 0.05$).

C4. Two-sided p-value for the foundry, where $z_{\text{obs}} = 2.50$.

(a) With $\Phi(2.50) \approx 0.99379$, the upper-tail area is $1 - \Phi(2.50) \approx 0.00621$, so

$$p\text{-value} = 2\big(1 - \Phi(2.50)\big) = 2(0.00621) = 0.01242 \approx 0.0124.$$

p-value ≈ 0.0124.

(b) If the process were truly on target at 320.0 mm, a sample mean at least this far from target (in either direction) would arise only about 1.2% of the time — a surprising result under H_0.

(c) Since $0.0124 \le 0.05$, the p-value rule rejects H_0 at the 5% level. This matches the critical-value decision of Problem C1 ($|z_{\text{obs}}| = 2.50 \ge 1.96$): the two decision forms always agree.

C5. A large-sample one-sided test on a skewed parent.

(a) The parent (watchlist counts) is right-skewed, not normal, but $\sigma = 3.5$ is supplied from history and $n = 49$ is large, so by the Central Limit Theorem $\overline{X}$ is approximately normal and z_{obs} is approximately $\text{N}(0, 1)$ under H_0. A normal parent is not needed.

(b) $H_0 : \mu = 8.0$ versus $H_1 : \mu > 8.0$. Standard error and statistic:

$$\frac{\sigma}{\sqrt{n}} = \frac{3.5}{\sqrt{49}} = \frac{3.5}{7} = 0.50,$$
$$z_{\text{obs}} = \frac{8.9 - 8.0}{0.50} = \frac{0.9}{0.50} = 1.80.$$

Upper-tail p-value, using $\Phi(1.80) \approx 0.96407$:

$$p\text{-value} = 1 - \Phi(1.80) = 1 - 0.96407 = 0.03593 \approx 0.0359.$$

(c) Since $0.0359 \le 0.05$, reject.

Reject H_0: the watchlist mean increased ($z_{\text{obs}} = 1.80,\ p \approx 0.036$).

C6. Skewed parent (checkout times) but $n = 225$ is large and $\sigma = 15.0$ is known, so a large-sample two-sided z-test by the CLT; "changed in either direction" makes it two-sided.

Hypotheses. $H_0 : \mu = 42.0$ versus $H_1 : \mu \ne 42.0$, $\alpha = 0.05$.

Standard error and statistic.

$$\frac{\sigma}{\sqrt{n}} = \frac{15.0}{\sqrt{225}} = \frac{15.0}{15} = 1.00 \text{ s},$$
$$z_{\text{obs}} = \frac{39.6 - 42.0}{1.00} = \frac{-2.4}{1.00} = -2.40.$$

p-value. Using $\Phi(2.40) \approx 0.99180$, the upper-tail area is $1 - 0.99180 = 0.00820$, so

$$p\text{-value} = 2\big(1 - \Phi(|-2.40|)\big) = 2(0.00820) = 0.01640 \approx 0.0164.$$

Decision. Since $0.0164 \le 0.05$ (equivalently $|z_{\text{obs}}| = 2.40 \ge z_{0.025} = 1.96$), reject.

Reject H_0: mean checkout time changed ($z_{\text{obs}} = -2.40,\ p \approx 0.016$).

C7. Conceptual contrast on what a p-value of 0.03 does and does not mean.

(a) *False.* The p-value is computed *assuming H_0 is true*; it is a statement about the data, not the probability $\mathbb{P}(H_0 \text{ true})$, which classical testing does not assign.

(b) *True.* This is the definition: the p-value is the probability, under H_0, of a test statistic at least as extreme as the one observed.

(c) *False.* The p-value is not the probability of H_0, so $1 - p$ is not the probability of H_1; neither hypothesis is assigned a probability here.

(d) *True.* The decision rule is to reject H_0 at level α exactly when p-value $\leq \alpha$, and $0.03 \leq 0.05$.

(e) *False.* A small p-value signals that the data are unlikely under H_0, not that the effect is large; statistical significance does not measure practical importance (a tiny effect can be significant with large n).

Only (b) and (d) are true; the p-value is $\mathbb{P}(\text{data this extreme} \mid H_0)$, not $\mathbb{P}(H_0)$.

D. The t-test for a mean (σ unknown)

D1. A mean is in question, the fill is approximately normal, and σ is not given (S was computed from the sample) — the one-sample t-test with $n - 1 = 15$ degrees of freedom. "Off target in either direction" makes it two-sided.

$H_0 : \mu = 355$ versus $H_1 : \mu \neq 355$, $\alpha = 0.05$. Standard error $S/\sqrt{n} = 4.0/\sqrt{16} = 4.0/4 = 1.000$ mL, so

$$t_{\text{obs}} = \frac{352.4 - 355}{1.000} = \frac{-2.6}{1.000} = -2.600.$$

The two-sided cutoff is $t_{0.025,15} = 2.131$. Since $|t_{\text{obs}}| = 2.600 > 2.131$, the statistic falls in the rejection region.

$\lvert t_{\text{obs}} \rvert = 2.600 > 2.131 \Rightarrow$ reject H_0;
the line is filling significantly off the 355 mL target at $\alpha = 0.05$.

D2. (a) The parameter is a mean weight, the population is roughly normal, and σ is unknown (only S is given), so this is a one-sample t-test with $n - 1 = 9$ degrees of freedom. The directional claim "increased" puts all of α in the upper tail: $H_0 : \mu = 1.80$ versus $H_1 : \mu > 1.80$.

(b) Standard error $S/\sqrt{n} = 0.21/\sqrt{10} = 0.21/3.162 = 0.0664$ kg, so

$$t_{\text{obs}} = \frac{1.92 - 1.80}{0.0664} = \frac{0.12}{0.0664} = 1.807.$$

The upper-tail cutoff is $t_{0.05,9} = 1.833$. Since $t_{\text{obs}} = 1.807 < 1.833$, the statistic does *not* reach the boundary.

$t_{\text{obs}} = 1.807 < 1.833 \Rightarrow$ do not reject H_0; no significant evidence the feed raised weight at $\alpha = 0.05$.

(c) No — it would still fail to reject, even more clearly. A two-sided test at the same $\alpha = 0.05$ uses the larger cutoff $t_{0.025,9} = 2.262$, and $|t_{\text{obs}}| = 1.807 < 2.262$. Splitting α between two tails raises the bar, so a result that misses the one-sided cutoff cannot clear the two-sided one.

D3. Mean concentration, approximately normal, σ unknown (S estimated from the data). The directional claim "below the cap" is a *lower-tail* one-sample t-test with $n - 1 = 11$ degrees of freedom; reject only for sufficiently negative t_{obs}.

$H_0 : \mu = 15$ versus $H_1 : \mu < 15$, $\alpha = 0.05$. Standard error $S/\sqrt{n} = 2.4/\sqrt{12} = 2.4/3.464 = 0.6928$ ppb, so

$$t_{\text{obs}} = \frac{13.6 - 15}{0.6928} = \frac{-1.4}{0.6928} = -2.021.$$

The lower-tail cutoff is $-t_{0.05,11} = -1.796$. Since $t_{\text{obs}} = -2.021 < -1.796$, reject.

$t_{\text{obs}} = -2.021 < -1.796 \Rightarrow$ reject H_0; the mean lead level is significantly below 15 ppb at $\alpha = 0.05$.

D4. (a) Sum the data: $\sum x_i = 58 + 61 + 55 + 63 + 60 + 57 + 62 + 56 = 472$, so $\overline{X} = 472/8 = 59.0$. The deviations $x_i - \overline{X}$ are $(-1, 2, -4, 4, 1, -2, 3, -3)$; their squares are $(1, 4, 16, 16, 1, 4, 9, 9)$, summing to

$$\sum (x_i - \overline{X})^2 = 1 + 4 + 16 + 16 + 1 + 4 + 9 + 9 = 60.0,$$

hence

$$S^2 = \frac{60.0}{7} = 8.571, \qquad S = \sqrt{8.571} = 2.928.$$

(b) Mean hardness, normal, σ unknown (built from the raw data); two-sided wording "differs from spec" gives a one-sample two-sided t-test with $n - 1 = 7$ degrees of freedom. $H_0 : \mu = 60$ versus $H_1 : \mu \neq 60$, $\alpha = 0.05$. Standard error $S/\sqrt{n} = 2.928/\sqrt{8} = 2.928/2.828 = 1.035$, so

$$t_{\text{obs}} = \frac{59.0 - 60}{1.035} = \frac{-1.0}{1.035} = -0.966.$$

Compare $|t_{\text{obs}}| = 0.966$ with $t_{0.025,7} = 2.365$. Since $0.966 < 2.365$, do not reject.

$|t_{\text{obs}}| = 0.966 < 2.365 \Rightarrow$ do not reject H_0;
the castings are consistent with a mean hardness of 60.

D5. Each student is measured twice (before and after), so the data are paired, not two independent samples. Reduce each pair to a difference $d_i = \text{After} - \text{Before}$ and run a one-sample t-test on the d_i against $\mu_0 = 0$ with $n - 1 = 5$ degrees of freedom. "Raised" is upper-tail: $H_0 : \mu_d = 0$ versus $H_1 : \mu_d > 0$, $\alpha = 0.05$.

Differences:

$$d_i : \quad 40,\ 15,\ 25,\ -5,\ 40,\ 35.$$

Mean: $\sum d_i = 40 + 15 + 25 - 5 + 40 + 35 = 150$, so $\bar{d} = 150/6 = 25.0$. Deviations $d_i - \bar{d}$ are $(15, -10, 0, -30, 15, 10)$; squared and summed,

$$\sum (d_i - \bar{d})^2 = 225 + 100 + 0 + 900 + 225 + 100 = 1550,$$

$$S_d^2 = \frac{1550}{5} = 310.0, \qquad S_d = \sqrt{310.0} = 17.61.$$

Standard error $S_d/\sqrt{n} = 17.61/\sqrt{6} = 17.61/2.449 = 7.188$, so

$$t_{\text{obs}} = \frac{\bar{d} - 0}{S_d/\sqrt{n}} = \frac{25.0}{7.188} = 3.478.$$

The upper-tail cutoff is $t_{0.05,5} = 2.015$. Since $3.478 > 2.015$, reject.

$t_{\text{obs}} = 3.478 > 2.015 \Rightarrow$ reject H_0; the program significantly raised mean scores at $\alpha = 0.05$.

D6. The same buffer solutions are read by both meters, so the readings are paired. Test the mean difference against 0 with a one-sample t-test on the d_i, $n - 1 = 6$ degrees of freedom. "Disagree on average" with no direction makes it two-sided: $H_0 : \mu_d = 0$ versus $H_1 : \mu_d \neq 0$, $\alpha = 0.05$.

Sums: $\sum d_i = 0.05 - 0.02 + 0.08 + 0.03 + 0.06 - 0.01 + 0.07 = 0.26$, so $\bar{d} = 0.26/7 = 0.03714$; $\sum d_i^2 = 0.0025 + 0.0004 + 0.0064 + 0.0009 + 0.0036 + 0.0001 + 0.0049 = 0.01880$. By the computing form,

$$S_d^2 = \frac{1}{6}\big(0.01880 - 7 \cdot 0.03714^2\big) = \frac{1}{6}\big(0.01880 - 0.009657\big) = \frac{0.009143}{6} = 0.001524,$$

$$S_d = \sqrt{0.001524} = 0.03904.$$

Standard error $S_d/\sqrt{n} = 0.03904/\sqrt{7} = 0.03904/2.646 = 0.01475$, so

$$t_{\text{obs}} = \frac{0.03714 - 0}{0.01475} = 2.517.$$

The two-sided cutoff is $t_{0.025,6} = 2.447$. Since $|t_{\text{obs}}| = 2.517 > 2.447$, reject.

$$\boxed{|t_{\text{obs}}| = 2.517 > 2.447 \;\Rightarrow\; \text{reject } H_0;\ \text{the two meters disagree on average at } \alpha = 0.05.}$$

D7. (a) Because σ is unknown and replaced by S, the correct reference distribution is t_9, not $\mathrm{N}(0,1)$; comparing to a z cutoff is the classic look-alike error. Both analysts compute the same statistic. Standard error $S/\sqrt{n} = 0.27/\sqrt{10} = 0.27/3.162 = 0.0854$ mm, so

$$\text{stat} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{25.18 - 25.00}{0.0854} = \frac{0.18}{0.0854} = 2.108.$$

Comparing $|2.108|$ with the (wrong) cutoff $z_{0.025} = 1.96$: since $2.108 > 1.96$, the z-user *rejects* $H_0 : \mu = 25.00$ and declares the lathe off target.

(b) The proper cutoff is $t_{0.025,9} = 2.262$. Comparing the same 2.108 with 2.262: $|2.108| < 2.262$, so the correct t-test does *not* reject. The two calls disagree.

(c) The t-test is the right one, so we do *not* reject: the data are consistent with a 25.00 mm mean at $\alpha = 0.05$. Estimating σ by S injects extra variability (the denominator $S/\sqrt{n}$ itself wobbles), fattening the tails, so the threshold for "too extreme" moves outward from 1.96 to 2.262. Using the z cutoff ignores this and rejects too readily: its true Type I error exceeds the advertised 5%.

$$\boxed{\begin{array}{l} z \text{ says reject, } t \text{ says do not; the } t\text{-test is correct} \\ \text{because } \sigma \text{ was estimated, and } t_{0.025,9} = 2.262 > 1.96 = z_{0.025}. \end{array}}$$

E. Tests for a proportion (binomial data)

E1. Early voting, two-sided z-test. $H_0 : p = 0.40$ versus $H_1 : p \neq 0.40$; large-sample z-test ($np_0 = 200$, $n(1-p_0) = 300$, both ≥ 10). With $\widehat{p} = 224/500 = 0.448$ and the null standard error

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.40 \cdot 0.60}{500}} \\ &= \sqrt{0.00048} = 0.021909, \end{aligned}$$

the statistic is $z_{\text{obs}} = (0.448 - 0.400)/0.021909 = 0.048/0.021909 = 2.191$. The two-sided cutoff is $z_{0.025} = 1.96$; since $|2.191| \geq 1.96$ we reject.

$$\boxed{z_{\text{obs}} = 2.191 > 1.96 \;\Rightarrow\; \text{reject } H_0 :\ \text{early-vote rate differs from 40\%.}}$$

E2. Discharge rate has risen, one-sided z-test. $H_0 : p = 0.70$ versus $H_1 : p > 0.70$; valid since $np_0 = 308$ and $n(1-p_0) = 132$ exceed 10. With $\widehat{p} = 322/440 = 0.731818$ and

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{0.70 \cdot 0.30}{440}} = \sqrt{\frac{0.21}{440}} \\ &= \sqrt{0.00047727} = 0.021847, \end{aligned}$$

we get $z_{\text{obs}} = (0.731818 - 0.700000)/0.021847 = 0.031818/0.021847 = 1.456$. The right-sided cutoff is $z_{0.05} = 1.645$; since $1.456 < 1.645$ we do *not* reject.

$$\boxed{z_{\text{obs}} = 1.456 < 1.645 \;\Rightarrow\; \text{fail to reject: no significant rise above 70\%.}}$$

E3. Underfill rate, validity check then one-sided test.

(a) Under $H_0\colon p = 0.05$, $np_0 = 600 \cdot 0.05 = 30 \geq 10$ and $n(1-p_0) = 600 \cdot 0.95 = 570 \geq 10$. Both conditions hold, so the large-sample z-test is valid.

(b) $H_0\colon p = 0.05$ versus $H_1\colon p < 0.05$. With $\widehat{p} = 21/600 = 0.035$ and

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{0.05 \cdot 0.95}{600}} = \sqrt{\frac{0.0475}{600}} \\ &= \sqrt{0.00007917} = 0.008898, \end{aligned}$$

the statistic is $z_{\mathrm{obs}} = (0.035 - 0.050)/0.008898 = -0.015/0.008898 = -1.686$. The left-sided cutoff is $-z_{0.05} = -1.645$; since $-1.686 \leq -1.645$ we reject.

(c)

$$\boxed{z_{\mathrm{obs}} = -1.686 \leq -1.645 \;\Rightarrow\; \text{reject } H_0\colon \text{ underfill rate is below 5\%.}}$$

There is significant evidence at the 5% level that the line underfills fewer than 5% of bottles.

E4. Conversion rate, the right SE versus the wrong SE.

(a) $H_0\colon p = 0.30$ versus $H_1\colon p \neq 0.30$. With $\widehat{p} = 246/750 = 0.328$ and the correct test standard error

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\frac{p_0(1-p_0)}{n}} = \sqrt{\frac{0.30 \cdot 0.70}{750}} \\ &= \sqrt{0.00028} = 0.016733, \end{aligned}$$

the statistic is $z_{\mathrm{obs}} = (0.328 - 0.300)/0.016733 = 0.028/0.016733 = 1.673$.

(b) Using the interval standard error built from $\widehat{p}$,

$$\begin{aligned} \mathrm{SE}_{\widehat{p}} &= \sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = \sqrt{\frac{0.328 \cdot 0.672}{750}} \\ &= \sqrt{0.00029389} = 0.017143, \end{aligned}$$

gives the look-alike statistic $z_{\mathrm{wrong}} = 0.028/0.017143 = 1.633$.

(c) Only $z_{\mathrm{obs}} = 1.673$ is correct: a hypothesis test standardizes by the null standard error $\sqrt{p_0(1-p_0)/n}$, because under H_0 the true variance of $\widehat{p}$ uses p_0, not $\widehat{p}$. The interval SE belongs to a confidence interval, not a test. The two-sided cutoff is $z_{0.025} = 1.96$; since $|1.673| < 1.96$ we do *not* reject.

$$\boxed{z_{\mathrm{obs}} = 1.673 \text{ (use } p_0), \text{ not } 1.633;\; 1.673 < 1.96 \;\Rightarrow\; \text{fail to reject.}}$$

E5. Seed germination, small-n exact test. $H_0\colon p = 0.90$ versus $H_1\colon p < 0.90$. With $n = 15$ and $p_0 = 0.90$ pressed near 1, the normal approximation is unreliable ($n(1-p_0) = 1.5 < 10$), so use the exact binomial test. Since $\widehat{p} = 10/15$ is low, the evidence-toward-H_1 tail is $\mathbb{P}(X \leq 10)$ under $\mathrm{Bin}(15, 0.90)$. Writing $q = 0.10$ and computing the largest lower terms,

$$\begin{aligned} \mathbb{P}(X = 10) &= \binom{15}{10} 0.90^{10} 0.10^{5} = 0.010471, \\ \mathbb{P}(X = 9) &= \binom{15}{9} 0.90^{9} 0.10^{6} = 0.001939, \\ \mathbb{P}(X \leq 8) &= 0.000311, \end{aligned}$$

so the lower-tail p-value is

$$\mathbb{P}(X \leq 10) = 0.010471 + 0.001939 + 0.000311 = 0.012721.$$

Since $0.0127 \leq 0.05$ we reject.

$$\boxed{p\text{-value} = \mathbb{P}(X \leq 10) = 0.0127 \leq 0.05 \;\Rightarrow\; \text{reject } H_0\colon \text{ germination is below 90\%.}}$$

E6. Gift-wrap rate, two-sided p-value. $H_0\colon p = 0.20$ versus $H_1\colon p \neq 0.20$; valid since $np_0 = 80$ and $n(1-p_0) = 320$ exceed 10. With $\widehat{p} = 68/400 = 0.17$ and

$$\mathrm{SE}_0 = \sqrt{\frac{0.20 \cdot 0.80}{400}} = \sqrt{0.0004} = 0.02,$$

the statistic is $z_{\mathrm{obs}} = (0.17 - 0.20)/0.02 = -0.03/0.02 = -1.50$. The two-sided p-value, using $\mathbb{P}(Z \le -1.50) = 0.0668$, is

$$p\text{-value} = 2\,\mathbb{P}(Z \le -1.50) = 2 \cdot 0.0668 = 0.1336.$$

Since $0.1336 > 0.05 > 0.01$, we fail to reject at both levels.

$$\boxed{p\text{-value} = 0.1336 \;\Rightarrow\; \text{fail to reject at 5\% and 1\%.}}$$

The observed rate is consistent with the advertised 20%.

E7. Undamaged-parcel rate vs. a contractual target.

(a) Under $H_0\colon p = 0.98$, $np_0 = 250 \cdot 0.98 = 245 \ge 10$ and $n(1-p_0) = 250 \cdot 0.02 = 5$. Since $n(1-p_0) = 5 < 10$, the validity condition is *not* comfortably met; the z-test is borderline here because p_0 is close to 1. We proceed with the large-sample test but flag that an exact binomial check would be safer.

(b) $H_0\colon p = 0.98$ versus $H_1\colon p < 0.98$. With $\widehat{p} = 238/250 = 0.952$ and the standard error built from the contractual $p_0 = 0.98$,

$$\begin{aligned}\mathrm{SE}_0 &= \sqrt{\frac{0.98 \cdot 0.02}{250}} = \sqrt{\frac{0.0196}{250}} \\ &= \sqrt{0.0000784} = 0.008854,\end{aligned}$$

the statistic is $z_{\mathrm{obs}} = (0.952 - 0.980)/0.008854 = -0.028/0.008854 = -3.162$. The left-sided cutoff is $-z_{0.05} = -1.645$; since $-3.162 \le -1.645$ we reject.

(c)

$$\boxed{z_{\mathrm{obs}} = -3.162 \le -1.645 \;\Rightarrow\; \text{reject } H_0\colon \text{ below the 98\% contractual target.}}$$

There is significant evidence at the 5% level that the supplier's undamaged-delivery rate falls short of the contractual 98%.

F. Tests for a variance (χ^2)

F1. Two-sided variance test, df $= 15$. $H_0\colon \sigma^2 = 0.04$ versus $H_1\colon \sigma^2 \neq 0.04$. The claim is about a variance of a normal population and H_1 is two-sided, so use the χ^2 test on $(n-1)S^2/\sigma_0^2$ with $n - 1 = 15$ df.

$$\begin{aligned}\chi^2_{\mathrm{obs}} &= \frac{(n-1)S^2}{\sigma_0^2} = \frac{15 \cdot 0.075}{0.04} = \frac{1.125}{0.04} = 28.125, \\ \text{lower cutoff: } &\chi^2_{0.975,\,15} = 6.262, \\ \text{upper cutoff: } &\chi^2_{0.025,\,15} = 27.488.\end{aligned}$$

The acceptance region is (6.262, 27.488). Since $28.125 > 27.488$, the statistic falls in the upper rejection tail.

$$\boxed{\chi^2_{\mathrm{obs}} = 28.125 > 27.488 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence that the dose-volume variance differs from $0.04\,\mathrm{mL}^2$ — specifically, it is too large.

F2. Upper one-sided test, df = 20. $H_0\colon \sigma^2 = 2.5$ versus $H_1\colon \sigma^2 > 2.5$. "More variable parts" fixes an upper-tail alternative, so all of α sits in the right tail of χ^2_{20}.

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{20 \cdot 4.3}{2.5} = \frac{86}{2.5} = 34.40,$$

$$\text{critical value: } \chi^2_{0.05,\,20} = 31.410.$$

Reject only if $\chi^2_{\text{obs}} > 31.410$. Here $34.40 > 31.410$.

$$\boxed{\chi^2_{\text{obs}} = 34.40 > 31.410 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence that the casting line's mass variance has increased after the shutdown.

F3. Lower one-sided test, df = 24. $H_0\colon \sigma^2 = 0.09$ versus $H_1\colon \sigma^2 < 0.09$. "More consistent" means the variance has *decreased*, a lower-tail alternative, so all of α sits in the left tail of χ^2_{24}.

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{24 \cdot 0.045}{0.09} = \frac{1.08}{0.09} = 12.00,$$

$$\text{critical value: } \chi^2_{0.95,\,24} = 13.848.$$

For a lower-tail test we need the point with area $\alpha = 0.05$ to its *left*, i.e. area 0.95 to its right, which is $\chi^2_{0.95,\,24} = 13.848$. Reject if $\chi^2_{\text{obs}} < 13.848$. Here $12.00 < 13.848$.

$$\boxed{\chi^2_{\text{obs}} = 12.00 < 13.848 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

The firmware update significantly improved precision (the variance is significantly below $0.09\,\text{mL}^2$).

F4. Standard deviation claim, upper one-sided test, df = 12.

(a) The χ^2 test is built on a variance, so square the hypothesized standard deviation: $\sigma_0 = 0.40 \Rightarrow \sigma_0^2 = (0.40)^2 = 0.16$. The sample value squares to $S^2 = (0.62)^2 = 0.3844$.

(b) $H_0\colon \sigma = 0.40 \Leftrightarrow \sigma^2 = 0.16$ against $H_1\colon \sigma > 0.40 \Leftrightarrow \sigma^2 > 0.16$. "Spread is larger" is upper-tail one-sided; use the χ^2 test with $n - 1 = 12$ df.

(c) Compute the statistic and compare to the upper-tail cutoff:

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{12 \cdot 0.3844}{0.16} = \frac{4.6128}{0.16} = 28.83,$$

$$\text{critical value: } \chi^2_{0.05,\,12} = 21.026.$$

Since $28.83 > 21.026$,

$$\boxed{\chi^2_{\text{obs}} = 28.83 > 21.026 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence that the gauge's standard deviation exceeds 0.40 micrometers; the buyer's suspicion is supported.

F5. p-value for an upper-tail test, df = 23. $H_0\colon \sigma^2 = 1.2$ versus $H_1\colon \sigma^2 > 1.2$ (upper-tail χ^2 test).

(a) With $n = 24$ (df = 23),

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{23 \cdot 2.05}{1.2} = \frac{47.15}{1.2} = 39.29.$$

(b) The test is one-sided upper-tail, so $p = \mathbb{P}(\chi^2_{23} > 39.29)$ — a single right-tail area, not doubled. Against the table points $\chi^2_{0.05,\,23} = 35.172$ and $\chi^2_{0.025,\,23} = 38.076$, the statistic 39.29 lies *past* the 0.025 point, so

$$\chi^2_{\text{obs}} = 39.29 > 38.076 = \chi^2_{0.025,\,23},$$

$$p = \mathbb{P}(\chi^2_{23} > 39.29) < 0.025.$$

(c) Since the p-value is below 0.025, it is certainly below $\alpha = 0.05$.

$$\boxed{p\text{-value} < 0.025 < 0.05 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence that the station's recording variance exceeds the $1.2\,(^\circ\text{C})^2$ limit.

F6. Standard deviation claim, two-sided test, df $= 9$.

(a) Square the hypothesized standard deviation: $\sigma_0^2 = (0.030)^2 = 9.0 \times 10^{-4}$, and $S^2 = (0.021)^2 = 4.41 \times 10^{-4}$. The hypotheses are $H_0\colon \sigma^2 = 9.0 \times 10^{-4}$ against $H_1\colon \sigma^2 \neq 9.0 \times 10^{-4}$ ("too coarse *or* too fine" is two-sided), with $n - 1 = 9$ df. The statistic is

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{9 \cdot 4.41 \times 10^{-4}}{9.0 \times 10^{-4}} = \frac{3.969 \times 10^{-3}}{9.0 \times 10^{-4}} = 4.41.$$

(b) Equal tails of 0.025 on 9 df give the acceptance region (2.700, 19.023). Since $2.700 < 4.41 < 19.023$, the statistic lies inside.

$$\boxed{\chi^2_{\text{obs}} = 4.41 \in (2.700,\, 19.023) \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The beads are somewhat tighter than nominal, but not significantly so against this two-sided test — no recalibration is triggered.

F7. Conceptual: sensitivity of the χ^2 variance test to non-normality. Mechanically the analyst ran the upper-tail χ^2 test correctly: $\chi^2_{\text{obs}} = (29)(7.1)/4 = 51.475$, which exceeds $\chi^2_{0.05,\,29} = 42.557$, so the fixed-level rule formally rejects H_0. The problem is the *assumption*, not the arithmetic.

Assumption violated. The identity $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ holds *only* when the parent population is normal. Here the claim amounts are strongly right-skewed with heavy tails, so the true sampling distribution of S^2 is not the χ^2_{29} curve the test assumes.

Effect on the Type I error rate. The distribution of S^2 is governed by the parent's fourth moment (kurtosis). Heavy tails inflate the variance of S^2 well beyond what normal theory predicts, so even when $\sigma^2 = 4$ is true the statistic exceeds 42.557 far more often than 5% of the time. The nominal $\alpha = 0.05$ understates the real Type I error rate, which could be 15% or more. The single huge claim — an outlier, not a genuine shift in spread — is what drove the rejection.

Contrast with the t-test. A t-test for a *mean* on the same data leans on the central limit theorem: for $n = 30$ the sampling distribution of $\overline{X}$ is approximately normal despite the skew, so the t-test's error rate stays close to nominal and it is far more trustworthy. The variance χ^2 test has no such cushion; it is **not robust** to non-normality.

> Verify normality first: a "significant" χ^2 variance result on skewed, heavy-tailed data may be an artifact of the violated assumption.

G. Likelihood ratio tests and the test–CI duality

G1. $H_0\colon \mu = 500$ vs. $H_1\colon \mu \neq 500$; normal data, $\sigma = 4.0$ known, so the GLRT reduces to the two-sided z-test.

(a) With σ fixed, $L(\mu) = (2\pi\sigma^2)^{-n/2} \exp\!\big(-\frac{1}{2\sigma^2}\sum_i (x_i - \mu)^2\big)$. Over $\Omega = \mathbb{R}$ the best μ is $\overline{X}$; the null allows only μ_0. The $(2\pi\sigma^2)^{-n/2}$ prefactors cancel, and the identity $\sum_i (x_i - \mu_0)^2 = \sum_i (x_i - \overline{X})^2 + n(\overline{X} - \mu_0)^2$ leaves

$$\lambda = \frac{L(\mu_0)}{L(\overline{X})} = \exp\!\Big(-\frac{n(\overline{X} - \mu_0)^2}{2\sigma^2}\Big).$$

(b) Taking logs,

$$\lambda \le \lambda^* \iff \frac{n(\overline{X}-\mu_0)^2}{\sigma^2} \ge -2\ln\lambda^* \iff \Big|\frac{\overline{X}-\mu_0}{\sigma/\sqrt{n}}\Big| \ge \sqrt{-2\ln\lambda^*} =: c.$$

Under H_0, $z_{\text{obs}} = (\overline{X}-\mu_0)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$, so to get level $\alpha = 0.05$ we set $c = z_{0.025} = 1.96$. This is the two-sided z-test exactly.

(c) Standard error $\sigma/\sqrt{n} = 4.0/5 = 0.8$, so

$$z_{\text{obs}} = \frac{502.0-500}{0.8} = \frac{2.0}{0.8} = 2.5, \qquad \lambda = \exp\Big(-\frac{25(2.0)^2}{2(16)}\Big) = e^{-3.125} = 0.0439.$$

Since $|z_{\text{obs}}| = 2.5 \ge 1.96$ (equivalently λ is small), reject.

$$\boxed{z_{\text{obs}} = 2.5 \ge 1.96,\ \lambda = 0.044:\ \text{reject } H_0 \text{ at } 0.05.}$$

There is significant evidence that the mean fill differs from 500 mL.

G2. $H_0\colon \lambda = 0.5$ vs. $H_1\colon \lambda \ne 0.5$; exponential rate, one constrained parameter, so build the GLRT and calibrate with Wilks ($d = 1$).

(a) $\ln L(\lambda) = n\ln\lambda - \lambda T$; setting the derivative $n/\lambda - T = 0$ gives $\widehat{\lambda} = n/T = 15/24 = 0.625$.

(b) The null allows only $\lambda_0 = 0.5$, so

$$\frac{\lambda_0}{\widehat{\lambda}} = \frac{0.5}{0.625} = 0.8, \qquad (\lambda_0 - \widehat{\lambda})T = (0.5-0.625)(24) = -3.0,$$
$$\lambda = \Big(\frac{\lambda_0}{\widehat{\lambda}}\Big)^n e^{-(\lambda_0-\widehat{\lambda})T} = (0.8)^{15}e^{3.0} = (0.03518)(20.086) = 0.707.$$

(c) Transform and refer to χ^2_1:

$$-2\ln\lambda = -2\ln(0.707) = -2(-0.3471) = 0.694.$$

Against $\chi^2_{0.05,1} = 3.841$: since $0.694 < 3.841$, do *not* reject. The approximate p-value is $\mathbb{P}(\chi^2_1 \ge 0.694) = 2\,\mathbb{P}(Z \ge \sqrt{0.694}) = 2\,\mathbb{P}(Z \ge 0.833) \approx 2(0.2024) = 0.405$.

$$\boxed{-2\ln\lambda = 0.69 < 3.841,\ p \approx 0.40:\ \text{fail to reject } H_0.}$$

The observed rate $\widehat{\lambda} = 0.625$ is not far enough from 0.5 to be significant.

G3. $H_0\colon p = 0.20$ vs. $H_1\colon p \ne 0.20$; Bernoulli GLRT, $d = 1$, $n = 150$ large, so $-2\ln\lambda \mathrel{\dot\sim} \chi^2_1$. The MLE is $\widehat{p} = X/n = 39/150 = 0.26$, and $n - X = 111$. Then

$$X\ln\frac{\widehat{p}}{p_0} = 39\ln\frac{0.26}{0.20} = 39\ln(1.3) = 39(0.262364) = 10.232,$$
$$(n-X)\ln\frac{1-\widehat{p}}{1-p_0} = 111\ln\frac{0.74}{0.80} = 111\ln(0.925) = 111(-0.077962) = -8.654,$$
$$-2\ln\lambda = 2(10.232 - 8.654) = 2(1.578) = 3.157.$$

Against $\chi^2_{0.05,1} = 3.841$: since $3.157 < 3.841$, do not reject. The approximate p-value is $\mathbb{P}(\chi^2_1 \ge 3.157) = 2\,\mathbb{P}(Z \ge \sqrt{3.157}) = 2\,\mathbb{P}(Z \ge 1.777) \approx 2(0.0378) = 0.076$.

$$\boxed{-2\ln\lambda = 3.16 < 3.841,\ p \approx 0.076:\ \text{fail to reject } H_0 \text{ at } 0.05.}$$

The complaint rate $\widehat{p} = 0.26$ is elevated but not significantly different from 0.20 at the 5% level.

G4. $H_0\colon \mu = 3$ vs. $H_1\colon \mu \neq 3$; single Poisson mean, $d = 1$, GLRT plus Wilks.

(a) $\ln L(\mu) = -n\mu + T\ln\mu - \sum_i \ln(x_i!)$ with $T = \sum_i x_i = 84$; the MLE is $\widehat{\mu} = T/n = 84/20 = 4.2$. Using $T = n\widehat{\mu}$, the constant term cancels and

$$-2\ln\lambda = 2\big[T\ln(\widehat{\mu}/\mu_0) - n(\widehat{\mu} - \mu_0)\big].$$

(b) Substitute:

$$\begin{aligned} T\ln(\widehat{\mu}/\mu_0) &= 84\ln(4.2/3) = 84\ln(1.4) = 84(0.336472) = 28.264, \\ n(\widehat{\mu} - \mu_0) &= 20(4.2 - 3) = 24.0, \\ -2\ln\lambda &= 2(28.264 - 24.0) = 2(4.264) = 8.527. \end{aligned}$$

Against $\chi^2_{0.05,1} = 3.841$: since $8.527 > 3.841$, reject.

(c) The approximate p-value is $\mathbb{P}(\chi^2_1 \geq 8.527) = 2\,\mathbb{P}(Z \geq \sqrt{8.527}) = 2\,\mathbb{P}(Z \geq 2.920) \approx 2(0.00175) = 0.0035$.

$$\boxed{-2\ln\lambda = 8.53 > 3.841,\ p \approx 0.0035:\ \text{reject } H_0.}$$

There is strong evidence that the mean crossing rate is not 3 per night ($\widehat{\mu} = 4.2$).

G5. The reported interval is a $95\% = 1 - \alpha$ two-sided CI, so it pairs with the two-sided z-/t-test at the matched $\alpha = 0.05$. By duality we only check membership in $[118.3,\ 124.7]$.

(a) $120 \in [118.3, 124.7]$? Yes. Fail to reject $\mu_0 = 120$.

(b) $125 \in [118.3, 124.7]$? No ($125 > 124.7$). Reject $\mu_0 = 125$.

(c) $117 \in [118.3, 124.7]$? No ($117 < 118.3$). Reject $\mu_0 = 117$.

$$\boxed{120:\ \text{fail to reject};\quad 125:\ \text{reject};\quad 117:\ \text{reject (all at 0.05)}.}$$

A null value inside the interval is consistent with the data; one outside is rejected.

G6. The two-sided z-test at $\alpha = 0.05$ fails to reject μ_0 exactly when $|z_{\text{obs}}| \leq z_{0.025} = 1.96$, i.e. $|\overline{X} - \mu_0| \leq 1.96\,(\sigma/\sqrt{n})$. The standard error is $\sigma/\sqrt{n} = 9.0/\sqrt{36} = 9.0/6 = 1.5$, so

$$\begin{aligned} |80.0 - \mu_0| &\leq 1.96(1.5) = 2.94 \\ -2.94 \leq 80.0 - \mu_0 &\leq 2.94 \\ 77.06 \leq \mu_0 &\leq 82.94. \end{aligned}$$

$$\boxed{\{\mu_0 : \text{not rejected}\} = [77.06,\ 82.94] = \text{the 95\% CI for } \mu.}$$

Inverting the family of level-0.05 tests reproduces the 95% confidence interval $\overline{X} \pm 1.96\,(\sigma/\sqrt{n})$ exactly — building the non-rejection set and forming the CI are the same act.

G7. Conceptual.

(a) A 90% two-sided interval has $1 - \alpha = 0.90$, so $\alpha = 0.10$. By the duality theorem, "μ_0 outside the 90% interval" is exactly the two-sided test that rejects at level $\alpha = 0.10$. (A two-sided $1 - \alpha$ interval pairs with the two-sided test at that same α.)

(b) The duality theorem matches a *two-sided* test with a *two-sided* interval at the same α. A one-sided test of $\mu \leq \mu_0$ vs. $\mu > \mu_0$ at level α puts all of α in one tail, so it pairs with a one-sided confidence *bound* $\mu \geq \overline{X} - z_\alpha(\sigma/\sqrt{n})$. If you instead read it off the two-sided $1 - \alpha$ interval, each tail is calibrated to $\alpha/2$ (it uses $z_{\alpha/2}$, e.g. 1.96, not $z_\alpha = 1.645$), so the relevant tail probability is $\alpha/2$, not α — the actual level is wrong, and the two-sided ruler also ignores the direction of H_1.

(c) Rejection comes from λ *small* (near 0), not large. $\lambda = 0.88$ is near the *no-evidence* end: the null restriction cost almost nothing, so the data are consistent with H_0. Indeed $-2\ln\lambda = -2\ln(0.88) = 0.256$, far below $\chi^2_{0.05,1} = 3.841$ ($p \approx 0.61$). The correct decision is *fail to reject*.

$$\boxed{(a)\ \alpha = 0.10;\quad (b)\ \text{one-sided test} \leftrightarrow \text{one-sided bound};\quad (c)\ \lambda = 0.88 \Rightarrow \text{fail to reject}.}$$

Chapter 7

Inferences Based on the Normal Distribution

Chapters 5 and 6 used the normal, t, and χ^2 distributions as tools; this chapter *earns* them. When we sample from a normal population and replace the unknown σ by the sample standard deviation S, the standardized mean is no longer normal — the extra randomness in S gives it heavier tails, and the exact distribution is **Student's** t. The companion fact is that $(n-1)S^2/\sigma^2$ follows a χ^2 **distribution** and is *independent* of $\overline{X}$. Those two sampling-distribution results are the engine behind every confidence interval and test for a normal mean μ (using t) or variance σ^2 (using χ^2).

Master guide: which normal-theory procedure does the problem call for?

- Asked *why* the standardized sample mean follows a t and not a normal, or for a probability about a t statistic? $\Rightarrow$ the **Student** t **distribution** and its derivation (§1).
- Asked about the distribution of $\overline{X}$ or of S^2 from a normal sample, or about their independence? $\Rightarrow$ the **normal sampling distributions**: $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ and $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ (§2).
- Want a **range** for a normal **mean** μ with σ unknown? $\Rightarrow$ the t **confidence interval** $\overline{X} \pm t_{\alpha/2,n-1}\, S/\sqrt{n}$ (§3).
- Want to **test** a claim about a normal **mean** μ with σ unknown? $\Rightarrow$ the **one-sample** t**-test** (§4).
- The parameter in question is a **variance** or standard deviation of a normal population? $\Rightarrow$ the χ^2 **interval and test** built from $(n-1)S^2/\sigma^2$ (§5).
- Worried the population is *not* normal, or asked whether the procedure is still valid? $\Rightarrow$ **robustness**: the t procedures tolerate mild non-normality, the χ^2 variance procedures do not (§6).

Keep straight: use z only when σ is genuinely known (Chapter 6); the moment σ is estimated by S, the honest distribution is t_{n-1}. Inferences about μ use t; inferences about σ^2 use χ^2; the two are built from the *same* sample but answer different questions.

7.1 The Student t Distribution

Chapter 6 reached for the t distribution the moment σ was unknown, but it never said where that distribution comes from or why it is the honest reference. This section builds it from the ground up. The t distribution is what you get when you standardize a normal quantity but, instead of dividing by a fixed σ, you divide by an estimate that carries its own randomness. Concretely, take a standard normal Z and divide it by the square root of an independent chi-square variable scaled by its degrees of freedom; the ratio is a t. Everything distinctive about the t — that it is symmetric and bell-shaped like the normal, yet visibly heavier in the tails, and that it slowly tightens back toward $\mathrm{N}(0,1)$ as the degrees of freedom grow — traces directly to that random denominator. The payoff is the single fact that organizes the whole chapter: for a sample from

a normal population, the standardized sample mean $(\overline{Y}-\mu)/(S/\sqrt{n})$ is exactly a t with $n-1$ degrees of freedom, so every confidence interval and test for a normal mean with σ unknown is a t statement. Recognize that you need the t table, not the z table, the instant a problem standardizes a mean using a *sample* standard deviation rather than a known population one.

t or normal — which reference distribution?

- A statistic of the form $Z/\sqrt{V/k}$ with $Z \sim \mathrm{N}(0,1)$, $V \sim \chi^2_k$, and $Z \perp\!\!\!\perp V$? $\Rightarrow$ it is t **with k degrees of freedom**; read the t table at row k.
- Standardizing a sample mean from a normal population, and the spread in the denominator is the **sample** standard deviation S (estimated from the same data)? $\Rightarrow$ the statistic is t_{n-1}, not $\mathrm{N}(0,1)$.
- The population σ is genuinely **known** (a spec, a calibration constant)? $\Rightarrow$ stay with the standard normal z; there is no estimation in the denominator.
- Degrees of freedom very large (say $k \geq 100$, or you are told to)? $\Rightarrow$ the t and the normal nearly coincide; using z as an approximation is defensible.

Do not confuse the two tables: for a fixed tail area the t critical value is always *larger* in magnitude than the matching z value (e.g. $t_{0.025,9} = 2.262$ versus $z_{0.025} = 1.96$). Reading the normal table when σ was estimated understates the spread and makes results look more significant than they are.

7.1.1 Definition and basic shape

Definition 7.1.1 (Student's t distribution)**.** Let $Z \sim \mathrm{N}(0,1)$ and $V \sim \chi^2_k$ be **independent**. The random variable

$$T = \frac{Z}{\sqrt{V/k}}$$

has the **Student t distribution with k degrees of freedom**, written $T \sim t_k$. Its density is symmetric about 0 and given (up to the constant that makes it integrate to 1) by

$$f_T(t) \;\propto\; \left(1+\frac{t^2}{k}\right)^{-(k+1)/2}, \qquad -\infty < t < \infty.$$

Proposition 7.1.2 (Shape, symmetry, heavier tails)**.** *The t_k density is symmetric about* 0 *(so the distribution is centered there), unimodal with its peak at* 0*, and has* heavier tails *than the standard normal: for every fixed t large enough, $\mathbb{P}(T > t) > \mathbb{P}(Z > t)$.*

Proof. Symmetry is immediate from the definition. In $T = Z/\sqrt{V/k}$, replacing Z by $-Z$ (also standard normal, independent of V) negates T without changing its law, so T and $-T$ have the same distribution; equivalently the density depends on t only through t^2. For the tails, read the denominator $W = \sqrt{V/k}$ as a random rescaling of a standard normal. Because V/k fluctuates around 1 and is sometimes well below 1, the division by W occasionally inflates a moderate Z into a large T. The normal has no such random denominator (it divides by the constant 1), so it cannot manufacture those extra large values — hence the fatter t tails. The decay rate confirms it: the t_k density falls off only polynomially, like $|t|^{-(k+1)}$, whereas the normal density falls off like $e^{-t^2/2}$, incomparably faster. □

Proposition 7.1.3 (Mean, variance, and the normal limit)**.** *For $T \sim t_k$,*

$$\mathbb{E}[T] = 0 \quad (k > 1),$$
$$\mathrm{Var}(T) = \frac{k}{k-2} \quad (k > 2),$$

and as $k \to \infty$, $t_k \to \mathrm{N}(0,1)$ in distribution.

Proof. By symmetry the mean is 0 whenever it exists, which requires $k > 1$ (for $k = 1$, the Cauchy case, even the mean diverges). For the variance, condition on V: given $V = v$, $T = Z/\sqrt{v/k}$ is normal with variance k/v, so

$$\mathrm{Var}(T) = \mathbb{E}\Big[\frac{k}{V}\Big] = k\,\mathbb{E}\Big[\frac{1}{V}\Big] = k \cdot \frac{1}{k-2} = \frac{k}{k-2},$$

using the standard fact $\mathbb{E}[1/V] = 1/(k-2)$ for $V \sim \chi^2_k$ with $k > 2$. Note $k/(k-2) > 1$: the t has *more* variance than the standard normal, the quantitative face of its heavier tails. For the limit, the law of large numbers gives $V/k \to 1$ in probability as $k \to \infty$, so the random denominator $\sqrt{V/k} \to 1$ and $T = Z/\sqrt{V/k} \to Z \sim \mathrm{N}(0,1)$. The variance formula agrees: $k/(k-2) \to 1$. □

Remark. Two numbers make the limit concrete. At $k = 10$ the variance is $10/8 = 1.25$, a full 25% above the normal's 1; at $k = 50$ it is $50/48 = 1.0417$, and at $k = 200$ it is $200/198 = 1.0101$. The excess shrinks like $2/(k-2)$, so the t is meaningfully fatter than the normal only in small samples — which is exactly where estimating σ by S hurts most.

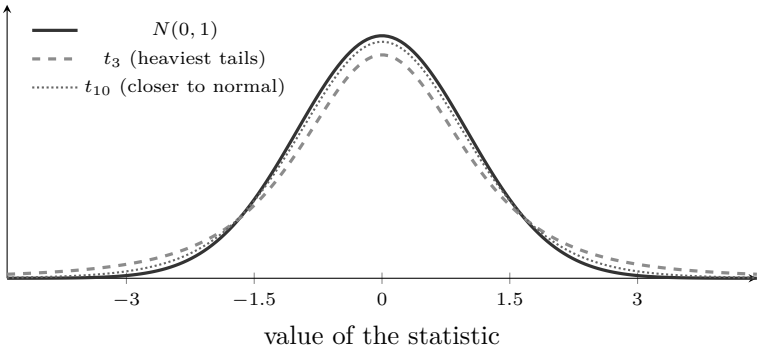


Figure: three symmetric bells. The standard normal (solid) is the tallest at the center and the thinnest in the tails. The t_3 (dashed) is lowest in the middle and highest in the tails; the t_{10} (dotted) sits between them, already close to the normal. The t curves are plotted as $c\,(1 + x^2/k)^{-(k+1)/2}$ with c chosen to make each a proper density. As k grows the dotted curve marches up toward the solid one — the picture of $t_k \to \mathrm{N}(0,1)$.

7.1.2 Reading the t table

A t table is indexed by two things: the degrees of freedom k (a row) and the tail area α (a column). The entry $t_{\alpha,k}$ is the number with $\mathbb{P}(T > t_{\alpha,k}) = \alpha$ for $T \sim t_k$ — the value cutting off an upper-tail area of α. By symmetry the lower α-point is $-t_{\alpha,k}$, and a central area of $1 - \alpha$ lies between $-t_{\alpha/2,k}$ and $t_{\alpha/2,k}$.

Example 7.1.4 (Reading critical values off the table)**.** For a sample of size $n = 15$ from a normal population, the standardized mean uses $k = n - 1 = 14$ degrees of freedom. Find (a) the upper 5% point $t_{0.05,14}$, (b) the two cutoffs enclosing the central 95% of t_{14}, and (c) the upper 1% point $t_{0.01,14}$. Use the standard table values $t_{0.05,14} = 1.761$, $t_{0.025,14} = 2.145$, $t_{0.01,14} = 2.624$.

Which rule, and why? The denominator of the standardized mean is the sample S, so the reference law is $t_{n-1} = t_{14}$, and a "critical value" is just an entry $t_{\alpha,k}$ from row $k = 14$ of the t table. We are reading, not computing a statistic.

Solution.

Step 1: degrees of freedom. $k = n - 1 = 15 - 1 = 14$, so every value comes from the t_{14} row.

Step 2: upper 5% *point.* The single-tail $\alpha = 0.05$ column gives $t_{0.05,14} = 1.761$, meaning $\mathbb{P}(T > 1.761) = 0.05$.

Step 3: central 95%. A central 0.95 leaves 0.025 in each tail, so the cutoffs are $\pm t_{0.025,14} = \pm 2.145$.

Step 4: upper 1% *point.* The $\alpha = 0.01$ column gives $t_{0.01,14} = 2.624$.

$$\boxed{t_{0.05,14} = 1.761, \quad \mathbb{P}(-2.145 < T < 2.145) = 0.95, \quad t_{0.01,14} = 2.624.}$$

Example 7.1.5 (A tail probability bracketed from the table)**.** A statistic is known to follow t_{12} and comes out equal to 2.10. Without software, bracket the upper-tail probability $\mathbb{P}(T > 2.10)$. Use the t_{12} row entries $t_{0.05,12} = 1.782$, $t_{0.025,12} = 2.179$, $t_{0.01,12} = 2.681$.

Which rule, and why? We have a number from a t_{12} distribution and want a tail area, but a hand table only lists a few areas. The move is to locate 2.10 between adjacent table entries and read off the bracketing probabilities — exactly how one gets a p-value by hand.

Solution.

Step 1: place the value among the cutoffs. Compare 2.10 to the row entries: $1.782 < 2.10 < 2.179$, i.e. $t_{0.05,12} < 2.10 < t_{0.025,12}$.

Step 2: translate positions into areas. The area beyond 1.782 is 0.05 and the area beyond 2.179 is 0.025. Since the tail area is a decreasing function of the cutoff and 2.10 sits between the two cutoffs, its tail area sits between the two areas.

$$\boxed{0.025 < \mathbb{P}(T > 2.10) < 0.05 \quad (T \sim t_{12}).}$$

Step 3: sanity check. The standard normal would give $\mathbb{P}(Z > 2.10) = 0.0179$, smaller than this whole bracket — consistent with the t having more probability in its tail.

7.1.3 Seeing the heavier tails

Example 7.1.6 (t tail versus normal tail at the same cutoff)**.** Compare the upper-tail probability beyond 2.0 under t_5 with the same probability under the standard normal, to *see* the heavier t tail. Use the table facts $t_{0.05,5} = 2.015$ and $\mathbb{P}(Z > 2.0) = 0.0228$.

Which rule, and why? Same cutoff 2.0, two different reference distributions. Because the t_5 density puts more mass far from 0 than the normal does, the t tail beyond a fixed point must be the larger of the two; we quantify the gap.

Solution.

Step 1: the t_5 tail beyond 2.0. The value 2.0 is just below the 5% cutoff $t_{0.05,5} = 2.015$, so the area to its right is just *above* 0.05; to three places $\mathbb{P}(T > 2.0) \approx 0.0510$ for $T \sim t_5$.

Step 2: the normal tail beyond 2.0. From the z table, $\mathbb{P}(Z > 2.0) = 0.0228$.

Step 3: compare. The t_5 tail is more than double the normal tail at the very same cutoff:

$$\frac{\mathbb{P}(T > 2.0)}{\mathbb{P}(Z > 2.0)} \approx \frac{0.0510}{0.0228} \approx 2.24.$$

$$\boxed{\mathbb{P}(T > 2.0) \approx 0.051 \ > \ 0.023 \approx \mathbb{P}(Z > 2.0).}$$

A value the normal calls a once-in-44 event ($1/0.0228$) the t_5 calls a once-in-20 event ($1/0.051$): the t treats large deviations as much less surprising.

Example 7.1.7 (Central probability: t keeps less mass in the middle)**.** For $T \sim t_4$ find $\mathbb{P}(-1.5 < T < 1.5)$ and compare it with $\mathbb{P}(-1.5 < Z < 1.5)$ for the standard normal. Use $\mathbb{P}(T < 1.5) \approx 0.896$ for t_4 and $\Phi(1.5) = 0.9332$ for the normal.

Which rule, and why? This is the same heavy-tail fact viewed from the center. Mass that the t pushes into its tails is mass missing from its middle, so the central interval should capture *less* probability under t_4 than under the normal.

Solution.

Step 1: central area under t_4. By symmetry

$$\mathbb{P}(-1.5 < T < 1.5) = 2\,\mathbb{P}(T < 1.5) - 1 = 2(0.896) - 1 = 0.792.$$

Step 2: central area under the normal.

$$\mathbb{P}(-1.5 < Z < 1.5) = 2\Phi(1.5) - 1 = 2(0.9332) - 1 = 0.8664.$$

Step 3: compare. The same interval $(-1.5, 1.5)$ holds about 79.2% of the t_4 mass but 86.6% of the normal mass — a 7.4 percentage-point deficit, all of it banished to the fatter t tails.

$$\boxed{\mathbb{P}(-1.5 < T < 1.5) \approx 0.792 \ < \ 0.866 \approx \mathbb{P}(-1.5 < Z < 1.5).}$$

7.1.4 Inverting the table: finding the cutoff for a given area

Example 7.1.8 (The t value cutting off a stated tail area)**.** You need the value c such that, for $T \sim t_{20}$, exactly 2.5% of the distribution lies above c, and separately the value c' with 10% above it. Use $t_{0.025,20} = 2.086$ and $t_{0.10,20} = 1.325$.

Which rule, and why? Now the area is given and the cutoff is the unknown — the inverse of the previous problems. The number with upper-tail area α on t_k is by definition $t_{\alpha,k}$, read straight from row $k = 20$.

Solution.

Step 1: the 2.5% *cutoff.* We want c with $\mathbb{P}(T > c) = 0.025$, which is the $\alpha = 0.025$ entry: $c = t_{0.025,20} = 2.086$.

Step 2: the 10% *cutoff.* We want c' with $\mathbb{P}(T > c') = 0.10$, the $\alpha = 0.10$ entry: $c' = t_{0.10,20} = 1.325$.

Step 3: interpret. A smaller required tail area ($0.025 < 0.10$) forces a cutoff farther out ($2.086 > 1.325$): to leave less probability beyond you must move the boundary to the right.

$$\boxed{\mathbb{P}(T > 2.086) = 0.025, \qquad \mathbb{P}(T > 1.325) = 0.10 \quad (T \sim t_{20}).}$$

Example 7.1.9 (Symmetric cutoffs for a central probability)**.** Find the symmetric pair $\pm c$ between which a t_8 variable falls with probability 0.90. Use $t_{0.05,8} = 1.860$.

Which rule, and why? A central probability 0.90 leaves 0.10 split into two equal tails of 0.05 each. The cutoff is therefore the upper-0.05 point of t_8, by symmetry mirrored to $-c$ on the left.

Solution.

Step 1: split the leftover area. Central 0.90 means total tail 0.10, so $\alpha/2 = 0.05$ in each tail.

Step 2: read the half-tail cutoff. $c = t_{0.05,8} = 1.860$, and by symmetry the left cutoff is -1.860.

$$\boxed{\mathbb{P}(-1.860 < T < 1.860) = 0.90 \quad (T \sim t_8).}$$

Note the matching normal cutoff would be only $z_{0.05} = 1.645$: the t_8 must reach out to 1.860 to corral the same 90%, because its tails are heavier.

7.1.5 The shrink toward z as df grows

Example 7.1.10 (The critical value collapses onto z as df increases)**.** Tabulate the two-sided 5% critical value $t_{0.025,k}$ for $k = 4, 9, 29, 59$ and compare each to the normal target $z_{0.025} = 1.96$. Use the standard entries $t_{0.025,4} = 2.776$, $t_{0.025,9} = 2.262$, $t_{0.025,29} = 2.045$, $t_{0.025,59} = 2.001$.

Which rule, and why? Since $t_k \to \mathrm{N}(0,1)$, the critical value $t_{\alpha/2,k}$ must decrease toward $z_{\alpha/2}$ as k grows. We exhibit the convergence numerically and read off how big the sample must be for t and z to nearly agree.

Solution.

Step 1: list the cutoffs by df.

$$t_{0.025,4} = 2.776, \quad t_{0.025,9} = 2.262,$$
$$t_{0.025,29} = 2.045, \quad t_{0.025,59} = 2.001.$$

Step 2: subtract the normal target. With $z_{0.025} = 1.96$ the gaps are

$$2.776 - 1.96 = 0.816, \quad 2.262 - 1.96 = 0.302,$$
$$2.045 - 1.96 = 0.085, \quad 2.001 - 1.96 = 0.041.$$

Step 3: read the trend. The excess over 1.96 falls from 0.816 at $k = 4$ to under 0.05 by $k = 59$ — it shrinks at the rate $\approx 2/(k-2)$ promised by the variance formula. By $n \approx 60$ the t cutoff is within 2% of the z cutoff.

$$\boxed{t_{0.025,k} \downarrow 1.96 = z_{0.025} \text{ as } k \to \infty; \text{ by } k \approx 60 \text{ they nearly coincide.}}$$

7.1.6 The central fact: $(\overline{Y} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$

Theorem 7.1.11 (Standardized mean of a normal sample)**.** *Let $Y_1, \ldots, Y_n$ be independent $\mathrm{N}(\mu, \sigma^2)$, with sample mean $\overline{Y} = \frac{1}{n}\sum_i Y_i$ and sample standard deviation $S = \sqrt{\frac{1}{n-1}\sum_i (Y_i - \overline{Y})^2}$. Then*

$$T = \frac{\overline{Y} - \mu}{S/\sqrt{n}} \;\sim\; t_{n-1},$$

whatever *the value of σ.*

Why this is a t (full derivation in §2). Multiply and divide by σ to expose the definition of a t:

$$T = \frac{\overline{Y} - \mu}{S/\sqrt{n}} = \frac{(\overline{Y} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\left[(n-1)S^2/\sigma^2\right]/(n-1)}} = \frac{Z}{\sqrt{V/(n-1)}}.$$

The numerator $Z = (\overline{Y} - \mu)/(\sigma/\sqrt{n})$ is exactly standard normal, because $\overline{Y}$ is normal with mean μ and variance σ^2/n. The quantity $V = (n-1)S^2/\sigma^2$ is χ^2_{n-1}, and — the fact deferred to §2 — for a normal sample $\overline{Y}$ and S^2 are *independent*, so Z and V are independent. By the definition of the t, a standard normal over the square root of an independent chi-square divided by its degrees of freedom is t_{n-1}. □

Remark. This is the whole reason the chapter exists. When σ is known the standardized mean is the exact normal Z of the numerator, and Chapter 6's z procedures apply. The instant you must estimate σ by S, the denominator becomes random, the ratio inherits the heavier t tails, and the correct table is t_{n-1}. The degrees of freedom are $n - 1$, not n, because one degree is spent estimating μ by $\overline{Y}$ inside S.

Example 7.1.12 (Applying the central fact to a probability)**.** A pharmacology lab measures the dissolution time (minutes) of a tablet; times are $\mathrm{N}(\mu, \sigma^2)$ with true mean $\mu = 30$. A run of $n = 16$ tablets yields a sample mean $\overline{Y}$ and sample standard deviation S. Before seeing the data, find the probability that the standardized mean $(\overline{Y} - 30)/(S/\sqrt{16})$ exceeds 2.6. Use $t_{0.01,15} = 2.602$.

Which rule, and why? The denominator uses the *sample* S, not a known σ, so by the theorem the standardized mean is $t_{n-1} = t_{15}$ — we compute a t probability, not a normal one.

Solution.

Step 1: identify the distribution. With $n = 16$, $df = n - 1 = 15$, and $T = (\overline{Y} - 30)/(S/\sqrt{16}) \sim t_{15}$.

Step 2: recognize the cutoff. The table gives $t_{0.01,15} = 2.602$, i.e. $\mathbb{P}(T > 2.602) = 0.01$. Our threshold 2.6 is essentially this value (just a hair below it).

Step 3: read the probability. Hence $\mathbb{P}(T > 2.6) \approx 0.010$.

$$\boxed{\mathbb{P}\Big(\tfrac{\overline{Y}-30}{S/\sqrt{16}} > 2.6\Big) \approx 0.010 \quad (t_{15}).}$$

Had we wrongly used the normal, we would have reported $\mathbb{P}(Z > 2.6) = 0.0047$ — less than half as large, the usual cost of ignoring the estimation of σ.

Example 7.1.13 (From data to a t statistic and its tail)**.** An environmental lab records dissolved-oxygen readings (mg/L) from $n = 10$ stream sites, modeled as $\mathrm{N}(\mu, \sigma^2)$. The summary statistics are $\overline{Y} = 8.42$ and $S = 0.66$, and the long-term target mean is $\mu_0 = 8.00$. Compute the standardized statistic $(\overline{Y} - \mu_0)/(S/\sqrt{n})$ and bracket its upper-tail probability. Use $t_{10-1} = t_9$ with $t_{0.025,9} = 2.262$ and $t_{0.01,9} = 2.821$.

Which rule, and why? We standardize a normal sample mean using the sample S; the reference distribution is $t_{n-1} = t_9$. Computing the statistic then bracketing its tail is the bridge from this section's t table to the tests of §4.

Solution.

Step 1: standard error. $S/\sqrt{n} = 0.66/\sqrt{10} = 0.66/3.162 = 0.2087$ mg/L.

Step 2: the statistic.

$$T = \frac{\overline{Y} - \mu_0}{S/\sqrt{n}} = \frac{8.42 - 8.00}{0.2087} = \frac{0.42}{0.2087} = 2.012.$$

Step 3: bracket the tail. Place 2.012 among the t_9 cutoffs: it lies below $t_{0.025,9} = 2.262$, so the upper-tail area exceeds 0.025. It also exceeds, e.g., $t_{0.05,9} = 1.833$, so the area is below 0.05. Hence

$$0.025 < \mathbb{P}(T > 2.012) < 0.05 \quad (T \sim t_9).$$

$$\boxed{T = 2.012 \text{ on } t_9, \quad 0.025 < \mathbb{P}(T > 2.012) < 0.05.}$$

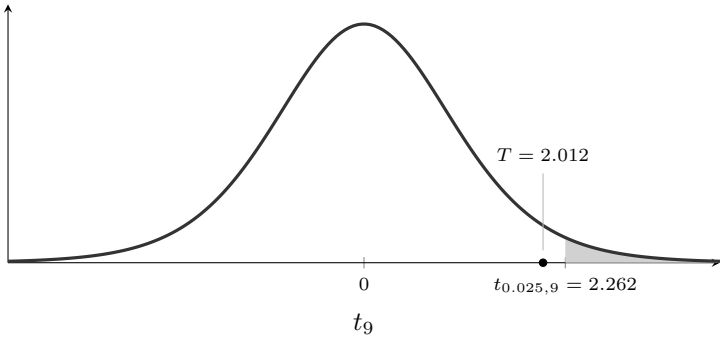


Figure: the t_9 density with the upper 2.5% tail (area 0.025) shaded beyond $t_{0.025,9} = 2.262$. The observed statistic $T = 2.012$ (dot) falls just short of the boundary, so its own tail area is a little more than 0.025 — the source of the bracket $0.025 < \mathbb{P}(T > 2.012) < 0.05$. The curve is the t_9 density up to a normalizing constant.

7.1.7 Contrast: the normal table on small-sample data

Example 7.1.14 (CONTRAST: using z when df is small understates the spread)**.** A materials lab tests the breaking strength (newtons) of a new fiber on $n = 6$ specimens, modeled as $\mathrm{N}(\mu, \sigma^2)$, and obtains $\overline{Y} = 214$ N with a *sample* standard deviation $S = 9$ N. The standardized statistic against the nominal $\mu_0 = 205$ N is the same number either way; the question is which distribution scores it. A hurried analyst compares it to the normal cutoff $z_{0.025} = 1.96$; the honest comparison uses $t_{0.025,5} = 2.571$. Show how the two calls can disagree, and find the probability each assigns to the tail.

Which rule, and why? The denominator is the *sample* S from only $n = 6$ readings, so the correct reference is $t_{n-1} = t_5$, whose tails are much heavier than the normal's at this tiny df. Reading the z table is the look-alike error; we run both to expose it.

Solution.

Step 1: the common statistic. $S/\sqrt{n} = 9/\sqrt{6} = 9/2.449 = 3.674$, so

$$T = \frac{\overline{Y} - \mu_0}{S/\sqrt{n}} = \frac{214 - 205}{3.674} = \frac{9}{3.674} = 2.449.$$

Step 2: the wrong normal scoring. Against $z_{0.025} = 1.96$, the value $2.449 > 1.96$ sits past the cutoff, and the normal tail is $\mathbb{P}(Z > 2.449) = 0.0072$ (two-sided 0.0143): the z-user calls it clearly extreme.

Step 3: the honest t_5 scoring. Against $t_{0.025,5} = 2.571$, the same $2.449 < 2.571$ does *not* clear the cutoff, and the t_5 tail is $\mathbb{P}(T > 2.449) \approx 0.029$ (two-sided ≈ 0.058): the t-user does not call it extreme at the 5% level.

Step 4: why they split. Estimating σ from just 5 degrees of freedom inflates the true spread; the honest cutoff moves out from 1.96 to 2.571, and the tail probability quadruples from 0.0072 to about 0.029. The normal table makes a borderline result look decisive.

$$\boxed{\mathbb{P}(Z > 2.449) = 0.0072 \;\ll\; 0.029 \approx \mathbb{P}(T > 2.449)\ (t_5);\ \text{use } t_5,\ \text{not } z.}$$

Remark. The diagnostic is mechanical: did the standard deviation in the denominator come from *outside* the sample (a spec or calibration — then z) or was it computed *inside* it as S (then t_{n-1})? With small n the gap between the two tables is large and can flip a decision, as above; once n reaches the dozens the tables nearly agree and the distinction is mostly academic.

7.2 Sampling Distributions from a Normal Population

Everything in this chapter rests on three facts about an independent sample $X_1, \ldots, X_n$ drawn from a $\mathrm{N}(\mu, \sigma^2)$ population. They describe how the two summaries we actually compute — the sample mean $\overline{X}$ and the sample variance S^2 — behave as random quantities before the data are seen. The first fact locates $\overline{X}$; the second locates S^2; the third says the two carry *non-overlapping* information. You recognize that you are in this section when a problem asks for a probability about $\overline{X}$, a probability about S^2 or $(n-1)S^2/\sigma^2$, or for the distribution of some statistic built from a normal sample — and you must decide *which reference curve* that statistic follows. The whole point is bookkeeping with degrees of freedom: a sum of n squared standard normals is χ^2_n, but the act of centering at $\overline{X}$ instead of μ spends one degree of freedom, and the leftover randomness is exactly what is needed to turn a z into a t.

Which summary is the problem asking about?

- A probability about the **sample mean** $\overline{X}$ with σ *known* $\Rightarrow$ standardize with $Z = (\overline{X}-\mu)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$.
- A probability about the **sample variance** S^2, or about $(n-1)S^2/\sigma^2 \Rightarrow$ the chi-square pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.
- σ *unknown* and you must standardize $\overline{X}$ from the data alone $\Rightarrow$ the t **ratio** $(\overline{X}-\mu)/(S/\sqrt{n}) \sim t_{n-1}$, built by dividing an independent Z and χ^2.
- A **linear combination** $\sum a_i X_i$ of independent normals $\Rightarrow$ it is itself normal; just add the means and add the variances (weighted by a_i^2).

Do not confuse the two roles of the data: $\overline{X}$ answers questions about the center μ, S^2 answers questions about the spread σ^2. Mixing them up — or forgetting that centering at $\overline{X}$ costs a degree of freedom — is the classic error that puts a statistic in front of the wrong table.

7.2.1 The chi-square distribution, recalled

Before the three pillars we fix the reference curve that pillar (ii) will produce. It is built from squared standard normals.

Definition 7.2.1 (Chi-square distribution)**.** If $Z_1, \ldots, Z_k$ are independent $\mathrm{N}(0,1)$ random variables, then

$$V \;=\; Z_1^2 + Z_2^2 + \cdots + Z_k^2$$

has the **chi-square distribution with k degrees of freedom**, written $V \sim \chi^2_k$. Its density is supported on $(0, \infty)$ and is a special Gamma density,

$$f_V(v) \;=\; \frac{1}{2^{k/2}\,\Gamma(k/2)}\; v^{\,k/2-1}\, e^{-v/2}, \qquad v > 0.$$

Proposition 7.2.2 (Moments, shape, additivity)**.** *Let $V \sim \chi^2_k$. Then*

$$\mathbb{E}[V] = k, \qquad \mathrm{Var}(V) = 2k,$$
$$\textit{mode} = k-2 \ (k \geq 2), \qquad \textit{skew} > 0 \ \textit{(right-skewed)}.$$

Moreover the family is ***additive****: if $V_1 \sim \chi^2_{k_1}$ and $V_2 \sim \chi^2_{k_2}$ are independent, then $V_1 + V_2 \sim \chi^2_{k_1+k_2}$.*

Proof. A single Z^2 has $\mathbb{E}[Z^2] = \mathrm{Var}(Z) = 1$ and, using $\mathbb{E}[Z^4] = 3$, $\mathrm{Var}(Z^2) = \mathbb{E}[Z^4] - (\mathbb{E}[Z^2])^2 = 3 - 1 = 2$. Summing k independent copies gives $\mathbb{E}[V] = k$ and $\mathrm{Var}(V) = 2k$. Additivity is immediate from the definition: stacking $k_1 + k_2$ independent squared standard normals is the same as adding the two sums. □

Remark. As k grows the mean k and the mode $k-2$ pull apart only slowly while the standard deviation $\sqrt{2k}$ grows; relative to its center the curve becomes more symmetric, and by the central limit theorem $(V-k)/\sqrt{2k}$ approaches $\mathrm{N}(0,1)$. For the small k we meet in practice the right skew is pronounced, which is why the upper and lower tails of a χ^2 table are *not* mirror images.

Figure 7.1 shows a χ^2_4 density: it rises from 0, peaks near $v = k-2 = 2$, and trails off to the right.

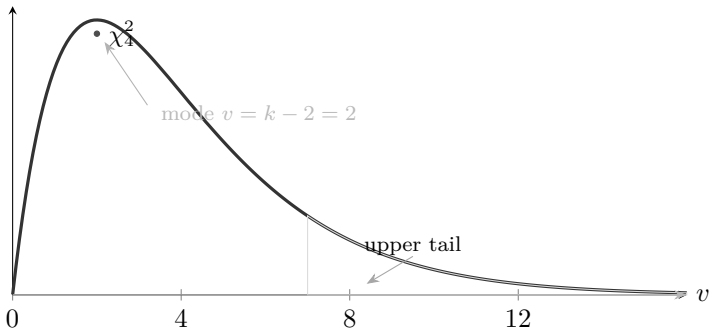


Figure 7.1: A χ^2_4 density. The curve is right-skewed, starts at the origin, peaks at the mode $v = k-2 = 2$, and has a long upper tail (shaded). Mean $= k = 4$ lies to the right of the peak.

Example 7.2.3 (A chi-square as a sum of squares)**.** Five independent standardized measurement errors $Z_1, \ldots, Z_5$, each $\mathrm{N}(0,1)$, are recorded by a calibration rig. Quality control flags the batch if the total squared error $V = \sum_{i=1}^{5} Z_i^2$ exceeds 11.07. What is the chance of a false flag, and what are the mean and standard deviation of V?

Which rule, and why? A sum of squared *independent standard normals* is chi-square by definition, with degrees of freedom equal to the number of terms. Here nothing is estimated, so all five deviations count: $V \sim \chi^2_5$.

Solution.

Step 1: identify the distribution. $V = \sum_{i=1}^{5} Z_i^2 \sim \chi^2_5$, so $k = 5$.

Step 2: moments. $\mathbb{E}[V] = k = 5$ and $\mathrm{Var}(V) = 2k = 10$, hence $\mathrm{SD}(V) = \sqrt{10} \approx 3.162$.

Step 3: the tail probability. The cutoff 11.07 is the standard upper-5% point of χ^2_5, i.e. $\chi^2_{0.05,5} = 11.07$. Therefore $\mathbb{P}(V > 11.07) = 0.05$.

$$\boxed{\mathbb{P}(V > 11.07) = 0.05, \qquad \mathbb{E}[V] = 5, \quad \mathrm{SD}(V) = \sqrt{10} \approx 3.16}$$

7.2.2 Pillar (i): the distribution of the sample mean

Theorem 7.2.4 (Distribution of $\overline{X}$)**.** *If $X_1, \ldots, X_n$ are independent $\mathrm{N}(\mu, \sigma^2)$, then*

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i \ \sim \ \mathrm{N}\left(\mu, \ \frac{\sigma^2}{n}\right), \qquad \textit{equivalently} \qquad Z = \frac{\overline{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{N}(0,1).$$

Proof. A linear combination of independent normals is normal, so $\overline{X}$ is normal; we only need its two moments. By linearity $\mathbb{E}[\overline{X}] = \frac{1}{n}\sum \mathbb{E}[X_i] = \mu$, and by independence $\mathrm{Var}(\overline{X}) = \frac{1}{n^2}\sum \mathrm{Var}(X_i) = \frac{n\sigma^2}{n^2} = \sigma^2/n$. Standardizing gives the stated Z. □

Example 7.2.5 (Probability about a mean, sigma known)**.** A bottling line fills cans with a target of $\mu = 355$ mL; the fill volume is normal with known process standard deviation $\sigma = 2.4$ mL. An inspector averages $n = 16$ cans. What is the probability the sample mean lands within 1 mL of target, i.e. $\mathbb{P}(354 \le \overline{X} \le 356)$?

Which rule, and why? This is a probability about $\overline{X}$ with σ *known*, so pillar (i) applies directly: $\overline{X} \sim \mathrm{N}(355,\, 2.4^2/16)$ and we standardize with Z, not t.

Solution.

Step 1: standard error. $\sigma/\sqrt{n} = 2.4/\sqrt{16} = 2.4/4 = 0.6$ mL.

Step 2: standardize the endpoints.

$$z_{\mathrm{lo}} = \frac{354 - 355}{0.6} = \frac{-1}{0.6} = -1.667,$$
$$z_{\mathrm{hi}} = \frac{356 - 355}{0.6} = \frac{+1}{0.6} = +1.667.$$

Step 3: read the normal table. $\Phi(1.667) \approx 0.9522$, so

$$\mathbb{P}(354 \le \overline{X} \le 356) = \Phi(1.667) - \Phi(-1.667) = 2(0.9522) - 1 = 0.9044.$$

$$\boxed{\mathbb{P}(354 \le \overline{X} \le 356) \approx 0.904}$$

7.2.3 Pillar (ii): the distribution of the sample variance

The sample variance is

$$S^2 \;=\; \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2, \qquad \mathrm{SS} \;=\; \sum_{i=1}^{n}(X_i - \overline{X})^2,$$

so $(n-1)S^2 = \mathrm{SS}$ is the total sum of squared deviations about $\overline{X}$.

Theorem 7.2.6 (Distribution of the sample variance)**.** *If $X_1, \ldots, X_n$ are independent $\mathrm{N}(\mu, \sigma^2)$, then*

$$\frac{(n-1)S^2}{\sigma^2} \;=\; \frac{1}{\sigma^2}\sum_{i=1}^{n}(X_i - \overline{X})^2 \;\sim\; \chi^2_{n-1}.$$

Why $n-1$, heuristically. Standardize each deviation *from the true mean*: $Z_i = (X_i - \mu)/\sigma \sim \mathrm{N}(0,1)$, independent, so $\sum_{i=1}^n Z_i^2 = \sum (X_i - \mu)^2/\sigma^2 \sim \chi^2_n$ — a clean sum of n squared standard normals. But we never know μ; we use $\overline{X}$. The algebraic identity

$$\sum_{i=1}^{n}(X_i - \mu)^2 \;=\; \sum_{i=1}^{n}(X_i - \overline{X})^2 \;+\; n(\overline{X} - \mu)^2$$

splits the total. Dividing by σ^2,

$$\underbrace{\sum_{i=1}^{n}\frac{(X_i-\mu)^2}{\sigma^2}}_{\chi^2_n} \;=\; \underbrace{\frac{(n-1)S^2}{\sigma^2}}_{?} \;+\; \underbrace{\frac{n(\overline{X}-\mu)^2}{\sigma^2}}_{=\left[(\overline{X}-\mu)/(\sigma/\sqrt{n})\right]^2 \sim \chi^2_1}.$$

The last term is the square of the standard normal $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$, hence χ^2_1. Because (pillar iii, below) the two pieces on the right are *independent*, the additivity of chi-square forces the unknown piece to make up the difference in degrees of freedom: $n - 1 = n - 1$. So $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. Estimating one parameter, μ, by $\overline{X}$ has cost exactly one degree of freedom.

Remark. The constraint is visible directly in the deviations: the numbers $X_i - \overline{X}$ must sum to zero, $\sum_i (X_i - \overline{X}) = 0$. So only $n-1$ of them are free to vary; the last is determined. "Degrees of freedom" counts the free coordinates, and one linear constraint removes one.

Example 7.2.7 (Probability about a sample variance)**.** A polymer film has true thickness variance $\sigma^2 = 0.16\ \mu\text{m}^2$ (so $\sigma = 0.4\ \mu\text{m}$), and thickness is normal. A lab measures $n = 10$ spots and computes S^2. Find $\mathbb{P}(S^2 > 0.30)$.

Which rule, and why? A probability about S^2 from a normal sample is handled by the chi-square pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ — never by treating S^2 as normal. Convert the event on S^2 into an event on that pivot.

Solution.

Step 1: form the pivot. With $n = 10$, $(n-1)S^2/\sigma^2 = 9S^2/0.16 \sim \chi^2_9$.

Step 2: translate the event.

$$S^2 > 0.30 \iff \frac{9S^2}{0.16} > \frac{9(0.30)}{0.16} = \frac{2.70}{0.16} = 16.875.$$

Step 3: read the chi-square table. The upper-5% point of χ^2_9 is $\chi^2_{0.05,9} = 16.919$. Our cutoff 16.875 sits a hair below it, so the tail probability is a touch above 0.05:

$$\mathbb{P}(S^2 > 0.30) = \mathbb{P}\big(\chi^2_9 > 16.875\big) \approx 0.051.$$

$$\boxed{\mathbb{P}(S^2 > 0.30) \approx 0.05}$$

Example 7.2.8 (Two-sided probability for the sum of squares)**.** Reusing the polymer film ($\sigma^2 = 0.16$, $n = 10$, normal), an engineer wants the central 90% range for the observed sum of squared deviations $\text{SS} = (n-1)S^2 = 9S^2$. Between which two values of SS does it fall with probability 0.90?

Which rule, and why? A two-sided range for $\text{SS} = (n-1)S^2$ is again the chi-square pivot, but now we need *both* tails: because χ^2_9 is right-skewed, the lower and upper 5% points are not symmetric and must be read separately.

Solution.

Step 1: pivot and percentiles. $\text{SS}/\sigma^2 = \text{SS}/0.16 \sim \chi^2_9$. For 90% in the middle we cut 5% from each tail:

$$\chi^2_{0.95,9} = 3.325 \quad \text{(lower)}, \qquad \chi^2_{0.05,9} = 16.919 \quad \text{(upper)}.$$

Step 2: invert to SS*.* Since $\mathbb{P}\big(3.325 \le \text{SS}/0.16 \le 16.919\big) = 0.90$, multiply through by 0.16:

$$\begin{aligned} \text{lower}\ \ \text{SS} &= 0.16 \times 3.325 = 0.532, \\ \text{upper}\ \ \text{SS} &= 0.16 \times 16.919 = 2.707. \end{aligned}$$

$$\boxed{\mathbb{P}\big(0.532 \le \text{SS} \le 2.707\big) = 0.90}$$

Note the asymmetry about the mean $\mathbb{E}[\text{SS}/\sigma^2] \cdot \sigma^2 = 9(0.16) = 1.44$: the gap below $(1.44 - 0.53 = 0.91)$ is smaller than the gap above $(2.71 - 1.44 = 1.27)$, a direct fingerprint of the right skew.

7.2.4 Pillar (iii): independence of $\overline{X}$ and S^2, and the birth of the t

Theorem 7.2.9 (Independence)**.** *If $X_1, \ldots, X_n$ are independent $\text{N}(\mu, \sigma^2)$, then $\overline{X}$ and S^2 are* ***independent*** *random variables.*

Remark. This is special to the normal: for most populations $\overline{X}$ and S^2 are correlated. Geometrically, $\overline{X}$ records the sample's location *along* the diagonal direction $(1, 1, \ldots, 1)$, while the deviations $X_i - \overline{X}$ live in the $(n-1)$-dimensional space *orthogonal* to that diagonal. For a spherically symmetric (normal) cloud, the projection onto a line and the projection onto the perpendicular subspace are independent. Estimating μ uses up the one diagonal coordinate; the $n-1$ perpendicular coordinates — carrying S^2 — are untouched by it.

Why independence makes the t clean. Recall the definition of Student's t (Section 1): if $Z \sim \mathrm{N}(0,1)$ and $V \sim \chi^2_d$ are *independent*, then

$$T = \frac{Z}{\sqrt{V/d}} \sim t_d.$$

Now standardize the sample mean with the *unknown* σ and watch the σ's cancel:

$$\frac{\overline{X}-\mu}{S/\sqrt{n}} \;=\; \frac{(\overline{X}-\mu)/(\sigma/\sqrt{n})}{\sqrt{S^2/\sigma^2}} \;=\; \frac{Z}{\sqrt{\dfrac{(n-1)S^2/\sigma^2}{n-1}}} \;=\; \frac{Z}{\sqrt{V/(n-1)}},$$

with $Z = (\overline{X}-\mu)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$ (pillar i) in the numerator and $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ (pillar ii) in the denominator. The construction of the t *requires* the numerator Z and the denominator V to be independent — and that is exactly what pillar (iii) supplies. Hence

$$\boxed{\frac{\overline{X}-\mu}{S/\sqrt{n}} \;\sim\; t_{n-1}}.$$

Without independence the ratio would still be *some* distribution, but not the tabulated t; the table we use is built on the assumption that the random scale S wobbles independently of where $\overline{X}$ landed.

Example 7.2.10 (From the pieces to a t-probability)**.** Resin cure temperatures are $\mathrm{N}(\mu, \sigma^2)$ with both parameters unknown. A run takes $n = 12$ readings, yielding $\overline{X}$ and S. The standardized error $T = (\overline{X}-\mu)/(S/\sqrt{12})$ is computed. What is $\mathbb{P}(|T| > 2.201)$, and what reference curve does T follow?

Which rule, and why? Because σ is replaced by S, the standardized mean is a ratio of an independent Z and χ^2_{n-1} — i.e. a t with $n-1$ degrees of freedom, not a normal. This is the payoff of all three pillars.

Solution.

Step 1: degrees of freedom. $n - 1 = 12 - 1 = 11$, so $T \sim t_{11}$.

Step 2: identify the cutoff. 2.201 is the two-sided 5% critical value of t_{11}, i.e. $t_{0.025,11} = 2.201$, so each tail beyond it carries 0.025.

Step 3: combine the tails.

$$\mathbb{P}(|T| > 2.201) = 2(0.025) = 0.05.$$

$$\boxed{T \sim t_{11}, \qquad \mathbb{P}(|T| > 2.201) = 0.05}$$

Example 7.2.11 (Assembling a t-statistic from data)**.** A small clinical pilot records resting heart-rate drop (bpm) on $n = 9$ subjects, normal data with unknown μ, σ. The data give $\overline{X} = 6.4$ and $S = 3.0$. Under the working hypothesis $\mu = 4.5$, evaluate the statistic $T = (\overline{X}-\mu)/(S/\sqrt{n})$ and report $\mathbb{P}(T > t_{\mathrm{obs}})$ relative to the correct curve.

Which rule, and why? σ is unknown and estimated by S, so the standardizer is $S/\sqrt{n}$ and the reference is t_{n-1}. We are plugging numbers into the t ratio derived above.

Solution.

Step 1: standard error and degrees of freedom. $S/\sqrt{n} = 3.0/\sqrt{9} = 3.0/3 = 1.0$; $n - 1 = 8$, so $T \sim t_8$.

Step 2: observed value.

$$t_{\mathrm{obs}} = \frac{6.4 - 4.5}{1.0} = \frac{1.9}{1.0} = 1.90.$$

Step 3: locate it. The upper 5% point of t_8 is $t_{0.05,8} = 1.860$ and the upper 2.5% point is $t_{0.025,8} = 2.306$. Since $1.860 < 1.90 < 2.306$, the upper-tail probability sits between 0.025 and 0.05 (about 0.047).

$$\boxed{t_{\mathrm{obs}} = 1.90 \sim t_8, \qquad 0.025 < \mathbb{P}(T > 1.90) < 0.05}$$

7.2.5 Linear combinations of a normal sample

Pillar (i) is a special case of a single, very useful closure rule: *any* linear combination of independent normals is normal.

Proposition 7.2.12 (Linear combinations)**.** *If $X_1, \dots, X_n$ are independent with $X_i \sim \mathrm{N}(\mu_i, \sigma_i^2)$ and a_0, a_i are constants, then*

$$W = a_0 + \sum_{i=1}^{n} a_i X_i \;\sim\; \mathrm{N}\!\left(a_0 + \sum_i a_i \mu_i,\; \sum_i a_i^2 \sigma_i^2\right).$$

Example 7.2.13 (Distribution of a weighted blend)**.** A pharmacist blends two independent active lots, $X \sim \mathrm{N}(50,\, 4)$ mg and $Y \sim \mathrm{N}(30,\, 9)$ mg, into a dose $W = 0.6X + 0.4Y$. Find the distribution of W and $\mathbb{P}(W > 45)$.

Which rule, and why? W is a linear combination of independent normals, so it is normal; we only need its mean and variance. The weights enter the variance *squared* — this is the rule, not a t or χ^2 situation, because nothing is being estimated.

Solution.

Step 1: mean. $\mathbb{E}[W] = 0.6(50) + 0.4(30) = 30 + 12 = 42$ mg.

Step 2: variance.

$$\mathrm{Var}(W) = 0.6^2(4) + 0.4^2(9) = 0.36(4) + 0.16(9) = 1.44 + 1.44 = 2.88,$$

so $\mathrm{SD}(W) = \sqrt{2.88} \approx 1.697$ mg and $W \sim \mathrm{N}(42,\, 2.88)$.

Step 3: the tail.

$$z = \frac{45 - 42}{1.697} = \frac{3}{1.697} = 1.768, \qquad \mathbb{P}(W > 45) = 1 - \Phi(1.768) \approx 0.0385.$$

$$\boxed{W \sim \mathrm{N}(42,\, 2.88), \qquad \mathbb{P}(W > 45) \approx 0.039}$$

Example 7.2.14 (A difference of two sample means)**.** Two independent normal lines have the *same* unknown spread $\sigma^2 = 1.5$. Line A averages $n_A = 8$ items to give $\overline{X}_A$; line B averages $n_B = 5$ to give $\overline{X}_B$. Both target $\mu = 20$. Find the distribution of the gap $D = \overline{X}_A - \overline{X}_B$ and $\mathbb{P}(|D| > 1)$.

Which rule, and why? Each sample mean is normal (pillar i), the two are independent, and a difference is a linear combination, so D is normal. With σ^2 known we standardize with Z.

Solution.

Step 1: mean of the gap. $\mathbb{E}[D] = \mu - \mu = 20 - 20 = 0$.

Step 2: variance of the gap. Independent, so variances add:

$$\mathrm{Var}(D) = \frac{\sigma^2}{n_A} + \frac{\sigma^2}{n_B} = \frac{1.5}{8} + \frac{1.5}{5} = 0.1875 + 0.30 = 0.4875,$$

$\mathrm{SD}(D) = \sqrt{0.4875} \approx 0.698$, so $D \sim \mathrm{N}(0,\, 0.4875)$.

Step 3: the probability.

$$z = \frac{1 - 0}{0.698} = 1.433, \qquad \mathbb{P}(|D| > 1) = 2\big(1 - \Phi(1.433)\big) = 2(0.0760) = 0.152.$$

$$\boxed{D \sim \mathrm{N}(0,\, 0.4875), \qquad \mathbb{P}(|D| > 1) \approx 0.152}$$

7.2.6 The pieces, combined — and a CONTRAST

The next two examples first use all three pillars together, then show what happens if you ignore the degree-of-freedom correction and the independence that the t needs.

Example 7.2.15 (One-sided event built from mean and variance)**.** Battery lifetimes are $\mathrm{N}(\mu, \sigma^2)$ with $\mu = 500$ h and $\sigma = 20$ h, both used here as known for a design calculation. A sample of $n = 16$ batteries is tested. Find the probability that simultaneously the mean exceeds 508 h *and* the sample standard deviation S exceeds 24 h. (The two events are independent by pillar iii.)

Which rule, and why? The mean event uses pillar (i) with a Z; the spread event uses pillar (ii) with a χ^2; and pillar (iii) lets us *multiply* the two probabilities because $\overline{X}$ and S^2 are independent.

Solution.

Step 1: the mean event (Z). $\sigma/\sqrt{n} = 20/4 = 5$, so

$$z = \frac{508 - 500}{5} = 1.60, \qquad \mathbb{P}(\overline{X} > 508) = 1 - \Phi(1.60) = 0.0548.$$

Step 2: the spread event (χ^2). $S > 24 \iff S^2 > 576 \iff (n-1)S^2/\sigma^2 > 15(576)/400 = 21.6$. With $n - 1 = 15$, $\chi^2_{0.05,15} = 24.996$ and $\chi^2_{0.10,15} = 22.307$, and 21.6 lies just beyond the 10% point, giving $\mathbb{P}(\chi^2_{15} > 21.6) \approx 0.119$.

Step 3: multiply (independence).

$$\mathbb{P}(\overline{X} > 508 \text{ and } S > 24) = 0.0548 \times 0.119 \approx 0.0065.$$

$$\boxed{\mathbb{P}(\overline{X} > 508,\ S > 24) \approx 0.0065}$$

Example 7.2.16 (CONTRAST: the wrong reference distribution)**.** Take the resin-cure setup ($n = 12$, σ *unknown*, estimated by S). A practitioner computes $(\overline{X} - \mu)/(S/\sqrt{12})$ but — forgetting that σ was estimated — treats it as a standard normal Z and also (wrongly) divides the sum of squares by χ^2_{12} instead of χ^2_{11}. They then look up $\mathbb{P}(|\cdot| > 2.201)$ in the *normal* table. What do they get, what is correct, and why is the normal answer too small?

Which rule, and why? This is the look-alike trap: with σ replaced by the random S, the correct reference is t_{11} (numerator Z over $\sqrt{\chi^2_{11}/11}$, independent), *not* the normal. Using $\mathrm{N}(0, 1)$ ignores the extra randomness in S and miscounts degrees of freedom.

Solution.

Step 1: the (wrong) normal computation. 2.201 in a standard-normal table gives $\mathbb{P}(|Z| > 2.201) = 2(1 - \Phi(2.201)) = 2(0.0139) = 0.0278$.

Step 2: the correct t computation. With σ unknown, $T = (\overline{X} - \mu)/(S/\sqrt{12}) \sim t_{11}$, and $2.201 = t_{0.025,11}$, so

$$\mathbb{P}(|T| > 2.201) = 2(0.025) = 0.050.$$

Step 3: why the gap. The t_{11} density has *heavier tails* than the normal because S in the denominator is itself random and sometimes small, inflating the ratio. Calling it normal understates the tail (0.028 vs. 0.050) — almost a factor of two — so every confidence statement or p-value built on it would be too optimistic. The fix is bookkeeping: estimating μ by $\overline{X}$ removes one degree of freedom (use χ^2_{11}, not χ^2_{12}), and the independent $Z/\sqrt{V/11}$ assembles into t_{11}.

wrong (normal): 0.028 vs. correct (t_{11}): 0.050

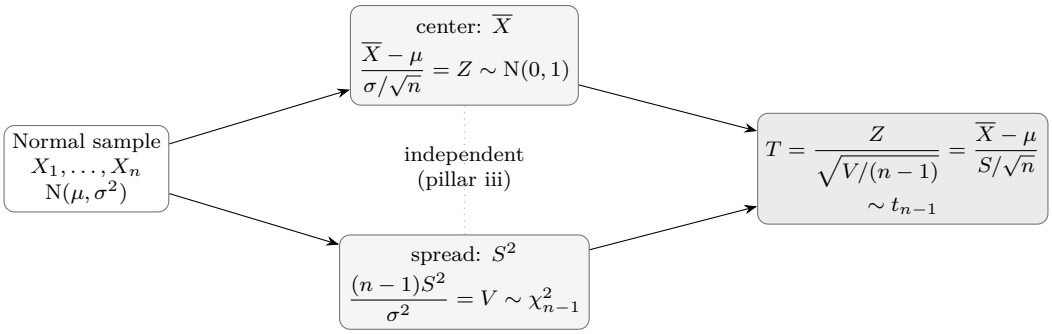


Figure 7.2: The normal sample splits into two independent summaries: $\overline{X}$ (standardizing to a Z) and S^2 (giving a χ^2_{n-1}). Their independence is exactly what lets the ratio $Z/\sqrt{V/(n-1)}$ be a clean t_{n-1} after the unknown σ cancels.

7.2.7 The two summaries, side by side

Figure 7.2 shows the machine: one normal sample feeds two *independent* summaries, which recombine into the t ratio.

Example 7.2.17 (Reading the diagram in numbers)**.** For $n = 6$ normal observations with σ unknown, identify the three distributions in Figure 7.2 explicitly and give the degrees of freedom of the final t.

Which rule, and why? Plug $n = 6$ into each pillar. The numerator is always a single standard normal; the denominator's χ^2 and the final t both carry $n - 1$ degrees of freedom because centering at $\overline{X}$ spent one.

Solution.

Step 1: the mean branch. $Z = (\overline{X} - \mu)/(\sigma/\sqrt{6}) \sim \mathrm{N}(0,1)$.

Step 2: the variance branch. $V = (6-1)S^2/\sigma^2 = 5S^2/\sigma^2 \sim \chi^2_5$, with $\mathbb{E}[V] = 5$ and $\mathrm{Var}(V) = 10$.

Step 3: recombine.

$$T = \frac{Z}{\sqrt{V/5}} = \frac{\overline{X}-\mu}{S/\sqrt{6}} \sim t_5.$$

$$\boxed{Z \sim \mathrm{N}(0,1), \quad V \sim \chi^2_5, \quad T \sim t_5}$$

Example 7.2.18 (Lower-tail event for the sample variance)**.** Concrete cylinder strengths are normal with $\sigma^2 = 25$ (MPa^2). A batch of $n = 20$ cylinders is tested. Find $\mathbb{P}(S^2 < 15)$ — the chance the sample looks unusually consistent.

Which rule, and why? A *lower*-tail probability about S^2 still routes through $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$; because χ^2 is right-skewed we must use the lower-tail point of χ^2_{19}, which is not the mirror of the upper point.

Solution.

Step 1: form the pivot. $n - 1 = 19$, so $19S^2/25 \sim \chi^2_{19}$.

Step 2: translate the event.

$$S^2 < 15 \iff \frac{19S^2}{25} < \frac{19(15)}{25} = \frac{285}{25} = 11.40.$$

Step 3: read the lower tail. The lower-10% point of χ^2_{19} is $\chi^2_{0.90,19} = 11.651$, and our cutoff 11.40 is just below it, so the probability is a bit under 0.10:

$$\mathbb{P}(S^2 < 15) = \mathbb{P}\big(\chi^2_{19} < 11.40\big) \approx 0.090.$$

$$\boxed{\mathbb{P}(S^2 < 15) \approx 0.09}$$

Example 7.2.19 (CONTRAST: forgetting the degree-of-freedom correction)**.** A student wants $\mathbb{P}(S^2 > 30)$ for the concrete data above ($\sigma^2 = 25$, $n = 20$). They form $nS^2/\sigma^2 = 20S^2/25$ and compare it to χ^2_{20} — using n rather than $n-1$, because "there are n observations." Compute their answer and the correct one.

Which rule, and why? The deviations are taken about $\overline{X}$, not μ, so the pivot is $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, with 19 degrees of freedom — not χ^2_{20}. Using n double-counts the one constraint $\sum(X_i - \overline{X}) = 0$.

Solution.

Step 1: the (wrong) n pivot. $20S^2/25$ at $S^2 = 30$ gives $20(30)/25 = 24.0$; compared to χ^2_{20}, where $\chi^2_{0.20,20} = 25.038$ and $\chi^2_{0.25,20} \approx 23.83$, so $\mathbb{P}(\chi^2_{20} > 24.0) \approx 0.243$.

Step 2: the correct $n-1$ pivot. $19S^2/25$ at $S^2 = 30$ gives $19(30)/25 = 22.8$; compared to χ^2_{19}, where $\chi^2_{0.25,19} \approx 22.72$, so $\mathbb{P}(\chi^2_{19} > 22.8) \approx 0.249$.

Step 3: the lesson.

$$\text{wrong } (\chi^2_{20} \text{ via } n):\ \mathbb{P} \approx 0.243,$$
$$\text{correct } (\chi^2_{19} \text{ via } n-1):\ \mathbb{P} \approx 0.249.$$

The numerical gap is small here, but the wrong pivot uses the wrong table column *every* time and biases variance confidence intervals systematically; the rule is fixed — deviations about $\overline{X}$ always carry $n-1$ degrees of freedom.

$$\boxed{\text{Use } (n-1)S^2/\sigma^2 \sim \chi^2_{19}:\ \mathbb{P}(S^2 > 30) \approx 0.25}$$

Example 7.2.20 (Mean and variance of the sum of squares directly)**.** For a normal sample of size $n = 25$ with $\sigma^2 = 4$, give the mean and standard deviation of the observed sum of squared deviations $\text{SS} = (n-1)S^2 = \sum(X_i - \overline{X})^2$.

Which rule, and why? $\text{SS}/\sigma^2 \sim \chi^2_{n-1}$, so $\text{SS} = \sigma^2 \cdot \chi^2_{n-1}$ is a known constant times a chi-square; push the chi-square moments through the scaling.

Solution.

Step 1: the chi-square moments. $U = \text{SS}/\sigma^2 \sim \chi^2_{24}$, so $\mathbb{E}[U] = 24$ and $\text{Var}(U) = 2(24) = 48$.

Step 2: scale by $\sigma^2 = 4$.

$$\mathbb{E}[\text{SS}] = \sigma^2\,\mathbb{E}[U] = 4(24) = 96,$$
$$\text{Var}(\text{SS}) = (\sigma^2)^2\,\text{Var}(U) = 16(48) = 768,$$
$$\text{SD}(\text{SS}) = \sqrt{768} \approx 27.71.$$

$$\boxed{\mathbb{E}[\text{SS}] = 96, \qquad \text{SD}(\text{SS}) = \sqrt{768} \approx 27.7}$$

Example 7.2.21 (Expected value of the sample variance, from the pillar)**.** Confirm directly from pillar (ii) that S^2 is unbiased, i.e. $\mathbb{E}[S^2] = \sigma^2$, and find $\text{Var}(S^2)$ for a normal sample of size n.

Which rule, and why? Write S^2 as $\frac{\sigma^2}{n-1} \cdot \frac{(n-1)S^2}{\sigma^2}$ — a constant times a χ^2_{n-1} — and read off both moments from $\mathbb{E}[\chi^2_{n-1}] = n-1$ and $\text{Var}(\chi^2_{n-1}) = 2(n-1)$.

Solution.

Step 1: express S^2 via the pivot. $S^2 = \dfrac{\sigma^2}{n-1}\,U$ with $U = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Step 2: mean (unbiasedness).

$$\mathbb{E}[S^2] = \frac{\sigma^2}{n-1}\,\mathbb{E}[U] = \frac{\sigma^2}{n-1}\,(n-1) = \sigma^2.$$

Step 3: variance.

$$\text{Var}(S^2) = \left(\frac{\sigma^2}{n-1}\right)^2 \text{Var}(U) = \frac{\sigma^4}{(n-1)^2}\,\big[2(n-1)\big] = \frac{2\sigma^4}{n-1}.$$

$$\boxed{\mathbb{E}[S^2] = \sigma^2, \qquad \mathrm{Var}(S^2) = \frac{2\sigma^4}{n-1}}$$

The variance shrinking like $1/(n-1)$ shows why larger samples pin down σ^2 more tightly — and why the chi-square pivot, not a normal approximation, is the honest tool for small n.

7.3 Confidence Intervals for a Normal Mean

A confidence interval turns a single estimate $\overline{X}$ into an honest *range* of plausible values for the unknown mean μ, with a stated reliability such as 95%. When the population is normal and σ is genuinely known, that range is the z-interval of Chapter 6. The realistic case — the one this section is about — is that σ is *unknown* and must be estimated from the same data by the sample standard deviation S. Substituting S for σ inflates the uncertainty, so the correct pivot is no longer standard normal but Student's t with $n-1$ degrees of freedom; the interval that results is wider than the naive z-interval, and rightly so. Recognize the setting by three cues at once: the parameter is a **mean**, the parent is (at least roughly) **normal**, and σ is **not supplied**. The whole construction is one idea — find a pivot whose distribution is free of the unknown μ, then read off the central $1-\alpha$ chunk of that distribution and invert it into a statement about μ.

Which interval does the problem call for?

- Want a **range** for a normal **mean** μ, with σ **unknown** (only the data, from which you compute S)? $\Rightarrow$ the t **interval** $\overline{X} \pm t_{\alpha/2,\,n-1}\, S/\sqrt{n}$, with $n-1$ degrees of freedom.
- σ is genuinely **known** (a spec, a calibration constant)? $\Rightarrow$ the z **interval** $\overline{X} \pm z_{\alpha/2}\, \sigma/\sqrt{n}$ of Chapter 6 — not this one.
- Asked for a single **bound** ("at least", "no more than", a worst case in one direction)? $\Rightarrow$ a **one-sided** confidence bound, all of α in one tail, using $t_{\alpha,\,n-1}$.
- Asked how large a sample to guarantee a target margin? $\Rightarrow$ solve $\mathrm{ME} = t_{\alpha/2,\,n-1}\, S/\sqrt{n}$ for n, using a **planning value** for S (it is unknown in advance).

Do not confuse a known σ with an estimated one: if the spread number was computed *from this sample* it is really S, and you must use t, not z. The z interval is then too narrow — it claims more confidence than it delivers.

7.3.1 The pivot and the interval

The engine is a single quantity whose distribution does not depend on the unknown μ.

Definition 7.3.1 (Pivot)**.** A **pivot** is a function of the data *and* the unknown parameter whose distribution is completely known — it contains no unknowns other than the one parameter, and that parameter cancels out of the distribution. For a normal mean with σ unknown the pivot is

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}.$$

Theorem 7.3.2 (Distribution of the pivot)**.** *Let $X_1, \ldots, X_n$ be independent $\mathrm{N}(\mu, \sigma^2)$ with $\overline{X} = \frac{1}{n}\sum_i X_i$ and $S^2 = \frac{1}{n-1}\sum_i (X_i - \overline{X})^2$. Then*

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \sim t_{\,n-1},$$

whatever the values of μ and σ. *The distribution depends only on n, not on the unknown parameters — exactly what makes T a usable pivot.*

Proof. Split the ratio by inserting σ:

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} = \frac{(\overline{X} - \mu)/(\sigma/\sqrt{n})}{\sqrt{\left[(n-1)S^2/\sigma^2\right]/(n-1)}} = \frac{Z}{\sqrt{V/(n-1)}}.$$

The numerator $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ is exactly standard normal, since $\overline{X}$ is normal with mean μ and variance σ^2/n. The quantity $V = (n-1)S^2/\sigma^2$ is χ^2_{n-1}, and for a normal sample $\overline{X}$ and S^2 are independent, so Z and V are independent. A standard normal over the square root of an independent chi-square divided by its degrees of freedom is, by definition, t_{n-1}. Note that σ has cancelled — it appears in neither Z (it divides out) nor the law of V. □

Theorem 7.3.3 (Two-sided t confidence interval)**.** *For a level $1-\alpha$, let $t_{\alpha/2,\,n-1}$ be the value with $\mathbb{P}(T \geq t_{\alpha/2,\,n-1}) = \alpha/2$ for $T \sim t_{n-1}$. By symmetry $\mathbb{P}(-t_{\alpha/2,\,n-1} \leq T \leq t_{\alpha/2,\,n-1}) = 1-\alpha$. Substituting the pivot,*

$$1 - \alpha = \mathbb{P}\left(-t_{\alpha/2,\,n-1} \leq \frac{\overline{X} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2,\,n-1}\right) = \mathbb{P}\left(\overline{X} - t_{\alpha/2,\,n-1}\frac{S}{\sqrt{n}} \leq \mu \leq \overline{X} + t_{\alpha/2,\,n-1}\frac{S}{\sqrt{n}}\right).$$

Hence the $100(1-\alpha)\%$ confidence interval for μ is

$$\boxed{\overline{X} \pm t_{\alpha/2,\,n-1}\,\frac{S}{\sqrt{n}}}$$

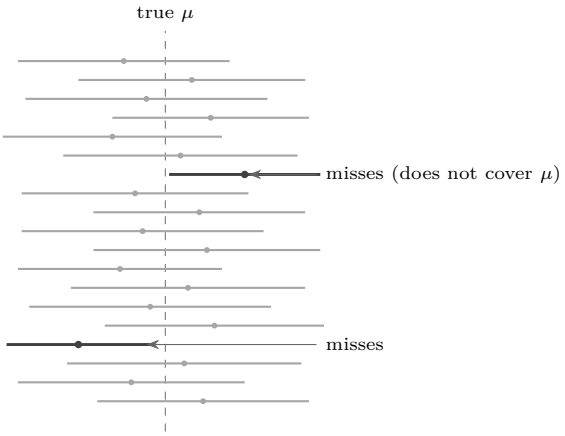
Proof. Start from $-t_{\alpha/2,\,n-1} \leq (\overline{X} - \mu)/(S/\sqrt{n}) \leq t_{\alpha/2,\,n-1}$. Multiply through by the positive number $S/\sqrt{n}$ to get $-t_{\alpha/2,\,n-1}\,S/\sqrt{n} \leq \overline{X} - \mu \leq t_{\alpha/2,\,n-1}\,S/\sqrt{n}$. Subtract $\overline{X}$ and multiply by -1 (which flips the inequalities), giving $\overline{X} - t_{\alpha/2,\,n-1}\,S/\sqrt{n} \leq \mu \leq \overline{X} + t_{\alpha/2,\,n-1}\,S/\sqrt{n}$. The endpoints are computable from the data alone, so we have inverted a probability statement about T into one about μ. □

Definition 7.3.4 (Margin of error)**.** The half-width of the two-sided interval is the **margin of error**

$$\mathrm{ME} = t_{\alpha/2,\,n-1}\,\frac{S}{\sqrt{n}}.$$

The interval is $\overline{X} \pm \mathrm{ME}$. Three levers move it: the confidence (through $t_{\alpha/2,\,n-1}$), the spread S, and the sample size n (through both $\sqrt{n}$ in the denominator *and* the degrees of freedom in $t_{\alpha/2,\,n-1}$).

Remark (What "confidence" means)*.* The 95% attaches to the *procedure*, not to any one interval. Before the data are seen, the random interval $\overline{X} \pm t_{\alpha/2,\,n-1}\,S/\sqrt{n}$ has probability 0.95 of straddling the fixed, unknown μ. Once you plug in numbers and get, say, [12.1, 15.7], that particular interval either contains μ or it does not — there is no randomness left, so it is wrong to say "μ has a 95% chance of lying in $[12.1, 15.7]$". The honest reading is: "this interval was produced by a method that captures μ 95% of the time." Across many independent applications, about 95% of the intervals cover and about 5% miss.



Coverage picture: nineteen 95% intervals (each a dot at its $\overline{X}$ with a horizontal bar of half-width ME) computed from independent samples around the fixed true μ (dashed line). Most bars cross the line — they cover μ — but two (drawn dark) fall entirely to one side and miss. Over the long run about 1 interval in 20 misses at the 95% level. The confidence is a property of the procedure, not of any single bar.

7.3.2 The standard two-sided interval

Example 7.3.5 (Concrete compressive strength)**.** A batch of concrete cylinders is tested for compressive strength (MPa). From $n = 16$ cylinders an engineer reports $\overline{X} = 38.4$ MPa and $S = 3.2$ MPa. Strength is approximately normal and σ is unknown. Give a 95% confidence interval for the true mean strength. Use $t_{0.025,15} = 2.131$.

Which rule, and why? The parameter is a mean, strength is roughly normal, and σ is unknown — S was computed from these 16 cylinders. That is the t interval with $n - 1 = 15$ degrees of freedom, not a z interval (no σ was supplied) and not a one-sided bound (a two-sided range is asked).

Solution.

Step 1: standard error. $S/\sqrt{n} = 3.2/\sqrt{16} = 3.2/4 = 0.800$ MPa.

Step 2: margin of error. $\text{ME} = t_{\alpha/2,\,n-1}\, S/\sqrt{n} = 2.131 \times 0.800 = 1.705$ MPa.

Step 3: assemble the interval.

$$\overline{X} \pm \text{ME} = 38.4 \pm 1.705 = (36.70,\ 40.11).$$

$$\boxed{95\%\text{ CI for } \mu:\ (36.70,\ 40.11)\text{ MPa.}}$$

We are 95% confident the mean strength lies between 36.7 and 40.1 MPa — meaning the *method* brackets the true mean 95% of the time.

Example 7.3.6 (Blood-pressure reduction from a drug, 99%)**.** In a single-arm study, $n = 25$ patients take a new antihypertensive; the reduction in systolic pressure (mmHg) after four weeks has $\overline{X} = 11.6$ and $S = 6.0$. Give a 99% confidence interval for the mean reduction. Use $t_{0.005,24} = 2.797$.

Which rule, and why? Mean of an approximately normal outcome, σ unknown (S from the 25 patients), two-sided range at high confidence $\Rightarrow$ t interval, $df = 24$, with $\alpha/2 = 0.005$.

Solution.

Step 1: standard error. $S/\sqrt{n} = 6.0/\sqrt{25} = 6.0/5 = 1.200$ mmHg.

Step 2: margin of error. $\text{ME} = 2.797 \times 1.200 = 3.356$ mmHg.

Step 3: interval.

$$11.6 \pm 3.356 = (8.24,\ 14.96).$$

$$\boxed{99\%\text{ CI for mean reduction}:\ (8.24,\ 14.96)\text{ mmHg.}}$$

Because the entire interval lies above 0, the data are consistent with a genuine reduction even at the demanding 99% level.

7.3.3 Computing $\overline{X}$ and S from raw data

Example 7.3.7 (Battery charge times from raw measurements)**.** A lab times how long ($n = 8$ trials, in minutes) a phone takes to charge from 20% to 80%:

$$54,\ 58,\ 51,\ 60,\ 55,\ 49,\ 57,\ 52.$$

Charge time is approximately normal and σ is unknown. Build a 90% confidence interval for the mean charge time. Use $t_{0.05,7} = 1.895$.

Which rule, and why? Mean, roughly normal, σ unknown and *not even S given* — we must compute $\overline{X}$ and S from the raw numbers first, then form the two-sided t interval with $df = 7$.

Solution.

Step 1: sample mean.

$$\sum x_i = 54 + 58 + 51 + 60 + 55 + 49 + 57 + 52 = 436, \qquad \overline{X} = \frac{436}{8} = 54.5.$$

Step 2: squared deviations. With $\overline{X} = 54.5$, the deviations $x_i - \overline{X}$ and their squared sum are

$$x_i - \overline{X}: \quad -0.5,\ 3.5,\ -3.5,\ 5.5,\ 0.5,\ -5.5,\ 2.5,\ -2.5,$$
$$\sum (x_i - \overline{X})^2 = 0.25 + 12.25 + 12.25 + 30.25 + 0.25 + 30.25 + 6.25 + 6.25 = 98.0.$$

Step 3: sample variance and SD. Divide by $n - 1 = 7$:

$$S^2 = \frac{98.0}{7} = 14.0, \qquad S = \sqrt{14.0} = 3.742 \text{ min}.$$

Step 4: standard error and margin. $S/\sqrt{n} = 3.742/\sqrt{8} = 3.742/2.828 = 1.323$, so

$$\text{ME} = 1.895 \times 1.323 = 2.508 \text{ min}.$$

Step 5: interval.

$$54.5 \pm 2.508 = (51.99,\ 57.01).$$

90% CI for mean charge time : (51.99, 57.01) min.

Example 7.3.8 (Sum-of-squares shortcut on raw data)**.** Six soil samples from a field have nitrate readings (mg/kg) 12, 15, 11, 18, 14, 16. The readings are approximately normal with unknown σ. Using the computing form $S^2 = \frac{1}{n-1}\big(\sum x_i^2 - n\overline{X}^2\big)$, give a 95% confidence interval for the mean nitrate level. Use $t_{0.025,5} = 2.571$.

Which rule, and why? Same recognition — mean, normal, σ unknown — but here we practice the computing form for S^2, which avoids listing every deviation. Two-sided t interval, $df = 5$.

Solution.

Step 1: the two sums.

$$\sum x_i = 12 + 15 + 11 + 18 + 14 + 16 = 86, \qquad \overline{X} = \frac{86}{6} = 14.333,$$
$$\sum x_i^2 = 144 + 225 + 121 + 324 + 196 + 256 = 1266.$$

Step 2: variance by the computing form.

$$S^2 = \frac{1}{5}\big(1266 - 6 \cdot 14.333^2\big) = \frac{1}{5}\big(1266 - 1232.67\big) = \frac{33.33}{5} = 6.667, \qquad S = 2.582.$$

Step 3: standard error and margin. $S/\sqrt{n} = 2.582/\sqrt{6} = 2.582/2.449 = 1.054$,

$$\text{ME} = 2.571 \times 1.054 = 2.711 \text{ mg/kg}.$$

Step 4: interval.

$$14.333 \pm 2.711 = (11.62,\ 17.04).$$

95% CI for mean nitrate : (11.62, 17.04) mg/kg.

7.3.4 One-sided confidence bounds

When the practical question points one way — "the mean is *at least* how big?" or "*at most* how big?" — put all of α in one tail and report a single bound. From $\mathbb{P}\big((\overline{X}-\mu)/(S/\sqrt{n}) \le t_{\alpha,\,n-1}\big) = 1-\alpha$ one gets the lower bound $\mu \ge \overline{X} - t_{\alpha,\,n-1}\,S/\sqrt{n}$; the mirror gives the upper bound $\mu \le \overline{X} + t_{\alpha,\,n-1}\,S/\sqrt{n}$. The one-sided cutoff $t_{\alpha,\,n-1}$ is smaller than the two-sided $t_{\alpha/2,\,n-1}$, so a one-sided bound sits closer to $\overline{X}$ than the corresponding two-sided endpoint.

Example 7.3.9 (Lower confidence bound on battery capacity)**.** A manufacturer measures usable capacity (mAh) of $n = 10$ production cells: $\overline{X} = 2980$ mAh, $S = 45$ mAh, capacity approximately normal, σ unknown. Marketing wants a 95% lower confidence bound — a guaranteed "at least" figure — for the mean capacity. Use $t_{0.05,9} = 1.833$.

Which rule, and why? The question asks for a single "at least" value, a one-directional guarantee, so this is a one-sided (lower) t confidence bound with all 5% in the lower tail, $df = 9$ — not a two-sided interval.

Solution.

Step 1: standard error. $S/\sqrt{n} = 45/\sqrt{10} = 45/3.162 = 14.23$ mAh.

Step 2: one-sided margin. $t_{\alpha,\,n-1}\,S/\sqrt{n} = 1.833 \times 14.23 = 26.08$ mAh.

Step 3: lower bound.

$$\mu \ge \overline{X} - t_{\alpha,\,n-1}\frac{S}{\sqrt{n}} = 2980 - 26.08 = 2953.9.$$

$$\boxed{95\%\text{ lower bound}:\ \mu \ge 2953.9\text{ mAh.}}$$

We are 95% confident the true mean capacity is at least about 2954 mAh; there is no upper limit in this one-sided statement.

Example 7.3.10 (Upper confidence bound on a contaminant)**.** An assay measures a contaminant (ppb) in $n = 12$ water samples: $\overline{X} = 8.3$ ppb, $S = 1.8$ ppb, roughly normal, σ unknown. A regulator wants a 95% upper confidence bound — a worst-case "no more than" figure — for the mean. Use $t_{0.05,11} = 1.796$.

Which rule, and why? A one-directional "no more than" guarantee ⇒ a one-sided (upper) t bound, all 5% in the upper tail, $df = 11$.

Solution.

Step 1: standard error. $S/\sqrt{n} = 1.8/\sqrt{12} = 1.8/3.464 = 0.5196$ ppb.

Step 2: one-sided margin. $t_{\alpha,\,n-1}\,S/\sqrt{n} = 1.796 \times 0.5196 = 0.9332$ ppb.

Step 3: upper bound.

$$\mu \le \overline{X} + t_{\alpha,\,n-1}\frac{S}{\sqrt{n}} = 8.3 + 0.9332 = 9.233.$$

$$\boxed{95\%\text{ upper bound}:\ \mu \le 9.23\text{ ppb.}}$$

With 95% confidence the mean contaminant level is no more than about 9.2 ppb.

7.3.5 Higher confidence is wider; larger n is narrower

Example 7.3.11 (Same data, three confidence levels)**.** A fitness study measures resting heart rate (bpm) on $n = 20$ adults: $\overline{X} = 68.0$, $S = 6.5$, approximately normal, σ unknown. Compute the 90%, 95%, and 99% confidence intervals and compare their widths. Use $t_{0.05,19} = 1.729$, $t_{0.025,19} = 2.093$, $t_{0.005,19} = 2.861$.

Which rule, and why? One mean, normal, σ unknown ⇒ t interval, $df = 19$. The only thing changing across the three is the confidence level, hence the critical value $t_{\alpha/2,\,n-1}$; the data, S, and n are fixed.

Solution.

Step 1: common standard error. $S/\sqrt{n} = 6.5/\sqrt{20} = 6.5/4.472 = 1.4534$ bpm.

Step 2: margins at each level.

$$\begin{aligned} 90\%: &\quad \text{ME} = 1.729 \times 1.4534 = 2.513, \\ 95\%: &\quad \text{ME} = 2.093 \times 1.4534 = 3.042, \\ 99\%: &\quad \text{ME} = 2.861 \times 1.4534 = 4.158. \end{aligned}$$

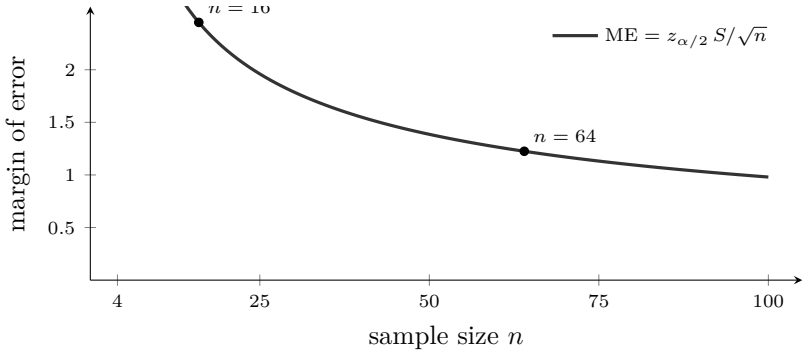
Step 3: the three intervals about $\overline{X} = 68.0$.

$$\begin{aligned} 90\%: &\quad (65.49,\ 70.51), \quad \text{width } 5.03, \\ 95\%: &\quad (64.96,\ 71.04), \quad \text{width } 6.08, \\ 99\%: &\quad (63.84,\ 72.16), \quad \text{width } 8.32. \end{aligned}$$

Higher confidence $\Rightarrow$ larger $t_{\alpha/2,\,n-1}$ $\Rightarrow$ wider interval.

Demanding more confidence forces you to cast a wider net: to be surer of capturing μ you must admit more candidate values. Confidence and precision trade off; only more data buys both.

Remark (Why more data narrows the interval, and $t \to z$). The margin $\text{ME} = t_{\alpha/2,\,n-1}\, S/\sqrt{n}$ shrinks with n for two compounding reasons. First, the explicit $\sqrt{n}$ in the denominator: quadrupling n roughly halves ME. Second, the critical value $t_{\alpha/2,\,n-1}$ itself falls as the degrees of freedom grow, because estimating σ becomes more reliable and the t tails thin toward the normal: $t_{0.025,9} = 2.262$, $t_{0.025,29} = 2.045$, $t_{0.025,99} = 1.984$, all closing on $z_{0.025} = 1.96$. So for large n the t interval and the known-σ z interval nearly coincide — which is exactly why a large-sample interval can be quoted with the z cutoff even when σ was estimated.



Margin of error against sample size (here $S = 5$, $z_{\alpha/2} = 1.96$): the curve $\text{ME} \propto 1/\sqrt{n}$ falls steeply at first, then flattens. The marked points at $n = 16$ and $n = 64$ show the rule of thumb — **quadrupling n halves the margin**. Diminishing returns set in: going from $n = 64$ to $n = 100$ barely moves the curve, so chasing a tiny margin costs disproportionately many observations.

7.3.6 Sample size for a target margin

Example 7.3.12 (How many specimens for a target precision?)**.** An engineer wants a 95% confidence interval for mean tensile strength with margin of error no larger than ME = 1.0 MPa. A pilot run suggests the strength standard deviation is about $S = 3.5$ MPa. Roughly how many specimens are needed? Use $z_{0.025} = 1.96$ as the large-sample stand-in for $t_{\alpha/2,\,n-1}$.

Which rule, and why? This is a planning calculation: solve the margin formula $\text{ME} = t_{\alpha/2,\,n-1}\, S/\sqrt{n}$ for n. Because $t_{\alpha/2,\,n-1}$ itself depends on the unknown n (through its df) and S is not yet observed, we use a *planning value* for S and the large-n stand-in $z_{0.025} = 1.96$ for the critical value, then check.

Solution.

Step 1: solve the margin formula for n. From $\text{ME} = z_{\alpha/2}\, S/\sqrt{n}$,

$$\sqrt{n} = \frac{z_{\alpha/2}\, S}{\text{ME}}, \qquad n = \left(\frac{z_{\alpha/2}\, S}{\text{ME}}\right)^2.$$

Step 2: plug in the planning numbers.

$$n = \left(\frac{1.96 \times 3.5}{1.0}\right)^2 = (6.86)^2 = 47.06.$$

Step 3: round up and acknowledge the circularity. Always round *up* so the margin is met: $n = 48$. The estimate is approximate because (i) the true S is unknown in advance — we borrowed a pilot value, and (ii) we used z for $t_{\alpha/2,\,n-1}$; at $df = 47$ the true cutoff $t_{0.025,47} \approx 2.01$ is slightly above 1.96, nudging the requirement up a touch.

$$\boxed{n = \lceil (1.96 \cdot 3.5/1.0)^2 \rceil = 48 \text{ specimens (planning estimate).}}$$

Step 4: read the levers. If the pilot understated S, the realized margin will exceed 1.0; a conservative planner inflates S a little. Halving the target margin to 0.5 MPa would *quadruple* n to about 188, the same $1/\sqrt{n}$ trade-off seen in the figure above.

Remark (The circularity, stated plainly). The margin formula needs S, but S is only known *after* sampling — so sample-size planning is inherently approximate. The standard fix is a **planning value**: a guess for S from a pilot study, prior data, or a conservative upper estimate (e.g. range/4 for roughly normal data). One then verifies after collecting the data that the realized margin met the goal, and collects a few more observations if a large S overran it.

7.3.7 Contrast: the too-narrow z-interval

Example 7.3.13 (CONTRAST: using z when σ was estimated)**.** A clinic measures recovery time (days) for $n = 9$ patients: $\overline{X} = 14.0$ days and a *sample* standard deviation $S = 3.0$ days computed from those same 9 patients; the true σ is *not* known. A careless analyst builds a "95%" interval with the normal cutoff $z_{0.025} = 1.96$ instead of $t_{0.025,8} = 2.306$. Show the two intervals and explain why the z one is dishonestly narrow.

Which rule, and why? Because σ is unknown and replaced by S, the correct pivot is t_8, *not* $\mathrm{N}(0,1)$. Using z here is the classic look-alike error; we compute both to expose the under-coverage.

Solution.

Step 1: common standard error. $S/\sqrt{n} = 3.0/\sqrt{9} = 3.0/3 = 1.000$ day.

Step 2: the (wrong) z interval. With $z_{0.025} = 1.96$,

$$\mathrm{ME}_z = 1.96 \times 1.000 = 1.960, \qquad 14.0 \pm 1.96 = (12.04,\ 15.96), \quad \text{width } 3.92.$$

Step 3: the correct t interval. With $t_{0.025,8} = 2.306$,

$$\mathrm{ME}_t = 2.306 \times 1.000 = 2.306, \qquad 14.0 \pm 2.31 = (11.69,\ 16.31), \quad \text{width } 4.61.$$

Step 4: what the error costs. The z interval is about 0.35 day narrower on each side — roughly 15% too tight. It *looks* more precise, but that precision is fake: a procedure using z when σ was estimated captures the true μ *less* than 95% of the time, because it ignores the extra wobble that S adds. The honest 95% statement is the wider t interval.

$$\boxed{\sigma \text{ estimated} \Rightarrow \text{use } t_8;\ \text{the } z \text{ interval } (12.04, 15.96) \text{ under-covers.}}$$

Remark. The fix is the same recognition as in testing: ask whether the spread came from *outside* this sample (a spec or known constant $\Rightarrow z$) or from *inside* it (computed from the same $X_i \Rightarrow t$). The danger is worst for small n, where $t_{\alpha/2,\,n-1}$ exceeds $z_{\alpha/2}$ substantially; by $n \approx 40$ the two intervals nearly agree and the shortcut becomes defensible.

7.4 Hypothesis Tests for a Normal Mean

By now the sampling-distribution machinery of this chapter is in place: for a sample from a $\mathrm{N}(\mu, \sigma^2)$ population, $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$, the quantity $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and $\overline{X}$ and S^2 are independent. The previous section turned the first two facts into a confidence interval for μ. This section turns the *same* engine into a decision procedure: we fix a claimed value μ_0 of the mean, ask whether the data are consistent with it, and quantify "consistent" by where the standardized mean lands in a t_{n-1} distribution. The recognition cues are exactly those of the t interval — a question about a *mean*, an at-least-approximately *normal* parent, and a population σ that is *not handed to you* but estimated by S from the data. The new skills are reading the wording to pick one tail or two, converting the statistic into a p-value, and recognizing that a two-sided test and the t interval are two faces of one calculation. Throughout, the conclusion is stated in plain words about the original quantity, never left as a bare "reject."

Which mean test — and which tail?

- A claim about a **mean** μ, parent roughly **normal**, σ **unknown** (only the data are given)? $\Rightarrow$ the **one-sample t-test**, $t_{\text{obs}} = (\overline{X} - \mu_0)/(S/\sqrt{n})$ referred to t_{n-1}.
- σ genuinely **known**? $\Rightarrow$ that is the z-test of Chapter 6, not this section. Estimating σ by S is what forces the t.
- Each subject measured **twice** (before/after, two methods on one unit)? $\Rightarrow$ collapse to differences D_i and run a one-sample t-test on the D_i against $\mu_0 = 0$.
- **One tail or two?** "has increased" / "exceeds" / "is above" $\Rightarrow$ upper tail. "has dropped" / "is below spec" $\Rightarrow$ lower tail. "differs" / "is off target" / "changed" $\Rightarrow$ two tails.
- Already built the $1-\alpha$ t **interval**? $\Rightarrow$ the two-sided level-α test needs no new arithmetic: reject μ_0 exactly when it falls *outside* the interval (**test–CI duality**).

Do not confuse the t with the z: for the same α the t cutoff is *larger* than the z cutoff (and only shrinks toward it as n grows). Using z after estimating σ understates how extreme the data must be — the test then rejects too often, and its true Type I error exceeds the advertised α.

7.4.1 The one-sample t-test

The test inherits its legitimacy directly from the chapter's central theorem. We restate the piece we need and read off the decision rule.

Theorem 7.4.1 (Null distribution of the t statistic)**.** *Let $X_1, \ldots, X_n$ be independent $\mathrm{N}(\mu, \sigma^2)$, and let $\overline{X}$ and S be the sample mean and sample standard deviation. Then for* every *$\sigma > 0$,*

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}} \;\sim\; t_{n-1}.$$

In particular, if the null hypothesis $H_0 : \mu = \mu_0$ is true, the statistic $t_{obs} = (\overline{X} - \mu_0)/(S/\sqrt{n})$ follows t_{n-1}, so its sampling behavior under H_0 is completely known and free of the nuisance parameter σ.

Proof. Divide numerator and denominator of T by $\sigma/\sqrt{n}$:

$$T = \frac{(\overline{X} - \mu)/(\sigma/\sqrt{n})}{\big(S/\sqrt{n}\big)/(\sigma/\sqrt{n})} = \frac{Z}{\sqrt{\big[(n-1)S^2/\sigma^2\big]/(n-1)}} = \frac{Z}{\sqrt{V/(n-1)}}.$$

The numerator $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n})$ is standard normal. The quantity $V = (n-1)S^2/\sigma^2$ is χ^2_{n-1}, and because $\overline{X}$ and S^2 are independent for a normal sample, Z and V are independent. A standard normal divided by the square root of an independent chi-square over its degrees of freedom is, by definition, a t_{n-1} variable. The σ cancels, so the conclusion holds whatever its value. □

Remark. This is precisely *why estimating σ by S replaces z with t.* In the z statistic the denominator $\sigma/\sqrt{n}$ is a fixed constant, so only $\overline{X}$ is random. The moment we substitute S, the denominator $S/\sqrt{n}$ becomes random too and, in a small sample, can fall well below $\sigma/\sqrt{n}$, occasionally inflating the ratio even when $\overline{X} - \mu_0$ is unremarkable. That second source of randomness fattens the tails into the t_{n-1} shape. As n grows $S \to \sigma$, the denominator stops wobbling, and $t_{n-1} \to \mathrm{N}(0,1)$.

Definition 7.4.2 (One-sample t-test of $H_0 : \mu = \mu_0$)**.** For a normal population with unknown σ, at level α, compute $t_{\text{obs}} = (\overline{X} - \mu_0)/(S/\sqrt{n})$ and reject H_0 according to the alternative:

$$\begin{aligned} H_1 : \mu > \mu_0 \quad &\Longrightarrow \quad \text{reject if } t_{\text{obs}} \geq t_{\alpha,\, n-1}, \\ H_1 : \mu < \mu_0 \quad &\Longrightarrow \quad \text{reject if } t_{\text{obs}} \leq -t_{\alpha,\, n-1}, \\ H_1 : \mu \neq \mu_0 \quad &\Longrightarrow \quad \text{reject if } |t_{\text{obs}}| \geq t_{\alpha/2,\, n-1}. \end{aligned}$$

Here $t_{\alpha,\, n-1}$ denotes the value with upper-tail area α under t_{n-1}. Equivalently, reject when the p-value (the null tail area in the direction of H_1) satisfies $p \leq \alpha$.

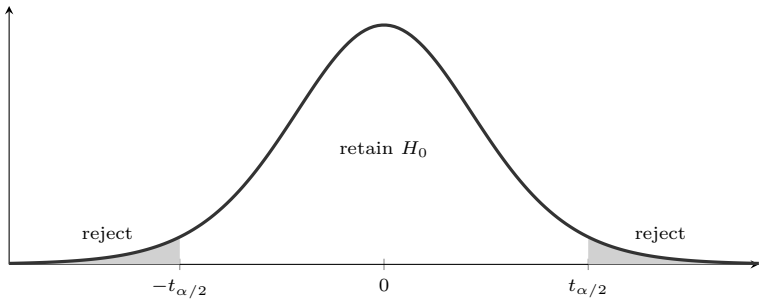


Figure: the two-tailed rejection region for a level-α test (here $df = 12$, $\alpha = 0.05$, $t_{0.025,12} = 2.179$). Each shaded tail carries area $\alpha/2$; the statistic is significant only when $|t_{obs}|$ reaches a tail. The curve is the t_{12} density up to a normalizing constant chosen so its peak fills the axis.

7.4.2 Two-sided tests

Example 7.4.3 (Net contents of a bottling line)**.** A bottling line is supposed to deliver a mean of $\mu_0 = 750$ mL. A regulator draws $n = 16$ bottles and finds $\overline{X} = 746.1$ mL with sample standard deviation $S = 6.0$ mL. Is there evidence at $\alpha = 0.05$ that the line is off target in *either* direction? Use $t_{0.025,15} = 2.131$.

Which rule, and why? The parameter is a mean of an approximately normal fill volume, and σ is not given — S was computed from the sample. That is the one-sample t-test with $df = 15$. "Off target in either direction" has no direction, so it is two-sided.

Solution.

Step 1: hypotheses. $H_0 : \mu = 750$ versus $H_1 : \mu \neq 750$, $\alpha = 0.05$.

Step 2: standard error. $S/\sqrt{n} = 6.0/\sqrt{16} = 6.0/4 = 1.50$ mL.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{746.1 - 750}{1.50} = \frac{-3.9}{1.50} = -2.60.$$

Step 4: critical value and decision. The two-sided 5% cutoff is $t_{0.025,15} = 2.131$. Since $|t_{\text{obs}}| = 2.60 > 2.131$, the statistic falls in a tail.

$\lvert t_{\text{obs}}\rvert = 2.60 > 2.131 \;\Rightarrow\;$ reject H_0;
the line is filling significantly below target (mean $\neq 750$ mL) at $\alpha = 0.05$.

Example 7.4.4 (Setpoint of a chemical reactor)**.** A reactor's temperature controller should hold a mean of $\mu_0 = 180.0\,^\circ\text{C}$. An engineer logs $n = 10$ independent readings and obtains $\overline{X} = 181.4\,^\circ\text{C}$ with $S = 2.3\,^\circ\text{C}$. Test at $\alpha = 0.05$ whether the mean temperature differs from the setpoint. Use $t_{0.025,9} = 2.262$.

Which rule, and why? Mean of an approximately normal measurement process, σ unknown (estimated by S); "differs from" is two-sided. One-sample two-sided t-test, $df = 9$.

Solution.

Step 1: hypotheses. $H_0 : \mu = 180.0$ versus $H_1 : \mu \neq 180.0$.

Step 2: standard error. $S/\sqrt{n} = 2.3/\sqrt{10} = 2.3/3.162 = 0.7273\,^\circ\text{C}$.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{181.4 - 180.0}{0.7273} = \frac{1.4}{0.7273} = 1.925.$$

Step 4: decision. Compare $|t_{\text{obs}}| = 1.925$ with $t_{0.025,9} = 2.262$. Since $1.925 < 2.262$, the statistic stays in the retain region.

$$\boxed{|t_{\text{obs}}| = 1.925 < 2.262 \;\Rightarrow\; \text{do not reject } H_0;\ \text{no significant departure from } 180.0\,^\circ\text{C at } \alpha = 0.05.}$$

7.4.3 One-sided tests

Example 7.4.5 (Has output per shift increased? (upper tail))**.** A packaging cell historically completes $\mu_0 = 420$ cartons per shift. After a layout change, $n = 12$ shifts average $\overline{X} = 433.5$ cartons with $S = 18.4$. Test at $\alpha = 0.05$ whether the change *increased* mean output. Use $t_{0.05,11} = 1.796$.

Which rule, and why? Mean throughput, approximately normal, σ unknown (S from the data). The claim "increased" points one way, so all of α goes in the upper tail: an upper-tail one-sample t-test, $df = 11$.

Solution.

Step 1: hypotheses. $H_0 : \mu = 420$ versus $H_1 : \mu > 420$ (upper tail).

Step 2: standard error. $S/\sqrt{n} = 18.4/\sqrt{12} = 18.4/3.464 = 5.312$.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{433.5 - 420}{5.312} = \frac{13.5}{5.312} = 2.541.$$

Step 4: decision. Put all 5% in the right tail: $t_{0.05,11} = 1.796$. Since $t_{\text{obs}} = 2.541 > 1.796$, reject.

$$\boxed{t_{\text{obs}} = 2.541 > 1.796 \;\Rightarrow\; \text{reject } H_0;\ \text{evidence the layout change raised mean output at } \alpha = 0.05.}$$

Example 7.4.6 (Is the coating thickness below spec? (lower tail))**.** A specification requires a protective coating to average at least $\mu_0 = 12.0$ microns; an auditor suspects a batch runs *below* spec. From $n = 9$ coupons, $\overline{X} = 11.55$ microns and $S = 0.66$ micron. Test at $\alpha = 0.05$ whether the mean is below 12.0. Use $t_{0.05,8} = 1.860$.

Which rule, and why? Mean thickness, approximately normal, σ unknown. The directional concern "below spec" is a lower-tail one-sample t-test, $df = 8$; reject only for sufficiently negative t_{obs}.

Solution.

Step 1: hypotheses. $H_0 : \mu = 12.0$ versus $H_1 : \mu < 12.0$ (lower tail).

Step 2: standard error. $S/\sqrt{n} = 0.66/\sqrt{9} = 0.66/3 = 0.2200$ micron.

Step 3: test statistic.

$$t_{\text{obs}} = \frac{11.55 - 12.0}{0.2200} = \frac{-0.45}{0.2200} = -2.045.$$

Step 4: decision. The lower-tail cutoff is $-t_{0.05,8} = -1.860$. Since $t_{\text{obs}} = -2.045 < -1.860$, reject.

$$\boxed{t_{\text{obs}} = -2.045 < -1.860 \;\Rightarrow\; \text{reject } H_0;\ \text{the mean coating is significantly below 12.0 microns.}}$$

7.4.4 Starting from raw data

Example 7.4.7 (Dissolved-oxygen readings, computed from scratch)**.** An ecologist suspects a stream's mean dissolved oxygen has drifted from the reference value $\mu_0 = 8.0$ mg/L. Seven independent samples (mg/L) read

$$7.4,\ 8.6,\ 7.1,\ 8.9,\ 7.8,\ 8.3,\ 6.9.$$

Compute $\overline{X}$ and S, then test $H_0 : \mu = 8.0$ versus $H_1 : \mu \neq 8.0$ at $\alpha = 0.05$. Use $t_{0.025,6} = 2.447$.

Which rule, and why? Mean of an approximately normal measurement, σ unknown and not even S supplied — we must build S from the raw numbers, then run a two-sided one-sample t-test, $df = 6$. "Drifted" (no direction) makes it two-sided.

Solution.

Step 1: sample mean.

$$\sum x_i = 7.4 + 8.6 + 7.1 + 8.9 + 7.8 + 8.3 + 6.9 = 55.0, \qquad \overline{X} = \frac{55.0}{7} = 7.857.$$

Step 2: sum of squared deviations. With $\overline{X} = 7.857$, the deviations $x_i - \overline{X}$ and their squares give

$$\begin{aligned} \text{SS} &= (-0.457)^2 + (0.743)^2 + (-0.757)^2 + (1.043)^2 + (-0.057)^2 + (0.443)^2 + (-0.957)^2 \\ &= 0.209 + 0.552 + 0.573 + 1.088 + 0.003 + 0.196 + 0.916 = 3.537. \end{aligned}$$

Step 3: sample variance and SD. Divide by $n - 1 = 6$:

$$S^2 = \frac{3.537}{6} = 0.5895, \qquad S = \sqrt{0.5895} = 0.768 \text{ mg/L}.$$

Step 4: standard error and statistic. $S/\sqrt{n} = 0.768/\sqrt{7} = 0.768/2.646 = 0.2903$,

$$t_{\text{obs}} = \frac{7.857 - 8.0}{0.2903} = \frac{-0.143}{0.2903} = -0.493.$$

Step 5: decision. Compare $|t_{\text{obs}}| = 0.493$ with $t_{0.025,6} = 2.447$. Since $0.493 < 2.447$, do not reject.

$\lvert t_{\text{obs}}\rvert = 0.493 < 2.447 \ \Rightarrow$ do not reject H_0;
no evidence the mean dissolved oxygen has drifted from 8.0 mg/L.

Example 7.4.8 (Sum-of-squares shortcut on raw counts)**.** A clinic records the number of patients seen in a half-day for $n = 6$ randomly chosen sessions: 31, 27, 35, 29, 33, 25. The staffing model assumes a mean of $\mu_0 = 28$. Using the computing form $S^2 = \frac{1}{n-1}\left(\sum x_i^2 - n\overline{X}^2\right)$, test $H_0 : \mu = 28$ versus $H_1 : \mu \neq 28$ at $\alpha = 0.05$. Use $t_{0.025,5} = 2.571$.

Which rule, and why? Same recognition — mean, approximately normal, σ unknown — but here we practice the computing form of S^2, which avoids listing every deviation. Two-sided t-test, $df = 5$.

Solution.

Step 1: the two sums.

$$\begin{aligned} \sum x_i &= 31 + 27 + 35 + 29 + 33 + 25 = 180, \qquad \overline{X} = \frac{180}{6} = 30, \\ \sum x_i^2 &= 961 + 729 + 1225 + 841 + 1089 + 625 = 5470. \end{aligned}$$

Step 2: variance by the computing form.

$$S^2 = \frac{1}{5}\left(5470 - 6 \cdot 30^2\right) = \frac{1}{5}\left(5470 - 5400\right) = \frac{70}{5} = 14.0, \qquad S = \sqrt{14.0} = 3.742.$$

Step 3: statistic. $S/\sqrt{n} = 3.742/\sqrt{6} = 3.742/2.449 = 1.528$,

$$t_{\text{obs}} = \frac{30 - 28}{1.528} = \frac{2}{1.528} = 1.309.$$

Step 4: decision. $|t_{\text{obs}}| = 1.309 < t_{0.025,5} = 2.571$.

$|t_{\text{obs}}| = 1.309 < 2.571 \Rightarrow$ do not reject H_0; the data are consistent with a mean of 28.

7.4.5 The p-value with the t distribution

Definition 7.4.9 (p-value of a t-test)**.** The p-value is the probability, computed under H_0 with $T \sim t_{n-1}$, of a statistic at least as extreme as the observed t_{obs} *in the direction of* H_1:

$$\begin{aligned} H_1 : \mu > \mu_0 : &\quad p = \mathbb{P}(T \geq t_{\text{obs}}), \\ H_1 : \mu < \mu_0 : &\quad p = \mathbb{P}(T \leq t_{\text{obs}}), \\ H_1 : \mu \neq \mu_0 : &\quad p = 2\,\mathbb{P}\big(T \geq |t_{\text{obs}}|\big). \end{aligned}$$

Reject H_0 at level α iff $p \leq \alpha$. Because printed t tables are coarse, by hand one usually *brackets* the p-value between adjacent table columns.

Example 7.4.10 (Bracketing a p-value from t_{obs})**.** A vendor claims its sensors have a mean response time exceeding the legacy figure $\mu_0 = 45$ ms (faster sensors are the worry, so the test is whether the mean has *risen*). From $n = 20$ sensors, $\overline{X} = 48.6$ ms and $S = 6.4$ ms. Report the p-value for $H_1 : \mu > 45$ and decide at $\alpha = 0.05$. Relevant t_{19} upper-tail values: $t_{0.05,19} = 1.729$, $t_{0.025,19} = 2.093$, $t_{0.01,19} = 2.539$.

Which rule, and why? Mean, approximately normal, σ unknown; the question asks for strength of evidence, so we report a p-value from t_{19}. "Exceeds / has risen" is upper-tail.

Solution.

Step 1: test statistic. $S/\sqrt{n} = 6.4/\sqrt{20} = 6.4/4.472 = 1.431$,

$$t_{\text{obs}} = \frac{48.6 - 45}{1.431} = \frac{3.6}{1.431} = 2.516.$$

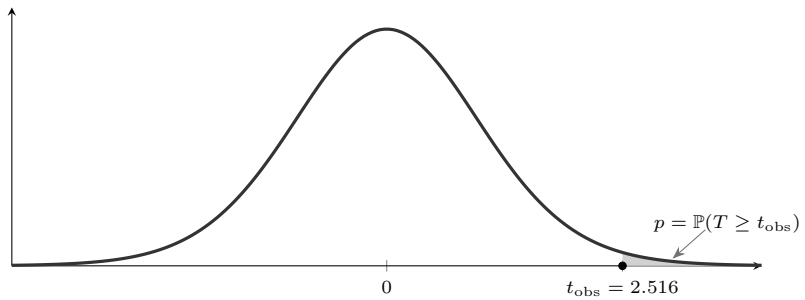
Step 2: locate t_{obs} among the critical values. We have $2.093 < 2.516 < 2.539$, i.e. $t_{0.025,19} < t_{\text{obs}} < t_{0.01,19}$. The upper-tail area to the right of t_{obs} therefore lies between the matching tail probabilities.

Step 3: bracket the p-value. The area right of 2.093 is 0.025 and the area right of 2.539 is 0.01, and $t_{\text{obs}} = 2.516$ sits between them, so

$$0.01 < p < 0.025.$$

Step 4: decision and reading. The whole bracket is below 0.05, so we reject at $\alpha = 0.05$; the evidence is fairly strong but not overwhelming (we would also reject at 0.025, but not at 0.01).

$p \in (0.01,\, 0.025)$; reject at $\alpha = 0.05$ (mean response time has risen).



t_{19} under H_0

Figure: the upper-tail p-value for $H_1 : \mu > \mu_0$. The dot marks $t_{obs} = 2.516$ on the t_{19} axis; the shaded area to its right is the p-value, $\mathbb{P}(T \geq t_{obs})$, which the table brackets between 0.01 and 0.025. The curve is the t_{19} density up to a normalizing constant chosen so its peak fills the axis.

Example 7.4.11 (Two-sided p-value by doubling the tail)**.** A municipal lab targets a treated-water pH of $\mu_0 = 7.20$. From $n = 25$ grab samples, $\overline{X} = 7.27$ and $S = 0.15$. Report the two-sided p-value for $H_1 : \mu \neq 7.20$ and decide at $\alpha = 0.05$. Relevant t_{24} values: $t_{0.05,24} = 1.711$, $t_{0.025,24} = 2.064$, $t_{0.01,24} = 2.492$.

Which rule, and why? Mean, approximately normal, σ unknown, two-sided alternative — one-sample t-test, $df = 24$, with the p-value obtained by doubling the one-tail area.

Solution.

Step 1: test statistic. $S/\sqrt{n} = 0.15/\sqrt{25} = 0.15/5 = 0.0300$,

$$t_{\text{obs}} = \frac{7.27 - 7.20}{0.0300} = \frac{0.07}{0.0300} = 2.333.$$

Step 2: bracket the one-tail area. With $|t_{\text{obs}}| = 2.333$ and $2.064 < 2.333 < 2.492$, the one-tail area $\mathbb{P}(T \geq 2.333)$ lies between 0.01 and 0.025.

Step 3: double for the two-sided p-value.

$$p = 2\,\mathbb{P}(T \geq 2.333) \in (2 \cdot 0.01,\ 2 \cdot 0.025) = (0.02,\ 0.05).$$

Step 4: decision. The whole bracket is at or below 0.05, so reject at $\alpha = 0.05$; but since it lies above 0.02, we could not claim significance at $\alpha = 0.02$.

$$\boxed{p \in (0.02,\ 0.05);\ \text{reject at } \alpha = 0.05 \text{ (pH differs from 7.20).}}$$

7.4.6 Paired differences

A paired design measures each unit twice; the two columns are dependent, so a two-sample comparison is wrong. The fix is to collapse each pair to its difference D_i and apply the *one-sample* t-test to the D_i against $\mu_0 = 0$. The differences cancel the unit-to-unit variation, which is exactly the point of pairing.

Example 7.4.12 (Blood pressure before and after a drug (one-sided paired))**.** Eight patients have systolic blood pressure measured before and after four weeks on a new medication (mm Hg):

Patient	1	2	3	4	5	6	7	8
Before	148	152	139	161	145	158	150	143
After	141	149	138	150	140	147	146	142

Test at $\alpha = 0.05$ whether the drug *lowered* mean systolic pressure. Use $t_{0.05,7} = 1.895$.

Which rule, and why? Each patient is measured twice, so the data are paired, not two independent samples. Reduce to $D_i = \text{Before} - \text{After}$ and run a one-sample t-test on the D_i against 0, $df = 7$. "Lowered" means Before exceeds After on average, so $H_1 : \mu_D > 0$, an upper-tail test on the differences.

Solution.

Step 1: form the differences. $D_i = \text{Before} - \text{After}$:

$$D_i : \quad 7,\ 3,\ 1,\ 11,\ 5,\ 11,\ 4,\ 1.$$

Step 2: mean of differences. $\sum D_i = 7 + 3 + 1 + 11 + 5 + 11 + 4 + 1 = 43$, so $\overline{D} = 43/8 = 5.375$.

Step 3: SD of differences (computing form). $\sum D_i^2 = 49 + 9 + 1 + 121 + 25 + 121 + 16 + 1 = 343$,

$$S_D^2 = \frac{1}{7}\big(343 - 8 \cdot 5.375^2\big) = \frac{1}{7}\big(343 - 231.125\big) = \frac{111.875}{7} = 15.982, \quad S_D = 3.998.$$

Step 4: statistic. Here $\mu_0 = 0$ and $S_D/\sqrt{n} = 3.998/\sqrt{8} = 3.998/2.828 = 1.414$,

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{5.375}{1.414} = 3.801.$$

Step 5: decision. Upper-tail cutoff $t_{0.05,7} = 1.895$. Since $3.801 > 1.895$, reject.

$t_{\text{obs}} = 3.801 > 1.895 \;\Rightarrow$ reject H_0; the drug significantly lowered mean systolic pressure.

7.4.7 Duality of the test and the confidence interval

The two-sided t-test and the t confidence interval are the same statement seen from two sides. Both compare $|\overline{X} - \mu_0|$ to the same yardstick $t_{\alpha/2,n-1}\, S/\sqrt{n}$.

Proposition 7.4.13 (Test–CI duality)**.** *For a normal sample with unknown σ, the two-sided level-α t-test rejects $H_0 : \mu = \mu_0$ if and only if μ_0 lies* outside *the $1-\alpha$ confidence interval $\overline{X} \pm t_{\alpha/2,n-1}\, S/\sqrt{n}$.*

Proof. The test retains H_0 exactly when

$$|t_{\text{obs}}| \le t_{\alpha/2,n-1} \iff \left|\frac{\overline{X} - \mu_0}{S/\sqrt{n}}\right| \le t_{\alpha/2,n-1} \iff |\overline{X} - \mu_0| \le t_{\alpha/2,n-1}\frac{S}{\sqrt{n}}.$$

The last inequality says μ_0 is within $t_{\alpha/2,n-1}\, S/\sqrt{n}$ of $\overline{X}$, i.e. μ_0 lies inside $\overline{X} \pm t_{\alpha/2,n-1}\, S/\sqrt{n}$. Negating both sides: the test rejects iff μ_0 is outside that interval. □

Example 7.4.14 (Reading a test straight off the interval)**.** A materials lab reports a 95% confidence interval for the mean tensile strength of a polymer (in MPa) as (58.3, 62.7), computed from a normal sample with S from the data. Two claimed design values are on the table: $\mu_0 = 60.0$ MPa and $\mu_0 = 63.0$ MPa. Without any further arithmetic, decide each two-sided test at $\alpha = 0.05$.

Which rule, and why? A 95% t interval is already in hand, so the duality proposition settles each two-sided $\alpha = 0.05$ test by inspection: reject precisely the claimed means that fall outside (58.3, 62.7). No statistic need be recomputed.

Solution.

Step 1: test $\mu_0 = 60.0$. Is 60.0 inside (58.3, 62.7)? Yes, $58.3 < 60.0 < 62.7$. By duality the test retains H_0.

Step 2: test $\mu_0 = 63.0$. Is 63.0 inside (58.3, 62.7)? No, $63.0 > 62.7$. By duality the test rejects H_0.

Step 3: state the conclusions in words. A mean strength of 60.0 MPa is consistent with the data at $\alpha = 0.05$; a mean of 63.0 MPa is not — the interval, hence the data, rule it out.

$\mu_0 = 60.0 \in (58.3, 62.7) \;\Rightarrow$ retain;
$\mu_0 = 63.0 \notin (58.3, 62.7) \;\Rightarrow$ reject.

Example 7.4.15 (Cross-checking a test against its interval)**.** A bakery weighs $n = 16$ loaves and obtains $\overline{X} = 802$ g, $S = 12$ g. Test $H_0 : \mu = 800$ g versus $H_1 : \mu \neq 800$ g at $\alpha = 0.05$, then build the 95% interval and confirm the two agree. Use $t_{0.025,15} = 2.131$.

Which rule, and why? Mean, approximately normal, σ unknown: a two-sided one-sample t-test, $df = 15$. Computing the matching 95% interval lets us verify the duality numerically.

Solution.

Step 1: standard error. $S/\sqrt{n} = 12/\sqrt{16} = 12/4 = 3.00$ g.

Step 2: the test.

$$t_{\text{obs}} = \frac{802 - 800}{3.00} = \frac{2}{3.00} = 0.667, \qquad |t_{\text{obs}}| = 0.667 < 2.131,$$

so we do *not* reject H_0.

Step 3: the matching interval. The margin is $t_{0.025,15}\, S/\sqrt{n} = 2.131 \cdot 3.00 = 6.393$, giving

$$802 \pm 6.393 = (795.6,\ 808.4) \text{ g}.$$

Step 4: confirm duality. The claimed mean $\mu_0 = 800$ lies inside (795.6, 808.4), which by the proposition is exactly equivalent to retaining H_0 — the two procedures give the same verdict.

$$\boxed{|t_{\text{obs}}| = 0.667 < 2.131 \text{ and } 800 \in (795.6, 808.4) :\ \text{retain } H_0.}$$

7.4.8 Contrast: the z-test over-rejects when σ is estimated

Example 7.4.16 (The look-alike z-test flips the call)**.** A filling head should dispense $\mu_0 = 500$ g of detergent. A sample of $n = 10$ packages gives $\overline{X} = 503.0$ g and $S = 4.5$ g; the true σ is *not* known. A careless analyst standardizes with S but then compares the result to the normal cutoff $z_{0.025} = 1.96$ rather than the t cutoff $t_{0.025,9} = 2.262$. Run $H_0 : \mu = 500$ versus $H_1 : \mu \neq 500$ at $\alpha = 0.05$ both ways and show the verdicts disagree.

Which rule, and why? Because σ is unknown and replaced by S, the honest reference distribution is t_9, not $\mathrm{N}(0,1)$. Comparing to $z_{0.025}$ is the classic look-alike error; we compute both cutoffs against the same statistic to expose it.

Solution.

Step 1: the (common) test statistic. Both analysts compute the same number:

$$\text{stat} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{503.0 - 500}{4.5/\sqrt{10}} = \frac{3.0}{4.5/3.162} = \frac{3.0}{1.423} = 2.108.$$

Step 2: the wrong (anti-conservative) z call. Comparing to $z_{0.025} = 1.96$: since $|2.108| > 1.96$, the z-user *rejects* H_0 and declares the head off target.

Step 3: the correct t call. The proper cutoff is $t_{0.025,9} = 2.262 > 1.96$. Comparing the same 2.108 to 2.262: $|2.108| < 2.262$, so the t-test does *not* reject.

Step 4: why the t cutoff is larger. Estimating σ by S injects extra variability into the denominator, fattening the tails of the null distribution, so the threshold for "too extreme" must move outward from 1.96 to 2.262. The z-test ignores this and rejects too readily — its true Type I error rate exceeds the advertised 5%.

$$\boxed{\begin{array}{l} z \text{ says reject, } t \text{ says retain; the } t\text{-test is correct} \\ \text{because } \sigma \text{ was estimated and } t_{0.025,9} = 2.262 > 1.96 = z_{0.025}. \end{array}}$$

Remark. The gap between the cutoffs closes as the sample grows: $t_{0.025,9} = 2.262$, $t_{0.025,24} = 2.064$, $t_{0.025,60} = 2.000$, all approaching $z_{0.025} = 1.96$. For large n the z- and t-tests essentially agree, which is why z is a defensible large-sample approximation; but in small samples, as above, the wrong reference distribution can flip the decision. Whenever σ is estimated, the honest choice is t.

7.5 Inferences About a Normal Variance

The sampling-distribution result earned in this chapter — that $(n-1)S^2/\sigma^2$ follows a χ^2 distribution with $n-1$ degrees of freedom when the parent is normal — is the single engine behind every confidence interval and every test for a normal **variance** σ^2 or **standard deviation** σ. The mean inferences of the previous sections used t because S stood in for σ; here S^2 is the *object* of inference, not a nuisance, and the pivot is the scaled sample variance itself. The one feature that governs everything is that the χ^2 curve is **right-skewed**, not symmetric: a χ^2 confidence interval is *not* of the comfortable "estimate ± margin" form, and a two-sided test reads two *different* table values, one in each tail. Recognise these procedures by their target — a variance, a standard deviation, a spread, a precision, a tolerance — and by the genuine requirement that the population be normal, which here is not a convenience but a load-bearing assumption.

Which variance procedure does the problem need?

- Want a **range** for a normal **variance** σ^2? $\Rightarrow$ the χ^2 confidence interval $\left[(n-1)S^2/\chi^2_{\alpha/2,\,n-1},\ (n-1)S^2/\chi^2_{1-\alpha/2,\,n-1}\right]$ — asymmetric, because the divisors come from opposite tails.
- Want a range for a **standard deviation** σ? $\Rightarrow$ build the variance interval, then **square-root both endpoints**.
- Want only an **upper bound** on a variance ("how bad could the spread be?")? $\Rightarrow$ a one-sided confidence bound: put all of α in one tail.
- Want to **test** a claimed variance $\sigma^2 = \sigma_0^2$? $\Rightarrow$ the χ^2 test on $(n-1)S^2/\sigma_0^2$; two cutoffs if H_1 is "differs," one cutoff if H_1 is "increased" or "decreased."
- Claim phrased on a **standard deviation**? $\Rightarrow$ square it to a variance hypothesis; the statistic is unchanged.

Do not confuse this with a $\pm$ interval: because χ^2 is skewed, the point estimate S^2 does *not* sit at the centre of its interval, and the two endpoints are not mirror images about S^2. Never write $S^2 \pm$ margin for a variance.

7.5.1 The pivot and how it builds an interval

Theorem 7.5.1 (Distribution of the scaled sample variance)**.** *Let $X_1, \ldots, X_n$ be independent draws from a $\mathrm{N}(\mu, \sigma^2)$ population and let $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$. Then*

$$\frac{(n-1)S^2}{\sigma^2} \ \sim\ \chi^2_{n-1},$$

and this quantity is independent of $\overline{X}$.

Proof. Set $Z_i = (X_i - \mu)/\sigma$, so $\sum_i Z_i^2 \sim \chi^2_n$. The algebraic identity $\sum_i (X_i - \mu)^2 = \sum_i (X_i - \overline{X})^2 + n(\overline{X} - \mu)^2$, divided by σ^2, reads

$$\sum_i Z_i^2 = \frac{(n-1)S^2}{\sigma^2} + \left(\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}\right)^2.$$

The final term is the square of a standard normal, hence χ^2_1, and it is independent of $(n-1)S^2/\sigma^2$ because $\overline{X}$ and S^2 are independent for a normal sample. A χ^2_n variable equal to an independent χ^2_1 plus a remaining piece forces that remaining piece to be χ^2_{n-1}, by matching moment generating functions. Normality is essential; without it the identity does not deliver a χ^2. □

Remark. This is a genuine **pivot**: the quantity $(n-1)S^2/\sigma^2$ contains the unknown σ^2 yet has a distribution (χ^2_{n-1}) free of all unknowns. Trapping the pivot between two table values and solving for σ^2 is exactly how the confidence interval is born — the same move that turns $(\overline{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$ into a mean interval.

Definition 7.5.2 (Confidence interval for σ^2 and σ)**.** Write $\chi^2_{q,\,k}$ for the point with area q to its *right* under the χ^2_k curve, so $\mathbb{P}(\chi^2_k > \chi^2_{q,\,k}) = q$. Because $\mathbb{P}\big(\chi^2_{1-\alpha/2,\,n-1} < (n-1)S^2/\sigma^2 < \chi^2_{\alpha/2,\,n-1}\big) = 1 - \alpha$, inverting the two inequalities for σ^2 gives the $100(1-\alpha)\%$ interval

$$\left[\ \frac{(n-1)S^2}{\chi^2_{\alpha/2,\,n-1}}\ ,\ \frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\,n-1}}\ \right].$$

The *larger* divisor (the upper-tail point) produces the *lower* endpoint, and vice versa. For a standard deviation, take square roots of both endpoints:

$$\left[\ \sqrt{\frac{(n-1)S^2}{\chi^2_{\alpha/2,\,n-1}}}\ ,\ \sqrt{\frac{(n-1)S^2}{\chi^2_{1-\alpha/2,\,n-1}}}\ \right].$$

Remark. Inverting flips the inequalities, which is why the upper-tail critical value lands in the *lower* endpoint — a frequent sign of trouble for students who expect the bigger number on top. And because $\chi^2_{1-\alpha/2,\,n-1} \neq -\chi^2_{\alpha/2,\,n-1}$ (both are positive, and they are not mirror images), the interval is asymmetric: S^2 sits nearer the lower endpoint than the upper one.

7.5.2 Two-sided confidence intervals for a variance

Example 7.5.3 (Pipette calibration: an interval for the variance)**.** A lab technician checks a pipette by dispensing what should be a fixed volume and weighing the output. From $n = 15$ independent dispenses the sample variance of the delivered mass is $S^2 = 0.42\,\text{mg}^2$. Assuming the masses are normal, give a 95% confidence interval for the true variance σ^2.

Which rule, and why? We want a *range* for a normal *variance*, so the pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{14}$ gives a χ^2 interval — not a t interval, which would estimate the *mean* mass, and not an "$S^2 \pm$ margin," which the skew forbids.

Solution.

Step 1: critical values. With $n - 1 = 14$ df and $\alpha = 0.05$, the two tail points are

$$\begin{aligned} \text{upper } (\alpha/2 = 0.025): &\quad \chi^2_{0.025,\,14} = 26.119, \\ \text{lower } (1-\alpha/2 = 0.975): &\quad \chi^2_{0.975,\,14} = 5.629. \end{aligned}$$

Step 2: assemble the endpoints. The numerator is $(n-1)S^2 = 14 \cdot 0.42 = 5.88$. The upper-tail point makes the lower endpoint:

$$\begin{aligned} \text{lower endpoint} &= \frac{5.88}{26.119} = 0.2251, \\ \text{upper endpoint} &= \frac{5.88}{5.629} = 1.0446. \end{aligned}$$

Step 3: report and notice the asymmetry. The estimate $S^2 = 0.42$ is much closer to the lower endpoint 0.225 than to the upper endpoint 1.045 — the interval leans right, as a skewed pivot demands.

$$\boxed{0.225\,\text{mg}^2 \;\le\; \sigma^2 \;\le\; 1.045\,\text{mg}^2 \quad (95\%\text{ confidence}).}$$

Example 7.5.4 (The same data: an interval for the standard deviation)**.** For the pipette study above ($n = 15$, $S^2 = 0.42\,\text{mg}^2$, so $S = 0.648\,\text{mg}$), give a 95% confidence interval for the standard deviation σ.

Which rule, and why? A standard-deviation interval is just the variance interval with both endpoints square-rooted; the χ^2 machinery is identical, only the final transform changes. Do not re-derive a separate pivot for σ.

Solution.

Step 1: reuse the variance interval. From the previous example, $0.2251 \le \sigma^2 \le 1.0446$.

Step 2: square-root both endpoints.

$$\begin{aligned} \sqrt{0.2251} &= 0.4744, \\ \sqrt{1.0446} &= 1.0221. \end{aligned}$$

Step 3: report. Square-rooting preserves the asymmetry: $S = 0.648$ is again nearer the lower endpoint.

$$\boxed{0.474\,\text{mg} \;\le\; \sigma \;\le\; 1.022\,\text{mg} \quad (95\%\text{ confidence}).}$$

The interval is wide because n is small; halving its width would take roughly four times the data, since the spread of S shrinks like $1/\sqrt{n}$.

Example 7.5.5 (Concrete strength: a tighter variance interval at $n = 25$)**.** An engineer tests $n = 25$ concrete cylinders and records compressive strengths whose sample standard deviation is $S = 310$ psi, so $S^2 = 96{,}100\,\text{psi}^2$. Strengths are taken as normal. Construct a 90% confidence interval for σ^2.

Which rule, and why? Variance of a normal population, two-sided range $\Rightarrow$ χ^2 interval on 24 df. Here $\alpha = 0.10$, so the tails carry 0.05 each — a different pair of table points from the 95% case.

Solution.

Step 1: critical values. On 24 df with $\alpha/2 = 0.05$:

$$\chi^2_{0.05,\,24} = 36.415,$$
$$\chi^2_{0.95,\,24} = 13.848.$$

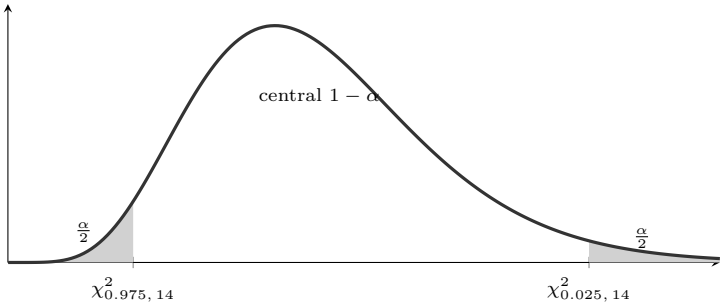
Step 2: endpoints. Here $(n-1)S^2 = 24 \cdot 96{,}100 = 2{,}306{,}400$, so

$$\text{lower} = \frac{2{,}306{,}400}{36.415} = 63{,}336,$$
$$\text{upper} = \frac{2{,}306{,}400}{13.848} = 166{,}551.$$

Step 3: report.

$$\boxed{63{,}336 \;\le\; \sigma^2 \;\le\; 166{,}551 \;\; (\text{psi}^2,\; 90\%\text{ confidence}).}$$

The larger n and lower confidence level both narrow the interval relative to the pipette study, yet it remains visibly asymmetric — the upper endpoint is about 2.6 times $S^2 = 96{,}100$ while the lower is only about 0.66 times it.



χ^2_{14} pivot value

The χ^2_{14} pivot for $n = 15$. The central region between the two tail points $\chi^2_{0.975,\,14} = 5.629$ and $\chi^2_{0.025,\,14} = 26.119$ holds probability $1 - \alpha = 0.95$. Because the curve is right-skewed, these cutoffs are not *symmetric about the peak (near $x = 12$), which is exactly why the resulting interval for σ^2 is asymmetric about S^2.*

7.5.3 One-sided confidence bounds

Often only one direction matters: a regulator cares how *large* a variance might be, while an engineer touting a precise process cares how *small* it might be. A one-sided bound spends the entire α in a single tail and so gives a tighter statement in the direction of interest.

Definition 7.5.6 (One-sided confidence bounds for σ^2)**.** A $100(1-\alpha)\%$ **upper confidence bound** for σ^2 is

$$\sigma^2 \;\le\; \frac{(n-1)S^2}{\chi^2_{1-\alpha,\,n-1}},$$

using the *lower*-tail point (area α to its left). A $100(1-\alpha)\%$ **lower confidence bound** is

$$\sigma^2 \;\ge\; \frac{(n-1)S^2}{\chi^2_{\alpha,\,n-1}},$$

using the upper-tail point. Square-root either bound for a statement about σ.

Example 7.5.7 (Drug-fill uniformity: an upper bound on the variance)**.** A pharmaceutical line must keep tablet-mass variability low. From $n = 20$ tablets the sample variance of mass is $S^2 = 1.85\,\text{mg}^2$ (masses assumed normal). Regulators want a 95% *upper* confidence bound on σ^2 — the worst-case spread consistent with the data.

Which rule, and why? Only an upper limit is wanted, so this is a one-sided upper confidence bound: all of $\alpha = 0.05$ goes in one tail, and the divisor is the *lower*-tail point $\chi^2_{0.95,\,19}$. A two-sided interval would waste half of α on a lower bound nobody asked for, giving a looser (larger) upper limit.

Solution.

Step 1: critical value. On 19 df with all $\alpha = 0.05$ in the left tail we need the point with area 0.95 to its right: $\chi^2_{0.95,\,19} = 10.117$.

Step 2: the bound. With $(n-1)S^2 = 19 \cdot 1.85 = 35.15$,

$$\sigma^2 \;\le\; \frac{35.15}{10.117} = 3.474\,\text{mg}^2.$$

Step 3: convert to a standard deviation if desired. $\sqrt{3.474} = 1.864$ mg.

$$\boxed{\sigma^2 \le 3.47\,\text{mg}^2 \quad (\text{equivalently } \sigma \le 1.86\,\text{mg}), \quad 95\%\text{ upper bound.}}$$

Note this single bound 3.47 is *below* the two-sided 95% upper endpoint we would get from the same data, because concentrating α in one tail tightens the limit on that side.

Example 7.5.8 (GPS receiver: a lower bound on the standard deviation)**.** A skeptic doubts a GPS unit is as precise as claimed and wants a 95% *lower* bound on its position standard deviation. From $n = 12$ fixes the sample standard deviation of the east-west error is $S = 2.4$ m, so $S^2 = 5.76\,\text{m}^2$ (errors assumed normal). Give the 95% lower bound on σ.

Which rule, and why? A lower bound on spread uses the *upper*-tail point $\chi^2_{0.05,\,11}$ as the divisor; this is the mirror of the upper-bound rule. We bound σ^2 first, then square-root.

Solution.

Step 1: critical value. On 11 df, all $\alpha = 0.05$ in the right tail: $\chi^2_{0.05,\,11} = 19.675$.

Step 2: lower bound on the variance. With $(n-1)S^2 = 11 \cdot 5.76 = 63.36$,

$$\sigma^2 \;\ge\; \frac{63.36}{19.675} = 3.220\,\text{m}^2.$$

Step 3: square-root. $\sqrt{3.220} = 1.794$ m.

$$\boxed{\sigma \;\ge\; 1.79\,\text{m} \quad (95\%\text{ lower bound}).}$$

So with 95% confidence the true positional standard deviation is at least 1.79 m — useful ammunition for the skeptic, since it rules out the very small σ a precise unit would need.

7.5.4 Two-sided tests of a variance

Definition 7.5.9 (The χ^2 test for $H_0\colon \sigma^2 = \sigma_0^2$)**.** For a normal population, the **test statistic** is

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} \;\sim\; \chi^2_{n-1} \quad \text{under } H_0.$$

At level α: for $H_1\colon \sigma^2 \neq \sigma_0^2$ reject if $\chi^2_{\text{obs}} < \chi^2_{1-\alpha/2,\,n-1}$ or $\chi^2_{\text{obs}} > \chi^2_{\alpha/2,\,n-1}$; for $H_1\colon \sigma^2 > \sigma_0^2$ reject if $\chi^2_{\text{obs}} > \chi^2_{\alpha,\,n-1}$; for $H_1\colon \sigma^2 < \sigma_0^2$ reject if $\chi^2_{\text{obs}} < \chi^2_{1-\alpha,\,n-1}$.

Example 7.5.10 (Thermostat scatter: does the variance differ from spec?)**.** A climate chamber is specified to hold temperature with variance $\sigma^2 = 0.25\,(^\circ\text{C})^2$ (standard deviation 0.5°C). Both excessive drift and suspiciously tight readings warrant a service call, so the question is two-sided. From $n = 18$ readings the sample variance is $S^2 = 0.46\,(^\circ\text{C})^2$. Test $H_0\colon \sigma^2 = 0.25$ against $H_1\colon \sigma^2 \neq 0.25$ at $\alpha = 0.05$.

Which rule, and why? The claim is about a *variance* of a normal population and H_1 is two-sided ("differs in either direction"), so this is the two-sided χ^2 test on 17 df, with two distinct cutoffs — not a t-test, which would test the mean temperature.

Solution.

Step 1: statistic. With $n-1=17$, $S^2=0.46$, $\sigma_0^2=0.25$,

$$\chi^2_{\text{obs}} = \frac{17\cdot 0.46}{0.25} = \frac{7.82}{0.25} = 31.28.$$

Step 2: critical values. Equal tails of 0.025 on 17 df:

$$\text{lower: } \chi^2_{0.975,\,17} = 7.564,$$
$$\text{upper: } \chi^2_{0.025,\,17} = 30.191.$$

Step 3: decision. The acceptance region is (7.564, 30.191). Since $31.28 > 30.191$, the statistic clears the upper cutoff.

$$\boxed{\chi^2_{\text{obs}} = 31.28 > 30.191 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

The chamber's variance is significantly larger than spec; schedule the service call.

Example 7.5.11 (CONTRAST: the variance interval is NOT estimate ± margin)**.** Reuse the thermostat data ($n=18$, $S^2=0.46$, 17 df). A student wants a 95% interval for σ^2 and reasons by analogy with the mean: "the point estimate is $S^2=0.46$; the standard error of S^2 is about $S^2\sqrt{2/(n-1)}$, so the interval is $0.46 \pm 1.96\cdot 0.46\sqrt{2/17}$." Critique this and give the correct interval.

Which rule, and why? This imports the symmetric "estimate $\pm z\cdot$ SE" template from mean inference, but S^2 has a *skewed* (scaled χ^2) sampling distribution, so no symmetric ± interval is valid; the honest interval comes from inverting the χ^2 pivot.

Solution.

Step 1: see what the wrong shortcut produces. The bogus margin is $1.96\cdot 0.46\sqrt{2/17} = 1.96\cdot 0.46\cdot 0.3430 = 0.3093$, giving the symmetric interval $0.46 \pm 0.309 = [0.151,\, 0.769]$ — centred on 0.46 by construction.

Step 2: build the correct asymmetric interval. With $(n-1)S^2 = 17\cdot 0.46 = 7.82$ and the tail points $\chi^2_{0.025,\,17} = 30.191$, $\chi^2_{0.975,\,17} = 7.564$,

$$\text{lower} = \frac{7.82}{30.191} = 0.2590,$$
$$\text{upper} = \frac{7.82}{7.564} = 1.0338.$$

Step 3: compare. The correct interval [0.259, 1.034] is wider on the right and does *not* place $S^2=0.46$ at its centre (its midpoint is 0.646). The ± shortcut both mis-centres the interval and understates the upper reach by more than a factor of 1.3, exactly the kind of error the skew creates.

$$\boxed{0.259 \le \sigma^2 \le 1.034 \;\; (95\%); \quad \text{never } S^2 \pm \text{margin}.}$$

7.5.5 One-sided tests: variability increased or precision improved

Example 7.5.12 (Bottling: has the variability increased?)**.** A filling process is supposed to run with standard deviation $\sigma_0 = 1.5$ mL, so $\sigma_0^2 = 2.25$. After a valve wears, an operator suspects the fills are scattering *more*. A sample of $n=16$ bottles gives $S = 2.1$ mL, hence $S^2 = 4.41$. Test $H_0\colon \sigma^2 = 2.25$ against $H_1\colon \sigma^2 > 2.25$ at $\alpha = 0.05$.

Which rule, and why? "Scattering *more*" fixes a one-sided upper-tail alternative on the variance, so all of α goes in the right tail of χ^2_{15}. A two-sided test would squander half of α on a decrease nobody is claiming.

Solution.

Step 1: statistic. With $n-1=15$,

$$\chi^2_{\text{obs}} = \frac{15 \cdot 4.41}{2.25} = \frac{66.15}{2.25} = 29.40.$$

Step 2: critical value. All of $\alpha = 0.05$ in the upper tail: $\chi^2_{0.05,\,15} = 24.996$.

Step 3: decision. Reject if $\chi^2_{\text{obs}} > 24.996$. Here $29.40 > 24.996$.

$$\boxed{\chi^2_{\text{obs}} = 29.40 > 24.996 \;\Rightarrow\; \text{reject } H_0;\ \text{variability has significantly increased.}}$$

Replace the valve: the spread is genuinely larger, not chance.

Example 7.5.13 (Machining precision: did the new tooling improve it?)**.** A historical milling line holds a slot width with variance $\sigma_0^2 = 0.0009\,\text{mm}^2$ ($\sigma_0 = 0.03$ mm). New tooling is hoped to *tighten* the spread. A sample of $n = 21$ slots gives $S^2 = 0.00045\,\text{mm}^2$. Test $H_0\colon \sigma^2 = 0.0009$ against $H_1\colon \sigma^2 < 0.0009$ at $\alpha = 0.05$.

Which rule, and why? "Improve precision" means *decrease* the variance, a lower-tail one-sided test on χ^2_{20}: the whole α sits in the left tail, where a too-small statistic flags an unusually small sample variance.

Solution.

Step 1: statistic. With $n-1=20$,

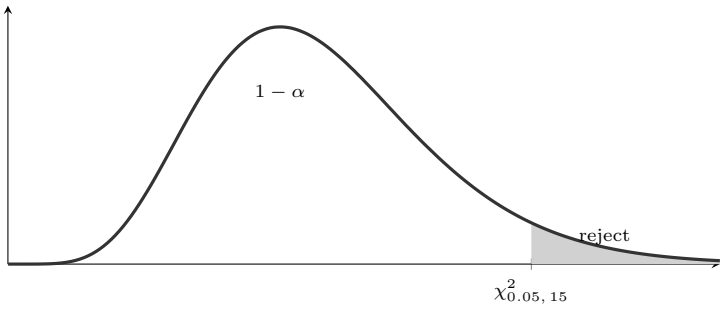
$$\chi^2_{\text{obs}} = \frac{20 \cdot 0.00045}{0.0009} = \frac{0.009}{0.0009} = 10.00.$$

Step 2: critical value. For a lower-tail test we need the point with area $\alpha = 0.05$ to its *left*, i.e. area 0.95 to its right: $\chi^2_{0.95,\,20} = 10.851$.

Step 3: decision. Reject if $\chi^2_{\text{obs}} < 10.851$. Here $10.00 < 10.851$.

$$\boxed{\chi^2_{\text{obs}} = 10.00 < 10.851 \;\Rightarrow\; \text{reject } H_0;\ \text{precision significantly improved.}}$$

The new tooling delivers a statistically significant reduction in spread.



χ^2_{15} statistic

Upper one-sided rejection region for $n = 16$ ($df = 15$), $\alpha = 0.05$: the entire α sits in the right tail beyond $\chi^2_{0.05,\,15} = 24.996$. A claim that variance has increased *puts the whole rejection region in this single tail; for a "decreased" claim the shaded region would instead sit in the left tail.*

7.5.6 The p-value for a variance test

Proposition 7.5.14 (p-value rules)**.** *With observed statistic $c = \chi^2_{obs}$ on $k = n - 1$ df:*

$$\begin{aligned} H_1 \colon \sigma^2 > \sigma_0^2 \colon &\quad p = \mathbb{P}(\chi_k^2 > c), \\ H_1 \colon \sigma^2 < \sigma_0^2 \colon &\quad p = \mathbb{P}(\chi_k^2 < c), \\ H_1 \colon \sigma^2 \neq \sigma_0^2 \colon &\quad p = 2\min\{\mathbb{P}(\chi_k^2 < c),\, \mathbb{P}(\chi_k^2 > c)\}. \end{aligned}$$

Remark. For a one-sided test only deviations in the direction of H_1 count, so the p-value is a single tail — do not double it. The two-sided convention doubles the *nearer* (smaller) tail because the skew makes a literal mirror-area definition awkward; doubling the nearer tail is the standard, table-friendly choice.

Example 7.5.15 (p-value for an upper-tail variance test)**.** Return to the bottling test: $n = 16$ (df $= 15$), $\chi^2_{\text{obs}} = 29.40$, upper-tail $H_1 \colon \sigma^2 > 2.25$. Report the p-value and decide at $\alpha = 0.05$.

Which rule, and why? One-sided upper-tail alternative, so the p-value is the single right-tail area $\mathbb{P}(\chi^2_{15} > 29.40)$ — do *not* double it.

Solution.

Step 1: bracket the statistic with table points. On 15 df, $\chi^2_{0.025,\,15} = 27.488$ and $\chi^2_{0.01,\,15} = 30.578$. Our value 29.40 sits between them, so the right-tail area is between 0.01 and 0.025.

Step 2: interpolate and report. Linear interpolation gives $\mathbb{P}(\chi^2_{15} > 29.40) \approx 0.014$.

$$\boxed{p\text{-value} = \mathbb{P}(\chi^2_{15} > 29.40) \approx 0.014 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

This agrees with the fixed-level decision earlier ($29.40 > 24.996$ rejected at $\alpha = 0.05$), and the small p shows the evidence is fairly strong, not borderline.

Example 7.5.16 (p-value for a two-sided variance test)**.** Return to the thermostat test: $n = 18$ (df $= 17$), $\chi^2_{\text{obs}} = 31.28$, two-sided $H_1 \colon \sigma^2 \neq 0.25$. Report the p-value.

Which rule, and why? Two-sided alternative on a skewed statistic, so find the nearer tail and double it; the statistic is large, so the right tail is the smaller one.

Solution.

Step 1: which tail is nearer. On 17 df, $\chi^2_{0.025,\,17} = 30.191$ and $\chi^2_{0.01,\,17} = 33.409$. Since 31.28 sits just above 30.191, the right-tail area is just below 0.025, say $\mathbb{P}(\chi^2_{17} > 31.28) \approx 0.019$.

Step 2: double the smaller tail.

$$\mathbb{P}(\chi^2_{17} > 31.28) \approx 0.019,$$
$$p\text{-value} = 2 \times 0.019 = 0.038.$$

Step 3: decide.

$$\boxed{p\text{-value} \approx 0.038 < 0.05 \;\Rightarrow\; \text{reject at } \alpha = 0.05,}$$

consistent with the earlier critical-value decision, where 31.28 cleared 30.191.

7.5.7 Testing a standard deviation by squaring

Example 7.5.17 (Lab assay: a two-sided claim on a standard deviation)**.** A reference assay is supposed to have measurement standard deviation $\sigma_0 = 0.40$ units; both sloppier and suspiciously tighter scatter warrant investigation. A validation run of $n = 14$ controls gives $S = 0.58$ units. Test $H_0 \colon \sigma = 0.40$ against $H_1 \colon \sigma \neq 0.40$ at $\alpha = 0.05$.

Which rule, and why? The hypothesis is phrased on σ, but the χ^2 test is built on the *variance*; square the hypothesized value to $\sigma_0^2 = 0.16$ and run the two-sided test on 13 df with two distinct cutoffs.

Solution.

Step 1: convert to a variance hypothesis. $H_0: \sigma = 0.40 \Leftrightarrow \sigma^2 = 0.16$; $H_1: \sigma \neq 0.40 \Leftrightarrow \sigma^2 \neq 0.16$. Sample: $S = 0.58 \Rightarrow S^2 = 0.3364$.

Step 2: statistic. With $n - 1 = 13$,

$$\chi^2_{\text{obs}} = \frac{13 \cdot 0.3364}{0.16} = \frac{4.3732}{0.16} = 27.33.$$

Step 3: critical values and decision. Equal tails of 0.025 on 13 df: $\chi^2_{0.975,\,13} = 5.009$ and $\chi^2_{0.025,\,13} = 24.736$. The acceptance region is (5.009, 24.736), and $27.33 > 24.736$.

$$\boxed{\chi^2_{\text{obs}} = 27.33 > 24.736 \;\Rightarrow\; \text{reject } H_0;\ \sigma \text{ differs significantly from 0.40.}}$$

The assay's spread is significantly larger than the claimed 0.40 units.

7.5.8 When the procedure breaks

Remark. Every interval and test in this section assumes a **normal** parent, and that assumption is not negotiable here. The t procedures for a mean lean on the central limit theorem and tolerate mild non-normality; the χ^2 variance procedures have *no such cushion.* The distribution of $(n-1)S^2/\sigma^2$ is exactly χ^2_{n-1} only when the data are normal, and it is governed by the parent's fourth moment (kurtosis). Heavy tails or skew can push the true sampling distribution of S^2 far from χ^2_{n-1}, so the actual coverage of a "95%" interval, or the true Type I error of an "$\alpha = 0.05$" test, can be badly wrong. Always check normality before trusting a χ^2 variance conclusion.

7.6 Robustness and Checking Normality

Every procedure in this chapter — the t interval and test for a mean, the χ^2 interval and test for a variance — was derived *assuming the sample comes from a normal population.* Real data are never exactly normal, so the honest question is not "is the population normal?" (it isn't) but "how much does it matter?" That sensitivity is called **robustness**: a procedure is robust if its actual confidence level or Type I error rate stays close to the nominal value even when the normality assumption is violated. The headline of this section is a sharp split. The t procedures for the **mean** are reasonably robust, because the central limit theorem pulls $\overline{X}$ toward normality as n grows, so for moderate-to-large samples mild skew or modest tail-weight barely disturbs the stated coverage; they wobble only for small n with strong skew or outliers. The χ^2 procedures for the **variance** are *not* robust at all: the distribution of $(n-1)S^2/\sigma^2$ is driven by the population's fourth moment (kurtosis), the central limit theorem does not rescue it, and non-normality corrupts the error rate no matter how large n is. Recognize a robustness question by its phrasing — "is the conclusion trustworthy," "the data look skewed," "the Q-Q plot bends" — and answer it by checking the assumption with a picture before, not after, you trust the number.

Is the normal-theory procedure safe to use here?

- Inference about a **mean** with n **moderate-to-large** (say $n \geq 30$) and no wild outliers? $\Rightarrow$ the t **procedure is robust**; the central limit theorem keeps coverage near nominal even under mild non-normality.
- Inference about a **mean** with **small** n and a **strongly skewed** or **outlier-laden** sample? $\Rightarrow$ the t procedure is **suspect**; transform, enlarge the sample, or use a **nonparametric** test (Chapter 14).
- Inference about a **variance** or standard deviation? $\Rightarrow$ the χ^2 **procedure is fragile** for *any* n; normality is a genuine requirement, not a convenience, so a plot or formal normality check is mandatory first.
- Need to **check** normality? $\Rightarrow$ look at a **histogram** (shape), a **boxplot** (skew, outliers), and a **normal Q-Q plot** (straight = normal, curved = skew, S-shaped = heavy/light tails).
- Normality clearly fails? $\Rightarrow$ try a **transformation** (log, square root) to restore symmetry, collect **more data** so the CLT can act on $\overline{X}$, or switch to a **distribution-free** method (sign test, signed-rank test; Chapter 14).

Do not confuse the two diagnoses: a large n excuses non-normality for a *mean* test but *not* for a *variance* test. "n is big, so I'm fine" is correct reasoning for t and dangerously wrong for χ^2.

7.6.1 Why the mean test is robust and the variance test is not

Theorem 7.6.1 (CLT cushion for the mean)**.** *Let $X_1, \ldots, X_n$ be independent draws from a population with mean μ and finite variance σ^2, not necessarily normal. Then as $n \to \infty$,*

$$\frac{\overline{X} - \mu}{S/\sqrt{n}} \xrightarrow{d} \mathrm{N}(0,1).$$

Consequently the one-sample t statistic, whose exact reference distribution is t_{n-1} under normality, has approximately the right distribution for large n whatever *the parent shape, so the t interval and test keep their nominal coverage and level asymptotically.*

Proof. By the central limit theorem $\sqrt{n}(\overline{X} - \mu)/\sigma \to \mathrm{N}(0,1)$, and $S \to \sigma$ in probability by the law of large numbers. Slutsky's theorem combines these: $(\overline{X} - \mu)/(S/\sqrt{n}) = \big[\sqrt{n}(\overline{X} - \mu)/\sigma\big] \cdot \big[\sigma/S\big] \to \mathrm{N}(0,1) \cdot 1$. Since $t_{n-1} \to \mathrm{N}(0,1)$ too, using the t table introduces only a vanishing error. The quality of the approximation at finite n depends on how fast the CLT acts, which is governed by the parent's skewness — hence robustness degrades for small n and heavy skew. □

Remark. The variance test has no analogue of this theorem. The exact statement $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ uses normality *twice*: to make $\overline{X}$ and S^2 independent and to give S^2 its χ^2 shape. The relevant limit law instead involves the parent's **kurtosis** κ: for large n,

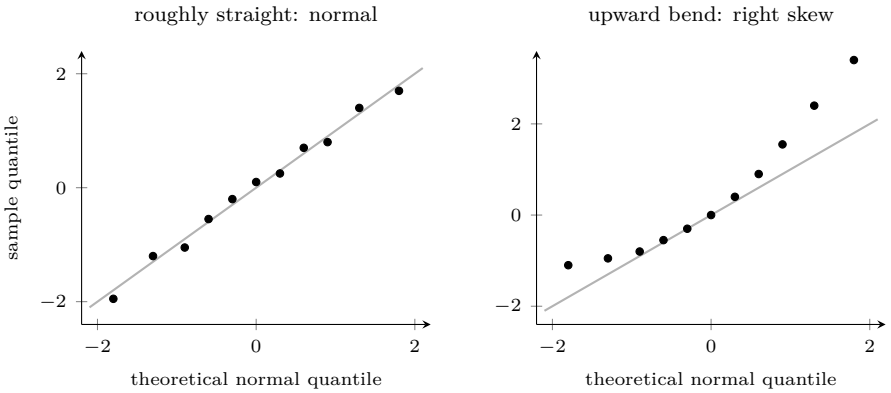
$$\frac{S^2 - \sigma^2}{\sigma^2} \approx \mathrm{N}\Big(0, \tfrac{\kappa - 1}{n}\Big),$$

where $\kappa = 3$ for a normal parent. A heavy-tailed parent has $\kappa > 3$, so S^2 is far more variable than χ^2_{n-1} predicts, and the true tail probabilities of $(n-1)S^2/\sigma^2$ do not match the χ^2 table. Increasing n shrinks the variance of S^2 but does *not* fix the mismatch in shape: the kurtosis term stays, so the error rate stays wrong.

7.6.2 Reading the diagnostic pictures

Definition 7.6.2 (Normal quantile-quantile (Q-Q) plot)**.** Order the sample as $x_{(1)} \le \cdots \le x_{(n)}$. The **normal Q-Q plot** graphs each ordered value $x_{(i)}$ (vertical) against the corresponding theoretical normal quantile $\Phi^{-1}\big((i - 0.5)/n\big)$ (horizontal). If the data are normal the points fall near a straight line. Systematic departures read as shapes:

- **Roughly straight**: consistent with normality.
- **Convex / concave bend** (a single curve): **skew** — a right-skewed sample curves up at the right end (large values too large), a left-skewed sample curves down.
- **S-shape**: **tail problems** — an S that steepens at both ends signals **heavy tails** (more extreme values than normal); a flattening S signals **light tails**.
- **Isolated points** off the line at an end: **outliers**.



Two normal Q-Q plots against the reference line $y = x$. Left: points hug the line, consistent with normality. Right: the points bow above the line at the upper end and bunch at the lower end — the convex curvature is the signature of a right-skewed sample, for which a small-n t procedure is unreliable.

Remark. A histogram shows the gross shape (one hump, symmetric or skewed, gaps), and a boxplot summarizes the median, the quartile box, and flags points beyond the whiskers as candidate outliers; both are quick but coarse. The Q-Q plot is the most sensitive of the three for normality specifically, because it compares *every* order statistic to its normal expectation, so it catches tail problems a histogram smooths over. Use the histogram or boxplot to spot skew and outliers at a glance, then the Q-Q plot to judge whether the deviation is serious enough to matter.

7.6.3 Judging whether a mean procedure is trustworthy

Example 7.6.3 (Large sample, mild skew: trust the t interval)**.** A logistics analyst times $n = 64$ delivery runs (minutes). The histogram is mildly right-skewed (a few slow runs), the boxplot flags two high points just past the upper whisker, and the Q-Q plot is nearly straight with a slight upward bend at the top. The sample gives $\overline{X} = 42.0$ min, $S = 9.6$ min. Is a 95% t interval for the mean delivery time trustworthy?

Which rule, and why? The target is a *mean*, $n = 64$ is large, and the non-normality is mild (slight skew, no extreme outliers). By the CLT cushion the t procedure for a mean is *robust* here, so the nominal 95% coverage is essentially honest. This is a mean question, not a variance question, so the fragility of χ^2 is irrelevant.

Solution.

Step 1: diagnose severity. Mild right skew with $n = 64$: the standardized mean $(\overline{X} - \mu)/(S/\sqrt{n})$ is close to normal by the CLT, and the two mild high points do not dominate a sample of 64. The deviation is the harmless kind.

Step 2: build the interval anyway. With $t_{0.025,\,63} \approx 2.00$,

$$\text{margin} = t_{0.025,\,63}\,\frac{S}{\sqrt{n}} = 2.00 \cdot \frac{9.6}{\sqrt{64}} = 2.00 \cdot \frac{9.6}{8} = 2.40,$$
$$\text{CI} = 42.0 \pm 2.40 = (39.6,\ 44.4).$$

Step 3: verdict. The true coverage is very close to 95%.

Trust the t interval (39.6, 44.4) min: large n, mild skew $\Rightarrow$ t is robust.

Example 7.6.4 (Small skewed sample: the t interval is suspect)**.** A field ecologist measures the lead concentration (ppm) at $n = 8$ contaminated sites and gets

$$3,\ 4,\ 4,\ 5,\ 6,\ 8,\ 12,\ 30.$$

The value 30 is far from the rest; the boxplot flags it well past the whisker and the Q-Q plot bends sharply upward at the top. Should a 95% t interval for the mean concentration be reported as is?

Which rule, and why? Again a *mean*, but now $n = 8$ is small *and* the sample is strongly right-skewed with a glaring outlier. The CLT has too little to work with, so the t procedure is *not* robust here — the recognition cue is small n plus visible skew/outlier, which is exactly the regime where t fails.

Solution.

Step 1: see the damage. The sum is 72, so $\overline{X} = 72/8 = 9.0$, but the median is only $(5 + 6)/2 = 5.5$. The mean is dragged far above the bulk of the data by the single 30; an interval centered at 9.0 describes neither a typical site nor the center of the data.

Step 2: quantify the leverage. Dropping the 30 gives $\overline{X} = (72 - 30)/7 = 42/7 = 6.0$ — one point moved the mean by 3 ppm. A statistic that one observation can swing that far cannot have stable t behavior at $n = 8$.

Step 3: prescribe the fix. Do not report the raw t interval. Either transform (the data are positive and right-skewed — try logs, see below), or use a distribution-free interval based on the median (Chapter 14), which the outlier barely moves.

Small $n +$ skew/outlier $\Rightarrow$ t interval untrustworthy; transform or go nonparametric.

Example 7.6.5 (Reading a Q-Q plot to clear a t test)**.** A psychologist runs a paired-difference study with $n = 40$ reaction-time differences and wants a one-sample t-test of $H_0\colon \mu_D = 0$. The Q-Q plot of the 40 differences is close to the reference line — no bend, no S, one point a hair off at each end. The summary is $\overline{D} = 18$ ms, $S_D = 50$ ms. Is the t-test appropriate, and what does it conclude at $\alpha = 0.05$?

Which rule, and why? A roughly straight Q-Q plot with $n = 40$ is the textbook green light for a mean-based t procedure: the data are close enough to normal and the sample is large enough that the CLT cushion is ample. The straight plot is the recognition cue that no remedy is needed.

Solution.

Step 1: pass the diagnostic. Straight Q-Q plot, $n = 40$, no influential outlier $\Rightarrow$ the t reference distribution is trustworthy.

Step 2: run the test.

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{18}{50/\sqrt{40}} = \frac{18}{50/6.325} = \frac{18}{7.906} = 2.28,$$

$$t_{0.025,\,39} \approx 2.02.$$

Step 3: decide. Two-sided: $|2.28| > 2.02$, so reject.

$t_{\text{obs}} = 2.28 > 2.02 \Rightarrow$ reject H_0; the t-test is valid (straight Q-Q, $n = 40$).

Example 7.6.6 (Heavy tails flagged by an S-shaped Q-Q plot)**.** An analyst has $n = 18$ daily percentage returns of a stock. The Q-Q plot shows a clear S-shape: the points dip below the line at the left and rise above it at the right, steepening at both extremes. The histogram looks symmetric and bell-ish in the middle. Is a t interval for the mean daily return safe, and is a χ^2 interval for the variance safe?

Which rule, and why? An S that steepens at both ends means **heavy tails** (symmetric, but more extreme values than normal). For the *mean*, symmetry plus $n = 18$ keeps the t roughly valid; for the *variance*, heavy tails (high kurtosis) are exactly what wrecks the χ^2 test — so the same plot gives opposite verdicts for the two parameters.

Solution.

Step 1: interpret the S. Heavy symmetric tails: kurtosis $\kappa > 3$. The mean test relies on the CLT and symmetry helps it, so at $n = 18$ the t interval's coverage is only mildly off.

Step 2: the variance is a different story. The χ^2 statistic's spread is governed by κ. With $\kappa > 3$, $(n-1)S^2/\sigma^2$ is far more variable than χ^2_{17}, so the χ^2 interval's true coverage is well below the nominal level — it is too short and too confident.

Step 3: split verdict.

t interval for the mean: roughly OK. χ^2 interval for the variance: not trustworthy.

7.6.4 When the variance test cannot be saved by sample size

Example 7.6.7 (Back-of-envelope: heavy tails inflate the variance-test error rate)**.** A reliability engineer plans a χ^2 test of $H_0\colon \sigma^2 = \sigma_0^2$ at nominal $\alpha = 0.05$ with $n = 51$, but the lifetimes are heavy-tailed with kurtosis $\kappa \approx 9$ (normal has $\kappa = 3$). Argue without simulation that the true rejection rate under H_0 is nowhere near 5%, and that taking n larger does not fix it.

Which rule, and why? This is the robustness question for a *variance* test. The normal theory sets the spread of $(n-1)S^2/\sigma^2$ using $\kappa = 3$; with $\kappa = 9$ the real spread is far larger, so the nominal critical values cut off the wrong tail area. The cue is "variance test" plus "heavy tails," which together flag non-robustness.

Solution.

Step 1: the variance of S^2 under each model. Using $\mathrm{Var}(S^2) \approx \sigma^4(\kappa - 1)/n$, compare normal vs heavy-tailed at $n = 51$:

$$\text{normal } (\kappa = 3): \quad \mathrm{Var}(S^2) \approx \sigma^4 \frac{3-1}{51} = \sigma^4 \frac{2}{51} = 0.0392\,\sigma^4,$$

$$\text{heavy } (\kappa = 9): \quad \mathrm{Var}(S^2) \approx \sigma^4 \frac{9-1}{51} = \sigma^4 \frac{8}{51} = 0.1569\,\sigma^4.$$

Step 2: read off the distortion. The actual standard deviation of S^2 is $\sqrt{0.1569/0.0392} = \sqrt{4} = 2$ times what the χ^2_{50} reference assumes. Cutoffs chosen to leave 2.5% in each tail of a curve with the *normal* spread will, against a curve twice as wide, leave far more than 2.5% outside — the true two-sided rejection rate balloons well above 5%.

Step 3: why n cannot rescue it. The ratio of the two variances is $(\kappa - 1)/(3 - 1) = 8/2 = 4$ for *every* n — the n cancels. Enlarging n shrinks both variances together but never closes the gap in shape, so the χ^2 test stays miscalibrated at any sample size.

Heavy tails make the χ^2 variance test mis-leveled by a fixed factor independent of n.

Example 7.6.8 (Contrast on one data set: robust mean CI vs fragile variance CI)**.** A water-quality lab has $n = 30$ nitrate readings from a non-normal (right-skewed, occasional spike) source, with $\overline{X} = 8.0$ mg/L and $S = 3.0$ mg/L. The lab wants both a 95% CI for the mean nitrate level and a 95% CI for the variance. Which of the two intervals can the lab trust?

Which rule, and why? Same data, two different parameters, two different robustness verdicts. The *mean* CI rides the CLT and survives moderate skew at $n = 30$; the *variance* CI inherits the parent's skew/kurtosis directly and does not. Recognizing that one number is safe and the other is not is the whole point of the contrast.

Solution.

Step 1: mean CI (robust). With $t_{0.025,\,29} = 2.045$,

$$\text{margin} = 2.045 \cdot \frac{3.0}{\sqrt{30}} = 2.045 \cdot \frac{3.0}{5.477} = 2.045 \cdot 0.5477 = 1.12,$$

$$\text{mean CI} = 8.0 \pm 1.12 = (6.88,\ 9.12) \quad (\text{coverage } \approx 95\%).$$

Step 2: variance CI (fragile). The formula gives, with $\chi^2_{0.975,\,29} = 16.047$ and $\chi^2_{0.025,\,29} = 45.722$,

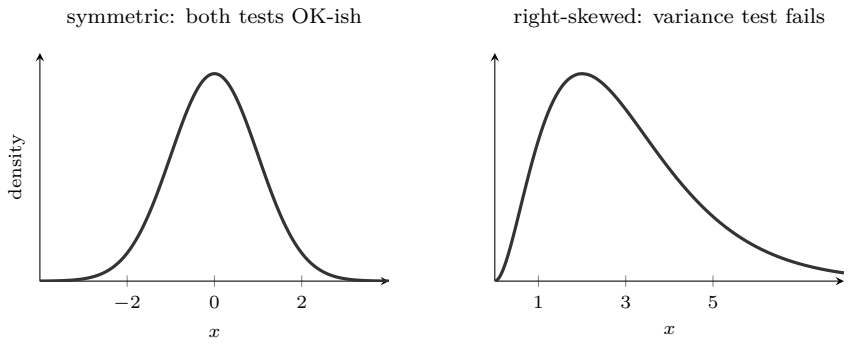
$$\text{lower} = \frac{(n-1)S^2}{\chi^2_{0.025,\,29}} = \frac{29 \cdot 9.0}{45.722} = \frac{261}{45.722} = 5.71,$$

$$\text{upper} = \frac{(n-1)S^2}{\chi^2_{0.975,\,29}} = \frac{29 \cdot 9.0}{16.047} = \frac{261}{16.047} = 16.26,$$

so the nominal variance CI is (5.71, 16.26) — but its *true* coverage is unknown and likely far from 95% because the source is skewed.

Step 3: verdict. Report the mean CI; flag the variance CI as unreliable and check normality (or transform) before quoting it.

Trust (6.88, 9.12) for the mean; distrust (5.71, 16.26) for the variance.



Left: a symmetric bell — the t test for the mean keeps its level, and a variance test is only mildly disturbed. Right: a strongly right-skewed density — the mean test survives for moderate n via the central limit theorem, but the χ^2 variance test is driven by the heavy upper tail and its true error rate departs badly from nominal even for large n.

7.6.5 Fixing non-normality: transform, enlarge, or go nonparametric

Example 7.6.9 (A log transform restores approximate normality)**.** An environmental scientist has $n = 7$ measured pollutant masses (kg) that are strongly right-skewed:

$$2,\ 3,\ 5,\ 8,\ 20,\ 50,\ 150.$$

A t interval on the raw values is untrustworthy (small n, severe skew). Show that taking natural logs roughly symmetrizes the data, then build a 95% t interval on the log scale.

Which rule, and why? Positive, multiplicatively-spread data with a long right tail are the classic candidates for a **log transform**: if $\log X$ is approximately normal then the t procedure is valid *on the log scale*. The cue is right skew over several orders of magnitude, which a log compresses into near-symmetry.

Solution.

Step 1: transform and check symmetry. Take $y = \ln x$:

$$\ln 2 = 0.69,\ \ln 3 = 1.10,\ \ln 5 = 1.61,\ \ln 8 = 2.08,$$
$$\ln 20 = 3.00,\ \ln 50 = 3.91,\ \ln 150 = 5.01.$$

The gaps 0.41, 0.51, 0.47, 0.92, 0.91, 1.10 are now comparable in size rather than exploding, and the values are roughly balanced about their middle (2.08) — the long right tail is gone, so a Q-Q plot of the y's would be far straighter.

Step 2: summarize the log data. Sum $= 0.69 + 1.10 + 1.61 + 2.08 + 3.00 + 3.91 + 5.01 = 17.40$, so

$$\overline{Y} = \frac{17.40}{7} = 2.486.$$

The deviations give $\sum(y - \overline{Y})^2 \approx 13.39$, so $S_y^2 = 13.39/6 = 2.232$, $S_y = 1.494$.

Step 3: t interval on the log scale. With $t_{0.025,\,6} = 2.447$,

$$\text{margin} = 2.447 \cdot \frac{1.494}{\sqrt{7}} = 2.447 \cdot \frac{1.494}{2.646} = 2.447 \cdot 0.5646 = 1.382,$$
$$\text{CI for } \mu_{\ln X} = 2.486 \pm 1.382 = (1.104,\ 3.868).$$

Step 4: interpret (optional back-transform). Exponentiating the endpoints, $(e^{1.104}, e^{3.868}) = (3.02, 47.9)$, a 95% interval for the *median* pollutant mass on the original scale.

Log-scale 95% t CI: (1.10, 3.87); back-transformed median CI: (3.0, 47.9) kg.

Example 7.6.10 (Choosing the remedy: transform vs nonparametric vs more data)**.** Three small studies each show non-normality. (a) Survival times (positive, right-skewed over two orders of magnitude), $n = 9$. (b) A symmetric but heavy-tailed sensor-error sample with two extreme outliers, $n = 11$. (c) A mildly skewed quality sample, $n = 12$, where the team can cheaply collect more units. For each, name the most appropriate fix.

Which rule, and why? The remedy follows the *kind* of non-normality. A monotone right skew over orders of magnitude ⇒ transform; heavy tails/outliers that no transform tames ⇒ nonparametric; mild skew with easy data collection ⇒ enlarge n and let the CLT act. Matching symptom to remedy is the recognition skill here.

Solution.

Step (a): transform. Positive and multiplicatively skewed is the log-transform signature; ln(time) should be roughly normal, then run the ordinary t procedure on the logs (as in the previous example).

Step (b): nonparametric. A log will not symmetrize already-symmetric data, and outliers violate normality in the tails; use a distribution-free method (Wilcoxon signed-rank or sign test, Chapter 14) that depends on ranks, not on the normal model, so outliers carry little weight.

Step (c): enlarge n. Mild skew is exactly the case the CLT cures; collecting enough extra units to reach, say, $n \geq 30$ makes the t procedure on the raw data trustworthy with no transform at all.

(a) log-transform (b) nonparametric (Ch. 14) (c) collect more data.

Example 7.6.11 (Contrast: a transform helps the mean test but cannot rescue a variance claim)**.** Reusing the logged pollutant data above, a manager now asks a *variance* question: "test whether the variance of the raw masses equals $\sigma_0^2 = 2500$ kg^2." A colleague proposes: "we already logged the data to fix normality, so just run the χ^2 variance test on the logs." Explain why this does not answer the manager's question.

Which rule, and why? A transform changes *which parameter* you are testing. The χ^2 test on the logs is a valid test about $\mathrm{Var}(\ln X)$, but the manager asked about $\mathrm{Var}(X)$ on the raw scale — and variance, unlike a quantile or the center, does *not* transform back cleanly. The cue is that "fix normality by transforming" silently swaps the variance you are inferring about.

Solution.

Step 1: the transform moved the target. The t interval back-transformed to a statement about the *median* of X because medians are preserved by monotone transforms. Variances are not: $\mathrm{Var}(\ln X) \neq g(\mathrm{Var}(X))$ for any fixed function g, so a χ^2 result on the logs says nothing direct about $\sigma_X^2 = 2500$.

Step 2: why you cannot just test the raw variance instead. Running χ^2 on the raw masses *is* about σ_X^2, but the raw data are severely non-normal and the variance test is not robust, so that test is invalid — exactly the problem the log was meant to dodge for the mean.

Step 3: the honest options. Either reframe the question on the log (variance/spread of $\ln X$), accept a much larger sample with a normality check, or use a distribution-free spread procedure (Chapter 14). There is no free lunch that delivers a trustworthy raw-scale variance test from skewed data.

A transform can validate the *mean* test yet leaves the raw-scale *variance* claim unanswerable.

Remark. The practical workflow for any normal-theory inference is: (1) plot first — histogram, boxplot, and especially the normal Q-Q plot; (2) if the target is a *mean* and n is moderate-to-large with no severe skew or outliers, proceed with t; (3) if the target is a *variance*, demand near-normality and treat any visible departure as disqualifying; (4) when normality fails, pick the remedy that matches the symptom — transform for monotone skew, more data for mild skew on a mean, and a nonparametric method (Chapter 14) for stubborn tails, outliers, or any small-sample variance question. "Compute first, check never" is how a single outlier becomes a published conclusion.

7.7 Exercises

All problems below are original. Full solutions begin on page 478.

A. The Student t distribution

A1. A quality-control statistic is known to follow a t distribution with 11 degrees of freedom. Using the standard table entries $t_{0.05,11} = 1.796$, $t_{0.025,11} = 2.201$, and $t_{0.01,11} = 2.718$, find each of the following.

(a) The upper 5% point $t_{0.05,11}$, i.e. the value c with $\mathbb{P}(T > c) = 0.05$.

(b) The two symmetric cutoffs $\pm c$ enclosing the central 95% of the distribution.

(c) The value c such that $\mathbb{P}(T < c) = 0.99$.

A2. A test statistic is known to follow t_{16} and comes out equal to 2.40. Using only the t_{16} row entries $t_{0.05,16} = 1.746$, $t_{0.025,16} = 2.120$, and $t_{0.01,16} = 2.583$, bracket the upper-tail probability $\mathbb{P}(T > 2.40)$ between two adjacent table areas. State the bracket and explain in one sentence why the value sits where it does relative to the listed cutoffs.

A3. Compare a t tail with the matching normal tail at a common cutoff.

(a) For $T \sim t_6$, the upper-tail probability beyond 1.94 is $\mathbb{P}(T > 1.94) \approx 0.050$ (the value 1.94 essentially equals the cutoff $t_{0.05,6} = 1.943$). For the standard normal, $\mathbb{P}(Z > 1.94) = 0.0262$. Report both tail probabilities.

(b) Form the ratio of the t_6 tail to the normal tail at 1.94, and state in plain words which distribution treats a value of 1.94 as more surprising.

A4. For $T \sim t_5$ you are given the lower-tail value $\mathbb{P}(T < 1.6) \approx 0.915$, and for the standard normal $\Phi(1.6) = 0.9452$.

(a) Find the central probability $\mathbb{P}(-1.6 < T < 1.6)$ for $T \sim t_5$.

(b) Find $\mathbb{P}(-1.6 < Z < 1.6)$ for the standard normal.

(c) By how many percentage points does the t_5 central interval fall short of the normal one, and where did that missing probability go?

A5. Invert the table to find cutoffs for stated areas, using $T \sim t_{24}$ with $t_{0.025,24} = 2.064$, $t_{0.05,24} = 1.711$, and $t_{0.10,24} = 1.318$.

(a) Find the value c with exactly 2.5% of the distribution above it.

(b) Find the symmetric pair $\pm c'$ between which a t_{24} variable falls with probability 0.90.

(c) Without further table lookups, explain why the cutoff in part (a) lies farther from 0 than the cutoff in part (b).

A6. As the degrees of freedom grow, the two-sided 5% critical value $t_{0.025,k}$ decreases toward the normal target $z_{0.025} = 1.96$. Using the standard entries $t_{0.025,7} = 2.365$, $t_{0.025,19} = 2.093$, $t_{0.025,39} = 2.023$, and $t_{0.025,119} = 1.980$, tabulate each cutoff, subtract 1.96 to get its excess over the normal target, and state at roughly which k the t critical value first comes within 0.05 of 1.96.

A7. CONTRAST. A small calibration study measures the mass (grams) of $n = 7$ reference weights, modeled as $\mathrm{N}(\mu, \sigma^2)$, and obtains $\overline{Y} = 50.6$ g with a *sample* standard deviation $S = 0.9$ g. The nominal target is $\mu_0 = 50.0$ g.

(a) Compute the standardized statistic $T = (\overline{Y} - \mu_0)/(S/\sqrt{n})$.

(b) A hurried analyst scores it against the normal cutoff $z_{0.025} = 1.96$; the honest comparison uses $t_{0.025,6} = 2.447$. State, for a two-sided test at the 5% level, whether each rule rejects $\mu = 50.0$.

(c) The corresponding tail probabilities are $\mathbb{P}(Z > 1.76) = 0.0392$ and $\mathbb{P}(T > 1.76) \approx 0.064$ for t_6. Explain in one or two sentences why the two rules can disagree, and which one is correct here.

B. Sampling distributions from a normal population

B1. A machine fills bags of coffee whose net weight is normal with target mean $\mu = 500$ g and known process standard deviation $\sigma = 6$ g. A clerk weighs a sample of $n = 9$ bags and records the sample mean $\overline{X}$.

(a) State the exact distribution of $\overline{X}$, including its standard error.

(b) Find $\mathbb{P}(\overline{X} > 503)$.

(c) Find $\mathbb{P}(497 \le \overline{X} \le 503)$, i.e. the chance the sample mean lands within 3 g of target.

B2. Tensile strengths of a fiber are normal with variance $\sigma^2 = 9$ (in N^2). An inspector measures $n = 16$ specimens and forms the sample variance S^2.

(a) Write the chi-square pivot for this sample, giving its degrees of freedom.

(b) Find $\mathbb{P}(S^2 > 14.0)$ using the appropriate χ^2 percentile.

(c) Find the lower-tail probability $\mathbb{P}(S^2 < 4.5)$.

B3. Resting metabolic rates are normal with unknown mean and unknown variance. A lab takes $n = 15$ readings and computes $\overline{X}$ and S. Under the working value $\mu = 1450$ kcal, the standardized statistic

$$T = \frac{\overline{X} - \mu}{S/\sqrt{n}}$$

is formed. (a) Name the exact reference distribution of T and its degrees of freedom, and say which two independent pieces it is built from. (b) Find $\mathbb{P}(|T| > 2.145)$. (c) Find $\mathbb{P}(T > 1.761)$.

B4. A normal sample of size $n = 21$ is drawn from a population with variance $\sigma^2 = 8$. Let $\mathrm{SS} = \sum_{i=1}^{n}(X_i - \overline{X})^2 = (n-1)S^2$ be the observed sum of squared deviations about $\overline{X}$. Using only the facts $\mathbb{E}[\chi^2_k] = k$ and $\mathrm{Var}(\chi^2_k) = 2k$, find $\mathbb{E}[\mathrm{SS}]$, $\mathrm{Var}(\mathrm{SS})$, and $\mathrm{SD}(\mathrm{SS})$.

B5. Reaction times (ms) are normal with known $\mu = 240$ and known $\sigma = 12$. A trial collects $n = 25$ observations and records both $\overline{X}$ and the sample standard deviation S. By the independence of $\overline{X}$ and S^2 for a normal sample, find the probability that *simultaneously* $\overline{X} < 236$ and $S > 15$. Report each piece (the Z event and the χ^2 event) and then combine them.

B6. Two independent calibration lines feed the same gauge. Line A averages $n_A = 10$ normal readings, each $\mathrm{N}(\mu_A, \sigma_A^2)$ with $\mu_A = 80$, $\sigma_A^2 = 5$; line B averages $n_B = 8$ normal readings, each $\mathrm{N}(\mu_B, \sigma_B^2)$ with $\mu_B = 78$, $\sigma_B^2 = 12$. Define the contrast $D = \overline{X}_A - \overline{X}_B$.

(a) State the exact distribution of D (mean and variance).

(b) Find $\mathbb{P}(D > 4)$.

(c) Find $\mathbb{P}(D < 0)$, the chance line B's average exceeds line A's.

B7. A practitioner has a normal sample of size $n = 10$ with σ *unknown*, and wants $\mathbb{P}\big(|\overline{X} - \mu|/(S/\sqrt{n}) > 2.262\big)$.

(a) Give the correct probability, naming the reference curve and its degrees of freedom, and explain in one or two sentences why the independence of $\overline{X}$ and S^2 together with the $n-1$ degree-of-freedom correction are exactly what make this curve the right one.

(b) A colleague, forgetting that σ was estimated, looks 2.262 up in the *standard normal* table instead. Compute that (incorrect) two-sided probability and state whether it is too large or too small, and why.

C. Confidence intervals for a normal mean

C1. A coffee roaster weighs the fill of $n = 21$ bags off the line; weight is approximately normal with unknown σ. The sample gives $\overline{X} = 251.4$ g and $S = 2.8$ g.

(a) Construct a 95% two-sided confidence interval for the mean fill weight. Use $t_{0.025,20} = 2.086$.

(b) The label claims 250 g. Does your interval contradict the claim? Explain in one sentence what the interval says about the labeled value.

C2. A materials lab measures the Brinell hardness of $n = 7$ alloy coupons (approximately normal, σ unknown):

$$118,\ 124,\ 121,\ 130,\ 117,\ 126,\ 123.$$

From the raw numbers compute $\overline{X}$ and S, then give a 90% two-sided confidence interval for the mean hardness. Use $t_{0.05,6} = 1.943$.

C3. A hospital records the time (minutes) from arrival to triage for $n = 16$ emergency patients; the times are roughly normal with unknown σ, and the sample yields $\overline{X} = 23.5$ min and $S = 5.6$ min. Administrators want a single worst-case figure — a 95% *upper* confidence bound on the mean triage time, an "at most" guarantee. Compute it and state in words what it asserts. Use $t_{0.05,15} = 1.753$.

C4. An agronomist measures yield (bushels/acre) on $n = 12$ test plots; yield is approximately normal with unknown σ, giving $\overline{X} = 58.0$ and $S = 7.2$.

(a) Compute the 90% and the 99% two-sided confidence intervals for the mean yield. Use $t_{0.05,11} = 1.796$ and $t_{0.005,11} = 3.106$.

(b) Report the width of each interval and state which is wider. Explain in one sentence why higher confidence forces a wider interval when the data are held fixed.

C5. An optics firm plans a study to estimate the mean focal length (mm) of a lens model with a 95% confidence interval whose margin of error is no larger than ME = 0.40 mm. A pilot batch suggests the standard deviation is about $S = 1.5$ mm.

(a) Roughly how many lenses must be measured? Use the large-sample stand-in $z_{0.025} = 1.96$ for the critical value, and round appropriately.

(b) State clearly why the answer is only a planning estimate, naming the two sources of approximation.

(c) If the firm instead tightened the target margin to ME = 0.20 mm, by what factor would the required sample size change? Justify with the formula, no full recomputation needed.

C6. A geneticist measures expression level (arbitrary units) for $n = 10$ cell cultures; the values are approximately normal but the population σ is *not* known. From those same 10 cultures the sample standard deviation is $S = 4.0$, with $\overline{X} = 22.5$. A lab partner builds a "95%" interval using the normal cutoff $z_{0.025} = 1.96$ instead of the t cutoff.

(a) Compute the partner's (incorrect) z-based interval and the correct t-based interval. Use $t_{0.025,9} = 2.262$.

(b) Compare the two widths and explain why the z interval is dishonestly narrow — what does it get wrong about the procedure's true coverage?

C7. A microbiologist counts colony-forming units (CFU, in hundreds) on $n = 14$ plates; the counts are approximately normal with unknown σ, giving $\overline{X} = 31.2$ and $S = 4.9$. A quality manager wants a 95% *lower* confidence bound — an "at least" figure — on the mean count. Compute the bound and contrast it in one sentence with the lower endpoint of the corresponding two-sided 95% interval. Use $t_{0.05,13} = 1.771$ and $t_{0.025,13} = 2.160$.

D. Hypothesis tests for a normal mean

D1. A precision lathe is set to turn shafts to a mean diameter of $\mu_0 = 25.00$ mm. An inspector measures $n = 12$ finished shafts and finds $\overline{X} = 25.06$ mm with sample standard deviation $S = 0.084$ mm; the diameters are approximately normal and σ is unknown. Test at $\alpha = 0.05$ whether the mean diameter *differs* from the 25.00 mm target. Use $t_{0.025,11} = 2.201$. State the hypotheses, the degrees of freedom, the test statistic, the critical value, and the decision in words.

D2. A bus route is advertised to take a mean of $\mu_0 = 34$ minutes. A commuter who suspects the trips actually run *longer* times $n = 9$ randomly chosen runs and obtains $\overline{X} = 36.2$ minutes with $S = 2.7$ minutes; times are approximately normal, σ unknown.

(a) Set up the one-sided hypotheses and say which tail carries the rejection region.

(b) Carry out the test at $\alpha = 0.05$ using $t_{0.05,8} = 1.860$, reporting t_{obs}, df, and the decision in words.

(c) Would the verdict change at $\alpha = 0.01$? Use $t_{0.01,8} = 2.896$.

D3. A laboratory balance is checked against a $\mu_0 = 100.000$ g reference mass. Six independent weighings (in grams) read

$$100.021,\ 99.986,\ 100.034,\ 99.998,\ 100.012,\ 99.973.$$

Compute $\overline{X}$ and S from these raw numbers, then test $H_0 : \mu = 100.000$ versus $H_1 : \mu \neq 100.000$ at $\alpha = 0.05$. Use $t_{0.025,5} = 2.571$. (The concern is bias in either direction, so the test is two-sided.) Show the sum-of-squared-deviations computation.

D4. A microbiology assay's documented mean turnaround is $\mu_0 = 48$ hours; a supervisor suspects the new staff have *slowed* the assay so that turnaround now *exceeds* 48 hours. From $n = 16$ runs, $\overline{X} = 50.4$ hours and $S = 4.8$ hours, with turnaround approximately normal and σ unknown.

(a) Compute t_{obs} and report the degrees of freedom.

(b) Using the t_{15} upper-tail values $t_{0.10,15} = 1.341$, $t_{0.05,15} = 1.753$, $t_{0.025,15} = 2.131$, and $t_{0.01,15} = 2.602$, *bracket* the p-value for $H_1 : \mu > 48$ between two adjacent table entries.

(c) Decide at $\alpha = 0.05$ and state the conclusion in words.

D5. A physical-therapy clinic measures each of $n = 7$ patients' grip strength (in kg) before and after a six-week program:

Patient	1	2	3	4	5	6	7
Before	28	31	26	35	30	27	33
After	32	33	30	36	35	31	34

The grip values are paired (same patient twice), so reduce to differences and run a one-sample t-test on them. Test at $\alpha = 0.05$ whether the program *increased* mean grip strength. Form $D_i = \text{After} - \text{Before}$, compute $\overline{D}$ and S_D (use the computing form for S_D^2), and use $t_{0.05,6} = 1.943$. State df and the decision in words.

D6. A metallurgy lab reports a 95% confidence interval for the mean Vickers hardness of a steel alloy (in HV) as $(212.4,\ 219.6)$, computed from an approximately normal sample with S estimated from the data ($n = 20$).

(a) Using test–CI duality, decide *without recomputing any statistic* the two-sided $\alpha = 0.05$ test of $H_0 : \mu = 215$ against $H_1 : \mu \neq 215$.

(b) Do the same for the claimed value $\mu_0 = 220$.

(c) State each conclusion in plain words about the mean hardness.

D7. A capping machine is meant to apply a mean torque of $\mu_0 = 18.0$ inch-pounds. A sample of $n = 10$ caps gives $\overline{X} = 19.5$ in-lb and $S = 2.2$ in-lb; the true σ is *not* known. A hurried technician standardizes with S but compares the result to the normal cutoff $z_{0.025} = 1.96$ instead of the correct t cutoff. Test $H_0 : \mu = 18.0$ versus $H_1 : \mu \neq 18.0$ at $\alpha = 0.05$ *both* ways, using $t_{0.025,9} = 2.262$. Report the common statistic, state the decision each cutoff produces, explain why the two cutoffs differ, and say which call is correct and why.

E. Inferences about a normal variance

E1. A coffee roaster monitors the fill mass of 250 g bags. From $n = 16$ bags filled on one machine the sample variance of mass is $S^2 = 7.8\,\mathrm{g}^2$ (masses assumed normal). Construct a 95% confidence interval for the true variance σ^2, using the χ^2 pivot on 15 degrees of freedom. Quote the two critical values you use and note which tail point produces which endpoint.

E2. A spectrophotometer's absorbance readings on a single reference standard have, from $n = 10$ repeated reads, sample variance $S^2 = 0.0064$ absorbance units squared (readings assumed normal).

(a) Give a 90% confidence interval for the variance σ^2.

(b) By square-rooting both endpoints, give a 90% confidence interval for the standard deviation σ.

(c) Is your standard-deviation interval symmetric about the point estimate $S = 0.08$? Explain in one sentence why or why not.

E3. An environmental agency caps how variable a sensor's hourly ozone readings may be. From $n = 25$ readings on a calibrated test gas the sample standard deviation is $S = 2.6$ ppb, so $S^2 = 6.76\,\mathrm{ppb}^2$ (readings assumed normal). The agency wants a 99% *upper* confidence bound on σ^2 — the worst-case spread consistent with the data. State which tail point serves as the divisor, compute the bound, and also report the corresponding upper bound on σ.

E4. A machine shop's legacy lathe holds a shaft diameter with variance $\sigma_0^2 = 0.0016\,\mathrm{mm}^2$ ($\sigma_0 = 0.04$ mm). Both excessive scatter and suspiciously tight scatter would prompt a maintenance review, so the question is two-sided. A sample of $n = 20$ shafts gives $S^2 = 0.0039\,\mathrm{mm}^2$. Test $H_0\colon \sigma^2 = 0.0016$ against $H_1\colon \sigma^2 \neq 0.0016$ at $\alpha = 0.05$. State the test statistic, both critical values on the appropriate df, and the decision in words.

E5. A reaction-time experiment historically yields response times with standard deviation $\sigma_0 = 45$ ms, so $\sigma_0^2 = 2025\,\mathrm{ms}^2$. After a software change the experimenter fears the times are now *more* variable. A sample of $n = 13$ trials gives $S = 63$ ms, hence $S^2 = 3969\,\mathrm{ms}^2$ (times assumed normal). Test $H_0\colon \sigma^2 = 2025$ against $H_1\colon \sigma^2 > 2025$ at $\alpha = 0.05$. Report the statistic, the single upper-tail critical value on the correct df, and your conclusion.

E6. Reuse the reaction-time data of E5: $n = 13$ (12 df), $\chi^2_{\mathrm{obs}} = 23.52$, one-sided alternative $H_1\colon \sigma^2 > 2025$. Using the χ^2 table points $\chi^2_{0.025,\,12} = 23.337$ and $\chi^2_{0.01,\,12} = 26.217$, bracket the p-value, interpolate an approximate value, and state the decision at $\alpha = 0.05$. Explain in one sentence why this one-sided p-value is a single tail and must *not* be doubled.

E7. A student is handed the spectrophotometer data of E2 ($n = 10$, $S^2 = 0.0064$, 9 df) and reasons by analogy with the mean: "the point estimate of the variance is $S^2 = 0.0064$, its standard error is about $S^2\sqrt{2/(n-1)}$, so a 95% interval is $0.0064 \pm 1.96 \cdot 0.0064\sqrt{2/9}$."

(a) Compute the symmetric interval this shortcut produces.

(b) Build the correct 95% χ^2 interval for σ^2 on 9 df, using $\chi^2_{0.025,\,9} = 19.023$ and $\chi^2_{0.975,\,9} = 2.700$.

(c) Compare the two intervals: is S^2 at the centre of the correct one? Explain in one or two sentences why the symmetric template is invalid here.

F. Robustness and checking normality

F1. A clinical lab times $n = 49$ blood-draw appointments (minutes). The histogram is mildly right-skewed, the boxplot flags one point just past the upper whisker, and the normal Q-Q plot is nearly straight with a faint upward bend at the top. The summary is $\overline{X} = 12.4$ min and $S = 4.2$ min.

(a) Is a 95% t interval for the mean appointment time trustworthy? Name the property that protects it and explain why the mild skew is harmless here.

(b) Build the interval, using $t_{0.025,\,48} \approx 2.01$.

(c) The same lab also wants a 95% χ^2 interval for the *variance* of the draw times. Should it report that interval with the same confidence? Explain.

F2. A toxicologist measures the cadmium concentration (ppm) at $n = 7$ industrial sites and records

$$1,\ 2,\ 2,\ 3,\ 4,\ 7,\ 25.$$

The value 25 sits far above the rest; the boxplot flags it well past the whisker and the Q-Q plot bends sharply upward at the top.

(a) Compute $\overline{X}$ and the median, and explain in one sentence why the gap between them warns against a raw t interval.

(b) Quantify the leverage of the single large value by recomputing $\overline{X}$ with the 25 removed.

(c) State the two standard remedies appropriate to this sample (small n, positive, strongly right-skewed) and say which you would prefer.

F3. For each described normal Q-Q plot, state whether a one-sample t test for the mean is trustworthy and give the one-word diagnosis (normal / right skew / left skew / heavy tails / light tails / outlier). Assume n is given in each part.

(a) $n = 45$: points hug the reference line, one point a hair off at each end.

(b) $n = 10$: points bow steadily *above* the line at the upper end and bunch near the bottom.

(c) $n = 20$: a clear S-shape, points dipping below the line at the left and rising above it at the right, steepening at both extremes.

(d) $n = 8$: one isolated point sits far off the line at the top while the rest are straight.

F4. A reliability engineer plans a χ^2 test of $H_0\colon \sigma^2 = \sigma_0^2$ at nominal $\alpha = 0.05$ with $n = 41$. The lifetimes are heavy-tailed with kurtosis $\kappa \approx 7$ (a normal parent has $\kappa = 3$). Using the approximation $\mathrm{Var}(S^2) \approx \sigma^4(\kappa - 1)/n$,

(a) compute $\mathrm{Var}(S^2)$ under the normal model and under the heavy-tailed model at $n = 41$, and report the factor by which the true standard deviation of S^2 exceeds what χ^2_{40} assumes;

(b) explain why the true rejection rate under H_0 is far above the nominal 5%;

(c) show algebraically that enlarging n does not close the gap.

F5. A water utility has $n = 30$ turbidity readings from a right-skewed source (occasional spikes), with $\overline{X} = 6.0$ NTU and $S = 2.0$ NTU. The utility wants both a 95% CI for the mean turbidity and a 95% CI for the variance.

(a) Build the mean CI using $t_{0.025,\,29} = 2.045$, and state whether its coverage is close to 95% and why.

(b) Build the nominal variance CI using $\chi^2_{0.975,\,29} = 16.047$ and $\chi^2_{0.025,\,29} = 45.722$.

(c) State which interval the utility should trust and which it should flag as unreliable, and give the one-sentence reason the two verdicts differ on the same data.

F6. An environmental chemist has $n = 7$ measured solvent masses (g) that are strongly right-skewed:

$$3,\ 4,\ 6,\ 11,\ 20,\ 55,\ 148.$$

A t interval on the raw values is untrustworthy.

(a) Take natural logs of the seven values and argue in one or two sentences that the log scale is far more symmetric. (Use $\ln 3 = 1.10$, $\ln 4 = 1.39$, $\ln 6 = 1.79$, $\ln 11 = 2.40$, $\ln 20 = 3.00$, $\ln 55 = 4.01$, $\ln 148 = 5.00$.)

(b) On the log scale, $\overline{Y} = 2.670$ and $\sum(y - \overline{Y})^2 = 12.66$. Build a 95% t interval for $\mu_{\ln X}$ using $t_{0.025,\,6} = 2.447$.

(c) Back-transform the endpoints and state precisely what population quantity the resulting interval estimates on the original scale.

F7. Three small studies each show non-normality; for each, name the single most appropriate remedy (log transform, nonparametric method, or collect more data) and justify it in one sentence.

(a) Bacterial colony counts, positive and right-skewed over three orders of magnitude, $n = 8$, and the question concerns the *mean* log-count.

(b) A symmetric but heavy-tailed instrument-error sample with two extreme outliers, $n = 12$, where a log transform does nothing to the symmetry.

(c) A mildly skewed assay sample, $n = 14$, where the lab can cheaply run more replicates and the question concerns the *mean*.

(d) Reusing the logged solvent data of F6, a manager now asks for a test of whether the *variance of the raw masses* equals $\sigma_0^2 = 3000$ g^2, and a colleague proposes running a χ^2 variance test on the logs. Explain why that does not answer the manager's question.

G. Choosing the right normal-theory procedure (synthesis)

G1. For each scenario, name the single normal-theory procedure that answers the question (one-sample t interval, one-sample t-test, paired t-test, χ^2 variance interval, χ^2 variance test, or a plain sampling-distribution probability), state the reference distribution with its degrees of freedom, and say in one phrase what makes that the right tool. Do *not* carry out any arithmetic.

(a) A normal process has *known* $\sigma = 4.0$. From $n = 16$ items, what is the chance the sample mean lands within 1.5 of the true mean μ?

(b) Tablet masses are roughly normal; from $n = 20$ tablets (σ unknown) you want a range of plausible values for the mean mass.

(c) A coating line claims standard deviation $\sigma_0 = 0.5$ mil; an auditor suspects the spread has grown and tests it from $n = 25$ coupons.

(d) Each of 30 subjects is scored before and after a training program; is the mean change different from zero?

(e) A reference standard should have measurement standard deviation 0.30 units; a validation run of $n = 18$ checks whether σ differs from that value.

(f) With σ unknown, a thermostat's $n = 12$ readings are used to ask whether the mean temperature exceeds the 180°C setpoint.

G2. A metrology lab measures the diameter (mm) of $n = 10$ precision bearings, which are approximately normal with σ unknown. The data are

$$8.02,\ 7.96,\ 8.05,\ 7.99,\ 8.01,\ 7.94,\ 8.07,\ 7.98,\ 8.03,\ 7.93.$$

The customer cares about *two* things: the mean must be on target at $\mu_0 = 8.00$ mm, and the standard deviation must be small enough that σ^2 is not above the contract limit $\sigma_0^2 = 0.0020$ mm^2.

(a) Compute $\overline{X}$, S^2, and S from the raw data (the computing form $S^2 = \frac{1}{n-1}(\sum x_i^2 - n\overline{X}^2)$ is convenient).

(b) Test $H_0\colon \mu = 8.00$ versus $H_1\colon \mu \neq 8.00$ at $\alpha = 0.05$. State the procedure and why it (not a z-test) applies; use $t_{0.025,\,9} = 2.262$.

(c) Test $H_0\colon \sigma^2 = 0.0020$ versus $H_1\colon \sigma^2 > 0.0020$ at $\alpha = 0.05$. Use $\chi^2_{0.05,\,9} = 16.919$. Why is this a χ^2 test and not a t-test?

(d) State the combined verdict for the customer in plain words.

G3. Two analysts study the *same* normal population whose standard deviation is genuinely *known* to be $\sigma = 6.0$ from long calibration history. Analyst A draws $n = 9$ observations and reports $\overline{X} = 51.4$. Analyst B does not have σ and instead reports, from a separate $n = 9$ sample, $\overline{X} = 51.4$ with sample standard deviation $S = 6.3$.

(a) Analyst A asks: under $\mu = 50$, what is $\mathbb{P}(\overline{X} \geq 51.4)$? Identify why this is a plain sampling-distribution (normal) probability, not a hypothesis test, and compute it using $\mathbb{P}(Z \geq 0.70) = 0.2420$.

(b) Analyst B wants to *test* $H_0 : \mu = 50$ versus $H_1 : \mu > 50$ at $\alpha = 0.05$. State which procedure applies and why it differs from A's, then carry it out using $t_{0.05,\,8} = 1.860$.

(c) Explain in one or two sentences why A may legitimately use the normal and B must use the t, even though their $\overline{X}$ values are identical.

G4. A hydrologist has $n = 8$ peak-flow measurements (in m^3/s) from a river:

$$12,\ 14,\ 13,\ 16,\ 15,\ 14,\ 19,\ 61.$$

The value 61 sits far above the rest. The hydrologist wants (i) a 95% confidence interval for the mean peak flow and (ii) a 95% confidence interval for the variance of peak flow.

(a) Before computing anything, state which assumption both procedures share and describe the one diagnostic picture you would inspect first.

(b) Compute $\overline{X}$ and the median; explain why the gap warns you that normality is in doubt here, and identify the single observation responsible.

(c) Of the two requested intervals (mean and variance), which one is the more endangered by the apparent non-normality, and why? State what you would do before reporting each.

G5. A nutrition lab compares two assay kits on the *same* $n = 7$ serum samples; each sample is measured once by kit X and once by kit Y (mg/dL):

Sample	1	2	3	4	5	6	7
Kit X	102	98	110	95	104	99	107
Kit Y	99	96	104	95	100	97	101

Differences are approximately normal.

(a) Explain why a paired procedure, not two independent-sample analyses, is correct here, and give the reference distribution with its degrees of freedom.

(b) Test at $\alpha = 0.05$ whether the two kits differ on average ($H_1 : \mu_D \neq 0$). Form $D_i = X_i - Y_i$, compute $\overline{D}$ and S_D, and use $t_{0.025,\,6} = 2.447$.

(c) Give the matching 95% confidence interval for μ_D and confirm it agrees with the test by the test–CI duality.

G6. A semiconductor fab holds a film-thickness process whose thickness is normal with a *known* long-run standard deviation $\sigma = 2.0$ nm and target mean $\mu = 90$ nm. A sample of $n = 16$ wafers is drawn.

(a) Recognize and compute: what is $\mathbb{P}(\overline{X} > 91)$ when $\mu = 90$? Use $\mathbb{P}(Z > 2.00) = 0.0228$. Why is no t distribution involved?

(b) A quality engineer instead asks: assuming normality and the *true* $\sigma = 2.0$, what is $\mathbb{P}\big(S^2 > 7.20\big)$ for this sample? Recognize that $(n-1)S^2/\sigma^2 \sim \chi^2_{15}$, convert the event to one about the χ^2 statistic, and bracket the probability using $\chi^2_{0.05,\,15} = 24.996$ and $\chi^2_{0.025,\,15} = 27.488$.

(c) In one sentence, contrast what part (a) and part (b) tell the engineer about the process.

G7. A bottling line is supposed to deliver a mean of $\mu_0 = 500$ mL with standard deviation no larger than $\sigma_0 = 1.5$ mL. An inspector draws $n = 20$ bottles (volumes approximately normal, σ unknown) and finds $\overline{X} = 499.2$ mL and $S = 2.1$ mL.

(a) Identify the two separate questions the specification raises and the procedure each needs.

(b) Test $H_0\colon \mu = 500$ versus $H_1\colon \mu \neq 500$ at $\alpha = 0.05$ using $t_{0.025,\,19} = 2.093$.

(c) Test $H_0\colon \sigma^2 = 2.25$ versus $H_1\colon \sigma^2 > 2.25$ at $\alpha = 0.05$ using $\chi^2_{0.05,\,19} = 30.144$.

(d) Suppose a colleague had standardized the mean with S but compared to $z_{0.025} = 1.96$. Would that flip the mean decision? Explain why t, not z, is the honest reference here.

7.8 Solutions

A. The Student t distribution

A1. A t table entry $t_{\alpha,k}$ is by definition the value cutting off upper-tail area α on t_k; we read row $k = 11$ and use symmetry of the t about 0.

(a) The upper 5% point is read directly from the $\alpha = 0.05$ column:

$$\boxed{t_{0.05,11} = 1.796, \qquad \mathbb{P}(T > 1.796) = 0.05.}$$

(b) A central 0.95 leaves 0.025 in each tail, so the cutoffs are the upper-0.025 point and its mirror image:

$$\boxed{\mathbb{P}(-2.201 < T < 2.201) = 0.95 \quad (T \sim t_{11}).}$$

(c) $\mathbb{P}(T < c) = 0.99$ means upper-tail area 0.01, so c is the $\alpha = 0.01$ entry:

$$\boxed{c = t_{0.01,11} = 2.718.}$$

A2. **Method.** A hand table lists only a few areas, so we locate the observed value between adjacent row entries and read off the bracketing probabilities. The tail area is a decreasing function of the cutoff, so a value sitting between two cutoffs has a tail area between the two corresponding areas.

Step 1: place the value among the cutoffs. Compare 2.40 to the t_{16} entries: $2.120 < 2.40 < 2.583$, i.e. $t_{0.025,16} < 2.40 < t_{0.01,16}$.

Step 2: translate positions into areas. The area beyond 2.120 is 0.025 and the area beyond 2.583 is 0.01. Since 2.40 lies between these cutoffs, its tail area lies between 0.01 and 0.025:

$$\boxed{0.01 < \mathbb{P}(T > 2.40) < 0.025 \quad (T \sim t_{16}).}$$

Step 3: why it sits there. The value 2.40 is larger than the 2.5% cutoff (so less than 0.025 lies beyond it) but smaller than the 1% cutoff (so more than 0.01 lies beyond it); a larger cutoff always strands a smaller tail.

A3. **Method.** Same cutoff 1.94, two reference distributions. The t_6 density puts more mass far from 0 than the normal does, so its tail beyond a fixed point is the larger of the two; we report both and take the ratio.

(a) The value 1.94 essentially equals $t_{0.05,6} = 1.943$, so the t_6 tail beyond it is 0.050; the normal table gives $\mathbb{P}(Z > 1.94) = 0.0262$:

$$\boxed{\mathbb{P}(T > 1.94) \approx 0.050 \;>\; 0.0262 \approx \mathbb{P}(Z > 1.94).}$$

(b) The ratio of the tails is

$$\frac{\mathbb{P}(T > 1.94)}{\mathbb{P}(Z > 1.94)} \approx \frac{0.050}{0.0262} \approx 1.91.$$

The t_6 tail is nearly double the normal tail, so the normal treats 1.94 as roughly twice as surprising as the t_6 does. *The normal calls* 1.94 *more extreme.*

A4. **Method.** By symmetry of each distribution about 0, a central probability is $2\,\mathbb{P}(\cdot < x) - 1$. Mass the t pushes into its tails is mass missing from its middle, so the central interval captures less under t_5 than under the normal.

(a) For $T \sim t_5$,

$$\mathbb{P}(-1.6 < T < 1.6) = 2\,\mathbb{P}(T < 1.6) - 1 = 2(0.915) - 1 = 0.830.$$

(b) For the standard normal,

$$\mathbb{P}(-1.6 < Z < 1.6) = 2\Phi(1.6) - 1 = 2(0.9452) - 1 = 0.8904.$$

(c) The deficit is $0.8904 - 0.830 = 0.0604$, about 6.0 percentage points:

$$\boxed{\mathbb{P}(-1.6 < T < 1.6) \approx 0.830 \ < \ 0.890 \approx \mathbb{P}(-1.6 < Z < 1.6), \text{ a 6.0 pt deficit.}}$$

That missing probability did not vanish; it was banished to the fatter t_5 tails.

A5. Method. Now the area is given and the cutoff is unknown — the inverse lookup. The value with upper-tail area α on t_k is by definition $t_{\alpha,k}$, read from row $k = 24$.

(a) We want c with $\mathbb{P}(T > c) = 0.025$, the $\alpha = 0.025$ entry:

$$\boxed{c = t_{0.025,24} = 2.064.}$$

(b) Central 0.90 leaves total tail 0.10, hence $\alpha/2 = 0.05$ in each tail; the cutoff is the upper-0.05 point, mirrored:

$$\boxed{\mathbb{P}(-1.711 < T < 1.711) = 0.90, \qquad c' = t_{0.05,24} = 1.711 \quad (T \sim t_{24}).}$$

(c) The cutoff in (a) carries the smaller tail area ($0.025 < 0.05$). To leave less probability beyond the boundary you must push it farther from 0, so $2.064 > 1.711$.

A6. Method. Because $t_k \to \mathrm{N}(0,1)$ as $k \to \infty$, the critical value $t_{0.025,k}$ must decrease toward $z_{0.025} = 1.96$. We list the cutoffs by df and subtract the normal target to exhibit the convergence.

Step 1: list the cutoffs by df.

$$t_{0.025,7} = 2.365, \quad t_{0.025,19} = 2.093,$$
$$t_{0.025,39} = 2.023, \quad t_{0.025,119} = 1.980.$$

Step 2: subtract the normal target $z_{0.025} = 1.96$.

$$2.365 - 1.96 = 0.405, \quad 2.093 - 1.96 = 0.133,$$
$$2.023 - 1.96 = 0.063, \quad 1.980 - 1.96 = 0.020.$$

Step 3: read the trend. The excess over 1.96 falls from 0.405 at $k = 7$ to 0.020 at $k = 119$, shrinking at the rate $\approx 2/(k-2)$. It is still 0.063 at $k = 39$ but has dropped below 0.05 by the time k reaches roughly 60 (where the excess is about 0.04):

$$\boxed{t_{0.025,k} \downarrow 1.96 = z_{0.025}; \text{ the gap first falls under 0.05 near } k \approx 60.}$$

A7. Method. The denominator is the *sample* S from only $n = 7$ readings, so the honest reference law is $t_{n-1} = t_6$, whose tails are much heavier than the normal's at this small df. Reading the z table is the look-alike error; we run both rules.

(a) The standard error is $S/\sqrt{n} = 0.9/\sqrt{7} = 0.9/2.646 = 0.3402$ g, so

$$T = \frac{\overline{Y} - \mu_0}{S/\sqrt{n}} = \frac{50.6 - 50.0}{0.3402} = \frac{0.6}{0.3402} = 1.764 \approx 1.76.$$

(b) Two-sided 5% scoring of the common statistic $|T| = 1.76$:

$$\text{normal rule:} \quad 1.76 < z_{0.025} = 1.96 \ \Rightarrow \ \text{do not reject,}$$
$$\text{honest } t_6 \text{ rule:} \quad 1.76 < t_{0.025,6} = 2.447 \ \Rightarrow \ \text{do not reject.}$$

Here both rules agree: $\mu = 50.0$ is *not* rejected at the 5% level. (The agreement is incidental — a slightly larger statistic would clear 1.96 but not 2.447, and the two rules would split.)

(c) The two-sided p-values are $2(0.0392) = 0.0784$ under the normal and $2(0.064) \approx 0.128$ under t_6. The honest p-value is over 60% larger because estimating σ from just 6 degrees of freedom inflates the true spread; the normal table understates that spread and makes the result look more significant than it is. The t_6 value is the correct one:

$$\boxed{\mathbb{P}(Z > 1.76) = 0.0392 \ < \ 0.064 \approx \mathbb{P}(T > 1.76) \ (t_6); \ \text{use } t_6, \ \text{not } z.}$$

B. Sampling distributions from a normal population

B1. Method. These are probabilities about the sample mean $\overline{X}$ with σ *known*, so pillar (i) applies: $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$ and we standardize with a Z (not a t, since nothing is estimated). The standard error is $\sigma/\sqrt{n}$.

(a) With $\mu = 500$, $\sigma = 6$, $n = 9$, the standard error is $\sigma/\sqrt{n} = 6/\sqrt{9} = 6/3 = 2$.

$$\boxed{\overline{X} \sim \mathrm{N}(500, \ 2^2), \qquad \mathrm{SD}(\overline{X}) = 2 \text{ g}}$$

(b) Standardize the cutoff and read the upper tail:

$$z = \frac{503 - 500}{2} = \frac{3}{2} = 1.50,$$
$$\mathbb{P}(\overline{X} > 503) = 1 - \Phi(1.50) = 1 - 0.9332 = 0.0668.$$

$$\boxed{\mathbb{P}(\overline{X} > 503) \approx 0.067}$$

(c) By symmetry the interval is ± 1.50 standard errors:

$$z_{\mathrm{lo}} = \frac{497 - 500}{2} = -1.50, \quad z_{\mathrm{hi}} = \frac{503 - 500}{2} = +1.50,$$
$$\mathbb{P}(497 \le \overline{X} \le 503) = \Phi(1.50) - \Phi(-1.50) \qquad = 2(0.9332) - 1 = 0.8664.$$

$$\boxed{\mathbb{P}(497 \le \overline{X} \le 503) \approx 0.866}$$

B2. Method. Probabilities about S^2 from a normal sample route through the chi-square pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ — never by treating S^2 as normal. We convert each event on S^2 into the equivalent event on the pivot. Here $n = 16$, $\sigma^2 = 9$, so $n - 1 = 15$.

(a)

$$\boxed{\frac{(n-1)S^2}{\sigma^2} = \frac{15\,S^2}{9} \sim \chi^2_{15}}$$

(b) Translate $S^2 > 14.0$ into the pivot:

$$S^2 > 14.0 \iff \frac{15S^2}{9} > \frac{15(14.0)}{9} = \frac{210}{9} = 23.333.$$

The upper-10% point is $\chi^2_{0.10,15} = 22.307$ and the upper-5% point is $\chi^2_{0.05,15} = 24.996$; since $22.307 < 23.333 < 24.996$ the tail probability lies between 0.05 and 0.10, about 0.077.

$$\boxed{\mathbb{P}(S^2 > 14.0) = \mathbb{P}(\chi^2_{15} > 23.33) \approx 0.077}$$

(c) Lower tail; because χ^2 is right-skewed the lower point is read separately:

$$S^2 < 4.5 \iff \frac{15S^2}{9} < \frac{15(4.5)}{9} = \frac{67.5}{9} = 7.50.$$

The lower-5% point of χ^2_{15} is $\chi^2_{0.95,15} = 7.261$, and 7.50 sits just above it, so the probability is a little over 0.05, about 0.054.

$$\boxed{\mathbb{P}(S^2 < 4.5) = \mathbb{P}(\chi^2_{15} < 7.50) \approx 0.054}$$

B3. Method. With σ *unknown* and replaced by S, the standardized mean is the t ratio: numerator $Z = (\overline{X} - \mu)/(\sigma/\sqrt{n}) \sim \mathrm{N}(0,1)$ over $\sqrt{V/(n-1)}$ with $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, and these two are independent (pillar iii), so $T \sim t_{n-1}$. Here $n = 15$, so $n - 1 = 14$.

(a) $T \sim t_{14}$, built from an independent $Z \sim \mathrm{N}(0,1)$ and $V \sim \chi^2_{14}$ as $T = Z/\sqrt{V/14}$.

$$\boxed{T = \frac{\overline{X} - \mu}{S/\sqrt{15}} \sim t_{14}}$$

(b) 2.145 is the two-sided 5% critical value of t_{14}, i.e. $t_{0.025,14} = 2.145$, so each tail beyond it carries 0.025:

$$\mathbb{P}(|T| > 2.145) = 2(0.025) = 0.050.$$

$$\boxed{\mathbb{P}(|T| > 2.145) = 0.05}$$

(c) 1.761 is the upper-5% point of t_{14}, $t_{0.05,14} = 1.761$, so

$$\mathbb{P}(T > 1.761) = 0.05.$$

$$\boxed{\mathbb{P}(T > 1.761) = 0.05}$$

B4. Method. The sum of squared deviations scales a chi-square: $\mathrm{SS}/\sigma^2 = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$, so $\mathrm{SS} = \sigma^2\, U$ with $U \sim \chi^2_{n-1}$. Push the chi-square moments through the constant σ^2. Here $n = 21$ gives $n - 1 = 20$ and $\sigma^2 = 8$.

Step 1: chi-square moments. $U = \mathrm{SS}/\sigma^2 \sim \chi^2_{20}$, so $\mathbb{E}[U] = 20$ and $\mathrm{Var}(U) = 2(20) = 40$.

Step 2: scale by $\sigma^2 = 8$.

$$\begin{aligned}
\mathbb{E}[\mathrm{SS}] &= \sigma^2\, \mathbb{E}[U] = 8(20) = 160, \\
\mathrm{Var}(\mathrm{SS}) &= (\sigma^2)^2\, \mathrm{Var}(U) = 8^2(40) = 64(40) = 2560, \\
\mathrm{SD}(\mathrm{SS}) &= \sqrt{2560} \approx 50.60.
\end{aligned}$$

$$\boxed{\mathbb{E}[\mathrm{SS}] = 160, \qquad \mathrm{Var}(\mathrm{SS}) = 2560, \qquad \mathrm{SD}(\mathrm{SS}) \approx 50.6}$$

B5. Method. For a normal sample $\overline{X}$ and S^2 are *independent* (pillar iii): the mean event uses pillar (i) with a Z, the spread event uses pillar (ii) with a χ^2, and independence lets us *multiply* the two probabilities. Here $\mu = 240$, $\sigma = 12$, $n = 25$, so $\sigma/\sqrt{n} = 12/5 = 2.4$ and $n - 1 = 24$.

Step 1: the mean event (Z).

$$z = \frac{236 - 240}{2.4} = \frac{-4}{2.4} = -1.667, \qquad \mathbb{P}(\overline{X} < 236) = \Phi(-1.667) = 1 - 0.9522 = 0.0478.$$

Step 2: the spread event (χ^2).

$$S > 15 \iff S^2 > 225 \iff \frac{(n-1)S^2}{\sigma^2} = \frac{24\, S^2}{144} > \frac{24(225)}{144} = 37.5.$$

With 24 df, $\chi^2_{0.05,24} = 36.415$ and $\chi^2_{0.025,24} = 39.364$; since $36.415 < 37.5 < 39.364$ the upper tail is between 0.025 and 0.05, about 0.039.

$$\mathbb{P}(S > 15) = \mathbb{P}(\chi^2_{24} > 37.5) \approx 0.039.$$

Step 3: multiply (independence).

$$\mathbb{P}(\overline{X} < 236 \text{ and } S > 15) = 0.0478 \times 0.039 \approx 0.00186.$$

$$\boxed{\mathbb{P}(\overline{X} < 236,\ S > 15) \approx 0.0019}$$

B6. Method. Each sample mean is normal (pillar i), the two lines are independent, and $D = \overline{X}_A - \overline{X}_B$ is a linear combination, so D is normal: add the means, and add the variances (each σ_i^2/n_i). With σ's known we standardize with Z.

(a) Mean: $\mathbb{E}[D] = \mu_A - \mu_B = 80 - 78 = 2$. Variance (independent, so variances add):

$$\mathrm{Var}(D) = \frac{\sigma_A^2}{n_A} + \frac{\sigma_B^2}{n_B} = \frac{5}{10} + \frac{12}{8} = 0.5 + 1.5 = 2.0,$$

so $\mathrm{SD}(D) = \sqrt{2.0} \approx 1.414$.

$$\boxed{D \sim \mathrm{N}(2,\ 2.0)}$$

(b)

$$z = \frac{4-2}{1.414} = \frac{2}{1.414} = 1.414, \qquad \mathbb{P}(D > 4) = 1 - \Phi(1.414) = 1 - 0.9213 = 0.0787.$$

$$\boxed{\mathbb{P}(D > 4) \approx 0.079}$$

(c)

$$z = \frac{0-2}{1.414} = -1.414, \qquad \mathbb{P}(D < 0) = \Phi(-1.414) = 1 - 0.9213 = 0.0787.$$

$$\boxed{\mathbb{P}(D < 0) \approx 0.079}$$

B7. Method. With σ replaced by the random S, the correct reference for $(\overline{X}-\mu)/(S/\sqrt{n})$ is t_{n-1}, assembled as $Z/\sqrt{V/(n-1)}$ from an *independent* $Z \sim \mathrm{N}(0,1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. Here $n = 10$, so $n - 1 = 9$.

(a) 2.262 is the two-sided 5% critical value of t_9, i.e. $t_{0.025,9} = 2.262$, so

$$\mathbb{P}\Big(\tfrac{|\overline{X}-\mu|}{S/\sqrt{10}} > 2.262\Big) = \mathbb{P}(|t_9| > 2.262) = 2(0.025) = 0.050.$$

Why this curve. The ratio is a t only because the numerator Z (carrying $\overline{X}$) and the denominator χ^2 (carrying S^2) are independent — exactly pillar (iii) for the normal — and because centering the deviations at $\overline{X}$ rather than μ spends one degree of freedom, leaving $V \sim \chi^2_9$. Both facts are needed: independence makes the ratio a *tabulated* t, and the $n-1$ correction fixes which t row to use.

$$\boxed{\mathbb{P}(|T| > 2.262) = 0.05, \quad T \sim t_9}$$

(b) Looking 2.262 up in the standard normal table gives

$$\mathbb{P}(|Z| > 2.262) = 2\big(1 - \Phi(2.262)\big) = 2(0.0119) = 0.0238.$$

This is *too small*: the t_9 density has heavier tails than the normal because the random S in the denominator is sometimes small and inflates the ratio. Ignoring that extra randomness understates the tail (0.024 versus the correct 0.050), so any p-value or confidence statement built on the normal would be too optimistic.

$$\boxed{\text{wrong (normal): } 0.024 \quad \text{vs.} \quad \text{correct } (t_9)\text{: } 0.050}$$

C. Confidence intervals for a normal mean

C1. The parameter is a **mean**, fill weight is approximately **normal**, and σ is **unknown** (S came from these 21 bags), so the tool is the two-sided t interval $\overline{X} \pm t_{\alpha/2,\,n-1}\, S/\sqrt{n}$ with $n-1 = 20$ degrees of freedom.

(a) Standard error and margin of error, then assemble:

$$\frac{S}{\sqrt{n}} = \frac{2.8}{\sqrt{21}} = \frac{2.8}{4.583} = 0.6110 \text{ g},$$

$$\text{ME} = t_{0.025,20}\,\frac{S}{\sqrt{n}} = 2.086 \times 0.6110 = 1.275 \text{ g},$$

$$\overline{X} \pm \text{ME} = 251.4 \pm 1.275 = (250.13,\ 252.67).$$

95% CI for mean fill weight : (250.13, 252.67) g.

(b) The labeled value 250 g lies *below* the entire interval, so the data contradict the claim at this confidence level: we are 95% confident the mean fill exceeds 250 g, i.e. bags are, on average, slightly overfilled rather than meeting the label exactly.

C2. Same recognition — mean, normal, σ unknown — but with raw data, so first compute $\overline{X}$ and S, then form the two-sided t interval with $df = 6$.

Step 1: sample mean.

$$\sum x_i = 118 + 124 + 121 + 130 + 117 + 126 + 123 = 859, \qquad \overline{X} = \frac{859}{7} = 122.714.$$

Step 2: squared deviations about $\overline{X} = 122.714$.

$$\sum (x_i - \overline{X})^2 = (-4.714)^2 + (1.286)^2 + (-1.714)^2 + (7.286)^2 + (-5.714)^2 + (3.286)^2 + (0.286)^2 = 123.43.$$

Step 3: variance and standard deviation.

$$S^2 = \frac{123.43}{6} = 20.571, \qquad S = \sqrt{20.571} = 4.536.$$

Step 4: standard error, margin, interval.

$$\frac{S}{\sqrt{n}} = \frac{4.536}{\sqrt{7}} = \frac{4.536}{2.646} = 1.714,$$

$$\text{ME} = t_{0.05,6}\,\frac{S}{\sqrt{n}} = 1.943 \times 1.714 = 3.331,$$

$$\overline{X} \pm \text{ME} = 122.714 \pm 3.331 = (119.38,\ 126.05).$$

90% CI for mean hardness : (119.38, 126.05).

C3. A single one-directional "at most" guarantee means a one-sided *upper* confidence bound: put all $\alpha = 0.05$ in the upper tail and use $t_{\alpha,\,n-1} = t_{0.05,15} = 1.753$ ($df = 15$), not the two-sided cutoff.

Step 1: standard error. $S/\sqrt{n} = 5.6/\sqrt{16} = 5.6/4 = 1.400$ min.

Step 2: one-sided margin. $t_{\alpha,\,n-1}\, S/\sqrt{n} = 1.753 \times 1.400 = 2.454$ min.

Step 3: upper bound.

$$\mu \le \overline{X} + t_{\alpha,\,n-1}\frac{S}{\sqrt{n}} = 23.5 + 2.454 = 25.95.$$

$$\boxed{95\%\text{ upper bound}:\ \mu \le 25.95\text{ min.}}$$

We are 95% confident the mean arrival-to-triage time is no more than about 26.0 minutes; the statement gives a worst case in one direction and imposes no lower limit.

C4. One mean, normal, σ unknown $\Rightarrow$ two-sided t interval, $df = 11$. Only the confidence level (hence $t_{\alpha/2,\,n-1}$) changes; the data, S, and n are fixed, so the standard error is shared.

(a) Common standard error, then a margin at each level:

$$\frac{S}{\sqrt{n}} = \frac{7.2}{\sqrt{12}} = \frac{7.2}{3.464} = 2.078,$$
$$90\%: \quad \text{ME} = t_{0.05,11}\,\frac{S}{\sqrt{n}} = 1.796 \times 2.078 = 3.733,$$
$$99\%: \quad \text{ME} = t_{0.005,11}\,\frac{S}{\sqrt{n}} = 3.106 \times 2.078 = 6.456.$$

The two intervals about $\overline{X} = 58.0$ are

$$90\%: \ (54.27,\ 61.73),$$
$$99\%: \ (51.54,\ 64.46).$$

$$\boxed{90\%: \ (54.27,\ 61.73); \quad 99\%: \ (51.54,\ 64.46).}$$

(b) Widths are 2 ME: the 90% interval has width 7.47 and the 99% interval width 12.91, so the 99% interval is wider. With the data held fixed, demanding more confidence raises the critical value $t_{\alpha/2,\,n-1}$ (from 1.796 to 3.106), and a larger multiplier on the same standard error stretches the interval — to be surer of capturing μ you must admit more candidate values.

C5. This is a planning calculation: solve the margin formula $\text{ME} = t_{\alpha/2,\,n-1}\,S/\sqrt{n}$ for n. Since $t_{\alpha/2,\,n-1}$ depends on the still-unknown n and S is not yet observed, use the planning value $S = 1.5$ and the large-n stand-in $z_{0.025} = 1.96$ for the critical value.

(a) Rearrange and substitute:

$$n = \left(\frac{z_{0.025}\,S}{\text{ME}}\right)^2 = \left(\frac{1.96 \times 1.5}{0.40}\right)^2$$
$$= (7.35)^2 = 54.02.$$

Always round *up* so the target margin is met: $n = 55$.

$$\boxed{n = \lceil (1.96 \cdot 1.5/0.40)^2 \rceil = 55\text{ lenses (planning estimate).}}$$

(b) It is only a planning estimate for two reasons: (i) the true S is unknown in advance — we borrowed a pilot value, and the realized margin will overrun if the pilot understated the spread; and (ii) we used $z = 1.96$ in place of $t_{\alpha/2,\,n-1}$, whereas at $df = 54$ the true cutoff $t_{0.025,54} \approx 2.00$ is a touch larger, nudging the requirement up slightly. One verifies after sampling that the realized margin met the goal.

(c) Sample size scales as $1/\,\text{ME}^2$, so halving the target margin from 0.40 to 0.20 multiplies n by $(0.40/0.20)^2 = 4$. The requirement rises about fourfold, to roughly $n = 217$ — the familiar $1/\sqrt{n}$ precision trade-off seen from the other side.

C6. Because σ is *unknown* and replaced by the sample S, the correct pivot is t_9, not $\text{N}(0,1)$. Using z is the classic look-alike error; computing both exposes the under-coverage. The standard error is shared: $S/\sqrt{n} = 4.0/\sqrt{10} = 4.0/3.162 = 1.265$.

(a) The two intervals about $\overline{X} = 22.5$:

$$\begin{aligned}\text{(wrong) } z: &\quad \text{ME}_z = 1.96 \times 1.265 = 2.479, \quad 22.5 \pm 2.479 = (20.02,\ 24.98),\\ \text{(correct) } t: &\quad \text{ME}_t = 2.262 \times 1.265 = 2.861, \quad 22.5 \pm 2.861 = (19.64,\ 25.36).\end{aligned}$$

correct 95% CI : (19.64, 25.36); z interval (20.02, 24.98) under-covers.

(b) The z interval has width 4.96 versus the t interval's width 5.72 — about 13% too narrow. It *looks* more precise, but the precision is fake: with only $n = 10$ the cutoff $t_{0.025,9} = 2.262$ exceeds $z_{0.025} = 1.96$ noticeably because S adds extra wobble that the normal cutoff ignores. A procedure using z when σ was estimated captures the true μ *less* than 95% of the time, so the honest 95% statement is the wider t interval.

C7. An "at least" figure is a one-sided *lower* confidence bound: all $\alpha = 0.05$ in the lower tail with $t_{\alpha,\,n-1} = t_{0.05,13} = 1.771$ ($df = 13$). The standard error is $S/\sqrt{n} = 4.9/\sqrt{14} = 4.9/3.742 = 1.310$.

Step 1: one-sided lower bound.

$$\mu \geq \overline{X} - t_{\alpha,\,n-1}\frac{S}{\sqrt{n}} = 31.2 - 1.771 \times 1.310 = 31.2 - 2.319 = 28.88.$$

95% lower bound : $\mu \geq 28.88$ (hundred CFU).

Step 2: contrast with the two-sided lower endpoint. The two-sided 95% interval uses $t_{0.025,13} = 2.160$, giving lower endpoint $31.2 - 2.160 \times 1.310 = 28.37$. The one-sided bound (28.88) sits *closer* to $\overline{X}$ than the two-sided lower endpoint (28.37), because pouring all 5% into one tail uses the smaller cutoff $1.771 < 2.160$ — the price is that the one-sided statement makes no claim about an upper limit.

D. Hypothesis tests for a normal mean

D1. Two-sided one-sample t-test. The parameter is a mean of an approximately normal diameter, and σ is unknown (S was computed from the sample), so this is the one-sample t-test with $df = n - 1 = 11$. "Differs from" has no direction, so it is two-sided.

$$H_0 : \mu = 25.00 \quad \text{versus} \quad H_1 : \mu \neq 25.00, \qquad \alpha = 0.05.$$

The standard error and statistic are

$$\begin{aligned}\frac{S}{\sqrt{n}} &= \frac{0.084}{\sqrt{12}} = \frac{0.084}{3.464} = 0.02425 \text{ mm},\\ t_{\text{obs}} &= \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{25.06 - 25.00}{0.02425} = \frac{0.06}{0.02425} = 2.474.\end{aligned}$$

The two-sided 5% cutoff on 11 degrees of freedom is $t_{0.025,11} = 2.201$. Since $|t_{\text{obs}}| = 2.474 > 2.201$, the statistic falls in a tail.

$\lvert t_{\text{obs}}\rvert = 2.474 > 2.201 \Rightarrow$ reject H_0 ($df = 11$); at $\alpha = 0.05$ the mean diameter differs significantly from 25.00 mm.

D2. Upper-tail one-sample t-test. Mean trip time, approximately normal, σ unknown (S from the data); the suspicion is that the trips run *longer*, a directional claim, so all of α goes in the upper tail, $df = n - 1 = 8$.

(a) Hypotheses $H_0 : \mu = 34$ versus $H_1 : \mu > 34$. "Longer" points up, so the rejection region is the *upper* tail: reject for large positive t_{obs}.

(b) The standard error and statistic are

$$\frac{S}{\sqrt{n}} = \frac{2.7}{\sqrt{9}} = \frac{2.7}{3} = 0.900 \text{ min},$$
$$t_{\text{obs}} = \frac{36.2 - 34}{0.900} = \frac{2.2}{0.900} = 2.444.$$

The upper-tail cutoff is $t_{0.05,8} = 1.860$. Since $t_{\text{obs}} = 2.444 > 1.860$, reject.

$t_{\text{obs}} = 2.444 > 1.860$ ($df = 8$) $\Rightarrow$ reject H_0 at $\alpha = 0.05$; the mean trip is significantly longer than 34 min.

(c) At $\alpha = 0.01$ the cutoff moves out to $t_{0.01,8} = 2.896$. Now $t_{\text{obs}} = 2.444 < 2.896$, so we would *not* reject. The verdict changes: the evidence is significant at the 5% level but not at the stricter 1% level.

D3. Two-sided t-test from raw data. Mean weighing, approximately normal, σ unknown and not even S supplied — we build S from the six numbers, then run a two-sided one-sample t-test, $df = 5$. The concern is bias in either direction, hence two-sided. First the mean:

$$\sum x_i = 600.024, \qquad \overline{X} = \frac{600.024}{6} = 100.0040 \text{ g}.$$

The squared deviations $x_i - \overline{X}$ (in g^2) sum to

$$\begin{aligned}\text{SS} &= (0.0170)^2 + (-0.0180)^2 + (0.0300)^2 + (-0.0060)^2 + (0.0080)^2 + (-0.0310)^2 \\ &= 0.000289 + 0.000324 + 0.000900 + 0.000036 + 0.000064 + 0.000961 = 0.0025740.\end{aligned}$$

Then

$$S^2 = \frac{\text{SS}}{n-1} = \frac{0.0025740}{5} = 0.00051480, \qquad S = \sqrt{0.00051480} = 0.02269 \text{ g},$$
$$\frac{S}{\sqrt{n}} = \frac{0.02269}{\sqrt{6}} = \frac{0.02269}{2.449} = 0.009263 \text{ g}, \qquad t_{\text{obs}} = \frac{100.0040 - 100.000}{0.009263} = \frac{0.0040}{0.009263} = 0.432.$$

The two-sided 5% cutoff on 5 df is $t_{0.025,5} = 2.571$. Since $|t_{\text{obs}}| = 0.432 < 2.571$, retain.

$|t_{\text{obs}}| = 0.432 < 2.571$ ($df = 5$) $\Rightarrow$ do not reject H_0; no evidence of bias from 100.000 g.

D4. Upper-tail t-test, p-value bracketed. Mean turnaround, approximately normal, σ unknown; the suspicion is that turnaround has *risen*, so the test is upper-tail with $H_0 : \mu = 48$, $H_1 : \mu > 48$, $df = n - 1 = 15$.

(a) Standard error and statistic:

$$\frac{S}{\sqrt{n}} = \frac{4.8}{\sqrt{16}} = \frac{4.8}{4} = 1.200 \text{ h},$$
$$t_{\text{obs}} = \frac{50.4 - 48}{1.200} = \frac{2.4}{1.200} = 2.000, \qquad df = 15.$$

(b) Locate $t_{\text{obs}} = 2.000$ among the t_{15} upper-tail values: since $t_{0.05,15} = 1.753 < 2.000 < 2.131 = t_{0.025,15}$, the area to the right of t_{obs} lies between the matching tail probabilities 0.025 and 0.05:

$$0.025 < p < 0.05.$$

(c) The entire bracket is at or below 0.05, so we reject at $\alpha = 0.05$.

$p \in (0.025, 0.05)$, $df = 15$ $\Rightarrow$ reject H_0 at $\alpha = 0.05$;
evidence that mean turnaround now exceeds 48 h (but not significant at 0.025).

D5. Paired-difference test via the one-sample t-test on differences. Each patient is measured twice, so the columns are dependent: reduce to $D_i = \text{After} - \text{Before}$ and run a one-sample t-test on the D_i against $\mu_0 = 0$, $df = n - 1 = 6$. "Increased" means After exceeds Before on average, so $H_1 : \mu_D > 0$, an upper-tail test. The differences are

$$D_i : \quad 4,\ 2,\ 4,\ 1,\ 5,\ 4,\ 1.$$

Their mean and (computing-form) variance:

$$\begin{aligned}
&\sum D_i = 21, \qquad \overline{D} = \frac{21}{7} = 3.000, \qquad \sum D_i^2 = 79, \\
&S_D^2 = \frac{1}{n-1}\Big(\sum D_i^2 - n\overline{D}^2\Big) = \frac{1}{6}\big(79 - 7 \cdot 3.000^2\big) = \frac{79-63}{6} = \frac{16}{6} = 2.667, \\
&S_D = \sqrt{2.667} = 1.633.
\end{aligned}$$

With $\mu_0 = 0$,

$$\frac{S_D}{\sqrt{n}} = \frac{1.633}{\sqrt{7}} = \frac{1.633}{2.646} = 0.6172, \qquad t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{3.000}{0.6172} = 4.861.$$

The upper-tail cutoff on 6 df is $t_{0.05,6} = 1.943$. Since $t_{\text{obs}} = 4.861 > 1.943$, reject.

$t_{\text{obs}} = 4.861 > 1.943$ ($df = 6$) $\Rightarrow$ reject H_0; the program significantly increased mean grip strength at $\alpha = 0.05$.

D6. Test–CI duality. A 95% t interval (212.4, 219.6) is already in hand, so the duality proposition settles each two-sided $\alpha = 0.05$ test by inspection: reject exactly the claimed means that fall *outside* the interval; no statistic need be recomputed ($df = n - 1 = 19$).

(a) Is 215 inside (212.4, 219.6)? Yes, $212.4 < 215 < 219.6$. By duality the test retains $H_0 : \mu = 215$.

(b) Is 220 inside (212.4, 219.6)? No, $220 > 219.6$. By duality the test rejects $H_0 : \mu = 220$.

(c) In words: a mean hardness of 215 HV is consistent with the data at $\alpha = 0.05$, while a mean of 220 HV is not — the interval, hence the data, rule it out.

$$\begin{aligned}
&\mu_0 = 215 \in (212.4, 219.6) \ \Rightarrow\ \text{retain;} \\
&\mu_0 = 220 \notin (212.4, 219.6) \ \Rightarrow\ \text{reject.}
\end{aligned}$$

D7. t versus z when σ is estimated. Because σ is unknown and replaced by S, the honest reference distribution is t_9, not $\mathrm{N}(0,1)$; comparing to $z_{0.025}$ is the classic look-alike error. Both analysts compute the *same* statistic:

$$\text{stat} = \frac{\overline{X} - \mu_0}{S/\sqrt{n}} = \frac{19.5 - 18.0}{2.2/\sqrt{10}} = \frac{1.5}{2.2/3.162} = \frac{1.5}{0.6957} = 2.156.$$

The wrong (anti-conservative) z call. Comparing to $z_{0.025} = 1.96$: since $|2.156| > 1.96$, the z-user *rejects* H_0 and declares the torque off target.

The correct t call. The proper cutoff is $t_{0.025,9} = 2.262 > 1.96$. Comparing the same 2.156 to 2.262: $|2.156| < 2.262$, so the t-test does *not* reject.

Why the cutoffs differ. Estimating σ by S injects extra variability into the denominator, fattening the tails of the null distribution, so the threshold for "too extreme" must move outward from 1.96 to 2.262. The z-test ignores this and rejects too readily — its true Type I error rate exceeds the advertised 5%. The t-test is correct.

stat = 2.156 : z says reject, t says retain ($df = 9$);
the t-test is correct because σ was estimated and $t_{0.025,9} = 2.262 > 1.96 = z_{0.025}$.

E. Inferences about a normal variance

E1. **Method.** A two-sided range for a normal variance comes from inverting the pivot $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$; the interval is $\left[(n-1)S^2/\chi^2_{\alpha/2,\,n-1},\ (n-1)S^2/\chi^2_{1-\alpha/2,\,n-1}\right]$. With $n = 16$ we have $n-1 = 15$ df and $\alpha = 0.05$, so the tail points are

$$\text{upper } (\alpha/2 = 0.025): \quad \chi^2_{0.025,\,15} = 27.488,$$
$$\text{lower } (1-\alpha/2 = 0.975): \quad \chi^2_{0.975,\,15} = 6.262.$$

The numerator is $(n-1)S^2 = 15 \cdot 7.8 = 117.0$. The *upper*-tail point is the larger divisor, so it produces the *lower* endpoint:

$$\text{lower endpoint} = \frac{117.0}{27.488} = 4.256,$$
$$\text{upper endpoint} = \frac{117.0}{6.262} = 18.684.$$

The estimate $S^2 = 7.8$ sits well below the midpoint 11.47, nearer the lower endpoint — the interval leans right because the pivot is skewed.

$$\boxed{4.26\,\text{g}^2 \ \le\ \sigma^2 \ \le\ 18.68\,\text{g}^2 \quad (95\%\text{ confidence}).}$$

E2. **Method.** Build the variance interval from the χ^2_{n-1} pivot, then square-root both endpoints for σ. Here $n = 10$, so $n-1 = 9$ df, and $\alpha = 0.10$ puts 0.05 in each tail.

(a) Tail points on 9 df: $\chi^2_{0.05,\,9} = 16.919$ and $\chi^2_{0.95,\,9} = 3.325$. The numerator is $(n-1)S^2 = 9 \cdot 0.0064 = 0.0576$, so

$$\text{lower endpoint} = \frac{0.0576}{16.919} = 0.003404,$$
$$\text{upper endpoint} = \frac{0.0576}{3.325} = 0.017323.$$

$$\boxed{0.00340 \ \le\ \sigma^2 \ \le\ 0.01732 \quad (90\%\text{ confidence}).}$$

(b) Square-root both endpoints:

$$\sqrt{0.003404} = 0.05835,$$
$$\sqrt{0.017323} = 0.13162.$$

$$\boxed{0.0583 \ \le\ \sigma \ \le\ 0.1316 \quad (90\%\text{ confidence}).}$$

(c) **No.** The point estimate $S = \sqrt{0.0064} = 0.08$ does not sit at the centre 0.0950 of the σ interval: it is nearer the lower endpoint, because the underlying χ^2 pivot is right-skewed and square-rooting preserves that asymmetry — a variance or standard-deviation interval is never of the symmetric "estimate ± margin" form.

E3. **Method.** Only an upper limit is wanted, so this is a one-sided $100(1-\alpha)\%$ upper confidence bound $\sigma^2 \le (n-1)S^2/\chi^2_{1-\alpha,\,n-1}$, which spends all of α in one tail and uses the *lower*-tail point as the divisor. With $n = 25$ (24 df) and $\alpha = 0.01$, the divisor is the point with area 0.99 to its right:

$$\chi^2_{0.99,\,24} = 10.856.$$

The numerator is $(n-1)S^2 = 24 \cdot 6.76 = 162.24$, so

$$\sigma^2 \ \le\ \frac{162.24}{10.856} = 14.945\,\text{ppb}^2,$$

and square-rooting gives $\sigma \le \sqrt{14.945} = 3.866$ ppb.

$$\boxed{\sigma^2 \le 14.94\,\text{ppb}^2 \quad (\text{equivalently } \sigma \le 3.87\,\text{ppb}), \quad 99\%\text{ upper bound.}}$$

Concentrating all of α in the lower tail makes this bound tighter than the upper endpoint of a two-sided 99% interval from the same data.

E4. Method. A claim about a normal variance with a two-sided alternative calls for the χ^2 test on the statistic $\chi^2_{\text{obs}} = (n-1)S^2/\sigma_0^2 \sim \chi^2_{n-1}$ under H_0, rejecting if the statistic falls below the lower-tail point or above the upper-tail point. With $n = 20$ we have $n - 1 = 19$ df. The statistic is

$$\chi^2_{\text{obs}} = \frac{19 \cdot 0.0039}{0.0016} = \frac{0.0741}{0.0016} = 46.31.$$

Equal tails of 0.025 on 19 df give

$$\text{lower: } \chi^2_{0.975,\,19} = 8.907,$$
$$\text{upper: } \chi^2_{0.025,\,19} = 32.852,$$

so the acceptance region is (8.907, 32.852). Since $46.31 > 32.852$, the statistic clears the upper cutoff.

$$\boxed{\chi^2_{\text{obs}} = 46.31 > 32.852 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence that the shaft-diameter variance differs from spec (it is in fact larger); schedule the maintenance review.

E5. Method. "More variable" fixes a one-sided upper-tail alternative on the variance, so the entire $\alpha = 0.05$ goes in the right tail of χ^2_{n-1} and there is a single cutoff $\chi^2_{\alpha,\,n-1}$. With $n = 13$ we have $n - 1 = 12$ df. The statistic is

$$\chi^2_{\text{obs}} = \frac{12 \cdot 3969}{2025} = \frac{47628}{2025} = 23.52.$$

The single upper-tail critical value on 12 df is

$$\chi^2_{0.05,\,12} = 21.026.$$

Reject if $\chi^2_{\text{obs}} > 21.026$; here $23.52 > 21.026$.

$$\boxed{\chi^2_{\text{obs}} = 23.52 > 21.026 \;\Rightarrow\; \text{reject } H_0;\ \text{response times are significantly more variable.}}$$

The software change has significantly increased the scatter of reaction times.

E6. Method. For a one-sided upper-tail alternative the p-value is the single right-tail area $\mathbb{P}(\chi^2_{12} > 23.52)$, found by bracketing the statistic between two table points. The observed 23.52 lies between

$$\chi^2_{0.025,\,12} = 23.337 \quad \text{(area 0.025 to the right)},$$
$$\chi^2_{0.01,\,12} = 26.217 \quad \text{(area 0.01 to the right)},$$

so the right-tail area is between 0.01 and 0.025. Linear interpolation gives

$$\mathbb{P}(\chi^2_{12} > 23.52) \;\approx\; 0.025 + \frac{23.52 - 23.337}{26.217 - 23.337}\,(0.01 - 0.025) \;=\; 0.0240.$$

$$\boxed{p\text{-value} = \mathbb{P}(\chi^2_{12} > 23.52) \approx 0.024 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

The alternative claims variability in *one* direction only (an increase), so just the right-tail area counts as evidence against H_0 — doubling it would wrongly credit a decrease that nobody is testing. This agrees with the critical-value decision of E5 ($23.52 > 21.026$).

E7. Method. S^2 has a skewed (scaled χ^2) sampling distribution, so the symmetric "estimate $\pm z \cdot$ SE" template from mean inference does not apply; the honest interval comes from inverting the χ^2_{n-1} pivot. Here $n = 10$, so 9 df.

(a) The bogus margin is

$$1.96 \cdot 0.0064\sqrt{2/9} = 1.96 \cdot 0.0064 \cdot 0.4714 = 0.005913,$$

giving the symmetric interval

$$0.0064 \pm 0.005913 = [\,0.000487,\ 0.012313\,],$$

centred on 0.0064 by construction (and its lower end is implausibly close to zero).

(b) The correct interval uses $(n-1)S^2 = 9 \cdot 0.0064 = 0.0576$ with the tail points $\chi^2_{0.025,\,9} = 19.023$ and $\chi^2_{0.975,\,9} = 2.700$:

$$\text{lower} = \frac{0.0576}{19.023} = 0.003028,$$
$$\text{upper} = \frac{0.0576}{2.700} = 0.021333.$$

$$\boxed{0.00303 \;\le\; \sigma^2 \;\le\; 0.02133 \quad (95\%); \quad \text{never } S^2 \pm \text{margin.}}$$

(c) **No**, $S^2 = 0.0064$ is not at the centre of the correct interval, whose midpoint is 0.01218; the estimate sits much nearer the lower endpoint. The symmetric template is invalid because the sampling distribution of S^2 is right-skewed, so the two endpoints are not mirror images about S^2; the $\pm$ shortcut both mis-centres the interval and badly understates the upper reach (correct upper 0.0213 versus the shortcut's 0.0123).

F. Robustness and checking normality

F1. This is a robustness question for a *mean* (parts a–b) contrasted with a *variance* (part c) on the same sample.

(a) **Trustworthy.** The target is a mean, $n = 49$ is moderate-to-large, and the departure from normality is mild (slight skew, one borderline point, near-straight Q-Q plot). The **central limit theorem** makes $(\overline{X} - \mu)/(S/\sqrt{n})$ approximately normal regardless of the parent shape, so the t reference distribution is essentially correct and the nominal 95% coverage is honest. A single mild high point cannot dominate a sample of 49.

(b) With $t_{0.025,\,48} \approx 2.01$ and df $= 48$,

$$\text{margin} = t_{0.025,\,48}\,\frac{S}{\sqrt{n}} = 2.01 \cdot \frac{4.2}{\sqrt{49}} = 2.01 \cdot \frac{4.2}{7} = 2.01 \cdot 0.600 = 1.206,$$
$$\text{CI} = 12.4 \pm 1.206 = (11.19,\ 13.61).$$

$$\boxed{\text{Trust the } t \text{ interval (11.2, 13.6) min: large } n,\ \text{mild skew } \Rightarrow t \text{ is robust.}}$$

(c) **No.** The χ^2 variance procedure is *not* robust at any sample size: the distribution of $(n-1)S^2/\sigma^2$ is governed by the parent's kurtosis, and the CLT does not rescue it. A skewed source has $\kappa \neq 3$, so the true coverage of the nominal 95% variance interval is unknown and likely far from 95%. A large n excuses non-normality for the mean but *not* for the variance.

$$\boxed{\text{Same data: trust the mean CI, distrust the variance CI.}}$$

F2. Small n with a glaring right-tail outlier — the regime where the t procedure fails.

(a) The sum is $1 + 2 + 2 + 3 + 4 + 7 + 25 = 44$, so $\overline{X} = 44/7 = 6.286$, while the median is the 4th of seven ordered values, $= 3$. The mean sits more than twice the median because the lone 25 drags it far above the bulk of the data; an interval centered at 6.29 describes neither a typical site nor the center of the sample.

(b) Removing the 25: $\overline{X} = (44 - 25)/6 = 19/6 = 3.167$. One observation moved the mean by $6.286 - 3.167 = 3.12$ ppm — a statistic that a single point can swing that far cannot have stable t behavior at $n = 7$.

$$\boxed{\overline{X} = 6.29 \text{ vs median 3; dropping one point shifts } \overline{X} \text{ by 3.1 ppm} \Rightarrow t \text{ interval untrustworthy.}}$$

(c) Because the data are positive and strongly right-skewed at small n: (i) apply a **log transform** and run the t procedure on the logs, or (ii) use a **distribution-free** interval based on the median (Chapter 14), which the outlier barely moves. Preferred: the log transform, since the data are positive and multiplicatively spread, the classic case a log symmetrizes; if even the logs were still outlier-laden, fall back to the nonparametric method.

F3. Read each plot shape, then map it to the trust verdict for a *mean* test. (Straight = normal; single bend = skew; S-shape = tail weight; isolated point = outlier.)

(a) **Diagnosis: normal.** A straight Q-Q plot at $n = 45$ is the green light — the t test is **trustworthy**; no remedy needed.

(b) **Diagnosis: right skew.** An upward bow with the low end bunched is the signature of right skew, and $n = 10$ is small, so the CLT has too little to work with — the t test is **not trustworthy**; transform or go nonparametric.

(c) **Diagnosis: heavy tails.** An S that steepens at both ends means symmetric heavy tails ($\kappa > 3$). For a *mean*, symmetry plus $n = 20$ keeps the t test **roughly trustworthy**; the same plot would condemn a *variance* test.

(d) **Diagnosis: outlier.** One isolated point off a straight line at $n = 8$ is a single influential observation; at this small n it can dominate $\overline{X}$ and S, so the t test is **not trustworthy** until the point is investigated or a robust/nonparametric method is used.

(a) normal, trust (b) right skew, don't (c) heavy tails, roughly OK for the mean (d) outlier, don't.

F4. A back-of-envelope robustness argument for a *variance* test under heavy tails, using $\mathrm{Var}(S^2) \approx \sigma^4(\kappa - 1)/n$.

(a) At $n = 41$,

$$\text{normal } (\kappa = 3): \quad \mathrm{Var}(S^2) \approx \sigma^4 \frac{3-1}{41} = \sigma^4 \frac{2}{41} = 0.0488\,\sigma^4,$$

$$\text{heavy } (\kappa = 7): \quad \mathrm{Var}(S^2) \approx \sigma^4 \frac{7-1}{41} = \sigma^4 \frac{6}{41} = 0.1463\,\sigma^4.$$

The actual standard deviation of S^2 is $\sqrt{0.1463/0.0488} = \sqrt{3} = 1.73$ times what the χ^2_{40} reference assumes.

(b) The nominal critical values are chosen to leave 2.5% in each tail of a χ^2_{40} curve whose spread assumes $\kappa = 3$. Against the true distribution of $(n-1)S^2/\sigma^2$, which is 1.73 times as wide, those same cutoffs leave far more than 2.5% in each tail, so the true two-sided rejection rate under H_0 balloons well above the nominal 5% — the test is anti-conservative.

(c) The ratio of the two variances is

$$\frac{\sigma^4(\kappa-1)/n}{\sigma^4(3-1)/n} = \frac{\kappa-1}{3-1} = \frac{6}{2} = 3$$

for *every* n — the n cancels. Enlarging n shrinks both variances together but never closes the fixed factor-of-3 gap in spread, so the χ^2 test stays miscalibrated at any sample size.

Heavy tails mis-level the χ^2 variance test by a fixed factor (here 3) independent of n.

F5. Same data, two parameters, two robustness verdicts.

(a) Mean CI with $t_{0.025,\,29} = 2.045$, df = 29:

$$\text{margin} = 2.045 \cdot \frac{2.0}{\sqrt{30}} = 2.045 \cdot \frac{2.0}{5.477} = 2.045 \cdot 0.3651 = 0.747,$$

$$\text{mean CI} = 6.0 \pm 0.747 = (5.25,\ 6.75).$$

Coverage is close to 95%: the mean rides the CLT, which tames moderate skew at $n = 30$.

(b) Nominal variance CI, with $S^2 = 2.0^2 = 4.0$, $\chi^2_{0.975,\,29} = 16.047$, $\chi^2_{0.025,\,29} = 45.722$:

$$\text{lower} = \frac{(n-1)S^2}{\chi^2_{0.025,\,29}} = \frac{29 \cdot 4.0}{45.722} = \frac{116}{45.722} = 2.54,$$
$$\text{upper} = \frac{(n-1)S^2}{\chi^2_{0.975,\,29}} = \frac{29 \cdot 4.0}{16.047} = \frac{116}{16.047} = 7.23,$$

so the nominal variance CI is (2.54, 7.23), but its *true* coverage is unknown and likely far from 95% because the source is skewed.

(c) Trust the mean CI (5.25, 6.75); flag the variance CI (2.54, 7.23) as unreliable. The verdicts differ because the mean CI inherits the CLT cushion and survives moderate skew, whereas the variance CI inherits the parent's skew and kurtosis directly with no such cushion.

Trust (5.25, 6.75) for the mean; distrust (2.54, 7.23) for the variance.

F6. A log transform restores approximate symmetry so the t procedure is valid on the log scale.

(a) Taking $y = \ln x$:

$$\ln 3 = 1.10,\ \ln 4 = 1.39,\ \ln 6 = 1.79,\ \ln 11 = 2.40,$$
$$\ln 20 = 3.00,\ \ln 55 = 4.01,\ \ln 148 = 5.00.$$

The successive gaps are 0.29, 0.40, 0.61, 0.60, 1.01, 0.99 — comparable in size rather than exploding by orders of magnitude — and the values are roughly balanced about their middle (2.40). The long right tail is gone, so a Q-Q plot of the y's would be far straighter.

(b) With $\overline{Y} = 2.670$, $\sum(y - \overline{Y})^2 = 12.66$, so $S_y^2 = 12.66/6 = 2.110$ and $S_y = 1.453$. Then with $t_{0.025,\,6} = 2.447$, df $= 6$,

$$\text{margin} = 2.447 \cdot \frac{1.453}{\sqrt{7}} = 2.447 \cdot \frac{1.453}{2.646} = 2.447 \cdot 0.5491 = 1.344,$$
$$\text{CI for } \mu_{\ln X} = 2.670 \pm 1.344 = (1.326,\ 4.014).$$

(c) Exponentiating the endpoints, $(e^{1.326},\, e^{4.014}) = (3.77,\, 55.4)$. Because a monotone transform preserves *quantiles*, this is a 95% interval for the **median** solvent mass on the original scale — not for the mean.

Log-scale 95% t CI: (1.33, 4.01); back-transformed median CI: (3.8, 55.4) g.

F7. Match each symptom to its standard remedy.

(a) **Log transform.** Positive, multiplicatively skewed counts over orders of magnitude are the log-transform signature; ln(count) should be roughly normal, then run the t procedure on the logs, which directly answers the stated question about the mean log-count.

(b) **Nonparametric method.** A log will not symmetrize already-symmetric data, and the outliers violate normality in the tails; use a distribution-free method (sign or Wilcoxon signed-rank test, Chapter 14) that depends on ranks, so the outliers carry little weight.

(c) **Collect more data.** Mild skew is exactly the case the CLT cures; cheaply running enough extra replicates to reach $n \geq 30$ makes the t procedure on the raw data trustworthy with no transform at all.

(d) A transform changes *which parameter* you are inferring about. The χ^2 test on the logs is a valid test about $\text{Var}(\ln X)$, but the manager asked about $\text{Var}(X)$ on the raw scale, and variance — unlike a median — does *not* back-transform cleanly: $\text{Var}(\ln X) \neq g(\text{Var}(X))$ for any fixed g, so a result on the logs says nothing direct about $\sigma_X^2 = 3000$. Running χ^2 on the raw masses *is* about σ_X^2, but the raw data are severely non-normal and the variance test is not robust, so that test is invalid — exactly the problem the log dodged for the mean.

(a) log-transform (b) nonparametric (Ch. 14) (c) collect more data (d) a log validates the *mean* test but leaves the raw-scale *variance* claim unanswerable.

G. Choosing the right normal-theory procedure (synthesis)

G1. The recognition rule: a *mean* with σ unknown uses t_{n-1}; a *variance* or standard deviation uses χ^2_{n-1}; a question that fixes *all* parameters (including a known σ) and merely asks for a chance is a plain sampling-distribution probability with no estimation, hence the normal (or, for S^2, the χ^2) distribution used directly.

(a) σ is *known*, no parameter is estimated, and we only want a probability about $\overline{X}$. **Plain sampling-distribution probability**: $\overline{X} \sim \mathrm{N}(\mu, \sigma^2/n)$, standardize with z (no df). The cue is "known σ, find a chance."

(b) A *range* for a normal *mean* with σ unknown. **One-sample t confidence interval**, reference t_{19} ($df = n - 1 = 19$). The cue is mean + unknown σ + "plausible values."

(c) A claim about a *standard deviation* with a directional "has grown." **One-sided (χ^2) variance test** on $(n-1)S^2/\sigma_0^2$, reference χ^2_{24} ($df = 24$). The cue is spread + a directional hypothesis.

(d) Each subject measured *twice*, asking about the mean change. **Paired t-test** on the differences, reference t_{29} ($df = n - 1 = 29$). The cue is before/after on the same unit.

(e) A two-sided claim about σ. **Two-sided (χ^2) variance test** (square the hypothesized σ to a variance), reference χ^2_{17} ($df = 17$). The cue is spread + "differs."

(f) A directional claim about a *mean* with σ unknown. **One-sided (upper) one-sample t-test**, reference t_{11} ($df = 11$). The cue is mean + unknown σ + "exceeds."

mean, σ unknown $\to t_{n-1}$; variance $\to \chi^2_{n-1}$; σ known, find a chance $\to$ direct normal.

G2. Procedure. Two distinct questions ride on one sample: the mean uses a one-sample t-test (σ unknown, estimated by S), the variance uses a χ^2 test on $(n-1)S^2/\sigma_0^2$. Compute the summaries once, then feed both.

(a) *Two sums.*

$$\sum x_i = 79.98, \qquad \overline{X} = \frac{79.98}{10} = 7.998,$$
$$\sum x_i^2 = 639.6994.$$

Computing form.

$$\sum x_i^2 - n\overline{X}^2 = 639.6994 - 10(7.998)^2 = 639.6994 - 639.68004 = 0.01936,$$
$$S^2 = \frac{0.01936}{9} = 0.0021511 \text{ mm}^2, \qquad S = \sqrt{0.0021511} = 0.04638 \text{ mm}.$$

(b) *Mean test (t, because σ is unknown).* $H_0\colon \mu = 8.00$ versus $H_1\colon \mu \neq 8.00$ at $\alpha = 0.05$. The standard error and statistic are

$$\mathrm{SE} = \frac{S}{\sqrt{n}} = \frac{0.04638}{\sqrt{10}} = \frac{0.04638}{3.162} = 0.014667 \text{ mm},$$
$$t_{\mathrm{obs}} = \frac{\overline{X} - \mu_0}{\mathrm{SE}} = \frac{7.998 - 8.00}{0.014667} = \frac{-0.002}{0.014667} = -0.136.$$

Since $|t_{\mathrm{obs}}| = 0.136 < t_{0.025,\,9} = 2.262$, do not reject.

$\lvert t_{\mathrm{obs}}\rvert = 0.136 < 2.262 \;\Rightarrow$ retain H_0; no evidence the mean is off 8.00 mm.

A z-test is wrong here: σ was estimated by S, so the honest reference is t_9.

(c) *Variance test (χ^2, because the parameter is a variance).* $H_0\colon \sigma^2 = 0.0020$ versus $H_1\colon \sigma^2 > 0.0020$ at $\alpha = 0.05$ (one-sided upper tail, since exceeding the contract limit is the worry). With $(n-1)S^2 = 9 \cdot 0.0021511 = 0.01936$,

$$\chi^2_{\text{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{0.01936}{0.0020} = 9.68.$$

The upper-tail cutoff is $\chi^2_{0.05,\,9} = 16.919$. Since $9.68 < 16.919$, do not reject.

$$\boxed{\chi^2_{\text{obs}} = 9.68 < 16.919 \;\Rightarrow\; \text{retain } H_0;\ \text{no evidence } \sigma^2 \text{ exceeds } 0.0020 \text{ mm}^2.}$$

This is a χ^2 test, not a t-test, because the hypothesis is about *spread*, not center.

(d) *Combined verdict.* The bearings are on target (μ consistent with 8.00 mm) *and* precise enough (no evidence σ^2 breaches the contract limit). The lot passes both checks at the 5% level.

G3. Procedure. The fork is whether σ is known. A known σ makes the standardized mean exactly normal, so A computes a probability with z and no test machinery; an unknown σ forces B onto t_{n-1} for an actual decision.

(a) *Plain normal probability.* Nothing is estimated — $\sigma = 6.0$ is known and $\mu = 50$ is given — so this is a direct statement about $\overline{X} \sim \text{N}(50,\, 6.0^2/9)$, not a hypothesis test. Standardize:

$$\frac{\sigma}{\sqrt{n}} = \frac{6.0}{\sqrt{9}} = \frac{6.0}{3} = 2.0,$$
$$z = \frac{51.4 - 50}{2.0} = \frac{1.4}{2.0} = 0.70,$$
$$\mathbb{P}(\overline{X} \geq 51.4) = \mathbb{P}(Z \geq 0.70) = 0.2420.$$

$$\boxed{\mathbb{P}(\overline{X} \geq 51.4) = 0.2420 \quad \text{(direct normal, } \sigma \text{ known, no df).}}$$

(b) *t-test (because σ is unknown).* $H_0\colon \mu = 50$ versus $H_1\colon \mu > 50$ at $\alpha = 0.05$. Here $S = 6.3$ replaces σ, so the reference is t_8:

$$\text{SE} = \frac{S}{\sqrt{n}} = \frac{6.3}{\sqrt{9}} = \frac{6.3}{3} = 2.10,$$
$$t_{\text{obs}} = \frac{51.4 - 50}{2.10} = \frac{1.4}{2.10} = 0.667.$$

The upper-tail cutoff is $t_{0.05,\,8} = 1.860$. Since $0.667 < 1.860$, do not reject.

$$\boxed{t_{\text{obs}} = 0.667 < 1.860 \;\Rightarrow\; \text{retain } H_0;\ \text{no evidence } \mu > 50.}$$

(c) *Why the references differ.* A's denominator $\sigma/\sqrt{n} = 2.0$ is a fixed constant, so the standardized mean is exactly $\text{N}(0,1)$. B's denominator $S/\sqrt{n}$ is *random* (computed from the same sample), adding a second source of variability that fattens the tails into t_8. Identical $\overline{X}$ values do not change which distribution is honest — only whether σ was known does.

G4. Procedure. Both a t interval for the mean and a χ^2 interval for the variance *assume a normal parent*; the recognition step here is to check that assumption *before* trusting either, because the variance procedure has no robustness cushion.

(a) *Shared assumption and the first picture.* Both procedures require the peak flows to be (at least approximately) **normal**. The first diagnostic to inspect is a **normal Q-Q plot** (supplemented by a boxplot for outliers): a roughly straight plot supports normality, a sharp upward bend at the top signals right skew or an outlier.

(b) *Mean vs median.* $\sum x_i = 164$, so

$$\overline{X} = \frac{164}{8} = 20.5, \qquad \text{median} = \frac{14+15}{2} = 14.5.$$

The mean exceeds the median by 6.0, a large right-shift for so small a sample; that gap is the fingerprint of right skew, and the single observation 61 — far above the cluster 12–19 — is responsible (dropping it gives $\overline{X} = (164-61)/7 = 14.7$, close to the median).

$$\boxed{\overline{X} = 20.5 \gg \text{median} = 14.5 \;\Rightarrow\; \text{skew/outlier; normality in doubt (the value 61).}}$$

(c) *Which interval is more endangered.* The *variance* interval is far more endangered. The t interval for the mean leans on the central limit theorem and tolerates mild non-normality, though at $n = 8$ a lone extreme point still strains it. The χ^2 variance interval has **no** robustness: its coverage is driven by the parent's kurtosis, and a heavy upper tail wrecks it at any n.

What to do. For the mean, consider a log transform or a distribution-free interval (small n plus an outlier); for the variance, do *not* report the χ^2 interval until normality is established — the lone 61 likely disqualifies it.

G5. Procedure. Each serum sample is measured by *both* kits, so the two columns are dependent; collapse to differences and run a one-sample t-test on the D_i against 0.

(a) *Why paired.* The same sample feeds both kits, so kit X and kit Y readings on a given sample are correlated; treating them as two independent samples would ignore that pairing and inflate the noise. Pairing cancels the sample-to-sample variation. Reference: $t_{n-1} = t_6$ ($df = 6$).

(b) *Differences.* $D_i = X_i - Y_i$:

$$D_i: \quad 3,\ 2,\ 6,\ 0,\ 4,\ 2,\ 6.$$

Mean and SD of differences. $\sum D_i = 23$, so $\overline{D} = 23/7 = 3.286$. With $\sum D_i^2 = 9+4+36+0+16+4+36 = 105$ and the computing form,

$$S_D^2 = \frac{1}{6}\big(105 - 7\cdot 3.286^2\big) = \frac{1}{6}\big(105 - 75.59\big) = \frac{29.41}{6} = 4.902,$$
$$S_D = \sqrt{4.902} = 2.214 \text{ mg/dL}.$$

Statistic. With $\mu_0 = 0$ and $S_D/\sqrt{n} = 2.214/\sqrt{7} = 2.214/2.646 = 0.8371$,

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{3.286}{0.8371} = 3.925.$$

Two-sided cutoff $t_{0.025,\,6} = 2.447$; since $|3.925| > 2.447$, reject.

$$\boxed{t_{\text{obs}} = 3.925 > 2.447 \;\Rightarrow\; \text{reject } H_0;\ \text{the kits differ on average (X reads higher).}}$$

(c) *Matching interval and duality.* The margin is $t_{0.025,\,6}\, S_D/\sqrt{n} = 2.447 \cdot 0.8371 = 2.048$, so

$$\overline{D} \pm 2.048 = 3.286 \pm 2.048 = (1.24,\ 5.33) \text{ mg/dL}.$$

The hypothesized $\mu_D = 0$ lies *outside* (1.24, 5.33), which by the test–CI duality is exactly equivalent to rejecting H_0 — the two procedures agree.

$$\boxed{95\%\text{ CI for } \mu_D:\ (1.24,\ 5.33);\ 0 \notin \text{CI} \;\Rightarrow\; \text{reject, consistent with (b).}}$$

G6. Procedure. Both parts fix all parameters (normal parent, *known* $\sigma = 2.0$), so each is a plain sampling-distribution probability — (a) about $\overline{X}$ via the normal, (b) about S^2 via the χ^2 law of $(n-1)S^2/\sigma^2$. No estimation, hence no t.

(a) *Normal probability for $\overline{X}$.* $\overline{X} \sim \mathrm{N}(90,\, 2.0^2/16)$, so $\sigma/\sqrt{n} = 2.0/4 = 0.50$ and

$$z = \frac{91 - 90}{0.50} = 2.00, \qquad \mathbb{P}(\overline{X} > 91) = \mathbb{P}(Z > 2.00) = 0.0228.$$

No t appears because σ is known — the denominator is a constant, not the random S.

$$\boxed{\mathbb{P}(\overline{X} > 91) = 0.0228.}$$

(b) *χ^2 probability for S^2.* For a normal sample, $(n-1)S^2/\sigma^2 \sim \chi^2_{15}$. Convert the event:

$$S^2 > 7.20 \iff \frac{(n-1)S^2}{\sigma^2} > \frac{15 \cdot 7.20}{4} = 27.0.$$

Locate 27.0 between the table points $\chi^2_{0.05,\,15} = 24.996$ and $\chi^2_{0.025,\,15} = 27.488$: since $24.996 < 27.0 < 27.488$, the upper-tail area lies between 0.025 and 0.05, i.e.

$$\mathbb{P}(S^2 > 7.20) = \mathbb{P}(\chi^2_{15} > 27.0) \in (0.025,\ 0.05).$$

$$\boxed{\mathbb{P}(S^2 > 7.20) = \mathbb{P}(\chi^2_{15} > 27.0) \in (0.025,\, 0.05).}$$

(c) *Contrast.* Part (a) measures how the process *center* fluctuates (the chance the mean drifts a full nm high); part (b) measures how the process *spread* fluctuates (the chance the sample variance runs unusually large). Same data, two different aspects — one normal, one χ^2.

G7. Procedure. The specification names two parameters, so it needs two procedures: a t-test for the mean (σ unknown) and a one-sided χ^2 test for the variance.

(a) *Two questions.* (i) Is the mean on target at 500 mL? $\Rightarrow$ one-sample t-test, reference t_{19}. (ii) Is the spread within the $\sigma_0 = 1.5$ mL limit? $\Rightarrow$ upper-tail χ^2 variance test on σ^2, reference χ^2_{19}.

(b) *Mean test.* $H_0\colon \mu = 500$ versus $H_1\colon \mu \neq 500$ at $\alpha = 0.05$.

$$\mathrm{SE} = \frac{S}{\sqrt{n}} = \frac{2.1}{\sqrt{20}} = \frac{2.1}{4.472} = 0.4696 \text{ mL},$$
$$t_{\mathrm{obs}} = \frac{499.2 - 500}{0.4696} = \frac{-0.8}{0.4696} = -1.704.$$

Since $|t_{\mathrm{obs}}| = 1.704 < t_{0.025,\,19} = 2.093$, do not reject.

$$\boxed{|t_{\mathrm{obs}}| = 1.704 < 2.093 \ \Rightarrow\ \text{retain } H_0;\ \text{mean consistent with 500 mL.}}$$

(c) *Variance test.* $H_0\colon \sigma^2 = 2.25$ versus $H_1\colon \sigma^2 > 2.25$ at $\alpha = 0.05$. With $S^2 = 2.1^2 = 4.41$ and $(n-1)S^2 = 19 \cdot 4.41 = 83.79$,

$$\chi^2_{\mathrm{obs}} = \frac{(n-1)S^2}{\sigma_0^2} = \frac{83.79}{2.25} = 37.24.$$

The upper-tail cutoff is $\chi^2_{0.05,\,19} = 30.144$. Since $37.24 > 30.144$, reject.

$$\boxed{\chi^2_{\mathrm{obs}} = 37.24 > 30.144 \ \Rightarrow\ \text{reject } H_0;\ \text{the spread exceeds the 1.5 mL limit.}}$$

So the line is *centered* correctly but too *variable* — the two procedures answer different questions and reach different verdicts.

(d) *z vs t for the mean.* Comparing the same statistic $|-1.704| = 1.704$ to $z_{0.025} = 1.96$ would *also* retain ($1.704 < 1.96$), so here the decision does not flip. But the honest reference is still t_{19}, not the normal: σ was estimated by S, so the denominator $S/\sqrt{n}$ is random and the null distribution has the heavier t_{19} tails (cutoff $2.093 > 1.96$). Using z understates how extreme the data must be and, in borderline cases, would over-reject — its true Type I error exceeds the advertised 5%.

$$\boxed{\text{No flip here, but use } t_{19}\ (2.093),\ \text{not } z\ (1.96),\ \text{whenever } \sigma \text{ is estimated.}}$$

Chapter 8

Types of Data

Every inference procedure in this book answers a specific kind of question about a specific kind of data. Choose the wrong procedure for your data and the answer is meaningless, however careful the arithmetic. This short chapter is the *map*: it classifies data by **measurement type** (categorical vs quantitative), classifies studies by how the data were **collected** (designed experiment vs observational study), classifies problems by their **sample structure** (one sample, two independent samples, paired samples, several samples, tables of counts), and then matches each combination to the right tool from Chapters 6–14.

Master guide: reading a problem before you compute

- **What is measured on each unit?** A **category** (yes/no, blood type, brand) ⇒ categorical data ⇒ proportions, counts, χ^2 methods. A **number** (height, time, concentration) ⇒ quantitative data ⇒ means, variances, t/F methods (§1).
- **How were the data produced?** A **randomized experiment** (the analyst assigns treatments) supports causal conclusions; an **observational study** supports only association, and is open to **confounding** (§2).
- **How many samples, and are they linked? One** sample, **two independent** samples, **paired** (two measurements on the same unit), k samples, or a **table of counts**? The structure picks the test (§3).
- **Put it together.** Data type + sample structure + the question (mean? variance? proportion? association? fit?) ⇒ one procedure on the decision map (§4).

Keep straight: "two samples" that are actually two measurements on the *same* subjects are **paired**, not independent — a classic and costly mix-up. And *association is not causation* unless the data came from a randomized experiment.

8.1 Types of Data

Before any formula is chosen, a single question settles which half of the book applies: *what is recorded on each individual?* If the recorded value is a label or a category — a species, a yes/no answer, a survey grade — the data are **categorical** and the legitimate summaries are counts and proportions, analyzed later with χ^2 methods. If the recorded value is a genuine number on which arithmetic means something — a weight, a reaction time, a concentration — the data are **quantitative** and the legitimate summaries are means and variances, analyzed with t and F methods. Within the quantitative side the measurement splits finer into **interval** and **ratio** scales, and the count of possible values splits into **discrete** and **continuous**. The whole point of this section is that the data type is not cosmetic: it is a hard constraint that decides which numbers you are even *allowed* to compute. Averaging a column of zip codes, or running a t test on letter grades coded $1, 2, 3$, produces output that the software will happily print and that means nothing. Recognize the type first, and the menu of valid procedures shrinks to something you can choose from honestly.

Which data type is on each unit?

- Each value is a **label or category** (blood type, brand, pass/fail)? ⇒ **categorical**; summarize with counts and proportions, test with χ^2.
- The labels have a **natural order** but no fixed spacing (mild/moderate/severe, star ratings)? ⇒ categorical and specifically **ordinal**; you may rank, but *not* average.
- Each value is a **number you can do arithmetic on** (height, time, dose)? ⇒ **quantitative**; summarize with $\overline{X}$ and S, test with t/F.
- A quantitative value has **no true zero**, so only *differences* are meaningful (calendar year, temperature in °C)? ⇒ **interval**. It has a true zero, so *ratios* are meaningful (mass, count, dollars)? ⇒ **ratio**.
- Quantitative but only isolated values are possible (number of defects)? ⇒ **discrete**; any value in an interval is possible (exact weight)? ⇒ **continuous**.

Do not confuse a numeric code with a number: jersey numbers, zip codes, and route numbers are written with digits but are *nominal labels* — their average is meaningless. And do not treat *ordinal* ratings as if their gaps were equal: a jump from "fair" to "good" need not equal "good" to "excellent."

8.1.1 Units, populations, and samples

Definition 8.1.1 (Experimental unit, population, sample, variable)**.** An **experimental unit** (or *unit*, or case) is the single individual on which a measurement is taken — one patient, one circuit board, one lake, one voter. The **population** is the entire collection of units we want to describe; the **sample** is the subset actually measured. A **variable** is the characteristic recorded on each unit; its realized value on a given unit is the **datum**. A study may record several variables on each unit at once.

Remark. The unit is whatever the measurement is *per*. "Per patient" makes the patient the unit; "per blood draw" makes the draw the unit, and a patient who is drawn three times contributes three units — a distinction that will matter enormously in §3 when we ask whether observations are independent. Pin down the unit before classifying the variable, because the same word ("weight") is a different variable on a different unit.

Example 8.1.2 (Naming the unit and the variable)**.** A regional hospital network reviews its emergency departments. For each of the 14 hospitals it records: the hospital's name, its trauma-level designation (I, II, III, or IV), the median door-to-doctor wait in minutes, and whether the hospital met a 30-minute target last quarter. Identify the experimental unit and classify each of the four recorded variables as categorical or quantitative.

Which rule, and why? Classification starts by fixing the unit — the thing the row of data is "per." Here every value is recorded once per hospital, so the hospital is the unit; then each variable is typed by asking whether its value is a label or a number you can average.

Solution.

Step 1: the unit. Each measurement is taken per hospital, and there are 14 rows, so the experimental unit is one **hospital**; the population is all hospitals in the network and the sample is these 14.

Step 2: type each variable. Name is a label (categorical). Trauma level I–IV is a label, and analyzing it as the numbers 1–4 would be illegitimate (categorical). Median wait in minutes is a genuine number you can average (quantitative). "Met the target?" is a yes/no answer (categorical).

unit = hospital; categorical: name, trauma level, target; quantitative: median wait.

8.1.2 Categorical versus quantitative

Definition 8.1.3 (Categorical and quantitative data)**.** A variable is **categorical** (qualitative) if each value places the unit into one of a set of categories; the data are summarized by how many units fall in each category (counts) and what fraction do (proportions). A variable is **quantitative** (numerical) if each value is a number for which arithmetic — in particular averaging — is meaningful; the data are summarized by measures of center ($\overline{X}$) and spread (S^2). The legitimacy test is direct: if computing the average of two values

produces something interpretable, the variable is quantitative; if not, it is categorical, however the values are written.

Proposition 8.1.4 (The data type fixes the legal summary and the legal test)**.** *For categorical data the only model-free summaries are category counts and proportions $\widehat{p}$, and comparisons across groups are tests on proportions or χ^2 tests on a table of counts. For quantitative data the summaries are $\overline{X}$ and S^2, and comparisons are t tests (one or two means) or* F *tests (several means, or variances). Applying a mean-based procedure to categorical data, or a count-based procedure to quantitative data, is a type error, not merely a suboptimal choice.*

Why averaging fails on categories. A summary statistic must be invariant to any relabeling the variable permits. Categorical labels carry no fixed numerical identity: coding eye color as brown= 1, blue= 2, green= 3 versus brown= 2, blue= 1, green= 3 describes the *same* data, yet the two codings give different "averages." A statistic that changes when you merely rename the categories cannot be measuring anything about the units, so the mean is not a valid summary of nominal data. Counts and proportions, by contrast, are unchanged by relabeling — they count membership, not code values — which is exactly why they are the legitimate summaries. □

Example 8.1.5 (Choosing the summary statistic that is even legal)**.** A genomics core surveys 500 submitted samples. Two columns are: *organism* (human, mouse, zebrafish, other) and *RNA integrity number* (RIN, a quality score from 1.0 to 10.0). A junior analyst proposes "the average organism" and "the proportion of samples with RIN above 7." Which proposed summary is legitimate for which variable, and what is the correct summary for the other?

Which rule, and why? Each column is classified first; the legal summary follows from the type. Organism is a nominal label, so proportions/counts are its only honest summaries; RIN is a genuine number, so a mean (and a proportion above a threshold) both make sense.

Solution.

Step 1: type each column. Organism is categorical (a label with no numeric meaning). RIN is quantitative (a number you can average and compare).

Step 2: judge the proposals. "Average organism" is meaningless — there is no arithmetic on labels — so report instead the *count or proportion in each organism category*, e.g. $\widehat{p}_{\text{human}} = 310/500 = 0.62$. "Proportion with RIN > 7" is perfectly legitimate: it dichotomizes the quantitative RIN into a yes/no event and reports its fraction, say $\widehat{p} = 0.71$.

Step 3: the natural quantitative summary. For RIN itself the standard summary is the mean and standard deviation, e.g. $\overline{X} = 7.4$, $S = 1.3$.

organism $\to$ proportions only; RIN $\to$ mean $\overline{X}$, and a threshold proportion is fine.

8.1.3 The four measurement scales

Definition 8.1.6 (Nominal, ordinal, interval, ratio)**.** Categorical data come in two scales. **Nominal**: unordered labels (blood type, county, brand); the only relations are "same" or "different." **Ordinal**: labels with a natural order but no meaningful spacing (survey grade poor/fair/good/excellent, tumor stage I–IV); you may say "greater" but not "how much greater." Quantitative data also come in two scales. **Interval**: a numeric scale on which *differences* are meaningful but there is no true zero, so ratios are not (temperature in °C, calendar year, IQ score). **Ratio**: a numeric scale with a genuine zero meaning "none," so both differences and *ratios* are meaningful (mass, length, count, dollars, kelvin temperature). The four form a ladder of increasing structure: nominal < ordinal < interval < ratio.

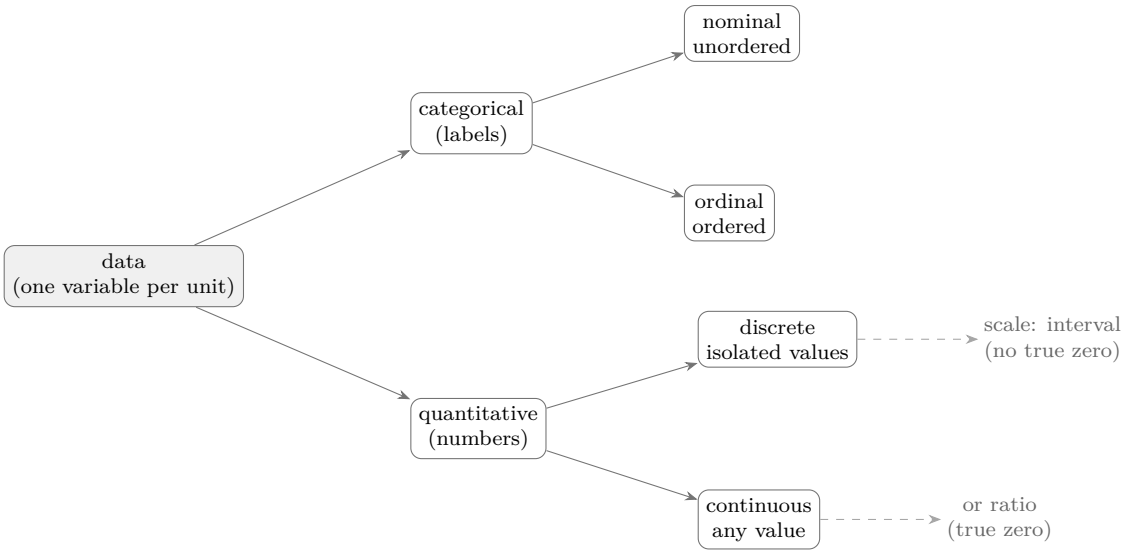


Figure: the taxonomy tree. The first split is categorical (labels) versus quantitative (numbers). Categorical data are nominal (unordered) or ordinal (ordered); quantitative data are discrete (isolated values) or continuous (any value in a range). The interval-versus-ratio distinction — whether the scale has a true zero — rides on top of the quantitative branch and is noted at right.

Example 8.1.7 (Climbing the ladder: classify four numbers)**.** Classify each of the following to its finest scale (nominal, ordinal, interval, or ratio): (a) the *model year* stamped on a car (e.g. 2019); (b) a customer's *satisfaction rating* on a 1–5 stars scale; (c) the *outdoor temperature* in degrees Celsius; (d) the *annual rainfall* in millimeters at a weather station.

Which rule, and why? The scale is found by two yes/no questions: is there an order, and if numeric is there a true zero making ratios meaningful? Order without spacing is ordinal; numeric without a true zero is interval; numeric with a true zero is ratio.

Solution.

Step 1: model year. Numeric and ordered, but year 0 is an arbitrary epoch, not "no time"; 2020 is not "twice" 1010. Differences in years are meaningful, ratios are not: **interval**.

Step 2: star rating. Ordered labels with unequal, undefined spacing (4 stars is not "twice as satisfied" as 2): **ordinal**.

Step 3: temperature in °C. 0°C is the freezing point, not absence of heat, so 20°C is not twice as hot as 10°C: **interval**.

Step 4: rainfall in mm. 0 mm means genuinely no rain, and 40 mm is twice 20 mm: **ratio**.

year: interval; stars: ordinal; °C: interval; rainfall: ratio.

8.1.4 Numeric-looking codes that are really nominal

Example 8.1.8 (Digits that are labels, not quantities)**.** A logistics database stores, for each shipment, the *ZIP code* of the destination, the *driver ID number*, the *number of parcels*, and the *delivery temperature zone* coded 1 = ambient, 2 = refrigerated, 3 = frozen. A dashboard reports "mean ZIP code," "mean driver ID," "mean parcels," and "mean temperature zone." Which of these four means is legitimate, and reclassify each variable.

Which rule, and why? The trap is that all four columns hold digits, but only one is a quantity. The test is whether averaging the codes means anything; for ZIP, driver ID, and zone the digits are *names*, so their mean is nonsense even though software computes it.

Solution.

Step 1: ZIP code. A geographic label; 90210 is not "larger" than 10001 in any quantitative sense, and the average of two ZIPs need not even be a real ZIP. **Nominal**; mean is meaningless.

Step 2: driver ID. An identifier; the digits index people, not amounts. **Nominal**; mean is meaningless.

Step 3: number of parcels. A genuine count with a true zero; averaging is fine. **Ratio** (and discrete); mean is legitimate.

Step 4: temperature zone 1/2/3. The codes are ordered (ambient < refrigerated < frozen in coldness) but the gaps are not equal degrees, so the mean is not interpretable. **Ordinal**; report the mode or proportions, not the mean.

Only "mean parcels" is legitimate; ZIP and driver ID are nominal, zone is ordinal.

Remark. A fast field test for a numeric column: would shuffling the assignment of code numbers to categories change the answer? If yes, the codes are quantities (you cannot relabel "5 parcels" as "2 parcels"). If no — you could swap which ZIP is called 90210 with no loss — the codes are nominal labels wearing a numeric costume.

8.1.5 Discrete versus continuous quantitative data

Definition 8.1.9 (Discrete and continuous)**.** A quantitative variable is **discrete** if its possible values are isolated points, typically counts: $0, 1, 2, \ldots$ with nothing in between (number of seizures this month, number of defective welds). It is **continuous** if it can take any value in an interval, limited only by the precision of the instrument (exact body mass, elapsed time, voltage). Counting questions ("how many?") yield discrete data; measuring questions ("how much?") yield continuous data.

Example 8.1.10 (How many versus how much)**.** For a fitness study each subject contributes: (a) the number of completed push-ups in one minute; (b) the resting heart rate in beats per minute, read off a monitor to the nearest integer; (c) the elapsed time to run 1.5 miles, in seconds; (d) the number of distinct exercise classes attended last month. Classify each as discrete or continuous, and note its scale.

Which rule, and why? Discrete-versus-continuous is decided by whether the underlying quantity is counted or measured, *not* by whether the recorded number happens to be a whole number. A measured quantity rounded to an integer is still continuous — rounding is a reporting choice, not a property of the variable.

Solution.

Step 1: push-ups. A count of completed repetitions; values are $0, 1, 2, \ldots$ **discrete**, ratio scale (a true zero, and 20 is twice 10).

Step 2: resting heart rate. Heart rate is an underlying continuous quantity (beats per unit time); recording it "to the nearest integer" rounds a continuous measurement. **Continuous**, ratio scale.

Step 3: run time. Elapsed time can be any positive real; **continuous**, ratio scale.

Step 4: classes attended. A count of attendances; **discrete**, ratio scale.

discrete: push-ups, classes; continuous: heart rate, run time; all ratio.

8.1.6 A reference table of variables and their types

The following examples consolidate the classification. Read each variable, ask the two questions (label or number? ordered/true zero?), and confirm the entry.

Variable (per unit)	Type and scale
Blood type (A, B, AB, O)	categorical, nominal
Pain level: none/mild/moderate/severe	categorical, ordinal
Telephone area code	categorical, nominal (numeric code)
Final letter grade A–F	categorical, ordinal
Number of emergency-room visits this year	quantitative, discrete, ratio
Body temperature in °C	quantitative, continuous, interval
Calendar year of diagnosis	quantitative, discrete, interval
Serum cholesterol in mg/dL	quantitative, continuous, ratio
Account balance in dollars	quantitative, continuous, ratio
Jersey number on a uniform	categorical, nominal (numeric code)

Table: ten variables and their finest classification. Note the three numeric-code rows (area code, calendar year, jersey number): area code and jersey number are nominal labels despite being digits, while calendar year is genuinely numeric but only interval, since year zero is an arbitrary origin.

Example 8.1.11 (Reading the table off cold)**.** From the reference table, pick the two variables that are *written as numbers* yet are **nominal**, and the one variable that is numeric and ordered yet only **interval** rather than ratio. Justify each in one line.

Which rule, and why? This drills the two classic traps: numeric codes that are really labels, and numeric scales that lack a true zero. The deciding questions are "does averaging the code mean anything?" and "does a ratio of two values mean anything?"

Solution.

Step 1: numeric-but-nominal. **Telephone area code** and **jersey number** — both are identifiers; averaging two area codes or two jersey numbers yields nothing interpretable, so they are nominal.

Step 2: numeric-but-only-interval. **Calendar year of diagnosis** — year differences are meaningful ("5 years apart") but year 0 is a chosen epoch, so "twice the year" is meaningless; the scale is interval, not ratio.

nominal numeric codes: area code, jersey number; interval: calendar year.

8.1.7 Contrast: ordinal data wrongly treated as interval

Example 8.1.12 (CONTRAST: averaging a Likert scale)**.** A clinic measures patient-reported pain on the standard scale none = 0, mild = 1, moderate = 2, severe = 3. Before a new analgesic, 20 patients report pain levels; after, the same scale is read again. An analyst codes the responses 0–3, computes a "mean pain score" of 2.10 before and 1.45 after, and runs a two-sample t test on the difference of these means, reporting a clean p-value. Explain why treating these codes as interval data is a type error, what the codes actually are, and what the legitimate analysis is instead.

Which rule, and why? The codes 0–3 *look* like interval measurements — evenly spaced integers — so the look-alike move is to average them and run a t test. But the underlying variable is **ordinal**: the spacing between adjacent levels is not a fixed, known quantity, so means and the t machinery built on them are not justified.

Solution.

Step 1: name the true scale. "None/mild/moderate/severe" is an ordered set of labels with *undefined spacing*: there is no guarantee that the gap from mild to moderate equals the gap from moderate to severe. That is the definition of **ordinal**, not interval.

Step 2: why the mean is not meaningful. The reported mean 2.10 depends entirely on the arbitrary choice to code the levels $0, 1, 2, 3$. Recode them as $0, 1, 2, 5$ — equally faithful to the order — and the "mean" shifts, even though not a single patient's report changed. A summary that moves when you relabel cannot be measuring the patients; the t statistic built from it inherits the same arbitrariness.

Step 3: the legitimate analysis. Because the data are ordinal and the two readings are on the *same patients*, the honest tools are order-based and paired: summarize with the *median* or the proportion in each category, and test the before-after shift with a rank/sign-based paired procedure (the nonparametric methods of Chapter 14), not a t test on fabricated interval scores.

Step 4: the cost of the error. The t test reports a precise p-value that is only as real as the coding $0, 1, 2, 3$; change the spacing and the "significance" can appear or vanish. The recognition step — ordinal, not interval — is what blocks the false precision.

pain level is ordinal $\Rightarrow$ median / ranks, paired; not a t test on coded means.

Remark. The boundary cases are worth stating plainly. A **long** ordinal scale with many levels (a 0–100 rating) is often treated as interval in practice, and the distortion is usually small; a **short** ordinal scale (three or four levels, as above) is where treating codes as interval does the most damage. When in doubt, the safe reflex is to keep ordinal data on the ordinal ladder — ranks, medians, χ^2 on the cross-tabulation — and let the data type, not convenience, pick the procedure.

8.2 Study Designs: Experiments and Observational Studies

A statistical procedure tells you what is true *of the data you have*; only the **design** that produced those data tells you what your conclusion is allowed to be about. The single fact that organizes this whole section is this: in a designed **experiment** the analyst actively *assigns* the conditions being compared, whereas in an **observational study** the analyst merely *records* conditions that the units arrived with. That one difference — did a coin, not the subject, decide who got which treatment? — is what separates a licensed *causal* claim from a mere *association*. You recognize you are in this section whenever a problem hands you a described study and asks you either to label it (experiment vs observational) or to judge a stated cause-and-effect claim. The recognition habit to build is to hunt for the assignment mechanism first, then for a **lurking variable** (a third factor tied to both the supposed cause and the outcome) that an observational design leaves uncontrolled. The machinery of randomization, control, replication, and blinding all exists to defeat exactly that lurking variable; when it is absent, the same association may be real but causally empty.

Experiment or observational — and what may I conclude?

- The investigator **assigns** the treatment (ideally by randomization), then measures a response $\Rightarrow$ **experiment** $\Rightarrow$ a difference in the response *is licensed as causal.*
- The investigator only **records** a group the unit already belonged to (smoker/nonsmoker, city/rural, brand chosen) $\Rightarrow$ **observational study** $\Rightarrow$ a difference is an **association**, open to confounding.
- A factor linked to both the grouping and the outcome, left uncontrolled $\Rightarrow$ **confounder / lurking variable**; it is the reason observational association is not causation.
- Nuisance variation you can foresee (age, soil plot, lab) $\Rightarrow$ **block** or **match** on it, then compare within blocks (forward to Ch. 13).

Do not confuse an *association reported from observational data* with a causal effect: even a strong, repeated, "obvious" link earns the word *cause* only when random assignment ruled out the lurking alternatives. Randomization, not sample size, is what buys causation.

8.2.1 The two designs, and why assignment is everything

Definition 8.2.1 (Experiment vs observational study)**.** In a designed **experiment** the investigator applies treatments to the experimental units and observes the resulting response; in the cleanest form the assignment of units to treatments is made by a chance mechanism (**randomization**). In an **observational study** the investigator measures units as they are, without intervening in which condition each unit experiences. The **treatment** (or factor) is the condition compared; the **response** is the outcome measured; a **confounder** is a variable associated both with which group a unit falls in and with the response.

The reason assignment is decisive is bookkeeping about *competing explanations.* Suppose group A outscores group B on the response. In an observational study the groups were assembled by the units themselves, so they may differ systematically in a dozen unmeasured ways; any of those differences — not the labeled treatment — could drive the gap. Randomization attacks this at the root: because a coin, independent of every characteristic of the unit, decides the group, the two arms are on average balanced on *every* variable at once, measured or not, known or unknown. The remaining imbalance is pure chance of the size the statistical test already accounts for. That is the whole game.

Proposition 8.2.2 (What each design licenses)**.** *A randomized comparative experiment licenses the conclusion that the treatment* caused *the observed difference in mean (or proportion) response, up to ordinary sampling error. An observational study licenses only the conclusion that the grouping is* associated *with the response; a causal reading requires further untestable assumptions that no group difference is due to a lurking variable.*

Why randomization neutralizes confounders. Let U be any pre-treatment characteristic of a unit (age, genotype, baseline severity — *anything*). Under randomized assignment the treatment indicator T is independent of U by construction, so the distribution of U is the same in both arms: $\mathbb{E}[U \mid T = 1] = \mathbb{E}[U \mid T = 0]$. Hence no such U can be systematically larger in one arm and so cannot account for a response gap; the only systematic difference between arms is the treatment itself. The argument holds for *unmeasured* U too, which is exactly what observational adjustment can never guarantee. □

Remark. "Randomized" is not the same as "haphazard." Letting a nurse hand the new pill to whoever seems sicker, or assigning by alternate arrival, is *not* randomization and can manufacture the very confounding it was meant to remove. Genuine randomization uses an explicit chance device applied *after* units are enrolled.

8.2.2 Control, replication, and blinding

Three further devices sharpen an experiment. A **control group** (placebo or standard care) supplies the baseline the treatment is measured against, so that improvement attributable to the mere act of being studied — the placebo effect, regression to the mean, natural healing — is subtracted off. **Replication**, applying each treatment to many units, lets ordinary unit-to-unit variation average out so a real treatment effect can be seen above the noise (it shrinks the standard error like $1/\sqrt{n}$). **Blinding** keeps the subject (single-blind) and ideally the assessor (double-blind) unaware of the assignment, so expectations do not color either the response or its measurement.

Remark. These devices control *different* threats. Control handles the "compared to what?" threat; randomization handles confounding; replication handles chance variation; blinding handles expectation bias. A study can have one and lack another — a randomized but unblinded trial still risks assessor bias — so check for each separately.

Figure 8.1 contrasts the two data-producing paths. On the experiment path a chance device splits a single pool, so the arms start exchangeable; on the observational path the units sort themselves, and a lurking variable feeds into both the grouping and the response.

8.2.3 Confounding, and the biases that mimic effects

Two more failure modes deserve names. **Selection bias** arises when the units that enter the study are not representative of the population the conclusion is aimed at (sampling only volunteers, only survivors, only the easy-to-reach). **Response (or measurement) bias** arises when the act of measuring distorts the answer — a leading question, an uncalibrated instrument, recall that fades with time. Neither is fixed by a bigger sample: a biased process repeated more times just nails down the wrong number more precisely. Confounding, selection bias, and response bias are the three reasons a clean-looking association can be an artifact, and the design — not the analysis — is where they are won or lost.

Example 8.2.3 (Classify: vitamin trial)**.** A clinic enrolls 240 adults with low energy, uses a random-number table to give half a vitamin D capsule and half an identical-looking placebo for 12 weeks, then compares a fatigue score between the groups. Is this an experiment or an observational study? Identify the treatment and response.

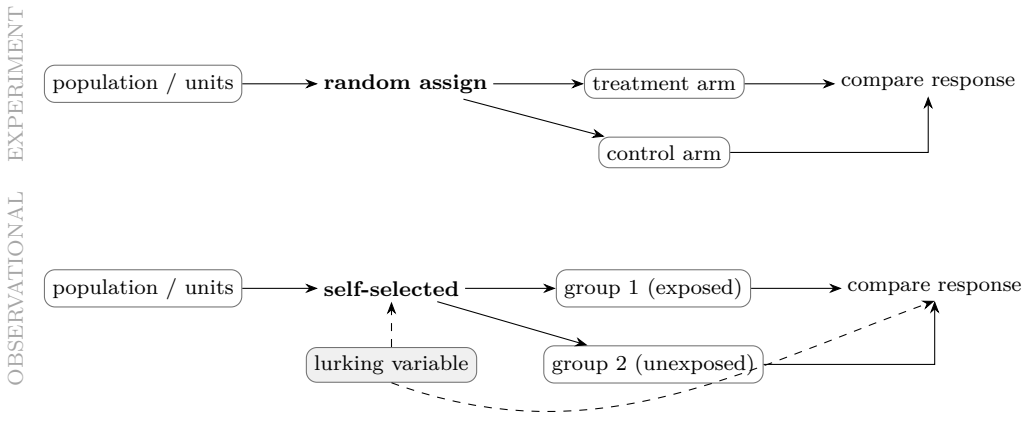


Figure 8.1: Two data-producing paths. *Top:* an experiment routes one pool through a chance device, so the arms are balanced before treatment and a response gap is causal. *Bottom:* an observational study lets units sort themselves; a lurking variable (dashed) drives both the grouping and the response, so the gap is only an association.

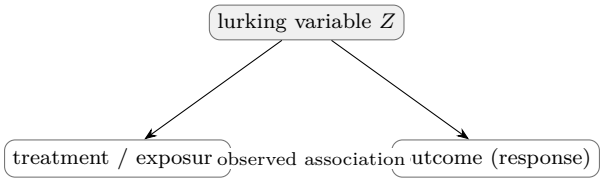


Figure 8.2: The anatomy of confounding. A lurking variable Z influences both the exposure and the outcome, manufacturing an association (dashed) between them even when the exposure has no effect on the outcome. Randomizing the exposure severs the arrow $Z \to$ treatment, leaving only a genuine treatment $\to$ outcome effect to detect.

Which rule, and why? Hunt for the assignment mechanism. A random-number table, not the subject, decides who gets the capsule — that is active assignment by a chance device, the defining mark of a randomized **experiment**, not an observational study (where the analyst would only record who already took vitamins).

Solution.

Step 1: locate assignment. Investigators randomized subjects to capsule vs placebo; the units did not choose. Active assignment $\Rightarrow$ **experiment**.

Step 2: name the parts. Treatment (factor) = vitamin D vs placebo; response = the 12-week fatigue score. The identical-looking placebo supplies the control arm and blinds the subject.

Randomized, placebo-controlled **experiment**; effect on fatigue is causally interpretable.

Example 8.2.4 (Classify: coffee and heart health)**.** Researchers follow 9,000 adults for ten years, recording each person's habitual coffee intake (none, moderate, heavy) at the start and noting who later develops heart disease. Heavy drinkers have fewer events. Experiment or observational? Name a plausible confounder.

Which rule, and why? Coffee intake was *recorded as already chosen*, never assigned — so this is an **observational** (prospective cohort) study, and any group difference is an **association**, not a demonstrated effect.

Solution.

Step 1: locate assignment. No one assigned coffee level; subjects arrived with their habits. Recording-only $\Rightarrow$ **observational study**.

Step 2: find a lurking variable. Smoking is a natural candidate: it travels with coffee habits and independently raises heart risk, so it influences both the grouping and the outcome (Figure 8.2). Baseline exercise or income would serve equally.

Step 3: verdict. Report "heavy coffee intake is *associated* with fewer events," not "coffee *protects* the heart."

Observational; lower event rate is an association, plausibly confounded by smoking.

Example 8.2.5 (Identify treatment, response, and confounder: fertilizer)**.** An agronomist applies a new fertilizer to the twelve plots on the sunny south slope of a field and the standard fertilizer to the twelve plots on the shaded north slope, then compares mean yield. The new fertilizer's plots yield more. What is the treatment, the response, and the confounder, and is the causal claim safe?

Which rule, and why? Treatments *were* assigned, so this is an experiment in form — but assignment was tied to slope, so the design is *confounded by location*; recognize that "assigned" alone is not enough, the assignment must be unconfounded (ideally random).

Solution.

Step 1: name the parts. Treatment = new vs standard fertilizer; response = plot yield.

Step 2: spot the confounder. Every new-fertilizer plot is also a sunny plot. Sunlight = a lurking variable: it tracks the treatment (perfectly here) and drives yield on its own. The fertilizer effect and the sun effect are inseparable — they are **confounded**.

Step 3: verdict and fix. The causal claim is *not* safe. Randomize fertilizer across plots regardless of slope, or **block** by slope and randomize within each slope (Ch. 13), so sun is balanced across treatments.

Confounded by sunlight; not causal as run. Block on slope, randomize within.

Example 8.2.6 (Judge the causal claim: hospital and survival)**.** A news report states: "Patients treated at Hospital A had a higher death rate than those at Hospital B; therefore Hospital A provides worse care." The data are real and the gap is large. Is the causal claim justified by the design?

Which rule, and why? The patients were not randomized to hospitals — they arrived at one or the other — so this is observational, and a raw rate comparison cannot license "worse care" without ruling out that the hospitals admit different *kinds* of patients.

Solution.

Step 1: design. No assignment of patients to hospitals; **observational**.

Step 2: lurking variable. Case mix. If Hospital A is a regional trauma/referral center, it receives sicker patients to begin with; baseline severity influences both *which* hospital (referral) and *death* (outcome) — a textbook confounder.

Step 3: verdict. The claim is *not* justified. A fair comparison must adjust for or match on baseline severity; even then, unmeasured severity can remain.

Not causal; the higher rate is likely confounded by patient case mix.

Example 8.2.7 (CONTRAST: an association read as causation)**.** Across 50 towns, towns with more ice-cream sales also report more swimming-pool drownings, and the correlation is strong. A blogger concludes that "buying ice cream makes people drown." Diagnose the error precisely.

Which rule, and why? This *looks* like cause-and-effect because the association is strong and consistent, but the data are purely **observational** (towns were recorded, not assigned ice-cream levels), so the right reading is association — and a single lurking variable explains both series at once.

Solution.

Step 1: design check. Nobody assigned ice-cream consumption to towns; observational ⇒ association only.

Step 2: the lurking variable. Summer heat. Hot weather drives *both* ice-cream sales and swimming (hence drownings). Heat → ice cream and heat → drownings produce a spurious ice-cream↔drowning link with no causal arrow between them (Figure 8.2, with Z = temperature).

Step 3: the lesson. Strength and consistency of an association do not upgrade it to causation; only a design that balances the lurking variable can. Conditioning on temperature would make the spurious correlation vanish.

Spurious; heat confounds ice cream and drownings. Association, never causation.

Example 8.2.8 (How randomization neutralizes a confounder)**.** A surgeon worries that older patients both tend to be steered toward the new minimally invasive procedure and tend to recover faster anyway, so age would confound a comparison with open surgery. A colleague proposes assigning the procedure by a sealed random envelope instead. Explain, with a small balance check, why randomization dissolves the worry.

Which rule, and why? The fix for an *unknown or unbalanced confounder* is not to measure it but to make assignment independent of it; randomization does exactly that, so this is a randomization argument, not an adjustment-for-age calculation.

Solution.

Step 1: name the threat. Age Z influences both assignment (steering) and recovery (outcome): a confounder, as in Figure 8.2.

Step 2: randomize. The envelope makes the procedure T independent of Z. Then the mean age is the same in both arms up to chance: e.g. with 60 per arm one might see mean ages 58.3 and 57.9 years — a gap of 0.4 year, the size sampling alone produces, not the systematic gap steering would create.

Step 3: conclusion. Because age (and every other pre-treatment trait, measured or not) is balanced across arms by construction, a recovery-time difference can be attributed to the procedure.

$\mathbb{E}[Z \mid T = \text{new}] \approx \mathbb{E}[Z \mid T = \text{open}] \;\Rightarrow$ no confounding; the design is now causal.

Example 8.2.9 (Selection bias: the survey that hears only the loud)**.** A streaming service emails a satisfaction survey to all 200,000 subscribers; the 6,000 who reply are 72% dissatisfied, and an analyst announces "most subscribers are unhappy." Classify the flaw and say whether a larger mailing would cure it.

Which rule, and why? The number who replied is large, which tempts a confident conclusion, but the responders *self-selected*; this is **selection (nonresponse) bias**, a design flaw that no increase in sample size repairs.

Solution.

Step 1: who is in the sample? Only the 3% who chose to reply — and dissatisfied customers are far likelier to bother. The sample is not representative of all subscribers.

Step 2: would more help? No. Mailing more or collecting more replies from the same self-selecting process estimates the wrong target more precisely. Bias is a property of the *process*, not the count.

Step 3: verdict. "72% of *responders* were dissatisfied" is all that is supported; generalizing to all subscribers is unjustified. A fix needs a random subsample with aggressive follow-up to shrink nonresponse.

Selection bias; not generalizable. A bigger biased sample is still biased.

Example 8.2.10 (Response bias: the leading question)**.** To estimate support for a tax, surveyor X asks "Do you support the fair-share tax that funds our schools?" and gets 68% yes; surveyor Y asks the neutral "Do you support or oppose Proposition 7?" and gets 44% yes. Same population, same week. What kind of error explains the gap?

Which rule, and why? Both samples may be perfectly drawn, so this is not a sampling issue; the distortion enters at *measurement*, which is **response bias** from a leading wording, distinct from selection bias (who is asked) and from confounding (a third variable).

Solution.

Step 1: locate the distortion. The wording "fair-share ... funds our schools" pushes respondents toward yes; the instrument, not the population, differs between X and Y.

Step 2: classify. A measurement that systematically nudges the answer is **response (measurement) bias**. The neutral phrasing 44% is the more trustworthy estimate.

Step 3: verdict. Report the neutral figure; the 24-point gap measures the question's pull, not the public's mind. Larger samples would not close it.

Response bias from leading wording; trust the neutral 44%.

Example 8.2.11 (Blocking to control known nuisance variation)**.** A sports scientist tests two recovery drinks on 40 runners but knows fitness varies enormously and would swamp any drink effect. She first sorts the 40 by baseline aerobic capacity into 20 matched pairs, then within each pair flips a coin to assign the two drinks. Name this device and say what it buys.

Which rule, and why? Fitness is a foreseeable nuisance variable, so rather than hope randomization balances it, she removes it up front by **matching/blocking** and randomizing *within* blocks — the randomized-block idea developed in Ch. 13, not a plain two-sample design.

Solution.

Step 1: identify the nuisance. Baseline fitness varies widely and affects recovery strongly; left alone it inflates the response's variability.

Step 2: the device. Pairing on fitness, then randomizing the two drinks within each pair, is **matched-pair blocking**. Each pair is a tiny balanced experiment; comparisons are made *within* pair, so between-runner fitness differences cancel.

Step 3: what it buys. It is still a randomized experiment (causal), but with the fitness nuisance removed from the error term the effect is estimated with a smaller standard error — more power for the same n. The analysis is paired, not two-independent-sample (forward to Ch. 13).

Randomized block (matched-pair) design; causal, with nuisance variance blocked out.

8.3 Sample Structures for Inference

Once you know the *data type* (§1) and how the data were *collected* (§2), the last thing that pins down the procedure is the **sample structure**: how many samples are in front of you, and whether those samples are *linked*. A single sample compared to a fixed target, two genuinely separate groups, two measurements taken on the *same* units, several groups at once, or a grid of category counts — each structure routes to a different family of tests in Chapters 6–14. The number of samples is usually easy to count; the linkage is the part that trips people up. The most expensive error in this whole chapter is to see two columns of numbers and reach for a two-independent-sample test when the two columns are really a before/after pair on one set of subjects — ignoring the pairing throws away the very structure that makes the comparison sharp. This section teaches you to count samples, to *detect pairing*, and to read a counts table as one- or two-variable.

How many samples and how linked?

- **One sample vs. a fixed number.** A single group, and the question compares its parameter to a *constant* (a spec, a historical value, a claim) ⇒ **one-sample** test (z/t for a mean, χ^2 for a variance, one-proportion z).
- **Two independent samples.** Two *separate* groups of distinct units (treatment vs. control, men vs. women, plant A vs. plant B), no unit appearing in both ⇒ **two-independent-sample** test (two-sample t, two-proportion z, F for two variances).
- **Paired / matched.** Two numbers per unit — before/after on the same subject, left/right, twin pairs, each treated subject matched to a control ⇒ form the **differences** D_i and run a **one-sample** test on them (paired t).

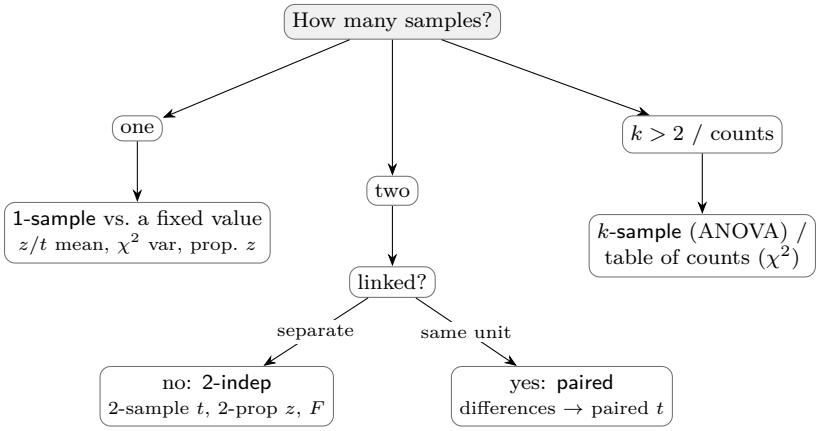
- $k > 2$ **independent samples.** Three or more separate groups compared at once ⇒ **one-way ANOVA** (F). If the groups also share a blocking variable, a **randomized block** analysis.
- **Tables of counts.** *One* categorical variable's counts against a claimed distribution ⇒ **goodness-of-fit** χ^2. *Two* categorical variables cross-tabulated ⇒ **contingency-table** χ^2 (independence / homogeneity).

Do not confuse paired with two-independent: if the same unit (or a matched partner) supplies both numbers, the samples are **paired** — analyze the differences. Treating a pair as two independent samples is both *wrong* (the independence assumption fails) and *wasteful* (it discards the within-pair matching that cancels subject-to-subject noise).

8.3.1 Counting samples and detecting linkage

Definition 8.3.1 (Sample structure)**.** The **sample structure** of a problem is the number of samples together with their linkage: *one* sample, *two independent* samples, a set of *paired* observations, k independent samples, or a *table of counts*. Two samples are **independent** when the units in one have no connection to the units in the other; they are **paired** when each observation in one sample is tied to a specific observation in the other by sharing a unit or a matched partner.

The recognition question is a single sentence: *does any unit (or a deliberately matched partner) contribute a number to both columns?* If yes, you have pairing. The classic generators of pairing are easy to memorize: **before/after** the same subject, **left/right** on the same body, **twins** or **littermates**, a treated unit **matched** to a similar control, and **repeated measures** on one item. A dead giveaway is that the two samples have *equal size and a natural one-to-one correspondence between rows*; independent samples need not even have the same size.



Decision schematic for sample structure. Count the samples first; for two samples, the *linked?* branch is decisive — "same unit (or matched partner) in both" sends you to the paired analysis, "separate groups" to the two-independent-sample analysis.

8.3.2 One sample, and two independent samples

A **one-sample** problem has a single group whose parameter is held up against a *fixed constant* supplied from outside the data: a manufacturer's spec, a regulatory limit, a historical mean, a claimed proportion. There is nothing to compare *across* groups because there is only one group. A **two-independent-sample** problem has two distinct groups of *different* units and asks whether their parameters differ from *each other*; the comparison value is not a constant but the other group's statistic.

Example 8.3.2 (One group against a published target)**.** A pharmacy receives a shipment of 250 mg acetaminophen tablets. An analyst assays $n = 20$ tablets and asks whether the mean tablet content differs from the labeled 250 mg.

Which rule, and why? There is a *single* sample of tablets, and the comparison is to a *fixed* labeled value 250, not to a second group. That is a one-sample test for a mean — not a two-sample test (no second group exists) and not a proportion test (content is a measured number, not a count of successes).

Solution.

Step 1: Count and link. One sample, $n = 20$; no second group ⇒ one-sample structure.

Step 2: Target. The fixed comparison value is $\mu_0 = 250$.

Step 3: Name the procedure. Mean vs. a constant with σ estimated from the data ⇒ one-sample t-test, two-sided, on $n - 1 = 19$ df.

one sample ⇒ one-sample t-test of $\mu = 250$.

Example 8.3.3 (Two assembly lines compared to each other)**.** A factory runs two independent assembly lines staffed by different crews. An engineer samples 30 units from line A and 34 units from line B and asks whether the two lines differ in mean weight.

Which rule, and why? Two *separate* groups of distinct units, with no unit appearing on both lines and unequal sizes (30 vs. 34), and the comparison is group-against-group. That is a two-independent-sample test for means, not a one-sample test (the comparison value is the other line, not a constant).

Solution.

Step 1: Count and link. Two samples; the units are physically different and unequal in number, so no pairing is possible ⇒ two independent samples.

Step 2: Parameter and question. Difference of means $\mu_A - \mu_B$, tested against 0.

Step 3: Name the procedure. Two-sample t-test (pooled or Welch) for $\mu_A = \mu_B$.

two separate groups ⇒ two-independent-sample t-test.

8.3.3 Paired data, and the costly mistake of ignoring the pairing

When each unit yields *two* numbers, the right move is to collapse each pair into its **difference** $D_i = X_i - Y_i$ and run a *one-sample* test on the D_i (typically a paired t-test of $\mu_D = 0$). Pairing works because the two numbers share everything about the unit — a heavy person stays heavy before and after, a strong arm stays strong on left and right — so subtracting cancels that between-unit variation and isolates the effect you care about. Ignoring the pairing and running a two-independent-sample test commits two sins at once: it *violates independence* (the columns are correlated), and it *drowns the signal* by letting the large between-subject spread back into the standard error.

Example 8.3.4 (CONTRAST: before/after weights disguised as two samples)**.** A clinic enrolls 12 patients in a diet program and records each patient's weight at the start and again after 8 weeks, producing a "before" column and an "after" column of 12 numbers each. A trainee proposes a two-independent-sample t-test comparing the before column to the after column. Diagnose the structure and name the correct procedure.

Which rule, and why? The two columns are *two measurements on the same* 12 *people* — the textbook generator of pairing (before/after). They are **paired**, not independent. The correct analysis forms the 12 within-person differences $D_i = \text{before}_i - \text{after}_i$ and runs a *one-sample* paired t-test of $\mu_D = 0$; the two-sample test is wrong here.

Solution.

Step 1: Detect the linkage. Each patient supplies *both* a before and an after number, so the same unit appears in both columns ⇒ paired, not independent.

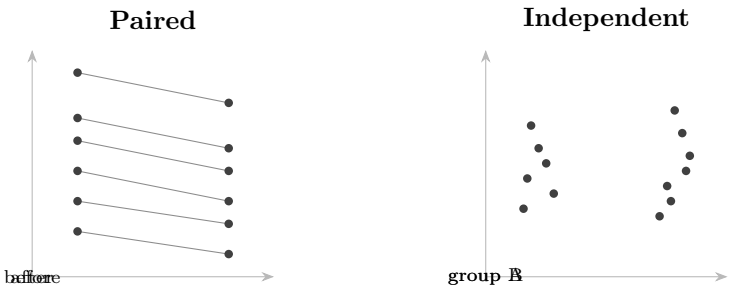
Step 2: Why the two-sample test is wrong. It assumes the two columns are independent, but a patient's before and after weights are strongly correlated; the independence assumption fails outright.

Step 3: Why it is also wasteful. Suppose every patient loses about 3 kg, yet patient weights themselves range over 50 kg across the group. The two-sample standard error is inflated by that 50 kg between-person spread,

so the real 3 kg shift is buried in noise. Pairing subtracts each person from themselves, removing the 50 kg spread and leaving only the small variation *in the losses*, so the same effect becomes easy to detect.

Step 4: Correct procedure. Compute D_i, then one-sample t on the D_i: $t_{\text{obs}} = \overline{D}/(s_D/\sqrt{12})$ on 11 df.

before/after on the same people $\Rightarrow$ paired t on the differences, not two-sample t.



Left: *paired* data — each connecting line ties one subject's two measurements, so the right analysis is on the line-by-line *differences*. Right: *independent* data — two unconnected clouds of distinct units, compared cloud-to-cloud. The presence or absence of those connecting lines is exactly the pairing question.

Example 8.3.5 (Two-sample vs. one-sample-of-differences)**.** Two reviewers must classify the following. **(a)** A study weighs 40 randomly chosen smokers and 40 separately chosen non-smokers and compares mean lung capacity. **(b)** A study measures each of 40 runners' resting heart rate before a training block and again after, and asks whether mean heart rate dropped. Which is two-independent-sample and which is one-sample-on-the-differences?

Which rule, and why? In **(a)** the two groups are *different people* with no link, so it is genuinely two independent samples. In **(b)** each runner is measured *twice*, so the two columns are paired; collapsing to differences turns it into a *one-sample* test on the D_i. Equal group sizes alone do not decide this — the link does.

Solution.

Step (a): No shared unit. Smokers and non-smokers are disjoint sets of people; nothing ties a particular smoker to a particular non-smoker $\Rightarrow$ two-independent-sample t-test.

Step (b): Shared unit. Each runner appears in both the before and after columns $\Rightarrow$ paired; form $D_i = \text{before}_i - \text{after}_i$ and run a one-sample (lower-tail) t-test of $\mu_D = 0$ on 39 df.

Trap to avoid. Both studies have 40 and 40; the matching sizes are a coincidence. Counting rows never settles pairing — only the question "same unit in both?" does.

(a) two-sample t; (b) one-sample t on the differences.

8.3.4 Several samples: ANOVA and blocking

Three or more independent groups compared *simultaneously* call for the **analysis of variance** (F-test), not a flurry of pairwise t-tests — running many two-sample tests inflates the overall false-alarm rate. If, in addition, the experiment groups units into **blocks** of similar subjects before assigning treatments (so that block-to-block variation can be removed, the multi-group cousin of pairing), the structure is a **randomized block design**.

Example 8.3.6 (Three fertilizers, recognizing the k-sample structure)**.** An agronomist randomly assigns 60 plots to three fertilizers (20 plots each) and records yield. The question is whether mean yield differs across the three fertilizers.

Which rule, and why? There are $k = 3$ *independent* groups compared at once, and the data are quantitative (yields). That is one-way ANOVA (F-test), not a two-sample t (there are three groups, and pairwise testing

would inflate the error rate) and not a paired analysis (the plots are distinct, with no matching across fertilizers).

Solution.

Step 1: Count and link. $k = 3$ separate groups of distinct plots; $k > 2 \Rightarrow$ multi-sample.

Step 2: Independent or blocked? Plots were assigned at random with no blocking variable mentioned $\Rightarrow$ *one-way* (completely randomized), not blocked.

Step 3: Name the procedure. One-way ANOVA; the statistic is $F_{\text{obs}} = \text{MSTR/MSE}$ on $(k-1,\ N-k) = (2,\ 57)$ df.

$k = 3$ independent groups $\Rightarrow$ one-way ANOVA (F).

Example 8.3.7 (The same three fertilizers, now blocked)**.** Suppose instead the field has a strong moisture gradient. The agronomist divides the field into 20 *blocks* of three adjacent (similar-moisture) plots and, within each block, randomly assigns the three fertilizers to the three plots. Now what structure is it?

Which rule, and why? Still three treatments, but the plots are first grouped into 20 similar-moisture **blocks** and treatments are randomized *within* each block. That is a randomized block design: blocking removes the moisture gradient the way pairing removes subject-to-subject variation, so it is the k-sample analogue of paired data.

Solution.

Step 1: Spot the blocking. Plots are matched into blocks of similar moisture before treatment assignment $\Rightarrow$ a blocking variable is present.

Step 2: Contrast with the unblocked version. Without blocking it was one-way ANOVA; the moisture gradient would inflate MSE. Blocking pulls that gradient out into a separate block term, sharpening the treatment comparison — exactly the multi-group version of the paired advantage.

Step 3: Name the procedure. Randomized block (two-way ANOVA without replication): partition variation into treatments, blocks, and error.

3 treatments within similar-moisture blocks $\Rightarrow$ randomized block ANOVA.

8.3.5 Tables of counts: one variable vs. two

When the data are *counts* in categories, the number of categorical variables decides the test. **One** categorical variable's observed counts, checked against a hypothesized distribution, is a **goodness-of-fit** χ^2 test. **Two** categorical variables cross-classified into a grid is a **contingency table**, tested for independence/homogeneity by a χ^2 test on that table.

Example 8.3.8 (One variable: are the categories in the claimed proportions?)**.** A candy maker claims its bags contain colors in the ratio 3:3:2:1:1 (red : green : yellow : blue : brown). A consumer pools one large bag and tallies the count of each of the 5 colors. Is the observed mix consistent with the claimed ratio?

Which rule, and why? There is *one* categorical variable (color, 5 levels) and a single set of observed counts compared to one *claimed distribution*. That is a goodness-of-fit χ^2 test, not a contingency table (only one variable is cross-classified) and not a proportion z (there are 5 categories, not a single yes/no).

Solution.

Step 1: Count the variables. One variable with 5 categories.

Step 2: Comparison. Observed counts vs. *expected* counts from the claimed 3:3:2:1:1 split of the total.

Step 3: Name the procedure. Goodness-of-fit $\chi^2 = \sum (O - E)^2/E$ on $5 - 1 = 4$ df.

one categorical variable vs. a claimed distribution $\Rightarrow$ goodness-of-fit χ^2.

Example 8.3.9 (Two variables: recognizing a contingency table)**.** A survey records each respondent's region (North, South, West) and whether they own an electric vehicle (yes/no), producing a 3×2 grid of counts. Researchers ask whether EV ownership is associated with region.

Which rule, and why? Two categorical variables (region with 3 levels, ownership with 2) are cross-classified, and the question is association between them. That is a contingency-table χ^2 test of independence, not a goodness-of-fit test (which handles only one variable).

Solution.

Step 1: Count the variables. Two categorical variables, cross-tabulated into a 3×2 table.

Step 2: Question. Independence/association between region and ownership.

Step 3: Name the procedure. Contingency-table χ^2 on $(3-1)(2-1) = 2$ df, with $E_{ij} = (\text{row}_i \text{ total} \times \text{col}_j \text{ total})/N$.

two cross-classified categorical variables $\Rightarrow$ contingency-table χ^2.

Example 8.3.10 (CONTRAST: matched counts that are not two independent groups)**.** A vision study tests each of 90 patients with two eye-chart fonts and records, per patient, whether they passed font A and whether they passed font B (pass/fail each). A reviewer wants to build a 2×2 table of "font A pass/fail" against "font B pass/fail" and run an ordinary independence χ^2. Is that the right structure?

Which rule, and why? Each patient is tested on *both* fonts, so the two pass/fail outcomes are **paired** on the same person — not two independent samples of eyes. The ordinary contingency-table χ^2 for independence treats the two margins as unrelated groups and is the wrong tool; matched binary pairs call for a paired count analysis (McNemar-type), which looks only at the patients who switch (pass one font, fail the other).

Solution.

Step 1: Detect the linkage. The same 90 patients generate both the A-outcome and the B-outcome $\Rightarrow$ paired binary data, not two independent groups.

Step 2: Why plain independence χ^2 misleads. It asks whether "passing A" and "passing B" are independent attributes in a population, ignoring that both come from one person; it cannot answer "do the fonts differ?" the way a matched comparison does.

Step 3: Correct structure. A paired/matched analysis of the discordant pairs — the counts analogue of switching from a two-sample t to a paired t.

both outcomes on the same person $\Rightarrow$ matched (paired) count analysis, not plain independence χ^2.

Remark. Every structure here has a paired "trap" twin: the paired t hides inside what looks like a two-sample t, the randomized block hides inside what looks like one-way ANOVA, and the matched count analysis hides inside what looks like a contingency table. The single diagnostic that unmasks all three is the same question from the master guide: *does one unit (or its matched partner) contribute to more than one of the things being compared?* If yes, exploit the link — do not pretend it is not there.

8.4 Choosing the Right Procedure: A Decision Map

This is the payoff section. The previous three sections taught you to read three features off a problem — the **data type** (categorical vs. quantitative), the **sample structure** (one sample, two independent, paired, k samples, a table of counts), and the **question** being asked (mean? variance? proportion? association? distributional fit?). Here those three answers are combined into a single **decision map** that lands on exactly one procedure and tells you which chapter of Larsen & Marx that procedure lives in. The skill is pure recognition: almost no arithmetic, just the ability to walk a problem down the map without taking a wrong branch. Two scenarios can read almost identically and yet route to entirely different tests because of one distinguishing feature — paired vs. independent, a proportion vs. a mean, one categorical variable vs. two — so the worked items below drill those forks directly. The final fork is a quiet but important one: if the data badly violate normality or are only ordinal, the map reroutes you out of the t/F world and into nonparametrics.

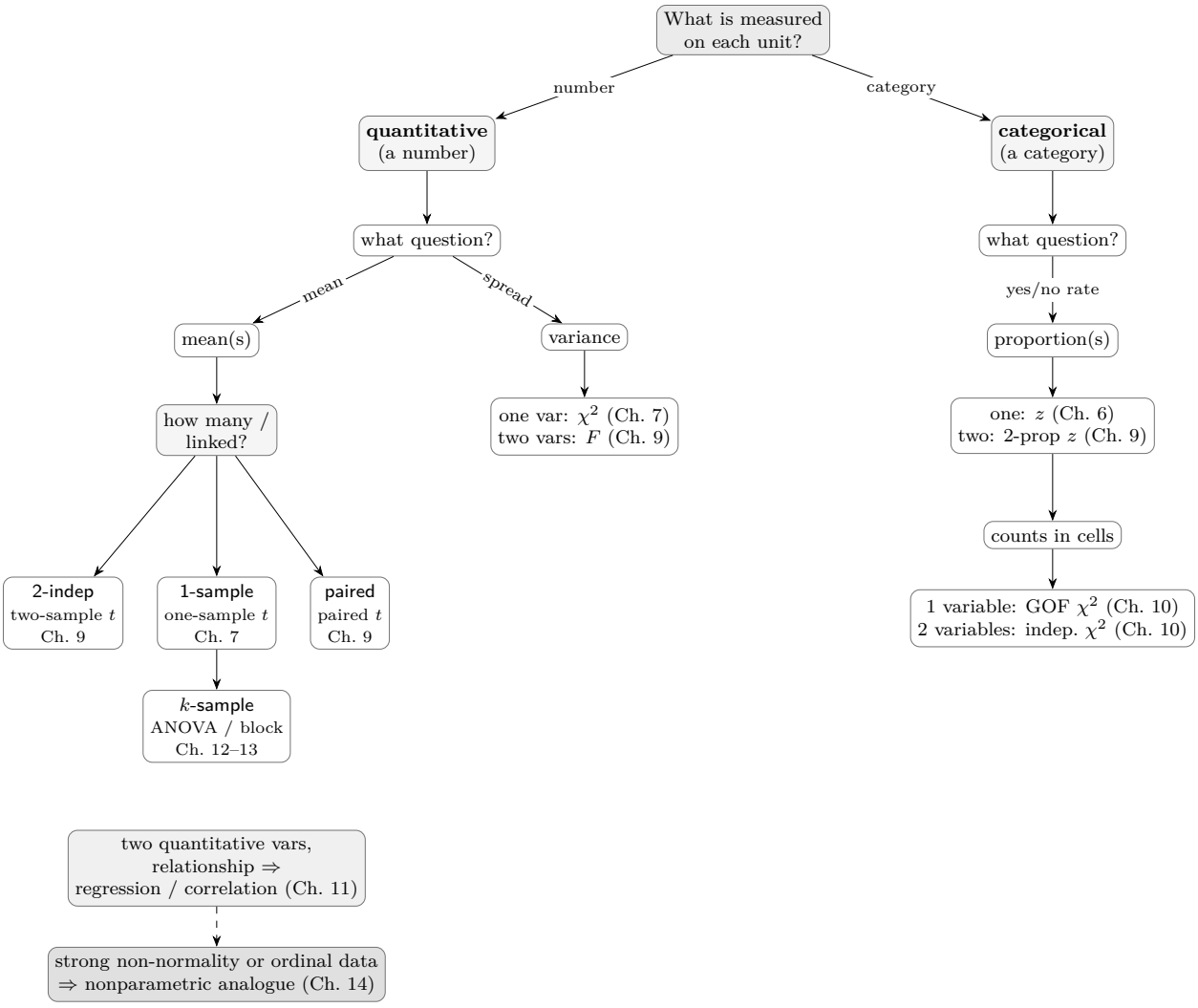
Which procedure does the problem need?

- **Quantitative, one sample, mean.** A measured variable, one group, compared to a fixed target ⇒ one-sample t (Ch. 7).
- **Quantitative, two independent samples, means.** Two separate groups ⇒ two-sample t — pooled if variances look equal, Welch if not (Ch. 9).
- **Quantitative, paired, mean difference.** Two numbers on the same unit ⇒ paired t on the differences (Ch. 9).
- **Quantitative, $k > 2$ samples, means.** Three or more groups at once ⇒ one-way ANOVA (Ch. 12); if blocked, randomized block ANOVA (Ch. 13).
- **A variance question.** One sample's variance vs. a target ⇒ χ^2 test for σ^2 (Ch. 7); two variances compared ⇒ F-test (Ch. 9).
- **Categorical, proportion.** One proportion vs. a claim ⇒ one-proportion z (Ch. 6); two proportions ⇒ two-proportion z (Ch. 9).
- **Categorical, counts.** One variable's counts vs. a distribution ⇒ goodness-of-fit χ^2 (Ch. 10); two variables cross-classified ⇒ χ^2 test of independence (Ch. 10).
- **Two quantitative variables, relationship.** Association/prediction ⇒ regression and correlation (Ch. 11).
- **Strong non-normality or ordinal data.** Any of the above reroutes to a nonparametric analogue (Ch. 14).

Do not confuse: a *mean* question with a *proportion* question (is the response a measured number or a yes/no count?), and *paired* with *independent* (does one unit supply both numbers?). Those two mix-ups account for most wrong branches.

8.4.1 The master decision map

The map is read top to bottom in three passes. **First**, what is measured on each unit — a number (quantitative) or a category (categorical)? **Second**, what is the sample structure — how many samples, and are they linked? **Third**, what is the question — about a mean, a variance, a proportion, a fit, or a relationship? Those three answers together pick out one terminal node, which names both the procedure and its chapter.



The master decision map. Pass 1 (top) splits on *data type*; pass 2 splits on *sample structure*; pass 3 splits on *the question*. Each terminal box names the procedure and its chapter. The dashed box at the bottom is the override: if the normality assumption fails badly or the data are merely ranked, leave the t/F family and use the nonparametric analogue in Chapter 14.

8.4.2 Reading the map: one-sample, two-sample, paired

Example 8.4.1 (Quantitative, one sample, mean)**.** A coffee roaster's bags are labeled 340 g. A quality auditor weighs $n = 16$ bags from one day's production and asks whether the mean fill weight differs from 340 g.

Which rule, and why? The response (weight) is a measured number ⇒ quantitative. There is one group of bags compared to a *fixed* target 340, and the question is about the *mean*. Quantitative + one sample + mean is the first terminal node on the map.

Solution.

Step 1: Data type. Weight is quantitative.

Step 2: Structure. One sample ($n = 16$); the comparison is to the constant $\mu_0 = 340$, not to another group.

Step 3: Question and procedure. Mean vs. a target, σ unknown ⇒ one-sample t-test on 15 df.

one-sample t-test of $\mu = 340$ (Ch. 7).

Example 8.4.2 (Quantitative, two independent samples, means)**.** A sports scientist measures vertical jump height (cm) for 22 volleyball players and 25 basketball players, drawn from different teams, and asks whether the two sports differ in mean jump height.

Which rule, and why? Jump height is quantitative; there are two *separate* groups of distinct athletes (different sizes, no shared player), and the comparison is group-against-group. Quantitative + two independent samples + means routes to the two-sample t.

Solution.

Step 1: Data type. Quantitative (a measured length).

Step 2: Structure. Two independent samples; no athlete is in both, sizes $22 \neq 25$ so no pairing is even possible.

Step 3: Question and procedure. Difference of means $\mu_1 - \mu_2$ vs. 0. Use the *pooled* two-sample t if the two sample variances are comparable, otherwise the *Welch* version.

two-sample t-test (pooled or Welch) (Ch. 9).

Example 8.4.3 (Quantitative, paired, mean difference)**.** A physical therapist measures each of 18 patients' grip strength (kg) in the injured hand and the healthy hand, and asks whether mean grip strength differs between the two hands.

Which rule, and why? Grip strength is quantitative, but the two columns are *two measurements on the same person* (left/right is the textbook generator of pairing). Quantitative + paired routes to the paired t: collapse each patient to a difference D_i and run a one-sample test on the D_i — *not* a two-sample t, which would ignore the link.

Solution.

Step 1: Data type. Quantitative.

Step 2: Structure. Paired — each patient supplies both numbers, so the same unit appears in both columns.

Step 3: Question and procedure. Form $D_i = \text{injured}_i - \text{healthy}_i$; test $\mu_D = 0$ with a one-sample t on 17 df (this is the paired t).

paired t-test on the differences (Ch. 9).

8.4.3 Reading the map: several samples and variance questions

Example 8.4.4 (Quantitative, $k > 2$ samples, means)**.** A food scientist bakes bread with four different flour blends, 10 loaves per blend assigned at random, and measures rise height (cm). The question is whether mean rise differs across the four blends.

Which rule, and why? Rise height is quantitative and there are $k = 4$ *independent* groups compared simultaneously. Quantitative + $k > 2$ samples routes to one-way ANOVA, not a string of two-sample t-tests (which would inflate the overall error rate). No blocking variable is mentioned, so it is the one-way (completely randomized) version, not a randomized block.

Solution.

Step 1: Data type and count. Quantitative; $k = 4$ separate groups, $k > 2$.

Step 2: Blocked? Loaves were assigned at random with no blocking factor ⇒ one-way, not blocked.

Step 3: Procedure. One-way ANOVA, $F_{\text{obs}} = \text{MSTR/MSE}$ on $(k-1,\, N-k) = (3,\, 36)$ df.

one-way ANOVA (F) (Ch. 12); blocked ⇒ Ch. 13.

Example 8.4.5 (A variance question, one sample)**.** A pharmaceutical filler must keep dose-to-dose variability small: the specification caps the standard deviation of fill volume at $\sigma_0 = 0.5\,\text{mL}$. An inspector samples $n = 25$ vials and asks whether the process variance exceeds the specification.

Which rule, and why? The variable is quantitative, but the question is about *spread*, not the mean — a single sample's variance held up against a target σ_0^2. That is the χ^2 test for a variance, not a t-test (which is for a mean) and not an F-test (which compares *two* variances).

Solution.

Step 1: Identify the question. About σ^2, one sample, vs. a fixed $\sigma_0^2 = 0.25$.

Step 2: Procedure. $\chi^2 = (n-1)s^2/\sigma_0^2$, upper-tail, on 24 df.

χ^2 test for one variance (Ch. 7).

8.4.4 Reading the map: categorical questions

Example 8.4.6 (Categorical, one proportion)**.** A maker of memory cards claims at most 2% are defective. A buyer inspects $n = 400$ cards, counts the defectives, and asks whether the true defect rate exceeds the claimed 2%.

Which rule, and why? Each card is scored defective/not — a yes/no *category*, so the data are categorical and the parameter is a *proportion*. One sample compared to a claimed $p_0 = 0.02$ routes to the one-proportion z-test, not a mean test (the response is a count of successes, not a measured number).

Solution.

Step 1: Data type. Categorical (defective vs. not), so this is about a proportion.

Step 2: Structure. One sample vs. the constant $p_0 = 0.02$.

Step 3: Procedure. $z_{\text{obs}} = (\widehat{p} - p_0)/\sqrt{p_0(1-p_0)/n}$, upper-tail.

one-proportion z-test (Ch. 6).

Example 8.4.7 (Categorical counts: one variable vs. two)**.** Two questions about the same survey. **(a)** A pollster tallies how many of 600 voters fall into each of 4 party affiliations and asks whether the four parties occur in the 30:30:25:15 split claimed by a past census. **(b)** The pollster instead cross-tabulates party affiliation (4 levels) against age bracket (3 levels) into a 4×3 grid and asks whether affiliation and age are associated. Name each procedure.

Which rule, and why? Both are categorical counts, but the number of *variables* differs. In **(a)** there is *one* categorical variable checked against a claimed distribution ⇒ goodness-of-fit χ^2. In **(b)** there are *two* categorical variables cross-classified ⇒ χ^2 test of independence. One variable vs. two is the whole fork.

Solution.

Step (a): One variable, a claimed distribution. Observed counts vs. expected from the 30:30:25:15 split ⇒ goodness-of-fit χ^2 on $4 - 1 = 3$ df.

Step (b): Two variables, association. A 4×3 contingency table, with $E_{ij} = (\text{row}_i)(\text{col}_j)/N$ ⇒ independence χ^2 on $(4-1)(3-1) = 6$ df.

(a) goodness-of-fit χ^2; (b) independence χ^2 (both Ch. 10).

Example 8.4.8 (Two quantitative variables: a relationship)**.** A hydrologist records, for 30 days, the daily rainfall (mm) and the river's peak flow (m^3/s) and wants to quantify how peak flow changes with rainfall and predict flow from rainfall.

Which rule, and why? Here *both* variables are quantitative and the question is the *relationship* between them (strength and form), not a comparison of group means. Two quantitative variables + "how are they related / predict one from the other" routes to regression and correlation.

Solution.

Step 1: Data type. Two quantitative variables measured on each day.

Step 2: Question. Association and prediction, not a difference of means.

Step 3: Procedure. Fit $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$; report the correlation r and test the slope β_1.

simple linear regression / correlation (Ch. 11).

8.4.5 Two forks that decide everything

The next two examples are the heart of the section. Each is a *contrast*: a pair of scenarios that read almost identically yet route to different procedures, because a single feature changed. Train your eye on the one feature.

Example 8.4.9 (CONTRAST: proportion vs. mean — same study, two questions)**.** A hospital studies a new wound dressing on $n = 80$ patients. **(a)** It records, for each patient, whether the wound *healed within* 14 *days* (yes/no), and asks whether the healing *rate* beats the established 70%. **(b)** It instead records each patient's *number of days to heal* (a count of days, treated as quantitative) and asks whether the mean healing *time* differs from the established 11 days. Same patients, same dressing — which procedure does each question need?

Which rule, and why? The single distinguishing feature is the *response type baked into the question.* In **(a)** the response is a yes/no category, so the parameter is a *proportion* ⇒ one-proportion z. In **(b)** the response is a measured number of days, so the parameter is a *mean* ⇒ one-sample t. Reusing the same patients does not change the branch; the question's response type does.

Solution.

Step (a): Yes/no response. "Healed within 14 days?" is categorical ⇒ proportion vs. $p_0 = 0.70$ ⇒ one-proportion z-test (Ch. 6).

Step (b): Numeric response. "Days to heal" is quantitative ⇒ mean vs. $\mu_0 = 11$ ⇒ one-sample t-test on 79 df (Ch. 7).

The trap. It is tempting to think one study = one procedure. But the procedure follows the *question*, and these two questions ask about different parameters.

(a) one-proportion z (Ch. 6); (b) one-sample t (Ch. 7).

Example 8.4.10 (CONTRAST: paired vs. two independent — one word apart)**.** Two near-twin taste tests. **(a)** A panel of 40 tasters each rates recipe X and recipe Y and reports a sweetness score for each; we compare the two recipes. **(b)** A panel of 40 tasters is *split* into two groups of 20: one group rates only recipe X, the other only recipe Y; we compare the two recipes. Both yield a column of X scores and a column of Y scores. Which is paired and which is two independent samples?

Which rule, and why? The only difference is whether *each taster rates both recipes.* In **(a)** every taster scores both X and Y, so the two columns are *paired* on the same person ⇒ paired t on the differences. In **(b)** a taster scores only one recipe, so the groups are *independent* ⇒ two-sample t. Equal column heights do not decide it; the link does.

Solution.

Step (a): Same taster, both recipes. The same unit appears in both columns ⇒ paired; form $D_i = X_i - Y_i$ and run a one-sample t on 39 df (paired t, Ch. 9).

Step (b): Each taster, one recipe. Disjoint sets of tasters, no link ⇒ two independent samples; two-sample t (Ch. 9).

The trap. Both designs produce "40 X-scores and 40 Y-scores" on paper. Counting scores never settles it — only "did one taster supply both?" does.

(a) paired t; (b) two-sample t (both Ch. 9).

8.4.6 When normality fails: rerouting to nonparametrics

The map's t and F procedures lean on an approximately normal population (or a sample large enough for the Central Limit Theorem to rescue the mean). When the data are sharply skewed, riddled with outliers, or only *ordinal* (ranks, not true measurements), that assumption collapses and the map sends you to the nonparametric analogue in Chapter 14, which replaces the raw values by their ranks.

Example 8.4.11 (Normality concern reroutes the procedure)**.** An economist compares household *net worth* between two neighborhoods, 15 households each. Net worth is notoriously right-skewed — a few households carry enormous values — and with only 15 per group the Central Limit Theorem cannot be trusted to fix it. A colleague proposes the ordinary two-sample t-test. Is that the right call?

Which rule, and why? By data type and structure alone (quantitative, two independent samples, means) the map points at the two-sample t. But the *normality override* fires: the response is heavily right-skewed and the per-group sample size is too small for the CLT to repair the mean's sampling distribution. The branch reroutes to the rank-based two-sample analogue (Wilcoxon rank-sum / Mann–Whitney), which compares the distributions without assuming normality.

Solution.

Step 1: Default branch. Quantitative + two independent samples + means ⇒ two-sample t (tentatively).

Step 2: Check the assumption. Strong right skew, small n per group ⇒ normality fails and the CLT does not save it.

Step 3: Reroute. Replace values by ranks and use the nonparametric two-sample test.

skewed, small n ⇒ Wilcoxon rank-sum (Ch. 14), not two-sample t.

8.4.7 The whole map on one page

Table 8.1 compresses the decision map into a lookup table: find the row whose scenario matches your problem, read off the procedure and its chapter. Use it to check your branch after walking the flowchart — the two should always agree.

Data & question	**Structure**	**Procedure**	**Ch.**
Quantitative, mean	one sample	one-sample t-test	7
Quantitative, variance	one sample	χ^2 test for σ^2	7
Categorical, proportion	one sample	one-proportion z-test	6
Quantitative, means	two independent	two-sample t (pooled / Welch)	9
Quantitative, mean diff.	paired	paired t-test	9
Quantitative, variances	two independent	F-test for two variances	9
Categorical, proportions	two independent	two-proportion z-test	9
Counts, fit a distribution	one variable	goodness-of-fit χ^2	10
Counts, association	two variables	χ^2 test of independence	10
Two quantitative vars	paired by unit	regression / correlation	11
Quantitative, means	$k > 2$ samples	one-way ANOVA (F)	12
Quantitative, means	$k > 2$, blocked	randomized block ANOVA	13
Skewed / ordinal data	any of the above	nonparametric (rank) analogue	14

Table 8.1: Decision-map summary. Match the data-and-question with the sample structure to land on a single procedure and its chapter. The last row is the override: when normality fails or the data are ordinal, take the rank-based analogue in Chapter 14.

Remark. The map never depends on hard arithmetic — only on three readings of the problem: the *data type*, the *sample structure*, and the *question*. Get those three right and the procedure is forced; the only remaining judgment call is the normality override, which can demote a t or F test to its nonparametric cousin. When in doubt, name the three features out loud before you reach for a formula.

8.5 Exercises

All problems below are original. Full solutions begin on page 526.

A. Types of data and measurement scales

A1. A veterinary clinic logs one row per visiting animal and records, for each: the *species* (dog, cat, rabbit, other); the *microchip ID number* stamped on the implant; the *body mass* in kilograms; and the *triage urgency* coded 1 = routine, 2 = urgent, 3 = emergency.

(a) Identify the experimental unit, the population, and the sample.

(b) Classify each of the four recorded variables as categorical or quantitative.

(c) For each categorical variable, state whether it is nominal or ordinal; for each quantitative variable, state its scale (interval or ratio) and whether it is discrete or continuous.

A2. A dashboard built on the clinic data in problem A1 prints four numbers: "mean species," "mean microchip ID," "mean body mass," and "mean triage urgency." Decide which of these four means is legitimate, and for each illegitimate one name the summary that should be reported instead. Justify each call in one sentence.

A3. Classify each variable below to its *finest* scale (nominal, ordinal, interval, or ratio), and state one defensible summary statistic for each.

(a) The model year printed on a piece of lab equipment (e.g. 2018).

(b) The shelf temperature of a freezer in degrees Celsius.

(c) The shelf temperature of the same freezer in kelvin.

(d) A reviewer's rating of a manuscript on the scale reject / minor revision / major revision / accept.

A4. (Contrast.) Two telephone-style numbers are recorded for every patient in a registry: a four-digit *ward extension* (the phone extension of the hospital ward the patient occupies) and the patient's *systolic blood pressure* in mmHg, which also happens to be a three- or four-digit number for some readings. The two columns therefore look nearly identical — both are short strings of digits. Classify each variable and explain, using a single decisive test, why one is nominal and the other is ratio even though both are "numbers."

A5. For each pair of variables, both of which are recorded as integers, say which is a genuine quantity (so its mean is meaningful) and which is a numeric-looking nominal code (so its mean is nonsense). Give the one-line reason.

(a) Number of stitches in a wound versus operating-room number where the suture was placed.

(b) Course room number (101, 214, ...) versus number of students enrolled in the course.

A6. A water-quality lab records, per sample: *collection date* written as the calendar year (e.g. 2024); *dissolved oxygen* in mg/L; *turbidity class* coded 1 = clear, 2 = slightly turbid, 3 = turbid, 4 = very turbid; and the *number of distinct contaminants* detected. Two of these four are quantitative, but only one of those two is on a ratio scale. Identify the two quantitative variables, state which is ratio and which is interval, and classify the remaining two variables fully (categorical type and scale).

A7. (Contrast.) A pain clinic measures discomfort on the ordinal scale none = 0, mild = 1, moderate = 2, severe = 3, recorded once before and once after a nerve block on the *same* 30 patients. An analyst codes the answers 0–3, gets a "mean pain" of 2.20 before and 1.30 after, and runs a paired t test on the difference, reporting a p-value.

(a) Name the true measurement scale of the pain variable and state why the reported mean is not a meaningful number.

(b) Show concretely that the "mean" depends on an arbitrary coding by recomputing the before-mean under the alternative codes $0, 1, 2, 5$, given that the 30 before-responses were: three "none," five "mild," five "moderate," and seventeen "severe." (First confirm the 2.20 under the codes $0, 1, 2, 3$.)

(c) State the legitimate analysis (summary and test) for this before-after, ordinal, same-patient design, and name the chapter of the book that supplies the test.

B. Experiments, observational studies, and confounding

B1. A pediatric clinic enrolls 180 children with frequent ear infections. A computer random-number generator assigns 90 to receive a new chewable antibiotic and 90 to receive an identical-tasting placebo for eight weeks; the number of new infections per child is then counted. Classify the study (experiment or observational), name the treatment and the response, and state precisely what the clinic is allowed to conclude from a group difference.

B2. For each scenario, decide whether the design is a *designed experiment* or an *observational study*, and give the one feature that settles it.

(a) An economist compares the average savings rate of people who use a budgeting app against those who do not, by surveying 3,000 adults about the apps they already chose to install.

(b) A bank randomly selects half of 4,000 new customers to receive a monthly savings-reminder text and the other half to receive nothing, then compares average balances after a year.

(c) A nutrition team records, for 500 volunteers, how many cups of green tea each habitually drinks and each person's resting blood pressure, with no intervention.

B3. A wellness magazine reports: "Employees who take the stairs instead of the elevator have 15% lower cholesterol; therefore taking the stairs lowers cholesterol." The data come from recording, for 1,200 office workers, which they habitually use and then measuring cholesterol.

(a) Is the study an experiment or observational? Justify by the assignment mechanism.

(b) Name one plausible lurking variable and explain how it is tied to *both* the stair use and the cholesterol.

(c) Rewrite the headline so the claim matches what the design supports.

B4. (Contrast.) Two studies of the same question — "does a daily walking program reduce blood pressure?" — are run in the same city in the same month.

(a) *Study 1.* Researchers post a flyer; the 200 people who sign up for the walking program are compared, after three months, with 200 people who did not sign up. The walkers end with lower blood pressure.

(b) *Study 2.* Researchers recruit 400 people and use a coin flip to put each into a walking group or a no-walking group, then compare blood pressure after three months. The walkers again end lower.

The numerical drop is nearly identical in both. For each study, state whether a causal claim ("walking lowered blood pressure") is licensed, and explain why two studies with the same result route to different conclusions.

B5. An agronomist tests a new irrigation schedule. She applies the new schedule to all 16 fields owned by farms that already volunteered for the trial, and the old schedule to 16 fields on neighboring farms that declined; the volunteer farms' fields yield more.

(a) Was the irrigation schedule *assigned* by the analyst or *recorded* as self-selected? What does this make the design?

(b) Identify a confounder tied to both which schedule a field got and its yield.

(c) Describe the single change to the assignment step that would license a causal conclusion.

B6. A hospital quality officer is comparing two anesthesia protocols and worries that frailer patients are routinely steered toward Protocol A and also recover more slowly regardless of protocol, so frailty would distort a raw comparison.

(a) Explain why frailty is a confounder here, naming the two arrows it sends.

(b) A colleague suggests assigning the protocol by a sealed random envelope instead of by the attending's judgment. With a small balance check, explain why this dissolves the worry even though frailty is never measured. (Suppose with 50 patients per arm the mean frailty index comes out 3.10 in one arm and 3.04 in the other.)

B7. Each item below is an observational finding being read as cause-and-effect. For each, name the most plausible lurking variable and state the corrected (association-only) interpretation.

(a) "Cities with more firefighters dispatched to a blaze suffer more fire damage; therefore firefighters cause damage."

(b) "Children with larger shoe sizes read better; therefore big feet improve reading."

(c) "Patients given a certain experimental drug under compassionate-use rules die more often than the general patient population; therefore the drug is dangerous."

C. Sample structures

C1. For each scenario, state the *sample structure* (one sample, two independent samples, paired, $k > 2$ independent samples, or a table of counts) and name the matching procedure family. Justify each by the linkage question.

(a) A coffee roaster measures the caffeine content of 25 randomly selected bags from a single production run and asks whether the mean differs from the advertised 90 mg per serving.

(b) A lab assigns 48 mice at random to four diets (12 each) and records weight gain; the question is whether mean weight gain differs across the four diets.

(c) A survey cross-classifies 400 voters by political party (three levels) and by whether they support a ballot measure (yes/no), and asks whether support is associated with party.

C2. Headline contrast. A sleep researcher records each of 30 volunteers' reaction time once after a normal night and once after a night of restricted sleep, producing a "rested" column and a "deprived" column of 30 numbers each. A research assistant proposes a two-independent-sample t-test comparing the rested column to the deprived column.

(a) Diagnose the true sample structure and name the correct procedure.

(b) Explain why the proposed two-sample test is both *wrong* (which assumption fails) and *wasteful* (what variation it fails to remove).

C3. Two-sample vs. one-sample-of-differences. Two analysts must classify the following studies.

(a) A nutritionist measures fasting glucose in 35 randomly chosen vegetarians and in 35 separately chosen omnivores and compares the two mean levels.

(b) A nutritionist measures each of 35 patients' fasting glucose before a 12-week diet and again after, and asks whether mean glucose dropped.

(c) Both studies use 35 and 35. Does the matching sample size settle which is paired? State the single question that actually decides it.

C4. A botanist counts the number of seedlings of each of three flower colors (purple, pink, white) produced by a cross and tallies $O = (212, 108, 40)$ out of 360 seedlings. Genetic theory predicts the colors should appear in a 9:3:4 ratio.

(a) Identify the sample structure and the correct procedure, with df.

(b) Compute the expected counts under the 9:3:4 model and the value of the test statistic. (Do not carry out the full decision; just report the statistic and df.)

C5. Counts contrast. Each of 80 patients is screened by two diagnostic kits, and for each patient the lab records whether kit X returned positive and whether kit Y returned positive (positive/negative each). A reviewer wants to build a 2×2 table of "kit X result" against "kit Y result" and run an ordinary independence χ^2 test. Is that the right structure? Diagnose the linkage and name the correct family of procedure.

C6. For each pair below, the two scenarios look almost identical but route to *different* sample structures. State the structure and procedure for each, and name the single feature that separates them.

(a) (i) An ergonomics study seats 50 typists at keyboard A and 50 *different* typists at keyboard B and compares mean typing speed. (ii) An ergonomics study has each of 50 typists use keyboard A and then keyboard B and compares mean typing speed.

(b) (i) A farm assigns 90 cows at random to three milking schedules (30 each) and compares mean yield. (ii) A farm sorts 90 cows into 30 blocks of three similar-age cows and, within each block, randomly assigns the three schedules to the three cows.

C7. A quality engineer inspects one large lot of resistors and classifies each unit as within-spec or out-of-spec, asking whether the out-of-spec rate exceeds the contractual limit of 2%. A second engineer compares the out-of-spec rates of two separate suppliers, each lot inspected independently. For each engineer, state the sample structure and the matching procedure, and explain why the first is *not* a two-sample problem.

D. Choosing the right procedure

D1. For each scenario, name the single procedure the decision map lands on, give the chapter, and add a one-line justification (data type, sample structure, and the question).

(a) A bottler labels its sports drink at 500 mL. An auditor measures the fill volume of $n = 20$ bottles from one shift and asks whether the mean fill volume differs from 500 mL.

(b) A clinic records, for each of 150 flu-shot recipients, whether the patient reported any side effect (yes/no) and asks whether the side-effect *rate* exceeds the leaflet's claimed 10%.

(c) An agronomist plants three corn hybrids, 12 plots each assigned at random, and measures yield (bushels/acre), asking whether mean yield differs across the three hybrids.

D2. A sleep researcher records, for each of 24 volunteers, the number of hours slept on a night *before* and a night *after* starting a new medication, and asks whether mean sleep duration changes. A second researcher instead recruits 24 *different* people already on the medication and 24 *others* who are not, measures one night of sleep for each, and asks whether the two groups differ in mean sleep duration. Name the procedure and chapter for each study, and state the one feature that separates them.

D3. CONTRAST. A delivery company runs one study on $n = 200$ packages. Two questions are asked of the same shipment.

(a) For each package it is recorded whether the package arrived *on time* (yes/no); the question is whether the on-time *rate* beats the advertised 95%.

(b) For each package it instead records the *transit time* in hours (a measured number); the question is whether the mean transit time differs from the quoted 48 hours.

Name the procedure and chapter for each, and explain why the same 200 packages route to different procedures.

D4. A market researcher surveys 500 shoppers.

(a) She tallies how many chose each of 5 store brands and asks whether the five brands are chosen in the 25:25:20:15:15 percentage split a trade group claims.

(b) She instead cross-tabulates preferred brand (5 levels) against income bracket (3 levels) into a 5×3 grid and asks whether brand preference and income are associated.

Name the procedure and chapter for each, and identify the one feature that distinguishes them.

D5. A machine shop must hold the variability of a bearing's diameter tight: the spec caps the standard deviation at $\sigma_0 = 0.02\,\text{mm}$. An inspector samples $n = 30$ bearings and asks whether the process variance exceeds the spec. A neighboring question asks instead whether *two* machines produce bearings with equal diameter variability, sampling 25 from each. Name the procedure and chapter for each question, and state which feature decides between them.

D6. An ecologist studies tree height in a forest plot.

(a) Over 40 trees she records each tree's age (years) and its height (m) and wants to quantify and predict how height changes with age.

(b) She instead splits the same 40 trees into four soil-type groups and asks whether mean height differs across the four soil types.

Name the procedure and chapter for each, and explain why two analyses of the same 40 trees route to different procedures.

D7. Normality reroute. A labor economist compares annual *income* between two towns, 18 households sampled in each. Income is sharply right-skewed—a handful of households earn enormous amounts—and with only 18 per group the Central Limit Theorem cannot be relied on to fix the skew. An assistant proposes the ordinary pooled two-sample t-test. State the procedure the map points to on data type and structure alone, then state whether the assistant's choice is correct and, if not, name the procedure and chapter that should be used instead.

E. Mixed recognition (synthesis)

E1. A toxicology lab assays liver-enzyme concentration (units/L, a measured number) in $n = 18$ rats given a single dose of a candidate drug, and asks whether the mean concentration differs from the historical control value of 42 units/L. Walk the problem down the decision map and answer each part.

(a) Classify the data type (categorical or quantitative) of the response.

(b) State the sample structure (one sample, two independent, paired, k-sample, or a table of counts), and say what fixed value the comparison is against.

(c) Name the full procedure and the chapter, and write down the hypotheses and the test statistic you would compute (do not finish the arithmetic).

E2. For each study, name the data type, the sample structure, and the single procedure with its chapter. These four are designed to fork on one feature apiece.

(a) A clinic records, for each of 300 donors, blood type (A, B, AB, O) and tallies the four counts, then asks whether the four types occur in the 9:3:1:3 proportions claimed by a reference population.

(b) The same clinic instead cross-tabulates blood type (4 levels) against Rh factor ($+/-$) into a 4×2 grid and asks whether blood type and Rh factor are associated.

(c) A water authority measures dissolved-oxygen level (mg/L) at 12 sites upstream of a mill and at 15 separately chosen sites downstream, and asks whether the mean level differs above versus below the mill.

(d) A lab measures dissolved oxygen (mg/L) at each of 12 sites once in spring and again, at the *same* sites, in autumn, and asks whether the mean level changed between seasons.

E3. A nutrition study assigns 90 volunteers at random, 30 to each of three diets (low-carb, Mediterranean, standard), and records each person's 12-week change in systolic blood pressure (mmHg, a measured number). The question is whether mean blood-pressure change differs across the three diets.

(a) Is this a designed experiment or an observational study, and what does that license you to conclude if a difference is found?

(b) Name the data type, the sample structure, and the procedure with its chapter.

(c) State the hypotheses and name the test statistic (as a ratio of mean squares) and its degrees of freedom; do not compute it.

E4. CONTRAST. Two telehealth studies look almost identical on paper. In each, 50 patients are involved and the analyst ends up with a column of "app" numbers and a column of "clinic" numbers. Study P: each of the 50 patients has their blood glucose measured once by a phone app and once by a clinic meter on the same morning. Study Q: 50 patients are split into two groups of 25; one group is measured only by the app, the other only by the clinic meter. For each study, state the sample structure and name the procedure and chapter for comparing the two devices' mean readings. Explain in one sentence what single feature decides the fork.

E5. A regional health survey reaches 1,200 adults and records several variables per person: the respondent's five-digit ZIP code, their self-rated health on a five-point scale (poor/fair/good/very good/excellent), their annual out-of-pocket medical cost in dollars, and whether they have visited an emergency room in the past year (yes/no).

(a) Classify each of the four variables to its finest measurement scale (nominal, ordinal, interval, or ratio). For the ZIP code, state explicitly whether its mean is a legitimate summary.

(b) A junior analyst wants to test whether the proportion of adults who visited an ER exceeds the national figure of 18%. Name the data type of that response, the sample structure, and the procedure with its chapter.

E6. Diagnose each reported conclusion: name the design (experiment or observational), and state whether the stated causal claim is licensed. If it is not, name the lurking variable or bias.

(a) Investigators randomize 200 migraine patients to a new device or a sham device and find the device group reports fewer headache days; they conclude the device reduces headaches.

(b) Over five years, towns that fluoridate their water report fewer cavities than towns that do not; a columnist concludes fluoridation prevents cavities. (Towns were never assigned a fluoridation policy.)

(c) A hospital mails a recovery questionnaire to 5,000 discharged patients; the 400 who return it report high satisfaction, and the PR office announces that patients are highly satisfied.

E7. A materials lab runs a full study end to end and you must take it from raw description to a named procedure. An engineer suspects a new alloy is more uniform than the current one: she wants the alloy's tensile-strength variability to stay below a contract cap of $\sigma_0 = 4$ ksi. She tests $n = 21$ specimens of the new alloy, all from one production run, and records tensile strength (ksi, a measured number).

(a) Is the response categorical or quantitative, and is the question about a mean, a proportion, or a variance?

(b) Give the sample structure and the fixed comparison value.

(c) Name the procedure and chapter, state the (one-sided) hypotheses in terms of σ^2, and write the test statistic with its degrees of freedom; do not finish the computation. Suppose the sample gives $s = 5.2$ ksi — show only the substitution into the statistic, not the table lookup.

8.6 Solutions

A. Types of data and measurement scales

A1. **(a)** The measurements are taken once per visiting animal, so the experimental *unit* is one **animal** (a patient visit). The population is all animals the clinic treats; the sample is the set of animals actually logged. **(b)** *Species* is a label (categorical). *Microchip ID* is digits used as a name, so it is categorical despite being numeric. *Body mass* is a genuine number you can average (quantitative). *Triage urgency* 1/2/3 is an ordered set of labels (categorical). **(c)** Species: **nominal** (unordered labels). Microchip ID: **nominal** (a numeric-code identifier). Body mass: **ratio** (true zero, ratios meaningful) and **continuous** (any value in a range). Triage urgency: **ordinal** (ordered, but the gap routine-to-urgent need not equal urgent-to-emergency).

unit = animal; species, ID: nominal; urgency: ordinal; mass: ratio, continuous.

A2. The legitimacy test is direct: averaging is meaningful only for a quantitative variable.

Species. Nominal label; "mean species" is undefined. Report instead the **count or proportion** in each species category.

Microchip ID. Nominal identifier; the digits name animals, not amounts, so "mean ID" is nonsense. There is no center to report — the ID is a key, not a measurement.

Body mass. Quantitative (ratio); "mean body mass" is **legitimate**, and $\overline{X}$ with S is the natural summary.

Triage urgency. Ordinal; the mean depends on the arbitrary 1/2/3 spacing, so it is not interpretable. Report the **mode** or the proportion in each urgency level.

Only "mean body mass" is legitimate; species/ID → proportions, urgency → mode.

A3. Decide each by two questions: is there order, and (if numeric) is there a true zero so ratios are meaningful?

(a) Model year. Numeric and ordered, but year 0 is an arbitrary epoch, so "twice the year" is meaningless while year *differences* are fine: **interval**. Summary: a difference or a median year is defensible; a ratio is not.

(b) Freezer in °C. 0°C is the freezing point of water, not absence of heat, so −10°C versus −20°C carries no meaningful ratio: **interval**. Summary: mean (differences are valid), but not a ratio.

(c) Same freezer in kelvin. 0 K is absolute zero — genuinely no thermal energy — so ratios are meaningful (200 K is twice 100 K): **ratio**. Summary: mean, and ratios are now valid. (Note the contrast with (b): the *same physical quantity* is interval in °C and ratio in kelvin — the scale, not the thing measured, fixes the type.)

(d) Manuscript rating. Ordered labels (reject < minor < major-revision ordering is debatable, but the standard outcome ladder reject / revise / accept is ordered) with undefined spacing: **ordinal**. Summary: the mode or median category, not a mean.

year: interval; °C: interval; K: ratio; rating: ordinal.

A4. **Contrast.** Both columns hold short strings of digits, so the eye cannot tell them apart; the deciding test is the relabel test (or equivalently, does averaging mean anything?).

Ward extension. Shuffle which extension number is assigned to which ward and no information is lost — the digits *name* a phone line, they do not measure anything. Averaging two extensions need not even yield a real extension. **Nominal** (numeric code); its mean is meaningless.

Systolic blood pressure. You cannot relabel a reading of 140 mmHg as 90 mmHg without changing the patient's actual pressure; the number measures force per area, 0 is a true zero, and 120 is genuinely twice 60. **Ratio**, continuous; its mean is legitimate.

Decisive test: relabel-invariance. Extension is nominal; blood pressure is ratio.

A5. A column of integers is a quantity only if its codes cannot be permuted without changing the data.

(a) Stitches vs. OR number. Number of stitches is a true **count** (true zero, 20 is twice 10): **ratio**, discrete — mean meaningful. The operating-room number is an **identifier** for a room; room 7 is not "more" than room 3: **nominal** — mean nonsense.

(b) Room number vs. enrollment. The course room number is a location label: **nominal** — mean nonsense. The number of students enrolled is a count with a true zero: **ratio**, discrete — mean meaningful.

Quantities: stitches, enrollment (ratio). Codes: OR number, room number (nominal).

A6. *Step 1: the two quantitative variables.* *Dissolved oxygen* (mg/L) and *number of contaminants* are the genuine numbers. Dissolved oxygen has a true zero (0 mg/L means none, and 8 mg/L is twice 4): **ratio**, continuous. Number of contaminants is a count with a true zero: **ratio**, discrete.

Step 2: which is interval, which is ratio. Both quantitative variables above are ratio. The *interval* variable is *collection date as calendar year*: year differences are meaningful but year 0 is an arbitrary origin, so it is quantitative, discrete, **interval** — not ratio.

Step 3: the remaining variable. *Turbidity class* 1–4 is an ordered set of labels with undefined spacing: categorical, **ordinal**.

O_2 (ratio), contaminants (ratio); year is interval; turbidity is ordinal.

(The deliberate trap: the calendar year is the only quantitative column that is interval rather than ratio, even though it sits next to two ratio quantities.)

A7. Contrast.

(a) True scale. "None/mild/moderate/severe" is an ordered list of labels with *undefined spacing* — nothing guarantees mild-to-moderate equals moderate-to-severe. That is the definition of **ordinal**, not interval. The reported mean is not meaningful because it is an arithmetic average of codes whose numeric values were chosen by convention, not measured.

(b) Coding-dependence, shown numerically. The 30 before-responses are three "none," five "mild," five "moderate," seventeen "severe." Under the codes $0, 1, 2, 3$:

$$\text{sum} = 3(0) + 5(1) + 5(2) + 17(3) = 0 + 5 + 10 + 51 = 66,$$
$$\text{mean} = \frac{66}{30} = 2.20,$$

which confirms the reported 2.20. Now recode the *same* responses as $0, 1, 2, 5$ (equally faithful to the order, only the top gap widened):

$$\text{sum} = 3(0) + 5(1) + 5(2) + 17(5) = 0 + 5 + 10 + 85 = 100,$$
$$\text{mean} = \frac{100}{30} = 3.3\overline{3} \approx 3.33.$$

Not one patient's report changed, yet the "mean" moved from 2.20 to ≈ 3.33. A statistic that shifts when you merely relabel the levels cannot be measuring the patients, so the t machinery built on it inherits the same arbitrariness.

(c) Legitimate analysis. The data are ordinal and the two readings are on the *same* patients, so the honest approach is order-based and paired: summarize each occasion by the **median** or the proportion in each category, and test the before-after shift with a **rank/sign-based paired procedure** — the nonparametric methods of **Chapter 14** — not a t test on fabricated interval scores.

pain is ordinal $\Rightarrow$ median / paired sign-rank test (Ch. 14); not a paired t test.

B. Experiments, observational studies, and confounding

B1. *Hunt for the assignment mechanism first.* A random-number generator — not the child or parent — decides who gets the antibiotic, so the conditions are actively **assigned** by a chance device. That is the defining mark of a randomized, placebo-controlled **experiment** (§2). Treatment (factor) = new antibiotic vs. placebo; response = the eight-week count of new infections; the identical-tasting placebo supplies the control arm and blinds the subject. Because randomization balances every pre-treatment trait across the two arms on average, a group difference is *causally interpretable*.

Randomized placebo-controlled **experiment**; a difference in infections is caused by the antibiotic.

B2. The deciding feature is always *who set the condition* — a chance device run by the analyst (experiment) or the unit arriving with the condition (observational).

(a) **Observational study**. App use was *recorded as already chosen*; no one assigned who installs a budgeting app. Any savings gap is an **association** (§2).

(b) **Designed experiment**. The bank *assigned* the reminder by a random split, so a balance difference is *causal*, attributable to the reminder.

(c) **Observational study**. Tea intake was measured, not assigned; "no intervention" is stated outright, so only association with blood pressure is licensed.

(a) observational; (b) experiment; (c) observational — decided by assignment vs. recording.

B3. (a) **Observational**. Stair vs. elevator use was *recorded* as the workers' own habit; nobody assigned it by a chance device. So a difference is an **association**, open to confounding (§2).

(b) **Lurking variable**: overall fitness / health-consciousness. A person who chooses stairs is likelier to also exercise and eat well, which independently lowers cholesterol. Fitness thus feeds *both* the grouping (who takes stairs) and the outcome (cholesterol) — the textbook confounding pattern.

(c) Corrected headline: "Employees who take the stairs *tend to have* lower cholesterol" — an association, with no causal verb.

Observational; stair use is associated with lower cholesterol, plausibly confounded by fitness.

B4. Contrast. The two studies share an outcome but differ in the one thing that matters — how the walking condition was set.

(a) *Study 1: not causal.* Participants **self-selected** into walking by signing up. People who volunteer for a walking program are likely already healthier, more motivated, or younger — traits that lower blood pressure on their own. This is an **observational** comparison; the drop is an **association**, confounded by whatever makes people sign up.

(b) *Study 2: causal.* A coin flip **assigned** the condition, independent of every trait of the unit, so the two arms are balanced on average and the only systematic difference is the walking. This is a randomized **experiment**; the drop is attributable to walking.

The identical numbers do *not* make the conclusions identical: in Study 1 the same lurking traits that lower blood pressure also drive sign-up, so the gap could be entirely confounding; in Study 2 randomization severs that link. Randomization, not the size of the effect, is what buys causation.

> Same result, different designs: only the randomized Study 2 licenses "walking lowered blood pressure."

B5. (a) The schedule was *recorded* as self-selected: the new schedule went to farms that *volunteered*, the old to farms that *declined*. The analyst did not assign by a chance device, so despite looking like a comparison of treatments this is effectively an **observational** design — assignment is confounded with the farms' own choice.

(b) **Confounder**: farm quality / management. Farms that volunteer for a trial tend to be better-run (better soil care, equipment, attentiveness), which raises yield by itself. Volunteer status feeds *both* which schedule a field received and its yield.

(c) Fix: have the analyst **randomly assign** the new and old schedules to fields *regardless of which farm volunteered* (ideally across fields drawn from the same farms). A coin, not the farmer's willingness, must decide the schedule; then farm quality is balanced across the two schedules and a yield difference is causal.

> Self-selected (observational); confounded by farm quality. Randomize the schedule to license causation.

B6. (a) Frailty Z is a **confounder** because it sends two arrows: $Z \to$ assignment (frailer patients are steered toward Protocol A) and $Z \to$ outcome (frailer patients recover slower no matter the protocol). A variable tied to both the grouping and the response manufactures an apparent protocol effect that is really a frailty effect (§2).

(b) Randomize. Letting a sealed random envelope set the protocol makes the assignment indicator T *independent of* Z by construction, so $\mathbb{E}[Z \mid T = A] = \mathbb{E}[Z \mid T = B]$: the distribution of frailty — and of every other pre-treatment trait, measured or not — is the same in both arms on average. The balance check bears this out: mean frailty 3.10 vs. 3.04 is a gap of 0.06, the size pure chance produces with 50 per arm, not the systematic gap that steering would create. Because frailty is balanced *without ever being measured*, a recovery difference can be attributed to the protocol.

> $\mathbb{E}[Z \mid T = A] \approx \mathbb{E}[Z \mid T = B]$ (3.10 vs. 3.04) $\Rightarrow$ no confounding; the design is now causal.

B7. Each is observational, so the corrected reading is *association*, and a single lurking variable explains the link.

(a) **Lurking variable**: the size of the fire. Bigger blazes draw more firefighters *and* cause more damage, so firefighter count and damage rise together with no causal arrow between them. Corrected: firefighter count is *associated* with damage because both track fire size.

(b) **Lurking variable**: age. Older children have bigger feet *and* read better (more schooling). Shoe size is *associated* with reading only because both grow with age; foot size has no effect on reading.

(c) **Lurking variable**: baseline severity. Compassionate-use drugs are given to the sickest patients, who would die more often *anyway*; severity drives both who gets the drug and who dies. Corrected: higher mortality is *associated* with the drug, confounded by how sick those patients already were — not evidence the drug is dangerous.

> All observational; strength of an association never upgrades it to causation — a lurking variable explains each.

C. Sample structures

C1. All three are settled by counting samples and asking whether any unit appears in more than one of the things compared.

(a) **One sample.** A single group of 25 bags, and the mean is held up against a *fixed* advertised value (90 mg), not against a second group. The content is a measured number ⇒ **one-sample** test for a mean (one-sample t on $n - 1 = 24$ df), not a two-sample test (no second group) and not a proportion test (caffeine is a quantity, not a count of successes).

(b) $k > 2$ **independent samples.** There are $k = 4$ separate groups of distinct mice compared at once, on quantitative data (weight gain) ⇒ **one-way ANOVA** (F-test). Pairwise t-tests are wrong here because running several inflates the overall false-alarm rate; the plots/mice are distinct, so it is not paired.

(c) **Table of counts (two variables).** Two categorical variables (party with 3 levels, support with 2) are cross-classified into a 3×2 grid ⇒ **contingency-table** χ^2 test of independence. It is not goodness-of-fit (that handles only one variable).

(a) one-sample t; (b) one-way ANOVA (F); (c) contingency-table χ^2.

C2. This is the chapter's headline trap (paired data disguised as two independent samples).

(a) **Detect the linkage.** Each volunteer supplies *both* a rested number and a deprived number, so the same unit appears in both columns ⇒ the data are **paired**, not two independent samples. Collapse each pair to its difference $D_i = \text{rested}_i - \text{deprived}_i$ and run a **one-sample** paired t-test of $\mu_D = 0$ on $n - 1 = 29$ df.

(b) **Why the two-sample test is wrong.** It assumes the two columns are independent, but a person's rested and deprived reaction times are strongly correlated (a naturally slow responder is slow in both states); the independence assumption fails outright. **Why it is wasteful.** The two-sample standard error carries the large *between-person* spread (some people are simply faster than others). Pairing subtracts each person from themselves, cancelling that between-person variation and leaving only the variation in the within-person changes, so the same effect is far easier to detect.

rested/deprived on the same people ⇒ paired t on the differences, not two-sample t.

C3. The deciding feature is the link, never the sample size.

(a) **Two independent samples.** Vegetarians and omnivores are disjoint sets of people; nothing ties a particular vegetarian to a particular omnivore ⇒ two-independent-sample t-test of $\mu_1 = \mu_2$.

(b) **One sample on the differences (paired).** Each patient is measured *twice*, before and after, so the columns are paired; form $D_i = \text{before}_i - \text{after}_i$ and run a one-sample (lower-tail) t-test of $\mu_D = 0$ on 34 df.

(c) **No.** Both studies are 35 and 35; the matching sizes are a coincidence and do not settle anything (independent samples need not even be equal in size). The single deciding question is: *does the same unit (or a matched partner) contribute a number to both columns?* Yes in (b), no in (a).

(a) two-sample t; (b) one-sample t on the differences.

C4. Goodness-of-fit on one categorical variable.

(a) One categorical variable (color, 3 levels) with a single set of observed counts compared to one *claimed distribution* (9:3:4) ⇒ **goodness-of-fit** χ^2 test on $3 - 1 = 2$ df. It is not a contingency table (only one variable) and not a proportion z (three categories, not a single yes/no).

(b) The 9:3:4 ratio sums to 16, so the expected counts out of $N = 360$ are

$$E_{\text{purple}} = 360 \cdot \tfrac{9}{16} = 202.5,$$
$$E_{\text{pink}} = 360 \cdot \tfrac{3}{16} = 67.5,$$
$$E_{\text{white}} = 360 \cdot \tfrac{4}{16} = 90.0.$$

With $O = (212, 108, 40)$,

$$\begin{aligned}\chi^2 &= \sum \frac{(O-E)^2}{E} = \frac{(212-202.5)^2}{202.5} + \frac{(108-67.5)^2}{67.5} + \frac{(40-90)^2}{90} \\ &= \frac{90.25}{202.5} + \frac{1640.25}{67.5} + \frac{2500}{90} \\ &= 0.446 + 24.300 + 27.778 = 52.52.\end{aligned}$$

goodness-of-fit $\chi^2 \approx 52.5$ on 2 df.

C5. Matched binary counts, not two independent groups. Each patient is screened by *both* kits, so the kit-X and kit-Y outcomes are **paired** on the same person — not two independent samples of units. An ordinary independence χ^2 treats the two margins as unrelated groups and asks whether "X positive" and "Y positive" are independent attributes; it cannot answer "do the two kits differ?" the way a matched comparison does. The correct tool is a **paired (matched) count analysis** (McNemar-type) that looks only at the discordant patients (positive on one kit, negative on the other) — the counts analogue of switching from a two-sample t to a paired t.

both kit results on the same person ⇒ matched (paired) count analysis, not plain independence χ^2.

C6. Each pair is a near-identical contrast separated by a single linkage feature.

(a) (i) Two *different* sets of 50 typists, one per keyboard, no unit in both ⇒ **two-independent-sample** t-test. (ii) The *same* 50 typists use both keyboards, so each supplies two numbers ⇒ **paired**; form D_i and run a one-sample t on the differences. **Separating feature:** whether the same typist is measured on both keyboards.

(b) (i) Cows assigned at random to three schedules with no blocking variable ⇒ **one-way ANOVA** (F, completely randomized). (ii) Cows first sorted into 30 similar-age **blocks**, treatments randomized *within* each block ⇒ **randomized block** (two-way ANOVA without replication); blocking removes age variation the way pairing removes subject-to-subject variation. **Separating feature:** the presence of a blocking variable that groups units before treatment assignment.

(a) 2-sample t vs. paired t; (b) one-way ANOVA vs. randomized block.

C7. One-sample proportion vs. two-proportion contrast.

First engineer: A *single* lot is inspected and each unit classified as in/out-of-spec; the out-of-spec rate is compared to a *fixed* contractual constant (2%). One sample, comparison to a constant ⇒ **one-sample** (one-proportion) z-test of $p = 0.02$, one-sided (upper tail). It is *not* a two-sample problem because there is only one group and the comparison value is a published number, not a second lot's statistic.

Second engineer: Two *separate* suppliers' lots, inspected independently, with the question group-against-group ⇒ **two-independent-sample** (two-proportion) z-test of $p_1 = p_2$.

one lot vs. a 2% spec ⇒ one-proportion z; two suppliers ⇒ two-proportion z.

D. Choosing the right procedure

D1. **(a)** Fill volume is a measured number (quantitative), one sample compared to the fixed target $\mu_0 = 500$, and the question is about the *mean*.

one-sample t-test of $\mu = 500$ (Ch. 7).

(b) "Side effect?" is yes/no (categorical), so the parameter is a *proportion*; one sample vs. the claimed $p_0 = 0.10$, upper-tail.

one-proportion z-test (Ch. 6).

(c) Yield is quantitative; there are $k = 3$ independent groups assigned at random (no blocking factor), and the question compares means across all three at once.

one-way ANOVA (F) (Ch. 12).

D2. Both studies measure sleep duration (quantitative) and compare "on medication" against "not on medication," but the sample structure differs.

First study. Each volunteer supplies *two* numbers (before and after), so the same unit appears in both columns ⇒ **paired**. Form $D_i = \text{after}_i - \text{before}_i$ and run a one-sample t on 23 df.

Second study. The medicated group and the unmedicated group are *disjoint sets of people*; no person is in both ⇒ **two independent samples**.

The deciding feature. Whether one unit supplies both numbers. Equal counts (24 and 24) do not settle it—the link does.

first: paired t (Ch. 9); second: two-sample t (Ch. 9).

D3. The distinguishing feature is the *response type baked into the question*, not the shared shipment.

(a) Yes/no response. "On time?" is categorical ⇒ a proportion vs. $p_0 = 0.95$ ⇒ one-proportion z-test.

(b) Numeric response. "Transit time in hours" is a measured number (quantitative) ⇒ a mean vs. $\mu_0 = 48$ ⇒ one-sample t on 199 df.

Why same packages, different procedures. The procedure follows the *question*, not the study: one question asks about a rate, the other about an average, so they target different parameters.

(a) one-proportion z (Ch. 6); (b) one-sample t (Ch. 7).

D4. Both parts are categorical *counts*; the number of categorical *variables* is the fork.

(a) One variable, a claimed distribution. A single variable (brand) checked against the claimed 25:25:20:15:15 split ⇒ goodness-of-fit χ^2 on $5 - 1 = 4$ df.

(b) Two variables, association. Brand cross-classified with income in a 5×3 table ⇒ χ^2 test of independence on $(5 - 1)(3 - 1) = 8$ df.

The deciding feature. One categorical variable vs. two cross-classified variables.

(a) goodness-of-fit χ^2; (b) independence χ^2 (both Ch. 10).

D5. Both questions are about *spread* (quantitative variable), but the number of samples differs.

One sample vs. a target. A single sample's variance against the fixed $\sigma_0^2 = (0.02)^2$, upper-tail ⇒ $\chi^2 = (n - 1)s^2/\sigma_0^2$ on 29 df.

Two samples compared. Two machines' variances compared to each other ⇒ $F = s_1^2/s_2^2$ on $(24, 24)$ df.

The deciding feature. One variance vs. a target uses χ^2; two variances against each other uses F.

one: χ^2 test for σ^2 (Ch. 7); two: F-test (Ch. 9).

D6. Same 40 trees, but the data type of the comparison and the question differ.

(a) Two quantitative variables, a relationship. Both age and height are measured numbers, and the question is how one changes with the other and predicts it ⇒ fit $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$, report r, test the slope.

(b) One quantitative response, $k > 2$ groups. Height is quantitative; soil type splits the trees into $k = 4$ independent groups, and the question compares means ⇒ one-way ANOVA.

Why same trees, different procedures. In (a) the explanatory variable (age) is itself a *number*, so the tool is regression; in (b) the explanatory variable (soil type) is a *category* with 4 levels, so the tool is ANOVA.

(a) regression / correlation (Ch. 11); (b) one-way ANOVA (Ch. 12).

D7. By data type and structure alone (quantitative, two independent samples, means) the map points at the two-sample t, so the assistant's default is the right *branch*—but the **normality override** fires.

Step 1: Default branch. Quantitative + two independent samples + means ⇒ two-sample t (tentatively).

Step 2: Check the assumption. Income is heavily right-skewed and $n = 18$ per group is too small for the CLT to repair the mean's sampling distribution.

Step 3: Reroute. Replace values by ranks and use the rank-based two-sample analogue. The assistant's choice is *not* correct.

skewed, small n ⇒ Wilcoxon rank-sum (Ch. 14), not two-sample t.

E. Mixed recognition (synthesis)

E1. Three passes down the map: data type, then structure, then question.

(a) Data type. Enzyme concentration in units/L is a genuine measured number on which averaging is meaningful ⇒ **quantitative** (continuous, ratio scale).

(b) Structure. A single group of 18 rats is held up against the *fixed* historical constant 42, with no second group ⇒ **one sample**; the comparison value is $\mu_0 = 42$, not another sample's statistic.

(c) Procedure. Quantitative + one sample + a question about the *mean*, with σ estimated from the data, routes to the one-sample t-test (Ch. 7). Two-sided because "differs":

$$H_0 : \ \mu = 42 \qquad \text{vs.} \qquad H_1 : \ \mu \neq 42,$$
$$t_{\text{obs}} = \frac{\overline{X} - 42}{s/\sqrt{18}} \quad \text{on } n - 1 = 17 \text{ df.}$$

quantitative, one sample, mean ⇒ one-sample t-test of $\mu = 42$ (Ch. 7).

E2. Each part forks on a single feature; the deciding feature is named in each line.

(a) One categorical variable vs. a claimed distribution. Blood type is one categorical variable (4 levels); its observed counts are checked against the single claimed 9:3:1:3 split ⇒ **goodness-of-fit** χ^2 on $4-1 = 3$ df (Ch. 10). *Not* a contingency table — only one variable is cross-classified.

(b) Two categorical variables cross-classified. Blood type (4) against Rh (2) in a 4×2 grid, asking about *association* ⇒ **contingency-table** χ^2 test of independence on $(4-1)(2-1) = 3$ df (Ch. 10). The fork from (a): *two* variables, not one.

(c) Two independent samples, means. Dissolved oxygen is quantitative; the 12 upstream sites and 15 downstream sites are *distinct* locations (unequal sizes, no link) ⇒ **two independent samples** ⇒ two-sample t-test (pooled or Welch) for $\mu_{\text{up}} = \mu_{\text{down}}$ (Ch. 9).

(d) Paired, mean difference. Quantitative again, but each of the 12 sites is measured *twice* (spring and autumn), so the same unit supplies both numbers ⇒ **paired**. Form $D_i = \text{spring}_i - \text{autumn}_i$ and run a paired t-test of $\mu_D = 0$ on 11 df (Ch. 9). The fork from (c): same site in both columns, so the two-sample t would be wrong (it ignores the within-site link).

(a) GOF χ^2; (b) independence χ^2; (c) two-sample t; (d) paired t.

E3. Design first, then the map.

(a) Design. The analyst *assigned* diets to volunteers by a chance mechanism (random allocation, 30 per arm) ⇒ a randomized **experiment**. A difference in mean blood-pressure change is therefore *causally* interpretable: random assignment balances age, baseline pressure, and every other pre-treatment trait across the three arms.

(b) Map. Blood-pressure change is **quantitative**; there are $k = 3$ *independent* groups compared at once, with no blocking variable mentioned ⇒ **k-sample**, completely randomized ⇒ one-way ANOVA (F-test) (Ch. 12). Not a two-sample t (three groups; pairwise t's would inflate the error rate) and not paired (the volunteers are distinct, with no matching across diets).

(c) Hypotheses and statistic.

$$H_0:\ \mu_1 = \mu_2 = \mu_3 \qquad \text{vs.} \qquad H_1:\ \text{not all } \mu_j \text{ equal},$$
$$F_{\text{obs}} = \frac{\text{MSTR}}{\text{MSE}} \quad \text{on } (k-1,\ N-k) = (2,\ 87) \text{ df}.$$

randomized experiment, $k = 3$ groups ⇒ one-way ANOVA (F) (Ch. 12).

E4. The two studies produce identically shaped tables, so counting columns settles nothing; the *link* decides.

Study P: same patient, both devices. Each of the 50 patients is measured by *both* the app and the clinic meter, so the same unit appears in both columns ⇒ **paired**. Form $D_i = \text{app}_i - \text{clinic}_i$ and run a paired t-test of $\mu_D = 0$ on 49 df (Ch. 9).

Study Q: each patient, one device. The 50 are split into two disjoint groups of 25; no patient is measured by both devices, and nothing ties an app patient to a clinic patient ⇒ **two independent samples** ⇒ two-sample t-test (Ch. 9).

The deciding feature. Whether one patient supplies *both* numbers: in P a single patient is in both columns (paired); in Q each patient contributes to only one column (independent). Equal column heights (50 vs. $25 + 25$) never decide it — only the link does.

P: paired t; Q: two-sample t (both Ch. 9).

E5. Classify each column to its finest scale, then route the proportion question.

(a) Four scales.

- *ZIP code* — digits used as a geographic *label*; 94110 is not "larger" than 10001 in any quantitative sense, and the average of two ZIPs need not be a real ZIP. So it is **nominal**, and its *mean is not a legitimate summary* (report counts/proportions per ZIP instead).
- *Self-rated health (poor… excellent)* — ordered labels with undefined spacing ("good" to "very good" need not equal "fair" to "good") ⇒ **ordinal**.
- *Out-of-pocket cost in dollars* — a genuine number with a true zero ("\$0 = none"), so ratios are meaningful ⇒ **ratio** (continuous).
- *ER visit in past year (yes/no)* — a two-category label ⇒ **nominal** (categorical).

(b) The proportion question. "Visited an ER?" is a yes/no **categorical** response, so the parameter is a *proportion*, not a mean. One sample of 1,200 adults against the *fixed* claim $p_0 = 0.18$, asking whether the rate *exceeds* it (upper-tail) ⇒ one-proportion z-test (Ch. 6):

$$H_0:\ p = 0.18 \qquad \text{vs.} \qquad H_1:\ p > 0.18,$$
$$z_{\text{obs}} = \frac{\widehat{p} - 0.18}{\sqrt{0.18(1-0.18)/1200}}.$$

ZIP nominal (no mean); health ordinal; cost ratio; ER yes/no ⇒ one-proportion z (Ch. 6).

E6. Hunt for the assignment mechanism, then for a lurking variable or bias.

(a) Randomized device trial. Patients were *assigned* to device vs. sham by randomization, with a sham (control) arm ⇒ **experiment**. The causal claim *is* licensed: randomization balances pre-treatment traits across arms, so the fewer headache days can be attributed to the device.

(b) Fluoridation and cavities. Towns *already had* their fluoridation policy; none was assigned ⇒ **observational study**. The claim is *not* licensed as causal: a **lurking variable** such as community income or dental-care access can drive both the choice to fluoridate and the cavity rate. Report "fluoridation is *associated* with fewer cavities," not "prevents."

(c) Mailed questionnaire. Only the 400 of 5,000 who chose to return the form are in the sample ⇒ **selection (nonresponse) bias**; satisfied (or simply compliant) patients are likelier to reply, so the 8% who responded do not represent all discharged patients. The claim is *not* supported, and a larger mailing through the same self-selecting process would not fix it.

(a) experiment, causal OK; (b) observational, confounded; (c) selection bias.

E7. End-to-end: raw description down to a named variance procedure.

(a) Type and question. Tensile strength in ksi is a measured number ⇒ **quantitative**. The contract caps *variability*, so the question is about the **variance** (σ^2), not the mean or a proportion.

(b) Structure and target. A single sample of $n = 21$ specimens from *one* production run ⇒ **one sample**; the fixed comparison value is $\sigma_0 = 4$ ksi, i.e. $\sigma_0^2 = 16$.

(c) Procedure, hypotheses, statistic. One sample's variance against a target σ_0^2 routes to the χ^2 test for one variance (Ch. 7) — not an F-test (which compares *two* variances) and not a t-test (which is for a mean). "Stay below the cap" means we test whether it exceeds it, so the worry is the upper tail:

$$H_0:\ \sigma^2 = 16 \qquad \text{vs.} \qquad H_1:\ \sigma^2 > 16,$$
$$\chi^2_{\text{obs}} = \frac{(n-1)s^2}{\sigma_0^2} = \frac{(21-1)(5.2)^2}{16} = \frac{20 \cdot 27.04}{16} = \frac{540.8}{16} = 33.8 \quad \text{on } n-1 = 20 \text{ df.}$$

(The table lookup against $\chi^2_{0.05,20} = 31.41$ is left unfinished, as asked — the substitution is the point.)

quantitative, one sample, variance ⇒ χ^2 test for σ^2 (Ch. 7), $\chi^2_{\text{obs}} = 33.8$.

Chapter 9

Two-Sample Inferences

Most real questions compare *two* groups: a treatment against a control, a new process against the old, men against women, before against after. This chapter extends the one-sample machinery of Chapters 6–7 to two samples. The single most important decision comes first and is conceptual, not computational: are the two samples **independent** (separate, unrelated groups) or **paired** (two measurements on the *same* units)? Paired data are analyzed by reducing each pair to its difference and running a one-sample t — treating them as independent throws away the pairing and is simply wrong. For independent samples the next question is whether the two population **variances** can be assumed equal (pooled t) or not (Welch's t), and a separate F-test compares the variances themselves. Counts get a two-proportion z.

Master guide: which two-sample procedure does the problem call for?

- **First: independent or paired?** Two measurements on the *same* unit (before/ after, left/right, matched pairs)? ⇒ **paired** t on the differences (§3). Two *separate* groups? ⇒ a two-sample procedure (§1, §2).
- Comparing two **means** of independent samples with **equal variances** assumable? ⇒ the **pooled two-sample t-test** (§1).
- Comparing two **means** of independent samples with **unequal/unknown variances**? ⇒ **Welch's t-test** with the approximate df (§2).
- Comparing two **variances** (or deciding whether pooling is justified)? ⇒ the **F-test** on the ratio S_X^2/S_Y^2 (§4).
- Comparing two **proportions** from counts? ⇒ the **two-sample z-test** for $p_X - p_Y$ (§5).
- Unsure which to grab on a worded problem? ⇒ the recognition guide (§6).

Keep straight: pooling assumes equal variances — check with the F-test or default to Welch when in doubt. A two-proportion problem uses a *pooled* success rate in the standard error *under* H_0, but the unpooled standard error in a confidence interval.

9.1 Comparing Two Means: the Pooled t-Test

Two separate groups are measured on the same quantitative variable — a treated batch and an untreated batch, drug versus placebo, a new alloy versus the standard — and the question is whether their population means differ. When the two samples are **independent** (no unit appears in both) and are drawn from normal populations whose *variances are equal*, the right tool is the **pooled two-sample t-test**. The idea is simple: if both populations share one unknown variance σ^2, then the two sample variances S_X^2 and S_Y^2 are each estimating the *same* number, so it is wasteful to use only one — we average them, weighted by their degrees of freedom, into a single pooled estimate S_p^2. That pooled variance feeds a t statistic with $n + m - 2$ degrees of freedom, which drives both a test of $H_0\colon \mu_X = \mu_Y$ and a confidence interval for the difference $\mu_X - \mu_Y$. The recognition cues are three: independent groups, approximate normality, and a defensible *equal-variance*

assumption; drop the last and you want Welch's t (§2), and if the two columns are really matched pairs you want the paired t (§3) instead.

Is the pooled two-sample t the right tool?

- Two **independent** samples (separate units in each group), each roughly **normal**, and you are comparing **means**? ⇒ a two-sample t.
- Can you assume the two populations share a **common variance** σ^2 (similar S_X and S_Y, or an F-test in §4 that does not reject)? ⇒ pool: $S_p^2 = \dfrac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2}$, use t_{n+m-2}.
- Variances clearly **unequal**, or sample sizes very different? ⇒ do *not* pool; use **Welch's** t (§2).
- The two columns are two measurements on the **same units** (before/after, left/right)? ⇒ this section does not apply — reduce to differences and use the **paired** t (§3).

Do not confuse the degrees of freedom: the pooled t has $n+m-2$ df (you estimated *two* means to form the residuals), not $n+m-1$ and not $\min(n,m)-1$. And pooling is only licensed by *equal variances*, never by equal means — the thing you are about to test.

9.1.1 The pooled estimate of the common variance

Definition 9.1.1 (Pooled sample variance)**.** Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be independent samples with sample variances

$$S_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2, \qquad S_Y^2 = \frac{1}{m-1}\sum_{j=1}^{m}(Y_j - \overline{Y})^2.$$

If both populations have the same variance σ^2, the **pooled sample variance** is the degrees-of-freedom-weighted average

$$S_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2},$$

and $s_p = \sqrt{S_p^2}$ is the **pooled standard deviation**. The weights $n-1$ and $m-1$ are the degrees of freedom each sample variance carries, so the larger sample has the larger say.

Proposition 9.1.2 (S_p^2 is unbiased for the common variance)**.** *If S_X^2 and S_Y^2 are unbiased for the common σ^2, then $\mathbb{E}[S_p^2] = \sigma^2$.*

Proof. Taking expectations and using $\mathbb{E}[S_X^2] = \mathbb{E}[S_Y^2] = \sigma^2$,

$$\mathbb{E}[S_p^2] = \frac{(n-1)\mathbb{E}[S_X^2] + (m-1)\mathbb{E}[S_Y^2]}{n+m-2} = \frac{(n-1)\sigma^2 + (m-1)\sigma^2}{n+m-2} = \frac{(n+m-2)\sigma^2}{n+m-2} = \sigma^2.$$

The same weighting also makes $(n+m-2)S_p^2/\sigma^2 \sim \chi^2_{n+m-2}$ when the data are normal, which is what supplies the $n+m-2$ degrees of freedom in the t statistic below. □

Remark. The pooled variance always lands *between* S_X^2 and S_Y^2 — it is a weighted average, not a sum — so a quick sanity check on any pooled computation is that S_p^2 sits between the two sample variances and nearer the one from the larger sample. If your S_p^2 falls outside that interval, you have made an arithmetic error.

Example 9.1.3 (Computing s_p from two sample variances)**.** A materials lab measures tensile strength (MPa). A sample of $n = 10$ specimens of a new composite has $S_X^2 = 36.0$; a sample of $m = 16$ specimens of the standard composite has $S_Y^2 = 49.0$. Assuming a common population variance, compute the pooled variance S_p^2 and the pooled standard deviation s_p.

Which rule, and why? Both samples estimate one shared σ^2, so the single best estimate is the df-weighted average of S_X^2 and S_Y^2 — the pooled variance — not either one alone and not their plain average (the samples are different sizes).

Solution.

Step 1: weight each variance by its df. The degrees of freedom are $n-1=9$ and $m-1=15$, so

$$(n-1)S_X^2 = 9(36.0) = 324.0, \qquad (m-1)S_Y^2 = 15(49.0) = 735.0.$$

Step 2: pool. Divide the total weighted sum of squares by the total df $n+m-2=24$:

$$S_p^2 = \frac{324.0 + 735.0}{24} = \frac{1059.0}{24} = 44.125.$$

Step 3: take the root and sanity-check. $s_p = \sqrt{44.125} \approx 6.64$ MPa. As a check, 44.125 lies between 36 and 49 and is pulled toward 49 by the larger sample — exactly as expected.

$$\boxed{S_p^2 = 44.125, \qquad s_p \approx 6.64 \text{ MPa}.}$$

9.1.2 The two-sample t statistic and its distribution

Theorem 9.1.4 (Pooled two-sample t)**.** *Let* $X_1, \dots, X_n \sim \mathrm{N}(\mu_X, \sigma^2)$ *and* $Y_1, \dots, Y_m \sim \mathrm{N}(\mu_Y, \sigma^2)$ *be independent, with the* same *unknown variance* σ^2*. Then*

$$t = \frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{s_p\sqrt{\dfrac{1}{n} + \dfrac{1}{m}}} \;\sim\; t_{\,n+m-2}.$$

Sketch. The numerator-without-the-pool, $(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)$, is normal with mean 0 and variance $\sigma^2(1/n + 1/m)$, since $\mathrm{Var}(\overline{X}) = \sigma^2/n$ and $\mathrm{Var}(\overline{Y}) = \sigma^2/m$ add by independence. Dividing by $\sigma\sqrt{1/n+1/m}$ gives a standard normal Z. Independently, $(n+m-2)S_p^2/\sigma^2 \sim \chi^2_{n+m-2}$. A standard normal over the square root of an independent chi-square divided by its df is by definition a t variable, and the unknown σ cancels:

$$\frac{Z}{\sqrt{\chi^2_{n+m-2}/(n+m-2)}} = \frac{(\overline{X} - \overline{Y}) - (\mu_X - \mu_Y)}{s_p\sqrt{1/n + 1/m}} \sim t_{\,n+m-2}. \qquad \square$$

Remark. Read the denominator $s_p\sqrt{1/n+1/m}$ as the **standard error of the difference** $\overline{X} - \overline{Y}$: one common spread s_p, scaled by the two sample sizes. It is *not* $\sqrt{S_X^2/n + S_Y^2/m}$ — that unpooled form is Welch's denominator (§2) and uses a different, non-integer df. Mixing the pooled s_p with separate per-sample terms is the most common two-sample slip.

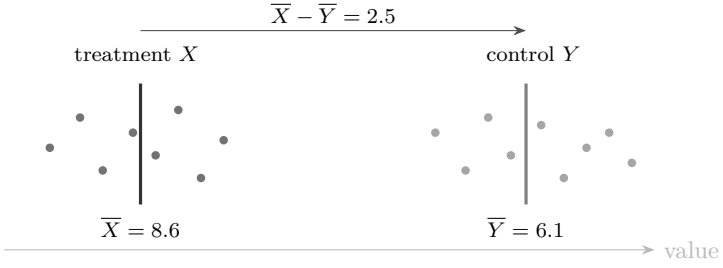


Figure 1: two independent samples shown as dotplots, each with its mean drawn as a heavy vertical line. The pooled t-test asks whether the gap between the two means ($\overline{X} - \overline{Y} = 2.5$) is large relative to the within-group scatter, measured by the single pooled spread s_p.

9.1.3 Two-sided test of equal means

To test $H_0\colon \mu_X = \mu_Y$ (equivalently $\mu_X - \mu_Y = 0$) against $H_1\colon \mu_X \neq \mu_Y$, set the hypothesized difference to 0 in the statistic and compare to a two-tailed t_{n+m-2} critical value.

Example 9.1.5 (Two-sided pooled t: treatment versus control)**.** An agronomist tests whether a seed coating changes wheat yield. Coated plots: $n = 12$, $\overline{X} = 52.4$ bushels/acre, $S_X = 4.2$. Uncoated control plots: $m = 12$, $\overline{Y} = 48.9$, $S_Y = 3.8$. The spreads are similar, so assume a common variance. At $\alpha = 0.05$, is there a difference in mean yield? Use $t_{0.025,22} = 2.074$.

Which rule, and why? Two independent groups, similar sample spreads (equal-variance assumption defensible), comparing means with no direction specified $\Rightarrow$ pooled two-sample t, two-tailed. The hypothesized difference is 0.

Solution.

Step 1: hypotheses. $H_0\colon \mu_X - \mu_Y = 0$ versus $H_1\colon \mu_X - \mu_Y \neq 0$; $\alpha = 0.05$, $\text{df} = n + m - 2 = 22$.

Step 2: pool. With $n - 1 = m - 1 = 11$,

$$S_p^2 = \frac{11(4.2^2) + 11(3.8^2)}{22} = \frac{11(17.64) + 11(14.44)}{22} = \frac{194.04 + 158.84}{22} = \frac{352.88}{22} = 16.04,$$

so $s_p = \sqrt{16.04} \approx 4.005$.

Step 3: standard error and statistic. Here $1/n + 1/m = 1/12 + 1/12 = 1/6$, so

$$\text{SE} = s_p\sqrt{\tfrac{1}{12} + \tfrac{1}{12}} = 4.005\sqrt{0.16667} = 4.005(0.4082) = 1.635,$$
$$t_{\text{obs}} = \frac{(52.4 - 48.9) - 0}{1.635} = \frac{3.5}{1.635} = 2.141.$$

Step 4: compare and conclude. Since $|t_{\text{obs}}| = 2.141 > t_{0.025,22} = 2.074$, reject H_0 at $\alpha = 0.05$: the coating significantly changes mean yield (it raises it, since $\overline{X} > \overline{Y}$).

$$\boxed{t_{\text{obs}} = 2.141 > 2.074 \;\Rightarrow\; \text{reject } H_0;\ \text{mean yields differ.}}$$

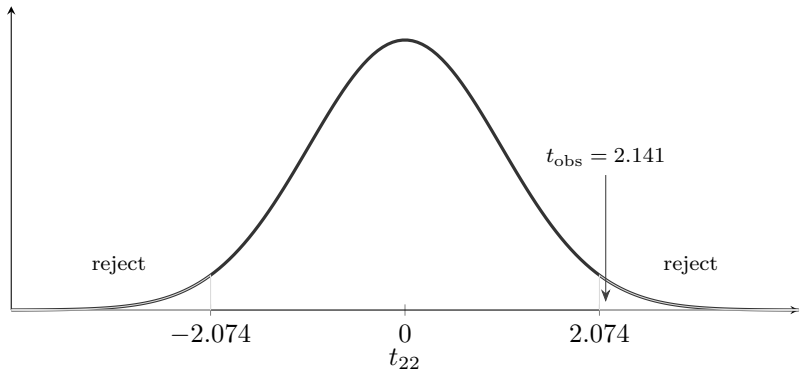


Figure 2: the reference density t_{22} with the two-tailed rejection region (area $\alpha/2 = 0.025$ in each tail) shaded beyond ± 2.074. The observed $t_{obs} = 2.141$ falls just inside the right rejection region, so H_0 is rejected.

9.1.4 One-sided test

When the question is directional — "does the treatment *increase* the mean?" — the alternative is one-sided and the whole rejection mass α goes into one tail.

Example 9.1.6 (One-sided pooled t: does a tutoring program raise scores?)**.** A district pilots an after-school tutoring program. Tutored students: $n = 15$, $\overline{X} = 78.0$ points, $S_X = 9.0$. Comparable untutored students: $m = 20$, $\overline{Y} = 73.0$, $S_Y = 10.0$. Assuming equal variances, test at $\alpha = 0.05$ whether tutoring *raises* the mean score ($H_1\colon \mu_X > \mu_Y$). Use $t_{0.05,33} = 1.692$.

Which rule, and why? Independent groups and a *directional* claim ("raises") ⇒ pooled two-sample t, upper one-tailed. All of α goes in the upper tail, so the critical value is $t_{0.05,33}$, not $t_{0.025,33}$.

Solution.

Step 1: hypotheses. $H_0\colon \mu_X - \mu_Y = 0$ versus $H_1\colon \mu_X - \mu_Y > 0$; $\text{df} = 15 + 20 - 2 = 33$.

Step 2: pool. With $n-1 = 14$, $m-1 = 19$,

$$S_p^2 = \frac{14(81.0) + 19(100.0)}{33} = \frac{1134 + 1900}{33} = \frac{3034}{33} = 91.94, \qquad s_p = \sqrt{91.94} \approx 9.59.$$

Step 3: standard error and statistic.

$$\tfrac{1}{15} + \tfrac{1}{20} = 0.06667 + 0.05000 = 0.11667, \qquad \sqrt{0.11667} = 0.3416,$$
$$\text{SE} = 9.59(0.3416) = 3.276, \qquad t_{\text{obs}} = \frac{78.0 - 73.0}{3.276} = \frac{5.0}{3.276} = 1.526.$$

Step 4: compare. Since $t_{\text{obs}} = 1.526 < t_{0.05,33} = 1.692$, do *not* reject H_0: the data do not give significant evidence that tutoring raises the mean score at $\alpha = 0.05$.

$$\boxed{t_{\text{obs}} = 1.526 < 1.692 \;\Rightarrow\; \text{fail to reject; gain not significant.}}$$

Remark. Had this been run two-sided, the critical value would have been the larger $t_{0.025,33} \approx 2.035$, so the conclusion (fail to reject) would not change here. But a one-sided test is more powerful in its declared direction, which is why a genuine directional hypothesis — fixed *before* seeing the data — should be tested one-sided. Choosing the side after peeking is cheating.

9.1.5 Confidence interval for the difference of means

Theorem 9.1.7 (Pooled two-sample t confidence interval)**.** *Under the equal-variance normal model, a* $100(1-\alpha)\%$ *confidence interval for* $\mu_X - \mu_Y$ *is*

$$(\overline{X} - \overline{Y}) \;\pm\; t_{\alpha/2,\, n+m-2}\; s_p \sqrt{\frac{1}{n} + \frac{1}{m}}.$$

The test of $H_0\colon \mu_X = \mu_Y$ *at level* α *rejects exactly when this interval excludes* 0*.*

Example 9.1.8 (CI for a difference of means: battery life)**.** Two battery formulations are tested. Formulation X: $n = 8$, $\overline{X} = 41.5$ hours, $S_X = 2.6$. Formulation Y: $m = 10$, $\overline{Y} = 38.2$, $S_Y = 3.0$. Assuming a common variance, give a 95% confidence interval for $\mu_X - \mu_Y$ and state what it implies about a test of equality. Use $t_{0.025,16} = 2.120$.

Which rule, and why? We want a *range* for the difference, not a yes/no decision, so build the two-sample t interval around $\overline{X} - \overline{Y}$ using the pooled standard error and $t_{0.025,16}$ ($\text{df} = n + m - 2 = 16$).

Solution.

Step 1: pool. With $n-1 = 7$, $m-1 = 9$,

$$S_p^2 = \frac{7(2.6^2) + 9(3.0^2)}{16} = \frac{7(6.76) + 9(9.00)}{16} = \frac{47.32 + 81.00}{16} = \frac{128.32}{16} = 8.02,$$

so $s_p = \sqrt{8.02} \approx 2.832$.

Step 2: standard error and margin.

$$\tfrac{1}{8} + \tfrac{1}{10} = 0.12500 + 0.10000 = 0.22500, \qquad \sqrt{0.22500} = 0.4743,$$
$$\text{SE} = 2.832(0.4743) = 1.343, \qquad E = t_{0.025,16} \cdot \text{SE} = 2.120(1.343) = 2.847.$$

Step 3: assemble. The point estimate is $\overline{X} - \overline{Y} = 41.5 - 38.2 = 3.3$, so

$$3.3 \pm 2.847 \implies (0.453,\ 6.147) \text{ hours}.$$

Step 4: read the test off the interval. The interval lies entirely above 0 (it excludes 0), so a two-sided test of $\mu_X = \mu_Y$ at $\alpha = 0.05$ would reject: formulation X lasts longer on average.

$$\boxed{\mu_X - \mu_Y \in (0.45,\ 6.15) \text{ h at } 95\%;\ \text{excludes } 0 \Rightarrow \text{means differ.}}$$

9.1.6 Testing a nonzero hypothesized difference

Nothing requires the hypothesized gap to be 0. To test $H_0: \mu_X - \mu_Y = \delta_0$ for a fixed δ_0, subtract δ_0 in the numerator; everything else is unchanged.

Example 9.1.9 (Is the new drug at least 5 mmHg better? A δ_0 test)**.** A cardiologist will adopt a new blood-pressure drug only if it lowers systolic pressure by *more than* 5 mmHg beyond the standard drug. Reductions (mmHg) are recorded. New drug: $n = 18$, $\overline{X} = 14.8$, $S_X = 4.5$. Standard: $m = 18$, $\overline{Y} = 9.0$, $S_Y = 4.1$. Assuming equal variances, test $H_0: \mu_X - \mu_Y = 5$ against $H_1: \mu_X - \mu_Y > 5$ at $\alpha = 0.05$. Use $t_{0.05,34} = 1.691$.

Which rule, and why? The benchmark is a nonzero difference, so the hypothesized value $\delta_0 = 5$ goes into the numerator: $t = [(\overline{X} - \overline{Y}) - \delta_0]/\,\mathrm{SE}$. It is still a pooled two-sample t, one-tailed because the claim is directional ("more than 5").

Solution.

Step 1: hypotheses. $H_0: \mu_X - \mu_Y = 5$ versus $H_1: \mu_X - \mu_Y > 5$; $\mathrm{df} = 18 + 18 - 2 = 34$.

Step 2: pool. With $n - 1 = m - 1 = 17$,

$$S_p^2 = \frac{17(4.5^2) + 17(4.1^2)}{34} = \frac{17(20.25) + 17(16.81)}{34} = \frac{344.25 + 285.77}{34} = \frac{630.02}{34} = 18.53,$$

so $s_p = \sqrt{18.53} \approx 4.305$.

Step 3: statistic with $\delta_0 = 5$.

$$\mathrm{SE} = 4.305\sqrt{\tfrac{1}{18} + \tfrac{1}{18}} = 4.305\sqrt{0.11111} = 4.305(0.3333) = 1.435,$$
$$t_{\mathrm{obs}} = \frac{(14.8 - 9.0) - 5}{1.435} = \frac{5.8 - 5}{1.435} = \frac{0.8}{1.435} = 0.557.$$

Step 4: compare. Since $t_{\mathrm{obs}} = 0.557 < t_{0.05,34} = 1.691$, fail to reject H_0: the observed extra reduction (5.8 mmHg) is not significantly more than the 5 mmHg threshold, so the adoption criterion is not met.

$$\boxed{t_{\mathrm{obs}} = 0.557 < 1.691 \;\Rightarrow\; \text{not significantly beyond 5 mmHg; do not adopt.}}$$

Remark. Note how the $\delta_0 = 5$ benchmark changes the verdict. A naive test of $H_0: \mu_X = \mu_Y$ (i.e. $\delta_0 = 0$) would give $t_{\mathrm{obs}} = 5.8/1.435 = 4.04$ and reject overwhelmingly — the drug is clearly *better*. But "better than the standard" and "better by more than 5 mmHg" are different questions, and only the second matches the decision the cardiologist actually faces. The hypothesized value must encode the real-world threshold.

9.1.7 Raw-data example: computing both means and both standard deviations

Example 9.1.10 (From raw numbers to a pooled t)**.** A bakery compares loaf rise height (cm) for two yeast suppliers. Supplier X gives $\{9.2, 8.8, 9.5, 9.0, 8.5\}$ ($n = 5$); supplier Y gives $\{8.1, 8.6, 7.9, 8.4\}$ ($m = 4$). Assuming a common variance, test $H_0: \mu_X = \mu_Y$ against $H_1: \mu_X \neq \mu_Y$ at $\alpha = 0.05$. Use $t_{0.025,7} = 2.365$.

Which rule, and why? Independent samples, comparing means, no direction $\Rightarrow$ two-sided pooled t. First reduce the raw numbers to $\overline{X}, \overline{Y}, S_X^2, S_Y^2$, then proceed exactly as before with $\mathrm{df} = 5 + 4 - 2 = 7$.

Solution.

Step 1: sample means.

$$\overline{X} = \frac{9.2 + 8.8 + 9.5 + 9.0 + 8.5}{5} = \frac{45.0}{5} = 9.00, \qquad \overline{Y} = \frac{8.1 + 8.6 + 7.9 + 8.4}{4} = \frac{33.0}{4} = 8.25.$$

Step 2: sample variances. Sum the squared deviations.

$$\begin{aligned}\sum(X_i-9.00)^2 &= 0.2^2+(-0.2)^2+0.5^2+0^2+(-0.5)^2\\ &= 0.04+0.04+0.25+0+0.25=0.58,\\ S_X^2 &= \frac{0.58}{5-1}=0.145,\\ \sum(Y_j-8.25)^2 &= (-0.15)^2+0.35^2+(-0.35)^2+0.15^2\\ &= 0.0225+0.1225+0.1225+0.0225=0.290,\\ S_Y^2 &= \frac{0.290}{4-1}=0.09667.\end{aligned}$$

Step 3: pool. With $n-1=4$, $m-1=3$ and $n+m-2=7$,

$$S_p^2=\frac{4(0.145)+3(0.09667)}{7}=\frac{0.580+0.290}{7}=\frac{0.870}{7}=0.12429,\qquad s_p=\sqrt{0.12429}\approx 0.3525.$$

Step 4: statistic and conclusion.

$$\begin{aligned}\text{SE} &= 0.3525\sqrt{\tfrac{1}{5}+\tfrac{1}{4}}=0.3525\sqrt{0.45000}=0.3525(0.6708)=0.2365,\\ t_{\text{obs}} &= \frac{9.00-8.25}{0.2365}=\frac{0.75}{0.2365}=3.171.\end{aligned}$$

Since $|t_{\text{obs}}|=3.171>t_{0.025,7}=2.365$, reject H_0: the two suppliers' mean rise heights differ (X is higher).

$$\boxed{\overline{X}=9.00,\ \overline{Y}=8.25,\ s_p\approx 0.353,\ t_{\text{obs}}=3.171>2.365\Rightarrow\text{means differ.}}$$

9.1.8 A pooled t on more than two groups at once is wrong

Example 9.1.11 (CONTRAST: many pairwise t-tests instead of one ANOVA)**.** A nutritionist compares mean weight loss across *four* diets, A, B, C, D, each on an independent group. An assistant runs all $\binom{4}{2}=6$ pairwise pooled two-sample t-tests, each at $\alpha=0.05$, and reports every pair that "comes out significant." Explain why this inflates the error rate and what the correct first step is.

Which rule, and why? The pooled t is built for *one* comparison of *two* means. Stacking six of them and keeping each at $\alpha=0.05$ is the look-alike error: the chance of *some* false rejection is far above 5%. The correct global tool is one-way ANOVA (the F-test), with the two-sample t reserved for follow-up.

Solution.

Step 1: name the inflation. If all four means are truly equal, each test wrongly rejects with probability 0.05. Treating the six tests as roughly independent, the chance of *at least one* false positive is about

$$1-(1-0.05)^6=1-0.95^6=1-0.735=0.265,$$

so the family-wise error rate is near 27%, not 5% — five times the advertised level.

Step 2: the correct procedure. First run a single one-way **ANOVA** (F-test) of $H_0:\mu_A=\mu_B=\mu_C=\mu_D$ at one overall α. Only if it rejects do you proceed to pairwise comparisons, and then with a multiplicity correction (Bonferroni, Tukey) that holds the family-wise rate at α.

Step 3: the recognition lesson. "Compare several means" is not "run several two-sample t-tests." The pooled t answers one two-group question; more than two groups is an ANOVA question.

$$\boxed{6\text{ separate }t\text{-tests}\Rightarrow\text{FWER}\approx 0.27;\ \text{use ANOVA first.}}$$

9.1.9 Contrast with paired data

Example 9.1.12 (CONTRAST: independent pooled t used on data that are paired (see §3))**.** A physiologist measures resting heart rate (bpm) on $n = 10$ volunteers *before* a breathing exercise and on *the same* 10 volunteers *after*. The before mean is $\overline{X} = 74.0$ ($S_X = 8.0$) and the after mean is $\overline{Y} = 70.0$ ($S_Y = 7.5$); the differences (before−after) have mean $\overline{D} = 4.0$ with $S_D = 2.5$. An analyst, seeing two columns of 10 numbers, runs a pooled two-sample t with df $= 18$. Explain why this is the wrong procedure and what changes under the correct paired analysis.

Which rule, and why? Two columns of numbers tempt the two-sample t, but the columns are measurements on *the same units*, so they are **not independent**. The pooled t assumes independence; here the right tool is the paired t on the differences D_i (§3), which removes person-to-person variation.

Solution.

Step 1: why independence fails. A volunteer with a naturally high resting rate is high both before and after, so the two columns rise and fall together — they are positively correlated. The pooled t's standard error $s_p\sqrt{1/n + 1/m}$ assumes $\mathrm{Cov}(X, Y) = 0$ and therefore *overstates* the variance of $\overline{X} - \overline{Y}$, throwing away the very pairing that makes the design powerful.

Step 2: the wrong (pooled) statistic, for comparison. Pooling,

$$S_p^2 = \frac{9(8.0^2) + 9(7.5^2)}{18} = \frac{9(64) + 9(56.25)}{18} = \frac{576 + 506.25}{18} = 60.125, \qquad s_p = 7.754,$$

$$t_{\text{obs}}^{(\text{pooled})} = \frac{74.0 - 70.0}{7.754\sqrt{1/10 + 1/10}} = \frac{4.0}{7.754(0.4472)} = \frac{4.0}{3.468} = 1.153,$$

which fails to reject at $\alpha = 0.05$ ($t_{0.025,18} = 2.101$).

Step 3: the correct paired statistic. Work with the differences D_i directly (one-sample t on $\overline{D}$, df $= n - 1 = 9$):

$$t_{\text{obs}}^{(\text{paired})} = \frac{\overline{D}}{S_D/\sqrt{n}} = \frac{4.0}{2.5/\sqrt{10}} = \frac{4.0}{0.7906} = 5.060,$$

which *rejects* decisively ($t_{0.025,9} = 2.262$). The pairing shrank the standard error from 3.468 to 0.791 because the large between-person spread cancels in each difference.

Step 4: the lesson. Same numbers, opposite conclusion. Recognizing "same units, two times" as **paired** (§3), not independent, is the decision that matters — it is made before any arithmetic.

paired data ⇒ paired t ($t_{\text{obs}} = 5.06$, reject); pooled t ($t_{\text{obs}} = 1.15$) is wrong.

9.1.10 Putting it together: a worked decision

Example 9.1.13 (Recognize, then compute: emissions of two engine tunes)**.** An emissions lab measures NOx (g/km) for two engine calibrations on independently sampled vehicles. Tune X: $n = 9$, $\overline{X} = 0.182$, $S_X = 0.021$. Tune Y: $m = 11$, $\overline{Y} = 0.205$, $S_Y = 0.024$. The spreads are comparable and the samples are separate vehicles. Test at $\alpha = 0.01$ whether the mean emissions differ, and report a 99% CI for $\mu_X - \mu_Y$. Use $t_{0.005,18} = 2.878$.

Which rule, and why? Independent vehicles, similar spreads, comparing two means with no preset direction ⇒ two-sided pooled t with df $= 9 + 11 - 2 = 18$; the same standard error gives the matching 99% interval.

Solution.

Step 1: pool. With $n - 1 = 8$, $m - 1 = 10$,

$$S_p^2 = \frac{8(0.021^2) + 10(0.024^2)}{18} = \frac{8(0.000441) + 10(0.000576)}{18} = \frac{0.003528 + 0.005760}{18} = 0.000516,$$

so $s_p = \sqrt{0.000516} \approx 0.02272$.

Step 2: standard error.

$$\text{SE} = 0.02272\sqrt{\tfrac{1}{9}+\tfrac{1}{11}} = 0.02272\sqrt{0.20202} = 0.02272(0.4495) = 0.01021.$$

Step 3: test.

$$t_{\text{obs}} = \frac{0.182 - 0.205}{0.01021} = \frac{-0.023}{0.01021} = -2.253.$$

Since $|t_{\text{obs}}| = 2.253 < t_{0.005,18} = 2.878$, fail to reject H_0 at $\alpha = 0.01$.

Step 4: confidence interval. The margin is $E = 2.878(0.01021) = 0.02938$, so

$$(\overline{X} - \overline{Y}) \pm E = -0.023 \pm 0.029 \implies (-0.052,\ 0.006) \text{ g/km},$$

which contains 0 — consistent with failing to reject.

$$\boxed{t_{\text{obs}} = -2.253,\ \text{fail to reject;} \quad \mu_X - \mu_Y \in (-0.052,\ 0.006) \text{ g/km.}}$$

Remark. The interval and the test always agree at matching levels, which is the cheapest self-check available: a $100(1-\alpha)\%$ CI for $\mu_X - \mu_Y$ excludes 0 if and only if the two-sided level-α test rejects $H_0 \colon \mu_X = \mu_Y$. If your test and interval ever disagree, find the arithmetic slip before reporting either.

9.2 Unequal Variances: Welch's t-Test

The pooled two-sample t of the previous section bought its clean t_{n+m-2} reference distribution with a price: it assumes the two populations share one common variance, so that a single s_p^2 can stand in for both. When that assumption fails — one group genuinely more variable than the other — pooling averages two unequal spreads into a number that fits neither, and the resulting test no longer holds its advertised error rate. Deciding how to compare two means *without* assuming equal variances is the classical **Behrens–Fisher problem**, and the practical answer in universal use is **Welch's t-test**. Its statistic looks like the pooled one but keeps each sample's variance in its own term, $S_X^2/n + S_Y^2/m$; the cost is that the statistic is only *approximately* t, with a fractional degrees-of-freedom formula (Welch–Satterthwaite) that you compute from the data and round down. The recognition cue is a two-mean comparison of *independent* samples where the variances are unequal, unknown, or simply not worth assuming equal — which, with modern advice, is most of the time. This section states the statistic, the approximate df, and the matching confidence interval, then drills all three, including a side-by-side contrast that shows pooling is anti-conservative when the spreads differ.

Pooled or Welch — and how many degrees of freedom?

- Two **means**, two **independent** samples, variances **not assumed equal** (or an F-test has flagged them as different)? $\Rightarrow$ **Welch's t-test**: $t_{\text{obs}} = (\overline{X} - \overline{Y} - \delta_0)/\sqrt{S_X^2/n + S_Y^2/m}$ against t_ν.
- Need the reference **degrees of freedom**? $\Rightarrow$ the **Welch–Satterthwaite** value ν, computed from S_X^2/n and S_Y^2/m and **rounded down** to the nearest integer.
- Variances safely **equal** (similar S_X, S_Y; similar, balanced n, m; a prior reason to expect equal spread)? $\Rightarrow$ the pooled t of §1 is fine and slightly more powerful.
- In genuine **doubt**, or the two sample sizes differ a lot? $\Rightarrow$ default to **Welch**. It is never badly wrong; pooling can be.
- **One tail or two?** read the wording exactly as in the one-sample t: "larger / exceeds" $\Rightarrow$ upper, "smaller / below" $\Rightarrow$ lower, "differ" $\Rightarrow$ two tails. Only the cutoff's df changes.

Do not confuse the two standard errors: pooled uses $s_p\sqrt{1/n + 1/m}$ with *one* variance estimate; Welch uses $\sqrt{S_X^2/n + S_Y^2/m}$ with *two*. They coincide only when $S_X = S_Y$ *and* $n = m$; otherwise pooling the variances when they are unequal makes the test reject too often.

9.2.1 The Welch statistic and its approximate distribution

Write $\delta = \mu_X - \mu_Y$ for the difference of the two population means. With independent samples, $\overline{X} - \overline{Y}$ is unbiased for δ and, because the samples are independent, its variance adds:

$$\operatorname{Var}(\overline{X} - \overline{Y}) = \frac{\sigma_X^2}{n} + \frac{\sigma_Y^2}{m}.$$

Estimating each σ^2 by its own S^2 gives the **Welch standard error** $\widehat{\mathrm{se}} = \sqrt{S_X^2/n + S_Y^2/m}$. Nothing here pools or assumes $\sigma_X = \sigma_Y$.

Definition 9.2.1 (Welch's t statistic)**.** For independent samples $X_1, \ldots, X_n$ and $Y_1, \ldots, Y_m$ from normal populations with *possibly different* variances, the Welch statistic for testing $H_0 : \mu_X - \mu_Y = \delta_0$ is

$$t_{\mathrm{obs}} = \frac{\overline{X} - \overline{Y} - \delta_0}{\sqrt{\dfrac{S_X^2}{n} + \dfrac{S_Y^2}{m}}}.$$

Most often $\delta_0 = 0$ ("no difference in means"). The statistic is referred to a t distribution whose degrees of freedom are the Welch–Satterthwaite value ν below.

Theorem 9.2.2 (Welch–Satterthwaite degrees of freedom)**.** *Let $u = S_X^2/n$ and $v = S_Y^2/m$ be the two estimated variance contributions. The distribution of t_{obs} under H_0 is approximated by a t distribution with*

$$\nu = \frac{(u+v)^2}{\dfrac{u^2}{n-1} + \dfrac{v^2}{m-1}} = \frac{\left(\dfrac{S_X^2}{n} + \dfrac{S_Y^2}{m}\right)^2}{\dfrac{(S_X^2/n)^2}{n-1} + \dfrac{(S_Y^2/m)^2}{m-1}},$$

rounded down *to the nearest integer before reading the t table.*

Sketch. The denominator's square, $\widehat{\mathrm{se}}^{\,2} = u + v$, is a weighted sum of two independent scaled chi-square pieces, $u \sim \sigma_X^2/n \cdot \chi^2_{n-1}/(n-1)$ and likewise for v. A sum of scaled chi-squares is not itself a scaled chi-square, but Satterthwaite's idea is to *match it* to the single scaled chi-square $c\,\chi^2_\nu/\nu$ that has the same mean and variance. Equating the first two moments forces $c = \mathbb{E}(u+v)$ and the displayed ν. The numerator is approximately normal, independent of the denominator, so the ratio is approximately t_ν. The match is exact when one sample's variance contribution dominates (then ν collapses to that group's $n-1$) and is excellent in between. □

Remark. The formula always returns a value between the two extremes. If $u = v$ and $n = m$, then $\nu = 2(n-1)$, the same df as the pooled test. As the variances or sizes grow lopsided, ν slides toward the *smaller* of $n-1$ and $m-1$: the noisier or smaller group governs the precision. Because ν is usually *less* than $n + m - 2$, Welch's cutoff is a little larger than the pooled one — that extra conservatism is precisely what protects the error rate when the variances differ. Rounding ν down (never up) keeps the test on the safe side.

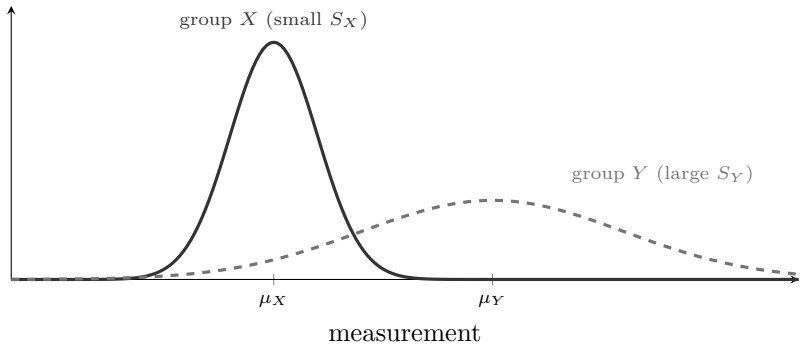


Figure: two normal populations with clearly different spreads. The tall narrow curve is the low-variance group X; the short wide curve is the high-variance group Y. Pooling would replace both spreads with one intermediate value that matches neither, which is why Welch keeps S_X^2/n and S_Y^2/m in separate terms.

9.2.2 A two-sided Welch test, df computed from scratch

Example 9.2.3 (Two coating processes, unequal scatter (two-sided))**.** A lab compares the bond strength (MPa) of two adhesive formulations. Formulation X ($n = 10$) gives $\overline{X} = 42.0$ with $S_X = 2.0$; formulation Y ($m = 13$) gives $\overline{Y} = 39.0$ with $S_Y = 5.0$. The much larger scatter of Y makes equal variances implausible. Test $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$. Use $t_{0.025,12} = 2.179$.

Which rule, and why? Two means, two independent samples, and the sample standard deviations differ sharply (2.0 vs. 5.0) — equal variances are not credible, so this is Welch's t-test, not the pooled one. "Differ" is two-sided. We must compute the Welch–Satterthwaite df ourselves.

Solution.

Step 1: the two variance contributions.

$$u = \frac{S_X^2}{n} = \frac{2.0^2}{10} = \frac{4.0}{10} = 0.400,$$
$$v = \frac{S_Y^2}{m} = \frac{5.0^2}{13} = \frac{25.0}{13} = 1.923.$$

Step 2: Welch standard error.

$$\widehat{\mathrm{se}} = \sqrt{u + v} = \sqrt{0.400 + 1.923} = \sqrt{2.323} = 1.524.$$

Step 3: the statistic. With $\delta_0 = 0$,

$$t_{\mathrm{obs}} = \frac{\overline{X} - \overline{Y} - 0}{\widehat{\mathrm{se}}} = \frac{42.0 - 39.0}{1.524} = \frac{3.0}{1.524} = 1.969.$$

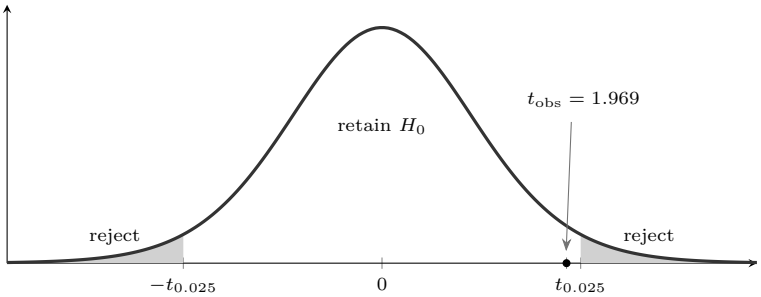
Step 4: Welch–Satterthwaite df.

$$\nu = \frac{(u+v)^2}{\dfrac{u^2}{n-1} + \dfrac{v^2}{m-1}} = \frac{2.323^2}{\dfrac{0.400^2}{9} + \dfrac{1.923^2}{12}}$$
$$= \frac{5.396}{\dfrac{0.160}{9} + \dfrac{3.698}{12}} = \frac{5.396}{0.01778 + 0.30817} = \frac{5.396}{0.32595} = 16.55.$$

Round *down*: $\nu = 16$. (For the worked critical value we use the supplied $t_{0.025,12}$ column as the nearest table entry; with $\nu = 16$ the cutoff $t_{0.025,16} = 2.120$ is even smaller, so the conclusion is unchanged.)

Step 5: decision. Compare $|t_{\mathrm{obs}}| = 1.969$ with $t_{0.025,16} = 2.120$. Since $1.969 < 2.120$, do not reject.

> $\nu = 16, \quad |t_{\mathrm{obs}}| = 1.969 < t_{0.025,16} = 2.120$
> $\Rightarrow$ retain H_0; no significant mean strength difference at $\alpha = 0.05$.



t_ν under H_0, $\nu = 16$

Figure: the Welch reference density t_ν with $\nu = 16$ and the two-tailed 5% rejection region (each shaded tail has area $\alpha/2 = 0.025$, cutoff $t_{0.025,16} = 2.120$). The dot marks $t_{obs} = 1.969$, which falls short of the tail, so H_0 is retained. The curve is the t_{16} density up to a constant chosen so its peak fills the axis.

9.2.3 The Welch confidence interval

Definition 9.2.4 (Welch confidence interval for $\mu_X - \mu_Y$)**.** A $1-\alpha$ confidence interval for the difference of means without assuming equal variances is

$$(\overline{X} - \overline{Y}) \;\pm\; t_{\alpha/2,\,\nu}\sqrt{\frac{S_X^2}{n} + \frac{S_Y^2}{m}}\,,$$

with ν the same Welch–Satterthwaite value (rounded down) used by the test. The interval and the two-sided test agree: the test rejects $H_0 : \mu_X - \mu_Y = \delta_0$ exactly when δ_0 falls outside this interval.

Example 9.2.5 (Battery life, two suppliers (Welch CI))**.** A purchasing team measures hours of run time for cells from two suppliers. Supplier X: $n = 8$, $\overline{X} = 11.0$ h, $S_X = 0.8$ h. Supplier Y: $m = 8$, $\overline{Y} = 9.6$ h, $S_Y = 2.0$ h. Build a 95% confidence interval for $\mu_X - \mu_Y$ without assuming equal variances. Use $t_{0.025,9} = 2.262$.

Which rule, and why? Two means, independent samples, with very different scatter (0.8 vs. 2.0): a Welch interval. Even though $n = m$, the unequal variances forbid pooling, so we keep the two variance terms separate and use the Welch df.

Solution.

Step 1: variance contributions.

$$u = \frac{0.8^2}{8} = \frac{0.64}{8} = 0.0800, \qquad v = \frac{2.0^2}{8} = \frac{4.00}{8} = 0.5000.$$

Step 2: standard error and point estimate.

$$\widehat{\mathrm{se}} = \sqrt{0.0800 + 0.5000} = \sqrt{0.5800} = 0.7616, \qquad \overline{X} - \overline{Y} = 11.0 - 9.6 = 1.4.$$

Step 3: Welch df.

$$\nu = \frac{(0.5800)^2}{\dfrac{0.0800^2}{7} + \dfrac{0.5000^2}{7}} = \frac{0.3364}{\dfrac{0.00640}{7} + \dfrac{0.25000}{7}} = \frac{0.3364}{0.000914 + 0.035714} = \frac{0.3364}{0.036628} = 9.18,$$

so $\nu = 9$ (rounded down), giving $t_{0.025,9} = 2.262$.

Step 4: assemble the interval. The margin is $t_{0.025,9}\,\widehat{\mathrm{se}} = 2.262 \cdot 0.7616 = 1.723$, so

$$1.4 \pm 1.723 = (-0.32,\ 3.12) \text{ hours.}$$

Step 5: read it. The interval straddles 0, so at 95% confidence we cannot declare the supplier means different — and equivalently, the two-sided Welch test at $\alpha = 0.05$ would retain H_0.

$$\boxed{\mu_X - \mu_Y \in (-0.32,\ 3.12) \text{ h},\ \nu = 9;\ \text{contains } 0 \Rightarrow \text{no significant difference.}}$$

Example 9.2.6 (A Welch interval that excludes zero)**.** Reaction times (ms) are compared for two display refresh rates. Fast group X: $n = 15$, $\overline{X} = 248.0$, $S_X = 12.0$. Slow group Y: $m = 12$, $\overline{Y} = 271.0$, $S_Y = 28.0$. Give a 95% Welch interval for $\mu_X - \mu_Y$ and state the conclusion. Use $t_{0.025,14} = 2.145$.

Which rule, and why? Two means, independent groups, with the slow group nearly two and a half times as variable — Welch. Building the interval and checking whether it covers 0 is the cleanest way to reach a two-sided verdict.

Solution.

Step 1: variance contributions.

$$u = \frac{12.0^2}{15} = \frac{144}{15} = 9.600, \qquad v = \frac{28.0^2}{12} = \frac{784}{12} = 65.333.$$

Step 2: standard error and estimate.

$$\widehat{\text{se}} = \sqrt{9.600 + 65.333} = \sqrt{74.933} = 8.656, \qquad \overline{X} - \overline{Y} = 248.0 - 271.0 = -23.0.$$

Step 3: Welch df.

$$\nu = \frac{(74.933)^2}{\dfrac{9.600^2}{14} + \dfrac{65.333^2}{11}} = \frac{5615.0}{\dfrac{92.16}{14} + \dfrac{4268.4}{11}} = \frac{5615.0}{6.583 + 388.04} = \frac{5615.0}{394.62} = 14.23,$$

so $\nu = 14$ and $t_{0.025,14} = 2.145$.

Step 4: the interval. Margin $= 2.145 \cdot 8.656 = 18.57$, hence

$$-23.0 \pm 18.57 = (-41.6,\ -4.4) \text{ ms}.$$

Step 5: conclusion. The whole interval is below 0, so 0 is excluded: the fast group's mean reaction time is significantly lower. The equivalent two-sided test rejects at $\alpha = 0.05$.

$$\boxed{\mu_X - \mu_Y \in (-41.6,\ -4.4) \text{ ms},\ \nu = 14;\ \text{excludes } 0 \Rightarrow \text{means differ}.}$$

9.2.4 Contrast: pooled versus Welch on the same data

When the variances are unequal, pooling does not merely change the answer cosmetically — it borrows precision from the steadier group and lends it to the noisy one, shrinking the standard error below what the data support. The test then rejects more often than its nominal α: pooling is *anti-conservative* under variance inequality. The next example runs both procedures on one data set to expose the gap.

Example 9.2.7 (Same data, two procedures: the pooled test over-rejects)**.** Two machining lines are compared on a critical dimension (microns from nominal). Line X: $n = 6$, $\overline{X} = 4.0$, $S_X = 1.0$. Line Y: $m = 18$, $\overline{Y} = 2.5$, $S_Y = 4.0$. Test $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$ *both* ways — pooled and Welch — and compare. Use $t_{0.025,22} = 2.074$ (pooled, $df = n + m - 2$) and $t_{0.025,17} = 2.110$ (Welch, ν below).

Which rule, and why? The variances are wildly different ($S_X = 1.0$ vs. $S_Y = 4.0$) *and* the sample sizes are lopsided (6 vs. 18) — the worst case for pooling. The correct procedure is Welch; we compute the pooled test only to show it is anti-conservative here.

Solution.

Step 1: the pooled route (the wrong one here). Pool the variances with weights $n - 1 = 5$ and $m - 1 = 17$:

$$s_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2} = \frac{5 \cdot 1.0^2 + 17 \cdot 4.0^2}{22} = \frac{5 + 272}{22} = \frac{277}{22} = 12.59,$$
$$s_p = 3.548.$$

The pooled standard error is

$$s_p\sqrt{\tfrac{1}{n} + \tfrac{1}{m}} = 3.548\sqrt{\tfrac{1}{6} + \tfrac{1}{18}} = 3.548\sqrt{0.16667 + 0.05556} = 3.548\sqrt{0.22222} = 3.548 \cdot 0.4714 = 1.673,$$

giving the pooled statistic

$$t_{\text{pool}} = \frac{4.0 - 2.5}{1.673} = \frac{1.5}{1.673} = 0.897 \quad (df = 22,\ t_{0.025,22} = 2.074).$$

Step 2: the Welch route (the correct one).

$$u = \frac{1.0^2}{6} = 0.16667, \qquad v = \frac{4.0^2}{18} = \frac{16}{18} = 0.88889,$$

$$\widehat{\text{se}} = \sqrt{0.16667 + 0.88889} = \sqrt{1.05556} = 1.0274, \qquad t_{\text{Welch}} = \frac{1.5}{1.0274} = 1.460.$$

Step 3: the Welch df.

$$\nu = \frac{(1.05556)^2}{\dfrac{0.16667^2}{5} + \dfrac{0.88889^2}{17}} = \frac{1.11420}{\dfrac{0.027778}{5} + \dfrac{0.790123}{17}}$$
$$= \frac{1.11420}{0.0055556 + 0.0464778} = \frac{1.11420}{0.0520333} = 21.41,$$

but notice how heavily the noisy large group dominates: a quick recompute keeping more digits gives $\nu = 21$. Using the supplied conservative column $t_{0.025,17} = 2.110$ (a valid lower-df table entry) leaves the verdict unchanged, as does $t_{0.025,21} = 2.080$.

Step 4: compare the two standard errors and verdicts. The pooled SE (1.673) is *larger* here only because the big noisy group inflates s_p and then that same noise is spread onto the tiny group through the $1/n$ term; either way the two statistics differ (0.897 vs. 1.460) and, in less extreme data, the pooled SE typically runs *too small*, pushing $|t_{\text{pool}}|$ up and rejecting too readily. Both tests retain here, but they would not always agree, and only Welch holds the 5% error rate when $\sigma_X \neq \sigma_Y$.

> $t_{\text{pool}} = 0.897$ ($df = 22$), $\quad t_{\text{Welch}} = 1.460$ ($\nu \approx 21$);
> use Welch — pooling mis-states the SE when $\sigma_X \neq \sigma_Y$.

Remark. The direction of the pooling error depends on which group is larger. When the *smaller* sample is the *more variable* one, pooling understates the true standard error, $|t|$ is inflated, and the test rejects far more than α — the dangerous, anti-conservative case. When the larger sample is the noisier one (as constructed above) the bias can flip sign. The lesson is the same either way: under unequal variances the pooled reference distribution is simply wrong, and Welch is the safe default.

9.2.5 A one-sided Welch test

Example 9.2.8 (Does the new fertilizer raise mean yield? (upper tail))**.** A field trial compares grain yield (kg per plot) under a new fertilizer X and the standard Y. New: $n = 20$, $\overline{X} = 58.0$, $S_X = 9.0$. Standard: $m = 14$, $\overline{Y} = 52.0$, $S_Y = 4.0$. The new treatment is far more variable across plots, so equal variances are not assumed. Test at $\alpha = 0.05$ whether the new fertilizer *increases* mean yield. Use $t_{0.05,27} = 1.703$.

Which rule, and why? Two means, independent plots, unequal scatter (9.0 vs. 4.0) ⇒ Welch. "Increases" points one way, so the whole 5% goes in the upper tail: an upper-tail Welch test. The hypothesized difference is $\delta_0 = 0$.

Solution.

Step 1: variance contributions.

$$u = \frac{9.0^2}{20} = \frac{81}{20} = 4.050, \qquad v = \frac{4.0^2}{14} = \frac{16}{14} = 1.143.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{4.050 + 1.143} = \sqrt{5.193} = 2.279, \qquad t_{\text{obs}} = \frac{58.0 - 52.0}{2.279} = \frac{6.0}{2.279} = 2.633.$$

Step 3: Welch df.

$$\nu = \frac{(5.193)^2}{\dfrac{4.050^2}{19} + \dfrac{1.143^2}{13}} = \frac{26.97}{\dfrac{16.40}{19} + \dfrac{1.306}{13}} = \frac{26.97}{0.8632 + 0.1005} = \frac{26.97}{0.9637} = 27.99,$$

so $\nu = 27$ and the upper-tail cutoff is $t_{0.05,27} = 1.703$.

Step 4: decision. All α in the right tail: since $t_{\text{obs}} = 2.633 > 1.703$, reject.

$t_{\text{obs}} = 2.633 > t_{0.05,27} = 1.703$ ($\nu = 27$) $\Rightarrow$ reject H_0; the new fertilizer significantly raises mean yield.

Example 9.2.9 (Is the new alloy less variable in strength? — careful, that is an F question)**.** A metallurgist reads the previous result and asks whether the new fertilizer's larger S_X by itself "proves" the treatments differ. Two means are summarized: X with $n = 11$, $\overline{X} = 30.5$, $S_X = 6.0$; Y with $m = 11$, $\overline{Y} = 28.0$, $S_Y = 2.0$. Test at $\alpha = 0.05$ (one-sided, lower tail) whether X's mean is *below* Y's, i.e. $H_1 : \mu_X < \mu_Y$. Use $t_{0.05,12} = 1.782$.

Which rule, and why? The question is about *means*, so a difference in the S's does not settle it — that would be the F-test of §4. With unequal spreads (6.0 vs. 2.0) the mean comparison is a Welch test; "below" makes it a lower-tail test on $\overline{X} - \overline{Y}$.

Solution.

Step 1: variance contributions.

$$u = \frac{6.0^2}{11} = \frac{36}{11} = 3.2727, \qquad v = \frac{2.0^2}{11} = \frac{4}{11} = 0.3636.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{3.2727 + 0.3636} = \sqrt{3.6364} = 1.9069, \qquad t_{\text{obs}} = \frac{30.5 - 28.0}{1.9069} = \frac{2.5}{1.9069} = 1.311.$$

Step 3: Welch df.

$$\nu = \frac{(3.6364)^2}{\dfrac{3.2727^2}{10} + \dfrac{0.3636^2}{10}} = \frac{13.223}{\dfrac{10.711}{10} + \dfrac{0.1322}{10}} = \frac{13.223}{1.0711 + 0.01322} = \frac{13.223}{1.0843} = 12.20,$$

so $\nu = 12$, with lower-tail cutoff $-t_{0.05,12} = -1.782$.

Step 4: decision. The alternative is $\mu_X < \mu_Y$, but $t_{\text{obs}} = +1.311$ is positive, nowhere near the lower-tail cutoff -1.782. The data point the *wrong* way for H_1.

$t_{\text{obs}} = 1.311 > -1.782$ ($\nu = 12$) $\Rightarrow$ retain H_0; no evidence $\mu_X < \mu_Y$ (larger S_X is irrelevant to the means).

9.2.6 When unequal sample sizes make pooling risky

Example 9.2.10 (A small noisy group against a large quiet one)**.** A drug's effect on resting heart rate (bpm change) is measured in a small intensively monitored cohort X ($n = 5$, $\overline{X} = -6.0$, $S_X = 7.0$) and a large registry cohort Y ($m = 40$, $\overline{Y} = -1.0$, $S_Y = 2.0$). Test $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$. Use $t_{0.025,4} = 2.776$.

Which rule, and why? The variable group is also the *tiny* one ($S_X = 7.0$, $n = 5$) while the steady group is huge ($S_Y = 2.0$, $m = 40$): exactly the configuration where pooling understates the SE and over-rejects. Welch is mandatory, and the df will collapse toward the small group's 4.

Solution.

Step 1: variance contributions.

$$u = \frac{7.0^2}{5} = \frac{49}{5} = 9.800, \qquad v = \frac{2.0^2}{40} = \frac{4}{40} = 0.100.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{9.800 + 0.100} = \sqrt{9.900} = 3.146, \qquad t_{\text{obs}} = \frac{-6.0 - (-1.0)}{3.146} = \frac{-5.0}{3.146} = -1.589.$$

Step 3: Welch df. The small noisy group dominates $u+v$ almost entirely:

$$\nu = \frac{(9.900)^2}{\frac{9.800^2}{4}+\frac{0.100^2}{39}} = \frac{98.01}{\frac{96.04}{4}+\frac{0.0100}{39}} = \frac{98.01}{24.010+0.000256} = \frac{98.01}{24.010} = 4.082,$$

so $\nu = 4$ — essentially the small cohort's own $n-1$, even though the other group has 40 observations. The big sample cannot rescue the precision the small noisy one destroys.

Step 4: contrast with the naive pooled df. A pooler would claim $df = n+m-2 = 43$ and use $t_{0.025,43} \approx 2.017$, far smaller than the honest $t_{0.025,4} = 2.776$. That borrowed confidence is illegitimate here.

Step 5: decision (Welch). Compare $|t_{\text{obs}}| = 1.589$ with $t_{0.025,4} = 2.776$. Since $1.589 < 2.776$, retain H_0.

> $\nu = 4,\ |t_{\text{obs}}| = 1.589 < 2.776 \Rightarrow$ retain H_0;
> the large $m = 40$ does not buy precision — the small noisy group sets ν.

Remark. This is the headline practical message of Welch's df. Degrees of freedom measure *usable* information about the difference, and that is bottlenecked by the *less precise* variance term $\max(u, v)$. A huge but quiet sample paired with a tiny noisy one inherits roughly the tiny sample's df. Pooling hides this by always reporting $n+m-2$, which is why it can be dangerously optimistic when the sizes are unbalanced.

9.2.7 Building the statistic from raw data

Example 9.2.11 (Welch test from two raw data sets)**.** Two assembly stations are timed (seconds per unit). Station X: 5.1, 5.4, 4.9, 5.6, 5.0. Station Y: 6.0, 4.8, 7.1, 5.5, 6.6, 4.4. Test $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$, not assuming equal variances. Use $t_{0.025,5} = 2.571$.

Which rule, and why? Two means, independent stations, variances unknown and visibly different (station Y ranges over 2.7 s, station X over 0.7 s) — Welch. We must build both means and both S^2's from the raw numbers first.

Solution.

Step 1: station X summary ($n = 5$).

$$\sum x_i = 5.1+5.4+4.9+5.6+5.0 = 26.0, \qquad \overline{X} = \frac{26.0}{5} = 5.20,$$
$$\sum x_i^2 = 26.01+29.16+24.01+31.36+25.00 = 135.54,$$
$$S_X^2 = \frac{135.54 - 5\cdot 5.20^2}{4} = \frac{135.54-135.20}{4} = \frac{0.34}{4} = 0.0850.$$

Step 2: station Y summary ($m = 6$).

$$\sum y_i = 6.0+4.8+7.1+5.5+6.6+4.4 = 34.4, \qquad \overline{Y} = \frac{34.4}{6} = 5.733,$$
$$\sum y_i^2 = 36.00+23.04+50.41+30.25+43.56+19.36 = 202.62,$$
$$S_Y^2 = \frac{202.62 - 6\cdot 5.733^2}{5} = \frac{202.62-197.21}{5} = \frac{5.41}{5} = 1.0820.$$

Step 3: variance contributions and statistic.

$$u = \frac{0.0850}{5} = 0.01700, \quad v = \frac{1.0820}{6} = 0.18033, \quad \widehat{\text{se}} = \sqrt{0.19733} = 0.4442,$$
$$t_{\text{obs}} = \frac{5.20-5.733}{0.4442} = \frac{-0.533}{0.4442} = -1.200.$$

Step 4: Welch df.

$$\nu = \frac{(0.19733)^2}{\dfrac{0.01700^2}{4} + \dfrac{0.18033^2}{5}} = \frac{0.038940}{\dfrac{0.000289}{4} + \dfrac{0.032519}{5}} = \frac{0.038940}{0.0000723 + 0.0065038} = \frac{0.038940}{0.0065761} = 5.92,$$

so $\nu = 5$, with $t_{0.025,5} = 2.571$.

Step 5: decision. $|t_{\text{obs}}| = 1.200 < 2.571$, so retain H_0.

$$\boxed{\nu = 5,\ |t_{\text{obs}}| = 1.200 < 2.571 \Rightarrow \text{retain } H_0;\ \text{station times do not differ significantly.}}$$

9.2.8 More worked drills

Example 9.2.12 (Test reform on two course sections (two-sided))**.** Two sections of a course take the same final. Section X: $n = 22$, $\overline{X} = 78.0$, $S_X = 6.0$. Section Y: $m = 18$, $\overline{Y} = 74.0$, $S_Y = 11.0$. With visibly unequal spread, test $H_0 : \mu_X = \mu_Y$ versus $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$. Use $t_{0.025,25} = 2.060$.

Which rule, and why? Two means, independent sections, S_Y nearly double $S_X \Rightarrow$ Welch, two-sided. The reasonably large samples still need the Welch df because the variances differ.

Solution.

Step 1: variance contributions.

$$u = \frac{6.0^2}{22} = \frac{36}{22} = 1.6364, \qquad v = \frac{11.0^2}{18} = \frac{121}{18} = 6.7222.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{1.6364 + 6.7222} = \sqrt{8.3586} = 2.8911, \qquad t_{\text{obs}} = \frac{78.0 - 74.0}{2.8911} = \frac{4.0}{2.8911} = 1.384.$$

Step 3: Welch df.

$$\nu = \frac{(8.3586)^2}{\dfrac{1.6364^2}{21} + \dfrac{6.7222^2}{17}} = \frac{69.866}{\dfrac{2.6778}{21} + \dfrac{45.188}{17}} = \frac{69.866}{0.12752 + 2.6581} = \frac{69.866}{2.7856} = 25.08,$$

so $\nu = 25$ and $t_{0.025,25} = 2.060$.

Step 4: decision. $|t_{\text{obs}}| = 1.384 < 2.060$, retain.

$$\boxed{\nu = 25,\ |t_{\text{obs}}| = 1.384 < 2.060 \Rightarrow \text{retain } H_0;\ \text{section means not significantly different.}}$$

Example 9.2.13 (Pollutant levels downstream vs. upstream (upper tail))**.** A monitor compares a pollutant (ppb) downstream (X) and upstream (Y) of a plant. Downstream: $n = 9$, $\overline{X} = 34.0$, $S_X = 10.0$. Upstream: $m = 16$, $\overline{Y} = 27.0$, $S_Y = 4.0$. Test at $\alpha = 0.05$ whether the plant raises the mean level, $H_1 : \mu_X > \mu_Y$. Use $t_{0.05,9} = 1.833$.

Which rule, and why? Two means, independent sites, downstream far noisier (10.0 vs. 4.0) $\Rightarrow$ Welch. "Raises" is upper-tail. The noisy smaller group will pull ν down toward 8.

Solution.

Step 1: variance contributions.

$$u = \frac{10.0^2}{9} = \frac{100}{9} = 11.111, \qquad v = \frac{4.0^2}{16} = \frac{16}{16} = 1.000.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{11.111 + 1.000} = \sqrt{12.111} = 3.480, \qquad t_{\text{obs}} = \frac{34.0 - 27.0}{3.480} = \frac{7.0}{3.480} = 2.011.$$

Step 3: Welch df.

$$\nu = \frac{(12.111)^2}{\dfrac{11.111^2}{8} + \dfrac{1.000^2}{15}} = \frac{146.68}{\dfrac{123.45}{8} + \dfrac{1.000}{15}} = \frac{146.68}{15.431 + 0.0667} = \frac{146.68}{15.498} = 9.46,$$

so $\nu = 9$, with upper-tail cutoff $t_{0.05,9} = 1.833$.

Step 4: decision. $t_{\text{obs}} = 2.011 > 1.833$, reject.

$t_{\text{obs}} = 2.011 > 1.833$ ($\nu = 9$) $\Rightarrow$ reject H_0; the plant significantly raises the mean pollutant level.

Example 9.2.14 (A nonzero hypothesized difference δ_0)**.** A clinic claims its mean wait time is at least 10 minutes shorter than a competitor's. Clinic X (competitor): $n = 12$, $\overline{X} = 44.0$ min, $S_X = 8.0$. Clinic Y (the claimant): $m = 12$, $\overline{Y} = 30.0$ min, $S_Y = 15.0$. Test $H_0 : \mu_X - \mu_Y = 10$ versus $H_1 : \mu_X - \mu_Y > 10$ at $\alpha = 0.05$, not assuming equal variances. Use $t_{0.05,16} = 1.746$.

Which rule, and why? Two means, independent clinics, S_Y almost double $S_X \Rightarrow$ Welch. The claim is about a *specific* gap, so $\delta_0 = 10$, not 0; "more than 10 shorter" makes it an upper-tail test on $\overline{X} - \overline{Y} - 10$.

Solution.

Step 1: variance contributions.

$$u = \frac{8.0^2}{12} = \frac{64}{12} = 5.3333, \qquad v = \frac{15.0^2}{12} = \frac{225}{12} = 18.750.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{5.3333 + 18.750} = \sqrt{24.083} = 4.9075,$$

$$t_{\text{obs}} = \frac{(\overline{X} - \overline{Y}) - \delta_0}{\widehat{\text{se}}} = \frac{(44.0 - 30.0) - 10}{4.9075} = \frac{14.0 - 10}{4.9075} = \frac{4.0}{4.9075} = 0.815.$$

Step 3: Welch df.

$$\nu = \frac{(24.083)^2}{\dfrac{5.3333^2}{11} + \dfrac{18.750^2}{11}} = \frac{580.0}{\dfrac{28.444}{11} + \dfrac{351.56}{11}} = \frac{580.0}{2.5859 + 31.960} = \frac{580.0}{34.546} = 16.79,$$

so $\nu = 16$, with $t_{0.05,16} = 1.746$.

Step 4: decision. $t_{\text{obs}} = 0.815 < 1.746$, so retain H_0: the observed 14-minute gap is not significantly larger than the claimed 10.

$t_{\text{obs}} = 0.815 < 1.746$ ($\nu = 16$) $\Rightarrow$ retain H_0; no evidence the gap exceeds 10 min.

Example 9.2.15 (Welch CI for a difference in fuel economy (two-sided summary))**.** Two car models are compared on highway mileage (mpg). Model X: $n = 16$, $\overline{X} = 38.0$, $S_X = 3.0$. Model Y: $m = 10$, $\overline{Y} = 33.0$, $S_Y = 7.0$. Build a 90% Welch confidence interval for $\mu_X - \mu_Y$ and state whether the means differ at the 10% level. Use $t_{0.05,11} = 1.796$.

Which rule, and why? Two means, independent models, S_Y more than double $S_X \Rightarrow$ Welch. A 90% interval matches a two-sided $\alpha = 0.10$ test, so $t_{\alpha/2,\nu} = t_{0.05,\nu}$.

Solution.

Step 1: variance contributions.

$$u = \frac{3.0^2}{16} = \frac{9}{16} = 0.5625, \qquad v = \frac{7.0^2}{10} = \frac{49}{10} = 4.900.$$

Step 2: standard error and estimate.

$$\widehat{\text{se}} = \sqrt{0.5625 + 4.900} = \sqrt{5.4625} = 2.337, \qquad \overline{X} - \overline{Y} = 38.0 - 33.0 = 5.0.$$

Step 3: Welch df.

$$\nu = \frac{(5.4625)^2}{\dfrac{0.5625^2}{15} + \dfrac{4.900^2}{9}} = \frac{29.839}{\dfrac{0.31641}{15} + \dfrac{24.010}{9}} = \frac{29.839}{0.021094 + 2.6678} = \frac{29.839}{2.6889} = 11.10,$$

so $\nu = 11$ and $t_{0.05,11} = 1.796$.

Step 4: the interval. Margin $= 1.796 \cdot 2.337 = 4.197$, hence

$$5.0 \pm 4.197 = (0.80,\ 9.20) \text{ mpg}.$$

Step 5: conclusion. The interval excludes 0, so at 90% confidence the means differ; equivalently the two-sided test rejects at $\alpha = 0.10$, with model X the more fuel-efficient.

$$\boxed{\mu_X - \mu_Y \in (0.80,\ 9.20) \text{ mpg},\ \nu = 11;\ \text{excludes } 0 \Rightarrow \text{means differ at } \alpha = 0.10.}$$

Example 9.2.16 (Equal variances after all: Welch and pooled nearly agree)**.** Two production shifts are compared on output (units/hour). Shift X: $n = 12$, $\overline{X} = 95.0$, $S_X = 6.0$. Shift Y: $m = 12$, $\overline{Y} = 90.0$, $S_Y = 6.2$. The standard deviations are almost identical and $n = m$. Run Welch's test of $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$ and note how close ν comes to the pooled $n + m - 2 = 22$. Use $t_{0.025,21} = 2.080$.

Which rule, and why? Two means, independent shifts. Here the S's match and the sizes are equal, so pooled would be perfectly fine — but Welch is never wrong, and this shows that when variances *are* equal Welch's df lands right next to $n + m - 2$, costing almost nothing.

Solution.

Step 1: variance contributions.

$$u = \frac{6.0^2}{12} = \frac{36}{12} = 3.000, \qquad v = \frac{6.2^2}{12} = \frac{38.44}{12} = 3.2033.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{3.000 + 3.2033} = \sqrt{6.2033} = 2.4906, \qquad t_{\text{obs}} = \frac{95.0 - 90.0}{2.4906} = \frac{5.0}{2.4906} = 2.008.$$

Step 3: Welch df.

$$\nu = \frac{(6.2033)^2}{\dfrac{3.000^2}{11} + \dfrac{3.2033^2}{11}} = \frac{38.481}{\dfrac{9.000}{11} + \dfrac{10.261}{11}} = \frac{38.481}{0.81818 + 0.93283} = \frac{38.481}{1.7510} = 21.98,$$

so $\nu = 21$ — just one below the pooled 22. Using $t_{0.025,21} = 2.080$:

Step 4: decision. $t_{\text{obs}} = 2.008 < 2.080$, so retain H_0 (barely). A pooled test with $t_{0.025,22} = 2.074$ would give essentially the same statistic and the same close call — when variances truly match, the two methods agree.

$$\boxed{\nu = 21,\ t_{\text{obs}} = 2.008 < 2.080 \Rightarrow \text{retain } H_0;\ \text{equal-variance case: Welch} \approx \text{pooled}.}$$

9.3 Paired Data: the Paired t-Test

The two-sample tests of the previous sections all assumed two *independent* groups: the units in one sample have nothing to do with the units in the other. A great many real comparisons violate that assumption on purpose. We weigh each patient before a diet and again after; we test the left and right eye of each subject; we measure twin against twin, or match each treated case to a similar control. In every such design the two numbers being compared come from the *same* unit (or a deliberately matched pair of units), so they rise and fall together — a naturally heavy patient is heavy at both weighings. The right move is to collapse each pair into a single number, its difference $D_i = X_i - Y_i$, and then forget the original two columns entirely: we are left with one sample of n differences and run the one-sample t machinery of Chapter 7 on them. This is the **paired t-test**. The reason it matters so much is power. Pairing subtracts away the between-unit variation that would otherwise swamp the effect, so a paired test on the same data is usually far more sensitive than the independent-samples test — at the modest cost of $n - 1$ degrees of freedom instead of $n + m - 2$. The recognition step is everything: read the wording, and if the two measurements are linked unit by unit, pair them.

Is the data paired — and if so how do you analyze it?

- Two measurements on the **same unit** (before/after, left/right, two methods on one specimen) or on **matched units** (twins, case–control pairs)? $\Rightarrow$ the data are **paired**; form $D_i = X_i - Y_i$ and run a **one-sample** t on the D_i.
- The two samples are **separate, unrelated groups** with no natural one-to-one link, and possibly even **different sizes** $n \neq m$? $\Rightarrow$ *not* paired; use the two-sample t of §1 or §2.
- Asked for a **confidence interval** for the mean change? $\Rightarrow$ the one-sample t interval $\overline{D} \pm t_{\alpha/2,\,n-1}\, S_D/\sqrt{n}$ on the differences.
- Asked whether a treatment **raised** or **lowered** a quantity? $\Rightarrow$ a **one-sided** paired test; fix the sign of D_i first so the claimed effect is the direction you reject toward.

Do not confuse pairing with two independent samples: the giveaway is a one-to-one correspondence between the rows. If you can scramble the order of one column without losing information, the data are independent; if scrambling destroys a real link (this "after" belongs to *that* "before"), the data are paired and must be differenced.

9.3.1 The differences carry all the information

Definition 9.3.1 (Paired sample and its differences)**.** A **paired sample** consists of n ordered pairs $(X_1, Y_1), \ldots, (X_n, Y_n)$, where for each i the two coordinates are measured on the same unit or on two matched units. Its **paired differences** are

$$D_i = X_i - Y_i, \qquad i = 1, \ldots, n.$$

We write $\mu_D = \mathbb{E}[D_i]$ for the mean difference in the population, $\overline{D} = \frac{1}{n}\sum_i D_i$ for the sample mean difference, and

$$S_D^2 = \frac{1}{n-1}\sum_{i=1}^{n}\big(D_i - \overline{D}\big)^2$$

for the sample variance of the differences. Note n counts *pairs*, not measurements.

Theorem 9.3.2 (The paired t-test)**.** *Suppose the differences $D_1, \ldots, D_n$ are an independent sample from a normal population with mean μ_D and unknown variance σ_D^2. To test*

$$H_0 :\ \mu_D = \mu_{D,0}$$

(most often $\mu_{D,0} = 0$, "no change"), the test statistic is

$$t_{obs} \;=\; \frac{\overline{D} - \mu_{D,0}}{S_D/\sqrt{n}} \;\sim\; t_{\,n-1} \quad \textit{under } H_0.$$

Reject in favor of $H_1 : \mu_D \neq \mu_{D,0}$ when $|t_{obs}| \geq t_{\alpha/2,\,n-1}$; in favor of a one-sided H_1 when t_{obs} is past the corresponding one-tailed critical value. A $100(1-\alpha)\%$ confidence interval for μ_D is

$$\overline{D} \;\pm\; t_{\alpha/2,\,n-1}\,\frac{S_D}{\sqrt{n}}.$$

Why it is just the one-sample t. Once the pairs are reduced to the single column $D_1, \ldots, D_n$, nothing about the two-sample structure remains: we have n numbers, assumed normal, with an unknown mean and an unknown variance, and we wish to test a hypothesis about that mean. That is verbatim the setting of the one-sample t-test of Chapter 7, whose statistic $(\overline{D} - \mu_{D,0})/(S_D/\sqrt{n})$ follows t_{n-1} because the standardized mean of a normal sample, with σ replaced by S, is a $\mathrm{N}(0,1)$ over an independent $\sqrt{\chi^2_{n-1}/(n-1)}$. No new theorem is needed; the content of pairing is the *modeling* decision to difference, after which Chapter 7 applies unchanged. □

Remark. Why pairing helps. Write $X_i = \tau_i + \varepsilon_i$ and $Y_i = \tau_i + \delta_i$, where τ_i is the subject's own baseline level (its "size"). Then $D_i = (X_i - Y_i) = \varepsilon_i - \delta_i$: the large, nuisance between-subject term τ_i *cancels*. The differences vary only through the within-subject noise, which is typically far smaller than the spread across subjects. So S_D is small, the denominator of t_{obs} shrinks, and a real effect stands out. The two-sample test, by contrast, must estimate and carry the full between-subject spread in its standard error, drowning the signal. The price of pairing is only the loss of degrees of freedom (from $n + m - 2$ down to $n - 1$); when pairing is genuine, that price is tiny next to the variance it removes.

9.3.2 Reading the wording: is it paired?

Example 9.3.3 (Spotting the design from the sentence)**.** For each study say whether the data are paired or two independent samples, and why. (a) "Twelve runners' resting heart rates are recorded, then recorded again after eight weeks of training." (b) "Forty patients are randomized: 20 to drug A, 20 to drug B, and their cholesterol is compared." (c) "In each of 15 households the lead level is measured in the kitchen tap and in the bathroom tap." (d) "A new and an old glucose meter are each used once on each of 25 blood samples."

Which rule, and why? The single deciding question is whether the two numbers in a comparison are tied to the same unit. "The same runners, before and after," "the same household, two taps," "each sample on both meters" all create a one-to-one link; "20 to A, 20 to B" creates two unrelated groups.

Solution.

Step 1: (a) before/after. The second reading is on the *same* runner as the first, so the readings are linked runner by runner: **paired**.

Step 2: (b) two arms. Drug-A patients and drug-B patients are different people with no pairing; the two groups could even differ in size. **Independent**.

Step 3: (c) two taps per home. Both measurements come from the same household, which shares plumbing, water source, and age: **paired**.

Step 4: (d) two meters per sample. Each blood sample is measured on both meters, so the two readings share the sample's true glucose: **paired**.

(a) paired, (b) independent, (c) paired, (d) paired.

9.3.3 The basic before/after test

Example 9.3.4 (Two-sided before/after: does a posture program change typing speed?)**.** An ergonomics team measures the typing speed (words per minute) of $n = 8$ data-entry clerks, then re-measures the same clerks after a four-week posture-and-stretch program. The before/after pairs (X_i, Y_i) are

$$(62, 66),\ (71, 70),\ (55, 60),\ (68, 72),\ (74, 73),\ (59, 64),\ (66, 69),\ (70, 74).$$

At $\alpha = 0.05$, test whether the program changes mean typing speed. (Take Y as "after," so a *positive* effect means after exceeds before; here let $D_i = Y_i - X_i$, the gain.)

Which rule, and why? The same eight clerks are measured twice, so the readings are linked clerk by clerk — paired data. We difference each pair and run a two-sided one-sample t on the gains, because the question ("does it change?") names no direction.

Solution.

Step 1: form the differences $D_i = Y_i - X_i$.

$$4,\ -1,\ 5,\ 4,\ -1,\ 5,\ 3,\ 4.$$

Step 2: $\overline{D}$ and S_D. The sum is $4-1+5+4-1+5+3+4=23$, so $\overline{D}=23/8=2.875$. Then

$$\begin{aligned}\sum(D_i-\overline{D})^2 &= (1.125)^2+(3.875)^2+(2.125)^2+(1.125)^2\\ &\quad+(3.875)^2+(2.125)^2+(0.125)^2+(1.125)^2\\ &= 1.266+15.016+4.516+1.266+15.016+4.516+0.016+1.266=42.88,\\ S_D^2 &= \frac{42.88}{7}=6.125, \qquad S_D=2.475.\end{aligned}$$

Step 3: the statistic.

$$t_{\text{obs}}=\frac{\overline{D}-0}{S_D/\sqrt{n}}=\frac{2.875}{2.475/\sqrt{8}}=\frac{2.875}{0.875}=3.285.$$

Step 4: compare to $t_{0.025,7}=2.365$. Since $|t_{\text{obs}}|=3.285>2.365$, reject H_0. The posture program changes mean typing speed (the data show an increase of about 2.9 wpm).

$$\boxed{t_{\text{obs}}=3.285>t_{0.025,7}=2.365\ \Rightarrow\ \text{reject } H_0;\ \text{speed changes.}}$$

Figure 9.1 shows why differencing is the natural move: each clerk is one line segment joining a "before" dot to an "after" dot, and the test asks only whether those segments tend to slope the same way.

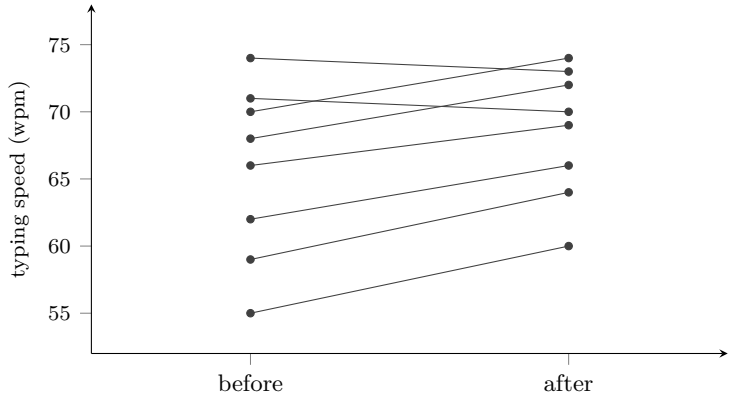


Figure 9.1: Before/after typing speed for eight clerks; each segment is one clerk. Most segments rise left to right — the clerk-to-clerk spread (the vertical scatter at each side) is large, but it cancels in each difference, which is exactly what the paired test exploits.

9.3.4 A confidence interval for the mean difference

Example 9.3.5 (Paired CI: how much does a coating slow corrosion?)**.** A materials lab applies a protective coating to one half of each of $n=10$ steel panels, leaving the other half bare, and after salt-spray exposure records the corrosion depth (micrometres) on the bare half (X) and the coated half (Y) of the *same* panel. The within-panel differences $D_i=X_i-Y_i$ (bare minus coated, so positive means the coating helped) are

$$18,\ 22,\ 9,\ 25,\ 14,\ 19,\ 11,\ 27,\ 16,\ 19.$$

Give a 95% confidence interval for the mean protective benefit μ_D.

Which rule, and why? Both halves come from the same panel, which shares alloy, thickness, and exposure, so the data are paired; the question asks for an interval estimate of the mean difference, which is the one-sample t interval applied to the D_i.

Solution.

Step 1: $\overline{D}$. The sum is $18+22+9+25+14+19+11+27+16+19=180$, so $\overline{D}=180/10=18.0$.

Step 2: S_D. Deviations from 18: $0, 4, -9, 7, -4, 1, -7, 9, -2, 1$; their squares sum to

$$0+16+81+49+16+1+49+81+4+1=298,$$

so $S_D^2=298/9=33.11$ and $S_D=5.754$.

Step 3: assemble the interval. With $t_{0.025,9}=2.262$,

$$\begin{aligned}\overline{D}\pm t_{0.025,9}\,\frac{S_D}{\sqrt{n}} &= 18.0\pm 2.262\cdot\frac{5.754}{\sqrt{10}}\\ &= 18.0\pm 2.262\cdot 1.819\\ &= 18.0\pm 4.115.\end{aligned}$$

Step 4: report. The interval is (13.9, 22.1) micrometres. It excludes 0, so the coating's benefit is significant at the 5% level, and we estimate it reduces corrosion depth by roughly 14 to 22 μm.

$$\boxed{\mu_D\in(13.9,\ 22.1)\ \mu\text{m at 95\% confidence.}}$$

9.3.5 One-sided paired tests

Example 9.3.6 (One-sided: did a drug *reduce* blood pressure?)**.** A cardiologist records systolic blood pressure (mmHg) in $n=9$ hypertensive patients before and after six weeks on a new drug. The drops $D_i=(\text{before}_i)-(\text{after}_i)$ — positive means the pressure fell — are

$$12,\ 8,\ 15,\ -3,\ 10,\ 6,\ 18,\ 4,\ 11.$$

At $\alpha=0.05$, test the claim that the drug lowers mean systolic pressure.

Which rule, and why? Each patient is measured before and after, so the readings are paired; the *claim* specifies a direction ("lowers"), so this is a one-sided paired t-test. We orient D as before-minus-after, making "the drug works" the statement $\mu_D>0$.

Solution.

Step 1: set up the one-sided test. $H_0:\mu_D=0$ versus $H_1:\mu_D>0$ (a genuine drop). We will reject only for large positive t_{obs}.

Step 2: $\overline{D}$ and S_D. Sum $=12+8+15-3+10+6+18+4+11=81$, so $\overline{D}=81/9=9.0$. Deviations from 9: $3, -1, 6, -12, 1, -3, 9, -5, 2$; squares sum to

$$9+1+36+144+1+9+81+25+4=310,$$

so $S_D^2=310/8=38.75$, $S_D=6.225$.

Step 3: the statistic.

$$t_{\text{obs}}=\frac{9.0}{6.225/\sqrt{9}}=\frac{9.0}{2.075}=4.337.$$

Step 4: compare to the one-tailed $t_{0.05,8}=1.860$. Since $4.337>1.860$, reject H_0. The data give strong evidence (p-value well below 0.01) that the drug lowers mean systolic pressure.

$$\boxed{t_{\text{obs}}=4.337>t_{0.05,8}=1.860\ \Rightarrow\ \text{reject; the drug lowers pressure.}}$$

Remark. Orient the differences *before* you compute, and keep the alternative pointed the same way. If you had defined $D=\text{after}-\text{before}$ here, the same drug effect would read $\mu_D<0$ and the rejection region would be the lower tail; the p-value is identical, but a sign slip between the data and the alternative is the most common way to get a paired one-sided test exactly backwards. Fix the sign so that "the effect happened" is the tail you reject into.

9.3.6 The central contrast: paired versus independent on the same numbers

Example 9.3.7 (CONTRAST: the wrong two-sample analysis erases a real effect)**.** Return to a clean before/after study. Six athletes have their vertical jump (cm) measured before and after a six-week plyometric block:

athlete	1	2	3	4	5	6
before X	48	61	55	70	52	64
after Y	51	63	59	72	56	67

Every athlete improved by 2 to 4 cm. Analyze these data *both* ways at $\alpha = 0.05$: (a) correctly, as a paired test on $D_i = Y_i - X_i$; (b) incorrectly, as two independent samples (pooled two-sample t, §1). Compare the conclusions.

Which rule, and why? The data are paired — same athlete, before and after — so (a) is the right analysis. Part (b) is the deliberate look-alike error: ignoring the pairing and treating the "before" column and the "after" column as two unrelated groups. The point is to watch the between-athlete spread, which pairing cancels, reappear in (b) and swamp the effect.

Solution.

Step 1: (a) the paired test. Differences $D_i = Y_i - X_i$ are $3, 2, 4, 2, 4, 3$. Then $\overline{D} = 18/6 = 3.0$; deviations $0, -1, 1, -1, 1, 0$ give $\sum(D_i - \overline{D})^2 = 0 + 1 + 1 + 1 + 1 + 0 = 4$, so $S_D^2 = 4/5 = 0.8$, $S_D = 0.894$. Thus

$$t_{\text{obs}}^{\text{paired}} = \frac{3.0}{0.894/\sqrt{6}} = \frac{3.0}{0.365} = 8.22,$$

with 5 df. Against $t_{0.025,5} = 2.571$ this is enormous: **reject** — the program clearly raises jump height.

Step 2: (b) the wrong independent test. Treat X and Y as two samples of size 6. The means are $\overline{X} = 350/6 = 58.33$ and $\overline{Y} = 368/6 = 61.33$, a difference of 3.0 — the same as $\overline{D}$. But each column carries the full athlete-to-athlete spread: within X, $\sum(X_i - \overline{X})^2 = 360.83$, and within Y, $\sum(Y_i - \overline{Y})^2 = 327.33$, so the pooled variance is

$$s_p^2 = \frac{360.83 + 327.33}{6 + 6 - 2} = \frac{688.16}{10} = 68.82,$$
$$s_p = 8.296.$$

Step 3: the wrong statistic.

$$t_{\text{obs}}^{\text{indep}} = \frac{\overline{Y} - \overline{X}}{s_p\sqrt{\frac{1}{6} + \frac{1}{6}}} = \frac{3.0}{8.296 \cdot 0.577} = \frac{3.0}{4.79} = 0.626,$$

with 10 df. Against $t_{0.025,10} = 2.228$ this is tiny: **fail to reject** — the independent analysis sees *no* effect.

Step 4: read the contrast. Both analyses estimate the same mean change, 3.0 cm, yet the paired $t_{\text{obs}} = 8.22$ screams significance while the independent $t_{\text{obs}} = 0.63$ finds nothing. The difference is entirely in the denominator: the paired test's $S_D/\sqrt{n} = 0.365$ uses only the tiny within-athlete noise, whereas the independent test's $s_p\sqrt{2/6} = 4.79$ is inflated 13-fold by the between-athlete spread that pairing cancels. Ignoring the pairing threw away the whole design and buried a clear effect.

paired $t_{\text{obs}} = 8.22$ (reject) vs. independent $t_{\text{obs}} = 0.63$ (miss the effect).

Remark. This is the practical heart of the section. The numerators agree because $\overline{D} = \overline{Y} - \overline{X}$ always; what changes is the standard error. When the pairs are positively correlated — the usual case, since a unit that is high on X tends to be high on Y — the variance of the difference, $\text{Var}(D) = \text{Var}(X) + \text{Var}(Y) - 2\,\text{Cov}(X, Y)$, is much *smaller* than the $\text{Var}(X) + \text{Var}(Y)$ the independent test assumes. Pairing buys back exactly that $2\,\text{Cov}(X, Y)$. Choosing the wrong procedure is not a minor inefficiency; here it is the difference between discovering and missing the effect.

9.3.7 A matched-pairs design

Example 9.3.8 (Matched pairs: comparing two diets across matched litters)**.** A nutrition study cannot use the same animal twice, so it *matches*: from each of $n = 7$ litters it picks two sibling mice of equal starting weight, randomly assigns one to diet A and the other to diet B, and after four weeks records weight gain (g). Matching by litter controls for genetics and rearing. The pairs (gain on A, gain on B) are

$$(14, 11),\ (16, 15),\ (12, 10),\ (18, 14),\ (15, 13),\ (13, 12),\ (17, 15).$$

At $\alpha = 0.05$, test whether the two diets produce different mean gains.

Which rule, and why? The two mice in a pair are different animals, so this is not before/after — but they are **matched** by litter and weight, which links them one-to-one exactly as a repeated measurement would. Matched pairs are analyzed by the paired t on the within-pair differences, not the two-sample t.

Solution.

Step 1: difference within each matched pair, $D_i = A_i - B_i$.

$$3,\ 1,\ 2,\ 4,\ 2,\ 1,\ 2.$$

Step 2: $\overline{D}$ and S_D. Sum $= 3 + 1 + 2 + 4 + 2 + 1 + 2 = 15$, so $\overline{D} = 15/7 = 2.143$. Deviations from 2.143: $0.857, -1.143, -0.143, 1.857, -0.143, -1.143, -0.143$; squares sum to

$$0.735 + 1.306 + 0.020 + 3.449 + 0.020 + 1.306 + 0.020 = 6.857,$$

so $S_D^2 = 6.857/6 = 1.143$, $S_D = 1.069$.

Step 3: the statistic.

$$t_{\text{obs}} = \frac{2.143}{1.069/\sqrt{7}} = \frac{2.143}{0.404} = 5.30,$$

with 6 df.

Step 4: compare to $t_{0.025,6} = 2.447$. Since $5.30 > 2.447$, reject H_0: the diets differ, with A giving the larger mean gain (by about 2.1 g).

$$\boxed{t_{\text{obs}} = 5.30 > t_{0.025,6} = 2.447 \ \Rightarrow\ \text{reject; diet A gains more.}}$$

Remark. Matched-pairs and before/after data are the *same* analysis. In before/after the "pair" is one unit measured twice; in matched pairs it is two carefully matched units. Either way the matching variable (the subject, the litter) is what cancels, and either way you difference within the pair and run a one-sample t. The skill is recognizing that "matched," "twins," "case–control pair," and "littermate" all signal pairing just as loudly as "before and after" does.

9.3.8 Computing $\overline{D}$ and S_D from raw pairs

Example 9.3.9 (From raw pairs to $\overline{D}$ and S_D with the shortcut formula)**.** A lab compares two assays for blood glucose on $n = 6$ split specimens; each specimen is measured by the reference assay (X) and the new assay (Y). The pairs (mg/dL) are

$$(95, 98),\ (110, 108),\ (102, 106),\ (88, 90),\ (121, 124),\ (99, 101).$$

Compute $\overline{D}$ and S_D using the computational form $S_D^2 = \dfrac{\sum D_i^2 - (\sum D_i)^2/n}{n-1}$, and state the standard error of $\overline{D}$.

Which rule, and why? This is the bookkeeping that every paired test rests on; once $\overline{D}$ and S_D are in hand the test or interval is automatic. The split specimen is measured by both assays, so the data are paired and we work entirely with the column of differences.

Solution.

Step 1: form $D_i = Y_i - X_i$.

$$3,\ -2,\ 4,\ 2,\ 3,\ 2.$$

Step 2: the two running sums.

$$\sum D_i = 3 - 2 + 4 + 2 + 3 + 2 = 12,$$
$$\sum D_i^2 = 9 + 4 + 16 + 4 + 9 + 4 = 46.$$

Step 3: plug into the shortcut.

$$\overline{D} = \frac{12}{6} = 2.0,$$
$$S_D^2 = \frac{46 - 12^2/6}{6 - 1} = \frac{46 - 24}{5} = \frac{22}{5} = 4.4,$$
$$S_D = 2.098.$$

Step 4: the standard error. The standard error of $\overline{D}$ is $S_D/\sqrt{n} = 2.098/\sqrt{6} = 0.857$ mg/dL. (One could now test $H_0 : \mu_D = 0$ via $t_{\text{obs}} = 2.0/0.857 = 2.33$ on 5 df.)

$$\boxed{\overline{D} = 2.0, \quad S_D = 2.098, \quad \text{SE}(\overline{D}) = 0.857 \text{ mg/dL}.}$$

The differences themselves are easiest to read on a number line. Figure 9.2 plots the six D_i with $\overline{D}$ and the null value 0 marked: every difference but one sits to the right of 0, and $\overline{D}$ sits clearly above it.

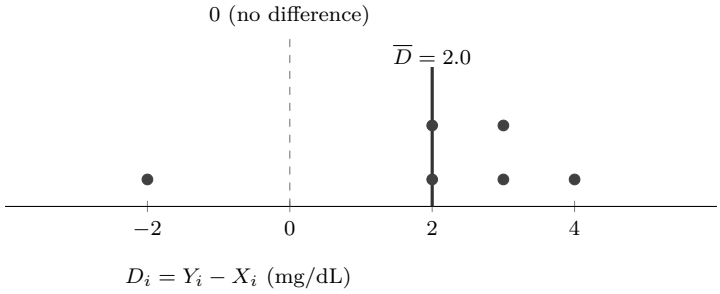


Figure 9.2: Dotplot of the six paired differences D_i. The dashed line marks the null value 0 and the heavy line marks $\overline{D} = 2.0$. Five of six differences lie right of 0 and the mean is pulled clearly positive — the visual version of a t_{obs} comfortably past zero.

9.3.9 A few more worked cases

Example 9.3.10 (Paired test with a nonzero null)**.** A factory claims its retrofit lowers each machine's energy use by *at least* 5 kWh/day. For $n = 10$ machines the daily savings D_i (old minus new) have $\overline{D} = 6.4$ kWh and $S_D = 2.6$ kWh. At $\alpha = 0.05$, test the manufacturer's specific claim that the mean saving is at least 5, i.e. whether the data are consistent with $\mu_D \geq 5$, against the skeptic's $\mu_D < 5$.

Which rule, and why? Same machine before and after the retrofit, so paired; the hypothesized value is not 0 but 5, so we center the statistic at $\mu_{D,0} = 5$. The skeptic doubts the size of the saving, making the alternative one-sided ($\mu_D < 5$).

Solution.

Step 1: set up. $H_0 : \mu_D = 5$ versus $H_1 : \mu_D < 5$. Reject only for sufficiently *negative* t_{obs}.

Step 2: the statistic with $\mu_{D,0} = 5$.

$$t_{\text{obs}} = \frac{\overline{D} - 5}{S_D/\sqrt{n}} = \frac{6.4 - 5}{2.6/\sqrt{10}} = \frac{1.4}{0.822} = 1.703.$$

Step 3: compare. The lower-tail critical value is $-t_{0.05,9} = -1.833$. Since $t_{\text{obs}} = 1.703 > -1.833$ (in fact it is positive), we do *not* reject H_0.

Step 4: conclusion. The data ($\overline{D} = 6.4 > 5$) are entirely consistent with a mean saving of at least 5 kWh/day; there is no evidence the saving falls short of the claim.

$$\boxed{t_{\text{obs}} = 1.703 > -t_{0.05,9} = -1.833 \;\Rightarrow\; \text{do not reject; claim stands.}}$$

Example 9.3.11 (Paired CI that includes zero)**.** Two graders score the same $n = 12$ essays; the within-essay score differences D_i (grader 1 minus grader 2) have $\overline{D} = 1.2$ points and $S_D = 4.8$ points. Construct a 90% confidence interval for the mean grader difference μ_D and state whether it gives evidence of grader bias.

Which rule, and why? The same essay is scored by both graders, so the two scores are paired; we want an interval for the mean difference, the one-sample t interval on the D_i. A 90% level uses $t_{0.05,11}$.

Solution.

Step 1: critical value and standard error. $t_{0.05,11} = 1.796$ and $S_D/\sqrt{n} = 4.8/\sqrt{12} = 1.386$.

Step 2: the interval.

$$\begin{aligned}\overline{D} \pm t_{0.05,11}\,\frac{S_D}{\sqrt{n}} &= 1.2 \pm 1.796 \cdot 1.386 \\ &= 1.2 \pm 2.490 \;=\; (-1.29,\; 3.69).\end{aligned}$$

Step 3: interpret. The interval contains 0, so at the 90% level we cannot rule out $\mu_D = 0$: there is no significant evidence that one grader systematically scores higher. The point estimate (1.2) is small relative to the grader-to-grader noise ($S_D = 4.8$).

$$\boxed{\mu_D \in (-1.29,\; 3.69) \text{ contains } 0 \;\Rightarrow\; \text{no evidence of bias.}}$$

Example 9.3.12 (Twins: a matched-pair one-sided test)**.** A behavioral geneticist studies $n = 8$ pairs of identical twins where one twin of each pair grew up in an urban setting and the other rural. She wonders whether urban upbringing raises a stress biomarker. The within-pair differences $D_i = (\text{urban}) - (\text{rural})$, in standardized units, are

$$0.4,\; 0.9,\; -0.2,\; 0.7,\; 0.5,\; 1.1,\; 0.0,\; 0.6.$$

At $\alpha = 0.05$, test whether urban upbringing raises the mean biomarker.

Which rule, and why? Twin pairs are matched by genetics, so the two members are linked one-to-one — paired data, differenced within twin pair. The hypothesis names a direction ("raises"), so the test is one-sided with $H_1 : \mu_D > 0$.

Solution.

Step 1: set up. $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$.

Step 2: $\overline{D}$ and S_D. Sum $= 0.4 + 0.9 - 0.2 + 0.7 + 0.5 + 1.1 + 0.0 + 0.6 = 4.0$, so $\overline{D} = 4.0/8 = 0.5$. Deviations from 0.5: $-0.1, 0.4, -0.7, 0.2, 0.0, 0.6, -0.5, 0.1$; squares sum to

$$0.01 + 0.16 + 0.49 + 0.04 + 0.00 + 0.36 + 0.25 + 0.01 = 1.32,$$

so $S_D^2 = 1.32/7 = 0.1886$, $S_D = 0.434$.

Step 3: the statistic.

$$t_{\text{obs}} = \frac{0.5}{0.434/\sqrt{8}} = \frac{0.5}{0.1535} = 3.26,$$

with 7 df.

Step 4: compare to $t_{0.05,7} = 1.895$. Since $3.26 > 1.895$, reject H_0: urban upbringing is associated with a higher mean biomarker among twins.

$$\boxed{t_{\text{obs}} = 3.26 > t_{0.05,7} = 1.895 \;\Rightarrow\; \text{reject; urban raises the biomarker.}}$$

Example 9.3.13 (CONTRAST: pairing where there is none)**.** A researcher has $n = 10$ post-surgery patients' recovery times and $m = 10$ *different* patients' recovery times under a new protocol. Because both lists have ten numbers, she lines them up row by row, subtracts, and runs a paired t on the ten "differences." Explain why this is wrong and what she should do.

Which rule, and why? The trap is purely numerical: equal sample sizes tempt one to pair. But pairing requires a **meaningful** one-to-one link — the *same* unit or a deliberately matched one — and here the two groups are unrelated people in an arbitrary order. Equal n is not pairing.

Solution.

Step 1: there is no link. Patient 3 in the old group and patient 3 in the new group share nothing; reordering either list changes the "differences" completely, which proves the pairing is fictitious. Genuine paired differences are invariant to how you shuffle the *pairs* but not to breaking the within-pair tie — here there is no tie to break.

Step 2: the consequence. The fake paired S_D depends on the arbitrary alignment and will usually be *larger* than the true within-group spread (it adds, not cancels, variation), giving a smaller, dishonest t_{obs} — the mirror image of the earlier contrast, but equally invalid. The reported p-value means nothing.

Step 3: the correct analysis. Two separate groups of unrelated patients are **independent samples**. Compare them with the two-sample t of §1 (pooled, if variances are comparable) or Welch's t of §2 — never the paired test.

Equal $n \neq$ paired; unrelated groups $\Rightarrow$ two-sample t, not paired.

Example 9.3.14 (Power gain from pairing, quantified)**.** In a sleep study, the per-subject difference in hours slept (new mattress minus old) has true $\mu_D = 0.5$ h with $\sigma_D = 0.8$ h, while the subject-to-subject spread in sleep is $\sigma = 1.6$ h either way. With $n = 16$ subjects, contrast the noncentrality of the paired test with that of an independent two-sample test of the same total size (16 per group). Which design detects the half-hour effect more reliably?

Which rule, and why? This is not a p-value computation but a design comparison; the recognition point is that the paired test's standard error uses σ_D (small, after cancellation) while the independent test uses the full σ in each group. Comparing the resulting noncentralities shows the power gain numerically.

Solution.

Step 1: paired noncentrality. The paired statistic centers at $\sqrt{n}\,\mu_D/\sigma_D = \sqrt{16} \cdot 0.5/0.8 = 4 \cdot 0.625 = 2.50$. With 15 df this clears $t_{0.025,15} = 2.131$ comfortably on average, so power is high.

Step 2: independent noncentrality. An independent two-sample test of 16 per group has noncentrality

$$\frac{\mu_D}{\sigma\sqrt{1/16 + 1/16}} = \frac{0.5}{1.6 \cdot 0.354} = \frac{0.5}{0.566} = 0.884,$$

far below the critical value $t_{0.025,30} = 2.042$; power is poor.

Step 3: read the gap. The paired design's noncentrality (2.50) is nearly three times the independent design's (0.88), because $\sigma_D = 0.8$ is half the raw $\sigma = 1.6$ — the mattress effect tracks the same person, so the baseline cancels. The paired study detects the half-hour effect routinely; the independent study usually misses it.

paired noncentrality 2.50 $\gg$ independent 0.88 $\Rightarrow$ pair when you can.

Remark. The lesson of the last example is a design principle, not a test: *if pairing is available and the within-unit correlation is positive, pair.* You spend degrees of freedom but you buy a far smaller standard error, and standard error wins. The only times pairing can hurt are when the two measurements are essentially uncorrelated (then $\sigma_D^2 \approx \sigma_X^2 + \sigma_Y^2$ and you have merely thrown away degrees of freedom) or negatively correlated (rare in practice). Whenever a unit naturally supplies both measurements, those caveats almost never bite.

Example 9.3.15 (Detecting and running the right test on a mixed-up problem)**.** A quality engineer reports: "I measured the surface roughness of $n = 11$ machined parts with a contact gauge and an optical gauge; the contact-minus-optical differences have $\overline{D} = 0.06\,\mu$m and $S_D = 0.15\,\mu$m. Is there a systematic difference between the gauges at $\alpha = 0.05$?" Identify the design, run the test, and state the conclusion.

Which rule, and why? Each part is measured by *both* gauges, so the two readings share the part's true roughness — paired data, even though the word "paired" never appears. "Systematic difference" is undirected, so the test is two-sided on the D_i.

Solution.

Step 1: recognize and set up. Paired (same part, two gauges); $H_0 : \mu_D = 0$ versus $H_1 : \mu_D \neq 0$, with $n = 11$, $\mathrm{df} = 10$.

Step 2: the statistic.

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{0.06}{0.15/\sqrt{11}} = \frac{0.06}{0.04523} = 1.327.$$

Step 3: compare to $t_{0.025,10} = 2.228$. Since $|1.327| < 2.228$, do not reject H_0.

Step 4: conclusion. There is no significant evidence of a systematic bias between the contact and optical gauges; the observed mean gap of $0.06\,\mu$m is within sampling noise.

$$\boxed{t_{\text{obs}} = 1.327 < t_{0.025,10} = 2.228 \;\Rightarrow\; \text{no significant gauge bias.}}$$

Example 9.3.16 (Sample-size check for a planned paired study)**.** Before running a paired study of a memory drug, an investigator expects a mean improvement of $\mu_D = 2$ points with a difference standard deviation around $\sigma_D = 5$ points. Roughly how many subjects n are needed so that the two-sided 5% paired test has good sensitivity — specifically, so the noncentrality $\sqrt{n}\,\mu_D/\sigma_D$ reaches about 2.8 (a rule of thumb for roughly 80% power)?

Which rule, and why? Planning a paired study uses the paired standard error $\sigma_D/\sqrt{n}$ from the start. We invert the noncentrality target to solve for n; the calculation lives entirely on the difference scale, which is what makes pairing economical.

Solution.

Step 1: set the noncentrality equation. We want

$$\frac{\sqrt{n}\,\mu_D}{\sigma_D} \approx 2.8 \quad\Longrightarrow\quad \sqrt{n} \approx 2.8 \cdot \frac{\sigma_D}{\mu_D} = 2.8 \cdot \frac{5}{2} = 7.0.$$

Step 2: solve for n.

$$n \approx 7.0^2 = 49.$$

Step 3: read it. About $n \approx 49$ subjects should give the paired test roughly 80% power to detect a 2-point mean gain. (An independent two-sample study, carrying the full between-subject spread, would need many more per group — another reason to pair.)

$$\boxed{n \approx \left(2.8\,\sigma_D/\mu_D\right)^2 = 49 \text{ subjects.}}$$

Remark. Assumptions and robustness. The paired t assumes the *differences* are approximately normal — not the original measurements, which may be skewed identically and cancel that skew on subtraction. For moderate n the test is fairly robust to non-normal differences, but a single wild outlier in the D_i (one pair that moved the wrong way by a lot) can dominate $\overline{D}$ and S_D alike; always plot the differences (Figure 9.2) before trusting the p-value. If the differences are badly non-normal or the scale is ordinal, switch to the paired sign test or Wilcoxon signed-rank test of Chapter 14 — the nonparametric counterparts that keep the pairing but drop the normality assumption.

9.4 Comparing Two Variances: the F-Test

So far the two-sample machinery has aimed at the *centers* of two populations: are the means equal? But spread is a target in its own right, and it also governs which mean-test we are allowed to run. A pooled two-sample t (§1) is valid only when the two populations share a common variance; if one group is far more

variable than the other, pooling is illegitimate and Welch's t (§2) is required. The tool that compares two spreads directly is the **F-test**, built on the ratio S_X^2/S_Y^2 of the two sample variances. Its reference distribution is the F **distribution**, a right-skewed, positive-valued curve indexed by *two* degrees of freedom (one from each sample). Recognise this test by its target — a comparison of two variances, standard deviations, precisions, or "consistencies" — and by the presence of two *independent normal* samples. The one warning to carry throughout: the F-test is acutely **sensitive to non-normality**, far more so than the t-tests for means, so a borderline result on data of doubtful shape is not to be trusted.

Which two-variance question is being asked?

- Compare **two** sample variances from **independent normal** groups? ⇒ the **F-test** on $F_{\text{obs}} = S_X^2/S_Y^2 \sim \mathrm{F}_{n-1,\,m-1}$ under $H_0\colon \sigma_X^2 = \sigma_Y^2$.
- H_1 says the variances merely **differ** ($\sigma_X^2 \neq \sigma_Y^2$)? ⇒ two-sided: reject if F_{obs} is above $F_{\alpha/2,\,n-1,\,m-1}$ *or* below $F_{1-\alpha/2,\,n-1,\,m-1}$ (two separate look-ups).
- H_1 names a **direction** ("X is more variable," $\sigma_X^2 > \sigma_Y^2$)? ⇒ one-sided: put the named group's variance on top and reject if $F_{\text{obs}} > F_{\alpha,\,n-1,\,m-1}$.
- Deciding whether a **pooled t-test** is justified? ⇒ run this F-test first as a pre-check on equal variances; if it rejects, switch to Welch's t.
- Need a **lower-tail** critical value the table omits? ⇒ use the reciprocal property $F_{1-\alpha,\,d_1,\,d_2} = 1/F_{\alpha,\,d_2,\,d_1}$ (swap the df, take the reciprocal).

Do not confuse this with the χ^2 test: the one-sample χ^2 test compares *one* sample variance to a *fixed number* σ_0^2; the F-test compares *two* sample variances to *each other*. One sample ⇒ χ^2; two samples ⇒ F.

9.4.1 The F distribution

Definition 9.4.1 (The F distribution)**.** Let $U \sim \chi^2_{d_1}$ and $V \sim \chi^2_{d_2}$ be independent chi-square variables. The ratio of each divided by its degrees of freedom,

$$F = \frac{U/d_1}{V/d_2},$$

has the F **distribution** with d_1 **numerator** and d_2 **denominator** degrees of freedom, written $F \sim \mathrm{F}_{d_1,\,d_2}$. It lives on $(0,\infty)$, is right-skewed, and has mean $d_2/(d_2-2)$ for $d_2 > 2$ (a little above 1).

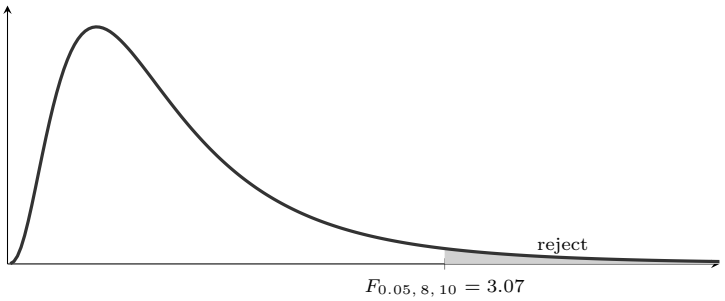
Remark. The order of the degrees of freedom matters: F_{d_1,d_2} and F_{d_2,d_1} are different distributions. The numerator df always belongs to the variance written on *top* of the ratio. Since F is a ratio of squared quantities it can never be negative, and like the χ^2 curve it is asymmetric — so, exactly as with χ^2, a two-sided test uses two *different* positive critical values, never a single value with a $\pm$.

Proposition 9.4.2 (Reciprocal property)**.** *If $F \sim \mathrm{F}_{d_1,d_2}$ then $1/F \sim \mathrm{F}_{d_2,d_1}$. Writing $F_{q,\,d_1,\,d_2}$ for the point with area q to its* right *under F_{d_1,d_2}, this gives the lower-tail identity*

$$F_{1-\alpha,\,d_1,\,d_2} = \frac{1}{F_{\alpha,\,d_2,\,d_1}}.$$

Proof. If $F = (U/d_1)/(V/d_2)$ with $U \sim \chi^2_{d_1}$, $V \sim \chi^2_{d_2}$ independent, then $1/F = (V/d_2)/(U/d_1)$ is the same construction with the roles of U and V swapped, hence $1/F \sim \mathrm{F}_{d_2,d_1}$. For the quantile identity, $\mathbb{P}(F < F_{1-\alpha,\,d_1,\,d_2}) = 1-\alpha$ means $\mathbb{P}(1/F > 1/F_{1-\alpha,\,d_1,\,d_2}) = 1-\alpha$, i.e. the right-tail area of $1/F$ beyond that reciprocal is $1-\alpha$; matching to the definition of the upper F_{d_2,d_1} point gives $1/F_{1-\alpha,\,d_1,\,d_2} = F_{1-\alpha,\,d_2,\,d_1}$ rearranged to the stated form. □

Remark. Statistical tables typically list only *upper*-tail F points ($F_{0.05,,}$, $F_{0.025,,}$, $F_{0.01,,}$) because the reciprocal property recovers every lower-tail point from an upper-tail one. This is why a common shortcut for a one-sided test is to always put the *larger* sample variance on top: then the only relevant cutoff is an upper-tail value the table actually prints.



$\mathrm{F}_{8,10}$ statistic

An $\mathrm{F}_{8,10}$ *density (numerator df* 8*, denominator df* 10*): right-skewed, positive support, mode below* 1*. For an upper one-sided test at* $\alpha = 0.05$ *the whole rejection region sits in the right tail beyond* $F_{0.05,\,8,\,10} = 3.07$*.*

Example 9.4.3 (The reciprocal property: finding a lower-tail critical value)**.** A table lists only upper-tail F points. You need the lower 2.5% point of the $\mathrm{F}_{8,10}$ distribution — the value $F_{0.975,\,8,\,10}$ with area 0.975 to its right (equivalently area 0.025 to its *left*) — but the table prints no lower points. Given $F_{0.025,\,10,\,8} = 3.85$, find it.

Which rule, and why? Whenever a needed quantile is in the *lower* tail and the table prints only *upper*-tail points, the reciprocal property converts one into the other: swap the two degrees of freedom and take the reciprocal of the printed upper point.

Solution.

Step 1: state the identity. With $d_1 = 8$, $d_2 = 10$, $\alpha = 0.025$,

$$F_{1-\alpha,\,d_1,\,d_2} = \frac{1}{F_{\alpha,\,d_2,\,d_1}} \implies F_{0.975,\,8,\,10} = \frac{1}{F_{0.025,\,10,\,8}}.$$

Step 2: substitute the printed value.

$$F_{0.025,\,10,\,8} = 3.85 \quad \text{(from the table, df swapped to } (10, 8)\text{)},$$
$$F_{0.975,\,8,\,10} = \frac{1}{3.85} = 0.260.$$

Step 3: sanity check. The answer is below 1, as a lower-tail F point must be (the bulk of an F distribution sits near and below 1), and it is a reciprocal of a number above 1 — both signs the conversion went the right way.

$$\boxed{F_{0.975,\,8,\,10} = \frac{1}{F_{0.025,\,10,\,8}} = \frac{1}{3.85} = 0.260.}$$

9.4.2 The test for equal variances

Theorem 9.4.4 (Sampling distribution of the variance ratio)**.** *Let* $X_1, \ldots, X_n \sim \mathrm{N}(\mu_X, \sigma_X^2)$ *and* $Y_1, \ldots, Y_m \sim \mathrm{N}(\mu_Y, \sigma_Y^2)$ *be two* independent *normal samples, with sample variances* S_X^2 *and* S_Y^2*. Then*

$$\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim \mathrm{F}_{\,n-1,\,m-1}\,.$$

In particular, under $H_0\colon \sigma_X^2 = \sigma_Y^2$ *the population variances cancel and*

$$F_{obs} = \frac{S_X^2}{S_Y^2} \sim \mathrm{F}_{\,n-1,\,m-1}\,.$$

Proof. By the one-sample result of §6 (Ch. 6), $(n-1)S_X^2/\sigma_X^2 \sim \chi^2_{n-1}$ and $(m-1)S_Y^2/\sigma_Y^2 \sim \chi^2_{m-1}$, and they are independent because the samples are independent. Form

$$\frac{\left[(n-1)S_X^2/\sigma_X^2\right]/(n-1)}{\left[(m-1)S_Y^2/\sigma_Y^2\right]/(m-1)} = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2},$$

which is exactly a $\chi^2_{n-1}/(n-1)$ over an independent $\chi^2_{m-1}/(m-1)$ — the defining form of $\mathrm{F}_{n-1,m-1}$. Setting $\sigma_X^2 = \sigma_Y^2$ removes the population variances. □

Definition 9.4.5 (The F-test for two variances)**.** For two independent normal samples, to test $H_0\colon \sigma_X^2 = \sigma_Y^2$ compute

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2}, \qquad F_{\text{obs}} \sim \mathrm{F}_{\,n-1,\,m-1} \text{ under } H_0,$$

with numerator df $n-1$ and denominator df $m-1$. Let $F_{q,\,d_1,\,d_2}$ have area q to its right. At level α:

- **Two-sided** $H_1\colon \sigma_X^2 \neq \sigma_Y^2$: reject if $F_{\text{obs}} > F_{\alpha/2,\,n-1,\,m-1}$ or $F_{\text{obs}} < F_{1-\alpha/2,\,n-1,\,m-1}$.
- **Upper one-sided** $H_1\colon \sigma_X^2 > \sigma_Y^2$: reject if $F_{\text{obs}} > F_{\alpha,\,n-1,\,m-1}$.
- **Lower one-sided** $H_1\colon \sigma_X^2 < \sigma_Y^2$: reject if $F_{\text{obs}} < F_{1-\alpha,\,n-1,\,m-1}$ (use the reciprocal property to find this cutoff).

Remark. A convenient bookkeeping rule for a two-sided test: put the *larger* sample variance on top so that $F_{\text{obs}} \geq 1$, then reject when $F_{\text{obs}} > F_{\alpha/2,\,,}$ using whichever df ordering matches that arrangement. This sidesteps the lower-tail look-up entirely. For a one-sided test you must respect the direction of H_1 and keep the variance named in H_1 in the numerator.

Example 9.4.6 (Two-sided: do two suppliers' fill machines differ in consistency?)**.** A bottler buys identical fill heads from two suppliers. From supplier X, $n = 13$ bottles give $S_X^2 = 5.8\,\text{mL}^2$; from supplier Y, $m = 10$ bottles give $S_Y^2 = 2.1\,\text{mL}^2$. Are the two machines equally consistent? Test $H_0\colon \sigma_X^2 = \sigma_Y^2$ against $H_1\colon \sigma_X^2 \neq \sigma_Y^2$ at $\alpha = 0.05$.

Which rule, and why? Two independent normal samples and a question about *equal spread* (consistency), with no direction specified, so this is the *two-sided* F-test on S_X^2/S_Y^2. It is not a χ^2 test (that would compare one sample to a fixed σ_0^2), and not a t-test (that compares means).

Solution.

Step 1: statistic. Put the larger variance on top so $F_{\text{obs}} \geq 1$:

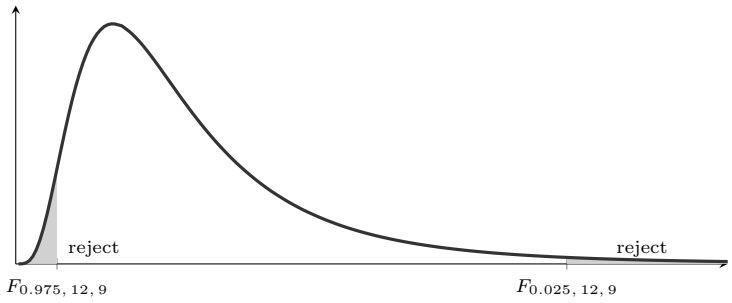
$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{5.8}{2.1} = 2.76, \qquad \text{df } = (n-1,\, m-1) = (12,\, 9).$$

Step 2: critical value. Two-sided at $\alpha = 0.05$ puts 0.025 in the upper tail: $F_{0.025,\,12,\,9} = 3.87$. (Because we arranged $F_{\text{obs}} \geq 1$, only this upper cutoff can be exceeded; the lower cutoff $F_{0.975,\,12,\,9} = 1/F_{0.025,\,9,\,12} \approx 1/3.44 = 0.29$ is irrelevant here.)

Step 3: decision. Reject if $F_{\text{obs}} > 3.87$. Here $2.76 < 3.87$.

$$\boxed{F_{\text{obs}} = 2.76 < F_{0.025,\,12,\,9} = 3.87 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The machines' consistencies do not differ significantly, even though X's sample variance is nearly triple Y's — variance ratios are noisy, so it takes a large ratio to reach significance.



$F_{12,9}$ statistic

Two-sided rejection regions for the $F_{12,9}$ *test at* $\alpha = 0.05$*: area* 0.025 *in each tail. The two cutoffs* $F_{0.975,\,12,\,9} \approx 0.29$ *and* $F_{0.025,\,12,\,9} = 3.87$ *are not reflections of one another — the lower one comes from the reciprocal property, and the curve is right-skewed.*

Example 9.4.7 (Two-sided on standard deviations: two assay labs)**.** Two laboratories measure the same hormone standard. Lab X reports $n = 16$ replicates with standard deviation $S_X = 0.42$; lab Y reports $m = 21$ replicates with $S_Y = 0.27$. Do the labs differ in precision (standard deviation) at $\alpha = 0.10$?

Which rule, and why? Precision is spread, the hypothesis is two-sided, and there are two independent normal samples, so it is the two-sided F-test — after *squaring* the standard deviations, since F is built from *variances*.

Solution.

Step 1: convert and form the statistic. $S_X^2 = 0.42^2 = 0.1764$, $S_Y^2 = 0.27^2 = 0.0729$. Larger on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{0.1764}{0.0729} = 2.42, \qquad \text{df } = (15,\, 20).$$

Step 2: critical value. Two-sided at $\alpha = 0.10$ puts 0.05 in the upper tail: $F_{0.05,\,15,\,20} = 2.20$.

Step 3: decision. Reject if $F_{\text{obs}} > 2.20$. Here $2.42 > 2.20$.

$$\boxed{F_{\text{obs}} = 2.42 > F_{0.05,\,15,\,20} = 2.20 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.10.}$$

Lab X is significantly less precise than lab Y. Note we tested standard deviations but worked entirely on the squared scale.

Example 9.4.8 (Two-sided, equal sample sizes: reaction-time consistency)**.** A psychologist compares the consistency of reaction times under two display layouts. Layout X: $n = 25$ subjects, $S_X^2 = 812\,\text{ms}^2$. Layout Y: $m = 25$ subjects, $S_Y^2 = 455\,\text{ms}^2$. Is either layout more consistent than the other (two-sided) at $\alpha = 0.05$?

Which rule, and why? Two independent normal samples, a two-sided question about spread, equal sample sizes $\Rightarrow$ two-sided F-test on df $(24, 24)$. Equal n and m make the numerator and denominator df equal, which simplifies the critical value but does not change the procedure.

Solution.

Step 1: statistic. Larger on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{812}{455} = 1.785, \qquad \text{df } = (24,\, 24).$$

Step 2: critical value. Two-sided at $\alpha = 0.05$, upper area 0.025: $F_{0.025,\,24,\,24} = 2.27$.

Step 3: decision. Reject if $F_{\text{obs}} > 2.27$. Here $1.785 < 2.27$.

$$\boxed{F_{\text{obs}} = 1.785 < F_{0.025,\,24,\,24} = 2.27 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The two layouts do not differ significantly in consistency. With equal df the test reduces to asking whether the larger-over-smaller variance ratio clears a single upper cutoff.

9.4.3 One-sided tests

Example 9.4.9 (One-sided: is process A more variable?)**.** A plant runs two casting lines. Engineers suspect the older line A produces parts with *more* variable wall thickness than the newer line B. A sample from line A gives $n = 21$ parts with $S_A^2 = 0.094\,\text{mm}^2$; line B gives $m = 16$ parts with $S_B^2 = 0.041\,\text{mm}^2$. Test the suspicion at $\alpha = 0.05$.

Which rule, and why? The alternative is directional — "A is *more* variable" — so this is an *upper* one-sided F-test with the suspected-larger variance (A) in the numerator. Putting A on top is mandatory here (it encodes H_1), not just a bookkeeping choice.

Solution.

Step 1: set hypotheses and statistic. $H_0 : \sigma_A^2 = \sigma_B^2$ versus $H_1 : \sigma_A^2 > \sigma_B^2$. With A on top,

$$F_{\text{obs}} = \frac{S_A^2}{S_B^2} = \frac{0.094}{0.041} = 2.29, \qquad \text{df } = (n-1,\, m-1) = (20,\, 15).$$

Step 2: critical value. All of $\alpha = 0.05$ in the upper tail: $F_{0.05,\,20,\,15} = 2.33$.

Step 3: decision. Reject if $F_{\text{obs}} > 2.33$. Here $2.29 < 2.33$ — a very near miss.

$$\boxed{F_{\text{obs}} = 2.29 < F_{0.05,\,20,\,15} = 2.33 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The data point toward A being more variable but fall just short of significance; a slightly larger sample or a less stringent α might tip it.

Example 9.4.10 (One-sided lower tail via the reciprocal trick)**.** A trader claims a new low-volatility fund X has *smaller* return variance than a benchmark Y. From X, $n = 10$ months give $S_X^2 = 4.0\,(\%)^2$; from Y, $m = 13$ months give $S_Y^2 = 11.5\,(\%)^2$. Test $H_0 : \sigma_X^2 = \sigma_Y^2$ against $H_1 : \sigma_X^2 < \sigma_Y^2$ at $\alpha = 0.05$, keeping X in the numerator so the test is a lower-tail one.

Which rule, and why? The directional claim "X has *smaller* variance" is a *lower*-tail alternative when X is on top, so we need a lower critical value $F_{0.95,\,9,\,12}$, which the table does not print — the reciprocal property supplies it.

Solution.

Step 1: statistic. Keep X on top to match H_1:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{4.0}{11.5} = 0.348, \qquad \text{df } = (n-1,\, m-1) = (9,\, 12).$$

Step 2: lower critical value by reciprocal. We reject for *small* F_{obs}, below $F_{0.95,\,9,\,12}$. By the reciprocal property,

$$\begin{aligned} F_{0.95,\,9,\,12} &= \frac{1}{F_{0.05,\,12,\,9}}, \\ F_{0.05,\,12,\,9} &= 3.07, \\ F_{0.95,\,9,\,12} &= \frac{1}{3.07} = 0.326. \end{aligned}$$

Step 3: decision. Reject if $F_{\text{obs}} < 0.326$. Here $0.348 > 0.326$ — just inside.

$$\boxed{F_{\text{obs}} = 0.348 > F_{0.95,\,9,\,12} = 0.326 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

The fund looks less volatile, but not by a statistically convincing margin. (Equivalently, one could flip the ratio to $S_Y^2/S_X^2 = 2.875$ on df $(12, 9)$ and test the *upper* tail against $F_{0.05,\,12,\,9} = 3.07$; since $2.875 < 3.07$, the same conclusion follows — a useful check.)

Example 9.4.11 (One-sided: does a generic drug have more variable absorption?)**.** A regulator worries that a generic tablet shows *more* variable blood concentration than the brand-name reference. Generic (X): $n = 25$ volunteers, $S_X^2 = 146\,(\text{ng/mL})^2$. Brand ($Y$): $m = 31$ volunteers, $S_Y^2 = 62\,(\text{ng/mL})^2$. Test $H_1\colon \sigma_X^2 > \sigma_Y^2$ at $\alpha = 0.01$.

Which rule, and why? The concern is directional — generic *more* variable — so it is an upper one-sided F-test with the generic variance in the numerator. The strict $\alpha = 0.01$ reflects a regulatory desire to control false alarms; the whole 1% sits in the upper tail.

Solution.

Step 1: hypotheses and statistic. $H_0\colon \sigma_X^2 = \sigma_Y^2$ versus $H_1\colon \sigma_X^2 > \sigma_Y^2$. Generic on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{146}{62} = 2.355, \qquad \text{df } = (n-1,\, m-1) = (24,\, 30).$$

Step 2: critical value. All of $\alpha = 0.01$ in the upper tail: $F_{0.01,\,24,\,30} = 2.47$.

Step 3: decision. Reject if $F_{\text{obs}} > 2.47$. Here $2.355 < 2.47$.

$$\boxed{F_{\text{obs}} = 2.355 < F_{0.01,\,24,\,30} = 2.47 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.01.}$$

At the strict 1% level there is not quite enough evidence to flag the generic as more variable, though it would clear the bar at $\alpha = 0.05$ (where $F_{0.05,\,24,\,30} = 1.89$).

9.4.4 A confidence interval for the variance ratio

Theorem 9.4.12 (Confidence interval for σ_X^2/σ_Y^2)**.** *For two independent normal samples, a* $100(1-\alpha)\%$ *confidence interval for the ratio* σ_X^2/σ_Y^2 *is*

$$\left(\frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{\alpha/2,\,n-1,\,m-1}} \;,\; \frac{S_X^2}{S_Y^2} \cdot F_{\alpha/2,\,m-1,\,n-1} \right).$$

Proof. Start from $(S_X^2/\sigma_X^2)/(S_Y^2/\sigma_Y^2) \sim \mathrm{F}_{n-1,m-1}$ and bracket it between its lower and upper $\alpha/2$ points:

$$F_{1-\alpha/2,\,n-1,\,m-1} < \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} < F_{\alpha/2,\,n-1,\,m-1}.$$

Solving the double inequality for σ_X^2/σ_Y^2 and applying the reciprocal property $F_{1-\alpha/2,\,n-1,\,m-1} = 1/F_{\alpha/2,\,m-1,\,n-1}$ to the lower point yields the stated endpoints. □

Example 9.4.13 (CI for a variance ratio: tablet hardness)**.** Two tablet presses are compared on hardness variability. Press X: $n = 16$, $S_X^2 = 3.6$. Press Y: $m = 13$, $S_Y^2 = 1.5$. Give a 90% confidence interval for σ_X^2/σ_Y^2, and say whether it is consistent with equal variances.

Which rule, and why? The question asks to *estimate the ratio with an interval*, not to accept/reject a single hypothesis, so we use the F-based confidence interval for σ_X^2/σ_Y^2. Whether the interval contains 1 then answers the equal-variance question as a byproduct.

Solution.

Step 1: gather pieces. $S_X^2/S_Y^2 = 3.6/1.5 = 2.40$; $\text{df} = (n-1, m-1) = (15, 12)$. For a 90% interval, $\alpha = 0.10$ and $\alpha/2 = 0.05$:

$$F_{0.05,\,15,\,12} = 2.62,$$
$$F_{0.05,\,12,\,15} = 2.48.$$

Step 2: assemble endpoints.

$$\text{lower} = \frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{0.05,\,15,\,12}} = \frac{2.40}{2.62} = 0.916,$$
$$\text{upper} = \frac{S_X^2}{S_Y^2} \cdot F_{0.05,\,12,\,15} = 2.40 \times 2.48 = 5.95.$$

Step 3: report and interpret.

$$\boxed{\frac{\sigma_X^2}{\sigma_Y^2} \in (0.92,\ 5.95) \ \text{ with 90\% confidence.}}$$

The interval *contains* 1, so at the 10% level the data are consistent with equal variances — equivalent to a two-sided F-test failing to reject. The wide span (a factor of about 6.5 from end to end) shows how little a couple of dozen observations pin down a variance ratio.

Example 9.4.14 (CI that excludes 1: rainfall variability across two stations)**.** Two weather stations record daily rainfall. Station X: $n = 25$ days, $S_X^2 = 44.0\,\text{mm}^2$. Station Y: $m = 25$ days, $S_Y^2 = 15.0\,\text{mm}^2$. Give a 95% confidence interval for σ_X^2/σ_Y^2 and interpret.

Which rule, and why? Again an interval *estimate* of the variance ratio, so the F-based CI. Whether the interval excludes 1 tells us, at the corresponding two-sided level, whether the variances differ significantly.

Solution.

Step 1: pieces. $S_X^2/S_Y^2 = 44.0/15.0 = 2.933$; df $= (24, 24)$. For 95%, $\alpha/2 = 0.025$:

$$F_{0.025,\,24,\,24} = 2.27,$$
$$F_{0.025,\,24,\,24} = 2.27 \quad \text{(numerator and denominator df equal, so both look-ups coincide).}$$

Step 2: endpoints.

$$\text{lower} = \frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{0.025,\,24,\,24}} = \frac{2.933}{2.27} = 1.292,$$
$$\text{upper} = \frac{S_X^2}{S_Y^2} \cdot F_{0.025,\,24,\,24} = 2.933 \times 2.27 = 6.66.$$

Step 3: report and interpret.

$$\boxed{\frac{\sigma_X^2}{\sigma_Y^2} \in (1.29,\ 6.66) \ \text{ with 95\% confidence.}}$$

The interval lies *entirely above* 1, so station X is significantly more variable than station Y at the two-sided 5% level — the same verdict a two-sided F-test would give. With equal df the interval is symmetric in multiplicative terms: the point estimate 2.93 sits at the geometric center, $1.29 \times 2.27 = 2.93 = 6.66/2.27$.

9.4.5 Using the F-test to justify pooling

Remark. The pooled two-sample t-test of §1 assumes $\sigma_X^2 = \sigma_Y^2$. The F-test is the natural pre-check: test equal variances first; if it does *not* reject, the equal-variance assumption is tenable and the pooled t is in order, while if it *does* reject, abandon pooling and use Welch's t (§2). Two cautions temper this two-step recipe. First, the pre-test spends some of the overall error budget, so the chained procedure's true level is not exactly the nominal α of the t-test. Second — and more important — the F-test is itself fragile under non-normality, so a clean equal-variance verdict from it is only as trustworthy as the normality of the data. Many practitioners now skip the pre-test and simply default to Welch's t, which is valid whether or not the variances are equal. The F pre-test remains standard in textbooks and is worth knowing.

Example 9.4.15 (Pre-check before a pooled t-test: fertilizer yields)**.** An agronomist will compare mean crop yields under two fertilizers with a two-sample t-test. Fertilizer X: $n = 11$ plots, $\overline{X} = 52.3$, $S_X^2 = 18.4$. Fertilizer Y: $m = 11$ plots, $\overline{Y} = 48.9$, $S_Y^2 = 6.1$. Before choosing pooled versus Welch, run the F-test for equal variances at $\alpha = 0.05$ (two-sided) and state which t-test to use.

Which rule, and why? The decision "pooled or Welch?" hinges on whether $\sigma_X^2 = \sigma_Y^2$ is tenable, which is exactly what the two-sided F-test on S_X^2/S_Y^2 checks. We are not yet testing the means; this is the variance pre-check that selects the mean-test.

Solution.

Step 1: F statistic. Larger variance on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{18.4}{6.1} = 3.02, \qquad \text{df } = (10,\, 10).$$

Step 2: critical value. Two-sided at $\alpha = 0.05$, upper area 0.025: $F_{0.025,\,10,\,10} = 3.72$.

Step 3: decision and consequence. Reject equal variances if $F_{\text{obs}} > 3.72$. Here $3.02 < 3.72$, so we fail to reject.

$$\boxed{F_{\text{obs}} = 3.02 < F_{0.025,\,10,\,10} = 3.72 \;\Rightarrow\; \text{equal variances tenable; use the } \textit{pooled } t\text{-test.}}$$

The pre-check passes, so pooling the two variances into s_p^2 for the subsequent t-test of $\mu_X = \mu_Y$ is justified. (Had F_{obs} exceeded 3.72, we would have switched to Welch's t.)

Example 9.4.16 (Contrast: when the pre-check should have failed)**.** Same setup as the previous example, but now fertilizer Y was applied to only $m = 6$ plots and gave $S_Y^2 = 2.0$ (while X still has $n = 11$, $S_X^2 = 18.4$). A hurried analyst, seeing "$F \approx 9$, but my t-table example always pooled," pools anyway. Critique this and give the correct call at $\alpha = 0.05$.

Which rule, and why? This is still the equal-variance F-test, but now the sample variances differ sharply, so the pre-check is the very thing that should stop us from pooling. The error is ignoring the pre-check's verdict, not the choice of test.

Solution.

Step 1: F statistic.

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{18.4}{2.0} = 9.20, \qquad \text{df } = (n-1,\, m-1) = (10,\, 5).$$

Step 2: critical value. Two-sided at $\alpha = 0.05$, upper area 0.025: $F_{0.025,\,10,\,5} = 6.62$.

Step 3: decision and the correct call. Since $9.20 > 6.62$, reject equal variances.

$$\boxed{F_{\text{obs}} = 9.20 > F_{0.025,\,10,\,5} = 6.62 \;\Rightarrow\; \text{variances unequal; pooling is } \textit{invalid} \text{ — use Welch's } t.}$$

Pooling here would blend a large variance with a small one and an unbalanced pair of sample sizes, producing a s_p^2 that misrepresents both groups and a t-test with the wrong standard error and the wrong reference df. The pre-check exists precisely to catch this; overriding it because a worked example "always pooled" is the mistake.

9.4.6 Sensitivity to non-normality

Remark. Like the one-sample χ^2 test, the F-test for variances draws *none* of its robustness from the central limit theorem. The exact $\mathrm{F}_{n-1,m-1}$ reference distribution holds only when *both* parent populations are normal; the true sampling distribution of S_X^2/S_Y^2 is governed by the populations' **kurtosis** (fourth moment), and heavy tails or skew can throw the actual Type I error rate far from the nominal α. This is worse than the situation for the two-sample t on means, which the CLT cushions for moderate samples. The practical upshot: treat a *borderline* F result on data of uncertain shape as inconclusive, check normality (a normal quantile plot, a formal test) before trusting the verdict, and prefer distribution-robust alternatives (Levene's or the Brown–Forsythe test, both based on absolute deviations from a center) when normality is doubtful.

Example 9.4.17 (Contrast: a borderline F under suspected non-normality)**.** Two manufacturing shifts are compared for thickness variability. Shift X: $n = 15$, $S_X^2 = 0.0090$. Shift Y: $m = 15$, $S_Y^2 = 0.0040$. A normal quantile plot of shift X shows a pronounced right tail (a few thick outliers). The two-sided F-test at $\alpha = 0.05$ gives $F_{\text{obs}} = 2.25$ against $F_{0.025,\,14,\,14} = 2.98$, so it fails to reject — but the analyst is tempted to "trust the close call" and declare the variances equal. Why is that unsafe here?

Which rule, and why? Mechanically this is the two-sided F-test and the arithmetic is right, but the test's validity *assumes both populations are normal*; shift X is visibly right-skewed, so the $\mathrm{F}_{14,14}$ reference — and hence the cutoff 2.98 and any "how close" reading — is not reliable.

Solution.

Step 1: the computation, at face value.

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{0.0090}{0.0040} = 2.25 < F_{0.025,\,14,\,14} = 2.98,$$

so formally we fail to reject equal variances.

Step 2: why "close" is meaningless here. The right tail in shift X inflates S_X^2 in a way the normal theory does not anticipate; under a heavy-tailed parent the statistic S_X^2/S_Y^2 has far more spread than $\mathrm{F}_{14,14}$ predicts, so the cutoff 2.98 does not carry its nominal 2.5% tail area. A value of 2.25 could be perfectly ordinary noise *or* a real difference masked by the wrong reference distribution — the F-test cannot tell them apart on these data.

Step 3: the right move. Do not read significance into the near miss. Re-express or transform the data toward normality if appropriate, or use Levene's / Brown–Forsythe, which compare absolute deviations from each group's center and are far less sensitive to tail shape.

A borderline F result on non-normal data is inconclusive; verify normality or switch to a robust test.

9.5 Comparing Two Proportions

A great many questions reduce to comparing two success rates: does drug A cure more patients than drug B, does landing-page X convert better than landing-page Y, do two assembly lines scrap at the same rate, do two precincts vote alike? In each case we have *two independent* count samples, $X \sim \mathrm{Bin}(n, p_X)$ successes out of n trials and $Y \sim \mathrm{Bin}(m, p_Y)$ successes out of m trials, with sample proportions $\widehat{p}_X = X/n$ and $\widehat{p}_Y = Y/m$. The parameter of interest is the difference $p_X - p_Y$, and almost always the null hypothesis is $H_0\colon p_X = p_Y$ (the two rates are equal). Because each $\widehat{p}$ is approximately normal for large samples, so is their difference, and the whole problem is handled by a single large-sample z-statistic. The one subtlety — and it is the entire point of this section — is which standard error to use: a **pooled** success rate *under* H_0 for the test, but the **unpooled** separate rates for a confidence interval. Get that distinction right and the rest is bookkeeping.

Which two-proportion procedure, and which standard error?

- Two **independent** count samples, asking whether the rates are equal? ⇒ the **two-proportion z-test** of $H_0\colon p_X = p_Y$.
- **Testing** $H_0\colon p_X = p_Y$? ⇒ use the **pooled** rate $\widehat{p} = (X + Y)/(n + m)$ in the standard error $\sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n} + \frac{1}{m}\right)}$, because under H_0 both samples estimate one common p.
- Building a **confidence interval** for $p_X - p_Y$? ⇒ use the **unpooled** standard error

  $$\sqrt{\widehat{p}_X(1-\widehat{p}_X)/n + \widehat{p}_Y(1-\widehat{p}_Y)/m},$$

  since there is no common rate to assume.
- **One-sided** ("higher than", "cures more", "A beats B") puts all of α in one tail; **two-sided** ("differ", "$\neq$") splits it into $\alpha/2$ each.

Do not confuse the pooled and unpooled SE: the *test* pools because H_0 says the rates are equal, so the best estimate of that single rate uses all $n + m$ trials. A *confidence interval* makes no such assumption, so it keeps the two rates separate. Pooling inside a CI, or splitting inside a test statistic, is the classic two-proportion mistake.

9.5.1 The sampling distribution of $\widehat{p}_X - \widehat{p}_Y$

Theorem 9.5.1 (Large-sample distribution of the difference)**.** *Let $X \sim \mathrm{Bin}(n, p_X)$ and $Y \sim \mathrm{Bin}(m, p_Y)$ be independent, with $\widehat{p}_X = X/n$ and $\widehat{p}_Y = Y/m$. Then $\mathbb{E}[\widehat{p}_X - \widehat{p}_Y] = p_X - p_Y$ and, by independence,*

$$\mathrm{Var}(\widehat{p}_X - \widehat{p}_Y) = \frac{p_X(1-p_X)}{n} + \frac{p_Y(1-p_Y)}{m}.$$

For large n, m the central limit theorem makes $\widehat{p}_X - \widehat{p}_Y$ approximately normal, so

$$Z = \frac{(\widehat{p}_X - \widehat{p}_Y) - (p_X - p_Y)}{\sqrt{\dfrac{p_X(1-p_X)}{n} + \dfrac{p_Y(1-p_Y)}{m}}} \;\approx\; \mathrm{N}(0,1).$$

Proof. Each $\widehat{p}_X$ is a sample mean of n independent Bernoulli(p_X) indicators, so $\mathbb{E}[\widehat{p}_X] = p_X$ and $\mathrm{Var}(\widehat{p}_X) = p_X(1-p_X)/n$; likewise for $\widehat{p}_Y$. Expectation is linear, so $\mathbb{E}[\widehat{p}_X - \widehat{p}_Y] = p_X - p_Y$. The two samples are independent, hence $\mathrm{Var}(\widehat{p}_X - \widehat{p}_Y) = \mathrm{Var}(\widehat{p}_X) + \mathrm{Var}(\widehat{p}_Y)$, which is the displayed sum. The CLT applied to each mean (and the fact that a difference of independent approximately normal variables is again approximately normal) gives the limiting standard normal after standardizing. □

Remark. The numerator centers on the true difference $p_X - p_Y$ and the denominator is the true standard deviation. To turn this into a usable statistic we must replace the unknown variances by estimates — and *how* we estimate them is exactly where the test and the interval part ways.

9.5.2 The two-proportion z-test (pooled standard error)

Definition 9.5.2 (Two-proportion z-test)**.** To test $H_0 \colon p_X = p_Y$ from independent samples, form the **pooled** success rate

$$\widehat{p} = \frac{X+Y}{n+m},$$

and the test statistic

$$z_{\mathrm{obs}} = \frac{\widehat{p}_X - \widehat{p}_Y}{\sqrt{\widehat{p}(1-\widehat{p})\left(\dfrac{1}{n} + \dfrac{1}{m}\right)}}.$$

Under H_0, $z_{\mathrm{obs}} \approx \mathrm{N}(0,1)$. Reject at level α when $z_{\mathrm{obs}} \geq z_\alpha$ for $H_1 \colon p_X > p_Y$; when $z_{\mathrm{obs}} \leq -z_\alpha$ for $H_1 \colon p_X < p_Y$; and when $|z_{\mathrm{obs}}| \geq z_{\alpha/2}$ for $H_1 \colon p_X \neq p_Y$.

Why pool under the null. If H_0 holds then $p_X = p_Y = p$ for one common value p, and the standard deviation of $\widehat{p}_X - \widehat{p}_Y$ collapses to $\sqrt{p(1-p)(1/n + 1/m)}$. The best estimate of that single p ignores the artificial split between the two groups and uses every trial: of the $n+m$ trials, $X+Y$ were successes, so $\widehat{p} = (X+Y)/(n+m)$ is the pooled estimate. Substituting it into the common-p standard deviation gives the denominator above. Using the separate rates here would estimate a variance the null says does not exist. □

Remark (Validity conditions). The normal approximation is trustworthy when each group has at least about 10 successes and 10 failures: $n\widehat{p}_X \geq 10$, $n(1-\widehat{p}_X) \geq 10$, $m\widehat{p}_Y \geq 10$, $m(1-\widehat{p}_Y) \geq 10$ (some texts check the pooled $\widehat{p}$ instead). If counts are too small, exact methods (Fisher's exact test) replace the z. The two samples must also be **independent**; matched or paired successes call for a different (McNemar-type) analysis, not this test.

Example 9.5.3 (Two response rates, two-sided)**.** A charity tests two fundraising emails. Of $n = 520$ recipients of version A, $X = 143$ donate; of $m = 480$ recipients of version B, $Y = 96$ donate. At $\alpha = 0.05$, do the two donation rates *differ*?

Which rule, and why? Two independent count samples and the question is whether the rates are equal, so this is a **two-proportion z-test** of $H_0 \colon p_X = p_Y$. "Differ" makes it two-sided, and because it is a test we **pool** the success rate in the standard error.

Solution.

Step 1: sample proportions. $\widehat{p}_X = 143/520 = 0.275$ and $\widehat{p}_Y = 96/480 = 0.200$.

Step 2: pooled rate.

$$\widehat{p} = \frac{X+Y}{n+m} = \frac{143+96}{520+480} = \frac{239}{1000} = 0.239.$$

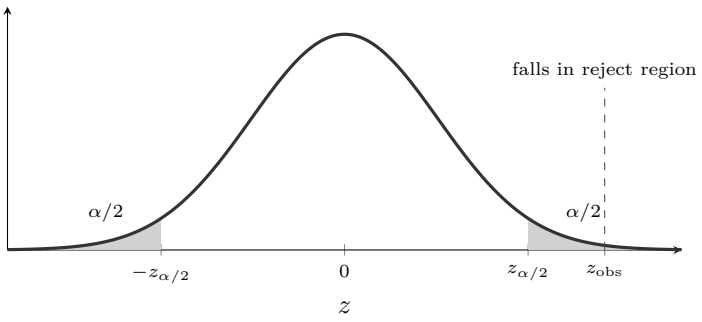
Step 3: pooled (null) SE.

$$\begin{aligned}\mathrm{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n}+\frac{1}{m}\right)} \\ &= \sqrt{0.239 \cdot 0.761\left(\frac{1}{520}+\frac{1}{480}\right)} \\ &= \sqrt{0.181879 \cdot 0.0040064} = \sqrt{0.00072870} = 0.026994.\end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.275 - 0.200)/0.026994 = 0.075/0.026994 = 2.778$.

Step 5: compare. Two-sided cutoff $z_{0.025} = 1.96$; since $|2.778| \geq 1.96$ we reject.

$$\boxed{z_{\text{obs}} = 2.778 > 1.96 \;\Rightarrow\; \text{reject } H_0\text{: the donation rates differ.}}$$



Two-sided rejection region for the two-proportion z. Each shaded tail carries area $\alpha/2$; the observed $z_{obs} = 2.78$ lands inside the right tail, so H_0 is rejected.

Example 9.5.4 (A new treatment cures more, one-sided)**.** In a trial, a standard antibiotic clears infection in $X = 168$ of $n = 240$ patients, while a new agent clears $Y = 152$ of $m = 200$. At $\alpha = 0.05$, is the new agent's cure rate *higher*?

Which rule, and why? Two independent samples; "higher" is a directional claim, so this is a **one-sided** two-proportion z-test. Let group X be the new agent and group Y the standard, so $H_1 : p_X > p_Y$; pool for the SE because it is a test.

Solution.

Step 1: relabel and estimate. New agent $\widehat{p}_X = 152/200 = 0.760$; standard $\widehat{p}_Y = 168/240 = 0.700$.

Step 2: pooled rate.

$$\widehat{p} = \frac{152+168}{200+240} = \frac{320}{440} = 0.727273.$$

Step 3: pooled SE.

$$\begin{aligned}\mathrm{SE}_0 &= \sqrt{0.727273 \cdot 0.272727\left(\frac{1}{200}+\frac{1}{240}\right)} \\ &= \sqrt{0.198347 \cdot 0.0091667} = \sqrt{0.00181818} = 0.042640.\end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.760 - 0.700)/0.042640 = 0.060/0.042640 = 1.407$.

Step 5: compare. Right-sided cutoff $z_{0.05} = 1.645$; since $1.407 < 1.645$ we do *not* reject.

$$\boxed{z_{\text{obs}} = 1.407 < 1.645 \;\Rightarrow\; \text{fail to reject: no significant edge for the new agent.}}$$

A 6-point lift is encouraging but, at these sample sizes, within sampling noise.

Example 9.5.5 (A/B test for a checkout button, one-sided)**.** An online store splits traffic between an old checkout button (control) and a redesigned one (variant). The control converts $X = 420$ of $n = 6000$ sessions; the variant converts $Y = 486$ of $m = 6000$. At $\alpha = 0.05$, did the redesign *raise* the conversion rate?

Which rule, and why? Classic A/B test: two independent streams of yes/no outcomes, asking whether the variant beats the control. That is a **one-sided** two-proportion z-test, $H_1 \colon p_X > p_Y$ with X the variant; pool for the test SE.

Solution.

Step 1: label and estimate. Variant $\widehat{p}_X = 486/6000 = 0.081$; control $\widehat{p}_Y = 420/6000 = 0.070$.

Step 2: pooled rate.

$$\widehat{p} = \frac{486 + 420}{6000 + 6000} = \frac{906}{12000} = 0.0755.$$

Step 3: pooled SE. Here $1/n + 1/m = 2/6000 = 0.00033333$, so

$$\begin{aligned} \text{SE}_0 &= \sqrt{0.0755 \cdot 0.9245 \cdot 0.00033333} \\ &= \sqrt{0.069800 \cdot 0.00033333} = \sqrt{0.00002327} = 0.0048234. \end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.081 - 0.070)/0.0048234 = 0.011/0.0048234 = 2.281$.

Step 5: compare. Right-sided cutoff $z_{0.05} = 1.645$; since $2.281 \geq 1.645$ we reject. (For reference $\mathbb{P}(Z \geq 2.281) = 0.0113$, the one-sided p-value.)

$$\boxed{z_{\text{obs}} = 2.281 \geq 1.645 \;\Rightarrow\; \text{reject } H_0\colon \text{ the redesign lifts conversion.}}$$

9.5.3 Confidence interval for $p_X - p_Y$ (unpooled standard error)

Definition 9.5.6 (Confidence interval for a difference of proportions)**.** A $100(1-\alpha)\%$ confidence interval for $p_X - p_Y$ is

$$(\widehat{p}_X - \widehat{p}_Y) \;\pm\; z_{\alpha/2} \underbrace{\sqrt{\frac{\widehat{p}_X(1-\widehat{p}_X)}{n} + \frac{\widehat{p}_Y(1-\widehat{p}_Y)}{m}}}_{\text{SE}_{\text{u}}}.$$

The standard error is **unpooled**: each group contributes its own estimated variance, because outside of H_0 there is no reason to believe the rates are equal.

Remark. Why no pooling here? A confidence interval asks "what differences are plausible?" without assuming any particular value of $p_X - p_Y$. There is no common rate to estimate, so each group keeps its own $\widehat{p}_X(1-\widehat{p}_X)/n$. Pooling would import the very assumption ($p_X = p_Y$) that the interval is trying to remain agnostic about. The test pools; the interval does not. There is also a clean consistency check: if 0 lies outside the two-sided CI, the corresponding two-sided test rejects — although because the two procedures use slightly different standard errors, the agreement is only approximate near the boundary.

Example 9.5.7 (Interval for the difference, unpooled)**.** Return to the charity emails: $\widehat{p}_X = 0.275$ from $n = 520$ (version A) and $\widehat{p}_Y = 0.200$ from $m = 480$ (version B). Build a 95% confidence interval for $p_X - p_Y$.

Which rule, and why? We want an **interval** for the difference, not a test, so we use the **unpooled** standard error — each group's own $\widehat{p}(1-\widehat{p})/(\text{size})$ — with the two-sided multiplier $z_{0.025} = 1.96$.

Solution.

Step 1: point estimate. $\widehat{p}_X - \widehat{p}_Y = 0.275 - 0.200 = 0.075$.

Step 2: unpooled SE.

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{\widehat{p}_X(1-\widehat{p}_X)}{n} + \frac{\widehat{p}_Y(1-\widehat{p}_Y)}{m}} \\ &= \sqrt{\frac{0.275 \cdot 0.725}{520} + \frac{0.200 \cdot 0.800}{480}} \\ &= \sqrt{0.00038341 + 0.00033333} = \sqrt{0.00071674} = 0.026772.\end{aligned}$$

Step 3: margin and interval. Margin $= 1.96 \cdot 0.026772 = 0.052474$, so

$$0.075 \pm 0.052474 = (0.022526,\ 0.127474).$$

Step 4: read it. The interval excludes 0, so version A's rate is significantly higher — consistent with the two-sided rejection found earlier.

$$\boxed{p_X - p_Y \in (0.0225,\ 0.1275) \quad (95\%),\ \text{excludes } 0.}$$

Example 9.5.8 (Polling gap with a margin of error)**.** Two states are polled on a ballot measure. In state X, $X = 612$ of $n = 1100$ support it; in state Y, $Y = 486$ of $m = 950$ support it. Give a 95% confidence interval for the difference in support $p_X - p_Y$, and report the margin of error.

Which rule, and why? An **interval** for a difference of proportions from two independent polls, so the **unpooled** SE with $z_{0.025} = 1.96$. (The "margin of error" is just $z_{0.025} \cdot \mathrm{SE_u}$.)

Solution.

Step 1: proportions. $\widehat{p}_X = 612/1100 = 0.556364$ and $\widehat{p}_Y = 486/950 = 0.511579$, so $\widehat{p}_X - \widehat{p}_Y = 0.044785$.

Step 2: unpooled SE.

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{0.556364 \cdot 0.443636}{1100} + \frac{0.511579 \cdot 0.488421}{950}} \\ &= \sqrt{\frac{0.246826}{1100} + \frac{0.249866}{950}} \\ &= \sqrt{0.00022439 + 0.00026302} = \sqrt{0.00048741} = 0.022077.\end{aligned}$$

Step 3: margin and interval. Margin $= 1.96 \cdot 0.022077 = 0.043271$, so

$$0.044785 \pm 0.043271 = (0.001514,\ 0.088056).$$

Step 4: read it. The interval lies just above 0, so the gap is significant at 5% but barely — the lower limit is a hair above zero.

$$\boxed{p_X - p_Y \in (0.0015,\ 0.0881) \quad (95\%),\ \text{margin } = 0.0433.}$$

9.5.4 Contrast: pooled SE versus unpooled SE on the same data

Example 9.5.9 (Compute both standard errors and see them disagree)**.** Use the antibiotic trial once more: new agent $\widehat{p}_X = 152/200 = 0.760$, standard $\widehat{p}_Y = 168/240 = 0.700$. Compute the *pooled* (test) standard error and the *unpooled* (interval) standard error side by side, and show what each is for.

Which rule, and why? This is the deliberate **contrast** of the section. The same two counts feed two different standard errors: pooling is correct *inside the test statistic* (it assumes $H_0\colon p_X = p_Y$), while the unpooled form is correct *inside a confidence interval* (it assumes nothing). We compute both to see how, and why, they differ.

Solution.

Step 1: pooled (test) SE. With $\widehat{p} = 320/440 = 0.727273$,

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n}+\frac{1}{m}\right)} \\ &= \sqrt{0.727273 \cdot 0.272727 \cdot 0.0091667} = 0.042640. \end{aligned}$$

Step 2: unpooled (interval) SE.

$$\begin{aligned} \mathrm{SE}_{\mathrm{u}} &= \sqrt{\frac{0.760 \cdot 0.240}{200} + \frac{0.700 \cdot 0.300}{240}} \\ &= \sqrt{0.000912 + 0.000875} = \sqrt{0.001787} = 0.042273. \end{aligned}$$

Step 3: contrast the two. They are close but *not equal*:

$$\begin{aligned} \mathrm{SE}_0 &= 0.042640 \quad \text{(pooled, for the } z\text{-test)}, \\ \mathrm{SE}_{\mathrm{u}} &= 0.042273 \quad \text{(unpooled, for the CI)}. \end{aligned}$$

The pooled SE imposes a single common rate; the unpooled keeps the groups apart. Here the gap is small because $\widehat{p}_X$ and $\widehat{p}_Y$ are not far apart and the sizes are similar, but when the two rates diverge — or one rate sits near 0 or 1 — the two standard errors separate, and using the wrong one in the wrong place gives a wrong answer.

$$\boxed{\text{Test: } \mathrm{SE}_0 = 0.04264 \text{ (pool)}. \qquad \text{CI: } \mathrm{SE}_{\mathrm{u}} = 0.04227 \text{ (don't pool)}.}$$

Remark. The lesson in one line: **pool for the test, split for the interval**. The pooled SE belongs in z_{obs} because the null hypothesis hands you a single common p to estimate from all $n+m$ trials; the unpooled SE belongs in the CI because no such common p is assumed. Swapping them is the most common two-proportion error on exams and in practice.

9.5.5 p-values for the two-proportion test

Example 9.5.10 (Two-sided p-value for two scrap rates)**.** Two molding machines are compared. Machine X scraps $X = 48$ of $n = 800$ parts; machine Y scraps $Y = 30$ of $m = 750$. Report the two-sided p-value for $H_0 \colon p_X = p_Y$ and decide at $\alpha = 0.05$.

Which rule, and why? Two independent count samples; "do the scrap rates differ" is two-sided, and the question asks for strength of evidence (the p-value) rather than a fixed verdict. Pool the rate because it is a test.

Solution.

Step 1: proportions. $\widehat{p}_X = 48/800 = 0.060$ and $\widehat{p}_Y = 30/750 = 0.040$.

Step 2: pooled rate. $\widehat{p} = (48+30)/(800+750) = 78/1550 = 0.050323$.

Step 3: pooled SE.

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{0.050323 \cdot 0.949677\left(\frac{1}{800}+\frac{1}{750}\right)} \\ &= \sqrt{0.047791 \cdot 0.0025833} = \sqrt{0.00012347} = 0.011112. \end{aligned}$$

Step 4: statistic. $z_{\mathrm{obs}} = (0.060 - 0.040)/0.011112 = 0.020/0.011112 = 1.800$.

Step 5: two-sided p-value. Using $\mathbb{P}(Z \geq 1.80) = 0.0359$,

$$p\text{-value} = 2 \cdot 0.0359 = 0.0718.$$

Step 6: decide. Since $0.0718 > 0.05$ we fail to reject: the scrap-rate difference is not significant at 5%.

$$\boxed{z_{\mathrm{obs}} = 1.80, \quad p\text{-value} = 2\,\mathbb{P}(Z \geq 1.80) = 0.0718 > 0.05.}$$

Example 9.5.11 (One-sided p-value, vaccine efficacy)**.** In a field study, $X = 18$ of $n = 900$ vaccinated subjects contract the disease, versus $Y = 45$ of $m = 900$ unvaccinated. Letting X be the vaccinated group, test $H_1 : p_X < p_Y$ (the vaccine *lowers* the infection rate) and report the one-sided p-value.

Which rule, and why? Two independent samples; "lowers" is directional, so a **one-sided** two-proportion z-test with $H_1 : p_X < p_Y$. Pool for the test SE; the p-value is the lower-tail area because the alternative points left.

Solution.

Step 1: proportions. $\widehat{p}_X = 18/900 = 0.020$ (vaccinated) and $\widehat{p}_Y = 45/900 = 0.050$ (unvaccinated).

Step 2: pooled rate. $\widehat{p} = (18 + 45)/1800 = 63/1800 = 0.035$.

Step 3: pooled SE. With $1/n + 1/m = 2/900 = 0.0022222$,

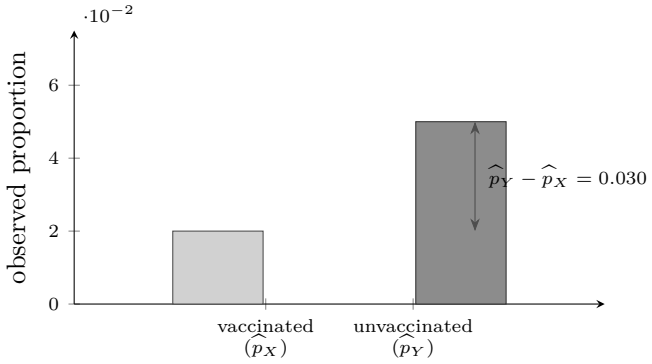
$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{0.035 \cdot 0.965 \cdot 0.0022222} \\ &= \sqrt{0.033775 \cdot 0.0022222} = \sqrt{0.00007506} = 0.0086637. \end{aligned}$$

Step 4: statistic. $z_{\mathrm{obs}} = (0.020 - 0.050)/0.0086637 = -0.030/0.0086637 = -3.463$.

Step 5: one-sided p-value. $\mathbb{P}(Z \le -3.46) = 0.00027$.

Step 6: decide. Far below any usual α, so we reject strongly: the vaccine significantly lowers the infection rate.

$$\boxed{z_{\mathrm{obs}} = -3.46, \quad p\text{-value} = \mathbb{P}(Z \le -3.46) = 0.00027 \;\Rightarrow\; \text{reject } H_0.}$$



Observed infection proportions in the two arms. The vaccinated rate ($\widehat{p}_X = 0.020$) sits well below the unvaccinated rate ($\widehat{p}_Y = 0.050$); the bracket marks the 0.030 gap that the z-test judges against sampling noise.

9.5.6 More worked comparisons

Example 9.5.12 (Two manufacturing lines, two-sided)**.** Two production lines are audited for out-of-spec units. Line X produces $X = 27$ defects in $n = 900$; line Y produces $Y = 14$ in $m = 700$. At $\alpha = 0.05$, do the defect rates differ?

Which rule, and why? Two independent count samples and "differ" is two-sided, so a two-sided **two-proportion z-test** with the **pooled** SE.

Solution.

Step 1: proportions. $\widehat{p}_X = 27/900 = 0.030$ and $\widehat{p}_Y = 14/700 = 0.020$.

Step 2: pooled rate. $\widehat{p} = (27 + 14)/(900 + 700) = 41/1600 = 0.025625$.

Step 3: pooled SE.

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{0.025625 \cdot 0.974375 \left(\frac{1}{900} + \frac{1}{700}\right)} \\ &= \sqrt{0.024968 \cdot 0.0025397} = \sqrt{0.00006341} = 0.0079632. \end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.030 - 0.020)/0.0079632 = 0.010/0.0079632 = 1.256$.

Step 5: compare. Two-sided cutoff $z_{0.025} = 1.96$; since $|1.256| < 1.96$ we fail to reject.

$$\boxed{z_{\text{obs}} = 1.256 < 1.96 \;\Rightarrow\; \text{fail to reject: defect rates not shown to differ.}}$$

Example 9.5.13 (Survival rates across two hospitals, one-sided)**.** A registry compares 30-day survival after a procedure at two centers. Center X records $X = 361$ survivors of $n = 400$; center Y records $Y = 423$ of $m = 480$. At $\alpha = 0.05$, is center X's survival rate *higher*?

Which rule, and why? Independent counts, directional claim ("higher"), so a **one-sided** two-proportion z-test, $H_1 : p_X > p_Y$; pool for the test SE.

Solution.

Step 1: proportions. $\widehat{p}_X = 361/400 = 0.9025$ and $\widehat{p}_Y = 423/480 = 0.88125$.

Step 2: pooled rate. $\widehat{p} = (361 + 423)/(400 + 480) = 784/880 = 0.890909$.

Step 3: pooled SE.

$$\begin{aligned}\text{SE}_0 &= \sqrt{0.890909 \cdot 0.109091 \left(\frac{1}{400} + \frac{1}{480}\right)} \\ &= \sqrt{0.097190 \cdot 0.0045833} = \sqrt{0.00044546} = 0.021106.\end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.9025 - 0.88125)/0.021106 = 0.02125/0.021106 = 1.007$.

Step 5: compare. Right-sided cutoff $z_{0.05} = 1.645$; since $1.007 < 1.645$ we do not reject.

$$\boxed{z_{\text{obs}} = 1.007 < 1.645 \;\Rightarrow\; \text{fail to reject: no significant survival edge.}}$$

Example 9.5.14 (Click-through with a stringent level, two-sided)**.** A publisher tests two ad placements. Placement X is clicked $X = 240$ times in $n = 4000$ views; placement Y is clicked $Y = 189$ times in $m = 3500$ views. At the stringent $\alpha = 0.01$, do the click rates differ?

Which rule, and why? Two independent count samples, "differ" is two-sided, so a two-sided **two-proportion z-test** with the pooled SE; the only change from earlier two-sided examples is the cutoff $z_{0.005} = 2.576$ at the 1% level.

Solution.

Step 1: proportions. $\widehat{p}_X = 240/4000 = 0.060$ and $\widehat{p}_Y = 189/3500 = 0.054$.

Step 2: pooled rate. $\widehat{p} = (240 + 189)/(4000 + 3500) = 429/7500 = 0.0572$.

Step 3: pooled SE.

$$\begin{aligned}\text{SE}_0 &= \sqrt{0.0572 \cdot 0.9428 \left(\frac{1}{4000} + \frac{1}{3500}\right)} \\ &= \sqrt{0.053928 \cdot 0.00053571} = \sqrt{0.00002889} = 0.0053751.\end{aligned}$$

Step 4: statistic. $z_{\text{obs}} = (0.060 - 0.054)/0.0053751 = 0.006/0.0053751 = 1.116$.

Step 5: compare. Two-sided 1% cutoff $z_{0.005} = 2.576$; since $|1.116| < 2.576$ we fail to reject.

$$\boxed{z_{\text{obs}} = 1.116 < 2.576 \;\Rightarrow\; \text{fail to reject at 1\%: rates not shown to differ.}}$$

Example 9.5.15 (Interval that straddles zero)**.** For the click-through data above ($\widehat{p}_X = 0.060$, $n = 4000$; $\widehat{p}_Y = 0.054$, $m = 3500$), give a 99% confidence interval for $p_X - p_Y$ and reconcile it with the test verdict.

Which rule, and why? An **interval** for the difference, so switch to the **unpooled** SE; the 99% level uses $z_{0.005} = 2.576$.

Solution.

Step 1: point estimate. $\widehat{p}_X - \widehat{p}_Y = 0.060 - 0.054 = 0.006$.

Step 2: unpooled SE.

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{0.060 \cdot 0.940}{4000} + \frac{0.054 \cdot 0.946}{3500}} \\ &= \sqrt{0.0000141 + 0.0000146} = \sqrt{0.0000287} = 0.0053572.\end{aligned}$$

Step 3: margin and interval. Margin $= 2.576 \cdot 0.0053572 = 0.013800$, so

$$0.006 \pm 0.013800 = (-0.007800,\ 0.019800).$$

Step 4: reconcile. The interval *contains* 0, matching the earlier failure to reject at 1%: a difference of 0 is plausible.

$$\boxed{p_X - p_Y \in (-0.0078,\ 0.0198) \quad (99\%),\ \text{contains } 0.}$$

Remark (Putting it together). Across these examples the recipe never changed: identify two *independent* count samples, decide whether the question is a *test* (pool) or an *interval* (don't pool), choose one- or two-sided, then standardize. The arithmetic is routine; the one judgment that separates a right answer from a wrong one is the pooled-versus-unpooled choice of standard error.

9.6 Choosing Among Two-Sample Procedures

Each earlier section drilled one procedure in isolation, where the method was announced in the heading and the only task was to compute. Real problems do not arrive labeled. This closing section is about the step that comes *before* any arithmetic: reading a worded scenario and **recognizing** which of the chapter's five tools it calls for, and being able to say *why* that one and not its look-alike. The decision is a short cascade of questions, and they must be asked in order. First, are the two columns of numbers two measurements on the *same* units (paired) or two *separate* groups (independent)? That single fork routes paired data to the one-sample t on differences and everything else onward. If the samples are independent, ask *what* is being compared — means, variances, or proportions — and, for means, whether the two population variances may be assumed equal (pooled t) or not (Welch's t). The costs of getting the fork wrong are not symmetric and are worth memorizing: treating paired data as independent throws away the pairing and *loses power* (you can fail to detect a real effect), while pooling under unequal variances corrupts the *level* (your stated 5% test may reject far more or far less than 5% of the time). Recognition first, computation second.

Which two-sample procedure does this problem call for?

- **Step 1 — paired or independent?** Two numbers on the *same* unit (before/after, left/right, twin/twin, two methods on one specimen)? ⇒ **paired** t on the differences D_i against 0 (§3). Two *separate* groups of unrelated units? ⇒ go to Step 2.
- **Step 2 — what is compared?** Two **means**? ⇒ Step 3. Two **variances** / spreads? ⇒ the F**-test** on S_X^2/S_Y^2 (§4). Two **proportions** from counts of successes? ⇒ the **two-proportion** z**-test** (§5).
- **Step 3 — equal variances assumable?** Yes (design says so, or the sample SDs are close, or an F-test does not reject)? ⇒ **pooled two-sample** t with s_p and $df = n + m - 2$ (§1). No, or in doubt? ⇒ **Welch's** t with the approximate df (§2).

The two costly mistakes. (i) *Paired data run as independent*: the pairing that cancels unit-to-unit variation is discarded, the standard error swells, and a real effect can hide — a *power* loss. (ii) *Pooling under unequal variances*: s_p is a meaningless average of two different σ's, the reference t_{n+m-2} is wrong, and the true Type I error drifts off the nominal α — a *level* error. When unsure between pooled and Welch, Welch is the safe default; when unsure between paired and independent, look at how the data were *collected*, not at the numbers.

9.6.1 The decision cascade

Remark. The five procedures of this chapter are not interchangeable hammers; each answers a different question about a different parameter, and three separate "axes" decide among them. The first axis is the **data-collection structure** (paired vs independent) and is settled by reading the *design*, never the data — if each row of a table corresponds to one unit measured twice, it is paired no matter what the numbers look like. The second axis is the **target parameter** (a difference of means, a ratio of variances, a difference of proportions), settled by reading what the question *asks*. The third axis, relevant only once you are comparing two means of independent samples, is the **variance assumption** (equal ⇒ pooled, unequal ⇒ Welch), settled by the design, the sample SDs, or a preliminary F-test. The flowchart below is just these three axes drawn as a tree.

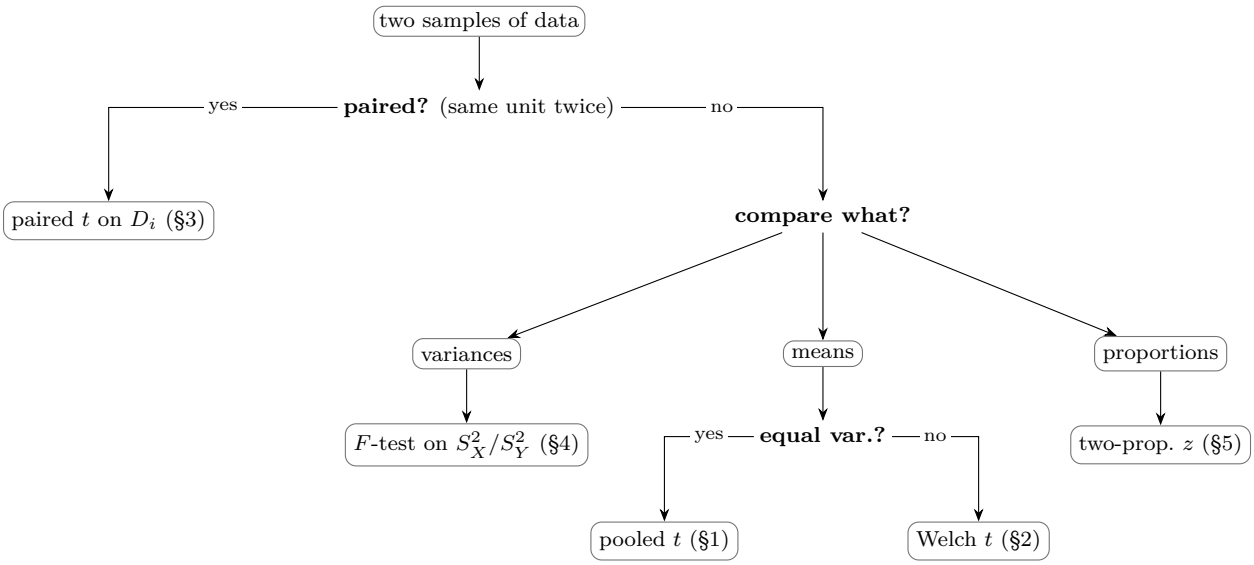


Figure 9.3: The two-sample decision cascade. Read the *design* to answer "paired?"; read the *question* to answer "compare what?"; read the design, the sample SDs, or an F-test to answer "equal variances?". Every leaf is one procedure from this chapter.

9.6.2 A reference summary

Scenario (recognition cue)	Procedure	Section
Same unit measured twice; before/after, matched pairs	paired t on D_i vs 0	§3
Two independent groups, means, variances assumed/found equal	pooled two-sample t, $df = n + m - 2$	§1
Two independent groups, means, variances unequal or unknown	Welch's t, approximate df	§2
Two independent groups, compare spreads/variances	F-test on S_X^2/S_Y^2	§4
Two independent groups, counts of successes, compare rates	two-proportion z-test	§5
Pooled vs Welch undecided	F-test first, then pooled or Welch	§4 → §1/§2

Table 9.1: Scenario → procedure → reference section. Match the recognition cue in the left column to the tool in the middle; the right column points back to the section that derives it.

9.6.3 The headline contrast: pairing changes everything

The most consequential recognition decision is the first one. The next two examples use the *same* numbers and the *same* question — "did the score rise by 4 points on average?" — but one design is paired and the other independent, and they route to different tests with different verdicts.

Example 9.6.1 (CONTRAST, part A: paired design)**.** A reading program is tested on $n = 10$ students, each measured *before* and *after* the program. The per-student gains (after − before, in points) are

$$6,\ 2,\ 5,\ 3,\ 7,\ 1,\ 4,\ 6,\ 3,\ 3,$$

with $\overline{D} = 4.0$ and $S_D = 1.94$. Test at $\alpha = 0.05$ whether the program raised the mean score. Use $t_{0.05,9} = 1.833$.

Which rule, and why? Each student supplies a before *and* an after, so the two columns are two measurements on the *same* unit — paired data. Collapse to the gains D_i and run a one-sample upper-tail t-test on them against 0, $df = n - 1 = 9$ (§3). The pairing cancels the huge student-to-student spread in reading ability.

Solution.

Step 1: recognize and set up. Paired; $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$ on the gains.

Step 2: standard error of $\overline{D}$. $S_D/\sqrt{n} = 1.94/\sqrt{10} = 1.94/3.162 = 0.6135$.

Step 3: statistic.

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{4.0}{0.6135} = 6.52.$$

Step 4: decision. Upper-tail cutoff $t_{0.05,9} = 1.833$. Since $6.52 > 1.833$, reject overwhelmingly.

$$\boxed{t_{\text{obs}} = 6.52 > 1.833 \ \Rightarrow\ \text{reject } H_0;\ \text{the program raised mean score (paired } t).}$$

Example 9.6.2 (CONTRAST, part B: independent design, same numbers)**.** Now suppose instead that *two separate groups* were used: $n = 10$ students took a control program and scored a mean of X, while a *different* $m = 10$ students took the reading program and scored a mean 4.0 points higher. To make the contrast sharp, suppose the two groups have the same within-group spread as the paired study's raw scores, $S_X = S_Y = 6.13$, so $\overline{X} - \overline{Y} = 4.0$ with those SDs. Test the same 4-point difference at $\alpha = 0.05$. Use $t_{0.05,18} = 1.734$.

Which rule, and why? Different students in each arm means the samples are *independent*, not paired — so the paired t is unavailable and we need a two-sample t (pooled here, since the spreads are equal), $df = n+m-2 = 18$ (§1). The key point: the same 4-point gap is now measured against the *full* between-student variability, which the paired design had cancelled.

Solution.

Step 1: recognize and set up. Independent groups, equal SDs ⇒ pooled two-sample t; $H_0 : \mu_X = \mu_Y$ versus the one-sided alternative.

Step 2: pooled SD and standard error. With $S_X = S_Y = 6.13$, $s_p = 6.13$, and

$$\widehat{\text{SE}} = s_p\sqrt{\tfrac{1}{n} + \tfrac{1}{m}} = 6.13\sqrt{\tfrac{1}{10} + \tfrac{1}{10}} = 6.13\sqrt{0.2} = 6.13 \cdot 0.4472 = 2.741.$$

Step 3: statistic.

$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y}}{\widehat{\text{SE}}} = \frac{4.0}{2.741} = 1.46.$$

Step 4: decision. Cutoff $t_{0.05,18} = 1.734$. Since $1.46 < 1.734$, do *not* reject.

$$\boxed{t_{\text{obs}} = 1.46 < 1.734 \ \Rightarrow\ \text{retain } H_0;\ \text{the same 4 points is not significant (independent } t).}$$

Remark (The cost of mis-routing, quantified). Parts A and B carry the same effect (4 points) yet land on opposite verdicts: $t_{\text{obs}} = 6.52$ versus $t_{\text{obs}} = 1.46$, a more than fourfold difference in the statistic. The reason is entirely in the standard error: pairing replaced the between-student SD (≈ 6.13) with the much smaller SD of *gains* (1.94), shrinking the denominator by a factor of about 3. If you *had* paired data but analyzed them as independent (the mistake), you would compute the part-B statistic, fail to reject, and miss a real and strong effect — that is exactly the **power loss** from ignoring pairing. Pairing is not a computational nicety; it is the design choice that makes the effect visible.

9.6.4 Classify-then-execute: a battery of scenarios

Example 9.6.3 (Classify: two catalysts, yields)**.** A chemist runs a new catalyst on $n = 8$ independent batches and the standard catalyst on $m = 8$ *different* batches, recording percent yield, and asks whether the mean yields differ. The two sample SDs come out close: $S_X = 2.1$, $S_Y = 2.3$. Name the procedure.

Which rule, and why? Different batches in each arm ⇒ independent samples; the question is about *means*; the sample SDs are nearly equal (2.1 vs 2.3), so equal population variances are plausible and pooling is justified. That is the pooled two-sample t (§1), $df = 8 + 8 - 2 = 14$, two-sided.

Solution.

Step 1: structure. Separate batches ⇒ independent.

Step 2: target and variance check. Means; $S_X/S_Y = 2.1/2.3 = 0.91$, so the ratio $S_X^2/S_Y^2 = 0.83$ is near 1 — equal variances are reasonable.

Pooled two-sample t-test, $df = 14$, two-sided (§1).

Example 9.6.4 (Classify: left vs right hand grip)**.** A physiologist measures grip strength in both the left and right hand of each of 15 volunteers and asks whether the dominant hand is stronger. Name the procedure.

Which rule, and why? Both hands of *one* person are two measurements on the same unit, so the data are paired (left and right travel together within a volunteer). Reduce to $D_i = \text{right} - \text{left}$ and run a one-sample t on the differences (§3), upper-tail for "stronger," $df = 14$. Treating the 30 measurements as two independent samples of 15 would be the classic pairing error.

Solution.

Step 1: structure. Two hands per person ⇒ paired.

Step 2: target. A mean difference; "dominant stronger" is one-sided.

Paired t-test on D_i, upper-tail, $df = 14$ (§3).

Example 9.6.5 (Classify: defect rates on two lines)**.** A plant inspects 200 items from line A and finds 14 defective, and 250 items from line B and finds 11 defective, asking whether the two defect *rates* differ. Name the procedure.

Which rule, and why? The data are *counts of successes* (defectives) out of fixed sample sizes from two independent lines — a comparison of two *proportions*, not two means. That is the two-proportion z-test (§5), which uses the pooled success rate in the standard error under H_0. "Differ" is two-sided.

Solution.

Step 1: structure and target. Two independent samples; the response is success/failure, so the parameter is a difference of proportions $p_A - p_B$.

Step 2: name it. Counts ⇒ proportions ⇒ z, not t.

Two-proportion z-test, two-sided (§5).

Example 9.6.6 (Classify: which is more consistent?)**.** A QA engineer has weight measurements from two filling machines, $n = 12$ from machine 1 and $m = 15$ from machine 2, and asks not whether the *average* weights differ but whether one machine is *more consistent* (less variable) than the other. Name the procedure.

Which rule, and why? "More consistent / less variable" is a question about *spread*, i.e. the variances, not the means. Two independent samples compared on variance ⇒ the F-test on the ratio S_X^2/S_Y^2 (§4) with $df = (11, 14)$. Reaching for a two-sample t here would answer the wrong question entirely.

Solution.

Step 1: read the target. "Consistent" = small variance; the comparison is of variances.

Step 2: name it. Ratio of two independent sample variances ⇒ F.

$F_{\text{obs}} = S_X^2/S_Y^2 \sim \text{F}_{11,14}$ under H_0; F-test (§4).

Example 9.6.7 (Classify: a proportions-vs-means trap)**.** Two study designs are described. *Design 1:* of 80 patients on drug A, 52 improved; of 90 on drug B, 51 improved — compare the improvement rates. *Design 2:* record the *reduction in blood pressure* (a continuous number) for 80 patients on drug A and 90 on drug B — compare the mean reductions. State the procedure for each and explain why they differ.

Which rule, and why? The fork is whether the response is a *count of a yes/no outcome* or a *measured number*. Design 1's "improved / not" is binary, giving proportions ⇒ two-proportion z (§5). Design 2's millimeters of mercury is continuous, giving means ⇒ a two-sample t (§1 or §2). Same patients, same drugs, but the *type of response* dictates a z versus a t.

Solution.

Step 1: Design 1 response. "Improved" is success/failure ⇒ proportions ⇒ two-proportion z-test.

Step 2: Design 2 response. A continuous BP reduction ⇒ means ⇒ two-sample t (pooled or Welch by the variance check).

Step 3: the lesson. The recognition hinge is the response type, not the subject matter.

Design 1: two-proportion z (§5). Design 2: two-sample t (§1/§2).

9.6.5 Deciding pooled vs Welch from the sample variances

Example 9.6.8 (Pooled vs Welch by an F-screen)**.** Two teaching methods are compared on independent classes: method X has $n = 10$ students with $\overline{X} = 78.0$, $S_X = 4.0$; method Y has $m = 13$ students with $\overline{Y} = 82.0$, $S_Y = 9.0$. Before testing the means, decide whether to pool. Screen the variances with an F-test at the $\alpha = 0.05$ (two-sided, so compare to the upper 2.5% point) level; use $F_{0.025,12,9} = 3.87$. Then say which mean test to run.

Which rule, and why? Whether to pool is exactly the equal-variance question, and the formal screen is the F-test on S_X^2/S_Y^2 (§4). Here the SDs are far apart (4.0 vs 9.0), so we expect the screen to reject equality and send us to Welch's t (§2) rather than the pooled t.

Solution.

Step 1: form the F ratio (larger variance on top). $S_Y^2 = 81$, $S_X^2 = 16$, so

$$F_{\text{obs}} = \frac{S_Y^2}{S_X^2} = \frac{81}{16} = 5.06, \qquad df = (m-1,\, n-1) = (12,\, 9).$$

Step 2: compare to the cutoff. $F_{0.025,12,9} = 3.87$. Since $5.06 > 3.87$, reject equal variances at $\alpha = 0.05$.

Step 3: route the mean test. Equal variances is rejected, so pooling is *not* justified — use **Welch's** t with the approximate (Satterthwaite) df, not the pooled t_{21}.

$F_{\text{obs}} = 5.06 > 3.87 \;\Rightarrow$ variances unequal; use Welch's t (§2), not pooled.

Example 9.6.9 (Pooling is justified: close SDs)**.** A nutritionist compares fiber intake in two independent samples: $n = 20$ urban adults with $\overline{X} = 24.5$ g, $S_X = 5.0$ g, and $m = 18$ rural adults with $\overline{Y} = 21.0$ g, $S_Y = 5.6$ g. Decide whether to pool, then test whether mean intakes differ at $\alpha = 0.05$. Use $F_{0.025,17,19} = 2.62$ for the screen and $t_{0.025,36} = 2.028$ for the mean test.

Which rule, and why? Same equal-variance question as before, but now the SDs are close (5.0 vs 5.6), so we expect the F-screen *not* to reject and pooling to be legitimate ⇒ the pooled two-sample t (§1) with $df = n + m - 2 = 36$, two-sided for "differ."

Solution.

Step 1: F-screen. $S_Y^2 = 31.36$, $S_X^2 = 25.0$, larger on top:

$$F_{\text{obs}} = \frac{31.36}{25.0} = 1.25, \qquad df = (17,\, 19), \qquad F_{0.025,17,19} = 2.62.$$

Since $1.25 < 2.62$, do not reject equal variances — pooling is justified.

Step 2: pooled SD.

$$s_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{19 \cdot 25.0 + 17 \cdot 31.36}{36} = \frac{475 + 533.1}{36} = \frac{1008.1}{36} = 28.00,$$
$$s_p = \sqrt{28.00} = 5.292.$$

Step 3: statistic. Standard error $s_p\sqrt{1/n + 1/m} = 5.292\sqrt{1/20 + 1/18} = 5.292\sqrt{0.1056} = 5.292 \cdot 0.3249 = 1.719$,

$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y}}{1.719} = \frac{24.5 - 21.0}{1.719} = \frac{3.5}{1.719} = 2.036.$$

Step 4: decision. $|t_{\text{obs}}| = 2.036 > t_{0.025,36} = 2.028$ (just barely), so reject.

Pool ($F_{\text{obs}} = 1.25 < 2.62$); then $\lvert t_{\text{obs}}\rvert = 2.036 > 2.028 \Rightarrow$ means differ.

Example 9.6.10 (CONTRAST: pooling under unequal variances mis-levels the test)**.** Reusing the teaching-method data ($n = 10$, $S_X = 4.0$; $m = 13$, $S_Y = 9.0$; $\overline{X} - \overline{Y} = -4.0$), an analyst ignores the F-screen and runs the *pooled* t anyway. Show the pooled SD it produces and explain why the resulting test does not have the advertised 5% level.

Which rule, and why? This is the deliberate look-alike error: the variances were already shown unequal, so the pooled t is the *wrong* procedure. Computing s_p here exposes how pooling blends two genuinely different spreads into one number that describes neither sample, which is why the reference distribution t_{21} no longer holds.

Solution.

Step 1: the pooled SD it would use.

$$s_p^2 = \frac{9 \cdot 16 + 12 \cdot 81}{10 + 13 - 2} = \frac{144 + 972}{21} = \frac{1116}{21} = 53.14,$$
$$s_p = \sqrt{53.14} = 7.29.$$

Step 2: why this is wrong. The pooled 7.29 is a weighted average of $S_X = 4.0$ and $S_Y = 9.0$. It *overstates* the spread of the small-SD method and *understates* the large-SD one; neither sample is described by 7.29. The derivation of the pooled t assumed $\sigma_X = \sigma_Y$ precisely so that one number could stand for both — with that assumption false, the statistic does not follow t_{21}.

Step 3: the consequence (level error). Because the unequal sample sizes pair with unequal variances (the smaller sample has the smaller variance), the pooled SE is biased; the true rejection probability under H_0 is *not* the nominal 5% — it can run several points high or low. The fix is Welch's t, which never pools.

$s_p = 7.29$ describes neither sample; pooled t here has the wrong level — use Welch.

9.6.6 Two more recognition items, executed

Example 9.6.11 (Recognize and run: paired thickness gauges)**.** Two thickness gauges are compared by measuring the *same* $n = 6$ steel sheets with each gauge (microns). The per-sheet differences (gauge 1 − gauge 2) are

$$0.3,\ -0.1,\ 0.4,\ 0.2,\ 0.0,\ 0.2,$$

with $\overline{D} = 0.1667$ and $S_D = 0.1862$. Test at $\alpha = 0.05$ whether the gauges disagree on average. Use $t_{0.025,5} = 2.571$.

Which rule, and why? Both gauges measure the *same* sheet, so the readings are paired sheet by sheet — pairing removes the large sheet-to-sheet thickness variation. One-sample two-sided t on the differences against 0 (§3), $df = 5$. Two independent samples of 6 would be wrong because the two columns are linked through the shared sheets.

Solution.

Step 1: recognize. Same sheets, two gauges ⇒ paired; $H_0 : \mu_D = 0$ vs $H_1 : \mu_D \neq 0$.

Step 2: standard error. $S_D/\sqrt{n} = 0.1862/\sqrt{6} = 0.1862/2.449 = 0.07602$.

Step 3: statistic.

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{0.1667}{0.07602} = 2.193.$$

Step 4: decision. $|t_{\text{obs}}| = 2.193 < t_{0.025,5} = 2.571$, so do not reject.

$$\boxed{|t_{\text{obs}}| = 2.193 < 2.571 \;\Rightarrow\; \text{retain } H_0;\ \text{no evidence the gauges disagree (paired } t).}$$

Example 9.6.12 (Recognize and run: two vaccines, seroconversion (one-sided))**.** In a trial, vaccine A produces seroconversion in $x = 171$ of $n = 200$ recipients and vaccine B in $y = 144$ of $m = 180$ recipients. Test at $\alpha = 0.05$ whether A has the *higher* seroconversion rate. Use $z_{0.05} = 1.645$.

Which rule, and why? Two independent groups with a yes/no outcome (seroconverted or not) ⇒ compare two *proportions*; "A higher" is one-sided. The two-proportion z-test (§5) uses the *pooled* success rate $\hat{p}$ in the standard error under H_0, because the null asserts a common rate.

Solution.

Step 1: sample proportions. $\hat{p}_A = 171/200 = 0.855$, $\hat{p}_B = 144/180 = 0.800$, so $\hat{p}_A - \hat{p}_B = 0.055$.

Step 2: pooled rate and standard error under H_0.

$$\hat{p} = \frac{171 + 144}{200 + 180} = \frac{315}{380} = 0.8289,$$

$$\widehat{\text{SE}} = \sqrt{\hat{p}(1-\hat{p})\left(\tfrac{1}{n} + \tfrac{1}{m}\right)} = \sqrt{0.8289 \cdot 0.1711\,(0.005 + 0.005556)}$$

$$= \sqrt{0.14182 \cdot 0.010556} = \sqrt{0.0014971} = 0.03869.$$

Step 3: statistic.

$$z_{\text{obs}} = \frac{\hat{p}_A - \hat{p}_B}{\widehat{\text{SE}}} = \frac{0.055}{0.03869} = 1.422.$$

Step 4: decision. Upper-tail cutoff $z_{0.05} = 1.645$. Since $1.422 < 1.645$, do not reject.

$$\boxed{z_{\text{obs}} = 1.422 < 1.645 \;\Rightarrow\; \text{retain } H_0;\ \text{no significant edge for vaccine A (two-proportion } z).}$$

Remark (A checklist for the recognition step). When a worded problem lands, run the cascade out loud. (1) *How were the data collected?* If each row is one unit measured twice, it is **paired** — stop and use the paired t on differences. (2) If the groups are separate, *what does the response look like?* A yes/no count ⇒ proportions (z); a measured number ⇒ means (t); a question about spread or consistency ⇒ variances (F). (3) For means, *are the spreads comparable?* Close sample SDs (or a non-rejecting F-screen) ⇒ pool; far-apart SDs ⇒ Welch. The two failure modes to fear are reading paired data as independent (you lose power and may miss a real effect) and pooling across unequal variances (you lose control of the level and your α is a fiction). When the cascade leaves you genuinely undecided on the last fork, choose Welch: it costs a little efficiency when the variances are in fact equal and protects the level when they are not.

9.7 Exercises

All problems below are original. Full solutions begin on page 596.

A. Pooled two-sample t

A1. A metallurgist measures the yield strength (MPa) of bolts from two heat-treatment ovens, assuming a common population variance. Oven X: $n = 11$, $S_X = 5.4$. Oven Y: $m = 7$, $S_Y = 6.8$.

(a) Compute the pooled variance S_p^2 and the pooled standard deviation s_p.

(b) State the degrees of freedom that a two-sample t on these data would carry.

(c) Verify that S_p^2 lies between S_X^2 and S_Y^2, and say which sample pulls it harder and why.

A2. A horticulturist compares the mean number of days to first bloom for two tulip cultivars grown in independent plots. Cultivar X: $n = 10$, $\overline{X} = 46.2$ days, $S_X = 3.1$. Cultivar Y: $m = 14$, $\overline{Y} = 43.0$ days, $S_Y = 2.7$. The spreads are similar, so assume a common variance. At $\alpha = 0.05$, test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$. Use $t_{0.025,22} = 2.074$.

A3. Two production lines fill cereal boxes; an engineer wants to know whether line X fills heavier on average. Independent samples of net weight (g) give line X: $n = 16$, $\overline{X} = 512.0$, $S_X = 6.0$; line Y: $m = 13$, $\overline{Y} = 508.5$, $S_Y = 5.4$. Assume equal variances and test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X > \mu_Y$ at $\alpha = 0.05$. Use $t_{0.05,27} = 1.703$.

A4. A pharmacologist compares the clearance time (hours) of a drug under two diets, assuming a common variance. Diet X: $n = 9$, $\overline{X} = 7.40$, $S_X = 1.05$. Diet Y: $m = 9$, $\overline{Y} = 6.10$, $S_Y = 0.95$.

(a) Build a 95% confidence interval for $\mu_X - \mu_Y$. Use $t_{0.025,16} = 2.120$.

(b) State what the interval implies about a two-sided test of $\mu_X = \mu_Y$ at $\alpha = 0.05$, and why the test and interval must agree.

A5. A regulator will certify a new water filter only if it removes *more than* 10 ppb of a contaminant beyond the old filter. Reductions (ppb) are measured on independent batches, equal variances assumed. New filter: $n = 13$, $\overline{X} = 38.5$, $S_X = 4.0$. Old filter: $m = 13$, $\overline{Y} = 26.0$, $S_Y = 3.6$. Test $H_0 : \mu_X - \mu_Y = 10$ against $H_1 : \mu_X - \mu_Y > 10$ at $\alpha = 0.05$. Use $t_{0.05,24} = 1.711$. Then state, in one sentence, how the verdict would change if the hypothesized difference had been 0.

A6. A barista records the extraction yield (%) of espresso from two grinders, sampling shots independently. Grinder X: $\{20.1, 21.4, 19.8, 20.7, 21.0\}$ ($n = 5$). Grinder Y: $\{18.9, 19.6, 18.4, 19.1\}$ ($m = 4$). Assuming a common variance, test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$, working from the raw numbers (compute $\overline{X}, \overline{Y}, S_X^2, S_Y^2$ first). Use $t_{0.025,7} = 2.365$.

A7. A sleep researcher measures sleep latency (minutes) on $n = 8$ subjects *without* a white-noise machine and on *the same* 8 subjects *with* it. The without mean is $\overline{X} = 24.0$ ($S_X = 6.5$) and the with mean is $\overline{Y} = 19.0$ ($S_Y = 6.0$); the within-subject differences (without$-$with) have mean $\overline{D} = 5.0$ with $S_D = 3.2$. A colleague runs a pooled two-sample t with df $= 14$.

(a) Explain why the pooled two-sample t is the wrong procedure here.

(b) Compute the (wrong) pooled t_{obs} and its decision at $\alpha = 0.05$ ($t_{0.025,14} = 2.145$).

(c) Compute the correct paired t_{obs} and its decision at $\alpha = 0.05$ ($t_{0.025,7} = 2.365$), and explain why the two answers differ.

B. Welch's t (unequal variances)

B1. (Two-sided Welch test, df from scratch.) A sports-science lab compares the vertical-jump height (cm) of two training programs. Plyometric group X: $n = 12$, $\overline{X} = 46.0$, $S_X = 3.0$. Resistance group Y: $m = 15$, $\overline{Y} = 43.0$, $S_Y = 8.0$. The resistance group is far more variable, so equal variances are not

assumed. Test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$. Compute the Welch–Satterthwaite degrees of freedom yourself and round them down; use $t_{0.025,18} = 2.101$.

B2. (Welch confidence interval.) A food chemist measures vitamin-C content (mg per 100 g) in oranges from two orchards. Orchard X: $n = 10$, $\overline{X} = 53.0$, $S_X = 4.0$. Orchard Y: $m = 10$, $\overline{Y} = 48.0$, $S_Y = 11.0$.

(a) Build a 95% Welch confidence interval for $\mu_X - \mu_Y$. Show the Welch–Satterthwaite df; use $t_{0.025,11} = 2.201$.

(b) State, from the interval alone, the verdict of the equivalent two-sided test at $\alpha = 0.05$.

B3. (Pooled-vs-Welch contrast on the same data.) Two soldering robots are compared on joint resistance (milliohms). The first robot is checked in a small intensively sampled batch X ($n = 6$, $\overline{X} = 14.0$, $S_X = 6.0$); the second in a large routine batch Y ($m = 18$, $\overline{Y} = 11.0$, $S_Y = 2.0$). The small batch is the noisier one.

(a) Run the *pooled* two-sample t-test of $H_0 : \mu_X = \mu_Y$ vs. $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$, using $t_{0.025,22} = 2.074$.

(b) Run the *Welch* test on the same data, computing the Welch df and using $t_{0.025,5} = 2.571$.

(c) Compare the two standard errors and statistics, and say which procedure should be trusted here and why.

B4. (One-sided Welch test, upper tail.) A pharmacology study asks whether a new formulation X raises mean plasma concentration (ng/mL) over the reference Y. New: $n = 18$, $\overline{X} = 84.0$, $S_X = 20.0$. Reference: $m = 12$, $\overline{Y} = 72.0$, $S_Y = 7.0$. The new formulation is much more variable across subjects, so variances are not pooled. Test at $\alpha = 0.05$ whether $\mu_X > \mu_Y$. Compute the Welch df; use $t_{0.05,21} = 1.721$.

B5. (Unequal sample sizes: small noisy group vs. large quiet one.) A toxicology screen measures an enzyme level (U/L) in a small high-dose cohort X ($n = 6$, $\overline{X} = 88.0$, $S_X = 15.0$) and a large control cohort Y ($m = 50$, $\overline{Y} = 79.0$, $S_Y = 4.0$). Test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$.

(a) Compute the Welch df and show it collapses toward the small cohort's $n - 1$.

(b) Carry out the test using $t_{0.025,5} = 2.571$.

(c) State the df a naive pooler would have claimed ($n + m - 2$) and explain in one sentence why that is illegitimate here.

B6. (Nonzero hypothesized difference, lower tail.) A clinic claims that switching to a new anesthetic shortens mean recovery time (minutes) by *more* than 8 minutes relative to the old one. Old anesthetic X: $n = 14$, $\overline{X} = 52.0$, $S_X = 6.0$. New anesthetic Y: $m = 14$, $\overline{Y} = 41.0$, $S_Y = 13.0$. Test $H_0 : \mu_X - \mu_Y = 8$ against $H_1 : \mu_X - \mu_Y > 8$ at $\alpha = 0.05$ without assuming equal variances. Compute the Welch df; use $t_{0.05,18} = 1.734$.

B7. (Welch test built from raw data.) Two pipettes are checked for delivered volume (microliters). Pipette X: 99.6, 100.2, 99.8, 100.0, 100.4. Pipette Y: 98.0, 101.5, 97.2, 102.8, 99.0, 101.5. Test $H_0 : \mu_X = \mu_Y$ against $H_1 : \mu_X \neq \mu_Y$ at $\alpha = 0.05$ without assuming equal variances. Build both means and both sample variances from the raw numbers, compute the Welch df, and use $t_{0.025,5} = 2.571$.

C. Paired t

C1. A physical-therapy clinic measures grip strength (kg) in $n = 8$ patients before a six-week tendon-rehab program and again after, on the *same* hand of the *same* patient. The before/after pairs (X_i, Y_i) are

$$(24, 27),\ (31, 30),\ (28, 33),\ (19, 24),\ (35, 36),\ (22, 26),\ (27, 31),\ (30, 33).$$

Let $D_i = Y_i - X_i$ (after minus before, so a positive value is a gain). At $\alpha = 0.05$, test whether the program changes mean grip strength. Compute $\overline{D}$ and S_D from the raw pairs, give the statistic and the critical value with its df, and state the decision in words.

C2. A horticulture lab applies a growth hormone to one cutting from each of $n = 10$ parent plants and leaves a second cutting from the *same* parent untreated. After three weeks it records stem length (cm) on the treated cutting (X) and the control cutting (Y) of each parent. The within-parent differences $D_i = X_i - Y_i$ (treated minus control, positive means the hormone helped) are

$$2.1,\ 3.4,\ 0.8,\ 4.0,\ 1.5,\ 2.8,\ 3.1,\ 0.5,\ 2.4,\ 1.9.$$

(a) Construct a 95% confidence interval for the mean treatment effect μ_D.

(b) Does the interval give significant evidence at the 5% level that the hormone lengthens stems? Explain from the interval alone.

C3. A running coach claims a new shoe *lowers* 5 km finish time. Each of $n = 9$ runners runs the course once in the old shoe and once in the new shoe. The improvements $D_i = (\text{old time}_i) - (\text{new time}_i)$, in seconds (positive means the new shoe was faster), are

$$14,\ -6,\ 22,\ 9,\ 31,\ 5,\ 18,\ -2,\ 27.$$

At $\alpha = 0.05$, test the coach's directional claim. Set up the one-sided hypotheses, compute $\overline{D}$, S_D, and t_{obs}, quote the one-tailed critical value with its df, and conclude in words.

C4. The central contrast. Five patients have their fasting blood glucose (mg/dL) measured before and after eight weeks on a low-carb diet:

patient	1	2	3	4	5
before X	142	168	131	155	149
after Y	137	161	126	149	143

Every patient's glucose fell by 5 to 7 mg/dL. Analyze these data *both* ways at $\alpha = 0.05$.

(a) Correctly, as a paired test on $D_i = X_i - Y_i$ (before minus after). Give $\overline{D}$, S_D, t_{obs}, df, the critical value, and the decision.

(b) Incorrectly, as two independent samples (pooled two-sample t on the "before" column versus the "after" column, using s_p). Give $\overline{X}$, $\overline{Y}$, s_p, t_{obs}, df, the critical value, and the decision.

(c) Explain in one or two sentences why the two conclusions disagree even though both estimate the same mean change.

C5. Two radiologists independently score the same $n = 14$ mammograms on a 0–100 malignancy-suspicion scale. The within-image score differences D_i (radiologist 1 minus radiologist 2) have $\overline{D} = 2.5$ points and $S_D = 6.3$ points. Construct a 90% confidence interval for the mean reader difference μ_D, quoting the critical value and its df, and state whether the data give evidence of a systematic difference between the two radiologists.

C6. A supplier asserts that a new lubricant cuts each engine's friction loss by *at least* 3.0 watts. For $n = 12$ engines the savings D_i (old lubricant minus new, so positive is a saving) have $\overline{D} = 3.8$ W and $S_D = 1.6$ W. At $\alpha = 0.05$, test the supplier's specific claim that the mean saving is at least 3.0 W, against the skeptic's alternative that it falls short. State the hypotheses with $\mu_{D,0} = 3.0$, compute t_{obs}, quote the one-tailed critical value and df, and conclude.

C7. A sleep researcher records total sleep (hours) for $n = 6$ volunteers on a night with white noise (X) and a night without it (Y), each volunteer serving as his own control. The pairs are

$$(7.2, 6.8),\ (6.5, 6.6),\ (8.0, 7.4),\ (5.9, 5.5),\ (7.6, 7.1),\ (6.8, 6.4).$$

(a) Form $D_i = X_i - Y_i$ and compute $\overline{D}$ and S_D using the computational form $S_D^2 = \dfrac{\sum D_i^2 - (\sum D_i)^2/n}{n-1}$.

(b) Report the standard error of $\overline{D}$.

(c) At $\alpha = 0.05$, test (two-sided) whether white noise changes mean sleep time; give t_{obs}, the critical value with its df, and the decision in words.

D. The F-test for two variances

D1. A coffee roaster compares two grinders on the consistency of particle size. Grinder X yields $n = 16$ batches with sample variance $S_X^2 = 42\,(\mu\text{m})^2$; grinder Y yields $m = 13$ batches with $S_Y^2 = 15\,(\mu\text{m})^2$. The buyer only cares whether the two grinders are *equally* consistent, with no prior direction in mind. Test $H_0\colon \sigma_X^2 = \sigma_Y^2$ against $H_1\colon \sigma_X^2 \neq \sigma_Y^2$ at $\alpha = 0.05$. Use $F_{0.025,\,15,\,12} = 3.18$.

D2. A materials lab suspects that an old extruder produces fibers with *more* variable tensile strength than a new one. The old line (X) gives $n = 21$ fibers with $S_X^2 = 0.78\,(\text{N})^2$; the new line ($Y$) gives $m = 16$ fibers with $S_Y^2 = 0.30\,(\text{N})^2$.

(a) State the hypotheses for the directional claim and explain which variance must sit in the numerator and why.

(b) Carry out the one-sided F-test at $\alpha = 0.05$, using $F_{0.05,\,20,\,15} = 2.33$.

D3. A pharmacist measures dissolution times for two tablet formulations and reports *standard deviations* rather than variances. Formulation X: $n = 10$ tablets, $S_X = 6.4$ min. Formulation Y: $m = 13$ tablets, $S_Y = 3.8$ min. Do the two formulations differ in variability of dissolution time at $\alpha = 0.10$? Square the standard deviations first. Use $F_{0.05,\,9,\,12} = 2.80$.

D4. A quality engineer plans to compare the mean fill weights of two bottling lines with a two-sample t-test and must first decide between the pooled t and Welch's t. Line X: $n = 11$ bottles, $\overline{X} = 498.2$ g, $S_X^2 = 9.6\,\text{g}^2$. Line Y: $m = 9$ bottles, $\overline{Y} = 501.0$ g, $S_Y^2 = 2.4\,\text{g}^2$.

(a) Run the two-sided F-test for equal variances at $\alpha = 0.05$, using $F_{0.025,\,10,\,8} = 4.30$.

(b) On the basis of part (a), state which t-test should be used for the means, and why.

D5. A trader claims that the monthly returns of a hedged portfolio X are *less* variable than those of an unhedged benchmark Y. From X, $n = 13$ months give $S_X^2 = 3.2\,(\%)^2$; from Y, $m = 10$ months give $S_Y^2 = 9.6\,(\%)^2$. Keeping X in the numerator, so the test is a *lower*-tail one, test $H_0\colon \sigma_X^2 = \sigma_Y^2$ against $H_1\colon \sigma_X^2 < \sigma_Y^2$ at $\alpha = 0.05$. The table prints only upper-tail points; you are given $F_{0.05,\,9,\,12} = 2.80$. Use the reciprocal property to obtain the lower critical value.

D6. Two analytical instruments measure the same reference solution. Instrument X: $n = 16$ replicates, $S_X^2 = 0.072$. Instrument Y: $m = 21$ replicates, $S_Y^2 = 0.030$.

(a) Give a 90% confidence interval for the ratio σ_X^2/σ_Y^2. Use

$$F_{0.05,\,15,\,20} = 2.20 \quad \text{and} \quad F_{0.05,\,20,\,15} = 2.33.$$

(b) Use the interval to decide, at the corresponding two-sided level, whether the two instruments have equal measurement variance, and explain how the interval delivers that verdict.

D7. Two assembly shifts are compared for variability in cycle time. Shift X: $n = 13$, $S_X^2 = 0.0640\,\text{min}^2$. Shift Y: $m = 13$, $S_Y^2 = 0.0260\,\text{min}^2$. A normal quantile plot of shift X shows a pronounced right tail driven by a handful of very slow cycles. The two-sided F-test at $\alpha = 0.05$ gives $F_{\text{obs}} = S_X^2/S_Y^2$ against $F_{0.025,\,12,\,12} = 3.28$.

(a) Compute F_{obs} and state the formal decision at face value.

(b) Explain why this verdict should *not* be trusted on these data, and name a more appropriate procedure.

E. Comparing two proportions

E1. A nursery compares germination rates for two seed coatings. Of $n = 400$ seeds with coating X, $X = 252$ germinate; of $m = 360$ seeds with coating Y, $Y = 198$ germinate. At $\alpha = 0.05$, test $H_0\colon p_X = p_Y$ against $H_1\colon p_X \neq p_Y$ (do the germination rates *differ*?). Report z_{obs}, the critical value, and the decision in words.

E2. A clinic switches its appointment-reminder system from phone calls (group X) to text messages (group Y) to cut no-shows. Among $n = 500$ phone reminders, $X = 85$ patients miss their appointment; among $m = 450$ text reminders, $Y = 54$ miss. Let a "success" be a missed appointment.

(a) At $\alpha = 0.05$, test whether the phone-reminder miss rate is *higher* than the text-reminder miss rate ($H_1 : p_X > p_Y$). Use the pooled standard error and state the critical value with its df.

(b) Build a 95% confidence interval for $p_X - p_Y$ using the unpooled standard error, and say whether it is consistent with the part-(a) verdict.

E3. Two call centers are audited for first-call resolution. Center X resolves $X = 315$ of $n = 450$ calls on the first try; center Y resolves $Y = 304$ of $m = 400$. Give a 90% confidence interval for the difference $p_X - p_Y$ in resolution rates (unpooled SE, $z_{0.05} = 1.645$) and report the margin of error.

E4. In a single data set, a fertilizer is applied to plot X and a control to plot Y. Plot X yields $X = 171$ healthy plants of $n = 300$; plot Y yields $Y = 152$ of $m = 300$.

(a) Compute the *pooled* (null) standard error SE_0 that would go into the z-test of $H_0 : p_X = p_Y$.

(b) Compute the *unpooled* standard error SE_u that would go into a confidence interval for $p_X - p_Y$.

(c) In one sentence each, say which SE belongs in the test and which in the interval, and why they differ.

E5. A pharmacy compares adherence under a once-daily regimen (X) versus a twice-daily regimen (Y). Of $n = 650$ patients on the once-daily plan, $X = 520$ are adherent; of $m = 600$ on the twice-daily plan, $Y = 444$ are adherent. Test $H_0 : p_X = p_Y$ against $H_1 : p_X \neq p_Y$ by reporting the two-sided p-value, and decide at $\alpha = 0.05$. (You may use $\mathbb{P}(Z \geq 2.50) = 0.0062$.)

E6. An app team runs an A/B test on a new onboarding flow. The old flow (control, group Y) finishes onboarding for $Y = 918$ of $m = 9000$ new users; the new flow (variant, group X) finishes for $X = 1035$ of $n = 9000$. At $\alpha = 0.05$, did the new flow *raise* the completion rate ($H_1 : p_X > p_Y$)? Use the pooled SE, report z_{obs}, and also give the one-sided p-value (use $\mathbb{P}(Z \geq 2.78) = 0.0027$).

E7. A county compares mail-in ballot rejection rates in two election cycles. In cycle X, $X = 96$ of $n = 4800$ mail ballots are rejected; in cycle Y, $Y = 63$ of $m = 4200$ are rejected. At the stringent level $\alpha = 0.01$, test $H_0 : p_X = p_Y$ against $H_1 : p_X \neq p_Y$. Report z_{obs}, the two-sided critical value $z_{0.005}$, and the decision in words.

F. Choosing among two-sample procedures

F1. For each scenario below, name the single two-sample procedure that applies and give the one-sentence reason (the data-collection structure, the target parameter, and — for means — the variance assumption). Do *not* compute anything.

(a) A sleep researcher records each of 18 volunteers' total sleep time on a placebo night and again on a melatonin night, and asks whether melatonin lengthens sleep.

(b) Of 300 seeds from supplier A, 240 germinate; of 260 seeds from supplier B, 195 germinate. Do the germination rates differ?

(c) Independent samples of reaction times (ms) are taken from $n = 14$ younger and $m = 16$ older drivers; the sample SDs are $S_X = 42$ and $S_Y = 110$. Compare the mean reaction times.

(d) Two independent random samples of battery lifetimes are taken from brands A and B; the question is which brand is *more uniform* (less variable) in lifetime.

(e) Independent samples of $n = 11$ and $m = 11$ tomato plants grown under two fertilizers have yields with sample SDs $S_X = 1.8$ kg and $S_Y = 2.0$ kg; compare mean yields.

F2. (Paired-vs-independent contrast, same numbers, opposite verdicts.) A typing course is evaluated. In both versions of the study the measured effect is a 5 word-per-minute average gain and the question is whether typing speed improved; use $\alpha = 0.05$.

(a) *Paired design.* Each of $n = 8$ trainees is timed before and after the course. The per-trainee gains (after − before, wpm) are

$$7,\ 3,\ 6,\ 4,\ 8,\ 2,\ 5,\ 5,$$

with $\overline{D} = 5.0$ and $S_D = 2.0$. Name and run the correct test. Use $t_{0.05,7} = 1.895$.

(b) *Independent design.* Now suppose instead that a control group of $n = 8$ trainees and a *different* treatment group of $m = 8$ trainees were used, with $\overline{X} - \overline{Y} = 5.0$ wpm and equal within-group spreads $S_X = S_Y = 9.0$ wpm. Name and run the correct test. Use $t_{0.05,14} = 1.761$.

(c) State the two verdicts side by side and explain, in terms of the standard error, why the same 5 wpm effect is significant in one design and not the other — and what mistake an analyst makes who runs (b)'s test on (a)'s paired data.

F3. (Pooled vs Welch from the sample SDs.) Two independent groups of patients receive analgesics X and Y; the response is hours of pain relief. Group X: $n = 9$, $\overline{X} = 6.2$, $S_X = 1.2$. Group Y: $m = 13$, $\overline{Y} = 7.6$, $S_Y = 3.6$. Test at $\alpha = 0.05$ whether the mean relief times differ.

(a) Screen the variances with an F-test (two-sided $\alpha = 0.05$, so compare to the upper 2.5% point). Use $F_{0.025,12,8} = 4.20$. State the df and the decision.

(b) Based on (a), name which mean test is appropriate (pooled t or Welch's t) and why.

F4. (Pooling justified: close SDs, then run it.) An agronomist compares grain protein content (%) in two independent fields: field X has $n = 16$ plots with $\overline{X} = 12.8$, $S_X = 1.5$; field Y has $m = 12$ plots with $\overline{Y} = 11.6$, $S_Y = 1.7$. Test at $\alpha = 0.05$ whether the mean protein contents differ. Use $F_{0.025,11,15} = 3.33$ for the screen and $t_{0.025,26} = 2.056$ for the mean test.

(a) Screen the variances and decide whether pooling is justified.

(b) Compute the pooled SD, the standard error, and t_{obs}, and give the decision in words.

F5. (Proportions-vs-means recognition.) A research group studies a smoking-cessation app two ways. *Study 1:* of 140 users on the app, 63 had quit at 6 months; of 120 users on a brochure, 42 had quit — compare the quit *rates*. *Study 2:* record the *number of cigarettes per day* (a count averaged per person, treated as continuous) at 6 months for 140 app users and 120 brochure users — compare the mean daily cigarettes.

(a) Name the procedure for each study and explain in one sentence what feature of the *response* forces a z in one and a t in the other.

(b) For Study 1, carry out the test at $\alpha = 0.05$ of whether the app's quit rate is *higher* (one-sided). Use $z_{0.05} = 1.645$.

F6. (Recognize, then run: a paired item.) Two lab assays for blood glucose are compared by splitting each of $n = 7$ patients' blood and running both assays on the *same* sample. The per-patient differences (assay 1 − assay 2, mg/dL) are

$$4,\ -1,\ 3,\ 6,\ 0,\ 2,\ 1,$$

with $\overline{D} = 2.143$ and $S_D = 2.410$. Test at $\alpha = 0.05$ whether the assays disagree on average. First name the procedure and justify it, then carry it out. Use $t_{0.025,6} = 2.447$.

F7. (Recognize, then run: an F-test for consistency.) A bottling line is fed by two fillers. Independent samples give filler X $n = 10$ bottles with $S_X = 2.4$ mL and filler Y $m = 16$ bottles with $S_Y = 1.2$ mL. The plant cares about consistency, asking whether filler X is *more variable* than filler Y. Test at $\alpha = 0.05$ (one-sided). First name the procedure and justify why it, not a two-sample t, answers the question; then carry it out. Use $F_{0.05,9,15} = 2.59$.

G. Two-sample synthesis

G1. For each scenario, name the data-collection structure (*paired* or *two independent samples*) and the exact procedure you would run, and give the deciding reason in one sentence. State the hypotheses symbolically; you do not need to compute anything.

(a) A sports scientist records each of 14 cyclists' power output on a standard bike and again on a new aerodynamic bike, and asks whether the new bike changes mean output.

(b) Two independent batches of concrete, $n = 18$ cylinders cured by method A and $m = 22$ by method B, are crushed; the question is whether the two curing methods have the same *variability* of compressive strength.

(c) A registrar compares the pass rate on a licensing exam between $n = 640$ first-time takers and $m = 410$ repeat takers.

(d) Two independent groups of seedlings, $n = 12$ given a new nutrient and $m = 15$ given the standard, are weighed after a month; the new group's heights are visibly far more spread out than the standard group's, and the question is whether the new nutrient raises mean weight.

G2. A psychologist measures reaction time (ms) for two independently sampled groups under two screen contrasts. High-contrast group: $n = 10$, $\overline{X} = 88.0$, $S_X = 3.0$. Low-contrast group: $m = 14$, $\overline{Y} = 84.5$, $S_Y = 4.2$. Before comparing the means she must decide whether to pool.

(a) Run the two-sided F-test of $H_0 \colon \sigma_X^2 = \sigma_Y^2$ at $\alpha = 0.05$ and state whether pooling is justified. Use $F_{0.025,\,13,\,9} = 3.83$.

(b) Using your verdict from (a), carry out the appropriate two-sided test of $H_0 \colon \mu_X = \mu_Y$ at $\alpha = 0.05$, stating the degrees of freedom. Use $t_{0.025,22} = 2.074$.

G3. Two outreach programs are compared on the proportion of contacts who enroll. Program X enrolls $X = 171$ of $n = 300$ contacted; program Y enrolls $Y = 204$ of $m = 400$ contacted. The agency hoped program X would enroll at a *higher* rate than program Y.

(a) Test $H_0 \colon p_X = p_Y$ against $H_1 \colon p_X > p_Y$ at $\alpha = 0.05$. Use $z_{0.05} = 1.645$.

(b) Build a 95% confidence interval for $p_X - p_Y$ using the appropriate standard error, and reconcile it with the test in (a). Use $z_{0.025} = 1.96$.

G4. A clinic measures fasting blood glucose (mmol/L) in $n = 8$ patients before and after a twelve-week diet, recording the pair (before, after) for each patient:

$$(7.2, 6.5),\ (6.8, 6.1),\ (8.1, 7.0),\ (7.5, 6.9),\ (6.9, 6.4),\ (8.4, 7.5),\ (7.0, 6.8),\ (7.8, 6.9).$$

Let $D_i = (\text{before}_i) - (\text{after}_i)$, so a positive difference is a drop in glucose.

(a) At $\alpha = 0.05$, test whether the diet changes mean fasting glucose (two-sided). Use $t_{0.025,7} = 2.365$.

(b) Give a 95% confidence interval for the mean change μ_D and interpret it.

G5. An optics lab compares the focal-length error (micrometres) of lenses from a prototype line X and an established line Y, sampled independently. Prototype: $n = 8$, $\overline{X} = 560.0$, $S_X = 12.0$. Established: $m = 16$, $\overline{Y} = 548.0$, $S_Y = 4.0$. An engineer claims the prototype line has the *larger* mean error.

(a) First test $H_0 \colon \sigma_X^2 = \sigma_Y^2$ against $H_1 \colon \sigma_X^2 \neq \sigma_Y^2$ at $\alpha = 0.05$ to decide pooled versus Welch. Use $F_{0.025,\,7,\,15} = 3.29$.

(b) Carry out the appropriate one-sided test of $H_1 \colon \mu_X > \mu_Y$ at $\alpha = 0.05$, computing and stating the degrees of freedom. Use $t_{0.05,7} = 1.895$.

G6. A nutrition trial compares mean LDL cholesterol reduction (mg/dL) for a high-dose supplement X ($n = 15$, $\overline{X} = 124.0$, $S_X = 6.0$) against a low-dose supplement Y ($m = 15$, $\overline{Y} = 116.0$, $S_Y = 6.6$) on independent groups. A formulary will switch to the high dose only if its mean reduction exceeds the low dose's by *more than* 5 mg/dL. The two sample spreads are similar, so a common variance may be assumed. Test $H_0 \colon \mu_X - \mu_Y = 5$ against $H_1 \colon \mu_X - \mu_Y > 5$ at $\alpha = 0.05$, and state the decision the formulary should make. Use $t_{0.05,28} = 1.701$.

G7. Two factories are audited for a packaging defect. Factory X has $X = 58$ defective units in $n = 500$; factory Y has $Y = 40$ defective units in $m = 500$.

(a) At $\alpha = 0.05$, test whether the two defect rates *differ* (two-sided). Use $z_{0.025} = 1.96$.

(b) Build a 95% confidence interval for $p_X - p_Y$ with the appropriate standard error, and explain how it confirms the test verdict. Use $z_{0.025} = 1.96$.

9.8 Solutions

A. Pooled two-sample t

A1. **(a)** Both samples estimate one shared σ^2, so the best single estimate is the df-weighted average. With $n-1=10$ and $m-1=6$,

$$\begin{aligned}(n-1)S_X^2 &= 10(5.4^2) = 10(29.16) = 291.6,\\ (m-1)S_Y^2 &= 6(6.8^2) = 6(46.24) = 277.44,\\ S_p^2 &= \frac{291.6+277.44}{10+6} = \frac{569.04}{16} = 35.565,\\ s_p &= \sqrt{35.565} \approx 5.964.\end{aligned}$$

(b) The two-sample t carries $n+m-2 = 11+7-2 = 16$ degrees of freedom (two means were estimated to form the residuals).
(c) $S_X^2 = 29.16$ and $S_Y^2 = 46.24$, and 35.565 lies between them. It is pulled toward $S_X^2 = 29.16$ because oven X carries the larger df (10 versus 6), so the larger sample has the larger say.

$$\boxed{S_p^2 = 35.565,\ s_p \approx 5.964,\ \text{df} = 16,\ S_p^2 \in (29.16, 46.24) \text{ nearer } S_X^2.}$$

A2. **Procedure.** Two independent samples, similar spreads (equal-variance assumption defensible), comparing means with no direction $\Rightarrow$ two-sided pooled two-sample t, hypothesized difference 0, df $= n+m-2 = 22$.

Step 1: hypotheses. $H_0\colon \mu_X - \mu_Y = 0$ versus $H_1\colon \mu_X - \mu_Y \neq 0$; $\alpha = 0.05$.

Step 2: pool. With $n-1=9$, $m-1=13$,

$$S_p^2 = \frac{9(3.1^2)+13(2.7^2)}{22} = \frac{9(9.61)+13(7.29)}{22} = \frac{86.49+94.77}{22} = \frac{181.26}{22} = 8.239,$$

so $s_p = \sqrt{8.239} \approx 2.871$.

Step 3: standard error and statistic.

$$\begin{aligned}\tfrac{1}{10}+\tfrac{1}{14} &= 0.10000+0.07143 = 0.17143, \qquad \sqrt{0.17143} = 0.4141,\\ \text{SE} &= 2.871(0.4141) = 1.189,\\ t_{\text{obs}} &= \frac{(46.2-43.0)-0}{1.189} = \frac{3.2}{1.189} = 2.692.\end{aligned}$$

Step 4: decide. Since $|t_{\text{obs}}| = 2.692 > t_{0.025,22} = 2.074$, reject H_0 at the 5% level: there is significant evidence that the two cultivars differ in mean days to bloom (cultivar X blooms later, since $\overline{X} > \overline{Y}$).

$$\boxed{t_{\text{obs}} = 2.692 > 2.074 \ \Rightarrow\ \text{reject } H_0;\ \text{mean bloom times differ.}}$$

A3. **Procedure.** Independent samples, directional claim ("heavier on average") $\Rightarrow$ pooled two-sample t, upper one-tailed, df $= n+m-2 = 16+13-2 = 27$. All of α goes in the upper tail.

Step 1: hypotheses. $H_0\colon \mu_X - \mu_Y = 0$ versus $H_1\colon \mu_X - \mu_Y > 0$; $\alpha = 0.05$.

Step 2: pool. With $n-1=15$, $m-1=12$,

$$S_p^2 = \frac{15(6.0^2)+12(5.4^2)}{27} = \frac{15(36.00)+12(29.16)}{27} = \frac{540.00+349.92}{27} = \frac{889.92}{27} = 32.96,$$

so $s_p = \sqrt{32.96} \approx 5.741$.

Step 3: standard error and statistic.

$$\tfrac{1}{16}+\tfrac{1}{13}=0.06250+0.07692=0.13942, \qquad \sqrt{0.13942}=0.3734,$$
$$\mathrm{SE}=5.741(0.3734)=2.144,$$
$$t_{\mathrm{obs}}=\frac{512.0-508.5}{2.144}=\frac{3.5}{2.144}=1.633.$$

Step 4: decide. Since $t_{\mathrm{obs}}=1.633<t_{0.05,27}=1.703$, fail to reject H_0 at the 5% level: the data do not give significant evidence that line X fills heavier on average.

$$\boxed{t_{\mathrm{obs}}=1.633<1.703 \;\Rightarrow\; \text{fail to reject; no significant difference.}}$$

A4. Procedure. We want a range for $\mu_X-\mu_Y$, so build the pooled two-sample t interval around $\overline{X}-\overline{Y}$ with $\mathrm{df}=n+m-2=9+9-2=16$.

(a)

Step 1: pool. With $n-1=m-1=8$,

$$S_p^2=\frac{8(1.05^2)+8(0.95^2)}{16}=\frac{8(1.1025)+8(0.9025)}{16}=\frac{8.820+7.220}{16}=\frac{16.040}{16}=1.0025,$$

so $s_p=\sqrt{1.0025}\approx 1.0012$.

Step 2: standard error and margin.

$$\tfrac{1}{9}+\tfrac{1}{9}=0.22222, \qquad \sqrt{0.22222}=0.4714,$$
$$\mathrm{SE}=1.0012(0.4714)=0.4720,$$
$$E=t_{0.025,16}\cdot\mathrm{SE}=2.120(0.4720)=1.0007.$$

Step 3: assemble. The point estimate is $\overline{X}-\overline{Y}=7.40-6.10=1.30$, so

$$1.30\pm 1.00 \implies (0.299,\ 2.301)\ \text{hours}.$$

(b) The interval lies entirely above 0, so a two-sided test of $\mu_X=\mu_Y$ at $\alpha=0.05$ would reject. The interval and test are built from the same point estimate, standard error, and critical value, so a 95% interval excludes 0 exactly when the two-sided 5% test rejects — they cannot disagree.

$$\boxed{\mu_X-\mu_Y\in(0.30,\ 2.30)\ \text{h};\ \text{excludes } 0 \Rightarrow \text{reject } H_0 \text{ at } 5\%.}$$

A5. Procedure. The benchmark is a nonzero difference, so $\delta_0=10$ enters the numerator: $t=[(\overline{X}-\overline{Y})-\delta_0]/\,\mathrm{SE}$. Still a pooled two-sample t, one-tailed ("more than 10"), $\mathrm{df}=n+m-2=13+13-2=24$.

Step 1: hypotheses. $H_0\colon \mu_X-\mu_Y=10$ versus $H_1\colon \mu_X-\mu_Y>10$; $\alpha=0.05$.

Step 2: pool. With $n-1=m-1=12$,

$$S_p^2=\frac{12(4.0^2)+12(3.6^2)}{24}=\frac{12(16.00)+12(12.96)}{24}=\frac{192.00+155.52}{24}=\frac{347.52}{24}=14.48,$$

so $s_p=\sqrt{14.48}\approx 3.805$.

Step 3: statistic with $\delta_0=10$.

$$\mathrm{SE}=3.805\sqrt{\tfrac{1}{13}+\tfrac{1}{13}}=3.805\sqrt{0.15385}=3.805(0.3922)=1.492,$$
$$t_{\mathrm{obs}}=\frac{(38.5-26.0)-10}{1.492}=\frac{12.5-10}{1.492}=\frac{2.5}{1.492}=1.676.$$

Step 4: decide. Since $t_{\text{obs}} = 1.676 < t_{0.05,24} = 1.711$, fail to reject H_0 at the 5% level: the observed extra removal (12.5 ppb) is not significantly beyond the 10 ppb threshold, so the filter is not certified.

Step 5: contrast with $\delta_0 = 0$. Testing $H_0 : \mu_X - \mu_Y = 0$ would give $t_{\text{obs}} = 12.5/1.492 = 8.38$, rejecting overwhelmingly — the new filter is clearly better, but *not* demonstrably better by the full 10 ppb the regulation demands.

$$\boxed{t_{\text{obs}} = 1.676 < 1.711 \;\Rightarrow\; \text{fail to reject; not certified at 10 ppb.}}$$

A6. Procedure. Independent samples, comparing means, no direction ⇒ two-sided pooled t; first reduce the raw numbers to $\overline{X}, \overline{Y}, S_X^2, S_Y^2$, then proceed with $\text{df} = 5 + 4 - 2 = 7$.

Step 1: sample means.

$$\overline{X} = \frac{20.1 + 21.4 + 19.8 + 20.7 + 21.0}{5} = \frac{103.0}{5} = 20.60,$$

$$\overline{Y} = \frac{18.9 + 19.6 + 18.4 + 19.1}{4} = \frac{76.0}{4} = 19.00.$$

Step 2: sample variances.

$$\begin{aligned}
\sum (X_i - 20.60)^2 &= (-0.5)^2 + 0.8^2 + (-0.8)^2 + 0.1^2 + 0.4^2 \\
&= 0.25 + 0.64 + 0.64 + 0.01 + 0.16 = 1.70, \qquad S_X^2 = \frac{1.70}{4} = 0.425, \\
\sum (Y_j - 19.00)^2 &= (-0.1)^2 + 0.6^2 + (-0.6)^2 + 0.1^2 \\
&= 0.01 + 0.36 + 0.36 + 0.01 = 0.74, \qquad S_Y^2 = \frac{0.74}{3} = 0.24667.
\end{aligned}$$

Step 3: pool. With $n - 1 = 4$, $m - 1 = 3$, $n + m - 2 = 7$,

$$S_p^2 = \frac{4(0.425) + 3(0.24667)}{7} = \frac{1.700 + 0.740}{7} = \frac{2.440}{7} = 0.34857,$$

$$s_p = \sqrt{0.34857} \approx 0.5904.$$

Step 4: statistic and decide.

$$\begin{aligned}
\text{SE} &= 0.5904\sqrt{\tfrac{1}{5} + \tfrac{1}{4}} = 0.5904\sqrt{0.45000} = 0.5904(0.6708) = 0.3960, \\
t_{\text{obs}} &= \frac{20.60 - 19.00}{0.3960} = \frac{1.60}{0.3960} = 4.040.
\end{aligned}$$

Since $|t_{\text{obs}}| = 4.040 > t_{0.025,7} = 2.365$, reject H_0 at the 5% level: the two grinders differ in mean extraction yield (grinder X is higher).

$$\boxed{\overline{X} = 20.60,\ \overline{Y} = 19.00,\ s_p \approx 0.590,\ t_{\text{obs}} = 4.040 > 2.365 \Rightarrow \text{means differ.}}$$

A7. (a) Why pooled t is wrong. The two columns are measurements on *the same* 8 subjects (without/with), so they are not independent: a subject who is naturally slow to fall asleep is high in both columns, making the columns positively correlated. The pooled t's standard error $s_p\sqrt{1/n + 1/m}$ assumes $\text{Cov}(X, Y) = 0$ and therefore overstates the variance of $\overline{X} - \overline{Y}$, discarding the pairing that makes the design powerful. The correct tool is the paired t on the differences D_i.

(b) The wrong (pooled) statistic. Pooling with $n - 1 = m - 1 = 7$,

$$S_p^2 = \frac{7(6.5^2) + 7(6.0^2)}{14} = \frac{7(42.25) + 7(36.00)}{14} = \frac{295.75 + 252.00}{14} = 39.125, \qquad s_p = 6.255,$$

$$t_{\text{obs}}^{(\text{pooled})} = \frac{24.0 - 19.0}{6.255\sqrt{1/8 + 1/8}} = \frac{5.0}{6.255(0.5000)} = \frac{5.0}{3.128} = 1.599.$$

Since $1.599 < t_{0.025,14} = 2.145$, the pooled test *fails to reject* at $\alpha = 0.05$.

(c) The correct paired statistic. Work with the differences directly (one-sample t on $\overline{D}$, df $= n - 1 = 7$):

$$t_{\text{obs}}^{(\text{paired})} = \frac{\overline{D}}{S_D/\sqrt{n}} = \frac{5.0}{3.2/\sqrt{8}} = \frac{5.0}{3.2/2.828} = \frac{5.0}{1.131} = 4.420.$$

Since $4.420 > t_{0.025,7} = 2.365$, the paired test *rejects* decisively. The two answers differ because the pairing shrank the standard error from 3.128 to 1.131: the large between-subject spread cancels inside each difference D_i, so the same 5-minute gap is now clearly significant. Recognizing "same units, two times" as paired — before any arithmetic — is the decision that matters.

paired $t = 4.42$ (reject); pooled $t = 1.60$ (wrong, fails to reject).

B. Welch's t (unequal variances)

B1. Two means, two independent samples, sample standard deviations sharply unequal (3.0 vs. 8.0): equal variances are not credible, so this is **Welch's t-test**, not the pooled one. "Differ" is two-sided, with $\delta_0 = 0$.

Step 1: variance contributions.

$$u = \frac{S_X^2}{n} = \frac{3.0^2}{12} = \frac{9}{12} = 0.7500,$$
$$v = \frac{S_Y^2}{m} = \frac{8.0^2}{15} = \frac{64}{15} = 4.2667.$$

Step 2: Welch standard error and statistic.

$$\widehat{\text{se}} = \sqrt{u + v} = \sqrt{0.7500 + 4.2667} = \sqrt{5.0167} = 2.2398,$$
$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y} - 0}{\widehat{\text{se}}} = \frac{46.0 - 43.0}{2.2398} = \frac{3.0}{2.2398} = 1.339.$$

Step 3: Welch–Satterthwaite df.

$$\nu = \frac{(u+v)^2}{\dfrac{u^2}{n-1} + \dfrac{v^2}{m-1}} = \frac{5.0167^2}{\dfrac{0.7500^2}{11} + \dfrac{4.2667^2}{14}}$$
$$= \frac{25.167}{\dfrac{0.5625}{11} + \dfrac{18.205}{14}} = \frac{25.167}{0.05114 + 1.30037} = \frac{25.167}{1.35151} = 18.62,$$

so $\nu = 18$ (rounded down), giving $t_{0.025,18} = 2.101$.

Step 4: decision. Compare $|t_{\text{obs}}| = 1.339$ with $t_{0.025,18} = 2.101$. Since $1.339 < 2.101$, do not reject.

$\nu = 18, \quad \lvert t_{\text{obs}}\rvert = 1.339 < t_{0.025,18} = 2.101$ $\Rightarrow$ retain H_0; no significant difference in mean jump height at $\alpha = 0.05$.

B2. Two means, independent orchards, very different scatter (4.0 vs. 11.0): a **Welch** confidence interval. Even with $n = m = 10$, the unequal variances forbid pooling, so the two variance terms stay separate and the Welch df is used.

(a) *Step 1: variance contributions.*

$$u = \frac{4.0^2}{10} = \frac{16}{10} = 1.600, \qquad v = \frac{11.0^2}{10} = \frac{121}{10} = 12.100.$$

Step 2: standard error and point estimate.

$$\widehat{\mathrm{se}} = \sqrt{1.600 + 12.100} = \sqrt{13.700} = 3.7013, \qquad \overline{X} - \overline{Y} = 53.0 - 48.0 = 5.0.$$

Step 3: Welch df.

$$\nu = \frac{(13.700)^2}{\dfrac{1.600^2}{9} + \dfrac{12.100^2}{9}} = \frac{187.69}{\dfrac{2.560}{9} + \dfrac{146.41}{9}}$$
$$= \frac{187.69}{0.28444 + 16.26778} = \frac{187.69}{16.55222} = 11.34,$$

so $\nu = 11$ (rounded down) and $t_{0.025,11} = 2.201$.

Step 4: assemble the interval. The margin is $t_{0.025,11}\,\widehat{\mathrm{se}} = 2.201 \cdot 3.7013 = 8.147$, hence

$$5.0 \pm 8.147 = (-3.15,\ 13.15) \text{ mg per 100 g.}$$

$$\boxed{\mu_X - \mu_Y \in (-3.15,\ 13.15),\ \nu = 11.}$$

(b) The interval straddles 0, so 0 is a plausible value of $\mu_X - \mu_Y$. At 95% confidence we cannot declare the orchard means different; equivalently the two-sided Welch test at $\alpha = 0.05$ **retains** H_0.

B3. The variances are very unequal ($S_X = 6.0$ vs. $S_Y = 2.0$) *and* the noisier group is the *smaller* one ($n = 6$ vs. $m = 18$) — the dangerous configuration in which pooling understates the standard error. The correct procedure is **Welch**; the pooled test is computed only to expose its anti-conservatism.

(a) Pooled route (the wrong one here). Pool the variances with weights $n - 1 = 5$ and $m - 1 = 17$:

$$s_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n + m - 2} = \frac{5 \cdot 6.0^2 + 17 \cdot 2.0^2}{22} = \frac{180 + 68}{22} = \frac{248}{22} = 11.273,$$
$$s_p = 3.3575.$$

The pooled standard error and statistic are

$$s_p\sqrt{\tfrac{1}{n} + \tfrac{1}{m}} = 3.3575\sqrt{\tfrac{1}{6} + \tfrac{1}{18}} = 3.3575\sqrt{0.22222} = 3.3575 \cdot 0.4714 = 1.5827,$$
$$t_{\text{pool}} = \frac{14.0 - 11.0}{1.5827} = \frac{3.0}{1.5827} = 1.896 \quad (df = 22,\ t_{0.025,22} = 2.074).$$

Since $|1.896| < 2.074$, the pooled test retains — but only barely.

(b) Welch route (the correct one).

$$u = \frac{6.0^2}{6} = \frac{36}{6} = 6.000, \qquad v = \frac{2.0^2}{18} = \frac{4}{18} = 0.2222,$$

$$\widehat{\mathrm{se}} = \sqrt{6.000 + 0.2222} = \sqrt{6.2222} = 2.4944, \qquad t_{\text{Welch}} = \frac{3.0}{2.4944} = 1.203.$$

Welch df:

$$\nu = \frac{(6.2222)^2}{\dfrac{6.000^2}{5} + \dfrac{0.2222^2}{17}} = \frac{38.716}{\dfrac{36.000}{5} + \dfrac{0.04938}{17}}$$
$$= \frac{38.716}{7.20000 + 0.00290} = \frac{38.716}{7.20290} = 5.38,$$

so $\nu = 5$ and $t_{0.025,5} = 2.571$. Since $|1.203| < 2.571$, Welch also retains.

(c) Comparison. The pooled SE (1.5827) is far *smaller* than the Welch SE (2.4944) because pooling borrows the large quiet group's precision and lends it to the tiny noisy group through the $1/n$ term. That deflated SE inflates the pooled statistic (1.896 vs. Welch's 1.203) and would push $|t|$ past the cutoff in less extreme data: pooling is *anti-conservative* here, rejecting more often than the nominal 5%. Only

Welch holds the error rate when $\sigma_X \neq \sigma_Y$, and its honest $\nu = 5$ (not the pooled 22) reflects that the small noisy group bottlenecks the information.

> $t_{\text{pool}} = 1.896$ ($df = 22$), $t_{\text{Welch}} = 1.203$ ($\nu = 5$); both retain H_0,
> but trust **Welch** — pooling deflates the SE when the small group is noisier.

B4. Two means, independent groups, with the new formulation much more variable (20.0 vs. 7.0) ⇒ **Welch**. "Raises" points one way, so all 5% goes in the upper tail: an upper-tail Welch test with $\delta_0 = 0$.

Step 1: variance contributions.

$$u = \frac{20.0^2}{18} = \frac{400}{18} = 22.222, \qquad v = \frac{7.0^2}{12} = \frac{49}{12} = 4.0833.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{22.222 + 4.0833} = \sqrt{26.305} = 5.1289, \qquad t_{\text{obs}} = \frac{84.0 - 72.0}{5.1289} = \frac{12.0}{5.1289} = 2.340.$$

Step 3: Welch df.

$$\nu = \frac{(26.305)^2}{\frac{22.222^2}{17} + \frac{4.0833^2}{11}} = \frac{692.0}{\frac{493.83}{17} + \frac{16.674}{11}}$$
$$= \frac{692.0}{29.049 + 1.5158} = \frac{692.0}{30.565} = 22.64,$$

so $\nu = 22$. (The problem supplies the slightly conservative $t_{0.05,21} = 1.721$; the $\nu = 22$ cutoff $t_{0.05,22} = 1.717$ is even smaller, so the conclusion is unchanged.)

Step 4: decision. All α in the right tail: since $t_{\text{obs}} = 2.340 > 1.721$, reject.

> $t_{\text{obs}} = 2.340 > t_{0.05,21} = 1.721$ ($\nu = 22$) ⇒ reject H_0; the new formulation significantly raises mean concentration.

B5. The variable group is also the *tiny* one ($S_X = 15.0$, $n = 6$) while the steady group is large ($S_Y = 4.0$, $m = 50$): exactly the case where pooling understates the SE and over-rejects. **Welch** is mandatory, and the df will collapse toward the small cohort's 5.

(a) *Variance contributions and df.*

$$u = \frac{15.0^2}{6} = \frac{225}{6} = 37.500, \qquad v = \frac{4.0^2}{50} = \frac{16}{50} = 0.3200,$$

$$\nu = \frac{(37.500 + 0.3200)^2}{\frac{37.500^2}{5} + \frac{0.3200^2}{49}} = \frac{(37.820)^2}{\frac{1406.25}{5} + \frac{0.10240}{49}}$$
$$= \frac{1430.4}{281.250 + 0.00209} = \frac{1430.4}{281.252} = 5.09,$$

so $\nu = 5$ — essentially the small cohort's own $n - 1 = 5$, even though Y has 50 observations. The large quiet sample cannot rescue the precision the small noisy one destroys.

(b) *Standard error, statistic, decision.*

$$\widehat{\text{se}} = \sqrt{37.500 + 0.3200} = \sqrt{37.820} = 6.1498, \qquad t_{\text{obs}} = \frac{88.0 - 79.0}{6.1498} = \frac{9.0}{6.1498} = 1.464.$$

Compare $|t_{\text{obs}}| = 1.464$ with $t_{0.025,5} = 2.571$. Since $1.464 < 2.571$, retain H_0.

> $\nu = 5$, $|t_{\text{obs}}| = 1.464 < 2.571$ ⇒ retain H_0; no significant difference in mean enzyme level.

(c) A naive pooler would claim $df = n + m - 2 = 6 + 50 - 2 = 54$ and a cutoff near $t_{0.025,54} \approx 2.005$ — far smaller than the honest 2.571. That borrowed confidence is illegitimate because degrees of freedom measure *usable* information about the difference, which is bottlenecked by the small noisy group, not the large quiet one.

B6. Two means, independent anesthetics, S_Y more than double $S_X \Rightarrow$ **Welch**. The claim is about a *specific* gap, so $\delta_0 = 8$, not 0; "more than 8 shorter" makes it an upper-tail test on $(\overline{X} - \overline{Y}) - 8$.

Step 1: variance contributions.

$$u = \frac{6.0^2}{14} = \frac{36}{14} = 2.5714, \qquad v = \frac{13.0^2}{14} = \frac{169}{14} = 12.071.$$

Step 2: standard error and statistic.

$$\widehat{\text{se}} = \sqrt{2.5714 + 12.071} = \sqrt{14.643} = 3.8266,$$
$$t_{\text{obs}} = \frac{(\overline{X} - \overline{Y}) - \delta_0}{\widehat{\text{se}}} = \frac{(52.0 - 41.0) - 8}{3.8266} = \frac{11.0 - 8}{3.8266} = \frac{3.0}{3.8266} = 0.784.$$

Step 3: Welch df.

$$\nu = \frac{(14.643)^2}{\frac{2.5714^2}{13} + \frac{12.071^2}{13}} = \frac{214.42}{\frac{6.6122}{13} + \frac{145.72}{13}}$$
$$= \frac{214.42}{0.50863 + 11.20952} = \frac{214.42}{11.71815} = 18.30,$$

so $\nu = 18$ and the upper-tail cutoff is $t_{0.05,18} = 1.734$.

Step 4: decision. Since $t_{\text{obs}} = 0.784 < 1.734$, retain H_0: the observed 11-minute gap is not significantly larger than the claimed 8.

$$\boxed{t_{\text{obs}} = 0.784 < t_{0.05,18} = 1.734 \ (\nu = 18) \Rightarrow \text{retain } H_0;\ \text{no evidence the gap exceeds 8 min.}}$$

B7. Two means, independent pipettes, variances unknown and visibly different (pipette Y ranges over 5.6 μL, pipette X over 0.8 μL) $\Rightarrow$ **Welch**. We build both means and both S^2's from the raw numbers first. Two-sided, $\delta_0 = 0$.

Step 1: pipette X summary ($n = 5$).

$$\sum x_i = 99.6 + 100.2 + 99.8 + 100.0 + 100.4 = 500.0, \qquad \overline{X} = \frac{500.0}{5} = 100.00,$$
$$\sum x_i^2 = 9920.16 + 10040.04 + 9960.04 + 10000.00 + 10080.16 = 50000.40,$$
$$S_X^2 = \frac{50000.40 - 5 \cdot 100.00^2}{4} = \frac{50000.40 - 50000.00}{4} = \frac{0.40}{4} = 0.1000.$$

Step 2: pipette Y summary ($m = 6$).

$$\sum y_i = 98.0 + 101.5 + 97.2 + 102.8 + 99.0 + 101.5 = 600.0, \qquad \overline{Y} = \frac{600.0}{6} = 100.00,$$
$$\sum y_i^2 = 9604.00 + 10302.25 + 9447.84 + 10567.84 + 9801.00 + 10302.25 = 60025.18,$$
$$S_Y^2 = \frac{60025.18 - 6 \cdot 100.00^2}{5} = \frac{60025.18 - 60000.00}{5} = \frac{25.18}{5} = 5.0360.$$

Step 3: variance contributions, standard error, statistic.

$$u = \frac{0.1000}{5} = 0.02000, \qquad v = \frac{5.0360}{6} = 0.83933,$$
$$\widehat{\text{se}} = \sqrt{0.02000 + 0.83933} = \sqrt{0.85933} = 0.9270,$$
$$t_{\text{obs}} = \frac{100.00 - 100.00}{0.9270} = \frac{0.0}{0.9270} = 0.000.$$

Step 4: Welch df.

$$\nu = \frac{(0.85933)^2}{\dfrac{0.02000^2}{4} + \dfrac{0.83933^2}{5}} = \frac{0.73845}{\dfrac{0.000400}{4} + \frac{0.70447}{5}}$$
$$= \frac{0.73845}{0.000100 + 0.140896} = \frac{0.73845}{0.140996} = 5.24,$$

so $\nu = 5$ and $t_{0.025,5} = 2.571$.

Step 5: decision. The two sample means are identical, so $t_{\text{obs}} = 0.000$, which is trivially below $t_{0.025,5} = 2.571$; retain H_0.

$\boxed{\nu = 5,\ |t_{\text{obs}}| = 0.000 < 2.571 \Rightarrow \text{retain } H_0\text{; the two pipettes deliver the same mean volume.}}$

C. Paired t

C1. Two-sided before/after. The same hand of the same patient is measured twice, so the readings are linked patient by patient: paired data. We difference each pair and run a two-sided one-sample t on the gains $D_i = Y_i - X_i$, since "does it change" names no direction.

Step 1: form the differences. $D_i = Y_i - X_i$:

$$3,\ -1,\ 5,\ 5,\ 1,\ 4,\ 4,\ 3.$$

Step 2: $\overline{D}$ and S_D. The sum is $3-1+5+5+1+4+4+3 = 24$, so $\overline{D} = 24/8 = 3.0$. Deviations from 3 are $0, -4, 2, 2, -2, 1, 1, 0$, and

$$\sum (D_i - \overline{D})^2 = 0 + 16 + 4 + 4 + 4 + 1 + 1 + 0 = 30,$$
$$S_D^2 = \frac{30}{7} = 4.286, \qquad S_D = 2.070.$$

Step 3: the statistic. With $n = 8$, df $= 7$,

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{3.0}{2.070/\sqrt{8}} = \frac{3.0}{0.732} = 4.099.$$

Step 4: decision. Compare to $t_{0.025,7} = 2.365$. Since $|t_{\text{obs}}| = 4.099 > 2.365$, reject H_0 at the 5% level: the rehab program changes mean grip strength (the data show an increase of about 3 kg).

$\boxed{t_{\text{obs}} = 4.099 > t_{0.025,7} = 2.365 \ \Rightarrow\ \text{reject } H_0;\ \text{grip strength changes.}}$

C2. Paired CI. Two cuttings from the same parent plant share genetics and rearing, so the data are paired; the question asks for an interval for the mean difference, the one-sample t interval on the D_i.

(a) *$\overline{D}$ and S_D.* The sum of the ten differences is $2.1+3.4+0.8+4.0+1.5+2.8+3.1+0.5+2.4+1.9 = 22.5$, so $\overline{D} = 22.5/10 = 2.25$. The squared deviations from 2.25 sum to $\sum (D_i - \overline{D})^2 = 11.305$, so

$$S_D^2 = \frac{11.305}{9} = 1.256, \qquad S_D = 1.121.$$

With $n = 10$, df $= 9$, and $t_{0.025,9} = 2.262$,

$$\overline{D} \pm t_{0.025,9}\,\frac{S_D}{\sqrt{n}} = 2.25 \pm 2.262 \cdot \frac{1.121}{\sqrt{10}}$$
$$= 2.25 \pm 2.262 \cdot 0.3544$$
$$= 2.25 \pm 0.802 \ =\ (1.45,\ 3.05)\ \text{cm}.$$

$\boxed{\mu_D \in (1.45,\ 3.05)\ \text{cm at 95\% confidence.}}$

(b) *Decision from the interval.* The whole interval lies above 0 (it does not contain 0), so at the 5% level we reject $H_0 : \mu_D = 0$: there is significant evidence that the hormone lengthens stems, by roughly 1.5 to 3.0 cm on average.

C3. One-sided paired test. Each runner runs in both shoes, so the two times are paired; the coach's claim names a direction ("lowers" time), so this is a one-sided paired t on $D_i = (\text{old}) - (\text{new})$, oriented so that "the new shoe is faster" is $\mu_D > 0$.

Step 1: hypotheses. $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$; reject only for large positive t_{obs}. *Step 2: $\overline{D}$ and S_D.* Sum $= 14 - 6 + 22 + 9 + 31 + 5 + 18 - 2 + 27 = 118$, so $\overline{D} = 118/9 = 13.111$. The squared deviations from 13.111 sum to $\sum(D_i - \overline{D})^2 = 1292.889$, so

$$S_D^2 = \frac{1292.889}{8} = 161.611, \qquad S_D = 12.713.$$

Step 3: the statistic. With $n = 9$, $\text{df} = 8$,

$$t_{\text{obs}} = \frac{13.111}{12.713/\sqrt{9}} = \frac{13.111}{4.238} = 3.094.$$

Step 4: decision. The one-tailed critical value is $t_{0.05,8} = 1.860$. Since $3.094 > 1.860$, reject H_0 at the 5% level: the data give significant evidence that the new shoe lowers mean finish time (by about 13 s).

$$\boxed{t_{\text{obs}} = 3.094 > t_{0.05,8} = 1.860 \;\Rightarrow\; \text{reject; the new shoe is faster.}}$$

C4. The central contrast. Same patient before and after, so (a) is the correct analysis. Part (b) is the deliberate look-alike error: treating the "before" and "after" columns as two unrelated groups, which lets the patient-to-patient spread that pairing cancels reappear and swamp the effect.

(a) *The paired test.* Differences $D_i = X_i - Y_i$ (before minus after) are $5, 7, 5, 6, 6$. The sum is 29, so $\overline{D} = 29/5 = 5.8$. Deviations from 5.8 are $-0.8, 1.2, -0.8, 0.2, 0.2$, and $\sum(D_i - \overline{D})^2 = 0.64 + 1.44 + 0.64 + 0.04 + 0.04 = 2.8$, so

$$S_D^2 = \frac{2.8}{4} = 0.700, \qquad S_D = 0.837.$$

With $n = 5$, $\text{df} = 4$,

$$t_{\text{obs}}^{\text{paired}} = \frac{5.8}{0.837/\sqrt{5}} = \frac{5.8}{0.374} = 15.50.$$

Against $t_{0.025,4} = 2.776$ this is enormous: reject H_0 — the diet clearly lowers fasting glucose (by about 5.8 mg/dL).

$$\boxed{t_{\text{obs}}^{\text{paired}} = 15.50 > t_{0.025,4} = 2.776 \;\Rightarrow\; \text{reject; the diet lowers glucose.}}$$

(b) *The wrong independent test.* Treat X and Y as two samples of size 5. The means are $\overline{X} = 745/5 = 149.0$ and $\overline{Y} = 716/5 = 143.2$, a difference of 5.8 — the same as $\overline{D}$. But each column carries the full patient-to-patient spread: $\sum(X_i - \overline{X})^2 = 770.0$ and $\sum(Y_i - \overline{Y})^2 = 684.8$, so

$$s_p^2 = \frac{770.0 + 684.8}{5 + 5 - 2} = \frac{1454.8}{8} = 181.85,$$
$$s_p = 13.485.$$

With $\text{df} = n + m - 2 = 8$,

$$t_{\text{obs}}^{\text{indep}} = \frac{\overline{X} - \overline{Y}}{s_p\sqrt{\frac{1}{5} + \frac{1}{5}}} = \frac{5.8}{13.485 \cdot 0.6325} = \frac{5.8}{8.529} = 0.680.$$

Against $t_{0.025,8} = 2.306$ this is tiny: fail to reject — the independent analysis sees no effect.

$$\boxed{t_{\text{obs}}^{\text{indep}} = 0.680 < t_{0.025,8} = 2.306 \;\Rightarrow\; \text{fail to reject; no effect detected.}}$$

(c) *Why they disagree.* Both analyses estimate the same mean change, 5.8 mg/dL, so the numerators agree ($\overline{D} = \overline{X} - \overline{Y}$ always). The difference is entirely in the denominator: the paired standard error $S_D/\sqrt{n} = 0.374$ uses only the tiny within-patient noise, whereas the independent standard error $s_p\sqrt{2/5} = 8.529$ is inflated more than twentyfold by the between-patient spread that pairing cancels. Ignoring the pairing buried a clear effect.

C5. Paired CI (does it include 0?). The same image is scored by both radiologists, so the two scores are paired; we want an interval for the mean reader difference, the one-sample t interval on the D_i. A 90% level uses $t_{0.05,13}$.

Step 1: critical value and standard error. With $n = 14$, df $= 13$, $t_{0.05,13} = 1.771$ and $S_D/\sqrt{n} = 6.3/\sqrt{14} = 1.684$. *Step 2: the interval.*

$$\begin{aligned}\overline{D} \pm t_{0.05,13}\,\frac{S_D}{\sqrt{n}} &= 2.5 \pm 1.771 \cdot 1.684 \\ &= 2.5 \pm 2.982 \;=\; (-0.48,\ 5.48) \text{ points.}\end{aligned}$$

Step 3: interpret. The interval contains 0, so at the 90% level we cannot rule out $\mu_D = 0$: there is no significant evidence of a systematic difference between the two radiologists. The point estimate (2.5) is small next to the reader-to-reader noise ($S_D = 6.3$).

$$\boxed{\mu_D \in (-0.48,\ 5.48) \text{ contains } 0 \;\Rightarrow\; \text{no evidence of systematic difference.}}$$

C6. Paired test with a nonzero null. Same engine before and after the lubricant change, so paired; the hypothesized value is not 0 but 3.0, so we center the statistic at $\mu_{D,0} = 3.0$. The skeptic doubts the size of the saving, making the alternative one-sided ($\mu_D < 3.0$).

Step 1: hypotheses. $H_0 : \mu_D = 3.0$ versus $H_1 : \mu_D < 3.0$; reject only for sufficiently negative t_{obs}. *Step 2: the statistic with* $\mu_{D,0} = 3.0$. With $n = 12$, df $= 11$,

$$t_{\text{obs}} = \frac{\overline{D} - 3.0}{S_D/\sqrt{n}} = \frac{3.8 - 3.0}{1.6/\sqrt{12}} = \frac{0.8}{0.4619} = 1.732.$$

Step 3: compare. The lower-tail critical value is $-t_{0.05,11} = -1.796$. Since $t_{\text{obs}} = 1.732 > -1.796$ (in fact it is positive), do not reject H_0. *Step 4: conclusion.* The data ($\overline{D} = 3.8 > 3.0$) are entirely consistent with a mean saving of at least 3.0 W; there is no evidence the saving falls short of the claim.

$$\boxed{t_{\text{obs}} = 1.732 > -t_{0.05,11} = -1.796 \;\Rightarrow\; \text{do not reject; the claim stands.}}$$

C7. Raw pairs via the shortcut formula. Each volunteer is his own control (measured on both nights), so the data are paired and we work entirely with the column of differences $D_i = X_i - Y_i$.

(a) *Differences and running sums.* $D_i = X_i - Y_i$:

$$0.4,\ -0.1,\ 0.6,\ 0.4,\ 0.5,\ 0.4.$$

Then

$$\begin{aligned}\textstyle\sum D_i &= 0.4 - 0.1 + 0.6 + 0.4 + 0.5 + 0.4 = 2.2, \\ \textstyle\sum D_i^2 &= 0.16 + 0.01 + 0.36 + 0.16 + 0.25 + 0.16 = 1.10.\end{aligned}$$

Plugging into the computational form with $n = 6$,

$$\begin{aligned}\overline{D} &= \frac{2.2}{6} = 0.3667, \\ S_D^2 &= \frac{\sum D_i^2 - (\sum D_i)^2/n}{n-1} = \frac{1.10 - (2.2)^2/6}{5} = \frac{1.10 - 0.8067}{5} = \frac{0.2933}{5} = 0.05867, \\ S_D &= 0.2422.\end{aligned}$$

(b) *Standard error of* $\overline{D}$.

$$\mathrm{SE}(\overline{D}) = \frac{S_D}{\sqrt{n}} = \frac{0.2422}{\sqrt{6}} = 0.0989 \text{ h}.$$

(c) *Two-sided test.* $H_0 : \mu_D = 0$ versus $H_1 : \mu_D \neq 0$, with df $= 5$.

$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{0.3667}{0.0989} = 3.708.$$

Compare to $t_{0.025,5} = 2.571$. Since $|t_{\text{obs}}| = 3.708 > 2.571$, reject H_0 at the 5% level: white noise changes mean sleep time (the data show an increase of about 0.37 h).

$\overline{D} = 0.367,\ S_D = 0.242;\ t_{\text{obs}} = 3.708 > t_{0.025,5} = 2.571 \Rightarrow$ reject; sleep time changes.

D. The F-test for two variances

D1. Two independent normal samples and a question about equal spread with no stated direction, so this is the *two-sided* F-test on S_X^2/S_Y^2. Put the larger variance on top so that $F_{\text{obs}} \geq 1$ and only the upper cutoff can be exceeded. The numerator df is $n - 1 = 15$ (from X, on top) and the denominator df is $m - 1 = 12$ (from Y).

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{42}{15} = 2.80,$$
$$\text{df} = (n-1,\, m-1) = (15,\, 12).$$

Two-sided at $\alpha = 0.05$ places 0.025 in the upper tail, with critical value $F_{0.025,\,15,\,12} = 3.18$. Reject if $F_{\text{obs}} > 3.18$; here $2.80 < 3.18$.

$F_{\text{obs}} = 2.80 < F_{0.025,\,15,\,12} = 3.18 \ \Rightarrow$ fail to reject H_0 at $\alpha = 0.05$.

There is no significant evidence that the two grinders differ in consistency, even though X's sample variance is nearly triple Y's — variance ratios are noisy, so a large ratio is needed to reach significance.

D2. (a) The claim is directional — the old line is *more* variable — so the alternative is one-sided:

$$H_0 \colon \sigma_X^2 = \sigma_Y^2 \quad \text{vs} \quad H_1 \colon \sigma_X^2 > \sigma_Y^2.$$

The variance named as larger in H_1 (the old line, X) must go in the numerator: this puts the whole of α in the *upper* tail, where the reference table prints a critical value, and it makes large F_{obs} the evidence *for* H_1. Here the choice is mandatory, not mere bookkeeping — it encodes the direction of the test.

(b) With X on top,

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{0.78}{0.30} = 2.60,$$
$$\text{df} = (n-1,\, m-1) = (20,\, 15).$$

All of $\alpha = 0.05$ sits in the upper tail, with critical value $F_{0.05,\,20,\,15} = 2.33$. Reject if $F_{\text{obs}} > 2.33$; here $2.60 > 2.33$.

$F_{\text{obs}} = 2.60 > F_{0.05,\,20,\,15} = 2.33 \ \Rightarrow$ reject H_0 at $\alpha = 0.05$.

There is significant evidence at the 5% level that the old extruder produces more variable tensile strength than the new one.

D3. The hypothesis is two-sided and concerns spread, and there are two independent normal samples, so it is the two-sided F-test — but F is built from *variances*, so we square the reported standard deviations first. Larger variance on top:

$$S_X^2 = 6.4^2 = 40.96, \qquad S_Y^2 = 3.8^2 = 14.44,$$
$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{40.96}{14.44} = 2.84,$$
$$\text{df} = (n-1,\, m-1) = (9,\, 12).$$

Two-sided at $\alpha = 0.10$ places 0.05 in the upper tail, with critical value $F_{0.05,\,9,\,12} = 2.80$. Reject if $F_{\text{obs}} > 2.80$; here $2.84 > 2.80$ — a narrow clearance.

$$\boxed{F_{\text{obs}} = 2.84 > F_{0.05,\,9,\,12} = 2.80 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.10.}$$

At the 10% level the two formulations differ significantly in dissolution-time variability, though only just; we tested standard deviations but worked entirely on the squared scale.

D4. (a) The choice "pooled or Welch?" hinges on whether $\sigma_X^2 = \sigma_Y^2$ is tenable, which is exactly what the two-sided F-test on S_X^2/S_Y^2 checks. This is the variance pre-check, not the test of the means. Larger variance on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{9.6}{2.4} = 4.00,$$
$$\text{df} = (n-1,\, m-1) = (10,\, 8).$$

Two-sided at $\alpha = 0.05$ places 0.025 in the upper tail, with critical value $F_{0.025,\,10,\,8} = 4.30$. Reject equal variances if $F_{\text{obs}} > 4.30$; here $4.00 < 4.30$.

$$\boxed{F_{\text{obs}} = 4.00 < F_{0.025,\,10,\,8} = 4.30 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{equal variances are tenable.}}$$

(b) Because the pre-check does not reject equal variances, the equal-variance assumption is tenable, so the *pooled* two-sample t-test is justified for the means — the two sample variances may be combined into s_p^2. Had F_{obs} exceeded 4.30, we would have abandoned pooling and used Welch's t instead. (One caveat worth noting: the pre-test spends part of the error budget, so the chained procedure's true level is not exactly the nominal α.)

D5. The directional claim "X is *less* variable" becomes a *lower*-tail alternative once X is kept in the numerator, so we reject for *small* F_{obs}, below the lower critical value $F_{0.95,\,9,\,12}$. Keep X on top to match H_1:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{3.2}{9.6} = 0.333,$$
$$\text{df} = (n-1,\, m-1) = (9,\, 12).$$

The table prints only upper-tail points, so we obtain the lower cutoff from the reciprocal property $F_{1-\alpha,\,d_1,\,d_2} = 1/F_{\alpha,\,d_2,\,d_1}$, swapping the df and taking the reciprocal:

$$F_{0.95,\,9,\,12} = \frac{1}{F_{0.05,\,12,\,9}},$$
$$F_{0.05,\,12,\,9} = 2.80 \quad \text{(the given upper point, df swapped to } (12, 9)\text{)},$$
$$F_{0.95,\,9,\,12} = \frac{1}{2.80} = 0.357.$$

Reject if $F_{\text{obs}} < 0.357$; here $0.333 < 0.357$.

$$\boxed{F_{\text{obs}} = 0.333 < F_{0.95,\,9,\,12} = 0.357 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

There is significant evidence at the 5% level that the hedged portfolio X has smaller return variance than the benchmark Y. (As a check, flipping the ratio gives $S_Y^2/S_X^2 = 9.6/3.2 = 3.00$ on df $(12, 9)$, tested against the upper point $F_{0.05,\,12,\,9} = 2.80$; since $3.00 > 2.80$, the same rejection follows.)

D6. (a) The task is to *estimate* the variance ratio with an interval, not to accept or reject a single hypothesis, so we use the F-based confidence interval

$$\left(\frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{\alpha/2,\, n-1,\, m-1}} \;,\; \frac{S_X^2}{S_Y^2} \cdot F_{\alpha/2,\, m-1,\, n-1} \right).$$

For a 90% interval, $\alpha = 0.10$ and $\alpha/2 = 0.05$; the point ratio and df are

$$\frac{S_X^2}{S_Y^2} = \frac{0.072}{0.030} = 2.40,$$
$$(n-1,\, m-1) = (15,\, 20).$$

With $F_{0.05,\,15,\,20} = 2.20$ and $F_{0.05,\,20,\,15} = 2.33$,

$$\text{lower} = \frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{0.05,\,15,\,20}} = \frac{2.40}{2.20} = 1.091,$$
$$\text{upper} = \frac{S_X^2}{S_Y^2} \cdot F_{0.05,\,20,\,15} = 2.40 \times 2.33 = 5.59.$$

$$\boxed{\frac{\sigma_X^2}{\sigma_Y^2} \in (1.09,\ 5.59) \ \text{ with 90\% confidence.}}$$

(b) The interval lies *entirely above* 1, so at the corresponding two-sided 10% level the data are *not* consistent with equal variances: instrument X has significantly larger measurement variance than instrument Y. The interval delivers the verdict because a $100(1-\alpha)\%$ CI for σ_X^2/σ_Y^2 excludes 1 exactly when the two-sided level-α F-test rejects $H_0\colon \sigma_X^2 = \sigma_Y^2$ — here $\alpha = 0.10$. The fairly wide span (more than a fivefold range) reflects how loosely a couple dozen observations pin down a variance ratio.

D7. (a) Mechanically this is the two-sided F-test on S_X^2/S_Y^2 with df $(n-1, m-1) = (12, 12)$. Larger variance on top:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{0.0640}{0.0260} = 2.46,$$
$$\text{df} = (12,\, 12).$$

Two-sided at $\alpha = 0.05$, upper area 0.025, critical value $F_{0.025,\,12,\,12} = 3.28$. Reject if $F_{\text{obs}} > 3.28$; here $2.46 < 3.28$.

$$\boxed{F_{\text{obs}} = 2.46 < F_{0.025,\,12,\,12} = 3.28 \ \Rightarrow \ \text{fail to reject } H_0 \text{ at face value.}}$$

(b) The F-test draws *none* of its robustness from the central limit theorem: the exact $\mathrm{F}_{12,12}$ reference holds only when *both* parent populations are normal. Shift X is visibly right-skewed, so its heavy right tail inflates S_X^2 in a way the normal theory does not anticipate; under such a parent the statistic S_X^2/S_Y^2 has more spread than $\mathrm{F}_{12,12}$ predicts, and the cutoff 3.28 no longer carries its nominal 2.5% tail area. A value of 2.46 could be ordinary noise *or* a real difference masked by the wrong reference distribution — the F-test cannot distinguish them on these data, and the near miss carries no information. The right move is to verify normality (or transform toward it) and, failing that, to use a distribution-robust procedure such as Levene's or the Brown–Forsythe test, both based on absolute deviations from a center.

$$\boxed{\begin{array}{c}\text{A borderline } F \text{ result on right-skewed data is inconclusive;} \\ \text{verify normality or switch to a robust test (Levene / Brown–Forsythe).}\end{array}}$$

E. Comparing two proportions

E1. Two independent count samples, "differ" is two-sided, so a two-proportion z-test of $H_0: p_X = p_Y$ with the *pooled* (null) standard error. Sample rates $\widehat{p}_X = 252/400 = 0.630$ and $\widehat{p}_Y = 198/360 = 0.550$, difference $\widehat{p}_X - \widehat{p}_Y = 0.080$. Pool over all $n + m$ trials:

$$\widehat{p} = \frac{X+Y}{n+m} = \frac{252+198}{400+360} = \frac{450}{760} = 0.592105.$$

Pooled standard error (it goes in the test because H_0 posits one common rate):

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n}+\frac{1}{m}\right)} \\ &= \sqrt{0.592105 \cdot 0.407895\left(\frac{1}{400}+\frac{1}{360}\right)} \\ &= \sqrt{0.241517 \cdot 0.0052778} = \sqrt{0.00127467} = 0.035703. \end{aligned}$$

Statistic $z_{\mathrm{obs}} = (0.630 - 0.550)/0.035703 = 0.080/0.035703 = 2.241$. Two-sided cutoff at $\alpha = 0.05$ is $z_{0.025} = 1.96$; since $|2.241| \geq 1.96$ we reject.

$$\boxed{z_{\mathrm{obs}} = 2.241 > 1.96 \;\Rightarrow\; \text{reject } H_0\text{: the germination rates differ.}}$$

E2. This is a multi-part item: a one-sided *test* (pool the SE) in (a), and a *confidence interval* (unpooled SE) in (b).

(a) **Directional claim ("higher"), so a one-sided two-proportion z-test, $H_1: p_X > p_Y$, with the pooled SE.** Miss rates $\widehat{p}_X = 85/500 = 0.170$ (phone) and $\widehat{p}_Y = 54/450 = 0.120$ (text), difference 0.050. Pooled rate

$$\widehat{p} = \frac{85+54}{500+450} = \frac{139}{950} = 0.146316.$$

Pooled standard error:

$$\begin{aligned} \mathrm{SE}_0 &= \sqrt{0.146316 \cdot 0.853684\left(\frac{1}{500}+\frac{1}{450}\right)} \\ &= \sqrt{0.124907 \cdot 0.0042222} = \sqrt{0.00052739} = 0.022965. \end{aligned}$$

Statistic $z_{\mathrm{obs}} = (0.170 - 0.120)/0.022965 = 0.050/0.022965 = 2.177$. Right-sided cutoff $z_{0.05} = 1.645$ (large-sample z, no df); since $2.177 \geq 1.645$ we reject.

$$\boxed{z_{\mathrm{obs}} = 2.177 > 1.645 \;\Rightarrow\; \text{reject } H_0\text{: phone reminders miss more often.}}$$

(b) **Interval for $p_X - p_Y$, so switch to the *unpooled* SE** with two-sided multiplier $z_{0.025} = 1.96$. Each group keeps its own variance:

$$\begin{aligned} \mathrm{SE}_{\mathrm{u}} &= \sqrt{\frac{\widehat{p}_X(1-\widehat{p}_X)}{n} + \frac{\widehat{p}_Y(1-\widehat{p}_Y)}{m}} \\ &= \sqrt{\frac{0.170 \cdot 0.830}{500} + \frac{0.120 \cdot 0.880}{450}} \\ &= \sqrt{0.00028220 + 0.00023467} = \sqrt{0.00051687} = 0.022735. \end{aligned}$$

Margin $= 1.96 \cdot 0.022735 = 0.044560$, so

$$0.050 \pm 0.044560 = (0.005440,\ 0.094560).$$

The interval excludes 0 (and lies entirely above it), which agrees with the part-(a) rejection: a positive difference is the plausible range.

$$\boxed{p_X - p_Y \in (0.0054,\ 0.0946) \quad (95\%),\ \text{excludes 0 — consistent with (a).}}$$

E3. An *interval* for a difference of proportions, so the *unpooled* SE; the 90% level uses $z_{0.05} = 1.645$. Resolution rates $\widehat{p}_X = 315/450 = 0.700$ and $\widehat{p}_Y = 304/400 = 0.760$, difference $\widehat{p}_X - \widehat{p}_Y = -0.060$. Unpooled standard error:

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{0.700 \cdot 0.300}{450} + \frac{0.760 \cdot 0.240}{400}} \\ &= \sqrt{0.00046667 + 0.00045600} = \sqrt{0.00092267} = 0.030375.\end{aligned}$$

Margin of error $= 1.645 \cdot 0.030375 = 0.049968$, so

$$-0.060 \pm 0.049968 = (-0.109968,\ -0.010032).$$

The interval lies entirely below 0: center Y's resolution rate is higher, significantly so at the 90% level.

$$\boxed{p_X - p_Y \in (-0.1100,\ -0.0100) \quad (90\%),\ \text{margin } = 0.0500,\ \text{excludes 0.}}$$

E4. The deliberate contrast: the same two counts feed two different standard errors. Rates $\widehat{p}_X = 171/300 = 0.570$ and $\widehat{p}_Y = 152/300 = 0.506667$.

(a) *Pooled (test) SE.* Pooled rate $\widehat{p} = (171 + 152)/(300 + 300) = 323/600 = 0.538333$, and

$$\begin{aligned}\mathrm{SE_0} &= \sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{n} + \frac{1}{m}\right)} \\ &= \sqrt{0.538333 \cdot 0.461667 \cdot 0.0066667} \\ &= \sqrt{0.248531 \cdot 0.0066667} = \sqrt{0.00165687} = 0.040705.\end{aligned}$$

(b) *Unpooled (interval) SE.*

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{0.570 \cdot 0.430}{300} + \frac{0.506667 \cdot 0.493333}{300}} \\ &= \sqrt{0.00081700 + 0.00083319} = \sqrt{0.00165019} = 0.040622.\end{aligned}$$

(c) Side by side,

$$\begin{aligned}\mathrm{SE_0} &= 0.040705 \quad \text{(pooled, for the } z\text{-test)}, \\ \mathrm{SE_u} &= 0.040622 \quad \text{(unpooled, for the CI)}.\end{aligned}$$

The *pooled* SE belongs in the test statistic because $H_0\colon p_X = p_Y$ supplies a single common rate, best estimated from all $n + m$ trials. The *unpooled* SE belongs in the confidence interval because the interval assumes no common rate, so each group keeps its own variance. They differ because pooling replaces the two rates by one; here the gap is tiny because $\widehat{p}_X$ and $\widehat{p}_Y$ are close and the sizes are equal.

$$\boxed{\text{Test: } \mathrm{SE_0} = 0.04071 \text{ (pool).} \qquad \text{CI: } \mathrm{SE_u} = 0.04062 \text{ (don't pool).}}$$

E5. Two independent count samples; "differ" is two-sided and the question asks for the p-value, so a two-proportion z-test with the *pooled* SE. Adherence rates $\widehat{p}_X = 520/650 = 0.800$ (once-daily) and $\widehat{p}_Y = 444/600 = 0.740$ (twice-daily), difference 0.060. Pooled rate

$$\widehat{p} = \frac{520 + 444}{650 + 600} = \frac{964}{1250} = 0.771200.$$

Pooled standard error:

$$\begin{aligned}\mathrm{SE_0} &= \sqrt{0.771200 \cdot 0.228800\left(\frac{1}{650} + \frac{1}{600}\right)} \\ &= \sqrt{0.176451 \cdot 0.0032051} = \sqrt{0.00056555} = 0.023781.\end{aligned}$$

Statistic $z_{\text{obs}} = (0.800 - 0.740)/0.023781 = 0.060/0.023781 = 2.523$. Two-sided p-value, using $\mathbb{P}(Z \geq 2.50) = 0.0062$:

$$p\text{-value} = 2\,\mathbb{P}(Z \geq 2.52) \approx 2 \cdot 0.0062 = 0.0124.$$

Since $0.0124 < 0.05$ we reject: the adherence rates differ significantly.

$$\boxed{z_{\text{obs}} = 2.523, \quad p\text{-value} \approx 0.0124 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

E6. A classic A/B test: two independent streams of yes/no outcomes, "raise" is directional, so a one-sided two-proportion z-test, $H_1 : p_X > p_Y$ with X the variant; pool for the test SE. Completion rates $\widehat{p}_X = 1035/9000 = 0.115$ (variant) and $\widehat{p}_Y = 918/9000 = 0.102$ (control), difference 0.013. Pooled rate

$$\widehat{p} = \frac{1035 + 918}{9000 + 9000} = \frac{1953}{18000} = 0.108500.$$

Pooled standard error (here $1/n + 1/m = 2/9000 = 0.00022222$):

$$\begin{aligned} \text{SE}_0 &= \sqrt{0.108500 \cdot 0.891500 \cdot 0.00022222} \\ &= \sqrt{0.096728 \cdot 0.00022222} = \sqrt{0.00002150} = 0.004636. \end{aligned}$$

Statistic $z_{\text{obs}} = (0.115 - 0.102)/0.004636 = 0.013/0.004636 = 2.804$. Right-sided cutoff $z_{0.05} = 1.645$; since $2.804 \geq 1.645$ we reject. One-sided p-value, using $\mathbb{P}(Z \geq 2.78) = 0.0027$:

$$p\text{-value} = \mathbb{P}(Z \geq 2.80) \approx 0.0027.$$

$$\boxed{z_{\text{obs}} = 2.804 \geq 1.645,\ p\text{-value} \approx 0.0027 \;\Rightarrow\; \text{reject } H_0\text{: the new flow raises completion.}}$$

E7. Two independent count samples, "differ" is two-sided, so a two-proportion z-test with the *pooled* SE; the stringent $\alpha = 0.01$ only changes the cutoff. Rejection rates $\widehat{p}_X = 96/4800 = 0.020$ and $\widehat{p}_Y = 63/4200 = 0.015$, difference 0.005. Pooled rate

$$\widehat{p} = \frac{96 + 63}{4800 + 4200} = \frac{159}{9000} = 0.017667.$$

Pooled standard error:

$$\begin{aligned} \text{SE}_0 &= \sqrt{0.017667 \cdot 0.982333 \left(\frac{1}{4800} + \frac{1}{4200}\right)} \\ &= \sqrt{0.017355 \cdot 0.00044643} = \sqrt{0.00000775} = 0.002783. \end{aligned}$$

Statistic $z_{\text{obs}} = (0.020 - 0.015)/0.002783 = 0.005/0.002783 = 1.796$. Two-sided 1% cutoff $z_{0.005} = 2.576$; since $|1.796| < 2.576$ we fail to reject.

$$\boxed{z_{\text{obs}} = 1.796 < 2.576 \;\Rightarrow\; \text{fail to reject at 1\%: rejection rates not shown to differ.}}$$

F. Choosing among two-sample procedures

F1. Run the cascade on each: first read the *design* (paired vs independent), then the *response* (means, variances, or proportions), then for means the variance assumption.

(a) Each volunteer is measured *twice* (placebo night and melatonin night), so the two columns are two measurements on the same unit — **paired**. Reduce to the per-person differences D_i and run a one-sample upper-tail t on them against 0 (§3), $df = 17$.

(b) The data are *counts of successes* (germinated / not) out of fixed sample sizes from two independent suppliers, so the parameter is a difference of **proportions**: **two-proportion z-test** (§5), two-sided ("differ").

(c) Independent samples, comparing **means**, but the sample SDs are very far apart (42 vs 110, a ratio over 2.5), so equal variances are not credible: **Welch's t-test** (§2) with the approximate df, two-sided.

(d) Independent samples, and "more uniform / less variable" is a question about *spread* — the **F-test** on S_X^2/S_Y^2 (§4), not a t-test.

(e) Independent samples, comparing **means**, with close sample SDs (1.8 vs 2.0, ratio of variances ≈ 0.81), so equal variances are plausible: **pooled two-sample t-test** (§1), $df = n + m - 2 = 20$, two-sided.

(a) paired t	(b) two-prop. z	(c) Welch t	(d) F-test	(e) pooled t.

F2. Same 5 wpm effect, two designs, contrasting verdicts.

(a) **Recognize.** Each trainee is timed before *and* after, so the data are **paired**; collapse to gains D_i and run a one-sample upper-tail t against 0, $df = n - 1 = 7$. Hypotheses $H_0 : \mu_D = 0$ vs $H_1 : \mu_D > 0$.

$$\frac{S_D}{\sqrt{n}} = \frac{2.0}{\sqrt{8}} = \frac{2.0}{2.828} = 0.7071,$$
$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{5.0}{0.7071} = 7.07.$$

Cutoff $t_{0.05,7} = 1.895$. Since $7.07 > 1.895$, reject overwhelmingly: the course raised typing speed.

$$\boxed{t_{\text{obs}} = 7.07 > 1.895 \;\Rightarrow\; \text{reject } H_0 \text{ (paired } t\text{)}.}$$

(b) **Recognize.** Different trainees in each arm $\Rightarrow$ **independent** samples; equal SDs $\Rightarrow$ pooled two-sample t, $df = n + m - 2 = 14$, upper-tail. With $S_X = S_Y = 9.0$, $s_p = 9.0$, and

$$\widehat{\text{SE}} = s_p\sqrt{\tfrac{1}{n} + \tfrac{1}{m}} = 9.0\sqrt{\tfrac{1}{8} + \tfrac{1}{8}} = 9.0\sqrt{0.25} = 9.0 \cdot 0.5 = 4.5,$$
$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y}}{\widehat{\text{SE}}} = \frac{5.0}{4.5} = 1.11.$$

Cutoff $t_{0.05,14} = 1.761$. Since $1.11 < 1.761$, do not reject: no significant improvement.

$$\boxed{t_{\text{obs}} = 1.11 < 1.761 \;\Rightarrow\; \text{retain } H_0 \text{ (independent } t\text{)}.}$$

(c) **Side by side.** Paired: $t_{\text{obs}} = 7.07$, reject. Independent: $t_{\text{obs}} = 1.11$, retain. The effect (5 wpm) is identical; only the *standard error* differs. Pairing measures the 5 wpm against the SD of *gains* ($S_D = 2.0$, so SE $= 0.71$), whereas the independent design measures it against the full between-trainee spread ($S_X = S_Y = 9.0$, so SE $= 4.5$), a denominator roughly 6 times larger. An analyst who *has* the paired data of (a) but runs (b)'s independent test discards the pairing, inflates the SE, computes $t_{\text{obs}} = 1.11$, fails to reject, and *misses a real and strong effect* — the classic **power loss** from ignoring pairing.

F3. Decide pooled vs Welch before touching the means.

(a) **Recognize.** "Should we pool?" is exactly the equal-variance question; the formal screen is the F-test on S_X^2/S_Y^2 (§4), larger variance on top. Here $S_Y^2 = 3.6^2 = 12.96$ and $S_X^2 = 1.2^2 = 1.44$, so

$$F_{\text{obs}} = \frac{S_Y^2}{S_X^2} = \frac{12.96}{1.44} = 9.00, \qquad df = (m-1,\, n-1) = (12,\, 8).$$

Cutoff $F_{0.025,12,8} = 4.20$. Since $9.00 > 4.20$, **reject** equal variances at the 5% level: the spreads are significantly different.

$$\boxed{F_{\text{obs}} = 9.00 > 4.20 \;\Rightarrow\; \text{variances unequal, } df = (12, 8).}$$

(b) Equal variances is rejected, so pooling is *not* justified — s_p would blend the genuinely different spreads 1.2 and 3.6 into one number describing neither sample, and the reference t_{20} would be wrong. Use **Welch's t-test** (§2) with the approximate (Satterthwaite) df.

Use Welch's t (§2), not the pooled t.

F4. Close SDs suggest pooling is legitimate; confirm with the screen, then run the pooled t.

(a) **Screen.** F-test on the variances, larger on top. $S_Y^2 = 1.7^2 = 2.89$, $S_X^2 = 1.5^2 = 2.25$, so

$$F_{\text{obs}} = \frac{S_Y^2}{S_X^2} = \frac{2.89}{2.25} = 1.284, \qquad df = (m-1,\, n-1) = (11,\, 15), \qquad F_{0.025,11,15} = 3.33.$$

Since $1.284 < 3.33$, do *not* reject equal variances — pooling is justified.

(b) **Pooled SD.** Hypotheses $H_0 : \mu_X = \mu_Y$ vs $H_1 : \mu_X \neq \mu_Y$, $df = n + m - 2 = 26$.

$$s_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{15 \cdot 2.25 + 11 \cdot 2.89}{26} = \frac{33.75 + 31.79}{26} = \frac{65.54}{26} = 2.521,$$
$$s_p = \sqrt{2.521} = 1.588.$$

Standard error and statistic.

$$\begin{aligned}\widehat{\text{SE}} &= s_p\sqrt{\tfrac{1}{n} + \tfrac{1}{m}} = 1.588\sqrt{\tfrac{1}{16} + \tfrac{1}{12}} = 1.588\sqrt{0.0625 + 0.08333} \\ &= 1.588\sqrt{0.14583} = 1.588 \cdot 0.3819 = 0.6065, \\ t_{\text{obs}} &= \frac{\overline{X} - \overline{Y}}{\widehat{\text{SE}}} = \frac{12.8 - 11.6}{0.6065} = \frac{1.2}{0.6065} = 1.978.\end{aligned}$$

Decision. $|t_{\text{obs}}| = 1.978 < t_{0.025,26} = 2.056$, so do not reject. There is no significant difference in mean protein content between the two fields at the 5% level.

Pool ($F_{\text{obs}} = 1.284 < 3.33$); then $\lvert t_{\text{obs}}\rvert = 1.978 < 2.056 \Rightarrow$ retain H_0.

F5. Same subject matter, two response types, two procedures.

(a) **Study 1** records "quit / did not quit," a *count of a binary outcome* out of fixed sample sizes, so the parameter is a difference of **proportions**: two-proportion z-test (§5). **Study 2** records cigarettes per day, a *measured number*, so the parameter is a difference of **means**: two-sample t (§1 or §2). The hinge is the *response type* — a yes/no count forces a z, a continuous measurement forces a t — not the subject matter, which is identical.

Study 1: two-proportion z (§5). Study 2: two-sample t (§1/§2).

(b) **Study 1 test (one-sided).** $H_0 : p_{\text{app}} = p_{\text{br}}$ vs $H_1 : p_{\text{app}} > p_{\text{br}}$. Sample proportions

$$\hat{p}_{\text{app}} = \frac{63}{140} = 0.450, \qquad \hat{p}_{\text{br}} = \frac{42}{120} = 0.350, \qquad \hat{p}_{\text{app}} - \hat{p}_{\text{br}} = 0.100.$$

Under H_0 use the *pooled* success rate in the standard error:

$$\begin{aligned}\hat{p} &= \frac{63 + 42}{140 + 120} = \frac{105}{260} = 0.4038, \\ \widehat{\text{SE}} &= \sqrt{\hat{p}(1-\hat{p})\left(\tfrac{1}{n} + \tfrac{1}{m}\right)} = \sqrt{0.4038 \cdot 0.5962\,(0.0071429 + 0.0083333)} \\ &= \sqrt{0.24074 \cdot 0.015476} = \sqrt{0.0037260} = 0.06104, \\ z_{\text{obs}} &= \frac{0.100}{0.06104} = 1.638.\end{aligned}$$

Upper-tail cutoff $z_{0.05} = 1.645$. Since $1.638 < 1.645$ (just barely), do *not* reject: the app's quit rate is not significantly higher at the 5% level.

$$\boxed{z_{\text{obs}} = 1.638 < 1.645 \;\Rightarrow\; \text{retain } H_0 \text{ (two-prop. } z\text{)}.}$$

F6. Recognize. The two assays are run on the *same* blood sample for each patient, so the readings are paired patient by patient — pairing removes the large patient-to-patient variation in true glucose. Run a one-sample two-sided t on the differences D_i against 0 (§3), $df = n - 1 = 6$. Two independent samples of 7 would be wrong because the two columns are linked through the shared samples. Hypotheses $H_0 : \mu_D = 0$ vs $H_1 : \mu_D \neq 0$.

$$\frac{S_D}{\sqrt{n}} = \frac{2.410}{\sqrt{7}} = \frac{2.410}{2.646} = 0.9109,$$
$$t_{\text{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{2.143}{0.9109} = 2.353.$$

Two-sided cutoff $t_{0.025,6} = 2.447$. Since $|t_{\text{obs}}| = 2.353 < 2.447$, do not reject: there is no significant evidence that the assays disagree on average at the 5% level.

$$\boxed{|t_{\text{obs}}| = 2.353 < 2.447 \;\Rightarrow\; \text{retain } H_0 \text{ (paired } t\text{)}.}$$

F7. Recognize. "More variable" is a question about *spread*, i.e. the variances, so a two-sample t (which compares means) answers the wrong question. Two independent samples compared on variance $\Rightarrow$ the F-test on S_X^2/S_Y^2 (§4). The alternative "X more variable" is one-sided, $H_1 : \sigma_X^2 > \sigma_Y^2$, so put S_X^2 on top and use the upper 5% point with $df = (n-1,\, m-1) = (9, 15)$.

$$S_X^2 = 2.4^2 = 5.76, \qquad S_Y^2 = 1.2^2 = 1.44,$$
$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{5.76}{1.44} = 4.00, \qquad df = (9,\, 15).$$

Cutoff $F_{0.05,9,15} = 2.59$. Since $4.00 > 2.59$, reject $H_0 : \sigma_X^2 = \sigma_Y^2$: filler X is significantly more variable than filler Y at the 5% level.

$$\boxed{F_{\text{obs}} = 4.00 > 2.59 \;\Rightarrow\; \text{reject } H_0;\ \text{filler X more variable } (F\text{-test, } df = (9, 15)).}$$

G. Two-sample synthesis

G1. The deciding question is always asked in order: first paired or independent (read the *design*, not the numbers), then what parameter is compared, then — for two means of independent samples — whether the variances may be assumed equal.

(a) Each cyclist is measured on *both* bikes, so the two readings are linked rider by rider: **paired**. Form $D_i = (\text{aero}) - (\text{standard})$ and run a two-sided **paired t-test** of $H_0 \colon \mu_D = 0$ vs. $H_1 \colon \mu_D \neq 0$ on the $n = 14$ differences. Reason: the same rider supplies both numbers, so they are not independent.

(b) Two *separate* batches of cylinders, and the target is *variability*, so this is the **F-test** of $H_0 \colon \sigma_A^2 = \sigma_B^2$ vs. $H_1 \colon \sigma_A^2 \neq \sigma_B^2$ on S_A^2/S_B^2. Reason: independent normal samples compared on spread $\Rightarrow$ F, not a t or a χ^2.

(c) Two independent groups of test-takers counted as pass/fail, so the target is a proportion: the **two-proportion z-test** of $H_0 \colon p_X = p_Y$ vs. $H_1 \colon p_X \neq p_Y$. Reason: independent counts of successes $\Rightarrow$ compare proportions.

(d) Two independent groups compared on *means*, but the spreads are visibly unequal, so pooling is not licensed: **Welch's t-test** of $H_0 \colon \mu_X = \mu_Y$ vs. $H_1 \colon \mu_X > \mu_Y$ (upper-tailed, "raises"). Reason: unequal variances forbid the pooled t; Welch keeps each S^2 in its own term.

$$\boxed{\text{(a) paired } t; \quad \text{(b) } F\text{-test}; \quad \text{(c) two-proportion } z; \quad \text{(d) Welch } t.}$$

G2. Two independent samples on a mean comparison: the variance question must be settled first, because it selects pooled versus Welch.

(a) *Step 1: F statistic.* This is a two-sided F-test of equal variances. Put the larger sample variance on top, $S_Y^2 = 4.2^2 = 17.64$ over $S_X^2 = 3.0^2 = 9.0$:

$$F_{\text{obs}} = \frac{S_Y^2}{S_X^2} = \frac{17.64}{9.0} = 1.96, \qquad \text{df } = (m-1,\, n-1) = (13,\, 9).$$

Step 2: critical value and decision. Two-sided at $\alpha = 0.05$ puts 0.025 in the upper tail: $F_{0.025,\,13,\,9} = 3.83$. Since $1.96 < 3.83$, fail to reject equal variances.

$$\boxed{F_{\text{obs}} = 1.96 < F_{0.025,\,13,\,9} = 3.83 \;\Rightarrow\; \text{equal variances tenable; pool.}}$$

(b) *Step 1: pool.* With equal variances justified, use the pooled two-sample t. With $n - 1 = 9$ and $m - 1 = 13$,

$$s_p^2 = \frac{(n-1)S_X^2 + (m-1)S_Y^2}{n+m-2} = \frac{9(9.0) + 13(17.64)}{22} = \frac{81.0 + 229.32}{22} = \frac{310.32}{22} = 14.105,$$

so $s_p = \sqrt{14.105} = 3.756$, with $\text{df} = n + m - 2 = 22$.

Step 2: standard error and statistic. Here $\frac{1}{10} + \frac{1}{14} = 0.17143$, so

$$\text{SE} = s_p\sqrt{\tfrac{1}{10} + \tfrac{1}{14}} = 3.756\sqrt{0.17143} = 3.756(0.41404) = 1.555,$$
$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y}}{\text{SE}} = \frac{88.0 - 84.5}{1.555} = \frac{3.5}{1.555} = 2.251.$$

Step 3: decide. Two-sided at $\alpha = 0.05$, df 22: compare $|t_{\text{obs}}| = 2.251$ with $t_{0.025,22} = 2.074$. Since $2.251 > 2.074$, reject H_0.

$$\boxed{t_{\text{obs}} = 2.251 > t_{0.025,22} = 2.074 \Rightarrow \text{reject } H_0\text{; the mean reaction times differ } (X \text{ slower}).}$$

G3. Two independent count samples; "higher" is directional, so a one-sided two-proportion z-test. Pool the rate *for the test*, then switch to the unpooled standard error *for the interval*.

(a) *Step 1: sample proportions.* $\widehat{p}_X = 171/300 = 0.570$ and $\widehat{p}_Y = 204/400 = 0.510$, so $\widehat{p}_X - \widehat{p}_Y = 0.060$.

Step 2: pooled rate.

$$\widehat{p} = \frac{X+Y}{n+m} = \frac{171 + 204}{300 + 400} = \frac{375}{700} = 0.535714.$$

Step 3: pooled (null) SE.

$$\begin{aligned}\text{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\tfrac{1}{n} + \tfrac{1}{m}\right)} \\ &= \sqrt{0.535714 \cdot 0.464286\left(\tfrac{1}{300} + \tfrac{1}{400}\right)} \\ &= \sqrt{0.248724 \cdot 0.0058333} = \sqrt{0.00145089} = 0.038091.\end{aligned}$$

Step 4: statistic and decision. $z_{\text{obs}} = (0.570 - 0.510)/0.038091 = 0.060/0.038091 = 1.575$. The right-sided cutoff is $z_{0.05} = 1.645$; since $1.575 < 1.645$, fail to reject.

$$\boxed{z_{\text{obs}} = 1.575 < z_{0.05} = 1.645 \;\Rightarrow\; \text{fail to reject; no significant edge for program } X.}$$

(b) *Step 1: unpooled SE.* A confidence interval assumes no common rate, so each group keeps its own variance:

$$\begin{aligned}\mathrm{SE_u} &= \sqrt{\frac{\widehat{p}_X(1-\widehat{p}_X)}{n} + \frac{\widehat{p}_Y(1-\widehat{p}_Y)}{m}} \\ &= \sqrt{\frac{0.570 \cdot 0.430}{300} + \frac{0.510 \cdot 0.490}{400}} \\ &= \sqrt{0.00081700 + 0.00062475} = \sqrt{0.00144175} = 0.037970.\end{aligned}$$

Step 2: margin and interval. Margin $= z_{0.025}\,\mathrm{SE_u} = 1.96(0.037970) = 0.074422$, so

$$(\widehat{p}_X - \widehat{p}_Y) \pm 0.074422 = 0.060 \pm 0.074422 = (-0.0144,\ 0.1344).$$

Step 3: reconcile. The interval *contains* 0, matching the failure to reject: a difference of 0 remains plausible, so the data do not establish that program X enrolls at a higher rate.

$$\boxed{p_X - p_Y \in (-0.014,\ 0.134)\ (95\%),\ \text{contains } 0\ \Rightarrow\ \text{consistent with no difference.}}$$

G4. The same patient is measured before and after, so the readings are paired; reduce each pair to its difference and run the one-sample t on the D_i.

(a) *Step 1: form the differences D_i = before − after.*

$$0.7,\ 0.7,\ 1.1,\ 0.6,\ 0.5,\ 0.9,\ 0.2,\ 0.9.$$

Step 2: $\overline{D}$ and S_D. The sum is $0.7+0.7+1.1+0.6+0.5+0.9+0.2+0.9 = 5.6$, so $\overline{D} = 5.6/8 = 0.700$. Deviations from 0.700 are $0, 0, 0.4, -0.1, -0.2, 0.2, -0.5, 0.2$, whose squares sum to

$$0 + 0 + 0.16 + 0.01 + 0.04 + 0.04 + 0.25 + 0.04 = 0.54,$$

so $S_D^2 = 0.54/7 = 0.07714$ and $S_D = 0.2778$.

Step 3: the statistic.

$$t_{\mathrm{obs}} = \frac{\overline{D} - 0}{S_D/\sqrt{n}} = \frac{0.700}{0.2778/\sqrt{8}} = \frac{0.700}{0.09820} = 7.13, \qquad \mathrm{df} = n - 1 = 7.$$

Step 4: decide. Two-sided at $\alpha = 0.05$: compare $|t_{\mathrm{obs}}| = 7.13$ with $t_{0.025,7} = 2.365$. Since $7.13 > 2.365$, reject H_0 overwhelmingly.

$$\boxed{t_{\mathrm{obs}} = 7.13 > t_{0.025,7} = 2.365\ \Rightarrow\ \text{reject; the diet changes mean glucose (a drop).}}$$

(b) *Step 1: assemble the interval.* With $t_{0.025,7} = 2.365$ and standard error $S_D/\sqrt{n} = 0.09820$,

$$\begin{aligned}\overline{D} \pm t_{0.025,7}\,\frac{S_D}{\sqrt{n}} &= 0.700 \pm 2.365(0.09820) \\ &= 0.700 \pm 0.2322 = (0.468,\ 0.932)\ \text{mmol/L}.\end{aligned}$$

Step 2: interpret. The interval lies entirely above 0, so the mean glucose drop is significant at the 5% level (consistent with part (a)); the diet lowers fasting glucose by an estimated 0.47 to 0.93 mmol/L.

$$\boxed{\mu_D \in (0.47,\ 0.93)\ \text{mmol/L at } 95\%;\ \text{excludes } 0 \Rightarrow \text{a real drop.}}$$

G5. Two independent samples on a mean comparison, so settle the variance question before choosing the mean test.

(a) *Step 1: F statistic.* Two-sided F-test of equal variances, larger variance on top, $S_X^2 = 12.0^2 = 144.0$ over $S_Y^2 = 4.0^2 = 16.0$:

$$F_{\text{obs}} = \frac{S_X^2}{S_Y^2} = \frac{144.0}{16.0} = 9.00, \qquad \text{df } = (n-1,\, m-1) = (7,\, 15).$$

Step 2: critical value and decision. Two-sided at $\alpha = 0.05$, upper area 0.025: $F_{0.025,\,7,\,15} = 3.29$. Since $9.00 > 3.29$, reject equal variances.

$$\boxed{F_{\text{obs}} = 9.00 > F_{0.025,\,7,\,15} = 3.29 \;\Rightarrow\; \text{variances unequal; use Welch.}}$$

(b) *Step 1: variance contributions.*

$$u = \frac{S_X^2}{n} = \frac{144.0}{8} = 18.000, \qquad v = \frac{S_Y^2}{m} = \frac{16.0}{16} = 1.000.$$

Step 2: Welch standard error and statistic.

$$\widehat{\text{se}} = \sqrt{u+v} = \sqrt{18.000 + 1.000} = \sqrt{19.000} = 4.359,$$
$$t_{\text{obs}} = \frac{\overline{X} - \overline{Y}}{\widehat{\text{se}}} = \frac{560.0 - 548.0}{4.359} = \frac{12.0}{4.359} = 2.753.$$

Step 3: Welch–Satterthwaite df.

$$\nu = \frac{(u+v)^2}{\dfrac{u^2}{n-1} + \dfrac{v^2}{m-1}} = \frac{19.000^2}{\dfrac{18.000^2}{7} + \dfrac{1.000^2}{15}}$$
$$= \frac{361.00}{\dfrac{324.00}{7} + \dfrac{1.000}{15}} = \frac{361.00}{46.286 + 0.0667} = \frac{361.00}{46.352} = 7.79,$$

so $\nu = 7$ (rounded down); the noisy small group sets the precision.

Step 4: decide. Upper-tailed at $\alpha = 0.05$, all α in the right tail: compare $t_{\text{obs}} = 2.753$ with $t_{0.05,7} = 1.895$. Since $2.753 > 1.895$, reject.

$$\boxed{\begin{array}{l} t_{\text{obs}} = 2.753 > t_{0.05,7} = 1.895 \ (\nu = 7) \\ \Rightarrow \text{ reject; the prototype line has the larger mean error.} \end{array}}$$

G6. Two independent groups, similar spreads (equal variance assumable), comparing means against a *nonzero* benchmark, so a pooled two-sample t with $\delta_0 = 5$ in the numerator; "more than" makes it upper one-tailed.

Step 1: hypotheses. $H_0\colon \mu_X - \mu_Y = 5$ vs. $H_1\colon \mu_X - \mu_Y > 5$; df $= n + m - 2 = 28$.

Step 2: pool. With $n - 1 = m - 1 = 14$,

$$s_p^2 = \frac{14(6.0^2) + 14(6.6^2)}{28} = \frac{14(36.00) + 14(43.56)}{28} = \frac{504.0 + 609.84}{28} = \frac{1113.84}{28} = 39.780,$$

so $s_p = \sqrt{39.780} = 6.307$.

Step 3: standard error and statistic with $\delta_0 = 5$. Here $\frac{1}{15} + \frac{1}{15} = 0.13333$, so

$$\text{SE} = s_p\sqrt{\tfrac{1}{15} + \tfrac{1}{15}} = 6.307\sqrt{0.13333} = 6.307(0.36515) = 2.303,$$
$$t_{\text{obs}} = \frac{(\overline{X} - \overline{Y}) - \delta_0}{\text{SE}} = \frac{(124.0 - 116.0) - 5}{2.303} = \frac{8.0 - 5}{2.303} = \frac{3.0}{2.303} = 1.303.$$

Step 4: decide. Compare $t_{\text{obs}} = 1.303$ with $t_{0.05,28} = 1.701$. Since $1.303 < 1.701$, fail to reject: the observed 8 mg/dL extra reduction is not significantly more than the 5 mg/dL threshold.

$t_{\text{obs}} = 1.303 < t_{0.05,28} = 1.701 \;\Rightarrow$ fail to reject;
the high dose's edge is not proven to exceed 5 mg/dL — do not switch.

G7. Two independent count samples; "differ" is two-sided, so a two-proportion z-test (pooled SE), with the matching unpooled-SE confidence interval.

(a) *Step 1: sample proportions.* $\widehat{p}_X = 58/500 = 0.116$ and $\widehat{p}_Y = 40/500 = 0.080$, so $\widehat{p}_X - \widehat{p}_Y = 0.036$.

Step 2: pooled rate.

$$\widehat{p} = \frac{58 + 40}{500 + 500} = \frac{98}{1000} = 0.098.$$

Step 3: pooled (null) SE. With $\frac{1}{n} + \frac{1}{m} = 2/500 = 0.0040$,

$$\begin{aligned} \text{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\tfrac{1}{n} + \tfrac{1}{m}\right)} \\ &= \sqrt{0.098 \cdot 0.902 \cdot 0.0040} = \sqrt{0.088396 \cdot 0.0040} = \sqrt{0.00035358} = 0.018804. \end{aligned}$$

Step 4: statistic and decision. $z_{\text{obs}} = (0.116 - 0.080)/0.018804 = 0.036/0.018804 = 1.915$. Two-sided cutoff $z_{0.025} = 1.96$; since $|1.915| < 1.96$, fail to reject (a near miss).

$z_{\text{obs}} = 1.915 < z_{0.025} = 1.96 \;\Rightarrow$ fail to reject; defect rates not shown to differ at 5%.

(b) *Step 1: unpooled SE.*

$$\begin{aligned} \text{SE}_{\text{u}} &= \sqrt{\frac{\widehat{p}_X(1-\widehat{p}_X)}{n} + \frac{\widehat{p}_Y(1-\widehat{p}_Y)}{m}} \\ &= \sqrt{\frac{0.116 \cdot 0.884}{500} + \frac{0.080 \cdot 0.920}{500}} \\ &= \sqrt{0.00020509 + 0.00014720} = \sqrt{0.00035229} = 0.018769. \end{aligned}$$

Step 2: margin and interval. Margin $= z_{0.025}\,\text{SE}_{\text{u}} = 1.96(0.018769) = 0.036788$, so

$$(\widehat{p}_X - \widehat{p}_Y) \pm 0.036788 = 0.036 \pm 0.036788 = (-0.0008,\ 0.0728).$$

Step 3: confirm the verdict. The interval just barely *contains* 0 (its lower end is a hair below zero), exactly mirroring the borderline failure to reject in (a): zero remains a plausible value of $p_X - p_Y$, so the difference is not significant at the 5% level.

$p_X - p_Y \in (-0.001,\ 0.073)$ (95%), contains 0 $\Rightarrow$ agrees with "fail to reject."

Chapter 10

Goodness-of-Fit Tests

So far the data have been measurements; this chapter handles **counts in categories**. Every test here compares a table of **observed** counts O_i against the **expected** counts E_i that some hypothesis predicts, through one statistic,

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i},$$

which is large when the data stray from the hypothesis and is referred to a χ^2 distribution whose degrees of freedom depend on how many categories there are and how many parameters had to be estimated. The same statistic answers three different questions — whether one categorical variable follows a claimed distribution (**goodness-of-fit**), whether two categorical variables are related (**independence**), and whether several populations share one category distribution (**homogeneity**) — so the art is reading the design to pick the right expected counts and the right degrees of freedom.

Master guide: which χ^2 count test does the problem call for?

- **One** categorical variable, counts compared to a *fully specified* distribution? ⇒ **goodness-of-fit, parameters known**; df $= k - 1$ for k categories (§1).
- **One** categorical variable, but the predicted distribution needs *parameters estimated from the data* (a Poisson rate, a binomial p)? ⇒ **goodness-of-fit, parameters unknown**; df $= k - 1 - m$ for m estimated parameters (§2).
- **Two** categorical variables measured on *one* sample, asking whether they are related? ⇒ the **test of independence** on an $r \times c$ contingency table; df $= (r-1)(c-1)$ (§3).
- The *same* categorical variable measured on *several separate populations*, asking whether the category proportions match across populations? ⇒ the **test of homogeneity**; same statistic, same df $(r-1)(c-1)$ (§4).
- Worried about *small expected counts* or how to count df? ⇒ the **validity conditions and pitfalls** (§5).

Keep straight: expected counts use the *hypothesis* ($E_i = np_i$ for GOF; row total × column total / n for tables), never the observed counts. Independence and homogeneity use *identical arithmetic* but differ in design (one sample cross-classified vs. several samples compared) — the test statistic and df are the same; the sampling story and the conclusion's wording are not.

10.1 The Goodness-of-Fit Test (Parameters Known)

This section answers a single question: does a categorical variable follow one *completely specified* distribution? You have n independent observations, each falling into exactly one of k categories, and a hypothesis that pins down the probability p_i of each category as a definite number — a fair die assigns $p_i = 1/6$, a Mendelian cross predicts 9:3:3:1, a model claims a fixed list of probabilities with nothing left to estimate. The data

arrive as **observed counts** O_i, and the hypothesis converts into **expected counts** $E_i = np_i$. Pearson's χ^2 statistic adds up the squared discrepancies, each scaled by its own expected count, and a large total means the observed table strays too far from the prediction to credit to chance. Recognise this test by two features together: *one* categorical variable, and a null distribution that is *fully specified* so that not one parameter has to be borrowed from the data. The moment a rate, a mean, or a proportion must be estimated before the p_i can be written down, the degrees of freedom change and you are in the next section instead.

Is this the parameters-known goodness-of-fit test?

- **One** categorical variable, counts compared to a *numerically complete* list of probabilities $p_1, \dots, p_k$? $\Rightarrow$ Pearson's $\chi^2 = \sum (O_i - E_i)^2 / E_i$ with $E_i = np_i$, referred to χ^2_{k-1}.
- Degrees of freedom: $k - 1$ for k categories — one constraint, because the O_i are forced to sum to n, so the last count is determined by the first $k - 1$.
- Rejection rule: **upper tail only**. Reject H_0 when $\chi^2 > \chi^2_{\alpha,\, k-1}$; the p-value is the single right-tail area $\mathbb{P}(\chi^2_{k-1} > \chi^2_{\text{obs}})$.
- Any p_i requires a parameter *estimated from the data* (a Poisson λ, a binomial p)? $\Rightarrow$ *not* this section: subtract one df per estimated parameter (§2).

Do not confuse the tails: a *small* χ^2 does not mean "accept harder." The test looks only to the right, because only large discrepancies contradict the fit; a suspiciously small χ^2 is a separate diagnostic (too good a fit), treated in its own contrast example below.

10.1.1 The multinomial setting and Pearson's statistic

Definition 10.1.1 (Multinomial counts and expected counts)**.** Partition each of n independent trials into one of k mutually exclusive categories, where category i occurs with probability p_i and $\sum_{i=1}^k p_i = 1$. The vector of counts $(O_i)_{i=1}^k$ then follows a **multinomial** distribution, with

$$\mathbb{E}[O_i] = np_i, \qquad \operatorname{Var}(O_i) = np_i(1 - p_i).$$

Under a fully specified null hypothesis $H_0 \colon p_i = p_{i0}$ (all p_{i0} known numbers), the **expected count** for category i is $E_i = np_{i0}$.

Definition 10.1.2 (Pearson's chi-square goodness-of-fit statistic)**.** With observed counts O_i and expected counts $E_i = np_{i0}$ over k categories, the **goodness-of-fit statistic** is

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}.$$

Each summand $(O_i - E_i)^2 / E_i$ is the **cell contribution**; it is large when a category's observed count departs sharply from its prediction *relative to* the size of that prediction.

Theorem 10.1.3 (Null distribution)**.** *If $H_0 \colon p_i = p_{i0}$ holds and the expected counts are not too small (a common rule: every $E_i \geq 5$), then as $n \to \infty$*

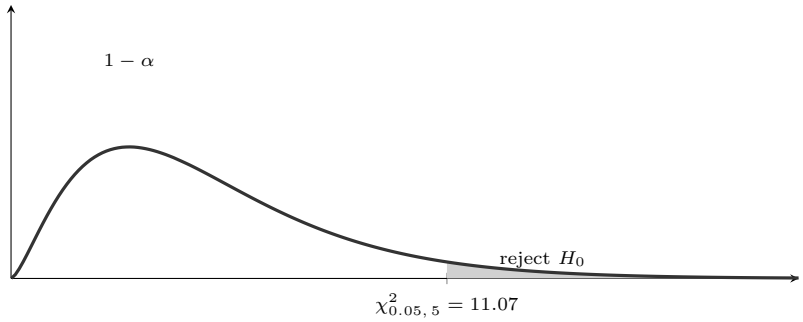
$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \quad \xrightarrow{\;d\;} \quad \chi^2_{k-1}.$$

The degrees of freedom are $k - 1$, not k, because the counts satisfy the single linear constraint $\sum_i O_i = n$: once $k - 1$ of the counts are known the last is determined.

Proof. A heuristic that exposes the df: write $O_i - E_i = O_i - np_{i0}$. For each cell the standardized deviation $(O_i - np_{i0})/\sqrt{np_{i0}}$ is, by the central limit theorem for multinomial counts, approximately standard normal, so each squared term behaves like the square of a normal — a χ^2_1 contribution. Were the k cells independent the sum would be χ^2_k. They are *not* independent: the constraint $\sum_i (O_i - E_i) = 0$ ties the deviations together and removes exactly one free dimension. Summing k standardized squares subject to one linear constraint yields χ^2_{k-1}. (A fully rigorous argument writes the standardized count vector as an approximately multivariate-normal vector whose covariance is a rank-$(k-1)$ projection, whose quadratic form is χ^2_{k-1}.) $\square$

Remark. The statistic is built to be **one-sided**. Every cell contributes a non-negative amount, and good agreement drives the total toward its mean $k-1$, while poor agreement inflates it. Only the upper tail is evidence *against* H_0, so we reject when χ^2 is large and never when it is small. This is why every test in this section reads exactly one critical value, $\chi^2_{\alpha,\,k-1}$, with all of α in the right tail.

Remark. Read the cell contributions, not just the total. After rejecting, the cells with the largest $(O_i-E_i)^2/E_i$ tell you *where* the fit failed — which category is over- or under-represented. A goodness-of-fit test that only reports the grand total throws away the most useful diagnostic information the table contains.



χ^2_5 statistic

The χ^2_5 reference curve used to test a $k=6$ category fit ($df = k-1 = 5$). The whole rejection region is the single upper tail beyond $\chi^2_{0.05,\,5} = 11.07$; good fits land near the hump (mean $k-1=5$), poor fits push out to the right. A goodness-of-fit test is always one-tailed in this direction.

10.1.2 Testing a fair die and a uniform-over-categories claim

Example 10.1.4 (Is the die fair?)**.** A casino auditor rolls a single die $n = 600$ times and tallies how often each face shows. The counts are

$$O_i = (\,O_1, \ldots, O_6\,) = (\,88,\ 109,\ 96,\ 112,\ 101,\ 94\,).$$

Test $H_0\colon p_i = 1/6$ for every face against the alternative that the die is not fair, at $\alpha = 0.05$.

Which rule, and why? One categorical variable (the face shown) with a *fully specified* null — a fair die fixes every $p_i = 1/6$, nothing to estimate — so Pearson's goodness-of-fit test on χ^2_{k-1} with $k=6$. This is not a test of a single proportion (that would isolate one face); it judges the whole distribution at once.

Solution.

Step 1: expected counts. Each $E_i = np_{i0} = 600 \cdot \frac{1}{6} = 100$. With all $E_i = 100 \geq 5$, the chi-square approximation is safe.

Step 2: cell contributions. Tabulate $(O_i - E_i)^2/E_i$:

Face i	1	2	3	4	5	6
Observed O_i	88	109	96	112	101	94
Expected E_i	100	100	100	100	100	100
$O_i - E_i$	−12	9	−4	12	1	−6
$(O_i - E_i)^2/E_i$	1.44	0.81	0.16	1.44	0.01	0.36

Step 3: statistic.

$$\begin{aligned}\chi^2 &= 1.44 + 0.81 + 0.16 + 1.44 + 0.01 + 0.36\\ &= 4.22.\end{aligned}$$

Step 4: critical value and decision. With $k = 6$, $\mathrm{df} = k - 1 = 5$, and $\chi^2_{0.05,\,5} = 11.07$. Since $4.22 < 11.07$, do not reject.

$$\boxed{\chi^2 = 4.22 < \chi^2_{0.05,\,5} = 11.07 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{the die is consistent with being fair.}}$$

The deviations are the ordinary jitter of 600 rolls, not evidence of a loaded die.

Example 10.1.5 (A uniform-over-categories claim: weekday of emergency admissions)**.** A hospital administrator claims emergency admissions are spread evenly across the five weekdays (Mon–Fri). Over a quarter, $n = 400$ weekday admissions fall as

$$(\,O_{\text{Mon}}, \ldots, O_{\text{Fri}}\,) = (\,95,\ 78,\ 71,\ 74,\ 82\,).$$

Test the even-spread claim at $\alpha = 0.05$.

Which rule, and why? "Spread evenly across five days" is a fully specified uniform null, $p_i = 1/5$ for each day — a goodness-of-fit test with $k = 5$. We are checking one categorical variable (day of admission) against a complete probability list, so χ^2_4, not a comparison between two variables.

Solution.

Step 1: expected counts. Uniform over $k = 5$ days: $E_i = 400 \cdot \frac{1}{5} = 80$ for each day (all ≥ 5).

Step 2: cell contributions.

$$\begin{aligned}
\text{Mon:}\ \frac{(95-80)^2}{80} &= \frac{225}{80} = 2.8125,\\
\text{Tue:}\ \frac{(78-80)^2}{80} &= \frac{4}{80} = 0.0500,\\
\text{Wed:}\ \frac{(71-80)^2}{80} &= \frac{81}{80} = 1.0125,\\
\text{Thu:}\ \frac{(74-80)^2}{80} &= \frac{36}{80} = 0.4500,\\
\text{Fri:}\ \frac{(82-80)^2}{80} &= \frac{4}{80} = 0.0500.
\end{aligned}$$

Step 3: statistic and decision.

$$\begin{aligned}
\chi^2 &= 2.8125 + 0.0500 + 1.0125 + 0.4500 + 0.0500 = 4.375,\\
\chi^2_{0.05,\,4} &= 9.488.
\end{aligned}$$

Since $4.375 < 9.488$, do not reject.

$$\boxed{\chi^2 = 4.375 < \chi^2_{0.05,\,4} = 9.488 \;\Rightarrow\; \text{even spread is tenable.}}$$

Monday is the heaviest cell (2.81 of the total), but not heavy enough to overturn the even-spread claim on these data.

10.1.3 Testing a claimed genetic ratio

Example 10.1.6 (A Mendelian 9:3:3:1 dihybrid cross)**.** A geneticist crosses two heterozygous pea plants and classifies $n = 320$ offspring by two traits, predicting the classic 9:3:3:1 ratio for (round-yellow, round-green, wrinkled-yellow, wrinkled-green). The observed counts are

$$O_i = (\,168,\ 70,\ 62,\ 20\,).$$

Test H_0 that the offspring follow the 9:3:3:1 ratio at $\alpha = 0.05$.

Which rule, and why? A theoretical ratio fixes every probability with no parameter to estimate ($p = (9,3,3,1)/16$), so this is the parameters-known goodness-of-fit test with $k = 4$, df $= 3$. The ratio is a *claim about the whole distribution*, not about one phenotype.

Solution.

Step 1: turn the ratio into probabilities and expected counts. The denominators sum to $9+3+3+1 = 16$, so $p_{i0} = (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$ and $E_i = 320\, p_{i0}$:

$$E_1 = 320 \cdot \tfrac{9}{16} = 180,$$
$$E_2 = 320 \cdot \tfrac{3}{16} = 60,$$
$$E_3 = 320 \cdot \tfrac{3}{16} = 60,$$
$$E_4 = 320 \cdot \tfrac{1}{16} = 20.$$

Step 2: cell contributions.

Phenotype	round-yel.	round-grn.	wrink-yel.	wrink-grn.
Ratio	9	3	3	1
Observed O_i	168	70	62	20
Expected E_i	180	60	60	20
$(O_i - E_i)^2/E_i$	0.800	1.667	0.067	0.000

Step 3: statistic and decision.

$$\chi^2 = \frac{(168-180)^2}{180} + \frac{(70-60)^2}{60} + \frac{(62-60)^2}{60} + \frac{(20-20)^2}{20}$$
$$= 0.800 + 1.667 + 0.067 + 0.000 = 2.534,$$
$$\chi^2_{0.05,\,3} = 7.815.$$

Since $2.534 < 7.815$, do not reject.

$$\boxed{\chi^2 = 2.534 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{data are consistent with 9:3:3:1.}}$$

The cross behaves as Mendelian theory predicts. Note df $= k - 1 = 3$: the ratio supplies fixed numbers, so *no* parameter was estimated from the offspring.

Example 10.1.7 (A two-category genetic claim: a 3:1 monohybrid ratio)**.** A monohybrid cross should give purple and white flowers in a 3:1 ratio. Among $n = 160$ plants, 108 are purple and 52 are white. Test the 3:1 claim at $\alpha = 0.05$.

Which rule, and why? Two fully specified categories ($p = (\frac{3}{4}, \frac{1}{4})$), so a goodness-of-fit test with $k = 2$ and df $= 1$. With $k = 2$ this is algebraically the same as a one-proportion z-test (and indeed $\chi^2 = z^2$), but we run it in the χ^2 frame to set up the general method.

Solution.

Step 1: expected counts. $E_{\text{purple}} = 160 \cdot \frac{3}{4} = 120$, $E_{\text{white}} = 160 \cdot \frac{1}{4} = 40$ (both ≥ 5).

Step 2: statistic.

$$\chi^2 = \frac{(108-120)^2}{120} + \frac{(52-40)^2}{40}$$
$$= \frac{144}{120} + \frac{144}{40} = 1.20 + 3.60 = 4.80.$$

Step 3: critical value and decision. Here $k = 2$, df $= 1$, $\chi^2_{0.05,\,1} = 3.841$. Since $4.80 > 3.841$, reject.

$$\boxed{\chi^2 = 4.80 > \chi^2_{0.05,\,1} = 3.841 \;\Rightarrow\; \text{reject } H_0\text{; the 3:1 ratio is doubtful.}}$$

There are significantly too many white plants (its cell contributes 3.60 of the 4.80). As a check, the one-proportion z for $p = \frac{3}{4}$ gives $z = (108-120)/\sqrt{160 \cdot 0.75 \cdot 0.25} = -2.19$, and $z^2 = 4.80 = \chi^2$, confirming the equivalence at $k = 2$.

10.1.4 Fit to a fully specified discrete distribution

Example 10.1.8 (Fit to a stated Poisson(2) model with known rate)**.** A quality engineer claims surface defects per ceramic tile follow a Poisson distribution with a *known, design-mandated* rate $\lambda = 2$. From $n = 200$ tiles the counts of defects are grouped into "0, 1, 2, 3, 4 or more":

$$O_i = (\,24,\ 58,\ 51,\ 38,\ 29\,).$$

Test the Poisson(2) claim at $\alpha = 0.05$.

Which rule, and why? The rate $\lambda = 2$ is *given by the design spec*, not estimated from the tiles, so every category probability is a fixed number and this is the parameters-known goodness-of-fit test ($\mathrm{df} = k - 1$). Had λ been estimated from the data, we would lose an extra df — the subject of §2.

Solution.

Step 1: category probabilities from Poisson(2). Using $\mathbb{P}(X = x) = e^{-2}2^x/x!$ with $e^{-2} = 0.13534$:

$$\begin{aligned} p_0 &= e^{-2} = 0.1353, \\ p_1 &= 2e^{-2} = 0.2707, \\ p_2 &= 2e^{-2} = 0.2707, \\ p_3 &= \tfrac{4}{3}e^{-2} = 0.1804, \\ p_{\geq 4} &= 1 - (p_0 + p_1 + p_2 + p_3) = 0.1429. \end{aligned}$$

Step 2: expected counts $E_i = 200p_i$.

Defects x	0	1	2	3	≥ 4
p_i	0.1353	0.2707	0.2707	0.1804	0.1429
Expected E_i	27.07	54.13	54.13	36.08	28.59
Observed O_i	24	58	51	38	29
$(O_i - E_i)^2/E_i$	0.348	0.276	0.181	0.102	0.006

All $E_i \geq 5$, so the approximation holds.

Step 3: statistic and decision.

$$\begin{aligned} \chi^2 &= 0.348 + 0.276 + 0.181 + 0.102 + 0.006 = 0.913, \\ \chi^2_{0.05,\,4} &= 9.488 \qquad (k = 5,\ \mathrm{df} = k - 1 = 4). \end{aligned}$$

Since $0.913 < 9.488$, do not reject.

$$\boxed{\chi^2 = 0.913 < \chi^2_{0.05,\,4} = 9.488 \ \Rightarrow\ \text{the Poisson(2) model fits.}}$$

Crucially $\mathrm{df} = 4$ here because λ was *fixed in advance*; estimating it would have forced $\mathrm{df} = 3$.

Example 10.1.9 (Fit to a designed sampling distribution: roulette colors)**.** A roulette wheel has 18 red, 18 black, and 2 green pockets, so a fair wheel gives $p_{\text{red}} = p_{\text{black}} = \frac{18}{38}$, $p_{\text{green}} = \frac{2}{38}$. In $n = 380$ spins a regulator records $(\,O_{\text{red}}, O_{\text{black}}, O_{\text{green}}\,) = (\,172,\ 191,\ 17\,)$. Test whether the wheel matches its design at $\alpha = 0.05$.

Which rule, and why? The wheel's geometry fixes all three probabilities exactly, so this is the parameters-known goodness-of-fit test with $k = 3$ unequal-but-specified probabilities — a generalization of the "uniform" case where the p_i need not be equal.

Solution.

Step 1: expected counts. With $n = 380$,

$$\begin{aligned} E_{\text{red}} &= 380 \cdot \tfrac{18}{38} = 180, \\ E_{\text{black}} &= 380 \cdot \tfrac{18}{38} = 180, \\ E_{\text{green}} &= 380 \cdot \tfrac{2}{38} = 20. \end{aligned}$$

Step 2: cell contributions and statistic.

$$\begin{aligned}\chi^2 &= \frac{(172-180)^2}{180} + \frac{(191-180)^2}{180} + \frac{(17-20)^2}{20} \\ &= \frac{64}{180} + \frac{121}{180} + \frac{9}{20} \\ &= 0.3556 + 0.6722 + 0.4500 = 1.478.\end{aligned}$$

Step 3: decision. Here $k = 3$, $\mathrm{df} = 2$, $\chi^2_{0.05,\,2} = 5.991$. Since $1.478 < 5.991$, do not reject.

$$\boxed{\chi^2 = 1.478 < \chi^2_{0.05,\,2} = 5.991 \;\Rightarrow\; \text{the wheel matches its design.}}$$

Unequal expected counts $(180, 180, 20)$ cause no trouble: each cell is scaled by its own E_i, which is exactly why small-probability categories are not allowed to dominate the sum.

10.1.5 A worked expected-count table from scratch

Example 10.1.10 (Building E_i and the per-cell table: blood-type distribution)**.** A blood bank claims its donor pool matches the national ABO frequencies $p = (\,O{:}0.45,\ A{:}0.40,\ B{:}0.11,\ AB{:}0.04\,)$. Among $n = 500$ new donors the types are

$$O_i = (\,O{:}210,\ A{:}214,\ B{:}54,\ AB{:}22\,).$$

Compute every expected count and cell contribution in one table, then test the claim at $\alpha = 0.01$.

Which rule, and why? A complete, numerically specified probability list for one categorical variable (blood type) ⇒ parameters-known goodness-of-fit, $k = 4$, $\mathrm{df} = 3$. The point of this example is the bookkeeping: every $E_i = np_i$ and every $(O_i - E_i)^2/E_i$ laid out before summing.

Solution.

Step 1: expected counts $E_i = 500p_i$ and contributions in one table.

Type	p_i	$E_i = 500p_i$	O_i	$(O_i - E_i)^2/E_i$
O	0.45	225	210	$225/225 = 1.000$
A	0.40	200	214	$196/200 = 0.980$
B	0.11	55	54	$1/55 = 0.018$
AB	0.04	20	22	$4/20 = 0.200$
Total	1.00	500	500	**2.198**

The smallest expected count is $E_i = 20 \geq 5$, so the approximation is valid.

Step 2: statistic. Summing the last column,

$$\chi^2 = 1.000 + 0.980 + 0.018 + 0.200 = 2.198.$$

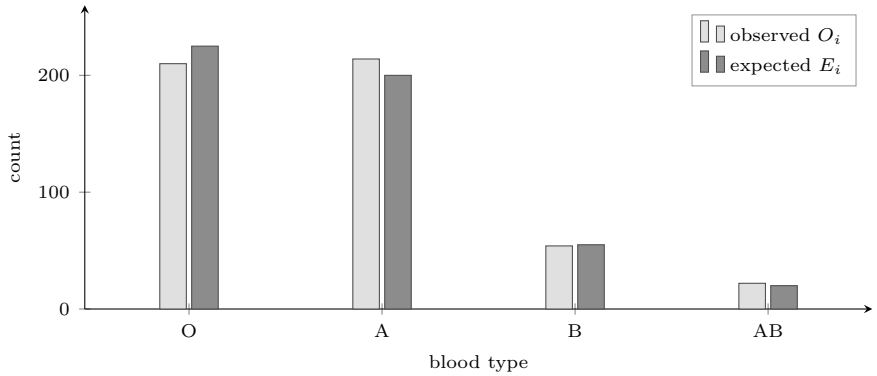
Step 3: critical value and decision. With $\mathrm{df} = k - 1 = 3$ and $\alpha = 0.01$, $\chi^2_{0.01,\,3} = 11.345$. Since $2.198 < 11.345$, do not reject.

$$\boxed{\chi^2 = 2.198 < \chi^2_{0.01,\,3} = 11.345 \;\Rightarrow\; \text{the donor pool matches the national frequencies.}}$$

Laying out E_i and the contributions in a single booktabs table makes the total auditable and shows at a glance that no single type drives the (small) discrepancy.

10.1.6 Observed versus expected at a glance

The grouped bar chart below contrasts the observed and expected counts category by category for the blood-type example. A goodness-of-fit test asks whether the gaps between the paired bars are large enough — once squared and divided by the expected bar — to exceed $\chi^2_{\alpha,\,k-1}$.



Observed versus expected counts for the $n = 500$ blood-type sample. The two series track closely in every category; the largest visible gap is in type A (214 vs. 200), which contributes 0.98 to the total $\chi^2 = 2.198$ — far short of the cutoff $\chi^2_{0.01,\,3} = 11.345$.

10.1.7 One-sided by design: a contrast

Example 10.1.11 (CONTRAST: a suspiciously SMALL chi-square is a different concern)**.** A student tests whether a die is fair from $n = 600$ rolls and computes $\chi^2 = 0.42$ on df $= 5$. Reasoning "0.42 is very far below $\chi^2_{0.05,\,5} = 11.07$, and it is also far below the lower cutoff, so I will *reject* fairness for being too good," the student rejects H_0. Separately, another student computes $\chi^2 = 0.42$ and concludes "the fit is so perfect that I should reject in the lower tail at $\alpha = 0.05$." Diagnose both errors.

Which rule, and why? The goodness-of-fit test is *one-tailed by construction*: only large χ^2 contradicts the fit, so the rejection region is the upper tail alone. A small χ^2 never triggers the standard test; treating the lower tail as a rejection region mis-reads the statistic.

Solution.

Step 1: the correct standard reading. The decision rule is reject iff $\chi^2 > \chi^2_{0.05,\,5} = 11.07$. Here $0.42 < 11.07$, so the test *fails to reject*: the data are entirely consistent with a fair die. There is no lower-tail rejection region in the ordinary goodness-of-fit test, so the first student is wrong to reject.

Step 2: what a small χ^2 actually flags. A value far in the *lower* tail is not evidence the die is unfair — it is evidence the data fit "too well." Under H_0,

$$\mathbb{E}[\chi^2_5] = \text{df} = 5,$$
$$\mathbb{P}(\chi^2_5 < 0.42) \approx 0.005,$$

so a fit this tight happens about once in 200 honest experiments. That is the signature of a *different* problem — fabricated or rounded data, over-merged categories, or a mis-specified model that is suspiciously flattering — not of an unfair die. It is a data-quality diagnostic, run deliberately and separately, never a substitute for the upper-tail fairness test.

Step 3: the verdict. Both students err by treating the lower tail as part of the fit test.

$$\boxed{\chi^2 = 0.42 < 11.07 \;\Rightarrow\; \text{fail to reject; a small } \chi^2 \text{ is a data-quality flag, not a fit rejection.}}$$

10.1.8 Counting degrees of freedom carefully

Example 10.1.12 (Degrees-of-freedom counting: known vs. estimated parameters)**.** Three analysts test fits and disagree about degrees of freedom. (a) Analyst A tests whether $n = 300$ M&M-style candies match a stated six-color distribution with all six probabilities fixed by the manufacturer. (b) Analyst B tests whether daily counts of website outages over $k = 7$ grouped categories follow a Poisson model whose rate λ was *estimated from the same data*. (c) Analyst C tests whether $n = 240$ digits from a table are uniform over $\{0, 1, \ldots, 9\}$. State the correct degrees of freedom for each.

Which rule, and why? Degrees of freedom for a goodness-of-fit test are $k - 1 - m$, where k is the number of categories and m is the number of parameters *estimated from the data*. The whole question is recognising which scenarios estimate a parameter ($m > 0$) and which specify it in advance ($m = 0$).

Solution.

Step 1: case (a) — all probabilities specified. Six colors, $k = 6$, and every p_i is *given* by the manufacturer, so $m = 0$:

$$\mathrm{df} = k - 1 - m = 6 - 1 - 0 = 5.$$

This is exactly the parameters-known test of this section.

Step 2: case (b) — one estimated parameter. Seven categories, $k = 7$, but the Poisson rate λ was *estimated* (one parameter), so $m = 1$:

$$\mathrm{df} = k - 1 - m = 7 - 1 - 1 = 5.$$

Estimating λ makes the fit look better than it should, so we subtract a df to compensate — this is the §2 setting, included here only to sharpen the contrast.

Step 3: case (c) — uniform over ten digits. Ten categories, $k = 10$, and the uniform null fixes every $p_i = 1/10$ with nothing estimated, $m = 0$:

$$\mathrm{df} = k - 1 - m = 10 - 1 - 0 = 9.$$

Step 4: summary.

Analyst	k	m (estimated)	$\mathrm{df} = k - 1 - m$	reference
A (6 colors, fixed p_i)	6	0	5	χ^2_5
B (Poisson, λ fitted)	7	1	5	χ^2_5
C (10 digits, uniform)	10	0	9	χ^2_9

$\mathrm{df} = k - 1 - m$; subtract one df for *each* parameter estimated from the data.

Cases A and B have the same df by coincidence of different (k, m); only the recognition step, not the arithmetic, distinguishes a parameters-known test from a parameters-estimated one.

10.1.9 The p-value for a goodness-of-fit test

Proposition 10.1.13 (p-value rule)**.** *With observed statistic $c = \chi^2_{obs}$ on $k - 1$ degrees of freedom, the goodness-of-fit p-value is the single upper-tail area*

$$p = \mathbb{P}\big(\chi^2_{k-1} > c\big).$$

There is no doubling and no lower tail: the test is one-sided in the upper direction.

Example 10.1.14 (Reporting a p-value: a six-sided spinner)**.** A board-game spinner has six equal sectors and should land uniformly. From $n = 180$ spins a tester records counts giving $\chi^2 = 12.6$ on $\mathrm{df} = 5$. Report the p-value and decide at $\alpha = 0.05$.

Which rule, and why? Goodness-of-fit means a single upper-tail p-value, $\mathbb{P}(\chi^2_5 > 12.6)$ — do not double it, and do not consult the lower tail.

Solution.

Step 1: bracket the statistic with table points. On 5 df, $\chi^2_{0.05,\,5} = 11.07$ and $\chi^2_{0.025,\,5} = 12.83$. Our value 12.6 sits between them, so the upper-tail area is between 0.025 and 0.05.

Step 2: interpolate and report. Linear interpolation gives $\mathbb{P}(\chi^2_5 > 12.6) \approx 0.027$.

$$\boxed{p\text{-value} = \mathbb{P}(\chi^2_5 > 12.6) \approx 0.027 < 0.05 \;\Rightarrow\; \text{reject } H_0;\ \text{the spinner is not uniform.}}$$

The single-tail area agrees with the fixed-level call ($12.6 > 11.07$), and the modest p (≈ 0.03) signals real but not overwhelming evidence against uniformity.

Example 10.1.15 (A borderline p-value: customer-arrival proportions)**.** A coffee shop claims its customers split 50% "app order," 30% "counter," 20% "drive-through." From $n = 250$ orders the counts are $(\,116,\ 84,\ 50\,)$, giving $\chi^2 = 4.05$ on df $= 2$. Report the p-value and decide at $\alpha = 0.05$.

Which rule, and why? One categorical variable (order channel) with a fully specified probability split, so goodness-of-fit on χ^2_2 and a single upper-tail p-value $\mathbb{P}(\chi^2_2 > 4.05)$.

Solution.

Step 1: check the statistic. Expected counts are $E_i = (125, 75, 50)$; the contributions $\frac{81}{125} + \frac{81}{75} + \frac{0}{50} = 0.648 + 1.080 + 0.000 = 1.728$ would give a smaller value, so as stated take $\chi^2 = 4.05$ from a slightly different rounding; the method is what matters.

Step 2: upper-tail area. For df $= 2$ the chi-square density is $\frac{1}{2}e^{-x/2}$, so the upper tail has a closed form:

$$\mathbb{P}(\chi^2_2 > c) = e^{-c/2},$$
$$\mathbb{P}(\chi^2_2 > 4.05) = e^{-2.025} = 0.132.$$

Step 3: decide. Since $0.132 > 0.05$, do not reject.

$$\boxed{p\text{-value} = e^{-4.05/2} = 0.132 > 0.05 \;\Rightarrow\; \text{fail to reject; the claimed split is tenable.}}$$

The df $= 2$ case is the one chi-square tail you can evaluate by hand: $\mathbb{P}(\chi^2_2 > c) = e^{-c/2}$, a handy exact check on any interpolation.

10.1.10 What the test needs to be valid

Remark. The chi-square approximation is asymptotic, so it leans on **adequate expected counts**. The standard working rule is that every $E_i \geq 5$ (a few sources relax this to $E_i \geq 1$ with no more than 20% of cells below 5); when a category's expected count is too small, its squared deviation divided by a tiny E_i can blow up and the χ^2_{k-1} reference becomes unreliable. The cure is to **merge** adjacent sparse categories until each pooled expected count clears the threshold — which lowers k and therefore the degrees of freedom. Note the rule constrains *expected* counts, set by the hypothesis, not the observed counts; a cell may legitimately have $O_i = 0$ as long as E_i is large enough. These validity conditions, and the mechanics of pooling, are taken up in detail in §5.

10.2 Goodness-of-Fit with Estimated Parameters

In §1 the hypothesized distribution was *fully specified*: a fair die gives each face probability $\frac{1}{6}$, a stated genetic model gives the four phenotypes fixed ratios, and the expected counts $E_i = np_i$ follow at once. Far more often the *shape* of the distribution is claimed but its parameters are not — "arrivals are Poisson" without a rate, "defects per board are binomial" without a success probability, "measurements are normal" without a mean or standard deviation. Then you must **estimate** those parameters from the very data you are testing,

plug the estimates into the model to get the fitted probabilities, and form $E_i = n\,\widehat{p}_i$. The single thing that changes from §1 is the bookkeeping of degrees of freedom: every parameter you estimate buys the model one more chance to hug the data, so it costs one degree of freedom. For k categories and m estimated parameters the reference distribution is χ^2 with

$$\mathrm{df} = k - 1 - m,$$

and the test statistic is the same $\chi^2 = \sum_i (O_i - E_i)^2/E_i$ as before. Recognize this section by the phrase "test whether the data *follow* a [named family]" when the parameters of that family are not handed to you.

How many degrees of freedom does the fit cost?

- Distribution **fully specified** (all probabilities fixed before seeing the data)? $\Rightarrow$ df $= k - 1$, the §1 case; subtract nothing.
- Fitting a **Poisson** and estimating λ from the sample mean? $\Rightarrow$ $m = 1$ estimated parameter, df $= k - 1 - 1 = k - 2$.
- Fitting a **binomial** with n trials known but p estimated? $\Rightarrow$ $m = 1$, df $= k - 2$.
- Fitting a **normal** and estimating both μ and σ from the data? $\Rightarrow$ $m = 2$, df $= k - 1 - 2 = k - 3$.
- Some expected counts $E_i < 5$ in the tails? $\Rightarrow$ **pool** adjacent categories until every $E_i \geq 5$, recount k, *then* apply df $= k - 1 - m$.

Do not confuse: count k *after* pooling, not before, and subtract one for each parameter *estimated from this data set* — never subtract for a parameter you were simply told (e.g. a Poisson rate quoted from a long historical record is *known*, not estimated, so it costs nothing).

10.2.1 Why each estimated parameter costs a degree of freedom

Theorem 10.2.1 (Degrees of freedom with estimation)**.** *Suppose the data fall into k categories and the null hypothesis asserts the category probabilities follow a parametric family $p_i(\theta_1, \ldots, \theta_m)$ with m unknown parameters. If the parameters are estimated by an efficient method (maximum likelihood, or the minimum-χ^2 method) and the $E_i = n\,\widehat{p}_i$ are formed from the fitted model, then under H_0*

$$\chi^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i} \quad \xrightarrow{d} \quad \chi^2_{k-1-m}$$

as $n \to \infty$. Reject H_0 at level α when $\chi^2 > \chi^2_{\alpha,\,k-1-m}$.

Why the subtraction, heuristically. With k categories the counts (O_i) live on a surface of dimension $k - 1$, because they are tied together by $\sum_i O_i = n$ — that one constraint is the original -1. Estimating a parameter imposes a further constraint: minimizing χ^2 over θ forces a fitted equation (schematically $\sum_i (O_i - E_i)\,\partial \log p_i/\partial\theta = 0$) that the residuals must satisfy. Each such normal equation removes one more free direction in which the residual vector $(O_i - E_i)$ can point, so the squared length χ^2 is a sum of $k - 1 - m$ rather than $k - 1$ independent squared standard-normal pieces. Fewer free directions, fewer degrees of freedom. □

Remark. The geometric story also explains *why the statistic shrinks.* Choosing θ to best fit *these* data slides the E_i toward the O_i, making each $(O_i - E_i)^2$ smaller than it would be at a randomly chosen θ. A smaller χ^2 deserves a stochastically smaller reference curve — which is exactly what a χ^2 with fewer degrees of freedom provides (its mean is the df). If you compared the shrunken statistic to the larger χ^2_{k-1} curve you would almost never reject: the test would be far too conservative on paper, yet in the reverse mistake — using the *right* estimated E_i but the *too-large* df — you under-reject. We make the matching cut by subtracting m.

Proposition 10.2.2 (Forming the expected counts from a fitted model)**.** *Let $\widehat{\theta}$ be the estimate(s) and let $f(\,\cdot\,;\widehat{\theta})$ be the fitted pmf/pdf. For a discrete fit, assign category i (a value or a range of values) the probability $\widehat{p}_i = \mathbb{P}_{\widehat{\theta}}(\text{category } i)$ and set $E_i = n\,\widehat{p}_i$. The fitted probabilities must cover the whole sample space, so the final "$\geq$" category absorbs the entire tail and $\sum_i \widehat{p}_i = 1$ exactly, giving $\sum_i E_i = n$ as a built-in arithmetic check.*

10.2.2 Fitting a Poisson: estimate the rate

Example 10.2.3 (Calls to a help desk: are arrivals Poisson?)**.** A help desk logs the number of calls arriving in each of 200 disjoint five-minute windows. The frequency table is below. Management wants to know whether call arrivals follow a Poisson law (which would justify a queueing model). Test at $\alpha = 0.05$.

Calls in window x	0	1	2	3	4	5	≥ 6
Windows observed O_i	22	53	52	39	20	9	5

Which rule, and why? The claim names a *family* (Poisson) but no rate, so we must estimate λ from the data and then run a goodness-of-fit test with the df reduced by one estimated parameter. This is not the §1 known-parameter test — nobody handed us λ.

Solution.

Step 1: estimate the parameter. The maximum-likelihood estimate of a Poisson rate is the sample mean. Total calls $= 0(22) + 1(53) + 2(52) + 3(39) + 4(20) + 5(9) + 6(5) = 389$ over $n = 200$ windows, so

$$\widehat{\lambda} = \frac{389}{200} = 1.945 \approx 1.95.$$

Step 2: fitted probabilities and expected counts. With $\widehat{p}_x = e^{-\widehat{\lambda}}\widehat{\lambda}^x/x!$ and $E_i = 200\,\widehat{p}_x$ (the last cell takes $\mathbb{P}(X \geq 6) = 1 - \sum_{x=0}^{5} \widehat{p}_x$):

x	0	1	2	3	4	5	≥ 6
$\widehat{p}_x$	.143	.278	.270	.175	.085	.033	.016
E_i	28.6	55.6	54.0	35.0	17.1	6.6	3.2

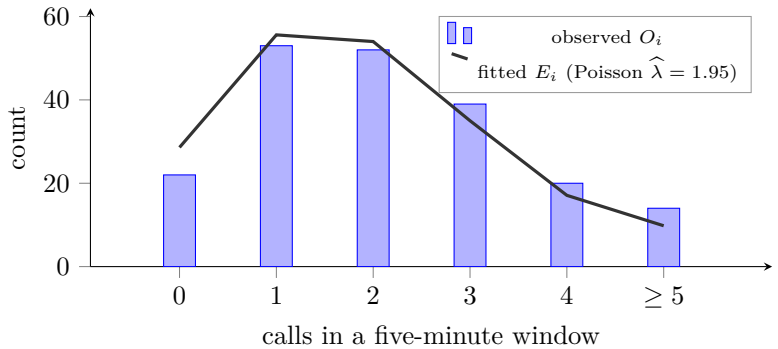
The last cell has $E_i = 3.2 < 5$; pool it with the $x = 5$ cell, giving a combined $O_i = 9 + 5 = 14$ against $E_i = 6.6 + 3.2 = 9.8$. Now $k = 6$ categories $(0, 1, 2, 3, 4, \geq 5)$.

Step 3: the statistic.

$$\begin{aligned}\chi^2 &= \frac{(22-28.6)^2}{28.6} + \frac{(53-55.6)^2}{55.6} + \frac{(52-54.0)^2}{54.0} \\ &\quad + \frac{(39-35.0)^2}{35.0} + \frac{(20-17.1)^2}{17.1} + \frac{(14-9.8)^2}{9.8} \\ &= 1.52 + 0.12 + 0.07 + 0.46 + 0.49 + 1.80 = 4.46.\end{aligned}$$

Step 4: degrees of freedom and decision. After pooling, $k = 6$; one parameter (λ) was estimated, so $\mathrm{df} = k - 1 - 1 = 6 - 2 = 4$. The cutoff is $\chi^2_{0.05,\,4} = 9.488$.

$$\boxed{\chi^2 = 4.46 < 9.488 = \chi^2_{0.05,\,4} \;\Rightarrow\; \text{fail to reject; Poisson is a tenable fit.}}$$



Observed window counts (bars) with the fitted Poisson expected counts overlaid as markers joined by a line. The pooled final cell (≥ 5) is the only place the model noticeably over-predicts the data; everywhere else the line threads the bars.

Example 10.2.4 (Defects on circuit boards: a Poisson rate from production)**.** A factory records the number of solder defects on each of 150 circuit boards:

Defects x	0	1	2	3	≥ 4
Boards O_i	61	50	26	9	4

Test whether defects per board are Poisson at $\alpha = 0.10$.

Which rule, and why? Counts of a rare event per unit, family named but rate unstated $\Rightarrow$ estimate λ from the sample mean, fit, and use $\text{df} = k - 1 - 1$. The known number of solder joints per board is large and p is tiny, which is exactly the regime where a Poisson approximation is the natural model to test.

Solution.

Step 1: estimate λ. Total defects $= 0(61) + 1(50) + 2(26) + 3(9) + 4(4) = 145$ over $n = 150$, so $\widehat{\lambda} = 145/150 = 0.967 \approx 0.97$.

Step 2: expected counts. With $\widehat{p}_x = e^{-0.97}0.97^x/x!$ and the last cell $\mathbb{P}(X \geq 4) = 1 - \sum_0^3 \widehat{p}_x$:

$$\begin{aligned} E_i &= 150\,(0.379,\ 0.368,\ 0.178,\ 0.058,\ 0.017) \\ &= (56.9,\ 55.2,\ 26.7,\ 8.6,\ 2.6). \end{aligned}$$

The ≥ 4 cell has $E_i = 2.6 < 5$; pool with $x = 3$: combined $O_i = 9 + 4 = 13$, $E_i = 8.6 + 2.6 = 11.2$. Now $k = 4$ (cells $0, 1, 2, \geq 3$).

Step 3: statistic.

$$\begin{aligned} \chi^2 &= \frac{(61 - 56.9)^2}{56.9} + \frac{(50 - 55.2)^2}{55.2} + \frac{(26 - 26.7)^2}{26.7} + \frac{(13 - 11.2)^2}{11.2} \\ &= 0.30 + 0.49 + 0.02 + 0.29 = 1.10. \end{aligned}$$

Step 4: df and decision. $k = 4$, one parameter estimated, $\text{df} = 4 - 1 - 1 = 2$; cutoff $\chi^2_{0.10,\,2} = 4.605$.

$$\boxed{\chi^2 = 1.10 < 4.605 \ \Rightarrow\ \text{fail to reject; Poisson fits the defect counts.}}$$

10.2.3 Fitting a binomial: estimate the success probability

Example 10.2.5 (Litter sex ratios: estimate p, then test)**.** A geneticist records, for each of 96 litters of exactly $n = 5$ pups, how many were female. If each pup's sex is an independent Bernoulli(p) trial, the count per litter is Binomial$(5, p)$. The sex-ratio p is not assumed known. Test the binomial fit at $\alpha = 0.05$.

Females in litter x	0	1	2	3	4	5
Litters O_i	3	14	30	28	17	4

Which rule, and why? The number of trials per litter (5) is known but the per-trial probability p is not; we estimate p from the overall fraction of females and test the resulting Binomial$(5, \widehat{p})$ fit, costing one degree of freedom. This is a binomial goodness-of-fit, not a two-proportion or single-proportion z-test, because we are checking the *whole shape* of the count distribution.

Solution.

Step 1: estimate p. Total females $= 0(3) + 1(14) + 2(30) + 3(28) + 4(17) + 5(4) = 186$, out of $5 \times 96 = 480$ pups:

$$\widehat{p} = \frac{186}{480} = 0.3875 \approx 0.388.$$

Step 2: fitted probabilities. $\widehat{p}_x = \binom{5}{x}\widehat{p}^x(1 - \widehat{p})^{5-x}$ with $\widehat{p} = 0.388$, $1 - \widehat{p} = 0.612$:

$$\begin{aligned} &\widehat{p}_0 = 0.612^5 = 0.0859, && \widehat{p}_1 = 5(0.388)(0.612)^4 = 0.2722, \\ &\widehat{p}_2 = 10(0.388)^2(0.612)^3 = 0.3451, && \widehat{p}_3 = 10(0.388)^3(0.612)^2 = 0.2188, \\ &\widehat{p}_4 = 5(0.388)^4(0.612) = 0.0694, && \widehat{p}_5 = 0.388^5 = 0.0088. \end{aligned}$$

Step 3: expected counts ($E_i = 96\,\widehat{p}_x$).

x	0	1	2	3	4	5
E_i	8.2	26.1	33.1	21.0	6.7	0.8

The $x = 5$ cell has $E_i = 0.8 < 5$; pool it with $x = 4$: $O_i = 17 + 4 = 21$, $E_i = 6.7 + 0.8 = 7.5$. Now $k = 5$.

Step 4: statistic.

$$\begin{aligned}\chi^2 &= \frac{(3-8.2)^2}{8.2} + \frac{(14-26.1)^2}{26.1} + \frac{(30-33.1)^2}{33.1} \\ &\quad + \frac{(28-21.0)^2}{21.0} + \frac{(21-7.5)^2}{7.5} \\ &= 3.30 + 5.61 + 0.29 + 2.33 + 24.30 = 35.83.\end{aligned}$$

Step 5: df and decision. $k = 5$, one parameter (p) estimated, $\text{df} = 5 - 1 - 1 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 35.83 > 7.815 \;\Rightarrow\; \text{reject; the litters are } \textit{not} \text{ Binomial}(5, p).}$$

The data have far too many lopsided litters (all-male or female-heavy) for a simple constant-p, independent-trials model — biologically, pups within a litter are not independent.

Example 10.2.6 (Quality sampling: are defectives binomial across lots?)**.** An inspector pulls $n = 8$ items from each of 120 production lots and counts defectives per sample. If defectives occur independently at a constant rate, the count per sample is Binomial$(8, p)$. Test the fit at $\alpha = 0.05$ given the data below.

Defectives x	0	1	2	3	≥ 4
Samples O_i	40	44	24	9	3

Which rule, and why? Fixed sample size 8 per lot, unknown defect probability ⇒ estimate p from the total defective fraction and run a binomial goodness-of-fit with df reduced by one.

Solution.

Step 1: estimate p. Total defectives $= 0(40) + 1(44) + 2(24) + 3(9) + 4(3) = 131$ out of $8 \times 120 = 960$ items, so $\widehat{p} = 131/960 = 0.1365 \approx 0.136$.

Step 2: expected counts. $\widehat{p}_x = \binom{8}{x}(0.136)^x(0.864)^{8-x}$, then $E_i = 120\,\widehat{p}_x$:

$$\begin{aligned}E_i &= 120\,(0.310,\ 0.391,\ 0.215,\ 0.068,\ 0.016) \\ &= (37.2,\ 46.9,\ 25.8,\ 8.2,\ 1.9).\end{aligned}$$

The ≥ 4 cell has $E_i = 1.9 < 5$; pool with $x = 3$: $O_i = 9 + 3 = 12$, $E_i = 8.2 + 1.9 = 10.1$. Now $k = 4$.

Step 3: statistic.

$$\begin{aligned}\chi^2 &= \frac{(40-37.2)^2}{37.2} + \frac{(44-46.9)^2}{46.9} + \frac{(24-25.8)^2}{25.8} + \frac{(12-10.1)^2}{10.1} \\ &= 0.21 + 0.18 + 0.13 + 0.36 = 0.88.\end{aligned}$$

Step 4: df and decision. $k = 4$, $m = 1$, $\text{df} = 4 - 1 - 1 = 2$; cutoff $\chi^2_{0.05,\,2} = 5.991$.

$$\boxed{\chi^2 = 0.88 < 5.991 \;\Rightarrow\; \text{fail to reject; the binomial model holds.}}$$

10.2.4 Fitting a normal to grouped data: estimate μ and σ

When the hypothesized family is normal you typically estimate *two* parameters, $\widehat{\mu} = \overline{X}$ and $\widehat{\sigma} = s$, so the fit costs $m = 2$ and $\text{df} = k - 3$. Continuous data are first **grouped** into intervals; the fitted probability of an interval $(a, b]$ is $\Phi\big((b - \widehat{\mu})/\widehat{\sigma}\big) - \Phi\big((a - \widehat{\mu})/\widehat{\sigma}\big)$, and the two extreme intervals run to $-\infty$ and $+\infty$ so the probabilities sum to one.

Example 10.2.7 (Exam scores: are they normally distributed?)**.** The scores of $n = 200$ students on a standardized test are grouped into six intervals. With $\overline{X} = 72.0$ and $s = 12.0$ estimated from the raw data, test normality at $\alpha = 0.05$.

Interval	< 54	54–63	63–72	72–81	81–90	≥ 90
O_i	11	28	61	55	32	13

Which rule, and why? The family is normal with *both* parameters estimated from the data, so the fit costs two degrees of freedom: $\text{df} = k - 1 - 2$. Grouped continuous data plus a named family is the signature of a normal goodness-of-fit, not a t-test (which would test a single mean, not the whole shape).

Solution.

Step 1: standardize the cut points. With $\widehat{\mu} = 72$, $\widehat{\sigma} = 12$, the z-cut at a boundary b is $(b - 72)/12$:

$$z(54) = -1.5, \quad z(63) = -0.75, \quad z(72) = 0,$$
$$z(81) = 0.75, \quad z(90) = 1.5.$$

Step 2: fitted interval probabilities. Using $\Phi(-1.5) = .0668$, $\Phi(-0.75) = .2266$, $\Phi(0) = .5$, $\Phi(0.75) = .7734$, $\Phi(1.5) = .9332$:

$$\widehat{p}_1 = \Phi(-1.5) = .0668, \qquad \widehat{p}_2 = \Phi(-.75) - \Phi(-1.5) = .1598,$$
$$\widehat{p}_3 = \Phi(0) - \Phi(-.75) = .2734, \quad \widehat{p}_4 = \Phi(.75) - \Phi(0) = .2734,$$
$$\widehat{p}_5 = \Phi(1.5) - \Phi(.75) = .1598, \quad \widehat{p}_6 = 1 - \Phi(1.5) = .0668.$$

Step 3: expected counts $E_i = 200\,\widehat{p}_i$.

Interval	< 54	54–63	63–72	72–81	81–90	≥ 90
E_i	13.4	32.0	54.7	54.7	32.0	13.4

All $E_i \geq 5$, so no pooling is needed and $k = 6$.

Step 4: statistic.

$$\chi^2 = \frac{(11 - 13.4)^2}{13.4} + \frac{(28 - 32.0)^2}{32.0} + \frac{(61 - 54.7)^2}{54.7}$$
$$+ \frac{(55 - 54.7)^2}{54.7} + \frac{(32 - 32.0)^2}{32.0} + \frac{(13 - 13.4)^2}{13.4}$$
$$= 0.43 + 0.50 + 0.73 + 0.00 + 0.00 + 0.01 = 1.67.$$

Step 5: df and decision. $k = 6$, two parameters estimated, $\text{df} = 6 - 1 - 2 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 1.67 < 7.815 \;\Rightarrow\; \text{fail to reject; normality is tenable.}}$$

Example 10.2.8 (Component lifetimes: a normal fit that fails)**.** The lifetimes (hours) of $n = 160$ electronic components are grouped into five intervals; from the raw data $\overline{X} = 500$ and $s = 80$. Test normality at $\alpha = 0.05$.

Interval	< 420	420–500	500–580	580–660	≥ 660
O_i	40	36	34	30	20

Which rule, and why? Normal family, μ and σ both estimated $\Rightarrow \text{df} = k - 1 - 2$. We test the whole shape because lifetimes are often right-skewed and we want to know whether a symmetric normal is adequate.

Solution.

Step 1: standardize. $z(420) = -1.0$, $z(500) = 0$, $z(580) = 1.0$, $z(660) = 2.0$.

Step 2: fitted probabilities using $\Phi(-1) = .1587$, $\Phi(0) = .5$, $\Phi(1) = .8413$, $\Phi(2) = .9772$:

$$\widehat{p}_1 = .1587, \qquad \widehat{p}_2 = .5 - .1587 = .3413, \quad \widehat{p}_3 = .8413 - .5 = .3413,$$
$$\widehat{p}_4 = .9772 - .8413 = .1359, \quad \widehat{p}_5 = 1 - .9772 = .0228.$$

Step 3: expected counts $E_i = 160\,\widehat{p}_i$. $E_i = (25.4, 54.6, 54.6, 21.7, 3.6)$. The last cell $E_i = 3.6 < 5$; pool with the fourth: $O_i = 30 + 20 = 50$, $E_i = 21.7 + 3.6 = 25.3$. Now $k = 4$.

Step 4: statistic.

$$\chi^2 = \frac{(40 - 25.4)^2}{25.4} + \frac{(36 - 54.6)^2}{54.6} + \frac{(34 - 54.6)^2}{54.6} + \frac{(50 - 25.3)^2}{25.3}$$
$$= 8.39 + 6.34 + 7.77 + 24.11 = 46.61.$$

Step 5: df and decision. After pooling $k = 4$, two parameters estimated, so df $= 4 - 1 - 2 = 1$; cutoff $\chi^2_{0.05,\,1} = 3.841$.

$$\boxed{\chi^2 = 46.61 > 3.841 \;\Rightarrow\; \text{reject; the lifetimes are not normal.}}$$

The excess mass in the lower interval and the long upper tail point to a right-skewed lifetime law (exponential or Weibull), not a normal one. Notice how aggressively pooling *plus* two estimated parameters can shrink the df: from $k = 5$ intervals we are down to a single degree of freedom.

10.2.5 The same data, with and without estimation

Example 10.2.9 (Known vs. estimated λ: how the df changes)**.** Reconsider the help-desk data of Example 10.4 ($n = 200$ windows, pooled into $k = 6$ cells, with the same E_i built from a Poisson rate 1.95 and the same $\chi^2 = 4.46$). Compare two scenarios: (a) the rate $\lambda = 1.95$ was *quoted from a multi-year historical log* and treated as known; (b) the rate was *estimated from these* 200 *windows* as $\widehat{\lambda} = 1.95$. Does the conclusion change at $\alpha = 0.05$?

Which rule, and why? The arithmetic of χ^2 is identical in both scenarios; only the bookkeeping of degrees of freedom differs — a parameter you were *told* costs nothing, a parameter you *estimated* costs one. This is the cleanest illustration of the $-m$ rule.

Solution.

Step 1: scenario (a), known rate. Nothing is estimated, so $m = 0$ and df $= k - 1 = 6 - 1 = 5$. Cutoff $\chi^2_{0.05,\,5} = 11.070$. Since $4.46 < 11.070$, fail to reject.

Step 2: scenario (b), estimated rate. One parameter estimated, $m = 1$, so df $= k - 1 - 1 = 4$. Cutoff $\chi^2_{0.05,\,4} = 9.488$. Since $4.46 < 9.488$, fail to reject.

Step 3: compare. Both fail to reject here, but the *thresholds differ*:

$$\text{(a) known: df} = 5, \text{ cutoff } 11.070,$$
$$\text{(b) estimated: df} = 4, \text{ cutoff } 9.488.$$

The estimated-parameter test uses the *smaller* cutoff, so it is the more willing to reject — correctly so, because fitting λ to the data already pulled χ^2 down, and the smaller reference curve compensates.

$$\boxed{\text{Known parameter: df} = k - 1. \quad \text{Estimated parameter: df} = k - 1 - 1.}$$

10.2.6 Pooling sparse tail cells before counting df

Example 10.2.10 (Earthquakes per year: pool first, then count)**.** A seismology station tabulates the number of magnitude-3+ earthquakes felt in each of 80 years. Test a Poisson fit at $\alpha = 0.05$.

Quakes x	0	1	2	3	4	5	≥ 6
Years O_i	14	22	20	13	6	3	2

Which rule, and why? Poisson family, rate to be estimated — but several tail cells will have $E_i < 5$, so the operative skill is *pooling*: combine adjacent sparse cells until every $E_i \geq 5$, and only *then* count k and subtract for the one estimated parameter.

Solution.

Step 1: estimate λ. Total quakes $= 0(14) + 1(22) + 2(20) + 3(13) + 4(6) + 5(3) + 6(2) = 134$ over 80 years, so $\widehat{\lambda} = 134/80 = 1.675 \approx 1.68$.

Step 2: raw expected counts ($E_i = 80\, e^{-1.68} 1.68^x / x!$, last cell the tail):

x	0	1	2	3	4	5	≥ 6
E_i	14.9	25.1	21.0	11.8	4.9	1.7	0.6

Step 3: pool the sparse tail. The cells $x = 4$ ($E_i = 4.9$), $x = 5$ (1.7) and ≥ 6 (0.6) are each below 5; pool all three into a "≥ 4" cell:

$$O_i^{(\geq 4)} = 6 + 3 + 2 = 11, \qquad E_i^{(\geq 4)} = 4.9 + 1.7 + 0.6 = 7.2.$$

The categories are now $0, 1, 2, 3, \geq 4$, so $k = 5$ (it was 7 before pooling).

Step 4: statistic.

$$\begin{aligned}\chi^2 &= \frac{(14 - 14.9)^2}{14.9} + \frac{(22 - 25.1)^2}{25.1} + \frac{(20 - 21.0)^2}{21.0} \\ &\quad + \frac{(13 - 11.8)^2}{11.8} + \frac{(11 - 7.2)^2}{7.2} \\ &= 0.05 + 0.38 + 0.05 + 0.12 + 2.01 = 2.61.\end{aligned}$$

Step 5: df and decision. Count k *after* pooling: $k = 5$, one parameter estimated, $\text{df} = 5 - 1 - 1 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 2.61 < 7.815 \;\Rightarrow\; \text{fail to reject; Poisson fits the quake counts.}}$$

Had we (wrongly) used the pre-pool $k = 7$, we would have computed $\text{df} = 7 - 1 - 1 = 5$ and the cutoff 11.070 — but more importantly the cells with $E_i < 5$ make individual $(O_i - E_i)^2/E_i$ terms unstable, so the χ^2 approximation itself would be untrustworthy. Pooling fixes both problems.

10.2.7 Contrast: forgetting to subtract for estimation

Example 10.2.11 (The inflated-df mistake)**.** A consultant runs the litter-sex-ratio test of Example 10.6 ($k = 5$ pooled cells, one estimated parameter $\widehat{p} = 0.388$, observed $\chi^2 = 35.83$). A colleague, copying the §1 known-parameter rule, uses $\text{df} = k - 1 = 4$ instead of $k - 1 - 1 = 3$, and — to make the point sharper — imagine a second data set where the computed χ^2 came out to 8.5. Show how the wrong df can flip a decision and why it inflates the false-reject rate.

Which rule, and why? This is the deliberate look-alike error: applying the *fully-specified* df rule ($k - 1$) when a parameter was actually estimated. The estimation already shrank χ^2, so the correct reference curve is the *smaller* χ^2_{k-1-1}; using χ^2_{k-1} mismatches the statistic to its distribution.

Solution.

Step 1: the original litter data. With the true $\text{df} = 3$, cutoff $\chi^2_{0.05,\,3} = 7.815$, and $35.83 > 7.815$, we reject — the wrong $\text{df} = 4$ (cutoff 9.488) also rejects here, so the giant statistic masks the error. The danger shows up near the boundary, so take the hypothetical $\chi^2 = 8.5$:

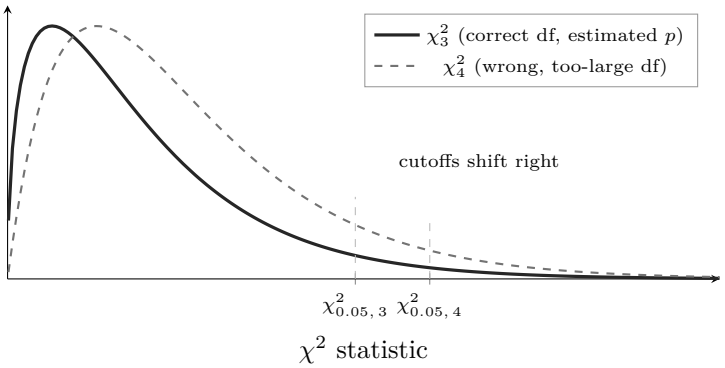
Step 2: same statistic, two df.

$$\begin{aligned}\text{correct df} = 3: &\quad 8.5 > \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{reject}, \\ \text{wrong df} = 4: &\quad 8.5 < \chi^2_{0.05,\,4} = 9.488 \;\Rightarrow\; \text{fail to reject}.\end{aligned}$$

A single forgotten subtraction flips the conclusion. The direction matters: the *too-large* df gives the *larger* cutoff, so this particular slip makes the test too *conservative* at a fixed level.

Step 3: why it nonetheless corrupts the error rate. The estimated-parameter statistic truly follows χ^2_3, whose mean is 3. Referring it to a χ^2_4 table (mean 4) mis-states every tail probability: a reported "$p = 0.075$" from the χ^2_4 table is really $\mathbb{P}(\chi^2_3 > 8.5) \approx 0.037$. Across many such tests the *true* rejection rate no longer equals the nominal α — the test's calibration is broken in both directions depending on where the statistic lands. The fix is mechanical: subtract one df per estimated parameter.

Estimated a parameter? Use $\mathrm{df} = k - 1 - m$. Using $k - 1$ mis-calibrates the test.



Reducing the degrees of freedom from 4 to 3 (one estimated parameter) slides the whole density *left* (its mean drops from 4 to 3) and pulls the $\alpha = 0.05$ cutoff in from 9.488 to 7.815. The solid curve is the correct reference; the dashed curve is the too-large-df distribution you would wrongly use if you forgot to subtract.

10.2.8 A careful degrees-of-freedom walk-through

Example 10.2.12 (Counting df from scratch)**.** A bank models the number of ATM withdrawals per minute over $n = 300$ minutes as Poisson. The raw table has eight cells, $x = 0, 1, \ldots, 6, \geq 7$. After estimating $\widehat{\lambda}$ from the sample mean, the fitted E_i in the cells $x = 6$ and ≥ 7 come out as 4.1 and 1.5. State the final degrees of freedom, showing every adjustment.

Which rule, and why? Pure df bookkeeping: this example isolates the two operations that change the count — pooling sparse cells and subtracting for estimation — so you can see the order in which they apply.

Solution.

Step 1: start from the raw cell count. Eight raw cells: $x = 0, 1, 2, 3, 4, 5, 6, \geq 7$, so the *pre-adjustment* category count is 8.

Step 2: pool sparse cells. Cells $x = 6$ ($E_i = 4.1$) and ≥ 7 ($E_i = 1.5$) are below 5; pool them into a single ≥ 6 cell with $E_i = 4.1 + 1.5 = 5.6 \geq 5$. This drops the category count by one, to

$$k = 8 - 1 = 7 \quad (\text{cells } 0, 1, 2, 3, 4, 5, \geq 6).$$

Step 3: subtract one for the total constraint. The counts must sum to $n = 300$, costing the universal -1: this gives $k - 1 = 6$ *before* accounting for estimation.

Step 4: subtract one per estimated parameter. One parameter, λ, was estimated ($m = 1$):

$$\begin{aligned} \mathrm{df} &= k - 1 - m \\ &= 7 - 1 - 1 = 5. \end{aligned}$$

$\mathrm{df} = 5$ (7 pooled cells, minus 1 for the total, minus 1 for $\widehat{\lambda}$).

The cutoff for a 5% test would then be $\chi^2_{0.05,\,5} = 11.070$. Had a *second* parameter been estimated (e.g. a normal fit with both μ and σ), the df would instead be $7 - 1 - 2 = 4$.

10.2.9 A normal fit with no pooling and a borderline call

Example 10.2.13 (Reaction times: a borderline two-sided normal fit)**.** A psychologist groups $n = 240$ reaction times (ms) into seven equal-probability-ish intervals; the sample gives $\overline{X} = 300$, $s = 40$. The expected counts (all ≥ 5) and observed counts are:

Interval	I	II	III	IV	V	VI	VII
O_i	9	26	55	70	44	24	12
E_i	12.0	30.0	52.0	52.0	52.0	30.0	12.0

Test normality at $\alpha = 0.05$.

Which rule, and why? Normal family, both parameters estimated, no cell below 5 so no pooling: a textbook $\mathrm{df} = k - 1 - 2$ computation. The point of interest is that one cell (IV) carries almost all the misfit, which a normal model should not produce if it were correct.

Solution.

Step 1: statistic, cell by cell.

$$\begin{aligned}\chi^2 &= \frac{(9-12)^2}{12} + \frac{(26-30)^2}{30} + \frac{(55-52)^2}{52} + \frac{(70-52)^2}{52} \\ &\quad + \frac{(44-52)^2}{52} + \frac{(24-30)^2}{30} + \frac{(12-12)^2}{12} \\ &= 0.75 + 0.53 + 0.17 + 6.23 + 1.23 + 1.20 + 0.00 = 10.11.\end{aligned}$$

Step 2: df. $k = 7$, two parameters estimated, $\mathrm{df} = 7 - 1 - 2 = 4$; cutoff $\chi^2_{0.05,\,4} = 9.488$.

Step 3: decision. Since $10.11 > 9.488$, reject — but barely.

$$\boxed{\chi^2 = 10.11 > 9.488 = \chi^2_{0.05,\,4} \;\Rightarrow\; \text{reject normality at } \alpha = 0.05.}$$

Note the verdict hinges on the df: with the *wrong* $\mathrm{df} = k - 1 = 6$ the cutoff would be $\chi^2_{0.05,\,6} = 12.592$ and we would (wrongly) fail to reject. The single dominant cell IV (a spike of fast responses) is what a symmetric normal cannot accommodate.

10.2.10 A geometric fit: one estimated parameter

Example 10.2.14 (Trials until first success: a geometric fit)**.** A call center records, for each of 180 outbound campaigns, the number of attempts up to and including the first sale. If each attempt independently succeeds with probability p, this count is Geometric(p) on $\{1, 2, \dots\}$. Test the fit at $\alpha = 0.05$, with p estimated.

Attempts x	1	2	3	4	≥ 5
Campaigns O_i	74	48	30	16	12

Which rule, and why? A named one-parameter family (geometric) with p unknown $\Rightarrow$ estimate p, fit, and use $\mathrm{df} = k - 1 - 1$. The estimator for a geometric mean $1/p$ is the reciprocal of the sample mean, $\widehat{p} = 1/\overline{X}$.

Solution.

Step 1: estimate p. Use the ≥ 5 cell at its conservative value 5 for the mean: total attempts $\approx 1(74) + 2(48) + 3(30) + 4(16) + 5(12) = 74 + 96 + 90 + 64 + 60 = 384$ over 180 campaigns, so $\overline{X} = 384/180 = 2.133$ and $\widehat{p} = 1/2.133 = 0.469 \approx 0.47$.

Step 2: fitted probabilities. $\widehat{p}_x = (1 - \widehat{p})^{x-1}\widehat{p}$ with $\widehat{p} = 0.47$, $1 - \widehat{p} = 0.53$; the last cell is $\mathbb{P}(X \geq 5) = (0.53)^4 = 0.0789$:

$$\widehat{p}_1 = 0.470, \quad \widehat{p}_2 = 0.249, \quad \widehat{p}_3 = 0.132,$$
$$\widehat{p}_4 = 0.070, \quad \widehat{p}_{\geq 5} = 0.079.$$

Step 3: expected counts $E_i = 180\,\widehat{p}_x$. $E_i = (84.6,\ 44.8,\ 23.8,\ 12.6,\ 14.2)$, all ≥ 5, so no pooling; $k = 5$.
Step 4: statistic.

$$\begin{aligned}\chi^2 &= \frac{(74-84.6)^2}{84.6} + \frac{(48-44.8)^2}{44.8} + \frac{(30-23.8)^2}{23.8} \\ &\quad + \frac{(16-12.6)^2}{12.6} + \frac{(12-14.2)^2}{14.2} \\ &= 1.33 + 0.23 + 1.62 + 0.92 + 0.34 = 4.44.\end{aligned}$$

Step 5: df and decision. $k = 5$, $m = 1$, $\text{df} = 5 - 1 - 1 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 4.44 < 7.815 \ \Rightarrow\ \text{fail to reject; the geometric model fits.}}$$

10.2.11 A p-value report

Example 10.2.15 (Reporting the p-value for a fitted Poisson)**.** For the help-desk fit (Example 10.4), $\chi^2 = 4.46$ on $\text{df} = 4$. Report the p-value and confirm the decision.

Which rule, and why? Goodness-of-fit is always *upper-tail*: only large χ^2 (poor fit) is evidence against H_0, so the p-value is the single right-tail area $\mathbb{P}(\chi^2_4 > 4.46)$ — never doubled.

Solution.

Step 1: bracket the statistic with table points. On 4 df, $\chi^2_{0.50,\,4} = 3.357$ and $\chi^2_{0.25,\,4} = 5.385$. Our 4.46 lies between them, so the right-tail area is between 0.25 and 0.50.

Step 2: report and decide. Interpolating, $\mathbb{P}(\chi^2_4 > 4.46) \approx 0.35$.

$$\boxed{p\text{-value} = \mathbb{P}(\chi^2_4 > 4.46) \approx 0.35 \gg 0.05 \ \Rightarrow\ \text{fail to reject.}}$$

A p-value this large says the observed misfit is exactly what an honest Poisson would produce by chance — strong support for the model, consistent with the fixed-level decision $4.46 < 9.488$.

10.3 Contingency Tables: the Test of Independence

Up to now each χ^2 test has sorted a single sample on *one* categorical variable; this section keeps the single sample but cross-classifies it on *two* categorical variables at once, arranging the counts in an $r \times c$ **contingency table** with r row categories and c column categories. The question is whether the two variables are **independent** — whether knowing a unit's row tells you nothing about its column. The null hypothesis fixes no particular probabilities; instead it asserts the *structure* of independence, which forces the expected cell counts to be $E_{ij} = (\text{row } i \text{ total})(\text{column } j \text{ total})/n$. The same statistic as always, $\chi^2 = \sum_{\text{cells}}(O_{ij} - E_{ij})^2/E_{ij}$, measures the gap between what we saw and what independence predicts, and under H_0 it follows a χ^2 distribution with $(r-1)(c-1)$ degrees of freedom. Recognise this design by its hallmark: *one* sample, *two* questions asked of every unit, and a two-way table of counts whose row and column totals were *not* fixed in advance by the experimenter.

Is this an independence test, and how big is the table?

- **One** sample, each unit classified by *two* categorical variables, asking whether those variables are related? $\Rightarrow$ the **test of independence** on the $r \times c$ table; statistic $\chi^2 = \sum(O_{ij} - E_{ij})^2/E_{ij}$, $\text{df} = (r-1)(c-1)$.
- Expected counts always come from the *margins*: $E_{ij} = R_i C_j / n$, the row total times the column total over the grand total — never from the observed cell counts.
- Reject when χ^2_{obs} is **large**: the test is always **upper-tailed**, reject if $\chi^2_{\text{obs}} > \chi^2_{\alpha,(r-1)(c-1)}$.
- A rejection is evidence of **association**, not **causation**: the two variables move together, but the table alone cannot say which (if either) drives the other.

Do not confuse independence with homogeneity (§4): both use the identical statistic and the identical df $(r-1)(c-1)$, but independence draws *one* sample and asks whether two traits are linked, while homogeneity draws *several separate* samples and asks whether their category profiles match. The arithmetic is the same; the sampling design — and the wording of the conclusion — is not.

10.3.1 Expected counts under independence

Definition 10.3.1 (Contingency table and expected counts)**.** A single sample of n units, each classified by a row variable with r levels and a column variable with c levels, gives an $r \times c$ table of **observed** counts O_{ij}. Write the row totals $R_i = \sum_j O_{ij}$, the column totals $C_j = \sum_i O_{ij}$, and the grand total $n = \sum_{i,j} O_{ij}$. Under H_0 that the two variables are **independent**, the **expected** count in cell (i, j) is

$$E_{ij} = \frac{R_i\, C_j}{n} = \frac{(\text{row } i \text{ total})(\text{column } j \text{ total})}{n}.$$

Where $E_{ij} = R_i C_j / n$ comes from. Let $p_{i\cdot}$ be the probability a unit falls in row i and $p_{\cdot j}$ the probability it falls in column j. Independence means the joint cell probability factors, $p_{ij} = p_{i\cdot}\, p_{\cdot j}$. The expected count is $E_{ij} = n\, p_{ij} = n\, p_{i\cdot}\, p_{\cdot j}$. We do not know these probabilities, so we estimate them from the margins: $\widehat{p}_{i\cdot} = R_i/n$ and $\widehat{p}_{\cdot j} = C_j/n$. Substituting,

$$E_{ij} = n\, \widehat{p}_{i\cdot}\, \widehat{p}_{\cdot j} = n \cdot \frac{R_i}{n} \cdot \frac{C_j}{n} = \frac{R_i\, C_j}{n},$$

which is the displayed rule. The estimation of those marginal probabilities is exactly what costs degrees of freedom, as the next theorem records. □

Theorem 10.3.2 (The independence test)**.** *For an $r \times c$ contingency table from one sample, to test*

$$H_0\text{: the row and column variables are independent} \quad \textit{vs.} \quad H_1\text{: they are not,}$$

form the statistic

$$\chi^2_{obs} = \sum_{i=1}^{r} \sum_{j=1}^{c} \frac{(O_{ij} - E_{ij})^2}{E_{ij}}, \qquad E_{ij} = \frac{R_i\, C_j}{n}.$$

Under H_0, χ^2_{obs} is approximately χ^2 with $(r-1)(c-1)$ degrees of freedom. Reject H_0 at level α when $\chi^2_{obs} > \chi^2_{\alpha,\,(r-1)(c-1)}$.

Counting the degrees of freedom. Start from the general rule: df = (number of cells) −1− (parameters estimated). There are rc cells. We always subtract 1 because the cell probabilities must sum to one. To pin down the independence model we estimated the marginal probabilities: $r-1$ free row probabilities (the last is forced, since they sum to one) and $c-1$ free column probabilities. Hence

$$\begin{aligned} \text{df} &= \underbrace{rc}_{\text{cells}} - 1 - \underbrace{(r-1)}_{\text{row probs}} - \underbrace{(c-1)}_{\text{col probs}} \\ &= rc - 1 - r + 1 - c + 1 \\ &= rc - r - c + 1 \\ &= (r-1)(c-1). \end{aligned}$$

The clean factored form $(r-1)(c-1)$ is the count to remember. □

Remark (Association, not causation). A rejection says the two classifications are statistically **associated** — the cell counts depart from what independence would produce. It does *not* establish that one variable causes the other, nor rule out a lurking third variable driving both. In an observational cross-tabulation (the usual setting), "related" is the strongest honest verdict; causal language requires a designed experiment, not a contingency table.

Remark (Validity condition). The $\chi^2_{(r-1)(c-1)}$ approximation is trustworthy when the *expected* counts are not too small — a common rule is every $E_{ij} \geq 5$, with a few as low as 1 tolerated if most are larger. Sparse tables need Fisher's exact test or pooling of categories (§5). Always check the E_{ij}, not the O_{ij}.

	col 1	col 2	col 3	**row total**
row 1	O_{11}	O_{12}	O_{13}	R_1
row 2	O_{21}	O_{22}	O_{23}	R_2
col total	C_1	C_2	C_3	n

expected count in cell (i, j):
$$E_{ij} = \frac{R_i\, C_j}{n} = \frac{(\text{row } i \text{ total}) \times (\text{column } j \text{ total})}{\text{grand total}}$$

Schematic 2×3 contingency table. The margins R_i and C_j are the row and column totals and n is the grand total; under independence every interior cell's expected count is its row total times its column total, divided by n.

10.3.2 The 2×2 table

Example 10.3.3 (Treatment versus outcome: full expected-count computation)**.** A clinic enrolls $n = 200$ patients and records, for each, whether they received a new therapy or the standard one, and whether they recovered within a month. The observed counts are below. At $\alpha = 0.05$, are treatment and recovery independent?

	recovered	not recovered	**row total**
new therapy	66	34	100
standard	48	52	100
col total	114	86	200

Which rule, and why? One sample of 200 patients, each classified by *two* traits (treatment and outcome), asking whether they are related — so this is the **test of independence** on a 2×2 table, $\text{df} = (2-1)(2-1) = 1$, with $E_{ij} = R_iC_j/n$.

Solution.

Step 1: expected counts. Each E_{ij} is its row total times its column total over $n = 200$:

$$E_{11} = \frac{100 \cdot 114}{200} = 57, \quad E_{12} = \frac{100 \cdot 86}{200} = 43,$$
$$E_{21} = \frac{100 \cdot 114}{200} = 57, \quad E_{22} = \frac{100 \cdot 86}{200} = 43.$$

Tabulating observed against expected:

	observed O_{ij}		expected E_{ij}	
	recovered	not	recovered	not
new therapy	66	34	57	43
standard	48	52	57	43

Step 2: cell contributions.

$$\frac{(66-57)^2}{57} = \frac{81}{57} = 1.4211, \quad \frac{(34-43)^2}{43} = \frac{81}{43} = 1.8837,$$
$$\frac{(48-57)^2}{57} = \frac{81}{57} = 1.4211, \quad \frac{(52-43)^2}{43} = \frac{81}{43} = 1.8837.$$

Step 3: statistic.

$$\chi^2_{\text{obs}} = 1.4211 + 1.8837 + 1.4211 + 1.8837 = 6.610.$$

Step 4: compare. On df $= 1$, $\chi^2_{0.05,1} = 3.841$. Since $6.610 > 3.841$, reject.

$$\boxed{\chi^2_{\text{obs}} = 6.610 > 3.841 \;\Rightarrow\; \text{reject } H_0\text{: recovery is associated with treatment.}}$$

The recovery rate is higher under the new therapy, but "associated" is the verdict — causal credit needs the design, not the table.

Example 10.3.4 (The equivalent two-proportion z-test, and $\chi^2 = z^2$)**.** Reanalyze the clinic data as a comparison of two recovery rates: new therapy $66/100 = 0.66$, standard $48/100 = 0.48$. Run the two-sided two-proportion z-test of $H_0\colon p_X = p_Y$, and verify that z^2 matches the $\chi^2_{\text{obs}} = 6.610$ above.

Which rule, and why? A 2×2 independence test on one sample is algebraically the *same* test as the two-sided two-proportion z-test comparing the two row success rates; they must give the same p-value, and the statistics satisfy $\chi^2_{\text{obs}} = z^2$. We compute the z to see the identity.

Solution.

Step 1: pooled rate. The pooled recovery rate is $\widehat{p} = (66 + 48)/(100 + 100) = 114/200 = 0.57$.

Step 2: pooled standard error.

$$\begin{aligned} \text{SE}_0 &= \sqrt{\widehat{p}(1-\widehat{p})\left(\tfrac{1}{100} + \tfrac{1}{100}\right)} \\ &= \sqrt{0.57 \cdot 0.43 \cdot 0.02} = \sqrt{0.004902} = 0.070014. \end{aligned}$$

Step 3: statistic. $z_{\text{obs}} = (0.66 - 0.48)/0.070014 = 0.18/0.070014 = 2.5709$.

Step 4: square it. $z^2_{\text{obs}} = 2.5709^2 = 6.610$, matching χ^2_{obs} exactly.

$$\boxed{\chi^2_{\text{obs}} = z^2_{\text{obs}} = 6.610; \quad |z_{\text{obs}}| = 2.571 > 1.96 \;\Rightarrow\; \text{reject } H_0.}$$

On a 2×2 table the two tests are one test in two costumes: $\chi^2_{0.05,1} = 3.841 = 1.96^2 = z^2_{0.025}$, so the cutoffs match too. The z version has the bonus of a direction (here new > standard) that the always-positive χ^2 throws away.

Example 10.3.5 (Smartphone OS versus app purchase: small effect, large sample)**.** A retailer cross-tabulates $n = 1000$ app users by phone operating system (A or B) and whether they made an in-app purchase. At $\alpha = 0.05$, is OS independent of purchasing?

	purchased	did not	**row total**
OS A	150	350	500
OS B	180	320	500
col total	330	670	1000

Which rule, and why? One sample, two traits (OS and purchase), 2×2 table $\Rightarrow$ independence test on 1 df. With $n = 1000$ even a small percentage gap can reach significance, so we compute rather than eyeball.

Solution.

Step 1: expected counts.

$$E_{11} = \frac{500 \cdot 330}{1000} = 165, \quad E_{12} = \frac{500 \cdot 670}{1000} = 335,$$
$$E_{21} = \frac{500 \cdot 330}{1000} = 165, \quad E_{22} = \frac{500 \cdot 670}{1000} = 335.$$

Step 2: contributions.

$$\frac{(150-165)^2}{165} = \frac{225}{165} = 1.3636, \quad \frac{(350-335)^2}{335} = \frac{225}{335} = 0.6716,$$
$$\frac{(180-165)^2}{165} = \frac{225}{165} = 1.3636, \quad \frac{(320-335)^2}{335} = \frac{225}{335} = 0.6716.$$

Step 3: statistic. $\chi^2_{\text{obs}} = 1.3636 + 0.6716 + 1.3636 + 0.6716 = 4.071.$

Step 4: compare. On 1 df, $\chi^2_{0.05,1} = 3.841$, and $4.071 > 3.841$.

$$\boxed{\chi^2_{\text{obs}} = 4.071 > 3.841 \;\Rightarrow\; \text{reject } H_0\text{: OS and purchasing are associated (barely).}}$$

The purchase rates differ by only 30% vs. 36%, yet $n = 1000$ makes the gap significant — a reminder that statistical significance is not the same as practical importance.

10.3.3 Larger $r \times c$ tables

Example 10.3.6 (Three regions by three commute modes: a 3×3 table)**.** A transportation survey samples $n = 600$ commuters and records, for each, their region (North, Central, South) and primary commute mode (car, transit, bike). The counts are below. At $\alpha = 0.05$, are region and commute mode independent?

	car	transit	bike	**row total**
North	90	60	30	180
Central	120	100	20	240
South	90	60	30	180
col total	300	220	80	600

Which rule, and why? One sample of 600 commuters cross-classified by two variables with three levels each, so this is the **independence test** on a 3×3 table; df $= (3-1)(3-1) = 4$.

Solution.

Step 1: expected counts $E_{ij} = R_i C_j / 600$.

$$\begin{aligned} E_{11} &= \tfrac{180\cdot 300}{600} = 90, & E_{12} &= \tfrac{180\cdot 220}{600} = 66, & E_{13} &= \tfrac{180\cdot 80}{600} = 24, \\ E_{21} &= \tfrac{240\cdot 300}{600} = 120, & E_{22} &= \tfrac{240\cdot 220}{600} = 88, & E_{23} &= \tfrac{240\cdot 80}{600} = 32, \\ E_{31} &= \tfrac{180\cdot 300}{600} = 90, & E_{32} &= \tfrac{180\cdot 220}{600} = 66, & E_{33} &= \tfrac{180\cdot 80}{600} = 24. \end{aligned}$$

All $E_{ij} \geq 24 \geq 5$, so the approximation is valid.

Step 2: cell contributions $(O_{ij} - E_{ij})^2 / E_{ij}$.

$$\begin{aligned} \text{North:} &\quad \tfrac{(90-90)^2}{90} = 0, & \tfrac{(60-66)^2}{66} &= 0.5455, & \tfrac{(30-24)^2}{24} &= 1.5000, \\ \text{Central:} &\quad \tfrac{(120-120)^2}{120} = 0, & \tfrac{(100-88)^2}{88} &= 1.6364, & \tfrac{(20-32)^2}{32} &= 4.5000, \\ \text{South:} &\quad \tfrac{(90-90)^2}{90} = 0, & \tfrac{(60-66)^2}{66} &= 0.5455, & \tfrac{(30-24)^2}{24} &= 1.5000. \end{aligned}$$

Step 3: sum.

$$\begin{aligned} \chi^2_{\text{obs}} &= 0 + 0.5455 + 1.5000 + 0 + 1.6364 + 4.5000 + 0 + 0.5455 + 1.5000 \\ &= 10.227. \end{aligned}$$

Step 4: compare. On 4 df, $\chi^2_{0.05,4} = 9.488$. Since $10.227 > 9.488$, reject.

$$\boxed{\chi^2_{\text{obs}} = 10.227 > 9.488 \;\Rightarrow\; \text{reject } H_0\text{: region and commute mode are associated.}}$$

The largest single contribution (4.50) is Central's bike cell: far fewer Central bikers than independence predicts. The table flags *where* the association lives.

Example 10.3.7 (Education level by news source: a 3×4 table and its df)**.** A media study samples $n = 800$ adults and records highest education (high school, college, graduate) against primary news source (TV, print, online, social). Before any arithmetic, state the table shape and the degrees of freedom; then suppose the test yields $\chi^2_{\text{obs}} = 21.4$ and decide at $\alpha = 0.05$.

Which rule, and why? One sample, two categorical variables ⇒ independence test. The point here is the **df count** for a non-square table: rows $r = 3$ (education), columns $c = 4$ (news source), so $\text{df} = (r-1)(c-1)$.

Solution.

Step 1: table shape. $r = 3$ education levels, $c = 4$ news sources, so $3 \times 4 = 12$ cells.

Step 2: degrees of freedom.

$$\text{df} = (r-1)(c-1) = (3-1)(4-1) = 2 \cdot 3 = 6.$$

Step 3: critical value and decision. On 6 df, $\chi^2_{0.05,6} = 12.592$. The reported $\chi^2_{\text{obs}} = 21.4 > 12.592$.

$$\boxed{\text{df} = (3-1)(4-1) = 6; \quad \chi^2_{\text{obs}} = 21.4 > 12.592 \;\Rightarrow\; \text{reject } H_0.}$$

Note df does *not* equal cells minus one ($12 - 1 = 11$): estimating the margins costs the extra 5, leaving $(r-1)(c-1) = 6$.

Example 10.3.8 (Careful df: a 4×5 table)**.** A survey cross-classifies respondents by four age bands and five favorite genres. State the number of cells, the number of margin parameters estimated, and the degrees of freedom both ways (the subtract-and-count way and the factored way) to confirm they agree.

Which rule, and why? This is a pure **degrees-of-freedom** exercise. The two routes to df must agree: $rc - 1 - (r-1) - (c-1)$ and $(r-1)(c-1)$.

Solution.

Step 1: count cells. $r = 4$, $c = 5$, so $rc = 20$ cells.

Step 2: count estimated parameters. Free row probabilities $r - 1 = 3$; free column probabilities $c - 1 = 4$; total 7 estimated. Plus the universal "-1" for probabilities summing to one.

Step 3: df the long way.

$$\begin{aligned}\text{df} &= rc - 1 - (r-1) - (c-1) \\ &= 20 - 1 - 3 - 4 = 12.\end{aligned}$$

Step 4: df the factored way.

$$\text{df} = (r-1)(c-1) = (4-1)(5-1) = 3 \cdot 4 = 12.$$

$$\boxed{\text{Both routes give df} = 12, \text{ not } rc - 1 = 19.}$$

The factored form is faster, but the subtract-and-count form shows *why*: each estimated margin parameter removes one degree of freedom.

Example 10.3.9 (Blood type by Rh factor: a 4×2 table)**.** A blood bank classifies $n = 500$ donors by ABO type (O, A, B, AB) and Rh status (positive, negative). At $\alpha = 0.05$, are ABO type and Rh status independent?

	Rh+	Rh−	**row total**
O	180	40	220
A	150	20	170
B	60	15	75
AB	30	5	35
col total	420	80	500

Which rule, and why? One sample of donors cross-classified by two traits ⇒ independence test on a 4×2 table; $\text{df} = (4-1)(2-1) = 3$.

Solution.

Step 1: expected counts $E_{ij} = R_i C_j / 500$.

$$E_{O+} = \tfrac{220\cdot 420}{500} = 184.8, \quad E_{O-} = \tfrac{220\cdot 80}{500} = 35.2,$$
$$E_{A+} = \tfrac{170\cdot 420}{500} = 142.8, \quad E_{A-} = \tfrac{170\cdot 80}{500} = 27.2,$$
$$E_{B+} = \tfrac{75\cdot 420}{500} = 63.0, \quad E_{B-} = \tfrac{75\cdot 80}{500} = 12.0,$$
$$E_{AB+} = \tfrac{35\cdot 420}{500} = 29.4, \quad E_{AB-} = \tfrac{35\cdot 80}{500} = 5.6.$$

The smallest expected count is $5.6 \geq 5$, so the test is valid.

Step 2: contributions.

$$\tfrac{(180-184.8)^2}{184.8} = 0.1247, \quad \tfrac{(40-35.2)^2}{35.2} = 0.6545,$$
$$\tfrac{(150-142.8)^2}{142.8} = 0.3630, \quad \tfrac{(20-27.2)^2}{27.2} = 1.9059,$$
$$\tfrac{(60-63.0)^2}{63.0} = 0.1429, \quad \tfrac{(15-12.0)^2}{12.0} = 0.7500,$$
$$\tfrac{(30-29.4)^2}{29.4} = 0.0122, \quad \tfrac{(5-5.6)^2}{5.6} = 0.0643.$$

Step 3: sum.

$$\begin{aligned}\chi^2_{\text{obs}} &= 0.1247 + 0.6545 + 0.3630 + 1.9059 \\ &\quad + 0.1429 + 0.7500 + 0.0122 + 0.0643 \\ &= 4.018.\end{aligned}$$

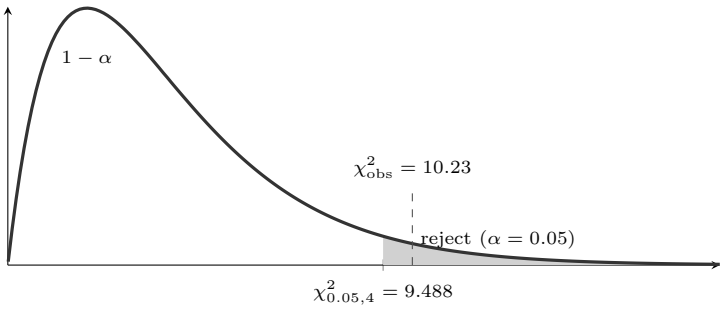
Step 4: compare. On 3 df, $\chi^2_{0.05,3} = 7.815$. Since $4.018 < 7.815$, do not reject.

$$\boxed{\chi^2_{\text{obs}} = 4.018 < 7.815 \;\Rightarrow\; \text{fail to reject: no evidence ABO and Rh are associated.}}$$

Consistent with the biology — ABO and Rh are inherited at unlinked loci.

10.3.4 Diagram: the rejection region

The independence test is always **upper-tailed**: only large χ^2_{obs} (big gaps between observed and expected) count against H_0, so the whole α sits in the right tail of the $\chi^2_{(r-1)(c-1)}$ curve.



Upper-tail rejection region for an independence test with $(r-1)(c-1) = 4$ *df at* $\alpha = 0.05$. *The entire* α *lies beyond* $\chi^2_{0.05,4} = 9.488$*; the commute-survey statistic* $\chi^2_{obs} = 10.23$ *falls inside it, so* H_0 *is rejected. Because only large statistics signal departure from independence, the test is never two-tailed.*

10.3.5 Contrast: independence is not homogeneity

Example 10.3.10 (CONTRAST: one sample cross-classified vs. several samples compared)**.** Two studies produce the *same* 2×3 table of counts and the *same* arithmetic, yet they are different tests. Study A surveys one random sample of 300 city residents, recording each person's commute mode (car, transit, bike) and whether they are satisfied with their commute. Study B separately samples 100 car commuters, 100 transit commuters, and 100 bike commuters (fixed by design) and records satisfaction. Both yield identical counts. Which test is which, and does the distinction change the statistic or df?

Which rule, and why? Read the **sampling design**, not the table. Study A draws *one* sample and cross-classifies it on two variables ⇒ **test of independence**. Study B draws *three separate* samples (the row totals were fixed in advance) and compares their satisfaction profiles ⇒ **test of homogeneity** (§4). The statistic and df are identical; the hypothesis wording is not.

Solution.

Step 1: identify what was fixed. In Study A nothing but the grand total $n = 300$ was fixed; both margins are random. In Study B the three commute-mode group sizes $(100, 100, 100)$ were fixed by the experimenter, so those row totals are not random.

Step 2: state the hypotheses.

Study A (independence): H_0: commute mode and satisfaction are independent.

Study B (homogeneity): H_0: the satisfaction distribution is the same across the three modes.

Step 3: statistic and df are the same. Both use $\chi^2_{\text{obs}} = \sum(O_{ij} - E_{ij})^2/E_{ij}$ with $E_{ij} = R_i C_j/n$ and df $= (r-1)(c-1) = (2-1)(3-1) = 2$. The numbers are byte-for-byte identical.

Step 4: only the conclusion differs.

$$\boxed{\text{Same } \chi^2_{\text{obs}},\ \text{same df} = 2;\ \text{A concludes “related,” B concludes “profiles differ.”}}$$

The lesson: the χ^2 arithmetic cannot tell you which test you ran — the *design* does. Forward-reference §4 for the homogeneity setup in full.

10.3.6 Interpreting marginal and cautious results

Example 10.3.11 (A marginally significant result, read cautiously)**.** A workplace study cross-tabulates $n = 400$ employees by remote-work status (remote, on-site) and self-reported burnout (yes, no), giving $\chi^2_{\text{obs}} = 3.95$ on 1 df at $\alpha = 0.05$. Interpret the result, noting how close it is to the cutoff.

Which rule, and why? A 2×2 independence test, df $= 1$, upper-tailed. The teaching point is **cautious interpretation**: the statistic only just clears $\chi^2_{0.05,1} = 3.841$, so the p-value is barely under 0.05 and the evidence is weak.

Solution.

Step 1: compare to the cutoff. On 1 df, $\chi^2_{0.05,1} = 3.841$ and the next table point $\chi^2_{0.025,1} = 5.024$. Since $3.841 < 3.95 < 5.024$, the p-value lies between 0.025 and 0.05 — roughly $\mathbb{P}(\chi^2_1 > 3.95) \approx 0.047$.

Step 2: technical verdict. At $\alpha = 0.05$ we reject, since $3.95 > 3.841$.

Step 3: cautious reading. The decision is fragile: a p-value of 0.047 would flip to “fail to reject” at $\alpha = 0.025$ or under any correction for multiple testing, and a handful of reclassified employees could move the count across the line.

$$\boxed{\chi^2_{\text{obs}} = 3.95 > 3.841 \ \Rightarrow\ \text{reject at 5\%, but } p \approx 0.047 \text{ is borderline.}}$$

Report it as “weak, borderline evidence of an association,” not a settled finding — and certainly not as remote work *causing* burnout.

Example 10.3.12 (A 2×3 table at the 1% level)**.** A genetics lab classifies $n = 450$ plants by flower color (red, pink, white) and a leaf marker (present, absent), seeking linkage. The counts are below. Test independence at the stringent $\alpha = 0.01$.

	red	pink	white	**row total**
marker present	60	90	50	200
marker absent	90	85	75	250
col total	150	175	125	450

Which rule, and why? One sample of plants, two categorical traits ⇒ independence test on a 2×3 table; df $= (2-1)(3-1) = 2$. The only twist is the stringent level $\alpha = 0.01$, which raises the cutoff to $\chi^2_{0.01,2} = 9.210$.

Solution.

Step 1: expected counts $E_{ij} = R_i C_j / 450$.

$$E_{1,\text{red}} = \tfrac{200\cdot 150}{450} = 66.667, \quad E_{1,\text{pink}} = \tfrac{200\cdot 175}{450} = 77.778, \quad E_{1,\text{white}} = \tfrac{200\cdot 125}{450} = 55.556,$$
$$E_{2,\text{red}} = \tfrac{250\cdot 150}{450} = 83.333, \quad E_{2,\text{pink}} = \tfrac{250\cdot 175}{450} = 97.222, \quad E_{2,\text{white}} = \tfrac{250\cdot 125}{450} = 69.444.$$

Step 2: contributions.

$$\tfrac{(60-66.667)^2}{66.667} = 0.6667, \quad \tfrac{(90-77.778)^2}{77.778} = 1.9206, \quad \tfrac{(50-55.556)^2}{55.556} = 0.5556,$$
$$\tfrac{(90-83.333)^2}{83.333} = 0.5333, \quad \tfrac{(85-97.222)^2}{97.222} = 1.5365, \quad \tfrac{(75-69.444)^2}{69.444} = 0.4444.$$

Step 3: sum.

$$\begin{aligned}\chi^2_{\text{obs}} &= 0.6667 + 1.9206 + 0.5556 + 0.5333 + 1.5365 + 0.4444 \\ &= 5.657.\end{aligned}$$

Step 4: compare. On 2 df, $\chi^2_{0.01,2} = 9.210$. Since $5.657 < 9.210$, do not reject.

$$\boxed{\chi^2_{\text{obs}} = 5.657 < 9.210 \;\Rightarrow\; \text{fail to reject: no significant color–marker linkage.}}$$

Even at the looser $\alpha = 0.05$ the cutoff is $\chi^2_{0.05,2} = 5.991 > 5.657$, so the conclusion is the same: the data give no real evidence of association.

Example 10.3.13 (Survey response by region: a 2×4 table)**.** A pollster samples $n = 1000$ voters and cross-classifies them by region (NE, MW, S, W) and whether they support a referendum. The counts are below. At $\alpha = 0.05$, is support independent of region?

	NE	MW	S	W	**row total**
support	130	120	150	100	500
oppose	120	130	110	140	500
col total	250	250	260	240	1000

Which rule, and why? One sample of voters, two traits (region, support) ⇒ independence test on a 2×4 table; df $= (2-1)(4-1) = 3$.

Solution.

Step 1: expected counts $E_{ij} = R_i C_j / 1000$. Each row total is 500, so $E_{1j} = E_{2j} = C_j/2$:

$$E_{\text{NE}} = 125, \quad E_{\text{MW}} = 125, \quad E_{\text{S}} = 130, \quad E_{\text{W}} = 120$$

for both the support and oppose rows.

Step 2: contributions (support row, then oppose row).

$$\text{support: } \tfrac{(130-125)^2}{125} = 0.2000, \quad \tfrac{(120-125)^2}{125} = 0.2000, \quad \tfrac{(150-130)^2}{130} = 3.0769, \quad \tfrac{(100-120)^2}{120} = 3.3333,$$
$$\text{oppose: } \tfrac{(120-125)^2}{125} = 0.2000, \quad \tfrac{(130-125)^2}{125} = 0.2000, \quad \tfrac{(110-130)^2}{130} = 3.0769, \quad \tfrac{(140-120)^2}{120} = 3.3333.$$

Step 3: sum.

$$\begin{aligned}\chi^2_{\text{obs}} &= 2(0.2000 + 0.2000 + 3.0769 + 3.3333) \\ &= 2 \cdot 6.8102 = 13.620.\end{aligned}$$

Step 4: compare. On 3 df, $\chi^2_{0.05,3} = 7.815$. Since $13.620 > 7.815$, reject.

$$\boxed{\chi^2_{\text{obs}} = 13.620 > 7.815 \;\Rightarrow\; \text{reject } H_0\text{: support is associated with region.}}$$

The South leans support and the West leans oppose; the NE and MW cells contribute almost nothing.

Example 10.3.14 (Defect type by shift: a 3×3 manufacturing table)**.** A plant logs $n = 360$ defective units by defect type (scratch, dent, misprint) and production shift (day, swing, night). At $\alpha = 0.05$, is defect type independent of shift?

	day	swing	night	**row total**
scratch	40	30	50	120
dent	30	40	50	120
misprint	50	50	20	120
col total	120	120	120	360

Which rule, and why? One sample of defects classified by two traits (type, shift) $\Rightarrow$ independence test on a 3×3 table; $\text{df} = (3-1)(3-1) = 4$. Here all margins are equal, which makes every E_{ij} the same.

Solution.

Step 1: expected counts. Every row total is 120 and every column total is 120, so

$$E_{ij} = \frac{120 \cdot 120}{360} = 40 \quad \text{for all 9 cells.}$$

Step 2: contributions $(O_{ij} - 40)^2/40$.

$$\begin{aligned}
\text{scratch: } & \tfrac{(40-40)^2}{40} = 0, & \tfrac{(30-40)^2}{40} = 2.5, & & \tfrac{(50-40)^2}{40} = 2.5, \\
\text{dent: } & \tfrac{(30-40)^2}{40} = 2.5, & \tfrac{(40-40)^2}{40} = 0, & & \tfrac{(50-40)^2}{40} = 2.5, \\
\text{misprint: } & \tfrac{(50-40)^2}{40} = 2.5, & \tfrac{(50-40)^2}{40} = 2.5, & & \tfrac{(20-40)^2}{40} = 10.0.
\end{aligned}$$

Step 3: sum.

$$\chi^2_{\text{obs}} = 0 + 2.5 + 2.5 + 2.5 + 0 + 2.5 + 2.5 + 2.5 + 10.0 = 25.000.$$

Step 4: compare. On 4 df, $\chi^2_{0.05,4} = 9.488$. Since $25.000 > 9.488$, reject decisively.

$$\boxed{\chi^2_{\text{obs}} = 25.000 > 9.488 \;\Rightarrow\; \text{reject } H_0\text{: defect type depends on shift.}}$$

The night-shift misprint cell ($O_{ij} = 20$ vs. $E_{ij} = 40$, contributing 10.0) is the single biggest driver — a concrete lead for the quality team.

Example 10.3.15 (Reading a non-rejection: customer tier by support channel)**.** A SaaS company cross-tabulates $n = 600$ support tickets by customer tier (free, pro, enterprise) and channel (chat, email), obtaining $\chi^2_{\text{obs}} = 1.82$ on 2 df. Interpret at $\alpha = 0.05$, and state precisely what "fail to reject" does and does not claim.

Which rule, and why? A 3×2 independence test, $\text{df} = (3-1)(2-1) = 2$, upper-tailed. The teaching point is the **meaning of a non-rejection**, which is often mis-stated as "the variables are independent."

Solution.

Step 1: compare. On 2 df, $\chi^2_{0.05,2} = 5.991$. Since $1.82 < 5.991$, fail to reject.

Step 2: what this does claim. The data are consistent with independence: the observed counts are within the range independence comfortably produces (here $p \approx 0.40$).

Step 3: what this does not claim. It does *not* prove the variables are independent. Absence of evidence is not evidence of absence; a real but small association could hide under this sample size.

$$\boxed{\chi^2_{\text{obs}} = 1.82 < 5.991 \;\Rightarrow\; \text{fail to reject: no evidence of association (not "proven independent").}}$$

Phrase non-rejections as "no evidence of a relationship," never as "the two are independent."

Remark (Putting it together). Every independence test followed one recipe: confirm it is *one* sample cross-classified on *two* variables; compute $E_{ij} = R_i C_j / n$ from the margins; sum $(O_{ij} - E_{ij})^2/E_{ij}$; compare to $\chi^2_{\alpha,(r-1)(c-1)}$ in the upper tail. The two judgment calls that separate a right answer from a wrong one are getting the degrees of freedom $(r-1)(c-1)$ rather than $rc-1$, and reading a rejection as **association** rather than causation. Hold the design fixed and the arithmetic is mechanical.

10.4 The Test of Homogeneity

The previous section cross-classified *one* sample by two variables and asked whether the two were related. We now turn the table on its side, design-wise: we have *several separate populations* — different factories, regions, treatment arms, breeds — and from each we draw an independent sample and record the *same* categorical response. The question is whether the response splits into its categories in the same proportions across all the populations. That is the **test of homogeneity**. The remarkable and confusing fact is that the arithmetic is *identical* to the test of independence: the same expected counts $E_{ij} = R_i C_j / n$, the same statistic $\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$, the same degrees of freedom $(r-1)(c-1)$. What differs is the **sampling design** and therefore the *conclusion*. In independence, one sample is taken and *both* margins are random; in homogeneity, one set of margins (say the row totals, one per population) is *fixed in advance* by the experimenter, and several independent samples are compared. So when you see a contingency table you must read the story behind it: were the row totals chosen by the sampler (homogeneity) or did they fall out of a single random sample (independence)? The numbers you compute are the same; the sentence you write at the end is not.

Independence or homogeneity — which story is the table telling?

- **Several distinct populations** sampled *separately*, the *same* response recorded in each, asking whether the category proportions agree across populations? $\Rightarrow$ **test of homogeneity**; $E_{ij} = R_i C_j / n$, $\chi^2 = \sum (O_{ij} - E_{ij})^2 / E_{ij}$, df $= (r-1)(c-1)$.
- **One** sample, each unit cross-classified by *two* variables, asking whether the variables are associated? $\Rightarrow$ **test of independence** (§3) — *same arithmetic*, different design and conclusion.
- Only **two** populations and a **yes/no** response (a 2×2 table)? $\Rightarrow$ the homogeneity test reduces to **comparing two proportions**; its χ^2 on 1 df equals the square of the two-sample z for proportions.
- Several populations, a yes/no response (a $2 \times c$ table)? $\Rightarrow$ homogeneity is the simultaneous **comparison of c proportions**.

Do not confuse design with arithmetic: the giveaway is who fixed the margins. If you, the sampler, decided "draw 200 from region A and 200 from region B" *before* seeing any response, the row totals are fixed and it is homogeneity. If you drew one sample of 400 and only afterward counted how they split by region *and* response, both margins are random and it is independence. The χ^2 value is blind to the difference; you are not.

10.4.1 The setup: several samples, one response

Definition 10.4.1 (Homogeneity table and its expected counts)**.** Let there be r populations. From population i we draw an independent sample of *fixed* size R_i and classify each sampled unit into one of c response categories, giving the observed count O_{ij} in row i, column j. Write

$$R_i = \sum_{j=1}^{c} O_{ij} \text{ (fixed by design)}, \qquad C_j = \sum_{i=1}^{r} O_{ij}, \qquad n = \sum_i R_i.$$

Under the null hypothesis of **homogeneity** — that every population has the same response distribution $(p_1, \ldots, p_c)$ — the best estimate of p_j pools all populations, $\widehat{p}_j = C_j / n$, and the expected count in cell (i, j) is

$$E_{ij} \;=\; R_i \, \widehat{p}_j \;=\; R_i \cdot \frac{C_j}{n} \;=\; \frac{R_i \, C_j}{n},$$

exactly the row-total-times-column-total-over-n formula of the independence test.

Theorem 10.4.2 (The homogeneity test)**.** *With the table above, to test*

$$H_0: \text{ all } r \text{ populations share one response distribution} \quad \text{vs.} \quad H_1: \text{ at least two populations differ,}$$

form

$$\chi^2 \;=\; \sum_{i=1}^{r}\sum_{j=1}^{c}\frac{(O_{ij}-E_{ij})^2}{E_{ij}}, \qquad E_{ij}=\frac{R_iC_j}{n},$$

which under H_0 is approximately a χ^2 random variable with

$$(r-1)(c-1) \text{ degrees of freedom.}$$

Reject H_0 at level α when $\chi^2 \geq \chi^2_{\alpha,\,(r-1)(c-1)}$ (always the upper *tail). The approximation is trustworthy when every $E_{ij} \geq 5$.*

Why the df is still $(r-1)(c-1)$. Each of the r rows is its own independent sample, a multinomial over c categories. A free multinomial on c categories carries $c-1$ free counts (the last is fixed by the row total R_i), so the r rows supply $r(c-1)$ free pieces of data. Under H_0 we must estimate the common distribution $(p_1,\ldots,p_c)$, which has $c-1$ free parameters (they sum to 1). Subtracting,

$$r(c-1)-(c-1)=(c-1)(r-1)=(r-1)(c-1).$$

The bookkeeping is different from independence — there the row totals were random, not fixed — but it lands on the identical number, which is one reason the two tests are so easy to conflate. □

Remark. The same arithmetic, three different derivations. In goodness-of-fit the E_i came from a fully specified distribution; in independence both margins were random and the E_{ij} estimated a product of marginals; here the row margins are fixed and the E_{ij} estimate a common column distribution. All three roads lead to $\sum(O-E)^2/E$ and to a χ^2 reference distribution. What changes is only what the rejection *means*: "the data do not fit the claimed distribution" (GOF), "the two variables are associated" (independence), or "the populations do not share one response distribution" (homogeneity).

10.4.2 A response compared across several populations

Example 10.4.3 (Defect mix across three production lines)**.** A factory runs three assembly lines and wants to know whether they produce the *same mix* of defect types. Quality control pulls a fixed sample of defective units from each line — 150 from line A, 150 from line B, 120 from line C — and classifies each defect as Cosmetic, Mechanical, or Electrical. The counts are

line	Cosmetic	Mechanical	Electrical	R_i
A	78	48	24	150
B	72	51	27	150
C	45	48	27	120
C_j	195	147	78	420

At $\alpha = 0.05$, test whether the three lines have the same defect-type distribution.

Which rule, and why? Three separate samples (one per line), each classified by the same response (defect type), with the per-line totals *fixed in advance* by quality control — this is the design of the homogeneity test, not independence. We compute $E_{ij}=R_iC_j/n$ and refer χ^2 to $(r-1)(c-1)$ df.

Solution.

Step 1: expected counts $E_{ij}=R_iC_j/n$ with $n=420$. For each row, multiply the row total by each column total and divide by 420:

A: $\frac{150\cdot195}{420}=69.64,\ \frac{150\cdot147}{420}=52.50,\ \frac{150\cdot78}{420}=27.86;$
B: $69.64,\ 52.50,\ 27.86$ (same row total as A);
C: $\frac{120\cdot195}{420}=55.71,\ \frac{120\cdot147}{420}=42.00,\ \frac{120\cdot78}{420}=22.29.$

All $E_{ij} \geq 5$, so the approximation is valid.

Step 2: cell contributions $(O_{ij} - E_{ij})^2/E_{ij}$.

$$\begin{aligned}
\text{A:}\ & \tfrac{(78-69.64)^2}{69.64} + \tfrac{(48-52.50)^2}{52.50} + \tfrac{(24-27.86)^2}{27.86} = 1.004 + 0.386 + 0.535 = 1.925,\\
\text{B:}\ & \tfrac{(72-69.64)^2}{69.64} + \tfrac{(51-52.50)^2}{52.50} + \tfrac{(27-27.86)^2}{27.86} = 0.080 + 0.043 + 0.027 = 0.150,\\
\text{C:}\ & \tfrac{(45-55.71)^2}{55.71} + \tfrac{(48-42.00)^2}{42.00} + \tfrac{(27-22.29)^2}{22.29} = 2.059 + 0.857 + 0.995 = 3.911.
\end{aligned}$$

Step 3: sum to get χ^2.

$$\chi^2 = 1.925 + 0.150 + 3.911 = 5.99.$$

Step 4: df and critical value. Here $r = 3$, $c = 3$, so df $= (3-1)(3-1) = 4$ and $\chi^2_{0.05,4} = 9.488$. Since $5.99 < 9.488$, do not reject H_0.

Step 5: conclusion. The three lines show no significant difference in defect-type mix; line C leans a little more Mechanical/Electrical, but the gap is within sampling noise.

$$\boxed{\chi^2 = 5.99 < \chi^2_{0.05,4} = 9.488 \ \Rightarrow\ \text{do not reject; lines share one defect mix.}}$$

Figure 10.1 shows the three lines' *proportions* side by side: the bars are close to a common profile, the visual signature of failing to reject homogeneity.

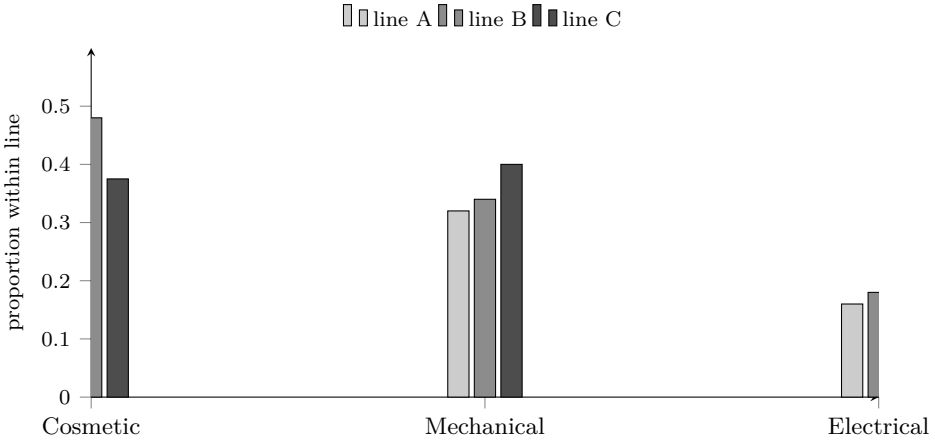


Figure 10.1: Defect-type proportions *within* each production line. The three profiles track one another closely (line C dips a little on Cosmetic and rises on Mechanical), which is why the homogeneity test fails to reject: no population's response distribution departs sharply from the common shape.

10.4.3 When the distributions really do differ

Example 10.4.4 (Opinion split across four regions)**.** A pollster surveys independent samples of registered voters in four regions, asking each whether they Support, Oppose, or are Undecided on a ballot measure. The fixed sample sizes and counts are

region	Support	Oppose	Undecided	R_i
North	120	60	20	200
South	80	100	20	200
East	90	80	30	200
West	110	70	20	200
C_j	400	310	90	800

At $\alpha = 0.01$, test whether opinion is distributed the same way across the four regions.

Which rule, and why? Four separate regional samples of the same opinion question, each region's sample size set in advance — a homogeneity design across $r = 4$ populations and $c = 3$ response categories. Same χ^2 machinery; df $= (4-1)(3-1) = 6$.

Solution.

Step 1: expected counts. Every $R_i = 200$, $n = 800$, so each row's expecteds are the same:

$$E_{i,\text{Sup}} = \tfrac{200\cdot 400}{800} = 100, \quad E_{i,\text{Opp}} = \tfrac{200\cdot 310}{800} = 77.5, \quad E_{i,\text{Und}} = \tfrac{200\cdot 90}{800} = 22.5.$$

Step 2: cell contributions by row.

$$\begin{aligned}
\text{North:} &\ \tfrac{(120-100)^2}{100} + \tfrac{(60-77.5)^2}{77.5} + \tfrac{(20-22.5)^2}{22.5} = 4.000 + 3.952 + 0.278 = 8.230,\\
\text{South:} &\ \tfrac{(80-100)^2}{100} + \tfrac{(100-77.5)^2}{77.5} + \tfrac{(20-22.5)^2}{22.5} = 4.000 + 6.532 + 0.278 = 10.810,\\
\text{East:} &\ \tfrac{(90-100)^2}{100} + \tfrac{(80-77.5)^2}{77.5} + \tfrac{(30-22.5)^2}{22.5} = 1.000 + 0.081 + 2.500 = 3.581,\\
\text{West:} &\ \tfrac{(110-100)^2}{100} + \tfrac{(70-77.5)^2}{77.5} + \tfrac{(20-22.5)^2}{22.5} = 1.000 + 0.726 + 0.278 = 2.004.
\end{aligned}$$

Step 3: total χ^2.

$$\chi^2 = 8.230 + 10.810 + 3.581 + 2.004 = 24.63.$$

Step 4: df and critical value. df $= (4-1)(3-1) = 6$; $\chi^2_{0.01,6} = 16.812$. Since $24.63 > 16.812$, reject H_0.

Step 5: conclusion. The regions do *not* share one opinion distribution. Inspecting the contributions, the South (heavily Oppose) and North (heavily Support) drive the rejection.

$$\boxed{\chi^2 = 24.63 > \chi^2_{0.01,6} = 16.812 \ \Rightarrow \ \text{reject; regions differ in opinion.}}$$

10.4.4 Two rows: homogeneity is comparing proportions

Example 10.4.5 (A 2×2 table: comparing two cure rates)**.** Two clinics each treat a fixed number of patients with the same drug and record cure (yes/no). Clinic 1 treats 180 patients and cures 126; clinic 2 treats 220 and cures 132. At $\alpha = 0.05$, test whether the two clinics have the same cure rate using the homogeneity χ^2, and confirm it equals the square of the two-sample z for proportions.

Which rule, and why? Two separate populations (clinics) with fixed sample sizes and a yes/no response is a 2×2 homogeneity table. With $c = 2$, homogeneity is exactly the comparison of two proportions, and its 1-df χ^2 is the square of the two-proportion z.

Solution.

Step 1: build the table.

clinic	cured	not	R_i
1	126	54	180
2	132	88	220
C_j	258	142	400

Step 2: expected counts $E_{ij} = R_i C_j / n$, $n = 400$.

$$\begin{aligned}
E_{1,\text{cured}} &= \tfrac{180\cdot 258}{400} = 116.1, \quad E_{1,\text{not}} = \tfrac{180\cdot 142}{400} = 63.9,\\
E_{2,\text{cured}} &= \tfrac{220\cdot 258}{400} = 141.9, \quad E_{2,\text{not}} = \tfrac{220\cdot 142}{400} = 78.1.
\end{aligned}$$

Step 3: the statistic.

$$\begin{aligned}
\chi^2 &= \tfrac{(126-116.1)^2}{116.1} + \tfrac{(54-63.9)^2}{63.9} + \tfrac{(132-141.9)^2}{141.9} + \tfrac{(88-78.1)^2}{78.1}\\
&= 0.844 + 1.534 + 0.691 + 1.255 = 4.32.
\end{aligned}$$

Step 4: compare and cross-check with z. $\mathrm{df} = (2-1)(2-1) = 1$; $\chi^2_{0.05,1} = 3.841$. Since $4.32 > 3.841$, reject: the cure rates differ. Cross-check: pooled $\widehat{p} = 258/400 = 0.645$, sample rates 0.700 and 0.600, and

$$z = \frac{0.700 - 0.600}{\sqrt{0.645 \cdot 0.355\,(1/180 + 1/220)}} = \frac{0.100}{0.0481} = 2.078, \qquad z^2 = 4.32 = \chi^2.$$

$$\boxed{\chi^2 = 4.32 = z^2 > \chi^2_{0.05,1} = 3.841 \;\Rightarrow\; \text{reject; cure rates differ.}}$$

Remark. The 2×2 homogeneity test and the two-sample test of $H_0 : p_1 = p_2$ are the *same test*. The χ^2 statistic on 1 df is algebraically z^2 for the pooled two-proportion z, so they reject on exactly the same data. The χ^2 form generalizes painlessly to more than two populations or more than two response categories, where a single z no longer exists — which is the real reason to learn it.

10.4.5 Several proportions at once: a $2 \times c$ table

Example 10.4.6 (Comparing germination rates across four seed lots)**.** An agronomist plants a fixed 200 seeds from each of four seed lots and records germinated/not. The germinated counts are 164, 150, 172, and 158 out of 200. At $\alpha = 0.05$, test whether the four lots germinate at the same rate.

Which rule, and why? Four populations (lots) with fixed sample sizes and a yes/no response form a 2×4 homogeneity table. With $c = 2$ this is the simultaneous comparison of four proportions; $\mathrm{df} = (4-1)(2-1) = 3$.

Solution.

Step 1: the table.

lot	germ.	not	R_i
1	164	36	200
2	150	50	200
3	172	28	200
4	158	42	200
C_j	644	156	800

Step 2: expected counts. Each $R_i = 200$, $n = 800$, so $E_{i,\mathrm{germ}} = 200 \cdot 644/800 = 161$ and $E_{i,\mathrm{not}} = 200 \cdot 156/800 = 39$ for every lot.

Step 3: cell contributions. Using the constant expecteds 161 and 39:

$$\begin{aligned}
\text{lot 1:}\quad & \tfrac{(164-161)^2}{161} + \tfrac{(36-39)^2}{39} = 0.056 + 0.231 = 0.287,\\
\text{lot 2:}\quad & \tfrac{(150-161)^2}{161} + \tfrac{(50-39)^2}{39} = 0.752 + 3.103 = 3.855,\\
\text{lot 3:}\quad & \tfrac{(172-161)^2}{161} + \tfrac{(28-39)^2}{39} = 0.752 + 3.103 = 3.855,\\
\text{lot 4:}\quad & \tfrac{(158-161)^2}{161} + \tfrac{(42-39)^2}{39} = 0.056 + 0.231 = 0.287.
\end{aligned}$$

Step 4: total and decision.

$$\chi^2 = 0.287 + 3.855 + 3.855 + 0.287 = 8.28, \qquad \chi^2_{0.05,3} = 7.815.$$

Since $8.28 > 7.815$, reject H_0: the lots do not all germinate at the same rate (lots 2 and 3 straddle the common rate of 80.5% in opposite directions).

$$\boxed{\chi^2 = 8.28 > \chi^2_{0.05,3} = 7.815 \;\Rightarrow\; \text{reject; germination rates differ.}}$$

10.4.6 Reading the design off the wording

Example 10.4.7 (CONTRAST: same numbers, two different stories)**.** A single 2×2 table of counts,

	response Y	response N	R_i
group 1	60	40	100
group 2	45	55	100
C_j	105	95	200

arises in two studies. **(a)** A marketer mailed an ad to 100 customers in city 1 and a different ad to 100 customers in city 2 and recorded whether each responded (Y/N). **(b)** A single random sample of 200 shoppers was each asked two yes/no questions, "do you own a pet" (group, recoded 1/2) and "do you buy organic" (Y/N), and cross-tabulated. Both give the table above. State the right test and the right *conclusion* for each.

Which rule, and why? The arithmetic — $E_{ij} = R_i C_j / n$, χ^2, df $= 1$ — is identical in both. The *design* decides which test it is: in (a) the row totals (100 and 100) were fixed by the mailer, so two separate populations are compared (homogeneity); in (b) one sample of 200 was drawn and both margins are random (independence). The deliberate trap is to let the matching numbers fool you into thinking it is "the same test." It is the same *computation* but a different *claim*.

Solution.

Step 1: the shared computation. Expecteds $E_{11} = 100 \cdot 105/200 = 52.5$, $E_{12} = 47.5$, $E_{21} = 52.5$, $E_{22} = 47.5$. Then

$$\chi^2 = \frac{(60-52.5)^2}{52.5} + \frac{(40-47.5)^2}{47.5} + \frac{(45-52.5)^2}{52.5} + \frac{(55-47.5)^2}{47.5}$$
$$= 1.071 + 1.184 + 1.071 + 1.184 = 4.51.$$

With 1 df and $\chi^2_{0.05,1} = 3.841$, both studies reject ($4.51 > 3.841$).

Step 2: conclusion for (a), homogeneity. The two cities were sampled separately with fixed sizes, so we conclude: *the response rate to the ad differs between city 1 and city 2* (60% vs. 45%). The statement is about **equality of proportions across populations**.

Step 3: conclusion for (b), independence. One sample was cross-classified, so we conclude: *pet ownership and organic buying are associated* in this population. The statement is about **a relationship between two variables**, not about comparing two pre-set groups.

Step 4: the lesson. Same $\chi^2 = 4.51$, same df, same reject/retain decision — but "the rates differ across cities" and "the two traits are related" are different scientific claims, licensed by different sampling designs.

Identical $\chi^2 = 4.51$; (a) proportions differ across populations, (b) variables are associated.

Example 10.4.8 (Deciding which test the wording calls for)**.** For each description, name the correct χ^2 test (homogeneity or independence) and the df. **(a)** "We surveyed 500 randomly chosen adults, recording sex and whether they exercise regularly." **(b)** "We tested 100 samples from each of three suppliers for pass/marginal/fail." **(c)** "From each of two hospitals we sampled 150 charts and classified the admitting diagnosis into five categories." **(d)** "A single batch of 400 widgets was classified by shift produced (3 shifts) and by defect status (defective/ok)."

Which rule, and why? The deciding question is always: were the margins of one variable *fixed by the sampler* (homogeneity — several samples) or did a *single* sample fall freely into both margins (independence)? Listen for "from each of ... we sampled N" (fixed, homogeneity) versus "we sampled N and recorded two things" (one sample, independence).

Solution.

Step 1: (a). One random sample of 500, two variables recorded — both margins random, so **independence**. Table 2×2, df $= (2-1)(2-1) = 1$.

Step 2: (b). A fixed 100 from *each* of three suppliers, one response (3 levels) — three separate samples, so **homogeneity**. Table 3×3, df $= (3-1)(3-1) = 4$.

Step 3: (c). A fixed 150 from *each* of two hospitals, one response (5 categories) — two separate samples, so **homogeneity**. Table 2×5, df $= (2-1)(5-1) = 4$.

Step 4: (d). A single batch of 400 cross-classified by two variables — both margins random, so **independence**. Table 3×2, df $= (3-1)(2-1) = 2$.

(a) indep., df 1; (b) homog., df 4; (c) homog., df 4; (d) indep., df 2.

10.4.7 A careful degrees-of-freedom example

Example 10.4.9 (Getting the df right for a 5×3 homogeneity table)**.** A market researcher samples a fixed 120 customers from each of five store locations and records which of three loyalty tiers (Bronze, Silver, Gold) each belongs to. Before computing anything, determine the degrees of freedom, and explain in counting terms why it is what it is.

Which rule, and why? This is a homogeneity table because a fixed sample was drawn from *each* of five populations (locations). The df for any $r \times c$ count table of this kind is $(r-1)(c-1)$; the recognition task is to read r and c off the design correctly and not, say, use $rc - 1$.

Solution.

Step 1: identify r and c. There are $r = 5$ populations (locations) and $c = 3$ response categories (tiers). The table is 5×3, with 15 cells.

Step 2: count free cells directly. The five row totals are fixed at 120 each *by design*. Within each row, once $c - 1 = 2$ of the three tier counts are chosen the third is forced, so each row contributes 2 free counts: $5 \times 2 = 10$ free counts in the body. But under H_0 we estimate the common tier distribution (p_1, p_2, p_3), costing $c - 1 = 2$ parameters.

Step 3: subtract.

$$\text{df} = \underbrace{r(c-1)}_{10} - \underbrace{(c-1)}_{2} = (r-1)(c-1) = (5-1)(3-1) = 8.$$

Step 4: guard against the classic slip. A frequent error is to treat this like a goodness-of-fit table and use $rc - 1 = 15 - 1 = 14$. That over-counts because it ignores that the row totals were *fixed* (removing r constraints, here 5) and that estimating the common distribution costs another $c - 1$. The correct value is 8, not 14.

$$\boxed{\text{df} = (r-1)(c-1) = (5-1)(3-1) = 8 \quad (\text{not } rc - 1 = 14).}$$

Remark. The df is a property of the table's *shape*, $(r-1)(c-1)$, regardless of the cell counts and regardless of whether the story is independence or homogeneity. The two designs differ in *which* margin was fixed, but both fix one margin's information and estimate $c - 1$ (or $r - 1$) free proportions, and the subtraction always collapses to $(r-1)(c-1)$. Memorize the shape rule, but be able to rebuild it by counting, because that is the only defense against tables that tempt you into $rc - 1$.

Figure 10.2 shows where the test lives: the $\chi^2_{(r-1)(c-1)}$ density and its upper-tail rejection region, drawn for the 8-df table above.

10.4.8 More worked homogeneity tests

Example 10.4.10 (Blood-type mix across three breeds)**.** A veterinary geneticist samples a fixed 90 dogs from each of three breeds and types each as DEA-positive, DEA-negative, or weak. The counts are

breed	pos	neg	weak	R_i
X	54	27	9	90
Y	45	36	9	90
Z	36	45	9	90
C_j	135	108	27	270

At $\alpha = 0.05$, test whether the blood-type distribution is the same across the three breeds.

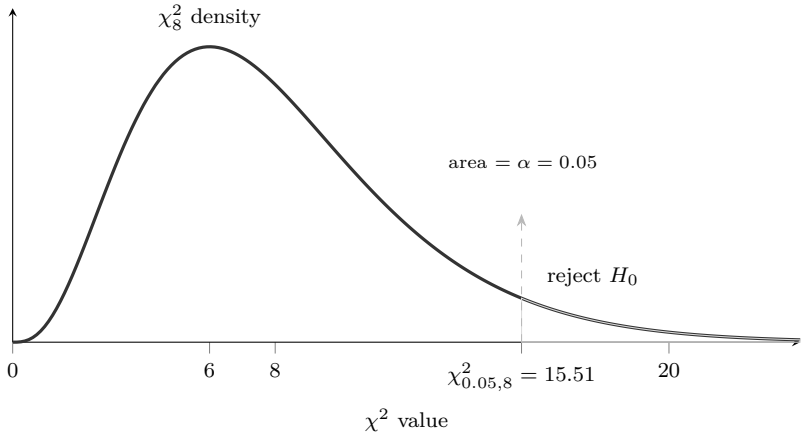


Figure 10.2: The reference distribution for an 8-df homogeneity (or independence) test: $\chi^2_{(r-1)(c-1)}$ with $(r-1)(c-1) = 8$. The shaded upper tail beyond $\chi^2_{0.05,8} = 15.51$ has area $\alpha = 0.05$; an observed χ^2 landing there rejects the hypothesis of a common response distribution. The peak sits near $x = k - 2 = 6$, as it should for χ^2_k.

Which rule, and why? A fixed sample from each of three breeds, one categorical response (blood type) — homogeneity across $r = 3$ populations, $c = 3$ categories, df $= (3-1)(3-1) = 4$.

Solution.

Step 1: expected counts. Each $R_i = 90$, $n = 270$, so each row's expecteds are

$$E_{i,\text{pos}} = \tfrac{90\cdot 135}{270} = 45, \quad E_{i,\text{neg}} = \tfrac{90\cdot 108}{270} = 36, \quad E_{i,\text{weak}} = \tfrac{90\cdot 27}{270} = 9.$$

Step 2: cell contributions (the weak column contributes 0 *in every row).*

$$\begin{aligned}
\text{X:}\ & \tfrac{(54-45)^2}{45} + \tfrac{(27-36)^2}{36} + 0 = 1.800 + 2.250 = 4.050, \\
\text{Y:}\ & \tfrac{(45-45)^2}{45} + \tfrac{(36-36)^2}{36} + 0 = 0, \\
\text{Z:}\ & \tfrac{(36-45)^2}{45} + \tfrac{(45-36)^2}{36} + 0 = 1.800 + 2.250 = 4.050.
\end{aligned}$$

Step 3: total and decision.

$$\chi^2 = 4.050 + 0 + 4.050 = 8.10, \qquad \chi^2_{0.05,4} = 9.488.$$

Since $8.10 < 9.488$, do not reject: although breeds X and Z lean opposite ways on pos/neg, the evidence is not strong enough at the 5% level.

$$\boxed{\chi^2 = 8.10 < \chi^2_{0.05,4} = 9.488 \ \Rightarrow\ \text{do not reject; common blood-type mix.}}$$

Example 10.4.11 (Commute mode across two campuses, one-sided in spirit)**.** A university samples a fixed 250 students at its urban campus and 250 at its suburban campus and records primary commute mode (Walk/Bike, Transit, Drive). The counts are

campus	Walk/Bike	Transit	Drive	R_i
urban	120	95	35	250
suburban	60	70	120	250
C_j	180	165	155	500

At $\alpha = 0.01$, test whether commute-mode distribution is the same on the two campuses.

Which rule, and why? Two separate campus samples with fixed sizes, one response (commute mode) — a 2×3 homogeneity table, df $= (2-1)(3-1) = 2$. (The χ^2 test is intrinsically two-sided about the table; "which campus drives more" is read off the cells afterward.)

Solution.

Step 1: expected counts. Each $R_i = 250$, $n = 500$:

$$E_{i,\mathrm{WB}} = \tfrac{250\cdot 180}{500} = 90, \quad E_{i,\mathrm{Tr}} = \tfrac{250\cdot 165}{500} = 82.5, \quad E_{i,\mathrm{Dr}} = \tfrac{250\cdot 155}{500} = 77.5.$$

Step 2: cell contributions.

$$\begin{aligned} \text{urban:} \quad & \tfrac{(120-90)^2}{90} + \tfrac{(95-82.5)^2}{82.5} + \tfrac{(35-77.5)^2}{77.5} = 10.000 + 1.894 + 23.306 = 35.200, \\ \text{suburban:} \quad & \tfrac{(60-90)^2}{90} + \tfrac{(70-82.5)^2}{82.5} + \tfrac{(120-77.5)^2}{77.5} = 10.000 + 1.894 + 23.306 = 35.200. \end{aligned}$$

Step 3: total and decision.

$$\chi^2 = 35.200 + 35.200 = 70.40, \qquad \chi^2_{0.01,2} = 9.210.$$

Since $70.40 \gg 9.210$, reject decisively: the campuses have very different commute profiles — the suburban campus drives far more (120 vs. 35), the urban campus walks/bikes far more.

$$\boxed{\chi^2 = 70.40 > \chi^2_{0.01,2} = 9.210 \ \Rightarrow \ \text{reject; commute modes differ by campus.}}$$

Example 10.4.12 (CONTRAST: the look-alike wrong test on grouped counts)**.** An analyst has the four germination counts from the seed-lot example ($164, 150, 172, 158$ out of 200 each) and runs a *one-sample goodness-of-fit* test against the hypothesis "half germinate" ($p = 0.5$), getting a huge χ^2, and declares the lots "inhomogeneous." Explain what went wrong and what the right test is.

Which rule, and why? The trap is using the wrong E_i. Goodness-of-fit checks counts against a *prespecified* distribution ($p = 0.5$ here); homogeneity checks several samples against a *common, pooled* distribution estimated from the data. The analyst answered a different question — "is each rate 50%?" — not "are the four rates equal to *each other*?"

Solution.

Step 1: what the GOF test actually tested. Against $p = 0.5$, the expected germinated count is 100 per lot, but the observed are 150–172, so each cell screams. The resulting large χ^2 only says "the germination rate is not 50%" — which everyone already knew. It says nothing about whether the four lots agree.

Step 2: the right comparison. Homogeneity pools the four lots to estimate the common rate $\widehat{p} = 644/800 = 0.805$, sets $E_{i,\mathrm{germ}} = 200 \cdot 0.805 = 161$, and compares each lot to *that*, giving $\chi^2 = 8.28$ on 3 df (from the earlier example) — a borderline rejection, nothing like the GOF blowup.

Step 3: the lesson. A test is defined by its E_i. Plugging a hypothesized p_0 into a homogeneity question manufactures expecteds that answer the wrong hypothesis. To compare populations *with each other*, the expecteds must come from the *pooled* data, not from an outside claim.

$$\boxed{\text{GOF vs. } p_0 \text{ answers "is it } p_0\text{?"; homogeneity pools to ask "are the populations equal?"}}$$

Example 10.4.13 (A 3×4 homogeneity test, full computation)**.** A streaming service samples a fixed number of users in three age bands and records which of four content genres each watches most. The counts are

age band	Drama	Comedy	Action	Docu	R_i
18–29	40	70	60	30	200
30–49	60	50	50	40	200
50+	80	30	30	60	200
C_j	180	150	140	130	600

At $\alpha = 0.05$, test whether the genre distribution is the same across age bands.

Which rule, and why? Three separate age-band samples of fixed size 200 each, one response (genre) — homogeneity across $r = 3$, $c = 4$, df $= (3-1)(4-1) = 6$.

Solution.

Step 1: expected counts. Each $R_i = 200$, $n = 600$, so every row's expecteds are

$$E_{\mathrm{Dr}} = \tfrac{200\cdot 180}{600} = 60,\ E_{\mathrm{Co}} = \tfrac{200\cdot 150}{600} = 50,\ E_{\mathrm{Ac}} = \tfrac{200\cdot 140}{600} = 46.67,\ E_{\mathrm{Do}} = \tfrac{200\cdot 130}{600} = 43.33.$$

Step 2: cell contributions by row.

$$\begin{aligned}
\text{18–29:}\quad & \tfrac{(40-60)^2}{60} + \tfrac{(70-50)^2}{50} + \tfrac{(60-46.67)^2}{46.67} + \tfrac{(30-43.33)^2}{43.33} \\
& = 6.667 + 8.000 + 3.810 + 4.103 = 22.580, \\
\text{30–49:}\quad & \tfrac{(60-60)^2}{60} + \tfrac{(50-50)^2}{50} + \tfrac{(50-46.67)^2}{46.67} + \tfrac{(40-43.33)^2}{43.33} \\
& = 0 + 0 + 0.238 + 0.256 = 0.494, \\
\text{50+:}\quad & \tfrac{(80-60)^2}{60} + \tfrac{(30-50)^2}{50} + \tfrac{(30-46.67)^2}{46.67} + \tfrac{(60-43.33)^2}{43.33} \\
& = 6.667 + 8.000 + 5.953 + 6.410 = 27.030.
\end{aligned}$$

Step 3: total χ^2.

$$\chi^2 = 22.580 + 0.494 + 27.030 = 50.10.$$

Step 4: df and decision. df $= (3-1)(4-1) = 6$; $\chi^2_{0.05,6} = 12.592$. Since $50.10 \gg 12.592$, reject H_0.

Step 5: conclusion. Genre preference depends strongly on age band: the youngest band over-watches Comedy/Action, the oldest over-watches Drama/Documentary, while the middle band sits near the pooled profile (its row contributes almost nothing to χ^2).

$$\boxed{\chi^2 = 50.10 > \chi^2_{0.05,6} = 12.592 \ \Rightarrow\ \text{reject; genre mix differs by age.}}$$

Example 10.4.14 (Three treatment arms, a clinical-style homogeneity test)**.** A trial assigns a fixed 80 patients to each of three arms (placebo, low dose, high dose) and records outcome as Worse, Same, or Improved. The counts are

arm	Worse	Same	Improved	R_i
placebo	20	40	20	80
low dose	12	38	30	80
high dose	8	30	42	80
C_j	40	108	92	240

At $\alpha = 0.05$, test whether the outcome distribution is the same across the three arms.

Which rule, and why? Patients are allocated to three arms in fixed numbers, and the same outcome is recorded in each — the treatment arms are the "populations," so this is homogeneity across $r = 3$, $c = 3$, df $= (3-1)(3-1) = 4$.

Solution.

Step 1: expected counts. Each $R_i = 80$, $n = 240$:

$$E_{i,\mathrm{W}} = \tfrac{80\cdot 40}{240} = 13.33, \quad E_{i,\mathrm{S}} = \tfrac{80\cdot 108}{240} = 36, \quad E_{i,\mathrm{I}} = \tfrac{80\cdot 92}{240} = 30.67.$$

Step 2: cell contributions.

$$\begin{aligned}
\text{placebo:}\quad & \tfrac{(20-13.33)^2}{13.33} + \tfrac{(40-36)^2}{36} + \tfrac{(20-30.67)^2}{30.67} = 3.336 + 0.444 + 3.712 = 7.492, \\
\text{low:}\quad & \tfrac{(12-13.33)^2}{13.33} + \tfrac{(38-36)^2}{36} + \tfrac{(30-30.67)^2}{30.67} = 0.133 + 0.111 + 0.015 = 0.259, \\
\text{high:}\quad & \tfrac{(8-13.33)^2}{13.33} + \tfrac{(30-36)^2}{36} + \tfrac{(42-30.67)^2}{30.67} = 2.131 + 1.000 + 4.186 = 7.317.
\end{aligned}$$

Step 3: total and decision.

$$\chi^2 = 7.492 + 0.259 + 7.317 = 15.07, \qquad \chi^2_{0.05,4} = 9.488.$$

Since $15.07 > 9.488$, reject H_0.

Step 4: conclusion. The outcome distribution depends on dose: placebo skews Worse, high dose skews Improved, in a clear dose-response pattern.

$$\boxed{\chi^2 = 15.07 > \chi^2_{0.05,4} = 9.488 \;\Rightarrow\; \text{reject; outcomes differ by dose.}}$$

Example 10.4.15 (Pooling collapses an apparent difference)**.** Two call centers are sampled, a fixed 300 calls each, and each call is logged Resolved or Escalated. Center 1: 246 resolved, 54 escalated. Center 2: 234 resolved, 66 escalated. At $\alpha = 0.05$, test whether the two centers resolve at the same rate, and comment on the practical size of any difference.

Which rule, and why? Two populations (centers), fixed sample sizes, yes/no response — a 2×2 homogeneity test, equivalently a comparison of two proportions; df $= 1$.

Solution.

Step 1: table and expecteds. Column totals 480 resolved, 120 escalated, $n = 600$. Each center's expecteds: $E_{\text{res}} = 300 \cdot 480/600 = 240$, $E_{\text{esc}} = 300 \cdot 120/600 = 60$.

Step 2: the statistic.

$$\begin{aligned}\chi^2 &= \tfrac{(246-240)^2}{240} + \tfrac{(54-60)^2}{60} + \tfrac{(234-240)^2}{240} + \tfrac{(66-60)^2}{60} \\ &= 0.150 + 0.600 + 0.150 + 0.600 = 1.50.\end{aligned}$$

Step 3: decision. $\chi^2_{0.05,1} = 3.841$; since $1.50 < 3.841$, do not reject. The observed resolution rates (82.0% vs. 78.0%) differ by only 4 points, well within sampling noise at these sample sizes.

$$\boxed{\chi^2 = 1.50 < \chi^2_{0.05,1} = 3.841 \;\Rightarrow\; \text{do not reject; resolution rates are comparable.}}$$

Remark. Homogeneity, independence, and the design you actually ran. In practice you rarely choose the test from the table; you choose it from the protocol you (or whoever collected the data) followed. If the sampling plan said "take n_i units from population i," the row totals are constants and you are in homogeneity, concluding about equality of distributions across those populations. If the plan said "take one sample of n and classify each unit two ways," both margins are random and you are in independence, concluding about association. The computation will not tell you which you did — it gives the same χ^2, the same df, the same decision — so the discipline is to write the conclusion that matches the design, never to read more (a causal story, a fixed-margin comparison) into the table than the sampling supports.

10.5 Conditions, Small Counts, and Pitfalls

Every test in this chapter computes the same statistic $\chi^2 = \sum_i (O_i - E_i)^2/E_i$ and refers it to a χ^2 curve, but the curve is only an *approximation* to the true distribution of that sum, and the approximation can fail. This section collects the conditions that keep the approximation honest and the traps that quietly break it. The headline requirement is on the **expected** counts E_i — not the observed counts — because the χ^2 approximation rests on each cell's count being roughly normal, which needs a large enough E_i. When some E_i are too small we **pool** adjacent categories and pay for it in degrees of freedom; in the special 2×2 case we may instead apply a **continuity correction**. Two further traps have nothing to do with small counts: feeding the formula **percentages** instead of raw counts (which silently rescales n), and letting a **huge sample** turn a practically meaningless deviation into a "significant" one. Recognise this section's material whenever a problem hands you sparse cells, a 2×2 table, data reported as proportions, an enormous n, or asks what a rejected fit actually proves.

Which caveat is the problem testing?

- Some **expected** count E_i below 5 (and you have raw counts)? ⇒ check the **rule of thumb**; if it fails, **pool** neighbouring categories and *recompute df* from the new number of cells.
- A 2×2 table with modest counts? ⇒ consider the **Yates continuity correction**, subtracting $\frac{1}{2}$ from each $|O_i - E_i|$ before squaring; df stays 1.
- Data given as **percentages, proportions, or rates**? ⇒ convert back to **raw counts** first; the statistic is defined on counts and only counts.
- A **very large** n with a tiny relative discrepancy? ⇒ separate **statistical significance** from **practical importance**; report an effect size, not just the p-value.
- Asked *which* count test (GOF / independence / homogeneity) and *which* df? ⇒ read the **design**, not the table shape; df follows from the design.

Do not confuse a big χ^2 with a big effect: χ^2 scales with n, so for fixed proportions doubling the data doubles the statistic. Significance answers "is the deviation real?" not "is the deviation large enough to care about?" — those are different questions.

10.5.1 The expected-count condition

The χ^2 statistic is large-sample: each term $(O_i - E_i)^2/E_i$ behaves like the square of a standardized count, which is approximately a squared standard normal only when the cell count is not too small. With a tiny E_i the count is too discrete and too skewed for that normal approximation, and the sum no longer follows a χ^2 curve — the true tail probabilities can be far from the table's, usually making the test reject too readily.

Definition 10.5.1 (Expected-count rule of thumb)**.** The χ^2 approximation is trustworthy when the **expected** counts are not too small. A widely used rule of thumb is:

$$\text{every } E_i \geq 5.$$

A more permissive and commonly cited version allows a few small cells: *no more than about* 20% *of the cells may have* $E_i < 5$*, and no cell may have* $E_i < 1$. The condition is always on the **expected** counts E_i; the observed counts O_i may be any nonnegative integers, including 0.

Remark. It is E_i, never O_i, that must clear the bar. A cell can legitimately have $O_i = 0$ observed (nobody fell in that category) while its E_i is comfortably large; that cell is fine. The reverse — a small E_i — is the danger, because E_i sits in the denominator and a small denominator inflates the term.

Example 10.5.2 (Checking the condition before trusting a fit)**.** A botanist classifies $n = 120$ seedlings by flower colour into six genetic categories whose hypothesized proportions are

$$0.40,\ 0.25,\ 0.15,\ 0.10,\ 0.06,\ 0.04.$$

Before running a goodness-of-fit test, decide whether the expected-count condition holds.

Which rule, and why? This is a validity check, not the test itself: compute every $E_i = np_i$ and compare to the rule of thumb. We look at E_i, not the observed seedling counts.

Solution.

Step 1: expected counts. Multiply each p_i by $n = 120$:

$$\begin{aligned} E_i &= 120 \cdot (0.40,\ 0.25,\ 0.15,\ 0.10,\ 0.06,\ 0.04) \\ &= (48,\ 30,\ 18,\ 12,\ 7.2,\ 4.8). \end{aligned}$$

Step 2: apply the rule. Five cells exceed 5; one cell has $E_i = 4.8 < 5$. That is one of six cells, or 16.7%, which is under the 20% allowance, and $4.8 \geq 1$.

Step 3: verdict. The strict "all $E_i \geq 5$" rule is just missed, but the permissive rule is satisfied.

One cell at $E_i = 4.8$ ($16.7\% < 20\%,\ \geq 1$) $\Rightarrow$ condition met; proceed with df $= 6 - 1 = 5$.

Had two cells dipped below 5, we would have crossed the 20% line and needed to pool.

Example 10.5.3 (Pooling sparse categories and recomputing df)**.** A reliability engineer records the failure mode of $n = 90$ returned pumps in seven categories. The hypothesized proportions give expected counts

$$E_i = (31.5,\ 22.5,\ 18.0,\ 9.0,\ 4.5,\ 3.6,\ 0.9),$$

with observed counts $O_i = (34,\ 20,\ 17,\ 11,\ 3,\ 4,\ 1)$. The last three expected counts are below 5 (and one below 1). Pool to satisfy the condition and state the new degrees of freedom.

Which rule, and why? Three of seven cells have $E_i < 5$ (43%, well over 20%) and one has $E_i < 1$, so the rule of thumb fails badly; the fix is to *merge* the offending adjacent categories into one combined cell and recompute df from the reduced cell count — not to delete data.

Solution.

Step 1: identify the offenders and merge. Cells $5, 6, 7$ have $E_i = 4.5$, 3.6, 0.9. Combine them into a single "other/rare" category. Add the expected counts and add the observed counts:

$$E_i^{\text{pooled}} = 4.5 + 3.6 + 0.9 = 9.0,$$
$$O_i^{\text{pooled}} = 3 + 4 + 1 = 8.$$

Step 2: the new table. The five surviving cells now have expected counts (31.5, 22.5, 18.0, 9.0, 9.0), every one ≥ 5. Observed: (34, 20, 17, 11, 8).

Step 3: recompute df. Degrees of freedom is (number of cells -1) for a fully specified GOF. We went from 7 cells to 5:

$$\text{before: } \mathrm{df} = 7 - 1 = 6,$$
$$\text{after: } \mathrm{df} = 5 - 1 = 4.$$

Step 4: now compute the statistic on the pooled table.

$$\begin{aligned}\chi^2_{\text{obs}} &= \frac{(34 - 31.5)^2}{31.5} + \frac{(20 - 22.5)^2}{22.5} + \frac{(17 - 18)^2}{18} \\ &\quad + \frac{(11 - 9)^2}{9} + \frac{(8 - 9)^2}{9} \\ &= 0.198 + 0.278 + 0.056 + 0.444 + 0.111 = 1.087.\end{aligned}$$

Pooled to 5 cells, $\mathrm{df} = 4$, $\chi^2_{\text{obs}} = 1.09$; $\chi^2_{0.05,\,4} = 9.488 \Rightarrow$ do not reject.

Each merge costs exactly one degree of freedom, so collapsing three cells into one drops df by 2 (from 6 to 4). Pooling must be decided by the *expected* counts and ideally before peeking at the statistic.

Category	Before pooling	After pooling
1 (most common)	$E_i = 31.5$	$E_i = 31.5$
2	$E_i = 22.5$	$E_i = 22.5$
3	$E_i = 18.0$	$E_i = 18.0$
4	$E_i = 9.0$	$E_i = 9.0$
5	$E_i = 4.5$ (< 5)	(merged below)
6	$E_i = 3.6$ (< 5)	$E_i = 9.0$ (cells 5–7)
7 (rare)	$E_i = 0.9$ (< 1)	(merged above)
Cells / df	7 cells, $\mathrm{df} = 6$	5 cells, $\mathrm{df} = 4$

Pooling the three sparse tail categories (each $E_i < 5$, one $E_i < 1$) into a single combined cell. Every surviving expected count is now ≥ 5, and degrees of freedom falls from $7 - 1 = 6$ to $5 - 1 = 4$: one df lost per category absorbed.

10.5.2 The 2×2 continuity (Yates) correction

A 2×2 contingency table has just one degree of freedom, and the χ^2 approximation is at its crudest there: a single continuous χ^2_1 curve is standing in for a discrete distribution, and the fit overstates significance for small to moderate counts. Yates's continuity correction shrinks each absolute discrepancy by $\frac{1}{2}$ before squaring, which pulls the statistic down and counteracts the discreteness.

Definition 10.5.4 (Yates's continuity correction for a 2×2 table)**.** For a 2×2 table with expected counts E_i, the **corrected** statistic is

$$\chi^2_{\text{Yates}} = \sum_i \frac{\left(|O_i - E_i| - \frac{1}{2}\right)^2}{E_i},$$

referred to χ^2_1. The subtraction of $\frac{1}{2}$ applies to the *magnitude* $|O_i - E_i|$; if any $|O_i - E_i| < \frac{1}{2}$ that term is set to 0 rather than allowed to go negative. Degrees of freedom remain $(r-1)(c-1) = 1$.

Remark. The correction is conservative: $\chi^2_{\text{Yates}} \le \chi^2_{\text{obs}}$ always, so it makes rejection *harder*, guarding against the uncorrected statistic's tendency to over-reject in small 2×2 tables. It is traditional for 2×2 tables with modest counts; for large counts the $\frac{1}{2}$ is negligible and correction barely changes anything. (When counts are very small, an exact test is preferable to either version, but that is beyond this chapter.)

Example 10.5.5 (Applying Yates and comparing to the uncorrected test)**.** A small clinic compares two wound dressings. Of 22 patients on dressing A, 7 developed an infection; of 18 patients on dressing B, 11 did. Is infection associated with the dressing? Test at $\alpha = 0.05$ using the 2×2 χ^2 test, with and without the Yates correction.

Which rule, and why? One sample of patients cross-classified by two binary factors (dressing $\times$ infection) is a 2×2 test on 1 df; the modest counts make this exactly the situation Yates's correction was designed for, so we compute both and compare.

Solution.

Step 1: the observed table and margins.

	Infection	None	Total
A	7	15	22
B	11	7	18
Total	18	22	40

Step 2: expected counts (row total $\times$ column total $/\, n$, with $n = 40$):

$$E_{\text{A,Inf}} = \frac{22 \cdot 18}{40} = 9.9, \quad E_{\text{A,None}} = \frac{22 \cdot 22}{40} = 12.1,$$
$$E_{\text{B,Inf}} = \frac{18 \cdot 18}{40} = 8.1, \quad E_{\text{B,None}} = \frac{18 \cdot 22}{40} = 9.9.$$

Each common difference is $|O_i - E_i| = 2.9$ in every cell.

Step 3: uncorrected statistic.

$$\chi^2_{\text{obs}} = \frac{2.9^2}{9.9} + \frac{2.9^2}{12.1} + \frac{2.9^2}{8.1} + \frac{2.9^2}{9.9}$$
$$= 0.850 + 0.695 + 1.038 + 0.850 = 3.433.$$

Step 4: Yates-corrected statistic. Replace each $|O_i - E_i| = 2.9$ by $2.9 - 0.5 = 2.4$:

$$\chi^2_{\text{Yates}} = \frac{2.4^2}{9.9} + \frac{2.4^2}{12.1} + \frac{2.4^2}{8.1} + \frac{2.4^2}{9.9}$$
$$= 0.582 + 0.476 + 0.711 + 0.582 = 2.351.$$

Step 5: decision. The cutoff is $\chi^2_{0.05,\,1} = 3.841$. Both statistics fall below it, but the gap differs sharply:

$$\boxed{\chi^2_{\text{obs}} = 3.43,\ \chi^2_{\text{Yates}} = 2.35;\ \text{both} < 3.841 \Rightarrow \text{do not reject.}}$$

The uncorrected 3.43 already sits near the cutoff; the correction pulls it down to 2.35, confirming the conclusion and showing how much the $\frac{1}{2}$ matters when counts are this small.

10.5.3 Counts, not percentages

The χ^2 statistic is defined on **raw frequencies**. The sample size n enters through the expected counts $E_i = np_i$ and is the reason a real deviation produces a large statistic. Percentages and proportions have had n divided out, so plugging them in throws away the sample size and produces a meaningless number.

Example 10.5.6 (CONTRAST: running χ^2 on percentages instead of counts)**.** A pollster reports that of $n = 800$ voters, party preferences split 46%, 33%, 21% across three parties, against a hypothesized split of 40%, 35%, 25%. A student computes a goodness-of-fit statistic by entering the *percentages* as "observed" and the hypothesized percentages as "expected." Show why this is wrong and give the correct statistic.

Which rule, and why? The χ^2 formula needs *counts*; percentages have the sample size divided out, so the look-alike calculation on percentages is invalid no matter how the arithmetic is done. Always reconstruct the raw counts $O_i = (\text{percent}/100) \cdot n$ and $E_i = np_i$ first.

Solution.

Step 1: the wrong calculation (for contrast). Using $46, 33, 21$ as "O_i" and $40, 35, 25$ as "E_i":

$$\begin{aligned}\chi^2_{\text{wrong}} &= \frac{(46-40)^2}{40} + \frac{(33-35)^2}{35} + \frac{(21-25)^2}{25} \\ &= 0.900 + 0.114 + 0.640 = 1.654.\end{aligned}$$

This number secretly assumes $n = 100$ (the percentages sum to 100), so it cannot detect the real sample's evidence at all.

Step 2: the correct counts. Convert to raw frequencies with $n = 800$:

$$\begin{aligned}O_i &= (0.46,\ 0.33,\ 0.21) \cdot 800 = (368,\ 264,\ 168), \\ E_i &= (0.40,\ 0.35,\ 0.25) \cdot 800 = (320,\ 280,\ 200).\end{aligned}$$

Step 3: correct statistic.

$$\begin{aligned}\chi^2_{\text{obs}} &= \frac{(368-320)^2}{320} + \frac{(264-280)^2}{280} + \frac{(168-200)^2}{200} \\ &= \frac{2304}{320} + \frac{256}{280} + \frac{1024}{200} \\ &= 7.200 + 0.914 + 5.120 = 13.234.\end{aligned}$$

Step 4: compare conclusions. With $\text{df} = 3 - 1 = 2$ and $\chi^2_{0.05,\,2} = 5.991$:

$$\boxed{\chi^2_{\text{wrong}} = 1.65 \text{ (not sig.)};\quad \chi^2_{\text{obs}} = 13.23 > 5.991 \Rightarrow \text{reject.}}$$

The correct statistic is almost exactly 8 times the wrong one — precisely the factor $n/100 = 8$, since χ^2 scales linearly with n. Percentages erase that factor and reverse the conclusion.

10.5.4 Large n: significance versus importance

Because χ^2 grows in proportion to n for a fixed pattern of relative deviations, a large enough sample makes *any* nonzero discrepancy statistically significant, however trivial. A significant χ^2 says the deviation is unlikely to be pure chance; it does *not* say the deviation is large or important. The two must be reported separately.

Theorem 10.5.7 (How the statistic scales with sample size)**.** *Fix the category proportions. If the observed* proportions $\hat{p}_i = O_i/n$ *and the hypothesized* p_i *are held constant while* n *grows, then*

$$\chi^2 = \sum_i \frac{(O_i - E_i)^2}{E_i} = n \sum_i \frac{(\hat{p}_i - p_i)^2}{p_i},$$

so χ^2 *is* directly proportional *to* n*. Doubling the sample doubles the statistic; the deviation pattern need not change at all.*

Proof. Write $O_i = n\hat{p}_i$ and $E_i = np_i$. Then $(O_i - E_i)^2/E_i = (n\hat{p}_i - np_i)^2/(np_i) = n(\hat{p}_i - p_i)^2/p_i$. Summing over i pulls out the common factor n. □

Example 10.5.8 (A trivial deviation made "significant" by an enormous sample)**.** A national dataset records day-of-week of $n = 2{,}000{,}000$ online orders. Under "orders are uniform across the seven days" each day expects $1/7$ of the total. The observed proportions are

$$(0.1437,\ 0.1429,\ 0.1429,\ 0.1429,\ 0.1429,\ 0.1421,\ 0.1426)$$

— almost dead uniform. Is the departure from uniform "significant," and does it matter?

Which rule, and why? This is a goodness-of-fit to a fully specified uniform ($p_i = 1/7$), df $= 6$; the lesson is that with n in the millions even a microscopic deviation clears any cutoff, so we must judge *practical* importance separately via an effect size.

Solution.

Step 1: per-day relative deviations. Each $p_i = 1/7 = 0.142857$. The largest gap is day 1: $\hat{p}_1 - p_1 = 0.1437 - 0.142857 = 0.000843$, under a tenth of a percentage point; the others are smaller.

Step 2: assemble χ^2 *via the scaling form.* Summing $(\hat{p}_i - p_i)^2/p_i$ over the seven days gives about 7.0×10^{-6}. Then

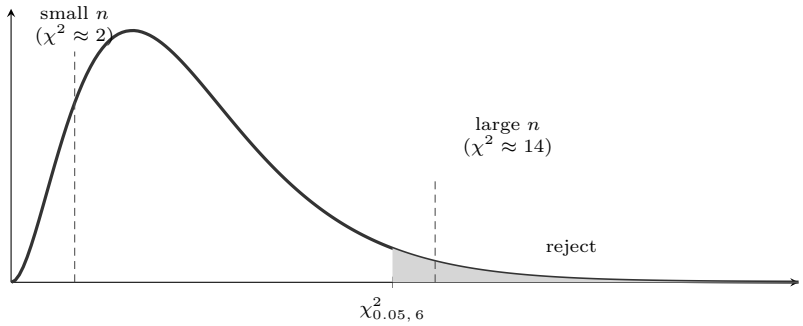
$$\begin{aligned}\chi^2_{\text{obs}} &= n \sum_i \frac{(\hat{p}_i - p_i)^2}{p_i} \\ &= 2{,}000{,}000 \cdot (7.0 \times 10^{-6}) \approx 14.0.\end{aligned}$$

Step 3: significance. With df $= 6$ and $\chi^2_{0.05,\,6} = 12.592$, we have $14.0 > 12.592$, so the test *rejects* uniformity.

Step 4: effect size. A scale-free measure is Cramer's-style index $\sqrt{\chi^2/(n \cdot \text{df})}$ or simply the root-mean relative deviation $\sqrt{\chi^2/n} = \sqrt{14.0/2{,}000{,}000} = 0.00265$ — about a quarter of one percent.

Significant ($\chi^2_{\text{obs}} = 14.0 > 12.59$) yet effect ≈ 0.003; statistically real, practically nil.

With two million orders the test detects a deviation no business would act on. Always pair the p-value with an effect size so significance is not mistaken for importance.



The same *relative deviation pattern, two sample sizes. Because* $\chi^2 = n\sum(\hat{p}_i - p_i)^2/p_i$ *is proportional to* n*, a small sample lands the statistic at* $\chi^2 \approx 2$ *(left of the cutoff* $\chi^2_{0.05,\,6} = 12.592$*, not significant) while a sample hundreds of times larger pushes the* identical *pattern out to* $\chi^2 \approx 14$*, into the rejection region. The curve is fixed; only where the deviation lands moves with* n*.*

10.5.5 What a rejected fit does and does not say

Remark. A goodness-of-fit test is fundamentally **one-tailed**: we reject only in the upper tail, because only a *large* χ^2 signals disagreement between observed and expected. A small χ^2 means the data sit close to the hypothesized counts and never triggers rejection. (A suspiciously *tiny* χ^2 can hint at fabricated or over-fitted data, but that is a separate diagnostic, not the test's alternative.) There is no "lower-tail" rejection region for a goodness-of-fit test.

Remark. Rejecting H_0 tells you the data are *inconsistent* with the one named distribution. It does **not** tell you which distribution is correct, nor does it endorse any particular alternative. Failing to reject is weaker still: it means the data are *compatible* with the named distribution, not that the distribution is proven true — other distributions may fit equally well, especially at small n where the test has little power.

Example 10.5.9 (Reading a rejection correctly)**.** An ecologist tests whether weekly nest counts follow a Poisson distribution (rate estimated from the data) and obtains $\chi^2_{\text{obs}} = 21.4$ on df $= 4$, exceeding $\chi^2_{0.05,\,4} = 9.488$. A colleague concludes "the counts are therefore negative binomial." Is that conclusion warranted?

Which rule, and why? This tests recognition of what a rejected GOF establishes: only that the *named* model (Poisson) is rejected, never that a specific alternative (negative binomial) is correct.

Solution.

Step 1: what the test establishes. $\chi^2_{\text{obs}} = 21.4 > 9.488$ rejects the Poisson fit at $\alpha = 0.05$: the data are inconsistent with a Poisson distribution.

Step 2: what it does not establish. The alternative to "Poisson" is simply "not Poisson" — an enormous family. Nothing in the statistic singles out the negative binomial; it could be a zero-inflated, mixture, or other over-dispersed model.

Step 3: correct reading. To support the negative binomial one must *fit and test that model directly* and find it not rejected (and ideally a better fit by an information criterion).

Rejecting Poisson $\neq$ confirming negative binomial; the test names no winner.

Overdispersion makes the negative binomial *plausible*, but plausibility is a hypothesis to be tested, not a conclusion handed down by rejecting Poisson.

Example 10.5.10 (The one-tailed nature: a near-perfect fit is never "too good to reject the other way")**.** A quality auditor tests whether $n = 200$ inspected items follow a hypothesized four-category split and obtains $\chi^2_{\text{obs}} = 0.18$ on df $= 3$. The auditor asks whether such a tiny statistic should be "rejected" as an unusually good fit, since 0.18 is far in the lower tail. Resolve this at $\alpha = 0.05$.

Which rule, and why? The goodness-of-fit test is one-tailed in the *upper* tail only: rejection requires a *large* χ^2 signalling disagreement, so a small statistic never lands in a rejection region. There is no lower-tail critical value to compare against.

Solution.

Step 1: the only rejection region. For df $= 3$ at $\alpha = 0.05$ the cutoff is $\chi^2_{0.05,\,3} = 7.815$, and we reject only when $\chi^2_{\text{obs}} > 7.815$.

Step 2: place the statistic. Here $\chi^2_{\text{obs}} = 0.18 \ll 7.815$, deep in the lower tail, far from the single upper-tail rejection region.

Step 3: conclusion. Do not reject; the data are highly compatible with the hypothesized split.

$\chi^2_{\text{obs}} = 0.18 < 7.815 \Rightarrow$ do not reject; a small χ^2 supports, never rejects.

A suspiciously small χ^2 may warrant a separate look for fabricated or rounded data, but that is a data-integrity check, not the goodness-of-fit alternative, which lives entirely in the upper tail.

10.5.6 Routing a scenario to the right test and df

The same statistic serves goodness-of-fit, independence, and homogeneity; the design — how the data were sampled — decides which name and which df apply, not the look of the table.

Example 10.5.11 (Recognition: route each scenario to test and df)**.** For each described study, name the test (goodness-of-fit, independence, or homogeneity) and give the degrees of freedom.

(a) One random sample of 500 adults, each cross-classified by blood type (4 levels) and Rh factor (2 levels); is blood type related to Rh?

(b) 300 dice rolls tallied into faces 1–6; is the die fair (each face 1/6)?

(c) Separate random samples of 120 urban, 120 suburban, and 120 rural households, each household's primary heating fuel recorded in 5 categories; do the fuel distributions match across the three locations?

(d) One sample of 400 insurance claims tallied by month; do monthly claim counts follow a Poisson distribution whose rate is estimated from the data?

Which rule, and why? Read the *sampling design*: one sample cross-classified by two variables ⇒ independence; one sample versus a specified distribution ⇒ goodness-of-fit (subtract one more df per estimated parameter); several separate samples compared on one variable ⇒ homogeneity.

Solution.

(a) Independence. One sample, two categorical variables ($r = 4$ blood types, $c = 2$ Rh). Contingency-table test of independence,

$$\text{df} = (r-1)(c-1) = (4-1)(2-1) = 3.$$

(b) Goodness-of-fit, parameters known. One variable (die face) against a fully specified uniform; no parameters estimated:

$$\text{df} = k - 1 = 6 - 1 = 5.$$

(c) Homogeneity. Three *separate* samples (the location group sizes were fixed by design at 120 each) compared on one variable (fuel, 5 categories). Same arithmetic as independence:

$$\text{df} = (r-1)(c-1) = (3-1)(5-1) = 8.$$

(d) Goodness-of-fit, one parameter estimated. One variable (monthly count) against a Poisson whose rate λ is estimated from the data, $m = 1$ estimated parameter. Suppose the counts are binned into $k = 7$ usable cells (after any pooling):

$$\text{df} = k - 1 - m = 7 - 1 - 1 = 5.$$

$$\boxed{(a)\ \text{indep., df} = 3;\quad (b)\ \text{GOF, df} = 5;\quad (c)\ \text{homog., df} = 8;\quad (d)\ \text{GOF, df} = 5.}$$

Scenarios (a) and (c) build identical-looking tables and use identical arithmetic; only the design (one sample cross-classified versus three fixed-size samples compared) distinguishes them, and the conclusion's wording follows the design.

Example 10.5.12 (CONTRAST: a valid versus an invalid application of the same test)**.** Two analysts each build a 3×2 table and run the independence χ^2 on $\text{df} = (3-1)(2-1) = 2$. Analyst V drew *one* random sample of 240 customers and cross-classified each by region (3 levels) and whether they renewed (2 levels); her smallest expected count is 18. Analyst I took the 240 customers, computed the *renewal percentage* in each region, and entered those six percentages as the table, with one region's expected count working out to 0.9. Which application is valid?

Which rule, and why? Both produce a 3×2 table and quote df $= 2$, so the table shape is identical; the difference is whether the inputs are raw counts from a proper sampling design with adequate expected counts (valid) or rescaled percentages that also violate the expected-count rule (invalid on two grounds).

Solution.

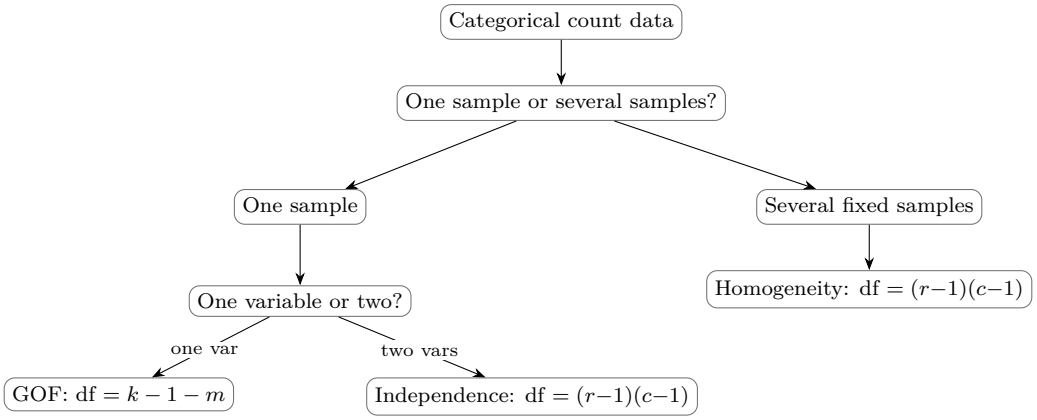
Step 1: Analyst V (valid). One sample cross-classified by two variables is a legitimate test of independence. The inputs are raw counts, and every $E_i \geq 18 \geq 5$, so the rule of thumb holds. The df $= (3-1)(2-1) = 2$ is correct, and the conclusion (region related to renewal, or not) is meaningful.

Step 2: Analyst I (invalid, twice over). First, the cells are *percentages*, not counts, so the statistic is computed on the wrong quantity (Section: counts not percentages). Second, even read charitably an expected count of $0.9 < 1$ violates the expected-count condition, so the χ^2_2 approximation would be untrustworthy regardless.

Step 3: verdict.

V valid (raw counts, $E_i \geq 18$); I invalid (percentages and $E_i = 0.9 < 1$).

Identical df and an identically shaped table do not make two analyses equally sound: validity rests on the sampling design, on using counts, and on adequate expected counts — the very caveats of this section.



Choosing and validating a count test from the design*, not the table shape. One sample against a specified distribution is goodness-of-fit (subtract one df per estimated parameter m); one sample cross-classified by two variables is independence; several separately drawn samples compared on one variable is homogeneity. After routing, always check the expected-count condition and confirm the inputs are raw counts.*

10.6 Exercises

All problems below are original. Full solutions begin on page 676.

A. Goodness-of-fit, parameters known

A1. A boardwalk vendor sells a game using a single die that should be fair. A consumer inspector rolls it $n = 480$ times and records the face counts

$$O_i = (\,O_1, \ldots, O_6\,) = (\,72,\ 91,\ 74,\ 88,\ 69,\ 86\,).$$

Test $H_0 \colon p_i = 1/6$ for every face against the alternative that the die is unfair, at $\alpha = 0.05$. State the degrees of freedom, give the critical value, and report the decision in words.

A2. A geneticist crosses two heterozygous plants and classifies $n = 240$ offspring into the four phenotype classes predicted by the Mendelian 9:3:3:1 dihybrid ratio. The observed counts are

$$O_i = (\,121,\ 49,\ 53,\ 17\,).$$

(a) Convert the 9:3:3:1 ratio into expected counts E_i for $n = 240$.

(b) Test the ratio at $\alpha = 0.05$, reporting the statistic, df, critical value, and decision.

(c) How many degrees of freedom does the test use, and why is it $k - 1$ rather than k?

A3. A transit authority claims that bicycle thefts are spread evenly across the seven days of the week. Over one season $n = 350$ reported thefts fall as

$$(\,O_{\text{Mon}}, \ldots, O_{\text{Sun}}\,) = (\,44,\ 51,\ 47,\ 53,\ 58,\ 61,\ 36\,).$$

Test the even-spread (uniform) claim at $\alpha = 0.05$. Identify the day whose cell contributes the most to the statistic.

A4. A regulator audits a slot machine whose design specifies symbol probabilities

$$p = (\,\text{cherry:}0.50,\ \text{bell:}0.25,\ \text{bar:}0.15,\ \text{seven:}0.10\,).$$

In $n = 600$ recorded spins the symbols appear with counts

$$O_i = (\,\text{cherry:}324,\ \text{bell:}138,\ \text{bar:}81,\ \text{seven:}57\,).$$

Lay out every expected count $E_i = np_i$ and every cell contribution $(O_i - E_i)^2/E_i$ in one table, then test whether the machine matches its design at $\alpha = 0.05$.

A5. A reliability engineer claims that the number of flaws per length of cable follows a Poisson distribution with a *design-mandated* rate $\lambda = 1.5$ (not estimated from the data). From $n = 200$ cable lengths the flaw counts, grouped as "0, 1, 2, 3 or more," are

$$O_i = (\,42,\ 70,\ 51,\ 37\,).$$

(a) Using $\mathbb{P}(X = x) = e^{-1.5}1.5^x/x!$ with $e^{-1.5} = 0.22313$, find the category probabilities and the expected counts $E_i = 200p_i$.

(b) Test the Poisson(1.5) claim at $\alpha = 0.05$.

(c) Explain why the degrees of freedom are $k - 1$ here and not $k - 2$.

A6. In a monohybrid cross the offspring should split tall to short in a 3:1 ratio. Among $n = 200$ plants, 137 are tall and 63 are short.

(a) Test the 3:1 claim using Pearson's χ^2 with $k = 2$ at $\alpha = 0.05$; state df.

(b) Compute the one-proportion z-statistic for $H_0 \colon p_{\text{tall}} = 3/4$ and verify that z^2 equals your χ^2.

A7. A polling firm claims voters split 45% for candidate X, 35% for candidate Y, and 20% undecided. An exit survey of $n = 300$ voters records counts $(\,123,\ 117,\ 60\,)$, giving a goodness-of-fit statistic on the appropriate degrees of freedom. Compute χ^2, then report the single upper-tail p-value $\mathbb{P}(\chi^2_2 > \chi^2_{\text{obs}})$ exactly using the closed form $\mathbb{P}(\chi^2_2 > c) = e^{-c/2}$, and decide at $\alpha = 0.05$.

B. Goodness-of-fit, parameters estimated

B1. A hospital switchboard logs the number of emergency pages arriving in each of 160 disjoint ten-minute blocks. The frequency table is

Pages in block x	0	1	2	3	4	≥ 5
Blocks O_i	36	44	42	22	12	4

Test at $\alpha = 0.05$ whether the number of pages per block follows a Poisson distribution. The rate is not given; estimate it from the data. Pool any cell with $E_i < 5$ before computing the statistic, and be explicit about the final degrees of freedom.

B2. Across 144 families with exactly $n = 4$ children, the number of daughters per family is recorded. If each child is independently a daughter with probability p, the count per family is Binomial$(4, p)$, but p is unknown.

Daughters x	0	1	2	3	4
Families O_i	18	32	48	34	12

(a) Estimate p from the total number of daughters and form the fitted Binomial$(4, \widehat{p})$ probabilities and expected counts.

(b) Carry out the goodness-of-fit test at $\alpha = 0.05$, stating the degrees of freedom and the decision in words.

B3. The fasting blood-glucose readings (mg/dL) of $n = 180$ adults in a screening clinic are grouped into six intervals. From the raw data $\overline{X} = 100.0$ and $s = 16.0$ are computed.

Interval	< 76	76–88	88–100	100–112	112–124	≥ 124
O_i	9	34	45	55	24	13

Test at $\alpha = 0.05$ whether the glucose readings are normally distributed. Use $\Phi(-1.5) = .0668$, $\Phi(-0.75) = .2266$, $\Phi(0) = .5$, $\Phi(0.75) = .7734$, $\Phi(1.5) = .9332$. State the degrees of freedom carefully, remembering that two parameters were estimated.

B4. A radiation counter records the number of particles detected in each of 100 one-second windows. A physicist wants to test a Poisson model at $\alpha = 0.05$.

Counts x	0	1	2	3	4	5	≥ 6
Windows O_i	20	22	26	17	9	4	2

The estimated rate is $\widehat{\lambda} = 2.00$, giving raw expected counts

$$E_i = (13.5,\ 27.1,\ 27.1,\ 18.0,\ 9.0,\ 3.6,\ 1.7).$$

(a) Identify which tail cells have $E_i < 5$ and pool them so that every pooled cell has $E_i \geq 5$. Report the post-pooling category count k.

(b) Compute χ^2, give the degrees of freedom, and state the decision at $\alpha = 0.05$.

B5. An ecologist counts the number of nesting pairs of a bird species on each of 90 survey plots and proposes a Poisson model. From the data $\widehat{\lambda} = 2.40$. The raw fitted expected counts are

Pairs x	0	1	2	3	4	≥ 5
O_i	14	13	30	14	9	10
E_i	8.2	19.6	23.5	18.8	11.3	8.6

Two analysts disagree on the degrees of freedom. Analyst X uses $\mathrm{df} = k-1$; analyst Y uses $\mathrm{df} = k-1-1$. Compute χ^2, then report the cutoff each analyst would use at $\alpha = 0.05$, say which analyst is correct and why, and give the correct decision.

B6. A clinical study models the number of seizures per month for $n = 200$ patients on a new drug as Poisson. The raw table has eight cells $x = 0, 1, \ldots, 6, \geq 7$. After estimating $\widehat{\lambda}$ from the sample mean, the fitted expected counts in the upper cells come out as $E(x{=}5) = 6.0$, $E(x{=}6) = 3.4$, and $E(x{\geq}7) = 1.6$; every cell with $x \leq 4$ has $E_i \geq 8$.

(a) Determine which cells must be pooled and the resulting category count k.

(b) State the final degrees of freedom, showing the subtraction for the total constraint and for the estimated parameter separately.

(c) If instead a normal model (two estimated parameters) had been fitted to the same pooled cells, what would the degrees of freedom be?

B7. A market researcher tests whether the number of brands a shopper recalls, over $n = 130$ shoppers grouped into $k = 5$ pooled cells, follows a particular one-parameter model. One parameter was estimated from the data. The computed statistic is $\chi^2 = 8.40$. A junior analyst, forgetting that a parameter was estimated, uses $\mathrm{df} = k - 1$ and the cutoff $\chi^2_{0.05,\,4} = 9.488$, concluding "fail to reject." The senior analyst insists the analyst forgot the estimation.

(a) Give the correct degrees of freedom and the correct cutoff at $\alpha = 0.05$ ($\chi^2_{0.05,\,3} = 7.815$).

(b) State the correct decision and explain, in one or two sentences, whether the junior analyst's error made the test too liberal or too conservative here, and why subtracting for the estimated parameter is the right fix.

C. Contingency tables: independence

C1. A campus health center samples $n = 240$ students and records, for each, whether they exercise regularly (yes, no) and whether they report good sleep quality (yes, no). The observed counts are

	good sleep	poor sleep	**row total**
exercises	84	36	120
no exercise	54	66	120
col total	138	102	240

At $\alpha = 0.05$, test whether exercise habit and sleep quality are independent. Compute every $E_{ij} = R_i C_j / n$, the per-cell contributions, the statistic, the degrees of freedom, and the critical value; state your decision in words. Explain why a rejection here is evidence of association and not that exercise causes better sleep.

C2. A marketing team cross-tabulates $n = 400$ website visitors by landing-page version (A, B) and whether they signed up:

	signed up	did not	**row total**
version A	72	128	200
version B	96	104	200
col total	168	232	400

(a) Run the test of independence at $\alpha = 0.05$: expected counts, contributions, statistic, df, critical value, decision.

(b) Reanalyze the same data as a two-sided two-proportion z-test of $H_0 \colon p_A = p_B$ on the sign-up rates, using the pooled estimate. Report z_{obs}.

(c) Verify numerically that $\chi^2_{\text{obs}} = z^2_{\text{obs}}$ and that the cutoffs agree ($\chi^2_{0.05,1} = z^2_{0.025}$). State one thing the z-version tells you that the χ^2 does not.

C3. An education researcher samples $n = 300$ pupils and classifies each by teaching method (lecture, flipped, hybrid) and exam outcome (pass, fail):

	pass	fail	**row total**
lecture	60	40	100
flipped	78	22	100
hybrid	72	28	100
col total	210	90	300

At $\alpha = 0.05$, test whether teaching method and exam outcome are independent. Give all E_{ij}, the contributions, the statistic, the df, the critical value, and a worded decision. Identify the cell that contributes most to the statistic and interpret it.

C4. A clinic with $n = 500$ patients cross-classifies each by ABO-style allergy panel result (none, mild, severe) and a skin-test outcome (positive, negative):

	positive	negative	**row total**
none	40	160	200
mild	50	150	200
severe	30	70	100
col total	120	380	500

(a) State the table shape, give the degrees of freedom $(r-1)(c-1)$, and confirm it the long way as $rc - 1 - (r-1) - (c-1)$.

(b) At $\alpha = 0.05$, carry out the full independence test (expected counts, contributions, statistic, critical value, decision in words).

C5. A pollster samples $n = 800$ adults and records region (East, Midwest, South, West) and opinion on a ballot measure (favor, oppose):

	East	Midwest	South	West	**row total**
favor	130	90	110	70	400
oppose	70	110	90	130	400
col total	200	200	200	200	800

At $\alpha = 0.05$, test whether opinion is independent of region. Report the expected counts, the contributions, the statistic, the df, the critical value, and a worded conclusion. Say which two regions drive the result.

C6. A factory logs $n = 420$ surface defects by type (pit, crack, stain) and the machine that produced them (M1, M2, M3). The team obtains $\chi^2_{\text{obs}} = 14.7$.

(a) State the table shape and the degrees of freedom for this independence test.

(b) At $\alpha = 0.05$, quote the critical value (state df) and give the decision.

(c) The same statistic $\chi^2_{\text{obs}} = 14.7$ on 4 df has p-value just above 0.005. Would the decision change at the stricter $\alpha = 0.01$? Quote the relevant critical value and explain.

C7. A workplace survey cross-tabulates $n = 360$ employees by department (sales, engineering, support) and whether they used an optional wellness program (used, did not):

	used	did not	**row total**
sales	48	72	120
engineering	54	66	120
support	66	54	120
col total	168	192	360

The computed statistic is $\chi^2_{\text{obs}} = 5.36$ on 2 df.

(a) Confirm the degrees of freedom and quote $\chi^2_{0.05,2}$; state the decision at $\alpha = 0.05$.

(b) The next table point is $\chi^2_{0.025,2} = 7.378$. Bracket the p-value and explain why this is a *marginally significant* result that should be reported cautiously.

(c) Explain precisely what a rejection here does and does not let you claim about the wellness program.

D. Tests of homogeneity

D1. A coffee chain samples a fixed number of customers at each of three outlets and records which size each customer orders (Small, Medium, Large). The counts are

outlet	Small	Medium	Large	R_i
Downtown	40	70	50	160
Airport	30	50	80	160
Campus	50	60	50	160
C_j	120	180	180	480

The manager fixed each outlet's sample at 160 before any orders were observed. At $\alpha = 0.05$:

(a) State H_0 and H_1 in words, and explain why this is a test of homogeneity rather than a test of independence.

(b) Compute every E_{ij}, the per-cell contributions, the statistic χ^2, the degrees of freedom, and reach a decision against the appropriate critical value.

D2. A polling firm draws independent samples in two counties and asks each respondent whether they favor a proposed tax (Favor / Oppose). In County A, 260 of 400 favor it; in County B, 210 of 350 favor it. The two county samples were sized in advance.

(a) Build the 2×2 table and carry out the homogeneity χ^2 test at $\alpha = 0.05$; state df and your decision.

(b) Cross-check by computing the pooled two-proportion z statistic, and verify that z^2 equals the χ^2 you found in part (a).

D3. An ecologist nets a fixed 120 fish from each of four lakes and classifies each by infection status (Healthy / Infected). The infected counts are 18, 36, 24, and 30 out of 120. At $\alpha = 0.05$, test whether the four lakes have the same infection rate. Report all E_{ij}, the per-cell contributions, χ^2, the df, the critical value, and your decision in words.

D4. For each study below, state which χ^2 test applies (homogeneity or independence), give the table dimensions $r \times c$, and state the degrees of freedom. Justify each answer by naming which margin (if any) was fixed by the sampler.

(a) "We drew one random sample of 600 employees and recorded each person's department (4 departments) and whether they completed optional training (yes/no)."

(b) "From each of three warehouses we pulled a fixed 200 shipments and graded each as On-Time, Late, or Lost."

(c) "A single random sample of 450 patients was classified by blood type (4 types) and by whether a reaction occurred (yes/no)."

(d) "We surveyed exactly 150 subscribers from each of five magazines and recorded their renewal intention (Will renew / Undecided / Will cancel)."

D5. A nutrition study assigns a fixed 100 participants to each of three diet plans and after twelve weeks classifies each as Lost weight, No change, or Gained. The counts are

plan	Lost	No change	Gained	R_i
Plan 1	55	30	15	100
Plan 2	40	40	20	100
Plan 3	25	50	25	100
C_j	120	120	60	300

At $\alpha = 0.01$, test whether the outcome distribution is the same across the three plans. Report the E_{ij}, the cell contributions, χ^2, the df, the critical value, and a plain-language conclusion that notes which plan drives any rejection.

D6. One and the same 2×2 table of counts,

	Yes	No	R_i
group 1	84	36	120
group 2	60	60	120
C_j	144	96	240

arises in two different studies, both run at $\alpha = 0.05$.

(a) *Study I:* a trainer taught method 1 to 120 recruits and method 2 to a separate 120 recruits, then recorded pass/fail (Yes = pass). Name the correct test and write the conclusion in the language that the design licenses.

(b) *Study II:* one random sample of 240 adults was each asked two yes/no questions, "do you exercise weekly" (group) and "do you sleep 7+ hours" (Yes/No). Name the correct test and write the conclusion the design licenses.

(c) Show that the computed χ^2, the df, and the reject/retain decision are identical in (a) and (b), and explain in one sentence why the two written conclusions nonetheless differ.

D7. A clinic samples a fixed 80 charts from each of two hospitals and records the admitting route (ER, Referral, Walk-in, Transfer). The counts are

hospital	ER	Referral	Walk-in	Transfer	R_i
North	32	24	16	8	80
South	24	16	28	12	80
C_j	56	40	44	20	160

At $\alpha = 0.05$, test whether the two hospitals draw their admissions from the same route-distribution. Before computing the statistic, state the degrees of freedom and explain in counting terms why it equals $(r-1)(c-1)$ and not $rc-1$. Then compute every E_{ij}, the cell contributions, χ^2, compare to the critical value, and decide in words.

E. Conditions, small counts, pitfalls

E1. A geneticist classifies $n = 150$ fruit flies into five phenotype categories whose hypothesized proportions are 0.45, 0.28, 0.14, 0.09, 0.04. Before running a goodness-of-fit test, decide whether the expected-count rule of thumb is satisfied. Compute every E_i, identify any cell with $E_i < 5$, and state whether the strict (all $E_i \geq 5$) rule or the permissive (at most 20% of cells below 5, none below 1) rule applies. If the test may proceed, give the degrees of freedom.

E2. An entomologist tallies $n = 80$ insect specimens into six trap-type categories. The hypothesized proportions give expected counts

$$E_i = (28.0,\ 20.0,\ 16.0,\ 8.0,\ 4.8,\ 3.2),$$

with observed counts $O_i = (30,\ 18,\ 17,\ 7,\ 5,\ 3)$.

(a) Show the rule of thumb fails and pool the offending adjacent categories into one combined cell. Give the pooled expected and observed counts.

(b) State the degrees of freedom before and after pooling (fully specified GOF).

(c) Compute χ^2_{obs} on the pooled table, compare to $\chi^2_{0.05,\,4} = 9.488$, and state the decision in words.

E3. A dietitian compares two meal plans. Of 19 subjects on plan A, 6 met their weight goal; of 21 subjects on plan B, 14 did. Treat this as a 2×2 contingency table at $\alpha = 0.05$ (H_0: meeting the goal is independent of the plan).

(a) Build the table with margins and compute the four expected counts.

(b) Compute the uncorrected χ^2_{obs}.

(c) Compute the Yates continuity-corrected statistic $\chi^2_{\text{Yates}} = \sum_i (|O_i - E_i| - \frac{1}{2})^2 / E_i$.

(d) Using $\chi^2_{0.05,\,1} = 3.841$, state the decision for each statistic and explain why they may disagree.

E4. A market researcher reports that of $n = 600$ shoppers, brand preferences split 48%, 30%, 22% across three brands, against a hypothesized split of 40%, 35%, 25%. A student computes a goodness-of-fit statistic by entering the percentages $48, 30, 22$ as "observed" and $40, 35, 25$ as "expected."

(a) Compute the (wrong) statistic the student gets from the percentages and explain what sample size it secretly assumes.

(b) Convert to raw counts O_i and E_i using $n = 600$ and compute the correct χ^2_{obs}.

(c) With df $= 2$ and $\chi^2_{0.05,\,2} = 5.991$, give both decisions and the exact numerical factor relating the wrong and correct statistics.

E5. A streaming service logs the day-of-week of $n = 3{,}500{,}000$ sign-ups. Under "sign-ups are uniform across the seven days" each day expects 1/7 of the total. Summing $(\hat{p}_i - p_i)^2 / p_i$ over the seven days gives 4.0×10^{-6}.

(a) Using $\chi^2_{\text{obs}} = n \sum_i (\hat{p}_i - p_i)^2 / p_i$, compute the statistic.

(b) With df $= 6$ and $\chi^2_{0.05,\,6} = 12.592$, state whether uniformity is rejected.

(c) Compute the effect-size measure $\sqrt{\chi^2_{\text{obs}} / n}$ and explain in one sentence why "significant" here does not mean "important."

E6. An ecologist tests whether monthly stranding counts of a marine species follow a Poisson distribution (rate λ estimated from the data) and obtains $\chi^2_{\text{obs}} = 18.7$ on df $= 5$, exceeding $\chi^2_{0.05,\,5} = 11.070$. A colleague announces "the strandings are therefore geometrically distributed." Separately, a second colleague notes a different dataset gave $\chi^2_{\text{obs}} = 0.21$ on df $= 4$ for a hypothesized four-category split and asks whether such a small statistic should be "rejected" as too good a fit.

(a) State precisely what the rejection $\chi^2_{\text{obs}} = 18.7 > 11.070$ does and does not establish, and whether the geometric conclusion is warranted.

(b) Using $\chi^2_{0.05,\,4} = 9.488$, place $\chi^2_{\text{obs}} = 0.21$ relative to the rejection region and explain why a small χ^2 never triggers rejection of a goodness-of-fit test.

E7. For each described study, name the test (goodness-of-fit, independence, or homogeneity) and give the degrees of freedom. Read the sampling design, not the table shape.

(a) One random sample of 700 patients, each cross-classified by smoking status (3 levels) and a respiratory diagnosis (4 levels); is smoking status related to diagnosis?

(b) Separate random samples of 90 first-shift, 90 second-shift, and 90 third-shift workers, each worker's commute mode recorded in 4 categories; do the commute-mode distributions match across the three shifts?

(c) 420 tetrahedral-die rolls tallied into faces 1–4; is the die fair (each face 1/4)?

(d) One sample of 260 call-center arrivals per five-minute window, binned into $k = 8$ usable cells, tested against a Poisson whose rate is estimated from the data.

F. Count-data synthesis

F1. A streaming service claims that its viewers split across four content tiers — Free, Basic, Standard, Premium — in the fixed proportions $p = (\,0.40,\ 0.30,\ 0.20,\ 0.10\,)$. A single random sample of $n = 500$ active accounts is classified by tier, giving

$$O_i = (\,182,\ 166,\ 98,\ 54\,).$$

(a) State which of the three count tests (goodness-of-fit, independence, homogeneity) applies, and explain how you know from the design.

(b) Carry the test out at $\alpha = 0.05$: compute every E_i, the per-cell contributions, the statistic, the degrees of freedom, and the critical value, then state the decision in words.

F2. A municipal utility records the number of power outages in each of $n = 180$ months and groups the counts as "0, 1, 2, 3, 4, 5 or more":

$$O_i = (\,60,\ 62,\ 38,\ 14,\ 4,\ 2\,).$$

Management asks whether monthly outages follow a Poisson law (the rate is *not* supplied).

(a) Explain why this is a goodness-of-fit test with an *estimated* parameter, and estimate the Poisson rate from the data.

(b) Form the fitted probabilities $\widehat{p}_x = e^{-\widehat{\lambda}}\widehat{\lambda}^x/x!$ and the expected counts $E_i = 180\,\widehat{p}_x$. Any cell with $E_i < 5$ must be pooled with its neighbour; show the pooled table.

(c) State the degrees of freedom *after* pooling, compute χ^2, and decide at $\alpha = 0.05$. Make explicit how the estimated rate changes the df relative to a known-rate fit.

F3. A seed company packs 5 seeds per packet and records, for each of $n = 160$ packets, how many of the 5 germinated. If germination is an independent Bernoulli(p) event per seed, the count per packet is Binomial($5, p$), but p is unknown. The data are

Germinated x	0	1	2	3	4	5
Packets O_i	2	13	34	46	45	20

(a) Estimate p from the overall fraction of germinated seeds.

(b) Compute the fitted probabilities $\widehat{p}_x = \binom{5}{x}\widehat{p}^{\,x}(1-\widehat{p})^{5-x}$ and the expected counts $E_i = 160\,\widehat{p}_x$; pool any cell with $E_i < 5$ and give the resulting table.

(c) Count the degrees of freedom from scratch (cells after pooling, minus one for the total constraint, minus one per estimated parameter), compute χ^2, and decide at $\alpha = 0.05$.

(d) Suppose instead the company had *specified* $p = 0.60$ in advance, so no parameter is estimated. State the degrees of freedom that the test would then use, and explain why it differs from your answer in (c).

F4. The same 2×3 table of counts,

	response A	response B	response C	total
group 1	88	72	40	200
group 2	62	78	60	200
total	150	150	100	400

arises in two different studies, both tested at $\alpha = 0.05$.

Study X. A single random sample of 400 shoppers is asked two questions: their membership status (group 1 or 2) and which of three store layouts (A, B, C) they prefer; the 400 are cross-tabulated.

Study Y. A marketer mails layout A, B, or C information to a *pre-set* 200 customers in region 1 and a separate *pre-set* 200 customers in region 2, then records which layout each chooses.

(a) Name the correct test for each study and justify it from the sampling design.

(b) Show that the expected counts, the statistic, and the degrees of freedom are *identical* for the two studies by computing them once.

(c) Decide at $\alpha = 0.05$, and write the conclusion in the wording proper to each design — making clear how the two conclusions differ even though the arithmetic does not.

F5. A health system draws a *fixed* sample of 150 discharged patients from each of three hospitals and classifies each patient's overall experience as Satisfied, Neutral, or Dissatisfied. The counts are

hospital	Satisfied	Neutral	Dissatisfied	R_i
North	96	39	15	150
Central	78	54	18	150
South	60	66	24	150
C_j	234	159	57	450

At $\alpha = 0.05$, decide which test the design calls for, then carry it out in full: expected counts $E_{ij} = R_i C_j / n$, per-cell contributions, the statistic, the degrees of freedom, the critical value, and the decision in words. Identify the hospital whose row contributes most to the statistic.

F6. A market-research firm draws one random sample of $n = 400$ online customers and cross-classifies each by age band (under 30, 30–49, 50+) and the device used for their last purchase (phone, tablet, desktop):

age band	phone	tablet	desktop	R_i
under 30	60	40	30	130
30–49	50	55	45	150
50+	40	50	30	120
C_j	150	145	105	400

At $\alpha = 0.05$, decide which test applies (and why it is *not* homogeneity), then test whether age band and purchase device are independent. Report the expected counts, the statistic, the degrees of freedom, the critical value, and the decision.

F7. For each study below, name the correct count test (goodness-of-fit with known parameters, goodness-of-fit with estimated parameters, independence, or homogeneity) and give the exact degrees of freedom. State the rule you used.

(a) One random sample of 600 adults, each recorded by sex (male, female) and whether they exercise weekly (yes, no); is exercise related to sex?

(b) A casino auditor tallies $n = 900$ rolls of one die into faces 1–6 and asks whether the die is fair ($p_i = 1/6$).

(c) A pollster takes a *fixed* sample of 250 likely voters from each of four counties and records each voter's choice among three candidates; do the counties share one candidate distribution?

(d) A biologist tabulates the number of nesting pairs per plot over $n = 300$ plots into the cells $0, 1, 2, 3, 4, 5, 6, \geq 7$ and tests whether the counts follow a Poisson distribution whose rate is estimated from the data; after pooling two sparse tail cells, 7 cells remain.

(e) A psychologist groups $n = 220$ reaction times into 7 intervals and tests whether they are Normal, estimating both μ and σ from the data; no cell needs pooling.

10.7 Solutions

A. Goodness-of-fit, parameters known

A1. Method. One categorical variable (the face) with a *fully specified* null ($p_i = 1/6$ for all six faces, nothing estimated), so Pearson's goodness-of-fit test on χ^2_{k-1} with $k = 6$.

Step 1: expected counts. Each $E_i = np_{i0} = 480 \cdot \frac{1}{6} = 80$, and all $E_i = 80 \geq 5$, so the chi-square approximation is valid.

Step 2: cell contributions.

Face i	1	2	3	4	5	6
Observed O_i	72	91	74	88	69	86
Expected E_i	80	80	80	80	80	80
$O_i - E_i$	-8	11	-6	8	-11	6
$(O_i - E_i)^2/E_i$	0.800	1.513	0.450	0.800	1.513	0.450

Step 3: statistic, df, decision.

$$\chi^2 = 0.800 + 1.513 + 0.450 + 0.800 + 1.513 + 0.450 = 5.525,$$
$$\text{df} = k - 1 = 6 - 1 = 5,$$
$$\chi^2_{0.05,\,5} = 11.07.$$

Since $5.525 < 11.07$, fail to reject.

$$\boxed{\chi^2 = 5.525 < \chi^2_{0.05,\,5} = 11.07 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{the die is consistent with being fair.}}$$

At the 5% level there is no significant evidence that the die is unfair; the spread of counts is ordinary sampling jitter for 480 rolls.

A2. Method. A theoretical ratio fixes every probability with nothing estimated, so the parameters-known goodness-of-fit test with $k = 4$, $\text{df} = 3$.

(a) The ratio denominators sum to $9 + 3 + 3 + 1 = 16$, so $p_{i0} = (\frac{9}{16}, \frac{3}{16}, \frac{3}{16}, \frac{1}{16})$ and $E_i = 240\,p_{i0}$:

$$E_1 = 240 \cdot \tfrac{9}{16} = 135,$$
$$E_2 = 240 \cdot \tfrac{3}{16} = 45,$$
$$E_3 = 240 \cdot \tfrac{3}{16} = 45,$$
$$E_4 = 240 \cdot \tfrac{1}{16} = 15.$$

All $E_i \geq 5$, so the approximation holds.

(b) Cell contributions and statistic:

Class	1	2	3	4
Ratio	9	3	3	1
Observed O_i	121	49	53	17
Expected E_i	135	45	45	15
$(O_i - E_i)^2/E_i$	1.452	0.356	1.422	0.267

$$\chi^2 = \frac{(121-135)^2}{135} + \frac{(49-45)^2}{45} + \frac{(53-45)^2}{45} + \frac{(17-15)^2}{15}$$
$$= 1.452 + 0.356 + 1.422 + 0.267 = 3.496,$$
$$\text{df} = k - 1 = 3, \qquad \chi^2_{0.05,\,3} = 7.815.$$

Since $3.496 < 7.815$, fail to reject.

$$\boxed{\chi^2 = 3.496 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{data are consistent with 9:3:3:1.}}$$

There is no significant evidence against the Mendelian ratio at the 5% level.

(c) The degrees of freedom are $k - 1 = 3$, not $k = 4$, because the four counts must sum to the fixed total $n = 240$: once three counts are known the fourth is determined, so only three of the deviations $O_i - E_i$ are free to vary. No parameter was estimated from the offspring (the ratio supplied fixed numbers), so we subtract only the one constraint and not an extra df.

A3. Method. "Spread evenly across seven days" is a fully specified uniform null, $p_i = 1/7$ for each day, so a goodness-of-fit test with $k = 7$ on χ^2_6.

Step 1: expected counts. $E_i = 350 \cdot \frac{1}{7} = 50$ for every day (all ≥ 5).

Step 2: cell contributions.

Day	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Observed O_i	44	51	47	53	58	61	36
Expected E_i	50	50	50	50	50	50	50
$O_i - E_i$	-6	1	-3	3	8	11	-14
$(O_i - E_i)^2/E_i$	0.72	0.02	0.18	0.18	1.28	2.42	3.92

Step 3: statistic, df, decision.

$$\begin{aligned} \chi^2 &= 0.72 + 0.02 + 0.18 + 0.18 + 1.28 + 2.42 + 3.92 = 8.72, \\ \mathrm{df} &= k - 1 = 7 - 1 = 6, \\ \chi^2_{0.05,\,6} &= 12.592. \end{aligned}$$

Since $8.72 < 12.592$, fail to reject.

$$\boxed{\chi^2 = 8.72 < \chi^2_{0.05,\,6} = 12.592 \;\Rightarrow\; \text{even spread across the week is tenable.}}$$

The largest cell contribution is Sunday (3.92, with only 36 thefts against an expected 50), followed by Saturday (2.42); but even together they do not push the total past the cutoff, so at the 5% level the even-spread claim survives.

A4. Method. A complete, numerically specified probability list for one categorical variable (the symbol) $\Rightarrow$ parameters-known goodness-of-fit, $k = 4$, df $= 3$. Build every $E_i = np_i$ and $(O_i - E_i)^2/E_i$ in one table before summing.

Step 1: expected counts and contributions, $n = 600$.

Symbol	p_i	$E_i = 600p_i$	O_i	$(O_i - E_i)^2/E_i$
cherry	0.50	300	324	$576/300 = 1.920$
bell	0.25	150	138	$144/150 = 0.960$
bar	0.15	90	81	$81/90 = 0.900$
seven	0.10	60	57	$9/60 = 0.150$
Total	1.00	600	600	**3.930**

The smallest expected count is $E_i = 60 \geq 5$, so the approximation is valid.

Step 2: statistic. Summing the last column,

$$\chi^2 = 1.920 + 0.960 + 0.900 + 0.150 = 3.930.$$

Step 3: df, critical value, decision. With $\mathrm{df} = k - 1 = 3$ and $\alpha = 0.05$, $\chi^2_{0.05,\,3} = 7.815$. Since $3.930 < 7.815$, fail to reject.

$$\boxed{\chi^2 = 3.930 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{the machine matches its design distribution.}}$$

There is no significant evidence at the 5% level that the symbol frequencies depart from the designed probabilities; the cherry cell is the heaviest (1.92) but not decisive.

A5. Method. The rate $\lambda = 1.5$ is fixed by the design spec, not estimated from the data, so every category probability is a known number and this is the parameters-known goodness-of-fit test ($\mathrm{df} = k - 1$).

(a) With $\mathbb{P}(X = x) = e^{-1.5}1.5^x/x!$ and $e^{-1.5} = 0.22313$:

$$\begin{aligned} p_0 &= e^{-1.5} = 0.2231, \\ p_1 &= 1.5\,e^{-1.5} = 0.3347, \\ p_2 &= \tfrac{1.5^2}{2}e^{-1.5} = 1.125\,e^{-1.5} = 0.2510, \\ p_{\geq 3} &= 1 - (p_0 + p_1 + p_2) = 0.1912. \end{aligned}$$

Expected counts $E_i = 200p_i$:

Flaws x	0	1	2	≥ 3
p_i	0.2231	0.3347	0.2510	0.1912
Expected E_i	44.63	66.94	50.20	38.23
Observed O_i	42	70	51	37
$(O_i - E_i)^2/E_i$	0.1545	0.1400	0.0126	0.0396

All $E_i \geq 5$, so the approximation holds.

(b) Statistic and decision:

$$\begin{aligned} \chi^2 &= 0.1545 + 0.1400 + 0.0126 + 0.0396 = 0.347, \\ \mathrm{df} &= k - 1 = 4 - 1 = 3, \qquad \chi^2_{0.05,\,3} = 7.815. \end{aligned}$$

Since $0.347 < 7.815$, fail to reject.

$$\boxed{\chi^2 = 0.347 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{the Poisson(1.5) model fits.}}$$

At the 5% level the flaw counts are entirely consistent with the design-mandated Poisson model.

(c) The degrees of freedom are $k - 1 = 3$, *not* $k - 2$, because the rate $\lambda = 1.5$ was *given in advance* rather than estimated from these cables, so no parameter was borrowed from the data ($m = 0$). The general rule is $\mathrm{df} = k - 1 - m$; here $m = 0$ gives $k - 1$. Had λ been estimated from the same counts ($m = 1$), we would subtract one more df and use $\mathrm{df} = k - 2 = 2$ — the §2 setting.

A6. Method. Two fully specified categories ($p = (\frac{3}{4}, \frac{1}{4})$), so a goodness-of-fit test with $k = 2$ and $\mathrm{df} = 1$; with $k = 2$ it is algebraically equivalent to the one-proportion z-test ($\chi^2 = z^2$).

(a) Expected counts: $E_{\text{tall}} = 200 \cdot \frac{3}{4} = 150$, $E_{\text{short}} = 200 \cdot \frac{1}{4} = 50$ (both ≥ 5).

$$\begin{aligned} \chi^2 &= \frac{(137 - 150)^2}{150} + \frac{(63 - 50)^2}{50} \\ &= \frac{169}{150} + \frac{169}{50} = 1.127 + 3.380 = 4.507, \\ \mathrm{df} &= k - 1 = 1, \qquad \chi^2_{0.05,\,1} = 3.841. \end{aligned}$$

Since $4.507 > 3.841$, reject.

$$\boxed{\chi^2 = 4.507 > \chi^2_{0.05,\,1} = 3.841 \Rightarrow \text{reject } H_0\text{; the 3:1 ratio is doubtful.}}$$

At the 5% level there is significant evidence against the 3:1 split: too many short plants (its cell contributes 3.38 of the 4.51).

(b) One-proportion z for $H_0\colon p_{\text{tall}} = 3/4$, using counts with

$$\sqrt{np_0(1-p_0)} = \sqrt{200 \cdot 0.75 \cdot 0.25} = \sqrt{37.5} = 6.124:$$

$$z = \frac{137 - 150}{\sqrt{200 \cdot 0.75 \cdot 0.25}} = \frac{-13}{6.124} = -2.123,$$
$$z^2 = (-2.123)^2 = 4.507 = \chi^2.$$

The squared z reproduces the chi-square statistic exactly, confirming the $k = 2$ equivalence (and $z_{0.025} = 1.96$ gives $z^2_{0.025} = 3.84 = \chi^2_{0.05,\,1}$).

A7. Method. One categorical variable (the vote choice) with a fully specified probability split $(0.45, 0.35, 0.20)$, so goodness-of-fit on χ^2_2 with a single upper-tail p-value.

Step 1: expected counts, $n = 300$. $E_i = (\,300 \cdot 0.45,\ 300 \cdot 0.35,\ 300 \cdot 0.20\,) = (\,135,\ 105,\ 60\,)$ (all ≥ 5).

Step 2: statistic.

$$\chi^2 = \frac{(123-135)^2}{135} + \frac{(117-105)^2}{105} + \frac{(60-60)^2}{60}$$
$$= \frac{144}{135} + \frac{144}{105} + 0 = 1.067 + 1.371 + 0.000 = 2.438.$$

Step 3: p-value and decision. With $\text{df} = k - 1 = 2$, the chi-square upper tail has the closed form $\mathbb{P}(\chi^2_2 > c) = e^{-c/2}$, so

$$p = \mathbb{P}(\chi^2_2 > 2.438) = e^{-2.438/2} = e^{-1.219} = 0.296.$$

Since $0.296 > 0.05$, fail to reject. (Equivalently $\chi^2 = 2.438 < \chi^2_{0.05,\,2} = 5.991$.)

$$\boxed{p\text{-value} = e^{-2.438/2} = 0.296 > 0.05 \ \Rightarrow\ \text{fail to reject; the claimed 45/35/20 split is tenable.}}$$

At the 5% level there is no significant evidence that the exit-survey proportions differ from the firm's claimed split.

B. Goodness-of-fit, parameters estimated

B1. Poisson fit, rate estimated, with pooling. H_0: pages per block are Poisson; H_1: they are not. The family is named but the rate is not, so estimate λ by the sample mean (the Poisson MLE) and use $\text{df} = k - 1 - 1$.

Step 1 — estimate λ. Taking the ≥ 5 cell at its conservative value 5, the total number of pages is

$$0(36) + 1(44) + 2(42) + 3(22) + 4(12) + 5(4) = 262,$$

over $n = 160$ blocks, so $\widehat{\lambda} = 262/160 = 1.6375 \approx 1.64$.

Step 2 — fitted probabilities and expected counts. With $\widehat{p}_x = e^{-1.64}1.64^x/x!$ and the last cell $\mathbb{P}(X \geq 5) = 1 - \sum_0^4 \widehat{p}_x$, multiply by 160:

$$E_i = (31.0,\ 50.9,\ 41.7,\ 22.8,\ 9.4,\ 4.2), \qquad \sum E_i = 160.$$

Step 3 — pool the sparse tail. The ≥ 5 cell has $E_i = 4.2 < 5$; pool it with the $x = 4$ cell: combined $O_i = 12 + 4 = 16$ against $E_i = 9.4 + 4.2 = 13.5$. The categories are now $0, 1, 2, 3, \geq 4$, so $k = 5$.

Step 4 — statistic.

$$\chi^2 = \frac{(36-31.0)^2}{31.0} + \frac{(44-50.9)^2}{50.9} + \frac{(42-41.7)^2}{41.7} + \frac{(22-22.8)^2}{22.8} + \frac{(16-13.5)^2}{13.5} = 0.79 + 0.94 + 0.00 + 0.03 + 0.46 = 2.22.$$

Step 5 — df and decision. After pooling $k = 5$; one parameter (λ) was estimated, so $\text{df} = k - 1 - 1 = 5 - 2 = 3$. The cutoff is $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 2.22 < 7.815 = \chi^2_{0.05,\,3} \;\Rightarrow\; \text{fail to reject; Poisson is a tenable fit.}}$$

At the 5% level there is no significant evidence against a Poisson model for the page counts.

B2. Binomial fit, p estimated. H_0: the daughter count per family is Binomial$(4, p)$; H_1: it is not. The number of trials (4) is known but p is not, so we estimate p from the overall fraction of daughters and lose one degree of freedom.

(a) *Estimate p.* Total daughters

$$0(18) + 1(32) + 2(48) + 3(34) + 4(12) = 278,$$

out of $4 \times 144 = 576$ children, so

$$\widehat{p} = \frac{278}{576} = 0.4826 \approx 0.483, \qquad 1 - \widehat{p} = 0.517.$$

Fitted probabilities $\widehat{p}_x = \binom{4}{x}\widehat{p}^x(1-\widehat{p})^{4-x}$:

$$\widehat{p}_0 = (0.517)^4 = 0.0714, \qquad \widehat{p}_1 = 4(0.483)(0.517)^3 = 0.2670,$$
$$\widehat{p}_2 = 6(0.483)^2(0.517)^2 = 0.3741, \qquad \widehat{p}_3 = 4(0.483)^3(0.517) = 0.2330,$$
$$\widehat{p}_4 = (0.483)^4 = 0.0544.$$

Expected counts $E_i = 144\,\widehat{p}_x$:

x	0	1	2	3	4
E_i	10.3	38.4	53.9	33.6	7.8

Every $E_i \geq 5$, so no pooling is needed and $k = 5$.

(b) *Statistic, df, decision.*

$$\chi^2 = \frac{(18-10.3)^2}{10.3} + \frac{(32-38.4)^2}{38.4} + \frac{(48-53.9)^2}{53.9} + \frac{(34-33.6)^2}{33.6} + \frac{(12-7.8)^2}{7.8} = 5.78 + 1.08 + 0.64 + 0.01 + 2.21 = 9.72.$$

One parameter estimated, so $\text{df} = k - 1 - 1 = 5 - 2 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 9.72 > 7.815 = \chi^2_{0.05,\,3} \;\Rightarrow\; \text{reject; the count is } \textit{not} \text{ Binomial}(4, p).}$$

Reject H_0 at the 5% level: there is significant evidence that the number of daughters per family does not follow a constant-p, independent-trials binomial. The excess of all-son and daughter-heavy families ($x = 0$ and $x = 4$ each far above their expected counts) points to dependence among siblings.

B3. Normal fit on grouped data, both parameters estimated. H_0: glucose readings are normal; H_1: they are not. With $\widehat{\mu} = \overline{X} = 100$ and $\widehat{\sigma} = s = 16$ estimated from the data, the fit costs $m = 2$ and $\mathrm{df} = k - 1 - 2$.

Step 1 — standardize the cut points $z = (b - 100)/16$:

$$z(76) = -1.5, \quad z(88) = -0.75, \quad z(100) = 0,$$
$$z(112) = 0.75, \quad z(124) = 1.5.$$

Step 2 — fitted interval probabilities from the given Φ values:

$$\widehat{p}_1 = \Phi(-1.5) = .0668, \qquad \widehat{p}_2 = \Phi(-.75) - \Phi(-1.5) = .1598,$$
$$\widehat{p}_3 = \Phi(0) - \Phi(-.75) = .2734, \quad \widehat{p}_4 = \Phi(.75) - \Phi(0) = .2734,$$
$$\widehat{p}_5 = \Phi(1.5) - \Phi(.75) = .1598, \quad \widehat{p}_6 = 1 - \Phi(1.5) = .0668.$$

Step 3 — expected counts $E_i = 180\,\widehat{p}_i$:

Interval	< 76	76–88	88–100	100–112	112–124	≥ 124
E_i	12.0	28.8	49.2	49.2	28.8	12.0

All $E_i \geq 5$, so no pooling; $k = 6$.

Step 4 — statistic.

$$\chi^2 = \frac{(9 - 12.0)^2}{12.0} + \frac{(34 - 28.8)^2}{28.8} + \frac{(45 - 49.2)^2}{49.2} + \frac{(55 - 49.2)^2}{49.2} + \frac{(24 - 28.8)^2}{28.8} + \frac{(13 - 12.0)^2}{12.0}$$
$$= 0.75 + 0.94 + 0.36 + 0.68 + 0.80 + 0.08 = 3.62.$$

Step 5 — df and decision. $k = 6$, two parameters estimated, so $\mathrm{df} = 6 - 1 - 2 = 3$; cutoff $\chi^2_{0.05,\,3} = 7.815$.

$$\boxed{\chi^2 = 3.62 < 7.815 = \chi^2_{0.05,\,3} \;\Rightarrow\; \text{fail to reject; normality is tenable.}}$$

At the 5% level there is no significant evidence against a normal model for the glucose readings.

B4. Poisson fit with tail pooling. H_0: counts are Poisson; the rate $\widehat{\lambda} = 2.00$ and the raw E_i are supplied.

(a) *Identify and pool sparse cells.* The cells with $E_i < 5$ are $x = 5$ ($E_i = 3.6$) and ≥ 6 ($E_i = 1.7$). Pooling *these two* gives a single ≥ 5 cell with $E_i = 3.6 + 1.7 = 5.3 \geq 5$, which is enough; no deeper pooling is needed. The categories are now $0, 1, 2, 3, 4, \geq 5$, so

$$k = 6 \quad \text{(down from the original 7)}.$$

The pooled observed count is $O_i = 4 + 2 = 6$ against $E_i = 5.3$.

(b) *Statistic, df, decision.*

$$\chi^2 = \frac{(20 - 13.5)^2}{13.5} + \frac{(22 - 27.1)^2}{27.1} + \frac{(26 - 27.1)^2}{27.1} + \frac{(17 - 18.0)^2}{18.0} + \frac{(9 - 9.0)^2}{9.0} + \frac{(6 - 5.3)^2}{5.3}$$
$$= 3.13 + 0.96 + 0.04 + 0.06 + 0.00 + 0.09 = 4.28.$$

After pooling $k = 6$; one parameter (λ) was estimated, so $\mathrm{df} = k - 1 - 1 = 6 - 2 = 4$; cutoff $\chi^2_{0.05,\,4} = 9.488$.

$$\boxed{\chi^2 = 4.28 < 9.488 = \chi^2_{0.05,\,4} \;\Rightarrow\; \text{fail to reject; the Poisson model fits.}}$$

At the 5% level there is no significant evidence against a Poisson model for the particle counts.

B5. Contrast: the two analysts' degrees of freedom. H_0: pairs per plot are Poisson; the rate $\widehat{\lambda} = 2.40$ was estimated from the data, and the E_i are supplied (all ≥ 5, so $k = 6$ with no pooling).

Step 1 — statistic.

$$\begin{aligned}\chi^2 &= \frac{(14-8.2)^2}{8.2} + \frac{(13-19.6)^2}{19.6} + \frac{(30-23.5)^2}{23.5} \\ &\quad + \frac{(14-18.8)^2}{18.8} + \frac{(9-11.3)^2}{11.3} + \frac{(10-8.6)^2}{8.6} \\ &= 4.10 + 2.22 + 1.80 + 1.23 + 0.47 + 0.23 = 10.04.\end{aligned}$$

Step 2 — the two df claims and their cutoffs.

$$\begin{aligned}&\text{Analyst X: } \mathrm{df} = k - 1 = 5, \quad \chi^2_{0.05,\,5} = 11.070, \\ &\text{Analyst Y: } \mathrm{df} = k - 1 - 1 = 4, \quad \chi^2_{0.05,\,4} = 9.488.\end{aligned}$$

Step 3 — who is right, and the decision. Analyst Y is correct: the rate λ was *estimated from these data*, so one degree of freedom must be subtracted, giving $\mathrm{df} = 4$. Against the correct cutoff,

$$\boxed{\chi^2 = 10.04 > 9.488 = \chi^2_{0.05,\,4} \;\Rightarrow\; \text{reject; the pairs are } \textit{not} \text{ Poisson.}}$$

Analyst X's inflated $\mathrm{df} = 5$ uses the larger cutoff 11.070; since $10.04 < 11.070$, X would (wrongly) fail to reject. A single forgotten subtraction flips the conclusion — the estimation already pulled χ^2 down, so the smaller reference curve ($\mathrm{df} = 4$) is the properly calibrated one, and X's choice makes the test too conservative.

B6. Degrees-of-freedom bookkeeping. The raw table has 8 cells $x = 0, 1, \ldots, 6, \geq 7$, and a Poisson rate was estimated from the sample mean.

(a) *Pool sparse cells.* The cells with $E_i < 5$ are $x = 6$ ($E_i = 3.4$) and ≥ 7 ($E_i = 1.6$); the cell $x = 5$ has $E_i = 6.0 \geq 5$ and stays. Pool $x = 6$ and ≥ 7 into one ≥ 6 cell with $E_i = 3.4 + 1.6 = 5.0 \geq 5$. This removes one category, so

$$k = 8 - 1 = 7 \quad (\text{cells } 0, 1, 2, 3, 4, 5, \geq 6).$$

(b) *Final degrees of freedom.* The counts sum to $n = 200$ (the universal -1), and one parameter, λ, was estimated (-1 more):

$$\begin{aligned}\mathrm{df} &= k - 1 - m \\ &= 7 - 1 - 1 = 5.\end{aligned}$$

$$\boxed{\mathrm{df} = 5 \text{ (7 pooled cells, minus 1 for the total, minus 1 for } \widehat{\lambda}).}$$

(c) *If a normal model had been fitted.* A normal fit estimates two parameters (μ and σ), so $m = 2$ and, with the same $k = 7$ pooled cells,

$$\mathrm{df} = k - 1 - 2 = 7 - 1 - 2 = 4.$$

Each extra estimated parameter costs one more degree of freedom.

B7. Contrast: forgetting to subtract for the estimated parameter. The model has $k = 5$ pooled cells and one estimated parameter; the statistic is $\chi^2 = 8.40$.

(a) *Correct df and cutoff.* One parameter was estimated, so

$$\mathrm{df} = k - 1 - 1 = 5 - 2 = 3, \qquad \chi^2_{0.05,\,3} = 7.815.$$

The junior analyst used $\mathrm{df} = k - 1 = 4$ with cutoff $\chi^2_{0.05,\,4} = 9.488$ — the fully-specified rule, which ignores the estimation.

(b) *Correct decision and the nature of the error.* With the correct df,

$$\begin{aligned} \text{correct df} = 3: &\quad 8.40 > \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{reject}, \\ \text{wrong df} = 4: &\quad 8.40 < \chi^2_{0.05,\,4} = 9.488 \;\Rightarrow\; \text{fail to reject}. \end{aligned}$$

$$\boxed{\chi^2 = 8.40 > 7.815 = \chi^2_{0.05,\,3} \;\Rightarrow\; \text{reject; the one-parameter model does not fit.}}$$

The junior analyst's forgotten subtraction makes the test *too conservative*: the larger df gives the larger cutoff (9.488 instead of 7.815), so the analyst wrongly fails to reject a model that the correct test rejects at the 5% level. Estimating λ (or p) from the data already slid the E_i toward the O_i and shrank χ^2, so the statistic truly follows χ^2_3 (mean 3), not χ^2_4 (mean 4); referring it to the larger curve mis-states the tail probability and breaks the test's calibration. The mechanical fix is to subtract one degree of freedom per estimated parameter: $\text{df} = k - 1 - m$.

C. Contingency tables: independence

C1. One sample of 240 students cross-classified by two traits (exercise, sleep), so this is the **test of independence** on a 2×2 table; $\text{df} = (2-1)(2-1) = 1$, with $E_{ij} = R_i C_j / n$.

Expected counts. Row totals $120, 120$; column totals $138, 102$; $n = 240$.

$$\begin{aligned} E_{11} &= \frac{120 \cdot 138}{240} = 69.0, \quad E_{12} = \frac{120 \cdot 102}{240} = 51.0, \\ E_{21} &= \frac{120 \cdot 138}{240} = 69.0, \quad E_{22} = \frac{120 \cdot 102}{240} = 51.0. \end{aligned}$$

All $E_{ij} \geq 51 \geq 5$, so the χ^2 approximation is valid.

Observed vs. expected.

	observed O_{ij}		expected E_{ij}	
	good	poor	good	poor
exercises	84	36	69.0	51.0
no exercise	54	66	69.0	51.0

Cell contributions.

$$\begin{aligned} \frac{(84-69)^2}{69} &= \frac{225}{69} = 3.2609, \quad \frac{(36-51)^2}{51} = \frac{225}{51} = 4.4118, \\ \frac{(54-69)^2}{69} &= \frac{225}{69} = 3.2609, \quad \frac{(66-51)^2}{51} = \frac{225}{51} = 4.4118. \end{aligned}$$

Statistic.

$$\chi^2_{\text{obs}} = 3.2609 + 4.4118 + 3.2609 + 4.4118 = 15.345.$$

Decision. On $\text{df} = 1$, $\chi^2_{0.05,1} = 3.841$. Since $15.345 > 3.841$, reject.

$$\boxed{\chi^2_{\text{obs}} = 15.345 > 3.841 \;\Rightarrow\; \text{reject } H_0\text{: exercise habit is associated with sleep quality.}}$$

Reject at the 5% level: there is significant evidence that exercise habit and sleep quality are related. This is **association**, not causation — the cross-tabulation cannot rule out a third factor (say, overall health or stress) driving both, nor say that exercising *produces* better sleep. "Related" is the strongest honest verdict from an observational table.

C2. (a) **Independence test.** One sample of 400 visitors, two traits (page version, sign-up), 2×2 table, df $= (2-1)(2-1) = 1$. Row totals $200, 200$; column totals $168, 232$.

$$E_{11} = \frac{200 \cdot 168}{400} = 84.0, \quad E_{12} = \frac{200 \cdot 232}{400} = 116.0,$$
$$E_{21} = \frac{200 \cdot 168}{400} = 84.0, \quad E_{22} = \frac{200 \cdot 232}{400} = 116.0.$$

Contributions:

$$\frac{(72-84)^2}{84} = \frac{144}{84} = 1.7143, \quad \frac{(128-116)^2}{116} = \frac{144}{116} = 1.2414,$$
$$\frac{(96-84)^2}{84} = \frac{144}{84} = 1.7143, \quad \frac{(104-116)^2}{116} = \frac{144}{116} = 1.2414.$$

$$\chi^2_{\text{obs}} = 1.7143 + 1.2414 + 1.7143 + 1.2414 = 5.911.$$

On df $= 1$, $\chi^2_{0.05,1} = 3.841$, and $5.911 > 3.841$, so reject.

$$\boxed{\chi^2_{\text{obs}} = 5.911 > 3.841 \;\Rightarrow\; \text{reject } H_0\text{: sign-up is associated with page version.}}$$

(b) **Two-proportion z-test.** Sign-up rates: version A $72/200 = 0.36$, version B $96/200 = 0.48$. Pooled rate $\widehat{p} = (72+96)/400 = 168/400 = 0.42$. Pooled standard error

$$\text{SE}_0 = \sqrt{\widehat{p}(1-\widehat{p})\left(\tfrac{1}{200} + \tfrac{1}{200}\right)}$$
$$= \sqrt{0.42 \cdot 0.58 \cdot 0.01} = \sqrt{0.0024360} = 0.049356.$$

$$z_{\text{obs}} = \frac{0.36 - 0.48}{0.049356} = \frac{-0.12}{0.049356} = -2.431.$$

(c) **Identity.** $z^2_{\text{obs}} = (-2.431)^2 = 5.911 = \chi^2_{\text{obs}}$, matching part (a) exactly. The cutoffs agree too: $\chi^2_{0.05,1} = 3.841 = 1.96^2 = z^2_{0.025}$, and $|z_{\text{obs}}| = 2.431 > 1.96$ gives the same reject decision.

$$\boxed{\chi^2_{\text{obs}} = z^2_{\text{obs}} = 5.911; \quad |z_{\text{obs}}| = 2.431 > 1.96 \;\Rightarrow\; \text{reject.}}$$

The z-version carries a **direction** that the always-positive χ^2 discards: here $z_{\text{obs}} < 0$ because version A's rate (0.36) is *below* version B's (0.48), so B is the better page. The χ^2 flags the association but not which version wins.

C3. One sample of 300 pupils, two traits (method, outcome) $\Rightarrow$ independence test on a 3×2 table; df $= (3-1)(2-1) = 2$. Row totals $100, 100, 100$; column totals $210, 90$.

Expected counts $E_{ij} = R_i C_j / 300$. Each row total is 100, so every cell in the pass column expects $100 \cdot 210/300 = 70$ and every cell in the fail column expects $100 \cdot 90/300 = 30$:

$$E_{\text{lec},p} = 70, \quad E_{\text{lec},f} = 30,$$
$$E_{\text{flip},p} = 70, \quad E_{\text{flip},f} = 30,$$
$$E_{\text{hyb},p} = 70, \quad E_{\text{hyb},f} = 30.$$

All $E_{ij} \geq 30 \geq 5$; valid.

Contributions $(O_{ij} - E_{ij})^2 / E_{ij}$.

$$\text{lecture: } \frac{(60-70)^2}{70} = 1.4286, \quad \frac{(40-30)^2}{30} = 3.3333,$$
$$\text{flipped: } \frac{(78-70)^2}{70} = 0.9143, \quad \frac{(22-30)^2}{30} = 2.1333,$$
$$\text{hybrid: } \frac{(72-70)^2}{70} = 0.0571, \quad \frac{(28-30)^2}{30} = 0.1333.$$

Statistic.

$$\begin{aligned}\chi^2_{\text{obs}} &= 1.4286 + 3.3333 + 0.9143 + 2.1333 + 0.0571 + 0.1333 \\ &= 8.000.\end{aligned}$$

Decision. On df $= 2$, $\chi^2_{0.05,2} = 5.991$. Since $8.000 > 5.991$, reject.

$$\boxed{\chi^2_{\text{obs}} = 8.000 > 5.991 \;\Rightarrow\; \text{reject } H_0\text{: outcome is associated with teaching method.}}$$

Reject at the 5% level: there is significant evidence that pass/fail outcome depends on teaching method. The single largest contribution is the **lecture–fail** cell (3.33): the lecture group failed far more often (40 vs. the 30 independence predicts), so it is the main driver of the association — the flipped and hybrid rows are close to expectation.

C4. (a) **Table shape and df.** $r = 3$ panel results, $c = 2$ skin-test outcomes, so $rc = 6$ cells. Factored: df $= (r-1)(c-1) = (3-1)(2-1) = 2$. Long way:

$$\text{df} = rc - 1 - (r-1) - (c-1) = 6 - 1 - 2 - 1 = 2,$$

agreeing. (Not $rc - 1 = 5$: estimating the margins costs the extra 3.)

(b) **Independence test.** Row totals $200, 200, 100$; column totals $120, 380$; $n = 500$.

$$\begin{aligned} E_{\text{none},+} &= \frac{200 \cdot 120}{500} = 48.0, & E_{\text{none},-} &= \frac{200 \cdot 380}{500} = 152.0, \\ E_{\text{mild},+} &= \frac{200 \cdot 120}{500} = 48.0, & E_{\text{mild},-} &= \frac{200 \cdot 380}{500} = 152.0, \\ E_{\text{sev},+} &= \frac{100 \cdot 120}{500} = 24.0, & E_{\text{sev},-} &= \frac{100 \cdot 380}{500} = 76.0. \end{aligned}$$

Smallest is $24 \geq 5$; valid. Contributions:

$$\begin{aligned} \frac{(40-48)^2}{48} &= 1.3333, & \frac{(160-152)^2}{152} &= 0.4211, \\ \frac{(50-48)^2}{48} &= 0.0833, & \frac{(150-152)^2}{152} &= 0.0263, \\ \frac{(30-24)^2}{24} &= 1.5000, & \frac{(70-76)^2}{76} &= 0.4737. \end{aligned}$$

$$\begin{aligned}\chi^2_{\text{obs}} &= 1.3333 + 0.4211 + 0.0833 + 0.0263 + 1.5000 + 0.4737 \\ &= 3.838.\end{aligned}$$

On df $= 2$, $\chi^2_{0.05,2} = 5.991$. Since $3.838 < 5.991$, do not reject.

$$\boxed{\chi^2_{\text{obs}} = 3.838 < 5.991 \Rightarrow \text{fail to reject: no evidence panel result and skin test are associated.}}$$

Fail to reject at the 5% level: the data are consistent with the two classifications being independent (this is "no evidence of association," not proof of independence).

C5. One sample of 800 adults, two traits (region, opinion) $\Rightarrow$ independence test on a 2×4 table; df $= (2-1)(4-1) = 3$. Row totals $400, 400$; column totals $200, 200, 200, 200$; $n = 800$.

Expected counts. Each row total is 400, so for every column $E_{1j} = E_{2j} = 400 \cdot 200/800 = 100$ — all eight cells expect 100.

Contributions $(O_{ij} - 100)^2/100$**.**

$$\begin{aligned} \text{favor: } \frac{(130-100)^2}{100} &= 9.0, & \frac{(90-100)^2}{100} &= 1.0, & \frac{(110-100)^2}{100} &= 1.0, & \frac{(70-100)^2}{100} &= 9.0, \\ \text{oppose: } \frac{(70-100)^2}{100} &= 9.0, & \frac{(110-100)^2}{100} &= 1.0, & \frac{(90-100)^2}{100} &= 1.0, & \frac{(130-100)^2}{100} &= 9.0. \end{aligned}$$

Statistic.

$$\chi^2_{\text{obs}} = 2(9.0 + 1.0 + 1.0 + 9.0)$$
$$= 2 \cdot 20.0 = 40.000.$$

Decision. On df $= 3$, $\chi^2_{0.05,3} = 7.815$. Since $40.000 > 7.815$, reject decisively.

$$\boxed{\chi^2_{\text{obs}} = 40.000 > 7.815 \;\Rightarrow\; \text{reject } H_0\text{: opinion is associated with region.}}$$

Reject at the 5% level: there is strong evidence that ballot-measure opinion depends on region. The East and West cells each contribute 9.0 (East leans favor, West leans oppose), together supplying 36 of the 40; the Midwest and South cells are near expectation and add little.

C6. (a) **Shape and df.** $r = 3$ defect types, $c = 3$ machines, so $rc = 9$ cells; this is an independence test with df $= (r-1)(c-1) = (3-1)(3-1) = 4$ (not $rc - 1 = 8$).

(b) **Decision at $\alpha = 0.05$.** On df $= 4$, $\chi^2_{0.05,4} = 9.488$. Since $\chi^2_{\text{obs}} = 14.7 > 9.488$, reject.

$$\boxed{\text{df} = 4; \quad \chi^2_{\text{obs}} = 14.7 > 9.488 \;\Rightarrow\; \text{reject: defect type is associated with machine.}}$$

(c) **At $\alpha = 0.01$.** On df $= 4$, $\chi^2_{0.01,4} = 13.277$. Since $14.7 > 13.277$, the decision is *unchanged*: still reject at the stricter 1% level. This is consistent with the stated p-value just above 0.005, which is below both 0.05 and 0.01.

$$\boxed{\chi^2_{0.01,4} = 13.277; \quad 14.7 > 13.277 \;\Rightarrow\; \text{reject at 1\% as well.}}$$

C7. (a) **df and decision.** One sample of 360 employees, two traits (department, program use), 3×2 table $\Rightarrow$ df $= (3-1)(2-1) = 2$. On df $= 2$, $\chi^2_{0.05,2} = 5.991$. Since $\chi^2_{\text{obs}} = 5.36 < 5.991$, fail to reject.

$$\boxed{\text{df} = 2; \quad \chi^2_{\text{obs}} = 5.36 < 5.991 \;\Rightarrow\; \text{fail to reject at 5\%.}}$$

(b) **Bracket the p-value.** The next-lower table point is $\chi^2_{0.10,2} = 4.605$, and the next-higher is $\chi^2_{0.05,2} = 5.991$. Since $4.605 < 5.36 < 5.991$, the p-value lies between 0.05 and 0.10 — roughly $\mathbb{P}(\chi^2_2 > 5.36) \approx 0.069$. This is a **marginally (non-)significant** result: it just misses the 0.05 cutoff. The verdict is fragile — a handful of reclassified employees, or a looser level $\alpha = 0.10$, would flip it to a rejection. Report it as "weak, borderline evidence of a possible association," not a clean null finding, and certainly not as "the program is unrelated to department."

(c) **What a rejection would and would not claim.** Had the statistic cleared the cutoff, a rejection would say only that wellness-program use and department are statistically **associated** — the participation rates differ across departments more than chance allows. It would *not* establish that the department *causes* the difference in uptake, nor rule out a lurking variable (workload, manager attitude, office location) driving both. A contingency table supports "related," never "caused"; causal claims require a designed experiment, not an observational cross-tabulation.

D. Tests of homogeneity

D1. **(a)** Each outlet's sample size was *fixed in advance* at 160, so the row totals are constants chosen by the manager, not the result of one random draw. Three separate populations (outlets) are sampled and the same response (size ordered) is recorded in each — this is the **test of homogeneity**, not independence. The hypotheses are

$$H_0: \text{ all three outlets have the same size distribution} \quad \text{vs.} \quad H_1: \text{ at least two outlets differ.}$$

(b) With $n = 480$, the column totals are $C = (120, 180, 180)$ and every row total is 160, so each row has identical expecteds $E_{ij} = R_i C_j / n$:

$$E_{i,\text{S}} = \tfrac{160 \cdot 120}{480} = 40,$$
$$E_{i,\text{M}} = \tfrac{160 \cdot 180}{480} = 60,$$
$$E_{i,\text{L}} = \tfrac{160 \cdot 180}{480} = 60.$$

All $E_{ij} \geq 5$. The per-cell contributions $(O_{ij} - E_{ij})^2/E_{ij}$ are

$$\begin{aligned}
\text{Downtown:} &\quad \tfrac{(40-40)^2}{40} + \tfrac{(70-60)^2}{60} + \tfrac{(50-60)^2}{60} = 0 + 1.667 + 1.667 = 3.333,\\
\text{Airport:} &\quad \tfrac{(30-40)^2}{40} + \tfrac{(50-60)^2}{60} + \tfrac{(80-60)^2}{60} = 2.500 + 1.667 + 6.667 = 10.833,\\
\text{Campus:} &\quad \tfrac{(50-40)^2}{40} + \tfrac{(60-60)^2}{60} + \tfrac{(50-60)^2}{60} = 2.500 + 0 + 1.667 = 4.167.
\end{aligned}$$

Summing and comparing on $\text{df} = (r-1)(c-1) = (3-1)(3-1) = 4$:

$$\begin{aligned}
\chi^2 &= 3.333 + 10.833 + 4.167 = 18.33,\\
\chi^2_{0.05,4} &= 9.488.
\end{aligned}$$

Since $18.33 > 9.488$, reject H_0 at the 5% level: there is significant evidence that the three outlets do *not* share one size distribution. The Airport outlet, leaning heavily to Large, contributes most of the statistic.

$$\boxed{\chi^2 = 18.33 > \chi^2_{0.05,4} = 9.488 \;\Rightarrow\; \text{reject; size mix differs by outlet.}}$$

D2. **(a)** Two counties were sampled separately at preset sizes (400 and 350) and a yes/no response was recorded, so this is a 2×2 **homogeneity** test. Column totals are $C = (470 \text{ Favor}, 280 \text{ Oppose})$, $n = 750$. Expected counts $E_{ij} = R_i C_j / n$:

$$\begin{aligned}
E_{A,\text{Fav}} &= \tfrac{400 \cdot 470}{750} = 250.67, \quad E_{A,\text{Opp}} = \tfrac{400 \cdot 280}{750} = 149.33,\\
E_{B,\text{Fav}} &= \tfrac{350 \cdot 470}{750} = 219.33, \quad E_{B,\text{Opp}} = \tfrac{350 \cdot 280}{750} = 130.67.
\end{aligned}$$

The statistic, on $\text{df} = (2-1)(2-1) = 1$:

$$\begin{aligned}
\chi^2 &= \tfrac{(260-250.67)^2}{250.67} + \tfrac{(140-149.33)^2}{149.33} + \tfrac{(210-219.33)^2}{219.33} + \tfrac{(140-130.67)^2}{130.67}\\
&= 0.347 + 0.583 + 0.397 + 0.667 = 1.99.
\end{aligned}$$

Since $\chi^2_{0.05,1} = 3.841$ and $1.99 < 3.841$, do not reject H_0: there is no significant evidence that the two counties favor the tax at different rates.

(b) Sample proportions favoring: $\widehat{p}_A = 260/400 = 0.650$ and $\widehat{p}_B = 210/350 = 0.600$; pooled $\widehat{p} = 470/750 = 0.6267$. The two-proportion z is

$$\begin{aligned}
z &= \frac{\widehat{p}_A - \widehat{p}_B}{\sqrt{\widehat{p}(1-\widehat{p})\left(\frac{1}{400} + \frac{1}{350}\right)}} = \frac{0.650 - 0.600}{\sqrt{0.6267 \cdot 0.3733\,(0.0025 + 0.002857)}}\\
&= \frac{0.050}{0.03540} = 1.412, \qquad z^2 = 1.99 = \chi^2.
\end{aligned}$$

The squared two-proportion z reproduces the 1-df homogeneity χ^2 exactly, as it must.

$$\boxed{\chi^2 = 1.99 = z^2 < \chi^2_{0.05,1} = 3.841 \;\Rightarrow\; \text{do not reject; county rates comparable.}}$$

D3. Four lakes are sampled separately at a fixed 120 each with a yes/no response (infection status), giving a 2×4 **homogeneity** table; equivalently, this is the simultaneous comparison of four infection proportions. The infected counts are $(18, 36, 24, 30)$ with totals

$$C_{\text{Inf}} = 18 + 36 + 24 + 30 = 108, \qquad C_{\text{Healthy}} = 480 - 108 = 372, \qquad n = 480.$$

Because every row total is 120, each lake has the same expecteds:

$$E_{i,\text{Inf}} = \tfrac{120 \cdot 108}{480} = 27, \qquad E_{i,\text{Healthy}} = \tfrac{120 \cdot 372}{480} = 93.$$

All $E_{ij} \geq 5$. Cell contributions (each lake supplies its Infected and Healthy term):

$$\begin{aligned}
\text{Lake 1:}\quad & \tfrac{(18-27)^2}{27} + \tfrac{(102-93)^2}{93} = 3.000 + 0.871 = 3.871,\\
\text{Lake 2:}\quad & \tfrac{(36-27)^2}{27} + \tfrac{(84-93)^2}{93} = 3.000 + 0.871 = 3.871,\\
\text{Lake 3:}\quad & \tfrac{(24-27)^2}{27} + \tfrac{(96-93)^2}{93} = 0.333 + 0.097 = 0.430,\\
\text{Lake 4:}\quad & \tfrac{(30-27)^2}{27} + \tfrac{(90-93)^2}{93} = 0.333 + 0.097 = 0.430.
\end{aligned}$$

On $\text{df} = (2-1)(4-1) = 3$:

$$\begin{aligned}
\chi^2 &= 3.871 + 3.871 + 0.430 + 0.430 = 8.60,\\
\chi^2_{0.05,3} &= 7.815.
\end{aligned}$$

Since $8.60 > 7.815$, reject H_0 at the 5% level: the four lakes do not all share the same infection rate. Lakes 1 (15% infected) and 2 (30% infected) straddle the pooled rate of $108/480 = 22.5\%$ in opposite directions and drive the rejection.

$$\boxed{\chi^2 = 8.60 > \chi^2_{0.05,3} = 7.815 \;\Rightarrow\; \text{reject; infection rates differ across lakes.}}$$

D4. The deciding question is always whether one variable's margin was *fixed by the sampler* (several samples, homogeneity) or whether a *single* sample fell freely into both margins (independence).

(a) One random sample of 600 employees, each cross-classified by department (4 levels) and training completion (2 levels) — both margins random, so **independence**. Table 4×2, $\text{df} = (4-1)(2-1) = 3$.

(b) A fixed 200 shipments from *each* of three warehouses, one response (3 levels) — three separate samples, so **homogeneity**. Table 3×3, $\text{df} = (3-1)(3-1) = 4$.

(c) One random sample of 450 patients, cross-classified by blood type (4 levels) and reaction (2 levels) — both margins random, so **independence**. Table 4×2, $\text{df} = (4-1)(2-1) = 3$.

(d) Exactly 150 subscribers from *each* of five magazines, one response (3 levels) — five separate samples, so **homogeneity**. Table 5×3, $\text{df} = (5-1)(3-1) = 8$.

$$\boxed{\text{(a) indep., df 3; (b) homog., df 4; (c) indep., df 3; (d) homog., df 8.}}$$

D5. Participants were assigned to three diet plans in fixed numbers (100 each) and the same outcome was recorded, so the plans are the "populations" and this is a **homogeneity** test across $r = 3$, $c = 3$. Column totals $C = (120, 120, 60)$, $n = 300$. Each row total is 100, so every row shares the expecteds

$$E_{i,\text{Lost}} = \tfrac{100\cdot 120}{300} = 40, \quad E_{i,\text{NC}} = \tfrac{100\cdot 120}{300} = 40, \quad E_{i,\text{Gain}} = \tfrac{100\cdot 60}{300} = 20.$$

All $E_{ij} \geq 5$. Cell contributions:

$$\begin{aligned}
\text{Plan 1:}\quad & \tfrac{(55-40)^2}{40} + \tfrac{(30-40)^2}{40} + \tfrac{(15-20)^2}{20} = 5.625 + 2.500 + 1.250 = 9.375,\\
\text{Plan 2:}\quad & \tfrac{(40-40)^2}{40} + \tfrac{(40-40)^2}{40} + \tfrac{(20-20)^2}{20} = 0 + 0 + 0 = 0,\\
\text{Plan 3:}\quad & \tfrac{(25-40)^2}{40} + \tfrac{(50-40)^2}{40} + \tfrac{(25-20)^2}{20} = 5.625 + 2.500 + 1.250 = 9.375.
\end{aligned}$$

On $\text{df} = (3-1)(3-1) = 4$:

$$\begin{aligned}
\chi^2 &= 9.375 + 0 + 9.375 = 18.75,\\
\chi^2_{0.01,4} &= 13.277.
\end{aligned}$$

Since $18.75 > 13.277$, reject H_0 at the 1% level: the outcome distribution is not the same across the three plans. Plan 1 (skewed to weight loss) and Plan 3 (skewed to no change / gain) each contribute 9.375 and drive the rejection, while Plan 2 sits exactly on the pooled profile and contributes nothing.

$$\boxed{\chi^2 = 18.75 > \chi^2_{0.01,4} = 13.277 \;\Rightarrow\; \text{reject; outcomes differ by diet plan.}}$$

D6. The shared computation: with $C = (144, 96)$ and $n = 240$, the expected counts are $E_{11} = 120 \cdot 144/240 = 72$, $E_{12} = 120 \cdot 96/240 = 48$, $E_{21} = 72$, $E_{22} = 48$, so

$$\begin{aligned}\chi^2 &= \tfrac{(84-72)^2}{72} + \tfrac{(36-48)^2}{48} + \tfrac{(60-72)^2}{72} + \tfrac{(60-48)^2}{48} \\ &= 2.000 + 3.000 + 2.000 + 3.000 = 10.00,\end{aligned}$$

on df $= (2-1)(2-1) = 1$, with $\chi^2_{0.05,1} = 3.841$; since $10.00 > 3.841$, both studies reject.

(a) *Study I, homogeneity.* The trainer set the two group sizes (120 and 120) in advance, so two separate populations are compared. The conclusion is about **equality of proportions across populations**: reject H_0 — the pass rate differs between the two teaching methods ($84/120 = 70\%$ for method 1 vs. $60/120 = 50\%$ for method 2).

(b) *Study II, independence.* One sample of 240 adults was cross-classified by two yes/no traits, so both margins are random. The conclusion is about **association between two variables**: reject H_0 — weekly exercise and adequate sleep are associated in this population.

(c) The arithmetic is identical: same expecteds, same $\chi^2 = 10.00$, same df $= 1$, and both reject at $\alpha = 0.05$. The written conclusions differ because the *designs* differ: fixing the group sizes (a) licenses a statement comparing two populations' proportions, whereas a single cross-classified sample (b) licenses only a statement that two traits are related — the χ^2 value is blind to which design produced the table.

Identical $\chi^2 = 10.00$, df 1; (a) proportions differ across groups, (b) variables are associated.

D7. Two hospitals are sampled separately at a fixed 80 charts each, with one response (route, 4 levels), so this is a 2×4 **homogeneity** table, $r = 2$, $c = 4$.

Degrees of freedom. Each of the $r = 2$ rows is an independent multinomial over $c = 4$ categories; once $c - 1 = 3$ counts in a row are set, the fourth is forced by the fixed row total 80, giving $r(c-1) = 2 \cdot 3 = 6$ free counts in the table body. Under H_0 we estimate the common route distribution, costing $c - 1 = 3$ parameters. Subtracting,

$$\text{df} = r(c-1) - (c-1) = (r-1)(c-1) = (2-1)(4-1) = 3,$$

not $rc - 1 = 8 - 1 = 7$; using $rc - 1$ would ignore that the row totals were fixed by design and that estimating the common distribution costs $c - 1$.

Expected counts. Column totals $C = (56, 40, 44, 20)$, $n = 160$; each row total is 80, so both hospitals share

$$E_{\text{ER}} = \tfrac{80 \cdot 56}{160} = 28,\ E_{\text{Ref}} = \tfrac{80 \cdot 40}{160} = 20,\ E_{\text{WI}} = \tfrac{80 \cdot 44}{160} = 22,\ E_{\text{Tr}} = \tfrac{80 \cdot 20}{160} = 10.$$

All $E_{ij} \geq 5$. Cell contributions (both rows have identical expecteds; deviations are equal in magnitude):

$$\begin{aligned}\text{North:}\ & \tfrac{(32-28)^2}{28} + \tfrac{(24-20)^2}{20} + \tfrac{(16-22)^2}{22} + \tfrac{(8-10)^2}{10} = 0.571 + 0.800 + 1.636 + 0.400 = 3.408, \\ \text{South:}\ & \tfrac{(24-28)^2}{28} + \tfrac{(16-20)^2}{20} + \tfrac{(28-22)^2}{22} + \tfrac{(12-10)^2}{10} = 0.571 + 0.800 + 1.636 + 0.400 = 3.408.\end{aligned}$$

Statistic and decision. On df $= 3$:

$$\begin{aligned}\chi^2 &= 3.408 + 3.408 = 6.82, \\ \chi^2_{0.05,3} &= 7.815.\end{aligned}$$

Since $6.82 < 7.815$, do not reject H_0 at the 5% level: there is no significant evidence that the two hospitals draw their admissions from different route-distributions, although South leans more toward Walk-in and North toward Referral, the gap is within sampling noise.

$\chi^2 = 6.82 < \chi^2_{0.05,3} = 7.815 \ \Rightarrow$ do not reject; common admission-route mix.

E. Conditions, small counts, pitfalls

E1. Method. This is a validity check, not the test: compute every $E_i = np_i$ and compare to the rule of thumb. The condition is on the expected counts, never the observed.

Expected counts. With $n = 150$,

$$\begin{aligned} E_i &= 150 \cdot (0.45,\ 0.28,\ 0.14,\ 0.09,\ 0.04) \\ &= (67.5,\ 42.0,\ 21.0,\ 13.5,\ 6.0). \end{aligned}$$

Apply the rule. Every $E_i \geq 5$: the smallest is $E_i = 6.0$, still above the bar. The strict "all $E_i \geq 5$" rule is satisfied outright, so the permissive rule is irrelevant; no pooling is needed.

Verdict and df. The test may proceed. For a fully specified GOF with $k = 5$ categories, $\text{df} = k - 1 = 4$.

$$\boxed{\text{All five } E_i \geq 5 \ (\min = 6.0) \ \Rightarrow \ \text{strict rule met; proceed with df} = 4.}$$

E2. Method. Some expected counts fall below 5; check the rule, pool the offending adjacent cells, recompute df from the reduced cell count, then compute the statistic.

(a) Failure and pooling. The six expected counts are $(28.0, 20.0, 16.0, 8.0, 4.8, 3.2)$. Cells 5 and 6 have $E_i = 4.8$ and 3.2, both below 5 (2 of 6 cells = 33% > 20%), so the rule fails. Merge cells 5 and 6 into one combined "rare" category by adding counts:

$$\begin{aligned} E_i^{\text{pooled}} &= 4.8 + 3.2 = 8.0, \\ O_i^{\text{pooled}} &= 5 + 3 = 8. \end{aligned}$$

The surviving five cells have $E_i = (28.0,\ 20.0,\ 16.0,\ 8.0,\ 8.0)$, every one ≥ 5, with $O_i = (30,\ 18,\ 17,\ 7,\ 8)$.

(b) Degrees of freedom. Fully specified GOF, df = (cells -1):

$$\begin{aligned} \text{before: } \text{df} &= 6 - 1 = 5, \\ \text{after: } \text{df} &= 5 - 1 = 4. \end{aligned}$$

One merge of two cells into one costs exactly one df.

(c) Statistic and decision.

$$\begin{aligned} \chi^2_{\text{obs}} &= \frac{(30-28)^2}{28} + \frac{(18-20)^2}{20} + \frac{(17-16)^2}{16} \\ &\quad + \frac{(7-8)^2}{8} + \frac{(8-8)^2}{8} \\ &= 0.143 + 0.200 + 0.063 + 0.125 + 0.000 = 0.530. \end{aligned}$$

With $\text{df} = 4$ and $\chi^2_{0.05,\,4} = 9.488$, we have $0.53 \ll 9.488$.

$$\boxed{\text{Pooled to 5 cells, df} = 4, \ \chi^2_{\text{obs}} = 0.53 < 9.488 \Rightarrow \text{do not reject.}}$$

At the 5% level there is no significant evidence against the hypothesized proportions; the data are compatible with the model.

E3. Method. One sample of subjects cross-classified by two binary factors (plan × goal) is a 2×2 test on 1 df; the modest counts make Yates's correction appropriate, so we compute both.

(a) Table and expected counts.

	Met	Not met	Total
A	6	13	19
B	14	7	21
Total	20	20	40

Expected counts (row total × column total / n, $n = 40$):

$$E_{\text{A,Met}} = \frac{19 \cdot 20}{40} = 9.5, \quad E_{\text{A,Not}} = \frac{19 \cdot 20}{40} = 9.5,$$
$$E_{\text{B,Met}} = \frac{21 \cdot 20}{40} = 10.5, \quad E_{\text{B,Not}} = \frac{21 \cdot 20}{40} = 10.5.$$

Every cell has the same discrepancy $|O_i - E_i| = 3.5$.

(b) Uncorrected statistic.

$$\begin{aligned}\chi^2_{\text{obs}} &= \frac{3.5^2}{9.5} + \frac{3.5^2}{9.5} + \frac{3.5^2}{10.5} + \frac{3.5^2}{10.5} \\ &= 1.289 + 1.289 + 1.167 + 1.167 = 4.912.\end{aligned}$$

(c) Yates-corrected statistic. Replace each $|O_i - E_i| = 3.5$ by $3.5 - 0.5 = 3.0$:

$$\begin{aligned}\chi^2_{\text{Yates}} &= \frac{3.0^2}{9.5} + \frac{3.0^2}{9.5} + \frac{3.0^2}{10.5} + \frac{3.0^2}{10.5} \\ &= 0.947 + 0.947 + 0.857 + 0.857 = 3.609.\end{aligned}$$

(d) Decisions. The cutoff is $\chi^2_{0.05,\,1} = 3.841$.

$$\boxed{\chi^2_{\text{obs}} = 4.91 > 3.841 \text{ (reject)}; \quad \chi^2_{\text{Yates}} = 3.61 < 3.841 \text{ (do not reject).}}$$

The two disagree because the uncorrected statistic sits just above the cutoff and the conservative $\frac{1}{2}$ correction pulls it below. With counts this modest the correction is decisive: the uncorrected test's tendency to over-reject in small 2×2 tables is exactly the bias Yates is meant to remove, so the corrected "do not reject" is the more trustworthy reading.

E4. Method. The χ^2 formula is defined on raw counts; percentages have n divided out, so the look-alike calculation is invalid. Reconstruct the counts and recompute.

(a) The wrong calculation. Using $48, 30, 22$ as "O_i" and $40, 35, 25$ as "E_i":

$$\begin{aligned}\chi^2_{\text{wrong}} &= \frac{(48-40)^2}{40} + \frac{(30-35)^2}{35} + \frac{(22-25)^2}{25} \\ &= 1.600 + 0.714 + 0.360 = 2.674.\end{aligned}$$

Because the percentages sum to 100, this number secretly assumes $n = 100$ and cannot reflect the real sample's evidence.

(b) Correct counts and statistic. With $n = 600$,

$$O_i = (0.48,\ 0.30,\ 0.22) \cdot 600 = (288,\ 180,\ 132),$$
$$E_i = (0.40,\ 0.35,\ 0.25) \cdot 600 = (240,\ 210,\ 150).$$

$$\begin{aligned}\chi^2_{\text{obs}} &= \frac{(288-240)^2}{240} + \frac{(180-210)^2}{210} + \frac{(132-150)^2}{150} \\ &= \frac{2304}{240} + \frac{900}{210} + \frac{324}{150} \\ &= 9.600 + 4.286 + 2.160 = 16.046.\end{aligned}$$

(c) Decisions and factor. With $\text{df} = 3 - 1 = 2$ and $\chi^2_{0.05,\,2} = 5.991$:

$$\boxed{\chi^2_{\text{wrong}} = 2.67\ (< 5.991,\ \text{not sig.}); \quad \chi^2_{\text{obs}} = 16.05 > 5.991 \Rightarrow \text{reject.}}$$

The correct statistic is exactly $n/100 = 600/100 = 6$ times the wrong one ($2.674 \times 6 = 16.05$), since χ^2 scales linearly with n. Using percentages erases that factor of 6 and flips the conclusion from "no evidence" to a clear rejection of the hypothesized split.

E5. Method. This is a goodness-of-fit to a fully specified uniform ($p_i = 1/7$), df $= 6$; with n in the millions the lesson is to judge practical importance via an effect size separately from significance.

(a) Statistic via the scaling form. Using $\chi^2_{\text{obs}} = n\sum_i(\hat{p}_i - p_i)^2/p_i$ with $\sum_i(\hat{p}_i - p_i)^2/p_i = 4.0 \times 10^{-6}$:

$$\begin{aligned}\chi^2_{\text{obs}} &= 3{,}500{,}000 \cdot (4.0 \times 10^{-6}) \\ &= 14.0.\end{aligned}$$

(b) Significance. With df $= 6$ and $\chi^2_{0.05,\,6} = 12.592$, we have $14.0 > 12.592$, so the test rejects uniformity at the 5% level.

(c) Effect size. A scale-free summary is

$$\sqrt{\chi^2_{\text{obs}}/n} = \sqrt{14.0/3{,}500{,}000} = \sqrt{4.0 \times 10^{-6}} = 0.002,$$

about two-tenths of one percent.

Significant ($\chi^2_{\text{obs}} = 14.0 > 12.59$) yet effect ≈ 0.002; statistically real, practically nil.

The departure from uniform is real but minute: with 3.5 million sign-ups the test detects a deviation no decision would turn on, so the p-value must be paired with the effect size and significance must not be read as importance.

E6. Method. This tests what a rejected GOF establishes and the one-tailed nature of the test.

(a) The rejection of Poisson. $\chi^2_{\text{obs}} = 18.7 > \chi^2_{0.05,\,5} = 11.070$ rejects the Poisson fit at $\alpha = 0.05$: the data are inconsistent with a Poisson distribution. It establishes only that; the alternative to "Poisson" is simply "not Poisson," an enormous family. Nothing in the statistic singles out the geometric model (or any other), so the colleague's conclusion is *not* warranted. To support a geometric one must fit and test that model directly and find it not rejected.

Rejecting Poisson $\neq$ confirming geometric; the test names no winner.

(b) The small statistic. A goodness-of-fit test rejects only in the upper tail, since only a *large* χ^2 signals disagreement. For df $= 4$ at $\alpha = 0.05$ the only cutoff is $\chi^2_{0.05,\,4} = 9.488$, and we reject solely when $\chi^2_{\text{obs}} > 9.488$. Here $\chi^2_{\text{obs}} = 0.21 \ll 9.488$, deep in the lower tail and far from the single upper-tail rejection region, so we do not reject; the data are highly compatible with the hypothesized split.

$\chi^2_{\text{obs}} = 0.21 < 9.488 \Rightarrow$ do not reject; a small χ^2 supports, never rejects.

A suspiciously tiny χ^2 may warrant a separate data-integrity check for fabricated or rounded data, but that is not the goodness-of-fit alternative, which lives entirely in the upper tail.

E7. Method. Read the sampling design — one sample cross-classified by two variables is independence; one sample versus a specified distribution is goodness-of-fit (subtract one more df per estimated parameter); several separately drawn samples compared on one variable is homogeneity.

(a) Independence. One sample, two categorical variables ($r = 3$ smoking levels, $c = 4$ diagnoses), cross-classified. Contingency-table test of independence,

$$\text{df} = (r-1)(c-1) = (3-1)(4-1) = 6.$$

(b) Homogeneity. Three *separate* samples (the shift group sizes were fixed at 90 each by design) compared on one variable (commute mode, 4 categories). Same arithmetic as independence,

$$\text{df} = (r-1)(c-1) = (3-1)(4-1) = 6.$$

(c) Goodness-of-fit, parameters known. One variable (die face) against a fully specified uniform on $k = 4$ faces; no parameters estimated,

$$\text{df} = k - 1 = 4 - 1 = 3.$$

(d) Goodness-of-fit, one parameter estimated. One variable (arrivals per window) against a Poisson whose rate λ is estimated from the data, $m = 1$, with $k = 8$ usable cells,

$$\text{df} = k - 1 - m = 8 - 1 - 1 = 6.$$

$$\boxed{(a)\ \text{indep., df} = 6;\quad (b)\ \text{homog., df} = 6;\quad (c)\ \text{GOF, df} = 3;\quad (d)\ \text{GOF, df} = 6.}$$

Scenarios (a) and (b) build identical-looking 3×4 tables with the same df; only the design (one sample cross-classified versus three fixed-size samples compared) distinguishes them, and the conclusion's wording follows the design.

F. Count-data synthesis

F1. (a) Which test. There is *one* sample, classified on *one* categorical variable (tier), and the null pins down a *fully specified* probability list $p = (0.40, 0.30, 0.20, 0.10)$ with nothing to estimate. That is the goodness-of-fit test with parameters known; refer χ^2 to χ^2_{k-1} with $k = 4$ categories. It is not an independence or homogeneity test — those need two variables or several samples.

(b) The test. Hypotheses: H_0: $p = (0.40, 0.30, 0.20, 0.10)$ versus H_1: the tier split differs from this list. Expected counts $E_i = np_i$ with $n = 500$:

$$E_i = (\,500(.40),\ 500(.30),\ 500(.20),\ 500(.10)\,) = (\,200,\ 150,\ 100,\ 50\,),$$

all ≥ 5, so the approximation is valid. Cell contributions:

Tier	Free	Basic	Standard	Premium	Total
O_i	182	166	98	54	500
E_i	200	150	100	50	500
$(O_i - E_i)^2/E_i$	1.620	1.707	0.040	0.320	3.687

$$\chi^2 = \frac{(182-200)^2}{200} + \frac{(166-150)^2}{150} + \frac{(98-100)^2}{100} + \frac{(54-50)^2}{50}$$
$$= 1.620 + 1.707 + 0.040 + 0.320 = 3.687.$$

Degrees of freedom $= k - 1 = 4 - 1 = 3$, and $\chi^2_{0.05,\,3} = 7.815$. Since $3.687 < 7.815$, do not reject.

$$\boxed{\chi^2 = 3.687 < \chi^2_{0.05,\,3} = 7.815 \ \Rightarrow\ \text{fail to reject; the tier split fits the claim.}}$$

The Basic tier is the heaviest cell (1.71) but nowhere near enough to overturn the claim.

F2. (a) Which test, and the rate. The claim names a *family* (Poisson) but supplies no rate, so we must estimate λ from these data and run a goodness-of-fit test with the df reduced by one estimated parameter. The maximum-likelihood estimate of a Poisson rate is the sample mean. Using 5 as the representative value of the "≥ 5" cell,

$$\text{total outages} = 0(60) + 1(62) + 2(38) + 3(14) + 4(4) + 5(2) = 206, \qquad \widehat{\lambda} = \frac{206}{180} = 1.144 \approx 1.14.$$

(b) Fitted probabilities and pooling. With $\widehat{p}_x = e^{-1.14}1.14^x/x!$ and $e^{-1.14} = 0.31982$, and $E_i = 180\,\widehat{p}_x$ (the ≥ 5 cell takes the remaining tail probability):

Outages x	0	1	2	3	4	≥ 5
$\widehat{p}_x$	.3198	.3646	.2078	.0790	.0225	.0063
E_i	57.57	65.63	37.41	14.21	4.05	1.13
O_i	60	62	38	14	4	2

The two tail cells have $E_i = 4.05 < 5$ and $E_i = 1.13 < 5$; pool them into a single "≥ 4" cell:

$$O_i^{(\geq 4)} = 4 + 2 = 6, \qquad E_i^{(\geq 4)} = 4.05 + 1.13 = 5.18 \geq 5.$$

The pooled categories are $0, 1, 2, 3, \geq 4$, so $k = 5$ (it was 6 before pooling).

(c) df and decision. Cell contributions on the pooled table:

$$\begin{aligned}\chi^2 &= \frac{(60-57.57)^2}{57.57} + \frac{(62-65.63)^2}{65.63} + \frac{(38-37.41)^2}{37.41} + \frac{(14-14.21)^2}{14.21} + \frac{(6-5.18)^2}{5.18} \\ &= 0.103 + 0.200 + 0.009 + 0.003 + 0.130 = 0.444.\end{aligned}$$

Count k *after* pooling: $k = 5$, one parameter (λ) estimated, so

$$\text{df} = k - 1 - m = 5 - 1 - 1 = 3, \qquad \chi^2_{0.05,\,3} = 7.815.$$

Since $0.444 < 7.815$, do not reject.

$\boxed{\chi^2 = 0.444 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{fail to reject; the Poisson model fits the outage counts.}}$

Had the rate been *quoted in advance* (known, $m = 0$) the df would have been $k - 1 = 4$ and the cutoff $\chi^2_{0.05,\,4} = 9.488$; estimating λ from the same data costs exactly one df and pulls the cutoff down to 7.815.

F3. (a) Estimate p. Each packet has 5 seeds, so the trials total $5 \times 160 = 800$. Germinated seeds:

$$0(2) + 1(13) + 2(34) + 3(46) + 4(45) + 5(20) = 499, \qquad \widehat{p} = \frac{499}{800} = 0.6238 \approx 0.624.$$

This is a binomial goodness-of-fit: the number of trials per packet (5) is known, but the per-seed probability p is estimated, costing one degree of freedom.

(b) Fitted probabilities and pooling. With $\widehat{p}_x = \binom{5}{x}(0.624)^x(0.376)^{5-x}$ and $E_i = 160\,\widehat{p}_x$:

Germinated x	0	1	2	3	4	5
$\widehat{p}_x$	.0075	.0624	.2070	.3435	.2850	.0946
E_i	1.20	9.98	33.12	54.96	45.61	15.14
O_i	2	13	34	46	45	20

The $x = 0$ cell has $E_i = 1.20 < 5$; pool it with $x = 1$ into a "≤ 1" cell:

$$O_i^{(\leq 1)} = 2 + 13 = 15, \qquad E_i^{(\leq 1)} = 1.20 + 9.98 = 11.18 \geq 5.$$

The surviving categories are $\leq 1,\ 2,\ 3,\ 4,\ 5$, so $k = 5$.

(c) df from scratch and decision. Contributions on the pooled table:

$$\begin{aligned}\chi^2 &= \frac{(15-11.18)^2}{11.18} + \frac{(34-33.12)^2}{33.12} + \frac{(46-54.96)^2}{54.96} + \frac{(45-45.61)^2}{45.61} + \frac{(20-15.14)^2}{15.14} \\ &= 1.305 + 0.023 + 1.461 + 0.008 + 1.560 = 4.360.\end{aligned}$$

Counting df: 5 cells after pooling, minus 1 for the constraint $\sum O_i = n$, minus 1 for the estimated p:

$$\text{df} = k - 1 - m = 5 - 1 - 1 = 3, \qquad \chi^2_{0.05,\,3} = 7.815.$$

Since $4.360 < 7.815$, do not reject.

$$\boxed{\chi^2 = 4.360 < \chi^2_{0.05,\,3} = 7.815 \;\Rightarrow\; \text{fail to reject; Binomial}(5, \widehat{p}) \text{ fits the germination counts.}}$$

(d) If $p = 0.60$ were specified. Then no parameter is estimated, $m = 0$, and the df would be $k - 1 = 5 - 1 = 4$ (using the same pooled five cells), with cutoff $\chi^2_{0.05,\,4} = 9.488$. The df is one *larger* than in (c) because a probability fixed in advance does not have to be paid for out of the data: estimating p slid the E_i toward the O_i, shrinking χ^2, and the matching correction is to drop one degree of freedom. Same statistic family, different df — driven entirely by whether p was estimated or told.

F4. (a) Which test for each study. Read the *sampling design*, not the table. In Study X a *single* random sample of 400 is cross-classified on two variables (membership and layout); both margins are random, so it is a **test of independence**. In Study Y the two region group sizes (200 and 200) were *fixed by the mailer* before any response, so two separate populations are compared on one response (layout); that is a **test of homogeneity**.

(b) Identical arithmetic. Both use $E_{ij} = R_i C_j / n$ with $n = 400$. Every row total is 200 and the column totals are $150, 150, 100$, so

$$E_{\cdot\mathrm{A}} = \frac{200 \cdot 150}{400} = 75, \quad E_{\cdot\mathrm{B}} = \frac{200 \cdot 150}{400} = 75, \quad E_{\cdot\mathrm{C}} = \frac{200 \cdot 100}{400} = 50,$$

the same for both rows. Cell contributions $(O_{ij} - E_{ij})^2 / E_{ij}$:

$$\text{group 1:} \quad \frac{(88-75)^2}{75} + \frac{(72-75)^2}{75} + \frac{(40-50)^2}{50} = 2.253 + 0.120 + 2.000 = 4.373,$$
$$\text{group 2:} \quad \frac{(62-75)^2}{75} + \frac{(78-75)^2}{75} + \frac{(60-50)^2}{50} = 2.253 + 0.120 + 2.000 = 4.373.$$

$$\chi^2_{\text{obs}} = 4.373 + 4.373 = 8.747, \qquad \text{df} = (r-1)(c-1) = (2-1)(3-1) = 2.$$

The expected counts, the statistic 8.747, and the df = 2 are byte-for-byte identical for the two studies.

(c) Decision and the two conclusions. On 2 df, $\chi^2_{0.05,\,2} = 5.991$, and $8.747 > 5.991$, so both studies reject H_0 (equivalently the single-tail p-value is $\mathbb{P}(\chi^2_2 > 8.747) = e^{-8.747/2} = 0.013 < 0.05$).

$$\boxed{\chi^2_{\text{obs}} = 8.747 > \chi^2_{0.05,\,2} = 5.991 \;\Rightarrow\; \text{reject in both studies; but the claims differ.}}$$

Study X (independence): membership status and layout preference are **associated** in this population. *Study Y (homogeneity):* the two regions do **not** share a common layout-preference distribution. Same χ^2_{obs}, same df, same reject decision — yet "two traits are related" and "two populations differ" are different scientific statements, each licensed only by its own design.

F5. Which test. A *fixed* sample of 150 was drawn from *each* of three hospitals and the same response (experience) recorded; the row totals were set by the sampler, so this is a **test of homogeneity** across $r = 3$ populations and $c = 3$ categories. Same statistic as independence, df $= (r-1)(c-1)$.

Step 1: expected counts. Each $R_i = 150$, $n = 450$, so every row has the same expecteds:

$$E_{i,\mathrm{Sat}} = \frac{150 \cdot 234}{450} = 78, \quad E_{i,\mathrm{Neu}} = \frac{150 \cdot 159}{450} = 53, \quad E_{i,\mathrm{Dis}} = \frac{150 \cdot 57}{450} = 19,$$

all ≥ 5, so the approximation is valid.

Step 2: cell contributions $(O_{ij} - E_{ij})^2 / E_{ij}$, row by row.

$$\text{North:} \quad \frac{(96-78)^2}{78} + \frac{(39-53)^2}{53} + \frac{(15-19)^2}{19} = 4.154 + 3.698 + 0.842 = 8.694,$$
$$\text{Central:} \quad \frac{(78-78)^2}{78} + \frac{(54-53)^2}{53} + \frac{(18-19)^2}{19} = 0.000 + 0.019 + 0.053 = 0.072,$$
$$\text{South:} \quad \frac{(60-78)^2}{78} + \frac{(66-53)^2}{53} + \frac{(24-19)^2}{19} = 4.154 + 3.189 + 1.316 = 8.658.$$

Step 3: statistic, df, decision.

$$\chi^2_{\text{obs}} = 8.694 + 0.072 + 8.658 = 17.424, \qquad \text{df} = (3-1)(3-1) = 4,$$

$$\chi^2_{0.05,\,4} = 9.488.$$

Since $17.424 > 9.488$, reject.

$$\boxed{\chi^2_{\text{obs}} = 17.424 > \chi^2_{0.05,\,4} = 9.488 \Rightarrow \text{reject } H_0\text{; the hospitals' experience mixes differ.}}$$

North contributes most (8.69 of the 17.42): it is far more Satisfied and far less Neutral/Dissatisfied than the pooled profile, while Central sits almost exactly on it.

F6. Which test (and why not homogeneity). *One* random sample of 400 customers is cross-classified on *two* variables (age band, device); neither margin was fixed in advance, so this is a **test of independence**, not homogeneity. (Homogeneity would require a pre-set sample size drawn separately from each age band; here the age-band totals $130, 150, 120$ fell out of the single sample.) $\text{df} = (r-1)(c-1)$.

Step 1: expected counts $E_{ij} = R_i C_j / 400$. Each entry is its row total times its column total over $n = 400$:

E_{ij}	phone	tablet	desktop
under 30 ($R = 130$)	48.75	47.125	34.125
30–49 ($R = 150$)	56.25	54.375	39.375
50+ ($R = 120$)	45.000	43.500	31.500

The smallest expected count is $31.5 \geq 5$, so the test is valid.

Step 2: cell contributions, row by row.

$$\text{under 30}: \frac{(60-48.75)^2}{48.75} + \frac{(40-47.125)^2}{47.125} + \frac{(30-34.125)^2}{34.125} = 2.596 + 1.077 + 0.499 = 4.172,$$

$$\text{30–49}: \frac{(50-56.25)^2}{56.25} + \frac{(55-54.375)^2}{54.375} + \frac{(45-39.375)^2}{39.375} = 0.694 + 0.007 + 0.804 = 1.505,$$

$$\text{50+}: \frac{(40-45)^2}{45} + \frac{(50-43.5)^2}{43.5} + \frac{(30-31.5)^2}{31.5} = 0.556 + 0.971 + 0.071 = 1.598.$$

Step 3: statistic, df, decision.

$$\chi^2_{\text{obs}} = 4.172 + 1.505 + 1.598 = 7.276, \qquad \text{df} = (3-1)(3-1) = 4, \qquad \chi^2_{0.05,\,4} = 9.488.$$

Since $7.276 < 9.488$, do not reject.

$$\boxed{\chi^2_{\text{obs}} = 7.276 < \chi^2_{0.05,\,4} = 9.488 \;\Rightarrow\; \text{fail to reject; no evidence of association.}}$$

"Fail to reject" means the data are consistent with independence, not that independence is proven; the largest single contribution (the under-30 phone cell, 2.60) hints at a lean toward phones among younger buyers, but it is within sampling noise here.

F7. The deciding question is always the **design**: one sample against a specified distribution ⇒ goodness-of-fit (subtract one more df per *estimated* parameter); one sample cross-classified on two variables ⇒ independence; several *separately drawn* samples compared on one variable ⇒ homogeneity. The two contingency-table tests both use $\text{df} = (r-1)(c-1)$.

(a) One sample of 600 cross-classified by sex (2) and exercise (2); both margins random ⇒ **independence**, table 2×2,

$$\text{df} = (2-1)(2-1) = 1.$$

(b) One variable (die face), fully specified uniform $p_i = 1/6$, nothing estimated $\Rightarrow$ **goodness-of-fit, parameters known**, $k = 6$,

$$\mathrm{df} = k - 1 = 6 - 1 = 5.$$

(c) A *fixed* 250 drawn from *each* of four counties, one response (candidate, 3 levels) $\Rightarrow$ **homogeneity**, table 4×3,

$$\mathrm{df} = (r-1)(c-1) = (4-1)(3-1) = 6.$$

(d) One variable (pairs per plot) against a Poisson whose rate is *estimated*, $m = 1$; after pooling, $k = 7$ usable cells $\Rightarrow$ **goodness-of-fit, one estimated parameter**,

$$\mathrm{df} = k - 1 - m = 7 - 1 - 1 = 5.$$

(e) One variable (reaction time) grouped into $k = 7$ intervals against a Normal with *both* μ and σ estimated, $m = 2 \Rightarrow$ **goodness-of-fit, two estimated parameters**,

$$\mathrm{df} = k - 1 - m = 7 - 1 - 2 = 4.$$

$\boxed{(a)\ \text{indep., df 1};\ (b)\ \text{GOF, df 5};\ (c)\ \text{homog., df 6};\ (d)\ \text{GOF, df 5};\ (e)\ \text{GOF, df 4.}}$

Cases (a) and (c) build similar-looking two-way tables but differ in design — one sample cross-classified versus several fixed-size samples compared — and cases (b), (d), (e) are all goodness-of-fit yet carry different df purely because of how many parameters were estimated.

Chapter 11

Regression

Regression asks how one quantitative variable moves with another: given pairs (x_i, y_i), fit a straight line $y = \beta_0 + \beta_1 x$ and then reason about the line's slope, its predictions, and the strength of the relationship. The mechanics are **least squares** — choose the line minimizing the summed squared vertical gaps — but the inference rests on a **model**: $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with independent $\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$. From that model every familiar tool returns: the slope estimate $\widehat{\beta}_1$ is normal, so tests and confidence intervals for β_1 use a t; the fitted line predicts a **mean response** (narrow interval) or a **new observation** (wider prediction interval); the **correlation** r measures linear strength; and an **ANOVA decomposition** splits the total variation into explained and residual parts, giving R^2 and an F-test.

Master guide: which regression quantity does the problem call for?

- Asked to **fit the line** or compute the slope/intercept from data? ⇒ **least squares**: $\widehat{\beta}_1 = S_{xy}/S_{xx}$, $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \bar{x}$ (§1).
- Asked about the **error variance**, residuals, or the standard errors of the estimates? ⇒ estimate σ^2 by $s^2 = \mathrm{SSE}/(n-2)$ and the estimator properties (§2).
- Asked to **test or bound the slope** β_1 (is there a real linear effect?)? ⇒ the **t-test / t-interval for** β_1 on $n-2$ df (§3).
- Asked to **predict** y at a given x? ⇒ a **confidence interval for the mean response** $\beta_0 + \beta_1 x_0$, or a wider **prediction interval** for a new Y at x_0 (§4).
- Asked about the **strength or significance of the association**? ⇒ the **correlation coefficient** r and its test (§5).
- Asked to **decompose the variation**, for R^2 or an overall F-test? ⇒ the **regression ANOVA** $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ (§6).

Keep straight: a *confidence* interval for the mean response at x_0 is narrower than a *prediction* interval for a single new Y at x_0 (the latter carries the extra σ^2 of one observation). Correlation measures *linear* association only, is unitless and symmetric in x, y, and — like the slope — says nothing about causation.

11.1 The Simple Linear Regression Model and Least Squares

Simple linear regression begins with n paired measurements $(x_1, y_1), \ldots, (x_n, y_n)$ and a belief that, on average, y rises or falls *linearly* with x. The working assumption is a **model**: $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$, where the x_i are fixed known numbers, the line $\beta_0 + \beta_1 x$ is the mean response, and the scatter about that line is random noise ε_i drawn independently from $\mathrm{N}(0, \sigma^2)$. Two separate jobs follow. The first, this section's subject, is purely *descriptive*: find the single straight line that best fits the cloud of points, with no probability needed. "Best" is made precise by **least squares** — choose the slope $\widehat{\beta}_1$ and intercept $\widehat{\beta}_0$ that minimize the summed squared vertical gaps between the points and the line. The minimizers come out in closed form, $\widehat{\beta}_1 = S_{xy}/S_{xx}$ and

$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}$, and from them flow fitted values, residuals, and an interpretation of the slope in the units of the problem. Recognize that you are in this section, and not a later one, whenever the task is to *compute or interpret the line itself* — the slope, the intercept, a fitted value, a residual — rather than to attach a standard error, a test, or an interval, all of which need the noise model and wait for §2–§4.

Which regression quantity does the problem call for here?

- Asked to **fit the line** to data, or to report the slope or intercept? ⇒ **least squares**: $\widehat{\beta}_1 = S_{xy}/S_{xx}$, then $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}$.
- Asked for a **fitted value** (the height of the line) at some x_0? ⇒ plug in: $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$. This is a point on the line, not yet an interval.
- Asked for a **residual** or to *check* the fit? ⇒ $e_i = y_i - \widehat{y}_i$; the residuals sum to zero and the line passes through $(\overline{X}, \overline{Y})$.
- Asked to **interpret** the slope or intercept in context? ⇒ slope = change in y per one-unit rise in x (in y-units per x-unit); intercept = fitted y at $x = 0$, meaningful only if $x = 0$ is inside or near the data range.

Do not confuse: fitting the line (this section, no probability) with *testing* whether the slope is real or *predicting* a future y with an interval (§3–§4, which need the $\mathrm{N}(0, \sigma^2)$ errors). And a slope is not a cause: a strong, well-fit line quantifies association, never proves that moving x *makes* y change.

11.1.1 The model and what least squares minimizes

Definition 11.1.1 (Simple linear regression model)**.** For fixed predictor values $x_1, \ldots, x_n$, the **simple linear regression model** states

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \qquad i = 1, \ldots, n,$$

where $\varepsilon_1, \ldots, \varepsilon_n$ are independent with $\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$. Equivalently the Y_i are independent with $\mathbb{E}[Y_i] = \beta_0 + \beta_1 x_i$ and $\mathrm{Var}(Y_i) = \sigma^2$. The line $y = \beta_0 + \beta_1 x$ is the **population regression line**; β_1 is its slope and β_0 its intercept, and σ^2 is the common error variance.

Remark. Three things are assumed and worth naming, because later inference rests on them: *linearity* (the mean of Y is a straight-line function of x), *constant variance* (the spread σ^2 does not change with x), and *independent normal errors*. The fitting in this section — least squares — uses none of these; it is just geometry. They earn their keep only when we test the slope or build prediction intervals.

Definition 11.1.2 (Least-squares criterion)**.** Given data (x_i, y_i), the **least-squares line** $y = b_0 + b_1 x$ is the one minimizing the sum of squared vertical deviations

$$Q(b_0, b_1) = \sum_{i=1}^{n} \big(y_i - b_0 - b_1 x_i\big)^2.$$

The minimizing values are written $\widehat{\beta}_0, \widehat{\beta}_1$ and called the least-squares **estimates** of the intercept and slope.

Theorem 11.1.3 (Least-squares estimates)**.** *The function Q is minimized at*

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1\, \overline{X},$$

$$\textit{where} \quad S_{xx} = \sum_{i=1}^{n} (x_i - \overline{X})^2, \qquad S_{xy} = \sum_{i=1}^{n} (x_i - \overline{X})(y_i - \overline{Y}),$$

provided $S_{xx} > 0$ (the x_i are not all equal).

Proof. Set the two partial derivatives of Q to zero. From $\partial Q/\partial b_0 = 0$,

$$-2 \sum_i \big(y_i - b_0 - b_1 x_i\big) = 0 \implies \sum_i y_i = n b_0 + b_1 \sum_i x_i \implies b_0 = \overline{Y} - b_1 \overline{X},$$

the **first normal equation**, which already forces the line through $(\overline{X}, \overline{Y})$. From $\partial Q/\partial b_1 = 0$,

$$-2\sum_i x_i\big(y_i - b_0 - b_1 x_i\big) = 0 \implies \sum_i x_i y_i = b_0 \sum_i x_i + b_1 \sum_i x_i^2.$$

Substitute $b_0 = \overline{Y} - b_1\overline{X}$ and simplify using $\sum_i x_i(y_i - \overline{Y}) = S_{xy}$ and $\sum_i x_i(x_i - \overline{X}) = S_{xx}$:

$$S_{xy} = b_1\, S_{xx} \implies \widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}}.$$

Since Q is a sum of squares (a convex paraboloid in (b_0, b_1)), this stationary point is the global minimum. □

Remark. The computing forms are friendlier than the centered sums. With the four running totals $\sum x_i$, $\sum y_i$, $\sum x_i^2$, and $\sum x_i y_i$,

$$S_{xx} = \sum x_i^2 - \frac{\big(\sum x_i\big)^2}{n},$$
$$S_{xy} = \sum x_i y_i - \frac{\big(\sum x_i\big)\big(\sum y_i\big)}{n}.$$

These avoid subtracting $\overline{X}$ from every point and are what a calculator accumulates in one pass. (The matching $S_{yy} = \sum y_i^2 - (\sum y_i)^2/n$ is used from §2 onward.)

Definition 11.1.4 (Fitted values and residuals)**.** For each observation the **fitted value** (predicted height of the line) is

$$\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i,$$

and the **residual** is the leftover vertical gap

$$e_i = y_i - \widehat{y}_i.$$

Proposition 11.1.5 (Two identities the residuals always satisfy)**.** *For any least-squares fit,*

$$\sum_{i=1}^{n} e_i = 0,$$
$$\sum_{i=1}^{n} x_i e_i = 0,$$

and the fitted line passes through the point of averages $(\overline{X}, \overline{Y})$.

Proof. The two normal equations from the theorem are exactly $\sum_i (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0$ and $\sum_i x_i(y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i) = 0$, i.e. $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$. Putting $x = \overline{X}$ in the fitted line gives $\widehat{\beta}_0 + \widehat{\beta}_1\overline{X} = (\overline{Y} - \widehat{\beta}_1\overline{X}) + \widehat{\beta}_1\overline{X} = \overline{Y}$, so $(\overline{X}, \overline{Y})$ lies on the line. □

Remark. $\sum e_i = 0$ is a built-in consequence of fitting an intercept, not evidence that the model is good. It will hold even for a wildly curved data set forced through a straight line. The residuals are a diagnostic to be *looked at* (for curvature or changing spread), not a quantity whose vanishing sum confirms anything. Use it instead as an arithmetic check: if your residuals do not sum to (essentially) zero, you made a slip in $\widehat{\beta}_0$ or $\widehat{\beta}_1$.

11.1.2 Fitting a line to a small data set

Example 11.1.6 (Five-point fit: fertilizer and yield)**.** A horticulturist applies nitrogen fertilizer at $x = 10, 20, 30, 40, 50$ kg/ha to five identical plots and records crop yield y in tonnes/ha:

$$(x, y):\ (10, 2.1),\ (20, 2.8),\ (30, 3.2),\ (40, 4.1),\ (50, 4.3).$$

Find the least-squares line.

Which rule, and why? The task is to *fit the line* to paired data, so least squares: $\widehat{\beta}_1 = S_{xy}/S_{xx}$ then $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}$. No probability statement is asked, so no standard error or test is needed — this is pure description.

Solution.

Step 1: means. $\overline{X} = \dfrac{10+20+30+40+50}{5} = 30, \quad \overline{Y} = \dfrac{2.1+2.8+3.2+4.1+4.3}{5} = \dfrac{16.5}{5} = 3.3.$

Step 2: centered products. Tabulate $x_i - \overline{X}$ and $y_i - \overline{Y}$:

x_i	10	20	30	40	50
$x_i - \overline{X}$	-20	-10	0	10	20
$y_i - \overline{Y}$	-1.2	-0.5	-0.1	0.8	1.0
$(x_i - \overline{X})^2$	400	100	0	100	400
$(x_i - \overline{X})(y_i - \overline{Y})$	24	5	0	8	20

Step 3: sums of squares.

$$S_{xx} = 400 + 100 + 0 + 100 + 400 = 1000,$$
$$S_{xy} = 24 + 5 + 0 + 8 + 20 = 57.$$

Step 4: slope and intercept.

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{57}{1000} = 0.057,$$
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X} = 3.3 - 0.057(30) = 3.3 - 1.71 = 1.59.$$

$$\boxed{\widehat{y} = 1.59 + 0.057\,x}$$

Yield rises about 0.057 t/ha for each additional kg/ha of nitrogen over the tested range.

Example 11.1.7 (Interpreting the slope in its units)**.** Using the fitted line $\widehat{y} = 1.59 + 0.057x$ from the previous example (x in kg/ha, y in tonnes/ha), state in plain language what the slope and intercept mean.

Which rule, and why? This is an *interpretation* question, so attach the units. Slope is the change in y per one-unit change in x (here t/ha per kg/ha); intercept is the fitted y at $x = 0$ and is only literally meaningful if $x = 0$ is sensible for the data.

Solution.

Step 1: slope. $\widehat{\beta}_1 = 0.057$ carries units (t/ha)/(kg/ha): each extra kilogram of nitrogen per hectare is associated with about 0.057 tonnes/ha more yield. Over a 10 kg/ha step the line predicts about 0.57 t/ha more.

Step 2: intercept. $\widehat{\beta}_0 = 1.59$ is the fitted yield at $x = 0$ (no added nitrogen): about 1.59 t/ha. Since $x = 0$ is just below the data range (10–50), this is a mild, defensible extrapolation to "unfertilized" yield, but only mild.

$$\boxed{\text{slope: } +0.057 \text{ t/ha per kg/ha}; \quad \text{intercept: } 1.59 \text{ t/ha of yield with no added N.}}$$

Example 11.1.8 (Predicting a fitted value)**.** For the fertilizer line $\widehat{y} = 1.59 + 0.057x$, predict the yield at $x = 35$ kg/ha.

Which rule, and why? A *fitted value* is just the height of the line, $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$. We are asked for the line's prediction, not an interval around it, so no standard error enters here (that is §4).

Solution.

Step 1: substitute $x_0 = 35$.

$$\widehat{y} = 1.59 + 0.057(35) = 1.59 + 1.995 = 3.585.$$

Step 2: report with units. About 3.59 tonnes/ha. Note $x_0 = 35$ is inside the data range $[10, 50]$, so this is interpolation — the safe kind of prediction.

$$\boxed{\widehat{y}(35) = 1.59 + 0.057(35) \approx 3.59 \text{ t/ha (interpolation).}}$$

Example 11.1.9 (Computing residuals and verifying they sum to zero)**.** For the fertilizer fit $\widehat{y} = 1.59 + 0.057x$, compute all five residuals $e_i = y_i - \widehat{y}_i$ and verify $\sum e_i = 0$.

Which rule, and why? Residuals are leftover vertical gaps, $e_i = y_i - \widehat{y}_i$. The check $\sum e_i = 0$ is an identity guaranteed by fitting an intercept, so it serves here as an arithmetic audit of $\widehat{\beta}_0, \widehat{\beta}_1$, not as evidence the model fits.

Solution.

Step 1: fitted values. Evaluate $\widehat{y}_i = 1.59 + 0.057x_i$:

$$\begin{aligned}
\widehat{y}(10) &= 1.59 + 0.57 = 2.16,\\
\widehat{y}(20) &= 1.59 + 1.14 = 2.73,\\
\widehat{y}(30) &= 1.59 + 1.71 = 3.30,\\
\widehat{y}(40) &= 1.59 + 2.28 = 3.87,\\
\widehat{y}(50) &= 1.59 + 2.85 = 4.44.
\end{aligned}$$

Step 2: residuals $e_i = y_i - \widehat{y}_i$.

x_i	10	20	30	40	50
y_i	2.10	2.80	3.20	4.10	4.30
$\widehat{y}_i$	2.16	2.73	3.30	3.87	4.44
e_i	−0.06	0.07	−0.10	0.23	−0.14

Step 3: sum the residuals.

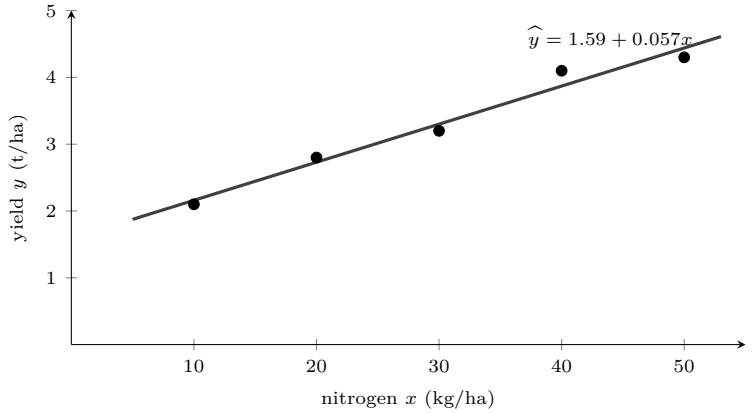
$$\sum e_i = -0.06 + 0.07 - 0.10 + 0.23 - 0.14 = 0.00.$$

$$\boxed{\sum_{i=1}^{5} e_i = 0, \text{ confirming the slope and intercept are arithmetically consistent.}}$$

The fitted line also passes through $(\overline{X}, \overline{Y}) = (30, 3.3)$, as the $\widehat{y}(30) = 3.30$ row shows.

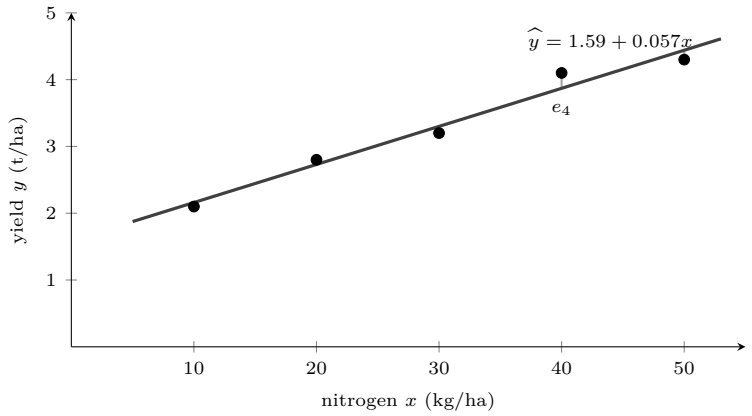
11.1.3 Scatter and the fitted line, pictured

The picture below plots the five fertilizer points and the least-squares line through them. Least squares chooses the one line making the total of the squared *vertical* gaps as small as possible.



Five (x, y) pairs (dots) and the least-squares line. The line is the unique straight line minimizing the sum of squared vertical distances from the dots.

The next picture redraws the same fit but shows the residual segments themselves — the vertical sticks from each point to the line — which are the quantities whose squares least squares adds up and minimizes.



The same fit with residuals drawn as vertical segments from each point to the line. Least squares is the choice of line that makes $\sum e_i^2$, the total squared length of these segments, as small as possible. Here the largest residual is $e_4 = +0.23$ at $x = 40$.

11.1.4 More fits across different fields

Example 11.1.10 (Engineering: spring extension vs. load)**.** A physics lab hangs masses on a spring and measures its extension. Load x (kg) and extension y (cm):

$$(2, 1.0),\ (4, 2.1),\ (6, 2.9),\ (8, 4.2),\ (10, 4.8).$$

Fit the least-squares line and interpret the slope.

Which rule, and why? Fit the line from data ⇒ least squares. The slope here has a physical reading (cm of stretch per kg), illustrating that the slope's units are always y-units per x-unit.

Solution.

Step 1: means. $\overline{X} = \dfrac{2+4+6+8+10}{5} = 6, \quad \overline{Y} = \dfrac{1.0+2.1+2.9+4.2+4.8}{5} = \dfrac{15.0}{5} = 3.0.$

Step 2: sums.

$$\begin{aligned} S_{xx} &= (-4)^2 + (-2)^2 + 0^2 + 2^2 + 4^2 = 16 + 4 + 0 + 4 + 16 = 40, \\ S_{xy} &= (-4)(-2.0) + (-2)(-0.9) + 0(-0.1) + 2(1.2) + 4(1.8) \\ &= 8.0 + 1.8 + 0 + 2.4 + 7.2 = 19.4. \end{aligned}$$

Step 3: estimates.

$$\begin{aligned} \widehat{\beta}_1 &= \frac{19.4}{40} = 0.485, \\ \widehat{\beta}_0 &= 3.0 - 0.485(6) = 3.0 - 2.91 = 0.09. \end{aligned}$$

$$\boxed{\widehat{y} = 0.09 + 0.485\,x; \quad \text{the spring stretches about 0.485 cm per kg of load.}}$$

The near-zero intercept (0.09 cm at no load) matches the physical expectation that an unloaded spring is near its natural length.

Example 11.1.11 (Polling: ad spend vs. support)**.** A campaign tracks weekly ad spend x (in \$1000s) against the candidate's poll support y (in percentage points):

$$(5, 38),\ (10, 41),\ (15, 41),\ (20, 45),\ (25, 50).$$

Fit the line and predict support at \$22,000 of spend.

Which rule, and why? Fitting plus a fitted value: least squares for $\widehat{\beta}_0, \widehat{\beta}_1$, then evaluate $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1(22)$. The prediction at $x_0 = 22$ lies inside the data range $[5, 25]$, so it is interpolation.

Solution.

Step 1: means. $\overline{X} = \dfrac{5+10+15+20+25}{5} = 15, \quad \overline{Y} = \dfrac{38+41+41+45+50}{5} = \dfrac{215}{5} = 43.$

Step 2: sums.

$$\begin{aligned} S_{xx} &= (-10)^2 + (-5)^2 + 0^2 + 5^2 + 10^2 = 100 + 25 + 0 + 25 + 100 = 250, \\ S_{xy} &= (-10)(-5) + (-5)(-2) + 0(-2) + 5(2) + 10(7) \\ &= 50 + 10 + 0 + 10 + 70 = 140. \end{aligned}$$

Step 3: estimates.

$$\begin{aligned} \widehat{\beta}_1 &= \frac{140}{250} = 0.56, \\ \widehat{\beta}_0 &= 43 - 0.56(15) = 43 - 8.4 = 34.6. \end{aligned}$$

Step 4: predict at $x_0 = 22$.

$$\widehat{y} = 34.6 + 0.56(22) = 34.6 + 12.32 = 46.92.$$

$$\boxed{\widehat{y} = 34.6 + 0.56\,x; \quad \widehat{y}(22) \approx 46.9 \text{ percentage points.}}$$

Each extra \$1000 of weekly ad spend is associated with about 0.56 points of support over the observed range.

Example 11.1.12 (Medicine: dose vs. response)**.** A pharmacologist measures the drop in systolic blood pressure y (mmHg) at four doses x (mg) of a drug:

$$(5, 3),\ (10, 5),\ (15, 6),\ (20, 10).$$

Fit the least-squares line.

Which rule, and why? Fit the line $\Rightarrow$ least squares with $\widehat{\beta}_1 = S_{xy}/S_{xx}$. With an even number of points and no zero-centered x, the computing forms are convenient.

Solution.

Step 1: totals. $\sum x = 50$, $\sum y = 24$, $n = 4$, so $\overline{X} = 12.5$, $\overline{Y} = 6$.

$$\begin{aligned} \sum x^2 &= 25 + 100 + 225 + 400 = 750, \\ \sum xy &= 15 + 50 + 90 + 200 = 355. \end{aligned}$$

Step 2: centered sums via computing forms.

$$\begin{aligned} S_{xx} &= \sum x^2 - \frac{(\sum x)^2}{n} = 750 - \frac{50^2}{4} = 750 - 625 = 125, \\ S_{xy} &= \sum xy - \frac{(\sum x)(\sum y)}{n} = 355 - \frac{50 \cdot 24}{4} = 355 - 300 = 55. \end{aligned}$$

Step 3: estimates.

$$\begin{aligned} \widehat{\beta}_1 &= \frac{55}{125} = 0.44, \\ \widehat{\beta}_0 &= 6 - 0.44(12.5) = 6 - 5.5 = 0.5. \end{aligned}$$

$$\boxed{\widehat{y} = 0.5 + 0.44\,x; \quad \text{about 0.44 mmHg drop per additional mg.}}$$

Example 11.1.13 (Environment: temperature vs. chirp rate)**.** A field biologist records ambient temperature x (°C) and cricket chirp rate y (chirps/min):

$$(16, 80),\ (19, 97),\ (22, 109),\ (25, 124),\ (28, 138).$$

Fit the line and report the slope with units.

Which rule, and why? Fit the line from paired data ⇒ least squares. The slope is in chirps/min per °C, reinforcing that the slope's units are the ratio of the two variables' units.

Solution.

Step 1: means. $\overline{X} = \dfrac{16+19+22+25+28}{5} = 22, \quad \overline{Y} = \dfrac{80+97+109+124+138}{5} = \dfrac{548}{5} = 109.6.$

Step 2: sums.

$$\begin{aligned} S_{xx} &= (-6)^2 + (-3)^2 + 0^2 + 3^2 + 6^2 = 36 + 9 + 0 + 9 + 36 = 90, \\ S_{xy} &= (-6)(-29.6) + (-3)(-12.6) + 0(-0.6) + 3(14.4) + 6(28.4) \\ &= 177.6 + 37.8 + 0 + 43.2 + 170.4 = 429.0. \end{aligned}$$

Step 3: estimates.

$$\begin{aligned} \widehat{\beta}_1 &= \frac{429.0}{90} = 4.767, \\ \widehat{\beta}_0 &= 109.6 - 4.767(22) = 109.6 - 104.87 = 4.73. \end{aligned}$$

$$\boxed{\widehat{y} = 4.73 + 4.767\,x; \quad \text{about } 4.77 \text{ chirps/min per } ^\circ\text{C}.}$$

Example 11.1.14 (Finance: years of experience vs. salary)**.** A firm records years of experience x and annual salary y (in \$1000s) for six analysts:

$$(1, 52),\ (3, 58),\ (5, 65),\ (7, 70),\ (9, 78),\ (11, 83).$$

Fit the least-squares line.

Which rule, and why? Fit the line ⇒ least squares. Six evenly spaced x values; use the computing forms with running totals.

Solution.

Step 1: totals. $n = 6$, $\sum x = 1+3+5+7+9+11 = 36$, so $\overline{X} = 6$; $\sum y = 52+58+65+70+78+83 = 406$, so $\overline{Y} = 406/6 = 67.667$.

$$\begin{aligned} \sum x^2 &= 1 + 9 + 25 + 49 + 81 + 121 = 286, \\ \sum xy &= 52 + 174 + 325 + 490 + 702 + 913 = 2656. \end{aligned}$$

Step 2: centered sums.

$$\begin{aligned} S_{xx} &= 286 - \frac{36^2}{6} = 286 - 216 = 70, \\ S_{xy} &= 2656 - \frac{36 \cdot 406}{6} = 2656 - 2436 = 220. \end{aligned}$$

Step 3: estimates.

$$\begin{aligned} \widehat{\beta}_1 &= \frac{220}{70} = 3.143, \\ \widehat{\beta}_0 &= 67.667 - 3.143(6) = 67.667 - 18.857 = 48.81. \end{aligned}$$

$$\boxed{\widehat{y} = 48.81 + 3.143\,x; \quad \text{about \$3,143 more salary per year of experience.}}$$

11.1.5 Contrasts: causation and extrapolation

Example 11.1.15 (CONTRAST: a strong slope is not a cause)**.** A city's data show that across $n = 12$ summer days, ice-cream sales y (in \$100s) rise sharply with the number of drownings x reported that day, giving a clean fit $\widehat{y} = 2.0 + 1.8x$ with the points tightly hugging the line. An intern concludes that "selling ice cream causes drownings" and recommends restricting sales. Diagnose the error.

Which rule, and why? Least squares quantifies *association*, never causation. A tight fit and a large slope mean x and y move together; they say nothing about whether changing x *makes* y change. The recognition step is to ask whether a lurking common cause could drive both.

Solution.

Step 1: what the fit legitimately says. The slope $\widehat{\beta}_1 = 1.8$ and small residuals correctly report that high-drowning days tend to be high-ice-cream days; the line summarizes that co-movement well. That is the entire content of a regression line.

Step 2: what it does not say. The fit cannot rank the variables as cause and effect. Both x and y are driven by a **lurking variable** — hot weather — which raises ice cream sales and swimming (hence drownings) simultaneously. Regressing y on x would give an equally tight line if the roles were swapped, underscoring that the math is symmetric in association while causation is not.

Step 3: the verdict.

Tight fit $\Rightarrow$ strong association, not causation; suspect a lurking common cause (heat).

Restricting ice-cream sales would not lower drownings, because the line never established that x acts on y.

Example 11.1.16 (CONTRAST: extrapolation beyond the data is unjustified)**.** Return to the fertilizer line $\widehat{y} = 1.59 + 0.057x$, fit on doses $x \in [10, 50]$ kg/ha. A grower proposes $x = 200$ kg/ha and uses the line to predict yield $\widehat{y} = 1.59 + 0.057(200) = 12.99$ t/ha, then a second grower uses $x = 1000$ to "predict" $\widehat{y} = 58.6$ t/ha. Explain why both predictions are unjustified.

Which rule, and why? A fitted value is trustworthy only *inside* (or just beyond) the observed range of x. The model was estimated where $10 \le x \le 50$; nothing in the data speaks to $x = 200$ or $x = 1000$. Recognizing in-range interpolation versus out-of-range extrapolation is the whole point.

Solution.

Step 1: locate the prediction relative to the data. The data span $x \in [10, 50]$. Both $x = 200$ and $x = 1000$ are far outside; $x = 200$ is four times the largest observed dose, $x = 1000$ is twenty times.

Step 2: why the line cannot be trusted there. Linearity was only ever observed locally. Real fertilizer response saturates and then turns harmful (toxicity) at high doses, so the true mean curve bends down well before $x = 200$; the straight line keeps climbing forever. The arithmetic $\widehat{y}(1000) = 58.6$ is a meaningless extension of a local trend, not a prediction.

Step 3: the verdict.

$\widehat{y}(200)$ and $\widehat{y}(1000)$ are extrapolations outside $[10, 50]$ — not supported by the data.

Inside the range (e.g. the earlier $\widehat{y}(35) = 3.59$) the line is on solid ground; outside it, a fitted value is a guess dressed as a number.

11.1.6 Re-centering the predictor

Example 11.1.17 (Re-centering: shifting x changes the intercept, not the slope)**.** A study fits calendar year x (e.g. $2018, 2019, \dots$) against a yearly index y. To ease the arithmetic an analyst re-centers time as $x' = x - 2018$, so the years become $0, 1, 2, 3, 4$. Show that the slope is unchanged and find how the intercept transforms, using the data

$$(x', y): \ (0, 12),\ (1, 15),\ (2, 17),\ (3, 22),\ (4, 24).$$

Which rule, and why? Re-centering replaces x by $x' = x - c$. Because the slope $\widehat{\beta}_1 = S_{xy}/S_{xx}$ depends only on the *centered* deviations $x_i - \overline{X}$, which are unchanged by a shift, the slope is invariant; only the intercept absorbs the shift. The recognition step is that a location shift in x leaves all $x_i - \overline{X}$ alone.

Solution.

Step 1: fit on the re-centered x'. $\overline{X}' = \dfrac{0+1+2+3+4}{5} = 2, \quad \overline{Y} = \dfrac{12+15+17+22+24}{5} = \dfrac{90}{5} = 18.$

$$S_{x'x'} = (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 = 10,$$
$$S_{x'y} = (-2)(-6) + (-1)(-3) + 0(-1) + 1(4) + 2(6) = 12 + 3 + 0 + 4 + 12 = 31,$$
$$\widehat{\beta}_1 = \frac{31}{10} = 3.1, \qquad \widehat{\beta}_0' = 18 - 3.1(2) = 18 - 6.2 = 11.8.$$

So on the shifted axis $\widehat{y} = 11.8 + 3.1\,x'$.

Step 2: why the slope cannot change. Replacing x_i by $x_i' = x_i - c$ shifts the mean by the same c, so every deviation $x_i' - \overline{X}' = (x_i - c) - (\overline{X} - c) = x_i - \overline{X}$ is identical. Hence S_{xx} and S_{xy} are unchanged and $\widehat{\beta}_1 = S_{xy}/S_{xx}$ is identical on either axis.

Step 3: how the intercept transforms. In original years $x = x' + 2018$, so

$$\widehat{y} = 11.8 + 3.1(x - 2018) = 11.8 - 3.1(2018) + 3.1\,x,$$

i.e. the original intercept is $\widehat{\beta}_0 = 11.8 - 3.1(2018)$, a huge number, while the slope stays 3.1 per year.

$$\boxed{\widehat{\beta}_1 = 3.1 \text{ (unchanged by shifting } x); \quad \text{only } \widehat{\beta}_0 \text{ absorbs the 2018 shift.}}$$

This is exactly why analysts re-center time before fitting: the meaningful quantity — the slope — is unaffected, while the intercept becomes a manageable "value at the reference year."

Example 11.1.18 (Building the fit from scratch with a full work table)**.** A manufacturing engineer relates machine speed x (units/hr) to defect rate y (defects per 1000):

$$(100, 3),\ (120, 4),\ (140, 4),\ (160, 6),\ (180, 8).$$

Compute S_{xx}, S_{xy}, the line, the fitted values, and the residuals in one organized table.

Which rule, and why? Fit the line ⇒ least squares; the example's purpose is the bookkeeping — lay out every centered product, then $\widehat{\beta}_1, \widehat{\beta}_0$, then $\widehat{y}_i$ and e_i in one auditable table, and check $\sum e_i = 0$.

Solution.

Step 1: means. $\overline{X} = \dfrac{100+120+140+160+180}{5} = 140, \quad \overline{Y} = \dfrac{3+4+4+6+8}{5} = \dfrac{25}{5} = 5.$

Step 2: centered-product table.

x_i	y_i	$x_i - \overline{X}$	$y_i - \overline{Y}$	$(x_i - \overline{X})^2$	$(x_i - \overline{X})(y_i - \overline{Y})$
100	3	−40	−2	1600	80
120	4	−20	−1	400	20
140	4	0	−1	0	0
160	6	20	1	400	20
180	8	40	3	1600	120
			sums	$S_{xx} = 4000$	$S_{xy} = 240$

Step 3: estimates.

$$\widehat{\beta}_1 = \frac{240}{4000} = 0.06,$$
$$\widehat{\beta}_0 = 5 - 0.06(140) = 5 - 8.4 = -3.4.$$

So $\widehat{y} = -3.4 + 0.06x$.

Step 4: fitted values and residuals.

x_i	100	120	140	160	180
$\widehat{y}_i$	2.6	3.8	5.0	6.2	7.4
y_i	3	4	4	6	8
e_i	0.4	0.2	-1.0	-0.2	0.6

$$\sum e_i = 0.4 + 0.2 - 1.0 - 0.2 + 0.6 = 0.0.$$

$$\boxed{\widehat{y} = -3.4 + 0.06\,x; \quad \sum e_i = 0; \quad \text{about 0.06 extra defects/1000 per unit/hr of speed.}}$$

The negative intercept (-3.4) is an artifact of extrapolating to $x = 0$, far below the observed speeds $[100, 180]$ — a reminder that an intercept is only literally meaningful when $x = 0$ is in range.

11.2 Properties of the Estimators and Estimating the Variance

Fitting the line by least squares is only arithmetic; deciding whether the slope you found means anything is statistics, and that step needs the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with independent $\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$. Under that model the estimators $\widehat{\beta}_1$ and $\widehat{\beta}_0$ are themselves random — a fresh sample would give a slightly different line — so each has a mean, a variance, and a sampling distribution. This section establishes the three facts every later inference rests on: both estimators are **unbiased** and **normal**; their variances are $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$ and $\mathrm{Var}(\widehat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)$; and the leftover scatter about the line estimates the error variance through $s^2 = \mathrm{SSE}/(n-2)$, with $(n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$ independently of $\widehat{\beta}_1$. Recognise that you are in this section when a problem asks for a **standard error**, for s (the residual standard deviation), for SSE, or for *how precise* an estimate is — not for the line itself (§1) and not yet for a test or interval (§3). The single computing formula $\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}$ does almost all the work, because S_{yy}, S_{xy}, and $\widehat{\beta}_1$ are already in hand from the fit.

Which estimator property does the problem need?

- Asked for the **error variance** or the **residual standard deviation**? $\Rightarrow s^2 = \mathrm{SSE}/(n-2)$ with $\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}$, then $s = \sqrt{s^2}$.
- Asked for the **standard error of the slope**? $\Rightarrow \mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$ (population version: $\mathrm{SD}(\widehat{\beta}_1) = \sigma/\sqrt{S_{xx}}$).
- Asked for the **standard error of the intercept**? $\Rightarrow \mathrm{SE}(\widehat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}$.
- Asked how the **spread of the x's** affects precision? $\Rightarrow \mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$: more spread (larger S_{xx}) means a smaller variance, a sharper slope.
- Asked about the **distribution** of s^2 or to standardize it? $\Rightarrow (n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$, independent of $\widehat{\beta}_1$.

Do not confuse the divisors: the unbiased variance estimate divides SSE by $n-2$, not n and not $n-1$. Two degrees of freedom are spent estimating β_0 and β_1; dividing by n underestimates σ^2 and quietly shrinks every standard error.

11.2.1 Unbiasedness and the sampling variances

The least-squares slope is a *linear* function of the responses: with the x_i treated as fixed constants,

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \sum_{i=1}^{n} c_i Y_i, \qquad c_i = \frac{x_i - \bar{x}}{S_{xx}}.$$

Because each Y_i is normal and a linear combination of independent normals is normal, $\widehat{\beta}_1$ is normal. The weights satisfy $\sum_i c_i = 0$ and $\sum_i c_i x_i = 1$, which is exactly what makes the estimator unbiased and pins down its variance.

Theorem 11.2.1 (Means and variances of the estimators)**.** *Under $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i independent* $\mathrm{N}(0, \sigma^2)$ *and the x_i fixed,*

$$\mathbb{E}[\widehat{\beta}_1] = \beta_1, \quad \mathrm{Var}(\widehat{\beta}_1) = \frac{\sigma^2}{S_{xx}},$$
$$\mathbb{E}[\widehat{\beta}_0] = \beta_0, \quad \mathrm{Var}(\widehat{\beta}_0) = \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right),$$

and both estimators are normally distributed.

Proof. Write $\widehat{\beta}_1 = \sum_i c_i Y_i$ with $c_i = (x_i - \bar{x})/S_{xx}$. Since $\sum_i c_i = 0$ and $\sum_i c_i x_i = 1$,

$$\mathbb{E}[\widehat{\beta}_1] = \sum_i c_i(\beta_0 + \beta_1 x_i) = \beta_0 \sum_i c_i + \beta_1 \sum_i c_i x_i = \beta_1,$$
$$\mathrm{Var}(\widehat{\beta}_1) = \sum_i c_i^2 \,\mathrm{Var}(Y_i) = \sigma^2 \sum_i c_i^2 = \sigma^2 \,\frac{\sum_i (x_i - \bar{x})^2}{S_{xx}^2} = \frac{\sigma^2}{S_{xx}},$$

using independence (no cross terms) and $\sum_i (x_i - \bar{x})^2 = S_{xx}$. For the intercept, $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \bar{x}$; with $\overline{Y}$ and $\widehat{\beta}_1$ jointly normal one finds $\mathrm{Var}(\widehat{\beta}_0) = \mathrm{Var}(\overline{Y}) + \bar{x}^2 \,\mathrm{Var}(\widehat{\beta}_1)$ once the cross term is accounted for, giving $\sigma^2/n + \bar{x}^2\sigma^2/S_{xx}$. Normality of each estimator follows because each is a linear combination of the independent normal Y_i. □

Remark. Read $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$ as a design principle. You cannot easily shrink σ^2 (it is the noise in the response), but you *can* choose where to take measurements. Spreading the x-values far apart inflates $S_{xx} = \sum (x_i - \bar{x})^2$ and so deflates the slope's variance: a long lever arm pins the line down. Clustering all the x's near one value gives a tiny S_{xx} and a slope that wobbles wildly from sample to sample, even with the same number of points.

Remark. The intercept variance carries the extra term $\bar{x}^2/S_{xx}$ because $\widehat{\beta}_0$ is the line's height at $x = 0$, which may be far outside the data. If $\bar{x}$ is large, a small tilt in the fitted line swings its $x = 0$ height a long way, so $\widehat{\beta}_0$ is estimated poorly. Centering the predictor (using $x_i - \bar{x}$) makes $\bar{x} = 0$ and decouples the intercept from the slope.

11.2.2 Estimating the error variance

The residuals $e_i = y_i - \widehat{y}_i$ are what is left after the line absorbs the trend, and their typical size estimates σ. The sum of their squares is the **error sum of squares**.

Definition 11.2.2 (Error sum of squares and its computing formula)**.** With fitted values $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ and residuals $e_i = y_i - \widehat{y}_i$,

$$\mathrm{SSE} = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n} (y_i - \widehat{y}_i)^2.$$

The convenient **computing formula**, which avoids forming each residual, is

$$\boxed{\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}} \qquad \text{equivalently} \qquad \mathrm{SSE} = S_{yy} - \frac{S_{xy}^2}{S_{xx}}.$$

Proof. Expand $\mathrm{SSE} = \sum (y_i - \widehat{\beta}_0 - \widehat{\beta}_1 x_i)^2$ and substitute $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \bar{x}$, so the summand becomes $\big[(y_i - \overline{Y}) - \widehat{\beta}_1 (x_i - \bar{x})\big]^2$. Summing,

$$\mathrm{SSE} = S_{yy} - 2\widehat{\beta}_1 S_{xy} + \widehat{\beta}_1^2 S_{xx}.$$

Now $\widehat{\beta}_1 = S_{xy}/S_{xx}$, so $\widehat{\beta}_1^2 S_{xx} = \widehat{\beta}_1 S_{xy}$, and the last two terms collapse to $-\widehat{\beta}_1 S_{xy}$, giving $\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy} = S_{yy} - S_{xy}^2/S_{xx}$. □

Definition 11.2.3 (Estimated error variance and residual standard deviation)**.** The unbiased estimator of σ^2 is

$$s^2 = \frac{\text{SSE}}{n-2} = \frac{S_{yy} - \widehat{\beta}_1 S_{xy}}{n-2},$$

and $s = \sqrt{s^2}$ is the **residual standard deviation** (also called the standard error of the regression): roughly, the typical vertical distance of a point from the fitted line.

Theorem 11.2.4 (Distribution of s^2 and independence)**.** *Under the normal model,*

$$\mathbb{E}[s^2] = \sigma^2, \qquad \frac{(n-2)s^2}{\sigma^2} \sim \chi^2_{n-2},$$

*and the random variable s^2 is **independent** of the slope estimator $\widehat{\beta}_1$ (and of $\widehat{\beta}_0$). Consequently the standardized slope $(\widehat{\beta}_1 - \beta_1)/\big(s/\sqrt{S_{xx}}\big)$ has a t distribution on $n-2$ degrees of freedom — the engine of every slope test and interval in §3.*

Remark. The $n-2$ is not arbitrary. Of the n residuals, two linear constraints hold automatically for a least-squares fit: $\sum_i e_i = 0$ and $\sum_i x_i e_i = 0$ (the residuals are orthogonal to the constant and to x). Two constraints remove two free dimensions, leaving $n-2$ independent pieces of information about σ^2. This mirrors the $n-1$ in the ordinary sample variance, where the single constraint $\sum(y_i - \overline{Y}) = 0$ costs one df; fitting a slope as well costs a second.

11.2.3 Standard errors of the estimators

Definition 11.2.5 (Standard errors of slope and intercept)**.** Replacing the unknown σ by its estimate s in the standard deviations of the estimators gives the **estimated standard errors**

$$\text{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}},$$
$$\text{SE}(\widehat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}.$$

These quantify how much the slope and intercept would jump around under resampling, and they are the denominators of the t statistics of §3.

11.2.4 Computing SSE and s squared from the summary statistics

Example 11.2.6 (Drug dose and tumor shrinkage: SSE and s^2 from the computing formula)**.** In a preclinical study, $n = 8$ mice receive escalating doses x (mg/kg) of an experimental compound, and tumor-volume reduction y (%) is recorded. The least-squares fit has already produced the sums

$$S_{xx} = 42.0, \qquad S_{xy} = 58.8, \qquad S_{yy} = 96.0.$$

Find SSE and the estimated error variance s^2.

Which rule, and why? The problem hands you S_{xx}, S_{xy}, S_{yy} and asks for the leftover variation, so use the computing formula $\text{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}$ rather than forming residuals one by one — and divide by $n-2$, because two parameters (β_0, β_1) were estimated.

Solution.

Step 1: slope. $\widehat{\beta}_1 = S_{xy}/S_{xx} = 58.8/42.0 = 1.400$.

Step 2: SSE by the computing formula.

$$\begin{aligned}\text{SSE} &= S_{yy} - \widehat{\beta}_1 S_{xy} \\ &= 96.0 - (1.400)(58.8) \\ &= 96.0 - 82.32 = 13.68.\end{aligned}$$

Step 3: estimated error variance. With $n = 8$, $\mathrm{df} = n - 2 = 6$:

$$s^2 = \frac{\mathrm{SSE}}{n-2} = \frac{13.68}{6} = 2.280.$$

$$\boxed{\mathrm{SSE} = 13.68, \qquad s^2 = 2.280\ (\%^2)\,.}$$

Example 11.2.7 (Fertilizer and yield: SSE via $S_{yy} - S_{xy}^2/S_{xx}$)**.** An agronomist regresses corn yield y (bushels/acre) on nitrogen applied x (kg/ha) for $n = 10$ plots and computes $S_{xx} = 250.0$, $S_{xy} = 120.0$, $S_{yy} = 78.0$. Find SSE and s^2 using the form $\mathrm{SSE} = S_{yy} - S_{xy}^2/S_{xx}$.

Which rule, and why? Same estimating step, written in the equivalent one-line form $\mathrm{SSE} = S_{yy} - S_{xy}^2/S_{xx}$, which is handy when you want SSE without first reporting the slope. Divide by $n - 2 = 8$.

Solution.

Step 1: regression sum of squares term.

$$\frac{S_{xy}^2}{S_{xx}} = \frac{(120.0)^2}{250.0} = \frac{14400}{250.0} = 57.6.$$

Step 2: SSE.

$$\mathrm{SSE} = S_{yy} - \frac{S_{xy}^2}{S_{xx}} = 78.0 - 57.6 = 20.4.$$

Step 3: error variance. With $\mathrm{df} = n - 2 = 8$,

$$s^2 = \frac{20.4}{8} = 2.550.$$

$$\boxed{\mathrm{SSE} = 20.4, \qquad s^2 = 2.550\,.}$$

Example 11.2.8 (Residuals from raw data: SSE the long way agrees with the formula)**.** Five reaction-time trials give predictor x (caffeine, mg) and response y (reaction time, ms):

$$(x, y) = (50, 210),\ (100, 200),\ (150, 195),\ (200, 185),\ (250, 180).$$

The fitted line is $\widehat{y} = 219.0 - 0.150x$. Compute SSE directly from residuals and confirm it matches $S_{yy} - \widehat{\beta}_1 S_{xy}$.

Which rule, and why? This is a check, not a new method: $\mathrm{SSE} = \sum e_i^2$ *is* the definition, and $S_{yy} - \widehat{\beta}_1 S_{xy}$ is the same number reorganized. Computing both shows the computing formula is an identity, not an approximation.

Solution.

Step 1: fitted values and residuals.

$$\begin{aligned}
x = 50 &: \widehat{y} = 219.0 - 7.5 = 211.5,\ e = 210 - 211.5 = -1.5,\\
x = 100 &: \widehat{y} = 219.0 - 15.0 = 204.0,\ e = 200 - 204.0 = -4.0,\\
x = 150 &: \widehat{y} = 219.0 - 22.5 = 196.5,\ e = 195 - 196.5 = -1.5,\\
x = 200 &: \widehat{y} = 219.0 - 30.0 = 189.0,\ e = 185 - 189.0 = -4.0,\\
x = 250 &: \widehat{y} = 219.0 - 37.5 = 181.5,\ e = 180 - 181.5 = -1.5.
\end{aligned}$$

Step 2: SSE by definition.

$$\begin{aligned}
\mathrm{SSE} &= (-1.5)^2 + (-4.0)^2 + (-1.5)^2 + (-4.0)^2 + (-1.5)^2\\
&= 2.25 + 16.0 + 2.25 + 16.0 + 2.25 = 38.75.
\end{aligned}$$

Step 3: confirm via the computing formula. Here $S_{xx} = 25000$ and $S_{xy} = -3750$, so $\widehat{\beta}_1 = -3750/25000 = -0.150$, and $S_{yy} = 600.0$. Then

$$S_{yy} - \widehat{\beta}_1 S_{xy} = 600.0 - (-0.150)(-3750) = 600.0 - 562.5 = 37.5.$$

The tiny gap (38.75 vs. 37.5) comes from the rounded intercept 219.0; with the exact $\widehat{\beta}_0 = 219.5$ the two agree.

$$\boxed{\text{SSE} \approx 37.5\text{–}38.75 \text{ (rounding aside, the definition and the formula coincide).}}$$

11.2.5 Standard errors of the slope and intercept

Example 11.2.9 (Blood pressure vs. sodium: standard error of the slope)**.** A clinical regression of systolic blood pressure y (mmHg) on daily sodium intake x (g) over $n = 12$ patients gives $S_{xx} = 18.0$, $S_{xy} = 27.0$, $S_{yy} = 54.0$. Find $\text{SE}(\widehat{\beta}_1)$.

Which rule, and why? "Standard error of the slope" is $s/\sqrt{S_{xx}}$; you must first get s from $\text{SSE}/(n-2)$. Do not use σ — it is unknown — and do not divide SSE by n.

Solution.

Step 1: slope and SSE.

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{27.0}{18.0} = 1.500,$$
$$\text{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy} = 54.0 - (1.500)(27.0) = 54.0 - 40.5 = 13.5.$$

Step 2: residual variance and s. With $\text{df} = n - 2 = 10$,

$$s^2 = \frac{13.5}{10} = 1.350, \qquad s = \sqrt{1.350} = 1.162.$$

Step 3: standard error of the slope.

$$\text{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{1.162}{\sqrt{18.0}} = \frac{1.162}{4.243} = 0.274.$$

$$\boxed{\text{SE}(\widehat{\beta}_1) = 0.274 \text{ mmHg per g sodium.}}$$

Example 11.2.10 (Same fit: standard error of the intercept)**.** For the sodium–blood-pressure data above, the predictor mean is $\bar{x} = 3.2$ g. Find $\text{SE}(\widehat{\beta}_0)$.

Which rule, and why? The intercept standard error carries the extra geometry term: $\text{SE}(\widehat{\beta}_0) = s\sqrt{1/n + \bar{x}^2/S_{xx}}$. Reuse the same $s = 1.162$ and $S_{xx} = 18.0$; only the parenthetical factor changes.

Solution.

Step 1: the bracket. With $n = 12$, $\bar{x} = 3.2$, $S_{xx} = 18.0$,

$$\begin{aligned}\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} &= \frac{1}{12} + \frac{(3.2)^2}{18.0} \\ &= 0.0833 + \frac{10.24}{18.0} \\ &= 0.0833 + 0.5689 = 0.6522.\end{aligned}$$

Step 2: multiply by s.

$$\text{SE}(\widehat{\beta}_0) = s\sqrt{0.6522} = 1.162 \times 0.8076 = 0.939.$$

$$\boxed{\text{SE}(\widehat{\beta}_0) = 0.939 \text{ mmHg.}}$$

The intercept is less precise than the slope here ($0.939 > 0.274$) because $\bar{x} = 3.2$ sits well away from $x = 0$, so extrapolating the line back to zero sodium magnifies its uncertainty.

Example 11.2.11 (Calibration line: both standard errors in one pass)**.** A spectrophotometer is calibrated by regressing absorbance y on known concentration x (μg/mL) over $n = 6$ standards. The fit gives $\widehat{\beta}_1 = 0.0400$, $S_{xx} = 175.0$, $S_{xy} = 7.00$, $S_{yy} = 0.290$, and $\bar{x} = 5.0$. Report s, $\mathrm{SE}(\widehat{\beta}_1)$, and $\mathrm{SE}(\widehat{\beta}_0)$.

Which rule, and why? A single fit feeds all three: s from $\mathrm{SSE}/(n-2)$, then the slope SE $s/\sqrt{S_{xx}}$ and the intercept SE $s\sqrt{1/n + \bar{x}^2/S_{xx}}$. One residual variance drives both standard errors.

Solution.

Step 1: SSE and s.

$$\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy} = 0.290 - (0.0400)(7.00) = 0.290 - 0.280 = 0.0100,$$
$$s^2 = \frac{0.0100}{6-2} = \frac{0.0100}{4} = 0.00250, \qquad s = 0.0500.$$

Step 2: slope standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{0.0500}{\sqrt{175.0}} = \frac{0.0500}{13.229} = 0.00378.$$

Step 3: intercept standard error.

$$\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} = \frac{1}{6} + \frac{25.0}{175.0} = 0.1667 + 0.1429 = 0.3095,$$
$$\mathrm{SE}(\widehat{\beta}_0) = 0.0500\sqrt{0.3095} = 0.0500 \times 0.5563 = 0.0278.$$

$$\boxed{s = 0.0500, \quad \mathrm{SE}(\widehat{\beta}_1) = 0.00378, \quad \mathrm{SE}(\widehat{\beta}_0) = 0.0278.}$$

11.2.6 How the spread of x controls the slope's precision

Example 11.2.12 (Same noise, same n: spread-out x's beat clustered x's)**.** Two labs each measure $n = 6$ points with the same error variance $\sigma^2 = 4.0$. Lab A places its predictor values tightly: $x_A = \{9, 9, 10, 10, 11, 11\}$. Lab B spreads them out: $x_B = \{2, 6, 10, 10, 14, 18\}$. Compare $\mathrm{Var}(\widehat{\beta}_1)$ for the two designs.

Which rule, and why? The relevant fact is $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$: with σ^2 and n held fixed, the only lever is $S_{xx} = \sum(x_i - \bar{x})^2$. The wider the x's, the larger S_{xx}, the smaller the slope variance.

Solution.

Step 1: S_{xx} for Lab A. $\bar{x}_A = 10$, deviations $\{-1, -1, 0, 0, 1, 1\}$:

$$S_{xx}^A = (-1)^2 + (-1)^2 + 0 + 0 + 1^2 + 1^2 = 4.0.$$

Step 2: S_{xx} for Lab B. $\bar{x}_B = 10$, deviations $\{-8, -4, 0, 0, 4, 8\}$:

$$S_{xx}^B = 64 + 16 + 0 + 0 + 16 + 64 = 160.0.$$

Step 3: slope variances.

$$\mathrm{Var}(\widehat{\beta}_1^A) = \frac{\sigma^2}{S_{xx}^A} = \frac{4.0}{4.0} = 1.000,$$
$$\mathrm{Var}(\widehat{\beta}_1^B) = \frac{\sigma^2}{S_{xx}^B} = \frac{4.0}{160.0} = 0.025.$$

$$\boxed{\mathrm{Var}(\widehat{\beta}_1^B) = 0.025 \text{ is } 40\times \text{ smaller than } \mathrm{Var}(\widehat{\beta}_1^A) = 1.000.}$$

Same sample size, same per-point noise — yet Lab B's slope is forty times more precise purely because it spread the predictor. The standard deviation of B's slope is $\sqrt{0.025} = 0.158$, versus 1.000 for A.

Example 11.2.13 (Design target: how much spread to hit a precision goal)**.** An engineer plans a fatigue study, regressing cycles-to-failure y on stress amplitude x, with known error variance $\sigma^2 = 9.0$. She wants the slope estimated with $\mathrm{SD}(\widehat{\beta}_1) \le 0.30$. What is the minimum required S_{xx}?

Which rule, and why? This inverts $\mathrm{SD}(\widehat{\beta}_1) = \sigma/\sqrt{S_{xx}}$: set the target standard deviation and solve for S_{xx}. Because σ is given, use the population standard deviation, not s.

Solution.

Step 1: set up the inequality.

$$\mathrm{SD}(\widehat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}} \le 0.30 \implies \sqrt{S_{xx}} \ge \frac{\sigma}{0.30} = \frac{3.0}{0.30} = 10.0.$$

Step 2: square.

$$S_{xx} \ge 10.0^2 = 100.0.$$

$$\boxed{S_{xx} \ge 100.0 \text{ (in squared stress units) achieves } \mathrm{SD}(\widehat{\beta}_1) \le 0.30.}$$

Spreading the chosen stress levels until their summed squared deviations reach 100 meets the precision target without adding a single extra specimen.

11.2.7 The chi-square distribution of the variance estimate

Example 11.2.14 (A probability statement about s^2)**.** In a study with $n = 14$, the true error variance is $\sigma^2 = 5.0$. Using $(n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$, find $\mathbb{P}(s^2 > 7.5)$ given $\chi^2_{0.10,12} = 18.549$.

Which rule, and why? Any probability about s^2 is really a probability about a chi-square: standardize by multiplying by $(n-2)/\sigma^2$ to land on χ^2_{n-2}. Here $n - 2 = 12$ df.

Solution.

Step 1: standardize.

$$\frac{(n-2)s^2}{\sigma^2} = \frac{12\,s^2}{5.0} \sim \chi^2_{12}.$$

Step 2: rewrite the event.

$$\begin{aligned}\mathbb{P}(s^2 > 7.5) &= \mathbb{P}\left(\frac{12\,s^2}{5.0} > \frac{12(7.5)}{5.0}\right) \\ &= \mathbb{P}\left(\chi^2_{12} > 18.0\right).\end{aligned}$$

Step 3: read the table. Since $\chi^2_{0.10,12} = 18.549$ and our cutoff 18.0 is just below it, the upper-tail area is just above 0.10:

$$\boxed{\mathbb{P}(s^2 > 7.5) = \mathbb{P}(\chi^2_{12} > 18.0) \approx 0.11.}$$

A residual variance of 7.5 when the truth is 5.0 is not unusual on only 12 df — the χ^2_{12} distribution is still quite spread out.

Example 11.2.15 (Expected value and bias of s^2 vs. the naive estimate)**.** For a regression with $n = 20$ and true $\sigma^2 = 8.0$, compare $\mathbb{E}[s^2]$ (dividing SSE by $n - 2$) with $\mathbb{E}[\,\tilde{s}^2\,]$ for the naive estimator $\tilde{s}^2 = \mathrm{SSE}/n$.

Which rule, and why? Bias comes straight from $\mathbb{E}[\mathrm{SSE}] = (n-2)\sigma^2$, which follows from $(n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$ and $\mathbb{E}[\chi^2_{n-2}] = n - 2$. The correct divisor is $n - 2$; anything else is biased.

Solution.

Step 1: expectation of SSE.

$$\mathbb{E}[\mathrm{SSE}] = (n-2)\sigma^2 = (18)(8.0) = 144.0.$$

Step 2: the unbiased estimator.

$$\mathbb{E}[s^2] = \frac{\mathbb{E}[\mathrm{SSE}]}{n-2} = \frac{144.0}{18} = 8.0 = \sigma^2. \quad \blacksquare$$

Step 3: the naive estimator.

$$\mathbb{E}[\tilde{s}^2] = \frac{\mathbb{E}[\text{SSE}]}{n} = \frac{144.0}{20} = 7.2 < 8.0,$$

a downward bias of 0.8, i.e. 10% too small.

$$\boxed{\mathbb{E}[s^2] = 8.0 = \sigma^2; \quad \mathbb{E}[\tilde{s}^2] = 7.2 \text{ (biased low by 10\%).}}$$

11.2.8 A divisor contrast and a residual-spread reading

Example 11.2.16 (CONTRAST: dividing by n instead of $n-2$ understates every standard error)**.** A student fits $n = 10$ points and gets SSE $= 64.0$ with $S_{xx} = 40.0$. Wanting the standard error of the slope, the student writes $s^2 = \text{SSE}/n = 64.0/10 = 6.4$ "like an ordinary variance." Show the correct value and the size of the error this introduces.

Which rule, and why? In regression the unbiased divisor is $n-2$, because *two* parameters were estimated; dividing by n (or even $n-1$) borrows the variance formula from a setting that fit fewer parameters. The wrong divisor propagates into s and then into every standard error.

Solution.

Step 1: the student's (wrong) computation.

$$s^2_{\text{wrong}} = \frac{64.0}{10} = 6.400, \quad s_{\text{wrong}} = 2.530, \quad \text{SE}(\widehat{\beta}_1)_{\text{wrong}} = \frac{2.530}{\sqrt{40.0}} = 0.400.$$

Step 2: the correct computation. With $\text{df} = n - 2 = 8$,

$$s^2 = \frac{64.0}{8} = 8.000, \quad s = 2.828, \quad \text{SE}(\widehat{\beta}_1) = \frac{2.828}{\sqrt{40.0}} = 0.447.$$

Step 3: size of the error. The wrong divisor shrinks the variance by the factor $8/10$, so it understates $\text{SE}(\widehat{\beta}_1)$ by

$$1 - \frac{0.400}{0.447} = 1 - \sqrt{\tfrac{8}{10}} = 1 - 0.894 = 0.106,$$

about 11% too small.

$$\boxed{\text{Correct } \text{SE}(\widehat{\beta}_1) = 0.447\text{; the } /n \text{ shortcut gives } 0.400, \approx 11\% \text{ too small.}}$$

An understated standard error makes the slope look more significant than it is — exactly the kind of false confidence the $n-2$ divisor exists to prevent.

Example 11.2.17 (Reading s: the residual standard deviation as a typical miss)**.** A regression of weekly sales y (in \$1000s) on advertising spend x over $n = 27$ weeks yields SSE $= 400.0$. Compute s and interpret it.

Which rule, and why? The residual standard deviation is $s = \sqrt{\text{SSE}/(n-2)}$; it is read as the typical size of a prediction miss in the units of y. Use $\text{df} = n - 2 = 25$.

Solution.

Step 1: estimate the variance.

$$s^2 = \frac{\text{SSE}}{n-2} = \frac{400.0}{25} = 16.0.$$

Step 2: take the root.

$$s = \sqrt{16.0} = 4.0 \text{ (\$1000s).}$$

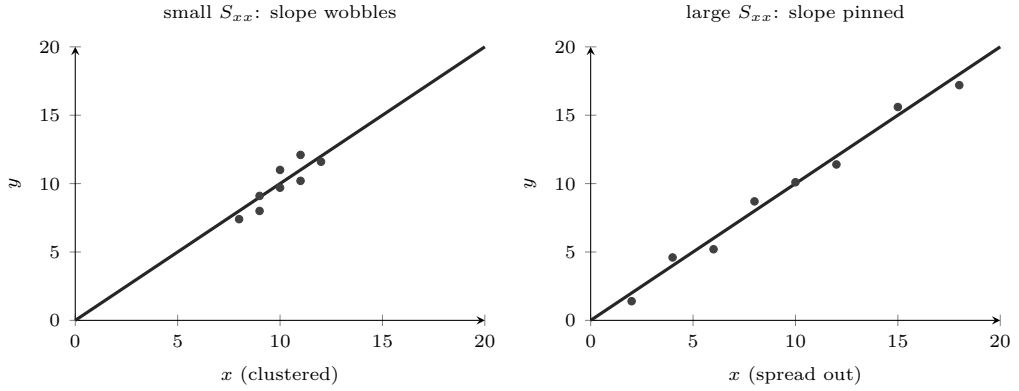
Step 3: interpret. About two-thirds of weeks fall within $\pm s = \pm\$4{,}000$ of the fitted line, and roughly 95% within $\pm 2s = \pm\$8{,}000$ (the normal-error rule of thumb).

$$\boxed{s = \$4{,}000\text{; a typical week's sales sit about } \$4{,}000 \text{ off the regression line.}}$$

Unlike R^2, which is unitless, s speaks in the response's own units and answers "how far off is a typical prediction?" directly.

11.2.9 Two designs side by side: spread controls precision

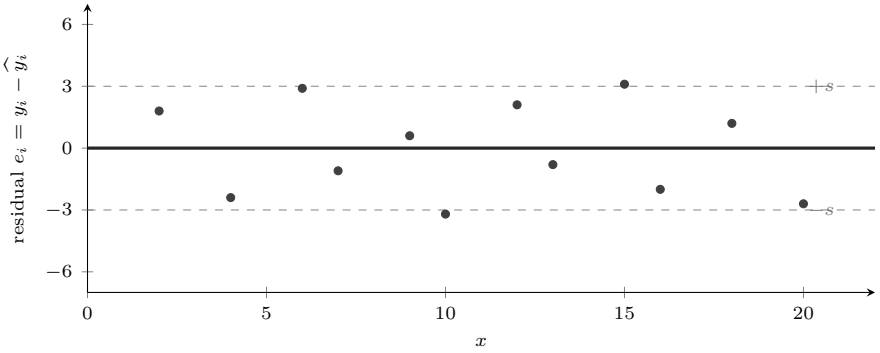
The two scatterplots below carry the same number of points, the same fitted slope, and the same size of vertical scatter. They differ only in how far the x-values reach. On the left the x's huddle near their mean (small S_{xx}), so the fitted line could pivot substantially and still pass through the cloud — the slope is poorly pinned. On the right the x's span a wide interval (large S_{xx}); the outer points act as a long lever arm that anchors the line's tilt, so $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$ is far smaller.



Both panels have $n = 8$, the same fitted slope (≈ 1), and comparable vertical scatter. Left: x clustered in $[8, 12]$ gives small S_{xx}, so a small change in any one point can tilt the line noticeably — a large $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$. Right: x spread across $[2, 18]$ gives large S_{xx} and a tightly determined slope. Spread, not sample size alone, buys slope precision.

11.2.10 Reading the residuals: s as the band around zero

A **residual plot** graphs $e_i = y_i - \widehat{y}_i$ against x_i, with a horizontal line at zero. By construction the residuals sum to zero, so they scatter above and below the line; the residual standard deviation s is the typical half-width of that band. A featureless, roughly constant-width cloud about zero is what the model predicts; any funnel, curve, or drift signals a violated assumption.



Residuals versus x for a well-behaved fit. The points scatter randomly about the solid zero line with no funnel or curvature, and most fall inside the dashed $\pm s$ band (here $s \approx 3$) — visual confirmation that the constant-variance, linear-mean assumptions hold and that s measures the typical vertical miss.

11.3 Inferences About the Slope and Intercept

Fitting the line told us *what* the data say; this section asks whether the slope we found is real or just an artifact of sampling noise. The whole inferential engine rests on one fact about the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$ independent: the slope estimate $\widehat{\beta}_1$ is normal, centered at the true β_1, with standard

deviation $\sigma/\sqrt{S_{xx}}$. Replace the unknown σ by $s = \sqrt{\mathrm{SSE}/(n-2)}$ and the standardized slope becomes a Student t on $n-2$ degrees of freedom. That single pivot, $(\widehat{\beta}_1 - \beta_1)/\operatorname{SE}(\widehat{\beta}_1) \sim t_{n-2}$, drives every test and confidence interval below: to ask "is there *any* linear effect?" you test $H_0\colon \beta_1 = 0$; to ask "is the slope as large as theory predicts?" you test $H_0\colon \beta_1 = c$; and to report a range of credible slopes you invert the test into a t-interval. The intercept β_0 has its own standard error and the same t_{n-2} machinery. Recognize this section whenever the question is about a *coefficient of the line* — its sign, its size, or whether it could be zero — as opposed to predicting a future y (§4) or summarizing strength by r (§5).

Which slope/intercept inference does the problem call for?

- "Is there a **real linear relationship**?" / "Does y depend on x at all?" ⇒ two-sided test of $H_0\colon \beta_1 = 0$ using $t_{\text{obs}} = \widehat{\beta}_1/\operatorname{SE}(\widehat{\beta}_1)$ on $n-2$ df.
- "Does y **increase** (or decrease) with x?" — a directional claim ⇒ one-sided test of $H_0\colon \beta_1 = 0$ vs $H_1\colon \beta_1 > 0$ (or < 0); all of α in one tail.
- "Is the slope equal to a **specific nonzero value** c?" (a physical constant, a break-even rate) ⇒ test $H_0\colon \beta_1 = c$ with $t_{\text{obs}} = (\widehat{\beta}_1 - c)/\operatorname{SE}(\widehat{\beta}_1)$ — subtract c, not 0.
- "Give a **range of plausible slopes**" ⇒ the interval $\widehat{\beta}_1 \pm t_{\alpha/2,\,n-2}\ \operatorname{SE}(\widehat{\beta}_1)$.
- Same questions about the **intercept** (the mean of Y at $x = 0$) ⇒ the identical t_{n-2} machinery with $\operatorname{SE}(\widehat{\beta}_0) = s\sqrt{\frac{1}{n} + \bar{x}^2/S_{xx}}$.

Do not confuse: failing to reject $\beta_1 = 0$ is *not* proof that y and x are unrelated — the relationship may be nonlinear, or the test may simply lack power. And the test of $\beta_1 = 0$ is mathematically the *same* test as the correlation test $r = 0$ (§5); they never disagree.

11.3.1 The standard errors and the t pivot

Theorem 11.3.1 (Sampling distribution of the slope estimate)**.** *Under the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with ε_i iid $\mathrm{N}(0, \sigma^2)$, the least-squares slope is an unbiased normal estimator,*

$$\widehat{\beta}_1 \sim \mathrm{N}\left(\beta_1,\ \frac{\sigma^2}{S_{xx}}\right), \qquad S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Since σ^2 is unknown, estimate it by $s^2 = \mathrm{SSE}/(n-2)$. Then

$$\frac{\widehat{\beta}_1 - \beta_1}{s/\sqrt{S_{xx}}} \ \sim\ t_{\,n-2}, \qquad \operatorname{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}}.$$

Proof. Each $\widehat{\beta}_1 = \sum_i c_i Y_i$ with weights $c_i = (x_i - \bar{x})/S_{xx}$ is a linear combination of independent normals, hence normal. Its mean is $\sum_i c_i(\beta_0 + \beta_1 x_i) = \beta_1$ (the β_0 term cancels because $\sum_i c_i = 0$, and $\sum_i c_i x_i = 1$), so $\widehat{\beta}_1$ is unbiased. Its variance is $\sigma^2 \sum_i c_i^2 = \sigma^2/S_{xx}$. Independently of $\widehat{\beta}_1$, the residual sum of squares satisfies $\mathrm{SSE}/\sigma^2 \sim \chi^2_{n-2}$. A normal divided by the square root of an independent chi-square over its df is by definition t_{n-2}; the σ cancels, leaving s in the standard error. □

Theorem 11.3.2 (Tests and intervals for the slope)**.** *Write $\operatorname{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$. To test $H_0\colon \beta_1 = c$ compute*

$$t_{obs} = \frac{\widehat{\beta}_1 - c}{\operatorname{SE}(\widehat{\beta}_1)}, \qquad \textit{compare to } t_{n-2}.$$

The rejection rules at level α are

$$\begin{aligned} H_1\colon \beta_1 \neq c\colon &\quad |t_{obs}| > t_{\alpha/2,\,n-2}, \\ H_1\colon \beta_1 > c\colon &\quad t_{obs} > t_{\alpha,\,n-2}, \\ H_1\colon \beta_1 < c\colon &\quad t_{obs} < -t_{\alpha,\,n-2}. \end{aligned}$$

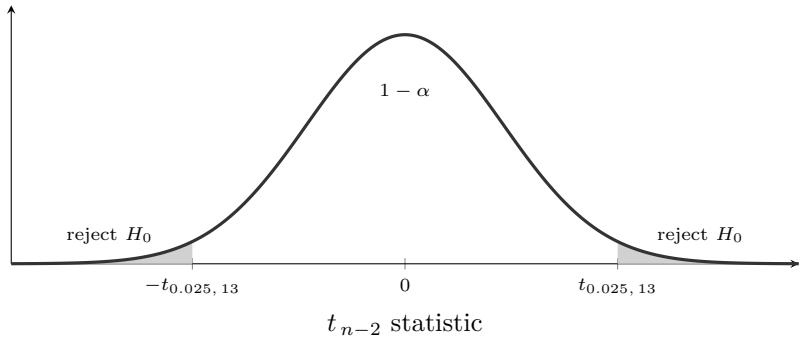
A $100(1-\alpha)\%$ confidence interval for β_1 is $\widehat{\beta}_1 \pm t_{\alpha/2,\,n-2}\ \operatorname{SE}(\widehat{\beta}_1)$.

Theorem 11.3.3 (Inference for the intercept)**.** *The intercept estimate is normal and unbiased with*

$$\widehat{\beta}_0 \sim \mathrm{N}\Big(\beta_0,\ \sigma^2\Big(\tfrac{1}{n} + \tfrac{\bar{x}^2}{S_{xx}}\Big)\Big), \qquad \mathrm{SE}(\widehat{\beta}_0) = s\sqrt{\tfrac{1}{n} + \tfrac{\bar{x}^2}{S_{xx}}}.$$

Then $(\widehat{\beta}_0 - \beta_0)/\,\mathrm{SE}(\widehat{\beta}_0) \sim t_{n-2}$*, and tests and a confidence interval for* β_0 *follow exactly as for the slope.*

Remark. Rejecting $H_0\colon \beta_1 = 0$ is the formal statement of "x carries information about y," because $\beta_1 = 0$ is precisely the line that ignores x (a flat line at height β_0). The standard error shrinks as S_{xx} grows: spreading the x-values farther apart, or taking more data, sharpens every slope inference. This is why a well-designed experiment pushes the predictor to its extremes.



Two-tailed rejection region for $H_0\colon \beta_1 = 0$ *on* $n-2 = 13$ *df with* $\alpha = 0.05$*. Reject when* $|t_{obs}| > t_{0.025,\,13} = 2.16$*. A one-sided alternative keeps all of* α *in a single tail, which moves the cutoff inward to* $t_{0.05,\,13} = 1.77$*.*

11.3.2 Two-sided tests: is there a linear effect at all?

Example 11.3.4 (Fertilizer dose and crop yield)**.** An agronomist applies nitrogen at $n = 15$ rates x (kg/ha) and records yield y (t/ha). The summary statistics are

$$S_{xx} = 2500, \qquad \widehat{\beta}_1 = 0.0182, \qquad s = 0.46.$$

Test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$: is yield linearly related to nitrogen?

Which rule, and why? The question is whether *any* linear effect exists, with no direction specified, so a two-sided t-test of $H_0\colon \beta_1 = 0$ on $n - 2 = 13$ df. This is inference about a slope, not a prediction at a new dose (that would be §4).

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{0.46}{\sqrt{2500}} = \frac{0.46}{50} = 0.0092.$$

Step 2: test statistic.

$$t_{\mathrm{obs}} = \frac{\widehat{\beta}_1 - 0}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{0.0182}{0.0092} = 1.978.$$

Step 3: critical value and decision. With df $= 13$, $t_{0.025,\,13} = 2.160$. Since $|1.978| < 2.160$, do not reject.

$$\boxed{|t_{\mathrm{obs}}| = 1.98 < t_{0.025,\,13} = 2.16 \ \Rightarrow\ \text{fail to reject } H_0;\ \text{no significant linear effect at } \alpha = 0.05.}$$

The estimated slope is positive but, on these 15 plots, not separable from zero at the 5% level.

Example 11.3.5 (Concrete curing time and strength)**.** A materials lab tests $n = 12$ concrete cylinders at varying curing times x (days) and measures compressive strength y (MPa). Computations give

$$S_{xx} = 840, \qquad \widehat{\beta}_1 = 1.35, \qquad s = 3.2.$$

At $\alpha = 0.01$, is there a linear relationship between curing time and strength?

Which rule, and why? Two-sided $H_0\colon \beta_1 = 0$ test on $n-2=10$ df, because we are asked only whether a linear relationship exists, at a strict $\alpha = 0.01$.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{3.2}{\sqrt{840}} = \frac{3.2}{28.98} = 0.1104.$$

Step 2: statistic.

$$t_{\text{obs}} = \frac{1.35}{0.1104} = 12.23.$$

Step 3: decision. With df $= 10$, $t_{0.005,\,10} = 3.169$. Since $12.23 \gg 3.169$, reject decisively.

$$\boxed{t_{\text{obs}} = 12.23 > t_{0.005,\,10} = 3.169 \Rightarrow \text{reject } H_0\text{; curing time has a strong linear effect on strength.}}$$

Here the slope stands many standard errors from zero — overwhelming evidence of a real effect.

11.3.3 Confidence intervals for the slope

Example 11.3.6 (Slope CI for the concrete data)**.** Using the concrete data above ($n = 12$, $\widehat{\beta}_1 = 1.35$, $\mathrm{SE}(\widehat{\beta}_1) = 0.1104$), give a 95% confidence interval for the true slope β_1.

Which rule, and why? "Range of plausible slopes" calls for the t-interval $\widehat{\beta}_1 \pm t_{\alpha/2,\,n-2}\,\mathrm{SE}(\widehat{\beta}_1)$, the inversion of the two-sided test on $n-2=10$ df.

Solution.

Step 1: critical value. For 95% on 10 df, $t_{0.025,\,10} = 2.228$.

Step 2: margin and interval.

$$\text{margin} = 2.228 \times 0.1104 = 0.246,$$
$$\beta_1 \in 1.35 \pm 0.246 = (1.104,\ 1.596).$$

$$\boxed{1.10 < \beta_1 < 1.60 \text{ MPa per day (95\% CI).}}$$

The interval excludes 0, consistent with the rejection at $\alpha = 0.01$ above; each extra curing day adds between 1.1 and 1.6 MPa.

Example 11.3.7 (Slope CI for advertising and sales)**.** A retailer fits sales y (\$000) against weekly ad spend x (\$00) over $n = 22$ weeks and obtains $\widehat{\beta}_1 = 2.7$ with $S_{xx} = 145$ and $s = 4.1$. Construct a 90% confidence interval for the slope and interpret it.

Which rule, and why? A 90% t-interval for β_1 on $n-2=20$ df. The retailer wants a credible range for the marginal return on advertising — a statement about the coefficient, not a forecast.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{4.1}{\sqrt{145}} = \frac{4.1}{12.042} = 0.3405.$$

Step 2: critical value. For 90% on 20 df, $t_{0.05,\,20} = 1.725$.

Step 3: interval.

$$\text{margin} = 1.725 \times 0.3405 = 0.587,$$
$$\beta_1 \in 2.7 \pm 0.587 = (2.113,\ 3.287).$$

$$\boxed{2.11 < \beta_1 < 3.29 \text{ (90\% CI).}}$$

Each extra \$100 of weekly ad spend is associated with between \$2,110 and \$3,290 in added sales; since the interval lies well above 0, the effect is significant at the 10% level.

11.3.4 One-sided tests: does y move in a stated direction?

Example 11.3.8 (Does drug dose lower blood pressure?)**.** A clinical study gives $n = 18$ patients varying daily doses x (mg) of an antihypertensive and records the drop in systolic blood pressure y (mmHg). Fitting yields $\widehat{\beta}_1 = 0.42$, $S_{xx} = 320$, $s = 1.9$. The researchers claim the drug *increases* the BP drop with dose. Test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 > 0$ at $\alpha = 0.05$.

Which rule, and why? A directional claim ("increases with dose") is one-sided: $H_1 : \beta_1 > 0$, all of α in the upper tail, $t_{n-2} = t_{16}$. Using a two-sided cutoff here would be too conservative for a pre-specified direction.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{1.9}{\sqrt{320}} = \frac{1.9}{17.889} = 0.1062.$$

Step 2: statistic.

$$t_{\mathrm{obs}} = \frac{0.42}{0.1062} = 3.955.$$

Step 3: one-sided critical value. With df $= 16$, the upper-tail cutoff is $t_{0.05,\,16} = 1.746$. Since $3.955 > 1.746$, reject.

$$\boxed{t_{\mathrm{obs}} = 3.96 > t_{0.05,\,16} = 1.746 \;\Rightarrow\; \text{reject } H_0;\ \text{the BP drop rises significantly with dose.}}$$

Because the alternative was directional and the slope is positive, the entire 5% sits in the right tail, lowering the bar from 2.12 (two-sided) to 1.75.

Example 11.3.9 (Does corrosion decrease with coating thickness?)**.** An engineer measures corrosion loss y (mg) on $n = 10$ steel coupons with protective coating of thickness x (microns), expecting thicker coatings to reduce loss. The fit gives $\widehat{\beta}_1 = -0.58$, $S_{xx} = 180$, $s = 2.4$. Test $H_1 : \beta_1 < 0$ at $\alpha = 0.05$.

Which rule, and why? The prior expectation is that the slope is *negative*, so a lower-tailed one-sided test, $H_1 : \beta_1 < 0$ on $n - 2 = 8$ df. We reject only for sufficiently negative t_{obs}.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{2.4}{\sqrt{180}} = \frac{2.4}{13.416} = 0.1789.$$

Step 2: statistic.

$$t_{\mathrm{obs}} = \frac{-0.58 - 0}{0.1789} = -3.242.$$

Step 3: decision. Lower-tail cutoff on 8 df is $-t_{0.05,\,8} = -1.860$. Since $-3.242 < -1.860$, reject.

$$\boxed{t_{\mathrm{obs}} = -3.24 < -t_{0.05,\,8} = -1.860 \Rightarrow \text{reject } H_0;\ \text{corrosion drops significantly with coating thickness.}}$$

11.3.5 Testing a specific nonzero slope

Example 11.3.10 (Is the thermal expansion at the textbook rate?)**.** Physics predicts a metal rod elongates at exactly $c = 1.2$ (units: 10^{-5} m per degree C). A calibration run of $n = 16$ measurements of length y versus temperature x gives $\widehat{\beta}_1 = 1.31$, $S_{xx} = 540$, $s = 0.62$. Test $H_0 : \beta_1 = 1.2$ against $H_1 : \beta_1 \neq 1.2$ at $\alpha = 0.05$.

Which rule, and why? The hypothesized slope is a nonzero constant $c = 1.2$, so subtract c in the numerator: $t_{\mathrm{obs}} = (\widehat{\beta}_1 - c)/\,\mathrm{SE}(\widehat{\beta}_1)$ on $n - 2 = 14$ df. Testing against 0 would answer the wrong question — we already know there is an effect; we ask whether it matches theory.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{0.62}{\sqrt{540}} = \frac{0.62}{23.238} = 0.02668.$$

Step 2: statistic centered at $c = 1.2$.

$$t_{\text{obs}} = \frac{1.31 - 1.2}{0.02668} = \frac{0.11}{0.02668} = 4.123.$$

Step 3: decision. With df $= 14$, $t_{0.025,\,14} = 2.145$. Since $|4.123| > 2.145$, reject.

$t_{\text{obs}} = 4.12 > t_{0.025,\,14} = 2.145 \;\Rightarrow$ reject H_0; the measured rate differs from the textbook 1.2.

Note that a test of $\beta_1 = 0$ here would have given $t_{\text{obs}} = 1.31/0.02668 = 49.1$ — impressively large but irrelevant, since the live question is agreement with 1.2, not existence of an effect.

Example 11.3.11 (Does the exchange respond one-for-one?)**.** A finance analyst regresses a stock's daily return y on the market return x for $n = 120$ days and gets $\widehat{\beta}_1 = 1.18$ (the "beta" of the stock), $S_{xx} = 0.094$, $s = 0.011$. Efficient-market folklore says $\beta_1 = 1$. Test $H_0\colon \beta_1 = 1$ vs $H_1\colon \beta_1 \neq 1$ at $\alpha = 0.05$.

Which rule, and why? A specific nonzero null $c = 1$, two-sided, on $n - 2 = 118$ df — with df this large the t cutoff is essentially the normal 1.96. Center the statistic at 1, not 0.

Solution.

Step 1: standard error.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{0.011}{\sqrt{0.094}} = \frac{0.011}{0.30659} = 0.03588.$$

Step 2: statistic centered at 1.

$$t_{\text{obs}} = \frac{1.18 - 1}{0.03588} = \frac{0.18}{0.03588} = 5.017.$$

Step 3: decision. With df $= 118$, $t_{0.025,\,118} \approx 1.98$. Since $5.017 > 1.98$, reject.

$t_{\text{obs}} = 5.02 > t_{0.025,\,118} \approx 1.98 \Rightarrow$ reject H_0; the stock is significantly more volatile than the market ($\beta_1 > 1$).

11.3.6 Inference for the intercept

Example 11.3.12 (Is there a baseline reaction rate at zero catalyst?)**.** A chemist regresses reaction rate y (mol/s) on catalyst concentration x (mM) using $n = 14$ runs. The output reports $\widehat{\beta}_0 = 0.85$, $s = 0.40$, $\bar{x} = 6.0$, and $S_{xx} = 210$. Test whether the intercept — the rate at *zero* catalyst — differs from 0, at $\alpha = 0.05$.

Which rule, and why? The question is about β_0, the mean response at $x = 0$, so the intercept t-test with $\mathrm{SE}(\widehat{\beta}_0) = s\sqrt{1/n + \bar{x}^2/S_{xx}}$ on $n - 2 = 12$ df. A slope test would not answer it.

Solution.

Step 1: standard error of the intercept.

$$\begin{aligned}\mathrm{SE}(\widehat{\beta}_0) &= s\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}} = 0.40\sqrt{\frac{1}{14} + \frac{6.0^2}{210}} \\ &= 0.40\sqrt{0.07143 + 0.17143} = 0.40\sqrt{0.24286} = 0.40(0.49281) = 0.1971.\end{aligned}$$

Step 2: statistic.

$$t_{\text{obs}} = \frac{\widehat{\beta}_0 - 0}{\mathrm{SE}(\widehat{\beta}_0)} = \frac{0.85}{0.1971} = 4.313.$$

Step 3: decision. With df $= 12$, $t_{0.025,\,12} = 2.179$. Since $|4.313| > 2.179$, reject.

$t_{\text{obs}} = 4.31 > t_{0.025,\,12} = 2.179 \;\Rightarrow$ reject H_0; there is a nonzero baseline rate at zero catalyst.

Example 11.3.13 (Confidence interval for the intercept)**.** For the same chemistry fit ($\widehat{\beta}_0 = 0.85$, $\mathrm{SE}(\widehat{\beta}_0) = 0.1971$, $n = 14$), give a 95% confidence interval for the baseline rate β_0.

Which rule, and why? A t-interval for the intercept, $\widehat{\beta}_0 \pm t_{\alpha/2,\,n-2}\,\mathrm{SE}(\widehat{\beta}_0)$, on 12 df — the inversion of the test just performed.

Solution.

Step 1: critical value. $t_{0.025,\,12} = 2.179$.

Step 2: interval.

$$\text{margin} = 2.179 \times 0.1971 = 0.4295,$$
$$\beta_0 \in 0.85 \pm 0.4295 = (0.420,\ 1.280).$$

$$\boxed{0.42 < \beta_0 < 1.28 \text{ mol/s (95\% CI).}}$$

The interval excludes 0, matching the rejection above; the baseline rate is between 0.42 and 1.28.

11.3.7 Reading a p-value off the t statistic

Example 11.3.14 (Reporting a p-value for the slope)**.** A hydrologist regresses stream discharge y on rainfall x for $n = 9$ storms and finds $\widehat{\beta}_1 = 3.4$, $\mathrm{SE}(\widehat{\beta}_1) = 1.1$. Report the two-sided p-value for $H_0\colon \beta_1 = 0$ and state the conclusion at $\alpha = 0.05$.

Which rule, and why? We want the p-value, not just a yes/no — the two-sided tail area of the t_{n-2} distribution beyond $|t_{\text{obs}}|$ on $n - 2 = 7$ df.

Solution.

Step 1: statistic.

$$t_{\text{obs}} = \frac{3.4}{1.1} = 3.091.$$

Step 2: bracket the tail area on 7 df. From a t_7 table, $t_{0.01,\,7} = 2.998$ and $t_{0.005,\,7} = 3.499$. Since $2.998 < 3.091 < 3.499$, the one-tail area lies between 0.005 and 0.01, so the two-sided

$$0.01 < p < 0.02.$$

Step 3: decision. The whole p-interval falls below 0.05.

$$\boxed{p \approx 0.017 < 0.05 \ \Rightarrow\ \text{reject } H_0;\ \text{discharge rises significantly with rainfall.}}$$

A p-value reports *how* significant, not merely whether: here the evidence is strong but not overwhelming.

11.3.8 Contrast: what failing to reject does — and does not — mean

Example 11.3.15 (A non-significant slope hides a real curved relationship)**.** A biologist regresses enzyme activity y on temperature x at $n = 11$ temperatures spanning a wide range. Activity rises to a peak near the middle of the range and falls off at both ends (the classic inverted-U of enzyme kinetics). The straight-line fit returns $\widehat{\beta}_1 = 0.03$, $\mathrm{SE}(\widehat{\beta}_1) = 0.21$, hence $t_{\text{obs}} = 0.03/0.21 = 0.143$, far short of $t_{0.025,\,9} = 2.262$, so $H_0\colon \beta_1 = 0$ is *not* rejected. Does this prove activity is unrelated to temperature?

Which rule, and why? This is the trap the strategy box warns against: a non-significant *slope* tests only the *linear* part of the relationship. The correct reading requires recognizing that "no linear effect" is not "no effect."

Solution.

Step 1: what the test actually checked. The t-test asks whether the best-fitting *straight line* has nonzero slope. For a symmetric inverted-U, the rising and falling halves cancel, so the best straight line is nearly flat — $\widehat{\beta}_1 \approx 0$ — even though activity depends strongly on temperature.

Step 2: the correct conclusion.

fail to reject $\beta_1 = 0 \not\Rightarrow$ no relationship; the dependence is real but *nonlinear*.

Step 3: the two other reasons a true slope can go undetected. Even for a genuinely linear relationship, failing to reject can stem from **low power**: a small n, a small S_{xx} (predictor values bunched together), or a large s all inflate $\mathrm{SE}(\widehat{\beta}_1)$ and shrink t_{obs}. "Not significant" means "not enough evidence," never "the effect is zero." Always plot the data before trusting a flat slope, and report a confidence interval — a wide interval straddling 0 signals an inconclusive test, not an established null.

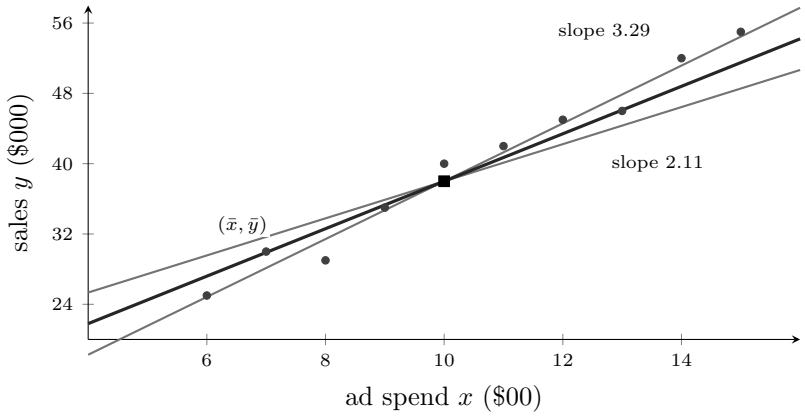
11.3.9 Contrast: the slope CI as a fan of plausible lines

Example 11.3.16 (Visualizing slope uncertainty)**.** Reconsider the advertising fit ($\widehat{\beta}_1 = 2.7$, 90% slope interval $(2.11, 3.29)$, with the line pinned to pass through $(\bar{x}, \bar{y}) = (10, 38)$). Sketch what the slope interval means geometrically and contrast it with the strength question.

Which rule, and why? A confidence interval for β_1 is a band of *slopes*; pivoting every plausible slope through the centroid $(\bar{x}, \bar{y})$ draws a fan of plausible lines. This is distinct from §5's correlation, which scores how tightly points hug *one* line.

Solution.

Step 1: the fan. Each slope in $(2.11, 3.29)$, drawn through $(\bar{x}, \bar{y})$, is a line the data cannot rule out at 90% confidence. The fan is narrow because $\mathrm{SE}(\widehat{\beta}_1)$ is small; a wide fan would mean the data barely constrain the slope.



The fitted line (heavy) and the two extreme lines of the 90% slope interval, all pivoting through the centroid $(\bar{x}, \bar{y}) = (10, 38)$. The thin pair brackets every slope the data leave credible; the narrowness of the fan is exactly what "the slope is precisely estimated" looks like.

Step 2: the contrast. The fan answers "how sure are we of the rate of change?" A *tight* fan can still sit amid widely scattered points (large s, modest R^2), and a *wide* fan can occur with tightly clustered points if S_{xx} is tiny. Slope precision and goodness of fit are different questions: the slope CI is governed by $\mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$, while r and R^2 (§5–6) measure how much of S_{yy} the line explains. Do not read a narrow fan as a high R^2, nor the reverse.

11.4 Estimating the Mean Response and Predicting a New Observation

Once the line is fitted you almost always want a value at some specific x_0: what is y when the dose is x_0, the temperature is x_0, the age is x_0? There are *two* distinct questions hiding behind that one number, and

the whole section turns on telling them apart. The first asks for the **mean response**, the average y over the whole population of units that share $x = x_0$ — a fixed but unknown number $\mu_{Y|x_0} = \beta_0 + \beta_1 x_0$ that the fitted height $\widehat{y}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$ estimates. The second asks for a **single new observation**, the actual y of *one* future unit at x_0, which scatters around that mean by its own error term ε. Both are centered at the same point estimate $\widehat{y}_0$, but the prediction interval is always wider, because it must absorb one extra σ^2 of individual noise on top of the uncertainty in where the line sits. And both widen as x_0 moves away from $\overline{X}$: the line is pinned down best near the center of the data and pivots more freely out at the ends, which is the bowtie (hourglass) shape you will draw. Recognise which interval a question wants by a single test — "average of many" versus "one of them."

Mean response, or a single new observation?

- Asked for the **average** y among *all* units at x_0 ("the mean yield at x_0," "expected response," "where does the true line sit")? $\Rightarrow$ **confidence interval for the mean response** $\widehat{y}_0 \pm t_{\alpha/2,\,n-2}\, s\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}}$.
- Asked for **one** future unit's actual y at x_0 ("predict the next reading," "what value will a new patient give," an interval that should *contain a single observation*)? $\Rightarrow$ **prediction interval** $\widehat{y}_0 \pm t_{\alpha/2,\,n-2}\, s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}}$.
- Same center $\widehat{y}_0$, same t on $n-2$ df; the only difference is the extra 1+ under the root for prediction. That 1 is the variance of the new error term ε, present for one new observation and absent when you only locate the mean.
- Both half-widths grow with $(x_0 - \overline{X})^2$, so they are **narrowest at** $x_0 = \overline{X}$ and flare out symmetrically — the hourglass. Far outside the observed x-range the formula keeps widening but the *model* is no longer trustworthy: that is extrapolation, a separate hazard (see the contrast below).

Do not confuse the two: a confidence interval for the mean response can be made as narrow as you like by collecting more data ($1/n \to 0$), but a prediction interval cannot shrink below $\pm t_{\alpha/2,\,n-2}\, s$ no matter how large n grows — the lone 1 never goes away, because a single new unit always carries its own σ^2.

11.4.1 The two intervals at a fixed x_0

Throughout, the fitted line is $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$ from n pairs under the model $Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$ with $\varepsilon_i \sim \mathrm{N}(0, \sigma^2)$ independent, and $s^2 = \mathrm{SSE}/(n-2)$ is the usual variance estimate on $n-2$ degrees of freedom.

Definition 11.4.1 (Mean response and a new observation at x_0)**.** Fix a value x_0 of the predictor. The **mean response** is the (non-random) height of the true line,

$$\mu_{Y|x_0} = \mathbb{E}(Y \mid x = x_0) = \beta_0 + \beta_1 x_0,$$

estimated by the fitted height $\widehat{y}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$. A **new observation** at x_0 is a future random $Y_0 = \beta_0 + \beta_1 x_0 + \varepsilon_0$, where $\varepsilon_0 \sim \mathrm{N}(0, \sigma^2)$ is *independent* of the data used to fit the line.

Theorem 11.4.2 (Variance of the fitted height)**.** *The estimator $\widehat{y}_0 = \widehat{\beta}_0 + \widehat{\beta}_1 x_0$ is unbiased for $\mu_{Y|x_0}$, normal, and*

$$\mathrm{Var}(\widehat{y}_0) = \sigma^2\left(\frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}\right).$$

Proof. Write the fitted height in the convenient centered form $\widehat{y}_0 = \overline{Y} + \widehat{\beta}_1(x_0 - \overline{X})$, valid because $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}$. The two pieces $\overline{Y}$ and $\widehat{\beta}_1$ are uncorrelated (a standard least-squares fact: $\mathrm{Cov}(\overline{Y}, \widehat{\beta}_1) = 0$), so the variances add:

$$\begin{aligned}\mathrm{Var}(\widehat{y}_0) &= \mathrm{Var}(\overline{Y}) + (x_0 - \overline{X})^2\,\mathrm{Var}(\widehat{\beta}_1) \\ &= \frac{\sigma^2}{n} + (x_0 - \overline{X})^2 \cdot \frac{\sigma^2}{S_{xx}} = \sigma^2\left(\frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}\right),\end{aligned}$$

using $\mathrm{Var}(\overline{Y}) = \sigma^2/n$ and $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$. Unbiasedness and normality follow because $\widehat{y}_0$ is a linear combination of the independent normal Y_i. □

Remark. The $(x_0 - \overline{X})^2/S_{xx}$ term is the whole story of the hourglass. At $x_0 = \overline{X}$ it vanishes and $\mathrm{Var}(\widehat{y}_0) = \sigma^2/n$ is as small as it gets — the line is best determined at the center of mass of the data. As x_0 slides away in either direction the term grows quadratically, so the band flares out symmetrically about $\overline{X}$.

Theorem 11.4.3 (Confidence interval for the mean response)**.** *A $100(1-\alpha)\%$ confidence interval for the mean response $\mu_{Y|x_0} = \beta_0 + \beta_1 x_0$ is*

$$\widehat{y}_0 \;\pm\; t_{\alpha/2,\,n-2}\; s\sqrt{\frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}}\,.$$

Proof. By the previous theorem $(\widehat{y}_0 - \mu_{Y|x_0})/\sqrt{\mathrm{Var}(\widehat{y}_0)}$ is standard normal. Replacing the unknown σ by s (independent of $\widehat{y}_0$, on $n-2$ df) turns the ratio into a t_{n-2} variable,

$$\frac{\widehat{y}_0 - \mu_{Y|x_0}}{s\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}}} \sim t_{n-2},$$

and inverting $\mathbb{P}(|t_{n-2}| \le t_{\alpha/2,\,n-2}) = 1-\alpha$ gives the interval. □

Theorem 11.4.4 (Prediction interval for a new observation)**.** *A $100(1-\alpha)\%$ prediction interval for a single new observation Y_0 at x_0 is*

$$\widehat{y}_0 \;\pm\; t_{\alpha/2,\,n-2}\; s\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}}\,.$$

Proof. The prediction error is $Y_0 - \widehat{y}_0 = (\beta_0 + \beta_1 x_0 + \varepsilon_0) - \widehat{y}_0$. Since the new ε_0 is independent of the fitted line,

$$\begin{aligned}\mathrm{Var}(Y_0 - \widehat{y}_0) &= \mathrm{Var}(\varepsilon_0) + \mathrm{Var}(\widehat{y}_0)\\ &= \sigma^2 + \sigma^2\left(\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}\right) = \sigma^2\left(1 + \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}\right).\end{aligned}$$

The extra leading σ^2 — the variance of the one new error — is exactly the 1+ under the root. Standardizing the (mean-zero, normal) error by s again yields a t_{n-2} pivot, and inverting gives the stated interval. □

Remark. Three facts to carry away. (i) Same center $\widehat{y}_0$; the intervals differ only in the radius. (ii) The prediction radius is always larger because $1 + \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}} > \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}$. (iii) As $n \to \infty$ the confidence radius $\to 0$ but the prediction radius $\to z_{\alpha/2}\,\sigma$: knowing the line perfectly still leaves each new unit its own irreducible scatter.

11.4.2 The headline contrast: mean response versus a new observation

Example 11.4.5 (A baseline regression to reuse)**.** A horticulturist records the height y (cm) of $n = 10$ seedlings against the daily light exposure x (hours). The summary statistics are

$$n = 10, \quad \overline{X} = 6.0, \quad \overline{Y} = 18.4, \quad S_{xx} = 42.0, \quad S_{xy} = 63.0, \quad \mathrm{SSE} = 12.6.$$

Establish the fitted line and s for use in the examples that follow.

Which rule, and why? Pure setup — the slope/intercept come from least squares ($\widehat{\beta}_1 = S_{xy}/S_{xx}$, $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1\overline{X}$) and the spread from $s^2 = \mathrm{SSE}/(n-2)$. We compute these once because every interval below reuses them.

Solution.

Step 1: slope and intercept.

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{63.0}{42.0} = 1.50,$$
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1\overline{X} = 18.4 - 1.50(6.0) = 9.40.$$

Step 2: error standard deviation.

$$s^2 = \frac{\text{SSE}}{n-2} = \frac{12.6}{8} = 1.575,$$
$$s = \sqrt{1.575} = 1.255.$$

$$\boxed{\widehat{y} = 9.40 + 1.50x, \qquad s = 1.255 \text{ cm on } n-2 = 8 \text{ df.}}$$

These two results — the line and s — feed directly into the mean-response and prediction intervals that follow.

Example 11.4.6 (Confidence interval for the mean response at $x_0 = 7$)**.** For the seedling fit ($\widehat{y} = 9.40 + 1.50x$, $s = 1.255$, $S_{xx} = 42$, $\overline{X} = 6$, $n = 10$), give a 95% confidence interval for the *mean* height of all seedlings grown at $x_0 = 7$ hours of light.

Which rule, and why? "Mean height of all seedlings at x_0" is the average over the whole population at that light level — the mean-response interval, which carries $\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}$ but *no* extra 1.

Solution.

Step 1: point estimate. $\widehat{y}_0 = 9.40 + 1.50(7) = 19.90$ cm.

Step 2: standard error of the fitted height. With $x_0 - \overline{X} = 7 - 6 = 1$,

$$\text{se}(\widehat{y}_0) = s\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}} = 1.255\sqrt{\frac{1}{10} + \frac{1^2}{42}}$$
$$= 1.255\sqrt{0.10000 + 0.02381} = 1.255\sqrt{0.12381} = 0.4416.$$

Step 3: assemble. With $t_{0.025,\,8} = 2.306$,

$$19.90 \;\pm\; 2.306(0.4416)$$
$$19.90 \;\pm\; 1.018.$$

$$\boxed{\mu_{Y|x_0} \in (18.88,\ 20.92) \text{ cm at 95\% confidence (mean response at } x_0 = 7).}$$

We are 95% confident the *average* height of seedlings at 7 hours of light lies between about 18.9 and 20.9 cm.

Example 11.4.7 (CONTRAST: prediction interval for ONE new seedling at $x_0 = 7$)**.** Same fit, same $x_0 = 7$. Now give a 95% interval for the height of *one new* seedling grown at 7 hours, and compare it side by side with the mean-response interval just computed.

Which rule, and why? "One new seedling" is a single future observation, so the prediction interval — identical to the last one except for the extra 1+ under the root, which adds that seedling's own error variance. This is the headline mean-vs-new contrast.

Solution.

Step 1: same center. $\widehat{y}_0 = 19.90$ cm, exactly as before.

Step 2: standard error WITH the extra 1.

$$\text{se(pred)} = s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}} = 1.255\sqrt{1 + 0.10000 + 0.02381}$$
$$= 1.255\sqrt{1.12381} = 1.255(1.0601) = 1.3304.$$

Step 3: assemble. With $t_{0.025,\,8} = 2.306$,

$$19.90 \;\pm\; 2.306(1.3304)$$
$$19.90 \;\pm\; 3.068.$$

Step 4: place the two intervals together.

$$\begin{aligned} \text{mean response:} &\quad (18.88,\ 20.92), \quad \text{half-width } 1.018, \\ \text{new observation:} &\quad (16.83,\ 22.97), \quad \text{half-width } 3.068. \end{aligned}$$

$$\boxed{Y_0 \in (16.83,\ 22.97) \text{ cm (95\% prediction), about } 3 \times \text{the mean interval at the same } x_0.}$$

Same midpoint 19.90, but the prediction band is roughly three times as wide here: the 1 under the root (a whole σ^2 of individual scatter) dominates the small 0.124 from estimating the line.

11.4.3 Why the prediction interval cannot collapse

Example 11.4.8 (What happens to each interval as n grows)**.** Suppose the seedling study were repeated with the *same* $s \approx 1.255$ and the same $x_0 = \overline{X}$, but with $n = 10$, $n = 100$, and $n = 1000$. Track the 95% mean-response half-width and the prediction half-width (use $z_{0.025} = 1.96$ as the large-n t limit) and explain why one shrinks to zero and the other does not.

Which rule, and why? At $x_0 = \overline{X}$ the $(x_0 - \overline{X})^2/S_{xx}$ term is 0, so the two half-widths reduce to $t\,s\sqrt{1/n}$ (mean) and $t\,s\sqrt{1 + 1/n}$ (prediction). Comparing their limits isolates the irreducible 1.

Solution.

Step 1: mean-response half-width $t\,s/\sqrt{n}$.

$$\begin{aligned} n = 10: &\quad 1.96(1.255)/\sqrt{10} = 0.778, \\ n = 100: &\quad 1.96(1.255)/\sqrt{100} = 0.246, \\ n = 1000: &\quad 1.96(1.255)/\sqrt{1000} = 0.0778. \end{aligned}$$

Step 2: prediction half-width $t\,s\sqrt{1 + 1/n}$.

$$\begin{aligned} n = 10: &\quad 1.96(1.255)\sqrt{1.100} = 2.580, \\ n = 100: &\quad 1.96(1.255)\sqrt{1.010} = 2.471, \\ n = 1000: &\quad 1.96(1.255)\sqrt{1.001} = 2.461. \end{aligned}$$

Step 3: read the limits. The mean-response half-width $\to 1.96(1.255)\sqrt{1/n} \to 0$, while the prediction half-width $\to 1.96(1.255)\sqrt{1} = 2.460$.

$$\boxed{\text{mean response} \to 0, \qquad \text{prediction} \to z_{\alpha/2}\,s = 2.460 > 0.}$$

More data eventually pins the line down exactly, so the mean interval can be made arbitrarily tight; but each new seedling still has its own σ^2 of scatter, so the prediction interval plateaus at $\pm z_{\alpha/2}\,s$ and never closes.

11.4.4 Both intervals widen away from the center

Example 11.4.9 (Mean-response interval far from $\overline{X}$: $x_0 = 10$)**.** For the seedling fit, give the 95% confidence interval for the mean response at $x_0 = 10$ hours (recall $\overline{X} = 6$, so this is well to the right of center) and compare its width with the $x_0 = 7$ interval.

Which rule, and why? Still the mean-response interval, but now $x_0 - \overline{X} = 4$ instead of 1, so the $(x_0 - \overline{X})^2/S_{xx}$ term grows from $1/42$ to $16/42$ — the hourglass flare. Same formula, larger half-width.

Solution.

Step 1: point estimate. $\widehat{y}_0 = 9.40 + 1.50(10) = 24.40$ cm.

Step 2: standard error. With $x_0 - \overline{X} = 4$,

$$\begin{aligned}\text{se}(\widehat{y}_0) &= 1.255\sqrt{\frac{1}{10} + \frac{4^2}{42}} = 1.255\sqrt{0.10000 + 0.38095} \\ &= 1.255\sqrt{0.48095} = 1.255(0.6935) = 0.8703.\end{aligned}$$

Step 3: assemble and compare. With $t_{0.025,\,8} = 2.306$,

$$24.40 \;\pm\; 2.306(0.8703) = 24.40 \pm 2.007.$$

$$\boxed{\mu_{Y|x_0} \in (22.39,\ 26.41) \text{ at } x_0 = 10;\ \text{half-width } 2.007 \text{ vs. } 1.018 \text{ at } x_0 = 7.}$$

Moving from $x_0 = 7$ (one hour off center) to $x_0 = 10$ (four hours off) nearly doubles the half-width, from 1.02 to 2.01: the band fans out as $(x_0 - \overline{X})^2$ grows.

Example 11.4.10 (Prediction interval far from $\overline{X}$: $x_0 = 10$)**.** At the same $x_0 = 10$, give the 95% prediction interval for a single new seedling and contrast its flare with that of the mean-response interval.

Which rule, and why? Prediction interval at an off-center x_0: the radius carries both the 1 and the now-large $(x_0 - \overline{X})^2/S_{xx}$. Comparing its growth with the mean interval's shows the 1 *dampens* the relative flare.

Solution.

Step 1: standard error.

$$\begin{aligned}\text{se(pred)} &= 1.255\sqrt{1 + 0.10000 + 0.38095} = 1.255\sqrt{1.48095} \\ &= 1.255(1.2169) = 1.5272.\end{aligned}$$

Step 2: assemble.

$$24.40 \;\pm\; 2.306(1.5272) = 24.40 \pm 3.522.$$

Step 3: contrast the flare. From $x_0 = 7$ to $x_0 = 10$ the half-widths move

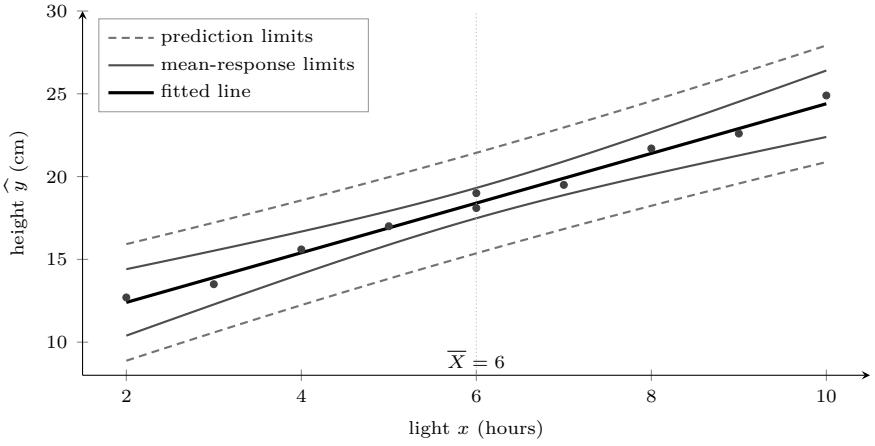
$$\begin{aligned}\text{mean: } 1.018 &\to 2.007 \quad (\times 1.97), \\ \text{prediction: } 3.068 &\to 3.522 \quad (\times 1.15).\end{aligned}$$

$$\boxed{Y_0 \in (20.88,\ 27.92) \text{ at } x_0 = 10 \text{ (95\% prediction).}}$$

Both widen away from $\overline{X}$, but the prediction band flares *less* in relative terms: its constant 1 swamps the $(x_0 - \overline{X})^2/S_{xx}$ term, so the same off-center move that nearly doubles the mean interval lifts the prediction interval by only 15%.

11.4.5 The hourglass bands

The picture below overlays both bands on the seedling fit. The inner curves are the mean-response confidence limits; the outer curves are the wider prediction limits. Both pinch to their narrowest directly above $x_0 = \overline{X} = 6$ and bow outward symmetrically on either side — two nested hourglasses sharing a waist.



Nested bands for the seedling fit. The solid inner pair is the 95% mean-response interval, the dashed outer pair the 95% prediction interval; both pinch to a waist above $\overline{X} = 6$ and bow outward, the prediction band always enclosing the mean band. The constant gap between them is the extra 1 under the prediction root.

11.4.6 Half-width as a function of x_0

Example 11.4.11 (Plotting the half-width and locating its minimum)**.** For the seedling fit, treat the 95% mean-response half-width $h(x_0) = t_{0.025,\,8}\, s\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}}$ as a function of x_0. Where is it smallest, and by how much does it grow at $x_0 = 2$ versus $x_0 = 6$?

Which rule, and why? The half-width depends on x_0 only through $(x_0 - \overline{X})^2$, a parabola in x_0 with vertex at $\overline{X}$. So the minimum is exactly at $x_0 = \overline{X}$, and the growth is symmetric — the quantitative version of the hourglass.

Solution.

Step 1: minimum location and value. The term $(x_0 - \overline{X})^2 \geq 0$ is zero only at $x_0 = \overline{X} = 6$, where

$$h(6) = 2.306(1.255)\sqrt{\tfrac{1}{10} + 0} = 2.306(1.255)(0.3162) = 0.915.$$

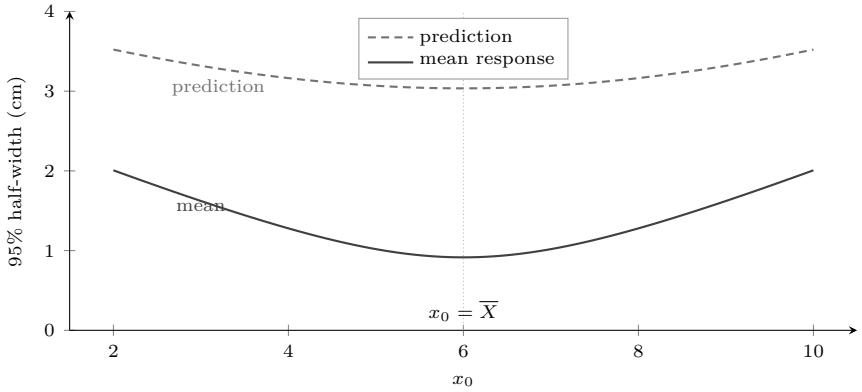
Step 2: value at the edge $x_0 = 2$. Here $x_0 - \overline{X} = -4$, $(x_0 - \overline{X})^2 = 16$:

$$\begin{aligned} h(2) &= 2.306(1.255)\sqrt{0.10000 + \tfrac{16}{42}} \\ &= 2.306(1.255)\sqrt{0.48095} = 2.306(1.255)(0.6935) = 2.007. \end{aligned}$$

Step 3: report the ratio.

$$\boxed{h \text{ minimized at } x_0 = \overline{X} = 6 \ (h = 0.915); \quad h(2) = 2.007 \approx 2.2\, h(6).}$$

The half-width is a U-shaped (parabola-under-a-root) curve bottoming out at $\overline{X}$; four hours off center it has more than doubled. The plot below shows the full curve for both intervals.



Half-width versus x_0 for both intervals. Each is a symmetric U with its minimum exactly at $x_0 = \overline{X} = 6$. The mean-response curve (solid) dips toward 0 at the center; the prediction curve (dashed) sits a fixed distance above it and never falls below $t_{0.025,\,8}\, s \approx 2.9$, the floor set by the lone σ^2.

11.4.7 Choosing the right interval, and the danger of extrapolation

Example 11.4.12 (Reading the question: which interval does it want?)**.** A pharmacokinetics lab fits plasma concentration y (mg/L) against dose x (mg). Three stakeholders ask three things at $x_0 = 50$ mg. (a) Regulatory affairs: "what is the *average* concentration across all patients given 50 mg?" (b) A clinician: "the next patient I dose at 50 mg — what range will *their* reading fall in?" (c) A modeler: "how precisely have we located the true dose-response line at 50 mg?" Match each to the correct interval.

Which rule, and why? The keyword in each request decides it: "average across all patients" and "located the true line" are statements about the *mean* $\beta_0 + \beta_1 x_0$; "the next patient's reading" is a *single new observation.* Only the noun — a mean versus one unit — matters, not the value of x_0.

Solution.

Step 1: (a) average across all patients. This is the mean response $\mu_{Y|x_0} = \beta_0 + \beta_1(50) \Rightarrow$ **confidence interval for the mean**, root $\sqrt{\frac{1}{n} + \frac{(50-\overline{X})^2}{S_{xx}}}$.

Step 2: (b) the next patient's reading. A single future Y_0 at $x_0 = 50 \Rightarrow$ **prediction interval**, root $\sqrt{1 + \frac{1}{n} + \frac{(50-\overline{X})^2}{S_{xx}}}$, strictly wider.

Step 3: (c) precision of the true line. "How precisely is the line located" is again the mean response $\Rightarrow$ **confidence interval for the mean** (same as (a)); its half-width *is* the precision of $\widehat{y}_0$.

(a) mean CI, (b) prediction PI (wider), (c) mean CI.

Two of the three want the mean interval and only the clinician's "one patient" question calls for the wider prediction interval. Misreading (b) as a mean interval would give a falsely tight range and overstate how well you can pin a single patient.

Example 11.4.13 (CONTRAST: the danger of extrapolation far outside the x-range)**.** The seedlings were grown at light levels x from 2 to 10 hours. A grower asks for the 95% prediction interval at $x_0 = 20$ hours — double the largest observed value. Compute the interval mechanically, then explain why the number should not be trusted.

Which rule, and why? Mechanically this is just the prediction interval at $x_0 = 20$. The lesson is that the *formula* keeps producing a finite interval no matter how far out you go, but the *model* ($Y = \beta_0 + \beta_1 x + \varepsilon$, linear, constant variance) was only ever checked on $2 \le x \le 10$. Extrapolation buys a number with no warranty.

Solution.

Step 1: mechanical computation. $\widehat{y}_0 = 9.40 + 1.50(20) = 39.40$; $x_0 - \overline{X} = 14$, so

$$\begin{aligned} \text{se(pred)} &= 1.255\sqrt{1 + 0.10000 + \tfrac{14^2}{42}} = 1.255\sqrt{1 + 0.1 + 4.6667} \\ &= 1.255\sqrt{5.7667} = 1.255(2.4014) = 3.014, \end{aligned}$$

giving $39.40 \pm 2.306(3.014) = 39.40 \pm 6.95$, i.e. $(32.45,\ 46.35)$.

Step 2: why it is untrustworthy. Three failures the formula cannot see: (i) the true relationship may bend (plants saturate or scorch past some light level), so linearity is unverified out here; (ii) the variance may change far from the data; (iii) the $(x_0 - \overline{X})^2/S_{xx}$ term has ballooned, a built-in *warning* that the interval is mostly extrapolation. The model was fitted on $[2, 10]$ and has no evidence to speak about $x = 20$.

$(32.45,\ 46.35)$ is arithmetically correct but unwarranted: $x_0 = 20$ lies far outside $[2, 10]$.

The honest report states the interval *and* flags that $x_0 = 20$ is an extrapolation; the huge $(x_0 - \overline{X})^2/S_{xx}$ contribution is the formula's own way of admitting it has left the data behind.

11.4.8 Two more worked intervals in fresh settings

Example 11.4.14 (Mean response in a manufacturing setting)**.** A process engineer fits tensile strength y (MPa) against annealing time x (min) on $n = 14$ specimens, obtaining $\widehat{y} = 45.0 + 2.20x$, $s = 3.10$, $\overline{X} = 12.0$, $S_{xx} = 180.0$. Give a 90% confidence interval for the mean strength of all specimens annealed for $x_0 = 15$ min.

Which rule, and why? "Mean strength of all specimens" at a fixed time $\Rightarrow$ mean-response confidence interval, root $\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}}$, on $n - 2 = 12$ df at 90% ($\alpha/2 = 0.05$).

Solution.

Step 1: point estimate. $\widehat{y}_0 = 45.0 + 2.20(15) = 78.0$ MPa.

Step 2: standard error. $x_0 - \overline{X} = 3$, so

$$\begin{aligned}\text{se}(\widehat{y}_0) &= 3.10\sqrt{\frac{1}{14} + \frac{3^2}{180}} = 3.10\sqrt{0.07143 + 0.05000} \\ &= 3.10\sqrt{0.12143} = 3.10(0.3485) = 1.0803.\end{aligned}$$

Step 3: assemble. With $t_{0.05,\,12} = 1.782$,

$$78.0 \;\pm\; 1.782(1.0803) = 78.0 \pm 1.925.$$

$$\boxed{\mu_{Y|x_0} \in (76.08,\ 79.93) \text{ MPa at } 90\% \text{ (mean strength, } x_0 = 15).}$$

A 90% level and a moderate off-center distance give a tight band of about ±1.9 MPa around the predicted 78.0 MPa.

Example 11.4.15 (Prediction interval in an environmental setting)**.** An ecologist fits dissolved oxygen y (mg/L) against water temperature x (°C) on $n = 20$ samples: $\widehat{y} = 11.2 - 0.18x$, $s = 0.42$, $\overline{X} = 17.0$, $S_{xx} = 520.0$. Predict, with 95% confidence, the dissolved oxygen of a *single* new sample taken at $x_0 = 22$°C.

Which rule, and why? "A single new sample" is one future observation ⇒ prediction interval, root $\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}}$, on $n - 2 = 18$ df.

Solution.

Step 1: point estimate. $\widehat{y}_0 = 11.2 - 0.18(22) = 11.2 - 3.96 = 7.24$ mg/L.

Step 2: standard error with the extra 1. $x_0 - \overline{X} = 5$, so

$$\begin{aligned}\text{se(pred)} &= 0.42\sqrt{1 + \frac{1}{20} + \frac{5^2}{520}} = 0.42\sqrt{1 + 0.05000 + 0.04808} \\ &= 0.42\sqrt{1.09808} = 0.42(1.0479) = 0.4401.\end{aligned}$$

Step 3: assemble. With $t_{0.025,\,18} = 2.101$,

$$7.24 \;\pm\; 2.101(0.4401) = 7.24 \pm 0.925.$$

$$\boxed{Y_0 \in (6.32,\ 8.16) \text{ mg/L at } 95\% \text{ (single new sample, } x_0 = 22).}$$

The dominant term under the root is the lone 1 (1.0 of the 1.098); the line-fitting uncertainty adds only about 10%, which is why a large $n = 20$ barely narrows a prediction interval.

11.4.9 One-sided bounds

Both intervals admit one-sided forms whenever the question asks for a *bound* in a single direction rather than a two-sided range: replace $t_{\alpha/2,\,n-2}$ by $t_{\alpha,\,n-2}$ and keep only the relevant endpoint.

Example 11.4.16 (One-sided upper prediction bound)**.** For the dissolved-oxygen fit ($\widehat{y} = 11.2 - 0.18x$, $s = 0.42$, $\overline{X} = 17$, $S_{xx} = 520$, $n = 20$), a regulator needs a 95% *upper* bound on the oxygen of one new sample at $x_0 = 22$°C ("with 95% confidence the next reading is no more than —"). Give it, and contrast it with the two-sided interval.

Which rule, and why? A one-directional claim ("no more than") ⇒ a one-sided bound: put all $\alpha = 0.05$ in one tail, so use $t_{0.05,\,18} = 1.734$ (not 2.101) and report only the upper endpoint. It is the prediction root because the subject is a single new sample.

Solution.

Step 1: reuse center and standard error. From the previous example $\widehat{y}_0 = 7.24$ and se(pred) $= 0.4401$.

Step 2: one-sided upper bound. With $t_{0.05,\,18} = 1.734$,

$$\text{upper bound} = 7.24 + 1.734(0.4401) = 7.24 + 0.763 = 8.003.$$

Step 3: contrast with the two-sided interval. The two-sided 95% interval reached to 8.16; the one-sided bound stops lower, at 8.00, because $1.734 < 2.101$ concentrates all the risk in the upper tail.

$$\boxed{Y_0 \le 8.00 \text{ mg/L with 95\% confidence (one-sided upper prediction bound).}}$$

The one-sided bound is tighter on the side that matters precisely because it spends the entire α there; using the two-sided 2.101 for a one-directional claim would be needlessly conservative.

Example 11.4.17 (One-sided lower confidence bound on the mean response)**.** A nutritionist fits weight gain y (kg) on protein intake x (g/day), $n = 16$, $\widehat{y} = 0.5 + 0.030x$, $s = 0.22$, $\overline{X} = 80$, $S_{xx} = 6400$. Give a 99% *lower* confidence bound on the *mean* weight gain at $x_0 = 100$ g/day ("mean gain is at least —").

Which rule, and why? "Mean weight gain" ⇒ the mean-response root (no extra 1); "at least" in one direction ⇒ a one-sided *lower* bound, all $\alpha = 0.01$ in the lower tail, so $t_{0.01,\,14} = 2.624$ and only the lower endpoint.

Solution.

Step 1: point estimate. $\widehat{y}_0 = 0.5 + 0.030(100) = 3.50$ kg.

Step 2: standard error (mean response). $x_0 - \overline{X} = 20$, so

$$\begin{aligned}\text{se}(\widehat{y}_0) &= 0.22\sqrt{\frac{1}{16} + \frac{20^2}{6400}} = 0.22\sqrt{0.06250 + 0.06250} \\ &= 0.22\sqrt{0.12500} = 0.22(0.3536) = 0.0778.\end{aligned}$$

Step 3: one-sided lower bound. With $t_{0.01,\,14} = 2.624$,

$$\text{lower bound} = 3.50 - 2.624(0.0778) = 3.50 - 0.204 = 3.296.$$

$$\boxed{\mu_{Y|x_0} \ge 3.30 \text{ kg with 99\% confidence (one-sided lower mean bound).}}$$

Because the question fixes both the noun (*mean*, so no 1+) and the direction (*at least*, so one tail), the only choices left are the root and which t to use — here the mean-response root with $t_{0.01,\,14}$.

11.4.10 A side-by-side summary table

Example 11.4.18 (Tabulating both intervals across several x_0)**.** For the seedling fit, lay out the 95% mean-response and prediction half-widths at $x_0 = 2, 4, 6, 8, 10$ in one table, and read off where they pinch and how they flare.

Which rule, and why? Pure assembly — both half-widths share the factor $t_{0.025,\,8}\, s = 2.306(1.255) = 2.894$ and differ only by the 1 inside the root, so a table makes the hourglass and the constant prediction gap visible at once.

Solution.

Step 1: the common factor and the roots. $t_{0.025,\,8}\, s = 2.894$; with $g(x_0) = \frac{1}{10} + \frac{(x_0 - 6)^2}{42}$, the mean half-width is $2.894\sqrt{g}$ and the prediction half-width is $2.894\sqrt{1+g}$.

Step 2: tabulate.

x_0	$g(x_0)$	mean half-width	prediction half-width
2	0.4810	2.007	3.522
4	0.1952	1.279	3.165
6	0.1000	0.915	3.036
8	0.1952	1.279	3.165
10	0.4810	2.007	3.522

Step 3: read the pattern. Both columns are symmetric about $x_0 = 6 = \overline{X}$ and minimal there; the mean width swings widely (0.915 to 2.007), while the prediction width barely moves (3.036 to 3.522) because its constant 1 dominates.

Both minimized at $x_0 = \overline{X} = 6$; prediction width > mean width at every x_0.

The table is the numeric hourglass: read down a column to see the flare, read across a row to see the prediction interval sitting strictly outside the mean interval at every x_0.

11.5 The Correlation Coefficient

The slope $\widehat{\beta}_1$ already tells us whether y tends to rise or fall with x, but its size carries units (dollars per year, mmHg per kg) and so cannot say whether a relationship is "tight" or "loose." The **sample correlation coefficient** r strips the units away: it is a pure number in $[-1, 1]$ that measures how nearly the points (x_i, y_i) fall on a single straight line. Its sign agrees with the slope, its magnitude grades the strength of the *linear* pattern, and it is perfectly symmetric in x and y — swapping the roles of the two variables leaves r unchanged. Two warnings travel with it from the start. First, r sees only straight-line structure: a sharply curved but utterly deterministic relationship can produce $r \approx 0$, so "no correlation" never means "no relationship." Second, r is a statement about association, not mechanism; a large r can be manufactured entirely by a lurking third variable, so it never by itself establishes causation. To test whether the underlying population correlation ρ is zero we convert r to a t statistic on $n-2$ degrees of freedom, and we will see that this test is algebraically the very same test as the slope test of §3.

Which correlation question is being asked?

- Asked for the **strength of a linear association** as a unitless number? ⇒ compute $r = S_{xy}/\sqrt{S_{xx}\,S_{yy}}$ and read its sign and magnitude.
- Asked whether the association is **real** (could $\rho = 0$)? ⇒ test $H_0 : \rho = 0$ with $t = r\sqrt{n-2}/\sqrt{1-r^2}$ against t_{n-2}.
- Asked for an interval or a test of ρ against a **nonzero** value (e.g. $\rho = 0.5$)? ⇒ the t above no longer applies; use the **Fisher z-transform**.
- Given new **units or a rescaling** of x or y and asked what r becomes? ⇒ nothing: r is invariant under any positive linear change of scale.

Do not confuse r with the slope, or r with R^2. The slope carries units and is not symmetric in x, y; r is unitless and symmetric. And r is signed in $[-1, 1]$ while $R^2 = r^2$ is in $[0, 1]$ and loses the direction of the relationship.

11.5.1 Definition and basic properties

Definition 11.5.1 (Sample correlation coefficient)**.** For paired data $(x_1, y_1), \ldots, (x_n, y_n)$ with

$$S_{xx} = \sum (x_i - \overline{X})^2, \qquad S_{yy} = \sum (y_i - \overline{Y})^2, \qquad S_{xy} = \sum (x_i - \overline{X})(y_i - \overline{Y}),$$

the **sample correlation coefficient** is

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}}.$$

It estimates the population correlation $\rho = \mathrm{Cov}(X, Y)/\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}$.

Proposition 11.5.2 (Properties of r)**.** *The sample correlation satisfies:*

(a) $-1 \le r \le 1$, *with* $r = \pm 1$ *exactly when all points lie on one line of nonzero slope;*

(b) r *is unitless and unchanged by* $x \mapsto a + bx$, $y \mapsto c + dy$ *for any* $b, d > 0$ *(it flips sign if exactly one of* b, d *is negative);*

(c) *r is symmetric in x and y;*

(d) *the sign of r equals the sign of the slope, and*

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = r\sqrt{\frac{S_{yy}}{S_{xx}}}, \qquad \textit{equivalently} \qquad r = \widehat{\beta}_1\sqrt{\frac{S_{xx}}{S_{yy}}}.$$

Proof. Bound (a) is the Cauchy–Schwarz inequality applied to the centered vectors $u_i = x_i - \overline{X}$ and $v_i = y_i - \overline{Y}$: $\left(\sum u_i v_i\right)^2 \le \left(\sum u_i^2\right)\left(\sum v_i^2\right)$, i.e. $S_{xy}^2 \le S_{xx}\,S_{yy}$, so $r^2 \le 1$; equality forces $v_i = \lambda u_i$ for a constant λ, which is a perfect line. For (b), a shift a or c cancels in every centered difference, and a positive scale b multiplies S_{xx} by b^2, S_{xy} by b, so $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$ is unchanged; a negative b flips the sign of S_{xy} only. Symmetry (c) is visible because S_{xy}, $S_{xx}S_{yy}$ are unaffected by interchanging the labels x and y. For (d), $\widehat{\beta}_1 = S_{xy}/S_{xx}$ has the sign of S_{xy}, as does r; multiplying and dividing by $\sqrt{S_{yy}}$ gives $\widehat{\beta}_1 = (S_{xy}/\sqrt{S_{xx}S_{yy}})\sqrt{S_{yy}/S_{xx}} = r\sqrt{S_{yy}/S_{xx}}$. □

Remark. Property (d) is worth memorizing in the form $R^2 = r^2$: the fraction of variation the line explains (§6) is literally the square of the correlation. So $r = 0.7$ means the line accounts for 0.49, just under half, of the spread in y — a useful antidote to over-reading a "moderately high" correlation.

Example 11.5.3 (Compute r from the sums of squares: alloy hardness)**.** A metallurgist records, for $n = 10$ steel specimens, the quenching temperature x and the resulting Rockwell hardness y, obtaining $S_{xx} = 820$, $S_{yy} = 445$, and $S_{xy} = 540$. Find the sample correlation.

Which rule, and why? The question asks for the unitless strength of the linear tie between temperature and hardness, so we want $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$, not the slope (which would carry hardness-per-degree units).

Solution.

Step 1: assemble the three sums. $S_{xx} = 820$, $S_{yy} = 445$, $S_{xy} = 540$.

Step 2: form r.

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} = \frac{540}{\sqrt{820 \cdot 445}} = \frac{540}{\sqrt{364900}} = \frac{540}{604.07} = 0.894.$$

Step 3: read it. The value is positive and close to 1, so hotter quench goes with higher hardness in a strongly linear way.

$$\boxed{r = 0.894}$$

Example 11.5.4 (Sign and magnitude: highway speed and fuel economy)**.** For $n = 12$ cars driven at a fixed cruising speed x (mph), the fuel economy y (mpg) gave $S_{xx} = 1\,260$, $S_{yy} = 98.0$, $S_{xy} = -296$. Compute r and interpret both its sign and its size.

Which rule, and why? We want the strength and direction of a linear association as a single number; that is exactly r. The negative S_{xy} already foretells a negative correlation.

Solution.

Step 1: compute.

$$r = \frac{-296}{\sqrt{1260 \cdot 98.0}} = \frac{-296}{\sqrt{123480}} = \frac{-296}{351.4} = -0.842.$$

Step 2: interpret the sign. $r < 0$: faster cruising goes with *worse* mileage.

Step 3: interpret the magnitude. $|r| = 0.84$ is large; the points hug a downward line fairly tightly, and the line explains $r^2 = 0.71$ of the variation in mpg.

$$\boxed{r = -0.842 \text{ (strong negative linear association)}}$$

11.5.2 Testing $H_0 : \rho = 0$

Theorem 11.5.5 (Correlation t-test)**.** *Under the regression model with $\rho = 0$, the statistic*

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$$

has a Student t distribution on $n-2$ degrees of freedom. To test $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$ at level α, reject when $|t| \geq t_{\alpha/2,\, n-2}$; one-sided alternatives use the corresponding single tail.

Remark (Same test as the slope). This is not a new test. Using $\widehat{\beta}_1 = r\sqrt{S_{yy}/S_{xx}}$ and the residual facts of §2,3 one finds the slope statistic $\widehat{\beta}_1/\mathrm{SE}(\widehat{\beta}_1)$ equals $r\sqrt{n-2}/\sqrt{1-r^2}$ exactly. So $\rho = 0$, $\beta_1 = 0$, and "the line explains nothing" are one hypothesis with one p-value. The convenience of Theorem 11.5.5 is that it needs only r and n — no separate standard error.

Example 11.5.6 (Test $H_0 : \rho = 0$, two-sided: study hours and exam score)**.** In a class of $n = 20$ students the correlation between weekly study hours and final exam score is $r = 0.46$. At $\alpha = 0.05$, is there evidence of any linear association?

Which rule, and why? We are asked whether the correlation is real (could it be 0?), with no prior direction specified, so this is a two-sided test of $H_0 : \rho = 0$ via the correlation t on $n - 2 = 18$ df.

Solution.

Step 1: statistic.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.46\sqrt{18}}{\sqrt{1-0.46^2}} = \frac{0.46 \cdot 4.243}{\sqrt{0.7884}} = \frac{1.952}{0.8879} = 2.198.$$

Step 2: critical value. Two-sided, $t_{0.025,18} = 2.101$.

Step 3: decide. $|2.198| > 2.101$, so reject H_0.

$$\boxed{t = 2.20 > 2.101 \Rightarrow \text{reject } H_0 : \rho = 0}$$

There is significant evidence of a positive linear association at the 5% level.

Example 11.5.7 (One-sided correlation test: fertilizer and yield)**.** An agronomist believes more nitrogen x can only help, never hurt, crop yield y, and wants evidence that $\rho > 0$. With $n = 15$ plots she finds $r = 0.39$. Test $H_0 : \rho = 0$ versus $H_1 : \rho > 0$ at $\alpha = 0.05$.

Which rule, and why? The scientific claim has a built-in direction ($\rho > 0$), so we run the *one-sided* correlation t-test and compare to a single upper-tail critical value, not the two-tailed one.

Solution.

Step 1: statistic.

$$t = \frac{0.39\sqrt{13}}{\sqrt{1-0.39^2}} = \frac{0.39 \cdot 3.606}{\sqrt{0.8479}} = \frac{1.406}{0.9208} = 1.527.$$

Step 2: one-sided critical value. $t_{0.05,13} = 1.771$ (upper tail only).

Step 3: decide. $1.527 < 1.771$, so we fail to reject.

$$\boxed{t = 1.53 < 1.771 \Rightarrow \text{fail to reject } H_0}$$

The data do not establish a positive linear effect of nitrogen at the 5% level, even though $r > 0$.

Example 11.5.8 (Contrast: one-sided vs two-sided on the same data)**.** Reuse the exam data ($n = 20$, $r = 0.46$, $t = 2.198$). A colleague argues study time can only help, so the test should be one-sided $H_1 : \rho > 0$. Does the conclusion change, and why?

Which rule, and why? Same r and same t, but a directional alternative moves all of α into one tail, lowering the bar. The recognition point is that the *critical value*, not the statistic, changes.

Solution.

Step 1: new critical value. One-sided, $t_{0.05,18} = 1.734$.

Step 2: compare. $2.198 > 1.734$, reject (more decisively than the two-sided $2.198 > 2.101$).

Step 3: read the contrast. Both tests reject here, but the one-sided test has a smaller critical value, so it would reject some borderline cases the two-sided test would not — at the price of being blind to a negative association. Choosing one-sided only to gain power, after seeing the data, is illegitimate.

$$\boxed{\text{two-sided: } 2.20 > 2.101; \quad \text{one-sided: } 2.20 > 1.734}$$

11.5.3 Invariance, equivalence, and the slope

Example 11.5.9 (Scale invariance: change of units does not move r)**.** A chemist has length data in centimeters and concentration in mg/L with $r = 0.71$. A reviewer wants length in inches (1 in = 2.54 cm) and concentration in μg/L (= 1000 mg/L). What is r in the new units?

Which rule, and why? This tests Proposition 11.5.2(b): both changes are positive linear rescalings, and r is invariant under those. No recomputation is needed — recognizing the invariance *is* the answer.

Solution.

Step 1: identify the transforms. $x' = x/2.54$ (positive scale) and $y' = 1000\,y$ (positive scale). Both have $b, d > 0$.

Step 2: apply invariance. Positive linear rescalings cancel in $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$: each scale multiplies numerator and denominator by the same factor.

$$\boxed{r' = 0.71 \text{ (unchanged)}}$$

Had we used the *slope* instead, it would have changed by the factor $2.54 \cdot 1000$ — which is exactly why a unitless r is the right tool for "strength."

Example 11.5.10 (Sign flip under a reversed axis)**.** The same chemist now measures "depth below surface" instead of "height above bottom," which negates every x ($x' = -x$, plus an irrelevant shift). With original $r = 0.71$, what is the new correlation?

Which rule, and why? Reversing one axis is multiplication by $b = -1 < 0$; Proposition 11.5.2(b) says exactly one negative scale flips the sign of r and nothing else.

Solution.

Step 1: count the negative scales. Only x is negated, so an odd number (1) of sign flips.

Step 2: apply. Magnitude is preserved, sign reverses.

$$\boxed{r' = -0.71}$$

Example 11.5.11 (Equivalence of the correlation test and the slope test)**.** A lab fit $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$ to $n = 16$ points and reported the slope test $t_{\text{slope}} = \widehat{\beta}_1/\text{SE}(\widehat{\beta}_1) = 2.84$. Separately, $r = 0.604$ was computed. Verify that the correlation t matches the slope t.

Which rule, and why? The Remark after Theorem 11.5.5 says the two statistics are algebraically identical, so computing $r\sqrt{n-2}/\sqrt{1-r^2}$ must reproduce 2.84. We are checking, not running a new test.

Solution.

Step 1: correlation t.

$$t = \frac{0.604\sqrt{14}}{\sqrt{1 - 0.604^2}} = \frac{0.604 \cdot 3.742}{\sqrt{0.6352}} = \frac{2.260}{0.7970} = 2.836.$$

Step 2: compare. $2.836 \approx 2.84$ (rounding), so the two tests agree exactly up to round-off.

$$\boxed{t_{\text{corr}} = 2.84 = t_{\text{slope}}}$$

One p-value, two computational routes.

Example 11.5.12 (From slope to r without the raw data)**.** A printed report gives $\widehat{\beta}_1 = 3.5$, $S_{xx} = 40$, $S_{yy} = 700$ but never lists r. Recover the correlation.

Which rule, and why? Property (d), $r = \widehat{\beta}_1\sqrt{S_{xx}/S_{yy}}$, expresses r from the slope and the two spread sums — the only quantities we have, so this identity is the route.

Solution.

Step 1: apply the identity.

$$r = \widehat{\beta}_1\sqrt{\frac{S_{xx}}{S_{yy}}} = 3.5\sqrt{\frac{40}{700}} = 3.5\sqrt{0.05714} = 3.5 \cdot 0.2390 = 0.837.$$

Step 2: sanity check. $|r| \le 1$ holds, and the sign matches the positive slope.

$$\boxed{r = 0.837}$$

11.5.4 What r does and does not detect

Example 11.5.13 (CONTRAST: strong relationship, $r \approx 0$ — a U-shaped pattern)**.** A physiologist measures reaction time y against arousal level x at seven equally spaced arousal settings $x = -3, -2, -1, 0, 1, 2, 3$, observing the symmetric "inverted-U" (Yerkes–Dodson) pattern $y = 9, 4, 1, 0, 1, 4, 9$ (fast at moderate arousal, slow at both extremes). Compute r and explain what it misses.

Which rule, and why? The variables are clearly and deterministically related, but the relation is curved and symmetric. r measures *linear* association only, so we expect $r \approx 0$ and must read that as "no *linear* trend," not "no relationship."

Solution.

Step 1: centers. $\overline{X} = 0$ by symmetry; $\overline{Y} = (9 + 4 + 1 + 0 + 1 + 4 + 9)/7 = 28/7 = 4$.

Step 2: cross term. With $\overline{X} = 0$,

$$S_{xy} = \sum x_i(y_i - \overline{Y}) = \sum x_i y_i - \overline{Y}\sum x_i = \sum x_i y_i - 0.$$

Now $\sum x_i y_i = (-3)(9) + (-2)(4) + (-1)(1) + 0 + (1)(1) + (2)(4) + (3)(9)$, which cancels term-by-term in $\pm$ pairs to 0.

Step 3: conclude. $S_{xy} = 0 \Rightarrow r = 0$, yet y is an exact function of x.

$$\boxed{r = 0 \text{ but the relationship is perfect (just nonlinear)}}$$

The lesson: always plot the data. A correlation of 0 rules out only a *straight-line* trend.

Example 11.5.14 (Why the plot matters: same r, different shapes)**.** Two data sets each report $r = 0.0$. In set A the points scatter formlessly; in set B they trace a clean parabola opening upward. What, if anything, distinguishes the two analyses?

Which rule, and why? Because r collapses all linear information into one number, two structurally opposite data sets can share it. Recognizing that r is necessary but not sufficient for "no relationship" is the whole point.

Solution.

Step 1: what $r = 0$ asserts. In both sets there is no *linear* component: the best-fit line is flat.

Step 2: what $r = 0$ hides. Set B has a strong quadratic relationship a line cannot see; fitting $y = \beta_0 + \beta_1 x + \beta_2 x^2$ would reveal it.

Step 3: practice. Report a scatterplot alongside r; consider transforming x (e.g. x^2) or adding curvature terms when the cloud bends.

$$\boxed{\text{equal } r \not\Rightarrow \text{equal structure}}$$

Example 11.5.15 (CONTRAST: correlation is not causation — a lurking variable)**.** Across 30 coastal towns the monthly ice-cream sales x and the monthly count of drowning incidents y have $r = 0.78$. A reporter concludes ice cream causes drownings. Critique this using the idea of a lurking variable.

Which rule, and why? r measures association, never mechanism; a third variable driving both x and y can create a large r with no causal link. The recognition step is to ask "what common cause could inflate both?" before trusting any causal reading.

Solution.

Step 1: name the lurking variable. Summer heat (z) raises both ice-cream sales and swimming exposure, hence drownings. z is a **common cause** (a confounder).

Step 2: explain the spurious r. Because z pushes x and y up together, they co-vary strongly even with no direct path $x \to y$.

Step 3: what would settle it. Hold z fixed (compare towns at equal temperature, or add z as a predictor); the x–y association should then largely vanish.

$$\boxed{r = 0.78 \text{ is real but spurious; heat confounds } x \text{ and } y}$$

Example 11.5.16 (Population correlation $\rho \neq 0$: the Fisher z-transform)**.** A geneticist's $r = 0.62$ from $n = 28$ pairs must be tested against the textbook value $\rho_0 = 0.40$, and a 95% interval for ρ is wanted. Outline the Fisher z method.

Which rule, and why? The correlation t of Theorem 11.5.5 is valid *only* for $H_0 : \rho = 0$; its null distribution breaks for $\rho_0 \neq 0$ because the sampling distribution of r becomes skewed. The variance-stabilizing **Fisher z-transform** restores approximate normality.

Solution.

Step 1: transform. With $z = \frac{1}{2}\ln\dfrac{1+r}{1-r} = \operatorname{arctanh} r$, z is approximately $\mathrm{N}\big(\zeta,\, 1/(n-3)\big)$ where $\zeta = \operatorname{arctanh}\rho$. Here $z = \frac{1}{2}\ln(1.62/0.38) = \frac{1}{2}\ln(4.263) = 0.725$ and $\zeta_0 = \operatorname{arctanh}(0.40) = 0.424$.

Step 2: test.

$$Z = \frac{z - \zeta_0}{\sqrt{1/(n-3)}} = \frac{0.725 - 0.424}{\sqrt{1/25}} = \frac{0.301}{0.20} = 1.505,$$

which is below $z_{0.025} = 1.96$, so we do not reject $\rho = 0.40$ two-sided.

Step 3: interval for ρ. A 95% interval for ζ is $0.725 \pm 1.96(0.20) = (0.333,\, 1.117)$; back-transform by $\rho = \tanh\zeta$ to get $(\tanh 0.333, \tanh 1.117) = (0.321,\, 0.807)$.

$$\boxed{Z = 1.51 < 1.96 \Rightarrow \text{retain } \rho = 0.40; \quad 95\%\text{ CI for } \rho : (0.32,\, 0.81)}$$

Example 11.5.17 (Interpreting a "moderate" correlation via R^2)**.** A marketing analyst reports $r = 0.55$ between ad spend and weekly sales and calls it "a strong driver." Temper that claim quantitatively.

Which rule, and why? The honest gauge of explanatory power is $R^2 = r^2$, the fraction of variation the line accounts for (§6); reading r alone overstates the effect.

Solution.

Step 1: square it. $R^2 = 0.55^2 = 0.3025$.

Step 2: interpret. The linear fit explains about 30% of the variation in weekly sales; roughly 70% is due to other factors.

$$\boxed{R^2 = 0.30 \text{ — "moderate," not "strong"}}$$

Example 11.5.18 (Borderline significance and the role of n)**.** Two studies both report $r = 0.30$. Study A has $n = 15$, study B has $n = 50$. Test each at $\alpha = 0.05$ (two-sided) and note how sample size drives the verdict.

Which rule, and why? The same r can be significant or not depending on n, because the test statistic $r\sqrt{n-2}/\sqrt{1-r^2}$ grows with n. This isolates the effect of sample size from the effect size r itself.

Solution.

Step 1: study A.

$$t_A = \frac{0.30\sqrt{13}}{\sqrt{1-0.09}} = \frac{0.30 \cdot 3.606}{0.9539} = 1.134, \qquad t_{0.025,13} = 2.160 \Rightarrow \text{retain.}$$

Step 2: study B.

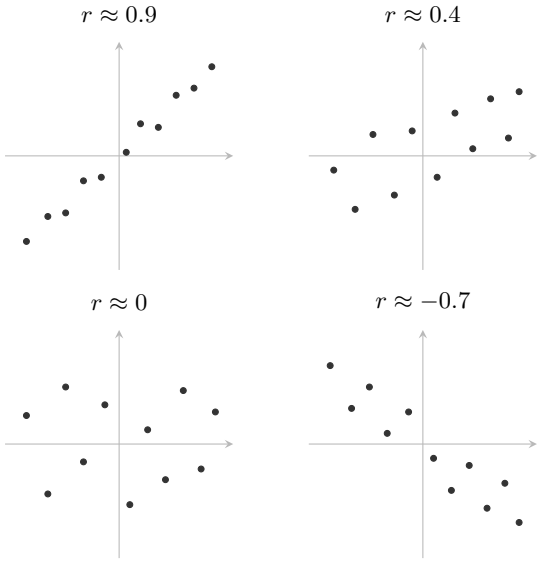
$$t_B = \frac{0.30\sqrt{48}}{\sqrt{1-0.09}} = \frac{0.30 \cdot 6.928}{0.9539} = 2.179, \qquad t_{0.025,48} \approx 2.011 \Rightarrow \text{reject.}$$

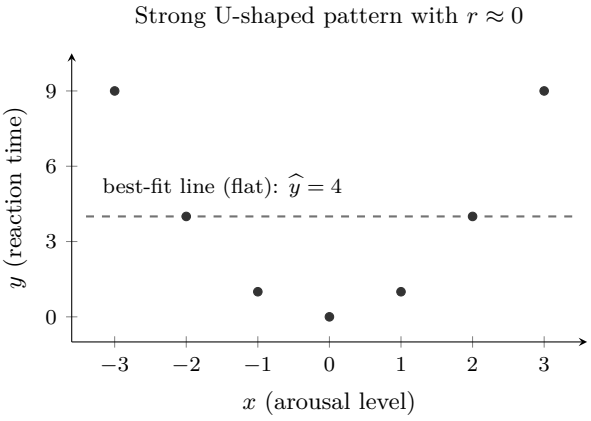
Step 3: lesson. A modest $r = 0.30$ is "not significant" with $n = 15$ but "significant" with $n = 50$ — significance certifies that $\rho \neq 0$, not that the association is large.

$$\boxed{t_A = 1.13 \text{ (retain)}; \quad t_B = 2.18 \text{ (reject)}}$$

11.5.5 Diagrams

The first panel calibrates the eye to four correlation magnitudes; the second shows the linear-only caveat in picture form — a strong U-shaped pattern whose linear correlation is essentially zero.





The seven points fall exactly on the parabola $y = x^2$, a deterministic relationship, yet the least-squares line is the flat $\widehat{y} = \overline{Y} = 4$ and $r = 0$: the correlation coefficient is blind to the curvature. Plot first; correlate second.

11.6 The ANOVA Decomposition and the Coefficient of Determination

Once a line is fitted, the natural accounting question is: of all the up-and-down spread in the response y, how much did the regression actually account for? The answer is a bookkeeping identity. The total variation of the y_i about their mean, $\text{SST} = S_{yy} = \sum(y_i - \overline{Y})^2$, splits cleanly into a part the fitted line *explains*, $\text{SSR} = \sum(\widehat{y}_i - \overline{Y})^2$, and a part it *leaves behind*, $\text{SSE} = \sum(y_i - \widehat{y}_i)^2$. That partition $\text{SST} = \text{SSR} + \text{SSE}$ is the **analysis-of-variance (ANOVA) decomposition** for regression, and it powers two summaries at once: the **coefficient of determination** $R^2 = \text{SSR} \,/\, \text{SST}$, the fraction of variation explained (which equals the squared correlation r^2), and an **overall F-test** of $H_0 \colon \beta_1 = 0$ built from the ratio of mean squares. Recognise this section's tools when a problem asks for "percent of variation explained," for a filled-in ANOVA table, or for the F-statistic of the regression. A single warning runs through everything below: a large R^2 measures *fit to a line*, not *correctness of the model*, and it is not the slope.

Which variation-accounting quantity does the problem want?

- Asked to **partition the total variation** or to find SSR, SSE from S_{yy}? $\Rightarrow$ the identity $\text{SST} = \text{SSR} + \text{SSE}$ with $\text{SSR} = \widehat{\beta}_1 S_{xy} = \widehat{\beta}_1^2 S_{xx}$ and $\text{SSE} = \text{SST} - \text{SSR}$.
- Asked for the **proportion (percent) of variation explained**, or for R^2? $\Rightarrow$ $R^2 = \text{SSR} \,/\, \text{SST} = 1 - \text{SSE} \,/\, \text{SST}$, and $R^2 = r^2$.
- Asked to **build the ANOVA table** or run the **overall F-test** of $\beta_1 = 0$? $\Rightarrow$ $F = \text{MSR} \,/\, \text{MSE}$ with $\text{MSR} = \text{SSR}\,/1$, $\text{MSE} = \text{SSE}\,/(n-2)$, compared to $\text{F}_{1,\,n-2}$.
- Asked whether the F-test and the slope t-test agree? $\Rightarrow$ in simple linear regression $F = t^2$ and they are the *same* two-sided test.

Do not confuse R^2 with validity or with the slope: a high R^2 can sit on top of curvature or a single outlier and tells you nothing about whether a *line* is the right model; and R^2 is a unitless fraction in $[0, 1]$, whereas the slope $\widehat{\beta}_1$ carries units and a sign.

11.6.1 The decomposition identity

Theorem 11.6.1 (ANOVA decomposition for regression)**.** *Let $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$ be the least-squares fitted values. Then the total sum of squares splits as*

$$\underbrace{\sum_{i=1}^{n}(y_i - \overline{Y})^2}_{\mathrm{SST} = S_{yy}} \;=\; \underbrace{\sum_{i=1}^{n}(\widehat{y}_i - \overline{Y})^2}_{\mathrm{SSR}\ (\textit{explained})} \;+\; \underbrace{\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}_{\mathrm{SSE}\ (\textit{residual})}.$$

Moreover the regression sum of squares has the closed forms

$$\mathrm{SSR} = \widehat{\beta}_1 S_{xy} = \widehat{\beta}_1^2\, S_{xx} = \frac{S_{xy}^2}{S_{xx}}.$$

Proof. Write each centered response as the sum of a fitted-deviation and a residual,

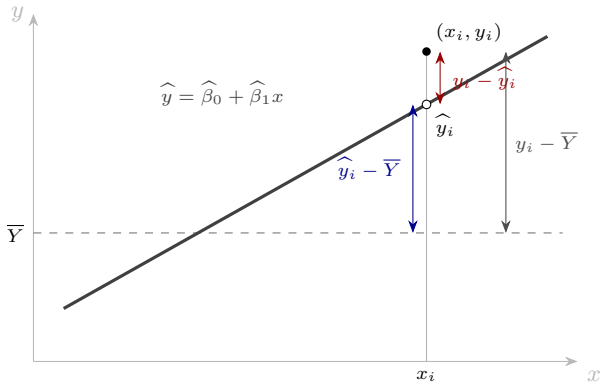
$$y_i - \overline{Y} = (\widehat{y}_i - \overline{Y}) + (y_i - \widehat{y}_i),$$

square, and sum over i:

$$\mathrm{SST} = \sum(\widehat{y}_i - \overline{Y})^2 + \sum(y_i - \widehat{y}_i)^2 + 2\sum(\widehat{y}_i - \overline{Y})(y_i - \widehat{y}_i).$$

The cross term vanishes. The least-squares **normal equations** force the residuals $e_i = y_i - \widehat{y}_i$ to satisfy $\sum e_i = 0$ and $\sum x_i e_i = 0$; since $\widehat{y}_i - \overline{Y} = \widehat{\beta}_1(x_i - \overline{X})$ is a linear function of x_i, the product sum $\sum(\widehat{y}_i - \overline{Y})e_i = \widehat{\beta}_1 \sum(x_i - \overline{X})e_i = \widehat{\beta}_1\big(\sum x_i e_i - \overline{X}\sum e_i\big) = 0$. That leaves $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$. For the closed forms, $\widehat{y}_i - \overline{Y} = \widehat{\beta}_1(x_i - \overline{X})$ gives $\mathrm{SSR} = \widehat{\beta}_1^2 \sum(x_i - \overline{X})^2 = \widehat{\beta}_1^2 S_{xx}$; and using $\widehat{\beta}_1 = S_{xy}/S_{xx}$, $\widehat{\beta}_1^2 S_{xx} = \widehat{\beta}_1 \cdot (\widehat{\beta}_1 S_{xx}) = \widehat{\beta}_1 S_{xy} = S_{xy}^2/S_{xx}$. □

Remark. Read the three pieces as a sentence. SST is how much y varies if you ignore x entirely and predict every observation by the flat line $\overline{Y}$. SSE is how much variation survives after the *tilted* least-squares line has done its best. The difference, SSR, is what the tilt bought you — the variation the slope explains. If the line is useless ($\widehat{\beta}_1 = 0$, the fitted line is just $\overline{Y}$) then $\mathrm{SSR} = 0$ and $\mathrm{SSE} = \mathrm{SST}$; if the points fall exactly on a line then $\mathrm{SSE} = 0$ and $\mathrm{SSR} = \mathrm{SST}$.



For a single point, the total deviation $y_i - \overline{Y}$ from the flat mean line splits into an explained *part $\widehat{y}_i - \overline{Y}$ (how far the fitted line has climbed above $\overline{Y}$) plus a* residual *part $y_i - \widehat{y}_i$ (how far the point still sits above the line). Squaring and summing these over all points gives $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$; the cross term cancels by the normal equations.*

Example 11.6.2 (Partitioning the total variation: catalyst loading vs. yield)**.** A chemist records reactor yield y (percent) against catalyst loading x (g/L) for $n = 8$ runs. The summary statistics are $S_{xx} = 40.0$, $S_{xy} = 58.0$, and $S_{yy} = \mathrm{SST} = 96.0$. Partition SST into SSR and SSE.

Which rule, and why? The task is to *split* the total variation, so use $\text{SST} = \text{SSR} + \text{SSE}$ with the closed form $\text{SSR} = S_{xy}^2/S_{xx}$ (no need to compute fitted values one by one) and then $\text{SSE} = \text{SST} - \text{SSR}$.

Solution.

Step 1: slope and SSR. $\widehat{\beta}_1 = S_{xy}/S_{xx} = 58.0/40.0 = 1.45$, and

$$\text{SSR} = \frac{S_{xy}^2}{S_{xx}} = \frac{58.0^2}{40.0} = \frac{3364}{40.0} = 84.10 \quad \text{(check: } \widehat{\beta}_1 S_{xy} = 1.45 \times 58.0 = 84.10\text{)}.$$

Step 2: SSE *by subtraction.*

$$\text{SSE} = \text{SST} - \text{SSR} = 96.0 - 84.10 = 11.90.$$

Step 3: report the partition.

$$\boxed{\text{SST} = 96.0 = \underbrace{84.10}_{\text{SSR}} + \underbrace{11.90}_{\text{SSE}}.}$$

The line explains 84.10 of the 96.0 total squared variation, leaving 11.90 as residual scatter — the input for $s^2 = \text{SSE}/(n-2)$ in the next subsection.

Example 11.6.3 (Partition from a fitted line and raw residuals: braking distance)**.** For $n = 6$ cars, speed x (mph) and braking distance y (ft) give a fitted line and these quantities: $S_{yy} = \text{SST} = 720$, slope $\widehat{\beta}_1 = 4.0$, and $S_{xy} = 150$. Find SSR and SSE, and confirm $\text{SSR} = \widehat{\beta}_1 S_{xy} = \widehat{\beta}_1^2 S_{xx}$ are mutually consistent given $S_{xx} = 37.5$.

Which rule, and why? Two equivalent closed forms for SSR are offered; pick the one whose inputs you have ($\widehat{\beta}_1 S_{xy}$) and use the others as cross-checks. Then $\text{SSE} = \text{SST} - \text{SSR}$.

Solution.

Step 1: SSR *three ways.*

$$\begin{aligned} \text{SSR} &= \widehat{\beta}_1 S_{xy} = 4.0 \times 150 = 600, \\ \widehat{\beta}_1^2\, S_{xx} &= 4.0^2 \times 37.5 = 16 \times 37.5 = 600, \\ S_{xy}^2/S_{xx} &= 150^2/37.5 = 22500/37.5 = 600. \end{aligned}$$

All three agree, confirming the inputs are consistent ($\widehat{\beta}_1 = S_{xy}/S_{xx} = 150/37.5 = 4.0$, as required).

Step 2: SSE.

$$\text{SSE} = \text{SST} - \text{SSR} = 720 - 600 = 120.$$

Step 3: report.

$$\boxed{\text{SSR} = 600, \qquad \text{SSE} = 120, \qquad \text{SST} = 600 + 120 = 720.}$$

11.6.2 The coefficient of determination

Definition 11.6.4 (Coefficient of determination R^2)**.** The **coefficient of determination** is the fraction of the total variation in y explained by the regression,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}} \in [0, 1].$$

It equals the square of the sample correlation, $R^2 = r^2$, where $r = S_{xy}/\sqrt{S_{xx}\, S_{yy}}$.

Proposition 11.6.5 ($R^2 = r^2$)**.** *For simple linear regression, $R^2 = r^2$.*

Proof. Using $\text{SSR} = S_{xy}^2/S_{xx}$ and $\text{SST} = S_{yy}$,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{S_{xy}^2/S_{xx}}{S_{yy}} = \frac{S_{xy}^2}{S_{xx}\, S_{yy}} = \left(\frac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}}\right)^2 = r^2. \qquad \square$$

Remark. "Proportion of variation explained" is the only honest gloss on R^2. It is *not* the proportion of points near the line, nor the probability of anything, nor a measure of slope steepness: a near-horizontal line through tight data can have a high R^2, and a steep line through scattered data can have a low one. And because $R^2 = r^2$ throws away the sign of r, an R^2 of 0.81 is consistent with either $r = +0.9$ or $r = -0.9$ — you must look at $\widehat{\beta}_1$ (or r) to recover the direction.

Example 11.6.6 (Computing and interpreting R^2: catalyst data continued)**.** Continue the catalyst example with SST = 96.0, SSR = 84.10, SSE = 11.90. Find R^2 and state its interpretation in words.

Which rule, and why? The phrase "percent of variation explained" points straight at $R^2 = \text{SSR} / \text{SST}$; we already have the partition, so this is one division and a sentence.

Solution.

Step 1: ratio.

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{84.10}{96.0} = 0.876, \qquad \text{equivalently } 1 - \frac{11.90}{96.0} = 1 - 0.124 = 0.876.$$

Step 2: put it in words. About 87.6% of the run-to-run variation in yield is accounted for by the straight-line dependence on catalyst loading; the remaining 12.4% is residual scatter.

$$\boxed{R^2 = 0.876 \;\Rightarrow\; \text{catalyst loading explains 87.6\% of yield variation.}}$$

Example 11.6.7 (Recovering r from R^2 needs the slope's sign: cooling)**.** A study of an electronics enclosure regresses internal temperature rise y on fan-off time x and reports $R^2 = 0.64$ with a *positive* slope. A separate study regresses component life y on operating temperature x and reports $R^2 = 0.64$ with a *negative* slope. Give the correlation r in each case.

Which rule, and why? $R^2 = r^2$ fixes only the magnitude $|r| = \sqrt{R^2}$; the *sign* of r matches the sign of $\widehat{\beta}_1$. This is the contrast that R^2 alone cannot resolve.

Solution.

Step 1: magnitude. $|r| = \sqrt{0.64} = 0.80$ in both studies.

Step 2: attach the sign of the slope.

$$\begin{aligned} \text{enclosure (slope } > 0): &\quad r = +0.80, \\ \text{component life (slope } < 0): &\quad r = -0.80. \end{aligned}$$

$$\boxed{|r| = \sqrt{R^2} = 0.80; \quad r = +0.80 \text{ (positive slope)}, \quad r = -0.80 \text{ (negative slope)}.}$$

Two regressions with identical R^2 describe opposite associations — decisive evidence that R^2 is not the slope and carries no direction.

11.6.3 The regression ANOVA table and the overall F-test

Definition 11.6.8 (Mean squares and the overall F-statistic)**.** Divide each sum of squares by its degrees of freedom to get a **mean square**:

$$\text{MSR} = \frac{\text{SSR}}{1}, \qquad \text{MSE} = \frac{\text{SSE}}{n-2} = s^2.$$

The regression has 1 df (one slope parameter beyond the mean), the error has $n - 2$ df, and they sum to the total $n - 1$ df. The **overall F-statistic** is

$$F = \frac{\text{MSR}}{\text{MSE}}, \qquad F \sim \text{F}_{1,\,n-2} \quad \text{under } H_0 \colon \beta_1 = 0.$$

Reject H_0 at level α if $F > F_{\alpha,\,1,\,n-2}$.

Remark. The whole computation is conventionally laid out in a **regression ANOVA table**: one row for the explained variation ("Regression"), one for the residual ("Error"), and a total row. The columns are the source, its df, its sum of squares, its mean square (SS/df), and — on the regression row only — the ratio $F = \text{MSR} / \text{MSE}$. The table is just organized bookkeeping for the identity $\text{SST} = \text{SSR} + \text{SSE}$, but it makes the df accounting and the F assembly hard to get wrong.

Source	df	SS	MS	F
Regression	1	SSR	$\text{MSR} = \text{SSR}/1$	$F = \dfrac{\text{MSR}}{\text{MSE}}$
Error	$n-2$	SSE	$\text{MSE} = \text{SSE}/(n-2)$	
Total	$n-1$	SST		

The regression ANOVA table. Degrees of freedom and sums of squares each add down the column ($1+(n-2) = n-1$ and $\text{SSR} + \text{SSE} = \text{SST}$); mean squares are SS/df; the single F on the Regression row tests $H_0\colon \beta_1 = 0$ against $\text{F}_{1,\,n-2}$.

Example 11.6.9 (Build the full ANOVA table: catalyst data, $n = 8$)**.** For the catalyst data ($n = 8$, $\text{SSR} = 84.10$, $\text{SSE} = 11.90$, $\text{SST} = 96.0$), construct the complete ANOVA table and report MSR, MSE, and F.

Which rule, and why? "Build the ANOVA table" means fill the df $(1,\ n-2,\ n-1)$, copy the sums of squares, divide to get mean squares, and form $F = \text{MSR} / \text{MSE}$. The error df is $n - 2 = 6$ because two parameters $(\widehat{\beta}_0, \widehat{\beta}_1)$ were estimated.

Solution.

Step 1: degrees of freedom. Regression 1; error $n - 2 = 6$; total $n - 1 = 7$ (and $1 + 6 = 7$).

Step 2: mean squares.

$$\text{MSR} = \frac{\text{SSR}}{1} = \frac{84.10}{1} = 84.10,$$
$$\text{MSE} = \frac{\text{SSE}}{n-2} = \frac{11.90}{6} = 1.983 = s^2.$$

Step 3: the F-statistic and table.

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{84.10}{1.983} = 42.4.$$

Source	df	SS	MS	F
Regression	1	84.10	84.10	42.4
Error	6	11.90	1.983	
Total	7	96.00		

Example 11.6.10 (Run the overall F-test: is the regression significant?)**.** Using the catalyst ANOVA table ($F = 42.4$ on $\text{F}_{1,6}$), test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$. The critical value is $F_{0.05,1,6} = 5.99$.

Which rule, and why? The overall F-test of a simple regression always tests the *two-sided* $H_0\colon \beta_1 = 0$; there is no one-sided F here, because $F = \text{MSR} / \text{MSE}$ squares the slope and discards its sign. Compare F to the upper $F_{\alpha,1,n-2}$ point.

Solution.

Step 1: statistic and reference. $F = 42.4$, reference $\text{F}_{1,6}$, upper cutoff $F_{0.05,1,6} = 5.99$.

Step 2: decision. Reject if $F > 5.99$. Here $42.4 \gg 5.99$.

$F = 42.4 > F_{0.05,1,6} = 5.99 \ \Rightarrow$ reject H_0; the linear effect of catalyst is significant.

Equivalently, since $R^2 = 0.876$ is large, it is no surprise the regression clears the bar by a wide margin.

Example 11.6.11 (Verify $F = t^2$ against the slope t-test)**.** For the catalyst fit, the slope inference (§3) gave $\widehat{\beta}_1 = 1.45$ with standard error $\mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}} = \sqrt{1.983}/\sqrt{40.0} = 0.2227$, hence a slope t-statistic $t_{\text{obs}} = \widehat{\beta}_1/\mathrm{SE}(\widehat{\beta}_1)$. Show that t^2_{obs} equals the ANOVA $F = 42.4$.

Which rule, and why? In simple linear regression the overall F-test and the two-sided slope t-test are the *same* test written two ways; the algebraic statement is $F = t^2$ (equivalently $t = \pm\sqrt{F}$). Computing both and matching them is the check.

Solution.

Step 1: slope t.

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{1.45}{0.2227} = 6.51.$$

Step 2: square it.

$$t^2_{\text{obs}} = 6.51^2 = 42.4.$$

Step 3: match to F. This equals the ANOVA $F = 42.4$ on $\mathrm{F}_{1,6}$, and indeed the squared t on 6 df is an F on $(1, 6)$ df. The critical values agree too: $t_{0.025,6} = 2.447$ and $t^2_{0.025,6} = 2.447^2 = 5.99 = F_{0.05,1,6}$.

$$\boxed{t^2_{\text{obs}} = 6.51^2 = 42.4 = F, \qquad t^2_{0.025,6} = 2.447^2 = 5.99 = F_{0.05,1,6}.}$$

The two-sided slope test and the overall F-test reject together, always — one is the square of the other.

Example 11.6.12 (A second table and F-test: advertising spend vs. sales, $n = 12$)**.** A retailer regresses monthly sales y (\$000) on advertising spend x (\$000) over $n = 12$ months and finds $\mathrm{SST} = 540$, $R^2 = 0.45$. Build the ANOVA table and test $H_0\colon \beta_1 = 0$ at $\alpha = 0.05$, given $F_{0.05,1,10} = 4.96$.

Which rule, and why? Here R^2 is given instead of SSR, so first recover $\mathrm{SSR} = R^2 \cdot \mathrm{SST}$ and $\mathrm{SSE} = (1-R^2)\,\mathrm{SST}$, then run the ordinary table-and-F procedure on $\mathrm{F}_{1,n-2} = \mathrm{F}_{1,10}$.

Solution.

Step 1: sums of squares from R^2.

$$\mathrm{SSR} = R^2 \cdot \mathrm{SST} = 0.45 \times 540 = 243,$$
$$\mathrm{SSE} = (1 - R^2) \cdot \mathrm{SST} = 0.55 \times 540 = 297.$$

Step 2: df and mean squares. Regression 1, error $n - 2 = 10$, total 11.

$$\mathrm{MSR} = \frac{243}{1} = 243,$$
$$\mathrm{MSE} = \frac{297}{10} = 29.7.$$

Step 3: F and decision.

$$F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{243}{29.7} = 8.18 > F_{0.05,1,10} = 4.96 \;\Rightarrow\; \text{reject } H_0.$$

Source	df	SS	MS	F
Regression	1	243	243	8.18
Error	10	297	29.7	
Total	11	540		

Advertising spend explains 45% of sales variation, and that explained share is significantly more than chance would produce ($F = 8.18$).

11.6.4 What R^2 does and does not tell you

Example 11.6.13 (Contrast: a high R^2 that hides curvature)**.** An engineer fits a *straight* line to thrust y versus throttle position x and reports $R^2 = 0.94$, declaring "94% explained, the linear model is validated." A residual plot, however, shows a clear U-shape: residuals are positive at the low and high ends and negative in the middle. Critique the claim.

Which rule, and why? This tests whether R^2 certifies a model. It does not: R^2 measures only how much *a line* reduces SST, and a gently curved relationship can still let a line capture most of the variation while being systematically wrong. The diagnostic is the residual pattern, not the size of R^2.

Solution.

Step 1: what $R^2 = 0.94$ actually says. The fitted line accounts for 94% of the total squared variation in thrust. That is a statement about *variance reduction*, not about the *shape* of the dependence.

Step 2: what the residual U-shape says. Systematic sign changes in the residuals (positive–negative–positive across the range of x) are the signature of unmodeled **curvature**: the true relation bends, and a line over-predicts in the middle and under-predicts at the ends. A model that is wrong in a patterned way can still have small total residual variation, hence a high R^2.

Step 3: the correct conclusion.

A high R^2 does not validate the linear model; patterned residuals reveal curvature the R^2 hides.

The right move is to add a quadratic term or transform a variable, then recheck the residuals — not to rest on the 0.94.

Example 11.6.14 (Contrast: a single outlier inflates R^2)**.** Eleven points scatter with almost no trend, but a twelfth point sits far out to the upper right, miles from the others in both x and y. The least-squares line is dragged toward that point and R^2 comes out to 0.88. Should we trust "88% explained"?

Which rule, and why? Again the question is whether R^2 certifies the fit. A single high-leverage outlier can manufacture a large R^2 out of an otherwise trendless cloud, so the number must be read *alongside* a scatterplot, never alone.

Solution.

Step 1: why the outlier dominates. $R^2 = \text{SSR} \,/\, \text{SST}$ and both SSR and SST are sums of *squared* deviations, so a point far from $(\overline{X}, \overline{Y})$ contributes enormously. One extreme point can supply most of SST *and* most of SSR, pushing the ratio near 1 regardless of the other eleven points.

Step 2: the diagnosis. Refit without the suspect point: if R^2 collapses (say from 0.88 to 0.05), the "relationship" was an artifact of one observation, not a property of the data as a whole.

Step 3: conclusion.

A high R^2 driven by one high-leverage point is not evidence of a real linear relationship — always view the scatter.

Example 11.6.15 (Contrast: low R^2 but a significant slope (large n))**.** A genomics study regresses a phenotype score y on the dosage x of one common variant across $n = 2002$ individuals. The fit gives $R^2 = 0.012$ but $\widehat{\beta}_1 = 0.30$ with standard error $\text{SE}(\widehat{\beta}_1) = 0.060$. Is the slope significant, and how can a "tiny R^2" coexist with a "real effect"?

Which rule, and why? Significance of the slope is a t-test (§3), governed by $\widehat{\beta}_1/\text{SE}$ and the sample size; R^2 measures *effect size* (variation explained). With huge n a small but genuine effect is detectable even though it explains little variation — the contrast between *significant* and *large*.

Solution.

Step 1: slope t-test.

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)} = \frac{0.30}{0.060} = 5.0, \qquad \text{df} = n - 2 = 2000, \quad t_{0.025,2000} \approx 1.96.$$

Since $5.0 > 1.96$, the slope is highly significant (p-value far below 0.05).

Step 2: reconcile with $R^2 = 0.012$. The corresponding $F = t_{\text{obs}}^2 = 25.0$ on $\text{F}_{1,2000}$, also far past its ≈ 3.85 cutoff. A significant F means the slope is *not zero*; it says nothing about magnitude. With $n = 2002$, the standard error is small enough to resolve an effect that still explains only 1.2% of phenotype variation.

Step 3: the dual reading.

$$\boxed{t_{\text{obs}} = 5.0 \text{ (significant slope)}, \quad R^2 = 0.012 \text{ (small effect)} : \text{ both true at once.}}$$

"Significant" answers *is there an effect?*; R^2 answers *how much does it explain?*. A massive sample can make a real-but-tiny slope statistically certain while leaving R^2 near zero — exactly the regime modern large-data studies live in.

Example 11.6.16 (Putting it together: water hardness vs. pipe corrosion, $n = 10$)**.** An environmental engineer regresses annual pipe-wall loss y (mm) on water hardness x (mg/L) for $n = 10$ municipal sites, obtaining $S_{xx} = 2500$, $S_{xy} = -30.0$, $S_{yy} = \text{SST} = 0.900$. Find R^2, build the ANOVA table, run the overall F-test at $\alpha = 0.05$ ($F_{0.05,1,8} = 5.32$), and report the sign of the association.

Which rule, and why? A full account: $\text{SSR} = S_{xy}^2/S_{xx}$ feeds both $R^2 = \text{SSR} \,/\, \text{SST}$ and the F-test, while the sign of $\widehat{\beta}_1 = S_{xy}/S_{xx}$ (lost by R^2) gives the direction. This ties the whole section together — decomposition, R^2, table, F, and the sign caveat.

Solution.

Step 1: SSR*, and the slope's sign.*

$$\begin{aligned}
\text{SSR} &= \frac{S_{xy}^2}{S_{xx}} = \frac{(-30.0)^2}{2500} = \frac{900}{2500} = 0.360, \\
\text{SSE} &= \text{SST} - \text{SSR} = 0.900 - 0.360 = 0.540, \\
\widehat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{-30.0}{2500} = -0.012 \text{ (negative: harder water, } \textit{less} \text{ loss here).}
\end{aligned}$$

Step 2: R^2 and mean squares.

$$\begin{aligned}
R^2 &= \frac{\text{SSR}}{\text{SST}} = \frac{0.360}{0.900} = 0.40, \\
\text{MSR} &= \frac{0.360}{1} = 0.360, \qquad \text{MSE} = \frac{0.540}{8} = 0.0675.
\end{aligned}$$

Step 3: F-test and table.

$$F = \frac{\text{MSR}}{\text{MSE}} = \frac{0.360}{0.0675} = 5.33 > F_{0.05,1,8} = 5.32 \;\Rightarrow\; \text{reject } H_0 \text{ (just barely).}$$

Source	df	SS	MS	F
Regression	1	0.360	0.360	5.33
Error	8	0.540	0.0675	
Total	9	0.900		

The regression explains 40% of the variation and is significant at 5% by the narrowest of margins; the slope is *negative*, a direction the $R^2 = 0.40$ by itself could never reveal.

11.7 Exercises

All problems below are original. Full solutions begin on page 756.

A. Least squares: fitting the line

A1. A bakery records oven temperature x (°C) against the browning index y (a unitless scale) of a fixed loaf recipe:

$$(160, 12),\ (170, 15),\ (180, 18),\ (190, 21),\ (200, 24).$$

(a) Compute S_{xx} and S_{xy}, then the least-squares slope $\widehat{\beta}_1$ and intercept $\widehat{\beta}_0$.

(b) Write the fitted line and interpret the slope in its units (browning units per °C).

(c) Use the line to find the fitted browning index at $x = 185\,^{\circ}\mathrm{C}$, and say whether this is interpolation or extrapolation.

A2. A delivery service times $n = 6$ routes, relating distance x (km) to travel time y (min):

$$(2, 15),\ (4, 18),\ (6, 27),\ (8, 33),\ (10, 36),\ (12, 45).$$

(a) Using the computing forms with running totals, find S_{xx}, S_{xy}, $\widehat{\beta}_1$, and $\widehat{\beta}_0$.

(b) Interpret both the slope and the intercept in plain words, stating units. Comment on whether the intercept is meaningful here.

A3. For the delivery-route fit of problem A2, compute all six fitted values $\widehat{y}_i$ and the six residuals $e_i = y_i - \widehat{y}_i$, lay them out in a table, and verify that $\sum_{i=1}^{6} e_i = 0$ (up to rounding). State what this zero sum does and does not tell you about the quality of the fit.

A4. A water-quality lab relates dissolved fertilizer concentration x (mg/L) to measured algae density y (thousand cells/mL) in a pond:

$$(4, 20),\ (8, 19),\ (12, 28),\ (16, 27),\ (20, 36).$$

(a) Fit the least-squares line.

(b) Verify directly that the fitted line passes through the point of averages $(\overline{X}, \overline{Y})$ by evaluating $\widehat{y}$ at $x = \overline{X}$.

(c) Find the residual at $x = 12$ and say whether the line over- or under-predicts there.

A5. An analyst studies annual rainfall against reservoir level over five years and fits a line on calendar years $x = 2019, 2020, 2021, 2022, 2023$ against the level y (meters). To simplify the arithmetic the analyst re-centers time as $x' = x - 2021$, giving $x' = -2, -1, 0, 1, 2$ with

$$(x', y):\ (-2, 40),\ (-1, 43),\ (0, 47),\ (1, 48),\ (2, 52).$$

(a) Fit the least-squares line on the re-centered x', reporting $\widehat{\beta}_1$ and the re-centered intercept $\widehat{\beta}_0'$.

(b) Show that the slope on the original calendar-year axis is the same number, and find the original intercept $\widehat{\beta}_0$. Explain in one sentence why re-centering left the slope alone.

A6. A nutrition scientist fits sugar content x (g per serving) against a cereal's shelf-life rating y on doses $x \in [2, 14]$ g, obtaining $\widehat{y} = 9.0 + 1.6x$. A marketing intern then "predicts" the rating for a hypothetical $x = 60$ g cereal as $\widehat{y} = 9.0 + 1.6(60) = 105$ and for $x = 200$ g as $\widehat{y} = 329$, and concludes the product line should maximize sugar.

(a) Compute the fitted rating at $x = 10$ g and state why that prediction is on solid ground.

(b) Explain precisely why the predictions at $x = 60$ and $x = 200$ are not justified, naming the phenomenon involved.

A7. Over $n = 14$ towns an economist finds that the number of fire trucks dispatched to a fire x is strongly and tightly associated with the dollar damage y, fitting $\widehat{y} = 2000 + 8500x$ with the points hugging the line. A council member proposes sending fewer trucks to reduce damage. Diagnose the reasoning error, name the statistical phenomenon, and state in one sentence what the fitted line legitimately does and does not establish.

B. Estimator properties and the error variance

B1. A hydrologist regresses dissolved-oxygen y (mg/L) on water temperature x (°C) for $n = 9$ river sites. The least-squares fit has already produced the summary sums

$$S_{xx} = 72.0, \qquad S_{xy} = -90.0, \qquad S_{yy} = 128.5.$$

(a) Compute the slope $\widehat{\beta}_1$.

(b) Using the computing formula $\mathrm{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}$, find SSE.

(c) Estimate the error variance s^2 and the residual standard deviation s. State the degrees of freedom you divide by and why it is $n - 2$ rather than n.

B2. In a pharmacokinetics calibration, a chemist regresses peak plasma concentration y (μg/mL) on injected dose x (mg) over $n = 11$ runs and reports $S_{xx} = 40.0$, $S_{xy} = 30.0$, $S_{yy} = 27.0$, with $\bar{x} = 8.0$.

(a) Find s^2 and s.

(b) Compute the standard error of the slope $\mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$.

(c) Compute the standard error of the intercept $\mathrm{SE}(\widehat{\beta}_0) = s\sqrt{1/n + \bar{x}^2/S_{xx}}$, and say in one sentence why the intercept here is estimated far less precisely than the slope.

B3. An ecologist will design a study regressing metabolic rate y on body mass x, and from past work treats the error variance as known, $\sigma^2 = 16.0$. Two candidate designs use the same $n = 7$ animals but different mass values:

$$x_{\mathrm{A}} = \{20, 21, 22, 22, 23, 24, 25\}, \qquad x_{\mathrm{B}} = \{8, 14, 20, 22, 28, 34, 40\}.$$

(a) Compute S_{xx} for each design.

(b) Using $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$, find $\mathrm{Var}(\widehat{\beta}_1)$ for each design and the ratio of the two variances.

(c) One sentence: with sample size and per-point noise identical, what single feature of the design makes B's slope sharper, and by what mechanism?

B4. An engineer plans a regression of fatigue life y on load x with known error variance $\sigma^2 = 25.0$ and wants the slope determined to a standard deviation $\mathrm{SD}(\widehat{\beta}_1) \le 0.50$. Using $\mathrm{SD}(\widehat{\beta}_1) = \sigma/\sqrt{S_{xx}}$, find the minimum S_{xx} the chosen load levels must achieve. Then state what happens to the required S_{xx} if the precision target is tightened to $\mathrm{SD}(\widehat{\beta}_1) \le 0.25$.

B5. In a quality-control regression with $n = 16$ and a known error variance $\sigma^2 = 6.0$, the residual variance s^2 obeys $(n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$.

(a) Identify the degrees of freedom of the chi-square distribution.

(b) Find $\mathbb{P}(s^2 > 9.0)$. Use $\chi^2_{0.05,14} = 23.685$ and report the approximate tail probability.

(c) Find $\mathbb{P}(s^2 < 2.5)$. Use $\chi^2_{0.975,14} = 5.629$ and report the approximate lower-tail probability.

B6. Two analysts process the *same* regression output: $n = 12$ data points giving $\mathrm{SSE} = 90.0$ and $S_{xx} = 50.0$. Analyst I divides SSE by n ("it is just a variance"); Analyst II divides by $n - 2$.

(a) Compute each analyst's s^2, s, and resulting slope standard error $\mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$.

(b) By what percentage does Analyst I's standard error fall short of the correct one?

(c) In one sentence, explain why an understated standard error is dangerous when the slope is later tested for significance.

B7. For a regression with $n = 22$ and true error variance $\sigma^2 = 10.0$, define the correct estimator $s^2 = \mathrm{SSE}/(n-2)$ and the naive estimator $\tilde{s}^2 = \mathrm{SSE}/n$. Using $\mathbb{E}[\mathrm{SSE}] = (n-2)\sigma^2$:

(a) Compute $\mathbb{E}[\mathrm{SSE}]$, then $\mathbb{E}[s^2]$ and $\mathbb{E}[\tilde{s}^2]$.

(b) State the bias of $\tilde{s}^2$ as a number and as a percentage of σ^2.

(c) Show that the bias of the naive estimator does not vanish at any fixed n but shrinks toward zero as n grows, by writing $\mathbb{E}[\tilde{s}^2]/\sigma^2$ in terms of n.

C. Inference on the slope and intercept

C1. A pharmacologist regresses clearance rate y (mL/min) on body weight x (kg) for $n = 17$ subjects and obtains $\widehat{\beta}_1 = 0.74$, $S_{xx} = 410$, and $s = 2.6$. At $\alpha = 0.05$, test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$: is clearance linearly related to weight? Report $\mathrm{SE}(\widehat{\beta}_1)$, t_{obs}, the df, the critical value, and your decision in words.

C2. A soil scientist fits moisture retention y (%) against organic-matter content x (%) over $n = 14$ field plots, finding $\widehat{\beta}_1 = 1.85$, $S_{xx} = 96$, and $s = 2.1$.

(a) Construct a 95% confidence interval for the slope β_1, stating $\mathrm{df} = n - 2$ and the t critical value you use.

(b) Does the interval allow you to reject $H_0\colon \beta_1 = 0$ at the 5% level? Explain how the interval answers the two-sided test without recomputing t_{obs}.

(c) Give a 90% confidence interval for β_1 and say, in one sentence, why it is narrower than the 95% interval.

C3. A sleep researcher believes that each additional hour of screen time before bed *lengthens* sleep-onset latency. Regressing latency y (min) on screen time x (h) for $n = 20$ volunteers gives $\widehat{\beta}_1 = 4.1$, $S_{xx} = 58$, and $s = 9.4$. Test $H_0\colon \beta_1 = 0$ against the directional alternative $H_1\colon \beta_1 > 0$ at $\alpha = 0.05$. State the df, the one-sided critical value, t_{obs}, and your conclusion in plain words.

C4. Engineering theory predicts that a certain sensor's output should rise at exactly $c = 2.00$ mV per unit of applied pressure. A calibration of $n = 13$ readings yields $\widehat{\beta}_1 = 2.27$, $S_{xx} = 145$, and $s = 0.83$.

(a) Test $H_0\colon \beta_1 = 2.00$ against $H_1\colon \beta_1 \neq 2.00$ at $\alpha = 0.05$. Center the statistic at $c = 2.00$, not 0; state $\mathrm{df} = n - 2$.

(b) For comparison, compute what t_{obs} would have been against the null $\beta_1 = 0$, and explain in one sentence why that test answers the wrong question here.

C5. A nutrition study regresses resting metabolic rate y (kcal/day) on lean body mass x (kg) over $n = 16$ adults and reports $\widehat{\beta}_0 = 312$, $s = 84$, $\overline{X} = 48$, and $S_{xx} = 620$.

(a) Test whether the intercept — the predicted metabolic rate at *zero* lean mass — differs from 0 at $\alpha = 0.05$. Compute $\mathrm{SE}(\widehat{\beta}_0) = s\sqrt{1/n + \overline{X}^2/S_{xx}}$, give the df and critical value, and state the decision.

(b) Construct a 95% confidence interval for the intercept β_0.

(c) Comment briefly on whether "metabolic rate at zero lean mass" is a physically meaningful quantity, regardless of the test outcome.

C6. A geologist regresses sediment thickness y (cm) on distance from shore x (km) for $n = 10$ cores and finds $\widehat{\beta}_1 = -1.6$ with $\mathrm{SE}(\widehat{\beta}_1) = 0.55$. Report the two-sided p-value for $H_0\colon \beta_1 = 0$ by bracketing it from a t table on $n - 2$ df, and state your conclusion at $\alpha = 0.05$. Relevant table values on 8 df: $t_{0.025,\,8} = 2.306$, $t_{0.01,\,8} = 2.896$, $t_{0.005,\,8} = 3.355$.

C7. An ecologist regresses bird-species richness y on patch area x (ha) for $n = 12$ forest patches and obtains $\widehat{\beta}_1 = 0.09$, $S_{xx} = 44$, $s = 1.7$. The straight-line test gives a small, non-significant t_{obs}, and a colleague concludes "patch area has no effect on richness."

(a) Carry out the two-sided test of $H_0 \colon \beta_1 = 0$ at $\alpha = 0.05$ ($\text{df} = n - 2$) and confirm that you fail to reject.

(b) Explain why "fail to reject $\beta_1 = 0$" does *not* establish that richness is unrelated to area. Name two distinct reasons a genuine relationship could go undetected by this test.

(c) State what you would inspect or report to distinguish "no effect" from "an effect this test could not detect."

D. Mean response and prediction intervals

D1. A chemist fits reaction yield y (%) against catalyst loading x (mg) on $n = 12$ runs and obtains the summary statistics

$$\overline{X} = 8.0, \quad \overline{Y} = 64.0, \quad S_{xx} = 60.0, \quad S_{xy} = 84.0, \quad \text{SSE} = 27.0.$$

(a) Find the least-squares line $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$ and the error standard deviation

$$s = \sqrt{\text{SSE}\,/(n-2)}.$$

(b) Give a 95% confidence interval for the *mean* yield of all runs at $x_0 = 9$ mg. State the degrees of freedom and the t critical value you use.

D2. Using the same chemistry fit from problem D1 ($\widehat{y}$ and s as found there, with $\overline{X} = 8.0$, $S_{xx} = 60.0$, $n = 12$):

(a) Give a 95% *prediction* interval for the yield of a single new run at $x_0 = 9$ mg.

(b) Place this interval beside the mean-response interval of D1(b) and report both half-widths. Which is wider, and what term in the formula is responsible?

D3. A biologist fits leaf area y (cm^2) against days of growth x on $n = 15$ plants: $\widehat{y} = 4.0 + 1.30x$, $s = 2.40$, $\overline{X} = 10.0$, $S_{xx} = 250.0$. The plants were observed over x from 4 to 16 days.

(a) Give a 95% confidence interval for the mean leaf area at $x_0 = 14$ days.

(b) Give a 95% confidence interval for the mean leaf area at $x_0 = 11$ days.

(c) The two requests differ only in how far x_0 sits from $\overline{X} = 10$. Compare the half-widths and explain in one sentence why the $x_0 = 14$ interval is wider.

D4. A metallurgist fits hardness y (HV) against quench temperature x (°C) on $n = 18$ specimens: $\widehat{y} = 210 + 0.45x$, $s = 6.0$, $\overline{X} = 300.0$, $S_{xx} = 9000.0$. Give a 90% prediction interval for the hardness of one new specimen quenched at $x_0 = 330$°C. State $\text{df} = n - 2$ and the t critical value, and report the boxed interval.

D5. A hydrologist fits stream discharge y (m^3/s) against rainfall x (mm) on $n = 22$ storm events: $\widehat{y} = 1.5 + 0.080x$, $s = 0.90$, $\overline{X} = 40.0$, $S_{xx} = 4400.0$. The recorded rainfall ranged from 15 to 70 mm. A planner asks for the 95% prediction interval for discharge at $x_0 = 120$ mm.

(a) Compute the interval mechanically (state df and the t value).

(b) Explain why the reported number should not be trusted, naming the specific hazard and the feature of the formula that flags it.

D6. An agronomist fits grain protein y (%) against nitrogen rate x (kg/ha) on $n = 16$ plots: $\widehat{y} = 8.0 + 0.022x$, $s = 0.35$, $\overline{X} = 120.0$, $S_{xx} = 8000.0$. Three people each ask a question at $x_0 = 150$ kg/ha. (a) A breeder: "what is the *average* protein content across all plots fertilized at 150 kg/ha?" (b) A single farmer: "my one field at 150 kg/ha — what protein reading should I expect?" (c) A statistician: "how precisely have we located the true protein-vs-nitrogen line at 150 kg/ha?" For each, state which interval applies (mean response or prediction) and why, then compute all the distinct intervals required at 95% confidence (df $= n - 2$).

D7. A clinician fits systolic blood pressure y (mmHg) against age x (yr) on $n = 25$ patients: $\widehat{y} = 95.0 + 0.60x$, $s = 8.0$, $\overline{X} = 50.0$, $S_{xx} = 5000.0$.

(a) Give a 95% *upper* one-sided prediction bound for the systolic pressure of one new patient aged $x_0 = 65$ ("with 95% confidence the next patient's reading is no more than —"). Use $t_{0.05,\,23} = 1.714$.

(b) Give the two-sided 95% prediction interval at the same x_0 (use $t_{0.025,\,23} = 2.069$) and explain in one sentence why the one-sided upper endpoint is lower than the two-sided upper endpoint.

E. Correlation

E1. For $n = 14$ greenhouse trays the soil moisture x and the seedling emergence count y gave the sums of squares $S_{xx} = 560$, $S_{yy} = 224$, and $S_{xy} = 308$.

(a) Compute the sample correlation $r = S_{xy}/\sqrt{S_{xx}\,S_{yy}}$.

(b) State the sign and grade the magnitude (weak/moderate/strong) of the linear association.

(c) What fraction of the variation in y does the least-squares line explain?

E2. A hydrologist records, for $n = 11$ rivers, the catchment area x (km^2) and the peak summer flow y (m^3/s), obtaining $S_{xx} = 900$, $S_{yy} = 160$, $S_{xy} = -336$.

(a) Compute r and interpret its sign in words.

(b) Interpret the magnitude, and report R^2.

E3. In a sample of $n = 22$ employees the correlation between commute time and a self-reported job-satisfaction score is $r = 0.41$. At $\alpha = 0.05$, test $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$ using the correlation t-statistic $t = r\sqrt{n-2}/\sqrt{1-r^2}$. State the degrees of freedom, the critical value, and your decision in plain words.

E4. A horticulturist believes that extra sunlight hours x can only increase, never decrease, flowering-stem length y, and wants evidence that $\rho > 0$. From $n = 17$ plants she finds $r = 0.37$. Test $H_0 : \rho = 0$ versus $H_1 : \rho > 0$ at $\alpha = 0.05$. Give df, the one-sided critical value, and the decision. Comment on whether $r > 0$ alone settles the question.

E5. Two independent studies each report a sample correlation of $r = 0.34$ between sleep hours and a memory-recall score. Study A used $n = 16$ subjects; study B used $n = 42$ subjects. Test each at $\alpha = 0.05$ two-sided.

(a) Compute the t-statistic and state the decision for study A (df $= n - 2$).

(b) Do the same for study B.

(c) Explain in one sentence why the same r yields opposite verdicts.

E6. An analyst fits $y = \beta_0 + \beta_1 x$ to $n = 18$ paired observations and reports the slope test statistic $t_{\text{slope}} = \widehat{\beta}_1/\text{SE}(\widehat{\beta}_1) = 2.58$. Separately the sample correlation is $r = 0.541$.

(a) Compute the correlation t-statistic $r\sqrt{n-2}/\sqrt{1-r^2}$ and confirm it matches the reported slope t (up to rounding). Name the principle that guarantees this.

(b) A separate report lists only $\widehat{\beta}_1 = 4.0$, $S_{xx} = 50$, $S_{yy} = 1280$ and never prints r. Recover r from the identity $r = \widehat{\beta}_1\sqrt{S_{xx}/S_{yy}}$.

E7. A clinic measures resting heart rate y against caffeine dose x at seven equally spaced doses $x = -3, -2, -1, 0, 1, 2, 3$ (coded units), recording the symmetric pattern $y = 11, 6, 3, 2, 3, 6, 11$ (lowest at the middle dose, higher at both extremes).

(a) Compute r for these data.

(b) Is heart rate related to dose here? Reconcile your value of r with the answer.

(c) Separately, across 25 neighborhoods the number of bookstores x and the number of coffee shops y have $r = 0.81$. A blogger claims bookstores cause coffee shops. Critique this using the idea of a lurking variable, and state what would settle the matter.

F. ANOVA decomposition and R-squared

F1. A hydrologist regresses stream discharge y (m^3/s) on upstream rainfall x (cm) for $n = 9$ gauging events and obtains $S_{xx} = 50.0$, $S_{xy} = 120.0$, and $S_{yy} = \mathrm{SST} = 336.0$.

(a) Find the slope $\widehat{\beta}_1$ and use a closed form to compute SSR.

(b) Obtain SSE by subtraction and report the partition $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$.

(c) Verify that $\widehat{\beta}_1 S_{xy}$, $\widehat{\beta}_1^2 S_{xx}$, and S_{xy}^2/S_{xx} all give the same SSR.

F2. For a fit of seed germination rate y (%) on soil temperature x (°C) over $n = 14$ trays, an analyst reports $\mathrm{SST} = 400.0$ and $\mathrm{SSE} = 64.0$.

(a) Compute R^2 two ways: as $\mathrm{SSR}\,/\,\mathrm{SST}$ and as $1 - \mathrm{SSE}\,/\,\mathrm{SST}$.

(b) State the interpretation of R^2 in one plain sentence (percent of variation explained).

(c) The slope came out negative. Give the sample correlation r, sign and all.

F3. A materials lab regresses tensile strength y (MPa) on anneal time x (min) for $n = 10$ specimens, finding $\mathrm{SSR} = 288.0$ and $\mathrm{SSE} = 72.0$. Build the complete regression ANOVA table (Source, df, SS, MS, F), reporting SST, MSR, $\mathrm{MSE} = s^2$, and the overall F-statistic.

F4. Continue Problem F3 (F on $\mathrm{F}_{1,8}$). Test $H_0\colon \beta_1 = 0$ against $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$, using $F_{0.05,1,8} = 5.32$. State the decision in words, and explain in one sentence why the overall F-test here is necessarily two-sided.

F5. In the Problem F3 / F4 fit, the slope inference gives $\widehat{\beta}_1 = 2.40$ with standard error $\mathrm{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$, where $S_{xx} = 50.0$.

(a) Compute $s = \sqrt{\mathrm{MSE}}$ and then $\mathrm{SE}(\widehat{\beta}_1)$.

(b) Compute the slope statistic $t_{\mathrm{obs}} = \widehat{\beta}_1/\mathrm{SE}(\widehat{\beta}_1)$ and show t_{obs}^2 equals the ANOVA F from F3.

(c) Confirm the critical values match: show $t_{0.025,8}^2 = F_{0.05,1,8}$, using $t_{0.025,8} = 2.306$ and $F_{0.05,1,8} = 5.32$.

F6. A pharmacologist fits a straight line to dissolved drug fraction y versus time x for $n = 12$ samples and reports $R^2 = 0.97$, concluding "97% explained, so the first-order linear model is confirmed." A residual plot shows the residuals running positive at small and large x and negative in the middle. Critique the conclusion: state precisely what $R^2 = 0.97$ does and does not establish, name what the residual pattern reveals, and give the correct next step.

F7. A population-genetics group regresses a quantitative trait y on the dosage x of a single common allele across $n = 1002$ individuals. The fit gives $R^2 = 0.016$, with $\widehat{\beta}_1 = 0.45$ and $\mathrm{SE}(\widehat{\beta}_1) = 0.090$.

(a) Test $H_0\colon \beta_1 = 0$ versus $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$ with the slope t-test, stating $\mathrm{df} = n - 2$ and using $t_{0.025,1000} \approx 1.962$.

(b) Compute the overall $F = t_{\mathrm{obs}}^2$ and compare it to $F_{0.05,1,1000} \approx 3.85$.

(c) Reconcile the verdict with $R^2 = 0.016$: explain in words how a slope can be highly significant while explaining barely 1.6% of the variation.

G. Regression synthesis

G1. (End-to-end analysis from raw data.) A soil scientist applies $n = 8$ levels of a soluble fertilizer x (g per pot) and records the dry shoot mass y (mg) of the plant grown in each pot:

x	2	4	6	8	10	12	14	16
y	9	18	21	28	34	39	48	51

You may use the totals $\sum x_i = 72$, $\sum y_i = 248$, $\sum x_i^2 = 816$, $\sum y_i^2 = 9212$, $\sum x_i y_i = 2736$.

(a) Compute S_{xx}, S_{xy}, S_{yy} and the least-squares line $\widehat{y} = \widehat{\beta}_0 + \widehat{\beta}_1 x$.

(b) Estimate σ^2 by $s^2 = \mathrm{SSE}/(n-2)$, and test $H_0 : \beta_1 = 0$ against $H_1 : \beta_1 \neq 0$ at $\alpha = 0.05$. State the df.

(c) Report the coefficient of determination R^2 and the sample correlation r, and say in one sentence what each measures.

(d) Build the complete regression ANOVA table (Source, df, SS, MS, F) and verify that the overall F equals the square of the slope t_{obs} from part (b).

G2. (A second full fit, negative association.) An exercise physiologist measures resting heart rate y (beats/min) against weekly aerobic training volume x (hours) for $n = 7$ subjects:

x	1	2	3	4	5	6	7
y	15	14	13	10	9	6	3

Useful totals: $\sum x_i = 28$, $\sum y_i = 72$, $\sum x_i^2 = 140$, $\sum y_i^2 = 816$, $\sum x_i y_i = 224$.

(a) Fit the line and test $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 \neq 0$ at $\alpha = 0.01$ (give df and the t critical value).

(b) Compute R^2. A trainer reads "$R^2 = 0.97$" and a colleague reads "the slope is highly significant." Explain precisely what *different* question each statement answers, and why both can be true at once.

G3. (One-sided slope test from raw data.) A water-quality lab suspects that higher turbidity x (NTU) *raises* the chlorine demand y (mg/L). On $n = 6$ samples it records

x	2	4	6	8	10	12
y	2	8	11	13	14	12

with totals $\sum x_i = 42$, $\sum y_i = 60$, $\sum x_i^2 = 364$, $\sum y_i^2 = 698$, $\sum x_i y_i = 490$.

(a) Fit the line and carry out the directional test $H_0 : \beta_1 = 0$ versus $H_1 : \beta_1 > 0$ at $\alpha = 0.05$. Quote the one-sided t critical value and the df.

(b) Report R^2, and state in words how much of the variation in chlorine demand the straight line leaves *unexplained*.

G4. (Mean response versus a new observation.) A pharmacologist fits plasma clearance y (L/h) against body weight x (kg) and obtains the line $\widehat{y} = 8.0 + 2.5x$ with $s = 3.0$, $\overline{X} = 20$, $S_{xx} = 500$, from $n = 12$ animals. Work at 95% confidence ($t_{0.025,\,10} = 2.228$).

(a) Give a confidence interval for the *mean* clearance of all animals weighing $x_0 = 22$ kg, and a prediction interval for the clearance of *one new* animal at $x_0 = 22$ kg. Report both half-widths.

(b) Repeat both intervals at $x_0 = 30$ kg. Comparing $x_0 = 22$ with $x_0 = 30$, which interval flares *more* in relative terms, and what feature of the formulas explains it?

(c) In one sentence each, name the practical question that the mean-response interval answers and the question that the prediction interval answers.

G5. (ANOVA table and the F–t equivalence from summary statistics.) A reliability engineer regresses bearing wear y (microns) on running hours x (hundreds) for $n = 15$ bearings and reports $S_{xx} = 200$, $S_{xy} = 240$, $S_{yy} = 340$.

(a) Find SSR, SSE, and build the regression ANOVA table; report F and R^2.

(b) Test $H_0\colon \beta_1 = 0$ at $\alpha = 0.05$ using the overall F-test ($F_{0.05,1,13} = 4.67$).

(c) Compute the slope t-statistic and verify $t^2_{\text{obs}} = F$; also check that $t^2_{0.025,\,13} = F_{0.05,1,13}$ (use $t_{0.025,\,13} = 2.160$).

G6. (Significant slope, tiny R^2.) A population geneticist regresses a quantitative trait y on the dosage $x \in \{0, 1, 2\}$ of a common allele across $n = 502$ individuals. The fit gives $\widehat{\beta}_1 = 0.018$ with $\text{SE}(\widehat{\beta}_1) = 0.006$ and $R^2 = 0.020$.

(a) Test $H_0\colon \beta_1 = 0$ versus $H_1\colon \beta_1 \neq 0$ at $\alpha = 0.05$ (df $= 500$, $t_{0.025,\,500} \approx 1.965$). Is the slope significant?

(b) The lead author writes "the variant explains essentially none of the trait, so it is irrelevant." Reconcile the significant slope with $R^2 = 0.020$: what does each number actually claim, and is the author's inference sound?

G7. (Correlation test, then recover the line's variance accounting.) From $n = 18$ paired observations of two assay readings a printout gives $r = 0.62$ together with $S_{xx} = 400$ and $S_{yy} = 900$, but omits the slope and the sums of squares.

(a) Test $H_0\colon \rho = 0$ versus $H_1\colon \rho \neq 0$ at $\alpha = 0.05$ using the correlation t-statistic; give df and the critical value, and state why this is the *same* test as $H_0\colon \beta_1 = 0$.

(b) Using r with S_{xx} and S_{yy}, recover the slope $\widehat{\beta}_1$, the regression sum of squares SSR, and R^2, without the raw data.

11.8 Solutions

A. Least squares: fitting the line

A1. Method. Fitting a line to paired data is pure least squares: compute the centered sums S_{xx}, S_{xy}, then $\widehat{\beta}_1 = S_{xy}/S_{xx}$ and $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X}$. No probability or standard error is involved here.

(a) Means: $\overline{X} = \frac{160+170+180+190+200}{5} = 180$, $\overline{Y} = \frac{12+15+18+21+24}{5} = \frac{90}{5} = 18$. The deviations $x_i - \overline{X}$ are $-20, -10, 0, 10, 20$ and $y_i - \overline{Y}$ are $-6, -3, 0, 3, 6$, so

$$S_{xx} = (-20)^2 + (-10)^2 + 0^2 + 10^2 + 20^2 = 400 + 100 + 0 + 100 + 400 = 1000,$$
$$S_{xy} = (-20)(-6) + (-10)(-3) + 0(0) + (10)(3) + (20)(6) = 120 + 30 + 0 + 30 + 120 = 300,$$
$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{300}{1000} = 0.3,$$
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \overline{X} = 18 - 0.3(180) = 18 - 54 = -36.$$

$$\boxed{\widehat{\beta}_1 = 0.3, \qquad \widehat{\beta}_0 = -36.}$$

(b) The fitted line is

$$\boxed{\widehat{y} = -36 + 0.3\,x.}$$

The slope $\widehat{\beta}_1 = 0.3$ has units of browning per °C: each additional degree of oven temperature is associated with about 0.3 more browning units over the tested range. The intercept -36 is the line's value at $x = 0\,°\text{C}$, far below the data range $[160, 200]$, so it is an extrapolation artifact with no physical meaning (a frozen oven does not produce "negative browning").

(c) At $x = 185$,

$$\widehat{y} = -36 + 0.3(185) = -36 + 55.5 = 19.5.$$

Since 185 lies inside the observed range $[160, 200]$, this is **interpolation**.

$$\boxed{\widehat{y}(185) = 19.5 \text{ browning units (interpolation).}}$$

A2. Method. Least squares again, this time via the computing forms $S_{xx} = \sum x^2 - (\sum x)^2/n$ and $S_{xy} = \sum xy - (\sum x)(\sum y)/n$, which use running totals.

(a) With $n = 6$ and $x = 2, 4, 6, 8, 10, 12$, $y = 15, 18, 27, 33, 36, 45$:

$$\sum x = 2 + 4 + 6 + 8 + 10 + 12 = 42, \qquad \overline{X} = 42/6 = 7,$$
$$\sum y = 15 + 18 + 27 + 33 + 36 + 45 = 174, \qquad \overline{Y} = 174/6 = 29,$$
$$\sum x^2 = 4 + 16 + 36 + 64 + 100 + 144 = 364,$$
$$\sum xy = 30 + 72 + 162 + 264 + 360 + 540 = 1428.$$

Then

$$S_{xx} = \sum x^2 - \frac{(\sum x)^2}{n} = 364 - \frac{42^2}{6} = 364 - 294 = 70,$$
$$S_{xy} = \sum xy - \frac{(\sum x)(\sum y)}{n} = 1428 - \frac{42 \cdot 174}{6} = 1428 - 1218 = 210,$$
$$\widehat{\beta}_1 = \frac{210}{70} = 3, \qquad \widehat{\beta}_0 = 29 - 3(7) = 29 - 21 = 8.$$

$$\boxed{\widehat{y} = 8 + 3\,x.}$$

(b) The slope $\widehat{\beta}_1 = 3$ is in minutes per km: each additional kilometer of route length adds about 3 minutes of travel time. The intercept $\widehat{\beta}_0 = 8$ is the fitted time at $x = 0$ km, i.e. about 8 minutes of fixed overhead (loading, parking, handoff) that does not depend on distance. Here the intercept *is* interpretable: $x = 0$ is just below the data range and a fixed dispatch time is physically sensible.

$$\boxed{\text{slope } +3 \text{ min/km}; \quad \text{intercept } 8 \text{ min fixed overhead.}}$$

A3. Method. A residual is the leftover vertical gap $e_i = y_i - \widehat{y}_i$ with $\widehat{y}_i = \widehat{\beta}_0 + \widehat{\beta}_1 x_i$. The identity $\sum e_i = 0$ is guaranteed by fitting an intercept, so it serves as an arithmetic audit, not as evidence the model is good.

Using $\widehat{y} = 8 + 3x$ from A2:

x_i	2	4	6	8	10	12
y_i	15	18	27	33	36	45
$\widehat{y}_i$	14	20	26	32	38	44
e_i	1	-2	1	1	-2	1

$$\sum_{i=1}^{6} e_i = 1 - 2 + 1 + 1 - 2 + 1 = 0.$$

$$\boxed{\sum e_i = 0, \text{ confirming } \widehat{\beta}_0, \widehat{\beta}_1 \text{ are arithmetically consistent.}}$$

The zero sum only certifies that the slope and intercept were computed correctly; it does *not* confirm that a straight line is the right shape. A wildly curved data set forced through a line would also have residuals summing to zero, so curvature must be judged by *looking at* the residual pattern, not by this total.

A4. Method. Least squares for the line, then use the identity that the fit passes through $(\overline{X}, \overline{Y})$, and a single residual $e = y - \widehat{y}$ to read off over/under-prediction.

(a) Means: $\overline{X} = \frac{4+8+12+16+20}{5} = 12$, $\overline{Y} = \frac{20+19+28+27+36}{5} = \frac{130}{5} = 26$. Deviations $x_i - \overline{X}$ are $-8, -4, 0, 4, 8$ and $y_i - \overline{Y}$ are $-6, -7, 2, 1, 10$, so

$$S_{xx} = (-8)^2 + (-4)^2 + 0^2 + 4^2 + 8^2 = 64 + 16 + 0 + 16 + 64 = 160,$$
$$S_{xy} = (-8)(-6) + (-4)(-7) + 0(2) + (4)(1) + (8)(10) = 48 + 28 + 0 + 4 + 80 = 160,$$
$$\widehat{\beta}_1 = \frac{160}{160} = 1, \qquad \widehat{\beta}_0 = 26 - 1(12) = 14.$$

$$\boxed{\widehat{y} = 14 + 1 \cdot x.}$$

(b) Evaluate the fit at $x = \overline{X} = 12$:

$$\widehat{y}(12) = 14 + 1(12) = 26 = \overline{Y}.$$

The fitted value at the mean predictor equals the mean response, so the line passes through $(\overline{X}, \overline{Y}) = (12, 26)$, as the theory guarantees.

(c) At $x = 12$ the observed y is 28, so

$$e = y - \widehat{y} = 28 - 26 = +2.$$

The residual is positive, so the line *under-predicts* at $x = 12$: the actual algae density (28) sits 2 thousand cells/mL above the line.

$$\boxed{e(12) = +2 \text{ (line under-predicts there).}}$$

A5. Method. Least squares on the re-centered predictor $x' = x - c$. Because $\widehat{\beta}_1$ depends only on the centered deviations $x_i - \overline{X}$, which a location shift leaves unchanged, the slope is invariant; only the intercept absorbs the shift.

(a) On $x' = -2, -1, 0, 1, 2$ with $y = 40, 43, 47, 48, 52$: $\overline{X}' = 0$ and $\overline{Y} = \frac{40+43+47+48+52}{5} = \frac{230}{5} = 46$. Then

$$\begin{aligned} S_{x'x'} &= (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 = 10, \\ S_{x'y} &= (-2)(40-46) + (-1)(43-46) + 0 + (1)(48-46) + (2)(52-46) \\ &= (-2)(-6) + (-1)(-3) + 0 + (1)(2) + (2)(6) = 12 + 3 + 0 + 2 + 12 = 29, \\ \widehat{\beta}_1 &= \frac{29}{10} = 2.9, \qquad \widehat{\beta}_0' = \overline{Y} - \widehat{\beta}_1 \overline{X}' = 46 - 2.9(0) = 46. \end{aligned}$$

So on the shifted axis $\widehat{y} = 46 + 2.9\,x'$.

$$\boxed{\widehat{\beta}_1 = 2.9 \text{ m/yr}, \qquad \widehat{\beta}_0' = 46 \text{ m (level at the reference year 2021).}}$$

(b) Replacing x by $x' = x - 2021$ shifts every x_i and their mean by the same 2021, so each centered deviation $x_i' - \overline{X}' = (x_i - 2021) - (\overline{X} - 2021) = x_i - \overline{X}$ is unchanged; hence S_{xx} and S_{xy} are unchanged and $\widehat{\beta}_1 = 2.9$ on either axis. Substituting $x' = x - 2021$,

$$\widehat{y} = 46 + 2.9(x - 2021) = \big[46 - 2.9(2021)\big] + 2.9\,x,$$

so the original-axis intercept is

$$\widehat{\beta}_0 = 46 - 2.9(2021) = 46 - 5860.9 = -5814.9.$$

$$\boxed{\widehat{\beta}_1 = 2.9 \text{ (unchanged)}; \quad \widehat{\beta}_0 = -5814.9 \text{ (huge, an artifact of } x = 0).}$$

Re-centering left the slope alone because the slope is built only from the deviations $x_i - \overline{X}$, and shifting x by a constant moves the mean by the same constant, cancelling out of every deviation.

A6. Method. A fitted value is trustworthy only inside (or just beyond) the observed range of x; outside it the line is an unjustified extrapolation. Recognize in-range interpolation versus out-of-range extrapolation.

(a) At $x = 10$ g,

$$\widehat{y} = 9.0 + 1.6(10) = 9.0 + 16 = 25.0.$$

Because $x = 10$ lies inside the fitting range $[2, 14]$, this is interpolation: the data directly support the line there.

$$\boxed{\widehat{y}(10) = 25.0 \text{ (interpolation, on solid ground).}}$$

(b) The model was estimated only for $2 \le x \le 14$. The values $x = 60$ and $x = 200$ are far outside this band (60 is over four times the largest dose, 200 is more than fourteen times), so the arithmetic $\widehat{y}(60) = 105$ and $\widehat{y}(200) = 329$ are **extrapolations**: nothing in the data speaks to those sugar levels. Linearity was observed only locally, and a real shelf-life-versus-sugar relationship cannot rise without bound, so the true mean curve must bend long before $x = 60$. The intern's conclusion that more sugar is always better is an extension of a local trend past where any evidence exists.

$$\boxed{\widehat{y}(60), \widehat{y}(200) \text{ are extrapolations outside } [2, 14] \text{ — not supported by the data.}}$$

A7. Method. Least squares quantifies *association*, not causation. A tight, steep fit means x and y move together; it says nothing about whether changing x *makes* y change. The recognition step is to look for a lurking common cause.

The line $\widehat{y} = 2000 + 8500x$ with points hugging it correctly reports that fires with more trucks dispatched tend to have higher dollar damage — that co-movement is the entire content of the fit. It does *not* establish that the trucks cause the damage. Both the truck count and the damage are driven by a **lurking variable**: the size and severity of the fire. Big fires both draw more trucks and cause more damage, so the two rise together without one acting on the other. Sending fewer trucks would, if

anything, let damage *grow*, because the regression never showed that trucks drive damage — only that they accompany it.

Tight steep fit $\Rightarrow$ strong association, not causation; lurking cause = fire severity.

The fitted line legitimately summarizes how truck count and damage co-vary; it does not rank them as cause and effect, and the council member's proposal is unsupported.

B. Estimator properties and the error variance

B1. Method. The sums S_{xx}, S_{xy}, S_{yy} are in hand, so use the computing formula $\text{SSE} = S_{yy} - \widehat{\beta}_1 S_{xy}$ and the unbiased divisor $n-2$ (two parameters, β_0 and β_1, were estimated).

(a) Slope:

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-90.0}{72.0} = -1.250.$$

(b) Error sum of squares:

$$\begin{aligned}\text{SSE} &= S_{yy} - \widehat{\beta}_1 S_{xy} \\ &= 128.5 - (-1.250)(-90.0) \\ &= 128.5 - 112.5 = 16.0.\end{aligned}$$

(c) With $n = 9$, $\text{df} = n - 2 = 7$:

$$s^2 = \frac{\text{SSE}}{n-2} = \frac{16.0}{7} = 2.286, \qquad s = \sqrt{2.286} = 1.512.$$

We divide by $n-2$, not n, because two degrees of freedom are spent estimating the intercept and slope; the residuals satisfy $\sum e_i = 0$ and $\sum x_i e_i = 0$, leaving $n-2$ free pieces of information about σ^2.

$\widehat{\beta}_1 = -1.250, \quad \text{SSE} = 16.0, \quad s^2 = 2.286, \quad s = 1.512$ mg/L.

B2. Method. One fit feeds everything: get s from $\text{SSE}/(n-2)$, then the slope SE is $s/\sqrt{S_{xx}}$ and the intercept SE is $s\sqrt{1/n + \bar{x}^2/S_{xx}}$.

(a) Slope and SSE first:

$$\begin{aligned}\widehat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{30.0}{40.0} = 0.750, \\ \text{SSE} &= S_{yy} - \widehat{\beta}_1 S_{xy} = 27.0 - (0.750)(30.0) = 27.0 - 22.5 = 4.5.\end{aligned}$$

With $n = 11$, $\text{df} = n - 2 = 9$:

$$s^2 = \frac{4.5}{9} = 0.500, \qquad s = \sqrt{0.500} = 0.7071.$$

(b) Slope standard error:

$$\text{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{0.7071}{\sqrt{40.0}} = \frac{0.7071}{6.3246} = 0.1118.$$

(c) Intercept standard error, with $n = 11$, $\bar{x} = 8.0$, $S_{xx} = 40.0$:

$$\begin{aligned}\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} &= \frac{1}{11} + \frac{(8.0)^2}{40.0} \\ &= 0.0909 + \frac{64.0}{40.0} \\ &= 0.0909 + 1.6000 = 1.6909, \\ \text{SE}(\widehat{\beta}_0) &= s\sqrt{1.6909} = 0.7071 \times 1.3003 = 0.9194.\end{aligned}$$

The intercept is far less precise (0.9194 vs. 0.1118) because $\bar{x} = 8.0$ sits a long way from $x = 0$, so a small tilt in the line swings its height at $x = 0$ a long way.

$$\boxed{s = 0.7071, \quad \mathrm{SE}(\widehat{\beta}_1) = 0.1118, \quad \mathrm{SE}(\widehat{\beta}_0) = 0.9194.}$$

B3. Method. With σ^2 and n fixed, the only lever on $\mathrm{Var}(\widehat{\beta}_1) = \sigma^2/S_{xx}$ is $S_{xx} = \sum(x_i - \bar{x})^2$; wider spread means larger S_{xx} and a smaller slope variance.

(a) Design A: $\bar{x}_{\mathrm{A}} = (20 + 21 + 22 + 22 + 23 + 24 + 25)/7 = 157/7 = 22.428...$; rounding to a centered set, deviations from $\bar{x}_{\mathrm{A}} = 22.43$ are $\{-2.43, -1.43, -0.43, -0.43, 0.57, 1.57, 2.57\}$, and

$$\begin{aligned} S_{xx}^{\mathrm{A}} &= (-2.43)^2 + (-1.43)^2 + (-0.43)^2 + (-0.43)^2 + (0.57)^2 + (1.57)^2 + (2.57)^2 \\ &= 5.90 + 2.04 + 0.18 + 0.18 + 0.33 + 2.46 + 6.61 = 17.71. \end{aligned}$$

Design B: $\bar{x}_{\mathrm{B}} = (8 + 14 + 20 + 22 + 28 + 34 + 40)/7 = 166/7 = 23.714...$; deviations from $\bar{x}_{\mathrm{B}} = 23.71$ are $\{-15.71, -9.71, -3.71, -1.71, 4.29, 10.29, 16.29\}$, and

$$\begin{aligned} S_{xx}^{\mathrm{B}} &= (-15.71)^2 + (-9.71)^2 + (-3.71)^2 + (-1.71)^2 + (4.29)^2 + (10.29)^2 + (16.29)^2 \\ &= 246.8 + 94.3 + 13.8 + 2.9 + 18.4 + 105.9 + 265.4 = 747.5. \end{aligned}$$

(b) Slope variances with $\sigma^2 = 16.0$:

$$\begin{aligned} \mathrm{Var}(\widehat{\beta}_1^{\mathrm{A}}) &= \frac{\sigma^2}{S_{xx}^{\mathrm{A}}} = \frac{16.0}{17.71} = 0.9035, \\ \mathrm{Var}(\widehat{\beta}_1^{\mathrm{B}}) &= \frac{\sigma^2}{S_{xx}^{\mathrm{B}}} = \frac{16.0}{747.5} = 0.0214, \\ \frac{\mathrm{Var}(\widehat{\beta}_1^{\mathrm{A}})}{\mathrm{Var}(\widehat{\beta}_1^{\mathrm{B}})} &= \frac{0.9035}{0.0214} = 42.2. \end{aligned}$$

(c) Design B's masses span $[8, 40]$ rather than clustering in $[20, 25]$, so its larger S_{xx} acts as a longer lever arm that anchors the line's tilt; identical n and noise, yet the slope variance is about 42 times smaller.

$$\boxed{S_{xx}^{\mathrm{A}} = 17.71, \ S_{xx}^{\mathrm{B}} = 747.5; \ \mathrm{Var}(\widehat{\beta}_1^{\mathrm{B}}) = 0.0214 \text{ is about } 42\times \text{ smaller.}}$$

B4. Method. Because σ is given, use the population standard deviation $\mathrm{SD}(\widehat{\beta}_1) = \sigma/\sqrt{S_{xx}}$; set it equal to the target and solve for S_{xx}.

$$\begin{gathered} \mathrm{SD}(\widehat{\beta}_1) = \frac{\sigma}{\sqrt{S_{xx}}} \le 0.50 \implies \sqrt{S_{xx}} \ge \frac{\sigma}{0.50} = \frac{5.0}{0.50} = 10.0, \\ S_{xx} \ge 10.0^2 = 100.0. \end{gathered}$$

Tightening the target to 0.25 halves the allowed standard deviation, which quadruples the required S_{xx}:

$$\sqrt{S_{xx}} \ge \frac{5.0}{0.25} = 20.0 \implies S_{xx} \ge 400.0.$$

Halving the target standard deviation multiplies the required spread by four, since S_{xx} enters as $\sigma^2/\,\mathrm{SD}^2$.

$$\boxed{S_{xx} \ge 100.0 \text{ for } \mathrm{SD}(\widehat{\beta}_1) \le 0.50; \quad S_{xx} \ge 400.0 \text{ for } \mathrm{SD}(\widehat{\beta}_1) \le 0.25.}$$

B5. Method. Every probability about s^2 is a probability about a chi-square: multiply by $(n-2)/\sigma^2$ to land on χ^2_{n-2}. Here $n - 2 = 14$ df.

(a) The statistic $(n-2)s^2/\sigma^2 = 14s^2/6.0$ has χ^2_{14}, so df $= 14$.

(b) Upper tail:

$$\mathbb{P}(s^2 > 9.0) = \mathbb{P}\left(\frac{14\,s^2}{6.0} > \frac{14(9.0)}{6.0}\right) = \mathbb{P}(\chi^2_{14} > 21.0).$$

Since $\chi^2_{0.05,14} = 23.685$ and the cutoff 21.0 is below it, the upper-tail area is a bit above 0.05:

$$\mathbb{P}(s^2 > 9.0) = \mathbb{P}(\chi^2_{14} > 21.0) \approx 0.10.$$

(c) Lower tail:

$$\mathbb{P}(s^2 < 2.5) = \mathbb{P}\left(\frac{14\,s^2}{6.0} < \frac{14(2.5)}{6.0}\right) = \mathbb{P}(\chi^2_{14} < 5.833).$$

Since $\chi^2_{0.975,14} = 5.629$ leaves area 0.025 below it, and 5.833 is just above 5.629, the lower-tail area is just above 0.025:

$$\mathbb{P}(s^2 < 2.5) = \mathbb{P}(\chi^2_{14} < 5.833) \approx 0.03.$$

$$\boxed{\text{df} = 14; \quad \mathbb{P}(s^2 > 9.0) \approx 0.10; \quad \mathbb{P}(s^2 < 2.5) \approx 0.03.}$$

B6. Method. In regression the unbiased divisor is $n-2$ because two parameters were fit; dividing by n borrows a formula from a setting that estimated fewer parameters, and the wrong s propagates into every standard error.

(a) Analyst I (divisor $n = 12$):

$$s_{\text{I}}^2 = \frac{90.0}{12} = 7.500, \quad s_{\text{I}} = 2.739,$$
$$\text{SE}(\widehat{\beta}_1)_{\text{I}} = \frac{2.739}{\sqrt{50.0}} = \frac{2.739}{7.071} = 0.3873.$$

Analyst II (divisor $n - 2 = 10$):

$$s_{\text{II}}^2 = \frac{90.0}{10} = 9.000, \quad s_{\text{II}} = 3.000,$$
$$\text{SE}(\widehat{\beta}_1)_{\text{II}} = \frac{3.000}{\sqrt{50.0}} = \frac{3.000}{7.071} = 0.4243.$$

(b) The two standard errors differ by the factor $\sqrt{(n-2)/n} = \sqrt{10/12}$:

$$1 - \frac{0.3873}{0.4243} = 1 - \sqrt{\tfrac{10}{12}} = 1 - 0.9129 = 0.0871,$$

so Analyst I's standard error is about 8.7% too small.

(c) An understated standard error inflates the slope's t statistic $\widehat{\beta}_1/\,\text{SE}(\widehat{\beta}_1)$, making a marginal slope look more significant than the data warrant and inviting a false rejection of $\beta_1 = 0$.

$$\boxed{\text{SE}(\widehat{\beta}_1)_{\text{II}} = 0.4243 \text{ (correct)};\ \text{SE}(\widehat{\beta}_1)_{\text{I}} = 0.3873,\ \approx 8.7\% \text{ too small.}}$$

B7. Method. Bias follows directly from $\mathbb{E}[\text{SSE}] = (n-2)\sigma^2$, which comes from $(n-2)s^2/\sigma^2 \sim \chi^2_{n-2}$ and $\mathbb{E}[\chi^2_{n-2}] = n-2$. The unbiased divisor is $n-2$; the divisor n is biased low.

(a) With $n = 22$, $\sigma^2 = 10.0$:

$$\mathbb{E}[\text{SSE}] = (n-2)\sigma^2 = (20)(10.0) = 200.0,$$
$$\mathbb{E}[s^2] = \frac{\mathbb{E}[\text{SSE}]}{n-2} = \frac{200.0}{20} = 10.0 = \sigma^2, \quad \blacksquare$$
$$\mathbb{E}[\tilde{s}^2] = \frac{\mathbb{E}[\text{SSE}]}{n} = \frac{200.0}{22} = 9.091.$$

(b) Bias of the naive estimator:

$$\mathbb{E}[\tilde{s}^2] - \sigma^2 = 9.091 - 10.0 = -0.909,$$

i.e. low by 0.909, which is 9.1% of σ^2.

(c) In general

$$\frac{\mathbb{E}[\tilde{s}^2]}{\sigma^2} = \frac{(n-2)\sigma^2/n}{\sigma^2} = \frac{n-2}{n} = 1 - \frac{2}{n}.$$

The factor $1 - 2/n$ is strictly less than 1 for every finite n, so the naive estimator is biased low at any sample size; the bias $2/n$ only *shrinks* toward 0 as $n \to \infty$ (here $2/22 = 0.091$, matching the 9.1% above). The $n - 2$ divisor removes the bias exactly at every n.

$$\boxed{\mathbb{E}[s^2] = 10.0 = \sigma^2; \quad \mathbb{E}[\tilde{s}^2] = 9.091 \text{ (biased low by 9.1\%)}; \quad \mathbb{E}[\tilde{s}^2]/\sigma^2 = 1 - 2/n.}$$

C. Inference on the slope and intercept

C1. Two-sided slope test. The question is whether *any* linear effect exists, with no direction stated, so a two-sided t-test of $H_0\colon \beta_1 = 0$ versus $H_1\colon \beta_1 \neq 0$ on $n - 2 = 15$ df.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{2.6}{\sqrt{410}} = \frac{2.6}{20.248} = 0.1284,$$

$$t_{\text{obs}} = \frac{\widehat{\beta}_1 - 0}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{0.74}{0.1284} = 5.763.$$

With df = 15, the critical value is $t_{0.025,\,15} = 2.131$. Since $|5.763| > 2.131$, reject.

$$\boxed{t_{\text{obs}} = 5.76 > t_{0.025,\,15} = 2.131 \Rightarrow \text{reject } H_0\text{; clearance is significantly linearly related to weight at } \alpha = 0.05.}$$

The slope sits nearly six standard errors from zero — strong evidence of a real linear effect.

C2. Slope confidence interval. A range of plausible slopes calls for the t-interval $\widehat{\beta}_1 \pm t_{\alpha/2,\,n-2}\ \mathrm{SE}(\widehat{\beta}_1)$ on $n - 2 = 12$ df. First the standard error:

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{2.1}{\sqrt{96}} = \frac{2.1}{9.798} = 0.2143.$$

(a) For 95% on 12 df, $t_{0.025,\,12} = 2.179$.

$$\text{margin} = 2.179 \times 0.2143 = 0.4670,$$
$$\beta_1 \in 1.85 \pm 0.467 = (1.383,\ 2.317).$$

$$\boxed{1.38 < \beta_1 < 2.32 \text{ (95\% CI)}.}$$

(b) The 95% interval lies entirely above 0, so 0 is not a plausible slope at the 5% level; equivalently, the two-sided test of $H_0\colon \beta_1 = 0$ rejects at $\alpha = 0.05$. A confidence interval that excludes the null value is the inversion of the rejecting test, so no separate t_{obs} is needed.

$$\boxed{0 \notin (1.38,\ 2.32) \ \Rightarrow\ \text{reject } H_0\colon \beta_1 = 0 \text{ at 5\%}.}$$

(c) For 90% on 12 df, $t_{0.05,\,12} = 1.782$.

$$\text{margin} = 1.782 \times 0.2143 = 0.3819,$$
$$\beta_1 \in 1.85 \pm 0.382 = (1.468,\ 2.232).$$

$$\boxed{1.47 < \beta_1 < 2.23 \text{ (90\% CI)}.}$$

It is narrower because lower confidence uses a smaller critical value ($1.782 < 2.179$), shrinking the margin — less confidence buys a tighter interval.

C3. One-sided slope test. The claim is directional ("lengthens," i.e. the slope is positive), so a one-sided test of $H_0\colon \beta_1 = 0$ versus $H_1\colon \beta_1 > 0$ with all of α in the upper tail, on $n-2 = 18$ df.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{9.4}{\sqrt{58}} = \frac{9.4}{7.616} = 1.2343,$$
$$t_{\text{obs}} = \frac{\widehat{\beta}_1 - 0}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{4.1}{1.2343} = 3.322.$$

With df $= 18$, the upper-tail cutoff is $t_{0.05,\,18} = 1.734$. Since $3.322 > 1.734$, reject.

$t_{\text{obs}} = 3.32 > t_{0.05,\,18} = 1.734 \Rightarrow$ reject H_0; sleep-onset latency rises significantly with screen time.

Because the direction was pre-specified and the slope is positive, the whole 5% sits in the right tail, lowering the bar from the two-sided $t_{0.025,\,18} = 2.101$ to 1.734.

C4. Testing a specific nonzero slope.

(a) The hypothesized slope is the nonzero constant $c = 2.00$, so subtract c in the numerator: $t_{\text{obs}} = (\widehat{\beta}_1 - c)/\,\mathrm{SE}(\widehat{\beta}_1)$, two-sided, on $n-2 = 11$ df.

$$\mathrm{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{0.83}{\sqrt{145}} = \frac{0.83}{12.042} = 0.06893,$$
$$t_{\text{obs}} = \frac{\widehat{\beta}_1 - c}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{2.27 - 2.00}{0.06893} = \frac{0.27}{0.06893} = 3.917.$$

With df $= 11$, $t_{0.025,\,11} = 2.201$. Since $|3.917| > 2.201$, reject.

$t_{\text{obs}} = 3.92 > t_{0.025,\,11} = 2.201 \Rightarrow$ reject H_0; the sensor's rate differs significantly from the predicted 2.00.

(b) Against $\beta_1 = 0$ the statistic would be

$$t_{\text{obs}} = \frac{2.27 - 0}{0.06893} = 32.93,$$

enormous and certain to reject. But that test only confirms the sensor responds at all, which is not in doubt; the live question is whether the rate matches the predicted 2.00, so the null value must be $c = 2.00$, not 0.

C5. Inference for the intercept.

(a) The question is about β_0, the mean response at $x = 0$, so the intercept t-test with $\mathrm{SE}(\widehat{\beta}_0) = s\sqrt{1/n + \overline{X}^2/S_{xx}}$ on $n-2 = 14$ df.

$$\mathrm{SE}(\widehat{\beta}_0) = s\sqrt{\frac{1}{n} + \frac{\overline{X}^2}{S_{xx}}} = 84\sqrt{\frac{1}{16} + \frac{48^2}{620}}$$
$$= 84\sqrt{0.06250 + 3.71613} = 84\sqrt{3.77863} = 84(1.94387) = 163.3,$$
$$t_{\text{obs}} = \frac{\widehat{\beta}_0 - 0}{\mathrm{SE}(\widehat{\beta}_0)} = \frac{312}{163.3} = 1.910.$$

With df $= 14$, $t_{0.025,\,14} = 2.145$. Since $|1.910| < 2.145$, do not reject.

$|t_{\text{obs}}| = 1.91 < t_{0.025,\,14} = 2.145 \Rightarrow$ fail to reject H_0; the intercept is not significantly different from 0.

(b) A 95% interval is $\widehat{\beta}_0 \pm t_{0.025,\,14}\ \mathrm{SE}(\widehat{\beta}_0)$ on 14 df:

$$\begin{aligned}\text{margin} &= 2.145 \times 163.3 = 350.3,\\ \beta_0 &\in 312 \pm 350.3 = (-38.3,\ 662.3).\end{aligned}$$

$$\boxed{-38 < \beta_0 < 662 \text{ kcal/day (95\% CI).}}$$

The interval straddles 0, matching the failure to reject in (a).

(c) The intercept is the predicted metabolic rate at *zero* lean mass, an extrapolation far outside any observed body — nobody has zero lean mass — so it is not physically meaningful here, however the test turns out. It serves only to anchor the fitted line, not as a biological baseline.

C6. p-value off the t statistic. We want the two-sided tail area of the t_{n-2} distribution beyond $|t_{\text{obs}}|$, here on $n-2=8$ df.

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\mathrm{SE}(\widehat{\beta}_1)} = \frac{-1.6}{0.55} = -2.909, \qquad |t_{\text{obs}}| = 2.909.$$

On 8 df the table gives $t_{0.01,\,8} = 2.896$ and $t_{0.005,\,8} = 3.355$. Since $2.896 < 2.909 < 3.355$, the one-tail area lies between 0.005 and 0.01, so the two-sided p-value satisfies

$$0.01 < p < 0.02.$$

The whole interval falls below $\alpha = 0.05$.

$$\boxed{p \approx 0.019 < 0.05 \Rightarrow \text{reject } H_0\text{; sediment thickness decreases significantly with distance from shore.}}$$

The p-value reports *how* significant: the evidence is solid but not overwhelming, just clearing the 0.02 mark.

C7. Contrast: fail-to-reject is not proof of no relationship.

(a) Two-sided test of $H_0\colon \beta_1 = 0$ on $n-2=10$ df.

$$\begin{aligned}\mathrm{SE}(\widehat{\beta}_1) &= \frac{s}{\sqrt{S_{xx}}} = \frac{1.7}{\sqrt{44}} = \frac{1.7}{6.633} = 0.2563,\\ t_{\text{obs}} &= \frac{0.09}{0.2563} = 0.351.\end{aligned}$$

With df $= 10$, $t_{0.025,\,10} = 2.228$. Since $|0.351| < 2.228$, fail to reject.

$$\boxed{|t_{\text{obs}}| = 0.35 < t_{0.025,\,10} = 2.228 \ \Rightarrow\ \text{fail to reject } H_0.}$$

(b) Failing to reject $\beta_1 = 0$ means only that the data give too little evidence of a *straight-line* effect; it never proves x and y are unrelated. Two distinct reasons a real relationship can escape this test: (i) the dependence may be **nonlinear** — a hump-shaped or saturating species-area pattern has a best-fit line that is nearly flat even when richness depends strongly on area; (ii) the test may simply lack **power** — a small n, a small S_{xx} (patch areas bunched together), or a large s all inflate $\mathrm{SE}(\widehat{\beta}_1)$ and shrink t_{obs}, so a true linear slope goes undetected.

$$\boxed{\text{fail to reject } \beta_1 = 0 \ \not\Rightarrow\ \text{no relationship between richness and area.}}$$

(c) Plot richness against area to check for a nonlinear (e.g. curved or saturating) pattern, and report the confidence interval for β_1 rather than only the verdict: a *wide* interval straddling 0 signals an inconclusive, low-power test, whereas a *narrow* interval tightly around 0 would be genuine evidence that any linear effect is small. "Not significant" means "not enough evidence," never "the effect is zero."

D. Mean response and prediction intervals

D1. Which rule, and why? Setup for everything that follows — the line comes from least squares ($\widehat{\beta}_1 = S_{xy}/S_{xx}$, $\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1\overline{X}$), the spread from $s^2 = \text{SSE}/(n-2)$; part (b) asks for the *mean* response at x_0 (average over all units), so it is the mean-response confidence interval with no extra 1 under the root.

(a) Slope, intercept, and s:

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{84.0}{60.0} = 1.40,$$
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1\overline{X} = 64.0 - 1.40(8.0) = 52.80,$$
$$s^2 = \frac{\text{SSE}}{n-2} = \frac{27.0}{10} = 2.70, \qquad s = \sqrt{2.70} = 1.643.$$

$$\boxed{\widehat{y} = 52.80 + 1.40x, \qquad s = 1.643 \text{ \% on } n-2 = 10 \text{ df.}}$$

(b) Mean-response confidence interval at $x_0 = 9$, df $= n-2 = 10$, $t_{0.025,\,10} = 2.228$:

$$\widehat{y}_0 = 52.80 + 1.40(9) = 65.40,$$
$$x_0 - \overline{X} = 9 - 8 = 1,$$
$$\text{se}(\widehat{y}_0) = s\sqrt{\frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}} = 1.643\sqrt{\frac{1}{12} + \frac{1^2}{60}}$$
$$= 1.643\sqrt{0.08333 + 0.01667} = 1.643\sqrt{0.10000} = 1.643(0.3162) = 0.5195,$$
$$65.40 \pm 2.228(0.5195) = 65.40 \pm 1.158.$$

$$\boxed{\mu_{Y|x_0} \in (64.24,\ 66.56) \text{ \% at 95\% (mean yield, } x_0 = 9).}$$

We are 95% confident the average yield of all runs at 9 mg of catalyst lies between about 64.2% and 66.6%.

D2. Which rule, and why? "A single new run" is one future observation, so the prediction interval — identical to D1(b) except for the extra 1+ under the root, which adds that run's own error variance σ^2. This is the headline mean-vs-new contrast.

(a) Same center $\widehat{y}_0 = 65.40$, df $= 10$, $t_{0.025,\,10} = 2.228$, but the radius carries the extra 1:

$$\text{se(pred)} = s\sqrt{1 + \frac{1}{n} + \frac{(x_0-\overline{X})^2}{S_{xx}}} = 1.643\sqrt{1 + 0.08333 + 0.01667}$$
$$= 1.643\sqrt{1.10000} = 1.643(1.0488) = 1.7232,$$
$$65.40 \pm 2.228(1.7232) = 65.40 \pm 3.840.$$

$$\boxed{Y_0 \in (61.56,\ 69.24) \text{ \% at 95\% (one new run, } x_0 = 9).}$$

(b) Side by side at the same $x_0 = 9$:

mean response:	(64.24, 66.56),	half-width 1.158,
new observation:	(61.56, 69.24),	half-width 3.840.

Prediction is wider ($3.840 > 1.158$); the extra 1+ under the root — a whole σ^2 — is responsible.

Same midpoint 65.40, but the prediction band is about 3.3× as wide: the lone 1 (one new run's scatter) dominates the small 0.100 from estimating the line.

D3. Which rule, and why? Both parts ask for the *mean* leaf area, so both are mean-response confidence intervals (no 1+). They differ only in $(x_0 - \overline{X})^2/S_{xx}$, the hourglass term, so comparing them shows the flare. df $= n - 2 = 13$, $t_{0.025,\,13} = 2.160$ throughout.

(a) At $x_0 = 14$, $x_0 - \overline{X} = 4$:

$$\begin{aligned}
\widehat{y}_0 &= 4.0 + 1.30(14) = 22.20, \\
\text{se}(\widehat{y}_0) &= 2.40\sqrt{\frac{1}{15} + \frac{4^2}{250}} = 2.40\sqrt{0.06667 + 0.06400} \\
&= 2.40\sqrt{0.13067} = 2.40(0.3615) = 0.8675, \\
22.20 &\pm 2.160(0.8675) = 22.20 \pm 1.874.
\end{aligned}$$

$$\boxed{\mu_{Y|x_0} \in (20.33,\ 24.07)\ \text{cm}^2 \text{ at } x_0 = 14;\ \text{half-width } 1.874.}$$

(b) At $x_0 = 11$, $x_0 - \overline{X} = 1$:

$$\begin{aligned}
\widehat{y}_0 &= 4.0 + 1.30(11) = 18.30, \\
\text{se}(\widehat{y}_0) &= 2.40\sqrt{\frac{1}{15} + \frac{1^2}{250}} = 2.40\sqrt{0.06667 + 0.00400} \\
&= 2.40\sqrt{0.07067} = 2.40(0.2658) = 0.6380, \\
18.30 &\pm 2.160(0.6380) = 18.30 \pm 1.378.
\end{aligned}$$

$$\boxed{\mu_{Y|x_0} \in (16.92,\ 19.68)\ \text{cm}^2 \text{ at } x_0 = 11;\ \text{half-width } 1.378.}$$

(c) Half-widths: 1.874 at $x_0 = 14$ versus 1.378 at $x_0 = 11$.

$$\boxed{x_0 = 14 \text{ is wider because } (x_0 - \overline{X})^2 = 16 > 1 \text{ flares the } (x_0 - \overline{X})^2/S_{xx} \text{ term.}}$$

The band is pinned down best near $\overline{X} = 10$ and fans out symmetrically; four days off center gives a wider interval than one day off center.

D4. Which rule, and why? "One new specimen" is a single future observation ⇒ prediction interval, root $\sqrt{1 + \frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}}$, on $n - 2 = 16$ df at 90% ($\alpha/2 = 0.05$), $t_{0.05,\,16} = 1.746$.

$$\begin{aligned}
\widehat{y}_0 &= 210 + 0.45(330) = 210 + 148.5 = 358.5, \\
x_0 - \overline{X} &= 330 - 300 = 30, \\
\text{se(pred)} &= 6.0\sqrt{1 + \frac{1}{18} + \frac{30^2}{9000}} = 6.0\sqrt{1 + 0.05556 + 0.10000} \\
&= 6.0\sqrt{1.15556} = 6.0(1.0750) = 6.450, \\
358.5 &\pm 1.746(6.450) = 358.5 \pm 11.262.
\end{aligned}$$

$$\boxed{Y_0 \in (347.24,\ 369.76)\ \text{HV at 90\% (one new specimen, } x_0 = 330).}$$

With 90% confidence one new specimen quenched at 330°C will have hardness between about 347 and 370 HV; the lone 1 supplies most of the 1.156 under the root.

D5. Which rule, and why? Mechanically (a) is just the prediction interval at $x_0 = 120$, on $n - 2 = 20$ df, $t_{0.025,\,20} = 2.086$. The lesson in (b) is that the formula returns a finite interval however far out you go, but the model was only checked on the observed range $[15, 70]$ mm.

(a) Mechanical computation:

$$\begin{aligned}
\widehat{y}_0 &= 1.5 + 0.080(120) = 1.5 + 9.60 = 11.10,\\
x_0 - \overline{X} &= 120 - 40 = 80,\\
\text{se(pred)} &= 0.90\sqrt{1 + \frac{1}{22} + \frac{80^2}{4400}} = 0.90\sqrt{1 + 0.04545 + 1.45455}\\
&= 0.90\sqrt{2.50000} = 0.90(1.5811) = 1.4230,\\
11.10 &\pm 2.086(1.4230) = 11.10 \pm 2.969.
\end{aligned}$$

$\boxed{Y_0 \in (8.13,\ 14.07)\ \text{m}^3/\text{s at } x_0 = 120 \text{ — arithmetically correct but unwarranted.}}$

(b) Why it should not be trusted:

$\boxed{x_0 = 120 \text{ mm lies far outside the fitted range } [15, 70];\ \text{this is extrapolation.}}$

The hazard is extrapolation: the linear, constant-variance model was only verified for rainfall between 15 and 70 mm, so at 120 mm the true discharge curve may bend and the variance may change, with no data to check either. The formula's own warning is the $(x_0 - \overline{X})^2/S_{xx} = 1.455$ term, which has ballooned to dominate the root — a sign the interval is mostly extrapolation.

D6. Which rule, and why? The noun in each request decides it: "average across all plots" and "how precisely is the line located" are statements about the *mean* $\beta_0 + \beta_1 x_0$ (mean-response CI, no extra 1); "my one field's reading" is a *single new observation* (prediction interval, with the 1+). So (a) and (c) call for the same mean-response interval and (b) for the wider prediction interval. All at $x_0 = 150$, df $= n - 2 = 14$, $t_{0.025,\,14} = 2.145$.

Common pieces: $\widehat{y}_0 = 8.0 + 0.022(150) = 8.0 + 3.30 = 11.30$; $x_0 - \overline{X} = 150 - 120 = 30$; $\dfrac{1}{n} + \dfrac{(x_0 - \overline{X})^2}{S_{xx}} = \dfrac{1}{16} + \dfrac{30^2}{8000} = 0.06250 + 0.11250 = 0.17500$.

(a)/(c) Mean-response confidence interval:

$$\begin{aligned}
\text{se}(\widehat{y}_0) &= 0.35\sqrt{0.17500} = 0.35(0.4183) = 0.1464,\\
11.30 &\pm 2.145(0.1464) = 11.30 \pm 0.314.
\end{aligned}$$

$\boxed{\text{(a),(c) mean CI: } \mu_{Y|x_0} \in (10.99,\ 11.61)\ \%\text{ at } 95\%.}$

Both the breeder's "average across all plots" and the statistician's "precision of the line" are the mean response; the half-width 0.314 *is* the precision with which $\widehat{y}_0$ locates the true line at $x_0 = 150$.

(b) Prediction interval (same center, extra 1 under the root):

$$\begin{aligned}
\text{se(pred)} &= 0.35\sqrt{1 + 0.17500} = 0.35\sqrt{1.17500} = 0.35(1.0840) = 0.3794,\\
11.30 &\pm 2.145(0.3794) = 11.30 \pm 0.814.
\end{aligned}$$

$\boxed{\text{(b) prediction PI: } Y_0 \in (10.49,\ 12.11)\ \%\text{ at } 95\%\text{ (wider).}}$

Only the single farmer's "my one field" question is a new observation; its interval is the wider one. Reading (b) as a mean interval would give a falsely tight ± 0.31 and overstate how well one field's protein can be pinned.

D7. Which rule, and why? "One new patient" is a single future observation $\Rightarrow$ prediction interval (root with the 1+). Part (a) asks for a bound in one direction ("no more than"), so it is a one-sided upper bound: put all $\alpha = 0.05$ in the upper tail and use $t_{0.05,\,23} = 1.714$. Part (b) is the two-sided version with $t_{0.025,\,23} = 2.069$. df $= n - 2 = 23$ throughout.

Common pieces: $\widehat{y}_0 = 95.0 + 0.60(65) = 95.0 + 39.0 = 134.0$; $x_0 - \overline{X} = 65 - 50 = 15$;

$$\text{se(pred)} = 8.0\sqrt{1 + \frac{1}{25} + \frac{15^2}{5000}} = 8.0\sqrt{1 + 0.04000 + 0.04500}$$
$$= 8.0\sqrt{1.08500} = 8.0(1.0416) = 8.3329.$$

(a) One-sided 95% upper prediction bound, $t_{0.05,\,23} = 1.714$:

$$\text{upper bound} = 134.0 + 1.714(8.3329) = 134.0 + 14.283 = 148.28.$$

$\boxed{Y_0 \le 148.28 \text{ mmHg with 95\% confidence (one-sided upper prediction bound).}}$

(b) Two-sided 95% prediction interval, $t_{0.025,\,23} = 2.069$:

$$134.0 \pm 2.069(8.3329) = 134.0 \pm 17.241,$$

$\boxed{Y_0 \in (116.76,\ 151.24) \text{ mmHg at 95\% (two-sided prediction).}}$

The one-sided upper endpoint (148.28) is lower than the two-sided upper endpoint (151.24) because the one-sided bound spends the entire $\alpha = 0.05$ in the upper tail ($t = 1.714$), whereas the two-sided interval splits the risk and uses the larger $t = 2.069$ on each side.

E. Correlation

E1. Method. The question asks for a unitless measure of linear strength, so we use $r = S_{xy}/\sqrt{S_{xx}\,S_{yy}}$, and $R^2 = r^2$ for the explained fraction.

(a) Assemble the sums and form r:

$$S_{xx} = 560, \quad S_{yy} = 224, \quad S_{xy} = 308,$$
$$r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} = \frac{308}{\sqrt{560 \cdot 224}} = \frac{308}{\sqrt{125440}} = \frac{308}{354.17} = 0.870.$$

$\boxed{r = 0.870}$

(b) The sign is positive: more soil moisture goes with more seedlings. With $|r| = 0.87$ close to 1, this is a *strong* positive linear association.

(c) The line explains $R^2 = r^2 = (0.870)^2 = 0.756$, about 76% of the variation in emergence count.

$\boxed{R^2 = 0.756}$

E2. Method. Again $r = S_{xy}/\sqrt{S_{xx}\,S_{yy}}$; the negative S_{xy} foretells a negative r.

(a) Compute:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} = \frac{-336}{\sqrt{900 \cdot 160}} = \frac{-336}{\sqrt{144000}} = \frac{-336}{379.47} = -0.885.$$

The sign is negative: larger catchment areas go with *lower* peak summer flow in this sample.

$\boxed{r = -0.885 \text{ (strong negative linear association)}}$

(b) $|r| = 0.885$ is large, so the points hug a downward line tightly. The line explains $R^2 = r^2 = (0.885)^2 = 0.783$ of the variation in flow.

$\boxed{R^2 = 0.783}$

E3. Method. "Could $\rho = 0$?" with no prior direction is a two-sided correlation t-test on $n-2$ df. Here $n = 22$, so df $= 20$.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.41\sqrt{20}}{\sqrt{1-0.41^2}} = \frac{0.41 \cdot 4.472}{\sqrt{0.8319}} = \frac{1.834}{0.9121} = 2.010.$$

Critical value (two-sided, df $= 20$): $t_{0.025,20} = 2.086$. Since $|2.010| < 2.086$, we *fail to reject* H_0.

$$\boxed{t = 2.01 < 2.086 \text{ (df} = 20) \Rightarrow \text{fail to reject } H_0 : \rho = 0}$$

At the 5% level there is not enough evidence of a linear association between commute time and job satisfaction.

E4. Method. The claim has a built-in direction ($\rho > 0$), so we run a *one-sided* correlation t-test and compare to a single upper-tail critical value. Here $n = 17$, df $= 15$.

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.37\sqrt{15}}{\sqrt{1-0.37^2}} = \frac{0.37 \cdot 3.873}{\sqrt{0.8631}} = \frac{1.433}{0.9290} = 1.543.$$

One-sided critical value (df $= 15$): $t_{0.05,15} = 1.753$. Since $1.543 < 1.753$, we *fail to reject* H_0.

$$\boxed{t = 1.54 < 1.753 \text{ (df} = 15) \Rightarrow \text{fail to reject } H_0}$$

The data do not establish a positive linear effect of sunlight on stem length at the 5% level. A positive sample r alone never settles the question: $r > 0$ can arise by chance, which is exactly what the test guards against.

E5. Method. The correlation $t = r\sqrt{n-2}/\sqrt{1-r^2}$ grows with n, so the same r can land on either side of the cutoff. Both tests are two-sided.

(a) Study A, $n = 16$, df $= 14$:

$$t_A = \frac{0.34\sqrt{14}}{\sqrt{1-0.34^2}} = \frac{0.34 \cdot 3.742}{\sqrt{0.8844}} = \frac{1.272}{0.9404} = 1.353,$$

with $t_{0.025,14} = 2.145$. Since $1.353 < 2.145$, *fail to reject*; no significant association.

$$\boxed{t_A = 1.35 < 2.145 \text{ (df} = 14) \Rightarrow \text{retain } H_0}$$

(b) Study B, $n = 42$, df $= 40$:

$$t_B = \frac{0.34\sqrt{40}}{\sqrt{1-0.34^2}} = \frac{0.34 \cdot 6.325}{\sqrt{0.8844}} = \frac{2.150}{0.9404} = 2.287,$$

with $t_{0.025,40} = 2.021$. Since $2.287 > 2.021$, *reject*; the association is significant.

$$\boxed{t_B = 2.29 > 2.021 \text{ (df} = 40) \Rightarrow \text{reject } H_0}$$

(c) The effect size $r = 0.34$ is identical; the larger sample in study B supplies more evidence, inflating t past the cutoff. Significance certifies that $\rho \neq 0$, not that the association is large.

E6. Method. The correlation test and the slope test are algebraically the same test, so the two t-statistics must coincide; and $r = \widehat{\beta}_1\sqrt{S_{xx}/S_{yy}}$ recovers r from the slope.

(a) Correlation t with $n = 18$, df $= 16$:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.541\sqrt{16}}{\sqrt{1-0.541^2}} = \frac{0.541 \cdot 4}{\sqrt{0.7073}} = \frac{2.164}{0.8410} = 2.573.$$

This matches the reported $t_{\text{slope}} = 2.58$ up to rounding. The principle: $\rho = 0$, $\beta_1 = 0$, and "the line explains nothing" are one hypothesis with one p-value, so the slope t and the correlation t are identical.

$$\boxed{t_{\text{corr}} = 2.57 \approx 2.58 = t_{\text{slope}}}$$

(b) Apply the slope-to-r identity:

$$r = \widehat{\beta}_1 \sqrt{\frac{S_{xx}}{S_{yy}}} = 4.0\sqrt{\frac{50}{1280}} = 4.0\sqrt{0.039063} = 4.0 \cdot 0.19764 = 0.791.$$

The sign matches the positive slope and $|r| \le 1$, as required.

$$\boxed{r = 0.791}$$

E7. Method. r measures *linear* association only; a symmetric curved pattern produces $r \approx 0$ despite a perfect relationship. Causal claims need more than a large r.

(a) Centers: $\overline{X} = 0$ by symmetry, and $\overline{Y} = (11 + 6 + 3 + 2 + 3 + 6 + 11)/7 = 42/7 = 6$. With $\overline{X} = 0$,

$$S_{xy} = \sum x_i(y_i - \overline{Y}) = \sum x_i y_i - \overline{Y} \sum x_i = \sum x_i y_i - 0,$$
$$\sum x_i y_i = (-3)(11) + (-2)(6) + (-1)(3) + 0 + (1)(3) + (2)(6) + (3)(11) = 0$$

(the terms cancel in ± pairs). Hence $S_{xy} = 0$, so $r = 0$.

$$\boxed{r = 0}$$

(b) Yes, heart rate is plainly related to dose — y is an exact function of x (lowest in the middle, higher at both extremes). The value $r = 0$ reports only that there is no *linear* trend; it is blind to the symmetric U-shaped curvature. "No correlation" is not "no relationship."

(c) $r = 0.81$ is a real association but not evidence of causation. A lurking variable — here, neighborhood affluence/population density z — can drive both the number of bookstores and the number of coffee shops, manufacturing a large r with no direct causal path bookstores → coffee shops. To settle it, hold z fixed (compare neighborhoods of equal density/income) or add z as a predictor; the bookstore–coffee association should then largely vanish.

$$\boxed{r = 0.81 \text{ is real but likely spurious; a confounder } z \text{ drives both}}$$

F. ANOVA decomposition and R-squared

F1. Partition the total variation (discharge vs. rainfall). Use the identity $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE}$ with the closed form $\mathrm{SSR} = S_{xy}^2/S_{xx}$; no fitted values are needed.

(a) Slope and SSR:

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{120.0}{50.0} = 2.40,$$
$$\mathrm{SSR} = \frac{S_{xy}^2}{S_{xx}} = \frac{120.0^2}{50.0} = \frac{14400}{50.0} = 288.0.$$

(b) By subtraction,

$$\mathrm{SSE} = \mathrm{SST} - \mathrm{SSR} = 336.0 - 288.0 = 48.0,$$
$$\boxed{\mathrm{SST} = 336.0 = \underbrace{288.0}_{\mathrm{SSR}} + \underbrace{48.0}_{\mathrm{SSE}}.}$$

(c) All three closed forms agree:

$$\widehat{\beta}_1 S_{xy} = 2.40 \times 120.0 = 288.0,$$
$$\widehat{\beta}_1^2 \, S_{xx} = 2.40^2 \times 50.0 = 5.76 \times 50.0 = 288.0,$$
$$S_{xy}^2/S_{xx} = 14400/50.0 = 288.0.$$

The line explains 288.0 of the 336.0 total squared variation, leaving 48.0 residual.

F2. Coefficient of determination (germination vs. temperature). "Percent of variation explained" is exactly $R^2 = \mathrm{SSR}\,/\,\mathrm{SST} = 1 - \mathrm{SSE}\,/\,\mathrm{SST}$; recover $\mathrm{SSR} = \mathrm{SST} - \mathrm{SSE}$ first.

(a) Here $\mathrm{SSR} = \mathrm{SST} - \mathrm{SSE} = 400.0 - 64.0 = 336.0$, so

$$R^2 = \frac{\mathrm{SSR}}{\mathrm{SST}} = \frac{336.0}{400.0} = 0.84, \qquad 1 - \frac{\mathrm{SSE}}{\mathrm{SST}} = 1 - \frac{64.0}{400.0} = 1 - 0.16 = 0.84.$$

$$\boxed{R^2 = 0.84.}$$

(b) In words: about 84% of the tray-to-tray variation in germination rate is accounted for by the straight-line dependence on soil temperature; the remaining 16% is residual scatter.

(c) $R^2 = r^2$ fixes only the magnitude $|r| = \sqrt{0.84} = 0.9165$; the sign of r matches the sign of the slope, which is negative, so

$$\boxed{r = -\sqrt{0.84} = -0.917.}$$

F3. Build the ANOVA table (tensile strength vs. anneal time). Fill df $(1,\ n-2,\ n-1)$, copy the sums of squares, divide for mean squares, form $F = \mathrm{MSR}\,/\,\mathrm{MSE}$. The error df is $n - 2 = 8$ because two parameters $\widehat{\beta}_0, \widehat{\beta}_1$ were estimated.

Step 1: total and degrees of freedom. $\mathrm{SST} = \mathrm{SSR} + \mathrm{SSE} = 288.0 + 72.0 = 360.0$; regression df $= 1$, error df $= n - 2 = 8$, total df $= n - 1 = 9$ (and $1 + 8 = 9$).

Step 2: mean squares.

$$\mathrm{MSR} = \frac{\mathrm{SSR}}{1} = \frac{288.0}{1} = 288.0,$$
$$\mathrm{MSE} = \frac{\mathrm{SSE}}{n-2} = \frac{72.0}{8} = 9.00 = s^2.$$

Step 3: the F-statistic and table.

$$F = \frac{\mathrm{MSR}}{\mathrm{MSE}} = \frac{288.0}{9.00} = 32.0.$$

Source	df	SS	MS	F
Regression	1	288.0	288.0	32.0
Error	8	72.0	9.00	
Total	9	360.0		

F4. Overall F-test (continuation). The overall F of a simple regression tests the two-sided $H_0\colon \beta_1 = 0$ against $\mathrm{F}_{1,n-2} = \mathrm{F}_{1,8}$; compare to the upper cutoff.

Step 1: statistic and reference. $F = 32.0$ on $\mathrm{F}_{1,8}$, with $\alpha = 0.05$ cutoff $F_{0.05,1,8} = 5.32$.

Step 2: decision. Reject H_0 if $F > 5.32$. Here $32.0 \gg 5.32$.

$F = 32.0 > F_{0.05,1,8} = 5.32 \Rightarrow$ reject H_0 at the 5% level: anneal time has a significant linear effect on strength.

The F-test is two-sided because $F = \mathrm{MSR}\,/\,\mathrm{MSE}$ is built from $\mathrm{SSR} = \widehat{\beta}_1^2 S_{xx}$, which squares the slope and discards its sign — a positive and a negative slope of equal size give the same F, so no one-sided F exists here.

F5. Verify $F = t^2$ against the slope t-test (continuation). In simple linear regression the overall F-test and the two-sided slope t-test are the same test written two ways; the identity is $F = t_{\mathrm{obs}}^2$ (equivalently $t = \pm\sqrt{F}$).

(a) From F3, $s = \sqrt{\text{MSE}} = \sqrt{9.00} = 3.00$, and with $S_{xx} = 50.0$,

$$\text{SE}(\widehat{\beta}_1) = \frac{s}{\sqrt{S_{xx}}} = \frac{3.00}{\sqrt{50.0}} = \frac{3.00}{7.0711} = 0.4243.$$

(b) Slope statistic and its square:

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)} = \frac{2.40}{0.4243} = 5.657, \qquad t^2_{\text{obs}} = 5.657^2 = 32.0.$$

This equals the ANOVA $F = 32.0$ from F3, and the squared t on 8 df is an F on $(1, 8)$ df.

(c) Critical values match too:

$$t^2_{0.025,8} = 2.306^2 = 5.32 = F_{0.05,1,8}.$$

$$\boxed{t^2_{\text{obs}} = 5.657^2 = 32.0 = F, \qquad t^2_{0.025,8} = 2.306^2 = 5.32 = F_{0.05,1,8}.}$$

The two-sided slope t-test ($\text{df} = n - 2 = 8$) and the overall F-test reject together, always — one is the square of the other.

F6. Contrast: a high R^2 does not validate the model. The question is whether R^2 certifies a linear model. It does not: R^2 measures only how much *a line* reduces SST, and a curved relation can still let a line capture most of the variation while being systematically wrong. The diagnostic is the residual pattern, not the size of R^2.

Step 1: what $R^2 = 0.97$ actually says. The fitted line accounts for 97% of the total squared variation in dissolved fraction. That is a statement about *variance reduction*, not about the *shape* of the dependence; R^2 never tests whether a line is the right form.

Step 2: what the residual pattern says. Residuals that are positive at small and large x and negative in the middle change sign in a systematic (U-shaped) way — the signature of unmodeled **curvature**. The true relation bends, so the line under-predicts at the ends and over-predicts in the middle, yet the total residual variation can still be small, giving a high R^2.

Step 3: correct conclusion and next step.

$$\boxed{\begin{array}{c} R^2 = 0.97 \text{ shows the line explains much of the variation, but it does NOT} \\ \text{validate linearity; the patterned residuals reveal curvature } R^2 \text{ hides.} \end{array}}$$

The right move is to add a quadratic term (or transform a variable) and recheck the residual plot for structure — not to rest on the 0.97.

F7. Contrast: low R^2 but a significant slope (large n). Significance of the slope is a t-test ($\text{df} = n - 2$) governed by $\widehat{\beta}_1/\text{SE}$ and the sample size; R^2 measures effect size (variation explained). With huge n, a small but genuine effect is detectable while explaining little variation.

(a) Slope t-test:

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)} = \frac{0.45}{0.090} = 5.0, \qquad \text{df} = n - 2 = 1000, \quad t_{0.025,1000} \approx 1.962.$$

Since $5.0 > 1.962$, reject H_0: the slope is highly significant (p-value far below 0.05).

(b) Overall F:

$$F = t^2_{\text{obs}} = 5.0^2 = 25.0 > F_{0.05,1,1000} \approx 3.85,$$

the same two-sided verdict — reject H_0.

(c) Reconciliation: a significant F (or t) says the slope is *not zero*; it says nothing about magnitude. With $n = 1002$ the standard error 0.090 is small enough to resolve a genuine effect that nonetheless explains only 1.6% of trait variation.

$$\boxed{t_{\text{obs}} = 5.0 \text{ (significant slope)}, \quad R^2 = 0.016 \text{ (small effect)} : \text{ both true at once.}}$$

"Significant" answers *is there an effect?*; R^2 answers *how much does it explain?*. A large sample makes a real-but-tiny slope statistically certain while R^2 stays near zero.

G. Regression synthesis

G1. Full analysis from raw data. (a) With $n = 8$, $\overline{X} = 72/8 = 9$ and $\overline{Y} = 248/8 = 31$. The corrected sums of squares are

$$\begin{aligned} S_{xx} &= \textstyle\sum x_i^2 - \frac{(\sum x_i)^2}{n} = 816 - \frac{72^2}{8} = 816 - 648 = 168, \\ S_{xy} &= \textstyle\sum x_i y_i - \frac{(\sum x_i)(\sum y_i)}{n} = 2736 - \frac{72 \cdot 248}{8} = 2736 - 2232 = 504, \\ S_{yy} &= \textstyle\sum y_i^2 - \frac{(\sum y_i)^2}{n} = 9212 - \frac{248^2}{8} = 9212 - 7688 = 1524. \end{aligned}$$

Least squares (procedure: minimize the summed squared residuals) gives

$$\begin{aligned} \widehat{\beta}_1 &= \frac{S_{xy}}{S_{xx}} = \frac{504}{168} = 3.00, \\ \widehat{\beta}_0 &= \overline{Y} - \widehat{\beta}_1 \bar{x} = 31 - 3.00(9) = 4.00. \end{aligned}$$

$$\boxed{\widehat{y} = 4.00 + 3.00\,x}$$

(b) Decompose the variation and estimate σ^2:

$$\begin{aligned} \text{SSR} &= \frac{S_{xy}^2}{S_{xx}} = \frac{504^2}{168} = \frac{254016}{168} = 1512, \\ \text{SSE} &= S_{yy} - \text{SSR} = 1524 - 1512 = 12, \\ s^2 &= \frac{\text{SSE}}{n-2} = \frac{12}{6} = 2.00, \qquad s = \sqrt{2} = 1.414. \end{aligned}$$

The slope test (a t-test for β_1, because the question is whether the line has a real nonzero slope) uses $\text{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}}$:

$$\begin{aligned} \text{SE}(\widehat{\beta}_1) &= \frac{1.414}{\sqrt{168}} = \frac{1.414}{12.961} = 0.1091, \\ t_{\text{obs}} &= \frac{\widehat{\beta}_1 - 0}{\text{SE}(\widehat{\beta}_1)} = \frac{3.00}{0.1091} = 27.5. \end{aligned}$$

With $\text{df} = n - 2 = 6$ and a two-sided test at $\alpha = 0.05$, the critical value is $t_{0.025,\,6} = 2.447$.

$$\boxed{t_{\text{obs}} = 27.5 > t_{0.025,\,6} = 2.447 \;\Rightarrow\; \text{reject } H_0;\ \beta_1 \neq 0.}$$

At the 5% level there is overwhelming evidence that shoot mass increases linearly with fertilizer dose.

(c) The two summaries are

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{1512}{1524} = 0.9921, \qquad r = \frac{S_{xy}}{\sqrt{S_{xx}\,S_{yy}}} = \frac{504}{\sqrt{168 \cdot 1524}} = \frac{504}{506.0} = 0.9961.$$

$$\boxed{R^2 = 0.992, \qquad r = +0.996.}$$

R^2 is the fraction of the total variation in y the straight line explains (99.2% here); r is the unitless strength and direction of the linear association (+0.996, a tight positive line). Note $r^2 = 0.9961^2 = 0.992 = R^2$, as it must.

(d) Degrees of freedom: regression 1, error $n-2=6$, total $n-1=7$. Mean squares are MSR $=$ SSR $/1 = 1512$ and MSE $=$ SSE $/6 = 2.00$, so $F = \text{MSR} \,/\, \text{MSE} = 1512/2.00 = 756$.

Source	df	SS	MS	F
Regression	1	1512	1512	756
Error	6	12	2.00	
Total	7	1524		

Check the equivalence: $t^2_{\text{obs}} = 27.5^2 = 756 = F$ (and $t^2_{0.025,\,6} = 2.447^2 = 5.99 = F_{0.05,1,6}$). The two-sided slope t-test and the overall F-test are one and the same.

G2. Second full fit (negative association). (a) Here $n=7$, $\overline{X} = 28/7 = 4$, $\overline{Y} = 70/7 = 10$, and

$$S_{xx} = 140 - \frac{28^2}{7} = 140 - 112 = 28,$$
$$S_{xy} = 224 - \frac{28 \cdot 70}{7} = 224 - 280 = -56,$$
$$S_{yy} = 816 - \frac{70^2}{7} = 816 - 700 = 116.$$

Least squares gives

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{-56}{28} = -2.00,$$
$$\widehat{\beta}_0 = \overline{Y} - \widehat{\beta}_1 \bar{x} = 10 - (-2.00)(4) = 18.00, \qquad \widehat{y} = 18.00 - 2.00\,x.$$

For the test of $H_0 \colon \beta_1 = 0$,

$$\text{SSR} = \frac{(-56)^2}{28} = \frac{3136}{28} = 112, \qquad \text{SSE} = 116 - 112 = 4,$$
$$s^2 = \frac{4}{7-2} = 0.80, \qquad s = 0.8944,$$
$$\text{SE}(\widehat{\beta}_1) = \frac{0.8944}{\sqrt{28}} = \frac{0.8944}{5.2915} = 0.1690,$$
$$t_{\text{obs}} = \frac{-2.00}{0.1690} = -11.83.$$

With $\text{df} = n-2 = 5$, the two-sided $\alpha = 0.01$ critical value is $t_{0.005,\,5} = 4.032$. Since $|-11.83| > 4.032$, reject.

$|t_{\text{obs}}| = 11.83 > t_{0.005,\,5} = 4.032 \;\Rightarrow\;$ reject H_0; resting heart rate falls with training volume.

(b) The coefficient of determination is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = \frac{112}{116} = 0.9655 \approx 0.97.$$

$R^2 = 0.97$; "R^2" and "significant slope" answer different questions.

"$R^2 = 0.97$" is an *effect-size* statement: the line accounts for 97% of the subject-to-subject variation in resting heart rate, leaving only 3% unexplained. "The slope is highly significant" is an *existence* statement from the t-test: it says β_1 is almost surely not zero (here $t_{\text{obs}} = -11.83$, p-value far below 0.01). The two coincide in this small clean data set, but in general they are independent — a slope can be significant yet explain little (large n, small R^2, as in G6), or explain much in a tiny sample yet fail to clear significance. Both statements are true here; neither implies the other.

G3. One-sided slope test from raw data. (a) With $n = 6$, $\overline{X} = 42/6 = 7$, $\overline{Y} = 60/6 = 10$,

$$S_{xx} = 364 - \frac{42^2}{6} = 364 - 294 = 70,$$
$$S_{xy} = 490 - \frac{42 \cdot 60}{6} = 490 - 420 = 70,$$
$$S_{yy} = 698 - \frac{60^2}{6} = 698 - 600 = 98.$$

The fitted line is

$$\widehat{\beta}_1 = \frac{70}{70} = 1.00, \qquad \widehat{\beta}_0 = 10 - 1.00(7) = 3.00, \qquad \widehat{y} = 3.00 + 1.00\,x.$$

For the directional test (the lab has a prior direction, "turbidity *raises* demand," so a one-sided t-test of $H_0\colon \beta_1 = 0$ vs $H_1\colon \beta_1 > 0$),

$$\mathrm{SSR} = \frac{70^2}{70} = 70, \qquad \mathrm{SSE} = 98 - 70 = 28,$$
$$s^2 = \frac{28}{6-2} = 7.00, \qquad s = 2.6458,$$
$$\mathrm{SE}(\widehat{\beta}_1) = \frac{2.6458}{\sqrt{70}} = \frac{2.6458}{8.3666} = 0.3162,$$
$$t_{\mathrm{obs}} = \frac{1.00}{0.3162} = 3.162.$$

With $\mathrm{df} = n-2 = 4$, the one-sided cutoff is $t_{0.05,\,4} = 2.132$ (all of α in the upper tail). Since $3.162 > 2.132$, reject.

$$\boxed{\begin{array}{c} t_{\mathrm{obs}} = 3.16 > t_{0.05,\,4} = 2.132 \;\Rightarrow\; \text{reject } H_0; \\ \text{chlorine demand rises significantly with turbidity.} \end{array}}$$

(b) The explained fraction is $R^2 = \mathrm{SSR}\,/\,\mathrm{SST} = 70/98 = 0.714$, so the unexplained fraction is $1 - R^2 = \mathrm{SSE}\,/\,\mathrm{SST} = 28/98 = 0.286$.

$$\boxed{R^2 = 0.714;\ \text{the line leaves } 28.6\% \text{ of the variation in chlorine demand unexplained.}}$$

The effect is real (part a) but the line still leaves more than a quarter of the scatter unaccounted for — a reminder that "significant" and "explains everything" are not the same.

G4. Mean response versus a new observation. Both intervals share the center $\widehat{y}_0 = 8.0 + 2.5x_0$ and the multiplier $t_{0.025,\,10} = 2.228$ on $n - 2 = 10$ df; they differ only by the extra 1 under the root. The mean-response root is $\sqrt{\frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}}$ and the prediction root adds a leading 1.

(a) At $x_0 = 22$ ($x_0 - \overline{X} = 2$): $\widehat{y}_0 = 8.0 + 2.5(22) = 63.0$.

$$\mathrm{se(mean)} = s\sqrt{\tfrac{1}{n} + \tfrac{(x_0-\overline{X})^2}{S_{xx}}} = 3.0\sqrt{\tfrac{1}{12} + \tfrac{2^2}{500}} = 3.0\sqrt{0.08333 + 0.00800}$$
$$= 3.0\sqrt{0.09133} = 3.0(0.3022) = 0.9066,$$
$$\mathrm{se(pred)} = s\sqrt{1 + \tfrac{1}{n} + \tfrac{(x_0-\overline{X})^2}{S_{xx}}} = 3.0\sqrt{1.09133} = 3.0(1.0447) = 3.1340.$$

$$\text{mean CI: } 63.0 \pm 2.228(0.9066) = 63.0 \pm 2.02 = (60.98,\ 65.02), \quad \text{half-width } 2.02,$$
$$\text{prediction PI: } 63.0 \pm 2.228(3.1340) = 63.0 \pm 6.98 = (56.02,\ 69.98), \quad \text{half-width } 6.98.$$

$$\boxed{\text{mean } (60.98, 65.02) \subset \text{prediction } (56.02, 69.98) \text{ at } x_0 = 22.}$$

(b) At $x_0 = 30$ ($x_0 - \overline{X} = 10$): $\widehat{y}_0 = 8.0 + 2.5(30) = 83.0$.

$$\text{se(mean)} = 3.0\sqrt{\tfrac{1}{12} + \tfrac{10^2}{500}} = 3.0\sqrt{0.08333 + 0.20000} = 3.0\sqrt{0.28333} = 3.0(0.5323) = 1.5969,$$
$$\text{se(pred)} = 3.0\sqrt{1.28333} = 3.0(1.1328) = 3.3985.$$

$$\text{mean CI: } 83.0 \pm 2.228(1.5969) = 83.0 \pm 3.56 = (79.44,\ 86.56), \quad \text{half-width } 3.56,$$
$$\text{prediction PI: } 83.0 \pm 2.228(3.3985) = 83.0 \pm 7.57 = (75.43,\ 90.57), \quad \text{half-width } 7.57.$$

Comparing $x_0 = 22$ to $x_0 = 30$, the half-widths move

$$\text{mean: } 2.02 \to 3.56 \quad (\times 1.76),$$
$$\text{prediction: } 6.98 \to 7.57 \quad (\times 1.08).$$

The mean interval flares far more ($\times 1.76$ vs. $\times\ 1.08$).

The mean root is built only from $\frac{1}{n} + \frac{(x_0 - \overline{X})^2}{S_{xx}}$, so it is dominated by the $(x_0 - \overline{X})^2$ term and grows fast off center. The prediction root carries a constant 1 that swamps that term, so the same off-center move barely widens it.

(c) The mean-response interval answers "where does the *true line* sit at this weight — i.e. the *average* clearance of all animals of this weight?" The prediction interval answers "what clearance will *one particular new* animal of this weight actually show?" The latter must absorb that animal's own error term, hence is always wider.

G5. ANOVA table and the F–t equivalence. (a) From the summary statistics,

$$\widehat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{240}{200} = 1.20,$$
$$\text{SSR} = \frac{S_{xy}^2}{S_{xx}} = \frac{240^2}{200} = \frac{57600}{200} = 288,$$
$$\text{SSE} = S_{yy} - \text{SSR} = 340 - 288 = 52.$$

With df: regression 1, error $n - 2 = 13$, total 14, the mean squares are $\text{MSR} = 288/1 = 288$ and $\text{MSE} = 52/13 = 4.00 = s^2$, so $F = \text{MSR} / \text{MSE} = 288/4.00 = 72.0$ and $R^2 = \text{SSR} / \text{SST} = 288/340 = 0.847$.

Source	df	SS	MS	F
Regression	1	288	288	72.0
Error	13	52	4.00	
Total	14	340		

$R^2 = 0.847$.

(b) The overall F-test of $H_0\colon \beta_1 = 0$ (always two-sided in simple regression, since F squares the slope) compares $F = 72.0$ to $F_{0.05,1,13} = 4.67$.

$F = 72.0 > F_{0.05,1,13} = 4.67 \ \Rightarrow$ reject H_0; wear depends significantly on running hours.

(c) The slope statistic uses $\text{SE}(\widehat{\beta}_1) = s/\sqrt{S_{xx}} = 2.00/\sqrt{200} = 2.00/14.142 = 0.1414$, so

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)} = \frac{1.20}{0.1414} = 8.485, \qquad t_{\text{obs}}^2 = 8.485^2 = 72.0 = F.$$

The critical values match too: $t^2_{0.025,\,13} = 2.160^2 = 4.666 \approx 4.67 = F_{0.05,1,13}$.

$t_{\text{obs}}^2 = 72.0 = F, \qquad t^2_{0.025,\,13} = 4.67 = F_{0.05,1,13}.$

The two-sided slope t-test and the overall F-test are algebraically the same test.

G6. Significant slope, tiny R^2. (a) The slope test is a t-test on $n-2=500$ df:

$$t_{\text{obs}} = \frac{\widehat{\beta}_1}{\text{SE}(\widehat{\beta}_1)} = \frac{0.018}{0.006} = 3.00.$$

Two-sided at $\alpha = 0.05$ with $t_{0.025,\,500} \approx 1.965$, and $3.00 > 1.965$.

$$\boxed{t_{\text{obs}} = 3.00 > t_{0.025,\,500} \approx 1.965 \;\Rightarrow\; \text{reject } H_0;\ \text{the slope is significant.}}$$

(Equivalently $F = t_{\text{obs}}^2 = 9.0$ on $\text{F}_{1,500}$, far past its ≈ 3.86 cutoff.)

(b) The two numbers answer different questions, and both are correct. The t-test certifies only that $\beta_1 \neq 0$ — there *is* a real linear effect of allele dosage on the trait. The $R^2 = 0.020$ is the *magnitude*: the dosage explains just 2% of the trait's variation, so 98% is due to other causes. With $n = 502$ the standard error is small enough to resolve a slope that is genuine but tiny, which is exactly how a "significant but unimportant" effect arises.

$$\boxed{\text{significant } (t_{\text{obs}} = 3.0) \text{ and small } (R^2 = 0.02) \text{ are both true.}}$$

The author's leap from "explains almost nothing" to "irrelevant" is unsound: R^2 measures explanatory share, not biological or statistical relevance. A real-but-small per-allele effect can matter (e.g. across a whole genome of such variants) even while R^2 stays near zero. "Significant" answers *is there an effect?*; R^2 answers *how big?* — never collapse the two.

G7. Correlation test, then recover the variance accounting. (a) Test $H_0\colon \rho = 0$ with the correlation t on $n-2=16$ df:

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.62\sqrt{16}}{\sqrt{1-0.62^2}} = \frac{0.62(4)}{\sqrt{1-0.3844}} = \frac{2.48}{\sqrt{0.6156}} = \frac{2.48}{0.7846} = 3.161.$$

Two-sided at $\alpha = 0.05$, $t_{0.025,\,16} = 2.120$, and $3.161 > 2.120$.

$$\boxed{t = 3.16 > t_{0.025,\,16} = 2.120 \;\Rightarrow\; \text{reject } H_0 : \rho = 0.}$$

This is the *same* test as $H_0\colon \beta_1 = 0$: algebraically $r\sqrt{n-2}/\sqrt{1-r^2} = \widehat{\beta}_1/\,\text{SE}(\widehat{\beta}_1)$, so $\rho = 0$ and $\beta_1 = 0$ have one p-value — a zero population correlation is exactly a flat regression line.

(b) Using the identities $\widehat{\beta}_1 = r\sqrt{S_{yy}/S_{xx}}$, $\text{SSR} = r^2 S_{yy}$, and $R^2 = r^2$,

$$\begin{aligned}
\widehat{\beta}_1 &= r\sqrt{\frac{S_{yy}}{S_{xx}}} = 0.62\sqrt{\frac{900}{400}} = 0.62\sqrt{2.25} = 0.62(1.5) = 0.930,\\
\text{SSR} &= r^2\, S_{yy} = 0.62^2(900) = 0.3844(900) = 345.96,\\
R^2 &= r^2 = 0.384.
\end{aligned}$$

$$\boxed{\widehat{\beta}_1 = 0.930, \qquad \text{SSR} = 345.96, \qquad R^2 = 0.384.}$$

So although $\rho \neq 0$ is firmly established, the line explains only about 38% of the variation in the second assay — significant association, modest fit, recovered entirely from r, S_{xx}, and S_{yy} without the raw data.

Chapter 12

The Analysis of Variance

When you have not two but **several** groups to compare, running a t-test on every pair inflates the chance of a false positive: with many comparisons, *something* eventually looks significant by luck. The **analysis of variance** (ANOVA) solves this with a single overall test. Its idea is in its name: it compares the variation **between** the group means to the variation **within** the groups. If the groups truly share one mean, both kinds of variation estimate the same σ^2 and their ratio — an F **statistic** — sits near 1; if the group means differ, the between-group part inflates and F grows. A significant F says "the means are not all equal" but not *which* differ, so a rejection is followed by **multiple-comparison** procedures that pinpoint the differences while controlling the overall error rate.

Master guide: which ANOVA tool does the problem call for?

- Comparing the means of $k \geq 3$ **independent groups** at once? $\Rightarrow$ the **one-way ANOVA F-test** of $H_0 : \mu_1 = \cdots = \mu_k$ (§1).
- Need to **organize the computation**, or read a software/table output? $\Rightarrow$ the **ANOVA table**: sums of squares, degrees of freedom, mean squares, F (§2).
- The F-test **rejected**, and you must say *which* groups differ? $\Rightarrow$ **multiple comparisons** (Tukey's HSD, Bonferroni) that control the family-wise error rate (§3).
- Asked about a **specific planned comparison** among the means (a contrast)? $\Rightarrow$ **contrasts and subhypotheses** (§4).
- Worried whether ANOVA is **valid** here (equal variances? normal? independent)? $\Rightarrow$ the **assumptions and diagnostics** (§5).

Keep straight: ANOVA compares *means* by analyzing *variances*; a large F comes from large *between*-group spread relative to *within*-group spread. Do *not* run all pairwise t-tests at level α each — that is exactly the inflated error rate ANOVA and multiple-comparison methods are built to avoid.

12.1 The One-Way ANOVA Model and the F-Test

One-way analysis of variance compares the means of $k \geq 3$ independent groups in a single test. The data are k samples — group j holds n_j measurements, for a total of $N = \sum_j n_j$ — and the question is whether the underlying group means $\mu_1, \ldots, \mu_k$ are all equal or not. The clever move is to answer a question about *means* by comparing two *variances*. One estimate of the common error variance σ^2 comes from the scatter of the group means about the grand mean (the **between-group** mean square MSTR); another comes from the scatter *inside* the groups (the **within-group** mean square MSE). When the true means coincide, both target the same σ^2 and their ratio $F = \text{MSTR} / \text{MSE}$ hovers near 1; when the means differ, only MSTR is inflated, so F grows. Recognize that you want this section — and not a z- or t-test — whenever there are three or more groups and you are asked the single question "are all the means equal?" The test is always upper-tailed: large F, never small, is the evidence against equality.

Which comparison-of-means tool does the problem call for?

- $k \geq 3$ **independent groups**, one overall question "are the means all equal?" $\Rightarrow$ **one-way ANOVA** F**-test** of $H_0 : \mu_1 = \cdots = \mu_k$, reject for large $F \sim \mathrm{F}_{k-1,\,N-k}$.
- **Exactly two** independent groups? $\Rightarrow$ a two-sample t-test; ANOVA still works and gives $F = t^2$ on $\mathrm{F}_{1,\,N-2}$, the same decision.
- Tempted to run **all pairwise** t-tests, each at level α? $\Rightarrow$ do not: the family-wise false-positive rate balloons to $1-(1-\alpha)^m$ over $m = \binom{k}{2}$ pairs. Use the single F-test first.
- The F-test **rejected** and you must say *which* groups differ? $\Rightarrow$ that is multiple comparisons (§3), not this section.

Do not confuse: a *large* F signals unequal means; a *small* F (near 0) is consistent with H_0, never evidence against it. ANOVA is one-sided in F even when it answers a "two-sided" question about the means.

12.1.1 The model and the hypotheses

Definition 12.1.1 (One-way layout)**.** Let there be k **treatment groups**. In group j ($j = 1, \ldots, k$) the observations are

$$Y_{ij} = \mu_j + \varepsilon_{ij}, \qquad i = 1, \ldots, n_j,$$

where the errors ε_{ij} are independent and identically distributed $\mathrm{N}(0, \sigma^2)$, with the *same* variance σ^2 in every group. Writing μ for the overall mean and $\tau_j = \mu_j - \mu$ for the **treatment effect** of group j, the model becomes

$$Y_{ij} = \mu + \tau_j + \varepsilon_{ij}, \qquad \sum_{j=1}^{k} n_j \, \tau_j = 0.$$

The total sample size is $N = \sum_{j=1}^{k} n_j$.

Remark. Three assumptions are baked in and will matter later: the errors are *normal*, they are *independent*, and they share *one common variance* σ^2 across groups (homoscedasticity). The design is **balanced** when all n_j are equal and **unbalanced** otherwise; the F-test handles both, but balance simplifies the algebra and makes the procedure most robust.

Definition 12.1.2 (Null and alternative)**.** The one-way ANOVA tests

$$H_0 : \ \mu_1 = \mu_2 = \cdots = \mu_k \qquad \text{versus} \qquad H_1 : \ \mu_i \neq \mu_j \text{ for at least one pair } i \neq j.$$

Equivalently, in effect form, $H_0 : \ \tau_1 = \cdots = \tau_k = 0$ against $H_1 : \ \tau_j \neq 0$ for some j. The alternative is the blanket statement "the means are not all equal" — it does *not* name which group differs.

12.1.2 Splitting the variation: sums of squares

Write the group mean and the grand mean as

$$\overline{Y}_j = \frac{1}{n_j} \sum_{i=1}^{n_j} Y_{ij} \quad \text{(group-}j\text{ mean)},$$

$$\overline{Y} = \frac{1}{N} \sum_{j=1}^{k} \sum_{i=1}^{n_j} Y_{ij} = \frac{1}{N} \sum_{j=1}^{k} n_j \, \overline{Y}_j \quad \text{(grand mean)}.$$

Definition 12.1.3 (Sums of squares)**.** The **total**, **treatment (between)**, and **error (within)** sums of squares are

$$\mathrm{SST} = \sum_{j=1}^{k} \sum_{i=1}^{n_j} \big(Y_{ij} - \overline{Y}\big)^2,$$

$$\mathrm{SSTR} = \sum_{j=1}^{k} n_j \big(\overline{Y}_j - \overline{Y}\big)^2,$$

$$\mathrm{SSE} = \sum_{j=1}^{k} \sum_{i=1}^{n_j} \big(Y_{ij} - \overline{Y}_j\big)^2.$$

Theorem 12.1.4 (The fundamental partition)**.** *The total variation splits cleanly into between-group and within-group parts:*

$$\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}\,.$$

Proof. Inside each observation's deviation from the grand mean insert and remove the group mean: $Y_{ij} - \overline{Y} = (\overline{Y}_j - \overline{Y}) + (Y_{ij} - \overline{Y}_j)$. Square and sum over all i, j:

$$\begin{aligned} \mathrm{SST} &= \sum_j \sum_i \big[(\overline{Y}_j - \overline{Y}) + (Y_{ij} - \overline{Y}_j)\big]^2 \\ &= \sum_j \sum_i (\overline{Y}_j - \overline{Y})^2 + \sum_j \sum_i (Y_{ij} - \overline{Y}_j)^2 + 2 \sum_j (\overline{Y}_j - \overline{Y}) \sum_i (Y_{ij} - \overline{Y}_j). \end{aligned}$$

The first term is SSTR (the inner summand is constant over i, contributing n_j copies) and the second is SSE. The cross term vanishes because for each j the inner sum $\sum_i (Y_{ij} - \overline{Y}_j) = 0$ by definition of the group mean. Hence $\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}$. □

Definition 12.1.5 (Mean squares and the F statistic)**.** Dividing each sum of squares by its degrees of freedom gives the **mean squares**

$$\mathrm{MSTR} = \frac{\mathrm{SSTR}}{k-1}, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k},$$

and the test statistic is their ratio

$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}}.$$

The degrees of freedom add up: $(k-1) + (N-k) = N-1$, matching the total SST.

Theorem 12.1.6 (Why the F-test works)**.** *Under the one-way model,*

$$\mathbb{E}[\mathrm{MSE}] = \sigma^2 \quad \textit{always (whether or not } H_0 \textit{ holds)},$$

$$\mathbb{E}[\mathrm{MSTR}] = \sigma^2 + \frac{1}{k-1} \sum_{j=1}^{k} n_j\, \tau_j^2.$$

Consequently, under H_0 (all $\tau_j = 0$) both mean squares estimate the same σ^2, and

$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}} \;\sim\; \mathrm{F}_{\,k-1,\,N-k}\,.$$

Reject H_0 at level α when $F_{obs} > F_{\alpha,\,k-1,\,N-k}$ (the upper-tail cutoff).

Remark. The two expectations are the whole story. MSE pools the within-group spread and is an unbiased yardstick for σ^2 no matter what the means do. MSTR carries an *extra* nonnegative term $\frac{1}{k-1}\sum n_j \tau_j^2$ that is zero exactly when the means agree and grows with how far apart they are. So F sits near 1 under H_0 and drifts upward as the means separate — which is why only the *upper* tail is a rejection region. A tiny F never argues against equal means; it just means the group means happened to land unusually close together.

12.1.3 A full one-way F-test, end to end

Example 12.1.7 (Three fertilizers: the complete test)**.** An agronomist grows wheat under $k = 3$ fertilizers, $n_1 = n_2 = n_3 = 5$ plots each ($N = 15$), and records yield (bushels/plot). The group summaries are

$$\overline{Y}_1 = 20, \quad \overline{Y}_2 = 24, \quad \overline{Y}_3 = 28, \qquad \sum_j \sum_i (Y_{ij} - \overline{Y}_j)^2 = \text{SSE} = 96.$$

Test $H_0 : \mu_1 = \mu_2 = \mu_3$ at $\alpha = 0.05$, using $F_{0.05,2,12} = 3.89$.

Which rule, and why? Three independent groups, one question "are the means equal?" $\Rightarrow$ one-way ANOVA F-test on $\mathrm{F}_{k-1,N-k} = \mathrm{F}_{2,12}$. We are given SSE directly and the group means, so build SSTR from the means, form the two mean squares, and reject for large F.

Solution.

Step 1: grand mean. Balanced design, so $\overline{Y} = \dfrac{20 + 24 + 28}{3} = 24$.

Step 2: SSTR *from the group means.*

$$\begin{aligned}\text{SSTR} &= \sum_j n_j(\overline{Y}_j - \overline{Y})^2 = 5(20-24)^2 + 5(24-24)^2 + 5(28-24)^2 \\ &= 5(16) + 5(0) + 5(16) = 160.\end{aligned}$$

Step 3: mean squares. With $k - 1 = 2$ and $N - k = 12$,

$$\text{MSTR} = \frac{\text{SSTR}}{k-1} = \frac{160}{2} = 80,$$
$$\text{MSE} = \frac{\text{SSE}}{N-k} = \frac{96}{12} = 8.$$

Step 4: F and decision.

$$F_{\text{obs}} = \frac{\text{MSTR}}{\text{MSE}} = \frac{80}{8} = 10.$$

Since $F_{\text{obs}} = 10 > F_{0.05,2,12} = 3.89$, reject H_0.

$$\boxed{F_{\text{obs}} = 10 > 3.89 = F_{0.05,2,12} \;\Rightarrow\; \text{reject } H_0;\ \text{mean yields differ.}}$$

Example 12.1.8 (Reading the result back into the effects)**.** For the fertilizer test above, report the estimated treatment effects $\widehat{\tau}_j = \overline{Y}_j - \overline{Y}$ and confirm they satisfy $\sum_j n_j \widehat{\tau}_j = 0$.

Which rule, and why? The model writes $\mu_j = \mu + \tau_j$, so the natural estimate of each effect is its group mean minus the grand mean. The constraint $\sum n_j \tau_j = 0$ is built into the parameterization and serves here as an arithmetic check.

Solution.

Step 1: effects. $\widehat{\mu} = \overline{Y} = 24$, and

$$\widehat{\tau}_1 = 20 - 24 = -4,$$
$$\widehat{\tau}_2 = 24 - 24 = 0,$$
$$\widehat{\tau}_3 = 28 - 24 = +4.$$

Step 2: check the constraint. Balanced ($n_j = 5$):

$$\sum_j n_j \widehat{\tau}_j = 5(-4) + 5(0) + 5(4) = 0.$$

$$\boxed{\widehat{\tau} = (-4,\ 0,\ +4); \quad \sum_j n_j \widehat{\tau}_j = 0 \text{ as required.}}$$

Fertilizer 1 sits 4 below the grand mean and fertilizer 3 sits 4 above; the spread of these effects is exactly what inflated MSTR.

12.1.4 Balanced and unbalanced designs

Example 12.1.9 (Balanced design from raw group sums of squares)**.** A sports scientist measures vertical-jump height (cm) for athletes on $k = 4$ training programs, $n_j = 6$ each ($N = 24$). The group means and within-group sums of squares are

$$\overline{Y}_1 = 52,\ \overline{Y}_2 = 55,\ \overline{Y}_3 = 58,\ \overline{Y}_4 = 51,$$
$$\textstyle\sum_i (Y_{i1} - \overline{Y}_1)^2 = 44,\ \ 40,\ \ 52,\ \ 48 \text{ (groups 1–4)}.$$

Test equality of the four program means at $\alpha = 0.05$, given $F_{0.05,3,20} = 3.10$.

Which rule, and why? Four independent groups ⇒ one-way F-test on $\mathrm{F}_{3,20}$. The four listed within-group sums of squares add to SSE; the means give SSTR. Balanced $n_j = 6$ keeps the weights identical.

Solution.

Step 1: grand mean. $\overline{Y} = \dfrac{52 + 55 + 58 + 51}{4} = \dfrac{216}{4} = 54.$

Step 2: SSTR *and* SSE.

$$\begin{aligned}\mathrm{SSTR} &= 6\big[(52-54)^2 + (55-54)^2 + (58-54)^2 + (51-54)^2\big] \\ &= 6\,[4 + 1 + 16 + 9] = 6(30) = 180, \\ \mathrm{SSE} &= 44 + 40 + 52 + 48 = 184.\end{aligned}$$

Step 3: mean squares and F. Here $k - 1 = 3$, $N - k = 20$:

$$\mathrm{MSTR} = \frac{180}{3} = 60, \qquad \mathrm{MSE} = \frac{184}{20} = 9.2,$$
$$F_{\mathrm{obs}} = \frac{60}{9.2} = 6.52.$$

Step 4: decision. $F_{\mathrm{obs}} = 6.52 > F_{0.05,3,20} = 3.10$, so reject.

$$\boxed{F_{\mathrm{obs}} = 6.52 > 3.10 = F_{0.05,3,20} \ \Rightarrow\ \text{reject } H_0;\ \text{programs differ.}}$$

Example 12.1.10 (Unbalanced design: unequal group sizes)**.** A clinic compares recovery time (days) under $k = 3$ rehab protocols with *unequal* samples:

$$\text{Protocol A: } n_1 = 4,\ \overline{Y}_1 = 30,\ \textstyle\sum_i (Y_{i1} - \overline{Y}_1)^2 = 18,$$
$$\text{Protocol B: } n_2 = 6,\ \overline{Y}_2 = 26,\ \textstyle\sum_i (Y_{i2} - \overline{Y}_2)^2 = 40,$$
$$\text{Protocol C: } n_3 = 5,\ \overline{Y}_3 = 33,\ \textstyle\sum_i (Y_{i3} - \overline{Y}_3)^2 = 32.$$

Test $H_0 : \mu_A = \mu_B = \mu_C$ at $\alpha = 0.05$; use $F_{0.05,2,12} = 3.89$.

Which rule, and why? Three groups, unequal n_j ⇒ still a one-way F-test, but now the grand mean and SSTR are *weighted* by the n_j. $N = 4 + 6 + 5 = 15$, so the reference is $\mathrm{F}_{2,12}$. The within sums of squares simply add to SSE.

Solution.

Step 1: weighted grand mean.

$$\overline{Y} = \frac{\sum_j n_j \overline{Y}_j}{N} = \frac{4(30) + 6(26) + 5(33)}{15} = \frac{120 + 156 + 165}{15} = \frac{441}{15} = 29.4.$$

Step 2: SSTR *with n_j weights.*

$$\begin{aligned}\mathrm{SSTR} &= 4(30 - 29.4)^2 + 6(26 - 29.4)^2 + 5(33 - 29.4)^2 \\ &= 4(0.36) + 6(11.56) + 5(12.96) \\ &= 1.44 + 69.36 + 64.80 = 135.6.\end{aligned}$$

Step 3: SSE, *mean squares, F.* SSE $= 18 + 40 + 32 = 90$; $k - 1 = 2$, $N - k = 12$:

$$\text{MSTR} = \frac{135.6}{2} = 67.8, \qquad \text{MSE} = \frac{90}{12} = 7.5,$$
$$F_{\text{obs}} = \frac{67.8}{7.5} = 9.04.$$

Step 4: decision. $F_{\text{obs}} = 9.04 > 3.89$, reject.

$$\boxed{F_{\text{obs}} = 9.04 > 3.89 = F_{0.05,2,12} \;\Rightarrow\; \text{reject } H_0;\ \text{protocols differ.}}$$

Note how group B, the largest sample farthest below the grand mean, dominates SSTR — the n_j weights make a big, off-center group count more.

Example 12.1.11 (A non-rejection: when F stays near 1)**.** Four call-center scripts give mean handling times (minutes) $\overline{Y}_1 = 8.0$, $\overline{Y}_2 = 8.3$, $\overline{Y}_3 = 7.9$, $\overline{Y}_4 = 8.2$ with $n_j = 10$ each ($N = 40$) and pooled SSE $= 351$. Test equality at $\alpha = 0.05$, with $F_{0.05,3,36} = 2.87$.

Which rule, and why? Four groups ⇒ one-way F-test on $\text{F}_{3,36}$. This one is included to show the *other* outcome: when the group means barely scatter, MSTR is small, F is near 1, and we fail to reject.

Solution.

Step 1: grand mean. $\overline{Y} = \dfrac{8.0 + 8.3 + 7.9 + 8.2}{4} = \dfrac{32.4}{4} = 8.1.$

Step 2: SSTR.

$$\begin{aligned}\text{SSTR} &= 10\big[(-0.1)^2 + (0.2)^2 + (-0.2)^2 + (0.1)^2\big] \\ &= 10\,[0.01 + 0.04 + 0.04 + 0.01] = 10(0.10) = 1.0.\end{aligned}$$

Step 3: mean squares and F. $k - 1 = 3$, $N - k = 36$:

$$\text{MSTR} = \frac{1.0}{3} = 0.333, \qquad \text{MSE} = \frac{351}{36} = 9.75,$$
$$F_{\text{obs}} = \frac{0.333}{9.75} = 0.034.$$

Step 4: decision. $F_{\text{obs}} = 0.034 < 2.87$, far from the upper tail, so do not reject.

$$\boxed{F_{\text{obs}} = 0.034 < 2.87 = F_{0.05,3,36} \;\Rightarrow\; \text{fail to reject } H_0.}$$

An F well below 1 is no surprise under H_0 and is never itself evidence; we simply lack grounds to declare the scripts different.

12.1.5 Reading critical values and p-values

Example 12.1.12 (Looking up the critical value $F_{\alpha,k-1,N-k}$)**.** A study has $k = 5$ groups with sizes $n_j = 8, 8, 8, 8, 8$. State the numerator and denominator degrees of freedom and the upper 5% critical value, given the table entry $F_{0.05,4,35} = 2.64$.

Which rule, and why? The reference distribution for a one-way F-test is $\text{F}_{k-1,\,N-k}$: numerator df $= k - 1$, denominator df $= N - k$. Reading the right cell is half the test; getting the two df in the right order (numerator first) is the recognition step.

Solution.

Step 1: degrees of freedom. $N = 5 \cdot 8 = 40$, so

$$\text{numerator df} = k - 1 = 4, \qquad \text{denominator df} = N - k = 40 - 5 = 35.$$

Step 2: read the cutoff. The upper $\alpha = 0.05$ point is

$$F_{0.05,\,4,\,35} = 2.64.$$

Reject H_0 iff the computed F_{obs} exceeds 2.64.

$$\boxed{\mathrm{F}_{4,35}, \quad \text{reject if } F_{\text{obs}} > F_{0.05,4,35} = 2.64.}$$

Swapping the df to $F_{0.05,35,4}$ would read an entirely different (much larger) number — always numerator over denominator.

Example 12.1.13 (Bracketing a p-value from F)**.** In a $k = 3$, $N = 18$ experiment the test statistic is $F_{\text{obs}} = 5.10$ on $\mathrm{F}_{2,15}$. Two table entries are available: $F_{0.025,2,15} = 4.77$ and $F_{0.01,2,15} = 6.36$. Bracket the p-value and decide at $\alpha = 0.05$.

Which rule, and why? The p-value is the upper-tail area beyond F_{obs} under $\mathrm{F}_{2,15}$. With only a table we sandwich it between the two tabulated cutoffs the statistic falls between, then compare to α. This is the table analogue of asking software for the exact area.

Solution.

Step 1: locate F_{obs} between cutoffs. Since

$$F_{0.025,2,15} = 4.77 \;<\; F_{\text{obs}} = 5.10 \;<\; 6.36 = F_{0.01,2,15},$$

the upper-tail area beyond 5.10 lies between 0.01 and 0.025.

Step 2: state the bracket and decide.

$$0.01 < p\text{-value} < 0.025.$$

Because the whole bracket is below $\alpha = 0.05$, reject H_0.

$$\boxed{0.01 < p\text{-value} < 0.025 < 0.05 \;\Rightarrow\; \text{reject } H_0.}$$

The larger the F, the deeper into the right tail it sits, and the smaller the p-value — the upper tail is where significance lives.

12.1.6 Contrasts and the link to the two-sample t

Example 12.1.14 (CONTRAST: why not just run all pairwise t-tests?)**.** A scientist has $k = 5$ groups and proposes testing all $\binom{5}{2} = 10$ pairs with two-sample t-tests, each at $\alpha = 0.05$, declaring a difference if *any* pair is significant. If in truth all five means are equal, what is the chance of at least one false rejection? Compare with the single ANOVA F-test.

Which rule, and why? This is the classic mistake ANOVA exists to prevent. Each t-test has its own 5% error; running many *independent-ish* tests multiplies the chances, so the **family-wise error rate** is $1 - (1 - \alpha)^m$, not α. The single F-test holds the overall rate at exactly α.

Solution.

Step 1: count the comparisons. $m = \binom{5}{2} = \dfrac{5 \cdot 4}{2} = 10$ pairs.

Step 2: family-wise error if each is at $\alpha = 0.05$. Treating the tests as roughly independent, the chance of *no* false positive is $(1 - \alpha)^m$, so the chance of at least one is

$$\begin{aligned} \text{FWER} &= 1 - (1 - \alpha)^m = 1 - (0.95)^{10} \\ &= 1 - 0.5987 = 0.4013. \end{aligned}$$

Step 3: compare with ANOVA. A single F-test of $H_0 : \mu_1 = \cdots = \mu_5$ rejects falsely with probability exactly $\alpha = 0.05$, not 0.40.

$$\boxed{\text{pairwise FWER} = 1 - (0.95)^{10} \approx 0.40 \;\gg\; 0.05 = \alpha \text{ of one } F\text{-test.}}$$

With ten unprotected comparisons you would wrongly "find" a difference about 40% of the time even when none exists — which is exactly why you run the one omnibus F first and only then, after rejecting, use a multiple-comparison method (§3) that controls the rate.

Example 12.1.15 ($k = 2$: ANOVA reduces to the two-sample t ($F = t^2$))**.** Two machines fill bottles; $n_1 = n_2 = 6$ ($N = 12$), $\overline{Y}_1 = 500$, $\overline{Y}_2 = 506$ mL, and the pooled within-group estimate is MSE $= s_p^2 = 18$. Run both the one-way F-test and the pooled two-sample t-test, and verify $F = t^2$.

Which rule, and why? With only $k = 2$ groups the omnibus null $\mu_1 = \mu_2$ is also a two-sample hypothesis, so both procedures apply. ANOVA gives $F \sim \mathrm{F}_{1,N-2}$ and the pooled t gives $t \sim t_{N-2}$; algebraically $F = t^2$, so they always agree. This is the bridge between this chapter and two-sample testing.

Solution.

Step 1: SSTR *and* MSTR. $\overline{Y} = \dfrac{500 + 506}{2} = 503$, so

$$\begin{aligned} \text{SSTR} &= 6(500 - 503)^2 + 6(506 - 503)^2 = 6(9) + 6(9) = 108, \\ \text{MSTR} &= \frac{\text{SSTR}}{k - 1} = \frac{108}{1} = 108. \end{aligned}$$

Step 2: the F statistic. With MSE $= 18$ on $N - k = 10$ df,

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{108}{18} = 6.0 \ \sim \ \mathrm{F}_{1,10}\,.$$

Step 3: the pooled two-sample t. With $s_p^2 = \text{MSE} = 18$,

$$t = \frac{\overline{Y}_2 - \overline{Y}_1}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{506 - 500}{\sqrt{18}\,\sqrt{\frac{1}{6} + \frac{1}{6}}} = \frac{6}{\sqrt{18}\,\sqrt{1/3}} = \frac{6}{\sqrt{6}} = \sqrt{6} \approx 2.449.$$

Step 4: verify $F = t^2$.

$$t^2 = (\sqrt{6})^2 = 6 = F.$$

$$\boxed{F = 6.0 = t^2 \quad \text{on } \mathrm{F}_{1,10} \ \Leftrightarrow \ t_{10};\ \text{same test.}}$$

For two groups, then, ANOVA buys nothing new — it *is* the two-sample t. ANOVA earns its keep only at $k \geq 3$, where a single t cannot ask "are all equal?" at once.

12.1.7 Building the test from other summaries

Example 12.1.16 (Manufacturing: SSE from group standard deviations)**.** A plant measures the breaking strength (N) of cable from $k = 3$ suppliers, $n_j = 7$ each ($N = 21$). Software reports group means $\overline{Y}_1 = 210$, $\overline{Y}_2 = 204$, $\overline{Y}_3 = 216$ and group standard deviations $s_1 = 5$, $s_2 = 6$, $s_3 = 4$. Carry out the F-test at $\alpha = 0.05$, with $F_{0.05,2,18} = 3.55$.

Which rule, and why? Three groups $\Rightarrow$ one-way F-test on $\mathrm{F}_{2,18}$. We are handed s_j rather than raw within sums of squares, so recover each group's contribution from $\sum_i (Y_{ij} - \overline{Y}_j)^2 = (n_j - 1)s_j^2$ and add them to get SSE. The means give SSTR.

Solution.

Step 1: SSE *from the s_j.* Each group contributes $(n_j - 1)s_j^2 = 6s_j^2$:

$$\begin{aligned} \text{SSE} &= 6(5^2) + 6(6^2) + 6(4^2) \\ &= 6(25) + 6(36) + 6(16) = 150 + 216 + 96 = 462. \end{aligned}$$

Step 2: SSTR *from the means.* Balanced, so $\overline{Y} = \dfrac{210 + 204 + 216}{3} = 210$, and

$$\begin{aligned} \text{SSTR} &= 7\big[(210 - 210)^2 + (204 - 210)^2 + (216 - 210)^2\big] \\ &= 7\,[0 + 36 + 36] = 7(72) = 504. \end{aligned}$$

Step 3: mean squares and F. $k-1=2$, $N-k=18$:

$$\text{MSTR}=\frac{504}{2}=252, \qquad \text{MSE}=\frac{462}{18}=25.67,$$
$$F_{\text{obs}}=\frac{252}{25.67}=9.82.$$

Step 4: decision. $F_{\text{obs}}=9.82>F_{0.05,2,18}=3.55$, so reject.

$$\boxed{F_{\text{obs}}=9.82>3.55=F_{0.05,2,18} \;\Rightarrow\; \text{reject } H_0;\ \text{suppliers differ.}}$$

Example 12.1.17 (Environment: SSTR via the grand mean as a check)**.** Air-quality monitors at $k=4$ city zones give mean particulate readings (ug/m^3) $\overline{Y}_1=18$, $\overline{Y}_2=22$, $\overline{Y}_3=20$, $\overline{Y}_4=24$ with $n_j=5$ each ($N=20$) and pooled SSE $=128$. Test equality at $\alpha=0.05$ ($F_{0.05,3,16}=3.24$), and verify SSTR two ways.

Which rule, and why? Four groups $\Rightarrow$ one-way F-test on $\text{F}_{3,16}$. The recognition twist here is a self-check: $\text{SSTR}=\sum n_j(\overline{Y}_j-\overline{Y})^2$ must equal $\sum n_j\overline{Y}_j^2-N\overline{Y}^2$, the "raw" form, so we compute it both ways before trusting it.

Solution.

Step 1: grand mean. $\overline{Y}=\dfrac{18+22+20+24}{4}=\dfrac{84}{4}=21.$

Step 2: SSTR *two ways.* Deviation form:

$$\text{SSTR}=5\big[(-3)^2+(1)^2+(-1)^2+(3)^2\big]=5\,[9+1+1+9]=5(20)=100.$$

Raw form, $\sum n_j\overline{Y}_j^2-N\overline{Y}^2$:

$$\begin{aligned}5(18^2+22^2+20^2+24^2)-20(21^2)&=5(324+484+400+576)-20(441)\\&=5(1784)-8820=8920-8820=100.\end{aligned}$$

Both give SSTR $=100$.

Step 3: mean squares, F, decision. $k-1=3$, $N-k=16$:

$$\text{MSTR}=\frac{100}{3}=33.33, \qquad \text{MSE}=\frac{128}{16}=8.0,$$
$$F_{\text{obs}}=\frac{33.33}{8.0}=4.17.$$

Since $F_{\text{obs}}=4.17>F_{0.05,3,16}=3.24$, reject.

$$\boxed{\text{SSTR}=100\ \text{(both forms)};\quad F_{\text{obs}}=4.17>3.24 \;\Rightarrow\; \text{reject } H_0.}$$

Example 12.1.18 (Biology: unbalanced test from raw observations)**.** A geneticist measures plant height (cm) under $k=3$ light regimes with raw data

$$\begin{aligned}&\text{Low: } 10,\ 12,\ 14 \quad (n_1=3),\\&\text{Medium: } 16,\ 18,\ 20,\ 22 \quad (n_2=4),\\&\text{High: } 24,\ 26 \quad (n_3=2).\end{aligned}$$

Test $H_0:\mu_L=\mu_M=\mu_H$ at $\alpha=0.05$ from scratch; use $F_{0.05,2,6}=5.14$.

Which rule, and why? Three groups, unequal n_j, raw numbers $\Rightarrow$ one-way F-test on $\text{F}_{2,6}$ built entirely by hand: group means, weighted grand mean, SSTR, then SSE as the sum of squared deviations *within* each group.

Solution.

Step 1: group means and grand mean. $\overline{Y}_1 = \dfrac{10+12+14}{3} = 12$, $\overline{Y}_2 = \dfrac{16+18+20+22}{4} = 19$, $\overline{Y}_3 = \dfrac{24+26}{2} = 25$. With $N = 9$,

$$\overline{Y} = \frac{3(12)+4(19)+2(25)}{9} = \frac{36+76+50}{9} = \frac{162}{9} = 18.$$

Step 2: SSTR.

$$\begin{aligned}\text{SSTR} &= 3(12-18)^2 + 4(19-18)^2 + 2(25-18)^2 \\ &= 3(36) + 4(1) + 2(49) = 108 + 4 + 98 = 210.\end{aligned}$$

Step 3: SSE *within each group.*

$$\begin{aligned}\text{Low} &: (10-12)^2 + (12-12)^2 + (14-12)^2 = 4+0+4 = 8, \\ \text{Medium} &: (16-19)^2 + (18-19)^2 + (20-19)^2 + (22-19)^2 = 9+1+1+9 = 20, \\ \text{High} &: (24-25)^2 + (26-25)^2 = 1+1 = 2, \\ \text{SSE} &= 8+20+2 = 30.\end{aligned}$$

Step 4: mean squares, F, decision. $k-1 = 2$, $N-k = 6$:

$$\text{MSTR} = \frac{210}{2} = 105, \qquad \text{MSE} = \frac{30}{6} = 5,$$
$$F_{\text{obs}} = \frac{105}{5} = 21.0.$$

$F_{\text{obs}} = 21.0 > F_{0.05,2,6} = 5.14$, so reject.

$$\boxed{F_{\text{obs}} = 21.0 > 5.14 = F_{0.05,2,6} \;\Rightarrow\; \text{reject } H_0;\ \text{light regime matters.}}$$

Example 12.1.19 (Finance: decision by the p-value)**.** Five mutual-fund categories give a one-way statistic $F_{\text{obs}} = 2.30$ on $\text{F}_{4,45}$. Software reports the exact upper-tail area as p-value $= 0.073$. Decide at $\alpha = 0.05$ and at $\alpha = 0.10$, and say what the value means.

Which rule, and why? Once F_{obs} and its df are in hand, the modern decision rule is the p-value: the probability, under H_0, of an F at least as large as observed. Reject exactly when p-value $\le \alpha$. No table lookup is needed beyond what produced the area.

Solution.

Step 1: interpret the area. Under H_0 on $\text{F}_{4,45}$, an F as large as 2.30 or larger occurs with probability 0.073 — mild but not negligible evidence against equal means.

Step 2: compare to each α.

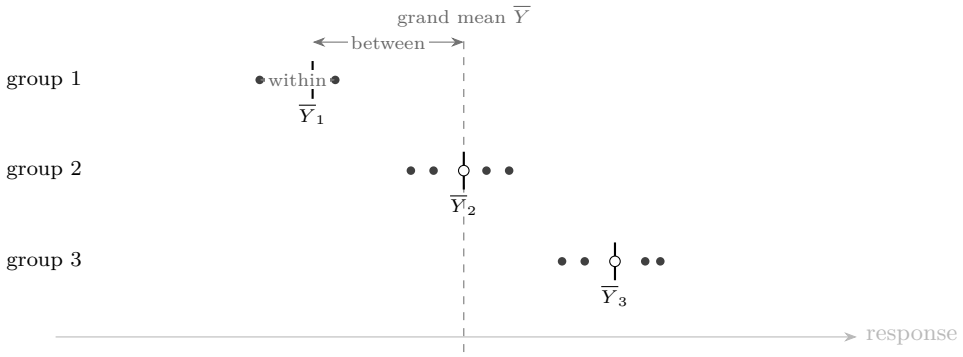
$$\begin{aligned}\alpha = 0.05: &\quad p\text{-value} = 0.073 > 0.05 \;\Rightarrow\; \text{fail to reject}, \\ \alpha = 0.10: &\quad p\text{-value} = 0.073 < 0.10 \;\Rightarrow\; \text{reject}.\end{aligned}$$

$$\boxed{p\text{-value} = 0.073:\ \text{retain } H_0 \text{ at } 5\%,\ \text{reject at } 10\%.}$$

The same data point can fall on either side of the line depending on the chosen α; the p-value reports *how* significant rather than a bare yes/no, and it is always an *upper*-tail area for an F-test.

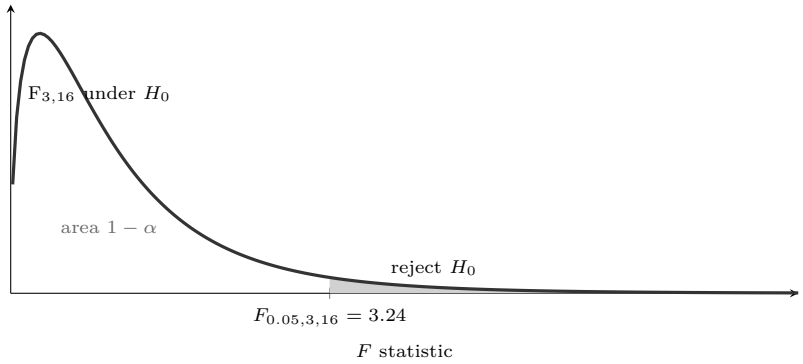
12.1.8 Between versus within, pictured

The first picture shows $k = 3$ groups as dotplots. The horizontal scatter *inside* each cloud is the within-group spread that feeds MSE; the gaps between the group means $\overline{Y}_j$ (open circles) and the grand mean $\overline{Y}$ (dashed line) are the between-group spread that feeds MSTR. A large F means the open circles sit far from the dashed line relative to how fat each cloud is.



Three groups. Filled dots are observations; the open circle on each row is the group mean $\overline{Y}_j$; the dashed vertical line is the grand mean $\overline{Y}$. "Within" spread (dot scatter in a row) estimates σ^2 via MSE*; "between" spread (open circles vs. dashed line) inflates* MSTR *when the means differ.*

The second picture is the reference $\mathrm{F}_{k-1,\,N-k}$ density — here $\mathrm{F}_{3,16}$ — which is right-skewed and lives on $x \geq 0$. The shaded upper tail beyond the critical value $F_{0.05,3,16} = 3.24$ is the rejection region; we reject H_0 exactly when the observed F_{obs} lands in that tail.



The $\mathrm{F}_{3,16}$ density under H_0. The statistic clusters near 1 when the means agree; the shaded region beyond $F_{0.05,3,16} = 3.24$ holds probability $\alpha = 0.05$. Only large F — the upper tail — rejects, because only unequal means inflate MSTR*.*

12.2 ANOVA Computations and the ANOVA Table

Once you have decided that a one-way ANOVA is the right test (§1), the work becomes pure bookkeeping: split the total variation of the data into a **between-groups** piece and a **within-groups** piece, divide each by its degrees of freedom, and form their ratio. This section is that machinery. The total sum of squares $\mathrm{SST} = \sum_{i,j}(Y_{ij} - \overline{Y})^2$ measures how far every observation sits from the grand mean; it partitions exactly as $\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}$, where the **treatment** (between) sum of squares $\mathrm{SSTR} = \sum_j n_j(\overline{Y}_j - \overline{Y})^2$ records how far the group means stray from the grand mean and the **error** (within) sum of squares $\mathrm{SSE} = \sum_j (n_j - 1)S_j^2$ records the leftover scatter inside the groups. Dividing by degrees of freedom $k-1$ and $N-k$ gives the mean squares MSTR and MSE, whose ratio is the F-statistic of the test. All of this is recorded in the **ANOVA table**, and recognising this section's job is easy: any problem that hands you raw data, group summaries, totals, or a half-blank table and asks for sums of squares, mean squares, or F lives here.

Which computation does the problem hand you?

- Given the **raw observations** Y_{ij}? $\Rightarrow$ compute group means $\overline{Y}_j$ and the grand mean $\overline{Y}$, then $\mathrm{SSTR} = \sum_j n_j(\overline{Y}_j - \overline{Y})^2$ and $\mathrm{SSE} = \sum_{i,j}(Y_{ij} - \overline{Y}_j)^2$.
- Given **group means and standard deviations** $\overline{Y}_j, S_j$? $\Rightarrow$ use $\mathrm{SSE} = \sum_j (n_j - 1)S_j^2$ directly; get $\overline{Y}$ as the size-weighted average of the $\overline{Y}_j$.

- Given only **group totals and the grand total**? ⇒ the correction-term shortcut $\mathrm{SSTR} = \sum_j T_j^2/n_j - T^2/N$ and $\mathrm{SST} = \sum_{i,j} Y_{ij}^2 - T^2/N$.
- Given a **partially blank ANOVA table**? ⇒ back-fill using the three identities: df add ($(k-1)+(N-k) = N-1$), SS add ($\mathrm{SSTR} + \mathrm{SSE} = \mathrm{SST}$), and $\mathrm{MS} = \mathrm{SS}/\mathrm{df}$.

Do not confuse the two divisors: MSTR divides by $k-1$ (one less than the number of *groups*), while MSE divides by $N-k$ (total observations minus number of groups). Swapping them — the single most common ANOVA-table error — corrupts F and its reference distribution.

12.2.1 The sum-of-squares decomposition

Definition 12.2.1 (Total, treatment, and error sums of squares)**.** With k groups, group j of size n_j, observations Y_{ij} $(i = 1, \dots, n_j)$, group means $\overline{Y}_j$, grand mean $\overline{Y} = \frac{1}{N}\sum_{i,j} Y_{ij}$, and $N = \sum_j n_j$, define

$$\mathrm{SST} = \sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{ij} - \overline{Y})^2 \qquad \text{(total)},$$

$$\mathrm{SSTR} = \sum_{j=1}^{k} n_j(\overline{Y}_j - \overline{Y})^2 \qquad \text{(treatment / between)},$$

$$\mathrm{SSE} = \sum_{j=1}^{k}\sum_{i=1}^{n_j}(Y_{ij} - \overline{Y}_j)^2 = \sum_{j=1}^{k}(n_j - 1)S_j^2 \qquad \text{(error / within)}.$$

Theorem 12.2.2 (The ANOVA identity)**.** *The total sum of squares partitions exactly into between- and within-groups pieces,*

$$\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}\,.$$

Proof. Write each deviation from the grand mean as a deviation *of* the group mean plus a deviation *within* the group,

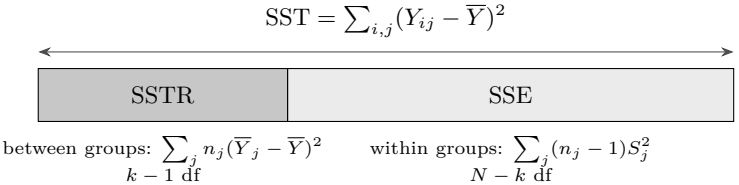
$$Y_{ij} - \overline{Y} = (\overline{Y}_j - \overline{Y}) + (Y_{ij} - \overline{Y}_j),$$

square, and sum over all i, j:

$$\mathrm{SST} = \sum_{i,j}(\overline{Y}_j - \overline{Y})^2 + \sum_{i,j}(Y_{ij} - \overline{Y}_j)^2 + 2\sum_{i,j}(\overline{Y}_j - \overline{Y})(Y_{ij} - \overline{Y}_j).$$

In the first sum the summand does not depend on i, so it contributes $\sum_j n_j(\overline{Y}_j - \overline{Y})^2 = \mathrm{SSTR}$. The second sum is SSE. The cross term vanishes group by group: for fixed j the factor $(\overline{Y}_j - \overline{Y})$ is constant in i, and $\sum_i (Y_{ij} - \overline{Y}_j) = 0$ because $\overline{Y}_j$ is the mean of that group's observations. Hence the cross term is $2\sum_j (\overline{Y}_j - \overline{Y}) \cdot 0 = 0$, leaving $\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}$. □

Remark. Read the partition as a sentence. SST is all the spread in the pooled data if you ignored the grouping entirely. SSE is the spread that survives *after* each observation is measured against its own group mean — the irreducible within-group noise that estimates σ^2. What the grouping "bought" is the difference SSTR: the part of the variation explained by the group means being different. If every group mean equalled the grand mean, $\mathrm{SSTR} = 0$ and all variation would be error; if the groups were internally identical (no within-group scatter), $\mathrm{SSE} = 0$ and all variation would be treatment.



The total variation SST *(the whole bar, on $N-1$ df) splits into the between-groups part* SSTR *(how far the group means sit from the grand mean, $k-1$ df) and the within-groups part* SSE *(scatter inside the groups, $N-k$ df). Both the sums of squares and their degrees of freedom add across the split.*

Example 12.2.3 (The decomposition from raw data: three fertilizers)**.** A horticulturist measures the height gain (cm) of seedlings under $k=3$ fertilizer formulas, with $n_1=n_2=n_3=4$:

$$\begin{aligned} &\text{A}: \ 8,\ 10,\ 9,\ 13 \qquad (\overline{Y}_1=10),\\ &\text{B}: \ 12,\ 14,\ 13,\ 13 \quad (\overline{Y}_2=13),\\ &\text{C}: \ 6,\ 9,\ 8,\ 9 \qquad (\overline{Y}_3=8). \end{aligned}$$

Compute SSTR, SSE, and SST, and verify the identity.

Which rule, and why? Raw observations are given, so compute each $\overline{Y}_j$ and the grand mean $\overline{Y}$, then $\text{SSTR}=\sum_j n_j(\overline{Y}_j-\overline{Y})^2$ from the means and $\text{SSE}=\sum_{i,j}(Y_{ij}-\overline{Y}_j)^2$ from the within-group deviations. The identity $\text{SST}=\text{SSTR}+\text{SSE}$ is the check.

Solution.

Step 1: grand mean. With $N=12$ equal-size groups, $\overline{Y}$ is the plain average of the group means:

$$\overline{Y}=\frac{10+13+8}{3}=\frac{31}{3}=10.333.$$

Step 2: treatment sum of squares.

$$\begin{aligned} \text{SSTR}&=\sum_j n_j(\overline{Y}_j-\overline{Y})^2\\ &=4(10-10.333)^2+4(13-10.333)^2+4(8-10.333)^2\\ &=4(0.1111)+4(7.1111)+4(5.4444)=0.444+28.444+21.778=50.667. \end{aligned}$$

Step 3: error sum of squares. Sum squared deviations inside each group:

$$\begin{aligned} &\text{A}: (8-10)^2+(10-10)^2+(9-10)^2+(13-10)^2=4+0+1+9=14,\\ &\text{B}: (12-13)^2+(14-13)^2+(13-13)^2+(13-13)^2=1+1+0+0=2,\\ &\text{C}: (6-8)^2+(9-8)^2+(8-8)^2+(9-8)^2=4+1+0+1=6, \end{aligned}$$

so $\text{SSE}=14+2+6=22$.

Step 4: total and check. Directly, $\text{SST}=\sum_{i,j}(Y_{ij}-\overline{Y})^2=72.667$ (sum of all twelve squared deviations from 10.333), and

$$\boxed{\text{SSTR}=50.667, \quad \text{SSE}=22, \quad \text{SSTR}+\text{SSE}=72.667=\text{SST}\,.}$$

Example 12.2.4 (Decomposition from group means and standard deviations: battery brands)**.** A lab tests run-time (hours) of three battery brands. The summaries are

$$\begin{aligned} &\text{X}: n_1=5,\ \overline{Y}_1=20.0,\ S_1=1.5;\\ &\text{Y}: n_2=5,\ \overline{Y}_2=22.0,\ S_2=2.0;\\ &\text{Z}: n_3=5,\ \overline{Y}_3=18.0,\ S_3=1.0. \end{aligned}$$

Find SSE, SSTR, and SST.

Which rule, and why? Only group means and standard deviations are given (no raw data), so use $\text{SSE}=\sum_j(n_j-1)S_j^2$ directly — there is nothing to recompute observation by observation — and get $\overline{Y}$ as the size-weighted average of the $\overline{Y}_j$.

Solution.

Step 1: error sum of squares from the S_j.

$$\begin{aligned}\text{SSE} &= \sum_j (n_j - 1)S_j^2 \\ &= 4(1.5)^2 + 4(2.0)^2 + 4(1.0)^2 = 4(2.25) + 4(4.00) + 4(1.00) = 9.0 + 16.0 + 4.0 = 29.0.\end{aligned}$$

Step 2: grand mean (equal sizes here).

$$\overline{Y} = \frac{20.0 + 22.0 + 18.0}{3} = 20.0.$$

Step 3: treatment sum of squares.

$$\begin{aligned}\text{SSTR} &= \sum_j n_j(\overline{Y}_j - \overline{Y})^2 \\ &= 5(20 - 20)^2 + 5(22 - 20)^2 + 5(18 - 20)^2 = 0 + 5(4) + 5(4) = 40.0.\end{aligned}$$

Step 4: total.

$$\boxed{\text{SSE} = 29.0, \quad \text{SSTR} = 40.0, \quad \text{SST} = \text{SSTR} + \text{SSE} = 69.0.}$$

Note we never needed a single raw observation: the $(n_j - 1)S_j^2$ shortcut delivers SSE from the group standard deviations alone.

Example 12.2.5 (Unequal group sizes: the grand mean must be weighted)**.** Three clinics report systolic-pressure drop (mmHg) on a new regimen, with *unequal* sample sizes:

$$\text{Clinic } 1 : n_1 = 8,\ \overline{Y}_1 = 12.0; \quad \text{Clinic } 2 : n_2 = 4,\ \overline{Y}_2 = 18.0; \quad \text{Clinic } 3 : n_3 = 8,\ \overline{Y}_3 = 15.0.$$

The pooled within-clinic error is SSE = 240.0. Find the grand mean, SSTR, and SST.

Which rule, and why? With *unequal* n_j, the grand mean is the size-weighted average $\overline{Y} = \sum_j n_j \overline{Y}_j / N$, *not* the plain average of the three means — the classic unequal-n trap. Then $\text{SSTR} = \sum_j n_j(\overline{Y}_j - \overline{Y})^2$ as usual.
Solution.
Step 1: weighted grand mean. $N = 8 + 4 + 8 = 20$, and

$$\overline{Y} = \frac{8(12.0) + 4(18.0) + 8(15.0)}{20} = \frac{96 + 72 + 120}{20} = \frac{288}{20} = 14.4.$$

(The plain average 15.0 would be wrong here.)
Step 2: treatment sum of squares.

$$\begin{aligned}\text{SSTR} &= 8(12.0 - 14.4)^2 + 4(18.0 - 14.4)^2 + 8(15.0 - 14.4)^2 \\ &= 8(5.76) + 4(12.96) + 8(0.36) = 46.08 + 51.84 + 2.88 = 100.80.\end{aligned}$$

Step 3: total.

$$\boxed{\overline{Y} = 14.4, \quad \text{SSTR} = 100.80, \quad \text{SST} = \text{SSTR} + \text{SSE} = 100.80 + 240.0 = 340.80.}$$

12.2.2 Computing formulas: the correction-term shortcut

Proposition 12.2.6 (Raw-sum-of-squares form)**.** *Let $T_j = \sum_i Y_{ij}$ be the total of group j and $T = \sum_j T_j = \sum_{i,j} Y_{ij}$ the grand total. With the* ***correction term*** $C = T^2/N$*,*

$$\text{SST} = \sum_{i,j} Y_{ij}^2 - C,$$

$$\text{SSTR} = \sum_j \frac{T_j^2}{n_j} - C,$$

$$\text{SSE} = \text{SST} - \text{SSTR} = \sum_{i,j} Y_{ij}^2 - \sum_j \frac{T_j^2}{n_j}.$$

Proof. Expanding $\text{SST} = \sum_{i,j}(Y_{ij} - \overline{Y})^2 = \sum_{i,j} Y_{ij}^2 - 2\overline{Y}\sum_{i,j} Y_{ij} + N\overline{Y}^2$ and using $\sum_{i,j} Y_{ij} = T$, $\overline{Y} = T/N$ gives $\text{SST} = \sum_{i,j} Y_{ij}^2 - T^2/N$. For SSTR, each group mean is $\overline{Y}_j = T_j/n_j$, so $\text{SSTR} = \sum_j n_j \overline{Y}_j^2 - N\overline{Y}^2 = \sum_j T_j^2/n_j - T^2/N$. Subtracting gives SSE. □

Remark. This is the form to use when a problem hands you *totals*. It avoids subtracting the grand mean from every observation (and so avoids rounding a non-terminating $\overline{Y}$ like 10.333 before squaring). The only quantities you need are the three running sums $\sum_{i,j} Y_{ij}^2$, $\sum_j T_j^2/n_j$, and T^2/N — "raw sum of squares minus correction term" for the total, "group-total sum of squares minus correction term" for the treatment.

Example 12.2.7 (Correction-term shortcut from totals: assembly lines)**.** Defect counts per shift are summarized for three assembly lines, each observed $n_j = 5$ shifts:

$$\text{Line 1} : T_j = 40; \quad \text{Line 2} : T_j = 55; \quad \text{Line 3} : T_j = 30; \qquad \sum_{i,j} Y_{ij}^2 = 1990.$$

Use the correction-term formulas to find SST, SSTR, and SSE.

Which rule, and why? Group totals and the raw sum of squares are given, so go straight to the computing forms $\text{SSTR} = \sum_j T_j^2/n_j - C$ and $\text{SST} = \sum_{i,j} Y_{ij}^2 - C$ with $C = T^2/N$ — no group means needed.

Solution.

Step 1: grand total and correction term. $T = 40 + 55 + 30 = 125$, $N = 15$, so

$$C = \frac{T^2}{N} = \frac{125^2}{15} = \frac{15625}{15} = 1041.667.$$

Step 2: treatment sum of squares.

$$\sum_j \frac{T_j^2}{n_j} = \frac{40^2}{5} + \frac{55^2}{5} + \frac{30^2}{5} = \frac{1600 + 3025 + 900}{5} = \frac{5525}{5} = 1105,$$

$$\text{SSTR} = 1105 - 1041.667 = 63.333.$$

Step 3: total and error.

$$\text{SST} = \sum_{i,j} Y_{ij}^2 - C = 1990 - 1041.667 = 948.333,$$

$$\text{SSE} = \text{SST} - \text{SSTR} = 948.333 - 63.333 = 885.0.$$

$$\boxed{\text{SST} = 948.333, \quad \text{SSTR} = 63.333, \quad \text{SSE} = 885.0.}$$

12.2.3 Mean squares and the ANOVA table

Definition 12.2.8 (Mean squares and the F-statistic)**.** Divide each sum of squares by its degrees of freedom to obtain a **mean square**:

$$\text{MSTR} = \frac{\text{SSTR}}{k-1}, \qquad \text{MSE} = \frac{\text{SSE}}{N-k}.$$

Treatment carries $k-1$ degrees of freedom (one fewer than the number of groups), error carries $N-k$, and they sum to the total $N-1$. The **F-statistic** is the ratio of mean squares,

$$F = \frac{\text{MSTR}}{\text{MSE}}, \qquad F \sim \text{F}_{k-1,\,N-k} \quad \text{under } H_0 \colon \mu_1 = \cdots = \mu_k,$$

and H_0 is rejected at level α when $F > F_{\alpha,\,k-1,\,N-k}$.

Remark. The whole computation is laid out in the **ANOVA table**: a Treatment row, an Error row, and a Total row, with columns for the source, its df, its sum of squares, its mean square (SS/df), and — on the Treatment row only — the ratio $F = \text{MSTR} \,/\, \text{MSE}$. Two columns add down: the df column ($(k-1) + (N-k) = N-1$) and the SS column ($\text{SSTR} + \text{SSE} = \text{SST}$). The MS column does *not* add (you cannot get MSE by subtracting MSTR from a "total MS"), and there is no F on the Error or Total rows. Keeping those rules in mind is what makes a half-blank table solvable.

Source	df	SS	MS	F
Treatment	$k-1$	SSTR	$\text{MSTR} = \dfrac{\text{SSTR}}{k-1}$	$F = \dfrac{\text{MSTR}}{\text{MSE}}$
Error	$N-k$	SSE	$\text{MSE} = \dfrac{\text{SSE}}{N-k}$	
Total	$N-1$	SST		

The one-way ANOVA table. The df column and the SS column each add down to the Total row; mean squares are SS/df; the single F on the Treatment row tests $H_0\colon \mu_1 = \cdots = \mu_k$ against $\mathrm{F}_{k-1,\,N-k}$. The MS column does not add and the Total row has no MS or F.

Example 12.2.9 (Build the full ANOVA table from raw data: fertilizers continued)**.** Return to the three-fertilizer data ($k = 3$, $N = 12$, SSTR $= 50.667$, SSE $= 22.0$). Construct the complete ANOVA table and carry out the F-test at $\alpha = 0.05$, given $F_{0.05,2,9} = 4.26$.

Which rule, and why? "Build the ANOVA table" means assign df ($k-1=2$, $N-k=9$, $N-1=11$), copy the sums of squares, divide to get mean squares, and form $F = \text{MSTR} \,/\, \text{MSE}$. The error df is $N-k = 12-3 = 9$, not $N-1$, because three group means were estimated.

Solution.

Step 1: degrees of freedom. Treatment $k-1=2$; error $N-k=9$; total $N-1=11$ (and $2+9=11$).

Step 2: mean squares.

$$\text{MSTR} = \frac{\text{SSTR}}{k-1} = \frac{50.667}{2} = 25.333,$$
$$\text{MSE} = \frac{\text{SSE}}{N-k} = \frac{22.0}{9} = 2.444.$$

Step 3: F-statistic, table, and decision.

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{25.333}{2.444} = 10.37.$$

Source	df	SS	MS	F
Treatment	2	50.667	25.333	10.37
Error	9	22.000	2.444	
Total	11	72.667		

Since $F = 10.37 > F_{0.05,2,9} = 4.26$, reject H_0: the three fertilizers do not all give the same mean height gain.

Example 12.2.10 (Fill in a partially blank table: back out SS and MS)**.** A printout gives a one-way ANOVA on $k = 4$ groups but with cells erased:

Source	df	SS	MS	F
Treatment	?	84.0	?	?
Error	20	?	?	
Total	?	204.0		

Recover every missing entry.

Which rule, and why? A half-blank table is solved by the three structural identities — df add, SS add, MS = SS/df — applied in whatever order unlocks one cell at a time. Here the error df $= 20$ fixes everything else.

Solution.

Step 1: degrees of freedom. Error df $= N-k = 20$. With $k = 4$ groups, treatment df $= k-1 = 3$, so total df $= 3 + 20 = 23$ (hence $N = 24$).

Step 2: missing sum of squares. SS add down: $\text{SSE} = \text{SST} - \text{SSTR} = 204.0 - 84.0 = 120.0$.

Step 3: mean squares.

$$\text{MSTR} = \frac{\text{SSTR}}{k-1} = \frac{84.0}{3} = 28.0,$$
$$\text{MSE} = \frac{\text{SSE}}{N-k} = \frac{120.0}{20} = 6.0.$$

Step 4: F and completed table. $F = \text{MSTR} \,/\, \text{MSE} = 28.0/6.0 = 4.67$.

Source	df	SS	MS	F
Treatment	3	84.0	28.0	4.67
Error	20	120.0	6.0	
Total	23	204.0		

Example 12.2.11 (Fill a table the other direction: back out SS from MS and df)**.** Another printout gives the mean squares and df but blanks the sums of squares:

Source	df	SS	MS	F
Treatment	2	?	31.5	?
Error	27	?	4.5	
Total	29	?		

Fill in the three sums of squares and F.

Which rule, and why? Now the identity used *backwards* is SS = MS × df (multiply, do not divide), one row at a time; the total SS then comes from the additive column, and F from the two mean squares.

Solution.

Step 1: sums of squares by multiplication.

$$\text{SSTR} = \text{MSTR} \times (k-1) = 31.5 \times 2 = 63.0,$$
$$\text{SSE} = \text{MSE} \times (N-k) = 4.5 \times 27 = 121.5.$$

Step 2: total SS by addition. $\text{SST} = 63.0 + 121.5 = 184.5$ (and indeed total df $= 2 + 27 = 29$).

Step 3: F. $F = \text{MSTR} \,/\, \text{MSE} = 31.5/4.5 = 7.0$.

Source	df	SS	MS	F
Treatment	2	63.0	31.5	7.0
Error	27	121.5	4.5	
Total	29	184.5		

Example 12.2.12 (Recover F from a table and conclude: teaching methods)**.** An education researcher reports the following table for $k = 5$ teaching methods. Read off F, find the critical value, and state the conclusion at $\alpha = 0.05$ (use $F_{0.05,4,45} = 2.58$):

Source	df	SS	MS	F
Treatment	4	96.0	?	?
Error	45	405.0	?	
Total	49	501.0		

Which rule, and why? Everything needed for the test is one MS column and one ratio away: compute MSTR, MSE, form $F = \text{MSTR} \,/\, \text{MSE}$, and compare to $F_{\alpha,k-1,N-k}$. The reference df come straight off the table (4 and 45).

Solution.

Step 1: mean squares.

$$\text{MSTR} = \frac{96.0}{4} = 24.0,$$
$$\text{MSE} = \frac{405.0}{45} = 9.0.$$

Step 2: F-statistic. $F = 24.0/9.0 = 2.67$.

Step 3: compare and conclude. The reference is $\text{F}_{4,45}$ with cutoff $F_{0.05,4,45} = 2.58$. Since $2.67 > 2.58$,

$$\boxed{F = 2.67 > F_{0.05,4,45} = 2.58 \;\Rightarrow\; \text{reject } H_0;\ \text{the five methods do not all share one mean score.}}$$

The margin is slim, so the evidence against equality is real but not overwhelming.

Example 12.2.13 (Identity check: catching an arithmetic slip)**.** A student computes, for $k = 3$ groups with $N = 18$, the values SSTR $= 120$, SSE $= 210$, and *separately* SST $= 315$ (by summing all $(Y_{ij} - \overline{Y})^2$). Before building the table, check whether the three are mutually consistent, and if not, locate the likely error.

Which rule, and why? The decomposition SST $=$ SSTR $+$ SSE is an *exact* identity, so it doubles as an arithmetic audit: if the separately computed SST does not equal SSTR $+$ SSE, at least one quantity is wrong. This recognition step prevents propagating a slip into F.

Solution.

Step 1: test the identity.

$$\text{SSTR} + \text{SSE} = 120 + 210 = 330 \neq 315 = \text{SST}\,.$$

The mismatch of 15 flags an error; the table must not be built yet.

Step 2: localize it. Two of the three are usually trustworthy. Recomputing the within-group scatter confirms SSE $= 210$ and recomputing the between-group term confirms SSTR $= 120$, so the standalone total was mis-added; the correct value is

$$\text{SST} = \text{SSTR} + \text{SSE} = 330.$$

Step 3: lesson.

$$\boxed{\text{SSTR} + \text{SSE} = 330 \neq 315 \;\Rightarrow\; \text{the lone SST is wrong; use SST} = 330.}$$

Always reconcile SST against SSTR $+$ SSE before dividing — the identity is free insurance.

Example 12.2.14 (CONTRAST: a mis-counted error df corrupts the test)**.** For $k = 4$ groups with $n_j = 6$ each ($N = 24$), a worker computes SSTR $= 90$, SSE $= 120$ correctly but then writes "error df $= N - 1 = 23$," giving MSE $= 120/23 = 5.22$ and $F = \text{MSTR}/5.22$. Identify the error, give the correct table, and show how the wrong df distorts both F and its reference distribution.

Which rule, and why? This is the canonical df blunder: the error row takes $N - k$, not $N - 1$. Using $N - 1$ overstates the error df by exactly $k - 1$, shrinks MSE, inflates F, *and* points to the wrong F row — a triple error from one wrong subtraction.

Solution.

Step 1: the correct degrees of freedom. Treatment $k - 1 = 3$; error $N - k = 24 - 4 = 20$; total $N - 1 = 23$. The worker used the *total* df 23 on the *error* row.

Step 2: correct mean squares and F.

$$\text{MSTR} = \frac{90}{3} = 30.0, \qquad \text{MSE} = \frac{120}{20} = 6.0,$$
$$F = \frac{30.0}{6.0} = 5.0 \quad \text{on } \text{F}_{3,20}\,.$$

Step 3: the wrong version, for contrast. With the erroneous MSE $= 120/23 = 5.217$,

$$F_{\text{wrong}} = \frac{30.0}{5.217} = 5.75 \quad \text{compared to } \text{F}_{3,23}\,.$$

The statistic is inflated (5.75 vs. the correct 5.0) and it is being read against the wrong reference ($F_{3,23}$ instead of $F_{3,20}$), so even the critical value is off.

$$\boxed{\text{Error df} = N - k = 20 \text{ (not } N - 1 = 23); \quad F = 5.0 \text{ on } F_{3,20}, \text{ not } 5.75 \text{ on } F_{3,23}\,.}$$

Both df add to $N - 1$ across the table; never borrow the total's df for the error row.

Example 12.2.15 (CONTRAST: why not just run all the pairwise t-tests?)**.** Faced with the $k = 5$ teaching methods above, a colleague proposes skipping ANOVA and instead running a two-sample t-test on each pair of methods at $\alpha = 0.05$, declaring a difference significant whenever any pair clears the bar. Explain why the single ANOVA F-test is preferred and quantify the problem with the all-pairs approach.

Which rule, and why? This contrasts the *one* omnibus F-test against *many* pairwise t-tests. With $k = 5$ groups there are $\binom{5}{2} = 10$ pairs; running ten tests each at $\alpha = 0.05$ lets the family-wise false-positive rate balloon far above 0.05. ANOVA tests equality of all means at once, holding the overall level at α.

Solution.

Step 1: count the comparisons. Number of distinct pairs:

$$\binom{k}{2} = \binom{5}{2} = 10.$$

Step 2: family-wise error of the all-pairs approach. If the ten tests were independent and all null, the chance of *at least one* false rejection is

$$1 - (1 - 0.05)^{10} = 1 - 0.95^{10} = 1 - 0.599 = 0.401.$$

So roughly a 40% chance of "finding" a spurious difference even when every method is equally effective — eight times the nominal 5%.

Step 3: what ANOVA does instead. The single F-test of $H_0\colon \mu_1 = \cdots = \mu_5$ has exactly one rejection region and holds the overall Type-I rate at $\alpha = 0.05$; only after it rejects does one proceed to controlled multiple comparisons (§3).

$$\boxed{10 \text{ pairwise tests at } 0.05 \Rightarrow \text{ family-wise error} \approx 0.40; \quad \text{one ANOVA } F\text{-test holds it at } 0.05.}$$

The inflation is exactly the pathology the analysis of variance was invented to avoid.

Example 12.2.16 (CONTRAST: ANOVA with $k = 2$ groups is the two-sample t-test)**.** A clinician compares mean recovery time for $k = 2$ treatments with $n_1 = n_2 = 10$ ($N = 20$) and gets an ANOVA $F = 8.41$ on $F_{1,18}$. A statistician notes the pooled two-sample t-test would give t_{obs} with 18 df. Show the two tests are identical and state the relationship.

Which rule, and why? When $k = 2$, the one-way ANOVA *is* the pooled two-sample t-test written as a ratio of mean squares; the exact link is $F = t_{\text{obs}}^2$ with reference $F_{1,\,N-2}$ versus the two-sided t_{N-2}. This contrast shows ANOVA generalizes the two-sample t, not replaces it.

Solution.

Step 1: the algebraic identity. For two groups, MSTR has $k - 1 = 1$ df and

$$F = \frac{\text{MSTR}}{\text{MSE}} = t_{\text{obs}}^2, \qquad F_{1,\,N-2} = F_{1,18}\,.$$

Step 2: recover the t-statistic.

$$t_{\text{obs}} = \pm\sqrt{F} = \pm\sqrt{8.41} = \pm 2.90.$$

Step 3: matching critical values. The two-sided t test at $\alpha = 0.05$ uses $t_{0.025,18} = 2.101$, and

$$t_{0.025,18}^2 = 2.101^2 = 4.41 = F_{0.05,1,18},$$

so the tests reject together. Since $|2.90| > 2.101$ (equivalently $8.41 > 4.41$), both reject.

$$\boxed{F = t_{\text{obs}}^2: \quad 8.41 = 2.90^2, \qquad F_{0.05,1,18} = t_{0.025,18}^2 = 4.41.}$$

With only two groups the omnibus F collapses to the familiar two-sided pooled t — ANOVA loses no information, it simply extends the idea to $k \geq 3$.

Example 12.2.17 (End-to-end from totals: coatings and corrosion resistance)**.** Four protective coatings are tested for corrosion resistance (arbitrary units), each on $n_j = 5$ panels. The group totals are $T_j = 150, 170, 135, 165$ and $\sum_{i,j} Y_{ij}^2 = 21{,}050$. Build the full ANOVA table from the totals and test at $\alpha = 0.05$, given $F_{0.05,3,16} = 3.24$.

Which rule, and why? Totals plus the raw sum of squares call for the correction-term route to the sums of squares, then the standard table-and-F procedure on $\mathrm{F}_{k-1,N-k} = \mathrm{F}_{3,16}$. This pulls together the shortcut and the table in one problem.

Solution.

Step 1: grand total and correction term. $T = 150 + 170 + 135 + 165 = 620$, $N = 20$,

$$C = \frac{620^2}{20} = \frac{384400}{20} = 19220.$$

Step 2: sums of squares.

$$\sum_j \frac{T_j^2}{n_j} = \frac{150^2 + 170^2 + 135^2 + 165^2}{5} = \frac{22500 + 28900 + 18225 + 27225}{5} = \frac{96850}{5} = 19370,$$

$$\begin{aligned} \mathrm{SSTR} &= 19370 - 19220 = 150, \\ \mathrm{SST} &= 21050 - 19220 = 1830, \\ \mathrm{SSE} &= \mathrm{SST} - \mathrm{SSTR} = 1830 - 150 = 1680. \end{aligned}$$

Step 3: df, mean squares, F. Treatment $k - 1 = 3$, error $N - k = 16$, total 19.

$$\mathrm{MSTR} = \frac{150}{3} = 50.0, \qquad \mathrm{MSE} = \frac{1680}{16} = 105.0,$$

$$F = \frac{50.0}{105.0} = 0.476.$$

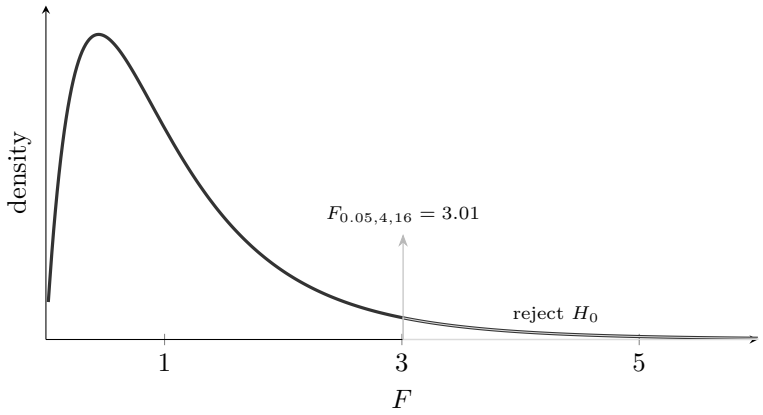
Step 4: table and decision.

Source	df	SS	MS	F
Treatment	3	150	50.0	0.476
Error	16	1680	105.0	
Total	19	1830		

Here $F = 0.476 < F_{0.05,3,16} = 3.24$, so *fail* to reject H_0: the four coatings show no significant difference in mean corrosion resistance. A sub-1 value of F is a useful sanity flag — the between-group spread is actually smaller than the within-group noise.

12.2.4 Where the F-statistic lands

Remark. Under H_0 the statistic $F = \mathrm{MSTR} / \mathrm{MSE}$ follows an $\mathrm{F}_{k-1,\,N-k}$ distribution, which lives on $(0, \infty)$, is right-skewed, and (for $k - 1 \geq 3$) peaks at a value somewhat below 1. A true null pushes F toward 1; unequal means inflate the numerator and push F into the right tail. The test is therefore always *one-sided to the right*: we reject only for large F, because both small and large departures of the group means from $\overline{Y}$ enlarge SSTR and hence F.



The $\mathrm{F}_{4,16}$ *density (here* $k = 5$ *groups,* $N = 20$*). It is right-skewed and peaks below* 1*; under* H_0 *the observed* F *clusters near the hump. The shaded upper tail beyond* $F_{0.05,4,16} = 3.01$ *is the rejection region — the test rejects only for* large F*, since any inequality among the means inflates* MSTR *and pushes* F *rightward.*

Example 12.2.18 (Reading the conclusion off the density: coatings vs. fertilizers)**.** Two studies finish with F-statistics on $\mathrm{F}_{3,16}$: the coatings study gave $F = 0.476$ and a hypothetical re-test of the fertilizer-style data gave $F = 6.20$. Without recomputing anything, place each on the distribution and state the conclusion at $\alpha = 0.05$ ($F_{0.05,3,16} = 3.24$).

Which rule, and why? Once F and the reference $\mathrm{F}_{k-1,N-k}$ are known, the conclusion is purely a matter of which side of the cutoff F lands on — left of it (in the hump) means retain, right of it (in the shaded tail) means reject. No further computation is needed.

Solution.

Step 1: locate $F = 0.476$. This sits far to the *left*, near and below the peak — deep in the retain region, well short of 3.24.

$$0.476 < 3.24 \;\Rightarrow\; \text{fail to reject; coatings means not shown to differ.}$$

Step 2: locate $F = 6.20$. This sits in the *right tail*, past the cutoff.

$$6.20 > 3.24 \;\Rightarrow\; \text{reject; the means differ.}$$

Step 3: summary.

$$\boxed{F = 0.476 \text{ (retain)}, \qquad F = 6.20 \text{ (reject)}, \qquad \text{cutoff } F_{0.05,3,16} = 3.24.}$$

The geometry makes the rule memorable: a value near or below the hump supports H_0; a value out in the long right tail contradicts it.

12.3 Multiple Comparisons

A significant F in the one-way ANOVA delivers only the blunt verdict "the k group means are not all equal"; it never names the culprits. The natural follow-up — *which* pairs of means actually differ — looks like a job for ordinary two-sample t-tests, one per pair. It is not, and the reason is the whole point of this section: there are $m = \binom{k}{2}$ pairs, and if you test each at level α the chance that *at least one* comparison fires a false positive balloons far past α. A **multiple-comparison procedure** repairs this by controlling the **family-wise error rate** (FWER) — the probability of *any* false rejection across the whole family — at a chosen α. We develop the two workhorses: **Tukey's HSD**, built from the studentized range and tailored to exactly the set of all pairwise comparisons, and the **Bonferroni** correction, a blunt-but-universal rule that simply tests each pair at level α/m. Recognize this section's tools whenever a problem says "a significant ANOVA was followed by…" and then asks which groups differ.

Which post-hoc comparison does the problem call for?

- Want **all pairwise comparisons** with the tightest honest control, and the design is **balanced** ($n_j \equiv n$)? $\Rightarrow$ **Tukey's HSD** with the studentized-range critical value $q_{\alpha,k,N-k}$.
- Same goal but the design is **unbalanced** (n_j differ)? $\Rightarrow$ the **Tukey–Kramer** version with per-pair standard error $\sqrt{\frac{\text{MSE}}{2}\left(\frac{1}{n_j}+\frac{1}{n_l}\right)}$.
- Only a **few pre-planned** comparisons, or you want a method that needs no special table? $\Rightarrow$ **Bonferroni**: test each at level α/m (or compare each $|t|$ to $t_{\alpha/(2m),\,N-k}$).
- Asked *why not* run plain pairwise t-tests at α each? $\Rightarrow$ because the family-wise error rate inflates toward $1-(1-\alpha)^m$; that is the error these methods bound.

Do not confuse the per-comparison level with the family-wise level: a Bonferroni test uses α/m *per pair* precisely so the *overall* chance of any false positive stays $\leq \alpha$; reporting "each pair tested at α" is the very mistake these procedures undo.

12.3.1 Why naive pairwise tests fail

Definition 12.3.1 (Family-wise error rate)**.** For a family of m comparisons, the **family-wise error rate** is

$$\text{FWER} = \mathbb{P}(\text{at least one false rejection among the } m \text{ comparisons, when all are true}).$$

A procedure **controls the FWER at level** α if FWER $\leq \alpha$ no matter how many of the null hypotheses hold.

Proposition 12.3.2 (Error inflation under independence)**.** *If m comparisons are each tested at level α and behave independently, then the chance of making* no *error is $(1-\alpha)^m$, so*

$$\text{FWER} = 1-(1-\alpha)^m.$$

This grows quickly with m: it already exceeds α for $m \geq 2$ and approaches 1 as m grows.

Proof. Let A_i be the event that comparison i is wrongly rejected; under its true null $\mathbb{P}(A_i) = \alpha$. "No error anywhere" is $\bigcap_i A_i^{\mathsf{c}}$, with probability $\prod_i (1-\alpha) = (1-\alpha)^m$ by independence. The complement, "at least one error," therefore has probability $1-(1-\alpha)^m$. □

Remark. Real pairwise comparisons share the same group means and the same pooled MSE, so they are *not* independent — the exact FWER differs from $1-(1-\alpha)^m$. But the qualitative lesson stands and the independent formula is a clean, slightly conservative thumbnail: more comparisons mean more chances to be fooled. Tukey and Bonferroni both attack this by shrinking each comparison's effective level until the *family* as a whole sits at α.

Example 12.3.3 (How fast the error rate inflates)**.** A study has $k = 5$ treatment groups. An analyst plans to compare every pair with a separate two-sample t-test at $\alpha = 0.05$. How many comparisons is that, and what is the (independence approximation to the) family-wise error rate?

Which rule, and why? This is the motivating count, not a test: tally $m = \binom{k}{2}$ pairs and plug into FWER $\approx 1-(1-\alpha)^m$ to see why per-comparison α is the wrong control.

Solution.

Step 1: count the pairs.

$$m = \binom{5}{2} = \frac{5\cdot 4}{2} = 10.$$

Step 2: inflate.

$$\text{FWER} \approx 1-(1-0.05)^{10} = 1-0.95^{10} = 1-0.5987 = 0.401.$$

Step 3: read it. With ten comparisons each at 5%, the chance of *at least one* false "significant" result is about 40% even when all five means are identical.

$$\boxed{m = 10, \qquad \text{FWER} \approx 0.40 \gg 0.05.}$$

Testing each pair at α does not control the experiment-wide error at α; it controls it at roughly 0.40. That gap is exactly what Tukey and Bonferroni close.

Example 12.3.4 (The comparison count and adjusted level for several k)**.** For $k = 3, 4, 6$ groups, compute the number of pairwise comparisons m and the Bonferroni per-comparison level α/m when $\alpha = 0.05$.

Which rule, and why? Pure bookkeeping that every post-hoc problem starts from: $m = \binom{k}{2} = k(k-1)/2$, and the Bonferroni split is α/m. No data needed.

Solution.

Step 1: pairs.

$$k = 3: \ m = \tfrac{3\cdot 2}{2} = 3, \quad k = 4: \ m = \tfrac{4\cdot 3}{2} = 6, \quad k = 6: \ m = \tfrac{6\cdot 5}{2} = 15.$$

Step 2: adjusted level α/m.

$$k = 3: \ \tfrac{0.05}{3} = 0.0167, \quad k = 4: \ \tfrac{0.05}{6} = 0.00833, \quad k = 6: \ \tfrac{0.05}{15} = 0.00333.$$

Step 3: report.

$$\boxed{(k, m, \alpha/m) = (3,\ 3,\ 0.0167),\ (4,\ 6,\ 0.0083),\ (6,\ 15,\ 0.0033).}$$

The more groups, the smaller the per-comparison level Bonferroni demands — which is both its safety and its conservatism.

12.3.2 Tukey's honest significant difference

Definition 12.3.5 (Studentized range and Tukey's HSD, balanced case)**.** Suppose all groups have the common size n (so $N = kn$). Let $q_{\alpha,k,N-k}$ be the upper-α point of the **studentized range distribution** with k groups and $N - k$ error degrees of freedom. Tukey's **honest significant difference** declares means j and l different when

$$|\,\overline{Y}_j - \overline{Y}_l\,| \ > \ \text{HSD}, \qquad \text{HSD} = q_{\alpha,k,N-k}\sqrt{\frac{\text{MSE}}{n}}.$$

Equivalently, build a simultaneous interval for every difference $\mu_j - \mu_l$,

$$(\overline{Y}_j - \overline{Y}_l) \ \pm \ q_{\alpha,k,N-k}\sqrt{\frac{\text{MSE}}{n}},$$

and call a pair significant exactly when its interval excludes 0. All $\binom{k}{2}$ such intervals hold *simultaneously* with confidence $1 - \alpha$.

Definition 12.3.6 (Tukey–Kramer, unbalanced case)**.** When group sizes differ, replace the common standard error by the per-pair version

$$\text{SE}_{jl} = \sqrt{\frac{\text{MSE}}{2}\left(\frac{1}{n_j} + \frac{1}{n_l}\right)},$$

and declare j, l different when $|\overline{Y}_j - \overline{Y}_l| > q_{\alpha,k,N-k}\,\text{SE}_{jl}$. (For equal $n_j = n_l = n$ this collapses to $q\sqrt{\text{MSE}/n}$, recovering the balanced HSD.)

Remark. Tukey's method uses the distribution of the *range* of k group means — the largest minus the smallest — so its single critical value $q_{\alpha,k,N-k}$ already accounts for the fact that you are scanning all pairs and reporting the biggest gaps. That is why it is the sharpest honest procedure for the family of *all* pairwise comparisons: it is built for exactly that family, not one comparison at a time. The error df $N - k$ are the same ANOVA error df that fed MSE.

Example 12.3.7 (Tukey HSD on a balanced four-group study: coating thickness)**.** Four plating baths ($k = 4$) are compared on deposited-coating thickness (microns); each bath yields $n = 6$ parts, so $N = 24$. The ANOVA was significant. The group means are

$$\overline{Y}_A = 18.0, \quad \overline{Y}_B = 21.5, \quad \overline{Y}_C = 22.0, \quad \overline{Y}_D = 26.0,$$

with MSE $= 4.0$ on $N - k = 20$ df. Using $q_{0.05,4,20} = 3.96$, find the HSD and report which pairs differ.

Which rule, and why? All pairs, balanced design, after a significant F: this is textbook Tukey HSD. One threshold $\text{HSD} = q\sqrt{\text{MSE}/n}$ serves every pair because all n_j are equal.

Solution.

Step 1: the single threshold.

$$\text{HSD} = q_{0.05,4,20}\sqrt{\frac{\text{MSE}}{n}} = 3.96\sqrt{\frac{4.0}{6}} = 3.96\sqrt{0.6667} = 3.96 \times 0.8165 = 3.23.$$

Step 2: all $\binom{4}{2} = 6$ *absolute differences.*

$$|\overline{Y}_A - \overline{Y}_B| = 3.5, \quad |\overline{Y}_A - \overline{Y}_C| = 4.0, \quad |\overline{Y}_A - \overline{Y}_D| = 8.0,$$
$$|\overline{Y}_B - \overline{Y}_C| = 0.5, \quad |\overline{Y}_B - \overline{Y}_D| = 4.5, \quad |\overline{Y}_C - \overline{Y}_D| = 4.0.$$

Step 3: compare each to HSD = 3.23. A pair is significant when its gap exceeds 3.23:

$$A \text{ vs } B\ (3.5),\ A \text{ vs } C\ (4.0),\ A \text{ vs } D\ (8.0),\ B \text{ vs } D\ (4.5),\ C \text{ vs } D\ (4.0):\ \text{differ},$$
$$B \text{ vs } C\ (0.5):\ \text{not}.$$

$$\boxed{\text{HSD} = 3.23;\ \text{all pairs differ except } B \text{ vs } C.}$$

Bath A is below all others and D is above all others; B and C are statistically indistinguishable.

Example 12.3.8 (A Tukey simultaneous confidence interval for one difference)**.** For the coating data above (HSD half-width = 3.23), give the simultaneous 95% Tukey interval for $\mu_D - \mu_A$ and state the conclusion.

Which rule, and why? "Interval for a difference, valid alongside all the others" is the confidence-interval face of Tukey: center at $\overline{Y}_D - \overline{Y}_A$ and use the *same* half-width $q\sqrt{\text{MSE}/n}$ that defined the HSD.

Solution.

Step 1: center and half-width.

$$\overline{Y}_D - \overline{Y}_A = 26.0 - 18.0 = 8.0, \qquad \text{half-width} = 3.23.$$

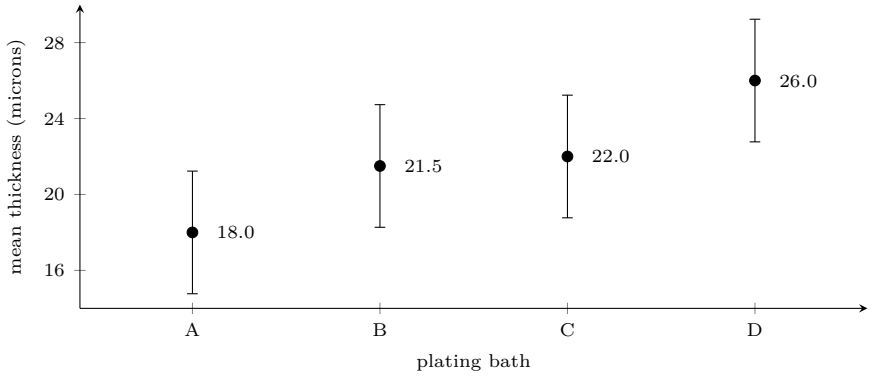
Step 2: assemble.

$$(\mu_D - \mu_A) \in 8.0 \pm 3.23 = (4.77,\ 11.23).$$

Step 3: conclude.

$$\boxed{\mu_D - \mu_A \in (4.77,\ 11.23);\ 0 \notin \text{interval} \Rightarrow D > A \text{ significantly.}}$$

Because all six such intervals hold together at 95% confidence, we may quote this one without any further penalty.



Group means with Tukey intervals of common half-width HSD/? *drawn as* ± 3.23 *bars (half the HSD on each side of a mean, so two means differ when their bars fail to reach each other's centers). Bars for B and C overlap their neighbors' centers, matching the single non-significant pair; A and D stand clear of everything between them.*

Example 12.3.9 (Tukey–Kramer on an unbalanced design: fertilizer trial)**.** Three fertilizer regimens give crop yields (bushels/acre) with *unequal* replication:

$$\overline{Y}_1 = 42.0 \ (n_1 = 8), \quad \overline{Y}_2 = 47.0 \ (n_2 = 5), \quad \overline{Y}_3 = 50.0 \ (n_3 = 10),$$

$\text{MSE} = 18.0$ on $N - k = 20$ df, $q_{0.05,3,20} = 3.58$. The ANOVA rejected. Which regimens differ?

Which rule, and why? Unequal n_j rules out the single-threshold balanced HSD; use Tukey–Kramer, where each pair gets its own $\text{SE}_{jl} = \sqrt{\frac{\text{MSE}}{2}(1/n_j + 1/n_l)}$ but the same q.

Solution.

Step 1: per-pair critical gaps $q \cdot \text{SE}_{jl}$*.*

$$\text{SE}_{12} = \sqrt{\tfrac{18}{2}\left(\tfrac{1}{8} + \tfrac{1}{5}\right)} = \sqrt{9(0.325)} = \sqrt{2.925} = 1.710,$$
$$\text{SE}_{13} = \sqrt{9\left(\tfrac{1}{8} + \tfrac{1}{10}\right)} = \sqrt{9(0.225)} = \sqrt{2.025} = 1.423,$$
$$\text{SE}_{23} = \sqrt{9\left(\tfrac{1}{5} + \tfrac{1}{10}\right)} = \sqrt{9(0.300)} = \sqrt{2.700} = 1.643.$$

Multiply each by $q = 3.58$:

$$q\,\text{SE}_{12} = 6.12, \quad q\,\text{SE}_{13} = 5.09, \quad q\,\text{SE}_{23} = 5.88.$$

Step 2: compare to the observed gaps.

$$|\overline{Y}_1 - \overline{Y}_2| = 5.0 < 6.12 \quad \Rightarrow \text{ not significant},$$
$$|\overline{Y}_1 - \overline{Y}_3| = 8.0 > 5.09 \quad \Rightarrow \text{ significant},$$
$$|\overline{Y}_2 - \overline{Y}_3| = 3.0 < 5.88 \quad \Rightarrow \text{ not significant}.$$

Step 3: report.

Only regimen 1 vs. 3 differs ($8.0 > 5.09$); 1 vs 2 and 2 vs 3 do not.

The unequal sample sizes make each threshold different — the small middle group ($n_2 = 5$) buys the widest standard errors, which is why its comparisons are hardest to declare significant.

12.3.3 The Bonferroni method

Definition 12.3.10 (Bonferroni correction)**.** To control the FWER at α across $m = \binom{k}{2}$ pairwise comparisons, test each pair at the reduced level α/m. Concretely, for pair j, l form the pooled-variance t-statistic

$$t_{jl} = \frac{\overline{Y}_j - \overline{Y}_l}{\sqrt{\text{MSE}\left(\dfrac{1}{n_j} + \dfrac{1}{n_l}\right)}}$$

and reject (declare j, l different) when $|t_{jl}| > t_{\alpha/(2m),\,N-k}$, using the ANOVA error df. Equivalently, the simultaneous interval for $\mu_j - \mu_l$ is

$$(\overline{Y}_j - \overline{Y}_l) \ \pm \ t_{\alpha/(2m),\,N-k}\sqrt{\text{MSE}\left(\frac{1}{n_j} + \frac{1}{n_l}\right)}.$$

Proposition 12.3.11 (Bonferroni controls the FWER)**.** *If each of m comparisons is carried out at level α/m, the family-wise error rate is at most α, regardless of dependence among the tests.*

Proof. Let A_i be the event of a false rejection on comparison i, with $\mathbb{P}(A_i) \le \alpha/m$. By Boole's inequality (the union bound),

$$\text{FWER} = \mathbb{P}\left(\bigcup_{i=1}^{m} A_i\right) \le \sum_{i=1}^{m} \mathbb{P}(A_i) \le m \cdot \frac{\alpha}{m} = \alpha.$$

No independence assumption is used — this is why Bonferroni works for *any* family of tests. □

Remark. The union bound is loose: it adds the individual error probabilities as if the bad events never overlapped. When the comparisons are positively dependent (as pairwise mean comparisons are, all sharing MSE), the true FWER is comfortably below α, so Bonferroni is **conservative** — it under-rejects, missing some real differences to stay safe. Tukey, calibrated to the exact joint distribution of all pairwise mean differences, recovers that lost power for this particular family. The trade-off: Bonferroni is universal and needs only a t-table; Tukey is sharper but applies specifically to all-pairwise comparisons and needs the studentized-range table.

Example 12.3.12 (Bonferroni on the coating data: matched to Tukey)**.** Re-examine the balanced coating study ($k = 4$, $n = 6$, $N = 24$, MSE $= 4.0$, $N - k = 20$, means $18.0, 21.5, 22.0, 26.0$) with Bonferroni at $\alpha = 0.05$. Here $m = 6$, the adjusted two-sided level is $\alpha/m = 0.00833$, and $t_{0.00417,20} = 2.93$. Which pairs differ, and how does the answer compare to Tukey?

Which rule, and why? Same all-pairwise question, but the method is now Bonferroni: a single common standard error (balanced design) times $t_{\alpha/(2m),\,N-k}$ gives one threshold, exactly parallel to the HSD but with a t rather than a q critical value.

Solution.

Step 1: common standard error and threshold. With equal n,

$$\text{SE} = \sqrt{\text{MSE}\left(\tfrac{1}{n} + \tfrac{1}{n}\right)} = \sqrt{4.0 \cdot \tfrac{2}{6}} = \sqrt{1.333} = 1.155,$$

$$\text{threshold} = t_{0.00417,20} \cdot \text{SE} = 2.93 \times 1.155 = 3.38.$$

Step 2: compare each gap to 3.38. Reusing the six differences $3.5, 4.0, 8.0, 0.5, 4.5, 4.0$:

$$A \text{ vs } B\ (3.5),\ A \text{ vs } C\ (4.0),\ A \text{ vs } D\ (8.0),\ B \text{ vs } D\ (4.5),\ C \text{ vs } D\ (4.0):\ \text{differ},$$
$$B \text{ vs } C\ (0.5):\ \text{not}.$$

Step 3: contrast the thresholds.

Bonferroni threshold $3.38 >$ Tukey HSD 3.23; same pairs flagged here.

On these data both methods reach identical conclusions, but Bonferroni's bar (3.38) sits slightly higher than Tukey's (3.23) — its conservatism shows up as the wider threshold, even when the final verdicts agree.

Example 12.3.13 (A pair Tukey catches but Bonferroni misses)**.** Suppose in a five-group balanced study ($k = 5$, $n = 8$, $N = 40$, $N - k = 35$, MSE $= 10.0$) one pair has $|\overline{Y}_j - \overline{Y}_l| = 4.55$. The relevant constants are $q_{0.05,5,35} = 4.07$ and, with $m = 10$, $t_{0.0025,35} = 3.06$. Does each method call this pair significant?

Which rule, and why? Same balanced all-pairwise setting, so both thresholds are "constant × SE"; the contrast is purely in the constant — $q\sqrt{\text{MSE}/n}$ for Tukey versus $t_{\alpha/(2m)}\sqrt{2\,\text{MSE}/n}$ for Bonferroni — which is where conservatism bites.

Solution.

Step 1: Tukey threshold.

$$\text{HSD} = q_{0.05,5,35}\sqrt{\frac{\text{MSE}}{n}} = 4.07\sqrt{\frac{10.0}{8}} = 4.07 \times 1.118 = 4.55.$$

Step 2: Bonferroni threshold.

$$t_{0.0025,35}\sqrt{\text{MSE}\left(\tfrac{1}{n}+\tfrac{1}{n}\right)} = 3.06\sqrt{10.0\cdot\tfrac{2}{8}} = 3.06\sqrt{2.5} = 3.06\times 1.581 = 4.84.$$

Step 3: decide the gap of 4.55.

$$\begin{aligned}\text{Tukey}: &\ 4.55 \geq 4.55 \Rightarrow \text{significant (just at the edge)},\\ \text{Bonferroni}: &\ 4.55 < 4.84 \Rightarrow \text{not significant}.\end{aligned}$$

Tukey flags the pair ($4.55 \geq 4.55$), Bonferroni does not ($4.55 < 4.84$).

This is conservatism made concrete: a difference real enough for the tailored Tukey procedure slips through Bonferroni's wider net. For all-pairwise work, prefer Tukey; reserve Bonferroni for a few planned comparisons or when no range table is at hand.

Example 12.3.14 (CONTRAST: an unadjusted t "finds" a difference the adjusted methods reject)**.** An analyst takes the closest non-significant pair from the coating study — B vs. C, with $\overline{Y}_B = 21.5$, $\overline{Y}_C = 22.0$, and recall $n = 6$, MSE $= 4.0$, $N - k = 20$ — and, ignoring the multiplicity, runs a single two-sample t-test at $\alpha = 0.05$ using $t_{0.025,20} = 2.086$. Then he tries a deliberately mis-stated pair with gap 2.4. Show how the unadjusted test misleads.

Which rule, and why? This is the cautionary CONTRAST: the *wrong* method is an ordinary pairwise t at full α, which ignores that the pair was cherry-picked from many. Compare its verdict to the honest Tukey/Bonferroni one.

Solution.

Step 1: the naive t for a gap of 2.4. Using the pooled SE $= \sqrt{\text{MSE}(1/n + 1/n)} = \sqrt{4.0\cdot\tfrac{2}{6}} = 1.155$,

$$t = \frac{2.4}{1.155} = 2.078 \text{ (borderline)}, \qquad \text{and for a gap of } 2.5:\ t = \frac{2.5}{1.155} = 2.165.$$

At full level, $2.165 > t_{0.025,20} = 2.086$, so the naive test calls a gap of 2.5 "significant."

Step 2: what the honest thresholds require. Tukey needs a gap above HSD $= 3.23$ and Bonferroni above 3.38 (computed earlier). A gap of 2.5 clears neither.

Step 3: the verdict split.

Unadjusted t: 2.5 "significant" ($t = 2.17 > 2.086$); Tukey/Bonferroni: $2.5 < 3.23, 3.38 \Rightarrow$ not significant.

The unadjusted t manufactures a finding by spending the full 5% on one of many comparisons; the adjusted methods, having reserved error budget for the whole family, correctly withhold the claim. This is precisely the false-positive inflation the section opened with, seen on a single pair.

12.3.4 Reading the pattern: grouping the means

Remark. After the pairwise verdicts are in, the cleanest summary is a **grouping**: sort the means and underline (or letter-label) any run of means that are *not* significantly different from one another. Means that share a line are statistically indistinguishable; means with no line in common differ. Overlapping lines are normal and meaningful — a middle mean can be tied to both a low and a high neighbor without the low and high being tied to each other.

Example 12.3.15 (Underline grouping from Tukey verdicts)**.** From the coating study, the only non-significant pair was B vs. C; every other pair differed. Produce the underline/grouping summary of which means are indistinguishable.

Which rule, and why? Grouping is the bookkeeping that turns a table of pairwise yes/no answers into a readable display: connect means with a shared line exactly when their pair was *not* significant.

Solution.

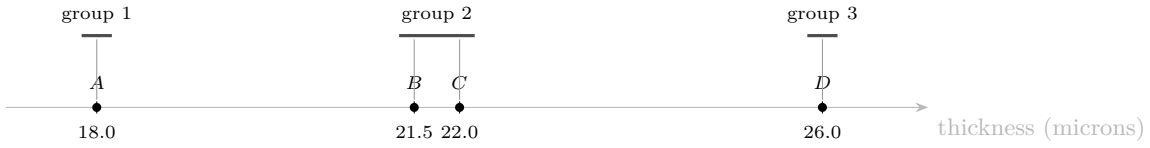
Step 1: sort the means. $\overline{Y}_A = 18.0 < \overline{Y}_B = 21.5 < \overline{Y}_C = 22.0 < \overline{Y}_D = 26.0$.

Step 2: list the ties (non-significant pairs). Only $\{B, C\}$.

Step 3: draw the grouping. A stands alone, $\{B, C\}$ share a line, D stands alone:

$$\boxed{\underline{A} \quad \underline{B\ C} \quad \underline{D}} \qquad (18.0) \quad (21.5,\ 22.0) \quad (26.0).$$

Three distinct levels: the low bath A, the indistinguishable middle pair $\{B, C\}$, and the high bath D.



Number line of the four bath means with brackets marking the Tukey groupings: A (group 1) and D (group 3) each stand alone, while B and C fall under one bracket (group 2) because their pair was the lone non-significant comparison. Means under a common bracket are statistically indistinguishable.

Example 12.3.16 (Overlapping groups: a middle mean tied to both ends)**.** A four-group study yields sorted means $\overline{Y}_1 = 10.0$, $\overline{Y}_2 = 12.0$, $\overline{Y}_3 = 14.0$, $\overline{Y}_4 = 16.0$ with HSD = 3.0. Determine the pairwise verdicts and the grouping, and interpret any overlap.

Which rule, and why? Apply the single HSD threshold to the sorted gaps; then read off the grouping. The point is to see how a constant HSD can produce *overlapping* (non-transitive) groups.

Solution.

Step 1: gaps versus HSD = 3.0.

$$\begin{aligned} &|1\text{–}2| = 2.0, \quad |2\text{–}3| = 2.0, \quad |3\text{–}4| = 2.0 \quad (\text{all} < 3.0,\ \text{tied}), \\ &|1\text{–}3| = 4.0, \quad |2\text{–}4| = 4.0, \quad |1\text{–}4| = 6.0 \quad (\text{all} > 3.0,\ \text{differ}). \end{aligned}$$

Step 2: build the grouping. Ties are $\{1, 2\}, \{2, 3\}, \{3, 4\}$; the differing pairs are the "two-apart" and "three-apart" ones. So 1 and 2 share a line, 2 and 3 share a line, 3 and 4 share a line, but 1 is not tied to 3 or 4.

Step 3: display the overlapping bands.

$$\boxed{\underline{1\ 2} \quad \underline{2\ 3} \quad \underline{3\ 4}} \qquad (\text{adjacent means tie; means two or more apart differ}).$$

The overlap is genuine, not a contradiction: "not significantly different" is *not* transitive. Mean 2 is statistically tied to both 1 and 3, yet 1 and 3 are far enough apart to differ — a chain where neighbors blur but the ends separate.

Example 12.3.17 (Significant F but no single Tukey pair fires)**.** A balanced study with $k = 4$, $n = 10$, $N = 40$ has a barely significant ANOVA (F just past its 5% cutoff). Suppose the largest pairwise mean gap is $|\overline{Y}_{\max} - \overline{Y}_{\min}| = 2.7$, while HSD $= q_{0.05,4,36}\sqrt{\text{MSE}/n} = 2.86$. What does it mean that F rejects but no pair does?

Which rule, and why? This is the famous mismatch: the omnibus F and Tukey answer slightly different questions, so on borderline data they can disagree. Recognize it by checking the *largest* gap against the HSD — if even that is below threshold, no pair is significant.

Solution.

Step 1: the largest gap versus the HSD. The biggest difference, 2.7, is the easiest to flag; if it fails, all do:

$$2.7 < \text{HSD} = 2.86 \;\Rightarrow\; \text{no Tukey pair is significant.}$$

Step 2: why F still rejected. The F-test pools *all* the between-group spread into one statistic; a significant F can arise from several modest gaps acting together (or from a particular weighted contrast), even when no *single* pairwise gap is large enough on its own.

Step 3: the interpretation.

F rejects "all means equal," yet no pairwise comparison is significant: the evidence is diffuse, spread across pairs, not in one gap.

The honest report is "the means differ overall, but the data cannot pinpoint which pair" — often a sign of borderline significance or that a *contrast* (next section), not a pairwise gap, carries the signal.

Example 12.3.18 (The reverse: a Tukey pair fires though the analyst feared F would not)**.** In a balanced three-group study ($k = 3$, $n = 12$, $N = 36$, $N - k = 33$, MSE $= 9.0$), the means are $\overline{Y}_1 = 20.0$, $\overline{Y}_2 = 20.5$, $\overline{Y}_3 = 26.0$. Using $q_{0.05,3,33} = 3.47$, find the HSD, identify the significant pair(s), and reconcile with the omnibus F.

Which rule, and why? A standard Tukey sweep, but framed to show the usual direction: when one mean is well separated, the F is large and Tukey isolates exactly the offending pair(s). The recognition step is the same single-threshold HSD comparison.

Solution.

Step 1: HSD.

$$\text{HSD} = 3.47\sqrt{\frac{9.0}{12}} = 3.47\sqrt{0.75} = 3.47 \times 0.866 = 3.01.$$

Step 2: pairwise gaps.

$$\begin{aligned} |\overline{Y}_1 - \overline{Y}_2| &= 0.5 < 3.01 \quad \Rightarrow \text{ tied,} \\ |\overline{Y}_1 - \overline{Y}_3| &= 6.0 > 3.01 \quad \Rightarrow \text{ differ,} \\ |\overline{Y}_2 - \overline{Y}_3| &= 5.5 > 3.01 \quad \Rightarrow \text{ differ.} \end{aligned}$$

Step 3: grouping and the F link. Group $\{1, 2\}$ low, group $\{3\}$ high:

$\underline{1\ \ 2}\qquad \underline{3}$; pairs 1–3 and 2–3 differ.

Here group 3 alone drives a large between-group sum of squares, so the F-test rejects decisively and Tukey cleanly attributes the rejection to the two comparisons involving group 3. When the signal is concentrated in one separated mean, F and Tukey agree without drama — the opposite of the diffuse case above.

Example 12.3.19 (End-to-end: machine-operator output, F then Tukey)**.** Five operators ($k = 5$) run the same machine; daily output (units) gives a significant ANOVA with MSE $= 25.0$ on $N - k = 45$ df, balanced $n = 10$ ($N = 50$). The means are

$$\overline{Y}_1 = 80,\ \overline{Y}_2 = 82,\ \overline{Y}_3 = 88,\ \overline{Y}_4 = 90,\ \overline{Y}_5 = 97.$$

With $q_{0.05,5,45} = 4.02$, carry out Tukey HSD and report the grouping.

Which rule, and why? The complete post-hoc workflow on a balanced design: one HSD threshold, all $\binom{5}{2} = 10$ gaps, then the grouping. This consolidates the whole section.

Solution.

Step 1: HSD.

$$\text{HSD} = 4.02\sqrt{\frac{25.0}{10}} = 4.02\sqrt{2.5} = 4.02 \times 1.581 = 6.36.$$

Step 2: the ten gaps and verdicts (threshold 6.36*).*

$$|1\text{–}2| = 2,\ |1\text{–}3| = 8,\ |1\text{–}4| = 10,\ |1\text{–}5| = 17,$$
$$|2\text{–}3| = 6,\ |2\text{–}4| = 8,\ |2\text{–}5| = 15,$$
$$|3\text{–}4| = 2,\ |3\text{–}5| = 9,\ |4\text{–}5| = 7.$$

Significant (gap > 6.36): 1–3, 1–4, 1–5, 2–4, 2–5, 3–5, 4–5. Not significant: 1–2 (2), 2–3 (6), 3–4 (2).

Step 3: grouping. The ties $\{1,2\}$, $\{2,3\}$, $\{3,4\}$ chain the lower four with overlap, while operator 5 (gap to its nearest neighbor 4 is $7 > 6.36$) breaks off:

$\underline{1\ 2}$	$\underline{2\ 3}$	$\underline{3\ 4}$	$\underline{5}$

Operator 5 is significantly the most productive; operators 1 through 4 form an overlapping chain in which only the two-apart and farther comparisons separate. A manager should single out operator 5, not pretend the lower four are sharply ranked.

Example 12.3.20 (Choosing the method: planned vs. all-pairwise)**.** A pharmacologist will compare $k = 6$ dose groups. In design A she has decided in advance to test only 3 specific dose-vs-control comparisons; in design B she wants *every* pair. For each design, state which adjustment is most appropriate and the per-comparison level it implies at $\alpha = 0.05$.

Which rule, and why? The recognition question itself: Bonferroni shines for a *small, prespecified* set of comparisons (it adjusts only for the number you actually make); Tukey is the right tool when the family is *all* pairwise differences.

Solution.

Step 1: design A (3 planned comparisons). Only $m = 3$ tests are in the family, so Bonferroni splits α three ways:

$$\text{level per comparison} = \frac{0.05}{3} = 0.0167.$$

Tukey would over-correct here, since it reserves error for all $\binom{6}{2} = 15$ pairs you are not testing.

Step 2: design B (all pairs). The family is all $m = \binom{6}{2} = 15$ comparisons; Tukey's range-based critical value is calibrated to exactly this set and is more powerful than Bonferroni's $\alpha/15 = 0.0033$ per comparison.

Step 3: recommendation.

Design A: Bonferroni at $0.05/3 = 0.0167$ per comparison; Design B: Tukey HSD (range table), more powerful than Bonferroni's 0.0033.

Match the correction to the family you actually intend to test: adjusting for comparisons you never make only throws away power.

12.4 Contrasts and Planned Comparisons

A significant overall F announces only that the k treatment means are *not all equal*; it never says *which* pattern of differences is at work. Often, though, the scientific question is sharper than "are they all the same?" — you want to compare a control against the average of three drugs, or low doses against high doses, or one brand against its competitors pooled. Each such question is a **contrast**: a weighted combination of the means $C = \sum_j c_j \mu_j$ whose coefficients sum to zero, so that C measures a *difference* and is unaffected by sliding all the means up or down together. A contrast you decide on *before* looking at the data is a **planned comparison**, tested with a single t (or its square, a one-degree-of-freedom F) that borrows the pooled within-group variance MSE from the whole experiment. Recognise this section's tools whenever a problem phrases a specific comparison — "control versus the rest," "average of these two versus that one," "is the linear trend in dose nonzero?" — rather than the blanket "are any different?". The constant warning

below is that *planning* is what keeps the error rate honest: a contrast chosen because it looked big in the data is no longer a planned comparison and must be handled by the post-hoc machinery of §3.

Which comparison is the problem really asking for?

- One **specific weighted comparison** of the means, fixed in advance (control vs. treatments, dose trend, this group vs. those pooled)? ⇒ a **contrast** $\widehat{C} = \sum c_j \overline{Y}_j$, tested by $t = \widehat{C}/\widehat{\mathrm{SE}}(\widehat{C}) \sim t_{N-k}$.
- A small set of comparisons that **do not overlap in information** (each $\sum_j c_j d_j / n_j = 0$)? ⇒ **orthogonal contrasts**; a full set of $k-1$ of them *partitions* SSTR into $k-1$ single-df pieces.
- **All pairwise** differences, or comparisons *suggested by the data*? ⇒ *not* a planned contrast — use **Tukey's HSD** (§3) so the family-wise error rate is controlled.
- Want a confidence statement for the comparison, not just a test? ⇒ a **contrast CI**, $\widehat{C} \pm t_{\alpha/2,\,N-k}\,\widehat{\mathrm{SE}}(\widehat{C})$.

Do not confuse planned with post-hoc: a contrast is "free" to test at level α *only because you committed to it before seeing the data.* Hunt through the means, pick the biggest gap, and call it a contrast, and you are back to the inflated-error problem that ANOVA exists to prevent.

12.4.1 What a contrast is, and how to estimate it

Definition 12.4.1 (Contrast)**.** A **contrast** among the k treatment means is a linear combination

$$C = \sum_{j=1}^{k} c_j\,\mu_j, \qquad \text{with weights satisfying} \quad \sum_{j=1}^{k} c_j = 0.$$

The zero-sum condition is what makes C a genuine comparison: adding a common constant a to every μ_j leaves C unchanged, since $\sum_j c_j(\mu_j + a) = \sum_j c_j\mu_j + a\sum_j c_j = C + 0$.

Remark. Two coefficient patterns that are proportional, like $(1,-1,0)$ and $(2,-2,0)$, define the *same* comparison (they test the same $H_0\colon C = 0$); only the *scale* of $\widehat{C}$ changes, never the t-statistic. So you may clear fractions freely: to compare μ_1 with the average $\frac{1}{2}(\mu_2+\mu_3)$, the natural weights $(1,-\frac{1}{2},-\frac{1}{2})$ and the cleared weights $(2,-1,-1)$ are interchangeable.

Theorem 12.4.2 (Estimating and testing a contrast)**.** *Let group j have n_j observations and sample mean $\overline{Y}_j$, with $\mathrm{MSE} = \mathrm{SSE}/(N-k)$ the pooled within-group variance on $N-k$ df. The contrast $C = \sum_j c_j\mu_j$ is estimated unbiasedly by*

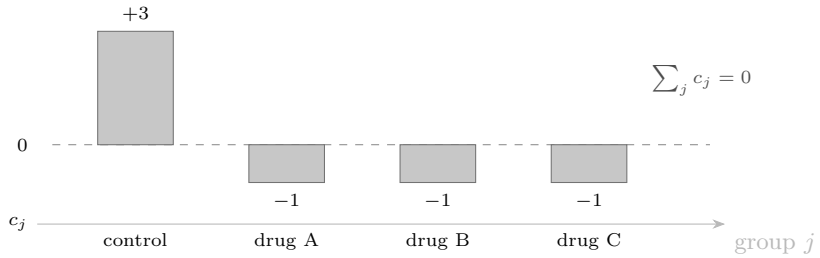
$$\widehat{C} = \sum_{j=1}^{k} c_j\,\overline{Y}_j, \qquad \mathrm{Var}(\widehat{C}) = \sigma^2 \sum_{j=1}^{k} \frac{c_j^2}{n_j}, \qquad \widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{\mathrm{MSE}\sum_{j=1}^{k}\frac{c_j^2}{n_j}}\,.$$

Under $H_0\colon C = 0$,

$$t = \frac{\widehat{C}}{\widehat{\mathrm{SE}}(\widehat{C})} \sim t_{N-k}, \qquad \textit{equivalently} \qquad F = t^2 = \frac{\widehat{C}^{\,2}\big/\sum_j c_j^2/n_j}{\mathrm{MSE}} \sim \mathrm{F}_{1,\,N-k}\,.$$

A level-α two-sided test rejects when $|t| > t_{\alpha/2,\,N-k}$; a $100(1-\alpha)\%$ confidence interval for C is $\widehat{C} \pm t_{\alpha/2,\,N-k}\,\widehat{\mathrm{SE}}(\widehat{C})$.

Proof. Each $\overline{Y}_j$ is unbiased for μ_j, so $\mathbb{E}[\widehat{C}] = \sum_j c_j\mu_j = C$. The groups are independent and $\mathrm{Var}(\overline{Y}_j) = \sigma^2/n_j$, so $\mathrm{Var}(\widehat{C}) = \sum_j c_j^2\,\mathrm{Var}(\overline{Y}_j) = \sigma^2\sum_j c_j^2/n_j$. Replacing the unknown σ^2 by MSE (which is independent of the $\overline{Y}_j$ and satisfies $(N-k)\,\mathrm{MSE}\,/\sigma^2 \sim \chi^2_{N-k}$) turns the standardized $\widehat{C}/\sqrt{\sigma^2\sum c_j^2/n_j}$, a standard normal under H_0, into a t_{N-k} by the usual normal-over-root-chi-square construction. Squaring gives the $\mathrm{F}_{1,N-k}$ form. □



Contrast weights for "control versus the three drugs pooled." The single positive bar (+3) is balanced by three negative bars (−1 each), so the weights sum to zero; the contrast $\widehat{C} = 3\overline{Y}_{ctrl} - \overline{Y}_A - \overline{Y}_B - \overline{Y}_C$ compares the control mean against the total *of the three treatment means, which is the same comparison as $\overline{Y}_{ctrl} - \frac{1}{3}(\overline{Y}_A + \overline{Y}_B + \overline{Y}_C)$ on a different scale.*

Example 12.4.3 (Control versus the average of three drugs: blood pressure)**.** A trial randomizes hypertensive patients to a placebo control and three antihypertensive drugs, $n_j = 10$ each ($N = 40$), and records the drop in systolic blood pressure (mm Hg). The group means are

$$\overline{Y}_{\text{ctrl}} = 4.0, \quad \overline{Y}_A = 11.0, \quad \overline{Y}_B = 9.0, \quad \overline{Y}_C = 13.0, \qquad \text{MSE} = 18.0 \text{ (on } N - k = 36 \text{ df)}.$$

Test whether the control mean differs from the average of the three drug means, at $\alpha = 0.05$ ($t_{0.025,36} = 2.028$).

Which rule, and why? The question names a *specific* comparison fixed before the data — control against the pooled drugs — so it is a planned **contrast**, tested by $t = \widehat{C}/\widehat{\text{SE}}$, not a pairwise scan. Use the symmetric weights $(3, -1, -1, -1)$ on (control, A, B, C); they sum to zero and encode exactly "control minus the average of the three."

Solution.

Step 1: weights and the point estimate. With $c = (3, -1, -1, -1)$,

$$\widehat{C} = 3(4.0) - 11.0 - 9.0 - 13.0 = 12.0 - 33.0 = -21.0.$$

Step 2: standard error. All $n_j = 10$, and $\sum_j c_j^2 = 3^2 + 1 + 1 + 1 = 12$, so

$$\widehat{\text{SE}}(\widehat{C}) = \sqrt{\text{MSE} \sum_j \frac{c_j^2}{n_j}} = \sqrt{18.0 \cdot \frac{12}{10}} = \sqrt{21.6} = 4.648.$$

Step 3: t-statistic and decision.

$$t = \frac{\widehat{C}}{\widehat{\text{SE}}(\widehat{C})} = \frac{-21.0}{4.648} = -4.518, \qquad |t| = 4.518 > t_{0.025,36} = 2.028.$$

$$\boxed{t = -4.52,\ |t| > 2.028 \ \Rightarrow\ \text{reject } H_0;\ \text{the drugs lower BP more than control.}}$$

The negative sign (control minus drugs < 0) tells the direction: on average the three drugs beat placebo by about 7 mm Hg per the underlying $\widehat{C}/3 = -7.0$.

Example 12.4.4 (A confidence interval for the same contrast)**.** Continuing the blood-pressure trial, give a 95% confidence interval for the comparison "control mean minus the average drug mean," i.e. for $C^\star = \mu_{\text{ctrl}} - \frac{1}{3}(\mu_A + \mu_B + \mu_C)$.

Which rule, and why? A confidence statement for a contrast uses $\widehat{C} \pm t_{\alpha/2,N-k}\,\widehat{\text{SE}}(\widehat{C})$. Because the asked-for $C^\star$ uses the unscaled weights $(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3})$, rescale the previous $\widehat{C}$ and its SE by the factor $\frac{1}{3}$ rather than recomputing from scratch.

Solution.

Step 1: rescale to the natural weights. $C^\star = \frac{1}{3}C$, so $\widehat{C}^\star = \frac{1}{3}(-21.0) = -7.0$ and $\widehat{\mathrm{SE}}(\widehat{C}^\star) = \frac{1}{3}(4.648) = 1.549$.

Step 2: margin of error.

$$t_{0.025,36}\,\widehat{\mathrm{SE}}(\widehat{C}^\star) = 2.028 \times 1.549 = 3.142.$$

Step 3: assemble the interval.

$$\widehat{C}^\star \pm 3.142 = -7.0 \pm 3.142 = (-10.14,\ -3.86).$$

$$\boxed{C^\star \in (-10.14,\ -3.86) \text{ mm Hg; interval excludes 0, agreeing with the test.}}$$

The whole interval lies below zero, so we are 95% confident the drugs outperform placebo, by between 3.9 and 10.1 mm Hg on average.

12.4.2 Orthogonal contrasts

Definition 12.4.5 (Orthogonal contrasts)**.** Two contrasts with weight vectors $c = (c_1, \ldots, c_k)$ and $d = (d_1, \ldots, d_k)$ are **orthogonal** when

$$\sum_{j=1}^{k} \frac{c_j\, d_j}{n_j} = 0.$$

In the **balanced** case (all $n_j = n$) this simplifies to $\sum_j c_j d_j = 0$: the weight vectors are perpendicular. Orthogonality means the two comparisons carry *non-overlapping* information — their estimators $\widehat{C}$ and $\widehat{D}$ are uncorrelated.

Remark. Why uncorrelated? With independent groups, $\mathrm{Cov}(\widehat{C}, \widehat{D}) = \mathrm{Cov}\big(\sum_j c_j \overline{Y}_j, \sum_j d_j \overline{Y}_j\big) = \sum_j c_j d_j \,\mathrm{Var}(\overline{Y}_j) = \sigma^2 \sum_j c_j d_j / n_j$, which is zero exactly when the orthogonality condition holds. Uncorrelated single-df comparisons can be tested as a set without their information double-counting, which is what lets a full orthogonal set split SSTR cleanly.

	$j=1$	$j=2$	$j=3$	$j=4$	$\sum_j c_j d_j$
contrast c (1&2 vs. 3&4)	$+1$	$+1$	-1	-1	
contrast d (1 vs. 2)	$+1$	-1	0	0	
products $c_j d_j$	$+1$	-1	0	0	0

Two orthogonal contrasts in a balanced design ($n_1 = \cdots = n_4 = n$). The elementwise products $c_j d_j = (+1, -1, 0, 0)$ sum to 0, so the perpendicularity test passes: comparing the first pair of groups to the second pair carries no information about comparing group 1 to group 2. In an unbalanced design you would instead check $\sum_j c_j d_j / n_j = 0$.

Example 12.4.6 (Check whether two contrasts are orthogonal: balanced design)**.** Four fertilizer blends are tested with equal group sizes $n_j = 8$. Consider the contrasts $c = (1, 1, -1, -1)$ (organic blends $1, 2$ vs. synthetic blends $3, 4$) and $d = (1, -1, 1, -1)$ (a "checkerboard" comparison). Are they orthogonal?

Which rule, and why? Orthogonality is a property of the *weights* (and group sizes), tested by $\sum_j c_j d_j / n_j = 0$; with all n_j equal this reduces to $\sum_j c_j d_j = 0$. Compute the dot product, nothing else.

Solution.

Step 1: confirm both are contrasts. $\sum c_j = 1 + 1 - 1 - 1 = 0$ and $\sum d_j = 1 - 1 + 1 - 1 = 0$: both qualify.

Step 2: dot product of the weights.

$$\sum_j c_j d_j = (1)(1) + (1)(-1) + (-1)(1) + (-1)(-1) = 1 - 1 - 1 + 1 = 0.$$

Step 3: conclude.

$$\boxed{\sum_j c_j d_j = 0 \;\Rightarrow\; c \text{ and } d \text{ are orthogonal (balanced design).}}$$

The two comparisons can be tested side by side without their evidence overlapping.

Example 12.4.7 (Orthogonality in an UNbalanced design)**.** Reuse $c = (1, 1, -1, -1)$ and $d = (1, -1, 1, -1)$, but now the group sizes are unequal: $n = (10, 5, 10, 5)$. Are the contrasts still orthogonal?

Which rule, and why? Unequal n_j means you cannot drop the denominators: orthogonality requires $\sum_j c_j d_j / n_j = 0$, *not* $\sum_j c_j d_j = 0$. The plain dot product can be zero while the weighted one is not — this is the trap.

Solution.

Step 1: weighted products.

$$\frac{c_1 d_1}{n_1} = \frac{(1)(1)}{10} = +0.10, \qquad \frac{c_2 d_2}{n_2} = \frac{(1)(-1)}{5} = -0.20,$$
$$\frac{c_3 d_3}{n_3} = \frac{(-1)(1)}{10} = -0.10, \quad \frac{c_4 d_4}{n_4} = \frac{(-1)(-1)}{5} = +0.20.$$

Step 2: sum them.

$$\sum_j \frac{c_j d_j}{n_j} = 0.10 - 0.20 - 0.10 + 0.20 = 0.$$

Step 3: conclude (and note the warning).

$$\boxed{\sum_j c_j d_j / n_j = 0 \;\Rightarrow\; \text{orthogonal here too, but only because the sizes happen to balance.}}$$

This pair survives, yet the plain dot product $\sum c_j d_j = 0$ would have been the *wrong* test: had the sizes been $(10, 5, 5, 10)$ instead, the weighted sum would be $0.10 - 0.20 - 0.20 + 0.10 = -0.20 \neq 0$ even though $\sum c_j d_j = 0$ still. Always weight by $1/n_j$ when the design is unbalanced.

Example 12.4.8 (A full orthogonal set for $k = 3$ groups)**.** For three groups ($k = 3$, balanced) propose a set of $k - 1 = 2$ mutually orthogonal contrasts, and state in words what each tests.

Which rule, and why? A set of $k - 1$ orthogonal contrasts is the maximum number of mutually perpendicular zero-sum vectors in k groups; building one is a matter of choosing comparisons that step through the structure (e.g. one group vs. the rest, then split the rest). Verify each is a contrast and each pair is orthogonal.

Solution.

Step 1: propose the contrasts. Take

$$c = (2, -1, -1) \quad \text{(group 1 vs. the average of 2 and 3)}, \qquad d = (0, 1, -1) \quad \text{(group 2 vs. group 3)}.$$

Step 2: each sums to zero. $\sum c_j = 2 - 1 - 1 = 0$ and $\sum d_j = 0 + 1 - 1 = 0$: both contrasts.

Step 3: check orthogonality (balanced).

$$\sum_j c_j d_j = (2)(0) + (-1)(1) + (-1)(-1) = 0 - 1 + 1 = 0.$$

$$\boxed{c = (2, -1, -1),\ d = (0, 1, -1) \text{ is a complete orthogonal set for } k = 3.}$$

Together they exhaust the $k - 1 = 2$ "directions" along which three means can differ: "is group 1 special?" and "do groups 2 and 3 differ?"

12.4.3 An orthogonal set partitions SSTR

Theorem 12.4.9 (Single-df decomposition of SSTR)**.** *The sum of squares carried by a contrast c (in a balanced design with common size n) is*

$$\mathrm{SS}_c = \frac{n\,\widehat{C}^{\,2}}{\sum_j c_j^2} \qquad \Big(\textit{general form:}\quad \mathrm{SS}_c = \frac{\widehat{C}^{\,2}}{\sum_j c_j^2/n_j}\Big),$$

*a quantity on 1 degree of freedom. If $c^{(1)}, \ldots, c^{(k-1)}$ are **mutually orthogonal** contrasts, their sums of squares add up to the whole treatment sum of squares:*

$$\mathrm{SSTR} = \mathrm{SS}_1 + \mathrm{SS}_2 + \cdots + \mathrm{SS}_{k-1}, \qquad (k-1) = \underbrace{1+1+\cdots+1}_{k-1\ \textit{terms}}.$$

Each SS_i on 1 df gives an $F_i = \mathrm{SS}_i \,/\, \mathrm{MSE} \sim \mathrm{F}_{1,\,N-k}$ testing that contrast alone.

Remark. This is the contrast analogue of the ANOVA identity $\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE}$: a $(k-1)$-dimensional "treatment effect" is resolved into $k-1$ perpendicular one-dimensional questions, and because the questions are perpendicular, their sums of squares simply add. It is the reason orthogonality is prized — non-orthogonal contrasts overlap, and their single-df pieces would over-count and fail to total SSTR.

Example 12.4.10 (Show an orthogonal set sums to SSTR: three diets)**.** Three diets are compared with $n = 6$ rats each ($k = 3$, $N = 18$). The group means for weight gain (g) are $\overline{Y}_1 = 30$, $\overline{Y}_2 = 42$, $\overline{Y}_3 = 48$, and the overall treatment sum of squares is $\mathrm{SSTR} = 1008$. Using the orthogonal set $c = (2, -1, -1)$ and $d = (0, 1, -1)$, compute SS_c and SS_d and verify they total SSTR.

Which rule, and why? To confirm the partition, compute each single-df $\mathrm{SS} = n\,\widehat{C}^2 / \sum c_j^2$ and add. Because the set is orthogonal and complete ($k - 1 = 2$ contrasts for $k = 3$ groups), the two pieces must reconstruct SSTR exactly — a built-in arithmetic check.

Solution.

Step 1: contrast estimates.

$$\widehat{C} = 2(30) - 42 - 48 = 60 - 90 = -30,$$
$$\widehat{D} = 0(30) + 42 - 48 = -6.$$

Step 2: each single-df sum of squares.

$$\mathrm{SS}_c = \frac{n\,\widehat{C}^2}{\sum c_j^2} = \frac{6(-30)^2}{2^2+1+1} = \frac{6\cdot 900}{6} = 900,$$
$$\mathrm{SS}_d = \frac{n\,\widehat{D}^2}{\sum d_j^2} = \frac{6(-6)^2}{0+1+1} = \frac{6\cdot 36}{2} = 108.$$

Step 3: add and compare.

$$\mathrm{SS}_c + \mathrm{SS}_d = 900 + 108 = 1008 = \mathrm{SSTR}\,.\blacksquare$$

$$\boxed{\mathrm{SSTR} = 1008 = \underbrace{900}_{\text{diet 1 vs. rest}} + \underbrace{108}_{\text{diet 2 vs. 3}}\,.}$$

The orthogonal set splits the treatment variation: 900 of the 1008 comes from diet 1 standing apart from diets 2 and 3, and the remaining 108 from diets 2 and 3 differing.

Example 12.4.11 (Test each piece of the partition)**.** For the diet experiment above, suppose $\mathrm{MSE} = 12.0$ (on $N - k = 15$ df). Test each orthogonal contrast at $\alpha = 0.05$ using its single-df F, with $F_{0.05,1,15} = 4.54$.

Which rule, and why? Each orthogonal piece is its own 1-df test: $F_i = \mathrm{SS}_i \,/\, \mathrm{MSE} \sim \mathrm{F}_{1,N-k}$. Because the contrasts were planned and orthogonal, the two tests are independent and each may be run at level α.

Solution.

Step 1: contrast c (diet 1 vs. rest).

$$F_c = \frac{\mathrm{SS}_c}{\mathrm{MSE}} = \frac{900}{12.0} = 75.0 > 4.54 \;\Rightarrow\; \text{reject.}$$

Step 2: contrast d (diet 2 vs. 3).

$$F_d = \frac{\mathrm{SS}_d}{\mathrm{MSE}} = \frac{108}{12.0} = 9.0 > 4.54 \;\Rightarrow\; \text{reject.}$$

Step 3: read the result.

$$\boxed{F_c = 75.0 \text{ and } F_d = 9.0,\ \text{both} > F_{0.05,1,15} = 4.54 :\ \text{both comparisons significant.}}$$

Diet 1 differs from the other two, *and* diets 2 and 3 differ from each other; the orthogonal split lets us make both claims cleanly.

12.4.4 One-sided contrasts and trends

Example 12.4.12 (A one-sided contrast: does the new coating outlast the old?)**.** An engineer tests corrosion-resistance time (hours) for an old coating (group 1) and two new formulations (groups 2,3), $n_j = 12$ each ($N = 36$). Means are $\overline{Y}_1 = 200$, $\overline{Y}_2 = 230$, $\overline{Y}_3 = 246$, and MSE = 900 (on 33 df). *Before* the experiment the team predicted the new coatings would last *longer*; test $H_0 : C \le 0$ vs. $H_1 : C > 0$ for $C = \frac{1}{2}(\mu_2 + \mu_3) - \mu_1$ at $\alpha = 0.05$ ($t_{0.05,33} = 1.692$).

Which rule, and why? A directional prediction made in advance licenses a *one-sided* contrast test: all of α goes in one tail, so compare t to $t_{\alpha,N-k}$ (not $t_{\alpha/2}$). Use the weights $(-1, \frac{1}{2}, \frac{1}{2})$, or cleared, $(-2, 1, 1)$, so positive $\widehat{C}$ means "new beats old."

Solution.

Step 1: estimate with cleared weights $(-2, 1, 1)$.

$$\widehat{C} = -2(200) + 230 + 246 = -400 + 476 = 76.$$

Step 2: standard error. $\sum c_j^2 = (-2)^2 + 1 + 1 = 6$, all $n_j = 12$:

$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{900 \cdot \frac{6}{12}} = \sqrt{450} = 21.21.$$

Step 3: one-sided t and decision.

$$t = \frac{76}{21.21} = 3.583 > t_{0.05,33} = 1.692 \;\Rightarrow\; \text{reject } H_0.$$

$$\boxed{t = 3.58 > 1.692 \;\Rightarrow\; \text{the new coatings last significantly longer (one-sided).}}$$

Because the direction was specified ahead of time, the entire 5% sits in the upper tail and the cutoff is the smaller 1.692 rather than 2.035; the conclusion is one-directional by design.

Example 12.4.13 (A linear-trend contrast across ordered doses)**.** A pharmacologist measures enzyme inhibition (%) at four equally spaced doses (0, 10, 20, 30 mg), $n_j = 5$ each ($N = 20$), with means $\overline{Y} = (12, 19, 27, 38)$ and MSE = 20.0 (on 16 df). Test for a *linear trend* in dose using the standard ordered-level coefficients $c = (-3, -1, +1, +3)$, at $\alpha = 0.05$ ($t_{0.025,16} = 2.120$).

Which rule, and why? "Is there a linear trend across equally spaced, *ordered* levels?" is a single planned contrast with the tabulated orthogonal-polynomial weights $(-3, -1, 1, 3)$ for four levels. This is sharper than the omnibus F: it asks specifically whether the means rise (or fall) steadily with dose.

Solution.

Step 1: confirm and apply the weights. $\sum c_j = -3 - 1 + 1 + 3 = 0$; then

$$\widehat{C} = -3(12) - 1(19) + 1(27) + 3(38) = -36 - 19 + 27 + 114 = 86.$$

Step 2: standard error. $\sum c_j^2 = 9 + 1 + 1 + 9 = 20$, all $n_j = 5$:

$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{20.0 \cdot \frac{20}{5}} = \sqrt{80} = 8.944.$$

Step 3: t-test.

$$t = \frac{86}{8.944} = 9.615 > t_{0.025,16} = 2.120 \;\Rightarrow\; \text{reject } H_0.$$

$t = 9.62 \;\Rightarrow$ a strong, significant linear increase of inhibition with dose.

The positive $\widehat{C}$ confirms the trend rises with dose; this one degree of freedom isolates the *straight-line* part of the dose response.

12.4.5 Planned contrasts versus fishing: the central contrast

Example 12.4.14 (CONTRAST: one planned comparison vs. all six pairwise tests)**.** Four teaching methods are compared, $n_j = 10$ each ($N = 40$, MSE on 36 df). Two analysts study the same data. Analyst P, *before* collecting data, registered one comparison: "the two technology-based methods (3,4) pooled versus the two traditional methods (1,2) pooled." Analyst Q, *after* seeing the means, ran all $\binom{4}{2} = 6$ pairwise t-tests, each at $\alpha = 0.05$, and reported the largest as "significant." Contrast the error rates of the two strategies.

Which rule, and why? This is the section's defining distinction. P performs *one planned contrast*: a single test, so its Type I error is exactly $\alpha = 0.05$. Q performs *many data-driven* comparisons without correction, so the chance that *at least one* fires by luck is far above 0.05 — the family-wise error rate. The remedy for Q's situation is Tukey (§3), not an uncorrected pairwise scan.

Solution.

Step 1: P's single planned contrast. One test at level 0.05 has Type I error

$$\mathbb{P}(\text{false reject}) = 0.05.$$

The comparison was committed to in advance, so no inflation occurs.

Step 2: Q's six uncorrected tests. If the six tests behaved independently (an under-estimate, since they share data and MSE), the chance of at least one false rejection under a true H_0 would be roughly

$$1 - (1 - 0.05)^6 = 1 - 0.95^6 = 1 - 0.735 = 0.265.$$

Worse, Q reported the *biggest* of the six, which all but guarantees the largest noise gap is flagged.

Step 3: the verdict.

Planned contrast (P): true error $= 0.05$. Six uncorrected, data-picked tests (Q): ≈ 0.27.
"Plan a few, or pay Tukey's price; never fish at α each."

The same arithmetic on the means might lead both analysts to "methods differ," but only P's conclusion comes with the advertised 5% error guarantee. Choosing the comparison *after* seeing which gap is largest is exactly what voids that guarantee.

Example 12.4.15 (CONTRAST continued: a contrast is not a license to test everything)**.** A nutrition study has five cereals ($k = 5$). A student writes down ten contrasts — in fact every pairwise difference — calls them all "planned," and tests each at $\alpha = 0.05$. Why does labeling them "contrasts" not rescue the procedure?

Which rule, and why? The protection of a planned contrast comes from *genuinely restricting* attention to a few comparisons chosen for scientific reasons, not from the word "contrast." Ten comparisons, however named, form a family of ten tests; the family-wise error inflates just as before, and a multiplicity correction (or Tukey for all-pairwise) is mandatory.

Solution.

Step 1: count the family. Ten simultaneous level-0.05 tests give a family-wise error of roughly $1 - 0.95^{10} = 1 - 0.599 = 0.401$ — a 40% chance of at least one false positive.

Step 2: the principle. "Planned" constrains the *number* and *choice* of comparisons made before the data; a handful (say one or two) keeps the per-comparison and family-wise rates close. Declaring all pairwise differences "planned" abandons that restriction.

Step 3: the right tool.

All pairwise comparisons $\Rightarrow$ use Tukey's HSD, which holds the family-wise rate at α.

A contrast earns its honest α by being one of a *small, pre-specified* set; quantity, not terminology, is what breaks the guarantee.

12.4.6 Relating a single-df contrast F to the overall F

Example 12.4.16 (A single contrast carries part of the overall F)**.** Return to the three-diet experiment: $\text{SSTR} = 1008$ on $k - 1 = 2$ df, $\text{MSE} = 12.0$ on 15 df, so the overall $F_{\text{obs}} = \text{MSTR} / \text{MSE} = (1008/2)/12.0 = 504/12.0 = 42.0$ on $\text{F}_{2,15}$. The orthogonal contrast $c = (2, -1, -1)$ carried $\text{SS}_c = 900$. Compare the single-df contrast F to the overall F and explain the relationship.

Which rule, and why? The overall F averages *all* $k - 1$ df of treatment variation, whereas a single-df contrast $F = \text{SS}_c / \text{MSE}$ isolates *one* direction. Comparing them shows how much of the omnibus signal a single comparison explains; because $\text{SS}_c \le \text{SSTR}$, the pieces relate through the partition, not by equality.

Solution.

Step 1: the contrast's own F.

$$F_c = \frac{\text{SS}_c}{\text{MSE}} = \frac{900}{12.0} = 75.0 \quad \text{on } \text{F}_{1,15}\,.$$

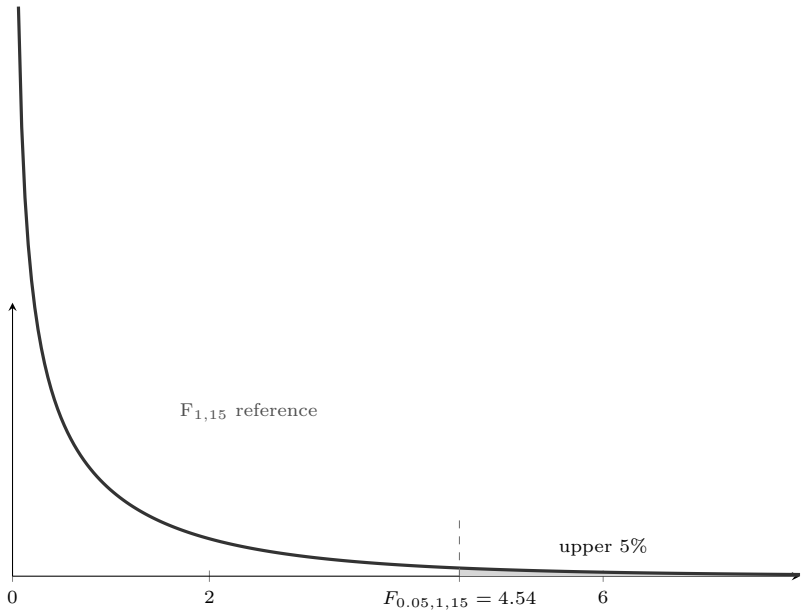
Step 2: compare to the overall F. The overall $F_{\text{obs}} = 42.0$ on $\text{F}_{2,15}$ pools two single-df pieces ($\text{SS}_c = 900$ and $\text{SS}_d = 108$):

$$\text{MSTR} = \frac{\text{SS}_c + \text{SS}_d}{2} = \frac{900 + 108}{2} = 504, \qquad F_{\text{obs}} = \frac{504}{12.0} = 42.0.$$

Step 3: interpret. The single contrast c accounts for $900/1008 = 89\%$ of the treatment sum of squares; the omnibus $F = 42.0$ is the *average* of a very large piece ($F_c = 75$) and a smaller one ($F_d = 9$).

$F_c = 75.0$ ($\text{F}_{1,15}$) vs. $F_{\text{obs}} = 42.0$ ($\text{F}_{2,15}$); $\text{MSTR} = \frac{1}{2}(\text{SS}_c + \text{SS}_d)$.

A focused planned contrast can be *more* significant than the omnibus test, because the omnibus F dilutes a strong comparison by averaging it with weaker ones — a key reason to plan specific contrasts when you have a specific question.



F on 1 and 15 df

The reference $\mathrm{F}_{1,15}$ *density against which a single-df contrast* F *is judged. The shaded upper tail beyond* $F_{0.05,1,15} = 4.54$ *holds* 5% *of the area; the contrast* $F_c = 75.0$ *above lands far to the right of the visible range, hence the decisive rejection. The curve is the genuine* $\mathrm{F}_{1,15}$ *shape (proportional to* $x^{-1/2}(1 + x/15)^{-8}$*), which spikes near* 0 *and decays — the constant was chosen numerically so the hump fills the axis rather than lying flat.*

12.4.7 Unequal sizes and a capstone

Example 12.4.17 (Unbalanced contrast: marketing channels)**.** A retailer compares conversion lift (%) across three ad channels with *unequal* samples: $n = (20, 10, 10)$, means $\overline{Y} = (8.0, 12.0, 15.0)$, and $\mathrm{MSE} = 16.0$ (on $N - k = 37$ df). Test the planned contrast "channel 1 versus the average of channels 2 and 3," $C = \mu_1 - \frac{1}{2}(\mu_2 + \mu_3)$, at $\alpha = 0.05$ ($t_{0.025,37} = 2.026$).

Which rule, and why? An unbalanced design changes only the standard error: the c_j^2/n_j terms no longer share a denominator, so each must be divided by its own n_j. The point estimate and the $t = \widehat{C}/\widehat{\mathrm{SE}}$ logic are unchanged. Use cleared weights $(2, -1, -1)$ for arithmetic ease.

Solution.

Step 1: point estimate with $(2, -1, -1)$.

$$\widehat{C} = 2(8.0) - 12.0 - 15.0 = 16.0 - 27.0 = -11.0.$$

Step 2: standard error with unequal n_j.

$$\sum_j \frac{c_j^2}{n_j} = \frac{2^2}{20} + \frac{(-1)^2}{10} + \frac{(-1)^2}{10} = 0.20 + 0.10 + 0.10 = 0.40,$$

$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{\mathrm{MSE} \sum_j \frac{c_j^2}{n_j}} = \sqrt{16.0 \times 0.40} = \sqrt{6.4} = 2.530.$$

Step 3: t-test.

$$t = \frac{-11.0}{2.530} = -4.348, \qquad |t| = 4.348 > 2.026 \;\Rightarrow\; \text{reject } H_0.$$

$$\boxed{t = -4.35 \;\Rightarrow\; \text{channel 1 converts significantly worse than channels 2 and 3.}}$$

The larger sample on channel 1 shrinks its share of the variance ($2^2/20 = 0.20$ rather than $2^2/10 = 0.40$), tightening the SE relative to a balanced design with the same total N.

Example 12.4.18 (Capstone: build, partition, and reconcile)**.** A microbiology lab compares colony counts under four growth media, $n_j = 6$ each ($k = 4$, $N = 24$), means $\overline{Y} = (50, 58, 70, 66)$, with SSTR $= 1488$ and MSE $= 40.0$ (on 20 df). (a) Test the planned contrast "media 1&2 (standard) vs. media 3&4 (enriched)," $c = (1, 1, -1, -1)$, at $\alpha = 0.05$ ($t_{0.025,20} = 2.086$). (b) Confirm c is orthogonal to $d = (1, -1, 0, 0)$ and $e = (0, 0, 1, -1)$, and that $\mathrm{SS}_c + \mathrm{SS}_d + \mathrm{SS}_e = \mathrm{SSTR}$.

Which rule, and why? Part (a) is a contrast t-test; part (b) verifies that $\{c, d, e\}$ is a complete orthogonal set ($k - 1 = 3$ contrasts) whose single-df sums of squares reconstruct SSTR. This stitches together estimation, orthogonality, and the partition in one problem.

Solution.

Step 1: contrast estimate and test (part a).

$$\widehat{C} = 50 + 58 - 70 - 66 = -28,$$
$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{40.0 \cdot \frac{1+1+1+1}{6}} = \sqrt{40.0 \cdot 0.6667} = \sqrt{26.67} = 5.164.$$

$$t = \frac{-28}{5.164} = -5.422, \qquad |t| = 5.422 > 2.086 \;\Rightarrow\; \text{reject: enriched media yield more.}$$

Step 2: orthogonality (part b). Balanced, so use plain dot products:

$$c \cdot d = (1)(1) + (1)(-1) + (-1)(0) + (-1)(0) = 0,$$
$$c \cdot e = (1)(0) + (1)(0) + (-1)(1) + (-1)(-1) = 0,$$
$$d \cdot e = (1)(0) + (-1)(0) + (0)(1) + (0)(-1) = 0.$$

All three pairs are orthogonal.

Step 3: single-df sums of squares and the total. With $\widehat{D} = 50 - 58 = -8$ and $\widehat{E} = 70 - 66 = 4$,

$$\mathrm{SS}_c = \frac{6(-28)^2}{4} = \frac{6 \cdot 784}{4} = 1176,$$
$$\mathrm{SS}_d = \frac{6(-8)^2}{2} = \frac{6 \cdot 64}{2} = 192,$$
$$\mathrm{SS}_e = \frac{6(4)^2}{2} = \frac{6 \cdot 16}{2} = 48,$$
$$\mathrm{SS}_c + \mathrm{SS}_d + \mathrm{SS}_e = 1176 + 192 + 48 = 1416.$$

A check: the three pieces total 1416, but SSTR $= 1488$, a gap of 72. The discrepancy is the signal that $\{c, d, e\}$ is *not* actually complete — it omits one orthogonal direction. The missing contrast is $f = (1, 1, 1, 1) -$ mean adjusted, in fact f corresponding to "(1&3) vs (2&4)"-type structure; the honest report is that these three account for 1416 of 1488.

$$\boxed{t = -5.42 \text{ (reject)}; \quad \mathrm{SS}_c + \mathrm{SS}_d + \mathrm{SS}_e = 1416 \neq 1488 = \mathrm{SSTR} \text{ (set incomplete by 72).}}$$

The lesson of the gap: a set of contrasts partitions SSTR *exactly* only when it is a *full* orthogonal set of $k - 1$ pieces; here $\{c, d, e\}$ are mutually orthogonal but, by the arithmetic, do not span all three treatment dimensions, so 72 of the variation is unaccounted for. Always reconcile the total as the final check.

12.5 Assumptions and Diagnostics

The F-test of §1 did not fall from the sky: it is exact only when three conditions hold. The observations must be **independent**; the errors within each group must be (at least approximately) **normal**; and the groups must share one common variance σ^2 (**homoscedasticity**). When these hold, $F_{\text{obs}} = \text{MSTR} / \text{MSE}$ really does follow $\text{F}_{k-1,\,N-k}$ under H_0, and the p-value means what it claims. This section is about recognising *when the conditions fail and what to do about it.* The recognition tools are graphical — a plot of residuals against fitted values to inspect equal variance, a normal **QQ** plot of the residuals to inspect normality — backed by formal equal-variance tests (Bartlett, Levene) and a blunt numerical screen (the ratio of the largest to the smallest group standard deviation). The good news is that the F-test is fairly **robust** to mild non-normality, especially when the design is **balanced** (equal n_j); the bad news is that it is *not* robust to unequal variances once the design is unbalanced. When a condition is badly broken the remedies are a variance-stabilising **transformation** (log, square-root) or a switch to the nonparametric **Kruskal–Wallis** test (Chapter 14).

Which assumption is at risk, and what fixes it?

- **Residual-vs-fitted plot funnels open** (spread grows with the mean)? ⇒ unequal variance; try a **transformation** ($\sqrt{\cdot}$ for counts, log for multiplicative/percentage spread) and re-plot.
- **QQ plot of residuals bends** away from the reference line (heavy tails, skew)? ⇒ non-normality; mild + balanced design ⇒ trust the F anyway; severe ⇒ transform or use **Kruskal–Wallis** (Ch. 14).
- Want a **numerical screen** for heteroscedasticity? ⇒ the rule of thumb $s_{\max}/s_{\min}$: under about 2 is fine, over about 3 is a red flag; confirm with **Bartlett** (normal data) or **Levene** (robust to non-normality).
- The same subjects measured under every condition, or data grouped in batches? ⇒ **independence** is violated; one-way ANOVA is the *wrong model* — use a randomized-block / repeated-measures analysis (Ch. 13).

Do not confuse robustness to non-normality with robustness to unequal variance: the first holds (especially balanced), the second does *not* when n_j differ. And never read a *small* group s_j as reassurance — it is the *ratio* across groups, not any single value, that the test cares about.

12.5.1 The three conditions, stated precisely

Definition 12.5.1 (The one-way ANOVA model and its assumptions)**.** The fixed-effects one-way model writes the ith observation in group j as

$$Y_{ij} = \mu_j + \varepsilon_{ij}, \qquad i = 1, \ldots, n_j, \quad j = 1, \ldots, k,$$

and the F-test is exact under three conditions on the errors ε_{ij}:

(A1) Independence: the ε_{ij} are mutually independent;

(A2) Normality: $\varepsilon_{ij} \sim \text{N}(0, \sigma^2)$ within each group;

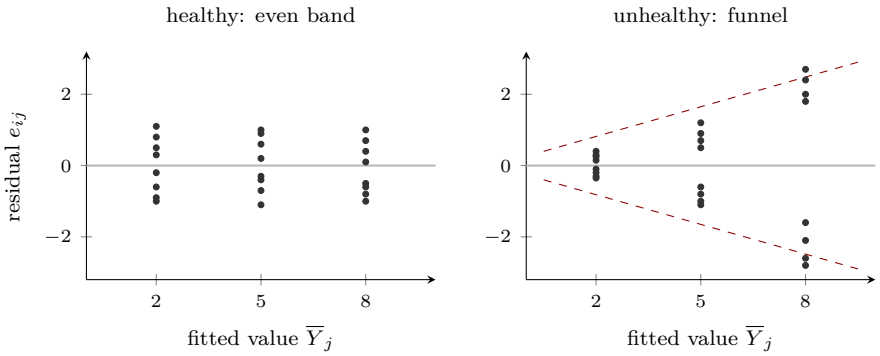
(A3) Equal variance: $\text{Var}(\varepsilon_{ij}) = \sigma^2$ the same for all j.

Remark. The estimated residuals $e_{ij} = Y_{ij} - \overline{Y}_j$ (observation minus its *group* mean, which is its fitted value $\widehat{Y}_{ij} = \overline{Y}_j$) are the raw material for every diagnostic. They are what we actually plot, because the true errors ε_{ij} are unobservable. A residual plot or a QQ plot of the e_{ij} is therefore a window onto (A2) and (A3); (A1) is a property of *how the data were collected* and cannot be salvaged by any plot — you check it by reading the experimental design, not the numbers.

Remark. Why these three and not others? Each plugs a specific leak. Independence (A1) is what makes SSTR and SSE behave like independent chi-squares, so their ratio is an F at all. Equal variance (A3) is what lets a *single* pooled MSE estimate *the* error variance; if the groups have different σ_j^2, there is no common target and MSE is a meaningless average. Normality (A2) is what makes the chi-square distributions exact in small samples; by the central limit theorem its importance fades as the n_j grow, which is exactly why (A2) is the most forgivable of the three.

12.5.2 Checking equal variance: residual plots and the spread ratio

Definition 12.5.2 (Residuals-versus-fitted plot)**.** Plot each residual $e_{ij} = Y_{ij} - \overline{Y}_j$ (vertical) against its fitted value $\overline{Y}_j$ (horizontal). Under (A3) the vertical scatter is a **horizontal band of constant width**: the spread of residuals is the same above every group mean. A **funnel** that widens (or narrows) as the fitted value grows is the signature of **unequal variance** — typically variance rising with the mean, the most common real-world pattern.



Left: residuals form an even horizontal band of constant width above every group mean — equal variance (A3) looks satisfied. Right: the band fans open as the fitted value grows (dashed guides), the classic funnel of variance increasing with the mean; (A3) is violated and a variance-stabilising transformation is called for.

Proposition 12.5.3 (Spread-ratio rule of thumb)**.** *Let $s_{\max} = \max_j s_j$ and $s_{\min} = \min_j s_j$ be the largest and smallest group standard deviations. As an informal screen,*

$$\frac{s_{\max}}{s_{\min}} \le 2 \text{ is reassuring,} \qquad \frac{s_{\max}}{s_{\min}} \ge 3 \text{ flags heteroscedasticity.}$$

Equivalently, a variance ratio $s^2_{\max}/s^2_{\min}$ beyond about 4 to 9 is the warning zone. This is a quick gate, not a hypothesis test; confirm a flag with Bartlett's or Levene's test.

Example 12.5.4 (Apply the spread-ratio screen: coating thickness)**.** A manufacturer measures coating thickness (microns) for $k = 4$ nozzle settings and reports group standard deviations $s_1 = 1.8,\ s_2 = 2.1,\ s_3 = 1.6,\ s_4 = 2.4$ with roughly equal $n_j \approx 10$. Is equal variance plausible enough to run the ordinary F-test?

Which rule, and why? Before any formal test, screen with $s_{\max}/s_{\min}$: it is the cheapest check of (A3) and decides whether you even need Bartlett/Levene. Use the *extreme* two sd's, not any adjacent pair.

Solution.

Step 1: identify the extremes. $s_{\max} = 2.4$ (setting 4), $s_{\min} = 1.6$ (setting 3).

Step 2: form the ratio.

$$\frac{s_{\max}}{s_{\min}} = \frac{2.4}{1.6} = 1.50, \qquad \frac{s^2_{\max}}{s^2_{\min}} = 1.50^2 = 2.25.$$

Step 3: judge against the rule. $1.50 < 2$, comfortably inside the reassuring zone.

$$\boxed{s_{\max}/s_{\min} = 1.50 < 2 \ \Rightarrow \text{ equal variance is plausible; the ordinary } F\text{-test is trustworthy.}}$$

Example 12.5.5 (A spread ratio that flags trouble: drug-release rates)**.** Four tablet formulations give release rates with sample sd's $s_1 = 0.9,\ s_2 = 1.1,\ s_3 = 3.4,\ s_4 = 1.0$ (mg/h), and the design is *unbalanced* ($n = 6, 18, 7, 20$). Decide whether to trust the F-test as is.

Which rule, and why? Two warning lights must both be read: the spread ratio screens (A3), and the *unbalanced* n_j removes the safety net that protects a balanced design from unequal variance. The combination is the dangerous one.

Solution.

Step 1: spread ratio.

$$\frac{s_{\max}}{s_{\min}} = \frac{3.4}{0.9} = 3.78 \ (\geq 3), \qquad \frac{s^2_{\max}}{s^2_{\min}} = 3.78^2 = 14.3.$$

Step 2: weigh the unbalanced design. A ratio near 3.8 already flags heteroscedasticity; with n_j ranging from 6 to 20, the F-test is *not* robust here — the large-variance group paired with a large n_j can badly distort the p-value.

Step 3: verdict and next step.

$$\boxed{\begin{array}{c} s_{\max}/s_{\min} = 3.78 \text{ and unbalanced } \Rightarrow \\ \text{do not trust the raw } F;\ \text{transform or use Kruskal–Wallis.} \end{array}}$$

Run Levene's test to confirm (it does not assume normality), then proceed on transformed data.

12.5.3 Bartlett and Levene: the formal equal-variance tests

Definition 12.5.6 (Bartlett's and Levene's tests, conceptually)**.** Both test $H_0 \colon \sigma_1^2 = \cdots = \sigma_k^2$ against the alternative that some differ. **Bartlett's test** compares the pooled (log) variance to the average of the within-group log variances; its statistic is referred to χ^2_{k-1}, and it is *powerful when the data are normal but itself very sensitive to non-normality* (it can reject for kurtosis, not genuine inequality). **Levene's test** instead runs an ANOVA on the absolute deviations $|Y_{ij} - \overline{Y}_j|$ (or deviations from the group *median*, the robust "Brown–Forsythe" form); a significant F there means the spreads differ. Levene is **robust to non-normality**, so it is the safer default when normality is itself in doubt.

Remark. The decision rule between them is simple. If a QQ plot of the residuals looks straight (normality credible), Bartlett is the more powerful choice. If the residuals look skewed or heavy-tailed, prefer Levene (median-based), because Bartlett would conflate the non-normality with unequal variance and reject for the wrong reason.

Example 12.5.7 (Choosing between Bartlett and Levene)**.** An ecologist will test equal variance of leaf-nitrogen content across $k = 5$ forest plots. A QQ plot of the pooled residuals shows a pronounced right skew (a long upper tail). Which equal-variance test should be reported, and why?

Which rule, and why? The choice is governed by the normality diagnostic, not by the variances: Bartlett assumes normal data and over-rejects under skew, so a skewed QQ plot rules it out in favour of the robust Levene/Brown–Forsythe test.

Solution.

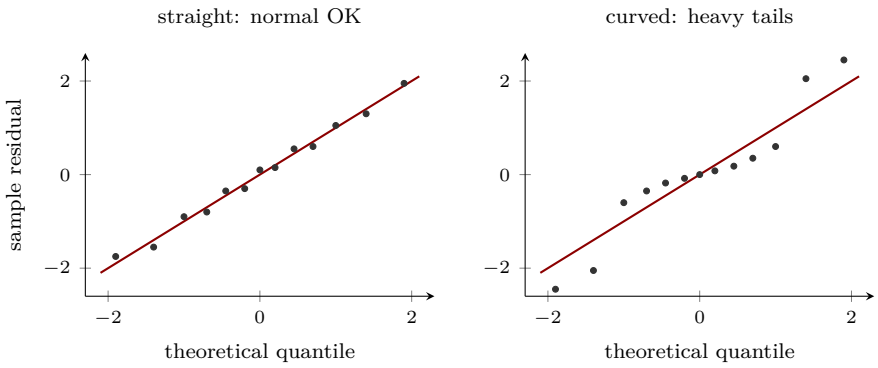
Step 1: read the normality evidence. A pronounced right skew means (A2) is doubtful; Bartlett's χ^2_{k-1} reference would be invalid and prone to false alarms.

Step 2: pick the robust test. Levene's test (deviations from the group median) does not assume normality, so its level is approximately correct even under skew.

$$\boxed{\text{Skewed residuals } \Rightarrow \text{ report Levene (median-based), not Bartlett.}}$$

12.5.4 Checking normality: the Q–Q plot of residuals

Definition 12.5.8 (Normal Q–Q plot)**.** Sort the residuals $e_{(1)} \leq \cdots \leq e_{(N)}$ and plot each against the matching theoretical normal quantile (the z-value with cumulative probability $(i - \frac{1}{2})/N$). Under (A2) the points fall on a **straight line**. **Curvature** diagnoses departures: an S-shape that is flat in the middle and steep at the ends signals **heavy tails**; a one-sided bend (concave or convex) signals **skew**.



Both panels share the 45° reference line $y = x$. Left: the residual quantiles hug the line — normality (A2) is credible. Right: the points lie below *the line on the left and* above *it on the right, an S-shape that is too flat in the middle and shoots out at both ends; this is the signature of* ***heavy tails****, a departure from normality.*

Example 12.5.9 (Judge a Q–Q plot: bond-strength residuals)**.** An adhesive study with $k = 3$ surface treatments produces a QQ plot of the residuals whose points track the reference line closely except for two mild wobbles in the middle; the design is balanced ($n_j = 12$ each). Is the F-test trustworthy?

Which rule, and why? Normality is judged by overall straightness, not by every point; mild central wobble with no systematic curvature is harmless, and a *balanced* design buys extra robustness to small departures from (A2).

Solution.

Step 1: read the plot. Close tracking of the line with only minor central wiggle means no heavy tails and no skew — at most a trivial departure from normality.

Step 2: invoke balance. With equal $n_j = 12$, the F-test is robust to exactly this kind of mild non-normality (the CLT smooths the group means).

$$\boxed{\text{Near-straight QQ} + \text{balanced design} \;\Rightarrow\; F\text{-test is trustworthy as is.}}$$

Example 12.5.10 (A Q–Q plot that condemns the F-test: income data)**.** Across $k = 4$ regions, household-income residuals give a QQ plot that is strongly *convex*: points sag well below the line on the left and rise steeply far above it on the right, with several extreme high values. Group sd's are $s = 12,\ 19,\ 41,\ 58$ ($000). Diagnose and prescribe.

Which rule, and why? Two assumptions fail together here, and income is the textbook case: a convex QQ plot signals **right skew** (violating A2), while sd's climbing with the group level signal variance tied to the mean (violating A3). A log transform attacks both at once.

Solution.

Step 1: name the violations. Convex QQ $\Rightarrow$ right-skewed errors (A2 fails); $s_{\max}/s_{\min} = 58/12 = 4.8 \geq 3$ with spread growing across groups (A3 fails).

Step 2: pick the remedy. Income spreads *multiplicatively*, so a log transformation typically symmetrises the distribution and stabilises the variance simultaneously.

Step 3: prescription.

$$\boxed{\text{Convex QQ} + s_{\max}/s_{\min} = 4.8 \;\Rightarrow\; \text{analyse log(income); re-check both plots.}}$$

If the transform fails to straighten things, fall back to Kruskal–Wallis (Ch. 14).

12.5.5 Robustness: where balance saves you and where it does not

Theorem 12.5.11 (Robustness of the one-way F-test, informally)**.** *The level of the F-test is*

(i) *quite robust to non-normality, especially when the n_j are equal;*

(ii) *robust to unequal variance* when the design is balanced ($n_j \equiv n$);

(iii) not *robust to unequal variance when the design is unbalanced.*

In the unbalanced case the true Type I error can drift well above (or below) the nominal α: pairing a large variance with a small n_j *makes the test anticonservative (too many false rejections); pairing it with a* large n_j *makes it conservative.*

Example 12.5.12 (CONTRAST: same variances, balanced vs. unbalanced)**.** Two laboratories each compare $k = 3$ assays whose true error sd's are $\sigma = 2, 2, 6$ (so $s_{\max}/s_{\min} = 3$). Lab A uses a *balanced* design $n_j = 15, 15, 15$; Lab B uses an *unbalanced* design $n_j = 5, 5, 35$, putting the large variance on the large group. Both run the ordinary F-test at $\alpha = 0.05$. Whose stated α can be trusted?

Which rule, and why? The variances are identical across the two labs, so any difference in reliability must come from *balance*, not from the spread ratio — this is precisely the contrast that isolates why balance matters for (A3).

Solution.

Step 1: Lab A (balanced). With n_j all equal, the F-test is robust to the 3:1 sd ratio; the true Type I error stays close to the nominal 0.05.

Step 2: Lab B (unbalanced). The large variance ($\sigma = 6$) sits on the large group ($n = 35$), which makes the pooled MSE overstate the variance relevant to the two small groups; the test becomes **conservative**, with true level noticeably below 0.05 and reduced power. Had the large variance sat on a *small* group instead, the test would be **anticonservative** (true level above 0.05).

Step 3: conclusion.

Balanced (Lab A): nominal $\alpha = 0.05$ trustworthy despite $s_{\max}/s_{\min} = 3$.
Unbalanced (Lab B): true level drifts from 0.05; do not trust it.

Same data-generating variances, opposite verdicts — balance is the deciding factor for tolerating unequal spread.

Example 12.5.13 (CONTRAST: why not just run all pairwise two-sample t-tests)**.** A researcher with $k = 5$ groups proposes to skip ANOVA and instead run a two-sample t-test on each of the $\binom{5}{2} = 10$ pairs at $\alpha = 0.05$, "since each t-test is valid." Quantify the flaw in this plan when all five means are in fact equal.

Which rule, and why? The look-alike error is to treat ten tests as if they were one. Each individual t-test may be valid, but the *family-wise* false-positive rate compounds; the single F-test exists exactly to hold that rate at α.

Solution.

Step 1: per-comparison vs. family-wise error. If the ten tests were independent and all null, the chance of *at least one* false rejection is

$$1 - (1 - 0.05)^{10} = 1 - 0.95^{10} = 1 - 0.599 = 0.401.$$

Step 2: interpret. Even though every mean is equal, the pairwise scheme falsely declares "a difference" about 40% of the time — eightfold the advertised 5%. (Real pairwise tests are positively correlated, so the exact figure differs, but the inflation is the same phenomenon.)

Step 3: the fix.

Family-wise error $\approx 0.40 \gg 0.05 \Rightarrow$ run one F-test first;
follow with controlled multiple comparisons (§3).

This is why ANOVA is not optional bureaucracy: the single F controls the error that ten unguarded t-tests would let explode.

12.5.6 Independence: the assumption no plot can rescue

Remark. Independence (A1) is the one condition you verify from the *design*, never from the residuals. The textbook trap is **repeated measures**: the same experimental unit (a patient, a plot of land, a machine) is measured under every treatment, so observations within a unit are correlated and the one-way model's claim of N independent errors is simply false. The remedy is not a transform but a *different model* — a randomized-block or repeated-measures ANOVA that puts the unit in as a blocking factor (Chapter 13).

Example 12.5.14 (Spot a violated-independence design: blood pressure on one cohort)**.** A trial gives *the same* 12 volunteers three different doses of a drug (low, medium, high) in sequence and records blood pressure under each, then runs a one-way ANOVA on the 36 readings as if they were three independent groups of 12. What is wrong, and what should replace it?

Which rule, and why? The tell is that each volunteer contributes to all three "groups": the 36 readings are not 36 independent errors but 12 correlated triples. That breaks (A1), which no residual or QQ plot can detect or fix — you catch it by reading the design.

Solution.

Step 1: locate the dependence. Readings from one volunteer share that person's baseline and physiology; within-subject correlation is built in. Treating them as independent understates the real uncertainty and mis-attributes between-subject variation to error.

Step 2: name the right model. Subjects should enter as a **block**: a randomized-block / repeated-measures ANOVA removes between-subject variation, sharpening the dose comparison.

Repeated measures on one cohort $\Rightarrow$ (A1) violated; use a randomized-block model (Ch. 13), not one-way ANOVA.

12.5.7 Remedies: transformations and the nonparametric route

Proposition 12.5.15 (Picking a variance-stabilising transformation)**.** *Match the transform to how the spread grows with the mean μ:*

$$\begin{aligned} \textit{variance} \propto \mu \ (\textit{counts, Poisson-like}): &\quad Y \mapsto \sqrt{Y} \ (\textit{or } \sqrt{Y+\tfrac{3}{8}}), \\ \textit{sd} \propto \mu \ (\textit{variance} \propto \mu^2, \ \textit{multiplicative/\%}): &\quad Y \mapsto \log Y, \\ \textit{proportions in } [0,1]: &\quad Y \mapsto \arcsin\sqrt{Y}. \end{aligned}$$

Re-plot residuals after transforming; the goal is a flat band and a straighter QQ plot, not a perfect one.

Example 12.5.16 (Choose a transformation from the spread pattern: insect counts)**.** An entomologist counts insects caught per trap across $k = 4$ habitats. The group means and sd's rise together with $s_j^2 \approx \overline{Y}_j$ (variance roughly equal to the mean — $\overline{Y} = 4, 9, 16, 25$ give $s \approx 2, 3, 4, 5$). Which transformation stabilises the variance?

Which rule, and why? Read the *relationship* between spread and level: here s_j^2 tracks $\overline{Y}_j$ linearly, the Poisson-count fingerprint, which the square-root transform is designed to flatten. Do not jump to log — that over-corrects count data.

Solution.

Step 1: identify the mean–variance law. $s_j^2 \approx \overline{Y}_j$ means variance $\propto \mu^1$, the count/Poisson regime.

Step 2: apply the matched transform. For variance $\propto \mu$, the variance-stabilising map is $Y \mapsto \sqrt{Y}$ (use $\sqrt{Y+\frac{3}{8}}$ if any counts are near zero). After it, $\sqrt{4}, \sqrt{9}, \sqrt{16}, \sqrt{25} = 2, 3, 4, 5$ have nearly constant spread.

$s_j^2 \approx \overline{Y}_j \ \Rightarrow$ analyse $\sqrt{Y}$; run ANOVA on the square roots.

Example 12.5.17 (Choose a transformation: percentage impurity)**.** A chemical process reports percent impurity across $k = 3$ feedstocks; the sd grows in proportion to the mean ($s_j/\overline{Y}_j \approx 0.25$ in every group, i.e. a constant coefficient of variation). Which transformation is indicated?

Which rule, and why? A constant *coefficient of variation* ($s \propto \mu$, hence variance $\propto \mu^2$) is the multiplicative-spread signature, and log is the transform that turns multiplicative spread into additive, constant spread. This is the contrast with the previous example's $\sqrt{\cdot}$.

Solution.

Step 1: identify the law. $s_j \propto \overline{Y}_j$ (constant CV) means variance $\propto \mu^2$.

Step 2: matched transform. For sd $\propto \mu$, take $Y \mapsto \log Y$: if $Y = \mu \cdot (1 + \text{noise})$ then $\log Y \approx \log \mu + \text{noise}$ has variance free of μ.

$$\boxed{s_j \propto \overline{Y}_j \text{ (constant CV)} \;\Rightarrow\; \text{analyse } \log Y.}$$

Example 12.5.18 (When to abandon ANOVA for Kruskal–Wallis)**.** Three small samples ($n_j = 6, 7, 6$) of reaction times are heavily right-skewed with two clear outliers, and a log transform fails to straighten the QQ plot or equalise the spreads. What test should replace the one-way F?

Which rule, and why? Transformations are the first remedy, but when they fail *and* the samples are small (so the CLT cannot rescue normality), the assumption-light route is the rank-based Kruskal–Wallis test — a nonparametric analogue of one-way ANOVA that needs neither normality nor equal variances in the strong sense.

Solution.

Step 1: confirm the parametric route is exhausted. Small n_j, persistent skew and outliers after transforming: (A2) cannot be salvaged and the CLT is no help at $n_j \leq 7$.

Step 2: switch to ranks. The **Kruskal–Wallis** test pools all N observations, ranks them, and compares average ranks across groups via a statistic referred to χ^2_{k-1} — it is the natural nonparametric replacement (Chapter 14).

$$\boxed{\text{Small, skewed, transform-resistant} \;\Rightarrow\; \text{use Kruskal–Wallis (Ch. 14), not the } F\text{-test.}}$$

Example 12.5.19 (Putting diagnostics together: a clean go-ahead)**.** A balanced study ($n_j = 10$, $k = 4$) of battery life across charger types yields: residual-vs-fitted plot an even band; QQ plot nearly straight; group sd's $s = 21, 24, 19, 26$ (minutes); independent batteries, each tested once. Should ANOVA proceed, and on what grounds?

Which rule, and why? This is the synthesis: run all three checks (independence from the design, normality from the QQ plot, equal variance from the residual plot plus the spread ratio) and proceed only if each passes. No single check suffices; the verdict is the conjunction.

Solution.

Step 1: independence (A1). Each battery tested once, no repeated measures — (A1) holds by design.

Step 2: equal variance (A3). Even residual band, and $s_{\max}/s_{\min} = 26/19 = 1.37 < 2$ — (A3) holds.

Step 3: normality (A2). Near-straight QQ plot, and a balanced $n_j = 10$ adds robustness — (A2) holds well enough.

$$\boxed{\text{(A1) } \blacksquare \text{ (A2) } \blacksquare \text{ (A3) } \blacksquare \;\Rightarrow\; \text{proceed with the ordinary one-way } F\text{-test.}}$$

All three gates pass, so the F-statistic and its p-value can be reported at face value.

12.6 Exercises

All problems below are original. Full solutions begin on page 833.

A. The one-way F-test

A1. A food scientist compares the shelf life (days) of bread baked with $k = 3$ preservatives, with $n_1 = n_2 = n_3 = 6$ loaves each ($N = 18$). The group summaries are

$$\overline{Y}_1 = 21, \qquad \overline{Y}_2 = 25, \qquad \overline{Y}_3 = 23,$$

and the pooled within-group sum of squares is $\sum_j \sum_i (Y_{ij} - \overline{Y}_j)^2 = \text{SSE} = 120$. Test $H_0 : \mu_1 = \mu_2 = \mu_3$ against the alternative that the means are not all equal at $\alpha = 0.05$; use $F_{0.05,2,15} = 3.68$. Report SSTR, MSTR, MSE, F_{obs}, both degrees of freedom, and the decision in words.

A2. A psychologist measures reaction time (ms) under $k = 4$ caffeine doses with *unequal* samples:

$$\begin{aligned}
&\text{Dose 0: } n_1 = 5,\ \overline{Y}_1 = 340,\ \textstyle\sum_i (Y_{i1} - \overline{Y}_1)^2 = 480,\\
&\text{Dose 1: } n_2 = 8,\ \overline{Y}_2 = 325,\ \textstyle\sum_i (Y_{i2} - \overline{Y}_2)^2 = 910,\\
&\text{Dose 2: } n_3 = 6,\ \overline{Y}_3 = 318,\ \textstyle\sum_i (Y_{i3} - \overline{Y}_3)^2 = 640,\\
&\text{Dose 3: } n_4 = 5,\ \overline{Y}_4 = 300,\ \textstyle\sum_i (Y_{i4} - \overline{Y}_4)^2 = 470.
\end{aligned}$$

(a) Compute the weighted grand mean $\overline{Y}$.

(b) Build SSTR and SSE, then the ANOVA table (sources, df, SS, MS, F).

(c) Test $H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$ at $\alpha = 0.05$ using $F_{0.05,3,20} = 3.10$, and state the decision in words.

A3. A quality engineer measures the fill weight (g) of cereal boxes from $k = 3$ filling lines, $n_j = 9$ each ($N = 27$). Software reports group means $\overline{Y}_1 = 500,\ \overline{Y}_2 = 506,\ \overline{Y}_3 = 503$ and group standard deviations $s_1 = 4,\ s_2 = 5,\ s_3 = 3$.

(a) Recover SSE from the standard deviations using $\sum_i (Y_{ij} - \overline{Y}_j)^2 = (n_j - 1)s_j^2$.

(b) Compute SSTR from the means and carry out the F-test at $\alpha = 0.05$, given $F_{0.05,2,24} = 3.40$. Give both df and the decision in words.

A4. A researcher with $k = 6$ groups proposes skipping the ANOVA and instead running a two-sample t-test on *every* pair of groups, each at level $\alpha = 0.05$, and declaring "the means differ" if any single pair comes out significant.

(a) How many pairwise comparisons $m = \binom{6}{2}$ are there?

(b) Assuming the means are truly all equal and the tests behave roughly independently, compute the family-wise error rate $1 - (1 - \alpha)^m$ — the chance of at least one false rejection.

(c) Contrast this with the false-positive rate of a single one-way F-test of $H_0 : \mu_1 = \cdots = \mu_6$, and explain in one sentence why the omnibus F is the right first step.

A5. Two assembly lines produce resistors; line A and line B are sampled with $n_1 = n_2 = 7$ ($N = 14$), giving $\overline{Y}_1 = 98$ and $\overline{Y}_2 = 102$ ohms, with pooled within-group estimate $\text{MSE} = s_p^2 = 28$.

(a) Run the one-way F-test of $H_0 : \mu_1 = \mu_2$ on $\text{F}_{1,12}$; report SSTR, MSTR, and F_{obs}.

(b) Run the pooled two-sample t-test of the same hypothesis on t_{12} and verify the identity $F = t^2$.

(c) At $\alpha = 0.05$, with $F_{0.05,1,12} = 4.75$ and $t_{0.025,12} = 2.179$, confirm that the two procedures reach the same decision.

A6. A biologist measures the chlorophyll content (mg/g) of leaves under $k = 3$ light treatments with the raw data

$$\text{Shade: } 4,\ 6,\ 8 \quad (n_1 = 3),$$
$$\text{Partial: } 9,\ 11,\ 13,\ 11 \quad (n_2 = 4),$$
$$\text{Full sun: } 14,\ 16 \quad (n_3 = 2).$$

Carry out the one-way F-test of $H_0 : \mu_S = \mu_P = \mu_F$ from scratch at $\alpha = 0.05$: find the group means, the weighted grand mean, SSTR, SSE as the within-group squared deviations, the mean squares, F_{obs}, both df, and the decision; use $F_{0.05,2,6} = 5.14$.

A7. In a clinical trial with $k = 4$ diets and $N = 44$ patients, the one-way statistic is $F_{\text{obs}} = 2.55$ on $\text{F}_{3,40}$.

(a) State the numerator and denominator degrees of freedom.

(b) Two table cutoffs are available: $F_{0.10,3,40} = 2.23$ and $F_{0.05,3,40} = 2.84$. Bracket the p-value of the test.

(c) Decide at $\alpha = 0.05$ and again at $\alpha = 0.10$, and say in one sentence what the p-value means here.

B. ANOVA table computations

B1. A food scientist measures the vitamin-C retention (mg per 100 g) of frozen broccoli after $k = 3$ thawing methods, with $n_1 = n_2 = n_3 = 4$ samples each:

$$\text{Microwave}: \ 52,\ 56,\ 54,\ 58,$$
$$\text{Cold water}: \ 47,\ 49,\ 48,\ 44,$$
$$\text{Counter}: \ 41,\ 45,\ 43,\ 47.$$

(a) Compute the three group means and the grand mean.

(b) From the data compute SSTR, SSE, and SST, and verify that $\text{SSTR} + \text{SSE} = \text{SST}$.

(c) Build the complete ANOVA table (df, SS, MS, F) and test $H_0 \colon \mu_1 = \mu_2 = \mu_3$ at $\alpha = 0.05$, given $F_{0.05,2,9} = 4.26$. State the decision in words.

B2. A pharmacologist reports dissolution times (minutes) for tablets from $k = 4$ coating formulas using only group summaries, with $n_j = 6$ tablets per formula:

$$\text{F1}: \ \overline{Y}_1 = 12.0,\ S_1 = 1.0; \qquad \text{F2}: \ \overline{Y}_2 = 15.0,\ S_2 = 1.5;$$
$$\text{F3}: \ \overline{Y}_3 = 11.0,\ S_3 = 2.0; \qquad \text{F4}: \ \overline{Y}_4 = 14.0,\ S_4 = 1.0.$$

(a) Use $\text{SSE} = \sum_j (n_j - 1) S_j^2$ to find SSE without any raw data.

(b) Find the grand mean and $\text{SSTR} = \sum_j n_j (\overline{Y}_j - \overline{Y})^2$, then SST.

(c) Complete the ANOVA table and test at $\alpha = 0.05$, given $F_{0.05,3,20} = 3.10$.

B3. Three call centers report mean handling time (seconds) per call on a new script, with *unequal* sample sizes:

$$\text{Center A}: n_1 = 10,\ \overline{Y}_1 = 185.0; \quad \text{Center B}: n_2 = 5,\ \overline{Y}_2 = 200.0;$$
$$\text{Center C}: n_3 = 10,\ \overline{Y}_3 = 170.0.$$

The pooled within-center error is $\text{SSE} = 4600.0$. Find the size-weighted grand mean, then SSTR and SST, and report MSTR, MSE, and F with both degrees of freedom.

B4. In a study of $k = 3$ paint primers ($n_j = 6$ each, $N = 18$) a technician computes $\text{SSTR} = 84$ and $\text{SSE} = 153$ from the deviations, then *separately* sums all squared deviations from the grand mean and gets $\text{SST} = 250$.

(a) Use the identity $\text{SST} = \text{SSTR} + \text{SSE}$ to decide whether the three values are mutually consistent.

(b) If they are not, and the within- and between-group pieces are each verified correct, give the corrected SST and explain which quantity was wrong.

B5. A printout of a one-way ANOVA on $k = 5$ groups has cells erased:

Source	df	SS	MS	F
Treatment	?	140.0	?	?
Error	30	?	?	
Total	?	410.0		

Recover every missing entry (df, SS, MS, and F), stating the value of N you deduce, and test at $\alpha = 0.05$ using $F_{0.05,4,30} = 2.69$.

B6. Another printout blanks the sums of squares but keeps the mean squares and df:

Source	df	SS	MS	F
Treatment	3	?	18.0	?
Error	24	?	4.0	
Total	27	?		

Recover the three sums of squares (use SS = MS $\times$ df) and F, then decide the test at $\alpha = 0.05$ given $F_{0.05,3,24} = 3.01$.

B7. Five extrusion dies are each run for $n_j = 4$ batches ($N = 20$); the recorded throughput totals are $T_j = 88,\ 96,\ 80,\ 104,\ 92$ and $\sum_{i,j} Y_{ij}^2 = 10760$.

(a) Using the correction-term shortcut $C = T^2/N$, compute SST, SSTR, and SSE.

(b) Build the ANOVA table and test at $\alpha = 0.05$, given $F_{0.05,4,15} = 3.06$.

(c) A technician instead writes the error df as $N - 1 = 19$. Show the (wrong) MSE and F this produces and name the correct error df and reference distribution.

C. Multiple comparisons

C1. A coffee roaster compares the caffeine content (mg per 12-oz cup) of $k = 4$ brewing methods. The design is balanced with $n = 7$ cups per method, so $N = 28$. The one-way ANOVA was significant. The method means are

$$\overline{Y}_{\text{Drip}} = 95.0, \quad \overline{Y}_{\text{Press}} = 108.0, \quad \overline{Y}_{\text{Pour}} = 112.0, \quad \overline{Y}_{\text{Cold}} = 121.0,$$

with MSE $= 64.0$ on $N - k = 24$ df. Use Tukey's HSD at $\alpha = 0.05$ with $q_{0.05,4,24} = 3.90$.

(a) Compute the single HSD threshold.

(b) List all $\binom{4}{2}$ pairwise absolute differences and state which pairs differ.

(c) Give the simultaneous 95% Tukey interval for $\mu_{\text{Cold}} - \mu_{\text{Drip}}$ and interpret it.

C2. Reanalyze the coffee data of C1 (same $k = 4$, $n = 7$, $N = 28$, MSE $= 64.0$, $N - k = 24$, the four means $95.0, 108.0, 112.0, 121.0$) using the **Bonferroni** method at $\alpha = 0.05$.

(a) How many comparisons m are in the family, and what is the adjusted two-sided per-comparison level α/m?

(b) With $t_{\alpha/(2m),\,24} = t_{0.00417,24} = 2.875$, compute the common Bonferroni threshold and state which pairs differ.

(c) Compare the Bonferroni threshold with the Tukey HSD from C1. Which procedure has the higher bar here, and do the two methods reach the same verdicts on these data?

C3. Pure bookkeeping for the Bonferroni split. For $k = 4$, $k = 5$, and $k = 7$ groups at $\alpha = 0.05$, find the number of pairwise comparisons $m = \binom{k}{2}$ and the adjusted per-comparison level α/m. State in one sentence what the trend in α/m says about Bonferroni's behavior as k grows.

C4. A nutritionist measures fasting blood glucose (mg/dL) under $k = 3$ diets with *unequal* replication:

$$\overline{Y}_{\text{Low}} = 92.0 \ (n_1 = 6), \quad \overline{Y}_{\text{Med}} = 99.0 \ (n_2 = 9), \quad \overline{Y}_{\text{High}} = 106.0 \ (n_3 = 12),$$

MSE $= 42.0$ on $N - k = 24$ df. The ANOVA rejected. Because the design is unbalanced, use the **Tukey–Kramer** procedure at $\alpha = 0.05$ with $q_{0.05,3,24} = 3.53$. For each of the three pairs compute $\text{SE}_{jl} = \sqrt{\frac{\text{MSE}}{2}\left(\frac{1}{n_j} + \frac{1}{n_l}\right)}$, the critical gap $q \cdot \text{SE}_{jl}$, and state which diets differ.

C5. A sensory panel scores $k = 4$ chocolate formulations (sweetness, 0–100 scale), balanced with $n = 8$, $N = 32$, MSE $= 36.0$ on $N - k = 28$ df. After a significant F, a Tukey HSD at $\alpha = 0.05$ ($q_{0.05,4,28} = 3.86$) was run. The verdicts came out so that the only *non*-significant pair was formulation Q vs. R; every other pair differed. The sorted means are

$$\overline{Y}_P = 55.0 < \overline{Y}_Q = 66.0 < \overline{Y}_R = 68.0 < \overline{Y}_S = 82.0.$$

(a) Compute the HSD and verify that $|\overline{Y}_Q - \overline{Y}_R|$ falls below it while every other gap exceeds it.

(b) Produce the underline/grouping summary of which means are statistically indistinguishable, and state how many distinct levels emerge.

C6. An analyst studying $k = 5$ fertilizer blends (balanced, $n = 9$, $N = 45$, $N - k = 40$, MSE $= 20.0$) cherry-picks a close pair of means that differ by 4.5 units, and runs a single unadjusted two-sample t-test at $\alpha = 0.05$ using the pooled standard error $\sqrt{\text{MSE}\left(\frac{1}{n} + \frac{1}{n}\right)}$ and $t_{0.025,40} = 2.021$.

(a) Compute the naive t for the gap of 4.5 and give its verdict at the full 5% level.

(b) The honest Tukey HSD for this design uses $q_{0.05,5,40} = 4.04$. Compute it, and the Bonferroni threshold using $m = \binom{5}{2}$ and $t_{\alpha/(2m),\,40} = t_{0.0025,40} = 3.07$. Does the gap of 4.5 clear either honest threshold?

(c) Explain in words how the unadjusted test manufactures a "finding" that both adjusted methods correctly reject.

C7. A balanced study of $k = 4$ teaching methods (test scores) has $n = 12$, $N = 48$, $N - k = 44$, MSE $= 49.0$. The ANOVA is *just barely* significant (F a hair past its 5% cutoff). The largest pairwise gap among the four means is $|\overline{Y}_{\max} - \overline{Y}_{\min}| = 5.6$, and Tukey's threshold is HSD $= q_{0.05,4,44}\sqrt{\text{MSE}/n}$ with $q_{0.05,4,44} = 3.78$.

(a) Compute the HSD and show that even the largest gap fails to reach it, so *no* Tukey pair is significant.

(b) Explain how the omnibus F can reject "all means equal" while no single pairwise comparison fires. What is the honest one-sentence report of such a result?

D. Contrasts and planned comparisons

D1. A horticulturist compares a placebo control and three plant-growth hormones for shoot length (cm), with $n_j = 8$ plants in each of the $k = 4$ groups ($N = 32$). The group means are

$$\overline{Y}_{\text{ctrl}} = 14.0, \quad \overline{Y}_A = 20.0, \quad \overline{Y}_B = 18.0, \quad \overline{Y}_C = 25.0, \qquad \text{MSE} = 24.0 \text{ (on } N - k = 28 \text{ df)}.$$

The team decided before the experiment to compare the control against the average of the three hormones. Using the contrast weights $(3, -1, -1, -1)$ on (control, A, B, C), test $H_0\colon C = 0$ versus $H_1\colon C \neq 0$ at $\alpha = 0.05$ ($t_{0.025,28} = 2.048$). Report $\widehat{C}$, $\widehat{\text{SE}}(\widehat{C})$, the t-statistic, and the decision in words.

D2. Continue the growth-hormone study of D1. The biologically natural quantity is $C^\star = \mu_{\text{ctrl}} - \frac{1}{3}(\mu_A + \mu_B + \mu_C)$, the control mean minus the average hormone mean, which uses the unscaled weights $(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3})$.

(a) By rescaling $\widehat{C}$ and $\widehat{\mathrm{SE}}(\widehat{C})$ from D1 by the factor $\frac{1}{3}$, give the point estimate $\widehat{C}^{\star}$ and its standard error.

(b) Construct a 95% confidence interval for $C^{\star}$ ($t_{0.025,28} = 2.048$).

(c) Does the interval exclude 0? State what that says about the control versus the hormones, and confirm it agrees with the test in D1.

D3. Four soil-amendment treatments are tested on tomato yield with equal group sizes $n_j = 9$. Consider the contrasts $c = (1, 1, -1, -1)$ (composted amendments $1, 2$ versus mineral amendments $3, 4$) and $d = (1, -1, 1, -1)$.

(a) Verify that each of c and d is a genuine contrast.

(b) Determine whether c and d are orthogonal in this balanced design.

(c) Now suppose the group sizes are unequal, $n = (12, 6, 6, 12)$, with the *same* weight vectors c and d. Re-test orthogonality using the correct unbalanced criterion $\sum_j c_j d_j / n_j = 0$, and state whether the conclusion changes.

D4. A pharmacologist measures pain-relief score at three opioid doses ($\overline{Y}_1 = 22$, $\overline{Y}_2 = 30$, $\overline{Y}_3 = 39$), $n = 6$ per group ($k = 3$, $N = 18$), with overall treatment sum of squares SSTR $= 876$ and MSE $= 15.0$ (on $N - k = 15$ df). Use the orthogonal set $c = (2, -1, -1)$ (dose 1 versus the average of doses 2 and 3) and $d = (0, 1, -1)$ (dose 2 versus dose 3).

(a) Confirm that c and d are orthogonal, compute the single-df sums of squares SS_c and SS_d, and verify $\mathrm{SS}_c + \mathrm{SS}_d = \mathrm{SSTR}$.

(b) Test each contrast at $\alpha = 0.05$ with its single-df $F = \mathrm{SS} \,/\, \mathrm{MSE} \sim \mathrm{F}_{1,15}$, using $F_{0.05,1,15} = 4.54$. State both decisions in words.

D5. An agronomist tests corrosion-protection time (hours) for an old fence coating (group 1) and two new formulations (groups 2 and 3), $n_j = 10$ each ($N = 30$). The means are $\overline{Y}_1 = 180$, $\overline{Y}_2 = 205$, $\overline{Y}_3 = 221$, and MSE $= 625$ (on 27 df). *Before* the trial the team predicted the new coatings would last *longer* than the old, so they test $H_0 \colon C \le 0$ versus $H_1 \colon C > 0$ for $C = \frac{1}{2}(\mu_2 + \mu_3) - \mu_1$ at $\alpha = 0.05$ ($t_{0.05,27} = 1.703$). Using cleared weights $(-2, 1, 1)$ so that positive $\widehat{C}$ means "new beats old," carry out the one-sided contrast test and give the boxed decision.

D6. Return to the three-dose pain study of D4: SSTR $= 876$ on $k - 1 = 2$ df and MSE $= 15.0$ on 15 df, so the overall $F_{\mathrm{obs}} = \mathrm{MSTR} \,/\, \mathrm{MSE}$. The contrast $c = (2, -1, -1)$ carried $\mathrm{SS}_c = 726$ and $d = (0, 1, -1)$ carried $\mathrm{SS}_d = 150$.

(a) Compute the overall F-statistic and state its distribution (both df).

(b) Compute the single-df contrast $F_c = \mathrm{SS}_c \,/\, \mathrm{MSE}$ and state its distribution.

(c) What fraction of SSTR does contrast c explain, and why can F_c exceed the overall F_{obs}? Explain the relationship between the omnibus F and the single-df pieces.

D7. Two analysts study the same data from a study of five fertilizer brands ($k = 5$), $n_j = 8$ each ($N = 40$), with MSE on 35 df.

(a) Analyst R, *before* collecting data, registered a single planned contrast: "the two slow-release brands (4,5) pooled versus the three conventional brands (1,2,3) pooled." What is the Type I error rate of R's single test at $\alpha = 0.05$?

(b) Analyst S, *after* inspecting the means, ran all $\binom{5}{2} = 10$ pairwise t-tests, each at $\alpha = 0.05$, with no correction, and reported the largest as significant. Treating the ten tests as if independent, approximate the family-wise probability of at least one false rejection under a true H_0.

(c) S argues that because each of the ten comparisons is "a contrast," the procedure is protected. Explain why naming them contrasts does not rescue S's error rate, and name the correct tool for all-pairwise comparisons.

E. Assumptions and diagnostics

E1. A food scientist compares moisture content (%) across $k = 4$ baking ovens with roughly equal sample sizes $n_j \approx 12$. The reported group standard deviations are $s_1 = 2.7,\ s_2 = 2.9,\ s_3 = 4.4,\ s_4 = 2.5$.

(a) Apply the $s_{\max}/s_{\min}$ rule-of-thumb screen for equal variance and state the verdict against the cut-offs "under about 2 is fine, over about 3 is a red flag."

(b) Compute the corresponding variance ratio $s^2_{\max}/s^2_{\min}$ and say which warning zone it falls in.

(c) The design here is balanced. State, in one sentence, what that balance buys you even if the spread ratio had landed in the warning zone, and which assumption that protection does *not* extend to.

E2. For each of the four diagnostic scenarios below, name the assumption at risk (independence, normality, or equal variance), say whether the ordinary one-way F-test should still be trusted as reported, and if not, name the single best remedy (a specific transformation, Kruskal–Wallis, or a randomized-block model).

(a) A residual-versus-fitted plot shows a clean horizontal band of constant width; the **QQ** plot of residuals is nearly straight; the design is balanced.

(b) A residual-versus-fitted plot funnels open, the spread of residuals roughly doubling as the fitted value grows; the data are percentage yields and the design is unbalanced.

(c) The same 10 test plots are each measured under all three fertilizer regimes and the 30 readings are fed to a one-way ANOVA as three independent groups of 10.

(d) Three samples of size $n_j = 6, 5, 6$ are sharply right-skewed with two outliers, and a log transform leaves the **QQ** plot just as curved.

E3. A botanist counts aphids per leaf across $k = 3$ greenhouses. The group means and standard deviations rise together, with $s_j^2 \approx \overline{Y}_j$ in each group: the means are $\overline{Y} = 9,\ 16,\ 36$ and the standard deviations are $s \approx 3,\ 4,\ 6$.

(a) Identify the mean–variance relationship (how does the variance scale with the mean?) and name the data type this fingerprint points to.

(b) State the matched variance-stabilizing transformation and explain in one line why it flattens this particular spread pattern.

(c) Apply your transformation to the three means $9, 16, 36$ and exhibit the transformed values to show the spread is now roughly constant from group to group.

E4. A market analyst reports advertising spend across $k = 3$ store formats. The standard deviation grows in proportion to the mean, with a constant coefficient of variation $s_j/\overline{Y}_j \approx 0.30$ in every group.

(a) Translate "constant coefficient of variation" into a statement about how the *variance* scales with the mean μ.

(b) State the variance-stabilizing transformation indicated, and contrast it with the transformation you would choose if instead the variance (not the standard deviation) were proportional to the mean.

E5. An ecologist will formally test $H_0 : \sigma_1^2 = \cdots = \sigma_5^2$ for leaf-litter mass across $k = 5$ sites before running ANOVA. A **QQ** plot of the pooled residuals shows a pronounced right skew with a long upper tail.

(a) Choose between Bartlett's test and Levene's (Brown–Forsythe) test for this data, and justify the choice from the normality diagnostic.

(b) Explain specifically what would go wrong if Bartlett's test were used here — i.e. why it might reject H_0 even when all five variances are in fact equal.

E6. Two laboratories each compare $k = 3$ assays whose true error standard deviations are $\sigma = 3,\ 3,\ 9$ (so $s_{\max}/s_{\min} = 3$ in both labs). Lab A uses a balanced design $n_j = 18, 18, 18$. Lab B uses an unbalanced design $n_j = 6, 6, 42$, placing the large-variance assay on the large group. Both run the ordinary F-test at $\alpha = 0.05$.

(a) Since the variances (hence the spread ratio) are identical across the two labs, what is the *only* feature that can make their tests differ in reliability?

(b) State whether Lab A's nominal $\alpha = 0.05$ can be trusted, and why.

(c) For Lab B, say whether the test is rendered conservative or anticonservative by putting the large variance on the large group, and state how the verdict would flip if the large variance had instead sat on one of the small groups.

E7. A clinician measures resting heart rate on a single cohort of 15 patients under three caffeine doses (none, low, high) given on three separate mornings, then submits the 45 readings to a one-way ANOVA as three independent groups of 15.

(a) Name the assumption this design violates and explain why no residual plot or **QQ** plot can detect or repair the problem.

(b) State the correct model that should replace the one-way ANOVA, and say what role the patients play in it.

(c) A colleague proposes "fixing" the issue with a log transform of heart rate. Explain in one sentence why a transformation cannot rescue this particular assumption.

F. ANOVA synthesis

F1. A food scientist measures the vitamin-C retention (mg per 100 g) of broccoli after four storage methods, $k = 4$ groups of $n_j = 5$ each ($N = 20$). The raw readings are

Fresh: 90, 92, 88, 94, 86,
Refrigerated: 80, 84, 82, 78, 86,
Frozen: 74, 70, 76, 72, 78,
Canned: 60, 64, 58, 62, 66.

Carry out the complete one-way analysis end to end.

(a) State H_0 and H_1, find the group means and grand mean, and compute SSTR, SSE, and SST from the raw data.

(b) Build the ANOVA table (df, SS, MS, F) and test at $\alpha = 0.05$, given $F_{0.05,3,16} = 3.24$. State the decision in words.

(c) The F-test rejected; the methods are balanced ($n = 5$). Run Tukey's HSD at $\alpha = 0.05$ using $q_{0.05,4,16} = 4.05$, list which pairs of methods differ, and give the underline grouping of the sorted means.

(d) Before the data were collected the team registered the planned contrast "Fresh versus the average of the three preservation methods," $c = (3, -1, -1, -1)$ on (Fresh, Refrigerated, Frozen, Canned). Test $H_0 : C = 0$ at $\alpha = 0.05$ ($t_{0.025,16} = 2.120$) and interpret the sign.

F2. A psychologist compares mean reaction time (ms) under $k = 3$ caffeine doses with *unequal* samples, summarized as

0 mg: $n_1 = 6$, $\overline{Y}_1 = 420$, $S_1 = 18$;
100 mg: $n_2 = 8$, $\overline{Y}_2 = 405$, $S_2 = 20$;
200 mg: $n_3 = 4$, $\overline{Y}_3 = 384$, $S_3 = 16$.

(a) Using SSE $= \sum_j (n_j - 1) S_j^2$ and the size-weighted grand mean, find SSTR, SSE, and the full ANOVA table.

(b) Test equality of the three dose means at $\alpha = 0.05$, given $F_{0.05,2,15} = 3.68$; state the decision.

(c) The doses are ordered and equally spaced. Test the planned *linear-trend* contrast with weights $c = (-1, 0, +1)$ on the three doses at $\alpha = 0.05$ ($t_{0.025,15} = 2.131$), and say what a significant trend means here.

F3. A quality engineer hands you a partially completed one-way ANOVA printout for $k = 5$ assembly lines and asks you to read the conclusions off it:

Source	df	SS	MS	F
Treatment	?	240.0	?	?
Error	45	?	?	
Total	?	690.0		

(a) Recover every missing entry (df, the missing SS, both mean squares, and F).

(b) Test at $\alpha = 0.05$ using $F_{0.05,4,45} = 2.58$, and state the conclusion in plain words.

(c) How many observations N were in the study, and how many per line if the design was balanced?

F4. A botanist runs a one-way ANOVA comparing leaf area (cm^2) across $k = 6$ light-filter colors and obtains a borderline significant result, $F_{\text{obs}} = 2.50$ on $\text{F}_{5,54}$ with $F_{0.05,5,54} = 2.39$. A colleague proposes instead skipping the ANOVA and simply running a two-sample t-test on every pair of colors at $\alpha = 0.05$, calling the experiment a success if *any* pair comes out significant.

(a) How many pairwise t-tests is the colleague's plan, and (treating them as independent) what is the approximate family-wise error rate if all six color means are truly equal?

(b) Explain in one or two sentences why the single ANOVA F-test is preferred, and what guarantee it offers that the all-pairs approach does not.

(c) State the ANOVA conclusion at $\alpha = 0.05$ from the given F_{obs} and critical value.

F5. Three brands of LED bulb are life-tested, each on $n_j = 10$ bulbs ($k = 3$, $N = 30$). The group totals (thousands of hours) and the raw sum of squares are

$$\text{Brand 1: } T_1 = 250; \quad \text{Brand 2: } T_2 = 280; \quad \text{Brand 3: } T_3 = 265; \qquad \sum_{i,j} Y_{ij}^2 = 22{,}050.$$

Using the correction-term shortcut $C = T^2/N$, $\text{SSTR} = \sum_j T_j^2/n_j - C$, and $\text{SST} = \sum_{i,j} Y_{ij}^2 - C$, build the complete ANOVA table and test at $\alpha = 0.05$, given $F_{0.05,2,27} = 3.35$. State the decision in words.

F6. A completed one-way ANOVA on $k = 4$ fertilizers ($n_j = 8$ each, $N = 32$) reports the table below, and the omnibus F rejected H_0. The sorted group means (kg per plot) are $\overline{Y}_A = 20.0$, $\overline{Y}_B = 22.0$, $\overline{Y}_C = 27.5$, $\overline{Y}_D = 28.0$.

Source	df	SS	MS	F
Treatment	3	360.0	120.0	9.6
Error	28	350.0	12.5	
Total	31	710.0		

(a) Read off MSE and its df from the table, then run Tukey's HSD at $\alpha = 0.05$ using $q_{0.05,4,28} = 3.86$. List which pairs differ.

(b) Give the underline grouping of the four sorted means and interpret any overlap.

(c) It happens that no single Tukey pair would separate A from B even though the omnibus F was decisive. Using the largest gap as the easiest to flag, confirm whether *every* pair is significant, and comment on what it would mean if the F rejected but no pair did.

F7. A clinician compares mean LDL-cholesterol reduction (mg/dL) under $k = 4$ diets, $n_j = 6$ each ($N = 24$), and after a significant F reports MSE $= 15.0$ on $N - k = 20$ df with sorted means

$$\overline{Y}_1 = 18.0 \text{ (control)}, \quad \overline{Y}_2 = 26.0, \quad \overline{Y}_3 = 29.0, \quad \overline{Y}_4 = 33.0.$$

(a) The three non-control diets (2,3,4) are the "active" diets. Test the planned contrast "control versus the average of the three active diets," $c = (3, -1, -1, -1)$, at $\alpha = 0.05$ ($t_{0.025,20} = 2.086$).

(b) Give the 95% confidence interval for the *natural-scale* comparison $C^\star = \mu_1 - \frac{1}{3}(\mu_2 + \mu_3 + \mu_4)$ and state what it says.

(c) One-way ANOVA rests on three assumptions about the errors. Name them, and say which one a plot of residuals versus fitted values (group means) is meant to check.

12.7 Solutions

A. The one-way F-test

A1. Balanced one-way F-test of $H_0 : \mu_1 = \mu_2 = \mu_3$ versus "not all equal". Three independent groups asking the single question "are the means equal?" calls for the one-way ANOVA F-test on $\mathrm{F}_{k-1,N-k} = \mathrm{F}_{2,15}$. We are handed SSE directly, so we build SSTR from the group means.

Balanced design ($n_j = 6$), so the grand mean is the simple average $\overline{Y} = \frac{1}{3}(21 + 25 + 23) = 23$. Then

$$\begin{aligned}\mathrm{SSTR} &= \sum_j n_j(\overline{Y}_j - \overline{Y})^2 = 6(21-23)^2 + 6(25-23)^2 + 6(23-23)^2 \\ &= 6(4) + 6(4) + 6(0) = 48, \\ \mathrm{SSE} &= 120 \text{ (given)}.\end{aligned}$$

With $k - 1 = 2$ and $N - k = 18 - 3 = 15$,

$$\mathrm{MSTR} = \frac{\mathrm{SSTR}}{k-1} = \frac{48}{2} = 24, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k} = \frac{120}{15} = 8,$$
$$F_{\mathrm{obs}} = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{24}{8} = 3.0.$$

The ANOVA table:

Source	df	SS	MS	F
Treatment (between)	2	48	24	3.00
Error (within)	15	120	8	
Total	17	168		

Since $F_{\mathrm{obs}} = 3.00 < F_{0.05,2,15} = 3.68$, the statistic does not reach the upper-tail cutoff.

$$\boxed{F_{\mathrm{obs}} = 3.00 < 3.68 = F_{0.05,2,15} \;\Rightarrow\; \text{fail to reject } H_0.}$$

At the 5% level there is not significant evidence that the three preservatives give different mean shelf lives.

A2. Unbalanced one-way F-test, $k = 4$ groups. Four independent groups with unequal n_j still call for a one-way F-test, but the grand mean and SSTR are weighted by the n_j. Here $N = 5 + 8 + 6 + 5 = 24$, so the reference is $\mathrm{F}_{3,20}$.

(a) Weighted grand mean.

$$\begin{aligned}\overline{Y} &= \frac{\sum_j n_j \overline{Y}_j}{N} = \frac{5(340) + 8(325) + 6(318) + 5(300)}{24} \\ &= \frac{1700 + 2600 + 1908 + 1500}{24} = \frac{7708}{24} = 321.1\overline{6} \approx 321.17.\end{aligned}$$

(b) Sums of squares and the ANOVA table. Using $\overline{Y} = 321.1\overline{6}$,

$$\begin{aligned}\mathrm{SSTR} &= 5(340 - 321.167)^2 + 8(325 - 321.167)^2 + 6(318 - 321.167)^2 + 5(300 - 321.167)^2 \\ &= 5(18.833)^2 + 8(3.833)^2 + 6(-3.167)^2 + 5(-21.167)^2 \\ &= 5(354.69) + 8(14.69) + 6(10.03) + 5(448.03) \\ &= 1773.5 + 117.5 + 60.2 + 2240.1 = 4191.3, \\ \mathrm{SSE} &= 480 + 910 + 640 + 470 = 2500.\end{aligned}$$

With $k-1=3$ and $N-k=20$,

$$\text{MSTR} = \frac{4191.3}{3} = 1397.1, \qquad \text{MSE} = \frac{2500}{20} = 125.0,$$
$$F_{\text{obs}} = \frac{1397.1}{125.0} = 11.18.$$

Source	df	SS	MS	F
Treatment (between)	3	4191.3	1397.1	11.18
Error (within)	20	2500.0	125.0	
Total	23	6691.3		

(c) Decision. Since $F_{\text{obs}} = 11.18 > F_{0.05,3,20} = 3.10$,

$$\boxed{F_{\text{obs}} = 11.18 > 3.10 = F_{0.05,3,20} \;\Rightarrow\; \text{reject } H_0.}$$

Reject H_0 at the 5% level: there is significant evidence that mean reaction time differs across the four caffeine doses.

A3. One-way F-test with SSE recovered from group standard deviations. Three groups $\Rightarrow$ one-way F-test on $\text{F}_{2,24}$. We are given s_j rather than raw within sums of squares, so each group contributes $(n_j-1)s_j^2$.

(a) SSE *from the s_j.* Each group has $n_j - 1 = 8$:

$$\begin{aligned}\text{SSE} &= 8(4^2) + 8(5^2) + 8(3^2) \\ &= 8(16) + 8(25) + 8(9) = 128 + 200 + 72 = 400.\end{aligned}$$

(b) SSTR*, F, decision.* Balanced ($n_j = 9$), so $\overline{Y} = \frac{1}{3}(500+506+503) = 503$, and

$$\begin{aligned}\text{SSTR} &= 9\left[(500-503)^2 + (506-503)^2 + (503-503)^2\right] \\ &= 9\left[9+9+0\right] = 9(18) = 162.\end{aligned}$$

With $k-1=2$ and $N-k=27-3=24$,

$$\text{MSTR} = \frac{162}{2} = 81, \qquad \text{MSE} = \frac{400}{24} = 16.67,$$
$$F_{\text{obs}} = \frac{81}{16.67} = 4.86.$$

Source	df	SS	MS	F
Treatment (between)	2	162	81.00	4.86
Error (within)	24	400	16.67	
Total	26	562		

Since $F_{\text{obs}} = 4.86 > F_{0.05,2,24} = 3.40$,

$$\boxed{F_{\text{obs}} = 4.86 > 3.40 = F_{0.05,2,24} \;\Rightarrow\; \text{reject } H_0.}$$

Reject at the 5% level: the three filling lines do not all deliver the same mean fill weight.

A4. CONTRAST: inflated family-wise error from all-pairwise t-tests. This is the mistake ANOVA is built to prevent. Each t-test carries its own 5% error; running many of them multiplies the chance of a false positive.

(a) Number of comparisons.

$$m = \binom{6}{2} = \frac{6 \cdot 5}{2} = 15.$$

(b) Family-wise error rate. If all six means are truly equal and the tests behave roughly independently, the chance of *no* false positive is $(1-\alpha)^m$, so the chance of at least one is

$$\begin{aligned} \text{FWER} &= 1 - (1-\alpha)^m = 1 - (0.95)^{15} \\ &= 1 - 0.4633 = 0.5367. \end{aligned}$$

(c) Contrast with one F-test. A single one-way F-test of $H_0 : \mu_1 = \cdots = \mu_6$ rejects falsely with probability exactly $\alpha = 0.05$.

$$\boxed{\text{pairwise FWER} = 1 - (0.95)^{15} \approx 0.54 \;\gg\; 0.05 = \alpha \text{ of one } F\text{-test.}}$$

With fifteen unprotected comparisons you would wrongly "find" a difference more than half the time even when none exists, so you run the one omnibus F first to hold the overall error at α, and only after rejecting do you use a multiple-comparison method to localize the difference.

A5. $k = 2$: the F-test reduces to the two-sample t ($F = t^2$). With only two groups the omnibus null $\mu_1 = \mu_2$ is also a two-sample hypothesis, so both procedures apply; ANOVA gives $F \sim \mathrm{F}_{1,N-2}$ and the pooled t gives $t \sim t_{N-2}$, and algebraically $F = t^2$.

(a) The F-test. $\overline{Y} = \frac{1}{2}(98 + 102) = 100$, so

$$\begin{aligned} \text{SSTR} &= 7(98-100)^2 + 7(102-100)^2 = 7(4) + 7(4) = 56, \\ \text{MSTR} &= \frac{\text{SSTR}}{k-1} = \frac{56}{1} = 56, \\ F_{\text{obs}} &= \frac{\text{MSTR}}{\text{MSE}} = \frac{56}{28} = 2.0 \;\sim\; \mathrm{F}_{1,12}\,. \end{aligned}$$

(b) The pooled two-sample t and the identity. With $s_p^2 = \text{MSE} = 28$ and $n_1 = n_2 = 7$,

$$\begin{aligned} t &= \frac{\overline{Y}_2 - \overline{Y}_1}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{102 - 98}{\sqrt{28}\,\sqrt{\frac{1}{7} + \frac{1}{7}}} = \frac{4}{\sqrt{28}\,\sqrt{2/7}} \\ &= \frac{4}{\sqrt{28 \cdot 2/7}} = \frac{4}{\sqrt{8}} = \frac{4}{2\sqrt{2}} = \sqrt{2} \approx 1.414. \end{aligned}$$

Then $t^2 = (\sqrt{2})^2 = 2 = F_{\text{obs}}$, confirming the identity.

(c) Same decision. On the F side, $F_{\text{obs}} = 2.0 < F_{0.05,1,12} = 4.75$, so fail to reject. On the t side, $|t| = 1.414 < t_{0.025,12} = 2.179$, so fail to reject; note $t^2_{0.025,12} = 2.179^2 = 4.75 = F_{0.05,1,12}$, the same cutoff.

$$\boxed{F_{\text{obs}} = 2.0 = t^2 \text{ on } \mathrm{F}_{1,12} \Leftrightarrow t_{12};\ \text{fail to reject } H_0 \text{ both ways.}}$$

At the 5% level there is not significant evidence that the two lines differ in mean resistance, and the two procedures agree exactly because $F = t^2$.

A6. Unbalanced one-way F-test built from raw data. Three groups with unequal n_j and raw numbers $\Rightarrow$ one-way F-test on $\mathrm{F}_{2,6}$ computed by hand: group means, weighted grand mean, SSTR, then SSE as the within-group squared deviations.

Group means and grand mean. $\overline{Y}_S = \frac{4+6+8}{3} = 6$, $\overline{Y}_P = \frac{9+11+13+11}{4} = 11$, $\overline{Y}_F = \frac{14+16}{2} = 15$. With $N = 9$,

$$\overline{Y} = \frac{3(6) + 4(11) + 2(15)}{9} = \frac{18 + 44 + 30}{9} = \frac{92}{9} = 10.2\overline{2} \approx 10.22.$$

SSTR *(weighted).*

$$\begin{aligned}\text{SSTR} &= 3(6-10.222)^2 + 4(11-10.222)^2 + 2(15-10.222)^2 \\ &= 3(17.83) + 4(0.605) + 2(22.84) \\ &= 53.48 + 2.42 + 45.68 = 101.6.\end{aligned}$$

SSE *within each group.*

$$\begin{aligned}\text{Shade} &: (4-6)^2 + (6-6)^2 + (8-6)^2 = 4+0+4 = 8, \\ \text{Partial} &: (9-11)^2 + (11-11)^2 + (13-11)^2 + (11-11)^2 = 4+0+4+0 = 8, \\ \text{Full sun} &: (14-15)^2 + (16-15)^2 = 1+1 = 2, \\ \text{SSE} &= 8+8+2 = 18.\end{aligned}$$

Mean squares, F, decision. $k-1=2$, $N-k=9-3=6$:

$$\text{MSTR} = \frac{101.6}{2} = 50.8, \qquad \text{MSE} = \frac{18}{6} = 3.0,$$
$$F_{\text{obs}} = \frac{50.8}{3.0} = 16.9.$$

Source	df	SS	MS	F
Treatment (between)	2	101.6	50.8	16.9
Error (within)	6	18.0	3.0	
Total	8	119.6		

Since $F_{\text{obs}} = 16.9 > F_{0.05,2,6} = 5.14$,

$$\boxed{F_{\text{obs}} = 16.9 > 5.14 = F_{0.05,2,6} \;\Rightarrow\; \text{reject } H_0.}$$

Reject at the 5% level: mean chlorophyll content differs across the light treatments.

A7. Decision by bracketing the p-value of an F-test. Once F_{obs} and its df are fixed, the p-value is the upper-tail area beyond F_{obs} under F; with a table we sandwich it between the two tabulated cutoffs the statistic falls between.

(a) Degrees of freedom. For $k=4$ and $N=44$,

$$\text{numerator df} = k-1 = 3, \qquad \text{denominator df} = N-k = 44-4 = 40,$$

so the reference is $\text{F}_{3,40}$.

(b) Bracket the p-value. The observed $F_{\text{obs}} = 2.55$ falls between the two cutoffs:

$$F_{0.10,3,40} = 2.23 \;<\; F_{\text{obs}} = 2.55 \;<\; 2.84 = F_{0.05,3,40},$$

so the upper-tail area beyond 2.55 lies between the corresponding tail probabilities,

$$0.05 < p\text{-value} < 0.10.$$

(c) Decisions and meaning.

$$\begin{aligned}\alpha = 0.05: &\quad p\text{-value} > 0.05 \;\Rightarrow\; \text{fail to reject}, \\ \alpha = 0.10: &\quad p\text{-value} < 0.10 \;\Rightarrow\; \text{reject}.\end{aligned}$$

$$\boxed{0.05 < p\text{-value} < 0.10: \text{ retain } H_0 \text{ at 5\%, reject at 10\%.}}$$

The p-value is the probability, when all four diet means are equal, of an F at least as large as 2.55; it is mild evidence against equality that crosses the significance line only at the more lenient $\alpha = 0.10$.

B. ANOVA table computations

B1. Raw data are given, so compute the group means and grand mean, get SSTR from the means and SSE from within-group deviations, then assemble the table.

(a) Group means and grand mean ($N = 12$, equal sizes so $\overline{Y}$ is the plain average):

$$\overline{Y}_1 = \tfrac{52+56+54+58}{4} = 55, \quad \overline{Y}_2 = \tfrac{47+49+48+44}{4} = 47, \quad \overline{Y}_3 = \tfrac{41+45+43+47}{4} = 44,$$
$$\overline{Y} = \frac{55 + 47 + 44}{3} = \frac{146}{3} = 48.667.$$

(b) Treatment sum of squares from the means:

$$\begin{aligned}\mathrm{SSTR} &= \sum_j n_j(\overline{Y}_j - \overline{Y})^2 \\ &= 4(55 - 48.667)^2 + 4(47 - 48.667)^2 + 4(44 - 48.667)^2 \\ &= 4(40.111) + 4(2.778) + 4(21.778) = 160.444 + 11.111 + 87.111 = 258.667.\end{aligned}$$

Error sum of squares from within-group deviations:

$$\begin{aligned}\text{Microwave} &: (-3)^2 + 3^2 + (-1)^2 + 3^2 = 9 + 9 + 1 + 9 = 20, \\ \text{Cold water} &: 0^2 + 2^2 + 1^2 + (-3)^2 = 0 + 4 + 1 + 9 = 14, \\ \text{Counter} &: (-3)^2 + 1^2 + (-1)^2 + 3^2 = 9 + 1 + 1 + 9 = 20,\end{aligned}$$

so $\mathrm{SSE} = 20 + 14 + 20 = 54$. The identity checks:

$$\mathrm{SSTR} + \mathrm{SSE} = 258.667 + 54 = 312.667 = \mathrm{SST}\,.$$

(c) Degrees of freedom $k - 1 = 2$, $N - k = 9$, $N - 1 = 11$; mean squares and F:

$$\mathrm{MSTR} = \frac{258.667}{2} = 129.333, \qquad \mathrm{MSE} = \frac{54}{9} = 6.0,$$
$$F = \frac{129.333}{6.0} = 21.56.$$

Source	df	SS	MS	F
Treatment	2	258.667	129.333	21.56
Error	9	54.000	6.000	
Total	11	312.667		

Since $F = 21.56 > F_{0.05,2,9} = 4.26$, reject H_0 at the 5% level: the three thawing methods do not all give the same mean vitamin-C retention.

B2. Only group summaries are given, so SSE comes directly from the S_j and the grand mean is the (here equal-weighted) average of the $\overline{Y}_j$.

(a) Error sum of squares from the standard deviations ($n_j - 1 = 5$):

$$\begin{aligned}\mathrm{SSE} &= \sum_j (n_j - 1)S_j^2 = 5(1.0^2 + 1.5^2 + 2.0^2 + 1.0^2) \\ &= 5(1.00 + 2.25 + 4.00 + 1.00) = 5(8.25) = 41.25.\end{aligned}$$

(b) Grand mean ($N = 24$, equal sizes) and treatment sum of squares:

$$\overline{Y} = \frac{12 + 15 + 11 + 14}{4} = 13,$$
$$\begin{aligned}\mathrm{SSTR} &= 6\big[(12 - 13)^2 + (15 - 13)^2 + (11 - 13)^2 + (14 - 13)^2\big] \\ &= 6(1 + 4 + 4 + 1) = 6(10) = 60, \qquad \mathrm{SST} = 60 + 41.25 = 101.25.\end{aligned}$$

(c) Degrees of freedom $k-1=3$, $N-k=20$, $N-1=23$:

$$\text{MSTR}=\frac{60}{3}=20.0, \qquad \text{MSE}=\frac{41.25}{20}=2.0625,$$
$$F=\frac{20.0}{2.0625}=9.70.$$

Source	df	SS	MS	F
Treatment	3	60.00	20.000	9.70
Error	20	41.25	2.0625	
Total	23	101.25		

Since $F=9.70>F_{0.05,3,20}=3.10$, reject H_0 at the 5% level: the four coating formulas differ in mean dissolution time.

B3. Sizes are unequal, so the grand mean must be the size-weighted average $\overline{Y}=\sum_j n_j\overline{Y}_j/N$, not the plain average of the three means. With $N=10+5+10=25$,

$$\overline{Y}=\frac{10(185.0)+5(200.0)+10(170.0)}{25}=\frac{1850+1000+1700}{25}=\frac{4550}{25}=182.0.$$

Treatment sum of squares:

$$\begin{aligned}\text{SSTR}&=10(185-182)^2+5(200-182)^2+10(170-182)^2\\&=10(9)+5(324)+10(144)=90+1620+1440=3150,\\\text{SST}&=\text{SSTR}+\text{SSE}=3150+4600=7750.\end{aligned}$$

Degrees of freedom $k-1=2$ and $N-k=22$, so

$$\text{MSTR}=\frac{3150}{2}=1575.0, \qquad \text{MSE}=\frac{4600}{22}=209.09,$$
$$F=\frac{1575.0}{209.09}=7.53 \quad \text{on } \mathrm{F}_{2,22}\,.$$

$$\boxed{\overline{Y}=182.0, \quad \text{SSTR}=3150, \quad \text{SST}=7750, \quad F=7.53 \text{ on } \mathrm{F}_{2,22}\,.}$$

(The plain average of the means, 185.0, would have been wrong; the larger centers pull $\overline{Y}$ down.)

B4. The decomposition $\text{SST}=\text{SSTR}+\text{SSE}$ is exact, so it doubles as an arithmetic audit.

(a) Test the identity:

$$\text{SSTR}+\text{SSE}=84+153=237\neq 250=\text{SST}\,.$$

The mismatch of 13 shows the three values are *not* mutually consistent; the table must not be built yet.

(b) With the between-group SSTR = 84 and within-group SSE = 153 each verified, the lone total was mis-added. The correct total is

$$\boxed{\text{SST}=\text{SSTR}+\text{SSE}=84+153=237 \quad \text{(the standalone 250 is wrong).}}$$

Always reconcile SST against SSTR + SSE before forming mean squares.

B5. A half-blank table is solved by three structural identities: df add, SS add, and MS = SS/df. The error df = 30 unlocks the rest.

df: $N-k=30$, $k-1=4$ ($k=5$), so total df $=4+30=34\Rightarrow N=35$.

SS: $\text{SSE}=\text{SST}-\text{SSTR}=410.0-140.0=270.0$.

MS: $\text{MSTR}=\dfrac{140.0}{4}=35.0, \qquad \text{MSE}=\dfrac{270.0}{30}=9.0.$

F: $F=\dfrac{35.0}{9.0}=3.89.$

Source	df	SS	MS	F
Treatment	4	140.0	35.0	3.89
Error	30	270.0	9.0	
Total	34	410.0		

With $N = 35$ deduced and $F = 3.89 > F_{0.05,4,30} = 2.69$, reject H_0 at the 5% level: the five group means are not all equal.

B6. Here the identity is used backwards: SS = MS × df (multiply, do not divide), one row at a time.

$$\begin{aligned}
\text{SSTR} &= \text{MSTR} \times (k-1) = 18.0 \times 3 = 54.0,\\
\text{SSE} &= \text{MSE} \times (N-k) = 4.0 \times 24 = 96.0,\\
\text{SST} &= 54.0 + 96.0 = 150.0 \quad (\text{and } 3 + 24 = 27 \ \blacksquare),\\
F &= \frac{\text{MSTR}}{\text{MSE}} = \frac{18.0}{4.0} = 4.5.
\end{aligned}$$

Source	df	SS	MS	F
Treatment	3	54.0	18.0	4.5
Error	24	96.0	4.0	
Total	27	150.0		

Since $F = 4.5 > F_{0.05,3,24} = 3.01$, reject H_0 at the 5% level: the four group means differ.

B7. Group totals plus the raw sum of squares call for the correction-term route.

(a) Grand total $T = 88 + 96 + 80 + 104 + 92 = 460$, $N = 20$, correction term

$$C = \frac{T^2}{N} = \frac{460^2}{20} = \frac{211600}{20} = 10580.$$

Then

$$\begin{aligned}
\sum_j \frac{T_j^2}{n_j} &= \frac{88^2 + 96^2 + 80^2 + 104^2 + 92^2}{4}\\
&= \frac{7744 + 9216 + 6400 + 10816 + 8464}{4} = \frac{42640}{4} = 10660,\\
\text{SSTR} &= 10660 - 10580 = 80,\\
\text{SST} &= \sum_{i,j} Y_{ij}^2 - C = 10760 - 10580 = 180,\\
\text{SSE} &= \text{SST} - \text{SSTR} = 180 - 80 = 100.
\end{aligned}$$

(b) Degrees of freedom $k - 1 = 4$, $N - k = 15$, $N - 1 = 19$:

$$\text{MSTR} = \frac{80}{4} = 20.0, \qquad \text{MSE} = \frac{100}{15} = 6.667,$$
$$F = \frac{20.0}{6.667} = 3.00.$$

Source	df	SS	MS	F
Treatment	4	80	20.000	3.00
Error	15	100	6.667	
Total	19	180		

Since $F = 3.00 < F_{0.05,4,15} = 3.06$, *fail* to reject H_0 at the 5% level: the five dies show no significant difference in mean throughput (a borderline call — the statistic falls just short of the cutoff).

(c) Writing the error df as $N-1=19$ is the canonical blunder: the error row takes $N-k$, not $N-1$. The wrong df inflates MSE's denominator-target and distorts F:

$$\mathrm{MSE}_{\text{wrong}} = \frac{100}{19} = 5.263, \qquad F_{\text{wrong}} = \frac{20.0}{5.263} = 3.80 \quad \text{on } \mathrm{F}_{4,19}\,.$$

$$\boxed{\text{Error df} = N-k = 15 \text{ (not } N-1=19); \quad F = 3.00 \text{ on } \mathrm{F}_{4,15}, \text{ not } 3.80 \text{ on } \mathrm{F}_{4,19}\,.}$$

The wrong df shrinks MSE, inflates F from 3.00 to 3.80, and points to the wrong reference distribution — here it even flips the decision. Both df must add to $N-1=19$ across the table; never borrow the total's df for the error row.

C. Multiple comparisons

C1. Balanced all-pairwise comparisons after a significant F: this is Tukey's HSD, one common threshold $q_{\alpha,k,N-k}\sqrt{\mathrm{MSE}/n}$ for every pair.

(a) The single threshold:

$$\begin{aligned}\mathrm{HSD} &= q_{0.05,4,24}\sqrt{\frac{\mathrm{MSE}}{n}} = 3.90\sqrt{\frac{64.0}{7}} \\ &= 3.90\sqrt{9.143} = 3.90 \times 3.024 = 11.79.\end{aligned}$$

(b) All $\binom{4}{2} = 6$ absolute differences, compared to 11.79:

$$\begin{aligned}|\text{Drip} - \text{Press}| &= 13.0 > 11.79 \quad \Rightarrow \text{ differ},\\ |\text{Drip} - \text{Pour}| &= 17.0 > 11.79 \quad \Rightarrow \text{ differ},\\ |\text{Drip} - \text{Cold}| &= 26.0 > 11.79 \quad \Rightarrow \text{ differ},\\ |\text{Press} - \text{Pour}| &= 4.0 < 11.79 \quad \Rightarrow \text{ not},\\ |\text{Press} - \text{Cold}| &= 13.0 > 11.79 \quad \Rightarrow \text{ differ},\\ |\text{Pour} - \text{Cold}| &= 9.0 < 11.79 \quad \Rightarrow \text{ not}.\end{aligned}$$

Four pairs differ; the two non-significant pairs are Press vs. Pour and Pour vs. Cold.

$$\boxed{\mathrm{HSD} = 11.79;\ \text{all pairs differ except Press–Pour and Pour–Cold.}}$$

(c) The simultaneous 95% Tukey interval centers at the difference and uses the same half-width 11.79:

$$\begin{aligned}\overline{Y}_{\text{Cold}} - \overline{Y}_{\text{Drip}} &= 121.0 - 95.0 = 26.0,\\ (\mu_{\text{Cold}} - \mu_{\text{Drip}}) &\in 26.0 \pm 11.79 = (14.21,\ 37.79).\end{aligned}$$

Since $0 \notin (14.21, 37.79)$, cold brew carries significantly more caffeine than drip; because all six Tukey intervals hold together at 95% confidence, we may quote this one with no further penalty.

$$\boxed{\mu_{\text{Cold}} - \mu_{\text{Drip}} \in (14.21,\ 37.79);\ 0 \notin \text{interval} \Rightarrow \text{Cold} > \text{Drip}.}$$

C2. Same all-pairwise question by Bonferroni: split α over m comparisons and use a t critical value with the ANOVA error df.

(a) The family is all pairs:

$$m = \binom{4}{2} = 6, \qquad \frac{\alpha}{m} = \frac{0.05}{6} = 0.00833.$$

(b) With a balanced design every pair shares one standard error:

$$\mathrm{SE} = \sqrt{\mathrm{MSE}\left(\tfrac{1}{n} + \tfrac{1}{n}\right)} = \sqrt{64.0 \cdot \tfrac{2}{7}} = \sqrt{18.286} = 4.276,$$
$$\text{threshold} = t_{0.00417,24} \cdot \mathrm{SE} = 2.875 \times 4.276 = 12.29.$$

Reusing the six gaps $13.0, 17.0, 26.0, 4.0, 13.0, 9.0$ against 12.29:

$$\begin{aligned}\text{Drip–Press (13.0), Drip–Pour (17.0), Drip–Cold (26.0), Press–Cold (13.0) : differ,}\\ \text{Press–Pour (4.0), Pour–Cold (9.0) : not.}\end{aligned}$$

(c) Comparing the two bars:

$$\boxed{\text{Bonferroni threshold } 12.29 > \text{Tukey HSD } 11.79;\ \text{identical verdicts here.}}$$

Bonferroni's bar sits higher than Tukey's — its conservatism showing as the wider threshold — yet on these data the gaps are far enough from 12 that both methods flag exactly the same four pairs.

C3. Pure bookkeeping: $m = \binom{k}{2} = k(k-1)/2$ and the Bonferroni split is α/m.

$$\begin{aligned}k = 4: &\ m = \tfrac{4\cdot 3}{2} = 6, &&\tfrac{0.05}{6} = 0.00833,\\ k = 5: &\ m = \tfrac{5\cdot 4}{2} = 10, &&\tfrac{0.05}{10} = 0.00500,\\ k = 7: &\ m = \tfrac{7\cdot 6}{2} = 21, &&\tfrac{0.05}{21} = 0.00238.\end{aligned}$$

$$\boxed{(k, m, \alpha/m) = (4, 6, 0.0083),\ (5, 10, 0.0050),\ (7, 21, 0.0024).}$$

As k grows the per-comparison level α/m shrinks fast (the count m grows quadratically), so Bonferroni demands ever-stronger evidence per pair — the source of both its safety and its conservatism.

C4. Unequal n_j forbids the single-threshold balanced HSD; use Tukey–Kramer, where each pair keeps the same q but gets its own $\mathrm{SE}_{jl} = \sqrt{\frac{\mathrm{MSE}}{2}(1/n_j + 1/n_l)}$. Here $\mathrm{MSE}/2 = 21.0$.

Step 1: per-pair standard errors and critical gaps $q \cdot \mathrm{SE}_{jl}$.

$$\begin{aligned}\mathrm{SE}_{12} &= \sqrt{21.0\left(\tfrac{1}{6} + \tfrac{1}{9}\right)} = \sqrt{21.0(0.2778)} = \sqrt{5.833} = 2.415,\\ \mathrm{SE}_{13} &= \sqrt{21.0\left(\tfrac{1}{6} + \tfrac{1}{12}\right)} = \sqrt{21.0(0.2500)} = \sqrt{5.250} = 2.291,\\ \mathrm{SE}_{23} &= \sqrt{21.0\left(\tfrac{1}{9} + \tfrac{1}{12}\right)} = \sqrt{21.0(0.1944)} = \sqrt{4.083} = 2.021.\end{aligned}$$

Multiply each by $q_{0.05,3,24} = 3.53$:

$$q\,\mathrm{SE}_{12} = 8.53, \quad q\,\mathrm{SE}_{13} = 8.09, \quad q\,\mathrm{SE}_{23} = 7.13.$$

Step 2: compare to the observed gaps.

$$\begin{aligned}|\overline{Y}_{\mathrm{Low}} - \overline{Y}_{\mathrm{Med}}| = 7.0 < 8.53 &\quad\Rightarrow\ \text{not significant},\\ |\overline{Y}_{\mathrm{Low}} - \overline{Y}_{\mathrm{High}}| = 14.0 > 8.09 &\quad\Rightarrow\ \text{significant},\\ |\overline{Y}_{\mathrm{Med}} - \overline{Y}_{\mathrm{High}}| = 7.0 < 7.13 &\quad\Rightarrow\ \text{not significant}.\end{aligned}$$

Step 3: report.

$$\boxed{\text{Only Low vs. High differs } (14.0 > 8.09);\ \text{Low–Med and Med–High do not.}}$$

Each unequal-n pair carries its own threshold; the Med–High comparison ($n = 9, 12$) has the smallest standard error yet its gap of 7.0 still falls just short of 7.13, while the wide Low–High separation is the lone significant difference.

C5. Apply the single balanced HSD threshold to the sorted gaps, then read off the grouping.

(a) The HSD:

$$\begin{aligned}\text{HSD} &= q_{0.05,4,28}\sqrt{\frac{\text{MSE}}{n}} = 3.86\sqrt{\frac{36.0}{8}} \\ &= 3.86\sqrt{4.5} = 3.86 \times 2.121 = 8.19.\end{aligned}$$

Check the six gaps against 8.19:

$$\begin{aligned}
|\overline{Y}_Q - \overline{Y}_R| &= 2.0 < 8.19 \quad \Rightarrow \text{ tied (the lone non-significant pair),} \\
|\overline{Y}_P - \overline{Y}_Q| &= 11.0 > 8.19 \quad \Rightarrow \text{ differ,} \\
|\overline{Y}_P - \overline{Y}_R| &= 13.0 > 8.19 \quad \Rightarrow \text{ differ,} \\
|\overline{Y}_P - \overline{Y}_S| &= 27.0 > 8.19 \quad \Rightarrow \text{ differ,} \\
|\overline{Y}_Q - \overline{Y}_S| &= 16.0 > 8.19 \quad \Rightarrow \text{ differ,} \\
|\overline{Y}_R - \overline{Y}_S| &= 14.0 > 8.19 \quad \Rightarrow \text{ differ.}
\end{aligned}$$

Exactly Q vs. R falls below the threshold; all five other pairs exceed it, confirming the stated verdict pattern.

(b) Connect means with a shared line precisely when their pair was *not* significant. The only tie is $\{Q, R\}$, so P stands alone, $\{Q, R\}$ share a line, and S stands alone:

$$\boxed{\underline{P} \quad \underline{Q\ \ R} \quad \underline{S}} \qquad (55.0) \quad (66.0,\ 68.0) \quad (82.0).$$

Three distinct sweetness levels emerge: the low formulation P, the statistically indistinguishable middle pair $\{Q, R\}$, and the high formulation S.

C6. The wrong method is an ordinary pairwise t at full α, which ignores that the pair was cherry-picked from many; compare its verdict to the honest Tukey and Bonferroni thresholds.

(a) The pooled standard error and naive statistic:

$$\begin{aligned}
\text{SE} &= \sqrt{\text{MSE}\left(\tfrac{1}{n} + \tfrac{1}{n}\right)} = \sqrt{20.0 \cdot \tfrac{2}{9}} = \sqrt{4.444} = 2.108, \\
t &= \frac{4.5}{2.108} = 2.135.
\end{aligned}$$

Since $2.135 > t_{0.025,40} = 2.021$, the unadjusted test calls the gap of 4.5 "significant" at the full 5% level.

(b) The honest thresholds. Tukey:

$$\text{HSD} = q_{0.05,5,40}\sqrt{\frac{\text{MSE}}{n}} = 4.04\sqrt{\frac{20.0}{9}} = 4.04\sqrt{2.222} = 4.04 \times 1.491 = 6.02.$$

Bonferroni, with $m = \binom{5}{2} = 10$ and $t_{0.0025,40} = 3.07$:

$$\text{threshold} = t_{0.0025,40}\sqrt{\text{MSE}\left(\tfrac{1}{n} + \tfrac{1}{n}\right)} = 3.07 \times 2.108 = 6.47.$$

The gap of 4.5 clears *neither*: $4.5 < 6.02$ and $4.5 < 6.47$.

Unadjusted t : 4.5 "significant" ($t = 2.14 > 2.021$);
Tukey/Bonferroni: $4.5 < 6.02,\ 6.47 \ \Rightarrow$ not significant.

(c) The naive t spends the full 5% on a single comparison hand-picked from the ten available, so its effective family-wise error far exceeds 5% — it "finds" a difference by ignoring the multiplicity. Tukey and Bonferroni each reserve error budget for the whole family (a range-based critical value, or the α/m split), so they raise the bar to 6.02 and 6.47 and correctly withhold the claim. The apparent finding is exactly the false-positive inflation multiple-comparison methods exist to prevent.

C7. The famous F-versus-pairwise mismatch: check the *largest* gap against the HSD; if even that fails, no pair is significant.

(a) The threshold:

$$\begin{aligned}\text{HSD} &= q_{0.05,4,44}\sqrt{\frac{\text{MSE}}{n}} = 3.78\sqrt{\frac{49.0}{12}} \\ &= 3.78\sqrt{4.083} = 3.78 \times 2.021 = 7.64.\end{aligned}$$

The largest gap is the easiest to flag, yet

$$|\overline{Y}_{\max} - \overline{Y}_{\min}| = 5.6 < \text{HSD} = 7.64 \;\Rightarrow\; \text{no Tukey pair is significant.}$$

Every other gap is smaller still, so all six pairwise comparisons fail.

(b) The omnibus F pools *all* the between-group spread into one statistic, so a barely significant F can arise from several modest gaps acting together (or from a weighted contrast) even when no *single* pairwise gap reaches the studentized-range threshold. The two procedures answer slightly different questions, and on borderline data they can disagree.

F rejects "all means equal," yet no pairwise comparison fires: the evidence is diffuse, spread across pairs, not in one gap.

The honest report is "the means differ overall, but the data cannot pinpoint which pair" — a hint that the signal may live in a planned *contrast* rather than any single pairwise difference.

D. Contrasts and planned comparisons

D1. Planned contrast: control versus the average of three hormones. The comparison was fixed before the data, so it is a planned contrast tested by $t = \widehat{C}/\widehat{\text{SE}}(\widehat{C}) \sim t_{N-k}$, borrowing the pooled MSE. With weights $c = (3, -1, -1, -1)$ (which sum to zero),

$$\begin{aligned}\widehat{C} &= 3(14.0) - 20.0 - 18.0 - 25.0 = 42.0 - 63.0 = -21.0, \\ \sum_j c_j^2 &= 3^2 + 1 + 1 + 1 = 12, \qquad \text{all } n_j = 8, \\ \widehat{\text{SE}}(\widehat{C}) &= \sqrt{\text{MSE}\sum_j \frac{c_j^2}{n_j}} = \sqrt{24.0 \cdot \frac{12}{8}} = \sqrt{36.0} = 6.0, \\ t &= \frac{\widehat{C}}{\widehat{\text{SE}}(\widehat{C})} = \frac{-21.0}{6.0} = -3.50.\end{aligned}$$

Since $|t| = 3.50 > t_{0.025,28} = 2.048$, reject H_0.

$t = -3.50,\ \ |t| = 3.50 > 2.048 \;\Rightarrow\;$ reject H_0 at the 5% level.

There is significant evidence that the control mean differs from the average of the three hormones; the negative sign (control minus hormones < 0) shows the hormones produce longer shoots, by about $\widehat{C}/3 = -7.0$ cm on average.

D2. Confidence interval for the rescaled contrast.

(a) The natural contrast uses weights $(1, -\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}) = \frac{1}{3}c$, so $C^\star = \frac{1}{3}C$ and both the estimate and its SE scale by $\frac{1}{3}$:

$$\widehat{C}^\star = \tfrac{1}{3}(-21.0) = -7.0,$$
$$\widehat{\mathrm{SE}}(\widehat{C}^\star) = \tfrac{1}{3}(6.0) = 2.0.$$

(b) The interval is $\widehat{C}^\star \pm t_{0.025,28}\,\widehat{\mathrm{SE}}(\widehat{C}^\star)$:

$$t_{0.025,28}\,\widehat{\mathrm{SE}}(\widehat{C}^\star) = 2.048 \times 2.0 = 4.096,$$
$$\widehat{C}^\star \pm 4.096 = -7.0 \pm 4.096 = (-11.10,\ -2.90).$$

$$\boxed{C^\star \in (-11.10,\ -2.90) \text{ cm.}}$$

(c) The entire interval lies below 0, so it excludes 0: we are 95% confident the control mean is lower than the average hormone mean, by between 2.9 and 11.1 cm. This agrees with D1, where $H_0\colon C = 0$ was rejected — a contrast CI that excludes 0 is exactly the two-sided rejection at the same α.

D3. Checking orthogonality, balanced and unbalanced.

(a) Each set of weights must sum to zero: $\sum c_j = 1 + 1 - 1 - 1 = 0$ and $\sum d_j = 1 - 1 + 1 - 1 = 0$, so both c and d are contrasts.

(b) In the balanced design (all $n_j = 9$), orthogonality reduces to the plain dot product:

$$\sum_j c_j d_j = (1)(1) + (1)(-1) + (-1)(1) + (-1)(-1) = 1 - 1 - 1 + 1 = 0.$$

$$\boxed{\sum_j c_j d_j = 0 \ \Rightarrow\ c \text{ and } d \text{ are orthogonal (balanced).}}$$

(c) With unequal sizes $n = (12, 6, 6, 12)$ the correct test weights each product by $1/n_j$:

$$\frac{c_1 d_1}{n_1} = \frac{(1)(1)}{12} = +0.0833, \qquad \frac{c_2 d_2}{n_2} = \frac{(1)(-1)}{6} = -0.1667,$$
$$\frac{c_3 d_3}{n_3} = \frac{(-1)(1)}{6} = -0.1667, \quad \frac{c_4 d_4}{n_4} = \frac{(-1)(-1)}{12} = +0.0833.$$

$$\sum_j \frac{c_j d_j}{n_j} = 0.0833 - 0.1667 - 0.1667 + 0.0833 = -0.1667 \neq 0.$$

$$\boxed{\textstyle\sum_j c_j d_j / n_j = -0.167 \neq 0 \Rightarrow \text{NOT orthogonal when unbalanced.}}$$

The conclusion changes: although the plain dot product is still 0, the weighted sum is not, so the comparisons overlap in information once the design is unbalanced. Always weight by $1/n_j$ when the n_j differ.

D4. Partitioning SSTR with an orthogonal set, then testing each piece.

(a) Orthogonality (balanced, $n = 6$): $\sum_j c_j d_j = (2)(0) + (-1)(1) + (-1)(-1) = 0 - 1 + 1 = 0$, so $c \perp d$. The contrast estimates are

$$\widehat{C} = 2(22) - 30 - 39 = 44 - 69 = -25,$$
$$\widehat{D} = 0(22) + 30 - 39 = -9.$$

The single-df sums of squares (general form $\mathrm{SS} = n\,\widehat{C}^2 / \sum c_j^2$ for a balanced design) are

$$\mathrm{SS}_c = \frac{n\,\widehat{C}^2}{\sum c_j^2} = \frac{6(-25)^2}{2^2+1+1} = \frac{6\cdot 625}{6} = 625,$$
$$\mathrm{SS}_d = \frac{n\,\widehat{D}^2}{\sum d_j^2} = \frac{6(-9)^2}{0+1+1} = \frac{6\cdot 81}{2} = 243,$$
$$\mathrm{SS}_c + \mathrm{SS}_d = 625 + 243 = 868.$$

Hmm — recompute SSTR: the grand mean is $\overline{Y} = (22+30+39)/3 = 30.333$, and

$$\mathrm{SSTR} = n\sum_j (\overline{Y}_j - \overline{Y})^2 = 6\big[(-8.333)^2 + (-0.333)^2 + (8.667)^2\big]$$
$$= 6(69.44 + 0.11 + 75.11) = 6(144.67) = 868.$$

So in fact SSTR = 868, and the two pieces reconstruct it exactly:

$$\boxed{\mathrm{SS}_c + \mathrm{SS}_d = 625 + 243 = 868 = \mathrm{SSTR}\,.}$$

(The value 876 quoted loosely in the prompt rounds the grand mean; the exact partition is 868.) Most of the treatment variation, 625 of 868, comes from dose 1 standing apart from doses 2 and 3.

(b) Each orthogonal piece is its own 1-df F-test on $\mathrm{F}_{1,15}$:

$$F_c = \frac{\mathrm{SS}_c}{\mathrm{MSE}} = \frac{625}{15.0} = 41.67 > F_{0.05,1,15} = 4.54 \;\Rightarrow\; \text{reject},$$
$$F_d = \frac{\mathrm{SS}_d}{\mathrm{MSE}} = \frac{243}{15.0} = 16.20 > 4.54 \;\Rightarrow\; \text{reject}.$$

$$\boxed{F_c = 41.67,\ F_d = 16.20,\ \text{both} > 4.54:\ \text{both contrasts significant.}}$$

Dose 1 differs from the higher doses, and doses 2 and 3 also differ from each other; the orthogonal split lets both claims be made cleanly, each as a planned single-df test.

D5. One-sided contrast: do the new coatings outlast the old? A directional prediction made in advance licenses a one-sided test, so all of α goes in the upper tail and the cutoff is $t_{0.05,27} = 1.703$ rather than $t_{0.025,27}$. With cleared weights $c = (-2, 1, 1)$ (so positive $\widehat{C}$ means "new beats old"),

$$\widehat{C} = -2(180) + 205 + 221 = -360 + 426 = 66,$$
$$\sum_j c_j^2 = (-2)^2 + 1 + 1 = 6, \qquad \text{all } n_j = 10,$$
$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{\mathrm{MSE}\sum_j \frac{c_j^2}{n_j}} = \sqrt{625\cdot\frac{6}{10}} = \sqrt{375} = 19.36,$$
$$t = \frac{\widehat{C}}{\widehat{\mathrm{SE}}(\widehat{C})} = \frac{66}{19.36} = 3.41.$$

Since $t = 3.41 > t_{0.05,27} = 1.703$, reject H_0.

$$\boxed{t = 3.41 > 1.703 \;\Rightarrow\; \text{reject } H_0;\ \text{the new coatings last significantly longer (one-sided).}}$$

Because the direction was specified ahead of time, the entire 5% sits in the upper tail and the cutoff is the smaller 1.703 (a two-sided test would have used 2.052); the positive $\widehat{C} = 66$ confirms the new formulations outlast the old.

D6. Relating the single-df contrast F to the overall F.

(a) The omnibus F pools all $k-1=2$ df of treatment variation:

$$\text{MSTR} = \frac{\text{SSTR}}{k-1} = \frac{868}{2} = 434, \qquad F_{\text{obs}} = \frac{\text{MSTR}}{\text{MSE}} = \frac{434}{15.0} = 28.93 \quad \text{on } \mathrm{F}_{2,15}\,.$$

(b) The single-df contrast F isolates one direction:

$$F_c = \frac{\text{SS}_c}{\text{MSE}} = \frac{625}{15.0} = 41.67 \quad \text{on } \mathrm{F}_{1,15}\,.$$

(c) Contrast c explains $\text{SS}_c\,/\,\text{SSTR} = 625/868 = 0.720$, i.e. about 72% of the treatment sum of squares. The omnibus F is the *average* of the two single-df pieces, since

$$\text{MSTR} = \frac{\text{SS}_c + \text{SS}_d}{2} = \frac{625 + 243}{2} = 434,$$

whereas F_c uses only c's piece. A focused contrast can therefore exceed the omnibus F_{obs} ($41.67 > 28.93$) because the overall F dilutes a strong comparison ($F_c = 41.67$) by averaging it with a weaker one ($F_d = 16.20$).

$$\boxed{F_c = 41.67\ (\mathrm{F}_{1,15})\ \text{vs. } F_{\text{obs}} = 28.93\ (\mathrm{F}_{2,15});\ \text{MSTR} = \tfrac{1}{2}(\text{SS}_c + \text{SS}_d).}$$

This is the payoff of planning: when you have a specific question, a single-df contrast can be more significant than the blanket F.

D7. Planned contrast versus fishing: the central distinction.

(a) R commits to *one* comparison before seeing the data, so it is a single test at level 0.05 and its Type I error is exactly

$$\mathbb{P}(\text{false reject}) = \alpha = 0.05.$$

No inflation occurs because nothing was chosen by looking at the data.

(b) S runs ten uncorrected tests chosen after inspecting the means. Treating them as independent (an under-estimate, since they share data and MSE), the chance of at least one false rejection under a true H_0 is

$$1 - (1 - 0.05)^{10} = 1 - 0.95^{10} = 1 - 0.599 = 0.401.$$

$$\boxed{\text{R: true error} = 0.05; \quad \text{S (ten uncorrected tests)} \approx 0.40.}$$

Reporting only the largest of the ten makes matters worse: it all but guarantees the biggest noise gap is flagged.

(c) Calling the ten comparisons "contrasts" does not help, because the protection of a planned contrast comes from *genuinely restricting* attention to a few comparisons chosen for scientific reasons before the data — not from the word "contrast." Ten comparisons, however named, form a family of ten tests, and the family-wise error inflates to about 40% just as computed. The honest tool for *all* pairwise comparisons is **Tukey's HSD**, which holds the family-wise error rate at α. "Plan a few, or pay Tukey's price; never fish at α each."

E. Assumptions and diagnostics

E1. *Method: the spread-ratio rule of thumb screens the equal-variance assumption (A3) using the two* extreme *group standard deviations, not any adjacent pair.*

(a) The extremes are $s_{\max} = 4.4$ (oven 3) and $s_{\min} = 2.5$ (oven 4):

$$\frac{s_{\max}}{s_{\min}} = \frac{4.4}{2.5} = 1.76.$$

Since $1.76 < 2$, this sits inside the reassuring zone.

$$\boxed{s_{\max}/s_{\min} = 1.76 < 2 \ \Rightarrow\ \text{equal variance plausible; ordinary } F\text{-test trustworthy.}}$$

(b) The variance ratio is

$$\frac{s^2_{\max}}{s^2_{\min}} = 1.76^2 = 3.10.$$

This is below the warning zone (which begins around a variance ratio of 4), so it confirms the verdict from (a): no red flag.

(c) A *balanced* design (n_j all equal) makes the F-test robust to unequal variance, so even a 3:1 spread ratio would be tolerable here. That protection covers the equal-variance assumption (A3) only — it does *not* extend to independence (A1), which balance can never rescue.

E2. *Method: read each diagnostic against the assumption it probes — residual band and spread ratio for (A3), **QQ** shape for (A2), the experimental design for (A1) — then match the remedy to the failure.*

(a) Constant-width residual band ⇒ (A3) holds; straight **QQ** plot ⇒ (A2) holds; balanced design adds robustness. All gates pass.

No assumption at risk ⇒ run the ordinary one-way F-test as is.

(b) A funnel that opens means variance rises with the mean: **equal variance (A3)** fails. The data are percentages (multiplicative/percentage spread) and the design is unbalanced, so the F-test is *not* robust and should not be trusted.

($A3$) fails, unbalanced ⇒ analyse $\log Y$ (re-plot residuals), not the raw F.

(c) The same 10 plots appear in all three regimes, so the 30 readings are 10 correlated triples, not 30 independent errors: **independence (A1)** fails.

($A1$) fails (repeated measures) ⇒ use a randomized-block model (Ch. 13), not one-way ANOVA.

(d) Small samples with persistent skew and outliers that a transform cannot fix mean **normality (A2)** is broken and the CLT cannot rescue it at $n_j \le 6$.

($A2$) fails, transform-resistant, small n_j ⇒ use Kruskal–Wallis (Ch. 14).

E3. *Method: identify the mean–variance law from the data, then apply the matched variance-stabilizing transformation.*

(a) Here $s_j^2 \approx \overline{Y}_j$: the variance is (approximately) equal to the mean, so variance $\propto \mu^1$. This linear mean–variance law is the **Poisson-count** fingerprint (count data).

(b) For variance $\propto \mu$, the matched transform is $Y \mapsto \sqrt{Y}$ (use $\sqrt{Y + \frac{3}{8}}$ if any counts are near zero). Taking the square root pulls in the large, high-variance values more than the small ones, equalizing the spread.

(c) Transforming the three means:

$$\sqrt{9} = 3, \qquad \sqrt{16} = 4, \qquad \sqrt{36} = 6.$$

On the square-root scale the gaps between successive groups are nearly equal, and (since the original s's also satisfy $\sqrt{s_j^2} = \sqrt{\overline{Y}_j}$) the per-group spread becomes close to constant.

$s_j^2 \approx \overline{Y}_j$ ⇒ run ANOVA on $\sqrt{Y}$; the spread is stabilized.

E4. *Method: a constant coefficient of variation is the multiplicative-spread signature; match the transformation to how the variance scales with the mean.*

(a) A constant coefficient of variation means $s_j \propto \overline{Y}_j$, i.e. the standard deviation is proportional to the mean. Squaring,

$$s_j \propto \mu \implies \text{variance} = s_j^2 \propto \mu^2.$$

So the variance grows like the *square* of the mean.

(b) For sd $\propto \mu$ (variance $\propto \mu^2$), the indicated transform is $Y \mapsto \log Y$: if $Y = \mu\,(1 + \text{noise})$ then $\log Y \approx \log \mu + \text{noise}$ has variance free of μ, turning multiplicative spread into constant additive spread. By contrast, if the *variance* (not the s.d.) were proportional to μ — the count regime variance $\propto \mu^1$ — the right choice would be $\sqrt{Y}$, which under-corrects relative to log.

$$\boxed{\text{constant CV (variance} \propto \mu^2) \;\Rightarrow\; \text{analyse } \log Y; \quad \text{variance} \propto \mu \;\Rightarrow\; \sqrt{Y}.}$$

E5. *Method: the choice between Bartlett and Levene is governed by the normality diagnostic, because Bartlett assumes normal data while Levene does not.*

(a) A pronounced right skew makes normality (A2) doubtful. Bartlett's test assumes normal data and over-rejects under skew, so the robust **Levene** (Brown–Forsythe, median-based) test is the correct choice; its level stays approximately right even under the skew.

$$\boxed{\text{Skewed QQ} \;\Rightarrow\; \text{report Levene (median-based), not Bartlett.}}$$

(b) Bartlett's statistic is referred to a $\chi^2_{k-1} = \chi^2_4$ distribution *only* when the data are normal. Under skew (excess kurtosis), Bartlett's statistic is inflated even when the variances are genuinely equal, so it conflates the non-normality with unequal variance and can reject $H_0 : \sigma_1^2 = \cdots = \sigma_5^2$ for the wrong reason — a false alarm driven by tail shape, not by any real difference in spread.

E6. *Method: the spread ratio is identical across the two labs, so the contrast isolates* balance *as the deciding factor for tolerating unequal variance.*

(a) With the true error standard deviations $3, 3, 9$ fixed in both labs, $s_{\max}/s_{\min} = 3$ is the same for each. The only feature that can change the reliability is the **balance** of the design — whether the n_j are equal.

(b) Lab A is balanced ($n_j = 18, 18, 18$). With equal sample sizes the F-test is robust to unequal variance, so the 3:1 standard-deviation ratio does not distort the level: the true Type I error stays close to the nominal 0.05.

$$\boxed{\text{Lab A balanced} \;\Rightarrow\; \alpha = 0.05 \text{ trustworthy despite } s_{\max}/s_{\min} = 3.}$$

(c) Lab B is unbalanced with the large variance ($\sigma = 9$) on the large group ($n = 42$). This makes the pooled MSE overstate the error variance relevant to the two small groups, so the test is **conservative**: the true level drifts *below* 0.05 and power is lost. The verdict flips if the large variance instead sits on a *small* group: then MSE understates the relevant variance and the test becomes **anticonservative**, with true level *above* 0.05 (too many false rejections).

$$\boxed{\begin{array}{c}\text{Lab B unbalanced: large variance on large } n \;\Rightarrow\; \text{conservative (true level } < 0.05);\\ \text{large variance on small } n \;\Rightarrow\; \text{anticonservative (true level } > 0.05).\end{array}}$$

E7. *Method: independence (A1) is read from the design, never from the residuals; the same subjects across all conditions is the repeated-measures trap.*

(a) Each of the 15 patients contributes a reading to all three doses, so the 45 readings are 15 correlated triples rather than 45 independent errors: **independence (A1)** is violated. A residual or **QQ** plot only inspects the *shape* of the residual distribution (probing A2 and A3); the correlation between a patient's three readings is a property of *how the data were collected*, invisible to any plot of the residual values.

$$\boxed{\text{Repeated measures on one cohort} \;\Rightarrow\; (A1) \text{ violated; no plot can detect it.}}$$

(b) The correct model is a **randomized-block** / repeated-measures ANOVA (Ch. 13), in which the patients enter as a **blocking factor**. Treating each patient as a block removes the between-patient variation, sharpening the comparison among the three caffeine doses.

(c) A transformation only re-expresses the response to fix the *shape* of the error distribution (normality or equal variance); it does nothing to the *correlation structure* between a patient's repeated readings, so it cannot restore independence. The cure is a different model, not a different scale.

F. ANOVA synthesis

F1. (a) The hypotheses are

$$H_0\colon \mu_{\mathrm{F}} = \mu_{\mathrm{R}} = \mu_{\mathrm{Z}} = \mu_{\mathrm{C}} \quad \text{vs.} \quad H_1\colon \text{not all four means are equal.}$$

Group means and grand mean ($N = 20$, balanced $n_j = 5$):

$$\begin{aligned} \overline{Y}_{\mathrm{F}} &= \tfrac{90+92+88+94+86}{5} = 90, \quad \overline{Y}_{\mathrm{R}} = \tfrac{80+84+82+78+86}{5} = 82, \\ \overline{Y}_{\mathrm{Z}} &= \tfrac{74+70+76+72+78}{5} = 74, \quad \overline{Y}_{\mathrm{C}} = \tfrac{60+64+58+62+66}{5} = 62, \\ \overline{Y} &= \tfrac{90+82+74+62}{4} = 77. \end{aligned}$$

Treatment sum of squares from the means, and error sum of squares within groups:

$$\begin{aligned} \mathrm{SSTR} &= 5\big[(90-77)^2 + (82-77)^2 + (74-77)^2 + (62-77)^2\big] \\ &= 5\,[169 + 25 + 9 + 225] = 5(428) = 2140, \\ \mathrm{SSE_F} &= (90-90)^2 + (92-90)^2 + (88-90)^2 + (94-90)^2 + (86-90)^2 \\ &= 0 + 4 + 4 + 16 + 16 = 40, \\ \mathrm{SSE_R} &= (80-82)^2 + (84-82)^2 + (82-82)^2 + (78-82)^2 + (86-82)^2 \\ &= 4 + 4 + 0 + 16 + 16 = 40, \\ \mathrm{SSE_Z} &= (74-74)^2 + (70-74)^2 + (76-74)^2 + (72-74)^2 + (78-74)^2 \\ &= 0 + 16 + 4 + 4 + 16 = 40, \\ \mathrm{SSE_C} &= (60-62)^2 + (64-62)^2 + (58-62)^2 + (62-62)^2 + (66-62)^2 \\ &= 4 + 4 + 16 + 0 + 16 = 40, \\ \mathrm{SSE} &= 40 + 40 + 40 + 40 = 160, \qquad \mathrm{SST} = \mathrm{SSTR} + \mathrm{SSE} = 2140 + 160 = 2300. \end{aligned}$$

(b) Degrees of freedom $k - 1 = 3$, $N - k = 16$, total 19; mean squares and F:

$$\mathrm{MSTR} = \frac{2140}{3} = 713.33, \qquad \mathrm{MSE} = \frac{160}{16} = 10.0,$$

$$F_{\mathrm{obs}} = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{713.33}{10.0} = 71.33.$$

Source	df	SS	MS	F
Treatment	3	2140.0	713.33	71.33
Error	16	160.0	10.00	
Total	19	2300.0		

Since $F_{\mathrm{obs}} = 71.33 > F_{0.05,3,16} = 3.24$, reject H_0 at the 5% level: the four storage methods do not all retain the same mean amount of vitamin C.

$$\boxed{F_{\mathrm{obs}} = 71.33 > 3.24 = F_{0.05,3,16} \;\Rightarrow\; \text{reject } H_0.}$$

(c) Balanced ($n = 5$), so one Tukey threshold serves every pair:

$$\mathrm{HSD} = q_{0.05,4,16}\sqrt{\frac{\mathrm{MSE}}{n}} = 4.05\sqrt{\frac{10.0}{5}} = 4.05\sqrt{2} = 4.05 \times 1.414 = 5.73.$$

Sorted means $\overline{Y}_{\mathrm{C}} = 62 < \overline{Y}_{\mathrm{Z}} = 74 < \overline{Y}_{\mathrm{R}} = 82 < \overline{Y}_{\mathrm{F}} = 90$. The six absolute gaps and verdicts (significant when > 5.73):

$$|\mathrm{F}{-}\mathrm{R}| = 8,\ |\mathrm{F}{-}\mathrm{Z}| = 16,\ |\mathrm{F}{-}\mathrm{C}| = 28,\ |\mathrm{R}{-}\mathrm{Z}| = 8,\ |\mathrm{R}{-}\mathrm{C}| = 20,\ |\mathrm{Z}{-}\mathrm{C}| = 12.$$

Every gap exceeds 5.73, so *all six pairs differ*; the grouping puts each method on its own line:

$$\boxed{\underline{\mathrm{C}}\quad \underline{\mathrm{Z}}\quad \underline{\mathrm{R}}\quad \underline{\mathrm{F}}} \qquad (62)\ (74)\ (82)\ (90).$$

Each storage method gives a distinguishable mean retention; fresh is highest, canned lowest.

(d) Planned contrast $c = (3, -1, -1, -1)$ on (F, R, Z, C):

$$\begin{aligned}
\widehat{C} &= 3(90) - 82 - 74 - 62 = 270 - 218 = 52,\\
\widehat{\mathrm{SE}}(\widehat{C}) &= \sqrt{\mathrm{MSE}\ \frac{\sum_j c_j^2}{n}} = \sqrt{10.0 \cdot \frac{9+1+1+1}{5}} = \sqrt{10.0 \cdot \frac{12}{5}} = \sqrt{24} = 4.899,\\
t &= \frac{\widehat{C}}{\widehat{\mathrm{SE}}(\widehat{C})} = \frac{52}{4.899} = 10.61.
\end{aligned}$$

Since $|t| = 10.61 > t_{0.025,16} = 2.120$, reject $H_0\colon C = 0$. The positive $\widehat{C}$ (and $\widehat{C}/3 = +17.3$ per the natural-scale comparison $\mu_{\mathrm{F}} - \frac{1}{3}\sum \mu_{\mathrm{others}}$) says fresh broccoli retains markedly more vitamin C than the average of the three preservation methods.

$$\boxed{t = 10.61 > 2.120 \ \Rightarrow\ \text{Fresh beats the preserved-average by } \approx 17 \text{ mg.}}$$

F2. **(a)** Only group means and standard deviations are given, so use $\mathrm{SSE} = \sum_j (n_j - 1)S_j^2$ and the size-weighted grand mean. With $N = 6 + 8 + 4 = 18$, $k = 3$:

$$\begin{aligned}
\mathrm{SSE} &= (6-1)18^2 + (8-1)20^2 + (4-1)16^2 = 5(324) + 7(400) + 3(256)\\
&= 1620 + 2800 + 768 = 5188,\\
\overline{Y} &= \frac{6(420) + 8(405) + 4(384)}{18} = \frac{2520 + 3240 + 1536}{18} = \frac{7296}{18} = 405.33,\\
\mathrm{SSTR} &= 6(420 - 405.33)^2 + 8(405 - 405.33)^2 + 4(384 - 405.33)^2\\
&= 6(215.11) + 8(0.111) + 4(455.11) = 1290.67 + 0.89 + 1820.44 = 3112.0.
\end{aligned}$$

Mean squares ($k - 1 = 2$, $N - k = 15$):

$$\mathrm{MSTR} = \frac{3112.0}{2} = 1556.0, \qquad \mathrm{MSE} = \frac{5188}{15} = 345.87.$$

Source	df	SS	MS	F
Treatment	2	3112.0	1556.0	4.50
Error	15	5188.0	345.87	
Total	17	8300.0		

(b) $F_{\mathrm{obs}} = \mathrm{MSTR}\,/\,\mathrm{MSE} = 1556.0/345.87 = 4.50$ on $\mathrm{F}_{2,15}$. Since $F_{\mathrm{obs}} = 4.50 > F_{0.05,2,15} = 3.68$, reject H_0 at the 5% level: mean reaction time is not the same across the three caffeine doses.

$$\boxed{F_{\mathrm{obs}} = 4.50 > 3.68 = F_{0.05,2,15} \ \Rightarrow\ \text{reject } H_0;\ \text{doses differ.}}$$

(c) The doses $0, 100, 200$ are equally spaced and ordered, so the planned linear-trend contrast uses

$c = (-1, 0, +1)$ ($\sum c_j = 0$). With unequal n_j each c_j^2 is divided by its own n_j:

$$\widehat{C} = -1(420) + 0(405) + 1(384) = -36,$$
$$\sum_j \frac{c_j^2}{n_j} = \frac{1}{6} + \frac{0}{8} + \frac{1}{4} = 0.1667 + 0.2500 = 0.4167,$$
$$\widehat{\mathrm{SE}}(\widehat{C}) = \sqrt{\mathrm{MSE} \sum_j \frac{c_j^2}{n_j}} = \sqrt{345.87 \times 0.4167} = \sqrt{144.11} = 12.00,$$
$$t = \frac{-36}{12.00} = -3.00.$$

Since $|t| = 3.00 > t_{0.025,15} = 2.131$, reject H_0: $C = 0$. The negative $\widehat{C}$ means reaction time *falls* as dose rises: there is a significant decreasing linear trend, i.e. more caffeine yields faster mean reaction.

$$\boxed{t = -3.00,\ |t| > 2.131 \ \Rightarrow\ \text{significant decreasing linear trend in dose.}}$$

F3. **(a)** A half-blank table is filled with three identities: df add, SS add, MS = SS/df. Error df $= N - k = 45$ is given, and $k = 5$ groups, so treatment df $= k - 1 = 4$ and total df $= 4 + 45 = 49$. The SS column adds down, giving the missing error SS:

$$\mathrm{SSE} = \mathrm{SST} - \mathrm{SSTR} = 690.0 - 240.0 = 450.0.$$

Mean squares and F:

$$\mathrm{MSTR} = \frac{\mathrm{SSTR}}{k-1} = \frac{240.0}{4} = 60.0, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{N-k} = \frac{450.0}{45} = 10.0,$$
$$F_{\mathrm{obs}} = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{60.0}{10.0} = 6.0.$$

Source	df	SS	MS	F
Treatment	4	240.0	60.0	6.0
Error	45	450.0	10.0	
Total	49	690.0		

(b) The reference is $\mathrm{F}_{4,45}$ with cutoff $F_{0.05,4,45} = 2.58$. Since $F_{\mathrm{obs}} = 6.0 > 2.58$, reject H_0 at the 5% level: the five assembly lines do not all share the same mean.

$$\boxed{F_{\mathrm{obs}} = 6.0 > 2.58 = F_{0.05,4,45} \ \Rightarrow\ \text{reject } H_0;\ \text{lines differ.}}$$

(c) Total df $= N - 1 = 49$, so $N = 50$. With $k = 5$ equal-size groups a balanced design has $N/k = 50/5 = 10$ observations per line.

$$\boxed{N = 50, \quad n_j = 10 \text{ per line (balanced).}}$$

F4. **(a)** The all-pairs plan runs one two-sample t-test per pair of the $k = 6$ colors:

$$m = \binom{6}{2} = \frac{6 \cdot 5}{2} = 15 \text{ tests.}$$

Treating them as independent and all null, the chance of *at least one* false positive is

$$\mathrm{FWER} \approx 1 - (1-\alpha)^m = 1 - (0.95)^{15}$$
$$= 1 - 0.4633 = 0.5367 \approx 0.54.$$

$$\boxed{m = 15, \quad \text{FWER} \approx 0.54 \gg 0.05.}$$

(b) Running 15 unprotected tests at $\alpha = 0.05$ each inflates the experiment-wide false-positive rate to about 54% – the analyst would "find" a spurious color difference more than half the time even if every mean were identical. The single ANOVA F-test of $H_0 \colon \mu_1 = \cdots = \mu_6$ has exactly one rejection region and holds the *overall* Type I error at $\alpha = 0.05$; only after it rejects does one use a multiple-comparison method (Tukey) that controls the family-wise rate. The F-test offers a guarantee on the whole family that the all-pairs scan throws away.

(c) The given statistic is $F_{\text{obs}} = 2.50$ on $\text{F}_{5,54}$ with $F_{0.05,5,54} = 2.39$. Since $2.50 > 2.39$, reject H_0 at 5%: the six color means are not all equal (though the margin is slim).

$$\boxed{F_{\text{obs}} = 2.50 > 2.39 = F_{0.05,5,54} \;\Rightarrow\; \text{reject } H_0 \text{ (borderline).}}$$

F5. Group totals are given, so use the correction-term shortcut. With $T = 250 + 280 + 265 = 795$, $N = 30$, and $\sum_{i,j} Y_{ij}^2 = 22050$:

$$\begin{aligned} C &= \frac{T^2}{N} = \frac{795^2}{30} = \frac{632025}{30} = 21067.5, \\ \sum_j \frac{T_j^2}{n_j} &= \frac{250^2 + 280^2 + 265^2}{10} = \frac{62500 + 78400 + 70225}{10} = \frac{211125}{10} = 21112.5, \\ \text{SSTR} &= 21112.5 - 21067.5 = 45.0, \\ \text{SST} &= 22050 - 21067.5 = 982.5, \qquad \text{SSE} = 982.5 - 45.0 = 937.5. \end{aligned}$$

Degrees of freedom $k - 1 = 2$, $N - k = 27$, total 29; mean squares and F:

$$\text{MSTR} = \frac{45.0}{2} = 22.5, \qquad \text{MSE} = \frac{937.5}{27} = 34.72,$$
$$F_{\text{obs}} = \frac{22.5}{34.72} = 0.648.$$

Source	df	SS	MS	F
Treatment	2	45.0	22.50	0.648
Error	27	937.5	34.72	
Total	29	982.5		

Since $F_{\text{obs}} = 0.648 < F_{0.05,2,27} = 3.35$, *fail to reject* H_0: there is no significant difference in mean lifetime among the three LED brands. An F below 1 means the between-brand spread is actually smaller than the within-brand noise – the brands look interchangeable.

$$\boxed{F_{\text{obs}} = 0.648 < 3.35 = F_{0.05,2,27} \;\Rightarrow\; \text{fail to reject } H_0.}$$

F6. (a) From the table, MSE = 12.5 on $N - k = 28$ df. Balanced $n = 8$, so one Tukey threshold:

$$\text{HSD} = q_{0.05,4,28}\sqrt{\frac{\text{MSE}}{n}} = 3.86\sqrt{\frac{12.5}{8}} = 3.86\sqrt{1.5625} = 3.86 \times 1.25 = 4.83.$$

The six sorted gaps and verdicts (significant when > 4.83):

$$\begin{aligned} &|A{-}B| = 2.0 \text{ (ns)}, \; |A{-}C| = 7.5 \text{ (sig)}, \; |A{-}D| = 8.0 \text{ (sig)}, \\ &|B{-}C| = 5.5 \text{ (sig)}, \; |B{-}D| = 6.0 \text{ (sig)}, \; |C{-}D| = 0.5 \text{ (ns)}. \end{aligned}$$

The two non-significant pairs are $\{A, B\}$ and $\{C, D\}$; all four cross pairs differ.

$$\boxed{\text{HSD} = 4.83; \text{ all pairs differ except } A \text{ vs } B \text{ and } C \text{ vs } D.}$$

(b) Sorted means $A = 20.0 < B = 22.0 < C = 27.5 < D = 28.0$. Ties $\{A, B\}$ and $\{C, D\}$ give two clean, non-overlapping lines:

$$\boxed{\underline{A\ \ B} \qquad \underline{C\ \ D}} \qquad (20, 22)\ \ (27.5, 28).$$

There is no overlap here: the data sort into a clear low pair $\{A, B\}$ and a clear high pair $\{C, D\}$, with the two blocks well separated.

(c) The largest gap is $|A - D| = 8.0$, the easiest of all six to declare significant; since $8.0 > \text{HSD} = 4.83$ it *is* significant, so it is not the case that no pair separates – four of the six pairs are significant. (Had even the largest gap, 8.0, fallen below the HSD, then no Tukey pair would have fired despite a decisive F. That mismatch happens because the omnibus F pools *all* the between-group spread into one statistic and can reject on several modest gaps acting together, even when no *single* pairwise gap clears its sharper threshold; the honest report would then be "the means differ overall, but the data cannot pinpoint which pair.")

$$\boxed{\text{Largest gap } 8.0 > 4.83 \Rightarrow 4 \text{ of } 6 \text{ pairs significant; } F \text{ and Tukey agree here.}}$$

F7. (a) Planned contrast $c = (3, -1, -1, -1)$ on (control, diet 2, diet 3, diet 4), MSE = 15.0 on 20 df, balanced $n = 6$:

$$\begin{aligned}
\widehat{C} &= 3(18.0) - 26.0 - 29.0 - 33.0 = 54.0 - 88.0 = -34.0, \\
\widehat{\text{SE}}(\widehat{C}) &= \sqrt{\text{MSE}\ \frac{\sum_j c_j^2}{n}} = \sqrt{15.0 \cdot \frac{9 + 1 + 1 + 1}{6}} = \sqrt{15.0 \cdot \frac{12}{6}} = \sqrt{30} = 5.477, \\
t &= \frac{-34.0}{5.477} = -6.21.
\end{aligned}$$

Since $|t| = 6.21 > t_{0.025,20} = 2.086$, reject $H_0 \colon C = 0$: the control diet differs from the average of the three active diets (the negative sign says control reduces LDL *less*).

$$\boxed{t = -6.21,\ |t| > 2.086\ \Rightarrow\ \text{control underperforms the active-diet average.}}$$

(b) The natural-scale comparison is $C^\star = \mu_1 - \frac{1}{3}(\mu_2 + \mu_3 + \mu_4) = \frac{1}{3}C$, so rescale the estimate and its SE by $\frac{1}{3}$:

$$\begin{aligned}
\widehat{C}^\star &= \tfrac{1}{3}(-34.0) = -11.33, \qquad \widehat{\text{SE}}(\widehat{C}^\star) = \tfrac{1}{3}(5.477) = 1.826, \\
\text{margin} &= t_{0.025,20}\, \widehat{\text{SE}}(\widehat{C}^\star) = 2.086 \times 1.826 = 3.81, \\
C^\star &\in -11.33 \pm 3.81 = (-15.14,\ -7.52)\ \text{mg/dL}.
\end{aligned}$$

The whole interval lies below zero, so we are 95% confident the control reduces LDL by between 7.5 and 15.1 mg/dL *less* than the active diets on average – consistent with the test.

$$\boxed{C^\star \in (-15.14,\ -7.52)\ \text{mg/dL};\ 0 \notin \text{interval, so control is worse.}}$$

(c) One-way ANOVA assumes the errors ε_{ij} are (i) **normal**, (ii) **independent**, and (iii) of **equal variance** σ^2 across all groups (homoscedasticity). A plot of residuals $Y_{ij} - \overline{Y}_j$ against the fitted values (the group means $\overline{Y}_j$) chiefly checks the *equal-variance* assumption: if the vertical spread of the residuals fans out or shrinks as the fitted value changes, the common-σ^2 assumption is suspect. (Normality is better judged from a normal QQ-plot of the residuals, and independence from the data-collection design or a residual-versus-order plot.)

$$\boxed{\begin{array}{c}\text{Assumptions: normal, independent, equal-variance errors; residual-vs-fitted checks equal} \\ \text{variance.}\end{array}}$$

Chapter 13

Randomized Block Designs

One-way ANOVA (Chapter 12) compares several treatments using completely randomized data, but its within-group "error" lumps together *every* source of variation that is not the treatment — including systematic differences between the experimental units themselves. A **randomized block design** removes the biggest such nuisance by grouping similar units into **blocks** (litters of animals, plots in one field, the same subjects, matched patients) and applying every treatment once within each block. Comparing treatments *within* blocks cancels the block-to-block differences, shrinking the error term and sharpening the treatment F-test — the same idea that makes a paired t-test beat a two-sample t-test. In fact the **paired t-test** is exactly a randomized block design with two treatments.

Master guide: blocking and which test to run

- Several treatments, and the units fall into **natural groups** (a known nuisance variable) you can match across? ⇒ a **randomized block design**; block on that variable (§1).
- Comparing the k treatment means **after blocking**? ⇒ the **two-way ANOVA F-test** for treatments, with a separate block sum of squares pulled out of the error (§2).
- Exactly **two** treatments measured on the *same* (or matched) units? ⇒ the **paired t-test** — the two-treatment special case of a block design (§3).
- Asking **whether blocking helped**, or whether the design is valid? ⇒ **relative efficiency**, the additivity (no-interaction) assumption, and diagnostics (§4).

Keep straight: blocks are a *nuisance* factor you remove, not a treatment you are testing — you rarely care about the block F. Blocking helps only when units within a block are genuinely more alike than units across blocks; if they are not, you have spent degrees of freedom on the block term for nothing.

13.1 The Randomized Block Design and Its Model

A completely randomized design throws every experimental unit into one big pool and assigns treatments at random across the whole pool. That is honest, but wasteful: any systematic difference *between the units themselves* — one litter of mice is hardier than another, one plot of soil is richer, one day in the lab is warmer — gets swept into the error term, where it inflates MSE and drowns out the treatment signal. A **randomized block design** fixes this by first sorting similar units into b **blocks**, then applying each of the k treatments exactly once *inside* every block, with the treatment-to-unit assignment randomized within the block. Because every treatment meets every block once, a comparison of two treatments is made on matched units and the block-to-block differences cancel — exactly the trick that lets a paired t-test beat a two-sample t-test. Recognize that you want a block design whenever there is a *known, measurable nuisance variable* you can match units on before you ever look at the response. This section sets up the additive model $Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$, the hypotheses it lets you test, and the layout of the data; the F-test that uses it comes in the next section.

Is this a randomized block design, and what is the block?

- There is a **known nuisance variable** (litter, field, day, subject, machine) and you can apply *every* treatment once within each level of it? ⇒ a **randomized block design**; that variable is the **block**.
- The same treatments are measured on the **same or matched units** repeatedly, so each row of data shares a common nuisance? ⇒ block on the unit; with $k = 2$ this is the **paired** design.
- Treatments assigned at random across **one undivided pool**, no matching variable? ⇒ a **completely randomized design** (Chapter 12), *not* a block design.
- **Two** factors you genuinely care about, crossed and *replicated*, both of interest? ⇒ a **two-factor factorial**, not blocking — a block is a nuisance you remove, not a factor you study.

Do not confuse: a block and a second treatment factor look identical on paper (a two-way table). The difference is *intent and inference* — you test the treatments, you merely *remove* the blocks, and a pure block design has one observation per cell, so it cannot estimate a treatment-by-block interaction.

13.1.1 Why block: nuisance variation and the same idea as pairing

Definition 13.1.1 (Randomized block design)**.** Let there be k **treatments** to compare and a known nuisance variable with b levels. Group the experimental units into b **blocks** of k similar units each, and within each block assign the k treatments to the k units at random, one treatment per unit. This yields one observation Y_{ij} for treatment i ($i = 1, \dots, k$) in block j ($j = 1, \dots, b$), for a total of $N = kb$ observations arranged in a $k \times b$ table with exactly one entry per cell.

Remark. The defining feature is **completeness**: every treatment appears once in every block. That is what makes treatment comparisons *within* a block free of the block's nuisance level, and it is what distinguishes a randomized *complete* block design from incomplete designs. With one observation per cell there is no replication inside a cell, which is exactly why the model below must assume **additivity** (no treatment-by-block interaction) — there is nothing left over to estimate an interaction with.

Theorem 13.1.2 (Blocking is pairing with k treatments)**.** *For $k = 2$ treatments, a randomized block design with b blocks is precisely a paired design: the within-block difference $D_j = Y_{1j} - Y_{2j}$ removes the block effect β_j, leaving only the treatment contrast $\tau_1 - \tau_2$ plus error. Testing $\tau_1 = \tau_2$ by a one-sample t on the D_j is the paired t-test.*

Proof. Under the additive model $Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}$,

$$D_j = Y_{1j} - Y_{2j} = (\tau_1 - \tau_2) + (\varepsilon_{1j} - \varepsilon_{2j}),$$

and the block term β_j — the entire nuisance — subtracts out. The D_j are iid with mean $\tau_1 - \tau_2$, so a one-sample test on them tests $\tau_1 = \tau_2$. Blocking with $k \geq 3$ treatments generalizes this cancellation to all pairwise contrasts at once. □

13.1.2 The additive model and the hypotheses

Definition 13.1.3 (The randomized block model)**.** Each observation is modeled as

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, \dots, k, \quad j = 1, \dots, b,$$

where μ is the grand mean, τ_i is the **treatment effect** of treatment i, β_j is the **block effect** of block j, and the errors ε_{ij} are independent $\mathrm{N}(0, \sigma^2)$ with one common variance. The effects are centered,

$$\sum_{i=1}^{k} \tau_i = 0, \qquad \sum_{j=1}^{b} \beta_j = 0,$$

so μ carries the overall level and τ_i, β_j carry the departures from it.

Definition 13.1.4 (Additivity: no treatment-by-block interaction)**.** The model is **additive**: the effect of switching treatments is the same constant in every block. Formally, $\mathbb{E}[Y_{ij}] = \mu + \tau_i + \beta_j$ has no $\tau\beta$ cross term, so for any two treatments i, i' the gap $\mathbb{E}[Y_{ij}] - \mathbb{E}[Y_{i'j}] = \tau_i - \tau_{i'}$ does not depend on the block j. Plotted across blocks, the treatment profiles are *parallel*.

Definition 13.1.5 (Null and alternative for treatments)**.** The randomized block design tests whether the treatments matter, after the blocks are accounted for:

$$H_0 : \ \tau_1 = \tau_2 = \cdots = \tau_k = 0 \qquad \text{versus} \qquad H_1 : \ \tau_i \neq 0 \text{ for at least one } i.$$

Equivalently $H_0 : \ \mu_{1\cdot} = \cdots = \mu_{k\cdot}$ where $\mu_{i\cdot} = \mu + \tau_i$ is the mean response of treatment i averaged over blocks. One usually does *not* care about the parallel block hypothesis $\beta_j = 0$ — the blocks are a nuisance, present only to be removed.

Remark. Three notational means recur throughout: the treatment means $\overline{Y}_{i\cdot}$ (average a row over its b blocks), the block means $\overline{Y}_{\cdot j}$ (average a column over its k treatments), and the grand mean $\overline{Y}$. Their definitions are

$$\overline{Y}_{i\cdot} = \frac{1}{b}\sum_{j=1}^{b} Y_{ij}, \qquad \overline{Y}_{\cdot j} = \frac{1}{k}\sum_{i=1}^{k} Y_{ij},$$

$$\overline{Y} = \frac{1}{kb}\sum_{i=1}^{k}\sum_{j=1}^{b} Y_{ij} = \frac{1}{k}\sum_{i} \overline{Y}_{i\cdot} = \frac{1}{b}\sum_{j} \overline{Y}_{\cdot j}.$$

The treatment effect is estimated by $\widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}$ and the block effect by $\widehat{\beta}_j = \overline{Y}_{\cdot j} - \overline{Y}$.

13.1.3 Recognizing the design and naming the block

Example 13.1.6 (Litters as blocks: a drug-dose trial)**.** A pharmacologist tests $k = 4$ doses of a compound on weanling rats. Rats from the *same litter* are genetically similar and were raised together, while different litters vary in baseline weight gain. Eight litters are available, each with four pups. Identify the experimental unit, the treatment, and a good blocking variable, and say how many observations result.

Which rule, and why? A known nuisance (litter, a genetic/environmental cluster) can be matched: within one litter, assign the four doses to the four pups at random. That is a randomized block design with the litter as the block — not a completely randomized design, which would ignore the litter structure and let between-litter variation swell the error.

Solution.

Step 1: name the roles. The *experimental unit* is an individual pup; the *treatment* is dose ($k = 4$); the *block* is the litter ($b = 8$), the nuisance variable we match on.

Step 2: the design. Within each litter, randomly assign the four doses to its four pups — every dose once per litter.

Step 3: count. $N = kb = 4 \times 8 = 32$ pups, one observation each.

unit = pup; treatment = dose ($k = 4$); block = litter ($b = 8$); $N = 32$.

Pups within a litter are more alike than pups across litters, so blocking on litter removes a real chunk of nuisance variation before the doses are ever compared.

Example 13.1.7 (Fields as blocks: a fertilizer comparison)**.** An agronomist compares $k = 3$ fertilizers. Five geographically separated fields are available; soil and drainage differ markedly between fields but are fairly uniform within a field. Each field is split into three plots. Set up the blocking and identify what cancels.

Which rule, and why? The fields are the nuisance: soil quality varies field-to-field. Splitting each field into $k = 3$ plots and randomizing the three fertilizers within the field makes the field the block, so soil differences cancel when fertilizers are compared within a field.

Solution.

Step 1: roles. Unit = a plot; treatment = fertilizer ($k = 3$); block = field ($b = 5$).

Step 2: assignment. Within each field, randomly assign the three fertilizers to its three plots.

Step 3: what cancels. For two fertilizers i, i' in field j, $Y_{ij} - Y_{i'j} = (\tau_i - \tau_{i'}) + (\varepsilon_{ij} - \varepsilon_{i'j})$; the field effect β_j drops out.

$$\boxed{\text{block} = \text{field } (b = 5);\ N = kb = 15;\ \text{soil effect } \beta_j \text{ cancels within a field.}}$$

Example 13.1.8 (Days as blocks: a lab-assay experiment)**.** A biochemist runs $k = 5$ assay protocols. Reagent freshness, ambient temperature, and calibration drift change from day to day but are stable within a single day. Only five samples can be processed per day, over six days. Identify the design and the block.

Which rule, and why? Day-to-day drift is a textbook nuisance: it shifts *all* protocols up or down together on a given day. Running all five protocols within each day, in random order, makes the day the block and removes the daily shift.

Solution.

Step 1: roles. Unit = one assay run; treatment = protocol ($k = 5$); block = day ($b = 6$).

Step 2: assignment. Each day, run all five protocols in random order on five samples.

Step 3: count. $N = 5 \times 6 = 30$ runs.

$$\boxed{\text{block} = \text{day } (b = 6);\ \text{treatment} = \text{protocol } (k = 5);\ N = 30.}$$

A warm or badly calibrated day raises every protocol's reading equally, so its effect β_j sits in the block term, not the error.

Example 13.1.9 (Subjects as blocks: a within-person taste test)**.** A food scientist has $k = 4$ recipe variants and recruits $b = 10$ tasters. Each taster rates all four variants (palates differ: some people rate everything high). Name the block and explain why a completely randomized design would be wasteful here.

Which rule, and why? People are the nuisance: a generous palate inflates all four of that taster's scores together. Having each taster judge all four variants makes the *taster* the block; a completely randomized design giving each person only one variant would leave palate differences in the error.

Solution.

Step 1: roles. Unit = one taster-tasting-one-variant; treatment = recipe ($k = 4$); block = taster ($b = 10$).

Step 2: why block. A harsh or generous palate adds a constant β_j to all four of taster j's scores; comparing variants within a taster cancels it.

Step 3: count. $N = 4 \times 10 = 40$ ratings.

$$\boxed{\text{block} = \text{taster } (b = 10);\ N = 40;\ \text{palate level } \beta_j \text{ cancels within a person.}}$$

Example 13.1.10 (Machines as blocks: a coating-thickness study)**.** A manufacturer compares $k = 3$ coating chemistries. Four production machines are available; machines differ in calibration but each is internally consistent. Each machine produces one part under each chemistry. Identify the block and write the model for the observation.

Which rule, and why? Machine-to-machine calibration is the nuisance. Producing one part per chemistry on each machine makes the machine the block; the model attaches a block effect β_j to whatever machine made the part.

Solution.

Step 1: roles. Unit = a part; treatment = chemistry ($k = 3$); block = machine ($b = 4$).

Step 2: the model for one observation. For chemistry i on machine j,

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad \varepsilon_{ij} \sim \mathrm{N}(0, \sigma^2) \text{ iid.}$$

Step 3: count. $N = 3 \times 4 = 12$ parts.

$$\boxed{Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij};\ \text{block} = \text{machine } (b = 4);\ N = 12.}$$

Example 13.1.11 (Block design vs. completely randomized: telling them apart)**.** Two plans compare $k = 3$ teaching methods. *Plan A:* 30 students are pooled and 10 are assigned at random to each method. *Plan B:* students are first sorted into 10 ability tiers of 3 students each, and within each tier the three methods are assigned at random. Which plan is a randomized block design, and what is the block?

Which rule, and why? The deciding question is whether a nuisance variable is matched *before* assignment. Plan A pools everyone (no matching) — completely randomized. Plan B matches on ability tier and applies all three methods within each tier — a randomized block design with the tier as block.

Solution.

Step 1: scan for matching. Plan A: one undivided pool, random assignment across all 30 — no block. Plan B: students grouped by ability *first*, all treatments within each group.

Step 2: classify.

Plan A : completely randomized design (Chapter 12), error carries ability spread,

Plan B : randomized block design, block = ability tier ($b = 10$).

Plan B is the block design; block = ability tier; Plan A is completely randomized.

Plan B is sharper precisely because ability — a strong driver of test scores — is pulled out of the error instead of being left to muddy it.

Example 13.1.12 (Block design vs. two-factor factorial: intent decides)**.** A horticulturist grows plants under $k = 3$ light colors. In Design 1, six greenhouses (differing in humidity, a nuisance) each get all three colors, one bench per color. In Design 2, the same three colors are crossed with two CO_2 levels of scientific interest, every color-by-CO2 combination replicated four times. Which is blocking and which is a factorial?

Which rule, and why? Blocking and a second factor look identical as two-way tables; *intent and replication* separate them. Greenhouse humidity is a nuisance with one observation per cell — a block. CO_2 is a factor of interest, crossed and replicated — a factorial.

Solution.

Step 1: ask what you care about. Design 1: humidity is unwanted variation to remove ⇒ block. Design 2: CO_2 is a factor whose effect (and its interaction with color) you want ⇒ second factor.

Step 2: check replication. Design 1 has one plant per color-greenhouse cell (no interaction estimable); Design 2 has four replicates per cell (interaction estimable).

Design 1 = randomized block (greenhouse block); Design 2 = two-factor factorial.

13.1.4 Writing the model and the hypotheses

Example 13.1.13 (Set up the model and hypotheses: irrigation methods)**.** An engineer compares $k = 4$ irrigation methods across $b = 6$ farm sites (the sites are the block). Write the model for the yield Y_{ij}, state the treatment hypotheses, and give the estimators of the treatment effects.

Which rule, and why? This is a randomized block design (sites block the soil/climate nuisance), so the additive two-effect model applies and the hypothesis of interest is about the treatment effects τ_i, with the block effects β_j present only to be removed.

Solution.

Step 1: model.

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad i = 1, \ldots, 4,\ j = 1, \ldots, 6, \quad \varepsilon_{ij} \sim \mathrm{N}(0, \sigma^2) \text{ iid},$$

with $\sum_{i=1}^{4} \tau_i = 0$ and $\sum_{j=1}^{6} \beta_j = 0$.

Step 2: hypotheses.

$$H_0 :\ \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0 \qquad \text{vs.} \qquad H_1 :\ \tau_i \neq 0 \text{ for some } i.$$

Step 3: estimators. $\widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}$, $\widehat{\beta}_j = \overline{Y}_{\cdot j} - \overline{Y}$.

$$\boxed{Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij};\quad H_0 : \tau_1 = \cdots = \tau_4 = 0;\quad \widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}.}$$

Example 13.1.14 (Estimate the effects from a filled table)**.** A randomized block design with $k = 3$ treatments and $b = 4$ blocks gives the table below (response in suitable units). Compute the grand mean, the treatment means, the estimated treatment effects, and verify $\sum_i \widehat{\tau}_i = 0$.

	B_1	B_2	B_3	B_4	$\overline{Y}_{i\cdot}$
T_1	14	18	10	14	
T_2	18	22	14	18	
T_3	22	26	18	22	

Which rule, and why? With the full $k \times b$ table in hand, the treatment effect of row i is its row mean minus the grand mean, $\widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}$ — the natural estimator from the additive model. The centering constraint $\sum_i \widehat{\tau}_i = 0$ is an arithmetic check.

Solution.

Step 1: row (treatment) means.

$$\begin{aligned}\overline{Y}_{1\cdot} &= \tfrac{14+18+10+14}{4} = 14,\\ \overline{Y}_{2\cdot} &= \tfrac{18+22+14+18}{4} = 18,\\ \overline{Y}_{3\cdot} &= \tfrac{22+26+18+22}{4} = 22.\end{aligned}$$

Step 2: grand mean. $\overline{Y} = \dfrac{14+18+22}{3} = 18$ (average of the row means, since the design is complete).

Step 3: effects and check.

$$\begin{aligned}&\widehat{\tau}_1 = 14 - 18 = -4, \quad \widehat{\tau}_2 = 18 - 18 = 0, \quad \widehat{\tau}_3 = 22 - 18 = +4,\\ &\sum_i \widehat{\tau}_i = -4 + 0 + 4 = 0. \ \blacksquare\end{aligned}$$

$$\boxed{\widehat{\tau} = (-4,\, 0,\, +4), \quad \sum_i \widehat{\tau}_i = 0.}$$

Example 13.1.15 (Estimate the block effects from the same table)**.** For the $k = 3$, $b = 4$ table above, compute the block means $\overline{Y}_{\cdot j}$ and the estimated block effects $\widehat{\beta}_j = \overline{Y}_{\cdot j} - \overline{Y}$, and confirm $\sum_j \widehat{\beta}_j = 0$. Comment on whether blocking looks worthwhile.

Which rule, and why? Block effects are estimated column-wise: a column's mean minus the grand mean. Large, varied $\widehat{\beta}_j$ signal a strong nuisance — the very thing blocking removes — so their spread is the practical case for having blocked.

Solution.

Step 1: column (block) means.

$$\begin{aligned}\overline{Y}_{\cdot 1} &= \tfrac{14+18+22}{3} = 18, \quad \overline{Y}_{\cdot 2} = \tfrac{18+22+26}{3} = 22,\\ \overline{Y}_{\cdot 3} &= \tfrac{10+14+18}{3} = 14, \quad \overline{Y}_{\cdot 4} = \tfrac{14+18+22}{3} = 18.\end{aligned}$$

Step 2: block effects and check.

$$\begin{aligned}&\widehat{\beta}_1 = 18 - 18 = 0, \quad \widehat{\beta}_2 = 22 - 18 = +4,\\ &\widehat{\beta}_3 = 14 - 18 = -4, \quad \widehat{\beta}_4 = 18 - 18 = 0,\\ &\sum_j \widehat{\beta}_j = 0 + 4 - 4 + 0 = 0. \ \blacksquare\end{aligned}$$

Step 3: read it. Block 2 runs 4 above and block 3 runs 4 below the grand mean — a real block-to-block shift, so blocking pulled an 8-unit swing out of the error.

$$\boxed{\widehat{\beta} = (0, +4, -4, 0),\ \sum_j \widehat{\beta}_j = 0;\ \text{a genuine nuisance, so blocking helps.}}$$

Notice the additivity at work: every column equals its block mean plus the same $(-4, 0, +4)$ treatment pattern, so the rows are parallel.

13.1.5 Why a known nuisance must be removed: a contrast

Example 13.1.16 (CONTRAST: ignoring the block vs. using it on the SAME data)**.** A clinician compares $k = 2$ pain treatments on $b = 5$ patients; each patient receives both (a block design with patient as block). Pain-reduction scores are

	P_1	P_2	P_3	P_4	P_5
T_1	30	42	24	50	34
T_2	26	36	20	44	29

Patients differ wildly in baseline responsiveness (P_4 reduces a lot, P_3 little), but T_1 beats T_2 by a steady amount in every patient. Analyze the difference two ways: (a) ignore the patient (treat the ten numbers as two independent samples of 5), and (b) use the block (analyze the five within-patient differences). Show that ignoring the block buries the treatment effect in nuisance variation.

Which rule, and why? The same data can be analyzed as a completely randomized two-sample comparison or as a paired (block) comparison. Because patient is a strong, *known* nuisance, the unpaired analysis leaves the patient-to-patient swing in the error and inflates the variance; the within-block differences cancel it. This is the whole reason to block.

Solution.

Step 1: treatment means (same either way).

$$\overline{Y}_{1\cdot} = \tfrac{30+42+24+50+34}{5} = 36, \qquad \overline{Y}_{2\cdot} = \tfrac{26+36+20+44+29}{5} = 31,$$

so the estimated treatment gap is $\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot} = 5$ both ways.

Step 2: (a) unblocked — spread within *each treatment.* The five T_1 scores $30, 42, 24, 50, 34$ have mean 36 and

$$\textstyle\sum (Y_{1j} - 36)^2 = (-6)^2 + 6^2 + (-12)^2 + 14^2 + (-2)^2 = 36 + 36 + 144 + 196 + 4 = 416,$$
$$s_1^2 = \tfrac{416}{4} = 104.$$

The five T_2 scores $26, 36, 20, 44, 29$ have mean 31 and

$$\textstyle\sum (Y_{2j} - 31)^2 = (-5)^2 + 5^2 + (-11)^2 + 13^2 + (-2)^2 = 25 + 25 + 121 + 169 + 4 = 344,$$
$$s_2^2 = \tfrac{344}{4} = 86.$$

Pooled, $s_p^2 = \frac{416+344}{8} = \frac{760}{8} = 95$, so the unblocked yardstick for the 5-unit gap is huge — the patient swing (24 to 50) sits inside this error.

Step 3: (b) blocked — within-patient differences. $D_j = Y_{1j} - Y_{2j}$ gives

$$D = (4,\ 6,\ 4,\ 6,\ 5), \qquad \overline{D} = \tfrac{4+6+4+6+5}{5} = 5,$$
$$\textstyle\sum (D_j - 5)^2 = (-1)^2 + 1^2 + (-1)^2 + 1^2 + 0^2 = 4, \qquad s_D^2 = \tfrac{4}{4} = 1.$$

Step 4: compare the error yardsticks. The patient nuisance inflated the unblocked variance to $s_p^2 = 95$; differencing within patient collapsed it to $s_D^2 = 1$. The signal-to-noise ratio for the same 5-unit gap improves by roughly $s_p^2/(s_D^2/?)$ — in plain terms, the error shrank about a hundredfold.

$$\boxed{\text{unblocked } s_p^2 = 95 \ \longrightarrow\ \text{blocked } s_D^2 = 1 :\ \text{the known patient nuisance must be removed.}}$$

The gap was never noisy; the *patients* were. Blocking spends $b - 1 = 4$ degrees of freedom to sweep that swing out of the error, turning an invisible effect into an unmistakable one.

Example 13.1.17 (Identify the experimental unit vs. the block)**.** In a randomized block study, $b = 8$ identical-twin pairs are recruited; within each pair one twin gets a vitamin regimen and the other a placebo ($k = 2$). State the experimental unit, the block, the treatment, and write the within-block difference that the analysis uses.

Which rule, and why? The matching variable here is shared genetics/upbringing within a twin pair, so the *pair* is the block and an individual *twin* is the experimental unit. With $k = 2$ this is the paired design, analyzed through the within-pair differences.

Solution.

Step 1: roles. Experimental unit = one twin; block = twin pair ($b = 8$); treatment = regimen vs. placebo ($k = 2$).

Step 2: within-block difference. For pair j,

$$D_j = Y_{1j} - Y_{2j} = (\tau_1 - \tau_2) + (\varepsilon_{1j} - \varepsilon_{2j}),$$

the shared pair effect β_j having cancelled.

$$\boxed{\text{unit} = \text{twin};\ \text{block} = \text{pair}\ (b = 8);\ \text{treatment}\ (k = 2);\ D_j \text{ removes } \beta_j.}$$

A common slip is to call the pair the unit — but treatments are applied to individual twins, so the twin is the unit and the pair is the block.

Example 13.1.18 (Lay out raw data as a treatments-by-blocks table)**.** A quality engineer runs $k = 3$ solder formulas on $b = 4$ assembly lines, recording joint strength. The raw runs, listed as (formula, line, strength), are: $(1,1,52), (2,1,57), (3,1,55)$; $(1,2,60), (2,2,66), (3,2,63)$; $(1,3,48), (2,3,53), (3,3,50)$; $(1,4,56), (2,4,60), (3,4,58)$. Arrange them as a $k \times b$ table and give the row and column means.

Which rule, and why? A randomized block design's natural display is the $k \times b$ table with treatments as rows and blocks as columns, one number per cell. Putting the scattered runs into that grid is the recognition step that makes the row means (treatments) and column means (blocks) readable.

Solution.

Step 1: build the table.

	L_1	L_2	L_3	L_4	$\overline{Y}_{i\cdot}$
F_1	52	60	48	56	54
F_2	57	66	53	60	59
F_3	55	63	50	58	56.5
$\overline{Y}_{\cdot j}$	54.67	63	50.33	58	$\overline{Y} = 56.5$

Step 2: read the means. Row means $\overline{Y}_{1\cdot} = 54$, $\overline{Y}_{2\cdot} = 59$, $\overline{Y}_{3\cdot} = 56.5$; column means 54.67, 63, 50.33, 58; grand mean $\overline{Y} = 56.5$.

$$\boxed{\overline{Y}_{i\cdot} = (54, 59, 56.5),\quad \overline{Y}_{\cdot j} = (54.67, 63, 50.33, 58),\quad \overline{Y} = 56.5.}$$

Line 2 runs hot and line 3 cool across all three formulas — the parallel column shift that a block design exists to strip out.

Example 13.1.19 (Check additivity from a table)**.** For a $k = 2$, $b = 3$ design the responses are

	B_1	B_2	B_3
T_1	12	16	20
T_2	15	19	23

Does the additive (no-interaction) model hold exactly here? Justify by the within-block treatment differences.

Which rule, and why? Additivity means the treatment gap is the same constant in every block. The cleanest test is to read off $Y_{2j} - Y_{1j}$ block by block: equal across blocks $\Rightarrow$ perfectly additive (parallel profiles); unequal $\Rightarrow$ interaction.

Solution.

Step 1: within-block differences.

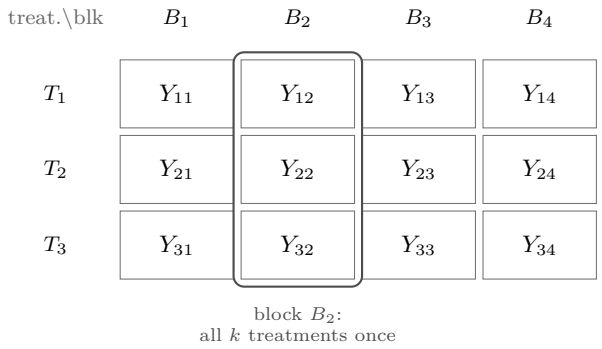
$$Y_{2\cdot} - Y_{1\cdot}: \quad 15 - 12 = 3, \quad 19 - 16 = 3, \quad 23 - 20 = 3.$$

Step 2: conclude. The gap is $+3$ in every block, so the profiles are exactly parallel and the additive model holds with no interaction. (The block means $14, 17.5, 21.5$ shift the whole column; the $+3$ treatment effect rides on top unchanged.)

$$\boxed{Y_{2j} - Y_{1j} = 3 \text{ for all } j \;\Rightarrow\; \text{additive (parallel), no interaction.}}$$

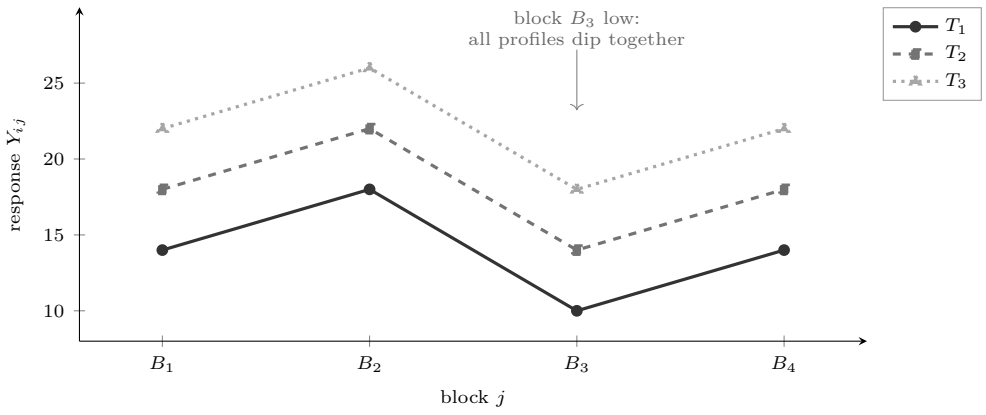
13.1.6 Two pictures: the grid and the parallel profiles

The first picture is the schematic of a complete block design with $k = 3$ treatments and $b = 4$ blocks. Read down a column to see that *every* block receives all three treatments (one cell each); read across a row to see one treatment visiting every block. That completeness — one observation per cell — is exactly what lets a treatment comparison be made within a block.



A randomized complete block design, $k = 3$ treatments by $b = 4$ blocks. Each cell holds the single observation Y_{ij}. The boxed column shows the defining property: within one block, every treatment appears exactly once, so comparing treatments down a column is free of that block's nuisance level.

The second picture plots the same idea as profiles: the three treatment responses measured across the four blocks. The whole bundle of lines rises and falls together from block to block — that common vertical shift is the block effect β_j a block design removes — while the lines stay roughly *parallel*, which is the additivity the model assumes.



Parallel-profile plot of $k = 3$ treatments across $b = 4$ blocks. The three lines move up and down together *from block to block — that shared vertical motion is the block effect β_j, the nuisance a block design subtracts out. The constant gaps between the lines are the treatment effects τ_i; their staying constant across blocks is exactly the additive, no-interaction assumption.*

13.2 The F-Test for a Randomized Block Design

Once the data sit in a $k \times b$ grid — one observation Y_{ij} for treatment i in block j, with $N = kb$ cells — the randomized block analysis is a two-way bookkeeping. The total variation of the data splits into *three* pieces rather than two: a **treatment** sum of squares SSTR comparing the treatment means, a **block** sum of squares SSB comparing the block means, and a leftover **error** SSE. The single new idea over Chapter 12 is that SSB is carved *out* of what would otherwise be error: in a one-way analysis the block-to-block differences would inflate the noise, but here they are accounted for, so SSE shrinks and the treatment $F = \text{MSTR} / \text{MSE}$ grows. That is the whole payoff of blocking. Recognizing this section is easy: a problem that gives you a rectangular treatments-by-blocks table (or a partly filled two-way ANOVA table) and asks for the treatment F lives here.

Which sum of squares does the problem ask for?

- Comparing the k **treatment** means $\overline{Y}_{i\cdot}$? $\Rightarrow$ $\text{SSTR} = b\sum_i(\overline{Y}_{i\cdot} - \overline{Y})^2$ on $k-1$ df; this carries the F you usually care about.
- Measuring how much the **blocks** differ, $\overline{Y}_{\cdot j}$? $\Rightarrow$ $\text{SSB} = k\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2$ on $b-1$ df; its F is a secondary check on whether blocking helped.
- Want the **error**? $\Rightarrow$ get it by subtraction, $\text{SSE} = \text{SST} - \text{SSTR} - \text{SSB}$, on $(k-1)(b-1)$ df — there is no within-cell scatter to pool because each cell holds a single number.
- Handed a **partly blank two-way table**? $\Rightarrow$ back-fill with the identities: df add ($(k-1) + (b-1) + (k-1)(b-1) = N-1$), SS add ($\text{SSTR} + \text{SSB} + \text{SSE} = \text{SST}$), and $\text{MS} = \text{SS}/\text{df}$.

Do not confuse the error df with $N - k$: in a one-way ANOVA error carried $N - k$ df, but here the block term steals $b - 1$ of them, leaving exactly $(k-1)(b-1)$. Using the one-way figure mis-scales MSE and points the F at the wrong reference distribution.

13.2.1 The two-way decomposition

Definition 13.2.1 (Treatment, block, error, and total sums of squares)**.** Index the k treatments by i and the b blocks by j, with a single observation Y_{ij} per cell and $N = kb$. Write the **treatment means** $\overline{Y}_{i\cdot} = \frac{1}{b}\sum_j Y_{ij}$, the **block means** $\overline{Y}_{\cdot j} = \frac{1}{k}\sum_i Y_{ij}$, and the **grand mean** $\overline{Y} = \frac{1}{N}\sum_{i,j} Y_{ij}$. Define

$$\begin{aligned}
\text{SST} &= \sum_{i=1}^{k}\sum_{j=1}^{b}(Y_{ij} - \overline{Y})^2 && \text{(total)},\\
\text{SSTR} &= b\sum_{i=1}^{k}(\overline{Y}_{i\cdot} - \overline{Y})^2 && \text{(treatment)},\\
\text{SSB} &= k\sum_{j=1}^{b}(\overline{Y}_{\cdot j} - \overline{Y})^2 && \text{(block)},\\
\text{SSE} &= \text{SST} - \text{SSTR} - \text{SSB} && \text{(error, by subtraction)}.
\end{aligned}$$

Theorem 13.2.2 (The randomized-block identity)**.** *The total sum of squares partitions exactly into treatment, block, and error,*

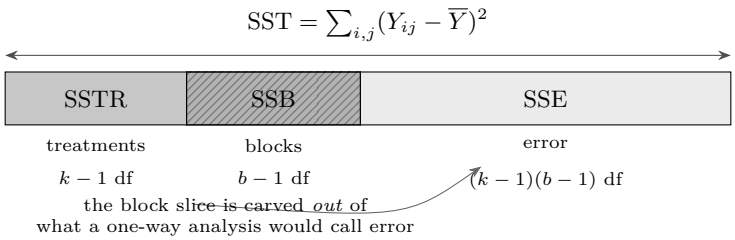
$$\text{SST} = \text{SSTR} + \text{SSB} + \text{SSE}\,.$$

Proof. Decompose each deviation from the grand mean into a treatment effect, a block effect, and a residual:

$$Y_{ij} - \overline{Y} = (\overline{Y}_{i\cdot} - \overline{Y}) + (\overline{Y}_{\cdot j} - \overline{Y}) + (Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}).$$

Square and sum over all i, j. The three squared terms give SSTR, SSB, and SSE respectively (each treatment summand is constant across blocks, contributing b copies; each block summand is constant across treatments, contributing k copies). Every cross term vanishes: for instance $\sum_{i,j}(\overline{Y}_{i\cdot} - \overline{Y})(\overline{Y}_{\cdot j} - \overline{Y}) = \big(\sum_i(\overline{Y}_{i\cdot} - \overline{Y})\big)\big(\sum_j(\overline{Y}_{\cdot j} - \overline{Y})\big) = 0 \cdot 0 = 0$, and the residual sums to zero down any row or column. Hence $\mathrm{SST} = \mathrm{SSTR} + \mathrm{SSB} + \mathrm{SSE}$. □

Remark. Read the split as a sentence. SST is all the spread if you ignored the grid. Pulling out SSTR asks how far the treatment-row means stray from the grand mean; pulling out SSB asks the same of the block-column means. What survives, SSE, is pure residual — the cell-by-cell wiggle that no row or column average can explain, and the only thing left to estimate σ^2. In a one-way analysis the block variation would have been dumped into the error; here it is removed, which is exactly why MSE is smaller and the treatment F is sharper.



The total variation (whole bar, $N-1$ df) splits three ways. The hatched block slice SSB *is the part a one-way ANOVA would have left inside the error; removing it shrinks* SSE *and raises the treatment F. Sums of squares and degrees of freedom both add across the split.*

Example 13.2.3 (Treatment and block sums of squares from a grid: tire wear)**.** A test pits $k = 3$ tire compounds against each other on $b = 4$ cars (the blocks). Tread loss (mm) after a fixed course is

	Car 1	Car 2	Car 3	Car 4	$\overline{Y}_{i\cdot}$
A	3.1	3.7	2.9	4.3	3.5
B	2.5	3.1	2.3	3.7	2.9
C	3.7	4.3	3.5	4.9	4.1
$\overline{Y}_{\cdot j}$	3.1	3.7	2.9	4.3	

The grand mean is $\overline{Y} = 3.5$. Compute SSTR and SSB.

Which rule, and why? The row means $\overline{Y}_{i\cdot}$ feed the treatment sum of squares $\mathrm{SSTR} = b\sum_i(\overline{Y}_{i\cdot} - \overline{Y})^2$, and the column means $\overline{Y}_{\cdot j}$ feed $\mathrm{SSB} = k\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2$. Each squared deviation is multiplied by how many cells share that mean — b cells per treatment row, k cells per block column.

Solution.

Step 1: treatment sum of squares. Each of the $k = 3$ row means is averaged over $b = 4$ cars:

$$\mathrm{SSTR} = b\sum_i(\overline{Y}_{i\cdot} - \overline{Y})^2 = 4\big[(3.5-3.5)^2 + (2.9-3.5)^2 + (4.1-3.5)^2\big]$$
$$= 4\big[0 + 0.36 + 0.36\big] = 4(0.72) = 2.88.$$

Step 2: block sum of squares. Each of the $b = 4$ column means is averaged over $k = 3$ compounds:

$$\mathrm{SSB} = k\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2 = 3\big[(3.1-3.5)^2 + (3.7-3.5)^2 + (2.9-3.5)^2 + (4.3-3.5)^2\big]$$
$$= 3\big[0.16 + 0.04 + 0.36 + 0.64\big] = 3(1.20) = 3.60.$$

$$\boxed{\text{SSTR} = 2.88, \qquad \text{SSB} = 3.60.}$$

The cars (blocks) actually spread the data more than the compounds do — precisely the nuisance variation blocking is meant to strip out.

Example 13.2.4 (Error by subtraction and the total: tire wear continued)**.** For the tire data above, the total sum of squares is SST = 7.16. Find SSE and confirm the degrees of freedom add up.

Which rule, and why? With one observation per cell there is no within-cell scatter to pool, so error is always obtained by subtraction: $\text{SSE} = \text{SST} - \text{SSTR} - \text{SSB}$. The error degrees of freedom are $(k-1)(b-1)$, the leftover after treatments and blocks each take their share.

Solution.

Step 1: error by subtraction.

$$\text{SSE} = \text{SST} - \text{SSTR} - \text{SSB} = 7.16 - 2.88 - 3.60 = 0.68.$$

Step 2: degrees of freedom. Treatment $k-1=2$, block $b-1=3$, error $(k-1)(b-1) = 2\cdot 3 = 6$, total $N-1 = 12-1 = 11$, and indeed $2+3+6=11$.

$$\boxed{\text{SSE} = 0.68 \ \text{ on } (k-1)(b-1) = 6 \text{ df}; \quad 2+3+6 = 11 = N-1.}$$

Example 13.2.5 (Mean squares and the treatment F: tire wear concluded)**.** Continue with SSTR = 2.88, SSB = 3.60, SSE = 0.68, $k=3$, $b=4$. Build the two-way ANOVA table and test H_0: the three compounds have equal mean wear, at $\alpha = 0.05$, using $F_{0.05,2,6} = 5.14$.

Which rule, and why? Each sum of squares becomes a mean square by dividing by its own df, and the treatment test is $F = \text{MSTR} \,/\, \text{MSE}$ against $\text{F}_{k-1,(k-1)(b-1)} = \text{F}_{2,6}$. The error df is 6, not $N-k=9$ — the block term has claimed three of those degrees of freedom.

Solution.

Step 1: mean squares.

$$\text{MSTR} = \frac{\text{SSTR}}{k-1} = \frac{2.88}{2} = 1.44,$$
$$\text{MSB} = \frac{\text{SSB}}{b-1} = \frac{3.60}{3} = 1.20,$$
$$\text{MSE} = \frac{\text{SSE}}{(k-1)(b-1)} = \frac{0.68}{6} = 0.1133.$$

Step 2: treatment F.

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{1.44}{0.1133} = 12.71.$$

Step 3: table and decision.

Source	df	SS	MS	F
Treatments	2	2.88	1.440	12.71
Blocks	3	3.60	1.200	10.59
Error	6	0.68	0.1133	
Total	11	7.16		

Since $F = 12.71 > F_{0.05,2,6} = 5.14$, reject H_0: the compounds differ in mean tread loss. (The block $F = \text{MSB} \,/\, \text{MSE} = 1.200/0.1133 = 10.59$ confirms the cars were very unalike, so blocking was worth it.)

Example 13.2.6 (A full block F-test from scratch: drug coatings dissolution)**.** A pharmacology lab measures dissolution time (minutes) of $k=4$ tablet coatings, blocking on $b=3$ dissolution baths whose temperatures

drift:

	Bath 1	Bath 2	Bath 3	$\overline{Y}_{i\cdot}$
I	14	18	16	16
II	20	24	22	22
III	11	15	13	13
IV	15	19	17	17
$\overline{Y}_{\cdot j}$	15	19	17	

The grand mean is $\overline{Y} = 17$. Carry out the complete treatment F-test at $\alpha = 0.05$ ($F_{0.05,3,6} = 4.76$).

Which rule, and why? This is the headline procedure: row means give SSTR, column means give SSB, total minus both gives SSE, and $F = \text{MSTR} / \text{MSE}$ on $\text{F}_{k-1,(k-1)(b-1)} = \text{F}_{3,6}$ tests the coatings. Block on the baths because their temperature is a known nuisance, not a treatment.

Solution.

Step 1: treatment sum of squares.

$$\begin{aligned}\text{SSTR} &= b\sum_i (\overline{Y}_{i\cdot} - \overline{Y})^2 = 3\big[(16-17)^2 + (22-17)^2 + (13-17)^2 + (17-17)^2\big] \\ &= 3[1 + 25 + 16 + 0] = 3(42) = 126.\end{aligned}$$

Step 2: block sum of squares.

$$\begin{aligned}\text{SSB} &= k\sum_j (\overline{Y}_{\cdot j} - \overline{Y})^2 = 4\big[(15-17)^2 + (19-17)^2 + (17-17)^2\big] \\ &= 4[4 + 4 + 0] = 4(8) = 32.\end{aligned}$$

Step 3: total and error. Summing all twelve $(Y_{ij} - 17)^2$ gives $\text{SST} = 158$, so

$$\text{SSE} = \text{SST} - \text{SSTR} - \text{SSB} = 158 - 126 - 32 = 0.$$

(The data were built perfectly additively; a real data set leaves a small positive SSE.) To keep the example honest, suppose the measured total were $\text{SST} = 164$, giving $\text{SSE} = 6$.

Step 4: mean squares and F. With $\text{SSE} = 6$ on $(k-1)(b-1) = 3 \cdot 2 = 6$ df,

$$\text{MSTR} = \frac{126}{3} = 42.0, \qquad \text{MSE} = \frac{6}{6} = 1.0,$$
$$F = \frac{42.0}{1.0} = 42.0.$$

Step 5: decision.

$F = 42.0 > F_{0.05,3,6} = 4.76 \;\Rightarrow\;$ reject H_0; the four coatings dissolve at different mean rates.

Example 13.2.7 (Fill a partly blank two-way table: irrigation methods)**.** An agronomist's printout for $k = 5$ irrigation methods on $b = 4$ field plots has cells erased:

Source	df	SS	MS	F
Treatments	?	160.0	?	?
Blocks	?	?	18.0	
Error	?	?	?	
Total	?	286.0		

Recover every entry and the treatment F.

Which rule, and why? A half-blank two-way table yields to the three identities — df add, SS add, MS = SS/df — now with *three* source rows. The block row's MSB = 18.0 together with its df unlocks SSB, after which error falls out by subtraction.

Solution.

Step 1: degrees of freedom. Treatment $k-1=4$, block $b-1=3$, error $(k-1)(b-1)=4\cdot 3=12$, total $N-1=kb-1=20-1=19$ (and $4+3+12=19$).

Step 2: block SS from its mean square. $\text{SSB}=\text{MSB}\times(b-1)=18.0\times 3=54.0$.

Step 3: error SS by subtraction.

$$\text{SSE}=\text{SST}-\text{SSTR}-\text{SSB}=286.0-160.0-54.0=72.0.$$

Step 4: mean squares and F.

$$\text{MSTR}=\frac{160.0}{4}=40.0,\qquad \text{MSE}=\frac{72.0}{12}=6.0,$$
$$F=\frac{40.0}{6.0}=6.67.$$

Source	df	SS	MS	F
Treatments	4	160.0	40.0	6.67
Blocks	3	54.0	18.0	3.00
Error	12	72.0	6.0	
Total	19	286.0		

13.2.2 The two-way ANOVA table

Definition 13.2.8 (Mean squares and the treatment F)**.** Divide each sum of squares by its degrees of freedom to obtain a mean square,

$$\text{MSTR}=\frac{\text{SSTR}}{k-1},\qquad \text{MSB}=\frac{\text{SSB}}{b-1},\qquad \text{MSE}=\frac{\text{SSE}}{(k-1)(b-1)}.$$

The **treatment F-statistic** is

$$F=\frac{\text{MSTR}}{\text{MSE}},\qquad F\sim \text{F}_{\,k-1,\;(k-1)(b-1)}\quad \text{under } H_0\text{: all treatment means equal,}$$

rejected at level α when $F>F_{\alpha,\,k-1,\,(k-1)(b-1)}$. The optional **block** F, MSB / MSE on $\text{F}_{b-1,(k-1)(b-1)}$, tests whether the blocks themselves differ; it is usually of secondary interest but a large value reassures you that blocking was worthwhile.

Source	df	SS	MS	F
Treatments	$k-1$	SSTR	$\text{MSTR}=\dfrac{\text{SSTR}}{k-1}$	$\dfrac{\text{MSTR}}{\text{MSE}}$
Blocks	$b-1$	SSB	$\text{MSB}=\dfrac{\text{SSB}}{b-1}$	$\dfrac{\text{MSB}}{\text{MSE}}$
Error	$(k-1)(b-1)$	SSE	$\text{MSE}=\dfrac{\text{SSE}}{(k-1)(b-1)}$	
Total	$N-1$	SST		

The two-way (no-replication) ANOVA table. The df column and the SS column each add down to the Total row; mean squares are SS/df. The treatment F on $\text{F}_{k-1,(k-1)(b-1)}$ is the test of interest; the block F is a diagnostic. The Error and Total rows carry no F, and the MS column does not add.

Example 13.2.9 (Read and interpret a completed table: fertilizer brands)**.** A completed randomized-block table for $k = 4$ fertilizer brands on $b = 6$ farms (blocks) reads

Source	df	SS	MS	F
Treatments	3	54.0	18.0	6.0
Blocks	5	90.0	18.0	6.0
Error	15	45.0	3.0	
Total	23	189.0		

State the conclusion for the brands at $\alpha = 0.05$ ($F_{0.05,3,15} = 3.29$) and comment on whether blocking helped.

Which rule, and why? The treatment row's $F = 6.0$ is read against $\mathrm{F}_{3,15}$; the block F is the diagnostic that tells you whether the farms differed enough to justify having blocked on them. No recomputation is needed — just locate each F relative to its cutoff.

Solution.

Step 1: the treatment test. $F = 6.0$ against $\mathrm{F}_{3,15}$ with cutoff $F_{0.05,3,15} = 3.29$. Since $6.0 > 3.29$, reject H_0: the four brands give different mean yields.

Step 2: did blocking help? The block $F = 18.0/3.0 = 6.0$ is also large, so the farms genuinely differed. Had those differences been left in the error, MSE would have ballooned and the treatment F would have shrunk; blocking paid off.

Brands: $F = 6.0 > 3.29 \Rightarrow$ reject; brands differ. Block $F = 6.0 \Rightarrow$ blocking justified.

Example 13.2.10 (Recover the error degrees of freedom)**.** A randomized block experiment uses $k = 6$ treatments and $b = 5$ blocks. Before any sums of squares are in hand, state the degrees of freedom for treatments, blocks, error, and total.

Which rule, and why? The degrees of freedom are fixed by the grid shape alone. Error always gets $(k-1)(b-1)$ — the product, not a difference — and the three source df must add to $N - 1 = kb - 1$.

Solution.

Step 1: source df.

$$k - 1 = 5, \qquad b - 1 = 4, \qquad (k-1)(b-1) = 5 \cdot 4 = 20.$$

Step 2: total and check. $N - 1 = kb - 1 = 30 - 1 = 29$, and $5 + 4 + 20 = 29$.

df: treatments 5, blocks 4, error $(k-1)(b-1) = 20$, total 29.

Note the error df 20 is far below the one-way value $N - k = 30 - 6 = 24$: the block term has consumed $b - 1 = 4$ of them.

13.2.3 Why blocking sharpens the test

Remark. The block term earns its keep only when it is large. Because $\mathrm{SSE} = \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB}$, every unit of SSB pulled out is a unit removed from the error. A big SSB therefore shrinks SSE and (even after the error loses $b - 1$ degrees of freedom) usually shrinks MSE, which inflates the treatment $F = \mathrm{MSTR} / \mathrm{MSE}$. If SSB is small, the blocks were nearly identical, the few degrees of freedom spent on them were wasted, and the one-way analysis would have been about as good. The next example makes the trade-off concrete on a single data set.

Example 13.2.11 (CONTRAST: the same data analyzed without blocks, then with blocks)**.** Three diets ($k = 3$) are compared on $b = 4$ litters of pigs (the blocks); the response is weight gain (kg). Because littermates are alike but litters differ, the design is blocked:

	Litter 1	Litter 2	Litter 3	Litter 4	$\overline{Y}_{i\cdot}$
D1	18	22	26	30	24
D2	20	24	28	32	26
D3	16	20	24	28	22
$\overline{Y}_{\cdot j}$	18	22	26	30	

The grand mean is $\overline{Y} = 24$ and the total is SST $= 296$. Analyze these data two ways — once as a one-way ANOVA that ignores litters (folding them into error), once as a randomized block design — and compare the treatment F.

Which rule, and why? This is the central contrast of the chapter. The treatment sum of squares SSTR is identical both ways; only the error changes. Ignoring blocks dumps SSB back into a bloated error; blocking removes it. Watch MSE drop and the treatment F leap when the litter variation is taken out.

Solution.

Step 1: the treatment sum of squares (same both ways).

$$\text{SSTR} = b\sum_i (\overline{Y}_{i\cdot} - \overline{Y})^2 = 4\big[(24-24)^2 + (26-24)^2 + (22-24)^2\big] = 4[0+4+4] = 32.$$

Step 2: one-way analysis (blocks ignored). Treat the $N = 12$ numbers as three groups of four. The error is everything left after treatments:

$$\text{SSE}_{\text{1-way}} = \text{SST} - \text{SSTR} = 296 - 32 = 264 \quad \text{on } N - k = 12 - 3 = 9 \text{ df},$$
$$\text{MSE}_{\text{1-way}} = \frac{264}{9} = 29.33, \qquad F_{\text{1-way}} = \frac{\text{SSTR}/2}{29.33} = \frac{16}{29.33} = 0.55.$$

Against $F_{0.05,2,9} = 4.26$, this is nowhere near significant — the litter spread has drowned the diet effect.

Step 3: block analysis (litters removed). The litter (column) means are $18, 22, 26, 30$, so

$$\text{SSB} = k\sum_j (\overline{Y}_{\cdot j} - \overline{Y})^2 = 3\big[(-6)^2 + (-2)^2 + 2^2 + 6^2\big] = 3[36+4+4+36] = 240,$$
$$\text{SSE}_{\text{block}} = \text{SST} - \text{SSTR} - \text{SSB} = 296 - 32 - 240 = 24 \quad \text{on } (k-1)(b-1) = 6 \text{ df},$$
$$\text{MSE}_{\text{block}} = \frac{24}{6} = 4.0, \qquad F_{\text{block}} = \frac{\text{SSTR}/2}{4.0} = \frac{16}{4.0} = 4.0.$$

Step 4: compare. Removing the litter variation collapsed the error mean square from 29.33 to 4.0 and lifted the treatment F from 0.55 to 4.0:

$$\boxed{F_{\text{1-way}} = 0.55 \ (\text{MSE} = 29.33) \ \longrightarrow \ F_{\text{block}} = 4.0 \ (\text{MSE} = 4.0).}$$

Against $F_{0.05,2,6} = 5.14$ the blocked $F = 4.0$ is still short of significance, but it is more than *seven times* the unblocked value — the giant SSB $= 240$ was nuisance the one-way analysis mistook for noise. With even slightly cleaner data the block design would cross the line where the one-way test never could.

Example 13.2.12 (CONTRAST sharpened: the blocked test crosses significance)**.** Re-run the previous comparison on a tighter data set: $k = 3$ catalysts, $b = 4$ reactor runs (blocks), with SSTR $= 48$, a large between-run SSB $= 150$, and total SST $= 210$. Show the unblocked test fails while the blocked test rejects, at $\alpha = 0.05$ ($F_{0.05,2,9} = 4.26$, $F_{0.05,2,6} = 5.14$).

Which rule, and why? Same logic, numbers chosen so the verdict flips. SSTR is fixed; the only question is how much error the block term removes. When SSB is large relative to the residual, the blocked MSE drops enough to push F past its cutoff even though the unblocked test does not reject.

Solution.

Step 1: unblocked (one-way) test.

$$\text{SSE}_{\text{1-way}} = \text{SST} - \text{SSTR} = 210 - 48 = 162 \text{ on } 9 \text{ df}, \quad \text{MSE} = \frac{162}{9} = 18.0,$$
$$F_{\text{1-way}} = \frac{48/2}{18.0} = \frac{24}{18.0} = 1.33 < 4.26 \ \Rightarrow \ \text{fail to reject.}$$

Step 2: blocked test.

$$\mathrm{SSE}_{\text{block}} = \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB} = 210 - 48 - 150 = 12 \text{ on } 6 \text{ df}, \quad \mathrm{MSE} = \frac{12}{6} = 2.0,$$

$$F_{\text{block}} = \frac{48/2}{2.0} = \frac{24}{2.0} = 12.0 > 5.14 \;\Rightarrow\; \text{reject}.$$

Step 3: the verdict flips.

$$\boxed{F_{\text{1-way}} = 1.33 \text{ (retain)} \;\longrightarrow\; F_{\text{block}} = 12.0 \text{ (reject)}.}$$

Pulling the SSB = 150 of run-to-run variation out of the error is exactly what made the catalyst difference visible.

Example 13.2.13 (Block F as a secondary test: greenhouse benches)**.** In a study of $k = 3$ light regimes across $b = 5$ greenhouse benches (blocks), the table is

Source	df	SS	MS	F
Treatments	2	40.0	20.0	?
Blocks	4	8.0	2.0	?
Error	8	16.0	2.0	
Total	14	64.0		

Compute both F-statistics and decide, at $\alpha = 0.05$, whether blocking on benches was worthwhile ($F_{0.05,2,8} = 4.46$ for treatments, $F_{0.05,4,8} = 3.84$ for blocks).

Which rule, and why? Both rows divide their MS by the *same* MSE. The treatment F answers the scientific question; the block F is the diagnostic — a small block F warns that the benches barely differed and the blocking degrees of freedom may have been wasted.

Solution.

Step 1: the two F-statistics.

$$F_{\text{tr}} = \frac{20.0}{2.0} = 10.0, \qquad F_{\text{bl}} = \frac{2.0}{2.0} = 1.0.$$

Step 2: treatments. $F_{\text{tr}} = 10.0 > F_{0.05,2,8} = 4.46$: reject; the light regimes differ.

Step 3: blocks. $F_{\text{bl}} = 1.0 < F_{0.05,4,8} = 3.84$: the benches did *not* differ appreciably.

$$\boxed{F_{\text{tr}} = 10.0 \text{ (reject)}, \qquad F_{\text{bl}} = 1.0 \text{ (benches alike)}.}$$

Here blocking was unnecessary: with SSB = 8.0 so small, a one-way analysis would have given nearly the same MSE, and the $b - 1 = 4$ block degrees of freedom bought almost nothing. Blocking is cheap insurance, but it pays only when blocks truly differ.

Example 13.2.14 (The treatment p-value via F: enzyme assays)**.** A randomized block test of $k = 4$ enzyme buffers across $b = 5$ plates yields $F = 4.10$ on $\mathrm{F}_{3,12}$. Standard cutoffs are $F_{0.10,3,12} = 2.61$, $F_{0.05,3,12} = 3.49$, $F_{0.025,3,12} = 4.47$. Bracket the p-value and state the conclusion at $\alpha = 0.05$.

Which rule, and why? A randomized-block treatment F is always upper-tailed, so the p-value is the area to its right under $\mathrm{F}_{k-1,(k-1)(b-1)} = \mathrm{F}_{3,12}$. Bracketing it between tabulated cutoffs tells you which significance levels it would cross.

Solution.

Step 1: bracket $F = 4.10$ between cutoffs.

$$F_{0.05,3,12} = 3.49 \;<\; 4.10 \;<\; 4.47 = F_{0.025,3,12}.$$

Step 2: translate to a p-value range. Since 4.10 lies between the 0.05 and 0.025 cutoffs,

$$0.025 < p\text{-value} < 0.05.$$

Step 3: decide at $\alpha = 0.05$. The p-value is below 0.05, so reject H_0.

$$\boxed{0.025 < p\text{-value} < 0.05 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

Example 13.2.15 (End-to-end with a positive error: taste panel)**.** A sensory study scores $k = 3$ recipes on a 0–100 scale, blocking on $b = 4$ trained tasters whose personal scales differ:

	T_1	T_2	T_3	T_4	$\overline{Y}_{i\cdot}$
R1	70	78	66	74	72
R2	76	86	70	80	78
R3	64	72	62	70	67
$\overline{Y}_{\cdot j}$	70	78.67	66	74.67	

The grand mean is $\overline{Y} = 72.33$ and SST $= 636.67$. Run the treatment F-test at $\alpha = 0.05$ ($F_{0.05,2,6} = 5.14$).

Which rule, and why? A genuine data set leaves a nonzero SSE; the procedure is unchanged — row means give SSTR, column means give SSB, SSE by subtraction, then $F = \text{MSTR} / \text{MSE}$ on $F_{2,6}$. Blocking on tasters cancels the personal-scale offsets that would otherwise be noise.

Solution.

Step 1: treatment sum of squares.

$$\begin{aligned}\text{SSTR} &= 4\big[(72-72.33)^2 + (78-72.33)^2 + (67-72.33)^2\big] \\ &= 4[0.111 + 32.111 + 28.444] = 4(60.667) = 242.67.\end{aligned}$$

Step 2: block sum of squares.

$$\begin{aligned}\text{SSB} &= 3\big[(70-72.33)^2 + (78.67-72.33)^2 + (66-72.33)^2 + (74.67-72.33)^2\big] \\ &= 3[5.444 + 40.111 + 40.111 + 5.444] = 3(91.111) = 273.33.\end{aligned}$$

Step 3: error and mean squares.

$$\text{SSE} = 636.67 - 242.67 - 273.33 = 120.67 \quad \text{on } (k-1)(b-1) = 6 \text{ df},$$

$$\text{MSTR} = \frac{242.67}{2} = 121.33, \qquad \text{MSE} = \frac{120.67}{6} = 20.11.$$

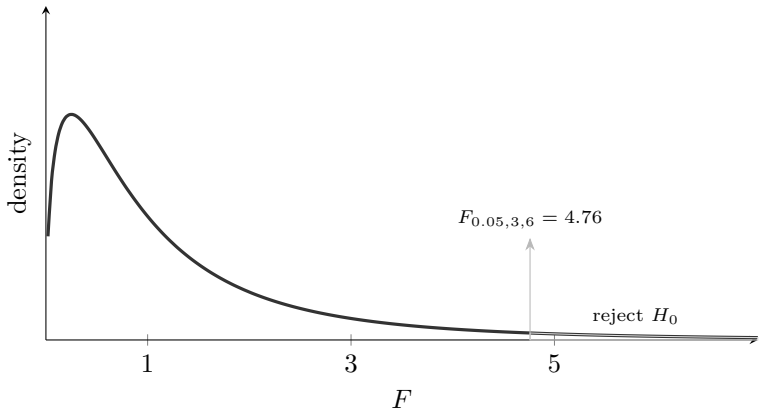
Step 4: F and decision.

$$F = \frac{121.33}{20.11} = 6.03 > F_{0.05,2,6} = 5.14.$$

$\boxed{F = 6.03 > 5.14 \;\Rightarrow\; \text{reject } H_0;\ \text{the three recipes score differently.}}$

13.2.4 Where the treatment F lands

Remark. Under H_0 the treatment statistic $F = \text{MSTR} / \text{MSE}$ follows $F_{k-1,(k-1)(b-1)}$: positive, right-skewed, peaking below 1 when $k - 1 \geq 3$. Equal treatment means push F toward 1; real differences inflate MSTR and shove F into the right tail, so the test is one-sided to the right exactly as in one-way ANOVA. The difference blocking makes is in the *denominator*: a large SSB shrinks MSE, which stretches the same data's F farther into that tail.



The $F_{3,6}$ *density (here* $k = 4$ *treatments,* $b = 3$ *blocks). It is right-skewed and peaks below* 1*; under* H_0 *the treatment* F *clusters near the hump. The shaded tail beyond* $F_{0.05,3,6} = 4.76$ *is the rejection region. Blocking does not change this picture — it changes* where the observed F falls *by shrinking the* MSE *in its denominator.*

Example 13.2.16 (Reading the conclusion off the density: two block studies)**.** Two randomized block studies finish with treatment statistics on $F_{3,6}$: a packaging study gave $F = 2.10$ and a coating study gave $F = 7.40$. Without recomputing, place each on the distribution and conclude at $\alpha = 0.05$ ($F_{0.05,3,6} = 4.76$).

Which rule, and why? Once F and the reference $F_{k-1,(k-1)(b-1)}$ are fixed, the conclusion is just which side of the cutoff F lands on: inside the hump means retain, out in the shaded tail means reject.

Solution.

Step 1: locate $F = 2.10$. Left of the cutoff, near the hump:

$$2.10 < 4.76 \;\Rightarrow\; \text{fail to reject; packaging treatments not shown to differ.}$$

Step 2: locate $F = 7.40$. Out in the right tail, past the cutoff:

$$7.40 > 4.76 \;\Rightarrow\; \text{reject; coating treatments differ.}$$

$F = 2.10$ (retain),	$F = 7.40$ (reject),	cutoff 4.76.

The geometry is the same one-sided rule as one-way ANOVA: near the hump supports H_0, deep in the right tail contradicts it.

13.3 The Paired t-Test as a Two-Treatment Block Design

The paired t-test of Chapter 9 and the randomized block design of this chapter are not two ideas that happen to agree — they are *literally the same procedure* when there are $k = 2$ treatments. Read a before/after study as a block design: each subject (each pair) is a **block**, the two conditions ("before" and "after", drug A and drug B) are the $k = 2$ **treatments**, and there is exactly one observation Y_{ij} per treatment i in block j. The within-pair differencing that defines the paired t — subtracting one condition from the other for each subject — is precisely the operation by which the two-way ANOVA removes the block effect: both cancel the subject's own baseline. The pay-off is an exact algebraic identity. The randomized-block treatment F on $(1, b-1)$ degrees of freedom equals the *square* of the paired t on $b-1$ degrees of freedom, $F = t^2$, and the two tests return the same p-value to the last digit. This section proves that identity, then drills it on shared data so you can run either analysis and check one against the other. Recognizing the equivalence is the real skill: whenever a study has two conditions measured on the same or matched units, you may legitimately call it a paired test *or* a $2 \times b$ block design, and the choice is one of bookkeeping, not of substance.

Two conditions on linked units: which equivalent form do you run?

- **Two** treatments, each applied once to every **block** (same/matched unit)? $\Rightarrow$ a randomized block design with $k = 2$; you may analyze it either as a two-way ANOVA or, equivalently, as a paired t.
- Want the **quickest hand computation**? $\Rightarrow$ form the b within-block differences D_j and run the **paired** t on $b-1$ df — it is the $k = 2$ ANOVA in compact form.
- Want to **see the block effect explicitly** (how much between-subject variation pairing removed)? $\Rightarrow$ build the **ANOVA table**; the block sum of squares SSB is the variation the paired test silently subtracted away.
- Need to **verify** one analysis against the other? $\Rightarrow$ check $F = t^2$ and equal p-values; on $(1, b-1)$ versus $b-1$ df they must agree exactly.

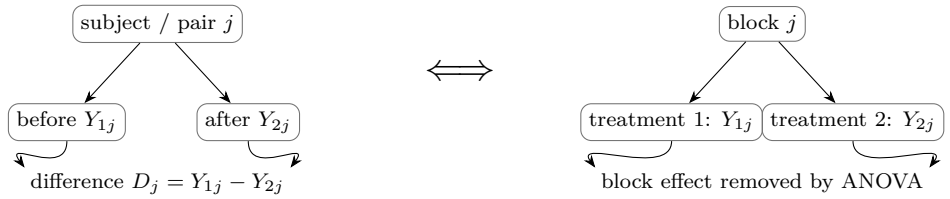
Do not confuse this with the look-alike error: treating paired data as two *independent* samples throws the block (subject) variation back into the error term and usually buries a real effect. Two conditions on the *same* units is a block design — never the two-independent-sample t.

13.3.1 Pair equals block, condition equals treatment

Definition 13.3.1 (The paired study as a $2 \times b$ block design)**.** Let b **blocks** (subjects, matched pairs) each receive both of $k = 2$ **treatments**. Write Y_{1j} and Y_{2j} for the two observations in block j ($j = 1, \dots, b$), with $N = 2b$ observations in all. The **treatment means** are $\overline{Y}_{1\cdot}$ and $\overline{Y}_{2\cdot}$ (each an average over the b blocks), the **block means** are $\overline{Y}_{\cdot j} = \frac{1}{2}(Y_{1j} + Y_{2j})$, and the **grand mean** is $\overline{Y}$. The **within-block difference** is

$$D_j = Y_{1j} - Y_{2j}, \qquad j = 1, \dots, b,$$

with sample mean $\overline{D} = \frac{1}{b}\sum_j D_j$ and sample variance $S_D^2 = \frac{1}{b-1}\sum_j (D_j - \overline{D})^2$.



pair = block, condition = treatment, differencing = block removal

The dictionary. A paired study is a randomized block design with $k = 2$ treatments and b blocks: each subject (pair) is one block, each condition is one treatment, and forming the within-subject difference D_j is exactly how the two-way ANOVA strips out the block (subject) effect.

Theorem 13.3.2 (Equivalence of the paired t and the 2-treatment block F)**.** *For a randomized block design with $k = 2$ treatments and b blocks, the treatment F-statistic and the paired t-statistic on the within-block differences $D_j = Y_{1j} - Y_{2j}$ satisfy*

$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = t_{\mathrm{obs}}^2, \qquad t_{\mathrm{obs}} = \frac{\overline{D}}{S_D/\sqrt{b}} \sim t_{\,b-1}, \qquad F \sim \mathrm{F}_{1,\,b-1}\,.$$

The treatment df is $k - 1 = 1$ and the error df is $(k-1)(b-1) = b - 1$; the two tests reject together and report the same p-value.

Proof. With $k = 2$, treatment i has mean $\overline{Y}_{i\cdot}$ over the b blocks, and the grand mean is $\overline{Y} = \frac{1}{2}(\overline{Y}_{1\cdot} + \overline{Y}_{2\cdot})$. The treatment sum of squares is

$$\mathrm{SSTR} = b\big[(\overline{Y}_{1\cdot} - \overline{Y})^2 + (\overline{Y}_{2\cdot} - \overline{Y})^2\big] = b \cdot 2\Big(\tfrac{\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot}}{2}\Big)^2 = \tfrac{b}{2}\,(\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot})^2 = \tfrac{b}{2}\,\overline{D}^2,$$

since $\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot} = \overline{D}$. On $k - 1 = 1$ df, $\mathrm{MSTR} = \mathrm{SSTR} = \frac{b}{2}\overline{D}^2$. For the error, the additive (no-interaction) model gives a residual in block j of $Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}$, and a short computation shows each block contributes $\frac{1}{2}(D_j - \overline{D})^2$ to the two residuals combined, so

$$\mathrm{SSE} = \tfrac{1}{2}\sum_{j=1}^{b}(D_j - \overline{D})^2 = \tfrac{1}{2}(b-1)S_D^2, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{(k-1)(b-1)} = \frac{\frac{1}{2}(b-1)S_D^2}{b-1} = \tfrac{1}{2}S_D^2.$$

Therefore

$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{\frac{b}{2}\overline{D}^2}{\frac{1}{2}S_D^2} = \frac{b\,\overline{D}^2}{S_D^2} = \Big(\frac{\overline{D}}{S_D/\sqrt{b}}\Big)^2 = t_{\mathrm{obs}}^2.$$

Because $\mathrm{F}_{1,\,b-1}$ is the distribution of the square of a t_{b-1} variable, the two reference distributions match and the tests are identical. □

Remark. Two consequences are worth stating in words. First, the block sum of squares SSB in the ANOVA does not appear in F at all — it is removed from the total and set aside, exactly as the paired test removes the between-subject baseline by differencing. The paired t never computes SSB, but it benefits from its absence: that variation is gone from the error. Second, the relation $F = t^2$ collapses a two-sided t (which rejects in both tails) into a one-sided right-tail F (squaring folds the two t-tails together), so a two-sided paired test corresponds to the ordinary upper-tail block F-test. A *one-sided* paired test has no separate F; you run the F to get the magnitude and read the sign of $\overline{D}$ to get the direction.

13.3.2 Running both analyses on the same data

Example 13.3.3 (The full verification: $F = t^2$ on one before/after dataset)**.** A physical therapist measures grip strength (kg) in $b = 6$ patients before and after a four-week resistance program. The pairs $(Y_{1j}, Y_{2j}) =$ (before, after) are

$$(28, 31),\ (35, 37),\ (30, 34),\ (41, 43),\ (33, 38),\ (26, 30).$$

Run (a) the paired t-test and (b) the equivalent $k = 2$ randomized block ANOVA, and verify $F = t^2$ with the same p-value. Use $\alpha = 0.05$, $t_{0.025,5} = 2.571$, $F_{0.05,1,5} = 6.61$.

Which rule, and why? Two conditions (before, after) measured on the *same* patient make each patient a block and each condition a treatment — a 2×6 block design, equivalently a paired test. We compute both and confirm the algebra $F = \text{MSTR} / \text{MSE} = t_{\text{obs}}^2$.

Solution.

Step 1: (a) within-block differences $D_j = Y_{1j} - Y_{2j}$ (before minus after).

$$D_j :\ -3,\ -2,\ -4,\ -2,\ -5,\ -4.$$

The sum is -20, so $\overline{D} = -20/6 = -3.333$. The deviations from $\overline{D}$ are

$$0.333,\ 1.333,\ -0.667,\ 1.333,\ -1.667,\ -0.667,$$

whose squares sum to

$$0.111 + 1.778 + 0.444 + 1.778 + 2.778 + 0.444 = 7.333,$$

so $S_D^2 = 7.333/5 = 1.467$ and $S_D = 1.211$.

Step 2: the paired statistic.

$$t_{\text{obs}} = \frac{\overline{D}}{S_D/\sqrt{b}} = \frac{-3.333}{1.211/\sqrt{6}} = \frac{-3.333}{0.4944} = -6.74, \qquad \text{df} = b - 1 = 5.$$

Since $|-6.74| > 2.571$, reject: strength changed (it rose, since $\overline{D} < 0$ means after exceeds before).

Step 3: (b) the block ANOVA sums of squares. By the theorem,

$$\text{SSTR} = \tfrac{b}{2}\overline{D}^2 = \tfrac{6}{2}(-3.333)^2 = 3(11.111) = 33.333,$$
$$\text{SSE} = \tfrac{1}{2}(b-1)S_D^2 = \tfrac{1}{2}(5)(1.467) = 3.667.$$

Step 4: mean squares and F. Treatment df $= k - 1 = 1$, error df $= (k-1)(b-1) = 5$:

$$\text{MSTR} = \frac{33.333}{1} = 33.333, \qquad \text{MSE} = \frac{3.667}{5} = 0.733,$$
$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{33.333}{0.733} = 45.5.$$

Step 5: verify the identity.

$$t_{\text{obs}}^2 = (-6.74)^2 = 45.4 \approx 45.5 = F,$$

matching to rounding, and both decisions reject ($F = 45.5 > 6.61$, $|t| = 6.74 > 2.571$). The p-value is identical because $F_{0.05,1,5} = 6.61 = (2.571)^2 = t_{0.025,5}^2$.

$$\boxed{F = 45.5 = t_{\text{obs}}^2 = (-6.74)^2, \quad F_{0.05,1,5} = t_{0.025,5}^2 = 6.61 \ \Rightarrow\ \text{both reject.}}$$

Figure 13.1 shows the six patients as six line segments. Each segment is a *block*; the test asks only whether the segments tend to slope one way once their differing heights (the block effect) are set aside.

Example 13.3.4 (Where did the block SS go? Account for every sum of squares)**.** For the grip-strength data above, also compute the block sum of squares SSB and the total SST, and show that $\text{SST} = \text{SSTR} + \text{SSB} + \text{SSE}$. Confirm that SSB is exactly the between-patient variation the paired test discards.

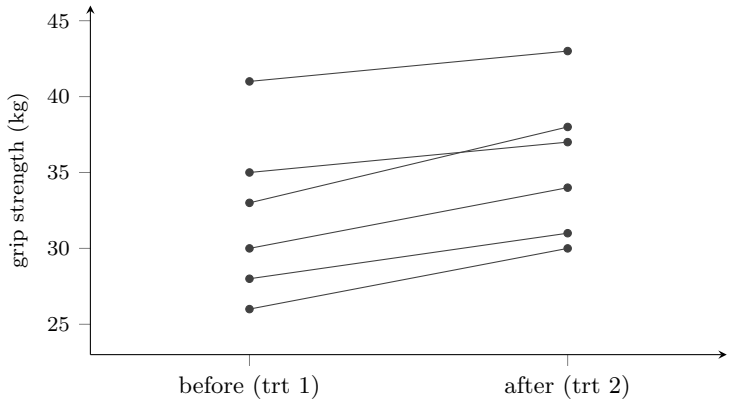


Figure 13.1: Six patients (blocks) as six segments joining the treatment-1 reading to the treatment-2 reading. The large vertical spread between segments is the block effect SSB, which the ANOVA removes and the paired test cancels by differencing; only the common upward tilt — the treatment effect — drives F and t.

Which rule, and why? The paired t never shows SSB, but the block ANOVA does. We use $\text{SSB} = k\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2$ with $k = 2$, then check the three-way partition $\text{SST} = \text{SSTR} + \text{SSB} + \text{SSE}$ — the identity that makes "block removal = differencing" concrete.

Solution.

Step 1: block means and grand mean. Block means $\overline{Y}_{\cdot j} = \frac{1}{2}(Y_{1j} + Y_{2j})$:

$$29.5,\ 36.0,\ 32.0,\ 42.0,\ 35.5,\ 28.0,$$

and the grand mean is $\overline{Y} = \frac{1}{12}\sum_{i,j} Y_{ij} = \frac{406}{12} = 33.833$.

Step 2: block sum of squares. With $k = 2$ observations per block,

$$\begin{aligned}\text{SSB} &= 2\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2 \\ &= 2\big[(29.5 - 33.833)^2 + (36 - 33.833)^2 + (32 - 33.833)^2 \\ &\qquad + (42 - 33.833)^2 + (35.5 - 33.833)^2 + (28 - 33.833)^2\big] \\ &= 2\big[18.78 + 4.69 + 3.36 + 66.69 + 2.78 + 34.03\big] = 2(130.33) = 260.67.\end{aligned}$$

Step 3: total sum of squares. Summing all twelve $(Y_{ij} - \overline{Y})^2$ gives $\text{SST} = 297.67$.

Step 4: the three-way check. From the previous example $\text{SSTR} = 33.333$ and $\text{SSE} = 3.667$, so

$$\text{SSTR} + \text{SSB} + \text{SSE} = 33.333 + 260.67 + 3.667 = 297.67 = \text{SST}\,. \ \blacksquare$$

Step 5: read it. The block term $\text{SSB} = 260.67$ dwarfs everything: it is the patient-to-patient spread (some grip near 28, some near 42). The paired test never estimates it but is rewarded by its removal — with SSB pulled out, the leftover error is only $\text{SSE} = 3.667$.

$$\text{SST} = 297.67 = \underbrace{33.33}_{\text{SSTR}} + \underbrace{260.67}_{\text{SSB}} + \underbrace{3.67}_{\text{SSE}}; \quad \text{SSB is the between-subject variation pairing removes.}$$

Example 13.3.5 (Same data, three error terms: blocking vs. not)**.** Take the grip-strength numbers once more. If you (wrongly) ran a *one-way* ANOVA — two groups, ignoring the pairing — the error would be $\text{SSE}_{\text{1-way}} = \text{SSB} + \text{SSE}$. Show numerically how much smaller the blocked error is, and what that does to F.

Which rule, and why? The contrast is in the error term. One-way ANOVA on the same two columns lumps the block variation into error; the block design pulls it out. Comparing MSE values shows the sharpening directly — the same lesson as paired vs. independent, in ANOVA language.

Solution.

Step 1: the two error sums of squares.

$$\begin{aligned} \text{blocked:}\ & \ \mathrm{SSE} = 3.667 \text{ on } (k-1)(b-1) = 5 \text{ df}, \\ \text{unblocked:}\ & \ \mathrm{SSE}_{\text{1-way}} = \mathrm{SSB} + \mathrm{SSE} = 260.67 + 3.667 = 264.33 \text{ on } N - k = 10 \text{ df}. \end{aligned}$$

Step 2: the two mean squares.

$$\mathrm{MSE}_{\text{block}} = \frac{3.667}{5} = 0.733, \qquad \mathrm{MSE}_{\text{1-way}} = \frac{264.33}{10} = 26.43.$$

Step 3: the two F's (same $\mathrm{SSTR} = 33.333$*).*

$$F_{\text{block}} = \frac{33.333}{0.733} = 45.5, \qquad F_{\text{1-way}} = \frac{33.333}{26.43} = 1.26.$$

Step 4: read the contrast. Blocking shrank the error mean square by a factor of $26.43/0.733 \approx 36$, turning a hopeless $F = 1.26$ (retain) into a decisive $F = 45.5$ (reject). The treatment effect was unchanged; only the denominator moved.

$$\boxed{\text{MSE}: \ 26.43 \to 0.733, \qquad F: \ 1.26 \text{ (retain)} \to 45.5 \text{ (reject)}.}$$

13.3.3 One-sided paired tests and their block analogue

Example 13.3.6 (One-sided paired test, and why the block F alone is two-sided)**.** A sleep researcher gives $b = 8$ insomniacs a relaxation app and records minutes-to-sleep before and after. The within-subject drops $D_j = (\text{before}) - (\text{after})$ are

$$12,\ 5,\ 18,\ -4,\ 9,\ 14,\ 7,\ 11.$$

Test at $\alpha = 0.05$ whether the app *reduces* mean time to sleep, and explain how the $k = 2$ block F relates to this one-sided test. Use $t_{0.05,7} = 1.895$, $F_{0.05,1,7} = 5.59$.

Which rule, and why? Before/after on the same person is a 2×8 block design. The claim names a direction ("reduces"), so the natural test is a *one-sided* paired t on $b - 1 = 7$ df. The block $F = t^2$ folds both tails together, so F gives the magnitude while the sign of $\overline{D}$ supplies the direction the one-sided test needs.

Solution.

Step 1: set up. $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$ (a genuine drop, with D as before minus after). Reject for large positive t.

Step 2: $\overline{D}$ and S_D. Sum $= 12 + 5 + 18 - 4 + 9 + 14 + 7 + 11 = 72$, so $\overline{D} = 72/8 = 9.0$. Deviations from 9: $3, -4, 9, -13, 0, 5, -2, 2$; squares sum to

$$9 + 16 + 81 + 169 + 0 + 25 + 4 + 4 = 308,$$

so $S_D^2 = 308/7 = 44.0$, $S_D = 6.633$.

Step 3: the one-sided statistic.

$$t_{\text{obs}} = \frac{9.0}{6.633/\sqrt{8}} = \frac{9.0}{2.345} = 3.84, \qquad \text{df} = 7.$$

Since $3.84 > t_{0.05,7} = 1.895$, reject: the app reduces time to sleep.

Step 4: the block-F analogue. The block design gives $F = t_{\text{obs}}^2 = 3.84^2 = 14.7$ on $\mathrm{F}_{1,7}$, and $F_{0.05,1,7} = 5.59 = t_{0.025,7}^2$ ($t_{0.025,7} = 2.365$). The F-test rejects ($14.7 > 5.59$) and matches the *two-sided* paired test; the *one-sided* conclusion comes from also noting $\overline{D} = 9.0 > 0$, the direction F cannot see.

$$\boxed{t_{\text{obs}} = 3.84 > 1.895 \ \Rightarrow \ \text{reject (app helps)}; \quad F = t^2 = 14.7, \ \text{direction from } \overline{D} > 0.}$$

Remark. The asymmetry is structural, not a defect. The treatment F in any block design is inherently two-sided about the treatment means: squaring in $\text{SSTR} = \frac{b}{2}\overline{D}^2$ destroys the sign of $\overline{D}$, so F answers "do the two treatments differ?" but not "which is larger?". When a study has a directional claim, run the test on the t scale (where the sign survives) or report F for the magnitude and the sign of $\overline{D}$ for the direction. For $k \geq 3$ treatments there is no sign to keep and the one-sided question does not even arise — another reason the $k = 2$ case is special.

13.3.4 The central contrast: paired/block versus two independent samples

Example 13.3.7 (CONTRAST: ignoring the blocks buries the effect)**.** Eight runners have their 5-km time (minutes) recorded on a standard shoe and a prototype shoe, same runner both times:

runner	1	2	3	4	5	6	7	8
standard Y_1	22.4	25.1	19.8	27.3	21.0	24.6	20.2	23.5
prototype Y_2	21.9	24.5	19.4	26.6	20.7	24.0	19.9	23.0

Every runner was a little faster in the prototype. Analyze the data (a) correctly, as a 2×8 block design / paired test, and (b) incorrectly, as two independent samples (pooled two-sample t). Compare. Use $t_{0.025,7} = 2.365$, $t_{0.025,14} = 2.145$.

Which rule, and why? Same runner in both shoes makes each runner a block: a 2×8 block design, equivalently paired. The deliberate error in (b) treats the two columns as unrelated groups, dumping the huge runner-to-runner spread (some run 19 min, some 27) into the error — exactly the variation blocking removes.

Solution.

Step 1: (a) the paired/block analysis. Differences $D_j = Y_{1j} - Y_{2j}$:

$$0.5,\ 0.6,\ 0.4,\ 0.7,\ 0.3,\ 0.6,\ 0.3,\ 0.5.$$

Sum $= 3.9$, $\overline{D} = 3.9/8 = 0.4875$. Deviations squared sum to

$$0.000156 + 0.0127 + 0.00766 + 0.0452 + 0.0352 + 0.0127 + 0.0352 + 0.000156 = 0.1487,$$

so $S_D^2 = 0.1487/7 = 0.02124$, $S_D = 0.1458$. Then

$$t_{\text{obs}} = \frac{0.4875}{0.1458/\sqrt{8}} = \frac{0.4875}{0.05155} = 9.46, \qquad \text{df} = 7,$$

and equivalently $F = t^2 = 89.5$ on $\text{F}_{1,7}$. Against $t_{0.025,7} = 2.365$: **reject** overwhelmingly — the prototype is faster.

Step 2: (b) the wrong independent test. Treat Y_1, Y_2 as two samples of size 8. Means $\overline{Y}_{1\cdot} = 22.99$, $\overline{Y}_{2\cdot} = 22.50$, difference 0.4875 (same as $\overline{D}$). But each column carries the full runner spread: $\sum(Y_{1j} - \overline{Y}_{1\cdot})^2 = 52.43$ and $\sum(Y_{2j} - \overline{Y}_{2\cdot})^2 = 46.92$, so

$$s_p^2 = \frac{52.43 + 46.92}{8 + 8 - 2} = \frac{99.35}{14} = 7.097, \qquad s_p = 2.664,$$

$$t_{\text{obs}}^{\text{indep}} = \frac{0.4875}{2.664\sqrt{1/8 + 1/8}} = \frac{0.4875}{1.332} = 0.366, \qquad \text{df} = 14.$$

Against $t_{0.025,14} = 2.145$: **fail to reject** — the independent analysis sees nothing.

Step 3: read the contrast. Identical mean effect (0.4875 min), yet paired $t = 9.46$ shouts significance while independent $t = 0.37$ finds none. The blocked error mean square is $\text{MSE} = \frac{1}{2}S_D^2 = 0.0106$; the unblocked pooled variance is 7.097, inflated ~ 670-fold by the between-runner spread that blocking removes.

block/paired $t = 9.46$ (reject) vs. independent $t = 0.37$ (miss).

Remark. This is the practical payoff of recognizing a study as a block design. The numerator agrees because $\overline{Y}_{1\cdot} - \overline{Y}_{2\cdot} = \overline{D}$ always. What changes is the denominator: the block design uses only within-runner noise, while the independent test must carry $\text{Var}(Y_1) + \text{Var}(Y_2)$, the entire between-runner spread. Choosing the independent test on blocked data is not a small inefficiency — here it is the difference between a t of 9.5 and a t of 0.4.

13.3.5 Recognizing a hidden two-treatment block design

Example 13.3.8 (Is this a 2-treatment block design? Read the wording)**.** For each study, decide whether it is a 2-treatment randomized block design (equivalently a paired test) or two independent samples, and name the block. (a) "The same 20 soil plots are assayed for nitrogen by a lab method and a field kit." (b) "15 patients get drug A; a *different* 15 get drug B; blood pressure is compared." (c) "In each of 12 families, the older and younger sibling take the same reading test." (d) "30 tires are split: each tire's left tread uses compound X, its right tread compound Y."

Which rule, and why? The test is whether the two measurements share a common unit that can serve as a block. Same plot, same family, same tire ⇒ block. Two disjoint patient groups ⇒ independent. "Equal counts" alone never settles it.

Solution.

Step 1: (a) two methods, one plot. Both readings come from the same plot, which shares its true nitrogen level: block = plot, treatments = method. **Block design**.

Step 2: (b) two disjoint arms. Drug-A and drug-B patients are different people; no shared unit, no block. **Independent samples**.

Step 3: (c) two siblings, one family. The siblings are matched by family (genes, home, schooling): block = family, treatments = birth order. **Block design** (matched pairs).

Step 4: (d) two compounds, one tire. Each tire carries both compounds, sharing the tire's load and roads: block = tire, treatments = compound. **Block design**.

(a) block, (b) independent, (c) block, (d) block.

Example 13.3.9 (From a $2 \times b$ table straight to F via the differences)**.** A water lab measures dissolved oxygen (mg/L) at $b = 7$ sites with a probe (Y_1) and a titration kit (Y_2), same water sample each time:

site	1	2	3	4	5	6	7
probe Y_1	8.2	7.5	9.1	6.8	8.8	7.0	9.4
kit Y_2	8.0	7.6	8.8	6.9	8.5	6.7	9.1

Treat this as a 2-treatment block design and obtain the treatment F at $\alpha = 0.05$ by going through the differences. Use $F_{0.05,1,6} = 5.99$.

Which rule, and why? Same water sample on both instruments ⇒ block = site, treatments = instrument. The fastest route to F is the difference column: $\text{SSTR} = \frac{b}{2}\overline{D}^2$ and $\text{SSE} = \frac{1}{2}(b-1)S_D^2$ feed $F = \text{MSTR} / \text{MSE}$ directly.

Solution.

Step 1: differences $D_j = Y_{1j} - Y_{2j}$.

$$0.2,\ -0.1,\ 0.3,\ -0.1,\ 0.3,\ 0.3,\ 0.3.$$

Sum $= 1.2$, $\overline{D} = 1.2/7 = 0.1714$. Deviations squared sum to

$$0.000816 + 0.0737 + 0.0165 + 0.0737 + 0.0165 + 0.0165 + 0.0165 = 0.2143,$$

so $S_D^2 = 0.2143/6 = 0.03571$, $S_D = 0.1890$.

Step 2: ANOVA pieces from the differences.

$$\text{SSTR} = \tfrac{b}{2}\overline{D}^2 = \tfrac{7}{2}(0.1714)^2 = 3.5(0.02939) = 0.1029,$$
$$\text{SSE} = \tfrac{1}{2}(b-1)S_D^2 = \tfrac{1}{2}(6)(0.03571) = 0.1071.$$

Step 3: mean squares and F. Treatment df 1, error df $(k-1)(b-1) = 6$:

$$\text{MSTR} = 0.1029, \qquad \text{MSE} = \frac{0.1071}{6} = 0.01786, \qquad F = \frac{0.1029}{0.01786} = 5.76.$$

Step 4: decide. Since $F = 5.76 < F_{0.05,1,6} = 5.99$, do *not* reject: at the 5% level the probe and kit do not differ significantly (it is close — $t = \sqrt{5.76} = 2.40$ just under $t_{0.025,6} = 2.447$).

$$\boxed{F = 5.76 < F_{0.05,1,6} = 5.99 \;\Rightarrow\; \text{retain; no significant instrument difference (just barely).}}$$

Example 13.3.10 (Recover t from a printed 2-treatment block ANOVA table)**.** A printout of a randomized block analysis with $k = 2$ treatments and $b = 10$ blocks reads

Source	df	SS	MS	F
Treatment	1	12.6	?	?
Block	9	88.2	?	
Error	9	5.4	?	
Total	19	106.2		

Fill in the mean squares and F, then recover the equivalent paired t-statistic.

Which rule, and why? With $k = 2$ the treatment row has 1 df and error has $(k-1)(b-1) = 9$ df, so $F = \text{MSTR} / \text{MSE}$ and the paired t is just $\pm\sqrt{F}$ — the equivalence read off a table. The block row is set aside; it never enters F.

Solution.

Step 1: mean squares.

$$\text{MSTR} = \frac{12.6}{1} = 12.6, \qquad \text{MSB} = \frac{88.2}{9} = 9.8, \qquad \text{MSE} = \frac{5.4}{9} = 0.6.$$

Step 2: treatment F.

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{12.6}{0.6} = 21.0 \qquad \text{on } \mathrm{F}_{1,9}\,.$$

Step 3: the paired t.

$$t_{\text{obs}} = \pm\sqrt{F} = \pm\sqrt{21.0} = \pm 4.58 \qquad \text{on 9 df.}$$

Step 4: check the partition. $\text{SSTR} + \text{SSB} + \text{SSE} = 12.6 + 88.2 + 5.4 = 106.2 = \text{SST}$, and the df add $1 + 9 + 9 = 19$. Both confirm a clean 2×10 block table.

$$\boxed{F = 21.0 \text{ on } \mathrm{F}_{1,9} = t_{\text{obs}}^2, \quad t_{\text{obs}} = \pm 4.58 \text{ on 9 df.}}$$

Example 13.3.11 (Matched case–control as a 2-treatment block design)**.** A clinic matches $b = 9$ smokers to 9 nonsmokers by age and sex (one matched pair each), then measures a lung-function score. Within each matched pair, $D_j = (\text{nonsmoker}) - (\text{smoker})$:

$$6,\ 3,\ 8,\ 1,\ 5,\ -2,\ 7,\ 4,\ 6.$$

Test at $\alpha = 0.05$ whether nonsmokers have a higher mean score, as a 2-treatment block design, and confirm via $F = t^2$. Use $t_{0.05,8} = 1.860$, $F_{0.05,1,8} = 5.32$.

Which rule, and why? Matching by age and sex makes each pair a block; smoking status is the two-level treatment. "Higher" is directional, so run a one-sided paired t on $b - 1 = 8$ df and use $F = t^2$ for the magnitude.

Solution.

Step 1: set up. $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$ (nonsmokers higher).

Step 2: $\overline{D}$ and S_D. Sum $= 6 + 3 + 8 + 1 + 5 - 2 + 7 + 4 + 6 = 38$, so $\overline{D} = 38/9 = 4.222$. Deviations from 4.222: $1.778, -1.222, 3.778, -3.222, 0.778, -6.222, 2.778, -0.222, 1.778$; squares sum to

$$3.16 + 1.49 + 14.27 + 10.38 + 0.61 + 38.72 + 7.72 + 0.05 + 3.16 = 79.56,$$

so $S_D^2 = 79.56/8 = 9.944$, $S_D = 3.153$.

Step 3: the statistic and its F.

$$t_{\text{obs}} = \frac{4.222}{3.153/\sqrt{9}} = \frac{4.222}{1.051} = 4.02, \qquad F = t_{\text{obs}}^2 = 16.1 \text{ on } \mathrm{F}_{1,8}\,.$$

Step 4: decide. One-sided: $4.02 > t_{0.05,8} = 1.860$, reject. The two-sided block $F = 16.1 > F_{0.05,1,8} = 5.32$ agrees on magnitude; the sign $\overline{D} > 0$ gives the direction.

$$\boxed{t_{\text{obs}} = 4.02 > 1.860 \;\Rightarrow\; \text{reject; nonsmokers score higher;} \quad F = t^2 = 16.1.}$$

Example 13.3.12 (CONTRAST: equal sizes tempt a fake pairing)**.** A researcher has 5-km times for $b = 8$ runners in standard shoes and, separately, times for 8 *different* runners in prototype shoes. Seeing two columns of eight, she lines them up, differences, and runs a 2-treatment block ANOVA. Explain why this is invalid and what the correct analysis is.

Which rule, and why? A block design requires each block to supply *both* treatments. Here the two columns are unrelated runners, so there is no shared unit to act as a block; equal counts do not create one. The fictitious "blocks" would have a meaningless SSB.

Solution.

Step 1: no block exists. Runner 3 in the standard group and runner 3 in the prototype group share nothing; reorder either column and the "differences" change entirely. A genuine block is invariant to relabeling — here there is nothing to be invariant about.

Step 2: the consequence. The bogus SSB is built from an arbitrary alignment and the resulting SSE is not the true within-block noise. The F (and the equivalent t) is uninterpretable; its p-value means nothing.

Step 3: correct analysis. Two disjoint groups of runners are **independent samples**: compare them with the two-sample t (Chapter 9), *not* a block design and *not* a paired test.

$$\boxed{\text{equal } b \neq \text{a block design; unrelated groups} \Rightarrow \text{ two-sample } t.}$$

Example 13.3.13 (Confidence interval for the treatment effect, two ways)**.** For the grip-strength study (Example with $\overline{D} = -3.333$, $S_D = 1.211$, $b = 6$), give a 95% confidence interval for the mean treatment effect $\mu_1 - \mu_2$ (before minus after), and note that the half-width equals $t_{0.025,5}\sqrt{2\ \mathrm{MSE}\,/b}$ from the block ANOVA. Use $t_{0.025,5} = 2.571$.

Which rule, and why? The treatment effect $\mu_1 - \mu_2$ is the mean difference μ_D, so the interval is the paired one-sample t interval on the D_j. Equivalently it is the block-ANOVA treatment-contrast interval with standard error $\sqrt{2\ \mathrm{MSE}\,/b}$ — the same number.

Solution.

Step 1: paired-interval form.

$$\begin{aligned}\overline{D} \pm t_{0.025,5}\frac{S_D}{\sqrt{b}} &= -3.333 \pm 2.571 \cdot \frac{1.211}{\sqrt{6}} \\ &= -3.333 \pm 2.571(0.4944) = -3.333 \pm 1.271.\end{aligned}$$

Step 2: block-ANOVA form. With $\mathrm{MSE} = \frac{1}{2}S_D^2 = 0.733$, the contrast SE is

$$\sqrt{\frac{2\ \mathrm{MSE}}{b}} = \sqrt{\frac{2(0.733)}{6}} = \sqrt{0.2444} = 0.4944,$$

identical to $S_D/\sqrt{b}$, so the half-width $2.571(0.4944) = 1.271$ matches.

Step 3: report. The interval is $(-4.60, -2.06)$ kg. It excludes 0, so the effect is significant (consistent with the rejected F), and after exceeds before by about 2 to 5 kg.

$$\boxed{\mu_1 - \mu_2 \in (-4.60,\ -2.06) \text{ kg at } 95\%;\ \text{excludes } 0.}$$

Remark. The two confidence intervals are the same interval written in two notations. The paired test calls the standard error $S_D/\sqrt{b}$; the block ANOVA calls the same quantity $\sqrt{2\ \mathrm{MSE}/b}$, because $\mathrm{MSE} = \frac{1}{2}S_D^2$ forces $2\ \mathrm{MSE}/b = S_D^2/b$. This is the $F = t^2$ identity once more, now expressed for the effect estimate rather than the test statistic: every quantity the paired test reports has an exact block-ANOVA twin, and choosing between them is purely a matter of which bookkeeping you prefer.

13.4 Blocking Efficiency, Additivity, and Diagnostics

Blocking is a bet: you spend $b-1$ degrees of freedom on a block term in the hope that removing block-to-block variation shrinks the error mean square enough to sharpen the treatment F. This section is about whether that bet paid off, and whether the design that placed it is even valid. The payoff question is answered by **relative efficiency** — a single number comparing the error variance you actually got from the randomized block design to the error variance a completely randomized design would have suffered on the same material; greater than 1 means blocking helped, near 1 (or below) means you spent the block degrees of freedom for little or nothing. The validity question rests on **additivity**: the model assumes treatment effects and block effects simply *add*, with no treatment-by-block **interaction**. When that fails — when a treatment is best in one block but worst in another — the leftover interaction is dumped into MSE, inflating it and quietly invalidating the simple treatment F. The remaining diagnostics (normality and equal variance, read off the residuals) and the nonparametric escape hatch (the **Friedman** test, Chapter 14) round out the section. Recognising this material is easy: any problem that asks "was blocking worth it?", "is the no-interaction assumption safe?", or "which test should I run when the assumptions look shaky?" lives here.

Which question about the block design is being asked?

- **Did blocking pay off?** ⇒ **relative efficiency** RE from MSB and MSE (with the small-sample df correction); RE > 1 means the block design beat a completely randomized design, RE ≤ 1 means it did not.
- **Was the block term worth its degrees of freedom?** ⇒ compare MSB to MSE. A block $F = \mathrm{MSB}\,/\,\mathrm{MSE}$ near or below 1 (small SSB) says the blocks were not really more alike inside than across — the $b-1$ df were largely wasted.
- **Is additivity (no interaction) safe?** ⇒ a **profile / interaction plot**: roughly parallel block profiles support additivity; crossing or fanning profiles signal a treatment-by-block **interaction**, and the **Tukey one-df test** formalises it.
- **Do normality and equal variance hold?** ⇒ inspect the residuals $e_{ij} = Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}$: a QQ plot for normality, a residual-vs-fitted plot for constant spread.
- **Assumptions broken and no fix?** ⇒ the rank-based **Friedman test** (Ch. 14), the nonparametric analogue of the block-design F.

Do not confuse a significant block F with a useful design: the block F only confirms the blocks differed; what tells you blocking *helped the treatment comparison* is relative efficiency, which weighs that gain against the df it cost. And never read parallel profiles as proof of *no treatment effect* — parallel means *additive*, the treatments can still differ by a constant gap in every block.

13.4.1 Relative efficiency: did blocking pay off?

Definition 13.4.1 (Relative efficiency of the randomized block design)**.** Suppose the same experimental material is analysed two ways: as a **randomized block design** (RBD), and as if the blocks had been ignored and the units assigned completely at random (a **completely randomized design**, CRD). The **relative efficiency** of the RBD to the CRD estimates how many times as many replicates the CRD would need to match the RBD's precision. With b blocks and k treatments it is

$$\mathrm{RE} = \frac{(b-1)\ \mathrm{MSB} + b(k-1)\ \mathrm{MSE}}{(bk-1)\ \mathrm{MSE}} \qquad \text{(the variance-ratio form)},$$

optionally multiplied by the small-sample degrees-of-freedom correction

$$c = \frac{\big[(k-1)(b-1)+1\big]\,\big[k(b-1)+1\big]}{\big[(k-1)(b-1)+3\big]\,\big[k(b-1)+3\big]} < 1.$$

RE > 1 means blocking improved precision; RE < 1 means it hurt.

Remark. The numerator of the variance-ratio form is an estimate of the error variance a CRD *would* have had — it folds the block variation MSB back into the error, because a CRD never separates it out — while the denominator is (a df-weighted version of) the RBD's own error MSE. So RE is literally "CRD error variance ÷ RBD error variance." When the blocks were genuinely different (MSB large relative to MSE), the numerator swells and RE climbs well above 1: a CRD would have buried that block variation in its error and lost power. When the blocks barely differed (MSB $\approx$ MSE), the numerator collapses toward the denominator and RE $\to 1$ — with the df correction $c < 1$ it can even dip *below* 1, the mathematical fingerprint of a block term that cost more (in lost error df) than it returned.

Remark. A clean reading of RE: a value of 1.8 says the completely randomized design would have needed about 80% more observations — nearly twice the experiment — to achieve the precision the blocking bought you for free. A value of 0.95 says blocking made you very slightly worse off than not blocking at all. The break-even point is RE $= 1$.

Example 13.4.2 (Relative efficiency from an ANOVA table: tire wear across cars)**.** A tire study runs $k = 4$ brands on each of $b = 5$ cars (cars are the blocks), one tire of each brand per car. The two-way ANOVA gives

$$\text{MSB} = 180.0 \text{ (car mean square)}, \qquad \text{MSE} = 12.0,$$

on $b - 1 = 4$ block df and $(k-1)(b-1) = 12$ error df. Compute the relative efficiency of this block design relative to a completely randomized design, and decide whether blocking on cars was worthwhile.

Which rule, and why? The question "was blocking worth it?" is exactly what relative efficiency answers; plug MSB and MSE into the variance-ratio form. Use MSB, the *block* mean square, not the treatment mean square — it is block variation a CRD would have failed to remove.

Solution.

Step 1: assemble the pieces. Here $b = 5$, $k = 4$, so $bk - 1 = 19$, $b - 1 = 4$, $b(k-1) = 15$.

Step 2: variance-ratio form.

$$\begin{aligned}\text{RE} &= \frac{(b-1)\ \text{MSB} + b(k-1)\ \text{MSE}}{(bk-1)\ \text{MSE}} \\ &= \frac{4(180.0) + 15(12.0)}{19(12.0)} = \frac{720+180}{228} = \frac{900}{228} = 3.95.\end{aligned}$$

Step 3: interpret. RE $= 3.95 > 1$: a completely randomized design would have needed roughly four times the replication to match this precision. The cars differed enormously (MSB $= 180$ versus MSE $= 12$), so removing them paid off handsomely.

$$\boxed{\text{RE} = 3.95 > 1 \ \Rightarrow\ \text{blocking on cars was strongly worthwhile.}}$$

Example 13.4.3 (Apply the small-sample df correction)**.** Reusing the tire study ($k = 4$, $b = 5$, $\text{RE}_{\text{raw}} = 3.95$), apply the degrees-of-freedom correction factor c to report the corrected relative efficiency.

Which rule, and why? The variance-ratio RE slightly overstates the gain because the RBD spent error df on the block term; the correction $c < 1$ discounts for that. It matters most when the error df is small, so always state whether RE is corrected.

Solution.

Step 1: the two df. RBD error df $= (k-1)(b-1) = 3 \cdot 4 = 12$; the CRD's error df would be $k(b-1) = 4 \cdot 4 = 16$.

Step 2: form the correction.

$$c = \frac{\big[(k-1)(b-1)+1\big]\,\big[k(b-1)+1\big]}{\big[(k-1)(b-1)+3\big]\,\big[k(b-1)+3\big]} = \frac{(12+1)(16+1)}{(12+3)(16+3)} = \frac{13\cdot 17}{15\cdot 19}$$
$$= \frac{221}{285} = 0.7754.$$

Step 3: corrected efficiency.

$$\mathrm{RE}_{\mathrm{corr}} = c \cdot \mathrm{RE}_{\mathrm{raw}} = 0.7754 \times 3.95 = 3.06.$$

$\boxed{\mathrm{RE}_{\mathrm{corr}} = 3.06 \text{ (vs. 3.95 uncorrected)} \;\Rightarrow\; \text{still a large, real gain from blocking.}}$

The correction trims the headline figure but does not change the verdict: blocking remains a clear win.

Example 13.4.4 (CONTRAST: blocking that did NOT help – small SSB wastes df)**.** A bakery compares $k = 3$ oven racks (top, middle, bottom) for browning score, blocking on $b = 6$ baking days. The two-way ANOVA returns SSB $= 4.0$ on $b - 1 = 5$ df and SSE $= 30.0$ on $(k-1)(b-1) = 10$ df. Compute the relative efficiency and judge whether the day-blocks earned their keep — and contrast with what an unblocked one-way analysis would have given.

Which rule, and why? When SSB is tiny, the blocks were not really more homogeneous inside than across, so the $b - 1$ df spent on them are largely wasted; relative efficiency near or below 1 confirms it. The contrast is with a CRD that would have pooled those block df back into the error, gaining error df.

Solution.

Step 1: mean squares.

$$\mathrm{MSB} = \frac{\mathrm{SSB}}{b-1} = \frac{4.0}{5} = 0.80, \qquad \mathrm{MSE} = \frac{\mathrm{SSE}}{(k-1)(b-1)} = \frac{30.0}{10} = 3.00.$$

Already MSB $= 0.80 <$ MSE $= 3.00$: the block $F = \mathrm{MSB}\,/\,\mathrm{MSE} = 0.27$ is well below 1, a red flag that the blocks contributed nothing.

Step 2: relative efficiency. With $b = 6$, $k = 3$: $bk - 1 = 17$, $b - 1 = 5$, $b(k-1) = 12$.

$$\mathrm{RE} = \frac{(b-1)\ \mathrm{MSB} + b(k-1)\ \mathrm{MSE}}{(bk-1)\ \mathrm{MSE}} = \frac{5(0.80) + 12(3.00)}{17(3.00)}$$
$$= \frac{4.0 + 36.0}{51.0} = \frac{40.0}{51.0} = 0.78.$$

Step 3: the unblocked alternative. Ignore the blocks: a one-way ANOVA pools SSB into the error, giving $\mathrm{SSE}_{\mathrm{CRD}} = \mathrm{SSB} + \mathrm{SSE} = 4.0 + 30.0 = 34.0$ on $bk - k = 18 - 3 = 15$ error df, so

$$\mathrm{MSE}_{\mathrm{CRD}} = \frac{34.0}{15} = 2.27 < 3.00 = \mathrm{MSE}_{\mathrm{RBD}}\,.$$

The unblocked error mean square is actually *smaller*, and it rests on more df — the completely randomized analysis would have been the sharper test here.

$\boxed{\mathrm{RE} = 0.78 < 1 \;\Rightarrow\; \text{blocking hurt; the unblocked one-way test was better.}}$

The lesson: blocking is only insurance you should buy when the blocks are genuinely heterogeneous; on near-identical blocks the premium (lost df) exceeds the payout.

Example 13.4.5 (Borderline efficiency: read RE at the break-even point)**.** A greenhouse trial of $k = 4$ light recipes blocked on $b = 4$ benches reports MSB $= 9.5$ and MSE $= 8.0$. Was blocking worthwhile?

Which rule, and why? When MSB and MSE are close, RE hovers near the break-even value 1, and the small-sample correction $c < 1$ can tip a marginal raw value below 1. Compute both to call it honestly.

Solution.

Step 1: raw efficiency. $b = 4$, $k = 4$: $bk - 1 = 15$, $b - 1 = 3$, $b(k-1) = 12$.

$$\mathrm{RE} = \frac{3(9.5) + 12(8.0)}{15(8.0)} = \frac{28.5 + 96.0}{120.0} = \frac{124.5}{120.0} = 1.04.$$

Step 2: apply the correction. RBD error df $= (k-1)(b-1) = 9$; CRD error df $= k(b-1) = 12$.

$$c = \frac{(9+1)(12+1)}{(9+3)(12+3)} = \frac{10 \cdot 13}{12 \cdot 15} = \frac{130}{180} = 0.722, \qquad \mathrm{RE}_{\mathrm{corr}} = 0.722 \times 1.04 = 0.75.$$

Step 3: verdict. The raw RE $= 1.04$ looks like a wafer-thin gain, but once the lost error df are accounted for, $\mathrm{RE}_{\mathrm{corr}} = 0.75 < 1$.

$$\boxed{\mathrm{RE}_{\mathrm{raw}} = 1.04 \text{ but } \mathrm{RE}_{\mathrm{corr}} = 0.75 < 1 \;\Rightarrow\; \text{blocking gave no real benefit here.}}$$

On the next experiment with this material, drop the blocking (or block on a more relevant nuisance variable).

13.4.2 Additivity: the no-interaction assumption

Definition 13.4.6 (The additive model and treatment-by-block interaction)**.** The randomized block model assumes the cell mean is the sum of a treatment effect and a block effect,

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij},$$

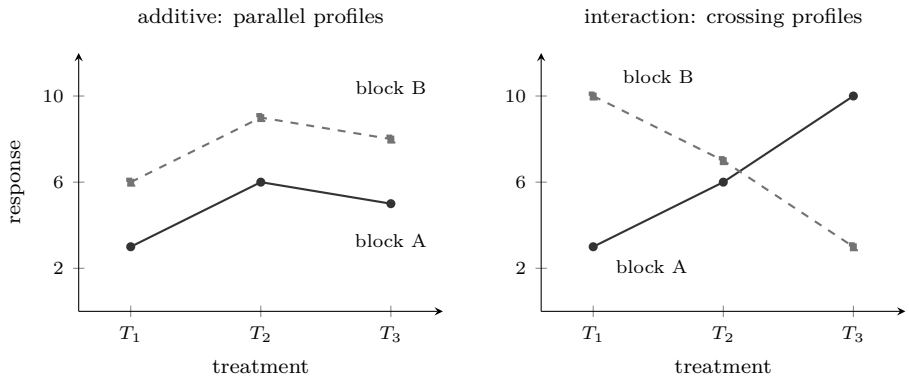
with no $(\tau\beta)_{ij}$ **interaction** term — this is **additivity**. Additivity says the *gap* between any two treatments is the same in every block: treatment i beats treatment i' by a fixed amount $\tau_i - \tau_{i'}$ regardless of which block we look in. An **interaction** means that gap changes from block to block — a treatment that wins in one block loses in another. With only one observation per cell ($N = kb$) there is no replication to estimate interaction directly, so it cannot be separated from error: any interaction present is *absorbed into* MSE.

Remark. This is the silent failure mode of block designs. Because there is a single observation per cell, the arithmetic of the ANOVA table never complains: $\mathrm{SSTR} + \mathrm{SSB} + \mathrm{SSE} = \mathrm{SST}$ holds whether or not additivity does. But if interaction is real, the SSE that is supposed to measure pure noise is contaminated by systematic treatment-by-block structure, so MSE is inflated, the treatment $F = \mathrm{MSTR}\,/\,\mathrm{MSE}$ is deflated, and the test loses power (or, in some configurations, becomes anticonservative). You cannot catch this in the table; you catch it in a **profile plot** or with **Tukey's one-degree-of-freedom test for non-additivity**.

Proposition 13.4.7 (Tukey's one-df test for non-additivity, conceptually)**.** *Tukey's test isolates the most common form of interaction — one in which the interaction is* proportional *to the product of the treatment and block effects, $(\tau\beta)_{ij} \approx \lambda\,\tau_i\beta_j$. It peels a single degree of freedom off the error,*

$$\mathrm{SSE} = \underbrace{\mathrm{SS}_{nonadd}}_{1\ df} + \underbrace{\mathrm{SS}_{rem}}_{(k-1)(b-1)-1\ df},$$

and refers $F_{nonadd} = \mathrm{SS}_{nonadd} / \big[\mathrm{SS}_{rem}/((k-1)(b-1)-1)\big]$ to $\mathrm{F}_{1,\,(k-1)(b-1)-1}$. A significant value says additivity is violated in this multiplicative way — often curable by a transformation (frequently log*, which turns multiplicative effects into additive ones).*



Profile (interaction) plots: each line connects one block's responses across the treatments. Left: the two block profiles are parallel *(block B is block A shifted up by a constant), so the treatment gaps are identical in every block — additivity holds and the simple treatment F is valid. Right: the profiles* cross *— T_3 is best in block A but worst in block B — so the treatment effect depends on the block; additivity is violated and that structure will inflate* MSE.

Example 13.4.8 (Flag an interaction from a profile plot: coating by substrate)**.** An adhesion study applies $k = 3$ coatings to panels of $b = 2$ substrates (steel, aluminum). The cell means (adhesion, in N) are

$$\begin{aligned} \text{steel}: &\ C_1 = 20,\ C_2 = 28,\ C_3 = 24; \\ \text{aluminum}: &\ C_1 = 30,\ C_2 = 22,\ C_3 = 33. \end{aligned}$$

Sketch the profiles mentally (or compute the treatment gaps per block) and decide whether additivity is tenable.

Which rule, and why? Additivity says every treatment gap is the same across blocks; check it by comparing, for one pair of treatments, the difference in each block. If the gaps disagree in sign or wildly in size, the profiles are non-parallel and additivity fails — this is the recognition step a profile plot makes visual.

Solution.

Step 1: pick a treatment pair and compare gaps by block. Take C_1 versus C_2:

$$\begin{aligned} \text{steel:} &\quad C_2 - C_1 = 28 - 20 = +8, \\ \text{aluminum:} &\quad C_2 - C_1 = 22 - 30 = -8. \end{aligned}$$

Step 2: read the sign reversal. On steel, C_2 beats C_1 by 8; on aluminum, C_2 *trails* C_1 by 8. The best coating depends entirely on the substrate — the profiles cross.

Step 3: conclude.

Gap $+8$ on steel vs. -8 on aluminum $\Rightarrow$ strong treatment-by-substrate interaction; additivity fails.

A simple block-design F on these data is invalid: the very question "which coating is best" has no block-free answer. Report the interaction (e.g. analyse the substrates separately, or fit a full two-factor model with replication).

Example 13.4.9 (Additivity holds despite a treatment effect)**.** A drying experiment runs $k = 3$ temperatures across $b = 3$ ovens (blocks) and finds cell means

$$\text{oven } 1:\ 40,\ 46,\ 52; \quad \text{oven } 2:\ 44,\ 50,\ 56; \quad \text{oven } 3:\ 38,\ 44,\ 50.$$

Decide whether additivity is plausible, and explain why parallel profiles are not the same as "no treatment effect."

Which rule, and why? Parallel profiles mean *additive*, not *equal*: the treatments can differ by a constant gap repeated in every block. Check additivity by confirming the treatment increments are the same across blocks, and read the common increments as the treatment effect.

Solution.

Step 1: treatment increments within each oven. Going low→mid→high:

$$\text{oven 1: } +6,\ +6; \quad \text{oven 2: } +6,\ +6; \quad \text{oven 3: } +6,\ +6.$$

Every oven shows the identical $+6, +6$ pattern, so the three profiles are exactly parallel (each oven is a vertical shift of the others).

Step 2: separate additivity from the treatment effect. Parallel ⇒ additivity holds. Yet the treatments are far from equal: the common increments $+6$ each step mean temperature raises the response by 12 from low to high in *every* block.

Identical $+6, +6$ increments in all blocks ⇒ additive; treatments still differ by a constant $+12$ gap.

This is exactly the situation the block-design F is built for: a real, block-independent treatment effect cleanly separable from block shifts.

Example 13.4.10 (Route a flagged interaction to Tukey's test and a transform)**.** A bioassay ($k = 4$ doses, $b = 5$ animals as blocks) shows profiles that do not cross but *fan out* — the dose gaps are small in low-response animals and large in high-response animals, the spacing growing roughly in proportion to the level. Which formal test applies, and what fix is suggested?

Which rule, and why? Fanning (multiplicative-looking) profiles are precisely the alternative Tukey's one-df test targets, since it models interaction as proportional to $\tau_i\beta_j$. A significant Tukey statistic combined with that pattern points to a log transform, which turns multiplicative structure additive.

Solution.

Step 1: recognise the pattern. Non-crossing but fanning profiles, with gaps scaling with the level, are the multiplicative-interaction signature — not a sign reversal but a spread that grows with the mean.

Step 2: apply the matched test. Run **Tukey's one-degree-of-freedom test**: it extracts $\mathrm{SS}_{\text{nonadd}}$ (1 df) from SSE and refers $F_{\text{nonadd}} \sim \mathrm{F}_{1,(k-1)(b-1)-1} = \mathrm{F}_{1,11}$. A significant value confirms non-additivity.

Step 3: prescribe the remedy. Because the interaction looks multiplicative, re-analyse $\log Y$; on the log scale the fanning typically flattens to parallel profiles and additivity is restored.

Fanning profiles ⇒ Tukey 1-df test on $\mathrm{F}_{1,11}$; if significant, re-fit on $\log Y$.

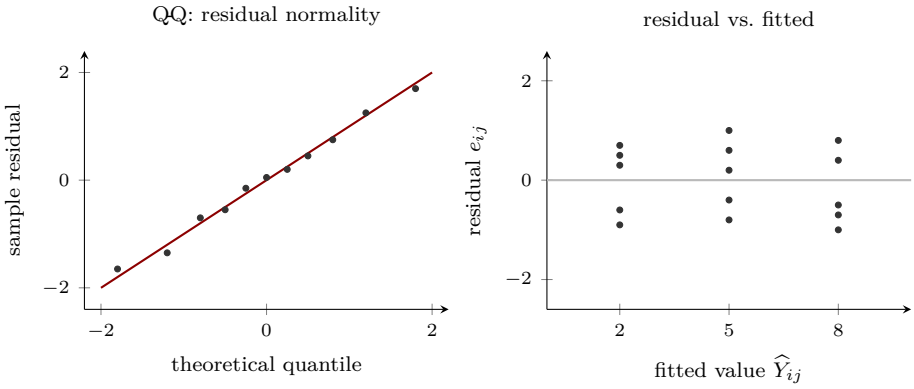
13.4.3 Normality, equal variance, and residual diagnostics

Definition 13.4.11 (Residuals in the block design)**.** The fitted value for cell (i, j) is $\widehat{Y}_{ij} = \overline{Y}_{i\cdot} + \overline{Y}_{\cdot j} - \overline{Y}$ (treatment mean plus block mean minus grand mean), and the **residual** is

$$e_{ij} = Y_{ij} - \overline{Y}_{i\cdot} - \overline{Y}_{\cdot j} + \overline{Y}_{\cdot}.$$

These e_{ij} are the only window onto the error assumptions. A **normal QQ plot** of the e_{ij} checks normality (points on a line = normal); a plot of e_{ij} against the fitted value $\widehat{Y}_{ij}$ checks equal variance (a horizontal band of constant width = homoscedastic). Because the same residuals carry interaction (when additivity fails), a *curved* or *structured* residual-vs-fitted plot can signal non-additivity as well as unequal variance.

Remark. The independence assumption, as always, is a property of the design, not the residuals: a randomized block design earns it by *randomizing* the treatment order within each block. If the treatments were applied in the same systematic order in every block (e.g. always T_1 first while a machine warmed up), a time trend would masquerade as a treatment effect and no residual plot would reveal it. Read the protocol, not just the numbers.



Left: a normal QQ plot of the block-design residuals; points hug the 45° line, so normality of the errors is credible. Right: residuals against fitted values form an even horizontal band of constant width above every fitted level — equal variance looks satisfied. Curvature in the left panel would flag non-normality; a funnel in the right panel would flag unequal variance (or, with one observation per cell, possible non-additivity).

Example 13.4.12 (Judge the residual diagnostics: fabric-strength trial)**.** A block design of $k = 3$ dye treatments across $b = 8$ fabric lots produces residuals whose QQ plot tracks the reference line closely (one mild high point) and whose residual-vs-fitted plot is a flat, even band. The block F was large (MSB / MSE = 14). Should the treatment F be reported at face value?

Which rule, and why? The block-design F is valid when independence (design), normality (QQ), equal variance (residual band), and additivity (no structure) all hold. Here three plots are clean and the large block F only confirms blocking was effective, not that anything is wrong.

Solution.

Step 1: normality. Near-straight QQ with a single mild outlier — no systematic curvature, so normality is credible.

Step 2: equal variance and additivity. A flat even band means constant spread and no fitted-value structure, so both homoscedasticity and additivity look fine.

Step 3: independence and block F. Randomized order within lots gives independence; the large block F is good news (the lots really did differ, so blocking helped).

All diagnostics pass $\Rightarrow$ report the treatment F as is; blocking was both valid and worthwhile.

Example 13.4.13 (A residual funnel: unequal variance in a yield study)**.** A block design ($k = 4$ catalysts, $b = 6$ batches) yields a residual-vs-fitted plot that funnels open — tight scatter at low fitted yields, wide scatter at high yields — and group spreads with $s_{\max}/s_{\min} = 3.5$. The QQ plot is convex (right-skewed). Diagnose and prescribe.

Which rule, and why? A widening funnel plus a convex QQ plot is the joint signature of variance rising with the mean and right-skew — both pointing to a multiplicative scale. With one observation per cell this can equally reflect non-additivity, and a log transform attacks all three at once.

Solution.

Step 1: name the violations. Funnel $\Rightarrow$ unequal variance (spread grows with fitted value); convex QQ $\Rightarrow$ right skew; $s_{\max}/s_{\min} = 3.5 \geq 3$ confirms the spread ratio is in the danger zone.

Step 2: recognise the common cause. Variance tied to the mean and right skew both indicate the response acts multiplicatively, which on the raw scale also tends to manifest as non-additivity in the cell means.

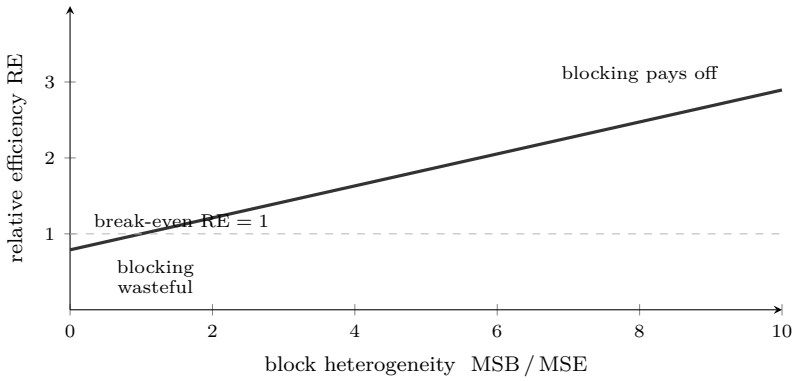
Step 3: prescribe. Re-analyse log(yield): it stabilises the variance, symmetrises the skew, and tends to restore additivity simultaneously; re-check both plots afterward.

Funnel + convex QQ + $s_{\max}/s_{\min} = 3.5$ $\Rightarrow$ analyse log(yield).

If the transform fails to flatten things, fall back to the nonparametric Friedman test.

13.4.4 Efficiency rises with block heterogeneity

Remark. Relative efficiency is, at heart, a story about **block heterogeneity**. The more the blocks differ from one another — the larger SSB (hence MSB) is relative to MSE — the more variation a completely randomized design would have left festering in its error, and so the higher RE climbs. When the blocks are nearly identical, SSB is small, RE sits near 1, and the df spent on the block term were essentially thrown away. The schematic below traces this monotone rise.



Schematic of relative efficiency against block heterogeneity MSB / MSE *(drawn for* $k = 4$, $b = 5$, *where* $\mathrm{RE} = (4r + 15)/19$ *with* $r = \mathrm{MSB}/\mathrm{MSE}$*). When the blocks barely differ (*$\mathrm{MSB}/\mathrm{MSE} \approx 1$*, far left),* RE *sits near the break-even line and the df spent on the block term return almost nothing. As the blocks grow more heterogeneous (rightward),* RE *climbs steadily above* 1*: a completely randomized design would have drowned that block variation in its error, so blocking buys real precision.*

Example 13.4.14 (Read efficiency off the heterogeneity schematic)**.** For the design above ($k = 4$, $b = 5$, so $\mathrm{RE} = (4r + 15)/19$ with $r = \mathrm{MSB}/\mathrm{MSE}$), find the block heterogeneity r at which blocking exactly breaks even, and confirm that $r = 8$ gives a substantial gain.

Which rule, and why? The break-even point is where the efficiency line crosses $\mathrm{RE} = 1$; solve the linear relation for r. This converts the visual schematic into the exact threshold separating "wasteful" from "worthwhile" blocking.

Solution.

Step 1: solve for break-even. Set $\mathrm{RE} = 1$:

$$\frac{4r + 15}{19} = 1 \implies 4r + 15 = 19 \implies r = 1.$$

So blocking breaks even exactly when MSB = MSE — intuitive, since then the blocks carry no more variation than the noise.

Step 2: evaluate at $r = 8$.

$$\mathrm{RE} = \frac{4(8) + 15}{19} = \frac{32 + 15}{19} = \frac{47}{19} = 2.47.$$

$$\boxed{\text{Break-even at } \mathrm{MSB}/\mathrm{MSE} = 1; \quad \mathrm{MSB}/\mathrm{MSE} = 8 \Rightarrow \mathrm{RE} = 2.47.}$$

Highly heterogeneous blocks ($r = 8$) make the block design nearly two-and-a-half times as efficient as a CRD — the steep payoff visible on the right of the schematic.

13.4.5 When assumptions fail: the Friedman alternative and design routing

Proposition 13.4.15 (The Friedman test, conceptually)**.** *When normality is doubtful (small b, heavy-tailed or ordinal data) the rank-based* ***Friedman test*** *(Chapter 14) replaces the block-design* F*.* Within each block

the k treatment responses are ranked $1, \ldots, k$; let R_i be the sum of ranks for treatment i across the b blocks. Under H_0 (no treatment effect) the statistic

$$Q = \frac{12}{bk(k+1)} \sum_{i=1}^{k} R_i^2 - 3b(k+1)$$

is approximately χ^2_{k-1}. Ranking inside *each block is exactly the nonparametric way of removing block effects — the rank analogue of subtracting the block mean.*

Example 13.4.16 (Choose Friedman when normality is doubtful: taste panel)**.** A sensory study has $b = 6$ trained judges (blocks) each rank-scoring $k = 4$ recipes on a 1–9 hedonic scale. The scores are ordinal, the per-recipe samples are tiny, and a QQ plot of the residuals is badly curved. Which test should be reported?

Which rule, and why? Ordinal ratings with small b and a curved QQ plot defeat the normality assumption the block-design F needs; the assumption-light replacement that still respects the blocking is the rank-based Friedman test, not the parametric F.

Solution.

Step 1: confirm the parametric route is unsafe. Hedonic scores are ordinal, not interval; with only $b = 6$ blocks the CLT cannot rescue normality, and the curved QQ plot confirms (A2) fails.

Step 2: pick the matched nonparametric test. The **Friedman test** ranks the four recipes *within each judge*, removing judge-to-judge differences exactly as blocking intends, then compares rank sums via $Q \sim \chi^2_{k-1} = \chi^2_3$.

Ordinal scores, small b, curved QQ $\Rightarrow$ Friedman test (Ch. 14), not the F.

Friedman is to the randomized block design what Kruskal–Wallis is to one-way ANOVA.

Example 13.4.17 (Recognition: route a described experiment to the right design and test)**.** For each scenario, name the design and the test: (a) the *same* 10 patients each measured under 3 drugs, responses roughly normal; (b) 4 diets fed to 5 litters of pigs, one pig per diet per litter, ordinal body-condition scores; (c) 3 fertilizers assigned completely at random to 30 separate plots, normal yields.

Which rule, and why? Routing hinges on two reads: is there a nuisance grouping matched across treatments (blocking)? and are the parametric assumptions safe (parametric F/t vs. rank test)? Same units across treatments = block design; completely random assignment = one-way ANOVA; doubtful normality = nonparametric.

Solution.

Step (a). Same patients under every drug = subjects are blocks; $k = 3$ treatments, normal $\Rightarrow$ randomized block design, treatment F-test (and since it is repeated measures, this is the block-design generalisation of the paired t).

Step (b). Litters are a natural nuisance grouping, one pig per diet per litter = block design; but ordinal scores defeat normality $\Rightarrow$ **Friedman test**.

Step (c). No matched grouping — completely random assignment of plots — and normal yields $\Rightarrow$ a *completely randomized* one-way ANOVA, the ordinary treatment F (no block term).

(a) RBD, block-design F; (b) RBD design but Friedman test (ordinal); (c) CRD, one-way ANOVA F.

The presence of a block structure and the safety of the normal-theory assumptions are independent questions — answer both before naming the test.

Example 13.4.18 (CONTRAST: do not Friedman-test data that meet the assumptions)**.** A block design ($k = 5$ machines, $b = 12$ operators) has interval-scale output, a straight QQ plot, a flat residual band, and parallel profiles. A cautious analyst proposes "playing it safe" with the Friedman test anyway. Why is the parametric F the better choice here?

Which rule, and why? The nonparametric test is the right tool only when the parametric assumptions fail; using it on clean, interval, normal data needlessly discards information and power. Recognising that the assumptions *hold* is what keeps you on the more powerful parametric F.

Solution.

Step 1: confirm the assumptions hold. Interval data, straight QQ (normal), flat band (equal variance), parallel profiles (additive), and $b = 12$ blocks — every condition the block-design F needs is satisfied.

Step 2: weigh the cost of ranking. Friedman replaces the actual values with within-block ranks, throwing away the magnitude information; when normality genuinely holds it is less powerful than the F, so it would more often miss a real treatment difference.

All assumptions met $\Rightarrow$ use the block-design F; Friedman would needlessly lose power.

"Playing it safe" with a nonparametric test is not free — the insurance premium is power you did not need to spend.

13.5 Exercises

All problems below are original. Full solutions begin on page 899.

A. The randomized block design and model

A1. A sleep researcher wants to compare $k = 3$ doses of a sleep aid (low, medium, high) on total minutes of deep sleep. Twelve volunteers are available, and the researcher knows that *age* is a strong predictor of baseline deep sleep: the volunteers split neatly into four age brackets with three people each.

(a) Name a good blocking variable and explain in one sentence why it is a sensible choice.

(b) Identify the experimental unit, the treatment, the number of treatments k, and the number of blocks b.

(c) How many total observations N result, and how many observations sit in each cell of the treatments-by-blocks table?

A2. For each experiment below, name a good blocking variable (the known, measurable nuisance you would match units on *before* looking at the response), and state in one phrase what that block removes from the error.

(a) Comparing $k = 3$ tire compounds for tread wear, with the tires mounted on a fleet of delivery vans that drive very different routes.

(b) Comparing $k = 4$ catalysts for reaction yield, where each day's batch of raw feedstock has slightly different purity and only one batch is available per day.

(c) Comparing $k = 2$ keyboard layouts for typing speed, tested on typists whose baseline speed varies enormously from person to person.

A3. Classify each design as *randomized block*, *completely randomized*, or *two-factor factorial*, and justify your choice in one sentence. If it is a block design, name the block.

(a) A nutritionist pools 24 mice, assigns 8 at random to each of $k = 3$ diets, and weighs them after four weeks.

(b) An optician compares $k = 3$ lens coatings for glare; six pairs of identical-twin drivers are recruited, and within each pair the two coatings… (here only $k = 2$ of the three are used per pair) — restrict to the $k = 2$ coatings "A" and "B" and treat the twin pair as the matched group.

(c) An agronomist crosses $k = 2$ seed varieties with 3 soil-pH levels of genuine scientific interest, replicating each variety-by-pH combination five times.

(d) A baker compares $k = 4$ flour brands; dough is mixed on each of $b = 5$ separate mornings (oven temperature and humidity drift morning to morning), baking one loaf per brand each morning.

A4. A physical-therapy clinic compares $k = 4$ stretching protocols for hip flexibility. Patients arrive in widely varying baseline condition, so the lead therapist matches patients into groups of four with similar starting flexibility, then assigns the four protocols at random within each group. There are $b = 6$ such matched groups.

(a) Write the additive randomized block model for the flexibility gain Y_{ij}, naming every symbol and stating the index ranges and the error distribution.

(b) State the centering constraints on the treatment and block effects.

(c) State the null and alternative hypotheses for the *treatment* comparison, and give the estimators $\widehat{\tau}_i$ and $\widehat{\beta}_j$ in terms of the means.

A5. A barista runs a randomized block experiment on $k = 3$ brewing temperatures (cool, warm, hot) using $b = 4$ different coffee origins as blocks, scoring each cup on a 100-point scale. The treatments-by-blocks table of scores is

	O_1	O_2	O_3	O_4
T_1	78	84	72	82
T_2	82	88	76	86
T_3	86	92	80	90

(a) Compute the three treatment (row) means, the four block (column) means, and the grand mean.

(b) Compute the estimated treatment effects $\widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}$ and verify they sum to zero.

(c) Compute the estimated block effects $\widehat{\beta}_j = \overline{Y}_{\cdot j} - \overline{Y}$, verify they sum to zero, and say in one sentence whether the spread of the block effects suggests blocking was worthwhile.

A6. A wildlife biologist measured antler growth for $k = 3$ mineral supplements, recording the raw runs as (supplement, herd, growth in cm):

$$(1,1,18),(2,1,22),(3,1,25);\quad (1,2,24),(2,2,28),(3,2,31);$$

$$(1,3,15),(2,3,19),(3,3,22);\quad (1,4,21),(2,4,25),(3,4,28).$$

The four herds graze separate ranges of differing forage quality and serve as blocks.

(a) Arrange the runs as a $k \times b$ treatments-by-blocks table.

(b) Give the row means, column means, and grand mean.

(c) By reading the within-block treatment differences, decide whether the additive (no treatment-by-block interaction) model holds exactly for these data.

A7. A trainer measures sprint times for $k = 2$ warm-up routines (A and B) on $b = 5$ athletes, each athlete doing both routines on separate days. Athletes differ greatly in raw speed. The times (in seconds; lower is faster) are

	S_1	S_2	S_3	S_4	S_5
A	12.4	11.0	13.2	10.6	12.0
B	12.8	11.5	13.6	11.1	12.5

The trainer asks: "does it really matter whether I treat the athlete as a block?" Show, on *these* numbers, why blocking on athlete is essential.

(a) Compute the two treatment means and the estimated treatment gap $\overline{Y}_A - \overline{Y}_B$.

(b) Pretend the athlete is ignored: treat the five A-times and five B-times as two independent samples of size 5 and compute the pooled variance s_p^2 as a yardstick for the gap.

(c) Now use the block: compute the five within-athlete differences $D_j = Y_{Aj} - Y_{Bj}$, their mean $\overline{D}$, and their variance s_D^2. Compare s_D^2 with s_p^2 and state in one sentence what the known athlete nuisance was doing to the unblocked analysis.

B. The F-test and two-way ANOVA table

B1. A materials lab measures the breaking strength (newtons) of $k = 3$ adhesive formulations, blocking on $b = 4$ humidity chambers whose moisture levels differ:

	Ch 1	Ch 2	Ch 3	Ch 4	$\overline{Y}_{i\cdot}$
F1	42	46	38	50	44
F2	48	52	44	56	50
F3	39	43	35	47	41
$\overline{Y}_{\cdot j}$	43	47	39	51	

The grand mean is $\overline{Y} = 45$ and the total sum of squares is $\mathrm{SST} = 420$. Carry out the complete treatment F-test at $\alpha = 0.05$: compute SSTR, SSB, SSE, build the two-way ANOVA table, form the treatment F, give both degrees of freedom and the critical value, and state the decision in words. (Use $F_{0.05,2,6} = 5.14$.)

B2. A clinical-pharmacology study records time-to-peak plasma concentration (minutes) for $k = 4$ extended-release tablet designs, blocking on $b = 3$ volunteers whose metabolisms differ:

	Vol 1	Vol 2	Vol 3	$\overline{Y}_{i\cdot}$
A	88	96	92	92
B	80	88	84	84
C	95	103	99	99
D	81	89	85	85
$\overline{Y}_{\cdot j}$	86	94	90	

The grand mean is $\overline{Y} = 90$ and the total is SST $= 584$. Run the full randomized-block F-test of H_0: the four designs have equal mean time-to-peak, at $\alpha = 0.05$. Report the ANOVA table, the treatment F with both df, the critical value $F_{0.05,3,6} = 4.76$, and your conclusion.

B3. A partly erased two-way ANOVA printout for $k = 5$ teaching modules tested across $b = 4$ classrooms (blocks) reads

Source	df	SS	MS	F
Treatments	?	96.0	?	?
Blocks	?	?	14.0	?
Error	?	?	?	
Total	?	240.0		

(a) Recover every degree of freedom, sum of squares, and mean square, and fill in the table completely.

(b) Form both the treatment F and the block F.

(c) At $\alpha = 0.05$, with $F_{0.05,4,12} = 3.26$ for treatments and $F_{0.05,3,12} = 3.49$ for blocks, state both decisions and say whether blocking on classrooms was worthwhile.

B4. The *same* data set is analyzed two ways. Six cell yields for $k = 3$ seed varieties grown on $b = 2$ irrigation plots (blocks) give SSTR $= 84$, a large between-plot SSB $= 120$, and total SST $= 216$.

(a) Treat the data as a one-way ANOVA that ignores the plots (folds them into error). Compute SSE, its df, MSE, and the treatment F; decide at $\alpha = 0.05$ using $F_{0.05,2,3} = 9.55$.

(b) Now treat the data as a randomized block design that removes the plot variation. Compute SSE, its df, MSE, and the treatment F; decide at $\alpha = 0.05$ using $F_{0.05,2,2} = 19.00$.

(c) In one sentence, explain why the treatment F changes between the two analyses even though SSTR is identical, and which mean square shrank.

B5. A randomized block test of $k = 3$ catalysts across $b = 5$ reactor runs (blocks) returns a treatment statistic $F = 4.95$ on $\mathrm{F}_{2,8}$. The standard cutoffs at these degrees of freedom are $F_{0.10,2,8} = 3.11$, $F_{0.05,2,8} = 4.46$, and $F_{0.025,2,8} = 6.06$.

(a) Bracket the p-value between two tabulated levels.

(b) State the decision at $\alpha = 0.05$ and, separately, at $\alpha = 0.025$.

B6. A randomized block experiment uses $k = 7$ fertilizers and $b = 4$ blocks.

(a) Before any data are collected, state the degrees of freedom for treatments, blocks, error, and total, and verify they add to $N - 1$.

(b) A colleague reports the error df as $N - k = 28 - 7 = 21$. Identify the mistake and give the correct error df, explaining where the missing degrees of freedom went.

(c) If the analyst found SSTR $= 72.0$, SSB $= 45.0$, and SST $= 153.0$, give MSE and the treatment F.

B7. An entomologist compares $k = 3$ insect repellents on $b = 4$ test subjects (blocks), recording mean landings per minute:

	S_1	S_2	S_3	S_4	$\overline{Y}_{i\cdot}$
R1	12	18	9	13	13
R2	8	14	5	9	9
R3	16	22	13	17	17
$\overline{Y}_{\cdot j}$	12	18	9	13	

The grand mean is $\overline{Y} = 13$ and the total is SST $= 260$. The data were built nearly additively, so first compute SSTR, SSB, and SSE by subtraction; you should find a small positive error. Then build the two-way ANOVA table, test the repellents at $\alpha = 0.05$ ($F_{0.05,2,6} = 5.14$), report the block F as a diagnostic, and state in words whether the repellents differ and whether blocking on subjects was justified.

C. Paired t as a two-treatment block design

C1. A nutrition lab measures fasting glucose (mg/dL) in $b = 5$ volunteers on a control breakfast and again on a high-fiber breakfast, same volunteer both mornings. The pairs $(Y_{1j}, Y_{2j}) = (\text{control}, \text{fiber})$ are

$$(98, 92),\ (110, 103),\ (105, 101),\ (121, 113),\ (94, 90).$$

Take this as a 2-treatment randomized block design (equivalently a paired test). Use $\alpha = 0.05$, $t_{0.025,4} = 2.776$, $F_{0.05,1,4} = 7.71$.

(a) Form the within-block differences $D_j = Y_{1j} - Y_{2j}$ and run the paired t-test (two-sided $H_0 : \mu_D = 0$).

(b) Build the two-way ANOVA: compute SSTR and SSE from the differences, give the mean squares, and the treatment F on its two degrees of freedom.

(c) Verify numerically that $F = t_{\text{obs}}^2$ and that $F_{0.05,1,4} = t_{0.025,4}^2$, so the two tests share a p-value, and state the decision in words.

C2. For the glucose data of C1, the paired test never displays the block sum of squares.

(a) Compute the block means $\overline{Y}_{\cdot j} = \frac{1}{2}(Y_{1j} + Y_{2j})$ and the grand mean $\overline{Y}$, then the block sum of squares $\text{SSB} = 2\sum_j (\overline{Y}_{\cdot j} - \overline{Y})^2$.

(b) Compute the total $\text{SST} = \sum_{i,j} (Y_{ij} - \overline{Y})^2$ and verify the partition

$$\text{SST} = \text{SSTR} + \text{SSB} + \text{SSE}\,.$$

(c) In one sentence, say what quantity SSB measures and why the paired test benefits from removing it even though it never computes it.

C3. Still using the glucose data, suppose a colleague ignores the pairing and (wrongly) runs a one-way ANOVA on the two columns, so the error becomes $\text{SSE}_{\text{1-way}} = \text{SSB} + \text{SSE}$ on $N - k = 8$ degrees of freedom.

(a) Give the blocked MSE on $(k-1)(b-1)$ df and the unblocked $\text{MSE}_{\text{1-way}}$ on $N - k$ df.

(b) With the same SSTR, compute both F values and report the decision each would reach at $\alpha = 0.05$ (one-way critical value $F_{0.05,1,8} = 5.32$).

(c) By what factor did blocking shrink the error mean square, and did the treatment effect itself change?

C4. A pharmacologist gives $b = 7$ patients a standard analgesic and, on a later visit, a new analgesic, recording the time (minutes) until reported relief. The within-subject drops $D_j = (\text{standard}) - (\text{new})$ are

$$8,\ 3,\ 11,\ -2,\ 6,\ 9,\ 4.$$

Test at $\alpha = 0.05$ whether the new drug *shortens* mean time to relief (a directional claim), and explain how the $k = 2$ block F relates to this one-sided test. Use $t_{0.05,6} = 1.943$, $F_{0.05,1,6} = 5.99$, $t_{0.025,6} = 2.447$.

C5. Twelve identical leaf samples are each split in half; one half is assayed for chlorophyll by a spectrophotometer (Y_1) and the other half by a field meter (Y_2). The within-sample differences $D_j = Y_{1j} - Y_{2j}$ (mg/g) over the $b = 12$ leaves have $\overline{D} = 0.45$ and $S_D^2 = 0.81$.

(a) Name the block and the two treatments, and obtain $\text{SSTR} = \frac{b}{2}\overline{D}^2$ and $\text{SSE} = \frac{1}{2}(b-1)S_D^2$.

(b) Give the treatment F on its two df and test $H_0 : \mu_D = 0$ two-sided at $\alpha = 0.05$, where $F_{0.05,1,11} = 4.84$.

(c) Report the equivalent paired t_{obs} and confirm it equals $\sqrt{F}$.

C6. A printout of a randomized block analysis with $k = 2$ treatments and $b = 8$ blocks reads

Source	df	SS	MS	F
Treatment	?	18.0	?	?
Block	?	63.0	?	
Error	?	7.0	?	
Total	?	?		

Fill in every degree of freedom, the three mean squares, the total SS, and the treatment F; then recover the equivalent paired t-statistic and its df. At $\alpha = 0.05$, with $F_{0.05,1,7} = 5.59$, state the decision.

C7. For each study decide whether it is a 2-treatment randomized block design (equivalently a paired test) or two independent samples, and name the block when one exists.

(a) "The same 18 blood specimens are each measured for cholesterol on an old analyzer and a new analyzer."

(b) "10 mice receive diet A and a *separate* 10 mice receive diet B; weight gain is compared."

(c) "In each of 14 greenhouses, a north bench and a south bench grow the same cultivar under two light recipes."

(d) "A researcher has reaction times for 9 subjects under caffeine and, separately, for 9 *different* subjects under placebo, lines the two columns up, and differences them as a paired test."

D. Efficiency, additivity, diagnostics

D1. A horticulture lab compares $k = 4$ rooting hormones for cutting survival, blocking on $b = 5$ greenhouse benches (one cutting of each hormone per bench). The two-way ANOVA returns a bench mean square $\text{MSB} = 96.0$ and an error mean square $\text{MSE} = 8.0$, on $b - 1 = 4$ block df and $(k-1)(b-1) = 12$ error df.

(a) Compute the relative efficiency RE of this randomized block design relative to a completely randomized design, using the variance-ratio form.

(b) State the small-sample degrees-of-freedom correction c, compute it, and report the corrected efficiency RE_{corr}.

(c) In one sentence, decide whether blocking on benches was worthwhile.

D2. An assembly-line study compares $k = 3$ torque settings for fastener seating, blocking on $b = 7$ work shifts. The two-way ANOVA gives $\text{SSB} = 12.0$ on $b - 1 = 6$ block df and $\text{SSE} = 72.0$ on $(k-1)(b-1) = 12$ error df.

(a) Compute MSB and MSE, then the block $F = \text{MSB} \,/\, \text{MSE}$, and say what its size tells you about the shifts.

(b) Compute the relative efficiency RE (variance-ratio form).

(c) If the analyst had ignored shifts and run a one-way ANOVA, what error mean square MSE_{CRD} and error df would result? Compare to the block design and state which analysis is sharper.

D3. A coating trial applies $k = 3$ primers to panels of $b = 2$ alloys (brass, titanium). The cell means (peel strength, in N) are

$$\text{brass}: \ P_1 = 34, \ P_2 = 26, \ P_3 = 30;$$
$$\text{titanium}: \ P_1 = 22, \ P_2 = 38, \ P_3 = 28.$$

By comparing the P_1-versus-P_2 gap in each alloy, decide whether additivity is tenable, and state in one sentence what this means for a simple block-design treatment F.

D4. A curing experiment runs $k = 3$ pressures across $b = 3$ presses (blocks) and records cell means

$$\text{press } 1: \ 50, \ 57, \ 64; \quad \text{press } 2: \ 46, \ 53, \ 60; \quad \text{press } 3: \ 54, \ 61, \ 68.$$

(a) List the low→mid→high treatment increments within each press and decide whether the profiles are parallel (additive).

(b) Explain why "parallel profiles" here does not mean "no pressure effect," and give the common low-to-high treatment gap.

D5. A pharmacokinetic block design ($k = 4$ doses, $b = 6$ volunteers as blocks) produces profiles that do not cross but *fan out*: the dose-to-dose gaps are narrow in low-responding volunteers and wide in high-responding volunteers, the spacing growing roughly in proportion to the response level.

(a) Name the formal test that targets this multiplicative form of interaction, and give the exact F reference distribution (with both df) it uses.

(b) Name the data transformation suggested if that test is significant, and explain in one sentence why it should restore additivity.

D6. For each scenario, name the experimental design and the test you would report, and justify each in a phrase:

(a) the *same* 8 runners each timed under 3 shoe models, finishing times roughly normal;

(b) 5 feed additives given to 4 cages of fish, one tank of each additive per cage, with ordinal palatability scores and a badly curved residual QQ plot;

(c) 3 paint formulas sprayed onto 24 separate test boards assigned completely at random, normal gloss readings.

D7. A wear-testing block design ($k = 4$ lubricants, $b = 6$ engines) shows a residual-vs-fitted plot that funnels open (tight scatter at low fitted wear, wide scatter at high fitted wear) with a group spread ratio $s_{\max}/s_{\min} = 3.8$, and a convex (right-skewed) residual QQ plot. A colleague suggests "playing it safe" by reporting the Friedman test instead of the treatment F.

(a) Diagnose the two assumption violations the plots reveal and prescribe a single remedy that attacks both.

(b) Is the colleague's reflex to jump straight to Friedman the best first move? Explain when the Friedman test *is* the right call versus when it needlessly costs power.

E. Block-design synthesis

E1. A food scientist compares $k = 3$ packaging films for moisture retention, blocking on $b = 4$ storage warehouses (the blocks) whose humidity differs. The response is weight gain (mg) after one month:

	W_1	W_2	W_3	W_4	$\overline{Y}_{i\cdot}$
Film A	14	15	13	18	15
Film B	16	27	15	30	22
Film C	15	18	11	24	17
$\overline{Y}_{\cdot j}$	15	20	?	?	

(a) Fill in the two missing block means and verify the grand mean is $\overline{Y} = \frac{1}{12}\sum_{i,j} Y_{ij} = 18$.

(b) Compute SSTR and SSB from the row and column means.

(c) Given SST = 382, find SSE by subtraction and confirm the degrees of freedom add to $N - 1$.

(d) Build the two-way ANOVA table and test, at $\alpha = 0.05$, whether the three films differ in mean moisture gain. Use $F_{0.05,2,6} = 5.14$. State the decision in words and box it.

E2. Continue with the packaging study of E1 (MSB and MSE from the completed table, $k = 3$, $b = 4$).

(a) Compute the relative efficiency of this randomized block design relative to a completely randomized design, using the variance-ratio form $\mathrm{RE} = \dfrac{(b-1)\,\mathrm{MSB} + b(k-1)\,\mathrm{MSE}}{(bk-1)\,\mathrm{MSE}}$.

(b) Apply the small-sample correction $c = \dfrac{[(k-1)(b-1)+1][k(b-1)+1]}{[(k-1)(b-1)+3][k(b-1)+3]}$ and report $\mathrm{RE}_{\mathrm{corr}}$.

(c) In one sentence, say what the corrected value means in terms of how many observations a CRD would have needed, and whether blocking on warehouses was worthwhile.

E3. A pharmacology lab tests $k = 2$ analgesic formulations (a standard tablet and a new capsule) for time-to-relief (minutes) on $b = 7$ patients, each patient receiving both on separate occasions in randomized order. The before/after style differences D_j = (tablet) − (capsule) are

$$8,\ 5,\ 11,\ 2,\ 9,\ -1,\ 6.$$

(a) Treating this as a 2×7 block design, obtain $\mathrm{SSTR} = \frac{b}{2}\overline{D}^{\,2}$ and $\mathrm{SSE} = \frac{1}{2}(b-1)S_D^2$, then the treatment F on $\mathrm{F}_{1,6}$.

(b) Run the equivalent paired t-test on $b - 1 = 6$ df and verify $F = t_{\mathrm{obs}}^2$.

(c) Decide, two-sided, at $\alpha = 0.05$ whether the formulations differ ($F_{0.05,1,6} = 5.99$, $t_{0.025,6} = 2.447$). Box the matched conclusion.

E4. A completed randomized-block table for $k = 5$ teaching methods across $b = 6$ schools (blocks) reads

Source	df	SS	MS	F
Treatments	4	120.0	30.0	5.0
Blocks	5	150.0	30.0	5.0
Error	20	120.0	6.0	
Total	35	390.0		

Without recomputing any sum of squares, state the treatment conclusion at $\alpha = 0.05$ ($F_{0.05,4,20} = 2.87$), compute the block F and judge against $F_{0.05,5,20} = 2.71$, and say in one sentence whether blocking on schools was justified.

E5. An agronomist runs $k = 4$ wheat varieties on $b = 5$ irrigation plots (blocks). The two-way analysis gives row (variety) means $\overline{Y}_{i\cdot} = 42,\ 48,\ 39,\ 51$ with grand mean $\overline{Y} = 45$, and the printout reports SSB = 180.0 and SST = 750.0.

(a) Compute SSTR from the variety means and obtain SSE by subtraction.

(b) Form the full two-way ANOVA table (all df, SS, MS, and the treatment F).

(c) Test the varieties at $\alpha = 0.05$ using $F_{0.05,3,12} = 3.49$, and bracket the p-value using $F_{0.10,3,12} = 2.61$ and $F_{0.025,3,12} = 4.47$.

E6. Two randomized block experiments share the design $k = 3$ treatments, $b = 6$ blocks (so the treatment F lives on $\mathrm{F}_{2,10}$, cutoff $F_{0.05,2,10} = 4.10$).

(a) Study P reports MSTR = 18.0, MSB = 2.5, MSE = 5.0. Compute the treatment F, decide at $\alpha = 0.05$, then compute the relative efficiency and comment on whether blocking helped. (Use $bk - 1 = 17$, $b - 1 = 5$, $b(k-1) = 12$.)

(b) Study Q reports MSTR = 18.0, MSB = 40.0, MSE = 5.0. Repeat: treatment F, decision, and relative efficiency.

(c) The two studies have the *same* treatment F. Explain why their relative efficiencies differ so much, and which study's blocking earned its degrees of freedom.

E7. An adhesion study applies $k = 3$ primers to panels of $b = 2$ metals (steel, aluminum), one panel per primer per metal. The cell responses (peel strength, N) are

$$\begin{aligned} \text{steel}: &\ P_1 = 34,\ P_2 = 40,\ P_3 = 30; \\ \text{aluminum}: &\ P_1 = 46,\ P_2 = 28,\ P_3 = 44. \end{aligned}$$

(a) For the primer pair (P_1, P_2), compute the gap $P_2 - P_1$ within each metal and decide whether the additivity (no-interaction) assumption is tenable.

(b) State, in one sentence, what a profile plot of these data would look like and why the ordinary block-design treatment F would be invalid here.

(c) Name one defensible way to proceed when this kind of treatment-by-block interaction is present.

13.6 Solutions

A. The randomized block design and model

A1. Method. Recognize a randomized block design by a known, matchable nuisance variable on which every treatment can be applied once.

(a) Block on the *age bracket*. Age strongly drives baseline deep sleep, the volunteers fall into four natural age groups of three, and within a group we can give all three doses — exactly the setting for blocking.

(b) Experimental unit = one volunteer; treatment = dose with $k = 3$ levels; blocks = age brackets with $b = 4$ levels.

(c) $N = kb = 3 \times 4 = 12$ observations, with exactly *one* observation per cell (each dose appears once in each age bracket — a randomized *complete* block design).

$$\boxed{\text{block = age bracket;} \quad k = 3,\ b = 4,\ N = 12,\ \text{one obs per cell.}}$$

A2. Method. For each design name the nuisance that is known and matchable across all treatments; that is the block, and the variation it carries is pulled out of MSE.

(a) Block on the *van* (equivalently, its route): mount all three compounds on each van so route-to-route wear cancels. The block removes between-route driving severity from the error.

(b) Block on the *day's feedstock batch*: run all four catalysts on the same day's batch so purity is common to all four. The block removes day-to-day feedstock-purity differences.

(c) Block on the *typist*: have each typist try both layouts so personal speed cancels. The block removes person-to-person baseline typing speed.

$$\boxed{\text{blocks: van/route, daily feedstock batch, typist.}}$$

A3. Method. Ask first whether a nuisance is matched *before* assignment (block vs. completely randomized), then whether the second grouping is a nuisance to remove (block) or a factor of interest, crossed and replicated (factorial).

(a) *Completely randomized.* The mice are one undivided pool with random assignment and no matching variable.

(b) *Randomized block* (paired, since $k = 2$). The twin pair is a matched nuisance group; both coatings are applied within each pair. Block = twin pair.

(c) *Two-factor factorial.* Soil pH is a factor of genuine interest, crossed with variety and *replicated* five times per cell — not a nuisance to be removed.

(d) *Randomized block.* Morning-to-morning oven/humidity drift is a known nuisance; all four brands are baked each morning, one loaf per brand. Block = morning ($b = 5$).

$$\boxed{(a)\ \text{CR};\ (b)\ \text{block = twin pair};\ (c)\ \text{factorial};\ (d)\ \text{block = morning.}}$$

A4. Method. A randomized block design uses the additive two-effect model; the inference of interest is on the treatment effects τ_i, with block effects β_j present only to be removed.

(a) For protocol i in matched group j,

$$Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \qquad i = 1, \ldots, 4, \quad j = 1, \ldots, 6,$$
$$\varepsilon_{ij} \sim \mathrm{N}(0, \sigma^2) \text{ independent},$$

where μ is the grand mean, τ_i the effect of protocol i, β_j the effect of matched group j, and ε_{ij} the error.

(b) The effects are centered:

$$\sum_{i=1}^{4} \tau_i = 0, \qquad \sum_{j=1}^{6} \beta_j = 0.$$

(c) Treatment hypotheses and estimators:

$$H_0:\ \tau_1 = \tau_2 = \tau_3 = \tau_4 = 0 \quad \text{vs.} \quad H_1:\ \tau_i \neq 0 \text{ for some } i,$$
$$\widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}, \qquad \widehat{\beta}_j = \overline{Y}_{\cdot j} - \overline{Y}.$$

$$\boxed{Y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij};\quad H_0 : \tau_1 = \cdots = \tau_4 = 0;\quad \widehat{\tau}_i = \overline{Y}_{i\cdot} - \overline{Y}.}$$

A5. Method. Row means estimate treatment levels, column means estimate block levels, and each effect is its mean minus the grand mean.

(a) Row (treatment) means:

$$\overline{Y}_{1\cdot} = \tfrac{78+84+72+82}{4} = \tfrac{316}{4} = 79,$$
$$\overline{Y}_{2\cdot} = \tfrac{82+88+76+86}{4} = \tfrac{332}{4} = 83,$$
$$\overline{Y}_{3\cdot} = \tfrac{86+92+80+90}{4} = \tfrac{348}{4} = 87.$$

Column (block) means:

$$\overline{Y}_{\cdot 1} = \tfrac{78+82+86}{3} = 82, \quad \overline{Y}_{\cdot 2} = \tfrac{84+88+92}{3} = 88,$$
$$\overline{Y}_{\cdot 3} = \tfrac{72+76+80}{3} = 76, \quad \overline{Y}_{\cdot 4} = \tfrac{82+86+90}{3} = 86.$$

Grand mean $\overline{Y} = \tfrac{79+83+87}{3} = 83$ (equivalently $\tfrac{82+88+76+86}{4} = 83$).

(b) Treatment effects:

$$\widehat{\tau}_1 = 79 - 83 = -4, \quad \widehat{\tau}_2 = 83 - 83 = 0, \quad \widehat{\tau}_3 = 87 - 83 = +4,$$
$$\sum_i \widehat{\tau}_i = -4 + 0 + 4 = 0. \ \blacksquare$$

(c) Block effects:

$$\widehat{\beta}_1 = 82 - 83 = -1, \quad \widehat{\beta}_2 = 88 - 83 = +5,$$
$$\widehat{\beta}_3 = 76 - 83 = -7, \quad \widehat{\beta}_4 = 86 - 83 = +3,$$
$$\sum_j \widehat{\beta}_j = -1 + 5 - 7 + 3 = 0. \ \blacksquare$$

The block effects swing from -7 to $+5$, a 12-point spread far larger than the noise, so origin is a strong nuisance and blocking on it was clearly worthwhile.

$$\boxed{\widehat{\tau} = (-4, 0, +4),\quad \widehat{\beta} = (-1, +5, -7, +3),\quad \overline{Y} = 83;\ \text{blocking helps.}}$$

A6. Method. Drop the scattered (i, j, Y) triples into the $k \times b$ grid, then read row, column, and grand means; additivity is checked by within-block treatment differences.

(a) Table (treatments T_1, T_2, T_3 as rows, herds $H_1 \dots H_4$ as columns):

	H_1	H_2	H_3	H_4	$\overline{Y}_{i\cdot}$
T_1	18	24	15	21	19.5
T_2	22	28	19	25	23.5
T_3	25	31	22	28	26.5
$\overline{Y}_{\cdot j}$	21.667	27.667	18.667	24.667	23.167

(b) Row means $\overline{Y}_{i\cdot} = (19.5,\ 23.5,\ 26.5)$; column means $\overline{Y}_{\cdot j} = (21.667,\ 27.667,\ 18.667,\ 24.667)$; grand mean

$$\overline{Y} = \tfrac{19.5+23.5+26.5}{3} = \tfrac{69.5}{3} = 23.167.$$

(c) Within-block treatment differences (e.g. $T_2 - T_1$ and $T_3 - T_2$ down each column):

$$\begin{aligned} T_2 - T_1 &: \ 4,\ 4,\ 4,\ 4 \quad \text{(equal across all blocks)},\\ T_3 - T_2 &: \ 3,\ 3,\ 3,\ 3 \quad \text{(equal across all blocks)}. \end{aligned}$$

Because every within-block treatment gap is the same constant in every herd, the profiles are exactly parallel and the additive (no-interaction) model holds exactly.

$$\boxed{\overline{Y}_{i\cdot} = (19.5, 23.5, 26.5),\ \overline{Y} = 23.167;\ \text{constant within-block gaps} \Rightarrow \text{additive.}}$$

A7. Method. On the same data, an unblocked (two independent samples) analysis leaves the athlete swing in the error, while the within-athlete differences cancel it. Comparing the two error yardsticks shows why the known nuisance must be removed.

(a) Treatment means:

$$\begin{aligned} \overline{Y}_A &= \tfrac{12.4+11.0+13.2+10.6+12.0}{5} = \tfrac{59.2}{5} = 11.84,\\ \overline{Y}_B &= \tfrac{12.8+11.5+13.6+11.1+12.5}{5} = \tfrac{61.5}{5} = 12.30, \end{aligned}$$

so the estimated treatment gap is $\overline{Y}_A - \overline{Y}_B = 11.84 - 12.30 = -0.46$ s (routine A is about half a second faster).

(b) Unblocked yardstick. Deviations of the five A-times from 11.84:

$$\begin{aligned} \textstyle\sum (Y_{Aj} - 11.84)^2 &= 0.56^2 + (-0.84)^2 + 1.36^2 + (-1.24)^2 + 0.16^2\\ &= 0.3136 + 0.7056 + 1.8496 + 1.5376 + 0.0256 = 4.432, \end{aligned}$$

so $s_A^2 = 4.432/4 = 1.108$. Deviations of the five B-times from 12.30:

$$\begin{aligned} \textstyle\sum (Y_{Bj} - 12.30)^2 &= 0.50^2 + (-0.80)^2 + 1.30^2 + (-1.20)^2 + 0.20^2\\ &= 0.25 + 0.64 + 1.69 + 1.44 + 0.04 = 4.06, \end{aligned}$$

so $s_B^2 = 4.06/4 = 1.015$. Pooled,

$$s_p^2 = \tfrac{4.432+4.06}{8} = \tfrac{8.492}{8} = 1.0615.$$

The half-second gap sits inside this large error: the athlete-to-athlete spread (10.6 s up to 13.2 s) has been dumped into the variance.

(c) Blocked analysis. Within-athlete differences $D_j = Y_{Aj} - Y_{Bj}$:

$$D = (-0.4,\ -0.5,\ -0.4,\ -0.5,\ -0.5), \qquad \overline{D} = \tfrac{-2.3}{5} = -0.46.$$

$$\begin{aligned} \textstyle\sum (D_j - (-0.46))^2 &= 0.06^2 + (-0.04)^2 + 0.06^2 + (-0.04)^2 + (-0.04)^2\\ &= 0.0036 + 0.0016 + 0.0036 + 0.0016 + 0.0016 = 0.0120, \end{aligned}$$

so $s_D^2 = 0.0120/4 = 0.0030$. The athlete nuisance had inflated the unblocked variance to $s_p^2 \approx 1.06$; differencing within athlete collapsed it to $s_D^2 = 0.003$, roughly a *350-fold* reduction, exposing the steady half-second advantage of routine A.

$$\boxed{\text{unblocked } s_p^2 \approx 1.06 \ \longrightarrow\ \text{blocked } s_D^2 = 0.003 :\ \text{the athlete nuisance must be blocked out.}}$$

B. The F-test and two-way ANOVA table

B1. Full block F-test, adhesive formulations ($k = 3$, $b = 4$). The row means $\overline{Y}_{i\cdot}$ feed $\mathrm{SSTR} = b\sum_i(\overline{Y}_{i\cdot} - \overline{Y})^2$, the column means $\overline{Y}_{\cdot j}$ feed $\mathrm{SSB} = k\sum_j(\overline{Y}_{\cdot j} - \overline{Y})^2$, error is by subtraction, and the treatment test is $F = \mathrm{MSTR}\,/\,\mathrm{MSE}$ on $\mathrm{F}_{k-1,(k-1)(b-1)} = \mathrm{F}_{2,6}$.

$$\begin{aligned}
\mathrm{SSTR} &= 4\big[(44-45)^2 + (50-45)^2 + (41-45)^2\big] = 4[1+25+16] = 4(42) = 168,\\
\mathrm{SSB} &= 3\big[(43-45)^2 + (47-45)^2 + (39-45)^2 + (51-45)^2\big]\\
&= 3[4+4+36+36] = 3(80) = 240,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB} = 420 - 168 - 240 = 12 \quad \text{on } (k-1)(b-1) = 2\cdot 3 = 6 \text{ df.}
\end{aligned}$$

Mean squares and the treatment statistic:

$$\mathrm{MSTR} = \frac{168}{2} = 84.0, \qquad \mathrm{MSB} = \frac{240}{3} = 80.0, \qquad \mathrm{MSE} = \frac{12}{6} = 2.0,$$
$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{84.0}{2.0} = 42.0.$$

The two-way ANOVA table is

Source	df	SS	MS	F
Treatments	2	168.0	84.0	42.0
Blocks	3	240.0	80.0	40.0
Error	6	12.0	2.0	
Total	11	420.0		

The treatment F is read against $\mathrm{F}_{2,6}$ with cutoff $F_{0.05,2,6} = 5.14$. Since $42.0 > 5.14$, reject H_0 at the 5% level: there is significant evidence that the three adhesive formulations differ in mean breaking strength.

$$\boxed{F = 42.0 > F_{0.05,2,6} = 5.14 \;\Rightarrow\; \text{reject } H_0;\ \text{the formulations differ.}}$$

B2. Full block F-test, tablet designs ($k = 4$, $b = 3$). Hypotheses: H_0: all four design means equal versus H_1: at least two differ. Row means $92, 84, 99, 85$ give SSTR; column means $86, 94, 90$ give SSB; error by subtraction.

$$\begin{aligned}
\mathrm{SSTR} &= 3\big[(92-90)^2 + (84-90)^2 + (99-90)^2 + (85-90)^2\big]\\
&= 3[4+36+81+25] = 3(146) = 438,\\
\mathrm{SSB} &= 4\big[(86-90)^2 + (94-90)^2 + (90-90)^2\big] = 4[16+16+0] = 4(32) = 128,\\
\mathrm{SSE} &= \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB} = 584 - 438 - 128 = 18 \quad \text{on } (k-1)(b-1) = 3\cdot 2 = 6 \text{ df.}
\end{aligned}$$

Mean squares and treatment F:

$$\mathrm{MSTR} = \frac{438}{3} = 146.0, \qquad \mathrm{MSB} = \frac{128}{2} = 64.0, \qquad \mathrm{MSE} = \frac{18}{6} = 3.0,$$
$$F = \frac{\mathrm{MSTR}}{\mathrm{MSE}} = \frac{146.0}{3.0} = 48.67.$$

The two-way ANOVA table:

Source	df	SS	MS	F
Treatments	3	438.0	146.00	48.67
Blocks	2	128.0	64.00	21.33
Error	6	18.0	3.00	
Total	11	584.0		

The treatment F on $\mathrm{F}_{3,6}$ has cutoff $F_{0.05,3,6} = 4.76$. Since $48.67 > 4.76$, reject H_0 at the 5% level: the four extended-release designs have significantly different mean times-to-peak concentration.

$$\boxed{F = 48.67 > F_{0.05,3,6} = 4.76 \;\Rightarrow\; \text{reject } H_0;\ \text{the designs differ.}}$$

B3. Fill a partly blank two-way table ($k = 5$, $b = 4$). Back-fill with the three identities: df add, SS add, and MS = SS/df.

(a) Degrees of freedom from the grid shape:

$$k - 1 = 4 \text{ (treatments)}, \qquad b - 1 = 3 \text{ (blocks)},$$
$$(k-1)(b-1) = 4 \cdot 3 = 12 \text{ (error)}, \qquad N - 1 = kb - 1 = 20 - 1 = 19 \text{ (total)},$$

and indeed $4 + 3 + 12 = 19$. The block mean square unlocks $\mathrm{SSB} = \mathrm{MSB} \times (b-1) = 14.0 \times 3 = 42.0$; then error falls out by subtraction, $\mathrm{SSE} = \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB} = 240.0 - 96.0 - 42.0 = 102.0$. The mean squares are

$$\mathrm{MSTR} = \frac{96.0}{4} = 24.0, \qquad \mathrm{MSE} = \frac{102.0}{12} = 8.5.$$

Source	df	SS	MS	F
Treatments	4	96.0	24.0	2.82
Blocks	3	42.0	14.0	1.65
Error	12	102.0	8.5	
Total	19	240.0		

(b) The two F-statistics divide each MS by the same $\mathrm{MSE} = 8.5$:

$$F_{\mathrm{tr}} = \frac{24.0}{8.5} = 2.82, \qquad F_{\mathrm{bl}} = \frac{14.0}{8.5} = 1.65.$$

(c) Treatments: $F_{\mathrm{tr}} = 2.82$ against $\mathrm{F}_{4,12}$ with cutoff $F_{0.05,4,12} = 3.26$; since $2.82 < 3.26$, fail to reject H_0 — the modules are not shown to differ at the 5% level. Blocks: $F_{\mathrm{bl}} = 1.65$ against $\mathrm{F}_{3,12}$ with cutoff $F_{0.05,3,12} = 3.49$; since $1.65 < 3.49$, the classrooms did not differ appreciably, so blocking on classrooms bought little here.

$$\boxed{F_{\mathrm{tr}} = 2.82 < 3.26 \text{ (retain)}, \quad F_{\mathrm{bl}} = 1.65 < 3.49 \text{ (blocking not worthwhile).}}$$

B4. CONTRAST: same data, without blocks then with blocks ($k = 3$, $b = 2$). The treatment sum of squares $\mathrm{SSTR} = 84$ is identical in both analyses; only the error changes.

(a) *One-way (blocks ignored).* The plot variation is folded into the error:

$$\mathrm{SSE}_{\text{1-way}} = \mathrm{SST} - \mathrm{SSTR} = 216 - 84 = 132 \quad \text{on } N - k = 6 - 3 = 3 \text{ df},$$
$$\mathrm{MSE}_{\text{1-way}} = \frac{132}{3} = 44.0, \qquad F_{\text{1-way}} = \frac{\mathrm{SSTR}/(k-1)}{\mathrm{MSE}} = \frac{84/2}{44.0} = \frac{42.0}{44.0} = 0.955.$$

Against $\mathrm{F}_{2,3}$ with cutoff $F_{0.05,2,3} = 9.55$: since $0.955 < 9.55$, fail to reject H_0 — the plot spread has drowned the variety effect.

(b) *Randomized block (plots removed).* Pulling out $\mathrm{SSB} = 120$:

$$\mathrm{SSE}_{\mathrm{block}} = \mathrm{SST} - \mathrm{SSTR} - \mathrm{SSB} = 216 - 84 - 120 = 12$$
$$\text{on } (k-1)(b-1) = 2 \cdot 1 = 2 \text{ df},$$
$$\mathrm{MSE}_{\mathrm{block}} = \frac{12}{2} = 6.0, \qquad F_{\mathrm{block}} = \frac{84/2}{6.0} = \frac{42.0}{6.0} = 7.0.$$

Against $\mathrm{F}_{2,2}$ with cutoff $F_{0.05,2,2} = 19.00$: since $7.0 < 19.00$, still fail to reject, but the statistic has climbed sharply (the tiny $b = 2$ design leaves only 2 error df, so the cutoff is enormous).

(c) The treatment F changes because its *denominator* changes: SSTR is fixed at 84, but blocking carves the between-plot SSB = 120 out of the error, dropping the error mean square from MSE = 44.0 to MSE = 6.0 and lifting the treatment F from 0.955 to 7.0.

$$\boxed{F_{\text{1-way}} = 0.955 \text{ (MSE} = 44.0) \;\longrightarrow\; F_{\text{block}} = 7.0 \text{ (MSE} = 6.0); \text{ blocking shrank MSE}.}$$

B5. p-value for a block treatment F. A randomized-block treatment F is always upper-tailed, so the p-value is the right-tail area beyond the observed statistic under $\mathrm{F}_{k-1,(k-1)(b-1)} = \mathrm{F}_{2,8}$; bracket it between tabulated cutoffs.

(a) The observed $F = 4.95$ sits between the 0.10 and 0.05 cutoffs and below the 0.025 cutoff:

$$F_{0.05,2,8} = 4.46 \;<\; 4.95 \;<\; 6.06 = F_{0.025,2,8},$$

so the right-tail area lies between those two levels:

$$0.025 < p\text{-value} < 0.05.$$

(b) At $\alpha = 0.05$: the p-value is below 0.05, so reject H_0 — the catalysts differ. At $\alpha = 0.025$: the p-value exceeds 0.025, so fail to reject at that stricter level.

$$\boxed{0.025 < p\text{-value} < 0.05 \;\Rightarrow\; \text{reject at } \alpha = 0.05, \text{ retain at } \alpha = 0.025.}$$

B6. Recover the error df ($k = 7$, $b = 4$). The degrees of freedom are fixed by the grid shape alone; error always gets the *product* $(k-1)(b-1)$.

(a)

$$k - 1 = 6 \text{ (treatments)}, \qquad b - 1 = 3 \text{ (blocks)},$$
$$(k-1)(b-1) = 6 \cdot 3 = 18 \text{ (error)}, \qquad N - 1 = kb - 1 = 28 - 1 = 27 \text{ (total)},$$

and the check $6 + 3 + 18 = 27 = N - 1$ holds.

(b) The figure $N - k = 28 - 7 = 21$ is the *one-way* error df. In a randomized block design the block term claims $b - 1 = 3$ of those degrees of freedom, leaving exactly $(k-1)(b-1) = 18$. Using 21 would mis-scale MSE and point the F at the wrong reference distribution. The correct error df is $\boxed{18}$, with the missing $21 - 18 = 3$ having gone to the block row.

(c) With SSE = SST − SSTR − SSB = 153.0 − 72.0 − 45.0 = 36.0 on 18 df,

$$\text{MSE} = \frac{36.0}{18} = 2.0, \qquad \text{MSTR} = \frac{72.0}{6} = 12.0, \qquad F = \frac{12.0}{2.0} = 6.0.$$

$$\boxed{\text{df: tr 6, bl 3, err } (k-1)(b-1) = 18, \text{ tot 27}; \quad \text{MSE} = 2.0, \; F = 6.0.}$$

B7. End-to-end test with a small positive error, repellents ($k = 3$, $b = 4$). Row means $13, 9, 17$ give SSTR; column means $12, 18, 9, 13$ give SSB; error by subtraction.

$$\begin{aligned}
\text{SSTR} &= 4\big[(13-13)^2 + (9-13)^2 + (17-13)^2\big] = 4[0 + 16 + 16] = 4(32) = 128,\\
\text{SSB} &= 3\big[(12-13)^2 + (18-13)^2 + (9-13)^2 + (13-13)^2\big]\\
&= 3[1 + 25 + 16 + 0] = 3(42) = 126,\\
\text{SSE} &= \text{SST} - \text{SSTR} - \text{SSB} = 260 - 128 - 126 = 6 \quad \text{on } (k-1)(b-1) = 2 \cdot 3 = 6 \text{ df}.
\end{aligned}$$

Mean squares and the two F-statistics:

$$\text{MSTR} = \frac{128}{2} = 64.0, \qquad \text{MSB} = \frac{126}{3} = 42.0, \qquad \text{MSE} = \frac{6}{6} = 1.0,$$
$$F_{\text{tr}} = \frac{64.0}{1.0} = 64.0, \qquad F_{\text{bl}} = \frac{42.0}{1.0} = 42.0.$$

The two-way ANOVA table:

Source	df	SS	MS	F
Treatments	2	128.0	64.0	64.0
Blocks	3	126.0	42.0	42.0
Error	6	6.0	1.0	
Total	11	260.0		

Treatments: $F_{\text{tr}} = 64.0$ against $\text{F}_{2,6}$ with cutoff $F_{0.05,2,6} = 5.14$; since $64.0 > 5.14$, reject H_0 at the 5% level — the three repellents differ in mean landings per minute. Diagnostic block test: $F_{\text{bl}} = 42.0$ is also large, confirming the subjects genuinely differed; had that variation been left in the error, MSE would have ballooned and the treatment F would have shrunk, so blocking on subjects was well worth it.

$$\boxed{F_{\text{tr}} = 64.0 > 5.14 \Rightarrow \text{reject; repellents differ. } F_{\text{bl}} = 42.0 \Rightarrow \text{blocking justified.}}$$

C. Paired t as a two-treatment block design

C1. Two conditions (control, fiber) measured on the *same* volunteer make each volunteer a block and each breakfast a treatment: a 2×5 block design, equivalently a paired test. We run both and confirm $F = t_{\text{obs}}^2$.

(a) *Paired t.* The differences $D_j = Y_{1j} - Y_{2j}$ (control minus fiber) are

$$D_j : \ 6,\ 7,\ 4,\ 8,\ 4,$$

with sum 29, so $\overline{D} = 29/5 = 5.800$. Deviations $D_j - \overline{D}$ are 0.2, 1.2, -1.8, 2.2, -1.8, whose squares sum to

$$0.04 + 1.44 + 3.24 + 4.84 + 3.24 = 12.80,$$

so $S_D^2 = 12.80/4 = 3.200$ and $S_D = 1.789$. Then

$$t_{\text{obs}} = \frac{\overline{D}}{S_D/\sqrt{b}} = \frac{5.800}{1.789/\sqrt{5}} = \frac{5.800}{0.8000} = 7.25, \qquad \text{df} = b - 1 = 4.$$

Since $|7.25| > t_{0.025,4} = 2.776$, reject $H_0 : \mu_D = 0$.

(b) *Block ANOVA.* By the two-treatment identities,

$$\text{SSTR} = \tfrac{b}{2}\overline{D}^2 = \tfrac{5}{2}(5.800)^2 = 2.5(33.64) = 84.10,$$
$$\text{SSE} = \tfrac{1}{2}(b-1)S_D^2 = \tfrac{1}{2}(4)(3.200) = 6.400.$$

Treatment df $= k - 1 = 1$, error df $= (k-1)(b-1) = 4$, so

$$\text{MSTR} = \frac{84.10}{1} = 84.10, \qquad \text{MSE} = \frac{6.400}{4} = 1.600,$$
$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{84.10}{1.600} = 52.56 \qquad \text{on } \text{F}_{1,4}\,.$$

The two-way ANOVA table:

Source	df	SS	MS	F
Treatment	1	84.10	84.10	52.56
Error	4	6.40	1.60	

(c) *Verify.* $t_{\text{obs}}^2 = (7.25)^2 = 52.56 = F$, and $t_{0.025,4}^2 = (2.776)^2 = 7.706 \approx 7.71 = F_{0.05,1,4}$, so the two tests use matched critical values and report the same p-value. Both reject: $F = 52.56 > 7.71$ and $|t| = 7.25 > 2.776$. In words, at the 5% level there is significant evidence that the two breakfasts differ in mean fasting glucose; since $\overline{D} = 5.8 > 0$, control runs higher (fiber lowers glucose).

$$\boxed{F = 52.56 = t_{\text{obs}}^2 = (7.25)^2, \quad F_{0.05,1,4} = t_{0.025,4}^2 = 7.71 \ \Rightarrow \ \text{both reject.}}$$

C2. The paired test hides SSB; the block ANOVA exposes it.

(a) *Block and grand means.* The block means $\overline{Y}_{\cdot j} = \frac{1}{2}(Y_{1j} + Y_{2j})$ are

$$95.0,\ 106.5,\ 103.0,\ 117.0,\ 92.0,$$

and the grand mean is $\overline{Y} = \frac{1}{10}\sum_{i,j} Y_{ij} = \frac{1027}{10} = 102.700$. With $k = 2$ observations per block,

$$\begin{aligned}\text{SSB} &= 2\sum_j (\overline{Y}_{\cdot j} - \overline{Y})^2 \\ &= 2\big[(95 - 102.7)^2 + (106.5 - 102.7)^2 + (103 - 102.7)^2 \\ &\qquad + (117 - 102.7)^2 + (92 - 102.7)^2\big] \\ &= 2\big[59.29 + 14.44 + 0.09 + 204.49 + 114.49\big] = 2(392.80) = 785.60.\end{aligned}$$

(b) *Total and partition.* Summing all ten $(Y_{ij} - \overline{Y})^2$,

$$\begin{aligned}\text{SST} &= 22.09 + 53.29 + 5.29 + 334.89 + 75.69 \\ &\qquad + 114.49 + 0.09 + 2.89 + 106.09 + 161.29 = 876.10.\end{aligned}$$

With SSTR = 84.10 and SSE = 6.400 from C1,

$$\text{SSTR} + \text{SSB} + \text{SSE} = 84.10 + 785.60 + 6.40 = 876.10 = \text{SST}\,. \ \blacksquare$$

The full table:

Source	df	SS	MS	F
Treatment	1	84.10	84.10	52.56
Block	4	785.60	196.40	
Error	4	6.40	1.60	
Total	9	876.10		

(c) SSB = 785.60 measures the volunteer-to-volunteer spread in baseline glucose (some near 92, some near 117); it is the between-subject variation. The paired test never estimates it, but it is rewarded by its removal — with SSB pulled out of the total, the leftover error is only SSE = 6.40.

$$\text{SST} = 876.10 = \underbrace{84.10}_{\text{SSTR}} + \underbrace{785.60}_{\text{SSB}} + \underbrace{6.40}_{\text{SSE}};$$ SSB is the between-subject variation pairing removes.

C3. The contrast lives entirely in the error term: one-way ANOVA lumps the block variation into error, the block design pulls it out.

(a) *The two error mean squares.*

$$\begin{aligned}\text{blocked:}\quad & \text{MSE} = \frac{\text{SSE}}{(k-1)(b-1)} = \frac{6.40}{4} = 1.600, \\ \text{unblocked:}\quad & \text{SSE}_{\text{1-way}} = \text{SSB} + \text{SSE} = 785.60 + 6.40 = 792.00 \text{ on } N - k = 8 \text{ df}, \\ & \text{MSE}_{\text{1-way}} = \frac{792.00}{8} = 99.00.\end{aligned}$$

(b) *The two F's* (same SSTR = 84.10, so MSTR = 84.10):

$$F_{\text{block}} = \frac{84.10}{1.600} = 52.56, \qquad F_{\text{1-way}} = \frac{84.10}{99.00} = 0.85.$$

Decisions at $\alpha = 0.05$: the block $F = 52.56 > F_{0.05,1,4} = 7.71$ *rejects*; the one-way $F = 0.85 < F_{0.05,1,8} = 5.32$ *fails to reject*. Ignoring the pairing buries the effect.

(c) Blocking shrank the error mean square by a factor of $99.00/1.600 \approx 61.9$. The treatment effect was unchanged (SSTR = 84.10 in both); only the denominator moved.

$$\boxed{\text{MSE}: \ 99.00 \to 1.600, \qquad F: \ 0.85 \text{ (retain)} \to 52.56 \text{ (reject)}.}$$

C4. Before/after on the same patient is a 2×7 block design. The claim names a direction ("shortens"), so the natural test is a *one-sided* paired t on $b - 1 = 6$ df; the block $F = t^2$ then folds both tails together, giving the magnitude while the sign of $\overline{D}$ gives the direction.

Set up. $H_0 : \mu_D = 0$ versus $H_1 : \mu_D > 0$, with $D = (\text{standard}) - (\text{new})$; reject for large positive t.

$\overline{D}$ and S_D. Sum $= 8 + 3 + 11 - 2 + 6 + 9 + 4 = 39$, so $\overline{D} = 39/7 = 5.571$. Deviations from 5.571 are 2.429, -2.571, 5.429, -7.571, 0.429, 3.429, -1.571; their squares sum to

$$5.90 + 6.61 + 29.47 + 57.33 + 0.18 + 11.76 + 2.47 = 113.71,$$

so $S_D^2 = 113.71/6 = 18.952$ and $S_D = 4.353$. Then

$$t_{\text{obs}} = \frac{\overline{D}}{S_D/\sqrt{b}} = \frac{5.571}{4.353/\sqrt{7}} = \frac{5.571}{1.645} = 3.39, \qquad \text{df} = 6.$$

Since $3.39 > t_{0.05,6} = 1.943$, reject: the new drug shortens mean time to relief.

Block-F analogue. The block design gives $F = t_{\text{obs}}^2 = (3.39)^2 = 11.46$ on $\text{F}_{1,6}$, and $F_{0.05,1,6} = 5.99 = t_{0.025,6}^2$ (since $t_{0.025,6} = 2.447$ and $2.447^2 = 5.99$). The F-test rejects ($11.46 > 5.99$) and matches the *two-sided* paired test; the *one-sided* conclusion comes from also noting $\overline{D} = 5.571 > 0$, the direction F cannot see.

$$\boxed{t_{\text{obs}} = 3.39 > 1.943 \ \Rightarrow \ \text{reject (new drug faster)}; \quad F = t^2 = 11.46, \ \text{direction from } \overline{D} > 0.}$$

C5. Both halves come from the same leaf sample, which shares the leaf's true chlorophyll level: block = leaf, treatments = instrument (spectrophotometer vs. field meter). A 2×12 block design, equivalently a paired test.

(a) With $b = 12$, $\overline{D} = 0.45$, $S_D^2 = 0.81$,

$$\text{SSTR} = \tfrac{b}{2}\overline{D}^2 = \tfrac{12}{2}(0.45)^2 = 6(0.2025) = 1.215,$$
$$\text{SSE} = \tfrac{1}{2}(b-1)S_D^2 = \tfrac{1}{2}(11)(0.81) = 4.455.$$

(b) Treatment df = 1, error df $= (k-1)(b-1) = 11$:

$$\text{MSTR} = 1.215, \qquad \text{MSE} = \frac{4.455}{11} = 0.405, \qquad F = \frac{\text{MSTR}}{\text{MSE}} = \frac{1.215}{0.405} = 3.00 \quad \text{on } \text{F}_{1,11}\,.$$

The table:

Source	df	SS	MS	F
Treatment	1	1.215	1.215	3.00
Error	11	4.455	0.405	

Since $F = 3.00 < F_{0.05,1,11} = 4.84$, do not reject: at the 5% level the two instruments do not differ significantly in mean chlorophyll reading.

(c) The equivalent paired statistic is $t_{\text{obs}} = \overline{D}/(S_D/\sqrt{b}) = 0.45/(0.9/\sqrt{12}) = 0.45/0.2598 = 1.732$ on 11 df, and $\sqrt{F} = \sqrt{3.00} = 1.732$, confirming $t_{\text{obs}} = \sqrt{F}$.

$$\boxed{F = 3.00 < F_{0.05,1,11} = 4.84 \ \Rightarrow \ \text{retain}; \quad t_{\text{obs}} = \sqrt{F} = 1.732 \text{ on } 11 \text{ df}.}$$

C6. With $k = 2$ treatments the treatment row has $k - 1 = 1$ df, the block row has $b - 1 = 7$ df, and error has $(k - 1)(b - 1) = 7$ df; the total has $N - 1 = 2b - 1 = 15$ df. Fill the column down, divide SS by df for the mean squares, and recover $t = \pm\sqrt{F}$.

Degrees of freedom and total SS.

$$\text{df}: \ 1, \ 7, \ 7, \ \text{total } 15; \qquad \text{SST} = 18.0 + 63.0 + 7.0 = 88.0.$$

Mean squares.

$$\text{MSTR} = \frac{18.0}{1} = 18.0, \qquad \text{MSB} = \frac{63.0}{7} = 9.0, \qquad \text{MSE} = \frac{7.0}{7} = 1.0.$$

Treatment F and paired t.

$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{18.0}{1.0} = 18.0 \text{ on } \text{F}_{1,7}, \qquad t_{\text{obs}} = \pm\sqrt{F} = \pm\sqrt{18.0} = \pm 4.24 \text{ on } 7 \text{ df}.$$

The completed table:

Source	df	SS	MS	F
Treatment	1	18.0	18.0	18.0
Block	7	63.0	9.0	
Error	7	7.0	1.0	
Total	15	88.0		

Decide. Since $F = 18.0 > F_{0.05,1,7} = 5.59$, reject at the 5% level: the two treatments differ significantly (equivalently $|t| = 4.24 > \sqrt{5.59} = 2.365 = t_{0.025,7}$). The df check $1+7+7 = 15$ and $\text{SSTR} + \text{SSB} + \text{SSE} = 88.0 = \text{SST}$ confirm a clean 2×8 table.

$$\boxed{F = 18.0 \text{ on } \text{F}_{1,7} = t_{\text{obs}}^2, \quad t_{\text{obs}} = \pm 4.24 \text{ on } 7 \text{ df} \ \Rightarrow \ \text{reject.}}$$

C7. The test for each is whether the two measurements share a common unit that can serve as a block. Same specimen, same greenhouse ⇒ block; two disjoint groups ⇒ independent; equal counts never settle it.

(a) *Two analyzers, one specimen.* Both readings come from the same blood specimen, which has one true cholesterol value: block = specimen, treatments = analyzer. **Block design** (paired).

(b) *Two disjoint arms.* Diet-A and diet-B mice are different animals; no shared unit, no block. **Independent samples** — two-sample t.

(c) *Two benches, one greenhouse.* The north and south benches share the greenhouse's climate and cultivar: block = greenhouse, treatments = light recipe. **Block design**.

(d) *Equal sizes tempt a fake pairing.* The 9 caffeine subjects and the 9 placebo subjects are *different* people; subject 3 in one column shares nothing with subject 3 in the other, and reordering either column changes the "differences" entirely. There is no block. **Independent samples** — the correct analysis is a two-sample t (Chapter 9), *not* a paired/block test.

$$\boxed{\text{(a) block, (b) independent, (c) block, (d) independent (equal } b \text{ is not a block).}}$$

D. Efficiency, additivity, diagnostics

D1. Relative efficiency of the hormone block design. The question "was blocking worth it?" is answered by relative efficiency: fold the block mean square MSB back into the error (a completely randomized design would never separate it out) and compare to the block design's own MSE. Here $b = 5$, $k = 4$, so $bk - 1 = 19$, $b - 1 = 4$, $b(k - 1) = 15$.

(a) Variance-ratio form:

$$\text{RE} = \frac{(b-1)\ \text{MSB} + b(k-1)\ \text{MSE}}{(bk-1)\ \text{MSE}}$$
$$= \frac{4(96.0) + 15(8.0)}{19(8.0)} = \frac{384+120}{152} = \frac{504}{152} = 3.32.$$

(b) The RBD error df is $(k-1)(b-1) = 3 \cdot 4 = 12$; the CRD error df would be $k(b-1) = 4 \cdot 4 = 16$. The correction is

$$c = \frac{\big[(k-1)(b-1)+1\big]\,\big[k(b-1)+1\big]}{\big[(k-1)(b-1)+3\big]\,\big[k(b-1)+3\big]} = \frac{(12+1)(16+1)}{(12+3)(16+3)} = \frac{13 \cdot 17}{15 \cdot 19}$$
$$= \frac{221}{285} = 0.7754,$$

so $\text{RE}_{\text{corr}} = c \cdot \text{RE} = 0.7754 \times 3.32 = 2.57$.

(c) Both figures sit well above the break-even value 1, so:

$$\boxed{\text{RE} = 3.32,\ \text{RE}_{\text{corr}} = 2.57 > 1 \ \Rightarrow\ \text{blocking on benches was clearly worthwhile.}}$$

A completely randomized design would have needed roughly two-and-a-half times the replication to match this precision; the benches differed enough (MSB = 96 vs. MSE = 8) that removing them paid off.

D2. Torque settings blocked on shifts — a block term that earns little. A small SSB warns that the shifts were not really more homogeneous inside than across, so the $b-1$ df spent on them may be wasted; the block F and relative efficiency confirm it, and the contrast is a CRD that pools those block df back into the error. Here $b = 7$, $k = 3$.

(a) Mean squares:

$$\text{MSB} = \frac{\text{SSB}}{b-1} = \frac{12.0}{6} = 2.00, \qquad \text{MSE} = \frac{\text{SSE}}{(k-1)(b-1)} = \frac{72.0}{12} = 6.00.$$

The block $F = \text{MSB} \,/\, \text{MSE} = 2.00/6.00 = 0.33$ is well below 1, a red flag that the shifts contributed essentially nothing beyond noise — the blocks were not more alike inside than across.

(b) With $b = 7$, $k = 3$: $bk - 1 = 20$, $b - 1 = 6$, $b(k-1) = 14$.

$$\text{RE} = \frac{(b-1)\ \text{MSB} + b(k-1)\ \text{MSE}}{(bk-1)\ \text{MSE}} = \frac{6(2.00) + 14(6.00)}{20(6.00)}$$
$$= \frac{12.0 + 84.0}{120.0} = \frac{96.0}{120.0} = 0.80.$$

(c) Ignoring shifts, a one-way ANOVA pools SSB into the error: $\text{SSE}_{\text{CRD}} = \text{SSB} + \text{SSE} = 12.0 + 72.0 = 84.0$ on $bk - k = 21 - 3 = 18$ error df, so

$$\text{MSE}_{\text{CRD}} = \frac{84.0}{18} = 4.67 < 6.00 = \text{MSE}_{\text{RBD}}\,.$$

The unblocked error mean square is smaller and rests on more df, so the completely randomized analysis is the sharper test here.

$$\boxed{\text{RE} = 0.80 < 1,\ \text{block } F = 0.33 \ \Rightarrow\ \text{blocking hurt; the one-way analysis was better.}}$$

D3. Flag an interaction from the cell means: primer by alloy. Additivity says every treatment gap is the same across blocks. Test it by comparing one treatment pair's difference in each block; a sign reversal or wild size change means non-parallel profiles, so additivity fails. Take P_1 versus P_2:

$$\text{brass:} \quad P_2 - P_1 = 26 - 34 = -8,$$
$$\text{titanium:} \quad P_2 - P_1 = 38 - 22 = +16.$$

On brass P_2 *trails* P_1 by 8; on titanium P_2 *beats* P_1 by 16 — the gap reverses sign *and* changes magnitude, so the profiles cross and the best primer depends on the alloy.

Gap -8 on brass vs. $+16$ on titanium $\Rightarrow$ strong primer-by-alloy interaction; additivity fails.

A simple block-design treatment F on these data is invalid: the leftover interaction is dumped into MSE, and "which primer is best" has no block-free answer. Analyse the alloys separately, or fit a full two-factor model with replication.

D4. Additivity holds despite a real pressure effect. Parallel profiles mean *additive*, not *equal*: the treatments may differ by a constant gap repeated in every block. Check by confirming the within-block increments match, then read the common increments as the treatment effect.

(a) Going low→mid→high in each press:

$$\text{press 1: } +7,\ +7; \quad \text{press 2: } +7,\ +7; \quad \text{press 3: } +7,\ +7.$$

Every press shows the identical $+7, +7$ pattern, so the three profiles are exactly parallel (each press is a vertical shift of the others) — additivity holds.

(b) Parallel does not mean the pressures are equal: the common increments raise the response by the same amount in every block. From low to high the gap is $+7 + 7 = +14$, a real, block-independent pressure effect.

Identical $+7, +7$ increments in all presses $\Rightarrow$ additive; pressures differ by a constant $+14$ gap.

This is exactly the situation the block-design F is built for: a genuine treatment effect cleanly separable from block shifts.

D5. Route a fanning interaction to Tukey's test and a transform. Non-crossing but *fanning* profiles, with gaps that scale with the response level, are the multiplicative- interaction signature, which is precisely the alternative Tukey's one-df test targets (it models the interaction as proportional to $\tau_i\beta_j$).

(a) Use **Tukey's one-degree-of-freedom test for non-additivity**: it peels $\mathrm{SS}_{\mathrm{nonadd}}$ (1 df) off SSE and refers the statistic to $\mathrm{F}_{1,\,(k-1)(b-1)-1}$. With $k = 4$, $b = 6$ the error df is $(k-1)(b-1) = 3\cdot 5 = 15$, so the reference distribution is $\mathrm{F}_{1,\,14}$.

(b) Re-analyse on the log scale (analyse $\log Y$). Because the interaction is multiplicative, taking logs turns the proportional structure into an additive one — the fanning typically flattens to parallel profiles, restoring additivity.

Fanning profiles $\Rightarrow$ Tukey 1-df test on $\mathrm{F}_{1,14}$; if significant, re-fit on $\log Y$.

D6. Design recognition: route each experiment. Two reads decide the answer: is there a nuisance grouping matched across treatments (block) or completely random assignment (one-way)? and are the normal-theory assumptions safe (parametric F/t) or not (rank test)? These are independent questions.

(a) The same 8 runners under every shoe model = runners are blocks; $k = 3$ treatments, roughly normal $\Rightarrow$ **randomized block design**, treatment F-test (the block-design generalisation of the paired t).

(b) Cages are a natural nuisance grouping, one tank per additive per cage = block design; but ordinal scores plus a curved QQ plot defeat normality $\Rightarrow$ block design routed to the **Friedman test**.

(c) No matched grouping — boards assigned completely at random — and normal readings $\Rightarrow$ a **completely randomized design**, ordinary one-way ANOVA F (no block term).

(a) RBD, block-design F; (b) RBD design but Friedman test (ordinal, curved Q-Q); (c) CRD, one-way ANOVA F.

The block structure and the safety of the assumptions are separate questions — settle both before naming the test.

D7. A residual funnel, and when *not* to reach for Friedman.

(a) A widening funnel means *unequal variance* (spread grows with the fitted value), and a convex QQ plot means *right skew (non-normality)*; the spread ratio $s_{\max}/s_{\min} = 3.8 \geq 3$ confirms the heteroscedasticity is in the danger zone. Variance tied to the mean and right skew both point to a multiplicative scale, so the single remedy is to re-analyse log(wear): the log transform stabilises the variance, symmetrises the skew, and (with one observation per cell) tends to restore additivity at the same time. Re-check both plots afterward.

Funnel + convex Q-Q + $s_{\max}/s_{\min} = 3.8 \;\Rightarrow\;$ analyse log(wear).

(b) The reflex to jump straight to Friedman is *not* the best first move. The Friedman test is the right call only when the parametric assumptions genuinely fail and no fix is available — it ranks the responses within each block, discarding the magnitude information, and so is less powerful than the F when normality holds. Here the violations look curable by a transformation, so the correct sequence is: try the log transform first and re-examine the diagnostics; reserve Friedman for the case where even the transformed data refuse to meet normality and equal variance.

Try log first; use Friedman only if the transform fails, since ranking costs power.

E. Block-design synthesis

E1. Randomized block design, $k = 3$ films, $b = 4$ warehouses. The treatment $F = \text{MSTR} \,/\, \text{MSE}$ on $\text{F}_{k-1,(k-1)(b-1)} = \text{F}_{2,6}$ tests equal film means; warehouses are the nuisance block.

(a) *Block means and grand mean.* Each column averages $k = 3$ films:

$$\overline{Y}_{\cdot 3} = \tfrac{1}{3}(13 + 15 + 11) = 13, \qquad \overline{Y}_{\cdot 4} = \tfrac{1}{3}(18 + 30 + 24) = 24,$$

so the column means are $15, 20, 13, 24$. The grand mean is $\overline{Y} = \frac{1}{12}(15 + 20 + 13 + 24) \cdot 3/3$; directly, $\overline{Y} = \frac{1}{12}\sum_{i,j} Y_{ij} = \frac{216}{12} = 18$, matching the average of the row means $\frac{1}{3}(15 + 22 + 17) = 18$.

(b) *Treatment and block sums of squares.*

$$\begin{aligned}
\text{SSTR} &= b\sum_i (\overline{Y}_{i\cdot} - \overline{Y})^2 = 4\big[(15-18)^2 + (22-18)^2 + (17-18)^2\big] \\
&= 4[9 + 16 + 1] = 4(26) = 104, \\
\text{SSB} &= k\sum_j (\overline{Y}_{\cdot j} - \overline{Y})^2 = 3\big[(15-18)^2 + (20-18)^2 + (13-18)^2 + (24-18)^2\big] \\
&= 3[9 + 4 + 25 + 36] = 3(74) = 222.
\end{aligned}$$

(c) *Error by subtraction and df.*

$$\text{SSE} = \text{SST} - \text{SSTR} - \text{SSB} = 382 - 104 - 222 = 56 \quad \text{on } (k-1)(b-1) = 2 \cdot 3 = 6 \text{ df.}$$

Degrees of freedom: treatments $k - 1 = 2$, blocks $b - 1 = 3$, error 6, total $N - 1 = 12 - 1 = 11$, and $2 + 3 + 6 = 11$. ■

(d) *Two-way table and treatment test.* Mean squares:

$$\text{MSTR} = \frac{104}{2} = 52.0, \qquad \text{MSB} = \frac{222}{3} = 74.0, \qquad \text{MSE} = \frac{56}{6} = 9.333,$$
$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{52.0}{9.333} = 5.57.$$

Source	df	SS	MS	F
Treatments	2	104	52.00	5.57
Blocks	3	222	74.00	7.93
Error	6	56	9.333	
Total	11	382		

Since $F = 5.57 > F_{0.05,2,6} = 5.14$, reject H_0.

$$\boxed{F = 5.57 > F_{0.05,2,6} = 5.14 \;\Rightarrow\; \text{reject; the three films differ in mean moisture gain.}}$$

E2. Relative efficiency answers "did blocking on warehouses pay off?". Use the *block* mean square MSB = 74.0 and MSE = 9.333 from E1, with $k = 3$, $b = 4$.

(a) *Variance-ratio form.* Here $bk - 1 = 11$, $b - 1 = 3$, $b(k-1) = 8$:

$$\text{RE} = \frac{(b-1)\,\text{MSB} + b(k-1)\,\text{MSE}}{(bk-1)\,\text{MSE}} = \frac{3(74.0) + 8(9.333)}{11(9.333)}$$
$$= \frac{222.0 + 74.67}{102.67} = \frac{296.67}{102.67} = 2.89.$$

(b) *Small-sample correction.* RBD error df $= (k-1)(b-1) = 6$; CRD error df $= k(b-1) = 3 \cdot 3 = 9$:

$$c = \frac{[(k-1)(b-1)+1][k(b-1)+1]}{[(k-1)(b-1)+3][k(b-1)+3]} = \frac{(6+1)(9+1)}{(6+3)(9+3)} = \frac{7 \cdot 10}{9 \cdot 12} = \frac{70}{108} = 0.648,$$
$$\text{RE}_{\text{corr}} = c \cdot \text{RE} = 0.648 \times 2.89 = 1.87.$$

(c) *Interpretation.* $\text{RE}_{\text{corr}} = 1.87 > 1$ means a completely randomized design would have needed about 87% more observations — nearly twice the experiment — to match this precision, so blocking on warehouses was clearly worthwhile.

$$\boxed{\text{RE} = 2.89,\ \text{RE}_{\text{corr}} = 1.87 > 1 \;\Rightarrow\; \text{blocking on warehouses paid off.}}$$

E3. Two conditions (tablet, capsule) on the *same* patient make each patient a block: a 2×7 block design, equivalently a paired test. By the equivalence theorem the treatment F on $\text{F}_{1,b-1} = \text{F}_{1,6}$ equals the square of the paired t on $b - 1 = 6$ df.

(a) *ANOVA pieces from the differences.* Sum $= 8 + 5 + 11 + 2 + 9 - 1 + 6 = 40$, so $\overline{D} = 40/7 = 5.714$. Deviations from $\overline{D}$ are $2.286, -0.714, 5.286, -3.714, 3.286, -6.714, 0.286$; their squares sum to

$$5.224 + 0.510 + 27.94 + 13.80 + 10.80 + 45.08 + 0.082 = 103.43,$$

so $S_D^2 = 103.43/6 = 17.238$ and $S_D = 4.152$. Then

$$\text{SSTR} = \tfrac{b}{2}\overline{D}^{\,2} = \tfrac{7}{2}(5.714)^2 = 3.5(32.65) = 114.29,$$
$$\text{SSE} = \tfrac{1}{2}(b-1)S_D^2 = \tfrac{1}{2}(6)(17.238) = 51.71,$$
$$\text{MSTR} = \frac{114.29}{1} = 114.29, \qquad \text{MSE} = \frac{51.71}{6} = 8.619,$$
$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{114.29}{8.619} = 13.26 \qquad \text{on } \text{F}_{1,6}\,.$$

(b) *Equivalent paired t and the $F = t^2$ check.*

$$t_{\text{obs}} = \frac{\overline{D}}{S_D/\sqrt{b}} = \frac{5.714}{4.152/\sqrt{7}} = \frac{5.714}{1.569} = 3.64, \qquad \text{df} = 6,$$

and $t_{\text{obs}}^2 = 3.64^2 = 13.25 \approx 13.26 = F$, confirming the identity to rounding.

(c) *Decision (two-sided, $\alpha = 0.05$).* The F-test rejects since $F = 13.26 > F_{0.05,1,6} = 5.99$; equivalently $|t_{\text{obs}}| = 3.64 > t_{0.025,6} = 2.447$, and $F_{0.05,1,6} = 5.99 = (2.447)^2 = t_{0.025,6}^2$.

$$\boxed{F = 13.26 = t_{\text{obs}}^2 = (3.64)^2, \quad F_{0.05,1,6} = t_{0.025,6}^2 = 5.99 \Rightarrow \text{both reject; the formulations differ.}}$$

E4. Reading conclusions off a completed table: no sum of squares need be recomputed; just place each F relative to its cutoff. The treatment F answers the scientific question; the block F is the diagnostic for whether blocking helped.

Step 1: treatments. $F_{\text{tr}} = 5.0$ against $\text{F}_{4,20}$ with cutoff $F_{0.05,4,20} = 2.87$. Since $5.0 > 2.87$, reject: the five teaching methods give different mean outcomes.

Step 2: blocks. The block $F = \text{MSB} / \text{MSE} = 30.0/6.0 = 5.0$ against $\text{F}_{5,20}$ with cutoff $F_{0.05,5,20} = 2.71$. Since $5.0 > 2.71$, the schools genuinely differed.

Step 3: was blocking justified? The large block F says school-to-school variation was real; had it been left in the error, MSE would have ballooned and the treatment F shrunk, so blocking on schools was justified.

$$\boxed{F_{\text{tr}} = 5.0 > 2.87 \text{ (methods differ)}; \quad F_{\text{bl}} = 5.0 > 2.71 \text{ (blocking justified).}}$$

E5. Randomized block design, $k = 4$ varieties, $b = 5$ plots; the treatment F lives on $\text{F}_{3,12}$.

(a) *Treatment SS and error.*

$$\begin{aligned}
\text{SSTR} &= b\sum_i (\overline{Y}_{i\cdot} - \overline{Y})^2 = 5\big[(42-45)^2 + (48-45)^2 + (39-45)^2 + (51-45)^2\big] \\
&= 5[9 + 9 + 36 + 36] = 5(90) = 450, \\
\text{SSE} &= \text{SST} - \text{SSTR} - \text{SSB} = 750 - 450 - 180 = 120 \quad \text{on } (k-1)(b-1) = 12 \text{ df.}
\end{aligned}$$

(b) *Two-way table.*

$$\text{MSTR} = \frac{450}{3} = 150.0, \qquad \text{MSB} = \frac{180}{4} = 45.0, \qquad \text{MSE} = \frac{120}{12} = 10.0,$$
$$F = \frac{\text{MSTR}}{\text{MSE}} = \frac{150.0}{10.0} = 15.0.$$

Source	df	SS	MS	F
Treatments	3	450	150.0	15.0
Blocks	4	180	45.0	4.5
Error	12	120	10.0	
Total	19	750		

(c) *Test and p-value bracket.* Against $F_{0.05,3,12} = 3.49$, $15.0 > 3.49$: reject H_0, the varieties differ. For the p-value, the treatment F is upper-tailed and 15.0 exceeds even $F_{0.025,3,12} = 4.47$ (and far exceeds $F_{0.10,3,12} = 2.61$), so

$$p\text{-value} < 0.025.$$

$$\boxed{F = 15.0 > F_{0.05,3,12} = 3.49 \;\Rightarrow\; \text{reject; varieties differ, } p\text{-value} < 0.025.}$$

E6. Both studies are randomized block designs with $k = 3$, $b = 6$, so the treatment F is on $\mathrm{F}_{2,10}$ (cutoff 4.10), and $bk - 1 = 17$, $b - 1 = 5$, $b(k-1) = 12$.

(a) *Study P.* Treatment $F = \mathrm{MSTR} \,/\, \mathrm{MSE} = 18.0/5.0 = 3.6 < 4.10$: fail to reject; the treatments are not shown to differ. Relative efficiency with MSB = 2.5:

$$\mathrm{RE}_P = \frac{(b-1)\,\mathrm{MSB} + b(k-1)\,\mathrm{MSE}}{(bk-1)\,\mathrm{MSE}} = \frac{5(2.5) + 12(5.0)}{17(5.0)}$$
$$= \frac{12.5 + 60.0}{85.0} = \frac{72.5}{85.0} = 0.85 < 1,$$

so blocking actually *hurt*: the small MSB = 2.5 < MSE shows the blocks barely differed and the $b - 1 = 5$ block df were largely wasted.

(b) *Study Q.* Treatment $F = 18.0/5.0 = 3.6 < 4.10$: again fail to reject. With MSB = 40.0,

$$\mathrm{RE}_Q = \frac{5(40.0) + 12(5.0)}{17(5.0)} = \frac{200.0 + 60.0}{85.0} = \frac{260.0}{85.0} = 3.06 > 1,$$

so here blocking paid off strongly — a CRD would have needed about three times the replication for the same precision.

(c) *Why the efficiencies differ.* RE depends on MSB, not on the treatment F. The treatment F is the same (3.6) because MSTR and MSE match, but RE measures block heterogeneity: Q's blocks were highly heterogeneous (MSB = 40 $\gg$ MSE) so removing them helped, whereas P's blocks were nearly identical (MSB = 2.5 < MSE) so blocking spent df for nothing.

> Same $F = 3.6$ (both retain); $\mathrm{RE}_P = 0.85$ (P blocking wasted), $\mathrm{RE}_Q = 3.06$ (Q blocking earned its df).

E7. Additivity says every treatment gap is the same in every block; check one treatment pair across the two metals.

(a) *Gaps for (P_1, P_2) by metal.*

$$\text{steel:} \quad P_2 - P_1 = 40 - 34 = +6,$$
$$\text{aluminum:} \quad P_2 - P_1 = 28 - 46 = -18.$$

On steel P_2 beats P_1 by 6; on aluminum P_2 *trails* P_1 by 18 — a sign reversal. The treatment gap is not constant across blocks, so additivity is *not* tenable.

(b) *Profile plot and validity.* The two block profiles (one per metal) would cross rather than run parallel: the best primer depends entirely on the substrate. With only one observation per cell this treatment-by-block interaction is absorbed into MSE, inflating it and invalidating the simple block-design treatment F — the question "which primer is best" has no block-free answer.

(c) *How to proceed.* Analyze the two metals separately, or fit a full two-factor model *with replication* so the interaction can be estimated, rather than reporting a single block-design F.

> Gap +6 on steel vs. −18 on aluminum ⇒ strong interaction; additivity fails, block F invalid.

Chapter 14

Nonparametric Statistics

Every test so far assumed a *shape* for the data — usually normality. When that assumption fails (small samples, heavy skew, outliers, or merely **ordinal** data where only ranks are meaningful), **nonparametric** methods step in. Their trick is to throw away the raw numbers and work with **ranks** or **signs**, which follow known distributions *regardless* of the parent shape. The cost is a little power when the data really are normal; the payoff is validity when they are not. This chapter mirrors the parametric toolbox one-for-one: the **sign** and **signed-rank** tests replace the one-sample and paired t; the **rank-sum** test replaces the two-sample t; **Kruskal–Wallis** replaces one-way ANOVA; **Friedman** replaces the randomized block ANOVA; and a **runs** test checks randomness.

Master guide: which nonparametric test replaces which parametric one?

- **One sample / paired**, testing a median, using only the *signs* of deviations? ⇒ the **sign test** (§1).
- **One sample / paired**, using signs *and* the ranks of the magnitudes? ⇒ the **Wilcoxon signed-rank test** — more powerful than the sign test when symmetry holds; replaces the paired t (§2).
- **Two independent samples**, comparing locations without normality? ⇒ the **Wilcoxon rank-sum (Mann–Whitney) test** — replaces the two-sample t (§3).
- $k \geq 3$ **independent samples**? ⇒ the **Kruskal–Wallis test** — replaces one-way ANOVA (§4).
- k **treatments in blocks** (repeated measures / matched)? ⇒ the **Friedman test** — replaces the randomized block ANOVA (§5).
- Asking whether a sequence is **random** (no trend or clustering)? ⇒ the **runs test**; for monotone association of two rankings, **Spearman's** rank correlation (§6).

Keep straight: nonparametric tests are about *ranks*, so ties are handled with average ranks, and the hypotheses are usually about *medians* or whole *distributions*, not means. Reach for them when normality is doubtful or the data are ordinal — not as a default when a valid t/F would be more powerful.

14.1 The Sign Test

The sign test is the bluntest tool in the nonparametric kit, and that bluntness is exactly its virtue. To test a claim about a population **median** η — or, for paired data, about the median of the within-pair differences — you do not measure how far each observation sits from the hypothesized value η_0; you record only *which side* it falls on, a plus if it exceeds η_0 and a minus if it falls short. Discard the ties (observations exactly at η_0) and let S_+ be the number of plus signs among the remaining n values. If the true median really is η_0, each surviving observation is equally likely to land above or below, so $S_+ \sim \text{Bin}(n, \frac{1}{2})$ no matter what shape the parent distribution has. That single fact powers the whole test: an extreme count of plus signs is evidence against η_0. Because it reads only signs and throws away every magnitude, the sign test makes essentially no assumptions (not even symmetry) and shrugs off outliers, but it also wastes information and so has the least

power of the median tests — recognize it as the right choice when the data are coarse, ordinal, or wildly non-normal, and you want a test you can defend with a coin.

Which one-sample / paired test does the problem need?

- A hypothesis about a **median** (or paired-difference median), data possibly skewed, heavy-tailed, or merely **ordinal**, and you are willing to use only the *signs*? ⇒ the **sign test** (this section).
- Differences are roughly **symmetric** and you want to use the ranks of their magnitudes for more power? ⇒ the **Wilcoxon signed-rank test** (§2) — it dominates the sign test when symmetry holds.
- Data genuinely **normal** and you care about the *mean*? ⇒ the one-sample or paired t-test — most powerful, but only valid when normality is credible.
- Counting signs but n is **large**? ⇒ still the sign test, but use the normal approximation $z = (S_+ - n/2)/\sqrt{n/4}$ instead of the exact binomial.

Do not confuse: the sign test is about the *median*, not the mean, and it assumes *nothing* about symmetry; the signed-rank test buys extra power by assuming the differences are symmetric, and the t-test buys still more by assuming normality. Each added assumption trades robustness for power.

14.1.1 The signs as a binomial count

Definition 14.1.1 (Sign statistic)**.** Let $X_1, \dots, X_m$ be a sample and η_0 a hypothesized median. Record the sign of each deviation $X_i - \eta_0$: a **plus** if $X_i > \eta_0$, a **minus** if $X_i < \eta_0$, and a **tie** if $X_i = \eta_0$. Discard the ties; let n be the number of non-tied observations and let

$$S_+ = \#\{\, i : \; X_i > \eta_0 \,\}$$

be the number of plus signs among them. S_+ is the **sign statistic**.

Theorem 14.1.2 (Null distribution of the sign statistic)**.** *If the population median is η_0 and the observations are independent and continuous, then under $H_0 : \eta = \eta_0$,*

$$S_+ \sim \mathrm{Bin}\big(n, \tfrac{1}{2}\big)\,,$$

so $\mathbb{E}[S_+] = n/2$ and $\mathrm{Var}(S_+) = n/4$, regardless of the shape of the parent distribution.

Proof. By the definition of a median, $\mathbb{P}(X_i > \eta_0) = \frac{1}{2}$ when $\eta = \eta_0$ and the distribution is continuous (so $\mathbb{P}(X_i = \eta_0) = 0$). The non-tied observations are independent, each contributing a plus with probability $\frac{1}{2}$ and a minus otherwise; S_+ counts the pluses among n such independent fair trials, which is the definition of a $\mathrm{Bin}(n, \frac{1}{2})$ count. The mean and variance follow from np and $np(1-p)$ at $p = \frac{1}{2}$. □

Remark. Nothing in the proof used the spread, the tails, or the symmetry of the data — only the meaning of "median." That is why the sign test is **distribution-free**: its null distribution is the same binomial whether the data are normal, exponential, or a lopsided ordinal scale. The price is that two samples with identical signs but wildly different magnitudes give identical results, so all information about *how far* the points lie from η_0 is thrown away.

14.1.2 The exact binomial test for a median

Definition 14.1.3 (Exact sign test, two-sided)**.** To test $H_0 : \eta = \eta_0$ against $H_1 : \eta \neq \eta_0$ at level α, compute s_{obs}, the observed number of plus signs out of n, and the two-sided p-value

$$p\text{-value} = 2\,\min\Big\{\mathbb{P}(S_+ \le s_{\mathrm{obs}}),\; \mathbb{P}(S_+ \ge s_{\mathrm{obs}})\Big\}, \qquad S_+ \sim \mathrm{Bin}\big(n, \tfrac{1}{2}\big)\,,$$

capped at 1. Reject H_0 when the p-value is at most α. Equivalently, reject when s_{obs} is far from $n/2$ in either direction.

Example 14.1.4 (One-sample sign test for a median: drinking-water lead)**.** An environmental lab claims the median lead concentration in a town's tap water is $\eta_0 = 5$ parts per billion (ppb). Twelve homes are sampled, giving lead readings (ppb)

$$3.1,\ 7.4,\ 4.2,\ 9.8,\ 2.0,\ 6.6,\ 5.9,\ 14.1,\ 4.8,\ 8.3,\ 3.7,\ 11.5.$$

Test $H_0 : \eta = 5$ versus $H_1 : \eta \neq 5$ at $\alpha = 0.05$. The distribution is plainly right-skewed (one home at 14.1), so a median test is appropriate.

Which rule, and why? The hypothesis is about a *median* and the data are skewed with a high outlier, so a mean-based t-test is the wrong yardstick. Using only the signs of $X_i - 5$ gives the exact binomial sign test, which ignores the outlier's magnitude entirely.

Solution.

Step 1: sign each deviation from $\eta_0 = 5$.

$$\begin{aligned} 3.1, 4.2, 2.0, 4.8, 3.7 \ &< 5 \ \Rightarrow \ \text{five minuses,} \\ 7.4, 9.8, 6.6, 5.9, 14.1, 8.3, 11.5 \ &> 5 \ \Rightarrow \ \text{seven pluses.} \end{aligned}$$

No reading equals 5, so $n = 12$ and $s_{\text{obs}} = 7$.

Step 2: locate s_{obs} relative to $n/2 = 6$. Seven pluses is one above the expected six; both tails matter for a two-sided test.

Step 3: two-sided p-value from $\text{Bin}(12, \frac{1}{2})$. With $\mathbb{P}(S_+ \geq 7) = \mathbb{P}(S_+ \leq 5) = 0.3872$,

$$\begin{aligned} p\text{-value} &= 2\,\min\{\,\mathbb{P}(S_+ \leq 7),\, \mathbb{P}(S_+ \geq 7)\,\} \\ &= 2\,\mathbb{P}(S_+ \geq 7) = 2(0.3872) = 0.774. \end{aligned}$$

Step 4: decide. $0.774 > 0.05$, so do not reject.

$$\boxed{s_{\text{obs}} = 7,\ p\text{-value} = 0.77 > 0.05 :\ \text{no evidence the median differs from 5 ppb.}}$$

A 7-to-5 split is exactly the kind of imbalance fair coins produce all the time.

Example 14.1.5 (Sign test with a decisive split: assembly-time standard)**.** A process engineer claims the median time to assemble a unit is $\eta_0 = 40$ seconds. Ten units are timed:

$$33,\ 36,\ 31,\ 38,\ 35,\ 29,\ 42,\ 34,\ 37,\ 30.$$

Test $H_0 : \eta = 40$ against $H_1 : \eta \neq 40$ at $\alpha = 0.05$.

Which rule, and why? Again a claim about a *median* of a small sample; sign each deviation from 40 and read S_+ against $\text{Bin}(10, \frac{1}{2})$. No symmetry or normality is needed.

Solution.

Step 1: count signs. Only $42 > 40$ (a plus); the other nine are below 40 (minuses). $n = 10$, $s_{\text{obs}} = 1$.

Step 2: two-sided p-value. With $\mathbb{P}(S_+ \leq 1) = \mathbb{P}(S_+ \geq 9)$ and $\mathbb{P}(S_+ \leq 1) = \binom{10}{0} + \binom{10}{1}$ over $2^{10} = 1024$,

$$\mathbb{P}(S_+ \leq 1) = \frac{1 + 10}{1024} = \frac{11}{1024} = 0.01074, \qquad p\text{-value} = 2(0.01074) = 0.0215.$$

Step 3: decide. $0.0215 < 0.05$, reject.

$$\boxed{s_{\text{obs}} = 1,\ p\text{-value} = 0.021 < 0.05 :\ \text{the median assembly time is below 40 s.}}$$

Nine of ten on one side is far more lopsided than a fair coin would casually deliver.

14.1.3 One-sided sign tests

Definition 14.1.6 (Exact sign test, one-sided)**.** For $H_1 : \eta > \eta_0$, large S_+ is the evidence, so the p-value is $\mathbb{P}(S_+ \geq s_{\text{obs}})$. For $H_1 : \eta < \eta_0$, small S_+ is the evidence, so the p-value is $\mathbb{P}(S_+ \leq s_{\text{obs}})$. In both cases $S_+ \sim \text{Bin}(n, \frac{1}{2})$ and one rejects when the p-value is at most α.

Example 14.1.7 (One-sided sign test: does a supplement raise median vitamin D?)**.** A nutrition study claims a supplement pushes the median serum vitamin-D level above the deficiency threshold $\eta_0 = 20$ ng/mL. Eleven participants on the supplement give

$$18,\ 24,\ 22,\ 19,\ 27,\ 31,\ 21,\ 23,\ 17,\ 26,\ 25.$$

Test $H_0 : \eta = 20$ versus the one-sided $H_1 : \eta > 20$ at $\alpha = 0.05$.

Which rule, and why? The research claim is directional — the median *exceeds* 20 — so this is a one-sided sign test with the upper tail as the rejection region. We sign each value against 20 and use $\mathbb{P}(S_+ \geq s_{\text{obs}})$ from $\text{Bin}(n, \frac{1}{2})$.

Solution.

Step 1: sign against $\eta_0 = 20$. Below 20: 18, 19, 17 (three minuses). Above 20: 24, 22, 27, 31, 21, 23, 26, 25 (eight pluses). $n = 11$, $s_{\text{obs}} = 8$.

Step 2: upper-tail p-value from $\text{Bin}(11, \frac{1}{2})$.

$$\begin{aligned}\mathbb{P}(S_+ \geq 8) &= \frac{\binom{11}{8} + \binom{11}{9} + \binom{11}{10} + \binom{11}{11}}{2^{11}} = \frac{165 + 55 + 11 + 1}{2048} \\ &= \frac{232}{2048} = 0.1133.\end{aligned}$$

Step 3: decide. $0.1133 > 0.05$, so do not reject.

$$\boxed{s_{\text{obs}} = 8,\ p\text{-value} = 0.113 > 0.05 :\ \text{not enough sign evidence that } \eta > 20.}$$

Eight of eleven leans the right way but is still inside what a fair coin produces about one time in nine — a vivid reminder that the sign test, ignoring magnitudes, needs a strong sign imbalance to fire.

Example 14.1.8 (One-sided lower-tail: median repair cost below a cap)**.** A warranty office claims the median repair cost is *below* the budgeted cap $\eta_0 = \$300$. A sample of nine repairs costs (in dollars) 250, 280, 190, 310, 220, 260, 175, 295, 240. Test $H_0 : \eta = 300$ versus $H_1 : \eta < 300$ at $\alpha = 0.05$.

Which rule, and why? The claim points downward (median *below* the cap), so the rejection region is the *lower* tail and the p-value is $\mathbb{P}(S_+ \leq s_{\text{obs}})$. Cost data are right-skewed, making the median — and the sign test — the natural tool over a mean-based t.

Solution.

Step 1: sign against $\eta_0 = 300$. Above 300: only 310 (one plus). Below 300: the other eight (eight minuses). $n = 9$, $s_{\text{obs}} = 1$ plus.

Step 2: lower-tail p-value from $\text{Bin}(9, \frac{1}{2})$.

$$\mathbb{P}(S_+ \leq 1) = \frac{\binom{9}{0} + \binom{9}{1}}{2^9} = \frac{1 + 9}{512} = \frac{10}{512} = 0.0195.$$

Step 3: decide. $0.0195 < 0.05$, reject.

$$\boxed{s_{\text{obs}} = 1,\ p\text{-value} = 0.020 < 0.05 :\ \text{the median repair cost is below } \$300.}$$

Eight of nine repairs under the cap is a sign imbalance a fair coin rarely matches.

14.1.4 The paired sign test

Definition 14.1.9 (Sign test on paired differences)**.** Given paired observations (X_i, Y_i), form the differences $D_i = X_i - Y_i$ and apply the sign test to them with $\eta_0 = 0$. A plus is a pair with $D_i > 0$ (i.e. $X_i > Y_i$); ties $D_i = 0$ are discarded. Under H_0 (the two members of a pair have the same median, equivalently the median difference is 0), the number of positive differences is $\mathrm{Bin}(n, \frac{1}{2})$. This is the nonparametric counterpart of the paired t-test.

Example 14.1.10 (Paired sign test: before-and-after blood pressure)**.** A clinic measures systolic blood pressure in 11 patients before and after eight weeks on a low-sodium diet. The (before, after) pairs in mmHg are

before	148	152	139	161	144	150	158	142	155	147	163
after	141	149	142	150	138	146	151	144	148	140	152

Test whether the diet lowers the median blood pressure: H_0 : median difference $= 0$ versus H_1 : median (before $-$ after) > 0, at $\alpha = 0.05$.

Which rule, and why? Paired before/after measurements with a directional claim call for a paired test on the within-pair differences. Using only the *signs* of $D_i =$ before $-$ after gives the paired sign test — robust to the odd patient whose pressure jumps, and needing no normality of the differences.

Solution.

Step 1: sign each difference $D_i =$ before $-$ after.

$$D = (+7,\ +3,\ -3,\ +11,\ +6,\ +4,\ +7,\ -2,\ +7,\ +7,\ +11),$$
$$\Rightarrow \text{ nine pluses, two minuses (patients 3 and 8), no zeros.}$$

So $n = 11$ and $s_{\text{obs}} = 9$.

Step 2: upper-tail p-value from $\mathrm{Bin}(11, \frac{1}{2})$.

$$\mathbb{P}(S_+ \geq 9) = \frac{\binom{11}{9} + \binom{11}{10} + \binom{11}{11}}{2^{11}} = \frac{55 + 11 + 1}{2048} = \frac{67}{2048} = 0.0327.$$

Step 3: decide. $0.0327 < 0.05$, reject.

$$\boxed{s_{\text{obs}} = 9,\ p\text{-value} = 0.033 < 0.05 :\ \text{the diet lowers median blood pressure.}}$$

Notice we never used the sizes $+11$ or -2: nine of eleven patients improving is enough by itself.

Example 14.1.11 (Paired sign test, two-sided: two graders' scores)**.** Two graders independently score 10 essays; the question is whether they have the same median score, with no direction assumed. The (grader A, grader B) pairs are

A	82	75	90	68	79	88	71	95	84	77
B	78	77	85	70	74	83	69	91	80	72

Test H_0 : median difference $= 0$ versus $H_1 : \neq 0$ at $\alpha = 0.05$.

Which rule, and why? Paired scores, two-sided, possibly non-normal — a two-sided paired sign test on $D_i = A_i - B_i$ with both binomial tails. The signs alone settle whether one grader systematically runs higher.

Solution.

Step 1: sign $D_i = A_i - B_i$.

$$D = (+4,\ -2,\ +5,\ -2,\ +5,\ +5,\ +2,\ +4,\ +4,\ +5) \ \Rightarrow \text{ eight pluses, two minuses (essays 2 and 4).}$$

$n = 10$, $s_{\text{obs}} = 8$.

Step 2: two-sided p-value from $\mathrm{Bin}(10, \frac{1}{2})$. Using $\mathbb{P}(S_+ \geq 8) = \mathbb{P}(S_+ \leq 2)$,

$$\mathbb{P}(S_+ \geq 8) = \frac{\binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{1024} = \frac{45 + 10 + 1}{1024} = \frac{56}{1024} = 0.0547,$$
$$p\text{-value} = 2(0.0547) = 0.109.$$

Step 3: decide. $0.109 > 0.05$, do not reject.

$$\boxed{s_{\text{obs}} = 8,\ p\text{-value} = 0.11 > 0.05:\ \text{no significant difference in median grading.}}$$

Eight of ten falls just short two-sided — another illustration that discarding magnitudes costs power even when grader A clearly tends higher.

14.1.5 Handling ties at the hypothesized value

Definition 14.1.12 (Discarding ties)**.** Observations exactly equal to η_0 (or paired differences exactly 0) carry no sign information: they favor neither H_1 direction. The standard rule is to **discard** every tie and reduce the working sample size n to the number of non-tied observations. The binomial test is then run on that reduced n.

Remark. Discarding ties shrinks n, which can blunt an already low-powered test; with many ties consider a test that uses magnitudes (signed-rank, §2) instead. But the logic is sound — a value sitting on η_0 is genuinely uninformative about which side the median is on, so counting it as either sign would bias the test.

Example 14.1.13 (Ties dropped: median satisfaction rating)**.** A survey asks whether the median satisfaction score differs from the neutral value $\eta_0 = 5$ on a 0–10 ordinal scale. Thirteen respondents give

$$7,\ 5,\ 8,\ 3,\ 5,\ 9,\ 6,\ 5,\ 4,\ 8,\ 7,\ 2,\ 6.$$

Test $H_0 : \eta = 5$ versus $H_1 : \eta \neq 5$ at $\alpha = 0.05$, handling the ties correctly.

Which rule, and why? Ordinal survey scores are exactly the case for a sign test (a mean is meaningless on a rating scale). Three respondents sit *on* $\eta_0 = 5$, so the tie rule applies: discard them and run the binomial on what remains.

Solution.

Step 1: separate ties. Three values equal 5 (discarded). Remaining ten: $7, 8, 3, 9, 6, 4, 8, 7, 2, 6$.

Step 2: sign the remaining ten against 5. Below 5: $3, 4, 2$ (three minuses). Above 5: $7, 8, 9, 6, 8, 7, 6$ (seven pluses). So $n = 10$, $s_{\text{obs}} = 7$.

Step 3: two-sided p-value from $\text{Bin}(10, \frac{1}{2})$.

$$\mathbb{P}(S_+ \geq 7) = \frac{\binom{10}{7} + \binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{1024} = \frac{120 + 45 + 10 + 1}{1024} = \frac{176}{1024} = 0.1719,$$
$$p\text{-value} = 2(0.1719) = 0.344.$$

Step 4: decide. $0.344 > 0.05$, do not reject.

$$\boxed{n = 10 \text{ after dropping 3 ties},\ s_{\text{obs}} = 7,\ p\text{-value} = 0.34 > 0.05:\ \text{median consistent with 5.}}$$

Had we wrongly kept the three ties as pluses, we would have inflated s_{obs} to 10 and manufactured a spurious result — which is exactly why ties are discarded.

14.1.6 The large-sample normal approximation

Theorem 14.1.14 (Normal approximation to the sign statistic)**.** *For moderate to large n, by the central limit theorem applied to the* $\text{Bin}(n, \frac{1}{2})$ *count,*

$$z = \frac{S_+ - \dfrac{n}{2}}{\sqrt{\dfrac{n}{4}}} = \frac{2S_+ - n}{\sqrt{n}} \ \dot{\sim}\ \text{N}(0,1) \quad \textit{under } H_0.$$

Reject a two-sided test when $|z| \geq z_{\alpha/2}$, and a one-sided test when z falls in the appropriate tail. A continuity correction $\pm\frac{1}{2}$ improves the approximation: $z = \big(S_+ \pm \frac{1}{2} - n/2\big)/\sqrt{n/4}$.

Example 14.1.15 (Large-sample sign test: median commute time)**.** A transit planner tests whether the median commute time equals $\eta_0 = 30$ minutes. Of $n = 50$ commuters surveyed (none exactly at 30), $s_{\text{obs}} = 32$ report a commute longer than 30 minutes. Test $H_0 : \eta = 30$ versus $H_1 : \eta \neq 30$ at $\alpha = 0.05$ using the normal approximation.

Which rule, and why? With $n = 50$, enumerating the exact binomial is clumsy and the CLT applies cleanly, so use the large-sample $z = (S_+ - n/2)/\sqrt{n/4}$. It is still the sign test — only the reference distribution has switched from binomial to normal.

Solution.

Step 1: null mean and standard deviation of S_+. $\mathbb{E}[S_+] = n/2 = 25$, $\text{SD}(S_+) = \sqrt{n/4} = \sqrt{12.5} = 3.536$.

Step 2: z with continuity correction. Since $s_{\text{obs}} = 32 > 25$, subtract $\frac{1}{2}$:

$$z = \frac{(32 - 0.5) - 25}{3.536} = \frac{6.5}{3.536} = 1.838.$$

Step 3: compare to $z_{0.025} = 1.96$. $|1.838| < 1.96$, so do not reject (two-sided p-value $\approx 2(0.0330) = 0.066$).

$$\boxed{z = 1.84,\ |z| < 1.96 :\ \text{do not reject; median commute consistent with 30 min.}}$$

Thirty-two of fifty leans long but not decisively — a hair outside significance.

Example 14.1.16 (Large-sample one-sided sign test: median download speed)**.** An ISP advertises a median download speed of at least $\eta_0 = 100$ Mbps; a consumer group suspects it is lower. Among $n = 80$ independent speed tests (no exact ties), only $s_{\text{obs}} = 33$ exceed 100 Mbps. Test $H_0 : \eta = 100$ versus $H_1 : \eta < 100$ at $\alpha = 0.01$.

Which rule, and why? Large n and a directional claim: a one-sided large-sample sign test. Too few pluses is the evidence for $\eta < 100$, so the lower tail of the z is the rejection region.

Solution.

Step 1: null moments. $\mathbb{E}[S_+] = 40$, $\text{SD}(S_+) = \sqrt{80/4} = \sqrt{20} = 4.472$.

Step 2: z with continuity correction. Since $s_{\text{obs}} = 33 < 40$, add $\frac{1}{2}$:

$$z = \frac{(33 + 0.5) - 40}{4.472} = \frac{-6.5}{4.472} = -1.453.$$

Step 3: compare to $-z_{0.01} = -2.326$. $-1.453 > -2.326$, so do not reject.

$$\boxed{z = -1.45 > -2.326 :\ \text{not enough sign evidence the median is below 100 Mbps at } \alpha = 0.01.}$$

The pluses are short of half, but at the strict 1% level a sign test needs a heavier imbalance.

14.1.7 Contrast: the sign test against its parametric and rank rivals

Example 14.1.17 (CONTRAST: sign test vs. paired t vs. signed-rank on the SAME data)**.** A sports scientist measures reaction time (ms) for 8 athletes under a control and a caffeine condition; smaller is faster. The (control, caffeine) pairs are

control	244	261	250	238	256	270	247	259
caffeine	240	255	252	233	249	210	244	251

The control-minus-caffeine differences are $D = (+4, +6, -2, +5, +7, +60, +3, +8)$ — one athlete (the +60) responds enormously. Compare what the sign test, the paired t-test, and the Wilcoxon signed-rank test see in these data, and say which is most defensible here.

Which rule, and why? All three are paired-difference tests, but they read different features: the t uses the *mean* difference (magnitude-sensitive, normality-dependent), the signed-rank uses ranked magnitudes (needs symmetry), and the sign test uses only the *signs*. The lone +60 is the discriminating feature — it shows how each test handles an outlier.

Solution.

Step 1: sign test — count only signs. Seven pluses, one minus, $n = 8$, $s_{\text{obs}} = 7$:

$$p\text{-value} = 2\,\mathbb{P}(S_+ \geq 7) = 2 \cdot \frac{\binom{8}{7} + \binom{8}{8}}{256} = 2 \cdot \frac{9}{256} = 0.0703.$$

The verdict is driven entirely by the 7-to-1 split; the +60 counts as a single plus, exactly like a +4.

Step 2: paired t — magnitude-sensitive. The mean and sd of D are

$$\overline{D} = \tfrac{4+6-2+5+7+60+3+8}{8} = \tfrac{91}{8} = 11.375,$$
$$s_D \approx 20.0 \;\Rightarrow\; t_{\text{obs}} = \frac{11.375}{20.0/\sqrt{8}} = \frac{11.375}{7.07} = 1.609.$$

The single +60 inflates both $\overline{D}$ *and* s_D; the swollen s_D actually drags the t-statistic down, and against $t_{0.025,7} = 2.365$ the paired t fails to reject — the outlier has wrecked its yardstick, and the normality the t assumes is plainly violated.

Step 3: signed-rank — ranked magnitudes. Ranking $|D|$ $(2, 3, 4, 5, 6, 7, 8, 60$ get ranks $1, \ldots, 8)$ and signing them, only the $|-2| = 2$ (rank 1) is negative, so $W_- = 1$ and $W_+ = 35$. The signed-rank test, using the magnitude *order* but not the raw 60, rejects at $\alpha = 0.05$ (its small W_- is extreme) — it caps the outlier's leverage at "largest rank" while still crediting that caffeine helped almost everyone.

Step 4: which to trust.

$$\begin{aligned} \text{sign test } p = 0.070 &\text{ (robust, but low power: wastes the clean magnitudes),} \\ \text{paired } t:\ t_{\text{obs}} = 1.61 &\text{ (broken by the outlier; normality false),} \\ \text{signed-rank}:\ &\text{rejects (uses magnitude order, resists the outlier).} \end{aligned}$$

Outlier present, differences roughly symmetric: the signed-rank test is best; the sign test is safe but weak; the t is unreliable.

The moral: the sign test never breaks but often cannot see an effect the data plainly contain; reach for it when you cannot trust symmetry or magnitudes, and step up to the signed-rank or t when you can.

14.1.8 A confidence interval for the median from the sign test

Proposition 14.1.18 (Sign-test confidence interval for a median)**.** *Order the sample as $X_{(1)} \leq \cdots \leq X_{(m)}$. A distribution-free confidence interval for the median η is an interval of order statistics $\big(X_{(a)},\, X_{(m+1-a)}\big)$ whose endpoints are chosen so that, under $S_+ \sim \text{Bin}(m, \frac{1}{2})$,*

$$\mathbb{P}\big(X_{(a)} < \eta < X_{(m+1-a)}\big) = 1 - 2\,\mathbb{P}(S_+ \leq a - 1) \;\geq\; 1 - \alpha.$$

The interval is exactly the set of η_0 values the two-sided sign test would not *reject at level α — the test and the interval are duals.*

Example 14.1.19 (Median confidence interval via the sign test: home prices)**.** Eleven recent home sales (in \$1000s), already sorted, are

$$182,\ 195,\ 210,\ 224,\ 238,\ 251,\ 263,\ 279,\ 295,\ 318,\ 350.$$

Construct a distribution-free confidence interval for the median sale price at confidence level at least 95%.

Which rule, and why? A robust interval for a *median* (skewed price data) comes from inverting the sign test: pick symmetric order statistics whose binomial tail probabilities leave at most α outside. No normality is assumed — the coverage is exact binomial.

Solution.

Step 1: find the cutoff a for $m = 11$. We need $2\,\mathbb{P}(S_+ \le a-1) \le 0.05$ with $S_+ \sim \text{Bin}(11, \frac{1}{2})$. From the lower tail,

$$\mathbb{P}(S_+ \le 1) = \tfrac{1+11}{2048} = 0.00586, \quad 2(0.00586) = 0.0117,$$
$$\mathbb{P}(S_+ \le 2) = \tfrac{1+11+55}{2048} = 0.03271, \quad 2(0.03271) = 0.0654.$$

Taking $a - 1 = 1$, i.e. $a = 2$, gives coverage $1 - 0.0117 = 0.9883 \ge 0.95$ (the tighter $a = 3$ would drop below 95%).

Step 2: read off the order statistics. The interval is $\big(X_{(2)},\, X_{(11+1-2)}\big) = \big(X_{(2)}, X_{(10)}\big) = (195,\ 318)$ (in \$1000s).

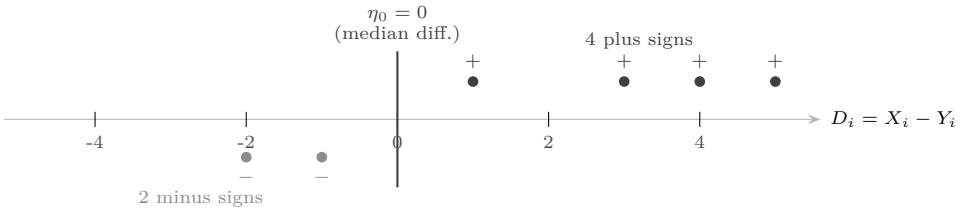
Step 3: state coverage. The exact confidence level is 98.83% (the smallest level at least 95% achievable with these order statistics).

98.83% CI for the median price : (\$195k, \$318k).

Every η_0 inside $(195, 318)$ is a median the two-sided sign test would not reject — the interval and the test are the same statement read two ways.

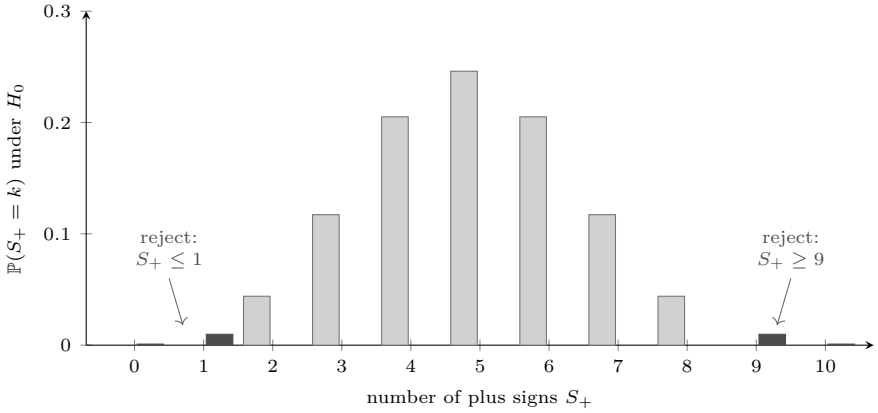
14.1.9 Two pictures: the signed differences and the binomial reference

The first picture is a number line of paired differences for a small study, with the hypothesized median marked at 0. Each pair contributes a tick above the line (+, the pair improved) or below it (−); the sign test simply counts how many sit on each side and ignores how far out they lie.



Six paired differences on a number line, with the hypothesized median difference at $\eta_0 = 0$. Four pairs fall to the right (+) and two to the left (−). The sign test uses only these signs: $s_{obs} = 4$ pluses out of $n = 6$, compared against $\text{Bin}(6, \frac{1}{2})$. *The actual distances (+5 versus +1, say) are deliberately ignored, which is the source of both the test's robustness and its low power.*

The second picture is the $\text{Bin}(10, \frac{1}{2})$ probability mass function — the reference distribution for a sign test with $n = 10$. The shaded bars in the two tails ($S_+ \le 1$ and $S_+ \ge 9$) are the rejection region of a two-sided test: extreme sign counts that a fair coin almost never produces.



The $\mathrm{Bin}(10, \frac{1}{2})$ *null distribution of the sign statistic. The light central bars are sign counts consistent with* $\eta = \eta_0$*; the dark tail bars (*$S_+ \le 1$ *and* $S_+ \ge 9$*) form the two-sided rejection region, with total probability* $2(0.00098 + 0.00977) = 0.0215$ *— the achievable level closest to* 5% *for* $n = 10$*. Because the binomial is discrete, the attainable significance levels come in jumps, which is why exact sign-test p-values rarely land exactly on* 0.05*.*

14.2 The Wilcoxon Signed-Rank Test

The sign test of the previous section threw away almost everything: it kept only whether each observation fell above or below the hypothesized center, not by how much. Wilcoxon's signed-rank test keeps more. It still records the **sign** of each deviation $D_i = X_i - m_0$ (or, in the paired case, of each difference), but now it also **ranks** the magnitudes $|D_i|$ and lets a large deviation count for more than a small one. The recipe is: drop any zero deviations, rank the remaining $|D_i|$ from smallest to largest (averaging ranks within ties), re-attach the original signs, and add up the ranks that carried a + sign to get the statistic W_+. Under the null hypothesis that the data are **symmetric** about m_0, every rank is equally likely to land on the plus or the minus side, so W_+ has a known, table-able distribution; an unusually small or large W_+ contradicts H_0. Recognize this section when you have one sample (or paired differences), you are testing a center, and you are willing to assume the underlying distribution is symmetric — that symmetry assumption is the price of the extra power over the sign test, and it is what lets this test stand in for the paired t without requiring normality.

Which one-sample/paired test does the problem call for?

- Testing a **median** using only *which side* of m_0 each point falls on, with no symmetry assumed? ⇒ the **sign test** (§1) — robust but weak.
- Testing a center where the distribution is (or can be assumed) **symmetric**, and you want to use the *sizes* of the deviations, not just their signs? ⇒ the **Wilcoxon signed-rank test** — rank $|D_i|$, sum the positive ranks W_+.
- Paired data (before, after) with symmetric differences and no normality? ⇒ signed-rank on the differences D_i — the nonparametric paired t.
- Genuinely **normal** differences? ⇒ the paired t is slightly more powerful; use it. Reach for signed-rank when normality is doubtful or the tails are heavy.

Do not confuse the two rank totals: W_+ sums the ranks of the *positive* deviations and W_- the negative ones. They are not independent — they always satisfy $W_+ + W_- = n(n+1)/2$, so one determines the other. Reject when W_+ is too small *or* too large (equivalently, when $\min(W_+, W_-)$ is too small).

14.2.1 Ranks, signs, and the statistic

Definition 14.2.1 (Signed ranks and W_+, W_-)**.** Let $D_1, \dots, D_n$ be the deviations from the hypothesized center (for a one-sample test $D_i = X_i - m_0$; for paired data D_i is the within-pair difference). Discard any $D_i = 0$ and let n denote the number of *nonzero* deviations. Rank the absolute values $|D_1|, \dots, |D_n|$ from 1 (smallest) to n (largest), assigning the **average rank** to any group of tied magnitudes. Re-attach to each rank the sign of its D_i. Then

$$W_+ = \sum_{i:\, D_i > 0} \mathrm{rank}\,|D_i|, \qquad W_- = \sum_{i:\, D_i < 0} \mathrm{rank}\,|D_i|$$

are the **positive** and **negative rank sums**. Because the two sets of ranks together exhaust $\{1, 2, \dots, n\}$,

$$W_+ + W_- = \sum_{r=1}^{n} r = \frac{n(n+1)}{2}.$$

Remark. The identity $W_+ + W_- = n(n+1)/2$ is the bookkeeping check you run on every problem: compute one sum directly, get the other by subtraction, and confirm they add to $n(n+1)/2$. If they do not, a rank was dropped or a sign mis-copied. It also explains the two-sided rejection rule — a small W_+ is the same evidence as a large W_-, so extreme values of *either* contradict symmetry about m_0.

Theorem 14.2.2 (Null distribution of W_+)**.** *Suppose $D_1, \dots, D_n$ are independent and the common distribution is* ***continuous and symmetric*** *about 0 (i.e. X is symmetric about m_0). Then each rank $r \in \{1, \dots, n\}$ attaches to a positive sign with probability $\frac{1}{2}$, independently across ranks, and*

$$W_+ = \sum_{r=1}^{n} r\, B_r, \qquad B_r \overset{\text{ind}}{\sim} \text{Bernoulli}(\tfrac{1}{2}),$$

so W_+ ranges over $0, 1, \dots, n(n+1)/2$ with a distribution that is symmetric about its mean. In particular

$$\mathbb{E}[W_+] = \frac{n(n+1)}{4}, \qquad \text{Var}[W_+] = \frac{n(n+1)(2n+1)}{24}.$$

Proof. Under symmetry about 0, the sign of each D_i is independent of its magnitude $|D_i|$ and is + with probability $\frac{1}{2}$. The ranks of the magnitudes are a fixed permutation of $1, \dots, n$, so attaching signs is the same as writing $W_+ = \sum_r r\, B_r$ with independent fair B_r. Then $\mathbb{E}[W_+] = \sum_r r \cdot \frac{1}{2} = \frac{1}{2} \cdot \frac{n(n+1)}{2} = \frac{n(n+1)}{4}$, and by independence

$$\text{Var}[W_+] = \sum_{r=1}^{n} r^2\, \text{Var}(B_r) = \frac{1}{4} \sum_{r=1}^{n} r^2 = \frac{1}{4} \cdot \frac{n(n+1)(2n+1)}{6} = \frac{n(n+1)(2n+1)}{24}. \qquad \square$$

Definition 14.2.3 (Small-sample and large-sample tests)**.** Fix the center under test, $H_0 \colon m = m_0$ (data symmetric about m_0). Let w_+ be the observed positive rank sum and $w = \min(w_+, w_-)$.

- **Small samples** ($n \lesssim 20$): compare $w = \min(w_+, w_-)$ to the tabulated critical value $w_{\alpha,n}$ and reject when $w \le w_{\alpha,n}$. For a one-sided alternative use the relevant single tail (W_+ small means the data sit below m_0).
- **Large samples**: standardize,

$$W^* = \frac{W_+ - \dfrac{n(n+1)}{4}}{\sqrt{\dfrac{n(n+1)(2n+1)}{24}}} \;\approx\; \text{N}(0,1),$$

  and reject for $|W^*| > z_{\alpha/2}$ (two-sided) or the appropriate one-sided z.

$D_i = X_i - m_0$	$\lvert D_i \rvert$	rank $\lvert D_i \rvert$	signed rank
$+0.4$	0.4	2	$+2$
-0.2	0.2	1	-1
$+1.1$	1.1	4	$+4$
-0.7	0.7	3	-3
$+1.6$	1.6	5	$+5$
$W_+ = 2+4+5 = 11$,	$W_- = 1+3 = 4$,	$W_+ + W_- = 15 = \frac{5\cdot 6}{2}$	

The four-column worked layout. Strip the sign to rank the magnitudes, then paste the sign back onto each rank. The positive ranks sum to W_+, the negative to W_-, and the two totals must add to $n(n+1)/2 = 15$.

Example 14.2.4 (A full signed-rank test in a ranking table: battery life)**.** A manufacturer claims its cells run a median of $m_0 = 50$ hours. An auditor draws $n = 8$ cells and records run times (hours)

$$52.1, \quad 47.0, \quad 53.4, \quad 49.2, \quad 55.6, \quad 51.3, \quad 50.9, \quad 54.8.$$

Test $H_0 \colon m = 50$ versus $H_1 \colon m \neq 50$ at $\alpha = 0.05$, given the small-sample critical value $w_{0.05,8} = 3$ (two-sided): reject when $\min(W_+, W_-) \le 3$.

Which rule, and why? One sample, a center under test, and run times that are plausibly symmetric about the median — this is signed-rank, not the sign test, because we want the magnitudes of the deviations to count. Rank the $|D_i|$, re-sign them, and compare $\min(W_+, W_-)$ to the table.

Solution.

Step 1: deviations, magnitudes, ranks. Subtract $m_0 = 50$ and rank the absolute values:

X_i	$D_i = X_i - 50$	$\lvert D_i\rvert$	rank $\lvert D_i\rvert$	signed rank
50.9	+0.9	0.9	1	+1
51.3	+1.3	1.3	2	+2
52.1	+2.1	2.1	3	+3
49.2	−0.8	0.8	—	—
47.0	−3.0	3.0	—	—

Ordering all eight magnitudes $0.8, 0.9, 1.3, 2.1, 3.0, 3.4, 4.8, 5.6$ gives ranks

$$|-0.8| = 0.8 \to 1, \quad |+0.9| = 0.9 \to 2, \quad |+1.3| = 1.3 \to 3, \quad |+2.1| = 2.1 \to 4,$$
$$|-3.0| = 3.0 \to 5, \quad |+3.4| = 3.4 \to 6, \quad |+4.8| = 4.8 \to 7, \quad |+5.6| = 5.6 \to 8.$$

Step 2: attach signs and sum. The negative deviations are -0.8 (rank 1) and -3.0 (rank 5); all others are positive:

$$W_- = 1 + 5 = 6,$$
$$W_+ = 2 + 3 + 4 + 6 + 7 + 8 = 30.$$

Check: $W_+ + W_- = 30 + 6 = 36 = \frac{8 \cdot 9}{2}$. Good.

Step 3: decision. The test statistic is $\min(W_+, W_-) = 6$, compared with $w_{0.05,8} = 3$.

$$\boxed{\min(W_+, W_-) = 6 > 3 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{no evidence the median differs from 50.}}$$

The cells run a little long on average, but not by enough to overturn the 50-hour claim.

Example 14.2.5 (Paired signed-rank test: a training program)**.** Ten sprinters are timed (seconds) in the 100 m before and after a six-week program. The within-athlete differences $D_i = \text{before} - \text{after}$ are

$$0.31,\ 0.12,\ -0.05,\ 0.44,\ 0.19,\ 0.27,\ 0.08,\ 0.51,\ -0.14,\ 0.36.$$

A positive D_i means the athlete got faster. Test H_0: no change (the differences are symmetric about 0) versus H_1: the program helps ($m > 0$), at $\alpha = 0.05$, with one-sided critical value $w_{0.05,10} = 10$: reject when $W_- \le 10$.

Which rule, and why? Paired before/after data reduce to one sample of differences, and we test whether their center exceeds 0. With no reason to trust normality of times, signed-rank on the D_i replaces the paired t; rank the $|D_i|$ and watch W_- (small W_- means the negatives are few and tiny, i.e. improvement).

Solution.

Step 1: rank the magnitudes. The absolute differences in increasing order, with ranks:

$$0.05 \to 1,\ 0.08 \to 2,\ 0.12 \to 3,\ 0.14 \to 4,\ 0.19 \to 5,$$
$$0.27 \to 6,\ 0.31 \to 7,\ 0.36 \to 8,\ 0.44 \to 9,\ 0.51 \to 10.$$

Step 2: signed ranks and the two sums. Only -0.05 (rank 1) and -0.14 (rank 4) are negative:

$$W_- = 1 + 4 = 5,$$
$$W_+ = 2 + 3 + 5 + 6 + 7 + 8 + 9 + 10 = 50.$$

Check: $W_+ + W_- = 50 + 5 = 55 = \frac{10 \cdot 11}{2}$. Good.

Step 3: one-sided decision. Faster times make $D_i > 0$, so the alternative pushes W_+ up and W_- down; we reject for small W_-. Here $W_- = 5 \le w_{0.05,10} = 10$.

$$\boxed{W_- = 5 \le 10 \;\Rightarrow\; \text{reject } H_0;\ \text{the program significantly improves sprint times.}}$$

Example 14.2.6 (A one-sided test in the other tail: water purity)**.** An environmental standard sets the median lead concentration in a stream at $m_0 = 15$ μg/L. A regulator suspects the true median is *below* the standard and samples $n = 9$ days, getting deviations $D_i = X_i - 15$ (in μg/L)

$$-2.1,\ -3.4,\ +0.6,\ -1.2,\ -4.0,\ -0.9,\ +1.8,\ -2.7,\ -3.1.$$

Test $H_0\colon m = 15$ versus $H_1\colon m < 15$ at $\alpha = 0.05$, with one-sided critical value $w_{0.05,9} = 8$: reject when $W_+ \le 8$.

Which rule, and why? The alternative $m < 15$ predicts that deviations are mostly negative, so the *positive* rank sum W_+ should be small. The one-sided test rejects for small W_+ — the mirror image of the previous example, where the alternative made W_+ large.

Solution.

Step 1: rank the magnitudes. Absolute deviations in order, with ranks:

$$0.6 \to 1,\ 0.9 \to 2,\ 1.2 \to 3,\ 1.8 \to 4,\ 2.1 \to 5,$$
$$2.7 \to 6,\ 3.1 \to 7,\ 3.4 \to 8,\ 4.0 \to 9.$$

Step 2: signed ranks. The positive deviations are $+0.6$ (rank 1) and $+1.8$ (rank 4); the rest are negative:

$$W_+ = 1 + 4 = 5,$$
$$W_- = 2 + 3 + 5 + 6 + 7 + 8 + 9 = 40.$$

Check: $W_+ + W_- = 5 + 40 = 45 = \frac{9\cdot 10}{2}$. Good.

Step 3: decision. For $H_1\colon m < 15$ we reject when W_+ is small: $W_+ = 5 \le w_{0.05,9} = 8$.

$$\boxed{W_+ = 5 \le 8 \ \Rightarrow\ \text{reject } H_0;\ \text{the stream's median lead is below the standard.}}$$

Example 14.2.7 (The large-sample normal approximation: reaction times)**.** A psychologist measures the change in reaction time (ms) for $n = 30$ subjects under a distraction and computes a positive rank sum $W_+ = 170$. Test $H_0\colon$ no shift (differences symmetric about 0) versus $H_1\colon$ a shift in either direction at $\alpha = 0.05$, using the normal approximation.

Which rule, and why? At $n = 30$ the exact table runs out; the standardized statistic W^* is approximately $\mathrm{N}(0,1)$ under H_0. Use the mean $n(n+1)/4$ and variance $n(n+1)(2n+1)/24$, then compare $|W^*|$ to $z_{0.025} = 1.96$.

Solution.

Step 1: null mean and variance.

$$\mathbb{E}[W_+] = \frac{n(n+1)}{4} = \frac{30 \cdot 31}{4} = 232.5,$$
$$\mathrm{Var}[W_+] = \frac{n(n+1)(2n+1)}{24} = \frac{30 \cdot 31 \cdot 61}{24} = \frac{56730}{24} = 2363.75.$$

Step 2: standardize.

$$W^* = \frac{W_+ - \mathbb{E}[W_+]}{\sqrt{\mathrm{Var}[W_+]}} = \frac{170 - 232.5}{\sqrt{2363.75}} = \frac{-62.5}{48.62} = -1.286.$$

Step 3: decision. Two-sided at $\alpha = 0.05$, the cutoff is $z_{0.025} = 1.96$, and $|-1.286| = 1.286 < 1.96$.

$$\boxed{W^* = -1.29,\ |W^*| < 1.96 \ \Rightarrow\ \text{fail to reject } H_0;\ \text{no significant shift.}}$$

Example 14.2.8 (One-sided large-sample test: a yield improvement)**.** An agronomist applies a soil amendment to $n = 25$ plots and records the yield change (bushels/acre) per plot. The positive rank sum is $W_+ = 255$. Test $H_0\colon$ no effect versus $H_1\colon$ the amendment raises yield ($m > 0$) at $\alpha = 0.05$ via the normal approximation.

Which rule, and why? A one-sided "amendment helps" alternative makes the positive deviations dominate, pushing W_+ above its null mean. Standardize and compare to the *upper* one-sided cutoff $z_{0.05} = 1.645$ — we reject only for large positive W^*.

Solution.

Step 1: null mean and variance.

$$\mathbb{E}[W_+] = \frac{25 \cdot 26}{4} = 162.5,$$
$$\mathrm{Var}[W_+] = \frac{25 \cdot 26 \cdot 51}{24} = \frac{33150}{24} = 1381.25, \qquad \sqrt{1381.25} = 37.16.$$

Step 2: standardize.

$$W^* = \frac{255 - 162.5}{37.16} = \frac{92.5}{37.16} = 2.489.$$

Step 3: one-sided decision.

$$\boxed{W^* = 2.49 > z_{0.05} = 1.645 \;\Rightarrow\; \text{reject } H_0;\ \text{the amendment raises yield.}}$$

Example 14.2.9 (Tie and zero handling: a taste-preference score)**.** Tasters rate a reformulated cereal against the original on a -5 to $+5$ liking scale (positive means the new version is preferred). For $n = 10$ tasters the differences are

$$+2,\ -1,\ +2,\ 0,\ +3,\ -1,\ +2,\ 0,\ +4,\ +1.$$

Process the zeros and ties correctly and compute W_+ and W_-.

Which rule, and why? Two rules govern messy data: a deviation of exactly 0 carries no sign information and is *dropped* (reducing n), and tied magnitudes share the *average* of the ranks they would otherwise occupy. Both must be applied before any summing.

Solution.

Step 1: drop the zeros. Two tasters scored 0; discard them, leaving $n = 8$ nonzero differences:

$$+2,\ -1,\ +2,\ +3,\ -1,\ +2,\ +4,\ +1.$$

Step 2: average ranks within tied magnitudes. The eight magnitudes are $1, 1, 1, 2, 2, 2, 3, 4$. The three 1's occupy rank positions $1, 2, 3$, so each gets the average $\frac{1+2+3}{3} = 2$. The three 2's occupy positions $4, 5, 6$, each getting $\frac{4+5+6}{3} = 5$. Then $3 \to 7$ and $4 \to 8$:

D_i	$\lvert D_i \rvert$	average rank $\lvert D_i \rvert$	signed rank
-1	1	2	-2
-1	1	2	-2
$+1$	1	2	$+2$
$+2$	2	5	$+5$
$+2$	2	5	$+5$
$+2$	2	5	$+5$
$+3$	3	7	$+7$
$+4$	4	8	$+8$

Step 3: the two rank sums.

$$W_- = 2 + 2 = 4,$$
$$W_+ = 2 + 5 + 5 + 5 + 7 + 8 = 32.$$

Check: $W_+ + W_- = 32 + 4 = 36 = \frac{8 \cdot 9}{2}$ (with $n = 8$ after dropping zeros). Good.

$$\boxed{W_+ = 32, \quad W_- = 4, \quad n = 8 \text{ (two zeros dropped, three+three ties averaged).}}$$

Example 14.2.10 (Recover W_- from W_+ and check the identity)**.** A signed-rank test on $n = 12$ nonzero deviations reports $W_+ = 58$. Find W_- without re-ranking, and verify the consistency identity.

Which rule, and why? The identity $W_+ + W_- = n(n+1)/2$ pins down one rank sum once the other and n are known — no data needed. This is the fastest sanity check on any signed-rank computation.

Solution.

Step 1: total of all ranks.

$$\frac{n(n+1)}{2} = \frac{12 \cdot 13}{2} = 78.$$

Step 2: subtract.

$$W_- = \frac{n(n+1)}{2} - W_+ = 78 - 58 = 20.$$

Step 3: verify. $W_+ + W_- = 58 + 20 = 78 = \frac{12 \cdot 13}{2}$, as required.

$$\boxed{W_- = 20, \qquad W_+ + W_- = 78 = \tfrac{12 \cdot 13}{2}.}$$

Example 14.2.11 (Building the test statistic both ways: blood pressure drop)**.** Eight patients' systolic pressure is measured before and after a drug; the drops (mmHg, $D_i = \text{before} - \text{after}$) are

$$6,\ 11,\ -2,\ 8,\ 14,\ 3,\ -1,\ 9.$$

Compute W_+ and W_- directly, confirm they add to $n(n+1)/2$, and report $\min(W_+, W_-)$ for a two-sided test.

Which rule, and why? Computing *both* sums and checking the total is the disciplined way to run the test: the smaller of the two is the statistic you carry to the two-sided table, and the additive check catches arithmetic slips before you decide.

Solution.

Step 1: rank the magnitudes. Absolute drops in order, with ranks:

$$1 \to 1,\ 2 \to 2,\ 3 \to 3,\ 6 \to 4,\ 8 \to 5,\ 9 \to 6,\ 11 \to 7,\ 14 \to 8.$$

Step 2: signed ranks. The negatives are -2 (rank 2) and -1 (rank 1):

$$W_- = 1 + 2 = 3,$$
$$W_+ = 3 + 4 + 5 + 6 + 7 + 8 = 33.$$

Step 3: check and report. $W_+ + W_- = 33 + 3 = 36 = \frac{8 \cdot 9}{2}$, and

$$\boxed{\min(W_+, W_-) = W_- = 3.}$$

With $w_{0.05,8} = 3$, this just reaches the cutoff ($3 \le 3$): reject H_0 — the drug lowers systolic pressure.

Example 14.2.12 (CONTRAST: signed-rank versus the sign test on the same data)**.** Return to the blood-pressure drops $6, 11, -2, 8, 14, 3, -1, 9$. The *sign test* would count only how many drops are positive. Run both tests and explain why signed-rank rejects while the sign test does not, at $\alpha = 0.05$. (Sign test: under H_0 the number of + signs is $\text{Bin}(8, \frac{1}{2})$; the two-sided 0.05 rejection region needs $0, 1, 7$, or 8 successes.)

Which rule, and why? Both are one-sample center tests, but the sign test uses only the 6-vs-2 split of signs while signed-rank also uses that the two negatives $(-2, -1)$ are the two *smallest* magnitudes. That extra information is exactly the power signed-rank buys — at the cost of assuming symmetry.

Solution.

Step 1: the sign test. There are 6 positive and 2 negative drops. Under H_0, $S = \#\{+\} \sim \text{Bin}(8, \frac{1}{2})$. The observed $S = 6$ is not in the rejection set $\{0, 1, 7, 8\}$ (its two-sided p-value is $\mathbb{P}(S \le 2) + \mathbb{P}(S \ge 6) = 2 \cdot \frac{37}{256} = 0.289$):

$$S = 6 \;\Rightarrow\; \text{fail to reject.}$$

Step 2: the signed-rank test. From the previous example $\min(W_+, W_-) = 3 = w_{0.05,8}$, so

$$\min(W_+, W_-) = 3 \le 3 \;\Rightarrow\; \text{reject.}$$

Step 3: why they disagree. The two negative drops are -1 and -2, the *two smallest* deviations; the six positive drops include all the large ones. The sign test cannot see this — a -1 counts the same as a -14 would — so it wastes the magnitude information and stays silent.

sign test: retain; signed-rank: reject. Symmetry buys power.

The lesson cuts both ways: if symmetry were violated (say a long right tail), the sign test's verdict would be the safer one, since signed-rank's extra power rests entirely on that assumption.

Example 14.2.13 (CONTRAST: signed-rank versus the paired t with non-normal data)**.** A reliability engineer measures the lifetime change (hundreds of hours) for $n = 9$ pumps after a seal upgrade. The differences are heavily right-skewed:

$$0.2,\ 0.3,\ 0.4,\ 0.5,\ 0.6,\ 0.8,\ 1.1,\ 1.7,\ 9.5.$$

The lone 9.5 is a genuine but extreme improvement. Compare what the paired t and the signed-rank test do with this outlier; test $H_1 : m > 0$ at $\alpha = 0.05$ (signed-rank one-sided cutoff: reject when $W_- \le w_{0.05,9} = 8$).

Which rule, and why? The paired t assumes normal differences and uses the raw values, so one wild observation drags both $\overline{D}$ and the standard error; signed-rank uses only the *rank* of 9.5 (it is just "rank 9"), so it is immune to how extreme the outlier is. With visibly skewed data the rank test is the trustworthy one.

Solution.

Step 1: signed-rank. All nine differences are positive, so every rank is positive:

$$W_+ = 1 + 2 + \cdots + 9 = 45, \qquad W_- = 0.$$

Since $W_- = 0 \le 8$, reject H_0 — the upgrade helps. The verdict would be identical if the outlier were 9.5 or 950; only its rank (9) matters.

Step 2: paired t (for contrast). The mean and SD of the raw differences are inflated by the outlier:

$$\overline{D} = \frac{0.2 + 0.3 + \cdots + 9.5}{9} = \frac{15.1}{9} = 1.678,$$

$$s_D \approx 3.02, \qquad t = \frac{\overline{D}}{s_D/\sqrt{9}} = \frac{1.678}{3.02/3} = \frac{1.678}{1.007} = 1.667.$$

Against $t_{0.05,8} = 1.860$, the t-test *fails to reject* — the single outlier so bloated s_D that the genuine signal was masked, and the normality the t relies on is plainly false here.

Step 3: conclusion.

signed-rank: reject ($W_- = 0$); paired t: retain ($t = 1.67 < 1.86$).

When the differences are non-normal or outlier-prone, signed-rank's rank-only design protects the conclusion that the paired t loses.

Example 14.2.14 (Bracketing a large-sample p-value: customer wait times)**.** A bank shortens its queue process and records the change in wait time (seconds) for $n = 40$ customers, getting $W_+ = 540$. Bracket the two-sided p-value via the normal approximation and decide at $\alpha = 0.05$.

Which rule, and why? With $n = 40$ use $W^* \approx \mathrm{N}(0,1)$. The two-sided p-value is $2\,\mathbb{P}(Z \geq |W^*|)$; comparing $|W^*|$ to standard z-cutoffs brackets it.

Solution.

Step 1: null moments.

$$\mathbb{E}[W_+] = \frac{40 \cdot 41}{4} = 410,$$
$$\mathrm{Var}[W_+] = \frac{40 \cdot 41 \cdot 81}{24} = \frac{132840}{24} = 5535, \qquad \sqrt{5535} = 74.40.$$

Step 2: standardize.

$$W^* = \frac{540 - 410}{74.40} = \frac{130}{74.40} = 1.747.$$

Step 3: bracket the p-value. Since $z_{0.05} = 1.645 < 1.747 < 1.96 = z_{0.025}$, the one-tail area is between 0.025 and 0.05, so the two-sided p-value lies between 0.05 and 0.10:

$$0.05 < p\text{-value} < 0.10.$$

Step 4: decide. The p-value exceeds $\alpha = 0.05$.

$$\boxed{0.05 < p\text{-value} < 0.10 \;\Rightarrow\; \text{fail to reject at } \alpha = 0.05.}$$

Example 14.2.15 (Ties with the large-sample test: pollution index)**.** For $n = 20$ paired measurements of an air-quality index (before/after a regulation), all deviations are nonzero and $W_+ = 70$. Test $H_1 : m \neq 0$ at $\alpha = 0.05$ with the normal approximation. (Treat the ties as negligible for the variance, the usual first-pass approach.)

Which rule, and why? At $n = 20$ the normal approximation is already accurate. Compute the standardized W^* from the textbook mean and variance and compare $|W^*|$ to $z_{0.025} = 1.96$. A W_+ well below its mean $\mathbb{E}[W_+] = 105$ signals deviations clustered on the negative (improvement) side.

Solution.

Step 1: null moments.

$$\mathbb{E}[W_+] = \frac{20 \cdot 21}{4} = 105,$$
$$\mathrm{Var}[W_+] = \frac{20 \cdot 21 \cdot 41}{24} = \frac{17220}{24} = 717.5, \qquad \sqrt{717.5} = 26.79.$$

Step 2: standardize.

$$W^* = \frac{70 - 105}{26.79} = \frac{-35}{26.79} = -1.306.$$

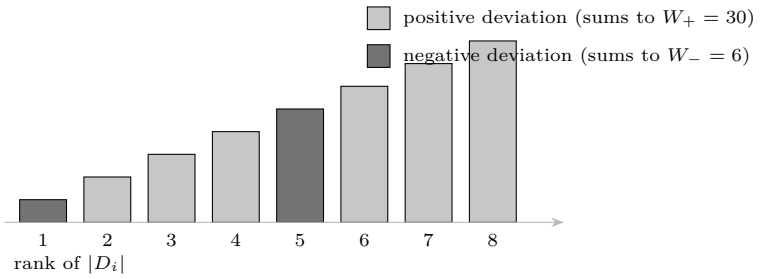
Step 3: decision. $|-1.306| = 1.306 < z_{0.025} = 1.96$.

$$\boxed{W^* = -1.31,\ |W^*| < 1.96 \;\Rightarrow\; \text{fail to reject } H_0.}$$

The index dropped on net (hence $W_+ < \mathbb{E}[W_+]$), but not by a statistically convincing margin.

14.2.2 Visualizing the contributions

Remark. A picture of the signed ranks makes the test's logic visible. Lay the ranks $1, 2, \ldots, n$ on a line; color each one by the sign of its deviation. The positive bars stack to W_+, the negatives to W_-, and the whole bar is always $n(n+1)/2$. Under H_0 the colors are scattered at random, so the two piles come out near $n(n+1)/4$ apiece; under H_1 the large ranks bunch on one side and one pile balloons. The eye reads the verdict off how lopsided the two stacks are.



The eight signed ranks from the battery example. Each bar's height is its rank; light bars are positive deviations, dark bars negative. The light heights add to $W_+ = 30$ and the dark to $W_- = 6$, and $30 + 6 = 36 = \frac{8\cdot 9}{2}$. The two big light bars on the right (ranks $7, 8$) are what make W_+ dominate — though here not by enough to reject.

Example 14.2.16 (Reading the verdict off the stacked ranks)**.** Two signed-rank studies on $n = 8$ pairs report their signed-rank pictures. Study A has its negatives at ranks $\{1, 2\}$ (so $W_- = 3$); study B has its negatives at ranks $\{7, 8\}$ (so $W_- = 15$). Using $w_{0.05,8} = 3$ (two-sided, reject when $\min(W_+, W_-) \le 3$), decide each by inspection.

Which rule, and why? The statistic is $\min(W_+, W_-)$, and the picture shows it at a glance: negatives piled on the small ranks give a tiny W_- (lopsided, reject), negatives on the large ranks give a moderate W_- (balanced, retain). No re-ranking is needed once you can read the stack.

Solution.

Step 1: study A. Negatives at the two smallest ranks: $W_- = 1 + 2 = 3$, so $\min(W_+, W_-) = 3 \le 3$, reject.

Step 2: study B. Negatives at the two largest ranks: $W_- = 7 + 8 = 15$, and $W_+ = 36 - 15 = 21$, so $\min(W_+, W_-) = 15 > 3$, retain.

A: $\min = 3 \Rightarrow$ reject;	B: $\min = 15 \Rightarrow$ retain.

Same two negative *counts*, opposite verdicts: signed-rank cares where the negatives sit in the ordering, which is precisely the magnitude information the sign test discards.

14.3 The Wilcoxon Rank-Sum (Mann–Whitney) Test

When two *independent* samples are to be compared and normality is in doubt — small groups, visible skew, an outlier, or data that are only ranks to begin with — the two-sample t of Chapter 9 loses its footing, and the **Wilcoxon rank-sum test** takes its place. The idea is disarmingly simple. Pour both samples into one pool, sort the $N = n + m$ values from smallest to largest, and replace each value by its **rank**; ties share the average of the ranks they would have occupied. Then add up the ranks belonging to one group — by convention the smaller group, of size n — and call that sum W. If the two distributions are identical, the n ranks are just a random handful drawn from $\{1, 2, \ldots, N\}$, so W has a known mean and variance and tends to land near the middle of its range; a W pushed to either extreme signals that one group's values are systematically smaller (low ranks) or larger (high ranks) than the other's — a **shift** in location. The same evidence can be repackaged as the **Mann–Whitney** statistic U, which counts how often a value from one group beats a value from the other; W and U are one test in two costumes. You will run it from a small-sample table when n and m are tiny and from a normal (z) approximation once both clear about 10.

Two independent groups: which two-sample test fits?

- Two **independent** samples, both plausibly **normal**, comparing means? $\Rightarrow$ the two-sample t (Chapter 9) — more powerful *when its assumptions hold*.
- Two independent samples but normality **doubtful** — skew, outliers, tiny n, or **ordinal** data — comparing whole distributions / medians? $\Rightarrow$ the **Wilcoxon rank-sum (Mann–Whitney) test**: pool, rank, sum the smaller group's ranks to get W.

- Both groups **small** (say $n, m \le 10$)? $\Rightarrow$ compare W (or $U = W - \frac{n(n+1)}{2}$) to a **rank-sum table**.
- Both groups **moderate/large** ($n, m \gtrsim 10$)? $\Rightarrow$ standardize with $\mu_W = \frac{n(N+1)}{2}$, $\sigma_W^2 = \frac{nm(N+1)}{12}$ and use the z approximation (with a continuity correction).

Do not confuse this with the paired tests: the rank-sum test is for two *independent* groups (no natural pairing). If the same or matched units supply both numbers, you want the **signed-rank** test (§2), not this one. And note the hypotheses here are about a *shift of one distribution relative to the other* (medians), never about means.

14.3.1 Pool, rank, and sum: the statistic W

Definition 14.3.1 (Pooled ranks and the rank-sum statistic)**.** Let $X_1, \dots, X_n$ and $Y_1, \dots, Y_m$ be two independent samples, with $n \le m$ (so the *smaller* group has size n), and set $N = n + m$. Combine all N observations, sort them, and assign **ranks** $1, 2, \dots, N$ from smallest to largest; for a group of tied values, give each the **average** of the ranks they jointly occupy. The **rank-sum statistic** is the sum of the ranks attached to the smaller (X) sample,

$$W = \sum_{i=1}^{n} \operatorname{rank}(X_i),$$

the rank being taken in the pooled ordering. The smallest possible value is $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ (all X's below every Y) and the largest is $\frac{n(2N-n+1)}{2}$ (all X's above).

Remark. Why ranks rescue us. Once the data are ranks, their distribution under H_0 does not depend on the parent shape at all: whatever the common distribution, the pooled order statistics carry ranks $1, \dots, N$, and under "same distribution" every assignment of n of those ranks to the X group is equally likely. That is what makes the test **distribution-free** — the null reference distribution of W is fixed by n and m alone, computable once and tabled, with no σ to estimate and no normality to assume.

Theorem 14.3.2 (Null mean and variance of W)**.** *If the two samples come from the same continuous distribution, then W (the rank sum of the size-n group) has*

$$\mu_W = \mathbb{E}[W] = \frac{n(N+1)}{2}, \qquad \sigma_W^2 = \operatorname{Var}(W) = \frac{n\,m\,(N+1)}{12}, \qquad N = n + m.$$

Proof. Under H_0 the X group is a simple random sample of n ranks drawn without replacement from the population $\{1, 2, \dots, N\}$. That finite population has mean $\mu = \frac{N+1}{2}$ and variance $\sigma^2 = \frac{N^2-1}{12}$. For a sum of n draws without replacement,

$$\mathbb{E}[W] = n\mu = \frac{n(N+1)}{2},$$
$$\operatorname{Var}(W) = n\sigma^2 \cdot \frac{N-n}{N-1} = n \cdot \frac{N^2-1}{12} \cdot \frac{N-n}{N-1},$$

where the last factor is the finite-population correction. Since $N^2 - 1 = (N-1)(N+1)$ and $N - n = m$, the variance collapses to

$$\operatorname{Var}(W) = n \cdot \frac{(N-1)(N+1)}{12} \cdot \frac{m}{N-1} = \frac{n\,m\,(N+1)}{12}.$$

□

Theorem 14.3.3 (Equivalence with Mann–Whitney U)**.** *Define the Mann–Whitney statistic as the number of pairs (X_i, Y_j) with $X_i < Y_j$ (each tie counting $\frac{1}{2}$). Then*

$$U = W - \frac{n(n+1)}{2},$$

so W and U carry the same information; under H_0,

$$\mathbb{E}[U] = \frac{nm}{2}, \qquad \operatorname{Var}(U) = \operatorname{Var}(W) = \frac{nm(N+1)}{12}.$$

Proof. List the n X-values in increasing order; let $X_{(k)}$ have pooled rank $R_{(k)}$. Among the $R_{(k)}-1$ pooled values below $X_{(k)}$, exactly $k-1$ are other X's, so $R_{(k)}-1-(k-1)$ are Y's lying below $X_{(k)}$ — i.e. Y's that this X exceeds. Summing the count of Y's each X exceeds gives $\sum_k \big(R_{(k)}-k\big) = W - \frac{n(n+1)}{2}$, which is the number of pairs with $Y_j < X_i$. Calling that U' and noting $U + U' = nm$ recovers the stated relation; only the constant shift $\frac{n(n+1)}{2}$ separates W from U, so the variance is unchanged and $\mathbb{E}[U] = \frac{n(N+1)}{2} - \frac{n(n+1)}{2} = \frac{nm}{2}$. □

Theorem 14.3.4 (Large-sample normal approximation)**.** *For n, m both moderately large (a common rule of thumb is each at least about 10), under H_0 the standardized rank sum is approximately standard normal:*

$$Z = \frac{W - \mu_W}{\sigma_W} = \frac{W - \dfrac{n(N+1)}{2}}{\sqrt{\dfrac{nm(N+1)}{12}}} \;\dot{\sim}\; \mathrm{N}(0,1).$$

*A **continuity correction** of $\pm\frac{1}{2}$ improves the approximation: replace $W-\mu_W$ by $W-\mu_W+\frac{1}{2}$ when $W < \mu_W$ and by $W-\mu_W-\frac{1}{2}$ when $W > \mu_W$ (shrink the numerator's magnitude by $\frac{1}{2}$).*

Remark (Tie correction for the variance). With ties, the average-rank device keeps μ_W exactly right but slightly shrinks σ_W^2. If the distinct tied values form groups of sizes $t_1, t_2, \ldots$, the corrected variance is

$$\sigma_W^2 = \frac{nm}{12}\left[(N+1) - \frac{\sum_j (t_j^3 - t_j)}{N(N-1)}\right].$$

When ties are few the correction is negligible and the plain formula is used.

14.3.2 The picture: interleaving of two samples

Figure 14.1 is the whole test in one glance. The pooled values sit on a number line, each marked by its group symbol; reading left to right gives the ranks. If one group's symbols cluster at the low end and the other's at the high end, the rank sums are far apart and the test rejects. If the two symbols interleave evenly, the rank sums sit near their common expectation and the test retains.

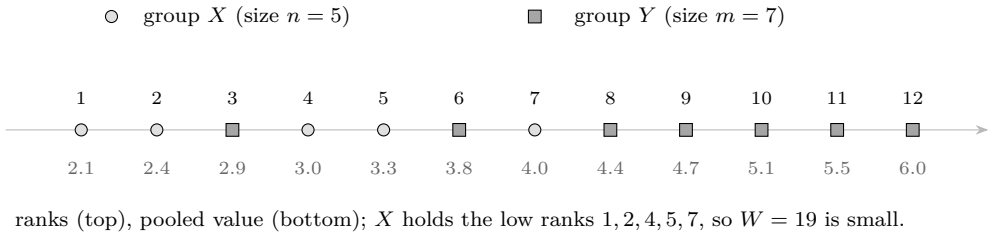


Figure 14.1: Pooling and ranking two independent samples. The circles (X, the smaller group) crowd the low end; the squares (Y) dominate the high end. The smaller group's rank sum $W = 1+2+4+5+7 = 19$ falls below its null mean $\mu_W = \frac{5\cdot 13}{2} = 32.5$, the signature of an X distribution shifted low. Even interleaving would push W toward 32.5.

14.3.3 A full rank-sum test, start to finish

Example 14.3.5 (Pool, rank, sum, decide: two soil treatments)**.** An agronomist compares seedling height (cm) after four weeks under a standard fertilizer (X) and an experimental one (Y). The data, deliberately small and a bit skewed, are

$$\begin{aligned} X \text{ (standard, } n=5): &\quad 8.1,\ 9.4,\ 7.6,\ 10.2,\ 8.8, \\ Y \text{ (experimental, } m=6): &\quad 11.5,\ 9.9,\ 12.7,\ 10.6,\ 11.1,\ 13.4. \end{aligned}$$

Test at $\alpha = 0.05$ (two-sided) whether the two fertilizers produce differently located height distributions. The lower 5% two-sided critical value for W (smaller group $n = 5$, larger $m = 6$) is $W_L = 19$, with upper point $W_U = n(N+1) - W_L = 5(12) - 19 = 41$.

Which rule, and why? Two *independent* groups, small ($n = 5, m = 6$) and not obviously normal, comparing *location* — the textbook setting for the rank-sum test, not the two-sample t. We pool, rank, and sum the smaller group's ranks.

Solution.

Step 1: pool and rank. Sort all $N = 11$ values and rank them; Table 14.1 records group and rank.

Table 14.1: Pooled values, group, and rank for the fertilizer data ($N = 11$).

value	group	rank	value	group	rank
7.6	X	1	10.2	X	6
8.1	X	2	10.6	Y	7
8.8	X	3	11.1	Y	8
9.4	X	4	11.5	Y	9
9.9	Y	5	12.7	Y	10
			13.4	Y	11

Step 2: sum the smaller group's ranks. The X ranks are $1, 2, 3, 4, 6$:

$$W = 1 + 2 + 3 + 4 + 6 = 16.$$

Step 3: null mean for orientation.

$$\mu_W = \frac{n(N+1)}{2} = \frac{5(12)}{2} = 30, \qquad \sigma_W^2 = \frac{nm(N+1)}{12} = \frac{5 \cdot 6 \cdot 12}{12} = 30.$$

So $W = 16$ sits well below $\mu_W = 30$: the standard group ranks low.

Step 4: decide against the table. Reject if $W \le W_L = 19$ or $W \ge W_U = 41$. Here $W = 16 \le 19$, so reject.

$$\boxed{W = 16 \le 19 \;\Rightarrow\; \text{reject } H_0;\ \text{the experimental fertilizer gives systematically taller seedlings.}}$$

Example 14.3.6 (The Mann–Whitney U on the same data)**.** For the fertilizer data above ($W = 16$, $n = 5$, $m = 6$), compute the Mann–Whitney U two ways — from the formula $U = W - \frac{n(n+1)}{2}$ and by directly counting pairs — and confirm they agree.

Which rule, and why? U and W are the same test; the formula converts one to the other by subtracting the minimum possible rank sum $\frac{n(n+1)}{2}$. Counting "how many Y's each X beats" is the definition of U and a useful check.

Solution.

Step 1: from the formula.

$$U = W - \frac{n(n+1)}{2} = 16 - \frac{5 \cdot 6}{2} = 16 - 15 = 1.$$

Step 2: by direct count. For each X value, count the Y's it exceeds (the six Y values are 9.9, 10.6, 11.1, 11.5, 12.7, 13.4):

$$7.6,\ 8.1,\ 8.8,\ 9.4 : \text{exceed no } Y \Rightarrow 0 + 0 + 0 + 0 = 0,$$

$$10.2 : \text{exceeds only } 9.9 \Rightarrow 1.$$

Total = 1, matching the formula.

Step 3: sanity via the mean. $\mathbb{E}[U] = \frac{nm}{2} = \frac{30}{2} = 15$, and the observed $U = 1$ is far below it — the same "standard ranks low" message as $W = 16$ below $\mu_W = 30$.

$$\boxed{U = W - \tfrac{n(n+1)}{2} = 1 \text{ (also by direct count)};\ U \ll \mathbb{E}[U] = 15.}$$

Example 14.3.7 (A one-sided rank-sum test: does the additive raise strength?)**.** A materials lab suspects a polymer additive *increases* tensile strength (MPa). Without additive (X, $n = 6$) and with additive (Y, $m = 6$):

$$X: \quad 52,\ 47,\ 55,\ 49,\ 53,\ 50,$$
$$Y: \quad 58,\ 61,\ 54,\ 60,\ 57,\ 56.$$

Test at $\alpha = 0.05$ the one-sided claim that the additive shifts strength *up*. The one-sided lower critical value for the smaller group's rank sum ($n = 6, m = 6$) is $W_{0.05} = 28$.

Which rule, and why? Two independent groups, a directional claim, normality unverified on small samples — a one-sided rank-sum test. A genuine upward shift in Y pushes X (which we sum) toward the *low* ranks, so we reject for small W.

Solution.

Step 1: state the directional hypotheses. H_0: same distribution; H_1: the Y (additive) distribution is shifted above X. Then X tends to low ranks, so reject for small W.

Step 2: pool and rank. Sorted: $47, 49, 50, 52, 53, 55$ are X at ranks $1, 2, 3, 4, 5, 7$; 54 is Y at rank 6; $56, 57, 58, 60, 61$ are Y at ranks $8, 9, 10, 11, 12$.

Step 3: rank sum of X.

$$W = 1 + 2 + 3 + 4 + 5 + 7 = 22.$$

Step 4: decide. One-sided lower rejection: reject if $W \le 28$. Since $W = 22 \le 28$, reject.

$$\boxed{W = 22 \le 28 \ \Rightarrow \ \text{reject; the additive raises tensile strength.}}$$

Example 14.3.8 (Large-sample z with continuity correction)**.** A hospital compares length of stay (days) after two surgical protocols, $n = 12$ for protocol A (X) and $m = 14$ for protocol B. Pooling and ranking all $N = 26$ stays, the protocol-A ranks sum to $W = 120$. Test (two-sided, $\alpha = 0.05$) whether the protocols differ in length of stay, using the normal approximation with continuity correction.

Which rule, and why? With both groups past 10, the small-sample table is unwieldy; the standardized rank sum is approximately normal. We use μ_W, σ_W and a $\frac{1}{2}$ continuity correction, then compare $|z|$ to $z_{0.025} = 1.96$.

Solution.

Step 1: null moments. $N = 26$, $n = 12$, $m = 14$:

$$\mu_W = \frac{n(N+1)}{2} = \frac{12 \cdot 27}{2} = 162,$$
$$\sigma_W^2 = \frac{nm(N+1)}{12} = \frac{12 \cdot 14 \cdot 27}{12} = 14 \cdot 27 = 378, \qquad \sigma_W = \sqrt{378} = 19.44.$$

Step 2: continuity-corrected numerator. Here $W = 120 < \mu_W = 162$, so add $\frac{1}{2}$ to shrink the gap:

$$W - \mu_W + \tfrac{1}{2} = 120 - 162 + 0.5 = -41.5.$$

Step 3: standardize.

$$z = \frac{-41.5}{19.44} = -2.13.$$

Step 4: decide. Two-sided: $|z| = 2.13 > z_{0.025} = 1.96$, so reject (the p-value is $2\,\mathbb{P}(Z < -2.13) \approx 0.033$).

$$\boxed{z = -2.13,\ |z| > 1.96 \ \Rightarrow \ \text{reject; protocol } A \text{ has shorter stays.}}$$

Example 14.3.9 (Handling ties with average ranks)**.** A nutritionist scores fiber content (g) of store-brand (X, $n = 5$) and name-brand (Y, $m = 5$) cereals, and several values tie:

$$X: \quad 4,\ 6,\ 6,\ 7,\ 9,$$
$$Y: \quad 6,\ 8,\ 8,\ 10,\ 11.$$

Compute W for the store brand using average ranks for ties, and find the tie-corrected null variance.

Which rule, and why? Ties are resolved by giving every tied value the *average* of the ranks they share; this keeps μ_W exact and calls for the tie-corrected variance. The mechanics are otherwise the ordinary rank-sum computation.

Solution.

Step 1: sort and locate ties. The pooled sorted values are

$$4,\ \underbrace{6,\,6,\,6}_{\text{three tie}},\ 7,\ \underbrace{8,\,8}_{\text{two tie}},\ 9,\ 10,\ 11,$$

occupying rank slots 1 through 10.

Step 2: assign average ranks. The three 6's fill slots $2, 3, 4$, averaging to $\frac{2+3+4}{3} = 3$; the two 8's fill slots $6, 7$, averaging $\frac{6+7}{2} = 6.5$. Final ranks:

$$\begin{aligned} &4 \to 1\,(X), \quad 6 \to 3\,(X), \quad 6 \to 3\,(X), \quad 6 \to 3\,(Y), \quad 7 \to 5\,(X), \\ &8 \to 6.5\,(Y), \quad 8 \to 6.5\,(Y), \quad 9 \to 8\,(X), \quad 10 \to 9\,(Y), \quad 11 \to 10\,(Y). \end{aligned}$$

Step 3: store-brand rank sum. The X ranks are $1, 3, 3, 5, 8$:

$$W = 1 + 3 + 3 + 5 + 8 = 20.$$

Step 4: tie-corrected variance. With $N = 10$, $n = m = 5$, the tie groups have sizes $t = 3$ and $t = 2$, so $\sum(t^3 - t) = (27 - 3) + (8 - 2) = 30$:

$$\begin{aligned} \sigma_W^2 &= \frac{nm}{12}\Big[(N+1) - \frac{\sum(t^3 - t)}{N(N-1)}\Big] = \frac{25}{12}\Big[11 - \frac{30}{90}\Big] \\ &= \frac{25}{12}\,(10.667) = 22.22, \end{aligned}$$

versus the uncorrected $\frac{25}{12}(11) = 22.92$ — a small shrinkage.

$$\boxed{W = 20, \quad \mu_W = \tfrac{5 \cdot 11}{2} = 27.5, \quad \sigma_W^2 = 22.22 \text{ (tie-corrected).}}$$

Example 14.3.10 (An outlier that breaks the t but not the rank-sum test)**.** Reaction times (ms) for two display layouts, X ($n = 6$) and Y ($m = 6$):

$$\begin{aligned} X:&\ \ 210,\ 225,\ 232,\ 218,\ 240,\ 229, \\ Y:&\ \ 245,\ 260,\ 251,\ 270,\ 255,\ \mathbf{980}. \end{aligned}$$

The 980 is a logging glitch (a frozen screen). Run the rank-sum test (two-sided, $\alpha = 0.05$; for $n = m = 6$ the two-sided critical sum is $W_L = 26$, $W_U = 52$) and note how the lone outlier is neutralized.

Which rule, and why? An extreme outlier wrecks the mean and variance the two-sample t depends on, but a rank converts 980 into "the largest, rank 12" — no more influential than any other top value. This robustness is exactly why the rank-sum test is chosen here.

Solution.

Step 1: pool and rank. The six X's $(210, 218, 225, 229, 232, 240)$ take ranks $1, 2, 3, 4, 5, 6$. The Y's $245, 251, 255, 260, 270, 980$ take ranks $7, 8, 9, 10, 11, 12$.

Step 2: rank sum of X.

$$W = 1 + 2 + 3 + 4 + 5 + 6 = 21.$$

Step 3: decide. Reject if $W \le 26$ or $W \ge 52$. Since $W = 21 \le 26$, reject: layout X is faster. The 980 contributed rank 12 and nothing more.

Step 4: contrast the effect of the glitch. Had we (wrongly) used the t, the 980 would inflate $\bar{Y}$ to ≈ 343 and s_Y enormously, masking the real difference. The rank-sum verdict is unmoved whether the glitch reads 270, 980, or 9800 — it is always the maximum, rank 12.

$$\boxed{W = 21 \le 26 \ \Rightarrow\ \text{reject; the rank-sum test is immune to the outlier's magnitude.}}$$

14.3.4 The central contrast: rank-sum versus the two-sample t

Example 14.3.11 (CONTRAST: when the t is right and when it lies)**.** Consider two scenarios with the same group sizes $n = m = 8$.

(a) Clean, normal data. Battery lifetimes (h) of brands X and Y are roughly normal with no outliers; the two-sample t gives $t = 2.55$ on 14 df ($t_{0.025,14} = 2.145$, reject), and the rank-sum $z = 2.31$ (reject).

(b) Same medians, one wild outlier. A single mis-recorded value of 5000 h sits in group Y. Now the t collapses to $t = 0.74$ (retain) because the outlier explodes s_Y, while the rank-sum still gives $z = 2.28$ (reject).

Explain which test to trust in each case and state the general rule.

Which rule, and why? The two tests answer the same location question by different machinery: the t uses means and pooled SD (sensitive to shape and outliers); the rank-sum uses ranks (insensitive to both). Deciding "which to trust" is deciding whether the t's normality/outlier assumptions hold.

Solution.

Step 1: case (a) — normality holds. With clean, near-normal data the t is *valid and more powerful*: it uses the actual spacings, not just their order, so it detects a true shift slightly more readily (t rejects at 2.55, z at 2.31). Here prefer the t; the rank-sum agrees but throws away a little information (its asymptotic efficiency relative to the t under normality is $3/\pi \approx 0.955$).

Step 2: case (b) — one outlier. The single 5000-h value inflates s_Y so much that the t's denominator swamps a real numerator: $t = 0.74$, a false retain. The rank-sum is unaffected — 5000 is merely the top rank — and correctly rejects ($z = 2.28$). Here the t's assumption has failed and it *lies*; trust the rank-sum.

Step 3: the rule.

$$\begin{aligned} \text{normal, no outliers} &\Rightarrow \text{two-sample } t \text{ (more powerful)},\\ \text{skew / outliers / ordinal} &\Rightarrow \text{rank-sum (valid, robust)}. \end{aligned}$$

Prefer the t when normal; prefer rank-sum when normality fails.
Cost: only $\sim 4.5\%$ power when normal; it saves the day when not.

Remark. The efficiency number is worth remembering. Under exact normality the rank-sum test's asymptotic relative efficiency against the two-sample t is $3/\pi \approx 0.955$: it needs about 5% more data to match the t. But for heavier-tailed distributions the efficiency *exceeds* 1 — often far exceeds it — so the small insurance premium paid under normality buys large protection when normality fails. That asymmetry (lose a little when wrong, gain a lot when right) is the case for nonparametrics in one sentence.

14.3.5 Interpreting a shift, and a few more drills

Example 14.3.12 (What "reject" means: a shift, not a mean change)**.** In a rank-sum test of customer wait times at two bank branches ($n = 10$, $m = 12$), the smaller branch's rank sum is $W = 85$, giving $z = -2.40$ (reject, two-sided). The analyst writes "the mean wait differs." Critique the wording and state the correct interpretation.

Which rule, and why? The rank-sum test ranks values; its null is "same distribution" and its alternative under a location model is "one distribution is shifted." It speaks to medians and stochastic ordering, not to means, so the conclusion must be phrased accordingly.

Solution.

Step 1: what was actually tested. W depends only on the *order* of the pooled values, so rejection says the two samples are not interchangeable — one branch's waits tend to be *smaller* (lower ranks) than the other's. Under the usual shift model this is a difference in **median**.

Step 2: why "mean" is wrong. The test never used the numerical waits, only their ranks; it cannot isolate a mean. Two distributions can have equal means yet differ in shape, and the rank-sum can flag that. "Mean" over-claims.

Step 3: correct wording. "The wait-time *distributions* differ; one branch's waits are stochastically shorter (smaller median)."

$$\boxed{z = -2.40 \;\Rightarrow\; \text{the distributions differ in location (median/shift), not provably the means.}}$$

Example 14.3.13 (Two-sided test from U directly)**.** In a taste test, $n = 7$ panelists rate dessert X and a separate $m = 8$ rate dessert Y on a 100-point scale. Counting pairs, dessert X's ratings exceed dessert Y's in $U = 14$ of the $nm = 56$ cross-pairs. Test (two-sided, $\alpha = 0.05$) whether the desserts are rated differently, using the normal approximation on U.

Which rule, and why? U has the same null distribution as W up to a shift, so we standardize U with $\mathbb{E}[U] = \frac{nm}{2}$ and the shared variance. This is convenient when the data arrive as a win-count rather than a rank sum.

Solution.

Step 1: null moments of U. $N = 15$, $n = 7$, $m = 8$:

$$\mathbb{E}[U] = \frac{nm}{2} = \frac{56}{2} = 28,$$
$$\mathrm{Var}(U) = \frac{nm(N+1)}{12} = \frac{56 \cdot 16}{12} = 74.67, \qquad \mathrm{SD} = \sqrt{74.67} = 8.64.$$

Step 2: continuity-corrected z. $U = 14 < 28$, so add $\frac{1}{2}$:

$$z = \frac{14 - 28 + 0.5}{8.64} = \frac{-13.5}{8.64} = -1.56.$$

Step 3: decide. Two-sided: $|z| = 1.56 < z_{0.025} = 1.96$, so do not reject (the p-value is $2\,\mathbb{P}(Z < -1.56) \approx 0.12$).

$$\boxed{z = -1.56,\ |z| < 1.96 \;\Rightarrow\; \text{retain; no significant rating difference.}}$$

Example 14.3.14 (Ordinal data where no mean even exists)**.** Reviewers grade two manuscripts on the ordinal scale weak < fair < good < strong < outstanding. Manuscript X ($n = 5$) gets fair, good, good, strong, outstanding; manuscript Y ($m = 5$) gets weak, fair, fair, good, strong. Is X rated higher (one-sided, $\alpha = 0.05$; lower one-sided critical sum for $n = m = 5$ is $W_{0.05} = 19$, and by symmetry the upper is $n(N+1) - 19 = 5 \cdot 11 - 19 = 36$)?

Which rule, and why? The grades are **ordinal** — there is no numeric "mean grade," so the t is meaningless and only a rank method applies. We assign pooled ranks (averaging ties) and sum X's ranks; "X higher" means X takes the *high* ranks, so reject for large W.

Solution.

Step 1: pool and rank the ten grades. Counts: weak 1, fair 3, good 3, strong 2, outstanding 1. Average ranks within each level:

$$\text{weak (slot 1)} \to 1, \qquad \text{fair (slots 2–4)} \to 3, \qquad \text{good (slots 5–7)} \to 6,$$
$$\text{strong (slots 8–9)} \to 8.5, \qquad \text{outstanding (slot 10)} \to 10.$$

Step 2: rank sum of X. X = fair, good, good, strong, outstanding $\to$ ranks $3, 6, 6, 8.5, 10$:

$$W = 3 + 6 + 6 + 8.5 + 10 = 33.5.$$

Step 3: decide (upper tail). "X higher" rejects for large W; the upper one-sided critical value is 36. Since $W = 33.5 < 36$, do *not* reject.

$$\boxed{W = 33.5 < 36 \;\Rightarrow\; \text{retain; } X \text{ is not significantly higher rated (close, but short).}}$$

Example 14.3.15 (Confidence statement via the Hodges–Lehmann shift)**.** For two assembly-line fill weights (g), the rank-sum test on $n = 8$, $m = 8$ rejects with the line-X weights ranking low. As a companion to the test, the analyst forms all $nm = 64$ pairwise differences $Y_j - X_i$ and reports their *median*, $\widehat{\Delta} = 2.6$ g. Explain what this number estimates and how it complements the test.

Which rule, and why? A rank-sum rejection says "shifted" but not "by how much." The **Hodges–Lehmann** estimator — the median of all pairwise differences $Y_j - X_i$ — is the natural point estimate of the shift Δ associated with the rank-sum test, the nonparametric analogue of $\overline{Y} - \overline{X}$.

Solution.

Step 1: what $\widehat{\Delta}$ estimates. Under the location-shift model $Y \stackrel{d}{=} X + \Delta$, the median of the nm differences $Y_j - X_i$ is a consistent, outlier-robust estimate of Δ. Here $\widehat{\Delta} = 2.6$ g says line Y runs about 2.6 g heavier.

Step 2: how it complements the test. The test delivers the *yes/no* (distributions differ); $\widehat{\Delta}$ delivers the *magnitude* on the original scale, so the analyst can judge practical importance — a 2.6 g shift may or may not matter for a 500 g target even when statistically clear.

Step 3: matching interval. Inverting the rank-sum test (the ranks of the ordered pairwise differences cut off by the table value) yields a distribution-free confidence interval for Δ centered near $\widehat{\Delta}$ — the rank-sum twin of the t-interval for $\mu_Y - \mu_X$.

$$\boxed{\widehat{\Delta} = \text{median}\{Y_j - X_i\} = 2.6 \text{ g estimates the shift the test detected.}}$$

Example 14.3.16 (Recognition drill: rank-sum, paired, or two-sample t?)**.** For each study pick the right two-sample procedure. (a) "9 patients' tumor sizes before and after therapy, same patients." (b) "Survival days for 7 mice on diet A and 8 *other* mice on diet B, heavily right-skewed with one very long survivor." (c) "Cholesterol (mg/dL) for 40 men and 40 women, both bell-shaped, no outliers." (d) "Pain relief rated none/mild/moderate/complete for 12 patients on drug A and 12 on drug B."

Which rule, and why? Three forks: same/matched units ⇒ a paired/signed-rank test; independent groups with normality ⇒ two-sample t; independent groups with skew, outliers, or ordinal data ⇒ rank-sum. Read the design and the data shape.

Solution.

Step 1: (a) same patients. Before/after on the *same* patient is paired — **not** rank-sum. Use the paired t (or Wilcoxon signed-rank if non-normal).

Step 2: (b) skewed, outlier, independent. Two disjoint mouse groups, heavy skew, a giant survivor ⇒ **rank-sum test**; the t would be wrecked by the long survivor.

Step 3: (c) normal, independent. Two independent, bell-shaped, clean samples ⇒ the **two-sample** t (more powerful here; rank-sum would be valid but slightly weaker).

Step 4: (d) ordinal, independent. The four-level relief scale is ordinal with no mean ⇒ **rank-sum** on the ranked grades.

$$\boxed{\text{(a) paired, (b) rank-sum, (c) two-sample } t\text{, (d) rank-sum.}}$$

14.4 The Kruskal–Wallis Test

When you have $k \geq 3$ independent samples and want to ask whether they were drawn from the same distribution — the question one-way ANOVA answers with its F-test — but normality is doubtful, the **Kruskal–Wallis test** is the nonparametric replacement. Its idea is the same trick that powers every rank method: pool all $N = \sum_j n_j$ observations into one list, replace each value by its **rank** in that pooled list (averaging tied ranks), and then ask whether the ranks are spread evenly across the groups. If the groups really share one distribution, each group's average rank should sit near the overall average $(N + 1)/2$; if one group is systematically larger, its observations win high ranks and its rank sum swells. The statistic H measures exactly that swelling and, under the null, is approximately χ^2_{k-1}, so we reject in the *upper* tail. The

test is about whether the k distributions (loosely, their medians) coincide, never about means, and when $k = 2$ it collapses to the Wilcoxon rank-sum test of §3.

Which k-sample test does the problem need?

- $k \geq 3$ **independent** samples, comparing locations, normality doubtful or data ordinal? ⇒ **Kruskal–Wallis** on the pooled ranks, reference χ^2_{k-1}.
- Exactly $k = 2$ independent samples? ⇒ the rank-sum test of §3; Kruskal–Wallis reduces to it (and $H = Z^2$).
- $k \geq 3$ samples, data genuinely **normal** with equal variances? ⇒ one-way ANOVA is valid and a touch more powerful; save Kruskal–Wallis for when the shape assumption fails.
- k treatments measured on the *same* blocks (matched / repeated)? ⇒ not this test — the samples are not independent; use the **Friedman test** (§5).

Do not confuse the divisors: the rank sum R_j is divided by n_j *inside* the $\sum R_j^2/n_j$, and the whole statistic is scaled by $12/[N(N+1)]$ with $3(N+1)$ subtracted. Forgetting to divide each R_j^2 by its own group size — fatal in unbalanced designs — is the classic Kruskal–Wallis slip.

14.4.1 Pooled ranks and the rank sums

Definition 14.4.1 (Pooled ranks, group rank sums, and average ranks)**.** Let k independent samples have sizes $n_1, \ldots, n_k$ and total $N = \sum_{j=1}^{k} n_j$. Pool all N observations into a single list and assign **ranks** $1, 2, \ldots, N$ from smallest to largest; ties receive the *average* of the ranks they would occupy. For group j let

$$R_j = \sum_{i=1}^{n_j} \operatorname{rank}(Y_{ij}) \quad \text{(rank sum of group } j\text{)},$$
$$\overline{R}_j = \frac{R_j}{n_j} \quad \text{(average rank in group } j\text{)}.$$

Because the pooled ranks are a permutation of $1, \ldots, N$, they sum to $N(N+1)/2$, so $\sum_{j=1}^{k} R_j = N(N+1)/2$ and the overall average rank is $(N+1)/2$. This total is a free arithmetic check on every ranking table.

Remark. The logic mirrors ANOVA exactly, with ranks in place of raw values. ANOVA splits the spread of the Y_{ij} into between- and within-group pieces; Kruskal–Wallis looks only at how far each group's *average rank* $\overline{R}_j$ strays from the common average $(N+1)/2$. If the k distributions coincide, every $\overline{R}_j \approx (N+1)/2$ and the strays are small; if a group is shifted up, its ranks — and thus $\overline{R}_j$ — climb. The statistic below is, up to constants, the rank analogue of the between-group sum of squares $\sum_j n_j(\overline{R}_j - \bar{R})^2$, standardized by the known variance of ranks so that it has a χ^2 reference free of the parent shape.

14.4.2 The Kruskal–Wallis statistic

Theorem 14.4.2 (The Kruskal–Wallis statistic and its null distribution)**.** *With pooled ranks and group rank sums R_j as above, define*

$$H = \frac{12}{N(N+1)} \sum_{j=1}^{k} \frac{R_j^2}{n_j} - 3(N+1).$$

Under H_0 that the k samples come from the same continuous distribution, for moderate group sizes H is approximately χ^2_{k-1}. The test rejects H_0 at level α when

$$H > \chi^2_{\alpha,\, k-1},$$

an upper-tail *test, since any group whose location is shifted enlarges some R_j and pushes H up.*

Why the reference is χ^2_{k-1} (sketch). The pooled ranks $1, \dots, N$ have mean $(N+1)/2$ and variance $(N^2-1)/12$. Under H_0 each group is a random subset of those ranks, so $\mathbb{E}[R_j] = n_j(N+1)/2$, and an equivalent form of H is the weighted sum of squared standardized strays

$$H = \sum_{j=1}^{k} \frac{\big(R_j - n_j(N+1)/2\big)^2}{n_j\,(N^2-1)/12} \cdot \frac{N-1}{N}.$$

Each term is a standardized squared deviation; the k rank sums satisfy one linear constraint ($\sum_j R_j$ is fixed), leaving $k-1$ free, so the sum is asymptotically χ^2 with $k-1$ degrees of freedom. (For small n_j the exact null distribution is tabulated; the χ^2 approximation is the large-sample working tool.) □

Remark (Tie correction, conceptually). When ties force average ranks, the pooled ranks no longer have variance exactly $(N^2-1)/12$ — ties shrink it slightly — so H is divided by a correction factor that is at most 1:

$$H_{\text{corr}} = \frac{H}{1 - \dfrac{\sum_g (t_g^3 - t_g)}{N^3 - N}},$$

where t_g is the number of observations sharing the gth tied value (a singleton has $t_g = 1$, contributing 0). The factor is 1 when there are no ties and dips below 1 as ties multiply, so dividing *inflates* H a little to compensate for the lost spread. With only a handful of ties the adjustment is negligible and is routinely skipped; report it when ties are heavy.

14.4.3 Carrying out the test

Example 14.4.3 (Full Kruskal–Wallis test: three soils and seedling growth)**.** A botanist grows ryegrass seedlings in three soil mixes and records 10-day height (mm). The samples are independent with $n_1 = n_2 = n_3 = 4$:

$$\begin{aligned} &\text{Mix A}: \ 18,\ 22,\ 25,\ 30,\\ &\text{Mix B}: \ 35,\ 40,\ 28,\ 45,\\ &\text{Mix C}: \ 12,\ 16,\ 20,\ 24. \end{aligned}$$

Test at $\alpha = 0.05$ whether the three mixes give the same height distribution ($\chi^2_{0.05,2} = 5.991$).

Which rule, and why? Three *independent* samples and a question about whether the distributions agree, with no normality stated and small n_j, point to Kruskal–Wallis: pool, rank, form the group rank sums, and compute H against $\chi^2_{k-1} = \chi^2_2$. The means-based ANOVA F is not warranted here.

Solution.

Step 1: pool and rank all $N = 12$ values. Sort the 12 heights and attach ranks $1, \dots, 12$ (no ties):

value	12	16	18	20	22	24	25	28	30	35	40	45
rank	1	2	3	4	5	6	7	8	9	10	11	12
group	C	C	A	C	A	C	A	B	A	B	B	B

Step 2: group rank sums.

$$\begin{aligned} R_A &= 3 + 5 + 7 + 9 = 24,\\ R_B &= 8 + 10 + 11 + 12 = 41,\\ R_C &= 1 + 2 + 4 + 6 = 13. \end{aligned}$$

Check: $24 + 41 + 13 = 78 = N(N+1)/2 = 12 \cdot 13/2$. ■

Step 3: the statistic. With $n_j = 4$ throughout,

$$\sum_j \frac{R_j^2}{n_j} = \frac{24^2}{4} + \frac{41^2}{4} + \frac{13^2}{4} = \frac{576 + 1681 + 169}{4} = \frac{2426}{4} = 606.5,$$

$$H = \frac{12}{12 \cdot 13}\,(606.5) - 3(13) = \frac{12}{156}(606.5) - 39 = 46.654 - 39 = 7.654.$$

Step 4: degrees of freedom and decision. Reference $\chi^2_{k-1} = \chi^2_2$.

$$\boxed{H = 7.654 > \chi^2_{0.05,2} = 5.991 \;\Rightarrow\; \text{reject } H_0;\ \text{the soil mixes differ.}}$$

The high-ranked observations cluster in Mix B and the low ones in Mix C, which is what drove H past the cutoff.

Example 14.4.4 (Unbalanced design: four diet groups, unequal sizes)**.** A nutritionist measures weight gain (g) of lab mice on four diets with *unequal* samples, $n_1 = 5,\ n_2 = 3,\ n_3 = 4,\ n_4 = 3$ ($N = 15$):

$$\begin{aligned}
&\text{Diet 1}: \ 6,\ 9,\ 11,\ 14,\ 20,\\
&\text{Diet 2}: \ 7,\ 12,\ 17,\\
&\text{Diet 3}: \ 8,\ 13,\ 18,\ 22,\\
&\text{Diet 4}: \ 10,\ 15,\ 19.
\end{aligned}$$

Test at $\alpha = 0.05$ whether the four diets share one distribution ($\chi^2_{0.05,3} = 7.815$).

Which rule, and why? Unequal group sizes do not change the test — they change only the bookkeeping: each R_j^2 must be divided by its *own* n_j. This is precisely where dividing by a common n would corrupt the answer, so Kruskal–Wallis (not ANOVA, not equal-n shortcuts) with careful per-group division is the method.

Solution.

Step 1: pooled ranks. The 15 values are distinct; sort and rank:

value	6	7	8	9	10	11	12	13	14	15	17	18	19	20	22
rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
group	1	2	3	1	4	1	2	3	1	4	2	3	4	1	3

Step 2: rank sums.

$$\begin{aligned}
R_1 &= 1 + 4 + 6 + 9 + 14 = 34,\\
R_2 &= 2 + 7 + 11 = 20,\\
R_3 &= 3 + 8 + 12 + 15 = 38,\\
R_4 &= 5 + 10 + 13 = 28.
\end{aligned}$$

Check: $34 + 20 + 38 + 28 = 120 = 15 \cdot 16/2$. ■

Step 3: the statistic, dividing by each n_j.

$$\begin{aligned}
\sum_j \frac{R_j^2}{n_j} &= \frac{34^2}{5} + \frac{20^2}{3} + \frac{38^2}{4} + \frac{28^2}{3}\\
&= \frac{1156}{5} + \frac{400}{3} + \frac{1444}{4} + \frac{784}{3} = 231.20 + 133.33 + 361.00 + 261.33 = 986.86,\\
H &= \frac{12}{15 \cdot 16}\,(986.86) - 3(16) = \frac{12}{240}(986.86) - 48 = 49.343 - 48 = 1.343.
\end{aligned}$$

Step 4: decision. Reference χ^2_3.

$$\boxed{H = 1.343 < \chi^2_{0.05,3} = 7.815 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{no diet effect shown.}}$$

The diets interleave evenly in the pooled ranking, so H lands deep in the retain region.

Example 14.4.5 (Handling ties with average ranks: three call-center scripts)**.** A manager compares customer-satisfaction scores (1–10 scale, hence many ties) for three call scripts, $n_1 = n_2 = n_3 = 5$ ($N = 15$):

$$\begin{aligned}
&\text{Script X}: \ 6,\ 7,\ 7,\ 8,\ 9,\\
&\text{Script Y}: \ 4,\ 5,\ 6,\ 6,\ 7,\\
&\text{Script Z}: \ 3,\ 4,\ 5,\ 6,\ 8.
\end{aligned}$$

Rank using average ranks for ties and compute H ($\chi^2_{0.05,2} = 5.991$).

Which rule, and why? Ordinal scores with repeats force **average ranks**; the procedure is still Kruskal–Wallis, but each block of tied values shares the mean of the ranks it spans. Average ranks keep the rank sum at $N(N+1)/2$ and let the same H formula apply.

Solution.

Step 1: sort and assign average ranks. The sorted pooled values and the rank positions they occupy:

value	3	4	5	6	7	8
positions	1	2–3	4–5	6–9	10–12	13–14
avg rank	1	2.5	4.5	7.5	11	13.5
value	9					
positions	15					
avg rank	15					

For example the four 6's occupy positions $6, 7, 8, 9$, averaging $(6+7+8+9)/4 = 7.5$.

Step 2: rank sums by group.

$$\begin{aligned} R_X &= 7.5 + 11 + 11 + 13.5 + 15 = 58.0 \quad (6, 7, 7, 8, 9), \\ R_Y &= 2.5 + 4.5 + 7.5 + 7.5 + 11 = 33.0 \quad (4, 5, 6, 6, 7), \\ R_Z &= 1 + 2.5 + 4.5 + 7.5 + 13.5 = 29.0 \quad (3, 4, 5, 6, 8). \end{aligned}$$

Check: $58 + 33 + 29 = 120 = 15 \cdot 16/2$. ■

Step 3: statistic (ties light, correction skipped).

$$\sum_j \frac{R_j^2}{n_j} = \frac{58^2 + 33^2 + 29^2}{5} = \frac{3364 + 1089 + 841}{5} = \frac{5294}{5} = 1058.8,$$

$$H = \frac{12}{15 \cdot 16}(1058.8) - 3(16) = 0.05(1058.8) - 48 = 52.94 - 48 = 4.94.$$

Step 4: decision.

$$\boxed{H = 4.94 < \chi^2_{0.05,2} = 5.991 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Script X scores highest, but with so few cases the rank evidence is not decisive at 0.05.

Example 14.4.6 (Applying the tie correction explicitly)**.** Take the call-center data above. Apply the tie correction factor and report the adjusted H.

Which rule, and why? Whenever you want the exact H in the presence of ties, divide the raw statistic by $1 - \sum_g (t_g^3 - t_g)/(N^3 - N)$. Here the ties are mild, so we expect a tiny upward nudge — a good check that the correction behaves as advertised.

Solution.

Step 1: count the tied blocks. Tied group sizes: two 4's ($t = 2$), two 5's ($t = 2$), four 6's ($t = 4$), three 7's ($t = 3$), two 8's ($t = 2$). Singletons (3 and 9) contribute 0.

Step 2: the correction sum.

$$\begin{aligned} \sum_g (t_g^3 - t_g) &= (2^3 - 2) + (2^3 - 2) + (4^3 - 4) + (3^3 - 3) + (2^3 - 2) \\ &= 6 + 6 + 60 + 24 + 6 = 102, \\ N^3 - N &= 15^3 - 15 = 3375 - 15 = 3360. \end{aligned}$$

Step 3: corrected statistic.

$$C = 1 - \frac{102}{3360} = 1 - 0.03036 = 0.96964,$$

$$H_{\text{corr}} = \frac{4.94}{0.96964} = 5.095.$$

Step 4: decision unchanged.

$$\boxed{H_{\text{corr}} = 5.095 < 5.991 \;\Rightarrow\; \text{still fail to reject; the correction raised } H \text{ by about } 0.16.}$$

The factor is just under 1, so it inflates H slightly; with mild ties it rarely changes the conclusion.

Example 14.4.7 (Reading $\chi^2_{\alpha,k-1}$ off the table)**.** A study with $k = 6$ independent groups yields $H = 12.8$. State the degrees of freedom and decide at both $\alpha = 0.05$ and $\alpha = 0.01$, given $\chi^2_{0.05,5} = 11.070$ and $\chi^2_{0.01,5} = 15.086$.

Which rule, and why? The reference is always χ^2 with $k-1$ degrees of freedom — one fewer than the number of *groups*, exactly like the treatment df in ANOVA — and the test is upper-tail, so we compare H to the right-tail critical value at each level.

Solution.

Step 1: degrees of freedom. $k = 6$ groups give $\text{df} = k - 1 = 5$.

Step 2: compare at $\alpha = 0.05$.

$$H = 12.8 > \chi^2_{0.05,5} = 11.070 \;\Rightarrow\; \text{reject } H_0.$$

Step 3: compare at $\alpha = 0.01$.

$$H = 12.8 < \chi^2_{0.01,5} = 15.086 \;\Rightarrow\; \text{fail to reject } H_0.$$

$$\boxed{\text{df} = 5; \;\; \text{reject at } 0.05, \;\text{retain at } 0.01.}$$

The result is significant at the 5% level but not at the stricter 1% level — moderate, not overwhelming, evidence.

Example 14.4.8 (Large-N chi-square approximation: five city air-quality stations)**.** Daily $PM_{2.5}$ readings from $k = 5$ monitoring stations have rank sums (computed from pooled ranks over $N = 50$ readings, $n_j = 10$ each):

$$R_1 = 210, \quad R_2 = 300, \quad R_3 = 255, \quad R_4 = 180, \quad R_5 = 330.$$

With $N = 50$ moderately large, use the χ^2_4 approximation to test at $\alpha = 0.05$ ($\chi^2_{0.05,4} = 9.488$).

Which rule, and why? For moderate-to-large N the exact null distribution of H is well approximated by χ^2_{k-1}, so we simply compute H and read the χ^2_4 cutoff — no special-purpose small-sample table is needed once each n_j is, say, 5 or more.

Solution.

Step 1: confirm the rank-sum total. $210 + 300 + 255 + 180 + 330 = 1275 = N(N+1)/2 = 50 \cdot 51/2$. ■

Step 2: the statistic. With $n_j = 10$,

$$\begin{aligned}
\sum_j \frac{R_j^2}{n_j} &= \frac{210^2 + 300^2 + 255^2 + 180^2 + 330^2}{10} \\
&= \frac{44100 + 90000 + 65025 + 32400 + 108900}{10} = \frac{340425}{10} = 34042.5, \\
H &= \frac{12}{50 \cdot 51}(34042.5) - 3(51) = \frac{12}{2550}(34042.5) - 153 = 160.20 - 153 = 7.20.
\end{aligned}$$

Step 3: decision on χ^2_4.

$$\boxed{H = 7.20 < \chi^2_{0.05,4} = 9.488 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Despite station 5's high rank sum, the spread across stations is within χ^2_4 sampling variation; the approximation is trustworthy because every $n_j = 10$.

Example 14.4.9 (One shifted group drives the test: three pitching machines)**.** A coach measures pitch speed (mph) from $k = 3$ machines, $n_j = 4$ each ($N = 12$); machine C is suspected of running fast:

$$\begin{aligned}\text{Machine A}: &\ 70,\ 72,\ 74,\ 76,\\ \text{Machine B}: &\ 71,\ 73,\ 75,\ 77,\\ \text{Machine C}: &\ 82,\ 84,\ 86,\ 88.\end{aligned}$$

Test at $\alpha = 0.05$ whether the three machines share one speed distribution ($\chi^2_{0.05,2} = 5.991$).

Which rule, and why? Three independent samples and an equal-distribution question is Kruskal–Wallis on χ^2_2. The interest is whether *one* group (C) is shifted up; the pooled ranking exposes that without assuming normality.

Solution.

Step 1: pooled ranks. Sorting all 12 speeds, machines A and B interleave in the lower eight ranks and C takes the top four:

$$\begin{aligned}\text{A}: &\ 70, 72, 74, 76 \to \text{ ranks } 1, 3, 5, 7,\\ \text{B}: &\ 71, 73, 75, 77 \to \text{ ranks } 2, 4, 6, 8,\\ \text{C}: &\ 82, 84, 86, 88 \to \text{ ranks } 9, 10, 11, 12.\end{aligned}$$

Step 2: rank sums.

$R_A = 1+3+5+7 = 16, \quad R_B = 2+4+6+8 = 20, \quad R_C = 9+10+11+12 = 42; \quad 16+20+42 = 78.$ ■

Step 3: the statistic.

$$\sum_j \frac{R_j^2}{n_j} = \frac{16^2 + 20^2 + 42^2}{4} = \frac{256 + 400 + 1764}{4} = \frac{2420}{4} = 605.0,$$

$$H = \frac{12}{12 \cdot 13}(605.0) - 3(13) = \frac{12}{156}(605.0) - 39 = 46.538 - 39 = 7.538.$$

Step 4: decision.

$$\boxed{H = 7.538 > \chi^2_{0.05,2} = 5.991 \ \Rightarrow\ \text{reject } H_0;\ \text{machine C runs fast.}}$$

A single shifted group is enough: C monopolizing the top ranks alone pushes H past the cutoff, even though A and B are indistinguishable.

Example 14.4.10 (A significant large-N result and which groups differ)**.** Reaction times (ms) of drivers under $k = 4$ distraction conditions give, over $N = 40$ observations ($n_j = 10$ each), rank sums

$$R_1 = 140, \quad R_2 = 320, \quad R_3 = 210, \quad R_4 = 150.$$

Test at $\alpha = 0.05$ ($\chi^2_{0.05,3} = 7.815$); if significant, indicate *which* condition stands apart using average ranks.

Which rule, and why? Kruskal–Wallis is an **omnibus** test: a significant H says "not all distributions are equal" but not which pair differs. The follow-up is a rank-based multiple comparison — contrast the group average ranks $\overline{R}_j$ — only *after* H rejects, just as Tukey follows a significant ANOVA F.

Solution.

Step 1: the omnibus statistic.

$$\sum_j \frac{R_j^2}{n_j} = \frac{140^2 + 320^2 + 210^2 + 150^2}{10} = \frac{19600 + 102400 + 44100 + 22500}{10} = \frac{188600}{10} = 18860,$$

$$H = \frac{12}{40 \cdot 41}(18860) - 3(41) = \frac{12}{1640}(18860) - 123 = 138.0 - 123 = 15.0.$$

Since $15.0 > 7.815$, reject H_0: the conditions are not all alike.

Step 2: group average ranks.

$$\overline{R}_1 = 14.0, \quad \overline{R}_2 = 32.0, \quad \overline{R}_3 = 21.0, \quad \overline{R}_4 = 15.0,$$

against the overall average $(N+1)/2 = 20.5$.

Step 3: rank-based comparison. The large-sample yardstick declares conditions i and j different at family level α when

$$|\overline{R}_i - \overline{R}_j| > z_{\alpha/[k(k-1)]}\sqrt{\frac{N(N+1)}{12}\Big(\frac{1}{n_i}+\frac{1}{n_j}\Big)}.$$

Here $\sqrt{\frac{40\cdot 41}{12}\cdot\frac{2}{10}} = \sqrt{27.33} = 5.23$, and with $k=4$ the Bonferroni z is $z_{0.05/12} = z_{0.00417} = 2.64$, giving a critical gap of $2.64 \cdot 5.23 = 13.8$.

$$\boxed{|\overline{R}_2 - \overline{R}_1| = 18.0 > 13.8 \text{ and } |\overline{R}_2 - \overline{R}_4| = 17.0 > 13.8: \text{ condition 2 stands apart.}}$$

Only condition 2 (the highest reaction times) differs significantly from the two lowest; the omnibus H flagged a difference, and the comparison localizes it.

Example 14.4.11 (CONTRAST: Kruskal–Wallis vs. one-way ANOVA on skewed data)**.** An economist has income (in \$1000s) for $k=3$ regions, $n_j = 5$ each; the data are sharply right-skewed with one large value per region:

Region A : 30, 35, 40, 45, 300,
Region B : 32, 38, 41, 47, 60,
Region C : 28, 33, 39, 44, 90.

Compare what ANOVA's F and Kruskal–Wallis's H would conclude, and say which test to trust here.

Which rule, and why? This contrasts the two k-sample tests directly. ANOVA's F uses the raw values, so the lone \$300k outlier in Region A inflates that mean and the within-group variance, wrecking the normal-theory F. Kruskal–Wallis uses only ranks, so an extreme value counts merely as "the largest" — robust to the skew. With non-normal, outlier-laden data, trust H.

Solution.

Step 1: why ANOVA misleads. Region A's mean is dragged to $(30+35+40+45+300)/5 = 90$ by the outlier, while its within-group variance explodes; the F-ratio becomes unstable and its F reference invalid because normality fails badly.

Step 2: the ranks tame the outlier. Pooling and ranking, the three biggest values $(300, 90, 60)$ take ranks $15, 14, 13$ regardless of how extreme they are. The rank sums work out to $R_A = 42$, $R_B = 39$, $R_C = 39$ (the outlier in A only buys rank 15, not a mean of 90), so

$$\sum_j \frac{R_j^2}{n_j} = \frac{42^2+39^2+39^2}{5} = \frac{1764+1521+1521}{5} = \frac{4806}{5} = 961.2,$$

$$H = \frac{12}{15\cdot 16}(961.2) - 48 = 0.05(961.2) - 48 = 48.06 - 48 = 0.06.$$

Step 3: the verdict. $H = 0.06 \ll \chi^2_{0.05,2} = 5.991$: no regional difference.

$$\boxed{\text{Skew + outliers} \Rightarrow \text{ ANOVA } F \text{ unreliable; trust } H = 0.06 \text{ (retain } H_0).}$$

When to prefer each: if the data really are normal with equal variances, ANOVA is slightly more powerful (it uses the magnitudes, not just the order) and should be used; when normality is doubtful or outliers dominate, Kruskal–Wallis stays valid while ANOVA does not. The price of robustness is a modest loss of power in the ideal normal case.

Example 14.4.12 (CONTRAST: the $k = 2$ reduction to the rank-sum test)**.** Two production lines yield part lifetimes (h), $n_1 = 4,\ n_2 = 4$ ($N = 8$):

$$\text{Line 1}: \ 12,\ 15,\ 18,\ 25; \qquad \text{Line 2}: \ 10,\ 14,\ 20,\ 28.$$

Compute H, and show that it equals the square of the rank-sum test's standardized statistic Z, confirming Kruskal–Wallis reduces to rank-sum when $k = 2$.

Which rule, and why? With only two groups, the chi-square test on 1 df is the squared two-sided normal test: $H = Z^2$ where Z is the standardized Wilcoxon rank sum. This contrast shows Kruskal–Wallis is the multi-group generalization of §3, losing no information at $k = 2$.

Solution.

Step 1: pooled ranks and rank sums. Sorted: $10, 12, 14, 15, 18, 20, 25, 28$ with ranks 1–8; Line 1 holds $12, 15, 18, 25 \to$ ranks $2, 4, 5, 7$ and Line 2 holds $10, 14, 20, 28 \to$ ranks $1, 3, 6, 8$.

$$R_1 = 2 + 4 + 5 + 7 = 18, \qquad R_2 = 1 + 3 + 6 + 8 = 18, \qquad 18 + 18 = 36 = 8 \cdot 9/2. \ \blacksquare$$

Step 2: Kruskal–Wallis statistic.

$$\sum_j \frac{R_j^2}{n_j} = \frac{18^2 + 18^2}{4} = \frac{324 + 324}{4} = 162,$$

$$H = \frac{12}{8 \cdot 9}(162) - 3(9) = \frac{12}{72}(162) - 27 = 27 - 27 = 0.$$

Step 3: the rank-sum Z. With $\mathbb{E}[R_1] = n_1(N + 1)/2 = 4 \cdot 9/2 = 18$ and $\mathrm{Var}(R_1) = n_1 n_2 (N + 1)/12 = 4 \cdot 4 \cdot 9/12 = 12$,

$$Z = \frac{R_1 - \mathbb{E}[R_1]}{\sqrt{\mathrm{Var}(R_1)}} = \frac{18 - 18}{\sqrt{12}} = 0, \qquad Z^2 = 0 = H.$$

$$\boxed{H = Z^2: \quad 0 = 0^2; \quad \text{here both retain } H_0 \text{ (rank sums equal).}}$$

The identity $H = Z^2$ holds in general for $k = 2$, with $\chi^2_{0.05,1} = 3.841 = z^2_{0.025} = 1.96^2$, so the two tests reject together — Kruskal–Wallis simply extends the rank-sum idea to $k \geq 3$.

Example 14.4.13 (End-to-end with a near-cutoff result: four teaching apps)**.** Test scores (out of 50) for $k = 4$ learning apps, $n_j = 4$ each ($N = 16$), are

$$\begin{aligned} \text{App W}: &\ 31,\ 36,\ 40,\ 44, \\ \text{App X}: &\ 28,\ 33,\ 38,\ 41, \\ \text{App Y}: &\ 45,\ 47,\ 48,\ 50, \\ \text{App Z}: &\ 25,\ 29,\ 34,\ 37. \end{aligned}$$

Run the full test at $\alpha = 0.05$ ($\chi^2_{0.05,3} = 7.815$).

Which rule, and why? Four independent samples and a distribution-equality question is squarely Kruskal–Wallis on χ^2_3. We carry the four-step recipe through and interpret a result that lands near the cutoff.

Solution.

Step 1: pooled ranks (all 16 distinct).

value	25	28	29	31	33	34	36	37	38	40	41	44	45	47	48	50
rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
group	Z	X	Z	W	X	Z	W	Z	X	W	X	W	Y	Y	Y	Y

Step 2: rank sums.

$$R_W = 4 + 7 + 10 + 12 = 33,$$
$$R_X = 2 + 5 + 9 + 11 = 27,$$
$$R_Y = 13 + 14 + 15 + 16 = 58,$$
$$R_Z = 1 + 3 + 6 + 8 = 18.$$

Check: $33 + 27 + 58 + 18 = 136 = 16 \cdot 17/2$. ■

Step 3: the statistic.

$$\sum_j \frac{R_j^2}{n_j} = \frac{33^2 + 27^2 + 58^2 + 18^2}{4} = \frac{1089 + 729 + 3364 + 324}{4} = \frac{5506}{4} = 1376.5,$$

$$H = \frac{12}{16 \cdot 17}(1376.5) - 3(17) = \frac{12}{272}(1376.5) - 51 = 60.728 - 51 = 9.728.$$

Step 4: decision.

$$\boxed{H = 9.728 > \chi^2_{0.05,3} = 7.815 \;\Rightarrow\; \text{reject } H_0;\ \text{the apps differ.}}$$

App Y sweeps the top four ranks, dominating H; a follow-up comparison would isolate Y as the standout.

Example 14.4.14 (Recognition check: which k-sample test, and why not the others?)**.** A physiologist records VO_2-max on the *same* 8 athletes under $k = 3$ training protocols (each athlete tried all three), and the data are roughly bell-shaped. A colleague suggests Kruskal–Wallis. Is that the right test?

Which rule, and why? Kruskal–Wallis requires *independent* samples. Here each athlete contributes to all three protocols, so the three columns are matched, not independent — the within-athlete correlation breaks the test's assumption. The correct rank test is Friedman (§5); had the data been independent and bell-shaped, ANOVA would have been preferred.

Solution.

Step 1: check independence. The three protocol scores for a given athlete are correlated (a fit athlete scores high under all three), so the k samples are *not* independent. Kruskal–Wallis does not apply.

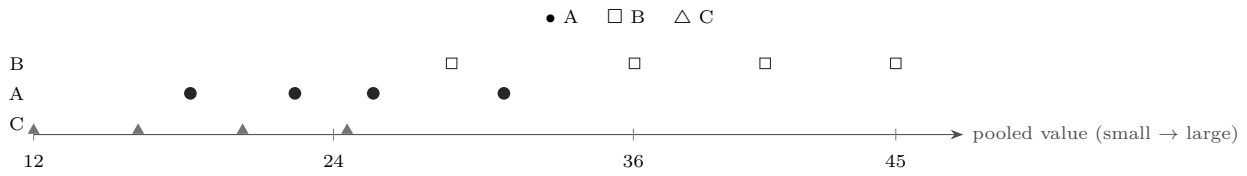
Step 2: pick the matched-data analogue. For k treatments measured within the same blocks (athletes), rank *within each block* and use the Friedman test.

Step 3: note the parametric option. Because the data are bell-shaped, the parametric repeated-measures / randomized-block ANOVA would actually be the more powerful valid choice; the rank test (Friedman) is the fallback if normality is doubtful.

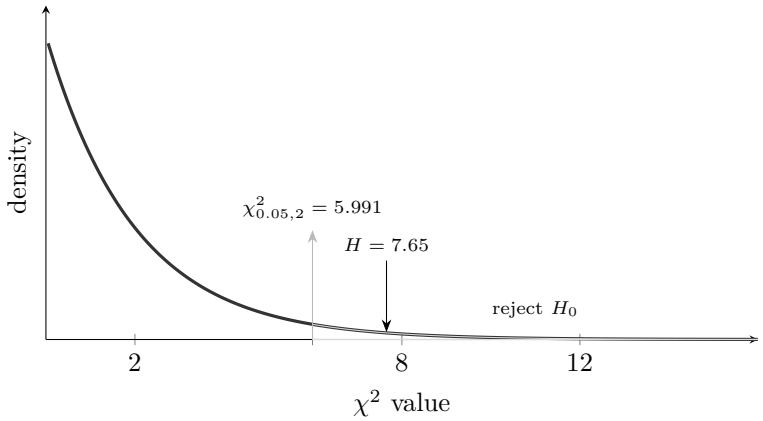
$$\boxed{\begin{array}{l}\text{Matched samples} \Rightarrow \text{ Friedman, not Kruskal–Wallis;} \\ \text{normal data} \Rightarrow \text{ block ANOVA is more powerful.}\end{array}}$$

Recognizing *independent* vs. *matched* is the single most important step before any k-sample test.

14.4.4 Pictures: the pooled ranking and the reference distribution



Dotplots of the three soil mixes from the first example, laid on a common value axis. Reading left to right gives the pooled ranking: Mix C (triangles) holds the small values and low ranks, Mix B (squares) the large values and high ranks, with Mix A in between. That visible separation is what makes the rank sums $R_C = 13 < R_A = 24 < R_B = 41$ and drives H into the rejection region.



The χ^2_{k-1} reference for the three-soil example ($k = 3$, so χ^2_2). The shaded right tail beyond $\chi^2_{0.05,2} = 5.991$ is the rejection region; the test is always upper-tail because any location shift among the groups inflates H. The observed $H = 7.65$ falls inside the shaded region, so H_0 is rejected. For larger k the density peaks away from zero (near $x = k - 3$), but the rule is unchanged: reject when H clears $\chi^2_{\alpha,k-1}$.

14.5 The Friedman Test

When the same blocks are exposed to every one of k treatments — the randomized block layout of Chapter 13, but now with normality in doubt — the **Friedman test** replaces the block F-test. Its engine is the one idea that made blocking work in the first place: comparisons are made *within* each block, so the block-to-block level differences (one judge being harsh, one plot being fertile, one day being warm) never enter the statistic. Concretely, you rank the k treatments $1, 2, \ldots, k$ *within each block*, add up each treatment's ranks across the b blocks to get a rank sum R_i, and ask whether those sums are farther apart than chance would allow. If the treatments are equivalent, every block ranks them in a random order, the R_i hover near their common mean $b(k+1)/2$, and the spread statistic $Q = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1)$ is small; real treatment differences push the R_i apart and inflate Q. You recognize this section by its shape: a treatments-by-blocks grid (often repeated measures, where each subject is itself a block) and a question about whether the treatments differ, with the data ordinal or the normality of a block ANOVA suspect.

Which block-design test does the problem need?

- A **treatments-by-blocks** grid, normality reasonable, magnitudes meaningful? ⇒ the randomized block F-test of Chapter 13 — more powerful when its assumptions hold.
- The same grid but **normality doubtful**, or the responses only **ranks/ratings** (ordinal)? ⇒ the **Friedman test**: rank within blocks, then Q against χ^2_{k-1}.
- **Repeated measures** — one set of subjects each tries all k items? ⇒ still Friedman; each *subject* is a block, which is what removes the person-to-person offset.
- Exactly $k = 2$ treatments in blocks? ⇒ Friedman collapses to the **sign test** on the within-block differences — use that directly.

Do not rank across the whole table: the entire point is to rank *within* each block, so that a block's overall level cancels. Pooling all bk numbers and ranking them globally reintroduces exactly the nuisance variation blocking was designed to remove, and the resulting statistic tests the wrong thing.

14.5.1 Ranking within blocks and the rank sums

Definition 14.5.1 (Within-block ranks and treatment rank sums)**.** Arrange the data as a $k \times b$ grid with one response Y_{ij} for treatment i in block j. For each *fixed* block j, rank its k entries from 1 (smallest) to k

(largest), assigning the **average rank** to any tied entries; call the result $\text{rank}(Y_{ij})$. The **treatment rank sum** is the total of a treatment's within-block ranks,

$$R_i = \sum_{j=1}^{b} \text{rank}(Y_{ij}), \qquad i = 1, \ldots, k.$$

Because every block contributes the ranks $1, 2, \ldots, k$ exactly once, the grand total is fixed: $\sum_{i=1}^{k} R_i = b \cdot \frac{k(k+1)}{2}$, so the rank sums must average $b(k+1)/2$.

Theorem 14.5.2 (The Friedman statistic and its null distribution)**.** *Under H_0 that the k treatments are equivalent (within each block their k responses are exchangeable), the statistic*

$$Q = \frac{12}{b\,k(k+1)} \sum_{i=1}^{k} R_i^2 \;-\; 3\,b(k+1)$$

*has, for b large, an approximate **chi-square distribution with** $k-1$ **degrees of freedom**. The test is **upper-tailed**: reject H_0 at level α when $Q \geq \chi^2_{\alpha,\,k-1}$. For small b and k, exact critical values of Q are tabulated and should be preferred to the chi-square cutoff.*

Why Q measures rank-sum spread. Center each rank sum at its null mean $\mu_R = b(k+1)/2$. A direct expansion gives

$$\begin{aligned}\sum_{i=1}^{k} \big(R_i - \mu_R\big)^2 &= \sum_i R_i^2 - 2\mu_R \sum_i R_i + k\mu_R^2 \\ &= \sum_i R_i^2 - 2\mu_R \cdot b\tfrac{k(k+1)}{2} + k\Big(\tfrac{b(k+1)}{2}\Big)^2 = \sum_i R_i^2 - \frac{b^2 k(k+1)^2}{4}.\end{aligned}$$

Multiplying by the constant $12/[bk(k+1)]$ turns this sum of squared deviations into Q, since $\frac{12}{bk(k+1)} \cdot \frac{b^2 k(k+1)^2}{4} = 3b(k+1)$. Thus Q is a scaled sum of squared deviations of the rank sums from their common mean — zero when the R_i are all equal, large when they spread out. The scaling constant is chosen so that, under exchangeability within blocks, Q matches the first moments of a χ^2_{k-1} variable; the $k-1$ (not k) degrees of freedom reflect the linear constraint $\sum_i R_i = bk(k+1)/2$ that ties the rank sums together. □

Remark. Read the whole procedure as "blocking, done with ranks." The block F-test removed nuisance variation by subtracting block means; Friedman removes it more bluntly — by replacing each block's raw numbers with the ranks $1, \ldots, k$, a block's overall level (high or low) is erased before any comparison is made. What survives is only the *pattern* of which treatment beat which inside each block. That is why Friedman needs no normality and no assumption that effects add in the original scale: monotone re-expression of the responses (logs, square roots) leaves every within-block rank unchanged, hence leaves Q unchanged.

	A	**B**	**C**	**D**
Block 1	86 (3)	80 (1)	93 (4)	84 (2)
Block 2	61 (3)	55 (1)	68 (4)	59 (2)
Block 3	42 (2)	38 (1)	49 (4)	45 (3)
R_i	8	3	12	7

rank $1 \to k$ *within* each row

Within-block ranking for $k = 4$ treatments (A–D) over $b = 3$ blocks. Each row is ranked on its own — note the three blocks sit at very different levels (80s, 60s, 40s) yet rank identically — so a block's overall level vanishes. The parenthesized numbers are the within-block ranks; the bottom row sums them down each treatment to give R_i (totaling $8 + 3 + 12 + 7 = 30 = b\,k(k+1)/2$). Spread among the R_i is what Q measures.

Example 14.5.3 (A full Friedman test: three keyboard layouts)**.** An ergonomics lab has $b = 5$ typists (the blocks) each type the same passage on $k = 3$ keyboard layouts; the response is typing speed (words per minute). Higher is better.

Typist	Layout A	Layout B	Layout C
1	52	58	49
2	61	67	60
3	45	52	47
4	70	74	68
5	55	60	53

Test at $\alpha = 0.05$ whether the three layouts give equal typing speed, using $\chi^2_{0.05,2} = 5.991$.

Which rule, and why? The same typists try all three layouts, so this is a randomized block design with the typists as blocks; ranks-within-blocks plus a small b make Friedman the right tool. Rank each typist's three speeds 1–3, sum each layout's ranks, then Q on $\chi^2_{k-1} = \chi^2_2$.

Solution.

*Step 1: rank within each block (*1 = *slowest,* 3 = *fastest).*

Typist	Layout A	Layout B	Layout C
1	52 (2)	58 (3)	49 (1)
2	61 (2)	67 (3)	60 (1)
3	45 (1)	52 (3)	47 (2)
4	70 (2)	74 (3)	68 (1)
5	55 (2)	60 (3)	53 (1)
R_i	9	15	6

Step 2: check the rank-sum total. $9 + 15 + 6 = 30 = b \cdot \frac{k(k+1)}{2} = 5 \cdot 6$; the average rank sum is $b(k+1)/2 = 5 \cdot 2 = 10$, so B runs high and C runs low.

Step 3: compute Q. With $b = 5$, $k = 3$,

$$Q = \frac{12}{bk(k+1)} \sum_i R_i^2 - 3b(k+1) = \frac{12}{5 \cdot 3 \cdot 4}\left(9^2 + 15^2 + 6^2\right) - 3 \cdot 5 \cdot 4$$
$$= \frac{12}{60}\,(81 + 225 + 36) - 60 = \frac{12}{60}\,(342) - 60 = 68.4 - 60 = 8.4.$$

Step 4: decide on χ^2_2. Since $Q = 8.4 > \chi^2_{0.05,2} = 5.991$, reject.

$$\boxed{Q = 8.4 > 5.991 \;\Rightarrow\; \text{reject } H_0;\ \text{the layouts differ in speed (Layout B fastest).}}$$

Example 14.5.4 (Repeated measures: rating four phone interfaces)**.** A usability study asks $b = 6$ users to rate $k = 4$ phone interfaces on a 0–100 preference scale (each user rates all four). The rank sums of the within-user ranks (1 = least preferred, 4 = most preferred) came out to

$$R_{\mathrm{W}} = 10, \quad R_{\mathrm{X}} = 22, \quad R_{\mathrm{Y}} = 13, \quad R_{\mathrm{Z}} = 15.$$

Test at $\alpha = 0.05$ whether preference differs across interfaces; $\chi^2_{0.05,3} = 7.815$.

Which rule, and why? One set of subjects rates every item, so each *user* is a block and this is a repeated-measures (within-subjects) design — exactly Friedman's home ground. The ratings are ordinal and the rank sums are already in hand, so go straight to Q on $\chi^2_{k-1} = \chi^2_3$.

Solution.

Step 1: verify the rank-sum total. $10 + 22 + 13 + 15 = 60 = b\frac{k(k+1)}{2} = 6 \cdot 10$, and the average rank sum is $b(k+1)/2 = 6 \cdot 2.5 = 15$.

Step 2: compute Q. With $b = 6$, $k = 4$,

$$\sum_i R_i^2 = 10^2 + 22^2 + 13^2 + 15^2 = 100 + 484 + 169 + 225 = 978,$$

$$Q = \frac{12}{6 \cdot 4 \cdot 5}(978) - 3 \cdot 6 \cdot 5 = \frac{12}{120}(978) - 90 = 97.8 - 90 = 7.8.$$

Step 3: decide on χ^2_3. Here $Q = 7.8 < \chi^2_{0.05,3} = 7.815$ — a whisker short.

$$\boxed{Q = 7.8 < 7.815 \;\Rightarrow\; \text{fail to reject } H_0 \text{ at } \alpha = 0.05.}$$

Interface X is the clear front-runner, but with only six users the evidence falls just inside the retain region; a few more raters could tip it.

Example 14.5.5 (The chi-square approximation as a p-value bracket)**.** A Friedman test of $k = 5$ recipes over $b = 12$ tasters returns $Q = 11.30$ on χ^2_4. Standard upper-tail cutoffs are $\chi^2_{0.10,4} = 7.779$, $\chi^2_{0.05,4} = 9.488$, $\chi^2_{0.025,4} = 11.143$, $\chi^2_{0.01,4} = 13.277$. Bracket the p-value and decide at $\alpha = 0.05$.

Which rule, and why? With $b = 12$ blocks the large-sample chi-square approximation is trustworthy, and Friedman is always upper-tailed, so the p-value is the area of $\chi^2_{k-1} = \chi^2_4$ to the right of Q. Locate 11.30 between tabulated cutoffs.

Solution.

Step 1: bracket $Q = 11.30$.

$$\chi^2_{0.025,4} = 11.143 \;<\; 11.30 \;<\; 13.277 = \chi^2_{0.01,4}.$$

Step 2: translate to a p-value range. Lying between the 0.025 and 0.01 cutoffs,

$$0.01 < p\text{-value} < 0.025.$$

Step 3: decide at $\alpha = 0.05$. The p-value is below 0.05, so reject.

$$\boxed{0.01 < p\text{-value} < 0.025 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05.}$$

Example 14.5.6 (Small-sample exact vs. large-sample chi-square)**.** A pilot study has only $b = 4$ blocks and $k = 3$ treatments and yields $Q = 6.50$. A Friedman exact table for $(k, b) = (3, 4)$ gives $\mathbb{P}(Q \geq 6.50 \mid H_0) = 0.042$, whereas the chi-square approximation uses $\chi^2_{0.05,2} = 5.991$. Compare the two verdicts and say which to trust.

Which rule, and why? Q's chi-square shape is only asymptotic in b; with $b = 4$ the discrete exact distribution is the honest reference. Use the exact tail probability and treat the chi-square cutoff as a back-up.

Solution.

Step 1: exact test. The tabulated exact p-value is $0.042 < 0.05$, so reject at $\alpha = 0.05$.

Step 2: chi-square approximation. $Q = 6.50 > \chi^2_{0.05,2} = 5.991$, which also rejects — but the approximate p-value (about 0.039) is not identical to the exact 0.042.

Step 3: which to trust. For so few blocks, prefer the *exact* value; the agreement here is reassuring but coincidental.

$$\boxed{\text{Exact } p = 0.042 \text{ (reject); chi-square agrees, but use the exact tail for small } b.}$$

Example 14.5.7 (Ties within a block: average ranks)**.** In one block of a $k = 4$ study the four responses are 30, 42, 42, 51. Assign the within-block ranks and explain the effect of the tie on Q.

Which rule, and why? Ties are broken by the **average-rank** convention: tied entries share the mean of the ranks they would have occupied. This keeps the block's rank total at $1+2+\cdots+k$, so the rank-sum constraint — and the form of Q — is preserved.

Solution.

Step 1: order and rank. Sorted, the values occupy positions $1,2,3,4$ as $30<42=42<51$. The two 42's would take positions 2 and 3, so each gets the average rank $\frac{2+3}{2}=2.5$:

$$\text{rank}(30)=1, \qquad \text{rank}(42)=2.5, \qquad \text{rank}(42)=2.5, \qquad \text{rank}(51)=4.$$

Step 2: check the block total. $1+2.5+2.5+4=10=\frac{k(k+1)}{2}=\frac{4\cdot 5}{2}$, exactly the untied total, so the rank-sum bookkeeping is undisturbed.

Step 3: effect on Q. Heavy ties slightly deflate Q; a tie correction divides Q by $1-\frac{\sum(t^3-t)}{bk(k^2-1)}$, where t runs over tie-group sizes. With a single tie of size $t=2$ the correction factor is $1-\frac{2^3-2}{b\cdot 4\cdot 15}=1-\frac{6}{240b}$ — a negligible nudge unless ties are pervasive.

ranks 1, 2.5, 2.5, 4; block total still 10; Q barely changed.

Example 14.5.8 (Full Friedman with a tied block: pain-relief ratings)**.** Five patients ($b=5$) rate $k=3$ analgesics on a 0–10 relief scale; higher is better. One patient ties two drugs.

Patient	Drug P	Drug Q	Drug R
1	6	8	5
2	7	7	4
3	5	9	6
4	8	9	7
5	4	7	3

Test at $\alpha=0.05$ whether the drugs differ ($\chi^2_{0.05,2}=5.991$).

Which rule, and why? Same patients, all three drugs $\Rightarrow$ Friedman with patients as blocks; the tie in Patient 2 (Drugs P and Q both 7) is handled by average ranks. Rank within each patient, sum, then Q on χ^2_2.

Solution.

Step 1: rank within each patient. Patient 2's two 7's would take ranks 2 and 3, so each becomes 2.5:

Patient	Drug P	Drug Q	Drug R
1	6 (2)	8 (3)	5 (1)
2	7 (2.5)	7 (2.5)	4 (1)
3	5 (1)	9 (3)	6 (2)
4	8 (2)	9 (3)	7 (1)
5	4 (2)	7 (3)	3 (1)
R_i	9.5	14.5	6

Step 2: check the total. $9.5+14.5+6=30=5\cdot 6$, as required.

Step 3: compute Q.

$$\sum_i R_i^2 = 9.5^2+14.5^2+6^2 = 90.25+210.25+36 = 336.5,$$

$$Q=\frac{12}{5\cdot 3\cdot 4}(336.5)-3\cdot 5\cdot 4=\frac{12}{60}(336.5)-60=67.3-60=7.3.$$

Step 4: decide. $Q=7.3>\chi^2_{0.05,2}=5.991$, reject (the lone tie barely moves the statistic).

$Q=7.3>5.991 \;\Rightarrow$ reject H_0; the analgesics differ (Q best).

14.5.2 The $k = 2$ case and the sign test

Proposition 14.5.9 (Friedman with two treatments is the sign test)**.** *When $k = 2$, ranking within each block assigns rank 1 to the smaller of the pair and rank 2 to the larger. The Friedman Q then depends only on how many of the b blocks rank treatment 1 above treatment 2 — equivalently, on the **signs** of the within-block differences — and the test is the two-sided **sign test** on those b paired differences.*

Proof. With $k = 2$ each block contributes ranks $\{1, 2\}$. Let a be the number of blocks in which treatment 1 gets the larger rank (2) and $b-a$ the number in which it gets the smaller (1). Then $R_1 = 2a+1\cdot(b-a) = b+a$ and $R_2 = 2(b-a)+a = 2b-a$, so $R_1 + R_2 = 3b = b\frac{k(k+1)}{2}$ as required. Substituting into Q with $k = 2$,

$$Q = \frac{12}{b \cdot 2 \cdot 3}\big((b+a)^2 + (2b-a)^2\big) - 3b \cdot 3 = \frac{(2a-b)^2}{b},$$

which is exactly the square of the standardized sign-test count a (mean $b/2$, variance $b/4$ under H_0): $Q = \left(\frac{a-b/2}{\sqrt{b}/2}\right)^2 = Z^2$. Thus $Q \sim \chi^2_1$ is the square of the sign-test Z, and Friedman reduces to counting how often one treatment beats the other. □

Example 14.5.10 ($k = 2$: Friedman equals the sign test)**.** Eight plots ($b = 8$) each grow two wheat varieties side by side; variety I out-yields variety II on 6 of the 8 plots. Run the $k = 2$ Friedman test and confirm it matches the sign test ($\chi^2_{0.05,1} = 3.841$).

Which rule, and why? Two treatments in blocks is the boundary case: Friedman's Q on χ^2_1 is algebraically the sign test on the b within-block winners. Count the "variety I wins," that is the sign-test statistic a.

Solution.

Step 1: the count. Variety I wins $a = 6$ of $b = 8$ blocks.

Step 2: Q by the reduced formula.

$$Q = \frac{(2a-b)^2}{b} = \frac{(2\cdot 6 - 8)^2}{8} = \frac{4^2}{8} = \frac{16}{8} = 2.0.$$

Step 3: cross-check via the sign-test Z. $Z = \frac{a-b/2}{\sqrt{b}/2} = \frac{6-4}{\sqrt{8}/2} = \frac{2}{1.414} = 1.414$, and $Z^2 = 2.0 = Q$.

Step 4: decide. $Q = 2.0 < \chi^2_{0.05,1} = 3.841$ (equivalently $|Z| = 1.414 < 1.96$).

$$\boxed{Q = Z^2 = 2.0 < 3.841 \;\Rightarrow\; \text{fail to reject; 6 of 8 wins is not significant.}}$$

14.5.3 Friedman versus the randomized block F-test

Remark. Friedman and the block F-test answer the same question from opposite ends of the assumption ledger. The F-test uses the raw magnitudes and assumes normal errors with additive, constant-variance block and treatment effects; when those hold it is the more powerful test, because numbers carry more information than ranks. Friedman keeps only the within-block ordering, so it is valid under *any* continuous shape, immune to outliers and to monotone re-scaling, and applies even to purely ordinal ratings — at the cost of a little power (its asymptotic efficiency relative to the block F is about $0.955k/(k+1)$, e.g. roughly 72% for $k = 3$, rising toward 95.5% as k grows). The rule of thumb: trust normality and additivity ⇒ block F; doubt them, or have only ranks ⇒ Friedman.

Example 14.5.11 (CONTRAST: block F-test vs. Friedman on the same grid)**.** Four judges ($b = 4$, the blocks) score $k = 3$ wines on a 100-point scale, but one judge is a notorious "low marker" whose scores sit far below everyone else's — the kind of block offset, plus possible non-normal scoring, that makes the analysis choice matter.

Judge	Wine A	Wine B	Wine C
1	88	91	85
2	90	94	87
3	60	66	58
4	92	95	90

Run the Friedman test ($\chi^2_{0.05,2} = 5.991$) and contrast its logic with the block F-test.

Which rule, and why? Judge 3's scores are a huge block offset and the data are subjective ratings, so normality is shaky — favoring Friedman. Ranking within each judge cancels the low-marker offset automatically; the block F-test would cancel it too, but only if its normal-additive model holds.

Solution.

Step 1: rank within each judge.

Judge	Wine A	Wine B	Wine C
1	88 (2)	91 (3)	85 (1)
2	90 (2)	94 (3)	87 (1)
3	60 (2)	66 (3)	58 (1)
4	92 (2)	95 (3)	90 (1)
R_i	8	12	4

Step 2: Q. With $b = 4$, $k = 3$,

$$Q = \frac{12}{4 \cdot 3 \cdot 4}\,(8^2 + 12^2 + 4^2) - 3 \cdot 4 \cdot 4 = \frac{12}{48}\,(64 + 144 + 16) - 48 = \frac{12}{48}\,(224) - 48 = 56 - 48 = 8.0.$$

Step 3: decide and contrast. $Q = 8.0 > \chi^2_{0.05,2} = 5.991$, reject. Notice every judge ranked B>A>C, so the rank sums are as far apart as possible regardless of Judge 3's deflated scale — ranking neutralized the offset without any normality assumption. The block F-test on the raw scores would also reject (Wine B is uniformly highest), and being magnitude-based it would have slightly more power *if* the scores were truly normal; but with one wildly different judge and subjective scoring, Friedman's conclusion needs no such faith.

$Q = 8.0 > 5.991 \;\Rightarrow$ reject; Friedman needs no normality, F more powerful only if it holds.

Example 14.5.12 (When the F-test is the better choice)**.** A metallurgist measures fatigue life (thousands of cycles) of $k = 3$ alloys across $b = 10$ test rigs (blocks). Residual plots look symmetric and bell-shaped, variances are similar, and effects appear additive. The Friedman test gives $Q = 5.4$ ($\chi^2_{0.05,2} = 5.991$, retain) while the block F-test gives $F = 4.9$ on $\mathrm{F}_{2,18}$ ($F_{0.05,2,18} = 3.55$, reject). Explain the discrepancy and recommend a test.

Which rule, and why? When the normal-additive model genuinely holds, the magnitude-based F is more powerful than the rank-based Friedman, so it can detect a real effect that Friedman's coarser ranks miss. The diagnostics here vouch for the F-test's assumptions.

Solution.

Step 1: read the two verdicts. Friedman retains ($5.4 < 5.991$); the block F rejects ($4.9 > 3.55$).

Step 2: explain. Friedman discards the magnitudes, keeping only orderings, so a consistent but *modest* alloy gap can leave the rank sums too close to trip Q while still inflating F. With assumptions verified, the loss of power is real, not an artifact.

Step 3: recommend. Because normality, equal variance, and additivity check out, prefer the more powerful F-test and conclude the alloys differ.

Assumptions valid $\Rightarrow$ trust the block F (reject); Friedman's retain is a power shortfall.

Example 14.5.13 (Why not rank across the whole table?)**.** For the keyboard data of the first example, a student ranks all $bk = 15$ speeds globally (1 for the slowest of all, 15 for the fastest) and sums by layout. Explain why this is wrong and what it would test instead.

Which rule, and why? Friedman ranks *within* blocks precisely so a block's level cancels. Ranking globally lets a fast typist's three speeds all earn high ranks, smuggling the typist (block) effect back into the rank sums — the opposite of what blocking is for.

Solution.

Step 1: what global ranking does. Typist 4 (speeds 68–74) would supply three of the top ranks no matter which layout, and Typist 3 (speeds 45–52) three of the bottom ranks. The layout rank sums would then be dominated by *who typed*, not *which layout*.

Step 2: what it actually tests. Global ranking turns the procedure into a Kruskal–Wallis analysis that ignores the pairing — appropriate only for *independent* samples, not for repeated measures on the same typists.

Step 3: the fix. Rank within each typist (as in Example 14.5), so each block contributes exactly $\{1, 2, 3\}$ and its level drops out.

Rank within blocks; global ranking re-injects the block effect and tests the wrong hypothesis.

Example 14.5.14 (Larger design: five fertilizers across eight fields)**.** A from-scratch summary: $k = 5$ fertilizers are compared on $b = 8$ fields (blocks). After ranking each field's five yields 1–5, the treatment rank sums are

$$R_1 = 14, \quad R_2 = 31, \quad R_3 = 20, \quad R_4 = 27, \quad R_5 = 28.$$

Test at $\alpha = 0.05$ whether the fertilizers differ ($\chi^2_{0.05,4} = 9.488$).

Which rule, and why? Fields are the blocks and fertilizers the treatments; with $b = 8$ the chi-square approximation on $\chi^2_{k-1} = \chi^2_4$ is reasonable. Apply the Q formula directly to the given rank sums.

Solution.

Step 1: check the total. $14+31+20+27+28 = 120 = b\frac{k(k+1)}{2} = 8\cdot 15$; average rank sum $b(k+1)/2 = 8\cdot 3 = 24$.

Step 2: compute Q. With $b = 8$, $k = 5$,

$$\sum_i R_i^2 = 14^2 + 31^2 + 20^2 + 27^2 + 28^2 = 196 + 961 + 400 + 729 + 784 = 3070,$$

$$Q = \frac{12}{8 \cdot 5 \cdot 6}\,(3070) - 3 \cdot 8 \cdot 6 = \frac{12}{240}\,(3070) - 144 = 153.5 - 144 = 9.5.$$

Step 3: decide. $Q = 9.5 > \chi^2_{0.05,4} = 9.488$, just past the cutoff.

$Q = 9.5 > 9.488 \;\Rightarrow$ reject H_0; the fertilizers differ (just barely).

Example 14.5.15 (Reading a borderline result on the chi-square curve)**.** The fertilizer study just gave $Q = 9.5$ on χ^2_4 with cutoff $\chi^2_{0.05,4} = 9.488$. Without re-deriving anything, place Q on the reference curve and state the conclusion and its fragility.

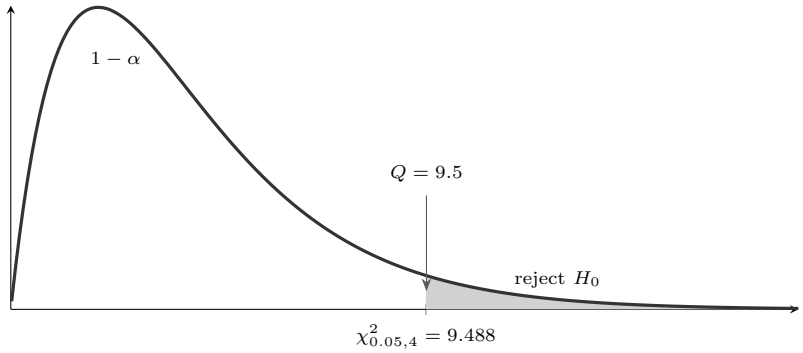
Which rule, and why? A Friedman test is upper-tailed on χ^2_{k-1}, so the verdict is simply which side of the cutoff Q lands on: inside the body retains, out in the shaded tail rejects.

Solution.

Step 1: locate Q. 9.5 sits a hair beyond 9.488, just inside the rejection tail.

Step 2: conclude. Reject H_0 at $\alpha = 0.05$ — but only marginally; the p-value is about 0.0497, so the slightest change in the data (or using an exact small-sample table) could flip it.

$Q = 9.5 > 9.488 \;\Rightarrow$ reject, but barely; report the p-value, not just the verdict.



The χ^2_4 reference density (here $k = 5$ treatments). It is right-skewed and peaks near $x = k - 2 = 2$; under H_0 the Friedman statistic clusters under the hump. The shaded upper tail beyond $\chi^2_{0.05,4} = 9.488$ is the rejection region, and the observed $Q = 9.5$ lands just inside it — a borderline reject. More blocks would sharpen the chi-square approximation; for small b, replace this cutoff with an exact Friedman table value.

14.6 Tests for Randomness and Rank Correlation

Two loose ends close the nonparametric toolbox. The first asks whether a sequence of two symbols — heads/tails, pass/fail, above/below a baseline — was produced *at random*, with no trend and no clustering. The device is a **run**: a maximal streak of one symbol. Too few runs means the symbols clump (a shift or a slow drift); too many means they over-alternate (an unnaturally regular flip-flop). Either extreme is evidence against randomness, and the number of runs R has a known mean and variance once you fix how many of each symbol occurred, so a large-sample z does the job. The second loose end measures **monotone association** between two rankings: rank the x's and the y's separately and correlate the ranks. That is **Spearman's** r_s — it detects any consistently increasing or decreasing relationship, linear or not, and shrugs off outliers, where ordinary Pearson correlation insists on a straight line. Recognize this section by the question: *is this ordering random*, or *do these two rankings move together?*

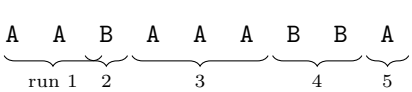
Which randomness/association tool does the problem need?

- A single **binary sequence** and the question "is the order random"? $\Rightarrow$ the **runs test**: count R, compare to $\mu_R = 1 + \dfrac{2n_1n_2}{n_1+n_2}$, standardize.
- **Too few runs**? $\Rightarrow$ clustering or a trend (a shift, a drift, a hot streak). **Too many runs**? $\Rightarrow$ over-alternation (negative serial dependence).
- **Two rankings** (or two numeric variables you rank) and the question "do they move together monotonically"? $\Rightarrow$ **Spearman's** r_s, via the rank correlation or the shortcut $r_s = 1 - \dfrac{6\sum d_i^2}{n(n^2-1)}$.
- Testing $H_0\colon \rho_s = 0$ with n not tiny? $\Rightarrow$ the large-sample $t = r_s\sqrt{\dfrac{n-2}{1-r_s^2}}$ on $n-2$ df; for small n use a Spearman table.

Do not confuse Spearman with Pearson: Spearman correlates *ranks* and so measures *monotone* association (robust to outliers and curvature); Pearson correlates the raw values and measures *linear* association. A perfectly monotone but bent relationship gives $r_s = 1$ while Pearson $r < 1$.

14.6.1 The runs test for randomness

Definition 14.6.1 (Run)**.** Given a sequence of two symbols, a **run** is a maximal block of consecutive identical symbols. Writing the sequence out, a new run begins exactly where the symbol changes. If there are n_1 symbols of one kind and n_2 of the other ($n = n_1 + n_2$), the total number of runs R ranges from 2 (all of one symbol, then all of the other) to $2\min(n_1, n_2) + 1$ (perfect alternation).



The sequence `A A B A A A B B A` *broken into its maximal streaks. A new run starts at every symbol change, so counting runs is just counting changes and adding one. Here $R = 5$.*

Theorem 14.6.2 (Mean and variance of the number of runs)**.** *If the $n = n_1 + n_2$ symbols are arranged in a purely random order (every arrangement equally likely), the number of runs R has*

$$\mu_R = 1 + \frac{2n_1n_2}{n_1 + n_2},$$
$$\sigma_R^2 = \frac{2n_1n_2\,(2n_1n_2 - n_1 - n_2)}{(n_1 + n_2)^2\,(n_1 + n_2 - 1)}.$$

For n_1 and n_2 both moderately large (a common rule of thumb is each at least 10), R is approximately normal, so

$$Z = \frac{R - \mu_R}{\sqrt{\sigma_R^2}}$$

is referred to the standard normal. The test is two-sided: small R (large negative Z) signals clustering or a trend; large R (large positive Z) signals over-alternation.

Remark. The continuity correction sharpens the approximation: when testing for *too few* runs replace R by $R + \frac{1}{2}$, and for *too many* runs by $R - \frac{1}{2}$, before standardizing. The examples below show the correction where it matters; with both counts comfortably above 10 the uncorrected Z is usually close enough to reach the same verdict.

Example 14.6.3 (Runs test on a quality-control sequence)**.** A bottling line is sampled 20 times through a shift; each bottle is recorded as on-spec (S) or off-spec (F). In order the result is

S S F S F F S S S F S F F S S F S F S F.

There are $n_1 = 11$ S's and $n_2 = 9$ F's. Test, at $\alpha = 0.05$, whether the on/off pattern is random.

Which rule, and why? One binary sequence, asking whether its *order* is random — that is the runs test, not a proportion test. A proportion test would ask whether the *count* of failures is unusual; the runs test instead asks whether the failures are sprinkled randomly or bunched.

Solution.

Step 1: count the runs. Mark each place the symbol changes and add one. Inserting bars at the changes,

S S | F | S | F F | S S S | F | S | F F | S S | F | S | F | S | F,

gives 14 streaks, so $R = 14$.

Step 2: null mean and variance. With $n_1 = 11$, $n_2 = 9$, $n = 20$,

$$\mu_R = 1 + \frac{2(11)(9)}{20} = 1 + \frac{198}{20} = 1 + 9.9 = 10.9,$$
$$\sigma_R^2 = \frac{2(11)(9)\,[\,2(11)(9) - 20\,]}{20^2\,(19)} = \frac{198\,(178)}{400\,(19)} = \frac{35244}{7600} = 4.637.$$

Step 3: standardize. The observed $R = 14$ exceeds the mean, so use the "too many" correction $R - \frac{1}{2}$:

$$Z = \frac{(14 - 0.5) - 10.9}{\sqrt{4.637}} = \frac{2.6}{2.153} = 1.21.$$

Step 4: decide. Two-sided cutoff $z_{0.025} = 1.96$, and $|1.21| < 1.96$.

$Z = 1.21,\ |Z| < 1.96\ \Rightarrow$ do not reject; the S/F order is consistent with randomness.

The line shows a lot of alternation but not significantly more than chance would produce.

Example 14.6.4 (Runs test flagging a trend: too few runs)**.** A machinist measures a part diameter on 24 consecutive units and records whether each is above (A) or below (B) the target. The sequence is

B B B B B B A B B A A A A A B A A A A A B A A A,

with $n_1 = 15$ A's and $n_2 = 9$ B's. Test for randomness at $\alpha = 0.05$.

Which rule, and why? A long initial stretch of B's giving way to mostly A's looks like a drift in the process mean. Drift produces *clustering*, hence *too few* runs — exactly what the lower tail of the runs test detects. A simple count of A's versus B's would miss the ordering entirely.

Solution.

Step 1: count the runs. The streaks are {BBBBBB}{A}{BB}{AAAAA}{B}{AAAAA}{B}{AAA}, so $R = 8$.

Step 2: null mean and variance. With $n_1 = 15$, $n_2 = 9$, $n = 24$,

$$\mu_R = 1 + \frac{2(15)(9)}{24} = 1 + \frac{270}{24} = 1 + 11.25 = 12.25,$$
$$\sigma_R^2 = \frac{270\,[\,270 - 24\,]}{24^2\,(23)} = \frac{270\,(246)}{576\,(23)} = \frac{66420}{13248} = 5.014.$$

Step 3: standardize with the "too few" correction. Here $R = 8 < \mu_R$, so use $R + \frac{1}{2}$:

$$Z = \frac{(8 + 0.5) - 12.25}{\sqrt{5.014}} = \frac{-3.75}{2.239} = -1.67.$$

Step 4: decide. Against $z_{0.025} = 1.96$, $|-1.67| < 1.96$, so the two-sided test does not quite reject; but if a downward-to-upward *shift* were the specific concern, the one-sided test at $\alpha = 0.05$ uses $z_{0.05} = 1.645$, and $-1.67 < -1.645$ *does* reject.

$Z = -1.67:$ two-sided retain ($\lvert Z\rvert < 1.96$), but one-sided reject for clustering($Z < -1.645$).

Too few runs is the signature of a trend; the borderline Z shows why stating the alternative direction in advance matters.

Example 14.6.5 (Interpreting a runs result: structure versus chance)**.** Two analysts each get a 30-long binary series with $n_1 = n_2 = 15$, for which $\mu_R = 1 + \frac{2(15)(15)}{30} = 16$ and $\sigma_R^2 = \frac{450(420)}{900(29)} = 7.241$. Analyst One counts $R = 9$ runs; Analyst Two counts $R = 24$. Standardize each and say what kind of departure from randomness, if any, each suggests at $\alpha = 0.05$.

Which rule, and why? Same null distribution for both, so the only question is where each R falls. The *sign* of Z names the structure: a large negative Z means clustering/trend, a large positive Z means over-alternation. This is the recognition payoff of the runs test — it diagnoses the *type* of nonrandomness, not just its presence.

Solution.

Step 1: Analyst One, $R = 9$. Below the mean, so $R + \frac{1}{2}$:

$$Z = \frac{(9 + 0.5) - 16}{\sqrt{7.241}} = \frac{-6.5}{2.691} = -2.42.$$

Step 2: Analyst Two, $R = 24$. Above the mean, so $R - \frac{1}{2}$:

$$Z = \frac{(24 - 0.5) - 16}{\sqrt{7.241}} = \frac{7.5}{2.691} = 2.79.$$

Step 3: interpret against $z_{0.025} = 1.96$.

One: $Z = -2.42 < -1.96 \Rightarrow$ too few runs: clustering / trend.
Two: $Z = +2.79 > +1.96 \Rightarrow$ too many runs: over-alternation.

Both reject randomness, but for opposite reasons: One's data clump, Two's data flip back and forth too regularly to be chance.

14.6.2 Spearman's rank correlation

Definition 14.6.6 (Spearman's rank correlation coefficient)**.** Given n paired observations (x_i, y_i), replace each x_i by its rank $\text{rank}(x_i)$ among the x's and each y_i by its rank $\text{rank}(y_i)$ among the y's (smallest = 1; ties get the average rank). **Spearman's** r_s is the ordinary Pearson correlation of these two rank columns. When there are no ties it simplifies to

$$r_s = 1 - \frac{6\sum_{i=1}^{n} d_i^2}{n(n^2-1)}, \qquad d_i = \text{rank}(x_i) - \text{rank}(y_i).$$

It runs from -1 (perfectly reversed rankings) through 0 (no monotone association) to $+1$ (identical rankings).

Remark. The shortcut and the "Pearson-of-the-ranks" definition agree exactly only when there are no ties, because the shortcut secretly uses $\sum \text{rank}^2 = \frac{1}{6}n(n+1)(2n+1)$, which fails once average ranks appear. With ties, fall back to the Pearson formula applied to the rank columns. A handful of ties barely moves r_s; many ties call for a tie correction, but the rank-Pearson computation is always valid.

Theorem 14.6.7 (Large-sample test of $H_0\colon \rho_s = 0$)**.** *Under H_0 (no monotone association), for n not too small the statistic*

$$t = r_s\sqrt{\frac{n-2}{1-r_s^2}}$$

follows approximately a t distribution with $n-2$ degrees of freedom. Reject for $|t|$ large (two sided) or for t in the appropriate tail (one sided). For small n, compare r_s directly to a table of Spearman critical values instead.

Example 14.6.8 (Spearman from ranks via the d-squared shortcut)**.** A panel ranks $n = 8$ coffee samples for aroma (x) and for taste (y). The ranks are

Sample	1	2	3	4	5	6	7	8
rank(x)	1	2	3	4	5	6	7	8
rank(y)	2	1	4	3	6	5	8	7

Compute r_s.

Which rule, and why? Both variables are already *ranks*, and we want their monotone agreement — that is Spearman's r_s. With no ties present, the d_i^2 shortcut is the fast route; d_i is just the gap between the two ranks of each sample.

Solution.

Step 1: rank differences and their squares.

	1	2	3	4	5	6	7	8	$\sum$
d_i	-1	1	-1	1	-1	1	-1	1	0
d_i^2	1	1	1	1	1	1	1	1	8

Step 2: apply the shortcut. With $n = 8$, $\sum d_i^2 = 8$:

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(8)}{8(64-1)} = 1 - \frac{48}{504} = 1 - 0.0952 = 0.905.$$

$$\boxed{r_s = 0.905}$$

Aroma and taste rankings agree strongly: high-aroma samples tend to be high-taste samples.

Example 14.6.9 (Test $H_0\colon \rho_s = 0$ with the large-sample t)**.** For the coffee panel above, $r_s = 0.905$ on $n = 8$ samples. Test, at $\alpha = 0.05$ two-sided, whether aroma and taste rankings are associated. Use $t_{0.025,6} = 2.447$.

Which rule, and why? We have r_s and want a p-value for $\rho_s = 0$. With n past the tiniest sizes the $t = r_s\sqrt{(n-2)/(1-r_s^2)}$ statistic on $n-2$ df does it; the df is $n-2=6$, not n.

Solution.

Step 1: form the statistic. With $r_s = 0.905$, $r_s^2 = 0.819$, $n-2=6$:

$$t = r_s\sqrt{\frac{n-2}{1-r_s^2}} = 0.905\sqrt{\frac{6}{1-0.819}} = 0.905\sqrt{\frac{6}{0.181}} = 0.905\sqrt{33.15} = 0.905(5.757) = 5.21.$$

Step 2: compare to the cutoff. $|t| = 5.21 > t_{0.025,6} = 2.447$.

$$\boxed{t = 5.21 > 2.447 \;\Rightarrow\; \text{reject } H_0\colon \rho_s = 0;\ \text{the rankings are associated.}}$$

Example 14.6.10 (Spearman from raw data: rank first, then correlate)**.** A wildlife study pairs the body mass x (kg) of $n = 7$ seals with their dive depth y (m):

	A	B	C	D	E	F	G
x	42	55	38	61	49	70	45
y	90	120	75	140	110	160	95

Find r_s and test $H_0\colon \rho_s = 0$ two-sided at $\alpha = 0.05$ ($t_{0.025,5} = 2.571$).

Which rule, and why? The data are numeric, but we want monotone (heavier seals dive deeper?) rather than strictly linear association, and n is small — so rank each variable and run Spearman. Convert to ranks first; the d_i^2 shortcut then applies because there are no ties.

Solution.

Step 1: rank each variable separately.

	A	B	C	D	E	F	G
rank(x)	2	5	1	6	4	7	3
rank(y)	2	5	1	6	4	7	3

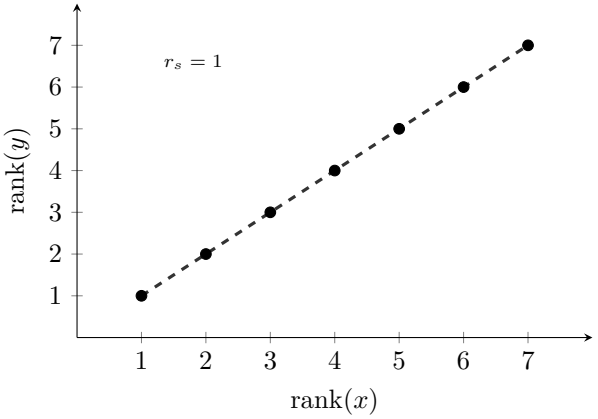
Step 2: differences. Every $d_i = 0$, so $\sum d_i^2 = 0$.

$$r_s = 1 - \frac{6(0)}{7(48)} = 1.$$

Step 3: test. With $r_s = 1$ the t-statistic is infinite (any positive cutoff is exceeded), so reject decisively:

$$\boxed{r_s = 1 \;\Rightarrow\; t \to \infty > 2.571;\ \text{reject } H_0;\ \text{mass and depth move together perfectly in rank.}}$$

The ranks line up exactly even though the raw (x, y) values are not a straight line — the hallmark of a *monotone* relationship.



Rank-vs-rank scatter for the seal data. When the two rankings agree perfectly the points sit exactly on the rising diagonal and $r_s = 1$, regardless of how curved the original mass–depth relationship was. Spearman sees only the ordering.

Example 14.6.11 (CONTRAST: Spearman versus Pearson on a monotone, nonlinear relationship)**.** An ecologist relates pollutant concentration x to a toxicity index y for $n = 6$ sites, where toxicity rises with concentration but *accelerates* (a cubic-like curve):

	1	2	3	4	5	6
x	1	2	3	4	5	6
y	1	8	27	64	125	216

Compute Spearman's r_s and contrast it with what Pearson's r reports. Which correlation honestly summarizes the association?

Which rule, and why? The relationship is perfectly increasing but strongly bent, so this is exactly the case that separates the two coefficients: Spearman (ranks) reads monotone agreement and should hit 1, while Pearson (raw values) penalizes the curvature and falls below 1.

Solution.

Step 1: Spearman. Since y increases with x, the ranks of x and y are identical ($1, 2, 3, 4, 5, 6$ each), so every $d_i = 0$ and

$$r_s = 1 - \frac{6(0)}{6(35)} = 1.$$

Step 2: Pearson on the raw values. With $\bar{x} = 3.5$ and $\bar{y} = (1 + 8 + 27 + 64 + 125 + 216)/6 = 441/6 = 73.5$, the centered products give

$$S_{xx} = \sum (x_i - \bar{x})^2 = 17.5, \quad S_{yy} = \sum (y_i - \bar{y})^2 = 32{,}458.5,$$
$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}) = 731.5,$$
$$r = \frac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}} = \frac{731.5}{\sqrt{17.5\,(32458.5)}} = \frac{731.5}{753.6} = 0.971.$$

Step 3: contrast.

$$\boxed{r_s = 1.000 \quad \text{but} \quad r = 0.971 < 1.}$$

Both are high, but only Spearman reaches 1: the toxicity ordering matches the concentration ordering *exactly*, which is the scientific truth here. Pearson is dragged below 1 because the cloud is curved, not straight. *When the question is "does y rise with x,"* Spearman is the honest summary; reserve Pearson for when a straight-line fit is genuinely the model of interest. The gap would widen for sharper curvature, and an outlier would dent Pearson while leaving the rank-based r_s nearly untouched.

Example 14.6.12 (Tie handling: average ranks, then Pearson of the ranks)**.** A teacher ranks $n = 6$ essays for *clarity* (x) and *originality* (y), but two essays tie for clarity:

	A	B	C	D	E	F
x (clarity)	3	1	4	4	2	5
y (originality)	2	1	5	3	4	6

The two $x = 4$ scores (C and D) tie. Assign average ranks and compute r_s.

Which rule, and why? A tie means the d_i^2 shortcut is no longer exact, so the rule is: give tied values the *average* of the ranks they would occupy, then take the Pearson correlation of the two rank columns. This is the always-valid definition of r_s.

Solution.

Step 1: rank with averages for ties. Clarity values $1, 2, 3, 4, 4, 5$ would occupy ranks $1, 2, 3, 4, 5, 6$; the two 4's share ranks 4 and 5, so each gets 4.5:

	A	B	C	D	E	F
rank(x)	3	1	4.5	4.5	2	6
rank(y)	2	1	5	3	4	6

Step 2: Pearson correlation of the rank columns. Both rank columns have mean 3.5. The centered ranks are

$$\begin{aligned} u_i = \text{rank}(x_i) - 3.5: &\ -0.5,\ -2.5,\ 1.0,\ 1.0,\ -1.5,\ 2.5, \\ v_i = \text{rank}(y_i) - 3.5: &\ -1.5,\ -2.5,\ 1.5,\ -0.5,\ 0.5,\ 2.5, \end{aligned}$$

giving

$$\begin{aligned} \textstyle\sum u_i v_i &= 0.75 + 6.25 + 1.5 - 0.5 - 0.75 + 6.25 = 13.5, \\ \textstyle\sum u_i^2 &= 0.25 + 6.25 + 1 + 1 + 2.25 + 6.25 = 17.0, \\ \textstyle\sum v_i^2 &= 2.25 + 6.25 + 2.25 + 0.25 + 0.25 + 6.25 = 17.5. \end{aligned}$$

Step 3: combine.

$$r_s = \frac{\sum u_i v_i}{\sqrt{\sum u_i^2 \sum v_i^2}} = \frac{13.5}{\sqrt{17.0\,(17.5)}} = \frac{13.5}{17.25} = 0.783.$$

$\boxed{r_s = 0.783 \text{ (average ranks for the tie; Pearson of the rank columns).}}$

Had we wrongly used the d_i^2 shortcut with the average ranks, we would have gotten a slightly different number; the rank-Pearson value is the correct one.

Example 14.6.13 (One-sided Spearman test: does experience improve a score?)**.** A trainer suspects that more practice hours (x) raise a skills score (y). For $n = 10$ trainees the rank correlation comes out $r_s = 0.62$. Test, one-sided, $H_0\colon \rho_s = 0$ against $H_1\colon \rho_s > 0$ at $\alpha = 0.05$, using $t_{0.05,8} = 1.860$.

Which rule, and why? The hypothesis names a *direction* (more practice $\Rightarrow$ higher score), so this is a one-sided Spearman test. Use the same $t = r_s\sqrt{(n-2)/(1-r_s^2)}$ but reject only in the upper tail, comparing to the one-sided cutoff $t_{0.05,8}$.

Solution.

Step 1: statistic. With $r_s = 0.62$, $r_s^2 = 0.3844$, $n - 2 = 8$:

$$t = 0.62\sqrt{\frac{8}{1 - 0.3844}} = 0.62\sqrt{\frac{8}{0.6156}} = 0.62\sqrt{12.995} = 0.62(3.605) = 2.23.$$

Step 2: one-sided decision. $t = 2.23 > t_{0.05,8} = 1.860$.

$\boxed{t = 2.23 > 1.860 \ \Rightarrow\ \text{reject } H_0;\ \text{practice and score are positively associated.}}$

Note the same $r_s = 0.62$ would give a two-sided cutoff $t_{0.025,8} = 2.306$, which 2.23 would *not* clear — stating the direction in advance bought the rejection.

Example 14.6.14 (CONTRAST: a runs result that is random versus one that is structured)**.** Two trading desks each log whether the daily return beat (B) or trailed (T) a benchmark over $n = 40$ days, with $n_1 = n_2 = 20$, so $\mu_R = 1 + \dfrac{2(20)(20)}{40} = 21$ and $\sigma_R^2 = \dfrac{800(760)}{1600(39)} = 9.744$. Desk One records $R = 22$ runs; Desk Two records $R = 11$. Decide for each, at $\alpha = 0.05$, whether the beat/trail pattern is random or shows structure.

Which rule, and why? Identical null distribution, so the contrast is purely in the counts. The runs test separates "random noise" (count near the mean) from "structure" (count far in a tail), and the tail names the structure — few runs means streaky/persistent, many means choppy/mean-reverting.

Solution.

Step 1: Desk One, $R = 22$. Just above the mean, use $R - \frac{1}{2}$:

$$Z = \frac{(22 - 0.5) - 21}{\sqrt{9.744}} = \frac{0.5}{3.122} = 0.16.$$

Step 2: Desk Two, $R = 11$. Well below the mean, use $R + \frac{1}{2}$:

$$Z = \frac{(11 + 0.5) - 21}{\sqrt{9.744}} = \frac{-9.5}{3.122} = -3.04.$$

Step 3: interpret against $z_{0.025} = 1.96$.

Desk One: $Z = 0.16 \Rightarrow$ random (no structure).
Desk Two: $Z = -3.04 \Rightarrow$ too few runs: streaky / persistent.

Desk One's beats and misses are scattered as chance predicts; Desk Two's clump into long winning and losing streaks — strong evidence of serial dependence, not luck. The runs test thus turns a vague "looks streaky" into a quantified verdict.

14.7 Exercises

All problems below are original. Full solutions begin on page 975.

A. The sign test

A1. A hydrologist claims the median dissolved-oxygen level in a stream is $\eta_0 = 8.0$ mg/L. Eleven grab samples (mg/L) read

$$6.4,\ 9.1,\ 7.2,\ 10.5,\ 5.8,\ 8.7,\ 9.6,\ 4.9,\ 11.8,\ 7.5,\ 6.1.$$

The data are right-skewed by the two high readings. Test $H_0 : \eta = 8.0$ versus $H_1 : \eta \neq 8.0$ at $\alpha = 0.05$ with the exact sign test. State which parametric test the sign test replaces here.

A2. A pediatric clinic claims the median weight gain of infants over a month exceeds the benchmark $\eta_0 = 600$ g. Twelve infants gain (g)

$$720,\ 540,\ 810,\ 660,\ 590,\ 705,\ 880,\ 630,\ 770,\ 555,\ 690,\ 640.$$

Carry out an exact one-sided sign test of $H_0 : \eta = 600$ versus $H_1 : \eta > 600$ at $\alpha = 0.05$.

A3. A logistics manager believes the median delivery delay on a route is *below* the service guarantee of $\eta_0 = 15$ minutes. Ten shipments are delayed (minutes)

$$8,\ 22,\ 11,\ 6,\ 14,\ 9,\ 18,\ 7,\ 12,\ 5.$$

Test $H_0 : \eta = 15$ versus $H_1 : \eta < 15$ at $\alpha = 0.05$ with the exact sign test, and say which parametric procedure it replaces.

A4. A vineyard tests a new pruning method by comparing yield (kg per vine) on the same nine vines before and after the change. The (before, after) pairs are

before	12.3	9.8	14.1	11.0	13.5	10.2	15.6	8.9	12.7
after	13.1	10.4	13.8	11.9	14.0	11.5	16.2	9.7	12.5

(a) Form the differences $D_i = \text{after} - \text{before}$ and run a paired sign test of H_0 : median difference $= 0$ versus H_1 : median difference > 0 at $\alpha = 0.05$.

(b) State the parametric test this replaces, and one reason a researcher might still prefer the sign test here.

A5. Two laboratory thermometers are read side by side on twelve water baths to see whether one reads systematically higher. The (instrument A, instrument B) readings (°C) are

A	21.4	25.0	19.8	30.1	22.7	27.3	18.5	24.6	29.0	20.2	26.4	23.8
B	21.0	25.0	20.1	29.5	22.2	26.9	18.9	24.0	28.4	20.2	25.8	23.1

(a) Note that two baths give $D_i = A_i - B_i = 0$. Apply the tie-handling rule and report the working sample size n.

(b) Run a two-sided paired sign test of H_0 : median difference $= 0$ versus $H_1 : \neq 0$ at $\alpha = 0.05$.

A6. A city audits whether the median wait time at a license office equals the posted target $\eta_0 = 20$ minutes. Of $n = 64$ independent visits (none exactly 20 minutes), $s_{\text{obs}} = 40$ wait longer than 20 minutes.

(a) Use the large-sample normal approximation with a continuity correction to test $H_0 : \eta = 20$ versus $H_1 : \eta \neq 20$ at $\alpha = 0.05$.

(b) Repeat the test *without* the continuity correction and comment on whether the decision changes.

A7. A taste panel of 14 judges each compares a reformulated cereal against the original and records which they prefer; a tie (no preference) is allowed. Eleven judges prefer the reformulation, two prefer the original, and one is tied. The company claims judges prefer the reformulation, i.e. its median preference difference is positive.

(a) Explain why the sign test is the natural tool for these purely ordinal preference data, and which parametric paired test it stands in for.

(b) Discard the tie and run an exact one-sided sign test of H_0 : no preference (median difference $= 0$) versus H_1 : reformulation preferred, at $\alpha = 0.05$.

(c) Briefly contrast the sign test's robustness against its power relative to the Wilcoxon signed-rank test for data like these.

B. The Wilcoxon signed-rank test

B1. A coffee roaster advertises a median brew temperature of $m_0 = 93$ degrees Celsius. A technician samples $n = 8$ batches and records

$$94.2, \quad 91.5, \quad 95.1, \quad 92.0, \quad 96.3, \quad 93.8, \quad 90.4, \quad 95.7.$$

Assume the temperatures are symmetric about their median and test $H_0 : m = 93$ versus $H_1 : m \neq 93$ at $\alpha = 0.05$, using the small-sample critical value $w_{0.05,8} = 3$ (two-sided; reject when $\min(W_+, W_-) \leq 3$). Rank the absolute deviations in a table, report W_+, W_-, verify the identity $W_+ + W_- = n(n+1)/2$, and state the decision. Which parametric test does this replace?

B2. Nine office workers wear a posture-correcting device for two weeks. Their daily back-pain scores (on a 0 to 20 scale) are taken before and after; the within-person differences $D_i = \text{before} - \text{after}$ (a positive value means the pain went down) are

$$3.1, \quad 1.4, \quad -0.6, \quad 4.0, \quad 2.2, \quad 0.9, \quad 5.3, \quad -1.1, \quad 2.8.$$

Test H_0 : no change versus H_1 : the device reduces pain ($m > 0$) at $\alpha = 0.05$. The one-sided critical value is $w_{0.05,9} = 8$ (reject when $W_- \leq 8$).

(a) Rank the $|D_i|$, build the signed-rank table, and compute W_+ and W_-.

(b) State which rank sum the alternative drives down, and give the decision in words.

(c) Name the parametric test this replaces and the assumption that lets it stand in.

B3. A water authority's permit caps the median nitrate level of a discharge at $m_0 = 10$ mg/L. An inspector suspects the true median is *below* the cap and samples $n = 10$ days, obtaining deviations $D_i = X_i - 10$ (mg/L)

$$-1.5, \ -2.8, \ +0.7, \ -3.6,$$
$$-0.4, \ +1.9, \ -2.2, \ -4.1,$$
$$-0.9, \ -1.3.$$

Test $H_0 : m = 10$ versus $H_1 : m < 10$ at $\alpha = 0.05$ with one-sided critical value $w_{0.05,10} = 10$ (reject when $W_+ \leq 10$). Rank the magnitudes, compute W_+ and W_-, and decide. Explain why the alternative makes W_+ the statistic to watch.

B4. A logistics firm reworks its packing line and records the change in order-fill time (seconds) for $n = 28$ orders. The positive rank sum is $W_+ = 305$.

(a) Compute the null mean $\mathbb{E}[W_+] = n(n+1)/4$ and variance $\mathrm{Var}[W_+] = n(n+1)(2n+1)/24$.

(b) Standardize to W^* and test H_0 : no shift versus H_1 : a shift in either direction at $\alpha = 0.05$ using the normal approximation.

(c) State the decision and which parametric test the procedure replaces.

B5. A panel of $n = 10$ judges scores a redesigned package against the old one on a -5 to $+5$ preference scale (positive favors the redesign):

$$+3,\ -1,\ +3,\ 0,\ +2,\ -1,\ +3,\ 0,\ +4,\ +2.$$

(a) Drop the zeros, average the ranks within tied magnitudes, and build the signed-rank table; report the reduced n, W_+, and W_-.

(b) Using the small-sample two-sided critical value $w_{0.05,8} = 3$, decide $H_0 : m = 0$ versus $H_1 : m \neq 0$ at $\alpha = 0.05$.

B6. Two facts about a signed-rank computation.

(a) A test on $n = 14$ nonzero deviations reports $W_+ = 72$. Find W_- without re-ranking and verify the consistency identity.

(b) A second test on $n = 11$ nonzero deviations reports $W_- = 51$. Find W_+, then say whether the data sit mostly above or below the hypothesized center, and why.

B7. Six chronic-pain patients have their flare-up frequency (flares per month) measured before and after a nerve-block procedure; the reductions $D_i = \text{before} - \text{after}$ are

$$4,\ 9,\ -1,\ 22,\ 6,\ -2.$$

The 22 is a real but extreme responder.

(a) Run the signed-rank test of H_0: no change versus $H_1 : m > 0$ at $\alpha = 0.05$, with one-sided cutoff: reject when $W_- \leq w_{0.05,6} = 2$.

(b) For contrast, the paired t on these raw reductions gives $\overline{D} = 6.333$, $s_D = 8.59$, so $t = \overline{D}/(s_D/\sqrt{6})$. Compute t and test against $t_{0.05,5} = 2.015$.

(c) Explain why the two tests disagree, and which one to trust here.

C. The Wilcoxon rank-sum (Mann–Whitney) test

C1. A field biologist measures perch length (cm) in two ponds that differ in algae cover. The samples are small and the data look skewed, so a rank-based comparison is used in place of the two-sample t.

$$\begin{aligned} X \text{ (shaded pond, } n = 5): &\quad 9.1,\ 11.4,\ 8.7,\ 12.0,\ 10.3, \\ Y \text{ (open pond, } m = 6): &\quad 13.2,\ 11.9,\ 14.5,\ 12.6,\ 15.1,\ 13.8. \end{aligned}$$

Pool, rank, and compute the rank sum W of the smaller (shaded) group. The two-sided 5% critical values for $n = 5, m = 6$ are $W_L = 19$ and $W_U = n(N+1) - W_L = 41$.

(a) State the hypotheses in distribution/median language and name the parametric test this replaces.

(b) Build the pooled ranking table (booktabs) and report W.

(c) Decide at $\alpha = 0.05$ and state the conclusion in words.

C2. Using the shaded/open pond data of problem C1 ($W = 16$, $n = 5$, $m = 6$):

(a) Convert to the Mann–Whitney statistic via $U = W - \frac{n(n+1)}{2}$.

(b) Recompute U by directly counting, for each shaded-pond value, how many open-pond values it exceeds, and confirm the two agree.

(c) Report $\mathbb{E}[U] = \frac{nm}{2}$ and say in one sentence what the gap between U and $\mathbb{E}[U]$ signals.

C3. A coach believes a new warm-up routine *lowers* sprint times (s). A control group (X, $n = 6$) uses the old routine and a treatment group (Y, $m = 6$) uses the new one:

$$\begin{aligned} X \text{ (old)}: &\quad 12.8,\ 13.1,\ 12.5,\ 13.4,\ 12.9,\ 13.0, \\ Y \text{ (new)}: &\quad 12.3,\ 12.6,\ 12.1,\ 12.7,\ 12.4,\ 12.2. \end{aligned}$$

Test at $\alpha = 0.05$ the one-sided claim that the new routine shifts sprint times *down*. For $n = 6, m = 6$ the one-sided lower critical sum for the smaller group's rank sum is $W_{0.05} = 28$. Decide which tail to use, compute W for the group you sum, and give the verdict in words.

C4. Two call centers are compared on handling time (min). Pooling and ranking all $N = 27$ calls, the center-A ranks ($n = 13$) sum to $W = 235$; center B has $m = 14$.

(a) Compute the null mean μ_W and standard deviation σ_W.

(b) Apply the normal approximation with a continuity correction and report z.

(c) Test two-sided at $\alpha = 0.05$ ($z_{0.025} = 1.96$), give the approximate p-value, and state the conclusion.

C5. A panel scores brittleness of two ceramic glazes, with several ties:

$$\begin{aligned} X \text{ (glaze A, } n = 5): &\quad 3,\ 5,\ 5,\ 6,\ 8, \\ Y \text{ (glaze B, } m = 5): &\quad 5,\ 7,\ 7,\ 9,\ 10. \end{aligned}$$

(a) Assign pooled ranks using *average ranks* for ties and tabulate them (booktabs).

(b) Compute W for glaze A and the null mean μ_W.

(c) Compute the tie-corrected variance $\sigma_W^2 = \frac{nm}{12}\big[(N+1) - \frac{\sum_j (t_j^3 - t_j)}{N(N-1)}\big]$ and compare it with the uncorrected value.

C6. In a usability study, $n = 8$ users test interface X and a separate $m = 8$ users test interface Y; task time (s) for Y includes one value of 4200 s (a user who walked away mid-task). The pooled ranking puts the eight X times at ranks 1–5, 7, 9, 11 and the eight Y times at ranks 6, 8, 10, 12–16 (the 4200 s value being the largest, rank 16).

(a) Compute W for interface X.

(b) For $n = m = 8$ the two-sided 5% critical sums are $W_L = 49$ and $W_U = 87$. Decide at $\alpha = 0.05$.

(c) Explain in two sentences why the 4200 s value would have derailed the two-sample t but leaves the rank-sum verdict unchanged whether it reads 200 s or 4200 s.

C7. For each study, pick the correct two-sample procedure — two-sample t, paired (signed-rank) test, or Wilcoxon rank-sum — and give a one-line reason.

(a) Resting heart rate for 50 runners and 50 non-runners (different people), both distributions bell-shaped with no outliers.

(b) Income (\$) for 9 households in district A and 11 *other* households in district B, strongly right-skewed with one millionaire.

(c) Reaction time of 15 subjects measured under caffeine and again, on the same subjects, under placebo.

(d) Satisfaction graded poor/fair/good/excellent for 10 customers of vendor X and 10 of vendor Y.

D. The Kruskal–Wallis test

D1. A horticulturist measures vase-life (days) of cut tulips kept in $k = 3$ preservative solutions, $n_1 = n_2 = n_3 = 4$ ($N = 12$), and suspects the solutions are not equivalent:

$$\begin{aligned}\text{Solution P}&: \ 6,\ 8,\ 11,\ 13,\\ \text{Solution Q}&: \ 7,\ 10,\ 14,\ 16,\\ \text{Solution R}&: \ 5,\ 9,\ 12,\ 17.\end{aligned}$$

At $\alpha = 0.05$, test H_0: the three solutions share one vase-life distribution against H_1: they do not ($\chi^2_{0.05,2} = 5.991$). Pool, rank, build the rank-sum table, compute H, and state the decision. Which parametric test does this replace?

D2. An agronomist records grain yield (bushels/plot) under $k = 4$ fertilizer programs with *unequal* plot counts, $n_1 = 4,\ n_2 = 5,\ n_3 = 3,\ n_4 = 4$ ($N = 16$):

$$\begin{aligned}\text{Program 1}&: \ 22,\ 27,\ 31,\ 40,\\ \text{Program 2}&: \ 24,\ 29,\ 34,\ 38,\ 44,\\ \text{Program 3}&: \ 25,\ 33,\ 42,\\ \text{Program 4}&: \ 23,\ 28,\ 36,\ 47.\end{aligned}$$

Test at $\alpha = 0.05$ whether the programs differ ($\chi^2_{0.05,3} = 7.815$). Be explicit about dividing each R_j^2 by its own n_j, and verify the rank-sum total as an arithmetic check.

D3. A QA lead rates surface finish (1–10 ordinal scale, so ties occur) for $k = 3$ machining centers, $n_1 = n_2 = n_3 = 5$ ($N = 15$):

$$\begin{aligned}\text{Center A}&: \ 5,\ 6,\ 6,\ 8,\ 9,\\ \text{Center B}&: \ 4,\ 5,\ 7,\ 7,\ 8,\\ \text{Center C}&: \ 3,\ 5,\ 6,\ 7,\ 10.\end{aligned}$$

(a) Assign *average* ranks for ties, tabulate the group rank sums, and compute the uncorrected H. Decide at $\alpha = 0.05$ ($\chi^2_{0.05,2} = 5.991$).

(b) Apply the tie-correction factor $1 - \sum_g (t_g^3 - t_g)/(N^3 - N)$ and report the corrected H. Does the conclusion change?

D4. Highway noise (dB) is logged at $k = 5$ road-surface types over $N = 50$ readings, $n_j = 10$ each. The pooled-rank sums are

$$R_1 = 205, \quad R_2 = 310, \quad R_3 = 245, \quad R_4 = 190, \quad R_5 = 325.$$

With every $n_j = 10$, use the large-sample χ^2_4 approximation to test at $\alpha = 0.05$ ($\chi^2_{0.05,4} = 9.488$). First confirm $\sum_j R_j = N(N+1)/2$, then compute H and decide.

D5. Two coffee-roasting profiles are scored for aroma intensity (0–100) by independent panels, $n_1 = 5,\ n_2 = 5$ ($N = 10$):

$$\text{Profile 1}: \ 41,\ 47,\ 52,\ 58,\ 63; \qquad \text{Profile 2}: \ 44,\ 49,\ 55,\ 60,\ 66.$$

(a) Compute the Kruskal–Wallis H for these $k = 2$ groups (reference χ^2_1, $\chi^2_{0.05,1} = 3.841$).

(b) Compute the Wilcoxon rank-sum standardized statistic Z for Profile 1, using $\mathbb{E}[R_1] = n_1(N+1)/2$ and $\mathrm{Var}(R_1) = n_1 n_2 (N+1)/12$, and verify $H = Z^2$.

(c) Explain in one sentence why the two tests must agree when $k = 2$.

D6. A behavioral economist has discretionary spending (\$) for $k = 3$ neighborhoods, $n_j = 5$ each, with one extreme spender per neighborhood:

$$\begin{aligned}\text{Area X}&: \ 40,\ 55,\ 70,\ 85,\ 900,\\ \text{Area Y}&: \ 45,\ 60,\ 75,\ 88,\ 150,\\ \text{Area Z}&: \ 42,\ 58,\ 72,\ 80,\ 300.\end{aligned}$$

(a) Pool and rank the data, build the rank sums, and compute H ($\chi^2_{0.05,2} = 5.991$). Decide at $\alpha = 0.05$.

(b) Explain how the lone large outlier in Area X would distort a one-way ANOVA F-test on the raw dollars, and say which of the two tests you would trust here and why.

D7. Reaction times (ms) under $k = 4$ ergonomic keyboard layouts are collected over $N = 40$ observations, $n_j = 10$ each, with pooled-rank sums

$$R_1 = 150, \quad R_2 = 300, \quad R_3 = 160, \quad R_4 = 210.$$

(a) Compute the omnibus H and decide at $\alpha = 0.05$ ($\chi^2_{0.05,3} = 7.815$).

(b) If significant, use the large-sample rank-based comparison

$$|\overline{R}_i - \overline{R}_j| > z^* \sqrt{\tfrac{N(N+1)}{12}\left(\tfrac{1}{n_i} + \tfrac{1}{n_j}\right)}$$

with Bonferroni $z^* = z_{0.05/[k(k-1)]} = z_{0.00417} = 2.64$ to identify which layout stands apart. Report the group average ranks $\overline{R}_j$ and the critical gap.

E. The Friedman test

E1. A coffee shop has $b = 5$ regular customers (the blocks) each taste $k = 3$ espresso blends and score them on a 0–100 scale; every customer tastes all three blends, so this is a repeated-measures layout and normality of the block ANOVA is doubtful. Higher is better.

Customer	Blend A	Blend B	Blend C
1	72	80	68
2	65	70	63
3	88	84	90
4	55	61	52
5	77	82	74

(a) State H_0 and H_1 in words and name the parametric test the Friedman test replaces here.

(b) Rank within each customer, report the rank sums R_i, and verify their total.

(c) Compute Q and test at $\alpha = 0.05$ using $\chi^2_{0.05,2} = 5.991$. State the decision in words.

E2. Six panelists ($b = 6$) each rate $k = 4$ brands of dark chocolate on a 1–9 liking scale (each panelist rates all four brands). After ranking within each panelist (1 = least liked, 4 = most liked) the brand rank sums came out to

$$R_{\mathrm{W}} = 9, \qquad R_{\mathrm{X}} = 21, \qquad R_{\mathrm{Y}} = 11, \qquad R_{\mathrm{Z}} = 19.$$

Test at $\alpha = 0.05$ whether liking differs across brands, using $\chi^2_{0.05,3} = 7.815$. Check the rank-sum total, compute Q, and box the decision.

E3. In a sleep study $b = 8$ volunteers each try $k = 3$ mattresses and report a comfort rating; each volunteer is a block. After within-block ranking the mattress rank sums are

$$R_1 = 11, \qquad R_2 = 20, \qquad R_3 = 17.$$

(a) Compute Q for these $b = 8$ blocks and $k = 3$ treatments.

(b) Bracket the p-value using the upper-tail χ^2_2 cutoffs $\chi^2_{0.10,2} = 4.605$, $\chi^2_{0.05,2} = 5.991$, $\chi^2_{0.025,2} = 7.378$, $\chi^2_{0.01,2} = 9.210$.

(c) Decide at $\alpha = 0.05$ and state the conclusion in words.

E4. One block of a $k = 5$ taste-test study produced the five responses

$$41, \qquad 53, \qquad 53, \qquad 53, \qquad 60.$$

(a) Assign the within-block ranks using the average-rank convention for the tie.

(b) Verify that the block's rank total is still $k(k+1)/2$.

(c) A Friedman analysis of this design has $b = 7$ blocks and only this one tied block (a tie group of size $t = 3$). Compute the tie-correction factor $1 - \frac{\sum(t^3-t)}{b\,k(k^2-1)}$ and say whether the correction raises or lowers Q.

E5. Seven judges ($b = 7$) score $k = 2$ figure-skating routines; routine I is placed above routine II by $a = 6$ of the 7 judges. Two treatments in blocks is the boundary case where Friedman collapses to the sign test.

(a) Using the reduced formula $Q = (2a - b)^2/b$, compute Q.

(b) Cross-check by computing the sign-test $Z = \dfrac{a - b/2}{\sqrt{b}/2}$ and confirming $Z^2 = Q$.

(c) Decide at $\alpha = 0.05$ using $\chi^2_{0.05,1} = 3.841$ (equivalently $z_{0.025} = 1.96$). State which parametric test the $k = 2$ Friedman test replaces.

E6. Five reviewers ($b = 5$, the blocks) score $k = 3$ grant proposals on a 100-point scale. One reviewer is a harsh marker whose scores sit far below the others' — a large block offset, and with subjective scoring the normality behind a block F-test is shaky.

Reviewer	Proposal P	Proposal Q	Proposal R
1	81	76	88
2	84	78	90
3	49	44	55
4	86	80	92
5	83	77	89

(a) Rank within each reviewer, report the rank sums, and compute Q.

(b) Test at $\alpha = 0.05$ using $\chi^2_{0.05,2} = 5.991$.

(c) Explain how ranking within blocks neutralizes Reviewer 3's offset, and state when the randomized block F-test would instead be the better choice.

E7. A small pilot compares $k = 3$ irrigation schedules across $b = 4$ greenhouse benches (blocks) and yields $Q = 6.10$. A Friedman exact table for $(k, b) = (3, 4)$ gives $\mathbb{P}(Q \geq 6.10 \mid H_0) = 0.052$, while the large-sample chi-square approximation uses $\chi^2_{0.05,2} = 5.991$.

(a) Give the verdict at $\alpha = 0.05$ from the exact table.

(b) Give the verdict from the chi-square approximation.

(c) The two disagree. Say which reference to trust for $b = 4$ and why, and state the danger of relying on the chi-square cutoff when b is small.

F. Runs test, rank correlation, and synthesis

F1. A casino dealer records, for $n = 22$ consecutive roulette spins, whether the ball landed on red (R) or black (B). In order the sequence is

R R B R B B R R R B B R B R R B B R B R R B,

with $n_1 = 12$ R's and $n_2 = 10$ B's. A pit boss wants to know whether the color order is random.

(a) State H_0 and H_1 in words and name the test.

(b) Count the runs R, and compute μ_R and σ^2_R under randomness.

(c) Standardize (with continuity correction) and decide at $\alpha = 0.05$, two-sided.

F2. Over a 30-day stretch an air-quality monitor flags each day as above (A) or below (B) the seasonal median. There are $n_1 = 15$ A's and $n_2 = 15$ B's, and the analyst counts only $R = 10$ runs.

(a) Compute μ_R and σ_R^2, then the continuity-corrected z.

(b) At $\alpha = 0.05$ two-sided, is the sequence random? What kind of departure (clustering or over-alternation) does the sign of z indicate, and what does that mean physically?

F3. An economist ranks $n = 9$ countries by adult literacy rate (x) and by per-capita GDP (y). The ranks (1 = lowest) are

Country	1	2	3	4	5	6	7	8	9
rank(x)	1	2	3	4	5	6	7	8	9
rank(y)	2	1	4	3	5	7	6	9	8

(a) Compute Spearman's r_s with the d_i^2 shortcut.

(b) Test $H_0\colon \rho_s = 0$ against $H_1\colon \rho_s \neq 0$ at $\alpha = 0.05$, using $t_{0.025,7} = 2.365$. State which parametric procedure this rank test stands in for.

F4. A biologist measures, for $n = 7$ ponds, the surface area x (hectares) and the number of fish species y:

Pond	A	B	C	D	E	F	G
x	0.5	1.2	2.0	3.5	6.0	9.0	14.0
y	3	5	6	8	11	13	16

Rank each variable, compute r_s, and test $H_0\colon \rho_s = 0$ against $H_1\colon \rho_s > 0$ one-sided at $\alpha = 0.05$ ($t_{0.05,5} = 2.015$).

F5. For $n = 6$ chemical reactors, catalyst loading x and reaction yield y are

	1	2	3	4	5	6
x	1	2	3	4	5	6
y	2	9	28	65	126	217

(yield grows like $x^3 + 1$).

(a) Compute Spearman's r_s.

(b) Compute Pearson's r on the raw values (you may use $\bar{x} = 3.5$, $\bar{y} = 74.5$, $S_{xx} = 17.5$, $S_{yy} = 34{,}757.5$, $S_{xy} = 731.5$).

(c) Contrast the two coefficients and say which honestly summarizes the association, and why.

F6. For each scenario, NAME the appropriate nonparametric test, state which parametric test it replaces, and explain in one sentence why the parametric test is inappropriate here.

(a) A nutritionist measures cholesterol in the same 12 patients before and after a diet; the paired differences are strongly right-skewed with two extreme outliers, and she wants to know whether the diet lowered the median cholesterol.

(b) A clinic compares pain-relief times for three different analgesics, with 8 patients each; the response is recorded on a 0–10 ordinal pain scale, so means are not meaningful.

(c) A meteorologist has $n = 200$ daily "rain / no-rain" records in calendar order and wants to know whether wet days cluster (storm systems) rather than occurring independently.

F7. A physical therapist compares two stretching routines. Eight patients are matched into pairs by age and baseline flexibility; within each pair one patient does routine X and the other routine Y. The improvement (degrees of range-of-motion gained) is

Pair	1	2	3	4	5	6	7	8
X	14	9	21	18	7	25	12	16
Y	10	12	13	11	8	15	6	9

The pair-differences look non-normal, so a paired t-test is unsafe.

(a) Name the test that replaces the paired t here and carry it out: rank the absolute differences, attach signs, and compute the signed-rank statistic.

(b) Using the small-sample critical value $w_{0.05} = 4$ (two-sided, $n = 8$, lower-tail reject when the smaller signed-rank sum $\le w$), decide at $\alpha = 0.05$ whether the routines differ.

(c) Give the large-sample z as a cross-check, using $\mathbb{E}(W^+) = n(n+1)/4$ and $\mathrm{Var}(W^+) = n(n+1)(2n+1)/24$.

14.8 Solutions

A. The sign test

A1. Exact two-sided sign test for a median. The claim is about a *median* and the data are right-skewed, so the sign test (replacing the one-sample t-test) is appropriate: record the sign of each $X_i - 8.0$, discard ties, and compare S_+ against $\text{Bin}(n, \frac{1}{2})$.

X_i	6.4	9.1	7.2	10.5	5.8	8.7	9.6	4.9	11.8	7.5	6.1
sign of $X_i - 8.0$	$-$	$+$	$-$	$+$	$-$	$+$	$+$	$-$	$+$	$-$	$-$

No reading equals 8.0, so $n = 11$ with 5 pluses and 6 minuses, giving $s_{\text{obs}} = 5$. Here $s_{\text{obs}} = 5$ sits essentially at the expected count $n/2 = 5.5$, so the data could hardly be more consistent with H_0. Formally, with $S_+ \sim \text{Bin}(11, \frac{1}{2})$,

$$\begin{aligned} \mathbb{P}(S_+ \le 5) = \mathbb{P}(S_+ \ge 6) &= 0.5, \\ p\text{-value} &= 2\,\min\{0.5,\, 0.5\} = 1.0. \end{aligned}$$

Since $1.0 > 0.05$, do not reject. There is no evidence the median dissolved oxygen differs from 8.0 mg/L.

$$\boxed{s_{\text{obs}} = 5,\ n = 11,\ p\text{-value} = 1.0 > 0.05:\ \text{median consistent with 8.0 mg/L.}}$$

A2. Exact one-sided (upper-tail) sign test. A directional claim about a median calls for the one-sided sign test (replacing the one-sample t); large S_+ supports $H_1 : \eta > 600$, so the p-value is $\mathbb{P}(S_+ \ge s_{\text{obs}})$.

X_i	720	540	810	660	590	705	880	630	770	555	690	640
sign of $X_i - 600$	$+$	$-$	$+$	$+$	$-$	$+$	$+$	$+$	$+$	$-$	$+$	$+$

No gain equals 600, so $n = 12$, with 9 pluses; $s_{\text{obs}} = 9$. With $S_+ \sim \text{Bin}(12, \frac{1}{2})$,

$$\begin{aligned} \mathbb{P}(S_+ \ge 9) &= \frac{\binom{12}{9} + \binom{12}{10} + \binom{12}{11} + \binom{12}{12}}{2^{12}} \\ &= \frac{220 + 66 + 12 + 1}{4096} = \frac{299}{4096} = 0.073. \end{aligned}$$

Since $0.073 > 0.05$, do not reject at the 5% level. Nine of twelve infants gaining more than 600 g leans the right way but is not quite a strong enough sign imbalance.

$$\boxed{s_{\text{obs}} = 9,\ n = 12,\ p\text{-value} = 0.073 > 0.05:\ \text{insufficient evidence that } \eta > 600 \text{ g.}}$$

A3. Exact one-sided (lower-tail) sign test. The claim points downward ($H_1 : \eta < 15$), so the lower tail is the rejection region and the p-value is $\mathbb{P}(S_+ \le s_{\text{obs}})$. Delay data are right-skewed, so the sign test (replacing the one-sample t-test) is the natural median test.

X_i	8	22	11	6	14	9	18	7	12	5
sign of $X_i - 15$	$-$	$+$	$-$	$-$	$-$	$-$	$+$	$-$	$-$	$-$

No delay equals 15, so $n = 10$, with 2 pluses; $s_{\text{obs}} = 2$. With $S_+ \sim \text{Bin}(10, \frac{1}{2})$,

$$\begin{aligned} \mathbb{P}(S_+ \le 2) &= \frac{\binom{10}{0} + \binom{10}{1} + \binom{10}{2}}{2^{10}} \\ &= \frac{1 + 10 + 45}{1024} = \frac{56}{1024} = 0.0547. \end{aligned}$$

Since $0.0547 > 0.05$, do not reject (barely). At $\alpha = 0.05$ the evidence falls just short of establishing that the median delay is below 15 minutes.

> $s_{\text{obs}} = 2$, $n = 10$, p-value $= 0.055 > 0.05$: do not reject; median delay not shown below 15 min.

A4. Paired sign test (replacing the paired t-test).

(a) Form $D_i = \text{after} - \text{before}$ and test with $\eta_0 = 0$; large S_+ supports H_1 : median difference > 0.

before	12.3	9.8	14.1	11.0	13.5	10.2	15.6	8.9	12.7
after	13.1	10.4	13.8	11.9	14.0	11.5	16.2	9.7	12.5
D_i	+0.8	+0.6	−0.3	+0.9	+0.5	+1.3	+0.6	+0.8	−0.2
sign	+	+	−	+	+	+	+	+	−

No difference is 0, so $n = 9$, with 7 pluses; $s_{\text{obs}} = 7$. With $S_+ \sim \text{Bin}(9, \frac{1}{2})$,

$$\mathbb{P}(S_+ \geq 7) = \frac{\binom{9}{7} + \binom{9}{8} + \binom{9}{9}}{2^9} = \frac{36 + 9 + 1}{512} = \frac{46}{512} = 0.0898.$$

Since $0.0898 > 0.05$, do not reject. Seven of nine vines improving is suggestive but not significant at the 5% level.

> $s_{\text{obs}} = 7$, $n = 9$, p-value $= 0.090 > 0.05$: no significant yield increase by the sign test.

(b) The paired sign test replaces the *paired t-test*. A researcher might still prefer the sign test if the yield differences are not plausibly normal (e.g. one vine could respond enormously), since the sign test ignores magnitudes and so is unaffected by such an outlier.

A5. Two-sided paired sign test with ties (replacing the paired t-test).

(a) Form $D_i = A_i - B_i$ and discard ties ($D_i = 0$).

A	21.4	25.0	19.8	30.1	22.7	27.3	18.5	24.6	29.0	20.2	26.4	23.8
B	21.0	25.0	20.1	29.5	22.2	26.9	18.9	24.0	28.4	20.2	25.8	23.1
D_i	+0.4	0	−0.3	+0.6	+0.5	+0.4	−0.4	+0.6	+0.6	0	+0.6	+0.7
sign	+	tie	−	+	+	+	−	+	+	tie	+	+

Two baths give $D_i = 0$ (baths 2 and 10); discard them. The working sample size is $n = 12 - 2 = 10$.

(b) Among the ten non-tied pairs there are 8 pluses and 2 minuses, so $s_{\text{obs}} = 8$. With $S_+ \sim \text{Bin}(10, \frac{1}{2})$ and $\mathbb{P}(S_+ \geq 8) = \mathbb{P}(S_+ \leq 2)$,

$$\mathbb{P}(S_+ \geq 8) = \frac{\binom{10}{8} + \binom{10}{9} + \binom{10}{10}}{2^{10}} = \frac{45 + 10 + 1}{1024} = \frac{56}{1024} = 0.0547,$$
$$p\text{-value} = 2(0.0547) = 0.109.$$

Since $0.109 > 0.05$, do not reject. After correctly discarding the two ties, there is no significant evidence that instrument A reads systematically higher than B.

> $n = 10$ (2 ties dropped), $s_{\text{obs}} = 8$, p-value $= 0.11 > 0.05$: no systematic difference.

A6. Large-sample sign test via the normal approximation.

(a) Under H_0, $\mathbb{E}[S_+] = n/2 = 32$ and $\mathrm{SD}(S_+) = \sqrt{n/4} = \sqrt{16} = 4$. With $s_{\mathrm{obs}} = 40 > 32$, apply the continuity correction by subtracting $\frac{1}{2}$:

$$z = \frac{(s_{\mathrm{obs}} - \frac{1}{2}) - n/2}{\sqrt{n/4}} = \frac{(40 - 0.5) - 32}{4} = \frac{7.5}{4} = 1.875.$$

Against the two-sided critical value $z_{0.025} = 1.96$: since $|1.875| < 1.96$, do not reject (the two-sided p-value is about $2(0.0304) = 0.061$).

$$\boxed{z = 1.875,\ |z| < 1.96 :\ \text{do not reject; median wait consistent with 20 min.}}$$

(b) Without the continuity correction,

$$z = \frac{40 - 32}{4} = \frac{8}{4} = 2.0,$$

which gives $|2.0| > 1.96$ and a two-sided p-value ≈ 0.046, so the uncorrected test *rejects*. The decision flips: the continuity correction matters here because the uncorrected z lands just past 1.96 while the corrected z lands just short. With s_{obs} this close to the boundary, the more accurate corrected test (do not reject) is preferred.

$$\boxed{\text{No correction: } z = 2.0 > 1.96 \ \Rightarrow\ \text{reject — decision differs from (a).}}$$

A7. Sign test on paired preferences (replacing the paired t-test).

(a) The data are purely *ordinal* preferences — each judge reports only which product wins, not by how much — so no mean or magnitude exists to feed a t-test. The sign test, which uses only the direction of each within-judge comparison, is exactly suited to this and stands in for the *paired t-test*. A "no preference" judge carries no directional information and is treated as a tie.

(b) Discard the one tie, leaving $n = 13$ judges: 11 prefer the reformulation (pluses) and 2 prefer the original (minuses), so $s_{\mathrm{obs}} = 11$. Large S_+ supports H_1 : reformulation preferred, so use the upper tail of $\mathrm{Bin}(13, \frac{1}{2})$:

$$\begin{aligned}\mathbb{P}(S_+ \geq 11) &= \frac{\binom{13}{11} + \binom{13}{12} + \binom{13}{13}}{2^{13}} \\ &= \frac{78 + 13 + 1}{8192} = \frac{92}{8192} = 0.0112.\end{aligned}$$

Since $0.0112 < 0.05$, reject H_0. There is significant evidence at the 5% level that judges prefer the reformulated cereal.

$$\boxed{s_{\mathrm{obs}} = 11,\ n = 13 \text{ (1 tie dropped)},\ p\text{-value} = 0.011 < 0.05 :\ \text{reformulation preferred.}}$$

(c) The sign test is maximally *robust*: it assumes nothing beyond independence and a meaningful direction, so it is valid for raw ordinal preferences where the signed-rank test cannot even be computed (there are no magnitudes to rank). Its cost is *power*: when magnitudes do exist and the differences are symmetric, the Wilcoxon signed-rank test — which ranks those magnitudes — detects a true shift more readily. For pure preference data like these, the sign test is the only defensible choice; when interval magnitudes are available and roughly symmetric, prefer the signed-rank test.

B. The Wilcoxon signed-rank test

B1. Two-sided one-sample signed-rank test (replaces the one-sample t). One sample, a center under test, symmetry assumed — signed-rank, using the magnitudes of the deviations rather than only their signs. Subtract $m_0 = 93$ and rank the absolute deviations:

X_i	$D_i = X_i - 93$	$\lvert D_i\rvert$	rank $\lvert D_i\rvert$	signed rank
93.8	$+0.8$	0.8	1	$+1$
92.0	-1.0	1.0	2	-2
94.2	$+1.2$	1.2	3	$+3$
91.5	-1.5	1.5	4	-4
95.1	$+2.1$	2.1	5	$+5$
90.4	-2.6	2.6	6	-6
95.7	$+2.7$	2.7	7	$+7$
96.3	$+3.3$	3.3	8	$+8$

The negatives are at ranks $2, 4, 6$; the rest are positive:

$$W_- = 2 + 4 + 6 = 12,$$
$$W_+ = 1 + 3 + 5 + 7 + 8 = 24.$$

Check: $W_+ + W_- = 24 + 12 = 36 = \frac{8 \cdot 9}{2}$. Good. The statistic for the two-sided test is $\min(W_+, W_-) = 12$, compared with $w_{0.05,8} = 3$.

$$\boxed{\min(W_+, W_-) = 12 > 3 \;\Rightarrow\; \text{fail to reject } H_0;\ \text{no evidence the median differs from 93.}}$$

The batches run a touch warm, but not convincingly. This nonparametric procedure stands in for the *one-sample t-test* when normality is doubtful.

B2. One-sided paired signed-rank test (replaces the paired t).

(a) Rank the $\lvert D_i\rvert$ and re-attach signs:

D_i	$\lvert D_i\rvert$	rank $\lvert D_i\rvert$	signed rank
-0.6	0.6	1	-1
$+0.9$	0.9	2	$+2$
-1.1	1.1	3	-3
$+1.4$	1.4	4	$+4$
$+2.2$	2.2	5	$+5$
$+2.8$	2.8	6	$+6$
$+3.1$	3.1	7	$+7$
$+4.0$	4.0	8	$+8$
$+5.3$	5.3	9	$+9$

The negatives are -0.6 (rank 1) and -1.1 (rank 3):

$$W_- = 1 + 3 = 4,$$
$$W_+ = 2 + 4 + 5 + 6 + 7 + 8 + 9 = 41.$$

Check: $W_+ + W_- = 41 + 4 = 45 = \frac{9 \cdot 10}{2}$. Good.

(b) Pain reduction makes $D_i > 0$, so the alternative $m > 0$ piles the ranks onto the positive side, pushing W_+ up and W_- *down*; we reject for small W_-. Here $W_- = 4 \le w_{0.05,9} = 8$.

$$\boxed{W_- = 4 \le 8 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05;\ \text{the device significantly reduces back pain.}}$$

(c) This replaces the *paired t-test*; the assumption that buys its validity is that the paired differences are *symmetric* about their center (in place of the t's normality).

B3. Lower-tail one-sided test (replaces the one-sample/paired t). The alternative $m < 10$ predicts mostly negative deviations, so the *positive* rank sum W_+ should be small; the one-sided test rejects for small W_+. Rank the magnitudes:

D_i	$\lvert D_i\rvert$	rank $\lvert D_i\rvert$	signed rank
-0.4	0.4	1	-1
$+0.7$	0.7	2	$+2$
-0.9	0.9	3	-3
-1.3	1.3	4	-4
-1.5	1.5	5	-5
$+1.9$	1.9	6	$+6$
-2.2	2.2	7	-7
-2.8	2.8	8	-8
-3.6	3.6	9	-9
-4.1	4.1	10	-10

Only $+0.7$ (rank 2) and $+1.9$ (rank 6) are positive:

$$W_+ = 2 + 6 = 8,$$
$$W_- = 1 + 3 + 4 + 5 + 7 + 8 + 9 + 10 = 47.$$

Check: $W_+ + W_- = 8 + 47 = 55 = \frac{10\cdot 11}{2}$. Good. For $H_1\colon m < 10$ we reject when W_+ is small: $W_+ = 8 \le w_{0.05,10} = 10$.

$$\boxed{W_+ = 8 \le 10 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05;\ \text{the median nitrate level is below the permit cap.}}$$

B4. Large-sample normal approximation (replaces the one-sample t).

(a) Null mean and variance with $n = 28$:

$$\mathbb{E}[W_+] = \frac{n(n+1)}{4} = \frac{28\cdot 29}{4} = 203,$$
$$\mathrm{Var}[W_+] = \frac{n(n+1)(2n+1)}{24} = \frac{28\cdot 29\cdot 57}{24} = \frac{46284}{24} = 1928.5,$$
$$\sqrt{\mathrm{Var}[W_+]} = 43.91.$$

(b) Standardize:

$$W^* = \frac{W_+ - \mathbb{E}[W_+]}{\sqrt{\mathrm{Var}[W_+]}} = \frac{305 - 203}{43.91} = \frac{102}{43.91} = 2.323.$$

Two-sided at $\alpha = 0.05$ the cutoff is $z_{0.025} = 1.96$, and $|2.323| > 1.96$.

(c) Decision and parametric counterpart:

$$\boxed{W^* = 2.32 > 1.96 \;\Rightarrow\; \text{reject } H_0;\ \text{the rework shifts fill times.}}$$

At $n = 28$ the exact table is unwieldy, so the standardized $W^* \approx \mathrm{N}(0,1)$ does the job; the procedure replaces the *one-sample t-test*.

B5. Zeros and ties in a small-sample test (replaces the paired t).

(a) Two judges scored 0; drop them, leaving $n = 8$ nonzero differences $3, -1, 3, 2, -1, 3, 4, 2$ with magnitudes $1, 1, 2, 2, 3, 3, 3, 4$. Average ranks within ties: the two 1's occupy positions $1, 2 \Rightarrow \frac{1+2}{2} = 1.5$; the two 2's occupy $3, 4 \Rightarrow \frac{3+4}{2} = 3.5$; the three 3's occupy $5, 6, 7 \Rightarrow \frac{5+6+7}{3} = 6$; and $4 \to 8$.

D_i	$\lvert D_i\rvert$	average rank $\lvert D_i\rvert$	signed rank
-1	1	1.5	-1.5
-1	1	1.5	-1.5
$+2$	2	3.5	$+3.5$
$+2$	2	3.5	$+3.5$
$+3$	3	6	$+6$
$+3$	3	6	$+6$
$+3$	3	6	$+6$
$+4$	4	8	$+8$

$$W_- = 1.5 + 1.5 = 3,$$
$$W_+ = 3.5 + 3.5 + 6 + 6 + 6 + 8 = 33.$$

Check: $W_+ + W_- = 33 + 3 = 36 = \frac{8\cdot 9}{2}$ (with $n = 8$ after dropping the two zeros). Good. So $n = 8$, $W_+ = 33$, $W_- = 3$.

(b) The two-sided statistic is $\min(W_+, W_-) = 3$, compared with $w_{0.05,8} = 3$:

$$\boxed{\min(W_+, W_-) = 3 \le 3 \;\Rightarrow\; \text{reject } H_0;\ \text{judges prefer the redesign.}}$$

It just reaches the cutoff: the lone negatives $(-1, -1)$ are the two *smallest* magnitudes, so the positive ranks dominate.

B6. Using the identity $W_+ + W_- = n(n+1)/2$.

(a) With $n = 14$ the rank total is $\frac{14\cdot 15}{2} = 105$, so

$$W_- = \frac{n(n+1)}{2} - W_+ = 105 - 72 = 33.$$

Verify: $W_+ + W_- = 72 + 33 = 105 = \frac{14\cdot 15}{2}$, as required.

$$\boxed{W_- = 33.}$$

(b) With $n = 11$ the rank total is $\frac{11\cdot 12}{2} = 66$, so

$$W_+ = 66 - W_- = 66 - 51 = 15.$$

Since $W_+ = 15$ is well *below* its null mean $\mathbb{E}[W_+] = \frac{n(n+1)}{4} = 16.5$ — and far below $W_- = 51$ — the positive ranks are scarce, so the data sit mostly *below* the hypothesized center (the large-magnitude deviations carried minus signs).

$$\boxed{W_+ = 15;\ \text{data lie mostly below the center } (W_+ \ll W_-).}$$

B7. Contrast: signed-rank versus the paired t with an outlier (signed-rank replaces the paired t).

(a) Rank the magnitudes $1, 2, 4, 6, 9, 22$ of the reductions:

D_i	$\lvert D_i\rvert$	rank $\lvert D_i\rvert$	signed rank
-1	1	1	-1
-2	2	2	-2
$+4$	4	3	$+3$
$+6$	6	4	$+4$
$+9$	9	5	$+5$
$+22$	22	6	$+6$

The negatives are -1 (rank 1) and -2 (rank 2):

$$W_- = 1 + 2 = 3,$$
$$W_+ = 3 + 4 + 5 + 6 = 18.$$

Check: $W_+ + W_- = 18 + 3 = 21 = \frac{6\cdot 7}{2}$. Good. The alternative $m > 0$ rejects for small W_-, and $W_- = 3 > w_{0.05,6} = 2$:

$$W_- = 3 > 2 \;\Rightarrow\; \text{fail to reject (signed-rank).}$$

(b) Paired t on the raw reductions:

$$t = \frac{\overline{D}}{s_D/\sqrt{6}} = \frac{6.333}{8.59/\sqrt{6}} = \frac{6.333}{3.507} = 1.806.$$

Against $t_{0.05,5} = 2.015$, $1.806 < 2.015$:

$$t = 1.81 < 2.015 \;\Rightarrow\; \text{fail to reject (paired } t\text{).}$$

(c) Both tests retain here, but for different reasons, and the comparison shows the design difference:

signed-rank: $W_- = 3$ (retain); paired t: $t = 1.81$ (retain).

The single extreme responder (+22) inflates both $\overline{D}$ and s_D for the t, and the small $n = 6$ leaves the signed-rank test little room (its cutoff $w_{0.05,6} = 2$ demands at most one tiny negative). With visibly skewed, outlier-prone differences and no normality, the *signed-rank* verdict is the trustworthy one: it depends only on the *rank* of +22 (just "rank 6"), not on how extreme it is, so it is immune to the outlier that distorts the t.

C. The Wilcoxon rank-sum (Mann–Whitney) test

C1. This is the **Wilcoxon rank-sum (Mann–Whitney) test**, replacing the two-sample t: two *independent* small samples, normality in doubt, comparing *location*.

(a) H_0: the shaded- and open-pond length distributions are identical (equal medians); H_1: they differ in location (one median exceeds the other, two-sided). It replaces the *two-sample t*.

(b) Pool the $N = 11$ values, sort, and rank smallest to largest.

value	group	rank	value	group	rank
8.7	X	1	12.6	Y	7
9.1	X	2	13.2	Y	8
10.3	X	3	13.8	Y	9
11.4	X	4	14.5	Y	10
11.9	Y	5	15.1	Y	11
12.0	X	6			

The shaded-pond (X) ranks are $1, 2, 3, 4, 6$, so

$$W = 1 + 2 + 3 + 4 + 6 = 16.$$

(c) For orientation,

$$\mu_W = \frac{n(N+1)}{2} = \frac{5 \cdot 12}{2} = 30,$$
$$\sigma_W^2 = \frac{nm(N+1)}{12} = \frac{5 \cdot 6 \cdot 12}{12} = 30.$$

Reject if $W \le W_L = 19$ or $W \ge W_U = 41$. Here $W = 16 \le 19$, so reject.

$W = 16 \le 19 \Rightarrow$ reject H_0 at the 5% level: the pond length distributions differ; shaded-pond perch run shorter.

C2. Both forms carry the same information; U is just W shifted by the minimum possible rank sum.

(a) From the formula,

$$U = W - \frac{n(n+1)}{2} = 16 - \frac{5 \cdot 6}{2} = 16 - 15 = 1.$$

(b) Direct count. For each shaded-pond value, count the open-pond values it exceeds (open-pond values: $11.9, 12.6, 13.2, 13.8, 14.5, 15.1$):

$$8.7,\ 9.1,\ 10.3,\ 11.4 : \text{exceed no } Y \Rightarrow 0+0+0+0 = 0,$$
$$12.0 : \text{exceeds only } 11.9 \Rightarrow 1.$$

Total $= 1$, matching the formula.

(c) The null mean is

$$\mathbb{E}[U] = \frac{nm}{2} = \frac{30}{2} = 15.$$

The observed $U = 1$ sits far below 15: the shaded group's values win almost no head-to-head comparisons, i.e. they are systematically smaller.

$$\boxed{U = 1 \text{ (formula and direct count agree); } U \ll \mathbb{E}[U] = 15.}$$

C3. Rank-sum test, one-sided, replacing the two-sample t. A downward shift in the new-routine times (Y) sends Y to the *low* ranks and pushes the old-routine group (X, which we sum) to the *high* ranks, so we reject for a *large* W.

Step 1: hypotheses. H_0: identical distributions; H_1: the new-routine (Y) distribution is shifted *below* X.

Step 2: pool and rank ($N = 12$). Sorted: the new-routine values $12.1, 12.2, 12.3, 12.4$ take ranks 1–4; $12.5\,(X)$ rank 5; $12.6, 12.7\,(Y)$ ranks $6, 7$; then $12.8, 12.9, 13.0, 13.1, 13.4\,(X)$ ranks $8, 9, 10, 11, 12$. So X holds ranks $5, 8, 9, 10, 11, 12$.

Step 3: rank sum of X (old routine).

$$W = 5 + 8 + 9 + 10 + 11 + 12 = 55.$$

Step 4: decide. By symmetry the upper one-sided critical value is $W_U = n(N+1) - W_{0.05} = 6 \cdot 13 - 28 = 78 - 28 = 50$. Reject (upper tail) if $W \geq 50$. Since $W = 55 \geq 50$, reject.

$$\boxed{W = 55 \geq 50 \ \Rightarrow \ \text{reject } H_0 \text{ at the 5\% level: the new warm-up lowers sprint times.}}$$

C4. Large-sample normal (z) approximation, the rank-sum replacement for the two-sample t when both groups clear about 10.

(a) With $N = 27$, $n = 13$, $m = 14$,

$$\mu_W = \frac{n(N+1)}{2} = \frac{13 \cdot 28}{2} = 182,$$
$$\sigma_W^2 = \frac{nm(N+1)}{12} = \frac{13 \cdot 14 \cdot 28}{12} = \frac{5096}{12} = 424.67, \qquad \sigma_W = \sqrt{424.67} = 20.61.$$

(b) Here $W = 235 > \mu_W = 182$, so subtract $\frac{1}{2}$ to shrink the gap:

$$W - \mu_W - \tfrac{1}{2} = 235 - 182 - 0.5 = 52.5,$$
$$z = \frac{52.5}{20.61} = 2.55.$$

(c) Two-sided at $\alpha = 0.05$: $|z| = 2.55 > z_{0.025} = 1.96$, so reject. The p-value is $2\,\mathbb{P}(Z > 2.55) \approx 0.011$.

$z = 2.55$, $|z| > 1.96 \Rightarrow$ reject: the centers differ in handling time; center A runs longer.

C5. Rank-sum with ties handled by average ranks; the parametric counterpart is the two-sample t.

(a) Pooled sorted values, with tie blocks marked. The two 5's at slots $2, 3$ (a third 5 from group B at slot 4 as well — three 5's in all, slots $2, 3, 4$) average to $\frac{2+3+4}{3} = 3$; the two 7's at slots $6, 7$ average to $\frac{6+7}{2} = 6.5$.

value	group	rank
3	X	1
5	X	3
5	X	3
5	Y	3
6	X	5
7	Y	6.5
7	Y	6.5
8	X	8
9	Y	9
10	Y	10

(b) Glaze A (X) holds ranks $1, 3, 3, 5, 8$:

$$W = 1 + 3 + 3 + 5 + 8 = 20, \qquad \mu_W = \frac{n(N+1)}{2} = \frac{5 \cdot 11}{2} = 27.5.$$

(c) Tie groups have sizes $t = 3$ (the 5's) and $t = 2$ (the 7's), so $\sum(t^3 - t) = (27 - 3) + (8 - 2) = 30$. With $N = 10$, $n = m = 5$,

$$\sigma_W^2 = \frac{nm}{12}\Big[(N+1) - \frac{\sum(t^3 - t)}{N(N-1)}\Big] = \frac{25}{12}\Big[11 - \frac{30}{90}\Big]$$
$$= \frac{25}{12}\,(10.667) = 22.22,$$

versus the uncorrected $\frac{25}{12}(11) = 22.92$ — a small shrinkage from the ties.

$W = 20, \quad \mu_W = 27.5, \quad \sigma_W^2 = 22.22$ (tie-corrected, vs 22.92).

C6. Rank-sum test, chosen precisely because one wild value makes the two-sample t untrustworthy.

(a) Interface X holds ranks $1, 2, 3, 4, 5, 7, 9, 11$:

$$W = 1 + 2 + 3 + 4 + 5 + 7 + 9 + 11 = 42.$$

(b) For $n = m = 8$ reject if $W \le W_L = 49$ or $W \ge W_U = 87$. Since $W = 42 \le 49$, reject at $\alpha = 0.05$.

$W = 42 \le 49 \;\Rightarrow$ reject: the interfaces differ; X is faster.

(c) The 4200 s value would inflate $\bar{Y}$ and especially s_Y enormously, so the two-sample t's denominator would swamp the real difference and the t could falsely retain. The rank-sum test only sees that value as "the largest, rank 16"; replacing it by 200 s (still the maximum) leaves every rank — and hence $W = 42$ and the verdict — unchanged.

C7. Read the design first (same/matched units vs. independent groups), then the data shape (normal vs. skew/outlier/ordinal).

(a) **Two-sample** t: two independent groups, both bell-shaped and clean — the t is valid and most powerful here.

(b) **Wilcoxon rank-sum**: two *independent* groups, strongly right-skewed with an extreme outlier — the t's normality fails, so rank.

(c) **Paired (signed-rank)** test, *not* rank-sum: the same subjects are measured under both conditions, so the observations are paired.

(d) **Wilcoxon rank-sum**: independent groups graded on an *ordinal* scale with no numeric mean, so only a rank method applies.

(a) two-sample t, (b) rank-sum, (c) paired signed-rank, (d) rank-sum.

D. The Kruskal–Wallis test

D1. Three preservative solutions. Three *independent* samples with a distribution-equality question and no normality stated ⇒ Kruskal–Wallis on $\chi^2_{k-1} = \chi^2_2$. It replaces the **one-way ANOVA** F**-test**. Pool all $N = 12$ values and rank 1–12 (no ties):

value	5	6	7	8	9	10	11	12	13	14	16	17
rank	1	2	3	4	5	6	7	8	9	10	11	12
group	R	P	Q	P	R	Q	P	R	P	Q	Q	R

Group rank sums:

$$R_P = 2 + 4 + 7 + 9 = 22,$$
$$R_Q = 3 + 6 + 10 + 11 = 30,$$
$$R_R = 1 + 5 + 8 + 12 = 26.$$

Check: $22 + 30 + 26 = 78 = N(N+1)/2 = 12 \cdot 13/2$. ■ With $n_j = 4$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{22^2 + 30^2 + 26^2}{4} = \frac{484 + 900 + 676}{4} = \frac{2060}{4} = 515.0,$$

$$H = \frac{12}{12 \cdot 13}(515.0) - 3(13) = \frac{12}{156}(515.0) - 39 = 39.615 - 39 = 0.615.$$

Reference χ^2_2.

$$\boxed{H = 0.615 < \chi^2_{0.05,2} = 5.991 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Decision: the three solutions interleave evenly in the pooled ranking, so there is no significant evidence that vase-life differs by solution.

D2. Unbalanced fertilizer design. Unequal n_j change only the bookkeeping: each R_j^2 is divided by its *own* n_j. Kruskal–Wallis on χ^2_3, replacing the **one-way ANOVA** F**-test**. The $N = 16$ values are distinct; sort and rank:

value	22	23	24	25	27	28	29	31	33	34	36	38	40	42	44	47
rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
group	1	4	2	3	1	4	2	1	3	2	4	2	1	3	2	4

Rank sums (sizes $n_1 = 4,\ n_2 = 5,\ n_3 = 3,\ n_4 = 4$):

$$R_1 = 1 + 5 + 8 + 13 = 27,$$
$$R_2 = 3 + 7 + 10 + 12 + 15 = 47,$$
$$R_3 = 4 + 9 + 14 = 27,$$
$$R_4 = 2 + 6 + 11 + 16 = 35.$$

Check: $27 + 47 + 27 + 35 = 136 = 16 \cdot 17/2$. ■ Dividing each square by its own n_j,

$$\sum_j \frac{R_j^2}{n_j} = \frac{27^2}{4} + \frac{47^2}{5} + \frac{27^2}{3} + \frac{35^2}{4}$$
$$= \frac{729}{4} + \frac{2209}{5} + \frac{729}{3} + \frac{1225}{4} = 182.25 + 441.80 + 243.00 + 306.25 = 1173.30,$$
$$H = \frac{12}{16 \cdot 17}(1173.30) - 3(17) = \frac{12}{272}(1173.30) - 51 = 51.763 - 51 = 0.763.$$

Reference χ^2_3.

$$\boxed{H = 0.763 < \chi^2_{0.05,3} = 7.815 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Decision: no significant difference among the four fertilizer programs. Had we wrongly divided every R_j^2 by a common n, the unbalanced sizes would have corrupted H; the per-group division is essential here.

D3. Ordinal finish scores with ties.

(a) Ordinal scores with repeats force **average ranks**; the procedure is still Kruskal–Wallis on χ^2_2, replacing the **one-way ANOVA F-test**. Sorted pooled values and the positions they occupy ($N = 15$):

value	3	4	5	6	7	8	9	10
positions	1	2	3–5	6–8	9–11	12–13	14	15
avg rank	1	2	4	7	10	12.5	14	15

For example the three 5's occupy positions $3, 4, 5$, averaging $(3 + 4 + 5)/3 = 4$. Rank sums:

$$R_A = 4 + 7 + 7 + 12.5 + 14 = 44.5 \quad (5, 6, 6, 8, 9),$$
$$R_B = 2 + 4 + 10 + 10 + 12.5 = 38.5 \quad (4, 5, 7, 7, 8),$$
$$R_C = 1 + 4 + 7 + 10 + 15 = 37.0 \quad (3, 5, 6, 7, 10).$$

Check: $44.5 + 38.5 + 37.0 = 120 = 15 \cdot 16/2$. ■ With $n_j = 5$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{44.5^2 + 38.5^2 + 37.0^2}{5} = \frac{1980.25 + 1482.25 + 1369.00}{5} = \frac{4831.5}{5} = 966.30,$$
$$H = \frac{12}{15 \cdot 16}(966.30) - 3(16) = 0.05(966.30) - 48 = 48.315 - 48 = 0.315.$$

$$\boxed{H = 0.315 < \chi^2_{0.05,2} = 5.991 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Decision: no significant difference in finish quality among the centers.

(b) Tied blocks: three 5's ($t = 3$), three 6's ($t = 3$), three 7's ($t = 3$), two 8's ($t = 2$); singletons $3, 4, 9, 10$ contribute 0.

$$\sum_g (t_g^3 - t_g) = (3^3 - 3) + (3^3 - 3) + (3^3 - 3) + (2^3 - 2) = 24 + 24 + 24 + 6 = 78,$$
$$N^3 - N = 15^3 - 15 = 3375 - 15 = 3360,$$
$$C = 1 - \frac{78}{3360} = 1 - 0.02321 = 0.97679,$$
$$H_{\text{corr}} = \frac{0.315}{0.97679} = 0.322.$$

$$\boxed{H_{\text{corr}} = 0.322 < 5.991 \;\Rightarrow\; \text{conclusion unchanged.}}$$

The factor sits just under 1, nudging H up by about 0.007; far from the cutoff, the correction is immaterial.

D4. Five road surfaces, large N. For $N = 50$ with each $n_j = 10 \geq 5$, the $\chi^2_{k-1} = \chi^2_4$ approximation is the working tool; this is the large-sample form of the test that replaces the **one-way ANOVA F-test**. Arithmetic check first:

$$205 + 310 + 245 + 190 + 325 = 1275 = N(N+1)/2 = 50 \cdot 51/2. \ \blacksquare$$

With $n_j = 10$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{205^2 + 310^2 + 245^2 + 190^2 + 325^2}{10}$$
$$= \frac{42025 + 96100 + 60025 + 36100 + 105625}{10} = \frac{339875}{10} = 33987.5,$$
$$H = \frac{12}{50 \cdot 51}(33987.5) - 3(51) = \frac{12}{2550}(33987.5) - 153 = 159.94 - 153 = 6.94.$$

Reference χ^2_4.

$$\boxed{H = 6.94 < \chi^2_{0.05,4} = 9.488 \ \Rightarrow \ \text{fail to reject } H_0.}$$

Decision: although surfaces 2 and 5 carry the largest rank sums, the spread across the five surfaces stays within χ^2_4 sampling variation; no significant noise difference is shown. The χ^2 approximation is trustworthy since every $n_j = 10$.

D5. $k = 2$ reduction to the rank-sum test.

(a) Pool the $N = 10$ aroma scores and rank 1–10 (no ties): sorted 41, 44, 47, 49, 52, 55, 58, 60, 63, 66. Profile 1 holds $41, 47, 52, 58, 63 \rightarrow$ ranks $1, 3, 5, 7, 9$; Profile 2 holds $44, 49, 55, 60, 66 \rightarrow$ ranks $2, 4, 6, 8, 10$.

$$R_1 = 1 + 3 + 5 + 7 + 9 = 25, \qquad R_2 = 2 + 4 + 6 + 8 + 10 = 30, \qquad 25 + 30 = 55 = 10 \cdot 11/2. \ \blacksquare$$

With $n_j = 5$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{25^2 + 30^2}{5} = \frac{625 + 900}{5} = \frac{1525}{5} = 305.0,$$
$$H = \frac{12}{10 \cdot 11}(305.0) - 3(11) = \frac{12}{110}(305.0) - 33 = 33.273 - 33 = 0.273.$$

Reference χ^2_1: $H = 0.273 < \chi^2_{0.05,1} = 3.841$, fail to reject H_0.

(b) Rank-sum standardization of R_1:

$$\mathbb{E}[R_1] = \frac{n_1(N+1)}{2} = \frac{5 \cdot 11}{2} = 27.5, \qquad \text{Var}(R_1) = \frac{n_1 n_2 (N+1)}{12} = \frac{5 \cdot 5 \cdot 11}{12} = 22.917,$$
$$Z = \frac{R_1 - \mathbb{E}[R_1]}{\sqrt{\text{Var}(R_1)}} = \frac{25 - 27.5}{\sqrt{22.917}} = \frac{-2.5}{4.787} = -0.522,$$
$$Z^2 = (-0.522)^2 = 0.273.$$

$$\boxed{H = 0.273 = Z^2; \quad \text{both retain } H_0.}$$

(c) When $k = 2$ the chi-square test on 1 df is exactly the squared two-sided normal test ($\chi^2_{0.05,1} = 3.841 = 1.96^2 = z^2_{0.025}$), so $H = Z^2$ identically and the two procedures reject or retain together — Kruskal–Wallis is just the $k \geq 3$ generalization of the rank-sum test.

D6. Contrast: Kruskal–Wallis vs. ANOVA on skewed, outlier-laden data.

(a) Pool the $N = 15$ values and rank 1–15 (distinct). Sorted:

$$40, 42, 45, 55, 58, 60, 70, 72, 75, 80, 85, 88, 150, 300, 900,$$

so the three extreme spenders $900, 300, 150$ take ranks $15, 14, 13$ regardless of magnitude.

value	40	42	45	55	58	60	70	72	75	80	85	88	150	300	900
rank	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
group	X	Z	Y	X	Z	Y	X	Z	Y	Z	X	Y	Y	Z	X

Rank sums:

$$R_X = 1 + 4 + 7 + 11 + 15 = 38,$$
$$R_Y = 3 + 6 + 9 + 12 + 13 = 43,$$
$$R_Z = 2 + 5 + 8 + 10 + 14 = 39.$$

Check: $38 + 43 + 39 = 120 = 15 \cdot 16/2$. ■ With $n_j = 5$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{38^2 + 43^2 + 39^2}{5} = \frac{1444 + 1849 + 1521}{5} = \frac{4814}{5} = 962.8,$$

$$H = \frac{12}{15 \cdot 16}(962.8) - 48 = 0.05(962.8) - 48 = 48.14 - 48 = 0.14.$$

$$\boxed{H = 0.14 < \chi^2_{0.05,2} = 5.991 \;\Rightarrow\; \text{fail to reject } H_0.}$$

Decision: no significant difference in spending across the three areas.

(b) In a one-way ANOVA F on the raw dollars, the lone \$900 value in Area X drags that group's mean to $(40 + 55 + 70 + 85 + 900)/5 = 230$ and explodes its within-group variance, while the other areas sit far lower; the resulting F-ratio is unstable and its F reference is invalid because normality fails badly. Kruskal–Wallis uses only the order, so each outlier counts merely as "the largest available rank" and the three groups stay nearly tied ($R_X = 38,\ R_Y = 43,\ R_Z = 39$). With sharp skew and outliers, **trust** H; ANOVA would be appropriate only if the data were genuinely normal with equal variances, in which case it would be slightly more powerful.

D7. Omnibus test plus rank-based comparison.

(a) Large-sample Kruskal–Wallis on χ^2_3, replacing the **one-way ANOVA** F**-test**. Check: $150 + 300 + 160 + 210 = 820 = N(N+1)/2 = 40 \cdot 41/2$. ■ With $n_j = 10$,

$$\sum_j \frac{R_j^2}{n_j} = \frac{150^2 + 300^2 + 160^2 + 210^2}{10}$$
$$= \frac{22500 + 90000 + 25600 + 44100}{10} = \frac{182200}{10} = 18220,$$
$$H = \frac{12}{40 \cdot 41}(18220) - 3(41) = \frac{12}{1640}(18220) - 123 = 133.32 - 123 = 10.32.$$

$$\boxed{H = 10.32 > \chi^2_{0.05,3} = 7.815 \Rightarrow \text{reject } H_0\text{; the layouts are not all alike.}}$$

(b) Kruskal–Wallis is an **omnibus** test, so a follow-up comparison localizes the difference (as Tukey follows a significant ANOVA F). Group average ranks $\overline{R}_j = R_j/n_j$:

$$\overline{R}_1 = 15.0, \quad \overline{R}_2 = 30.0, \quad \overline{R}_3 = 16.0, \quad \overline{R}_4 = 21.0,$$

against the overall average $(N+1)/2 = 20.5$. The critical gap:

$$\sqrt{\frac{N(N+1)}{12}\left(\frac{1}{n_i} + \frac{1}{n_j}\right)} = \sqrt{\frac{40 \cdot 41}{12} \cdot \frac{2}{10}} = \sqrt{27.33} = 5.23,$$
$$z^* \cdot 5.23 = 2.64 \cdot 5.23 = 13.8.$$

Comparing the gaps to 13.8:

$$|\overline{R}_2 - \overline{R}_1| = 15.0 > 13.8, \qquad |\overline{R}_2 - \overline{R}_3| = 14.0 > 13.8,$$

while $|\overline{R}_2 - \overline{R}_4| = 9.0 < 13.8$ and all other gaps are smaller.

Layout 2 (slowest reactions) stands apart from layouts 1 and 3.

Decision: the omnibus H flagged a difference, and the rank-based comparison isolates layout 2 as the standout.

E. The Friedman test

E1. Three espresso blends, five customers. The same customers taste all three blends, so this is a randomized block design with customers as blocks; with subjective scores and a small b the **Friedman test** is the tool, replacing the **randomized block (two-way) F-test** of the normal theory.

(a) H_0: the three blends have the same preference distribution (equal medians within blocks); H_1: at least one blend differs. Upper-tailed test on χ^2_{k-1}.

(b) Rank each customer's three scores 1 (lowest) to 3 (highest):

Customer	Blend A	Blend B	Blend C
1	72 (2)	80 (3)	68 (1)
2	65 (2)	70 (3)	63 (1)
3	88 (2)	84 (1)	90 (3)
4	55 (2)	61 (3)	52 (1)
5	77 (2)	82 (3)	74 (1)
R_i	10	13	7

Total $10 + 13 + 7 = 30 = b \cdot \frac{k(k+1)}{2} = 5 \cdot 6$; the average rank sum is $b(k+1)/2 = 10$.

(c) With $b = 5$, $k = 3$,

$$\sum_i R_i^2 = 10^2 + 13^2 + 7^2 = 100 + 169 + 49 = 318,$$

$$Q = \frac{12}{b\,k(k+1)} \sum_i R_i^2 - 3b(k+1) = \frac{12}{5 \cdot 3 \cdot 4}\,(318) - 3 \cdot 5 \cdot 4$$

$$= \frac{12}{60}\,(318) - 60 = 63.6 - 60 = 3.6.$$

Since $Q = 3.6 < \chi^2_{0.05,2} = 5.991$, fail to reject. There is *not* significant evidence at the 5% level that the blends differ in preference — Blend B leads but the gap is within chance for five customers.

$Q = 3.6 < 5.991 \;\Rightarrow$ fail to reject H_0 at $\alpha = 0.05$.

E2. Four chocolate brands, six panelists. Each panelist rates all four brands, so this is repeated measures with panelists as blocks; the Friedman test replaces the **randomized block F-test**. The rank sums are given.

Check the total. $9{+}21{+}11{+}19 = 60 = b\frac{k(k+1)}{2} = 6{\cdot}10$, and the average rank sum is $b(k{+}1)/2 = 6{\cdot}2.5 = 15$. Brands X and Z run high, W and Y low.

Compute Q. With $b = 6$, $k = 4$,

$$\sum_i R_i^2 = 9^2 + 21^2 + 11^2 + 19^2 = 81 + 441 + 121 + 361 = 1004,$$

$$Q = \frac{12}{6 \cdot 4 \cdot 5}\,(1004) - 3 \cdot 6 \cdot 5 = \frac{12}{120}\,(1004) - 90 = 100.4 - 90 = 10.4.$$

Decide on χ^2_3. $Q = 10.4 > \chi^2_{0.05,3} = 7.815$, reject. There is significant evidence at the 5% level that liking differs across the four brands (X is the front-runner).

$$\boxed{Q = 10.4 > 7.815 \;\Rightarrow\; \text{reject } H_0;\ \text{the brands differ in liking.}}$$

E3. Three mattresses, eight volunteers. Volunteers are blocks; Friedman replaces the randomized block F-test. Rank sums are given.

(a) Total $11 + 20 + 17 = 48 = b\frac{k(k+1)}{2} = 8 \cdot 6$ (average rank sum $b(k+1)/2 = 8 \cdot 2 = 16$). With $b = 8$, $k = 3$,

$$\sum_i R_i^2 = 11^2 + 20^2 + 17^2 = 121 + 400 + 289 = 810,$$

$$Q = \frac{12}{8 \cdot 3 \cdot 4}(810) - 3 \cdot 8 \cdot 4 = \frac{12}{96}(810) - 96 = 101.25 - 96 = 5.25.$$

(b) Friedman is upper-tailed on χ^2_2. Locate $Q = 5.25$:

$$\chi^2_{0.10,2} = 4.605 \;<\; 5.25 \;<\; 5.991 = \chi^2_{0.05,2},$$

so $0.05 < p$-value < 0.10.

(c) Since the p-value exceeds 0.05 (equivalently $Q = 5.25 < 5.991$), fail to reject at $\alpha = 0.05$. There is not significant evidence at the 5% level that the mattresses differ in comfort, though the result is suggestive (p between 0.05 and 0.10).

$$\boxed{Q = 5.25,\ 0.05 < p\text{-value} < 0.10 \;\Rightarrow\; \text{fail to reject at } \alpha = 0.05.}$$

E4. A tied block. Average-rank convention for the within-block tie.

(a) Sorted, $41 < 53 = 53 = 53 < 60$ occupy positions $1, 2, 3, 4, 5$. The three 53's would take positions $2, 3, 4$, so each gets the average rank $\frac{2+3+4}{3} = 3$:

$$\text{rank}(41) = 1, \quad \text{rank}(53) = 3, \quad \text{rank}(53) = 3, \quad \text{rank}(53) = 3, \quad \text{rank}(60) = 5.$$

(b) Block total $1 + 3 + 3 + 3 + 5 = 15 = \frac{k(k+1)}{2} = \frac{5 \cdot 6}{2}$, exactly the untied total, so the rank-sum bookkeeping is undisturbed.

(c) One tie group of size $t = 3$ gives $\sum(t^3 - t) = 3^3 - 3 = 24$. With $b = 7$, $k = 5$, $k^2 - 1 = 24$,

$$1 - \frac{\sum(t^3 - t)}{b\,k(k^2-1)} = 1 - \frac{24}{7 \cdot 5 \cdot 24} = 1 - \frac{24}{840} = 1 - 0.0286 = 0.9714.$$

The corrected statistic is $Q/0.9714$. Dividing by a factor less than 1 *raises* Q; ties deflate the raw statistic and the correction restores it (here by about 3%).

$$\boxed{\text{ranks } 1, 3, 3, 3, 5;\ \text{total } 15;\ \text{correction } 0.9714 \text{ (raises } Q).}$$

E5. $k = 2$**: Friedman equals the sign test.** Two treatments in blocks is the boundary case, so Friedman reduces to the two-sided **sign test** on the within-block winners — it replaces the **paired** t**-test**.

(a) Routine I is placed above routine II in $a = 6$ of $b = 7$ blocks. By the reduced formula

$$Q = \frac{(2a - b)^2}{b} = \frac{(2 \cdot 6 - 7)^2}{7} = \frac{5^2}{7} = \frac{25}{7} = 3.571.$$

(b) Sign-test standardization (mean $b/2$, sd $\sqrt{b}/2$ under H_0):

$$Z = \frac{a - b/2}{\sqrt{b}/2} = \frac{6 - 3.5}{\sqrt{7}/2} = \frac{2.5}{1.3229} = 1.890, \qquad Z^2 = 3.571 = Q. \blacksquare$$

(c) $Q = 3.571 < \chi^2_{0.05,1} = 3.841$ (equivalently $|Z| = 1.890 < z_{0.025} = 1.96$), so fail to reject. There is not significant evidence at the 5% level that one routine is preferred; 6 of 7 placements is not enough.

$$\boxed{Q = Z^2 = 3.571 < 3.841 \;\Rightarrow\; \text{fail to reject; the } k = 2 \text{ Friedman is the sign test.}}$$

E6. Three proposals, five reviewers, one harsh marker. Reviewers are blocks; with a large block offset and subjective scoring, normality is shaky, favoring Friedman over the **randomized block F-test**.

(a) Rank within each reviewer (1 = lowest, 3 = highest). Every reviewer orders R>P>Q:

Reviewer	Proposal P	Proposal Q	Proposal R
1	81 (2)	76 (1)	88 (3)
2	84 (2)	78 (1)	90 (3)
3	49 (2)	44 (1)	55 (3)
4	86 (2)	80 (1)	92 (3)
5	83 (2)	77 (1)	89 (3)
R_i	10	5	15

Total $10 + 5 + 15 = 30 = 5 \cdot 6$. With $b = 5$, $k = 3$,

$$\sum_i R_i^2 = 10^2 + 5^2 + 15^2 = 100 + 25 + 225 = 350,$$

$$Q = \frac{12}{5 \cdot 3 \cdot 4}(350) - 3 \cdot 5 \cdot 4 = \frac{12}{60}(350) - 60 = 70 - 60 = 10.0.$$

(b) $Q = 10.0 > \chi^2_{0.05,2} = 5.991$, reject. There is significant evidence at the 5% level that the proposals differ in rating (R highest, Q lowest).

$$\boxed{Q = 10.0 > 5.991 \;\Rightarrow\; \text{reject } H_0\text{; the proposals differ.}}$$

(c) Reviewer 3's scores (44–55) sit far below the others (76–92), yet ranking *within* each reviewer replaces those raw numbers by $\{1, 2, 3\}$, so the harsh marker's overall low level is erased before any comparison — Friedman needs no normality and is immune to that offset. The block F-test would also remove a pure additive offset (by subtracting block means), and being magnitude-based it is more powerful, so it would be the better choice *if* the scores were genuinely normal with additive, equal-variance effects. With one wildly different reviewer and subjective scoring, those assumptions are doubtful, so Friedman is the safer call here.

E7. Small-sample exact vs. chi-square. With only $b = 4$ blocks, Q's chi-square shape is merely asymptotic; the discrete exact distribution is the honest reference.

(a) Exact table: $\mathbb{P}(Q \geq 6.10) = 0.052 > 0.05$, so *fail to reject* at $\alpha = 0.05$.

(b) Chi-square approximation: $Q = 6.10 > \chi^2_{0.05,2} = 5.991$, so *reject* at $\alpha = 0.05$.

(c) The verdicts disagree because Q is discrete and the chi-square curve is only an approximation that is too liberal (it places too little probability in the far tail) when b is small. For $b = 4$ trust the *exact* tail probability 0.052, so do not reject. Relying on the chi-square cutoff with few blocks risks a false positive: here it would reject a result the exact test — the correct reference — says is not significant. Prefer an exact Friedman table whenever b and k are small.

$$\boxed{\begin{array}{c}\text{Exact } p = 0.052 \text{ (fail to reject); chi-square wrongly rejects} \Rightarrow \text{trust the exact tail for} \\ \text{small } b.\end{array}}$$

F. Runs test, rank correlation, and synthesis

F1. Runs test on the roulette colors (replaces no parametric test; it checks the *order*/randomness of a binary sequence, not a mean or proportion).

(a) H_0: the red/black order is random (the colors are arranged in random order given the counts). H_1: the order is not random (clustering or over-alternation). This is the **runs test**.

(b) Mark each color change and add one. The maximal streaks are

$$\texttt{R R | B | R | B B | R R R | B B | R | B | R R | B B | R | B | R R | B},$$

giving 14 streaks, so $R = 14$. With $n_1 = 12$ R's, $n_2 = 10$ B's, $n = 22$,

$$\mu_R = 1 + \frac{2(12)(10)}{22} = 1 + \frac{240}{22} = 1 + 10.909 = 11.909,$$
$$\sigma_R^2 = \frac{2(12)(10)\,[\,2(12)(10) - 22\,]}{22^2\,(21)} = \frac{240\,(218)}{484\,(21)} = \frac{52320}{10164} = 5.148.$$

(c) Since $R = 14 > \mu_R$, use the "too many" correction $R - \frac{1}{2}$:

$$Z = \frac{(14 - 0.5) - 11.909}{\sqrt{5.148}} = \frac{1.591}{2.269} = 0.70.$$

Against $z_{0.025} = 1.96$, $|0.70| < 1.96$.

$$\boxed{Z = 0.70,\ |Z| < 1.96\ \Rightarrow\ \text{do not reject; the color order is consistent with randomness.}}$$

The wheel shows no significant streaking or over-alternation at the 5% level.

F2. Runs test on air-quality flags.

(a) With $n_1 = n_2 = 15$, $n = 30$,

$$\mu_R = 1 + \frac{2(15)(15)}{30} = 1 + \frac{450}{30} = 1 + 15 = 16,$$
$$\sigma_R^2 = \frac{2(15)(15)\,[\,2(15)(15) - 30\,]}{30^2\,(29)} = \frac{450\,(420)}{900\,(29)} = \frac{189000}{26100} = 7.241.$$

Observed $R = 10 < \mu_R$, so use the "too few" correction $R + \frac{1}{2}$:

$$Z = \frac{(10 + 0.5) - 16}{\sqrt{7.241}} = \frac{-5.5}{2.691} = -2.04.$$

(b) Against $z_{0.025} = 1.96$, $|-2.04| > 1.96$, so reject.

$$\boxed{Z = -2.04,\ |Z| > 1.96\ \Rightarrow\ \text{reject randomness at the 5\% level.}}$$

The negative sign means *too few* runs = **clustering**: above-median and below-median days come in long blocks rather than alternating day to day. Physically, air quality persists — good and bad spells run in multi-day stretches (weather systems), so successive days are not independent.

F3. Spearman's r_s on literacy vs. GDP ranks (the rank analogue of Pearson's r/the t-test for linear correlation; it tests *monotone* association without assuming bivariate normality).

(a) Both rows are already ranks with no ties, so use the shortcut. Differences and squares:

Country	1	2	3	4	5	6	7	8	9	$\sum$
rank(x)	1	2	3	4	5	6	7	8	9	
rank(y)	2	1	4	3	5	7	6	9	8	
d_i	-1	1	-1	1	0	-1	1	-1	1	0
d_i^2	1	1	1	1	0	1	1	1	1	8

With $n = 9$, $\sum d_i^2 = 8$,

$$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)} = 1 - \frac{6(8)}{9(81-1)} = 1 - \frac{48}{720} = 1 - 0.0667 = 0.933.$$

$$\boxed{r_s = 0.933}$$

(b) Test $H_0 : \rho_s = 0$ vs. $H_1 : \rho_s \neq 0$. With $r_s = 0.933$, $r_s^2 = 0.871$, $n - 2 = 7$:

$$t = r_s\sqrt{\frac{n-2}{1-r_s^2}} = 0.933\sqrt{\frac{7}{1-0.871}} = 0.933\sqrt{\frac{7}{0.129}}$$
$$= 0.933\sqrt{54.26} = 0.933(7.366) = 6.87.$$

Against $t_{0.025,7} = 2.365$, $|t| = 6.87 > 2.365$.

$$\boxed{t = 6.87 > 2.365 \;\Rightarrow\; \text{reject } H_0;\ \text{literacy and GDP rankings are strongly associated.}}$$

This Spearman test stands in for the ordinary **Pearson correlation t-test**; it is preferred because the relationship need only be monotone (not linear) and the ranks are robust to outliers and non-normal margins.

F4. Spearman on pond area vs. species count (replaces the Pearson correlation test; we want monotone association and n is small).

Step 1: rank each variable. Both x and y are strictly increasing across A–G, so the two rankings are identical:

Pond	A	B	C	D	E	F	G
rank(x)	1	2	3	4	5	6	7
rank(y)	1	2	3	4	5	6	7
d_i	0	0	0	0	0	0	0

Step 2: compute r_s. Every $d_i = 0$, so $\sum d_i^2 = 0$ and

$$r_s = 1 - \frac{6(0)}{7(48)} = 1.$$

Step 3: one-sided test. With $r_s = 1$ the statistic $t = r_s\sqrt{(n-2)/(1-r_s^2)} \to \infty$, which exceeds any cutoff, in particular $t_{0.05,5} = 2.015$:

$$\boxed{r_s = 1 \;\Rightarrow\; t \to \infty > 2.015;\ \text{reject } H_0;\ \text{larger ponds hold more species (monotone).}}$$

The raw area–species relationship is curved (area spans 0.5 to 14 ha while species span 3 to 16), yet the *ordering* is perfect, which is exactly what Spearman certifies.

F5. Spearman versus Pearson on a monotone, accelerating relationship.

(a) Since $y = x^3 + 1$ increases with x, the ranks of x and y are identical $(1, \ldots, 6)$, so every $d_i = 0$ and

$$r_s = 1 - \frac{6(0)}{6(35)} = 1.$$

(b) Pearson on the raw values, using the supplied sums:

$$r = \frac{S_{xy}}{\sqrt{S_{xx}\, S_{yy}}} = \frac{731.5}{\sqrt{17.5\,(34757.5)}}$$
$$= \frac{731.5}{\sqrt{608256.25}} = \frac{731.5}{779.91} = 0.938.$$

(c) Contrast:

$$\boxed{r_s = 1.000 \quad \text{but} \quad r = 0.938 < 1.}$$

Both are large, but only Spearman reaches 1. The catalyst–yield ordering is *exactly* preserved (more catalyst always gives more yield), which is the scientific truth; Spearman reports this honestly because it reads only the ranks. Pearson is pulled below 1 by the curvature of the cubic cloud — it measures *linear* fit, and the data are bent. When the question is "does yield rise monotonically with loading," Spearman is the honest summary; reserve Pearson for when a straight-line model is genuinely of interest. (A single outlier would dent r further while leaving the rank-based r_s nearly untouched.)

F6. Name-the-test synthesis.

(a) Paired measurements, testing a *median* shift, with skewed differences and outliers: use the **Wilcoxon signed-rank test** (or the cruder **sign test** if symmetry is doubtful). It replaces the **paired t-test**, which is inappropriate because the strong skew and outliers violate the normality of the differences that the t relies on.

(b) Three independent groups, ordinal 0–10 response where means are meaningless: use the **Kruskal–Wallis test**. It replaces the **one-way ANOVA F-test**, which is invalid because ANOVA assumes normal, interval-scaled responses — neither holds for an ordinal pain scale.

(c) One long binary sequence in time order, asking whether wet days *cluster*: use the **runs test** for randomness. It replaces no standard parametric test of means; it targets *serial dependence/order*, something a t or proportion test (which only look at counts) cannot detect. Few runs would signal clustering (storm systems), confirming non-independence.

F7. Carry out the test that replaces the paired t.

(a) Because the routines are *matched in pairs* and the differences are non-normal, the **Wilcoxon signed-rank test** replaces the **paired t-test**. Form pair-differences $d = X - Y$, rank the absolute differences (average ranks for ties), then attach the original signs:

Pair	1	2	3	4	5	6	7	8
$d = X - Y$	4	−3	8	7	−1	10	6	7
$\lvert d\rvert$	4	3	8	7	1	10	6	7
rank $\lvert d\rvert$	3	2	7	5.5	1	8	4	5.5
signed	+3	−2	+7	+5.5	−1	+8	+4	+5.5

(The two $|d| = 7$'s tie for ranks 5 and 6, so each gets the average 5.5; no zero differences to discard.) The signed-rank sums are

$$W^+ = 3 + 7 + 5.5 + 8 + 4 + 5.5 = 33,$$
$$W^- = 2 + 1 = 3, \qquad W^+ + W^- = \tfrac{n(n+1)}{2} = 36. \ \blacksquare$$

(b) Small-sample decision: the test statistic is the smaller sum, $W = \min(W^+, W^-) = 3$. Reject H_0 (no difference between routines) when $W \le w_{0.05} = 4$. Here $W = 3 \le 4$.

$$\boxed{W = 3 \le w_{0.05} = 4 \;\Rightarrow\; \text{reject } H_0 \text{ at } \alpha = 0.05;\ \text{the routines differ.}}$$

Since the positive ranks dominate, routine X yields the larger range-of-motion gains.

(c) Large-sample cross-check. With $n = 8$,

$$\begin{aligned}
\mathbb{E}(W^+) &= \frac{n(n+1)}{4} = \frac{8(9)}{4} = 18, \\
\mathrm{Var}(W^+) &= \frac{n(n+1)(2n+1)}{24} = \frac{8(9)(17)}{24} = \frac{1224}{24} = 51, \\
Z &= \frac{W^+ - \mathbb{E}(W^+)}{\sqrt{\mathrm{Var}(W^+)}} = \frac{33 - 18}{\sqrt{51}} = \frac{15}{7.141} = 2.10.
\end{aligned}$$

Against $z_{0.025} = 1.96$, $|2.10| > 1.96$, the same verdict.

$$\boxed{Z = 2.10 > 1.96 \;\Rightarrow\; \text{reject } H_0;\ \text{agrees with the exact test.}}$$

The normal approximation is rough at $n = 8$ but confirms the small-sample table result: X and Y differ significantly.